# The Foreman and the serif family

Handbook of the general definitions, constructions and theorems

2 November 2026

# How this handbook is written

This handbook states the general theory of the Foreman and the serif family: the members of the serif family and their relations; the tables and families they evaluate; the Foreman's classification of a family and his tree; the generalized integrals the Foreman sets; the constructions and compositions of families; the motive of a family and the arithmetic of its values; rays of data and the laws their values obey; the relations among values; and the family read as a model of data. Every definition, construction, theorem and procedure of the program is stated here, once, at the place where it is first needed.

*The order of the book.* The serif family comes first (Part I), because every value the theory produces is closed in it. Then the path of a computation: a list of marks and the tables and families built on it (Part II); the classification of a family by its realizations, its reading, its bunches and the Foreman tree (Part III); the integrals the Foreman procedure sets, and their evaluation along the tree (Part IV); the constructions that build families from cells, holds and bunches, and price them (Part V); the motive of a family and the coaction along its tree, positivity and the gap (Part VI); rays of data, their modules, their two ends and the reading of their values (Part VII); relations among values and the content they live in (Part VIII); and the family as a model of data (Part IX). No statement and no proof uses a statement of a later chapter, and a later chapter is mentioned only as a pointer; inside a chapter a statement may point ahead to a result proved below it, and then says so.

*Proofs.* A statement that is more general than its counterpart in the reference *The Foreman and the serif family* (16 October 2026), or that has no counterpart there, is proved in Part X, one section per chapter in the order of the chapters; the other statements are proved in the reference. In the source file every statement is preceded by a comment that says which: either the label and line of the statement in the reference, or that it is proved in the last part.

*Exercises.* Computations on particular lists, tables and rays are the exercises of Part XI. Each is a guided example with its solution worked in full: every step names the statement or procedure it uses, and every quantity the next step needs is displayed in exact arithmetic.

*Names.* The text uses the program's own names. The classical names of its objects, and the classical names of results that the program reaches by its own route, are collected in the Aliases at the end.

*One governing definition per object.* Each object of the theory is defined once, with all of its slots. Every later appearance of the object is that definition with particular data in its slots, written through the full signature of its governing symbol: a specialization displays the data it chooses, and it never receives a symbol of its own. Thus a Breta on one scale is a Breta with one bunch, a Trailmoment on a line is a Trailmoment whose cut region is a half-line, and a value of interest at particular exponents is the interest with those exponents in its slots.

*Abbreviations are local.* Inside one statement an abbreviation may be declared, once and in words, for data that the statement holds fixed; the declaration displays the full symbol it

abbreviates, and it lapses at the end of the statement.

*Relations are stated at their height.* A relation holds on the values of the slots on which its proof establishes it (Definition 5.6). A hypothesis that restricts a slot is written into the statement that needs it, and nothing outside that statement restricts the object.

*Final answers* are written with exp, log, the stacked Gamma and the serif family (Definition 1.1(vi)).

# Contents

**How this handbook is written** **1**

**I The serif family** **14**

**1 Conventions, the stacked Gamma and the finite part** **15**
1.1 The stacked Gamma . . . . . 16
1.2 The finite part . . . . . 17

**2 The Gramma and the Zamma** **19**
2.1 The Gramma . . . . . 19
2.2 The Zamma . . . . . 21

**3 The Breta** **24**
3.1 Definition and the tower . . . . . 24
3.2 Letters . . . . . 26
3.3 The shape integral . . . . . 27
3.4 Products of shape integrals . . . . . 28
3.5 The Beta port . . . . . 28

**4 Fibre integrals and closing** **31**

**5 The Trailmoment** **34**
5.1 Definition . . . . . 34
5.2 Slots, corners and heights . . . . . 35
5.3 Relations at free slots . . . . . 36
5.4 Lattices . . . . . 38
5.5 The Gamma read by Trailmoments . . . . . 38
5.6 Reflections of the index . . . . . 41
5.7 Joint sources . . . . . 42
5.8 Trails . . . . . 43
5.9 Products of Trailmoments . . . . . 44
5.10 Convolutions of inputs . . . . . 46

**6 Rational words and live inputs** **48**
6.1 Live inputs . . . . . 51

**II Tables and families** **54**

**7 Tables** **55**
7.1 Characters, tables, lists . . . . . 55
7.2 Circuits and molecules . . . . . 55
7.3 The Skeleton . . . . . 56
7.4 Multivectors of any tensor rank . . . . . 58

**8 Families** **60**
8.1 The interest and its walls . . . . . 60
8.2 The native gauge . . . . . 65

**III Classification** **67**

**9 Realizations and the still** **68**
9.1 Realizations . . . . . 68
9.2 The potential and the still . . . . . 69

**10 The reading, classes, atoms** **71**
10.1 Views and the reading . . . . . 71
10.2 Identifying writings . . . . . 72
10.3 What a class holds . . . . . 73

**11 Lean, gauge and label** **74**
11.1 The lean . . . . . 74
11.2 The gauge . . . . . 74
11.3 Answers and the label . . . . . 75
11.4 Maps as certificates . . . . . 76

**12 Restriction to a bunch** **77**
12.1 Patterns and the Taylor cube . . . . . 77
12.2 The Skeleton polynomials of a bunch . . . . . 78
12.3 Levels . . . . . 80
12.4 Shares and the native split . . . . . 83

**13 The order of classes and the Foreman bunch** **84**
13.1 The rises of a class . . . . . 84
13.2 The order . . . . . 85
13.3 What the first stages read . . . . . 86

**14 Peel, children and the Foreman tree** **88**
14.1 The peel and the tree . . . . . 88
14.2 Patterns of one kind . . . . . 89
14.3 What the room keeps . . . . . 90

**IV Evaluation** **93**

**15 The integral at a bunch** **94**
15.1 The scale of a bunch . . . . . 94
15.2 Closing . . . . . 96
15.3 The port . . . . . 99
15.4 Signed letters . . . . . 100

**16 Integrals, expectations and the Foreman procedure** **101**
16.1 Forms, sections, completions . . . . . 101
16.2 Three integrals and their expectations . . . . . 103
16.3 Quasi-integrals as Laurent data . . . . . 107
16.4 The pseudointegral . . . . . 109

**17 The family as one Laplace transform** **114**
17.1 Questions and the law of the tilt . . . . . 114

**18 The Breta law and the port on the balance wall** **119**
18.1 The laws . . . . . 119
18.2 The port on the balance wall . . . . . 122
18.3 Cycles . . . . . 126

**19 Bunches of width one: pooling and cleaning** **128**
19.1 Bunches of width one . . . . . 128
19.2 Pooling . . . . . 130
19.3 Walls glue . . . . . 131
19.4 Full leans . . . . . 131
19.5 Partial leans . . . . . 132
19.6 Closing at a wall and closing at the gauge . . . . . 135
19.7 The scale on the totals . . . . . 141
19.8 Width zero . . . . . 146
19.9 Pooling is a move . . . . . 147
19.10Cleaning a molecule . . . . . 147

**20 Words of a tree** **149**
20.1 The two letters of a table at once . . . . . 149
20.2 The junction in pattern form . . . . . 150
20.3 The word of a tree . . . . . 151
20.4 Closing the tree by draws . . . . . 153
20.5 Devices at the centre . . . . . 157
20.6 The junction under level form . . . . . 159
20.7 The node in one scale . . . . . 160
20.8 The reach read on the law . . . . . 168
20.9 One tree . . . . . 173

**V Construction and composition** **177**

**21 Gluing and what crosses it** **178**
21.1 Gluing along a mark . . . 178
21.2 Splicing hinges in pairs . . . 183
21.3 What crosses a gluing . . . 184

**22 Periods of constructions** **187**
22.1 The core reads its cells . . . 187
22.2 Pinned cells . . . 189
22.3 Chains and transfer . . . 190
22.4 A bunch of any width . . . 191

**23 The law of a cell** **193**
23.1 Cells and their laws . . . 193
23.2 Absorbing a cell . . . 194
23.3 Closures . . . 195
23.4 Cores and their markers . . . 196
23.5 Two cells glued . . . 197
23.6 What a cell retains . . . 200
23.7 Series and parallel nodes . . . 202
23.8 Hinge trees . . . 203

**24 Repetition and the change of query** **207**
24.1 Repetition at a node . . . 207
24.2 Chains . . . 209
24.3 The weight of a hinge . . . 210
24.4 Changing masses and weights . . . 211

**25 Holds and the lean convolution** **213**
25.1 Newton words . . . 213
25.2 Holds and the periods at the full leans . . . 215
25.3 The lean convolution . . . 217
25.4 Holding some marks . . . 218

**26 Periods at the full leans and networks along a hinge tree** **220**
26.1 Molecules and marker loads . . . 220
26.2 The periods of a substitution at its full leans . . . 221
26.3 What a node costs . . . 222
26.4 Blocks aggregate . . . 223
26.5 The network theorem . . . 224
26.6 Nodes in hold form . . . 226
26.7 The procedure and its cost . . . 227
26.8 Changing the query in hold form . . . 228

**27 Finite interfaces and certified enclosures** **231**
27.1 The joint law of the channels and its moments . . . 231
27.2 Repetition is a finite matrix . . . 233
27.3 A certified enclosure of the period . . . 234
27.4 Changing the query with finite data . . . 235
27.5 Certified parameter banks . . . 235
27.6 Procedures . . . 237

**28 Bunches of any width** **239**
28.1 The seam of a bunch . . . 239
28.2 Pieces, their laws, and the glue . . . 242
28.3 The Gramma of a set of pairs . . . 248
28.4 What a piece retains . . . 251
28.5 Storage, and what replaces it . . . 252

**29 Composing bunches** **256**
29.1 Cores, nested pieces and chains of bunches . . . 256
29.2 The procedure . . . 261
29.3 The correspondence . . . 262

**30 Foreman fusion** **264**
30.1 Quotients, lifts and the level cell of a pair . . . 265
30.2 The fused table . . . 265
30.3 The seam of the fusion bunch and the glue . . . 268
30.4 The price of width . . . 271
30.5 Cousins and hidden symmetry . . . 273
30.6 Two-sums and repair . . . 274
30.7 The procedure . . . 276

**31 Closing a glue along a direction** **278**
31.1 Closing along a direction . . . 278
31.2 Closing along a path . . . 279
31.3 Retained crossings as joint words . . . 289
31.3.1 Completing the retained fibre before taking its size . . . 289
31.3.2 Coincident crossings and the boundary at root rate one . . . 296

**32 Laws at their ends: regions of equality and reach** **299**
32.1 Regions on which a law is represented . . . 299
32.2 The reach along a direction . . . 304
32.3 What the calculus of bunches settles . . . 313

**33 Price and complexity** **316**
33.1 Constructions and their price . . . 316

**VI The arithmetic of the output** **319**

**34 The motive of a family and the coaction along a tree** **320**
34.1 The model . . . 320

34.2 Motivic periods . . . 322
34.3 Steps along a tree of bunches . . . 324
34.4 The weights of the coaction . . . 324
34.5 The node . . . 326
34.6 Pinned cells . . . 327
34.7 Rational data . . . 329
34.8 Separation by conjugates . . . 330

**35 Positive words** **331**
35.1 Coefficient jets and basis contrasts . . . 331
35.2 Manufactured tables and their port words . . . 333
35.3 Mark relations with the room integrated . . . 335

**36 The critical-line law** **339**
36.1 Critical-line words . . . 340

**37 The gap and the shape words** **343**
37.1 Clearing and the gap . . . 343
37.2 Shape words with polynomial insertions . . . 343
37.3 Rows . . . 346
37.4 The Beta port ladder and the filters . . . 346
37.5 Growing Beta ports . . . 348

**VII Rays** **350**

**38 Rays, native gauges and tiers** **351**
38.1 The relations along a ray . . . 351
38.2 Native gauges carry solutions . . . 352
38.3 The centre ray of a family . . . 354
38.4 What a wall carries . . . 357
38.5 Corners of a wall of width one . . . 359
38.6 What a value can contain . . . 361

**39 The two ends of a ray** **363**
39.1 The module of a ray . . . 364
39.2 The motive of a ray . . . 366
39.3 Symmetries of a ray . . . 367
39.4 The box of the walls . . . 369
39.5 Sources at the high end . . . 373
39.6 Closings at the low end . . . 376
39.7 Separating the sources and gluing them . . . 379
39.8 The pencil of a ray . . . 381
39.9 The pencil along a hinge tree . . . 383
39.10What the closings see . . . 385
39.11Wheels: the letters as a holonomy . . . 388
39.12The procedure . . . 390

**40 The stride's law and the gauge end** **392**
40.1 The stride's law . . . 392
40.2 The gauge end . . . 394
40.3 The torus value . . . 397

**41 Reading a ray** **399**
41.1 The low end and the slit . . . 399
41.2 Positive selection . . . 401
41.3 Positivity as a moment problem . . . 401
41.4 The procedure . . . 403
41.5 What a reading costs . . . 406

**42 Inputs along a ray** **409**
42.1 Inputs read from a ray . . . 410
42.2 Closing an input . . . 410
42.3 A mark opened at an integer strength . . . 411
42.4 Boundaries and minimality . . . 412
42.5 The two ends paired . . . 413
42.6 The centre ray of $\binom{\mathcal{E}}{1}$ . . . 413
42.7 The centre ray of a substitution . . . 418
42.8 The value of a table as a closed serif word . . . 421
42.9 The class at silence . . . 423
42.10The sources of a substitution of classes . . . 425

**43 Arithmetic along a ray** **427**
43.1 Closing a free share along a ray . . . 427
43.2 The clearing along a ray . . . 429
43.3 The merit of a ray . . . 430
43.4 Laws at the low end and their closings . . . 431
43.5 The hinge closing as a row over the room . . . 432
43.6 Critical primes . . . 433

**VIII Relations and the content** **435**

**44 Relations and separation** **436**
44.1 Words and relations . . . 436
44.2 Relation blocks . . . 437
44.3 Separators . . . 438
44.4 Completeness along a ray . . . 439
44.5 Separating generators . . . 440

**45 The content and its coaction** **441**
45.1 The content . . . 441
45.2 Chains . . . 443
45.3 The cutting rule . . . 443
45.4 The letter of a live input . . . 445
45.5 Separation of the content . . . 446

45.6 The closing in any number of shares . . . 448
45.7 The group of a period . . . 451
45.8 The group of a word along the tree . . . 455

**46 Finite algebra for scale reads** **458**
46.1 Complete relations for linear letters and unit columns . . . 458
46.2 Polynomial letters on one scale . . . 462

**47 The module of a family** **466**
47.1 The data of a family . . . 466
47.2 The module and its shifts . . . 467
47.3 Boxes in all the data . . . 469
47.4 Closings and the elements of value zero . . . 470
47.5 Masters from a box . . . 471
47.6 Value zero on a subspace of the data . . . 472
47.7 The values separate the module . . . 474

**48 The module along the tree** **476**
48.1 Families of positive letters . . . 477
48.2 Share readings . . . 479
48.3 Paths through a chain of bunches . . . 481
48.4 Shares and scales . . . 483
48.5 Trees of nested bunches . . . 485
48.6 What the word along a tree reads . . . 490
48.7 The module of a family from its nodes . . . 492
48.8 The module of a transform from its block . . . 496
48.9 The procedure in all the data . . . 501

**IX The family as inference** **508**

**49 The family as inference** **509**
49.1 Samples of bases . . . 509
49.2 Samples of tilts . . . 512
49.3 Draws, orders and races . . . 518
49.4 The interaction of a bunch . . . 520

**X Proofs of the general forms** **522**

**50 Proofs of the general forms** **523**
50.1 Chapter 1: conventions, the stacked Gamma and the finite part . . . 523
50.2 Chapter 2: the Gramma and the Zamma . . . 526
50.3 Chapter 3: the Breta . . . 529
50.4 Chapter 4: fibre integrals and closing . . . 533
50.5 Chapter 5: the Trailmoment . . . 534
50.6 Chapter 6: rational words and live inputs . . . 538
50.7 Chapter 7: tables . . . 541
50.8 Chapter 8: families . . . 543

50.9 Chapter 9: realizations and the still . . . 548
50.10Chapter 10: the reading, classes, atoms . . . 549
50.11Chapter 11: lean, gauge and label . . . 550
50.12Chapter 12: restriction to a bunch . . . 552
50.13Chapter 13: the order of classes and the Foreman bunch . . . 555
50.14Chapter 14: peel, children and the Foreman tree . . . 556
50.15Chapter 15: the integral at a bunch . . . 558
50.16Chapter 16: integrals, expectations and the Foreman procedure . . . 563
50.17Chapter 17: the family as one Laplace transform . . . 569
50.17.1 What the one-table proofs read, and what a family reads . . . 569
50.17.2 The exponent of a question . . . 569
50.17.3 The two ends of strength . . . 571
50.17.4 How far a question can lean, and the tails . . . 573
50.18Chapter 18: the Breta law and the port on the balance wall . . . 575
50.19Chapter 19: bunches of width one: pooling and cleaning . . . 582
50.20Chapter 20: words of a tree . . . 595
50.21Chapter 21: gluing and what crosses it . . . 608
50.22Chapter 22: periods of constructions . . . 612
50.23Chapter 23: the law of a cell . . . 618
50.24Chapter 24: repetition and the change of query . . . 626
50.25Chapter 25: holds and the lean convolution . . . 629
50.26Chapter 26: periods at the full leans and networks along a hinge tree . . . 636
50.27Chapter 27: finite interfaces and certified enclosures . . . 645
50.28Chapter 28: bunches of any width . . . 647
50.29Chapter 29: composing bunches . . . 658
50.30Chapter 30: Foreman fusion . . . 659
50.31Chapter 31: closing a glue along a direction . . . 671
50.32Chapter 32: laws at their ends: regions of equality and reach . . . 682
50.33Chapter 34: the motive of a family and the coaction along a tree . . . 684
50.34Chapter 35: positive words . . . 700
50.35Chapter 36: the critical-line law . . . 704
50.36Chapter 37: the gap and the shape words . . . 705
50.37Chapter 38: rays, native gauges and tiers . . . 707
50.38Chapter 39: the two ends of a ray . . . 719
50.39Chapter 40: the stride's law and the gauge end . . . 749
50.40Chapter 41: reading a ray . . . 757
50.41Chapter 42: inputs along a ray . . . 767
50.42Chapter 43: arithmetic along a ray . . . 789
50.43Chapter 44: relations and separation . . . 796
50.44Chapter 45: the content and its coaction . . . 798
50.45Chapter 46: finite algebra for scale reads . . . 815
50.46Chapter 47: the module of a family . . . 818
50.47Chapter 48: the module along the tree . . . 828
50.48Chapter 49: the family as inference . . . 868
50.48.1 Samples of bases . . . 868
50.48.2 Statistics from a centre . . . 870
50.48.3 The Grey law, the dial and the sorted draw . . . 871

50.48.4 The interaction of a bunch . . . 873
50.49 Receipts . . . 875
50.49.1 Chapters 35 to 46 . . . 875
50.49.2 Chapters 17 and 49 . . . 875

**XI Exercises** **876**

**51 Exercises on tables, families and classification** **878**
51.1 The list on two marks . . . 878
51.2 The pins of the atom formula . . . 881
51.3 Foreman trees . . . 884
51.4 Skeleton values at width two . . . 889
51.5 The record of a peel . . . 889

**52 Exercises on evaluation and construction** **892**
52.1 One share per station . . . 892
52.2 The width of the marks left in place . . . 895
52.3 Fusing a uniform list . . . 896

**53 Exercises on the motive and the arithmetic of values** **898**
53.1 Uniform lists at their centres . . . 898
53.2 Substitutions of dual pairs . . . 906
53.3 Full leans . . . 908

**54 Exercises on rays: the small centres** **911**
54.1 The centre ray, bracketed . . . 911
54.2 The centre ray by its box . . . 916
54.3 The centre ray from its gauge end . . . 919
54.4 The torus at three marks . . . 921
54.5 A lean of three marks . . . 925
54.6 The balanced directions . . . 927
54.7 The centre of $\binom{[4]}{1}$ . . . 929
54.8 The centre of $\binom{[5]}{1}$ . . . 938
54.9 $W_3$ at $(\mathbf{1}; 0, 2)$ . . . 945
54.10 Positivity against the gauge end . . . 948
54.11 What the closings see . . . 949

**55 Exercises on rays: other lists** **957**
55.1 The list $\mathrm{H}_4$ . . . 957
55.2 The five-mark lists . . . 966
55.3 Larger lists and the cost of a reading . . . 970
55.4 Triangles of classes . . . 987

**56 Exercises on the content** **994**
56.1 Two centre values in the content . . . 994
56.2 The cutting rule on two trails . . . 996
56.3 Separation of the content . . . 999

56.4 The closing of a period in any number of shares . . . . . 1002

**Aliases** **1004**

# Part I

# The serif family

# Chapter 1

# Conventions, the stacked Gamma and the finite part

**Definition 1.1** (Conventions)**.** (i) $\mathcal{E}$ and $B$ are finite nonempty sets and $n = |\mathcal{E}|$. For $x \in \mathbb{R}^{\mathcal{E}}_{>0}$ and $v \in \mathbb{C}^{\mathcal{E}}$, $x^v = \prod_e x_e^{v_e}$ with $x_e^{v_e} = \exp(v_e \log x_e)$. For a vector $v$, $|v| = \sum_e v_e$. $\mathbf{1}$ is the all-ones vector, $\mathbf{1}_S$ the indicator of $S$, $x^S = x^{\mathbf{1}_S}$, $v_S = \sum_{e \in S} v_e$, $v|_S$ the restriction of $v$ to $S$, and $x^{-1} = (x_e^{-1})_e$. For a set $X$ and elements $a, b$, $X - a = X \setminus \{a\}$ and $X + b = X \cup \{b\}$. $\mathbb{N} = \{0, 1, 2, \dots\}$.

(ii) For $\alpha \in \mathbb{R}^{\mathcal{E}}_{>0}$ the *logarithmic coordinates* are $y = \log \alpha$ and the *signed coordinates* are $z = \frac{1}{2} y$, coordinatewise, so $\alpha = e^y = e^{2z}$.

(iii) $\Delta_B = \{u \in \mathbb{R}^B_{>0} : |u| = 1\}$ is the *shape simplex* of $B$, and $d\omega_B(u)$ is Lebesgue measure in the coordinates $(u_e)_{e \neq e_0}$ for a fixed $e_0 \in B$; a change of $e_0$ is affine of determinant $\pm 1$. For $|B| = 1$, $\omega_B$ is the Dirac measure at the point of $\Delta_B$.

(iv) An integral *converges* when it converges absolutely.

(v) $(a)_j = \Gamma(a + j)/\Gamma(a)$ for $j \in \mathbb{N}$; $d_n = \operatorname{lcm}(1, \dots, n)$ with $d_0 = 1$; $e_k$ is the elementary symmetric polynomial of degree $k$; for a coordinate $x$ of the unit interval, $\bar{x} = 1 - x$; $[X]\, Y$ is the coefficient of $X$ in $Y$.

(vi) *Final answers* are written with exp, log, the stacked Gamma and the serif family: the Gramma $\mathsf{G}$ and the Zamma $\mathsf{Z}$ (Chapter 2), the Breta $\mathsf{B}$ (Chapter 3) and the Trailmoment $\mathsf{T}$ (Chapter 5). A value of interest is left in another form only with a proof that it has no such expression. A *word* is an expression built from rationals, the data and the variables by sums, products, powers with rational exponents or exponents among the data, these functions, integrals over simplices, half-lines and fibres, and the iterate $\mathcal{Q}^M$ of an integral operator whose kernel is a word. A finite sequence $(w_1, \dots, w_k)$ is a *string* of length $k$. An *insertion* is a function multiplied into an integrand.

(vii) For masses $\mathbf{m}$, $d\alpha_{\mathbf{m}} := \alpha^{\mathbf{m}-\mathbf{1}} d\alpha$ is a measure on the orthant, and a form $f\, d\alpha_{\mathbf{m}}$ unchanged by $\alpha \mapsto \lambda\alpha$, $f$ homogeneous of degree $-m_{\mathcal{E}}$, is integrated over the rays of the orthant, that is over any section (Chapter 16), as the form $W\, d\alpha_{\mathbf{m}}$ of a family word $W$ (Chapter 8) is on its balance wall; a mark or a set of marks as a subscript, as in $d\alpha_B$, denotes coordinates. The law of a random variable, or of a measure under a map, is its image measure; an identity between members of the serif family is a relation (Chapter 5). An expectation names its random variable and that variable's law, $\mathbb{E}_{U \sim P}[g(U)] = \int g(u)\, P(du)$; inside a statement that has named them once, the subscript may be dropped for the rest of that

statement. The *Gamma law* of index $\nu > 0$ is $\mathrm{Gamma}_\nu(dx) = x^{\nu-1}e^{-x}dx/\Gamma(\nu)$ on $(0,\infty)$, and for $\mathbf{m} \in \mathbb{R}^B_{>0}$, $\mathrm{Gamma}_{\mathbf{m}} = \bigotimes_{e\in B} \mathrm{Gamma}_{m_e}$ is the law on $\mathbb{R}^B_{>0}$ of independent Gamma coordinates of indices $m_e$. The generalized integrals of Chapter 16 are unnormalized, and their expectations divide them by the period of the centre: on a connected family every expectation of a form is taken under the law of its centre, $\mathbb{P}_{\mathrm{c}}$, of the departure that its evaluation sets.

(viii) The *root rate* of terms $a_\eta$ indexed by sizes $\eta \to \infty$, as the terms of a lattice by the real parts of their exponents or the terms of a closing by its steps, is $\limsup |a_\eta|^{1/\eta}$. When the number of terms of size at most $N$ grows at most polynomially in $N$, terms of root rate below one are absolutely summable, and terms of root rate above one do not tend to zero.

## 1.1 The stacked Gamma

**Definition 1.2** (The stacked Gamma)**.** For $\alpha \in \mathbb{C}^p$ and $\beta \in \mathbb{C}^q$,

$$\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\alpha_1,\ \dots,\ \alpha_p\\ \beta_1,\ \dots,\ \beta_q\end{matrix}\right] := \frac{\Gamma(\alpha_1)\cdots\Gamma(\alpha_p)}{\Gamma(\beta_1)\cdots\Gamma(\beta_q)},$$

wherever no $\alpha_i$ is a pole of $\Gamma$; it is a meromorphic function of its entries. The stacked Gamma is *balanced* when $\sum_i \alpha_i = \sum_j \beta_j$.

The subscripts of the bracket record the numbers of its upper and lower entries, so a quotient of Gammas is read off the symbol with no translation. The stacked Gamma $\Gamma_{\substack{|B|\\1}}\!\left[\begin{matrix}\mathbf{m}\\ m_B\end{matrix}\right] = \prod_{e\in B}\Gamma(m_e)/\Gamma(m_B)$ of masses $\mathbf{m} \in \mathbb{C}^B$ over their total $m_B$ is, for $\mathrm{Re}\, m_e > 0$, the integral of $u^{\mathbf{m}-\mathbf{1}}$ over the shape simplex $\Delta_B$ (Theorem 3.9).

**Proposition 1.3** (Relations of the stacked Gamma)**.** *(i)* Concatenation. $\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right]\Gamma_{\substack{p'\\q'}}\!\left[\begin{matrix}\boldsymbol{\alpha}'\\ \boldsymbol{\beta}'\end{matrix}\right] = \Gamma_{\substack{p+p'\\q+q'}}\!\left[\begin{matrix}\boldsymbol{\alpha},\ \boldsymbol{\alpha}'\\ \boldsymbol{\beta},\ \boldsymbol{\beta}'\end{matrix}\right]$.

*(ii)* Cancellation. $\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha},\ c\\ \boldsymbol{\beta},\ c\end{matrix}\right] = \Gamma_{\substack{p-1\\q-1}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right]$.

*(iii)* Shift. $\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\alpha_1+1,\ \boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right] = \alpha_1\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\alpha_1,\ \boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right]$ *and* $\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \beta_1+1,\ \boldsymbol{\beta}\end{matrix}\right] = \beta_1^{-1}\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \beta_1,\ \boldsymbol{\beta}\end{matrix}\right]$.

*(iv)* Residue. *For* $j \in \mathbb{N}$, $\operatorname*{Res}_{\alpha_1=-j}\Gamma_{\substack{p\\q}}\!\left[\begin{matrix}\alpha_1,\ \boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right] = \frac{(-1)^j}{j!}\Gamma_{\substack{p-1\\q}}\!\left[\begin{matrix}\boldsymbol{\alpha}\\ \boldsymbol{\beta}\end{matrix}\right]\Big|_{\alpha_1=-j}$.

*(v)* Reflection. $\Gamma_{\substack{2\\0}}\big[a,\ 1-a\big] = \pi/\sin\pi a$.

*(vi)* Laurent data at a pole. *For* $j \in \mathbb{N}$,

$$\Gamma(-j+\varepsilon) = \frac{(-1)^j}{j!}\Big(\frac{1}{\varepsilon} + (\log\Gamma)'(j+1) + O(\varepsilon)\Big), \qquad (\log\Gamma)'(\nu+1) = (\log\Gamma)'(\nu) + \frac{1}{\nu}.$$

*Remark* 1.4 (Balance is a property of the written stacked Gamma). Balance is not a property of the value of a stacked Gamma: the shift (iii) and the reflection (v) change the sums of the entries,

multiplying the value by an algebraic number when the entries are rational. For rational $a \notin \mathbb{Z}$, (v) at $a$ and at $\frac{1}{2}$, with $\Gamma(1) = 1$, gives

$$\Gamma_{\substack{2\\0}}\big[a,\ 1-a\big] = \frac{1}{\sin \pi a}\,\Gamma_{\substack{2\\1}}\left[\begin{matrix}\frac{1}{2},\ \frac{1}{2}\\ 1\end{matrix}\right],$$

the left side with the sums 1 and 0, the right side balanced, and $\sin \pi a$ algebraic.

## 1.2 The finite part

**Definition 1.5** (The finite part)**.** A function $h$ on $(0,\infty)$ has a *log-power expansion* at 0 if for every $N$ there are finitely many coefficients $c_{k,j}$, with exponents $k \in \mathbb{C}$ of real part below $N$ and $j \in \mathbb{N}$, such that $h(x) - \sum c_{k,j} x^k \log^j x = O(x^N)$ as $x \to 0$; at $\infty$ the same holds with $x^{-k}$ and $O(x^{-N})$. The coefficients are determined by $h$. Then $\operatorname{Reg}_{x\to 0} h$ and $\operatorname{Reg}_{x\to\infty} h$ are the coefficients of $x^0 \log^0 x$ in these expansions, zero if absent. If $f$ has a primitive $h$ with log-power expansions at both ends,

$$\operatorname{Reg}\int_0^\infty f\,dx := \operatorname{Reg}_{x\to\infty} h - \operatorname{Reg}_{x\to 0} h,$$

which does not depend on the primitive; on $(0,b]$, $\operatorname{Reg}\int_0^b f := h(b) - \operatorname{Reg}_{x\to 0} h$. A sum is regularized the same way, $\operatorname{Reg}_{N\to\infty}$ of its partial sums. The coordinate is part of the data of a finite part, and for iterated finite parts so is the order.

**Proposition 1.6** (A convergent integral is its own finite part)**.** *If $\int_0^\infty f\,dx$ converges and $f$ has a primitive with log-power expansions at both ends, then $\operatorname{Reg}\int_0^\infty f\,dx = \int_0^\infty f\,dx$.*

**Theorem 1.7** (A finite part is a constant term)**.** *If $f$ has log-power expansions at 0 and at $\infty$, then*

$$\operatorname{Reg}\int_0^\infty f\,dx = \operatorname*{CT}_{\varepsilon=0}\int_0^\infty x^\varepsilon f(x)\,dx,$$

*the right side continued meromorphically in $\varepsilon$ from $\int_0^1$ and from $\int_1^\infty$ separately. Termwise, $x^b \log^j x$ in the expansion at 0 contributes $\int_0^1 x^{b+\varepsilon}\log^j x\,dx = (-1)^j j!/(b+1+\varepsilon)^{j+1}$, whose constant term is $(-1)^j j!/(b+1)^{j+1}$ for $b \neq -1$ and 0 for $b = -1$; the end at $\infty$ is the same after $x \mapsto 1/x$.*

**Corollary 1.8** (Finite parts at a pole of a Mellin integral)**.** *Let $f$ have a log-power expansion at 0 and be integrable on $[\delta,1]$ for every $\delta > 0$, and let $I(a) = \int_0^1 x^{a-1} f(x)\,dx$, continued meromorphically. Then $\operatorname{Reg}\int_0^1 x^{a-1} f(x)\,dx = \operatorname{CT}_{a'=a} I(a')$ for every $a \in \mathbb{C}$: it is $I(a)$ at every $a$ that is not a pole, and the poles lie at the points $a = -k$ for the exponents $k$ of the expansion, a term $x^k \log^j x$ giving a pole of order at most $j+1$.*

**Lemma 1.9** (A finite part depends on the coordinate)**.** *Let $h$ have a log-power expansion at $\infty$, and let $a_j$ be its coefficient of $x^0 \log^j x$, $j \in \mathbb{N}$. For $c,p > 0$, $h(cx^p)$ has a log-power expansion at $\infty$, and*

$$\operatorname{Reg}_{x\to\infty} h(cx^p) = \sum_{j\geq 0} a_j\,(\log c)^j;$$

*the same holds at 0. Consequently, if $f$ has log-power expansions at 0 and at $\infty$, with coefficients $f_j^0$ and $f_j^\infty$ of $x^{-1}\log^j x$, then for $c,p > 0$*

$$\operatorname{Reg}\int_0^\infty f(cx^p)\,c\,p\,x^{p-1}\,dx = \operatorname{Reg}\int_0^\infty f(x)\,dx + \sum_{j\geq 0}\frac{(\log c)^{j+1}}{j+1}\big(f_j^\infty - f_j^0\big)\,:$$

*a power of the coordinate leaves a finite part unchanged, and a dilation moves it by a polynomial in* $\log c$ *read from the terms* $x^{-1}\log^j x$ *of the integrand at its two ends.*

A finite part is the difference of the regularized limits of one primitive at the two ends, not the integral of a measure. It is linear, it moves with the coordinate (Lemma 1.9), and iterated in several variables it moves with the order. The generalized integrals of Chapter 16 are built from it: a quasi-integral is an iterated finite part along a chart and an order, and the pseudointegral is the value of the native completion (Definition 16.29).

# Chapter 2

# The Gramma and the Zamma

## 2.1 The Gramma

**Definition 2.1** (Gramma)**.** Let $\mathcal{E}$ be a finite set of *scales*. A *configuration* on $\mathcal{E}$ is a finite family $t = (t_1, \dots, t_N)$ of *exponents* $t_k \in \mathbb{R}^{\mathcal{E}}$; $t$ also denotes the $\mathcal{E} \times N$ matrix with these columns. For *coefficients* $\boldsymbol{\xi} \in (0,\infty)^N$ and an *index* $\nu \in \mathbb{C}^{\mathcal{E}}$, the *Gramma* is

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) := \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\nu-\mathbf{1}} \exp\Big( -\sum_{k=1}^{N} \xi_k\, x^{t_k} \Big)\, dx$$

wherever the integral converges. Put $\operatorname{cone}(t) = \{\sum_k c_k t_k :\ c \in [0,\infty)^N\}$ and $\mathcal{T}_t = \{\nu : \operatorname{Re}\nu \in \operatorname{int}\operatorname{cone}(t)\}$. In $y = \log x$,

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \int_{\mathbb{R}^{\mathcal{E}}} e^{\langle \nu, y\rangle} \exp\Big( -\sum_{k} \xi_k e^{\langle t_k, y\rangle} \Big)\, dy,$$

the moment generating function, in the index, of the density $\exp\big(-\sum_k \xi_k e^{\langle t_k,y\rangle}\big)$. Exponents and coefficients combine by concatenation, $t \oplus t'$ and $\boldsymbol{\xi} \oplus \boldsymbol{\xi}'$ (configurations on two sets of scales extended by zero to their union), and a shape $u$ dresses coefficients by $u^t\boldsymbol{\xi} := (u^{t_k}\xi_k)_k$. The *unit configuration* on $\mathcal{E}$ is $I_{\mathcal{E}} = (\mathbf{e}_e)_{e\in\mathcal{E}}$, one unit column per scale, and on one scale the configuration of the single exponent 1 is written (1).

The three slots of the Gramma (Definition 5.6) are its index, its configuration and its coefficients. Every structural relation of the Gramma is a statement about how these slots move: the index moves along the exponents, the coefficients are rescaled by the torus of the scales, and the configuration passes to its facets at its walls.

**Theorem 2.2** (Domain)**.** $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ *converges if and only if* $\nu \in \mathcal{T}_t$*, and it is holomorphic on* $\mathcal{T}_t$*. If* $\operatorname{cone}(t) = \mathbb{R}^{\mathcal{E}}$ *it is entire; if* $\operatorname{cone}(t)$ *lies in a hyperplane it converges nowhere.*

**Proposition 2.3** (Affine changes of the logarithmic coordinates)**.** *For* $A \in GL_{\mathcal{E}}(\mathbb{R})$ *and* $\mathbf{h} \in \mathbb{R}^{\mathcal{E}}$*:* $\mathcal{T}_{A^{\mathsf{T}}t} = A^{\mathsf{T}}\mathcal{T}_t$*, and for* $\nu \in \mathcal{T}_t$

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = |\det A|\, e^{\langle \nu, \mathbf{h}\rangle}\, \mathsf{G}_{A^{\mathsf{T}}\nu;A^{\mathsf{T}}t}\big((e^{\langle t_k,\mathbf{h}\rangle}\xi_k)_k\big).$$

*At* $A = -I_{\mathcal{E}}$ *and* $\mathbf{h} = 0$ *this is the reflection,* $\mathcal{T}_{-t} = -\mathcal{T}_t$ *and* $\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \mathsf{G}_{-\nu;-t}(\boldsymbol{\xi})$*.*

**Theorem 2.4** (Relations of the Gramma)**.** *Let* $\nu \in \mathcal{T}_t$*; then* $\nu + t_k \in \mathcal{T}_t$ *for every* $k$*.*

*(i)* Ladder. $\partial_{\xi_k}\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = -\mathsf{G}_{\nu+t_k;t}(\boldsymbol{\xi})$*; for* $\mathbf{w}\in\mathbb{N}^N$, $\partial^{\mathbf{w}}\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = (-1)^{|\mathbf{w}|}\mathsf{G}_{\nu+t\mathbf{w};t}(\boldsymbol{\xi})$.

*(ii)* Exchange, one relation per scale. *For each* $e\in\mathcal{E}$: $\nu_e\,\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \sum_k t_{k,e}\,\xi_k\,\mathsf{G}_{\nu+t_k;t}(\boldsymbol{\xi})$.

*(iii)* Torus homogeneity. *For* $\lambda\in\mathbb{R}^{\mathcal{E}}_{>0}$: $\mathsf{G}_{\nu;t}\big((\lambda^{t_k}\xi_k)_k\big) = \lambda^{-\nu}\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ *(Proposition 2.3 at* $A = I_{\mathcal{E}}$, $\mathbf{h} = \log\lambda$*).*

*(iv)* Toric relation. *For every* $\mathbf{v}\in\mathbb{Z}^N$ *with* $t\mathbf{v} = 0$*, written* $\mathbf{v} = \mathbf{v}^+ - \mathbf{v}^-$ *with* $\mathbf{v}^{\pm}\in\mathbb{N}^N$ *of disjoint supports,*

$$\partial^{\mathbf{v}^+}\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = (-1)^{|\mathbf{v}^+|-|\mathbf{v}^-|}\,\partial^{\mathbf{v}^-}\mathsf{G}_{\nu;t}(\boldsymbol{\xi}).$$

The ladder moves the index along one exponent; the exchange relation, one for each scale, is linear in the Grammas at the index and at its $N$ neighbours $\nu + t_k$; the toric relation records the linear dependencies among the exponents themselves. On one scale with integer exponents the exchange relation is a recurrence on the indices $\nu + \mathbb{Z}$ linking the indices from $\nu + \min(0, \min_k t_k)$ to $\nu + \max(0, \max_k t_k)$.

**Theorem 2.5** (Product)**.** *If* $\mathcal{E} = \mathcal{E}_1 \sqcup \cdots \sqcup \mathcal{E}_q$ *and each* $t_k$ *is supported in one* $\mathcal{E}_i$ *(zero vectors assigned to* $\mathcal{E}_1$*), then* $\mathcal{T}_t = \mathcal{T}_{t^{(1)}} \times \cdots \times \mathcal{T}_{t^{(q)}}$ *and*

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \prod_{i=1}^{q} \mathsf{G}_{\nu|_{\mathcal{E}_i};t^{(i)}}(\boldsymbol{\xi}^{(i)}),$$

*with* $t^{(i)}$ *the exponents supported in* $\mathcal{E}_i$ *and* $\boldsymbol{\xi}^{(i)}$ *their coefficients.*

**Corollary 2.6** (Square configurations)**.** *Let* $N = |\mathcal{E}|$ *and let the matrix* $t$ *be invertible, and put* $\mathbf{b} = t^{-1}\nu$*. Then* $\mathcal{T}_t = \{\nu : \operatorname{Re}\mathbf{b}\in(0,\infty)^N\}$ *and*

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = |\det t|^{-1}\prod_{k=1}^{N}\Gamma(b_k)\,\xi_k^{-b_k}.$$

*In particular* $\mathsf{G}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi}) = \prod_e \Gamma(\nu_e)\,\xi_e^{-\nu_e}$*; on one scale* $\mathsf{G}_{\nu;(t_1)}(\xi) = t_1^{-1}\Gamma(\nu/t_1)\,\xi^{-\nu/t_1}$ *for* $t_1 > 0$*; and* $\Gamma(\nu) = \mathsf{G}_{\nu;(1)}(1)$*, whose exchange relation is* $\nu\Gamma(\nu) = \Gamma(\nu+1)$*.*

The Gamma function is thus the Gramma of one scale and one unit exponent, and at every invertible square configuration the Gramma is a stacked Gamma times powers, divided by $|\det t|$. A configuration with more exponents than scales is where the Gramma becomes more than a stacked Gamma times powers: the difference $N - |\mathcal{E}|$ is the dimension of the share side of Proposition 4.2.

**Lemma 2.7** (Scale and shape)**.** *The map* $(0,\infty)\times\Delta_B\to\mathbb{R}^B_{>0}$, $(\rho,u)\mapsto\rho u$*, is a bijection with inverse* $x\mapsto(|x|, x/|x|)$*, and* $dx = \rho^{|B|-1}\,d\rho\,d\omega_B(u)$*. Consequently* $(\rho u)^{\pi} = \rho^{\langle\pi,\mathbf{1}_B\rangle}u^{\pi}$ *for* $\pi\in\mathbb{R}^B$*, and* $x^{\mathbf{m}-\mathbf{1}}dx = \rho^{m_B-1}u^{\mathbf{m}-\mathbf{1}}\,d\rho\,d\omega_B(u)$ *for* $\mathbf{m}\in\mathbb{C}^B$*.*

**Theorem 2.8** (Scale and shape along sets)**.** *Let* $B_1,\dots,B_q$ *be pairwise disjoint nonempty subsets of* $\mathcal{E}$*,* $R = \mathcal{E}\setminus(B_1\cup\cdots\cup B_q)$ *and* $\nu\in\mathcal{T}_t$*. With the* levels $\ell_{k,i} = \langle t_k, \mathbf{1}_{B_i}\rangle$*, the configuration* $\ell = (\ell_1,\dots,\ell_N)$ *on* $q$ *scales and the coefficients* $\mathbf{b}(u,x_R) = \big(\xi_k\prod_i u_i^{t_k|_{B_i}}x_R^{t_k|_R}\big)_k$ *for* $u = (u_i)_i\in\prod_i\Delta_{B_i}$*,*

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \int_{\mathbb{R}^R_{>0}} x_R^{\nu|_R-\mathbf{1}}\int_{\prod_i\Delta_{B_i}}\prod_i u_i^{\nu|_{B_i}-\mathbf{1}}\;\mathsf{G}_{(\nu_{B_i})_i;\ell}\big(\mathbf{b}(u,x_R)\big)\,d\omega(u)\,dx_R,$$

*with* $d\omega = \prod_i d\omega_{B_i}$*, the inner Gramma, on* $q$ *scales at the index* $(\nu_{B_i})_i$*,* $\nu_{B_i} = \sum_{e\in B_i}\nu_e$*, converging for almost every* $(u,x_R)$*; when* $R$ *is empty there is no outer integral.*

Passing to scale and shape on disjoint sets of scales replaces each set by one scale, its total, carrying the levels of the exponents on the set; what is left of the exponents dresses the coefficients by the shapes. The Breta of Chapter 3 is this passage made into a symbol.

**Theorem 2.9** (The walls of a Gramma)**.** *Let $\mathcal{E} = \mathcal{E}' \sqcup \{e\}$, write $t_k = (t'_k, t_{k,e})$ and $\nu = (\nu', \nu_e)$, and suppose $t_{k,e} \geq 0$ for every $k$, with some $t_{k,e} > 0$. Let $H = \{k : t_{k,e} = 0\}$ be the* facet terms*, suppose* $\operatorname{cone}(t'_H)$ *has nonempty interior in $\mathbb{R}^{\mathcal{E}'}$ and contains $t'_k$ for every $k \notin H$, and let $\operatorname{Re}\nu' \in \operatorname{int}\operatorname{cone}(t'_H)$. Then $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ converges for $0 < \operatorname{Re}\nu_e < c$, for some $c \in (0, \infty]$, and extends to a meromorphic function of $\nu_e \in \mathbb{C}$ with at most simple poles at $\nu_e = -p$, $p \in P = \{\sum_{k\notin H} n_k t_{k,e} : \mathbf{n} \in \mathbb{N}^{H^c}\}$, where*

$$\operatorname*{Res}_{\nu_e=-p} \mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \sum_{\mathbf{n}\in\mathbb{N}^{H^c}:\ \sum_k n_k t_{k,e} = p} \prod_{k\notin H} \frac{(-\xi_k)^{n_k}}{n_k!}\ \mathsf{G}_{\nu' + \sum_{k\notin H} n_k t'_k;\ t'_H}(\boldsymbol{\xi}_H).$$

*At $p = 0$ the residue is the Gramma of the facet terms: at its first wall a Gramma factors onto its facet. A linear change of the logarithmic coordinates, $y = Ay'$ with $A \in GL_{\mathcal{E}}(\mathbb{R})$, gives $\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = |\det A|\, \mathsf{G}_{A^{\mathsf{T}}\nu;A^{\mathsf{T}}t}(\boldsymbol{\xi})$ (Proposition 2.3 at $\mathbf{h} = 0$), and with the last column of $A$ the inward normal of a facet of* $\operatorname{cone}(t)$ *it puts that facet in the position of $H$.*

The poles of a Gramma sit on the walls of its cone, and each residue is a Gramma of the facet, of one scale fewer, at an index moved by the exponents that leave the facet. On one scale with positive exponents this is the continuation of the Gramma to the whole index line, with at most simple poles at the points $-p$, $p = \langle \mathbf{n}, t\rangle$ for $\mathbf{n} \in \mathbb{N}^N$, the residue at $-p$ being $\sum_{\langle \mathbf{n},t\rangle = p} \prod_k (-\xi_k)^{n_k}/n_k!$.

**Proposition 2.10** (The expansion lattice)**.** *Let $S$ be a set of terms such that the integral with the signs of those terms flipped converges,*

$$\int_{\mathbb{R}^{\mathcal{E}}} \exp\Big(\langle \operatorname{Re}\nu, y\rangle + \sum_{k\in S} \xi_k e^{\langle t_k, y\rangle} - \sum_{k\notin S} \xi_k e^{\langle t_k, y\rangle}\Big) dy < \infty.$$

*Then, absolutely,*

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \sum_{\mathbf{n}\in\mathbb{N}^S} \prod_{k\in S} \frac{(-\xi_k)^{n_k}}{n_k!}\ \mathsf{G}_{\nu + \sum_{k\in S} n_k t_k;\ t_{S^c}}(\boldsymbol{\xi}_{S^c}).$$

**Theorem 2.11** (The decay polytope)**.** *Let $\nu_0 \in \operatorname{int}\operatorname{cone}(t)$ be real, $\eta \in \mathbb{R}^{\mathcal{E}}$, and $D_t = \{y \in \mathbb{R}^{\mathcal{E}} : |\langle t_k, y\rangle| < \frac{\pi}{2}\ \forall k\}$. For every $y \in D_t$,*

$$\big|\mathsf{G}_{\nu_0 + i\eta;t}(\boldsymbol{\xi})\big| \leq e^{-\langle \eta, y\rangle}\ \mathsf{G}_{\nu_0;t}\big((\xi_k \cos\langle t_k, y\rangle)_k\big).$$

*Consequently, with $h_t(\eta) = \sup_{y\in D_t}\langle \eta, y\rangle$, for every $\varepsilon \in (0,1)$*

$$\big|\mathsf{G}_{\nu_0 + i\eta;t}(\boldsymbol{\xi})\big| \leq \mathsf{G}_{\nu_0;t}\big(\sin(\tfrac{\pi\varepsilon}{2})\boldsymbol{\xi}\big)\, e^{-(1-\varepsilon)h_t(\eta)},$$

*and $h_t$ is a norm when the $t_k$ span $\mathbb{R}^{\mathcal{E}}$.*

## 2.2 The Zamma

**Definition 2.12** (Zamma)**.** Wherever $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ converges and no $\nu_e$ lies in $-\mathbb{N}$, the *Zamma* is the Gramma divided by the Gammas of its own index,

$$\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}) := \frac{\mathsf{G}_{\nu;t}(\boldsymbol{\xi})}{\prod_{e\in\mathcal{E}} \Gamma(\nu_e)}.$$

At the unit configuration it is $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi}) := \prod_e \xi_e^{-\nu_e}$ for every $\nu \in \mathbb{C}^{\mathcal{E}}$; where $\operatorname{Re}\nu_e > 0$ for every $e$ this is the quotient $\mathsf{G}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi})/\prod_e \Gamma(\nu_e)$. The Zamma carries the continuations of the Gramma wherever the Gramma has them.

The Zamma divides out of the Gramma what is first order in its index, the Gammas that the unit configuration produces on its own; what remains is the content of the configuration. It has the three slots of the Gramma.

**Proposition 2.13** (First facts about the Zamma)**.** *(i)* First order. *At the unit configuration the Zamma is the monomial of Definition 2.12; at an invertible square configuration $t$, $N = |\mathcal{E}|$, with $\mathbf{b} = t^{-1}\nu$,*

$$\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}) = |\det t|^{-1}\,\Gamma_{\!N}\!\begin{bmatrix}\mathbf{b}\\ \nu\end{bmatrix}\prod_k \xi_k^{-b_k}.$$

*(ii)* A power of a sum is a Zamma. *With $N$ unit exponents on one scale, $\mathsf{Z}_{\nu;(1,\dots,1)}(\boldsymbol{\xi}) = \big(\sum_k \xi_k\big)^{-\nu}$.*

*(iii)* Relations. *Where the terms converge, with quotients of Gammas in place of the factors the index moves,*

$$\partial_{\xi_k}\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}) = -\prod_e \Gamma_1\!\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e\end{bmatrix}\mathsf{Z}_{\nu+t_k;t}(\boldsymbol{\xi}),$$

$$\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}) = \sum_k t_{k,e}\,\xi_k\,\Gamma_1\!\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e + 1\end{bmatrix}\prod_{f\neq e}\Gamma_1\!\begin{bmatrix}\nu_f + t_{k,f}\\ \nu_f\end{bmatrix}\mathsf{Z}_{\nu+t_k;t}(\boldsymbol{\xi}) \qquad (e \in \mathcal{E}).$$

*The other relations of Theorem 2.4 are read with the same normalization, and identities persist on the meromorphic continuations.*

*(iv)* As an expectation. *If a configuration on $\mathcal{E}$ contains the unit configuration $I_{\mathcal{E}}$ with unit coefficients, then for real $\nu \in \mathbb{R}^{\mathcal{E}}_{>0}$*

$$\mathsf{Z}_{\nu;I_{\mathcal{E}}\oplus t'}(\mathbf{1}\oplus\boldsymbol{\xi}') = \mathbb{E}_{U\sim\mathrm{Gamma}_\nu}\Big[\exp\Big(-\sum_k \xi'_k U^{t'_k}\Big)\Big];$$

*it lies in $(0,1]$, decreases in every coefficient, and, where every $\nu + t'_k$ lies in $\mathbb{R}^{\mathcal{E}}_{>0}$,*

$$\mathsf{Z}_{\nu;I_{\mathcal{E}}\oplus t'}(\mathbf{1}\oplus\boldsymbol{\xi}') \;\geq\; \exp\Big(-\sum_k \xi'_k \prod_{e\in\mathcal{E}}\Gamma_1\!\begin{bmatrix}\nu_e + t'_{k,e}\\ \nu_e\end{bmatrix}\Big).$$

**Proposition 2.14** (Powers of a polynomial in the scales)**.** *Let $t$ be a configuration and $\nu \in \mathcal{T}_t$ with no $\nu_e$ in $-\mathbb{N}$, or $t = I_{\mathcal{E}}$ and $\nu \in \mathbb{C}^{\mathcal{E}}$; let $\xi_k(\rho) = \sum_{l\in\mathbb{N}^d}\rho^l\xi_k^{(l)}$ be polynomials in $\rho \in \mathbb{R}^d$ with real coefficients and $\boldsymbol{\xi}^{(0)} \in (0,\infty)^N$, and let $c_q^\nu = [\rho^q]\,\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}(\rho))$ be the Taylor coefficients at $\rho = 0$. At the unit configuration, where $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi}) = \boldsymbol{\xi}^{-\nu}$ is read with the principal branch near $\boldsymbol{\xi}^{(0)}$, the coefficients $\xi_k^{(l)}$ with $l \neq 0$ may be complex and $\rho \in \mathbb{C}^d$.*

*(i)* $c_q^\nu = \sum_{\mathbf{n}}(-1)^{|\mathbf{n}|}\prod_{e\in\mathcal{E}}\Gamma_1\!\begin{bmatrix}\nu_e + (t\mathbf{w})_e\\ \nu_e\end{bmatrix}\mathsf{Z}_{\nu+t\mathbf{w};t}(\boldsymbol{\xi}^{(0)})\prod_{l\neq0,\,k}\frac{(\xi_k^{(l)})^{n_{l,k}}}{n_{l,k}!}$, *the sum over the arrays $\mathbf{n} = (n_{l,k})_{l\neq0,\,k}$ in $\mathbb{N}$ with $\sum_{l,k} l\,n_{l,k} = q$, where $\mathbf{w} = (\sum_l n_{l,k})_k$; at the unit configuration the factor $\prod_e\Gamma_1\!\begin{bmatrix}\nu_e + (t\mathbf{w})_e\\ \nu_e\end{bmatrix}$ is the polynomial $\prod_e(\nu_e)_{w_e}$, and away from it each product $\prod_e\Gamma_1\!\begin{bmatrix}\nu_e + (t\mathbf{w})_e\\ \nu_e\end{bmatrix}\mathsf{Z}_{\nu+t\mathbf{w};t}(\boldsymbol{\xi}^{(0)})$ is $\mathsf{G}_{\nu+t\mathbf{w};t}(\boldsymbol{\xi}^{(0)})/\prod_e\Gamma(\nu_e)$.*

*(ii)* *At one unit exponent, $t = (1)$, with $\boldsymbol{\xi}(\rho) = L(\rho)$ and the index $\lambda$: for every $i$, $\sum_l L^{(l)}\,(q_i - l_i + 1 + \lambda l_i)\,c^{\lambda}_{q-l+\mathbf{e}_i} = 0$, coefficients with a negative index being $0$.*

*(iii)* $\partial_{\xi_k^{(l)}} c^{\nu}_q = -\prod_e \Gamma_1\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e\end{bmatrix} c^{\nu+t_k}_{q-l}$, *read in the same way.*

*(iv)* *For a configuration $t$, an index $\nu \in \mathbb{C}^{\mathcal{E}}$ and coefficients $\xi_k(\rho)$ depending smoothly on $\rho$ (Proposition 2.13(iii) and the chain rule),*

$$\rho_i \partial_{\rho_i} \mathsf{Z}_{\nu;t}(\boldsymbol{\xi}(\rho)) = -\sum_k \prod_e \Gamma_1\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e\end{bmatrix} \big(\rho_i \partial_{\rho_i} \xi_k\big) \mathsf{Z}_{\nu+t_k;t}(\boldsymbol{\xi}(\rho)).$$

# Chapter 3

# The Breta

## 3.1 Definition and the tower

**Definition 3.1** (Breta)**.** The data of a Breta are:

- a finite set $\mathcal{E}$ of scales and a partition $\mathcal{P}$ of $\mathcal{E}$ into *bunches* $Q$; each bunch is read in scale and shape, $x_Q = \rho_Q u_Q$ with $\rho_Q > 0$ and $u_Q \in \Delta_Q$;
- a *shape index* $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ and a *scale index* $\boldsymbol{\nu} = (\nu_Q)_{Q\in\mathcal{P}}$;
- terms $k = 1, \dots, N$, each with a *shape exponent* $\pi_k \in \mathbb{R}^{\mathcal{E}}$, a *level* $\ell_k = (\ell_{k,Q})_Q \in \mathbb{R}^{\mathcal{P}}$ and a coefficient $\xi_k > 0$.

With $u^{\pi_k} = \prod_Q u_Q^{\pi_k|_Q}$ and $d\omega = \prod_Q d\omega_Q$, the *Breta* is

$$\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) := \int_{\prod_Q \Delta_Q} u^{\mathbf{m}-\mathbf{1}}\ \mathsf{Z}_{\boldsymbol{\nu};\ell}\big((u^{\pi_k}\xi_k)_k\big)\, d\omega(u),$$

the Zamma on the bunch scales with configuration $\ell$, wherever the integral converges. The data are *homogeneous* when every level is the degree of its shape exponent on each bunch, $\ell_{k,Q} = \langle \pi_k, \mathbf{1}_Q\rangle$; then the shape exponents form a configuration $t = (\pi_k)_k$ on $\mathcal{E}$ and the Breta is written $\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};t}(\boldsymbol{\xi})$. When $\mathcal{P} = \{B\}$ has one bunch, the scale index has one entry and is written as a number, $\mathsf{B}^{\{B\}}_{\mathbf{m},\nu;t}(\boldsymbol{\xi})$.

The Breta has five slots: the partition, the shape index, the scale index, the terms and the coefficients. It is the Gramma read in scale and shape on every bunch at once (Theorem 2.8), with the scale integrals closed into a Zamma and the shapes left as an integral; the scale index is free, and for homogeneous data fixing it at the bunch totals of the shape index, $\nu_Q = m_Q$, makes the Breta the Gramma itself up to Gammas.

**Theorem 3.2** (The tower)**.** *Let the data be homogeneous and* $\mathbf{m} \in \mathcal{T}_t$*. For every partition* $\mathcal{P}$ *of the scales,*

$$\mathsf{G}_{\mathbf{m};t}(\boldsymbol{\xi}) = \prod_{Q\in\mathcal{P}} \Gamma(m_Q)\ \mathsf{B}^{\mathcal{P}}_{\mathbf{m},(m_Q)_Q;t}(\boldsymbol{\xi}).$$

*With the finest partition the Breta is* $\mathsf{Z}_{\mathbf{m};t}(\boldsymbol{\xi})$*; with the coarsest,* $\Gamma(m_{\mathcal{E}})$ *times the Breta is the integral of Theorem 2.8 with* $R = \varnothing$*. At the unit configuration* $I_B$ *of a bunch* $B$ *and the coarsest partition* $\{B\}$*, the Zamma inside is* $\big(\sum_e \xi_e u_e\big)^{-m_B}$*, and*

$$\prod_{e\in B} \Gamma(m_e)\, \xi_e^{-m_e} = \Gamma(m_B)\, \mathsf{B}^{\{B\}}_{\mathbf{m},m_B;I_B}(\boldsymbol{\xi}) \qquad (\operatorname{Re} m_e > 0).$$

**Corollary 3.3** (Coarsening)**.** *Let the data be homogeneous, and let $\mathcal{P}$ refine $\mathcal{Q}$. Then*

$$\mathsf{B}^{\mathcal{Q}}_{\mathbf{m},(m_Q)_Q;t}(\boldsymbol{\xi}) = \prod_{Q\in\mathcal{Q}} \Gamma_{1}^{k_Q}\begin{bmatrix}(m_P)_{P\subseteq Q}\\ m_Q\end{bmatrix}\ \mathsf{B}^{\mathcal{P}}_{\mathbf{m},(m_P)_P;t}(\boldsymbol{\xi}),$$

*with $k_Q$ the number of bunches of $\mathcal{P}$ inside $Q$.*

**Proposition 3.4** (The Breta on the orthant)**.** *For any data,*

$$\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = \frac{1}{\prod_Q \Gamma(\nu_Q)} \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\mathbf{m}-\mathbf{1}} \prod_Q |x_Q|^{\nu_Q - m_Q} \exp\Big(-\sum_k \xi_k\, x^{\pi_k} \prod_Q |x_Q|^{\ell_{k,Q}-\langle \pi_k,\mathbf{1}_Q\rangle}\Big)\, dx,$$

*either side converging when the other does: a free scale index is one more power of each bunch's total, a level that is not the degree of its shape exponent on a bunch is one more power of that bunch's total inside its term, and for homogeneous data at $\nu_Q = m_Q$ this is Theorem 3.2.*

**Theorem 3.5** (Relations of the Breta)**.** *Each identity holds wherever every Breta in it converges. The terms are $(\pi,\ell)$; homogeneous data have $t_k = \pi_k$ and $\ell_{k,Q} = \langle t_k, \mathbf{1}_Q\rangle$. In this statement write $g_k = \prod_Q \Gamma_1\begin{bmatrix}\nu_Q+\ell_{k,Q}\\ \nu_Q\end{bmatrix}$.*

*(i)* Shape split*, per bunch $Q$:*

$$\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = \sum_{e\in Q} \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_e,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}).$$

*(ii)* Ladder*, per term $k$:*

$$\partial_{\xi_k} \mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = -g_k\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi}).$$

*(iii)* Scale exchange*, per bunch $Q$:*

$$\mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = \sum_k \ell_{k,Q}\,\xi_k\,\Gamma_1\begin{bmatrix}\nu_Q+\ell_{k,Q}\\ \nu_Q+1\end{bmatrix} \prod_{Q'\neq Q} \Gamma_1\begin{bmatrix}\nu_{Q'}+\ell_{k,Q'}\\ \nu_{Q'}\end{bmatrix}\ \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi}).$$

*(iv)* Shape exchange*, for scales $e\neq f$ of one bunch, with $\operatorname{Re} m_e, \operatorname{Re} m_f > 0$:*

$$m_e\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_f,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) - m_f\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_e,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) = \sum_k \xi_k\, g_k \Big[\pi_{k,e}\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_f,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi}) - \pi_{k,f}\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_e,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})\Big].$$

*(v)* Shape exchange on the orthant*, for scales $e\neq f$ of one bunch $Q$:*

$$\begin{aligned}(m_e - m_f)\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) + (\nu_Q - m_Q)\Big(\mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_e,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi}) - \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_f,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi})\Big)\\ = \sum_k \xi_k\, g_k \Big[(\pi_{k,e}-\pi_{k,f})\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})\\ + \big(\ell_{k,Q} - \langle \pi_k,\mathbf{1}_Q\rangle\big)\Big(\mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_e,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi}) - \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_f,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})\Big)\Big];\end{aligned}$$

*for homogeneous data the second line of the right side is absent.*

*For homogeneous data a term raises the shape total of each bunch and its scale index by the same level, so the ladder, the scale exchange and the shape exchange (v) keep the condition $\nu_Q = m_Q$; in (v) the shifts of the shape index alone carry the factor $\nu_Q - m_Q$, which vanishes there. The shape split and the shape exchange (iv) shift the shape index alone.*

## 3.2 Letters

A *letter* is a function that enters an integrand through a power $P^{-s}$, $s$ its *letter exponent*. On bunches read in scale and shape, a polynomial letter with positive coefficients is a sum of patterns, each a monomial in the scales and the shapes; the letter presentation writes the integral of a kernel against a product of letters as one Breta, by giving every letter a scale of its own.

**Proposition 3.6** (Letter presentation)**.** *Let the scales $B$ carry a partition $\mathcal{Q}$ into bunches, each read in scale and shape, $x_Q = \rho_Q u_Q$, with $\rho = (\rho_Q)_{Q\in\mathcal{Q}}$ and $u = (u_Q)_{Q\in\mathcal{Q}}$; let* kernel terms *$g_j \rho^{\lambda_j} u^{\pi_j}$ ($g_j > 0$, $\lambda_j \in \mathbb{R}^{\mathcal{Q}}$, $\pi_j \in \mathbb{R}^B$) and letters*

$$P_i(\rho, u) = \sum_p C_{i,p}\, \rho^{\ell_{i,p}} u^{\pi_{i,p}}, \qquad C_{i,p} > 0, \qquad i = 1, \dots, N,$$

*be given, the sum over the* patterns *$p$ of the letter $P_i$, with $\ell_{i,p} \in \mathbb{R}^{\mathcal{Q}}$, $\pi_{i,p} \in \mathbb{R}^B$ and letter exponents $s_i$, $\operatorname{Re} s_i > 0$. The array $(C_{i,p})$ lists the coefficients of the patterns. Introduce one scale $v_i$ per letter, put $V = \{v_1, \dots, v_N\}$, and take the partition of the scales $B \sqcup V$*

$$\mathcal{P} = \mathcal{Q} \cup \{\{v_1\}, \dots, \{v_N\}\}.$$

*On $B \sqcup V$ the shape index is $(\mathbf{m}, \mathbf{s})$, and on the bunches the scale index is $(\boldsymbol{\nu}, \mathbf{s})$, $\boldsymbol{\nu} \in \mathbb{C}^{\mathcal{Q}}$. The term arrays $(\widetilde{\pi}, \widetilde{\ell})$ have one column per kernel term and one per pattern:*

| *column* | *shape exponent on $B \sqcup V$* | *level on $\mathcal{P}$* | *coefficient* |
|---|---|---|---|
| $j$ | $(\pi_j, 0_V)$ | $(\lambda_j, 0_V)$ | $g_j$ |
| $(i,p)$ | $(\pi_{i,p}, \mathbf{e}_i)$ | $(\ell_{i,p}, \mathbf{e}_i)$ | $C_{i,p}$ |

*with $\mathbf{e}_i$ the unit vector at $v_i$ in the shape column and at the singleton bunch $\{v_i\}$ in the level column, and $\mathbf{g} \oplus \mathbf{C}$ lists the coefficients in this column order. Then*

$$\int_{\prod_Q \Delta_Q} u^{\mathbf{m}-\mathbf{1}} \int_{\mathbb{R}^{\mathcal{Q}}_{>0}} \rho^{\boldsymbol{\nu}-\mathbf{1}} \exp\Big(-\sum_j g_j \rho^{\lambda_j} u^{\pi_j}\Big) \prod_i P_i(\rho, u)^{-s_i}\, d\rho\, d\omega(u) = \prod_{Q\in\mathcal{Q}} \Gamma(\nu_Q)\, \mathsf{B}^{\mathcal{P}}_{(\mathbf{m},\mathbf{s}),(\boldsymbol{\nu},\mathbf{s});(\widetilde{\pi},\widetilde{\ell})}(\mathbf{g} \oplus \mathbf{C}).$$

*The shapes of the singleton bunches are points, so the entries of the shape index on $V$ do not enter the integral; they move with the scale entries on $V$, which keeps the relations of Theorem 3.5 literal. In the rest of this statement the partition, the term arrays and the coefficients are fixed, and*

$$b(\mathbf{m}, \boldsymbol{\nu}; \mathbf{s}) \quad \textit{stands for} \quad \mathsf{B}^{\mathcal{P}}_{(\mathbf{m},\mathbf{s}),(\boldsymbol{\nu},\mathbf{s});(\widetilde{\pi},\widetilde{\ell})}(\mathbf{g} \oplus \mathbf{C}).$$

*The relations of Theorem 3.5 read: the scale exchange at $\{v_i\}$ is the* letter split

$$b(\mathbf{m}, \boldsymbol{\nu}; \mathbf{s}) = \sum_p C_{i,p} \prod_{Q\in\mathcal{Q}} \Gamma_1\begin{bmatrix} \nu_Q + \ell_{i,p,Q} \\ \nu_Q \end{bmatrix} b(\mathbf{m} + \pi_{i,p}, \boldsymbol{\nu} + \ell_{i,p}; \mathbf{s} + \mathbf{e}_i);$$

*the ladders are*

$$\partial_{g_j} b(\mathbf{m}, \boldsymbol{\nu}; \mathbf{s}) = -\prod_{Q\in\mathcal{Q}} \Gamma_1\begin{bmatrix} \nu_Q + \lambda_{j,Q} \\ \nu_Q \end{bmatrix} b(\mathbf{m} + \pi_j, \boldsymbol{\nu} + \lambda_j; \mathbf{s}),$$

$$\partial_{C_{i,p}} b(\mathbf{m},\boldsymbol{\nu};\mathbf{s}) = -s_i \prod_{Q\in\mathcal{Q}} \Gamma_1\!\begin{bmatrix}\nu_Q+\ell_{i,p,Q}\\ \nu_Q\end{bmatrix}\, b(\mathbf{m}+\pi_{i,p},\boldsymbol{\nu}+\ell_{i,p};\mathbf{s}+\mathbf{e}_i);$$

*and the scale exchange on each $Q\in\mathcal{Q}$ is*

$$\begin{aligned}\nu_Q\, b(\mathbf{m},\boldsymbol{\nu};\mathbf{s}) = \sum_j \lambda_{j,Q}\, g_j \prod_{Q'\in\mathcal{Q}} \Gamma_1\!\begin{bmatrix}\nu_{Q'}+\lambda_{j,Q'}\\ \nu_{Q'}\end{bmatrix}\, b(\mathbf{m}+\pi_j,\boldsymbol{\nu}+\lambda_j;\mathbf{s})\\ + \sum_{i,p} s_i\, \ell_{i,p,Q}\, C_{i,p} \prod_{Q'\in\mathcal{Q}} \Gamma_1\!\begin{bmatrix}\nu_{Q'}+\ell_{i,p,Q'}\\ \nu_{Q'}\end{bmatrix}\, b(\mathbf{m}+\pi_{i,p},\boldsymbol{\nu}+\ell_{i,p};\mathbf{s}+\mathbf{e}_i).\end{aligned}$$

**Proposition 3.7** (Letter data on the orthant)**.** *Use the partition and the term arrays of Proposition 3.6, and for $\alpha\in\mathbb{R}^B_{>0}$ write $P_i(\alpha)$ for the letter at $\rho_Q=|\alpha_Q|$, $u_Q=\alpha_Q/|\alpha_Q|$:*

$$P_i(\alpha) = \sum_p C_{i,p}\,\alpha^{\pi_{i,p}} \prod_{Q\in\mathcal{Q}} |\alpha_Q|^{\ell_{i,p,Q}-\langle\pi_{i,p},\mathbf{1}_Q\rangle}.$$

*Then, either side converging when the other does,*

$$\begin{aligned}&\prod_{Q\in\mathcal{Q}} \Gamma(\nu_Q)\, \mathsf{B}^{\mathcal{P}}_{(\mathbf{m},\mathbf{s}),(\boldsymbol{\nu},\mathbf{s});(\widetilde{\pi},\widetilde{\ell})}(\mathbf{g}\oplus\mathbf{C})\\ &\qquad = \int_{\mathbb{R}^B_{>0}} \alpha^{\mathbf{m}-\mathbf{1}} \prod_{Q\in\mathcal{Q}} |\alpha_Q|^{\nu_Q-m_Q} \exp\Big(-\sum_j g_j\,\alpha^{\pi_j} \prod_{Q\in\mathcal{Q}} |\alpha_Q|^{\lambda_{j,Q}-\langle\pi_j,\mathbf{1}_Q\rangle}\Big) \prod_i P_i(\alpha)^{-s_i}\, d\alpha :\end{aligned}$$

*each bunch total $|\alpha_Q|$ is one more letter, at the exponent $m_Q-\nu_Q$, and it enters every kernel term and pattern whose level on $Q$ is not its degree there. For homogeneous letter data, $\lambda_{j,Q}=\langle\pi_j,\mathbf{1}_Q\rangle$ and $\ell_{i,p,Q}=\langle\pi_{i,p},\mathbf{1}_Q\rangle$, the letters are $P_i(\alpha)=\sum_p C_{i,p}\alpha^{\pi_{i,p}}$, and at $\nu_Q=m_Q$ the totals are absent.*

**Proposition 3.8** (Equal levels collapse)**.** *In a Breta the dressed coefficients of all terms with one level add: the Zamma inside has one term per distinct level, whose coefficient is that level's polynomial in the shape. For homogeneous letters of a bunch the level of a pattern is its degree.*

## 3.3 The shape integral

**Theorem 3.9** (The shape integral)**.** *For $\operatorname{Re} m_e>0$ ($e\in B$),*

$$\int_{\Delta_B} u^{\mathbf{m}-\mathbf{1}}\, d\omega_B(u) = \mathsf{B}^{\{B\}}_{\mathbf{m},m_B;I_B}(\mathbf{1}) = \Gamma_{|B|}\!\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix} :$$

*the tower (Theorem 3.2) at the unit configuration $I_B$ with unit coefficients, where the Zamma inside the Breta is $\big(\sum_e u_e\big)^{-m_B}=1$.*

The stacked Gamma of the masses over their total is therefore the integral of the shape density $u^{\mathbf{m}-\mathbf{1}}$ on the simplex, and the Dirichlet law is that density normalised.

**Definition 3.10** (Dirichlet law)**.** For $\mathbf{m}\in\mathbb{R}^B_{>0}$, $\operatorname{Dir}_{\mathbf{m}}(du) := u^{\mathbf{m}-\mathbf{1}}\, d\omega_B(u)/\Gamma_{|B|}\!\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}$, a probability measure on $\Delta_B$.

**Theorem 3.11** (Moments and steps)**.** *For* $\mathbf{m} \in \mathbb{R}^B_{>0}$ *and* $\mathbf{k} \in \mathbb{C}^B$ *with* $\mathrm{Re}(m_e + k_e) > 0$ *for every* $e \in B$*,*

$$\mathbb{E}_{U\sim \mathrm{Dir}_{\mathbf{m}}}\big[U^{\mathbf{k}}\big] = \frac{\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}+\mathbf{k}\\ m_B+|\mathbf{k}|\end{matrix}\right]}{\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}\\ m_B\end{matrix}\right]},$$

*which for* $\mathbf{k} \in \mathbb{N}^B$ *is* $\prod_e (m_e)_{k_e}/(m_B)_{|\mathbf{k}|}$*. For every* $\mathbf{m} \in \mathbb{C}^B$ *off the poles the stacked Gamma* $\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}\\ m_B\end{matrix}\right]$ *satisfies the shape split and the shape exchange of its Breta* $\mathsf{B}^{\{B\}}_{\mathbf{m},m_B;I_B}(\mathbf{1})$*:*

$$\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}\\ m_B\end{matrix}\right] = \sum_{e\in B}\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}+\mathbf{e}_e\\ m_B+1\end{matrix}\right], \qquad m_e\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}+\mathbf{e}_f\\ m_B+1\end{matrix}\right] = m_f\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}+\mathbf{e}_e\\ m_B+1\end{matrix}\right],$$

$$\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}+\mathbf{e}_e\\ m_B+1\end{matrix}\right] = \frac{m_e}{m_B}\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}\\ m_B\end{matrix}\right].$$

**Theorem 3.12** (Aggregation)**.** *Let* $B = B_1 \sqcup \cdots \sqcup B_q$ *with nonempty blocks and* $J = \{1, \dots, q\}$*. The map* $(w, v^{(1)}, \dots, v^{(q)}) \mapsto u$*,* $u|_{B_j} = w_j v^{(j)}$*, is a bijection* $\Delta_J \times \prod_j \Delta_{B_j} \to \Delta_B$*, and*

$$u^{\mathbf{m}-\mathbf{1}}d\omega_B = \prod_j w_j^{m_{B_j}-1}\, d\omega_J(w) \prod_j (v^{(j)})^{\mathbf{m}|_{B_j}-\mathbf{1}} d\omega_{B_j}(v^{(j)}).$$

*Consequently* $\Gamma_{\substack{|B|\\1}}\left[\begin{matrix}\mathbf{m}\\ m_B\end{matrix}\right] = \Gamma_{\substack{q\\1}}\left[\begin{matrix}(m_{B_j})_j\\ m_B\end{matrix}\right]\prod_j \Gamma_{\substack{|B_j|\\1}}\left[\begin{matrix}\mathbf{m}|_{B_j}\\ m_{B_j}\end{matrix}\right]$*, and under* $\mathrm{Dir}_{\mathbf{m}}$ *the block totals and the block shapes are independent, with laws* $\mathrm{Dir}_{(m_{B_j})_j}$ *and* $\mathrm{Dir}_{\mathbf{m}|_{B_j}}$*.*

## 3.4 Products of shape integrals

**Theorem 3.13** (The Shruffle)**.** *For a bunch* $Q$ *and a total order* $\varsigma$ *on* $Q$ *let* $\Delta^{\varsigma}_Q = \{u \in \Delta_Q : u_{\varsigma(1)} > u_{\varsigma(2)} > \cdots\}$ *and* $I^{\varsigma}_Q(f;\mathbf{m}) = \int_{\Delta^{\varsigma}_Q} u^{\mathbf{m}-\mathbf{1}} f(u)\, d\omega_Q$*. For pairwise disjoint bunches* $A_1, \dots, A_q$ *with total orders* $\varsigma_i$ *on* $A_i$*, functions* $f_i$ *on* $\Delta_{A_i}$*,* $A = A_1 \sqcup \cdots \sqcup A_q$ *and* $\bigotimes_i f_i(w) = \prod_i f_i(w_{A_i}/|w_{A_i}|)$ *on* $\Delta_A$*, with* $\mathrm{Re}\, m_{A_i} > 0$ *and the integrals absolutely convergent,*

$$\prod_{i=1}^{q} I^{\varsigma_i}_{A_i}(f_i;\mathbf{m}|_{A_i}) = \Gamma_{\substack{1\\q}}\left[\begin{matrix} & m_A & \\ m_{A_1},\ & \dots,\ & m_{A_q}\end{matrix}\right]\sum_{\varsigma} I^{\varsigma}_A\Big(\bigotimes_i f_i;\mathbf{m}\Big),$$

*the sum over the orders* $\varsigma$ *of* $A$ *that restrict to every* $\varsigma_i$*.*

The product of shape integrals on disjoint bunches $A_1, \dots, A_q$ is $\Gamma_{\substack{1\\q}}\left[\begin{matrix} & m_A & \\ m_{A_1},\ & \dots,\ & m_{A_q}\end{matrix}\right]$ times the sum of the shape integrals on the joined bunch over the interleavings of their orders: this is the Shruffle, the product rule of the shape side, whose counterpart for Trailmoments at a common cut is the Struffle (Theorem 5.25).

## 3.5 The Beta port

**Definition 3.14** (Beta port)**.** For $q \in \mathbb{N}$ and $a, b \in \mathbb{C}$,

$$\Pi^{a,b}_q(x) = \frac{(-1)^q x^{1-a}(1-x)^{1-b}}{q!}\,\partial^q_x\big[x^{a+q-1}(1-x)^{b+q-1}\big]$$

$$= \sum_{k=0}^{q} (-1)^{q+k} \binom{a+q-1}{q-k} \binom{b+q-1}{k} x^k (1-x)^{q-k},$$

a polynomial of degree at most $q$. For positive integers $a, b$ its coefficients are integers; for fixed $q$ and $a, b$ affine in an integer parameter, its coefficients are integers of polynomial growth in that parameter.

**Theorem 3.15** (Relations of the Beta port)**.** *(i)* Action on a monomial. *For* $\operatorname{Re} c > 0$ *and* $\operatorname{Re} b > 0$*:* $\int_0^1 x^{c-1}(1-x)^{b-1}\,\Pi_q^{a,b}(x)\,dx = \binom{c-a}{q}\Gamma\,{}_2{}_1\!\left[\begin{matrix} c,\ b+q \\ c+b+q \end{matrix}\right]$*. With* $c = a + j$*,* $j \in \mathbb{N}$*, the binomial vanishes for* $j < q$*: the Beta port is orthogonal, for the density* $x^{a-1}(1-x)^{b-1}$*, to every polynomial of degree below* $q$*.*

*(ii)* Integration by parts. *For* $\operatorname{Re} a, \operatorname{Re} b > 0$ *and* $g$ *with* $q$ *continuous derivatives on* $[0,1]$*,*

$$\int_0^1 x^{a-1}(1-x)^{b-1}\,\Pi_q^{a,b}(x)\,g(x)\,dx = \frac{1}{q!}\int_0^1 x^{a+q-1}(1-x)^{b+q-1}\,g^{(q)}(x)\,dx;$$

*with one Beta port per scale, of orders* $\mathbf{n} = (n_e)_{e\in\mathcal{E}}$*,*

$$\int_{[0,1]^{\mathcal{E}}} \prod_e x_e^{a_e-1}(1-x_e)^{b_e-1}\Pi_{n_e}^{a_e,b_e}(x_e)\,g(\mathbf{x})\,d\mathbf{x} \\ = \frac{1}{\prod_e n_e!}\int_{[0,1]^{\mathcal{E}}} \prod_e x_e^{a_e+n_e-1}(1-x_e)^{b_e+n_e-1}\,\partial^{\mathbf{n}} g(\mathbf{x})\,d\mathbf{x}.$$

*(iii)* Letter ladder. *For letters* $L_i = c_i + d_i x$ *positive on* $[0,1]$ *(*$c_i > 0$ *and* $c_i + d_i > 0$*) and exponents* $w_i \in \mathbb{C}$*,*

$$\frac{(-1)^q}{q!}\,\partial_x^q \prod_i L_i^{-w_i} = \prod_i L_i^{-w_i} \sum_{|v|=q} \prod_i \frac{(w_i)_{v_i}}{v_i!}\Big(\frac{d_i}{L_i}\Big)^{v_i},$$

*so that, for* $\operatorname{Re} a, \operatorname{Re} b > 0$*,*

$$\int_0^1 x^{a-1}(1-x)^{b-1}\,(-1)^q \Pi_q^{a,b} \prod_i L_i^{-w_i}\,dx \\ = \sum_{|v|=q} \prod_i \frac{(w_i)_{v_i}}{v_i!}\int_0^1 x^{a-1}(1-x)^{b-1}\prod_i L_i^{-w_i}\prod_i \Big(\frac{x(1-x)d_i}{L_i}\Big)^{v_i}dx.$$

*Let* $a, b > 0$ *and every* $w_i \geq 0$*. If every* $d_i \geq 0$ *the value is nonnegative, and if every* $d_i \leq 0$ *it is* $(-1)^q$ *times a nonnegative number; whatever the signs of the* $d_i$*, if* $x(1-x)|d_i|/L_i \leq \varepsilon$ *on* $[0,1]$ *for every* $i$*, its modulus is at most* $\frac{(|\mathbf{w}|)_q}{q!}\varepsilon^q \int_0^1 x^{a-1}(1-x)^{b-1}\prod_i L_i^{-w_i}dx$*.*

*(iv)* Reflection. $\Pi_q^{a,b}(1-x) = (-1)^q \Pi_q^{b,a}(x)$*.*

*(v)* Three-term and derivative relations. *For* $q \geq 1$ *and* $a, b \in \mathbb{C}$ *with* $a + b + 2q - i \neq 0$ *for* $i = 0, 1, 2$*,*

$$x\,\Pi_q^{a,b} = \frac{(q+1)(a+b+q-1)}{(a+b+2q)(a+b+2q-1)}\,\Pi_{q+1}^{a,b} + \Big(\frac{(q+1)(a+q)}{a+b+2q} - \frac{q(a+q-1)}{a+b+2q-2}\Big)\Pi_q^{a,b}$$

$$+\frac{(a+q-1)(b+q-1)}{(a+b+2q-1)(a+b+2q-2)}\,\Pi^{a,b}_{q-1},$$

$$x(1-x)\,\partial_x\Pi^{a,b}_q=\Big(\frac{q(b+q-1)}{a+b+2q-2}-q\,x\Big)\Pi^{a,b}_q+\frac{(a+q-1)(b+q-1)}{a+b+2q-2}\,\Pi^{a,b}_{q-1};$$

*for every $q\in\mathbb{N}$ the coefficient of $x^j$ in $\Pi^{a,b}_q$ is $(-1)^{q+j}\binom{a+q-1}{q-j}\binom{a+b+q+j-2}{j}$, and that of $x^q$ is $\binom{a+b+2q-2}{q}$.*

The Beta port moves derivatives of an integrand onto its shape density: integrating against $\Pi^{a,b}_q$ is integrating the $q$-th derivative against the density with both exponents raised by $q$. Against a product of linear letters whose slopes have one sign it keeps the sign of the integrand, up to $(-1)^q$ when the slopes are negative, which is what the arithmetic of Part VI uses. The Beta port of order $q$ is orthogonal, for the density $x^{a-1}(1-x)^{b-1}$, to every polynomial of lower degree (Theorem 3.15(i)), and at every pair of exponents multiplication by $x$ and the operator $x(1-x)\partial_x$ move each Beta port by at most one step in its order (Theorem 3.15(v)): finite coefficient rules, which is what turns a mark relation (Theorem 35.10) into a recurrence of bounded width.

# Chapter 4

# Fibre integrals and closing

**Definition 4.1** (Fibre integral)**.** Let $\mathbf{b} \in \mathbb{C}^N$ be *shares*, pinned by $P\mathbf{b} = \boldsymbol{\pi}$ with $P$ a real $q \times N$ matrix with linearly independent rows and $\boldsymbol{\pi} \in \mathbb{C}^q$. Let $f$ be holomorphic on a tube $\{\operatorname{Re}\mathbf{b} \in U\}$, and let $\mathbf{c} \in U$ be an *admissible real point*, $P\mathbf{c} = \operatorname{Re}\boldsymbol{\pi}$. The *fibre integral* is

$$\int_{P\mathbf{b}=\boldsymbol{\pi}} f := (2\pi)^{q-N} \int_{\mathbb{R}^N} f(\mathbf{c}+i\mathbf{y})\,\delta^q\big(P\mathbf{y} - \operatorname{Im}\boldsymbol{\pi}\big)\,d\mathbf{y} = \frac{(2\pi)^{q-N}}{\sqrt{\det PP^{\mathsf{T}}}} \int_{P\mathbf{y}=\operatorname{Im}\boldsymbol{\pi}} f(\mathbf{c}+i\mathbf{y})\,d\mathrm{vol}_{N-q}(\mathbf{y})$$

whenever it converges, $\mathrm{vol}_{N-q}$ being the $(N-q)$-dimensional volume on the affine subspace; by Cauchy's theorem it does not depend on $\mathbf{c}$ within one connected set of admissible real points. Its *dimension* is $N-q$. With one pin, $P = (1,\dots,1)$ and $\boldsymbol{\pi} = s$, it is the *share integral*

$$\int_{a_1+\cdots+a_k=s} f(\mathbf{a}) = \frac{1}{(2\pi i)^{k-1}} \int_{c_1+i\mathbb{R}} \cdots \int_{c_{k-1}+i\mathbb{R}} f\Big(a_1,\dots,a_{k-1},\, s - \sum_{i<k} a_i\Big)\,da_1\cdots da_{k-1}.$$

At dimension zero, $q = N$, the value is $|\det P|^{-1} f(P^{-1}\boldsymbol{\pi})$. When $\boldsymbol{\pi}$ is real, with $V = \{\mathbf{y} :\ P\mathbf{y} = 0\}$ and the measure $d\boldsymbol{\vartheta} = (\det PP^{\mathsf{T}})^{-1/2}\,d\mathrm{vol}_{N-q}$ on it, the fibre integral is written $\int_{\mathbf{c}+iV} f = (2\pi)^{q-N}\int_V f(\mathbf{c}+i\boldsymbol{\vartheta})\,d\boldsymbol{\vartheta}$, its admissible real point shown.

**Proposition 4.2** (The share side of the Gramma)**.** *For $\nu \in \mathcal{T}_t$,*

$$\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = \int_{t\mathbf{b}=\nu} \prod_{k=1}^{N} \Gamma(b_k)\,\xi_k^{-b_k},$$

*the fibre integral with one share per term, pinned by the configuration ($|\mathcal{E}|$ pins), of dimension $N - |\mathcal{E}|$; its admissible real points are the $\mathbf{c} \in (0,\infty)^N$ with $t\mathbf{c} = \operatorname{Re}\nu$, and they exist exactly on the domain. At dimension zero it is the value of Corollary 2.6.*

On the share side the configuration is the matrix of the pins and the index their values, one pin per scale; each term carries one share, with its Gamma and its coefficient's power as factors.

**Theorem 4.3** (The native gauge of a configuration)**.** *Let the $t_k$ span $\mathbb{R}^{\mathcal{E}}$, $\nu \in \operatorname{int}\operatorname{cone}(t)$ real and $\boldsymbol{\xi} > 0$.*

*(i) $\Phi_{\boldsymbol{\xi}}(y) = \sum_k \xi_k e^{\langle t_k, y\rangle} - \langle \nu, y\rangle$ is strictly convex and tends to $+\infty$ along every ray; its unique critical point $y^\star$ solves $\sum_k \xi_k e^{\langle t_k, y^\star\rangle} t_k = \nu$, and $e^{y^\star}$ is the* native gauge *of the configuration.*

*(ii) The shares $w_k^\star = \xi_k e^{\langle t_k, y^\star\rangle}$ are an admissible real point of the share side (Proposition 4.2). On the real fibre $\{\mathbf{w} > 0 :\ t\mathbf{w} = \nu\}$ the function $S(\mathbf{w}) = \sum_k \big(w_k - w_k \log(w_k/\xi_k)\big)$ is strictly concave with unique critical point $\mathbf{w}^\star$, and $S(\mathbf{w}^\star) = \Phi_{\boldsymbol{\xi}}(y^\star)$.*

*(iii) For $\nu' \in \mathbb{C}^{\mathcal{E}}$, as $\lambda \to \infty$,*

$$\mathsf{G}_{\lambda\nu+\nu';t}(\lambda\boldsymbol{\xi}) = e^{\langle \nu', y^\star\rangle}\, e^{-\lambda\Phi_{\boldsymbol{\xi}}(y^\star)} \Big(\frac{2\pi}{\lambda}\Big)^{|\mathcal{E}|/2} \det\Big(\sum_k w_k^\star t_k t_k^{\mathsf{T}}\Big)^{-1/2} \big(1 + O(\lambda^{-1})\big).$$

The native gauge of a configuration is read on both of its sides at once: on the orthant side its logarithmic coordinate $y^\star$ minimizes the convex function $\Phi_{\boldsymbol{\xi}}$, on the share side the shares $\mathbf{w}^\star$ it determines maximize the entropy-like function $S$ on the real fibre, and the minimum equals the maximum.

**Corollary 4.4** (Splitting a coefficient)**.** *Let $k \geq 2$, $X \in (0,\infty)^k$, $\operatorname{Re} s > 0$ and $\mathbf{c} \in (0,\infty)^{k-1}$ with $\sum_i c_i < \operatorname{Re} s$. On this contour, every share having positive real part,*

$$\mathsf{Z}_{s;(1)}(X_1 + \cdots + X_k) = \Big(\sum_{i=1}^k X_i\Big)^{-s} = \int_{a_1+\cdots+a_k=s} {}_1\Gamma_k\begin{bmatrix}\mathbf{a}\\ s\end{bmatrix} \prod_{i=1}^k X_i^{-a_i},$$

*equivalently $\Gamma(s)(X_1+\cdots+X_k)^{-s} = \int_{|\mathbf{a}|=s} \prod_i \Gamma(a_i) X_i^{-a_i}$: the share side of the one-scale Gamma $\mathsf{G}_{s;(1,\dots,1)}(X)$ (Proposition 4.2). The stacked Gamma ${}_1\Gamma_k\begin{bmatrix}\mathbf{a}\\ s\end{bmatrix}$ is the* share kernel *of the letter $X_1 + \cdots + X_k$ at the exponent $s$.*

**Theorem 4.5** (Closing)**.** *Let a fibre integral have the form $\int_{P\mathbf{b}=\boldsymbol{\pi}} \Gamma\big(\ell(\mathbf{b})\big) G(\mathbf{b})$ with $\ell$ affine and not constant on the fibre. Suppose that, moving the contour along a direction of the fibre in which $\ell$ decreases, $\Gamma(\ell)$ is the only factor with poles crossed, and the shifted contour's contribution tends to zero. Then*

$$\int_{P\mathbf{b}=\boldsymbol{\pi}} \Gamma\big(\ell(\mathbf{b})\big)\, G(\mathbf{b}) = \sum_{j\geq 0} \frac{(-1)^j}{j!} \int_{P\mathbf{b}=\boldsymbol{\pi},\ \ell(\mathbf{b})=-j} G(\mathbf{b}),$$

*each term a fibre integral with one more pin. When several factors have poles crossed, each contributes its own lattice; where several share a pole hyperplane, the residue is a Laurent coefficient of the Gamma factors (Chapter 5).*

Closing trades one dimension of the fibre for a sum over the poles crossed: the Gamma whose argument falls is replaced by its residues. Where closing can be repeated until the fibre has dimension zero, every term is the value $|\det P|^{-1} G(P^{-1}\boldsymbol{\pi})$ of the remaining factors at one point.

**Theorem 4.6** (The share kernels form a semigroup)**.** *For a letter $\sum_{i=1}^k X_i$, shares $\mathbf{a}, \mathbf{b}, \mathbf{g} \in \mathbb{C}^k$ and $|\mathbf{g}| = s + s'$, on admissible contours,*

$$\int_{\mathbf{a}+\mathbf{b}=\mathbf{g},\ |\mathbf{a}|=s} {}_1\Gamma_k\begin{bmatrix}\mathbf{a}\\ s\end{bmatrix}\, {}_1\Gamma_k\begin{bmatrix}\mathbf{b}\\ s'\end{bmatrix} = {}_1\Gamma_k\begin{bmatrix}\mathbf{g}\\ s+s'\end{bmatrix}.$$

Splitting the letter at the exponent $s$ and at the exponent $s'$ and merging the shares is splitting it at $s + s'$: the share kernels of Corollary 4.4 compose like the powers they open.

**Corollary 4.7** (Four Gammas on a line)**.** *On a contour separating the poles of $\Gamma(a+z)\Gamma(b+z)$ from those of $\Gamma(c-z)\Gamma(d-z)$,*

$$\frac{1}{2\pi i}\int \Gamma(a+z)\,\Gamma(b+z)\,\Gamma(c-z)\,\Gamma(d-z)\,dz = {}_1\Gamma_4\begin{bmatrix} a+c,\ b+d,\ a+d,\ b+c \\ a+b+c+d \end{bmatrix}:$$

*Theorem 4.6 at $k = 2$, multiplied by $\Gamma(a+c)\Gamma(b+d)$, with the shares $(a+z,\ c-z)$ at the exponent $a+c$ and $(d-z,\ b+z)$ at the exponent $b+d$, merged to $(a+d,\ b+c)$.*

The four Gammas are the Gamma factors of the four shares of a Gramma on three scales, pinned by the three scales, and the one free share is the contour variable; integrating the orthant side scale by scale leaves $\Gamma(a+c)\Gamma(b+d)$ times the Beta corner ${}_{2}\Gamma_{1}\left[\begin{matrix} a+d,\ b+c \\ a+b+c+d \end{matrix}\right]$, the shape integral of two scales (Theorem 3.9).

# Chapter 5

# The Trailmoment

## 5.1 Definition

**Definition 5.1** (Trailmoment)**.** The data of a Trailmoment are an index $\nu$ and a configuration $t$, those of a Zamma; a measure $\phi$, the *input*, on a space $Y$, the *source*; a measurable *coefficient function* $\mathbf{c} : Y \to (0,\infty)^N$; and a measurable *cut region* $Y' \subseteq Y$. The *Trailmoment* is

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \int_{Y'} \mathsf{Z}_{\nu;t}\big(\mathbf{c}(y)\big)\,\phi(dy)$$

wherever it converges: a normalized Zamma kernel read against the input over the cut region.

The Trailmoment has five slots: its index and configuration, which are the kernel's, its coefficient function, its input and its cut region. On a line, $Y = \mathbb{R}$, the cut region $(N,\infty)$ is the *cut* at $N$, and the region $(N, N']$ between two cuts is a *head*; on $\mathbb{R}^d$ the region $X_N = \{N < y_1 < \cdots < y_d\}$ is a *trail region*, the nested tails that give the Trailmoment its name; the trails of Section 5.8 are read over it. The input is written as a density with its domain marked by an indicator, so that the measure is part of the data.

**Definition 5.2** (Lattice inputs; the index-one value)**.** The *lattice input* on the coset $b+q\mathbb{Z}$, $b \in \mathbb{R}$, of step $q > 0$ and frequency $a \in \mathbb{C}$ is $e^{2\pi i a x}\,\mathbb{I}_{b+q\mathbb{Z}} = \sum_{m\in b+q\mathbb{Z}} e^{2\pi i a m}\,\delta_m$; at frequency 0 it is counting measure on $b+q\mathbb{Z}$, written $\mathbb{I}_{b+q\mathbb{Z}}$, and for $\operatorname{Im} a > 0$ the values $e^{2\pi i a m}$ decay geometrically toward $+\infty$. The input $f\,\mathbb{I}_{b+q\mathbb{Z}}$ has the density $f$ on the lattice, and $\mathbb{I}_{\mathbb{Z}}$ is the *counting input*. At one unit exponent and the coefficient function $x$ the kernel is $\mathsf{Z}_{\nu;(1)}(x) = x^{-\nu}$, so at a cut beyond which every lattice point is positive

$$(\mathsf{T}_{\nu;(1)}[x]\,f\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty)) = \sum_{m\in b+\mathbb{Z},\ m>N} f(m)\,m^{-\nu}.$$

At index one the lattice input is regularized at its upper end:

$$(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty)) := \lim_{M\to\infty}\Big((\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,N+M]) - \log M\Big),$$

which exists, the heads being $\log M + (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty)) + O(M^{-1})$: the regularized value is the coefficient of $M^0 \log^0 M$ in the heads.

**Proposition 5.3** (The cut relation)**.** *The Trailmoment is additive in its cut region: for disjoint measurable $Y_1', Y_2' \subseteq Y$, $(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y_1' \sqcup Y_2') = (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y_1') + (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y_2')$, either side converging when the other does; on a line, for $N < N'$,*

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((N,\infty)) = (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((N,N']) + (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((N',\infty)),$$

*and for a lattice input at index one it holds with the regularized values.*

**Definition 5.4** (Inputs)**.** An input is a nonnegative $\sigma$-finite measure $\mu$, or such a measure with a complex density, $\phi = f\mu$ with $|\phi| = |f|\mu$; a Trailmoment converges when $\int |\mathsf{Z}_{\nu;t}(\mathbf{c})|\, d|\phi|$ over the cut region is finite. The image $\tau_*\phi$ of an input with a complex density under a measurable map $\tau$ is used only in integrals $\int g\, d(\tau_*\phi) := \int g \circ \tau\, d\phi$ with $\int |g \circ \tau|\, d|\phi| < \infty$; $|\tau_*\phi|$ denotes $\tau_*|\phi|$, and tameness and convergence of $\tau_*\phi$ are meant with it. The *Lebesgue input* $\mathbb{I}_{\mathbb{R}}$ is Lebesgue measure on $\mathbb{R}$, $f\,\mathbb{I}_{\mathbb{R}}$ is the input with density $f$, $\mathbb{I}_D$ is Lebesgue measure on an interval $D$, and $\mathbb{I}_{\mathbb{R}^{\mathcal{E}}}$ is Lebesgue measure on $\mathbb{R}^{\mathcal{E}}$; the lattice inputs are those of Definition 5.2, and at integer frequency the lattice input on $\mathbb{Z}$ is the counting input.

An input is *tame beyond the cut $N$* when its part beyond $N$ is carried by $[x_0,\infty)$ for some $x_0 > 0$ and $\int_{(N,\infty)} x^{-\beta}\, |\phi|(dx) < \infty$ for some real $\beta$. Its Laplace transform beyond the cut is

$$\mathcal{F}_N(u) = \int_{(N,\infty)} e^{-ux}\, \phi(dx),$$

which converges for $\operatorname{Re} u > 0$, with $|\mathcal{F}_N(u)| \le e^{-(u-1)x_0} \int_{(N,\infty)} e^{-x}\, |\phi|(dx)$ for $u \ge 1$. When $\mathcal{F}_N$ is moreover meromorphic at $u = 0$, $(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$ denotes, for every $\nu$ off its poles, the continuation in which $\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$ is the meromorphic continuation of $\int_0^\infty u^{\nu-1}\mathcal{F}_N(u)\, du$, the Laplace side of Theorem 5.11 below at one unit exponent. At index one on a lattice input it is the regularized value of Definition 5.2, which is the constant term of the continuation at $\nu = 1$.

**Proposition 5.5** (The continued Trailmoment on a line)**.** *Let $\phi$ be tame beyond the cut $N$ with $\mathcal{F}_N(u) = \sum_{k \ge -k_0} c_k u^k$ near $u = 0$. Then $\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$ is meromorphic in $\nu$ with at most simple poles, at $\nu = -k$, $k \ge -k_0$, with residue $c_k$, and at the nonpositive integers*

$$(\mathsf{T}_{-j;(1)}[x]\,\phi)((N,\infty)) = (-1)^j j!\, c_j \qquad (j \in \mathbb{N}).$$

At a nonpositive integer index the Trailmoment of a tame input is a Taylor coefficient of its Laplace transform, so a cut that releases a head changes these values by the head read at the same index.

## 5.2 Slots, corners and heights

**Definition 5.6** (Slots, corners, height)**.** A *slot* is a place of a symbol of the serif family where one of its data stands. A Gramma or a Zamma has three slots: its index, its configuration and its coefficients. A Breta has five: its partition into bunches, its shape index, its scale index, its terms (shape exponents and levels) and its coefficients. A Trailmoment has five: its index $\nu$, its configuration $t$, its coefficient function $\mathbf{c}$, its input $\phi$ and its cut region. A *corner* of a member of the serif family is that member with some of its slots given particular values. A *relation* is an identity between members of the serif family. Its *height* is the set of values of the slots on which one proof establishes it; a slot is *free* in a relation when the height does not restrict it and *pinned* otherwise. For a pinned slot, the *mechanism* is the step of the proof that uses the pinned value.

A relation recited at a corner is understood once its height is known: which slots the proof leaves free, which it uses, and at which step. A relation whose height is every value of every slot is a property of the operator. A pin records what a proof uses; it does not say that the relation fails at the other values of the slot, and where the converse is proved, so that the relation is a property of the pinned value, it is stated with the relation.

## 5.3 Relations at free slots

**Theorem 5.7** (The gauge of the coefficient function)**.** *Let $(\nu, t, \mathbf{c}, \phi, Y')$ be the data of a Trailmoment on a source $Y$ and let $\ell : Y \to (0,\infty)^{\mathcal{E}}$ be measurable. Then*

$$\big(\mathsf{T}_{\nu;t}[(\ell^{t_k} c_k)_k]\,\phi\big)(Y') = \big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\ell^{-\nu}\phi\big)(Y'),$$

*either side converging when the other does. If the columns $t_k$, $k \in S$, form a basis of $\mathbb{R}^{\mathcal{E}}$, the choice $\log \ell = -(t_S^{\mathsf{T}})^{-1} \log \mathbf{c}_S$ makes $\ell^{t_k} c_k = 1$ for every $k \in S$: the gauge moves $|\mathcal{E}|$ of the $N$ coefficient functions into the input, and the kernel then depends on the source point only through the $N - |\mathcal{E}|$ functions $\ell^{t_k} c_k$, $k \notin S$. At the unit configuration, $(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,\phi)(Y') = \int_{Y'} \mathbf{c}^{-\nu}\, d\phi$.*

At the unit configuration the kernel is a density on the input: index, coefficient function and input enter only through the measure $\mathbf{c}^{-\nu}\phi$. With $N > |\mathcal{E}|$ terms the kernel depends on the source point through $N - |\mathcal{E}|$ gauge-invariant functions as well as through a density, and this count is the dimension of the share side (Theorem 5.10).

**Theorem 5.8** (Exchange and ladder of the Trailmoment)**.** *Each identity holds wherever every Trailmoment in it converges.*

*(i)* Exchange, one relation per scale. *For each $e \in \mathcal{E}$,*

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \sum_{k=1}^{N} t_{k,e}\ \Gamma_1^{\,1}\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e + 1\end{bmatrix} \prod_{f \neq e} \Gamma_1^{\,1}\begin{bmatrix}\nu_f + t_{k,f}\\ \nu_f\end{bmatrix}\ \big(\mathsf{T}_{\nu + t_k;t}[\mathbf{c}]\,c_k\phi\big)(Y').$$

*(ii)* Ladder. *If $\mathbf{c} = \mathbf{c}_\zeta$ depends differentiably on a real parameter $\zeta$ and the differentiated integrand is dominated by an integrable function locally uniformly in $\zeta$,*

$$\partial_\zeta(\mathsf{T}_{\nu;t}[\mathbf{c}_\zeta]\,\phi)(Y') = -\sum_{k=1}^{N} \prod_{e} \Gamma_1^{\,1}\begin{bmatrix}\nu_e + t_{k,e}\\ \nu_e\end{bmatrix}\ \big(\mathsf{T}_{\nu + t_k;t}[\mathbf{c}_\zeta]\,(\partial_\zeta c_k)\,\phi\big)(Y').$$

*(iii)* The unit configuration. *At $t = I_{\mathcal{E}}$ the exchange relation at each scale has one term, and on the interior of the set of indices where the Trailmoment converges the index is differentiated through the input: for every $e \in \mathcal{E}$,*

$$(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,\phi)(Y') = (\mathsf{T}_{\nu + \mathbf{e}_e;I_{\mathcal{E}}}[\mathbf{c}]\,c_e\phi)(Y'), \qquad \partial_{\nu_e}(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,\phi)(Y') = -(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,(\log c_e)\,\phi)(Y').$$

At the unit configuration a step of the index at a scale is a factor of that coordinate of the coefficient function on the input, in both directions. With several terms on a scale the exchange relation is linear in $N + 1$ Trailmoments whose inputs differ by the factors $c_k$, one relation per scale, and a step of the index is no longer a density.

**Theorem 5.9** (Transport)**.** *For a measurable map $\tau : Y \to Y_2$, an input $\phi$ on $Y$, a coefficient function $\mathbf{c}$ on $Y_2$ and a cut region $Y_2' \subseteq Y_2$,*

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\tau_*\phi)(Y_2') = (\mathsf{T}_{\nu;t}[\mathbf{c}\circ\tau]\,\phi)(\tau^{-1}Y_2'),$$

*whenever the right side converges; for a nonnegative input either side converges when the other does. In particular, transporting by the coefficient function itself, a convergent Trailmoment is*

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \big(\mathsf{T}_{\nu;t}[\mathrm{id}]\,\mathbf{c}_*(\phi|_{Y'})\big)\big((0,\infty)^N\big)\,:$$

*coefficient function, input and cut region enter only through the image $\mathbf{c}_*(\phi|_{Y'})$ of the input on the orthant of coefficients.*

A Trailmoment of a nonnegative input is therefore a function of three things: an index, a configuration, and a measure on the orthant of coefficients. Two descriptions of that measure, by different coefficient functions, inputs and cut regions, give the same member of the serif family. An increasing bijection of the line carries half-lines to half-lines: with $\tau(x) = x + h$ it is the *shift* of the cut, and with $\tau(x) = qx$ at one unit exponent it gives $(\mathsf{T}_{\nu;(1)}[x]\,\tau_*\phi)((qN,\infty)) = q^{-\nu}(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$.

**Theorem 5.10** (The share side of the Trailmoment)**.** *Let $\operatorname{Re}\nu \in \operatorname{int}\operatorname{cone}(t)$, and let $\mathbf{s} \in (0,\infty)^N$ be an admissible real point, $t\mathbf{s} = \operatorname{Re}\nu$, with $\int_{Y'}\prod_k c_k(y)^{-s_k}\,|\phi|(dy) < \infty$. On the contour through $\mathbf{s}$,*

$$\prod_{e\in\mathcal{E}}\Gamma(\nu_e)\ (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \int_{t\mathbf{b}=\nu}\ \prod_{k=1}^{N}\Gamma(b_k)\ (\mathsf{T}_{\mathbf{b};I_N}[\mathbf{c}]\,\phi)(Y'),$$

$$(\mathsf{T}_{\mathbf{b};I_N}[\mathbf{c}]\,\phi)(Y') = \int_{Y'}\prod_{k=1}^{N} c_k(y)^{-b_k}\,\phi(dy),$$

*a fibre integral of dimension $N - |\mathcal{E}|$ of Trailmoments at the unit configuration $I_N$, with one scale per term of $t$. Where some $\nu_e$ lies in $-\mathbb{N}$ the left side is read as $\int_{Y'}\mathsf{G}_{\nu;t}(\mathbf{c})\,d\phi$.*

Every configuration is written through the unit configuration: a Trailmoment at any configuration is a superposition of Trailmoments at the unit configuration over the fibre of its configuration, each multiplied by the Gammas of its shares. Closing the fibre integral (Theorem 4.5) replaces the superposition by residues, and the residues are Trailmoments at the unit configuration at shifted indices; that is why their values appear in the closings of every configuration.

**Theorem 5.11** (The Laplace side of the Trailmoment)**.** *The joint Laplace transform of the data $(\mathbf{c},\phi,Y')$ is $\mathcal{F}(v) = \int_{Y'} e^{-\langle v,\mathbf{c}(y)\rangle}\,\phi(dy)$ for $v \in (0,\infty)^N$. For $\nu \in \mathcal{T}_t$, wherever the double integral converges,*

$$\prod_{e\in\mathcal{E}}\Gamma(\nu_e)\ (\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)(Y') = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\nu-\mathbf{1}}\,\mathcal{F}\big(x^{t_1},\dots,x^{t_N}\big)\,dx = \big(\mathsf{T}_{\mathbf{1}-\nu;I_{\mathcal{E}}}[x]\ \mathcal{F}(x^t)\,\mathbb{I}_{\mathbb{R}^{\mathcal{E}}}\big)\big(\mathbb{R}^{\mathcal{E}}_{>0}\big),$$

*where $\mathcal{F}(x^t) = \mathcal{F}(x^{t_1},\dots,x^{t_N})$ and the kernel of the last Trailmoment is $\prod_e x_e^{\nu_e-1}$, the Zamma of the unit configuration at the index $\mathbf{1}-\nu$; where some $\nu_e$ lies in $-\mathbb{N}$ the left side is read as $\int_{Y'}\mathsf{G}_{\nu;t}(\mathbf{c})\,d\phi$. The exponential is the Zamma of a term at exponent zero, $\mathsf{Z}_{1;(1,0)}(1,\xi) = e^{-\xi}$, so the joint Laplace transform is a Trailmoment of the same input: $\mathcal{F}(v) = (\mathsf{T}_{1;(1,0)}[(1,\langle v,\mathbf{c}\rangle)]\,\phi)(Y')$.*

Index $\nu$ on the input side is index $\mathbf{1}-\nu$ on the Laplace side. Besides the input itself, a Trailmoment has two sides: the share side, which resolves the configuration into unit configurations, and the Laplace side, which replaces the input by its Laplace transform; the reflections of the index (Theorems 5.18 and 5.19), the Shruffle (Theorem 5.26) and the Cremona (Theorem 5.29) are read on the Laplace side, and the lattice relations and the Struffle on the input itself.

## 5.4 Lattices

**Theorem 5.12** (The lattice relations)**.** *Let $b\in\mathbb{R}$, let $q\geq 1$ be an integer, let $(\nu,t,\mathbf{c})$ be the data of a kernel on $\mathbb{R}$, $f$ a density and $N$ a cut.*

*(i)* Cosets. *Wherever the right side converges,*

$$\sum_{r=0}^{q-1}\big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,f\,\mathbb{I}_{\frac{b+r}{q}+\mathbb{Z}}\big)((N,\infty))=\big(\mathsf{T}_{\nu;t}[\mathbf{c}(\,\cdot\,/q)]\,f(\,\cdot\,/q)\,\mathbb{I}_{b+\mathbb{Z}}\big)((qN,\infty)).$$

*(ii)* Frequencies. *For $a\in\mathbb{R}$, wherever the Trailmoments on the left converge,*

$$\sum_{r=0}^{q-1}\big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,e^{2\pi i(a+\frac{r}{q})(x-b)}f\,\mathbb{I}_{b+\mathbb{Z}}\big)((N,\infty))=q\,\big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,e^{2\pi ia(x-b)}f\,\mathbb{I}_{b+q\mathbb{Z}}\big)((N,\infty)),$$

*the right side summing over the lattice of step $q$.*

A lattice is assembled from a sublattice in two ways: by cosets, the $q$ cosets of $\mathbb{Z}$ in $q^{-1}\mathbb{Z}$; by frequencies, the $q$ exponentials $e^{2\pi irx/q}$, $0\leq r<q$. Both statements are free in the index, the configuration, the coefficient function, the density, the coset and the cut. At one unit exponent with the coefficient function $x$ the rescaled coefficient function of (i) multiplies the kernel by the scalar $q^{\nu}$, $\mathsf{Z}_{\nu;(1)}(x/q)=q^{\nu}x^{-\nu}$; at index one, with the regularized values, the same assembly carries the further term $q\log q$, the finite part read in two coordinates (Lemma 1.9).

## 5.5 The Gamma read by Trailmoments

**Proposition 5.13** (Heads and higher orders at lattice data)**.** *For real $b>-1$ and $|z|<1+b$,*

$$\log\Gamma(1+b-z)=\log\Gamma(1+b)+\sum_{i\geq 1}\frac{z^i}{i}\,(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((b,\infty)),$$

*the index-one Trailmoment regularized; equivalently, for $j\in\mathbb{N}$,*

$$(\log\Gamma)^{(j+1)}(1+b)=(-1)^{j+1}j!\,(\mathsf{T}_{j+1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((b,\infty)).$$

*Further $\Gamma(-k+z)=\dfrac{(-1)^k\,\Gamma(1+z)\,\Gamma(1-z)}{z\,\Gamma(k+1-z)}$ for $k\in\mathbb{N}$. In this statement write $h=\prod_a\Gamma(1+b_a-z_a)/\prod_c\Gamma(1+b'_c-z'_c)$, a stacked Gamma whose constants $b_a,b'_c$ lie in one translate $b_0+\mathbb{N}$ with $b_0>-1$, with $\sum_a z_a=\sum_c z'_c$, $|z_a|<1+b_a$ and $|z'_c|<1+b'_c$. Then*

$$\log\frac{h}{h|_{z=0}}=\sum_c z'_c\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b_0+\mathbb{Z}})((b_0,b'_c])$$

$$- \sum_a z_a \, (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{b_0+\mathbb{Z}})((b_0, b_a])$$
$$+ \sum_{i\geq 2} \frac{1}{i} \Big( \sum_a z_a^i \, (\mathsf{T}_{i;(1)}[x]\, \mathbb{I}_{b_0+\mathbb{Z}})((b_a, \infty))$$
$$- \sum_c z_c'^i \, (\mathsf{T}_{i;(1)}[x]\, \mathbb{I}_{b_0+\mathbb{Z}})((b_c', \infty))\Big) :$$

*the regularized index-one values cancel, the first order is a finite sum over a head of the lattice, rational when $b_0$ is, and the orders $i \geq 2$ are Trailmoments of the lattice input cut at the data. At integer data, $b_0 = 0$, the first order is rational and the higher orders are Trailmoments of the counting input.*

**Proposition 5.14** (Laurent data of the Gamma at its poles)**.** *Let $j \in \mathbb{N}$. For $|\varepsilon| < 1$, with $(-1)^j \varepsilon\, \Gamma(-j+\varepsilon) = \Gamma(1+\varepsilon)\, \Gamma(1-\varepsilon)/\Gamma(1+j-\varepsilon)$,*

$$\log\big((-1)^j \varepsilon\, \Gamma(-j+\varepsilon)\big) = -\log j! - (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((j,\infty))\, \varepsilon$$
$$- \sum_{k\geq 2} \frac{(\mathsf{T}_{k;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((j,\infty))}{k}\, \varepsilon^k + \sum_{k\geq 1} \frac{(\mathsf{T}_{2k;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))}{k}\, \varepsilon^{2k},$$

*and $(\log\Gamma)'(j+1) = -(\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((j,\infty))$. Let $h$ be holomorphic near $b = -j$, and let $k_1, \ldots, k_p$ be integers with $j_i = j - k_i \in \mathbb{N}$, so that every $\Gamma(b+k_i)$ has a pole at $-j$. Then*

$$\operatorname*{Res}_{b=-j} \Big( h(b) \prod_{i=1}^{p} \Gamma(b+k_i) \Big) = (-1)^{j_1+\cdots+j_p} \sum_{l=0}^{p-1} \frac{h^{(l)}(-j)}{l!}\, [\varepsilon^{p-1-l}] \prod_{i=1}^{p} \big((-1)^{j_i} \varepsilon\, \Gamma(-j_i+\varepsilon)\big),$$

*each factor of the product being the exponential of the display at $j_i$; at $p = 2$ and $k_1 = k_2 = 0$ it is $\frac{1}{(j!)^2}\big(h'(-j) + 2\,(\log\Gamma)'(j+1)\, h(-j)\big)$. Every Laurent coefficient of $\prod_i \Gamma(b+k_i)$ at $-j$ is $\prod_i (j_i!)^{-1}$ times a polynomial in Trailmoments of the counting input cut at the $j_i$ and at $0$.*

**Proposition 5.15** (Selecting an index inside a word)**.** *Let $w \geq 2$ be an integer and $h(z) = \prod_i \Gamma(1 + b_i - a_i z)^{n_i}$, a finite product with real $b_i > -1$, $a_i \in \mathbb{C}$ and $n_i \in \mathbb{Z}$. Its $w$-th multiplicative difference satisfies*

$$\prod_{j=0}^{w} h(jz)^{(-1)^{w-j}\binom{w}{j}} = 1 + (w-1)! \Big[ \sum_i n_i a_i^w \, (\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{b_i+\mathbb{Z}})((b_i,\infty)) \Big] z^w + O(z^{w+1}).$$

*At integer values $b_i = k_i \in \mathbb{N}$ with rational $a_i$, an integer $\ell \geq 1$ with every $\ell a_i \in \mathbb{Z}$ and $k_* = \max_i k_i$, the coefficient of $z^w$ is $A\,(\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) - B$ with $A = (w-1)! \sum_i n_i a_i^w$ and $B = (w-1)! \sum_i n_i a_i^w \, (\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,k_i])$, both cleared by $\ell^w d_{k_*}^w$. For the one factor $\Gamma(k+1-z)$, the coefficient of $z^w$ in $\prod_{j=0}^{w} \Gamma(k+1-jz)^{(-1)^{w-j}\binom{w}{j}}$ is $(w-1)!\, (\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((k,\infty))$, and passing from $k$ to $k+1$ multiplies this product by $\prod_{j=0}^{w} (k+1-jz)^{(-1)^{w-j}\binom{w}{j}}$. The difference acts multiplicatively on one word, not on the sums of a closing; along a ray from the origin whose law of $[1]$ is reflexive of order two, the coefficient of the coordinate is read without any sum, from the closings of the ray (Chapter 43).*

The $w$-th difference annihilates every power of $z$ below the $w$-th in the logarithm of a Gamma word, so it removes the regularized index-one values together with all Trailmoments of lower index and keeps exactly those of index $w$: from a word in the Gammas of the data it selects one Trailmoment index, with a rational head.

**Lemma 5.16** (Two Gammas and their reflections). *The stacked Gamma* $\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big] = \pi^2/\sin^2\pi a$ *has at every integer* $m$ *the principal part* $(a-m)^{-2}$ *and no simple part, it equals* $\sum_{m\in\mathbb{Z}}(a+m)^{-2}$*, and on the line* $a=\frac12+iy$ *it equals* $4\pi^2/(e^{\pi y}+e^{-\pi y})^2$*.*

*(i)* Right closing. *For* $0<c<1$ *and* $g$ *rational, holomorphic on* $\{\operatorname{Re} a\ge c\}$*, with* $g(a)=O(|a|^{-2})$*:* $\dfrac{1}{2\pi i}\displaystyle\int_{(c)}\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]\ g(a)\,da=-\sum_{m\ge1}g'(m)$.

*(ii)* Left closing. *For* $0<X<1$*:* $\dfrac{1}{2\pi i}\displaystyle\int_{(1/2)}\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]\ X^{-a}\,da=\frac{\log X}{X-1}$.

*(iii)* Fourier read. *For real* $t$*,*

$$\frac{1}{2\pi}\int_{\mathbb{R}}\Gamma_{\substack{4\\0}}\big[\tfrac12+iy,\ \tfrac12+iy,\ \tfrac12-iy,\ \tfrac12-iy\big]\ e^{ity}\,dy=\Gamma_{\substack{2\\0}}\big[1+\tfrac{it}{2\pi},\ 1-\tfrac{it}{2\pi}\big]=\frac{t}{e^{t/2}-e^{-t/2}}.$$

A closing whose poles coincide (Theorem 4.5) takes Laurent coefficients of its Gammas, and a balanced word near lattice data has heads and higher orders (Proposition 5.13): in both cases the Gammas contribute words in Trailmoments of lattice inputs, read at the cuts that the data set. The stacked Gamma of Lemma 5.16 is the simplest coincidence, two Gammas and their two reflections meeting at every integer: its double poles have no simple parts, so its right closing reads the derivative of the rest of the integrand at the positive integers, and on the line $\operatorname{Re} a=\frac12$ it is $2\pi$ times the density of the critical-line law (Chapter 36).

**Theorem 5.17** (Reflected pairs and matched parity). *(i)* Cosets. *For* $0<b<1$ *and every integer* $n\ge1$*,*

$$(\mathsf{T}_{n;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((b-1,\infty))+(-1)^n\,(\mathsf{T}_{n;(1)}[x]\,\mathbb{I}_{-b+\mathbb{Z}})((-b,\infty))=\frac{(-1)^{n-1}}{(n-1)!}\,\partial_b^{n-1}\big(\pi\cot\pi b\big),$$

*the index-one values regularized. The left side is* $\sum_{m\in b+\mathbb{Z}}m^{-n}$ *over the whole lattice, in symmetric order at* $n=1$*, and the right side is* $\pi^n$ *times a polynomial of degree* $n$ *in* $\cot\pi b$ *with rational coefficients.*

*(ii)* Frequencies. *For* $0<b\le1$ *and every integer* $n\ge2$*, the Trailmoment* $(\mathsf{T}_{n;(1)}[x]\,\sin\big(2\pi bx-\frac{\pi(n-1)}{2}\big)\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ *is* $\frac{(-1)^{n-1}(2\pi)^n}{2}$ *times the coefficient of* $u^{n-1}$ *in* $e^{-bu}/(1-e^{-u})$*. The density is* $(-1)^{n/2}\cos(2\pi bx)$ *at even* $n$ *and* $(-1)^{(n-1)/2}\sin(2\pi bx)$ *at odd* $n$*, and the value is* $\pi^n$ *times a polynomial of degree* $n$ *in* $b$ *with rational coefficients.*

*(iii)* Periodic densities. *For integers* $0<r<q$ *and* $n\ge1$*,*

$$(\mathsf{T}_{n;(1)}[x]\,\big(\mathbb{I}_{r+q\mathbb{Z}}+(-1)^n\,\mathbb{I}_{-r+q\mathbb{Z}}\big))((0,\infty))=\frac{(-1)^{n-1}}{q^n(n-1)!}\,\partial_b^{n-1}\big(\pi\cot\pi b\big)\Big|_{b=r/q},$$

*at* $n=1$ *with the points of the two lattices taken in increasing order.*

*(iv)* Folding. *At* $b=\frac12$ *the two lattices of (i) are one, and (i) reads*

$$\big(1+(-1)^n\big)(\mathsf{T}_{n;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty))=\frac{(-1)^{n-1}}{(n-1)!}\,\partial_b^{n-1}(\pi\cot\pi b)\Big|_{b=1/2}:$$

*at odd* $n$ *both sides vanish, and at even* $n=2j$*, with the lattice relation at the step* 2 *(Theorem 5.12(i)),*

$$(\mathsf{T}_{2j;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))=\frac{-1}{2\,(2^{2j}-1)\,(2j-1)!}\ \partial_b^{2j-1}\big(\pi\cot\pi b\big)\Big|_{b=1/2}.$$

*At $b = 1$, (ii) gives $(\mathsf{T}_{2j;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (-1)^{j+1}\frac{(2\pi)^{2j}}{2}$ times the coefficient of $u^{2j-1}$ in $1/(e^u-1)$, and the identity $0=0$ at odd $n$.*

At an integer index $n$ the kernel $\mathsf{Z}_{n;(1)}(x) = x^{-n}$ has the parity of $n$ under $x \mapsto -x$. What closes, for every coset $b+\mathbb{Z}$ and every integer index, is the value of the whole coset, points of both signs: the reflection of the Gamma evaluates it, and at $n=2$ and $0<b<1$ it is $\Gamma_{\substack{4\\0}}\big[b,\ b,\ 1-b,\ 1-b\big]$ (Lemma 5.16). A Trailmoment reads the points beyond its cut only, so this relation gives it when the input, continued to the whole lattice, has the parity of the index: an even periodic density closes at even indices and an odd one at odd indices, by (iii), and on the coset $q\mathbb{Z}$ by (iv). On a lattice equal to its own reflection the pair of (i) folds onto one Trailmoment with the factor $1+(-1)^n$; at even $n$ that Trailmoment closes, and at odd $n$ the relation is the identity $0=0$ and carries no information. Statement (ii) is the same rule read on frequencies. The mismatched parities, the sine at an even index and the cosine at an odd index, are reached by neither statement on any coset or at any frequency; the counting input at an odd index is the cosine at frequency zero, and its Trailmoments are the ones that the positive words of Part VI carry.

## 5.6 Reflections of the index

**Theorem 5.18** (Reflection through the poles of the Laplace transform)**.** *Let $\mathcal{F}$ be a function on $(0,\infty)$ that extends to a meromorphic function on $\mathbb{C}$ with no pole on $(0,\infty)$, such that*

*(a) its poles other than $u=0$ are simple, with residue $r_p$ at the pole $p$, and its pole at $0$ has order $k_0 \geq 0$, with $k_0 = 0$ when there is none;*

*(b) there are radii $R_m \to \infty$ and a real $A$ with $\sup_{|u|=R_m} |\mathcal{F}(u)| = O(R_m^A)$.*

*Write $\Sigma_m(\nu) = \sum_{0<|p|<R_m} r_p\,(-p)^{\nu-1}$, with $(-p)^{\nu-1} = \exp\big((\nu-1)\log(-p)\big)$ and $|\arg(-p)| < \pi$.*

*(i)* On the strip. *If moreover $\mathcal{F}(u) = O(u^A)$ as $u \to +\infty$, then for $k_0 < \operatorname{Re}\nu < -A$ with $\nu \notin \mathbb{Z}$,*

$$\int_0^\infty u^{\nu-1}\mathcal{F}(u)\,du = \frac{\pi}{\sin\pi\nu}\lim_{m\to\infty}\Sigma_m(\nu).$$

*(ii)* Beyond the strip. *If $\mathcal{F} = \mathcal{F}_N$ is the Laplace transform of an input $\phi$ tame beyond the cut $N$, its poles lie in $\operatorname{Re} u \leq 0$, and for $\operatorname{Re}\nu < -A$ with $\nu \notin \{1,2,3,\dots\}$,*

$$(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty)) = \Gamma(1-\nu)\lim_{m\to\infty}\Sigma_m(\nu).$$

Reflection of the index replaces an input by the poles of its Laplace transform: each simple pole $p$ becomes the point measure $r_p\,\delta_{-p}$, read by the kernel $\mathsf{Z}_{1-\nu;(1)}$. The relation is free in the input within (a) and (b), and pinned at one unit exponent with the coefficient function $x$; the mechanism is the Laplace side of the kernel $\mathsf{Z}_{\nu;(1)}$, where the index is the exponent of the scale and a full turn of the scale multiplies the integrand by $e^{2\pi i\nu}$. For the input $e^{-x}\,\mathbb{I}_{\mathbb{R}}$ cut at 0, whose Laplace transform $1/(1+u)$ has the one pole $-1$, statement (i) is the reflection of the Gamma (Proposition 1.3(v)); for a lattice input the strip of (i) is empty, and the reflection is a statement about the continued Trailmoment, which is (ii).

**Theorem 5.19** (Reflection by inversion of the Laplace transform)**.** *Let $\phi$ and $\psi$ be tame beyond the cut $0$, with Laplace transforms $\mathcal{F}$ and $\mathcal{G}$, and suppose that for some $k \in \mathbb{R}$, $\lambda > 0$, $C \neq 0$ and $f_0, g_0 \in \mathbb{C}$,*

$$\mathcal{F}(u) + f_0 = C\,u^{-k}\Big(\mathcal{G}\big(\tfrac{\lambda}{u}\big) + g_0\Big) \qquad (u>0).$$

*Then* $\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((0,\infty))$*, defined for large* $\operatorname{Re}\nu$ *by the Laplace side, continues meromorphically to* $\mathbb{C}$*,*

$$\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((0,\infty)) = \int_{\sqrt{\lambda}}^{\infty} u^{\nu-1}\mathcal{F}(u)\,du + C\lambda^{\nu-k}\int_{\sqrt{\lambda}}^{\infty} u^{k-\nu-1}\mathcal{G}(u)\,du + \frac{C\,g_0\,\lambda^{(\nu-k)/2}}{\nu-k} - \frac{f_0\,\lambda^{\nu/2}}{\nu},$$

*whose only possible poles are simple, at* $\nu = k$ *and* $\nu = 0$*. The same holds with* $\psi$ *in the place of* $\phi$*, and*

$$\Gamma(\nu)\,(\mathsf{T}_{\nu;(1)}[x]\,\phi)((0,\infty)) = C\,\lambda^{\nu-k}\,\Gamma(k-\nu)\,(\mathsf{T}_{k-\nu;(1)}[x]\,\psi)((0,\infty)).$$

The inversion $u \mapsto \lambda/u$ of the scale is the reflection of the Gramma (Proposition 2.3) on the Laplace side. The relation is free in the two inputs, which need not be lattices and whose Laplace transforms need not be meromorphic; it uses one property of the pair, that the inversion of the scale carries the Laplace transform of one to that of the other. The constants $f_0$ and $g_0$ are the multiples of $\delta_0$ that the two inputs would carry at the origin, where the kernel $\mathsf{Z}_{\nu;(1)}$ is not defined, and for $k \neq 0$ they give the residues of the continuation, $-f_0$ at $\nu = 0$ and $Cg_0$ at $\nu = k$.

**Theorem 5.20** (Reflection at the coefficient function $x+\mu$)**.** *Let* $\phi$*,* $\psi$*,* $\mathcal{F}$*,* $\mathcal{G}$*,* $k$*,* $\lambda$*,* $C$*,* $f_0$ *and* $g_0$ *be as in Theorem 5.19, and let* $\mu > 0$*. For* $\operatorname{Re} s > \max(k,0)$ *with* $\int_{(0,\infty)}(x+\mu)^{-\operatorname{Re} s}\,|\phi|(dx) < \infty$*,*

$$\Gamma(s)\Big[(\mathsf{T}_{s;(1)}[x+\mu]\,\phi)((0,\infty)) + f_0\,\mu^{-s}\Big] = C\,\lambda^{s-k}\int_{(0,\infty)}\mathsf{G}_{k-s;(1,-1)}(y,\lambda\mu)\,\psi(dy) + C\,g_0\,\Gamma(s-k)\,\mu^{k-s}.$$

*The integral on the right converges for every* $s \in \mathbb{C}$ *and is entire in* $s$*.*

## 5.7 Joint sources

**Theorem 5.21** (Closure with the source kept joint)**.** *(i)* $\mathsf{Z}_{\nu;t}(\mathbf{c})\,\mathsf{Z}_{\eta;s}(\mathbf{d}) = \mathsf{Z}_{(\nu,\eta);\,t\oplus s}(\mathbf{c}\oplus\mathbf{d})$*, the configuration* $t \oplus s$ *acting on the disjoint union of the two sets of scales.*

*(ii) A finite expression built from Zammas with positive indices by products and by nested reads against nonnegative measures is one Trailmoment on a source: its configuration is the concatenation* $\oplus$ *of the factors' configurations, and its input is the joint measure* $\phi(dy)\,\psi_y(dw)$ *of the source variables, every shared variable, conditional dependence and cut retained. The same holds for complex data where every integral converges absolutely.*

**Proposition 5.22** (The joint Laplace transform of Dirichlet inputs)**.** *Let* $U = (U_Q)_{Q\in\mathcal{P}}$ *have independent blocks* $U_Q \sim \operatorname{Dir}_{\mathbf{m}|_Q}$ *on a partition* $\mathcal{P}$ *of the scales, let* $P_1,\dots,P_q$ *be finite sums* $P_i(u) = \sum_{\mathbf{a}}[u^{\mathbf{a}}]P_i\,u^{\mathbf{a}}$ *over exponents* $\mathbf{a} \in [0,\infty)^{\mathcal{E}}$*, polynomials when the exponents are integers, and let* $t$ *be a configuration of* $q$ *columns on* $d$ *scales and* $\nu \in \mathbb{C}^d$*. Then* $\mathcal{F}(v) = \mathbb{E}_{U\sim\bigotimes_Q \operatorname{Dir}_{\mathbf{m}|_Q}}\big[\exp\big(-\sum_i v_i P_i(U)\big)\big]$ *is entire and*

$$\mathcal{F}(v) = \sum_{\mathbf{k}\in\mathbb{N}^q}\frac{(-v)^{\mathbf{k}}}{\mathbf{k}!}M(\mathbf{k}),$$

$$M(\mathbf{k}) = \sum_{\mathbf{a}}\big[u^{\mathbf{a}}\big]\big(P_1(u)^{k_1}\cdots P_q(u)^{k_q}\big)\prod_e \Gamma_1\begin{bmatrix} m_e + a_e \\ m_e\end{bmatrix}\prod_Q \Gamma_1\begin{bmatrix} m_Q \\ m_Q + a_Q\end{bmatrix}, \qquad a_Q = \sum_{e\in Q} a_e;$$

*for integer exponents the stacked Gammas are* $\prod_e (m_e)_{a_e} / \prod_Q (m_Q)_{a_Q}$. *If every* $P_i$ *is positive on the interior of the source and* $\operatorname{Re}\nu_j > 0$, *then, wherever the read converges,*

$$\mathbb{E}_{U\sim\bigotimes_Q \mathrm{Dir}_{\mathbf{m}|_Q}}\big[\mathsf{Z}_{\nu;t}\big(P_1(U),\dots,P_q(U)\big)\big] = \frac{1}{\prod_j \Gamma(\nu_j)} \int_{(0,\infty)^d} x^{\nu-\mathbf{1}}\, \mathcal{F}\big(x^{t_1},\dots,x^{t_q}\big)\, dx,$$

*the Laplace side of the Trailmoment* $(\mathsf{T}_{\nu;t}[(P_i)_i] \otimes_Q \mathrm{Dir}_{\mathbf{m}|_Q})\big(\prod_Q \Delta_Q\big)$. *For real coefficients, two choices of the data with equal joint moment arrays* $M$ *give the vectors* $(P_i(U))_i$ *equal laws, hence equal reads of every kernel.*

## 5.8 Trails

**Definition 5.23** (Trail data)**.** Take $d$ scales, an index $\boldsymbol{\nu} \in \mathbb{C}^d$, the *trail columns* $I_d = (\mathbf{e}_1,\dots,\mathbf{e}_d)$, and *constant columns* $(t_k)_{k\in C}$ with constant coefficients $\lambda_k > 0$ (a constant column of coefficient zero is absent). Put

$$t = I_d \oplus (t_k)_{k\in C}, \qquad \mathbf{c}(y) = (y_1,\dots,y_d,(\lambda_k)_{k\in C}), \qquad X_N = \{N < y_1 < \cdots < y_d\}.$$

The *trail* at the cut $N$ of an input $\phi$ on $\mathbb{R}^d$, carried where every trail coefficient is positive, is the Trailmoment of Definition 5.1 with this coefficient function and cut region,

$$(\mathsf{T}_{\boldsymbol{\nu};\, I_d\oplus(t_k)_{k\in C}}[(y_1,\dots,y_d,(\lambda_k)_{k\in C})]\,\phi)(X_N);$$

with no constant columns it is $(\mathsf{T}_{\boldsymbol{\nu};I_d}[y]\,\phi)(X_N)$. For lattices $\mathbb{L}_1,\dots,\mathbb{L}_d \subset \mathbb{R}$ the input $\mathbb{I}_{\mathbb{L}_1\times\cdots\times\mathbb{L}_d}$ is counting measure on their product, of which the cut region selects the points in $X_N$. The input is *given by increments* $\psi_1,\dots,\psi_d$, measures on $(0,\infty)$, when it is the image $\phi_N$ of $\bigotimes_j \psi_j$ under $s \mapsto y$, $y_i = N + s_1 + \cdots + s_i$; their Laplace transforms are $L_j(U) = \int e^{-sU}\psi_j(ds)$. By transport (Theorem 5.9) the trail of such an input is the Trailmoment on the increment source $(0,\infty)^d$ with the input $\bigotimes_j \psi_j$ and the coefficient function $s \mapsto (N+s_1,\dots,N+s_1+\cdots+s_d,(\lambda_k)_{k\in C})$, over the whole source: there the cut $N$ is carried by the coefficient function.

**Theorem 5.24** (The trail by its increments)**.** *Let the input be given by increments, put* $t = I_d \oplus (t_k)_{k\in C}$, *and let* $U_i = x_i + \cdots + x_d$ *be the sums of the scales from the* $i$*-th on,* $U_{d+1} = 0$. *Wherever the right side converges absolutely, with* $x_i = U_i - U_{i+1}$,

$$\begin{aligned}&(\mathsf{T}_{\boldsymbol{\nu};t}[(y,(\lambda_k)_{k\in C})]\,\phi_N)(X_N)\\ &\qquad = \int_{U_1>\cdots>U_d>0} \prod_{i=1}^{d} \frac{(U_i-U_{i+1})^{\nu_i-1}}{\Gamma(\nu_i)}\, \exp\Big(-\sum_{k\in C}\lambda_k x^{t_k}\Big) e^{-NU_1} \prod_{j=1}^{d} L_j(U_j)\, dU.\end{aligned}$$

*Its relations hold wherever the reads in them converge, the derivatives under the integral being dominated:*

*(i)* Moving the cut, *with the increments fixed:*

$$\partial_N(\mathsf{T}_{\boldsymbol{\nu};t}[(y,(\lambda_k)_{k\in C})]\,\phi_N)(X_N) = -\sum_i \nu_i\,(\mathsf{T}_{\boldsymbol{\nu}+\mathbf{e}_i;t}[(y,(\lambda_k)_{k\in C})]\,\phi_N)(X_N).$$

*(ii)* Ladder.

$$\partial_{\lambda_k}(\mathsf{T}_{\boldsymbol{\nu};t}[(y,(\lambda_h)_{h\in C})]\,\phi_N)(X_N) = -\Gamma_d\begin{bmatrix}\boldsymbol{\nu}+t_k\\ \boldsymbol{\nu}\end{bmatrix}\,(\mathsf{T}_{\boldsymbol{\nu}+t_k;t}[(y,(\lambda_h)_{h\in C})]\,\phi_N)(X_N).$$

*(iii)* Exchange, per scale.

$$(\mathsf{T}_{\boldsymbol{\nu};t}[(y,(\lambda_k)_{k\in C})]\,\phi_N)(X_N) = (\mathsf{T}_{\boldsymbol{\nu}+\mathbf{e}_i;t}[(y,(\lambda_k)_{k\in C})]\,y_i\phi_N)(X_N) + \frac{1}{\nu_i}\sum_{k\in C} t_{k,i}\lambda_k \Gamma_d{}_d\begin{bmatrix}\boldsymbol{\nu}+t_k\\ \boldsymbol{\nu}\end{bmatrix}\,(\mathsf{T}_{\boldsymbol{\nu}+t_k;t}[(y,(\lambda_h)_{h\in C})]\,\phi_N)(X_N).$$

*(iv)* Composition. *If every constant column is supported on one scale, partition $C$ into sets $C_i$ with $\mathrm{supp}(t_k)\subseteq\{i\}$ for $k\in C_i$, assigning every zero column to one scale, and put $t^{(i)} = (1)\oplus(t_{k,i})_{k\in C_i}$. Then*

$$\mathsf{Z}_{\boldsymbol{\nu};t}(y,(\lambda_k)_{k\in C}) = \prod_i \mathsf{Z}_{\nu_i;t^{(i)}}(y_i,(\lambda_k)_{k\in C_i}),$$

*and the trail is the iterated read over $s_1,\dots,s_d>0$, in that order, of $\prod_i \mathsf{Z}_{\nu_i;t^{(i)}}(N+s_1+\cdots+s_i,(\lambda_k)_{k\in C_i})$ against $\bigotimes_i\psi_i$: each one-scale Trailmoment has the next read in its input. With counting increments, every $\psi_j$ the counting input on the positive integers, no constant columns and $N\in b+\mathbb{Z}$, this is the nest of one-scale Trailmoments at successive running cuts, with indices $\nu_1,\dots,\nu_d$ and lattice input $\mathbb{I}_{b+\mathbb{Z}}$, and the trail is $(\mathsf{T}_{\boldsymbol{\nu};I_d}[y]\,\mathbb{I}_{(b+\mathbb{Z})^d})(\{N<y_1<\cdots<y_d\})$.*

## 5.9 Products of Trailmoments

**Theorem 5.25** (The Struffle)**.** *(i) Let $\phi,\psi$ be inputs on $\mathbb{R}$ with any indices, configurations and coefficient functions. If the product below converges with $|\phi|$, $|\psi|$ and the moduli of the kernels, then*

$$\begin{aligned}&(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((N,\infty))\,(\mathsf{T}_{\nu';t'}[\mathbf{c}']\,\psi)((N,\infty))\\ &\quad=\Big(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\big[x\mapsto(\mathsf{T}_{\nu';t'}[\mathbf{c}']\,\psi)((x,\infty))\big]\phi\Big)((N,\infty))\\ &\qquad+\Big(\mathsf{T}_{\nu';t'}[\mathbf{c}']\,\big[x\mapsto(\mathsf{T}_{\nu;t}[\mathbf{c}]\,\phi)((x,\infty))\big]\psi\Big)((N,\infty))\\ &\qquad+\big(\mathsf{T}_{(\nu,\nu');\,t\oplus t'}[\mathbf{c}\oplus\mathbf{c}']\,\phi\wedge\psi\big)((N,\infty)),\end{aligned}$$

*where the* diagonal input *is $\phi\wedge\psi=\sum_x\phi(\{x\})\psi(\{x\})\delta_x$, the sum over the points that both inputs charge, and the diagonal kernel is the Zamma of the concatenation $t\oplus t'$ (Theorem 5.21(i)).*

*(ii) For inputs $\phi_1,\dots,\phi_r$ on $\mathbb{R}$ with kernel data $(\nu_i,t_i,\mathbf{c}_i)$, if $\prod_i(\mathsf{T}_{\nu_i;t_i}[\mathbf{c}_i]\,\phi_i)((N,\infty))$ converges with the total variations and the moduli of the kernels, it is the sum over the ordered partitions $(B_1,\dots,B_s)$ of $\{1,\dots,r\}$ into nonempty blocks of*

$$\int_{N<x_1<\cdots<x_s}\prod_{j=1}^{s}\Big(\prod_{i\in B_j}\mathsf{Z}_{\nu_i;t_i}(\mathbf{c}_i(x_j))\Big)\Big(\bigwedge_{i\in B_j}\phi_i\Big)(dx_j),$$

*with $\bigwedge_{i\in B}\phi_i=\sum_x\prod_{i\in B}\phi_i(\{x\})\delta_x$ for a block of at least two elements and $\phi_i$ for the block $\{i\}$.*

The Struffle orders the points of the inputs along the line; the diagonal is there exactly when the inputs charge common points beyond the cut. It is free in the index, the configuration and

the coefficient function of both factors and in their inputs on the line; the cut region is pinned at one common cut. At one unit exponent with one coefficient function on both factors the diagonal kernel is $\mathsf{Z}_{\nu+\nu';(1)}$.

**Theorem 5.26** (The Shruffle on a line)**.** *For a string $w = (w_1, \dots, w_n)$ of measurable functions on $(0,\infty)$ put* $\operatorname{chain}(w) = \int_{x_1>\cdots>x_n>0} \prod_i w_i(x_i)\,dx$*, defined when* $\operatorname{chain}(|w|) < \infty$*.*

*(i) If* $\operatorname{chain}(|w^{(1)}|), \dots, \operatorname{chain}(|w^{(r)}|)$ *are finite,* $\prod_{i=1}^{r} \operatorname{chain}(w^{(i)}) = \sum_{\varsigma} \operatorname{chain}(\varsigma)$*, the sum over the $(n_1 + \cdots + n_r)!/(n_1! \cdots n_r!)$ interleavings of the $r$ strings, $n_i$ the length of $w^{(i)}$.*

*(ii) Let a trail have its input given by nonnegative increments, with Laplace transforms $L_1, \dots, L_d$, and a cut $N \in \mathbb{R}$ such that the image of the first increment $\psi_1$ under $s \mapsto s + N$ is carried by $(0,\infty)$, as it is when $N \geq 0$; let it have no constant columns and positive integer indices $a_1, \dots, a_d$. Then, both sides finite or infinite together, with $1^a$ for a copies of the constant function $1$, the trail is the chain integral of its string:*

$$(\mathsf{T}_{\mathbf{a};I_d}[y]\,\phi_N)(X_N) = \operatorname{chain}\big(e^{-N\cdot}L_1,\ 1^{a_1-1},\ L_2,\ 1^{a_2-1},\ \dots,\ L_d,\ 1^{a_d-1}\big).$$

*(iii) The product of finitely many such trails is the sum, over the interleavings of their strings, of trails of the same kind at the cut $0$: every interleaving begins with a Laplace transform and is the string of the trail whose increments have the Laplace transforms of the interleaving in their order, each index being one more than the number of constant functions $1$ that follow its Laplace transform, $e^{-N\cdot}L$ being the Laplace transform of the increment moved by $N$.*

The Shruffle orders the scales on the Laplace side. It is free in the nonnegative increments, in the number of scales, and in the cut as long as the first point beyond it is positive; it is pinned at positive integer indices without constant columns, where $(U_i - U_{i+1})^{a-1}/\Gamma(a)$ is the volume of a chain of points. Where a nest of Trailmoments on the input side is a trail given by increments all equal to one measure at the cut 0, as for the counting input cut at 0, the Struffle and the Shruffle expand one product into the same members, and comparing the two expansions gives linear relations among the trails.

**Theorem 5.27** (Trails at frequencies are iterated integrals)**.** *Let $d \geq 1$, $\boldsymbol{\nu} \in \{1, 2, \dots\}^d$ and $\mathbf{a} \in \mathbb{R}^d$, put $\zeta_i = e^{2\pi i a_i}$, and let $\zeta_d \neq 1$ when $\nu_d = 1$. Put $c_i = \zeta_i\zeta_{i+1}\cdots\zeta_d$, $w = |\boldsymbol{\nu}|$, and let $(\ell_1, \dots, \ell_w)$ be the word $c_1^{-1}, 0^{\nu_1-1}, c_2^{-1}, 0^{\nu_2-1}, \dots, c_d^{-1}, 0^{\nu_d-1}$, $0^r$ standing for $r$ letters $0$. Then the heads of the trail over $\{0 < y_1 < \cdots < y_d \leq N\}$ of the lattice input $e^{2\pi i\langle \mathbf{a}, y\rangle}\,\mathbb{I}_{\mathbb{Z}^d}$ at the indices $\boldsymbol{\nu}$ converge as $N \to \infty$, to the trail itself when $\nu_d \geq 2$, and with the trail read as this limit*

$$\big(\mathsf{T}_{\boldsymbol{\nu};I_d}[y]\ e^{2\pi i\langle \mathbf{a}, y\rangle}\,\mathbb{I}_{\mathbb{Z}^d}\big)\big(\{0 < y_1 < \cdots < y_d\}\big) = (-1)^d \int_{0<t_1<\cdots<t_w<1} \prod_{j=1}^{w} \frac{dt_j}{t_j - \ell_j},$$

*an absolutely convergent integral. For $|z| < 1$ the trail with the factor $z^{y_d}$ in its density is $(-1)^d$ times the same integral over $0 < t_1 < \cdots < t_w < z$ along the segment from $0$ to $z$.*

On the Laplace side the input of the trail, given by its increments, has the letters $c_i z/(1 - c_i z)$ in the variable $z = e^{-U}$, singular at the $c_i^{-1}$, and the index of each scale adds the letters 0 of the factor $1^{\nu_i-1}$ of the Shruffle (Theorem 5.26(ii)): the theorem is the chain integral of that string, read in $z$, for inputs with complex densities.

**Theorem 5.28** (The sum over the splittings of an index)**.** *Let $b \in \mathbb{R}$, let $N$ be a cut, and let $m_0 > 0$ be the first point of $b + \mathbb{Z}$ beyond $N$. For an integer $w \geq 3$,*

$$\sum_{a=1}^{w-2} (\mathsf{T}_{(a,w-a);I_2}[y]\, \mathbb{I}_{(b+\mathbb{Z})^2})(\{N < y_1 < \cdots < y_2\}) = (\mathsf{T}_{w-1;(1)}[x]\, \eta\, \mathbb{I}_{b+\mathbb{Z}})((N,\infty)),$$

$$\eta(m) = (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((m-m_0,\infty)) - (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{b+\mathbb{Z}})((m,\infty)) = \sum_{k\geq 1} \Big(\frac{1}{m-m_0+k} - \frac{1}{m+k}\Big),$$

*the index-one values regularized. At every point $m$ of the lattice beyond the cut,*

$$\eta(m) - \frac{1}{m} = (\log\Gamma)'(m) - (\log\Gamma)'(m-m_0+1),$$

*which is positive for $m_0 > 1$, zero for $m_0 = 1$ and negative for $m_0 < 1$. So the sum over the splittings is $(\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{b+\mathbb{Z}})((N,\infty))$ if and only if $m_0 = 1$, that is, for the counting input at a cut in $[0,1)$; it is larger for $m_0 > 1$ and smaller for $m_0 < 1$. For the counting input cut at an integer $c \geq 0$, $m_0 = c+1$ and $\eta(m) = (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((m-c-1,m])$, a head of $c+1$ terms. For a lattice $b + q\mathbb{Z}$ of step $q > 0$ with a positive first point $m_0$ beyond the cut, the sum over $1 \leq a \leq w-2$ of the trails with the indices $(a, w-a)$ on its points is $(\mathsf{T}_{w;(1)}[x]\, \mathbb{I}_{b+q\mathbb{Z}})((N,\infty))$ if and only if $m_0 = q$.*

Summing the trails on two scales over the splittings of an index telescopes in the inner variable, and what is left is one Trailmoment of index one less, with the density $\eta$. The index rises back by one exactly when $\eta(m) = \frac{1}{m}$ at every point, which is exactly when the first point beyond the cut is 1.

**Theorem 5.29** (The Cremona of an increment)**.** *Let $L > 0$ be continuous on $(0,\infty)$ with $\int_x^\infty L < \infty$ for $x > 0$ and $\int_0^\infty L = \infty$, and let $\mathrm{tail}_L(x) = \int_x^\infty L(u)\, du$, the* tail function*, a decreasing bijection of $(0,\infty)$. For a string $w$ whose entries are $L(x)\, dx$ or $dx$, let $\mathrm{chain}_L(w)$ be its chain integral over $x_1 > \cdots > x_n > 0$, largest variable first, and $w^*$ the reversed string with $L(x)\, dx$ and $dx$ swapped. Then $\mathrm{tail}_L \circ \mathrm{tail}_L = \mathrm{id}$ if and only if $L(\mathrm{tail}_L(x))\, L(x) = 1$ for every $x$, and in that case $\mathrm{chain}_L(w) = \mathrm{chain}_L(w^*)$ for every string, finite or infinite together.*

The Cremona uses one property of the increment: its tail function $\mathrm{tail}_L$ is an involution of the scale, so that the substitution $y = \mathrm{tail}_L(x)$ carries $dx$ to $L(y)\, dy$ as well as $L(x)\, dx$ to $dy$. For the counting increments, $L(u) = 1/(e^u - 1)$ and the involution is the curve $e^{-x} + e^{-y} = 1$.

## 5.10 Convolutions of inputs

**Definition 5.30** (The two convolutions)**.** Let $\phi$ and $\psi$ be inputs carried by $[0,\infty)$, and $\phi \otimes \psi$ their product on $[0,\infty)^2$. The *additive convolution* $\phi \boxplus \psi$ is the image of $\phi \otimes \psi$ under $(x,y) \mapsto x + y$. When both inputs are carried by $(0,\infty)$, the *multiplicative convolution* $\phi \boxtimes \psi$ is the image of $\phi \otimes \psi$ under $(x,y) \mapsto xy$. Both are images in the sense of Definition 5.4: for inputs with complex densities they are used only in integrals that converge absolutely on the product $\phi \otimes \psi$. The multiplicative convolution of two $\sigma$-finite measures need not be $\sigma$-finite; it is used only where an integral of Theorem 5.31(ii) below converges absolutely, and there it is $\sigma$-finite, its kernel being positive and finite on $(0,\infty)$. For lattice inputs with densities, $(f\, \mathbb{I}_{\mathbb{Z}}) \boxplus (g\, \mathbb{I}_{\mathbb{Z}})$ and $(f\, \mathbb{I}_{\mathbb{Z}}) \boxtimes (g\, \mathbb{I}_{\mathbb{Z}})$ denote the convolutions of the parts beyond the cut 0: for densities $f$ and $g$ on the positive integers, the first has the density $\sum_{j+k=n} f(j)g(k)$ at $n$ and the second the density $\sum_{jk=n} f(j)g(k)$; likewise for lattice inputs on other cosets. For an input carried by $[0,\infty)$ write $\mathcal{F}_\phi(u) = \int_{[0,\infty)} e^{-ux}\, \phi(dx)$, the kernel being 1 at the point $x = 0$: it is the Laplace transform beyond every cut $N < 0$, and for an input carried by $(0,\infty)$ the Laplace transform beyond the cut 0.

**Theorem 5.31** (Two convolutions)**.** *Let $\phi$ on $Y_1$ and $\psi$ on $Y_2$ be inputs with coefficient functions $c$ and $c'$ and cut regions $Y_1'$ and $Y_2'$, and write $c+c'$ and $c\,c'$ for the functions $(y_1, y_2) \mapsto c(y_1)+c'(y_2)$ and $(y_1, y_2) \mapsto c(y_1)\,c'(y_2)$ on $Y_1 \times Y_2$.*

*(i)* The kernel $\mathsf{Z}_{1;(1,0)}(1, \cdot)$ on a sum. *If the two Trailmoments on the right converge,*

$$\big(\mathsf{T}_{1;(1,0)}[(1, c+c')]\ \phi \otimes \psi\big)(Y_1' \times Y_2') = (\mathsf{T}_{1;(1,0)}[(1, c)]\,\phi)(Y_1')\ (\mathsf{T}_{1;(1,0)}[(1, c')]\,\psi)(Y_2').$$

*For inputs carried by $[0, \infty)$ and $u \in \mathbb{C}$ such that $e^{-x \operatorname{Re} u}$ is integrable for $|\phi|$ and for $|\psi|$, $\mathcal{F}_{\phi \boxplus \psi}(u) = \mathcal{F}_\phi(u)\,\mathcal{F}_\psi(u)$.*

*(ii)* The kernel $\mathsf{Z}_{\nu;I_\mathcal{E}}$ on a product. *For coefficient functions $\mathbf{c}$ and $\mathbf{c}'$ with values in $(0, \infty)^\mathcal{E}$, $\mathbf{c}\,\mathbf{c}'$ their coordinatewise product and $\nu \in \mathbb{C}^\mathcal{E}$, if the two Trailmoments on the right converge,*

$$\big(\mathsf{T}_{\nu;I_\mathcal{E}}[\mathbf{c}\,\mathbf{c}']\ \phi \otimes \psi\big)(Y_1' \times Y_2') = (\mathsf{T}_{\nu;I_\mathcal{E}}[\mathbf{c}]\,\phi)(Y_1')\ (\mathsf{T}_{\nu;I_\mathcal{E}}[\mathbf{c}']\,\psi)(Y_2').$$

*For inputs carried by $(0, \infty)$,*

$$(\mathsf{T}_{\nu;(1)}[x]\,\phi \boxtimes \psi)((0,\infty)) = (\mathsf{T}_{\nu;(1)}[x]\,\phi)((0,\infty))\ (\mathsf{T}_{\nu;(1)}[x]\,\psi)((0,\infty)),$$
$$\mathcal{F}_{\phi \boxtimes \psi}(u) = \int_{(0,\infty)} \mathcal{F}_\psi(ux)\,\phi(dx),$$

*the first when its right side converges and the second when $\iint e^{-xy \operatorname{Re} u}\,|\phi|(dx)\,|\psi|(dy) < \infty$.*

*(iii)* Converses. *Let $g$ be a continuous complex-valued function on $(0, \infty)^\mathcal{E}$, not identically zero, with $g(s+s') = g(s)\,g(s')$ for all $s, s' \in (0, \infty)^\mathcal{E}$. Then $g(s) = e^{-\langle u, s\rangle}$ for one $u \in \mathbb{C}^\mathcal{E}$. Let $h$ be a continuous complex-valued function on $(0, \infty)^\mathcal{E}$, not identically zero, with $h(x\,x') = h(x)\,h(x')$ for the coordinatewise product. Then $h = \mathsf{Z}_{\nu;I_\mathcal{E}}$ for one $\nu \in \mathbb{C}^\mathcal{E}$. The two hypotheses are the product relations of (i) and of (ii), for a kernel $g$ or $h$ in the place of the kernel there, on every pair of Dirac measures.*

The two convolutions add and multiply the points of two inputs, and each has one kernel that turns it into a product: on sums the kernel $\mathsf{Z}_{1;(1,0)}(1, \cdot)$ of a term at exponent zero beside a unit term, which is $e^{-c}$; on products the kernel $\mathsf{Z}_{\nu;I_\mathcal{E}}$ of the unit configuration, the power $\mathbf{c} \mapsto \mathbf{c}^{-\nu}$. Both statements are free in the two inputs, the two coefficient functions and the two cut regions, and each is pinned in its kernel; (iii) is the converse of that pin.

# Chapter 6

# Rational words and live inputs

A rational word acting on a Trailmoment through its input is moved, by partial fractions and the shift of the cut, onto Trailmoments of shifted inputs at the unit configuration; the pole columns of the word then separate what is a Trailmoment from what is a finite difference. An input is *live* when the values of its density are themselves read from the serif family, a Breta or a Gramma at a shifting slot, and the relations of that member give the input a recurrence of finite order. Every identity for rational words and live inputs is a finite identity first, at finite cuts; its infinite form uses the displayed convergence, or the common finite upper cut of the closing it comes from.

**Definition 6.1** (Rational words and their transport)**.** A *rational word* is a function

$$R(x) = \sum_{k=0}^{k_{\max}} \sum_{j=1}^{J_k} c_{k,j}(x+k)^{-j},$$

kept with its partial-fraction presentation. For the data $(\nu, t, \mathbf{c}, \phi)$ of a Trailmoment on a line and the cut region $(N, \infty)$, the rational word acts through the input,

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,(R\phi))((N,\infty)) = \int_{(N,\infty)} R(x)\,\mathsf{Z}_{\nu;t}(\mathbf{c}(x))\,\phi(dx).$$

Write $(\mathrm{Sh}^h f)(m) = f(m+h)$, $\Delta H(m) = H(m+1) - H(m)$, $\tau_k(x) = x + k$ and $\phi_k = (\tau_k)_*\phi$. On the counting input at index zero, where $\mathsf{Z}_{0;(1)}(x) = 1$, this is $(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{m>N} R(m)f(m)$ for an integer cut $N \geq 0$.

**Theorem 6.2** (Shifted inputs)**.** *With* $g_k(y) = \mathsf{Z}_{\nu;t}(\mathbf{c}(y-k))f(y-k)$,

$$(\mathsf{T}_{\nu;t}[\mathbf{c}]\,(Rf\phi))((N,\infty)) = \sum_{k,j} c_{k,j}\,(\mathsf{T}_{j;(1)}[x]\,(g_k\phi_k))((N+k,\infty)).$$

*On the counting input at integer cuts* $M > N \geq 0$*,*

$$(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,M]) = \sum_{k,j} c_{k,j}\,(\mathsf{T}_{j;(1)}[x]\,(\mathrm{Sh}^{-k} f)\,\mathbb{I}_{\mathbb{Z}})((N+k,M+k]),$$

*and when the Trailmoments converge individually*

$$(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{k,j} c_{k,j}\,(\mathsf{T}_{j;(1)}[x]\,(\mathrm{Sh}^{-k} f)\,\mathbb{I}_{\mathbb{Z}})((N+k,\infty)).$$

*When only the combination converges, the common finite upper cut is kept and the limit is taken in the combination. For $f = \mathbf{1}$, $\sum_k c_{k,1} = 0$ and an integer $m_0 \geq 1$, with the index-one Trailmoments regularized,*

$$\begin{aligned}(\mathsf{T}_{0;(1)}[x]\, R\, \mathbb{I}_{\mathbb{Z}})((m_0-1,\infty)) &= \sum_{k,s} c_{k,s}\, (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((m_0-1+k,\infty)) \\ &= \sum_{s\geq 2}\sum_{k} c_{k,s}\, (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((m_0-1+k,\infty)) - \sum_{k} c_{k,1}\, (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0, m_0-1+k]) \\ &= \int_0^\infty \Big[\sum_{k,s} \frac{c_{k,s}}{\Gamma(s)} u^{s-1} e^{-ku}\Big] \frac{e^{-(m_0-1)u}}{e^u-1}\, du.\end{aligned}$$

**Theorem 6.3** (Common cut and heads)**.** *For a common cut $N' \geq N + k_{\max}$,*

$$\begin{aligned}(\mathsf{T}_{j;(1)}[x]\, (\mathrm{Sh}^{-k} f)\mathbb{I}_{\mathbb{Z}})((N+k,\infty)) &= (\mathsf{T}_{j;(1)}[x]\, (\mathrm{Sh}^{-k} f)\mathbb{I}_{\mathbb{Z}})((N',\infty)) \\ &\quad + (\mathsf{T}_{j;(1)}[x]\, (\mathrm{Sh}^{-k} f)\mathbb{I}_{\mathbb{Z}})((N+k,N']),\end{aligned}$$

*the head being $\sum_{N+k<\ell\leq N'} \ell^{-j} f(\ell-k)$ (Proposition 5.3). For $R = -g'$ with $g(a) = \sum_{h,s} c_{h,s}(a+h)^{-s} = O(a^{-2})$ and $C_s = \sum_h c_{h,s}$, under individual convergence,*

$$\begin{aligned}(\mathsf{T}_{0;(1)}[x]\, (-g')f\, \mathbb{I}_{\mathbb{Z}})((N,\infty)) &= \sum_{s} s\, C_s\, (\mathsf{T}_{s+1;(1)}[x]\, f\, \mathbb{I}_{\mathbb{Z}})((N',\infty)) \\ &\quad + \sum_{h,s} s\, c_{h,s}\, (\mathsf{T}_{s+1;(1)}[x]\, (\mathrm{Sh}^{-h} f - f)\, \mathbb{I}_{\mathbb{Z}})((N',\infty)) \\ &\quad + \sum_{h,s} s\, c_{h,s}\, (\mathsf{T}_{s+1;(1)}[x]\, (\mathrm{Sh}^{-h} f)\, \mathbb{I}_{\mathbb{Z}})((N+h,N']).\end{aligned}$$

**Theorem 6.4** (Pole columns are a rational difference quotient)**.** *Let $R(t) = \sum_{k=0}^{k_{\max}} \sum_{s=1}^{w} c_{k,s}(t+k)^{-s}$ with coefficients in a subfield $\mathbb{K}$ of $\mathbb{C}$, $C_s = \sum_k c_{k,s}$, and $H(t) = \sum_{k,s} c_{k,s} \sum_{j=0}^{k-1} (t+j)^{-s}$, a rational function with coefficients in $\mathbb{K}$. Then*

$$R(t) = \sum_{s=1}^{w} C_s t^{-s} + \Delta H(t),$$

*and $R(t) = O(t^{-2})$ exactly when $C_1 = 0$. Conversely a difference $\Delta H$ of a proper rational function with poles on the nonpositive integer lattice has every column sum zero; modulo such differences, the rational words of maximum order $w$ have the basis $t^{-1}, \ldots, t^{-w}$, and those that are $O(t^{-2})$ the basis $t^{-2}, \ldots, t^{-w}$. If the poles fall in several cosets of $\mathbb{Z}$, each coset has its own columns.*

**Theorem 6.5** (Summation by parts with a live input)**.** *Let $H$ be a function on the integers above $N$ and $f$ a density on them.*

*(i)* $\Delta(Hf)(m) = \Delta H(m) f(m) + H(m+1)\Delta f(m)$, *so at finite cuts $M > N$*

$$\sum_{N<m\leq M} \Delta H(m) f(m) = H(M+1)f(M+1) - H(N+1)f(N+1) - \sum_{N<m\leq M} H(m+1)\Delta f(m).$$

*(ii) If the upper boundary vanishes and the series converges, $\sum_{m>N} \Delta H(m) f(m) = -H(N+1)f(N+1) - \sum_{m>N} H(m+1)\Delta f(m)$.*

*(iii) For a Trailmoment read as a function of its cut,* $\Delta\,(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((m,\infty)) = -(m+1)^{-p}g(m+1)$ *(Proposition 5.3), and hence*

$$\sum_{m>N}\Delta H(m)\,(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((m,\infty)) \\ = -H(N+1)\,(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((N+1,\infty)) + (\mathsf{T}_{p;(1)}[x]\,Hg\,\mathbb{I}_{\mathbb{Z}})((N+1,\infty)),$$

*with one fewer Trailmoment in the input.*

*(iv) With* $R$, $C_s$, $H$ *as in Theorem 6.4 and the index-one Trailmoment regularized (Definition 5.2),*

$$\lim_{M\to\infty}\Big((\mathsf{T}_{0;(1)}[x]\,R\,\mathbb{I}_{\mathbb{Z}})((N,N+M]) - C_1\log M\Big) = \sum_{s=1}^{w} C_s\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) - H(N+1);$$

*when* $C_1 = 0$*, that is when* $R = O(t^{-2})$*, the left side is* $(\mathsf{T}_{0;(1)}[x]\,R\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$ *and the sum starts at* $s = 2$*. For a general density* $f$*, when the terms converge and the upper boundary vanishes,*

$$(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{s=1}^{w} C_s\,(\mathsf{T}_{s;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) \\ - H(N+1)f(N+1) - \sum_{m>N} H(m+1)\Delta f(m).$$

**Theorem 6.6** (Translation and the terminating descent)**.** *For the counting input, each inner Trailmoment below is read at its displayed running cut.*

*(i)* $(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((y-k,\infty)) = (\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((y,\infty)) + (\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((y-k,y])$*, the head being* $\sum_{a=0}^{k-1}(y-a)^{-p}g(y-a)$ *(Proposition 5.3).*

*(ii)*

$$(\mathsf{T}_{j;(1)}[x]\,\big[x\mapsto(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((x-k,\infty))\big]\,\mathbb{I}_{\mathbb{Z}})((N+k,\infty)) \\ = (\mathsf{T}_{j;(1)}[x]\,\big[x\mapsto(\mathsf{T}_{p;(1)}[x]\,g\,\mathbb{I}_{\mathbb{Z}})((x,\infty))\big]\,\mathbb{I}_{\mathbb{Z}})((N+k,\infty)) \\ + \sum_{a=0}^{k-1}(\mathsf{T}_{p;(1)}[x]\,(x+a)^{-j}g\,\mathbb{I}_{\mathbb{Z}})((N+k-a,\infty)).$$

*(iii) For positive integers* $p, j$*, each product* $z^{-p}(z+a)^{-j}$ *has finite rational partial fractions with poles at* $0$ *and* $-a$ *only. The recursion therefore terminates on every finite nest of Trailmoments of the counting input, each taking the next as a function of its cut: it returns finitely many such nests, finite shifted copies of* $g$ *and their heads, keeps the selected cuts, lowers at each correction the number of Trailmoments nested in the remaining input, and uses only rational operations on the pole data.*

*(iv) If* $R(-L-t) = -R(t)$*, then* $\sum_k c_{k,j} = 0$ *for every even* $j$*, and for a density* $f$ *that is a nest of* $d$ *Trailmoments of the counting input, at a common cut* $C$*,*

$$(\mathsf{T}_{0;(1)}[x]\,Rf\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{j}\Big(\sum_{k} c_{k,j}\Big)(\mathsf{T}_{j;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((C,\infty)) + E_d,$$

*with $E_d$ a sum of terms with at most $d$ nested Trailmoments, so the terms with $d+1$ nested Trailmoments carry only odd outer indices, starting at three when $\sum_k c_{k,1} = 0$. For $f = \mathbf{1}$, an integer $m_0 \geq 1$ and $R(-k_{\max} - t) = -R(t)$, $c_{k_{\max}-k,s} = -(-1)^s c_{k,s}$, and at every even index $s$ the Trailmoments combine into the finite sum*

$$\sum_k c_{k,s}\Big[(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((m_0 - 1 + k, \infty)) - (\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((m_0 - 1, \infty))\Big] \\ = -\sum_k c_{k,s}\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((m_0 - 1, m_0 - 1 + k]).$$

**Proposition 6.7** (Zamma kernels and general measures)**.** *For counting measure the product rule applies to the kernel times the density, $g(m) = \mathsf{Z}_{\nu;t}(\mathbf{c}(m))f(m)$, with $\Delta g(m) = \mathsf{Z}_{\nu;t}(\mathbf{c}(m+1))\Delta f(m) + f(m)\Delta[\mathsf{Z}_{\nu;t}(\mathbf{c}(m))]$. For a general measure $\mu$ restricted to $(N, M]$, $\int \Delta H\, g\, d\mu_{N,M} = \int H\, d\big((\tau_1)_*(g\mu_{N,M}) - g\mu_{N,M}\big)$.*

## 6.1 Live inputs

**Theorem 6.8** (A Breta input with a recurrence of finite order)**.** *For $\operatorname{Re} a, \operatorname{Re} b > 0$, real $q_i < 1$ with $q_i \neq 0$, and $s_i \in \mathbb{C}$, let*

$$f(m) = \int_0^1 x^{m+a-1}(1-x)^{b-1} \prod_{i=1}^{r} (1 - q_i x)^{-s_i}\, dx \qquad (m \geq 0).$$

*For $\operatorname{Re} s_i > 0$ this is a Breta in which $m$ sits in the shape index: on the bunch $\{1, 2\}$ with the shape $u = (x, 1-x)$, the kernel terms $\rho u_1$ and $\rho u_2$ with unit coefficients and the letters $P_i(\rho, u) = \rho\big((1-q_i)u_1 + u_2\big)$, Proposition 3.6 at the scale index $1 + |\mathbf{s}|$ gives*

$$f(m) = \Gamma(1 + |\mathbf{s}|)\, \mathsf{B}^{\mathcal{P}}_{((m+a,\,b),\,\mathbf{s}),\,(1+|\mathbf{s}|,\,\mathbf{s});\,(\widetilde{\pi},\widetilde{\ell})}(\mathbf{g} \oplus \mathbf{C}).$$

*Then $f(m) = O(m^{-\operatorname{Re} b})$, and with $Q = \prod_i (1 - q_i x)$, $Q_i = Q/(1 - q_i x)$ and*

$$B_m(x) = \big[(m+a)(1-x) - bx\big]Q(x) + x(1-x)\sum_i (s_i - 1) q_i Q_i(x) = \sum_{\ell=0}^{r+1} B_\ell(m) x^\ell,$$

*the density obeys $\sum_{\ell=0}^{r+1} B_\ell(m) f(m+\ell) = 0$, with $B_0(m) = m + a$ and $B_{r+1}(m) = (-1)^{r+1}(\prod_i q_i)(m + a + b + r - \sum_i s_i)$. If $a + b + r - \sum_i s_i \notin -\mathbb{N}$, both extreme coefficients are nonzero for every $m \geq 0$, and all shifts reduce to $(f(m), \ldots, f(m+r))$ with coefficients rational in $m$.*

**Theorem 6.9** (A fixed set of input-dependent evaluations)**.** *With $f$, $a$, $b$, $r$, $q_i$ and $s_i$ as in Theorem 6.8, write in this statement $Q(x) = \prod_i (1 - q_i x)$, $C(x) = (1-x)Q(x)$, $\eta(x) = x^{a-1}(1-x)^{b-1}\prod_i (1 - q_i x)^{-s_i}$ and*

$$B(x) = \big[a(1-x) - bx\big]Q(x) + x(1-x)\sum_i (s_i - 1)\, q_i\, \frac{Q(x)}{1 - q_i x},$$

*so that $(xC\eta)' = B\eta$; let $B_\ell$, $C_\ell$ and $Q_\ell$ be the coefficients of $x^\ell$.*

*(i) For each maximum index $A$, every shift reduces to $f(N+1),\dots,f(N+r+1)$ and the Trailmoments $(\mathsf{T}_{p;(1)}[x+k]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{m>N} f(m)(m+k)^{-p}$, $1\le p\le A$, $0\le k\le r$, together with $(\mathsf{T}_{p;(1)}[x+a]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$ when $a$ is an integer at least $r+1$.*

*(ii) For $k\ge r+1$ and $p\ge 2$,*

$$\sum_{\ell=0}^{r+1}(B_\ell - kC_\ell)\,(\mathsf{T}_{p;(1)}[x+k-\ell]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) + \sum_{\ell=0}^{r+1} C_\ell\,(\mathsf{T}_{p-1;(1)}[x+k-\ell]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$$
$$= \sum_{\ell=0}^{r+1}\sum_{v=1}^{\ell}\Big[\frac{B_\ell - kC_\ell}{(N+v+k-\ell)^p} + \frac{C_\ell}{(N+v+k-\ell)^{p-1}}\Big]f(N+v),$$

*and at $p=1$,*

$$\sum_{\ell=0}^{r+1}(B_\ell - kC_\ell)\,(\mathsf{T}_{1;(1)}[x+k-\ell]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$$
$$= \sum_{\ell=0}^{r+1}\sum_{v=1}^{\ell}\frac{B_\ell - kC_\ell}{N+v+k-\ell}f(N+v) - \sum_{\ell=0}^{r} Q_\ell f(N+1+\ell).$$

*(iii) Cuts move by $(\mathsf{T}_{p;(1)}[x+k]\,f\,\mathbb{I}_{\mathbb{Z}})((N+1,\infty)) = (\mathsf{T}_{p;(1)}[x+k]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) - f(N+1)/(N+1+k)^p$ (Proposition 5.3).*

*(iv) For rational data, if $c_*$ clears the coefficients of $B$, $C$ and $Q$, then $d^A_{N+J}\prod_{r+1\le k\le J,\ k\ne a}|c_*(a-k)|^A$ clears all shifts through $J$ and indices through $A$.*

**Theorem 6.10** (The row of a rational function against an input with a recurrence)**.** *Let $f$ be a density on the integers $m>N$, $N\in\mathbb{Z}$, with $\sum_{m>N}|f(m)|(1+|m|)^d<\infty$, that satisfies $\sum_{i=0}^{\tau} p_i(m)\,f(m+i)=0$ for every integer $m>N$, with $\tau\ge 1$ and $p_i\in\mathbb{Z}[m]$ of degree at most $d$. Let $u\ge 1$ be an integer and $h_*\in\frac{1}{u}\mathbb{Z}$ with $N+1+h_*>0$. In this statement write, for integers $s\ge 1$, $r\ge 0$ and $h\in h_*+\mathbb{N}$,*

$$T_s(h) = (\mathsf{T}_{s;(1)}[x+h]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{m>N}\frac{f(m)}{(m+h)^s},$$
$$M_r = (\mathsf{T}_{-r;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{m>N} m^r f(m).$$

*(i) For a rational function $R(m) = P(m) + \sum_{h,s} c_{h,s}(m+h)^{-s}$, $P$ a polynomial of degree at most $d$ and finitely many $h\in h_*+\mathbb{N}$, $\sum_{m>N} R(m)f(m) = \sum_r([m^r]P)\,M_r + \sum_{h,s} c_{h,s}\,T_s(h)$: one coordinate for each index and shift.*

*(ii) Let $J\ge\tau$ with $p_0(-h_*-j)\ne 0$ for $\tau\le j\le J$. Then for $s\ge 1$ and $\tau\le j\le J$, $T_s(h_*+j)$ is a rational combination of the $T_{s'}(h_*+i)$, $1\le s'\le s$, $0\le i<\tau$, of the moments $M_r$, $0\le r\le d-1$, and of the values $f(N+1),\dots,f(N+\tau)$.*

*(iii) For every $w\ge 1$ the coefficients of (ii) of all the $T_s(h_*+j)$, $1\le s\le w$, $\tau\le j\le J$, are cleared by*

$$d^w_{u(N+h_*+J)}\prod_{j=\tau}^{J}\big|u^d p_0(-h_*-j)\big|^w.$$

*A rational function with poles in several cosets of $\mathbb{Z}$ takes one such reduction for each coset.*

Theorem 6.9 is the case of the Breta input of Theorem 6.8, read through its differential equation; here the input is any density with a recurrence, the live inputs of the closings along a ray among them (Theorem 43.1). The clearing has two parts: the heads released by the moves of the cut, cleared by a power of $d_n$ on the scale $u$ of the coset, and the values of the first coefficient of the recurrence at the shifts used, raised to the power of the index, which along a ray is the frame of the recurrence (Proposition 38.11).

**Theorem 6.11** (A Gramma input with a recurrence of finite order)**.** *Let a one-scale configuration $t$ have positive integer exponents with greatest exponent $L$, the term at exponent $L$ present. For $\lambda, c > 0$ put $g(m) = \mathsf{G}_{\lambda+m;t}(\boldsymbol{\xi})/\Gamma(c+m)$. Then*

$$(m+\lambda)g(m) = \sum_k t_k \xi_k (m+c)_{t_k}\, g(m+t_k),$$

*a shift relation of order $L$, the exchange relation of the Gramma (Theorem 2.4(ii)) divided by $\Gamma(c+m)$; for $L \geq 2$, $\log g(m) \leq -(1-1/L)m\log m + O(m)$, so $g$ decays faster than every inverse power.*

# Part II

# Tables and families

# Chapter 7

# Tables

## 7.1 Characters, tables, lists

**Definition 7.1** (Characters, tables, lists)**.** The *characters* are a finite set $\mathcal{E}$, whose elements are the *marks*; each mark $e$ carries a positive variable $\alpha_e$, and its name carries nothing. A *table* on $\mathcal{E}$ is a nonempty family $\mathcal{B}$ of $r$-element subsets of $\mathcal{E}$, the *bases*, with the *exchange property*: for bases $T, T'$ and $a \in T \setminus T'$ there is $b \in T' \setminus T$ with $(T \setminus \{a\}) \cup \{b\}$ a basis. Each basis carries a weight $\theta_T > 0$; $r$ is the *rank* and $r^* = n - r$ the *corank*. A *list* is a table whose weights are all 1; it is specified by its characters and its bases, $(\mathcal{E}, \mathcal{B})$, and by nothing else; for a set $S$ and $0 \le k \le |S|$, $\binom{S}{k}$ is the list on $S$ whose bases are the $k$-subsets of $S$. A mark in no basis is a *loop*, a mark in every basis a *coloop*. The *dual table* has the bases $\mathcal{E} \setminus T$ with the weights $\theta_T$. The *Foreman polynomials* of a table are its basis and cobasis polynomials,

$$K(\alpha) = \sum_{T \in \mathcal{B}} \theta_T \, \alpha^T, \qquad \Psi(\alpha) = \sum_{T \in \mathcal{B}} \theta_T \, \alpha^{\mathcal{E} \setminus T} \qquad (\alpha \in \mathbb{R}^{\mathcal{E}}_{>0});$$

$K(\mathbf{1}) = \Psi(\mathbf{1}) = \sum_T \theta_T$ is the total weight. $K$ and $\Psi$ are the two letters of the table (Section 3.2): they enter a family word through powers (Chapter 8). The *table's gauge* is $\alpha = \mathbf{1}$.

A table carries two descriptors of one object, what is there (the bases, read by $K$) and what is not (their complements, read by $\Psi$), and the marks on which both sit; any two of these determine the third, and the theory keeps both polynomials at every step.

**Proposition 7.2** (One function read twice)**.** *For every table, $\Psi(\alpha) = \alpha^{\mathcal{E}} K(\alpha^{-1})$, and the Foreman polynomials of the dual table are $K^* = \Psi$, $\Psi^* = K$. Inverting the gauge swaps $K$ and $\Psi$ up to the factor $\alpha^{\mathcal{E}}$, and up to scale the gauges fixed by $\alpha \mapsto \alpha^{-1}$ are $\alpha \propto \mathbf{1}$.*

**Definition 7.3** (Deletion and contraction)**.** For a mark $e$ and $R = \mathcal{E} \setminus \{e\}$, the *deletion* has the bases $T \not\ni e$ and the *contraction* the bases $T \setminus \{e\}$, $T \ni e$, both on $R$ with the inherited weights; the deletion is a table when $e$ is not a coloop, and the contraction when $e$ is not a loop. With $K_{\setminus e}, \Psi_{\setminus e}$ and $K_{/e}, \Psi_{/e}$ the sums over these bases, a sum without terms being 0,

$$K = K_{\setminus e} + \alpha_e K_{/e}, \qquad \Psi = \alpha_e \Psi_{\setminus e} + \Psi_{/e}.$$

## 7.2 Circuits and molecules

**Definition 7.4** (Independence, circuits, linking, molecules)**.** A set $S \subseteq \mathcal{E}$ is *independent* in a table if it lies in a basis; its *rank* $r(S)$ is the largest size of an independent subset. A *circuit* is a minimal

set that is not independent, and a *cocircuit* is a minimal set that meets every basis. Two marks are *linked* if they lie in a common circuit of some table of a family on $\mathcal{E}$, a finite sequence of tables on $\mathcal{E}$ (Chapter 8). The *molecules* are the smallest nonempty sets of marks that contain, with each mark, every mark linked to it; a molecule with one mark is a *unit*, and the set of molecules is the *polycule*. A family whose marks form more than one molecule is a *compound*.

**Definition 7.5** (Minors, restriction, contraction)**.** For disjoint $A, D \subseteq \mathcal{E}$ such that some basis contains $A$ and avoids $D$, the *minor* contracting $A$ and deleting $D$ is the table on $\mathcal{E} \setminus (A \cup D)$ with the bases $T \setminus A$, $T \supseteq A$, $T \cap D = \varnothing$, and the weights $\theta_T$; deletion and contraction of one mark are the minors with $D = \{e\}$ and with $A = \{e\}$. For $X \subseteq \mathcal{E}$ the *restriction* $\mathcal{B}|X$ is the table on $X$ with the bases $T \cap X$, $|T \cap X| = r(X)$, and the *contraction* $\mathcal{B}/X$ is the table on $\mathcal{E} \setminus X$ with the bases $T \setminus X$ of the same $T$; when the weights of these $T$ factor as $\theta_T = \theta'_{T\cap X}\theta''_{T\setminus X}$, $\mathcal{B}|X$ and $\mathcal{B}/X$ carry $\theta'$ and $\theta''$. When $e$ is not a loop, $\mathcal{B}/\{e\}$ is the contraction of $e$ (Definition 7.3); at a loop $\mathcal{B}/\{e\}$ is the deletion of $e$, while $K_{/e} = \Psi_{/e} = 0$.

**Proposition 7.6** (Duality of minors)**.** *For a table $(\mathcal{B}, \theta)$ on $\mathcal{E}$ with the dual table $\mathcal{B}^*$ and $X \subseteq \mathcal{E}$, $X' = \mathcal{E} \setminus X$,*

$$(\mathcal{B}|X)^* = \mathcal{B}^*/X', \qquad (\mathcal{B}/X)^* = \mathcal{B}^*|X',$$

*the duals taken on $X$ and on $X'$: a basis is at the top of $X$ exactly when its complement is at the top of $X'$ in the dual table. When the weights of the bases at the top of $X$ factor, $\theta_T = \theta'_{T\cap X}\theta''_{T\setminus X}$, the minors of $\mathcal{B}^*$ carry the same weights, $\mathcal{B}^*/X'$ those of $\mathcal{B}|X$ and $\mathcal{B}^*|X'$ those of $\mathcal{B}/X$ on the complemented bases. So $K_{\mathcal{B}^*/X'} = \Psi_{\mathcal{B}|X}$, $\Psi_{\mathcal{B}^*/X'} = K_{\mathcal{B}|X}$, $K_{\mathcal{B}^*|X'} = \Psi_{\mathcal{B}/X}$ and $\Psi_{\mathcal{B}^*|X'} = K_{\mathcal{B}/X}$.*

**Lemma 7.7** (Linking read on the bases)**.** *Marks $e \neq f$ lie in a common circuit of a table exactly when some basis $T$ has $e \in T$, $f \notin T$ and $(T \setminus \{e\}) \cup \{f\}$ a basis.*

**Lemma 7.8** (The circuit lemma)**.** *If $w \in \mathbb{R}^{\mathcal{E}}$ has $\langle w, \mathbf{1}_T\rangle$ constant over the bases of a table, then $w_a = w_b$ for any two marks $a, b$ linked by a circuit of that table. If $\langle w, \mathbf{1}_T\rangle$ is constant over the bases of every table of a family, $w$ is constant on every molecule.*

**Proposition 7.9** (Molecules split every table)**.** *Each molecule is a union of molecules of every table, those of a table being the molecules of that table alone, and every table is the* direct sum *of its restrictions to the molecules: its bases are exactly the unions of one basis of each restriction. When the weights factor across the molecules, $\theta_T = \prod_M \theta_M(T \cap \mathcal{E}_M)$, as they do for a list, the Foreman polynomials factor, $K(\alpha) = \prod_M K_M(\alpha_{\mathcal{E}_M})$ and $\Psi(\alpha) = \prod_M \Psi_M(\alpha_{\mathcal{E}_M})$.*

**Lemma 7.10** (Unions of molecules)**.** *For a table on $\mathcal{E}$ of rank $r$ and $S \subseteq \mathcal{E}$ the following are equivalent: (a) $r(S) + r(\mathcal{E} \setminus S) = r$; (b) every basis meets $S$ in the same number of marks; (c) $S$ is a union of molecules of the table (Definition 7.4).*

**Theorem 7.11** (The Foreman polynomials of a linked table do not factor)**.** *If the circuits of a table link all of its $n \geq 2$ marks, then $K$ and $\Psi$ are irreducible polynomials.*

## 7.3 The Skeleton

**Definition 7.12** (Skeleton counts)**.** The *cocircuit layer* of a table is its set of cocircuits and the *circuit layer* its set of circuits; the two layers together are the table's *Skeleton*. For $S \subseteq \mathcal{E}$, with $r^*(S) = |S| + r(\mathcal{E} \setminus S) - r$ the dual rank,

$$d(S) = |S| - r^*(S), \qquad d^*(S) = |S| - r(S), \qquad c(S) = |S| - d(S) - d^*(S), \qquad \varepsilon(S) = d(S) - d^*(S) :$$

$c(S)$ is the *width* and $\varepsilon(S)$ the *slant* of $S$.

**Theorem 7.13** (The window)**.** *For every $S \subseteq \mathcal{E}$: $d(S) = \min_T |T \cap S|$, $d^*(S) = \min_T |S \setminus T|$, $r(S) = \max_T |T \cap S|$, and $\{|T \cap S| : T \in \mathcal{B}\} = \{d(S), d(S)+1, \dots, r(S)\}$. Hence $c(S) = r(S) - d(S)$ is the width of this window. Moreover $c(\mathcal{E} \setminus S) = c(S)$ and $\varepsilon(S) + \varepsilon(\mathcal{E} \setminus S) = 2r - n$.*

The number of marks of $S$ that a basis occupies ranges over a window of consecutive integers; its floor is $d(S)$, its width is the width $c(S)$, and the same window read on the complements has the floor $d^*(S)$. Every count that the integrals of a family need is read from these two layers.

**Definition 7.14** (Signed indicators and the centred polynomial)**.** For $T \subseteq \mathcal{E}$ the *signed indicator* is $\chi_T = \mathbf{1}_T - \mathbf{1}_{\mathcal{E}\setminus T}$. For $\alpha = e^{2z}$ the *centred polynomial* is

$$\Lambda(z) = \sum_{T \in \mathcal{B}} \theta_T \, e^{\langle z, \chi_T \rangle}.$$

**Theorem 7.15** (One function, two Foreman polynomials)**.** *For $\alpha = e^{2z}$:*

*(i) $K(\alpha) = e^{\langle \mathbf{1}, z \rangle} \Lambda(z)$ and $\Psi(\alpha) = e^{\langle \mathbf{1}, z \rangle} \Lambda(-z)$;*

*(ii) the dual table has the centred polynomial $\Lambda^*(z) = \Lambda(-z)$: duality and inversion of the gauge are $z \mapsto -z$;*

*(iii) for every $v \in \mathbb{R}^{\mathcal{E}}$ constant on each molecule of the table alone, with value $v_M$ on the molecule $M$, $\langle v, \chi_T \rangle = \sum_M v_M (2r(M) - |M|)$ for every basis and*

$$\Lambda(z + v) = \exp\Big( \sum_M v_M \big(2r(M) - |M|\big) \Big) \Lambda(z);$$

*these are the only $v$ for which $\Lambda(z + \lambda v)/\Lambda(z)$ does not depend on $z$ for every $\lambda$, and at $v = \lambda \mathbf{1}$ the factor is $e^{\lambda(2r-n)}$.*

**Theorem 7.16** (The inner moment generating function)**.** *For a gauge $\alpha = e^{2z}$ the* basis law *and the* cobasis law *at $\alpha$ are the laws on $\mathcal{B}$*

$$\mathbb{P}_\alpha(T) = \frac{\theta_T \alpha^T}{K(\alpha)} = \frac{\theta_T e^{\langle z, \chi_T \rangle}}{\Lambda(z)}, \qquad \mathbb{P}^*_\alpha(T) = \frac{\theta_T \alpha^{\mathcal{E}\setminus T}}{\Psi(\alpha)} = \mathbb{P}_{\alpha^{-1}}(T).$$

*Then*

$$\frac{\Lambda(z+w)}{\Lambda(z)} = \mathbb{E}_{T \sim \mathbb{P}_\alpha}\big[e^{\langle w, \chi_T \rangle}\big], \qquad \nabla \log \Lambda(z) = \mathbb{E}_{T \sim \mathbb{P}_\alpha}[\chi_T] = 2R(\alpha) - \mathbf{1},$$

*where $R_e(\alpha) = \alpha_e \partial_e \log K(\alpha) = \mathbb{P}_\alpha(e \in T)$ is the* occupation*, and the derivatives of $\log \Lambda$ of order $k$ at $z$ are the joint cumulants of order $k$ of $\chi_T$ under $\mathbb{P}_\alpha$. The* even *and* odd statistics *$E(z) = \log \Lambda(z) + \log \Lambda(-z)$ and $O(z) = \log \Lambda(z) - \log \Lambda(-z)$ satisfy $e^{E(z)} = K\Psi/\alpha^{\mathcal{E}}$, $e^{O(z)} = K/\Psi$, and, for every $w \in \mathbb{R}^{\mathcal{E}}$,*

$$e^{E(z+w)} = e^{E(z)} \, \mathbb{E}_{(T,T') \sim \mathbb{P}_\alpha \otimes \mathbb{P}^*_\alpha}\big[e^{\langle 2w, \mathbf{1}_T - \mathbf{1}_{T'} \rangle}\big] :$$

*$E(z + \cdot) - E(z)$ is the cumulant generating function of $\chi_T - \chi_{T'}$ for independent $T \sim \mathbb{P}_\alpha$ and $T' \sim \mathbb{P}^*_\alpha$; at the table's gauge $e^{E(0)} = K(\mathbf{1})^2$. For table $j$ of a family the two laws at $\alpha$ are $\mathbb{P}_{j,\alpha}$ and $\mathbb{P}^*_{j,\alpha} = \mathbb{P}_{j,\alpha^{-1}}$, the occupation is $R_j(\alpha)$, and the* dual occupation *$R^*_j(\alpha)$, with $R^*_{j,e}(\alpha) = \mathbb{P}^*_{j,\alpha}(e \notin T) = 1 - R_{j,e}(\alpha^{-1})$, is the occupation of the dual table. At the table's gauge the two laws coincide, $\mathbb{P}^*_{j,\mathbf{1}} = \mathbb{P}_{j,\mathbf{1}}$.*

The centred polynomial is a moment generating function: its terms at $z$, divided by $\Lambda(z)$, are the law of a random basis, and its logarithm generates the cumulants of the signed indicator. The basis law and the cobasis law at a gauge are one law read at $\alpha$ and at $\alpha^{-1}$, and at every gauge the product of the two evaluations generates the difference of a basis drawn from the basis law and an independent basis drawn from the cobasis law.

**Definition 7.17** (Weight law, indicators, levels, cumulants)**.** The *weight law* of table $j$ is its basis law at the table's gauge, $\mathbb{P}_{j,\mathbf{1}}(T) = \theta_{j,T}/K_j(\mathbf{1})$ on $\mathcal{B}_j$; for a list it is uniform. For a random basis $T$ the *indicators* are $X_e = \mathbf{1}[e \in T]$, and for a set $C$ of marks the *level* is $N_C = |T \cap C| = \sum_{e\in C} X_e$. For functions $Y_1, \dots, Y_k$ of $T$, $\kappa_{j,\alpha}(Y_1, \dots, Y_k)$ is their joint cumulant under $\mathbb{P}_{j,\alpha}$, and $Y^{\times a}$ in its argument lists $Y$ $a$ times; along a string $w = (e_1, \dots, e_k)$ of marks, $\kappa_{j,\alpha}(w) = \kappa_{j,\alpha}(X_{e_1}, \dots, X_{e_k}) = \partial_{y_{e_1}} \cdots \partial_{y_{e_k}} \log K_j(e^y)$ at $y = \log \alpha$. With one table the index $j$ is dropped: $\mathbb{P}_\alpha$, $R(\alpha)$, $\kappa_\alpha$.

**Theorem 7.18** (The Skeleton is the Newton data of $\Lambda$)**.** *For every $S \subseteq \mathcal{E}$ the values $\langle \mathbf{1}_S, \chi_T \rangle$, $T \in \mathcal{B}$, fill $\{2j - |S| : \ d(S) \le j \le r(S)\}$. The Newton polytope of $\Lambda$ is $2P - \mathbf{1}$ with $P = \operatorname{conv}\{\mathbf{1}_T\}$ the base polytope; its width in the direction $\mathbf{1}_S$ is $2c(S)$ and the midpoint of its extent in that direction is $\varepsilon(S)$.*

**Proposition 7.19** (Loops and coloops)**.** *If $e$ is a loop, $\Lambda(z) = e^{-z_e}\Lambda'(z|_{\mathcal{E}\setminus e})$ with $\Lambda'$ the centred polynomial of the same bases on $\mathcal{E} \setminus e$; if $e$ is a coloop, $\Lambda(z) = e^{z_e}\Lambda'(z|_{\mathcal{E}\setminus e})$ with $\Lambda'$ that of the bases $T \setminus e$.*

## 7.4 Multivectors of any tensor rank

**Definition 7.20** (Multivectors and tensor rank)**.** Let $\mathbb{F}$ be $\mathbb{R}$ or $\mathbb{C}$. A *multivector* of a table of rank $r$ is an $\boldsymbol{\omega} \in \bigwedge^r \mathbb{F}^{\mathcal{E}}$ with $|\boldsymbol{\omega}_T|^2 = \theta_T$ for $T \in \mathcal{B}$ and $\boldsymbol{\omega}_T = 0$ otherwise; $\boldsymbol{\omega}_T = \sqrt{\theta_T}$ is one. Its *tensor rank* $\operatorname{rk}(\boldsymbol{\omega})$ is the least number of decomposable elements of $\bigwedge^r \mathbb{F}^{\mathcal{E}}$ with sum $\boldsymbol{\omega}$, written $\boldsymbol{\omega} = \sum_{k=1}^{\operatorname{rk}(\boldsymbol{\omega})} \boldsymbol{\omega}^{(k)}$, $\boldsymbol{\omega}^{(k)}_T = \det A^{(k)}_T$ with *slices* $A^{(k)} \in \mathbb{F}^{r\times n}$. The table is *tensorminental of tensor rank $q$* over $\mathbb{F}$ when $q$ is the least tensor rank of its multivectors, and *determinantal* when $q = 1$. The *dual multivector* is the Hodge complement $\boldsymbol{\omega}^* \in \bigwedge^{n-r} \mathbb{F}^{\mathcal{E}}$, $|\boldsymbol{\omega}^*_{\mathcal{E}\setminus T}| = |\boldsymbol{\omega}_T|$, and $\operatorname{rk}(\boldsymbol{\omega}^*) = \operatorname{rk}(\boldsymbol{\omega})$.

Every table has multivectors, and every multivector has a tensor rank; tensor rank one, the determinantal case, is where a single matrix carries the table, and it is one case of the statements on multivectors, which hold at every tensor rank.

**Proposition 7.21** (The letters of a tensorminental table)**.** *For a decomposition $\boldsymbol{\omega} = \sum_k \boldsymbol{\omega}^{(k)}$ into decomposable elements with slices $A^{(k)}$ and $D = \operatorname{diag}(\alpha)$,*

$$K(\alpha) = \sum_{k,l} \det\left(A^{(k)} D A^{(l)\mathsf{H}}\right) = \mathbf{1}^{\mathsf{H}} \mathbf{G}(\alpha) \mathbf{1}, \qquad \mathbf{G}_{kl}(\alpha) = \det\left(A^{(k)} D A^{(l)\mathsf{H}}\right),$$

*and $\Psi$ is the same expression for the slices of the dual multivector. $\mathbf{G}(\alpha)$ is the Gram matrix of the $\boldsymbol{\omega}^{(k)}$ for $\langle \eta, \xi \rangle_\alpha = \sum_T \alpha^T \eta_T \overline{\xi_T}$, Hermitian and positive semidefinite.*

**Theorem 7.22** (Tensor rank is hereditary)**.** *For disjoint $A, D \subseteq \mathcal{E}$ with $A$ extending to a basis avoiding $D$, $\boldsymbol{\omega}_{A|D} := (\mathbf{e}_A \lrcorner \boldsymbol{\omega})|_{\mathcal{E}\setminus(A\cup D)}$, the restriction of the interior product of $\boldsymbol{\omega}$ with the wedge $\mathbf{e}_A$ of the unit vectors of the marks of $A$, $(\mathbf{e}_A \lrcorner \boldsymbol{\omega})_X = \pm \boldsymbol{\omega}_{X\cup A}$, is a multivector of the minor that contracts $A$ and deletes $D$, with the weight $\theta_{X\cup A}$ on each basis $X$ of the minor, and $\operatorname{rk}(\boldsymbol{\omega}_{A|D}) \le \operatorname{rk}(\boldsymbol{\omega})$.*

**Theorem 7.23** (The pattern law of a tensorminental table is a finite interference)**.** *Let the table have a multivector of tensor rank $q$ with slices $A^{(1)},\dots,A^{(q)}$, let $B\subseteq\mathcal{E}$ and $R=\mathcal{E}\setminus B$, and let $\alpha$ be a gauge at which every $M_{kl}=A^{(k)}D_\alpha A^{(l)\mathsf{H}}$, $D_\alpha=\operatorname{diag}(\alpha)$, with $\mathbf{G}_{kl}\not\equiv 0$ is invertible; pairs with $\mathbf{G}_{kl}\equiv 0$ as a polynomial in the gauge are omitted (Proposition 7.21). For $x\in\mathbb{C}^B$, with $x\circ\alpha_B=(x_e\alpha_e)_{e\in B}$,*

$$\mathbb{E}_{T\sim\mathbb{P}_\alpha}\Big[\prod_{e\in B}x_e^{\mathbf{1}[e\in T]}\Big]=\frac{K(x\circ\alpha_B,\alpha_R)}{K(\alpha)}=\sum_{k,l}w_{kl}\det\big(I+(D_x-I)\Pi^{(kl)}_{BB}\big),$$

*where $\mathbb{P}_\alpha$ is the basis law at $\alpha$, $D_x=\operatorname{diag}(x)$, $w_{kl}=\mathbf{G}_{kl}(\alpha)/K(\alpha)$ and $\Pi^{(kl)}=A^{(l)\mathsf{H}}M_{kl}^{-1}A^{(k)}D_\alpha$. Here $\sum_{k,l}w_{kl}=1$, $w_{lk}=\overline{w_{kl}}$, every $\Pi^{(kl)}$ is idempotent, and $\Pi^{(kk)}$ is the $D_\alpha$-orthogonal projection onto the row space of $A^{(k)}$. The hypothesis holds off a proper algebraic set of gauges.*

**Corollary 7.24** (Moments and cumulants)**.** *In the setting of Theorem 7.23:*

*(i)* $\mathbb{P}_\alpha(S\subseteq T)=\sum_{k,l}w_{kl}\det\Pi^{(kl)}_{SS}$*;*

*(ii) the law of $T\cap B$ is the $w$-mixture of $q^2$ determinantal components, whose cumulants are $\kappa^{(kl)}(e)=\Pi^{(kl)}_{ee}$ and, for $p\geq 2$ distinct marks,*

$$\kappa^{(kl)}(e_1,\dots,e_p)=(-1)^{p-1}\sum_{\varsigma}\prod_{i=1}^{p}\Pi^{(kl)}_{e_{\varsigma(i)}e_{\varsigma(i+1)}},$$

*the sum over the cyclic orders $\varsigma$ of $\{1,\dots,p\}$, with $\varsigma(p+1)=\varsigma(1)$; the joint cumulants $\kappa_\alpha$ of the indicators $X_e=\mathbf{1}[e\in T]$ under $\mathbb{P}_\alpha$ (Definition 7.17) are their total cumulants over the mixing index: for distinct marks $e_1,\dots,e_p$ of $B$,*

$$\kappa_\alpha(e_1,\dots,e_p)=\sum_{\mathfrak{p}}\kappa_w\Big(\big(\kappa^{(kl)}(e_C)\big)_{C\in\mathfrak{p}}\Big),$$

*the sum over the partitions $\mathfrak{p}$ of $\{1,\dots,p\}$ into nonempty blocks, $e_C=(e_c)_{c\in C}$, where for finitely many functions $(Y_i)_{i\in I}$ of the index $(k,l)$*

$$\kappa_w\big((Y_i)_{i\in I}\big)=\sum_{\mathfrak{q}}(-1)^{|\mathfrak{q}|-1}(|\mathfrak{q}|-1)!\prod_{D\in\mathfrak{q}}\sum_{k,l}w_{kl}\prod_{i\in D}Y_i(k,l),$$

*the sum over the partitions $\mathfrak{q}$ of $I$, is their joint cumulant over the index with the mixing coefficients $w_{kl}$; at one mark this is the occupation $R_e(\alpha)=\sum_{k,l}w_{kl}\Pi^{(kl)}_{ee}$. This is an identity of generating functions, valid for complex mixing coefficients summing to one.*

A table with a multivector of tensor rank $q$ reads the law of its patterns as an interference of $q^2$ determinantal components, mixed with complex coefficients that sum to one. At tensor rank one there is a single component and the mixture is a determinantal law; at every higher tensor rank the joint cumulants over the mixing index of the components' cumulants are the part that no single component carries.

# Chapter 8

# Families

## 8.1 The interest and its walls

**Definition 8.1** (Family, interest, walls)**.** A *family* on $\mathcal{E}$ is a finite sequence $F = \big((\mathcal{B}_j, \theta_j);\, s_j, t_j\big)_{j=1}^{N}$ of tables on $\mathcal{E}$ with exponents $s_j, t_j \in \mathbb{C}$; $K_j, \Psi_j, \Lambda_j, E_j, O_j$, $r_j$ and $d_j, d_j^*, c_j, \varepsilon_j$ belong to table $j$. A table is *active* when its exponents are not both zero, and a letter when its exponent is not zero. For *masses* $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ the *interest* is

$$J_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\mathbf{m}-\mathbf{1}} e^{-|\alpha|} \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j}\, d\alpha,$$

with its slots: the family's tables, the masses $\mathbf{m}$ and the exponents $\mathbf{s} = (s_j)_j$, $\mathbf{t} = (t_j)_j$. The *family word* $W = \prod_j K_j^{-s_j} \Psi_j^{-t_j}$ is homogeneous of degree $-\sum_j (s_j r_j + t_j r_j^*)$, and with the measure $d\alpha_{\mathbf{m}} = \alpha^{\mathbf{m}-\mathbf{1}} d\alpha$ (Definition 1.1(vii)), $J_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int W\, e^{-|\alpha|} d\alpha_{\mathbf{m}}$. The *pair coordinates* are the *strengths* $\sigma_j = \frac{s_j + t_j}{2}$ and the *leans* $\delta_j = \frac{s_j - t_j}{2}$, with the total strength $\sigma = \sum_j \sigma_j$, so that $s_j = \sigma_j + \delta_j$ and $t_j = \sigma_j - \delta_j$; the *offsets* are $\boldsymbol{\mu} = \mathbf{m} - \sigma \mathbf{1}$. The *walls* and *top walls* are

$$\kappa_S = m_S - \sum_j \big(s_j d_j(S) + t_j d_j^*(S)\big),$$

$$\kappa_S^{\mathrm{top}} = \sum_j \big(s_j r_j(S) + t_j(|S| - d_j(S))\big) - m_S \qquad (\varnothing \neq S \subseteq \mathcal{E}).$$

The *imbalance* is $\kappa_{\mathcal{E}} = m_{\mathcal{E}} - \sum_j (s_j r_j + t_j r_j^*)$, and the *balance wall* is the hyperplane $\kappa_{\mathcal{E}} = 0$. On it the data are fixed by their pair coordinates $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$, with $\boldsymbol{\mu}_\perp = \boldsymbol{\mu} - \frac{\mu_{\mathcal{E}}}{n}\mathbf{1}$ the projection of the offsets to $\mathbf{1}^\perp$: the exponents are $(\boldsymbol{\sigma} + \boldsymbol{\delta}, \boldsymbol{\sigma} - \boldsymbol{\delta})$ and the masses are the *balanced masses*

$$\mathbf{m}_F(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta}) = \boldsymbol{\mu}_\perp + \Big(\sigma + \frac{1}{n} \sum_j \delta_j (2r_j - n)\Big)\mathbf{1},$$

linear in $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$. $F$ is *connected* if every nonempty proper $S$ has $c_j(S) > 0$ for some active table $j$. A family of one table is named by its table: for a table $\mathcal{A}$ with its weights $\theta^{\mathcal{A}}$, every object of the family $((\mathcal{A}, \theta^{\mathcal{A}}); s, t)$ carries $\mathcal{A}$ in place of $F$, as $J_{\mathcal{A}}(\mathbf{m}; s, t)$. A table may carry two systems of positive weights, $\theta^K$ for its letter $K$ and $\theta^\Psi$ for its letter $\Psi$; at $(s, t)$ it names the family of the two tables $((\mathcal{B}, \theta^K); s, 0)$ and $((\mathcal{B}, \theta^\Psi); 0, t)$, whose word is $K_{\theta^K}^{-s} \Psi_{\theta^\Psi}^{-t}$.

The exponents of a table enter in two coordinates: $\sigma_j$, how much the family weighs the table, and $\delta_j$, which side it leans to, toward the bases $(K)$ or toward their complements $(\Psi)$. The pair $(K_j \Psi_j, K_j / \Psi_j)$ carries $\sigma_j$ and $\delta_j$ separately, and the walls are read in either coordinates.

**Theorem 8.2** (The interest is a partition function in the pair coordinates)**.** *(i) In the offsets,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\boldsymbol{\mu}-\mathbf{1}} e^{-|\alpha|} \prod_j \Lambda_j(\tfrac{1}{2}\log\alpha)^{-s_j} \Lambda_j(-\tfrac{1}{2}\log\alpha)^{-t_j}\, d\alpha.$$

*(ii) In signed coordinates,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = 2^n \int_{\mathbb{R}^{\mathcal{E}}} \exp\Big(2\langle\boldsymbol{\mu}, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big) - \sum_e e^{2z_e}\Big)\, dz,$$

*the partition function of the exponential family with natural parameters $(\boldsymbol{\mu};\sigma_j,\delta_j)$ and sufficient statistics $(2z;-E_j,-O_j)$.*

*(iii)* $\kappa_S = \mu_S + \sum_j \big(\sigma_j\, c_j(S) - \delta_j\, \varepsilon_j(S)\big)$ *and* $\kappa_{\mathcal{E}} = \mu_{\mathcal{E}} - \sum_j \delta_j(2r_j - n)$.

**Lemma 8.3** (Greedy bases on a sector)**.** *Let $\pi$ be a total order of $\mathcal{E}$ with tails $U_k = \{\pi(k),\dots,\pi(n)\}$. There are bases $T^{\pm}_{\pi}$ with $|T^{+}_{\pi}\cap U_k| = d(U_k)$ and $|U_k \setminus T^{-}_{\pi}| = d^*(U_k)$ for all $k$, and on the sector $H_\pi = \{\alpha_{\pi(1)} \ge \cdots \ge \alpha_{\pi(n)}\}$ every basis has $\alpha^T \le \alpha^{T^+_\pi}$ and $\alpha^{\mathcal{E}\setminus T} \le \alpha^{\mathcal{E}\setminus T^-_\pi}$. For a nonempty set of $r$-element subsets of $\mathcal{E}$ this conclusion holds exactly when the set has the exchange property.*

The sectors are where the exchange property is used: on each sector one basis dominates every monomial of $K$ and one cobasis every monomial of $\Psi$, and that is what gives the walls their form. What does not use the exchange property is the opening of every monomial by its shares (Corollary 4.4).

**Lemma 8.4** (Cube integrals)**.** *Let $h(q,\lambda)$ be smooth in $q$ near $[0,1]^m$, with every $q$-derivative continuous in $(q,\lambda)$ and holomorphic in $\lambda$ on an open set. Then* $\mathrm{cube}(x,\lambda;h) = \int_{[0,1]^m} \prod_k q_k^{x_k-1} h(q,\lambda)\, dq$*, defined for* $\operatorname{Re} x_k > 0$*, extends meromorphically to $\mathbb{C}^m$, jointly in $\lambda$, with poles on the hyperplanes $x_k = -j$, $j\in\mathbb{N}$: with $h = \sum_{j<M} h_j(q',\lambda) q_1^j + q_1^M r_M$,*

$$\mathrm{cube}(x,\lambda;h) = \sum_{j<M} \frac{\mathrm{cube}(x',\lambda;h_j)}{x_1 + j} + \mathrm{cube}\big((x_1+M,x'),\lambda;r_M\big),$$

*repeated in each coordinate.*

**Theorem 8.5** (Existence and continuation)**.** *(i) $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ converges if and only if* $\operatorname{Re}\kappa_S > 0$ *for every nonempty $S\subseteq\mathcal{E}$, whatever $\mathbf{s},\mathbf{t}\in\mathbb{C}^N$; it is holomorphic there.*

*(ii) On that region*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma(\kappa_{\mathcal{E}}) \sum_\pi \int_{[0,1]^{n-1}} \prod_{k=2}^{n} q_k^{\kappa_{U_k}-1} h_\pi(q)\, dq, \qquad h_\pi = L_\pi^{-\kappa_{\mathcal{E}}} \prod_j k_{j,\pi}^{-s_j} \psi_{j,\pi}^{-t_j},$$

*where in the sector coordinates $\alpha_{\pi(1)} = \rho$, $\alpha_{\pi(k)} = \rho q_2\cdots q_k$ one has*

$$K_j = \rho^{r_j} \prod_k q_k^{d_j(U_k)} k_{j,\pi}(q), \qquad \Psi_j = \rho^{n-r_j} \prod_k q_k^{d^*_j(U_k)} \psi_{j,\pi}(q), \qquad |\alpha| = \rho L_\pi(q);$$

*the* sector letters *$k_{j,\pi},\psi_{j,\pi},L_\pi$ have nonnegative coefficients and positive constant terms $\theta_{j,T^+_{j,\pi}}$, $\theta_{j,T^-_{j,\pi}}$, $T^{\pm}_{j,\pi}$ the greedy bases of table $j$ (Lemma 8.3), and* 1*, and $L_\pi \in [1,n]$.*

*(iii)* $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *extends to a meromorphic function of* $(\mathbf{m},\mathbf{s},\mathbf{t})$ *with poles on the hyperplanes* $\kappa_S = -j$, $S \neq \varnothing$, $j \in \mathbb{N}$*; identities valid on an open subset of the region persist.*

**Corollary 8.6** (Past several walls at once)**.** *For an order* $\pi$ *write* $\kappa_k = \kappa_{U_k}$ *and let*

$$I_\pi = \int_{[0,1]^{n-1}} \prod_k q_k^{\kappa_k - 1} h_\pi(q)\, dq = \int_{[0,\infty)^{n-1}} e^{-\sum_k \kappa_k u_k} h_\pi(e^{-u})\, du$$

*be the term of Theorem 8.5(ii), written also in the gaps* $u_k = -\log q_k$*; in the log-ratios* $\eta_k = \log(\alpha_{\pi(1)}/\alpha_{\pi(k)}) = u_2 + \cdots + u_k$ *the sector is the trail region* $0 < \eta_2 < \cdots < \eta_n$*. Let no* $\kappa_k$ *be a nonpositive integer and* $L_k \in \mathbb{N}$ *with* $\operatorname{Re}\kappa_k + L_k > 0$*. Then the continuation of* $I_\pi$ *in the masses is*

$$\sum_{A \subseteq \{2,\dots,n\}} \sum_{\mathbf{l}:\ l_k < L_k\ (k \in A)} \prod_{k \in A} \frac{1}{\kappa_k + l_k} \int_{[0,1]^{A^c}} \prod_{k \notin A} q_k^{\kappa_k - 1} \Big[ \prod_{k \notin A} (1 - \mathrm{Tay}_k)\, h_\pi^{(A,\mathbf{l})} \Big] dq_{A^c},$$

*where* $h_\pi^{(A,\mathbf{l})} = \partial_{q_A}^{\mathbf{l}} h_\pi / \mathbf{l}!$ *at* $q_A = 0$ *and* $\mathrm{Tay}_k$ *takes the Taylor polynomial in* $q_k$ *of degree below* $L_k$*: the term* $A$ *collapses the tails* $U_k$*,* $k \in A$*, together, at the levels* $\mathbf{l}$*.*

**Theorem 8.7** (Projective form; the period)**.** *Put*

$$\begin{aligned} P_F(\mathbf{m};\mathbf{s},\mathbf{t}) &= \int_{\Delta_\mathcal{E}} u^{\boldsymbol{\mu}-\mathbf{1}} \prod_j \Lambda_j(\tfrac{1}{2}\log u)^{-s_j} \Lambda_j(-\tfrac{1}{2}\log u)^{-t_j}\, d\omega_\mathcal{E}(u) \\ &= \int_{\Delta_\mathcal{E}} u^{\mathbf{m}-\mathbf{1}} \prod_j K_j(u)^{-s_j} \Psi_j(u)^{-t_j}\, d\omega_\mathcal{E}(u). \end{aligned}$$

*It converges if and only if* $\operatorname{Re}\kappa_S > 0$ *for every nonempty proper* $S$*, and* $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma(\kappa_\mathcal{E}) P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *as meromorphic functions;* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *is the* period*. On the balance wall the period is the residue of* $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *there; for every positive* $h$ *homogeneous of degree one, the integral of* $\alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j} d\alpha$ *over* $\{h = 1\}$ *(Definition 1.1(vii)) is the period.*

**Definition 8.8** (The continued interest)**.** The continuation of Theorem 8.5(iii) keeps the symbol $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ away from its poles. Among them is the whole balance wall, where $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ has the residue $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ (Theorem 8.7). The factor $e^{-|\alpha|}$ damps the scaling mode, so no completion enters the continued interest.

**Theorem 8.9** (The interest is a Gramma; its letter and mark relations)**.** *For* $\operatorname{Re} s_j, \operatorname{Re} t_j > 0$*, on the scales* $\mathcal{E} \sqcup \{v_j\} \sqcup \{w_j\}$ *let* $t_F$ *be the configuration of the columns* $(\mathbf{e}_e; 0; 0)$ *with coefficient* $1$*, and* $(\chi_T/2; \mathbf{e}_j; 0)$*,* $(-\chi_T/2; 0; \mathbf{e}_j)$ *with coefficient* $\theta_{j,T}$ *for* $T \in \mathcal{B}_j$*. Then*

$$\prod_j \Gamma(s_j)\Gamma(t_j)\, J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \mathsf{G}_{(\boldsymbol{\mu};\mathbf{s};\mathbf{t});t_F}(\mathbf{1} \oplus \boldsymbol{\theta} \oplus \boldsymbol{\theta}),$$

*either side converging with the other. As identities of meromorphic functions:*

*(i) (letter relations) for each* $j$*,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_{T \in \mathcal{B}_j} \theta_{j,T} J_F(\mathbf{m} + \mathbf{1}_T; \mathbf{s} + \mathbf{e}_j, \mathbf{t}) = \sum_{T \in \mathcal{B}_j} \theta_{j,T} J_F(\mathbf{m} + \mathbf{1}_{\mathcal{E}\setminus T}; \mathbf{s}, \mathbf{t} + \mathbf{e}_j);$$

*(ii) (mark relations) for each $e$,*

$$m_e J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = J_F(\mathbf{m}+\mathbf{e}_e;\mathbf{s},\mathbf{t}) + \sum_j s_j \sum_{T\ni e} \theta_{j,T} J_F(\mathbf{m}+\mathbf{1}_T;\mathbf{s}+\mathbf{e}_j,\mathbf{t}) + \sum_j t_j \sum_{T\not\ni e} \theta_{j,T} J_F(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T};\mathbf{s},\mathbf{t}+\mathbf{e}_j);$$

*(iii) the same for the period $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ of Theorem 8.7, with $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma(\kappa_{\mathcal{E}})P_F(\mathbf{m};\mathbf{s},\mathbf{t})$: the letter relations hold verbatim, and in the mark relation the pure mass shift carries the factor $\kappa_{\mathcal{E}}$, $m_e P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \kappa_{\mathcal{E}} P_F(\mathbf{m}+\mathbf{e}_e;\mathbf{s},\mathbf{t}) + \cdots$.*

The interest is not a new kind of object: it is one Gramma, on the marks and one scale for each letter, and its relations are that Gramma's exchange relations. The letter relation is the exchange relation on a letter's scale, and the mark relation the exchange relation on a mark's scale, rewritten with the letter relations.

**Definition 8.10** (Breta shape law; Breta law)**.** For real data with every proper wall positive, the *Breta shape law* $\mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ is the probability measure

$$u^{\mathbf{m}-\mathbf{1}} \prod_j K_j(u)^{-s_j} \Psi_j(u)^{-t_j}\, d\omega_{\mathcal{E}}(u) / P_F(\mathbf{m};\mathbf{s},\mathbf{t}) \quad \text{on } \Delta_{\mathcal{E}}.$$

With every wall positive, the *Breta law* $\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ is the law on $\mathbb{R}^{\mathcal{E}}_{>0}$ with density $\alpha^{\mathbf{m}-\mathbf{1}} e^{-|\alpha|} \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j} / J_F(\mathbf{m};\mathbf{s},\mathbf{t})$.

**Proposition 8.11** (The interest over latent vectors)**.** *Let every table $j$ of a family be determinantal over $\mathbb{F}_j \in \{\mathbb{R},\mathbb{C}\}$ (Definition 7.20), put $\beta_j = \dim_{\mathbb{R}} \mathbb{F}_j$, and let $2s_j/\beta_j$ and $2t_j/\beta_j$ be nonnegative integers. Let $V_j \subseteq \mathbb{F}_j^{\mathcal{E}}$ be the row space of a slice of table $j$ and $V_j^{\perp}$ its orthogonal complement, each with its Lebesgue measure, and for the* latent vectors *$x_{j,i} \in V_j$, $1 \le i \le 2s_j/\beta_j$, and $x^*_{j,i} \in V_j^{\perp}$, $1 \le i \le 2t_j/\beta_j$, put $\mathfrak{q}_e = \sum_j \big( \sum_i |(x_{j,i})_e|^2 + \sum_i |(x^*_{j,i})_e|^2 \big)$. Then*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \frac{\prod_e \Gamma(m_e)}{\prod_j K_j(\mathbf{1})^{s_j+t_j} \pi^{r_j s_j + r_j^* t_j}} \int \prod_{e\in\mathcal{E}} \big(1+\mathfrak{q}_e\big)^{-m_e} \prod_{j,i} dx_{j,i}\, dx^*_{j,i},$$

*both sides finite or infinite together. Where they are finite, the Breta law $\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ (Definition 8.10) is the law of $\alpha$ under the joint law of $\alpha$ and the latent vectors with density proportional to $\alpha^{\mathbf{m}-\mathbf{1}} \exp\big(-\sum_e \alpha_e(1+\mathfrak{q}_e)\big)$, under which, given the latent vectors, the scales $\alpha_e$ are independent and $\alpha_e(1+\mathfrak{q}_e)$ has the law $\mathrm{Gamma}_{m_e}$. Over $\mathbb{R}$ with $s_j = t_j$ the sums $u_{j,i} = x_{j,i} + x^*_{j,i}$ run over $\mathbb{R}^{\mathcal{E}}$, with $|(x_{j,i})_e|^2 + |(x^*_{j,i})_e|^2 = (\mathrm{pr}_j\, u_{j,i})_e^2 + \big((I-\mathrm{pr}_j)u_{j,i}\big)_e^2$, $\mathrm{pr}_j$ the orthogonal projection onto $V_j$.*

The letters of a determinantal table open as Gaussian integrals: its basis polynomial is a Gram determinant, whose inverse powers are Gaussian integrals over the row space of its slice, real or complex, and its cobasis polynomial is the same over the orthogonal complement, each letter with as many latent vectors as its exponent asks; over $\mathbb{R}$ at equal exponents the two pair into Gaussian vectors on $\mathbb{R}^{\mathcal{E}}$ read through the two projections. The integrals over the scales are then Gamma integrals, and the interest becomes a rational integral in the latent vectors.

**Proposition 8.12** (The interest under Gamma draws)**.** *Let the data be real with $\mathbf{m} > 0$.*

*(i) Where every proper wall is positive,* $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\,\mathbb{E}_{U\sim \mathrm{Dir}_{\mathbf{m}}}[W(U)]$. *With* $\phi_{\mathbf{m}}$ *the joint law of* $\big(K_1(U),\dots,K_N(U),\Psi_1(U),\dots,\Psi_N(U)\big)$ *for* $U\sim \mathrm{Dir}_{\mathbf{m}}$, *an input on the source* $(0,\infty)^{2N}$, $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *is a Trailmoment at the unit configuration* $I_{2N}$*:*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\,\big(\mathsf{T}_{(\mathbf{s},\mathbf{t});I_{2N}}[\mathrm{id}]\,\phi_{\mathbf{m}}\big)\big((0,\infty)^{2N}\big).$$

*(ii) Where moreover* $\kappa_{\mathcal{E}}>0$, *so that every wall is positive, let* $\alpha\sim \mathrm{Gamma}_{\mathbf{m}}$, *with total* $|\alpha|$ *and shape* $U=\alpha/|\alpha|$. *The scale and the shape separate:* $|\alpha|\sim \mathrm{Gamma}_{m_{\mathcal{E}}}$ *and* $U\sim \mathrm{Dir}_{\mathbf{m}}$ *are independent, and*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_e \Gamma(m_e)\,\mathbb{E}_{\alpha\sim\mathrm{Gamma}_{\mathbf{m}}}[W(\alpha)],$$

$$\mathbb{E}_{\alpha\sim\mathrm{Gamma}_{\mathbf{m}}}[W(\alpha)] = \frac{\Gamma(\kappa_{\mathcal{E}})}{\Gamma(m_{\mathcal{E}})}\,\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[W(U)].$$

*(iii) In the exponents,* $J_F(\mathbf{m};\mathbf{s},\mathbf{t})/\prod_e\Gamma(m_e)$ *is the moment generating function of the letters' logarithms* $-\log K_j(\alpha)$, $-\log\Psi_j(\alpha)$ *for* $\alpha\sim\mathrm{Gamma}_{\mathbf{m}}$, *and* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ *that of* $-\log K_j(U)$, $-\log\Psi_j(U)$ *for* $U\sim\mathrm{Dir}_{\mathbf{m}}$. *Their ratio* $\Gamma(\kappa_{\mathcal{E}})/\Gamma(m_{\mathcal{E}})$ *has its pole on the balance wall, where the moment generating function under* $\mathrm{Gamma}_{\mathbf{m}}$ *diverges and that under* $\mathrm{Dir}_{\mathbf{m}}$ *is the period over* $\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$.

**Theorem 8.13** (Walls under Cremona)**.** *As functions of* $(\boldsymbol{\mu};\boldsymbol{\sigma},\boldsymbol{\delta})$: $\kappa_S^{\mathrm{top}}(\boldsymbol{\mu};\boldsymbol{\sigma},\boldsymbol{\delta}) = \kappa_S(-\boldsymbol{\mu};\boldsymbol{\sigma},-\boldsymbol{\delta})$, $\kappa_S+\kappa_S^{\mathrm{top}} = 2\sum_j\sigma_j c_j(S)$, *and* $\kappa_{\mathcal{E}\setminus S} = \kappa_{\mathcal{E}}+\kappa_S^{\mathrm{top}}$. *In pair coordinates* $\kappa_S^{\mathrm{top}} = \sum_j\big(\sigma_j c_j(S)+\delta_j\varepsilon_j(S)\big)-\mu_S$.

**Theorem 8.14** (Cremona on the balance wall)**.** *For a connected family, on the hyperplane* $\kappa_{\mathcal{E}}=0$,

$$P_F(\boldsymbol{\mu}+\sigma\mathbf{1};\boldsymbol{\sigma}+\boldsymbol{\delta},\boldsymbol{\sigma}-\boldsymbol{\delta}) = P_F(-\boldsymbol{\mu}+\sigma\mathbf{1};\boldsymbol{\sigma}-\boldsymbol{\delta},\boldsymbol{\sigma}+\boldsymbol{\delta}),$$

*the walls of the right side at* $S$ *being those of the left at* $\mathcal{E}\setminus S$*; the identity continues meromorphically along the hyperplane.*

On the balance wall the period is unchanged when the offsets and the leans change sign together: Cremona (Definition 9.1) swaps the two exponents of each table, $\delta_j\mapsto-\delta_j$, negates the offsets, and swaps every wall with the wall of the complement.

**Proposition 8.15** (Dual tables)**.** *Replacing every table by its dual, with the same exponents, gives the interest at the exponents* $(t_j,s_j)$*: in pair coordinates* $\delta_j\mapsto-\delta_j$.

**Proposition 8.16** (The wall of a compound)**.** *For every nonempty proper* $S$, $\kappa_S+\kappa_{\mathcal{E}\setminus S} = \kappa_{\mathcal{E}}+2\sum_j\sigma_j c_j(S)$. *So if* $c_j(S)=0$ *in every table with* $\sigma_j\neq0$, *then* $\kappa_S+\kappa_{\mathcal{E}\setminus S}=\kappa_{\mathcal{E}}$, *and on the balance wall a compound split along* $S$ *has a wall with real part at most zero, and its period diverges.*

## 8.2 The native gauge

**Definition 8.17** (Target; native gauge)**.** For data with $m_{\mathcal{E}} \neq 0$ the *target* is $\boldsymbol{\tau} = \frac{\sum_j (s_j r_j + t_j r_j^*)}{m_{\mathcal{E}}} \mathbf{m}$; on the balance wall it is $\mathbf{m}$. A point $\alpha^\star \in (\mathbb{C}^\times)^{\mathcal{E}}$ with $K_j(\alpha^\star)\Psi_j(\alpha^\star) \neq 0$ for every $j$ is *native* when its *usage* equals the target,

$$\sum_j \big(s_j\, R_j(\alpha^\star) + t_j\, R_j^*(\alpha^\star)\big) = \boldsymbol{\tau},$$

where $R_{j,e} = \alpha_e \partial_e K_j / K_j$ is the occupation of table $j$ and $R^*_{j,e} = \alpha_e \partial_e \Psi_j / \Psi_j$ that of its dual (Theorem 7.16), rational functions homogeneous of degree zero; so nativity is a property of the point up to scale. A native point in $\mathbb{R}^{\mathcal{E}}_{>0}$, written $\alpha^\star = e^{2z^\star}$, is a *native gauge*; the other native points are the *native gauges at complex points.*

**Definition 8.18** (Potential; its even and odd parts)**.** In logarithmic coordinates the *potential* of a family with its exponent pairs is

$$\Phi(y) = \sum_j \Big(s_j \log K_j(e^y) + t_j \log \Psi_j(e^y)\Big) = \Phi_{\mathbf{E}}(y) + \Phi_{\mathbf{O}}(y),$$

$$\Phi_{\mathbf{E}}(y) = \sum_j \sigma_j \log \mathbf{E}_j(e^y), \qquad \Phi_{\mathbf{O}}(y) = \sum_j \delta_j \log \mathbf{O}_j(e^y), \qquad \mathbf{E}_j = K_j\Psi_j, \quad \mathbf{O}_j = K_j/\Psi_j,$$

the *even* and *odd letters* of table $j$, so that $\mathbf{E}_j(e^{2z}) = e^{2\langle \mathbf{1}, z\rangle} e^{E_j(z)}$ and $\mathbf{O}_j(e^{2z}) = e^{O_j(z)}$. The pair $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$, either part written $\Phi_{\mathbf{X}}$ with $\mathbf{X} \in \{\mathbf{E}, \mathbf{O}\}$, is read whole: neither part is switched off at any lean, and the two are never added before they are read.

**Theorem 8.19** (The native gauge is the saddle)**.** *The potential satisfies* $\partial_{y_e}\Phi = \sum_j \big(s_j R_{j,e} + t_j R^*_{j,e}\big)$*, the usage at* $\alpha = e^y$*, and the integrand of the interest is* $e^{\langle \mathbf{m}, y\rangle - \Phi(y) - |\alpha|}$ *in* $y$*. In the signed coordinates* $z = \frac{1}{2}y$*,* $\Phi(y) = \sigma\langle \mathbf{1}, y\rangle + \Phi^\circ(z)$ *with the* centred potential $\Phi^\circ(z) = \sum_j \big(s_j \log \Lambda_j(z) + t_j \log \Lambda_j(-z)\big) = \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)$*.*

*(i) A gauge* $\alpha^\star$ *is native iff* $\nabla\Phi(\log \alpha^\star) = \boldsymbol{\tau}$*; on the balance wall, iff* $\log \alpha^\star$ *is a critical point of* $\langle \mathbf{m}, y\rangle - \Phi(y)$*.*

*(ii) For real data with* $s_j, t_j \geq 0$*, not all zero, and* $\mathbf{m} > 0$*,* $\Phi$ *is convex, and a native gauge exists iff* $\boldsymbol{\tau}$ *lies in the relative interior of the Newton sum* $\mathcal{N}_F = \sum_j \big(s_j P_j + t_j P_j^*\big)$*, with* $P_j$ *the base polytope of table* $j$ *(Theorem 7.18) and* $P_j^*$ *that of its dual; it is then unique modulo the space of directions along which* $\Phi$ *is affine.*

*(iii) For such data* $\mathcal{N}_F = \{x :\ x_{\mathcal{E}} = g(\mathcal{E}),\ x_S \geq g(S)\ \forall S\}$ *with the set function* $g = \sum_j (s_j d_j + t_j d_j^*)$*. For a connected family that space is* $\mathbb{R}\mathbf{1}$*, and the condition is* $\tau_S - g(S) > 0$ *for every nonempty proper* $S$*: the walls at the target are positive. The native gauge is then the unique point, up to scale, at which* $\alpha^{\boldsymbol{\tau}} W(\alpha)$*, constant along each ray, is largest; on the balance wall* $\boldsymbol{\tau} = \mathbf{m}$*.*

A native gauge is a gauge at which the family uses each mark exactly as much as the target asks, and the same equation at a complex point gives the native gauges at complex points that the rays of Chapter 38 use. With real data, nonnegative exponents and positive masses, for a connected family it is unique up to scale and exists exactly when every proper wall at the target is positive (Theorem 8.19(iii)); on a compound it exists exactly when every molecule of the active family is balanced at the target with its proper walls positive there, with one free scale for each such molecule (Proposition 8.20). It is a point at which $\alpha^{\boldsymbol{\tau}} W(\alpha)$ is largest, the saddle of the potential, and it is the measuring stick of every computation on the family.

**Proposition 8.20** (The native gauges of a compound)**.** *Let $F$ have real data, $s_j, t_j \geq 0$ not all zero, $\mathbf{m} > 0$, and target $\boldsymbol{\tau}$, and let $\mathcal{E}_1, \ldots, \mathcal{E}_q$ be the molecules of its active family. A native gauge exists exactly when every $\mathcal{E}_i$ is balanced at the target, $\tau_{\mathcal{E}_i} = \sum_j \big(s_j r_j(\mathcal{E}_i) + t_j r_j^*(\mathcal{E}_i)\big)$, and every proper wall of every $\mathcal{E}_i$ is positive there, $\tau_S > \sum_j \big(s_j d_j(S) + t_j d_j^*(S)\big)$ for $\varnothing \neq S \subsetneq \mathcal{E}_i$. The native gauges are then the gauges $\beta$ with $\beta/\alpha^\star$ constant on every $\mathcal{E}_i$, $\alpha^\star$ any one of them: one free scale for each molecule. When the weights factor across the $\mathcal{E}_i$, the native gauges are the products of native gauges of the molecules. A mark that is a loop or a coloop in every table is a unit, whose letters are monomials.*

**Proposition 8.21** (The centre of an arbitrary family)**.** *Take tables on $\mathcal{E}$, $s_j = t_j = \sigma_j \geq 0$ with $\sigma = \sum_j \sigma_j > 0$, and $\mathbf{m} = \sigma\mathbf{1}$. Then $\delta_j = 0$, $\boldsymbol{\mu} = 0$, the centred potential $\Phi^\circ(z) = \sum_j \sigma_j E_j(z)$ is even, and:*

*(i) the gauge $\mathbf{1}$ is native, unique modulo one scale for each molecule of the active family (Proposition 8.20);*

*(ii) the walls are $\kappa_S = \sum_j \sigma_j c_j(S)$, positive on every proper nonempty $S$ exactly when the active family is connected;*

*(iii) every odd derivative of $\Phi^\circ$ vanishes at $z = 0$, so every odd derivative of order at least three of the potential $\Phi$ vanishes at $y = 0$.*

The family at the centre, every $\delta_j = 0$ (Definition 9.1), with the masses $\sigma\mathbf{1}$ has its offsets zero. It exists for every family, its native gauge is the table's gauge, and its walls are the combinations $\kappa_S = \sum_j \sigma_j c_j(S)$ of the widths; it is where the family is read before any lean is asked of it.

**Proposition 8.22** (Prescribing a gauge)**.** *For any positive gauge $a$ and positive exponent pairs, the usage $\tau_e = \sum_j \big(s_j R_{j,e}(a) + t_j R_{j,e}^*(a)\big)$ has total $\sum_j (s_j r_j + t_j r_j^*)$, and every mass vector on the ray $c\boldsymbol{\tau}$, $c > 0$, makes a native. Fixing values $K_j^0 > 0$ and $\Psi_j^0 > 0$ of the letters at $a$, the constraints*

$$\sum_T \theta_{j,T} a^T = K_j^0, \qquad \sum_T \theta_{j,T} a^{\mathcal{E}\setminus T} = \Psi_j^0, \qquad \sum_{j,T} \theta_{j,T} \Big[ \frac{s_j a^T}{K_j^0} \mathbf{1}_T + \frac{t_j a^{\mathcal{E}\setminus T}}{\Psi_j^0} \mathbf{1}_{\mathcal{E}\setminus T} \Big] = \boldsymbol{\tau}$$

*are linear in the weights, and every positive deformation in their homogeneous kernel preserves them exactly.*

# Part III

# Classification

# Chapter 9

# Realizations and the still

## 9.1 Realizations

**Definition 9.1** (Realization, lean, carrier)**.** Let a family carry tables $j$ on its marks $\mathcal{E}$, $n = |\mathcal{E}|$, its potential read as the pair $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$ of Definition 8.18. A *realization* of the family is $\Upsilon = \big((s_j, t_j)_j;\, \alpha\big)$: exponents $s_j, t_j \geq 0$ with $s_j + t_j > 0$, and a gauge $\alpha \in \mathbb{R}^{\mathcal{E}}_{>0}$. Its pair coordinates are the strengths $\sigma_j = \frac{1}{2}(s_j + t_j) > 0$ and the leans $\delta_j = \frac{1}{2}(s_j - t_j)$ of Definition 8.1, $|\delta_j| \leq \sigma_j$, and in them the realization is written

$$\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha} = \big((\sigma_j + \delta_j,\ \sigma_j - \delta_j)_j;\ \alpha\big).$$

$\boldsymbol{\delta} = (\delta_j)_j$ is the *lean*, and the *centre* is $\boldsymbol{\delta} = 0$; at the table's gauge the realization at the centre has the masses $\sigma\mathbf{1}$ (Proposition 8.21), and a family at the centre is taken with these masses. Table $j$ has a *basis lean* when $\delta_j > 0$, a *cobasis lean* when $\delta_j < 0$, and *rides the line* when $\delta_j = 0$; its lean is *full* when $|\delta_j| = \sigma_j$ (then the factor of table $j$ in the family word is $K_j^{-s_j}$ alone, or $\Psi_j^{-t_j}$ alone) and *partial* when $0 < |\delta_j| < \sigma_j$. The family has a basis lean when every $\delta_j \geq 0$ and some $\delta_j > 0$, a cobasis lean when every $\delta_j \leq 0$ and some $\delta_j < 0$, and otherwise, off the centre, a mixed lean. The realization is *at one pair* $(\sigma, \delta)$, $\sigma = \sum_j \sigma_j$ and $\delta = \sum_j \delta_j$, when its pairs lie on one ray, $(\sigma_j, \delta_j) = \frac{\sigma_j}{\sigma}(\sigma, \delta)$ for every $j$; its *lean sign* is the sign of $\delta$. A family with one table is at one pair. The *masses* of $\Upsilon$ are its usage,

$$\tau_e(\Upsilon) = \sum_j \Big( s_j \, \frac{\alpha_e \partial_e K_j}{K_j}(\alpha) + t_j \, \frac{\alpha_e \partial_e \Psi_j}{\Psi_j}(\alpha) \Big),$$

for which $\alpha$ is the native gauge (Proposition 8.22). The *carrier* of $\Upsilon$ is $(\alpha;\, K_j, \Psi_j;\, \boldsymbol{\tau};\, s_j, t_j)$. A gauge is written as a deviation from the table's gauge, $\alpha = e^{y_0}$. Inverting the gauge and swapping the two exponents of every table, $\big((s_j, t_j)_j; \alpha\big) \mapsto \big((t_j, s_j)_j; \alpha^{-1}\big)$, sends the offsets and the lean to $-\boldsymbol{\mu}$ and $-\boldsymbol{\delta}$, the change of data of Theorem 8.14; so *Cremona* swaps basis and cobasis leans, and the table's gauge is its fixed point up to scale (Proposition 7.2).

A realization is read off its exponents: the masses are the usage at its gauge, so the gauge is native by definition, and a realization sits on a wall only along a union of molecules, so that a realization of a molecule never does (Proposition 9.2). A molecule is read at all of its realizations, and a particular integral is one of them, fixed by its exponents and its native gauge.

**Proposition 9.2** (A realization sits on a wall only along its molecules)**.** *Let a family carry real exponents $s_j, t_j \geq 0$, $s_j + t_j > 0$, and a gauge $\alpha$, with masses its usage $\mathbf{m} = \sum_j \big(s_j R_j(\alpha) + t_j R_j^*(\alpha)\big)$, that is, a realization (Definition 9.1). With $N_S = |T \cap S|$,*

$$\kappa_S = \sum_j \Big( s_j \big( \mathbb{E}_{T \sim \mathbb{P}_{j,\alpha}}[N_S] - d_j(S) \big) + t_j \big( r_j(S) - \mathbb{E}_{T \sim \mathbb{P}^*_{j,\alpha}}[N_S] \big) \Big) \;\geq\; 0 \qquad (\varnothing \neq S \subseteq \mathcal{E}),$$

*with equality exactly when $S$ is a union of molecules (Definition 7.4). In particular $\kappa_{\mathcal{E}} = 0$, and every proper wall of a connected family is positive.*

**Proposition 9.3** (Masses at a gauge)**.** *At $\alpha = e^{y_0}$, in this statement write $R^{\pm}_{j,e} = \frac{1}{2}\big(R_{j,e}(e^{y_0}) \pm R_{j,e}(e^{-y_0})\big)$; then*

$$\tau_e = \sum_j \big( s_j R_{j,e}(e^{y_0}) + t_j (1 - R_{j,e}(e^{-y_0})) \big) = \sum_j \big( \sigma_j (1 + 2R^-_{j,e}) + \delta_j (2R^+_{j,e} - 1) \big),$$

*and at $\alpha = \mathbf{1}$, where $R^-_{j,e} = 0$ and $R^+_{j,e} = R_{j,e}(\mathbf{1})$, $\tau_e = \sum_j \big( \sigma_j + \delta_j (2R_{j,e}(\mathbf{1}) - 1) \big)$.*

## 9.2 The potential and the still

**Definition 9.4** (Still, rungs, tallies, line functions)**.** The *still* of $\Upsilon$ is the sequence of symmetric tensors

$$\kappa_\Upsilon(e_1, \ldots, e_k) = 2^k \, \partial_{y_{e_1}} \cdots \partial_{y_{e_k}} \Phi(\log \alpha), \qquad k \geq 2,$$

the Taylor tensor of $\Phi$ in the signed coordinates $z = \frac{1}{2} y$ from order two on; its entries of order $k$ form *rung $k$*. *Rung one* is the gradient $\partial_{y_e} \Phi(\log \alpha) = \tau_e(\Upsilon)$, the masses (Theorem 8.19), taken without the factor 2. Each part has its own still, $\kappa^{\mathbf{E}}_\Upsilon$ and $\kappa^{\mathbf{O}}_\Upsilon$, with $\kappa_\Upsilon = \kappa^{\mathbf{E}}_\Upsilon + \kappa^{\mathbf{O}}_\Upsilon$. The *tally* of a mark $e$ at rung $k$ is the multiset $\mathrm{tally}^{\mathbf{X}}_k(e) = \{\!\{ \kappa^{\mathbf{X}}_\Upsilon(e, w) : \; w \in \mathcal{E}^{k-1} \}\!\}$. The *self-line* and *partner-line functions* of $e$ are

$$\varphi^{\mathbf{X}}_e(v) = \partial_{y_e} \Phi_{\mathbf{X}}\big( \log \alpha + v \mathbf{1}_e \big), \qquad \varphi^{\mathbf{X}}_{e,g}(v) = \partial_{y_e} \Phi_{\mathbf{X}}\big( \log \alpha + v \mathbf{1}_g \big) \quad (g \neq e).$$

The still is the Taylor tensor of the potential at the realization from rung two on, one object, recording how the marks covary under the two laws; beside it, rung one is the masses. The letters are affine in every variable, so they have finitely many nontrivial derivatives, and every rung is read from those through the logarithm; the readings of Chapter 10 take the potential as one function, of which the still is the expansion.

**Proposition 9.5** (The potential of a compound splits by molecules)**.** *Let the weights factor across the molecules. Then $\Phi_{\mathbf{X}}(y) = \sum_M \Phi_{\mathbf{X},M}(y_{\mathcal{E}_M})$ with $\Phi_{\mathbf{X},M}$ the potential of the restriction to $M$, and every entry of the still on a string that meets two molecules is $0$. A mark that is a loop or a coloop in every table is a unit whose letters are monomials (Proposition 8.20): every entry of the still on a string through it is $0$, and its rung one is its mass.*

**Proposition 9.6** (A deviation reweights the bases)**.** *At the realization $\Upsilon = \big((s_j, t_j)_j; e^{y_0}\big)$, for $k \geq 2$ and every string $w$ of length $k$,*

$$\kappa_\Upsilon(w) = 2^k \sum_j \Big( s_j \, \kappa_{j, e^{y_0}}(w) + t_j (-1)^k \kappa_{j, e^{-y_0}}(w) \Big).$$

*In this statement write* $\kappa_j^{\pm}(w) = \frac{1}{2}\big(\kappa_{j,e^{y_0}}(w) \pm \kappa_{j,e^{-y_0}}(w)\big)$*; then*

$$\kappa_\Upsilon(w) = \begin{cases} 2^{k+1} \sum_j \big(\sigma_j \kappa_j^+(w) + \delta_j \kappa_j^-(w)\big), & k \text{ even}, \\ 2^{k+1} \sum_j \big(\sigma_j \kappa_j^-(w) + \delta_j \kappa_j^+(w)\big), & k \text{ odd}, \end{cases}$$

*and the cumulants at the deviation are read from those at the table's gauge,* $\kappa_{j,e^{y_0}}(w) = \sum_{i\geq 0} \frac{1}{i!} \sum_{u\in\mathcal{E}^i} \kappa_{j,\mathbf{1}}(w,u) \prod_{l=1}^i y_{0,u_l}$*, for* $y_0$ *in the domain of convergence of the Taylor series of* $y \mapsto \log K_j(e^y)$ *at* 0*.*

**Proposition 9.7** (The still at the table's gauge)**.** *At every realization* $\Upsilon = \big((s_j,t_j)_j;\mathbf{1}\big)$*, for* $k \geq 2$ *and every string* $w$ *of length* $k$*,*

$$\kappa_\Upsilon(w) = 2^k \sum_j \big(s_j + (-1)^k t_j\big)\kappa_{j,\mathbf{1}}(w) = \begin{cases} 2^{k+1} \sum_j \sigma_j\, \kappa_{j,\mathbf{1}}(w), & k \text{ even}, \\ 2^{k+1} \sum_j \delta_j\, \kappa_{j,\mathbf{1}}(w), & k \text{ odd}. \end{cases}$$

*This is Proposition 9.6 at* $y_0 = 0$*. At the table's gauge* $\Phi_{\mathbf{E}}$ *is even in* $y$ *and* $\Phi_{\mathbf{O}}$ *is odd, each up to a linear term; so the even rungs are those of* $\kappa_\Upsilon^{\mathbf{E}}$*, independent of the lean, and the odd rungs are those of* $\kappa_\Upsilon^{\mathbf{O}}$*, linear in the lean and zero at the centre.*

**Proposition 9.8** (Balanced tallies)**.** *At every realization* $\Upsilon$*, for each part, every rung* $k \geq 2$ *and every mark* $e$*, the tally of* $e$ *sums to* 0*:* $\sum_{w\in\mathcal{E}^{k-1}} \kappa_\Upsilon^{\mathbf{X}}(e,w) = 0$*.*

**Proposition 9.9** (The two occupations of a mark)**.** *For a mark* $e$ *of table* $j$*, neither a loop nor a coloop, with room* $R = \mathcal{E} - e$*, let* $K_{j,/e}(\alpha_R)$ *and* $K_{j,\setminus e}(\alpha_R)$ *be the basis polynomials of its contraction and deletion at* $e$ *(Definition 7.3), and*

$$\mathrm{D}_{j,e}(y_R) = \log K_{j,/e}(e^{y_R}) - \log K_{j,\setminus e}(e^{y_R}), \qquad \mathrm{D}_{j,e}^{\pm}(y_R) = \tfrac{1}{2}\big(\mathrm{D}_{j,e}(y_R) \pm \mathrm{D}_{j,e}(-y_R)\big).$$

*At the gauge* $e^y$ *the occupation and the dual occupation of* $e$ *(Theorem 7.16) are*

$$\frac{\alpha_e \partial_e K_j}{K_j} = \frac{1}{1+e^{-\lambda_{j,e}-\eta_{j,e}}}, \qquad \frac{\alpha_e \partial_e \Psi_j}{\Psi_j} = \frac{1}{1+e^{-\lambda_{j,e}+\eta_{j,e}}},$$
$$\lambda_{j,e} = y_e + \mathrm{D}_{j,e}^-(y_R), \qquad \eta_{j,e} = \mathrm{D}_{j,e}^+(y_R),$$

*and at the table's gauge* $\lambda_{j,e} = 0$*: the two occupations are* $(1+e^{-\eta_{j,e}})^{-1}$ *and* $(1+e^{\eta_{j,e}})^{-1}$*.*

**Proposition 9.10** (Occupations at a gauge)**.** *At* $\alpha = e^{y_0}$*, with* $R_{j,e}^{\pm}$ *as in Proposition 9.3,* $\varphi_e^{\mathbf{E}}(0) = \sum_j \sigma_j(1 + 2R_{j,e}^-)$ *and* $\varphi_e^{\mathbf{O}}(0) = \sum_j \delta_j(2R_{j,e}^+ - 1)$*. At* $\alpha = \mathbf{1}$*,* $\varphi_e^{\mathbf{E}}(0) = \sum_j \sigma_j$ *at every mark and* $\varphi_e^{\mathbf{O}}(0) = \sum_j \delta_j(2R_{j,e}(\mathbf{1}) - 1)$*.*

# Chapter 10

# The reading, classes, atoms

## 10.1 Views and the reading

**Definition 10.1** (Views; the reading; classes; atoms)**.** Let $M$ be a molecule with $n$ marks carrying tables $j$, and $\Upsilon$ a realization of $M$ (Definition 9.1). The room of a mark $e$ is $\mathcal{E} \setminus \{e\}$ (Chapter 12 calls it the room of the bunch $\{e\}$), and an *order* of a set is a listing of its elements. For a mark $e$ and distinct marks $\mathbf{g} = (g_1, \dots, g_k)$ of its room, $0 \le k \le n-1$, the *view* of $e$ with the partners $\mathbf{g}$ is the pair $(\Phi^{\mathbf{E}}_{e,\mathbf{g}}, \Phi^{\mathbf{O}}_{e,\mathbf{g}})$,

$$\Phi^{\mathbf{X}}_{e,\mathbf{g}}(v_0, v_1, \dots, v_k) = \Phi_{\mathbf{X}}\Big( \log\alpha + v_0 \mathbf{1}_e + \sum_{i=1}^{k} v_i \mathbf{1}_{g_i} \Big),$$

the potential on the scales of $e$ and its partners with every other mark at the gauge; the variable $v_i$ sits at *position* $i$. The *reading of* $e$ *with* $k$ *partners* is the multiset of its views over all $k$-tuples of distinct partners,

$$\mathrm{read}^{(k)}_{\Upsilon}(e) = \{\{ (\Phi^{\mathbf{E}}_{e,\mathbf{g}}, \Phi^{\mathbf{O}}_{e,\mathbf{g}}) : \ \mathbf{g} \in (\mathcal{E} \setminus e)^{(k)} \}\} :$$

the functions are kept and the names of the partners are not. With $k = n-1$, $\mathbf{g}$ is an order of the room and the pair $(\Phi^{\mathbf{E}}_{e,\mathbf{g}}, \Phi^{\mathbf{O}}_{e,\mathbf{g}})$ is the potential *written from* $e$ in that order; $\mathrm{read}^{(n-1)}_{\Upsilon}(e)$ is the *reading* of $e$. The *classes* of $\Upsilon$ are the level sets of $e \mapsto \mathrm{read}^{(n-1)}_{\Upsilon}(e)$; they form the partition $\mathrm{Cl}(\Upsilon)$. The realization is *atomic*, an *atom*, when $\mathrm{Cl}(\Upsilon) = \{\mathcal{E}\}$. For a family whose weights factor across its molecules, the classes of a realization are, by definition, the classes of the realizations of its molecules, each read on its own marks. With the gauge $\alpha$ and $\boldsymbol{\sigma}$ fixed, the partition at the lean $\boldsymbol{\delta}$ is $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha})$.

The reading of a mark is the whole potential seen from that mark, its partners unnamed, and the reading is a function on the marks whose values are functions. Classes are its level sets, and an atom is a realization at which it is constant, so that the integral cannot tell its marks apart at any order.

**Proposition 10.2** (A view through the patterns)**.** *For table $j$ and positions $0, \dots, k$ held by $e, g_1, \dots, g_k$, let the pattern of a basis be the set $p \subseteq \{0, \dots, k\}$ of positions it holds (its trace $T \cap \{e, g_1, \dots, g_k\}$ read in positions, the pattern of the basis on the bunch $\{e, g_1, \dots, g_k\}$ in Chapter 12), $\bar{p}$ its complement, and*

$$K_{j;e,\mathbf{g}}(x) = \sum_{p} \vartheta^{K}_{j,p}\, x^{p}, \qquad \Psi_{j;e,\mathbf{g}}(x) = \sum_{p} \vartheta^{\Psi}_{j,p}\, x^{\bar{p}}, \qquad \vartheta^{K}_{j,p} = \sum_{T:\, p} \theta_{j,T} \alpha^{T}, \quad \vartheta^{\Psi}_{j,p} = \sum_{T:\, p} \theta_{j,T} \alpha^{\mathcal{E}\setminus T},$$

*sums over the bases of pattern $p$. With $x_i = e^{v_i}$,*

$$\Phi^{\mathbf{E}}_{e,\mathbf{g}} = \sum_j \sigma_j \log\big(K_{j;e,\mathbf{g}}\Psi_{j;e,\mathbf{g}}\big), \qquad \Phi^{\mathbf{O}}_{e,\mathbf{g}} = \sum_j \delta_j \log\big(K_{j;e,\mathbf{g}}/\Psi_{j;e,\mathbf{g}}\big).$$

*At every gauge*

$$K_{j;e,\mathbf{g}}(\mathbf{1}+x) = K_j(\alpha) \sum_{A\subseteq\{0,\dots,k\}} \mathbb{P}_{j,\alpha}\big(p(T)\supseteq A\big)\, x^A,$$
$$\Psi_{j;e,\mathbf{g}}(\mathbf{1}+x) = \Psi_j(\alpha) \sum_{A\subseteq\{0,\dots,k\}} \mathbb{P}^*_{j,\alpha}\big(p(T)\cap A=\varnothing\big)\, x^A,$$

*with $p(T)$ the pattern of $T$, so the view with $k$ partners is read from the joint inclusion laws of $e, g_1, \dots, g_k$ under the basis and cobasis laws. At the table's gauge $\vartheta^K_{j,p} = \vartheta^\Psi_{j,p} = \vartheta_{j,p}$, the weight of the bases of pattern $p$, both laws are the weight law $\mathbb{P}_{j,\mathbf{1}}$, and $\Psi_{j;e,\mathbf{g}}(x) = x^{\{0,\dots,k\}}K_{j;e,\mathbf{g}}(x^{-1})$. For one table, two views of one realization agree at the centre exactly when the polynomials $K_{e,\mathbf{g}}\Psi_{e,\mathbf{g}}$ agree, and at a lean exactly when $K_{e,\mathbf{g}}$ and $\Psi_{e,\mathbf{g}}$ both agree; at the table's gauge $K_{e,\mathbf{g}}$ alone decides at a lean.*

**Proposition 10.3** (More partners refine; the tallies are inside)**.** *(i) $\mathrm{read}^{(k)}_\Upsilon(e)$ determines $\mathrm{read}^{(k')}_\Upsilon(e)$ for every $k' \le k$; so the level sets of the reading with $k$ partners refine those with $k'$.*

*(ii) For $k \ge 1$, $\mathrm{read}^{(k)}_\Upsilon(e)$ determines the self-line functions $\varphi^{\mathbf{X}}_e$, the multisets $\{\{\varphi^{\mathbf{X}}_{e,g}\}\}_{g\ne e}$ of partner-line functions, and for every rung the multiset of the entries $\kappa^{\mathbf{X}}_\Upsilon(e,w)$ over the strings $w$ of marks with at most $k$ distinct marks other than $e$; in particular the tallies of $e$ at the rungs $2,\dots,k+1$.*

*(iii) The reading determines the tallies of $e$ at every rung $k \ge 2$, its self-line and partner-line functions for both parts, and its mass $\tau_e(\Upsilon)$.*

**Proposition 10.4** (One writing determines the reading)**.** *The reading of $e$ is determined by any one writing of the potential from $e$. So the reading is constant on a set $C$ of marks exactly when one pair of functions is the potential written from every mark of $C$: there are orders $\mathbf{g}_e$ of the rooms, $e \in C$, with $\Phi^{\mathbf{X}}_{e,\mathbf{g}_e} = \Phi^{\mathbf{X}}_{f,\mathbf{g}_f}$ for all $e, f \in C$ and both parts. In particular $\Upsilon$ is an atom exactly when one pair of functions is the potential written from every mark.*

## 10.2 Identifying writings

**Theorem 10.5** (One table)**.** *Let $M$ carry one table, $n \ge 2$, at the table's gauge, and let $C$ be a set of marks. The reading is constant on $C$*

*(i) at a lean, exactly when one polynomial is $K$ written from every mark of $C$;*

*(ii) at the centre, exactly when one polynomial is, from every mark of $C$, either $K$ or $\Psi$ written from it.*

*Here $K$ written from $e$ in the order $\mathbf{g}$ is $K_{e,\mathbf{g}}$ with $k = n-1$, a polynomial in the position variables.*

**Theorem 10.6** (Several tables: the potential is read factor by factor)**.** *At a realization $\Upsilon$ write every $K_j(\alpha\circ x)$ and $\Psi_j(\alpha\circ x)$ as a positive constant times a product of irreducible polynomials in*

*$x$, each positive on the orthant and equal to $1$ at $x = \mathbf{1}$. Give each distinct such polynomial $L$ the exponents*

$$w_{\mathbf{E}}(L) = \sum \sigma_j > 0, \qquad w_{\mathbf{O}}(L) = \sum \pm\delta_j,$$

*summed over its occurrences with multiplicity, with $+$ in a $K_j$ and $-$ in a $\Psi_j$. Then $\Phi_{\mathbf{X}}(\log\alpha + \log x) = \Phi_{\mathbf{X}}(\log\alpha) + \sum_L w_{\mathbf{X}}(L)\log L(x)$, and two pairs $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$ of this form, in the same variables, are equal exactly when they have the same polynomials with the same pairs of exponents $(w_{\mathbf{E}}(L), w_{\mathbf{O}}(L))$. So the reading is constant on $C$ exactly when one set of polynomials with exponents $\{(L, w_{\mathbf{E}}(L), w_{\mathbf{O}}(L))\}$, with the variables placed in positions, is the family written from every mark of $C$. For tables whose weights factor across their molecules the polynomials $L$ are the normalized Foreman polynomials of their restrictions to those molecules.*

**Corollary 10.7** (Moving blocks of partners reads alike)**.** *Let $\mathbf{g}, \mathbf{g}'$ be orders of the rooms of $e$ and $f$ with $(\Phi^{\mathbf{E}}_{e,\mathbf{g}}, \Phi^{\mathbf{O}}_{e,\mathbf{g}}) = (\Phi^{\mathbf{E}}_{f,\mathbf{g}'}, \Phi^{\mathbf{O}}_{f,\mathbf{g}'})$, and let $\beta$ be the bijection of $\mathcal{E}$ with $\beta(e) = f$ and $\beta(g_i) = g'_i$. Then for all disjoint sets $S_1, \dots, S_m$ of the room of $e$,*

$$\Phi_{\mathbf{X}}\Big(\log\alpha + v_0\mathbf{1}_e + \sum_{i=1}^{m} v_i\mathbf{1}_{S_i}\Big) = \Phi_{\mathbf{X}}\Big(\log\alpha + v_0\mathbf{1}_f + \sum_{i=1}^{m} v_i\mathbf{1}_{\beta(S_i)}\Big).$$

*So on a class every move of the gauge that shifts a mark and several sets of partners, each by one amount, reads alike: for every partition $S_1, \dots, S_m$ of the room, the potential moved by $v_0$ on the mark and by $v_i$ on $S_i$ reads alike from every mark of the class.*

**Proposition 10.8** (Computing the classes)**.** *Read every mark with $k = 0, 1, 2, \dots$ partners and group the marks by value. The level sets only refine as $k$ grows (Proposition 10.3). A level set is a class as soon as one pair of functions is written from each of its marks (Proposition 10.4); the orders are built one partner at a time, keeping every view equal to the restriction of that one function, which loses nothing since a view with fewer partners is a restriction of a writing. Where no one function serves a whole level set, the marks from which a given function can be written form one class and the rest are read again. The computation ends by $k = n - 1$, and at a gauge $\alpha$ every step is an identity of polynomials with coefficients in the field generated by the weights and the coordinates of $\alpha$; at the table's gauge that is the field generated by the weights, rational for rational weights and integers for lists (Proposition 10.2, Theorem 10.6).*

## 10.3 What a class holds

**Proposition 10.9** (Masses and occupations on a class)**.** *At every realization $\Upsilon$ the masses $\tau_e(\Upsilon)$ and each of their two parts $\varphi^{\mathbf{E}}_e(0)$ and $\varphi^{\mathbf{O}}_e(0)$ (Proposition 9.10) are constant on every class. At the table's gauge:*

*(i) $\sum_j \delta_j(2R_{j,e}(\mathbf{1}) - 1)$ is constant on every class, and $\kappa^{\mathbf{E}}_{\Upsilon}(e,e) = 8\sum_j \sigma_j R_{j,e}(\mathbf{1})(1 - R_{j,e}(\mathbf{1}))$;*

*(ii) with one table, the occupation $R_e(\mathbf{1})$ is constant on every class at a lean, and at the centre two marks $e, f$ of one class have $R_f(\mathbf{1}) = R_e(\mathbf{1})$ or $R_f(\mathbf{1}) = 1 - R_e(\mathbf{1})$.*

# Chapter 11

# Lean, gauge and label

## 11.1 The lean

**Theorem 11.1** (The lean only splits)**.** *At every gauge and every $\boldsymbol{\sigma}$, the partition at a lean $\boldsymbol{\delta}$ refines the partition at the centre. An atom at a lean is an atom at the centre, and a realization that is not an atom at the centre is an atom at no lean.*

**Proposition 11.2** (What of the lean is read)**.** *With the gauge $\alpha$ fixed, $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha})$ depends on $(\boldsymbol{\sigma},\boldsymbol{\delta})$ only through the rays $\{c\boldsymbol{\sigma}\}$ and $\{c'\boldsymbol{\delta}\}$, $c,c'>0$, and $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},-\boldsymbol{\delta};\alpha})=\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha})$. So at one pair, $\boldsymbol{\delta}=\frac{\delta}{\sigma}\boldsymbol{\sigma}$, there are two partitions: $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},0;\alpha})$ at the centre, and $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha})$, the same for every $\delta\neq 0$ and every $\boldsymbol{\sigma}$ on its ray, which refines it.*

**Proposition 11.3** (A lean reads one side)**.** *Let $\Upsilon$ be at one pair $(\sigma,\delta)$, at the table's gauge, and put*

$$\Phi_K(y)=\frac{1}{\sigma}\sum_j \sigma_j \log K_j(e^y), \qquad \Phi_\Psi(y)=\frac{1}{\sigma}\sum_j \sigma_j \log \Psi_j(e^y)=\Phi_K(-y)+\langle \mathbf{1},y\rangle.$$

*Then $\Phi_{\mathbf{E}}=\sigma(\Phi_K+\Phi_\Psi)$ and $\Phi_{\mathbf{O}}=\delta(\Phi_K-\Phi_\Psi)$, and:*

*(i) at every lean the reading of a mark determines, and is determined by, the multiset of writings of $\Phi_K$ from it; so for $\delta\neq 0$, $\mathrm{Cl}(\Upsilon)$ is the partition of the marks into the level sets of the multiset of writings of $\Phi_K$, and equally of $\Phi_\Psi$;*

*(ii) with the tensors $\mathcal{K}_k(w)=\frac{1}{\sigma}\sum_j \sigma_j \kappa_{j,\mathbf{1}}(w)$ on the strings of length $k$, rung $k$ is $2^{k+1}\sigma\mathcal{K}_k$ for even $k$ and $2^{k+1}\delta\mathcal{K}_k$ for odd $k$: at a basis lean every rung is a positive multiple of the cumulant sequence $(\mathcal{K}_k)_k$ of the bases, at a cobasis lean of the cumulant sequence $((-1)^k\mathcal{K}_k)_k$ of the cobases, and at the centre only the even rungs, common to the two sequences, are read.*

A lean reads the molecule with a preference: at a basis lean it reads the cumulant sequence of the bases, at a cobasis lean that of the cobases, and at the centre only what the two sequences share. The lean can only split the classes of the centre, never join them.

## 11.2 The gauge

**Theorem 11.4** (The gauge step)**.** *Let $\Upsilon=((s_j,t_j)_j;\alpha)$ be a realization of a molecule. If the self-line functions $\varphi_e^{\mathbf{E}}$ of all marks agree up to additive constants, in particular if $\Upsilon$ is an atom, then $\alpha\propto\mathbf{1}$.*

**Corollary 11.5** (Scale changes nothing)**.** *At $\alpha = c\mathbf{1}$ every writing is the writing at $\mathbf{1}$ plus a constant that depends on neither the mark nor the order. So $\mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};c\mathbf{1}}) = \mathrm{Cl}(\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}})$, and every atom is an atom at the table's gauge.*

## 11.3 Answers and the label

**Proposition 11.6** (Assembly from arbitrary seed tables)**.** *Let $H$ be a set of bijections of $\mathcal{E}$ closed under composition and inverses that takes every mark to every other. Start from seed tables on $\mathcal{E}$, and for each seed include its image under every $h \in H$, weights transported with the bases and repetitions kept. Give every image of seed $a$ the exponents $(s_a, t_a)$; let $q_a$ be the number of its images with multiplicity, $r_a$ its rank, $D = \sum_a q_a(s_a r_a + t_a(n - r_a))$, $\mathbf{m} = \frac{D}{n}\mathbf{1}$, and suppose the active assembled family is connected. Then for every positive exponent pairs the gauge $\mathbf{1}$ is native, every proper wall is positive, and the realization $((s_a, t_a)_a; \mathbf{1})$ of the assembled family is an atom.*

*(i) At $\mathbf{1}$ the occupations summed over the images of seed $a$ are the same at every mark, $q_a r_a/n$, and the dual occupations summed likewise are $q_a(n - r_a)/n$; their sum with the coefficients $s_a$ and $t_a$ is $D/n$.*

*(ii) The images under all permutations of a seed with at least two bases form a connected family.*

*(iii) The seed weights may vary over any positive values before assembly, giving atoms at the one gauge $\mathbf{1}$.*

**Definition 11.7** (Answers and the label)**.** A molecule *answers yes* when some realization of it is atomic, and *no* otherwise. Let $k$ be the number of molecules of a family with at least two marks and $i$ the number of them that answer yes; units are atomic at every realization and are not counted. The family is *Foreman $k$-symmetric* when $i = k$ and *Foreman asymmetric* when $i < k$; an asymmetric family with $i > 0$ is *sub-symmetric ($i$ of $k$)*. The *label* of the family is the pair $(i, k)$.

**Theorem 11.8** (The answer is read at the centre)**.** *A molecule answers yes if and only if some centre $\Upsilon_{\boldsymbol{\sigma},0;\mathbf{1}}$ is atomic. With one table this holds if and only if one polynomial is, from every mark, either $K$ or $\Psi$ written from it; the condition does not involve $\sigma$.*

**Corollary 11.9** (The label is read at the centres)**.** *By Theorem 11.8 the label of a family is read at the centres $\Upsilon_{\boldsymbol{\sigma},0;\mathbf{1}}$ of the table's gauge of its molecules, and a molecule carrying one table is decided by reading it once, at any one such centre.*

**Corollary 11.10** (The leaning realizations at one pair)**.** *Let a molecule be read at one pair at the table's gauge, with the strengths $\boldsymbol{\sigma}$. Its leaning realizations $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$, $\boldsymbol{\delta} = \frac{\delta}{\sigma}\boldsymbol{\sigma}$, $\delta \neq 0$, are atomic all together or not at all; if they are, so is the centre, and an atomic centre need not make them atomic.*

Whether a molecule can be atomic and whether a particular realization of it is atomic are read differently. The first is read at the centres of the table's gauge, at any one centre when the molecule carries one table; the second at the realization itself, and a lean may split what the centre held together.

**Lemma 11.11** (No degenerate pair)**.** *At every realization $\Upsilon$ of a family and for marks $e \neq f$, $\kappa^{\mathbf{E}}_{\Upsilon}(e, f)^2 \leq \kappa^{\mathbf{E}}_{\Upsilon}(e, e)\,\kappa^{\mathbf{E}}_{\Upsilon}(f, f)$, with equality exactly when $e$ or $f$ is a unit or $\{e, f\}$ is a molecule. In particular the inequality is strict for any two marks of a molecule with at least three marks.*

**Lemma 11.12** (Equal diagonals on a class)**.** *At every realization $\Upsilon$, for each part and every $k \geq 2$, the diagonal entry $\kappa^{\mathbf{X}}_{\Upsilon}(e,\dots,e) = 2^k(\varphi^{\mathbf{X}}_e)^{(k-1)}(0)$ of rung $k$ is the same at every mark of a class, the reading determining the self-line functions (Proposition 10.3(iii)); in particular $\kappa^{\mathbf{E}}_{\Upsilon}(e,e) = 4\,(\varphi^{\mathbf{E}}_e)'(0)$ is constant on every class, and at an atom it is the same at every mark.*

## 11.4 Maps as certificates

**Definition 11.13** (Carrying and exchanging)**.** Let $F$ and $F'$ be families on $\mathcal{E}$ and $\mathcal{E}'$ with tables indexed by $\{1,\dots,N\}$ and $\{1,\dots,N'\}$. A bijection $g : \mathcal{E} \to \mathcal{E}'$ *carries* $F$ to $F'$ when some bijection $h : \{1,\dots,N\} \to \{1,\dots,N'\}$ has $g(\mathcal{B}_j) = \mathcal{B}'_{h(j)}$, $\theta'_{h(j),g(T)} = \theta_{j,T}$ and $(\sigma'_{h(j)}, \delta'_{h(j)}) = (\sigma_j, \delta_j)$; it *exchanges* $F$ with $F'$ when instead $T \mapsto \mathcal{E}' \setminus g(T)$ maps $\mathcal{B}_j$ onto $\mathcal{B}'_{h(j)}$ with $\theta'_{h(j),\mathcal{E}'\setminus g(T)} = \theta_{j,T}$ and the same condition on the pairs. With $F' = F$, $g$ carries or exchanges the family. Write $(g^*y)_e = y_{g(e)}$ for $y \in \mathbb{R}^{\mathcal{E}'}$.

**Proposition 11.14** (What a map certifies)**.** *At gauges $\alpha$ of $F$ and $\alpha'$ of $F'$ with $g^* \log \alpha' = \log \alpha$, the tables' gauges in particular, and in this statement writing $\Phi^{\boldsymbol{\delta}}_{\mathbf{O}}$ for the odd potential at the lean $\boldsymbol{\delta}$: if $g$ carries $F$ to $F'$ then $\Phi_{\mathbf{X}} \circ g^* = \Phi'_{\mathbf{X}}$, and the reading of $g(e)$ in $F'$ equals that of $e$ in $F$; if $g$ exchanges $F$ with $F'$ then $\Phi_{\mathbf{E}} \circ g^* = \Phi'_{\mathbf{E}}$ and $\Phi^{-\boldsymbol{\delta}}_{\mathbf{O}} \circ g^* = \Phi'^{\boldsymbol{\delta}}_{\mathbf{O}}$, and the reading of $g(e)$ in $F'$ at $-\boldsymbol{\delta}$ equals that of $e$ in $F$ at $\boldsymbol{\delta}$. So a map carrying the family takes classes onto classes, and an exchange of the family takes the classes at $\boldsymbol{\delta}$ onto those at $-\boldsymbol{\delta}$ and the classes of the centre onto the classes of the centre.*

A map is a certificate, not a definition: classes are the level sets of the reading, and a map carrying or exchanging the family is one way to exhibit that two marks read alike. With one table the writings and the maps say the same thing (Corollary 11.16); with several tables the reading is the definition.

**Corollary 11.15** (An exchange negates odd cumulants)**.** *If $g$ exchanges a family, with $h$ the bijection of its tables (Definition 11.13), then under the weight laws $\kappa_{h(j),\mathbf{1}}(g(w)) = (-1)^k \kappa_{j,\mathbf{1}}(w)$ for every table $j$ and every string $w$ of length $k \geq 2$; so the stills at the table's gauge satisfy $\kappa_{\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}}(g(w)) = \kappa_{\Upsilon_{\boldsymbol{\sigma},-\boldsymbol{\delta};\mathbf{1}}}(w)$, and $g$ maps every class of the centre onto itself.*

**Corollary 11.16** (With one table the writings are certificates)**.** *In a one-table molecule at the table's gauge, $e$ and $f$ lie in one class at a lean exactly when some bijection taking $e$ to $f$ carries the table, and in one class at the centre exactly when some bijection taking $e$ to $f$ carries or exchanges it.*

**Theorem 11.17** (A class across the centre)**.** *In a one-table molecule at the table's gauge, a class $C$ of the centre is, at every lean, one class or two classes with equally many marks. In the second case the two are $C_K$ and $C_\Psi$, the marks of $C$ from which $K$, respectively $\Psi$, is written as one and the same polynomial, and an exchange $\beta$ built from two such writings has $\beta(C) = C$ and $\beta(C_K) = C_\Psi$.*

Leaning away from the centre, a class of a one-table molecule either stays whole or splits into the half from which the basis polynomial is written and the half from which the cobasis polynomial is: the lean sign decides which half the Foreman meets first (Corollary 13.11).

# Chapter 12

# Restriction to a bunch

## 12.1 Patterns and the Taylor cube

**Definition 12.1** (Patterns and children)**.** Let $(\mathcal{B},\theta)$ be a table of rank $r$ on $\mathcal{E}$. A bunch (Definition 3.1) is here any nonempty set $B\subseteq\mathcal{E}$, and $R=\mathcal{E}\setminus B$ is its *room*; for a family the objects of a bunch in table $j$ carry the index $j$. The *patterns* of $B$ are the traces of the bases on it, $\mathrm{pat}(B)=\{T\cap B:\ T\in\mathcal{B}\}$, and $\mathrm{pat}_j(B)$ for table $j$ of a family. For $p\in\mathrm{pat}(B)$ the *signed pattern* is $\chi_p=\mathbf{1}_p-\mathbf{1}_{B\setminus p}\in\mathbb{R}^B$ and the *signed level* is $\ell_p=|p|-\frac{|B|}{2}$. The *child* at $p$ is the table on $R$ with the bases $\mathcal{B}_{(p)}=\{T\setminus p:\ T\cap B=p\}$ and the weights $\theta_{(p),X}=\theta_{X\cup p}$; its Foreman polynomials and its centred polynomial are $K_{(p)}$, $\Psi_{(p)}$ and $\Lambda_{(p)}$.

**Lemma 12.2** (Children are tables)**.** *$\mathcal{B}_{(p)}$ is the set of bases of the minor contracting $p$ and deleting $B\setminus p$ (Definition 7.5); it has the exchange property and rank $r-|p|$.*

**Lemma 12.3** (Children do not widen)**.** *Let $B$ be a bunch of a table with the room $R$, $p$ a pattern of $B$, $A\subseteq R$ and $W\subseteq B$. The width of $A$ in the child at $p$ is at most the width of $A\cup W$ in the table: $c_{(p)}(A)\le c(A\cup W)$.*

**Definition 12.4** (Communicatroid; interchange)**.** The *communicatroid* of $B$ is the indexed family of its children $(\mathcal{B}_{(p)},\theta_{(p)})_{p\in\mathrm{pat}(B)}$, tables on the room indexed by their patterns; its *gluing* is $\{U\sqcup p:\ p\in\mathrm{pat}(B),\ U\in\mathcal{B}_{(p)}\}$ with the weights $\theta_{(p),U}$. An indexed family $(\mathcal{A}_p)_p$ of tables on $R$, indexed by sets $p\subseteq B$ with $\mathcal{A}_p$ of rank $r-|p|$, has *interchange* when, for $U\sqcup p$ and $U'\sqcup p'$ in its gluing and $x\in(U\sqcup p)\setminus(U'\sqcup p')$, there is $y\in(U'\sqcup p')\setminus(U\sqcup p)$ with

$$U-x+y\in\mathcal{A}_p\ \ (x,y\in R),\qquad U+y\in\mathcal{A}_{p-x}\ \ (x\in B,\ y\in R),$$
$$U-x\in\mathcal{A}_{p+y}\ \ (x\in R,\ y\in B),\qquad U\in\mathcal{A}_{p-x+y}\ \ (x,y\in B):$$

a step of the exchange property inside a child, one level down, one level up, or along a level.

**Proposition 12.5** (Interchange is the parent's exchange property)**.** *The gluing of the communicatroid of $B$ is the table, weights included, and the communicatroid has interchange. Conversely, an indexed family of tables on $R$ with interchange is the communicatroid of $B$ in its gluing, which is a table: the children remember the parent because the parent is the glued children.*

**Proposition 12.6** (Adjacent siblings)**.** *If $p$ and $p+b$ are patterns of $B$, the children at $p$ and at $p+b$ are the deletion and the contraction of $b$ in one table on $R+b$, the minor contracting $p$ and deleting $B\setminus(p+b)$. So the child at $p+b$ is a quotient of the child at $p$ of rank one less: every*

*set of room marks that is maximal for its rank in the child at $p+b$ is maximal for its rank in the child at $p$, every basis of the child at $p+b$ extends by one room mark to a basis of the child at $p$, and every basis of the child at $p$ less one suitable mark is a basis of the child at $p+b$.*

**Theorem 12.7** (The Taylor cube)**.** *For $\alpha=(\alpha_B,\alpha_R)$ and $z=(z_B,z_R)$,*

$$K(\alpha)=\sum_{p\in\mathrm{pat}(B)}\alpha_B^{p}K_{(p)}(\alpha_R),\qquad \Psi(\alpha)=\sum_{p\in\mathrm{pat}(B)}\alpha_B^{B\setminus p}\Psi_{(p)}(\alpha_R),$$
$$\Lambda(z)=\sum_{p\in\mathrm{pat}(B)}e^{\langle z_B,\chi_p\rangle}\Lambda_{(p)}(z_R),$$

*with $K_{(p)}=(\partial^p K)|_{\alpha_B=0}$. Along the scale of $B$, $\alpha_B=e^{2z_B}=\rho u$ with $u\in\Delta_B$, the factor of $p$ in $\Lambda$ is $e^{\langle z_B,\chi_p\rangle}=\rho^{\ell_p}u^{\chi_p/2}$; the values $|p|$ fill $\{d(B),\dots,r(B)\}$, and the signed levels fill a window of width $c(B)$ centred at $\varepsilon(B)/2$. For $B=\{g\}$ this is the expansion $K=K_{\setminus g}+\alpha_g K_{/g}$, $\Psi=\alpha_g\Psi_{\setminus g}+\Psi_{/g}$ of Definition 7.3.*

The Taylor cube is the bunch read on its own coordinates: each pattern contributes a monomial on the bunch, at the level $|p|$, times the polynomial of its child on the room. The levels that occur fill the window of the bunch (Theorem 7.13), and nothing about the room is lost, since the children glued back are the table.

**Theorem 12.8** (The pattern tensor)**.** *Under the basis law at $\alpha$, $\mathbb{P}_\alpha(T\cap B=p)=\alpha_B^{p}K_{(p)}(\alpha_R)/K(\alpha)$, and under the cobasis law $\mathbb{P}^*_\alpha(T\cap B=p)=\alpha_B^{B\setminus p}\Psi_{(p)}(\alpha_R)/\Psi(\alpha)$. The moments of the bunch are the upper sums $\mathbb{P}_\alpha(T\cap B\supseteq S)=\sum_{p\supseteq S}\mathbb{P}_\alpha(T\cap B=p)$, and the joint cumulants $\kappa_\alpha(w)$ on the strings $w$ of marks of $B$ are their image under the moment–cumulant relation. Scaling $\alpha_B$ multiplies the entry of $p$ by its own monomial, and inverting the gauge swaps the two laws pattern by pattern (Proposition 12.28(ii) below).*

**Proposition 12.9** (Tinkers along a bunch; selecting patterns)**.** *For a* tinker *$v=(v_p)_{p\in\mathrm{pat}(B)}$ of positive numbers, the weights $v_{T\cap B}\theta_T$ define the* tinkered table*, with the same bases and the Foreman polynomials*

$$K_v(\alpha)=\sum_p v_p\,\alpha_B^{p}K_{(p)}(\alpha_R),\qquad \Psi_v(\alpha)=\sum_p v_p\,\alpha_B^{B\setminus p}\Psi_{(p)}(\alpha_R).$$

*Selecting the pattern $p$ on $B$, that is keeping the bases with $T\cap B=p$, and then $p'$ on $B'$ selects $p\cup p'$ on $B\cup B'$ when $p\cap B'=p'\cap B$, and nothing otherwise; so the selections on a table with $n$ marks are at most the $3^n$ conditions that require each mark present, absent, or nothing. At the centre every tinkered table keeps $\mathbf{1}$ native (Proposition 8.21).*

## 12.2 The Skeleton polynomials of a bunch

**Definition 12.10** (Skeleton polynomials, values and coins)**.** At a gauge $\alpha$, with the level $N_B=|T\cap B|$ and the *colevel* $N_B^*=|B\setminus T|$, the *Skeleton polynomials* of $B$ are

$$\mathsf{Q}_B^{K}(x;\alpha)=\mathbb{E}_{T\sim\mathbb{P}_\alpha}\big[x^{N_B-d(B)}\big]=x^{-d(B)}\frac{K(x\alpha_B,\alpha_R)}{K(\alpha)},$$
$$\mathsf{Q}_B^{\Psi}(x;\alpha)=\mathbb{E}_{T\sim\mathbb{P}^*_\alpha}\big[x^{N_B^*-d^*(B)}\big]=x^{-d^*(B)}\frac{\Psi(x\alpha_B,\alpha_R)}{\Psi(\alpha)},$$

and $\mathsf{Q}^K_{j,B}$, $\mathsf{Q}^\Psi_{j,B}$ for table $j$ of a family. Each has degree $c(B)$, positive coefficients and value 1 at $x=1$, so its roots avoid $[0,\infty)$; writing them $-\xi_1,\dots,-\xi_{c(B)}$,

$$\mathsf{Q}^K_B(x;\alpha)=\prod_{i=1}^{c(B)}\frac{x+\xi_i}{1+\xi_i}=\prod_{i=1}^{c(B)}(1-p_i+p_ix),\qquad p_i=\frac{1}{1+\xi_i}.$$

The $\xi_i$ are the *Skeleton values*, their multiset $\operatorname{spec}^K_B(\alpha)\subset\mathbb{C}\setminus(-\infty,0]$ is the *Skeleton spectrum*, and the $p_i$ are the *coins*; likewise $\operatorname{spec}^\Psi_B(\alpha)$ and the coins of $\mathsf{Q}^\Psi_B$. The *cumulants of a coin* are $\varkappa_1(p)=p$ and $\varkappa_{k+1}(p)=p(1-p)\,\varkappa_k'(p)$.

The Skeleton polynomial is the law of the level of the bunch, shifted to start at its floor: its degree is the width of the bunch, and its roots are the negatives of the Skeleton values. When the spectrum is positive the level less its floor $d(B)$ is a sum of $c(B)$ independent $\{0,1\}$-valued variables whose means are the coins.

**Proposition 12.11** (Relations of the Skeleton polynomials)**.** *(i)* Scaling. *Scaling $\alpha_B$ by $\lambda>0$ divides the spectrum by $\lambda$:*

$$\mathsf{Q}^K_B\big(x;(\lambda\alpha_B,\alpha_R)\big)=\mathsf{Q}^K_B(\lambda x;\alpha)/\mathsf{Q}^K_B(\lambda;\alpha);$$

*likewise for $\Psi$; scaling all of $\alpha$ changes nothing.*

*(ii)* Complement. $\mathsf{Q}^K_R(x;\alpha)=x^{c(B)}\mathsf{Q}^K_B(x^{-1};\alpha)$*: the spectrum of the room is the reciprocal of that of the bunch, and each coin $p_i$ becomes $1-p_i$; likewise for $\Psi$.*

*(iii)* Cremona. $\mathsf{Q}^\Psi_B(x;\alpha)=x^{c(B)}\mathsf{Q}^K_B(x^{-1};\alpha^{-1})$*; at the table's gauge,* $\mathsf{Q}^\Psi_B(x;\mathbf{1})=x^{c(B)}\mathsf{Q}^K_B(x^{-1};\mathbf{1})$.

*(iv)* Coins. $\log\mathsf{Q}^K_B(e^v;\alpha)=\sum_i\log(1-p_i+p_ie^v)$*; under $\mathbb{P}_\alpha$ the level $N_B$ has mean $d(B)+\sum_i p_i=\sum_{e\in B}R_e(\alpha)$ and $k$-th cumulant $\sum_i\varkappa_k(p_i)$ for $k\ge2$. When the spectrum is positive the coins lie in $(0,1)$ and $N_B-d(B)$ is distributed as a sum of independent $\{0,1\}$-valued variables whose means are the coins.*

**Proposition 12.12** (The still read along a bunch)**.** *Let a family be at a gauge $\alpha$, let $p_{j,i}$ be the coins of $\mathsf{Q}^K_{j,B}(\cdot;\alpha)$ and $p^*_{j,i}$ those of $\mathsf{Q}^\Psi_{j,B}(\cdot;\alpha)$, and put $\partial_B=\sum_{e\in B}\partial_{y_e}$. Then*

$$\sum_j\Big(s_j\sum_i p_{j,i}+t_j\sum_i p^*_{j,i}\Big)=\partial_B\Phi(\log\alpha)-\sum_j\big(s_jd_j(B)+t_jd^*_j(B)\big),$$
$$\partial_B^k\Phi(\log\alpha)=\sum_j\Big(s_j\sum_i\varkappa_k(p_{j,i})+t_j\sum_i\varkappa_k(p^*_{j,i})\Big)\qquad(k\ge2).$$

*At a native gauge the first total is the wall $\kappa_B$ at the masses $\boldsymbol{\tau}$, and on the balance wall it is $\kappa_B$; the entries of rung $k$ of the still of $\big((s_j,t_j)_j;\alpha\big)$, summed over $B^k$, are $2^k$ times the second total.*

**Proposition 12.13** (Coins of a tensorminental table)**.** *In the setting of Theorem 7.23, let $p_i^{(kl)}$, $1\le i\le|B|$, be the eigenvalues of $\Pi^{(kl)}_{BB}$. Then*

$$x^{d(B)}\mathsf{Q}^K_B(x;\alpha)=\sum_{k,l}w_{kl}\prod_{i=1}^{|B|}\big(1-p_i^{(kl)}+p_i^{(kl)}x\big)\,:$$

*the Skeleton polynomial is the $w$-mixture of the coin products of the components. At tensor rank one over $\mathbb{F}\in\{\mathbb{R},\mathbb{C}\}$ there is one component, and $\Pi^{(11)}_{BB}$ is similar to the compression to $\mathbb{F}^B$ of the*

*orthogonal projection onto the row space of $AD_\alpha^{1/2}$, $A$ the slice; its eigenvalues are $\cos^2\vartheta_i$ for the principal angles $\vartheta_i$ between $\mathbb{F}^B$ and that row space. Of these, $d(B)$ are 1 and $d^*(B)$ are 0, and the $c(B)$ coins are the values $\cos^2\vartheta_i \in (0,1)$, with the positive Skeleton values $\tan^2\vartheta_i$.*

**Definition 12.14** (The pair of Skeleton functions)**.** The *even* and *odd Skeleton functions* of $B$ are $\mathsf{Q}^{\mathbf{E}}_B(x;\alpha) = \mathsf{Q}^K_B(x;\alpha)\,\mathsf{Q}^\Psi_B(x;\alpha)$ and $\mathsf{Q}^{\mathbf{O}}_B(x;\alpha) = \mathsf{Q}^K_B(x;\alpha)/\mathsf{Q}^\Psi_B(x;\alpha)$. So

$$\mathsf{Q}^{\mathbf{E}}_B(x;\alpha) = \mathbb{E}_{(T,T')\sim\mathbb{P}_\alpha\otimes\mathbb{P}^*_\alpha}\Big[x^{|T\cap B|-d(B)+|B\setminus T'|-d^*(B)}\Big],$$

and its roots are the negatives of the *joint spectrum* $\operatorname{spec}^K_B(\alpha)\cup\operatorname{spec}^\Psi_B(\alpha)$.

**Proposition 12.15** (A set on its own scale is read by its Skeleton functions)**.** *For a family at a gauge $\alpha$, a set $C$ of marks and $x = e^v$,*

$$\Phi_{\mathbf{E}}(\log\alpha + v\mathbf{1}_C) - \Phi_{\mathbf{E}}(\log\alpha) = \sum_j \sigma_j\Big(\log\mathsf{Q}^{\mathbf{E}}_{j,C}(x;\alpha) + \big(d_j(C)+d_j^*(C)\big)v\Big),$$
$$\Phi_{\mathbf{O}}(\log\alpha + v\mathbf{1}_C) - \Phi_{\mathbf{O}}(\log\alpha) = \sum_j \delta_j\Big(\log\mathsf{Q}^{\mathbf{O}}_{j,C}(x;\alpha) + \big(d_j(C)-d_j^*(C)\big)v\Big).$$

*Along the scale of a set the $K$-half of each table's joint spectrum carries the exponent $s_j = \sigma_j + \delta_j$ and the $\Psi$-half carries $t_j = \sigma_j - \delta_j$.*

## 12.3 Levels

**Definition 12.16** (Levels)**.** For $d(B) \le k \le r(B)$ the *level-$k$ patterns* are $\operatorname{pat}^{(k)}(B) = \{p \in \operatorname{pat}(B) :\ |p| = k\}$, the *level-$k$ bases* are $\mathcal{B}^{(k)}_B = \{T \in \mathcal{B} :\ |T\cap B| = k\}$, and the *level letter* is

$$K^{(k)}_B(\alpha) = \sum_{T\in\mathcal{B}^{(k)}_B}\theta_T\alpha^T = \sum_{p\in\operatorname{pat}^{(k)}(B)}\alpha^p_B K_{(p)}(\alpha_R).$$

For $d^*(B) \le k \le |B| - d(B)$ the *colevel letter* is $\Psi^{(k)}_B(\alpha) = \sum_{T:\,|B\setminus T|=k}\theta_T\alpha^{\mathcal{E}\setminus T}$. The *excess* of the level $k$, of its patterns and of its bases is $k - d(B)$, and that of the colevel $k$ is $k - d^*(B)$. Then $K = \sum_k K^{(k)}_B$, $\Psi = \sum_k \Psi^{(k)}_B$,

$$x^{d(B)}\mathsf{Q}^K_B(x;\alpha) = \sum_k x^k\,\frac{K^{(k)}_B(\alpha)}{K(\alpha)}, \qquad x^{d^*(B)}\mathsf{Q}^\Psi_B(x;\alpha) = \sum_k x^k\,\frac{\Psi^{(k)}_B(\alpha)}{\Psi(\alpha)}.$$

**Proposition 12.17** (Pattern levels are lifts)**.** *Let $\mathcal{B}|B$ be the restriction to $B$ and $\mathcal{B}/R$ the contraction of the room, a table on $B$ with rank function $r_{\mathcal{B}/R}(X) = r(X\cup R) - r(R)$ and rank $d(B)$.*

*(i) $p \subseteq B$ is a pattern if and only if $p$ is independent and $r_{\mathcal{B}/R}(p) = d(B)$.*

*(ii) For $i = k - d(B)$, the independent sets $X \subseteq B$ with $|X| - r_{\mathcal{B}/R}(X) \le i$ are the independent sets of a table on $B$ of rank $k$, the $i$-th lift of $\mathcal{B}/R$ toward $\mathcal{B}|B$, whose bases are the level-$k$ patterns. So every pattern level $\operatorname{pat}^{(k)}(B)$ is a table on $B$.*

*(iii) The room sides $T\cap R$ of the level-$k$ bases are the bases of the $(r(B)-k)$-th lift of $\mathcal{B}/B$ toward $\mathcal{B}|R$.*

**Definition 12.18** (Interaction)**.** Let a table have a bunch $B$ with room $R$ and $\mathrm{pat}(B) = 2^B$: every $p \subseteq B$ is a pattern, equivalently $c(B) = |B|$, that is $B$ is independent and some basis avoids it (Proposition 12.17(i)). The *interaction* and the *dual interaction* of $B$ are

$$\iota_B(\alpha_R) = \sum_{p\subseteq B} (-1)^{|B\setminus p|} \log K_{(p)}(\alpha_R), \qquad \iota^*_B(\alpha_R) = \sum_{p\subseteq B} (-1)^{|p|} \log \Psi_{(p)}(\alpha_R),$$

the second the interaction of the dual table, and for a split of $B$ into blocks $B_1, \dots, B_k$,

$$\iota_{(B_1,\dots,B_k)}(\alpha_R) = \sum_{A\subseteq\{1,\dots,k\}} (-1)^{k-|A|} \log K_{(\cup_{a\in A} B_a)}(\alpha_R).$$

For two marks, $\iota_{\{e,f\}} = \log \frac{K_{(\{e,f\})} K_{(\varnothing)}}{K_{(\{e\})} K_{(\{f\})}}$, the logarithm of the odds ratio of the presence of the two marks. For a family with $\mathrm{pat}_j(B) = 2^B$ in every active table $j$, the *interaction* of $B$ is the pair

$$\iota^{\mathbf{E}}_B = \sum_j \sigma_j\big(\iota_{j,B} + \iota^*_{j,B}\big), \qquad \iota^{\mathbf{O}}_B = \sum_j \delta_j\big(\iota_{j,B} - \iota^*_{j,B}\big), \qquad \iota^{\mathbf{E}}_B + \iota^{\mathbf{O}}_B = \sum_j \big(s_j \iota_{j,B} + t_j \iota^*_{j,B}\big),$$

the interactions of the even and the odd part of the potential (Definition 8.18).

**Theorem 12.19** (Levels are sums of children)**.** $\mathcal{B}^{(k)}_B = \bigsqcup_{p\in\mathrm{pat}^{(k)}(B)} \mathcal{B}_{[p]}$, *where* $\mathcal{B}_{[p]} = \{T \in \mathcal{B} : T \cap B = p\}$ *is the set of bases of the table on* $\mathcal{E}$ *that has* $p$ *as coloops,* $B \setminus p$ *as loops and the child at* $p$ *on* $R$*, with the parent's weights; its basis polynomial is* $\alpha^p_B K_{(p)}(\alpha_R)$*. A bijection* $g$ *of* $\mathcal{E}$ *that carries the table and keeps* $B$ *takes* $\mathcal{B}_{[p]}$ *to* $\mathcal{B}_{[g(p)]}$*. The extreme levels have constant children:*

$$\mathcal{B}^{(d(B))}_B = \{p \sqcup X :\ p \in \mathcal{B}/R,\ X \in \mathcal{B}|R\}, \qquad \mathcal{B}^{(r(B))}_B = \{p \sqcup X :\ p \in \mathcal{B}|B,\ X \in \mathcal{B}/B\};$$

*for a table determinantal over* $\mathbb{R}$ *or* $\mathbb{C}$ *the weights of the extreme levels factor along these products, and with them the extreme letters,* $K^{(d(B))}_B(\alpha) = K_{\mathcal{B}/R}(\alpha_B)\, K_{\mathcal{B}|R}(\alpha_R)$ *and* $K^{(r(B))}_B(\alpha) = K_{\mathcal{B}|B}(\alpha_B)\, K_{\mathcal{B}/B}(\alpha_R)$*.*

The bases of one level need not form a table: a step of the exchange property may leave the level, and it then moves to an adjacent one (Proposition 12.5). What a level keeps is its patterns, which form a table (Proposition 12.17); its bases are the glued children of those patterns, and children on adjacent levels are linked by Proposition 12.6. The level is the coordinate the bunch keeps for itself, and the children hand the rest down to the room.

**Proposition 12.20** (The bottom of a chain factors over its steps)**.** *Let* $(\mathcal{B}, \theta)$ *be a table on* $\mathcal{E}$ *and* $\varnothing = S_0 \subsetneq S_1 \subsetneq \cdots \subsetneq S_\ell \subsetneq S_{\ell+1} = \mathcal{E}$*,* $\ell \geq 1$*, a chain with the steps* $Z_i = S_i \setminus S_{i-1}$*,* $1 \leq i \leq \ell+1$*. Some bases are at the bottom of every* $S_i$*,* $|T \cap S_i| = d(S_i)$ *for* $1 \leq i \leq \ell$*. For any one of them,* $T_0$*, and every* $i$*, the traces on* $Z_i$ *of these bases are the bases of the child* $\mathcal{B}_{(T_0\setminus Z_i)}$ *at the pattern* $T_0 \setminus Z_i$ *of* $\mathcal{E} \setminus Z_i$ *(Definition 12.1), the* step table at the bottom *at* $Z_i$*, which depends on* $S_{i-1}$ *and* $S_i$ *alone; and the bases at the bottom of every* $S_i$ *are exactly the unions* $\bigsqcup_i X_i$ *of one basis* $X_i$ *of each step table. Likewise the bases at the top of every* $S_i$*,* $|T \cap S_i| = r(S_i)$*, with one of them* $T_1$*, give the* step tables at the top $\mathcal{B}_{(T_1\setminus Z_i)}$*.*

*For a list, the terms of* $K$ *of the least degree in* $\alpha_{S_i}$ *for every* $i$ *at once are the product* $\prod_i K_{\mathcal{B}_{(T_0\setminus Z_i)}}(\alpha_{Z_i})$*, and those of* $\Psi$ *the product* $\prod_i \Psi_{\mathcal{B}_{(T_1\setminus Z_i)}}(\alpha_{Z_i})$*; the factors are the* step letters *of* $K$ *and of* $\Psi$*, the same products holding for a table whose weights on these bases are products of one factor for each step. For the linear letter* $\alpha_J$*, the basis polynomial of* $\binom{J}{1}$ *read on* $\mathcal{E}$ *with the marks outside* $J$ *as loops, the step letter is* $\alpha_{J\cap Z_i}$ *at the first step* $Z_i$ *with* $J \subseteq S_i$ *and* $1$ *at the other*

*steps. Along a maximal chain, every step one mark, exactly one basis is at the bottom of every $S_i$, and for a list the terms of $K$ of the least degree in every $\alpha_{S_i}$ at once are one monomial with the coefficient* 1.

**Theorem 12.21** (Shared walls)**.** *Let $L_i = \sum_h K_{i,h}$, $i = 1, \dots, q$, each a sum of the basis polynomials $K_{i,h}$ of tables $\mathcal{B}_{i,h}$ on $\mathcal{E}$, with exponents $w_i > 0$, and put $d_{i,h}(S) = \min_{T \in \mathcal{B}_{i,h}} |T \cap S|$. Then*

$$\int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\mathbf{m}-\mathbf{1}} e^{-|\alpha|} \prod_i L_i(\alpha)^{-w_i}\, d\alpha$$

*converges if and only if there are $w_{i,h} > 0$ with $\sum_h w_{i,h} = w_i$ and $m_S > \sum_{i,h} w_{i,h}\, d_{i,h}(S)$ for every nonempty $S \subseteq \mathcal{E}$; equivalently $\mathbf{m} \in \operatorname{int}\big(\sum_i w_i \operatorname{conv} \bigcup_h P_{i,h} + \mathbb{R}^{\mathcal{E}}_{\geq 0}\big)$, with $P_{i,h}$ the base polytope of $\mathcal{B}_{i,h}$.*

**Proposition 12.22** (The walls of one level are exact)**.** *With $P$ the base polytope, $P_B^{(k)} = \operatorname{conv}\{\mathbf{1}_T : T \in \mathcal{B}_B^{(k)}\}$ and $d_B^{(k)}(S) = \min_{T \in \mathcal{B}_B^{(k)}} |T \cap S|$:*

$$P_B^{(k)} = P \cap \{x_B = k\}, \qquad P_B^{(k)} + \mathbb{R}^{\mathcal{E}}_{\geq 0} = \{x :\ x_S \geq d_B^{(k)}(S) \text{ for all } S\}.$$

*So for $w > 0$ the integral $\int_{\mathbb{R}^{\mathcal{E}}_{>0}} \alpha^{\mathbf{m}-\mathbf{1}} e^{-|\alpha|} K_B^{(k)}(\alpha)^{-w} d\alpha$ converges exactly when $m_S > w\, d_B^{(k)}(S)$ for every nonempty $S$. The one-dimensional faces of $P_B^{(k)}$ that do not join two bases differing by a single step of the exchange property are diagonals of square two-dimensional faces of $P$, one step into $B$ and one out of it.*

For several levels together the 0/1 walls need not be exact: the facets of $\sum_k w_k P_B^{(k)} + \mathbb{R}^{\mathcal{E}}_{\geq 0}$ are normal to rays of the common refinement of the normal fans of the $P_B^{(k)}$, which can lie outside $\{0,1\}^{\mathcal{E}}$, and the exact domain of several levels together is then the one of Theorem 12.21.

**Corollary 12.23** (The native level split is interior)**.** *Let a family with real data, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, have a native gauge $\alpha^\star$, and give the level letters $K_{j,B}^{(k)}$ of table $j$ the exponents $s_j\, \mathbb{P}_{j,\alpha^\star}(N_B = k)$ and its colevel letters $\Psi_{j,B}^{(k)}$ the exponents $t_j\, \mathbb{P}^*_{j,\alpha^\star}(N_B^* = k)$. Then $\boldsymbol{\tau}$ lies in the relative interior of the Minkowski sum of the level and colevel polytopes, each multiplied by the exponent of its letter, and every wall of these letters at the masses $\boldsymbol{\tau}$,*

$$\tau_S - \sum_{j,k} \Big( s_j\, \mathbb{P}_{j,\alpha^\star}(N_B = k)\, d_{j,B}^{(k)}(S) + t_j\, \mathbb{P}^*_{j,\alpha^\star}(N_B^* = k)\, d_{j,B}^{*(k)}(S) \Big),$$

*$d_{j,B}^{(k)}$ the level counts of Proposition 12.22 and $d_{j,B}^{*(k)}(S)$ the least of $|S \setminus T|$ over the colevel-$k$ bases, is positive except on the sets $S$ on which $x_S$ is constant over every level and colevel polytope, among them $B$ and $R$. If the marks of $B$ are joined to one another, and the marks of $R$ to one another, by chains in which consecutive marks lie in a common circuit of a table with a positive exponent, as they are when such a table is connected, then the directions $v$ with $\langle v, x\rangle$ constant on that Minkowski sum are spanned by $\mathbf{1}_B$ and $\mathbf{1}_R$, there are no exceptions besides these and $\varnothing, \mathcal{E}$, and*

$$\int_{\Delta_B \times \Delta_R} u^{\boldsymbol{\tau}|_B - \mathbf{1}} v^{\boldsymbol{\tau}|_R - \mathbf{1}} \prod_{j,k} K_{j,B}^{(k)}(u,v)^{-s_j \mathbb{P}_{j,\alpha^\star}(N_B = k)} \Psi_{j,B}^{(k)}(u,v)^{-t_j \mathbb{P}^*_{j,\alpha^\star}(N_B^* = k)}\, d\omega_B(u)\, d\omega_R(v)$$

*converges.*

## 12.4 Shares and the native split

**Definition 12.24** (Shares, loads, children of a family)**.** For a family and a bunch $B$, *shares* are $\mathbf{a}_j = (a_{j,p})_{p\in \mathrm{pat}_j(B)}$ with $|\mathbf{a}_j| = s_j$ and $\mathbf{b}_j = (b_{j,p})_{p\in \mathrm{pat}_j(B)}$ with $|\mathbf{b}_j| = t_j$; the *pair shares* are $\sigma_{j,p} = \frac{a_{j,p}+b_{j,p}}{2}$ and $\delta_{j,p} = \frac{a_{j,p}-b_{j,p}}{2}$. A $K$-share of pattern $p$ loads the marks of $p$, and a $\Psi$-share loads those of $B \setminus p$: the *load* and the *odd load* on $e \in B$ are

$$\mathrm{load}_e = \sum_j \Big( \sum_{p\ni e} a_{j,p} + \sum_{p\not\ni e} b_{j,p} \Big) = \sigma + \langle \boldsymbol{\delta} \rangle_e, \qquad \langle \boldsymbol{\delta} \rangle_e = \sum_{j,p} \delta_{j,p} \chi_p(e).$$

The *children* are the family

$$F_B(\mathbf{a}, \mathbf{b}) = \big( (\mathcal{B}_{j,(p)}, \theta_{j,(p)});\, a_{j,p}, b_{j,p} \big)_{j,p} \quad \text{on } R.$$

**Definition 12.25** (Pattern probabilities; the native split)**.** At a gauge $\beta$ the *pattern probabilities* of table $j$ are $\pi_{j,p}(\beta) = \mathbb{P}_{j,\beta}(T \cap B = p)$ and $\pi^*_{j,p}(\beta) = \mathbb{P}^*_{j,\beta}(T \cap B = p)$. At a native gauge $\alpha^\star$ the *native split* is the choice of shares $\mathbf{a}^\star$, $\mathbf{b}^\star$ with $a^\star_{j,p} = s_j \pi_{j,p}(\alpha^\star)$ and $b^\star_{j,p} = t_j \pi^*_{j,p}(\alpha^\star)$, and $F_B(\mathbf{a}^\star, \mathbf{b}^\star)$ are the children at the native split.

**Theorem 12.26** (Nativity is hereditary)**.** *$\alpha^\star|_R$ is a native gauge for the children $F_B(\mathbf{a}^\star, \mathbf{b}^\star)$ with target $\boldsymbol{\tau}|_R$, and they lie on their own balance wall at the masses $\boldsymbol{\tau}|_R$; every set $S \subseteq R$ with $c_{j,(p)}(S) = 0$ for every child table is balanced, $\tau_S = \sum_{j,p} \big( a^\star_{j,p} r_{j,(p)}(S) + b^\star_{j,p} r^*_{j,(p)}(S) \big)$.*

**Proposition 12.27** (The centre is hereditary)**.** *Let $\Upsilon = ((s_j, t_j)_j; \mathbf{1})$ be at the table's gauge, with its masses. The native split gives the child $(j,p)$ the pair*

$$\sigma_{j,p} = \pi_{j,p}(\mathbf{1})\, \sigma_j, \qquad \delta_{j,p} = \pi_{j,p}(\mathbf{1})\, \delta_j, \qquad \pi_{j,p}(\mathbf{1}) = \mathbb{P}_{j,\mathbf{1}}(T \cap B = p) > 0,$$

*and the gauge $\mathbf{1}$ on $R$, the children's own table's gauge, native for them. At a centre every child sits at a centre; at one pair the children are at the same pair; a full lean is handed down full.*

**Proposition 12.28** (The split is forced)**.** *(i) For every gauge $\beta$ and every probability vector $w$ on $\mathrm{pat}_j(B)$ with every $w_p > 0$,*

$$K_j(\beta) \;\geq\; \prod_p \Big( \frac{\beta_B^p K_{j,(p)}(\beta_R)}{w_p} \Big)^{w_p},$$

*with equality if and only if $\pi_{j,p}(\beta) = w_p$ for every $p$; likewise for $\Psi_j$ with the pattern terms $\beta_B^{B\setminus p} \Psi_{j,(p)}(\beta_R)$ and $\pi^*_{j,p}(\beta)$. So the shares $a_{j,p} = s_j w_p$ replace $K_j^{-s_j}$ by the product of its pattern terms exactly at the gauges where the basis law puts the proportions $w$ on the patterns; at the gauge of a realization the only such shares are its native split.*

*(ii) $\beta_B^{-p} K_{j,(p)}(\beta_R^{-1}) = \beta^{-\mathcal{E}} \beta_B^{B\setminus p} \Psi_{j,(p)}(\beta_R)$ and $\pi_{j,p}(\beta^{-1}) = \pi^*_{j,p}(\beta)$: Cremona turns the $K$-term of each pattern into its $\Psi$-term, acting on each child as the child's own Cremona. At the table's gauge $\pi^*_{j,p}(\mathbf{1}) = \pi_{j,p}(\mathbf{1})$: the forced shares split $s_j$ and $t_j$ in the same proportions.*

A bunch hands its room a family of children, one for each pattern of each table, and the native split is the one choice of shares under which the room inherits everything: the gauge stays native, the masses restrict, and a centre hands down centres. The inequality of (i) is why: at a gauge, the pattern terms of a letter can stand in for the letter with no loss only in the proportions in which the law at that gauge already uses them.

# Chapter 13

# The order of classes and the Foreman bunch

## 13.1 The rises of a class

**Definition 13.1** (The rises of a class)**.** Let a molecule $M$ with $n$ marks be read at realizations $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$ at the table's gauge with $\boldsymbol{\sigma}$ fixed, and let $C$ be a class of the realization at hand, with $m$ marks, and $R$ its room. For a function $f$ of $v = (v_0, \dots, v_k)$ analytic at 0, $\mathring{f}(v) = f(v) - f(0) - \sum_i v_i\, \partial_{v_i} f(0)$ is $f$ less its value and its tangent.

(i) For $0 \le k \le n-m$ and distinct partners $\mathbf{g} = (g_1, \dots, g_k)$ in $R$, the *rise of $C$ with the partners* $\mathbf{g}$ is the increase of the potential along the class, each partner on its own scale:

$$V^{\mathbf{X}}_{C,\mathbf{g}}(v_0, \dots, v_k) = \Phi_{\mathbf{X}}\Big(v_0 \mathbf{1}_C + \sum_{i=1}^{k} v_i \mathbf{1}_{g_i}\Big) - \Phi_{\mathbf{X}}\Big(\sum_{i=1}^{k} v_i \mathbf{1}_{g_i}\Big).$$

*Stage $k$* is the multiset $\mathcal{V}_k(C)$ of the pairs $(\mathring{V}^{\mathbf{E}}_{C,\mathbf{g}}, \mathring{V}^{\mathbf{O}}_{C,\mathbf{g}})$ over the $k$-tuples of distinct marks of $R$; stage 0 is the class on its own scale.

(ii) For an order $w = (c_1, \dots, c_m; r_1, \dots, r_{n-m})$ of the marks that lists those of $C$ first and then those of $R$, the *rise from the class* is

$$\overline{V}^{\mathbf{X}}_{C,w}(u_1, \dots, u_m, v_1, \dots, v_{n-m}) = \Phi_{\mathbf{X}}\Big(\sum_i u_i \mathbf{1}_{c_i} + \sum_j v_j \mathbf{1}_{r_j}\Big) - \Phi_{\mathbf{X}}(0),$$

with positions $0, \dots, n-1$ in the order of $w$, and the *last stage* is the multiset $\mathcal{V}_*(C)$ of the pairs $(\overline{V}^{\mathbf{E}}_{C,w}, \overline{V}^{\mathbf{O}}_{C,w})$ over the $m!\,(n-m)!$ such orders.

$\mathcal{V}_k(C)$ has $(n-m)!/(n-m-k)!$ elements; $v_k = 0$ turns the rise with $(g_1, \dots, g_k)$ into the rise with $(g_1, \dots, g_{k-1})$; for a single mark an element of $\mathcal{V}_*(\{e\})$ is a writing of the potential from $e$, less $\Phi_{\mathbf{X}}(0)$.

A class is first read on its own scale, then with one partner of the room at a time, then with more, and last from the class with every mark on its own scale. Every stage before the last is taken less its value and tangent; the last stage holds the whole potential.

**Proposition 13.2** (Every rise is read by the Skeleton functions of the class)**.** *With $x = e^{v_0}$ and $\alpha = e^{\sum_i v_i \mathbf{1}_{g_i}}$, the gauge the partners move to,*

$$V^{\mathbf{E}}_{C,\mathbf{g}}(v_0, \dots, v_k) = \sum_j \sigma_j \Big( \log \mathsf{Q}^{\mathbf{E}}_{j,C}(x;\alpha) + \big(d_j(C) + d^*_j(C)\big) v_0 \Big),$$

$$V^{\mathbf{O}}_{C,\mathbf{g}}(v_0,\dots,v_k)=\sum_j \delta_j\Big(\log \mathsf{Q}^{\mathbf{O}}_{j,C}(x;\alpha)+\big(d_j(C)-d^*_j(C)\big)v_0\Big).$$

*Stage 0 reads the Skeleton functions of $C$ at the table's gauge, and stage $k$ reads them at every gauge that $k$ partners move to.*

**Proposition 13.3** (The rises hold the cumulants of the class with its partners)**.** *For $\mathbf{a}=(a_0,\dots,a_k)\in\mathbb{N}^{k+1}$ write, in this statement, $\kappa_j(\mathbf{a})=\kappa_{j,\mathbf{1}}\big(N_C^{\times a_0},X_{g_1}^{\times a_1},\dots,X_{g_k}^{\times a_k}\big)$, the joint cumulant, under the weight law $\mathbb{P}_{j,\mathbf{1}}$, of the sequence of variables in which the level $N_C$ appears $a_0$ times and the indicator $X_{g_i}$ appears $a_i$ times. For $a_0\geq 1$ and $|\mathbf{a}|\geq 2$,*

$$[v^{\mathbf{a}}]\,V^{\mathbf{E}}_{C,\mathbf{g}}=\frac{1+(-1)^{|\mathbf{a}|}}{\mathbf{a}!}\sum_j\sigma_j\kappa_j(\mathbf{a}),\qquad [v^{\mathbf{a}}]\,V^{\mathbf{O}}_{C,\mathbf{g}}=\frac{1-(-1)^{|\mathbf{a}|}}{\mathbf{a}!}\sum_j\delta_j\kappa_j(\mathbf{a}),$$

*and the coefficients with $a_0=0$ vanish. By multilinearity $\kappa_j(\mathbf{a})$ is the sum, over the strings $u\in C^{a_0}$, of the cumulants $\kappa_{j,\mathbf{1}}$ along the strings that continue $u$ with each $g_i$ taken $a_i$ times: the entries of the still on those strings, summed over the class and read table by table.*

**Corollary 13.4** (The first two stages in coins)**.** *(i) For $a\geq 2$, $[v_0^a]\,\mathring{V}^{\mathbf{E}}_{C,()}=\frac{2}{a!}\sum_j\sigma_j\kappa_{j,\mathbf{1}}(N_C^{\times a})$ for even $a$ and $[v_0^a]\,\mathring{V}^{\mathbf{O}}_{C,()}=\frac{2}{a!}\sum_j\delta_j\kappa_{j,\mathbf{1}}(N_C^{\times a})$ for odd $a$, the other coefficients being zero; and $\kappa_{j,\mathbf{1}}(N_C^{\times a})=\sum_i\varkappa_a(p_{j,i})$ over the coins $p_{j,i}$ of $\mathsf{Q}^K_{j,C}(\cdot;\mathbf{1})$.*

*(ii) For a partner $g$ and $q\geq 1$, with $R_{j,g}(e^{\pm v_0\mathbf{1}_C})$ the occupation of $g$ in table $j$ at the gauge $e^{\pm v_0\mathbf{1}_C}$,*

$$[v_1^q]\,V^{\mathbf{E}}_{C,(g)}=\frac{1}{q!}\sum_j\sigma_j\Big(\varkappa_q\big(R_{j,g}(e^{v_0\mathbf{1}_C})\big)+(-1)^q\varkappa_q\big(R_{j,g}(e^{-v_0\mathbf{1}_C})\big)-\big(1+(-1)^q\big)\varkappa_q\big(R_{j,g}(\mathbf{1})\big)\Big),$$

$$[v_1^q]\,V^{\mathbf{O}}_{C,(g)}=\frac{1}{q!}\sum_j\delta_j\Big(\varkappa_q\big(R_{j,g}(e^{v_0\mathbf{1}_C})\big)-(-1)^q\varkappa_q\big(R_{j,g}(e^{-v_0\mathbf{1}_C})\big)-\big(1-(-1)^q\big)\varkappa_q\big(R_{j,g}(\mathbf{1})\big)\Big).$$

## 13.2 The order

**Definition 13.5** (Comparing)**.** (i) *Exponents.* For $\mathbf{a},\mathbf{a}'\in\mathbb{N}^{k+1}$, $\mathbf{a}\sqsubset\mathbf{a}'$ when $a_i<a'_i$ at the largest position $i$ with $a_i\neq a'_i$.

(ii) *Functions.* For functions $f\neq f'$ of the positions $0,\dots,k$, analytic at 0: $f\lhd f'$ when $[v^{\mathbf{a}}]f<[v^{\mathbf{a}}]f'$ at the $\sqsubset$-least exponent at which their coefficients differ.

(iii) *Pairs.* $(f^{\mathbf{E}},f^{\mathbf{O}})\lhd(f'^{\mathbf{E}},f'^{\mathbf{O}})$ when $f^{\mathbf{E}}\lhd f'^{\mathbf{E}}$, or $f^{\mathbf{E}}=f'^{\mathbf{E}}$ and $f^{\mathbf{O}}\lhd f'^{\mathbf{O}}$.

(iv) *Multisets.* Two different multisets of pairs with equally many elements are read from the least element upward in $\lhd$, and the one that first holds the smaller element comes first.

(v) *Classes.* For classes $C\neq D$ of one realization with $m$ marks each, $C\prec D$ when $\mathcal{V}_s(C)\lhd\mathcal{V}_s(D)$ at the first $s$ in $0,1,\dots,n-m,*$ with $\mathcal{V}_s(C)\neq\mathcal{V}_s(D)$.

**Definition 13.6** (The Foreman bunch)**.** At the centre, the *Foreman bunch* $B(M;\boldsymbol{\sigma},0)$ is the class of $\Upsilon_{\boldsymbol{\sigma},0;\mathbf{1}}$ with the fewest marks and, among those, the $\prec$-least. At a lean $\boldsymbol{\delta}$ each class $C$ of $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$ lies in one class $C^\circ$ of the centre (Theorem 11.1), and $B(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ is the class with the least

$$\mathrm{key}_{\boldsymbol{\sigma},\boldsymbol{\delta}}(C)=\big(|C|,\ |C^\circ|,\ C^\circ,\ C\big),$$

compared entry by entry: sizes as integers, $C^\circ$ by $\prec$ at the centre, $C$ by $\prec$ at the lean. At every realization the Foreman bunch is a class with the fewest marks among the classes of that realization; the classes of the centre and the order only choose among those.

**Theorem 13.7** (What the order is)**.** *(i)* Total. *For every $m$, $\prec$ is a strict total order on the classes with $m$ marks; $B(M;\boldsymbol{\sigma},0)$ and $B(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ are each one class.*

*(ii)* Directions. *With the classes given, $\prec$ depends on the realization only through the rays of $\boldsymbol{\sigma}$ and $\boldsymbol{\delta}$; at one pair only through the lean sign.*

*(iii)* Mirror. *If $g$ exchanges the family (Definition 11.13), every stage of $g(C)$ at $-\boldsymbol{\delta}$ equals that stage of $C$ at $\boldsymbol{\delta}$; so $C \prec D$ at $\boldsymbol{\delta}$ exactly when $g(C) \prec g(D)$ at $-\boldsymbol{\delta}$, and $B(M;\boldsymbol{\sigma},-\boldsymbol{\delta}) = g\big(B(M;\boldsymbol{\sigma},\boldsymbol{\delta})\big)$. If $g$ carries the family, the same holds at one lean.*

*(iv)* Nesting. *When some class at $\boldsymbol{\delta}$ inside $B(M;\boldsymbol{\sigma},0)$ has the fewest marks among the classes at $\boldsymbol{\delta}$, $B(M;\boldsymbol{\sigma},\boldsymbol{\delta}) \subseteq B(M;\boldsymbol{\sigma},0)$, and it is the class at $\boldsymbol{\delta}$ inside $B(M;\boldsymbol{\sigma},0)$ with the fewest marks and, among those, the $\prec$-least. In a one-table molecule this is so whenever $B(M;\boldsymbol{\sigma},0)$ splits at $\boldsymbol{\delta}$. Otherwise $B(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ need not lie in $B(M;\boldsymbol{\sigma},0)$.*

Every rise is a linear combination of logarithms of polynomials in $e^{v_0},\dots,e^{v_k}$, less the same combination at $v_0 = 0$ (Proposition 13.2), so equality of two rises is an identity of polynomials with exponents (Theorem 10.6). When two rises differ, the first differing exponent is found column by column: $v_k = 0$ is the rise with one partner fewer, the coefficient of $v_k^b$ is a rational function of the scales at the other positions (Corollary 13.4(ii) for $k = 1$), each column is certified equal by a polynomial identity, and the deciding coefficient is a joint cumulant of Proposition 13.3, multiplied by the $\sigma_j$ or the $\delta_j$, and rational when the weights of the tables are.

**Proposition 13.8** (Rung one enters only at the last stage)**.** *Every stage before the last reads the still from rung two on; the masses, rung one, enter only at the last stage, and at a centre they are the same at every mark.*

## 13.3 What the first stages read

**Proposition 13.9** (At the centre the first number is how far the class is from a scale of its own)**.** $\operatorname{Var}_{T\sim\mathbb{P}_{j,\mathbf{1}}} |T\cap U| = 0$ *in every table exactly when $U$ is a union of molecules. So for a class $C$ of a molecule other than the whole,*

$$[v_0^2]\,\mathring{V}^{\mathbf{E}}_{C,()} = \sum_j \sigma_j \operatorname{Var}_{T\sim\mathbb{P}_{j,\mathbf{1}}} |T\cap C| > 0.$$

**Proposition 13.10** (Mirror classes)**.** *Let $C, D$ be classes with equally many marks whose levels are mirrored across the tables: there are a bijection $h$ of the tables with $(\sigma_{h(j)},\delta_{h(j)}) = (\sigma_j,\delta_j)$ and integers $k_j$ with $\mathbb{P}_{h(j),\mathbf{1}}(|T\cap D| = a) = \mathbb{P}_{j,\mathbf{1}}(|T\cap C| = k_j - a)$ for every $j$ and every $a$. This holds with $h$ the identity for two classes that together make up the molecule ($k_j = r_j$), and for $C$ and $g(C)$ under an exchange $g$ of the family, $h$ the bijection of its tables ($k_j = |C|$). Then:*

*(i)* $\kappa_{h(j),\mathbf{1}}(N_D^{\times a}) = (-1)^a \kappa_{j,\mathbf{1}}(N_C^{\times a})$ *for $a \geq 2$, so $\sum_j \sigma_j \kappa_{j,\mathbf{1}}(N_D^{\times a}) = (-1)^a \sum_j \sigma_j \kappa_{j,\mathbf{1}}(N_C^{\times a})$, and likewise with the $\delta_j$: the $\mathbf{E}$ parts of $\mathcal{V}_0(C)$ and $\mathcal{V}_0(D)$ are equal and their $\mathbf{O}$ parts are negatives;*

*(ii) at a lean, if some odd $a \geq 3$ has $\sum_j \delta_j \kappa_{j,\mathbf{1}}(N_C^{\times a}) \neq 0$ and $a$ is the least such, then $C \prec D$ exactly when $\sum_j \delta_j \kappa_{j,\mathbf{1}}(N_C^{\times a}) < 0$, and at the opposite lean the order is the other one;*

*(iii) at the centre stage* $0$ *does not separate* $C$ *and* $D$*, and a later stage does.*

**Corollary 13.11** (A bunch across the centre)**.** *In a one-table molecule, if* $B(M;\sigma,0)$ *splits at a lean* $\delta$ *into* $C_K$ *and* $C_\Psi$ *(Theorem 11.17), then* $B(M;\sigma,\delta) \in \{C_K, C_\Psi\}$ *and* $B(M;\sigma,-\delta) = \beta\big(B(M;\sigma,\delta)\big)$ *is the other one,* $\beta$ *the exchange of Theorem 11.17; the two are mirror classes, ordered at a lean by Proposition 13.10(ii).*

The Foreman bunch is the least of the smallest classes, and the order that chooses it reads the class first on its own scale. At the centre the first number read is the spread of the level of the class under the weight laws, which vanishes only for a union of molecules; a lean adds the odd cumulants of that level, and two mirror classes are told apart by the first of them that does not vanish, the opposite lean reversing the choice.

# Chapter 14

# Peel, children and the Foreman tree

## 14.1 The peel and the tree

**Definition 14.1** (Peel; childcules)**.** Families are read at realizations at the table's gauge, and for a molecule $M$ of a family $F$, $F|M$ is the family of the restrictions of its tables to $M$. The *peel* of a family $F$ at a bunch $B$ is the family of its children at the native split of the table's gauge (Proposition 12.27),

$$F_B(\mathbf{a}^\star, \mathbf{b}^\star), \qquad a^\star_{j,p} = \pi_{j,p}(\mathbf{1})\, s_j, \quad b^\star_{j,p} = \pi_{j,p}(\mathbf{1})\, t_j,$$

on the room $R$; children with the same bases and weights are one table, with their exponents added. A *childcule* is a molecule of the peel.

**Definition 14.2** (The record of a peel)**.** The *record* of the peel of $F$ at $B$ is the bunch $B$ with, for every table $j$, the communicatroid of $B$ (Definition 12.4) and the pattern probabilities $\pi_{j,p}(\mathbf{1})$: the indexed children $\big(j,p;\ \mathcal{B}_{j,(p)}, \theta_{j,(p)};\ \pi_{j,p}(\mathbf{1})s_j,\ \pi_{j,p}(\mathbf{1})t_j\big)$. The peel is the record with the indices $(j,p)$ forgotten and equal children merged.

**Proposition 14.3** (The record keeps the parent)**.** *The record determines $F$: table $j$ is the gluing of its communicatroid (Proposition 12.5), and its exponents are the sums of its children's,* $\sum_p \pi_{j,p}(\mathbf{1})(s_j, t_j) = (s_j, t_j)$.

Merging equal children forgets which pattern each share came from, and the Gammas $\Gamma\big(\mu_e - \langle\boldsymbol{\delta}\rangle_e\big)$ of the port (Theorem 15.15) depend on it. The record keeps it. A bunch of width one whose communicatroid is that of one mark, two children on two levels, may instead be pooled into one mark of the room, its shape integrated and its scale kept (Theorem 19.8).

**Definition 14.4** (The Foreman tree)**.** For a family $F$ read at $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$, the *Foreman tree* $\mathrm{Tree}(F;\boldsymbol{\sigma},\boldsymbol{\delta})$ splits $F$ into molecules. A molecule with one mark is a unit (Definition 7.4), and a molecule $M$ whose realization is atomic is an atom (Definition 10.1); otherwise the tree records the Foreman bunch $B(M;\boldsymbol{\sigma}_M,\boldsymbol{\delta}_M)$, $(\boldsymbol{\sigma}_M,\boldsymbol{\delta}_M)$ the pair coordinates of the tables of $M$, and continues with the peel of $F|M$ at that bunch, on its room.

The tree asks each molecule whether it is an atom; where it is not, the tree takes the least of its smallest classes as the bunch, hands the room the children of the bunch at the native split, and asks again. Nothing is lost on the way down: the record keeps the parent, and the native split keeps the gauge native and the centre a centre.

**Proposition 14.5** (Peeling in two steps is peeling at once)**.** *For $B = B_1 \sqcup B_2$, the peel at $B_2$ of the peel of $F$ at $B_1$ is the peel of $F$ at $B$: the child of $F$ at $(j,p)$ and the child at $\big((j, p\cap B_1), p\cap B_2\big)$ of the peel at $B_1$ are one table on $\mathcal{E}\setminus B$ with the same exponents. So if a molecule is peeled at $B_1$, its room is one childcule and the bunch there is $B_2$, the family reached is the peel at $B_1 \sqcup B_2$.*

**Theorem 14.6** (Three trees)**.** $\mathrm{Tree}(F;\boldsymbol{\sigma},\boldsymbol{\delta})$ *depends on $(\boldsymbol{\sigma},\boldsymbol{\delta})$ only through their rays; at one pair only through the lean sign. So a molecule at one pair has three Foreman trees,* $\mathrm{Tree}(F;\boldsymbol{\sigma},0)$ *and* $\mathrm{Tree}(F;\boldsymbol{\sigma},\pm\boldsymbol{\delta})$ *for a basis lean $\boldsymbol{\delta}$, fixed by its tables.*

**Lemma 14.7** (Maps descend)**.** *Let $g:\mathcal{E}\to\mathcal{E}'$ be a bijection, $B\subseteq\mathcal{E}$ with room $R$, and $B'=g(B)$.*

*(i) If $g$ carries $F$ to $F'$, then $g|_R$ carries the peel of $F$ at $B$ to the peel of $F'$ at $B'$, the child $(j,p)$ going to $(h(j),g(p))$, $h$ the bijection of the tables of Definition 11.13.*

*(ii) If $g$ exchanges $F$ with $F'$, then $g|_R$ exchanges the peel of $F$ at $B$ with the peel of $F'$ at $B'$, the child $(j,p)$ going to $(h(j),B'\setminus g(p))$.*

*(iii) In both cases $g$ maps the molecules of $F$ onto those of $F'$, and on a molecule $M$ its restriction carries, respectively exchanges, $F|M$ with $F'|g(M)$.*

**Theorem 14.8** (The mirror)**.** *If $g$ exchanges $F$ with $F'$, then $g\big(\mathrm{Tree}(F;\boldsymbol{\sigma},\boldsymbol{\delta})\big)=\mathrm{Tree}(F';\boldsymbol{\sigma},-\boldsymbol{\delta})$: molecules go onto molecules, units onto units, atoms onto atoms, each recorded bunch onto the bunch recorded at the same place, rooms onto rooms. If $g$ carries $F$ to $F'$ the same holds at the same lean.*

**Corollary 14.9** (A molecule with an exchange)**.** *If $g$ exchanges a molecule $M$ (Definition 11.13), then $g$ maps* $\mathrm{Tree}(M;\boldsymbol{\sigma},0)$ *onto itself, every class of the centre onto itself (Corollary 11.15), and* $\mathrm{Tree}(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ *onto* $\mathrm{Tree}(M;\boldsymbol{\sigma},-\boldsymbol{\delta})$*: $B(M;\boldsymbol{\sigma},-\boldsymbol{\delta})=g\big(B(M;\boldsymbol{\sigma},\boldsymbol{\delta})\big)$ and $g\big(B(M;\boldsymbol{\sigma},0)\big)=B(M;\boldsymbol{\sigma},0)$. If $g$ carries $M$, it maps every class of $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$ onto itself and each of the trees at 0, $\boldsymbol{\delta}$ and $-\boldsymbol{\delta}$ onto itself.*

**Lemma 14.10** (An exchange turns a window over)**.** *If $g$ exchanges a family $F$ with $F'$, with the bijection $h$ of the tables (Definition 11.13), and $S'=g(S)$, then for every table $j$ the window of $S'$ in table $h(j)$ of $F'$ is $\{|S|-r_j(S),\dots,|S|-d_j(S)\}$: $S$ in table $j$ and $S'$ in table $h(j)$ have the same width, and the bases of table $j$ at level $k$ on $S$ correspond to those of table $h(j)$ at level $|S|-k$ on $S'$.*

**Corollary 14.11** (Below a no)**.** *A molecule that answers no is peeled at its Foreman bunch read at a centre of the table's gauge; each childcule is a molecule at a centre of the table's gauge, and whether it answers yes is read there in turn.*

## 14.2 Patterns of one kind

**Proposition 14.12** (Level form)**.** *A function of the patterns of a bunch $B$ has* level form *when it depends on $|p|$ alone, and $B$ is in level form in a table when its children are, and in a family when it is in level form in every table. $B$ is in level form in table $j$, $K_{j,(p)}=K_{j,(p')}$ whenever $|p|=|p'|$, if and only if every transposition of two marks of $B$, fixing every other mark, maps the bases of table $j$ onto themselves with their weights; so $B$ is in level form in a family exactly when every such transposition carries the family with the identity on its tables.*

**Definition 14.13** (Kinds of pattern)**.** Let a one-table molecule be read at the table's gauge and $B$ a set of marks. A *writing from a pattern $p$* is $K$ written in an order whose first $|p|$ positions hold

$p$, the next $|B \setminus p|$ hold $B \setminus p$, and the rest the room; a writing of $\Psi$ from $B \setminus p$ likewise. Patterns $p, p'$ are of one *kind at a lean* when one polynomial is written from $p$ and from $p'$, and of one *kind at the centre* when moreover a writing from $p$ may equal a writing of $\Psi$ from $B \setminus p'$.

**Theorem 14.14** (One child per kind)**.** *(i) $p, p'$ are of one kind at a lean exactly when a bijection keeping $B$ and carrying the table maps $p$ to $p'$, and of one kind at the centre exactly when, in addition, an exchanging bijection keeping $B$ may map $p$ to $B \setminus p'$. Being of one kind is reflexive, symmetric and transitive.*

*(ii) Patterns of one kind have one child between them: the child at $p'$ is the image of the child at $p$ under the bijection restricted to the room, or the dual of that image when the bijection exchanges; their pattern probabilities are equal.*

*(iii) So the level letters $K_B^{(k)}(\alpha) = \sum_{p \in \mathrm{pat}^{(k)}(B)} \alpha_B^p K_{(p)}(\alpha_R)$ are determined by one child per kind, the patterns of each kind, and the bijections.*

**Corollary 14.15** (A family with maps in hand)**.** *Let $\mathcal{G}$ be any set of bijections keeping $B$ that carry or exchange a family. Patterns reached from one another by the moves $p \mapsto g(p)$, $g \in \mathcal{G}$ carrying, and $p \mapsto B \setminus g(p)$, $g \in \mathcal{G}$ exchanging, have children that are one table up to the bijections on the room and duality, with equal pattern probabilities.*

Kinds are read on the writings of the two polynomials of one table; with several tables there is no single pair of polynomials to write, and the corollary takes the bijections as given.

**Theorem 14.16** (The pattern tensor of a class)**.** *Let $B$ be a class of a one-table molecule at a realization at the table's gauge, $e, f \in B$, and $\beta$ the bijection taking an order in which the class's one function is written from $e$ to one in which it is written from $f$ (Theorem 10.5). Then:*

*(i) $\beta$ carries the table, or at the centre possibly exchanges it, and maps every class onto itself; in particular $\beta(B) = B$ and $\beta(R) = R$;*

*(ii) the child at $\beta(p)$ is the child at $p$ with the room mapped by $\beta$, or, when $\beta$ exchanges, the child at $B \setminus \beta(p)$ is the dual of that image; so $p$ and $\beta(p)$, respectively $p$ and $B \setminus \beta(p)$, are of one kind;*

*(iii) when $\beta$ carries, the patterns through $e$ correspond to those through $f$; when it exchanges, the patterns through $e$ correspond to those avoiding $f$; in both cases with equal pattern probabilities and with children equal up to the bijection on the room and duality.*

*So the pattern tensor of a class is the same from each of its marks, at the centre up to complements.*

## 14.3 What the room keeps

**Definition 14.17** (Keeping the potential; twins; descents; copies)**.** Let a family be read at a realization $\Upsilon = ((s_j, t_j)_j; \alpha)$, with the potential $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$ of Definition 8.18 and $(g^* y)_e = y_{g(e)}$ as in Definition 11.13. A bijection $g$ of the marks of a molecule $M$ *keeps the potential* when $\Phi_{\mathbf{X}}(\log \alpha + g^* y) = \Phi_{\mathbf{X}}(\log \alpha + y)$ for every $y$ and both parts; at the table's gauge this is $\Phi_{\mathbf{X}} \circ g^* = \Phi_{\mathbf{X}}$. A bijection $g : M \to M'$ onto the marks of a molecule $M'$, read at a realization with the gauge $\alpha'$, *takes the potential* of $M$ to that of $M'$ when $\Phi_{\mathbf{X}}(\log \alpha + g^* y) = \Phi'_{\mathbf{X}}(\log \alpha' + y) + c_{\mathbf{X}}$ for every $y$ and both parts, with constants $c_{\mathbf{X}}$. Twins, descents and copies are read at the table's gauge, $\alpha = \mathbf{1}$: two childcules of the peel of $M$ are *twins* when a bijection keeping the potential of $M$ maps one onto the other. A *descent* of a Foreman tree is a sequence $M_0 \supset M_1 \supset \cdots \supset M_d$ of its nodes from the molecule $M_0$ to a leaf $M_d$, each $M_{k+1}$ a childcule of the peel of $M_k$ at its bunch.

On a descent, with $B_i$ the bunch of $M_i$, the *copies inside* $M_k$ of $Y = M_i$ or $Y = B_i$, $k \le i$, are the sets $g_k(g_{k+1}(\cdots g_{i-1}(Y)\cdots))$, each $g_l$ a bijection keeping the potential of $M_l$; for $k = i$ they are $Y$ alone.

**Lemma 14.18** (The writings certify the potential)**.** *Let a molecule be read at a realization* $\Upsilon = ((s_j, t_j)_j; \alpha)$.

*(i) If $e$ and $f$ read alike, the bijection $\beta$ that takes, position by position, an order with $e$ first to an order with $f$ first in which one pair of functions is written from both (Proposition 10.4) keeps the potential, and $\beta(e) = f$.*

*(ii) A bijection keeping the potential maps every class onto itself, and a bijection taking the potential of $M$ to that of $M'$ maps the classes of $M$ onto those of $M'$, at the centre and at the realization's lean.*

*So two marks read alike exactly when a bijection keeping the potential takes one to the other.*

**Lemma 14.19** (The peel reads the factors of the letters)**.** *Write the potential of a family by its irreducible factors with their exponents $(L, w_{\mathbf{E}}(L), w_{\mathbf{O}}(L))$, each normalized by $L(\mathbf{1}) = 1$ (Theorem 10.6), and for a bunch $B$ let $L_{[q]}$ be the coefficient of $\alpha_B^q$ in $L$, a polynomial on the room $R$. In this statement write $\Phi_{\mathbf{X}}^{\mathrm{peel}}$ for the potential of the peel $F_B(\mathbf{a}^\star, \mathbf{b}^\star)$. Up to a constant,*

$$\Phi_{\mathbf{X}}^{\mathrm{peel}}(y_R) = \sum_L w_{\mathbf{X}}(L) \sum_q L_{[q]}(\mathbf{1}) \log L_{[q]}\big(e^{y_R}\big),$$

*the inner sum over the $q$ with $L_{[q]} \neq 0$, whose values $L_{[q]}(\mathbf{1})$ add to 1.*

**Theorem 14.20** (The room keeps its classes)**.** *Let a molecule $M$ be read at a realization at the table's gauge, let $B$ be a union of its classes, and let the children's weights factor across the childcules of the peel. Every bijection $g$ keeping the potential of $M$ keeps $B$; its restriction to the room keeps the potential of the peel, maps each childcule $Y$ onto a childcule $g(Y)$, and takes the potential of $Y$ to that of $g(Y)$, the constant being 0 when $g(Y) = Y$. Consequently, for a class $C$ of $M$ inside the room:*

*(i) the childcules that $C$ meets are twins of one another, and $C$ meets each of them in equally many marks;*

*(ii) inside each childcule, $C$ lies in one class.*

**Proposition 14.21** (The tree is read off the potential)**.** *Let a molecule $M$ of $F$ and a molecule $M'$ of $F'$ be read at realizations at the table's gauge, with pair coordinates $(\boldsymbol{\sigma}, \boldsymbol{\delta})$ and $(\boldsymbol{\sigma}', \boldsymbol{\delta}')$, the children's weights factoring across the childcules at every node of their trees, and let $g : M \to M'$ take the potential of $M$ to that of $M'$. Then $g$ maps* $\mathrm{Tree}(F|M; \boldsymbol{\sigma}, \boldsymbol{\delta})$ *onto* $\mathrm{Tree}(F'|M'; \boldsymbol{\sigma}', \boldsymbol{\delta}')$*: classes onto classes, units onto units, atoms onto atoms, each bunch onto the bunch recorded at the same place, and each childcule onto a childcule, its potential to that of its image.*

**Theorem 14.22** (The leaf bounds the bunches)**.** *Let a molecule $F$ be read at a realization at the table's gauge, the centre or a lean, the children's weights factoring across the childcules at every node, and let $M_0 \supset M_1 \supset \cdots \supset M_d$ be a descent of its Foreman tree at that realization, $B_k$ the bunch of $M_k$ and $R_k = M_k \setminus B_k$. Let $h_k$ be the number of twins of $M_{k+1}$ among the childcules of $M_k$, $M_{k+1}$ itself included, and $h_{<k} = h_0 h_1 \cdots h_{k-1}$.*

*(i) A class of $M_k$ inside $R_k$ that meets $M_{k+1}$ meets the twins of $M_{k+1}$ and no other childcule, each in equally many marks, and lies in one class of each. The copies inside $M_k$ of $M_i$,*

*$k \le i \le d$, respectively of $B_i$, $k \le i < d$, are $h_{<i}/h_{<k}$ disjoint nodes, respectively bunches, of the subtree at $M_k$, leaves when $i = d$; and $M_d$ together with its copies inside $M_k$ is a union of whole classes of $M_k$.*

*(ii) A class $C$ of $M_k$ that meets $M_{k+1}$ has at least $|B_k|$ marks and meets $M_{k+1}$ in $|C|/h_k \ge |B_k|/h_k$ of them, all in one class of $M_{k+1}$; the bunch $B_{k+1}$, and the leaf when $k+1 = d$, contains such a part. So*

$$|B_0| \le h_{<1}|B_1| \le h_{<2}|B_2| \le \cdots \le h_{<d-1}|B_{d-1}| \le h_{<d}|M_d|,$$

*the $k$-th term being the number of marks of the copies of $B_k$ inside $M_0$, and the last that of the copies of $M_d$: down the descent the bunches with their copies never shrink, and the leaf with its copies has at least as many marks as every bunch with its copies.*

Theorem 14.22 rests on two things: the peel splits a class only across twin childcules, each part inside one class (Theorem 14.20), and the bunch is a class with the fewest marks of the realization at which its node is read. Peeled at another bunch, a molecule can end at a leaf smaller than the bunch above it; at a lean the count is read on the classes of the lean, which can split a class of the centre into parts smaller than the least class of the centre; and the copies are needed when a room falls into twins.

# Part IV

# Evaluation

# Chapter 15

# The integral at a bunch

## 15.1 The scale of a bunch

**Proposition 15.1** (A scale integral is a Breta of one bunch)**.** *Let $J = \{0, \dots, L\}$ carry coefficients $\boldsymbol{\xi} = (\xi_\ell)_{\ell \in J}$ and letter exponents $\mathbf{w} = (w_\ell)_{\ell \in J}$ with total $|\mathbf{w}| = \sum_\ell w_\ell$, and let $I_J$ be the unit configuration on $J$. For $\boldsymbol{\xi} > 0$, $\operatorname{Re} w_\ell > 0$ and $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$,*

$$\int_0^\infty x^{\kappa-1} \prod_{\ell \in J} (x + \xi_\ell)^{-w_\ell}\, dx = \Gamma_{\,|J|}^{\,2}\begin{bmatrix} \kappa, & |\mathbf{w}| - \kappa \\ & \mathbf{w} \end{bmatrix} \mathsf{B}^{\{J\}}_{\mathbf{w},\, |\mathbf{w}|-\kappa;\, I_J}(\boldsymbol{\xi}) :$$

*the Breta of Definition 3.1 with the one bunch $J$, the shape index $\mathbf{w}$, the scale index $|\mathbf{w}| - \kappa$, the unit configuration as its terms and $\boldsymbol{\xi}$ as its coefficients. For $\xi_\ell \in \mathbb{C} \setminus (-\infty, 0]$ the left side, with principal branches, converges exactly when $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$ and is holomorphic in its data there, and it continues the right side from positive coefficients and from letter exponents with positive real part; wherever the left side converges, the product on the right denotes its value. Coincident coefficients merge into one with the summed letter exponent.*

The scale of a bunch is integrated against a product of linear letters, one for each point of the joint spectrum, and that integral is a Breta: the points of the spectrum are the coefficients, the letter exponents the shape index, and the index of the scale integral sits in the scale index as $|\mathbf{w}| - \kappa$. Nothing beyond the general Breta is needed to write it. With one coefficient the Breta is the power $\xi_0^{\kappa - w_0}$, and the identity is the integral of a linear letter: for $c_0, c_1 > 0$ and $0 < \operatorname{Re}\nu < \operatorname{Re} s$, $\int_0^\infty \rho^{\nu-1}(c_0 + c_1\rho)^{-s}\, d\rho = \Gamma_{\,1}^{\,2}\begin{bmatrix} \nu, & s - \nu \\ & s \end{bmatrix} c_0^{\nu - s} c_1^{-\nu}$. A stacked Gamma $\Gamma_{\,1}^{\,2}\begin{bmatrix} a, & b \\ & a+b \end{bmatrix}$ is a *Beta corner*.

**Proposition 15.2** (Where the Breta of one bunch converges)**.** *In Proposition 15.1 let $\operatorname{Re} w_\ell > 0$ for every $\ell$ and $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$, and let $\Xi_J$ be the set of the $\boldsymbol{\xi} \in (\mathbb{C} \setminus (-\infty, 0])^J$ whose convex hull avoids $(-\infty, 0]$. It is open, star-shaped about $\mathbf{1}$, and contains every $\boldsymbol{\xi}$ whose coordinates have positive real parts. On $\Xi_J$ the integral that defines the Breta at positive coefficients (Definition 3.1),*

$$\int_{\Delta_J} u^{\mathbf{w}-\mathbf{1}} \Big( \sum_{\ell \in J} \xi_\ell u_\ell \Big)^{-(|\mathbf{w}|-\kappa)} d\omega_J(u),$$

*with the principal branch, converges, is holomorphic in $\boldsymbol{\xi}$, and continues the Breta $\mathsf{B}^{\{J\}}_{\mathbf{w},\, |\mathbf{w}|-\kappa;\, I_J}(\boldsymbol{\xi})$ from positive coefficients; so the identity of Proposition 15.1 holds on $\Xi_J$ with both sides convergent integrals.*

**Theorem 15.3** (The scale of a bunch is one Breta)**.** *Let a family $F$ on $\mathcal{E}$ be at a gauge $\alpha$ with masses $\mathbf{m}$, let $B$ be a proper bunch with room $R$, its Skeleton polynomials, values and coins those of Definition 12.10, and let $\xi_0, \dots, \xi_L$ be the distinct points of the joint spectrum of the letters with nonzero exponent, the union of the spectra $\operatorname{spec}^K_{j,B}(\alpha)$ with $s_j \neq 0$ and $\operatorname{spec}^\Psi_{j,B}(\alpha)$ with $t_j \neq 0$, with the letter exponents*

$$w_\ell = \sum_j \big(s_j \operatorname{mult}^K_{j,\ell} + t_j \operatorname{mult}^\Psi_{j,\ell}\big) = \sum_j \big(\sigma_j(\operatorname{mult}^K_{j,\ell} + \operatorname{mult}^\Psi_{j,\ell}) + \delta_j(\operatorname{mult}^K_{j,\ell} - \operatorname{mult}^\Psi_{j,\ell})\big),$$

*$\operatorname{mult}^K_{j,\ell}$ and $\operatorname{mult}^\Psi_{j,\ell}$ the multiplicities of $\xi_\ell$ in $\operatorname{spec}^K_{j,B}(\alpha)$ and $\operatorname{spec}^\Psi_{j,B}(\alpha)$, and $J = \{0, \dots, L\}$. The* scale integral

$$\operatorname{Sc}_B(\alpha; \mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_0^\infty x^{m_B - 1} \prod_j K_j(x\alpha_B, \alpha_R)^{-s_j} \Psi_j(x\alpha_B, \alpha_R)^{-t_j}\, dx$$

*converges exactly when $\operatorname{Re}\kappa_B > 0$ and $\operatorname{Re}\kappa_B^{\text{top}} > 0$, and then*

$$\operatorname{Sc}_B(\alpha; \mathbf{m}; \mathbf{s}, \mathbf{t}) = \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j} \prod_\ell (1 + \xi_\ell)^{w_\ell}\ \Gamma_{\,2\,|J|}\begin{bmatrix} \kappa_B, & \kappa_B^{\text{top}} \\ & \mathbf{w} \end{bmatrix} \mathsf{B}^{\{J\}}_{\mathbf{w},\, \kappa_B^{\text{top}};\, I_J}(\boldsymbol{\xi}),$$

*with $|\mathbf{w}| = \sum_j (s_j + t_j) c_j(B) = \kappa_B + \kappa_B^{\text{top}}$, so that the scale index $|\mathbf{w}| - \kappa_B$ of Proposition 15.1 is the top wall; the Breta is read by that proposition at complex spectra.*

**Corollary 15.4** (The scale of a bunch of width one)**.** *Let $B$ be a bunch with room $R$. Along $B$ each letter of a table $j$ with $c_j(B) = 1$ has one Skeleton value, the level letter at the floor over the level letter at the largest level, colevel letters for $\Psi_j$ (Definition 12.16), and its coin is the probability of the largest level, respectively colevel; the letters of a table with $c_j(B) = 0$ have none:*

$$\operatorname{spec}^K_{j,B}(\alpha) = \Big\{ \frac{K^{(d_j(B))}_{j,B}(\alpha)}{K^{(d_j(B)+1)}_{j,B}(\alpha)} \Big\},\ \text{coin } \mathbb{P}_{j,\alpha}\big(N_B = d_j(B) + 1\big);$$

$$\operatorname{spec}^\Psi_{j,B}(\alpha) = \Big\{ \frac{\Psi^{(d^*_j(B))}_{j,B}(\alpha)}{\Psi^{(d^*_j(B)+1)}_{j,B}(\alpha)} \Big\},\ \text{coin } \mathbb{P}^*_{j,\alpha}\big(N^*_B = d^*_j(B) + 1\big).$$

*At $B = \{e\}$, $e$ neither a loop nor a coloop of table $j$, the two levels are $K_{j,\setminus e}(\alpha_R)$ and $\alpha_e K_{j,/e}(\alpha_R)$ and the two colevels $\Psi_{j,/e}(\alpha_R)$ and $\alpha_e \Psi_{j,\setminus e}(\alpha_R)$: each Skeleton value is a deletion–contraction ratio, and the coins are the occupation $R_{j,e}(\alpha)$ and the dual occupation $R^*_{j,e}(\alpha)$.*

**Corollary 15.5** (One share per coefficient beyond the first)**.** *Let $\operatorname{Re} w_\ell > 0$ for all $\ell$, $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$, and choose $c_\ell \in (0, \operatorname{Re} w_\ell)$, $\ell = 1, \dots, L$, with $\operatorname{Re}\kappa - \operatorname{Re} w_0 < \sum_\ell c_\ell < \operatorname{Re}\kappa$, possible exactly under these conditions. With the share integrals on $\operatorname{Re} a_\ell = c_\ell$ and $|\mathbf{a}| = a_1 + \cdots + a_L$,*

$$\Gamma_{\,2\,|J|}\begin{bmatrix} \kappa, & |\mathbf{w}| - \kappa \\ & \mathbf{w} \end{bmatrix} \mathsf{B}^{\{J\}}_{\mathbf{w},\, |\mathbf{w}| - \kappa;\, I_J}(\boldsymbol{\xi})$$
$$= \int_{a_1 + b_1 = w_1} \cdots \int_{a_L + b_L = w_L} \prod_{\ell=1}^{L} \Gamma_{\,2\,1}\begin{bmatrix} a_\ell, & b_\ell \\ & w_\ell \end{bmatrix} \xi_\ell^{-b_\ell} \cdot \xi_0^{\kappa - |\mathbf{a}| - w_0}\, \Gamma_{\,2\,1}\begin{bmatrix} \kappa - |\mathbf{a}|, & w_0 - \kappa + |\mathbf{a}| \\ & w_0 \end{bmatrix}.$$

*Any coefficient may serve as $\xi_0$; $L + 1$ distinct values need $L$ shares.*

**Theorem 15.6** (The scale in its spectral gaps)**.** *Let $\xi_\ell \in \mathbb{C}\setminus(-\infty,0]$ and $w_\ell \in \mathbb{C}$, $\ell \in J$, with $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$, and let $r>0$ have $|\xi_\ell - r| < r$ for every $\ell$; such an $r$ exists exactly when every $\xi_\ell$ has positive real part, since $|\xi_\ell - r| < r$ reads $\operatorname{Re}\xi_\ell > |\xi_\ell|^2/(2r)$. Put $g_\ell = 1-\xi_\ell/r$, the* spectral gaps *at $r$, so that $|g_\ell|<1$. Then*

$$\Gamma_{\,2\,|J|}\!\begin{bmatrix}\kappa,\ |\mathbf{w}|-\kappa\\ \mathbf{w}\end{bmatrix}\,\mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}(\boldsymbol{\xi}) = r^{\kappa-|\mathbf{w}|}\sum_{\mathbf{n}\in\mathbb{N}^J}\prod_{\ell\in J}\frac{(w_\ell)_{n_\ell}}{n_\ell!}g_\ell^{\,n_\ell}\;\Gamma_{\,2\,1}\!\begin{bmatrix}\kappa,\ |\mathbf{w}|+|\mathbf{n}|-\kappa\\ |\mathbf{w}|+|\mathbf{n}|\end{bmatrix},$$

*a convergent series whose terms, grouped by the total order $|\mathbf{n}|$, have root rate at most $\max_\ell|g_\ell|$. A coefficient equal to $r$ has spectral gap zero and drops out. For $0<\xi_\ell\le r$, $w_\ell>0$ and $0<\kappa<|\mathbf{w}|$ the gaps lie in $[0,1)$ and every term is positive. For positive coefficients the bound $\max_\ell|g_\ell|$ is least at $r=\frac12(\max_\ell\xi_\ell+\min_\ell\xi_\ell)$, where it is $\frac{\max_\ell\xi_\ell-\min_\ell\xi_\ell}{\max_\ell\xi_\ell+\min_\ell\xi_\ell}$, against $1-\frac{\min_\ell\xi_\ell}{\max_\ell\xi_\ell}$ at $r=\max_\ell\xi_\ell$.*

**Corollary 15.7** (Inversion is Cremona on the scale)**.** *In Theorem 15.3:*

$$\mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}(\boldsymbol{\xi}) = \prod_\ell \xi_\ell^{-w_\ell}\,\mathsf{B}^{\{J\}}_{\mathbf{w},\,\kappa;\,I_J}(\boldsymbol{\xi}^{-1}),$$

$$\mathrm{Sc}_B(\alpha;\mathbf{m};\mathbf{s},\mathbf{t}) = (\alpha^{\mathcal{E}})^{-2\sigma}\;\mathrm{Sc}_B(\alpha^{-1};2\sigma\mathbf{1}-\mathbf{m};\mathbf{t},\mathbf{s})\,:$$

*the joint spectrum at $(\alpha^{-1};\mathbf{t},\mathbf{s})$ is the reciprocal of that at $(\alpha;\mathbf{s},\mathbf{t})$ with the same letter exponents, and the index of the scale integral moves from $\kappa_B$ to $\kappa_B^{\mathrm{top}}=|\mathbf{w}|-\kappa_B$.*

## 15.2 Closing

**Definition 15.8** (Exceedance polynomials)**.** For a bunch $B$ of a table the *exceedance polynomials* are $\mathsf{Q}^K_{B,>}(x;\alpha)=\sum_{k\ge0}\mathbb{P}_\alpha(N_B-d(B)>k)\,x^k$ and $\mathsf{Q}^\Psi_{B,>}(x;\alpha)=\sum_{k\ge0}\mathbb{P}^*_\alpha(N^*_B-d^*(B)>k)\,x^k$. They have nonnegative coefficients and degree $c(B)-1$, and $1-\mathsf{Q}^K_B=(1-x)\mathsf{Q}^K_{B,>}$, $1-\mathsf{Q}^\Psi_B=(1-x)\mathsf{Q}^\Psi_{B,>}$.

**Theorem 15.9** (The scale closes on its two sectors)**.** *Let $B$ be a proper bunch of a family with room $R$, and let the data be real with $s_j,t_j\ge0$, $\kappa_B>0$ and $\kappa_B^{\mathrm{top}}>0$. For $S\in\{B,R\}$ write the polynomials $\mathsf{Q}^K_{j,S}(\cdot;\alpha)$, with the exponents $s_j$, and $\mathsf{Q}^\Psi_{j,S}(\cdot;\alpha)$, with the exponents $t_j$, as $\mathsf{Q}_{S,i}(\cdot;\alpha)$ with exponents $w_i$, $1\le i\le 2N$ for a family of $N$ tables, $\mathbf{w}=(w_i)_i$, and their exceedance polynomials as $\mathsf{Q}_{S,i,>}$.*

*(i) The scale integral is the sum of the Trailmoments of its two sectors, with the coefficient functions $x\mapsto(\mathsf{Q}_{S,i}(x;\alpha))_i$ and the inputs $x^{\kappa-1}\mathbb{I}_{(0,1)}$:*

$$\mathrm{Sc}_B(\alpha;\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_j K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}\Big(\big(\mathsf{T}_{\mathbf{w};I_{2N}}[(\mathsf{Q}_{B,i}(x;\alpha))_i]\,x^{\kappa_B-1}\mathbb{I}_{(0,1)}\big)\big((0,1)\big) + \big(\mathsf{T}_{\mathbf{w};I_{2N}}[(\mathsf{Q}_{R,i}(x;\alpha))_i]\,x^{\kappa_B^{\mathrm{top}}-1}\mathbb{I}_{(0,1)}\big)\big((0,1)\big)\Big).$$

*(ii) With $\lambda_S=\max_i\big(1-\mathsf{Q}_{S,i}(0;\alpha)\big)<1$ and $\kappa>0$,*

$$\big(\mathsf{T}_{\mathbf{w};I_{2N}}[(\mathsf{Q}_{S,i}(x;\alpha))_i]\,x^{\kappa-1}\mathbb{I}_{(0,1)}\big)\big((0,1)\big) = \int_0^1 x^{\kappa-1}\prod_i\mathsf{Q}_{S,i}(x;\alpha)^{-w_i}\,dx = \sum_{\mathbf{n}\in\mathbb{N}^{2N}}\prod_i\frac{(w_i)_{n_i}}{n_i!}\sum_{k\ge0}\Big([x^k]\prod_i\mathsf{Q}_{S,i,>}(x;\alpha)^{n_i}\Big)\Gamma_{\,2\,1}\!\begin{bmatrix}\kappa+k,\ |\mathbf{n}|+1\\ \kappa+k+|\mathbf{n}|+1\end{bmatrix},$$

*a series of nonnegative terms, those with $|\mathbf{n}|=M$ summing to at most $\frac{(|\mathbf{w}|)_M}{M!}\lambda_S^M\kappa^{-1}$.*

*For complex exponents with* $\operatorname{Re}\kappa > 0$ *the series converges absolutely to the same value. No Skeleton value is used, and a complex spectrum closes as a real one does.*

**Proposition 15.10** (Closing around a point of the axis)**.** *The expansion* $x+\xi = (x+r)\big(1-\frac{r-\xi}{x+r}\big)$, $r>0$, *converges uniformly on* $x \ge 0$ *exactly when* $|\xi - r| < r$*; one* $r$ *serves every value of a finite spectrum exactly when every value has positive real part, that is, when every irreducible real factor of every Skeleton polynomial has positive coefficients.*

**Corollary 15.11** (The period closes on the ordering sectors)**.** *Let the data be complex with* $\kappa_{\mathcal{E}} = 0$ *and* $\operatorname{Re}\kappa_S > 0$ *for every nonempty proper* $S$*. For an order* $\pi$ *with tails* $U_k$ *and sector letters* $k_{j,\pi}, \psi_{j,\pi}$ *(Theorem 8.5), with* $N_S = |T \cap S|$*, on* $[0,1]^{n-1}$

$$1 - \frac{k_{j,\pi}(q)}{K_j(\mathbf{1})} = \sum_{k=2}^{n} (1-q_k)\, \mathbb{E}_{T\sim\mathbb{P}_{j,\mathbf{1}}}\Big[ \prod_{k'<k} q_{k'}^{\,N_{U_{k'}} - d_j(U_{k'})} \sum_{0 \le i < N_{U_k} - d_j(U_k)} q_k^{\,i} \Big],$$

*the same for* $\psi_{j,\pi}$ *with the colevels and* $d_j^*$*; both sides lie in* $[0,\lambda]$*,* $\lambda < 1$ *the largest of the numbers* $1 - \theta_{j,T^{\pm}_{j,\pi}}/K_j(\mathbf{1})$*. Consequently*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_j K_j(\mathbf{1})^{-s_j-t_j} \sum_{\pi} \sum_{\mathbf{n},\mathbf{n}'} \prod_j \frac{(s_j)_{n_j}(t_j)_{n_j'}}{n_j!\, n_j'!} \times \int_{[0,1]^{n-1}} \prod_k q_k^{\,\kappa_{U_k}-1} \prod_j \Big(1 - \frac{k_{j,\pi}(q)}{K_j(\mathbf{1})}\Big)^{n_j} \Big(1 - \frac{\psi_{j,\pi}(q)}{K_j(\mathbf{1})}\Big)^{n_j'} dq,$$

*a convergent series, each integral a finite positive combination of products*

$$\prod_k \Gamma \begin{bmatrix} \kappa_{U_k} + \beta_k,\ \gamma_k + 1 \\ \kappa_{U_k} + \beta_k + \gamma_k + 1 \end{bmatrix}, \qquad \beta_k, \gamma_k \in \mathbb{N};$$

*for real data with* $s_j, t_j \ge 0$ *every term is nonnegative. With* $w = \sum_j (|s_j| + |t_j|)$*, the sum of the moduli of the exponents, the terms with* $|\mathbf{n}| + |\mathbf{n}'| = M$ *have moduli summing to at most* $\frac{(w)_M}{M!} \lambda^M \prod_j K_j(\mathbf{1})^{-\operatorname{Re}(s_j+t_j)} \sum_\pi \prod_k (\operatorname{Re}\kappa_{U_k})^{-1}$*.*

**Proposition 15.12** (An enclosure of the period in the field of its data)**.** *In Corollary 15.11 let the data be real, and let the weights, masses and exponents lie in a subfield* $k$ *of* $\mathbb{R}$*. For an order* $\pi$ *put* $a_{j,\pi} = 1 - k_{j,\pi}/K_j(\mathbf{1})$ *and* $b_{j,\pi} = 1 - \psi_{j,\pi}/K_j(\mathbf{1})$*, let* $\lambda_\pi < 1$ *be the largest of the numbers* $1 - \theta_{j,T^{\pm}_{j,\pi}}/K_j(\mathbf{1})$*, and let* $f_{\pi,M}(q)$ *be the coefficient of* $\zeta^M$ *in the power series* $\prod_j (1-\zeta a_{j,\pi})^{-s_j}(1-\zeta b_{j,\pi})^{-t_j}$*.*

*(i)* $\prod_j K_j(\mathbf{1})^{s_j+t_j}\, P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_\pi \sum_{M\ge 0} I_{\pi,M}$ *with* $I_{\pi,M} = \int_{[0,1]^{n-1}} \prod_k q_k^{\,\kappa_{U_k}-1} f_{\pi,M}(q)\, dq$*, an element of* $k$*, nonnegative when* $s_j, t_j \ge 0$*, by* $\int \prod_k q_k^{\,\kappa_{U_k}-1+\beta_k} dq = \prod_k (\kappa_{U_k}+\beta_k)^{-1}$*.*

*(ii) With* $w = \sum_j (|s_j| + |t_j|)$*, if* $r_{\pi,Q} = \lambda_\pi \max\big(1, \frac{w+Q+1}{Q+2}\big) < 1$*, then* $\big|\sum_{M>Q} I_{\pi,M}\big| \le \operatorname{err}_{\pi,Q} = \prod_k \kappa_{U_k}^{-1} \frac{(w)_{Q+1}}{(Q+1)!} \frac{\lambda_\pi^{Q+1}}{1-r_{\pi,Q}}$*, the sum being nonnegative when* $s_j, t_j \ge 0$*, and* $\operatorname{err}_{\pi,Q} \to 0$ *as* $Q \to \infty$*: the partial sums minus and plus* $\sum_\pi \operatorname{err}_{\pi,Q}$*, and for* $s_j, t_j \ge 0$ *the partial sums and the partial sums plus* $\sum_\pi \operatorname{err}_{\pi,Q}$*, are elements of* $k$ *enclosing the left side of (i) within any prescribed width.*

*(iii) The coefficients $f_{\pi,M}$ satisfy a linear recurrence in $M$ with coefficients polynomial in $M$ and in the $a_{j,\pi}, b_{j,\pi}$: the power series $f(\zeta)=\prod_i(1-\zeta A_i)^{-w_i}$, with $(A_i,w_i)$ running over the pairs $(a_{j,\pi},s_j)$ and $(b_{j,\pi},t_j)$, satisfies $\prod_i(1-\zeta A_i)\,f'(\zeta)=\big(\sum_i w_iA_i\prod_{i'\neq i}(1-\zeta A_{i'})\big)f(\zeta)$.*

**Theorem 15.13** (The peak closing of a period)**.** *Let the data be real with $s_j,t_j\geq 0$, $\kappa_{\mathcal{E}}=0$ and every proper wall positive. Write the letters $K_j$ (exponent $s_j$, degree $r_j$, counts $d_j$) and $\Psi_j$ (exponent $t_j$, degree $n-r_j$, counts $d_j^*$) as $L_i$ with exponents $w_i$, degrees $g_i$ and counts $d_i(S)$, choose references $M_i\geq\max_{\Delta_{\mathcal{E}}}L_i$, and put $b_{\mathbf{n}}=\int_{\Delta_{\mathcal{E}}}u^{\mathbf{m-1}}\prod_i(M_i-L_i(u))^{n_i}d\omega_{\mathcal{E}}(u)\geq 0$.*

*(i) $P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\sum_{\mathbf{n}}\prod_i\frac{(w_i)_{n_i}}{n_i!}M_i^{-w_i-n_i}b_{\mathbf{n}}$, a series of nonnegative terms. With the references $M_i=\max_T\theta_{j,T}/g_i!$, $j$ the table of the letter $L_i$, every $b_{\mathbf{n}}$ is a finite positive combination of the stacked Gammas $\Gamma^{\,n}_{\,1}\!\left[\begin{matrix}\mathbf{m}+\boldsymbol{\gamma}\\ m_{\mathcal{E}}+|\boldsymbol{\gamma}|\end{matrix}\right]$, $\boldsymbol{\gamma}\in\mathbb{N}^{\mathcal{E}}$; the same series, with every $b_{\mathbf{n}}$ such a finite positive combination, holds for $\int_{\Delta_{\mathcal{E}}}u^{\mathbf{m-1}}\prod_i L_i^{-w_i}\,d\omega_{\mathcal{E}}$ with any letters $L_i$ homogeneous of degree $g_i$ with nonnegative coefficients and squarefree monomials, $M_i$ being the largest coefficient of $L_i$ over $g_i!$.*

*(ii) For each letter let $\varpi_i$ be the least of $\kappa_S/d_i(S)$ over nonempty proper $S$ with $d_i(S)>0$, and $k_i$ the largest length of a chain $S_1\subsetneq\cdots\subsetneq S_{k_i}$ with $\kappa_{S_l}=\varpi_i d_i(S_l)$; put $\varpi=\min_i\varpi_i$ and $k=\max\{k_i:\varpi_i=\varpi\}$. The partial sums $P^{\leq Q}$ of (i) over $\max_i n_i\leq Q$ satisfy $P_F(\mathbf{m};\mathbf{s},\mathbf{t})-P^{\leq Q}=\Theta\big(Q^{-\varpi}(\log Q)^{k-1}\big)$, and if a single letter attains $\varpi$,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})-P^{\leq Q}\sim\frac{c_i\,\Gamma(w_i+\varpi)}{\varpi\,\Gamma(k)\,\Gamma(w_i)}\,Q^{-\varpi}(\log Q)^{k-1},$$

*with $c_i>0$ the coefficient of $(z+\varpi_i)^{-k_i}$ in $M_i^{-z}\int_{\Delta_{\mathcal{E}}}u^{\mathbf{m-1}}L_i^z\prod_{i'}L_{i'}^{-w_{i'}}d\omega_{\mathcal{E}}$, whose pole at $-\varpi_i$ has order exactly $k_i$. A letter that vanishes on no proper face closes geometrically.*

*(iii) The fastest start is the sharp reference $M_i=\max_{\Delta_{\mathcal{E}}}L_i$; where it is attained inside the simplex, $u_e^\star\partial_e\log L_i(u^\star)=g_iu_e^\star$ for every mark. It can be attained on a proper face of the simplex.*

*(iv) At the target of a molecule with a native gauge every proper wall is positive (Theorem 8.19(iii)), so the series applies there and its rate $\varpi$ is positive.*

**Theorem 15.14** (Continued Beta corners)**.** *In Theorem 15.9 and Corollary 15.11 let the data be complex and let no index $\kappa$, respectively no $\kappa_{U_k}$, be a nonpositive integer; for an index with $\operatorname{Re}\kappa\leq 0$ let $h\in\mathbb{N}$ have $-h-1<\operatorname{Re}\kappa\leq -h$. Then the series converge, with each Beta corner $\Gamma^{\,2}_{\,1}\!\left[\begin{matrix}\kappa+k,\ |\mathbf{n}|+1\\ \kappa+k+|\mathbf{n}|+1\end{matrix}\right]$ in Theorem 15.9 and $\Gamma^{\,2}_{\,1}\!\left[\begin{matrix}\kappa_{U_k}+\beta_k,\ \gamma_k+1\\ \kappa_{U_k}+\beta_k+\gamma_k+1\end{matrix}\right]$ in Corollary 15.11 read as the meromorphic stacked Gamma, to the* Reg *of the integral, its continuation in the indices; the terms of total order $M$ are at most $\frac{(w)_M}{M!}\lambda^M$ times a polynomial in $M$ whose degree is the sum of the integers $h+1$ over the indices with real part at most $0$ (over the proper walls with $\operatorname{Re}\kappa_S\leq 0$ in Corollary 15.11), with $w=\sum_j(|s_j|+|t_j|)$ the sum of the moduli of the exponents. The sector series needs no positivity of the walls and computes the continued value at every point off the pole hyperplanes.*

The closings expand each letter around a reference and integrate term by term: on the two sectors of a bunch's scale the reference is the letter's value at the end of the sector, on the ordering sectors of the simplex it is the total weight, and at the peak it is a bound on the simplex. For real data with nonnegative exponents every term is a positive combination of stacked Gammas, nonnegative where the walls are positive, and the rate at which they fall is read from the walls.

## 15.3 The port

**Theorem 15.15** (The port)**.** *Let $B$ be a bunch of a family $F$ on $\mathcal{E}$, with room $R$, and suppose there are reals $c_{j,p}, c'_{j,p} > 0$, one for each pattern $p$ of $B$ in table $j$, with $\sum_p c_{j,p} = \operatorname{Re} s_j$ and $\sum_p c'_{j,p} = \operatorname{Re} t_j$, such that (a) $\operatorname{Re}\mu_e - \sum_{j,p} \frac{c_{j,p}-c'_{j,p}}{2}\chi_p(e) > 0$ for $e \in B$, and (b) the children $F_B(\mathbf{c},\mathbf{c}')$ converge at the masses $\operatorname{Re}\mathbf{m}|_R$. Then $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ converges and, with the share integrals on $\operatorname{Re} a_{j,p} = c_{j,p}$, $\operatorname{Re} b_{j,p} = c'_{j,p}$,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j} \int_{|\mathbf{b}_j|=t_j} \prod_j \Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix} \Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix} \times \prod_{e\in B} \Gamma\big(\mu_e - \langle\boldsymbol{\delta}\rangle_e\big)\, J_{F_B(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b}),$$

*with the odd loads $\langle\boldsymbol{\delta}\rangle_e$ of the shares (Definition 12.24). For $B = \mathcal{E}$ this is the* atom formula*: every child is $\{\varnothing\}$ with the weight $\theta_{j,T}$, and $J_{F_{\mathcal{E}}(\mathbf{a},\mathbf{b})}(\varnothing;\mathbf{a},\mathbf{b}) = \prod_{j,T} \theta_{j,T}^{-a_{j,T}-b_{j,T}}$.*

The port opens every letter over the patterns of the bunch, one share for each pattern and letter. On the bunch the shares leave a product of Gammas, one per mark, whose arguments are the offsets less the odd loads; on the room they leave the interest of the children at the shares as their exponents.

**Corollary 15.16** (A loop or coloop in every table)**.** *Let every mark of a set $A$ be in every table a loop or a coloop, with $\eta_{j,e} = -1$ for a loop and $\eta_{j,e} = +1$ for a coloop. Then $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_{e\in A}\Gamma\big(\mu_e - \sum_j \eta_{j,e}\delta_j\big)\, J_{F'}(\mathbf{m}|_{\mathcal{E}\setminus A};\mathbf{s},\mathbf{t})$, with $F'$ the family on $\mathcal{E}\setminus A$ of the deletions of its tables at the loops and the contractions at the coloops (Definition 7.3).*

**Corollary 15.17** (The damped bunch integral is a Breta)**.** *In Theorem 15.15 fix $\alpha_R$, let $\operatorname{Re} s_j, \operatorname{Re} t_j > 0$ (a letter with exponent zero is 1 and gets no scale), put $z = \frac{1}{2}\log\alpha$, introduce one singleton scale per letter, $V = \{v_{j,+}, v_{j,-}\}_j$, put $\mathbf{q} = (\mathbf{s},\mathbf{t})$ on $V$, and take the partition $\mathcal{P}_{\text{port}} = \{B\} \cup \{\{v\} : v \in V\}$. The term arrays $(\pi^{\text{port}}, \ell^{\text{port}})$ of the letter presentation (Proposition 3.6) have the columns:*

- *for each $e \in B$, the kernel column with shape exponent $(\mathbf{e}_e, 0_V)$, level $(1, 0_V)$ and coefficient 1;*
- *for every $(j,p)$, a column with shape exponent $(\chi_p/2, \mathbf{e}_{j,+})$, level $(\ell_p, \mathbf{e}_{j,+})$ and coefficient $\Lambda_{j,(p)}(z_R)$;*
- *for every $(j,p)$, a column with shape exponent $(-\chi_p/2, \mathbf{e}_{j,-})$, level $(-\ell_p, \mathbf{e}_{j,-})$ and coefficient $\Lambda_{j,(p)}(-z_R)$;*

*and $\boldsymbol{\xi}_{\text{port}}(\alpha_R)$ lists the coefficients in this order. Then, either side converging with the other,*

$$\int_{\mathbb{R}^B_{>0}} \alpha_B^{\boldsymbol{\mu}|_B - \mathbf{1}} e^{-|\alpha_B|} \prod_j \Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j}\, d\alpha_B = \Gamma(\mu_B)\, \mathsf{B}^{\mathcal{P}_{\text{port}}}_{(\boldsymbol{\mu}|_B,\mathbf{q}),\,(\mu_B,\mathbf{q});\,(\pi^{\text{port}},\ell^{\text{port}})}\big(\boldsymbol{\xi}_{\text{port}}(\alpha_R)\big).$$

The scale of Theorem 15.3 keeps the letters closed and integrates the scale alone, one share per distinct Skeleton value beyond the first (Corollary 15.5).

**Theorem 15.18** (The native split is an admissible contour)**.** *For real data with $s_j, t_j \geq 0$, not all zero, $\mathbf{m} > 0$ and $\kappa_{\mathcal{E}} > 0$, a native gauge $\alpha^\star$ and a bunch $B$, the real parts $c_{j,p} = s_j\pi_{j,p}(\alpha^\star)$, $c'_{j,p} = t_j\pi^*_{j,p}(\alpha^\star)$ of the native split satisfy the hypotheses of Theorem 15.15; a letter with exponent zero is 1 and is not opened.*

**Corollary 15.19** (The port decides convergence)**.** *For real data with $s_j, t_j \geq 0$ and any bunch $B$, $J_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ converges if and only if some real parts satisfy the hypotheses of Theorem 15.15 at $B$.*

## 15.4 Signed letters

**Definition 15.20** (Signed letters)**.** A *signed letter* is $L = L_+ - L_-$ with $L_\pm$ positive combinations of monomials of one degree $g$; adjoining $\zeta|u|^g$ gives $L+\zeta$ on the simplex, positive for $\zeta > -\min_{\Delta_\mathcal{E}} L$.

**Proposition 15.21** (A signed letter is a boundary value)**.** *Let a signed letter $L$ enter a form with exponent $s$ beside positive letters $L_i$ with exponents $w_i$, and for $\zeta > -\min_{\Delta_\mathcal{E}} L$ put $G(\zeta) = \int_{\Delta_\mathcal{E}} u^{\mathbf{m-1}} (L+\zeta)^{-s} \prod_i L_i^{-w_i}\, d\omega_\mathcal{E}$.*

*(i) With $\nu$ the image of $u^{\mathbf{m-1}} \prod_i L_i^{-w_i} d\omega_\mathcal{E}$ under $u \mapsto L(u)$, $G(\zeta) = \int (y+\zeta)^{-s}\, \nu(dy)$ is holomorphic off $(-\infty, -\min_{\Delta_\mathcal{E}} L]$, continues across that half-line except at the negatives of the critical values of $L$ on the simplex and its faces, and satisfies a linear differential equation in $\zeta$ with polynomial coefficients.*

*(ii) $G(0 \pm i0) = \int_{L>0} u^{\mathbf{m-1}} L^{-s} \prod_i L_i^{-w_i} d\omega_\mathcal{E} + e^{\mp i\pi s} \int_{L<0} u^{\mathbf{m-1}} |L|^{-s} \prod_i L_i^{-w_i} d\omega_\mathcal{E}$, the region integrals convergent for $\operatorname{Re} s < 1$ near regular zeros and taken by Lemma 8.4 in the coordinate $y = L(u)$ otherwise.*

*(iii) $G(0+i0) - G(0-i0) = -2i \sin(\pi s) \int_{L<0} u^{\mathbf{m-1}} |L|^{-s} \prod_i L_i^{-w_i} d\omega_\mathcal{E}$.*

On the scale of a bunch a zero of a signed letter is a split point: on each of the intervals it bounds, rescaled to $[0,1]$, the zero is at an endpoint, where it is a wall of the integral over that interval and Theorem 15.14 applies, and the phase $e^{\mp i\pi s}$ attaches to the interval on which the letter is negative.

# Chapter 16

# Integrals, expectations and the Foreman procedure

## 16.1 Forms, sections, completions

**Definition 16.1** (Differential forms, charts, schedules)**.** The *differential form* of a family at masses $\mathbf{m}$ is

$$\omega = W\, d\alpha_{\mathbf{m}} = \alpha^{\mathbf{m-1}} \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j} d\alpha,$$

the family word against the measure $d\alpha_{\mathbf{m}}$ (Definition 8.1). It is *projective* when $\kappa_{\mathcal{E}} = 0$, and then it is integrated over the rays of the orthant, over any section (Lemma 16.2, next). The *chart* at a mark $g$ is the section $\alpha_g = 1$ with the coordinates $\alpha_e$, $e \neq g$, and the chart integral is $\int_{\alpha_g=1} \omega = \int_{\mathbb{R}^{\mathcal{E}-g}_{>0}} (\cdots)|_{\alpha_g=1} \prod_{e\neq g} d\alpha_e$. Passing to the chart at $g$ is the *snip* at $g$. A *schedule* $\mathfrak{s} = (g; \varsigma)$ is the chart at a mark $g$ and an order $\varsigma(1), \dots, \varsigma(n-1)$ of the other marks.

**Lemma 16.2** (Every section gives the period)**.** *If $\omega$ is projective with convergent period, $f$ its integrand, $h > 0$ homogeneous of degree one and $\varphi$ a function on $(0,\infty)$ with $\int_0^\infty |\varphi(\rho)|\, d\rho/\rho < \infty$, then $\int_{\mathbb{R}^{\mathcal{E}}_{>0}} f\, \varphi(h)\, d\alpha = \big(\int_0^\infty \varphi(\rho) \frac{d\rho}{\rho}\big) P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$. In particular every chart integral converges and equals $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$.*

**Proposition 16.3** (A section is a table)**.** *Let $W = \prod_j K_j^{-s_j} \Psi_j^{-t_j}$ be a family word, in this statement of the degree $-d_W$, $d_W = \sum_j (s_j r_j + t_j r_j^*)$, and $\mathbf{m}$ masses with every proper wall positive. Then $\int_{\Delta_{\mathcal{E}}} u^{\mathbf{m-1}} W\, d\omega_{\mathcal{E}}$ is the value on its balance wall of the family with the list $\binom{\mathcal{E}}{1}$, whose bases are the singletons and whose letter is $K_0 = |\alpha|$, added at the exponents $(m_{\mathcal{E}} - d_W, 0)$; its proper walls are those of the family. In particular, for the family $F_0$ of the one table $\binom{\mathcal{E}}{1}$, $\mathrm{Dir}_{\mathbf{m}} = \mathrm{Breta}^\circ(F_0; \mathbf{m}; m_{\mathcal{E}}, 0)$, the Breta shape law of $\binom{\mathcal{E}}{1}$ at a full basis lean; and for two families $F_1, F_2$ on $\mathcal{E}$ with words $W_1, W_2$, the logarithms of the letters of $F_2$ under the law on $\Delta_{\mathcal{E}}$ with density proportional to $u^{\mathbf{m-1}} W_1$ have the Laplace transform*

$$\int_{\Delta_{\mathcal{E}}} u^{\mathbf{m-1}} W_1 W_2\, d\omega_{\mathcal{E}} \Big/ \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m-1}} W_1\, d\omega_{\mathcal{E}}$$

*in the exponents of $W_2$, a ratio of two values of the family $F_1 \cup F_2 \cup F_0$.*

**Definition 16.4** (Completions)**.** For $w \in \mathbb{R}^{\mathcal{E}}$ with $w_{\mathcal{E}} = 1$ the *completion* of $\omega = W\, d\alpha_{\mathbf{m}}$ along $w$ is $\omega_{\mathbf{m}'} = \alpha^{-\kappa_{\mathcal{E}} w} \omega = W\, d\alpha_{\mathbf{m}'}$, $\mathbf{m}' = \mathbf{m} - \kappa_{\mathcal{E}} w$, which is projective: a completion acts on the measure

and leaves the word. The *chart completion* at $g$ has $w = \mathbf{e}_g$ and the masses $\mathbf{m}^{(g)} = \mathbf{m} - \kappa_{\mathcal{E}}\mathbf{e}_g$. For real data the *native completion* has $w = \mathbf{m}/m_{\mathcal{E}}$; its measure is the differential

$$d\alpha_{\boldsymbol{\tau}} = \alpha^{-(\kappa_{\mathcal{E}}/m_{\mathcal{E}})\mathbf{m}}\, d\alpha_{\mathbf{m}}, \qquad \boldsymbol{\tau} = \frac{n\sigma + \sum_j \delta_j(2r_j - n)}{n\sigma + \mu_{\mathcal{E}}}\,\mathbf{m},$$

at the target, with $\kappa_{\mathcal{E}} = \mu_{\mathcal{E}} - \sum_j \delta_j(2r_j - n)$ and $m_{\mathcal{E}} = n\sigma + \mu_{\mathcal{E}}$ in pair coordinates (Theorem 8.2(iii)), and the completed form is $\omega_{\boldsymbol{\tau}} = W\, d\alpha_{\boldsymbol{\tau}}$.

**Proposition 16.5** (Snips are completions)**.** *(i)* $\int_{\alpha_g=1}\omega = \int_{\alpha_g=1}\omega_{\mathbf{m}^{(g)}}$*, the period* $P_F(\mathbf{m}^{(g)};\mathbf{s},\mathbf{t})$ *wherever it converges.*

*(ii)* $K_j|_{\alpha_g=1} = K_{j,\backslash g} + K_{j,/g}$ *and* $\Psi_j|_{\alpha_g=1} = \Psi_{j,\backslash g} + \Psi_{j,/g}$*, and for* $\operatorname{Re} s_j, \operatorname{Re} t_j > 0$*, on the contours of Corollary 4.4,*

$$\prod_j K_j^{-s_j}\Psi_j^{-t_j}\Big|_{\alpha_g=1} = \int\cdots\int \prod_j \Gamma_1^2\begin{bmatrix} a_j,\ a_j' \\ s_j \end{bmatrix}\Gamma_1^2\begin{bmatrix} b_j,\ b_j' \\ t_j \end{bmatrix} K_{j,\backslash g}^{-a_j}K_{j,/g}^{-a_j'}\Psi_{j,\backslash g}^{-b_j}\Psi_{j,/g}^{-b_j'}$$

*over* $a_j + a_j' = s_j$*,* $b_j + b_j' = t_j$*: a snip opens every letter into its deletion and its contraction, with stacked-Gamma shares.*

*(iii) For a bunch* $B$*, the integral of* $\omega$ *over the bunch section* $|\alpha_B| = 1$ *is the integral over the rays of the projective form* $|\alpha_B|^{-\kappa_{\mathcal{E}}}\omega$*. In this statement write* $F'$ *for* $F$ *with one table appended at the exponents* $(\kappa_{\mathcal{E}}, 0)$*: the list* $\binom{B}{1}$ *read on* $\mathcal{E}$*, whose bases are the singletons of* $B$*, with the marks outside* $B$ *as loops, and whose letter is* $|\alpha_B|$*. Then* $|\alpha_B|^{-\kappa_{\mathcal{E}}}\omega$ *is the form of* $F'$ *at the masses* $\mathbf{m}$*,* $F'$ *is balanced there, its proper walls are* $\kappa_S - \kappa_{\mathcal{E}}\,\mathbf{1}[B \subseteq S]$*, and*

$$\int_{|\alpha_B|=1}\omega = P_{F'}\big(\mathbf{m};(\mathbf{s},\kappa_{\mathcal{E}}),(\mathbf{t},0)\big)$$

*wherever* $\operatorname{Re}\kappa_S - \operatorname{Re}\kappa_{\mathcal{E}}\,\mathbf{1}[B \subseteq S] > 0$ *for every nonempty proper* $S$*. At* $B = \{g\}$ *the appended letter is the monomial* $\alpha_g$*, and this is (i); at* $B = \mathcal{E}$ *the appended table is the list* $\binom{\mathcal{E}}{1}$ *(Proposition 16.3). For* $k = -\kappa_{\mathcal{E}} \in \mathbb{N}$*,* $k \geq 1$*,* $|\alpha_B|^k = \sum_{|\mathbf{n}|=k}\binom{k}{\mathbf{n}}\alpha_B^{\mathbf{n}}$*, so* $\int_{|\alpha_B|=1}\omega = \sum_{\mathbf{n}\in\mathbb{N}^B,\,|\mathbf{n}|=k}\binom{k}{\mathbf{n}}\int_{|\alpha_B|=1}\omega_{\mathbf{m}+\mathbf{n}}$*, each term the period* $P_F(\mathbf{m}+\mathbf{n};\mathbf{s},\mathbf{t})$ *wherever it converges.*

**Corollary 16.6** (The period fibres over the scale of a bunch)**.** *Let* $B$ *be a proper bunch of a family* $F$ *on* $\mathcal{E}$*, with room* $R$*, and let* $h > 0$ *be a function of* $\alpha_R$ *homogeneous of degree one. Wherever the integral of* $W\, d\alpha_{\mathbf{m}}$ *over the section* $\{h(\alpha_R) = 1\}$ *of the orthant converges, both sections taken with the measures of Lemma 16.2,*

$$\int_{h(\alpha_R)=1} W\, d\alpha_{\mathbf{m}} = \int_{\{\alpha_R\in\mathbb{R}^R_{>0}:\ h(\alpha_R)=1\}}\int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}}\alpha_R^{\mathbf{m}|_R - \mathbf{1}}\ \mathrm{Sc}_B(u,\alpha_R;\mathbf{m};\mathbf{s},\mathbf{t})\, d\omega_B(u),$$

*and at every point* $\mathrm{Sc}_B$ *is the product of Theorem 15.3, a stacked Gamma times a Breta of one bunch. On the balance wall the left side is* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*. At the chart* $h = \alpha_g$*,* $g \in R$*, the left side is* $P_F(\mathbf{m}^{(g)};\mathbf{s},\mathbf{t})$ *(Proposition 16.5(i)) and the outer integral is over* $\alpha_{R-g}$ *at* $\alpha_g = 1$*; at* $h = |\alpha_R|$ *it is over* $\Delta_R$ *against* $d\omega_R$*.*

**Proposition 16.7** (The scaling mode)**.** *Along $z \mapsto z + c\mathbf{1}$, $E_j(z + c\mathbf{1}) = E_j(z)$, $O_j(z + c\mathbf{1}) = O_j(z) + 2c(2r_j - n)$, and the exponent $2\langle \boldsymbol{\mu}, z\rangle - \sum_j (\sigma_j E_j + \delta_j O_j)$ of the pair form moves by $2c\kappa_{\mathcal{E}}$. Off the balance wall the form is not integrable along this direction; it is* damped *by $e^{-|\alpha|}$ (the interest, $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma(\kappa_{\mathcal{E}})P_F(\mathbf{m};\mathbf{s},\mathbf{t})$),* conditioned *by a section (at a chart a completion, Proposition 16.5; in general another projective form, Proposition 16.32 below), or* slid *by moving the masses along their ray to the balance wall (the native completion).*

**Theorem 16.8** (The slide)**.** *Let $\omega$ have real data on a connected family, with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, native gauge $\alpha^\star$ and target $\boldsymbol{\tau}$.*

*(i) If $\kappa_{\mathcal{E}} \neq 0$, exactly one completion of $\omega$ has $\alpha^\star$ as a native gauge: $\omega_{\boldsymbol{\tau}}$. If $\kappa_{\mathcal{E}} = 0$, every completion is $\omega$ and $\boldsymbol{\tau} = \mathbf{m}$.*

*(ii) As the masses slide along their ray, $\mathbf{m} \mapsto \lambda\mathbf{m}$, the data $(\lambda\mathbf{m};\mathbf{s},\mathbf{t})$ lie on the balance wall exactly when $\lambda\mathbf{m} = \boldsymbol{\tau}$.*

*(iii) Every proper wall of $\omega_{\boldsymbol{\tau}}$ is positive, so its period converges.*

*(iv) $\boldsymbol{\tau}$, $\omega_{\boldsymbol{\tau}}$ and $\alpha^\star$ are the same at every point of the ray of $\mathbf{m}$.*

Each of the three treatments of the free scaling mode (Proposition 16.7) is an evaluation, and sliding keeps the native gauge.

## 16.2 Three integrals and their expectations

**Lemma 16.9** (The period is meromorphic in the data)**.** *For every family, $P_F$ is meromorphic in the masses and the exponents, with poles only where a proper wall is a nonpositive integer, and so is its restriction to the balance wall.*

**Definition 16.10** (Three integrals and their expectations)**.** Let $\omega = W\, d\alpha_{\mathbf{m}}$ have real data, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, so that $W$ is homogeneous of degree $-\tau_{\mathcal{E}}$, $\tau_{\mathcal{E}} = n\sigma + \sum_j \delta_j(2r_j - n)$ with $\sigma = \sum_j \sigma_j$. On a connected family the centre of $\omega$ is the form of the family at the centre (Definition 9.1), at the exponents $(\boldsymbol{\sigma}, \boldsymbol{\sigma})$ and the masses $\sigma\mathbf{1}$,

$$\omega_{\mathrm{c}} = \prod_j \big(K_j \Psi_j\big)^{-\sigma_j} d\alpha_{\sigma\mathbf{1}}.$$

It is projective, its proper walls $\sum_j \sigma_j c_j(S)$ are positive and the table's gauge is native for it (Proposition 8.21), so its period $P_{\mathrm{c}} = P_F(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$ is a positive convergent integral, and $\mathbb{P}_{\mathrm{c}} = \mathrm{Breta}^\circ(F;\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$, the measure $\omega_{\mathrm{c}}/P_{\mathrm{c}}$ read on the rays, is the *law of the centre*. For masses $\mathbf{m}'$ of total $\tau_{\mathcal{E}}$ the *departure* of $W\, d\alpha_{\mathbf{m}'}$ from the centre is the ratio of the two forms,

$$\frac{W\, d\alpha_{\mathbf{m}'}}{\omega_{\mathrm{c}}}(\alpha) = \alpha^{\mathbf{m}' - \sigma\mathbf{1}} \prod_j \Big(\frac{K_j(\alpha)}{\Psi_j(\alpha)}\Big)^{-\delta_j},$$

homogeneous of degree 0: an *offset factor* $\alpha^{\mathbf{m}'-\sigma\mathbf{1}}$ and a *lean factor* $\prod_j (K_j/\Psi_j)^{-\delta_j}$. Each integral below is unnormalized. On a connected family its expectation divides it by $P_{\mathrm{c}}$: it is the expectation, under the law of the centre, of the departure that its evaluation sets.

(i) The *integral* $\int W\, d\alpha_{\mathbf{m}}$ is defined when $\omega$ is projective with convergent period: it is the integral of $\omega$ over any section, $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$, and $\int W\, d\alpha_{\mathbf{m}} = P_{\mathrm{c}}\, \mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\big[(\omega/\omega_{\mathrm{c}})(U)\big]$.

(ii) For a schedule $\mathfrak{s} = (g; \varsigma)$ the *quasi-integral* is the iterated finite part

$$\operatorname{Reg}^{\mathfrak{s}}\!\int W\, d\alpha_{\mathbf{m}} = \operatorname{Reg}\!\int_0^\infty dx_{n-1} \cdots \operatorname{Reg}\!\int_0^\infty dx_1 \Big(\alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\Big)\Big|_{\alpha_g = 1}, \qquad x_k = \alpha_{\varsigma(k)},$$

defined whenever the integrand of every one of these iterated integrals has log-power expansions at 0 and at $\infty$. It sets the departure of the chart completion, and the *quasi-expectation* is $\mathfrak{E}^{\mathfrak{s}}_{U\sim\mathbb{P}_{\mathrm{c}}}\big[(\omega_{\mathbf{m}^{(g)}}/\omega_{\mathrm{c}})(U)\big] = \operatorname{Reg}^{\mathfrak{s}}\!\int W\, d\alpha_{\mathbf{m}} \big/ P_{\mathrm{c}}$.

(iii) For a connected family of at least two marks, the *pseudointegral* is

$$\Big\backslash\!\!\!\int W\, d\alpha_{\mathbf{m}} = \operatorname*{CT}_{\eta=0} P_F(\boldsymbol{\tau} + \eta\mathbf{m}; \mathbf{s}, \mathbf{t}),$$

the constant term at the target along the ray of the masses, $P_F$ being meromorphic in the masses with poles only where a proper wall is a nonpositive integer, on the *pole hyperplanes* $\kappa_S = -q$, $S$ proper, $q \in \mathbb{N}$ (Lemma 16.9). Off those hyperplanes it is $P_F(\boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$, the value at the target of the meromorphic function $P_F$ on the balance wall. With a native gauge every proper wall is positive at $\boldsymbol{\tau}$ and $\backslash\!\!\int W\, d\alpha_{\mathbf{m}} = \int W\, d\alpha_{\boldsymbol{\tau}}$: the Foreman sets the measure to the native completion. It sets the departure of the native completion, and the *pseudoexpectation* is

$$\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\Big[\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(U)\Big] = \Big\backslash\!\!\!\int W\, d\alpha_{\mathbf{m}} \Big/ P_{\mathrm{c}};$$

with a native gauge it is the expectation $\mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$: the Foreman sets the departure and the centre sets the law. On a family whose weights factor across its molecules, the pseudointegral and the pseudoexpectation are products over the molecules, each at its own target and with its own centre; a one-mark factor contributes the value of its integrand at $\alpha = \mathbf{1}$ to the pseudointegral and 1 to the pseudoexpectation, its integrand and its centre's agreeing there.

A point $(\boldsymbol{\mu}; \boldsymbol{\sigma}, \boldsymbol{\delta})$ splits into an even part, the strengths $\sigma_j$, and an odd part, the offsets and the leans; the fold (Theorem 16.18) negates the odd part and keeps the even part. The centre is the point with the same even part and no odd part. It lies on the balance wall, it is inside its walls at every strength on a connected family, and its native gauge is the table's gauge, so its period is a convergent integral of the family itself, the same for every evaluation. Dividing by it prices the odd part of a point against its even part: with a native gauge the pseudoexpectation is the exponential of a divergence from the law of the centre, at least 1, equal to 1 exactly when the native completion is the centre's form, and unchanged by the fold (Proposition 16.34).

**Theorem 16.11** (Quasi-integrals extend the integral)**.** *Each* $\operatorname{Reg}^{\mathfrak{s}}\!\int$ *is linear in the integrand, and* $\operatorname{Reg}^{\mathfrak{s}}\!\int W\, d\alpha_{\mathbf{m}} = \int W\, d\alpha_{\mathbf{m}}$ *for every schedule wherever* $\int W\, d\alpha_{\mathbf{m}}$ *is defined; there every chart completion is* $\omega$*,* $\mathbf{m}^{(g)} = \mathbf{m}$*, and on a connected family* $\mathfrak{E}^{\mathfrak{s}}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\mathbf{m}^{(g)}}/\omega_{\mathrm{c}})(U)] = \mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega/\omega_{\mathrm{c}})(U)]$*. Elsewhere the value can depend on the chart, on the order and on the coordinates (Lemma 1.9).*

**Proposition 16.12** (The form of a compound at one point)**.** *Let* $F$ *be the direct sum of families* $F_1, \dots, F_q$ *on* $\mathcal{E}_1, \dots, \mathcal{E}_q$*,* $q \geq 2$*,* $\mathcal{E} = \mathcal{E}_1 \sqcup \cdots \sqcup \mathcal{E}_q$*, every table a direct sum across the split with factoring weights, so that* $K_j = \prod_i K_{j,i}$ *and* $\Psi_j = \prod_i \Psi_{j,i}$*; let the data be real, the masses* $\mathbf{m} = (\mathbf{m}_1, \dots, \mathbf{m}_q)$ *on the balance wall of* $F$*, and* $\kappa_i$ *the imbalance of* $F_i$ *at* $\mathbf{m}_i$*, so* $\sum_i \kappa_i = 0$*. On the section* $|\alpha_q| = 1$ *with* $\alpha_i = \rho_i u_i$*,* $u_i \in \Delta_{\mathcal{E}_i}$*, for* $i < q$ *and* $u_q = \alpha_q$*, the form of* $F$ *is*

$$\prod_{i<q} \rho_i^{\kappa_i - 1} d\rho_i \;\cdot\; \prod_{i=1}^{q} u_i^{\mathbf{m}_i - \mathbf{1}} \prod_j K_{j,i}^{-s_j} \Psi_{j,i}^{-t_j}\, d\omega_{\mathcal{E}_i}.$$

*(i) Its integral diverges for all $\kappa_1, \dots, \kappa_q$.*

*(ii) The finite part of each $\rho_i$-integral is 0 for every $\kappa_i$.*

*(iii) On the masses $\mathbf{m}(\boldsymbol{\eta}) = (\mathbf{m}_i^0 + i\eta_i w_i)_{i=1}^q$, $\boldsymbol{\eta} \in \mathbb{R}^q$ with $\sum_i \eta_i = 0$, with $\mathbf{m}_i^0$ balanced for $F_i$ inside its walls and $w_i \in \mathbb{R}^{\mathcal{E}_i}$ of total 1, so that $\kappa_i = i\eta_i$, the integral of the form is, as a distribution in $(\eta_1, \dots, \eta_{q-1})$, $(2\pi)^{q-1} \prod_{i=1}^q P_{F_i}(\mathbf{m}_i^0; \mathbf{s}, \mathbf{t})$ times the Dirac measure at 0.*

*So the compound carries nothing beyond the points of its parts, and only where each molecule is balanced on its own.*

**Theorem 16.13** (Presentations)**.** *Let $\omega = W\, d\alpha_{\mathbf{m}}$ have real data on a connected family, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$. A* presentation *of $\omega$ is a bijection $\varphi$ of the marks together with coordinates $x = \lambda^{-1}\alpha$, $\lambda \in \mathbb{R}^{\mathcal{E}}_{>0}$, in which the letters are $K_j(\lambda x)$, $\Psi_j(\lambda x)$ and the form carries the constant $\lambda^{\mathbf{m}}$. The split, the targets, the native gauges as points of the torus, the stills, the classes, the native splits and the children computed from a presentation are the images under $\varphi$ of those computed from $\omega$, and, when the target lies on no pole hyperplane, the pseudointegral and the pseudoexpectation of the presentation, its centre presented with it, are $(\lambda^{\mathbf{m}})^{\kappa\varepsilon/m_{\mathcal{E}}}$ times those of $\omega$, the factor being 1 on the balance wall; on a pole hyperplane the constant term moves with the coordinates as well (Proposition 16.31 below). In the coordinates $x$ the table's gauge is $x = \lambda^{-1}$, and the Foreman tree is the same. On a family whose weights factor across its molecules the factor is the product of the molecules' factors.*

**Definition 16.14** (The Foreman procedure)**.** *Classification*, from the tables alone:

(1) *Split* the marks into molecules and units (Proposition 9.5).

(2) *Read* each molecule $M$ at a centre $\Upsilon_{\boldsymbol{\sigma},0;\mathbf{1}}$ of the table's gauge (Chapter 10).

(3) *Answer*: a constant reading is an atom; otherwise take the Foreman bunch $B(M; \boldsymbol{\sigma}, 0)$ (Definition 13.6).

(4) *Peel* at $B(M; \boldsymbol{\sigma}, 0)$; the childcules sit at the centres they inherit (Proposition 12.27).

(5) *Recurse* to atoms and units: the output is the Foreman tree and, at the root, the label.

*Evaluation* at a real point $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$:

(6) *Native gauge* of each molecule, and its target $\boldsymbol{\tau}$ (Theorem 8.19).

(7) *Slide* the masses to the target (Theorem 16.8).

(8) *Ports* along the tree, every share contour through the native split (Theorems 15.15, 15.18 and 12.26); on the balance wall the port of Chapter 18.

(9) *Atoms*: the atom formula (Theorem 15.15 with $B = \mathcal{E}$; on the balance wall, Chapter 18).

On the orthant side the ports integrate the Foreman bunch of each node inside the integral over its room, all its marks at once in scale and shape (Corollary 15.17): the root's bunch first and the atoms last. This is the *Foreman's order*.

**Theorem 16.15** (The Foreman computes the pseudointegral)**.** *Let $\omega = W\, d\alpha_{\mathbf{m}}$ have real data on a connected family, with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, and a native gauge. For every $\eta > 0$ the procedure at the masses $\boldsymbol{\tau} + \eta\mathbf{m}$ runs on admissible contours and computes $J_F(\boldsymbol{\tau} + \eta\mathbf{m}; \mathbf{s}, \mathbf{t})$, and*

$$\int W\, d\alpha_{\mathbf{m}} = \lim_{\eta \to 0^+} \frac{J_F(\boldsymbol{\tau} + \eta\mathbf{m}; \mathbf{s}, \mathbf{t})}{\Gamma(\eta m_{\mathcal{E}})}.$$

*No finite part, chart, order or closing rule enters: the regulator $\eta$ leaves by a limit.*

**Corollary 16.16** (One member among the generalized integrals)**.** *Wherever $\int W\, d\alpha_{\mathbf{m}}$ is defined for real data on a connected family, $⨍ W\, d\alpha_{\mathbf{m}} = \int W\, d\alpha_{\mathbf{m}}$, and with Theorem 16.11 every evaluation sets the one departure $\omega/\omega_{\mathrm{c}}$ on the balance wall. Off the balance wall, with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, the period at the target converges exactly when the family has a native gauge (Theorem 8.19(iii)), and otherwise $⨍ W\, d\alpha_{\mathbf{m}}$ is its continuation in the masses, on a pole hyperplane the constant term of the continuation along the ray.*

**Theorem 16.17** (The pseudointegral off the domain)**.** *For every family, if some nonempty proper $S$ has $c_j(S) = 0$ in every active table, then, first where every proper wall has positive real part and then as meromorphic functions (Lemma 16.9),*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma{\scriptstyle\begin{bmatrix}\kappa_S,\ \kappa_{\mathcal{E}\setminus S}\\ \kappa_{\mathcal{E}}\end{bmatrix}}_{\!2}^{\!1} \int_{\Delta_S\times\Delta_{\mathcal{E}\setminus S}} \alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}\, d\omega_S\, d\omega_{\mathcal{E}\setminus S},$$

*and the restriction of $P_F$ to the balance wall is $0$. Let now $F$ be connected with real data, $s_j, t_j \geq 0$ not all zero; on the balance wall the value of $P_F$ at a target depends on no path.*

*(i) The continuation in the masses of $f \mapsto \int_{\Delta_{\mathcal{E}}} f\, u^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j}\Psi_j^{-t_j} d\omega_{\mathcal{E}}$, taken by Lemma 8.4 on each ordering sector, is a distribution on $\Delta_{\mathcal{E}}$ with normaliser $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ and monomial moments $P_F(\mathbf{m}+\boldsymbol{\gamma};\mathbf{s},\mathbf{t})/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$; where every proper wall is positive it is $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ times the Breta shape law. Just past the first wall crossed it is not a positive measure: $P_F$ passes from $+\infty$ to $-\infty$ there, while $P_F(\mathbf{m}+\boldsymbol{\gamma};\mathbf{s},\mathbf{t})$ is a positive convergent integral for $\gamma_S$ large.*

*(ii) Theorem 8.14 and the recurrences of Theorem 8.9 hold for these values as identities of meromorphic functions.*

**Theorem 16.18** (The fold: Cremona off the balance wall)**.** *Let $\omega = W\, d\alpha_{\mathbf{m}}$, with the exponents $(\mathbf{s},\mathbf{t})$, have real data on a connected family, with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, native gauge $\alpha^\star$ and target $\boldsymbol{\tau}$. Then $2\sigma\mathbf{1} - \boldsymbol{\tau}$ is positive and balanced for the exponents $(\mathbf{t},\mathbf{s})$, with native gauge $(\alpha^\star)^{-1}$, and for every $\lambda > 0$*

$$⨍ \prod_j K_j^{-s_j}\Psi_j^{-t_j}\, d\alpha_{\mathbf{m}} = ⨍ \prod_j K_j^{-t_j}\Psi_j^{-s_j}\, d\alpha_{\lambda(2\sigma\mathbf{1}-\boldsymbol{\tau})}.$$

*The map from the ray of $\mathbf{m}$ to the ray of $2\sigma\mathbf{1} - \boldsymbol{\tau}$ is an involution on rays.*

The fold swaps the two letters of every table and reflects the target through $\sigma\mathbf{1}$, negating the offsets and the leans; it keeps the strengths, so the two forms have one centre, and it carries uniform masses to uniform masses.

**Proposition 16.19** (Points and realizations)**.** *A real point $(\mathbf{m};\mathbf{s},\mathbf{t})$ of a connected family, with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$ and native gauge $\alpha^\star$, determines the realization $((s_j,t_j)_j;\alpha^\star)$, whose masses are the target; every realization arises so (Proposition 8.22). The reading of a point is the reading of its realization and may differ from point to point; the label is read at centres of the table's gauge, the Foreman tree at the realizations of the table's gauge at the centre and at the leans (Theorem 14.6), and both are functions of the tables. An atomic point is at the table's gauge up to scale (Theorem 11.4), and its centre is atomic (Theorem 11.1).*

**Theorem 16.20** (Parity)**.** *Let $g$ be an exchange of a connected family, with the bijection $h$ of its tables (Definition 11.13); then $r_{h(j)} = n - r_j$ for every $j$. With $(g^*z)_e = z_{g(e)}$ (Definition 11.13), let $\Phi^\circ$ be the centred potential (Theorem 8.19).*

*(i)* $\Lambda_j(g^*z) = \Lambda_{h(j)}(-z)$*; so* $E_j(g^*z) = E_{h(j)}(z)$*,* $O_j(g^*z) = -O_{h(j)}(z)$ *and* $\Phi^\circ(-g^*z) = \Phi^\circ(z)$*.*

*(ii) If the data are real with* $s_j, t_j \geq 0$*, not all zero, and* $\mathbf{m} > 0$*, have a native gauge, and* $\tau_{g(e)} = 2\sigma - \tau_e$ *for every* $e$*, the native gauge* $e^{2z^\star}$ *with* $\langle \mathbf{1}, z^\star \rangle = 0$ *has* $-g^*z^\star = z^\star$*, and the still of the realization* $\Upsilon = ((s_j, t_j)_j; e^{2z^\star})$ *has* $\kappa_\Upsilon(g(e_1), \dots, g(e_k)) = (-1)^k \kappa_\Upsilon(e_1, \dots, e_k)$ *for every* $k$*.*

*(iii) Under (ii) even rungs never separate* $e$ *from* $g(e)$*, and at odd rungs their tallies (Definition 9.4) are negatives of each other.*

*(iv) If moreover* $\boldsymbol{\delta} = 0$ *and* $\boldsymbol{\tau} = \sigma\mathbf{1}$*, then* $z^\star = 0$*, every odd rung vanishes, and no rung separates* $e$ *from* $g(e)$*.*

*(v) At* $(\mathbf{t}, \mathbf{s})$ *and the target* $2\sigma\mathbf{1} - \boldsymbol{\tau}$ *the native gauge is* $e^{-2z^\star}$ *and rung* $k$ *is* $(-1)^k$ *times rung* $k$ *at* $(\mathbf{s}, \mathbf{t})$*; under (ii) this is the still at* $(\mathbf{s}, \mathbf{t})$ *carried by* $g$*, and the Foreman bunch at* $(\mathbf{t}, \mathbf{s})$ *is the image under* $g^{-1}$ *of the Foreman bunch at* $(\mathbf{s}, \mathbf{t})$*.*

## 16.3 Quasi-integrals as Laurent data

**Definition 16.21** (Regular schedules)**.** For a schedule $\mathfrak{s} = (g; \varsigma)$ put $x_k = \alpha_{\varsigma(k)}$ and $v_k = \mathbf{e}_{\varsigma(k)} - \mathbf{e}_g$. Let $\mathcal{J}_0$ be the chart integrand and $\mathcal{J}_k(\varepsilon_1, \dots, \varepsilon_k; x_{>k}) = \int_0^\infty x_k^{\varepsilon_k} \mathcal{J}_{k-1}\, dx_k$, with $\int_0^1$ and $\int_1^\infty$ each continued meromorphically in the *regulator* $\varepsilon_k$. The schedule is *regular* when for every $k$, as $x_k \to 0$ and as $x_k \to \infty$, $\mathcal{J}_{k-1}$ is a finite sum of terms $c(\varepsilon, x_{>k})\, x_k^{a(\varepsilon)}$ plus $O(x_k^{\pm M})$ for every $M$, the exponents affine in $\varepsilon_{<k}$ and independent of $x_{>k}$, uniformly for the regulators in a small polydisc away from their poles.

**Theorem 16.22** (Bridge)**.** *For real data on a connected family and a regular schedule,* $\mathrm{Reg}^{\mathfrak{s}} \int W\, d\alpha_{\mathbf{m}}$ *is defined and*

$$\mathrm{Reg}^{\mathfrak{s}} \int W\, d\alpha_{\mathbf{m}} = \operatorname*{CT}_{\varepsilon_{n-1}} \cdots \operatorname*{CT}_{\varepsilon_1} P_F\Big(\mathbf{m}^{(g)} + \sum_{k=1}^{n-1} \varepsilon_k v_k; \mathbf{s}, \mathbf{t}\Big),$$

*the constant terms taken innermost first,* $|\varepsilon_1| \ll \cdots \ll |\varepsilon_{n-1}|$*.*

**Theorem 16.23** (A wall is closed by its last variable)**.** *For a schedule* $\mathfrak{s} = (g; \varsigma)$ *put* $\mathbf{p} = \mathbf{m}^{(g)}$ *and* $I(\varepsilon) = P_F\big(\mathbf{p} + \sum_k \varepsilon_k v_k; \mathbf{s}, \mathbf{t}\big)$*. A proper* $U$ *is* polar *when* $\kappa_U(\mathbf{p}) \in -\mathbb{N}$*; its* members *are the* $k$ *with* $\varsigma(k) \in U$ *if* $g \notin U$*, and with* $\varsigma(k) \notin U$ *if* $g \in U$*;* $\mathrm{last}(U)$ *is its last member.*

*(i) Near* $0$*,* $I(\varepsilon) = \sum_C f_C(\varepsilon) \big/ \prod_{U \in C} \ell_U(\varepsilon) + H(\varepsilon)$*,* $\ell_U = \pm \sum_{k\ member} \varepsilon_k$*, over the nonempty sets* $C$ *of polar walls that are tails of one order, with* $f_C, H$ *holomorphic; the member sets in one* $C$ *are nested or disjoint.*

*(ii) Under* $\mathrm{CT}_{\varepsilon_k}$*,* $1/\ell_U$ *is holomorphic at* $\varepsilon_k = 0$ *unless* $k = \mathrm{last}(U)$*; for one polar wall with numerator* $f_C$ *the iterated constant term of* $f_C/\ell_U$ *is* $\pm\partial_{\varepsilon_a} f_C(0)$*,* $a = \mathrm{last}(U)$*.*

*(iii) For a fixed chart, the quasi-integral of a regular schedule depends on the order only through* $U \mapsto \mathrm{last}(U)$ *on the polar walls.*

With one regulator $\varepsilon_D$ for each class $D$ of a realization, taken in some order of the classes, $\ell_U = \pm \sum_D |U \cap D|\, \varepsilon_D$; a wall is closed by the last class it meets, which contributes $|U \cap D|^{-1} \partial_{\varepsilon_D}$, and the value depends only on the order of the classes, which can be taken from the order of classes (Theorem 13.7).

**Proposition 16.24** (The coordinates set the log-scale)**.** *In the coordinates $u=\alpha/\beta$, $\beta_g=1$, with the regulators $\varepsilon_k$ entering as $u_{\varsigma(k)}^{\varepsilon_k}$, the regulated function is $e^{-\sum_k \varepsilon_k\lambda_k}I(\varepsilon)$, $\lambda_k=\log\beta_{\varsigma(k)}$, and for one polar wall with numerator $f_C$ and last member a the iterated constant term becomes $\pm\big(\partial_{\varepsilon_a}f_C(0)-\lambda_a f_C(0)\big)$: changing coordinates shifts the finite part by $-\log\beta_{\varsigma(a)}$ times the signed residue of each wall closed by a mark whose gauge differs from the chart's.*

**Theorem 16.25** (Residue at a wall)**.** *Let $\omega$ be projective, $S$ a proper bunch with room $R$, $g\in R$ and $\alpha_S=\rho u$. In this statement write $K_j^{[l]}=K_{j,S}^{(d_j(S)+l)}(u,\alpha_R)$ and $\Psi_j^{[l]}=\Psi_{j,S}^{(d_j^*(S)+l)}(u,\alpha_R)$ for the level and colevel letters of Definition 12.16 counted from their floors, so that $K_j(\rho u,\alpha_R)=\rho^{d_j(S)}\sum_{l\ge0}\rho^l K_j^{[l]}$ and $\Psi_j(\rho u,\alpha_R)=\rho^{d_j^*(S)}\sum_{l\ge0}\rho^l\Psi_j^{[l]}$. At points of $\kappa_S=-q$, $q\in\mathbb{N}$, where every other proper wall is positive,*

$$\operatorname*{Res}_{\kappa_S=-q}P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\alpha_g=1}\int_{\Delta_S}u^{\mathbf{m}|_S-\mathbf{1}}\alpha_R^{\mathbf{m}|_R-\mathbf{1}}\prod_j\big(K_j^{[0]}\big)^{-s_j}\big(\Psi_j^{[0]}\big)^{-t_j}\,[\rho^q]h\;d\omega_S\,d\alpha_{R-g},$$

*where*

$$h=\prod_j\Big(\sum_l\rho^lK_j^{[l]}/K_j^{[0]}\Big)^{-s_j}\Big(\sum_l\rho^l\Psi_j^{[l]}/\Psi_j^{[0]}\Big)^{-t_j},$$

$$[\rho^q]\Big(1+\sum_{l\ge1}\rho^lX_l\Big)^{-s}=\sum_{\sum_l ln_l=q}(-1)^{|\mathbf{n}|}\frac{\Gamma(s+|\mathbf{n}|)}{\Gamma(s)}\prod_l\frac{X_l^{n_l}}{n_l!}.$$

*If the weights factor on the bottom levels, then with the* faces *of the wall, the families*

$$F^S=\big((\mathcal{B}_j/R;\,s_j,0),\;(\mathcal{B}_j|S;\,0,t_j)\big)_j\ \text{on } S,\qquad F^R=\big((\mathcal{B}_j|R;\,s_j,0),\;(\mathcal{B}_j/S;\,0,t_j)\big)_j\ \text{on } R$$

*(Definition 7.5), the residue at $\kappa_S=0$ is the product of their periods at the masses $\mathbf{m}|_S$ and $\mathbf{m}|_R$; both faces are balanced, the walls of $F^S$ are the $\kappa_U$, $U\subseteq S$, and those of $F^R$ are the $\kappa_{S\cup U}$, $U\subseteq R$, on the wall. Here $P_F$ is the meromorphic function of Lemma 16.9; on a connected family with real data and $s_j,t_j\ge0$, not all zero, it is the normaliser of the distribution of Theorem 16.17(i).*

**Proposition 16.26** (A molecule at a nonpositive wall)**.** *Let a molecule carry real data with $s_j,t_j\ge0$ and target $\boldsymbol{\tau}$, and let $S$ be a proper bunch with room $R$, $\alpha_S=\rho u$ on a chart $\alpha_g=1$, $g\in R$, with $H(\rho)=H(\rho;u,\alpha_R)$ the native completion $W\,d\alpha_{\boldsymbol{\tau}}$ divided by $\rho^{\kappa_S(\boldsymbol{\tau})-1}d\rho\,d\omega_S\,d\alpha_{R-g}$ and $H_l=[\rho^l]H$ as in Theorem 16.25.*

*(i) Along the collapse of $S$, $\Phi(y_0-\lambda\mathbf{1}_S)-\langle\boldsymbol{\tau},y_0-\lambda\mathbf{1}_S\rangle=\lambda\,\kappa_S(\boldsymbol{\tau})+O(1)$ as $\lambda\to\infty$, the left side being the negative logarithm of the density in $y=\log\alpha$ (Theorem 8.19); so the density does not decay along the collapse exactly when $\kappa_S(\boldsymbol{\tau})\le0$, and then the form is not integrable at $\rho=0$.*

*(ii) With $\langle\cdot\rangle$ the integral over $u$ and $\alpha_R$, if every other proper wall at $\boldsymbol{\tau}$ is positive, $\kappa_S(\boldsymbol{\tau})$ is not a nonpositive integer, and $L\in\mathbb{N}$ has $\kappa_S(\boldsymbol{\tau})+L>0$,*

$$P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})=\sum_{l<L}\frac{\langle H_l\rangle}{\kappa_S(\boldsymbol{\tau})+l}+\int_0^1\rho^{\kappa_S(\boldsymbol{\tau})-1}\Big(\langle H\rangle-\sum_{l<L}\rho^l\langle H_l\rangle\Big)d\rho+\int_1^\infty\rho^{\kappa_S(\boldsymbol{\tau})-1}\langle H\rangle\,d\rho,$$

*the continuation of the period past the wall of $S$; its poles at $\kappa_S=-q$ have the residues $\langle H_q\rangle$ of Theorem 16.25. When the weights factor on the bottom levels, $\langle H_0\rangle$ is the product of the integrals of the faces $F^S$ at $\boldsymbol{\tau}|_S$ and $F^R$ at $\boldsymbol{\tau}|_R$, whose imbalances are $\kappa_S(\boldsymbol{\tau})$ and $-\kappa_S(\boldsymbol{\tau})$: the leading term is the compound of Proposition 16.12.*

**Corollary 16.27** (The vertex)**.** *At $S = \mathcal{E} - e$, with $e$ neither a loop nor a coloop in any table, the face $F^R$ is the single mark $e$ and contributes* 1*, and*

$$\langle H_0 \rangle = \int_{\Delta_{\mathcal{E}-e}} u^{\boldsymbol{\tau}|_{\mathcal{E}-e}-\mathbf{1}} \prod_j K_{j,/e}(u)^{-s_j} \Psi_{j,\backslash e}(u)^{-t_j} \, d\omega_{\mathcal{E}-e}(u),$$

*the integral of the* vertex child*: the contractions at $e$ on the basis side and the deletions on the cobasis side. Proposition 16.26(ii) at $S = \mathcal{E} - e$ removes the leading terms of the vertex from the period exactly.*

## 16.4 The pseudointegral

**Definition 16.28** (Period datum; moves)**.** Let a family of tables with positive rational weights have rational exponents and masses and lie on the balance wall. Its *period datum* is the positive orthant of $\mathbb{R}^{\mathcal{E}}$, read projectively, with the form $W \, d\alpha_{\mathbf{m}} = \alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j} d\alpha$, algebraic over $\mathbb{Q}$ and homogeneous of degree zero, the hypersurface $\{\prod_j K_j \Psi_j = 0\}$ and the coordinate hyperplanes; the face $\{\alpha_S = 0\}$ carries the wall $\kappa_S$ (Theorem 8.5). A *move* between period data is a change of variables by a birational map defined over $\mathbb{Q}$ carrying the open domain bijectively onto the open domain, faces to faces and form to form; a product of data; a linear combination; or the boundary term of an exact form.

**Definition 16.29** (The pseudointegral over a region)**.** Let $F$ be a family with real data, $s_j, t_j \geq 0$ not all zero, masses $\mathbf{m} > 0$ and weights factoring across its molecules, with form $\omega = W \, d\alpha_{\mathbf{m}}$.

(i) A *region* of $\omega$ is a domain $\mathcal{R}$ in coordinates $x$ with a change of variables mapping it bijectively onto a section $\{h = 1\}$ of the orthant, $h > 0$ homogeneous of degree one: a chart $\alpha_g = 1$, the simplex, a bunch section $|\alpha_B| = 1$. The *density* of $\omega$ on $\mathcal{R}$ is the function $f$ for which $f(x)\,dx$ is the form that $\omega$ induces on the section (Lemma 16.2). A region of the damped form $e^{-|\alpha|}\omega$ is the same with the orthant in place of the section. For a family of tables with positive rational weights and rational data on the balance wall the regions reached from these by moves (Definition 16.28) are domains cut out by polynomial inequalities over $\mathbb{Q}$.

(ii) The *pseudointegral* of $\omega$ over a region is

$$\int\!\!\cdots\!\!\int_{\mathcal{R}} f(x)\,dx := \int W \, d\alpha_{\mathbf{m}}$$

(Definition 16.10(iii)), and over a region of the damped form, away from the poles of $J_F$, it is the continued interest, $J_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int W \, e^{-|\alpha|} d\alpha_{\mathbf{m}}$ (Definition 8.8), written $\int \alpha^{\mathbf{m}-\mathbf{1}} e^{-|\alpha|} \prod_j K_j^{-s_j} \Psi_j^{-t_j} d\alpha$ when the integrand is displayed. The value belongs to $\omega$: the native completion and the ray of Definition 16.10(iii) are taken in the table's coordinates $\alpha$, and a presentation (Theorem 16.13) declares them anew and is another form; the region and its coordinates say how the integration is presented, and a density determines $\omega$ only together with the family, a chart $\alpha_g = 1$ not seeing the mass $m_g$ and the simplex not seeing a power of $|\alpha|$, the letter of the list $\binom{\mathcal{E}}{1}$ (Proposition 16.3). Divided by the period of the centre, $P_{\mathrm{c}}$ on a molecule, it is the pseudoexpectation $\mathsf{E}_{U \sim \mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$.

(iii) The pseudointegral of $\omega$ is read in the Foreman's order (Definition 16.14). The tree is drawn from the tables at the centre of the table's gauge; on each molecule the measure $d\alpha_{\mathbf{m}}$ is set to $d\alpha_{\boldsymbol{\tau}}$; the bunches are integrated in the Foreman's order, each at once in scale and shape; where the target lies outside a wall the value is the continuation in the masses, and on a

pole hyperplane its constant term along the ray of the masses. The damped form takes no completion, and the Foreman computes it on the ray beyond the target (Theorem 16.15).

**Theorem 16.30** (The pseudointegral and the native completion)**.** *Let $\omega = W\,d\alpha_{\mathbf{m}}$ have real data on a connected family, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, with target $\boldsymbol{\tau}$.*

*(i) With a native gauge $\alpha^\star$ the pseudointegral is read at the realization $\Upsilon = ((s_j,t_j)_j; \alpha^\star)$, whose masses are $\boldsymbol{\tau}$ and whose proper walls are positive. The form $W\,d\alpha_{\boldsymbol{\tau}}$ is positive with convergent period, and every iterated integral of it over every section, in any coordinates, in the Foreman's order and in every other, converges absolutely to*

$$\Big\backslash W\,d\alpha_{\mathbf{m}} = \int W\,d\alpha_{\boldsymbol{\tau}} = P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t}) :$$

*no finite part is taken, and the pseudoexpectation is the expectation of a positive function, $\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)] = \mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$.*

*(ii) If $\boldsymbol{\tau}$ lies on no pole hyperplane, every regular schedule of the native completion returns the pseudointegral,*

$$\mathrm{Reg}^{\mathfrak{s}}\int W\,d\alpha_{\boldsymbol{\tau}} = \Big\backslash W\,d\alpha_{\mathbf{m}} = P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t}),$$

*each iterated integral a finite part (Definition 1.5), and the value does not depend on the order; its quasi-expectation is the pseudoexpectation, $\mathfrak{E}^{\mathfrak{s}}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)] = \mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$.*

*(iii) On a pole hyperplane $\Big\backslash W\,d\alpha_{\mathbf{m}} = \mathrm{CT}_{\eta=0}\,P_F(\boldsymbol{\tau}+\eta\mathbf{m};\mathbf{s},\mathbf{t})$ exists and depends on $\mathbf{m}$ only through its ray. It takes no chart and no order: its regulator is $\eta$, along the ray of the masses, and its section is the simplex that the damping singles out, $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = J_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma(\kappa_{\mathcal{E}})$ (Theorem 8.7).*

**Proposition 16.31** (A pole target in other coordinates)**.** *In Theorem 16.30(iii) let $x = \lambda^{-1}\alpha$ be the coordinates of a presentation (Theorem 16.13), $W^\lambda(x) = W(\lambda x)$, and*

$$G_k(\eta) = \int_{\Delta_{\mathcal{E}}} \big(\log|\lambda u|\big)^k\, u^{\boldsymbol{\tau}+\eta\mathbf{m}-\mathbf{1}} W^\lambda(u)\,d\omega_{\mathcal{E}}(u) \qquad (k\in\mathbb{N}),$$

*continued in $\eta$, so that $G_0(\eta)$ is the integral over the simplex of the presented word $W^\lambda$ at the masses $\boldsymbol{\tau}+\eta\mathbf{m}$. Each $G_k$ has at $\eta = 0$ a pole of order at most $n-1$, and the presentation has pseudointegral*

$$(\lambda^{\mathbf{m}})^{\kappa_{\mathcal{E}}/m_{\mathcal{E}}}\Big(\Big\backslash W\,d\alpha_{\mathbf{m}} - \lambda^{\boldsymbol{\tau}}\sum_{1\leq i+k\leq n-1}\frac{\langle\mathbf{m},\log\lambda\rangle^i}{i!}\,\frac{(-m_{\mathcal{E}})^k}{k!}\,[\eta^{-i-k}]\,G_k\Big),$$

*$[\eta^{-i-k}]$ the Laurent coefficient at $\eta = 0$. Off the pole hyperplanes every $G_k$ is holomorphic at $0$ and this is Theorem 16.13; on them the finite part is taken in the coordinates of the table, as every finite part is taken in its own coordinate (Lemma 1.9).*

**Proposition 16.32** (Sections off the balance wall)**.** *Let $\omega = W\,d\alpha_{\mathbf{m}}$ have real data, $s_j, t_j \geq 0$ not all zero, $\mathbf{m} > 0$ and imbalance $\kappa_{\mathcal{E}} \neq 0$.*

*(i) The integral of $\omega$ over a section $\{h = 1\}$ is the integral over every section of the projective form $h^{-\kappa_{\mathcal{E}}}\omega$, wherever it converges. Over a chart it is the period at the chart completion (Proposition 16.5(i)), over a bunch section $|\alpha_B| = 1$ the period of the family with the list $\binom{B}{1}$, the marks outside $B$ as loops, appended at the exponents $(\kappa_{\mathcal{E}}, 0)$, a multinomial sum of*

*completions when $-\kappa_{\mathcal{E}} \in \mathbb{N}$ (Proposition 16.5(iii)), and over the simplex, where the proper walls at $\mathbf{m}$ are positive, $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} \mathbb{E}_{U\sim \mathrm{Dir}_{\mathbf{m}}}[W(U)]$ (Proposition 8.12(i)). Different sections integrate different projective forms; the pseudointegral integrates the native completion over all of them.*

*(ii) The walls of the chart completion at $g$ and of the native completion are*

$$\kappa_S(\mathbf{m}^{(g)}) = \kappa_S(\mathbf{m}) - \kappa_{\mathcal{E}}\,\mathbf{1}_S(g), \qquad \kappa_S(\boldsymbol{\tau}) = \kappa_S(\mathbf{m}) - \frac{\kappa_{\mathcal{E}}}{m_{\mathcal{E}}}\, m_S :$$

*the snip moves the walls through $g$ by the whole imbalance and leaves the others, and the native completion moves each wall by its share. For $\kappa_{\mathcal{E}} > 0$ a proper wall through $g$ is lower at the chart completion than at the target by $\kappa_{\mathcal{E}}(1 - m_S/m_{\mathcal{E}})$, and one avoiding $g$ is higher by $\kappa_{\mathcal{E}} m_S/m_{\mathcal{E}}$. So a snip can put the walls through $g$ on pole hyperplanes, or past them, while the target is inside every wall.*

**Theorem 16.33** (Integrals over sections)**.** *Let $\omega = W\, d\alpha_{\mathbf{m}}$ have real data, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, and for $h > 0$ homogeneous of degree one let $\int_{h=1}\omega$ denote the integral over the section $\{h = 1\}$ with the measure of Lemma 16.2.*

*(i) By Proposition 16.32(i) the integral of $\omega$ over a section $\{h = 1\}$ is the integral over the rays of $h^{-\kappa_{\mathcal{E}}}\omega$, and for $\kappa_{\mathcal{E}} \neq 0$, $h \mapsto h^{-\kappa_{\mathcal{E}}}$ carries the sections bijectively onto the positive multipliers of degree $-\kappa_{\mathcal{E}}$. The monomial sections $\{\alpha^w = 1\}$, $w_{\mathcal{E}} = 1$, are the completions, $(\alpha^w)^{-\kappa_{\mathcal{E}}}\omega = W\, d\alpha_{\mathbf{m}-\kappa_{\mathcal{E}} w}$: they act on the measure and keep the word.*

*(ii) The native completion is the section $\{\alpha^{\mathbf{m}} = 1\}$, the hyperplane $\langle \mathbf{m}, y\rangle = 0$ in $y = \log\alpha$, where the density $e^{\langle \mathbf{m}, y\rangle}$ of $d\alpha_{\mathbf{m}}$ against $dy$ is 1; it is the one monomial section with this property. With a native gauge on a connected family of at least two marks, for every mark $g$,*

$$\int\!\!\!\!\!\!- W\, d\alpha_{\mathbf{m}} = \frac{m_{\mathcal{E}}}{m_g}\int_{\mathbb{R}^{\mathcal{E}-g}} W(e^y)\, dy_{\mathcal{E}-g}, \qquad y_g = -\frac{1}{m_g}\sum_{e\neq g} m_e y_e,$$

*and $\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$ is this integral over the integral of $\omega_{\mathrm{c}}$ over the same section, which is $P_{\mathrm{c}}$.*

*(iii) The simplex $\{|\alpha| = 1\}$ is not a monomial section for $n \geq 2$: its multiplier is $|\alpha|^{-\kappa_{\mathcal{E}}}$, which keeps the measure and multiplies the word by a power of the letter $K_0 = |\alpha|$ of the list $\binom{\mathcal{E}}{1}$. The integral over it is the integral of the projective form $K_0^{-\kappa_{\mathcal{E}}} W\, d\alpha_{\mathbf{m}}$, the family with that table added at the exponents $(\kappa_{\mathcal{E}}, 0)$ (Proposition 16.3, Proposition 16.5(iii) at $B = \mathcal{E}$), and wherever every proper wall at $\mathbf{m}$ is positive it is the period $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ of Proposition 8.12(i), the law $\mathrm{Dir}_{\mathbf{m}}$ there being the law of $\binom{\mathcal{E}}{1}$ at a full basis lean.*

*(iv) On the balance wall every section reads $\int W\, d\alpha_{\mathbf{m}}$ (Theorem 8.7).*

**Proposition 16.34** (The centre sets the law)**.** *Let $\omega = W\, d\alpha_{\mathbf{m}}$ have real data on a connected family of $n \geq 2$ marks, $s_j, t_j \geq 0$ not all zero and $\mathbf{m} > 0$, with target $\boldsymbol{\tau}$, centre $\omega_{\mathrm{c}}$ and law of the centre $\mathbb{P}_{\mathrm{c}}$ (Definition 16.10). For laws $\mathbb{P}, \mathbb{P}'$ with $\mathbb{P}$ absolutely continuous with respect to $\mathbb{P}'$ write $\mathrm{KL}(\mathbb{P}\,\|\,\mathbb{P}') = \mathbb{E}_{U\sim\mathbb{P}}\big[\log\frac{d\mathbb{P}}{d\mathbb{P}'}(U)\big]$, the Kullback–Leibler divergence.*

*(i) The native completion is the centre times the departure, $\omega_{\boldsymbol{\tau}} = \frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}\,\omega_{\mathrm{c}}$, and with a native gauge the Foreman's law, the shape law $\mathrm{Breta}^{\circ}(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ of the native completion, is the law of the centre reweighted by the departure:*

$$\mathrm{Breta}^{\circ}(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t})(du) = \frac{(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(u)}{\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]}\,\mathbb{P}_{\mathrm{c}}(du).$$

*(ii) With a native gauge,*

$$\log \mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\Big[\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(U)\Big] = \mathrm{KL}\big(\mathbb{P}_{\mathrm{c}} \,\big\|\, \mathrm{Breta}^{\circ}(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t})\big) \;\geq\; 0,$$

*with equality exactly when* $\mathrm{Breta}^{\circ}(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t}) = \mathbb{P}_{\mathrm{c}}$. *So every pseudointegral with a native gauge is at least the period of its centre,* $\int W\, d\alpha_{\mathbf{m}} \geq P_{\mathrm{c}}$, *and among the forms of given strengths* $\sigma_j$ *with a native gauge the centre has the least value.*

*(iii) The pseudoexpectation is kept by the fold: the two forms of Theorem 16.18, at the masses* $\mathbf{m}$ *with the exponents* $(\mathbf{s},\mathbf{t})$ *and at the masses* $\lambda(2\sigma\mathbf{1}-\boldsymbol{\tau})$ *with* $(\mathbf{t},\mathbf{s})$, *have one centre and one pseudoexpectation, and the same holds for continued values off the pole hyperplanes where* $2\sigma\mathbf{1}-\boldsymbol{\tau} > 0$.

*(iv) Let* $\omega$ *be the form of a realization at the table's gauge,* $\alpha^{\star} = \mathbf{1}$ *(Definition 9.1). In signed coordinates on* $\mathbf{1}^{\perp}$ *the departure is*

$$\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(e^{2z}) = \exp\Big(-\sum_j \delta_j \tilde{O}_j(z)\Big), \qquad \tilde{O}_j(z) = O_j(z) - \langle \nabla O_j(0), z\rangle,$$

*each* $\tilde{O}_j$ *odd with value, gradient and Hessian* $0$ *at* $0$. *The Foreman's law and the law of the centre have their mode at the table's gauge and the same Hessian of the logarithm of the density there, and the logarithm of their ratio is, up to a constant,* $-\sum_j \delta_j \tilde{O}_j$*: minus the odd part of the centred potential less its linear term, whose Taylor coefficients at the table's gauge are minus the odd rungs of order at least three of the potential, the rungs that vanish at the centre (Proposition 8.21(iii)).*

**Proposition 16.35** (The list $\binom{\mathcal{E}}{1}$ as a normaliser)**.** *Let* $\omega = W\, d\alpha_{\mathbf{m}}$ *have real data on a connected family of* $n \geq 2$ *marks,* $s_j, t_j \geq 0$ *not all zero and* $\mathbf{m} > 0$, *so that* $W$ *is homogeneous of degree* $-\tau_{\mathcal{E}}$*; let* $K_0 = |\alpha|$ *be the letter of the list* $\binom{\mathcal{E}}{1}$, *whose bases are the singletons.*

*(i) Every completion* $W\, d\alpha_{\mathbf{m}'}$, $\mathbf{m}' = \mathbf{m} - \kappa_{\mathcal{E}} w$, *has total mass* $m'_{\mathcal{E}} = \tau_{\mathcal{E}}$. *The native completion is the one whose masses keep the proportions of* $\mathbf{m}$, $\boldsymbol{\tau}/\tau_{\mathcal{E}} = \mathbf{m}/m_{\mathcal{E}}$, *and they are positive, while the chart completion at* $g$ *has the mass* $m_g - \kappa_{\mathcal{E}}$ *at* $g$.

*(ii) For* $\mathbf{m}' > 0$ *with* $m'_{\mathcal{E}} = \tau_{\mathcal{E}}$ *the form* $K_0^{-\tau_{\mathcal{E}}} d\alpha_{\mathbf{m}'}$ *is projective, with integral* $\Gamma_1^n \begin{bmatrix} \mathbf{m}' \\ \tau_{\mathcal{E}} \end{bmatrix}$ *over every section: the period of* $\binom{\mathcal{E}}{1}$ *at the full basis lean* $(\tau_{\mathcal{E}}, 0)$, *whose law is* $\mathrm{Dir}_{\mathbf{m}'}$ *(Proposition 16.3). So* $\Gamma_1^n \begin{bmatrix} \boldsymbol{\tau} \\ \tau_{\mathcal{E}} \end{bmatrix} = \int K_0^{-\tau_{\mathcal{E}}} d\alpha_{\mathbf{m}}$*; for a schedule with* $\mathbf{m}^{(g)} > 0$, $\Gamma_1^n \begin{bmatrix} \mathbf{m}^{(g)} \\ \tau_{\mathcal{E}} \end{bmatrix} = \mathrm{Reg}^{\mathfrak{s}} \int K_0^{-\tau_{\mathcal{E}}} d\alpha_{\mathbf{m}}$, *a convergent integral; and with a native gauge*

$$\int W\, d\alpha_{\mathbf{m}} \Big/ \int K_0^{-\tau_{\mathcal{E}}} d\alpha_{\mathbf{m}} = \mathbb{E}_{U\sim\mathrm{Dir}_{\boldsymbol{\tau}}}[W(U)].$$

*(iii) With a native gauge, and for continued values off the pole hyperplanes where* $2\sigma\mathbf{1}-\boldsymbol{\tau} > 0$, *the fold keeps the pseudointegral and carries the target* $\boldsymbol{\tau}$ *to* $2\sigma\mathbf{1}-\boldsymbol{\tau}$ *(Theorem 16.18), so the quotients of (ii) for* $\omega$ *and for its image are* $P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ *divided by* $\Gamma_1^n \begin{bmatrix} \boldsymbol{\tau} \\ \tau_{\mathcal{E}} \end{bmatrix}$ *and by* $\Gamma_1^n \begin{bmatrix} 2\sigma\mathbf{1}-\boldsymbol{\tau} \\ \tau^*_{\mathcal{E}} \end{bmatrix}$, $\tau^*_{\mathcal{E}} = 2n\sigma - \tau_{\mathcal{E}}$. *The second quotient over the first is* $\Gamma_1^n \begin{bmatrix} \boldsymbol{\tau} \\ \tau_{\mathcal{E}} \end{bmatrix} / \Gamma_1^n \begin{bmatrix} 2\sigma\mathbf{1}-\boldsymbol{\tau} \\ \tau^*_{\mathcal{E}} \end{bmatrix}$, *which is not* $1$ *in general. The period of* $\binom{\mathcal{E}}{1}$ *is that of a full basis lean of a table outside the family, and it prices a lean and its mirror differently; the centre, which the fold fixes, prices them alike (Proposition 16.34(iii)).*

*(iv) Along the ray of the masses, wherever the proper walls at $\lambda\mathbf{m}$ are positive, the integral of the projective form $K_0^{-(\lambda m_{\mathcal{E}}-\tau_{\mathcal{E}})}W\,d\alpha_{\lambda\mathbf{m}}$ is $\Gamma_1^n\begin{bmatrix}\lambda\mathbf{m}\\ \lambda m_{\mathcal{E}}\end{bmatrix}\,\mathbb{E}_{U\sim\mathrm{Dir}_{\lambda\mathbf{m}}}[W(U)]$ (Theorem 16.33(iii), Proposition 8.12(i)). The power of $K_0$ vanishes exactly at $\lambda=\tau_{\mathcal{E}}/m_{\mathcal{E}}$, where the form is the native completion $W\,d\alpha_{\boldsymbol{\tau}}$.*

**Corollary 16.36** (The pseudointegral is the quasi-integral that always exists)**.** *For real data with $s_j,t_j\geq 0$ not all zero, $\mathbf{m}>0$ and weights factoring across the molecules, the pseudointegral $\int W\,d\alpha_{\mathbf{m}}$ and the pseudoexpectation $\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$ are defined. On a molecule whose target lies on no pole hyperplane the pseudointegral is the quasi-integral of the native completion and the pseudoexpectation its quasi-expectation, the same along every regular schedule; on a pole hyperplane, where the regular schedules of the native completion can depend on the order (Theorem 16.23), it is the one constant term along the ray of the masses. On each molecule every generalized expectation divides by the one period of its centre, a positive convergent integral (Proposition 8.21); when the molecule has a native gauge, the native completion is the one completion that keeps it (Theorem 16.8(i)). The quasi-integrals of $\omega$ itself are Laurent data at its chart completions (Theorem 16.22), which off the balance wall can depend on the chart and on the order.*

The pseudointegral names a procedure as well as a value. Written for a family whose realization is specified later, $\int\cdots\int_{\mathcal{R}}$ commits to the Foreman's order at every realization: the tree is drawn once from the tables, and the realization $((s_j,t_j)_j;\alpha^\star)$ of the point fixes the measure $d\alpha_{\boldsymbol{\tau}}$ and where the contours of the ports run. An iterated integral in another order, or after a snip, computes a quasi-integral and not the pseudointegral, even where the numbers agree; when they agree, that is Theorem 16.30(i)–(ii), not the order that was followed.

On a pole hyperplane the regulator $\eta$ of the pseudointegral enters as $u^{\eta\mathbf{m}}$, as in Theorem 1.7. Past a wall the pseudointegral normalizes the distribution of Theorem 16.17(i).

# Chapter 17

# The family as one Laplace transform

On its balance wall a family is one Laplace transform. In the pair coordinates of its data and the signed coordinates of the shape, its period is the Laplace transform, over the tilts, of the even and odd statistics against the offsets (Definition 17.1; for the interest, Theorem 8.2); the Breta shape laws at the points of the balance wall form an exponential family in these coordinates, and every ratio of periods is a moment generating function of the statistics. On the sector of every order the exponent of a question is minus twice its wall at the tilt, the walls of the tails weighted by the differences of the tilt along the order, less the logarithms of the sector letters, which are bounded. The Laplace transform is read through that one reading of the walls: where the exponent is largest and how it falls, how the period behaves at low and at high strength, how far a question can lean, and how fast the tails of the statistics fall.

## 17.1 Questions and the law of the tilt

**Definition 17.1** (Tilts; questions)**.** The space of *tilts* is $\mathcal{T} = \mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$, with the Lebesgue measure $dz$ of a chart $z_g = 0$, which does not depend on $g$. The *tilt* of a shape $u \in \Delta_{\mathcal{E}}$ is the image in $\mathcal{T}$ of its signed coordinates $\frac{1}{2}\log u$, and a tilt is read at its representative in $\mathbf{1}^{\perp}$ when an odd statistic is applied to it, since $O_j(z + c\mathbf{1}) = O_j(z) + 2c(2r_j - n)$ (Theorem 7.15(iii)). A *question* to a family $F$ is a point of its balance wall with real data, written in its pair coordinates $(\boldsymbol{\mu}_{\perp}; \boldsymbol{\sigma}, \boldsymbol{\delta})$: the data $(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \big(\mathbf{m}_F(\boldsymbol{\mu}_{\perp}; \boldsymbol{\sigma}, \boldsymbol{\delta});\ \boldsymbol{\sigma} + \boldsymbol{\delta},\ \boldsymbol{\sigma} - \boldsymbol{\delta}\big)$ (Definition 8.1). In the coordinates $z = \frac{1}{2}\log u$ of Theorem 8.7,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = 2^{n-1} \int_{\mathcal{T}} \exp\Big(2\langle \boldsymbol{\mu}_{\perp}, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)\Big)\, dz.$$

When every proper wall is positive, the tilt $Z$ of $U \sim \mathrm{Breta}^{\circ}(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$ has the density

$$2^{n-1} \exp\Big(2\langle \boldsymbol{\mu}_{\perp}, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)\Big) \Big/ P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$$

with respect to $dz$: this is the *law of the question*. A direction $v = (\dot{\mathbf{m}}; \dot{\mathbf{s}}, \dot{\mathbf{t}})$ of the data is *balanced* when $\dot{m}_{\mathcal{E}} = \sum_j (\dot{s}_j r_j + \dot{t}_j r_j^*)$; its pair coordinates $(c; \mathbf{a}, \mathbf{b})$ have $\dot{\mathbf{m}} = \mathbf{m}_F(c; \mathbf{a}, \mathbf{b})$, $\dot{\mathbf{s}} = \mathbf{a} + \mathbf{b}$ and $\dot{\mathbf{t}} = \mathbf{a} - \mathbf{b}$, and $D_v$ is the derivative along $v$. The derivatives $\partial_{\sigma_j}$, $\partial_{\delta_j}$ and $\nabla_{\boldsymbol{\mu}_{\perp}}$ of a function of the question are those along the balanced directions with the pair coordinates $(0; \mathbf{e}_j, \mathbf{0})$, $(0; \mathbf{0}, \mathbf{e}_j)$ and $(\mathbf{e}_e - \frac{1}{n}\mathbf{1}; \mathbf{0}, \mathbf{0})$, that is $(\mathbf{1}; \mathbf{e}_j, \mathbf{e}_j)$, $\big(\frac{2r_j - n}{n}\mathbf{1}; \mathbf{e}_j, -\mathbf{e}_j\big)$ and $\big(\mathbf{e}_e - \frac{1}{n}\mathbf{1}; \mathbf{0}, \mathbf{0}\big)$. The centre of a question is the question $(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$ with the same strengths, the data of the family at the centre

(Definition 9.1), of pair coordinates $(0;\boldsymbol{\sigma},\mathbf{0})$; on a connected family its law is the law of the centre $\mathbb{P}_{\mathrm{c}}$ of Definition 16.10 read on the tilts.

**Corollary 17.2** (The Laplace transform of the tilt and the statistics)**.** *Let a question* $(\mathbf{m};\mathbf{s},\mathbf{t})$ *have every proper wall positive. In this statement* $\mathbb{E}$ *is the expectation of* $U\sim\mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ *and* $Z$ *is the tilt of* $U$*. For* $c\in\mathbf{1}^{\perp}\otimes\mathbb{C}$ *and* $\mathbf{a},\mathbf{b}\in\mathbb{C}^N$ *such that the real parts of the shifted data* $\big(\mathbf{m}+\mathbf{m}_F(c;\mathbf{a},\mathbf{b});\,\mathbf{s}+\mathbf{a}+\mathbf{b},\,\mathbf{t}+\mathbf{a}-\mathbf{b}\big)$ *have every proper wall positive,*

$$\mathbb{E}\Big[e^{2\langle c,Z\rangle-\sum_j(a_jE_j(Z)+b_jO_j(Z))}\Big]=\frac{P_F\big(\mathbf{m}+\mathbf{m}_F(c;\mathbf{a},\mathbf{b});\,\mathbf{s}+\mathbf{a}+\mathbf{b},\,\mathbf{t}+\mathbf{a}-\mathbf{b}\big)}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})},$$

*analytic in* $(c,\mathbf{a},\mathbf{b})$ *there: the shift moves the pair coordinates of the question by* $(c;\mathbf{a},\mathbf{b})$*. So the joint cumulants of order* $k$ *of* $(E_j(Z),O_j(Z))_j$ *are* $(-1)^k$ *times the derivatives of order* $k$ *of* $\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *in* $(\sigma_j,\delta_j)_j$*, and their characteristic function is the same ratio at* $c=0$ *and purely imaginary* $(\mathbf{a},\mathbf{b})$*: the family on the fibre of imaginary exponents over the question.*

**Definition 17.3** (The wall at a tilt)**.** For an order $\pi$ of $\mathcal{E}$ with tails $U_k=\{\pi(k),\dots,\pi(n)\}$, the sector $H_\pi$ of Lemma 8.3 is read in the tilts as $\{z\in\mathcal{T}:\ z_{\pi(1)}\geq\cdots\geq z_{\pi(n)}\}$. On it the measure $dz$ of Definition 17.1 is the Lebesgue measure in the differences $z_{\pi(k-1)}-z_{\pi(k)}\geq 0$, $2\leq k\leq n$, which at the gauge $\alpha=e^{2z}$ are half the gaps $\log(\alpha_{\pi(k-1)}/\alpha_{\pi(k)})$ of Corollary 8.6. For data on the balance wall the *wall at a tilt* $z\in H_\pi$ is

$$\kappa(z)=\sum_{k=2}^{n}\kappa_{U_k}\big(z_{\pi(k-1)}-z_{\pi(k)}\big),$$

the walls of the tails of $\pi$ weighted by the differences of $z$ along $\pi$, half the walls weighted by the gaps. It does not depend on the order whose sector holds $z$, the tails with a positive difference being the sets $\{e:\ z_e<z_f\}$ for the $f\in\mathcal{E}$ with $z_f>\min_e z_e$; it is positively homogeneous of degree one and linear on every sector, and $\kappa(-\mathbf{1}_S)=\kappa_S$ for every nonempty proper $S$, with $-\mathbf{1}_S$ read in $\mathcal{T}$, the direction along which $S$ collapses (Proposition 16.26(i)). In pair coordinates $\kappa(z)$ is linear in the question.

**Proposition 17.4** (The exponent of a question reads its walls; the even statistic is least at the table's gauge)**.** *Let a family carry a question* $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$*, its exponents of any signs, and in this statement write* $X(z)=2\langle\boldsymbol{\mu}_\perp,z\rangle-\sum_j\big(\sigma_jE_j(z)+\delta_jO_j(z)\big)$ *for the exponent of the law of the question (Definition 17.1) and* $\kappa(z)$ *for its wall at the tilt* $z$ *(Definition 17.3).*

*(i) On the sector* $H_\pi$ *of every order* $\pi$*, with the sector letters* $k_{j,\pi},\psi_{j,\pi}$ *of Theorem 8.5(ii) read at the sector coordinates* $q_k=\alpha_{\pi(k)}/\alpha_{\pi(k-1)}=e^{-2(z_{\pi(k-1)}-z_{\pi(k)})}$ *of* $\alpha=e^{2z}$*,*

$$X(z)=-2\kappa(z)-\sum_j\big(s_j\log k_{j,\pi}(q)+t_j\log\psi_{j,\pi}(q)\big).$$

*With* $h_j(z)=\max_{T\in\mathcal{B}_j}\langle\chi_T,z\rangle$*, the support function of the Newton polytope* $2P_j-\mathbf{1}$ *of* $\Lambda_j$ *(Theorem 7.18),* $2\kappa(z)=\sum_j\big(s_jh_j(z)+t_jh_j(-z)\big)-2\langle\boldsymbol{\mu}_\perp,z\rangle$ *on* $\mathbf{1}^{\perp}$*, and for* $s_j,t_j\geq 0$ *the wall at a tilt is the support function of* $\mathcal{N}_F-\mathbf{m}$*,* $\mathcal{N}_F$ *the Newton sum of Theorem 8.19(ii). The sector letters are polynomials with nonnegative coefficients, with the constant terms* $\theta_{j,T^+_{j,\pi}}$ *and* $\theta_{j,T^-_{j,\pi}}$ *and the value* $K_j(\mathbf{1})$ *at* $q=\mathbf{1}$ *(Theorem 8.5(ii)), so on the cube they lie in* $[\theta_{j,T^+_{j,\pi}},K_j(\mathbf{1})]$ *and* $[\theta_{j,T^-_{j,\pi}},K_j(\mathbf{1})]$*; with* $\theta_{j,\min}=\min_T\theta_{j,T}$*,* $|X+2\kappa|\leq\sum_j(|s_j|+|t_j|)\max\big(|\log\theta_{j,\min}|,|\log K_j(\mathbf{1})|\big)$*, and for* $s_j,t_j\geq 0$

$$2\kappa(z)+\sum_j(s_j+t_j)\log\theta_{j,\min}\leq-X(z)\leq 2\kappa(z)+\sum_j(s_j+t_j)\log K_j(\mathbf{1}).$$

*(ii)* $\kappa > 0$ *on* $\mathcal{T} \setminus 0$ *exactly when every proper wall is positive, and then* $\kappa(z) \geq \kappa_{\min}\big(\max_e z_e - \min_e z_e\big)$, $\kappa_{\min}$ *the least proper wall.*

*(iii) At the centre of strengths* $\boldsymbol{\sigma}$, $\sigma_j \geq 0$, *where* $\boldsymbol{\mu}_\perp = 0$ *and* $\boldsymbol{\delta} = 0$, *the exponent is minus the centred potential* $\sum_j \sigma_j E_j$ *(Proposition 8.21) and* $2\kappa(z) = \sum_j \sigma_j \operatorname{wd}_j(z)$, *with* $\operatorname{wd}_j(z) = h_j(z) + h_j(-z)$ *the width of* $2P_j - \mathbf{1}$ *in the direction* $z$, *twice the width of* $P_j$, *so that* $\operatorname{wd}_j(\mathbf{1}_S) = 2c_j(S)$. *Every* $E_j$ *is convex and even,* $E_j(z) \geq E_j(0) = 2\log K_j(\mathbf{1})$, *and*

$$\sum_j \sigma_j \operatorname{wd}_j(z) + 2\sum_j \sigma_j \log\theta_{j,\min} \leq \sum_j \sigma_j E_j(z) \leq \sum_j \sigma_j \operatorname{wd}_j(z) + 2\sum_j \sigma_j \log K_j(\mathbf{1}).$$

*If the family is connected,* $\sum_j \sigma_j E_j$ *is strictly convex on* $\mathcal{T}$ *and* $\sum_j \sigma_j \operatorname{wd}_j > 0$ *on* $\mathcal{T} \setminus 0$.

**Theorem 17.5** (The two ends of strength)**.** *Let a question* $q = (\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$ *of a family have every proper wall positive, and for* $\lambda \in \mathbb{C}$ *with* $\operatorname{Re}\lambda > 0$ *read the data* $\lambda q$ *of pair coordinates* $(\lambda\boldsymbol{\mu}_\perp; \lambda\boldsymbol{\sigma}, \lambda\boldsymbol{\delta})$, *a question when* $\lambda$ *is real, whose walls* $\lambda\kappa_S$ *have positive real parts; in this statement* $P_F(\lambda q)$ *is its period,* $2^{n-1}\int_{\mathcal{T}} e^{\lambda X(z)}dz$ *by the change of variables of Definition 17.1 with the powers of the positive letters principal (Theorem 8.7), and* $X$ *and* $\kappa$ *are those of* $q$ *(Proposition 17.4).*

*(i)* Low strength. *As* $\lambda \downarrow 0$,

$$\lambda^{n-1}P_F(\lambda q) \to \sum_\pi \prod_{k=2}^{n} \kappa_{U_k}^{-1} = (n-1)!\,\operatorname{vol}\{z \in \mathcal{T} :\ \kappa(z) \leq 1\},$$

*and*

$$\lambda^{n-1}P_F(\lambda q) = \sum_\pi \prod_{k=2}^{n} \kappa_{U_k}^{-1}\Big(1 - \lambda\sum_j \big(s_j \log\theta_{j,T^+_{j,\pi}} + t_j \log\theta_{j,T^-_{j,\pi}}\big)\Big) + o(\lambda),$$

*with the greedy bases* $T^\pm_{j,\pi}$ *of Lemma 8.3: the walls set the period at low strength, and the weights enter at the next order, through the greedy bases.*

*(ii)* High strength, in the half-plane. *Let* $s_j, t_j \geq 0$. *Then* $-X$ *is strictly convex on* $\mathcal{T}$, *its unique critical point* $z^\star$ *is the native gauge of every question* $\lambda q$, $\lambda > 0$, *and as* $|\lambda| \to \infty$ *with* $\operatorname{Re}\lambda \geq \lambda_0$ *for a fixed* $\lambda_0 > 0$, *in particular as* $\lambda \to +\infty$ *and along* $\lambda = \lambda_0 + i\eta$, $|\eta| \to \infty$,

$$P_F(\lambda q) = 2^{n-1}e^{\lambda X(z^\star)}\Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}} \det\big(-\nabla^2 X(z^\star)\big)^{-\frac{1}{2}}\big(1 + O(|\lambda|^{-1})\big),$$

*the power principal and the Hessian taken on a chart* $z_g = 0$, *where it is rung two of the still of the realization* $\big((s_j, t_j)_j; e^{2z^\star}\big)$ *of the active tables (Definition 9.4) with the row and the column of* $g$ *deleted. At the centre of strengths* $\boldsymbol{\sigma}$, $z^\star = 0$, $X(0) = -2\sum_j \sigma_j \log K_j(\mathbf{1})$, *and on the chart* $z_g = 0$, $-\nabla^2 X(0) = 8\sum_j \sigma_j \big(\kappa_{j,\mathbf{1}}(X_e, X_f)\big)_{e,f\neq g}$.

*Remark* 17.6 (The volume of the walls is the volume of a polar body). At the centre of strengths $\boldsymbol{\sigma}$, $\sigma_j \geq 0$, Proposition 17.4(iii) gives $2\kappa(z) = \sum_j \sigma_j \operatorname{wd}_j(z)$, and $\operatorname{wd}_j(z) = h_j(z) + h_j(-z)$, with $h_j$ the support function of $2P_j - \mathbf{1}$ and $P_j$ the base polytope of table $j$ (Theorem 7.18), is twice the support function of $P_j - P_j$ at $z$. So $\kappa$ is the support function of the centrally symmetric polytope $\mathcal{Q} = \sum_j \sigma_j(P_j - P_j)$, a Minkowski sum in $\mathbf{1}^\perp$, read on $\mathcal{T} = \mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$, and $\{z \in \mathcal{T} :\ \kappa(z) \leq 1\}$ is its polar body $\mathcal{Q}^\circ = \{z :\ \langle x, z\rangle \leq 1 \text{ for every } x \in \mathcal{Q}\}$, bounded when every proper wall is positive (Proposition 17.4(ii)). By Theorem 17.5(i), $\sum_\pi \prod_{k=2}^n \kappa_{U_k}^{-1} = (n-1)!\operatorname{vol}(\mathcal{Q}^\circ)$: the volume $V_1$ of the direction of the centre ray at the strengths $\boldsymbol{\sigma}$ (Proposition 39.34), whose walls are those of this question, is $(n-1)!$ times the volume of the polar body of $\sum_j \sigma_j(P_j - P_j)$.

At low strength the walls alone set the period, through the volume of the region where the wall at a tilt is at most one, and the weights enter at the next order through the greedy bases. At high strength the period concentrates at the native gauge, where it is a Gaussian integral whose Hessian is rung two of the still.

**Proposition 17.7** (How far a question can lean; the values of the statistics)**.** *For every family and every nonempty proper $S$ put $\varepsilon_j^{\perp}(S) = \varepsilon_j(S) - |S|\frac{2r_j - n}{n}$, so that $\varepsilon_j^{\perp}(\mathcal{E} \setminus S) = -\varepsilon_j^{\perp}(S)$, and for a question $(\boldsymbol{\mu}_{\perp}; \boldsymbol{\sigma}, \boldsymbol{\delta})$ let*

$$\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma}) = \sum_j \sigma_j c_j(S), \qquad \kappa_S^{\mathbf{O}}(\boldsymbol{\mu}_{\perp}, \boldsymbol{\delta}) = \mu_{\perp,S} - \sum_j \delta_j \varepsilon_j^{\perp}(S),$$

*the even and the odd part of the wall at $S$: on the balance wall $\kappa_S = \kappa_S^{\mathbf{E}} + \kappa_S^{\mathbf{O}}$ and $\kappa_{\mathcal{E}\setminus S} = \kappa_S^{\mathbf{E}} - \kappa_S^{\mathbf{O}}$, and the fold, which negates the odd part (Theorem 16.18), swaps the two. So a question is inside its walls exactly when $|\kappa_S^{\mathbf{O}}(\boldsymbol{\mu}_{\perp}, \boldsymbol{\delta})| < \kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})$ for every $S$. When the centre of strengths $\boldsymbol{\sigma}$ is inside its walls, the* slope *of an odd direction $(c, \mathbf{b}) \in \mathbf{1}^{\perp} \times \mathbb{R}^N$ at $\boldsymbol{\sigma}$ is*

$$\ell_{\boldsymbol{\sigma}}^*(c, \mathbf{b}) = \min\Big\{ \frac{\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})}{|\kappa_S^{\mathbf{O}}(c, \mathbf{b})|} : \ \kappa_S^{\mathbf{O}}(c, \mathbf{b}) \neq 0 \Big\},$$

*$\ell_{\boldsymbol{\sigma}}^* = \infty$ if every $\kappa_S^{\mathbf{O}}(c, \mathbf{b})$ vanishes; it is homogeneous of degree one in $\boldsymbol{\sigma}$, and even and positively homogeneous of degree $-1$ in $(c, \mathbf{b})$. The questions of strengths $\boldsymbol{\sigma}$ inside their walls are those whose odd part has $\ell_{\boldsymbol{\sigma}}^*(\boldsymbol{\mu}_{\perp}, \boldsymbol{\delta}) > 1$, an open convex polyhedron symmetric under the fold, bounded exactly when no nonzero odd direction has $\kappa_S^{\mathbf{O}}(c, \mathbf{b}) = 0$ for every $S$. For one connected table at offsets zero they form the wedge $\sigma > 0$, $|\delta| < \ell_1^*(0,1)\,\sigma$, with $\ell_1^*(0,1) = \min\{c(S)/|\varepsilon^{\perp}(S)| : \ \varepsilon^{\perp}(S) \neq 0\}$.*

*(i) For the tilt $Z$ of $U \sim \mathrm{Breta}^{\circ}(F; \sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$, the centre of strengths $\boldsymbol{\sigma}$, and real $c \in \mathbf{1}^{\perp}$, $\mathbf{a}, \mathbf{b} \in \mathbb{R}^N$, the expectation of $e^{2\langle c, Z\rangle - \sum_j (a_j E_j(Z) + b_j O_j(Z))}$ is finite exactly when every proper wall of the question $(c; \boldsymbol{\sigma} + \mathbf{a}, \mathbf{b})$ is positive, $|\kappa_S^{\mathbf{O}}(c, \mathbf{b})| < \kappa_S^{\mathbf{E}}(\boldsymbol{\sigma} + \mathbf{a})$ for every $S$, and it is then the ratio of the periods of that question and of the centre (Corollary 17.2); in particular $\mathbb{E}_{U \sim \mathrm{Breta}^{\circ}(F; \sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})}\big[e^{2\langle c, Z\rangle - \sum_j b_j O_j(Z)}\big]$ is finite exactly when $\ell_{\boldsymbol{\sigma}}^*(c, \mathbf{b}) > 1$.*

*(ii) For a question inside its walls with $|\delta_j| \leq \sigma_j$ the native gauge $z^{\star}$ minimizes $-2\langle \boldsymbol{\mu}_{\perp}, z\rangle + \sum_j (\sigma_j E_j + \delta_j O_j)$ on $\mathbf{1}^{\perp}$, so $\big(z^{\star}; E_j(z^{\star}), O_j(z^{\star})\big)_j$ is the point of the set of values $\{(z; E_j(z), O_j(z))_j : \ z \in \mathbf{1}^{\perp}\}$ of the statistics at which $-2\langle \boldsymbol{\mu}_{\perp}, x\rangle + \sum_j (\sigma_j y_j + \delta_j w_j)$ is least; at the centre it is $\big(0; 2\log K_j(\mathbf{1}), 0\big)_j$.*

**Proposition 17.8** (The tails of the statistics are read at the walls)**.** *Let a question $q$ of a family have every proper wall positive, let $Z$ be the tilt of $U \sim \mathrm{Breta}^{\circ}(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$ at its data, and for a real balanced direction $v$ of pair coordinates $(c; \mathbf{a}, \mathbf{b})$ (Definition 17.1) let $Y = 2\langle c, Z\rangle - \sum_j \big(a_j E_j(Z) + b_j O_j(Z)\big)$. On the balance wall the walls are linear in the pair coordinates, $\kappa_S(q + \lambda v) = \kappa_S(q) + \lambda \kappa_S(v)$ with $\kappa_S(v) = \kappa_S^{\mathbf{E}}(\mathbf{a}) + \kappa_S^{\mathbf{O}}(c, \mathbf{b})$ (Proposition 17.7). Let*

$$\lambda^* = \min\Big\{ \frac{\kappa_S(q)}{-\kappa_S(v)} : \ \kappa_S(v) < 0 \Big\},$$

*the value of $\lambda$ at which the ray $q + \lambda v$, $\lambda > 0$, first meets a wall, with $\lambda^* = \infty$ if no $\kappa_S(v)$ is negative. For real $\lambda \geq 0$, $\mathbb{E}[e^{\lambda Y}]$ is finite exactly for $\lambda < \lambda^*$, where it is the ratio of the periods of the questions $q + \lambda v$ and $q$ (Corollary 17.2); if $\lambda^* < \infty$,*

$$\mathbb{P}(Y > x) = e^{-\lambda^* x + o(x)} \qquad (x \to \infty),$$

*and if $\lambda^* = \infty$, $Y$ is bounded above. Under the law of the centre of strengths $\boldsymbol{\sigma}$ an odd direction, of pair coordinates $(c; \mathbf{0}, \mathbf{b})$, has $\lambda^* = \ell_{\boldsymbol{\sigma}}^*(c, \mathbf{b})$ for $Y$ and for $-Y$ alike.*

The period of a question is a Laplace transform in the tilts, so the shape laws form an exponential family whose natural parameters are the offsets and the pair coordinates of the exponents and whose statistics are the tilt and the even and odd statistics $E_j$, $O_j$. On the sector of each order the exponent of a question is minus twice its wall at the tilt, the walls of the tails weighted by the differences of the tilt along the order, up to the bounded sector letters. So the walls alone fix how the period blows up as the strength goes to zero, the weights entering only at the next order and only through the greedy bases; at high strength, real or complex with its real part bounded below, the period is read at the native gauge of the ray. At a centre the exponent is minus the centred potential $\sum_j \sigma_j E_j$, which is convex, smallest at the table's gauge and within a constant of the widths of the Newton polytopes. A question can lean only as far as its strengths allow: the odd part of every wall is bounded by its even part, the slope of its odd part exceeds one, and the same slope bounds the exponential moments of the odd statistics at the centre; along every direction the tail of a statistic falls at the rate of the first wall the direction meets. Integrating over the simplex is integrating against the list $\binom{\mathcal{E}}{1}$ at a full basis lean, so every Dirichlet expectation of letters is a ratio of periods (Proposition 16.3).

# Chapter 18

# The Breta law and the port on the balance wall

## 18.1 The laws

**Theorem 18.1** (The pseudointegral is a normalizer)**.** *Let $F$ be connected on $n \geq 2$ marks with real data, $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, a native gauge $\alpha^\star$, target $\boldsymbol{\tau}$ and form $\omega = W\, d\alpha_{\mathbf{m}}$.*

*(i) $\int W\, d\alpha_{\mathbf{m}} = P_F(\boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ is the normalizer of* $\mathrm{Breta}^\circ(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$*, and the pseudoexpectation $\mathsf{E}_{U \sim \mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)] = \int W\, d\alpha_{\mathbf{m}}/P_{\mathrm{c}}$ is the normalizer of its density against the law of the centre (Proposition 16.34(i)). With $\phi_{\mathbf{m}'}$ the joint law of $\big(K_1(U), \dots, K_N(U), \Psi_1(U), \dots, \Psi_N(U)\big)$ for $U \sim \mathrm{Dir}_{\mathbf{m}'}$, both periods are stacked Gammas of masses over their total times Trailmoments (Proposition 8.12(i)):*

$$\mathsf{E}_{U \sim \mathbb{P}_{\mathrm{c}}}\big[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)\big] = \frac{\Gamma_1^n \begin{bmatrix} \boldsymbol{\tau} \\ \tau_{\mathcal{E}} \end{bmatrix} \big(\mathsf{T}_{(\mathbf{s},\mathbf{t});I_{2N}}[\mathrm{id}]\, \phi_{\boldsymbol{\tau}}\big)\big((0,\infty)^{2N}\big)}{\Gamma_1^n \begin{bmatrix} \sigma\mathbf{1} \\ n\sigma \end{bmatrix} \big(\mathsf{T}_{(\boldsymbol{\sigma},\boldsymbol{\sigma});I_{2N}}[\mathrm{id}]\, \phi_{\sigma\mathbf{1}}\big)\big((0,\infty)^{2N}\big)}.$$

*(ii) In the coordinates $y = \log u$ modulo $\mathbb{R}\mathbf{1}$ the law* $\mathrm{Breta}^\circ(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ *has density proportional to $e^{\langle \boldsymbol{\tau}, y\rangle - \Phi(y)}$, whose unique mode is $\log \alpha^\star$.*

*(iii) (Cremona) If $U \sim \mathrm{Breta}^\circ(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$ on the balance wall, then $U^{-1}/|U^{-1}| \sim \mathrm{Breta}^\circ(F; 2\sigma\mathbf{1} - \mathbf{m}; \mathbf{t}, \mathbf{s})$.*

*(iv) (The even–odd pair) On the balance wall the inversion of (iii) carries* $\mathrm{Breta}^\circ(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$ *at the data $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$ to the law at $(-\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, -\boldsymbol{\delta})$; at $(0; \boldsymbol{\sigma}, \mathbf{0})$ it fixes the law, and with $Z$ the signed coordinate of $U$ the odd statistics $(O_j(Z))_j$ have a law symmetric under $O \mapsto -O$. The joint Laplace transform of the even and odd statistics is Corollary 17.2.*

**Proposition 18.2** (The mark relation of the Breta shape law)**.** *Let $F$ have real data, masses $\mathbf{m}$ on the balance wall and every proper wall positive, and in this statement write $\mathbf{u}(\alpha) = \sum_j \big(s_j R_j(\alpha) + t_j R_j^*(\alpha)\big)$ for the usage at $\alpha$ and $\mathbb{E}$ for the expectation of $U \sim \mathrm{Breta}^\circ(F; \mathbf{m}; \mathbf{s}, \mathbf{t})$. For $Y = \log U$ modulo $\mathbb{R}\mathbf{1}$, every mark $e$ and every observable $g$, a smooth function of degree $0$ in $\alpha$ such that $g$ and $\partial_{y_e} g$ grow at most exponentially at a rate below the walls,*

$$\mathbb{E}\big[(m_e - \mathbf{u}_e(U))\, g(U) + \partial_{y_e} g(U)\big] = 0.$$

*At $g=1$ the masses are the mean usage, $\mathbb{E}[\mathbf{u}(U)]=\mathbf{m}$; with a native gauge $\alpha^\star$ the usage equals the target $\boldsymbol{\tau}=\mathbf{m}$ at the mode $\log\alpha^\star$ (Theorem 18.1(ii)).*

**Proposition 18.3** (Scale and shapes along a bunch)**.** *For a proper bunch $B$ with room $R$, masses $\mathbf{m}>0$ and every proper wall of positive real part,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\Gamma_{1}^{|B|}\begin{bmatrix}\mathbf{m}|_B\\ m_B\end{bmatrix}\Gamma_{1}^{|R|}\begin{bmatrix}\mathbf{m}|_R\\ m_R\end{bmatrix}$$
$$\times\,\mathbb{E}_{(U'_B,U'_R)\sim \mathrm{Dir}_{\mathbf{m}|_B}\otimes \mathrm{Dir}_{\mathbf{m}|_R}}\Big[\int_0^\infty x^{m_B-1}(1+x)^{-\kappa_{\mathcal{E}}}\prod_j K_j(xU'_B,U'_R)^{-s_j}\Psi_j(xU'_B,U'_R)^{-t_j}\,dx\Big],$$

*the inner integral being the scale integral of the family with the list $\binom{\mathcal{E}}{1}$ appended at the exponents $(\kappa_{\mathcal{E}},0)$, whose letter at $(xU'_B,U'_R)$ is $1+x$ (Proposition 16.3); on the balance wall it is $\mathrm{Sc}_B(U'_B,U'_R;\mathbf{m};\mathbf{s},\mathbf{t})$ (Theorem 15.3), and the display is Corollary 16.6 at $h=|\alpha_R|$. For real data with every proper wall positive and $U\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$, with the internal shapes $V_B=U_B/|U_B|$ and $V_R=U_R/|U_R|$ and the totals $|U_B|+|U_R|=1$ (Theorem 3.12), the scale $\rho=|U_B|/|U_R|$ of $B$ given the internal shapes has density proportional to $\rho^{m_B-1}(1+\rho)^{-\kappa_{\mathcal{E}}}\prod_j K_j(\rho V_B,V_R)^{-s_j}\Psi_j(\rho V_B,V_R)^{-t_j}$.*

**Theorem 18.4** (Structure of the Breta law)**.** *Let $X\sim\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$; in this statement $\mathbb{E}$ is the expectation of $X$.*

*(i)* $|X|\sim\mathrm{Gamma}_{\kappa_{\mathcal{E}}}$ *and* $X/|X|\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$*, independently.*

*(ii) The laws form an exponential family with respect to $e^{-|\alpha|}d\alpha/\alpha^{\mathbf{1}}$, with statistics $(\log X_e)_e$ and $(-\log K_j(X),-\log\Psi_j(X))_j$, natural parameters $(\mathbf{m},\mathbf{s},\mathbf{t})$ and log-partition $\log J_F(\mathbf{m};\mathbf{s},\mathbf{t})$: $\mathbb{E}\log X_e=\partial_{m_e}\log J_F(\mathbf{m};\mathbf{s},\mathbf{t})$, $\mathbb{E}\log(K_j\Psi_j)(X)=-D_{(\mathbf{0};\mathbf{e}_j,\mathbf{e}_j)}\log J_F(\mathbf{m};\mathbf{s},\mathbf{t})$, $\mathbb{E}\log(K_j/\Psi_j)(X)=-D_{(\mathbf{0};\mathbf{e}_j,-\mathbf{e}_j)}\log J_F(\mathbf{m};\mathbf{s},\mathbf{t})$, $D_v$ the derivative along the direction $v$ of the data, the masses fixed, and the covariances are second derivatives.*

*(iii)* $\mathbb{E}[X^{\mathbf{k}}]=J_F(\mathbf{m}+\mathbf{k};\mathbf{s},\mathbf{t})/J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *and* $\mathbb{E}\big[\prod_j K_j(X)^{-a_j}\Psi_j(X)^{-b_j}\big]=J_F(\mathbf{m};\mathbf{s}+\mathbf{a},\mathbf{t}+\mathbf{b})/J_F(\mathbf{m};\mathbf{s},\mathbf{t})$.

**Theorem 18.5** (The race representation)**.** *Let $X\sim\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$. Given $X=\alpha$, run for each table $j$ a* basis race*, the basis $T$ ringing with intensity $\theta_{j,T}\alpha^T$, and a* cobasis race*, with intensity $\theta_{j,T}\alpha^{\mathcal{E}\setminus T}$, all clocks independent; let $Y_j$ and $Y_j^*$ be the first ringing times and $T_j$, $T'_j$ the winners. In this statement $\mathbb{E}$ and $\mathbb{P}$ are taken over the clocks of the races at $\alpha$.*

*(i) In each race the time and the winner are independent: $K_j(\alpha)Y_j\sim\mathrm{Gamma}_1$ and $T_j\sim\mathbb{P}_{j,\alpha}$; $\Psi_j(\alpha)Y_j^*\sim\mathrm{Gamma}_1$ and $T'_j\sim\mathbb{P}^*_{j,\alpha}$.*

*(ii)* $K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}=\mathbb{E}\big[Y_j^{s_j}(Y_j^*)^{t_j}\big]/\big(\Gamma(1+s_j)\Gamma(1+t_j)\big)$.

*(iii) With $\mathrm{g}_j=-\log(K_jY_j)$ and $\mathrm{g}'_j=-\log(\Psi_jY_j^*)$, independent, $-\log(Y_jY_j^*)=\log(K_j\Psi_j)+\mathrm{g}_j+\mathrm{g}'_j$ and $-\log(Y_j/Y_j^*)=\log(K_j/\Psi_j)+\mathrm{g}_j-\mathrm{g}'_j$, and, for real $a,b$ with $1+a\pm b>0$, $\mathbb{E}\big[e^{-a(\mathrm{g}_j+\mathrm{g}'_j)-b(\mathrm{g}_j-\mathrm{g}'_j)}\big]=\Gamma(1+a+b)\,\Gamma(1+a-b)$.*

*(iv)* $\mathbb{P}(Y_j<Y_j^*)=K_j/(K_j+\Psi_j)=1/(1+e^{-O_j(z)})$, $\alpha=e^{2z}$.

*(v) $D=\chi_{T_j}-\chi_{T'_j}$ has $\mathbb{E}[e^{\langle w,D\rangle}]=e^{E_j(z+w)-E_j(z)}$: the even statistic $E_j$ is the cumulant function of the difference of the two winners (by (i), Theorem 7.16 for table $j$).*

The two exponents of a table are the two clocks of its races: its even letter reads the sum $\mathrm{g}_j+\mathrm{g}'_j$ of the two logarithmic clocks of (iii) and its odd letter their difference, so that the pair coordinates $(\sigma_j,\delta_j)$ are the exponents of $\mathrm{g}_j+\mathrm{g}'_j$ and $\mathrm{g}_j-\mathrm{g}'_j$.

**Theorem 18.6** (Conjugacy)**.** *Let* $X \sim \mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$.

*(i) Given the winner* $T$ *of the* $j$*-th basis race, its time unobserved,* $X \sim \mathrm{Breta}(F;\mathbf{m}+\mathbf{1}_T;\mathbf{s}+\mathbf{e}_j,\mathbf{t})$*; given the winner* $T'$ *of the* $j$*-th cobasis race,* $X \sim \mathrm{Breta}(F;\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T'};\mathbf{s},\mathbf{t}+\mathbf{e}_j)$*. Both keep* $\kappa_{\mathcal{E}}$*.*

*(ii) Under the joint law of* $X$ *and the races of Theorem 18.5,* $\mathbb{P}(T_j = T) = \theta_{j,T} P_F(\mathbf{m}+\mathbf{1}_T;\mathbf{s}+\mathbf{e}_j,\mathbf{t})/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *and* $\mathbb{P}(T'_j = T') = \theta_{j,T'} P_F(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T'};\mathbf{s},\mathbf{t}+\mathbf{e}_j)/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*, each summing to one.*

*(iii) Given the time* $Y_j = \lambda$ *as well,* $X$ *has the density of* $\mathrm{Breta}(F;\mathbf{m}+\mathbf{1}_T;\mathbf{s},\mathbf{t})$ *multiplied by* $e^{-\lambda K_j(\alpha)}$ *and renormalized: the kernel terms* $\lambda\theta_{j,T}\alpha^T$ *join those of* $e^{-|\alpha|}$*, and the law exists only on the orthant.*

**Corollary 18.7** (Conjugacy for the continued period)**.** *For every family and every table* $j$ *of it, with its bases* $\mathcal{B}_j$ *and weights* $\theta_j$*,*

$$\sum_{T\in\mathcal{B}_j} \theta_{j,T}\, P_F(\mathbf{m}+\mathbf{1}_T;\mathbf{s}+\mathbf{e}_j,\mathbf{t}) = P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_{T'\in\mathcal{B}_j} \theta_{j,T'}\, P_F(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T'};\mathbf{s},\mathbf{t}+\mathbf{e}_j)$$

*as identities of meromorphic functions of the masses and the exponents (Lemma 16.9), and so on the balance wall: the identities of Theorem 18.6(ii) hold for the continued values of Theorem 16.17.*

**Theorem 18.8** (The winners of a tensorminental table)**.** *If table* $j$ *is tensorminental of tensor rank* $q$ *with slices* $A^{(k)}$*, then for a bunch* $B$ *with room* $R$ *and at a gauge as in Theorem 7.23, the pattern* $T_j \cap B$ *of the basis winner has the generating function* $\sum_{k,l} w_{kl} \det\big(I+(D_x-I)\Pi^{(kl)}_{BB}\big)$ *of that theorem, and the cobasis winner's is the same with the slices of the dual multivector. Conditioning on the winner's pattern leaves tables with multivectors of tensor rank at most* $q$ *on the room.*

**Proposition 18.9** (The period along a ray of data)**.** *Let* $F$ *be connected, let* $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ *be real data on its balance wall with* $\boldsymbol{\tau} > 0$*,* $s_j, t_j \geq 0$ *not all zero, and a native gauge* $\alpha^\star$*, let* $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ *be complex data on the balance wall, and in this statement write* $W_0 = \prod_j K_j^{-s_{0,j}} \Psi_j^{-t_{0,j}}$*. As* $\lambda \to +\infty$ *the data* $(\mathbf{m}_0+\lambda\boldsymbol{\tau};\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})$ *have every proper wall of positive real part, and, with* $\Phi$ *the potential at* $(\mathbf{s},\mathbf{t})$ *and a chart* $g$*,*

$$P_F(\mathbf{m}_0+\lambda\boldsymbol{\tau};\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t}) \sim (\alpha^\star)^{\mathbf{m}_0} W_0(\alpha^\star)\Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}} \det\big(\nabla^2\Phi(\log\alpha^\star)\big)^{-1/2}_{(g)}\, e^{-\lambda(\Phi(\log\alpha^\star)-\langle\boldsymbol{\tau},\log\alpha^\star\rangle)},$$

*(*$g$*) deleting the row and the column of* $g$*, where* $\nabla^2\Phi$ *is* $\frac{1}{4}$ *times rung two of the still and* $(\alpha^\star)^{\mathbf{m}_0}W_0(\alpha^\star)$ *does not depend on the scale of* $\alpha^\star$*. At* $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0) = 0$ *the display is the leading term of Theorem 17.5(ii) at real* $\lambda$ *for the question of the data* $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$*. For real* $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$*, under* $\mathrm{Breta}^\circ(F;\mathbf{m}_0+\lambda\boldsymbol{\tau};\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})$*,* $\sqrt{\lambda}(Y-\log\alpha^\star)$ *tends to the centred normal law on* $\mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$ *with covariance* $(\nabla^2\Phi)^{-1}$*,* $Y = \log U$*.*

A base point added to the data moves the ray off the origin: along $(\mathbf{m}_0+\lambda\boldsymbol{\tau};\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})$, with complex $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ on the balance wall, the leading term of Theorem 17.5(ii) gains the factor $(\alpha^\star)^{\mathbf{m}_0}\prod_j K_j(\alpha^\star)^{-s_{0,j}}\Psi_j(\alpha^\star)^{-t_{0,j}}$ at the native gauge $\alpha^\star = e^{2z^\star}$.

**Definition 18.10** (Skeleton law)**.** For coefficients $\xi_\ell \in \mathbb{C}\setminus(-\infty,0]$, $\ell \in J = \{0,\dots,L\}$, closed under conjugation, real nonzero letter exponents $w_\ell$, equal on conjugate coefficients, with total

$|\mathbf{w}|$, and $0<\kappa<|\mathbf{w}|$, the *Skeleton law* $\mathrm{Skel}_{\mathbf{w},\kappa}(\boldsymbol{\xi})$ is the law on $(0,\infty)$ with density

$$y^{\kappa-1}\prod_{\ell}(y+\xi_\ell)^{-w_\ell}\Big/\Gamma_{\,|J|}^{\,2}\begin{bmatrix}\kappa,\ |\mathbf{w}|-\kappa\\ \mathbf{w}\end{bmatrix}\ \mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}(\boldsymbol{\xi}),$$

its normalizer being Proposition 15.1; the density is positive, and integrable exactly when $0<\kappa<|\mathbf{w}|$.

**Theorem 18.11** (The Skeleton law)**.** *(i) Under* $\mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})$*, the scale of a proper bunch* $B$ *given the internal shapes of* $B$ *and of its room has the law* $\mathrm{Skel}_{\mathbf{w},\kappa_B}(\boldsymbol{\xi})$*:* $\boldsymbol{\xi}$ *the joint spectrum of* $B$ *at that point and* $\mathbf{w}$ *its letter exponents of Theorem 15.3, and off the balance wall the coefficient* 1 *besides, with the letter exponent* $\kappa_{\mathcal{E}}$*, added to the letter exponent of* 1 *when* 1 *is a point of the joint spectrum; coefficients of letter exponent zero are discarded.*

*(ii) For* $Y\sim\mathrm{Skel}_{\mathbf{w},\kappa}(\boldsymbol{\xi})$ *and* $-\kappa<\operatorname{Re}k<|\mathbf{w}|-\kappa$*,*

$$\mathbb{E}_{Y\sim\mathrm{Skel}_{\mathbf{w},\kappa}(\boldsymbol{\xi})}[Y^k]=\Gamma_{\,2}^{\,2}\begin{bmatrix}\kappa+k,\ |\mathbf{w}|-\kappa-k\\ \kappa,\ |\mathbf{w}|-\kappa\end{bmatrix}\frac{\mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa-k;\,I_J}(\boldsymbol{\xi})}{\mathsf{B}^{\{J\}}_{\mathbf{w},\,|\mathbf{w}|-\kappa;\,I_J}(\boldsymbol{\xi})},$$

*the quotient of the values of Proposition 15.1 at the indices* $\kappa+k$ *and* $\kappa$*, read as that quotient when a letter exponent is not positive; and* $1/Y\sim\mathrm{Skel}_{\mathbf{w},|\mathbf{w}|-\kappa}(\boldsymbol{\xi}^{-1})$*.*

*(iii) With real coefficients, positive letter exponents and a point* $r\geq\max_\ell\xi_\ell$ *of the axis,* $Y$ *is the mixture over* $\mathbf{n}\in\mathbb{N}^J$ *of* $rY_{\mathbf{n}}$*,* $Y_{\mathbf{n}}\sim\mathrm{Skel}_{|\mathbf{w}|+|\mathbf{n}|,\kappa}(1)$ *the law with the one coefficient* 1*, mixed in proportion to* $\prod_{\ell\in J}\frac{(w_\ell)_{n_\ell}}{n_\ell!}g_\ell^{n_\ell}\Gamma_{\,1}^{\,2}\begin{bmatrix}\kappa,\ |\mathbf{w}|+|\mathbf{n}|-\kappa\\ |\mathbf{w}|+|\mathbf{n}|\end{bmatrix}$*,* $g_\ell=1-\xi_\ell/r$ *the spectral gaps at* $r$ *of Theorem 15.6; at* $r=\max_\ell\xi_\ell$ *the largest coefficient drops out.*

*(iv) In the situation of (i) on the balance wall, with* $s_j,t_j\geq0$*: given* $Y\leq1$*,* $Y$ *is the mixture of the first coordinates of the laws* $\mathrm{Dir}_{(\kappa_B+k,|\mathbf{n}|+1)}$*, and given* $Y\geq1$*,* $1/Y$ *is the mixture of the first coordinates of* $\mathrm{Dir}_{(\kappa_B^{\mathrm{top}}+k,|\mathbf{n}|+1)}$*, mixed in proportion to the terms of Theorem 15.9(ii); the two sectors carry probabilities proportional to the sector Trailmoments of that theorem at the indices* $\kappa_B$ *and* $\kappa_B^{\mathrm{top}}$*.*

## 18.2 The port on the balance wall

**Definition 18.12** (Wall contour; joint period)**.** On the balance wall, with $\operatorname{Re}s_j>0$ or $s_j=0$, and $\operatorname{Re}t_j>0$ or $t_j=0$, for every $j$, a *wall contour* at a bunch $B$ puts the shares of Theorem 15.15 of the letters with nonzero exponent on real parts $c_{j,p},c'_{j,p}>0$ with $\sum_p c_{j,p}=\operatorname{Re}s_j$, $\sum_p c'_{j,p}=\operatorname{Re}t_j$ and real odd load equal to the real part of the offset on the bunch, $\sum_{j,p}\frac{c_{j,p}-c'_{j,p}}{2}\chi_p(e)=\operatorname{Re}\mu_e$; equivalently the real part of every mass of the bunch is carried exactly, $\operatorname{Re}\mathrm{load}_e=\operatorname{Re}m_e$. The *fibre integral at* $B$ is the fibre integral (Definition 4.1) with the pins $|\mathbf{a}_j|=s_j$, $|\mathbf{b}_j|=t_j$ and $\langle\boldsymbol{\delta}\rangle_e=\mu_e$, $e\in B$. For a family $G=((\mathcal{B}_i,\theta_i);a_i,b_i)_i$ whose molecules are $R_1,\dots,R_q$, the *joint period* $P_G^{\otimes}(\mathbf{m};\mathbf{a},\mathbf{b})$ is the integral of $\alpha^{\mathbf{m}-\mathbf{1}}\prod_i K_i(\alpha)^{-a_i}\Psi_i(\alpha)^{-b_i}$ over $\alpha\in\prod_{k=1}^q\Delta_{R_k}$ against $\prod_k d\omega_{R_k}$, wherever it converges; when the weights factor across the molecules it is the product of the molecules' periods.

**Theorem 18.13** (The port on the balance wall)**.** *Let* $F$ *have data on the balance wall with* $\operatorname{Re}s_j>0$ *or* $s_j=0$*, and* $\operatorname{Re}t_j>0$ *or* $t_j=0$*, for every* $j$*, let* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *converge, and let* $B$ *be a proper bunch with room* $R$*; put* $z=\frac12\log\alpha$*.*

*(i) The* bunch factor $\Xi_B(\alpha_R;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t}) = \int_{\mathbb{R}^B_{>0}} \alpha_B^{\boldsymbol{\mu}|_B-\mathbf{1}} \prod_j \Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j}\, d\alpha_B$ *converges for every* $\alpha_R$, $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\Delta_R} v^{\boldsymbol{\mu}|_R-\mathbf{1}}\, \Xi_B(v;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t})\, d\omega_R(v)$, *and on every wall contour*

$$\Xi_B(\alpha_R;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j,\ |\mathbf{b}_j|=t_j,\ \langle\boldsymbol{\delta}\rangle=\boldsymbol{\mu}|_B} \prod_j \Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix} \times \prod_{j,p}\Lambda_{j,(p)}(z_R)^{-a_{j,p}}\Lambda_{j,(p)}(-z_R)^{-b_{j,p}}.$$

*(ii) Let* $R_1,\dots,R_q$ *be the molecules of the children and* $\kappa_{R_i}(\mathbf{a},\mathbf{b})$ *the wall at* $R_i$ *of the children* $F_B(\mathbf{a},\mathbf{b})$ *with the masses* $\mathbf{m}|_R$ *(Definition 8.1); as* $R_i$ *is a molecule,* $\kappa_{R_i}(\mathbf{a},\mathbf{b}) = m_{R_i} - \sum_{j,p}\big(a_{j,p}r_{j,(p)}(R_i) + b_{j,p}(|R_i| - r_{j,(p)}(R_i))\big)$, *with* $r_{j,(p)}$ *the rank function of the child* $(j,p)$. *Let* $\mathcal{X}$ *be the fibre of the shares cut by the pins* $|\mathbf{a}_j| = s_j$, $|\mathbf{b}_j| = t_j$, $\langle\boldsymbol{\delta}\rangle = \boldsymbol{\mu}|_B$ *and* $\kappa_{R_1} = \cdots = \kappa_{R_{q-1}} = 0$, *with* $|B|+q-1$ *pins beyond the share sums;* $\kappa_{R_q} = 0$ *holds on it. On a wall contour with* $\operatorname{Re}\kappa_{R_i} = 0$ *for* $i < q$ *at which* $P^{\otimes}_{F_B(\mathbf{c},\mathbf{c}')}(\operatorname{Re}\mathbf{m}|_R;\mathbf{c},\mathbf{c}')$ *converges,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\mathcal{X}} \prod_j \Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\, P^{\otimes}_{F_B(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b}).$$

*(iii) The native split of the real parts* $(\operatorname{Re}\mathbf{m};\operatorname{Re}\mathbf{s},\operatorname{Re}\mathbf{t})$, *which have a native gauge, is a wall contour whose real parts are those of the points of* $\mathcal{X}$, *and* $P^{\otimes}$ *converges there; for real data it is the native split, and it lies on the fibre.*

The bunch is integrated on its orthant, scale and shape at once: writing $\alpha_B = \rho u$ before opening the letters would split the $|B|$ pins into a pin on the scale and a stacked Gamma over the shape whose indices sum to zero, carrying $1/\Gamma(0) = 0$, and neither has a value alone. For the same reason $B = \mathcal{E}$ is excluded: the imaginary part of the odd load then lies in the hyperplane $\sum_{e\in\mathcal{E}}\operatorname{Im}\langle\boldsymbol{\delta}\rangle_e = \operatorname{Im}\mu_{\mathcal{E}}$, the scaling mode, and one mark stays in the room.

**Lemma 18.14** (Connected tables have connected children)**.** *If some table with* $s_j + t_j > 0$ *is connected, then at every bunch the children at a split with positive shares form a connected family.*

At the masses $\boldsymbol{\tau}+\eta\mathbf{m}$ the port on the orthant carries the stacked Gamma with the upper entries $\nu_e = \mu_e - \langle\boldsymbol{\delta}\rangle_e$, $e \in B$, and $\kappa_{R_i}$, and the lower entry $\eta m_{\mathcal{E}}$; its $|B|+q$ upper entries sum to $\eta m_{\mathcal{E}}$ and have real parts positive multiples of $\eta$ on the wall contour of the native split (Theorem 18.13(iii)); as $\eta \to 0$ it tends to $(2\pi)^{|B|+q-1}$ times the delta of the pins. So the port on the balance wall is the limit of the Foreman procedure on the ray (Theorem 16.15): the Gammas are the pins.

**Proposition 18.15** (Peeling a molecule)**.** *In Theorem 18.13 with a native gauge, let* $R_1$ *be a molecule of the children and* $g \in R_1$. *Each child letter is* $\sum_X \alpha_{R_1}^X K_{[X]}(\alpha_{R\setminus R_1})$ *over the bases* $X$ *of its restriction to* $R_1$, *with* $K_{[X]}$ *the letter on* $R\setminus R_1$ *with the bases* $Y$ *and the weights* $\theta_{X\cup Y}$ *(for* $\Psi$, $\alpha_{R_1}^{R_1\setminus X}\Psi_{[X]}$*). Opening these sums with one share per part,* $P^{\otimes}$ *is the fibre integral with the* $|R_1|-1$ *pins* $\operatorname{load}_e = m_e$, $e \in R_1 - g$, *of the stacked Gammas of these shares against the joint period of the new children on the remaining molecules; the pin at* $g$ *holds by the balance of* $R_1$, *the native split lies on the fibre, and the joint period converges there. With weights multiplicative across* $R_1$ *every* $K_{[X]}$ *is a multiple of one letter and this is the factorization of Theorem 18.13(ii).*

**Corollary 18.16** (Atoms on the balance wall)**.** *In Theorem 18.13, fix $g$. On a wall contour for $\mathcal{E}-g$, with one share per basis and letter,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j,\ |\mathbf{b}_j|=t_j,\ \langle\boldsymbol{\delta}\rangle=\boldsymbol{\mu}|_{\mathcal{E}-g}} \prod_j \Gamma_{1}^{|\mathcal{B}_j|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{1}^{|\mathcal{B}_j|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\prod_{j,T}\theta_{j,T}^{-a_{j,T}-b_{j,T}},$$

*independent of $g$; the pin at $g$ holds by itself.*

Running the ports on the balance wall down to rooms of one mark writes $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ as an iterated fibre integral with $n-1$ pins in all; with one share per pattern and letter its dimension is $\sum(|\mathcal{B}_j|-1)-(n-1)$, one term for each active letter, $j$ its table, independent of the tree when no letter is split across molecules.

**Proposition 18.17** (The pins reach every integer point)**.** *In Corollary 18.16 take one share for each basis and each active letter, the $K_j$ with $s_j\neq 0$ and the $\Psi_j$ with $t_j\neq 0$, and write the pins in the shares as the share sums $|\mathbf{a}_j|=s_j$, $|\mathbf{b}_j|=t_j$ of the active letters and the loads $\mathrm{load}_e=m_e$, $e\in\mathcal{E}-g$ (Definition 12.24). Let every two marks be joined by a sequence of marks in which any two consecutive marks lie in a common circuit of a table with an active letter. Then the pins map the integer points of the share space onto the integer points of their target. So at integer exponents and masses the fibre has integer points, its integer points are $x_0+\Lambda$ for one of them $x_0$ and the lattice $\Lambda$ of the integer points of its directions, which has the rank of the dimension of the fibre, and a basis of $\Lambda$ gives integral coordinates on the fibre in which the integer points are exactly the points with integer coordinates.*

**Corollary 18.18** (Closure)**.** *If the weights of a form $W\,d\alpha_{\mathbf{m}}$ with real data, $s_j,t_j\geq 0$ not all zero, factor across its molecules and every molecule has a native gauge, $\int W\,d\alpha_{\mathbf{m}}$ is a product over the molecules of iterated fibre integrals over affine fibres of share space, with integrands products of stacked Gammas of the shares and monomials in the weights, $n-1$ pins on a molecule of $n$ marks. The period of each molecule at its target is also the nonnegative series of Corollary 15.11 and of Theorem 15.13. Every quasi-integral is an iterated constant term of the continuation of the share integral of Theorem 15.15 at its chart completion.*

*What a closed form is.* Definition 1.1(vi) writes final answers as words, and Closure writes the period of every family as one, in the stacked Gamma, monomials in the weights and fibre integrals alone. That such a word is a closed form is checked against the criterion by which any function is counted as one.

**Definition 18.19** (Closed form)**.** A function is *named* when it is given by an integral or a series of fixed shape whose calculus is known: where it converges, how it continues in its data, where its poles lie, how it grows, and how it is evaluated to any precision. An *alphabet* is a finite set of named functions with fixed operations. A value is *in closed form* relative to an alphabet when it is a finite expression in it, and an alphabet is *closed on a class* of values when every value of the class is in closed form relative to it. The *serif alphabet* is exp, log, the stacked Gamma and the serif family, with the operations of a word, and a *serif word* is a word in it (Definition 1.1(vi)).

This is the criterion on which $\Gamma$ and the Beta integral are counted as closed forms, and with them the classical functions listed among the Aliases: each is an integral or a series of fixed shape whose calculus is known. An integral counts as unevaluated when its shape varies with its input, as the letters $K_j$ and $\Psi_j$ of a family vary with its tables, and it is evaluated when it is carried to an integral of fixed shape, which is what the atom formula does (Corollary 18.16).

**Proposition 18.20** (The serif word is the closed form of the periods)**.** *In the terms of Definition 18.19:*

*(i) The stacked Gamma is named: it is a quotient of values of* $\Gamma$*, which is the Gramma of one scale,* $\Gamma(\nu) = \mathsf{G}_{\nu;(1)}(1)$*, and it is meromorphic in its entries. The Gramma converges exactly on* $\mathcal{T}_t$ *and is holomorphic there. The fibre integral is a fixed operation: it is fixed by its pins and does not depend on its admissible real point.*

*(ii) The alphabet of the stacked Gamma, with monomials in the data and fibre integrals over affine fibres of share space, is closed on the periods of the convergent families on the balance wall with* $\operatorname{Re} s_j > 0$ *or* $s_j = 0$ *and* $\operatorname{Re} t_j > 0$ *or* $t_j = 0$ *for every* $j$*, and on the pseudointegrals of Corollary 18.18: each is a fibre integral, or a product of iterated fibre integrals, of products of stacked Gammas of the shares and monomials in the weights.*

*(iii) On the word the calculus of every such period is read: its convergence at the walls (Theorem 8.7), its continuation in all its data (Corollary 18.18), its native gauge, where its integrand is largest (Theorem 8.19), the integer points of its fibre and its closings (Proposition 18.17, Theorem 4.5), and its nonnegative series (Corollary 15.11, Theorem 15.13).*

*(iv) The series of stacked Gammas are readings of words. For* $k > 0$ *and* $0 < \mathsf{m} < 2$ *let* $F_{\mathsf{m}}$ *be the family on* $\{1,2\}$ *of two copies of* $\binom{[2]}{1}$*, the first with the weights* $1$ *at the exponents* $(k,0)$*, the second with the weight* $1$ *on the basis* $\{1\}$ *and* $\mathsf{m}$ *on the basis* $\{2\}$ *at the exponents* $(0,k)$*. At the masses* $(k,k)$*,*

$$P_{F_{\mathsf{m}}}\big((k,k);(k,0),(0,k)\big) = \frac{1}{2\pi i}\int_{\operatorname{Re} x = k/2} \Gamma_{\substack{4\\2}}\begin{bmatrix} x,\ x,\ k-x,\ k-x \\ k,\ k \end{bmatrix}\, \mathsf{m}^{x-k}\, dx$$
$$= \sum_{j\ge 0} \Gamma_{\substack{2\\1}}\begin{bmatrix} k+j,\ k+j \\ 2k+j \end{bmatrix} \frac{(1-\mathsf{m})^j}{j!}.$$

*For* $n \ge 1$*,* $-1 < z < 0$ *and* $-1 < c < 0$*,*

$$\sum_{j\ge 1} \frac{z^j}{j^n} = \frac{z}{2\pi i}\int_{\operatorname{Re} s = c} \Gamma_{\substack{2\\0}}\big[-s,\ 1+s\big]\, \Big(\Gamma_{\substack{1\\1}}\begin{bmatrix} 1+s \\ 2+s \end{bmatrix}\Big)^n (-z)^s\, ds,$$

*and the integral, which converges for every* $z < 0$*, continues the series to every* $z < 0$*.*

So the serif word is the closed form of the periods by the criterion that makes $\Gamma$ one: its functions are named on the grounds on which $\Gamma$ is, the alphabet is closed on the periods, their calculus is read on the word, and the classical closed forms are among its readings (Aliases). Along a ray the values of a family satisfy one recurrence and span a space of finite dimension (Theorem 38.31); twists and product weights multiply the integrand by a monomial in the loads (Theorem 25.12(ii), Proposition 26.20); and down the hinge tree each cell enters through the period of its own hinge minors at the loads of its marker (Theorem 22.4), on the centre ray of a table with norms on its marks through its own word (Proposition 42.26). Smaller alphabets fail already on a class: no function whose continuations span a local system with a unipotent subgroup of finite index in its monodromy is a nonzero period of the pencil of a class of at least three marks (Corollary 42.30). That a value has no closed form in a smaller alphabet is a statement about that alphabet.

**Corollary 18.21** (The marginal law of the shape)**.** *In Theorem 18.13 let* $U \sim \mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})$*, with the internal shapes* $V_B = U_B/|U_B|$ *and* $V_R = U_R/|U_R|$ *of* $B$ *and* $R$ *(Proposition 18.3).*

*(i)* $\mathbb{E}_{U'\sim \mathrm{Dir}_{\mathbf{m}|_B}}\big[\,\mathrm{Sc}_B(U',v;\mathbf{m};\mathbf{s},\mathbf{t})\big] = v^{-\sigma\mathbf{1}}\,\Xi_B(v;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t})/\Gamma_{|B|\atop 1}\begin{bmatrix}\mathbf{m}|_B\\ m_B\end{bmatrix}$.

*(ii)* *Given* $V_R = v$, *for* $\mathbf{k}\in\mathbb{C}^B$ *with* $|\mathbf{k}| = 0$, $\mathbb{E}_{U\sim \mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})}[V_B^{\mathbf{k}} \mid V_R = v] = \Xi_B(v;\boldsymbol{\mu}|_B + \mathbf{k};\mathbf{s},\mathbf{t})/\Xi_B(v;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t})$, *the bunch factor at shifted offsets, whose fibre is* $\langle\boldsymbol{\delta}\rangle = \boldsymbol{\mu}|_B + \mathbf{k}$.

*(iii)* $V_R$ *has density proportional to* $v^{\boldsymbol{\mu}|_R-\mathbf{1}}\,\Xi_B(v;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t})$, *the fibre mixture of the children's shape laws in proportion to the stacked Gammas times* $P_{F_B(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})$ *when the children are connected (Lemma 18.14).*

*(iv)* *Given both shapes, the scale of* $B$ *has the Skeleton law of Theorem 18.11(i).*

## 18.3 Cycles

**Definition 18.22** (Cycle of a family)**.** A *cycle* of a family is a bunch $C$ that is a circuit of one of its tables. That table's cobasis polynomial has floor one on $C$, and on its bottom colevel the letter is linear on $C$; a letter linear on a bunch gives the bunch the *kernel* $e^{-h}$, $h$ the letter divided by the sum of its coefficients (Theorem 18.23(ii) below).

**Theorem 18.23** (A cycle carries its own kernel)**.** *Let* $F$ *have real data with* $s_j, t_j \geq 0$, $\kappa_{\mathcal{E}} = 0$, $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *convergent, and* $C \neq \mathcal{E}$ *a circuit of table* $j_0$ *with* $t_{j_0} > 0$ *and room* $R$.

*(i)* *The patterns of table* $j_0$ *on* $C$ *of colevel one are exactly* $p_e = C - e$, $e \in C$, *and the child at* $p_e$ *is the contraction* $\mathcal{B}_{j_0}/C$ *on* $R$ *with the weights* $\theta_{j_0,p_e\cup X}$. *So* $\Psi_{j_0} = \Psi^\flat + \Psi^\sharp$ *with* $\Psi^\flat = \sum_{e\in C}\alpha_e\Psi_{j_0,(p_e)}(\alpha_R)$ *linear in* $\alpha_C$. *The circuits are the only bunches whose colevel-one patterns are one per mark.*

*(ii)* *With* $G(\alpha_R) = \sum_{e\in C}\Psi_{j_0,(p_e)}(\alpha_R)$ *and* $h = \Psi^\flat/G$, *if the family with* $\Psi_{j_0}^{-t_{j_0}}$ *replaced by* $(\Psi^\flat)^{-c}(\Psi^\sharp)^{-(t_{j_0}-c)}$ *converges, then on* $\mathrm{Re}\,\beta = c$

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \frac{1}{\Gamma(t_{j_0})}\int_{\beta+\beta'=t_{j_0}}\Gamma(\beta') \times\int_{\mathbb{R}^{\mathcal{E}}_{>0}}\alpha^{\mathbf{m}-\mathbf{1}}e^{-h(\alpha)}G(\alpha_R)^{-\beta}\Psi^\sharp(\alpha)^{-\beta'}K_{j_0}^{-s_{j_0}}\prod_{j\neq j_0}K_j^{-s_j}\Psi_j^{-t_j}\,d\alpha.$$

*When the colevel-one children are multiples* $\vartheta_e$ *of one letter, as for unit weights and for determinantal tables, the kernel is* $e^{-\langle\vartheta,\alpha_C\rangle/|\vartheta|}$.

*(iii)* *On a wall contour at* $C$ *the pins at the marks of* $C$ *hold exactly when* $b_{j_0,p_e} = m_e - \lambda_e$, $\lambda_e$ *the load on* $e$ *of all the other shares, and* $\sum_{j,p}\big(a_{j,p}(|p| - d_j(C)) + b_{j,p}(|C\setminus p| - d_j^*(C))\big) = \kappa_C$, *the coefficients being the excesses of the level and the colevel of* $p$ *in table* $j$ *(Definition 12.16); the fibre integral at* $C$ *is over the other shares with the single pin that their total excess is the wall of the cycle, the Gammas* $\Gamma(m_e - \lambda_e)$ *of the cycle standing in the places of the* $\Gamma(b_{j_0,p_e})$.

*Dually, a cocircuit has floor one in the basis polynomial, and its bottom level gives its kernel.*

**Corollary 18.24** (A table that is one circuit)**.** *Let* $F = ((\mathcal{B},\theta);s,t)$ *with* $s,t > 0$, *where the bases of* $\mathcal{B}$ *are the sets* $\mathcal{E} - e$, $e\in\mathcal{E}$, *with the weights* $\theta_e$; *let the masses lie on the balance wall,* $m_{\mathcal{E}} = (n-1)s + t$, *with* $m_S > s(|S|-1)$ *for every nonempty proper* $S$, *and put* $\nu_e = m_e - s$. *Then*

$$P_F(\mathbf{m};s,t) = \frac{\prod_e\theta_e^{-\nu_e}}{\Gamma(s)\Gamma(t)}\int_0^\infty y^{s-1}\prod_{e\in\mathcal{E}}\mathsf{G}_{\nu_e;(1,-1)}\big(1,\theta_e^2 y\big)\,dy$$

$$= \frac{\prod_e \theta_e^{-\nu_e}}{\Gamma(s)\Gamma(t)} \int_{\sum_e a_e = s} \prod_{e \in \mathcal{E}} \Gamma(a_e)\, \Gamma(a_e + \nu_e)\, \theta_e^{-2a_e} \, :$$

*the cobasis polynomial $\Psi(\alpha) = \langle \theta, \alpha \rangle$ is linear on the whole set, which so carries its own kernel (Definition 18.22), and the fibre of the atom formula (Corollary 18.16), of dimension $n-1$, closes to one scale.*

# Chapter 19

# Bunches of width one: pooling and cleaning

A bunch whose bases meet it on two levels only is seen by its room as one mark: the room reads the bunch through two ratios of polynomials on its marks, one for each Foreman polynomial, and the bunch is a table on its own marks and one more. A bunch of width one is read as a mark of its room, its shape integrated while its scale is kept, its walls glued, and its scale closed at a wall, at the gauge and on the totals of the bunch and the room. Cleaning, the repeated pooling of such bunches, is a move between period data and has one result.

## 19.1 Bunches of width one

**Definition 19.1** (Parallel and series bunches)**.** A *parallel bunch* of a table (a parallel class) is a set of at least two marks, none a loop, no two of which lie in a common basis; a *series bunch* (a series class) is a set of at least two marks, none a coloop, every basis containing all but at most one of them. A bunch $B$ of width $c(B) = r(B) - d(B)$ (Definition 7.12, Theorem 7.13) has its patterns on the $c(B)+1$ levels $d(B), \dots, r(B)$ (Theorem 12.7), and its room has the same width.

**Definition 19.2** (Hinge values; their ratio)**.** Let a bunch $B$ of a table $(\mathcal{B}, \theta)$ have width one, the weights factoring across $B$ and its room $R$, $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$, so that $\mathcal{B}|B$ and $\mathcal{B}/R$ carry $\theta'$ and $\mathcal{B}|R$ and $\mathcal{B}/B$ carry $\theta''$ (Definition 7.5). The *hinge value of $B$ by $K$* and its *hinge value by $\Psi$* are

$$\gamma_B^K = \frac{K_{\mathcal{B}|B}}{K_{\mathcal{B}/R}}, \qquad \gamma_B^\Psi = \frac{\Psi_{\mathcal{B}/R}}{\Psi_{\mathcal{B}|B}},$$

functions of $\alpha_B$ homogeneous of degree one, the values at which $K$ and $\Psi$ see the bunch replaced by one mark, and their *ratio*

$$\varrho_B = \frac{\gamma_B^\Psi}{\gamma_B^K} = \frac{\mathbf{E}_{\mathcal{B}/R}}{\mathbf{E}_{\mathcal{B}|B}},$$

the quotient of the even letters $\mathbf{E} = K\Psi$ (Definition 8.18) of the two tables on $B$, is homogeneous of degree zero, a function of the shape $u = \alpha_B/|\alpha_B|$ alone. The room has width one as well, and its hinge values are those of $R$ as a bunch: $\gamma_R^K = K_{\mathcal{B}|R}/K_{\mathcal{B}/B}$, $\gamma_R^\Psi = \Psi_{\mathcal{B}/B}/\Psi_{\mathcal{B}|R}$ and $\varrho_R = \mathbf{E}_{\mathcal{B}/B}/\mathbf{E}_{\mathcal{B}|R}$. Statements about hinge values and their ratios are read at one fixed factorization. For table $j$ of a family they are $\gamma_{j,B}^K$, $\gamma_{j,B}^\Psi$ and $\varrho_{j,B}$, each table with its own factorization.

**Proposition 19.3** (Hinge values of parallel and series bunches)**.** *A parallel or series bunch has width at most one, and width zero exactly when it is a union of molecules. Let a parallel bunch $B$ of width one have its weights factoring across $B$ and $R$ (Definition 19.2); then $\mathcal{B}/R$ has the single basis $\varnothing$, with the weight $\theta'_{\varnothing}$, and $\mathcal{B}|B$ the singletons, with the weights $\theta'_{\{e\}}$, and in this statement write $w_e = \theta'_{\{e\}}/\theta'_{\varnothing}$. So $K_{\mathcal{B}/R} = \theta'_{\varnothing}$, $K_{\mathcal{B}|B} = \theta'_{\varnothing} \sum_{e\in B} w_e\alpha_e$, $\Psi_{\mathcal{B}|B} = \theta'_{\varnothing}\, e_{|B|}(\alpha_B) \sum_{e\in B} w_e\alpha_e^{-1}$ and $\Psi_{\mathcal{B}/R} = \theta'_{\varnothing}\, e_{|B|}(\alpha_B)$, and*

$$\gamma_B^K = \sum_{e\in B} w_e\alpha_e, \qquad \gamma_B^\Psi = \Big(\sum_{e\in B} w_e\alpha_e^{-1}\Big)^{-1} :$$

*$K$ adds the marks of the class with the factors $w_e$ and $\Psi$ combines them as reciprocals with the same factors. A series bunch $B$ of width one, its weights factoring, swaps the two roles: $\mathcal{B}|B$ has the single basis $B$, with the weight $\theta'_B$, and $\mathcal{B}/R$ the sets $B-e$, with the weights $\theta'_{B-e}$; with $w_e = \theta'_{B-e}/\theta'_B$, $K_{\mathcal{B}|B} = \theta'_B\, e_{|B|}(\alpha_B)$, $K_{\mathcal{B}/R} = \theta'_B\, e_{|B|}(\alpha_B) \sum_e w_e\alpha_e^{-1}$, $\Psi_{\mathcal{B}|B} = \theta'_B$ and $\Psi_{\mathcal{B}/R} = \theta'_B \sum_e w_e\alpha_e$, so $\gamma_B^\Psi = \sum_e w_e\alpha_e$ and $\gamma_B^K = \big(\sum_e w_e\alpha_e^{-1}\big)^{-1}$. In a list, the case used for cleaning (Definition 19.35 below), with the factorization $\theta' = \theta'' = 1$, every $w_e$ is 1, and a pair with the shape $(x, 1-x)$ has $\varrho_B = x(1-x)$ when it is parallel and $\varrho_B = 1/(x(1-x))$ when it is in series.*

**Theorem 19.4** (A bunch of width one is one mark to its room)**.** *Let a bunch $B$ of a table $(\mathcal{B},\theta)$ have width one, the weights factoring across $B$ and $R$, $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$, and let $g$ be a new mark. The* cell of the bunch $\mathcal{B}_B^+$ *is the table on $B+g$ whose bases avoiding $g$ are those of $\mathcal{B}|B$ and whose bases through $g$ are the $X+g$, $X\in\mathcal{B}/R$, with the weights $\theta'$; the* cell of the room $\mathcal{B}_R^+$ *is the table on $R+g$ whose bases avoiding $g$ are those of $\mathcal{B}|R$ and whose bases through $g$ are the $Y+g$, $Y\in\mathcal{B}/B$, with the weights $\theta''$. Both are tables, lists when $(\mathcal{B},\theta)$ is one, in which $g$, the* hinge, *is neither a loop nor a coloop, and the bases of $\mathcal{B}$ are theirs glued along $g$ with $g$ on exactly one side:*

$$\mathcal{B} = \{X\sqcup Y:\ X\in\mathcal{B}|B,\ Y\in\mathcal{B}/B\}\ \sqcup\ \{X\sqcup Y:\ X\in\mathcal{B}/R,\ Y\in\mathcal{B}|R\}.$$

*With the first argument of the polynomials of a cell the variable of its hinge and the hinge values of Definition 19.2,*

$$K(\alpha) = K_{\mathcal{B}/R}(\alpha_B)\, K_{\mathcal{B}_R^+}\big(\gamma_B^K(\alpha_B),\alpha_R\big), \qquad \Psi(\alpha) = \Psi_{\mathcal{B}|B}(\alpha_B)\, \Psi_{\mathcal{B}_R^+}\big(\gamma_B^\Psi(\alpha_B),\alpha_R\big).$$

*Conversely, if $K(\alpha) = \sum_{l=0}^k A_l(\alpha_B)\, C_l(\alpha_R)$ with every $C_l$ homogeneous, then $c(B)\le k$; in particular $K(\alpha) = A(\alpha_B)\big(C_0(\alpha_R) + G(\alpha_B)\, C_1(\alpha_R)\big)$ with $C_0$ and $C_1$ homogeneous gives $c(B)\le 1$.*

The deletions and the contractions of the hinge are the four minors $\mathcal{B}_B^+\setminus g = \mathcal{B}|B$ and $\mathcal{B}_B^+/g = \mathcal{B}/R$, $\mathcal{B}_R^+\setminus g = \mathcal{B}|R$ and $\mathcal{B}_R^+/g = \mathcal{B}/B$ of the table. So the hinge values of $B$ are those of $B$ in its own cell, and across a bunch of width one the bunch and the room see each other as one mark: the room reads the hinge at $\alpha_g = \gamma_B^K(\alpha_B)$ in $K$ and at $\alpha_g = \gamma_B^\Psi(\alpha_B)$ in $\Psi$.

**Corollary 19.5** (The scale of a bunch of width one is that of one mark)**.** *Along the scale of a bunch $B$ of width one of a table whose weights factor across $B$ and $R$, each Foreman polynomial has one Skeleton value, the room's hinge value over the bunch's:*

$$\operatorname{spec}_B^K(\alpha) = \Big\{\frac{\gamma_R^K(\alpha_R)}{\gamma_B^K(\alpha_B)}\Big\} = \Big\{\frac{K_{\mathcal{B}/R}(\alpha_B)\, K_{\mathcal{B}|R}(\alpha_R)}{K_{\mathcal{B}|B}(\alpha_B)\, K_{\mathcal{B}/B}(\alpha_R)}\Big\},$$
$$\operatorname{spec}_B^\Psi(\alpha) = \Big\{\frac{\gamma_R^\Psi(\alpha_R)}{\gamma_B^\Psi(\alpha_B)}\Big\} = \Big\{\frac{\Psi_{\mathcal{B}|B}(\alpha_B)\, \Psi_{\mathcal{B}/B}(\alpha_R)}{\Psi_{\mathcal{B}/R}(\alpha_B)\, \Psi_{\mathcal{B}|R}(\alpha_R)}\Big\},$$

*the Skeleton value of the hinge as a one-mark bunch of the cell of the room (Corollary 15.4), read at $\alpha_g = \gamma_B^K(\alpha_B)$, respectively at $\alpha_g = \gamma_B^\Psi(\alpha_B)$. The value of $K$ over that of $\Psi$ is $\varrho_B(\alpha_B)/\varrho_R(\alpha_R)$, the even letters of the minors of the level $d(B)$ over those of the level $r(B)$, and their product is the same quotient of the odd letters.*

**Lemma 19.6** (One hinge value of any degree is width at most one)**.** *Let $B$ be a bunch of a table $(\mathcal{B}, \theta)$ with the room $R$, and suppose that on the open orthant*

$$K(\alpha) = A(\alpha_B)\, P\big(\alpha_R, \gamma(\alpha_B)\big),$$

*with $A$ and $\gamma$ positive functions of $\alpha_B$, homogeneous, $\gamma$ of degree one, and $P$ a polynomial in $\alpha_R$ and one more variable with real coefficients, of any degree in that variable. Then $c(B) \le 1$. Conversely, when the weights factor across $B$ and $R$ and $c(B) \le 1$, $K$ has this form: at width zero with $P$ free of its last variable, and at width one with $A = K_{\mathcal{B}/R}$, $\gamma = \gamma_B^K$ and $P(\alpha_R, y) = K_{\mathcal{B}_R^+}(y, \alpha_R)$ (Theorem 19.4). The same holds for $\Psi$, the basis polynomial of the dual table, in which $B$ has the same width.*

**Proposition 19.7** (The freedom of the factorization)**.** *Let a bunch $B$ of a table $(\mathcal{B}, \theta)$ have width one, the weights factoring across $B$ and $R$, $\theta_T = \theta'_{T\cap B}\, \theta''_{T\cap R}$. The factorization is fixed up to one positive constant on each of the two levels: multiplying $\theta'$ by $\lambda_d$ on the patterns of the level $d(B)$ and by $\lambda_r$ on those of the level $r(B)$, and dividing $\theta''$ by $\lambda_d$ and by $\lambda_r$ on the room sides of the bases of these levels, multiplies the hinge values $\gamma_B^K$ and $\gamma_R^K$ by $c = \lambda_r/\lambda_d$, $\gamma_B^\Psi$ and $\gamma_R^\Psi$ by $c^{-1}$, and the ratios $\varrho_B$ and $\varrho_R$ by $c^{-2}$ (Definition 19.2), and leaves $\gamma_R^K/\gamma_B^K$, $\gamma_R^\Psi/\gamma_B^\Psi$ and $\varrho_B/\varrho_R$ unchanged. In a family each table has its own two constants.*

## 19.2 Pooling

**Theorem 19.8** (Pooling integrates the shape and keeps the scale)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a family and $B$ a bunch of width one in every table, the weights of every table factoring across $B$ and $R$, and read $\alpha_B = \rho u$, $\rho > 0$, $u \in \Delta_B$ (Lemma 2.7). For $u \in \Delta_B$ the* pooled family *$F_{B\to g}(u)$ is the family on $R + g$ with, for every $j$, two tables with the bases of the cell of the room $\mathcal{B}_{j,R}^+$ (Theorem 19.4): the first with its weights multiplied by $\gamma_{j,B}^K(u)$ on its bases through $g$ and the exponents $(s_j, 0)$, the second with its weights multiplied by $\gamma_{j,B}^\Psi(u)$ on its bases avoiding $g$ and the exponents $(0, t_j)$. Its letters are $K_{\mathcal{B}_{j,R}^+}\big(\gamma_{j,B}^K(u)\alpha_g, \alpha_R\big)^{-s_j}$ and $\Psi_{\mathcal{B}_{j,R}^+}\big(\gamma_{j,B}^\Psi(u)\alpha_g, \alpha_R\big)^{-t_j}$; listing the $N$ first tables and then the $N$ second ones, its exponents are $(\mathbf{s}, \mathbf{0})$ and $(\mathbf{0}, \mathbf{t})$. To* pool *$B$ is to pass from $F$ to its pooled families $F_{B\to g}(u)$, keeping the letters $K_{\mathcal{B}_j/R}$ and $\Psi_{\mathcal{B}_j|B}$ of the shape of $B$, its cell kernel (Chapter 22), and the hinge values of every table. Give the hinge the mass $\kappa_B$, the wall of $B$. Then $F_{B\to g}(u)$ has the imbalance of $F$, and*

$$J_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} \prod_j K_{\mathcal{B}_j/R}(u)^{-s_j} \Psi_{\mathcal{B}_j|B}(u)^{-t_j} \times J_{F_{B\to g}(u)}\big((\kappa_B, \mathbf{m}|_R); (\mathbf{s}, \mathbf{0}), (\mathbf{0}, \mathbf{t})\big)\, d\omega_B(u),$$

*both sides converging together; the same holds with $P$ for $J$, on the balance wall as well.*

The port (Theorem 15.15) integrates the scale and the shape of a bunch together; pooling integrates the shape of a bunch of width one and keeps its scale, which becomes the hinge of the room. For a list the pooled families again have weights that factor across every bunch of $R$ and

across every bunch containing $g$, so poolings can be repeated. Proposition 18.3 is the same split with the scale integral left whole. Read backwards, pooling is gluing: the table is its cell of the bunch substituted into its cell of the room at the hinge (Proposition 21.3), and every cell of a substitution is pooled at once by Theorem 22.3.

## 19.3 Walls glue

**Theorem 19.9** (Walls glue)**.** *Let a bunch $B$ of a table $(\mathcal{B},\theta)$ have width one, the weights factoring across $B$ and $R$, with the cells $\mathcal{B}_B^+$ and $\mathcal{B}_R^+$ of Theorem 19.4. For $S_B \subseteq B$ and $S_R \subseteq R$ the counts of the table glue from those of the cells, the hinge counted on exactly one side, as in the gluing of the bases:*

$$d(S_B \sqcup S_R) = \min\left\{d_{\mathcal{B}_B^+}(S_B+g) + d_{\mathcal{B}_R^+}(S_R),\ d_{\mathcal{B}_B^+}(S_B) + d_{\mathcal{B}_R^+}(S_R+g)\right\},$$
$$d^*(S_B \sqcup S_R) = \min\left\{d^*_{\mathcal{B}_B^+}(S_B+g) + d^*_{\mathcal{B}_R^+}(S_R),\ d^*_{\mathcal{B}_B^+}(S_B) + d^*_{\mathcal{B}_R^+}(S_R+g)\right\}.$$

*For a family $F = ((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ with masses $\mathbf{m}$, $B$ having width one in every table and the weights of every table factoring across $B$ and $R$:*

*(i) the walls of the pooled family $F_{B\to g}(u)$ at the masses $(\kappa_B,\mathbf{m}|_R)$ are walls of $F$: its wall at $S_R$ is $\kappa_{S_R}$ and its wall at $S_R+g$ is $\kappa_{B\sqcup S_R}$;*

*(ii) the density $u^{\mathbf{m}|_B-\mathbf{1}}\prod_j K_{\mathcal{B}_j/R}(u)^{-s_j}\Psi_{\mathcal{B}_j|B}(u)^{-t_j}$ of the cell kernel of $B$ has at the face $u_{S_B}=0$ of $\Delta_B$ the wall of $F$, $m_{S_B} - \sum_j \left(s_j\, d_{\mathcal{B}_j/R}(S_B) + t_j\, d^*_{\mathcal{B}_j|B}(S_B)\right) = \kappa_{S_B}$;*

*(iii) on the balance wall, with the cell families $G_B = ((\mathcal{B}_{j,B}^+,\theta_j');s_j,t_j)_j$ at the masses $(\mathbf{m}|_B, 2\sigma-\kappa_B)$ and $G_R = ((\mathcal{B}_{j,R}^+,\theta_j'');s_j,t_j)_j$ at the masses $(\kappa_B,\mathbf{m}|_R)$, $\sigma$ the total strength, whose walls are written $\kappa_S^{G_B}$ and $\kappa_S^{G_R}$, both cell families lie on their balance walls and every wall of a cell family is a wall of $F$,*

$$\kappa_{S_B}^{G_B} = \kappa_{S_B},\qquad \kappa_{S_B+g}^{G_B} = \kappa_{S_B\sqcup R},\qquad \kappa_{S_R}^{G_R} = \kappa_{S_R},\qquad \kappa_{S_R+g}^{G_R} = \kappa_{B\sqcup S_R};$$

*if moreover $s_j,t_j\geq 0$ and every active letter reads $B$ alike, its hinge value a constant multiple of one function $\gamma$ on $\Delta_B$, as when one letter is active, every wall of $F$ is glued from walls of the cell families,*

$$\kappa_{S_B\sqcup S_R} = \min\left\{\kappa_{S_B}^{G_B} + \kappa_{S_R}^{G_R},\ \kappa_{S_B+g}^{G_B} + \kappa_{S_R+g}^{G_R}\right\},$$

*and $F$ has every proper wall positive exactly when both cell families do.*

## 19.4 Full leans

**Theorem 19.10** (Pooling factors at one hinge value)**.** *Let $F = ((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ have real exponents and be on the balance wall with every proper wall positive, and let a bunch $B$ have width one in every table, the weights of every table factoring across $B$ and $R$. Suppose that every active letter reads $B$ alike, its hinge value a constant multiple of one function $\gamma$ on $\Delta_B$, the factorizations being chosen so that every constant is 1: every active letter has the one hinge value $\gamma$. (A change of the factorization of table $j$ multiplies $\gamma_{j,B}^K$ by a positive constant and $\gamma_{j,B}^\Psi$ by its inverse (Proposition 19.7). So a table with one active letter can always be so normalized, and when every table is at a full lean the hypothesis is a condition across the tables alone; one table with both letters*

*active can be so normalized exactly when $\varrho_B$ is constant.) Then $0 < \kappa_B < 2\sigma$, $\sigma$ the total strength (Definition 8.1), and, with the cell families $G_B = ((\mathcal{B}^+_{j,B}, \theta'_j); s_j, t_j)_j$ and $G_R = ((\mathcal{B}^+_{j,R}, \theta''_j); s_j, t_j)_j$,*

$$\Gamma_{\!1}^{\,2}\begin{bmatrix}\kappa_B, & 2\sigma-\kappa_B\\ & 2\sigma\end{bmatrix} P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{G_B}\big((\mathbf{m}|_B, 2\sigma-\kappa_B);\mathbf{s},\mathbf{t}\big)\, P_{G_R}\big((\kappa_B, \mathbf{m}|_R);\mathbf{s},\mathbf{t}\big),$$

*both cell families on their balance walls with every proper wall positive. One table at the exponents $(0,t)$ or $(s,0)$ has one active letter, and then $2\sigma$ is $t$, respectively $s$.*

When every active letter has one hinge value, as for one table at a full lean, where one letter is active, the hinge carries the mass $\kappa_B$ in the room and $2\sigma - \kappa_B$ in the bunch, and the period of the family is the product of the periods of its two cell families divided by the Beta corner of the hinge.

**Proposition 19.11** (Pooling a realization)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a family, $B$ a bunch of width one in every table, the weights of every table factoring across $B$ and $R$, and let $\Upsilon = ((s_j, t_j)_j; \alpha)$ be a realization of $F$ with masses $\boldsymbol{\tau}$ (Definition 9.1); write $\alpha_B = \rho u$, $\rho > 0$, $u \in \Delta_B$ (Lemma 2.7), and read the two tables of index $j$ of the pooled family $F_{B\to g}(u)$ of Theorem 19.8 as one table with two systems of weights at the exponents $(s_j, t_j)$ (Definition 8.1). Then $((s_j, t_j)_j; (\rho, \alpha_R))$ is a realization of $F_{B\to g}(u)$, with the masses $\big(\tau_B - \sum_j (s_j d_j(B) + t_j d^*_j(B)),\ \boldsymbol{\tau}|_R\big)$, the first at the hinge: pooling sends a realization to a realization, and the mass of the hinge is the wall $\kappa_B$ at the masses $\boldsymbol{\tau}$. With one table at the exponents $(0,t)$ the letter $\Psi$ of $F_{B\to g}(u)$ at $(\rho, \alpha_R)$ is that of the cell of the room $(\mathcal{B}^+_R, \theta'')$ at $(\gamma^\Psi_B(\alpha_B), \alpha_R)$, and $((0,t); (\gamma^\Psi_B(\alpha_B), \alpha_R))$ is a realization of $(\mathcal{B}^+_R, \theta'')$ with the masses $(\tau_B - t\, d^*(B),\ \boldsymbol{\tau}|_R)$.*

**Proposition 19.12** (Hinge values that agree)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have real exponents and be on the balance wall with every proper wall positive, and let a bunch $B$ have width one in every table, the weights of every table factoring across $B$ and $R$. Suppose that every active letter reads $B$ alike, its hinge value a constant multiple of one function $\gamma$ on $\Delta_B$: $\gamma^K_{j,B} = c_j\gamma$ whenever $s_j \neq 0$ and $\gamma^\Psi_{j,B} = c^*_j\gamma$ whenever $t_j \neq 0$, with constants $c_j, c^*_j > 0$; put $c_j = 1$ when $s_j = 0$ and $c^*_j = 1$ when $t_j = 0$. Let $G^{\mathbf{c}}_R$ be the pooled family of Theorem 19.8 with these constants in place of the hinge values: for every $j$ the table $\mathcal{B}^+_{j,R}$ with its weights multiplied by $c_j$ on its bases through $g$, at the exponents $(s_j, 0)$, and with its weights multiplied by $c^*_j$ on its bases avoiding $g$, at the exponents $(0, t_j)$. Then*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{G^{\mathbf{c}}_R}\big((\kappa_B, \mathbf{m}|_R); (\mathbf{s},\mathbf{0}), (\mathbf{0},\mathbf{t})\big) \times \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} \prod_j K_{\mathcal{B}_j/R}(u)^{-s_j}\, \Psi_{\mathcal{B}_j|B}(u)^{-t_j}\, \gamma(u)^{-\kappa_B}\, d\omega_B(u) :$$

*the shape and the room separate at every pair of exponents. For one table with $\varrho_B$ constant, a factorization with $\varrho_B = 1$ (Proposition 19.7) gives $c = c^* = 1$ and $\gamma = \gamma^K_B$, and $G^{\mathbf{c}}_R$ is then $((\mathcal{B}^+_R, \theta''); s, t)$.*

## 19.5 Partial leans

**Theorem 19.13** (Gluing across a bunch of width one)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have real data with $s_j, t_j \geq 0$, $s_j + t_j > 0$, and masses $\mathbf{m}$ on the balance wall with every proper wall positive, and let a bunch $B$ have width one in every table, the weights factoring across $B$ and $R$. In table $j$ call*

$d_j(B)$ *the* level $d$ *and* $r_j(B) = d_j(B) + 1$ *the* level $r$. Level shares $\mathbf{a} = (a_j^d, a_j^r)_j$ *and* $\mathbf{b} = (b_j^d, b_j^r)_j$ *split the exponents over the two levels,* $a_j^d + a_j^r = s_j$ *and* $b_j^d + b_j^r = t_j$, *a letter of exponent zero being opened by nothing, its two level shares* $0$*; they are the shares of Definition 12.24 summed over the patterns of each level, and their pair shares (Definition 12.24) are the* even share $\sigma_j^\ell = \frac{1}{2}(a_j^\ell + b_j^\ell)$ *and the* odd share $\delta_j^\ell = \frac{1}{2}(a_j^\ell - b_j^\ell)$, $\ell \in \{d, r\}$. *The* bunch family *and the* room family *are*

$$F^{\text{bunch}}(\mathbf{a}, \mathbf{b}) = \big((\mathcal{B}_j/R;\, a_j^d, b_j^d),\ (\mathcal{B}_j|B;\, a_j^r, b_j^r)\big)_j \quad \text{on } B,$$
$$F^{\text{room}}(\mathbf{a}, \mathbf{b}) = \big((\mathcal{B}_j|R;\, a_j^d, b_j^d),\ (\mathcal{B}_j/B;\, a_j^r, b_j^r)\big)_j \quad \text{on } R:$$

*each level carries one table of the bunch and one of the room, at the same shares. The scale of* $B$ *pins the shares of the minors it multiplies,*

$$\sum_j \big(a_j^r + b_j^d\big) = \kappa_B, \qquad \textit{equivalently} \qquad \sum_j \delta_j^d = \tfrac{1}{2}\Big(\sum_j s_j - \kappa_B\Big),$$

*and, for pattern shares with these level sums, equivalently* $\sum_{e\in B}\langle\boldsymbol{\delta}\rangle_e = \mu_B$, *the sum over the bunch of the pins* $\langle\boldsymbol{\delta}\rangle_e = \mu_e$ *of a wall contour (Definition 18.12). On this pin both families lie on their balance walls at* $\mathbf{m}|_B$ *and* $\mathbf{m}|_R$, *and on every contour on which the stacked Gammas below have arguments of positive real part and both periods converge absolutely,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = 2^{N-1} \int_{\sum_j \delta_j^d = \frac{1}{2}(\sum_j s_j - \kappa_B)} \prod_j \Gamma_{\!\!\begin{smallmatrix}4\\2\end{smallmatrix}}\!\begin{bmatrix} a_j^d,\ b_j^d,\ a_j^r,\ b_j^r \\ s_j,\ t_j \end{bmatrix} \times P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b})\, P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}),$$

*the fibre integral (Definition 4.1) in the pair coordinates* $(\sigma_j^d, \delta_j^d)_j$, *with* $\sigma_j^r = \sigma_j - \sigma_j^d$ *and* $\delta_j^r = \delta_j - \delta_j^d$*: its dimension is* $2N - 1$, *and its* $N$ *even shares* $\sigma_j^d$ *are free. This display needs every* $s_j$ *and* $t_j$ *positive; in general, with* $p$ *the number of active letters,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\sum_j (a_j^r + b_j^d) = \kappa_B} \prod_{s_j \neq 0} \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix} a_j^d,\ a_j^r \\ s_j \end{bmatrix} \prod_{t_j \neq 0} \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix} b_j^d,\ b_j^r \\ t_j \end{bmatrix} \times P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b})\, P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}),$$

*the share integral (Definition 4.1) over the shares* $a_j^r$, $s_j \neq 0$, *and* $b_j^d$, $t_j \neq 0$, *of the minors that the scale of* $B$ *multiplies, pinned by their sum* $\kappa_B$, *of dimension* $p - 1$*; with every exponent positive it is the display above, and at* $p = 1$ *it is the value at the one admissible point. For one table with* $s, t > 0$ *the odd share is fixed,* $2\delta^d = s - \kappa_B = \delta\,(1 + \varepsilon(B)) - \mu_B$, *and the even share alone is free: on a line* $\operatorname{Re}\sigma^d = \sigma_*^d$ *of such a contour,*

$$P_F(\mathbf{m}; s, t) = \frac{1}{2\pi i} \int_{\operatorname{Re}\sigma^d = \sigma_*^d} \Gamma_{\!\!\begin{smallmatrix}4\\2\end{smallmatrix}}\!\begin{bmatrix} a^d,\ b^d,\ a^r,\ b^r \\ s,\ t \end{bmatrix} P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b})\, P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b})\, d\sigma^d.$$

*Swapping* $B$ *and* $R$ *swaps the two levels and leaves the formula unchanged.*

The scale of the bunch is opened letter by letter, one share on each level of each letter; the scale integrates to the pin, and what is left is the form of the bunch family on its simplex times the form of the room family on its own. The level shares are the pattern shares of the port (Theorem 15.15) summed over each level: at width one the children of the level $d$ are all $\mathcal{B}_j|R$ and those of the level

$r$ all $\mathcal{B}_j/B$, up to constant factors of their weights (Theorem 12.19), so the room family is the children of the port with equal children merged. At the level shares $(a_j^d, b_j^d, a_j^r, b_j^r) = (s_j, 0, 0, t_j)$, on the wall $\kappa_B = 0$, the two families are the faces of Theorem 16.25. Left unopened on the room, as in Theorem 19.8, the shape enters the pooled family only through the hinge values of $B$ by its letters, and a rescaling of the hinge variable absorbs one of them: the shares that separate the shape from the room are one fewer than the hinge values of $B$ by the active letters counted up to constant factors, none when every active letter reads $B$ alike (Proposition 19.12), as for one table at a full lean (Theorem 19.10).

**Corollary 19.14** (One Skeleton value pins the share)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ have real exponents and be on the balance wall with every proper wall positive, and let a bunch $B$ have width one in every table, the weights factoring across $B$ and $R$. Suppose that every active letter reads $B$ alike, its hinge value a constant multiple of one function $\gamma$ on $\Delta_B$, and reads $R$ alike with the same constant, its hinge value of $R$ that constant times one function $\gamma_R$ on $\Delta_R$: the active letters have the one Skeleton value $\gamma_R(v)/\gamma(u)$ on the scale of $B$ at every gauge $(u, v) \in \Delta_B \times \Delta_R$ (Corollary 19.5). Then $0 < \kappa_B < 2\sigma$, $\sigma$ the total strength, and for all level shares $(\mathbf{a}, \mathbf{b})$ of Theorem 19.13, those of the inactive letters 0, with $a_j^d + a_j^r = s_j$, $b_j^d + b_j^r = t_j$ and $\sum_j (a_j^r + b_j^d) = \kappa_B$,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix}\kappa_B, & 2\sigma - \kappa_B\\ & 2\sigma\end{bmatrix} P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b})\, P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}),$$

*the product of the two periods not depending on the level shares: the Beta corner of the hinge times the periods of the bunch family and the room family. One active letter has one Skeleton value: for one table at $(0, t)$, at the level shares $(a^d, b^d, a^r, b^r) = (0, \kappa_B, 0, t - \kappa_B)$, the two periods are those of the forms $u^{\mathbf{m}|_B - \mathbf{1}} \Psi_{\mathcal{B}/R}^{-\kappa_B} \Psi_{\mathcal{B}|B}^{-(t-\kappa_B)}$ on $\Delta_B$ and $v^{\mathbf{m}|_R - \mathbf{1}} \Psi_{\mathcal{B}|R}^{-\kappa_B} \Psi_{\mathcal{B}/B}^{-(t-\kappa_B)}$ on $\Delta_R$, and at $(s, 0)$, at the level shares $(s - \kappa_B, 0, \kappa_B, 0)$, the same holds with the letters of $K$.*

**Lemma 19.15** (Slopes and the range of the ratios)**.** *In Theorem 19.13 with every $s_j$ and $t_j$ positive, the even share $\sigma_j^d$ of table $j$ enters the two families only through $\big(\varrho_{j,B}(u)/\varrho_{j,R}(v)\big)^{-\sigma_j^d}$, $(u, v) \in \Delta_B \times \Delta_R$, the ratio of the hinge values of the bunch over that of the room in table $j$ (Definition 19.2). The wall at $S$ of either family is affine in the shares, with the* slope *in $\sigma_j^d$ the width of $S$ in the table of the level $d$ of table $j$ less its width in the table of the level $r$ of table $j$: with $r_j^+$ the rank function of the cell of table $j$ on the side of $S$,*

$$\begin{aligned} &\big[r_j^+(S + g) - r_j^+(S)\big] + \big[r_j^+(B \setminus S + g) - r_j^+(B \setminus S)\big] - 1 \qquad (S \subseteq B),\\ &1 - \big[r_j^+(S + g) - r_j^+(S)\big] - \big[r_j^+(R \setminus S + g) - r_j^+(R \setminus S)\big] \qquad (S \subseteq R). \end{aligned}$$

*$\sup_{\Delta_B} \varrho_{j,B} < \infty$ exactly when no slope of table $j$ on $B$ is positive and $\inf_{\Delta_B} \varrho_{j,B} > 0$ exactly when none on $B$ is negative; $\inf_{\Delta_R} \varrho_{j,R} > 0$ exactly when no slope of table $j$ on $R$ is positive and $\sup_{\Delta_R} \varrho_{j,R} < \infty$ exactly when none on $R$ is negative. So a wall of either family falls as $\sigma_j^d$ moves toward the level $d$ only if $\sup \varrho_{j,B} / \inf \varrho_{j,R} = \infty$, and as it moves toward the level $r$ only if $\inf \varrho_{j,B} / \sup \varrho_{j,R} = 0$. These conditions, like the quotient $\varrho_{j,B}/\varrho_{j,R}$, do not depend on the factorization. For one table the index $j$ is omitted.*

A wall that falls as the even share moves toward the level being closed on is therefore the same thing as the two levels coming to overlap at that wall (Proposition 19.25): the quotient $\varrho_B/\varrho_R$ of the even letters of the two levels, below one where the closing starts, is unbounded near the face where the wall sits.

## 19.6 Closing at a wall and closing at the gauge

A share between the levels of a letter can be closed in two places. *At the gauge* every letter is expanded about its value at a point of the scale; the terms are nonnegative and nothing is asked of the Skeleton values. That closing is Theorem 15.9, and it closes every letter on the scale at once. *At a wall* a letter is expanded about one of its two extreme coefficients, the other coefficients entering as integer powers; the terms then separate over the bunch and the room, and the expansion holds as far as the letters are separated along the scale. The closing at a wall holds for any number of letters and any width; across a bunch of width one it is a lattice of products of periods of the bunch family and the room family, and it stops where the values no longer stand in one order, which for one table is where the two levels overlap (Proposition 19.25).

**Definition 19.16** (Letters on the scale)**.** Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ have real masses and real exponents of either sign, let $B$ be a proper bunch with room $R$, and read $\alpha_B = \rho u$ with $u \in \Delta_B$. At the gauge $(u, \alpha_R)$ the *letters on the scale* of $B$ are the $2N$ polynomials in $\rho$

$$\rho^{-d_j(B)} K_j(\rho u, \alpha_R) = \sum_{l=0}^{c_j(B)} \rho^l \, K_{j,B}^{(d_j(B)+l)}(u, \alpha_R), \qquad \rho^{-d_j^*(B)} \Psi_j(\rho u, \alpha_R) = \sum_{l=0}^{c_j(B)} \rho^l \, \Psi_{j,B}^{(d_j^*(B)+l)}(u, \alpha_R),$$

with the letter exponents $s_j$ and $t_j$: their coefficients are the level and colevel letters of Definition 12.16 counted from their floors, and divided by their values at $\rho = 1$ they are the Skeleton polynomials $\mathsf{Q}_{j,B}^{K}(\rho; (u, \alpha_R))$ and $\mathsf{Q}_{j,B}^{\Psi}(\rho; (u, \alpha_R))$ (Definition 12.10). Listed as $L_i(\rho) = \sum_{l=0}^{c_i} \rho^l L_i^{(l)}$, $i = 1, \dots, 2N$, with the letter exponent $w_i$ and the width $c_i = c_j(B)$ of their table, every coefficient is positive (Theorem 12.7), a letter of width zero is constant, and the scale integral of Theorem 15.3 at the gauge $(u, \alpha_R)$ is

$$\mathrm{Sc}_B(u, \alpha_R; \mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_0^\infty \rho^{\kappa_B - 1} \prod_i L_i(\rho)^{-w_i} \, d\rho,$$

convergent exactly when $0 < \kappa_B < \sum_i w_i c_i$, where $\sum_i w_i c_i = \kappa_B + \kappa_B^{\mathrm{top}}$.

**Definition 19.17** (Letters read from an end of the scale; separated groups)**.** Fix $\lambda \in (0, 1)$. A letter $L(\rho) = \sum_{l=0}^{c} \rho^l L^{(l)}$ with positive coefficients is *read from the bottom up to $a$* when $\sum_{l \ge 1} a^l L^{(l)} \le \lambda \, L^{(0)}$, and *read from the top down to $a$* when $\sum_{l < c} a^l L^{(l)} \le \lambda \, a^c L^{(c)}$. A partition of the letters on the scale of $B$ into *groups* $\mathcal{G}_1, \dots, \mathcal{G}_q$ is *separated* by the scales $0 < a_1 < \cdots < a_{q-1}$ when every letter of $\mathcal{G}_k$ is read from the bottom up to $a_{k-1}$ (for $k \ge 2$) and from the top down to $a_k$ (for $k \le q-1$); a letter whose exponent is a nonpositive integer is a polynomial factor of the integrand, may be put in any group, and nothing is asked of it. A group $\mathcal{G}$ has the product $f_{\mathcal{G}} = \prod_{i \in \mathcal{G}} L_i^{-w_i}$, the *scale degree* $w_{\mathcal{G}} = \sum_{i \in \mathcal{G}} w_i c_i$, and the *bottom* and *top coefficients* $\beta_{\mathcal{G},n}^{\downarrow}$ and $\beta_{\mathcal{G},n}^{\uparrow}$,

$$f_{\mathcal{G}}(\rho) = \sum_{n \ge 0} \beta_{\mathcal{G},n}^{\downarrow} \, \rho^n \quad (\rho \text{ small}), \qquad f_{\mathcal{G}}(\rho) = \rho^{-w_{\mathcal{G}}} \sum_{n \ge 0} \beta_{\mathcal{G},n}^{\uparrow} \, \rho^{-n} \quad (\rho \text{ large}).$$

Below the scale $a$ a letter read from the bottom is its lowest coefficient up to the factor $1 + \lambda$, and above $a$ a letter read from the top is its top term $\rho^c L^{(c)}$ up to the same factor. At width one, $L(\rho) = L^{(0)} + \rho L^{(1)}$ has the single Skeleton value $\xi = L^{(0)}/L^{(1)}$ (Corollary 19.5); it is read from the bottom up to $a$ exactly when $a \le \lambda \xi$ and from the top down to $a$ exactly when $a \ge \xi/\lambda$. So single letters of width one with the values $\xi_1, \dots, \xi_q$ are separated, by $a_k = \sqrt{\xi_k \xi_{k+1}}$, exactly when $\xi_k \le \lambda^2 \xi_{k+1}$ for every $k$: their values stand in one order with room between them.

**Theorem 19.18** (The scale closes between separated groups)**.** *At the gauge $(u, \alpha_R)$ let the letters on the scale of $B$ be separated into groups $\mathcal{G}_1, \dots, \mathcal{G}_q$, let $0 < \kappa < \sum_i w_i c_i$, and suppose $\kappa - \sum_{k' \le k} w_{\mathcal{G}_{k'}} \notin \mathbb{Z}$ for $1 \le k < q$. For $1 \le k \le q$ and $\mathbf{n} = (n_{k'})_{k' \ne k} \in \mathbb{N}^{q-1}$ put*

$$\nu_k(\mathbf{n}) = \kappa - \sum_{k'<k} \left( w_{\mathcal{G}_{k'}} + n_{k'} \right) + \sum_{k'>k} n_{k'}.$$

*(i)*

$$\int_0^\infty \rho^{\kappa-1} \prod_i L_i(\rho)^{-w_i} \, d\rho = \sum_{k=1}^{q} \sum_{\mathbf{n}} \prod_{k'<k} \beta^{\uparrow}_{\mathcal{G}_{k'}, n_{k'}} \prod_{k'>k} \beta^{\downarrow}_{\mathcal{G}_{k'}, n_{k'}} \ \mathrm{Reg} \int_0^\infty \rho^{\nu_k(\mathbf{n})-1} f_{\mathcal{G}_k}(\rho) \, d\rho;$$

*at $\kappa = \kappa_B$ the left side is $\mathrm{Sc}_B(u, \alpha_R; \mathbf{m}; \mathbf{s}, \mathbf{t})$.*

*(ii) The series converges absolutely, and the sum of the absolute values of its terms is at most $C$ times the left side, where $C$ depends only on $\lambda$, the letter exponents, the widths, $\kappa$ and the distances of the numbers $\kappa - \sum_{k' \le k} w_{\mathcal{G}_{k'}}$ from $\mathbb{Z}$.*

*(iii) $\beta^{\downarrow}_{\mathcal{G},n} = \prod_{i \in \mathcal{G}} \left( L_i^{(0)} \right)^{-w_i} [\rho^n] \prod_{i \in \mathcal{G}} \left( 1 + \sum_{l \ge 1} \rho^l L_i^{(l)} / L_i^{(0)} \right)^{-w_i}$: the bottom coefficients of a group are the integrand of Theorem 16.25 for the residue at $\kappa_B = -n$, taken for the letters of the group alone; the top coefficients are the same with the coefficients $L_i^{(c_i)}$ in place of $L_i^{(0)}$, for the residue at $\kappa_B^{\mathrm{top}} = -n$.*

Each group in turn keeps the scale. The groups that lie above it on the scale are read from the bottom, the groups below it from the top, and what is left to integrate is the keeping group alone, at an index shifted by the integers of the others, and as a finite part when that index has left the interval $(0, w_{\mathcal{G}})$ in which the group's own integral converges. The coefficients are the wall residues the family already has: for each letter read, a power of its extreme coefficient times a polynomial in its other coefficients. Every coefficient of $K_j$ is a sum of the pattern terms $u^p K_{j,(p)}(\alpha_R)$ and every coefficient of $\Psi_j$ of the terms $u^{B \setminus p} \Psi_{j,(p)}(\alpha_R)$, so the children enter with their pattern index, as in the record of a peel (Definition 14.2).

**Corollary 19.19** (Two groups: the walls of one against the continuation of the other)**.** *Let the letters be separated into a lower group $\mathcal{L}$ and an upper group $\mathcal{U}$, with $0 < \kappa < \sum_i w_i c_i$ and $\kappa - w_{\mathcal{L}} \notin \mathbb{Z}$. Then*

$$\int_0^\infty \rho^{\kappa-1} f_{\mathcal{L}} f_{\mathcal{U}} \, d\rho = \sum_{n \ge 0} \beta^{\downarrow}_{\mathcal{U},n} \, \mathrm{Reg} \int_0^\infty \rho^{\kappa+n-1} f_{\mathcal{L}} \, d\rho + \sum_{n \ge 0} \beta^{\uparrow}_{\mathcal{L},n} \, \mathrm{Reg} \int_0^\infty \rho^{\kappa - w_{\mathcal{L}} - n - 1} f_{\mathcal{U}} \, d\rho.$$

*For any group, $\mathrm{Reg} \int_0^\infty \rho^{\nu-1} f_{\mathcal{G}} \, d\rho$ is the value at $\nu$ of a meromorphic function, and at a pole of that function its constant term there (Theorem 1.7); the poles are at most simple and lie among the points $\nu = -n$, of residue $\beta^{\downarrow}_{\mathcal{G},n}$, and $\nu = w_{\mathcal{G}} + n$, of residue $-\beta^{\uparrow}_{\mathcal{G},n}$, $n \in \mathbb{N}$; the two families of points meet only when $w_{\mathcal{G}} \in -\mathbb{N}$, and at a common point the two residues add.*

The scale integral of a product closes on the walls of each factor. The residues of the upper group at the wall of $B$ are paired with the scale integral of the lower group continued past its top, and the residues of the lower group at the top wall with the scale integral of the upper group continued past its bottom. No bound on a moved contour is used: both expansions hold on the stretch of the scale between the groups, and the scale is divided there.

**Theorem 19.20** (Joint closing across a bunch of width one)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ have real data, its masses on the balance wall with $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ convergent, and let a bunch $B$ have width at most one in every table, the weights factoring across $B$ and $R$. A table in which $B$ has width zero is the direct sum of its restrictions to $B$ and to $R$ (Proposition 7.9); its two letters on the scale are constants, and it enters the bunch family and the room family of Theorem 19.13 as those restrictions with its own exponents. For the $N'$ tables in which $B$ has width one the letters on the scale (Definition 19.16) are, on $\Delta_B \times \Delta_R$,*

$$K_{\mathcal{B}_j/R}(u)K_{\mathcal{B}_j|R}(v) + \rho\, K_{\mathcal{B}_j|B}(u)K_{\mathcal{B}_j/B}(v), \qquad \Psi_{\mathcal{B}_j|B}(u)\Psi_{\mathcal{B}_j/B}(v) + \rho\, \Psi_{\mathcal{B}_j/R}(u)\Psi_{\mathcal{B}_j|R}(v),$$

*of the letter exponents $s_j$ and $t_j$, with the Skeleton values of Corollary 19.5; in each the coefficient multiplying $\rho$ is its* scale coefficient*, the level letter of the level $r$ for $K_j$ and the colevel letter of the level $d$ for $\Psi_j$, and the other its* constant coefficient*. Suppose the $2N'$ letters can be listed as $\ell = 1, \dots, 2N'$, with letter exponents $w_\ell$ and Skeleton values $\xi_\ell$, so that for some $\vartheta < 1$*

$$\xi_\ell \le \vartheta\, \xi_{\ell+1} \quad \text{on } \Delta_B \times \Delta_R \qquad (1 \le \ell < 2N'),$$

*and that $\kappa_B - \sum_{\ell' \le \ell} w_{\ell'} \notin \mathbb{Z}$ for $1 \le \ell < 2N'$. For a letter $i$ and $\mathbf{n} = (n_\ell)_{\ell \ne i} \in \mathbb{N}^{2N'-1}$ put $\nu_i(\mathbf{n}) = \kappa_B - \sum_{\ell<i}(w_\ell + n_\ell) + \sum_{\ell>i} n_\ell$, and let $(\mathbf{a}, \mathbf{b})_{i,\mathbf{n}}$ be the level shares in which*

- *a letter $\ell < i$ has its constant coefficient at $-n_\ell$ and its scale coefficient at $w_\ell + n_\ell$,*
- *a letter $\ell > i$ has its scale coefficient at $-n_\ell$ and its constant coefficient at $w_\ell + n_\ell$,*
- *the letter $i$ has its scale coefficient at $\nu_i(\mathbf{n})$ and its constant coefficient at $w_i - \nu_i(\mathbf{n})$.*

*Then:*

*(i) at every $(\mathbf{a}, \mathbf{b})_{i,\mathbf{n}}$ the bunch family and the room family lie on their balance walls and their periods converge;*

*(ii) with both periods at $(\mathbf{a}, \mathbf{b}) = (\mathbf{a}, \mathbf{b})_{i,\mathbf{n}}$,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \sum_{i=1}^{2N'} \sum_{\mathbf{n} \in \mathbb{N}^{2N'-1}} \prod_{\ell \ne i} \frac{(-1)^{n_\ell}(w_\ell)_{n_\ell}}{n_\ell!}\, \Gamma^{\,2}_{\,1}\!\begin{bmatrix} \nu_i(\mathbf{n}),\ w_i - \nu_i(\mathbf{n}) \\ w_i \end{bmatrix} \times P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b})\, P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}),$$

*where the stacked Gamma is 0 when $w_i \in -\mathbb{N}$; when no letter exponent lies in $-\mathbb{N}$ the coefficient is $\prod_{\ell \ne i} \frac{(-1)^{n_\ell}}{n_\ell!}\, \Gamma^{\,2N'+1}_{\,2N'}\!\begin{bmatrix} (w_\ell + n_\ell)_{\ell \ne i},\ \nu_i(\mathbf{n}),\ w_i - \nu_i(\mathbf{n}) \\ (w_\ell)_\ell \end{bmatrix}$;*

*(iii) the series converges absolutely, the absolute values of all its terms add up to at most $C\, P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$, and those with $|\mathbf{n}| = M$ to at most $C'(1+M)^D \vartheta^M\, P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$, where $C$, $C'$ and $D$ depend only on $\vartheta$, the exponents and $\kappa_B$.*

The $2N'$ Skeleton values stand in one order on the whole shape domain, and each letter in turn keeps the scale. A letter whose value lies below the keeper's is read from the top of the scale, so its constant coefficient is opened; a letter whose value lies above is read from the bottom, so its scale coefficient is opened: each letter opens the one of its two terms that is the smaller at the keeper's place on the scale. An opened coefficient sits at the share $-n$, which puts the $n$-th power of that coefficient into the forms of the two families, a moment; the keeper carries the Beta corner $\Gamma^{\,2}_{\,1}\!\left[\begin{smallmatrix} \nu,\ w-\nu \\ w \end{smallmatrix}\right]$ of a linear letter (Proposition 15.1 with one coefficient), continued when $\nu$

has left $(0, w)$. So closing across the bunch pairs moments of the bunch family with moments of the room family on $2N'$ lattices of dimension $2N'-1$, the free shares of Theorem 19.13 closed at once. Nothing is asked of the walls of the two families, (i) being a conclusion, and nothing uniform in $\mathbf{n}$ is needed beyond (iii), which bounds the whole lattice by the period itself.

The values are quotients of hinge values, the room's over the bunch's by each letter (Corollary 19.5). The value of $K_j$ over that of $\Psi_j$ is $\varrho_{j,B}(u)/\varrho_{j,R}(v)$, the even letters of its two levels, and their product is the corresponding quotient of the odd letters: the two values of a table sit on either side of the geometric mean set by its odd letters, at the distance set by its even letters. The $N'$ quotients of even letters and the $N'-1$ quotients of the products of one table by those of another are what the $N'$ even shares and the $N'-1$ free odd shares of Theorem 19.13 carry; for one table only $\varrho_B/\varrho_R$ is left.

**Corollary 19.21** (Closing the even share)**.** *For one table in Theorem 19.13 let $f(\sigma^d)$ be the integrand of its one-table display.*

*(i) If $\sup_{\Delta_B}\varrho_B < \inf_{\Delta_R}\varrho_R$, then $P_F(\mathbf{m};s,t) = \sum \operatorname{Res} f$ over the poles of the Gamma pair $\Gamma(a^d)\Gamma(b^d)$ of the level $d$, at $\sigma^d=\delta^d-n$ and $\sigma^d=-\delta^d-n$, $n\in\mathbb{N}$, the $n$-th terms of the size $n^{s+t-2}\big(\sup\varrho_B/\inf\varrho_R\big)^n$; every period at these points converges, and when $2\delta^d\notin\mathbb{Z}$ the $n$-th terms are at most $C(1+n)^D\big(\sup\varrho_B/\inf\varrho_R\big)^n P_F(\mathbf{m};s,t)$, with $C$ and $D$ depending only on that ratio, the exponents and $\kappa_B$ (Theorem 19.20(i),(iii) with one table, its letters listed as $K$, then $\Psi$). If $\inf_{\Delta_B}\varrho_B > \sup_{\Delta_R}\varrho_R$, then $P_F(\mathbf{m};s,t) = -\sum\operatorname{Res} f$ over the poles of the Gamma pair of the level $r$, at $\sigma^r=\pm\delta^r-n$. The share closes toward the level whose even letter is the smaller.*

*(ii) At $\sigma^d=\delta^d-n$ the level shares are $(a^d,b^d,a^r,b^r)=(2\delta^d-n,\,-n,\,s-2\delta^d+n,\,t+n)$, and the residue is*

$$\frac{(-1)^n}{n!}\Gamma^{\,3}_{\,2}\!\left[\begin{matrix}2\delta^d-n,\ s-2\delta^d+n,\ t+n\\ s,\ t\end{matrix}\right] P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})\,P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b}),$$

*where $P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})} = \int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}} K_{\mathcal{B}/R}^{-2\delta^d} K_{\mathcal{B}|B}^{-(s-2\delta^d)} \Psi_{\mathcal{B}|B}^{-t}\,\varrho_B^n\,d\omega_B$ is the $n$-th moment of $\varrho_B$ against a form that does not depend on $n$, and $P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}$ likewise the $n$-th moment of $\varrho_R^{-1}$. At $\sigma^d=-\delta^d-n$ the level shares are $(-n,\,-2\delta^d-n,\,s+n,\,t+2\delta^d+n)$, and the same holds with the form $\Psi_{\mathcal{B}/R}^{2\delta^d}K_{\mathcal{B}|B}^{-s}\Psi_{\mathcal{B}|B}^{-t-2\delta^d}$ on $B$ and its counterpart on $R$. Closing pairs the moments of the ratios $\varrho_B$ and $\varrho_R$ of the bunch and of the room. The two lattices of a level meet exactly when $2\delta^\ell\in\mathbb{Z}$, at the centre exactly when $\mu_B\in\mathbb{Z}$. At a common point, where $\Gamma(a^d)$ sits at $-k$ and $\Gamma(b^d)$ at $-n$, the pole is double: with $h=\Gamma^{\,2}_{\,2}\!\left[\begin{matrix}s+k,\ t+n\\ s,\ t\end{matrix}\right] P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}$,*

$$\operatorname{Res} f = \frac{(-1)^{n+k}}{n!\,k!}\,h\,\Big((\log\Gamma)'(n+1)-(\log\Gamma)'(t+n)+(\log\Gamma)'(k+1)-(\log\Gamma)'(s+k) \\ +\partial_{\sigma^d}\log\big(P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}\big)\Big).$$

*At integer exponents the two differences of $(\log\Gamma)'$ are heads, finite sums of reciprocals (Proposition 5.13), and $(\log\Gamma)'(j+1)=-(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))$ for $j\in\mathbb{N}$ (Proposition 5.14).*

Closed on a lattice, the even share writes the period as a row over finitely many data of the room whenever the room's periods along the lattice satisfy a recurrence of finite order: the room's initial periods, and at a meeting of the two lattices their derivatives, against sums along the lattice of the bunch's data and the room's fundamental solutions (Proposition 43.10).

Off resonance every pole is simple and the closed value is a sum over two lattices of products of moments of the two ratios $\varrho_B$ and $\varrho_R$, with stacked Gamma coefficients: no heads. Each lattice's series has a pole where the two lattices of the level meet, from $\Gamma(2\delta^d - n)$ and $\Gamma(-2\delta^d - n)$, and their sum tends to the double-pole residues of (ii); the heads at the centre are this meeting. When the ranges of the even letters of the two levels overlap, the quotient $\varrho_B(u)/\varrho_R(v)$ taking values on both sides of 1, neither closing of (i) applies; the residues of a family at one of its walls are those of Theorem 16.25, periods of the family's faces.

**Corollary 19.22** (Joint closing, nested)**.** *In Theorem 19.20 let $B'$ be a bunch of the bunch family, of width at most one in each of its tables, with factoring weights, and let the letters of the bunch family on the scale of $B'$ have their Skeleton values in one order with a ratio $\vartheta' < 1$. Then at every $(\mathbf{a},\mathbf{b})_{i,\mathbf{n}}$ at which the non-integrality condition of Theorem 19.20 holds for the bunch family at $B'$, $P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})$ is given by Theorem 19.20 for the family $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ at $B'$; likewise for the room family at a bunch of $R$. Continuing down a tree of splits of width one, each node splitting its family at a bunch of width one into the bunch family and the room family, with the values ordered at every node, $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is a nested lattice sum of stacked Gammas and of the periods of the families at the leaves, each leaf at exponents shifted by the integers of the lattices above it. The sum is taken from the leaves up, and at each node it converges absolutely at that node's own root rate.*

The tree of the nesting is a tree of splits of width one and not the Foreman tree of Definition 14.4: a node may be any bunch of width one at which the values are ordered. At a lattice point of the node above, a letter of the family below sits at an exponent $-n$ or $w+n$; the letters below are the same letters at shifted powers, and the next split applies to them because whether the values are in one order is read off the tables. Nothing uniform in the outer integers is used: at each lattice point the inner sum is an identity. The constant $C$ of Theorem 19.20(iii) at an inner node depends on the exponents there and grows at most geometrically in the outer integers; that matters only for summing in another order. The multiple series converges absolutely when the outer ratio $\vartheta$ beats that growth, and otherwise the sum is the iterated one.

**Proposition 19.23** (Values in one order without room)**.** *In Theorem 19.20 let $\xi_\ell < \xi_{\ell+1}$ on $\Delta_B \times \Delta_R$ for every $\ell$, with no $\vartheta < 1$. For $0 < \zeta < 1$ let $P^{(\zeta)}$ be the integral obtained from $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ by multiplying the constant coefficient of the $\ell$-th letter by $\zeta^{-\ell}$; it is the period of the family in which $K_j$ and $\Psi_j$ are read from two copies of table $j$ whose level weights are scaled accordingly. Then every period of Theorem 19.20(i) converges, and with $A_{i,\mathbf{n}}$ the term of index $(i,\mathbf{n})$ of the series of Theorem 19.20(ii) and $x_i = \sum_{\ell>i} \ell w_\ell + i\sum_{\ell\le i} w_\ell - i\kappa_B$,*

$$P^{(\zeta)} = \sum_{i,\mathbf{n}} \zeta^{\,x_i + \sum_\ell |\ell - i|\, n_\ell} A_{i,\mathbf{n}} \quad \textit{absolutely}, \qquad P^{(\zeta)} \to P_F(\mathbf{m};\mathbf{s},\mathbf{t}) \ \ (\zeta \uparrow 1).$$

*So the lattice sum of Theorem 19.20(ii) is summable to the period by scaling the level weights, and equals the period whenever it converges absolutely.*

**Proposition 19.24** (The same shares closed at the gauge)**.** *Let $F = ((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ have complex exponents and masses $\mathbf{m}$ and let a bunch $B$ have width one in every table, the weights factoring across $B$ and $R$, with $\operatorname{Re}\kappa_B > 0$ and $\operatorname{Re}\kappa_B^{\mathrm{top}} > 0$. Write the Skeleton polynomials at the gauge $(u,v) \in \Delta_B \times \Delta_R$ as*

$$\mathsf{Q}^K_{j,B}\big(x;(u,v)\big) = \pi_{j,d} + \pi_{j,r}\,x, \qquad \mathsf{Q}^\Psi_{j,B}\big(x;(u,v)\big) = \pi^*_{j,r} + \pi^*_{j,d}\,x,$$

*the chances of the two levels under the basis law and the cobasis law at $(u,v)$, each the sum over its level of the pattern probabilities of Definition 12.25. Then, with $M=|\mathbf{n}|+|\mathbf{n}'|$,*

$$\mathrm{Sc}_B(u,v;\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_j K_j(u,v)^{-s_j}\Psi_j(u,v)^{-t_j} \sum_{\mathbf{n},\mathbf{n}'\in\mathbb{N}^N} \prod_j \frac{(s_j)_{n_j}(t_j)_{n'_j}}{n_j!\,n'_j!} \\ \times \Big( \prod_j \pi_{j,r}^{n_j}\pi_{j,d}^{*\,n'_j}\ \Gamma_{\,1}^{\,2}\begin{bmatrix}\kappa_B,\ M+1\\ \kappa_B+M+1\end{bmatrix} + \prod_j \pi_{j,d}^{n_j}\pi_{j,r}^{*\,n'_j}\ \Gamma_{\,1}^{\,2}\begin{bmatrix}\kappa_B^{\mathrm{top}},\ M+1\\ \kappa_B^{\mathrm{top}}+M+1\end{bmatrix}\Big),$$

*a series converging absolutely, of nonnegative terms when the data are real with $s_j,t_j\ge 0$, whatever the order of the Skeleton values: it is Theorem 15.9 at width one, whose exceedance polynomials are the constants $\pi_{j,r}$, $\pi^*_{j,d}$ on the bunch and $\pi_{j,d}$, $\pi^*_{j,r}$ on the room. Without the prefactor the terms with $M$ fixed add up in absolute value to at most $\frac{(\sum_j(|s_j|+|t_j|))_M}{M!}\big(\lambda_B^M/\operatorname{Re}\kappa_B+\lambda_R^M/\operatorname{Re}\kappa_B^{\mathrm{top}}\big)$, with $\lambda_B=\max_j(\pi_{j,r},\pi^*_{j,d})<1$ and $\lambda_R=\max_j(\pi_{j,d},\pi^*_{j,r})<1$.*

At the gauge every letter is closed about its value at one point of the scale, both levels together. The lattice has one index for each letter, and what is raised to the powers are the chances of the levels at that point. Nothing is asked of the order of the values, and the root rate is the largest chance of a level. The price is that the closed letters $K_j(u,v)$ and $\Psi_j(u,v)$ stay in front, so the terms do not separate over the bunch and the room; integrated over the shapes on their ordering sectors the same closing is Corollary 15.11, with Theorem 15.14 past the walls.

**Proposition 19.25** (Overlapping levels)**.** *In Theorem 19.13 with one table let $s,t>0$ and $2\delta^d = s-\kappa_B\notin\mathbb{Z}$, and let $\mathcal{M}(x)$ be the integral over $\Delta_B\times\Delta_R$ of*

$$u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}\big(K_{\mathcal{B}/R}(u)K_{\mathcal{B}|R}(v)\big)^{\kappa_B-s}\big(K_{\mathcal{B}|B}(u)K_{\mathcal{B}/B}(v)\big)^{-\kappa_B}\big(\Psi_{\mathcal{B}|B}(u)\Psi_{\mathcal{B}/B}(v)\big)^{-t}\Big(\frac{\varrho_B(u)}{\varrho_R(v)}\Big)^x.$$

*The terms of the two lattices of the level $d$ in Corollary 19.21(ii) are $A_n=a_n\mathcal{M}(n)$ and $A'_n=a'_n\mathcal{M}(n+2\delta^d)$, with*

$$a_n = \Gamma_{\,2}^{\,2}\begin{bmatrix}2\delta^d,\ 1-2\delta^d\\ s,\ t\end{bmatrix}\Gamma_{\,2}^{\,2}\begin{bmatrix}t+n,\ \kappa_B+n\\ 1+n,\ 1-2\delta^d+n\end{bmatrix},$$
$$a'_n = -\Gamma_{\,2}^{\,2}\begin{bmatrix}2\delta^d,\ 1-2\delta^d\\ s,\ t\end{bmatrix}\Gamma_{\,2}^{\,2}\begin{bmatrix}s+n,\ t+2\delta^d+n\\ 1+n,\ 1+2\delta^d+n\end{bmatrix}.$$

*Suppose these integrals converge for every $n$. If $\varrho_B(u)>\varrho_R(v)$ on a set of positive measure, then $|A_n+A'_n|\to\infty$; likewise for the level $r$ if $\varrho_B(u)<\varrho_R(v)$ on a set of positive measure. So when the two levels* overlap*, both sets having positive measure, the lattices of neither level sum to the period by convergent periods.*

The two lattices of a level have opposite signs and coefficients whose ratio tends to $-1$. When $\sup\varrho_B<\inf\varrho_R$ each of them converges by itself. When $\varrho_B(u)<\varrho_R(v)$ throughout with $\sup\varrho_B=\inf\varrho_R$ neither need converge; the moments at $n$ and at $n+2\delta^d$ then agree to leading order, so the two lattices cancel to that order, and Proposition 19.23 sums them to the period by scaling the level weights. Past that, the two lattices part and the series diverges. By Lemma 19.15 the overlap reaches a wall when $\varrho_B/\varrho_R$ is unbounded, and then the periods themselves stop converging along the lattice. An overlap is closed on the totals of the bunch and the room (Section 19.7), and, read in one scale, on the lattice of continued periods of a level whenever the reach of the bunch on its side lies below the decay of the other (Theorem 20.39).

**Proposition 19.26** (Between groups at the wall, inside a group at the gauge)**.** *In Theorem 19.18 fix a group $\mathcal{G}$, an index $\nu$ with $\nu \notin -\mathbb{N}$ and $\nu - w_{\mathcal{G}} \notin \mathbb{N}$, and a point $\rho_0$ of the scale. For the letter $i$ of $K_j$ in $\mathcal{G}$ put $\mathsf{Q}_i = \mathsf{Q}^K_{j,B}(\cdot;(\rho_0 u, \alpha_R))$ and $\mathsf{Q}_i^\top = \mathsf{Q}^K_{j,R}(\cdot;(\rho_0 u, \alpha_R))$, and for that of $\Psi_j$ the same with $\Psi$; then $\mathsf{Q}_i(x) = L_i(x\rho_0)/L_i(\rho_0)$ and $\mathsf{Q}_i^\top(x) = x^{c_i}\mathsf{Q}_i(1/x)$ (Proposition 12.11(i),(ii)), and*

$$\operatorname{Reg}\int_0^\infty \rho^{\nu-1} f_{\mathcal{G}}(\rho)\,d\rho = \rho_0^\nu f_{\mathcal{G}}(\rho_0)\Big(\operatorname{Reg}\int_0^1 x^{\nu-1}\prod_{i\in\mathcal{G}}\mathsf{Q}_i(x)^{-w_i}dx + \operatorname{Reg}\int_0^1 x^{w_{\mathcal{G}}-\nu-1}\prod_{i\in\mathcal{G}}\mathsf{Q}_i^\top(x)^{-w_i}dx\Big).$$

*Each of the two finite parts is the series of Theorem 15.9(ii) for these polynomials, absolutely convergent, with its Beta corners continued as in Theorem 15.14. So for every separated partition every share of the scale is closed: between the groups on the lattices of Theorem 19.18, inside a group at the gauge. With one group this is Theorem 15.9; with single letters of width one in one order it is Theorem 19.20.*

A group that cannot be split, because the values of its letters do not stand in one order, is kept as one Breta of one bunch (Proposition 15.1); it is closed about a point of the scale, where no order is needed, and the letters outside it are closed about its ends. The two closings do not compete: the wall closing does as much as the scale separates, and the gauge closing does the rest. When its index lies in $(0, w_{\mathcal{G}})$ and its letter exponents are nonnegative, a group read on the totals of the bunch and the room closes about constant references, its terms pairing the bunch with the room at width one (Section 19.7).

## 19.7 The scale on the totals

The scale of a bunch is the quotient of two totals, that of the bunch and that of the room, and the two totals lie on the simplex $\Delta_{\{B,R\}}$ of the two blocks (Theorem 3.12). Read there, a letter on the scale is homogeneous in the totals, with its level or colevel letters as coefficients. At width one it is linear, with its constant coefficient at one vertex of the simplex and its scale coefficient at the other, and its Skeleton value is the point of the scale where its two terms are equal. The free shares of Theorem 19.13 are the share side of this integral over the totals, the pin being the shape integral over the totals (Theorem 3.9), and the lattices of Section 19.6 close them. The integral over the totals can also be closed as it stands. On the simplex a linear letter is at most the larger of its two coefficients, whatever their ratio, so it closes about a constant reference with nonnegative terms, which pair moments of the bunch with moments of the room. No order of the values is asked, and the overlap that stops the lattices of a level (Proposition 19.25) is not seen.

**Definition 19.27** (The totals and their letters)**.** The *totals* of a bunch $B$ and its room $R$ are the point $x = (x_B, x_R)$ of the shape simplex $\Delta_{\{B,R\}}$ of the two blocks. Let $\mathbf{m}^{\mathrm{tot}} = (m_B^{\mathrm{tot}}, m_R^{\mathrm{tot}})$ have positive entries, let $\phi$ be a nonnegative measure on $\Delta_B \times \Delta_R$, and let finitely many *letters on the totals*

$$\widehat{L}_i(x) = \sum_{l=0}^{c_i} x_B^{\,l}\, x_R^{\,c_i-l} L_i^{(l)}, \qquad i = 1, \dots, q,$$

with letter exponents $w_i \in \mathbb{R}$, have coefficients $L_i^{(l)} \geq 0$ that are measurable functions of the shapes $(u, v) \in \Delta_B \times \Delta_R$, each letter positive on the interior of $\Delta_{\{B,R\}}$ at $\phi$-almost every shape. The

*integral over the totals* at the shapes $(u, v)$ is the inner integral below, over $\Delta_{\{B,R\}}$, and its integral against $\phi$ is the Trailmoment on the source $\Delta_{\{B,R\}} \times \Delta_B \times \Delta_R$ (Definition 5.1) with the coefficient function $(x, u, v) \mapsto (\widehat{L}_i(x))_i$ and the input $x^{\mathbf{m}^{\mathrm{tot}}-\mathbf{1}} d\omega(x)\, \phi(du, dv)$, read over the whole source,

$$\begin{aligned}\big(\mathsf{T}_{\mathbf{w};I_q}[(\widehat{L}_i)_i]\; x^{\mathbf{m}^{\mathrm{tot}}-\mathbf{1}} d\omega(x)\, \phi(du, dv)\big)\big(\Delta_{\{B,R\}} \times \Delta_B \times \Delta_R\big)& \\ = \int_{\Delta_B \times \Delta_R} \int_{\Delta_{\{B,R\}}} x^{\mathbf{m}^{\mathrm{tot}}-\mathbf{1}} \prod_i \widehat{L}_i(x)^{-w_i}\, d\omega(x)\, \phi(du, dv),&\end{aligned}$$

wherever it converges; its kernel $\mathsf{Z}_{\mathbf{w};I_q}((\widehat{L}_i)_i) = \prod_i \widehat{L}_i^{-w_i}$ is the product of the one-scale kernels $\mathsf{Z}_{w_i;(1)}(\widehat{L}_i)$ (Definition 2.12, Theorem 5.21(i)). For one linear letter $x_R L^{(0)} + x_B L^{(1)}$ the integral over the totals is the Breta with the one bunch $\{B, R\}$, the shape index $\mathbf{m}^{\mathrm{tot}}$, the scale index $w$ and the unit configuration (Definition 3.1):

$$\int_{\Delta_{\{B,R\}}} x^{\mathbf{m}^{\mathrm{tot}}-\mathbf{1}} \big(x_B L^{(1)} + x_R L^{(0)}\big)^{-w} d\omega(x) = \mathsf{B}^{\{\{B,R\}\}}_{\mathbf{m}^{\mathrm{tot}},\, w;\, I_{\{B,R\}}}\big(L^{(1)}, L^{(0)}\big).$$

**Theorem 19.28** (The scale on the totals)**.** *At the gauge $(u, \alpha_R)$ let $\kappa_B > 0$ and $\kappa_B^{\mathrm{top}} > 0$, and read every letter on the scale of $B$ (Definition 19.16) on the totals,*

$$\widehat{L}_i(x) = x_R^{c_i} L_i(x_B/x_R) = \sum_{l=0}^{c_i} x_B^{\,l}\, x_R^{\,c_i - l} L_i^{(l)},$$

*a letter of width zero being a constant on the totals.*

*(i) At every point $\rho_0 > 0$ of the scale, which may depend on the shapes,*

$$\mathrm{Sc}_B(u, \alpha_R; \mathbf{m}; \mathbf{s}, \mathbf{t}) = \rho_0^{\kappa_B} \int_{\Delta_{\{B,R\}}} x_B^{\kappa_B - 1} x_R^{\kappa_B^{\mathrm{top}} - 1} \prod_i \widehat{L}_i(\rho_0 x_B,\ x_R)^{-w_i}\, d\omega(x),$$

*both sides converging together. At $\rho_0 = 1$ and $\alpha_R = v \in \Delta_R$ the point $(\rho u, v)$ is projectively $(x_B u, x_R v) \in \Delta_{\mathcal{E}}$, whose totals on the blocks $B$ and $R$ are $x_B$ and $x_R$ (Theorem 3.12), with $\rho = x_B/x_R$. On the balance wall $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ is the Trailmoment of Definition 19.27 for the letters $\widehat{L}_i$ at $\mathbf{m}^{\mathrm{tot}} = (\kappa_B, \kappa_B^{\mathrm{top}})$ with $\phi = u^{\mathbf{m}|_B - \mathbf{1}} v^{\mathbf{m}|_R - \mathbf{1}} d\omega_B(u)\, d\omega_R(v)$; both sides converge together.*

*(ii) Let $B$ have width at most one in every table, the weights factoring across $B$ and $R$; write the letters of width one $L_i(\rho) = L_i^{(0)} + \rho L_i^{(1)}$, with their constant and scale coefficients and their Skeleton values $\xi_i = L_i^{(0)}/L_i^{(1)}$ (Theorem 19.20), and fix one of them, $i_0$. At the point $\rho_0 = \xi_{i_0}$ the two terms of $i_0$ are equal and its letter on the totals is the constant $L_{i_0}^{(0)}$. The products in the displays run over the letters of width one, and both right sides carry the further factor $\prod L_i^{-w_i}$ over the constant letters of the tables of width zero:*

$$\begin{aligned}\mathrm{Sc}_B(u, v; \mathbf{m}; \mathbf{s}, \mathbf{t}) &= \big(L_{i_0}^{(0)}\big)^{\kappa_B - w_{i_0}} \big(L_{i_0}^{(1)}\big)^{\kappa_B^{\mathrm{top}} - w_{i_0}} \\ &\qquad \times \int_{\Delta_{\{B,R\}}} x_B^{\kappa_B - 1} x_R^{\kappa_B^{\mathrm{top}} - 1} \prod_{i \neq i_0} \big(x_R L_i^{(0)} L_{i_0}^{(1)} + x_B L_{i_0}^{(0)} L_i^{(1)}\big)^{-w_i}\, d\omega(x) \\ &= \xi_{i_0}^{\kappa_B} \prod_i \big(L_i^{(0)}\big)^{-w_i} \int_{\Delta_{\{B,R\}}} x_B^{\kappa_B - 1} x_R^{\kappa_B^{\mathrm{top}} - 1} \prod_{i \neq i_0} \Big(x_R + x_B \frac{\xi_{i_0}}{\xi_i}\Big)^{-w_i} d\omega(x).\end{aligned}$$

*On the balance wall $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the Trailmoment of Definition 19.27 for these linear letters at $\mathbf{m}^{\rm tot}=(\kappa_B,\kappa_B^{\rm top})$, with $\phi$ of (i) multiplied by $(L_{i_0}^{(0)})^{\kappa_B-w_{i_0}}(L_{i_0}^{(1)})^{\kappa_B^{\rm top}-w_{i_0}}$ and by the further factor of the constant letters: the coefficients of each linear letter are the constant coefficient of one letter times the scale coefficient of the other, and their ratio is the ratio of the two Skeleton values.*

*(iii) In (ii) let every $w_i>0$, and open each linear letter by Corollary 4.4, with the share $\nu_i$ on $x_BL_{i_0}^{(0)}L_i^{(1)}$ and $w_i-\nu_i$ on $x_RL_i^{(0)}L_{i_0}^{(1)}$, on a contour with $0<\operatorname{Re}\nu_i<w_i$ and $\kappa_B-w_{i_0}<\sum_{i\neq i_0}\operatorname{Re}\nu_i<\kappa_B$. The integral over the totals is then the Beta corner $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}\nu_{i_0},&w_{i_0}-\nu_{i_0}\\&w_{i_0}\end{smallmatrix}\right]$ of the letter $i_0$ at $\nu_{i_0}=\kappa_B-\sum_{i\neq i_0}\nu_i$, so that the shares on the scale coefficients add to $\kappa_B$ identically, and, with the further factor of (ii) on the right,*

$$\mathrm{Sc}_B(u,v;\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\sum_i\nu_i=\kappa_B}\prod_i\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}\nu_i,&w_i-\nu_i\\&w_i\end{smallmatrix}\right]\big(L_i^{(0)}\big)^{-(w_i-\nu_i)}\big(L_i^{(1)}\big)^{-\nu_i},$$

*the share side of the scale integral with one share per letter (Corollary 15.5). Integrated over the shapes it is Theorem 19.13: the pin of the shares is the shape integral over the totals.*

The scale integral is the integral over the totals, and the shares of Theorem 19.13 are its share side: opening every letter but one leaves over the totals the Beta corner of the last, and the pin is what that integral asks of the shares. The point $\rho_0$ is free. At $\rho_0=1$ the totals are those of Theorem 3.12, the coordinates of a node in Theorem 20.15; at the value of a letter that letter leaves the integral, and for one table a single linear letter is left.

**Corollary 19.29** (One table: the even share on the totals)**.** *For one table in Theorem 19.13 write, on $\Delta_B\times\Delta_R$,*

$$\mathbf{E}^d=\mathbf{E}_{\mathcal{B}/R}(u)\,\mathbf{E}_{\mathcal{B}|R}(v),\qquad \mathbf{E}^r=\mathbf{E}_{\mathcal{B}|B}(u)\,\mathbf{E}_{\mathcal{B}/B}(v)$$

*for the even letters of the two levels, so that $\mathbf{E}^d/\mathbf{E}^r=\varrho_B(u)/\varrho_R(v)$, and let $2\delta^d=s-\kappa_B$ and $2\delta^r=\kappa_B-t$; here $\kappa_B^{\rm top}=s+t-\kappa_B$. Then*

$$\begin{aligned}\mathrm{Sc}_B(u,v;\mathbf{m};s,t)&=\big(K_{\mathcal{B}/R}(u)K_{\mathcal{B}|R}(v)\big)^{-2\delta^d}\big(K_{\mathcal{B}|B}(u)K_{\mathcal{B}/B}(v)\big)^{-2\delta^r}\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\rm top}),\,t;\,I_{\{B,R\}}}\big(\mathbf{E}^d,\mathbf{E}^r\big)\\&=\big(\Psi_{\mathcal{B}/R}(u)\Psi_{\mathcal{B}|R}(v)\big)^{2\delta^d}\big(\Psi_{\mathcal{B}|B}(u)\Psi_{\mathcal{B}/B}(v)\big)^{2\delta^r}\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\rm top}),\,s;\,I_{\{B,R\}}}\big(\mathbf{E}^r,\mathbf{E}^d\big),\end{aligned}$$

*and $\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\rm top}),t;I_{\{B,R\}}}(\mathbf{E}^d,\mathbf{E}^r)=(\mathbf{E}^r)^{-t}\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\rm top}),t;I_{\{B,R\}}}\big(\varrho_B(u)/\varrho_R(v),1\big)$. So $P_F(\mathbf{m};s,t)$ is the Trailmoment of Definition 19.27 for the single linear letter $x_R\mathbf{E}^r+x_B\mathbf{E}^d$ at the exponent $t$ and at $\mathbf{m}^{\rm tot}=(\kappa_B,\kappa_B^{\rm top})$, with $\phi=\big(K_{\mathcal{B}/R}K_{\mathcal{B}|R}\big)^{-2\delta^d}\big(K_{\mathcal{B}|B}K_{\mathcal{B}/B}\big)^{-2\delta^r}u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}d\omega_B\,d\omega_R$, and the shapes enter the Breta only through the ratio $\varrho_B(u)/\varrho_R(v)$ of its coefficients. Opened by Corollary 4.4, the linear letter has the share $b^d=\sigma^d-\delta^d$ on $x_B\mathbf{E}^d$, and the integrand is that of Theorem 19.13: the free even share is the share of this letter, and the integral over the totals is the even share integrated.*

**Theorem 19.30** (Closing at the peaks, one reference per coefficient)**.** *In the Trailmoment of Definition 19.27 let every $w_i>0$, and choose constants $\bar L_{i,l}\geq L_i^{(l)}$, one for each coefficient, at $\phi$-almost every shape; the* reference letters *are $\bar L_i(x)=\sum_{l=0}^{c_i}x_B^lx_R^{c_i-l}\bar L_{i,l}$. For an array $\mathbf{k}=(k_{i,l})$ with the rows $\mathbf{k}_i=(k_{i,l})_{0\leq l\leq c_i}$ put*

$$\boldsymbol{\beta}(\mathbf{k})=\big(\beta_B(\mathbf{k}),\beta_R(\mathbf{k})\big)=\Big(m_B^{\rm tot}+\sum_{i,l}l\,k_{i,l},\ m_R^{\rm tot}+\sum_{i,l}(c_i-l)\,k_{i,l}\Big),\qquad \mathbf{w}(\mathbf{k})=\big(w_i+|\mathbf{k}_i|\big)_i.$$

*Introduce one singleton scale $\mathfrak{v}_i$ per letter, $\mathfrak{V} = \{\mathfrak{v}_1, \dots, \mathfrak{v}_q\}$, take the partition $\mathcal{P}_{\mathrm{tot}} = \{\{B, R\}\} \cup \{\{\mathfrak{v}_i\} : 1 \le i \le q\}$, and let $(\pi^{\mathrm{tot}}, \ell^{\mathrm{tot}})$ be the term arrays of the letter presentation (Proposition 3.6) with the kernel column of shape exponent $0$, level $(1, 0_{\mathfrak{V}})$ and coefficient $1$, and for every $(i, l)$ with $\bar{L}_{i,l} > 0$ a column of shape exponent $\big((l, c_i - l), \mathbf{e}_i\big)$, level $(0, \mathbf{e}_i)$ and coefficient $\bar{L}_{i,l}$; $1 \oplus \bar{\mathbf{L}}$ lists the coefficients in this order. Then, with the arrays $\mathbf{k}$ supported on these columns,*

$$\int_{\Delta_B \times \Delta_R} \int_{\Delta_{\{B,R\}}} x^{\mathbf{m}^{\mathrm{tot}} - \mathbf{1}} \prod_i \widehat{L}_i(x)^{-w_i} \, d\omega(x) \, \phi(du, dv)$$
$$= \sum_{\mathbf{k}} \prod_i \frac{(w_i)_{|\mathbf{k}_i|}}{\mathbf{k}_i!} \; \mathsf{B}^{\mathcal{P}_{\mathrm{tot}}}_{(\boldsymbol{\beta}(\mathbf{k}), \mathbf{w}(\mathbf{k})), \, (1, \mathbf{w}(\mathbf{k})); \, (\pi^{\mathrm{tot}}, \ell^{\mathrm{tot}})}\big(1 \oplus \bar{\mathbf{L}}\big) \int \prod_{i,l} \big(\bar{L}_{i,l} - L_i^{(l)}\big)^{k_{i,l}} \, d\phi,$$

*a series of nonnegative terms, both sides finite or infinite together. The Breta coefficient is a number: the kernel integrates its scale to $1$, and*

$$\mathsf{B}^{\mathcal{P}_{\mathrm{tot}}}_{(\boldsymbol{\beta}(\mathbf{k}), \mathbf{w}(\mathbf{k})), \, (1, \mathbf{w}(\mathbf{k})); \, (\pi^{\mathrm{tot}}, \ell^{\mathrm{tot}})}\big(1 \oplus \bar{\mathbf{L}}\big) = \int_{\Delta_{\{B,R\}}} x^{\boldsymbol{\beta}(\mathbf{k}) - \mathbf{1}} \prod_i \bar{L}_i(x)^{-w_i - |\mathbf{k}_i|} \, d\omega(x).$$

*Each moment is a read of $\phi$ against the one-scale kernels $\mathsf{Z}_{-k_{i,l};(1)}\big(\bar{L}_{i,l} - L_i^{(l)}\big) = \big(\bar{L}_{i,l} - L_i^{(l)}\big)^{k_{i,l}}$.*

**Corollary 19.31** (Closing at the peak on the totals)**.** *In Theorem 19.30 take the references $\bar{L}_{i,l} = \binom{c_i}{l} \bar{L}_i$ with constants $\bar{L}_i > 0$ such that the deviations $h_{i,l} = \binom{c_i}{l} \bar{L}_i - L_i^{(l)}$ are nonnegative. The reference letter is then the constant $\bar{L}_i (x_B + x_R)^{c_i} = \bar{L}_i$ on the totals, the Breta coefficient is $\prod_i \bar{L}_i^{-w_i - |\mathbf{k}_i|}$ times the Beta corner $\Gamma^{\,2}_{\,1}\!\begin{bmatrix} \beta_B(\mathbf{k}), & \beta_R(\mathbf{k}) \\ \beta_B(\mathbf{k}) + \beta_R(\mathbf{k}) & \end{bmatrix}$, and that Trailmoment is*

$$\sum_{\mathbf{k}} \prod_i \frac{(w_i)_{|\mathbf{k}_i|}}{\mathbf{k}_i!} \, \bar{L}_i^{-w_i - |\mathbf{k}_i|} \; \Gamma^{\,2}_{\,1}\!\begin{bmatrix} \beta_B(\mathbf{k}), & \beta_R(\mathbf{k}) \\ \beta_B(\mathbf{k}) + \beta_R(\mathbf{k}) & \end{bmatrix} \int \prod_{i,l} h_{i,l}^{k_{i,l}} \, d\phi.$$

*For linear letters $\widehat{L}_i(x) = x_R L_i^{(0)} + x_B L_i^{(1)}$ with $\bar{L}_i \ge L_i^{(0)}, L_i^{(1)}$, with $n_i$ counting the constant coefficient and $n_i'$ the scale coefficient of the letter $i$, it is*

$$\sum_{\mathbf{n}, \mathbf{n}'} \prod_i \frac{(w_i)_{n_i + n_i'}}{n_i! \, n_i'!} \, \bar{L}_i^{-w_i - n_i - n_i'} \; \Gamma^{\,2}_{\,1}\!\begin{bmatrix} m_B^{\mathrm{tot}} + |\mathbf{n}'|, & m_R^{\mathrm{tot}} + |\mathbf{n}| \\ |\mathbf{m}^{\mathrm{tot}}| + |\mathbf{n}| + |\mathbf{n}'| & \end{bmatrix} \int \prod_i \big(\bar{L}_i - L_i^{(0)}\big)^{n_i} \big(\bar{L}_i - L_i^{(1)}\big)^{n_i'} \, d\phi.$$

*If a coefficient is a product $f'(u) f''(v)$ and its reference a product $\bar{f}' \bar{f}''$ of constants $\bar{f}' \ge f'$ and $\bar{f}'' \ge f''$, then $\bar{f}' \bar{f}'' - f' f'' = \bar{f}'(\bar{f}'' - f'') + f''(\bar{f}' - f')$, both terms nonnegative and each a function on the bunch times a function on the room; so when every coefficient, every reference and $\phi$ are products, every moment is a finite nonnegative combination of products of moments over $\Delta_B$ and over $\Delta_R$.*

No order of the values is used, only bounds on the coefficients, and the terms are nonnegative. Applied to every letter on the whole simplex with the references $M_i = \max_T \theta_{j,T} / g_i!$, Theorem 15.13 is the draw series (Proposition 20.12); applied on the totals it closes the scale of the bunch alone, and the shapes of the bunch and the room enter only through the moments of the deviations of the coefficients from their references. Across a bunch of width one every coefficient is a letter of a minor on the bunch times a letter of a minor on the room (Theorem 19.20), so with references that are products the moments pair the bunch with the room, as the lattices of Theorem 19.20 do, with nonnegative terms. If every $\widehat{L}_i$ is at least $(1 - \lambda) \bar{L}_i$ on the totals at every shape, with $\lambda < 1$,

the terms of total order $\sum_i |\mathbf{k}_i| = k$ add up to at most $\frac{(|\mathbf{w}|)_k}{k!}\lambda^k$ times the term of order 0; when a coefficient vanishes on a face of a shape simplex the terms need not decay geometrically.

For one table read at $\rho_0 = 1$ the two letters on the totals are linear, and so are their reference letters, one constant for each coefficient. Rescaling the scale by the ratio of the references of the letter $K$ carries its reference letter to a multiple of $x_R + x_B$, and that of $\Psi$ to a multiple of the linear letter whose scale coefficient over its constant coefficient is the reference of the even letter of the level $d$ over that of the level $r$; the rescaling multiplies the Breta coefficient by a power of the ratio of the references of $K$. Two letters on the line of the totals have two zeros besides its two endpoints, and four points of a line have one modulus, the ratio of the references of the even letters of the two levels: the Breta coefficients depend on the references only through their scales and this modulus, defined up to its inverse by the naming of the two levels. What one reference for each coefficient adds is that every deviation can vanish at its own peak, at the price of the one modulus in the coefficients. When the modulus is 1 both reference letters are multiples of $x_R+x_B$, which is 1 on the simplex, and Corollary 19.31 applies. With $N$ tables there are $2N$ letters on the totals and $2N-1$ such moduli.

**Corollary 19.32** (The ladder on the totals)**.** *In Theorem 19.28(ii) let every $w_i > 0$, put $w' = \sum_{i\neq i_0} w_i$, and let $\zeta > 0$.*

*(i) At shapes where every ratio $\xi_{i_0}/\xi_i$ lies in $(0, 2\zeta)$,*

$$\int_{\Delta_{\{B,R\}}} x_B^{\kappa_B-1} x_R^{\kappa_B^{\mathrm{top}}-1} \prod_{i\neq i_0} \Big(x_R + x_B\,\frac{\xi_{i_0}}{\xi_i}\Big)^{-w_i} d\omega(x)$$
$$= \sum_{\mathbf{n}} \prod_{i\neq i_0} \frac{(w_i)_{n_i}}{n_i!} \Big(\zeta - \frac{\xi_{i_0}}{\xi_i}\Big)^{n_i}\ \mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B+|\mathbf{n}|,\kappa_B^{\mathrm{top}}),\, w'+|\mathbf{n}|;\, I_{\{B,R\}}}(\zeta, 1),$$

*absolutely: the linear letters are moved to the common ratio $\zeta$, where they merge into one, and the coefficients do not depend on the shapes. At $\zeta = 1$ they are the Beta corners $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}\kappa_B+|\mathbf{n}|,\ \kappa_B^{\mathrm{top}}\\ \kappa_B+\kappa_B^{\mathrm{top}}+|\mathbf{n}|\end{smallmatrix}\right]$. For one linear letter this is the ladder of the Breta in its first coefficient (Theorem 3.5(ii)), $\partial_\zeta \mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\mathrm{top}}),w;I_{\{B,R\}}}(\zeta,1) = -w\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B+1,\kappa_B^{\mathrm{top}}),w+1;I_{\{B,R\}}}(\zeta,1)$, summed as a Taylor series.*

*(ii) Let $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ converge and let every ratio $\xi_{i_0}/\xi_i$ lie in $[\zeta_-,\zeta_+] \subset (0,2\zeta)$ on $\Delta_B\times\Delta_R$. Then (i) integrates term by term against $\phi$ of Theorem 19.28(i) multiplied by $\xi_{i_0}^{\kappa_B}\prod_i (L_i^{(0)})^{-w_i}$ and by the further factor of Theorem 19.28(ii), and the terms of $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ with $|\mathbf{n}| = n$ add up in absolute value to at most $C\frac{(w')_n}{n!}\vartheta^n P_F(\mathbf{m};\mathbf{s},\mathbf{t})$, where $\vartheta = \max(1-\zeta_-/\zeta,\ \zeta_+/\zeta - 1) < 1$ and $C = \mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\mathrm{top}}),w';I_{\{B,R\}}}(\zeta,1)\big/\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B,\kappa_B^{\mathrm{top}}),w';I_{\{B,R\}}}(\zeta_+,1)$. If $\zeta \geq \sup \xi_{i_0}/\xi_i$ for every $i$ the terms are nonnegative and no lower bound on the ratios is needed; both sides are then finite or infinite together.*

*(iii) For one table with $i_0$ the letter of $K$, under the hypotheses of (ii) the only other ratio is $\varrho_B(u)/\varrho_R(v)$, and*

$$P_F(\mathbf{m};s,t) = \sum_{n\geq 0} \frac{(t)_n}{n!}\,\mathsf{B}^{\{\{B,R\}\}}_{(\kappa_B+n,\kappa_B^{\mathrm{top}}),\,t+n;\,I_{\{B,R\}}}(\zeta,1) \int_{\Delta_B\times\Delta_R} \big(\zeta\mathbf{E}^r - \mathbf{E}^d\big)^n\,(\mathbf{E}^r)^{-t-n}\,\phi(du,dv),$$

*with $\mathbf{E}^d$, $\mathbf{E}^r$ and $\phi$ of Corollary 19.29. With $i_0$ the letter of $\Psi$ the ratio is $\varrho_R(v)/\varrho_B(u)$, and the same holds with $s$ for $t$, the moments $\int(\zeta\mathbf{E}^d - \mathbf{E}^r)^n(\mathbf{E}^d)^{-s-n}$ and the measure*

$\phi = \big(\Psi_{\mathcal{B}/R}\Psi_{\mathcal{B}|R}\big)^{2\delta^d}\big(\Psi_{\mathcal{B}|B}\Psi_{\mathcal{B}/B}\big)^{2\delta^r} u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}d\omega_B\,d\omega_R$. *So one table closes on the totals with nonnegative terms about any* $\zeta \geq \sup\varrho_B/\inf\varrho_R$ *when no slope is positive, and with* $i_0$ *the letter of* $\Psi$ *about any point at least* $\sup\varrho_R/\inf\varrho_B$ *when no slope is negative (Lemma 19.15). With* $\zeta = \zeta'\zeta''$, $\zeta' \geq \sup_{\Delta_B}\varrho_B$ *and* $\zeta'' \geq 1/\inf_{\Delta_R}\varrho_R$,

$$\zeta\mathbf{E}^r - \mathbf{E}^d = \zeta'\mathbf{E}_{\mathcal{B}|B}(u)\big(\zeta''\mathbf{E}_{\mathcal{B}/B}(v) - \mathbf{E}_{\mathcal{B}|R}(v)\big) + \mathbf{E}_{\mathcal{B}|R}(v)\big(\zeta'\mathbf{E}_{\mathcal{B}|B}(u) - \mathbf{E}_{\mathcal{B}/R}(u)\big),$$

*both terms nonnegative and each a function on the bunch times a function on the room.*

*(iv) Since* $x_R + x_B\xi_{i_0}/\xi_i = (\xi_{i_0}/\xi_i)\big(x_B + x_R\xi_i/\xi_{i_0}\big)$, *the integral of (i) is* $\prod_{i\neq i_0}(\xi_i/\xi_{i_0})^{w_i}$ *times the same integral with* $\kappa_B$ *and* $\kappa_B^{\mathrm{top}}$ *swapped and every ratio replaced by its inverse. So (i) and (ii) hold in the inverse ratios, the factor* $\prod_{i\neq i_0}(\xi_i/\xi_{i_0})^{w_i}$ *joining* $\phi$, *with nonnegative terms about any common point at least* $\sup\xi_i/\xi_{i_0}$: *the ladder closes whenever the ratios are bounded on one side.*

The three closings of the even share read the same linear letters. At a wall the even share closes toward the level whose even letter is the smaller, which needs $\varrho_B/\varrho_R$ on one side of 1 with room (Corollary 19.21(i)); the ladder moves the linear letters to a common ratio, which needs the quotient bounded on one side, and its coefficients are constants of the scale; the peak closing moves each linear letter to a constant reference, which needs the coefficients bounded and asks nothing of their ratio. In all three the terms pair moments of the bunch with moments of the room. With $\varrho_B(u) < \varrho_R(v)$ throughout and $\sup\varrho_B = \inf\varrho_R$, the ladder about $\zeta = 1$ is a series of nonnegative terms whose coefficients are Beta corners, where Proposition 19.23 sums the lattices by scaling the level weights. Where the quotient takes every positive value, of the three only the peak closing applies; the gauge closing applies as well, its terms not separating over the bunch and the room (Proposition 19.24).

## 19.8 Width zero

**Proposition 19.33** (A loop or a coloop on the simplex)**.** *Let a mark* $e$ *be in every table a loop* ($\eta_j = -1$) *or a coloop* ($\eta_j = +1$), *let* $F'$ *be the family of Corollary 15.16 on* $\mathcal{E} - e$, *and put* $\nu_e = \mu_e - \sum_j \eta_j\delta_j$, *so that* $\kappa' = \kappa_{\mathcal{E}} - \nu_e$ *is the imbalance of* $F'$ *at the masses* $\mathbf{m}|_{\mathcal{E}-e}$. *Where* $\nu_e > 0$ *and* $\kappa' > 0$,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma_{\!1}^{\,2}\begin{bmatrix}\nu_e,\ \kappa'\\ \kappa_{\mathcal{E}}\end{bmatrix}\ P_{F'}(\mathbf{m}|_{\mathcal{E}-e};\mathbf{s},\mathbf{t}),$$

*the mark rescaling every other mark by the common factor* $1 - u_e$; *it is the compound formula of Theorem 16.17 at* $S = \{e\}$, *with the walls* $\kappa_{\{e\}} = \nu_e$ *and* $\kappa_{\mathcal{E}\setminus e} = \kappa'$. *Marks* $e_1,\dots,e_k$ *of this kind combine first among themselves and then with the rest,* $\Gamma_{\!1}^{\,k+1}\begin{bmatrix}(\nu_{e_i})_i,\ \kappa'\\ \kappa_{\mathcal{E}}\end{bmatrix} = \Gamma_{\!1}^{\,k}\begin{bmatrix}(\nu_{e_i})_i\\ a\end{bmatrix}\Gamma_{\!1}^{\,2}\begin{bmatrix}a,\ \kappa'\\ \kappa_{\mathcal{E}}\end{bmatrix}$ *with* $a = \sum_i \nu_{e_i}$ *and* $\kappa'$ *the imbalance once all of them are removed (Theorem 3.12). On the balance wall* $\kappa' = -\nu_e$ *and the period of the compound diverges (Proposition 8.16): the Foreman tree splits molecules first (Definition 14.4).*

The bunch $\{e\}$ has width zero: the room does not see it at all, and the period integrates it out as the Beta corner $\Gamma_{\!1}^{\,2}\begin{bmatrix}\nu_e,\ \kappa'\\ \kappa_{\mathcal{E}}\end{bmatrix}$, whose Gamma $\Gamma(\nu_e)$ is the Gamma $\Gamma(\mu_e - \langle\boldsymbol{\delta}\rangle_e)$ of the port at $e$ (Theorem 15.15).

## 19.9 Pooling is a move

**Theorem 19.34** (Pooling is a move)**.** *Let a bunch $B$ have width one in every table of a family of tables with positive rational weights, the weights of every table factoring across $B$ and $R$ with rational factors, at rational exponents and masses on the balance wall.*

*(i) Theorem 19.8 is one change of variables, $(\rho, u, \alpha_R) \mapsto (\rho u, \alpha_R)$, the scale and the shape of $B$: it is birational over $\mathbb{Q}$ with the inverse $\alpha \mapsto (|\alpha_B|, \alpha_B/|\alpha_B|, \alpha_R)$, bijective on the open domains, and it carries the face of $S \subseteq \mathcal{E}$ to $\{\rho = 0\} \times \{\alpha_{S\cap R} = 0\}$ if $S \supseteq B$ and to $\{u_{S\cap B} = 0\} \times \{\alpha_{S\cap R} = 0\}$ otherwise, with the walls matched by Theorem 19.9: by (i) there when $S \supseteq B$ or $S \subseteq R$, by (ii) there when $S \subseteq B$, and by the gluing of the counts otherwise.*

*(ii) When the factorizations, with rational factors, give every active letter the one hinge value $\gamma$ of Theorem 19.10, that theorem is a composite of moves: the change of variables (i); the change $\rho \mapsto \rho\,\gamma(u)$, birational over $\mathbb{Q}$ and positive on the open domain; the product of the datum of the shape with that of the cell family of the room; and the integral of the hinge, the Beta corner of a linear letter (Proposition 15.1 with one coefficient), itself a change of variables and a product with the datum of the list on two marks whose bases are the two singletons, at the exponents $(0, 2\sigma)$. Every integral in the chain converges exactly when $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ does (Theorem 19.10).*

*So the identities of Theorems 19.8 and 19.10 hold between period data, not only between their values: when every active letter has one hinge value the period datum of the family times that of the list on two marks is the product of the data of its two cell families.*

In the interior Theorem 19.8 writes the period as an integral over the shape simplex, whose faces carry the walls of Theorem 19.9(ii), of the periods of the pooled families $F_{B\to g}(u)$, whose walls are those of Theorem 19.9(i).

## 19.10 Cleaning a molecule

**Definition 19.35** (Cleaning)**.** A *cleaning* of a list is a sequence of poolings of parallel and series bunches of width one, each in the list left by the previous ones, until at most two marks or no such class remain.

**Proposition 19.36** (Poolings of disjoint classes commute)**.** *Let $X$ and $Y$ be disjoint parallel or series bunches of width one of a list. Pooling $X$ keeps $Y$ a class of the same kind and width, with the same restriction to $Y$ and the same contraction of its room, and the two orders of pooling give the same list on $(\mathcal{E} \setminus (X \cup Y)) \sqcup \{g_X, g_Y\}$, the same cell kernels and the same hinge values. A cleaning ends after at most $|\mathcal{E}| - 2$ poolings.*

**Proposition 19.37** (Cleaning commutes with maps)**.** *A bijection that carries one table to another (Definition 11.13) maps the parallel and series bunches of the first onto those of the second, with the same widths and, the weights factoring across a bunch and its room and the factorization carried along, the same hinge values. An exchange maps parallel bunches onto series bunches and swaps the hinge values by $K$ and by $\Psi$, the weight $\theta'_p$ of a pattern $p$ of a bunch $B$ becoming that of the pattern $g(B \setminus p)$ of its image. Between lists it maps the cleaning of the first list at the exponents $(s, t)$ onto the cleaning of the second at $(t, s)$.*

**Theorem 19.38** (Cleaning has one result)**.** *Let a list be one molecule. Every cleaning of it ends at the same list up to the names of its marks. If that list has more than two marks, every cleaning*

*ends with the same marks, each the same set of marks of the original list; a list that ends with two marks may divide its marks between them in more than one way. Pooling the parallel bunches first, then the series bunches, and repeating, is one cleaning, and every other order ends at the same place.*

Pooling keeps a molecule a molecule, and two poolings at one list can be continued to one list; so the end of a cleaning does not depend on the order of the poolings. The cleaned list has no parallel or series bunch of width one left, each of its marks gathers a set of marks of the molecule, and the period of the molecule is the integral, over the shape simplices of the pooled bunches against their cell kernels, of the periods of the pooled families of the cleaned list.

# Chapter 20

# Words of a tree

A period is read back from what the Foreman procedure leaves: the bunches peeled at the nodes of a tree of bunches and the leaves at its bottom. The period of a family is the word of such a tree, an iterated fibre integral of stacked Gammas and of the periods of the leaves; opened at the atoms along a chain, or wherever no table is opened on two childcules, the word is one Gramma on the marks and the letters. The word closes in three ways: by draws, with nonnegative terms at nonnegative exponents; at the centre, through the half-sum and the difference of the two letters of a table; and node by node in one scale, where each side of a node of width one is an interval from the decay of its transform to its reach. Read on one tree, the period shows what each node adds, what the leaves contain, and how the integrations of the simplex are shared out.

## 20.1 The two letters of a table at once

**Proposition 20.1** (The pair ladder is the expansion about the midpoint)**.** *For $a, b > 0$, $s, t > 0$ and $x = \frac{a-b}{a+b} \in (-1, 1)$,*

$$a^{-s} b^{-t} = \Big(\frac{a+b}{2}\Big)^{-s-t} (1+x)^{-s} (1-x)^{-t},$$

*and the coefficients of the power series $(1+x)^{-s}(1-x)^{-t}$ are*

$$[x^k](1+x)^{-s}(1-x)^{-t} = \sum_{i+i'=k} (-1)^i \frac{(s)_i\,(t)_{i'}}{i!\,i'!} = \frac{(s+t)_k}{k!} \Gamma_{\begin{smallmatrix}1\\2\end{smallmatrix}}\begin{bmatrix} s+t \\ s,\ t \end{bmatrix} \int_0^1 w^{s-1}(1-w)^{t-1}(1-2w)^k\, dw.$$

*So $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix} s,\ t \\ s+t+k \end{bmatrix} [x^k](1+x)^{-s}(1-x)^{-t} = \frac{1}{k!}\int_0^1 w^{s-1}(1-w)^{t-1}(1-2w)^k\, dw$; swapping $s$ and $t$ multiplies the $k$-th coefficient by $(-1)^k$; and at $s = t$ only even $k$ occur, $[x^{2l}](1-x^2)^{-s} = (s)_l/l!$.*

The two letters of a table enter a period as $\Lambda(z)^{-s}\Lambda(-z)^{-t}$. The pair ladder expands such a pair about its midpoint; under one power, the shape of a two-element bunch in the tower decides how much of each letter is taken (Proposition 20.3).

**Definition 20.2** (Side shape; half-sum and difference; pattern coefficients)**.** For table $j$ of a family and $w \in [0,1]$ the letter of the *side shape* $w$ is $L_{j,w}(z) = w\Lambda_j(z) + (1-w)\Lambda_j(-z)$; $L_{j,1/2}(z) = \frac{1}{2}\big(\Lambda_j(z) + \Lambda_j(-z)\big)$ is the *half-sum* and $\partial_w L_{j,w}(z) = \Lambda_j(z) - \Lambda_j(-z)$ the *difference* of the two letters of table $j$. For a bunch $B$ with room $R$ and the Taylor cube $\Lambda_j(z) = \sum_{p \subseteq B} e^{\langle z_B, \chi_p\rangle} \Lambda_{j,(p)}(z_R)$

(Theorem 12.7), $\Lambda_{j,(p)} = 0$ when $p$ is not a pattern of table $j$, the *pattern coefficients* of the side shape are

$$c_{j,p}(w; z_R) = w\,\Lambda_{j,(p)}(z_R) + (1-w)\,\Lambda^*_{j,(p)}(z_R), \qquad \Lambda^*_{j,(p)}(z_R) = \Lambda_{j,(B\setminus p)}(-z_R),$$

so that $L_{j,w}(z) = \sum_p e^{\langle z_B, \chi_p\rangle} c_{j,p}(w; z_R)$. For one table the index $j$ is omitted.

**Proposition 20.3** (The side shape, and what an exchange does to it)**.** *Let a bijection $g$ carry or exchange a family $F$, with the bijection $h$ of its tables (Definition 11.13). If $g$ carries $F$, then $L_{h(j),w}(z) = L_{j,w}(g^*z)$; if it exchanges $F$, then $L_{h(j),w}(z) = L_{j,1-w}(g^*z)$ (Definition 20.2). For $s_j, t_j > 0$, by the tower at the unit configuration on a two-element set $J$ (Theorem 3.2),*

$$\begin{aligned}\Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j} &= \Gamma_{\frac12}\begin{bmatrix} s_j+t_j \\ s_j,\ t_j\end{bmatrix}\, \mathsf{B}^{\{J\}}_{(s_j,t_j),\, s_j+t_j;\, I_J}\big(\Lambda_j(z), \Lambda_j(-z)\big) \\ &= \Gamma_{\frac12}\begin{bmatrix} s_j+t_j \\ s_j,\ t_j\end{bmatrix} \int_0^1 w^{s_j-1}(1-w)^{t_j-1}\, \mathsf{Z}_{s_j+t_j;(1)}\big(L_{j,w}(z)\big)\, dw,\end{aligned}$$

*and for one table an exchange together with $w \mapsto 1-w$ keeps $\mathsf{Z}_{s+t;(1)}(L_w(z))$, keeping the factor $w^{s-1}(1-w)^{t-1}$ exactly when $s = t$.*

The side shape weighs the basis letter against the cobasis letter, and an exchange of the table turns it over. The half-sum is even in $z$ and the difference odd: $\Lambda(z)\Lambda(-z) = L_{1/2}(z)^2 - \frac14\big(\partial_w L_w(z)\big)^2$.

## 20.2 The junction in pattern form

**Theorem 20.4** (The junction in pattern form)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a molecule on $n$ marks with real exponents $s_j, t_j \geq 0$, its masses on the balance wall with every proper wall positive, and let $B$ be a proper bunch with room $R$ and $g \in R$. For a table at a full lean put $w_j = 1$ when $t_j = 0$ and $w_j = 0$ when $s_j = 0$. With the pattern coefficients $c_{j,p}(w; z_R)$ of the side shapes (Definition 20.2),*

$$\begin{aligned}P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \frac{2^{n-1}}{\prod_{j:\, s_j>0}\Gamma(s_j)\prod_{j:\, t_j>0}\Gamma(t_j)} \int \prod_{j:\, s_j, t_j>0} w_j^{s_j-1}(1-w_j)^{t_j-1} \int_{\mathbb{R}^{R-g}} e^{2\langle \boldsymbol{\mu}|_R, z_R\rangle} \\ \times \int \prod_{j\ \mathit{active}} y_j^{s_j+t_j-1}\, \mathsf{G}_{2\boldsymbol{\mu}|_B;(\chi_p)_p}\Big(\Big(\sum_{j\ \mathit{active}} y_j\, c_{j,p}(w_j; z_R)\Big)_p\Big)\, d\mathbf{y}\, dz_{R-g}\Big|_{z_g=0}\, d\mathbf{w},\end{aligned}$$

*with $d\mathbf{y} = \prod_j dy_j$ over the active tables, $y_j \in (0,\infty)$, and $d\mathbf{w} = \prod_j dw_j$ over the tables with $s_j, t_j > 0$, $w_j \in [0,1]$; the Gramma on the scales $e^{z_B}$ with one term for each $p \subseteq B$ whose coefficient $\sum_j y_j c_{j,p}(w_j; z_R)$ is not 0. A table at a full lean has no integral over its side shape: at $(s_j, 0)$ its coefficients are $y_j\Lambda_{j,(p)}(z_R)$, at $(0, t_j)$ they are $y_j\Lambda^*_{j,(p)}(z_R)$.*

The two letters of every table become one letter, the side shapes are integrated last, and the bunch is integrated on its own orthant against the patterns of all the tables at once: each pattern is one term of one Gramma on the bunch, whose coefficient adds the pattern coefficients of the tables, each at its own scale $y_j$, and the room sits in the coefficients. A map that keeps the bunch moves the coefficients with the side shapes and the tables.

**Proposition 20.5** (What a map keeping the bunch gives the coefficients)**.** *In Theorem 20.4 let a bijection $\tilde{g}$ of $\mathcal{E}$ keep $B$, with $\tilde{g}_R$ its restriction to the room, and carry or exchange $F$ with the bijection $h$ of its tables. If $\tilde{g}$ carries $F$, then $c_{h(j),\tilde{g}(p)}(w; z_R) = c_{j,p}(w; \tilde{g}_R^* z_R)$ for every $j$, $p$ and $w$; if it exchanges $F$, then $c_{h(j),\tilde{g}(p)}(1-w; z_R) = c_{j,p}(w; \tilde{g}_R^* z_R)$.*

**Proposition 20.6** (Level form of the half-sum)**.** *In Theorem 20.4 the half-sum and the difference of table $j$ have the pattern coefficients $c_{j,p}(\frac{1}{2}; z_R) = \frac{1}{2}\big(\Lambda_{j,(p)} + \Lambda^*_{j,(p)}\big)(z_R)$ and $\partial_w c_{j,p}(w; z_R) = \big(\Lambda_{j,(p)} - \Lambda^*_{j,(p)}\big)(z_R)$.*

*(i) For every table, $c_{j,B\setminus p}(\frac{1}{2}; -z_R) = c_{j,p}(\frac{1}{2}; z_R)$ and $\partial_w c_{j,B\setminus p}(w; -z_R) = -\partial_w c_{j,p}(w; z_R)$.*

*(ii) The half-sum of table $j$ has level form on $B$ (Proposition 14.12), $c_{j,p}(\frac{1}{2}; \cdot)$ depending on $|p|$ alone, exactly when every transposition of two marks of $B$ fixing the rest carries the multiset $\mathcal{B}_j \uplus \{\mathcal{E} \setminus T : T \in \mathcal{B}_j\}$, with its weights, onto itself; if $2r_j \neq n$ this is the condition that every such transposition carries table $j$ (Proposition 14.12).*

*(iii) If every such transposition carries or exchanges table $j$, its half-sum has level form and the square of its difference is unchanged by every permutation of $B$.*

## 20.3 The word of a tree

**Definition 20.7** (Tree of bunches)**.** A *tree of bunches* of a family $F$ on $\mathcal{E}$ has *vertices* $v$, each with marks $M_v \subseteq \mathcal{E}$ and tables on $M_v$ with their exponents, the root having $M_v = \mathcal{E}$ and the active tables of $F$. A vertex whose tables form one molecule is a *node*. A node $v$ that is not a leaf carries a proper bunch $B_v \subsetneq M_v$; the tables at $v$ have their children at $B_v$, indexed by the table and the pattern as in the record of the peel (Definition 14.2) and not merged, the *tables made at* $v$, each carrying the letters its parent carries, its exponent in each letter a positive multiple of the parent's, as at every native split (Definition 12.25). The molecules $R_{v,1}, \dots, R_{v,q_v}$ of the tables made at $v$ on $M_v \setminus B_v$, in a fixed order, are the childcules of $v$ (Definition 14.1), each a node whose tables are the restrictions to it of the tables made at $v$. The *room* of $v$ is the one childcule when $q_v = 1$; when $q_v \geq 2$ it is a *split vertex*, with the marks $M_v \setminus B_v$ and the tables made at $v$, whose children are the childcules. The root of a family whose active tables form two or more molecules is likewise a split vertex, whose children are the molecules with the restrictions of its tables. A leaf $v$ has $B_v = M_v$. The *width* of a node is the largest width $c(B_v)$ of its bunch in the tables at $v$, zero at a leaf. Merging equal children with their exponents added, as the peel does (Definition 14.1), keeps every letter when the exponents are nonnegative, and can cancel a letter when exponents of both signs meet; the tables made at a node are therefore indexed. The tables at a node are thus the children of the tables of $F$ at the patterns shown on the bunches above it, restricted to its marks (Proposition 14.5); with a native gauge they carry as exponents the shares of the native splits at the nodes above (Theorem 12.26). Its descents are the sequences of nodes from the root to a leaf, each a childcule of the one before, as on the Foreman tree (Definition 14.17), which is a tree of bunches (Definition 14.4).

**Corollary 20.8** (The tree of a linked molecule is a chain)**.** *Let some active table of a molecule $F$ be connected, as for a molecule of one table. Every tree of bunches of $F$ is then a chain $M_0 \supset M_1 \supset \cdots \supset M_d$ with $M_{k+1} = M_k \setminus B_k$, above one leaf: the tables at $M_k$ are the children of the tables of $F$ at $S_k = B_0 \sqcup \cdots \sqcup B_{k-1}$ (Proposition 14.5), which form one molecule (Lemma 18.14). With a native gauge their exponents are the shares of the native split, and on the Foreman tree the family at $M_k$ is the peel $F_{S_k}(\mathbf{a}^\star, \mathbf{b}^\star)$ (Definition 14.1).*

**Corollary 20.9** (A linked molecule)**.** *Let some active table of $F$ be connected, as for a molecule of one table. Every peel then leaves one childcule, so the tree is a chain (Corollary 20.8), there are no twins, and the room of each node is the next node. For every chain that peels a class at each node, every class of a node other than its bunch lies inside one class of the next node, and the leaf is a union of whole classes of every node above it (Theorems 14.20(ii) and 14.22(i)). For the Foreman tree, which is then a chain, at every realization, the centre and every lean, Theorem 14.22(ii) with no twins gives*

$$|B_0| \le |B_1| \le \cdots \le |B_{d-1}| \le |M_d| :$$

*the leaf alone has at least as many marks as every bunch.*

**Corollary 20.10** (The word of a tree)**.** *Let $F$ have real data with $s_j, t_j \ge 0$, a native gauge and its masses on the balance wall, and let the children's weights factor across the childcules at every node of a tree of bunches. Run the port of Theorem 18.13(ii) at every node, on the native splits. A table $\mathcal{B}'$ made at a node, with the exponents $(a_{\mathcal{B}'}, b_{\mathcal{B}'})$, is the direct sum of its restrictions to the childcules there (Proposition 7.9); let $q_{\mathcal{B}'}$ be the number of those childcules, other than leaves, on which its letter is not a monomial. Then*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int_{\mathcal{X}} \prod_{\mathcal{B}'} \big(\Gamma(a_{\mathcal{B}'})\Gamma(b_{\mathcal{B}'})\big)^{1-q_{\mathcal{B}'}} \prod_{leaves\ v} P_{F_v}(\mathbf{m}|_{M_v};\mathbf{a}_v,\mathbf{b}_v),$$

*the* word of the tree*: an iterated fibre integral through the native splits, over the fibre $\mathcal{X}$ cut out by the share sums and by the $|B_v|+q_v-1$ pins of every node $v$ that is not a leaf (Theorem 18.13(ii)), a monomial restriction entering the pins of its node as a load; $F_v$ is the family of the tables at the leaf $v$ with their exponents $(\mathbf{a}_v,\mathbf{b}_v)$, and a letter with exponent zero carries no shares and no Gamma. A table opened on one childcule carries no factor, and when an active table is connected every $q_{\mathcal{B}'}$ is 0 or 1 (Corollary 20.8).*

**Theorem 20.11** (Building back up)**.** *In Corollary 20.10 open every leaf by the atom formula on the balance wall (Corollary 18.16), a unit being a monomial, and write the shares $(a,b)$ of one pattern of one table in pair coordinates, $\sigma = \frac{a+b}{2}$ and $\delta = \frac{a-b}{2}$.*

*(i)* Polycules multiply. *At every node and at every point of its fibre on the wall contour of the native split (Theorem 18.13(iii)), the integral over the shares below the node is the product over its childcules of their words: the polycule of the childcules is evaluated as the product of its molecules, a unit contributing a monomial in the weights.*

*(ii)* The pins are odd. *If every table has $s_j, t_j > 0$, every pin other than the share sums involves the odd shares alone: the pins $\langle \boldsymbol{\delta} \rangle_e = \mu_e$ of the bunch, the pins of the childcules,*

$$\kappa_{R_i} = \mu_{R_i} - \sum_{j,p} \delta_{j,p}\big(2r_{j,(p)}(R_i) - |R_i|\big) = 0,$$

*the sum over the tables made at the node, indexed as in Theorem 18.13 by the table $j$ at the node and the pattern $p$, and the pins of the atoms. The share sums are $\sum_p \sigma_{j,p} = \sigma_j$ and $\sum_p \delta_{j,p} = \delta_j$, so the fibre is the product of an even fibre, cut by the even sums alone, and an odd fibre.*

*(iii)* One Gramma. *If at every node every table made there is a monomial on all its childcules but at most one, in particular along a chain (Corollary 20.8), the leaf shares are one for each basis $T \in \mathcal{B}_j$ and letter, the pins of the tree are, for a fixed mark $g$, the pins $\langle \boldsymbol{\delta} \rangle_e = \mu_e$,*

*$e \in \mathcal{E}-g$, of Corollary 18.16 after a change of determinant $\pm1$, and the word is the atom formula, one Gramma on the marks and the letters:*

$$\prod_{j:\,s_j>0}\Gamma(s_j)\prod_{j:\,t_j>0}\Gamma(t_j)\,P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\mathsf{G}_{(\boldsymbol{\mu}|_{\mathcal{E}-g};\mathbf{s};\mathbf{t});\,t_F^{(g)}}(\boldsymbol{\theta}\oplus\boldsymbol{\theta}),$$

*with $t_F^{(g)}$ the configuration $t_F$ of Theorem 8.9 without its unit columns and without the row of $g$: on the scales $(\mathcal{E}-g)\sqcup\{v_j\}\sqcup\{w_j\}$, the columns $(\frac12\chi_T|_{\mathcal{E}-g};\mathbf{e}_j;0)$ and $(-\frac12\chi_T|_{\mathcal{E}-g};0;\mathbf{e}_j)$, $T\in\mathcal{B}_j$, each with the coefficient $\theta_{j,T}$, a letter with exponent zero omitted: one scale for each mark but $g$, which carries an odd pin, and one for each letter.*

*(iv)* The even shares close jointly. *In (iii) let every table have $s_j,t_j>0$, and write $c_{j,T}=b_{j,T}-a_{j,T}=-2\delta_{j,T}$. At fixed differences the $K$-shares of table $j$ close into one scale,*

$$\int_{\sum_T a_{j,T}=s_j}\prod_{T\in\mathcal{B}_j}\Gamma(a_{j,T})\,\Gamma(a_{j,T}+c_{j,T})\,\theta_{j,T}^{-2a_{j,T}}=\int_0^\infty y^{s_j-1}\prod_{T\in\mathcal{B}_j}\mathsf{G}_{c_{j,T};(1,-1)}\big(1,\theta_{j,T}^2y\big)\,dy,$$

*and $\prod_j\Gamma(s_j)\Gamma(t_j)P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the fibre integral over the differences, with the pins $\sum_T c_{j,T}=t_j-s_j$ and $-\frac12\sum_{j,T}c_{j,T}\chi_T(e)=\mu_e$, $e\in\mathcal{E}-g$, of $\prod_j\prod_T\theta_{j,T}^{-c_{j,T}}$ times these scale integrals; on the wall contour of the native split (Theorem 18.13(iii)) every integral converges absolutely. The fibre of differences has dimension $\sum_j(|\mathcal{B}_j|-1)-(n-1)$. At the centre, every table riding the line, $s_j=t_j$, with the masses $\sigma\mathbf{1}$, the pins are $\sum_{T\in\mathcal{B}_j}c_{j,T}=0$ for every $j$ and $\sum_j M_j^{\mathsf{T}}\mathbf{c}_j=0$, with $\mathbf{c}_j=(c_{j,T})_{T\in\mathcal{B}_j}$ and the incidence matrices $(M_j)_{Te}=\mathbf{1}_T(e)$, and the contour is $\operatorname{Re}\mathbf{c}=0$.*

Built back up, the word is the period read from its atoms. At every node the childcules multiply and the pins see only the odd shares; along a chain nothing is opened twice, and the word is the atom formula itself, one Gramma on the marks and the letters whose configuration is that of the interest with the masses' own columns removed. The even shares are free but for their sums, and they close table by table into one scale of Grammas $\mathsf{G}_{\nu;(1,-1)}$, as at the centre of a molecule (Theorem 20.11(iv)) and for the table that is one circuit (Corollary 18.24).

## 20.4 Closing the tree by draws

**Proposition 20.12** (The draw series)**.** *Let $U=(U_Q)_{Q\in\mathcal{P}}$ have independent blocks $U_Q\sim\operatorname{Dir}_{\mathbf{m}|_Q}$ on a partition $\mathcal{P}$ of the marks, and let $P_1,\dots,P_q$ be nonzero polynomials with nonnegative coefficients, $P_i$ homogeneous of degree $g_{i,Q}$ in the variables of each block $Q$. For an exponent vector $\mathbf{a}$ with block sums $g_{i,Q}$ put $b_{\mathbf{a}}=\prod_Q g_{i,Q}!/\prod_e a_e!$ and $C_i=\max_{\mathbf{a}}\big([u^{\mathbf{a}}]P_i\big)/b_{\mathbf{a}}$. Then $\prod_Q\big(\sum_{e\in Q}u_e\big)^{g_{i,Q}}-P_i/C_i$ has nonnegative coefficients and equals $1-P_i/C_i\in[0,1)$ on the interior of the source, and for $\nu_i\ge0$*

$$\mathbb{E}_{U\sim\bigotimes_Q\operatorname{Dir}_{\mathbf{m}|_Q}}\Big[\prod_i P_i(U)^{-\nu_i}\Big]=\prod_i C_i^{-\nu_i}\sum_{\mathbf{k}\in\mathbb{N}^q}\prod_i\frac{(\nu_i)_{k_i}}{k_i!}\times\mathbb{E}_{U\sim\bigotimes_Q\operatorname{Dir}_{\mathbf{m}|_Q}}\Big[\prod_i\Big(1-\frac{P_i(U)}{C_i}\Big)^{k_i}\Big],$$

*every term nonnegative, both sides finite or infinite together. The left side is the Trailmoment of Proposition 5.22,*

$$\Big(\mathsf{T}_{\boldsymbol{\nu};I_q}[(P_i)_i]\bigotimes_Q\operatorname{Dir}_{\mathbf{m}|_Q}\Big)\Big(\prod_Q\Delta_Q\Big),$$

*and each expectation on the right is a finite nonnegative combination of the moments of Theorem 3.11. For a family* $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ *with* $s_j, t_j \geq 0$*, one block,* $P_{2j-1} = K_j$ *and* $P_{2j} = \Psi_j$ *with the indices* $(s_j, t_j)$*, the constants are* $C_{2j-1} = \theta_{j,\max}/r_j!$ *and* $C_{2j} = \theta_{j,\max}/(n-r_j)!$*,* $\theta_{j,\max} = \max_T \theta_{j,T}$*; the left side times* $\Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}$ *is* $\int_{\Delta_\mathcal{E}} u^{\mathbf{m-1}} \prod_j K_j(u)^{-s_j} \Psi_j(u)^{-t_j}\, d\omega_\mathcal{E}(u)$*, which on the balance wall is* $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$*; and the expectation of the term* $\mathbf{k} = (k_j, l_j)_j$ *is the chance that, in the urn that draws a mark with probability proportional to its current mass and then adds one to that mass, for* $j = 1, \dots, N$ *in turn* $k_j$ *rounds of* $r_j$ *draws and then* $l_j$ *rounds of* $n - r_j$ *draws all fail for table* $j$*, a round failing when its marks are not a basis, respectively not a cobasis, of table* $j$*, and, when they are the basis* $T$*, respectively the cobasis* $\mathcal{E} \setminus T$*, with probability* $1 - \theta_{j,T}/\theta_{j,\max}$*: a finite nonnegative combination of Dirichlet moments, a finite sum of them for lists. This series is Theorem 15.13(i) with the references* $M_{2j-1} = C_{2j-1}$ *and* $M_{2j} = C_{2j}$*.*

**Corollary 20.13** (At $\delta = 0$ and at a full lean the period reads the law of one letter for each table)**.** *Let a family* $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$*, every table active, carry its masses* $\mathbf{m}$ *on the balance wall with every proper wall positive, every table riding the line,* $(s_j, t_j) = (\sigma_j, \sigma_j)$*, or at a full lean,* $(s_j, 0)$ *or* $(0, t_j)$*. Put* $L_j = \mathbf{E}_j = K_j \Psi_j$ *and* $\nu_j = \sigma_j$ *when table* $j$ *rides the line,* $L_j = K_j$ *and* $\nu_j = s_j$ *at* $(s_j, 0)$*, and* $L_j = \Psi_j$ *and* $\nu_j = t_j$ *at* $(0, t_j)$*, and let* $\phi_\mathbf{m}$ *be the joint law of* $(L_1(U), \dots, L_N(U))$ *for* $U \sim \mathrm{Dir}_\mathbf{m}$ *(Proposition 8.12). The kernel* $\prod_j a_j^{-s_j} b_j^{-t_j}$ *of the letters* $a_j = K_j$ *and* $b_j = \Psi_j$ *is the kernel* $\mathsf{Z}_{\boldsymbol{\nu}; I_N}\big((L_j)_j\big)$*, one scale for each table, so*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} \big(\mathsf{T}_{\boldsymbol{\nu}; I_N}[\mathrm{id}]\, \phi_\mathbf{m}\big)\big((0,\infty)^N\big);$$

*with one table* $\phi_\mathbf{m}$ *is a law on the half-line and the right side is* $\Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} (\mathsf{T}_{\nu;(1)}[x]\, \phi_\mathbf{m})((0,\infty))$*, at* $\delta = 0$ *the law* $\mathbf{E}_* \mathrm{Dir}_\mathbf{m}$ *of the even letter* $\mathbf{E}(U) = K(U)\Psi(U)$ *(Theorem 5.9). The law* $\phi_\mathbf{m}$ *is carried by* $\prod_j (0, \max_{\Delta_\mathcal{E}} L_j]$*, it is determined by its moments, and its moments are finite sums of Dirichlet moments (Theorem 3.11).*

**Theorem 20.14** (The draw series along a chain)**.** *Let a family* $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ *have the exponents* $s_j, t_j \geq 0$ *and masses* $\mathbf{m} \in \mathbb{R}^\mathcal{E}_{>0}$*, and let* $M_0 = \mathcal{E} \supset M_1 \supset \cdots \supset M_d$ *be a chain of sets with* $B_i = M_i \setminus M_{i+1}$*,* $i < d$*, and* $S_i = B_0 \sqcup \cdots \sqcup B_{i-1}$*,* $S_0 = \varnothing$*,* $S_{d+1} = \mathcal{E}$*. A round of draws* extends on $S$ *for table* $j$ *when its draws inside* $S$ *are distinct and are* $T \cap S$ *for some basis* $T$ *of table* $j$*. In this statement write* $\theta_j(S; P)$ *for the largest weight* $\theta_{j,T}$ *over the bases* $T$ *of table* $j$ *with* $T \cap S = P$*, and say that a round that extends on* $S_i$ *and on* $S_{i+1}$*, with the traces* $P_i$ *and* $P_{i+1}$*, is* accepted at $S_{i+1}$ *with probability* $\theta_j(S_{i+1}; P_{i+1})/\theta_j(S_i; P_i)$*, independently for each* $i$*. Let* $\mathfrak{f}_{j,i}(u)$*,* $0 \leq i \leq d$*, be the chance that a round of* $r_j$ *draws from* $u$ *extends on* $S_i$ *and is accepted at* $S_1, \dots, S_i$ *but does not both extend on* $S_{i+1}$ *and get accepted there, and* $\mathfrak{f}^*_{j,i}$ *the same for rounds of* $n - r_j$ *draws and the cobases* $\mathcal{E} \setminus T$ *with the weights* $\theta_{j,T}$*. Then, with* $\theta_{j,\max}$ *the largest weight of table* $j$*,* $1 - r_j!\, K_j/\theta_{j,\max} = \sum_i \mathfrak{f}_{j,i}$*,* $1 - (n-r_j)!\, \Psi_j/\theta_{j,\max} = \sum_i \mathfrak{f}^*_{j,i}$*, and*

$$\int_{\Delta_\mathcal{E}} u^{\mathbf{m-1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, d\omega_\mathcal{E} = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} \prod_j \Big(\frac{r_j!}{\theta_{j,\max}}\Big)^{s_j} \Big(\frac{(n-r_j)!}{\theta_{j,\max}}\Big)^{t_j} \times \sum_{\mathbf{k},\mathbf{l}} \prod_j \frac{(s_j)_{|\mathbf{k}_j|}\, (t_j)_{|\mathbf{l}_j|}}{\mathbf{k}_j!\, \mathbf{l}_j!}\ \mathbb{E}_{U \sim \mathrm{Dir}_\mathbf{m}}\Big[\prod_{j,i} \mathfrak{f}_{j,i}(U)^{k_{j,i}}\, \mathfrak{f}^*_{j,i}(U)^{l_{j,i}}\Big],$$

*over* $\mathbf{k}_j, \mathbf{l}_j \in \mathbb{N}^{d+1}$*, every term a finite nonnegative combination of Dirichlet moments, a finite sum of them for lists, whose rounds are always accepted. The numbers* $\sum_{i' \geq i} k_{j,i'}$ *of failing basis rounds of table* $j$ *that pass* $B_0, \dots, B_{i-1}$ *decrease with* $i$*: they are nested, as the coordinates of the region* $X_N$ *of a trail are (Definition 5.23). Let a bijection* $g$ *keep each* $B_i$ *and the masses, let* $h$ *be a bijection of the tables, and put* $(g_* u)_{g(e)} = u_e$*. If* $g$ *maps the bases of every table* $j$ *onto those of table* $h(j)$ *with their weights, then* $\mathfrak{f}_{h(j),i}(g_* u) = \mathfrak{f}_{j,i}(u)$ *and* $\mathfrak{f}^*_{h(j),i}(g_* u) = \mathfrak{f}^*_{j,i}(u)$*; if* $T \mapsto \mathcal{E} \setminus g(T)$ *does, then* $\mathfrak{f}^*_{h(j),i}(g_* u) = \mathfrak{f}_{j,i}(u)$ *and* $\mathfrak{f}_{h(j),i}(g_* u) = \mathfrak{f}^*_{j,i}(u)$*, and the array of expectations is unchanged when every* $(\mathbf{k}_j, \mathbf{l}_j)$ *is replaced by* $(\mathbf{l}_{h^{-1}(j)}, \mathbf{k}_{h^{-1}(j)})$*; for one table, where an exchange forces* $2r = n$*, this is the symmetry under* $\mathbf{k} \leftrightarrow \mathbf{l}$*.*

**Theorem 20.15** (The draw series along a tree)**.** *Let a molecule* $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_j$ *on* $\mathcal{E}$ *carry* $s_j, t_j \geq 0$ *and masses* $\mathbf{m} \in \mathbb{R}^{\mathcal{E}}_{>0}$*, and take a tree of bunches (Definition 20.7) at every node of which the children's weights factor across the childcules, as they do for lists: for a table* $\mathcal{B}'$ *at a node* $v$ *and a pattern* $p$ *of* $\mathcal{B}'$ *at* $B_v$*, the restrictions* $\mathcal{B}'_{(p)}|R_{v,i}$ *of the child to the childcules (Proposition 7.9) carry weights whose products are the weights of the child,* $\theta_{(p),X} = \prod_i \theta_{(p),i}(X \cap R_{v,i})$*. Write* $\theta_{\max}(\mathcal{B}')$ *for the largest weight of a table* $\mathcal{B}'$*, and say that a round of draws that is a basis* $T$ *of* $\mathcal{B}'$ *is* accepted *with probability* $\theta_T / \theta_{\max}(\mathcal{B}')$*. A cobasis round of table* $j$ *is a basis round of its dual table, whose children are the duals of the children, with the same weights (Lemma 14.7(ii) with the identity). The* coordinates of the tree *are, at each node* $v$*, the totals* $x^{(v)} \in \Delta_{\{B_v, R_{v,1}, \dots, R_{v,q_v}\}}$ *of the shape of* $M_v$ *on its blocks and the shape* $u^{(v)} \in \Delta_{B_v}$ *of the bunch; under* $\mathrm{Dir}_{\mathbf{m}}$ *they are independent, with the laws* $\mathrm{Dir}_{(m_{B_v}, m_{R_{v,1}}, \dots, m_{R_{v,q_v}})}$ *and* $\mathrm{Dir}_{\mathbf{m}|_{B_v}}$ *(Theorem 3.12). For a table* $\mathcal{B}'$ *at a node write its chances as functions of the coordinates of the subtree of that node.*

*(i)* A round passes a node, then each childcule. *For a table* $\mathcal{B}'$ *of rank* $r'$ *at a node* $v$*, the chance* $\mathfrak{s}_{\mathcal{B}'} = r'! \, K_{\mathcal{B}'} / \theta_{\max}(\mathcal{B}')$ *that* $r'$ *draws from the shape of* $M_v$ *are a basis of* $\mathcal{B}'$ *and are accepted is*

$$\mathfrak{s}_{\mathcal{B}'} = \sum_p \mathfrak{p}_{\mathcal{B}',p} \prod_i \mathfrak{s}_{\mathcal{B}'_{(p)}|R_{v,i}},$$

$$\mathfrak{p}_{\mathcal{B}',p} = \frac{\prod_i \theta_{\max}\big(\mathcal{B}'_{(p)}|R_{v,i}\big)}{\theta_{\max}(\mathcal{B}')} \, \frac{r'!}{\prod_i r'_{p,i}!} \, \big(x^{(v)}_{B_v}\big)^{|p|} \big(u^{(v)}\big)^p \prod_i \big(x^{(v)}_{R_{v,i}}\big)^{r'_{p,i}},$$

*over the patterns* $p$ *of* $\mathcal{B}'$ *at* $B_v$*, with* $r'_{p,i}$ *the rank on* $R_{v,i}$ *of the child* $\mathcal{B}'_{(p)}$*:* $\mathfrak{p}_{\mathcal{B}',p}$ *is the chance that the round* passes $v$ *showing* $p$*, its draws in* $B_v$ *being the marks of* $p$ *and its numbers of draws in the childcules the* $r'_{p,i}$*, and is accepted there, with the probability given by its first factor, which is at most* 1*; and* $\mathfrak{s}_{\mathcal{B}'_{(p)}|R_{v,i}}$ *is the chance that its draws in* $R_{v,i}$ *are a basis of the restriction of the child there and are accepted. At a leaf* $\mathfrak{s}_{\mathcal{B}'} = r'! \, K_{\mathcal{B}'} / \theta_{\max}(\mathcal{B}')$*.*

*(ii)* A failure is met at one node. *The chance* $1 - \mathfrak{s}_{\mathcal{B}'}$ *that the round fails, by not being a basis or by not being accepted, is* $\sum_w \mathfrak{f}_{\mathcal{B}' \to w}$ *over the nodes and leaves* $w$ *of the subtree of* $v$*, where* $\mathfrak{f}_{\mathcal{B}' \to v} = 1 - \sum_p \mathfrak{p}_{\mathcal{B}',p}$ *and, for* $w$ *in the subtree of* $R_{v,i}$*,*

$$\mathfrak{f}_{\mathcal{B}' \to w} = \sum_p \mathfrak{p}_{\mathcal{B}',p} \prod_{i' < i} \mathfrak{s}_{\mathcal{B}'_{(p)}|R_{v,i'}} \; \mathfrak{f}_{\mathcal{B}'_{(p)}|R_{v,i} \to w},$$

*the chance that the round first fails at* $w$*. Each* $\mathfrak{f}_{\mathcal{B}' \to w}$ *is a finite sum of products, over the blocks of the tree, of polynomials with nonnegative coefficients in one block each.*

*(iii)* The series. *With* $\mathfrak{f}_{j,w}$ *and* $\mathfrak{f}^*_{j,w}$ *these chances for the basis and the cobasis rounds of table* $j$ *of* $F$ *from the root, and* $\theta_{j,\max}$ *the largest weight of table* $j$*,*

$$\int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, d\omega_{\mathcal{E}} = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} \prod_j \Big(\frac{r_j!}{\theta_{j,\max}}\Big)^{s_j} \Big(\frac{(n-r_j)!}{\theta_{j,\max}}\Big)^{t_j}$$
$$\times \sum_{\mathbf{k},\mathbf{l}} \prod_j \frac{(s_j)_{|\mathbf{k}_j|}\,(t_j)_{|\mathbf{l}_j|}}{\mathbf{k}_j!\,\mathbf{l}_j!}\ \mathbb{E}_{U\sim \mathrm{Dir}_{\mathbf{m}}}\Big[\prod_{j,w} \mathfrak{f}_{j,w}(U)^{k_{j,w}}\, \mathfrak{f}^*_{j,w}(U)^{l_{j,w}}\Big],$$

*every term nonnegative, both sides finite or infinite together; on the balance wall the left side is $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$. Each expectation is a finite nonnegative combination of products over the blocks of Dirichlet moments, a finite sum of them for lists: at every node, the node's own moments times the product over its childcules of theirs.*

*(iv)* Nesting along the tree. *The numbers $k_{j,\geq w} = \sum_{w'} k_{j,w'}$, over the $w'$ in the subtree of $w$, of the basis rounds of table $j$ that first fail in the subtree of $w$, satisfy $k_{j,\geq w} = k_{j,w} + \sum_{w'} k_{j,\geq w'}$ over the childcules $w'$ of $w$, and likewise for the cobasis rounds: along every descent they are nested, as the coordinates of the region $X_N$ of a trail are (Definition 5.23). On a chain this is Theorem 20.14.*

At nonnegative exponents every share of the word closes at once. The draw series closes each letter about its largest coefficient on the simplex, and the tree only sorts its failures: a round that fails is charged to the first node at which it stops passing, the chances of passing are monomials in the coordinates of that node, and the coordinates of different nodes are independent. So every term is a nonnegative combination of products, over the blocks of the tree, of Dirichlet moments, and the counts of failures nest along the descents as the coordinates of the region $X_N$ of a trail do.

**Corollary 20.16** (What each node of a chain integrates)**.** *Let the tree of a molecule on $n$ marks be a chain $M_0 \supset M_1 \supset \cdots \supset M_d$ with bunches $B_k$ and rooms $R_k = M_{k+1}$. The simplex of the molecule is the product*

$$\Delta_{M_0} \cong \prod_{k=0}^{d-1} \Big(\Delta_{B_k} \times \Delta_{\{B_k,R_k\}}\Big) \times \Delta_{M_d}$$

*of the shape of each bunch, the totals of the bunch and its room at each node, and the shape of the leaf, and under $\mathrm{Dir}_{\mathbf{m}}$ these coordinates are independent: they are the coordinates of the tree of Theorem 20.15, which use no weights. The node of $B_k$ has $|B_k|$ coordinates, $|B_k|-1$ of shape and one total, the leaf has $|M_d|-1$, and they add up to $n-1$. When some active table is connected, the Foreman tree, which is then a chain, has, at the centre and at every lean, no node with more coordinates than the leaf has marks (Corollary 20.9).*

**Proposition 20.17** (Rates at the centre)**.** *Let a family on $n \geq 3$ marks be at the centre, every table with the exponents $(\sigma_j,\sigma_j)$, with the masses $\sigma\mathbf{1}$, $\sigma = \sum_j \sigma_j$, and let every active table be connected.*

*(i) The walls are $\kappa_S = \sum_j \sigma_j c_j(S)$ (Proposition 8.21(ii)), so $\kappa_S = \kappa_{\mathcal{E}\setminus S}$.*

*(ii) The rate of the peak closing (Theorem 15.13(ii)) is $\varpi = \sigma/\max_j \max(r_j-1, n-r_j-1)$, the maximum over the active tables, attained at the faces $S = \mathcal{E}-e$, where every active table has $c_j = 1$, $d_j = r_j - 1$ and $d_j^* = n - r_j - 1$.*

*(iii) Near the vertex of $e$, with $u_e = 1-x$ and $u_{\mathcal{E}-e} = xv$, $v \in \Delta_{\mathcal{E}-e}$, every active table has*

$$K_j(u) = (1-x)\,x^{r_j-1} K_{j,/e}(v) + x^{r_j} K_{j,\setminus e}(v),$$
$$\Psi_j(u) = (1-x)\,x^{n-r_j-1} \Psi_{j,\setminus e}(v) + x^{n-r_j} \Psi_{j,/e}(v);$$

*the* vertex product *of $e$ in table $j$ is $K_{j,/e}\Psi_{j,\setminus e}$, the product of the letters of the two minors that dominate as $x \to 0$.*

*(iv) For two marks $e, f$ and a bijection $\beta$ with $\beta(e) = f$ that carries or exchanges the family, with the bijection $h$ of its tables, the restriction $\beta : \mathcal{E} - e \to \mathcal{E} - f$ takes the vertex product of $e$ in table $j$ to that of $f$ in table $h(j)$. For one table such a $\beta$ exists exactly when $e$ and $f$ lie in one class at the centre (Corollary 11.16).*

## 20.5 Devices at the centre

**Proposition 20.18** (The half-sum and the difference of a table with $2r = n$)**.** *Let $F = ((\mathcal{B}, \theta); \sigma, \sigma)$, $\sigma > 0$, have $2r = n$ and masses $\mathbf{m}$ on its balance wall. The half-sum $\frac{1}{2}(K + \Psi) = \frac{1}{2}\sum_T \theta_T\big(\alpha^T + \alpha^{\mathcal{E}\setminus T}\big)$ is homogeneous of degree $r$ and unchanged by $T \mapsto \mathcal{E} \setminus T$; the difference $K - \Psi$ vanishes at the table's gauge; in signed coordinates they are $e^{\langle \mathbf{1}, z\rangle} L_{1/2}(z)$ and $e^{\langle \mathbf{1}, z\rangle}\partial_w L_w(z)$ (Definition 20.2). Then $K\Psi = \big(\frac{1}{2}(K + \Psi)\big)^2 - \frac{1}{4}(K - \Psi)^2$ with $|K - \Psi| < K + \Psi$ on the orthant, and by Proposition 20.1*

$$(K\Psi)^{-\sigma} = \sum_{l \geq 0} \frac{(\sigma)_l}{l!\, 4^l}\, (K - \Psi)^{2l} \Big(\frac{K + \Psi}{2}\Big)^{-2\sigma - 2l},$$

*a series of nonnegative terms. For a mark $e$ with room $R = \mathcal{E} - e$ write $\frac{1}{2}(K + \Psi) = h_0(\alpha_R) + \alpha_e h_1(\alpha_R)$, so that, by Definition 7.3, $h_0 = \frac{1}{2}(K_{\setminus e} + \Psi_{/e})$, $h_1 = \frac{1}{2}(K_{/e} + \Psi_{\setminus e})$ and $h_1(u) = u^R h_0(u^{-1})$. Expanding $(K - \Psi)^{2l} = \sum_{\mathbf{a}} \big([\alpha^{\mathbf{a}}](K - \Psi)^{2l}\big)\, \alpha^{\mathbf{a}}$ and integrating the mark $e$ of each term as the Beta corner of a linear letter (Proposition 15.1 with one coefficient), with $w_l = 2\sigma + 2l$,*

$$P_F(\mathbf{m}; \sigma, \sigma) = \sum_{l \geq 0} \frac{(\sigma)_l}{l!\, 4^l} \sum_{\mathbf{a}} \big([\alpha^{\mathbf{a}}](K - \Psi)^{2l}\big)\, \Gamma_{\substack{2\\1}}\begin{bmatrix} m_e + a_e, & w_l - m_e - a_e \\ & w_l \end{bmatrix} \times \int_{\Delta_R} u^{(\mathbf{m}+\mathbf{a})|_R - \mathbf{1}} h_0(u)^{-(w_l - m_e - a_e)} h_1(u)^{-(m_e + a_e)}\, d\omega_R(u),$$

*whenever the terms converge absolutely.*

The half-sum is a positive polynomial whose support is closed under complements, but it is not in general the basis polynomial of a table: the bases together with the cobases need not have the exchange property, even for connected tables, so the integrals left over the room are integrals of a positive polynomial against its complement and not periods of a family. When complementation maps the bases onto themselves the difference vanishes and the room is a period.

**Corollary 20.19** (Merged Skeleton values)**.** *If $T \mapsto \mathcal{E} \setminus T$ maps the bases of a table $(\mathcal{B}, \theta)$ onto themselves with $\theta_{\mathcal{E}\setminus T} = \theta_T$, then $2r = n$, no mark is a loop or a coloop, $\Psi = K$, $\Psi_{\setminus e} = K_{/e}$ and $\Psi_{/e} = K_{\setminus e}$ for every mark $e$, and the two Skeleton values of the scale of $e$ coincide at every gauge (Corollary 15.4). For $F = ((\mathcal{B}, \theta); s, t)$ with $s, t \geq 0$, $s + t > 0$, its masses on the balance wall with every proper wall positive, and a mark $e$, one has $0 < m_e < s + t$ and*

$$P_F(\mathbf{m}; s, t) = \Gamma_{\substack{2\\1}}\begin{bmatrix} m_e, & s + t - m_e \\ & s + t \end{bmatrix}\, P_{\mathcal{B}\setminus e}\big(\mathbf{m}|_{\mathcal{E}-e};\, s + t - m_e,\, m_e\big),$$

*the period of the deletion $(\mathcal{B} \setminus e, \theta)$ on $\mathcal{E} - e$ at the exponents $(s + t - m_e, m_e)$, on its balance wall; so $P_F(\mathbf{m}; s, t)$ depends on the exponents only through $s + t$, and at the centre, $s = t = \sigma$ and $\mathbf{m} = \sigma\mathbf{1}$, the identity reads $P_F(\sigma\mathbf{1}; \sigma, \sigma) = \Gamma_{\substack{2\\1}}\begin{bmatrix} \sigma, & \sigma \\ & 2\sigma \end{bmatrix}\, P_{\mathcal{B}\setminus e}(\sigma\mathbf{1}|_{\mathcal{E}-e}; \sigma, \sigma)$.*

**Theorem 20.20** (The scale integral of one mark)**.** *Let a family $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_j$ carry real exponents $s_j, t_j \geq 0$, its masses on the balance wall with every proper wall positive, and let $e$ be a mark that is neither a loop nor a coloop in any active table, with room $R = \mathcal{E} - e$; write $\Lambda_{j,\backslash e}$ and $\Lambda_{j,/e}$ for the centred polynomials of the deletion and the contraction of table $j$, so that $\Lambda_j(z) = e^{z_e}\Lambda_{j,/e}(z_R) + e^{-z_e}\Lambda_{j,\backslash e}(z_R)$ (Theorem 12.7).*

*(i) At the gauge $(1, e^{2z_R})$ the scale integral of the bunch $\{e\}$ (Theorem 15.3) is*

$$\begin{aligned}&\mathrm{Sc}_{\{e\}}\big((1, e^{2z_R}); \mathbf{m}; \mathbf{s}, \mathbf{t}\big)\\ &= e^{-2\sigma\langle \mathbf{1}, z_R\rangle}\int_0^\infty x^{m_e-1}\prod_j \big(\Lambda_{j,\backslash e}(z_R) + x\Lambda_{j,/e}(z_R)\big)^{-s_j}\big(\Lambda_{j,/e}(-z_R) + x\Lambda_{j,\backslash e}(-z_R)\big)^{-t_j}dx,\end{aligned}$$

*$\sigma = \sum_j \sigma_j$ the total strength: up to the factor $e^{-2\sigma\langle \mathbf{1}, z_R\rangle}$ it depends only on $m_e$, the exponents and the functions $\Lambda_{j,\backslash e}$ and $\Lambda_{j,/e}$ on the room, and Corollary 16.6 at the bunch $\{e\}$ reads, for every $g \in R$,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = 2^{n-2}\int_{\mathbb{R}^{R-g}} e^{2\langle \mathbf{m}|_R, z_R\rangle}\ \mathrm{Sc}_{\{e\}}\big((1, e^{2z_R}); \mathbf{m}; \mathbf{s}, \mathbf{t}\big)\, dz_{R-g}\Big|_{z_g=0}.$$

*(ii) At an atom $\Upsilon = ((s_j, t_j)_j; \alpha^\star)$ with its target $\boldsymbol{\tau}$, under $\mathrm{Breta}^\circ(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ the law of the log-ratios $(y_{g_k} - y_e)_{k=1}^{n-1}$, read in an order $\mathbf{g}_e$ of the room that writes the potential from $e$ as the one pair of functions of Proposition 10.4, does not depend on $e$.*

*(iii) For one table at the centre, $s = t = \sigma$ and $\boldsymbol{\mu} = 0$, with the even statistics $E_{\backslash e}$ and $E_{/e}$ of the deletion and the contraction (Theorem 7.16) and a two-element set $J$, the functions of $z_R$ understood,*

$$\begin{aligned}&\int_0^\infty x^{\sigma-1}\big(\Lambda_{\backslash e} + x\Lambda_{/e}\big)^{-\sigma}\big(\Lambda_{/e}(-\cdot) + x\Lambda_{\backslash e}(-\cdot)\big)^{-\sigma}dx\\ &\qquad = e^{-\frac{\sigma}{2}(E_{\backslash e}+E_{/e})}\,\mathsf{B}^{\{J\}}_{(\sigma,\sigma),\,\sigma;\,I_J}\big(e^{\frac{1}{2}(E_{\backslash e}-E_{/e})},\, e^{\frac{1}{2}(E_{/e}-E_{\backslash e})}\big),\end{aligned}$$

*and the Breta is ${}_1^{}\Gamma_2^{}\left[\begin{matrix}\sigma,\ \sigma\\ 2\sigma\end{matrix}\right]$ where the two even statistics agree.*

For one table at the centre the scale integral of a mark is the factor $e^{-\frac{\sigma}{2}(E_{\backslash e}+E_{/e})}$ times a Breta that sees the room through one number, the difference of the even statistics of the deletion and the contraction; at an atom the law of the log-ratios read from a mark does not depend on the mark. Proposition 20.21 reads the scale integral of the mark with the exponents in the shape index.

**Proposition 20.21** (A mark is the Breta of its children's even letters)**.** *For one table, a mark $e$ that is neither a loop nor a coloop, exponents $s, t > 0$, $0 < m_e < s+t$ and the top wall $\kappa^{\mathrm{top}}_{\{e\}} = s+t-m_e$, integrating $\alpha_e = x$ with the room fixed,*

$$\int_0^\infty x^{m_e-1}K^{-s}\Psi^{-t}\,dx = K_{/e}^{\,t-m_e}\Psi_{\backslash e}^{\,s-m_e}\,{}_2^{}\Gamma_2^{}\left[\begin{matrix}m_e,\ \kappa^{\mathrm{top}}_{\{e\}}\\ s,\ t\end{matrix}\right]\ \mathsf{B}^{\{J\}}_{(s,t),\,\kappa^{\mathrm{top}}_{\{e\}};\,I_J}\big(\mathbf{E}_{\backslash e}, \mathbf{E}_{/e}\big),$$

*with $\mathbf{E}_{\backslash e} = K_{\backslash e}\Psi_{\backslash e}$ and $\mathbf{E}_{/e} = K_{/e}\Psi_{/e}$ the even letters of the deletion and the contraction, $J$ a two-element set, and*

$$\mathsf{B}^{\{J\}}_{(s,t),\,\kappa;\,I_J}(a, b) = \int_0^1 w^{s-1}(1-w)^{t-1}\,\mathsf{Z}_{\kappa;(1)}\big(wa + (1-w)b\big)\,dw,$$

*the shape $w$ of $J$ weighing the deletion against the contraction. At the centre, $s = t = m_e = \sigma$, the prefactor and the stacked Gamma are 1.*

The same scale integral is the Breta of Corollary 19.29 for the bunch $\{e\}$, whose levels $d$ and $r$ have the children $\mathcal{B}\setminus e$ and $\mathcal{B}/e$: there the walls $(m_e, \kappa^{\mathrm{top}}_{\{e\}})$ fill the shape index and the exponent $t$ the scale index, here the exponents $(s,t)$ fill the shape index and the top wall the scale index. Both read the scale integral of the mark through the even letters of the two children, the deletion and the contraction.

## 20.6 The junction under level form

**Lemma 20.22** (An even power of $e^{v}-e^{-v}$ on a symmetric two-sided scale)**.** *For $\xi>0$ and $l\in\mathbb{N}$,*

$$\int_{\mathbb{R}} \big(e^{v}-e^{-v}\big)^{2l} e^{-\xi(e^{v}+e^{-v})}\,dv = 4^{l}\big(\tfrac{1}{2}\big)_{l}\,\xi^{-l}\,\mathsf{G}_{l;(1,-1)}(\xi,\xi).$$

**Theorem 20.23** (The junction under level form)**.** *Let $F=((\mathcal{B}_j,\theta_j);\sigma_j,\sigma_j)_{j=1}^{N}$, every $\sigma_j>0$, be a molecule on $n$ marks with the masses $\sigma\mathbf{1}$, $\sigma=\sum_j\sigma_j$, let $B$, $b=|B|$, be a proper bunch with room $R$ such that every transposition of two marks of $B$ fixing the rest carries or exchanges every table, and let $g\in R$. By Proposition 20.6 every half-sum has level form; in this statement write $q_{j,k}(z_R)$ for the common value of $c_{j,p}(\frac{1}{2};z_R)$ over $|p|=k$ and $\mathbf{q}_j=(q_{j,0},\dots,q_{j,b})$, so that $q_{j,b-k}(-z_R)=q_{j,k}(z_R)$ and*

$$L_{j,1/2}(z)=\sum_{k=0}^{b} e^{-\langle\mathbf{1},z_B\rangle}e_k\big(e^{2z_B}\big)\,q_{j,k}(z_R),\qquad e^{-\langle\mathbf{1},z_B\rangle}e_k\big(e^{2z_B}\big)=\sum_{|p|=k}e^{\langle z_B,\chi_p\rangle},$$

*the centred polynomial of the list on $B$ whose bases are the $k$-subsets of $B$; and for each $\mathbf{l}=(l_j)_j\in\mathbb{N}^N$*

$$\prod_j\big(\partial_w L_{j,w}(z)\big)^{2l_j}=e^{-2|\mathbf{l}|\langle\mathbf{1},z_B\rangle}\sum_{|\mathbf{h}|=2|\mathbf{l}|}\prod_{k=0}^{b}e_k\big(e^{2z_B}\big)^{h_k}\,c^{(\mathbf{l})}_{\mathbf{h}}(z_R),$$

*with $c^{(\mathbf{l})}_{\mathbf{h}}$ Laurent polynomials in $e^{z_R}$.*

*(i) With the Gramma $\mathsf{G}_{0;(\chi_p)_{p\subseteq B}}\big((\upsilon_{|p|})_{p\subseteq B}\big)$, $\boldsymbol{\upsilon}=(\upsilon_0,\dots,\upsilon_b)$, whose coefficients depend on $|p|$ alone and which depends on $B$ through $b$ alone,*

$$P_F(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})=2^{n-1}\sum_{\mathbf{l}\in\mathbb{N}^N}\prod_j\frac{(\sigma_j)_{l_j}}{l_j!\,4^{l_j}\,\Gamma(2\sigma_j+2l_j)}\int_{\mathbb{R}^{R-g}}\int_{(0,\infty)^N}\prod_j y_j^{2\sigma_j+2l_j-1}$$
$$\times\sum_{|\mathbf{h}|=2|\mathbf{l}|}c^{(\mathbf{l})}_{\mathbf{h}}(z_R)\Big((-\partial_{\boldsymbol{\upsilon}})^{\mathbf{h}}\,\mathsf{G}_{0;(\chi_p)_p}\big((\upsilon_{|p|})_p\big)\Big)\Big|_{\boldsymbol{\upsilon}=\sum_j y_j\mathbf{q}_j(z_R)}\,d\mathbf{y}\,dz_{R-g}\Big|_{z_g=0},$$

*the inner sum being $\int_{\mathbb{R}^B}\prod_j\big(\partial_w L_{j,w}(z)\big)^{2l_j}e^{-\sum_j y_j L_{j,1/2}(z)}\,dz_B\ge 0$, a level with $\sum_j y_j q_{j,k}=0$ absent.*

*(ii)* One mark at a time. *For $e\in B$, $B'=B-e$ and $\boldsymbol{\upsilon}>0$,*

$$\mathsf{G}_{0;(\chi_p)_{p\subseteq B}}\big((\upsilon_{|p|})_p\big)=\int_0^{\infty}\mathsf{G}_{0;(\chi_p)_{p\subseteq B'}}\big((x\,\upsilon_{|p|+1}+x^{-1}\upsilon_{|p|})_p\big)\,\frac{dx}{x},$$

*and for $B=\{e\}$ this Gramma is $\mathsf{G}_{0;(1,-1)}(\upsilon_1,\upsilon_0)$.*

*(iii)* The share side. *Let $G$ be the family on $B$ of the lists whose bases are the $k$-subsets of $B$, $1 \le k \le b-1$, with the exponents $(a'_k, 0)$, at the masses $\frac{1}{b}\sum_{k=1}^{b-1} k a'_k\, \mathbf{1}_B$ on its balance wall. Where every $\operatorname{Re} a'_k > 0$ its period converges, is unchanged when the exponents of the lists of ranks $k$ and $b-k$ are swapped, and is read on the hyperplane $\langle \mathbf{1}, w\rangle = 0$:*

$$P_G\Big(\tfrac{1}{b}\sum_{k=1}^{b-1} k a'_k\, \mathbf{1}_B; (a'_k)_{k=1}^{b-1}, \mathbf{0}\Big) = 2^{b-1}\sqrt{b}\int_{\langle \mathbf{1}, w\rangle = 0} \prod_{k=1}^{b-1} e_k\big(e^{2w}\big)^{-a'_k}\, d\mathrm{vol}_{b-1}(w).$$

*For $\boldsymbol{\upsilon} > 0$ and $\mathbf{h} \in \mathbb{N}^{b+1}$, on a contour with $\operatorname{Re} a_k > h_k$ for every $k$ and with $a'_k = a_k - h_k$,*

$$\Big((-\partial_{\boldsymbol{\upsilon}})^{\mathbf{h}}\, \mathsf{G}_{0;(\chi_p)_p}\big((\upsilon_{|p|})_p\big)\Big)(\boldsymbol{\upsilon}) = 2^{1-b}\int_{\sum_{k=0}^{b}(2k-b)(a_k-h_k)=0} \prod_{k=0}^{b} \Gamma(a_k)\, \upsilon_k^{-a_k}\; P_G\Big(\tfrac{1}{b}\sum_{k=1}^{b-1} k a'_k\, \mathbf{1}_B; (a'_k)_{k=1}^{b-1}, \mathbf{0}\Big).$$

*(iv)* Two marks. *If $B = \{e, f\}$ and the swap of $e$ and $f$ exchanges every table, then with $z_\pm = z_e \pm z_f$, for every $j$,*

$$L_{j,1/2}(z) = e^{-z_+} c_{j,\varnothing}(\tfrac{1}{2}; z_R) + e^{z_+} c_{j,\varnothing}(\tfrac{1}{2}; -z_R) + \big(e^{z_-} + e^{-z_-}\big) c_{j,\{e\}}(\tfrac{1}{2}; z_R),$$
$$\partial_w L_{j,w}(z) = \big(e^{z_-} - e^{-z_-}\big)\, \partial_w c_{j,\{e\}}(w; z_R),$$

*the functions $c_{j,\{e\}}(\frac{1}{2}; \cdot)$ and $\partial_w c_{j,\{e\}}$ are even, and, writing in this statement $C_\varnothing(z_R) = \sum_j y_j c_{j,\varnothing}(\frac{1}{2}; z_R)$ and $C_e(z_R) = \sum_j y_j c_{j,\{e\}}(\frac{1}{2}; z_R)$,*

$$P_F(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma}) = 2^{n-2} \sum_{\mathbf{l}\in\mathbb{N}^N} \big(\tfrac{1}{2}\big)_{|\mathbf{l}|} \prod_j \frac{(\sigma_j)_{l_j}}{l_j!\,\Gamma(2\sigma_j + 2l_j)} \int_{\mathbb{R}^{R-g}} \prod_j \big(\partial_w c_{j,\{e\}}\big)^{2l_j}$$
$$\times \int_{(0,\infty)^N} \prod_j y_j^{2\sigma_j + 2l_j - 1}\, C_e^{-|\mathbf{l}|}\, \mathsf{G}_{0;(1,-1)}\big(C_\varnothing(-z_R),\, C_\varnothing(z_R)\big)$$
$$\times\, \mathsf{G}_{|\mathbf{l}|;(1,-1)}\big(C_e(z_R),\, C_e(z_R)\big)\, d\mathbf{y}\, dz_{R-g}\Big|_{z_g = 0},$$

*a series of nonnegative terms.*

Under level form the bunch enters only through its levels: one Gramma with one coefficient for each level, which depends on the number of its marks alone and is computed one mark at a time, is differentiated at the level coefficients of the half-sums, added over the tables with their scales, and its share side is a period of the lists of $k$-subsets of the bunch's own marks. For a pair whose swap exchanges every table the bunch splits along $z_e + z_f$ and $z_e - z_f$ into two Grammas $\mathsf{G}_{\nu;(1,-1)}$.

## 20.7 The node in one scale

At a bunch of width one the integrand of Theorem 19.13 is, in the even share, the product of two Mellin transforms, one carrying the bunch and one the room. Read back on the half-line, the period is one integral over one scale of two functions, the bunch transform and the room transform, each an integral of a Gramma $\mathsf{G}_{\nu;(1,-1)}$ over its own shape. A closing toward a level is the expansion of one of them in powers of the scale, integrated term by term against the other. The expansion

of a bunch is measured by one number, its reach: every bunch has a reach, no reach lies below the least value of the bunch's ratio, and a node closes on its lattice toward every level at which the growths of its two families multiply to less than one. Each side of a node is an interval, from the decay of its transform to its reach, and, against a room and a bunch bounded below, a node closes on a geometric lattice exactly when the two intervals are disjoint.

**Definition 20.24** (Node of width one; its two transforms)**.** A *node of width one*, a node of a tree of bunches (Definition 20.7) whose bunch has width one, is a family $F = ((\mathcal{B}, \theta); s, t)$ with real data, $s, t > 0$, its masses on the balance wall with every proper wall positive, and a bunch $B$ of width one with room $R$, the weights factoring across them: the setting of Theorem 19.13 for one table. Along the level $d$ its level shares are $(a^d, b^d, a^r, b^r) = (s - \kappa_B + b^d,\ b^d,\ \kappa_B - b^d,\ t - b^d)$, the odd share being fixed at $2\delta^d = s - \kappa_B$; its ratios $\varrho_B$ on $\Delta_B$ and $\varrho_R$ on $\Delta_R$ are those of Definition 19.2; and $\sigma_*^d$ is a real value of the even share at which both families converge and the four arguments of its stacked Gamma are positive, so that the line $\operatorname{Re}\sigma^d = \sigma_*^d$ is a contour of Theorem 19.13, with $b_* = \sigma_*^d - \delta^d > 0$ the real part of $b^d$ there. The node carries the nonnegative measures

$$\phi_B(du) = u^{\mathbf{m}|_B - \mathbf{1}} K_{\mathcal{B}/R}^{-(s-\kappa_B)} K_{\mathcal{B}|B}^{-\kappa_B} \Psi_{\mathcal{B}|B}^{-t}\, d\omega_B(u) \quad \text{on } \Delta_B,$$
$$\phi_R(dv) = v^{\mathbf{m}|_R - \mathbf{1}} K_{\mathcal{B}|R}^{-(s-\kappa_B)} K_{\mathcal{B}/B}^{-\kappa_B} \Psi_{\mathcal{B}/B}^{-t}\, d\omega_R(v) \quad \text{on } \Delta_R,$$

the forms of the bunch family and of the room family at the level share $b^d = 0$; Chapter 23 calls $\phi_B$ the base form of the cell of the bunch $\mathcal{B}_B^+$ at the exponents $(s, t)$. At the share $b^d$ the periods of the two families are

$$P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B; \mathbf{a}, \mathbf{b}) = \int_{\Delta_B} \varrho_B^{-b^d}\, d\phi_B, \qquad P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R; \mathbf{a}, \mathbf{b}) = \int_{\Delta_R} \varrho_R^{b^d}\, d\phi_R,$$

continued meromorphically in $b^d$ by their sector forms (Theorem 8.5, Lemma 8.4): the *continued periods along the level d*. For $y > 0$ the *bunch transform* and the *room transform* are

$$f_B(y) = \int_{\Delta_B} \mathsf{G}_{s-\kappa_B;(1,-1)}\big(1, \varrho_B(u)\, y\big)\, \phi_B(du),$$
$$f_R(y) = \int_{\Delta_R} \big(\varrho_R(v)\, y\big)^{\kappa_B}\, \mathsf{G}_{t-\kappa_B;(1,-1)}\big(1, \varrho_R(v)\, y\big)\, \phi_R(dv),$$

with values in $[0, \infty]$. Swapping $B$ and $R$ gives the *swapped node*, with the bunch $R$, the room $B$ and the two levels swapped (Theorem 19.13).

The bunch family at the share $b^d$ has the form $\phi_B$ times $(\mathbf{E}_{\mathcal{B}/R}/\mathbf{E}_{\mathcal{B}|B})^{-b^d} = \varrho_B^{-b^d}$, and the room family the form $\phi_R$ times $(\mathbf{E}_{\mathcal{B}|R}/\mathbf{E}_{\mathcal{B}/B})^{-b^d} = \varrho_R^{b^d}$: the even share enters through the ratio of the hinge values of each side alone (Lemma 19.15).

**Lemma 20.25** (Two letters in one scale)**.** *For linear letters $L_i(\rho) = L_i^{(0)} + \rho L_i^{(1)}$ with positive coefficients and the Skeleton values $\xi_i = L_i^{(0)}/L_i^{(1)}$, $i = 1, 2$, exponents $s, t > 0$ and $0 < \kappa < s + t$,*

$$\int_0^\infty \rho^{\kappa-1} L_1(\rho)^{-s} L_2(\rho)^{-t}\, d\rho$$
$$= \frac{(L_1^{(0)})^{\kappa-s} (L_1^{(1)})^{-\kappa} (L_2^{(0)})^{-t}}{\Gamma(s)\Gamma(t)} \int_0^\infty x^{\kappa-1}\, \mathsf{G}_{s-\kappa;(1,-1)}\Big(1, \frac{\xi_1}{\xi_2}\, x\Big)\, \mathsf{G}_{t-\kappa;(1,-1)}(1, x)\, dx.$$

**Theorem 20.26** (The node in one scale)**.** *Let $F$ and $B$ be a node of width one (Definition 20.24).*

*(i) The period is one integral over one scale of the two transforms,*

$$P_F(\mathbf{m}; s, t) = \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty f_B(y)\, f_R(y)\, \frac{dy}{y},$$

*the integrand nonnegative.*

*(ii) With the hinge values of Definition 19.2,*

$$f_B(y) = \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} K_{\mathcal{B}|B}^{-s} \Psi_{\mathcal{B}|B}^{-t}\, \mathsf{G}_{s-\kappa_B;(1,-1)}\big(\gamma_B^K(u)^{-1},\, y\,\gamma_B^\Psi(u)\big)\, d\omega_B(u) :$$

*the bunch meets the room through its two hinge values, as in pooling (Theorem 19.8).*

*(iii) For real $b^d > 0$ with $a^d > 0$, and for real $b^d$ with $a^r > 0$ and $b^r > 0$, respectively, in $[0, \infty]$,*

$$\int_0^\infty y^{b^d - 1} f_B(y)\, dy = \Gamma(b^d)\, \Gamma(a^d) \int_{\Delta_B} \varrho_B^{-b^d}\, d\phi_B,$$
$$\int_0^\infty y^{-b^d - 1} f_R(y)\, dy = \Gamma(a^r)\, \Gamma(b^r) \int_{\Delta_R} \varrho_R^{b^d}\, d\phi_R.$$

*Where the integrals over the shapes are finite they are the continued periods of the bunch family and of the room family at the share $b^d$, with their Gamma pairs, and (i), written through these two integrals in the share $b^d$, is Theorem 19.13 for one table.*

*(iv) If no slope on the room is positive, so that $\inf \varrho_R > 0$ (Lemma 19.15), then $\phi_R(\Delta_R) < \infty$, and with $a = \frac{t+\kappa_B}{2} - \frac{1}{4}$*

$$f_R(y) \le \sqrt{\pi}\, e^{(t-\kappa_B)^2/4}\, \phi_R(\Delta_R)\, \big(y \inf \varrho_R\big)^a\, e^{-2\sqrt{y \inf \varrho_R}} \qquad \Big(y \ge \frac{\max(1, a^2)}{\inf \varrho_R}\Big) :$$

*the room transform decays at the least ratio of the room.*

**Lemma 20.27** (The bunch transform on the orthant)**.** *For a node of width one and $y > 0$,*

$$f_B(y) = \int_{\mathbb{R}^B_{>0}} x^{\mathbf{m}|_B - \mathbf{1}} K_{\mathcal{B}|B}(x)^{-s} \Psi_{\mathcal{B}|B}(x)^{-t} \exp\Big( - \frac{1}{\gamma_B^K(x)} - y\,\gamma_B^\Psi(x)\Big)\, dx.$$

The bunch transform is a Laplace transform in the hinge value $\gamma_B^\Psi$, damped by the hinge value $\gamma_B^K$: the room meets the bunch through its two hinge values and nothing else. Its expansion in powers of $y$ is measured by its reach.

**Definition 20.28** (Lattice expansion; reach)**.** A function $f$ on $(0, \infty)$ has a *lattice expansion of reach* $x \ge 0$ when there are finitely many reals $\eta_1, \dots, \eta_q$, an integer $J \ge 0$ and coefficients $\beta_{\eta,j}$, for $\eta$ in $A = \bigcup_i (\eta_i + \mathbb{N})$ and $0 \le j \le J$, such that

$$f(y) = \sum_{\eta \in A} \sum_{j=0}^{J} \beta_{\eta,j}\, y^\eta (-\log y)^j \quad (y > 0), \qquad |\beta_{\eta,j}| \le C(1+\eta)^D\, \frac{x^\eta}{\Gamma(1+\eta)^2} \quad (\eta \in A,\ \eta \ge 0),$$

with constants $C$ and $D$; the series then converges absolutely at every $y > 0$, and $\sum_{\eta,j} |\beta_{\eta,j}| < \infty$. Its *least exponent* is the least $\eta$ with some $\beta_{\eta,j} \ne 0$. A function has at most one lattice expansion. The *reach of the bunch* of a node of width one, $\mathrm{reach}(B)$, is the infimum of the $x$ for which the bunch transform has a lattice expansion of reach $x$, and the *reach of the room*, $\mathrm{reach}(R)$, is the reach of $R$ as the bunch of the swapped node.

**Lemma 20.29** (The lattice of one mark)**.** *Let $\nu$ be real.*

*(i) For $x > 0$, $\mathsf{G}_{\nu;(1,-1)}(1,x)$ is the sum of the residues of $\Gamma(z)\Gamma(z+\nu)x^{-z}$ at the poles of $\Gamma(z)\Gamma(z+\nu)$. Off the integers*

$$\mathsf{G}_{\nu;(1,-1)}(1,x) = \sum_{k\geq 0} \frac{(-1)^k}{k!}\Big(\Gamma(\nu-k)\,x^k + \Gamma(-\nu-k)\,x^{\nu+k}\Big);$$

*at $\nu = l \in \mathbb{N}$*

$$\mathsf{G}_{l;(1,-1)}(1,x) = \sum_{k<l} \frac{(-1)^k}{k!}\Gamma(l-k)\,x^k + (-1)^l \sum_{k\geq l} \frac{x^k}{k!\,(k-l)!}\Big((\log\Gamma)'(1+k) + (\log\Gamma)'(1+k-l) - \log x\Big),$$

*and $\mathsf{G}_{-l;(1,-1)}(1,x) = x^{-l}\mathsf{G}_{l;(1,-1)}(1,x)$.*

*(ii) For $x \geq 1$, $0 < \mathsf{G}_{\nu;(1,-1)}(1,x) \leq \sqrt{\pi}\, e^{\nu^2/4}\, x^{\nu/2-1/4} e^{-2\sqrt{x}}$.*

*(iii) For $w > 0$, $y \mapsto \mathsf{G}_{\nu;(1,-1)}(1,w^2y)$ has a lattice expansion of reach $w^2$, with exponents in $\mathbb{N} \cup (\nu+\mathbb{N})$, in $-l+\mathbb{N}$ at $\nu = -l$, and at most one power of $-\log y$.*

**Lemma 20.30** (Products of lattice expansions)**.** *If $f_1$ and $f_2$ have lattice expansions of reaches $x_1$ and $x_2$, then $f_1 f_2$ has one of reach $(\sqrt{x_1}+\sqrt{x_2})^2$, whose terms are the products of theirs collected by exponent, and for real $a$, $y^a f_1$ has one of reach $x_1$.*

**Lemma 20.31** (Closing a lattice expansion against a decaying function)**.** *Let $f$ have a lattice expansion $\sum_{\eta,j} \beta_{\eta,j} y^\eta (-\log y)^j$ of reach $x$ with least exponent $\eta_0$, and let $g$ be measurable on $(0,\infty)$ and integrable on its compact subsets, with $\int_0^1 y^{\eta_0-1-\epsilon}|g(y)|\,dy < \infty$ for some $\epsilon > 0$ and $|g(y)| \leq C' y^a e^{-2\sqrt{x'y}}$ for $y \geq y_0$, where $x < x'$. Then*

$$\int_0^\infty f(y)\,g(y)\,\frac{dy}{y} = \sum_{\eta,j} \beta_{\eta,j} \int_0^\infty y^{\eta-1}(-\log y)^j\, g(y)\,dy,$$

*every integral on the right and the series converging absolutely, and the terms with $\eta \geq N$ add up in absolute value to at most $C''(1+N)^{D''}(x/x')^N$, with $C''$ and $D''$ independent of $N$.*

**Lemma 20.32** (Continued cube integrals grow at most exponentially)**.** *Let $G$ and $h$ be analytic on a neighbourhood of $[0,1]^m$ in $\mathbb{C}^m$, with $h > 0$ on $[0,1]^m$, and let $a_k, b_k$ be real with $a_k > 0$ wherever $b_k = 0$. The integral*

$$I(z) = \int_{[0,1]^m} \prod_k q_k^{a_k + b_k z - 1}\, G(q)\, h(q)^{-z}\, dq,$$

*convergent where every $a_k + b_k \operatorname{Re} z > 0$, continues meromorphically to $z \in \mathbb{C}$, with poles only in the set $\mathcal{W}$ of the real points at which some $a_k + b_k z$ lies in $-\mathbb{N}$ (Lemma 8.4). For every $\vartheta > 0$ there is $M$, and then for every $\epsilon > 0$ there are $C$ and $D$, with*

$$|I(z)| \leq C(1+|z|)^D\, M^{|\operatorname{Re} z|}\, e^{\vartheta|\operatorname{Im} z|}$$

*at every $z$ at distance at least $\epsilon$ from $\mathcal{W}$.*

**Definition 20.33** (Growth along a level; growth at the integers)**.** For a node of width one, the *growth* $\operatorname{growth}(B)$ of the bunch family along the level $d$ is the infimum of the $M \geq 0$ for which, for every $\epsilon > 0$, there are $C$, $D$ and $\vartheta < \pi$ with

$$\left|P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})\right| \leq C(1+|b^d|)^D\, M^{-\operatorname{Re} b^d}\, e^{\vartheta|\operatorname{Im} b^d|}$$

for the continued period at the share $b^d$, at every $b^d$ with $\operatorname{Re} b^d \leq 0$ at distance at least $\epsilon$ from the set $\mathcal{W}_B$ of the real points at which a proper wall of the bunch family lies in $-\mathbb{N}$, which holds its poles; the growth $\operatorname{growth}(R)$ of the room family is the same with $P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})$ and the set $\mathcal{W}_R$ of the room family. Along the level $r$ the growths are those of the swapped node. The *growth at the integers* of the bunch is

$$\operatorname{growth}_{\mathbb{Z}}(B) = \limsup_{k\to\infty} \left|P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})\right|^{1/k},$$

the continued period of the bunch family taken at $b^d = -k$, $k \in \mathbb{N}$.

**Theorem 20.34** (A node closes at the growth of its two families)**.** *Let $F$ and $B$ be a node of width one.*

*(i)* Every bunch has a reach. *The growths are finite;* $\operatorname{growth}(B) \leq \sup \varrho_B$ *and* $\operatorname{growth}(R) \leq 1/\inf \varrho_R$ *when these are finite; and* $\operatorname{reach}(B) \leq \operatorname{growth}(B)$. *The exponents of the lattice expansion of $f_B$ are the $\eta$ for which $b^d = -\eta$ is a pole of $\Gamma(b^d)\Gamma(a^d)P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})$ to the left of the line $\operatorname{Re} b^d = b_*$, and $\sum_j j!\,\beta_{\eta,j}(b^d+\eta)^{-j-1}$ is its principal part there: the lattice of the bunch is the lattice of its continued period with its Gamma pair, the wall residues joining.*

*(ii)* Against a room bounded below. *Let no slope on the room be positive, and let* $\operatorname{reach}(B) < \inf \varrho_R$. *Then*

$$P_F(\mathbf{m};s,t) = \frac{1}{\Gamma(s)\Gamma(t)} \sum_{\eta,j} \beta_{\eta,j}\, \partial_{b^d}^j \Big[\Gamma(a^r)\Gamma(b^r)\, P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})\Big]_{b^d=-\eta},$$

*absolutely, every room period in it convergent, and for every $x > \operatorname{reach}(B)$ the terms with $\eta \geq N$ add up in absolute value to at most $C(1+N)^D(x/\inf\varrho_R)^N$. The terms are the residues of the integrand of Theorem 19.13 at its poles toward the level $d$: the even share closes on the lattice of continued periods of the bunch family at the root rate* $\operatorname{reach}(B)/\inf\varrho_R$.

*(iii)* Both families continued. *Whatever the slopes, if* $\operatorname{growth}(B)\operatorname{growth}(R) < 1$, *then $P_F(\mathbf{m};s,t)$ is the sum of the residues of the integrand*

$$\Gamma_{\!\!\begin{smallmatrix}4\\2\end{smallmatrix}}\!\begin{bmatrix} a^d,\ b^d,\ a^r,\ b^r \\ s,\ t \end{bmatrix} P_{F^{\text{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})\, P_{F^{\text{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})$$

*of Theorem 19.13, continued in $b^d$, at its poles with $\operatorname{Re} b^d < b_*$; for every $M_B > \operatorname{growth}(B)$ and $M_R > \operatorname{growth}(R)$ the residues at the poles with real part in $(-N-1,-N]$ add up in absolute value to at most $C(1+N)^D(M_BM_R)^N$. The even share closes on the lattices of continued periods of both families, the wall residues of both joining.*

*(iv)* Toward the level $r$. *Swapping $B$ and $R$ leaves the period and Theorem 19.13 unchanged; the swapped node has the bunch $R$ with the ratio $\varrho_R$ and the room $B$ with the ratio $\varrho_B$. Applied to it, (i) to (iii) close the even share toward the level $r$: against a bunch with* $\inf\varrho_B > 0$ *at the root rate* $\operatorname{reach}(R)/\inf\varrho_B$, *and in general at a root rate at most the product of the growths of the two families along the level $r$.*

A node is therefore closed on its lattice toward the level at which the growths of its two families multiply to less than one, each growth read on its family alone; where neither level has a product below one, the peak closing (Theorem 19.30) closes the node with nonnegative terms and needs no root rate. Against a room bounded below the bunch enters through its reach alone, and the domination is a reach against a decay.

**Proposition 20.35** (No reach is below the least ratio)**.** *Let $F$ and $B$ be a node of width one.*

*(i) If $\inf \varrho_B > 0$, there are $C$ and $a$ with $f_B(y) \le C y^a e^{-2\sqrt{y \inf \varrho_B}}$ for $y \ge 1$, and for every $x' > \inf \varrho_B$ there is $C_- > 0$ with $f_B(y) \ge C_- e^{-2\sqrt{x' y}}$ for $y \ge 1$: the bunch transform decays at the least ratio of the bunch.*

*(ii)* $\operatorname{reach}(B) \ge \inf \varrho_B$*: no lattice expansion of the bunch transform has a reach below the least ratio.*

*(iii) When $\varrho_B$ is bounded,* $\operatorname{reach}(B) = \operatorname{growth}(B) = \sup \varrho_B$*: the reach of a bounded bunch is its largest ratio.*

Against a room bounded below, the root rate $\operatorname{reach}(B)/\inf \varrho_R$ of Theorem 20.34(ii) therefore lies between $\inf \varrho_B/\inf \varrho_R$ and $\sup \varrho_B/\inf \varrho_R$. The series bunches take the lower end, and the bounded bunches the upper, which is $\sup(\varrho_B/\varrho_R)$, the root rate of Corollary 19.21(i).

**Proposition 20.36** (A series bunch is its marks)**.** *Let the bunch $B$ of a node of width one be a series bunch, $\theta'_B$ the weight of its pattern $B$ and $\theta'_{B-e}$ that of its pattern $B - e$ (Definition 19.2, Proposition 19.3), and in this statement write $w_e = \theta'_{B-e}/\theta'_B$.*

*(i) The bunch transform is the product over the marks of $B$:*

$$\begin{aligned} f_B(y) &= (\theta'_B)^{-s-t} \prod_{e \in B} \mathsf{G}_{m_e - s;(1,-1)}(w_e y,\, w_e) \\ &= (\theta'_B)^{-s-t}\, y^{s-\kappa_B} \prod_{e \in B} w_e^{s-m_e}\, \mathsf{G}_{m_e-s;(1,-1)}\big(1,\, w_e^2 y\big). \end{aligned}$$

*(ii) $\varrho_B(u) = \big(\sum_e w_e/u_e\big)\big(\sum_e w_e u_e\big) \ge \big(\sum_e w_e\big)^2$, with equality exactly at the point of equal coordinates of $\Delta_B$, and $\varrho_B$ is unbounded: whatever the weights, the least ratio of a series bunch is taken at the point of equal coordinates of its shape.*

*(iii) $f_B$ has the lattice expansion of reach $\big(\sum_e w_e\big)^2 = \inf \varrho_B$ that is the product of the lattices of its marks, so* $\operatorname{reach}(B) = \inf \varrho_B$*. With $\boldsymbol{\nu} = \mathbf{m}|_B - s\mathbf{1}_B$ and no $\nu_e$ an integer, its exponents lie in the lattices $-\nu_S + \mathbb{N}$, $S \subseteq B$: $S = \varnothing$ and $S = B$ are the lattices of the Gamma pair, and for every other $S$ the lattice is where the wall of the bunch family at $B \setminus S$, continued, meets $-\mathbb{N}$. Finally $f_B(y) \le C\, y^a\, e^{-2(\sum_e w_e)\sqrt{y}}$ for $y \ge \max_e w_e^{-2}$, with constants $C$ and $a$.*

**Proposition 20.37** (A bunch in series is the product of its parts)**.** *Let the bunch of a node of width one be $B = B_1 \sqcup \cdots \sqcup B_q$, and on each $B_i$ let $\mathcal{X}_i$ and $\mathcal{Y}_i$ be tables, $\mathcal{Y}_i$ of rank one less than $\mathcal{X}_i$, with the hinge values $\gamma_i^K = K_{\mathcal{X}_i}/K_{\mathcal{Y}_i}$ and $\gamma_i^\Psi = \Psi_{\mathcal{Y}_i}/\Psi_{\mathcal{X}_i}$ and the ratio $\varrho_i = \mathbf{E}_{\mathcal{Y}_i}/\mathbf{E}_{\mathcal{X}_i}$, such that for constants $C, C' > 0$*

$$K_{\mathcal{B}|B} = C \prod_i K_{\mathcal{X}_i}, \qquad \Psi_{\mathcal{B}|B} = C' \prod_i \Psi_{\mathcal{X}_i}, \qquad \frac{1}{\gamma_B^K} = \sum_i \frac{1}{\gamma_i^K}, \qquad \gamma_B^\Psi = \sum_i \gamma_i^\Psi :$$

*$B$ is the* series connection *of the parts $B_i$. A series bunch is the series connection of its marks, with $K_{\mathcal{X}_e} = \alpha_e$, $\Psi_{\mathcal{X}_e} = 1$, $K_{\mathcal{Y}_e} = w_e$, $\Psi_{\mathcal{Y}_e} = w_e \alpha_e$ and $\varrho_e = w_e^2$ (Proposition 20.36).*

*(i)* $f_B = C^{-s}C'^{-t}\prod_i f_i^\circ$, *with the* orthant transforms *of the parts*

$$f_i^\circ(y) = \int_{\mathbb{R}_{>0}^{B_i}} x^{\mathbf{m}|_{B_i}-\mathbf{1}} K_{\mathcal{X}_i}(x)^{-s}\Psi_{\mathcal{X}_i}(x)^{-t}\exp\Big(-\frac{1}{\gamma_i^K(x)} - y\,\gamma_i^\Psi(x)\Big)\,dx;$$

*and* $\sqrt{\inf\varrho_B} = \sum_i \sqrt{\inf\varrho_i}$.

*(ii) If every* $f_i^\circ$ *has a lattice expansion of reach* $x_i$, *then* $f_B$ *has one of reach* $\big(\sum_i\sqrt{x_i}\big)^2$.

*(iii) Let every part be a single mark, or have a bounded ratio and a finite measure* $u^{\mathbf{m}|_{B_i}-\mathbf{1}}K_{\mathcal{X}_i}^{-s}\Psi_{\mathcal{X}_i}^{-t}(\gamma_i^\Psi)^{-a_i}\,d\omega_{B_i}$ *on* $\Delta_{B_i}$, *where* $a_i = m_{B_i} - s\,r_i - t\,(|B_i| - r_i)$ *with* $r_i$ *the rank of* $\mathcal{X}_i$; *and let no sum of the* $a_i$ *over a nonempty set of parts be an integer. Then* $f_i^\circ$ *has the reach* $\sup\varrho_i$, *and* $\operatorname{reach}(B) = \big(\sum_i\sqrt{\sup\varrho_i}\big)^2$: *in a series connection the square roots of the reaches add.*

**Lemma 20.38** (The point of equal coordinates is stationary)**.** *Let* $\mathcal{X}$ *be a table on a set* $S$, *with any weights. At every point* $u$ *of* $(0,\infty)^S$ *with equal coordinates,* $u_e\,\partial_{u_e}\log\mathbf{E}_\mathcal{X}(u) = 1$ *for every* $e \in S$. *So a ratio of even letters of tables on* $S$ *is stationary at the point of equal coordinates of* $\Delta_S$: $\varrho_B$ *on* $\Delta_B$, $\varrho_R$ *on* $\Delta_R$, *and* $\varrho_B/\varrho_R$ *at the pair of these points.*

**Theorem 20.39** (The two intervals of a node)**.** *For a node of width one, the bunch transform of the swapped node is* $y^{-t}f_R(y)$, *so that* $\inf\varrho_B$ *and* $\inf\varrho_R$ *are the decays of the two transforms, and* $\operatorname{reach}(B)$ *and* $\operatorname{reach}(R)$ *their reaches (Proposition 20.35(i), Lemma 20.30).*

*(i)* No transform reaches less far than it decays: $\inf\varrho_B \le \operatorname{reach}(B)$ *and* $\inf\varrho_R \le \operatorname{reach}(R)$.

*(ii)* Toward the level $d$. *Let no slope on the room be positive, so that* $\inf\varrho_R > 0$, *and index the terms of the lattice of Theorem 20.34(ii) by* $(\eta, j)$. *Their root rate is* $\operatorname{reach}(B)/\inf\varrho_R$: *for every* $x > \operatorname{reach}(B)$ *they are at most* $C(1+\eta)^D(x/\inf\varrho_R)^\eta$ *in modulus, and for every* $\epsilon > 0$ *infinitely many of them exceed* $\big(\operatorname{reach}(B)/\inf\varrho_R - \epsilon\big)^\eta$. *So the lattice converges absolutely when* $\operatorname{reach}(B) < \inf\varrho_R$, *and its terms do not tend to zero when* $\operatorname{reach}(B) > \inf\varrho_R$.

*(iii)* Toward the level $r$. *Let no slope on the bunch be negative, so that* $\inf\varrho_B > 0$. *The terms of the lattice of continued periods toward the level* $r$ *have the root rate* $\operatorname{reach}(R)/\inf\varrho_B$ *in the same sense: the lattice converges absolutely when* $\operatorname{reach}(R) < \inf\varrho_B$, *and its terms do not tend to zero when* $\operatorname{reach}(R) > \inf\varrho_B$.

*(iv)* One level at most. $\operatorname{reach}(B) < \inf\varrho_R$ *and* $\operatorname{reach}(R) < \inf\varrho_B$ *never hold together. Under both hypotheses of (ii) and (iii) the node therefore closes on a geometrically convergent lattice exactly when the intervals* $[\inf\varrho_B, \operatorname{reach}(B)]$ *of the bunch and* $[\inf\varrho_R, \operatorname{reach}(R)]$ *of the room are disjoint, toward the level of the lower one, at the ratio of the upper end of the lower interval to the lower end of the upper. When each interval extends beyond the lower end of the other, neither lattice converges, and the share closes at its peaks (Theorem 19.30); when they touch, the root rate is one and domination decides nothing.*

Each side of a node is read by two numbers of its own transform, its decay and the reach of its expansion, and no expansion reaches less far than its transform decays. A series bunch has both at its least ratio, and its interval is the point $(\sum_e w_e)^2$ (Proposition 20.36); a bounded bunch spreads over its whole range $[\inf\varrho_B, \sup\varrho_B]$ (Proposition 20.35(iii)); in a series connection of marks and bounded parts the square roots of the reaches add (Proposition 20.37(iii)). When the bunch and the room are both series bunches the two intervals are points: the node closes toward the level of the smaller point at the ratio of the two points, its terms toward the other level grow at the inverse ratio, and where the points meet domination decides nothing. Where a far side is unbounded, the

level toward it can still close with both families continued, at a root rate at most the product of their growths (Theorem 20.34(iii)). By Lemma 20.38 the value of $\varrho_B/\varrho_R$ at the points of equal coordinates of $\Delta_B$ and $\Delta_R$ is a stationary value of the node's ratio, but the root rate is read at the extremes of the two transforms. The reach of a bunch that is neither bounded nor built from parallel and series bunches is read on the law of its ratio (Section 20.8).

**Theorem 20.40** (A series connection closes at its reach)**.** *(i) Let the bunch $B$ of a node of width one be the series connection of parts $B_1, \dots, B_q$ (Proposition 20.37), every orthant transform $f_i^\circ$ having a lattice expansion of reach $x_i$, and let no slope on the room be positive. If $\left(\sum_i \sqrt{x_i}\right)^2 < \inf \varrho_R$, the even share closes toward the level $d$ on the lattice of continued periods of the bunch family (Theorem 20.34(ii)): for every $x > \left(\sum_i \sqrt{x_i}\right)^2$ its terms with $\eta \geq N$ add up in absolute value to at most a constant times $(1+N)^D (x/\inf \varrho_R)^N$, and its coefficients $\beta_{\eta,j}$ are those of $C^{-s}C'^{-t}\prod_i f_i^\circ$, finite sums of products, over the parts, of the coefficients of their lattice expansions. Under the hypotheses of Proposition 20.37(iii) the root rate of the terms is $\left(\sum_i \sqrt{\sup \varrho_i}\right)^2 / \inf \varrho_R$ (Theorem 20.39(ii)). For a series bunch, the series connection of its marks with $w_e$ as in Proposition 20.36 and $x_e = w_e^2$, the root rate is $\inf \varrho_B / \inf \varrho_R = \left(\sum_e w_e\right)^2 / \inf \varrho_R$ and the coefficients are those of $(\theta'_B)^{-s-t} y^{s-\kappa_B} \prod_e w_e^{s-m_e} \mathsf{G}_{m_e-s;(1,-1)}(1, w_e^2 y)$: finite sums of products, over the marks of $B$, of the coefficients of Lemma 20.29, Laurent coefficients of the Gamma pairs $\Gamma(z)\Gamma(z+m_e-s)$.*

*(ii) Let $B$ be the series connection of parts $B_1, \dots, B_q$ in the dual table at the exponents $(t, s)$ (Proposition 8.15), its orthant transforms there having lattice expansions of reaches $x_i^*$, and let no slope on the room be negative. If $\left(\sum_i \sqrt{x_i^*}\right)^2 \sup \varrho_R < 1$, the even share closes toward the level $r$ on the lattice of continued periods of the bunch family, at a root rate at most $\left(\sum_i \sqrt{x_i^*}\right)^2 \sup \varrho_R$. A parallel bunch $B$ is a series bunch of the dual table (Proposition 19.37), with $w_e = \theta'_{\{e\}}/\theta'_\varnothing$, $\theta'_\varnothing$ and $\theta'_{\{e\}}$ the weights of its patterns $\varnothing$ and $\{e\}$; then $x_e^* = w_e^2$ and $\sup \varrho_B = \left(\sum_e w_e\right)^{-2}$, taken at the point of equal coordinates of $\Delta_B$, so the share closes when $\sup \varrho_R < \sup \varrho_B$, at the root rate $\sup \varrho_R / \sup \varrho_B$.*

No order of the values is asked: the room's ratio need only be bounded on the side away from the closing, and the bunch's ratio may be unbounded, its walls falling toward the level closed on. When the bunch and the room are both series bunches, the root rate $\inf \varrho_B / \inf \varrho_R$ is read at the points of equal coordinates of both shapes. When the extreme ratio of the room sits at a vertex of its shape simplex the root rate is not read at the points of equal coordinates, which are only stationary.

**Theorem 20.41** (Products of Grammas $\mathsf{G}_{\nu;(1,-1)}$ close on nested lattices)**.** *For a finite set $S$, numbers $w_e > 0$ and real indices $\nu_e$, put $w_S = \sum_{e\in S} w_e$ and $\nu_S^- = \sum_{e\in S} \min(0, \nu_e)$, and in this statement write, for $\operatorname{Re} z > -\nu_S^-$,*

$$\mathcal{M}_S(z) = \int_0^\infty y^{z-1} \prod_{e\in S} \mathsf{G}_{\nu_e;(1,-1)}\big(1, w_e^2 y\big)\, dy.$$

*(i) $\mathcal{M}_{\{e\}}(z) = w_e^{-2z}\, \Gamma(z)\Gamma(z+\nu_e)$, and for two marks with $w_e = w_f = w$*

$$\mathcal{M}_{\{e,f\}}(z) = w^{-2z}\, {}_4\Gamma_1\left[\begin{matrix} z,\ z+\nu_e,\ z+\nu_f,\ z+\nu_e+\nu_f \\ 2z+\nu_e+\nu_f \end{matrix}\right].$$

*(ii) For $S = S_1 \sqcup S_2$ with $w_{S_1} < w_{S_2}$, let $\prod_{e\in S_1} \mathsf{G}_{\nu_e;(1,-1)}(1, w_e^2 y) = \sum \beta_{\eta,j} y^\eta (-\log y)^j$ be the product of the lattices of the marks of $S_1$ (Lemmas 20.29 and 20.30). For $\operatorname{Re} z > -\nu_S^-$ and $i \in \mathbb{N}$,*

$$(-\partial_z)^i \mathcal{M}_S(z) = \sum_{\eta,j} \beta_{\eta,j}\, (-\partial_z)^{i+j} \mathcal{M}_{S_2}(z+\eta),$$

*every $z+\eta$ in the domain of $\mathcal{M}_{S_2}$, absolutely, the terms with $\eta \geq N$ adding up in absolute value to at most $C(1+N)^D (w_{S_1}/w_{S_2})^{2N}$.*

*(iii) Removing at each step a mark $e$ of least $w_e$, until one mark is left or two with equal $w_e$, writes $\mathcal{M}_S(z)$ and its derivatives as nested lattice sums of products of Laurent coefficients of Gamma pairs at the points of the lattices, ending at (i): at most $|S|-1$ nested lattices, summed from the inside out, the one that removes $e$ from $S'$ converging absolutely at every point of the lattices above it, at the root rate $\big(w_e/(w_{S'} - w_e)\big)^2$, its constant depending on that point.*

**Corollary 20.42** (A table that is one circuit is a nested lattice of Gamma data)**.** *In Corollary 18.24 the integral over $y$ is the integral of Theorem 20.41 with $S = \mathcal{E}$, $w_e = \theta_e$ and the indices $\nu_e = m_e - s$, taken at $z = s$, and $s > -\sum_e \min(0, \nu_e)$. So the period of every table that is one circuit, at all exponents $s, t > 0$, all masses of Corollary 18.24 and all weights, is a nested lattice sum of products of Laurent coefficients of Gamma pairs, with at most $n-1$ nested lattices, each converging absolutely at every point of the lattices above it and summed from the inside out. With unit weights, removing one mark at a time, the root rates are $(n-1)^{-2}, (n-2)^{-2}, \dots, \frac{1}{4}$, and the last two marks give four Gammas over one; splitting the marks into a lighter part $S_1$ and a heavier part $S_2$ converges at the root rate $(\theta_{S_1}/\theta_{S_2})^2$. At integer data with unit weights every coefficient is a rational combination of products of values $(\log\Gamma)^{(i)}(k)$, $i, k \geq 1$, which are Trailmoments of the counting input (Propositions 5.13 and 5.14).*

## 20.8 The reach read on the law

The reach of a bunch is defined by the expansion of its transform (Definition 20.28) and is bounded below by the least ratio and above by the growth of the continued family (Proposition 20.35, Theorem 20.34). Read on one measure on the half-line, the law of the ratio under the base form, the reach is the point beyond which the law has a density equal to its own wall series (Definition 20.43), the series whose terms are the poles of the continued family; every point at which the law fails to be analytic therefore bounds the reach from below, and the least ratio and the largest ratio of a bounded bunch are such points. From above the reach is bounded by the ceiling of a continuation (Definition 20.47): in each sector the falling variables are carried around zero on circles, as far as the sector letters allow, and the largest modulus of the ratio met there bounds the reach. From below it is bounded by the growth of the continued family at the integers; a continuation whose largest values sit at simple tops that do not cancel has that growth as its ceiling, so that the reach is at least the modulus of a stationary value of the ratio on a flat face, and equal to it when the ceilings come down to it.

**Definition 20.43** (The law of a bunch; wall series)**.** The *law* of the bunch of a node of width one is the image $(\varrho_B)_*\phi_B$ of its base form under its ratio, a measure on $(0,\infty)$ (Definition 5.4). The continued period of the bunch family at the share $b^d$ is $\int_0^\infty x^{-b^d}\, d\big((\varrho_B)_*\phi_B\big)(x)$ wherever the integral converges, which is for $\operatorname{Re} b^d$ in an open interval containing $b_*$, the *domain interval* of the law, the walls being affine in $b^d$ and the conditions of Theorem 8.7 strict. A function $g$ on $(x_0,\infty)$,

$x_0 \geq 0$, is a *wall series of radius* $x_0$ when there are finitely many reals $\eta_i$, an integer $J \geq 0$ and coefficients $a_{\eta,j}$, for $\eta \in A = \bigcup_i (\eta_i + \mathbb{N})$ and $0 \leq j \leq J$, with $\sum_{\eta,j} |a_{\eta,j}| x'^{-\eta} < \infty$ for every $x' > x_0$, such that

$$g(x) = \sum_{\eta,j} a_{\eta,j}\, x^{-\eta-1} (\log x)^j \qquad (x > x_0).$$

The series then converges absolutely and locally uniformly on $\{x \in \mathbb{C} : \ |x| > x_0, \ |\arg x| < \pi\}$, so a wall series is real-analytic.

**Theorem 20.44** (The reach is read on the law)**.** *Let $F$ and $B$ be a node of width one.*

*(i) On* $(\mathrm{reach}(B), \infty)$ *the law* $(\varrho_B)_* \phi_B$ *has a density, and the density is a wall series of radius* $\mathrm{reach}(B)$*. Its terms are the poles of the continued period of the bunch family to the left of the domain interval: a pole* $b^d = -\eta$ *with the principal part* $\sum_j j!\, a_{\eta,j} (b^d + \eta)^{-j-1}$ *gives the terms* $a_{\eta,j} x^{-\eta-1} (\log x)^j$*, and there are no others.*

*(ii) If on some* $(x_0, \infty)$ *the law has a density that is a wall series of radius* $x_0$*, then* $\mathrm{reach}(B) \leq x_0$*.*

*(iii) So the reach is the least* $x_0 \geq 0$ *for which the law has on* $(x_0, \infty)$ *a density that is a wall series of radius* $x_0$*. In particular* $\mathrm{reach}(B) \geq x_1$ *at every* $x_1 > 0$ *near which* $(\varrho_B)_* \phi_B$ *has no real-analytic density: at the least ratio when it is positive, and at the largest ratio of a bounded bunch.*

So no reach lies below a point where the law of its ratio is singular, and Proposition 20.35(ii), with the lower half of (iii), is the case of the two ends of the law. The law of a family at a node is likewise its own wall series beyond its growth, so that $\mathrm{reach}(B) \leq \mathrm{growth}(B)$ of Theorem 20.34(i) is an equality (Corollary 32.11); along a direction of the shares the law reads the reach of a piece of any width (Theorem 32.10). From above the reach is bounded by the ceilings of continuations.

**Lemma 20.45** (Finite parts on a keyhole)**.** *Let $0 < r \leq 1$, $r' > r$, a real, and let $G$ be holomorphic on $U \times \mathbb{C}$ for an open $U$ containing $\{|q| \leq r'\} \cup [r, 1]$. Then $I(z) = \int_0^1 q^{a+z-1} G(q, z)\, dq$, defined for $\mathrm{Re}\, z > -a$, continues meromorphically to $\mathbb{C}$ with poles only where $a + z \in -\mathbb{N}$, and*

$$I(z) = \sum_{l \geq 0} g_l(z)\, \frac{r^{a+z+l}}{a+z+l} + \int_r^1 q^{a+z-1} G(q, z)\, dq, \qquad g_l(z) = \frac{1}{2\pi i} \oint_{|\zeta| = r'} \zeta^{-l-1} G(\zeta, z)\, d\zeta.$$

*Where $|a + z + l| \geq \epsilon$ for every $l \in \mathbb{N}$,*

$$|I(z)| \leq \frac{r'}{\epsilon (r' - r)}\, r^{a + \mathrm{Re}\, z} \max_{|\zeta| = r'} |G(\zeta, z)| + \int_r^1 q^{a + \mathrm{Re}\, z - 1} |G(q, z)|\, dq.$$

*If $0 < q_0 \leq 1$ and $U$ contains the closed disc $D$ with diameter $[-q_0, 1]$, then off the points with $a + z \in \mathbb{Z}$*

$$I(z) = \frac{1}{e^{2\pi i (a+z)} - 1} \oint_{\partial D} q^{a+z-1} G(q, z)\, dq,$$

*the circle run once counterclockwise from 1, with $q^{a+z-1}$ continued along it from its positive value at 1.*

**Definition 20.46** (Falling and base variables; keyholes)**.** For a node of width one and an order $\pi$ of its bunch $B$ let $q = (q_k)_k$ be its sector coordinates and $U_k$ its tails, and let $k_{\mathcal{B}|B,\pi}$, $\psi_{\mathcal{B}|B,\pi}$, $k_{\mathcal{B}/R,\pi}$, $\psi_{\mathcal{B}/R,\pi}$ be the sector letters of the two tables of the bunch family (Theorem 8.5(ii)). Along the

level $d$ the wall of the bunch family at $U_k$ is $\kappa_k + \mathrm{sl}_k\, b^d$ at the share $b^d$, with its *slope* $\mathrm{sl}_k \in \{-1, 0, 1\}$ (Lemma 19.15). On the sector of $\pi$ the base form is

$$\phi_\pi = \prod_k q_k^{\kappa_k - 1}\, k_{\mathcal{B}/R,\pi}^{-(s-\kappa_B)}\, k_{\mathcal{B}|B,\pi}^{-\kappa_B}\, \psi_{\mathcal{B}|B,\pi}^{-t}\, dq,$$

and the ratio is

$$\varrho_\pi = \frac{k_{\mathcal{B}/R,\pi}\, \psi_{\mathcal{B}/R,\pi}}{k_{\mathcal{B}|B,\pi}\, \psi_{\mathcal{B}|B,\pi}} \prod_k q_k^{-\mathrm{sl}_k},$$

continued to the complex points of the coordinates at which the sector letters do not vanish; so the continued period of the bunch family at the share $b^d$ is $\sum_\pi \int_{[0,1]^{|B|-1}} \varrho_\pi^{-b^d}\, \phi_\pi$. The $q_k$ with $\mathrm{sl}_k = 1$ are the *falling variables* of $\pi$, whose walls fall as $\mathrm{Re}\, b^d$ decreases, and the others its *base variables*, *flat* when $\mathrm{sl}_k = 0$ and *rising* when $\mathrm{sl}_k = -1$. Where some flat variables vanish the sector reaches a *flat face* of the shape, on which every wall crossed has slope 0 and $\varrho_\pi$ has the finite limit given by the leading parts of the letters; where some falling variable vanishes it reaches a *falling face*. For $0 < r \le b \le 1$ the *keyhole* and its *outline* are $Y_r[0,b] = \{q \in \mathbb{C} : |q| \le r\} \cup [r,b]$ and $\mathcal{O}_r[0,b] = \{|q| = r\} \cup [r,b]$, with $Y_r = Y_r[0,1]$ and $\mathcal{O}_r = \mathcal{O}_r[0,1]$.

**Definition 20.47** (Keyhole continuations; ceiling)**.** A *keyhole continuation* of the bunch family of a node of width one assigns to every order $\pi$ of $B$ finitely many closed boxes $\Omega$ with disjoint interiors covering the cube of its base variables, and to every such box and falling variable $q_k$ a radius $r_{\pi,\Omega,k} \in (0,1]$, such that none of the four sector letters vanishes on the set $Y = \Omega \times \prod_k Y_{r_{\pi,\Omega,k}}$, the product over the falling variables. The boxes may also divide the sides of the falling variables: they then cover the whole cube of the order, a box being the product of its part in the base variables and of its sides in the falling variables, with radii of its own; a falling variable with the side $[0, b_k]$ carries a radius $r \le b_k$ and the sets $Y_r[0,b_k]$ and $\mathcal{O}_r[0,b_k]$, and one with the side $[a_k, b_k]$, $a_k > 0$, is kept real, its sets $Y$ and $\mathcal{O}$ being $[a_k, b_k]$. A box, divided or not, is written $\Omega$ with its radii and its sets $Y$ and $\mathcal{O} = \Omega \times \prod_k \mathcal{O}_{r_{\pi,\Omega,k}}$, where $\Omega$ stands for its part in the variables without keyholes and the products run over the falling variables with keyholes. The *ceiling* $H_\mathcal{O}$ of the continuation is the largest value of $|\varrho_\pi|$ on the sets $\mathcal{O}$ of its boxes, an order without falling variables contributing the largest value of $\varrho_\pi$ on its cube.

The sector letters are positive on the closed cube, with positive constant terms (Theorem 8.5(ii)), so small radii give keyhole continuations, and every bunch has one of finite ceiling. Whether a letter vanishes on a keyhole is decided on its outline and at one point.

**Lemma 20.48** (Zeros on a keyhole are decided on its outline and at one point)**.** *Let $\Omega$ be a connected set, $J$ a finite set, $0 < r_j \le b_j \le 1$ for $j \in J$, and $L$ a continuous function on $\Omega \times \mathbb{C}^J$, a polynomial in $q = (q_j)_{j \in J}$. If $L$ has no zero on $\Omega \times \prod_j \mathcal{O}_{r_j}[0,b_j]$, and for some $v_0 \in \Omega$ and each $j$ the polynomial in $q_j$ obtained from $L(v_0, \cdot)$ by putting $q_k = r_k$ for $k \ne j$ has no zero in the disc $|q_j| < r_j$, then $L$ has no zero on $\Omega \times \prod_j Y_{r_j}[0,b_j]$.*

So a box and its radii keep a letter from vanishing on the set $Y$ of the box when the letter has no zero on its set $\mathcal{O}$, where the ceiling is read, and, at one point of $\Omega$ with the other falling coordinates at their radii, none in the disc of each keyhole; the falling variables kept real count among the coordinates of $\Omega$.

**Theorem 20.49** (No reach exceeds the ceiling)**.** *For a node of width one, the reach of its bunch is at most the ceiling of every keyhole continuation:* $\inf \varrho_B \le \mathrm{reach}(B) \le H_\mathcal{O}$ *for the ceiling $H_\mathcal{O}$ of each. When $\varrho_B$ is bounded, every keyhole continuation has the ceiling* $\sup \varrho_B$*, and* $\mathrm{reach}(B) = \sup \varrho_B$.

The letters of the contraction enter the keyhole continuations because the family is continued at points off the integers, where $\varrho_\pi^{-b^d}$ branches at their zeros; at the integers $-k$, $k \in \mathbb{N}$, when $s - \kappa_B$ is zero or a negative integer, only the letters of the restriction $\mathcal{B}|B$ matter. When a zero of a letter of the contraction is met by the keyholes that would bring $|\varrho_\pi|$ down, the ceiling can exceed the reach. From below, the reach is read at the integers, where $\psi_{\mathcal{B}/R,\pi}$ does not enter, nor $k_{\mathcal{B}/R,\pi}$ when $s - \kappa_B$ is an integer, and a continuation whose largest values sit at simple tops that do not cancel cannot be lowered.

**Definition 20.50** (Integer keyhole continuations; simple tops)**.** For a node of width one let $\nu = s - \kappa_B$. An *integer keyhole continuation* is defined as in Definition 20.47 with only $k_{\mathcal{B}|B,\pi}$ and $\psi_{\mathcal{B}|B,\pi}$, and $k_{\mathcal{B}/R,\pi}$ when $\nu$ is not an integer, required not to vanish on the sets $Y$ of its boxes; its ceiling is defined in the same way. For an order $\pi$ and a box $\Omega$ write $J$ for its falling variables with keyholes and $r_j$ for their radii, and let $C_j$ be the circle $|q_j| = r_j$ run once counterclockwise from $r_j$. The *keyhole contour* of the continuation is the formal sum over the orders and their boxes of

$$\Omega \times \prod_{j \in J} \Big( \frac{C_j}{e^{2\pi i \kappa_j} - 1} + [r_j, 1] \Big),$$

the segment being $[r_j, b_j]$ on a divided side $[0, b_j]$, a falling variable kept real on $[a_j, b_j]$ entering with that segment, and an order without falling variables contributing its cube; on the keyhole contour of $\pi$ it carries the form $\phi_\pi$, the powers continued along the circles from their positive values at $r_j$. A *top* is a point of the keyhole contour at which $|\varrho_\pi|$ equals the ceiling. A top $p$ is *simple* when it lies on no face of its box other than faces $q_j = 0$ of flat variables, its falling coordinates lie on the circles away from $r_j$ or inside the segments, and, with $\mathbf{x}$ the coordinates that vanish at $p$ and $\mathbf{y}$ real coordinates of the keyhole contour for the others, vanishing at $p$ (the angle on a circle),

$$\log \frac{\varrho_\pi}{\varrho_\pi(p)} = -\langle \boldsymbol{\lambda}, \mathbf{x} \rangle - \tfrac{1}{2}\, \mathbf{y}^{\mathsf{T}} \mathbf{Q}\, \mathbf{y} + O\big(|\mathbf{x}|^2 + |\mathbf{x}|\,|\mathbf{y}| + |\mathbf{y}|^3\big)$$

on the keyhole contour near $p$, with $\operatorname{Re} \lambda_j > 0$ for every $j$ and $\operatorname{Re} \mathbf{Q}$ positive definite. The *exponent* of $p$ is $\operatorname{ex}(p) = \sum_j \kappa_j + \frac{1}{2} \dim \mathbf{y}$, the sum over the coordinates $\mathbf{x}$, and its *coefficient* is

$$A(p) = \phi_p \prod_j \Gamma(\kappa_j)\, \lambda_j^{-\kappa_j}\, (2\pi)^{\dim \mathbf{y}/2} (\det \mathbf{Q})^{-1/2},$$

$\phi_p$ the value at $p$ of the density of $\phi_\pi$ with the coefficient $1/(e^{2\pi i \kappa} - 1)$ of each circle of the keyhole contour through $p$, $\kappa$ the wall of its falling variable, against $\prod_j x_j^{\kappa_j - 1}\, d\mathbf{x}\, d\mathbf{y}$, the powers and the root continued from real $\boldsymbol{\lambda}$ and real $\mathbf{Q}$ along segments.

The coordinates $\mathbf{x}$ belong to flat variables, whose walls $\kappa_j$ are positive, so the Gammas are finite.

**Theorem 20.51** (A simple top is the growth at the integers)**.** *For a node of width one let the walls $\kappa_j$ of the falling variables of every order be off the integers, and let* $\operatorname{growth}_{\mathbb{Z}}(B)$ *be the growth at the integers (Definition 20.33).*

*(i) For every integer keyhole continuation and every integer $k \geq \max(0, s - \kappa_B)$, the continued period at $b^d = -k$ is the integral of $\varrho_\pi^k \phi_\pi$ over its keyhole contour. So its modulus is at most $C H_{\mathcal{O}}^k$, $H_{\mathcal{O}}$ the ceiling, and* $\operatorname{growth}_{\mathbb{Z}}(B) \leq H_{\mathcal{O}}$.

*(ii)* $\operatorname{reach}(B) \geq \operatorname{growth}_{\mathbb{Z}}(B)$.

*(iii) Let an integer keyhole continuation of ceiling $H_{\mathcal{O}}$ have finitely many tops, all simple, let* $\mathrm{ex}_0$ *be the least of their exponents, and for* $|\zeta| = 1$ *let* $A_\zeta$ *be the sum of the coefficients* $A(p)$ *of the tops with* $\mathrm{ex}(p) = \mathrm{ex}_0$ *and* $\varrho_\pi(p) = H_{\mathcal{O}}\zeta$*. Each* $A(p)$ *is nonzero, and each top is a stationary point of the restriction of* $\varrho_\pi$ *to the flat face* $\mathbf{x} = 0$*, the sector itself when there are no coordinates* $\mathbf{x}$*. If some* $A_\zeta \neq 0$*, then* $\mathrm{growth}_{\mathbb{Z}}(B) = H_{\mathcal{O}} \leq \mathrm{reach}(B)$*: the reach is at least the modulus of a stationary value of the ratio on a flat face.*

*(iv) If moreover there are keyhole continuations with ceilings arbitrarily close to* $H_{\mathcal{O}}$*, then* $\mathrm{reach}(B) = H_{\mathcal{O}}$*. In particular, if a keyhole continuation, with all four letters kept from vanishing, has finitely many tops, all simple, and some* $A_\zeta \neq 0$*, the reach is its ceiling.*

So $\mathrm{growth}_{\mathbb{Z}}(B) \leq \mathrm{reach}(B)$ always, with equality to the ceiling whenever the tops of an integer keyhole continuation are finitely many, all simple, and do not cancel, and the keyhole continuations come down to it: the reach is then the modulus of a stationary value of the ratio on a flat face, the one at which the best continuation is pinned. The stationary point need not be a positive point of the face.

The upper end needs the letters of the contraction only where they meet the falling faces. Elsewhere a zero of $k_{\mathcal{B}/R,\pi}$ or $\psi_{\mathcal{B}/R,\pi}$ is a zero of $\varrho_\pi$, and the law is read near infinity on the fibres $\varrho_\pi = x$ with $|x|$ large, which never pass through it.

**Lemma 20.52** (A falling variable takes each large value once)**.** *Let* $g$ *be meromorphic on a neighbourhood of the closed disc* $|q| \leq r$*, with a simple pole at* $0$ *as its only pole there, and* $|g| \leq M$ *on the circle* $|q| = r$*. Then* $g$ *takes every value* $x$ *with* $|x| > M$ *exactly once in the disc, with* $g' \neq 0$ *there, and maps the open set* $\{|q| < r : \ |g(q)| > M\} \cup \{0\}$ *biholomorphically onto* $\{|x| > M\} \cup \{\infty\}$*.*

**Lemma 20.53** (A wall series continued is a wall series)**.** *If the sum of a wall series of radius* $x_0$ *continues analytically along every path in* $\{|x| > x_1\}$*, with* $x_1 < x_0$*, then the series is a wall series of radius* $x_1$*.*

**Definition 20.54** (Fibre continuations; stationary bound)**.** A *fibre continuation* is defined as in Definition 20.47, divided sides allowed, with only $k_{\mathcal{B}|B,\pi}$ and $\psi_{\mathcal{B}|B,\pi}$ required not to vanish on the sets $Y$ of its boxes, and $k_{\mathcal{B}/R,\pi}$ and $\psi_{\mathcal{B}/R,\pi}$ only at their points with a falling coordinate 0; its ceiling $H_{\mathcal{O}}$ is defined in the same way. Its *stationary bound* $S$ is the supremum of $|\varrho_\pi|$ over the points of these sets with no falling coordinate 0 at which at least two falling coordinates lie inside their discs, $|q_j| < r_j$, and the derivatives of $\varrho_\pi$ in all the falling coordinates inside their discs vanish; $S = 0$ when there is no such point.

With one falling coordinate inside its disc and the others on their outlines, $\varrho_\pi$ is in that coordinate as in Lemma 20.52, with the bound $H_{\mathcal{O}}$ on the circle, or identically 0 when a rising base coordinate is 0; so it has no stationary point there with $|\varrho_\pi| > H_{\mathcal{O}}$, and the stationary bound concerns two or more falling coordinates at once.

**Theorem 20.55** (The contraction enters only on the falling faces)**.** *For a node of width one and every fibre continuation, of ceiling* $H_{\mathcal{O}}$ *and stationary bound* $S$*,* $\mathrm{reach}(B) \leq \max(H_{\mathcal{O}}, S)$*. So, under the hypotheses of Theorem 20.51, if a fibre continuation that is also an integer keyhole continuation has* $S \leq H_{\mathcal{O}}$ *and finitely many tops, all simple, with some* $A_\zeta \neq 0$*, then* $\mathrm{reach}(B) = \mathrm{growth}_{\mathbb{Z}}(B) = H_{\mathcal{O}}$*: the reach is read at the integers.*

The letters of the contraction, which Theorem 20.49 keeps from vanishing on whole keyholes, matter for the upper bound only on the falling faces, where the coefficient of the pole of the ratio vanishes, and through the stationary points of the ratio in two or more falling variables. The two bounds from above work box by box, so they can share one continuation.

**Theorem 20.56** (Keyhole boxes and fibre boxes mix)**.** *Let a continuation assign to every order $\pi$ of the bunch $B$ of a node of width one boxes and radii as in Definition 20.47, the sides of the falling variables divided or not, each box either a* keyhole box*, on whose sets $Y$ none of the four sector letters vanishes, or a* fibre box*, on whose sets $Y$ the conditions of Definition 20.54 hold. Let $H_{\mathcal{O}}$ be its ceiling and $S$ the stationary bound of Definition 20.54 taken over its fibre boxes alone. Then* $\operatorname{reach}(B) \leq \max(H_{\mathcal{O}}, S)$.

**Corollary 20.57** (One falling variable needs only the restriction)**.** *In an order with at most one falling variable, a box on whose sets $Y$ the letters of the restriction $\mathcal{B}|B$ do not vanish is a fibre box, and it adds nothing to the stationary bound. So in a continuation of Theorem 20.56 such boxes need only the letters of the restriction, and the reach is at most* $\max(H_{\mathcal{O}}, S)$ *with $S$ taken over the fibre boxes of the orders with two or more falling variables.*

So the stationary bound enters only where a letter of the contraction vanishes inside the keyholes of a sector with two or more falling variables. The zeros of the contraction on the falling faces can still keep every such bound above the reach.

These theorems bound the right end of each interval $[\inf \varrho_B, \operatorname{reach}(B)]$ of Theorem 20.39 from both sides. The reach need not be read on the positive shape: the ratio is real at the real points of a flat face whatever their signs, and a keyhole of a falling variable crosses the negative axis nearer to 0 than every pole of the ratio on it, so a simple top can sit at a point with a negative coordinate.

## 20.9 One tree

Read on one tree, the period of a point, built back up, is the periods of the leaves multiplied by the share kernels of the nodes and integrated over the free shares; its $n-1$ integrations are shared out among the bunches and the leaves; at integer data the share kernels bring the Trailmoments of the counting input order by order; and where the free shares close on integer points the period is the sum of the residues along the lattice of the tree.

**Definition 20.58** (Share kernels, contents, integration count)**.** Let a molecule of lists with a native gauge be read on a tree of bunches, its Foreman tree or another, with the port of the balance wall run at every node on the native splits (Corollary 20.10). The share kernels of the tree (Corollary 4.4) are what the ports place on the shares: one stacked Gamma $\Gamma_{\substack{k\\1}}\begin{bmatrix}\mathbf{a}\\ a\end{bmatrix}$ for each letter opened at a node, $k$ the number of its patterns, $\mathbf{a}$ its shares and $a$ its exponent, which in Corollary 20.10 leave the factors $\big(\Gamma(a_{\mathcal{B}'})\Gamma(b_{\mathcal{B}'})\big)^{1-q_{\mathcal{B}'}}$ and $\prod_j \Gamma(s_j)^{-1}\Gamma(t_j)^{-1}$; each is balanced (Definition 1.2), its shares adding to its exponent on the fibre. The *free shares* are coordinates of the fibre of the word. Let $\mathcal{Z}$ be the $\mathbb{Q}$-algebra generated by the Trailmoments $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $k \geq 2$, of the counting input; it maps into the content $\mathcal{C}^{(1)}$ of Chapter 45, the algebra of trails of lattice inputs, which also holds the sums along the lattices of paired words (Chapters 43 and 45). The *content* $\mathcal{C}_M$ of a childcule $M$ of a node, a leaf in particular, is the $\mathcal{Z}$-module spanned by the values, at the integer exponents and masses at which it converges, of the period of a family on $M$ whose tables are those of $M$, and of its derivatives in the exponents and masses along the balance wall; the content of the molecule is the same with its own tables. The content $\mathcal{C}_B$ of the bunch $B$ of a node with room $R$ is the same for the families on $B$ whose tables, each with both letters, are the minors $\mathcal{B}'|B$ and $\mathcal{B}'/R$ of the tables $\mathcal{B}'$ at the node: those of the face of $B$ (Theorem 16.25), at width one those of the bunch family (Theorem 19.13). The content $\mathcal{C}_R$ of the room is the same for the families on $R$ whose tables are the $\mathcal{B}'|R$ and $\mathcal{B}'/B$.

The *integration count* of an integral as written is the number of its integrations, assigned to the written expression by these rules. The period of a family on $k$ marks has the integration count $k-1$, its simplex. A trail $(\mathsf{T}_{\mathbf{a};I_d}[y]\,\mathbb{I}_{(b+\mathbb{Z})^d})(\{b<y_1<\cdots<y_d\})$ of a lattice input with integer indices, $b>-1$ (Definition 5.23), has the integration count $a_1+\cdots+a_d$, its total index; the counting trails are the case $b=0$. A Gamma has the integration count one, $\Gamma(a)=\int_0^\infty x^{a-1}e^{-x}\,dx$ being one integration, so a stacked Gamma with $k$ Gammas above and $l$ below has the integration count $k-l$. A derivative in an exponent or a mass adds one, the logarithm of a letter or of a mark. A product has the sum of the integration counts of its factors, and a fibre integral of dimension $D$ (Definition 4.1) the integration count of its integrand less $D$, each share integrated inverting one Mellin transform. A real number has *integration count at most* $w$ when it is a rational combination of values, at rational data with unit weights, of integrals of integration count at most $w$.

The openings keep the integration count. The power $(X_1+\cdots+X_k)^{-s}$ is the share integral of $\Gamma_{\substack{k\\1}}\begin{bmatrix}\mathbf{a}\\ s\end{bmatrix}\prod_i X_i^{-a_i}$, of integration count $(k-1)-(k-1)=0$ (Corollary 4.4), and $e^{-y}$ is the integral of $\Gamma(b)y^{-b}$ over one share, of integration count $1-1=0$. A Beta corner $\Gamma_{\substack{2\\1}}\begin{bmatrix}a,\ b\\ a+b\end{bmatrix}$ has the integration count one, as the period of the list on two marks whose bases are the two singletons; the stacked Gamma $\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}$ of the shape integral has the integration count $|B|-1$, its simplex (Theorem 3.9); the interest $J_F=\Gamma(\kappa_{\mathcal{E}})P_F$ has the integration count $n$, its orthant (Theorem 8.7). The atom formula of a family on $n$ marks (Corollary 18.16) has the integration count $\sum(|\mathcal{B}_j|-1)$ of its stacked Gammas, over the active letters, less its dimension $\sum(|\mathcal{B}_j|-1)-(n-1)$: $n-1$, the integration count of the period. Taking the residue at a pole of one Gamma removes that Gamma and adds one pin (Theorem 4.5), so every term of a closing has the integration count of the integral closed.

**Theorem 20.59** (One tree suffices)**.** *Let $F$ be a molecule of lists on $n$ marks with real exponents $s_j,t_j\geq 0$ and masses $\mathbf{m}$ on the balance wall, with $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ convergent, and let its Foreman tree be read at its exponent pairs, at the centre or at a lean (Proposition 16.19, Theorem 14.6). Run the port of the balance wall at every node on the native splits of the point, as in Corollary 20.10. Parts (i), (ii), (iv) and (v) hold as well for every tree of bunches in place of the Foreman tree.*

*(i)* The word starts at the leaves. *$P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the word of the tree (Corollary 20.10): the periods of the leaves, each read at its shares and each entering once on its descent, multiplied by the share kernels and integrated over the free shares. At a node the words of the childcules multiply, and along a chain, its leaf opened by the atom formula, the word is one Gramma on the marks and the letters (Theorem 20.11).*

*(ii)* The integrations are counted. *The word has the integration count $n-1$, that of the period, and at rational data $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ has integration count at most $n-1$. At a node with bunch $B$ and $q$ childcules the integrations of the simplex are the $|B|+q-1$ pins of the node, $|B|-1$ for the shape of the bunch and $q$ for the totals of the bunch and of the childcules, $q+1$ numbers that add to one; in a leaf $L$ they are the $|L|-1$ integrations of its period.*

*(iii)* The leaf bounds the bunches. *Along every descent of the Foreman tree, at the centre and at every lean, the copies of every bunch have at most as many marks as the copies of the leaf (Theorem 14.22). When some active list is connected, as for a molecule of one list, the tree is a chain and $|B_0|\leq\cdots\leq|B_{d-1}|\leq|L|$ (Corollary 20.9): the shape of every bunch has at most as many integrations as the leaf, and its node adds one, the total of the bunch against its room.*

*(iv)* At integer data the share kernels bring $\mathcal{Z}$. *Let the exponents and masses be integers and let $x$ be a point of the fibre at which every share is an integer. The pins and the share sums have integer coefficients, so the directions of the fibre have a rational basis; write the fibre near $x$ as $x + V\mathbf{v}$ with $V$ rational. Near $x$ the share kernels are*

$$c \prod_i \ell_i(\mathbf{v})^{-\epsilon_i} \exp h(\mathbf{v}),$$

*with $c$ rational, $\ell_i(\mathbf{v})$ the deviation from its value at $x$ of each argument of the share kernels that is a nonpositive integer at $x$, $\epsilon_i$ equal to $1$ for a Gamma above and to $-1$ for one below, and $h$ a power series without constant term whose part of degree $k$ has its coefficients in $\mathbb{Q} + \mathbb{Q}\,(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ for $k \geq 2$ and in $\mathbb{Q}$ for $k = 1$: the terms of order $k$ of the share kernels carry total index at most $k$. Where the periods of the leaves converge at $x$, every residue of the word at $x$, taken in turn along rational directions spanning the fibre, is therefore a finite $\mathcal{Z}$-combination of products, over the leaves, of elements of their contents.*

*(v)* Summed along the tree. *A free share closed on the poles of the share kernels, where they alone have poles on the way and the closing converges, writes the word as a sum over a lattice of terms of its own integration count, each a fibre integral with one more pin (Theorem 4.5); where every free share closes so, on points at which every share is an integer and the periods of the leaves converge, the period at integer data is the sum along that lattice of the residues of (iv), finite $\mathcal{Z}$-combinations of products of the contents of the leaves. Across a bunch of width one, grouping its patterns by level puts the period of the bunch family in the word in place of the bunch (Theorem 19.13); for one table the even share then closes on its lattice whenever $\sup \varrho_B < \inf \varrho_R$ or $\inf \varrho_B > \sup \varrho_R$, and at integer data its residues are finite $\mathcal{Z}$-combinations of products of an element of $\mathcal{C}_B$ and one of $\mathcal{C}_R$ (Corollary 19.21).*

Whatever the order of the even letters, with $s, t > 0$, the even share of a node of width one closes on the lattices of continued periods of the bunch family and the room family toward a level at which their growths multiply to less than one (Theorem 20.34); a series bunch closes toward the level $d$ against every room with $\inf \varrho_R > \inf \varrho_B$, and a series connection against every room with $\sqrt{\inf \varrho_R}$ above the sum of the square roots of the reaches of its parts, the room's ratio bounded only on the side away from the closing (Theorem 20.40); and against a room bounded below and a bunch bounded below the node closes on a geometric lattice exactly when the intervals of the bunch and of the room are disjoint, at its peaks otherwise (Theorem 20.39). Off the integers the level shares of several tables close in the same way across bunches of width one whose values stand in one order, node after node down a tree of splits of width one (Theorem 19.20, Corollary 19.22). At nonnegative exponents the period is in every case the sum, along the draws nested on the descents of the tree, of products over its blocks of Dirichlet moments (Theorem 20.15).

The pins need not be unimodular, so a closing can reach points at which some share is not an integer, even at integer data. When the active tables link all the marks the pins reach every integer point of their target (Proposition 18.17), and the fibre has integral coordinates through an integer point; the shares that a closing leaves after pinning the others at the poles of their Gammas are solved from a square block of the pins, whose determinant can exceed 1 in modulus: for the list $\binom{[4]}{2}$ at the data $(\mathbf{1}; 1, 1)$ (Exercise 51.3), the fibre has points at which every share off such a block is an integer and some share is $\frac{1}{2}$. There the share kernels have rational arguments, and their expansion runs in heads and Trailmoments of the lattice inputs of those arguments (Proposition 5.13); that is why (iv) and (v) speak of integer points. Where a node has no free share its sum has one term. For a list whose bunch has width one, at a full lean, $(s, 0)$ or $(0, t)$, the level shares are pinned and the

period is a Beta corner times the periods of the bunch family and the room family (Corollary 19.14); at rational data this is an identity of period data (Theorem 19.34), at integer data the Beta corner is rational, and with the room a leaf the period is a rational multiple of a product of an element of $\mathcal{C}_B$ and one of the content of the leaf. When the two hinge values of a bunch of width one agree the shape and the room separate at every pair of exponents (Proposition 19.12), and at the wall of a set, with the weights factoring on its bottom levels, the residue of the period is the product of the periods of its two faces (Theorem 16.25).

The integration count of a period bounds that of its value, and the theorem is stated for lists because the weights of a table can carry integrations of their own, which the integration count of a period leaves out. With unit weights and rational data nothing is carried in. On an atom the tree is the atom and the theorem says no more than its own content; its atom formula is a word of the same kind whose leaves are single marks, and the argument of (iv) puts each of its residues at integer points in $\mathcal{Z}$. The lattice can make what no bunch or leaf shows: a tree whose bunches and leaves have their contents in $\mathcal{Z}$ can sum, along the lattice of one free share, to a Trailmoment of a lattice input with a frequency (Definition 5.4) that none of them carries. The finite form of the theorem, the period in the $\mathcal{Z}$-algebra generated by the contents of the bunches and the leaves, holds where the shares are pinned at an integer point; where they are free the tree gives the sum along its lattice. For a word of one free share made of Gammas alone, as the atom formula leaves it at integer data, that sum lies in the content of Definition 45.1 when the word is paired (Propositions 43.2 and 45.3), and otherwise it is a combination of Trailmoments of live inputs (Theorem 43.1).

Theorem 20.59 is the Foreman procedure read, step by step, as a statement about numbers. *Split first*: a compound is split into its polycule before any molecule is asked whether it answers yes, and at every node the polycule of the childcules is evaluated as the product of its molecules (Theorem 20.11(i)). *Read at the table's gauge, then ask*: each molecule is read at its table's gauge, native for the masses it uses and inherited by its children (Proposition 12.27), whether it answers yes is read at the centre (Theorem 11.8), and a realization that is not atomic is peeled. *Peel small*: the peel is taken at a class with the fewest marks, it splits a class only across twin childcules, each part inside one class (Theorem 14.20), so the leaf with its copies holds whole classes of every node above it and is at least as large as every bunch with its copies (Theorem 14.22). *The share kernels reapply*: what a node adds is its port, stacked Gammas of the shares of its bunch and the pins of its shape and its totals; near integer data a balanced stacked Gamma brings $\mathcal{Z}$ order by order in total index, and in the draw series each node contributes its own Dirichlet moments to every product over the blocks. *One tree*: every tree of bunches computes the same period and shares out the same $n-1$ integrations, and the Foreman tree is one in which the copies of no bunch have more marks than the copies of the leaf, and when some active list is connected no bunch has more marks than the leaf. *Pinned shares*: where the shares of a node are pinned, as for one list at a full lean across a bunch of width one, the node contributes an identity of period data (Theorem 19.34).

Chapter 34 lifts the statements of this section from periods to motivic periods: at integer data the left factors of the coaction of the motivic period of a family are read along every tree of bunches, the Foreman tree among them, through the steps attached to its vertices (Theorem 34.13); at its nodes, at any width, through the joint spectrum of the bunch (Theorem 34.16); and where the shares of a node are pinned the identities of period data above are identities of motivic periods (Theorem 34.17, Corollary 34.18).

# Part V

# Construction and composition

# Chapter 21

# Gluing and what crosses it

Every table read at a bunch is the gluing of what its bunch and its room carry (Proposition 12.5), and at a bunch of width one the gluing is along one mark, the hinge (Theorem 19.4). Gluing is an operation with its own calculus. A table is built from cells, tables with one marked hinge, substituted into the marks of a core; the bases, the weights, the multivectors, the letters and the walls of the result are read off the cells and the core; the blocks of Theorem 21.5 and the series connections of Section 20.7 are the cells and cores of the simplest kinds; two cell families glued along their hinges have hinge walls that add to $2\sigma$ on the balance wall; and every connected table whose weights factor at its bunches of width one, every list and every determinantal table among them, is a tree of such substitutions whose nodes are prime tables and series and parallel nodes. Cells with several hinges splice in pairs along any graph of pairs, closed paths included, and stay tables, their channels composing multilinearly. What a room receives from its bunch, at any width, for any family and any multivector, is the joint law of finitely many bunch letters: as many as the ranks of the level weights, which for a multivector are bounded by the squares of its interface ranks and for a determinantal table by $\binom{c(B)}{i}^2$ at the level $d(B)+i$, whatever the numbers of marks; for a determinantal table the bunch crosses its gluing as one positive definite matrix of size $c(B)$, whose determinant at width one is the hinge value $\gamma_B^K$, and one scalar.

## 21.1 Gluing along a mark

**Definition 21.1** (Cells, cores, substitution)**.** A *cell* is a table $\mathcal{A}$ on $B \sqcup \{g\}$ in which the mark $g$, its *hinge*, is neither a loop nor a coloop; it may carry two systems of positive weights on its bases, $\theta^K$ for its letter $K_{\mathcal{A}}$ and $\theta^\Psi$ for its letter $\Psi_{\mathcal{A}}$ (Definition 8.1), and it is *plain* when they agree, as for one table. Its *hinge minors* are the deletion $\mathcal{A} \setminus g$ and the contraction $\mathcal{A}/g$, tables on $B$ of ranks $r_{\mathcal{A}}$ and $r_{\mathcal{A}} - 1$ with the inherited weights (Definition 7.3), so that

$$K_{\mathcal{A}} = K_{\mathcal{A}\setminus g} + \alpha_g K_{\mathcal{A}/g}, \qquad \Psi_{\mathcal{A}} = \alpha_g \Psi_{\mathcal{A}\setminus g} + \Psi_{\mathcal{A}/g};$$

its hinge values and its ratio, as for a bunch of width one (Definition 19.2), are

$$\gamma_{\mathcal{A}}^K = \frac{K_{\mathcal{A}\setminus g}}{K_{\mathcal{A}/g}}, \qquad \gamma_{\mathcal{A}}^\Psi = \frac{\Psi_{\mathcal{A}/g}}{\Psi_{\mathcal{A}\setminus g}}, \qquad \varrho_{\mathcal{A}} = \frac{\gamma_{\mathcal{A}}^\Psi}{\gamma_{\mathcal{A}}^K} = \frac{\mathbf{E}_{\mathcal{A}/g}}{\mathbf{E}_{\mathcal{A}\setminus g}},$$

functions of $\alpha_B$ homogeneous of the degrees 1, 1 and 0. The cell of a bunch $\mathcal{B}_B^+$ of Theorem 19.4 is a cell, with the hinge minors $\mathcal{B}|B$ and $\mathcal{B}/R$, the hinge values $\gamma_B^K$, $\gamma_B^\Psi$ and the ratio $\varrho_B$ of Definition 19.2. A *core* is a table $\mathcal{H}$ on marks $h_1, \dots, h_q$, none of them a loop or a coloop, plain or

with two systems of weights as a cell. For cells $\mathcal{A}_i$ on $B_i \sqcup \{g_i\}$, the $B_i$ disjoint, the *substitution* $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ is the family of sets $\bigsqcup_i X_i \subseteq \bigsqcup_i B_i$ for which some basis $T$ of $\mathcal{H}$ has $X_i \in \mathcal{A}_i \setminus g_i$ when $h_i \in T$ and $X_i \in \mathcal{A}_i/g_i$ when $h_i \notin T$, with the weight, in each letter,

$$\theta_{\sqcup_i X_i} = \theta^{\mathcal{H}}_T \prod_{h_i \in T} \theta^{\mathcal{A}_i}_{X_i} \prod_{h_i \notin T} \theta^{\mathcal{A}_i}_{X_i + g_i},$$

every $\theta$ read as the weight $\theta^K$ of the letter $K$ throughout, or as $\theta^\Psi$ throughout; $T$ is determined by the sizes, $h_i \in T$ exactly when $|X_i| = r_{\mathcal{A}_i}$. A mark $h$ is *left in place* by the cell $\binom{\{h,g\}}{1}$, the list on $\{h, g\}$ with the bases $\{h\}$ and $\{g\}$, whose hinge minors have the single bases $\{h\}$ and $\varnothing$ and whose hinge values are both $\alpha_h$. A plain cell is a *dual pair* when its contraction $\mathcal{A}/g$ is the dual table of its deletion $\mathcal{A} \setminus g$, weights included: the bases of $\mathcal{A}/g$ are the $B \setminus X$, $X$ a basis of $\mathcal{A} \setminus g$, with $\theta^{\mathcal{A}}_{(B\setminus X)+g} = \theta^{\mathcal{A}}_X$; a mark left in place is one. One cell $\mathcal{A}$ at one mark $h$ of $\mathcal{H}$, the other marks left in place, is the *gluing of $\mathcal{A}$ and $\mathcal{H}$ along $h$*: the gluing (Definition 12.4) of the communicatroid of $B$ whose children are $\theta^{\mathcal{A}}_X$ times $\mathcal{H}/h$ at the bases $X$ of $\mathcal{A} \setminus g$ and $\theta^{\mathcal{A}}_{X+g}$ times $\mathcal{H} \setminus h$ at the bases $X$ of $\mathcal{A}/g$. A family is *substituted* table by table, $(\mathcal{B}_j, \theta_j) = \mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{qj}\rangle$, the cells of all tables at the mark $h_i$ sharing the marks $B_i \sqcup \{g_i\}$; its masses are on $\mathcal{E} = \bigsqcup_i B_i$.

**Lemma 21.2** (Gluing along a mark gives a table)**.** *Let $\mathcal{A}$ be a cell on $B \sqcup \{g\}$ and $\mathcal{A}'$ a cell on $R \sqcup \{g'\}$. The sets $X \sqcup Y$ with $X \in \mathcal{A} \setminus g$ and $Y \in \mathcal{A}'/g'$, or $X \in \mathcal{A}/g$ and $Y \in \mathcal{A}' \setminus g'$, are the bases of a table on $B \sqcup R$ of rank $r_{\mathcal{A}} + r_{\mathcal{A}'} - 1$, in which $B$ is a bunch of width one whose cell of the bunch and cell of the room (Theorem 19.4) are $\mathcal{A}$ and $\mathcal{A}'$, the hinges read as the one mark $g$.*

**Proposition 21.3** (Substitution)**.** *(i) $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ is a table of rank $r_{\mathcal{H}} + \sum_i (r_{\mathcal{A}_i} - 1)$; it is the gluing of the cells to $\mathcal{H}$ along its marks one at a time, in any order; each $B_i$ is a bunch of it of width one whose cell of the bunch is $\mathcal{A}_i$, its weights factoring across $B_i$ and its room.*

*(ii) Substituting into a mark of a cell is substituting the substituted cell: for a mark $x$ of $B_1$ and a cell $\mathcal{A}'$ at $x$,*

$$\big(\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle\big)\langle \mathcal{A}' \text{ at } x\rangle = \mathcal{H}\big\langle \mathcal{A}_1\langle \mathcal{A}' \text{ at } x\rangle, \mathcal{A}_2, \dots, \mathcal{A}_q\big\rangle.$$

*(iii) Conversely, a table with a bunch $B$ of width one, the weights factoring across $B$ and $R$, is the gluing of its cell of the bunch $\mathcal{B}^+_B$ and its cell of the room $\mathcal{B}^+_R$ along $g$ (Theorem 19.4): every gluing along a mark is that decomposition read backwards, and every such decomposition is a gluing.*

*(iv) If $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ is connected, so are $\mathcal{H}$ and every cell. A connected table with a bunch $B$ of width one, $|B|, |R| \geq 2$, the weights factoring across $B$ and $R$, is the gluing along a mark of two connected tables on fewer marks.*

**Theorem 21.4** (The letters of a substitution)**.** *Let $K$ and $\Psi$ be the letters of $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$, each with its own weights, and read the hinge minors and the hinge values of each cell on its own marks. Then*

$$K(\alpha) = \prod_i K_{\mathcal{A}_i/g_i}(\alpha_{B_i})\ K_{\mathcal{H}}\big(\gamma^K_{\mathcal{A}_1}(\alpha_{B_1}), \dots, \gamma^K_{\mathcal{A}_q}(\alpha_{B_q})\big),$$
$$\Psi(\alpha) = \prod_i \Psi_{\mathcal{A}_i\setminus g_i}(\alpha_{B_i})\ \Psi_{\mathcal{H}}\big(\gamma^\Psi_{\mathcal{A}_1}(\alpha_{B_1}), \dots, \gamma^\Psi_{\mathcal{A}_q}(\alpha_{B_q})\big).$$

*For one cell, the other marks left in place, this is Theorem 19.4.*

The core reads each of its cells as one mark, at the hinge value of the cell by the letter: in $K$ the mark $h_i$ stands at $\gamma^K_{\mathcal{A}_i}$ and in $\Psi$ at $\gamma^\Psi_{\mathcal{A}_i}$, and what is left over is a product of one letter of a hinge minor of every cell. Nothing in the core sees more of a cell than its two hinge values.

A family can be generated rather than found: seed tables assembled under a set of bijections give atoms (Proposition 11.6), and a mark that is neither a loop nor a coloop, replaced by a block of new marks, gives a bunch of width one, which the room reads as the mark it replaced.

**Theorem 21.5** (Replacement of marks by blocks)**.** *Let $(\mathcal{B},\theta)$ be a table of rank $r$ on $\mathcal{E}=\mathcal{E}_1\sqcup\mathcal{E}_2$, and replace every mark $h$ by a* block *$X_h$ of $q_h\geq 1$ new marks, by the* first rule *at the marks of $\mathcal{E}_1$ and by the* complementary rule*, the first rule read on the dual table, at those of $\mathcal{E}_2$: a basis $T$ gives the sets that contain one mark of $X_h$ for each $h\in T\cap\mathcal{E}_1$, none for $h\in\mathcal{E}_1\setminus T$, all of $X_h$ for $h\in T\cap\mathcal{E}_2$ and all of $X_h$ but one for $h\in\mathcal{E}_2\setminus T$, each with the weight of $T$. They are the bases of a table on $\bigsqcup_h X_h$ of rank $r+\sum_{h\in\mathcal{E}_2}(q_h-1)$; a block of one mark leaves its mark in place, and when no mark is a loop or a coloop the table is the substitution into $(\mathcal{B},\theta)$ of the* parallel blocks $\binom{X_h\sqcup\{g_h\}}{1}$ *at $\mathcal{E}_1$ and the* series blocks $\binom{X_h\sqcup\{g_h\}}{q_h}$ *at $\mathcal{E}_2$ (Definition 21.1). Its Foreman polynomials are, each $e_k$ read on the marks of its block,*

$$K'=\prod_{h\in\mathcal{E}_2} e_{q_h-1}\cdot K\Big(\big(e_1\big)_{h\in\mathcal{E}_1},\big(\tfrac{e_{q_h}}{e_{q_h-1}}\big)_{h\in\mathcal{E}_2}\Big),\qquad \Psi'=\prod_{h\in\mathcal{E}_1} e_{q_h-1}\cdot\Psi\Big(\big(\tfrac{e_{q_h}}{e_{q_h-1}}\big)_{h\in\mathcal{E}_1},\big(e_1\big)_{h\in\mathcal{E}_2}\Big);$$

*for one block $X=\{x_1,\dots,x_q\}$ at the mark $h$, the other marks left in place, the first rule gives $K'=K_{\setminus h}+e_1(\alpha_X)\,K_{/h}$ and $\Psi'=e_q(\alpha_X)\,\Psi_{\setminus h}+e_{q-1}(\alpha_X)\,\Psi_{/h}$, and the complementary rule gives $K'=e_q(\alpha_X)\,K_{/h}+e_{q-1}(\alpha_X)\,K_{\setminus h}$ and $\Psi'=\Psi_{/h}+e_1(\alpha_X)\,\Psi_{\setminus h}$ (Definition 7.3). In scale and shape, $\alpha_X=\rho u$ with $u\in\Delta_X$: $e_1(\alpha_X)=\rho$, $e_q(\alpha_X)=\rho^q e_q(u)$, $e_{q-1}(\alpha_X)=\rho^{q-1}e_{q-1}(u)$, and $\alpha_X^{\mathbf{m}|_X-\mathbf{1}}d\alpha_X=\rho^{m_X-1}d\rho\;u^{\mathbf{m}|_X-\mathbf{1}}d\omega_X(u)$ (Lemma 2.7).*

**Theorem 21.6** (Counts, walls and atoms under replacement)**.** *Let $(\mathcal{B},\theta)$ be a table of rank $r$ on $\mathcal{E}$, $n=|\mathcal{E}|$, and replace every mark $h$ by a block $X_h$ of $q_h\geq 1$ new marks as in Theorem 21.5, by the first rule at the marks of $\mathcal{E}_1$ and by the complementary rule at those of $\mathcal{E}_2$, $\mathcal{E}=\mathcal{E}_1\sqcup\mathcal{E}_2$.*

*(i) The marks become $\sum_h q_h$; the rank becomes $r+\sum_{h\in\mathcal{E}_2}(q_h-1)$; the total weight becomes*

$$\sum_{T\in\mathcal{B}}\theta_T\prod_{h\in T\cap\mathcal{E}_1}q_h\prod_{h\in\mathcal{E}_2\setminus T}q_h=\prod_{h\in\mathcal{E}_2}q_h\cdot K\Big((q_h)_{h\in\mathcal{E}_1},(q_h^{-1})_{h\in\mathcal{E}_2}\Big),$$

*which is $K((q_h)_h)$ under the first rule alone and $\Psi((q_h)_h)$ under the complementary rule alone.*

*(ii) For a set $Y$ of new marks let $\lceil Y\rceil=\{h:\ X_h\cap Y\neq\varnothing\}$ and $\lfloor Y\rfloor=\{h:\ X_h\subseteq Y\}$, and let $r'$, $d'$, $d^{*\prime}$, $c'$ be the counts of the new table (Definition 7.12). Then*

$$d'(Y)=\sum_{h\in\mathcal{E}_2}\max\big(|X_h\cap Y|-1,0\big)+d\big((\lfloor Y\rfloor\cap\mathcal{E}_1)\sqcup(\lceil Y\rceil\cap\mathcal{E}_2)\big),$$
$$d^{*\prime}(Y)=\sum_{h\in\mathcal{E}_1}\max\big(|X_h\cap Y|-1,0\big)+d^*\big((\lceil Y\rceil\cap\mathcal{E}_1)\sqcup(\lfloor Y\rfloor\cap\mathcal{E}_2)\big),$$

*and $c'(Y)=|Y|-d'(Y)-d^{*\prime}(Y)$. Under the first rule alone $r'(Y)=r(\lceil Y\rceil)$, $d'(Y)=r-r(\mathcal{E}\setminus\lfloor Y\rfloor)$ and $c'(Y)=r(\lceil Y\rceil)+r(\mathcal{E}\setminus\lfloor Y\rfloor)-r$; under the complementary rule alone $c'(Y)=|\lceil Y\rceil|-|\lfloor Y\rfloor|+r(\mathcal{E}\setminus\lceil Y\rceil)+r(\lfloor Y\rfloor)-r$. If $Y=\bigcup_{h\in S}X_h$, then $c'(Y)=c(S)$; when the seed table is connected on at least two marks, a proper nonempty part of one block has width one.*

*(iii) Let a family of seed tables with positive exponent pairs be connected on at least two marks and carried to itself by a set $H$ of bijections as in Proposition 11.6, and replace every mark by a block of the same $q$ marks, all by the first rule or all by the complementary rule. Then the bijections of $H$ lift to the blocks, the permutations inside a block carry every replaced table to itself, the replaced family is connected, and for every $q$ its realization at the table's gauge is an atom.*

The shape of a block is integrated by Theorem 19.8, and its scale is the mark of the seed. A block $X_h$ of Theorem 21.5 with $q_h \geq 2$ whose mark is neither a loop nor a coloop of the seed is a parallel bunch of width one under the first rule and a series bunch of width one under the complementary rule (Definition 19.1).

**Corollary 21.7** (Blocks and closures)**.** *(i) The series block on a set $B$ of $\ell$ marks (Theorem 21.5) is the cell $\binom{B\sqcup\{g\}}{\ell}$, whose hinge minors have $K_{\backslash g} = e_\ell(\alpha_B)$, $K_{/g} = e_{\ell-1}(\alpha_B)$, $\Psi_{\backslash g} = 1$ and $\Psi_{/g} = e_1(\alpha_B)$; the parallel block is the cell $\binom{B\sqcup\{g\}}{1}$, with $K_{\backslash g} = e_1(\alpha_B)$, $K_{/g} = 1$, $\Psi_{\backslash g} = e_{\ell-1}(\alpha_B)$ and $\Psi_{/g} = e_\ell(\alpha_B)$. Series blocks at the marks $h_i$, $i \in S$, of a core $\mathcal{H}$ and parallel blocks at the others replace the marks of $\mathcal{H}$ by blocks as in Theorem 21.5, the series blocks by its complementary rule and the parallel blocks by its first rule, and give the letters of that theorem, each $e_k$ read on the marks of its block: a series block is seen by $\Psi$ through its total and by $K$ through the hinge value $e_\ell/e_{\ell-1} = \big(\sum_x \alpha_x^{-1}\big)^{-1}$, a parallel block the other way round.*

*(ii) The* series closure *of cells $\mathcal{A}_1, \dots, \mathcal{A}_q$ is the substitution $\binom{\{h_1,\dots,h_q\}}{q-1}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ into the core of unit weights; its letters are*

$$K = \prod_i K_{\mathcal{A}_i\backslash g_i} \cdot \sum_i \frac{K_{\mathcal{A}_i/g_i}}{K_{\mathcal{A}_i\backslash g_i}}, \qquad \Psi = \prod_i \Psi_{\mathcal{A}_i\backslash g_i} \cdot \sum_i \frac{\Psi_{\mathcal{A}_i/g_i}}{\Psi_{\mathcal{A}_i\backslash g_i}},$$

*so that the bunch $B_1 \sqcup \cdots \sqcup B_{q-1}$ is the series connection of the parts $B_i$ with the tables $\mathcal{A}_i \setminus g_i$ and $\mathcal{A}_i/g_i$ (Proposition 20.37, with $C = C' = 1$). The* parallel closure $\binom{\{h_1,\dots,h_q\}}{1}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ *has the letters $K = \prod_i K_{\mathcal{A}_i/g_i} \cdot \sum_i K_{\mathcal{A}_i\backslash g_i}/K_{\mathcal{A}_i/g_i}$ and $\Psi = \prod_i \Psi_{\mathcal{A}_i/g_i} \cdot \sum_i \Psi_{\mathcal{A}_i\backslash g_i}/\Psi_{\mathcal{A}_i/g_i}$; it is the dual of the series closure of the dual cells.*

*(iii) The gluing of two cells along their hinges is $\binom{\{h_1,h_2\}}{1}\langle \mathcal{A}_1, \mathcal{A}_2\rangle$, with the letters $K = K_{\mathcal{A}_1\backslash g_1} K_{\mathcal{A}_2/g_2} + K_{\mathcal{A}_1/g_1} K_{\mathcal{A}_2\backslash g_2}$ and $\Psi = \Psi_{\mathcal{A}_1/g_1}\Psi_{\mathcal{A}_2\backslash g_2} + \Psi_{\mathcal{A}_1\backslash g_1}\Psi_{\mathcal{A}_2/g_2}$.*

**Proposition 21.8** (Dual pairs)**.** *Let $\mathcal{A}$ be a plain cell on $B \sqcup \{g\}$ of rank $r_\mathcal{A}$ that is a dual pair (Definition 21.1), and put $k = K_{\mathcal{A}\backslash g}$ and $\psi = \Psi_{\mathcal{A}\backslash g}$.*

*(i) A plain cell is a dual pair exactly when its bases are closed under complement with equal weights; then $|B| = 2r_\mathcal{A} - 1$, so that $k$ has the degree $r_\mathcal{A}$ and $\psi$ the degree $r_\mathcal{A} - 1$.*

*(ii) $K_{\mathcal{A}/g} = \psi$ and $\Psi_{\mathcal{A}/g} = k$; the two hinge values agree, $\gamma_\mathcal{A}^K = \gamma_\mathcal{A}^\Psi = k/\psi$, and $\varrho_\mathcal{A} = 1$.*

*(iii) Let $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ be a substitution of dual pairs into a plain core $\mathcal{H}$, of rank $r_\mathcal{H}$, with $k_i$ and $\psi_i$ as above and $\pi = \psi_1 \cdots \psi_q$; a mark left in place is a dual pair with $k_i = \alpha_h$ and $\psi_i = 1$. Then, as identities of polynomials, with $\pi k_i/\psi_i = k_i \prod_{l\neq i} \psi_l$ and $K$, $\Psi$ the letters of the substitution,*

$$K_\mathcal{H}\big((\pi k_i/\psi_i)_i\big) = \pi^{\,r_\mathcal{H}-1} K, \qquad \Psi_\mathcal{H}\big((\pi k_i/\psi_i)_i\big) = \pi^{\,q-r_\mathcal{H}-1}\Psi.$$

**Proposition 21.9** (The multivector of a substitution)**.** *Let $\mathcal{H}$ and the cells carry multivectors $\boldsymbol{\omega}_{\mathcal{H}}$ and $\boldsymbol{\omega}_i$ (Definition 7.20), the hinge first in the order of the marks of each cell, and order the marks of the substitution by putting $B_i$, in its order in $\mathcal{A}_i$, in the place of $h_i$ in the order of $\mathcal{H}$. Then*

$$\boldsymbol{\omega}\Big(\bigsqcup_i X_i\Big) = \boldsymbol{\omega}_{\mathcal{H}}(T) \prod_{h_i \in T} \boldsymbol{\omega}_i(X_i) \prod_{h_i \notin T} \boldsymbol{\omega}_i(g_i + X_i)$$

*is a multivector of $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$; it is multilinear in $(\boldsymbol{\omega}_{\mathcal{H}}, \boldsymbol{\omega}_1, \dots, \boldsymbol{\omega}_q)$, a substitution of decomposable multivectors is decomposable, and*

$$\mathrm{rk}(\boldsymbol{\omega}) \le \mathrm{rk}(\boldsymbol{\omega}_{\mathcal{H}}) \prod_i \mathrm{rk}(\boldsymbol{\omega}_i).$$

*A substitution of determinantal tables is determinantal.*

**Theorem 21.10** (The walls of a substitution)**.** *For $S = \bigsqcup_i S_i$, $S_i \subseteq B_i$, let $D(S)$ be the set of the marks $h_i$ of the core with $d_{\mathcal{A}_i}(S_i + g_i) > d_{\mathcal{A}_i}(S_i)$ and $D^*(S)$ that of the $h_i$ with $d^*_{\mathcal{A}_i}(S_i + g_i) > d^*_{\mathcal{A}_i}(S_i)$, the two differences being 0 or 1. The counts of the substitution are*

$$d(S) = \sum_i d_{\mathcal{A}_i}(S_i) + d_{\mathcal{H}}\big(D(S)\big), \qquad d^*(S) = \sum_i d^*_{\mathcal{A}_i}(S_i) + d^*_{\mathcal{H}}\big(D^*(S)\big),$$

*and for a substituted family, with $D_j(S)$ and $D^*_j(S)$ read in table $j$, its walls are*

$$\kappa_S = \sum_i \Big(m_{S_i} - \sum_j \big(s_j d_{\mathcal{A}_{ij}}(S_i) + t_j d^*_{\mathcal{A}_{ij}}(S_i)\big)\Big) - \sum_j \Big(s_j d_{\mathcal{H}_j}\big(D_j(S)\big) + t_j d^*_{\mathcal{H}_j}\big(D^*_j(S)\big)\Big).$$

*At $S = B_i$, $D(S) = D^*(S) = \{h_i\}$ and the wall is $\kappa_{B_i} = m_{B_i} - \sum_j \big(s_j(r_{\mathcal{A}_{ij}} - 1) + t_j(|B_i| - r_{\mathcal{A}_{ij}})\big)$, the mass that pooling gives the hinge of $B_i$ (Theorem 19.8).*

**Corollary 21.11** (The width of the marks left in place)**.** *Let a substitution have the core $\mathcal{H}$ on $L \sqcup H'$, the marks of $L$ left in place and cells with at least two marks at the marks of $H'$. Every $S \subseteq L$ has the width it has in the core, and*

$$c(S) = c_{\mathcal{H}}(S) \le |L \setminus S| + |H'|.$$

*So a bunch of width $c$ that is all the marks left in place needs at least $c$ other cells, and every width occurs: in $\binom{B \sqcup \{h_1, \dots, h_c\}}{c}$ with $|B| = c$, the marks of $B$ left in place, $B$ has the width $c$, whatever the cells at $h_1, \dots, h_c$.*

**Lemma 21.12** (Two-sums of cells; hinge walls)**.** *Let $\mathcal{A}$ be a cell on $B \sqcup \{g\}$ and $\mathcal{A}'$ a cell on $R \sqcup \{g\}$ with the one hinge $g$, $B \cap R = \varnothing$, and let $\mathcal{A} \oplus_g \mathcal{A}'$ be their gluing along the hinge, $\binom{\{h_1, h_2\}}{1}\langle \mathcal{A}, \mathcal{A}'\rangle$ (Corollary 21.7(iii)), each basis carrying in each letter the product of the weights of the two bases of $\mathcal{A}$ and $\mathcal{A}'$ it is made of: a table on $B \sqcup R$ of rank $r_{\mathcal{A}} + r_{\mathcal{A}'} - 1$, plain when $\mathcal{A}$ and $\mathcal{A}'$ are, in which $B$ is a bunch of width one with the cell of the bunch $\mathcal{A}$ and the cell of the room $\mathcal{A}'$ (Lemma 21.2); every table with a bunch of width one, the weights factoring across it, is the two-sum of its two cells (Proposition 21.3(iii)). Let the cell families $F_B = ((\mathcal{A}_j, \theta^{\mathcal{A}_j}); s_j, t_j)_{j=1}^N$ on $B \sqcup \{g\}$ and $F_R = ((\mathcal{A}'_j, \theta^{\mathcal{A}'_j}); s_j, t_j)_{j=1}^N$ on $R \sqcup \{g\}$ be glued table by table into the family $F = ((\mathcal{A}_j \oplus_g \mathcal{A}'_j); s_j, t_j)_j$ on $B \sqcup R$, with masses $\mathbf{m}$. The walls of $F$ at $B$ and at $R$ are the hinge walls of the two cell families, for one table those of the two cells, $\kappa_{\mathcal{A}}$ and $\kappa_{\mathcal{A}'}$,*

$$\kappa_{F_B} = m_B - \sum_j \big(s_j(r_{\mathcal{A}_j} - 1) + t_j(|B| - r_{\mathcal{A}_j})\big), \qquad \kappa_{F_R} = m_R - \sum_j \big(s_j(r_{\mathcal{A}'_j} - 1) + t_j(|R| - r_{\mathcal{A}'_j})\big),$$

*and $\kappa_{F_B} + \kappa_{F_R} = 2\sigma + \kappa_{\mathcal{E}}$, $\sigma$ the total strength and $\kappa_{\mathcal{E}}$ the imbalance of $F$ (Definition 8.1): on the balance wall the two hinge walls add to $2\sigma = \sum_j (s_j + t_j)$, and the mass $2\sigma - \kappa_{F_B}$ on $g$ puts the cell family $F_B$ itself on its balance wall.*

Each letter of a two-sum is a letter of a hinge minor of one cell times the letter of the other cell read at one hinge value of the first (Theorem 21.4): $K$ sees $\mathcal{A}$ through $\gamma_{\mathcal{A}}^{K}$ and $\Psi$ through $\gamma_{\mathcal{A}}^{\Psi}$. The hinge wall of a cell is a number of the cell alone, its masses and its exponents, and balance makes the two hinge walls of a two-sum complementary; the period of a two-sum is read through them in Theorem 23.9.

## 21.2 Splicing hinges in pairs

A cell has one hinge. Gluing along several marks at once, or around a closed path of cells, needs cells with several hinges and a splicing of hinges in pairs. The splicing stays inside tables because the levels of a bunch are tables (Proposition 12.17); one pair joining two cells is the gluing of Lemma 21.2.

**Definition 21.13** (Cells with several hinges; channels)**.** A *cell with the hinges* $G$ is a table $\mathcal{A}$ on $B \sqcup G$ with two systems of positive weights $\theta^K$, $\theta^\Psi$ on its bases; $n_{\mathcal{A}} = |B|$ is its number of marks other than its hinges. Its *channels* are the letters of the children of the bunch $G$ (Definition 12.1): for a pattern $p \in \mathrm{pat}(G)$, the basis polynomial $K_{\mathcal{A},(p)}$ of the child at $p$ with the weights $\theta^K$ and its cobasis polynomial $\Psi_{\mathcal{A},(p)}$ with the weights $\theta^\Psi$, polynomials on $B$, so that by the Taylor cube (Theorem 12.7)

$$K_{\mathcal{A}} = \sum_{p \in \mathrm{pat}(G)} \alpha_G^p \, K_{\mathcal{A},(p)}(\alpha_B), \qquad \Psi_{\mathcal{A}} = \sum_{p \in \mathrm{pat}(G)} \alpha_G^{G \setminus p} \, \Psi_{\mathcal{A},(p)}(\alpha_B) :$$

$p$ is the set of hinges a basis occupies, in both letters. A channel has positive coefficients on its support and the degree

$$\deg K_{\mathcal{A},(p)} = r_{\mathcal{A}} - |p|, \qquad \deg \Psi_{\mathcal{A},(p)} = n_{\mathcal{A}} - r_{\mathcal{A}} + |p|.$$

The *channel vector* $\mathrm{ch}(\mathcal{A})$ lists the channels by their *tags*, the pairs of a letter and a set $p \subseteq G$, $2^{|G|+1}$ of them; a tag whose set is not a pattern is an absent channel, counted as zero. A cell of Definition 21.1 is the case of one hinge $g$ that is neither a loop nor a coloop, with $K_{\mathcal{A},(\varnothing)} = K_{\mathcal{A}\setminus g}$, $K_{\mathcal{A},(\{g\})} = K_{\mathcal{A}/g}$, $\Psi_{\mathcal{A},(\varnothing)} = \Psi_{\mathcal{A}\setminus g}$ and $\Psi_{\mathcal{A},(\{g\})} = \Psi_{\mathcal{A}/g}$. *Splicing* a pair $\{a, b\}$ of hinges keeps the bases that occupy exactly one of them and forgets the pair; on either letter it is

$$L \longmapsto (\partial_{\alpha_a} + \partial_{\alpha_b}) L \big|_{\alpha_a = \alpha_b = 0}.$$

Cells are spliced along pairs of their hinges by splicing the pairs in their direct sum (Proposition 7.9).

**Theorem 21.14** (Splicing hinges in pairs)**.** *(i) Splicing a pair $\{a, b\}$ of hinges of a cell $\mathcal{A}$ at which some basis occupies exactly one of $a$, $b$ gives a cell on $B \sqcup (G \setminus \{a, b\})$ of rank $r_{\mathcal{A}} - 1$: its bases are the sets $T \setminus \{a, b\}$ with $|T \cap \{a, b\}| = 1$, each carrying in each letter the sum of the weights of the bases $T$ that give it.*

*(ii) Let a direct sum of cells carry a set of disjoint pairs of hinges, forming any graph on the cells: closed paths, a pair inside one cell and several pairs between two cells are allowed. If*

*some choice of one basis in each cell occupies exactly one hinge of every pair, splicing the pairs one at a time, in any order, gives one cell; its rank is the sum of the ranks less the number of pairs, its bases are the unions of the unpaired parts of such choices, and its weight in each letter at a basis is the sum, over the choices that give it, of the products of the weights chosen. Plain cells give a plain cell.*

*(iii) Under the hypothesis of (ii), let the cells be $\mathcal{A}_1, \dots, \mathcal{A}_m$ with the hinges $G_1, \dots, G_m$, and in this statement let $G'$ be the set of the hinges that lie in no pair, $k$ the number of pairs and, for $p \subseteq G'$, let $J$ run over the sets of paired hinges that contain exactly one hinge of every pair. The spliced cell $\mathcal{A}$ has the channels*

$$K_{\mathcal{A},(p)} = \sum_J \prod_{i=1}^{m} K_{\mathcal{A}_i,((p \sqcup J)\cap G_i)}, \qquad \Psi_{\mathcal{A},(p)} = \sum_J \prod_{i=1}^{m} \Psi_{\mathcal{A}_i,((p \sqcup J)\cap G_i)},$$

*absent channels counting as zero; it has $\sum_i n_{\mathcal{A}_i}$ marks besides its hinges $G'$ and the rank $\sum_i r_{\mathcal{A}_i} - k$, and every product on the right has the degree of the channel on the left. The map $\big(\mathrm{ch}(\mathcal{A}_1), \dots, \mathrm{ch}(\mathcal{A}_m)\big) \mapsto \mathrm{ch}(\mathcal{A})$ is multilinear. For two cells $\mathcal{A}_1$ and $\mathcal{A}_2$ spliced along $k$ pairs, each pair named once, so that the paired hinges of either cell form one set $I$ of $k$ names and the other hinges form $G_1'$ and $G_2'$, it is the bilinear map*

$$K_{\mathcal{A}_{12},(p_1 \sqcup p_2)} = \sum_{I' \subseteq I} K_{\mathcal{A}_1,(p_1 \sqcup I')}\, K_{\mathcal{A}_2,(p_2 \sqcup (I \setminus I'))},$$
$$\Psi_{\mathcal{A}_{12},(p_1 \sqcup p_2)} = \sum_{I' \subseteq I} \Psi_{\mathcal{A}_1,(p_1 \sqcup I')}\, \Psi_{\mathcal{A}_2,(p_2 \sqcup (I \setminus I'))},$$

*$p_1 \subseteq G_1'$, $p_2 \subseteq G_2'$; with $k = 1$ and no other hinges it is Corollary 21.7(iii).*

Chapter 27 turns the bilinearity of Theorem 21.14(iii) into finite moment vectors and transfer matrices.

## 21.3 What crosses a gluing

**Definition 21.15** (Level weights, level factorizations, interface)**.** Let $(\mathcal{B}, \theta)$ be a table of rank $r$ with a bunch $B$, of any width, and room $R$. For a level $k$, $d(B) \le k \le r(B)$, the *level weights* are the matrix $\Theta_B^{(k)} = (\theta_{p \sqcup X})_{p,X}$ indexed by the $k$-sets $p \subseteq B$ and the $(r-k)$-sets $X \subseteq R$, with $\theta_{p \sqcup X} = 0$ when $p \sqcup X$ is not a basis. A *level factorization* writes every level letter (Definition 12.16) as a sum of products over a finite set $C_k$,

$$K_B^{(k)}(\alpha) = \sum_{c \in C_k} V_{k,c}(\alpha_B)\, L_{k,c}(\alpha_R),$$

each $V_{k,c}$ a real linear combination of the monomials $\alpha_B^p$, $|p| = k$, and each $L_{k,c}$ a real linear combination of the monomials $\alpha_R^X$, $|X| = r - k$: equivalently, a writing of $\Theta_B^{(k)}$ as a sum of $|C_k|$ products of a column and a row, so that $|C_k| \ge \operatorname{rank} \Theta_B^{(k)}$, with equality for some factorization. The $V_{k,c}$ are the *bunch letters* and the $L_{k,c}$ the *room letters*. With $V_{k,c}^*(\alpha_B) = \alpha_B^B V_{k,c}(\alpha_B^{-1})$ and $L_{k,c}^*(\alpha_R) = \alpha_R^R L_{k,c}(\alpha_R^{-1})$ the same factorization writes the colevel letter of the colevel $|B| - k$,

$$\Psi_B^{(|B|-k)}(\alpha) = \sum_{c \in C_k} V_{k,c}^*(\alpha_B)\, L_{k,c}^*(\alpha_R) :$$

each level is one function read twice (Proposition 7.2). The *stations* of $B$ are the maximal sets of patterns whose children are positive multiples of one another, each with a fixed representative, the *child of the station*; every station lies in one level. The *children factorization* lets $c$ run over the stations of level $k$, with $V_{k,c} = \sum_{p\in c} w_p\, \alpha_B^p$ for the child at $p$ equal to $w_p$ times the child of the station, and $L_{k,c}$ the basis polynomial of the child of the station: all its terms are nonnegative. The *interface dimension* of $B$ is $\sum_k \operatorname{rank} \Theta_B^{(k)}$; for a family it is summed over the tables.

Along the scale of $B$, $\alpha_B = \rho u$ with $u \in \Delta_B$ (Lemma 2.7), $V_{k,c}(\rho u) = \rho^k V_{k,c}(u)$ and $V^*_{k,c}(\rho u) = \rho^{|B|-k} V^*_{k,c}(u)$; since $|B| - r(B) = d^*(B)$,

$$K(\rho u, \alpha_R) = \rho^{d(B)} \sum_k \rho^{k-d(B)} \sum_c V_{k,c}(u)\, L_{k,c}(\alpha_R),$$
$$\Psi(\rho u, \alpha_R) = \rho^{d^*(B)} \sum_k \rho^{r(B)-k} \sum_c V^*_{k,c}(u)\, L^*_{k,c}(\alpha_R),$$

each letter counted from its own bottom level, both powers running over $\rho^0, \dots, \rho^{c(B)}$.

**Theorem 21.16** (What crosses a gluing)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a family on $\mathcal{E}$ with real data and masses $\mathbf{m}$ on its balance wall, $B$ a bunch with room $R$, and fix a level factorization of every table, with the bunch letters $V_{j,k,c}$, the room letters $L_{j,k,c}$ and the starred letters $V^*_{j,k,c}$, $L^*_{j,k,c}$. Let $\mathbf{V}(u) = \big(V_{j,k,c}(u), V^*_{j,k,c}(u)\big)_{j,k,c}$ be the* letter vector *of $u \in \Delta_B$. Then, in $[0,\infty]$,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\Delta_R} v^{\mathbf{m}|_R - \mathbf{1}} \int_{\Delta_B} u^{\mathbf{m}|_B - \mathbf{1}} \int_0^\infty \rho^{\kappa_B - 1} \prod_j \Big( \sum_{k,c} \rho^{k-d_j(B)} V_{j,k,c}(u)\, L_{j,k,c}(v) \Big)^{-s_j} \times \Big( \sum_{k,c} \rho^{r_j(B)-k} V^*_{j,k,c}(u)\, L^*_{j,k,c}(v) \Big)^{-t_j} d\rho\, d\omega_B(u)\, d\omega_R(v),$$

*$\kappa_B$ the wall of $B$, and the integrand, apart from the measure $u^{\mathbf{m}|_B-\mathbf{1}} d\omega_B(u)$, depends on $u$ only through $\mathbf{V}(u)$. So the period sees the bunch only through its wall $\kappa_B$, its widths $c_j(B)$ and the image $\mathbf{V}_*\big(u^{\mathbf{m}|_B-\mathbf{1}} d\omega_B(u)\big)$, the joint law of its bunch letters, unnormalized: two families that agree on the room, with the same room letters, and whose bunches have equal walls, widths and laws of their letter vectors, have equal periods, whatever the numbers of marks of the bunches; the bunches are interchangeable in every room. The same holds for the interest $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ at any real masses, with $\alpha_R \in \mathbb{R}^R_{>0}$ in place of $v \in \Delta_R$ and the factor $e^{-\rho - |\alpha_R|}$ in the integrand.*

At width one with factoring weights the children factorization has one set at each level, and the letter vector is $(K_{\mathcal{B}/R}, K_{\mathcal{B}|B}; \Psi_{\mathcal{B}|B}, \Psi_{\mathcal{B}/R})$, each letter from its bottom level: Theorem 19.8, which keeps the scale as the hinge $g$, keeps of it the cell kernel (Definition 22.2) and the two hinge values $\gamma_B^K$, $\gamma_B^\Psi$. The theorem holds whatever the weights; what the weights decide is how many bunch letters there are.

**Theorem 21.17** (Multivectors bound what crosses)**.** *Let the table carry a multivector $\boldsymbol{\omega}$ of any tensor rank (Definition 7.20), the marks of $B$ ordered before those of $R$, so that $\boldsymbol{\omega}_{p\sqcup X}$ is the coefficient of $\mathbf{e}_p \wedge \mathbf{e}_X$, $\mathbf{e}_p$ the wedge of the unit vectors of the marks of $p$ in their order. At the level $k$ let the* interface rank $\operatorname{ir}_k(\boldsymbol{\omega})$ *of $\boldsymbol{\omega}$ at $B$ be the rank of the matrix $(\boldsymbol{\omega}_{p\sqcup X})_{p,X}$, and write $\boldsymbol{\omega}_{p\sqcup X} = \sum_{i=1}^{\operatorname{ir}_k(\boldsymbol{\omega})} f_i(p)\, f'_i(X)$.*

*(i)* $\operatorname{rank}\Theta_B^{(k)} \le \operatorname{ir}_k(\boldsymbol{\omega})^2$, *and* $\operatorname{rank}\Theta_B^{(k)} \le \frac12 \operatorname{ir}_k(\boldsymbol{\omega})\big(\operatorname{ir}_k(\boldsymbol{\omega})+1\big)$ *when* $\boldsymbol{\omega}$ *is real. With the positive semidefinite Hermitian matrices*

$$\mathbf{C}_k(\alpha_B) = \sum_p \alpha_B^p\, f(p) f(p)^*, \qquad \mathbf{C}'_k(\alpha_R) = \sum_X \alpha_R^X\, f'(X) f'(X)^*,$$

*with* $f(p) = \big(f_i(p)\big)_i$ *and* $f'(X) = \big(f'_i(X)\big)_i$, *the level letter is* $K_B^{(k)}(\alpha) = \operatorname{tr}\big(\mathbf{C}_k(\alpha_B)\,\overline{\mathbf{C}'_k(\alpha_R)}\big)$, *and the real and imaginary parts of the entries of* $\mathbf{C}_k(\alpha_B)$ *are the bunch letters of a real level factorization with at most* $\operatorname{ir}_k(\boldsymbol{\omega})^2$ *terms.*

*(ii) If* $\boldsymbol{\omega} = \sum_{l=1}^{\operatorname{rk}(\boldsymbol{\omega})} \boldsymbol{\omega}^{(l)}$ *with slices* $A^{(l)}$, *let the* width of the slice $c_l$ *be the dimension of the intersection of the column spaces of the columns* $A_B^{(l)}$ *of* $B$ *and* $A_R^{(l)}$ *of* $R$, *and its* floor $d_l = r - \dim\big($*column space of* $A_R^{(l)}\big)$, *which at tensor rank one are* $c(B)$ *and* $d(B)$. *Then* $\operatorname{ir}_k(\boldsymbol{\omega}) \le \sum_l \binom{c_l}{k-d_l}$.

*(iii) So* $\operatorname{rank}\Theta_B^{(k)} \le \big(\sum_l \binom{c_l}{k-d_l}\big)^2$, *and the interface dimension of* $B$ *is at most* $\sum_k \big(\sum_l \binom{c_l}{k-d_l}\big)^2$, *a bound in the widths and floors of the slices alone, whatever the numbers of marks of* $B$ *and* $R$; *for a real multivector each square may be replaced by* $\frac12 i_k(i_k+1)$, $i_k = \sum_l \binom{c_l}{k-d_l}$. *For a determinantal table, of tensor rank one with* $c_1 = c(B)$ *and* $d_1 = d(B)$, $\operatorname{ir}_k(\boldsymbol{\omega}) \le \binom{c(B)}{k-d(B)}$, $\operatorname{rank}\Theta_B^{(k)} \le \binom{c(B)}{k-d(B)}^2$ *and the interface dimension is at most* $\binom{2c(B)}{c(B)}$; *at width one each level has one bunch letter, as Theorem 12.19 says.*

**Theorem 21.18** (The interface of a determinantal table)**.** *Let a table be determinantal over* $\mathbb{F}$ *with slice* $A$, $\theta_T = |\det A_T|^2$, *and let* $B$ *have width* $c(B) = w$, *the minors* $\mathcal{B}|B$ *and* $\mathcal{B}/R$ *on* $B$ *and* $\mathcal{B}|R$ *and* $\mathcal{B}/B$ *on* $R$ *carrying the factoring weights of Theorem 12.19. There are Hermitian* $w \times w$ *matrices* $\mathbf{S}_B(\alpha_B)$ *and* $\mathbf{S}_R(\alpha_R)$, *positive definite on the open orthant, each entry a rational function homogeneous of degree one, with*

$$K(\alpha) = K_{\mathcal{B}/R}(\alpha_B)\, K_{\mathcal{B}/B}(\alpha_R)\, \det\big(\mathbf{S}_B(\alpha_B) + \mathbf{S}_R(\alpha_R)\big), \qquad \det \mathbf{S}_B = \frac{K_{\mathcal{B}|B}}{K_{\mathcal{B}/R}}, \qquad \det \mathbf{S}_R = \frac{K_{\mathcal{B}|R}}{K_{\mathcal{B}/B}},$$

*and* $\Psi(\alpha) = \alpha^{\mathcal{E}} K(\alpha^{-1})$. *Along the scale of* $B$, $K(\rho\alpha_B, \alpha_R) = \rho^{d(B)} K_{\mathcal{B}/R}(\alpha_B) K_{\mathcal{B}/B}(\alpha_R) \det\big(\rho\,\mathbf{S}_B + \mathbf{S}_R\big)$, *so the Skeleton values of* $B$ *(Definition 12.10) are the eigenvalues of* $\mathbf{S}_B^{-1/2}\mathbf{S}_R\mathbf{S}_B^{-1/2}$, *real and positive. At width one* $\mathbf{S}_B = \gamma_B^K$, *the hinge value by* $K$ *(Definition 19.2). So a bunch of width* $w$ *of a determinantal table crosses its gluing as one positive definite* $w \times w$ *matrix and one scalar, and the same read at* $\alpha^{-1}$ *for* $\Psi$.

For a determinantal table a bunch $B$ is thus read by its room through one positive definite matrix, $\binom{c(B)+1}{2}$ real entries when it is symmetric and $c(B)^2$ real parameters when it is Hermitian, and one scalar for each letter, and the bunch letters of Theorem 21.17 are the compound determinants of that matrix. For a multivector of tensor rank $\operatorname{rk}(\boldsymbol{\omega})$ the level letters carry the entries of the matrices $\mathbf{C}_k$ of Theorem 21.17(i), whose size is fixed by the interface ranks. In every case Theorem 21.16 says what crosses, and Theorems 21.17 and 21.18 say how much.

# Chapter 22

# Periods of constructions

A construction (Definition 33.1) hands its period a structure: the period of a substitution is read from the core and the cells, for any family. The core reads each cell through the joint law of its hinge values, which is pooling done at every cell at once; opened on the core's own shares, the cells ride on the core's fibre, each entering through the period of its own family at the exponents the shares assign it; and where every active letter reads a cell alike, the cell's share is pinned; where the active letters read every cell by one hinge value, as at a full lean of one table, the period factors through the core. Chains of cells are the powers of one operator, and the letters of any chain of bunches, and their integer joint moments, are products of finite matrices. A bunch of any width opens on one share per term of its level factorization; in level form, one term per level, one table at a full lean keeps $c(B) - 1$ free shares. The same periods are read through the laws of the cell families, one measure on $(0,\infty)^{2N-1}$ for each (Chapter 23), and through the periods at the full leans of the nodes (Chapters 25 and 26).

## 22.1 The core reads its cells

**Definition 22.1** (Twists)**.** A *twist* of a family $F$ on $\mathcal{E}$ assigns to every table $j$ two vectors $\mathbf{x}_j^K, \mathbf{x}_j^\Psi \in (0,\infty)^{\mathcal{E}}$. The *twisted family* $F[\mathbf{x}]$, $\mathbf{x} = (\mathbf{x}_j^K, \mathbf{x}_j^\Psi)_j$, has the bases and the exponents of $F$, the basis weights $\theta^K_{j,T}(\mathbf{x}_j^K)^T$ and the cobasis weights $\theta^\Psi_{j,T}(\mathbf{x}_j^\Psi)^{\mathcal{E}\setminus T}$: its letters are $K_j(\mathbf{x}_j^K \circ \alpha)$ and $\Psi_j(\mathbf{x}_j^\Psi \circ \alpha)$, with $\mathbf{x} \circ \alpha = (x_e \alpha_e)_e$. A twist *at a mark* $g$ has all its coordinates 1 off $g$ and is written by its values $\mathbf{x} = (x_j^K, x_j^\Psi)_j$ at $g$; then $F[\mathbf{x}@g]$ multiplies, in table $j$, the weights in $K_j$ of the bases through $g$ by $x_j^K$ and the weights in $\Psi_j$ of the bases avoiding $g$ by $x_j^\Psi$. A twist of one table $\mathcal{H}$ in its cobasis letter alone, $\mathbf{x}^K = \mathbf{1}$, is written $\mathcal{H}[\mathbf{x}]$ with $\mathbf{x} = \mathbf{x}^\Psi$: $K_{\mathcal{H}[\mathbf{x}]} = K_{\mathcal{H}}$ and $\Psi_{\mathcal{H}[\mathbf{x}]}(\alpha) = \Psi_{\mathcal{H}}(\mathbf{x} \circ \alpha)$. On the balance wall the period obeys the *torus relation*: for every twist $\mathbf{x}$, every $\boldsymbol{\lambda} \in (0,\infty)^{\mathcal{E}}$ and all constants $c_j, c_j^* > 0$,

$$P_{F[(c_j \boldsymbol{\lambda} \circ \mathbf{x}_j^K,\ c_j^* \boldsymbol{\lambda} \circ \mathbf{x}_j^\Psi)_j]}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \boldsymbol{\lambda}^{-\mathbf{m}} \prod_j c_j^{-r_j s_j}\, c_j^{*\,-r_j^* t_j}\, P_{F[\mathbf{x}]}(\mathbf{m}; \mathbf{s}, \mathbf{t}),$$

both sides converging together, in $[0,\infty]$ for real data; at the scalar $\boldsymbol{\lambda} = \mu\mathbf{1}$ with $c_j = c_j^* = \mu^{-1}$ the twist is unchanged and the factor is 1, by balance. So a twist at a mark $g$, $\boldsymbol{\lambda}$ equal to 1 off $g$, has $P_{F[\lambda \mathbf{x}@g]}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \lambda^{-m_g} P_{F[\mathbf{x}@g]}(\mathbf{m}; \mathbf{s}, \mathbf{t})$ and is read with $x_1^K = 1$, a point of $(0,\infty)^{2N-1}$; for one table it is one number, $x = x_1^\Psi$, and only the cobasis letter moves. A twist of every mark of one letter by one factor multiplies the period by that factor to the power $-r_j s_j$ or $-r_j^* t_j$; for one table in its cobasis letter, $P_{\mathcal{H}[\mu\mathbf{x}]}(\mathbf{m}; s, t) = \mu^{-r^* t} P_{\mathcal{H}[\mathbf{x}]}(\mathbf{m}; s, t)$.

**Definition 22.2** (Cell kernels; the core family read at a shape)**.** Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a substituted family (Definition 21.1), table $j$ being $\mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{qj}\rangle$ with the cells $\mathcal{A}_{ij}$ on $B_i \sqcup \{g_i\}$. The *cell kernel* of $B_i$ is the measure

$$u^{\mathbf{m}|_{B_i}-\mathbf{1}} \prod_j K_{\mathcal{A}_{ij}/g_i}(u)^{-s_j} \Psi_{\mathcal{A}_{ij}\setminus g_i}(u)^{-t_j}\, d\omega_{B_i}(u) \qquad \text{on } \Delta_{B_i}.$$

The *core family* is $F_{\mathcal{H}} = ((\mathcal{H}_j, \theta^{\mathcal{H}_j}); s_j, t_j)_j$ on the marks $h_1, \dots, h_q$. For $u = (u_i)_i \in \prod_i \Delta_{B_i}$ the *core family read at* $u$ is its twist $F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]$ (Definition 22.1) by the hinge values, $\boldsymbol{\gamma}(u)$ putting $\gamma^K_{\mathcal{A}_{ij}}(u_i)$ in $K_j$ and $\gamma^\Psi_{\mathcal{A}_{ij}}(u_i)$ in $\Psi_j$ at the mark $h_i$: table $j$ has the basis weights $\theta_T^{\mathcal{H}_j} \prod_{h_i \in T} \gamma^K_{\mathcal{A}_{ij}}(u_i)$ and the cobasis weights $\theta_T^{\mathcal{H}_j} \prod_{h_i \notin T} \gamma^\Psi_{\mathcal{A}_{ij}}(u_i)$, and the letters $K_{\mathcal{H}_j}\big((\gamma^K_{\mathcal{A}_{ij}}(u_i)\,\alpha_{h_i})_i\big)$ and $\Psi_{\mathcal{H}_j}\big((\gamma^\Psi_{\mathcal{A}_{ij}}(u_i)\,\alpha_{h_i})_i\big)$.

**Theorem 22.3** (The core reads its cells)**.** *For a substituted family $F$ the core family read at $u$ has the imbalance of $F$, and*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\prod_i \Delta_{B_i}} \prod_i \Big( u_i^{\mathbf{m}|_{B_i}-\mathbf{1}} \prod_j K_{\mathcal{A}_{ij}/g_i}(u_i)^{-s_j} \Psi_{\mathcal{A}_{ij}\setminus g_i}(u_i)^{-t_j}\, d\omega_{B_i}(u_i)\Big) \\ \times P_{F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]}\big((\kappa_{B_1}, \dots, \kappa_{B_q});\mathbf{s},\mathbf{t}\big),$$

*both sides converging together; the same holds with $J$ for $P$, and on the balance wall. So the period of a substitution depends on each cell only through its wall $\kappa_{B_i}$ and the joint law, under its cell kernel, of its hinge values $\big(\gamma^K_{\mathcal{A}_{ij}}, \gamma^\Psi_{\mathcal{A}_{ij}}\big)_j$: two cells with equal walls and equal laws are interchangeable in every core. For one cell this is Theorem 19.8.*

Pooling, done at every cell at once, leaves the core: each cell is integrated over its shape with its kernel and collapses to one mark of the core, which keeps the mass $\kappa_{B_i}$ and is read through the cell's hinge values as weights. The cells meet only in the core.

**Theorem 22.4** (The cells ride on the core's fibre)**.** *Let $F$ be a substituted family with complex data, every exponent* 0 *or of positive real part. Open $K_{\mathcal{H}_j}$ with one share $a_{j,T}$ and $\Psi_{\mathcal{H}_j}$ with one share $b_{j,T}$ per basis $T$ of $\mathcal{H}_j$ (in this statement $|\mathcal{H}_j|$ is the number of bases of $\mathcal{H}_j$), $\sum_T a_{j,T} = s_j$ and $\sum_T b_{j,T} = t_j$, a letter with exponent zero opened by nothing, and let the* loads *of the mark $h_i$ be $a_j(h_i) = \sum_{T \ni h_i} a_{j,T}$ and $b_j(h_i) = \sum_{T \not\ni h_i} b_{j,T}$. The* load family $F_i(\mathbf{a},\mathbf{b})$ *is the family on $B_i$ with, for every $j$, the table $\mathcal{A}_{ij} \setminus g_i$ at the exponents $\big(a_j(h_i),\, t_j - b_j(h_i)\big)$ and the table $\mathcal{A}_{ij}/g_i$ at $\big(s_j - a_j(h_i),\, b_j(h_i)\big)$; its imbalance is $\kappa_{B_i} - \sum_j \big(a_j(h_i) + b_j(h_i)\big)$. In this statement write $\mathbf{s}^{(i)} = \big((a_j(h_i))_j, (s_j - a_j(h_i))_j\big)$ and $\mathbf{t}^{(i)} = \big((t_j - b_j(h_i))_j, (b_j(h_i))_j\big)$ for its exponents, the tables $\mathcal{A}_{ij} \setminus g_i$ listed first. On every contour of positive real parts of the shares at which all load families converge absolutely,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j,\ |\mathbf{b}_j|=t_j} \prod_j \Gamma_{|\mathcal{H}_j|}\begin{bmatrix}\mathbf{a}_j \\ s_j\end{bmatrix} \Gamma_{|\mathcal{H}_j|}\begin{bmatrix}\mathbf{b}_j \\ t_j\end{bmatrix} \prod_{j,T} \big(\theta_T^{\mathcal{H}_j}\big)^{-a_{j,T}-b_{j,T}} \\ \times \prod_i J_{F_i(\mathbf{a},\mathbf{b})}\big(\mathbf{m}|_{B_i};\mathbf{s}^{(i)},\mathbf{t}^{(i)}\big),$$

*with $J_{F_i(\mathbf{a},\mathbf{b})}\big(\mathbf{m}|_{B_i};\mathbf{s}^{(i)},\mathbf{t}^{(i)}\big) = \Gamma\big(\kappa_{B_i} - \sum_j (a_j(h_i) + b_j(h_i))\big) P_{F_i(\mathbf{a},\mathbf{b})}\big(\mathbf{m}|_{B_i};\mathbf{s}^{(i)},\mathbf{t}^{(i)}\big)$. The integrand without the periods of the load families is that of the interest $J_{F_{\mathcal{H}}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)$ of the core family (Definition 22.2) opened with one share per basis (Corollary 4.4).*

The cells ride on the core's share space: the core's letters are opened over its bases, every $K$-share of a basis loads the marks it occupies and every $\Psi$-share the marks it leaves out, and each cell enters through the period of its own two hinge minors at the exponents that the loads of its mark assign them.

## 22.2 Pinned cells

**Theorem 22.5** (A cell read alike is pinned)**.** *Let $F$ be a substituted family at every cell of which every active letter reads alike: for every $i$ there are a function $\gamma_i$ on $\Delta_{B_i}$ and constants $c_{ij}, c^*_{ij} > 0$ with $\gamma^K_{\mathcal{A}_{ij}} = c_{ij}\gamma_i$ whenever $s_j \neq 0$ and $\gamma^\Psi_{\mathcal{A}_{ij}} = c^*_{ij}\gamma_i$ whenever $t_j \neq 0$, and $c_{ij} = 1$ when $s_j = 0$, $c^*_{ij} = 1$ when $t_j = 0$. Let $F_\mathcal{H}[\mathbf{c}]$ be the core family twisted by these constants at the marks $h_i$, $c_{ij}$ in $K_j$ and $c^*_{ij}$ in $\Psi_j$ (Definitions 22.1 and 22.2). Then on the balance wall*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{F_\mathcal{H}[\mathbf{c}]}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i \int_{\Delta_{B_i}} \gamma_i^{-\kappa_{B_i}}\, u^{\mathbf{m}|_{B_i}-\mathbf{1}} \prod_j K^{-s_j}_{\mathcal{A}_{ij}/g_i}\Psi^{-t_j}_{\mathcal{A}_{ij}\setminus g_i}\, d\omega_{B_i}(u),$$

*the core at the walls of the cells times the* pinned shape integrals *of the cells. In general the free shares at the cell $i$ are one fewer than the distinct hinge values of $B_i$ by the active letters, as at a bunch of width one (Theorem 19.13).*

**Corollary 22.6** (Read by one hinge value, the period factors through the core)**.** *Let $F$ be a substituted family with real data on its balance wall (Definition 21.1), table $j$ being $\mathcal{H}_j\langle\mathcal{A}_{1j},\dots,\mathcal{A}_{qj}\rangle$, at every cell of which the active letters read the cell by one hinge value: for every $i$ there is a function $\gamma_i$ on $\Delta_{B_i}$ with $\gamma^K_{\mathcal{A}_{ij}} = \gamma_i$ whenever $s_j \neq 0$ and $\gamma^\Psi_{\mathcal{A}_{ij}} = \gamma_i$ whenever $t_j \neq 0$, as at a mark left in place, whose two hinge values are both $\alpha_h$. Let $\sigma$ be the total strength (Definition 8.1) and $F_i = ((\mathcal{A}_{ij}, \theta^{\mathcal{A}_{ij}}); s_j, t_j)_j$ the cell family of the cells at $B_i$, on $B_i \sqcup \{g_i\}$. Then, in $[0,\infty]$,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{F_\mathcal{H}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i \int_{\Delta_{B_i}} \gamma_i^{-\kappa_{B_i}}\, u^{\mathbf{m}|_{B_i}-\mathbf{1}} \prod_j K^{-s_j}_{\mathcal{A}_{ij}/g_i}\Psi^{-t_j}_{\mathcal{A}_{ij}\setminus g_i}\, d\omega_{B_i}(u),$$

*and at a cell with $0 < \kappa_{B_i} < 2\sigma$ the family $F_i$ at the masses $(\mathbf{m}|_{B_i}, 2\sigma - \kappa_{B_i})$ is on its balance wall and its pinned shape integral is $P_{F_i}\big((\mathbf{m}|_{B_i}, 2\sigma-\kappa_{B_i});\mathbf{s},\mathbf{t}\big)/\Gamma_1^2\begin{bmatrix}\kappa_{B_i}, & 2\sigma-\kappa_{B_i}\\ & 2\sigma\end{bmatrix}$. If $F$ has every proper wall positive, then $0 < \kappa_{B_i} < 2\sigma$ for every cell, and*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})\prod_i \Gamma_1^2\begin{bmatrix}\kappa_{B_i}, & 2\sigma-\kappa_{B_i}\\ & 2\sigma\end{bmatrix} = P_{F_\mathcal{H}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i P_{F_i}\big((\mathbf{m}|_{B_i}, 2\sigma-\kappa_{B_i});\mathbf{s},\mathbf{t}\big),$$

*every factor on its balance wall with every proper wall positive, the hinge $g_i$ at the mass $2\sigma - \kappa_{B_i}$ and the mark $h_i$ of the core family at $\kappa_{B_i}$. For one cell this is Theorem 19.10, and the first display is the shape-integral form of Proposition 19.12.*

**Corollary 22.7** (The pinned shape integral is the period of a load family)**.** *In Corollary 22.6, let the loads $a_{ij}, b_{ij}$ of the cell $i$ be real, with $a_{ij} = 0$ when $s_j = 0$, $b_{ij} = 0$ when $t_j = 0$ and $\sum_j (a_{ij} + b_{ij}) = \kappa_{B_i}$. The load family $F_i(\mathbf{a},\mathbf{b})$ of Theorem 22.4 at the loads $a_j(h_i) = a_{ij}$ and $b_j(h_i) = b_{ij}$, with the tables $\mathcal{A}_{ij} \setminus g_i$ at $(a_{ij}, t_j - b_{ij})$ and $\mathcal{A}_{ij}/g_i$ at $(s_j - a_{ij}, b_{ij})$ and the masses $\mathbf{m}|_{B_i}$, has the imbalance zero, and its form does not depend on the loads: in $[0,\infty]$ its period is the pinned shape integral of the cell,*

$$P_{F_i(\mathbf{a},\mathbf{b})}\big(\mathbf{m}|_{B_i};\mathbf{s}^{(i)},\mathbf{t}^{(i)}\big) = \int_{\Delta_{B_i}} \gamma_i^{-\kappa_{B_i}}\, u^{\mathbf{m}|_{B_i}-\mathbf{1}} \prod_j K^{-s_j}_{\mathcal{A}_{ij}/g_i}\Psi^{-t_j}_{\mathcal{A}_{ij}\setminus g_i}\, d\omega_{B_i}(u),$$

*and* $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{F_{\mathcal{H}}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i P_{F_i(\mathbf{a},\mathbf{b})}\big(\mathbf{m}|_{B_i};\mathbf{s}^{(i)},\mathbf{t}^{(i)}\big)$*. If every proper wall of* $F$ *is positive and* $s_j, t_j \geq 0$*, there are such loads with* $0 \leq a_{ij} \leq s_j$ *and* $0 \leq b_{ij} \leq t_j$*, integers when the data are; the load family then has nonnegative exponents, and every proper wall of it is positive when* $P_F$ *is finite.*

## 22.3 Chains and transfer

**Theorem 22.8** (A chain is the power of one operator)**.** *Let* $F = ((\mathcal{A}_j,\theta_j);s_j,t_j)_{j=1}^N$ *be a family on* $\mathcal{E}$ *with two marks* $g_{\text{in}} \neq g_{\text{out}}$*, each neither a loop nor a coloop in every table, real exponents, masses on the balance wall with* $m_{g_{\text{in}}} + m_{g_{\text{out}}} = \sum_j (s_j + t_j)$*, and let* $F^{\triangleright k}$ *be* $k$ *copies of* $F$ *glued table by table along marks,* $g_{\text{out}}^{(i)}$ *with* $g_{\text{in}}^{(i+1)}$ *(Lemma 21.12). For* $\mathbf{x} = (x_j^K, x_j^\Psi)_j \in (0,\infty)^{2N}$ *let* $F^{\triangleright k}[\mathbf{x}@g_{\text{in}}^{(1)}]$ *be* $F^{\triangleright k}$ *twisted at* $g_{\text{in}}^{(1)}$ *(Definition 22.1), and in this statement write* $P_k(\mathbf{x})$ *for its period at the masses of the copies and* $\mathcal{A}_j^{\mathbf{x}}$ *for table* $j$ *of* $F[\mathbf{x}@g_{\text{in}}]$*, read as a cell with the hinge* $g_{\text{out}}$*. With* $B = \mathcal{E}\setminus\{g_{\text{out}}\}$ *define on nonnegative functions* $H$ *of the twist*

$$(\mathcal{Q}H)(\mathbf{x}) = \int_{\Delta_B} H\Big(\big(\gamma^K_{\mathcal{A}_j^{\mathbf{x}}}(u),\gamma^\Psi_{\mathcal{A}_j^{\mathbf{x}}}(u)\big)_j\Big)\, u^{\mathbf{m}|_B - \mathbf{1}} \prod_j K_{\mathcal{A}_j^{\mathbf{x}}/g_{\text{out}}}(u)^{-s_j}\,\Psi_{\mathcal{A}_j^{\mathbf{x}}\setminus g_{\text{out}}}(u)^{-t_j}\, d\omega_B(u),$$

$$P_0(\mathbf{x}) = \int_0^1 v^{m_{g_{\text{in}}}-1}(1-v)^{m_{g_{\text{out}}}-1}\prod_j \big(1-v+x_j^K v\big)^{-s_j}\big(1-v+x_j^\Psi v\big)^{-t_j}\, dv.$$

*Then* $P_k = \mathcal{Q}P_{k-1}$*, and in* $[0,\infty]$

$$P_{F^{\triangleright k}}(\mathbf{m};\mathbf{s},\mathbf{t}) = (\mathcal{Q}^k P_0)(\mathbf{1}).$$

*The operator acts on functions of the* $2N$ *twists, of which one is removed by the scale of the hinge. Where the twist is pinned, the active letters reading the cell by one hinge value,* $\gamma^K_{\mathcal{A}_j}$ *for* $s_j \neq 0$ *and* $\gamma^\Psi_{\mathcal{A}_j}$ *for* $t_j \neq 0$ *all equal, as at a full lean of one table, every* $P_k$ *is a multiple of one power of the twist along the diagonal* $\mathbf{x} = \lambda\mathbf{1}$*,* $\mathcal{Q}$ *multiplies it by* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix} m_{g_{\text{in}}}, & m_{g_{\text{out}}} \\ m_{g_{\text{in}}}+m_{g_{\text{out}}} & \end{bmatrix}$*, and*

$$P_{F^{\triangleright k}}(\mathbf{m};\mathbf{s},\mathbf{t}) = \frac{P_F(\mathbf{m};\mathbf{s},\mathbf{t})^k}{\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix} m_{g_{\text{in}}}, & m_{g_{\text{out}}} \\ m_{g_{\text{in}}}+m_{g_{\text{out}}} & \end{bmatrix}^{k-1}}.$$

*Hinge values that are only proportional, with constants other than one, still pin each cell (Theorem 22.5) but carry the twist off the diagonal.*

A chain is pooled one copy at a time: the first copy, read as the cell of its bunch, collapses to its hinge and twists the next copy by its hinge values, so the period of the chain is one operator on functions of the twist applied once per copy, and the two-mark table left at the end supplies the starting function. Where the hinge values pin the twist the operator is a multiplication, and the chain costs a power.

**Theorem 22.9** (Letters by transfer)**.** *Let* $(\mathcal{B},\theta)$ *be a table with any weights and* $\varnothing = S_0 \subset S_1 \subset \cdots \subset S_L = \mathcal{E}$ *a chain of bunches, with the steps* $B_{l-1} = S_l \setminus S_{l-1}$*. Group the patterns of each* $S_l$ *into its stations (Definition 21.15), the children being tables on* $\mathcal{E}\setminus S_l$ *and the child at* $p$ *being* $w_p$ *times the child of its station* $c$*. For a station* $c$ *of* $S_{l-1}$ *and a station* $c'$ *of* $S_l$ *let*

$$\mathbf{M}_l^K(c,c') = \sum_{p'} w(c,p')\,\alpha^{p'}_{B_{l-1}}, \qquad \mathbf{M}_l^\Psi(c,c') = \sum_{p'} w(c,p')\,\alpha^{B_{l-1}\setminus p'}_{B_{l-1}},$$

*the sums over the patterns $p'$ of $B_{l-1}$ in the child of the station $c$ whose child is $w(c,p')$ times the child of the station $c'$. Then, with $\mathbf{e}$ the unit vector of the station of the empty pattern of $S_0$ and $\boldsymbol{\theta}_\varnothing$ the vector of the weights of the one basis $\varnothing$ of the children of the stations of $S_L$,*

$$K = \mathbf{e}^{\mathsf{T}}\mathbf{M}_1^K\mathbf{M}_2^K\cdots\mathbf{M}_L^K\,\boldsymbol{\theta}_\varnothing, \qquad \Psi = \mathbf{e}^{\mathsf{T}}\mathbf{M}_1^\Psi\cdots\mathbf{M}_L^\Psi\,\boldsymbol{\theta}_\varnothing.$$

*The number of states at $S_l$ is the number of its stations, at least the interface dimension of $S_l$ (Definition 21.15), and the states can be reduced to that many linearly independent ones; at a bunch $S_l$ of width one with factoring weights it is two. For a table carrying a multivector $\boldsymbol{\omega}$ of any tensor rank (Definition 7.20), with the widths $c_i$ and the floors $d_i$ of its slices at $S_l$ (Theorem 21.17(ii)), the interface dimension of $S_l$ is at most $\sum_k\big(\sum_i\binom{c_i}{k-d_i}\big)^2$, each square $q^2$ replaced by $\frac{1}{2}q(q+1)$ when $\boldsymbol{\omega}$ is real; for a determinantal table it is at most $\binom{2c(S_l)}{c(S_l)}$ (Theorem 21.17).*

**Theorem 22.10** (Integer joint moments by fixed matrices)**.** *Let the letters of a family $F = ((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ be products of matrix functions of the variables of the steps $B_0,\dots,B_{L-1}$ of one chain of bunches, $K_j = \mathbf{e}_j^{\mathsf{T}}\mathbf{M}_{j,0}^K(\alpha_{B_0})\cdots\mathbf{M}_{j,L-1}^K(\alpha_{B_{L-1}})\boldsymbol{\theta}_{j,\varnothing}$ and $\Psi_j = \mathbf{e}_j^{\mathsf{T}}\mathbf{M}_{j,0}^\Psi(\alpha_{B_0})\cdots\mathbf{M}_{j,L-1}^\Psi(\alpha_{B_{L-1}})\boldsymbol{\theta}_{j,\varnothing}$ with constant vectors $\mathbf{e}_j$ and $\boldsymbol{\theta}_{j,\varnothing}$, as Theorem 22.9 gives for every table along any chain of bunches; and let the steps carry independent positive measures $\phi_0,\dots,\phi_{L-1}$ with the required moments finite. For integers $a_j,b_j\ge 0$ put*

$$\mathbf{A}_l = \int\bigotimes_j\Big(\mathbf{M}_{j,l}^K(\alpha)^{\otimes a_j}\otimes\mathbf{M}_{j,l}^\Psi(\alpha)^{\otimes b_j}\Big)\,\phi_l(d\alpha),$$

*one $\alpha$ in all the factors. Then*

$$\int\prod_j K_j^{a_j}\Psi_j^{b_j}\prod_{l<L}\phi_l(d\alpha_{B_l}) = \Big(\bigotimes_j\mathbf{e}_j^{\otimes(a_j+b_j)}\Big)^{\mathsf{T}}\mathbf{A}_0\mathbf{A}_1\cdots\mathbf{A}_{L-1}\Big(\bigotimes_j\boldsymbol{\theta}_{j,\varnothing}^{\otimes(a_j+b_j)}\Big).$$

*For a chain of identical steps, with one pair of matrix functions $\mathbf{M}_j^K,\mathbf{M}_j^\Psi$ for each table, the states of all the $S_l$ indexed by one set, $\boldsymbol{\theta}_{j,\varnothing}$ vanishing at the states that do not occur at $S_L$, and one measure $\phi$, every $\mathbf{A}_l$ is one matrix $\mathbf{A}$: the moment sequence along $L$ is the power of one matrix, of size $\prod_j(\text{number of states of table } j)^{a_j+b_j}$, and satisfies a linear recurrence with constant coefficients of that order. With independent laws $\mathrm{Gamma}_{m_x}$ on the marks (Definition 1.1(vii)) the moment $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}\big[\prod_j K_j(U)^{a_j}\Psi_j(U)^{b_j}\big]$ is that integral divided by the rising factorial $(m_{\mathcal{E}})_{\sum_j(a_jr_j+b_jr_j^*)}$; with the law $\mathrm{Dir}_{\mathbf{m}|_{B_l}}$ on each step the entries are the Dirichlet moments $\prod_{x\in B_l}(m_x)_{k_x}/(m_{B_l})_{|\mathbf{k}|}$ of the monomials $\alpha_{B_l}^{\mathbf{k}}$ (Theorem 3.11).*

## 22.4 A bunch of any width

**Theorem 22.11** (Gluing across a bunch of any width)**.** *Let $F$ be a family with real data on its balance wall with every proper wall positive, $B$ a bunch of any width with room $R$, and fix level factorizations with nonnegative terms (Definition 21.15), the children factorization for instance, with the bunch letters $V_{j,k,c}$ and the room letters $L_{j,k,c}$; drop the terms whose bunch letter or room letter vanishes identically, and let $C_j$ be the set of the terms $(k,c)$ left for table $j$. Open every active letter with one share per term, $a_{j,k,c}$ for the term $V_{j,k,c}L_{j,k,c}$ of $K_j$ with $\sum_{k,c}a_{j,k,c}=s_j$ and $b_{j,k,c}$ for the term $V_{j,k,c}^*L_{j,k,c}^*$ of $\Psi_j$ with $\sum_{k,c}b_{j,k,c}=t_j$, pinned by the scale of $B$,*

$$\sum_{j,k,c}\Big(\big(k-d_j(B)\big)a_{j,k,c}+\big(r_j(B)-k\big)b_{j,k,c}\Big)=\kappa_B.$$

*On every contour of positive real parts at which the two integrals below converge,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{|\mathbf{a}_j|=s_j,\ |\mathbf{b}_j|=t_j,\ pin} \prod_j \Gamma_{|C_j|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|C_j|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}$$
$$\times \int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}} \prod_{j,k,c} V_{j,k,c}^{-a_{j,k,c}} {V^*_{j,k,c}}^{-b_{j,k,c}}\, d\omega_B(u) \int_{\Delta_R} v^{\mathbf{m}|_R-\mathbf{1}} \prod_{j,k,c} L_{j,k,c}^{-a_{j,k,c}} {L^*_{j,k,c}}^{-b_{j,k,c}}\, d\omega_R(v),$$

*the two inner integrals projective integrals of degree zero, the periods of the* bunch word *and the* room word *at the shares. Their letters are the bunch and room letters of the level factorization, which need not be the letters of tables, so their convergence is that of the integrals themselves. The free shares are the terms less the active letters less one. For one table at a full lean $(0,t)$ in level form, one station per level, with the share $b_k$ and the bunch and room letters $V_k^*$, $L_k^*$ of the station of the level $k$, $d(B)\le k\le r(B)$, the pins are $\sum_k b_k = t$ and $\sum_k (r(B)-k)\,b_k = \kappa_B$, the free shares are the $c(B)-1$ shares of the levels strictly between $d(B)$ and $r(B)$, and on a contour of such shares*

$$P_F(\mathbf{m};0,t) = \frac{1}{(2\pi i)^{c(B)-1}} \int \frac{\prod_{k=d(B)}^{r(B)}\Gamma(b_k)}{c(B)\,\Gamma(t)} \int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}} \prod_k V_k^{*-b_k}\, d\omega_B(u)$$
$$\times \int_{\Delta_R} v^{\mathbf{m}|_R-\mathbf{1}} \prod_k L_k^{*-b_k}\, d\omega_R(v) \prod_{d(B)<k<r(B)} db_k,$$

*with $b_{d(B)} = \frac{1}{c(B)}\big(\kappa_B - \sum_{d(B)<k<r(B)}(r(B)-k)\,b_k\big)$ and $b_{r(B)} = t - b_{d(B)} - \sum_{d(B)<k<r(B)} b_k$. At $c(B)=2$, with $b_{d(B)+1} = z$, the shares are $\frac{\kappa_B - z}{2}$, $z$ and $\frac{2t-\kappa_B-z}{2}$; at $c(B)=1$ no share is free, and the period is $\Gamma_{\substack{2\\1}}\begin{bmatrix}\kappa_B,\ t-\kappa_B\\ t\end{bmatrix}$ times the two inner periods (Corollary 19.14). At width one with factoring weights, every letter active, the free shares are $2N-1$, and this is Theorem 19.13.*

For one table in level form at a full lean $(0,t)$ the scale can also be integrated before the shapes. Writing $\Psi(\rho u, v) = \rho^{d^*(B)} \sum_{i=0}^{c(B)} \rho^i V^*_{r(B)-i}(u) L^*_{r(B)-i}(v)$ and rescaling $\rho$ so that the two extreme terms become 1 and $x^{c(B)}$, the period is the integral over $\Delta_B \times \Delta_R$ of a form that factors into a bunch part and a room part, times the scale integral $\int_0^\infty x^{\kappa_B-1}\big(1+\sum_{i=1}^{c(B)-1}\zeta_i x^i + x^{c(B)}\big)^{-t}dx$, whose coefficients

$$\zeta_i = \frac{V^*_{r(B)-i}(u)}{V^*_{r(B)}(u)^{1-i/c(B)}\, V^*_{d(B)}(u)^{i/c(B)}} \cdot \frac{L^*_{r(B)-i}(v)}{L^*_{r(B)}(v)^{1-i/c(B)}\, L^*_{d(B)}(v)^{i/c(B)}}$$

are each a bunch part times a room part: the room receives $c(B)-1$ ratios. At $c(B)=1$ the scale integral is $\Gamma_{\substack{2\\1}}\begin{bmatrix}\kappa_B,\ t-\kappa_B\\ t\end{bmatrix}$, which is Theorem 19.10; its Mellin transform in $(\zeta_1,\dots,\zeta_{c(B)-1})$ at $(z_1,\dots,z_{c(B)-1})$ is $\prod_{k=d(B)}^{r(B)}\Gamma(b_k)/\big(c(B)\Gamma(t)\big)$ at the shares $b_{r(B)-i} = z_i$ of Theorem 22.11, its share kernel with the factor $1/c(B)$ of the pin, at $c(B)=2$ the kernel $\Gamma(z)\Gamma\big(\frac{\kappa_B-z}{2}\big)\Gamma\big(\frac{2t-\kappa_B-z}{2}\big)/\big(2\Gamma(t)\big)$. With the factorization taken to be the canonical one of the seam of the bunch, the bunch word is written through one law, which with the wall is exactly what the bunch contributes (Chapter 28).

# Chapter 23

# The law of a cell

A table with a bunch of width one across which its weights factor is two cells glued along a hinge that neither of them shows to the other (Lemma 21.12). What a cell family must carry so that its contribution to every family built on it is fixed is its hinge wall and one measure on $(0,\infty)^{2N-1}$, the law of its ratio vector under its base form, for one cell a measure on the half-line, and nothing less (Theorem 23.11). A cell family enters a period only through that law (Theorem 23.9); the law is read in two ways, by the transform of the cell family and by its dual, and the two transforms expand one function on the two sides of one strip (Proposition 23.8). The step that makes all of this compose is Lemma 23.2: a cell sitting at a mark of any family can be integrated away, leaving the family with that mark twisted by the ratio vector of the cell family; pooling (Theorem 19.8) and Theorem 22.3, in which the core sees each cell as one mark, are its cases. The series closure of any cells, at any positive exponents, is one integral over the shared scales of the product of the transforms of the cells, so identical cells enter as one transform to a power (Theorem 23.3). A core with cells at its markers is read the same way: the law of a substitution is the mixture, over the laws of its cell families, of the laws of the twisted core family (Theorem 23.6); at a series node the transforms multiply and at a parallel node the dual transforms multiply (Theorem 23.14); and the period of a family built along a common hinge tree is read from the leaves up, with one law per edge, computed once for each distinct subtree (Theorem 23.17, Proposition 23.18). The calculus extends to bunches of every width (Chapter 28).

## 23.1 Cells and their laws

**Definition 23.1** (The law of a cell family)**.** A *cell family* is a family $F = ((\mathcal{A}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ on $B \sqcup \{g\}$ whose tables are cells with the one hinge $g$ (Definition 21.1), with real exponents, carrying masses $\mathbf{m}$ on $B$; one cell $\mathcal{A}$ at the exponents $(s,t)$ is the cell family $((\mathcal{A},\theta);s,t)$, and its objects carry the subscript $\mathcal{A}$. Its *hinge wall* is

$$\kappa_F = m_B - \sum_j \big(s_j(r_{\mathcal{A}_j} - 1) + t_j(|B| - r_{\mathcal{A}_j})\big),$$

for one cell the hinge wall $\kappa_{\mathcal{A}}$ of Lemma 21.12. Its *base form* and its *ratio vector* are

$$\phi_F(du) = u^{\mathbf{m-1}} \prod_j K_{\mathcal{A}_j/g}(u)^{-s_j}\, \Psi_{\mathcal{A}_j\setminus g}(u)^{-t_j}\, \gamma^K_{\mathcal{A}_1}(u)^{-\kappa_F}\, d\omega_B(u) \quad \text{on } \Delta_B,$$

$$\boldsymbol{\varrho}_F = \Big(\frac{\gamma^K_{\mathcal{A}_j}}{\gamma^K_{\mathcal{A}_1}},\ \frac{\gamma^\Psi_{\mathcal{A}_j}}{\gamma^K_{\mathcal{A}_1}}\Big)_j :\ \Delta_B \to (0,\infty)^{2N-1},$$

the coordinate $\gamma^K_{\mathcal{A}_1}/\gamma^K_{\mathcal{A}_1} = 1$ omitted, and its *law* is the image $(\boldsymbol{\varrho}_F)_*\phi_F$, a positive measure on $(0,\infty)^{2N-1}$. For $\mathbf{z} = (z_{j,K}, z_{j,\Psi})_j$, also written as the pair $(\mathbf{z}_K, \mathbf{z}_\Psi)$ of $\mathbf{z}_K = (z_{j,K})_j$ and $\mathbf{z}_\Psi = (z_{j,\Psi})_j$, write

$$\boldsymbol{\varrho}_F^{-\mathbf{z}} = \prod_{j\geq 2}\Big(\frac{\gamma^K_{\mathcal{A}_j}}{\gamma^K_{\mathcal{A}_1}}\Big)^{-z_{j,K}} \prod_j \Big(\frac{\gamma^\Psi_{\mathcal{A}_j}}{\gamma^K_{\mathcal{A}_1}}\Big)^{-z_{j,\Psi}},$$

the coordinates of the ratio vector to the powers $-z_{j,K}$ $(j \geq 2)$ and $-z_{j,\Psi}$; the *Mellin transform* of the law, $\int_{\Delta_B} \boldsymbol{\varrho}_F^{-\mathbf{z}}\, d\phi_F$, lies in $[0,\infty]$ for real $\mathbf{z}$ and is defined for complex $\mathbf{z}$ at whose real part it is finite, with modulus at most that value. For one cell the ratio vector is the ratio $\varrho_{\mathcal{A}}$ (Definition 21.1), the base form is

$$\phi_{\mathcal{A}}(du) = u^{\mathbf{m-1}} K_{\mathcal{A}/g}(u)^{\kappa_{\mathcal{A}}-s} K_{\mathcal{A}\setminus g}(u)^{-\kappa_{\mathcal{A}}} \Psi_{\mathcal{A}\setminus g}(u)^{-t}\, d\omega_B(u),$$

the law $(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}$ is a measure on $(0,\infty)$, and, with $z = z_{1,\Psi}$, the Mellin transform is $\int_{\Delta_B} \varrho_{\mathcal{A}}^{-z}\, d\phi_{\mathcal{A}} = \int_0^\infty x^{-z}\, d\big((\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}\big)(x)$. For a cell family with $s_j, t_j > 0$ the *transform*, the *dual transform* and the two *hinge measures* are, for $\mathbf{x}, \mathbf{y} \in (0,\infty)^N$,

$$f_F(\mathbf{x},\mathbf{y}) = \int_{\mathbb{R}^B_{>0}} \alpha^{\mathbf{m-1}} \prod_j K^{-s_j}_{\mathcal{A}_j\setminus g} \Psi^{-t_j}_{\mathcal{A}_j\setminus g} \exp\Big(-\sum_j \frac{x_j}{\gamma^K_{\mathcal{A}_j}} - \sum_j y_j\, \gamma^\Psi_{\mathcal{A}_j}\Big)\, d\alpha,$$

$$\check{f}_F(\mathbf{x},\mathbf{y}) = \int_{\mathbb{R}^B_{>0}} \alpha^{\mathbf{m-1}} \prod_j K^{-s_j}_{\mathcal{A}_j/g} \Psi^{-t_j}_{\mathcal{A}_j/g} \exp\Big(-\sum_j \frac{x_j}{\gamma^\Psi_{\mathcal{A}_j}} - \sum_j y_j\, \gamma^K_{\mathcal{A}_j}\Big)\, d\alpha,$$

with values in $[0,\infty]$, the measure $\mathfrak{r}_F$, the image of $\alpha^{\mathbf{m-1}} \prod_j K^{-s_j}_{\mathcal{A}_j\setminus g} \Psi^{-t_j}_{\mathcal{A}_j\setminus g}\, d\alpha$ under $\alpha \mapsto \big((1/\gamma^K_{\mathcal{A}_j}(\alpha))_j, (\gamma^\Psi_{\mathcal{A}_j}(\alpha))_j\big)$, and the measure $\check{\mathfrak{r}}_F$, the image of $\alpha^{\mathbf{m-1}} \prod_j K^{-s_j}_{\mathcal{A}_j/g} \Psi^{-t_j}_{\mathcal{A}_j/g}\, d\alpha$ under $\alpha \mapsto \big((1/\gamma^\Psi_{\mathcal{A}_j}(\alpha))_j, (\gamma^K_{\mathcal{A}_j}(\alpha))_j\big)$, positive measures on $(0,\infty)^{2N}$. For $c > 0$, $f_F(\mathbf{x},\mathbf{y}) = c^{\kappa_F - |\mathbf{s}|} f_F(\mathbf{x}/c, c\mathbf{y})$ and $\check{f}_F(\mathbf{x},\mathbf{y}) = c^{\kappa_F - |\mathbf{t}|} \check{f}_F(\mathbf{x}/c, c\mathbf{y})$, so each is fixed by its values at $x_1 = 1$. For one cell they are read there: the transform and the dual transform are $f_{\mathcal{A}}(y) = f_F(1,y)$ and $\check{f}_{\mathcal{A}}(y) = \check{f}_F(1,y)$, $y > 0$, and the hinge measures $\mathfrak{r}^\Psi_{\mathcal{A}}$ and $\mathfrak{r}^K_{\mathcal{A}}$ are the images of $e^{-\lambda}\mathfrak{r}_F(d\lambda, d\mu)$ and of $e^{-\lambda}\check{\mathfrak{r}}_F(d\lambda, d\mu)$ under $(\lambda,\mu) \mapsto \mu$, positive measures on $(0,\infty)$.

The base form of a cell is the restriction to $\Delta_B$ of the orthant form $x^{\mathbf{m-1}} K^{\kappa_{\mathcal{A}}-s}_{\mathcal{A}/g} K^{-\kappa_{\mathcal{A}}}_{\mathcal{A}\setminus g} \Psi^{-t}_{\mathcal{A}\setminus g} dx$, homogeneous of degree $m_B + (\kappa_{\mathcal{A}} - s)(r_{\mathcal{A}} - 1) - \kappa_{\mathcal{A}} r_{\mathcal{A}} - t(|B| - r_{\mathcal{A}}) = 0$, so it is read on any section of a positive function of degree one. When $\mathcal{A}$ is the cell of a bunch $B$ of width one in a table, its hinge minors are the restriction $\mathcal{B}|B$ and the contraction $\mathcal{B}/R$, $\kappa_{\mathcal{A}}$ is the wall $\kappa_B$, $\phi_{\mathcal{A}}$ is the base form $\phi_B$, $(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}$ is the law of the bunch, $\int \varrho_{\mathcal{A}}^{-b^d} d\phi_{\mathcal{A}}$ is the continued period of the bunch family at the share $b^d$, and $f_{\mathcal{A}}$ is the bunch transform $f_B$ (Definitions 20.24 and 20.43, Lemma 20.27); the cell kernel of Definition 22.2 is $(\gamma^K_{\mathcal{A}})^{\kappa_{\mathcal{A}}}\phi_{\mathcal{A}}$, and for a cell family $(\gamma^K_{\mathcal{A}_1})^{\kappa_F}\phi_F$. None of these objects mentions a room: a cell carries them by itself, and they are all that a room ever sees of it (Theorem 23.11). The hinge wall is read from the other side of the hinge: with the mass $s + t - \kappa_{\mathcal{A}}$ on $g$ the table $\mathcal{A}$ itself is on its balance wall, and in a two-sum on its balance wall that mass is the hinge wall of the other cell (Lemma 21.12).

## 23.2 Absorbing a cell

**Lemma 23.2** (Absorbing a cell)**.** *Let $F = ((\mathcal{A}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a cell family on $B \sqcup \{g\}$ with masses $\mathbf{m}$ on $B$, its hinge wall $\kappa_F$, base form $\phi_F$ and ratio vector $\boldsymbol{\varrho}_F$, whose coordinates are*

*written $x_{j,K}$ and $x_{j,\Psi}$, $x_{1,K}=1$. Let $Z$ be a finite set disjoint from $B\sqcup\{g\}$, $\mathbf{m}_Z$ masses on $Z$, and $f_1,\dots,f_p$ positive continuous functions on $(0,\infty)^Z\times(0,\infty)$, each assigned to one letter of $F$, with real exponents $a_i$ such that the exponents assigned to each letter add to its exponent, $s_j$ for $K_{\mathcal{A}_j}$ and $t_j$ for $\Psi_{\mathcal{A}_j}$. In this statement write $L_i$, $\gamma_i$ and $x_i$ for the letter of the hinge minor, the hinge value and the coordinate of the letter of $f_i$: $L_i=K_{\mathcal{A}_j/g}$, $\gamma_i=\gamma^K_{\mathcal{A}_j}$ and $x_i=x_{j,K}$ when it is $K_{\mathcal{A}_j}=K_{\mathcal{A}_j/g}\,(\gamma^K_{\mathcal{A}_j}+\alpha_g)$, and $L_i=\Psi_{\mathcal{A}_j\setminus g}$, $\gamma_i=\gamma^\Psi_{\mathcal{A}_j}$ and $x_i=x_{j,\Psi}$ when it is $\Psi_{\mathcal{A}_j}=\Psi_{\mathcal{A}_j\setminus g}\,(\gamma^\Psi_{\mathcal{A}_j}+\alpha_g)$. Let $H\ge 0$ be a Borel function on $(0,\infty)^p$. If the form*

$$\omega=\alpha_B^{\mathbf{m}-\mathbf{1}}\alpha_Z^{\mathbf{m}_Z-\mathbf{1}}\,H\Big(\big(f_i(\alpha_Z,\gamma_i(\alpha_B))\big)_i\Big)\prod_{i=1}^{p}\Big[L_i(\alpha_B)\,f_i\big(\alpha_Z,\gamma_i(\alpha_B)\big)\Big]^{-a_i}d\alpha_B\,d\alpha_Z$$

*is homogeneous of degree zero on $(0,\infty)^{B\sqcup Z}$, then, in $[0,\infty]$,*

$$\int_{\{h=1\}}\omega=\int_{(0,\infty)^{2N-1}}\Big(\int_{\{h'=1\}}\omega_{\mathbf{x}}\Big)\,d\big((\boldsymbol{\varrho}_F)_*\phi_F\big)(\mathbf{x}),$$
$$\omega_{\mathbf{x}}=\alpha_Z^{\mathbf{m}_Z-\mathbf{1}}\alpha_g^{\kappa_F-1}\,H\Big(\big(f_i(\alpha_Z,x_i\alpha_g)\big)_i\Big)\prod_i f_i(\alpha_Z,x_i\alpha_g)^{-a_i}\,d\alpha_Z\,d\alpha_g,$$

*the integrals taken over any sections of positive functions $h$, $h'$ of degree one on the two orthants.*

For one table the ratio vector is the ratio of the cell. The lemma says that whatever the rest of a family does with a cell, provided it sees the cell through the cell's letters at the hinge, it sees it only through the law: the cell collapses to its hinge, which keeps the mass $\kappa_F$, and the letters that see the hinge see it stretched by the coordinates of the ratio vector. Pooling (Theorem 19.8) is the case of one cell family read on the shape before the law is formed; Theorem 22.3, in which the core sees each cell as one mark, is the lemma at every cell of a core, with the hinge values of the cells kept as weights of the core instead of their ratios.

## 23.3 Closures

**Theorem 23.3** (The series closure at positive exponents)**.** *Let every table of $F$ be a series closure, $(\mathcal{B}_j,\theta_j)=\binom{\{h_1,\dots,h_q\}}{q-1}\langle\mathcal{A}_{1j},\dots,\mathcal{A}_{qj}\rangle$ (Corollary 21.7(ii)), with exponents $s_j,t_j>0$, and let $f_{F_i}$ be the transform of the cell family $F_i=((\mathcal{A}_{ij},\theta^{\mathcal{A}_{ij}});s_j,t_j)_j$ of the cells at $B_i$, on $B_i\sqcup\{g_i\}$ (Definition 23.1). On the balance wall, in $[0,\infty]$,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\prod_j\frac{1}{\Gamma(s_j)\Gamma(t_j)}\int\prod_j x_j^{s_j-1}\prod_{j\ge2}y_j^{t_j-1}\;\prod_{i=1}^{q}f_{F_i}\big(\mathbf{x},(1,y_2,\dots,y_N)\big)\;d\mathbf{x}\,dy_2\cdots dy_N,$$

*the integral over $(0,\infty)^N\times(0,\infty)^{N-1}$: the cells communicate only through $2N-1$ shared scales. A cell entering $k$ times enters as its transform to the power $k$: one transform per kind of cell, its multiplicity a power, the order of the cells immaterial; for one table the closure is $\frac{1}{\Gamma(s)\Gamma(t)}\int_0^\infty y^{t-1}\prod_i f_{\mathcal{A}_i}(y)\,dy$. With $|\mathbf{s}|=\sum_j s_j$ and the wall $\kappa_{B_i}$ of the cell $B_i$, its hinge wall $\kappa_{F_i}$ (Theorem 21.10), each transform is one Gramma integrated over the shape of the cell,*

$$f_{F_i}(\mathbf{x},\mathbf{y})=\int_{\Delta_{B_i}}\mathsf{G}_{|\mathbf{s}|-\kappa_{B_i};(1,-1)}\Big(\sum_j\frac{x_j}{\gamma^K_{\mathcal{A}_{ij}}(u)},\;\sum_j y_j\,\gamma^\Psi_{\mathcal{A}_{ij}}(u)\Big)$$
$$\times\,u^{\mathbf{m}|_{B_i}-\mathbf{1}}\prod_j K_{\mathcal{A}_{ij}\setminus g_i}(u)^{-s_j}\,\Psi_{\mathcal{A}_{ij}\setminus g_i}(u)^{-t_j}\,d\omega_{B_i}(u),$$

*against the form of the family of the hinge minors* $F_i \setminus g_i = ((\mathcal{A}_{ij} \setminus g_i, \theta^{\mathcal{A}_{ij}}); s_j, t_j)_j$ *at the masses* $\mathbf{m}|_{B_i}$*. When that family is on its balance wall,* $\kappa_{B_i} = |\mathbf{s}|$*, as at every cell of one table at the centre,* $s = t = \sigma$ *with every mass* $\sigma$*, the index is* $0$*, the Gramma depends on its two coefficients through their product,* $\mathsf{G}_{0;(1,-1)}(a,b) = \mathsf{G}_{0;(1,-1)}(1,ab)$ *(Theorem 2.4(iii)), and, when the family of the hinge minors has every proper wall positive,*

$$f_{F_i}(\mathbf{x},\mathbf{y}) = P_{F_i\setminus g_i}(\mathbf{m}|_{B_i};\mathbf{s},\mathbf{t})\ \mathbb{E}_{U\sim \mathrm{Breta}^\circ(F_i\setminus g_i;\mathbf{m}|_{B_i};\mathbf{s},\mathbf{t})}\Big[\mathsf{G}_{0;(1,-1)}\Big(1,\ \sum_j \frac{x_j}{\gamma^K_{\mathcal{A}_{ij}}(U)} \sum_j y_j\, \gamma^{\Psi}_{\mathcal{A}_{ij}}(U)\Big)\Big],$$

*for one table* $P_{\mathcal{A}_i\setminus g_i}(\mathbf{m}|_{B_i}; s, t)\, \mathbb{E}\big[\mathsf{G}_{0;(1,-1)}\big(1, xy\, \varrho_{\mathcal{A}_i}(U)\big)\big]$*: the image under the ratio of the form of the hinge minor, read through* $\mathsf{G}_{0;(1,-1)}$*. So* $k \geq 2$ *copies of one cell family closed in series have the period* $\prod_j(\Gamma(s_j)\Gamma(t_j))^{-1} \int \prod_j x_j^{s_j-1} \prod_{j\geq 2} y_j^{t_j-1} f_{F_1}\big(\mathbf{x}, (1, y_2, \ldots, y_N)\big)^k\, d\mathbf{x}\, dy_2 \cdots dy_N$*.*

For one table the closure of $q$ cells is a node of width one (Definition 20.24) with the bunch $B_1 \sqcup \cdots \sqcup B_{q-1}$, a series connection (Proposition 20.37), against the room $B_q$. Theorem 20.26 reads it as $\frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty f_B(y) f_R(y) \frac{dy}{y}$, and Theorem 23.3 is the same integral written symmetrically in all the cells, $\frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{t-1} \prod_i f_{\mathcal{A}_i}(y)\, dy$ with the transforms of the cells (Definition 23.1), which are the orthant transforms $f_i^\circ$ of Proposition 20.37; for $k$ equal cells at the centre the formula for $k$ copies is Theorem 24.2(i), with $k-1$ copies at a series node and one copy as the room. The table that is one circuit, $\binom{\mathcal{E}}{n-1}$, is the series closure of $n$ single marks, each with the transform $\mathsf{G}_{s-m_e;(1,-1)}(1,y)$ up to its weight (Corollary 18.24). The parallel closure is the dual statement (Proposition 8.15).

## 23.4 Cores and their markers

A table is built from cells by two-sums at the markers of cores.

**Definition 23.4** (Markers; the twisted core)**.** Let the exponents be $s, t > 0$. A *core with markers* is a plain table $\mathcal{H}$ on $\mathcal{E}_\mathcal{H} \sqcup G_\mathcal{H}$ none of whose marks is a loop or a coloop, a core of Definition 21.1, whose *markers* are the marks of $G_\mathcal{H}$, the marks at which cells may be substituted (the marks $h_i$ of Definition 21.1). *Cells at the markers* are plain cells $\mathcal{A}_1, \ldots, \mathcal{A}_k$ on $B_i \sqcup \{g_i\}$ whose hinges are identified with distinct markers $g_1, \ldots, g_k$, the sets $\mathcal{E}_\mathcal{H}$ and $B_i$ disjoint, and $\mathcal{H}\langle\mathcal{A}_1, \ldots, \mathcal{A}_k\rangle$ is their substitution (Definition 21.1), the marks of $\mathcal{E}_\mathcal{H}$ and the markers without a cell left in place. For $\mathbf{x} \in (0,\infty)^k$ the *twisted core* $\mathcal{H}[\mathbf{x}]$ is $\mathcal{H}$ twisted in its cobasis letter by $x_i$ at every $g_i$ (Definition 22.1), carrying the masses $\kappa_{\mathcal{A}_i}$ on the markers $g_i$ and the given masses $\mathbf{m}|_{\mathcal{E}_\mathcal{H}}$ on $\mathcal{E}_\mathcal{H}$; it has the basis weights $\theta_T$ and the cobasis weights $\theta_T \prod_{g_i \notin T} x_i$, and its period is that of a table with two systems of weights (Definition 8.1). With the cells at the shapes $u = (u_i)_i$, the twisted core $\mathcal{H}[(\varrho_{\mathcal{A}_i}(u_i))_i]$ is the core read at $u$ of Definition 22.2 with the scale of every marker $g_i$ divided by $\gamma^K_{\mathcal{A}_i}(u_i)$. A core with one marker $g_0$ left free is a *hinged core*; then $\mathcal{H}[\mathbf{x}]$ is a cell with the hinge $g_0$, not plain, and its hinge wall, base form, ratio and law are those of Definition 23.1.

**Lemma 23.5** (Substitution at the markers)**.** *The substitution* $\mathcal{H}\langle\mathcal{A}_1, \ldots, \mathcal{A}_k\rangle$ *does not depend on the order of the two-sums (Proposition 21.3(i)). Its bases are the sets* $(T \cap \mathcal{E}_\mathcal{H}) \sqcup \bigsqcup_i C_i$ *with* $T$ *a basis of* $\mathcal{H}$ *and, for each* $i$*,* $C_i$ *a basis of* $\mathcal{A}_i$ *avoiding* $g_i$ *when* $g_i \in T$ *and* $C_i + g_i$ *a basis of* $\mathcal{A}_i$ *when* $g_i \notin T$*, with the product weights. Its rank is* $r_\mathcal{H} + \sum_i (r_{\mathcal{A}_i} - 1)$*, and with* $G'$ *the markers left free (Theorem 21.4)*

$$K_{\mathcal{H}\langle\mathcal{A}_1,\ldots,\mathcal{A}_k\rangle} = \prod_i K_{\mathcal{A}_i/g_i} \cdot K_\mathcal{H}\big(\alpha_{\mathcal{E}_\mathcal{H}}, (\gamma^K_{\mathcal{A}_i})_i, \alpha_{G'}\big),$$

$$\Psi_{\mathcal{H}\langle\mathcal{A}_1,\dots,\mathcal{A}_k\rangle}=\prod_i \Psi_{\mathcal{A}_i\setminus g_i}\cdot\Psi_{\mathcal{H}}\big(\alpha_{\mathcal{E}_\mathcal{H}},(\gamma^{\Psi}_{\mathcal{A}_i})_i,\alpha_{G'}\big).$$

*For a hinged core the letters of the two hinge minors $\mathcal{H}\langle\mathcal{A}_1,\dots,\mathcal{A}_k\rangle\setminus g_0$ and $\mathcal{H}\langle\mathcal{A}_1,\dots,\mathcal{A}_k\rangle/g_0$ are the same products with the letters of $\mathcal{H}\setminus g_0$ and $\mathcal{H}/g_0$ in place of $K_\mathcal{H}$ and $\Psi_\mathcal{H}$.*

**Theorem 23.6** (The law of a substitution)**.** *Let $F$ be a family with real data whose table $j$ is $\mathcal{H}_j\langle\mathcal{A}_{1j},\dots,\mathcal{A}_{kj}\rangle$, the cores $\mathcal{H}_j$ plain on $\mathcal{E}_\mathcal{H}\sqcup G_\mathcal{H}$ with $\{g_1,\dots,g_k\}\subseteq G_\mathcal{H}$, the cells of all tables at $g_i$ forming the cell family $F_i=((\mathcal{A}_{ij},\theta^{\mathcal{A}_{ij}});s_j,t_j)_j$ on $B_i\sqcup\{g_i\}$ (Definition 23.1), with masses on $\mathcal{E}_\mathcal{H}$ and on the marks of the cells. For $\mathbf{x}=(\mathbf{x}_i)_i$, $\mathbf{x}_i=(x_{i,j,K},x_{i,j,\Psi})_j\in(0,\infty)^{2N-1}$ with $x_{i,1,K}=1$, let $F_\mathcal{H}[\mathbf{x}]$ be the family $((\mathcal{H}_j,\theta^{\mathcal{H}_j});s_j,t_j)_j$ twisted at every $g_i$ by $\mathbf{x}_i$, $x_{i,j,K}$ in $K_j$ and $x_{i,j,\Psi}$ in $\Psi_j$ (Definition 22.1), with the masses $\kappa_{F_i}$ on the markers $g_i$ and the given masses on $\mathcal{E}_\mathcal{H}$.*

*(i) If $G_\mathcal{H}=\{g_1,\dots,g_k\}$ and $F$ is on its balance wall, then, in $[0,\infty]$,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int P_{F_\mathcal{H}[\mathbf{x}]}\big((\mathbf{m}|_{\mathcal{E}_\mathcal{H}},\kappa_{F_1},\dots,\kappa_{F_k});\mathbf{s},\mathbf{t}\big)\prod_i d\big((\boldsymbol{\varrho}_{F_i})_*\phi_{F_i}\big)(\mathbf{x}_i).$$

*With the core $\binom{\{g_1,g_2\}}{1}$ in every table this is the gluing of two cell families $F_1$ and $F_2$, $\kappa_{F_1}+\kappa_{F_2}=2\sigma$, $\sigma$ the total strength (Definition 8.1): the period of the twisted core family is $\int_0^\infty v^{\kappa_{F_1}-1}\prod_j(x_{1,j,K}v+x_{2,j,K})^{-s_j}(x_{1,j,\Psi}v+x_{2,j,\Psi})^{-t_j}\,dv$, a Breta of one bunch with at most $2N$ coefficients (Proposition 15.1), unchanged when $(\kappa_{F_1},\mathbf{x}_1)$ and $(\kappa_{F_2},\mathbf{x}_2)$ are swapped.*

*(ii) If $G_\mathcal{H}=\{g_0,g_1,\dots,g_k\}$, $F$ is a cell family with the hinge $g_0$; its hinge wall is that of $F_\mathcal{H}[\mathbf{x}]$, which does not depend on $\mathbf{x}$,*

$$\kappa_F=m_{\mathcal{E}_\mathcal{H}}+\sum_i\kappa_{F_i}-\sum_j\big(s_j(r_{\mathcal{H}_j}-1)+t_j(|\mathcal{E}_\mathcal{H}|+k-r_{\mathcal{H}_j})\big),$$

*and its law is the mixture of the twisted laws: for every Borel $H\ge0$ on $(0,\infty)^{2N-1}$,*

$$\int H\,d\big((\boldsymbol{\varrho}_F)_*\phi_F\big)=\int\Big(\int H\,d\big((\boldsymbol{\varrho}_{F_\mathcal{H}[\mathbf{x}]})_*\phi_{F_\mathcal{H}[\mathbf{x}]}\big)\Big)\prod_i d\big((\boldsymbol{\varrho}_{F_i})_*\phi_{F_i}\big)(\mathbf{x}_i).$$

*In particular $\int\boldsymbol{\varrho}_F^{-\mathbf{z}}d\phi_F=\int\big(\int\boldsymbol{\varrho}_{F_\mathcal{H}[\mathbf{x}]}^{-\mathbf{z}}d\phi_{F_\mathcal{H}[\mathbf{x}]}\big)\prod_i d\big((\boldsymbol{\varrho}_{F_i})_*\phi_{F_i}\big)(\mathbf{x}_i)$ for every real $\mathbf{z}$.*

*For one table with $s,t>0$ the laws are measures on $(0,\infty)$ and the twisted core family is the twisted core $\mathcal{H}[\mathbf{x}]$ of Definition 23.4.*

The core is the only place where cells interact. Below a core each cell family is one measure on $(0,\infty)^{2N-1}$, for one table on a half-line; at the core the cell families are fed into the core family as twists of both letters of every table at the markers, of the cobasis letter alone for one table (Lemma 23.2), and the core family, with the masses $\kappa_{F_i}$ on its markers, is integrated once for each value of the twists. So the work at a core is governed by the core itself and by the number of its markers, not by the sizes of the cells: a cell with a thousand marks and a cell with one mark cost the same once their laws are known. Theorem 19.13 is the gluing of two cell families read through Mellin transforms (Theorem 23.9(iv)).

## 23.5 Two cells glued

**Lemma 23.7** (Two marks)**.** *For $s,t>0$, $\kappa,\kappa'>0$ with $\kappa+\kappa'=s+t$, and $x,x'>0$, let $\binom{\{g,g'\}}{1}[(x,x')]$ be the list $\binom{\{g,g'\}}{1}$ twisted in its cobasis letter by $x$ at $g$ and by $x'$ at $g'$ (Definition 22.1). With the masses $\kappa$ on $g$ and $\kappa'$ on $g'$ its period is*

$$P_{\binom{\{g,g'\}}{1}[(x,x')]}\big((\kappa,\kappa');s,t\big)=\int_0^\infty v^{\kappa-1}(1+v)^{-s}(xv+x')^{-t}\,dv$$

$$= x^{-t}\,\Gamma_{\!\overline{2}}\!\begin{bmatrix}\kappa,\ \kappa'\\ s,\ t\end{bmatrix}\ \mathsf{B}^{\{\{0,1\}\}}_{(s,t),\,\kappa';\,I_{\{0,1\}}}\big((1,x'/x)\big)$$
$$= \frac{1}{\Gamma(s)\Gamma(t)}\int_0^\infty y^{t-1}\,\mathsf{G}_{s-\kappa;(1,-1)}(1,xy)\,\mathsf{G}_{s-\kappa';(1,-1)}(1,x'y)\,dy$$

*(Proposition 15.1, the two coefficients merging into one of letter exponent $s+t$ when $x=x'$), unchanged when $(\kappa,x)$ and $(\kappa',x')$ are swapped. For a family of tables the kernel is the period of the twisted core family of Theorem 23.6(i) with the core $\binom{\{g_1,g_2\}}{1}$ in every table.*

**Proposition 23.8** (Faces of one law)**.** *Let $F=((\mathcal{A}_j,\theta_j);s_j,t_j)_{j=1}^N$ be a cell family on $B\sqcup\{g\}$ with plain tables, $s_j,t_j>0$, masses $\mathbf{m}$ on $B$ and the hinge wall $\kappa=\kappa_F$; in this statement the points of $(0,\infty)^{2N-1}$ are written $\mathbf{z}=(z_{j,K},z_{j,\Psi})_j$, $z_{1,K}=1$, and $\mathbf{x}'=(x_2,\dots,x_N)$.*

*(i)* Law forms. *For $\mathbf{x},\mathbf{y}\in(0,\infty)^N$,*

$$f_F(\mathbf{x},\mathbf{y})=\int \mathsf{G}_{|\mathbf{s}|-\kappa;(1,-1)}\Big(\sum_j\frac{x_j}{z_{j,K}},\ \sum_j y_jz_{j,\Psi}\Big)\prod_j z_{j,K}^{-s_j}\,d\big((\boldsymbol{\varrho}_F)_*\phi_F\big)(\mathbf{z}),$$
$$\check f_F(\mathbf{x},\mathbf{y})=\int \mathsf{G}_{|\mathbf{t}|-\kappa;(1,-1)}\Big(\sum_j x_j\frac{z_{1,\Psi}}{z_{j,\Psi}},\ \sum_j y_j\frac{z_{j,K}}{z_{1,\Psi}}\Big)\prod_j\Big(\frac{z_{j,\Psi}}{z_{1,\Psi}}\Big)^{-t_j}z_{1,\Psi}^{-\kappa}\,d\big((\boldsymbol{\varrho}_F)_*\phi_F\big)(\mathbf{z}):$$

*the two transforms depend on $F$ only through its hinge wall and its law. For one cell, for $y>0$,*

$$f_{\mathcal{A}}(y)=\int_{\Delta_B}\mathsf{G}_{s-\kappa;(1,-1)}\big(1,\varrho_{\mathcal{A}}y\big)\,d\phi_{\mathcal{A}},\qquad \check f_{\mathcal{A}}(y)=\int_{\Delta_B}\mathsf{G}_{t-\kappa;(1,-1)}\big(1,y/\varrho_{\mathcal{A}}\big)\,\varrho_{\mathcal{A}}^{-\kappa}\,d\phi_{\mathcal{A}}.$$

*(ii)* Hinge measures.

$$f_F(\mathbf{x},\mathbf{y})=\int e^{-\langle\mathbf{x},\boldsymbol{\lambda}\rangle-\langle\mathbf{y},\boldsymbol{\mu}\rangle}\,d\mathfrak{r}_F(\boldsymbol{\lambda},\boldsymbol{\mu}),\qquad \check f_F(\mathbf{x},\mathbf{y})=\int e^{-\langle\mathbf{x},\boldsymbol{\lambda}\rangle-\langle\mathbf{y},\boldsymbol{\mu}\rangle}\,d\check{\mathfrak{r}}_F(\boldsymbol{\lambda},\boldsymbol{\mu}),$$

*and $\mathfrak{r}_F$ is the image of the product of $\prod_j z_{j,K}^{-s_j}\,(\boldsymbol{\varrho}_F)_*\phi_F(d\mathbf{z})$ and $p^{\kappa-|\mathbf{s}|-1}dp$ under $(\mathbf{z},p)\mapsto\big((1/(pz_{j,K}))_j,(pz_{j,\Psi})_j\big)$. So $f_F$ and $\check f_F$ are completely monotone in their $2N$ variables, $(-1)^{|\mathbf{n}|}\partial^{\mathbf{n}}f_F\ge 0$ for every multi-index $\mathbf{n}$ on the interior of the set where $f_F$ is finite and the same for $\check f_F$, and log-convex along every segment on which they are finite. For one cell, $f_{\mathcal{A}}(y)=\int e^{-y\lambda}d\mathfrak{r}^{\Psi}_{\mathcal{A}}(\lambda)$ and $\check f_{\mathcal{A}}(y)=\int e^{-y\lambda}d\mathfrak{r}^{K}_{\mathcal{A}}(\lambda)$; $\mathfrak{r}^{\Psi}_{\mathcal{A}}$ is the image of $\alpha^{\mathbf{m}-\mathbf{1}}K_{\mathcal{A}\setminus g}^{-s}\Psi_{\mathcal{A}\setminus g}^{-t}e^{-1/\gamma^K_{\mathcal{A}}}d\alpha$ under $\gamma^{\Psi}_{\mathcal{A}}$ and of $(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dx)\otimes p^{s-\kappa-1}e^{-p}dp$ under $(x,p)\mapsto x/p$, and $\mathfrak{r}^{K}_{\mathcal{A}}$ the image of $\alpha^{\mathbf{m}-\mathbf{1}}K_{\mathcal{A}/g}^{-s}\Psi_{\mathcal{A}/g}^{-t}e^{-1/\gamma^{\Psi}_{\mathcal{A}}}d\alpha$ under $\gamma^{K}_{\mathcal{A}}$ and of $x^{-\kappa}(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dx)\otimes p^{t-\kappa-1}e^{-p}dp$ under $(x,p)\mapsto 1/(xp)$; so $f_{\mathcal{A}}$ and $\check f_{\mathcal{A}}$ are decreasing and log-convex wherever finite.*

*(iii)* Mellin transforms. *For real $\mathbf{a},\mathbf{b}\in(0,\infty)^N$ with $\sum_j b_j-\sum_j a_j=\kappa-|\mathbf{s}|$, in $[0,\infty]$,*

$$\int_{(0,\infty)^{2N-1}}\prod_{j\ge2}x_j^{a_j-1}\prod_j y_j^{b_j-1}\,f_F\big((1,\mathbf{x}'),\mathbf{y}\big)\,d\mathbf{x}'\,d\mathbf{y}$$
$$=\prod_j\Gamma(a_j)\Gamma(b_j)\int_{\Delta_B}\boldsymbol{\varrho}_F^{-(\mathbf{s}-\mathbf{a},\mathbf{b})}\,d\phi_F,$$

*and for real* $\mathbf{a},\mathbf{b}\in(0,\infty)^N$ *with* $\sum_j b_j-\sum_j a_j=\kappa-|\mathbf{t}|$ *the same holds for* $\check f_F$ *with* $\int_{\Delta_B}\boldsymbol{\varrho}_F^{-(\mathbf{b},\mathbf{t}-\mathbf{a})}\,d\phi_F$ *on the right; both Mellin transforms of the law are taken on the hyperplane* $\sum_j(z_{j,K}+z_{j,\Psi})=\kappa$ *of their arguments. For one cell, with* $z=b$*, these are*

$$\int_0^\infty y^{z-1}f_{\mathcal A}(y)\,dy=\Gamma(z)\Gamma(z+s-\kappa)\int_{\Delta_B}\varrho_{\mathcal A}^{-z}\,d\phi_{\mathcal A}\qquad\big(z>\max(0,\kappa-s)\big),$$
$$\int_0^\infty y^{z-1}\check f_{\mathcal A}(y)\,dy=\Gamma(z)\Gamma(z+t-\kappa)\int_{\Delta_B}\varrho_{\mathcal A}^{z-\kappa}\,d\phi_{\mathcal A}\qquad\big(z>\max(0,\kappa-t)\big).$$

*(iv)* The dual cell family. *The dual tables* $\mathcal A_j^*$*, with the hinge* $g$*, at the exponents* $(t_j,s_j)$ *form a cell family* $F^*$ *with the hinge wall* $\kappa$*, the hinge values* $\gamma^K_{\mathcal A_j^*}=\gamma^\Psi_{\mathcal A_j}$ *and* $\gamma^\Psi_{\mathcal A_j^*}=\gamma^K_{\mathcal A_j}$*, the base form* $(\gamma^\Psi_{\mathcal A_1}/\gamma^K_{\mathcal A_1})^{-\kappa}\phi_F$*, the law the image of* $z_{1,\Psi}^{-\kappa}\,(\boldsymbol{\varrho}_F)_*\phi_F(d\mathbf{z})$ *under* $\mathbf{z}\mapsto(z_{j,\Psi}/z_{1,\Psi},\,z_{j,K}/z_{1,\Psi})_j$*, the transform* $\check f_F$ *and the dual transform* $f_F$*. For one cell the ratio is* $1/\varrho_{\mathcal A}$*, the base form* $\varrho_{\mathcal A}^{-\kappa}\phi_{\mathcal A}$*, the law the image of* $x^{-\kappa}(\varrho_{\mathcal A})_*\phi_{\mathcal A}(dx)$ *under* $x\mapsto1/x$*, and* $\int\varrho_{\mathcal A^*}^{-z}d\phi_{\mathcal A^*}=\int\varrho_{\mathcal A}^{z-\kappa}d\phi_{\mathcal A}$*.*

*(v)* Change of transform. *For real* $\mathbf{a},\mathbf{b}$ *with* $0<a_j<s_j$*,* $0<b_j<t_j$ *and* $\sum_j b_j-\sum_j a_j=\kappa-|\mathbf{s}|$*, the left side of (iii) for* $\check f_F$ *at* $(\mathbf{t}-\mathbf{b},\mathbf{s}-\mathbf{a})$ *times* $\prod_j\Gamma(a_j)\Gamma(b_j)$ *equals that for* $f_F$ *at* $(\mathbf{a},\mathbf{b})$ *times* $\prod_j\Gamma(t_j-b_j)\Gamma(s_j-a_j)$*, in* $(0,\infty]$*. For one cell, with* $z=s-a$*: for real* $z$ *with* $\max(0,\kappa-t)<z<\min(s,\kappa)$*, in* $(0,\infty]$*,*

$$\int_0^\infty y^{z-1}\check f_{\mathcal A}(y)\,dy=\frac{\Gamma(z)\Gamma(z+t-\kappa)}{\Gamma(\kappa-z)\Gamma(s-z)}\int_0^\infty y^{\kappa-z-1}f_{\mathcal A}(y)\,dy.$$

*When* $\mathcal A$ *is the cell of the bunch of a node of width one (Definition 20.24) and* $\int\varrho_{\mathcal A}^{-b}d\phi_{\mathcal A}<\infty$ *at some* $b$ *with* $\max(0,\kappa_{\mathcal A}-s)<b<\min(t,\kappa_{\mathcal A})$*, the expansion of* $f_{\mathcal A}$ *at* $y\to0$ *is the sum of the residues of* $y^{-z}\Gamma(z)\Gamma(z+s-\kappa_{\mathcal A})\int\varrho_{\mathcal A}^{-z}d\phi_{\mathcal A}$ *to the left of the line* $\operatorname{Re}z=b$ *(Theorem 20.34(i)), and that of* $\check f_{\mathcal A}$*, the transform of* $\mathcal A^*$ *read as the bunch of the dual node of width one at the exponents* $(t,s)$ *(Proposition 8.15), is the sum of the residues of* $y^{-z}\Gamma(z)\Gamma(z+t-\kappa_{\mathcal A})\int\varrho_{\mathcal A}^{z-\kappa_{\mathcal A}}d\phi_{\mathcal A}$ *to the left of the line* $\operatorname{Re}z=\kappa_{\mathcal A}-b$*, at the poles of the Gamma pair and at the points* $\kappa_{\mathcal A}-w$*,* $w$ *a pole of the continued* $\int\varrho_{\mathcal A}^{-w}d\phi_{\mathcal A}$ *to the right of the line* $\operatorname{Re}w=b$*: the two transforms expand one function on the two sides of its strip.*

So one law has four faces. The base form and its image, the law, are where the cell family lives; the transform reads the law through Grammas of index $|\mathbf{s}|-\kappa_F$ and is the Laplace transform of the hinge measure $\mathfrak r_F$, carried by the reciprocals of the hinge values by $K$ and the hinge values by $\Psi$; the dual transform reads the law turned over through Grammas of index $|\mathbf{t}|-\kappa_F$ and is the Laplace transform of $\check{\mathfrak r}_F$, carried by the reciprocals of the hinge values by $\Psi$ and the hinge values by $K$; and on the Mellin side both are the one function $\int\boldsymbol{\varrho}_F^{-\mathbf{z}}d\phi_F$ with two Gamma products. Passing from one transform to the other costs nothing on the Mellin side, a reflection and a ratio of Gammas, and costs integral transforms on the side of the transforms; for one cell, on the half-line, the two expansions at $y\to0$ are read on opposite sides of the strip: the transform expands where the law is large, the dual transform where it is small.

**Theorem 23.9** (Two cell families glued)**.** *Let* $F_1$ *and* $F_2$ *be cell families with plain tables on* $B\sqcup\{g\}$ *and* $R\sqcup\{g\}$*, at the exponents* $(s_j,t_j)_j$ *with* $s_j,t_j>0$*, and let* $F_1\oplus_g F_2$*, whose table* $j$ *is the two-sum of the tables* $j$ *of* $F_1$ *and* $F_2$ *(Lemma 21.12), carry masses* $\mathbf{m}$ *on its balance wall, so that* $\kappa_{F_1}+\kappa_{F_2}=2\sigma$*,* $\sigma$ *the total strength (Definition 8.1); let* $0<\kappa_{F_1}<2\sigma$*. In this statement the points of the laws of* $F_1$ *and* $F_2$ *are written* $\mathbf{z}$ *and* $\mathbf{z}'$*,* $\mathcal G(\mathbf{z},\mathbf{z}')=\int_0^\infty v^{\kappa_{F_1}-1}\prod_j(z_{j,K}v+z'_{j,K})^{-s_j}(z_{j,\Psi}v+z'_{j,\Psi})^{-t_j}\,dv$ *is*

*the period of the twisted core family of Theorem 23.6(i) with the core* $\binom{\{g_1,g_2\}}{1}$ *in every table, the transforms are read at* $x_1 = 1$*, with* $\mathbf{x}' = (x_2, \dots, x_N)$*, and the hinge measures of* $F_1$ *act in the variables* $(\boldsymbol{\lambda}, \boldsymbol{\mu})$*, those of* $F_2$ *in* $(\boldsymbol{\lambda}', \boldsymbol{\mu}')$*. Then, in* $[0, \infty]$*,*

$$P_{F_1 \oplus_g F_2}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \iint \mathcal{G}(\mathbf{z}, \mathbf{z}')\, d\big((\boldsymbol{\varrho}_{F_1})_* \phi_{F_1}\big)(\mathbf{z})\, d\big((\boldsymbol{\varrho}_{F_2})_* \phi_{F_2}\big)(\mathbf{z}') \tag{i}$$

$$= \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j \geq 2} x_j^{s_j - 1} \prod_j y_j^{t_j - 1}\, f_{F_1} f_{F_2}\, d\mathbf{x}'\, d\mathbf{y}$$

$$= \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j \geq 2} x_j^{t_j - 1} \prod_j y_j^{s_j - 1}\, \check{f}_{F_1} \check{f}_{F_2}\, d\mathbf{x}'\, d\mathbf{y} \tag{ii}$$

$$= \frac{1}{\Gamma(s_1)} \iint e^{-\lambda_1 - \lambda_1'} \prod_{j \geq 2} (\lambda_j + \lambda_j')^{-s_j} \prod_j (\mu_j + \mu_j')^{-t_j}\, d\mathfrak{r}_{F_1}\, d\mathfrak{r}_{F_2}$$

$$= \frac{1}{\Gamma(t_1)} \iint e^{-\lambda_1 - \lambda_1'} \prod_{j \geq 2} (\lambda_j + \lambda_j')^{-t_j} \prod_j (\mu_j + \mu_j')^{-s_j}\, d\check{\mathfrak{r}}_{F_1}\, d\check{\mathfrak{r}}_{F_2}, \tag{iii}$$

*and these are finite exactly when every proper wall of* $F_1 \oplus_g F_2$ *is positive. On every contour of the level shares of Theorem 19.13 on which its stacked Gammas have arguments of positive real part and the two Mellin transforms below converge absolutely (Definition 23.1),*

$$P_{F_1 \oplus_g F_2}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = 2^{N-1} \int_{\sum_j \delta_j^d = \frac{1}{2}(|\mathbf{s}| - \kappa_{F_1})} \prod_j \Gamma_2 \begin{bmatrix} a_j^d, & b_j^d, & a_j^r, & b_j^r \\ & & s_j, & t_j \end{bmatrix}$$
$$\times \int_{\Delta_B} \boldsymbol{\varrho}_{F_1}^{-(\mathbf{a}^r, \mathbf{b}^d)}\, d\phi_{F_1} \int_{\Delta_R} \boldsymbol{\varrho}_{F_2}^{-(\mathbf{a}^d, \mathbf{b}^r)}\, d\phi_{F_2}, \tag{iv}$$

*the formula of Theorem 19.13, the periods of its bunch family and room family at the level shares being these two Mellin transforms. For one table the laws are* $(\varrho_{\mathcal{A}})_* \phi_{\mathcal{A}}$ *and* $(\varrho_{\mathcal{A}'})_* \phi_{\mathcal{A}'}$ *and the kernel of (i) is the period of Lemma 23.7; (ii) reads* $\frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{t-1} f_{\mathcal{A}} f_{\mathcal{A}'}\, dy = \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{s-1} \check{f}_{\mathcal{A}} \check{f}_{\mathcal{A}'}\, dy$*, its first form Theorem 20.26(i) with the room transform* $f_R(y) = y^t f_{\mathcal{A}'}(y)$*; (iii) reads* $\frac{1}{\Gamma(s)} \iint (\lambda + \lambda')^{-t}\, d\mathfrak{r}_{\mathcal{A}}^{\Psi}\, d\mathfrak{r}_{\mathcal{A}'}^{\Psi} = \frac{1}{\Gamma(t)} \iint (\lambda + \lambda')^{-s}\, d\mathfrak{r}_{\mathcal{A}}^{K}\, d\mathfrak{r}_{\mathcal{A}'}^{K}$*; and (iv) holds on a line* $\operatorname{Re} z = z_0$ *with* $\max(0, \kappa_{\mathcal{A}} - s) < z_0 < \min(t, \kappa_{\mathcal{A}})$ *on which* $\int \varrho_{\mathcal{A}}^{-z_0} d\phi_{\mathcal{A}}$ *and* $\int \varrho_{\mathcal{A}'}^{z_0 - t} d\phi_{\mathcal{A}'}$ *are finite, as*

$$P_{\mathcal{A} \oplus_g \mathcal{A}'}(\mathbf{m}; s, t) = \frac{1}{\Gamma(s)\Gamma(t)} \frac{1}{2\pi i} \int_{\operatorname{Re} z = z_0} \Gamma(z)\Gamma(z + s - \kappa_{\mathcal{A}})\Gamma(t - z)\Gamma(\kappa_{\mathcal{A}} - z)$$
$$\times \int_{\Delta_B} \varrho_{\mathcal{A}}^{-z}\, d\phi_{\mathcal{A}} \int_{\Delta_R} \varrho_{\mathcal{A}'}^{z - t}\, d\phi_{\mathcal{A}'}\, dz.$$

The period of a two-sum is a pairing of two laws by one kernel, the period of the twisted core family on two marks, for one table the period of two marks. Read through the Grammas the pairing is one integral over the scales of the two transforms, symmetric in the two sides; for one table, read through the hinge measures, it is the moment of order $-t$ of a sum: under $\mathfrak{r}_{\mathcal{A}}^{\Psi}$ the variable is the hinge value $\gamma_{\mathcal{A}}^{\Psi}$ of the cell (Proposition 23.8(ii)), and gluing the two cells into one table adds their hinge values by $\Psi$, as $\Psi = \Psi_{\mathcal{A} \setminus g} \Psi_{\mathcal{A}' \setminus g} (\gamma_{\mathcal{A}}^{\Psi} + \gamma_{\mathcal{A}'}^{\Psi})$ says.

## 23.6 What a cell retains

**Definition 23.10** (One weighted mark)**.** For $\mathbf{x}_0 = (x_{0,j,K}, x_{0,j,\Psi})_{j=1}^N$ with $x_{0,1,K} = 1$, a point of $(0, \infty)^{2N-1}$, and a mark $e$ with the mass $m_e$, the *one-mark cell family of ratio vector* $\mathbf{x}_0$ has in

table $j$ the bases $\{e\}$ and $\{g\}$, with the weights $x_{0,j,K}$ and 1 in its letter $K_j$ and the weights 1 and $x_{0,j,\Psi}$ in its letter $\Psi_j$ (Definition 21.1): $K_j = x_{0,j,K}\alpha_e + \alpha_g$ and $\Psi_j = \alpha_g + x_{0,j,\Psi}\alpha_e$. Its hinge values are $x_{0,j,K}\alpha_e$ and $x_{0,j,\Psi}\alpha_e$, its ratio vector is the constant $\mathbf{x}_0$, its hinge wall is $m_e$, and its law is $\delta_{\mathbf{x}_0}$. For one table and $x_0 > 0$, the *one-mark cell of ratio* $x_0$ is the plain table $\left(\binom{\{e,g\}}{1}, (1, x_0^{1/2})\right)$ on $\{e,g\}$, the bases $\{e\}$ and $\{g\}$ with the weights 1 and $x_0^{1/2}$. Its hinge minors have $K_{\setminus g} = \alpha_e$, $K_{/g} = x_0^{1/2}$, $\Psi_{\setminus g} = 1$ and $\Psi_{/g} = x_0^{1/2}\alpha_e$, its ratio is the constant $x_0$, its hinge wall is $m_e$, and its law is $x_0^{(m_e-s)/2}\,\delta_{x_0}$.

**Theorem 23.11** (The law is what a cell family retains)**.** *Let $F$ and $F'$ be cell families with plain tables, at the exponents $(s_j,t_j)_j$ with $s_j,t_j>0$, with masses on their marks other than the hinge and one hinge wall $\kappa \in (0,2\sigma)$, $\sigma$ the total strength (Definition 8.1), and let there be a real $\mathbf{z} = (z_{j,K}, z_{j,\Psi})_j$ with $0 < z_{j,K} < s_j$, $0 < z_{j,\Psi} < t_j$ and $\sum_j (z_{j,K} + z_{j,\Psi}) = \kappa$ at which $\int \boldsymbol{\varrho}_F^{-\mathbf{z}} d\phi_F$ and $\int \boldsymbol{\varrho}_{F'}^{-\mathbf{z}} d\phi_{F'}$ are finite (Definition 23.1). The following are equivalent.*

*(a)* $(\boldsymbol{\varrho}_F)_*\phi_F = (\boldsymbol{\varrho}_{F'})_*\phi_{F'}$.

*(b)* $P_{F\oplus_g F''} = P_{F'\oplus_g F''}$, *in* $[0,\infty]$, *for every cell family $F''$ with the hinge wall $2\sigma-\kappa$, the masses of $F$, $F'$ and $F''$ kept.*

*(c) (b) holds for the one-mark cell families $F''$ of ratio vector $\mathbf{x}_0$ and mass $2\sigma-\kappa$ (Definition 23.10), for every $\mathbf{x}_0$ in a nonempty open subset of $(0,\infty)^{2N-1}$.*

*(d)* $\int \boldsymbol{\varrho}_F^{-\mathbf{w}} d\phi_F = \int \boldsymbol{\varrho}_{F'}^{-\mathbf{w}} d\phi_{F'}$ *at every complex $\mathbf{w}$ with real part $\mathbf{z}$.*

*So the hinge wall and the law are the whole of what a cell family contributes to the families built on it, and the one-mark cell families, whose laws are single points, already separate any two laws. For one table the plain one-mark cells of ratio $x_0$ suffice in (c), and (d) is the agreement of $\int \varrho_{\mathcal{A}}^{-w} d\phi_{\mathcal{A}}$ and $\int \varrho_{\mathcal{A}'}^{-w} d\phi_{\mathcal{A}'}$ on the line* $\operatorname{Re} w = z_{1,\Psi}$, $\max(0,\kappa-s) < z_{1,\Psi} < \min(t,\kappa)$.

What determines the contribution of a cell family to a larger construction is its hinge wall, which balance hands to the other side of the hinge, and its law, one positive measure on $(0,\infty)^{2N-1}$, on the half-line for one cell. Anything finer about the cells, their marks, their masses one by one, their letters, is invisible from outside, and anything coarser fails already against one weighted mark. A cell family is therefore replaceable, in every family it can sit in, by any other cell family of the same wall and the same law, and a cell family whose law is one point is a mark. Theorem 21.16 says the same at any width for the joint law of the bunch letters; at width one the joint law collapses to the law of the ratio vector.

**Corollary 23.12** (Cells that are marks)**.** *Let $F$ be a cell family with plain tables and $s_j,t_j>0$, and $\mathcal{A}$ a plain cell with exponents $s,t>0$.*

*(i) The law of $F$ is carried by a single point, $(\boldsymbol{\varrho}_F)_*\phi_F = \phi_F(\Delta_B)\,\delta_{\mathbf{x}_0}$, exactly when the ratio vector is constant, $\boldsymbol{\varrho}_F \equiv \mathbf{x}_0$; then $\phi_F(\Delta_B) \in (0,\infty]$, finite whenever $F$ sits in a gluing of finite period, and in every gluing $F$ acts as $\phi_F(\Delta_B)$ times the one-mark cell family of ratio vector $\mathbf{x}_0$ and mass $\kappa_F$. For one cell with $\varrho_{\mathcal{A}} \equiv x_0$ this is $\phi_{\mathcal{A}}(\Delta_B)\,x_0^{(s-\kappa_{\mathcal{A}})/2}$ times the one-mark cell of ratio $x_0$ and mass $\kappa_{\mathcal{A}}$.*

*(ii) If $\mathcal{A}/g$ is the dual table of $\mathcal{A}\setminus g$ with the identity on $B$ and the same weights, then $\varrho_{\mathcal{A}} \equiv 1$ and $\phi_{\mathcal{A}}(\Delta_B) = P_{\mathcal{A}\setminus g}(\mathbf{m}; \kappa_{\mathcal{A}}, s+t-\kappa_{\mathcal{A}})$.*

*(iii) If a table $(\mathcal{B},\theta)$ is* identically self-dual*, its bases closed under complement with $\theta_{\mathcal{E}\setminus T} = \theta_T$, then for every mark $g$ and masses on the balance wall with every proper wall positive,*

$$P_F(\mathbf{m}; s, t) = \Gamma_{\binom{2}{1}}\begin{bmatrix} m_g, & s+t-m_g \\ & s+t \end{bmatrix} P_{F\setminus g}\big(\mathbf{m}|_{\mathcal{E}\setminus g}; s+t-m_g, m_g\big),$$

*$F = ((\mathcal{B},\theta); s, t)$ and $F \setminus g = ((\mathcal{B} \setminus g, \theta); s + t - m_g, m_g)$ the family of the deletion.*

So (iii) splits off any mark of an identically self-dual table, by one Beta corner, at every pair of exponents, and Proposition 19.12, every letter active, is (i) read across a bunch of width one, the ratio vector being constant. The relation is a consequence of structure alone: the mark $g$ and the rest of the table see each other through a ratio that is identically one, so the hinge integral separates. For a table that is its own dual only up to a relabelling of its marks the hypothesis of (iii) fails, and (iii) does not apply.

## 23.7 Series and parallel nodes

**Definition 23.13** (Series and parallel cores)**.** For $q \geq 2$ and the markers $G = \{g_0, g_1, \dots, g_q\}$ a *series core* is a table with the bases of the list $\binom{G}{q}$ and a *parallel core* a table with the bases of the list $\binom{G}{1}$, with any positive weights. With cells $\mathcal{A}_1, \dots, \mathcal{A}_q$ at $g_1, \dots, g_q$ and the hinge $g_0$, $\binom{G}{q}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ and $\binom{G}{1}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$, the cores with unit weights, are the *series connection* of the cells, whose bunch $\bigsqcup_i B_i$ is the series connection of the parts $B_i$ of Proposition 20.37, and their *parallel connection*; at the vertices of a hinge tree series and parallel cores are *series nodes* and *parallel nodes*.

A series bunch of marks is the series connection of its marks, each a one-mark cell, and a parallel bunch the parallel connection (Proposition 19.3); and the series closure $\binom{\{h_1,\dots,h_q\}}{q-1}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ of Corollary 21.7(ii) is the series connection of $\mathcal{A}_1, \dots, \mathcal{A}_{q-1}$ glued to $\mathcal{A}_q$ along $g_0$.

**Theorem 23.14** (Series and parallel nodes)**.** *Let $F_1, \dots, F_q$, $q \geq 2$, be cell families with plain tables at the exponents $(s_j, t_j)_j$, $s_j, t_j > 0$, the table $j$ of $F_i$ being $\mathcal{A}_{ij}$, and $G = \{g_0, \dots, g_q\}$.*

*(i)* Series. *The cell family $F$ whose table $j$ is $\mathcal{A}_j = \binom{G}{q}\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{qj}\rangle$ has $\kappa_F = \sum_i \kappa_{F_i} - (q-1)|\mathbf{s}|$, its hinge values combine in every table as $1/\gamma^K_{\mathcal{A}_j} = \sum_i 1/\gamma^K_{\mathcal{A}_{ij}}$ and $\gamma^\Psi_{\mathcal{A}_j} = \sum_i \gamma^\Psi_{\mathcal{A}_{ij}}$, and*

$$f_F = \prod_{i=1}^{q} f_{F_i}, \qquad \mathfrak{r}_F = \mathfrak{r}_{F_1} * \cdots * \mathfrak{r}_{F_q},$$

*the convolution on $(0,\infty)^{2N}$. For one table, $\mathcal{A} = \binom{G}{q}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ has $f_{\mathcal{A}} = \prod_i f_{\mathcal{A}_i}$ and $\mathfrak{r}^\Psi_{\mathcal{A}} = \mathfrak{r}^\Psi_{\mathcal{A}_1} * \cdots * \mathfrak{r}^\Psi_{\mathcal{A}_q}$, the convolution on $(0,\infty)$, and the series core twisted by $\mathbf{x}$ at $g_1, \dots, g_q$ has, on the simplex of its markers $g_1, \dots, g_q$ with the coordinates $v$, the ratio and the base form*

$$\Big(\sum_i \frac{1}{v_i}\Big)\Big(\sum_i x_i v_i\Big), \qquad \prod_i v_i^{\kappa_{\mathcal{A}_i}-1}\, e_{q-1}(v)^{\kappa_{\mathcal{A}}-s}\Big(\prod_i v_i\Big)^{-\kappa_{\mathcal{A}}} d\omega_{\{g_1,\dots,g_q\}}(v).$$

*(ii)* Parallel. *The cell family $F$ whose table $j$ is $\mathcal{A}_j = \binom{G}{1}\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{qj}\rangle$ has $\kappa_F = \sum_i \kappa_{F_i} - (q-1)|\mathbf{t}|$, its hinge values combine as $\gamma^K_{\mathcal{A}_j} = \sum_i \gamma^K_{\mathcal{A}_{ij}}$ and $1/\gamma^\Psi_{\mathcal{A}_j} = \sum_i 1/\gamma^\Psi_{\mathcal{A}_{ij}}$, and*

$$\check{f}_F = \prod_{i=1}^{q} \check{f}_{F_i}, \qquad \check{\mathfrak{r}}_F = \check{\mathfrak{r}}_{F_1} * \cdots * \check{\mathfrak{r}}_{F_q};$$

*for one table $\check{f}_{\mathcal{A}} = \prod_i \check{f}_{\mathcal{A}_i}$ and $\mathfrak{r}^K_{\mathcal{A}} = \mathfrak{r}^K_{\mathcal{A}_1} * \cdots * \mathfrak{r}^K_{\mathcal{A}_q}$.*

*(iii)* Extremes of the ratio. *For one table, with plain cells $\mathcal{A}_1, \dots, \mathcal{A}_q$ at the exponents $(s,t)$: in a series connection $\sqrt{\inf \varrho_{\mathcal{A}}} = \sum_i \sqrt{\inf \varrho_{\mathcal{A}_i}}$ (Proposition 20.37(i)) and $\sup \varrho_{\mathcal{A}} = \infty$; in a parallel connection $1/\sqrt{\sup \varrho_{\mathcal{A}}} = \sum_i 1/\sqrt{\sup \varrho_{\mathcal{A}_i}}$ and $\inf \varrho_{\mathcal{A}} = 0$, with $1/\sqrt{\infty} = 0$.*

*(iv)* Reaches. *For one table: if every $f_{\mathcal{A}_i}$ has a lattice expansion of reach $\zeta_i$ (Definition 20.28), the series connection has $f_{\mathcal{A}}$ with a lattice expansion of reach $\big(\sum_i \sqrt{\zeta_i}\big)^2$ (Proposition 20.37(ii)); if every $\check{f}_{\mathcal{A}_i}$ has one of reach $\zeta_i$, the parallel connection has $\check{f}_{\mathcal{A}}$ with one of reach $\big(\sum_i \sqrt{\zeta_i}\big)^2$.*

A series node adds the hinge values by $\Psi$ of its cells and a parallel node their hinge values by $K$; in the matching transform each is a product, and in the matching hinge measure a convolution. In the other transform neither node is a product: the dual orthant form of a series connection does not factor, since $K_{\mathcal{A}/g_0} = \prod_i K_{\mathcal{A}_i/g_i}\, e_{q-1}\big((\gamma^K_{\mathcal{A}_i})_i\big)$ and $1/\gamma^K_{\mathcal{A}} = \sum_i 1/\gamma^K_{\mathcal{A}_i}$ mix the cells. What changes at the edges where the kind of node changes is the transform, not the cells (Proposition 23.8(v)). The ceilings of Section 20.8 are read on a bunch as a whole, where the letters $K_{\mathcal{A}/g_0}$ and $\Psi_{\mathcal{A}/g_0}$ of the hinge minors of a series connection mix the cells and their zeros can raise a ceiling above the reach; at a node the cells are separate factors and the reach is read cell by cell (Lemma 20.30). So ceilings are to be applied to the prime nodes of a hinge tree, and reaches combined across series and parallel nodes by (iv).

## 23.8 Hinge trees

**Definition 23.15** (Hinge tree)**.** A *hinge tree* of a table on $\mathcal{E}$ is a finite tree $\mathfrak{T}$ with a core $\mathcal{H}_v$ on $\mathcal{E}_v \sqcup G_v$ at each vertex $v$, its *node*, where the $\mathcal{E}_v$ partition $\mathcal{E}$ and $G_v = \{g_{v,\mathfrak{e}}\}$ has one marker for each edge $\mathfrak{e}$ at $v$, such that the table is obtained from the $\mathcal{H}_v$ by the two-sums identifying $g_{v,\mathfrak{e}}$ with $g_{w,\mathfrak{e}}$ for every edge $\mathfrak{e} = \{v,w\}$. Rooted at a vertex $o$, every vertex $v \neq o$ has a parent edge $\mathfrak{e}_v$, and the *subtree cell* $\mathcal{A}_v$ is the substitution of the $\mathcal{H}_w$ over the vertices $w$ below $v$, $v$ included, with the hinge $g_{v,\mathfrak{e}_v}$; its cells are the subtree cells of the vertices directly below $v$. The tables of a family $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_j$ have a *common hinge tree* when one tree $\mathfrak{T}$, with the same sets $\mathcal{E}_v$ and markers $G_v$, is a hinge tree of every table, table $j$ with the nodes $\mathcal{H}_{v,j}$; the nodes then form the *node families* $F_{\mathcal{H}_v} = ((\mathcal{H}_{v,j}, \theta^{\mathcal{H}_{v,j}}); s_j, t_j)_j$ and, rooted at $o$, the subtree cells $\mathcal{A}_{v,j}$ the cell families $F_v = ((\mathcal{A}_{v,j}, \theta^{\mathcal{A}_{v,j}}); s_j, t_j)_j$ of the vertices $v \neq o$ (Definition 23.1).

**Theorem 23.16** (A connected table is a tree of prime tables)**.** *Call a connected table on at least four marks* prime *when it has no bunch of width one with at least two marks on each side. Every connected table on at least three marks whose weights factor across each bunch of width one at which it is decomposed, a list or a determinantal table for instance (Theorem 12.19), is obtained by substitutions along a tree from prime tables and the series and parallel nodes $\binom{S}{|S|-1}$ and $\binom{S}{1}$, $|S| \geq 3$ (Definition 23.13), the weights carried along. With no two series nodes and no two parallel nodes adjacent in the tree, the tree, the* canonical hinge tree *(Definition 23.15), is unique up to the names of the markers. The prime nodes of a list are therefore invariants of the list.*

By Lemma 23.5, applied leaf by leaf, the substitution along a tree does not depend on the order of the two-sums: its bases are the sets $\bigsqcup_v (T_v \cap \mathcal{E}_v)$ with $T_v$ a basis of $\mathcal{H}_v$ and every edge's marker in exactly one of the two bases at its ends. Every connected table has a hinge tree with one vertex, and decomposing at bunches of width one with at least two marks on each side, the weights factoring across each such bunch, markers counted as marks, refines it until no node has such a bunch. A node on at least four marks without one is prime (Theorem 23.16); the nodes on three marks, with the bases of $\binom{G}{2}$ or of $\binom{G}{1}$, $|G| = 3$, are series and parallel nodes, and merging adjacent

series nodes, and adjacent parallel nodes, gives the series and parallel nodes of any size; with no two series and no two parallel nodes adjacent this is the canonical hinge tree of Theorem 23.16.

**Theorem 23.17** (The period along a hinge tree)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ have real data on its balance wall and its tables a common hinge tree rooted at $o$, with the node families $F_{\mathcal{H}_v}$ and the cell families $F_v$ of the vertices $v \neq o$ (Definition 23.15), and let $F_{\mathcal{H}_v}[\mathbf{x}]$ be $F_{\mathcal{H}_v}$ twisted at the markers of the edges to the vertices $w$ directly below $v$ by $\mathbf{x}_w \in (0,\infty)^{2N-1}$, with the masses $\kappa_{F_w}$ there and $\mathbf{m}|_{\mathcal{E}_v}$ on $\mathcal{E}_v$ (Theorem 23.6).*

*(i) For $v \neq o$, $\kappa_{F_v}$ is the hinge wall of $F_{\mathcal{H}_v}[\mathbf{x}]$, and*

$$\int H\, d\big((\boldsymbol{\varrho}_{F_v})_*\phi_{F_v}\big) = \int \Big( \int H\, d\big((\boldsymbol{\varrho}_{F_{\mathcal{H}_v}[\mathbf{x}]})_*\phi_{F_{\mathcal{H}_v}[\mathbf{x}]}\big)\Big) \prod_w d\big((\boldsymbol{\varrho}_{F_w})_*\phi_{F_w}\big)(\mathbf{x}_w)$$

*for every Borel $H \geq 0$ on $(0,\infty)^{2N-1}$.*

*(ii) The period is*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int P_{F_{\mathcal{H}_o}[\mathbf{x}]}\big((\mathbf{m}|_{\mathcal{E}_o}, (\kappa_{F_w})_w);\mathbf{s},\mathbf{t}\big) \prod_w d\big((\boldsymbol{\varrho}_{F_w})_*\phi_{F_w}\big)(\mathbf{x}_w),$$

*the product over the vertices $w$ directly below the root.*

*(iii) The law of $F_v$ is determined by the tables $\mathcal{H}_{w,j}$ and the masses on $\mathcal{E}_w$ for the vertices $w$ below $v$, and by the exponents; nothing above $v$ enters it.*

*For one table the cell families are the subtree cells $\mathcal{A}_v$, with laws on $(0,\infty)$.*

Where the active letters read every subtree cell by one hinge value, as at a full lean of one table, every cell of the tree is pinned (Theorem 22.5), and Chapter 26 writes the period along the tree as a product and carries it to every load.

**Proposition 23.18** (What the work depends on)**.** *In Theorem 23.17:*

*(i) Two vertices whose subtree cells are carried onto each other, in every table, by one bijection keeping the tables, the weights, the masses and the hinge have equal hinge walls and equal laws. So the laws to be computed are one for each orbit of subtree cells under such bijections.*

*(ii) Let $s_j, t_j > 0$. At a series node $v$ whose cell families directly below fall into orbits $c_1, \dots, c_p$ with multiplicities $k_1, \dots, k_p$, $f_{F_v} = \prod_l f_{c_l}^{k_l}$ and $\kappa_{F_v} = \sum_l k_l \kappa_{c_l} - \big(\sum_l k_l - 1\big)|\mathbf{s}|$, $f_{c_l}$ and $\kappa_{c_l}$ the transform and the hinge wall of a member of $c_l$; at a parallel node $\check{f}_{F_v} = \prod_l \check{f}_{c_l}^{k_l}$ and $\kappa_{F_v} = \sum_l k_l \kappa_{c_l} - \big(\sum_l k_l - 1\big)|\mathbf{t}|$. The multiplicities enter as exponents.*

*(iii) A change at a vertex $v_0$ of the weights, of the masses keeping their total, or of the tables $\mathcal{H}_{v_0,j}$ keeping their ranks and their numbers of marks, changes no hinge wall and changes the laws of the cell families $F_v$ only for the vertices $v$ on the path from $v_0$ to the root. A change of masses at two vertices $v_0, v_1$ that keeps the total mass changes the hinge wall $\kappa_{F_v}$ exactly for the vertices $v$ whose subtree contains one of $v_0, v_1$ and not the other, the edges on the path between $v_0$ and $v_1$, by the change of the total mass below $\mathfrak{e}_v$.*

A hinge tree is read from the leaves up by a computation whose reusable data are the laws on the edges. Its cost is governed by the prime nodes, where cells interact through the node's own integral, and by the number of distinct subtrees, since equal subtrees have equal laws; series and parallel nodes cost no integration in the matching transform, and repeated subtrees enter as powers, as the hinge tree procedure makes explicit.

**Definition 23.19** (The hinge tree procedure)**.** *Given*: a common hinge tree of the tables of a family (Definition 23.15), the masses on the marks of its nodes, exponents $s_j, t_j > 0$, and a root. *Reusable data*: for every orbit of subtree cells (Proposition 23.18(i)) its hinge wall $\kappa$ and its law, kept in one of three representations: (a) the transform $f$ and the dual transform $\check f$ on a grid of $(\mathbf{x}, \mathbf{y})$ with $x_1 = 1$, of $y$ for one table; (b) the Mellin transform $\int \boldsymbol{\varrho}^{-\mathbf{z}} d\phi$ of the law on a vertical subspace of its domain of convergence, for one table $\int \varrho^{-z} d\phi$ on a vertical line of its strip; (c) for one table, the lattice expansion of $f$ at $y \to 0$, the residues of $y^{-z}\Gamma(z)\Gamma(z+s-\kappa)\int \varrho^{-z} d\phi$ to the left of the line (Theorem 20.34(i)). *Operations*, from the leaves up:

(1) a one-mark leaf, the one-mark cell family of ratio vector $\mathbf{x}_0$ (Definition 23.10), has $\kappa = m_e$, the law $\delta_{\mathbf{x}_0}$ and the transforms

$$f(\mathbf{x},\mathbf{y}) = \prod_j x_{0,j,K}^{-s_j}\, \mathsf{G}_{|\mathbf{s}|-m_e;(1,-1)}\Big(\sum_j \frac{x_j}{x_{0,j,K}},\ \sum_j y_j x_{0,j,\Psi}\Big),$$
$$\check f(\mathbf{x},\mathbf{y}) = \prod_j x_{0,j,\Psi}^{-t_j}\, \mathsf{G}_{|\mathbf{t}|-m_e;(1,-1)}\Big(\sum_j \frac{x_j}{x_{0,j,\Psi}},\ \sum_j y_j x_{0,j,K}\Big);$$

for one table the one-mark cell of ratio $x_0$ has the law $x_0^{(m_e-s)/2}\,\delta_{x_0}$ and

$$f(y) = x_0^{(m_e-s)/2}\mathsf{G}_{s-m_e;(1,-1)}(1, x_0 y), \qquad \check f(y) = x_0^{(m_e-s)/2-m_e}\mathsf{G}_{t-m_e;(1,-1)}(1, y/x_0);$$

(2) at a series node multiply the transforms of the vertices directly below, in (a) pointwise, in (b) by Mellin convolution of the Mellin transforms of Proposition 23.8(iii), for one table $\Gamma(z)\Gamma(z+s-\kappa)\int \varrho^{-z} d\phi$, in (c) by the product of lattice expansions (Lemma 20.30); at a parallel node the same with the dual transforms;

(3) along an edge where the kind of node changes, change the transform by Proposition 23.8(v): in (b) a reflection and a ratio of Gammas, in (a) a Mellin transform followed by an inverse Mellin transform, each one integral on $(0,\infty)^{2N-1}$, on the half-line for one table;

(4) at a prime node evaluate the Mellin transform of the law of the twisted node family, or its transform, as a period of the twisted node family, and integrate it against the laws of the vertices directly below (Theorem 23.6(ii));

(5) at the root evaluate Theorem 23.6(i); for a series root, with the cell families $F_i$ directly below and their transforms read at $x_1 = 1$, $\mathbf{x}' = (x_2, \dots, x_N)$, the period is

$$\prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j\ge 2} x_j^{s_j-1} \prod_j y_j^{t_j-1} \prod_i f_{F_i}\, d\mathbf{x}'\, d\mathbf{y},$$
$$\text{for one table}\quad \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{t-1} \prod_i f_{\mathcal{A}_i}(y)\, dy,$$

and for a parallel root the same with $s_j$ and $t_j$ swapped and the dual transforms (Theorem 23.9(ii)).

*Output*: the period, and with it every law on every edge, ready to be reused when the tree is changed (Chapter 24).

**Corollary 23.20** (Series-parallel families)**.** *Let the tables of a family with exponents $s_j, t_j > 0$ have a common hinge tree whose nodes are series and parallel nodes and whose leaves are one-mark cell families, and at every edge joining a series node to a parallel node let the law of the subtree below*

*have a Mellin transform finite at a real* $\mathbf{z}$ *with* $0 < z_{j,K} < s_j$ *and* $0 < z_{j,\Psi} < t_j$ *for every* $j$ *(Proposition 23.8(v)). Then its period is obtained from the transforms of the leaves, Grammas of the marks (Definition 23.19(1)), by pointwise products, by two integral transforms on* $(0,\infty)^{2N-1}$ *for each edge joining a series node to a parallel node, and by one final integral on* $(0,\infty)^{2N-1}$*; the number of integrations depends on the alternations of the tree and not on the number of marks, and subtrees repeated at a node enter as powers. For one table the integral transforms are on the half-line.*

# Chapter 24

# Repetition and the change of query

A construction that repeats a cell should not repeat the work, and a query that changes one weight or one mass should not restart the computation. A cell family repeated $k$ times at a series node enters its period as the $k$-th power of its transform, and repetition without bound forces its hinge wall to be $\sum_j s_j$; for one cell the growth in $k$ is then factorial, with the base $A_1/\min(s,t)$, $A_1$ the period of the cell with its hinge deleted, and a prefactor read from one further constant of the cell and one leading coefficient of the room (Theorems 24.1 and 24.2); a chain of repeated cell families is a power of one positive operator (Proposition 24.4); the weight of a hinge enters through a Mellin convolution whose two expansions are the two closings of the node of Section 20.7 (Theorem 24.5); and a change of data at one node moves only the laws on one path of the tree (Proposition 24.6).

## 24.1 Repetition at a node

For a cell family $F$ with plain tables $\mathcal{A}_j$ and $k \geq 1$, $F_k^{\mathrm{s}}$ and $F_k^{\mathrm{p}}$ are the cell families whose table $j$ is the series and the parallel connection of $k$ copies of $\mathcal{A}_j$ (Definition 23.13), with $F_1^{\mathrm{s}} = F_1^{\mathrm{p}} = F$; for one plain cell $\mathcal{A}$ they are written $\mathcal{A}_k^{\mathrm{s}}$ and $\mathcal{A}_k^{\mathrm{p}}$. The cell family is *series-balanced* when $\kappa_F = |\mathbf{s}|$ and *parallel-balanced* when $\kappa_F = |\mathbf{t}|$; by Definition 23.1 the first says that the deletions $\mathcal{A}_j \setminus g$, at the exponents $(s_j, t_j)$, form a family on its balance wall at the masses $\mathbf{m}$, $m_B = \sum_j \big(s_j r_{\mathcal{A}_j} + t_j(|B| - r_{\mathcal{A}_j})\big)$, and the second that the contractions $\mathcal{A}_j/g$ do; for one cell they say $\kappa_{\mathcal{A}} = s$ and $\kappa_{\mathcal{A}} = t$.

**Theorem 24.1** (How often a cell family repeats)**.** *Let $F$ be a cell family with plain tables and real exponents.*

*(i)* $\kappa_{F_k^{\mathrm{s}}} = |\mathbf{s}| + k(\kappa_F - |\mathbf{s}|)$ *and* $\kappa_{F_k^{\mathrm{p}}} = |\mathbf{t}| + k(\kappa_F - |\mathbf{t}|)$.

*(ii) If $F'$ is a cell family with $F_k^{\mathrm{s}} \oplus_g F'$ on its balance wall and of convergent period, then $0 < |\mathbf{s}| + k(\kappa_F - |\mathbf{s}|) < 2\sigma$, $\sigma$ the total strength (Definition 8.1). So series connections of any number of copies of $F$ occur in convergent periods only if $F$ is series-balanced; otherwise $k < |\mathbf{s}|/(|\mathbf{s}| - \kappa_F)$ when $\kappa_F < |\mathbf{s}|$ and $k < |\mathbf{t}|/(\kappa_F - |\mathbf{s}|)$ when $\kappa_F > |\mathbf{s}|$. For parallel connections the same holds with $|\mathbf{t}|$ in place of $|\mathbf{s}|$ and $|\mathbf{s}|$ in place of $|\mathbf{t}|$.*

The hinge wall of a series connection drifts by $\kappa_F - |\mathbf{s}|$ for every copy, and the room must absorb the drift; only a series-balanced cell family repeats without limit. It is the cell family whose deletions at the hinge form a balanced family, and for one cell the period of that deletion is the number that governs the repetition.

**Theorem 24.2** (Series repetition)**.** *Let $F$ be a series-balanced cell family on $B \sqcup \{g\}$ and $F' = ((\mathcal{A}'_j, \theta'_j); s_j, t_j)_j$ a cell family on $R \sqcup \{g\}$ with $\kappa_{F'} = |\mathbf{t}|$, plain tables, exponents $s_j, t_j > 0$, so that $F \overset{\mathrm{s}}{_k}\!\oplus_g F'$ is on its balance wall for every $k$; in this statement write $P(k)$ for its period, at the masses of the copies and of $F'$, and read the transforms at $x_1 = 1$, $\mathbf{x}' = (x_2, \dots, x_N)$.*

*(i) In $[0, \infty]$,*

$$P(k) = \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j \geq 2} x_j^{s_j - 1} \prod_j y_j^{t_j - 1} \, f_F^k f_{F'} \, d\mathbf{x}' \, d\mathbf{y},$$

*the $k$-th moment of the image under $f_F$ of the measure*

$$\prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \prod_{j \geq 2} x_j^{s_j - 1} \prod_j y_j^{t_j - 1} \, f_{F'}(\mathbf{x}, \mathbf{y}) \, d\mathbf{x}' \, d\mathbf{y}.$$

*Where finite, $P(k)^2 \leq P(k-1)P(k+1)$, and for $z \geq 0$, with $P(0)$ the period at $\mathbf{m}|_R$ of the family of the contractions $((\mathcal{A}'_j/g, \theta'_j); s_j, t_j)_j$, the hinge of $F'$ contracted,*

$$\sum_{k \geq 0} P(k) \frac{z^k}{k!} = \prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)} \int \prod_{j \geq 2} x_j^{s_j - 1} \prod_j y_j^{t_j - 1} \, e^{z f_F} f_{F'} \, d\mathbf{x}' \, d\mathbf{y}.$$

*For one cell, $P(k) = \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{t-1} f_{\mathcal{A}}(y)^k f_{\mathcal{A}'}(y) \, dy$ and $P(0) = \int \varrho_{\mathcal{A}'}^{-t} d\phi_{\mathcal{A}'} = P_{\mathcal{A}'/g}(\mathbf{m}|_R; s, t)$.*

*(ii) For one cell $\mathcal{A}$ and one cell $\mathcal{A}'$, at the exponents $(s, t)$, let $\mathcal{A} \backslash g$ have every proper wall positive at the masses of $\mathcal{A}$ and the exponents $(s, t)$, and put*

$$A_1 = \int_{\Delta_B} d\phi_{\mathcal{A}} = P_{\mathcal{A}\backslash g}(\mathbf{m}|_B; s, t), \qquad A_0 = 2A_1(\log\Gamma)'(1) - \int_{\Delta_B} \log \varrho_{\mathcal{A}} \, d\phi_{\mathcal{A}}.$$

*Then $\int \varrho_{\mathcal{A}}^{-z} d\phi_{\mathcal{A}}$ is finite on a neighbourhood of $z = 0$, and for every $\varepsilon \in (0, 1)$ at which $\int \varrho_{\mathcal{A}}^{\varepsilon} d\phi_{\mathcal{A}} < \infty$ there is $C$ with $\big| f_{\mathcal{A}}(y) - A_1(-\log y) - A_0 \big| \leq C y^{\varepsilon}$ for all $y > 0$.*

*(iii) Let moreover $\int \varrho_{\mathcal{A}'}^{-z} d\phi_{\mathcal{A}'}$ be finite on a neighbourhood of $z = \max(0, t - s)$, and let the* polynomial of the room *be*

$$\mathcal{R}(v) = \begin{cases} \Gamma(s-t) \int d\phi_{\mathcal{A}'} & (s > t), \\ \Gamma(t-s) \int \varrho_{\mathcal{A}'}^{s-t} d\phi_{\mathcal{A}'} & (s < t), \\ v \int d\phi_{\mathcal{A}'} + 2(\log\Gamma)'(1) \int d\phi_{\mathcal{A}'} - \int \log \varrho_{\mathcal{A}'} \, d\phi_{\mathcal{A}'} & (s = t), \end{cases}$$

*with the coefficients $\mathcal{R}(v - A_0/A_1) = \sum_i \mathcal{R}_i v^i$; let $P(1) < \infty$. Then there is $\varepsilon > 0$ such that for every $\vartheta \in \big(\min(s,t)/(\min(s,t) + \varepsilon), 1\big)$*

$$P(k) = \frac{A_1^k \, e^{\min(s,t) A_0/A_1}}{\Gamma(s)\Gamma(t)} \sum_i \mathcal{R}_i \, \frac{\Gamma(k + i + 1)}{\min(s,t)^{k+i+1}} \big(1 + O(\vartheta^k)\big).$$

*In particular $P(k) \sim \frac{c \, e^{\min(s,t) A_0/A_1}}{\Gamma(s)\Gamma(t)} \Gamma(k + 1 + j) A_1^k \min(s,t)^{-k-1-j}$, with $j = 1$ and $c = \int d\phi_{\mathcal{A}'}$ when $s = t$ and $j = 0$ and $c = \mathcal{R}$ otherwise: the growth is factorial, with the base $A_1/\min(s,t)$.*

*(iv) Under the hypotheses of (iii) the series of (i) converges for $0 \leq z < \min(s,t)/A_1$ and diverges at $z = \min(s,t)/A_1$.*

So a repeated cell family enters its construction as a power, and the whole sequence $P(k)$ is the sequence of moments of one positive measure: log-convex, with an exponential generating function that is one integral. For one cell the factorial growth is not a property of the room: the transform of a series-balanced cell grows like $A_1\log(1/y)$ at small scales, its $k$-th power concentrates where $\log(1/y)$ is of the order $k/\min(s,t)$, and there only two numbers of the cell survive, the period $A_1$ of the cell with its hinge deleted and the constant $A_0$, and only the first lattice term of the room, through $\min(s,t)$ and $\mathcal{R}$. The error is exponentially small, not a power of $1/k$: the next terms of both lattice expansions sit at distance $\varepsilon$ to the right and contribute in the ratio $\big(\min(s,t)/(\min(s,t)+\varepsilon)\big)^k$. Theorem 23.3 reads the series closure of any cells, for any family, through their transforms; for $k$ equal cells at the centre it is (i) with $k-1$ copies at the series node and one copy as the room.

**Corollary 24.3** (Parallel repetition)**.** *Let $F$ be a parallel-balanced cell family on $B\sqcup\{g\}$ and $F'=((\mathcal{A}'_j,\theta'_j);s_j,t_j)_j$ a cell family on $R\sqcup\{g\}$ with $\kappa_{F'}=|\mathbf{s}|$, plain tables, exponents $s_j,t_j>0$, and in this statement write $P(k)$ for the period of $F_k^{\mathrm{p}}\oplus_g F'$, the transforms read at $x_1=1$, $\mathbf{x}'=(x_2,\dots,x_N)$. Then*

$$P(k)=\prod_j\frac{1}{\Gamma(s_j)\Gamma(t_j)}\int\prod_{j\ge2}x_j^{t_j-1}\prod_j y_j^{s_j-1}\,\check f_F^{\,k}\check f_{F'}\,d\mathbf{x}'\,d\mathbf{y},$$

*with $P(0)$ the period at $\mathbf{m}|_R$ of the family of the deletions $((\mathcal{A}'_j\setminus g,\theta'_j);s_j,t_j)_j$; for one cell, $P(k)=\frac{1}{\Gamma(s)\Gamma(t)}\int_0^\infty y^{s-1}\check f_{\mathcal{A}}(y)^k\check f_{\mathcal{A}'}(y)\,dy$. For one cell $\mathcal{A}$ and one cell $\mathcal{A}'$ let $\mathcal{A}/g$ have every proper wall positive at the masses of $\mathcal{A}$ and the exponents $(s,t)$, and put*

$$A_1=\int\varrho_{\mathcal{A}}^{-t}d\phi_{\mathcal{A}}=P_{\mathcal{A}/g}(\mathbf{m}|_B;s,t),\qquad A_0=2A_1(\log\Gamma)'(1)+\int\varrho_{\mathcal{A}}^{-t}\log\varrho_{\mathcal{A}}\,d\phi_{\mathcal{A}}.$$

*Then Theorem 24.2(ii) to (iv) hold with $s$ and $t$ swapped and with these $A_1$ and $A_0$, the dual transform $\check f_{\mathcal{A}}$ in place of the transform $f_{\mathcal{A}}$, the finiteness of $\int\varrho_{\mathcal{A}}^{-z}d\phi_{\mathcal{A}}$ near $z=t$ in place of near $z=0$, the condition $\int\varrho_{\mathcal{A}}^{-t-\varepsilon}d\phi_{\mathcal{A}}<\infty$ in place of $\int\varrho_{\mathcal{A}}^{\varepsilon}d\phi_{\mathcal{A}}<\infty$, the finiteness of $\int\varrho_{\mathcal{A}'}^{-z}d\phi_{\mathcal{A}'}$ near $z=\min(s,t)$ as the hypothesis on $\mathcal{A}'$, and the polynomial of the room*

$$\mathcal{R}(v)=\begin{cases}\Gamma(t-s)\int\varrho_{\mathcal{A}'}^{-s}d\phi_{\mathcal{A}'} & (s<t),\\ \Gamma(s-t)\int\varrho_{\mathcal{A}'}^{-t}d\phi_{\mathcal{A}'} & (s>t),\\ v\int\varrho_{\mathcal{A}'}^{-s}d\phi_{\mathcal{A}'}+2(\log\Gamma)'(1)\int\varrho_{\mathcal{A}'}^{-s}d\phi_{\mathcal{A}'}+\int\varrho_{\mathcal{A}'}^{-s}\log\varrho_{\mathcal{A}'}\,d\phi_{\mathcal{A}'} & (s=t),\end{cases}$$

*whose leading coefficient is the constant $c$ of the equivalent in (iii).*

## 24.2 Chains

**Proposition 24.4** (A chain is a power of one operator)**.** *Let a* chain cell family *$U$ have in every table a hinged core $\mathcal{U}_j$ (Definition 23.4) with one further marker $g_{\mathrm{in}}$, its hinge $g_{\mathrm{out}}$, own marks $\mathcal{E}_{\mathcal{U}}$ with masses, and real exponents $(s_j,t_j)$; further fixed cell families may be substituted into $U$ beforehand. For a cell family $F_0$ let $F_{l+1}=U\langle F_l\rangle$, table by table, the cell at $g_{\mathrm{in}}$.*

*(i) $\kappa_{F_{l+1}}=\kappa_{F_l}+\kappa_U$, with $\kappa_U=m_{\mathcal{E}_{\mathcal{U}}}-\sum_j\big(s_j(r_{\mathcal{U}_j}-1)+t_j(|\mathcal{E}_{\mathcal{U}}|+1-r_{\mathcal{U}_j})\big)$ the hinge wall of $U$ with the mass $0$ on $g_{\mathrm{in}}$. For a fixed cell family $F'$, the families $F_l\oplus_g F'$ are all on their balance walls exactly when $F_0\oplus_g F'$ is and $\kappa_U=0$; then $\kappa_{F_l}=\kappa_{F_0}$.*

*(ii) Let $\kappa_U = 0$, and in this statement write $\lambda_l$ for the law of $F_l$ and $\lambda^U_{\mathbf{x}}$ for the law of $U$ twisted at $g_{\text{in}}$ by $\mathbf{x} \in (0,\infty)^{2N-1}$ (Definition 22.1) with the mass $\kappa_{F_0}$ there. On positive measures $\mu$ and nonnegative functions $H$ on $(0,\infty)^{2N-1}$ put*

$$\mathcal{Q}\mu = \int \lambda^U_{\mathbf{x}}\, d\mu(\mathbf{x}), \qquad (\mathcal{Q}^* H)(\mathbf{x}) = \int H\, d\lambda^U_{\mathbf{x}},$$

$$h_{F'}(\mathbf{x}) = \int \Big( \int_0^\infty v^{\kappa_{F_0}-1} \prod_j (x_{j,K} v + x'_{j,K})^{-s_j} (x_{j,\Psi} v + x'_{j,\Psi})^{-t_j}\, dv \Big)\, d\big((\varrho_{F'})_* \phi_{F'}\big)(\mathbf{x}'),$$

*the inner integral the period of the twisted core family of Theorem 23.6(i) with the core $\binom{\{g_1,g_2\}}{1}$ in every table. Then $\lambda_l = \mathcal{Q}^l \lambda_0$ and $P_{F_l \oplus_g F'} = \int (\mathcal{Q}^{*l} h_{F'})\, d\lambda_0$; for $z \geq 0$, $\sum_l z^l P_{F_l \oplus_g F'} = \int \sum_l z^l (\mathcal{Q}^{*l} h_{F'})\, d\lambda_0$, in $[0,\infty]$.*

*(iii) If $H > 0$ and $M > 0$ have $\mathcal{Q}^* H \leq MH$ and $h_{F'} \leq CH$, then $P_{F_l \oplus_g F'} \leq C M^l \int H\, d\lambda_0$; if $H' \geq 0$ and $M' > 0$ have $\mathcal{Q}^* H' \geq M' H'$ and $h_{F'} \geq c\, H'$, then $P_{F_l \oplus_g F'} \geq c\, M'^l \int H'\, d\lambda_0$.*

*(iv) Let the tables be plain and $s_j, t_j > 0$. If $U$ is the series core on $\{g_{\text{out}}, g_{\text{in}}, g'\}$ with a fixed cell family $F''$ at $g'$, then $f_{F_{l+1}} = f_{F''} f_{F_l}$ and $\kappa_U = \kappa_{F''} - |\mathbf{s}|$; if it is the parallel core, then $\check{f}_{F_{l+1}} = \check{f}_{F''} \check{f}_{F_l}$ and $\kappa_U = \kappa_{F''} - |\mathbf{t}|$.*

*For one table the laws are measures on $(0,\infty)$, the twist is one number, and the inner integral of $h_{F'}$ is the period of Lemma 23.7.*

A chain is one kernel applied $l$ times: the law of the chain cell family twisted at $g_{\text{in}}$, computed once, carries the law of the chain from one length to the next, and the period at length $l$ pairs the $l$-th iterate with the room. The work for all lengths up to $l$ is $l$ applications of one operator, and for one length $\lceil \log_2 l \rceil$ compositions of kernels. A series chain cell family is diagonal in the transform and a parallel one in the dual transform, so for them the operator is a multiplication and the length is an exponent; a chain cell family that alternates the two kinds mixes the two hinge values, and its operator is not diagonal in either. Theorem 22.8 is the same chain read on functions of the twists, a family glued to copies of itself at two marks, with the pinned case $P_F(\mathbf{m};\mathbf{s},\mathbf{t})^k / \Gamma_2 \binom{m_{g_{\text{in}}},\ m_{g_{\text{out}}}}{m_{g_{\text{in}}} + m_{g_{\text{out}}}}^{k-1}$; Proposition 24.4 reads it on laws, and positive test functions bound it from both sides. Every polynomial Gamma moment of a chain is the power of one finite matrix (Theorem 27.5).

## 24.3 The weight of a hinge

**Theorem 24.5** (The weight of a hinge)**.** *For plain cells $\mathcal{A}$ on $B \sqcup \{g\}$ and $\mathcal{A}'$ on $R \sqcup \{g\}$ with exponents $s, t > 0$, and $w > 0$, let $w\mathcal{A}$ be $\mathcal{A}$ with the weights of its bases through $g$ multiplied by $w$, so that in $(w\mathcal{A}) \oplus_g \mathcal{A}'$ the bases made of a basis of $\mathcal{A}/g$ and a basis of $\mathcal{A}' \setminus g$, the second kind of Lemma 21.2, carry the factor $w$; let it be on its balance wall, put $q = w^2$, and in this statement write $\Pi(q) = q^{(s-\kappa_{\mathcal{A}})/2} P_{(w\mathcal{A}) \oplus_g \mathcal{A}'}(\mathbf{m}; s, t)$.*

*(i) For a cell family $F$ with plain tables $\mathcal{A}_j$ and $\mathbf{w} \in (0,\infty)^N$, let $\mathbf{w}F$ be $F$ with the weights of the bases through $g$ multiplied by $w_j$ in table $j$. Its hinge wall is $\kappa_F$, its ratio vector has the coordinates $(w_1/w_j)\, \gamma^K_{\mathcal{A}_j} / \gamma^K_{\mathcal{A}_1}$ and $w_1 w_j\, \gamma^\Psi_{\mathcal{A}_j} / \gamma^K_{\mathcal{A}_1}$, the hinge values of $F$ being read, $\phi_{\mathbf{w}F} = w_1^{\kappa_F} \prod_j w_j^{-s_j}\, \phi_F$, and, for $s_j, t_j > 0$, $f_{\mathbf{w}F}(\mathbf{x},\mathbf{y}) = f_F(\mathbf{w} \circ \mathbf{x}, \mathbf{w} \circ \mathbf{y})$ and $\check{f}_{\mathbf{w}F}(\mathbf{x},\mathbf{y}) = \prod_j w_j^{-s_j - t_j}\, \check{f}_F(\mathbf{x}/\mathbf{w}, \mathbf{y}/\mathbf{w})$, the products and quotients taken coordinatewise. For one cell, $\varrho_{w\mathcal{A}} = w^2 \varrho_{\mathcal{A}}$, $\phi_{w\mathcal{A}} = w^{\kappa_{\mathcal{A}} - s} \phi_{\mathcal{A}}$, $f_{w\mathcal{A}}(y) = w^{\kappa_{\mathcal{A}} - s} f_{\mathcal{A}}(w^2 y)$ and $\check{f}_{w\mathcal{A}}(y) = w^{-\kappa_{\mathcal{A}} - s} \check{f}_{\mathcal{A}}(y/w^2)$.*

*(ii)* $\Pi(q) = \frac{1}{\Gamma(s)\Gamma(t)} \int_0^\infty y^{t-1} f_{\mathcal{A}}(qy) f_{\mathcal{A}'}(y)\, dy$, *a decreasing function of* $q$, *and for real* $z$ *with* $\max(0, \kappa_{\mathcal{A}} - s) < z < \min(t, \kappa_{\mathcal{A}})$, *in* $(0, \infty]$,

$$\int_0^\infty q^{z-1} \Pi(q)\, dq = \frac{\Gamma(z)\Gamma(z+s-\kappa_{\mathcal{A}}) \int \varrho_{\mathcal{A}}^{-z} d\phi_{\mathcal{A}} \cdot \Gamma(t-z)\Gamma(\kappa_{\mathcal{A}} - z) \int \varrho_{\mathcal{A}'}^{z-t} d\phi_{\mathcal{A}'}}{\Gamma(s)\Gamma(t)}.$$

*(iii) If this is finite at a real* $c$ *in that interval, then* $\Pi$ *is finite and continuous on* $(0, \infty)$ *and is the value there of the function* $\frac{1}{2\pi i} \int_{(c)} q^{-z} (\cdots)\, dz$, *with* $(\cdots)$ *the right side of the display in (ii), holomorphic on the sector* $|\arg q| < 2\pi$ *of the logarithmic cover of* $\mathbb{C} \setminus \{0\}$.

*(iv) In the setting of Section 20.7, with* $\mathcal{A}$ *the cell of the bunch and* $\mathcal{A}'$ *that of the room, let* $f_{\mathcal{A}}$ *have a lattice expansion* $\sum_{\eta,j} \beta_{\eta,j} y^\eta (-\log y)^j$ *of every reach above* $\operatorname{reach}(B)$ *(Definition 20.28), and let* $\inf \varrho_{\mathcal{A}'} > 0$. *Then for* $0 < q < \inf \varrho_{\mathcal{A}'} / \operatorname{reach}(B)$, *absolutely,*

$$\Pi(q) = \frac{1}{\Gamma(s)\Gamma(t)} \sum_{\eta,j} \beta_{\eta,j}\, q^\eta \sum_{i=0}^{j} \binom{j}{i} (-\log q)^{j-i} \times (-\partial_z)^i \Big[ \Gamma(z)\Gamma(z+s-\kappa_{\mathcal{A}'}) \int \varrho_{\mathcal{A}'}^{-z} d\phi_{\mathcal{A}'} \Big]_{z=t+\eta};$$

*at* $q = 1$ *this is the closing of Theorem 20.34(ii) toward the level* $d$. *Swapping* $\mathcal{A}$ *and* $\mathcal{A}'$, $q^{-t}$ *times the same expansion in* $1/q$ *holds for* $q > \operatorname{reach}(R)/\inf \varrho_{\mathcal{A}}$, *and is the closing toward the level* $r$. *The two ranges never overlap.*

The closings of a node of Section 20.7 are therefore not two separate devices but the two expansions, at $q \to 0$ and at $q \to \infty$, of one function of the weight of its hinge, holomorphic on two sheets around the origin and given everywhere by the contour integral of (iii), whose integrand is the product of the Mellin transforms of the two sides over $\Gamma(s)\Gamma(t)$ (Proposition 23.8(iii)). The unweighted period is the value at $q = 1$; a closing reaches it when 1 lies inside its range, its terms do not tend to zero when 1 lies beyond it, and at the common end of the two ranges the root rate is one and domination decides nothing (Theorem 20.39(ii) to (iv), the ranges being read at the reaches of the two sides); and the coefficients of each expansion are the reusable lattice data of one side paired with the Mellin data of the other, the same for every weight.

## 24.4 Changing masses and weights

**Proposition 24.6** (Carrying a calculation through a change)**.** *Let the tables of a family have a common hinge tree rooted at* $o$ *(Definition 23.15), with every law computed by Definition 23.19.*

*(i) A change at a vertex* $v'$ *of the weights, of the masses keeping their total, or of the tables* $\mathcal{H}_{v',j}$ *keeping their ranks and their numbers of marks, is carried through by recomputing the laws at the vertices on the path from* $v'$ *to* $o$, *each from the new law of the vertex directly below it on the path and the old laws of the other vertices directly below it, and then the root.*

*(ii) A change of the weight of the hinge of the edge* $\mathfrak{e}_v$ *by the factor* $w_j$ *in table* $j$ *replaces the law of* $F_v$ *by* $w_1^{\kappa_{F_v}} \prod_j w_j^{-s_j}$ *times its image under* $(x_{j,K}, x_{j,\Psi})_j \mapsto \big((w_1/w_j)\, x_{j,K},\, w_1 w_j\, x_{j,\Psi}\big)_j$ *(Theorem 24.5(i)), for one table by* $w^{\kappa_{\mathcal{A}_v} - s}$ *times its image under* $x \mapsto w^2 x$; *nothing below* $v$ *changes, and above* $v$ *(i) applies along the path from* $v$ *to* $o$.

*(iii) A change of masses at two vertices* $v', v''$ *keeping the total mass changes the hinge walls named in Proposition 23.18(iii), those on the edges of the path between* $v'$ *and* $v''$, *and is carried through by recomputing the laws on the paths from* $v'$ *and from* $v''$ *to* $o$.

A query about one part of a construction is answered by the part of the computation that sees it. The laws on the edges off the path are kept, and with them the work that made them; a construction with many equal subtrees, or one asked many queries about one of its nodes, pays for each subtree once and for each query along one path.

# Chapter 25

# Holds and the lean convolution

A table built along a hinge tree is read from the laws of its subtrees, and at a prime node the twisted core is integrated over its own simplex once for every value of the twists (Definition 23.19(4)). Holding the marks removes that integral, and with it the dependence of the reusable data on the query. Every mark of every letter of a family can be twisted, and a twist can be *held* at a load, by a Mellin transform. Twists separate any word in positive letters into its factors, its period becoming one fibre integral of the periods of the factors (Theorem 25.4), with the convergence read off the Newton sums (Theorem 25.2); the chapter opens with these words. Holding every mark separates the letters of the family completely: the hold is the product of the periods of its letters alone, the periods of its tables at their full leans, each read at its own load (Lemma 25.11). Undoing the hold writes the period at all exponents, every mass and every twist as one fibre integral of dimension $(2N-1)(n-1)$ over the splits of the masses among the letters (Theorem 25.12). The periods at the full leans of a table built along a hinge tree are the products of those of its nodes over Beta corners at its edges, with no integral at all (Theorem 26.5), so the period of a family of such tables is one fibre integral of node data (Theorem 26.13).

## 25.1 Newton words

Twists separate any product of powers of positive letters into its factors, and nothing in this uses the exchange property. The convergence criterion of Theorem 8.5, whose sectors rest on the greedy bases that need the exchange property (Lemma 8.3), extends to such products through the Gramma and its domain (Theorem 25.2), and on its domain the separation holds (Theorem 25.4). The holds of the next section are the case of the letters of a family (Lemma 25.11).

**Definition 25.1** (Positive letters; Newton words and sums)**.** A *positive letter* on $\mathcal{E}$ is a nonzero homogeneous polynomial $L = \sum_k c_k x^{t_k}$ of degree $\deg L \geq 1$ with positive coefficients $c_k$ on distinct exponents $t_k \in \mathbb{N}^{\mathcal{E}}$, its *support*; its Newton polytope is $\mathcal{N}_L = \operatorname{conv}\{t_k\}$, in the hyperplane $|x| = \deg L$. A *product of powers* of positive letters $W = \prod_i L_i^{-w_i}$, with exponents $w_i \in \mathbb{C}$, has the degree $D = \sum_i w_i \deg L_i$, and for masses $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ with $m_{\mathcal{E}} = D$ its period $P_W(\mathbf{m})$ is the integral of the form $\alpha^{\mathbf{m}-\mathbf{1}} W \, d\alpha$ over a section (Lemma 16.2), wherever it converges absolutely; it does not depend on the section. It is a *Newton word* when every $w_i > 0$, with the *Newton sum* $\mathcal{N}(W) = \sum_i w_i \mathcal{N}_{L_i}$, in the hyperplane $|x| = D$. A Newton word is *split into factors* when it is written $W = W_1 \cdots W_p$, each factor a product of some of its letters with their exponents. The family word of a family is a product of powers of its letters, with $P_W(\mathbf{m}) = P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$, and a Newton word in its active letters when its exponents are nonnegative; for a table $\mathcal{N}_K = P(\mathcal{H})$ and $\mathcal{N}_\Psi = P(\mathcal{H}^*)$, and for a family $\mathcal{N}(W)$ is the Newton sum $\mathcal{N}_F$ of Theorem 8.19.

**Theorem 25.2** (Convergence of a Newton word)**.** *$P_W(\mathbf{m})$ converges exactly when $\mathcal{N}(W)$ has dimension $n-1$ and $\operatorname{Re}\mathbf{m} \in \operatorname{ri}\mathcal{N}(W)$; it is then holomorphic in $\mathbf{m}$ on the balance hyperplane. For a family of tables this is the condition that every proper wall be positive (Theorems 8.7 and 8.19(iii)).*

**Proposition 25.3** (One positive letter)**.** *Let $L = \sum_k c_k x^{t_k}$ be a positive letter with $\dim\mathcal{N}_L = n-1$, $\mathbf{a} \in \mathbb{C}^{\mathcal{E}}$ with $\operatorname{Re}|\mathbf{a}| > 0$, and $w = |\mathbf{a}|/\deg L$.*

*(i)* $P_{L^{-w}}(\mathbf{a}) = \deg L\, \mathsf{G}_{\mathbf{a};(t_k)_k}\big((c_k)_k\big)/\Gamma(w)$*, both sides converging together.*

*(ii)* $P_{L^{-w}}(\mathbf{a})$ *converges exactly when* $\operatorname{Re}\mathbf{a} \in (\operatorname{Re} w)\operatorname{ri}\mathcal{N}_L$*.*

*(iii) For real $\mathbf{a}$ at which it converges, every $\boldsymbol{\eta} \in \mathbb{R}^{\mathcal{E}}$ with $|\boldsymbol{\eta}| = 0$ and every $\varepsilon \in (0,1)$,*

$$\big|P_{L^{-w}}(\mathbf{a}+i\boldsymbol{\eta})\big| \le \min\Big(1,\ \big(\sin\tfrac{\pi\varepsilon}{2}\big)^{-w} e^{-(1-\varepsilon)h_L(\boldsymbol{\eta})}\Big)\, P_{L^{-w}}(\mathbf{a}),$$

*with $h_L$ the support function of $\{y :\ |\langle t_k, y\rangle| < \frac{\pi}{2}$ for every $k\}$, a norm on $\mathbb{R}^{\mathcal{E}}$.*

**Theorem 25.4** (Separation into factors)**.** *Let $W = W_1 \cdots W_p$ be split into factors with $\dim\mathcal{N}(W_i) = n-1$ for every $i$, of degrees $D_i$, and let $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ have $m_{\mathcal{E}} = D$. Points $(\mathbf{c}_i)_i$ with $\mathbf{c}_i \in \operatorname{ri}\mathcal{N}(W_i)$ and $\sum_i \mathbf{c}_i = \operatorname{Re}\mathbf{m}$ exist exactly when $\operatorname{Re}\mathbf{m} \in \operatorname{ri}\mathcal{N}(W)$, and on each of them, for a fixed mark $e_0$ and all twists $\mathbf{x}_i \in (0,\infty)^{\mathcal{E}}$ of the factors,*

$$\int_{\alpha_{e_0}=1} \alpha^{\mathbf{m}-\mathbf{1}} \prod_i W_i(\mathbf{x}_i \circ \alpha)\, d\alpha' = \int_{\mathbf{a}_1+\cdots+\mathbf{a}_p=\mathbf{m},\ |\mathbf{a}_i|=D_i\ (i<p)} \ \prod_i P_{W_i}(\mathbf{a}_i)\, \mathbf{x}_i^{-\mathbf{a}_i},$$

*a fibre integral of dimension $(p-1)(n-1)$, absolutely convergent, $P_{W_i}$ the period of the factor $W_i$ at its own masses.*

**Corollary 25.5** (Exponents add on one letter)**.** *For a positive letter $L$ with $\dim\mathcal{N}_L = n-1$, exponents $w, w' > 0$ and $\operatorname{Re}\mathbf{m} \in (w+w')\operatorname{ri}\mathcal{N}_L$,*

$$P_{L^{-w-w'}}(\mathbf{m}) = \int_{\mathbf{a}+\mathbf{a}'=\mathbf{m},\ |\mathbf{a}|=w\deg L} P_{L^{-w}}(\mathbf{a})\, P_{L^{-w'}}(\mathbf{a}').$$

*At $L = e_1(x) = \sum_{e\in\mathcal{E}} x_e$, $P_{L^{-w}}(\mathbf{a}) = \Gamma_1^n \begin{bmatrix}\mathbf{a}\\ w\end{bmatrix}$ and this is Theorem 4.6; for an identically self-dual table $\Psi = K$, and the lean convolution is this identity at $L = K$, $w = s$, $w' = t$, $P_{\mathcal{H}}(\mathbf{m}; s, t) = P_{\mathcal{H}}(\mathbf{m}; s+t, 0)$.*

**Proposition 25.6** (Simplicial letters)**.** *If the support of a positive letter $L$ consists of $n$ linearly independent exponents $t_1, \dots, t_n$, so that $L$ is* simplicial*, with the configuration $t = (t_1, \dots, t_n)$, then for $\mathbf{a}$ with $|\mathbf{a}| = w\deg L$ and the shares $\mathbf{b} = t^{-1}\mathbf{a}$,*

$$P_{L^{-w}}(\mathbf{a}) = \frac{\deg L}{|\det t|}\, \Gamma_1^n \begin{bmatrix}\mathbf{b}\\ w\end{bmatrix} \prod_k c_k^{-b_k}, \qquad |\mathbf{b}| = w,$$

*converging exactly when $\operatorname{Re}\mathbf{b} > 0$: the Gramma of Proposition 25.3(i) at a square configuration (Corollary 2.6). A table that is one molecule has a simplicial basis letter exactly when it has $n$ bases.*

**Lemma 25.7** (Concatenation of Grammas)**.** *Let $t^{(1)}, \dots, t^{(p)}$, $p \ge 2$, be configurations on $\mathcal{E}$ with $t^{(2)}, \dots, t^{(p)}$ spanning $\mathbb{R}^{\mathcal{E}}$, $\boldsymbol{\xi}^{(1)}, \dots, \boldsymbol{\xi}^{(p)}$ coefficients, and $\mathbf{c}_i \in \operatorname{int}\operatorname{cone}(t^{(i)})$ for $i \ge 2$ with $\operatorname{Re}\nu - \sum_{i\ge 2}\mathbf{c}_i \in \operatorname{int}\operatorname{cone}(t^{(1)})$. Then*

$$\mathsf{G}_{\nu;t^{(1)}\oplus\cdots\oplus t^{(p)}}\big(\boldsymbol{\xi}^{(1)}\oplus\cdots\oplus\boldsymbol{\xi}^{(p)}\big)$$
$$=\frac{1}{(2\pi i)^{(p-1)n}}\int_{\mathbf{c}_2+i\mathbb{R}^{\mathcal{E}}}\cdots\int_{\mathbf{c}_p+i\mathbb{R}^{\mathcal{E}}}\mathsf{G}_{\nu-\sum_{i\geq 2}\mathbf{w}_i;t^{(1)}}(\boldsymbol{\xi}^{(1)})\prod_{i\geq 2}\mathsf{G}_{\mathbf{w}_i;t^{(i)}}(\boldsymbol{\xi}^{(i)})\,d\mathbf{w}_2\cdots d\mathbf{w}_p,$$

*absolutely convergent: a Gramma of a concatenated configuration is the Mellin convolution of the Grammas of its parts.*

For one table the concatenation $(\mathbf{1}_T)_T\oplus(\mathbf{1}_{\mathcal{E}\setminus T})_T$ with the coefficients $\boldsymbol{\theta}\oplus\lambda\boldsymbol{\theta}$ is the Gramma $\mathsf{G}_{\mathbf{m};(\mathbf{1}_T)_T\oplus(\mathbf{1}_{\mathcal{E}\setminus T})_T}(\boldsymbol{\theta}\oplus\lambda\boldsymbol{\theta})=\int x^{\mathbf{m}-\mathbf{1}}e^{-K(x)-\lambda\Psi(x)}dx$, and Lemma 25.7 writes it as $(2\pi i)^{-n}\int\mathsf{G}_{\mathbf{m}-\mathbf{b};(\mathbf{1}_T)_T}(\boldsymbol{\theta})\,\mathsf{G}_{\mathbf{b};(\mathbf{1}_{\mathcal{E}\setminus T})_T}(\boldsymbol{\theta})\,\lambda^{-|\mathbf{b}|/r^*}d\mathbf{b}$, by the torus homogeneity of $\mathsf{G}_{\mathbf{b};(\mathbf{1}_{\mathcal{E}\setminus T})_T}$. The Mellin transform in $\lambda$ at $t$ pins $|\mathbf{b}|=r^*t$, the pin contributing the factor $r^*$, and through Proposition 25.10(i) returns $\Gamma(s)\Gamma(t)/r$ times the lean convolution of Theorem 25.12 at $\mathbf{x}=\mathbf{1}$; Proposition 26.22 states the resulting identity $\int_0^\infty\lambda^{t-1}\mathsf{G}_{\mathbf{m};(\mathbf{1}_T)_T\oplus(\mathbf{1}_{\mathcal{E}\setminus T})_T}(\boldsymbol{\theta}\oplus\lambda\boldsymbol{\theta})\,d\lambda=\Gamma(s)\Gamma(t)P_{\mathcal{H}}(\mathbf{m};s,t)/r$. So at fixed masses the lean is the Mellin variable of the relative scale $\lambda$ of the two letters, and the two-step integration separates what the masses fix, the Gramma of the concatenated configuration, from what the exponents choose, one Mellin transform in one variable.

## 25.2 Holds and the periods at the full leans

**Definition 25.8** (Holding)**.** Let $F=((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ be a family on $\mathcal{E}$, $n=|\mathcal{E}|\geq 2$, with exponents $s_j,t_j>0$, whose tables are each one molecule, table $j$ of rank $r_j$ and corank $r_j^*=n-r_j$, so that $0<r_j<n$ and the bases of every table span $\mathbb{R}^{\mathcal{E}}$: a vector orthogonal to every $\mathbf{1}_T$, $T\in\mathcal{B}_j$, is constant by Lemma 7.8, and then zero since $r_j>0$. $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the period (Theorem 8.7) on the balance wall $m_{\mathcal{E}}=\sum_j(r_js_j+r_j^*t_j)$, and $F[\mathbf{x}]$ is $F$ twisted by $\mathbf{x}=(\mathbf{x}_j^K,\mathbf{x}_j^\Psi)_j$ (Definition 22.1). For $G\subseteq\mathcal{E}$, masses $\mathbf{m}$ and real loads $\boldsymbol{\beta}=(\boldsymbol{\beta}_j^K,\boldsymbol{\beta}_j^\Psi)_j$, $\boldsymbol{\beta}_j^K,\boldsymbol{\beta}_j^\Psi\in\mathbb{R}^G$, with $\boldsymbol{\beta}_1^K$ omitted, *holding the marks of $G$ at $\boldsymbol{\beta}$* is the Mellin transform in the twists of every letter but $K_1$ at the marks of $G$, the twist of $K_1$ and the twists at the other marks kept at $\mathbf{1}$,

$$\mathrm{Hold}_G(\boldsymbol{\beta})=\int_{((0,\infty)^G)^{2N-1}}\prod_{j\geq 2}(\mathbf{x}_j^K)^{\boldsymbol{\beta}_j^K-\mathbf{1}}\prod_{j=1}^N(\mathbf{x}_j^\Psi)^{\boldsymbol{\beta}_j^\Psi-\mathbf{1}}\,P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t})\,d\mathbf{x},$$

in $[0,\infty]$ for real masses and loads; for complex masses or loads it is taken where it converges absolutely, that is where it is finite at the real parts.

The twists of every letter but $K_1$ are held. By the torus relation of Definition 22.1, a twist of every mark of one letter by one factor multiplies the period by that factor to the power $-r_js_j$, or $-r_j^*t_j$ for a cobasis letter, so that a hold of every mark is taken with one twist of each held letter fixed; and a twist of every letter by one vector $\boldsymbol{\lambda}$ multiplies the period by $\boldsymbol{\lambda}^{-\mathbf{m}}$, so that the twist of $K_1$ is kept at $\mathbf{1}$. A twist is a change of weights: Proposition 26.20 carries product weights on the bases into twists.

**Definition 25.9** (Loads; admissible points)**.** For a split of the masses among the letters of $F$, $\mathbf{m}=\sum_j(\mathbf{a}_j+\mathbf{b}_j)$, $\mathbf{a}_j$ is the *$K_j$-load* and $\mathbf{b}_j$ the *$\Psi_j$-load*, as a $K$-share of a pattern loads its marks and a $\Psi$-share the marks it leaves out (Definition 12.24); for one table, $\mathbf{m}=\mathbf{a}+\mathbf{b}$ with the $K$-load $\mathbf{a}$ and the $\Psi$-load $\mathbf{b}$. The periods of a table $\mathcal{H}$ of $F$, with the weights $\boldsymbol{\theta}$, of rank $r$ and corank $r^*$, at its two full leans are read as functions of their loads, $P_{\mathcal{H}}(\mathbf{a};|\mathbf{a}|/r,0)$ and $P_{\mathcal{H}}(\mathbf{b};0,|\mathbf{b}|/r^*)$, the

exponent of the one active letter being the total load divided by its degree. Its letters give the Grammas (Definition 2.1)

$$\mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_{T\in\mathcal{H}}}(\boldsymbol{\theta})=\int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\mathbf{a}-\mathbf{1}}e^{-K_{\mathcal{H}}(x)}dx,\qquad \mathsf{G}_{\mathbf{b};(\mathbf{1}_{\mathcal{E}\setminus T})_{T\in\mathcal{H}}}(\boldsymbol{\theta})=\int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\mathbf{b}-\mathbf{1}}e^{-\Psi_{\mathcal{H}}(x)}dx.$$

A real load $\mathbf{a}$ with $|\mathbf{a}|=rs$, $s>0$, is *admissible for the full basis lean* when $\mathbf{a}\in s\cdot\operatorname{ri}P(\mathcal{H})$, $P(\mathcal{H})$ the base polytope of $\mathcal{H}$ (Theorem 7.18) and ri the relative interior, and a real $\mathbf{b}$ with $|\mathbf{b}|=r^*t$, $t>0$, is *admissible for the full cobasis lean* when $\mathbf{b}\in t\cdot\operatorname{ri}P(\mathcal{H}^*)$, $P(\mathcal{H}^*)$ the base polytope of the dual table. A real point $\mathbf{c}=(\mathbf{c}_j^K,\mathbf{c}_j^\Psi)_j$ is *admissible* for $(\mathbf{m};\mathbf{s},\mathbf{t})$ when $\sum_j(\mathbf{c}_j^K+\mathbf{c}_j^\Psi)=\operatorname{Re}\mathbf{m}$, every $\mathbf{c}_j^K$ is admissible for the full basis lean of table $j$ at $s_j$ and every $\mathbf{c}_j^\Psi$ for its full cobasis lean at $t_j$; for one table it is named by its $\Psi$-load $\mathbf{c}=\mathbf{c}_1^\Psi$, with $\operatorname{Re}\mathbf{m}-\mathbf{c}$ admissible for the full basis lean and $\mathbf{c}$ for the full cobasis lean.

**Proposition 25.10** (The periods at the two full leans)**.** *In (i), (ii) and (iv) $\mathcal{H}$ is a table of $F$, with the weights $\boldsymbol{\theta}$, of rank $r$ and corank $r^*$.*

*(i) For $\mathbf{a}\in\mathbb{C}^{\mathcal{E}}$ with $\operatorname{Re}|\mathbf{a}|>0$ and $s=|\mathbf{a}|/r$, $P_{\mathcal{H}}(\mathbf{a};s,0)=r\,\mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_T}(\boldsymbol{\theta})/\Gamma(s)$, both sides converging together; likewise $P_{\mathcal{H}}(\mathbf{b};0,t)=r^*\mathsf{G}_{\mathbf{b};(\mathbf{1}_{\mathcal{E}\setminus T})_T}(\boldsymbol{\theta})/\Gamma(t)$ with $t=|\mathbf{b}|/r^*$. By Cremona, $P_{\mathcal{H}}(\mathbf{b};0,t)=P_{\mathcal{H}}(t\mathbf{1}-\mathbf{b};t,0)$.*

*(ii) $P_{\mathcal{H}}(\mathbf{a};s,0)$ converges exactly when $\operatorname{Re}a_S>(\operatorname{Re}s)\,d(S)$ for every nonempty proper $S$, which for real $s>0$ is $\operatorname{Re}\mathbf{a}\in s\cdot\operatorname{ri}P(\mathcal{H})$; $P_{\mathcal{H}}(\mathbf{b};0,t)$ converges exactly when $\operatorname{Re}b_S>(\operatorname{Re}t)\,d^*(S)$ for every nonempty proper $S$, which for real $t>0$ is $\operatorname{Re}\mathbf{b}\in t\cdot\operatorname{ri}P(\mathcal{H}^*)$.*

*(iii) At every gauge $\alpha$ the occupation $R_j(\alpha)$ of table $j$ lies in $\operatorname{ri}P(\mathcal{B}_j)$ and its dual occupation $R_j^*(\alpha)$ in $\operatorname{ri}P(\mathcal{B}_j^*)$ (Theorem 7.16). At a native gauge $\alpha^\star$ of real masses $\mathbf{m}$ on the balance wall (Definition 8.17), which exists when every proper wall is positive, the* native loads *$\mathbf{a}_j^\star=s_jR_j(\alpha^\star)$ and $\mathbf{b}_j^\star=t_jR_j^*(\alpha^\star)$ are a split of $\mathbf{m}$ among the letters of $F$, and $\mathbf{c}=(\mathbf{a}_j^\star,\mathbf{b}_j^\star)_j$ is admissible.*

*(iv) On a vertical line the periods at the full leans decay exponentially: for admissible real $\mathbf{a}$, every $\mathbf{y}\in\mathbb{R}^{\mathcal{E}}$ with $|\mathbf{y}|=0$ and every $\varepsilon\in(0,1)$,*

$$\big|P_{\mathcal{H}}(\mathbf{a}+i\mathbf{y};s,0)\big|\le\min\Big(1,\ \big(\sin\tfrac{\pi\varepsilon}{2}\big)^{-s}e^{-(1-\varepsilon)h(\mathbf{y})}\Big)\,P_{\mathcal{H}}(\mathbf{a};s,0),$$

*with $h$ the support function of the set $\{y:\ |\langle\mathbf{1}_T,y\rangle|<\frac{\pi}{2}$ for every basis $T\}$, a norm on $\mathbb{R}^{\mathcal{E}}$; likewise at the full cobasis lean with the vectors $\mathbf{1}_{\mathcal{E}\setminus T}$.*

*Items (i), (ii) and (iv) hold for every positive letter with a Newton polytope of dimension $n-1$, in the form of Proposition 25.3.*

**Lemma 25.11** (Holding every mark separates the letters)**.** *Fix $e_0\in\mathcal{E}$ and let $\mathbf{m}\in\mathbb{C}^{\mathcal{E}}$ have $m_{\mathcal{E}}=\sum_j(r_js_j+r_j^*t_j)$. For real loads $\boldsymbol{\beta}=(\boldsymbol{\beta}_j^K,\boldsymbol{\beta}_j^\Psi)_j$ on $\mathcal{E}\setminus e_0$, $\boldsymbol{\beta}_1^K$ omitted, let $\mathbf{a}_j$ ($j\ge2$) extend $\boldsymbol{\beta}_j^K$ by $a_{j,e_0}=r_js_j-|\boldsymbol{\beta}_j^K|$, let $\mathbf{b}_j$ extend $\boldsymbol{\beta}_j^\Psi$ by $b_{j,e_0}=r_j^*t_j-|\boldsymbol{\beta}_j^\Psi|$, and put $\mathbf{a}_1=\mathbf{m}-\sum_{j\ge2}\mathbf{a}_j-\sum_j\mathbf{b}_j$. Then, in $[0,\infty]$ for real $\mathbf{m}$,*

$$\operatorname{Hold}_{\mathcal{E}\setminus e_0}(\boldsymbol{\beta})=\prod_{j=1}^{N}P_{\mathcal{B}_j}(\mathbf{a}_j;s_j,0)\,P_{\mathcal{B}_j}(\mathbf{b}_j;0,t_j),$$

*the twists at $e_0$ being 1; for complex $\mathbf{m}$ and complex $\boldsymbol{\beta}$ the same holds whenever the integral converges at the real parts.*

The lemma is the whole mechanism. Every letter but $K_1$ sees its own twists and $K_1$ sees none; a Mellin transform in all of them at once absorbs every mark of each twisted letter into its own load, and what is left of $K_1$ is its own period at the remaining load. Nothing is integrated against anything: the hold is a product.

## 25.3 The lean convolution

**Theorem 25.12** (The lean convolution)**.** *Let $\mathbf{m} \in \mathbb{C}^{\mathcal{E}}$ have $m_{\mathcal{E}} = \sum_j (r_j s_j + r_j^* t_j)$.*

*(i) Admissible points for $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ exist exactly when every proper wall of $\operatorname{Re}\mathbf{m}$ is positive; they form a convex set, and for real $\mathbf{m}$ the native loads (Proposition 25.10(iii)) are one of them.*

*(ii) On every admissible point and for every twist $\mathbf{x} = (\mathbf{x}_j^K, \mathbf{x}_j^\Psi)_j$ (Definition 22.1),*

$$P_{F[\mathbf{x}]}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\substack{\sum_j (\mathbf{a}_j + \mathbf{b}_j) = \mathbf{m} \\ |\mathbf{a}_j| = r_j s_j \ (j \geq 2),\ |\mathbf{b}_j| = r_j^* t_j}} \prod_{j=1}^{N} P_{\mathcal{B}_j}(\mathbf{a}_j; s_j, 0)\, P_{\mathcal{B}_j}(\mathbf{b}_j; 0, t_j)\, (\mathbf{x}_j^K)^{-\mathbf{a}_j} (\mathbf{x}_j^\Psi)^{-\mathbf{b}_j},$$

*the fibre integral of Definition 4.1, of dimension $(2N-1)(n-1)$, absolutely convergent; in the coordinates of the loads of every letter but $K_1$ at the marks other than a fixed mark $e_0$ it is $(2\pi i)^{-(2N-1)(n-1)} \int \cdots$ over the imaginary translate through the admissible point, the loads at $e_0$ and $\mathbf{a}_1$ being fixed by the pins. At $\mathbf{x} = \mathbf{1}$ it is the period.*

*(iii) The modulus of the integrand is at most its value at the real point,*

$$\prod_j P_{\mathcal{B}_j}(\mathbf{c}_j^K; s_j, 0)\, P_{\mathcal{B}_j}(\mathbf{c}_j^\Psi; 0, t_j)\, (\mathbf{x}_j^K)^{-\mathbf{c}_j^K} (\mathbf{x}_j^\Psi)^{-\mathbf{c}_j^\Psi},$$

*and it decays exponentially in the imaginary part of the load of every letter but $K_1$, at the rate of Proposition 25.10(iv) for its table.*

*With exponents $s_j, t_j \geq 0$, not all zero, the same holds over the active letters, $p$ of them, any one of them in the place of $K_1$, in dimension $(p-1)(n-1)$.*

For one table, at $\mathcal{H} = \binom{\{1,2\}}{1}$ the lean convolution is Corollary 4.7, and at the circuit $\binom{\mathcal{E}}{n-1}$ it is Corollary 18.24, the periods at the full leans being those of Proposition 26.9. Opening each of them on its share side (Proposition 4.2) gives the atom formula of Corollary 18.16, one share per basis and letter: the atom formula opens every letter over its bases and every mass over the letters at once, while the lean convolution keeps each letter closed, read at its full lean, and opens only the split of each mass among the letters.

**Corollary 25.13** (What a family retains)**.** *The Grammas $\mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_{T \in \mathcal{B}_j}}(\boldsymbol{\theta}_j)$, $j = 1, \ldots, N$, each one function of the $n$ variables $\mathbf{a}$, determine every period of $F$ and of all its twists: at all exponents $s_j, t_j \geq 0$, not all zero, at all masses on the balance wall with every proper wall positive, and at every twist. At the exponents $(\mathbf{s}, \mathbf{t})$ only the values of the Gramma of table $j$ on the hyperplanes $|\mathbf{a}| = r_j s_j$ and $|t_j \mathbf{1} - \mathbf{b}| = r_j t_j$ enter, through $P_{\mathcal{B}_j}(\mathbf{a}; s_j, 0) = r_j\, \mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_T}(\boldsymbol{\theta}_j)/\Gamma(s_j)$ and $P_{\mathcal{B}_j}(\mathbf{b}; 0, t_j) = r_j\, \mathsf{G}_{t_j \mathbf{1} - \mathbf{b};(\mathbf{1}_T)_T}(\boldsymbol{\theta}_j)/\Gamma(t_j)$. Conversely the periods of table $j$ at its full basis leans determine its Gramma, through Proposition 25.10(i).*

A cell of a hinge tree retains, for one pair of exponents, one measure on the half-line (Theorem 23.11); a family retains, for all exponents and all masses at once, one function of the loads for each of its tables. The two retentions serve different queries; Lemma 25.16 relates them, and Section 26.6 compares them.

## 25.4 Holding some marks

**Theorem 25.14** (Pinned holds)**.** *Let every proper wall of* $\operatorname{Re}\mathbf{m}$ *be positive,* $e_0$ *a fixed mark,* $\varnothing \neq G \subseteq \mathcal{E}\setminus e_0$*, and* $\boldsymbol{\beta}=(\boldsymbol{\beta}_j^K,\boldsymbol{\beta}_j^\Psi)_j$*,* $\boldsymbol{\beta}_1^K$ *omitted, the restriction to* $G$ *of the loads of an admissible point. Then for every real* $\boldsymbol{\eta}=(\boldsymbol{\eta}_j^K,\boldsymbol{\eta}_j^\Psi)_j$ *of the same shape the hold* $\operatorname{Hold}_G(\boldsymbol{\beta}+i\boldsymbol{\eta})$ *converges absolutely and equals the fibre integral of Theorem 25.12(ii) at* $\mathbf{x}=\mathbf{1}$ *with the* $(2N-1)|G|$ *further pins* $\mathbf{a}_j|_G=\boldsymbol{\beta}_j^K+i\boldsymbol{\eta}_j^K$ *(*$j\geq 2$*) and* $\mathbf{b}_j|_G=\boldsymbol{\beta}_j^\Psi+i\boldsymbol{\eta}_j^\Psi$*,*

$$\operatorname{Hold}_G(\boldsymbol{\beta}+i\boldsymbol{\eta})=\int_{\sum_j(\mathbf{a}_j+\mathbf{b}_j)=\mathbf{m},\ |\mathbf{a}_j|=r_js_j\ (j\geq 2),\ |\mathbf{b}_j|=r_j^*t_j,\ \text{the pins on } G}\ \prod_{j=1}^{N}P_{\mathcal{B}_j}(\mathbf{a}_j;s_j,0)\,P_{\mathcal{B}_j}(\mathbf{b}_j;0,t_j),$$

*of dimension* $(2N-1)(n-1-|G|)$*, in the coordinates of the loads of every letter but* $K_1$ *at the marks off* $G\cup e_0$ *the integral* $(2\pi i)^{-(2N-1)(n-1-|G|)}\int\cdots$*. At* $G=\mathcal{E}\setminus e_0$ *it is Lemma 25.11.*

For one mark the hold has a closed form in the minors at that mark. For a mark $g$ that is neither a loop nor a coloop of any table, the patterns of the bunch $\{g\}$ in table $j$ are $\varnothing$ and $\{g\}$, with the children the deletion $\mathcal{B}_j\setminus g$, of rank $r_j$ and corank $r_j^*-1$, and the contraction $\mathcal{B}_j/g$, of rank $r_j-1$ and corank $r_j^*$ (Definition 12.1). For $K$-loads $a_j$ and $\Psi$-loads $b_j$ at $g$ the children family of $\{g\}$ (Definition 12.24) at the $K$-shares $(s_j-a_j,a_j)$ and the $\Psi$-shares $(b_j,t_j-b_j)$ of the patterns $\varnothing$ and $\{g\}$ is

$$F_{\{g\}}\big((s_j-a_j,a_j),(b_j,t_j-b_j)\big)_j=\Big(\big((\mathcal{B}_j\setminus g,\theta_j);\ s_j-a_j,\ b_j\big),\ \big((\mathcal{B}_j/g,\theta_j);\ a_j,\ t_j-b_j\big)\Big)_j,$$

a family of $2N$ tables on $\mathcal{E}\setminus g$ whose word is $\prod_j K_{j\setminus g}^{a_j-s_j}\Psi_{j\setminus g}^{-b_j}K_{j/g}^{-a_j}\Psi_{j/g}^{b_j-t_j}$. At masses $\mathbf{m}|_{\mathcal{E}\setminus g}$ with $\sum_j(a_j+b_j)=m_g$ it is on its balance wall: its degree is $\sum_j\big((s_j-a_j)r_j+b_j(r_j^*-1)+a_j(r_j-1)+(t_j-b_j)r_j^*\big)=\sum_j(r_js_j+r_j^*t_j)-m_g$.

**Proposition 25.15** (Holding one mark)**.** *Let* $g$ *be a mark that is neither a loop nor a coloop of any table,* $\mathbf{m}$ *real masses on the balance wall,* $a_j$ *(*$j\geq 2$*) and* $b_j$ *(*$j\leq N$*) real, and* $a_1=m_g-\sum_{j\geq 2}a_j-\sum_j b_j$*. The hold*

$$\operatorname{Hold}_{\{g\}}\big((a_j)_{j\geq 2},(b_j)_j\big)=\int_{(0,\infty)^{2N-1}}\prod_{j\geq 2}(x_j^K)^{a_j-1}\prod_{j=1}^{N}(x_j^\Psi)^{b_j-1}\,P_{F[\mathbf{x}@g]}(\mathbf{m};\mathbf{s},\mathbf{t})\,d\mathbf{x},\qquad x_1^K=1,$$

*is finite exactly when* $0<a_j<s_j$ *and* $0<b_j<t_j$ *for every* $j$ *and every proper wall of* $F_{\{g\}}\big((s_j-a_j,a_j),(b_j,t_j-b_j)\big)_j$ *at* $\mathbf{m}|_{\mathcal{E}\setminus g}$ *is positive, and then*

$$\operatorname{Hold}_{\{g\}}\big((a_j)_{j\geq 2},(b_j)_j\big)=\prod_{j=1}^{N}\Gamma_1^2\begin{bmatrix}a_j,\ s_j-a_j\\ s_j\end{bmatrix}\Gamma_1^2\begin{bmatrix}b_j,\ t_j-b_j\\ t_j\end{bmatrix}\,P_{F_{\{g\}}((s_j-a_j,a_j),(b_j,t_j-b_j))_j}\big(\mathbf{m}|_{\mathcal{E}\setminus g}\big);$$

*the identity holds as well for complex loads and complex masses whose real parts are such data. Where the loads are those at* $g$ *of an admissible point, Theorem 25.14 writes the same value as the lean convolution with the pins at* $g$*. If* $m_g>0$*, then for every letter of* $F$ *whose exponent* $u$ *exceeds* $m_g$*, as the twists at* $g$ *of all the other letters decrease to* $0$*, the twist of that letter staying* $1$*,* $P_{F[\mathbf{x}@g]}(\mathbf{m};\mathbf{s},\mathbf{t})$ *increases, in* $(0,\infty]$*, to* $\Gamma_1^2\begin{bmatrix}m_g,\ u-m_g\\ u\end{bmatrix}$ *times the period at* $\mathbf{m}|_{\mathcal{E}\setminus g}$ *of the children family of* $\{g\}$ *at the shares that put the whole mass* $m_g$ *on that letter:* $(s_j-m_g,m_g)$ *if it is* $K_j$*,* $(m_g,t_j-m_g)$ *if it is* $\Psi_j$*,* $(s_{j'},0)$ *for every other letter* $K_{j'}$ *and* $(0,t_{j'})$ *for every other letter* $\Psi_{j'}$*.*

The ends of the twist are the faces of the children family at which every letter but one has the mark $g$ switched off: the chosen letter keeps both its minors and every other letter keeps one. For one table the two limits are those of $P_{\mathcal{H}[x@g]}$ as $x \downarrow 0$ and of $x^{m_g} P_{\mathcal{H}[x@g]}$ as $x \uparrow \infty$ (Definition 22.1), and a weight on a mark driven to one end of its range leaves the deletion and the contraction of the mark, two tables carrying three active letters between them. With Proposition 26.20 and Definition 22.1, a weight $w$ on the bases through $g$ is the twist $w^{-2}$ up to the factor $w^{-m_g-t}$, so the period behaves like $w^{-m_g-t}$ times the first limit as $w \to \infty$ and like $w^{m_g-t}$ times the second as $w \to 0$, where these are finite.

**Lemma 25.16** (The Mellin transform of a law is the period of a children family)**.** *Let $F = ((\mathcal{A}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a cell family on $B \sqcup \{g\}$ with masses $\mathbf{m}$ on $B$, hinge wall $\kappa_F$, base form $\phi_F$ and ratio vector $\boldsymbol{\varrho}_F$ (Definition 23.1). In this statement let $\mathbf{z} = (z_{j,K}, z_{j,\Psi})_j$ have $\sum_j (z_{j,K} + z_{j,\Psi}) = \kappa_F$, and write $\boldsymbol{\varrho}_F^{-\mathbf{z}} = \prod_{j \geq 2} \left(\gamma^K_{\mathcal{A}_j} / \gamma^K_{\mathcal{A}_1}\right)^{-z_{j,K}} \prod_j \left(\gamma^\Psi_{\mathcal{A}_j} / \gamma^K_{\mathcal{A}_1}\right)^{-z_{j,\Psi}}$, the coordinates of the ratio vector to the powers $-z_{j,K}$ $(j \geq 2)$ and $-z_{j,\Psi}$. For real $\mathbf{z}$, in $[0, \infty]$,*

$$\int_{\Delta_B} \boldsymbol{\varrho}_F^{-\mathbf{z}} \, d\phi_F = P_{F_{\{g\}}((z_{j,K},\, s_j - z_{j,K}),(t_j - z_{j,\Psi},\, z_{j,\Psi}))_j}(\mathbf{m}),$$

*the period of the children family of the bunch $\{g\}$ of $F$ (Definition 12.24), its tables $\mathcal{A}_j \setminus g$ at the exponents $(z_{j,K}, t_j - z_{j,\Psi})$ and $\mathcal{A}_j / g$ at $(s_j - z_{j,K}, z_{j,\Psi})$, on its balance wall at $\mathbf{m}$. Let moreover $s_j, t_j > 0$, and put the mass $\sum_j (s_j + t_j) - \kappa_F$ on $g$, which puts the family of the tables $\mathcal{A}_j$ on its balance wall. For complex $\mathbf{z}$ whose real part has $0 < \operatorname{Re} z_{j,K} < s_j$ and $0 < \operatorname{Re} z_{j,\Psi} < t_j$ for every $j$ and is a point at which $\int \boldsymbol{\varrho}_F^{-\operatorname{Re}\mathbf{z}} d\phi_F$ is finite,*

$$\int_{(0,\infty)^{2N-1}} \prod_{j \geq 2} (x_j^K)^{s_j - z_{j,K} - 1} \prod_{j=1}^N (x_j^\Psi)^{t_j - z_{j,\Psi} - 1} \, P_{F[\mathbf{x}@g]}\Big(\big(\mathbf{m}, \textstyle\sum_j (s_j + t_j) - \kappa_F\big); \mathbf{s}, \mathbf{t}\Big) \, d\mathbf{x}$$
$$= \prod_{j=1}^N \Gamma_{\binom{2}{1}}\begin{bmatrix} s_j - z_{j,K}, & z_{j,K} \\ & s_j \end{bmatrix} \Gamma_{\binom{2}{1}}\begin{bmatrix} t_j - z_{j,\Psi}, & z_{j,\Psi} \\ & t_j \end{bmatrix} \int_{\Delta_B} \boldsymbol{\varrho}_F^{-\mathbf{z}} \, d\phi_F,$$

*$F[\mathbf{x}@g]$ the family of the tables $\mathcal{A}_j$ twisted at $g$, $x_1^K = 1$ (Definition 22.1). So the Mellin transform of the law of a cell family is the Mellin transform of its own period in the twists of its hinge, divided by $2N$ Beta corners.*

**Corollary 25.17** (Identically self-dual tables twisted at one mark)**.** *Let the family be one table $\mathcal{H}$ at the exponents $(s, t)$, identically self-dual (Corollary 23.12(iii)), $g$ a mark, and $\mathbf{m}$ on the balance wall with every proper wall positive. For every $x > 0$,*

$$P_{\mathcal{H}[x@g]}(\mathbf{m}; s, t) = P_{\mathcal{H}\setminus g}\big(\mathbf{m}|_{\mathcal{E}\setminus g};\ s + t - m_g,\ m_g\big) \, P_{\binom{\{g,g'\}}{1}[(x,1)]}\big((m_g, s + t - m_g); s, t\big),$$

*the second factor the period $\int_0^\infty v^{m_g - 1}(1 + v)^{-s}(1 + xv)^{-t} dv$ of the twisted list on two marks (Lemma 23.7); and the hold at $g$ is $\Gamma_{\binom{2}{1}}\begin{bmatrix} m_g - b, & s - m_g + b \\ & s \end{bmatrix} \Gamma_{\binom{2}{1}}\begin{bmatrix} b, & t - b \\ & t \end{bmatrix}$ times the same period of $\mathcal{H} \setminus g$ for every $b$ in its strip: the children family does not depend on the split. At $x = 1$ this is Corollary 23.12(iii), and Corollary 4.7 recovers it from the hold,*

$$\frac{1}{2\pi i} \int \Gamma_{\binom{2}{1}}\begin{bmatrix} m_g - b, & s - m_g + b \\ & s \end{bmatrix} \Gamma_{\binom{2}{1}}\begin{bmatrix} b, & t - b \\ & t \end{bmatrix} db = \Gamma_{\binom{2}{1}}\begin{bmatrix} m_g, & s + t - m_g \\ & s + t \end{bmatrix}.$$

# Chapter 26

# Periods at the full leans and networks along a hinge tree

At a full lean the period of a two-sum is the product of the periods of its two cells over one Beta corner (Theorem 19.10), and along any hinge tree this is Corollary 26.11, which for one table at a full lean is Theorem 26.5 at the loads $\mathbf{m}$. Along a hinge tree this needs only bookkeeping, and the bookkeeping is affine: every marker receives a load that is a fixed affine function of the loads of the marks, the same function at every pair of exponents. So the periods of a substitution at its full leans are products of those of its nodes, read at affine loads, over Beta corners at its edges, with no integral. Where the active letters read every cell by one hinge value, the period along a hinge tree is the product of the periods of the nodes over Beta corners at the edges: at a full lean of a table whose weights factor at its bunches of width one, a product of stacked Gammas, monomials in the weights and the periods of the prime nodes of its canonical hinge tree, and $k$ copies glued along any tree cost one period and a power. Blocks aggregate exactly where every active letter reads them by one hinge value, their total or the inverse of the sum of the reciprocals of their marks (Theorem 26.12). The nodes whose periods at the full leans cost nothing are characterized by counting bases. The period of a family of such tables, at all exponents, every mass and every twist, is then one fibre integral of node data, of dimension $(2N-1)(n-1)$: the lean convolution of Chapter 25 with the periods of every letter at its full lean read along the hinge tree of its table (Theorem 26.13).

## 26.1 Molecules and marker loads

**Lemma 26.1** (Substitutions of molecules)**.** *For a core $\mathcal{H}$ and cells $\mathcal{A}_1, \dots, \mathcal{A}_q$ at its marks (Definition 21.1), the substitution $\mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q \rangle$ is one molecule exactly when $\mathcal{H}$ and the $\mathcal{A}_i$, as tables on their marks and hinges, are each one molecule. In particular the two-sum $\mathcal{A} \oplus_g \mathcal{A}'$ of two cells is one molecule exactly when $\mathcal{A}$ and $\mathcal{A}'$ are.*

So a substitution of nodes that are molecules is one molecule, and a table with every proper wall positive, which is one molecule by Proposition 8.16, has every cell of its hinge trees one molecule.

**Definition 26.2** (Branches of a hinge tree)**.** Let $\mathcal{B}$ be a table on $\mathcal{E}$, $|\mathcal{E}| = n$, of rank $r$ and corank $r^*$, with a hinge tree $\mathfrak{T}$ (Definition 23.15) whose nodes $\mathcal{H}_v$ on $\mathcal{E}_v \sqcup G_v$ are each one molecule, of rank $r_v$ and corank $r_v^* = |\mathcal{E}_v| + |G_v| - r_v$; the weights of $\mathcal{B}$ are the products of the weights of the nodes (Lemma 21.12). For an edge $\mathfrak{e} = \{v, w\}$ of $\mathfrak{T}$ and its end $v$, the *branch* $\mathfrak{T}_v^{\mathfrak{e}}$ is the component

of $\mathfrak{T}-\mathfrak{e}$ containing $v$, $\mathcal{E}_v^{\mathfrak{e}}$ the union of the $\mathcal{E}_u$ over its vertices, and $\mathcal{A}_v^{\mathfrak{e}}$ the substitution of its nodes, a cell on $\mathcal{E}_v^{\mathfrak{e}} \sqcup \{g_{v,\mathfrak{e}}\}$ of rank $r_v^{\mathfrak{e}}$ and corank $r_v^{*\mathfrak{e}} = |\mathcal{E}_v^{\mathfrak{e}}| + 1 - r_v^{\mathfrak{e}}$; $\mathcal{B}$ is the two-sum of $\mathcal{A}_v^{\mathfrak{e}}$ and $\mathcal{A}_w^{\mathfrak{e}}$.

**Definition 26.3** (Marker loads; node loads)**.** For a table with a hinge tree as in Definition 26.2 and loads $\mathbf{a}, \mathbf{b} \in \mathbb{C}^{\mathcal{E}}$ with $|\mathbf{a}| = rs$ and $|\mathbf{b}| = r^*t$, the *marker loads* at the end $v$ of the edge $\mathfrak{e}$ are

$$k_{v,\mathfrak{e}} = s\, r_v^{\mathfrak{e}} - a_{\mathcal{E}_v^{\mathfrak{e}}}, \qquad l_{v,\mathfrak{e}} = t\, r_v^{*\mathfrak{e}} - b_{\mathcal{E}_v^{\mathfrak{e}}},$$

and the *node loads* of $v$ are $\mathbf{a}_v = \big(\mathbf{a}|_{\mathcal{E}_v}, (k_{v,\mathfrak{e}})_{\mathfrak{e}\ni v}\big)$ and $\mathbf{b}_v = \big(\mathbf{b}|_{\mathcal{E}_v}, (l_{v,\mathfrak{e}})_{\mathfrak{e}\ni v}\big)$, the load of the marker $g_{v,\mathfrak{e}}$ being $k_{v,\mathfrak{e}}$, respectively $l_{v,\mathfrak{e}}$.

**Lemma 26.4** (Node loads)**.** *In Definition 26.3:*

*(i)* $k_{v,\mathfrak{e}} + k_{w,\mathfrak{e}} = s$ *and* $l_{v,\mathfrak{e}} + l_{w,\mathfrak{e}} = t$ *on every edge.*

*(ii)* $|\mathbf{a}_v| = r_v s$ *and* $|\mathbf{b}_v| = r_v^* t$ *at every node.*

*(iii) If* $\mathbf{a}+\mathbf{b} = \mathbf{m}$ *with* $\mathbf{m}$ *on the balance wall of* $\mathcal{B}$*, then* $k_{v,\mathfrak{e}} + l_{v,\mathfrak{e}} = s + t - \kappa_{\mathcal{A}_v^{\mathfrak{e}}} = \kappa_{\mathcal{A}_w^{\mathfrak{e}}}$*: the node masses* $\mathbf{a}_v + \mathbf{b}_v$ *are* $\mathbf{m}|_{\mathcal{E}_v}$ *on the marks and the hinge walls of the branches on the markers, the masses of the twisted core in Theorem 23.6.*

## 26.2 The periods of a substitution at its full leans

**Theorem 26.5** (Periods at the full leans along a hinge tree)**.** *In Definition 26.3, for real loads* $\mathbf{a}$ *with* $|\mathbf{a}| = rs$*, if every marker load* $k_{v,\mathfrak{e}}$ *lies in* $(0, s)$*, then, in* $(0, \infty]$*,*

$$P_{\mathcal{B}}(\mathbf{a}; s, 0) = \frac{\prod_v P_{\mathcal{H}_v}(\mathbf{a}_v; s, 0)}{\prod_{\mathfrak{e}=\{v,w\}} \Gamma_2^{\,1}\!\begin{bmatrix} k_{v,\mathfrak{e}}, & k_{w,\mathfrak{e}} \\ & s \end{bmatrix}},$$

*and if some marker load lies outside* $(0, s)$*, then* $P_{\mathcal{B}}(\mathbf{a}; s, 0) = +\infty$*. So* $P_{\mathcal{B}}(\mathbf{a}; s, 0)$ *is finite exactly when every marker load lies in* $(0, s)$ *and the period of every node at its full basis lean is finite at its node loads. The same holds at the full cobasis leans, with* $\mathbf{b}$*, the loads* $l_{v,\mathfrak{e}}$ *and* $t$*. For complex loads both identities hold on the tube over the real loads at which they are finite, both sides being holomorphic there.*

In the language of bunches this is Theorem 19.10 at every bunch of width one of the tree at once: the bunch $\mathcal{E}_v^{\mathfrak{e}}$ has, at $(s, 0)$, the wall $k_{w,\mathfrak{e}}$, and its two cells are $\mathcal{A}_v^{\mathfrak{e}}$ and $\mathcal{A}_w^{\mathfrak{e}}$ (Lemma 21.12). What the tree adds is that the loads of all markers are affine in the loads of the marks, fixed once for every pair of exponents, so that the periods of the substitution at its full leans are evaluated at any loads by evaluating each node once.

**Corollary 26.6** (Convergence is decided at the nodes)**.** *Let* $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ *have exponents* $s_j, t_j > 0$*, and let table* $j$ *carry a hinge tree* $\mathfrak{T}_j$ *as in Definition 26.2, its nodes* $\mathcal{H}_{j,v}$ *on* $\mathcal{E}_{j,v} \sqcup G_{j,v}$ *each one molecule, with the node loads of Definition 26.3 read in table* $j$*. For masses* $\mathbf{m}$ *on the balance wall of* $F$ *every proper wall of* $F$ *is positive exactly when there is a real split* $\mathbf{m} = \sum_j (\mathbf{a}_j + \mathbf{b}_j)$ *with* $|\mathbf{a}_j| = r_j s_j$ *and* $|\mathbf{b}_j| = r_j^* t_j$ *at which every node of every table has*

$$(\mathbf{a}_{j,v})_S > s_j\, d_{j,v}(S) \quad \text{and} \quad (\mathbf{b}_{j,v})_S > t_j\, d_{j,v}^*(S) \qquad (\varnothing \neq S \subsetneq \mathcal{E}_{j,v} \sqcup G_{j,v}),$$

*$d_{j,v}$ and $d_{j,v}^*$ the counts of $\mathcal{H}_{j,v}$. This is a system of* $\sum_j \sum_v 2(2^{|\mathcal{E}_{j,v}|+|G_{j,v}|} - 2)$ *strict linear inequalities in* $(2N-1)(n-1)$ *unknowns, a number linear in the number of nodes when the nodes have bounded size, against the* $2^n - 2$ *walls of* $F$*; the native loads of* $F$ *solve it.*

## 26.3 What a node costs

**Lemma 26.7** (Bases outnumber marks)**.** *A table on $n \geq 2$ marks that is one molecule has at least $n$ bases, and exactly $n$ only when its rank is $1$ or $n-1$, that is, as a list, when it is $\binom{\mathcal{E}}{1}$ or $\binom{\mathcal{E}}{n-1}$.*

**Corollary 26.8** (The cost of a node)**.** *For a node $\mathcal{H}_v$ that is one molecule on $n_v = |\mathcal{E}_v| + |G_v|$ marks with $|\mathcal{H}_v|$ bases, the share side of its period at each of its two full leans (Proposition 4.2) is a fibre integral of dimension $|\mathcal{H}_v| - n_v \geq 0$. The dimension is zero exactly when $\mathcal{H}_v$ has the bases of $\binom{\mathcal{E}_v \sqcup G_v}{n_v - 1}$ or of $\binom{\mathcal{E}_v \sqcup G_v}{1}$, with any weights: the series and the parallel nodes up to their weights, whose periods at the full leans are stacked Gammas times monomials in the weights (Proposition 25.6; Proposition 26.9, next); a prime node on at least four marks has $|\mathcal{H}_v| > n_v$.*

**Proposition 26.9** (Series and parallel nodes at their full leans)**.** *On a set $\mathcal{X}$ of $q+1$ marks let $\binom{\mathcal{X}}{q}$ carry the weight $\theta_e$ on its basis $\mathcal{X} \setminus e$ and $\binom{\mathcal{X}}{1}$ the weight $\theta_e$ on its basis $\{e\}$, the series and the parallel node with weights (Definition 23.13). Then*

$$P_{\binom{\mathcal{X}}{q}}(\mathbf{a}; s, 0) = \Gamma_{\substack{q+1\\1}}\begin{bmatrix}(s-a_e)_e\\ s\end{bmatrix}\prod_e \theta_e^{a_e - s}, \qquad P_{\binom{\mathcal{X}}{q}}(\mathbf{b}; 0, t) = \Gamma_{\substack{q+1\\1}}\begin{bmatrix}(b_e)_e\\ t\end{bmatrix}\prod_e \theta_e^{-b_e},$$

$$P_{\binom{\mathcal{X}}{1}}(\mathbf{a}; s, 0) = \Gamma_{\substack{q+1\\1}}\begin{bmatrix}(a_e)_e\\ s\end{bmatrix}\prod_e \theta_e^{-a_e}, \qquad P_{\binom{\mathcal{X}}{1}}(\mathbf{b}; 0, t) = \Gamma_{\substack{q+1\\1}}\begin{bmatrix}(t-b_e)_e\\ t\end{bmatrix}\prod_e \theta_e^{b_e - t},$$

*with $|\mathbf{a}| = qs$, $|\mathbf{b}| = t$ at the series node and $|\mathbf{a}| = s$, $|\mathbf{b}| = qt$ at the parallel node; each converges exactly when every Gamma above has an argument of positive real part. These are Proposition 25.6 at the configurations $(\mathbf{e}_x)_{x\in\mathcal{X}}$, of the letters $K$ of $\binom{\mathcal{X}}{1}$ and $\Psi$ of $\binom{\mathcal{X}}{q}$, and $(\mathbf{1} - \mathbf{e}_x)_{x\in\mathcal{X}}$, of the letters $K$ of $\binom{\mathcal{X}}{q}$ and $\Psi$ of $\binom{\mathcal{X}}{1}$.*

**Corollary 26.10** (Series-parallel tables at their full leans)**.** *In Definition 26.2 let every node of $\mathfrak{T}$ be a series or a parallel node (Definition 23.13), the node $v$ on $\mathcal{E}_v \sqcup G_v$ carrying the weight $\theta_{v,x}$ on its basis that omits $x$ if it is a series node and on its basis $\{x\}$ if it is a parallel node, and write $(\mathbf{a}_v)_x$ and $(\mathbf{b}_v)_x$ for the node loads at $x \in \mathcal{E}_v \sqcup G_v$ (Definition 26.3). Then*

$$P_{\mathcal{B}}(\mathbf{a}; s, 0) = \frac{1}{\Gamma(s)} \prod_{e \text{ at series}} \Gamma(s - a_e) \prod_{e \text{ at parallel}} \Gamma(a_e) \prod_{\mathfrak{e} \text{ series–parallel}} \frac{\Gamma(s - k_{\mathfrak{e}})}{\Gamma(k_{\mathfrak{e}})}$$
$$\times \prod_{v \text{ series}} \prod_x \theta_{v,x}^{(\mathbf{a}_v)_x - s} \prod_{v \text{ parallel}} \prod_x \theta_{v,x}^{-(\mathbf{a}_v)_x},$$

$$P_{\mathcal{B}}(\mathbf{b}; 0, t) = \frac{1}{\Gamma(t)} \prod_{e \text{ at series}} \Gamma(b_e) \prod_{e \text{ at parallel}} \Gamma(t - b_e) \prod_{\mathfrak{e} \text{ series–parallel}} \frac{\Gamma(l_{\mathfrak{e}})}{\Gamma(t - l_{\mathfrak{e}})}$$
$$\times \prod_{v \text{ series}} \prod_x \theta_{v,x}^{-(\mathbf{b}_v)_x} \prod_{v \text{ parallel}} \prod_x \theta_{v,x}^{(\mathbf{b}_v)_x - t},$$

*the products over the marks by the kind of their node, over the edges joining a series node to a parallel node, with $k_{\mathfrak{e}}$, $l_{\mathfrak{e}}$ the marker loads at the series end, and over the nodes and their marks and markers $x$; an edge joining two nodes of one kind contributes no Gamma. Each of these periods is finite exactly when every Gamma above its line has an argument with positive real part.*

The period of a series-parallel table at a full lean is written down from its tree in time linear in the size of the tree, one Gamma for each mark, two for each change of kind along an edge and a monomial in the weights of each node, while the table may have a number of bases exponential in $n$. The atom formula of the same period (Corollary 18.16) has one share for each basis.

**Corollary 26.11** (Periods along a hinge tree read by one hinge value)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ have real data on its balance wall with every proper wall positive, let its tables have a common hinge tree $\mathfrak{T}$ rooted at $o$, with the node families $F_{\mathcal{H}_v}$ (Definition 23.15), and let the active letters read every subtree cell by one hinge value: for every $v \neq o$ there is a function $\gamma_v$ with $\gamma^{K}_{\mathcal{A}_{v,j}} = \gamma_v$ whenever $s_j \neq 0$ and $\gamma^{\Psi}_{\mathcal{A}_{v,j}} = \gamma_v$ whenever $t_j \neq 0$. With $\sigma$ the total strength (Definition 8.1),*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \frac{\prod_v P_{F_{\mathcal{H}_v}}(\mathbf{m}_v; \mathbf{s}, \mathbf{t})}{\prod_{edges\ \mathfrak{e}} \Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}\kappa_{\mathfrak{e}}, & 2\sigma - \kappa_{\mathfrak{e}}\\ & 2\sigma\end{smallmatrix}\right]},$$

*each node family on its balance wall with every proper wall positive, its masses $\mathbf{m}_v$ those of its marks and, on the marker at an end of the edge $\mathfrak{e}$, the wall of the marks across $\mathfrak{e}$: $\kappa_{\mathfrak{e}}$ on one side and $2\sigma - \kappa_{\mathfrak{e}}$ on the other. For one table at a full lean, $(0, t)$ or $(s, 0)$, one letter is active, the hypothesis holds on every hinge tree, and this is Theorem 26.5 at the loads $\mathbf{m}$. With the canonical hinge tree of a connected table whose weights factor across each bunch of width one at which it is decomposed, a list or a determinantal table for instance (Theorem 23.16), a series or a parallel node, with any weights on its bases, contributes at a full lean a stacked Gamma times a monomial in its weights (Proposition 26.9), so the period at a full lean of such a table is a product of stacked Gammas, monomials in the weights and the periods of its prime nodes. In particular connected tables $\mathcal{A}_1, \ldots, \mathcal{A}_k$ at $(0, t)$, each on its balance wall with every proper wall positive, glued along any tree, every gluing joining a mark of mass $\kappa_{\mathfrak{e}}$ of one to a mark of mass $t - \kappa_{\mathfrak{e}}$ of another, have the period*

$$\frac{\prod_v P_{\mathcal{A}_v}(\mathbf{m}_v; 0, t)}{\prod_{\mathfrak{e}} \Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}\kappa_{\mathfrak{e}}, & t - \kappa_{\mathfrak{e}}\\ & t\end{smallmatrix}\right]},$$

*whatever the shape of the tree; for $k$ copies of one table glued at the masses $(\kappa, t - \kappa)$ this is $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}\kappa, & t - \kappa\\ & t\end{smallmatrix}\right] \big(P_{\mathcal{A}}(\mathbf{m}; 0, t)/\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}\kappa, & t - \kappa\\ & t\end{smallmatrix}\right]\big)^{k}$. On the chain $F^{\triangleright k}$ of Theorem 22.8, with every proper wall positive and the active letters reading each copy by one hinge value, the product is $P_F^k/\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{smallmatrix}m_{g_{\mathrm{in}}}, & m_{g_{\mathrm{out}}}\\ & 2\sigma\end{smallmatrix}\right]^{k-1}$, the display of that theorem.*

Where the active letters read every cell through one hinge value, as at a full lean of one table, where only one letter sees a hinge, $\Psi$ at $(0, t)$ and $K$ at $(s, 0)$, every cell is pinned, every hinge integral is the Beta corner of a linear letter (Proposition 15.1 with one coefficient), and the period of the whole construction is the product of the periods of its nodes over the Beta corners of its hinges. Nothing is integrated against anything.

## 26.4 Blocks aggregate

**Theorem 26.12** (Blocks read by one hinge value aggregate exactly)**.** *Let $F$ be a family substituted with blocks (Corollary 21.7(i)) into the tables $\mathcal{H}_j$ of a core family $F_{\mathcal{H}} = ((\mathcal{H}_j, \theta^{\mathcal{H}_j}); s_j, t_j)_j$, with unit weights in the blocks, the block $B_i$ of $\ell_i$ marks. The active letters read $B_i$* through its total *when every table with $s_j \neq 0$ has a parallel block and every table with $t_j \neq 0$ a series block at $B_i$, and through $e_{\ell_i}/e_{\ell_i - 1} = \big(\sum_{x \in B_i} \alpha_x^{-1}\big)^{-1}$ when every table with $s_j \neq 0$ has a series block and every table with $t_j \neq 0$ a parallel block (Corollary 21.7(i)). Let $U \sim \mathrm{Dir}_{\mathbf{m}}$ (Definition 3.10), with the block totals $|U_{B_i}|$.*

*(i) If the active letters read every block through its total, the joint law of all the active letters, $\big(K_j(U)\big)_{s_j\neq 0}$ and $\big(\Psi_j(U)\big)_{t_j\neq 0}$, with the totals $(|U_{B_i}|)_i$ themselves, is that of $\big(K_{\mathcal{H}_j}(U')\big)_{s_j\neq 0}$, $\big(\Psi_{\mathcal{H}_j}(U')\big)_{t_j\neq 0}$ and $U'$ for $U'\sim \mathrm{Dir}_{(m_{B_i})_i}$: every nonnegative function of these variables, and every absolutely integrable one, has the same expectation on both sides.*

*(ii) Let the active letters read $B_i$ through its total for $i\in I$ and through $e_{\ell_i}/e_{\ell_i-1}$ for $i\notin I$, and let $\sigma$ be the total strength (Definition 8.1). The wall of a block is $\kappa_{B_i}=m_{B_i}$ for $i\in I$ and $\kappa_{B_i}=m_{B_i}-(\ell_i-1)\,2\sigma$ for $i\notin I$, and on the balance wall, in $[0,\infty]$,*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\prod_{i\in I}\Gamma_{\substack{\ell_i\\1}}\begin{bmatrix}\mathbf{m}|_{B_i}\\ m_{B_i}\end{bmatrix}\prod_{i\notin I}\Gamma_{\substack{\ell_i\\1}}\begin{bmatrix}(2\sigma-m_x)_{x\in B_i}\\ 2\sigma-\kappa_{B_i}\end{bmatrix}\;P_{F_{\mathcal{H}}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big),$$

*a stacked Gamma read as $+\infty$ when one of its upper arguments is not positive; both sides are finite together when these are positive.*

*(iii) If the active letters read every block through its total, then at fixed pooled masses $m_{B_i}$, dividing each among any number of marks leaves the expectation $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[W(U)]$ of the family word unchanged and multiplies the period by the stacked Gammas $\Gamma_{\substack{|B_i|\\1}}\begin{bmatrix}\mathbf{m}|_{B_i}\\ m_{B_i}\end{bmatrix}$ alone: lengthening the blocks creates no new integral.*

*A block read through its total by some active letters and through $e_\ell/e_{\ell-1}$ by others keeps the block as one mark of Theorem 22.3, with its kernel and its hinge values.*

## 26.5 The network theorem

The lean convolution of a family of substitutions, the periods of its letters at their full leans written along the hinge trees of its tables, is one fibre integral whose integrand is a product of node data at affine loads. Read as one integral over all the loads, it has dimension $(2N-1)(n-1)$ for every family whose tables carry hinge trees, and for series-parallel tables its integrand is a written-down stacked Gamma times an exponential of the loads. Integrated from the leaves up, when the tables share one hinge tree, the same integral is computed node by node with one rule at every node, prime or not (Theorem 26.16): a node enters through the periods of its tables at their full leans and its cell families through the Mellin transforms of their laws; for one table this is the hinge tree procedure of Chapter 23.

**Theorem 26.13** (The network)**.** *Let $F=((\mathcal{B}_j,\theta_j);s_j,t_j)_{j=1}^N$ be a family on $\mathcal{E}$ with exponents $s_j,t_j>0$ whose table $j$ carries a hinge tree $\mathfrak{T}_j$ as in Definition 26.2, its nodes $\mathcal{H}_{j,v}$ each one molecule, and let $\mathbf{m}$ be on the balance wall of $F$ with every proper wall positive. Then every table is one molecule, and on every admissible point (Definition 25.9) and for every twist $\mathbf{x}$ (Definition 22.1),*

$$P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t})=\int\prod_{j=1}^{N}\frac{\prod_{v\in\mathfrak{T}_j}P_{\mathcal{H}_{j,v}}(\mathbf{a}_{j,v};s_j,0)\,P_{\mathcal{H}_{j,v}}(\mathbf{b}_{j,v};0,t_j)}{\prod_{\mathfrak{e}=\{v,w\}\in\mathfrak{T}_j}\Gamma_{\substack{2\\1}}\begin{bmatrix}k_{j;v,\mathfrak{e}},\;k_{j;w,\mathfrak{e}}\\ s_j\end{bmatrix}\Gamma_{\substack{2\\1}}\begin{bmatrix}l_{j;v,\mathfrak{e}},\;l_{j;w,\mathfrak{e}}\\ t_j\end{bmatrix}}\;(\mathbf{x}_j^K)^{-\mathbf{a}_j}(\mathbf{x}_j^\Psi)^{-\mathbf{b}_j},$$

*the fibre integral of Theorem 25.12 over the loads $(\mathbf{a}_j,\mathbf{b}_j)_j$ of the marks, of dimension $(2N-1)(n-1)$, with the node loads $\mathbf{a}_{j,v},\mathbf{b}_{j,v}$ and the marker loads $k_{j;v,\mathfrak{e}},l_{j;v,\mathfrak{e}}$ of Definition 26.3 read in table $j$ at its loads $(\mathbf{a}_j,\mathbf{b}_j)$ and its exponents $(s_j,t_j)$. At every admissible point the periods of every node at both full leans are finite and every marker load of table $j$ lies in $(0,s_j)$, respectively $(0,t_j)$;*

*the native loads of $F$ are an admissible point. The nodes enter only through their periods at the two full leans, as functions of their loads, the same functions for all exponents, all masses and all twists.*

**Corollary 26.14** (Series-parallel periods)**.** *In Theorem 26.13 let every node of every table be a series or a parallel node with the weights of Corollary 26.10, $\theta_{j,v,x}$ at the node $v$ of $\mathfrak{T}_j$. Then*

$$\prod_j \Gamma(s_j)\Gamma(t_j)\, P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t}) = \int \prod_j \Bigg[ \prod_{e \text{ at series}} \Gamma(s_j - a_{j,e})\Gamma(b_{j,e}) \prod_{e \text{ at parallel}} \Gamma(a_{j,e})\Gamma(t_j - b_{j,e}) \times \prod_{\mathfrak{e} \text{ series–parallel}} \frac{\Gamma(s_j - k_{j,\mathfrak{e}})\,\Gamma(l_{j,\mathfrak{e}})}{\Gamma(k_{j,\mathfrak{e}})\,\Gamma(t_j - l_{j,\mathfrak{e}})} \prod_{v \text{ series}} \prod_x \theta_{j,v,x}^{(\mathbf{a}_{j,v})_x - (\mathbf{b}_{j,v})_x - s_j} \times \prod_{v \text{ parallel}} \prod_x \theta_{j,v,x}^{(\mathbf{b}_{j,v})_x - (\mathbf{a}_{j,v})_x - t_j} \, (\mathbf{x}_j^K)^{-\mathbf{a}_j} (\mathbf{x}_j^\Psi)^{-\mathbf{b}_j} \Bigg],$$

*the kinds of the nodes and the marker loads $k_{j,\mathfrak{e}}$, $l_{j,\mathfrak{e}}$ at the series end read in $\mathfrak{T}_j$, the fibre integral of Theorem 26.13, of dimension $(2N-1)(n-1)$, on any point at which every Gamma above the line has an argument of positive real part. Every period of a family of series-parallel tables with any weights on their nodes, at all exponents, at all masses with positive walls and at every twist, is a fibre integral of dimension $(2N-1)(n-1)$ of an explicit stacked Gamma times an exponential of the loads, whatever the numbers of bases, and at a full lean of one table no integral is left (Corollary 26.10).*

**Proposition 26.15** (Pooling a pair)**.** *In Corollary 26.14, for one table twisted in its cobasis letter by $\mathbf{x}$ (Definition 22.1), let $e, f$ be two marks of one series node $v$ whose twists times the squares of their weights agree, $x_e\theta_{v,e}^2 = x_f\theta_{v,f}^2$, and $c_e = s - m_e$, $c_f = s - m_f$. Their loads enter the integrand through $\Gamma(c_e + b_e)\Gamma(b_e)\Gamma(c_f + b_f)\Gamma(b_f)$ and otherwise through $b_{\{e,f\}} = b_e + b_f$ alone, and at fixed $b_{\{e,f\}}$*

$$\frac{1}{2\pi i} \int \Gamma(c_e + b_e)\Gamma(b_e)\Gamma\big(c_f + b_{\{e,f\}} - b_e\big)\Gamma\big(b_{\{e,f\}} - b_e\big)\, db_e = \Gamma_{\,1}^{\,4}\!\left[\begin{matrix} c_e + c_f + b_{\{e,f\}},\ c_e + b_{\{e,f\}},\ c_f + b_{\{e,f\}},\ b_{\{e,f\}} \\ c_e + c_f + 2b_{\{e,f\}} \end{matrix}\right].$$

*For two marks of one parallel node $v$ with $x_e\theta_{v,e}^{-2} = x_f\theta_{v,f}^{-2}$, at fixed $b_{\{e,f\}}$, the pair gives*

$$\Gamma_{\,1}^{\,4}\!\left[\begin{matrix} m_e + m_f - b_{\{e,f\}},\ m_e + t - b_{\{e,f\}},\ m_f + t - b_{\{e,f\}},\ 2t - b_{\{e,f\}} \\ m_e + m_f + 2t - 2b_{\{e,f\}} \end{matrix}\right].$$

*Each pooled pair lowers the dimension of the fibre by one, and disjoint pairs can be pooled at once.*

The pooled pair is a series pair read as one cell: its factor is $\Gamma_{\,1}^{\,2}\left[\begin{matrix} c_e + b_{\{e,f\}},\ c_f + b_{\{e,f\}} \\ c_e + c_f + 2b_{\{e,f\}} \end{matrix}\right] \Gamma(c_e + c_f + b_{\{e,f\}})\Gamma(b_{\{e,f\}})$, the Beta corner being the Mellin transform of the law of the pair at $b_{\{e,f\}}$ and the two Gammas those of its hinge in the periods of the node at its full leans (Theorem 26.16). A series bunch of three or more marks at one node can be pooled pair by pair only once: the pooled factor of a pair is no longer of the form $\Gamma(c+b)\Gamma(b)$, and the class is a circuit of its marks (Corollary 18.24).

## 26.6 Nodes in hold form

**Theorem 26.16** (Nodes in hold form)**.** *Let $F$ be a substituted family (Definition 21.1) whose tables are each one molecule, with exponents $s_j, t_j > 0$, table $j$ being $\mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{kj}\rangle$ with the cores $\mathcal{H}_j$ on $\mathcal{E}_{\mathcal{H}} \sqcup G_{\mathcal{H}}$ and the cells $\mathcal{A}_{ij}$ on $B_i \sqcup \{g_i\}$ at the markers $g_1, \dots, g_k$ (Definition 23.4), and let $F_i = ((\mathcal{A}_{ij}, \theta^{\mathcal{A}_{ij}}); s_j, t_j)_j$ be the cell family at $g_i$ (Definition 23.1). For loads $(\mathbf{a}_j, \mathbf{b}_j)_j$ on the marks and markers of the cores write $\mathbf{z}_i = (a_{j,g_i}, b_{j,g_i})_j$ for their values at $g_i$, and in this statement write $\int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i} d\phi_{F_i}$ for the Mellin transform of the law of $F_i$ of Lemma 25.16 at $\mathbf{z}_i$.*

*(i)* Root. *If $G_{\mathcal{H}} = \{g_1, \dots, g_k\}$ and $F$ is on its balance wall with every proper wall positive, let $\mathcal{F}$ be the fibre of the loads of the cores with the pins $\sum_j (\mathbf{a}_j + \mathbf{b}_j) = (\mathbf{m}|_{\mathcal{E}_{\mathcal{H}}}, \kappa_{F_1}, \dots, \kappa_{F_k})$, $|\mathbf{a}_j| = r_{\mathcal{H}_j} s_j$ $(j \geq 2)$ and $|\mathbf{b}_j| = r^*_{\mathcal{H}_j} t_j$. On the node loads at the cores (Definition 26.3, in every table) of any admissible point of $F$ and for twists $\mathbf{x}$ of $F$ with every coordinate $1$ off $\mathcal{E}_{\mathcal{H}}$,*

$$P_{F[\mathbf{x}]}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\mathcal{F}} \prod_{j=1}^{N} P_{\mathcal{H}_j}(\mathbf{a}_j; s_j, 0)\, P_{\mathcal{H}_j}(\mathbf{b}_j; 0, t_j)\, (\mathbf{x}_j^K)^{-\mathbf{a}_j|_{\mathcal{E}_{\mathcal{H}}}} (\mathbf{x}_j^{\Psi})^{-\mathbf{b}_j|_{\mathcal{E}_{\mathcal{H}}}} \prod_{i=1}^{k} \int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i}\, d\phi_{F_i},$$

*of dimension $(2N-1)(|\mathcal{E}_{\mathcal{H}}| + k - 1)$, every $\int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i} d\phi_{F_i}$ finite at the real point.*

*(ii)* Hinged core. *If $G_{\mathcal{H}} = \{g_0, g_1, \dots, g_k\}$, let $F'$ be the cell family with the hinge $g_0$ whose table $j$ is $\mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{kj}\rangle$, with the hinge wall $\kappa_{F'}$, and for $\mathbf{z} = (z_{j,K}, z_{j,\Psi})_j$ with $\sum_j (z_{j,K} + z_{j,\Psi}) = \kappa_{F'}$ let $\mathcal{F}_{\mathbf{z}}$ be the fibre of the loads of the cores with the pins $\sum_j (\mathbf{a}_j + \mathbf{b}_j) = \big(\mathbf{m}|_{\mathcal{E}_{\mathcal{H}}}, \kappa_{F_1}, \dots, \kappa_{F_k}, \sum_j (s_j + t_j) - \kappa_{F'}\big)$, $|\mathbf{a}_j| = r_{\mathcal{H}_j} s_j$ $(j \geq 2)$, $|\mathbf{b}_j| = r^*_{\mathcal{H}_j} t_j$, $a_{j,g_0} = s_j - z_{j,K}$ $(j \geq 2)$ and $b_{j,g_0} = t_j - z_{j,\Psi}$. For $\mathbf{z}$ whose real part is $(s_j - c^K_{j,g_0}, t_j - c^{\Psi}_{j,g_0})_j$, $\mathbf{c}$ the node loads at the cores of an admissible point of the family of the tables of $F'$ with the mass $\sum_j (s_j + t_j) - \kappa_{F'}$ on $g_0$,*

$$\prod_{j=1}^{N} \Gamma_1^2 \begin{bmatrix} s_j - z_{j,K}, & z_{j,K} \\ & s_j \end{bmatrix} \Gamma_1^2 \begin{bmatrix} t_j - z_{j,\Psi}, & z_{j,\Psi} \\ & t_j \end{bmatrix} \int \boldsymbol{\varrho}_{F'}^{-\mathbf{z}}\, d\phi_{F'}$$
$$= \int_{\mathcal{F}_{\mathbf{z}}} \prod_{j=1}^{N} P_{\mathcal{H}_j}(\mathbf{a}_j; s_j, 0)\, P_{\mathcal{H}_j}(\mathbf{b}_j; 0, t_j) \prod_{i=1}^{k} \int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i}\, d\phi_{F_i},$$

*of dimension $(2N-1)(|\mathcal{E}_{\mathcal{H}}| + k - 1)$.*

*The cores and the cells are molecules by Lemma 26.1.*

**Corollary 26.17** (The glue and the series and parallel rules in hold form)**.** *On the lines of Theorem 26.16:*

*(i) For one table with $\mathcal{H} = \binom{\{g_1, g_2\}}{1}$ and the cells $\mathcal{A}$ and $\mathcal{A}'$, Theorem 26.16(i) is the integral of Theorem 23.9(iv), with the same integrand; for a family with the core $\binom{\{g_1, g_2\}}{1}$ in every table it is the integral of Theorem 19.13 over the scale shares.*

*(ii) For one table and a series node $\binom{G}{q}$, Theorem 26.16(ii) reads*

$$\Gamma(z)\Gamma(z + s - \kappa_{\mathcal{A}}) \int \varrho_{\mathcal{A}}^{-z} d\phi_{\mathcal{A}} = \int_{z_1 + \dots + z_q = z} \prod_{i=1}^{q} \Gamma(z_i)\Gamma(z_i + s - \kappa_{\mathcal{A}_i}) \int \varrho_{\mathcal{A}_i}^{-z_i} d\phi_{\mathcal{A}_i},$$

*the Mellin transform of $f_{\mathcal{A}}$ as the Mellin convolution of those of the $f_{\mathcal{A}_i}$ (Proposition 23.8(iii)): the Mellin transform, on these lines, of the product rule $f_{\mathcal{A}} = \prod_i f_{\mathcal{A}_i}$ of Theorem 23.14(i).*

*(iii) For one table and a parallel node* $\binom{G}{1}$ *it reads, with* $z' = \kappa_{\mathcal{A}} - z$ *and* $z'_i = \kappa_{\mathcal{A}_i} - z_i$,

$$\Gamma(z')\Gamma(z' + t - \kappa_{\mathcal{A}}) \int \varrho_{\mathcal{A}}^{z' - \kappa_{\mathcal{A}}} d\phi_{\mathcal{A}} = \int_{z'_1 + \cdots + z'_q = z'} \prod_{i=1}^{q} \Gamma(z'_i)\Gamma(z'_i + t - \kappa_{\mathcal{A}_i}) \int \varrho_{\mathcal{A}_i}^{z'_i - \kappa_{\mathcal{A}_i}} d\phi_{\mathcal{A}_i},$$

*the Mellin convolution of the transforms of the* $\check{f}_{\mathcal{A}_i}$*: the Mellin transform, on these lines, of the product rule* $\check{f}_{\mathcal{A}} = \prod_i \check{f}_{\mathcal{A}_i}$ *of Theorem 23.14(ii).*

So the three rules of the hinge tree procedure, the glue and the series and parallel products (Definition 23.19), are one rule read at three nodes, on the lines that the network certifies, the rules themselves holding on their own wider ranges; and the same rule applies at a prime node, where the procedure integrates the twisted core over its simplex for every value of the twists (Definition 23.19(4)). In Theorem 26.16 the prime node enters through the periods of its tables at their full leans, functions of its loads that do not depend on the twists, on the exponents or on the masses, and the cell families enter through the Mellin transforms of their laws on imaginary translates, for one table the representation (b) of Definition 23.19; Theorem 26.16(i) is Theorem 23.6(i) with the twisted core replaced by its lean convolution and the integration against the laws carried out through Mellin transforms.

## 26.7 The procedure and its cost

**Definition 26.18** (The network procedure)**.** *Given*: a hinge tree of every table of a family (Definition 23.15) with the weights and the marks of its nodes; exponents $s_j, t_j > 0$ and masses $\mathbf{m}$ on the balance wall, or several of each; optionally twists $\mathbf{x}$, or product weights (Proposition 26.20 below). *Reusable data*: for every orbit of nodes, over all the tables, under bijections keeping the table and its weights, the Gramma $\mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_T}(\boldsymbol{\theta})$ of its table, a function of the loads alone (Corollary 25.13), kept as (a) the closed form at series and parallel nodes (Proposition 26.9); (b) the share side, a fibre integral of dimension $|\mathcal{H}_v| - n_v$ (Corollary 26.8); (c) for an identically self-dual node, the reduction to the deletion of one mark (Corollaries 23.12(iii) and 25.17); (d) values on the imaginary translates in use. *Operations*:

(1) *Contour.* Find a split satisfying the node inequalities of Corollary 26.6, a linear feasibility problem with $\sum_j \sum_v 2(2^{|\mathcal{E}_{j,v}|+|G_{j,v}|} - 2)$ strict inequalities in $(2N-1)(n-1)$ unknowns; the native loads solve it, and infeasibility proves divergence.

(2) *Assemble.* The node loads of every table by Definition 26.3, the integrand of Theorem 26.13, at series-parallel nodes that of Corollary 26.14, with $\prod_j (\mathbf{x}_j^K)^{-\mathbf{a}_j} (\mathbf{x}_j^\Psi)^{-\mathbf{b}_j}$ for twists.

(3) *Pool*, for one table, disjoint pairs of marks of one series or parallel node whose twists and weights satisfy the condition of Proposition 26.15, by that proposition.

(4) *Integrate* the remaining loads on the contour; in the imaginary parts the integrand is analytic on a strip about the contour, the admissible points forming an open set, and decays exponentially there (Theorem 25.12(iii)), so the trapezoidal rule with step $\varsigma$ has an error of the order $e^{-2\pi d/\varsigma}$, $d$ the half-width of the strip; or close contours on lattices (Theorem 4.5).

(5) *Recursive Mellin evaluation*, an alternative to (2) to (4) when the hinge trees of the tables are one rooted tree with the same marks $\mathcal{E}_v$ at every vertex, so that at every vertex $v$ other than the root the subtree cells of the tables (Definition 23.15) form a cell family, the *subtree cell family* of $v$ (Definition 23.1): from the leaves up, compute the Mellin transform of the law of each orbit of subtree cell families on an imaginary translate by Theorem 26.16(ii),

one fibre integral of dimension $(2N-1)(|\mathcal{E}_v|+|G_v|-2)$ per point, and finish at the root by Theorem 26.16(i); for one table, at series and parallel nodes with repeated subtrees, equal and with equal masses, pass to the transforms, where repetition is a power (Proposition 23.18(ii)).

*Output*: the period, and with the same reusable data the period at other exponents, masses and twists (Proposition 26.21 below).

**Proposition 26.19** (What the network costs)**.** *In Definition 26.18:*

*(i) The fibre of the loads in (1)–(4) has dimension $(2N-1)(n-1)$, for one table less one for each pooled pair; opening every prime node on the share sides of the periods of its table at the two full leans adds $2(|\mathcal{H}_{j,v}|-n_{j,v})$ for each, and series and parallel nodes add nothing. The dimension does not depend on the numbers $|\mathcal{B}_j|$ of bases of the tables, against $\sum_j 2|\mathcal{B}_j|-2N-n+1$ for the atom formula (Corollary 18.16) and $\sum_j|\mathcal{B}_j|-n+1$ after its even shares close (Theorem 20.11(iv)).*

*(ii) The reusable data are one function of $n_v$ variables per orbit of nodes, the same for every node of the orbit, in every table, and for every query; a value of the period of a node at a full lean at given loads, once computed, serves every node of the orbit and every query at which those loads occur.*

*(iii) In the recursive Mellin evaluation the work is one Mellin transform on an imaginary translate for each orbit of subtree cell families (as in Proposition 23.18(i)), each point costing one fibre integral of dimension $(2N-1)(|\mathcal{E}_v|+|G_v|-2)$ at its top node, and at the root one of dimension $(2N-1)(|\mathcal{E}_o|+|G_o|-1)$, whose integrand needs the periods of the tables of the node at their full leans, at the loads of the fibre: closed forms at series and parallel nodes, and at a prime node an integral of its own, of dimension $|\mathcal{H}_{j,v}|-n_v$ on its share side, unless another representation of Definition 26.18 is cheaper.*

In the recursive Mellin evaluation equal subtrees at a series node enter as a power of one transform (Theorems 23.14(i) and 23.9(ii)), whose growth in the number of copies Theorem 24.2 reads when the subtree is series-balanced.

## 26.8 Changing the query in hold form

A query about a period is a choice of exponents, of masses, of weights and of the kind of answer asked for. In the hold form each of them enters at one place: exponents and masses move the pins and the affine marker loads, twists and product weights multiply the integrand by an exponential of the loads, and a replaced node, or a general change of the weights of one node, replaces that node's two functions and nothing else. Product weights are read as twists, and the exponents at fixed masses as the Mellin variables of the relative scales of the letters.

**Proposition 26.20** (Product weights are twists)**.** *For a family $F$ on $\mathcal{E}$ and $\boldsymbol{\zeta}=(\boldsymbol{\zeta}_j)_j$, $\boldsymbol{\zeta}_j\in(0,\infty)^{\mathcal{E}}$, let $F^{\boldsymbol{\zeta}}$ be $F$ with the weights $\theta_{j,T}\boldsymbol{\zeta}_j^T$ in table $j$. Then $K_{F^{\boldsymbol{\zeta}},j}(\alpha)=K_j(\boldsymbol{\zeta}_j\circ\alpha)$ and $\Psi_{F^{\boldsymbol{\zeta}},j}(\alpha)=\boldsymbol{\zeta}_j^{\mathcal{E}}\Psi_j(\alpha/\boldsymbol{\zeta}_j)$: $F^{\boldsymbol{\zeta}}$ has the letters of the twist $F[\mathbf{x}]$ with $\mathbf{x}_j^K=\boldsymbol{\zeta}_j$ and $\mathbf{x}_j^\Psi=\boldsymbol{\zeta}_j^{-1}$ (Definition 22.1), its cobasis letters multiplied by the constants $\boldsymbol{\zeta}_j^{\mathcal{E}}$, and*

$$P_{F^{\boldsymbol{\zeta}}}(\mathbf{m};\mathbf{s},\mathbf{t})=\prod_j\boldsymbol{\zeta}_j^{-t_j\mathbf{1}}\,P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t}),$$

$$P_{\mathcal{B}_j^{\boldsymbol{\zeta}_j}}(\mathbf{a};s_j,0)=\boldsymbol{\zeta}_j^{-\mathbf{a}}P_{\mathcal{B}_j}(\mathbf{a};s_j,0),\qquad P_{\mathcal{B}_j^{\boldsymbol{\zeta}_j}}(\mathbf{b};0,t_j)=\boldsymbol{\zeta}_j^{\mathbf{b}-t_j\mathbf{1}}P_{\mathcal{B}_j}(\mathbf{b};0,t_j),$$

*so that the lean convolution of $F^{\boldsymbol{\zeta}}$ is that of $F$ with the exponential $\prod_j \boldsymbol{\zeta}_j^{-t_j\mathbf{1}}\boldsymbol{\zeta}_j^{\mathbf{b}_j-\mathbf{a}_j}$ of the loads. With one $\boldsymbol{\zeta}$ for every table, $P_{F^{\boldsymbol{\zeta}}}(\mathbf{m};\mathbf{s},\mathbf{t}) = \boldsymbol{\zeta}^{-\mathbf{m}-|\mathbf{t}|\mathbf{1}}P_{F[\mathbf{x}']}(\mathbf{m};\mathbf{s},\mathbf{t})$, $\mathbf{x}'$ the twist of every cobasis letter by $\boldsymbol{\zeta}^{-2}$ and $|\mathbf{t}| = \sum_j t_j$. Weights $\zeta_{j,g}$ on the bases through one mark $g$, in each table $j$, are the twist at $g$ with $x_j^K = \zeta_{j,g}$ and $x_j^\Psi = \zeta_{j,g}^{-1}$, up to the factor $\prod_j \zeta_{j,g}^{-t_j}$: the weight of a hinge of Theorem 24.5, at any mark.*

**Proposition 26.21** (Carrying a calculation through a change, in hold form)**.** *In Theorem 26.13, with its node data computed:*

*(i) a change of the exponents $(s_j, t_j)$ changes the pins $|\mathbf{a}_j| = r_j s_j$, $|\mathbf{b}_j| = r_j^* t_j$ and the marker loads of table $j$, which are affine in $(s_j, t_j)$, and nothing else; the native loads of the new data are a new contour;*

*(ii) a change of the masses changes the load $\mathbf{a}_1 = \mathbf{m} - \sum_{j\geq 2}\mathbf{a}_j - \sum_j \mathbf{b}_j$ in the coordinates of Theorem 25.12(ii), and through it the marker loads $k_{1;v,\mathfrak{e}}$ of table 1, and in the integrand nothing else; in (i) and (ii) the contour must move when the old one is no longer admissible;*

*(iii) a change of twists, or of product weights on the bases of the tables, multiplies the integrand by an exponential of the loads (Proposition 26.20); product weights on the bases of one node multiply its periods at the two full leans by exponentials of its node loads;*

*(iv) replacing a node of a table by another table on the same marks and markers, of the same rank, one molecule with its markers neither loops nor coloops, or changing the weights of a node in any way, replaces the periods of that node at its two full leans and nothing else;*

*(v) in the recursive Mellin evaluation (Definition 26.18(5)), a change at one node changes the Mellin transforms of the laws of the subtree cell families on the path from it to the root only, each recomputed by Theorem 26.16(ii) from the new one of the vertex directly below it on the path and the kept ones of the other vertices directly below it, as Proposition 24.6 says of laws.*

The reusable data of Chapter 23, the laws of the subtrees, depend on the exponents and on the masses below each edge (Theorem 23.17(iii)); the reusable data of the network do not, and a sweep over exponents, over masses, over twists or over product weights recomputes no node. What does change is the contour, and Corollary 26.6 decides at the nodes whether the new query has an answer.

**Proposition 26.22** (The exponents are moment indices)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ be a family on $\mathcal{E}$, fix real masses $\mathbf{m}$, and let $I$ be the set of exponents $(\mathbf{s},\mathbf{t}) \in (0,\infty)^{2N}$ with $\sum_j (r_j s_j + r_j^* t_j) = m_{\mathcal{E}}$ at which every proper wall of $(\mathbf{m};\mathbf{s},\mathbf{t})$ is positive: a relatively open convex subset of that hyperplane, possibly empty. In this statement write $L_1, \dots, L_{2N}$ for the letters $K_j$ and $\Psi_j$ of $F$, $w_i$ for their exponents and $g_i$ for their degrees, and for each $i$ and $\boldsymbol{\lambda} \in (0,\infty)^{2N-1}$ put*

$$\mathsf{G}_i(\mathbf{m};\boldsymbol{\lambda}) = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\mathbf{m}-\mathbf{1}} \exp\Big( - L_i(x) - \sum_{i'\neq i} \lambda_{i'} L_{i'}(x)\Big)\, dx,$$

*the Gramma (Definition 2.1) of the concatenated configuration of all the letters, with the coefficients of $L_i$ and $\lambda_{i'}$ times those of $L_{i'}$. For $(\mathbf{s},\mathbf{t}) \in I$ and every $i$,*

$$\Gamma(w_i)\, P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = g_i \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\mathbf{m}-\mathbf{1}} e^{-L_i(x)} \prod_{i'\neq i} L_{i'}(x)^{-w_{i'}}\, dx$$

$$= \frac{g_i}{\prod_{i' \neq i} \Gamma(w_{i'})} \int_{(0,\infty)^{2N-1}} \prod_{i' \neq i} \lambda_{i'}^{w_{i'}-1} \, \mathsf{G}_i(\mathbf{m}; \boldsymbol{\lambda}) \, d\boldsymbol{\lambda}.$$

*At fixed masses the numbers* $\Gamma(w_i) P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})/g_i$ *on* $I$*, functions of the exponents of the other letters, are the integrals of* $\prod_{i' \neq i} L_{i'}^{-w_{i'}}$ *against the one positive measure* $x^{\mathbf{m}-\mathbf{1}} e^{-L_i(x)} dx$*, and* $\prod_{i'} \Gamma(w_{i'}) \, P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})/g_i$ *is the Mellin transform at* $(w_{i'})_{i' \neq i}$ *of the one function* $\mathsf{G}_i(\mathbf{m}; \cdot)$ *of the relative scales of the other letters to* $L_i$*, completely monotone in each of them. On* $I$ *the period* $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ *is log-convex, by Proposition 8.12(iii) and Hölder's inequality.*

So the period of a family at fixed masses is one integral over the orthant with any one of its letters closed: the closed letter leaves one positive measure on the orthant, the integrand of its Gramma at the index $\mathbf{m}$, and the period is a joint moment of the other letters under it, their exponents with the sign changed being its orders.

# Chapter 27

# Finite interfaces and certified enclosures

A cell enters every construction through its hinge wall and its law (Theorem 23.11), and a cell with several hinges through the joint law of its channels (Theorem 21.16, Definition 21.13). A law is not a finite list of numbers. Finitely many numbers are retained for every prescribed polynomial moment, the joint moments of the channels under independent Gamma draws on the marks, up to a degree $D$: they compose by the bilinear map of Theorem 21.14, a repeated chain cell acts on them by one finite matrix, so that every polynomial Gamma moment of a repetition satisfies a linear recurrence with constant coefficients, and the walls turn a moment of finite degree into a certified enclosure of the period. The period is retained as well over a neighbourhood of the query: the walls give the neighbourhood explicitly, the Taylor coefficients in the weights are periods with walls at least as high, and a finite parameter bank with two explicit errors answers every query inside it. Exact representation size, arithmetic work and accuracy are separate quantities.

## 27.1 The joint law of the channels and its moments

Let $\mathcal{A}$ be a cell with hinges $G$ on its marks $B$, or cells with the hinges $G$ on the same marks $B$ kept together (Definition 27.1), with channel vector $\mathrm{ch}(\mathcal{A})$ (Definition 21.13), with positive masses $\mathbf{m}$ on $B$; the hinges carry no mass. Its total mass $m_B$ and the product $\prod_{e\in B}\Gamma(m_e)$ are kept as data: at a splicing the total masses add and these products multiply. For a multi-index $\mathbf{k}$ on the channels, $\mathrm{ch}(\mathcal{A})^{\mathbf{k}}$ is the product of the channels with these multiplicities, a polynomial of the degree $\deg\mathbf{k}=\sum_c k_c\deg c$, the sum over the channels $c$.

**Definition 27.1** (The channel law; moment vectors)**.** Cells $\mathcal{A}_1,\dots,\mathcal{A}_N$ with the hinges $G$ on the same marks $B$ are *kept together* when their channels are read jointly: $\mathcal{A}=(\mathcal{A}_1,\dots,\mathcal{A}_N)$ has the channel vector $\mathrm{ch}(\mathcal{A})=(\mathrm{ch}(\mathcal{A}_j))_j$, listing the channels of all of them by the tags $(j,\mathrm{letter},p)$, $N2^{|G|+1}$ of them, absent channels counting as zero; one cell is the case $N=1$, and the tables of a family, with $G=\varnothing$, are kept together with their letters as channels. The *channel law* of $\mathcal{A}$ is the image $\mathrm{ch}(\mathcal{A})_*\,\mathrm{Dir}_{\mathbf{m}}$ of the Dirichlet law of the shape (Definition 3.10) under the channel vector. The *Gamma draws* on the marks are $\alpha\sim\mathrm{Gamma}_{\mathbf{m}}$, independent $\mathrm{Gamma}_{m_e}$ variables (Proposition 8.12), and the *moments* and the *moment vector of degree* $D$ of $\mathcal{A}$ are

$$\mathrm{Mom}_{\mathcal{A}}(\mathbf{k})=\mathbb{E}_{\alpha\sim\mathrm{Gamma}_{\mathbf{m}}}\big[\,\mathrm{ch}(\mathcal{A})(\alpha)^{\mathbf{k}}\big],\qquad \mathrm{Mom}_{\mathcal{A}}^{\le D}=\big(\mathrm{Mom}_{\mathcal{A}}(\mathbf{k})\big)_{|\mathbf{k}|\le D},$$

at most $\binom{q+D}{D}$ numbers, $q$ the number of channels. A *polynomial Gamma moment* of degree at

most $D$ of $\mathcal{A}$ is $\mathbb{E}_{\alpha\sim\mathrm{Gamma}_{\mathbf{m}}}\big[p(\mathrm{ch}(\mathcal{A})(\alpha))\big]$ for a polynomial $p$ of degree at most $D$ in the channels, and a *shape moment* is the same with $U\sim\mathrm{Dir}_{\mathbf{m}}$ in place of $\alpha$. A cell without marks has constant channels, a point as its law and the products of its channels as its moments, with total mass 0 and the empty product of Gammas.

The channels must be kept jointly: the moments of the channels one at a time do not determine a product of different channels exposed to the same room.

**Proposition 27.2** (The channel law composes)**.** *Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be cells, or cells kept together, with disjoint marks $B_1$, $B_2$ and masses $\mathbf{m}_1$, $\mathbf{m}_2$, spliced table by table along pairs of their hinges to $\mathcal{A}_{12}$ (Theorem 21.14), both with marks. Let $U_1\sim\mathrm{Dir}_{\mathbf{m}_1}$, $U_2\sim\mathrm{Dir}_{\mathbf{m}_2}$ and $(V_1,V_2)\sim\mathrm{Dir}_{(m_{B_1},m_{B_2})}$ be independent; replace every channel $c$ of $\mathcal{A}_1$ by $V_1^{\deg c}c(U_1)$ and every channel $c$ of $\mathcal{A}_2$ by $V_2^{\deg c}c(U_2)$. The law of the image of the two channel vectors so replaced under the bilinear map of Theorem 21.14(iii), applied table by table, is the channel law of $\mathcal{A}_{12}$. For the tables of a family, kept together without hinges, the channels are their letters, and the period of the family is Proposition 8.12(i) read on the channel law: one Trailmoment of the channel law on its source space (Definition 5.1, Theorem 5.21). For a cell family $F=((\mathcal{A}_j,\theta_j);s_j,t_j)_{j=1}^N$ with the one hinge $g$ (Definition 23.1), its cells kept together, with the coordinates $(x_{j,0},x_{j,1},x^*_{j,0},x^*_{j,1})$ of the channels $K_{\mathcal{A}_j\setminus g}$, $K_{\mathcal{A}_j/g}$, $\Psi_{\mathcal{A}_j\setminus g}$, $\Psi_{\mathcal{A}_j/g}$, the law of $F$ is the weighted image*

$$(\boldsymbol{\varrho}_F)_*\phi_F=\Gamma_{\substack{|B|\\1}}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}\ (R_F)_*\Big[\prod_j x_{j,1}^{-s_j}x_{j,0}^{*\,-t_j}\Big(\frac{x_{1,0}}{x_{1,1}}\Big)^{-\kappa_F}\ \mathrm{ch}(\mathcal{A})_*\,\mathrm{Dir}_{\mathbf{m}}\Big],$$

*with $R_F=\big((x_{j,0}x_{1,1}/(x_{j,1}x_{1,0}))_{j\ge2},(x^*_{j,1}x_{1,1}/(x^*_{j,0}x_{1,0}))_j\big)$, an equality of positive measures, possibly infinite: the channel law, which knows nothing of the exponents, gives the law of the cell family at all exponents by a weight and an image.*

**Theorem 27.3** (Gamma draws close the moments)**.** *(i)* Leaves. *If $\mathrm{ch}(\mathcal{A})(x)^{\mathbf{k}}=\sum_{\mathbf{h}}c_{\mathbf{h}}x^{\mathbf{h}}$, then $\mathrm{Mom}_{\mathcal{A}}(\mathbf{k})=\sum_{\mathbf{h}}c_{\mathbf{h}}\prod_e(m_e)_{h_e}$, rational at rational masses and weights.*

*(ii)* Assembly. *Let $\mathrm{ch}(\mathcal{A}_{12})=\mathrm{sp}\big(\mathrm{ch}(\mathcal{A}_1),\mathrm{ch}(\mathcal{A}_2)\big)$ with the bilinear map* sp *of Theorem 21.14(iii), applied table by table to cells kept together, and write $\mathrm{sp}(c,c')^{\mathbf{k}}=\sum_{\mathbf{k}_1,\mathbf{k}_2}b_{\mathbf{k};\mathbf{k}_1,\mathbf{k}_2}c^{\mathbf{k}_1}c'^{\mathbf{k}_2}$, a sum over $|\mathbf{k}_1|=|\mathbf{k}_2|=|\mathbf{k}|$. Then*

$$\mathrm{Mom}_{\mathcal{A}_{12}}(\mathbf{k})=\sum_{\mathbf{k}_1,\mathbf{k}_2}b_{\mathbf{k};\mathbf{k}_1,\mathbf{k}_2}\,\mathrm{Mom}_{\mathcal{A}_1}(\mathbf{k}_1)\,\mathrm{Mom}_{\mathcal{A}_2}(\mathbf{k}_2),$$

*and no moment of degree above $|\mathbf{k}|$ of either cell is needed; a splicing inside one cell is linear and keeps the degree.*

*(iii)* Return to the shape. *For a cell with marks, $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}\big[\mathrm{ch}(\mathcal{A})(U)^{\mathbf{k}}\big]=\mathrm{Mom}_{\mathcal{A}}(\mathbf{k})/(m_B)_{\deg\mathbf{k}}$: the joint moments of the channel law through degree $D$ are read off $\mathrm{Mom}_{\mathcal{A}}^{\le D}$, the rank, the sizes and the tags fixing the denominators.*

*(iv)* Sufficiency. *Two cells with the same hinges, channel tags and degrees and the same $\mathrm{Mom}^{\le D}$ give every final polynomial Gamma moment of degree at most $D$ in the channels, in any assembly with cells on disjoint marks, the same value, even where an intermediate cell exposes more hinges than the replaced one; with the same total mass as well, they give every final shape moment of degree at most $D$ of the channels of the root, under its Dirichlet law, the same value.*

*(v)* The whole law. *For a cell with marks, the* $\mathrm{Mom}_{\mathcal{A}}^{\le D}$ *for all* $D$*, with* $m_B$ *and the degrees, determine its channel law, and with it the expectation of every integrable function of the channels (Proposition 27.2).*

The draws are used for nonnegative moments of the channels. A negative moment of a period integrand under the draws diverges on the balance wall: the scale contributes $\int_0^\infty S^{-1}e^{-S}dS$, the pole of $\Gamma(\kappa_{\mathcal{E}})$ in Proposition 8.12. The return to the shape, followed by Proposition 27.2 or by the peak closing (Theorem 15.13(i)) with its stopping rule (Theorem 27.7), is the change of representation that the period needs.

**Corollary 27.4** (The work follows the distinct interfaces)**.** *Fix a degree* $D$*, an assembly that exposes at most* $h$ *hinges at every cell it keeps, and identifications of its repeated subassemblies keeping their data and hinges. Every requested moment of the root is computed exactly by keeping at most* $\binom{2^{h+1}+D}{D}$ *numbers for each distinct cell,* $\binom{N2^{h+1}+D}{D}$ *for* $N$ *tables kept together. The arithmetic is* $O\big(\sum_v E_v(D)\big)$ *operations once the local expansions and the leaf moments are built,* $E_v(D)$ *the number of nonzero coefficients* $b_{\mathbf{k};\mathbf{k}_1,\mathbf{k}_2}$ *at a distinct splicing* $v$ *of two cells, at most the product of the sizes of its three moment vectors, or of two for a splicing inside one cell. Equal copies share one moment vector and remain independent draws: their moments are multiplied as in Theorem 27.3(ii), never identified.*

This counts operations. Building the local expansions, finding an assembly of small width, the bit lengths of exact coefficients and the degree a moment needs are separate parts of the cost; finding the cheapest assembly is a search, as for constructions (Section 33.1).

## 27.2 Repetition is a finite matrix

**Theorem 27.5** (A finite matrix for every polynomial Gamma moment)**.** *Let a chain cell* $\mathcal{U}$ *(Proposition 24.4, here with several hinges) be a cell with two disjoint sets of hinges* $G_{\mathrm{in}}$ *and* $G_{\mathrm{out}}$*, each in a fixed bijection with the hinges of a cell* $\mathcal{A}_0$*, and let* $\mathcal{A}_{l+1}$ *be* $\mathcal{U}$*, with fresh marks and the same masses and weights at every use, spliced onto* $\mathcal{A}_l$ *by pairing* $G_{\mathrm{in}}$ *with the hinges of* $\mathcal{A}_l$*; so* $\mathcal{A}_{l+1}$ *exposes* $G_{\mathrm{out}}$*, named through the bijection as the hinges of* $\mathcal{A}_l$*, with the same channel tags, and its channel vector is the image of* $\mathrm{ch}(\mathcal{A}_l)$ *and that of the copy of* $\mathcal{U}$ *under the map* sp*. Keep one list of channel tags for every* $l$*, absent channels being zero. For every* $D$ *there is a matrix* $\mathbf{T}_D$*, independent of* $l$ *and computed from* $\mathcal{U}$ *by finite expansion and Theorem 27.3(i), with*

$$\mathrm{Mom}_{\mathcal{A}_l}^{\le D} = \mathbf{T}_D^{\,l}\ \mathrm{Mom}_{\mathcal{A}_0}^{\le D}\,.$$

*Every fixed polynomial Gamma moment of degree at most* $D$ *of* $\mathcal{A}_l$*, a fixed terminal cell spliced on it included, is* $\mathbf{e}^{\mathsf{T}}\mathbf{T}_D^{\,l}\,\mathrm{Mom}_{\mathcal{A}_0}^{\le D}$ *for a fixed row* $\mathbf{e}^{\mathsf{T}}$*; so its generating series in* $l$ *is* $\mathbf{e}^{\mathsf{T}}(I-\zeta\mathbf{T}_D)^{-1}\,\mathrm{Mom}_{\mathcal{A}_0}^{\le D}$*, rational in* $\zeta$*, the sequence satisfies the linear recurrence with constant coefficients given by the characteristic polynomial of* $\mathbf{T}_D$*, of order at most* $\binom{q+D}{D}$*, and one length* $l$ *costs* $O\big(\binom{q+D}{D}^3\log(l{+}1)\big)$ *operations by repeated squaring.* $\mathbf{T}_D$ *is block diagonal by the degree, and a moment of one degree uses its block alone; a repeated sequence of different chain cells is the power of the product of their matrices.*

On a cell with one hinge $g$, a chain cell $\mathcal{U}$ with the hinges $i$ and $o$, $i$ spliced to $g$, and the channels $K_{\mathcal{U},(P)}$, $\Psi_{\mathcal{U},(P)}$, $P\subseteq\{i,o\}$, acts by

$$\begin{pmatrix} K_{\mathcal{A}_{l+1},(\varnothing)} \\ K_{\mathcal{A}_{l+1},(\{o\})} \end{pmatrix} = \begin{pmatrix} K_{\mathcal{U},(\{i\})} & K_{\mathcal{U},(\varnothing)} \\ K_{\mathcal{U},(\{i,o\})} & K_{\mathcal{U},(\{o\})} \end{pmatrix} \begin{pmatrix} K_{\mathcal{A}_l,(\varnothing)} \\ K_{\mathcal{A}_l,(\{g\})} \end{pmatrix},$$

and by the same matrix of the $\Psi_{\mathcal{U},(P)}$ on the $\Psi$ channels, which is Theorem 21.14(iii) with one pair. A moment of degree $a$ in the $K$ channels and $b$ in the $\Psi$ channels lives on $(a+1)(b+1)$ monomials, and the block of $\mathbf{T}_D$ on them is the joint expectation over the draws of $\mathcal{U}$ of the expanded actions: the two matrices share the marks of $\mathcal{U}$ and are not independent. For a family at integer exponents, the joint moments of its letters along a chain of bunches with matrix letters are the case of Theorem 22.10. An iterate of an integral operator gives no finite closing relation by itself (Theorem 22.8, Proposition 24.4); every polynomial Gamma moment of a repetition does. The period is recovered from a sequence of such moments with certified tails, in Section 27.3.

**Corollary 27.6** (Shape moments have recurrences)**.** *For the repetition of Theorem 27.5 let the total masses be $m_{\mathcal{A}_l} = m_{\mathcal{A}_0} + l\, m_{\mathcal{U}}$, $m_{\mathcal{U}}$ that of $\mathcal{U}$, let $p$ be a polynomial in the channels, the same for every $l$, homogeneous of degree $d_l = d_0 + lv$ in the marks, $v$ an integer, and let $M_l$ be the polynomial Gamma moment of $\mathcal{A}_l$ for $p$. The shape moment for $p$ is $M_l/(m_{\mathcal{A}_l})_{d_l}$, and every recurrence $\sum_i c_i M_{l+i} = 0$ gives $\sum_i c_i \frac{(m_{\mathcal{A}_{l+i}})_{d_{l+i}}}{(m_{\mathcal{A}_l})_{d_l}} \frac{M_{l+i}}{(m_{\mathcal{A}_{l+i}})_{d_{l+i}}} = 0$. If the masses are rational and $h \geq 1$ is an integer with $h m_{\mathcal{U}} \in \mathbb{Z}$, then on each residue of $l$ modulo $h$, with $\mathbf{T}_D^h$ for $\mathbf{T}_D$, the coefficients are rational functions of $l$ over the field of the weights, polynomials after clearing denominators, in $\mathbb{Q}[l]$ at rational weights.*

## 27.3 A certified enclosure of the period

**Theorem 27.7** (Explicit stopping from the walls)**.** *Let a family $F$ on $n \geq 2$ marks have real masses on its balance wall and exponents $s_j, t_j \geq 0$, with every proper wall positive. Write its active letters, the $K_j$ with $s_j > 0$ and the $\Psi_j$ with $t_j > 0$, as $L_i$ with exponents $w_i > 0$, degrees $g_i$ and counts $d_i(S)$, as in Theorem 15.13, so that $m_{\mathcal{E}} = \sum_i g_i w_i$ and $\kappa_S = m_S - \sum_i w_i d_i(S)$. Choose references $M_i \geq \max_{\Delta_{\mathcal{E}}} L_i$, and in this statement put $Y_i = L_i/M_i \in (0,1]$ on the open simplex. For an order $\pi$ of the marks let $U_{\pi,k}$ be its tails, $L_{i,\pi}$ the sector letter of $L_i$, the $k_{j,\pi}$ or $\psi_{j,\pi}$ of Theorem 8.5(ii), and $\underline{L}_{i,\pi} > 0$ its constant term. For integers $Q_i \geq 0$ put*

$$P^{\leq \mathbf{Q}} = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix} \prod_i M_i^{-w_i} \sum_{0 \leq k_i \leq Q_i} \prod_i \frac{(w_i)_{k_i}}{k_i!}\ \mathbb{E}_{U \sim \mathrm{Dir}_{\mathbf{m}}} \Big[ \prod_i \big(1 - Y_i(U)\big)^{k_i} \Big],$$

*the partial sums of the peak closing (Theorem 15.13(i)), computed exactly from the moments of the letters through the degree $\sum_i Q_i$, each monomial with its own rising factorial (Theorem 27.3(iii)). For every choice of $\epsilon_i > 0$ with*

$$\kappa_S - \epsilon_i d_i(S) > 0 \qquad (\varnothing \neq S \subsetneq \mathcal{E}),$$

*which is exactly $\epsilon_i < \varpi_i$, $\varpi_i$ the least of the ratios $\kappa_S/d_i(S)$ in Theorem 15.13(ii) (any $\epsilon_i > 0$ when no proper $S$ has $d_i(S) > 0$), put*

$$C_i = M_i^{\epsilon_i}\, n^{g_i \epsilon_i} \sum_{\pi} \underline{L}_{i,\pi}^{-\epsilon_i} \prod_{i'} \underline{L}_{i',\pi}^{-w_{i'}} \prod_{k=2}^{n} \frac{1}{\kappa_{U_{\pi,k}} - \epsilon_i d_i(U_{\pi,k})}.$$

*Then*

$$0 \leq P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) - P^{\leq \mathbf{Q}} \leq \sum_i C_i \frac{\Gamma(w_i + \epsilon_i)\, \Gamma(w_i + Q_i + 1)}{\Gamma(w_i)\, \Gamma(w_i + Q_i + 1 + \epsilon_i)},$$

*and for $\epsilon_i \leq 1$ the last factor is at most $(w_i/(w_i + Q_i + 1))^{\epsilon_i}$. So finitely many moments and finitely many walls enclose the period to any prescribed accuracy; the same moments answer every other*

*choice of exponents $w_i$ with $\sum_i g_i w_i = m_{\mathcal{E}}$ at the same masses, only the coefficients of $P^{\le \mathbf{Q}}$, the walls and the certificate changing.*

At rational data the constants of the certificate are positive rationals and their rational powers, and no period other than the one sought enters it; the stacked Gamma $\Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}$ and the Gamma ratios stay exact products and quotients of values of the Gamma function, so the enclosure is exact without being rational. The certificate can itself be expensive, $C_i$ being a sum over all $n!$ orders, and it is crude: it replaces every sector letter by its least value. The true tail is of the order $Q^{-\varpi}(\log Q)^{k-1}$ (Theorem 15.13(ii)); the certified decay $Q_i^{-\epsilon_i}$ tends to it as $\epsilon_i \to \varpi_i$, at the cost of the factor $\prod_k (\kappa_{U_{\pi,k}} - \epsilon_i d_i(U_{\pi,k}))^{-1}$, which grows without bound there. Sharper references and sharper sector bounds lower the degree needed without changing the moments. The degree needed can grow with the size of the assembly, so this is a terminating certified procedure, not a uniform fast approximation.

## 27.4 Changing the query with finite data

**Proposition 27.8** (Parameters and insertions)**.** *Fix the local supports and the assembly.*

*(i) For each $D$ the leaf moments and the splicing and transfer coefficients are polynomials in the local weights and in the masses; they can be kept symbolic and specialized at any positive values.*

*(ii) Carrying the leaf moments, Theorem 27.3(ii) and Theorem 27.5 in the truncated ring $\mathbb{Q}(\boldsymbol{\theta}, \mathbf{m})[\boldsymbol{\zeta}]/(\boldsymbol{\zeta})^{h+1}$ of $p$ local parameters $\boldsymbol{\zeta}$, over the field generated by the weights $\boldsymbol{\theta}$ and the masses, gives every derivative of the moments through the order $h$, at most $\binom{p+h}{h}$ coefficients per entry.*

*(iii) A further polynomial observable, a mark total for instance, can be made a channel when every local update is separately affine in each independent cell vector, a constant channel adjoined.*

*(iv) Raising the masses by nonnegative integers $\boldsymbol{\gamma}$ is a polynomial insertion: for every observable $g$,*

$$\mathbb{E}_{\alpha \sim \mathrm{Gamma}_{\mathbf{m}+\boldsymbol{\gamma}}}[g(\alpha)] = \mathbb{E}_{\alpha \sim \mathrm{Gamma}_{\mathbf{m}}}\left[\alpha^{\boldsymbol{\gamma}} g(\alpha)\right] \Big/ \prod_e (m_e)_{\gamma_e}.$$

The moments do not depend on the exponents asked for: changing the exponents reuses every moment, and changing a local weight or mass changes the local data and their ancestors in the assembly only, as Proposition 24.6 says of laws. Shared parameters may appear in many copies, whose draws stay independent.

## 27.5 Certified parameter banks

A family $F$ on its balance wall, its letters $K_j$ and $\Psi_j$ written $L_i$, $i = 1, \dots, \ell$, with the exponents $w_i$ (the $s_j$ and $t_j$), the degrees $g_i$ and the counts $d_i(S) = \min\{\gamma_S : [u^\gamma] L_i > 0\}$, so that $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_i L_i(u)^{-w_i} d\omega_{\mathcal{E}}(u)$, with $m_{\mathcal{E}} = \sum_i g_i w_i$ and the walls $\kappa_S = m_S - \sum_i w_i d_i(S) > 0$ at the starting data, is asked a neighbourhood of queries: $p$ complex parameters $\boldsymbol{\zeta}$, and a bank of Taylor coefficients through the total order $h$, at most $\binom{h+p}{p}$ numbers. The order of a bank is the number of differentiations, not the degree of a polynomial moment of Section 27.1.

**Theorem 27.9** (Taylor coefficients in the weights are periods with walls at least as high)**.** *Let the exponents $w_i$ be real and change the coefficients of the letters, $L_i(\boldsymbol{\zeta}) = L_i + \sum_{j=1}^{p} \zeta_j D_{ij}$, each $D_{ij}$ homogeneous of degree $g_i$ with support in that of $L_i$ and coefficients of either sign, and in this statement write $P(\boldsymbol{\zeta}) = \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_i L_i(\boldsymbol{\zeta})(u)^{-w_i}\, d\omega_{\mathcal{E}}(u)$, so that $P(\mathbf{0}) = P_F(\mathbf{m};\mathbf{s},\mathbf{t})$. Then, with $l_i = \sum_j l_{ij}$,*

$$[\boldsymbol{\zeta}^{\mathbf{k}}]P = \sum_{\substack{l_{ij}\geq 0 \\ \sum_i l_{ij} = k_j}} \prod_i \frac{(-1)^{l_i}(w_i)_{l_i}}{\prod_j l_{ij}!} \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} \prod_i \Big( L_i^{-w_i - l_i} \prod_j D_{ij}^{l_{ij}} \Big)\, d\omega_{\mathcal{E}},$$

*and after the numerators are expanded every term is a convergent period of the family at masses $\mathbf{m} + \boldsymbol{\gamma}$ and exponents $w_i + l_i$, balanced, with every proper wall $\kappa'_S = \kappa_S + \gamma_S - \sum_i l_i d_i(S) \geq \kappa_S$.*

**Theorem 27.10** (A uniform certificate for weights)**.** *In the setting of Theorem 27.9, with $P(\boldsymbol{\zeta})$ as there, write $L_i = \sum_{\boldsymbol{\gamma}} c_{i,\boldsymbol{\gamma}} u^{\boldsymbol{\gamma}}$ with $c_{i,\boldsymbol{\gamma}} > 0$ and $D_{ij} = \sum_{\boldsymbol{\gamma}} c_{i,\boldsymbol{\gamma}} v_{i,\boldsymbol{\gamma},j} u^{\boldsymbol{\gamma}}$, $V_{ij} = \max_{\boldsymbol{\gamma}} |v_{i,\boldsymbol{\gamma},j}|$, and choose radii $R_j > 0$ with $r_i = \sum_j R_j V_{ij} < 1$. On $|\zeta_j| \leq R_j$, $|P(\boldsymbol{\zeta})| \leq P(\mathbf{0}) \prod_i \max\big((1-r_i)^{-w_i}, (1+r_i)^{-w_i}\big) =: M$, made explicit by $P(\mathbf{0}) \leq \sum_\pi \prod_i \max\big(L_{i,\pi}(\mathbf{0})^{-w_i}, L_{i,\pi}(\mathbf{1})^{-w_i}\big) \prod_{k=2}^{n} \kappa_{U_{\pi,k}}^{-1}$, with the sector letters $L_{i,\pi}$ of the letters on the sector of the order $\pi$, the $k_{j,\pi}$ or $\psi_{j,\pi}$ of Theorem 8.5(ii), or by the upper end of the enclosure of Proposition 15.12; $P$ is holomorphic near the closed polydisc, $|[\boldsymbol{\zeta}^{\mathbf{k}}]P| \leq M\mathbf{R}^{-\mathbf{k}}$, and its Taylor polynomial $P^{[h]}$ through the order $h$ has, on $|\zeta_j| \leq \vartheta R_j$ with $0 \leq \vartheta < 1$,*

$$|P(\boldsymbol{\zeta}) - P^{[h]}(\boldsymbol{\zeta})| \leq M \sum_{k>h} \binom{k+p-1}{p-1} \vartheta^k \leq M\, \frac{\binom{h+p}{p-1}\vartheta^{h+1}}{1 - \vartheta(h+p+1)/(h+2)},$$

*the second bound when $\vartheta(h+p+1) < h+2$.*

**Theorem 27.11** (Wall-certified parameter banks)**.** *Keep the letters and change the masses and exponents affinely, $m_e(\boldsymbol{\zeta}) = m_e^0 + \sum_j a_{ej}\zeta_j$ and $w_i(\boldsymbol{\zeta}) = w_i^0 + \sum_j b_{ij}\zeta_j$ with real slopes and balance in every direction, so that $\kappa_S(\boldsymbol{\zeta}) = \kappa_S^0 + \sum_j c_{Sj}\zeta_j$ with $c_{Sj} = \sum_{e\in S} a_{ej} - \sum_i b_{ij} d_i(S)$, and in this statement write $P(\boldsymbol{\zeta}) = P_F(\mathbf{m}(\boldsymbol{\zeta});\mathbf{s}(\boldsymbol{\zeta}),\mathbf{t}(\boldsymbol{\zeta}))$, the exponents $w_i(\boldsymbol{\zeta})$ on the letters. Choose $R_j > 0$ with*

$$\underline{\kappa}_S := \kappa_S^0 - \sum_j |c_{Sj}| R_j > 0 \qquad (\varnothing \neq S \subsetneq \mathcal{E}).$$

*On the sector of an order $\pi$ every letter is its monomial times its sector letter $L_{i,\pi}$, the $k_{j,\pi}$ or $\psi_{j,\pi}$ of Theorem 8.5(ii), with nonnegative coefficients in the sector coordinates; put $\underline{L}_{i,\pi} = L_{i,\pi}(\mathbf{0})$, $\overline{L}_{i,\pi} = L_{i,\pi}(\mathbf{1})$, $\eta_{i,\pi} = \max(|\log \underline{L}_{i,\pi}|, |\log \overline{L}_{i,\pi}|)$, $\nu_i = \sum_j |b_{ij}| R_j$,*

$$B_\pi = \prod_i \max\big(\underline{L}_{i,\pi}^{-w_i^0}, \overline{L}_{i,\pi}^{-w_i^0}\big) e^{\eta_{i,\pi}\nu_i}, \qquad M_{\mathbf{R}} = \sum_\pi B_\pi \prod_{k=2}^{n} \underline{\kappa}_{U_{\pi,k}}^{-1}.$$

*Then $P$ is holomorphic near $|\zeta_j| \leq R_j$ with $|P| \leq M_{\mathbf{R}}$, every $|[\boldsymbol{\zeta}^{\mathbf{k}}]P| \leq M_{\mathbf{R}}\mathbf{R}^{-\mathbf{k}}$, the bound of Theorem 27.10 holds with $M_{\mathbf{R}}$ for $M$, and*

$$[\boldsymbol{\zeta}^{\mathbf{k}}]P = \frac{1}{\mathbf{k}!} \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}^0-\mathbf{1}} \prod_i L_i^{-w_i^0} \prod_j \Big( \sum_e a_{ej} \log u_e - \sum_i b_{ij} \log L_i \Big)^{k_j} d\omega_{\mathcal{E}},$$

*absolutely integrable and of degree zero. The starting exponents may be any reals at which the proper walls are positive.*

**Theorem 27.12** (A finite compiler with two errors)**.** *In the setting of Theorem 27.11, with $P(\boldsymbol{\zeta})$ as there, put $y_{i,\pi} = 1 - L_{i,\pi}/\overline{L}_{i,\pi} \in [0, \lambda_\pi]$, $\lambda_\pi = \max_i(1 - \underline{L}_{i,\pi}/\overline{L}_{i,\pi}) < 1$, $\beta_i = |w_i^0| + \nu_i$ and $A_\pi = \prod_i \overline{L}_{i,\pi}^{-w_i^0} e^{|\log \overline{L}_{i,\pi}|\nu_i} \prod_{k=2}^n \underline{\kappa}_{U_{\pi,k}}^{-1}$. Expand $\prod_i L_{i,\pi}^{-w_i(\boldsymbol{\zeta})} = \prod_i \overline{L}_{i,\pi}^{-w_i(\boldsymbol{\zeta})} \sum_{\mathbf{l}} \prod_i \frac{(w_i(\boldsymbol{\zeta}))_{l_i}}{l_i!} y_{i,\pi}^{l_i}$, truncate at $|\mathbf{l}| \le Q$, and integrate the finitely many terms exactly, each $\prod_i y_{i,\pi}^{l_i} = \sum_{\boldsymbol{\gamma}} c_{\boldsymbol{\gamma}} q^{\boldsymbol{\gamma}}$ in the sector coordinates $q$ contributing*

$$\prod_i \overline{L}_{i,\pi}^{-w_i(\boldsymbol{\zeta})} \prod_i \frac{(w_i(\boldsymbol{\zeta}))_{l_i}}{l_i!} \sum_{\boldsymbol{\gamma}} c_{\boldsymbol{\gamma}} \prod_{k=2}^n \frac{1}{\kappa_{U_{\pi,k}}(\boldsymbol{\zeta}) + \gamma_k},$$

*to obtain $\widetilde{P}_Q(\boldsymbol{\zeta})$. Then on the whole polydisc*

$$|P(\boldsymbol{\zeta}) - \widetilde{P}_Q(\boldsymbol{\zeta})| \le \mathrm{err}_Q := \sum_\pi A_\pi \sum_{k>Q} \frac{(|\boldsymbol{\beta}|)_k}{k!} \lambda_\pi^k \le \sum_\pi A_\pi \frac{(|\boldsymbol{\beta}|)_{Q+1} \lambda_\pi^{Q+1}}{(Q+1)!\,(1 - r_{\pi,Q})},$$

*the second bound when $r_{\pi,Q} = \lambda_\pi \max\left(1, \frac{|\boldsymbol{\beta}|+Q+1}{Q+2}\right) < 1$ for every $\pi$; and the Taylor polynomial $\widetilde{P}_Q^{[h]}$ of $\widetilde{P}_Q$ through the order $h$, computed by finite arithmetic in a truncated ring, has on $|\zeta_j| \le \vartheta R_j$*

$$|P(\boldsymbol{\zeta}) - \widetilde{P}_Q^{[h]}(\boldsymbol{\zeta})| \le \frac{\mathrm{err}_Q}{(1-\vartheta)^p} + M_{\mathbf{R}} \sum_{k>h} \binom{k+p-1}{p-1} \vartheta^k.$$

*With no parameters, $p = 0$, so that $\nu_i = 0$ and $\underline{\kappa}_S = \kappa_S$, and with every $w_i^0 \ge 0$, the first bound is that of Proposition 15.12(ii): there $\overline{L}_{i,\pi} = L_i(\mathbf{1})$ for every $\pi$, $\widetilde{P}_Q = \prod_i L_i(\mathbf{1})^{-w_i^0} \sum_\pi \sum_{M \le Q} I_{\pi,M}$, and the explicit tail is $\prod_i L_i(\mathbf{1})^{-w_i^0} \sum_\pi \mathrm{err}_{\pi,Q}$, the letters being the $K_j$ and $\Psi_j$ with the exponents $s_j$ and $t_j$.*

For fixed data and $\vartheta < 1$ the orders $Q$ and $h$ grow like the logarithm of the inverse error, up to the polynomial factors of the scalar tails; expanding the sector polynomials still costs work that depends on the dimension and the supports. The walls also cover paths: along a segment on which every proper wall is positive, with $\kappa_-$ the least proper wall at the two ends and $a$ the largest change of a proper wall along it, every point of the segment, parametrized by $\lambda \in [0,1]$, has the disc of radius $\kappa_-/(2a)$ in the complex parameter on which every proper wall keeps a real part at least $\kappa_-/2$, the walls being affine, and a compact path has a finite cover by such discs. A bank at one point does not give the coefficients at another; what is reused everywhere is the compiled sector data, the coefficient procedure and the wall test.

## 27.6 Procedures

**Definition 27.13** (The interface procedure)**.** *Given*: cells with their weights and masses; the pairs of hinges to splice and an assembly order with its identifications of repeated parts; a polynomial moment, a peak truncation or an accuracy, with, for an accuracy, references and a wall certificate. *Steps*: (1) read the channels of the leaves by coefficient extraction, with their degrees, the sizes, the total masses and the products $\prod_e \Gamma(m_e)$; (2) choose $D$, the degree of the moment or $\sum_i Q_i$ from Theorem 27.7; (3) compute the leaf moments by Theorem 27.3(i), only at the multi-indices the moment pulls back to; (4) compile every distinct splicing by Theorem 27.3(ii), and every repeated chain cell once into $\mathbf{T}_D$ and its power (Theorem 27.5), reusing a vector only when its tags and data match; (5) splice the pairs at the root, return to the shape by Theorem 27.3(iii),

and evaluate $P^{\leq \mathbf{Q}}$ with its certificate. *Output*: an exact polynomial moment, a recurrence or rational generating series for a repetition, or a certified enclosure of the period, with the vectors, the splicing coefficients, the parameter expressions and the certificate kept for the next query.

**Definition 27.14** (The bank procedure)**.** *Given*: a family on its balance wall with positive walls, and affine directions of change of its masses and exponents, or of the weights of its letters. *Steps* along masses and exponents: compute the walls and a box of radii satisfying the strict inequalities of Theorem 27.11; form the sector constants and $M_{\mathbf{R}}$; choose $Q$ and $h$ until the bound of Theorem 27.12 is below the tolerance; compile the retained sector terms in the truncated ring. Along weights, where the sector letters move with $\boldsymbol{\zeta}$ and Theorem 27.12 does not apply: choose radii by Theorem 27.10, with its bound made explicit there; each coefficient through the order $h$ is the finite signed sum of periods with walls at least as high of Theorem 27.9, each enclosed by Proposition 15.12, and the tail beyond $h$ is bounded by Theorem 27.10. *Output*: the parameter bank, its domain and its certified error; every query inside the smaller box is a polynomial evaluation, and the exact coefficients remain available as the periods of Theorem 27.9 or the integrals of Theorem 27.11.

# Chapter 28

# Bunches of any width

A bunch of width one is one mark to its room (Theorem 19.4): its patterns fill two levels, each level is one child of the bunch against one table of the room, and what the room sees of the bunch is its wall and the law of its ratio vector, one ratio for one table (Lemma 23.2, Theorem 23.11). At every width a room sees its bunch through the wall, the widths and the joint law of the bunch letters of a level factorization (Theorem 21.16); that statement has a canonical form, and the calculus of width one carries to every width. At the width $c$ the patterns fill $c+1$ levels, and a level between the extremes pairs several stations of patterns with several stations of room sides: which pairs meet, and with which constants, is the seam of the bunch (Definition 28.1), graded by the excess, which no description by marks or by level totals contains. The room sees the bunch through the letters of the tables it presents to it, one for each station of room sides, and the law of these letters is what the bunch retains: it suffices against every partner, and the tinkers of one partner already separate any two laws (Theorem 28.19). A bunch of width $c$ is absorbed into its partner as one stand-in mark of degree $c$ (Lemma 28.8), a mark of a table exactly at width one (Theorem 28.24), and a nonempty family of sets of one size is the family of bases of a table exactly when, at one bunch, its children and room children are tables, adjacent ones are quotients, the extreme ones are one, and its levels are lifts (Theorem 28.30). The circuits of the seam are the directions of the shares of the compatible pairs that neither side sees, and the period integrates over them through the Gramma of the compatible pairs (Proposition 28.14). Width one is the case of the hinge throughout: the seam of a bunch of width one is its hinge (Proposition 28.3(ii)), and its law is the law of its ratio vector read through one fixed change of scale (Proposition 28.11). Bunches compose along chains and trees (Chapter 29), the glue closes along the directions of its shares (Chapter 31), and a law has regions of equality and a reach at its ends (Chapter 32).

## 28.1 The seam of a bunch

**Definition 28.1** (Seam)**.** Let a table $(\mathcal{B}, \theta)$ have the bunch $B$ and the room $R$. A level $k$ of $B$ is named by its excess $i = k - d(B)$ (Definition 12.16), $0 \le i \le c(B)$: its patterns have $d(B)+i$ marks and its room sides $d(R)+c(B)-i$, the room having the width of the bunch (Definition 19.1). Fix an excess $i$.

(i) Two patterns of excess $i$ lie in one station when their children (Definition 12.1) are positive multiples of one another, weights included, as at the bunches $S_l$ of Theorem 22.9. The *room child* at a room side $X$ is the child of the bunch $R$ at its pattern $X$: the table on $B$ with the bases $\mathcal{B}_{(X)} = \{p : \ p \sqcup X \in \mathcal{B}\}$ and the weights $\theta_{(X),p} = \theta_{p \sqcup X}$ (Lemma 12.2). Two room sides of excess $i$ lie in one station when their room children are positive multiples of one another.

The stations of patterns of excess $i$ form the set $\mathcal{K}_i$ and those of room sides the set $\mathcal{D}_i$: the stations of the bunch and of the room are $\mathcal{K} = \bigsqcup_i \mathcal{K}_i$ and $\mathcal{D} = \bigsqcup_i \mathcal{D}_i$, each station carrying its excess $i(C)$ or $i(D)$.

(ii) Within a station $C$ the child at $p \in C$ is $w_p$ times one weighted table on $R$, the *child of the station* $C$, with the bases $\mathcal{B}_{(C)}$ and the weights $\theta_{(C)}$, the *multipliers* $w_p > 0$ being normalized by $\max_{p\in C} w_p = 1$; likewise the room child at $X \in D$ is $w'_X$ times the *room child of the station* $D$, a table on $B$ with the bases $\mathcal{B}_{(D)}$ and the weights $\theta_{(D)}$, with $\max_{X\in D} w'_X = 1$. The *pair constant* of $C \in \mathcal{K}_i$ and $D \in \mathcal{D}_i$ is $h_{CD} = \theta_{p\sqcup X}/(w_p w'_X)$ for any $p \in C$ and $X \in D$ with $p \sqcup X \in \mathcal{B}$, which does not depend on the choice, and $h_{CD} = 0$ when there is none; $(C, D)$ is *compatible* when $h_{CD} > 0$. The *seam matrix* of the excess $i$ is $H_i = (h_{CD})_{C\in\mathcal{K}_i,\, D\in\mathcal{D}_i}$, and the *seam* of $B$ is $(H_0, \dots, H_{c(B)})$ with its stations; a family has one seam in each table.

(iii) The *station letters* are $\mathsf{x}_C = \sum_{p\in C} w_p\, \alpha_B^p$ and $\mathsf{x}^*_C = \sum_{p\in C} w_p\, \alpha_B^{B\setminus p}$ on $B$, and $\mathsf{y}_D = \sum_{X\in D} w'_X\, \alpha_R^X$ and $\mathsf{y}^*_D = \sum_{X\in D} w'_X\, \alpha_R^{R\setminus X}$ on $R$. The *seam letters* of $B$ are the letters $K_{(D)}$ and $\Psi_{(D)}$ of the room children of the stations $D \in \mathcal{D}$, polynomials on $B$; those of $R$ are the letters $K_{(C)}$ and $\Psi_{(C)}$ of the children of the stations $C \in \mathcal{K}$, polynomials on $R$.

For a list every multiplier and every pair constant is 1. The normalization by the largest multiplier chooses nothing and commutes with every bijection of the marks; any other normalization multiplies $\mathsf{x}_C$ and $\mathsf{y}_D$ by positive constants and $h_{CD}$ by their inverses, and leaves every product $h_{CD}\,\mathsf{x}_C\,\mathsf{y}_D$ unchanged. In table $j$ of a family the stations, the letters and the constants carry the index $j$: $\mathcal{K}_j$, $\mathcal{D}_j = \bigsqcup_i \mathcal{D}_{j,i}$, $\mathsf{x}_{jC}$, $\mathsf{y}_{jD}$, $h_{j,CD}$, $K_{j,(D)}$ and so on.

**Proposition 28.2** (The seam identity)**.** *In Definition 28.1:*

*(i) The seam letters are the station letters through the seam:*

$$K_{(D)} = \sum_C h_{CD}\,\mathsf{x}_C, \qquad \Psi_{(D)} = \sum_C h_{CD}\,\mathsf{x}^*_C,$$
$$K_{(C)} = \sum_D h_{CD}\,\mathsf{y}_D, \qquad \Psi_{(C)} = \sum_D h_{CD}\,\mathsf{y}^*_D :$$

*the seam letters of a side are the letters of the tables that the other side presents to it.*

*(ii) If $(C, D)$ is compatible, every $p \in C$ and $X \in D$ make a basis, with $\theta_{p\sqcup X} = h_{CD}\, w_p\, w'_X$, and the bases of excess $i$ are the disjoint union of these* rectangles $C \times D$ *over the compatible pairs.*

*(iii) With $\mathsf{x}_i$, $\mathsf{x}^*_i$, $\mathsf{y}_i$, $\mathsf{y}^*_i$ the vectors of the station letters of excess $i$,*

$$K = \sum_{i=0}^{c(B)} \mathsf{x}_i^{\mathsf{T}} H_i\, \mathsf{y}_i = \sum_{D\in\mathcal{D}} K_{(D)}\,\mathsf{y}_D = \sum_{C\in\mathcal{K}} \mathsf{x}_C\, K_{(C)},$$
$$\Psi = \sum_{i=0}^{c(B)} \mathsf{x}_i^{*\mathsf{T}} H_i\, \mathsf{y}^*_i = \sum_{D\in\mathcal{D}} \Psi_{(D)}\,\mathsf{y}^*_D = \sum_{C\in\mathcal{K}} \mathsf{x}^*_C\, \Psi_{(C)},$$

*and along the scale of $B$, $\alpha_B = \rho u$ with $u \in \Delta_B$ (Lemma 2.7),*

$$K(\rho u, \alpha_R) = \rho^{d(B)} \sum_{i=0}^{c(B)} \rho^i \sum_{D\in\mathcal{D}_i} K_{(D)}(u)\,\mathsf{y}_D(\alpha_R),$$

$$\Psi(\rho u, \alpha_R) = \rho^{d^*(B)} \sum_{i=0}^{c(B)} \rho^{c(B)-i} \sum_{D \in \mathcal{D}_i} \Psi_{(D)}(u)\, \mathsf{y}^*_D(\alpha_R).$$

*The writing by the stations of the room, with the bunch letters $K_{(D)}$ and the room letters $\mathsf{y}_D$, is a level factorization with nonnegative terms (Definition 21.15), the* room-station factorization*; the writing by the stations of the bunch is the children factorization.*

*(iv) Within one excess no two stations of $\mathcal{K}_i$ have proportional rows of $H_i$, and no two stations of $\mathcal{D}_i$ proportional columns.*

*(v) At the excesses 0 and $c(B)$ every pattern meets every room side in a basis (Theorem 12.19). $H_0$, respectively $H_{c(B)}$, is a single positive entry exactly when the weights of that level factor along these products, as for unit weights, for weights factoring across $B$ and $R$ and for tables determinantal over $\mathbb{R}$ or $\mathbb{C}$; when both do, the room child of the one station of excess 0 is $\mathcal{B}/R$ and that of excess $c(B)$ is $\mathcal{B}|B$ (Proposition 12.17), these tables carrying the weights of the room children of the two stations, and the seam letters at the two extremes are $K_{\mathcal{B}/R}$, $\Psi_{\mathcal{B}/R}$ and $K_{\mathcal{B}|B}$, $\Psi_{\mathcal{B}|B}$.*

*(vi) Swapping $B$ and $R$ transposes every $H_i$ and reverses the excesses, $i \mapsto c(B) - i$; the seam letters of $R$ are then those of the bunch $R$.*

A compatible pair fills its rectangle: inside it every tuple of the bunch meets every tuple of the room, with weights of rank one. What the marks do not show is the pattern of the rectangles, which stations of the bunch meet which stations of the room. At an extreme excess every station of the bunch meets every station of the room, in one rectangle when the weights of that level factor; between the extremes a station need not meet every station of the other side, and the seam matrix $H_i$, not the marks, says how they meet.

**Proposition 28.3** (The width ladder)**.** *In Definition 28.1:*

*(i) $c(B) = 0$ exactly when $B$ is a union of molecules of the table; there is one excess, every pattern meets every room side, and $K = \mathsf{x}_0^{\mathsf{T}} H_0\, \mathsf{y}_0$, $\Psi = \mathsf{x}_0^{*\mathsf{T}} H_0\, \mathsf{y}_0^*$. The seam is one rectangle $C \times D$ exactly when the weights factor across $B$ and $R$, as for lists; then $K = h_{CD}\, \mathsf{x}_C\, \mathsf{y}_D$ and $\Psi = h_{CD}\, \mathsf{x}_C^*\, \mathsf{y}_D^*$: the letters factor over the bunch and the room.*

*(ii) $c(B) = 1$ exactly when there are two excesses, both extreme. Each is one rectangle exactly when the weights factor across $B$ and $R$; then for the stations $D_0$ and $D_1$ of the excesses 0 and 1 the seam letters are $K_{(D_0)} = K_{\mathcal{B}/R}$, $K_{(D_1)} = K_{\mathcal{B}|B}$, $\Psi_{(D_0)} = \Psi_{\mathcal{B}/R}$ and $\Psi_{(D_1)} = \Psi_{\mathcal{B}|B}$, the room station letters are $\mathsf{y}_{D_0} = K_{\mathcal{B}|R}$ and $\mathsf{y}_{D_1} = K_{\mathcal{B}/B}$, these two tables carrying the multipliers, the largest 1, and the room-station factorization is the writing*

$$K = K_{\mathcal{B}/R}\, K_{\mathcal{B}|R} + K_{\mathcal{B}|B}\, K_{\mathcal{B}/B}$$

*of Theorem 19.4: the letter adds over two levels, and the seam is the hinge.*

*(iii) For $c(B) \geq 2$ the excesses $0 < i < c(B)$ can carry several rectangles.*

*At every width an excess with one station on one side has one station on the other, and the children at the patterns of each excess are equal, $B$ being in level form, exactly when every transposition of two marks of $B$, fixing every other mark, carries the table (Proposition 14.12); then every excess is one rectangle.*

## 28.2 Pieces, their laws, and the glue

The seam identity writes a family as a pairing of what the bunch presents to the room and what the room presents to the bunch, station by station. The pairing uses nothing of the exchange property and holds for pieces of any kind.

**Definition 28.4** (Pieces with stations; their glue)**.** Let $N \geq 1$ and $c_1, \dots, c_N \geq 0$ be integers. A set of *stations* for $N$ tables of the widths $c_j$ is a family $\mathcal{D} = (\mathcal{D}_j)_{j=1}^{N}$ of finite sets graded by excesses, $\mathcal{D}_j = \bigsqcup_{i=0}^{c_j} \mathcal{D}_{j,i}$, with $\mathcal{D}_{j,0}$ and $\mathcal{D}_{j,c_j}$ nonempty; a station $D \in \mathcal{D}_{j,i}$ has the excess $i(D) = i$. A *piece* with the stations $\mathcal{D}$ on a finite set $B$, at exponents $(s_j, t_j)_j$ with $s_j, t_j \geq 0$, carries for every $j$ and $D \in \mathcal{D}_j$ two nonzero polynomials $V_{jD}$ and $V^*_{jD}$ on $B$ with nonnegative coefficients, its *seam letters* (positive letters, Definition 25.1, where their degree is positive), homogeneous of the degrees $d_j + i(D)$ and $d^*_j + c_j - i(D)$ for integers $d_j, d^*_j \geq 0$, and masses $\mathbf{m}_B$; its *wall* and its *top wall* are

$$\kappa_B = m_B - \sum_j \big(s_j d_j + t_j d^*_j\big), \qquad \kappa_B^{\mathrm{top}} = \sum_j (s_j + t_j) c_j - \kappa_B.$$

A *partner* is a piece with the same stations on a set $R$ disjoint from $B$, read with the reversed excesses $c_j - i(D)$: its letters, the *room letters* $L_{jD}$ and $L^*_{jD}$ (as in Definition 21.15), have the degrees $e_j + c_j - i(D)$ and $e^*_j + i(D)$, and its wall is $\kappa_R = m_R - \sum_j (s_j e_j + t_j e^*_j)$. A *family of Newton words* is a sequence $(K_j, \Psi_j; s_j, t_j)_{j=1}^{N}$ of pairs of positive letters on one set with exponents $s_j, t_j \geq 0$, carrying masses; its word is $\prod_j K_j^{-s_j} \Psi_j^{-t_j}$ and its period is the period of that word (Definition 25.1). The *glue* of the piece and the partner is the family of Newton words on $B \sqcup R$ with the letters

$$K_j = \sum_{D \in \mathcal{D}_j} V_{jD}\, L_{jD}, \qquad \Psi_j = \sum_{D \in \mathcal{D}_j} V^*_{jD}\, L^*_{jD},$$

the exponents $(s_j, t_j)$ and the masses $(\mathbf{m}_B, \mathbf{m}_R)$; its period is written $P_{B \sqcup R}(\mathbf{m})$, and its balance wall is $\kappa_R = \kappa_B^{\mathrm{top}}$. A piece is named by its set, or by a calligraphic letter $\mathcal{X}$ when several pieces are compared, and its objects carry that name.

By Proposition 28.2(iii) a bunch of a family of tables, with its seam letters $V_{jD} = K_{j,(D)}$ and $V^*_{jD} = \Psi_{j,(D)}$, is a piece with the stations of its room, $d_j = d_j(B)$ and $d^*_j = d^*_j(B)$; its room, with the room station letters $L_{jD} = \mathsf{y}_{jD}$ and $L^*_{jD} = \mathsf{y}^*_{jD}$, is a partner, $e_j = d_j(R)$ and $e^*_j = d^*_j(R)$; and their glue is the family. By (vi) the same holds with $B$ and $R$ swapped, on the stations $\mathcal{K}$. The walls are the walls $\kappa_B$ and $\kappa_R$ of the bunch and of the room, and the top wall is the top wall $\kappa_B^{\mathrm{top}}$ of the bunch (Definition 8.1). A glue need not be a family of tables; its period converges exactly when its Newton sum has the dimension $|B \sqcup R| - 1$ and the masses lie in its relative interior (Theorem 25.2), and every statement about pieces is a statement about the period.

**Definition 28.5** (The law of a piece)**.** Let a piece on $B$ with the stations $\mathcal{D}$ have exponents with $\sum_j (s_j + t_j) c_j > 0$. Its active letters (Definition 8.1) are $K_j$ when $s_j > 0$ and $\Psi_j$ when $t_j > 0$; every coordinate, share, pin, product and count indexed by a letter runs over the active letters only, an inactive letter entering no period. The *excess sums* are $V_j^{(i)} = \sum_{D \in \mathcal{D}_{j,i}} V_{jD}$ and $V_j^{*(i)} = \sum_{D \in \mathcal{D}_{j,i}} V^*_{jD}$, the superscript counting excesses. On $\Delta_B$ put

$$\gamma_B^{\rho} = \prod_j \Big(\frac{V_j^{(c_j)}}{V_j^{(0)}}\Big)^{s_j/(\kappa_B + \kappa_B^{\mathrm{top}})} \Big(\frac{V_j^{*(0)}}{V_j^{*(c_j)}}\Big)^{t_j/(\kappa_B + \kappa_B^{\mathrm{top}})},$$

$$\phi_B^\rho(du) = u^{\mathbf{m}_B-\mathbf{1}} \prod_j \big(V_j^{(0)}\big)^{-s_j} \big(V_j^{*(c_j)}\big)^{-t_j} \, (\gamma_B^\rho)^{-\kappa_B} \, d\omega_B(u),$$

$$\varrho_{jD}^\rho = \frac{V_{jD}}{V_j^{(0)} \, (\gamma_B^\rho)^{i(D)}}, \qquad \varrho_{jD}^{\rho *} = \frac{V_{jD}^*}{V_j^{*(c_j)} \, (\gamma_B^\rho)^{c_j - i(D)}},$$

the *scale of the piece*, homogeneous of degree one, the *scale form*, and the coordinates of the *seam vector* $\boldsymbol{\varrho}_B^\rho = (\varrho_{jD}^\rho, \varrho_{jD}^{\rho *})_{j,D}$, homogeneous of degree zero, with a coordinate $\varrho_{jD}^\rho$ for each active $K_j$ and $D \in \mathcal{D}_j$ and a coordinate $\varrho_{jD}^{\rho *}$ for each active $\Psi_j$ and $D \in \mathcal{D}_j$. Its values, the *seam points*, lie on the *seam torus*: the set of the points $\mathbf{x} = (x_{jD}, x_{jD}^*)_{j,D}$ with positive coordinates on which $\sum_{D \in \mathcal{D}_{j,0}} x_{jD} = 1$ for every active $K_j$, $\sum_{D \in \mathcal{D}_{j,c_j}} x_{jD}^* = 1$ for every active $\Psi_j$, and

$$\prod_j \Big( \sum_{D \in \mathcal{D}_{j,c_j}} x_{jD} \Big)^{s_j} \Big( \sum_{D \in \mathcal{D}_{j,0}} x_{jD}^* \Big)^{t_j} = 1,$$

of dimension $n_{\mathcal{D}}$, the number of the coordinates less the number of the active letters less one, which is $2|\mathcal{D}| - 2N - 1$, $|\mathcal{D}| = \sum_j |\mathcal{D}_j|$, when every letter is active. The *law* of the piece is the image $(\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho$, a positive measure carried by the seam torus. Its Mellin transform, at shares $(\mathbf{a}, \mathbf{b}) = (a_{jD}, b_{jD})_{j,D}$ of the active letters,

$$\int_{\Delta_B} (\boldsymbol{\varrho}_B^\rho)^{-(\mathbf{a},\mathbf{b})} \, d\phi_B^\rho = \int \mathbf{x}^{-(\mathbf{a},\mathbf{b})} \, d\big((\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho\big)(\mathbf{x}), \qquad \mathbf{x}^{-(\mathbf{a},\mathbf{b})} = \prod_{j,D} x_{jD}^{-a_{jD}} \, {x_{jD}^*}^{-b_{jD}},$$

is the *seam function*, in $[0, \infty]$ at real shares and defined at complex shares wherever it converges absolutely. A partner has its scale, scale form, seam vector, law and seam function by the same definition, with its room letters and the reversed excesses.

The scale form is projective: its orthant form has the degree $m_B - \sum_j (s_j d_j + t_j d_j^*) - \kappa_B = 0$, so it is read on any section (Lemma 16.2). The seam points lie on the seam torus: the sums at the excesses 0 and $c_j$ are 1 by definition, and the product is $(\gamma_B^\rho)^{\kappa_B + \kappa_B^{\mathrm{top}}} (\gamma_B^\rho)^{-\sum_j (s_j + t_j) c_j} = 1$. The scale of the piece is the geometric mean of the oriented extreme ratios of the active letters, each taken to the root of its width: $K_j$ measures the scale by $(V_j^{(c_j)} / V_j^{(0)})^{1/c_j}$ and $\Psi_j$ by $(V_j^{*(0)} / V_j^{*(c_j)})^{1/c_j}$, in the proportions $s_j c_j$ and $t_j c_j$; a letter of width zero does not see the scale. Any other positive scale $q \, \gamma_B^\rho$ of degree one, $q$ a positive function of the seam vector, multiplies $\phi_B^\rho$ by $q^{-\kappa_B}$ and the coordinates of the seam vector by powers of $q$: a fixed bijection of the tori with a density, so the law is determined up to it. The scale named is canonical: it uses the seam letters of the active letters alone, treats the two letters alike, and is kept at the centre by the bijections exchanging the family (Proposition 28.6(iii)). With weights factoring at the extreme levels, $V_j^{(0)} = K_{\mathcal{B}_j/R}$, $V_j^{(c_j)} = K_{\mathcal{B}_j|B}$, $V_j^{*(0)} = \Psi_{\mathcal{B}_j/R}$ and $V_j^{*(c_j)} = \Psi_{\mathcal{B}_j|B}$ (Proposition 28.2(v)); so

$$\gamma_B^\rho = \prod_j \Big( \frac{K_{\mathcal{B}_j|B}}{K_{\mathcal{B}_j/R}} \Big)^{s_j/(\kappa_B + \kappa_B^{\mathrm{top}})} \Big( \frac{\Psi_{\mathcal{B}_j/R}}{\Psi_{\mathcal{B}_j|B}} \Big)^{t_j/(\kappa_B + \kappa_B^{\mathrm{top}})},$$

$$\phi_B^\rho(du) = u^{\mathbf{m}_B-\mathbf{1}} \prod_j K_{\mathcal{B}_j/R}^{-s_j} \Psi_{\mathcal{B}_j|B}^{-t_j} \, (\gamma_B^\rho)^{-\kappa_B} \, d\omega_B(u).$$

The law is read off the measure of Theorem 21.16. For the room-station factorization the letter vector is $\mathbf{V}(u) = \big(K_{j,(D)}(u), \Psi_{j,(D)}(u)\big)_{j,D}$; the density of $\phi_B^\rho$ against $u^{\mathbf{m}_B-\mathbf{1}} d\omega_B$ and the seam vector are functions of $\mathbf{V}(u)$, so the law is an image of the measure $\mathbf{V}_*\big(u^{\mathbf{m}_B-\mathbf{1}} d\omega_B(u)\big)$ with

a density. The seam vector forgets the seam letters of the inactive letters and, of the others, one constant for each active letter and the scale, $V_{jD} \mapsto \lambda^{i(D)} V_{jD}$ and $V^*_{jD} \mapsto \lambda^{c_j - i(D)} V^*_{jD}$: the directions that the pins of Lemma 28.7 fix, on which the scale form changes by constant factors.

**Proposition 28.6** (Symmetries of a seam)**.** *Let a bijection $g$ of the marks $\mathcal{E}$ of a family map $B$ onto $B'$ and $R$ onto $R'$.*

*(i) If $g$ carries table $j$ to table $j'$ (Definition 11.13), it maps the patterns and the room sides of $B$ of each excess in table $j$ onto those of $B'$ of the same excess in table $j'$, stations onto stations and compatible pairs onto compatible pairs, and the station letters and seam letters of the image stations are those of the stations read at $g^*\alpha$, the normalization of the multipliers commuting with $g$.*

*(ii) If $g$ exchanges table $j$ with table $j'$, the windows of $B$ and $B'$ correspond as in Lemma 14.10, and $p \mapsto B' \setminus g(p)$, $X \mapsto R' \setminus g(X)$ map the stations of $B$ of excess $i$ in table $j$ onto those of $B'$ of excess $c(B) - i$ in table $j'$ and compatible pairs onto compatible pairs, the station letter $\mathsf{x}_C$ and the seam letter $K_{(D)}$ going to the station letter $\mathsf{x}^*_{C'}$ and the seam letter $\Psi_{(D')}$ of the image stations $C'$, $D'$, and back.*

*(iii) So the bijections that keep $B$ and carry or exchange the family permute the stations of its seams, with their station letters and seam letters, up to the bijection. Let such a bijection also keep the masses. A carrying one maps the seam torus of $B$ onto itself, permuting the coordinates, and keeps the law $(\boldsymbol{\varrho}^\rho_B)_* \phi^\rho_B$. An exchanging one, sending table $j$ to table $j'$, maps the seam torus of $B$ at the lean $\boldsymbol{\delta}$ onto the seam torus of $B$ at the lean $-\boldsymbol{\delta}$, the coordinates $\varrho^\rho_{jD}$ of table $j$ going to the coordinates $\varrho^{\rho *}_{j'D'}$ of table $j'$ and back, and maps the law of $B$ at $\boldsymbol{\delta}$ onto the law of $B$ at $-\boldsymbol{\delta}$; at the centre it keeps the law. At the centre the potential is kept by the carrying and the exchanging maps alike, at a lean by the carrying maps (Proposition 11.14): the centre does not tell apart stations joined by a bijection carrying or exchanging the family, nor a lean stations joined by a carrying one.*

At the centre two stations can lie in one orbit through an exchange and in different orbits at every lean: the centre sees them as one, a lean separates them. These are the confusable stations of the seam, the analogue for stations of the halves $C_K$, $C_\Psi$ of a class of marks (Theorem 11.17); the seam carries them down the tree with the classes of marks (Theorem 14.20).

**Lemma 28.7** (The seam function is the period of the seam word)**.** *On the* pins of the seam*, the letter totals $\sum_D a_{jD} = s_j$ for every active $K_j$ and $\sum_D b_{jD} = t_j$ for every active $\Psi_j$ and the* scale pin

$$\sum_{j,D} \Big( i(D)\, a_{jD} + \big(c_j - i(D)\big)\, b_{jD} \Big) = \kappa_B,$$

*the seam function is the period at the masses $\mathbf{m}_B$ of the* seam word $W_B(\mathbf{a}, \mathbf{b}) = \prod_{j,D} V_{jD}^{-a_{jD}} {V^*_{jD}}^{-b_{jD}}$ *(Definition 25.1):*

$$\int_{\Delta_B} (\boldsymbol{\varrho}^\rho_B)^{-(\mathbf{a},\mathbf{b})}\, d\phi^\rho_B = P_{W_B(\mathbf{a},\mathbf{b})}(\mathbf{m}_B) = \int_{\Delta_B} u^{\mathbf{m}_B - \mathbf{1}} \prod_{j,D} V_{jD}(u)^{-a_{jD}} V^*_{jD}(u)^{-b_{jD}}\, d\omega_B(u).$$

For the piece of a bunch of a family the seam word is the bunch word of the room-station factorization (Theorem 22.11), and its period on the pins is the Mellin transform of one law.

**Lemma 28.8** (Absorbing a piece of any width)**.** *Let a piece on $B$ with the stations $\mathcal{D}$ carry masses $\mathbf{m}_B$ and exponents $(s_j, t_j)_j$ with $\sum_j (s_j + t_j) c_j > 0$, its wall $\kappa_B$ and its seam vector $\boldsymbol{\varrho}^\rho_B$, whose*

*values are written* $\mathbf{x} = (x_{jD}, x^*_{jD})_{j,D}$. *Let* $g$ *be a mark outside* $B$, *the* stand-in *of the piece,* $Z$ *a finite set disjoint from* $B \sqcup \{g\}$, $\mathbf{m}_Z$ *masses on* $Z$, *and* $f_1, \dots, f_p$ *positive continuous functions, each assigned to one active letter of the piece,* $f_l$ *defined on* $(0,\infty)^Z \times (0,\infty)^{\mathcal{D}_j}$ *when its letter is* $K_j$ *or* $\Psi_j$ *and homogeneous of degree one in its second argument,* $f_l(\alpha_Z, \lambda\mathbf{y}) = \lambda\, f_l(\alpha_Z, \mathbf{y})$ *for* $\lambda > 0$, *with real exponents* $a_l$ *such that the exponents assigned to each letter add to its exponent. In this statement write* $\mathbf{V}_l$ *for the vector of the seam letters of the letter of* $f_l$, $(V_{jD})_{D\in\mathcal{D}_j}$ *for* $K_j$ *and* $(V^*_{jD})_{D\in\mathcal{D}_j}$ *for* $\Psi_j$; $V_l$ *for its extreme excess sum,* $V^{(0)}_j$ *for* $K_j$ *and* $V^{*(c_j)}_j$ *for* $\Psi_j$; *and* $\mathbf{x}_l(\alpha_g)$ *for the vector* $(x_{jD}\,\alpha_g^{\,i(D)})_D$ *for* $K_j$ *and* $(x^*_{jD}\,\alpha_g^{\,c_j - i(D)})_D$ *for* $\Psi_j$. *Let* $H \ge 0$ *be a Borel function on* $(0,\infty)^p$. *If the form*

$$\omega = \alpha_B^{\mathbf{m}_B - \mathbf{1}} \alpha_Z^{\mathbf{m}_Z - 1}\, H\Big(\Big(\frac{f_l\big(\alpha_Z, \mathbf{V}_l(\alpha_B)\big)}{V_l(\alpha_B)}\Big)_l\Big) \prod_{l=1}^{p} f_l\big(\alpha_Z, \mathbf{V}_l(\alpha_B)\big)^{-a_l}\, d\alpha_B\, d\alpha_Z$$

*is homogeneous of degree zero on* $(0,\infty)^{B\sqcup Z}$, *then, in* $[0,\infty]$,

$$\int_{\{h=1\}} \omega = \int \Big( \int_{\{h'=1\}} \omega_{\mathbf{x}} \Big)\, d\big((\boldsymbol{\varrho}^\rho_B)_* \phi^\rho_B\big)(\mathbf{x}),$$
$$\omega_{\mathbf{x}} = \alpha_Z^{\mathbf{m}_Z - \mathbf{1}} \alpha_g^{\kappa_B - 1}\, H\Big(\big(f_l(\alpha_Z, \mathbf{x}_l(\alpha_g))\big)_l\Big) \prod_l f_l\big(\alpha_Z, \mathbf{x}_l(\alpha_g)\big)^{-a_l}\, d\alpha_Z\, d\alpha_g,$$

*the integrals taken over any sections of positive functions* $h$, $h'$ *of degree one on the two orthants.*

The piece collapses to its stand-in $g$, which carries the mass $\kappa_B$; a letter of table $j$ that sees the piece sees the stand-in with the degree $c_j$, at its stations twisted by the seam point. At width one this is Lemma 23.2 read through Proposition 28.11.

**Lemma 28.9** (The native split at the stations)**.** *Let a family of tables have real data on its balance wall, with* $s_j, t_j \ge 0$, *not all zero, and* $\mathbf{m} > 0$, *and a native gauge* $\alpha^\star$ *(Definition 8.17),* $B$ *a bunch with* $\sum_j (s_j + t_j) c_j(B) > 0$, *and put, for the active letters,*

$$a_{jD} = s_j\, \mathbb{P}_{j,\alpha^\star}(T \cap R \in D), \qquad b_{jD} = t_j\, \mathbb{P}^*_{j,\alpha^\star}(T \cap R \in D), \qquad D \in \mathcal{D}_j,$$

*the native split (Definition 12.25) aggregated to the stations of room sides. Then* $(\mathbf{a}, \mathbf{b})$ *is a point of the pins of the seam with positive coordinates, at which* $\mathbf{m}_B$ *and* $\mathbf{m}_R$ *lie in the relative interiors of the Newton sums of the bunch word and of the room word (Theorem 22.11); the two words converge there exactly when these Newton sums have the full dimensions* $|B|-1$ *and* $|R|-1$ *(Theorem 25.2).*

**Theorem 28.10** (The glue across a seam)**.** *Let a piece on* $B$ *and a partner on* $R$ *with the stations* $\mathcal{D}$ *be glued, with masses on the balance wall of the glue and* $\sum_j (s_j + t_j) c_j > 0$.

*(i)* The piece absorbed. *In* $[0,\infty]$,

$$P_{B\sqcup R}(\mathbf{m}) = \int P_{R^{(\mathbf{x})}}(\mathbf{m}_R, \kappa_B)\, d\big((\boldsymbol{\varrho}^\rho_B)_* \phi^\rho_B\big)(\mathbf{x}),$$

*with* $R^{(\mathbf{x})}$ *the* partner with its stand-in at $\mathbf{x}$*: the family of Newton words on* $R \sqcup \{g\}$ *with the letters* $\sum_D x_{jD}\, \alpha_g^{\,i(D)} L_{jD}$ *and* $\sum_D x^*_{jD}\, \alpha_g^{\,c_j - i(D)} L^*_{jD}$, *the exponents* $(s_j, t_j)$ *and the mass* $\kappa_B$ *on* $g$.

*(ii)* Both absorbed. *On the scales $\{g\}$ and one scale for each active letter let $t_{\mathcal{D}}$ be the configuration whose column at the coordinate $(j,D)$ of $K_j$ is $i(D)\,\mathbf{e}_g+\mathbf{e}_{K_j}$ and at that of $\Psi_j$ is $(c_j-i(D))\,\mathbf{e}_g+\mathbf{e}_{\Psi_j}$. In $[0,\infty]$,*

$$P_{B\sqcup R}(\mathbf{m})=\iint \Gamma(\kappa_B)\,\mathsf{Z}_{(\kappa_B,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{x}\circ\mathbf{x}')\,d\big((\boldsymbol{\varrho}_B^{\rho})_*\phi_B^{\rho}\big)(\mathbf{x})\,d\big((\boldsymbol{\varrho}_R^{\rho})_*\phi_R^{\rho}\big)(\mathbf{x}'),$$

*$\mathbf{x}\circ\mathbf{x}'$ the coordinatewise product and the index $(\kappa_B,\mathbf{s},\mathbf{t})$ read on the active letters (Definition 2.12). The kernel is the period of the* stand-in pair*, the marks $g$ and $g'$ with the masses $\kappa_B$ and $\kappa_R$ and the letters $\sum_D x_{jD}x'_{jD}\,\alpha_g^{i(D)}\alpha_{g'}^{c_j-i(D)}$ and $\sum_D x^*_{jD}x'^*_{jD}\,\alpha_g^{c_j-i(D)}\alpha_{g'}^{i(D)}$, read on one scale:*

$$\Gamma(\kappa)\,\mathsf{Z}_{(\kappa,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{x}\circ\mathbf{x}')=\int_0^{\infty}v^{\kappa-1}\prod_j\Big(\sum_{D\in\mathcal{D}_j}x_{jD}x'_{jD}\,v^{i(D)}\Big)^{-s_j}\Big(\sum_{D\in\mathcal{D}_j}x^*_{jD}x'^*_{jD}\,v^{c_j-i(D)}\Big)^{-t_j}dv.$$

*(iii)* One share per station. *On every contour of shares $a_{jD}$, $b_{jD}$ of the active letters with positive real parts, on the pins of the seam (Lemma 28.7), which on the balance wall are the pins of the partner as well, at which the seam functions of the piece and of the partner converge,*

$$P_{B\sqcup R}(\mathbf{m})=\int_{pins}\prod_{j:\,s_j>0}\Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\prod_{j:\,t_j>0}\Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\int_{\Delta_B}(\boldsymbol{\varrho}_B^{\rho})^{-(\mathbf{a},\mathbf{b})}\,d\phi_B^{\rho}\int_{\Delta_R}(\boldsymbol{\varrho}_R^{\rho})^{-(\mathbf{a},\mathbf{b})}\,d\phi_R^{\rho},$$

*a fibre integral of the dimension $n_{\mathcal{D}}$ of the seam torus: the two seam functions are evaluated at the same shares, one per station and letter.*

*Parts (i) and (ii) hold in every case; (iii) needs a point of the pins at which both seam functions are finite. Full dimension of the Newton sums of the seam word and of the room word is necessary for it (Theorem 25.2), and when the glue is a family of tables with a native gauge, every proper wall positive, it is sufficient, the native split at the stations being such a point (Lemma 28.9); without it the share side carries one pin more for each direction that a side does not see (Theorem 28.12 below).*

The glue is one pairing at every width: the bunch is a law on its seam torus, the room another, and the kernel is the period of two stand-in marks whose letters pair the stations. At width one the seam torus has one dimension less than the number of active letters, a half-line for one table with $s,t>0$ and $(0,\infty)^{2N-1}$ when every letter is active (Proposition 28.11), and the kernel is the period of two marks in every table (Lemma 23.7, Theorem 23.6(i)); at the width $c$ the seam torus has the dimension $n_{\mathcal{D}}$ and the kernel is one scale with at most $c_j+1$ levels in the letters of table $j$, exactly $c_j+1$ for a table. Opened on its shares, the glue is the product of the two seam functions against the share kernels of Corollary 4.4, one for each active letter.

**Proposition 28.11** (Width one is the hinge)**.** *Let $B$ have width one in every table of a family with exponents $s_j,t_j>0$, the weights factoring across $B$ and $R$, $\theta_{j,T}=\theta'_{j,T\cap B}\,\theta''_{j,T\cap R}$, with $\theta''_j$ of largest value $1$ on the room sides of each excess, so that $\theta''_j$ gives the multipliers of the room sides (Definition 28.1), with the stations $D_0$ and $D_1$ of the excesses $0$ and $1$ in every table (Proposition 28.3(ii)), the hinge values $\gamma^K_{j,B}$, $\gamma^{\Psi}_{j,B}$ of Definition 19.2, and the cell family $F_B^+=((\mathcal{B}^+_{j,B},\theta'_j);s_j,t_j)_j$ of the bunch (Definition 23.1), with its hinge wall $\kappa_B$, base form $\phi_{F_B^+}$ and ratio vector $\boldsymbol{\varrho}_{F_B^+}$, whose values are written $\mathbf{x}=(x_{j,K},x_{j,\Psi})_j$, $x_{1,K}=1$. In this statement write $\psi(\mathbf{x})=\prod_j x_{j,K}^{s_j/(2\sigma)}x_{j,\Psi}^{t_j/(2\sigma)}$, $\sigma=\sum_j\sigma_j$ the total strength (Definition 8.1).*

*(i)* $\gamma^{\rho}_{B} = \prod_j (\gamma^{K}_{j,B})^{s_j/(2\sigma)} (\gamma^{\Psi}_{j,B})^{t_j/(2\sigma)} = \gamma^{K}_{1,B}\,\psi(\boldsymbol{\varrho}_{F^+_B})$ *and* $\phi^{\rho}_{B} = \psi(\boldsymbol{\varrho}_{F^+_B})^{-\kappa_B}\,\phi_{F^+_B}$.

*(ii) The seam vector of the piece has* $\varrho^{\rho}_{jD_0} = \varrho^{\rho*}_{jD_1} = 1$, $\varrho^{\rho}_{jD_1} = x_{j,K}/\psi(\mathbf{x})$ *and* $\varrho^{\rho*}_{jD_0} = x_{j,\Psi}/\psi(\mathbf{x})$ *at* $\mathbf{x} = \boldsymbol{\varrho}_{F^+_B}$*; the map* $\mathbf{x} \mapsto \big(x_{j,K}/\psi(\mathbf{x}),\, x_{j,\Psi}/\psi(\mathbf{x})\big)_j$ *is a bijection of* $(0,\infty)^{2N-1}$ *onto the seam torus, and under it the law* $(\boldsymbol{\varrho}^{\rho}_{B})_*\phi^{\rho}_{B}$ *is the image of* $\psi^{-\kappa_B}$ *times the law* $(\boldsymbol{\varrho}_{F^+_B})_*\phi_{F^+_B}$ *of the cell family.*

*(iii) The partner with its stand-in at the seam point of* $\mathbf{x}$ *is, after* $\alpha_g \mapsto \psi(\mathbf{x})\,\alpha_g$*, the family* $((\mathcal{B}^{+}_{j,R}, \theta''_j); s_j, t_j)_j$ *of the cells of the room twisted at* $g$ *by* $\mathbf{x}$*,* $x_{j,K}$ *in* $K_j$ *and* $x_{j,\Psi}$ *in* $\Psi_j$ *(Definition 22.1); the change of variable multiplies* $\alpha_g^{\kappa_B - 1} d\alpha_g$ *by* $\psi(\mathbf{x})^{\kappa_B}$*, which cancels the density of (ii), and Lemma 28.8 is Lemma 23.2 for the cell family* $F^{+}_{B}$.

*(iv) The room, read as the partner, has the same description through the cell family* $F^{+}_{R} = ((\mathcal{B}^{+}_{j,R}, \theta''_j); s_j, t_j)_j$ *of the room, with its hinge wall* $\kappa_R$ *and ratio vector* $\boldsymbol{\varrho}_{F^+_R}$*, whose values are written* $\mathbf{x}'$*,* $x'_{1,K} = 1$*: its scale is* $\gamma^{K}_{1,R}\,\psi(\boldsymbol{\varrho}_{F^+_R})$*, its scale form* $\psi(\boldsymbol{\varrho}_{F^+_R})^{-\kappa_R}\phi_{F^+_R}$*, its seam vector has* $\varrho^{\rho}_{jD_1} = \varrho^{\rho*}_{jD_0} = 1$, $\varrho^{\rho}_{jD_0} = x'_{j,K}/\psi(\mathbf{x}')$ *and* $\varrho^{\rho*}_{jD_1} = x'_{j,\Psi}/\psi(\mathbf{x}')$ *at* $\mathbf{x}' = \boldsymbol{\varrho}_{F^+_R}$*, and its law is the image of* $\psi^{-\kappa_R}$ *times the law* $(\boldsymbol{\varrho}_{F^+_R})_*\phi_{F^+_R}$ *under the bijection of (ii). At the seam points the coordinate ratios* $\varrho^{\rho*}_{jD_0}/\varrho^{\rho}_{jD_1}$ *of the piece and* $\varrho^{\rho*}_{jD_1}/\varrho^{\rho}_{jD_0}$ *of the partner are the ratios* $\varrho_{j,B}$ *and* $\varrho_{j,R}$ *of Definition 19.2. At the seam points* $\mathbf{p}$ *of the piece and* $\mathbf{p}'$ *of the partner that are the images of* $\mathbf{x}$ *and* $\mathbf{x}'$*,*

$$\begin{aligned}\Gamma(\kappa_B)\,\mathsf{Z}_{(\kappa_B,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{p}\circ\mathbf{p}') &= \int_0^{\infty} v^{\kappa_B-1} \prod_j \Big(\frac{x_{j,K}\,v}{\psi(\mathbf{x})} + \frac{x'_{j,K}}{\psi(\mathbf{x}')}\Big)^{-s_j} \Big(\frac{x_{j,\Psi}\,v}{\psi(\mathbf{x})} + \frac{x'_{j,\Psi}}{\psi(\mathbf{x}')}\Big)^{-t_j} dv \\ &= \psi(\mathbf{x})^{\kappa_B}\psi(\mathbf{x}')^{\kappa_R}\,P_{F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]}\big((\kappa_B,\kappa_R);\mathbf{s},\mathbf{t}\big),\end{aligned}$$

*$F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]$ the core* $\binom{\{g,g'\}}{1}$ *in every table twisted at $g$ by $\mathbf{x}$ and at $g'$ by $\mathbf{x}'$, with the masses $\kappa_B$ and $\kappa_R$ (Theorem 23.6(i)). So Theorem 28.10(i) is Lemma 23.2 at the hinge, by part (iii); Theorem 28.10(ii) is Theorem 23.6(i) with that core and the cell families $F^{+}_{B}$ at $g$ and $F^{+}_{R}$ at $g'$, the factor $\psi(\mathbf{x})^{\kappa_B}\psi(\mathbf{x}')^{\kappa_R}$ cancelling the densities of the two laws; and Theorem 28.10(iii) is Theorem 19.13.*

At width one the scale of the piece is the geometric mean of the hinge values in the proportions of the exponents, and the general law is the law of the cell family of the bunch with a fixed power of its ratio vector as density. Every statement about pieces therefore specializes to its statement about cells through this one fixed change, and the calculus of cells (Chapter 23) is the case of width one of the calculus of pieces.

**Theorem 28.12** (The share side with transverse pins)**.** *In Theorem 28.10(iii), the period of the glue converging, so that its Newton sum has full dimension (Theorem 25.2), let* $\vec{\mathcal{N}}_B \subseteq \mathbb{R}^B$ *and* $\vec{\mathcal{N}}_R \subseteq \mathbb{R}^R$ *be the spaces of the directions of the Newton sums of the seam word of the piece and of the room word of the partner at shares with positive real parts, which do not depend on the shares, and* $M = \vec{\mathcal{N}}_B^{\perp} \oplus \vec{\mathcal{N}}_R^{\perp} \subseteq \mathbb{R}^{B \sqcup R}$*, which contains* $\mathbf{1}$*; let* $\mathbf{v}_1, \ldots, \mathbf{v}_q$ *complete* $\mathbf{1}$ *to a basis of the lattice* $M \cap \mathbb{Z}^{B\sqcup R}$*. For every active letter and station fix the exponents* $\mathbf{n}_{jD}$*,* $\mathbf{n}^{*}_{jD}$*,* $\mathbf{n}'_{jD}$ *and* $\mathbf{n}'^{*}_{jD}$ *of one monomial of each of* $V_{jD}$*,* $V^{*}_{jD}$*,* $L_{jD}$ *and* $L^{*}_{jD}$*, and in this statement write*

$$\mathbf{n}(\mathbf{a},\mathbf{b}) = \sum_{j,D} \Big( a_{jD}\big(\mathbf{n}_{jD} + \mathbf{n}'_{jD}\big) + b_{jD}\big(\mathbf{n}^{*}_{jD} + \mathbf{n}'^{*}_{jD}\big)\Big),$$

*which modulo* $\vec{\mathcal{N}}_B \oplus \vec{\mathcal{N}}_R$ *does not depend on the monomials fixed. For a word* $W$ *on* $B$ *whose Newton sum has the space of directions* $\vec{\mathcal{N}}_B$*, its* reduced period *is*

$$P^\flat_W(\mathbf{m}_B) = \int_{\mathbb{R}^B/\vec{\mathcal{N}}_B^\perp} e^{\langle \mathbf{m}_B, y\rangle}\, W(e^y)\, dy,$$

*dy the quotient of Lebesgue measure by the measure on* $\vec{\mathcal{N}}_B^\perp$ *that gives its integer lattice the covolume one; likewise on* $R$*. On every contour of shares of the active letters with positive real parts, pinned by the letter totals and by* $\langle \mathbf{n}(\mathbf{a},\mathbf{b}), \mathbf{v}_l\rangle = \langle \mathbf{m}, \mathbf{v}_l\rangle$*,* $l \le q$*, pins that are independent by the full dimension, at which the reduced periods of the two words converge,*

$$P_{B\sqcup R}(\mathbf{m}) = \int_{pins} \prod_{j:\, s_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix} \prod_{j:\, t_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\; P^\flat_{W_B(\mathbf{a},\mathbf{b})}(\mathbf{m}_B)\, P^\flat_{W_R(\mathbf{a},\mathbf{b})}(\mathbf{m}_R),$$

$W_R(\mathbf{a},\mathbf{b}) = \prod_{j,D} L_{jD}^{-a_{jD}} {L^*_{jD}}^{-b_{jD}}$ *the room word of the partner; on the pins the integrands of the reduced periods are invariant under* $\vec{\mathcal{N}}_B^\perp$ *and* $\vec{\mathcal{N}}_R^\perp$*. When* $\vec{\mathcal{N}}_B$ *and* $\vec{\mathcal{N}}_R$ *are the hyperplanes of sum zero,* $q = 1$*, the pin is the scale pin, the reduced periods are the periods, and this is Theorem 28.10(iii). When every room letter of the partner is a monomial, as for the room of a table whose stations of room sides are single room sides, with the coefficients* 1*,* $\vec{\mathcal{N}}_R = 0$*, the pins on* $R$ *say that the exponent of the partner's monomials is* $\mathbf{m}_R$*, and the reduced period of the partner is the product of the coefficients of these monomials raised to minus the shares.*

So a degenerate seam word or room word costs pins, not convergence: the directions that a side does not see are fixed by the masses, one pin each, and the integral over them is gone.

## 28.3 The Gramma of a set of pairs

The station letters are what the bunch is; the seam letters are what the room sees. The seam joins the two, and its compatible pairs enter the period through one Gramma.

**Definition 28.13** (The Gramma of a set of pairs)**.** Let $\mathfrak{R} \subseteq \mathcal{K}\times\mathcal{D}$ be a set of pairs of stations of two finite sets, every station in some pair, with coefficients $h_{CD} > 0$ on its pairs. Its *station map* sends $\mathbf{e} = (e_{CD}) \in \mathbb{C}^{\mathfrak{R}}$ to $\partial\mathbf{e}$, with $(\partial\mathbf{e})_C = \sum_D e_{CD}$ and $(\partial\mathbf{e})_D = \sum_C e_{CD}$. Two pairs lie in one *part* when a chain of pairs, each sharing a station with the next, joins them; a *circuit* is a minimal set of pairs whose columns $\mathbf{e}_C + \mathbf{e}_D$ are linearly dependent, which is the set of pairs of a closed chain $C_1D_1C_2D_2\cdots C_mD_mC_1$, $m \ge 2$, of distinct stations with $(C_l, D_l)$ and $(C_{l+1}, D_l)$ in $\mathfrak{R}$. Choose one station of $\mathcal{D}$ in each part, and call the other stations *kept*. For *loads* $(\mathbf{a},\mathbf{a}') \in \mathbb{C}^{\mathcal{K}}\oplus\mathbb{C}^{\mathcal{D}}$ whose totals on the two sides of every part agree, the *Gramma of* $\mathfrak{R}$ is the Gramma $\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h})$ (Definition 2.1) on the scales of the kept stations whose configuration $\partial\mathfrak{R}$ has at the pair $(C,D)$ the column $\mathbf{e}_C + \mathbf{e}_D$, a station that is not kept read as 0, whose index is the loads of the kept stations, and whose coefficients are the $h_{CD}$. By Proposition 4.2 it is the fibre integral

$$\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h}) = \int_{\partial\mathbf{e}=(\mathbf{a},\mathbf{a}')} \prod_{(C,D)\in\mathfrak{R}} \Gamma(e_{CD})\, h_{CD}^{-e_{CD}},$$

pinned by the sums at the kept stations, which imply the others. Another choice of the stations not kept changes the pins by an integer matrix of determinant $\pm1$, which leaves the fibre integral unchanged; its dimension is $|\mathfrak{R}| - |\mathcal{K}| - |\mathcal{D}|$ plus the number of parts. The *seam Gramma* of the excess $i$ is the Gramma of the set $\mathfrak{R}_i$ of the compatible pairs of $H_i$, with the pair constants as its coefficients.

**Proposition 28.14** (The seam transfers the station loads of the bunch to those of the room)**.** *For a bunch of a family let the* station word *of the bunch be* $\prod_{j,C} \mathsf{x}_{jC}^{-a_{jC}} {\mathsf{x}^*_{jC}}^{-b_{jC}}$*, with the station loads* $a_{jC}$ *of the active* $K_j$ *and* $b_{jC}$ *of the active* $\Psi_j$ *on the pins* $\sum_C a_{jC} = s_j$*,* $\sum_C b_{jC} = t_j$ *and* $\sum_{j,C} \big(i(C)a_{jC} + (c_j - i(C))b_{jC}\big) = \kappa_B$*, and let the station loads* $a_{jD}$ *and* $b_{jD}$ *of the room be shares on the pins of the seam (Lemma 28.7). On contours of positive real parts at which the station word of the bunch converges,*

$$\prod_{j:\, s_j>0} \prod_{D\in\mathcal{D}_j} \Gamma(a_{jD}) \prod_{j:\, t_j>0} \prod_{D\in\mathcal{D}_j} \Gamma(b_{jD})\; P_{W_B(\mathbf{a},\mathbf{b})}(\mathbf{m}_B)$$
$$= \int \prod_{j:\, s_j>0} \prod_{i=0}^{c_j} \mathsf{G}_{(\mathbf{a}_{j,\mathcal{K}_i},\mathbf{a}_{j,\mathcal{D}_i});\partial\mathfrak{R}_{j,i}}(\mathbf{h}_j) \prod_{j:\, t_j>0} \prod_{i=0}^{c_j} \mathsf{G}_{(\mathbf{b}_{j,\mathcal{K}_i},\mathbf{b}_{j,\mathcal{D}_i});\partial\mathfrak{R}_{j,i}}(\mathbf{h}_j)\; P_{\prod_{j,C} \mathsf{x}_{jC}^{-a_{jC}} {\mathsf{x}^*_{jC}}^{-b_{jC}}}(\mathbf{m}_B),$$

*the integral over the station loads of the bunch with equal totals on the two sides of every part of every set* $\mathfrak{R}_{j,i}$ *of the compatible pairs of table* $j$ *at the excess* $i$*,* $\mathbf{h}_j$ *its pair constants.*

Along a circuit the pair terms $h_{CD}\,\mathsf{x}_C\,\mathsf{y}_D$ of a level have the alternating product $\prod_l h_{C_lD_l}/h_{C_{l+1}D_l}$, a constant: a direction $\mathbf{e}$ of the pair shares with $\partial\mathbf{e} = 0$ moves neither the station loads of the bunch, which the bunch sees, nor those of the room, which the room sees, and the period integrates over it through the seam Gramma alone. A part without circuits has a fibre of dimension zero, and its Gramma is the product $\prod \Gamma(e_{CD}) h_{CD}^{-e_{CD}}$ at the one solution of $\partial\mathbf{e} = (\mathbf{a}, \mathbf{a}')$ (Corollary 2.6).

**Proposition 28.15** (The Gramma of a set of pairs in the Foreman's order)**.** *Let* $\mathfrak{R}$ *have the parts* $\mathfrak{R}_1, \dots, \mathfrak{R}_\nu$*, the part* $\mathfrak{R}_l$ *on the stations* $\mathcal{K}_l \sqcup \mathcal{D}_l$*, and loads with* $|\mathbf{a}_{\mathcal{K}_l}| = |\mathbf{a}'_{\mathcal{D}_l}|$ *for every* $l$*; in this statement write* $L_C(v) = \sum_{D:\,(C,D)\in\mathfrak{R}} h_{CD}\, v_D$ *for* $C \in \mathcal{K}_l$ *and* $v \in \Delta_{\mathcal{D}_l}$*. Then*

$$\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h}) = \prod_{C\in\mathcal{K}} \Gamma(a_C) \prod_{l=1}^{\nu} \int_{\Delta_{\mathcal{D}_l}} v^{\mathbf{a}'_{\mathcal{D}_l}-\mathbf{1}} \prod_{C\in\mathcal{K}_l} L_C(v)^{-a_C}\, d\omega_{\mathcal{D}_l}(v)$$
$$= \prod_{C\in\mathcal{K}} \Gamma(a_C) \prod_{l=1}^{\nu} P_{\prod_{C\in\mathcal{K}_l} L_C^{-a_C}}(\mathbf{a}'_{\mathcal{D}_l}),$$

*both sides converging together, the period of the part* $l$ *being that of the word* $\prod_{C\in\mathcal{K}_l} L_C^{-a_C}$ *on* $\mathcal{D}_l$ *at the masses* $\mathbf{a}'_{\mathcal{D}_l}$*; with* $\operatorname{Re} a_C > 0$ *it converges exactly when, for every* $l$*,* $\operatorname{Re}\mathbf{a}'_{\mathcal{D}_l}$ *lies in the relative interior of* $\sum_{C\in\mathcal{K}_l} \operatorname{Re} a_C \operatorname{conv}\{\mathbf{e}_D :\ (C,D)\in\mathfrak{R}\}$ *(Theorem 25.2).*

The stations of the bunch are integrated first, each on its half-line, and those of the room last, on the simplex of each part: the Foreman's order of Definition 16.14, the bunch inside the integral over its room. For a seam the inner step returns, at each station $C$ of the bunch, its child letter $K_{(C)} = \sum_D h_{CD}\,\mathsf{y}_D$ with every room station letter $\mathsf{y}_D$ replaced by a scale $v_D$. No station is singled out and no order is chosen.

**Proposition 28.16** (Closing the Gramma of a set of pairs)**.** *In Proposition 28.15 let the loads be real and positive in the domain for (i) and (iii); in this statement write, for a part* $\mathfrak{R}_l$*,* $L_C(v) = \sum_{D:\,(C,D)\in\mathfrak{R}} h_{CD}\, v_D$ *for* $C \in \mathcal{K}_l$ *and* $v \in \Delta_{\mathcal{D}_l}$*,* $M_C = \max_D h_{CD}$ *for its largest coefficient, its largest value on the simplex, and* $\mathsf{G}_{(\mathbf{a},\mathbf{a}')}$ *for* $\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h})$*.*

*(i)* Peak closing. *For every part* $\mathfrak{R}_l$*, with* $\bar{L}_C = 1 - L_C/M_C \geq 0$ *on* $\Delta_{\mathcal{D}_l}$*,*

$$\int_{\Delta_{\mathcal{D}_l}} v^{\mathbf{a}'_{\mathcal{D}_l}-\mathbf{1}} \prod_{C\in\mathcal{K}_l} L_C(v)^{-a_C}\, d\omega_{\mathcal{D}_l}(v)$$
$$= \prod_{C\in\mathcal{K}_l} M_C^{-a_C}\, \Gamma_{\substack{|\mathcal{D}_l|\\1}}\!\begin{bmatrix}\mathbf{a}'_{\mathcal{D}_l}\\ |\mathbf{a}'_{\mathcal{D}_l}|\end{bmatrix} \sum_{\mathbf{n}\in\mathbb{N}^{\mathcal{K}_l}} \prod_{C\in\mathcal{K}_l} \frac{(a_C)_{n_C}}{n_C!}\; \mathbb{E}_{V\sim \mathrm{Dir}_{\mathbf{a}'_{\mathcal{D}_l}}}\Big[\prod_{C\in\mathcal{K}_l} \bar{L}_C(V)^{n_C}\Big],$$

*a series of nonnegative terms, each a finite nonnegative combination of Dirichlet moments (Theorem 3.11), rational at rational loads and coefficients; it is Theorem 15.13(i) with the sharp references $M_C$, and $\mathsf{G}_{(\mathbf{a},\mathbf{a}')}$ is $\prod_C \Gamma(a_C)$ times the product of these right sides over the parts.*

*(ii)* Relations at shifted loads. *At all loads in the domain, real or complex, for every $C$ and $D$,*

$$a_C\, \mathsf{G}_{(\mathbf{a},\mathbf{a}')} = \sum_{D'\colon (C,D')\in\mathfrak{R}} h_{CD'}\, \mathsf{G}_{(\mathbf{a}+\mathbf{e}_C,\, \mathbf{a}'+\mathbf{e}_{D'})},$$
$$a'_D\, \mathsf{G}_{(\mathbf{a},\mathbf{a}')} = \sum_{C'\colon (C',D)\in\mathfrak{R}} h_{C'D}\, \mathsf{G}_{(\mathbf{a}+\mathbf{e}_{C'},\, \mathbf{a}'+\mathbf{e}_D)}.$$

*(iii)* Residues. *A part that is one circuit $C_1D_1\cdots C_mD_m$ has one share $z$ moving round the circuit with alternating signs: for a real point $\mathbf{q}$ of the fibre the pairs $(C_l, D_l)$ carry $q_l + z$ and the pairs $(C_{l+1}, D_l)$ carry $q'_l - z$. With $x = \prod_l h_{C_lD_l}/h_{C_{l+1}D_l}$ and $\Sigma = \sum_l (q_l + q'_l)$, the total load of the part, its Gramma is a constant times*

$$\frac{1}{2\pi i}\int_{(0)} x^{-z} \prod_{l=1}^{m} \Gamma(q_l + z)\, \Gamma(q'_l - z)\, dz.$$

*If $x < 1$, or $x = 1$ and $\Sigma < m$, the contour moves to the left across the poles of the $\Gamma(q_l + z)$, and if $x > 1$, or $x = 1$ and $\Sigma < m$, to the right across those of the $\Gamma(q'_l - z)$ (Theorem 4.5): the Gramma is the limit of the sums of the residues at the poles crossed, a series over $m$ lattices, absolutely convergent when $x \neq 1$ or $\Sigma < m - 1$. The residue at a simple pole is a product of the other Gammas at the shifted point, and where poles of several factors meet, the $q_l$ being congruent modulo $\mathbb{Z}$, it is a Laurent coefficient. At $x = 1$ with $\Sigma \geq m$ the shifted contours need not tend to zero and the residues need not sum.*

**Theorem 28.17** (Shares inside the rectangles close)**.** *Fix a table and an excess $i$, and give every basis $T$ of excess $i$ a share $a_T$. For pattern loads $\mathbf{a}^B = (a^B_p)_p$ and room-side loads $\mathbf{a}^R = (a^R_X)_X$ whose station loads $a^B_C = \sum_{p\in C} a^B_p$ and $a^R_D = \sum_{X\in D} a^R_X$ have equal totals on every part of $\mathfrak{R}_i$, on contours of positive real parts,*

$$\int_{\sum_X a_{p\sqcup X}=a^B_p,\ \sum_p a_{p\sqcup X}=a^R_X} \prod_T \Gamma(a_T)\, \theta_T^{-a_T}$$
$$= \prod_{C\in\mathcal{K}_i} \Gamma_{\substack{|C|\\1}}\!\begin{bmatrix}\mathbf{a}^B|_C\\ a^B_C\end{bmatrix} \prod_p w_p^{-a^B_p} \prod_{D\in\mathcal{D}_i} \Gamma_{\substack{|D|\\1}}\!\begin{bmatrix}\mathbf{a}^R|_D\\ a^R_D\end{bmatrix} \prod_X {w'_X}^{-a^R_X}\; \mathsf{G}_{(\mathbf{a}^B_{\mathcal{K}_i},\mathbf{a}^R_{\mathcal{D}_i});\partial\mathfrak{R}_i}(\mathbf{h}),$$

*$\mathbf{a}^B_{\mathcal{K}_i}$ and $\mathbf{a}^R_{\mathcal{D}_i}$ the vectors of the station loads, the fibre on the left pinned by these sums at all patterns and at all room sides but one in each part of $\mathfrak{R}_i$, which imply the others.*

Inside the rectangle of a compatible pair the shares spread freely and close to stacked Gammas, one for each station; what does not close is how the stations share one another, the seam Gramma.

## 28.4 What a piece retains

**Definition 28.18** (Tinkers)**.** A *tinker* of a partner $\mathcal{Y}$ with the stations $\mathcal{D}$ is a vector $\mathbf{v} = (v_{jD}, v^*_{jD})_{j,D}$ of positive numbers, one for each coordinate of the seam vector; the *tinkered partner* $\mathcal{Y}_{\mathbf{v}}$ has the room letters $v_{jD}L_{jD}$ and $v^*_{jD}L^*_{jD}$. A tinker is *plain* when $v_{jD} = v^*_{jD}$ for every coordinate. On the room of a table a plain tinker multiplies the weights of the room sides of each station by one constant, which keeps the bases and gives a table (Proposition 12.9, read for the bunch $R$); a tinker that is not plain gives the room two systems of weights, one for each letter, as a cell may carry (Definition 21.1).

**Theorem 28.19** (The law is what a piece retains, at every width)**.** *Let $\mathcal{X}$ and $\mathcal{X}'$ be pieces with the stations $\mathcal{D}$, on sets $B$ and $B'$ of any sizes, with the exponents $(s_j, t_j)_j$, $\sum_j (s_j + t_j)c_j > 0$, and one wall $\kappa$. Let $\mathcal{Y}_0$ be a partner of both with the wall $\sum_j (s_j + t_j)c_j - \kappa$, and let some real point $\mathbf{c}$ of the pins of the seam, with positive coordinates, lie in the interior, relative to the pins, of the domains of the seam functions of $\mathcal{X}$, $\mathcal{X}'$ and $\mathcal{Y}_0$. The following are equivalent.*

*(a)* $(\boldsymbol{\varrho}^{\rho}_{\mathcal{X}})_* \phi^{\rho}_{\mathcal{X}} = (\boldsymbol{\varrho}^{\rho}_{\mathcal{X}'})_* \phi^{\rho}_{\mathcal{X}'}$.

*(b)* $P_{\mathcal{X} \sqcup \mathcal{Y}} = P_{\mathcal{X}' \sqcup \mathcal{Y}}$, *in* $[0, \infty]$, *for every partner* $\mathcal{Y}$ *with the wall* $\sum_j (s_j + t_j)c_j - \kappa$.

*(c)* $P_{\mathcal{X} \sqcup \mathcal{Y}_{0,\mathbf{v}}} = P_{\mathcal{X}' \sqcup \mathcal{Y}_{0,\mathbf{v}}}$ *for every tinker* $\mathbf{v}$ *of* $\mathcal{Y}_0$.

*(d) The seam functions of $\mathcal{X}$ and $\mathcal{X}'$ agree on the contour of the pins through $\mathbf{c}$.*

*When every table $j$ with both letters active has $|\mathcal{D}_j| = 1$, except at most one, which has $c_j \geq 1$ and $|\mathcal{D}_j| = 2$, the plain tinkers suffice in (c). For one table at width one with factoring weights, $|\mathcal{D}| = 2$, and the theorem is Theorem 23.11 read through Proposition 28.11: the one-mark cell of ratio 1 (Definition 23.10), glued at the hinge, is a partner whose plain tinkers $(1, q^{1/2})$ are the one-mark cells of ratio $q$.*

So the wall and the law are the whole of what a piece contributes to the families glued on it, and the tinkers of one partner already separate any two laws. A plain tinker multiplies the two room letters of a station by one constant, so with every letter active, and some table of positive width, the plain tinkers of one partner move only $|\mathcal{D}| - N$ of the $2|\mathcal{D}| - 2N - 1$ directions of the seam torus. The other directions are reached by partners with one system of weights whose weights and masses vary.

**Definition 28.20** (Plain partners)**.** A partner on $R$ with the stations $\mathcal{D}$ is *plain* when for every table $j$ with an active letter and every $D \in \mathcal{D}_j$ there are a nonempty family $\mathcal{S}_{jD}$ of subsets of $R$ of $e_j + c_j - i(D)$ marks, its *room sides*, and positive weights $w'_{jD,X}$, $X \in \mathcal{S}_{jD}$, with

$$L_{jD} = \sum_{X \in \mathcal{S}_{jD}} w'_{jD,X}\, \alpha^X_R, \qquad L^*_{jD} = \sum_{X \in \mathcal{S}_{jD}} w'_{jD,X}\, \alpha^{R \setminus X}_R,$$

one system of weights for the two letters of a station; then $e^*_j = |R| - e_j - c_j$. The room of a family of tables is plain (Proposition 28.2(iii)), its stations of room sides being disjoint, with $w'_{jD,X}$ the multiplier of $X$ in table $j$ (Definition 28.1).

**Definition 28.21** (The station transform of a piece)**.** For a piece $\mathcal{X}$ on $B$ with the stations $\mathcal{D}$, $\sum_j (s_j + t_j)c_j > 0$, the wall $\kappa_{\mathcal{X}}$ and the masses $\mathbf{m}_B$, and for $\mathbf{p} = (p_{jD}, p^*_{jD})_{j,D}$ with positive coordinates, one for each coordinate of the seam vector, the *station transform* is

$$\mathcal{I}_{\mathcal{X}}(\mathbf{p}) = \int \Gamma(\kappa_{\mathcal{X}})\, \mathsf{Z}_{(\kappa_{\mathcal{X}}, \mathbf{s}, \mathbf{t}); t_{\mathcal{D}}}(\mathbf{x} \circ \mathbf{p})\, d\big((\boldsymbol{\varrho}^{\rho}_{\mathcal{X}})_* \phi^{\rho}_{\mathcal{X}}\big)(\mathbf{x})$$

$$= \int_{(0,\infty)^B} \alpha^{\mathbf{m}_B - \mathbf{1}} \prod_j \Big( \sum_D p_{jD} V_{jD} \Big)^{-s_j} \Big( \sum_D p^*_{jD} V^*_{jD} \Big)^{-t_j} d\alpha,$$

in $[0,\infty]$: the kernel of Theorem 28.10(ii) read against the law, the period of the piece against the constant room letters $\mathbf{p}$, the second expression by the computation of Lemma 28.8 with $\alpha = \rho u$ and $v = \rho\, \gamma^\rho_{\mathcal{X}}(u)$.

**Theorem 28.22** (Plain partners separate laws)**.** *Let $\mathcal{X}$ and $\mathcal{X}'$ be pieces as in Theorem 28.19, with a real point $\mathbf{c}$ of the pins, with positive coordinates, in the interior of the domains of their seam functions on the pins. Fix room sides $\mathcal{S}_{jD}$ on a set $R$, every family with at least two members, such that the room word of the plain partners with these room sides has at $\mathbf{c}$ a Newton sum of the dimension $|R|-1$, as when one family of an active letter has a Newton polytope of that dimension, and masses $\mathbf{m}^0_R$ in its relative interior, with $\kappa_R = \sum_j (s_j + t_j) c_j - \kappa$. Then $(\boldsymbol{\varrho}^\rho_{\mathcal{X}})_* \phi^\rho_{\mathcal{X}} = (\boldsymbol{\varrho}^\rho_{\mathcal{X}'})_* \phi^\rho_{\mathcal{X}'}$ exactly when $P_{\mathcal{X} \sqcup \mathcal{Y}} = P_{\mathcal{X}' \sqcup \mathcal{Y}}$ for every plain partner $\mathcal{Y}$ with these room sides, at every positive weights and every masses in a neighbourhood of $\mathbf{m}^0_R$ on the hyperplane $\kappa_R = \sum_j (s_j + t_j) c_j - \kappa$. Such room sides exist for all stations and degrees: $e_j = 1$, $|R| \geq \max_j c_j + 2$, and every $\mathcal{S}_{jD}$ all the subsets of $1 + c_j - i(D)$ marks.*

The room of a family of tables is a plain partner, and multiplying the weights of the bases of each table by positive numbers that depend on the room side alone keeps the bases and the stations, and multiplies the members of each station of room sides by these numbers. So when a bunch of a family has at least two room sides in every station of every table with an active letter and the room word of its room has a Newton sum of full dimension, these families, with the masses of the room near the family's and those of the bunch kept, separate its law from every other law whose seam function converges, as its own does, at the native split (Lemma 28.9, which puts the masses of the room in the relative interior there).

**Lemma 28.23** (Partners on one mark see a point mass)**.** *Let $\mathcal{X}$ be a piece on $B$ with the stations $\mathcal{D}$ (Definition 28.4), the exponents $(s_j, t_j)_j$ with $w = \sum_j (s_j + t_j) c_j > 0$, the masses $\mathbf{m}_B$ and the wall $\kappa = \kappa_{\mathcal{X}}$ with $0 < \kappa < w$. For a point $\mathbf{x}$ of its seam torus (Definition 28.5), the* one-mark piece at $\mathbf{x}$ *is the piece on one mark $g$ with the seam letters $x_{jD} \alpha_g^{i(D)}$ and $x^*_{jD} \alpha_g^{c_j - i(D)}$ and the mass $\kappa$; its scale is $\alpha_g$, its seam vector is $\mathbf{x}$, its law is $\delta_{\mathbf{x}}$, and its station transform (Definition 28.21) is $\Gamma(\kappa) \mathsf{Z}_{(\kappa,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{x} \circ \mathbf{p})$. If for some $c > 0$ and $\mathbf{x}_0$ on the seam torus*

$$\mathcal{I}_{\mathcal{X}}(\mathbf{p}) = c\, \Gamma(\kappa)\, \mathsf{Z}_{(\kappa,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{x}_0 \circ \mathbf{p}) \qquad \text{for every } \mathbf{p},$$

*then the law of $\mathcal{X}$ is $c\, \delta_{\mathbf{x}_0}$. The station transform at $\mathbf{p}$ is the period of the glue of $\mathcal{X}$ with the partner on one mark $b$ with the room letters $p_{jD} \alpha_b^{c_j - i(D)}$ and $p^*_{jD} \alpha_b^{i(D)}$ and the mass $w - \kappa$, the tinker $\mathbf{p}$ of one partner on one mark (Definition 28.18). So a piece that acts on these partners as $c$ times one mark has the law of $c$ times one mark, with no condition on the domains of the seam functions as in Theorem 28.19.*

## 28.5 Storage, and what replaces it

**Theorem 28.24** (A bunch is stored on one mark exactly at width one)**.** *In Theorem 28.10(i) let the partner be the room of a family of tables. The partner with its stand-in at $\mathbf{x}$, $R^{(\mathbf{x})}$, is a family of tables on $R \sqcup \{g\}$, with a system of weights for each letter as a cell may carry (Definition 21.1), at every point of the seam torus exactly when every table with an active letter has $c_j \leq 1$, and $c_j = 1$ when both of its letters are active. At width one with factoring weights, normalized as*

*in Proposition 28.11, it is the family of the cells $\mathcal{B}^{+}_{j,R}$ of the room with $g$ twisted in both letters, which the change of variable of Proposition 28.11(iii) turns into the twist of Definition 22.1, and Lemma 28.8 is Lemma 23.2 with the scale changed from $\gamma^{K}_{1,B}$ to $\gamma^{\rho}_{B}$, a fixed bijection of the tori with a density (Proposition 28.11). A table with $c_j = 0$ does not see the stand-in, in either letter, and at the width $c_j \geq 2$ the stand-in enters the letters of table $j$ with the degree $c_j$, the coefficient of $\alpha_g^{c_j}$ in $K_j$ being $\sum_{D \in \mathcal{D}_{j,c_j}} x_{jD} L_{jD} \neq 0$: the partner with its stand-in is a family of Newton words, whose period converges exactly when its Newton sum has the dimension $|R|$ and the masses lie in its relative interior (Theorem 25.2).*

So the single mark of width one is the case $c = 1$ of a stand-in of degree $c$. What replaces storage at every width is the stand-in itself, the station-wise pairing of Theorem 28.10: every statement about pieces is made on pieces with stations, whose glue is a family of Newton words, and table structure is never used to compose.

**Theorem 28.25** (A point mass is a pinned set)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ be a family on $\mathcal{E}$ with real exponents $s_j, t_j \geq 0$ and masses on its balance wall with every proper wall positive, and let $\varnothing \neq S \subsetneq \mathcal{E}$, with the room $S' = \mathcal{E} \setminus S$, be read as a piece with the stations of its room and $S'$ as a partner (Definition 28.4). Call $S$* pinned *when every active table has width at most one at $S$, with its weights factoring across $S$ and $S'$, and the active letters of the tables of width one read $S$ alike (Theorem 22.5): there are a function $\gamma$ on $\Delta_S$ and constants $c_j, c_j^* > 0$ with $\gamma^{K}_{j,S} = c_j \gamma$ whenever $s_j > 0$ and $\gamma^{\Psi}_{j,S} = c_j^* \gamma$ whenever $t_j > 0$ (Definition 19.2). Then $w = \sum_j (s_j + t_j) c_j(S) > 0$ and $0 < \kappa_S < w$, and the following are equivalent.*

- *(i) The law of $S$ is carried by one point: $(\boldsymbol{\varrho}^{\rho}_{S})_* \phi^{\rho}_{S} = c\, \delta_{\mathbf{x}_0}$ (Definition 28.5) with $c > 0$ and $\mathbf{x}_0$ on the seam torus.*
- *(ii) $S$ is pinned.*
- *(iii) $S$* splits off*: there are $c > 0$ and $\mathbf{x}_0$ on the seam torus such that every partner $\mathcal{Y}$ of $S$ with the wall $w - \kappa_S$ has $P_{S \sqcup \mathcal{Y}} = c\, P_{\mathcal{Y}^{(\mathbf{x}_0)}}$ in $[0, \infty]$, $\mathcal{Y}^{(\mathbf{x}_0)}$ the partner with its stand-in at $\mathbf{x}_0$ carrying the mass $\kappa_S$ (Theorem 28.10(i)).*
- *(iv) $S$ splits off against the partners on one mark: its station transform is $c$ times that of the one-mark piece at $\mathbf{x}_0$ with the mass $\kappa_S$ (Lemma 28.23).*

*When they hold, the seam vector is the constant $\mathbf{x}_0$, $c = \int_{\Delta_S} \phi^{\rho}_{S}$, and $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = c\, P_{S'^{(\mathbf{x}_0)}}$. For a family of lists, read at the factorization with the weights 1, the active letters of the tables of width one read $S$ alike exactly when their hinge values are equal, $c_j = c_j^* = 1$, and then $\mathbf{x}_0 = \mathbf{1}$. For one list $\mathcal{B}$ with $s, t > 0$ and $|S| \geq 2$, $S$ is pinned exactly when it has width one and $\varrho_S = 1$, that is $\mathbf{E}_{\mathcal{B}|S} = \mathbf{E}_{\mathcal{B}/S'}$.*

When $\mathcal{B}|S$ and $\mathcal{B}/S'$ are connected, Chapter 30 reads the last condition as a dual pair (Corollary 30.18).

**Corollary 28.26** (Which sets split off)**.** *Let $F$ and $S$ be as in Theorem 28.25.*

- *(i) If some active table has width two or more at $S$, then $S$ does not split off, and its law is carried by no point.*
- *(ii) If $F = ((\mathcal{B}, \theta); s, t)$ is one prime table (Theorem 23.16) with $s + t > 0$, no set with at least two marks on each side splits off.*

*(iii) If $F = (\mathcal{B}; s, 0)$ is one list with $s > 0$ and $e$ is a mark that is neither a loop nor a coloop, then $\mathcal{E} \setminus \{e\}$ splits off, and*

$$P_F(\mathbf{m}; s, 0) = \Gamma_{\substack{2\\1}}\begin{bmatrix} s - m_e, & m_e \\ & s \end{bmatrix} P_{((\mathcal{B}/e;\, m_e, 0),\, (\mathcal{B}\setminus e;\, s-m_e, 0))}\big(\mathbf{m}|_{\mathcal{E}\setminus\{e\}}\big),$$

*the period of the bunch family of Corollary 19.14; at $(0, t)$, $t > 0$, the same holds with $\Gamma_{\substack{2\\1}}\begin{bmatrix} t - m_e, & m_e \\ & t \end{bmatrix}$ and the family $((\mathcal{B}/e;\, 0, t - m_e), (\mathcal{B} \setminus e;\, 0, m_e))$.*

*(iv) If $F = (\mathcal{B}; s, t)$ is one list on at least three marks with $s, t > 0$ and $e$ is neither a loop nor a coloop, then $\mathcal{E} \setminus \{e\}$ splits off exactly when $\mathbf{E}_{\mathcal{B}\setminus e} = \mathbf{E}_{\mathcal{B}/e}$.*

**Lemma 28.27** (The level cell of a bunch)**.** *Let $B$ have the width $c = c(B) \geq 1$ in a table, and let $G$ be a set of $c$ new marks. The sets $p \sqcup F$, $F \subseteq G$ and $p$ a pattern of excess $c - |F|$, are the bases of a table $\mathcal{B}_B^\natural$ on $B \sqcup G$ of rank $r(B)$, the* level cell *of $B$, with any positive weights; $\mathcal{B}_B^\natural \setminus G = \mathcal{B}|B$ and $\mathcal{B}_B^\natural/G = \mathcal{B}/R$. At width one $\mathcal{B}_B^\natural$ is the cell of the bunch $\mathcal{B}_B^+$ of Theorem 19.4.*

**Theorem 28.28** (The level envelope)**.** *Let the weights of a table factor across $B$ and $R$, $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$, with $c = c(B) \geq 1$, and let $\mathcal{B}_B^\natural$ on $B \sqcup G$ and $\mathcal{B}_R^\natural$ on $R \sqcup G'$ be the level cells of the bunch and of the room, $G'$ a copy of $G$, with the weights $\theta'_p$ on $p \sqcup F$ and $\theta''_X\binom{c}{|F'|}^{-1}$ on $X \sqcup F'$. Splicing the $c$ pairs of a mark of $G$ and its copy (Theorem 21.14) gives the table $\mathcal{B}^{\mathrm{env}}$ on $\mathcal{E}$, the* level envelope *of the bunch, whose bases are the sets $p \sqcup X$ of a pattern $p$ and a room side $X$ of one excess, with the weights $\theta'_p\,\theta''_X$. It has the patterns and the room sides of the table at every excess, its seam has one rectangle at every excess, and $\mathcal{B} \subseteq \mathcal{B}^{\mathrm{env}}$, with equality exactly when every excess of the seam of $B$ is one rectangle.*

A bunch in level form is therefore stored on $c(B)$ pairs of hinges, the table being the splicing of its two level cells, and every bunch is read off its envelope by its seam: the table keeps, at each excess, the rectangles of the compatible pairs.

**Proposition 28.29** (When a glue of seamed pieces is a table: what is necessary)**.** *Let $\mathcal{B}$ be a nonempty family of $r$-sets of $B \sqcup R$, with children and room children as in Definition 28.1. If $\mathcal{B}$ is the family of bases of a table, then:*

*(a) every child and every room child is a table;*

*(b) the children at adjacent patterns $p$ and $p + b$ are quotients, the child at $p + b$ of the child at $p$, and likewise the room children at adjacent room sides;*

*(c) at the least and at the largest number of marks of a pattern all the children are one table, and likewise all the room children;*

*(d) with $d$ and $d + c$ the least and the largest numbers of marks of a pattern, for every $k$ with $d \leq k \leq d + c$ the patterns of $k$ marks are the $k$-sets independent in the table of the patterns of $d + c$ marks that span the table of the patterns of $d$ marks, the bases of the $(k - d)$-th lift of the latter toward the former (Proposition 12.17); and likewise for the room sides.*

**Theorem 28.30** (The gluing criterion)**.** *Conditions (a)–(d) of Proposition 28.29 suffice: a nonempty family of $r$-sets of $B \sqcup R$ satisfying them is the family of bases of a table. So a nonempty family of sets of one size is the family of bases of a table exactly when, at one bunch, its children and room children are tables, adjacent ones are quotients, the extreme ones are one, and its levels are lifts (Proposition 12.17).*

Each of (c) and (d) is needed, and (d) on each side. The criterion is what storage of a bunch on new marks of a table asks, and nothing in the periods depends on it: the glue of Theorem 28.10 composes pieces whether or not their glue is a table.

# Chapter 29

# Composing bunches

The pieces of Chapter 28 compose. A family with several bunches is read through a core in which every bunch is replaced by its stand-in at its seam point, and its period is the mixture of the periods of these cores over the laws of the bunches (Theorem 29.1); a bunch inside a bunch makes the law of the larger one a mixture of the laws of its nodes (Theorem 29.2); and along a chain of bunches the letters are products of transfer matrices indexed by the stations of room sides, the laws are carried from bunch to bunch by one transfer each, and the period opens as a flow of shares through the stations (Theorem 29.3, Corollary 29.5). Every operation is a rule on the seam functions on contours or on the station transforms (Theorem 29.6), a closed chain of cells is read the same way (Theorem 29.8), and the Foreman tree with nodes of any width is evaluated by one procedure whose reusable data are the laws on its bunches (Definition 29.9). Read backward, the period as a function of the masses determines the potential, so the classes of marks and the tree are read off the periods, and with retention at every node this is the correspondence between a realization and its word (Theorems 29.11 and 29.12).

## 29.1 Cores, nested pieces and chains of bunches

**Theorem 29.1** (The core reads its pieces, at any width)**.** *Let $B_1, \dots, B_q$ be disjoint bunches of a family $F$ on its balance wall, each with $\sum_j (s_j + t_j) c_j(B_l) > 0$, with their scales $\gamma^\rho_{B_l}$, scale forms $\phi^\rho_{B_l}$ and seam vectors $\boldsymbol{\varrho}^\rho_{B_l}$ (Definition 28.5), and let $Z$ be the marks outside them. For $u = (u_l)_l \in \prod_l \Delta_{B_l}$ read the letters of $F$ at $\alpha_{B_l} = \alpha_{g_l} u_l / \gamma^\rho_{B_l}(u_l)$, for new marks $g_l$, and divide each letter of table $j$ by the product over $l$ of the extreme excess sums of the bunches $B_l$ in that letter (Definition 28.5), $V_j^{(0)}$ for $K_j$ and $V_j^{*(c_j(B_l))}$ for $\Psi_j$, read at the same point. The letters obtained depend on $u$ only through the seam points $\mathbf{x}_l = \boldsymbol{\varrho}^\rho_{B_l}(u_l)$; with them, the exponents of $F$ and the masses $\kappa_{B_l}$ on $g_l$ and $\mathbf{m}_Z$ on $Z$ they form the* stand-in core $F^{(\mathbf{x}_1, \dots, \mathbf{x}_q)}$*, a family of Newton words on $Z \sqcup \{g_1, \dots, g_q\}$ (Definition 28.4) in which the stand-in $g_l$ enters the letters $K_j$ and $\Psi_j$ with the degree $c_j(B_l)$. In $[0, \infty]$,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{\prod_l \Delta_{B_l}} \prod_l \phi^\rho_{B_l}(du_l)\ P_{F^{(\boldsymbol{\varrho}^\rho_{B_1}(u_1), \dots, \boldsymbol{\varrho}^\rho_{B_q}(u_q))}}\big(\mathbf{m}_Z, \kappa_{B_1}, \dots, \kappa_{B_q}\big)$$
$$= \int P_{F^{(\mathbf{x}_1, \dots, \mathbf{x}_q)}}\big(\mathbf{m}_Z, \kappa_{B_1}, \dots, \kappa_{B_q}\big) \prod_l d\big((\boldsymbol{\varrho}^\rho_{B_l})_* \phi^\rho_{B_l}\big)(\mathbf{x}_l).$$

*At width one with factoring weights the stand-in core is the twisted family $F_{\mathcal{H}}[\mathbf{x}]$ of Theorem 23.6(i), up to the change of scale of Proposition 28.11, and this is Theorem 23.6(i); before the laws are*

*formed it is Theorem 22.3 with the scale of every marker divided by the scale $\gamma^{\rho}_{B_l}$ of its piece.*

The core meets its pieces only through their seam points: every bunch collapses to one stand-in mark carrying the mass of its wall, and enters each letter with the degree of its width, at its stations twisted by its seam point. The cores of Chapter 22 are the case in which every piece is a cell.

**Theorem 29.2** (The law of a piece from its sub-pieces)**.** *Let $S$ be a bunch of a family with $\sum_j (s_j + t_j) c_j(S) > 0$, and $S_1, \dots, S_k \subseteq S$ disjoint bunches with $\sum_j (s_j + t_j) c_j(S_l) > 0$.*

*(i) The seam letters of $S$ are linear in those of each $S_l$, with nonnegative coefficients: in each table, for $S_1$, every station $D$ of room sides of $S$ and any member $X \in D$,*

$$K_{(D)} = \sum_q \frac{w'_{q \sqcup X}}{w'_X}\, \alpha^{\,q}\, K_{(D_1(q))}, \qquad \Psi_{(D)} = \sum_q \frac{w'_{q \sqcup X}}{w'_X}\, \alpha^{(S \setminus S_1) \setminus q}\, \Psi_{(D_1(q))},$$

*the sums over the sets $q \subseteq S \setminus S_1$ for which $q \sqcup X$ is a room side of $S_1$, $D_1(q)$ its station, $w'_{q \sqcup X}$ its multiplier there and $w'_X$ that of $X$ at $S$; the coefficients and the stations $D_1(q)$ do not depend on the member $X$.*

*(ii) Let the* node $S^{(\mathbf{x}_1, \dots, \mathbf{x}_k)}$*, the node at $S$ of the tree of the bunches $S \supseteq S_1, \dots, S_k$, be the piece on $(S \setminus \bigcup_l S_l) \sqcup \{g_1, \dots, g_k\}$ whose seam letters are those of $S$ with each $S_l$ replaced by its stand-in at $\mathbf{x}_l$, as in Theorem 29.1, with the masses $\kappa_{S_l}$ on $g_l$. The node has the wall of $S$, and, in this statement writing $\lambda_S$, $\lambda_{S_l}$ and $\lambda_{\mathbf{x}}$ for the laws of $S$, of $S_l$ and of the node at $\mathbf{x} = (\mathbf{x}_l)_l$, the law of $S$ is the mixture of the laws of its nodes:*

$$\int H \, d\lambda_S = \int \Big( \int H \, d\lambda_{\mathbf{x}} \Big) \prod_l d\lambda_{S_l}(\mathbf{x}_l) \qquad \text{for every Borel } H \geq 0,$$

*and so are the seam functions, $\int \mathbf{z}^{-(\mathbf{a},\mathbf{b})}\, d\lambda_S(\mathbf{z}) = \int \big( \int \mathbf{z}^{-(\mathbf{a},\mathbf{b})}\, d\lambda_{\mathbf{x}}(\mathbf{z}) \big) \prod_l d\lambda_{S_l}(\mathbf{x}_l)$ at all shares $(\mathbf{a}, \mathbf{b})$.*

*At width one, with factoring weights, this is Theorem 23.6(ii), up to the change of scale of Proposition 28.11.*

**Theorem 29.3** (A chain of bunches)**.** *Let $\varnothing = S_0 \subset S_1 \subset \cdots \subset S_L = \mathcal{E}$, $L \geq 2$, be a chain of bunches of a family with the steps $B_{l-1} = S_l \setminus S_{l-1}$, and for $0 < l < L$ let $\mathcal{D}(S_l)$ be the stations of room sides of the bunch $S_l$, in each table, the index $j$ dropped.*

*(i)* Letters. *For $2 \leq l \leq L-1$, $D' \in \mathcal{D}(S_{l-1})$ and $D \in \mathcal{D}(S_l)$ put, for any $X \in D$,*

$$\mathbf{N}^K_l(D', D) = \sum_q \frac{w'_{q \sqcup X}}{w'_X}\, \alpha^{\,q}_{B_{l-1}}, \qquad \mathbf{N}^{\Psi}_l(D', D) = \sum_q \frac{w'_{q \sqcup X}}{w'_X}\, \alpha^{B_{l-1} \setminus q}_{B_{l-1}},$$

*the sums over the $q \subseteq B_{l-1}$ for which $q \sqcup X$ is a room side of $S_{l-1}$ in the station $D'$. With $\mathbf{k} = (K_{(D)})_{D \in \mathcal{D}(S_1)}$ and $\mathbf{k}^* = (\Psi_{(D)})_{D \in \mathcal{D}(S_1)}$ the seam letters of $S_1$ and $\mathsf{y} = (\mathsf{y}_D)_{D \in \mathcal{D}(S_{L-1})}$ and $\mathsf{y}^* = (\mathsf{y}^*_D)_{D \in \mathcal{D}(S_{L-1})}$ the room station letters of $S_{L-1}$,*

$$K = \mathbf{k}^{\mathsf{T}} \mathbf{N}^K_2 \cdots \mathbf{N}^K_{L-1}\, \mathsf{y}, \qquad \Psi = \mathbf{k}^{*\mathsf{T}} \mathbf{N}^{\Psi}_2 \cdots \mathbf{N}^{\Psi}_{L-1}\, \mathsf{y}^*,$$

*products of matrices of polynomials with nonnegative coefficients, indexed by the stations of room sides of the bunches $S_l$: Theorem 22.9 with these stations as the states.*

*(ii)* Laws. *Let $\sum_j (s_j + t_j) c_j(S_l) > 0$ for every $0 < l < L$, and in this statement write $\lambda_l$ for the law of $S_l$. For $2 \le l \le L-1$ let $S_l^{(\mathbf{x})}$ be the node of $S_l$ with its sub-bunch $S_{l-1}$ at $\mathbf{x}$ (Theorem 29.2(ii)), the piece on $B_{l-1} \sqcup \{g\}$ with the seam letters $\sum_{D'} x_{D'}\, \alpha_g^{i(D')} \mathbf{N}_l^K(D', D)$ and $\sum_{D'} x^*_{D'}\, \alpha_g^{c(S_{l-1}) - i(D')} \mathbf{N}_l^\Psi(D', D)$, $D \in \mathcal{D}(S_l)$, and the mass $\kappa_{S_{l-1}}$ on $g$, and let $\lambda_{\mathbf{x}}^{(l)}$ be its law. Then $\lambda_l$ is the image of $\lambda_{l-1}$ under the* transfer $\mathcal{Q}_l$,

$$\lambda_l = \mathcal{Q}_l \lambda_{l-1} = \int \lambda_{\mathbf{x}}^{(l)}\, d\lambda_{l-1}(\mathbf{x}),$$

*and $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int P_{B_{L-1}^{(\mathbf{x})}}(\mathbf{m}_{B_{L-1}}, \kappa_{S_{L-1}})\, d\lambda_{L-1}(\mathbf{x})$, the room $B_{L-1}$ of the last bunch $S_{L-1}$ with its stand-in at $\mathbf{x}$ (Theorem 28.10(i)).*

*(iii)* Repetition. *If the nodes $S_l^{(\cdot)}$, $2 \le l \le L-1$, are one piece up to bijections of marks keeping seam letters and masses, the transfers are one operator $\mathcal{Q}$ and $\lambda_l = \mathcal{Q}^{l-1} \lambda_1$: the chain is the power of one operator, whatever the widths of its bunches. At width one with factoring weights this is Theorem 22.8 and Proposition 24.4, the seam point being the twist up to the change of scale of Proposition 28.11.*

**Lemma 29.4** (The native split along a chain)**.** *Let a family of tables have real data on its balance wall, with $s_j, t_j \ge 0$, not all zero, and $\mathbf{m} > 0$, and a native gauge $\alpha^\star$ (Definition 8.17), and let $\varnothing = S_0 \subset S_1 \subset \cdots \subset S_L = \mathcal{E}$ be a chain of bunches as in Theorem 29.3, with the steps $B_{l-1} = S_l \setminus S_{l-1}$. For the active letters put*

$$e^{(l)}_{jD'D} = s_j\, \mathbb{P}_{j,\alpha^\star}\big(T \setminus S_{l-1} \in D',\ T \setminus S_l \in D\big), \qquad f^{(l)}_{jD'D} = t_j\, \mathbb{P}^*_{j,\alpha^\star}\big(T \setminus S_{l-1} \in D',\ T \setminus S_l \in D\big),$$

*and $a^{(l)}_{jD} = s_j\, \mathbb{P}_{j,\alpha^\star}(T \setminus S_l \in D)$, $b^{(l)}_{jD} = t_j\, \mathbb{P}^*_{j,\alpha^\star}(T \setminus S_l \in D)$, the native split at the stations of the bunch $S_l$ (Lemma 28.9). Then the entry shares conserve the loads, $\sum_{D'} e^{(l)}_{jD'D} = a^{(l)}_{jD} = \sum_{D''} e^{(l+1)}_{jDD''}$, and likewise with $f$ and $b$; and $\mathbf{m}_{B_{l-1}}$ lies in the relative interior of the Newton sum of the word of the node $l$ at its shares, which converges there exactly when that Newton sum has the dimension $|B_{l-1}| - 1$.*

**Corollary 29.5** (The word of a chain of bunches)**.** *In Theorem 29.3 read the room $B_{L-1}$ of the last bunch $S_{L-1}$ as a last node whose transfer matrix from $\mathcal{D}(S_{L-1})$ to one terminal station is the column $\mathsf{y}$, and $\mathsf{y}^*$ for $\Psi$, and open every active letter with one share per nonzero entry of every transfer matrix: $e^{(l)}_{jD'D}$ for the entry $(D', D)$ of $\mathbf{N}_l^K$ and $f^{(l)}_{jD'D}$ for that of $\mathbf{N}_l^\Psi$, $2 \le l \le L$. A station $D$ of a bunch $S_l$, $2 \le l \le L-1$, carries the load $a^{(l)}_{jD} = \sum_{D'} e^{(l)}_{jD'D}$ in and $\sum_{D''} e^{(l+1)}_{jDD''}$ out, and a station $D$ of the first bunch $S_1$ the load $a^{(1)}_{jD} = \sum_{D''} e^{(2)}_{jDD''}$; likewise $b$ with $f$. In this statement write $W_l(\mathbf{e}, \mathbf{f}) = \prod_{D',D} \mathbf{N}_l^K(D', D)^{-e_{D'D}}\, \mathbf{N}_l^\Psi(D', D)^{-f_{D'D}}$ for the word of the node $l$ on $B_{l-1}$, the products over the active letters of every table understood. On contours of positive real parts at which the seam function of $S_1$ and the words of the nodes converge, which needs their Newton sums to have full dimension and, for a family with a native gauge, holds as soon as they do (Lemmas 28.9 and 29.4),*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int \prod_{j:\, s_j > 0} \frac{\prod_{l=2}^{L} \prod_{D',D} \Gamma(e^{(l)}_{jD'D})}{\Gamma(s_j) \prod_{l=2}^{L-1} \prod_D \Gamma(a^{(l)}_{jD})} \prod_{j:\, t_j > 0} \frac{\prod_{l=2}^{L} \prod_{D',D} \Gamma(f^{(l)}_{jD'D})}{\Gamma(t_j) \prod_{l=2}^{L-1} \prod_D \Gamma(b^{(l)}_{jD})}$$
$$\times P_{W_{S_1}(\mathbf{a}^{(1)}, \mathbf{b}^{(1)})}(\mathbf{m}_{B_0}) \prod_{l=2}^{L} P_{W_l(\mathbf{e}^{(l)}, \mathbf{f}^{(l)})}(\mathbf{m}_{B_{l-1}}),$$

*$W_{S_1}$ the seam word of the first bunch $S_1$ (Lemma 28.7), the fibre pinned by the letter totals, the equality of the loads in and out at every station of the bunches $S_2, \dots, S_{L-1}$, and the homogeneity of the first bunch and of every node but one. At a bunch or a node whose Newton sum is degenerate the pins and the reduced period of Theorem 28.12 take the place of the period. At $L = 2$ this is Theorem 28.10(iii).*

The word of a chain of bunches is a flow: one share per transfer entry, conserved at every station, each node contributing the period of its own word at the loads of its entries. Read with insertions, the same flow carries the module of the family: every value of every insertion is this word with the periods of the nodes replaced by the values of the masters of their modules, and the chance of a basis is the probability of its path under the flow times the chances of its pieces in the entries (Theorem 48.7). Every station through which the flow passes is paid once by a Gamma in the denominator, except at the first bunch, where the seam function of the piece takes the loads. At width one with factoring weights every bunch $S_l$ has two stations, and this is the word of the tree (Corollary 20.10) grouped by the two stations of each bunch.

**Theorem 29.6** (Two transforms of a law, and their rules)**.** *Let a piece $\mathcal{X}$ on $B$ with the stations $\mathcal{D}$ have $\sum_j (s_j + t_j)c_j > 0$, and a real point $\mathbf{c}$ of the pins of the seam, with positive coordinates, in the interior of the domain of its seam function on the pins.*

*(i)* Two transforms. *The law $(\boldsymbol{\varrho}^{\rho}_{\mathcal{X}})_* \phi^{\rho}_{\mathcal{X}}$ is determined by its seam function on the contour through $\mathbf{c}$, and by its station transform $\mathcal{I}_{\mathcal{X}}$ (Definition 28.21) on any nonempty open set of points $\mathbf{p}$; the two are related by*

$$\mathcal{I}_{\mathcal{X}}(\mathbf{p}) = \int_{pins} \prod_{j:\, s_j > 0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix} \mathbf{a}_j \\ s_j \end{bmatrix} \prod_{j:\, t_j > 0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix} \mathbf{b}_j \\ t_j \end{bmatrix} \; \mathbf{p}^{-(\mathbf{a},\mathbf{b})} \int_{\Delta_B} (\boldsymbol{\varrho}^{\rho}_{\mathcal{X}})^{-(\mathbf{a},\mathbf{b})} \, d\phi^{\rho}_{\mathcal{X}},$$

$\mathbf{p}^{-(\mathbf{a},\mathbf{b})} = \prod_{j,D} p_{jD}^{-a_{jD}} {p^*_{jD}}^{-b_{jD}}$.

*(ii)* The glue. *For a partner $\mathcal{Y}$ with the wall $\sum_j (s_j + t_j)c_j - \kappa_{\mathcal{X}}$, $P_{\mathcal{X} \sqcup \mathcal{Y}} = \int \mathcal{I}_{\mathcal{X}}(\mathbf{x}') \, d\big((\boldsymbol{\varrho}^{\rho}_{\mathcal{Y}})_* \phi^{\rho}_{\mathcal{Y}}\big)(\mathbf{x}')$, the station transform read against the law of the partner; on the pins, at a point where the seam function of the partner converges too, it is the share side of Theorem 28.10(iii). It extends the node of width one in one scale (Theorem 20.26) to any width.*

*(iii)* Nesting. *In Theorem 29.2, $\mathcal{I}_S(\mathbf{p}) = \int \mathcal{I}_{S(\mathbf{x}_1, \dots, \mathbf{x}_k)}(\mathbf{p}) \prod_l d\big((\boldsymbol{\varrho}^{\rho}_{S_l})_* \phi^{\rho}_{S_l}\big)(\mathbf{x}_l)$. With one sub-piece $S_1$, in this statement write $\mathbf{N}^K_{2,j}(\alpha)$ and $\mathbf{N}^{\Psi}_{2,j}(\alpha)$, $\alpha = \alpha_{S \setminus S_1}$, for the matrices $\mathbf{N}^K_2$ and $\mathbf{N}^{\Psi}_2$ in table $j$ of the chain $\varnothing \subset S_1 \subset S \subset \mathcal{E}$ (Theorem 29.3(i)), the transfer matrices from the bunch $S_1$ to the bunch $S$. Then*

$$\mathcal{I}_S(\mathbf{p}) = \int_{(0,\infty)^{S \setminus S_1}} \alpha^{\mathbf{m}_{S \setminus S_1} - \mathbf{1}} \, \mathcal{I}_{S_1}\big(\big(\mathbf{N}^K_{2,j}(\alpha)\, \mathbf{p}_j\big)_j, \big(\mathbf{N}^{\Psi}_{2,j}(\alpha)\, \mathbf{p}^*_j\big)_j\big) \, d\alpha,$$

$\mathbf{p}_j = (p_{jD})_D$ *and* $\mathbf{p}^*_j = (p^*_{jD})_D$*; and at shares $(\mathbf{a}, \mathbf{b})$ on the pins of the seam of $S$, on contours of positive real parts at which the seam function of $S_1$ and the word of the node converge,*

$$\int_{\Delta_S} (\boldsymbol{\varrho}^{\rho}_S)^{-(\mathbf{a},\mathbf{b})} \, d\phi^{\rho}_S = \int \prod_{j:\, s_j > 0} \frac{\prod_{D_1, D} \Gamma(e_{jD_1D})}{\prod_D \Gamma(a_{jD})} \prod_{j:\, t_j > 0} \frac{\prod_{D_1, D} \Gamma(f_{jD_1D})}{\prod_D \Gamma(b_{jD})} \times \int_{\Delta_{S_1}} (\boldsymbol{\varrho}^{\rho}_{S_1})^{-(\mathbf{a}^{(1)}, \mathbf{b}^{(1)})} \, d\phi^{\rho}_{S_1} \; P_{W(\mathbf{e},\mathbf{f})}(\mathbf{m}_{S \setminus S_1}),$$

*the shares $e_{jD_1D}$ and $f_{jD_1D}$ one per nonzero entry of the transfer matrices, pinned by $\sum_{D_1} e_{jD_1D} = a_{jD}$, $\sum_{D_1} f_{jD_1D} = b_{jD}$ and the homogeneity of the node, with the loads $a^{(1)}_{jD_1} = \sum_D e_{jD_1D}$ and $b^{(1)}_{jD_1} = \sum_D f_{jD_1D}$, and $W(\mathbf{e},\mathbf{f})$ the word of the entries on $S \setminus S_1$ as in Corollary 29.5.*

*(iv)* Repetition. *The rule of (iii) for a node repeated along a chain is one operator on seam functions, and on station transforms one operator on functions of $\mathbf{p}$; a chain of repeated nodes is its power applied to the first bunch.*

So every operation along a chain is a rule on the seam functions on contours, or on the station transforms: the representations (b) and (a) of Definition 23.19, at every width.

**Definition 29.7** (Cyclic chains)**.** A family on $\mathcal{E} = B_1 \sqcup \cdots \sqcup B_k$, $k \geq 2$, the *segments* $B_i$ indexed by $\mathbb{Z}_k$, is a *cyclic chain* with the finite sets of *states* $\mathcal{J}_0, \ldots, \mathcal{J}_{k-1}$, $\mathcal{J}_k = \mathcal{J}_0$, and the *top layer* $h \in \mathbb{N}$, when every active letter $L$ is

$$L = [\varepsilon^h] \operatorname{tr} \prod_{i=1}^{k} \mathbf{M}_{L,i}(\alpha_{B_i}; \varepsilon),$$

the coefficient of $\varepsilon^h$ in the trace of a product of matrices $\mathbf{M}_{L,i}$ indexed by $\mathcal{J}_{i-1} \times \mathcal{J}_i$, the *segment matrices*, whose entries are polynomials in $\alpha_{B_i}$ and $\varepsilon$ with nonnegative coefficients.

**Theorem 29.8** (Cyclic chains)**.** *(i) Let every table $j$ of a family be a closed chain of cells $\mathcal{A}_{j,1}, \ldots, \mathcal{A}_{j,k}$ with several hinges spliced along hinge pairs (Theorem 21.14(ii)), the cells $\mathcal{A}_{j,i}$ of all tables having the marks $B_i$ besides their hinges, and the pairs $I_i$ joining the cell $i$ to the cell $i+1$ named alike in every table, each pair named once. The family is a cyclic chain with the top layer $0$: its states at the boundary $i$ are the sets $J \subseteq I_i$ of the pairs whose hinge on the side of the cell $i$ is occupied, and the segment matrices of the letters of table $j$ are the channels of its cells (Definition 21.13),*

$$\mathbf{M}_{K_j,i}[J, J'] = K_{\mathcal{A}_{j,i},((I_{i-1} \setminus J) \sqcup J')}, \qquad \mathbf{M}_{\Psi_j,i}[J, J'] = \Psi_{\mathcal{A}_{j,i},((I_{i-1} \setminus J) \sqcup J')}.$$

*(ii) Widen the states by layers, $\mathcal{J}_i \times \{0, \ldots, h\}$, a term in $\varepsilon^\mu$ of an entry raising the layer by $\mu$, and write $\widetilde{\mathbf{M}}_{L,i}$ for the widened segment matrices. Then $L = \sum_{J \in \mathcal{J}_0} \mathbf{e}^{\mathsf{T}}_{(J,0)} \prod_i \widetilde{\mathbf{M}}_{L,i}\, \mathbf{e}_{(J,h)}$: for every starting state a chain in the sense of Theorem 29.3(i), the segment matrices as transfer matrices. In this statement write $w_L$ for the exponent of the active letter $L$. Open every active letter first over the starting states $J$ at which its term $\mathbf{e}^{\mathsf{T}}_{(J,0)} \prod_i \widetilde{\mathbf{M}}_{L,i}\, \mathbf{e}_{(J,h)}$ is not zero, with shares $c_{L,J}$, $\sum_J c_{L,J} = w_L$, and then every chain as in Corollary 29.5: one share $e^{(J)}_{L,i}$ per starting state $J$ and per nonzero entry of a widened segment matrix that lies on a path from $(J,0)$ to $(J,h)$, the shares of each starting state a flow from $(J,0)$, with the total $c_{L,J}$, to $(J,h)$, conserved at every boundary, $a^{(J)}_{L,i}$ the loads it leaves at the boundary after the segment $i$, and $e^{(J)}_{L,1} = a^{(J)}_{L,1}$. On contours of positive real parts at which the words of the segments converge,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int \prod_L \frac{1}{\Gamma(w_L)} \prod_{L,J} \frac{\prod_{i=2}^{k} \prod \Gamma(e^{(J)}_{L,i})}{\prod_{i=2}^{k-1} \prod \Gamma(a^{(J)}_{L,i})} \prod_{i=1}^{k} P_{W_i}(\mathbf{m}_{B_i}),$$

*the inner products over the entries and over the states, $W_i$ the word on $B_i$ of the entries of the segment $i$ at their total shares $\sum_J e^{(J)}_{L,i}$, and the fibre pinned by the conservation and by the homogeneity of every segment but one.*

*(iii) For $k$ copies of one segment the integrand of (ii), read at each boundary as a function of the loads of all the starting states, is the kernel of one operator $\mathcal{C}$, restricted at each boundary to the loads of the states that lie on a path there, which integrates the loads out over the fibre pinned by the homogeneity of the segment. With $\mathcal{C}_\circ$ the same kernel without that pin, which for one segment is implied by those of the others, the letter totals and the balance, the period is*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int \prod_L \frac{\prod_J \Gamma(c_{L,J})}{\Gamma(w_L)} \left(\mathcal{C}^{k-1}\mathcal{C}_\circ\right)(\mathbf{c},\mathbf{c})\, d\mathbf{c},$$

*a power of one operator taken from the loads $\mathbf{c}$ at the starting states back to them. Its price is that of one segment and the binary digits of $k$ (Chapter 33); evaluated, the power takes at most $\lfloor \log_2 k \rfloor$ squarings of $\mathcal{C}$ and as many products.*

A closed chain is a chain whose last boundary is glued back to its first: the trace sums over the states at one boundary, and each starting state carries its own flow around the closed chain. Cells spliced along any graph of hinge pairs stay tables (Theorem 21.14); around a closed chain their letters are traces of products of channel matrices, and repeated cells give the power of one operator.

## 29.2 The procedure

The Newton sums of Lemmas 28.9 and 29.4 can be degenerate. When every station of room sides is a single room side, every room station letter is a monomial, the room word is a monomial, and its period diverges at every share; the period is then read by Theorem 28.10(i) or (ii), or on the share side of Theorem 28.12 with $\vec{\mathcal{N}}_R = 0$.

**Definition 29.9** (The seam procedure)**.** *Given*: a family $F$ with real data on its balance wall, every proper wall positive. *Classification*, from the tables alone: the molecules, the readings at the centre and the Foreman tree (Definition 16.14, steps (1)–(5)); for a molecule with a connected active table the tree is a chain whose bunches $B_0, \dots, B_{d-1}$ (Corollary 20.8) are the steps of the chain of bunches $S_k = B_0 \sqcup \cdots \sqcup B_{k-1}$, the leaf being the last step. *Seams*, from the tables alone, at every bunch $S_k$:

(1) group the patterns of $S_k$ of each excess by their weighted children up to positive multiples, and the room sides by their weighted room children, with the multipliers normalized by their largest value (Definition 28.1); the stations, the station letters and the pair constants are determined;

(2) record the seam matrices $H_i$ and the transfer matrices $\mathbf{N}_k^K$, $\mathbf{N}_k^\Psi$ of Theorem 29.3(i);

(3) record the orbits of the stations under the bijections that keep $S_k$ and keep the potential (Proposition 28.6, Lemma 14.18).

*Evaluation* at a point $(\mathbf{m};\mathbf{s},\mathbf{t})$:

(4) the law of $S_1 = B_0$ (Definition 28.5), with its canonical scale;

(5) the transfers of the laws from bunch to bunch (Theorem 29.3(ii)), one operator for repeated nodes (Theorem 29.3(iii)), the transfer entries of pairs of stations in one orbit being images of one another (Proposition 28.6(i));

(6) the period at the leaf with its stand-in at the seam point of the last bunch $S_d$, integrated against the law of that bunch (Theorem 28.10(i)); equivalently the word of Corollary 29.5

on the contour through the native shares of Lemma 29.4, with the transverse pins of Theorem 28.12 at every bunch or node whose Newton sum is degenerate, every node a Newton word with multilinear letters closed by its peak series with the references of Theorem 15.13(i) and every seam Gramma by its peak closing with the sharp references (Proposition 28.16(i)).

*Output*: the period, with the seams, the transfer matrices and the law on every bunch, ready to be reused when a node is changed. At a general Foreman tree the bunches of a node with several childcules are composed by Theorem 29.1, the childcules of a peel being the pieces of the core.

Nothing in the procedure is chosen. The tree is the Foreman tree (Definition 14.4); the stations are fixed by the children, and their multipliers by the normalization of the largest one; the scale of a piece is the geometric mean of the oriented extreme ratios of the active letters, each taken to the root of its width, kept by the symmetries of the family (Proposition 28.6(iii)); inside a seam Gramma the stations of the bunch are integrated before those of the room, the Foreman's order (Proposition 28.15), and another choice of the stations that are not kept changes the pins by a unimodular matrix; the real point of the share contour, where the word is used, is the native split at the stations and the entries (Lemmas 28.9 and 29.4); the references of the closings are those of Theorem 15.13(i) and the sharp ones, read off the coefficients. Repetition is paid once: stations in one orbit carry data that are images of one another, and repeated nodes one operator.

**Proposition 29.10** (What the work depends on)**.** *(i) The law of a piece depends on the piece only through the joint law of the seam letters of its active letters under the measure $u^{\mathbf{m}_B-\mathbf{1}}d\omega_B$, with its exponents and wall: two pieces, on sets of any sizes, whose seam letters have the same joint law have equal laws and are interchangeable in every glue (Theorem 28.19); in particular so are two pieces whose seam letters and masses correspond under a bijection of their marks.*

*(ii) Along a chain the letters cost one product of matrices of the sizes $|\mathcal{D}(S_l)|$, and a seam Gramma of a part without circuits costs no integration (Definition 28.13, Corollary 2.6).*

*(iii) The law of a bunch depends on the room only through the room children of the bunch, the tables on the bunch that the room presents to it: changing the room while keeping its room children and their stations changes no law below the bunch.*

## 29.3 The correspondence

The procedure reads a realization forward, from its tables to its word. The other direction asks what the periods distinguish, and it has two answers: as a function of the masses the period determines the potential, which carries the classes of marks and the tree; at one realization each bunch is seen by its room through its wall and its law, and only through them.

**Theorem 29.11** (The period function determines the potential)**.** *Let a family have real exponents $s_j, t_j \geq 0$ with $s_j + t_j > 0$ for every $j$ and a period that converges at some masses on its balance wall, and let $U$ be the set of the complex masses on the balance wall at which it converges.*

*(i) With the measure $dy$ on the hyperplane $\mathbf{1}^\perp = \{y \in \mathbb{R}^{\mathcal{E}} : \langle \mathbf{1}, y\rangle = 0\}$ of the section $\prod_e \alpha_e = 1$ (Lemma 16.2),*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\mathbf{1}^\perp} e^{\langle \mathbf{m},y\rangle - \Phi(y)}\, dy, \qquad \mathbf{m} \in U,$$

*$\Phi$ the potential at the table's gauge (Definition 8.18). The function $P_F(\cdot;\mathbf{s},\mathbf{t})$ on $U$ determines $\Phi$, and is determined by it.*

*(ii) Two families on the same marks with the same exponents have the same period function exactly when they have the same irreducible factors with their exponents* $(L, w_{\mathbf{E}}(L), w_{\mathbf{O}}(L))$ *(Theorem 10.6) and the same constant* $\Phi(0) = \sum_j (s_j + t_j) \log K_j(\mathbf{1})$*; at the centre, every* $\delta_j = 0$*, exactly when they have the same* $(L, w_{\mathbf{E}}(L))$ *and the same constant. In particular a family and the family of its dual tables (Proposition 8.15), the bases replaced by their complements, have the same period function at the centre.*

*(iii) A bijection* $g$ *of the marks keeps the period function,* $P_F(g^*\mathbf{m}; \mathbf{s}, \mathbf{t}) = P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ *on* $U$*, exactly when it keeps the potential, both parts. For a molecule, two marks lie in one class exactly when a bijection keeping the period function takes one to the other (Lemma 14.18), and when its children's weights factor across the childcules at every node, every such bijection carries the Foreman tree onto itself, with its classes, units, atoms, bunches and childcules (Proposition 14.21).*

*(iv) At the centre the bijections that carry the family and those that exchange it keep the period function; at a lean the carrying ones keep it, and an exchanging one sends it to the period function of the opposite lean (Proposition 11.14). For a molecule of one table two marks lie in one class at the centre exactly when a bijection carrying or exchanging the table takes one to the other, and at a lean exactly when a carrying one does (Corollary 11.16), and a class of the centre that is not a class at a lean is there two classes of equal size, which an exchange swaps (Theorem 11.17).*

So two marks of a molecule are told apart by the period function of the centre, or of a lean, exactly when no bijection keeping that potential joins them; for one table these are, at the centre, the bijections carrying or exchanging the table, and at a lean the carrying ones. The confusable classes, the classes of a lean that share a class of the centre, are exactly what the period function of the centre does not separate and that of a lean does. The tree is read off the period function but does not determine it; the irreducible factors with their exponents, and the constant $\Phi(0)$, do.

**Theorem 29.12** (The correspondence)**.** *Let a molecule be read at a realization with masses on its balance wall and every proper wall positive, its children's weights factoring across the childcules at every node.*

*(i)* Forward. *Its period is the reassembly of the laws of the bunches of its Foreman tree with the leaves, by the seam procedure: the law of the first bunch on its seam torus, transferred along the chain by the later bunches as nodes, the childcules of a node composed at its core, and glued to the leaves by the stand-ins; nothing in it is chosen (Definition 29.9, Theorems 29.1 and 29.3).*

*(ii)* Backward at the realization. *At the node reached after* $k$ *peels the room sees the bunch* $S_k$*, the union of the bunches peeled down to it, through its wall and its law, and only through them: bunches with the same stations, exponents, walls and laws are interchangeable in every family glued on that room, and under the hypotheses of Theorem 28.19 two different laws are told apart by the tinkers of one room and by plain partners (Theorems 28.19 and 28.22); the laws of nested bunches compose by Theorems 29.2 and 29.3.*

Whether the molecule has an atomic realization is read at a centre: it has one exactly when some centre is atomic, and an atom at a lean is an atom at the centre (Theorems 11.8 and 11.1). The word of the tree is the forward procedure written out, and retention is the statement that it loses nothing that a room could detect.

# Chapter 30

# Foreman fusion

A table of rank $r$ and corank $r^*$ is the contraction of $c = r^* - r$ new marks in an identically self-dual table, in which the new marks form a bunch in level form and their deletion leaves the dual table, exactly when every cocircuit of the table is a union of circuits and, at $c = 0$, the table has the weights of its dual. The bases of the larger table are then unique, and at $c \leq 1$ the larger table itself: they are those of the level cell of the pair formed by the dual table and the table, the new marks are its fusion bunch, and their children on the marks of the table are the lifts of the table toward its dual, one for each level, whose weights between the two extremes, the middle weights, are a choice paired by complement, all 1 unless others are named. This is Foreman fusion, and the table is Foreman-embedded in its fusion. The fused table is identically self-dual, so each of its marks comes off by one Beta corner at every pair of exponents (Corollary 20.19); its automorphisms contain those of the table that keep the middle weights, all of them when the middle weights are 1, times the permutations of the fusion bunch, and are often more; and every identically self-dual table is the fusion of each of its one-mark contractions. At the fusion width one the fusion bunch is one mark $g$, the letter of the fused table is $\Psi + \alpha_g K$, and the table is read off it by one Beta corner: the law of the fusion bunch is a point. At the width $c$ the seam of the fusion bunch is one rectangle at every excess, its seam letters are the elementary symmetric polynomials of the new marks up to constants, all 1 for a list, and the fused table is the glue (Theorem 28.10) of a universal piece of width $c$, which does not depend on the table, with the lift partner, whose room letters are the letters of the lifts. The law of the universal piece is the image of a measure on the torus of the fusion bunch, of dimension $c-1$, and the law of the lift partner is a point only when its lift ratios are constant, which they are not at $c \geq 2$: these two laws are the price of width. Down the partial fusions the first step is a Beta corner and every later step is a cell whose law is not a point. Every one-mark contraction of the fused table has the fusion width one and fuses back into it: these cousins share one period up to Beta corners, and an automorphism of the fused table carries the cousin at a mark onto the cousin at the image of that mark. Across a two-sum of lists the widths add, the condition is read at the hinge by four conditions on each cell, and at the total width one a two-sum fuses exactly when both cells fuse, the cell of the width zero being then identically self-dual. A table of the fusion width $c \geq 0$ that does not fuse has uncovered marks, or has the width zero and the bases but not the weights of its dual; doubling the uncovered marks in parallel, or one mark in the second case, gives a table that fuses. The chapter ends with the branch these statements set at every node of the Foreman tree.

## 30.1 Quotients, lifts and the level cell of a pair

**Definition 30.1** (Quotients; lifts; the level cell of a pair)**.** Let $\mathcal{M}$ and $\mathcal{N}$ be tables on one set of marks $\mathcal{E}$, of the ranks $r_{\mathcal{M}}$ and $r_{\mathcal{N}}$, with the rank functions $r_{\mathcal{M}}(Y)$ and $r_{\mathcal{N}}(Y)$ (Definition 7.4). $\mathcal{N}$ is a *quotient* of $\mathcal{M}$ when every set that is maximal for its rank in $\mathcal{N}$ is maximal for its rank in $\mathcal{M}$, as the child at $p+b$ is of the child at $p$ (Proposition 12.6); its *rank difference* is $c = r_{\mathcal{M}} - r_{\mathcal{N}}$. For $0 \le j \le c$ the *$j$-th lift of $\mathcal{N}$ toward $\mathcal{M}$* is the table $\mathcal{L}_j(\mathcal{M},\mathcal{N})$ on $\mathcal{E}$ with the rank function

$$r_{\mathcal{L}_j(\mathcal{M},\mathcal{N})}(Y) = \min\big(r_{\mathcal{M}}(Y),\ r_{\mathcal{N}}(Y)+j\big) \qquad (Y \subseteq \mathcal{E}),$$

a table by Proposition 30.2(iii) below; for a bunch $B$ of a table and $(\mathcal{M},\mathcal{N}) = (\mathcal{B}|B, \mathcal{B}/R)$ these are the lifts of Proposition 12.17. For a set $G$ of $c$ new marks the *level cell* of the pair $(\mathcal{M},\mathcal{N})$ is the table on $\mathcal{E} \sqcup G$ whose bases are the sets $X \sqcup Q$ with $Q \subseteq G$ and $X$ a basis of $\mathcal{L}_{c-|Q|}(\mathcal{M},\mathcal{N})$; for $(\mathcal{B}|B, \mathcal{B}/R)$ it is the level cell of the bunch $B$ (Lemma 28.27). It carries any positive weights, and $G$ is in level form in it (Proposition 14.12) when the weight of $X \sqcup Q$ depends on $X$ and $|Q|$ alone.

**Proposition 30.2** (The level cell of a pair)**.** *Let $\mathcal{N}$ be a quotient of $\mathcal{M}$ on $\mathcal{E}$ with the rank difference $c$, let $G$ be a set of $c$ new marks, and in this statement write $\mathcal{L}_j$ for $\mathcal{L}_j(\mathcal{M},\mathcal{N})$.*

*(i) For tables $\mathcal{M}$ and $\mathcal{N}$ on $\mathcal{E}$, $\mathcal{N}$ is a quotient of $\mathcal{M}$ exactly when $r_{\mathcal{M}}(Y) - r_{\mathcal{M}}(X) \ge r_{\mathcal{N}}(Y) - r_{\mathcal{N}}(X)$ for all $X \subseteq Y \subseteq \mathcal{E}$. Then $c \ge 0$, every set independent in $\mathcal{N}$ is independent in $\mathcal{M}$, every set spanning $\mathcal{M}$ spans $\mathcal{N}$, and a quotient of $\mathcal{N}$ is a quotient of $\mathcal{M}$.*

*(ii) The level cell of $(\mathcal{M},\mathcal{N})$ is a table of rank $r_{\mathcal{M}}$ with the rank function*

$$r^{\natural}(Y \sqcup P) = \min\big(r_{\mathcal{M}}(Y) + |P|,\ r_{\mathcal{N}}(Y) + c\big) \qquad (Y \subseteq \mathcal{E},\ P \subseteq G).$$

*In it every subset of $G$ is a pattern of the bunch $G$, $d(G) = 0$ and $c(G) = c$; the child at $Q \subseteq G$ is $\mathcal{L}_{c-|Q|}$; the deletion of $G$ is $\mathcal{M}$ and the contraction of $G$ is $\mathcal{N}$; and every permutation of $G$ maps its bases onto themselves.*

*(iii) Each $\mathcal{L}_j$ is a table of rank $r_{\mathcal{N}}+j$ whose bases are the sets of $r_{\mathcal{N}}+j$ marks that are independent in $\mathcal{M}$ and span $\mathcal{N}$; $\mathcal{L}_0 = \mathcal{N}$, $\mathcal{L}_c = \mathcal{M}$, and $\mathcal{L}_j$ is a quotient of $\mathcal{L}_{j'}$ for $j \le j'$. Lifts compose: for $0 \le a \le b \le c$ and $0 \le i \le b-a$,*

$$\mathcal{L}_i(\mathcal{L}_b, \mathcal{L}_a) = \mathcal{L}_{a+i}.$$

*(iv) For disjoint $C, D \subseteq G$ the minor of the level cell contracting $C$ and deleting $D$ is the level cell of the pair $(\mathcal{L}_{c-|C|}, \mathcal{L}_{|D|})$ on $\mathcal{E} \sqcup \big(G \setminus (C \cup D)\big)$.*

The level cell stores a whole chain of lifts on $c$ new marks: one level of the bunch $G$ for each lift, the deletion of $G$ at the top and its contraction at the bottom, and every interval of the chain as a minor at the marks of $G$. Nothing in its bases is chosen except the names of the new marks, which every permutation of $G$ moves without moving the bases; its weights are a choice.

## 30.2 The fused table

**Lemma 30.3** (Complements of the lifts toward the dual)**.** *Let $\mathcal{B}$ be a table of rank $r$ on $\mathcal{E}$, $n = |\mathcal{E}|$, which is a quotient of its dual table $\mathcal{B}^*$, of rank $r^* = n - r$ (Definition 30.1), let $c = r^* - r$, and let $\mathcal{L}_j = \mathcal{L}_j(\mathcal{B}^*, \mathcal{B})$, $0 \le j \le c$. Then $X$ is a basis of $\mathcal{L}_j$ exactly when $\mathcal{E} \setminus X$ is a basis of $\mathcal{L}_{c-j}$: the dual table of $\mathcal{L}_j$ has the bases of $\mathcal{L}_{c-j}$.*

**Definition 30.4** (Fusion width; Foreman fusion; majors)**.** Let $(\mathcal{B},\theta)$ be a table of rank $r$ and corank $r^*$ on $\mathcal{E}$, $n=|\mathcal{E}|$, with its dual table $(\mathcal{B}^*,\theta^*)$, $\theta^*_{\mathcal{E}\setminus T}=\theta_T$ (Definition 7.1), and with the rank function $r(Y)$ and the dual rank function $r^*(Y)$ (Definition 7.12). Its *fusion width* is $c=r^*-r=n-2r$, and $(\mathcal{B},\theta)$ is *fusable* when $\mathcal{B}$ is a quotient of $\mathcal{B}^*$ (Definition 30.1) and, at the fusion width zero, $\theta^*=\theta$. A quotient forces $c\ge 0$ (Proposition 30.2(i)), and a quotient of the same rank has the same bases, so a fusable table of the fusion width zero is identically self-dual (Corollary 23.12(iii)), and a list of the fusion width zero is fusable exactly when it is identically self-dual. Let $c\ge 0$, and let $G$ be a set of $c$ new marks, the *fusion bunch*.

(i) A *major* of $(\mathcal{B},\theta)$ is a table on $\mathcal{E}\sqcup G$ whose contraction of $G$ is $(\mathcal{B},\theta)$ and whose deletion of $G$ is $(\mathcal{B}^*,\theta^*)$.

(ii) The *lifts* of a fusable $(\mathcal{B},\theta)$ are $\mathcal{L}_j=\mathcal{L}_j(\mathcal{B}^*,\mathcal{B})$, $0\le j\le c$, with the weights $\theta^{(j)}$: $\theta^{(0)}=\theta$ on $\mathcal{L}_0=\mathcal{B}$, $\theta^{(c)}=\theta^*$ on $\mathcal{L}_c=\mathcal{B}^*$, and for $0<j<c$ the *middle weights*, positive weights with $\theta^{(c-j)}_{\mathcal{E}\setminus X}=\theta^{(j)}_X$, complement mapping the bases of $\mathcal{L}_j$ onto those of $\mathcal{L}_{c-j}$ (Lemma 30.3), a choice, all 1 unless others are named; so the lifts of a list are lists unless other middle weights are named. Its *lift letters* are $L_l=K_{\mathcal{L}_{c-l}}$, $0\le l\le c$, read with these weights, so that $L_0=\Psi$ and $L_c=K$, and its *lift ratios* are $z_j=K^2_{\mathcal{L}_j}/(K_{\mathcal{L}_{j-1}}K_{\mathcal{L}_{j+1}})$, $0<j<c$.

(iii) The *Foreman fusion* of a fusable $(\mathcal{B},\theta)$, with its middle weights, is the level cell of $(\mathcal{B}^*,\mathcal{B})$ on $\mathcal{E}\sqcup G$ with the weight $\theta^{(c-|Q|)}_X$ on $X\sqcup Q$, written $(\mathcal{B}^\sharp,\theta^\sharp)$; $(\mathcal{B},\theta)$, the contraction of the fusion bunch in it, is its *Foreman embedding*. For $G'\subseteq G$ the *partial fusion* at $G'$ is $\mathcal{B}^\sharp/(G\setminus G')$, written $\mathcal{B}^\sharp_i$ for $|G'|=i$ when the marks of $G'$ are not named; $\mathcal{B}^\sharp_0=\mathcal{B}$ and $\mathcal{B}^\sharp_c=\mathcal{B}^\sharp$.

At the fusion width zero $G$ is empty and $\mathcal{B}^\sharp=\mathcal{B}$. A table of negative fusion width $c$ has a dual of the fusion width $-c$, and the dual is the table that is fused.

**Theorem 30.5** (Foreman fusion)**.** *Let $(\mathcal{B},\theta)$ be fusable, of the fusion width $c\ge 0$, rank $r$ and corank $r^*$, with its Foreman fusion $(\mathcal{B}^\sharp,\theta^\sharp)$ on $\mathcal{E}\sqcup G$ and its lifts $\mathcal{L}_j$.*

*(i) $\mathcal{B}^\sharp$ is a table of rank $r^*$ on $n+c=2r^*$ marks with the rank function*

$$r^\sharp(Y\sqcup P)=\min\big(r^*(Y)+|P|,\ r(Y)+c\big)\qquad (Y\subseteq\mathcal{E},\ P\subseteq G),$$

*its contraction of $G$ is $(\mathcal{B},\theta)$ and its deletion of $G$ is $(\mathcal{B}^*,\theta^*)$: it is a major of $(\mathcal{B},\theta)$.*

*(ii) The dual of each lift is a lift, weights included: $\mathcal{L}^*_j=\mathcal{L}_{c-j}$. So $(\mathcal{B}^\sharp,\theta^\sharp)$ is identically self-dual (Corollary 23.12(iii)), and*

$$K^\sharp=\Psi^\sharp=\sum_{l=0}^{c}e_l(\alpha_G)\,L_l(\alpha_{\mathcal{E}}),\qquad L_l=K_{\mathcal{L}_{c-l}}=\Psi_{\mathcal{L}_l}.$$

*(iii) Every basis of every major of $(\mathcal{B},\theta)$ is a basis of $\mathcal{B}^\sharp$.*

*(iv) Every major of $(\mathcal{B},\theta)$ in which $G$ is in level form, equivalently that every permutation of $G$ carries, has the bases of $\mathcal{B}^\sharp$, the weights $\theta$ on the bases $X\sqcup G$ and $\theta^*$ on the bases $X\sqcup\varnothing$, and on the others positive weights depending on $X$ and $|Q|$ alone; it is identically self-dual exactly when these weights, read as middle weights $\theta^{(c-|Q|)}_X$ on the bases $X\sqcup Q$, are paired as in Definition 30.4(ii), and then it is the Foreman fusion with these middle weights. So the identically self-dual majors in level form are the Foreman fusions with all the choices of middle weights, and for a list $\mathcal{B}^\sharp$ is the only one that is a list. For every bijection $\beta$ of $\mathcal{E}$ carrying $(\mathcal{B},\theta)$ with $\theta^{(j)}\circ\beta=\theta^{(j)}$ for $0<j<c$, as for the middle weights 1, and every permutation $\pi$ of $G$, $\beta\sqcup\pi$ carries $(\mathcal{B}^\sharp,\theta^\sharp)$.*

(v) *At the fusion width one, $G = \{g\}$, the bases of $\mathcal{B}^\sharp$ are the sets $T + g$ and $\mathcal{E} \setminus T$, $T \in \mathcal{B}$, both with the weight $\theta_T$, $K^\sharp = \Psi + \alpha_g K$, and $(\mathcal{B}^\sharp, \theta^\sharp)$ is the only major of $(\mathcal{B}, \theta)$.*

The fusion stores the table and its dual as the two extreme children of one bunch, and fills the levels between them with the lifts, which is the only way to fill them in level form that the exchange property allows (Proposition 28.29(d)): the containment (iii) says that no major has a basis the fusion lacks, and level form at $G$ selects the bases of the fusion among the majors, its weights between the extremes being a choice paired by complement. Self-duality comes from the chain of lifts turning over under duality, the $j$-th lift onto the $(c - j)$-th. At the fusion width one there is no level between the extremes, and the fusion is the only major; at the width $c$ the middle levels admit majors that are not in level form, among them identically self-dual ones, and the bases of the fusion are the ones that every permutation of the new marks keeps.

**Theorem 30.6** (The tell)**.** *For a table $(\mathcal{B}, \theta)$ of the fusion width $c$, conditions (a), (b) and (c) are equivalent and imply $c \geq 0$. The table is fusable exactly when they hold and, at $c = 0$, $\theta^* = \theta$; so a list, or a table of the fusion width $c \geq 1$, is fusable exactly when (a) holds. For $c \geq 0$ the table is fusable exactly when (d) holds.*

(a) *$\mathcal{B}$ is a quotient of $\mathcal{B}^*$.*

(b) *Every cocircuit of $\mathcal{B}$ is a union of circuits of $\mathcal{B}$ (Definition 7.4).*

(c) *For every mark $e$ and every split $\mathcal{E} \setminus \{e\} = A \sqcup A'$, the rank of $A$ or the rank of $A'$ does not grow when $e$ is added to it.*

(d) *$(\mathcal{B}, \theta)$ has a major (Definition 30.4(i)); for $c \geq 1$, equivalently, some table on $\mathcal{E} \sqcup G$, $|G| = c$, has $\mathcal{B}^*$ and $\mathcal{B}$ as the children of the empty pattern and of the full pattern of its bunch $G$.*

*A table that satisfies (b) has no coloop. For $c < 0$ the dual table has the fusion width $-c$, and the theorem read on it has (b) in the form: every circuit of $\mathcal{B}$ is a union of cocircuits.*

The tell is read on the Skeleton of the table (Definition 7.12): the cocircuit layer must be covered by the circuit layer, cocircuit by cocircuit, before anything is integrated. In the form (c) it is a statement about one mark at a time: however the other marks are split in two, one half spans the mark. A coloop fails it at once, being a cocircuit of one mark; a loop lies in no cocircuit and never fails it. The weights enter only at the width zero, where the bases of a table that passes are those of its dual and the table fuses when its weights are those of its dual as well.

**Corollary 30.7** (The fused tables are the identically self-dual tables)**.** *(i) Let $(\mathcal{H}, \theta)$ be identically self-dual on $\mathcal{E}'$ and $C \subseteq \mathcal{E}'$ independent, every transposition of two marks of $C$ carrying $(\mathcal{H}, \theta)$. Then $\mathcal{H}/C$ is fusable of the fusion width $|C|$, and $(\mathcal{H}, \theta)$ is its Foreman fusion with the fusion bunch $C$ and the middle weights read off $\theta$. In particular every identically self-dual table on at least one mark is the Foreman fusion of each of its one-mark contractions, and the fused tables are exactly the identically self-dual tables.*

(ii) *If $\mathcal{B}$ is connected and fusable, $\mathcal{B}^\sharp$ is connected.*

**Proposition 30.8** (Partial fusions and inner lifts)**.** *Let $(\mathcal{B}, \theta)$ be fusable of the fusion width $c \geq 1$, with its lifts $\mathcal{L}_j$.*

(i) *For disjoint $C, D \subseteq G$, $\mathcal{B}^\sharp/C \setminus D$ is the level cell of $(\mathcal{L}_{c-|C|}, \mathcal{L}_{|D|})$ on $\mathcal{E} \sqcup \big(G \setminus (C \cup D)\big)$, with the weights of $\mathcal{B}^\sharp$.*

(ii) *The partial fusion at $G'$, $|G'| = i$, is the level cell of $(\mathcal{L}_i, \mathcal{B})$ on $\mathcal{E} \sqcup G'$. It has the fusion width $c - i$, it is fusable, and its Foreman fusion with the middle weights read off $\theta^\sharp$ is $(\mathcal{B}^\sharp, \theta^\sharp)$ with*

*the fusion bunch $G \setminus G'$. So $\mathcal{B} = \mathcal{B}_0^\sharp, \mathcal{B}_1^\sharp, \dots, \mathcal{B}_c^\sharp = \mathcal{B}^\sharp$, each the contraction of the next at one mark, all fuse into one table.*

*(iii) $\mathcal{L}_j$ has the fusion width $c - 2j$. For $2j \le c$ it is fusable, with the Foreman fusion $\mathcal{B}^\sharp/C \setminus D$, $|C| = |D| = j$, its middle weights read off $\theta^\sharp$. For even $c$ the middle lift $\mathcal{L}_{c/2}$ is identically self-dual; for odd $c$ the lift $\mathcal{L}_{(c-1)/2}$ has the fusion width one and fuses into $\mathcal{B}^\sharp/C \setminus D$ with $|C| = |D| = (c-1)/2$.*

So the fused table holds, at its minors on the fusion bunch, the fusion of every table in the chain between $\mathcal{B}$ and $\mathcal{B}^*$: each partial fusion fuses into it again, and each inner lift fuses into the minor that deletes as many marks of $G$ as it contracts.

*Remark* 30.9 (Examples). (i) The uniform list $\binom{\mathcal{E}}{r}$ with $n \ge 2r$ is fusable, its cocircuits having $r^* + 1 \ge r + 1$ marks; its lifts are the uniform lists $\binom{\mathcal{E}}{r+j}$, and its fusion is $\binom{\mathcal{E} \sqcup G}{r^*}$: the fusion of $\binom{[n]}{r}$ is $\binom{[2r^*]}{r^*}$, and the fusion of $\binom{[5]}{2}$ is $\binom{[6]}{3}$; for $\binom{[2r+1]}{r}$, $r \ge 2$, the fusion $\binom{[2r+2]}{r+1}$ is the table one mark larger of Proposition 39.64(ii).

(ii) A list of rank two with no loop has as its bases the pairs of marks in different classes of a partition of its marks, the classes of two or more marks being its parallel classes (Definition 19.1). With $k \ge 3$ classes it is fusable exactly when $k \ge 4$, or $k = 3$ and every class has at least two marks: its cocircuits are the complements of the classes, and a mark of a class $P'$ lies in a circuit inside the complement of another class $P$ exactly when $P'$ has a second mark or two classes other than $P$ and $P'$ exist. So the list $\binom{[3]}{2}$ and every such list with three classes one of which is a single mark fail, and those with four or more classes pass at every fusion width (receipt `t03_rank2_census.py`, every connected list of rank two on 5 to 9 marks).

(iii) The list of rank two on $n \ge 5$ marks with one parallel pair $P$ and every other mark in a class of its own has the fusion width $n - 4$, and its fusion has exactly two sets of $n - 2$ marks that are not bases, the complementary halves $P \sqcup G$ and $\mathcal{E} \setminus P$; its automorphisms are the $2((n-2)!)^2$ bijections keeping this pair of halves, transitive on the marks. At $n = 5$ the fusion is the two-sum of two copies of $\binom{[4]}{2}$.

(iv) Of the 108 lists of rank three on seven marks, up to the names of the marks, 22 are fusable, of the fusion width one, and they fuse into the 17 identically self-dual lists on eight marks; of the 18 that are prime (Theorem 23.16), 10 are fusable, those whose lines all have three marks and meet pairwise (receipts `census7.py`, `atoms7.py`).

(v) The majors of $\binom{[4]}{1}$ that are lists, of the fusion width two, are 79 lists on six marks; 7 of them are identically self-dual and one is in level form at $G$, and that one is the fusion $\binom{[6]}{3}$ (receipt `t02_majors_U14.py`). At rank three the following are fusable, of the fusion widths two and three: $\binom{[8]}{3}$ and $\binom{[9]}{3}$; the list $\mathrm{F}_7$ on the marks $\mathbb{Z}/7$ whose dependent sets of three marks are the seven sets $\{i, i+1, i+3\}$, with a free mark added; the list $\mathrm{A}_9$ on the marks $\mathbb{F}_3^2$ whose dependent sets of three marks are the twelve sets $\{a, a+d, a+2d\}$, $d \ne 0$, and $\mathrm{A}_9$ less a mark; the list $\mathrm{Z}_9$ on the marks $a_i, b_i, c_i$, $i = 1, 2, 3$, whose dependent sets of three marks are $\{a_1, a_2, a_3\}$, $\{b_1, b_2, b_3\}$, $\{c_1, c_2, c_3\}$ and the six sets $\{a_i, b_j, c_{i+j-2}\}$ with $i \ne j$; and five further lists of rank three whose dependent sets of three marks form configurations of lines; and their fusions have the seam of Proposition 30.10 (receipt `w01_seam_rank3.py`).

## 30.3 The seam of the fusion bunch and the glue

**Proposition 30.10** (The seam of the fusion bunch)**.** *Let $(\mathcal{B}, \theta)$ be fusable of the fusion width $c \ge 1$, with its fusion $(\mathcal{B}^\sharp, \theta^\sharp)$, its lifts $\mathcal{L}_j$ and its lift letters $L_l$.*

*(i) In $\mathcal{B}^\sharp$ the bunch $G$ has $\mathrm{pat}(G) = 2^G$, $d(G) = 0$ and $c(G) = c$, and it is in level form. At the excess $l$ (Definition 28.1) there is one station of patterns, all $Q \subseteq G$ with $|Q| = l$, with the multipliers 1 and the station letters $e_l(\alpha_G)$ and $e_{c-l}(\alpha_G)$, and one station $D_l$ of room sides, the bases of $\mathcal{L}_{c-l}$, whose room children are positive multiples of $\binom{G}{l}$. Every excess is one rectangle, $\mathcal{B}^\sharp$ is its own level envelope (Theorem 28.28), and the room-station factorization (Proposition 28.2(iii)) is*

$$K^\sharp = \sum_{l=0}^{c} e_l(\alpha_G)\, L_l(\alpha_\mathcal{E}), \qquad \Psi^\sharp = \sum_{l=0}^{c} e_{c-l}(\alpha_G)\, L_{c-l}(\alpha_\mathcal{E}).$$

*The seam letters of $G$ are $K_{(D_l)} = \mu_l e_l(\alpha_G)$ and $\Psi_{(D_l)} = \mu_l e_{c-l}(\alpha_G)$, with $\mu_l$ the largest weight of $\mathcal{L}_{c-l}$, the room-station letters are $L_l/\mu_l$ and $L_{c-l}/\mu_l$, and the pair constants are $\mu_l$; for a list every $\mu_l$ is 1.*

*(ii) The interaction of $G$ (Definition 12.18) and its dual interaction are*

$$\iota_G = \iota_G^* = \sum_{j=0}^{c} (-1)^j \binom{c}{j} \log K_{\mathcal{L}_j}(\alpha_\mathcal{E}),$$

*and $\iota_G = -\log z_1$ at $c = 2$.*

*(iii) At the table's gauge the pattern probabilities of the excess $l$ add up to $\Pi_l = \binom{c}{l} L_l(\mathbf{1})/K^\sharp(\mathbf{1})$, with $\Pi_l = \Pi_{c-l}$, and the peel of $((\mathcal{B}^\sharp, \theta^\sharp); s, t)$ at $G$ (Definition 14.1) is the family of the lifts $((\mathcal{L}_j, \theta^{(j)}); \Pi_j s, \Pi_j t)_{0 \le j \le c}$, whose word is $\prod_j K_{\mathcal{L}_j}^{-2\sigma\Pi_j}$.*

Between the extremes the seam of a bunch can carry several rectangles (Proposition 28.3(iii)); the fusion bunch carries one at every excess, with the multipliers 1 on its patterns and the pair constants $\mu_l$, and what it presents to its room at the excess $l$ is the elementary symmetric polynomial of degree $l$ of its own marks. These letters do not depend on the table. Everything the table contributes sits on the other side of the seam, in the letters of its lifts.

**Definition 30.11** (The universal piece; the lift partner)**.** For $c \ge 1$ the *universal piece of width* $c$ is the piece on a set $G$ of $c$ marks for one table (Definition 28.4) with the stations $D_0, \dots, D_c$, $D_l$ of the excess $l$, the seam letters $V_{D_l} = e_l(\alpha_G)$ and $V_{D_l}^* = e_{c-l}(\alpha_G)$, and $d = d^* = 0$. The *lift partner* of a fusable $(\mathcal{B}, \theta)$ of the fusion width $c$ is the partner on $\mathcal{E}$ with the same stations, the room letters $L_{D_l} = L_l$ and $L_{D_l}^* = L_{c-l}$, and $e = e^* = r$.

By Proposition 30.10(i) and Proposition 28.2(iii) the glue of the universal piece of width $c$ and the lift partner of $(\mathcal{B}, \theta)$ is the fusion $(\mathcal{B}^\sharp, \theta^\sharp)$. Since $\mathcal{B}^\sharp$ is identically self-dual its word at the exponents $(s, t)$ is $K^{\sharp\,-2\sigma}$, $\sigma = \frac{s+t}{2}$ the strength (Definition 8.1), and the fused table is read below as the family $((\mathcal{B}^\sharp, \theta^\sharp); 2\sigma, 0)$, with the one active letter $K^\sharp$; its walls are those at $(s, t)$, $d$ and $d^*$ agreeing on an identically self-dual table. Its balance wall is $m_\mathcal{E} + m_G = 2\sigma r^*$. In this section $H_G = \{w \in \mathbb{R}^G : \sum_g w_g = 0\}$ is the torus of $G$ in logarithmic coordinates, with the measure $d\lambda_G(w) = dw_{g_1} \cdots dw_{g_{c-1}}$ for an order $g_1, \dots, g_c$ of $G$, and $\mathbf{e}(y) = (e_0(y), \dots, e_c(y))$ for $y \in (0, \infty)^G$.

**Theorem 30.12** (The glue of a fused table)**.** *Let $(\mathcal{B}, \theta)$ be fusable of the fusion width $c \ge 1$, with its fusion $(\mathcal{B}^\sharp, \theta^\sharp)$ at a strength $\sigma > 0$ and masses $\mathbf{m}$ on $\mathcal{E} \sqcup G$ on its balance wall. For $\mathbf{x} = (x_0, \dots, x_c) \in (0, \infty)^{c+1}$ let $W[\mathbf{x}]$ be the Newton word on $\mathcal{E} \sqcup \{g\}$, $g$ the stand-in of the universal piece, with the one letter $\sum_{l=0}^{c} x_l\, \alpha_g^l\, L_l(\alpha_\mathcal{E})$ at the exponent $2\sigma$ (Definition 25.1).*

*(i)* The universal law. *The scale of the universal piece is* $\gamma_G^{\rho} = e_c(\alpha_G)^{1/c}$, *its seam vector at the excess* $l$ *is* $e_l(\alpha_G)/\gamma_G^{\rho}(\alpha_G)^l$, *and its law is the image of the measure* $c\, e^{\langle \mathbf{m}|_G, w\rangle}\, d\lambda_G(w)$ *on* $H_G$ *under* $w \mapsto \mathbf{e}(e^w)$.

*(ii)* The torus form. *In* $[0,\infty]$,

$$P_{\mathcal{B}^\sharp}(\mathbf{m}; s, t) = c \int_{H_G} e^{\langle \mathbf{m}|_G, w\rangle}\, P_{W[\mathbf{e}(e^w)]}\big(\mathbf{m}|_{\mathcal{E}}, m_G\big)\, d\lambda_G(w),$$

*the stand-in carrying the mass* $m_G$, *the wall of the universal piece.*

*(iii)* The scale of the stand-in. *At a fixed* $\alpha_{\mathcal{E}}$ *write* $\sum_l x_l L_l\, y^l = x_c L_c \prod_{i=1}^{c}(y + \xi_i)$; *the* $\xi_i$, *in* $\mathbb{C} \setminus (-\infty, 0]$, *are the Skeleton values of the stand-in, read on the letter of* $W[\mathbf{x}]$ *as in Definition 12.10. For* $0 < m_G < 2\sigma c$,

$$\int_0^\infty \rho^{m_G - 1} \Big( \sum_l x_l \rho^l L_l \Big)^{-2\sigma} d\rho = (x_c L_c)^{-2\sigma}\, \Gamma_c^2 \begin{bmatrix} m_G, & 2\sigma c - m_G \\ 2\sigma, & \dots, \ 2\sigma \end{bmatrix}\, \mathsf{B}^{\{J\}}_{(2\sigma)_{i\in J},\, 2\sigma c - m_G;\, I_J}(\boldsymbol{\xi}),$$

*the Breta of one bunch* $J = \{1, \dots, c\}$ *of Proposition 15.1, coincident Skeleton values merging.*

*(iv)* One share per level. *On every contour of shares* $\mathbf{a} = (a_0, \dots, a_c)$ *with positive real parts on the pins* $\sum_l a_l = 2\sigma$ *and* $\sum_l l\, a_l = m_G$ *at which the two periods below converge, as at the native split at the stations when every proper wall of* $\mathcal{B}^\sharp$ *is positive (Theorem 28.10(iii)),*

$$P_{\mathcal{B}^\sharp}(\mathbf{m}; s, t) = \int_{\sum_l a_l = 2\sigma,\ \sum_l l\, a_l = m_G} \Gamma_1^{c+1} \begin{bmatrix} \mathbf{a} \\ 2\sigma \end{bmatrix}\, P_{\prod_{l=1}^{c} e_l^{-a_l}}\big(\mathbf{m}|_G\big)\, P_{\prod_{l=0}^{c} L_l^{-a_l}}\big(\mathbf{m}|_{\mathcal{E}}\big),$$

*a fibre integral of the dimension* $c - 1$.

*(v)* The universal factor. *For* $c \geq 2$, $P_{\prod_{l=1}^{c} e_l^{-a_l}}(\mathbf{m}|_G) = P_{\prod_{l=1}^{c-1} e_l^{-a_l}}(\mathbf{m}|_G - a_c \mathbf{1})$, *the period on* $G$ *of the uniform lists* $\binom{G}{l}$, $0 < l < c$, *at the exponents* $(a_l, 0)$ *and the masses* $\mathbf{m}|_G - a_c \mathbf{1}$; *at* $c = 1$ *it is* $1$.

The universal piece is absorbed into its stand-in at its own scale, the geometric mean of the scales of $G$, and what is left of it is a point of the torus of $G$, entering the letter of the stand-in through the elementary symmetric polynomials of the exponentials of that point. The stand-in enters with the degree $c$, so $W[\mathbf{x}]$ is not the word of a table for $c \geq 2$ (Theorem 28.24), and integrating its scale is one Breta over the $c$ Skeleton values of the stand-in. Opened on its shares, the fused table carries one share per level and two pins, the strength and the scale, so $c - 1$ shares are free; the universal factor is the period of the family of its lifts, in which the table itself is the member with $a_1 = \dots = a_{c-1} = 0$, $P_{L_0^{-a_0} L_c^{-a_c}}(\mathbf{m}|_{\mathcal{E}}) = P_{\mathcal{B}}(\mathbf{m}|_{\mathcal{E}}; a_c, a_0)$.

**Corollary 30.13** (Width one)**.** *At the fusion width one, on the balance wall of* $\mathcal{B}^\sharp$ *with every proper wall positive,*

$$P_{\mathcal{B}^\sharp}(\mathbf{m}|_{\mathcal{E}}, m_g; s, t) = \Gamma_1^2 \begin{bmatrix} m_g, & 2\sigma - m_g \\ & 2\sigma \end{bmatrix}\, P_{\mathcal{B}}\big(\mathbf{m}|_{\mathcal{E}}; m_g, 2\sigma - m_g\big)\ :$$

*the exponents of the table are read off the mass of the fusion bunch. The universal piece is the mark* $g$ *itself, the laws of the universal piece and of the lift partner are points, and the share form of Theorem 30.12(iv) has no free share.*

**Corollary 30.14** (Width two). *Let $c=2$, $G=\{g_1,g_2\}$, with the masses $m_1,m_2$ on $g_1,g_2$.*

*(i) On a line of one free share $a_2$, with $a_1=m_1+m_2-2a_2$ and $a_0=2\sigma-m_1-m_2+a_2$,*

$$P_{\mathcal{B}^\sharp}(\mathbf{m};s,t)=\frac{1}{2\pi i}\int\frac{\Gamma(a_0)\,\Gamma(a_2)\,\Gamma(m_1-a_2)\,\Gamma(m_2-a_2)}{\Gamma(2\sigma)}\,P_{L_0^{-a_0}L_1^{-a_1}L_2^{-a_2}}\big(\mathbf{m}|\mathcal{E}\big)\,da_2:$$

*the universal factor is the Beta corner $\Gamma_{\,1}^{\,2}\Big[\begin{matrix}m_1-a_2,\ m_2-a_2\\ a_1\end{matrix}\Big]$, and the Gamma of the middle level cancels.*

*(ii) The universal law is the image of $2v^{m_1-m_2-1}dv$ on $(0,\infty)$ under $v\mapsto(1,v+v^{-1},1)$, and*

$$P_{\mathcal{B}^\sharp}(\mathbf{m};s,t)=2\int_0^\infty v^{m_1-m_2-1}\,P_{(L_0+(v+v^{-1})\alpha_gL_1+\alpha_g^2L_2)^{-2\sigma}}\big(\mathbf{m}|\mathcal{E},m_1+m_2\big)\,dv.$$

*(iii) At a fixed $\alpha_{\mathcal{E}}$, for $0<m_1<2\sigma$ and $0<m_2<2\sigma$,*

$$\int_{(0,\infty)^2}\alpha_{g_1}^{m_1-1}\alpha_{g_2}^{m_2-1}K^{\sharp\,-2\sigma}\,d\alpha_G$$
$$=\Gamma_{\,1}^{\,2}\Big[\begin{matrix}m_2,\ 2\sigma-m_2\\ 2\sigma\end{matrix}\Big]\,L_0^{m_1+m_2-2\sigma}L_1^{-m_1-m_2}\int_0^\infty x^{m_1-1}(1+x)^{m_2-2\sigma}\big(1+z_1^{-1}x\big)^{-m_2}dx,$$

*a Beta corner times a Breta of one bunch with the two coefficients 1 and $z_1$ (Proposition 15.1). At $m_1=m_2=1$, $2\sigma=2$ it is $\log z_1/(L_1^2-L_0L_2)$ where $z_1\neq1$ and $L_1^{-2}$ where $z_1=1$, and at integer $m_1,m_2,2\sigma$, where $z_1\neq1$, it is $L_0^{m_1+m_2-2\sigma}L_1^{-m_1-m_2}\big(R_1(z_1)+R_2(z_1)\log z_1\big)$ with rational functions $R_1,R_2$.*

## 30.4 The price of width

**Lemma 30.15** (Products of lifts). *Let $\mathcal{N}$ be a quotient of $\mathcal{M}$ with the rank difference $c$, its lifts $\mathcal{L}_j=\mathcal{L}_j(\mathcal{M},\mathcal{N})$ carrying any positive weights. For $0\le i'<i\le j<j'\le c$ with $i+j=i'+j'$, $K_{\mathcal{L}_i}K_{\mathcal{L}_j}$ has a monomial in which exactly $r_{\mathcal{N}}+i$ exponents equal 2, while in every monomial of $K_{\mathcal{L}_{i'}}K_{\mathcal{L}_{j'}}$ at most $r_{\mathcal{N}}+i'$ do; so the two products are not proportional. In particular, for $c\ge2$, no ratio $K_{\mathcal{L}_j}^2/(K_{\mathcal{L}_{j-1}}K_{\mathcal{L}_{j+1}})$, $0<j<c$, is constant on the orthant.*

**Theorem 30.16** (The price of width). *Let $(\mathcal{B},\theta)$ be fusable of the fusion width $c\ge1$, with its fusion read at a strength $\sigma>0$ as in Theorem 30.12, its lift letters $L_l$ and its lift ratios $z_j$.*

*(i) The law of the universal piece is carried by one point exactly when $c=1$. The law of the lift partner is carried by one point exactly when every lift ratio is constant, that is exactly when $c=1$.*

*(ii)* The Skeleton braid. *Write $\sum_l\binom{c}{l}L_l\,x^l=L_0\prod_{i=1}^c(1+\eta_ix)$; the $\eta_i$ are the reciprocals of the Skeleton values of $G$ at the gauge $(\mathbf{1}_G,\alpha_{\mathcal{E}})$ (Definition 12.10). Then*

$$K^\sharp=\frac{L_0}{c!}\sum_{\pi\in S_c}\prod_{i=1}^c\big(1+\alpha_{g_i}\eta_{\pi(i)}\big)=\frac{L_0}{c!}\,\mathrm{per}\,\big(\mathbf{1}\mathbf{1}^{\mathsf{T}}+\alpha_G\boldsymbol{\eta}^{\mathsf{T}}\big),$$

*$\mathbf{1}\mathbf{1}^{\mathsf{T}}$ the $c\times c$ matrix of ones and* per *the permanent: $K^\sharp$ is $L_0$ times the average, over the bijections from the marks of $G$ to the Skeleton values, of products of $c$ letters of one mark. At a gauge $\alpha_{\mathcal{E}}$ the $\eta_i$ coincide exactly when every lift ratio is 1 there, and for $c\ge2$ they coincide on no open set of the orthant: $K^\sharp$ is not $L_0\prod_{g\in G}(1+\alpha_g\eta)$ for any one function $\eta$.*

*(iii)* The partial fusions. *For $1 \le i \le c$, $G' \subseteq G$ with $|G'| = i$ and $g \in G'$, read the partial fusion at $G'$ as a cell with the hinge $g$ (Definition 21.1). For $i = c$ its ratio is* 1*, and on the balance wall of $\mathcal{B}^\sharp$ with every proper wall positive*

$$P_{\mathcal{B}^\sharp}(\mathbf{m}; s, t) = \Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix} m_g, & 2\sigma - m_g \\ & 2\sigma \end{bmatrix} P_{\mathcal{B}^\sharp/g}\big(\mathbf{m}|_{\mathcal{E}\sqcup G-g};\, m_g,\, 2\sigma - m_g\big);$$

*for $i < c$ its ratio is not constant, and at any exponents $s, t > 0$ and masses its law is not carried by one point (Corollary 23.12(i)).*

At the fusion width one both laws are points and the glue is a product: one Beta corner. At the width $c \ge 2$ neither is: the universal law spreads over the $(c-1)$-dimensional torus of $G$, and the lift partner's seam vector moves with the room because the lifts are not a geometric chain, the lift ratios being not constant. Read on the Skeleton values of the fusion bunch, the letter of the fused table is an average over the bijections from the marks of $G$ to the Skeleton values, and it is one product of $c$ letters of one mark on no open set, and at no gauge for a list (Remark 30.19(i)); one product is what $c$ independent fusions of width one would give. Read down the partial fusions, only the step at the top is free.

Among the plain extensions of a table by one mark with connected hinge minors, the fusion is the only one that is free in this way, up to one constant on the weights of the bases avoiding the new mark.

**Proposition 30.17** (The free extensions by one mark)**.** *Let $\mathcal{A}$ be a plain cell on $\mathcal{E} \sqcup \{g\}$, $|\mathcal{E}| \ge 2$, whose hinge minors $\mathcal{A}\setminus g$ and $\mathcal{A}/g$ are connected. Its ratio $\varrho_{\mathcal{A}}$ (Definition 21.1) is constant exactly when $\mathcal{A}/g$ is the dual of $\mathcal{A}\setminus g$ with its weights multiplied by one constant $\mu > 0$; then $\varrho_{\mathcal{A}} \equiv \mu^2$, $\mathcal{A}/g$ has the fusion width one, and $\mu K_{\mathcal{A}}(\alpha_{\mathcal{E}}, \alpha_g) = K^\sharp(\alpha_{\mathcal{E}}, \mu\alpha_g)$, $K^\sharp$ the letter of the Foreman fusion of $\mathcal{A}/g$: $\mathcal{A}$ is that fusion with the weights of its bases avoiding $g$ divided by $\mu$. For a list, $\varrho_{\mathcal{A}}$ is constant exactly when $\mathcal{A}$ is identically self-dual, that is when $\mathcal{A}$ is the Foreman fusion of $\mathcal{A}/g$. So, at any exponents $s, t > 0$ and masses, the law of $\mathcal{A}$ is a point (Corollary 23.12(i)) exactly when $\mathcal{A}$ is the Foreman fusion of $\mathcal{A}/g$ up to one constant on the weights of the bases avoiding $g$, and for a list exactly when it is that fusion.*

**Corollary 30.18** (A pinned set of one list is a dual pair)**.** *Let $\mathcal{B}$ be a list on $\mathcal{E}$, let $s, t > 0$, and let $S \subsetneq \mathcal{E}$, $|S| \ge 2$, with $S' = \mathcal{E} \setminus S$, be such that $\mathcal{B}|S$ and $\mathcal{B}/S'$ are connected. Then $S$ is pinned (Theorem 28.25) exactly when it has width one and $\mathcal{B}/S' = (\mathcal{B}|S)^*$: the cell of the bunch $\mathcal{B}_S^+$ (Theorem 19.4) is a dual pair.*

*Remark* 30.19 (The lift ratios at a gauge; width three). (i) Let $\mathcal{B}$ be a list. The room sides of $G$ in $\mathcal{B}^\sharp$ are the sets independent in $\mathcal{B}^*$ that span $\mathcal{B}$: the bases of the identity map from $\mathcal{B}^*$ to its quotient $\mathcal{B}$ read as a morphism of matroids, the arena property that Theorem 30.6(a) supplies. For such a morphism Eur and Huh (*Logarithmic concavity for morphisms of matroids*, Corollary 4.6) show that the basis generating polynomial, each basis counted once and homogenized by one more variable to the power $n$ less the size of the basis, is Lorentzian, and a linear change of the variables with nonnegative coefficients keeps that property (Brändén and Huh), so at every gauge $\alpha_{\mathcal{E}}$ the values $K_{\mathcal{L}_j}(\alpha_{\mathcal{E}})/\binom{n}{r+j}$ form a log-concave sequence:

$$z_j(\alpha_{\mathcal{E}}) \;\ge\; \frac{(k+1)(n-k+1)}{k(n-k)} > 1, \qquad k = r + j,\ 0 < j < c.$$

So for a list at $c \ge 2$ the lift ratios exceed one at every gauge and the $\eta_i$ of Theorem 30.16(ii) never coincide. The bound holds on the 6720 exact ratios of the receipt `t12_ratios_and_tell.py`,

with equality in 35 of them, and it is attained at the table's gauge whenever $\mathcal{L}_{j-1}$, $\mathcal{L}_j$ and $\mathcal{L}_{j+1}$ are uniform lists, as for every uniform list: there $K_{\mathcal{L}_i}(\mathbf{1}) = \binom{n}{r+i}$. The receipt `t08_logconcave.py` finds every lift ratio above one on 6720 exact ratios of its own. With weights the lift ratios can fall below one: $\binom{[4]}{1}$ with the weights 2 and the middle weights 1 has $K_{\mathcal{L}_0}(\mathbf{1}) = K_{\mathcal{L}_2}(\mathbf{1}) = 8$ and $K_{\mathcal{L}_1}(\mathbf{1}) = 6$, so $z_1 = \frac{9}{16}$ at the table's gauge.

(ii) At $c = 2$ the discriminant of $L_0 + 2L_1x + L_2x^2$ is $4L_0L_2(z_1 - 1)$, so the Skeleton values of $G$ at $(\mathbf{1}_G, \alpha_{\mathcal{E}})$ are real and distinct where $z_1 > 1$, at every gauge for a list, and complex conjugate where $z_1 < 1$.

(iii) At $c = 3$, with the scale $a$ of one mark of $G$ held fixed, $K^\sharp$ is the letter of width two with the lift letters $L'_l = L_l + aL_{l+1}$, $0 \le l \le 2$; so at the masses 1 on $G$ and $2\sigma = 2$ the integral over the scales of $G$ at a fixed room point is, by Corollary 30.14(iii),

$$\int_0^\infty \frac{\log\big(L_1'^2/(L'_0L'_2)\big)}{L_1'^2 - L'_0L'_2}\,da, \qquad L_1'^2 - L'_0L'_2 = -\big((L_1L_3 - L_2^2)a^2 + (L_0L_3 - L_1L_2)a + (L_0L_2 - L_1^2)\big),$$

the coefficients of the quadratic being the $2\times 2$ minors of the matrix with the rows $(L_0, L_1, L_2)$ and $(L_1, L_2, L_3)$, and its discriminant $-\frac{1}{27}$ times that of the diagonal level cubic $L_0+3L_1x+3L_2x^2+L_3x^3$ (receipts `t10_weight.py`, a rendering, and `t13_faces_and_cubic.py`). The cubic is real-rooted at all 560 gauges sampled on 8 lists of the fusion width three (receipts `w02_skeleton_braid.py`, `w03_real_c3.py`).

## 30.5 Cousins and hidden symmetry

**Proposition 30.20** (Cousins)**.** *The* cousins *of an identically self-dual table are its one-mark contractions. Let $(\mathcal{B}, \theta)$ be fusable of the fusion width $c \ge 1$, and $x$ a mark of $\mathcal{E} \sqcup G$.*

*(i) The cousin $\mathcal{B}^\sharp/x$ has the fusion width one and is fusable, $\mathcal{B}^\sharp \setminus x$ is its dual, and its Foreman fusion is $(\mathcal{B}^\sharp, \theta^\sharp)$ with the fusion bunch $\{x\}$. For $x \in G$ the cousin is the partial fusion $\mathcal{B}^\sharp_{c-1}$, and at $c = 1$ the table $\mathcal{B}$ is the cousin at $g$.*

*(ii) On the balance wall of $\mathcal{B}^\sharp$ with every proper wall positive, for every two marks $x, y$,*

$$\begin{aligned} P_{\mathcal{B}^\sharp}(\mathbf{m}; s, t) &= \Gamma\begin{bmatrix} m_x, & 2\sigma - m_x \\ 1 & 2\sigma \end{bmatrix} P_{\mathcal{B}^\sharp/x}\big(\mathbf{m}_{-x}; m_x, 2\sigma - m_x\big) \\ &= \Gamma\begin{bmatrix} m_y, & 2\sigma - m_y \\ 1 & 2\sigma \end{bmatrix} P_{\mathcal{B}^\sharp/y}\big(\mathbf{m}_{-y}; m_y, 2\sigma - m_y\big), \end{aligned}$$

*$\mathbf{m}_{-x}$ the masses of the marks other than $x$: the cousins share one period up to Beta corners, each reading its exponents off the mass of the mark it lacks.*

*(iii) For every bijection $\beta$ of $\mathcal{E} \sqcup G$ carrying $(\mathcal{B}^\sharp, \theta^\sharp)$, $P_{\mathcal{B}^\sharp}(\beta^*\mathbf{m}; s, t) = P_{\mathcal{B}^\sharp}(\mathbf{m}; s, t)$, with $(\beta^*\mathbf{m})_e = m_{\beta(e)}$. At $c = 1$ the function*

$$(\mathbf{m}|_{\mathcal{E}}, s') \;\mapsto\; \Gamma\begin{bmatrix} s', & 2\sigma - s' \\ 1 & 2\sigma \end{bmatrix} P_{\mathcal{B}}\big(\mathbf{m}|_{\mathcal{E}}; s', 2\sigma - s'\big)$$

*on the part of the hyperplane $m_{\mathcal{E}} + s' = 2\sigma r^*$ where every proper wall of $\mathcal{B}^\sharp$ is positive, which the automorphisms keep, is invariant under the automorphisms of $\mathcal{B}^\sharp$ acting on $\mathcal{E} \sqcup \{g\}$, $s'$ read as the mass of $g$: an automorphism $\beta$ with $\beta(g) = e \ne g$ makes the mass $m_e$ the basis exponent and gives $s'$ to the mark $\beta^{-1}(g)$ as its mass, and the transposition of $g$ and $e$, when it is an automorphism, exchanges $s'$ and $m_e$.*

What fusion buys at every width is one step below the fused table. The fused table is identically self-dual and its automorphisms permute its marks; each cousin is the fused table with one mark integrated by a Beta corner, so the cousins are tables of fusion width one, often not isomorphic to one another, whose periods are one function. At the fusion width one, the table itself is a cousin and the hidden symmetry acts on its exponent and its masses together; at the width $c \geq 2$ the partial fusion $\mathcal{B}^{\sharp}_{c-1}$ is the cousin, and the table lies $c-1$ cells below it.

*Remark* 30.21 (Hidden symmetry). The index of the automorphisms of $\mathcal{B}$ times the permutations of $G$ in those of $\mathcal{B}^{\sharp}$ measures the symmetry the fusion adds. For $\binom{\mathcal{E}}{r}$ it is $\binom{n+c}{c}$, all bijections of $\mathcal{E} \sqcup G$ carrying $\binom{\mathcal{E} \sqcup G}{r^*}$; for the list of rank two with one parallel pair of Remark 30.9(iii) it is $(n-2)(n-3)$; for the 10 fusable prime lists of rank three on seven marks the fusion is transitive and its group is 8 times that of the list (receipts `census_extra.py`, `atoms7.py`). At the fusion width two the index is 45 for $\binom{[8]}{3}$, 20 for one line of three marks and five free marks, 15 for one line of four marks and four free marks, 6 for two lines of three marks meeting in a mark and three free marks, 2 for two disjoint lines of three marks and two free marks, and 1 for $\mathrm{F}_7$ with a free mark, for $\mathrm{A}_9$ less a mark and for two disjoint lines of four marks; at the width three it is 220 for $\binom{[9]}{3}$ and 1 for $\mathrm{A}_9$ and for $\mathrm{Z}_9$ (receipt `w01_seam_rank3.py`). At $\binom{[2r+1]}{r}$, $r \geq 2$, the symmetric group of $\binom{[2r+2]}{r+1}$ accounts for a nonzero submodule that the closings miss at its centre, with every symmetry of the four kinds of its centre ray in the group (Proposition 39.64); at $\binom{[5]}{2}$ its part in the span of a box is a line (Exercise 54.20). Among 40 identically self-dual lists of rank five on ten marks drawn at random, in each of which the sets of five marks that are not bases form a family closed under complement whose members meet pairwise in at most three marks, 23 have ten cousins no two of which are isomorphic (receipt `random10.py`), and all the cousins of these 23 are prime (receipt `random10b.py`).

## 30.6 Two-sums and repair

**Definition 30.22** (Covered marks; the conditions at a hinge). Let $\mathcal{B}$ be a table on $\mathcal{E}$. A mark $e$ of a cocircuit $D$ is *covered in* $D$ when some circuit $C$ has $e \in C \subseteq D$, and $e$ is *uncovered* when some cocircuit contains it without covering it. For a cell $\mathcal{A}$ on $B \sqcup \{g\}$ with the hinge $g$ (Definition 21.1) the *conditions at the hinge* are:

(a) every cocircuit avoiding $g$ is a union of circuits;

(u) every cocircuit $D$ through $g$ covers every mark of $D \setminus \{g\}$;

(s) every cocircuit $D$ through $g$ covers every mark of $D \setminus \{g\}$ by circuits avoiding $g$;

(h) every cocircuit through $g$ covers $g$.

For two cells $\mathcal{A}_1$, $\mathcal{A}_2$ the conditions are written $a_1, u_1, s_1, h_1$ and $a_2, u_2, s_2, h_2$.

**Theorem 30.23** (Fusion across a two-sum). *Let $\mathcal{A}_1$ and $\mathcal{A}_2$ be cells that are lists, on $\mathcal{E}_1 \sqcup \{g\}$ and $\mathcal{E}_2 \sqcup \{g\}$ with $n_1$ and $n_2$ marks, of the ranks $r_1, r_2$ and the fusion widths $c_1, c_2$, and $\mathcal{S} = \mathcal{A}_1 \oplus_g \mathcal{A}_2$ their two-sum on $\mathcal{E}_1 \sqcup \mathcal{E}_2$ (Lemma 21.12).*

*(i) $\mathcal{S}$ has the fusion width $c_1 + c_2$. Its circuits are the circuits of $\mathcal{A}_1$ and of $\mathcal{A}_2$ that avoid $g$ and the sets $(C_1 \setminus g) \cup (C_2 \setminus g)$ with $C_i$ a circuit of $\mathcal{A}_i$ through $g$; its cocircuits are the same with cocircuits in place of circuits; and $\mathcal{S}^* = \mathcal{A}_1^* \oplus_g \mathcal{A}_2^*$.*

*(ii) $\mathcal{S}$ is fusable exactly when*

$$a_1 \text{ and } u_1 \text{ and } a_2 \text{ and } u_2 \text{ and } (s_1 \text{ or } h_2) \text{ and } (s_2 \text{ or } h_1),$$

*and a cell, read as a table, is fusable exactly when a, u and h hold. So two fusable cells give a fusable two-sum.*

*(iii) If $\mathcal{S}$ is fusable, then $c_1, c_2 \geq 0$, and in $\mathcal{S}^\sharp$ the sets $\mathcal{E}_1$ and $\mathcal{E}_2$ are bunches of the widths $c_1+1$ and $c_2+1$.*

*(iv) If both cells are fusable, the two-sum $\mathcal{A}_1^\sharp \oplus_g \mathcal{A}_2^\sharp$, with the fusion bunch $G_1 \sqcup G_2$, is an identically self-dual major of $\mathcal{S}$. It is $\mathcal{S}^\sharp$ when $c_1 = 0$ or $c_2 = 0$. When $c_1, c_2 \geq 1$ its children at one mark of $G_1$ and at one mark of $G_2$ have the ranks $r_1^* - 1$ and $r_1^*$ on $\mathcal{E}_1$, $r_1^* = n_1 - r_1$ the corank of $\mathcal{A}_1$, so $G_1 \sqcup G_2$ is not in level form in it and it is a major other than $\mathcal{S}^\sharp$.*

*(v) If $c_1 + c_2 = 1$, $\mathcal{S}$ is fusable exactly when both cells are fusable; then, for $c_2 = 0$, $\mathcal{A}_2$ is identically self-dual, $\mathcal{S}^\sharp = \mathcal{A}_1^\sharp \oplus_g \mathcal{A}_2$, and the hinge of $\mathcal{S}$ is a hinge of $\mathcal{S}^\sharp$.*

The condition is local at the hinge: away from it each cell answers for itself, and through it a cocircuit of the two-sum joins a cocircuit of each cell, so a mark is covered either by a circuit of its own cell avoiding the hinge or by a circuit through the hinge continued by a circuit through the hinge on the other side; that continuation exists exactly when the other cell covers its hinge. In the fused table the hinge of a cell of positive width thickens into a bunch of that width plus one. Fusing cell by cell gives a major, identically self-dual, that is not the fusion as soon as two cells have positive width: the canonical fusion of a two-sum is the global level cell. At the total width one the hinge survives the fusion, with the identically self-dual cell on one side and the fusion of the other cell on the other.

**Corollary 30.24** (A two-sum with an identically self-dual cell)**.** *Let $\mathcal{A}_1$ be identically self-dual and $s, t > 0$, with the masses $\mathbf{m}$ of $\mathcal{S} = \mathcal{A}_1 \oplus_g \mathcal{A}_2$ on its balance wall with every proper wall positive, and let $\kappa = m_{\mathcal{E}_1} - s(r_1 - 1) - t(|\mathcal{E}_1| - r_1)$ be the hinge wall of $\mathcal{A}_1$ (Definition 23.1). Then*

$$P_{\mathcal{S}}(\mathbf{m}; s, t) = P_{\mathcal{A}_1}\big(\mathbf{m}|_{\mathcal{E}_1}, 2\sigma - \kappa; s, t\big)\, P_{\mathcal{A}_2}\big(\mathbf{m}|_{\mathcal{E}_2}, \kappa; s, t\big) \Big/ \Gamma_1^2 \begin{bmatrix} 2\sigma - \kappa, & \kappa \\ & 2\sigma \end{bmatrix},$$

*the hinge carrying $2\sigma - \kappa$ in $\mathcal{A}_1$ and $\kappa$ in $\mathcal{A}_2$. When the cells are lists and $\mathcal{A}_2$ is fusable of the fusion width one, this is the period of $\mathcal{S}$ read across the hinge that $\mathcal{S}^\sharp = \mathcal{A}_1 \oplus_g \mathcal{A}_2^\sharp$ keeps.*

By Corollary 20.19 at the hinge, the first factor divided by the Beta corner is the period $P_{\mathcal{A}_1 \backslash g}(\mathbf{m}|_{\mathcal{E}_1}; \kappa, 2\sigma - \kappa)$ of the deletion at the hinge. So identically self-dual cells attached at their hinges to one cell that is not identically self-dual cost the period of that cell and, for each of them, the period of its deletion at the hinge; two cells that are not identically self-dual on the two sides of a hinge pair two laws, and the hinge is a convolution (Theorem 23.9).

**Proposition 30.25** (Uncovered marks and repair)**.** *Let $(\mathcal{B}, \theta)$ be a table on $\mathcal{E}$ of the fusion width $c$.*

*(i) $\mathcal{B}$ has no uncovered mark (Definition 30.22) exactly when every cocircuit of $\mathcal{B}$ is a union of circuits. So a list, or a table of the fusion width $c \neq 0$, is fusable exactly when it has no uncovered mark, and a table of the fusion width zero exactly when it is identically self-dual.*

*(ii) For a set $S$ of marks, none a loop, let $(\mathcal{B}_S, \theta)$ be the table in which each $h \in S$ is replaced by a block $\{h, h'\}$ of two marks by the first rule of Theorem 21.5, a parallel pair, every new basis keeping the weight of its basis, and $S' = \{h' : h \in S\}$. Then $\mathcal{B}_S$ has the fusion width $c + |S|$, $(\mathcal{B}_S, \theta) \setminus S' = (\mathcal{B}, \theta)$, and $\mathcal{B}_S$ has no uncovered mark exactly when $S$ contains every uncovered mark of $\mathcal{B}$; when $\mathcal{B}$ is a list or $c + |S| \neq 0$, $\mathcal{B}_S$ is fusable exactly then.*

*(iii) If $\mathcal{B}_S$ is fusable, $(\mathcal{B},\theta) = (\mathcal{B}_S)^\sharp/G \setminus S'$ is a minor of an identically self-dual table. If $c \geq 0$ and $(\mathcal{B},\theta)$ is not fusable, $\mathcal{B}_S$ is fusable for $S$ the set of the uncovered marks when there is one, and for $S$ any one mark when there is none, no mark being then a loop. Doubling every mark of a list that is not a loop gives a fusable list.*

*(iv) At the full basis lean, where every other proper wall is positive, the residue of $P_{\mathcal{B}_S}(\mathbf{m};s,0)$ at the wall $m_{h'}=0$ of one copy $h'$ is $P_{\mathcal{B}_{S-h}}(\mathbf{m}_{-h'};s,0)$ (Theorem 16.25); so $P_{\mathcal{B}}(\cdot;s,0)$ is the residue of $P_{\mathcal{B}_S}(\cdot;s,0)$ at the walls of the copies, taken one at a time.*

A mark is uncovered when a cocircuit through it has no circuit through it inside: the cocircuit is a set the bases must meet, and the mark is not tied by any circuit to the other marks there. A parallel copy is a circuit of two marks through it inside every cocircuit through it, so doubling an uncovered mark covers it, and each doubled mark adds one to the fusion width. The list $\binom{[3]}{2}$, of the fusion width $-1$, has every mark uncovered: one or two doubled marks do not fuse, all three do, at the width two (receipts `t04_twosums.py`, `w04_repair.py`); its dual $\binom{[3]}{1}$, which the procedure below reads first, fuses as it is, into $\binom{[4]}{2}$.

## 30.7 The procedure

The statements of this chapter set a branch at every node of the Foreman tree (Definition 14.4), read before anything is integrated.

(1) *Peel.* If the realization of the node is not an atom, peel it at its Foreman bunch and continue on the room (Definition 14.4).

(2) *Go down.* If the table of the node is identically self-dual, every mark comes off by one Beta corner at every pair of exponents (Corollary 20.19); its cousins share its period (Proposition 30.20).

(3) *Go up.* Otherwise read the fusion width $c = r^* - r$, passing to the dual table when it is negative, and the tell on the Skeleton (Theorem 30.6). If the table is fusable, fuse it (Theorem 30.5); at $c = 0$ it is not, a fusable table of the width zero being identically self-dual. At $c = 1$ the fusion is free, one Beta corner (Corollary 30.13): the automorphisms of the identically self-dual table $\mathcal{B}^\sharp$ act on the exponent and the masses of the table, its cousins share its period, and its module carries over (Chapter 39). At $c \geq 2$ the table is the child of the full pattern of the fusion bunch of $\mathcal{B}^\sharp$; the fused table is the universal piece of width $c$ glued to the lift partner (Theorem 30.12), both laws spread (Theorem 30.16), the step from $\mathcal{B}^\sharp$ to $\mathcal{B}^\sharp_{c-1}$ is a Beta corner and the $c-1$ steps below it are cells whose laws are not points.

(4) *Two-sums.* On a two-sum of lists the tell is read at the hinge (Theorem 30.23(ii)); at the total width one the hinge survives the fusion and the period is a product over one Beta corner (Corollary 30.24); when both cells are fusable and one has the width zero the fusion is the two-sum of the fusions of the cells, and when both cells have positive width it is the global level cell, not that two-sum (Theorem 30.23(iv)).

(5) *Repair.* If the table is not fusable and has uncovered marks, they are the obstruction: doubling exactly those in parallel gives a fusable table with the table as its deletion at the copies (Proposition 30.25), at the fusion width $c$ plus the number of uncovered marks; a table of the width zero with no uncovered mark that is not identically self-dual, which needs weights, becomes fusable when any one mark is doubled.

The fusion width is the price, and it is read off the rank and the number of marks before any integral is taken: at the width one the fusion bunch is a single mark with a point for its law; at the width $c \geq 2$ it is the universal piece, whose law spreads over a torus of the dimension $c - 1$ against a lift partner whose law is not a point.

# Chapter 31

# Closing a glue along a direction

A glue opened on its shares is a fibre integral of Gammas and two seam functions (Theorem 28.10(iii)). It is closed by moving the contour along a direction of the shares, collecting a fibre integral of residues at every polar hyperplane crossed and showing that what is left tends to zero. The glue closes on the lattices of its Gammas along every direction whose rate, read off the two laws through the largest values of the monomials of their letter vectors, is negative (Theorem 31.2), and past the walls of the piece, or of the piece and the partner both, at the growths of the continued seam functions (Theorem 31.4). Moving the contour is exact at every step, and the limit needs only a bound that travels with it (Lemma 31.3): the series depends on the direction only through its chamber (Theorem 31.6), the laws can be closed one source point at a time and averaged, at the rate read on the support of the source measure rather than on its convex hull (Theorem 31.7), and past the walls of a piece the bound is read on the outline of a keyhole continuation, the modulus and the phase of the sector letters taken at one point (Theorem 31.11). At simple nonresonant crossings of a rational ray, a crossing term kept with its falling shares integrated on their strips is a finite sum of joint Trailmoments of fixed functions, a falling Gamma becoming a positive share and one Beta variable and a continued source factor a segment and circle integrals (Theorem 31.18); for a positive kernel its root rate is the exponential of the least Gamma rate of its chamber in every fibre dimension and, averaged, of the essential supremum over the source of that least rate, whose sign, when it is not zero, decides whether the terms tend to zero (Theorem 31.26, Corollary 31.27); and an averaged closing can converge where no direction of its chamber certifies it, at a family of tables whose seam has single stations (Theorem 31.8). Coincident crossings are packets of factors polar on one crossed hyperplane, assembled before a sign or a top is read (Theorem 31.30), and at the root rate one a positive moment sequence is decided by its law near its top (Theorem 31.32).

## 31.1 Closing along a direction

**Definition 31.1** (Letter vectors)**.** Let a piece on $B$ and a partner on $R$ with the stations $\mathcal{D}$ be glued with masses on the balance wall of the glue (Definition 28.4), the glue read on its shares as in Theorem 28.10(iii) or Theorem 28.12. The *letter vectors* of the piece and of the partner are $\mathbf{V}_B = (V_{jD}, V_{jD}^*)_{j,D}$ and $\mathbf{V}_R = (L_{jD}, L_{jD}^*)_{j,D}$, and for a vector $\mathbf{v}$ of the coordinates of the shares $\mathbf{V}_B^{\mathbf{v}} = \prod_{j,D} V_{jD}^{v_{jD}} {V_{jD}^*}^{v_{jD}^*}$, and likewise $\mathbf{V}_R^{\mathbf{v}}$.

**Theorem 31.2** (Closing the glue along a direction)**.** *In Theorem 28.12 let $V$ be the space of the free directions of the shares, so that for $\mathbf{v} \in V$ the monomials $\mathbf{V}_B^{\mathbf{v}}$ and $\mathbf{V}_R^{\mathbf{v}}$ (Definition 31.1) are*

*homogeneous of degree zero, and their suprema over the orthants do not change under* $\vec{\mathcal{N}}_B^{\perp}$ *and* $\vec{\mathcal{N}}_R^{\perp}$. *When the seam word of the piece has a Newton sum of full dimension,* $\sup \mathbf{V}_B^{\mathbf{v}} = \sup_{u\in\Delta_B} \boldsymbol{\varrho}_B^{\rho}(u)^{\mathbf{v}}$, *the largest value of the monomial* $\mathbf{x}^{\mathbf{v}}$ *on the support of the law of the piece, and when the room word of the partner has a Newton sum of full dimension,* $\sup \mathbf{V}_R^{\mathbf{v}} = \sup_{v\in\Delta_R} \boldsymbol{\varrho}_R^{\rho}(v)^{\mathbf{v}}$, *the largest value on the support of the law of the partner; where the room letters of the partner are monomials,* $\sup \mathbf{V}_R^{\mathbf{v}} = \boldsymbol{\xi}^{\mathbf{v}}$, $\boldsymbol{\xi}$ *their coefficients. For* $\boldsymbol{\sigma} \in V$ *put*

$$h(\boldsymbol{\sigma}) = \sum_{j,D} \big(\sigma_{jD} \log|\sigma_{jD}| + \sigma^*_{jD} \log|\sigma^*_{jD}|\big) + \log\sup \mathbf{V}_B^{-\boldsymbol{\sigma}} + \log\sup \mathbf{V}_R^{-\boldsymbol{\sigma}},$$

*with* $0\log 0 = 0$, *the first sum over the shares of the active letters, the* rate *of the direction; it is homogeneous of degree one. Let* $h(\boldsymbol{\sigma}) < 0$, *and let the real point* $\mathbf{c}$ *of the pins lie off the null set of the points whose ray* $\mathbf{c} + \tau\boldsymbol{\sigma}$, $\tau \geq 0$, *meets two polar hyperplanes of the Gammas that are not parallel on the fibre at one point, or, for arbitrarily large* $\tau$, *at points less than* $\tau^{-3}$ *apart. Then the ray meets only the hyperplanes* $a_{jD} = -n$ *with* $\sigma_{jD} < 0$ *and* $b_{jD} = -n$ *with* $\sigma^*_{jD} < 0$, $n \in \mathbb{N}$, *and*

$$P_{B\sqcup R}(\mathbf{m}) = \sum_{\text{crossings}} \mathcal{R},$$

*the sum, in the order of crossing, of the fibre integrals* $\mathcal{R}$ *of the residues of the integrand at the hyperplanes crossed, each pinned also by its hyperplane, of the dimension* $\dim V - 1$ *and of the same kind, at the point where the ray meets it; a hyperplane of the fibre on which several Gammas have poles is crossed once, for every* $\mathbf{c}$, *and its residue is a Laurent coefficient (Theorem 4.5). The sum converges absolutely: the residue at a crossing at* $\tau$, *and the integral left after it, are* $O\big(e^{\tau(h(\boldsymbol{\sigma})+\epsilon)}\big)$ *for every* $\epsilon > 0$.

So the glue closes on the lattices of its Gammas along every direction of the cone $\{h < 0\}$ of $V$, the terms falling at least by the factor $e^{h(\boldsymbol{\sigma})}$ per unit of $\boldsymbol{\sigma}$. The cone and the rate are read off the laws through the largest values of the monomials of their letter vectors, and the term $\sum \sigma \log|\sigma|$ of the rate is the cost of moving shares between stations: it vanishes at width one with the stations of one excess merged, every letter then moving its two shares oppositely, and along the one direction of a node of width one the condition is that the ranges of the two ratios be separated, as in Section 20.7. A direction along which a law is unbounded, its supremum infinite, crosses the walls of its seam function as well. The seam function of a piece continues past its walls with a growth that its sector integrals bound, and the partner can be read in its monomials, which have no walls, or continued past its own walls in the same way.

## 31.2 Closing along a path

The closings of Theorems 31.2 and 31.4 move the contour of the share fibre along a ray, collect a fibre integral of residues at every polar hyperplane crossed, and show that what is left tends to zero. The two steps come apart. Moving the contour is exact at every step: the period is the sum of the crossing terms met so far and of the fibre integral at the current point, by Cauchy's theorem on bounded contours. Only the last step is a limit, and it asks for a bound on the remaining fibre integral at the points that the path passes through, not for a domination of the double family of crossings and fibres, which is what interchanging the sum over the crossings with the fibre integrals would ask, and what fails past the walls. The step is Lemma 31.3, with three consequences: a closing depends on its direction only through its chamber (Theorem 31.6); the laws can be closed one source point at a time and averaged (Theorem 31.7); and past the walls of a piece the bound

that travels is read on the outline of a keyhole continuation (Theorem 31.11). The bounds multiply over products, survive averaging and pass to the crossing terms (Proposition 31.13).

**Lemma 31.3** (Closing along a path)**.** *Let $A$ be the affine space of the real points of the real pins of a fibre integral and $V$ its space of directions, of dimension $n \geq 1$, with the measure $d\boldsymbol{\vartheta}$, the fibre integral through a real point $p$ being $\int_{p+iV} I$ (Definition 4.1) wherever it converges absolutely; and let $\Omega \subseteq A$ be open. Let $I$ be meromorphic on the tube $\Omega + iV$, with its poles on the complex hyperplanes $\{\ell_H = 0\}$ of a family $\mathfrak{H}$ of real hyperplanes $H = \{\ell_H = 0\}$ of $A$, locally finite in $\Omega$, each $\ell_H$ real, affine and not constant. Let $\gamma : [0, \infty) \to \Omega$ be continuous and piecewise linear, with finitely many breaks on every bounded interval and $\gamma(0) = \mathbf{c}$ off $\bigcup \mathfrak{H}$, and assume:*

*(H0)* Clean crossings. *$\gamma$ meets $\bigcup \mathfrak{H}$ at a discrete set of times, its* crossings*, none of them a break of $\gamma$; at a crossing $\tau^*$ the hyperplanes of $\mathfrak{H}$ through $\gamma(\tau^*)$ are one hyperplane $H$, and $\ell_H \circ \gamma$ changes sign at $\tau^*$ with a nonzero slope.*

*(H1)* Local decay. *For every compact $Q \subset \Omega$ and $\delta > 0$ there are $C, c > 0$ with $|I(x+i\boldsymbol{\vartheta})| \leq Ce^{-c|\boldsymbol{\vartheta}|}$ whenever $x \in Q$ and $x + i\boldsymbol{\vartheta}$ lies at distance at least $\delta$ from every hyperplane $\{\ell_H = 0\}$ of the tube.*

*(H2)* A travelling bound. *There are times $\tau_1 < \tau_2 < \cdots$ tending to infinity, none a crossing, at which*

$$\beta(\tau_k) = \int_V \big|I(\gamma(\tau_k) + i\boldsymbol{\vartheta})\big|\, d\boldsymbol{\vartheta}$$

*is finite and tends to zero.*

*Then for every $k$*

$$\int_{\mathbf{c}+iV} I = \sum_{\tau^* < \tau_k} \mathcal{R}_{\tau^*} + \int_{\gamma(\tau_k)+iV} I, \qquad \Big|\int_{\gamma(\tau_k)+iV} I\Big| \leq (2\pi)^{-n}\beta(\tau_k),$$

*so that $\int_{\mathbf{c}+iV} I = \lim_k \sum_{\tau^* < \tau_k} \mathcal{R}_{\tau^*}$. The* crossing term *at a crossing $\tau^*$ of the hyperplane $H$ is*

$$\mathcal{R}_{\tau^*} = \pm(2\pi)^{1-n} \int_{V_H} \mathrm{Res}_{w=0}\, I\big(\gamma(\tau^*) + w\,\mathbf{e} + i\boldsymbol{\vartheta}'\big)\, d\mu_H(\boldsymbol{\vartheta}'),$$

*with $V_H \subset V$ the kernel of the linear part $\ell_H^{\mathrm{lin}}$ of $\ell_H$, $\mu_H$ the measure on $V_H$ with $d\boldsymbol{\vartheta} = d\mu_H\, d\ell_H^{\mathrm{lin}}(\boldsymbol{\vartheta})$, $\mathbf{e} \in V$ any vector with $\ell_H^{\mathrm{lin}}(\mathbf{e}) = 1$, on which the term does not depend, and the sign $+$ when $\ell_H$ decreases along $\gamma$ and $-$ when it increases. The residue is that of the whole integrand, a Laurent coefficient where several of its factors are polar on $H$, and the crossing term is its fibre integral, pinned also by $\ell_H = 0$. If some $\tau_k$ lies between any two consecutive crossings and $\sum_k \beta(\tau_k) < \infty$, the crossing terms are absolutely summable, a crossing between $\tau_k$ and $\tau_{k+1}$, the only one there, having $|\mathcal{R}_{\tau^*}| \leq (2\pi)^{-n}\big(\beta(\tau_k) + \beta(\tau_{k+1})\big)$.*

**Theorem 31.4** (Closing past the walls)**.** *Let the seam function of the piece converge at a real point of the pins of the seam with positive shares, so that its seam word has a Newton sum of full dimension (Theorem 25.2). Read the partner in its monomials, $L_{jD} = \sum_k \xi_{jDk}\, \alpha_R^{\mathbf{n}_{jDk}}$ and $L^*_{jD} = \sum_k \xi^*_{jDk}\, \alpha_R^{\mathbf{n}^*_{jDk}}$, with one share $a_{jDk}$ or $b_{jDk}$ for each monomial, and write $\bar{a}_{jD} = \sum_k a_{jDk}$ and $\bar{b}_{jD} = \sum_k b_{jDk}$; bars mark sums over the monomials.*

*(i) On every contour of shares with positive real parts, pinned by the letter totals and by $\sum_{j,D,k}(a_{jDk}\mathbf{n}_{jDk} + b_{jDk}\mathbf{n}^*_{jDk}) = \mathbf{m}_R$, at which the seam function of the piece converges, the*

*period of the glue converging,*

$$P_{B\sqcup R}(\mathbf{m}) = \int_{pins} \prod_{j:\, s_j>0} \frac{\prod_{D,k} \Gamma(a_{jDk})\, \xi_{jDk}^{-a_{jDk}}}{\Gamma(s_j)} \prod_{j:\, t_j>0} \frac{\prod_{D,k} \Gamma(b_{jDk})\, {\xi^*_{jDk}}^{-b_{jDk}}}{\Gamma(t_j)}\; P_{W_B(\bar{\mathbf{a}},\bar{\mathbf{b}})}(\mathbf{m}_B).$$

*(ii) Fix $e_0 \in B$ and a* fan of the seam word*: finitely many rational simplicial cones of dimension $|B|-1$ with disjoint interiors covering $\mathbb{R}^{B\setminus e_0}$, each contained, for every seam letter $V$ of an active letter, in a set $\{\mathbf{v} :\ \langle \mathbf{t}, \mathbf{v}\rangle \le \langle \mathbf{t}', \mathbf{v}\rangle$ for every exponent $\mathbf{t}'$ of the support of $V\}$, $\mathbf{t}$ one exponent of that support, the pairings taken on $B \setminus e_0$. On a cone $\pi$ of it, a* sector*, with the primitive integral generators $\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1}$, write the logarithmic coordinates $x = \log \alpha_{B\setminus e_0}$ of the section $\alpha_{e_0} = 1$ as $x = -\sum_k u_k \mathbf{v}_{\pi,k}$, $u_k \ge 0$; the $q_k = e^{-u_k} \in (0,1]$ are the* sector coordinates*. On the section a seam letter $V$ of an active letter is $V = \prod_k q_k^{\langle \mathbf{t}, \mathbf{v}_{\pi,k}\rangle}\, k_{V,\pi}(q)$, $\mathbf{t}$ the exponent of its support at which every generator of $\pi$ pairs least, and its* sector letter *$k_{V,\pi}$ is a polynomial in $q$ with nonnegative coefficients and a positive constant term. Write $k_{jD,\pi}$ and $\psi_{jD,\pi}$ for the sector letters of $V_{jD}$ and $V^*_{jD}$, $\mathbf{t}_{jD,\pi}$ and $\mathbf{t}^*_{jD,\pi}$ for their exponents, and for a generator $\mathbf{v}$ of $\pi$ call*

$$\kappa_{\mathbf{v}}(\mathbf{z}) = \Big\langle \mathbf{m}_B - \sum_{j,D} \big(\bar{a}_{jD}\, \mathbf{t}_{jD,\pi} + \bar{b}_{jD}\, \mathbf{t}^*_{jD,\pi}\big),\ \mathbf{v} \Big\rangle, \qquad \mathbf{z} = (\bar{\mathbf{a}}, \bar{\mathbf{b}}),$$

*the pairing on $B \setminus e_0$, the* wall *of $\mathbf{v}$: affine in the shares with a rational linear part, and the same for every sector of which $\mathbf{v}$ is a generator. On the pins of the seam, at every real point at which every wall is positive,*

$$P_{W_B(\mathbf{z})}(\mathbf{m}_B) = \sum_{\pi} |\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})| \int_{[0,1]^{|B|-1}} \prod_k q_k^{\kappa_{\mathbf{v}_{\pi,k}}(\mathbf{z})-1} \prod_{j,D} k_{jD,\pi}(q)^{-\bar{a}_{jD}}\, \psi_{jD,\pi}(q)^{-\bar{b}_{jD}}\, dq,$$

*and the seam function converges at a real point exactly when every wall is positive there. It continues meromorphically to all complex shares, with poles, of order at most $|B|-1$, only on the hyperplanes on which a wall lies in $-\mathbb{N}$, and for every $\theta > 0$ there are $M$, $C_0$ and $n_1$ with*

$$\big|P_{W_B(\mathbf{z})}(\mathbf{m}_B)\big| \le C_0\, \delta^{-(|B|-1)} (1+|\mathbf{z}|)^{n_1} M^{|\operatorname{Re}\mathbf{z}|_1} e^{\theta |\operatorname{Im}\mathbf{z}|_1},$$

*$\delta$ the smaller of 1 and the distance of $\mathbf{z}$ from these hyperplanes. The piece is* of tables *when at every station the seam letters of the active letters are the basis and the cobasis polynomials of one table on $B$ with the bases $\mathcal{B}_{jD}$ and positive weights, as for a bunch of a family of tables (Proposition 28.2(i)). Its seam word then has a Newton sum of full dimension exactly when the family of these tables is connected; the sectors of the orders $\pi$ of $B$ (Theorem 8.5(ii)) form a fan of it, every determinant being 1, the sector letters being those of the tables $\mathcal{B}_{jD}$ and the generators $\mathbf{1}_{U_k} - (\mathbf{1}_{U_k})_{e_0}\mathbf{1}$ on $B \setminus e_0$, $U_k$ the tails; the wall of the generator of $U$ is the proper wall $\kappa_U = m_U - \sum_{j,D} \big(\bar{a}_{jD}\, d_{jD}(U) + \bar{b}_{jD}\, d^*_{jD}(U)\big)$, $d_{jD}$ and $d^*_{jD}$ the counts of $\mathcal{B}_{jD}$.*

*(iii) For a direction $\boldsymbol{\sigma}$ of the shares of (i) and $\theta \ge 0$ let $g(\boldsymbol{\sigma};\theta)$, at the real point $\bar{\mathbf{c}}$ of the pins of the seam, be the infimum of the $M$ for which there are $C$, $n_1$ and $n_2$ with*

$$\big|P_{W_B(\bar{\mathbf{c}}+\tau\bar{\boldsymbol{\sigma}}+i\bar{\boldsymbol{\vartheta}})}(\mathbf{m}_B)\big| \le C\, \delta^{-n_2} (1+\tau+|\boldsymbol{\vartheta}|)^{n_1} M^{\tau} e^{\theta|\boldsymbol{\vartheta}|_1}$$

*for $\tau \ge 0$ and $\boldsymbol{\vartheta}$ in the directions of the shares, $\delta$ the smaller of 1 and the distance from the polar hyperplanes: the* growth *of the continued seam function along $\boldsymbol{\sigma}$ at the angle $\theta$, the*

*growth of Definition 20.33 read along any direction with the angle fixed. It is finite for $\theta > 0$, and $g(\boldsymbol{\sigma}; 0) = \sup \mathbf{V}_B^{-\bar{\boldsymbol{\sigma}}}$ when no wall of the seam word falls along $\bar{\boldsymbol{\sigma}}$, which is exactly when that supremum is finite. Put*

$$h_\Gamma(\boldsymbol{\sigma}) = \sum_l \sigma_l \log|\sigma_l| - \sum_l \sigma_l \log \xi_l, \qquad h_\theta(\boldsymbol{\sigma}) = h_\Gamma(\boldsymbol{\sigma}) + \log g(\boldsymbol{\sigma}; \theta) + \frac{2\theta^2}{\pi} \sum_{l:\, \sigma_l > 0} \sigma_l,$$

*$l$ running over the shares and $\xi_l$ being the coefficient of the monomial of the share $l$; $h_\Gamma$ is the* Gamma rate *of the direction. Let the real point $\mathbf{c}$ of the pins lie off the null set of Theorem 31.2, taken with the polar hyperplanes of the seam function as well, and let $g$ and $h_\theta$ be taken at its sums $\bar{\mathbf{c}}$. If $h_\theta(\boldsymbol{\sigma}) < 0$ for some $\theta \in [0, \pi/8)$, then*

$$P_{B \sqcup R}(\mathbf{m}) = \sum_{crossings} \mathcal{R},$$

*the sum, in the order of crossing, of the fibre integrals of the residues of the integrand at the polar hyperplanes met by the ray $\mathbf{c} + \tau\boldsymbol{\sigma}$: the hyperplanes $a_l = -n$ of the shares with $\sigma_l < 0$, $n \in \mathbb{N}$, and the hyperplanes on which a wall of (ii) that falls along $\bar{\boldsymbol{\sigma}}$ lies in $-\mathbb{N}$, a hyperplane polar for several factors being crossed once, with its full residue. The sum converges absolutely, the terms at $\tau$ and the integrals left after them being $O\big(e^{\tau(h_\theta(\boldsymbol{\sigma})+\epsilon)}\big)$ for every $\epsilon > 0$.*

*(iv)* Both sides continued. *Let the glue be read on the share side of Theorem 28.10(iii), one share $a_{jD}$ or $b_{jD}$ per station, in place of the reading of (i), and let the seam functions of the piece and of the partner both converge at a real point $\mathbf{c}$ of the pins with positive shares. By (ii), applied to the piece and to the partner, a piece on $R$ read with the reversed excesses (Definition 28.4), both continue meromorphically to all complex shares of the pins, with their poles on the hyperplanes on which a wall of the seam word of the piece or of the room word of the partner lies in $-\mathbb{N}$, each with the bound of (ii). For a direction $\boldsymbol{\sigma}$ of the shares and $\theta \geq 0$ let $g_B(\boldsymbol{\sigma}; \theta)$ and $g_R(\boldsymbol{\sigma}; \theta)$ be the growths of the two continued seam functions along $\boldsymbol{\sigma}$ at $\mathbf{c}$, defined as in (iii) with the shares in place of their sums, and put*

$$h_{\theta_B, \theta_R}(\boldsymbol{\sigma}) = \sum_l \sigma_l \log|\sigma_l| + \log g_B(\boldsymbol{\sigma}; \theta_B) + \log g_R(\boldsymbol{\sigma}; \theta_R) + \frac{2(\theta_B + \theta_R)^2}{\pi} \sum_{l:\, \sigma_l > 0} \sigma_l.$$

*Let $\mathbf{c}$ lie off the null set of Theorem 31.2, taken with the polar hyperplanes of both seam functions as well. If $h_{\theta_B, \theta_R}(\boldsymbol{\sigma}) < 0$ for some $\theta_B, \theta_R \geq 0$ with $\theta_B + \theta_R < \pi/8$, then $P_{B \sqcup R}(\mathbf{m}) = \sum_{crossings} \mathcal{R}$, the sum, in the order of crossing, of the fibre integrals of the residues of the integrand at the polar hyperplanes met by the ray $\mathbf{c} + \tau\boldsymbol{\sigma}$: the hyperplanes $a_l = -n$ of the shares with $\sigma_l < 0$, $n \in \mathbb{N}$, and those on which a wall of either word that falls along $\boldsymbol{\sigma}$ lies in $-\mathbb{N}$, a hyperplane polar for several factors being crossed once, with its full residue. The sum converges absolutely, the terms at $\tau$ and the integrals left after them being $O\big(e^{\tau(h_{\theta_B, \theta_R}(\boldsymbol{\sigma})+\epsilon)}\big)$ for every $\epsilon > 0$. A side along whose direction no wall of its word falls has the growth $\sup \mathbf{V}^{-\boldsymbol{\sigma}}$ of its letter vector at every angle, the largest value of the monomial on the support of its law (Theorem 31.2).*

Along a direction in which the seam function converges, (iii) at $\theta = 0$ is the bound of Theorem 31.2 for the glue read in the monomials of the partner, and past a wall the residues of the seam function join those of the Gammas at the cost $\log g$ of its continuation, as the walls of the two families join the lattice at width one (Theorem 20.34(iii)). Read in its monomials as well,

any piece contributes only its coefficients: the integrand is then Gammas and coefficients, every supremum of a monomial of the letter vectors is a monomial in the coefficients, and every direction of the shares with $h_\Gamma(\boldsymbol{\sigma}) < 0$ closes, by Theorem 31.2. In which representation to close is a choice of rate: a law bounded along the direction closes at the largest value of its monomial, a side past its walls at the growth of its continuation, a side read in monomials at its coefficients, at the price of one pin for each of its marks. So (iii) and (iv) read each side at its own rate, the coefficients $-\sum_l \sigma_l \log \xi_l$ for a side read in its monomials and the logarithm of the growth of its continued seam function for a side read in its seam letters; at width one, where the share side has one free share, (iv) continues both families of the node, as Theorem 20.34(iii) does.

Expanding each active letter $L$, with its exponent $e_L$, in some of its terms, $L = Y_L + X_L$ with $X_L = \sum_k X_{L,k}$ the chosen terms, gives the multiple lattice whose terms are $(-1)^{|\mathbf{n}_L|}(e_L)_{|\mathbf{n}_L|}/\mathbf{n}_L!$ times the periods of $\alpha^{\mathbf{m}-\mathbf{1}} \prod_L Y_L^{-e_L-|\mathbf{n}_L|} \prod_k X_{L,k}^{n_{L,k}}$, positive words up to sign. Summed term by term in absolute value, it converges absolutely exactly when $X_L \le Y_L$ on the orthant for every active letter and the integral of $\alpha^{\mathbf{m}-\mathbf{1}} \prod_L (Y_L - X_L)^{-e_L}$ over a section is finite (Proposition 2.10). This needs the Newton polytope of $X_L$ inside that of $Y_L$, so it never holds for a letter of a table, all of whose monomials are vertices of its Newton polytope: for families of tables the lattices close only in the order of a direction.

Theorems 31.2 and 31.4 are the lemma along a ray: (H0) holds off a null set of starting points, (H1) from the exponential decay of the Gamma factors in the imaginary directions (Theorem 2.11 at one scale, carried to negative real parts by the shift of Proposition 1.3(iii)), and (H2) from the largest values of the monomials on the supports of the laws or from the growth of the continued seam function. Two closings of one fibre integral along two paths agree at all times up to their remainders,

$$\sum_{\tau^* < \tau_k} \mathcal{R}_{\tau^*} - \sum_{\tau'^* < \tau'_k} \mathcal{R}'_{\tau'^*} = \int_{\gamma'(\tau'_k)+iV} I - \int_{\gamma(\tau_k)+iV} I,$$

and one of them may converge while the other does not. Each crossing term is a fibre integral of one dimension less, of an integrand of the same kind, which the lemma can close again along a path of its own when a bound travels with it there; this is how multiple lattices arise.

**Definition 31.5** (Chambers of a closing)**.** Let $I$ be meromorphic on the tube $\Omega + iV$ of Lemma 31.3 over an open convex $\Omega \subseteq A$, with its poles on the complex hyperplanes on which one of finitely many real affine functions $\ell$ on $A$, not constant, lies in $-\mathbb{N}$: its *polar functions*. A direction $\boldsymbol{\sigma} \in V$ *falls* a polar function $\ell$ when $\partial_{\boldsymbol{\sigma}} \ell < 0$, at the *falling rate* $-\partial_{\boldsymbol{\sigma}} \ell$. For a real point $\mathbf{c} \in \Omega$ at which every polar function is positive, the *chamber* $C(\boldsymbol{\sigma})$ of a direction $\boldsymbol{\sigma}$ whose ray $\mathbf{c} + \tau\boldsymbol{\sigma}$, $\tau \ge 0$, lies in $\Omega$ is the set of the $\boldsymbol{\sigma}' \in V$ whose ray lies in $\Omega$ and which fall exactly the polar functions that $\boldsymbol{\sigma}$ falls, each at the same falling rate; it is convex. In Theorem 31.2 the polar functions are the shares of the active letters, and $\Omega$ is the interior of the set of the real points of the pins at which the seam functions of the piece and of the partner converge, their reduced periods in Theorem 28.12, which is convex by Hölder's inequality and contains the ray $\mathbf{c} + \tau\boldsymbol{\sigma}$, $\tau \ge 0$, from a point $\mathbf{c} \in \Omega$ whenever $\sup \mathbf{V}_B^{-\boldsymbol{\sigma}}$ and $\sup \mathbf{V}_R^{-\boldsymbol{\sigma}}$ are finite. In Theorem 31.4(iii) they are the shares and the walls of the seam word, read as affine functions of the shares through their sums (Theorem 31.4(ii)), and $\Omega = A$. In a fibre integral of Gammas and coefficients, as the share side of a Gramma (Proposition 4.2) and the kernel of Theorem 31.7 below, they are the shares and $\Omega = A$, so that

$$C(\boldsymbol{\sigma}) = \{\boldsymbol{\sigma}' \in V : \ \sigma'_l = \sigma_l \text{ where } \sigma_l < 0, \ \ \sigma'_l \ge 0 \text{ elsewhere}\},$$

a compact convex polytope when the pins fix the total of the shares of every letter; the walls of source factors that are cube integrals (Definition 31.16 below) are polar functions as well.

**Theorem 31.6** (A closing depends only on its chamber)**.** *Let $I$, $\mathbf{c}$ and $\boldsymbol{\sigma}$ be as in Definition 31.5, let $I$ have the local decay (H1) of Lemma 31.3 on $\Omega$, and let $\boldsymbol{\sigma}' \in C(\boldsymbol{\sigma})$. The rays $\mathbf{c}+\tau\boldsymbol{\sigma}$ and $\mathbf{c}+\tau\boldsymbol{\sigma}'$ meet the same polar hyperplanes at the same times, and (H0) holds along one of them exactly when it holds along the other; at every other time the fibre integrals through their two points are equal, and at a crossing their crossing terms are equal. So the closings along the two directions are one series, with the same remainders: the closing along $\boldsymbol{\sigma}$ converges, with the bounds of its terms and of its remainders, as soon as the closing along some direction of its chamber does. In Theorem 31.2, or in Theorem 31.4(iii), if $\mathbf{c}$ lies off the null set of Theorem 31.2, taken for Theorem 31.4(iii) with the polar hyperplanes of the seam function as well, the closing along $\boldsymbol{\sigma}$ converges absolutely as soon as*

$$\inf_{\boldsymbol{\sigma}' \in C(\boldsymbol{\sigma})} h(\boldsymbol{\sigma}') < 0,$$

*$h$ the rate of Theorem 31.2, the infimum being taken for Theorem 31.4(iii) over the rates $h_\theta$, $\theta \in [0, \pi/8)$, as well; its terms and the integrals left after them are then $O(e^{\tau(h'+\epsilon)})$ for every $\epsilon > 0$, $h'$ that infimum. On the chamber the rate $h$ of Theorem 31.2 is convex.*

So the closing along a direction converges as soon as some direction of its chamber is in the cone of Theorem 31.2, and its best certified rate is a convex problem in the free coordinates of the chamber. A chamber has positive dimension when the shares that do not fall have free directions on the pins, which stations of one excess supply. Theorem 31.2 reads both laws along one direction, through the largest values of the monomials of their letter vectors. The glue is also the average over the laws of the stand-in kernel (Theorem 28.10(ii)), whose integrand at one source pair has no walls, and, with the partner read in its monomials, the average over the law of the piece alone of an integrand without walls; each source point can therefore close along its own best direction of the chamber, and the averages of the crossing terms close the glue.

**Theorem 31.7** (Averaging the laws)**.** *Take a share side, on pins that fix the total of the shares of every active letter, with $V$ the space of the directions of the shares on the pins, whose integrand is*

$$\prod_{j\colon s_j>0} \frac{\prod \Gamma(a_l)}{\Gamma(s_j)} \prod_{j\colon t_j>0} \frac{\prod \Gamma(a_l)}{\Gamma(t_j)} \int e^{-\langle \mathbf{a}, y(\zeta)\rangle}\, d\lambda(\zeta),$$

*the products over the shares $l$ of $K_j$ and of $\Psi_j$, $\mathbf{a}$ the vector of all the shares, $\lambda$ a positive measure on a source, the* source measure*, whose points are the* source points *$\zeta$, and $y = (y_l)_l$ a continuous map of the source to the space of the shares. In Theorem 28.10 two share sides are of this kind, their shares $l$ those of the active letters:*

- *(A) that of Theorem 28.10(iii), with one share per station; $\lambda = (\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho \otimes (\boldsymbol{\varrho}_R^\rho)_* \phi_R^\rho$, a source point $\zeta = (\mathbf{x}, \mathbf{x}')$ is a seam point of the piece and one of the partner, and $y_l(\zeta) = \log(x_{jD} x'_{jD})$ or $\log(x^*_{jD} x'^*_{jD})$ for the share of the station $D$ of the letter;*
- *(B) that of Theorem 28.12 with the partner read in its monomials, one share per monomial, written as in Theorem 31.4(i), the seam word of the piece having a Newton sum of full dimension; $\lambda = (\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho$ is the law of the piece, a source point is a seam point $\zeta = \mathbf{x}$ of the piece, and $y_l(\zeta) = \log x_{jD} + \log \xi_l$ or $\log x^*_{jD} + \log \xi_l$ for the share of the monomial of the coefficient $\xi_l$ at the station $D$.*

*Let $S_\lambda$ be the support of the image of $\lambda$ under $y$, the* log-support *of the source measure, $\mathbf{c}$ a real point of the pins with positive coordinates at which $\int e^{-\langle \mathbf{c}, y\rangle}\, d\lambda$ converges (in (A) the seam functions of the piece and of the partner converge there, in (B) that of the piece), $\boldsymbol{\sigma} \in V$, and $C = C(\boldsymbol{\sigma})$ the*

*chamber of Definition 31.5 of the kernel below, whose polar functions are the shares:*

$$C = \{\boldsymbol{\sigma}' \in V :\ \sigma'_l = \sigma_l \text{ where } \sigma_l < 0,\quad \sigma'_l \geq 0 \text{ elsewhere}\},$$

*a compact convex polytope. Two shares are in one* bundle *when they belong to one letter and have one column in the pins: in (A) when their stations have one excess, in (B) when moreover they carry one monomial of the partner. For $y$ in the space of the shares put*

$$h(\boldsymbol{\sigma}'; y) = \sum_l \sigma'_l \log|\sigma'_l| - \langle \boldsymbol{\sigma}', y\rangle, \qquad h_C(y) = \min_{\boldsymbol{\sigma}' \in C} h(\boldsymbol{\sigma}'; y), \qquad \hat{h}_C = \sup_{y \in S_\lambda} h_C(y) \in (-\infty, \infty],$$

*the* rate *of $\boldsymbol{\sigma}'$ at $y$, the* chamber rate *of the kernel at $y$, concave in $y$, and the* averaged rate.

*Let $\mathbf{c}$ be interior, relative to the pins, to the set of the real points at which the Laplace transform $\int e^{-\langle \mathbf{a}, y\rangle}\, d(y_*\lambda)(y)$ of $y_*\lambda$ converges; equivalently, for a basis $\mathbf{v}_1, \dots, \mathbf{v}_d$ of $V$ there is $\eta > 0$ with*

$$\sum_{\boldsymbol{\epsilon} \in \{-1,1\}^d} \int \exp\Big( - \Big\langle \mathbf{c} - \eta \sum_i \epsilon_i \mathbf{v}_i,\, y \Big\rangle \Big)\, d(y_*\lambda)(y) < \infty, \tag{31.1}$$

*a finite certificate that $\mathbf{c}$ is interior to the set where that Laplace transform converges, that set being convex. Put $|y|_V = \sum_i |\langle \mathbf{v}_i, y\rangle|$, let $p$ be the largest total order of the Gamma factors polar on one hyperplane that the ray of $\boldsymbol{\sigma}$ crosses, at most their number, and in this statement write $w(y) = e^{-\langle \mathbf{c}, y\rangle}(1 + |y|_V)^{p-1}$. Every power of $1 + |y|_V$ is at most a constant times $e^{\eta |y|_V} \leq \sum_{\boldsymbol{\epsilon}} \exp(\eta \sum_i \epsilon_i \langle \mathbf{v}_i, y\rangle)$, so (31.1) gives $\int w\, d(y_*\lambda) < \infty$.*

*(i)* The kernel. *The* kernel *at a source point $\zeta$ is the fibre integral on the pins*

$$\int_{pins} \prod_{j:\, s_j > 0} \frac{\prod \Gamma(a_l)}{\Gamma(s_j)} \prod_{j:\, t_j > 0} \frac{\prod \Gamma(a_l)}{\Gamma(t_j)}\ e^{-\langle \mathbf{a}, y(\zeta)\rangle},$$

*of an integrand without walls, and the fibre integral of the share side is its integral against $\lambda$, which in (A) and (B) is $P_{B \sqcup R}(\mathbf{m})$; in (A) the kernel is $\Gamma(\kappa_B)\, \mathsf{Z}_{(\kappa_B, \mathbf{s}, \mathbf{t}); t_{\mathcal{D}}}(\mathbf{x} \circ \mathbf{x}')$ of Theorem 28.10(ii). Let $\mathbf{c}$ lie off the null set of Theorem 31.2 for the direction $\boldsymbol{\sigma}$. For every $\epsilon > 0$ there is $C_\epsilon$, not depending on the source point, such that the crossing terms $\mathcal{R}_k(\zeta)$ of the closing of the kernel along $\boldsymbol{\sigma}$, at their crossing times $\tau_k$, satisfy*

$$|\mathcal{R}_k(\zeta)| \leq C_\epsilon\, w(y)\, e^{\tau_k (h_C(y) + \epsilon)}, \qquad y = y(\zeta),$$

*and the integrals left at times separating the crossings satisfy the same bound at the time $\tau$ with $e^{-\langle \mathbf{c}, y\rangle}$ in place of $w(y)$. When every crossed hyperplane carries a simple pole, $p = 1$ and $w(y) = e^{-\langle \mathbf{c}, y\rangle}$.*

*(ii)* The averaged closing. *If $\hat{h}_C < 0$, the fibre integral of the share side is*

$$\sum_k \int \mathcal{R}_k\, d\lambda,$$

*the sum in the order of crossing, absolutely convergent, its terms $O(e^{\tau_k(\hat{h}_C + \epsilon)})$ for every $\epsilon > 0$. At a crossing whose point along some $\boldsymbol{\sigma}' \in C$ lies in the interior of the set where $\int e^{-\langle \mathbf{a}, y\rangle}\, d\lambda$ converges, the averaged term is the crossing term of the share side there; so when it converges along the ray of $\boldsymbol{\sigma}$, the averaged closing is the closing of the share side along $\boldsymbol{\sigma}$, in (A) and (B) that of Theorem 31.2.*

*(iii)* One direction reads the convex hull. $\min_{\boldsymbol{\sigma}'\in C}\sup_{y\in S_\lambda} h(\boldsymbol{\sigma}';y) = \sup_{y\in\operatorname{conv} S_\lambda} h_C(y) \geq \hat{h}_C$, *and* $\sup_{y\in S_\lambda} h(\boldsymbol{\sigma}';y)$ *is the rate of the closing of the share side along* $\boldsymbol{\sigma}'$ *read through the largest value of* $e^{-\langle\boldsymbol{\sigma}',y\rangle}$ *on the log-support, in (A) and (B) the rate* $h(\boldsymbol{\sigma}')$ *of Theorem 31.2: the best single direction of the chamber reads the convex hull of the log-support, the averaged closing the log-support itself. In a bundle* $G$ *of shares that* $\boldsymbol{\sigma}$ *does not fall, the least value of* $\sum_{l\in G}(\sigma'_l\log\sigma'_l - \sigma'_l y_l)$ *over the* $\sigma'_l \geq 0$ *with* $\sum_G \sigma'_l = g$ *is* $g\log g - g\log\sum_{l\in G} e^{y_l}$*, attained at* $\sigma'_l = g\,e^{y_l}/\sum_{l'\in G} e^{y_{l'}}$*: at each source point the best direction merges the bundle into the sum* $e^{y_G} = \sum_{l\in G} e^{y_l}$*, which in (A) is the sum* $\sum_D x_{jD}x'_{jD}$ *that the kernel reads at the excess, splitting* $g$ *in the proportions of its terms.*

At width one the share side of a node merged by its bundles has at most one free share and the seam torus is a half-line, so the log-support is an interval, its own hull, and one direction loses nothing. A gap between the averaged rate and the best single direction needs a chamber of positive dimension and a log-support that is not convex in its directions: the best split at each source point moves with it, and one direction must take one split for all of them. Averaging can reach a rate that no single direction certifies, and an averaged closing is in general no closing of one share side along one ray: its terms are averages of the crossing terms of the kernel, and it can converge past walls that every ray of the chamber meets. Where the averaged rate is infinite, a closing past the walls can still converge, at the rate of the best source point: past its walls the law is read through its expansion at its falling end, not through its moments, as at width one (Theorem 20.44). Past the walls of a piece the bound that travels is read on the sector letters (Theorem 31.11).

**Theorem 31.8** (An averaging gap at a family of tables)**.** *There are three tables with the bases* $\binom{\{1,2,3,4\}}{2}$*, at the exponents* $s_j = 1$*,* $t_j = 0$ *and the masses* $\frac{3}{2}$ *on every mark, whose bunch* $\{1,2\}$*, with the room* $\{3,4\}$*, read through its seam with the product of the two laws as the source measure, the kind (A) of Theorem 31.7, has one station at each excess of each letter, so that every bundle of Theorem 31.7 is a single share, and a chamber* $C$ *with*

$$\min_{\boldsymbol{\sigma}\in C}\sup_{y\in S_\lambda} h(\boldsymbol{\sigma};y) = 0, \qquad \hat{h}_C = \sup_{y\in S_\lambda}\min_{\boldsymbol{\sigma}\in C} h(\boldsymbol{\sigma};y) < 0 :$$

*no direction of the chamber certifies the closing (Theorem 31.2), and the averaged closing converges geometrically (Theorem 31.7(ii)). Two directions of the chamber, the better one taken at each source point, already give the bound.*

*Remark* 31.9 (What the gap separates). The gap lies between the averaged closing and the directions of one share side, the seam of the bunch with its bundles merged, and no merging of stations removes it, every bundle being a single share. It is not a gap against every closing of the period: closing each source point along its own direction, as the bound does with two directions, is itself a closing of the period, and so is completing another inner read before closing. A gap in every closing is a statement about a specified set of closings; for the closings of one share side by the directions of its chamber, the family of Theorem 31.8 has one, while at a seam whose two stations of one excess multiply the same monomial of the partner, merging the two stations can remove such a gap.

**Definition 31.10** (Keyhole continuations along a direction; ceiling and phase)**.** In Theorem 31.4 let $\boldsymbol{\sigma}$ be a direction of the shares and $\bar{\boldsymbol{\sigma}}$ its sums over the monomials of the partner, and fix a fan of the seam word (Theorem 31.4(ii)). For a sector $\pi$ of it let $q = (q_k)_k$ be its sector coordinates, $\kappa_k = \kappa_{\mathbf{v}_{\pi,k}}$ the walls of its generators and $\mathbf{k}_\pi = (k_{jD,\pi}, \psi_{jD,\pi})_{j,D}$ the vector of its sector letters, so

that the seam function on the pins is the sum over the sectors of

$$|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})|\int_{[0,1]^{|B|-1}}\prod_k q_k^{\kappa_k(\bar{\mathbf{a}},\bar{\mathbf{b}})-1}\,\mathbf{k}_\pi(q)^{-(\bar{\mathbf{a}},\bar{\mathbf{b}})}\,dq,$$

the walls affine in the shares; for a piece of tables and the sectors of the orders $\pi$ of $B$ these are the walls $\kappa_{U_k}$ of the tails and the sector letters of the tables $\mathcal{B}_{jD}$ (Theorem 8.5(ii)). Put $\dot\kappa_k=\partial_{\bar{\boldsymbol\sigma}}\kappa_k$, the derivative of the wall along $\bar{\boldsymbol\sigma}$, and

$$\varrho_{\pi,\bar{\boldsymbol\sigma}}(q)=\prod_k q_k^{\dot\kappa_k}\;\mathbf{k}_\pi(q)^{-\bar{\boldsymbol\sigma}},$$

the *directional ratio in the sector* of $\pi$; its *falling variables* along $\bar{\boldsymbol\sigma}$ are the $q_k$ with $\dot\kappa_k<0$. A *keyhole continuation along* $\bar{\boldsymbol\sigma}$ is defined as in Definition 20.47, with the sectors in place of the orders, with these falling variables, and with the sector letters of $\mathbf{k}_\pi$ required not to vanish on the sets $Y$ of its boxes; its *outline* $\mathcal{O}$ is the union of the sets $\mathcal{O}$ of its boxes, and its *ceiling along* $\bar{\boldsymbol\sigma}$ is the largest value of $|\varrho_{\pi,\bar{\boldsymbol\sigma}}|$ on the outline, $\pi$ the sector of the box. Each set $Y$ is a product of a box and of keyholes, simply connected, and on a neighbourhood of it the sector letters have logarithms real at the real points; the *phase* of the continuation at a point $q$ of the outline of a box of $\pi$ is the linear function $\boldsymbol\vartheta\mapsto\langle\bar{\boldsymbol\vartheta},\arg\mathbf{k}_\pi(q)\rangle$ on the directions of the shares, arg the imaginary part of the logarithm and the bars marking sums over the monomials of the partner.

The modulus $|\varrho_{\pi,\bar{\boldsymbol\sigma}}(q)|$ is the factor by which the part of the integrand that grows with $\tau$ along $\bar{\boldsymbol\sigma}$ is multiplied per unit of $\tau$ at the point; the phase is the linear function by which the imaginary parts of the shares enter it. With no falling variables the outline is the real cube, the letters are positive there, the phase vanishes, and the ceiling is $\sup\mathbf{V}_B^{-\bar{\boldsymbol\sigma}}$, the largest value of the monomial of Theorem 31.2; at width one, along the fall of the share $b^d$ of Section 20.7, the directional ratio in the sector is the ratio $\varrho_\pi$ of Definition 20.46.

**Theorem 31.11** (The travelling bound past the walls)**.** *In Theorem 31.4 let $\boldsymbol\sigma$ be a direction of the shares and take a keyhole continuation along $\bar{\boldsymbol\sigma}$ (Definition 31.10) with the outline $\mathcal{O}$.*

*(i) For every $\epsilon>0$ there is $C$ such that for $\tau\ge0$ and $\boldsymbol\vartheta$ in the directions of the shares*

$$\big|P_{W_B(\bar{\mathbf{c}}+\tau\bar{\boldsymbol\sigma}+i\bar{\boldsymbol\vartheta})}(\mathbf{m}_B)\big|\le C\,\delta^{-(|B|-1)}\sup_{q\in\mathcal{O}}\Big(|\varrho_{\pi,\bar{\boldsymbol\sigma}}(q)|^\tau\,e^{\tau\epsilon+\langle\bar{\boldsymbol\vartheta},\arg\mathbf{k}_\pi(q)\rangle}\Big),$$

*$\delta$ the smaller of 1 and the distance of $\bar{\mathbf{c}}+\tau\bar{\boldsymbol\sigma}$ from the hyperplanes on which a wall of the fan of Definition 31.10 lies in $-\mathbb{N}$, which hold the poles of the seam function.*

*(ii) Let $\boldsymbol\upsilon$ be a bounded function on $\mathcal{O}$ with values in the vectors indexed by the shares, with $\langle\bar{\boldsymbol\vartheta},\arg\mathbf{k}_\pi(q)\rangle=\sum_l\upsilon_l(q)\vartheta_l$ for $\boldsymbol\vartheta$ in the directions of the shares and $|\upsilon_l(q)|\le\frac{\pi}{2}-\epsilon_0$ for some $\epsilon_0>0$; representatives differing by a vector orthogonal to the directions give the same phase on the fibre. Put*

$$h_{\boldsymbol\upsilon}(\boldsymbol\sigma)=h_\Gamma(\boldsymbol\sigma)+\sup_{q\in\mathcal{O}}\Big(\log|\varrho_{\pi,\bar{\boldsymbol\sigma}}(q)|-\sum_{l:\,\sigma_l>0}\sigma_l\log\cos\upsilon_l(q)\Big),$$

*$h_\Gamma$ the Gamma rate of Theorem 31.4(iii). If $h_{\boldsymbol\upsilon}(\boldsymbol\sigma)<0$ and the real point $\mathbf{c}$ of the pins lies off the null set of Theorem 31.2, taken with these hyperplanes as well, the conclusion of Theorem 31.4(iii) holds along $\boldsymbol\sigma$, its terms and the integrals left after them being $O(e^{\tau(h_{\boldsymbol\upsilon}(\boldsymbol\sigma)+\epsilon)})$ for every $\epsilon>0$.*

**Corollary 31.12** (When the phase costs nothing)**.** *In Theorem 31.11(ii) let $H_{\mathcal{O}}$ be the ceiling of the continuation along $\bar{\boldsymbol{\sigma}}$. Then $h_{\boldsymbol{\upsilon}}(\boldsymbol{\sigma}) = h_\Gamma(\boldsymbol{\sigma}) + \log H_{\mathcal{O}}$ in each of two cases:*

*(i) $\upsilon_l(q) = 0$ for every $q \in \mathcal{O}$ and every share with $\sigma_l > 0$: the phase sits on shares that fall or stay, whose Gammas absorb it;*

*(ii) $-\sum_{l:\,\sigma_l>0} \sigma_l \log\cos\upsilon_l(q) \le \log H_{\mathcal{O}} - \log|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}(q)|$ for every $q \in \mathcal{O}$: the modulus lost away from the highest points pays for the phase.*

*With no falling variables (i) holds with $\boldsymbol{\upsilon} = 0$, and the rate is that of Theorem 31.2 for the glue read in the monomials of the partner.*

Theorem 31.4(iii) pays for an allowed growth $e^{\theta|\operatorname{Im}\mathbf{z}|_1}$ of the seam function with the term $\frac{2\theta^2}{\pi}\sum_{\sigma_l>0}\sigma_l$; Theorem 31.11 pays only for the phase that the letters have at the points where the ceiling is read, and at the exact price $-\log\cos\upsilon_l$ of the rotation. Enlarging the radii of the keyholes, as far as the sector letters do not vanish, can lower the ceiling; it comes down to the reach at simple tops that do not cancel (Theorem 32.15(iii)). The joint words of Section 31.3 complete the retained fibre on its strips before a modulus is taken, and their bound $r_G c_{\max}$ has no phase term; the ceiling $c_{\max}$ of the joint word need not be the ceiling of this outline.

**Proposition 31.13** (How travelling bounds compose)**.** *(i)* Products. *When the share integrand carries several continued factors, each bounded as in Theorem 31.11(i) on its own outline, their product is bounded on the product of the outlines by the products of their moduli and of the exponentials of their phases, and Theorem 31.11(ii) holds with these on the product outline: the phases of different factors at one point of it may cancel.*

*(ii)* Averaging. *Let integrands $I_\zeta$, for $\zeta$ in a source with a measure $\lambda$, have fibre integrals that close along one path with their crossings at the same times, with the crossing terms $\mathcal{R}_k(\zeta)$, and with crossing terms and remainders, at times $\tau_k$ separating the crossings, at most $M(\zeta)\beta(\tau_k)$, where $M$ is integrable against $\lambda$ and $\sum_k \beta(\tau_k) < \infty$. Then $\int \big(\int_{\mathbf{c}+iV} I_\zeta\big)\, d\lambda(\zeta) = \sum_k \int \mathcal{R}_k\, d\lambda$, absolutely convergent, the $k$-th term at most $\|M\|_1\beta(\tau_k)$. When for some $\hat{h} < 0$ the bound holds with $\beta(\tau) = e^{\tau(\hat{h}+\epsilon)}$ for every small $\epsilon > 0$, $M$ depending on $\epsilon$, the terms are $O(e^{\tau(\hat{h}+\epsilon)})$: the rate is the bound $\hat{h}$ shared by the source points, read on the support of $\lambda$, the uniformity over the source being the hypothesis on $M$. Where the fibre integral and the source integral of a crossing term converge absolutely together, $\int \mathcal{R}_k\, d\lambda$ is the crossing term of the averaged integrand $\int I_\zeta\, d\lambda$.*

*(iii)* Crossing. *The crossing terms of the closings of Theorems 31.4(iii) and 31.11 are fibre integrals of one dimension less of integrands of the same kind. At a share crossing $a_l = -n$ the Gamma becomes $(-1)^n/n!$ and the other factors are read on the hyperplane. At a wall crossing where the pole of the seam function is simple, its residue is the* face coefficient *of the walls through the hyperplane: the sum, over the generators $\mathbf{v}$ of a fan of the seam word (Theorem 31.4(ii)) whose wall $\kappa_{\mathbf{v}}$ takes a value $-n_{\mathbf{v}} \in -\mathbb{N}$ on the hyperplane and over the sectors $\pi$ of which they are generators, of $|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})|$ times the cube integral over the other coordinates of the sector of the $n_{\mathbf{v}}$-th Taylor coefficient at $q_{\mathbf{v}} = 0$ of the rest of the integrand (Lemma 8.4), $q_{\mathbf{v}}$ the sector coordinate of $\mathbf{v}$, each divided by the derivative of $\kappa_{\mathbf{v}}$ along the vector $\mathbf{e}$ of Lemma 31.3; its letters are the sector letters at $q_{\mathbf{v}} = 0$, positive on the face, and its numerators are polynomials in the shares and in the derivatives of the letters. For a piece of tables and the sectors of the orders the generators are those of the tails $U$, and the sectors the orders of which they are tails. Where the pole is multiple, because several factors, or two walls of one sector, are polar on the hyperplane, the residue is a*

*Laurent coefficient, a finite sum of derivatives in the shares of products of such terms, which brings powers of the logarithms of the letters and of the $q_k$. A keyhole continuation of the seam function restricts to one of each face coefficient, $q_{\mathbf{v}} = 0$ lying in the disc of the keyhole of $q_{\mathbf{v}}$, and Cauchy's estimate on its circle bounds the Taylor coefficients by the values on the outline; so each crossing term can be closed again by Lemma 31.3 whenever a travelling bound of its own tends to zero.*

Completing an inner read is not a crossing. Merging the stations of one excess by Corollary 4.4, integrating a source before closing, or finishing a transverse fibre replaces the integrand by a completed one, with a smaller fibre; the identity between the two integrands holds where the completed read converges and persists under continuation, but the travelling bound does not pass from one to the other, and the completed object can grow along the outer path where the uncompleted integrand does not. Completing an inner read is free when the completed object obeys, along the outer path, a bound already in hand. Merging the stations of one excess reads the law through the sum of their seam letters, whose expansion at the falling end of the law can stop at an edge of the support nearer than the far edge that the stations read, and the merged closing can then diverge where the station closing converges. The two closings agree at every step up to their remainders (Lemma 31.3), and only the remainder of the station closing need tend to zero. On the shares, merging the stations is conditioning on their merged seam letter: integration commutes with completing the fibre, and continuation commutes with it only while the level sets of the merged seam letter stay inside the region where the expansion of the law at its falling end represents it. Completing the inner fibre before closing and keeping its inherited cut are two representations, whose finite heads agree together with their boundary words; the regions on which a law equals its expansion, with their cuts and boundary words, are built in Section 32.1, and a convergent germ alone does not equal the law.

## 31.3 Retained crossings as joint words

### 31.3.1 Completing the retained fibre before taking its size

Theorem 31.11 bounds the share integrand before its fibre is completed, and pays for the phase of the letters where it reads the ceiling. The procedure for joint words completes the retained shares first. It keeps the sectors and coordinate cuts of every continued source and its finite heads, and its auxiliary variables are part of one joint input of a Trailmoment (Definition 5.1), their number not growing along a retained cohort (Definition 31.16): a falling Gamma kept inside a crossing term becomes a positive share and one Beta variable, and a continued source strip becomes a segment and a few circle integrals whose powers on the circles have integral exponents.

**Lemma 31.14** (A Gamma strip as a Beta input)**.** *Let $\delta \geq 1$ be an integer, $b = a + \delta$, and $0 < \operatorname{Re} b < 1$. For a nonzero coefficient $\xi$, with one fixed logarithm,*

$$\Gamma(a)\,\xi^{-a} = \frac{(-1)^{\delta}\xi^{\delta}}{\Gamma(\delta)} \int_0^1 (1-u)^{\delta-1}\,\Gamma(b)\,(\xi u)^{-b}\,du.$$

*The identity is pointwise in every remaining share. So on a fibre $P\mathbf{a} = \boldsymbol{\pi}$ replacing $a_l$ by $b_l = a_l + \delta_l$ changes the pins to $P\mathbf{b} = \boldsymbol{\pi} + P(\delta_l)_l$ and does not move the contour. A share with positive real part is left unchanged, with $\delta_l = 0$ and no auxiliary variable. In a family $\operatorname{Re} a_l(n) = a_l^0 - d_l n$ with $d_l \in \mathbb{N}$ and $a_l^0 \notin \mathbb{Z}$ one may take $\delta_l(n) = d_l n + \delta_l(0)$ with $0 < a_l^0 + \delta_l(0) < 1$, the lift being used at the $n$ with $\delta_l(n) \geq 1$, all but finitely many when $d_l \geq 1$: one Beta input serves at every $n$.*

**Lemma 31.15** (A continued source strip in six integrals)**.** *Let $g$ be holomorphic near $\{|q| \le r_2\} \cup [r,1]$, and choose $0 < r < r_1 < rT < r_2$ and $T > 1$. If $\ell \ge 1$ is an integer and $0 < \operatorname{Re}(\kappa + \ell) < 1$, the continued integral $I_\kappa = \operatorname{Reg} \int_0^1 q^{\kappa-1} g(q)\, dq$ (Definition 1.5) is*

$$I_\kappa = \int_r^1 q^{\kappa-1} g(q)\, dq + R_\ell + R'_\ell,$$

*all circles run counterclockwise, with*

$$R_\ell = \frac{1}{2\pi i} \int_0^r q^{\kappa+\ell-1} \oint_{|\zeta|=r_1} \frac{g(\zeta)\, \zeta^{-\ell}}{\zeta - q}\, d\zeta\, dq,$$
$$R'_\ell = -\frac{r^\kappa}{2\pi i} \Big( \int_1^T u^{\kappa-1} \oint_{|\zeta|=r_2} + \int_T^\infty u^{\kappa-1} \oint_{|\zeta|=r_1} \Big) g(\zeta)\, \frac{1 - (ru/\zeta)^\ell}{\zeta - ru}\, d\zeta\, du.$$

*Splitting the numerator of $R'_\ell$ gives four absolutely convergent integrals, six in all. This remains true with holomorphic parameters and other source variables on compact sets, uniformly when the strip and the radius inequalities have positive margins. For $\kappa(\mathbf{a}, n) = \kappa_0(\mathbf{a}) - dn$ and $\ell(n) = \ell_0 + dn$, $d \in \mathbb{N}$, each of the six has a fixed domain and is a fixed function times an $n$-th power; the exponent of $\zeta$ that contains $\ell(n)$ is integral and independent of the shares that move. At a simple wall $\kappa = -N$, $N \in \mathbb{N}$, the residue is the single coefficient input*

$$[q^N] g(q) = \frac{1}{2\pi i} \oint_{|\zeta|=r_1} g(\zeta)\, \zeta^{-N-1}\, d\zeta.$$

*A cut upper end $q_0 > 0$ is kept by applying the same formula after $q = q_0 v$, the factor $q_0^\kappa$ included; a positive lower cut is an ordinary segment throughout.*

Two of the six integrals simplify by Cauchy's formula: the integral of $R'_\ell$ over $[1, T]$ with the numerator 1 is $-\int_r^{rT} q^{\kappa-1} g(q)\, dq$, and the integral over $(T, \infty)$ with the numerator 1 vanishes, its pole $ru$ lying outside the smaller circle. So $I_\kappa$ is the segment $\int_{rT}^1 q^{\kappa-1} g$, the remainder $R_\ell$ on the circle of radius $r_1$, and the two integrals of $R'_\ell$ with the numerator $(ru/\zeta)^\ell$; along a family $\kappa_0 - dn$ the segment carries the $n$-th power of $q^{-d}$, at most $(rT)^{-d}$, and the circle integrals that of $\zeta^{-d}$, of modulus $r_1^{-d}$ or $r_2^{-d}$. The radius $r$ splits the contour, not the source: the sum does not depend on it.

**Definition 31.16** (A rational retained cohort)**.** A *retained cohort* is one sequence of simple crossing terms, or one sequence of travelling fibre integrals, of a fibre integral of Gammas, coefficients and source factors whose pins carry a *grading*, a rational vector $\mathbf{v}$ with $\langle t_l, \mathbf{v} \rangle = 1$ for the column $t_l$ of every share in the pins, such as the share side of Theorem 31.4 and the kernel of Theorem 31.7, graded by their letter totals, and the share side of a Gramma whose configuration has a grading (Proposition 4.2), with the following data retained: the pins; the inherited contour; the crossed hyperplane and its orientation, if any; the source sectors and their coordinate cuts. The contours have the real points $\mathbf{a}^0 + n\mathbf{d}$ with $\mathbf{d}$ rational. Nonresonance is tested against the full displayed arrangement of the polar hyperplanes of the Gammas and of the source walls, including candidate walls which might cancel after the sectors are added: the real points lie on none of these hyperplanes other than the one being crossed, and at a source crossing each contributing sector has just one wall on that hyperplane. The constant terms of the shares and walls and the indices are real. Finitely many initial terms may be removed and kept as a finite head. Passing to finitely many arithmetic subsequences makes the changes of all falling shares and walls integral; this is the *integer step* of

the cohort, and its index $n$ always refers to that step. The same definition permits a finite product of source factors that are cube integrals in their charts (Lemma 8.4)

$$\operatorname{Reg}\int_{(0,1)^k} q^{\boldsymbol{\kappa}(\mathbf{a})-\mathbf{1}}\prod_i P_i(q)^{-w_i(\mathbf{a})}\,dq,$$

the $P_i$ with nonnegative coefficients and positive constant terms and the affine functions $\kappa_i$, $w_i$ with rational linear parts. A growing wall is eventually positive; a stationary wall may be negative and off the integers, and is then continued with a constant integer $\ell$ in Lemma 31.15. Finite coordinate box cuts are part of the retained data. The sector integrals of a fan of a seam word (Theorem 31.4(ii)), those of tables among them (Corollary 8.6), have these properties.

**Proposition 31.17** (Computing the cohorts of a rational ray)**.** *Let the pins and the finitely many candidate polar functions $\ell_i$ be rational, and let a rational open box in the affine space of the pins be given on which the required initial walls and shares are positive. For a prescribed rational direction a rational starting point in that box can be chosen such that distinct polar hyperplanes never meet the ray at the same time; factors on the same hyperplane are kept together. After a finite head, the crossing order and the separating gaps are a finite repeated pattern, and its crossing cohorts, endpoint cohorts, the integers $\ell$ of Lemma 31.15 and a uniform positive separation are computed by finite rational arithmetic. More precisely, with the direction multiplied by a common denominator so that every falling polar function decreases at an integer falling rate $d_i > 0$, there are at most $\sum_i d_i$ crossings in each interval of length one of this parameter; neither that number nor the stored source sectors depend on how many such intervals are read.*

**Theorem 31.18** (A finite procedure for a retained cohort)**.** *For a rational retained cohort there is a finite procedure, using the pins and the monomials of the source factors, with the following output, the* joint word *of the cohort, after a finite head and a possible change to the integer step:*

$$\mathcal{R}_n = G_n\sum_i\int_{Y_i} c_i(x)^n\,\phi_i(dx) = G_n\sum_i\big(\mathsf{T}_{-n;(1)}[c_i]\,\phi_i\big)(Y_i),\qquad n\in\mathbb{N},\tag{31.2}$$

*a finite sum over $i$. Here $Y_i$ is a compact cube times finitely many circles, the coefficient function $c_i$ is continuous on it, and the input $\phi_i$ is a complex measure of finite total variation; the coefficients, their branches, the orientations and every cut are retained. On every $Y_i$ each coefficient of a letter is a real Laurent monomial times a* coefficient factor*, holomorphic there, with its real part bounded below by a positive constant and its modulus bounded above, uniformly on the compact variables; these two bounds are the* margins *of the coefficient factors, and the radii $r < r_1 < rT < r_2$ of Lemma 31.15 are admissible while the margins hold. At the integer index $-n$ the Zamma of the one-term configuration is the polynomial $\mathsf{Z}_{-n;(1)}(c) = c^n$, at $c = 0$ as well, so the second expression is the Trailmoment of Definition 5.1 extended to a continuous complex coefficient function and a complex input of finite variation, as Theorem 5.21(ii) extends its reads to complex data where every integral converges absolutely.*

*The scalar is explicit. Let $\mathbf{v}$ be the grading of the pins (Definition 31.16) and $w = \langle \mathbf{v}, \boldsymbol{\nu}\rangle$, $\boldsymbol{\nu}$ the values of the pins, the total of the shares on the pins, which on the share sides of Theorems 31.4 and 31.7 is $|\mathbf{s}| + |\mathbf{t}|$, the sum of the exponents of the active letters. Let $\mathcal{S}_-$ be the shares treated by Lemma 31.14, with $\delta_l(n) = \delta_l(0) + d_l n$. At a Gamma crossing let $-N(n)$, with $N(n) = N_0 + d_0 n$, be the pole of its Gamma that is crossed; at a source crossing or for a travelling fibre omit $N$ and the factor $\Gamma(N+1)$ of $G_n$, and put $d_0 = 0$. Up to fixed nonzero factors of the share side, as $1/\Gamma(s_j)$*

*and $1/\Gamma(t_j)$, and the inherited crossing sign,*

$$G_n = \frac{(-1)^{N(n)+\sum_{l\in\mathcal{S}_-}\delta_l(n)}\,\Gamma\big(w+N(n)+\sum_{l\in\mathcal{S}_-}\delta_l(n)\big)}{\Gamma(N(n)+1)\prod_{l\in\mathcal{S}_-}\Gamma(\delta_l(n))},$$

*fixed nonzero constants may equally be put in the inputs. With $\bar d = d_0+\sum_{l\in\mathcal{S}_-} d_l$,*

$$\lim_{n\to\infty}|G_n|^{1/n} = r_G = \frac{\bar d^{\,\bar d}}{d_0^{d_0}\prod_{l\in\mathcal{S}_-} d_l^{d_l}}, \qquad 0^0 = 1;$$

*when $\bar d = 0$ the scalar is constant and $r_G = 1$, and in all cases $|G_n| \le C(n+1)^p r_G^n$ for computable $C$ and $p$. The procedure introduces at most six integrals per continued source wall and one Beta variable per negative retained Gamma, whatever $n$. Its remaining work is a finite common Newton fan and simplicial subdivision; this fan, the inputs and the coefficient functions $c_i$ serve for every $n$, and the procedure does not expand the Taylor head of degree $\ell(n)-1$ of Lemma 31.15 or a residue coefficient of degree $N(n)$ into its monomials.*

**Corollary 31.19** (Finite bounds and the exact positive root rate)**.** *For the joint word of Theorem 31.18 put $c_{\max} = \max_i \max_{Y_i}|c_i|$ and $M = \sum_i \|\phi_i\|$. Then*

$$|\mathcal{R}_n| \le M\,|G_n|\,c_{\max}^n \le CM(n+1)^p (r_G c_{\max})^n,$$

*and $c_{\max}$ is the* ceiling *of the joint word. Every quantity in this bound has a finite certificate: the margins of the coefficient factors and the sector walls bound $M$, and compact subdivision bounds $c_{\max}$; with computable input coefficients a strict inequality $r_G c_{\max} < r < 1$ is certified by finite subdivision and gives a computable geometric tail. If the original source has a convergent positive representation on every contour of the cohort, no source continuation or coefficient input is needed, and the remaining constant terms are real, then the output of the procedure has, after its common scalar sign is removed, one positive input $\phi$ and a coefficient function $c \ge 0$, and*

$$\lim_{n\to\infty}|\mathcal{R}_n|^{1/n} = r_G \operatorname*{ess\,sup}_{\phi} c.$$

*For a positive read on an unbounded domain, before it is compactified, the same holds with an infinite essential supremum whenever the moments exist. So averaging over a positive source measure takes the essential supremum on the joint input of the source, Beta and shape variables; it does not reverse the order of an infimum over directions and a supremum over source points.*

**Definition 31.20** (Simple tops of the joint word)**.** For (31.2) with $c_{\max} > 0$, tops are read as in Definition 20.50, with the coefficient function $c_i$ in place of $\varrho_\pi$, the oriented $Y_i$ in place of the keyhole contour and the input $\phi_i$ in place of $\phi_\pi$: a simple top is a point $p$ with $|c_i(p)| = c_{\max}$ for which local real coordinates on the oriented $Y_i$ are $(\mathbf{x},\mathbf{y}) \in [0,\epsilon)^{b_1}\times(-\epsilon,\epsilon)^{b_2}$, and

$$\log\frac{c_i(\mathbf{x},\mathbf{y})}{c_i(p)} = -\langle\boldsymbol{\lambda},\mathbf{x}\rangle - \tfrac12\,\mathbf{y}^{\mathsf T}\mathbf{Q}\,\mathbf{y} + O\big(|\mathbf{x}|^2+|\mathbf{x}|\,|\mathbf{y}|+|\mathbf{y}|^3\big), \qquad \phi_i = \prod_{j=1}^{b_1} x_j^{\kappa_j-1}\,g(\mathbf{x},\mathbf{y})\,d\mathbf{x}\,d\mathbf{y},$$

with $\kappa_j > 0$, $g(0,0)\neq 0$, $\operatorname{Re}\lambda_j > 0$ and $\operatorname{Re}\mathbf{Q}$ positive definite, $\mathbf{Q}$ complex symmetric, and $g$, continuous and bounded near $(0,0)$, including the orientation and every factor of the circle integrals. Its exponent and its coefficient are

$$\operatorname{ex}(p) = \sum_j \kappa_j + \tfrac{b_2}{2}, \qquad A(p) = g(0,0)\,(2\pi)^{b_2/2}(\det\mathbf{Q})^{-1/2}\prod_j \Gamma(\kappa_j)\,\lambda_j^{-\kappa_j},$$

the square root continued from positive definite real matrices. The subdivisions of the procedure, the splits of the Beta intervals and of the scales and the cones of the fan, divide one domain into parts on which the integrand is the restriction of one function; tops are read with these parts reassembled, a point on a common boundary of the parts being a simple top when the expansion holds on its reassembled neighbourhood, in the coordinates before the subdivision. Parts that are different integrals, as the segment and the circle integrals of Lemma 31.15, keep their own coefficients, and the contributions at one point are added before noncancellation is tested.

**Theorem 31.21** (The joint tops give the root rate of a complex word)**.** *Let the largest values of $|c_i|$ on the joint word of Theorem 31.18 sit at finitely many simple tops. Let $\mathrm{ex}_0$ be the least of their exponents, and for $|\zeta| = 1$ let $A_\zeta$ be the sum of the coefficients $A(p)$ of the tops with $\mathrm{ex}(p) = \mathrm{ex}_0$ and $c_i(p) = c_{\max}\,\zeta$. If some $A_\zeta$ is not zero, then*

$$\limsup_{n\to\infty} |\mathcal{R}_n|^{1/n} = r_G\, c_{\max}.$$

*No separate price of the phase is added to $c_{\max}$: the completed fibre and its source phases belong to the one coefficient function and the oriented input. A largest value of $|c_i|$ alone is not the hypothesis; the expansion includes the stationarity of the complex phase in every tangential variable.*

This is Theorem 20.51 for joint words: the coordinates $\mathbf{x}$ that vanish at a top carry the Gammas of the boundary, the coordinates $\mathbf{y}$ the factor $(2\pi)^{b_2/2}(\det \mathbf{Q})^{-1/2}$, and the coefficients of the tops of least exponent and one phase add before the root rate is read.

*Remark* 31.22 (Where the tops of a continued source factor sit)*.* In the integrals of Lemma 31.15 a falling wall carries the $n$-th power of $q^{-d}$ on the segment and of $\zeta^{-d}$ on the circles. On a circle the modulus of $\zeta^{-d}$ is constant and its phase turns, so a largest value of $|c_i|$ there is a simple top only where the other factors make the phase stationary, a saddle in the radius. The radii $r$, $r_1$, $r_2$ and $T$ are free within the margins of the coefficient factors, and the sum does not depend on them; Theorem 31.21 is applied after the radii are chosen. When the largest values of $|c_i|$ sit at simple tops of the segment, of the shape or of the Beta inputs, or at saddles of the circles, and do not cancel, the root rate is $r_G c_{\max}$; a largest value spread over a circle, or one at the end $q = rT$ of the segment, which moves with the radii, certifies the upper bound only, and the radii may be moved, within the margins, to lower it. Where the law has log expansions the continuation is made on the law, and its bound has no phase and no margin (Proposition 32.14).

**Corollary 31.23** (Closing along a path under an endpoint condition)**.** *In Lemma 31.3 keep (H0) and (H1) and replace (H2) by the weaker condition $\int_{\gamma(\tau_k)+iV} I \to 0$ along times $\tau_k \to \infty$ off the poles. Then the sums of the signed crossing terms before $\tau_k$ converge to $\int_{\mathbf{c}+iV} I$. If the crossing terms are absolutely summable, this is their absolutely convergent ordered sum; (H2) remains a sufficient condition for the endpoint condition.*

**Corollary 31.24** (Closing with the joint words)**.** *For a rational ray with simple nonresonant crossings, compute the finitely many arithmetic cohorts of crossings and one cohort of endpoints off the poles in each repeating gap. If each crossing cohort and each required endpoint cohort has a certified upper root rate $r_G c_{\max} < 1$, the crossing terms are absolutely summable and their ordered sum is the initial fibre integral. The finite head is retained, and the bounds of Corollary 31.19 give an effective stopping rule. A cohort with the positive hypothesis of Corollary 31.19, or the noncancelling simple tops of Theorem 31.21, has the stated exact root rate and cannot tend to zero when it exceeds one.*

The procedure keeps the falling rates, which fix the crossed heads, and the cut source; within the chamber the growing shares can still be changed by Theorem 31.6, the completed word being the same. The bound $r_G c_{\max}$ has no phase term: $c_{\max}$ is the ceiling of this joint word, which need not equal the ceiling of a keyhole continuation read before the fibre is completed, nor the least ceiling over all joint words.

**Lemma 31.25** (Positive capacity and the growing shares)**.** *Let $t_g \in \mathbb{R}^r$, $g \in G$, be a finite nonempty configuration with a grading $\langle t_g, \mathbf{v}\rangle = 1$, let $\xi_g > 0$, $w > 0$ and $\langle \mathbf{v}, \nu\rangle = w$, and in this statement write $L(X) = \sum_{g\in G} \xi_g X^{t_g}$ and $\Pi(\nu, w) = \{\mathbf{b} \in [0,\infty)^G : \ \sum_g b_g = w,\ \sum_g b_g t_g = \nu\}$. With $0^0 = 1$ and the infimum of the empty set equal to $+\infty$,*

$$\sup_{X\in(0,\infty)^r} \frac{X^\nu}{L(X)^w} = \inf_{\mathbf{b}\in\Pi(\nu,w)} \prod_{g\in G} \Big(\frac{b_g}{w\,\xi_g}\Big)^{b_g},$$

*the left side being the* capacity *of $L$ at $(\nu, w)$. The supremum may equally be taken on any logarithmic section transverse to $\mathbf{v}$. When $\nu/w$ lies in the relative interior of the convex hull of the $t_g$, the infimum has a unique positive minimizer and is attained by a gauge $X$, unique modulo the directions constant on all the $t_g$; on a proper face the same computation on that face gives the value, approached by suppressing the other monomials.*

The supremum on the left is the largest value of a monomial of the letter vector, and the logarithm of the infimum on the right is the least Gamma rate over the polytope $\Pi(\nu, w)$ of shares, less $w \log w$: the lemma is the duality that turns the chamber rate of a positive kernel into a capacity of its letter.

**Theorem 31.26** (The exact positive rate of a retained chamber)**.** *Take a Gramma $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ with rational configuration and real index read on its share side (Proposition 4.2), the pins $t\mathbf{a} = \nu$ with the grading of the letter totals and positive coefficients $\boldsymbol{\xi}$. Let a nonzero rational ray start at a positive real point of the pins, and consider any simple nonresonant Gamma crossing cohort of that ray, or any cohort of travelling fibres off the poles, with its integer step as in Definition 31.16. Write its direction per step as $\boldsymbol{\sigma}$, and put*

$$\mathcal{S}_- = \{l :\ \sigma_l < 0\}, \qquad d_l = -\sigma_l \ \ (l \in \mathcal{S}_-), \qquad \mathcal{S}_+ = \mathcal{S}_-^c,$$

*$t_l$ the columns of $t$, with the chamber of $\boldsymbol{\sigma}$ (Definition 31.5), the polar functions of the share side being the shares,*

$$C = \{\boldsymbol{\sigma}' :\ t\boldsymbol{\sigma}' = 0,\ \sigma'_l = -d_l\ (l \in \mathcal{S}_-),\ \sigma'_l \geq 0\ (l \in \mathcal{S}_+)\}.$$

*Then the retained terms have a common sign at each index, independent of $\boldsymbol{\xi}$, after their crossing orientation is fixed, and*

$$\lim_{n\to\infty} |\mathcal{R}_n(\boldsymbol{\xi})|^{1/n} = \exp \min_{\boldsymbol{\sigma}'\in C} h_\Gamma(\boldsymbol{\sigma}'),$$

*$h_\Gamma$ the Gamma rate of Theorem 31.4(iii), the coefficients being those of the Gramma: the exponential of the least Gamma rate of the chamber. This minimizes over the growing shares, the falling rates $d_l$ being fixed; if the integer step has the length $\Delta$ in the parameter of the ray, the logarithm is $\Delta$ times the rate per unit of that parameter.*

**Corollary 31.27** (Exact positive averaging at every retained dimension)**.** *In Theorem 31.26 let the coefficients be measurable positive functions $\xi_l(\zeta)$ of a source with a nonzero positive measure $\lambda$, the configuration, the index, the chamber and the integer step fixed across the source, and put*

$$h_C(\zeta) = \min_{\boldsymbol{\sigma}' \in C} \Big( \sum_l \sigma'_l \log|\sigma'_l| - \sum_l \sigma'_l \log \xi_l(\zeta) \Big), \qquad \hat{h}_C = \operatorname*{ess\,sup}_\lambda h_C.$$

*Let one averaged retained term be absolutely integrable, so that after multiplication by $\prod_l \xi_l(\zeta)^{-a_l^0}$, $\mathbf{a}^0$ the real point at which the cohort starts (Definition 31.16), the joint positive input is finite. If $\hat{h}_C < \infty$, every later averaged term is integrable and*

$$\lim_{n\to\infty} \Big| \int \mathcal{R}_n(\zeta)\, d\lambda(\zeta) \Big|^{1/n} = e^{\hat{h}_C};$$

*if all these terms are finite and $\hat{h}_C = +\infty$, their root rate is $+\infty$. No estimate uniform over the source and no retained fibre of dimension one is needed. This applies to the kernel of Theorem 31.7 on rational pins along a rational ray whose crossings are simple and nonresonant, under a finite averaged rate and the certificate* (31.1)*, whose bound supplies the finite first term. For a continuous coefficient map the essential supremum is the supremum on the support of the image of $\lambda$; it is $\operatorname{ess\,sup}_\zeta \min_C$, with no interchange to $\min_C \operatorname{ess\,sup}_\zeta$.*

**Theorem 31.28** (Averaging is sharp)**.** *In the setting of Theorem 31.7, with its chamber $C$, its averaged rate $\hat{h}_C$, its source measure $\lambda$ and the crossing terms $\mathcal{R}_k$ of the closing of its kernel at the crossing times $\tau_k$, let the pins be rational, as in the kinds (A) and (B), let $\boldsymbol{\sigma}$ have rational coordinates, let the crossings of its ray be simple and nonresonant (Definition 31.16), and let $\hat{h}_C < \infty$. Along each cohort of crossings, with its integer step of length $\Delta$ in $\tau$, the averaged terms have one sign at each index and $\lim_{n\to\infty} \big| \int \mathcal{R}_n\, d\lambda \big|^{1/n} = e^{\Delta \hat{h}_C}$ (Theorem 31.26, Corollary 31.27); so for every $\epsilon > 0$, $\big| \int \mathcal{R}_k\, d\lambda \big| \geq e^{\tau_k(\hat{h}_C - \epsilon)}$ for all large $k$, and the terms of the averaged closing, which are those of the closing of the share side along $\boldsymbol{\sigma}$ when $\int e^{-\langle \mathbf{a}, y\rangle}\, d\lambda$ converges along it, tend to zero when $\hat{h}_C < 0$ and do not when $\hat{h}_C > 0$. When every share that $\boldsymbol{\sigma}$ falls is alone in its bundle and the share side read with one share per bundle (Corollary 4.4 merging each bundle) has a fibre of dimension one, a positive multiple of $\boldsymbol{\sigma}$ has a direction with rational coordinates in its chamber, along which the closing is the same series after a change of scale of $\tau$ (Theorem 31.6); the crossings are simple and nonresonant as soon as no two lattices of the falling shares meet; and every crossing term is*

$$\mathcal{R}_k(\zeta) = r_k \prod_G e^{-\bar{a}_{k,G}\, y_G(\zeta)}, \qquad y_G = \log \sum_{l \in G} e^{y_l},$$

*with $r_k$ real and not depending on the source point and $\bar{\mathbf{a}}_k$ the crossing point of the merged shares.*

So Theorem 31.28 holds in every retained dimension, with no condition on the bundles: the falling shares need not be alone in them, and the share side merged by its bundles may have any number of free shares. The chamber minimum is a finite convex problem, shared by all the cohorts of the ray once their common integer step is chosen, and the completed source average is one joint Trailmoment. The positivity belongs to the completed source average; a continued source factor, with its oriented circle integrals, is read by Theorem 31.21, its phases not being replaced by a supremum over a positive support. A completed crossing and the tail it inherits are different series: closing a retained share only through finitely many of its poles leaves an exact Taylor remainder whose rate depends on how many poles are taken with each crossing, and it can grow while the completed crossings fall.

### 31.3.2 Coincident crossings and the boundary at root rate one

A residue is taken after the factors on its hyperplane have been assembled. A deformation may separate those factors for computation, but its separate residues are then one packet. The finite operation of Theorem 31.30 retains that packet, including its vanishing factors, and extends Theorem 31.18 to coincident crossings.

**Lemma 31.29** (A source pole with a uniform parameter margin)**.** *Use the radii of Lemma 31.15. Near $\kappa = -N$, $N \in \mathbb{N}$, put $z = \kappa + N$ and fix $0 < \eta < 1$. On $|\operatorname{Re} z| \le \eta$ the same continued source factor is exactly*

$$I_\kappa = \int_r^1 q^{\kappa-1} g(q)\, dq + R_{N+1} + R'_N + g_N \frac{r^z}{z}, \qquad g_N = \frac{1}{2\pi i} \oint g(\zeta)\, \zeta^{-N-1}\, d\zeta,$$

*$R_{N+1}$ and $R'_N$ given by the formulas of that lemma with these indices, $R'_0 = 0$. The first three terms are uniformly absolutely convergent on the whole strip; splitting $R'_N$ gives at most seven integrals in all, the circle integral $g_N$ carrying the explicit pole $r^z/z$. If $N = N_0 + dn$, the domains stay fixed and their dependence on $n$ is again in fixed powers; in particular a small complex circle of the parameter about $z = 0$ has a common integrable source representation after its displayed pole is removed.*

**Theorem 31.30** (A finite jet for a coincident crossing)**.** *Consider a rational affine family of crossing fibres, indexed by $n$, with $n$ in one fixed arithmetic progression. Fix the inherited contour and the whole packet of factors polar on its crossed hyperplane, let the other polar hyperplanes be separated from this packet on the real contour by a positive constant, and let the total order of the packet be at most $p$, independent of $n$. Perturb the constant terms of all shares and walls by one parameter $\varepsilon$, with a fixed rational vector of perturbations, keeping the affine pins and their prescribed continuation; the vector can be chosen outside finitely many rational hyperplanes so that distinct members of the packet separate, factors forced to remain coincident being kept as one member. Let $\mathcal{R}_{n,j}(\varepsilon)$ be the inherited crossing terms of the separated members. Then*

$$\mathcal{R}_n = [\varepsilon^0] \sum_j \mathcal{R}_{n,j}(\varepsilon).$$

*Each member has a pole of order at most $p - 1$ in $\varepsilon$ if its separated pole is simple, and the same bound suffices after keeping forced multiple members intact; so the constant coefficient uses only the Taylor coefficients through the degree $p - 1$ of the holomorphic parts after their poles are removed. Further meromorphic normalizing factors raise this order by their known pole orders. Every coefficient is a jet of the Gammas, of the source or of the parameter; no limit in $n$ is part of the computation. More generally, whenever a finite exact procedure returns terms $\mathcal{R}_{n,j} = \varepsilon^{-o_j} \widetilde{\mathcal{R}}_{n,j}$ with $\widetilde{\mathcal{R}}_{n,j}$ holomorphic, the required coefficients are exactly $[\varepsilon^{o_j}]\widetilde{\mathcal{R}}_{n,j}$; positive orders of a factor that vanishes at $\varepsilon = 0$ are retained just as negative orders of a polar factor are. The local decay hypothesis of Lemma 31.3 is used on the common enclosing contour, uniformly on a small disc of the parameter. For the Gamma words and the continued sector integrals a uniform procedure is available: factors whose Gamma poles lie on the crossed hyperplane are constant on its residual fibre and are put in the scalar Gamma word, each source wall coincident on that hyperplane is treated by Lemma 31.29, and the other walls and Gamma strips have strict margins.*

The auxiliary parameter separates the poles of one assembled crossing; it does not identify the independent regulators of a product of words, which keep their separate constant terms (Proposition 44.2). If the value asked for is a finite part, its prescribed Mellin regulator is retained and its

constant coefficient taken as in Theorem 1.7 and Corollary 1.8; finite parts in different coordinates are not interchangeable (Lemma 1.9: a dilation of the coordinate moves a finite part by a polynomial in the logarithm of the dilation). Several such regulators give finitely many multivariate Laurent coefficients, with the coefficient order fixed by the value asked for, the retention of finitely many numbers at every corner (Theorem 46.10) applied to a whole crossing fibre. The separation hypothesis concerns the actual residual contour: a transverse meeting of different hyperplanes needs the finite ordered packet of the chosen crossing flag, with its iterated transverse contours, and does not allow an arbitrary order of point residues. Nor does a Laurent coefficient keep a sharp root rate by itself: the coefficient can vanish.

**Proposition 31.31** (Bounds and top coefficients pass to jets)**.** *In Theorem 31.30 let the procedure of Theorem 31.18 for the separated terms give a uniform bound on a circle of the parameter,* $|\varepsilon^{o_j}\mathcal{R}_{n,j}(\varepsilon)| \le C_\delta(1+n)^{p'}(q+\delta)^n$ *for every* $\delta > 0$*, the circle allowed to shrink with* $\delta$*. Then* $\limsup_n |\mathcal{R}_n|^{1/n} \le q$*. For perturbations of the constant terms only, the slopes in* $n$*, the linear parts of the pins, the letters and the coefficients fixed, Lemma 31.29 and the extraction of the scalar give more: one common fan serves on a small disc of the parameter, each coefficient function* $c_i$ *is independent of the parameter, only the fixed amplitudes, the explicit polar factors and the constant terms of the Gammas varying, and their endpoint powers have positive margins once the explicit poles of Lemma 31.29 are removed; so the finitely many Laurent coefficients of each order have the same upper root rate as their common collection of coefficient functions and Gamma slopes, with only polynomial and logarithmic losses in* $n$*. If the joint word has finitely many simple tops (Definition 31.20), uniformly separated from the rest of its domain, its finitely many Laurent coefficients supply every prescribed finite order of the expansion of* $\mathcal{R}_n$ *at the tops: differentiate the exact integrand to the required order in the parameter, expand its phase and amplitude to the required Taylor order at each top, use* $\mathbf{x} = \mathbf{X}/n$ *in its decaying boundary coordinates and* $\mathbf{y} = \mathbf{Y}/\sqrt{n}$ *in its coordinates* $\mathbf{y}$*, integrate the resulting monomials against the Gamma densities in* $\mathbf{X}$ *and the density* $e^{-\frac{1}{2}\mathbf{Y}^{\mathsf{T}}\mathbf{Q}\mathbf{Y}}$ *in* $\mathbf{Y}$*, and expand the Gamma prefactors to the same order. Equal top phases are added before any test of nonvanishing. A surviving coefficient at the root rate* $q$ *proves* $\limsup_n |\mathcal{R}_n|^{1/n} = q$*; if the tested coefficients cancel, the upper bound remains and no sharp lower bound follows.*

At the root rate one the boundary is decided exactly, without simple or isolated tops, by the law of the coefficient function that the procedure retains in the positive case; its geometry is computed once and answers several queries at the boundary.

**Theorem 31.32** (The test at root rate one for a positive moment sequence)**.** *Let* $\phi$ *be a finite positive measure,* $0 \le c \le 1$ *on its support,* $m_n = \int c^n \, d\phi$*, and*

$$b_n = G_n \, m_n, \qquad G_n > 0, \qquad G_n \sim C \, n^\beta, \quad C > 0, \ \beta \in \mathbb{R} :$$

*the form of Corollary 31.19 at the root rate one, the coefficient function divided by its positive largest value* $c_{\max}$ *and the scalar multiplied by* $c_{\max}^n$*, the exponent* $\beta$ *read from the explicit Gamma quotient and the scalar sign a fixed number of modulus one to the power* $n$*. Finitely many initial terms have no effect.*

*(i) The series* $\sum_n b_n$ *is finite exactly when:* $\beta < -1$*, always;* $\beta = -1$ *and* $\int \log \frac{1}{1-c} \, d\phi < \infty$*;* $\beta > -1$ *and* $\int (1-c)^{-\beta-1} \, d\phi < \infty$*; the integrands of the last two conditions having the value* $+\infty$ *at* $c = 1$*. These are also the exact conditions for absolute convergence after multiplication by any* $\zeta^n$*,* $|\zeta| = 1$*.*

*(ii) Let moreover $G_n$ be a balanced Gamma quotient with its exponential factor removed, or more generally let its forward differences satisfy $\Delta^k G_n = O(n^{\beta-k})$ for every fixed $k$. For $|\zeta| = 1$, $\zeta \neq 1$, the ordered series $\sum_n \zeta^n b_n$ converges exactly when $b_n \to 0$, and with $M(t) = \phi\{c > 1-t\}$ this is: $\beta < 0$, always; $\beta = 0$ and $\phi\{c = 1\} = 0$; $\beta > 0$ and $M(t) = o(t^\beta)$ as $t \downarrow 0$.*

*So (i) and (ii) tell absolute convergence, conditional convergence and the failure of the term test apart with one retained law.*

A certified law near its largest value of the form $M(t) \asymp t^a(\log(1/t))^j$ turns these criteria into comparisons of powers and logarithms; the criteria stay exact for nonisolated or degenerate maxima, Dirac measures and mixtures. They do not change the crossing order: a periodic family must be reassembled in its original order before a conditional verdict is used, since splitting an alternating series into its even and odd subseries loses exactly that information.

**Proposition 31.33** (Finite certificates at the boundary from the jets of the tops)**.** *Let the assembled terms at the root rate one have a certified finite expansion*

$$\mathcal{R}_n = \sum_{\zeta \in Z} \zeta^n \sum_{\beta \geq -1} \sum_{j=0}^{J_\beta} c_{\zeta,\beta,j}\, n^\beta (\log n)^j + O(n^{-1-\delta}), \qquad \delta > 0,$$

*$Z$ a finite subset of the unit circle, the real exponents $\beta$ a finite set and equal triples combined; the jets of the tops of Proposition 31.31 produce such a certificate to every order needed when its hypotheses on simple tops and uniformity hold. Then:*

*(i) the series is absolutely convergent if and only if every displayed coefficient vanishes;*

*(ii) it converges in its given order if and only if every coefficient with $\zeta = 1$ vanishes and every coefficient with $\beta \geq 0$ vanishes;*

*(iii) when (ii) holds and (i) does not, the convergence is conditional.*

*The verdict uses exact certificates of zero or nonzero for the finitely many assembled coefficients; where those are unavailable, the finitely many coefficients are the output, a numerical near cancellation not being a certificate of zero.*

# Chapter 32

# Laws at their ends: regions of equality and reach

A closing past the walls reads a law where its moments diverge, at its falling ends, through its expansion there. The regions on which a law equals its expansion at an end are built from the sector letters, with the map from the motion of the shares to the exponents, the coefficients, the truncation bounds and exact boundary words (Theorems 32.2 and 32.4); the domain where a series converges and the region where it equals the law are kept apart (Proposition 32.6). The reach is read on the law along every rational direction of the shares, as at width one: along a rational direction the seam function is the Mellin transform of one measure on the half-line, the image of the law, multiplied by $\mathbf{x}^{-\mathbf{w}_0}$ for the starting point $\mathbf{w}_0$, under the directional ratio, and the reach is the far boundary of the region on which that measure is its own wall series (Theorem 32.10). Run on the laws of the two families at a node of width one, the same argument reads the growth of each family on its law and makes that of the bunch its reach (Corollary 32.11). For a piece the reach is bounded by the ceilings of the keyhole continuations along the direction, with no phase, and attained at their simple tops that do not cancel (Theorem 32.15). A closing past the walls converges when the exponential of the Gamma rate times the reach of its tube, the imaginary translates of the line through its starting point, is less than one, its terms then having at most that root rate, and a law with log expansions bounds that reach in every direction with nothing paid for the phase; with one free share the tube is the line, and the terms reach that root rate under conditions that keep the poles apart (Theorems 32.16 and 32.17, Corollary 32.18).

## 32.1 Regions on which a law is represented

A region of equality belongs to an expansion of a law: to the coordinates advanced, the coordinates retained, and the grouping of its terms. A disc in a sector variable is not by itself a region of equality of the law: the map to the seam torus can identify points, have critical values, or have an image of smaller dimension than its torus. The procedure of Theorem 32.4 keeps that map, and gives a law, with its tangential source retained, before any assertion about a density on the whole torus.

**Definition 32.1** (A chart at an end of a law)**.** Let a part of the source of a law have the coordinates $(q, z) \in (0, r_1) \times \cdots \times (0, r_k) \times Z$ and the form

$$q^{\boldsymbol{\kappa}-\mathbf{1}} A(q, z)\, dq\, \phi(dz), \qquad \kappa_i > 0,$$

$\phi$ a finite positive measure retained as an input and $A$ holomorphic in $q$ on a neighbourhood of a closed polydisc, uniformly bounded there for $z \in Z$; complex densities are allowed, and a fixed

monomial in the seam coordinates can be included in the integrand and removed after integration. The selected seam coordinates of the law at the end are $\check{x}_i^{-1}$, with

$$\check{x}_i = \prod_{j=1}^{k} q_j^{N_{ij}}\, \mathfrak{u}_i(q,z), \qquad 1 \le i \le k,$$

$\mathbf{N} = (N_{ij})$ an invertible integer matrix and the *chart factors* $\mathfrak{u}_i$ positive on the real source and holomorphic and nonzero on the complex neighbourhood; the other coordinates of the seam point are a retained map $\mathbf{C}(q,z)$, and branches are continued from the positive source. The *chart coordinates* are

$$\mathfrak{t}_j = \prod_i \check{x}_i^{(\mathbf{N}^{-1})_{ji}} = q_j \prod_i \mathfrak{u}_i(q,z)^{(\mathbf{N}^{-1})_{ji}}.$$

A certificate of the chart supplies positive radii $R_i$ and a uniform analytic inverse $q_j = \mathfrak{t}_j\, g_j(\mathfrak{t},z)$ for $|\mathfrak{t}_i| < R_i$, with bounds for the inverse, its Jacobian and the holomorphic functions used in Theorem 32.2 and Corollary 32.3 below; the cuts on $Z$ are retained. Write $\mathbf{1}_{\mathrm{cut}}(\mathfrak{t},z)$ for the indicator of any further retained cut of the source, $\mathbf{1}_{\mathrm{cut}} = 1$ when there is none; if a cut of the source moves in these coordinates, its indicator is retained in the integrand and is neither differentiated nor included in a holomorphic coefficient.

**Theorem 32.2** (Computing and using an end chart)**.** *For a chart of Definition 32.1 the following computations are effective.*

*(i) A smaller certified polydisc exists whenever the holomorphy and nonvanishing conditions hold uniformly on a compact tangential source. In its coordinates the source form is*

$$\mathfrak{t}^{\boldsymbol{\kappa}-\mathbf{1}}\, \widetilde{A}(\mathfrak{t},z)\, d\mathfrak{t}\, \phi(dz), \qquad \widetilde{A} = A(\mathfrak{t}g,z) \prod_j g_j^{\kappa_j - 1} \det \frac{\partial(\mathfrak{t}g)}{\partial \mathfrak{t}},$$

*the determinant positive on a small enough positive chart. The seam map is $(\mathfrak{t},z) \mapsto \big(\mathfrak{t}^{\mathbf{N}}, \mathbf{C}(\mathfrak{t}g(\mathfrak{t},z),z)\big)$; with $\mathbf{C}_{\mathfrak{t}}(z) = \mathbf{C}(\mathfrak{t}g(\mathfrak{t},z),z)$, its density in the normal coordinates $\check{x}$, a measure in the tangential ones, is*

$$\frac{\check{x}^{-\mathbf{1}}\mathfrak{t}^{\boldsymbol{\kappa}}}{|\det \mathbf{N}|}\, (\mathbf{C}_{\mathfrak{t}})_* \big(\mathbf{1}_{\mathrm{cut}}(\mathfrak{t},\cdot)\widetilde{A}(\mathfrak{t},\cdot)\,\phi\big)\, d\check{x},$$

*a density only in the selected normal coordinates, whose tangential value may be a singular measure.*

*(ii) Write $\widetilde{A} = \sum_{\mathbf{n}\ge 0} \widetilde{A}_{\mathbf{n}}(z)\, \mathfrak{t}^{\mathbf{n}}$. The coefficients are tangential words computed by finite jet arithmetic on the original letters; the inverse certifies that they represent the source and is not evaluated. If $|\widetilde{A}| \le M$ on $|\mathfrak{t}_i| = R_i$, then $|\widetilde{A}_{\mathbf{n}}(z)| \le M \prod_i R_i^{-n_i}$. For every bounded measurable test function $H$, the law on the chart is, for real $0 < \mathfrak{t}_i < R_i$,*

$$\mathfrak{t}^{\boldsymbol{\kappa}-\mathbf{1}} \sum_{\mathbf{n}\ge 0} \mathfrak{t}^{\mathbf{n}} \int_Z \widetilde{A}_{\mathbf{n}}(z)\, \mathbf{1}_{\mathrm{cut}}(\mathfrak{t},z)\, H\big(\mathfrak{t}^{\mathbf{N}}, \mathbf{C}(\mathfrak{t}g(\mathfrak{t},z),z)\big)\, \phi(dz)\, d\mathfrak{t}. \tag{32.1}$$

*The same seam map and the same retained cut occur in every term. With rational data the inverse branch is algebraic, specified by the positive branch and the certificate; alternatively every exact pairing with a test function is kept on the original source $(q,z)$ with that cut pulled back, and an inverse function need not be added to the operations. The series converges in*

*total variation on compact subcharts, the normal power included. If the tangential map and the cut do not depend on* $\mathfrak{t}$*, its coefficients are the fixed face laws* $\mathbf{C}_*(\widetilde{A}_{\mathbf{n}}\phi)$*; otherwise* (32.1) *is a series of pairings on the one tangential source, not an assertion that those images are fixed.*

*(iii) With* $\vartheta_i = |\mathfrak{t}_i|/R_i < 1$*, truncating to* $0 \le n_i < k_i$ *has, before the normal power and for* $\|H\|_\infty \le 1$*, the error at most*

$$M\,\phi(Z)\prod_i (1-\vartheta_i)^{-1}\sum_i \vartheta_i^{k_i}. \tag{32.2}$$

*It also has an exact boundary word: with* $T_i$ *the Taylor operator through the degree* $k_i - 1$ *in* $\mathfrak{t}_i$*,*

$$\widetilde{A} = \prod_i T_i\widetilde{A} + \sum_{\varnothing\neq I\subseteq\{1,\dots,k\}}\prod_{i\in I}(1-T_i)\prod_{j\notin I}T_j\widetilde{A}, \tag{32.3}$$

$$(1-T_i)\widetilde{A} = \frac{\mathfrak{t}_i^{k_i}}{2\pi i}\oint_{|\zeta_i|=R_i}\frac{\widetilde{A}(\mathfrak{t}_1,\dots,\zeta_i,\dots,\mathfrak{t}_k,z)}{\zeta_i^{k_i}(\zeta_i-\mathfrak{t}_i)}\,d\zeta_i :$$

*there are* $2^k - 1$ *boundary words, whatever the truncation orders* $k_i$*.*

*(iv) The chart records the map from the motion of the shares to the exponents. The following statements about walls and continuation concern normal Mellin powers with an analytic tangential composite, or a further word with its own sector certificate; the bounded measurable test functions of (ii) have only the convergent identity and the error bound, and no continuation is asserted for them. For the Mellin power in the seam coordinates* $\breve{x}_i^{-1}$ *at the shares* $\mathbf{z}$*, the normal exponent of a term is* $\boldsymbol{\kappa} + \mathbf{N}^{\mathsf{T}}\mathbf{z} + \mathbf{n} - \mathbf{1}$*. For a fixed tangential cut and a normal Mellin power, integrating these powers gives the walls* $\kappa_j + (\mathbf{N}^{\mathsf{T}}\mathbf{z})_j + n_j = 0$*; a moving cut keeps its indicator in the integrand, with no such set of poles. A motion* $\boldsymbol{\sigma}$ *of the shares in the coordinates of these Mellin powers has the wall slopes* $\mathbf{N}^{\mathsf{T}}\boldsymbol{\sigma}$*; for the original coordinates of the shares one first applies to* $\boldsymbol{\sigma}$ *the recorded linear change of the exponents of the Mellin powers. Other requested coordinates remain in the tangential word, and are not replaced by their values on the face unless that replacement is part of the stated test function.*

**Corollary 32.3** (Face jets for an analytic test function)**.** *Let the tangential cut be fixed, the map* $\mathbf{C}(\mathfrak{t}g(\mathfrak{t},z),z)$ *holomorphic on a specified polydisc about* $\mathfrak{t} = 0$*, that point included, and* $H$ *a bounded holomorphic tangential test function on a neighbourhood of its image. Expand* $\widetilde{A}(\mathfrak{t},z)\,H\big(\mathbf{C}(\mathfrak{t}g(\mathfrak{t},z),z)\big)$ *instead of* $\widetilde{A}$ *alone: its coefficient of degree* $\mathbf{n}$ *is a finite sum of tangential source integrals of derivatives of* $H$ *at* $\mathbf{C}(0,z)$*, multiplied by jets of* $\widetilde{A}$*,* $\mathbf{C}$ *and* $g$ *of degrees at most* $\mathbf{n}$*. These are fixed face functionals, and their series and boundary words obey the same coefficient bounds as in Theorem 32.2(ii), with the bound of this composite. A change of analytic tangential test function needs its jets, not a new continuation of the source. The normal monomial* $\mathfrak{t}^{\mathbf{N}}$ *remains unexpanded in the integrand; a normal test function may be expanded as well when its composite is holomorphic and bounded on the polydisc and its jets are supplied by finite operations, a further condition, since negative entries of* $\mathbf{N}$ *can give poles at* $\mathfrak{t} = 0$*.*

**Theorem 32.4** (Certified regions of equality and composition)**.** *For rational positive polynomial letters and rational monomial seam coordinates the following is a procedure.*

*(i) The finite sector data determine the exponent matrices and the chart factors of Definition 32.1. Every invertible normal minor gives a proposed end chart, with the other variables*

*and test functions retained; enumerating rational tangential boxes and radii and verifying the inequalities of the certificate and of nonvanishing produces certified charts. Every compact tangential set on the end face $q = 0$ on which the proposed chart has nonvanishing chart factors and an invertible normal matrix has a small enough normal neighbourhood covered by finitely many charts produced in this way. Extending a chart away from that face needs its certificate, or a new certificate of inverse functions for the full Jacobian; invertibility of* $\mathbf{N}$ *alone does not supply it.*

*(ii) The region of equality supplied by one chart is the image of its certified polydisc in* $\mathfrak{t}$ *under* $\check{x} = \mathfrak{t}^{\mathbf{N}}$*, with the specified branches and cuts; its positive part satisfies* $\sum_i (\mathbf{N}^{-1})_{ji} \log \check{x}_i < \log R_j$*,* $1 \le j \le k$*, and need not be a rectangle in the original seam coordinates. A finite atlas is certified for a requested region of equality only after its source patches cover the inverse image of that region; a patch not yet covered is retained as its original source word. So at every step the output is an exact finite head together with the boundary words* (32.3) *and the uncovered source word, with bounds for the parts certified small; an uncovered word is never omitted.*

*(iii) Atlases may be restricted to common source patches, added, transported by their seam maps, and composed with a further test function. On compact certified subregions every order of the absolutely convergent reads has the same value. At finite truncations the equality uses the boundary words* (32.3)*, and moving a cut retains the source between the old and the new cut. For words whose sector certificate supplies a meromorphic continuation, the same finite identities hold there, the full Laurent coefficient taken before pairing with a test function at coincident poles; an arbitrary bounded test function is not thereby continued. An infinite read in another grouping is asserted only with its own bound of convergence or as the continuation of the retained word.*

The retained data are the sector syntax, the matrix $\mathbf{N}$, the cuts, the recursion for the inverse jets and its certificate, and the tangential coefficient words, computed once per distinct source chart. For a prescribed error on $|\mathfrak{t}_i| \le \vartheta_i R_i$ the orders $k_i$ are read off (32.2); the finite head has $\prod_i k_i$ coefficient words and its exact remainder $2^k - 1$ boundary words. A repeated chart reuses its coefficient words, a new analytic test function supplies its finite face jets, and the certificate and the retained boundary remain valid when only the truncation changes.

**Proposition 32.5** (When a retained chart carries a density)**.** *A retained chart is in general not a chart of an ambient density. It becomes one in the selected normal coordinates and further tangential coordinates* $\mathbf{y}$ *when their image has a certified local product parametrization over* $(\mathfrak{t}, \mathbf{y})$*, with compact fibres, fixed cuts, a holomorphic form and an integrable uniform bound; integration over those fibres then gives a holomorphic coefficient function, and the series of Theorem 32.2, or of Corollary 32.3, represents that density with the same bounds. For a source form made of integrable powers of the boundary coordinates of a cube times a holomorphic density, such parametrizations are built near every compact regular fibre of a rational map* $\mathbf{C}$ *on the cube, when the map is defined on a neighbourhood of the closed cube, its restriction to every face meeting the fibre has a surjective differential, and its letters are nonzero. To assert a density for the whole law, the target patch must have no other or uncovered preimages, and in particular no omitted pole face may supply further source patches; otherwise the procedure describes the covered part of the law only, and the further preimages remain uncovered source words under Theorem 32.4. With the cube coordinates* $v_i$ *put*

$$\mathbf{J}_{ji} = v_i(1 - v_i)\, \partial_{v_i} C_j, \qquad X_j = \sum_i v_i(1 - v_i) \big[\mathbf{J}^{\mathsf{T}} (\mathbf{J}\mathbf{J}^{\mathsf{T}})^{-1}\big]_{ij} \partial_{v_i};$$

*then* $D\mathbf{C}(X_j) = \mathbf{e}_j$*. On a small enough neighbourhood the flows of these vector fields carry the reference fibre to the neighbouring fibres, keep all faces and give the product parametrization. Uniform nonvanishing of* $\det(\mathbf{J}\mathbf{J}^{\mathsf{T}})$*, of the letters and of the relevant Jacobians, together with a bound for the vector fields on a larger neighbourhood, is a finite certificate; the iteration of the integral equations of the flows, certified to converge geometrically, computes the local flows and their holomorphic extensions.*

The possible failures of this procedure are explicit: zeros of letters, loss of properness at an omitted face of the source, or loss of surjectivity of the differential of the seam map on a face. Their images are candidate boundaries, not automatically singularities of the law, since different contributions can cancel. If the source has smaller dimension than the number of requested seam coordinates, the retained law and its atlas still exist but no ambient density is asserted. A genuine edge of the support is an obstruction: if the law vanishes on an open set on one side of an edge and has positive measure arbitrarily close on the other, it has no analytic density across that edge, by the identity theorem, as in Theorem 20.44(iii).

**Proposition 32.6** (Convergence domains and equality regions)**.** *Let the fixed face laws of a certified chart, or the face functionals for a fixed bounded set of analytic test functions, be* $\widetilde{A}_{\mathbf{n}}$ *in a Banach space. The interior* $\Omega_{\mathrm{abs}}$ *of*

$$\Big\{\mathfrak{t} : \sum_{\mathbf{n}\geq 0} \|\widetilde{A}_{\mathbf{n}}\|\,|\mathfrak{t}^{\mathbf{n}}| < \infty\Big\}$$

*is the logarithmically convex* domain of absolute convergence *of that series, which contains with each of its points* $\mathfrak{t}$ *every* $\mathfrak{t}'$ *with* $|\mathfrak{t}'_i| \leq |\mathfrak{t}_i|$ *for all* $i$*, and every certified polydisc lies in it. Its* region of equality *is retained separately: the chart patches covered by the source on which the series is proved to read the law, together with the patches reached by a chain of transports of charts that keep the equality. The series represents the law only on the intersection of the domain* $\Omega_{\mathrm{abs}}$ *and the region of equality; analytic continuation of the sum alone is not a transport of the equality across an edge of the support or a singular value of the law. For a positive integer vector* $\mathbf{d}$ *and a fixed point* $\mathfrak{t}$ *of the chart retain the grouping*

$$\widetilde{A}_{(N)}(\mathfrak{t}) = \sum_{\langle \mathbf{d},\mathbf{n}\rangle = N} \widetilde{A}_{\mathbf{n}}\,\mathfrak{t}^{\mathbf{n}};$$

*the grouped analytic germ in one variable* $v$ *on the curve* $(\mathfrak{t}_i v^{d_i})_i$ *has the exact radius* $R$ *with* $R^{-1} = \limsup_N \|\widetilde{A}_{(N)}(\mathfrak{t})\|^{1/N}$*: inside it the grouped series converges, and outside it its grouped terms do not tend to zero. These statements concern the germ, and they become statements about the law only where its region of equality is certified as well; on the boundary of convergence the coefficients decide. Completing another read replaces these grouped terms by their images, whose radius can be larger, the equality with the new expansion of the law still using the transported source and cuts. Neither radius is a reach of the law by definition, nor is it replaced by the supremum of an unintegrated modulus of the source.*

The comparison of radii for a bounded read coefficient by coefficient concerns the same normal coordinates and the same bank of coefficients. Merging a law and then choosing a new coordinate is another operation: the source is carried by that map and its expansion is regraded in the new coordinate, whose coefficients and grouped terms need not be bounded images of the old ones, and whose boundary of equality can come earlier. The old and the new expansion agree through their common source, with the cuts, the finite heads and the boundary words transported; their

domains of convergence and atlases of equality are recorded in their own coordinates. Maximal analytic continuation, and deciding every cancellation at a singular boundary, are neither used by the procedure nor concluded from a ceiling. In one selected coordinate with a fixed tangential test function the procedure certifies local patches of equality with the wall series, and the global inversion of Theorem 32.10 (Theorem 20.44 at width one) identifies the reach of the law along that coordinate; in several coordinates the certified atlas of equality, the domain of convergence of the series and its retained grouping are stored separately.

## 32.2 The reach along a direction

At width one the reach of a bunch is read on the law of its ratio, as the point beyond which the law is its own wall series (Theorem 20.44); it is bounded above by the ceilings of the keyhole continuations and attained at their simple tops that do not cancel (Theorems 20.49 and 20.51), and against a room bounded below the terms of the node have the root rate of the reach over the least ratio of the room, which infinitely many of them reach (Theorem 20.39). A closing past the walls at any width advances along one direction of the shares, and along a rational direction the seam function is the Mellin transform of one measure on the half-line, the directional law (Definition 32.9). The theory of width one applies to that measure line by line, and its counterparts are stated along a direction.

**Definition 32.7** (Walls and growth bounds)**.** Let a piece have its seam function $P_{W_B(\mathbf{w})}(\mathbf{m}_B)$ on the pins of the seam, an affine space $A_B$ of real shares $\mathbf{w} = (\mathbf{a}, \mathbf{b})$ of the active letters with the space of directions $\vec{A}_B$, of dimension $n_{\mathcal{D}}$; for $\mathbf{w} \in A_B + i\vec{A}_B$ it is $\int \mathbf{x}^{-\mathbf{w}}\, d\big((\boldsymbol{\varrho}_B^{\rho})_*\phi_B^{\rho}\big)(\mathbf{x})$ where this converges, the law being carried by the closure of the image of the open simplex under the seam vector. The seam function has *walls* when

(W1) there are finitely many affine functions $\kappa_U$ on $A_B$ with rational linear parts, its *walls*, such that it converges at a real point exactly when every wall is positive there, which happens on an open convex set $\Omega_B \subseteq A_B$, and it continues meromorphically to $A_B + i\vec{A}_B$ with its poles on the hyperplanes on which a wall lies in $-\mathbb{N}$;

and it has *growth bounds* when

(W2) for every $\theta > 0$ there are $M$, $C_0$, $n_1$ and $n_2$ with $|P_{W_B(\mathbf{z})}(\mathbf{m}_B)| \le C_0\, \delta^{-n_2}(1 + |\mathbf{z}|)^{n_1} M^{|\operatorname{Re}\mathbf{z}|_1} e^{\theta|\operatorname{Im}\mathbf{z}|_1}$, $\delta$ the smaller of 1 and the distance of $\mathbf{z}$ from these hyperplanes.

The same definitions apply to the period of any word whose exponents are affine functions of the shares.

Every piece whose seam function converges at a real point has walls and growth bounds, and so does the seam function of its seam letters merged by bundles: the seam word of a piece is a word in positive letters, and so is the word of the merged seam letters of Theorem 31.7, sums of seam letters of several stations of one excess, which are no longer letters of tables; such a word continues past its walls as the seam word of every piece does (Theorem 31.4(ii)).

**Lemma 32.8** (A word in positive letters continues past its walls)**.** *(i) Let $W = \prod_l L_l^{-w_l}$ be a product of powers of positive letters on a set $B$ (Definition 25.1), whose exponents $w_l$ and masses $\mathbf{m}_B$ are affine functions of $\mathbf{z} \in \mathbb{C}^p$ with rational linear parts, the masses on the balance hyperplane of $W$ for every $\mathbf{z}$, and let its period converge at a real point $\mathbf{z}_0$, the exponents there being of any sign. Then there are finitely many affine functions $\kappa_U$ on $\mathbb{C}^p$ with rational linear parts, its walls, such that the period converges at a real point exactly when every $\kappa_U$ is*

*positive there, and* $P_W(\mathbf{z})$ *continues meromorphically to* $\mathbb{C}^p$*, with its poles, of order at most* $|B|-1$*, on the hyperplanes* $\kappa_U(\mathbf{z}) \in -\mathbb{N}$*; and for every* $\theta > 0$ *there are* $M$*,* $C_0$ *and* $n_1$ *with*

$$|P_W(\mathbf{z})| \le C_0\, \delta^{-(|B|-1)} (1+|\mathbf{z}|)^{n_1} M^{|\operatorname{Re}\mathbf{z}|_1} e^{\theta |\operatorname{Im}\mathbf{z}|_1},$$

$\delta$ *the smaller of* 1 *and the distance of* $\mathbf{z}$ *from these hyperplanes.*

*(ii) For a piece and a partner read in its monomials, as in Theorem 31.7(B), merge the shares of each bundle* $G$ *of Theorem 31.7 into* $\bar{a}_G$*, and put* $L_G = \sum_{l\in G} \xi_l\, V_{jD(l)}$ *for a bundle of shares of* $K_j$*, and* $L_G$ *alike with the seam letters* $V^*_{jD}$ *for a bundle of shares of* $\Psi_j$*. On the pins of the merged shares the* merged seam word $W^{\mathrm{m}}_B(\bar{\mathbf{a}}) = \prod_G L_G^{-\bar{a}_G}$ *is such a product at the masses* $\mathbf{m}_B$*, and its period, the* merged seam function*,*

$$P_{W^{\mathrm{m}}_B(\bar{\mathbf{a}})}(\mathbf{m}_B) = \int \prod_G e^{-\bar{a}_G\, y_G}\, d\lambda, \qquad y_G = \log \sum_{l\in G} e^{y_l},$$

*reads the merged values* $y_G$ *at the source points, against the source measure* $\lambda$ *(Theorem 31.28); it converges at the sums over the bundles of every real point with positive coordinates at whose sums over the monomials the seam function of the piece converges.*

For the seam word of a piece, at the masses $\mathbf{m}_B$ and with the shares as its exponents, Lemma 32.8(i) gives walls and growth bounds whenever the seam function converges at a real point, the functions $\kappa_U$ of the lemma serving as the walls, with $n_2 = |B|-1$; when the shares are positive at that point the seam word has a Newton sum of full dimension (Theorem 25.2), and the walls of the generators of a fan of the seam word serve, with the sector integrals of Theorem 31.4(ii), the proper walls for a piece of tables. A piece whose seam function on the pins is a product of Beta corners $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix} x,\ y \\ x+y \end{bmatrix}$ of affine functions of the shares with rational linear parts and of real exponentials of the shares, converging exactly where those affine functions are positive, has walls, those functions, and growth bounds, by the reflection formula (Proposition 1.3(v)) and the asymptotic expansion of $\log\Gamma$ away from the negative real axis, and growth bounds even at $\theta = 0$ when for each Beta corner the linear parts of its two arguments on $\vec{A}_B$ are proportional, as when one argument does not depend on the shares.

**Definition 32.9** (Directional ratios and laws; the reach along a direction)**.** Let a piece have walls and growth bounds (Definition 32.7), and let $\bar{\boldsymbol{\sigma}} \in \vec{A}_B \setminus \{0\}$ have rational coordinates. Its *directional ratio* is the positive function $\mathbf{x}^{-\bar{\boldsymbol{\sigma}}}$ on the seam torus. For $\mathbf{w}_0 \in \Omega_B$ and $\bar{\boldsymbol{\vartheta}} \in \vec{A}_B$ the *directional law* at $\mathbf{w}_0 + i\bar{\boldsymbol{\vartheta}}$ is the image

$$\lambda_{\bar{\boldsymbol{\sigma}},\mathbf{w}_0+i\bar{\boldsymbol{\vartheta}}} = \left(\mathbf{x}^{-\bar{\boldsymbol{\sigma}}}\right)_* \Big( \mathbf{x}^{-\mathbf{w}_0 - i\bar{\boldsymbol{\vartheta}}}\, (\boldsymbol{\varrho}^{\rho}_B)_* \phi^{\rho}_B \Big),$$

a complex measure on $(0,\infty)$ whose variation is at most the positive directional law $\lambda_{\bar{\boldsymbol{\sigma}},\mathbf{w}_0}$, which integrates to the seam function at $\mathbf{w}_0$; it is *oscillating* when $\bar{\boldsymbol{\vartheta}} \neq 0$, and the line $\{\mathbf{w}_0 + i\bar{\boldsymbol{\vartheta}} + z\bar{\boldsymbol{\sigma}} : z \in \mathbb{R}\}$ is the *imaginary translate* by $\bar{\boldsymbol{\vartheta}}$ of the line through $\mathbf{w}_0$. Its *line function* is

$$\ell(z) = P_{W_B(\mathbf{w}_0+i\bar{\boldsymbol{\vartheta}}+z\bar{\boldsymbol{\sigma}})}(\mathbf{m}_B) = \int_0^\infty X^z\, d\lambda_{\bar{\boldsymbol{\sigma}},\mathbf{w}_0+i\bar{\boldsymbol{\vartheta}}}(X),$$

absolutely convergent for $\operatorname{Re} z$ in the open interval of the $x$ with $\mathbf{w}_0 + x\bar{\boldsymbol{\sigma}} \in \Omega_B$, which contains 0, and continued meromorphically by (W1). A wall falls along $\bar{\boldsymbol{\sigma}}$ when its falling rate $-\dot{\kappa}_U$,

$\dot\kappa_U = \partial_{\bar{\boldsymbol\sigma}}\kappa_U$, is positive (Definition 31.5). The poles of $\ell$ with $\operatorname{Re} z \ge 0$ lie among the *candidate poles*

$$z = \big(\kappa_U(\mathbf{w}_0) + n + i\,\partial_{\bar{\boldsymbol\vartheta}}\kappa_U\big)/(-\dot\kappa_U), \qquad n \in \mathbb{N},$$

of the falling walls: on finitely many horizontal lines, with real parts in finitely many cosets of one lattice $N_0^{-1}\mathbb{Z}$, the falling rates being rational. Let $\Sigma$ be the half-strip of the $z$ with $\operatorname{Re} z \ge 0$ and $\operatorname{Im} z$ within 1 of the interval spanned by 0 and these lines. The *reach along* $\bar{\boldsymbol\sigma}$ at $\mathbf{w}_0 + i\bar{\boldsymbol\vartheta}$ is

$$r(\bar{\boldsymbol\sigma}; \mathbf{w}_0 + i\bar{\boldsymbol\vartheta}) = \inf\big\{M \ge 0 :\ \text{for every } \delta > 0 \text{ there are } C, n' \text{ with } |\ell(z)| \le C(1+|z|)^{n'} M^{\operatorname{Re} z}\big\},$$

the bound asked at every $z \in \Sigma$ at distance at least $\delta$ from the candidate poles; it is finite by (W2), and by the maximum modulus principle on circles about the candidates it does not change when the actual poles are used. A *wall series* is as in Definition 20.43, the $\eta_i$ allowed to be complex and its condition read as $\sum_{\eta,j} |a_{\eta,j}| x'^{-\operatorname{Re}\eta} < \infty$ for every $x' > x_0$, $x_0$ its radius. At $\bar{\boldsymbol\vartheta} = 0$ the directional law is the *law along* $\bar{\boldsymbol\sigma}$. The same definitions apply to the merged seam function of Lemma 32.8(ii) on the pins of the merged shares, the merged values $e^{y_G}$ in place of the seam coordinates and the source measure in place of the law.

The falling end of the directional law is $X \to \infty$, where the walls that fall along $\bar{\boldsymbol\sigma}$ make its moments diverge. In an end chart of Theorem 32.2 the Mellin power at the shares $\mathbf{z}$ is $\mathfrak{t}^{\mathbf{N}^{\mathsf T}\mathbf{z}}$, so the directional ratio is $\mathfrak{t}^{-\mathbf{d}}$, $\mathbf{d} = -\mathbf{N}^{\mathsf T}\bar{\boldsymbol\sigma}$ the falling rates of the walls of the chart, times a power of the tangential coordinates: the wall series of the directional law groups the terms of the chart by their exponents $(\kappa_j + n_j)/d_j$ along the direction, and the reach is the far boundary, in $X$, of the region on which the law equals that expansion. At width one with one table and factoring weights, normalized as in Proposition 28.11, the seam torus is a curve, $\vec{A}_B$ is the line of $\bar{\boldsymbol\sigma} = (-1, 1, -1, 1)$ in the shares $(a_{D_0}, a_{D_1}, b_{D_0}, b_{D_1})$, and the directional ratio is $x^*_{D_0}/x_{D_1}$, which is $\varrho_B$ at the seam points (Proposition 28.11); every coordinate of the torus is a power of $\varrho_B$, so the directional law is a power of $\varrho_B$ times the law of Definition 20.43, and by Theorems 32.10 and 20.44 the reach along $\bar{\boldsymbol\sigma}$ is the reach $\operatorname{reach}(B)$ of the bunch.

**Theorem 32.10** (The reach along a direction is read on the law)**.** *Let $\bar{\boldsymbol\sigma}$, $\mathbf{w}_0$, $\bar{\boldsymbol\vartheta}$, the directional law $\lambda = \lambda_{\bar{\boldsymbol\sigma}, \mathbf{w}_0 + i\bar{\boldsymbol\vartheta}}$, its line function $\ell$ and its reach $r$ be as in Definition 32.9.*

*(i) On $(r, \infty)$ the measure $\lambda$ has a density which is a wall series of radius $r$. Its terms are the poles of $\ell$ with $\operatorname{Re} z > 0$: a pole $\eta$ with the principal part $\sum_j (-1)^{j+1} j!\, a_{\eta,j}(z-\eta)^{-j-1}$ gives the terms $a_{\eta,j} X^{-\eta-1}(\log X)^j$, and there are no others.*

*(ii) If on some $(x_0, \infty)$ the measure $\lambda$ has a density which is a wall series of radius $x_0$, then $r \le x_0$; and for every $x' > x_0$ there is $C$ with $|\ell(z)| \le C\delta^{-J-1} x'^{\operatorname{Re} z}$ at every $z$ with $\operatorname{Re} z \ge 0$ at distance at least $\delta \le 1$ from the poles, whatever $\operatorname{Im} z$, $J$ the largest power of the logarithm in the series.*

*(iii) So $r$ is the least $x_0 \ge 0$ for which $\lambda$ has on $(x_0, \infty)$ a density which is a wall series of radius $x_0$. It does not change when $\mathbf{w}_0$ moves along $\bar{\boldsymbol\sigma}$, $r(c\bar{\boldsymbol\sigma}; \cdot) = r(\bar{\boldsymbol\sigma}; \cdot)^c$ for rational $c > 0$, and $r$ is at least every point near which $\lambda$ has no real-analytic density. At $\bar{\boldsymbol\vartheta} = 0$ the law $\lambda$ is positive and carried by the closure of the values of $\mathbf{x}^{-\bar{\boldsymbol\sigma}}$ on the support of the law of the piece: $r \ge X_{\min}$, the least of these values, when it is positive, the least value of the directional ratio taking the place of the least ratio (Proposition 20.35); and no wall falls along $\bar{\boldsymbol\sigma}$ exactly when $\mathbf{x}^{-\bar{\boldsymbol\sigma}}$ is bounded on that support, and then $r = X_{\max}$, the largest. When the seam word has a Newton sum of full dimension, as for a piece of tables whose tables form a connected family (Theorem 31.4(ii)), $X_{\min} = (\sup \mathbf{V}_B^{\bar{\boldsymbol\sigma}})^{-1}$ and $X_{\max} = \sup \mathbf{V}_B^{-\bar{\boldsymbol\sigma}}$, the suprema of the monomials of the letter vector of Theorem 31.2 (Definition 31.1).*

*(iv)* *At $\bar{\boldsymbol{\vartheta}} = 0$ let $\mathbb{L} = \{z_0 + kz_1 : \ k \in \mathbb{N}\}$, $z_1 > 0$ rational, be a progression at positive distance from the poles of $\ell$, and put $r_{\mathrm{res}} = \limsup_\eta \max_j |a_{\eta,j}|^{1/\eta}$ over the poles and $r_{\mathbb{L}} = \limsup_k |\ell(z_0 + kz_1)|^{1/(z_0+kz_1)}$. Then $r = \max(r_{\mathrm{res}}, r_{\mathbb{L}})$: the reach is carried by the residues of the line or by its values on every progression.*

*(v)* *For every $\theta \geq 0$, $r$ is the infimum of the $M \geq 0$ for which, for every $\delta > 0$, there are $C$ and $n'$ with $|\ell(z)| \leq C(1 + |z|)^{n'} M^{\operatorname{Re} z} e^{\theta |\operatorname{Im} z|}$ at every $z$ with $\operatorname{Re} z \geq 0$ at distance at least $\delta$ from the candidate poles: the growth of the line function on its whole half-plane, at any allowance in the imaginary direction, is the reach.*

At width one the same argument, run on the law of a family at a node, reads the growth of the family on its law, and that of the bunch is its reach.

**Corollary 32.11** (At width one the growth of a family is read on its law)**.** *In Section 20.7 let $\lambda_B$ be the image of the base form $\phi_B$ under $\varrho_B$ and $\lambda_R$ the image of $\phi_R$ under $1/\varrho_R$. For $X \in \{B, R\}$ the growth* $\mathrm{growth}(X)$ *along the level $d$ (Definition 20.33) is the least $x_0 \geq 0$ for which $\lambda_X$ has on $(x_0, \infty)$ a density that is a wall series of radius $x_0$. In particular* $\mathrm{growth}(B) = \mathrm{reach}(B)$*: the inequality* $\mathrm{reach}(B) \leq \mathrm{growth}(B)$ *of Theorem 20.34(i) is an equality, and Theorem 20.34(iii) closes a node on the lattices of both families when the two points beyond which their laws are their own wall series multiply to less than one.*

A closing of the share side moves the imaginary parts of the shares over a whole fibre, so its bounds read the seam function on every imaginary translate of a line as well as on the line itself.

**Corollary 32.12** (The reach of a tube)**.** *Let $\bar{\boldsymbol{\sigma}}$ and $\mathbf{w}_0$ be as in Definition 32.9 and $V' \subseteq \vec{A}_B$ a subspace containing $\bar{\boldsymbol{\sigma}}$. For $\theta \geq 0$ let $g_{V'}(\bar{\boldsymbol{\sigma}}; \theta)$ be the infimum of the $M$ for which there are $C$, $n_1$ and $n_2$ with*

$$\left|P_{W_B(\mathbf{w}_0 + \tau\bar{\boldsymbol{\sigma}} + i\bar{\boldsymbol{\vartheta}})}(\mathbf{m}_B)\right| \leq C\,\delta^{-n_2}(1 + \tau + |\bar{\boldsymbol{\vartheta}}|)^{n_1} M^\tau e^{\theta|\bar{\boldsymbol{\vartheta}}|_1} \qquad (\tau \geq 0,\ \bar{\boldsymbol{\vartheta}} \in V'),$$

*$\delta$ the smaller of 1 and the distance from the hyperplanes of (W1), and let $r^\sharp_{V'}(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) = \lim_{\theta \downarrow 0} g_{V'}(\bar{\boldsymbol{\sigma}}; \theta)$ be the* reach of the tube *over $V'$.*

*(i)* *$r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0 + i\bar{\boldsymbol{\vartheta}}) \leq r^\sharp_{V'}$ for every $\bar{\boldsymbol{\vartheta}} \in V'$; in particular $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) \leq r^\sharp_{V'}$, with equality when $V' = \mathbb{R}\bar{\boldsymbol{\sigma}}$.*

*(ii)* A certificate on the real fibres. *Let $\mathbf{v}_1, \ldots, \mathbf{v}_{n_{\mathcal{D}}-1}$ complete $\bar{\boldsymbol{\sigma}}$ to a basis of $\vec{A}_B$, and $y_i = \langle \mathbf{v}_i, \log \mathbf{x} \rangle$ on the torus. If the image of the measure $\mathbf{x}^{-\mathbf{w}_0}(\boldsymbol{\varrho}^\rho_B)_* \phi^\rho_B$ under $(\mathbf{x}^{-\bar{\boldsymbol{\sigma}}}, \mathbf{y})$, on the part where $\mathbf{x}^{-\bar{\boldsymbol{\sigma}}} > x_0$, is $\sum_{\eta,j} X^{-\eta-1}(\log X)^j dX \otimes \phi_{\eta,j}(d\mathbf{y})$, $j \leq J$, with finite signed measures $\phi_{\eta,j}$, the $\eta$ in finitely many sets $\eta_i + \mathbb{N}$, and $\sum_{\eta,j} \|\phi_{\eta,j}\| x'^{-\eta} < \infty$ for every $x' > x_0$, then $g_{\vec{A}_B}(\bar{\boldsymbol{\sigma}}; 0) \leq x_0$, so that $r^\sharp_{\vec{A}_B} \leq x_0$: an expansion at the face over the real fibres certifies the tube, with no allowance for the phase.*

The certificate asks the expansion to hold over whole fibres of the directional ratio, with fixed measures on them, along one direction. Read in the coordinates in which the shares pair with the law linearly, expansions with fixed tangential measures at the falling ends certify the tube in every direction at once: in those coordinates a complex share multiplies the law by a factor of modulus one, and the continuation is made on the law itself.

**Definition 32.13** (Log expansions of a law)**.** For a piece, with the space $\vec{A}_B$ of the directions of its shares of Definition 32.7 and its dual $\vec{A}^*_B$, the *log-ratio map* sends a point $\mathbf{x}$ of the seam torus

to the linear function $\mathbf{z} \mapsto \langle \mathbf{z}, \log \mathbf{x} \rangle$ on $\vec{A}_B$, written $\log \mathbf{x}|_{\vec{A}_B}$; for a real point $\bar{\mathbf{c}}$ of the pins at which the seam function converges, $\lambda_{\bar{\mathbf{c}}} = (\log \mathbf{x}|_{\vec{A}_B})_* \big(\mathbf{x}^{-\bar{\mathbf{c}}} (\boldsymbol{\varrho}_B^{\rho})_* \phi_B^{\rho}\big)$ is a finite positive measure on $\vec{A}_B^*$, and the seam function at $\bar{\mathbf{c}} + \mathbf{z}$ is $\int e^{-\langle \mathbf{z}, f \rangle}\, d\lambda_{\bar{\mathbf{c}}}(f)$ for $\mathbf{z} \in \vec{A}_B + i\vec{A}_B$ wherever the integral converges absolutely. A *log expansion* is the image of the complex measure

$$\sum_{\boldsymbol{\eta}, \mathbf{j}} \mathbf{p}^{\mathbf{j}} e^{-\langle \boldsymbol{\eta}, \mathbf{p} \rangle}\, d\mathbf{p} \otimes \phi_{\boldsymbol{\eta}, \mathbf{j}}(df) \qquad \text{on } [0, \infty)^k \times Y_Q$$

under $(\mathbf{p}, f) \mapsto f - \sum_{h \le k} p_h u_h$, where $u_1, \dots, u_k \in \vec{A}_B^*$ are linearly independent, $\boldsymbol{\eta}$ runs over the union of finitely many translates of $\mathbb{N}^k$ by positive vectors, $\mathbf{j} \le \mathbf{J}$, and the $\phi_{\boldsymbol{\eta}, \mathbf{j}}$ are complex measures on a compact set $Y_Q \subset \vec{A}_B^*$, its *inner face*, with $\sum \|\phi_{\boldsymbol{\eta}, \mathbf{j}}\| < \infty$. The law *has log expansions at* $\bar{\mathbf{c}}$ when $\lambda_{\bar{\mathbf{c}}}$ is the sum of finitely many log expansions $Q$ and of a complex measure $\phi_0$ carried by a compact set $Y_0$. Its *inner bound* along a direction $\bar{\boldsymbol{\sigma}} \in \vec{A}_B$ is the largest value of the directional ratio $e^{-\langle \bar{\boldsymbol{\sigma}}, f \rangle}$ on $Y_0$ and on the inner faces,

$$X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}}) = \max \Big( \max_{f \in Y_0} e^{-\langle \bar{\boldsymbol{\sigma}}, f \rangle},\ \max_{Q} \max_{f \in Y_Q} e^{-\langle \bar{\boldsymbol{\sigma}}, f \rangle} \Big),$$

whose logarithm is convex in $\bar{\boldsymbol{\sigma}}$.

An end chart of Theorem 32.2 on a compact tangential source, whose tangential map and cut do not depend on $\mathfrak{t}$ and whose normal coordinates enter the log-ratio map through $k$ linearly independent elements of $\vec{A}_B^*$, gives finitely many log expansions and a compact part: the log-ratio map is linear in $\log \mathfrak{t}$ and in the logarithms of the tangential coordinates; with $p_h = \log(R_h'/\mathfrak{t}_h)$ on the polydisc of the radii $R_h' < R_h$ the exponents are $\boldsymbol{\kappa} + \mathbf{n}$ and the coefficients $\widetilde{A}_{\mathbf{n}} R'^{\boldsymbol{\kappa} + \mathbf{n}}$, summable by Theorem 32.2(ii); where some $\mathfrak{t}_h$ lie in $[R_h', R_h)$ these join the tangential coordinates, the others keeping their expansions, summable uniformly by Cauchy's estimate on the closed polydisc; and where all of them do, the part is compact. A law with independent coordinates has such log expansions from the expansions of its factors at their ends.

**Proposition 32.14** (A law with log expansions continues without phase)**.** *Let the law of a piece have log expansions at* $\bar{\mathbf{c}}$ *(Definition 32.13). Then for* $\mathbf{z} \in \vec{A}_B + i\vec{A}_B$ *off the hyperplanes* $\langle \mathbf{z}, u_h \rangle = \eta_h$ *of the log expansions,*

$$P_{W_B(\bar{\mathbf{c}} + \mathbf{z})}(\mathbf{m}_B) = \int e^{-\langle \mathbf{z}, f \rangle}\, d\phi_0(f) + \sum_{Q} \sum_{\boldsymbol{\eta}, \mathbf{j}} \prod_{h \le k} \frac{j_h!}{(\eta_h - \langle \mathbf{z}, u_h \rangle)^{j_h + 1}} \int_{Y_Q} e^{-\langle \mathbf{z}, f \rangle}\, d\phi_{\boldsymbol{\eta}, \mathbf{j}}(f),$$

*continued: the seam function at* $\bar{\mathbf{c}} + \cdot$ *is meromorphic on* $\vec{A}_B + i\vec{A}_B$ *with its poles on these hyperplanes, and there are* $C$ *and* $n$ *with*

$$\big| P_{W_B(\bar{\mathbf{c}} + \mathbf{z})}(\mathbf{m}_B) \big| \le C\, \delta^{-n} \max \Big( \max_{f \in Y_0} e^{-\langle \mathrm{Re}\, \mathbf{z}, f \rangle},\ \max_{Q} \max_{f \in Y_Q} e^{-\langle \mathrm{Re}\, \mathbf{z}, f \rangle} \Big),$$

$\delta$ *the smaller of* 1 *and the distance of* $\mathbf{z}$ *from these hyperplanes; the same holds, with another constant, for* $\delta$ *the smaller of* 1 *and the distance from the polar hyperplanes of the seam function alone. On the contour through a point* $\bar{\mathbf{c}} + \tau\bar{\boldsymbol{\sigma}}$, $\tau \ge 0$, *the bound is* $C\delta^{-n} X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}})^{\tau}$, *with no factor in the imaginary parts and no power of* $\tau$. *So, for a piece whose seam word has a Newton sum of full dimension and a rational* $\bar{\boldsymbol{\sigma}}$, $r(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}}) \le r^{\sharp}_{\vec{A}_B}(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}}) \le g_{\vec{A}_B}(\bar{\boldsymbol{\sigma}}; 0) \le X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}})$, *and the growth of Theorem 31.4(iii) is* $g(\boldsymbol{\sigma}; 0) \le X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}})$ *for every direction* $\boldsymbol{\sigma}$ *of the shares of Theorem 31.4(i) whose sums over the monomials are* $\bar{\boldsymbol{\sigma}}$.

The certificate of Corollary 32.12(ii) is one expansion along the direction itself, with fibre measures of any extent and no condition on the rest of the law: on the contours through the points of a ray of $\bar{\boldsymbol{\sigma}}$ the factor $\mathbf{x}^{-i\bar{\boldsymbol{\vartheta}}}$ has modulus one and only the values of the directional ratio enter, not the compactness of the faces. Neither certificate applies as stated where the tangential coordinates of a face move with the normal ones, as when the support of the law curves in its log-ratio coordinates, or where a cut of the fibres moves, as in Theorem 32.2: the measures on the fibres are then not fixed, and the imaginary translates are read one at a time by Theorem 32.10, or bounded by the ceilings of Theorem 32.15, which carry the phase of the letters on these lines through their constants.

Without such a certificate a tube can reach farther than every one of its lines. There are sector integrals in two shares, with a single wall and growth bounds, for which every line of the tube along the fall of the wall has a finite reach while the reach of the tube over all the directions of the shares is infinite: the factor $\mathbf{x}^{-i\bar{\boldsymbol{\vartheta}}}$ enters the Taylor coefficients of a letter at the falling face through the powers of $\bar{\boldsymbol{\vartheta}}$. The seam function of a piece can do the same: when its seam function on the pins is a Beta corner whose two arguments move independently on the pins, it has growth bounds only at positive $\theta$, and on an imaginary translate along which one argument alone moves in the imaginary direction the Beta corner grows like a power of the imaginary part whose degree grows with the real parts, so that its tube reaches infinitely far while each of its lines has a finite reach. The sector letters of a piece of tables are never monomials in the falling variables, through which alone the factor $\mathbf{x}^{-i\bar{\boldsymbol{\vartheta}}}$ would enter without reaching the Taylor coefficients at the falling face. Theorem 32.16(i) is the limit $\theta \to 0$ of Theorem 31.4(iii); when the reach of the tube is infinite, the bound of that theorem at a positive $\theta$, which pays for the phase, is the one that travels.

**Theorem 32.15** (Ceilings, simple tops and fibres along a direction)**.** *Let the seam function of the piece converge at a real point of the pins with positive shares, let $\bar{\boldsymbol{\sigma}}$ and $\mathbf{w}_0$ be as in Definition 32.9, the walls being those of a fan of the seam word (Theorem 31.4(ii)), and for a sector $\pi$ of it let $\varrho_{\pi,\bar{\boldsymbol{\sigma}}} = \prod_k q_k^{\dot{\kappa}_k}\, \mathbf{k}_\pi^{-\bar{\boldsymbol{\sigma}}}$ be the directional ratio in the sector (Definition 31.10), the sector letters $L$ of $\mathbf{k}_\pi$ entering with the coordinates $\bar{\sigma}_L$ of $\bar{\boldsymbol{\sigma}}$ at their stations and $w_{0,L}$ of $\mathbf{w}_0$. The keyhole continuations along a direction $\boldsymbol{\sigma}$ depend on $\boldsymbol{\sigma}$ only through $\bar{\boldsymbol{\sigma}}$, and the falling variables of $\pi$ are those with $\dot{\kappa}_k < 0$; for a piece of tables the sectors of the orders $\pi$ of $B$ may be taken.*

*(i)* No reach exceeds the ceiling. *For every keyhole continuation along $\bar{\boldsymbol{\sigma}}$, of ceiling $H_{\mathcal{O}}$, and every $\bar{\boldsymbol{\vartheta}} \in \vec{A}_B$, $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0 + i\bar{\boldsymbol{\vartheta}}) \le H_{\mathcal{O}}$: the phase of the letters does not enter the reach.*

*(ii)* The values at the integers. *Let $\bar{\boldsymbol{\sigma}}$ be scaled so that every $\dot{\kappa}_k$ is an integer, and let the walls of the falling variables of every sector be off the integers at $\mathbf{w}_0$. An* integer keyhole continuation along $\bar{\boldsymbol{\sigma}}$ *is defined as in Definition 31.10 with every sector letter kept from vanishing except those $L$ whose exponent $-w_{0,L} - m\bar{\sigma}_L$ at $\mathbf{w}_0 + m\bar{\boldsymbol{\sigma}}$ is a nonnegative integer for every large integer $m$, its ceiling defined in the same way; its keyhole contour, its tops and its simple tops, with their exponents* $\exp(p)$ *and coefficients $A(p)$, are those of Definition 20.50 with $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ in place of $\varrho_\pi$, the sector form $\phi_{\pi,\mathbf{w}_0} = |\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})| \prod_k q_k^{\kappa_k(\mathbf{w}_0)-1} \mathbf{k}_\pi^{-\mathbf{w}_0} dq$ in place of $\phi_\pi$, the walls $\kappa_j(\mathbf{w}_0)$ of the sector in place of $\kappa_j$, and the faces $q_j = 0$ with $\dot{\kappa}_j = 0$ in place of the faces of the level variables. Then for every large integer $m$ the seam function at $\mathbf{w}_0 + m\bar{\boldsymbol{\sigma}}$ is the integral of $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}^m\, \phi_{\pi,\mathbf{w}_0}$ over the keyhole contour, and the growth at the integers* $\mathrm{growth}_{\mathbb{Z}}(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) = \limsup_m |P_{W_B(\mathbf{w}_0 + m\bar{\boldsymbol{\sigma}})}(\mathbf{m}_B)|^{1/m}$ *is at most the ceiling of every integer keyhole continuation along $\bar{\boldsymbol{\sigma}}$, and at most $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0)$.*

*(iii)* Simple tops. *If an integer keyhole continuation along $\bar{\boldsymbol{\sigma}}$ of ceiling $H_{\mathcal{O}}$ has finitely many tops, all simple, and some $A_\zeta \neq 0$ (Theorem 20.51(iii)), then* $\mathrm{growth}_{\mathbb{Z}}(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) = H_{\mathcal{O}} \le r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0)$*; if moreover some keyhole continuation along $\bar{\boldsymbol{\sigma}}$ has the ceiling $H_{\mathcal{O}}$, then $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) = H_{\mathcal{O}}$.*

*(iv)* Fibres. *Let $\bar{\boldsymbol{\sigma}}$ be scaled so that its coordinates and every $\dot{\kappa}_k$ are integers. Call a box of a continuation along $\bar{\boldsymbol{\sigma}}$ a* fibre box *when on its sets $Y$ the letters with $\bar{\sigma}_L \geq 0$ do not vanish, and those with $\bar{\sigma}_L < 0$ do not vanish at the points with a falling coordinate 0; its stationary bound, as in Definition 20.54 with $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ in place of $\varrho_\pi$, is the supremum of $|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}|$ over the points of these sets with no falling coordinate 0 at which some falling coordinates lie inside their discs and the derivatives of $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ in them vanish, the points with only one such coordinate, of falling rate one, left out. For every continuation along $\bar{\boldsymbol{\sigma}}$ whose boxes are keyhole boxes or fibre boxes, of ceiling $H_{\mathcal{O}}$, with the stationary bound $S$ of its fibre boxes, $r(\bar{\boldsymbol{\sigma}}; \mathbf{w}_0) \leq \max(H_{\mathcal{O}}, S)$.*

So the reach along a direction sits between the growth at the integers and the ceiling of every keyhole continuation along it, and the two meet at simple tops that do not cancel when a keyhole continuation along the direction has the same ceiling, as at width one; and, by Theorem 32.10, without any top it is the far boundary of the region on which the directional law is its own wall series. The phase of the letters, which Theorem 31.11 pays for at the points where the ceiling is read, does not enter the reach: on a line the imaginary parts stay bounded. It enters only through the transverse fibre of a closing, which Theorem 32.16 reads.

The integer scaling in (iv) is needed: a sector with one falling variable of falling rate one and a letter entering with a coordinate that is not an integer can have a fibre box of stationary bound 0 and a reach above its ceiling, the directional ratio not being single valued in the disc, so that Lemma 20.52 does not apply; scaled to integers, the stationary point that the bound left out is counted.

**Theorem 32.16** (The closing past the walls is read along its direction)**.** *Read the glue as in Theorem 31.4(i), the partner in its monomials, for a piece with walls and growth bounds whose seam word has a Newton sum of full dimension. Let $V$ be the space of the directions of the shares, of dimension $n$, the number of free shares, $\boldsymbol{\sigma} \in V$ a direction with rational coordinates, $\bar{\boldsymbol{\sigma}}$ its sums, $V' \subseteq \vec{A}_B$ the space of the sums of the directions in $V$, and $\mathbf{c}$ the real starting point, with $\bar{\mathbf{c}} \in \Omega_B$, off the null set of Theorem 31.2 taken with the polar hyperplanes of the seam function. Let $h_\Gamma$ be the Gamma rate of Theorem 31.4(iii), so that the rate of that theorem, which holds for such a piece (Theorem 31.4, its seam function converging at the sums $\bar{\mathbf{c}} \in \Omega_B$ of the positive shares of the starting point), is $h_\theta(\boldsymbol{\sigma}) = h_\Gamma(\boldsymbol{\sigma}) + \log g(\boldsymbol{\sigma}; \theta) + \frac{2\theta^2}{\pi} \sum_{\sigma_l > 0} \sigma_l$, and let $I$ be the integrand of the share side.*

*(i)* The tube bounds the closing. *If $h_\Gamma(\boldsymbol{\sigma}) + \log r^\sharp_{V'}(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}}) < 0$, the closing along $\boldsymbol{\sigma}$ converges absolutely, its terms and the integrals left after them being $O\big(e^{\tau(h_\Gamma(\boldsymbol{\sigma}) + \log r^\sharp_{V'} + \epsilon)}\big)$ for every $\epsilon > 0$; the closing along $\boldsymbol{\sigma}$ is the closing along every direction of its chamber (Theorem 31.6), whose rates bound it as well. For the share side of Theorem 31.4(iv), with $\boldsymbol{\sigma}$ a direction of its shares with rational coordinates and $\mathbf{c}$ off the null set of that part, the closing along $\boldsymbol{\sigma}$ converges absolutely when $\sum_l \sigma_l \log|\sigma_l| + \log r^\sharp_B + \log r^\sharp_R < 0$, in this sentence $r^\sharp_B$ and $r^\sharp_R$ being the reaches of the tubes over the space of the directions of its shares (Corollary 32.12) of the seam functions of the piece and of the partner along $\boldsymbol{\sigma}$ at $\mathbf{c}$; its terms and the integrals left after them are then $O\big(e^{\tau(\sum_l \sigma_l \log|\sigma_l| + \log r^\sharp_B + \log r^\sharp_R + \epsilon)}\big)$ for every $\epsilon > 0$.*

*(ii)* One free share. *Let $n = 1$, and put $r = r(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}})$ and $\ell(z) = P_{W_B(\bar{\mathbf{c}} + z\bar{\boldsymbol{\sigma}})}(\mathbf{m}_B)$. Then $r^\sharp_{V'} = r$, and for every $r' > r$ the terms are $O\big((1+\tau)^{n'} e^{\tau(h_\Gamma(\boldsymbol{\sigma}) + \log r')}\big)$; so the closing converges absolutely when $h_\Gamma(\boldsymbol{\sigma}) + \log r < 0$. Let $r > 0$, let $\mathbb{L}$ be the progression of the crossings of a falling share $l_0$, and let the data keep the poles apart: from some point on, $\mathbb{L}$ meets no pole of $\ell$ and no pole of another Gamma, and no pole of $\ell$ is a pole of a Gamma. If the poles*

*of $\ell$ crossed by the ray are simple from some point on, or the values of $\ell$ on $\mathbb{L}$ carry the reach (Theorem 32.10(iv)), then for every $\epsilon > 0$ infinitely many terms have modulus at least $e^{\tau(h_\Gamma(\boldsymbol{\sigma})+\log r-\epsilon)}$: the rate $h_\Gamma(\boldsymbol{\sigma}) + \log r$ is exact, and the terms do not tend to zero when it is positive. Without either proviso, if those poles have order at most $p$, the same holds for all but at most $p-1$ of the partners obtained by multiplying the coefficients by $e^{\tau_0 v_l}$, $\tau_0 \in \mathbb{R}$, for a fixed $\mathbf{v}$ with $\langle \mathbf{v}, \boldsymbol{\sigma}\rangle = 1$, the rate of the partner of $\tau_0$ being $h_\Gamma(\boldsymbol{\sigma}) - \tau_0 + \log r$. The same holds for the share side of Theorem 28.10(iii) with one free share and a partner whose seam function converges along the ray, with $\sum_l \sigma_l \log|\sigma_l| + \log\sup \mathbf{V}_R^{-\bar{\boldsymbol{\sigma}}}$ in place of $h_\Gamma(\boldsymbol{\sigma})$, the partners being the tinkered partners that multiply the room letters of each station $D$ by $e^{\tau_0 v_D}$ (Definition 28.18); at a node of width one, where the share side has one free share, this rate is $\log\big(\operatorname{reach}(B)/\inf \varrho_R\big)$, the logarithm of the root rate of Theorem 20.39(ii).*

*(iii)* Several free shares. *Let $n \geq 2$. Fix a complement $V_\perp$ of $\mathbb{R}\boldsymbol{\sigma}$ in $V$, a Lebesgue measure $d\boldsymbol{\vartheta}_\perp$ on it and $k_\sigma > 0$ with $d\boldsymbol{\vartheta} = k_\sigma\, dy\, d\boldsymbol{\vartheta}_\perp$ for $\boldsymbol{\vartheta} = y\boldsymbol{\sigma} + \boldsymbol{\vartheta}_\perp$, $d\boldsymbol{\vartheta}$ the measure of Definition 4.1. Let $T_\sigma$ be the space of the* transverse tinkers, *the tinkers of the partner read in its monomials (Definition 28.18, with one multiplier $e^{v_l}$ for each monomial in place of each coordinate of the seam vector), multiplying the coefficients read on $V$ as $\xi_l \mapsto \xi_l e^{v_l}$, with $\langle \mathbf{v}, \boldsymbol{\sigma}\rangle = 0$, identified with the dual of $V_\perp$ and carrying the dual Lebesgue measure $d\mathbf{v}$. The crossings of the closing along $\boldsymbol{\sigma}$ are the same for every tinker, and at a crossing at the time $\tau^*$*

$$\int_{T_\sigma} e^{\langle \mathbf{v}, \mathbf{c}\rangle}\, \mathcal{R}_{\tau^*}\big(\boldsymbol{\xi} e^{\mathbf{v}}\big)\, d\mathbf{v} = -k_\sigma \operatorname*{Res}_{z=\tau^*} I(\mathbf{c} + z\boldsymbol{\sigma}),$$

*$\mathcal{R}_{\tau^*}(\boldsymbol{\xi} e^{\mathbf{v}})$ the crossing term of the partner with the coefficients $\boldsymbol{\xi} e^{\mathbf{v}}$. Integrated over the transverse tinkers against the factor $e^{\langle \mathbf{v}, \mathbf{c}\rangle}$, the terms of the closing are $-k_\sigma$ times the residues of the integrand on the real line through $\mathbf{c}$ in the direction $\boldsymbol{\sigma}$; these are $O\big((1+\tau)^{n'} e^{\tau(h_\Gamma(\boldsymbol{\sigma})+\log r')}\big)$ for every $r' > r = r(\bar{\boldsymbol{\sigma}}; \bar{\mathbf{c}})$, and under the conditions of (ii) on that line infinitely many of them have modulus at least $e^{\tau(h_\Gamma(\boldsymbol{\sigma})+\log r-\epsilon)}$.*

**Theorem 32.17** (A closing with one free share is read on the law along it)**.** *Take, in Theorem 31.7, the share side of the kind (B), its bundles merged by Corollary 4.4, the bars marking sums over the bundles, and in this statement write*

$$I(\bar{\mathbf{a}}) = \frac{\prod_G \Gamma(\bar{a}_G)}{\prod_j \Gamma(s_j) \prod_j \Gamma(t_j)}\; P_{W_B^{\mathrm{m}}(\bar{\mathbf{a}})}(\mathbf{m}_B),$$

*the products over the bundles and over the active letters, with the merged seam function of Lemma 32.8. Let the pins of the merged shares leave a fibre of dimension one, spanned by a rational direction $\bar{\boldsymbol{\sigma}}$, and let $\bar{\mathbf{c}}$ be a real point of the merged pins with positive coordinates at which the merged seam function converges. Let the merged law along $\bar{\boldsymbol{\sigma}}$, the law along $\bar{\boldsymbol{\sigma}}$ of the merged seam function (Definition 32.9), be the image of $\prod_G e^{-\bar{c}_G y_G}\lambda$ under $X = \prod_G e^{-\bar{\sigma}_G y_G}$, $\lambda$ the source measure, its line function $\ell(\zeta) = P_{W_B^{\mathrm{m}}(\bar{\mathbf{c}}+\zeta\bar{\boldsymbol{\sigma}})}(\mathbf{m}_B)$ and its reach $r$, the least $x_0 \geq 0$ for which it has on $(x_0, \infty)$ a density that is a wall series of radius $x_0$ (Definition 32.9, Theorem 32.10(iii)); and in this statement write $h_\Gamma^{\mathrm{m}}(\bar{\boldsymbol{\sigma}}) = \sum_G \bar{\sigma}_G \log|\bar{\sigma}_G|$, the Gamma rate of the merged share side, whose coefficients are inside the merged values.*

*(i) There is $k_P > 0$, depending only on the pins, with $P_{B\sqcup R}(\mathbf{m}) = \frac{k_P}{2\pi i}\int_{\operatorname{Re}\zeta=0} I(\bar{\mathbf{c}} + \zeta\bar{\boldsymbol{\sigma}})\, d\zeta$, and the crossing term (Lemma 31.3) at a crossing $\zeta^*$ of the ray is $-k_P \operatorname{Res}_{\zeta=\zeta^*} I(\bar{\mathbf{c}} + \zeta\bar{\boldsymbol{\sigma}})$: at the crossing $\bar{a}_f = -n$ of the falling share $f$, at which no other factor of $I$ is polar, it is*

$$k_P\, \frac{(-1)^n}{n!\,|\bar{\sigma}_f|}\, \frac{\prod_{G\neq f} \Gamma(\bar{a}_G)}{\prod_j \Gamma(s_j) \prod_j \Gamma(t_j)}\; P_{W_B^{\mathrm{m}}(\bar{\mathbf{a}})}(\mathbf{m}_B)$$

*at* $\bar{\mathbf{a}} = \bar{\mathbf{c}} + \zeta^*\bar{\boldsymbol{\sigma}}$*, and at a simple pole of the merged seam function at which no Gamma is polar it is* $-k_P \prod_G \Gamma(\bar{a}_G)/\big(\prod_j \Gamma(s_j)\prod_j \Gamma(t_j)\big)$ *there times* $\operatorname{Res}_{\zeta^*} \ell$*.*

*(ii) For every* $r' > r$ *there are* $C$ *and* $n'$ *such that the crossing terms at their times* $\tau$*, and the integrals left at the times at a fixed distance from the crossings, are at most* $C(1+\tau)^{n'} e^{\tau h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r'^{\tau}$*.*

*(iii) Let the ray* $\bar{\mathbf{c}} + \tau\bar{\boldsymbol{\sigma}}$*,* $\tau \geq 0$*, cross its polar points simply: no crossing point of a falling merged share, one with* $\bar{\sigma}_G < 0$*, is a pole of the merged seam function or a crossing point of another falling share, and the poles of the merged seam function on the ray are simple. Then for each falling share* $f$ *the crossing terms of* $f$ *and those at the poles of the merged seam function have together the root rate* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r$*: the upper limit of* $|\mathcal{R}|^{1/\tau}$ *over them is* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r$*. The same holds when the data keep the poles apart only from some point on, as in Theorem 32.16(ii), read on* $\ell$ *and on the progression of the crossings of* $f$*, and the poles of* $\ell$ *crossed by the ray are simple from some point on or the values of* $\ell$ *on that progression carry the reach; without either proviso, if those poles have order at most* $p$*, it holds for all but at most* $p-1$ *of the partners obtained by multiplying the coefficients of the shares of each bundle* $G$ *by* $e^{\tau_0 v_G}$*,* $\tau_0 \in \mathbb{R}$*, for a fixed* $\mathbf{v}$ *with* $\sum_G v_G \bar{\sigma}_G = 1$*, whose merged laws are constant multiples of the images of the merged law under* $X \mapsto e^{-\tau_0} X$*, with the root rates* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}}) - \tau_0} r$*.*

*(iv) So the closing of the merged share side along* $\bar{\boldsymbol{\sigma}}$*,* $P_{B \sqcup R}(\mathbf{m}) = \sum \mathcal{R}$ *over the crossings of the ray in their order, converges absolutely when* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r < 1$*, and under the conditions of (iii) its terms do not tend to zero when* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r > 1$*. No hypothesis on tops or on the phase of the letters enters, only the separations of the poles in (iii): the rate is read on the merged law along the direction. Where* $X$ *is bounded on the support of the law of the piece,* $r$ *is its largest value there, and when moreover the falling shares are alone in their bundles,* $h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}}) + \log r$ *is the averaged rate* $\hat{h}_C$ *of Theorem 31.28; where* $X$ *is unbounded,* $\hat{h}_C$ *is infinite and the averaged terms diverge past the walls, and (iii) carries the sharpness of that theorem past them, with the reach of the merged law in place of the largest value of* $X$*. When every bundle is a single share,* $\boldsymbol{\sigma} = \bar{\boldsymbol{\sigma}}$*, the merged seam function is the product of* $\boldsymbol{\xi}^{-\mathbf{a}}$ *and of the seam function at the sums over the monomials (Theorem 31.4(i)),* $r$ *is* $e^{-\sum_l \sigma_l \log \xi_l}$ *times the reach of the piece along the sums of* $\boldsymbol{\sigma}$ *(Definition 32.9), and* $e^{h^{\mathrm{m}}_\Gamma(\bar{\boldsymbol{\sigma}})} r = e^{h_0(\boldsymbol{\sigma})}$*,* $h_0$ *the rate of Theorem 31.4(iii) at* $\theta = 0$*, whose growth* $g(\boldsymbol{\sigma}; 0)$ *is on a fibre of dimension one that reach: on such a fibre the rate of Theorem 31.4(iii) at* $\theta = 0$ *is exact under the conditions of (iii), and (iii) is Theorem 32.16(ii).*

The merged share side and the share side at the stations, before the stations of one excess are merged (Corollary 4.4), are two representations of one period, with two laws along their directions; completing the splits of a bundle can move the edge of the support at which the law stops being its own wall series. The merged law can stop being its own wall series at the image of a nearer edge of the support, where a level set of the merged seam letter ends the expansion of the law at its falling end, while the station closings, with more free shares, can read a farther edge: the merged closing can diverge where the station closing converges.

**Corollary 32.18** (A law with log expansions bounds a closing of any fibre dimension)**.** *In Theorem 31.4(iii), with the starting point* $\mathbf{c}$ *off its null set, let the law of the piece have log expansions at the sums* $\bar{\mathbf{c}}$ *of* $\mathbf{c}$ *(Definition 32.13), with the inner bound* $X_{\mathrm{in}}$*. The function* $h_\Gamma(\boldsymbol{\sigma}') + \log X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}}')$ *is convex on the chamber* $C(\boldsymbol{\sigma})$ *of Definition 31.5. If its infimum* $h'$ *on the chamber is negative, the closing along* $\boldsymbol{\sigma}$ *converges absolutely to* $P_{B \sqcup R}(\mathbf{m})$*, its terms and the integrals left after them being* $O(e^{\tau(h'+\epsilon)})$ *for every* $\epsilon > 0$*. The fibre may have any dimension, and no margin of coefficient*

*factors and no phase enter: the rate is read on the compact part of the law and on the inner faces of its log expansions. In Theorem 31.4(iv), with* $\mathbf{c}$ *off the null set of that part, if the laws of the piece and of the partner have log expansions at* $\mathbf{c}$*, with the inner bounds written* $X^B_{\mathrm{in}}$ *and* $X^R_{\mathrm{in}}$ *in this sentence, the same holds with* $\sum_l \sigma'_l \log|\sigma'_l| + \log X^B_{\mathrm{in}}(\boldsymbol{\sigma'}) + \log X^R_{\mathrm{in}}(\boldsymbol{\sigma'})$*, convex on the chamber* $C(\boldsymbol{\sigma})$*, in place of* $h_\Gamma(\boldsymbol{\sigma'}) + \log X_{\mathrm{in}}(\bar{\boldsymbol{\sigma}}')$*.*

So the counterparts of the reach theorems of width one hold along every rational direction. The tinkers of Definition 28.18 play two parts: all the tinkers of one partner separate the laws (Theorem 28.19), and the integral over the transverse tinkers of a closing, the partner read in its monomials, reads the reach of the real line (Theorem 32.16(iii)).

For a positive kernel the root rate of a crossing term is the exponential of the least Gamma rate of its chamber in every fibre dimension (Theorem 31.26), and averaging reads the log-support of the source measure (Theorem 31.7); Corollary 32.18 reads, past the walls, the compact part and the inner faces of the log expansions. It is an upper bound: with two or more free shares a crossing term is a fibre integral of positive dimension, whose size the law along one direction does not fix, and the best direction of a chamber can lose against the averaged closing of the same source (Theorem 31.8); the exact rate is then read on a joint word at a noncancelling simple top (Theorem 31.21). One partner does not read the real line: it averages over its transverse fibre, and its terms can fall below the rate of every real line of its chamber, the integral of the terms over the transverse tinkers then being carried by the other tinkered partners. Whatever the closing, the reach along the direction of the closing is at least the least value of the directional ratio on the support of the law, and at least every point near which the directional law, or after merging the merged law, has no real-analytic density (Theorem 32.10(iii)).

## 32.3 What the calculus of bunches settles

*Remark* 32.19 (What the calculus of bunches settles, and what it leaves open)*.* A closing of the glue along a direction of its share fibre is exact at every step and depends only on its chamber (Lemma 31.3, Theorem 31.6); Theorems 31.2, 31.4 and 31.11 give sufficient conditions for it, and the closings can be averaged over the source (Theorem 31.7). The cases that these conditions leave are settled as follows.

(i) *Crossing terms of positive dimension.* Along a rational ray, at simple nonresonant crossings and after a finite head, a crossing term kept with its falling shares integrated on their strips is, along its cohort, a scalar $G_n$ times a finite sum $\sum_i \big(\mathsf{T}_{-n;(1)}[c_i]\,\phi_i\big)(Y_i)$ of joint Trailmoments of fixed functions on fixed compact domains, the joint word (Theorem 31.18): a falling Gamma becomes a positive share and one Beta variable (Lemma 31.14), a continued source strip a segment and circle integrals (Lemma 31.15), and nothing is extracted from the term. For a positive kernel its rate is the least Gamma rate of the chamber, in every fibre dimension and with no hypothesis on the tops (Theorem 31.26); averaged over a positive source it is the averaged rate $\hat{h}_C$ (Corollary 31.27), so a nonzero averaged rate decides by its sign whether the averaged terms tend to zero, with no condition on the bundles or the dimension (Theorem 31.28), and at $\hat{h}_C = 0$ Theorem 31.32 decides. The joint word has the root rate $r_G c_{\max}$ at noncancelling simple tops, $r_G$ the root rate of the scalar and $c_{\max}$ the ceiling of the word (Theorem 31.21). Coincident crossings are packets of factors polar on one crossed hyperplane, assembled before a sign or a top is read (Theorem 31.30, Proposition 31.31); at the root rate one a positive moment sequence is decided by its law near the top (Theorem 31.32), and an expansion by its coefficients (Proposition 31.33).

(ii) *The phase.* A joint word is bounded by $|G_n| \sum_i \|\phi_i\| \, c_{\max}^n$ (Corollary 31.19): the fibre is completed before a modulus is taken, and the phase of the letters enters the coefficient functions, where it can cancel, and costs no term of the bound. It confines the radii of the procedure to the margins of the coefficient factors (Theorem 31.18, Remark 31.22). Read on the law, the phase costs nothing where the law has log expansions: in its log-ratio coordinates the shares pair with the law linearly, and the compact part and the inner faces of its log expansions bound the continued seam function on whole contours (Proposition 32.14). Where the tangential coordinates move with the normal coordinates at a falling end, the phase enters the coefficients of the expansion there, and a tube can reach infinitely far while each of its lines has a finite reach.

(iii) *Regions of equality and bounds at any width.* The regions on which a law equals its expansion at a falling end are certified from the sector letters, with the map from the motion of the shares to the exponents, the coefficients, the truncation bounds and exact boundary words (Theorems 32.2 and 32.4); the domain where a series converges and the region where it equals the law are kept apart (Proposition 32.6), and density charts are built at regular fibres (Proposition 32.5). The terms of a closing have at most the root rate $r_G c_{\max}$ of every joint word whose radii keep within the holomorphy of its source and the margins of its coefficient factors, $c_{\max}$ the ceiling of the word, and have that root rate when the word has a noncancelling simple top.

(iv) *The reach at any width.* Along every rational direction of the shares the reach is read on the law: it is the far boundary of the region on which the directional law, the law multiplied by $\mathbf{x}^{-\mathbf{w}_0}$ for the starting point $\mathbf{w}_0$ and read through the directional ratio, is its own wall series, whose exponents lie among the values $(\kappa_U(\mathbf{w}_0) + n)/(-\dot{\kappa}_U)$ of the walls divided by their falling rates, and it is at least every point at which that law is not analytic, the least value of the directional ratio among them (Theorem 32.10). The walls and the growth bounds that this uses hold for every seam function of a piece that converges at a real point and for its seam letters merged by bundles (Lemma 32.8), and at width one the same argument reads the growth of a family on its law, that of the bunch being its reach (Corollary 32.11). For a piece the reach is at most the ceiling of every keyhole continuation along the direction, with no phase, equal to it at simple tops that do not cancel when a keyhole continuation has the same ceiling, and bounded by fibres where letters vanish inside the keyholes, the direction scaled to integers (Theorem 32.15). A closing past the walls converges when the exponential of the Gamma rate times the reach of its tube is less than one, its terms then having at most that root rate (Theorem 32.16(i)); an expansion at the face over the real fibres bounds the reach of the tube along one direction and log expansions in every direction, with no allowance for the phase (Corollary 32.12, Proposition 32.14), and otherwise the tube can reach farther than every one of its lines; a law with log expansions bounds a closing of any fibre dimension by the rate read on the compact part and the inner faces of its log expansions (Corollary 32.18). With one free share, as at a node of width one, the tube is the line, and the terms attain the rate that the reach along the direction gives when the poles of the line and of the Gammas keep apart and the poles of the line are eventually simple or its values on a progression of crossings carry the reach (Theorem 32.16); the closing of a share side merged by its bundles to one free share, a series other than the station closings, has under the same conditions the root rate of Theorem 32.17(iii), read on the merged law along its direction, its terms not tending to zero beyond it, which carries the sharpness of Theorem 31.28 past the walls (Theorem 32.17). In every number of free shares the terms integrated over the tinkers of the partner transverse to the direction are the residues of the

real line, which attain its rate under the same conditions (Theorem 32.16), while the terms of one partner average over its transverse fibre and can fall below every real line of its chamber.

(v) *A gap at a family of tables.* A family of three tables whose seam has single stations has a chamber in which no direction certifies the closing while the averaged closing converges geometrically (Theorem 31.8); the gap is relative to the closings of one share side by the directions of its chamber (Remark 31.9).

At width one the reach is read on the law, as the point beyond which the law is its own wall series, and it is the ceiling of a keyhole continuation whose tops are simple and do not cancel (Theorems 20.44 and 20.51). At any width the first holds along every rational direction (Theorem 32.10), and the second along every rational direction for every piece, under the conditions of Theorem 32.15(ii) and (iii), and for joint words (Theorem 31.21). What a node of width one does not have is a transverse fibre. With two or more free shares the crossing terms of one partner integrate over the transverse fibre the imaginary translates of the line through its starting point. Theorem 32.16(i) bounds them by the reach of the tube when the exponential of the Gamma rate times that reach is less than one; a law with log expansions bounds the tube by the directional ratio on the compact part and the inner faces of its log expansions, and a closing of any fibre dimension with nothing paid for the phase (Corollary 32.18), while without log expansions the reach of the tube can be infinite though every line of the tube has a finite reach, and the bound that travels pays for the phase at a positive $\theta$ (Theorem 31.4(iii)). Theorem 32.16(iii) gives the integral of the terms over the transverse tinkers the rate of the real line, under the conditions of its part (ii) on that line; Theorem 31.21 gives the rate of one partner at the noncancelling simple tops of its joint words. The real line does not give the rate of one partner, whose terms can fall below every real line of its chamber, and the best direction of a chamber can lose against averaging (Theorem 31.8). Before the walls, under the hypotheses of Theorem 31.28, the rate of one partner is averaged over the source, and past them the closing of a share side merged by its bundles to one free share has, under the conditions of Theorem 32.17(iii), the rate read on the merged law along its direction. Two problems are open. The first is the rate past the walls of one partner whose joint words have no noncancelling simple top: at two or more free shares, where it asks for an averaged closing of the continued law, and at one free share when the data do not keep the poles apart or no proviso of Theorems 32.16(ii) and 32.17(iii) holds for that partner. The second is to read the growth $g(\boldsymbol{\sigma};\theta)$ on the law, as Theorem 32.10 reads the reach of a line. At $\theta = 0$ the reach of the line bounds it below and log expansions bound it above by their inner bound (Corollary 32.12, Proposition 32.14), and the two meet when, for every $\epsilon > 0$, the law has log expansions whose compact part and inner faces lie where the directional ratio is at most the reach plus $\epsilon$; in general, and without log expansions, where it can be infinite at $\theta = 0$, it is known only through these bounds and those of Theorems 31.4(iii) and 31.11 and of Corollary 31.19, within the margins of the coefficient factors (Theorem 31.18). Degenerate and cancelling tops are open at width one as well.

# Chapter 33

# Price and complexity

A construction is a description of a family, and the exact theorems of Part V turn it into a description of the period; both have a price. The price of a construction is the length of its description, a repeated subtree described once and its multiplicity written in binary; the work of a word is the dimension of the integrals it asks for, an integral that recurs counted once. Repetition then costs what it should: $M$ copies of a cell glued along a tree at a full lean, or closed in series at any exponents, or chained, have periods whose words cost the word of the cell and the binary length of $M$, and a departure from the repetition costs the cells that depart. The laws of cells (Chapter 23), the networks of holds (Chapters 25 and 26) and the finite interfaces (Chapter 27) are priced the same way: a law is computed once for each distinct subtree, a node of a network costs its periods at the two full leans whatever the number of bases of the whole table, and a polynomial Gamma moment of a repetition is the power of one finite matrix. What a value can contain is bounded the same way along a ray (Theorem 38.31).

## 33.1 Constructions and their price

**Definition 33.1** (Construction)**.** A *construction* $\mathfrak{C}$ is a finite rooted tree with a family at each leaf, its marks, bases, weights, masses and exponents given, and at each inner node one of the operations: the direct sum of its subtrees; the substitution of its subtrees, each with a named hinge, into the marks of a core family named at the node, a mark left without a subtree being left in place (Definition 21.1); and the *powers* of a subtree $\mathfrak{C}_1$ with an integer $M$: $\mathfrak{C}_1^{\oplus M}$, $M$ copies in direct sum; $\circ(\mathfrak{C}_1^{M})$, the series closure of $M$ copies of the cell $\mathfrak{C}_1$ (Corollary 21.7(ii)); and $\mathfrak{C}_1^{\triangleright M}$, the chain of $M$ copies glued along two named marks (Theorem 22.8). A subtree may be given by a reference to one already described. A construction builds one family, by Proposition 21.3 at each substitution.

**Definition 33.2** (Price; work)**.** The *price* $\mathrm{price}(\mathfrak{C})$ of a construction is the number of bits of its description as written: for every leaf its marks, bases, weights, masses and exponents, the numbers written as rationals in binary; for every node its operation, its core and its attachments; for every power the $\lceil \log_2(M+1) \rceil$ bits of $M$; a subtree may be written once and referred to by an index wherever it recurs. The price of a word (Definition 1.1(vi)) is defined in the same way, and its *work* $\mathrm{work}(w)$ is the sum of the dimensions of its distinct integrals, an integral that recurs, under a power or an iterate in particular, counted once. The integration count of Definition 20.58 counts every occurrence: the work of $w^M$ is that of $w$, and its integration count is $M$ times that of $w$.

The work counts the distinct integrals a word is written with, and the price the length of what

has to be written down; neither counts the digits of an evaluated value, nor the cost of evaluating: a scalar power costs one exponentiation, a power of a $k \times k$ matrix costs $O(k^3 \log M)$ operations by repeated squaring, and an iterate $\mathcal{Q}^M$ of an operator costs $M$ applications of $\mathcal{Q}$, nested integrals of $M$ times its dimension, or $O(\log M)$ matrix products when $\mathcal{Q}$ leaves a finite-dimensional space of functions invariant. Every polynomial Gamma moment of degree $D$ of a repetition (Chapter 27) is such a matrix power, a fixed row times a power of the transfer matrix $\mathbf{T}_D$ applied to the moment vector of the start (Theorem 27.5), and the period itself is reached through such moments with the certified stopping rule of Theorem 27.7, at a degree that depends on the accuracy asked.

**Theorem 33.3** (Pricing by repetition)**.** *Let $\mathcal{A}$ be a family, a cell where a hinge is needed, described by a construction of price* $\mathrm{price}(\mathcal{A})$*, with the period $P_{\mathcal{A}}$ at its data, and let $M \geq 1$.*

*(i)* Direct sum. *The molecules of $\mathcal{A}^{\oplus M}$, $\mathcal{A}$ connected, are its $M$ copies, and the product of their periods is $P_{\mathcal{A}}^{M}$ (Proposition 7.9; the period of the direct sum itself diverges on the balance wall, Proposition 8.16): price at most* $\mathrm{price}(P_{\mathcal{A}}) + \lceil \log_2(M+1) \rceil + O(1)$*, work that of $P_{\mathcal{A}}$.*

*(ii)* A tree of copies at a full lean. *For a table $\mathcal{A}$ at the exponents $(0,t)$, $M$ copies glued along any tree at the masses $(\kappa, t-\kappa)$ on the two marks of every gluing have the word* $\Gamma_{1}^{2}\!\begin{bmatrix}\kappa, & t-\kappa \\ & t\end{bmatrix} \big(P_{\mathcal{A}}(\mathbf{m};0,t)/\Gamma_{1}^{2}\!\begin{bmatrix}\kappa, & t-\kappa \\ & t\end{bmatrix}\big)^{M}$ *(Corollary 26.11): price at most* $\mathrm{price}(P_{\mathcal{A}}) + \lceil \log_2(M+1) \rceil + O(1)$*, work that of $P_{\mathcal{A}}$, whatever the shape of the tree.*

*(iii)* A series closure. *The series closure of $n_k$ copies of cells $\mathcal{A}_k$, $k \leq d$, at positive exponents of $N$ tables on the balance wall has the word of Theorem 23.3,* $\prod_j (\Gamma(s_j)\Gamma(t_j))^{-1} \int \prod_j x_j^{s_j-1} \prod_{j \geq 2} y_j^{t_j-1} \prod_k \mathfrak{L}_k(x,y)^{n_k}\, dx\, dy$ *with the responses $\mathfrak{L}_k$ of the cells: price $O\big(\sum_k \mathrm{price}(\mathcal{A}_k)\big) + \sum_k \lceil \log_2(n_k+1) \rceil + O(N)$, the letters of a constructed cell being written through Theorem 21.4, and work at most $\sum_k |B_k| + 2N - 1$, $B_k$ the marks of $\mathcal{A}_k$ besides its hinge, independent of the multiplicities and of the order of the cells.*

*(iv)* A chain. *$\mathcal{A}^{\triangleright M}$ has the word $(\mathcal{Q}^M P_0)(\mathbf{1})$ of Theorem 22.8, with the operator $\mathcal{Q}$ and the function $P_0$ of that theorem: price at most* $\mathrm{price}(\mathcal{A}) + \lceil \log_2(M+1) \rceil + O(1)$*, work at most $|\mathcal{E}_{\mathcal{A}}| - 1$; evaluated, the iterate unrolls to nested integrals of total dimension $M(|\mathcal{E}_{\mathcal{A}}| - 2) + 1$, that of the expanded period, unless $\mathcal{Q}$ leaves a finite-dimensional space of functions invariant; where the twist is pinned the word is* $P_{\mathcal{A}}^{M} / \Gamma_{1}^{2}\!\begin{bmatrix} m_{g_{\mathrm{in}}}, & m_{g_{\mathrm{out}}} \\ m_{g_{\mathrm{in}}} + m_{g_{\mathrm{out}}} & \end{bmatrix}^{M-1}$*, $m_{g_{\mathrm{in}}}, m_{g_{\mathrm{out}}}$ the masses of the two named marks $g_{\mathrm{in}}, g_{\mathrm{out}}$, of work $|\mathcal{E}_{\mathcal{A}}| - 1$ evaluated as well.*

*(v)* Integer joint moments *of a repeated transfer matrix, with Gamma laws on the marks or Dirichlet laws on the steps, have as word a fixed row times the power $\mathbf{A}^M$ of the one matrix of Theorem 22.10 times a fixed column, with rational entries at rational masses (Theorem 22.10): price at most that of the matrices of the cell plus $\lceil \log_2(M+1) \rceil + O(1)$, work zero.*

*(vi)* Blocks *read through their totals cost one stacked Gamma each,* $\Gamma_{1}^{|B_i|}\!\begin{bmatrix}\mathbf{m}|_{B_i} \\ m_{B_i}\end{bmatrix}$ *at the block $B_i$ (Theorem 26.12): the work of the period of the family with blocks is that of the period of its core family, whatever the lengths of the blocks.*

*(vii)* A series repetition. *$M$ copies of a series-balanced cell $\mathcal{A}$ at a series node against a cell $\mathcal{A}'$, at the exponents $(s,t)$, have the word* $(\Gamma(s)\Gamma(t))^{-1} \int_0^\infty y^{t-1} f_{\mathcal{A}}(y)^M f_{\mathcal{A}'}(y)\, dy$*, with the transforms $f_{\mathcal{A}}, f_{\mathcal{A}'}$ of Theorem 24.2: price at most* $\mathrm{price}(\mathcal{A}) + \mathrm{price}(\mathcal{A}') + \lceil \log_2(M+1) \rceil + O(1)$*, work $|B| + |R| - 1$ for the marks $B$ of $\mathcal{A}$ and $R$ of $\mathcal{A}'$ besides the hinge, whatever $M$; at a*

*series or parallel node of a hinge tree the multiplicities of equal subtrees enter as exponents (Proposition 23.18).*

*(viii)* Series-parallel tables. *A table with a hinge tree of series and parallel nodes over marks has periods at the full leans that are stacked Gammas times monomials in the weights of its nodes, with one Gamma for each mark and two for each change of kind along an edge, and every period, at every lean and every twist, a fibre integral of dimension $n-1$ of an explicit stacked Gamma times an exponential of the loads, and a family of $N$ such tables one of dimension $(2N-1)(n-1)$ (Corollaries 26.10 and 26.14): price linear in the size of the tree, work at most $n-1$, less one for each pooled pair (Proposition 26.15), and at most $(2N-1)(n-1)$ for the family, whatever the numbers of bases, which can be exponential in $n$.*

*(ix)* Polynomial Gamma moments of a repetition *have as word a fixed row times a power of the transfer matrix $\mathbf{T}_D$ applied to a fixed vector, with rational entries at rational data (Theorem 27.5): price that of the matrix of the chain cell plus $\lceil \log_2(M+1) \rceil + O(1)$, work zero.*

*In the cases with a multiplicity $M$ the expanded family has of the order of $M$ times as many marks, and its period written as one integral over its simplex has the work of one less than its number of marks.*

**Theorem 33.4** (Repeated subtrees and departures)**.** *(i) In a construction, replacing identical subtrees, with matching hinges and data, by one subtree and references leaves the family built, the period and every exact word of Part V unchanged, and lowers the price by the price of the copies removed, less that of the references.*

*(ii) In (ii), (iii) and (iv) of Theorem 33.3, replacing $q$ of the $M$ copies by cells $\mathcal{A}'_1, \dots, \mathcal{A}'_q$ changes the word to $\Gamma_1^2 \begin{bmatrix} \kappa, & t-\kappa \\ & t \end{bmatrix} \left( P_{\mathcal{A}} / \Gamma_1^2 \begin{bmatrix} \kappa, & t-\kappa \\ & t \end{bmatrix} \right)^{M-q} \prod_i \left( P_{\mathcal{A}'_i} / \Gamma_1^2 \begin{bmatrix} \kappa_i, & t-\kappa_i \\ & t \end{bmatrix} \right)$ at a full lean, to the product of $\mathfrak{L}_{\mathcal{A}}^{M-q}$ and of the responses of the new cells in a series closure, and to the composite of $\mathcal{Q}^{M-q}$ with the $q$ operators of the new cells in a chain: the price grows by $\sum_i \operatorname{price}(\mathcal{A}'_i) + O(q \log M)$ and the work by at most the dimensions of the integrals of the new cells. A change of $q$ leaves in a tree with at most $h$ nodes on every path from its root to a leaf touches at most $qh$ of its nodes.*

Repetition is thus priced at the binary length of its count, as $2^M$ is written with $M$ rather than with its digits, and the work follows the distinct subtrees. The construction is the certificate of the compression: a family given expanded, without its construction, can still be read for its bunches of width one, its prime nodes (Theorem 23.16) and its twins (Theorem 14.22), but finding its cheapest construction is a search. The prime nodes of the canonical hinge tree of a list are invariants of the list (Theorem 23.16); at a full lean its period is a product of stacked Gammas and the periods at the full leans of those nodes (Corollary 26.11, Theorem 26.5), and at every lean a fibre integral of their periods at the full leans, of dimension $n-1$ (Theorem 26.13); a prime node $\mathcal{H}_v$ on $n_v$ marks with $|\mathcal{H}_v|$ bases has periods at the full leans of dimension $|\mathcal{H}_v| - n_v > 0$ on their share sides (Corollary 26.8). Symmetry is a second source of cheapness, distinct from repetition: a table assembled from a seed by a set of bijections (Proposition 11.6) is described by the seed and the bijections, and the table that is one circuit on $n$ marks, prime for no $n$, is a series closure of single marks whose period is one integral over one scale of a product of $n$ Grammas (Corollary 18.24), of work one whatever $n$; the list $\binom{\mathcal{E}}{r}$ with $2 \le r \le |\mathcal{E}| - 2$ is prime and cheap to describe, and its period at each full lean is one fibre integral of dimension $\binom{|\mathcal{E}|}{r} - |\mathcal{E}|$ on its share side (Corollary 26.8).

# Part VI

# The arithmetic of the output

# Chapter 34

# The motive of a family and the coaction along a tree

A family of lists at integer data whose proper walls are all at least one has as its period the period of a mixed motive: the relative cohomology of the toric variety of the braid fan, less the closures of the zero loci of its letters and of the linear letters, relative to its toric boundary. The walls are the orders of the form along the boundary divisors, and every boundary stratum is the product of the models of the steps of its flag. Brown's coaction acts on the motivic period, and birational changes of variables, Cremona among them, are identities of motivic periods. Along any tree of bunches the left factors of the coaction of weight at most $k$ are sums of products of open periods of steps of at most $k+1$ marks; each step is attached to one vertex of the tree, its letters are letters of children of the tables there, and its open periods are rational combinations of constant terms of continued periods of families of lists on the step, hence of convergent periods of such families and of their derivatives in the masses. At a node of width $c$ the bunch and its room meet only through the Skeleton values of the letters, at most $c$ for each letter: every open period of a node step is a shape integral of rational functions against finite parts of scale integrals, constant terms of stacked Gammas times Bretas of one bunch at the joint spectrum, and at the root so is every left factor; at width one each Skeleton value is a quotient of hinge values and the share side separates the bunch from the room. Cells read through one hinge value split off as identities of motivic periods, whatever the widths of the bunches of the core; so do the cells of dual pairs at every pair of exponents, and at a full lean the motivic period of a list is a rational multiple of the product of those of its prime nodes. A pinned set of at least two marks at which every active table has width one splits off motivically.

## 34.1 The model

The motive of a family lives on the toric variety of the braid fan. Its charts are the sector cubes of Theorem 8.5(ii), and its boundary strata are indexed by chains of sets of marks.

**Definition 34.1** (The model of a family)**.** Let $F = ((\mathcal{B}_j; s_j, t_j))_{j=1}^{N}$ be a family of lists on $\mathcal{E}$, $n = |\mathcal{E}| \geq 2$ (Definition 8.1), with integer exponents and integer masses $\mathbf{m}$ on its balance wall; it is *convergent* when $\kappa_S \geq 1$ for every $\varnothing \neq S \subsetneq \mathcal{E}$. Its *letters* are $\mathrm{Let}(F) = \{K_j : \ s_j \neq 0\} \cup \{\Psi_j : t_j \neq 0\}$; the *linear letters* are the $\alpha_J = \sum_{e \in J} \alpha_e$, $J \subseteq \mathcal{E}$, the basis polynomials of the lists $\binom{J}{1}$; and the *augmented letters* are $\mathrm{Let}^{+}(F) = \mathrm{Let}(F) \cup \{\alpha_J : \ J \subseteq \mathcal{E}, \ |J| \geq 2\}$. For an order $e_1, \ldots, e_n$ of

$\mathcal{E}$ put

$$\Omega_{\mathcal{E}} = \sum_{i=1}^{n} (-1)^{i-1} \alpha_{e_i}\, d\alpha_{e_1} \wedge \cdots \wedge \widehat{d\alpha_{e_i}} \wedge \cdots \wedge d\alpha_{e_n},$$

which another order changes by its sign; the *form* of $F$ is

$$\omega_F = \alpha^{\mathbf{m}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, \frac{\Omega_{\mathcal{E}}}{\alpha^{\mathcal{E}}},$$

a rational form of degree $n-1$ on $\mathbb{P}(\mathbb{Q}^{\mathcal{E}})$. Let $X_{\mathcal{E}}$ be the toric variety over $\mathbb{Q}$ of the *braid fan* in $\mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$, whose rays are the images $\rho_S$ of the $\mathbf{1}_S$, $\varnothing \neq S \subsetneq \mathcal{E}$, and whose cones are spanned by the rays of the chains $S_1 \subsetneq \cdots \subsetneq S_\ell$; $D_S$ is the divisor of $\rho_S$, $D = \bigcup_S D_S$ the *boundary*, and the *positive cell* $\widetilde{\sigma}_{\mathcal{E}}$ is the closure in $X_{\mathcal{E}}(\mathbb{R})$ of the set $\widetilde{\sigma}^\circ_{\mathcal{E}}$ of the points of the torus with every $\alpha_e > 0$. An order $e_1, \dots, e_n$ of $\mathcal{E}$ gives the *chart* $U \cong \mathbb{A}^{n-1}$ of the cone of the chain $S_i = \{e_1, \dots, e_i\}$, $i < n$, with the coordinates $\varepsilon_i = \alpha_{e_i}/\alpha_{e_{i+1}}$, so that $\alpha_{e_j}/\alpha_{e_n} = \prod_{i \geq j} \varepsilon_i$. For a polynomial $f$ on $\mathcal{E}$, $\widetilde{Y}_f$ is the closure in $X_{\mathcal{E}}$ of its zero locus in the torus. The *model* of $F$ is the pair

$$V_F = X_{\mathcal{E}} \setminus \bigcup_{f \in \mathrm{Let}^+(F)} \widetilde{Y}_f, \qquad D_F = D \cap V_F,$$

with the relative cohomology $H_F = H^{n-1}(V_F, D_F)$.

**Definition 34.2** (Flags and steps)**.** A *flag* of $\mathcal{E}$ is a chain $\mathfrak{f}: \varnothing = S_0 \subsetneq S_1 \subsetneq \cdots \subsetneq S_\ell \subsetneq S_{\ell+1} = \mathcal{E}$, $\ell \geq 0$. Its *steps* are the sets $Z_i = S_i \setminus S_{i-1}$, $1 \leq i \leq \ell+1$, as in Theorem 22.9, and its *dimension* is $\sum_i (|Z_i| - 1) = n - 1 - \ell$. In the model of a family $F$ (Definition 34.1) its *stratum* is $D_{\mathfrak{f}} = D_{S_1} \cap \cdots \cap D_{S_\ell} \cap V_F$, the model $V_F$ itself at $\ell = 0$. The *letters of the step* $(Z_i; \mathfrak{f})$ are the step letters at $Z_i$ (Proposition 12.20) of the augmented letters of $F$ that are not monomials: $K$ of the step list of $\mathcal{B}_j$ at $Z_i$ at the bottom of the chain for every $K_j \in \mathrm{Let}(F)$, $\Psi$ of its step list at the top for every $\Psi_j \in \mathrm{Let}(F)$, and the $\alpha_J$ with $J \subseteq Z_i$, $|J| \geq 2$; for $\ell = 0$ the step lists at the one step $\mathcal{E}$ are the lists of $F$ themselves. They form a set $\mathrm{Let}(Z_i; \mathfrak{f})$ of polynomials on $Z_i$, which depends on $S_{i-1}$ and $S_i$ alone. The *model of the step* is

$$V_{Z_i;\mathfrak{f}} = X_{Z_i} \setminus \bigcup_{g \in \mathrm{Let}(Z_i;\mathfrak{f})} \widetilde{Y}_g,$$

a point when $|Z_i| = 1$, with its toric boundary $\partial V_{Z_i;\mathfrak{f}}$.

**Lemma 34.3** (Charts, strata and the letters in a chart)**.** *In Definition 34.1:*

*(i) $X_{\mathcal{E}}$ is smooth and projective of dimension $n-1$, and the charts of the orders cover it. In the chart of the order $e_1, \dots, e_n$, $D_{S_i} \cap U = \{\varepsilon_i = 0\}$ for $i < n$ and $D_S \cap U = \varnothing$ for every other $S$; $D$ is a divisor with simple normal crossings and smooth components, and $D_{S_1} \cap \cdots \cap D_{S_\ell}$ is not empty exactly when the $S_i$ form a chain. In the chart $\widetilde{\sigma}_{\mathcal{E}} \cap U = [0, \infty)^{n-1}$, and the closure of the set of the points of $\widetilde{\sigma}^\circ_{\mathcal{E}}$ with $\alpha_{e_1} < \cdots < \alpha_{e_n}$, the* sector *of the order, is the cube $[0,1]^{n-1}$; the sectors of the orders cover $\widetilde{\sigma}_{\mathcal{E}}$, two of them meeting in a set of measure zero.*

*(ii) Let $f$ be a letter $K$ or $\Psi$ of a list $\mathcal{B}$ on $\mathcal{E}$, or a linear letter $\alpha_J$, of the degree $\delta$, with the* floor *$d_f(S)$: $d(S)$ for $K$, $d^*(S)$ for $\Psi$ (Theorem 7.13), and 1 or 0 for $\alpha_J$ as $J \subseteq S$ or not. In the chart of an order,*

$$f(\alpha)/\alpha_{e_n}^{\delta} = \prod_{i<n} \varepsilon_i^{d_f(S_i)}\, \widetilde{f}(\varepsilon),$$

*with $\widetilde{f}$ a polynomial with nonnegative integer coefficients and constant term 1, the sector letter of Theorem 8.5(ii); so $\widetilde{f} \geq 1$ where $\varepsilon \geq 0$. For $I \subseteq \{1, \dots, n-1\}$, the restriction of $\widetilde{f}$ to $\{\varepsilon_i = 0 : \ i \in I\}$ is the product, over the steps $Z$ of the chain $(S_i)_{i \in I}$, of the polynomials that this rule attaches to the step letters of $f$ at $Z$ (Proposition 12.20) in the chart of $X_Z$ of the order induced on $Z$, whose coordinates are the $\varepsilon_k$ with $e_k, e_{k+1} \in Z$. Moreover $\widetilde{Y}_f \cap U = \{\widetilde{f} = 0\}$.*

*(iii) In the chart, with the walls $\kappa_S$ of $F$,*

$$\omega_F = \pm \prod_{i<n} \varepsilon_i^{\kappa_{S_i}-1} \prod_j \widetilde{K}_j^{-s_j} \widetilde{\Psi}_j^{-t_j} \, d\varepsilon_1 \wedge \cdots \wedge d\varepsilon_{n-1}.$$

**Proposition 34.4** (The model)**.** *Let $F$ be a family as in Definition 34.1.*

*(i) $\widetilde{\sigma}_{\mathcal{E}}$ is compact, $\widetilde{\sigma}_{\mathcal{E}} \setminus \widetilde{\sigma}_{\mathcal{E}}^{\circ} \subseteq D$, and $\widetilde{\sigma}_{\mathcal{E}}$ misses $\widetilde{Y}_f$ for every letter $f$ of a list on $\mathcal{E}$ and every linear letter; so $\widetilde{\sigma}_{\mathcal{E}} \subseteq V_F(\mathbb{R})$.*

*(ii) $\omega_F$ has the order $\kappa_S - 1$ along $D_S$ for every $\varnothing \neq S \subsetneq \mathcal{E}$ and neither zeros nor poles on the points of the torus in $V_F$. So $\omega_F$ is a regular form on $V_F$ exactly when $F$ is convergent, and then its integral over $\widetilde{\sigma}_{\mathcal{E}}$, oriented so that it is positive, converges and is the period $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ (Theorem 8.7).*

*(iii) $V_F$ is a smooth affine variety, $D_F$ is a divisor in it with simple normal crossings and smooth components, and every stratum is smooth and affine. More generally $X_{\mathcal{E}} \setminus \bigcup_{g \in \mathcal{G}} \widetilde{Y}_g$ is affine for every finite set $\mathcal{G}$ of polynomials on $\mathcal{E}$ that contains the linear letters $\alpha_J$, $|J| \geq 2$.*

*(iv) The stratum of a flag $\mathfrak{f}$ with the steps $Z_1, \dots, Z_{\ell+1}$ is isomorphic to $\prod_i V_{Z_i;\mathfrak{f}}$ (Definition 34.2), compatibly with the boundaries, and $\widetilde{\sigma}_{\mathcal{E}}$ meets it in $\prod_i \widetilde{\sigma}_{Z_i}$.*

*(v) For the tables of $F$ and complex masses and exponents on the balance wall, wherever every proper wall has positive real part,*

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \sum_{\pi} \operatorname{cube}\big(\boldsymbol{\kappa}_\pi(\mathbf{m}); h_\pi\big), \qquad \boldsymbol{\kappa}_\pi(\mathbf{m}) = \big(\kappa_{S_i^\pi}(\mathbf{m})\big)_{i<n}, \quad h_\pi = \prod_j \widetilde{K}_j^{-s_j} \widetilde{\Psi}_j^{-t_j},$$

*the sum over the orders $\pi$ of $\mathcal{E}$, with the chains $S_i^\pi$ and the sector letters of their charts, and* cube *the cube integral of Lemma 8.4. The right side continues meromorphically along the balance wall, jointly in the exponents, with poles only on the hyperplanes $\kappa_S = -j$, $\varnothing \neq S \subsetneq \mathcal{E}$, $j \in \mathbb{N}$: it is the* continued period *of Theorems 8.5(iii) and 8.7.*

The walls are thus the orders of the form along the boundary, and convergence is regularity. The linear letters are removed so that the model is affine; they are the letters of the lists $\binom{J}{1}$, and the motivic period does not see them (Lemma 34.7).

## 34.2 Motivic periods

**Definition 34.5** (Motivic periods; Galois conjugates; open spaces)**.** An object $M$ of Brown's category of realizations over $\mathbb{Q}$ (F. Brown, *Notes on motivic periods*, Commun. Number Theory Phys. 11 (2017), 557–655, §3.1) has a Betti realization $M_{\mathrm{B}}$ and a de Rham realization $M_{\mathrm{dR}}$; its *motivic periods* $[M, \gamma, \omega]^{\mathrm{mot}}$, $\gamma \in M_{\mathrm{B}}^\vee$ and $\omega \in M_{\mathrm{dR}}$, span the ring $\mathcal{P}^{\mathrm{mot}}$, bilinear in $(\gamma, \omega)$,

with $[M,\phi^{\mathrm{t}}\gamma',\omega]^{\mathrm{mot}} = [M',\gamma',\phi(\omega)]^{\mathrm{mot}}$ for every morphism $\phi : M \to M'$; its de Rham periods $[M,f,\omega]^{\mathrm{dR}}$, $f \in M_{\mathrm{dR}}^{\vee}$, span the ring $\mathcal{P}^{\mathrm{dR}}$; and the *coaction* $\Delta : \mathcal{P}^{\mathrm{mot}} \to \mathcal{P}^{\mathrm{mot}} \otimes \mathcal{P}^{\mathrm{dR}}$ is

$$\Delta[M,\gamma,\omega]^{\mathrm{mot}} = \sum_i [M,\gamma,e_i]^{\mathrm{mot}} \otimes [M,e_i^{\vee},\omega]^{\mathrm{dR}}$$

for a basis $(e_i)$ of $M_{\mathrm{dR}}$ with the dual basis $(e_i^{\vee})$; the $[M,\gamma,e_i]^{\mathrm{mot}}$ are its *left factors*. For $\xi \in \mathcal{P}^{\mathrm{mot}}$ the space of its *Galois conjugates* is

$$\mathrm{C}(\xi) = \big\{(\mathrm{id}\otimes\varphi)\Delta\xi : \ \varphi \in \mathrm{Hom}_{\mathbb{Q}}(\mathcal{P}^{\mathrm{dR}},\mathbb{Q})\big\};$$

for an object $M$ and $\gamma \in M_{\mathrm{B}}^{\vee}$, $L(M,\gamma) = \{[M,\gamma,\omega]^{\mathrm{mot}} : \ \omega \in M_{\mathrm{dR}}\}$; and $\mathcal{A}\cdot\mathcal{A}'$ is the span of the products of the elements of two subspaces. For a convergent family $F$ (Definition 34.1) the *motivic period* is

$$P_F^{\mathrm{mot}} = [H_F, \widetilde{\sigma}_{\mathcal{E}}, \omega_F]^{\mathrm{mot}},$$

the positive cell, oriented so that $\Omega_{\mathcal{E}}/\alpha^{\mathcal{E}}$ is positive on $\widetilde{\sigma}_{\mathcal{E}}^{\circ}$, read as a class of relative homology, and the form as a class of relative de Rham cohomology; its period is $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ (Proposition 34.4(ii)). For a step $(Z;\mathfrak{f})$ of a flag of $\mathcal{E}$ (Definition 34.2) with $|Z| \geq 2$, its *open space* is the set of motivic periods

$$\mathrm{Open}(Z;\mathfrak{f}) = \Big\{ \big[H^{|Z|-1}(V_{Z;\mathfrak{f}}, \partial V_{Z;\mathfrak{f}}), \widetilde{\sigma}_Z, \eta\big]^{\mathrm{mot}} : \ \eta \text{ a regular form of degree } |Z|-1 \text{ on } V_{Z;\mathfrak{f}} \Big\},$$

$\widetilde{\sigma}_Z$ oriented as $\widetilde{\sigma}_{\mathcal{E}}$ is, and $\mathrm{Open}(Z;\mathfrak{f}) = \mathbb{Q}$ when $|Z| = 1$.

**Lemma 34.6** (Galois conjugates)**.** *For $\xi,\xi' \in \mathcal{P}^{\mathrm{mot}}$, an object $M$ and $\gamma \in M_{\mathrm{B}}^{\vee}$:*

*(i)* $\xi \in \mathrm{C}(\xi)$*;*

*(ii)* $\mathrm{C}([M,\gamma,\omega]^{\mathrm{mot}}) \subseteq L(M,\gamma)$*, and* $\Delta L(M,\gamma) \subseteq L(M,\gamma)\otimes\mathcal{P}^{\mathrm{dR}}$*;*

*(iii)* $\mathrm{C}(\xi+\xi') \subseteq \mathrm{C}(\xi)+\mathrm{C}(\xi')$*,* $\mathrm{C}(\xi\xi') \subseteq \mathrm{C}(\xi)\cdot\mathrm{C}(\xi')$*, and* $\mathrm{C}(q\,\xi) = \mathrm{C}(\xi)$ *for* $q \in \mathbb{Q}$*,* $q \neq 0$*.*

**Lemma 34.7** (Birational invariance)**.** *For $i = 1,2$ let $X_i$ be a smooth projective variety over $\mathbb{Q}$ of dimension $m$, $D_i \subset X_i$ a divisor with simple normal crossings and smooth components, $Y_i \subset X_i$ a closed set, $X_i^{\circ} = X_i \setminus Y_i$ and $D_i^{\circ} = D_i \setminus Y_i$. Let $\sigma_i \subset X_i(\mathbb{R})$ be compact, the closure of an oriented open semialgebraic set $\sigma_i^{\circ} \subseteq (X_i \setminus (D_i \cup Y_i))(\mathbb{R})$ of dimension $m$, with $\sigma_i \cap Y_i = \varnothing$ and $\sigma_i \setminus \sigma_i^{\circ} \subseteq D_i$, and let $\omega_i$ be a regular form of degree $m$ on $X_i^{\circ}$. If an isomorphism $\Phi : U_1 \to U_2$ of dense open sets $U_i \subseteq X_i \setminus (D_i \cup Y_i)$ with $\sigma_i^{\circ} \subseteq U_i(\mathbb{R})$ maps $\sigma_1^{\circ}$ onto $\sigma_2^{\circ}$ preserving the orientations and has $\Phi^*\omega_2 = \omega_1$, then*

$$\big[H^m(X_1^{\circ}, D_1^{\circ}), \sigma_1, \omega_1\big]^{\mathrm{mot}} = \big[H^m(X_2^{\circ}, D_2^{\circ}), \sigma_2, \omega_2\big]^{\mathrm{mot}}.$$

*In particular, with $\Phi$ the identity, removing a larger closed set that misses $\sigma_1$ leaves the motivic period unchanged; the motivic period of a convergent family is the same in the model without the linear letters, $X_{\mathcal{E}} \setminus \bigcup_{f\in\mathrm{Let}(F)} \widetilde{Y}_f$; and two convergent families on $\mathcal{E}$ with one form have one motivic period.*

**Corollary 34.8** (Cremona between motivic periods)**.** *Let $F = ((\mathcal{B}_j; s_j, t_j))_j$ be a convergent family with the masses $\mathbf{m}$ and the total strength $\sigma$, and $F^{\mathrm{c}}$ the family of the same tables with the exponents $(t_j, s_j)$ and the masses $2\sigma\mathbf{1} - \mathbf{m}$. Then $F^{\mathrm{c}}$ is convergent, its wall at $S$ is the wall of $F$ at $\mathcal{E} \setminus S$, and $P_{F^{\mathrm{c}}}^{\mathrm{mot}} = P_F^{\mathrm{mot}}$: Theorem 8.14 is an identity of motivic periods.*

## 34.3 Steps along a tree of bunches

The boundary strata are indexed by flags, and a stratum is a product over the steps of its flag (Proposition 34.4(iv)). A tree of bunches (Definition 20.7) assigns every step to one of its vertices.

**Definition 34.9** (Attachment)**.** Let $\mathcal{T}$ be a tree of bunches of a family $F$ (Definition 20.7) and $(Z;\mathfrak{f})$ a step of a flag of $\mathcal{E}$. Starting at the root, $Z$ descends through vertices whose marks contain it:

(i) at a split vertex, if $Z$ lies in one of its children it descends to that child; otherwise $Z$ is *attached* there as a *node step*;

(ii) at a node $v$ that is not a leaf, with its bunch $B_v$: if $Z \subseteq B_v$, $Z$ is attached to $v$ as a *bunch step*; if $Z$ meets $B_v$ and $M_v \setminus B_v$, as a *node step*; and if $Z \subseteq M_v \setminus B_v$, it descends to the room of $v$;

(iii) at a leaf, $Z$ is attached as a *leaf step*.

Every step is attached to exactly one vertex.

**Lemma 34.10** (The letters of an attached step)**.** *Let $\mathcal{T}$ be a tree of bunches of a family $F$ as in Definition 34.1, and let the step $(Z;\mathfrak{f})$ be attached to the vertex $v$, with the marks $M_v$. Every letter of the step other than the linear letters is $K$ or $\Psi$ of the child, at a pattern of $M_v \setminus Z$ (Definition 12.1), of a table at $v$ that carries that letter.*

Every family has trees of bunches: at every node that is not a leaf choose a bunch and pass to its room, which has fewer marks. For nonnegative exponents the Foreman tree is one of them (Definition 14.4); with nonnegative exponents merging equal children keeps every letter, so the record and the peel give the same tree, and it is fixed by the tables and the rays of the exponent pairs (Theorem 14.6). The statements below hold along every tree of bunches, and along the Foreman tree the vertex at which each step is attached, and the node form there, are fixed by the family.

## 34.4 The weights of the coaction

**Lemma 34.11** (One form carries every period)**.** *Let $V$ be a smooth affine variety over $\mathbb{Q}$ of dimension $m$ and $D \subset V$ a divisor with simple normal crossings and smooth components. Every class of the relative de Rham cohomology $H^m_{\mathrm{dR}}(V,D)$ is the class of a regular form of degree $m$ on $V$. So, for a convergent family $F$ (Definition 34.1), $L(H_F,\widetilde{\sigma}_{\mathcal{E}})$ is the open space of the one step $\mathcal{E}$ of the flag $\varnothing \subsetneq \mathcal{E}$, and for every step, $L\big(H^{|Z|-1}(V_{Z;\mathfrak{f}},\partial V_{Z;\mathfrak{f}}),\widetilde{\sigma}_Z\big) = \mathrm{Open}(Z;\mathfrak{f})$.*

So the open space of the whole set of marks already holds every period of $H_F$ on its cell, and a decomposition of the left factors into products over steps carries information only when it is graded. The steps enter through the weight filtration $W$ of the de Rham realization, by Brown's weight-graded face maps.

**Proposition 34.12** (Weight-graded faces)**.** *Let $F$ be convergent (Definition 34.1), $k < n-1$ and $e \in W_k(H_F)_{\mathrm{dR}}$. Then*

$$[H_F,\widetilde{\sigma}_{\mathcal{E}},e]^{\mathrm{mot}} \in \sum_{\mathfrak{f}\ \textit{of dimension } k}\ \prod_{\textit{steps } Z\ \textit{of } \mathfrak{f}} \mathrm{Open}(Z;\mathfrak{f}).$$

**Theorem 34.13** (Small steps along the tree)**.** *Let $F$ be convergent, $\mathcal{T}$ a tree of bunches of $F$, and $(e_i)$ a basis of $(H_F)_{\mathrm{dR}}$ adapted to the weight filtration $W$. For $k < n-1$ every left factor $[H_F, \widetilde{\sigma}_{\mathcal{E}}, e_i]^{\mathrm{mot}}$ of the coaction of $P_F^{\mathrm{mot}}$ with $e_i \in W_k$ satisfies*

$$[H_F, \widetilde{\sigma}_{\mathcal{E}}, e_i]^{\mathrm{mot}} \in \sum_{\mathfrak{f} \text{ of dimension } k} \ \prod_{\text{steps } Z \text{ of } \mathfrak{f}} \mathrm{Open}(Z; \mathfrak{f}).$$

*Only steps of at most $k+1$ marks occur. Each is attached to exactly one vertex $v$ of $\mathcal{T}$, and every letter of it other than the linear letters is $K$ or $\Psi$ of the child, at a pattern of $M_v \setminus Z$, of a table at $v$ that carries that letter.*

The theorem is read on Brown's affine model, which removes the linear letters $\alpha_J$ to make the complement affine. They are the letters of the lists $\binom{J}{1}$, and they enter the families of the steps below as tables. Each open period is a constant term of continued periods, and those constant terms are read on convergent data.

**Proposition 34.14** (Laurent coefficients are read on convergent data)**.** *Let $F = ((\mathcal{B}_j; s_j, t_j))_{j=1}^N$ be a family of lists on $\mathcal{E}$ with integer exponents and integer masses $\mathbf{m}$ on its balance wall, and $\mathbf{v} \in \mathbb{Q}^{\mathcal{E}}$ with $v_{\mathcal{E}} = 0$ and $v_S \neq 0$ for every $\varnothing \neq S \subsetneq \mathcal{E}$. Then $\varepsilon \mapsto P_F(\mathbf{m} + \varepsilon\mathbf{v}; \mathbf{s}, \mathbf{t})$, the continued period (Proposition 34.4(v)), is meromorphic near $\varepsilon = 0$, and there are finitely many convergent families $F_k = ((\mathcal{B}_j; s_j^{(k)}, t_j^{(k)}))_j$ on the same tables, with $\mathbf{s}^{(k)} - \mathbf{s},\ \mathbf{t}^{(k)} - \mathbf{t} \in \mathbb{N}^N$ and integer masses $\mathbf{m}^{(k)}$ on their balance walls, and rational functions $R_k \in \mathbb{Q}(\varepsilon)$, with*

$$P_F(\mathbf{m} + \varepsilon\mathbf{v}; \mathbf{s}, \mathbf{t}) = \sum_k R_k(\varepsilon)\, P_{F_k}\big(\mathbf{m}^{(k)} + \varepsilon\mathbf{v}; \mathbf{s}^{(k)}, \mathbf{t}^{(k)}\big)$$

*near $\varepsilon = 0$. So every coefficient of the Laurent expansion at $\varepsilon = 0$ is a rational combination of the absolutely convergent integrals*

$$\partial_\varepsilon^l P_{F_k}\big(\mathbf{m}^{(k)} + \varepsilon\mathbf{v}; \mathbf{s}^{(k)}, \mathbf{t}^{(k)}\big)\Big|_{\varepsilon=0} = \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}^{(k)} - \mathbf{1}} \langle \mathbf{v}, \log u \rangle^l \prod_j K_j(u)^{-s_j^{(k)}} \Psi_j(u)^{-t_j^{(k)}}\, d\omega_{\mathcal{E}}(u),$$

*$l \in \mathbb{N}$. If $\mathbf{s}, \mathbf{t} \geq 0$, every $F_k$ has nonnegative exponents and positive masses.*

**Corollary 34.15** (Open periods from families of lists)**.** *Let $(Z; \mathfrak{f})$ be a step of a flag of $\mathcal{E}$ with $|Z| \geq 2$ and $\eta$ a regular form of degree $|Z| - 1$ on $V_{Z;\mathfrak{f}}$. On the torus,*

$$\eta = \sum_t c_t\, \omega_{F_t}, \qquad c_t \in \mathbb{Q},$$

*a finite sum over families $F_t$ of lists on $Z$ at integer data on their balance walls: the tables of $F_t$ are the lists of the letters of the step, the step list for its $K$ or its $\Psi$ and $\binom{J}{1}$ for $\alpha_J$, each carrying its letter with an exponent $k_{t,g} \in \mathbb{N}$, and the masses are some $\mathbf{a}_t \in \mathbb{Z}^Z$. For every $\mathbf{v} \in \mathbb{Q}^Z$ with $v_Z = 0$ and $v_S \neq 0$ for every $\varnothing \neq S \subsetneq Z$, each continued period $P_{F_t}(\mathbf{a}_t + \varepsilon\mathbf{v})$ (Proposition 34.4(v)) is meromorphic in $\varepsilon$ near $0$, and*

$$\int_{\widetilde{\sigma}_Z} \eta = \sum_t c_t \operatorname*{CT}_{\varepsilon=0} P_{F_t}(\mathbf{a}_t + \varepsilon\mathbf{v}).$$

*By Proposition 34.14 each constant term is a rational combination of convergent periods of families of lists on $Z$ with nonnegative integer exponents and positive integer masses, values of the serif family (Proposition 8.12), and of their derivatives along $\mathbf{v}$ in the masses. So every element of $\mathrm{Open}(Z; \mathfrak{f})$ has such a number as its period, and by Theorem 34.13 every left factor of weight at most $k$ has as its period a sum of products of such numbers over the steps of the flags of dimension $k$, whose letters are letters of children of the tables at the vertices where the steps are attached.*

## 34.5 The node

**Theorem 34.16** (The node form)**.** *Let $F$ be convergent (Definition 34.1), $\mathcal{T}$ a tree of bunches of $F$, $v$ a node of $\mathcal{T}$ that is not a leaf, with its bunch $B_v$ of width $c$ (Definition 20.7), and $(Z;\mathfrak{f})$ a node step of $v$ (Definition 34.9), with $Z_B = Z \cap B_v$ and $Z_R = Z \cap (M_v \setminus B_v)$. Read the bunch side in scale and shape and the room side on its simplex, $\alpha_{Z_B} = \rho u$ with $\rho > 0$ and $u \in \Delta_{Z_B}$, and $\alpha_{Z_R} = u' \in \Delta_{Z_R}$ (Lemma 2.7).*

*(i) Every letter $g$ of the step is the basis polynomial of a list $\mathcal{N}_g$ on $Z$: the step list at the bottom for a letter $K$, the dual of the step list at the top for a letter $\Psi$ (Proposition 7.2), and $\binom{J}{1}$, its marks outside $J$ loops, for $\alpha_J$. The width $c_g$ of $Z_B$ in $\mathcal{N}_g$ is at most $c$, and $c \geq 1$.*

*(ii) With $d_g$ the floor of $Z_B$ in $\mathcal{N}_g$ and the Skeleton values $\xi_{g,1}, \dots, \xi_{g,c_g}$ of $K_{\mathcal{N}_g}$ along $Z_B$ at $(u,u')$ (Definition 12.10),*

$$g(\rho u, u') = \rho^{d_g}\, K_{\mathcal{N}_g|Z_B}(u)\, K_{\mathcal{N}_g/Z_B}(u') \prod_{i=1}^{c_g} \big(\rho + \xi_{g,i}(u,u')\big),$$

*the leading coefficient being the top level of $Z_B$ (Theorem 12.19); these Skeleton values, each with its letter, form the joint spectrum of $Z_B$ at $(u,u')$ (Definition 12.14).*

*(iii) For every regular form $\eta$ of degree $|Z|-1$ on $V_{Z;\mathfrak{f}}$ there are finitely many terms $t$, with rational functions $A_t$ on $\Delta_{Z_B}$ and $B_t$ on $\Delta_{Z_R}$, integers $\nu_t$ and letter exponents $k_{t,g} \in \mathbb{N}$, such that*

$$\int_{\widetilde{\sigma}_Z} \eta = \int_{\Delta_{Z_B} \times \Delta_{Z_R}} \sum_t A_t(u)\, B_t(u')\, \mathrm{Sc}_t(u,u')\, d\omega_{Z_B}(u)\, d\omega_{Z_R}(u'),$$

*the integrand absolutely integrable. Here $\mathrm{Sc}_t = 0$ when no letter with $k_{t,g} > 0$ has a Skeleton value, and otherwise*

$$\begin{aligned} \mathrm{Sc}_t &= \mathrm{Reg} \int_0^\infty \rho^{\nu_t - 1} \prod_{g,i} \big(\rho + \xi_{g,i}\big)^{-k_{t,g}}\, d\rho \\ &= \operatorname*{CT}_{\kappa=\nu_t} \Gamma_{|J_t|}^{\,2} \begin{bmatrix} \kappa, & |\mathbf{w}_t| - \kappa \\ & \mathbf{w}_t \end{bmatrix} \mathsf{B}^{\{J_t\}}_{\mathbf{w}_t, |\mathbf{w}_t|-\kappa;\, I_{J_t}}\big(\boldsymbol{\xi}(u,u')\big), \end{aligned}$$

*$J_t$ indexing the Skeleton values $\xi_{g,i}$ of the letters with $k_{t,g} > 0$, each with the letter exponent $w_{t,(g,i)} = k_{t,g}$, so that $|\mathbf{w}_t| = \sum_g k_{t,g} c_g$: the finite part in $\rho$ (Definition 1.5) is the constant term at $\nu_t$ of the stacked Gamma times the Breta of one bunch of Proposition 15.1, continued from the strip $0 < \mathrm{Re}\,\kappa < |\mathbf{w}_t|$. This is the* node form *of $\eta$ at $v$: the bunch and the room meet only through the joint spectrum, as the coefficients of Bretas of one bunch at the unit configuration.*

*(iv) (*Width one.*) If $c = 1$, every letter has at most one Skeleton value, the room's hinge value over the bunch's, $\xi_g = \gamma_R^g(u')/\gamma_B^g(u)$ with $\gamma_B^g = K_{\mathcal{N}_g|Z_B}/K_{\mathcal{N}_g/Z_R}$ and $\gamma_R^g = K_{\mathcal{N}_g|Z_R}/K_{\mathcal{N}_g/Z_B}$ (Corollary 19.5), positive. For $\kappa$ in the strip, the share side (Corollary 15.5) writes the stacked Gamma times the Breta of the term $t$ as an integral over $|J_t|-1$ shares of stacked Gammas times powers $\xi_g^{-b_g} = \gamma_R^g(u')^{-b_g} \gamma_B^g(u)^{\,b_g}$, the $b_g$ affine in the shares and in $\kappa$, with real parts constant along the contour. On every such contour on which these stacked Gammas have arguments with positive real parts and*

$$\int_{\Delta_{Z_B}} |A_t| \prod_g (\gamma_B^g)^{\mathrm{Re}\, b_g}\, d\omega_{Z_B} < \infty, \qquad \int_{\Delta_{Z_R}} |B_t| \prod_g (\gamma_R^g)^{-\mathrm{Re}\, b_g}\, d\omega_{Z_R} < \infty,$$

*the integral over $\Delta_{Z_B} \times \Delta_{Z_R}$ of $A_t B_t$ times the stacked Gamma times the Breta is the share integral of the stacked Gammas times $\int_{\Delta_{Z_B}} A_t \prod_g (\gamma_B^g)^{b_g} d\omega_{Z_B}$ times $\int_{\Delta_{Z_R}} B_t \prod_g (\gamma_R^g)^{-b_g} d\omega_{Z_R}$: the bunch and the room separate. When $0 < \nu_t < |\mathbf{w}_t|$, $\mathrm{Sc}_t$ is the convergent scale integral at $\kappa = \nu_t$, and the term itself separates.*

*(v) If the root of $\mathcal{T}$ is a node that is not a leaf, the period of every element of $L(H_F, \widetilde{\sigma}_{\mathcal{E}})$, every left factor of the coaction of $P_F^{\mathrm{mot}}$ among them, is given by the node form at the root.*

*(vi) At a split vertex and a node step $Z$ of it that meets the child $M_1$, the same holds with $Z_B = Z \cap M_1$, $Z_R = Z \setminus M_1$ and $c = 1$: the letters other than the linear letters have width zero at $Z_B$, and only the $\alpha_J$ with $J$ meeting $Z_B$ and $Z_R$ have a Skeleton value.*

At width two or more the Skeleton values are the negatives of the roots of Skeleton polynomials of degree $c_g$, whose extreme coefficients are products of a bunch letter and a room letter and whose middle coefficients are sums over the patterns of their levels (Theorem 12.19); the node form reads them as the coefficients of a Breta of one bunch at any width, while hinge values are defined at width one only (Definition 19.2). For the period itself, $\eta = \omega_F$ at the root, the glue reads the bunch and the room through their two laws (Theorem 28.10(ii)), opened with one share per station (Theorem 28.10(iii)) or with one more pin for every direction a side does not see (Theorem 28.12); at width one in one table this is Theorem 19.13, and with one Skeleton value Corollary 19.14. Theorem 34.16 extends the node form from the period to every form on every node step, and so to every left factor; width one enters it only in (iv).

## 34.6 Pinned cells

**Theorem 34.17** (Pinned cells split off motivically)**.** *Let $F = ((\mathcal{B}_j; s_j, t_j))_{j=1}^N$ be a convergent family (Definition 34.1) on $\mathcal{E} = L \sqcup R_1 \sqcup \cdots \sqcup R_q$, $q \geq 1$, $|R_i| \geq 2$, whose active tables are substituted table by table (Definition 21.1): $\mathcal{B}_j = \mathcal{H}_j\langle \mathcal{A}_{1j}, \ldots, \mathcal{A}_{qj} \rangle$, the marks of $L$ left in place, with the cores $\mathcal{H}_j$, lists on $L \sqcup \{h_1, \ldots, h_q\}$, and the cells $\mathcal{A}_{ij}$, lists on $R_i \sqcup \{g_i\}$; here and below $j$ runs over the active tables, the others carrying no letter. Let the active letters read every cell through one hinge value: for every $i$ there is a function $\gamma_i$ on $\Delta_{R_i}$ with $\gamma^K_{\mathcal{A}_{ij}} = \gamma_i$ whenever $s_j \neq 0$ and $\gamma^\Psi_{\mathcal{A}_{ij}} = \gamma_i$ whenever $t_j \neq 0$. Let $\kappa_i = \kappa_{R_i}$, the wall of the cell (Theorem 21.10), put $2\sigma = \sum_j (s_j + t_j)$, and fix integer loads $a_{ij}, b_{ij}$ with $a_{ij} = 0$ when $s_j = 0$, $b_{ij} = 0$ when $t_j = 0$ and $\sum_j (a_{ij} + b_{ij}) = \kappa_i$.*

*(i) $1 \leq \kappa_i \leq 2\sigma - 1$. The* core family *$F_{\mathcal{H}} = ((\mathcal{H}_j; s_j, t_j))_j$ on $L \sqcup \{h_1, \ldots, h_q\}$, with the masses $\mathbf{m}|_L$ and $\kappa_i$ at $h_i$, is convergent, and so is, for every $i$, the* pinned family *$G_i$ on $R_i$, the load family at zero imbalance of Corollary 22.7: the tables $\mathcal{A}_{ij} \setminus g_i$ at $(a_{ij}, t_j - b_{ij})$ and $\mathcal{A}_{ij}/g_i$ at $(s_j - a_{ij}, b_{ij})$, with the masses $\mathbf{m}|_{R_i}$.*

*(ii) The motivic period factors through the core:*

$$P_F^{\mathrm{mot}} = P_{F_{\mathcal{H}}}^{\mathrm{mot}} \prod_{i=1}^{q} P_{G_i}^{\mathrm{mot}}, \qquad \mathrm{C}(P_F^{\mathrm{mot}}) \subseteq \mathrm{C}(P_{F_{\mathcal{H}}}^{\mathrm{mot}}) \cdot \prod_{i=1}^{q} \mathrm{C}(P_{G_i}^{\mathrm{mot}}).$$

*(iii) The* cell family *$F_i = ((\mathcal{A}_{ij}; s_j, t_j))_j$ on $R_i \sqcup \{g_i\}$, with the masses $\mathbf{m}|_{R_i}$ and $2\sigma - \kappa_i$ at $g_i$, is convergent, and*

$$P_{F_i}^{\mathrm{mot}} = \Gamma_{\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}} \begin{bmatrix} \kappa_i, & 2\sigma - \kappa_i \\ & 2\sigma \end{bmatrix} P_{G_i}^{\mathrm{mot}},$$

*the Beta corner a positive rational number.*

*(iv) The form, the period and the motivic period of $G_i$ do not depend on the loads: its form is $\gamma_i^{-\kappa_i}\alpha^{\mathbf{m}|_{R_i}}\prod_j K_{\mathcal{A}_{ij}/g_i}^{-s_j}\Psi_{\mathcal{A}_{ij}\setminus g_i}^{-t_j}\,\Omega_{R_i}/\alpha^{R_i}$, and its period is the pinned shape integral of the cell (Theorem 22.5).*

*(v) Every $S \subseteq L$ has in $\mathcal{B}_j$ the width it has in $\mathcal{H}_j$ (Corollary 21.11): a bunch among the marks left in place keeps the width it has in the core, and no condition is placed on it.*

One hinge value confines the set it reads to width at most one (Lemma 19.6), and the marks of a cell always form a set of width one (Proposition 21.3(i)); so it is the cells that have width one, and a bunch among the marks left in place has any width up to the number of the other marks and cells (Corollary 21.11). With $q$ cells the letters read the room through $q$ hinge values, one for each cell.

**Corollary 34.18** (Dual pairs and full leans)**.** *Let $F = (\mathcal{B}; s, t)$ be a convergent family of one list (Definition 34.1).*

*(a) (*Dual pairs.*) Let $\mathcal{B} = \mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q\rangle$ as in Theorem 34.17, with every cell a dual pair (Definition 21.1). Then every cell is read through one hinge value at all integers $s, t$; the pinned family of the cell $i$ has the form of the family $(\mathcal{B}|R_i;\ \kappa_i,\ s+t-\kappa_i)$ at the masses $\mathbf{m}|_{R_i}$, with $\kappa_i = m_{R_i} - (s+t)(r(R_i)-1)$; and*

$$P_F^{\mathrm{mot}} = P_{(\mathcal{H};s,t)}^{\mathrm{mot}} \prod_{i=1}^{q} P_{(\mathcal{B}|R_i;\,\kappa_i,\,s+t-\kappa_i)}^{\mathrm{mot}},$$

*the core at the masses $\mathbf{m}|_L$ and $\kappa_i$ at $h_i$.*

*(b) (*Full leans.*) Let $t = 0$ or $s = 0$. Split $\mathcal{B}$ at a set of width one with at least two marks on each side into the substitution of its two cells (Proposition 21.3(iii)), and repeat at the nodes so obtained until no node has such a set. The nodes on at least four marks are then prime (Theorem 23.16), and those on two or three marks are the lists $\binom{S}{1}$ and $\binom{S}{|S|-1}$. Give each prime node $\mathcal{H}_v$ the family $F_v = (\mathcal{H}_v; s, t)$ with the masses of its marks and, at each marker, the wall of $F$ at the marks on the other side of the split that made it. Every $F_v$ is convergent, and*

$$P_F^{\mathrm{mot}} = c\prod_v P_{F_v}^{\mathrm{mot}}, \qquad c \in \mathbb{Q}_{>0};$$

*so $\mathrm{C}(P_F^{\mathrm{mot}}) \subseteq \prod_v \mathrm{C}(P_{F_v}^{\mathrm{mot}})$, and a list without prime nodes has a rational motivic period at its full leans.*

Part (b) is the motivic form of Corollary 26.11 and Theorem 26.5: the series and parallel nodes contribute stacked Gammas at integer arguments (Proposition 26.9), which are rational, and their motivic periods are these rationals. At a full lean the width of a bunch plays no part, and the two full leans of one list are exchanged by Cremona, as motivic periods (Corollary 34.8). In a prime list no set with two marks on each side splits off (Corollary 28.26(ii)), so the splitting of (b) ends at the prime nodes; their only cells are the complements of single marks, which at a full lean split off by one Beta corner and, with both letters active, exactly where the deletion and the contraction have one even letter (Corollary 28.26(iii),(iv)).

**Theorem 34.19** (A pinned set splits off motivically)**.** *Let $F = ((\mathcal{B}_j; s_j, t_j))_j$ be a convergent family (Definition 34.1) with $s_j, t_j \geq 0$, and let $S$, $|S| \geq 2$, be pinned (Theorem 28.25) with width one in every active table, $S' = \mathcal{E} \setminus S$. Every active table is then the substitution of its cell of the bunch $\mathcal{B}_{j,S}^{+}$ into its cell of the room $\mathcal{B}_{j,S'}^{+}$ (Theorem 19.4, Proposition 21.3(iii)), the marks of $S'$ left in*

*place, and the active letters read the cell through the one hinge value* $\gamma = \gamma^K_{j,S} = \gamma^\Psi_{j,S}$. *The core family of Theorem 34.17 is the room with its stand-in at* $\mathbf{1}$, $S'^{(\mathbf{1})}$ *(Theorem 28.10(i)), the family* $((\mathcal{B}^+_{j,S'}; s_j, t_j))_j$ *of the active tables with the masses* $(\mathbf{m}|_{S'}, \kappa_S)$*; the period of its pinned family* $G$ *is* $c = \int_{\Delta_S} \phi^\rho_S$, *the total mass of the law of* $S$*; and*

$$P^{\mathrm{mot}}_F = P^{\mathrm{mot}}_{S'^{(\mathbf{1})}}\, P^{\mathrm{mot}}_G, \qquad \mathrm{C}(P^{\mathrm{mot}}_F) \subseteq \mathrm{C}(P^{\mathrm{mot}}_{S'^{(\mathbf{1})}}) \cdot \mathrm{C}(P^{\mathrm{mot}}_G) :$$

*the identity* $P_F = c\, P_{S'^{(\mathbf{1})}}$ *of Theorem 28.25 is an identity of motivic periods.*

Every period of a family glued on a set is the integral, against the law of the set, of the period of the partner with a stand-in for it (Theorem 28.10(i)). Where the law is a point the integral is one evaluation (Theorem 28.25); where the set is moreover a cell, the change of variables behind the evaluation is birational and the evaluation is an identity of motivic periods. No other set splits off, and a set of width two or more enters every period glued on it through a law that is not a point (Corollary 28.26(i)); at every node, of every width, the open periods of node steps have the node form of Theorem 34.16.

## 34.7 Rational data

At rational masses and exponents the form of a family is multivalued on its model, a section of a local system of finite order. On the Kummer cover that adjoins the roots of the coordinate ratios and of the letters it becomes a single-valued regular form, and the period of the family is a period of the cover.

**Proposition 34.20** (Rational data on a Kummer cover)**.** *Let* $F = ((\mathcal{B}_j; s_j, t_j))_{j=1}^N$ *be a family of lists on* $\mathcal{E}$, $n = |\mathcal{E}| \geq 2$, *with masses* $\mathbf{m}$ *and exponents in* $\frac{1}{M}\mathbb{Z}$ *on its balance wall,* $M \geq 1$, *and* $\kappa_S > 0$ *for every nonempty proper* $S$. *Define* $\mathrm{Let}(F)$, $V_F$ *and* $D_F$ *as in Definition 34.1, the letters read as a set of polynomials, and for* $g \in \mathrm{Let}(F)$ *write* $e_g$ *for the sum of the exponents* $s_j$ *with* $K_j = g$ *and* $t_j$ *with* $\Psi_j = g$, *and* $\delta_g$ *for its degree.*

*(i) On the points of the torus in* $V_F$,

$$\omega_F = \prod_{e \neq e_0} \Big(\frac{\alpha_e}{\alpha_{e_0}}\Big)^{m_e} \prod_{g \in \mathrm{Let}(F)} \Big(\frac{g}{\alpha_{e_0}^{\delta_g}}\Big)^{-e_g} \frac{\Omega_{\mathcal{E}}}{\alpha^{\mathcal{E}}},$$

*a regular form times a branch of a function whose* $M$*-th power is a regular unit: a section of a local system of rank one and order dividing* $M$.

*(ii) Let* $V_F^{(M)}$ *be the normalization of* $V_F$ *in the algebra, a finite product of fields, obtained from its function field by adjoining the* $M$*-th roots* $r_e$ *of* $\alpha_e/\alpha_{e_0}$, $e \neq e_0$, *and* $w_g$ *of* $g/\alpha_{e_0}^{\delta_g}$, $g \in \mathrm{Let}(F)$, *and* $D^{(M)}$ *the preimage of* $D_F$. *In the chart of an order, with* $\eta_i^M = \varepsilon_i$ *and* $\widetilde{w}_g^M = \widetilde{g}$ *for the sector letters (Lemma 34.3),* $V_F^{(M)}$ *is smooth with the coordinates* $\eta$, $D^{(M)} = \{\prod_i \eta_i = 0\}$ *is a divisor with simple normal crossings, and the pullback of* $\omega_F$ *is*

$$\omega_F^{(M)} = \pm M^{n-1} \prod_{i<n} \eta_i^{M\kappa_{S_i}-1} \prod_{g \in \mathrm{Let}(F)} \widetilde{w}_g^{-Me_g}\, d\eta_1 \wedge \cdots \wedge d\eta_{n-1},$$

*a single-valued form, regular on* $V_F^{(M)}$.

*(iii) The positive cell lifts to the closure* $\widetilde{\sigma}_{\mathcal{E}}^{(M)}$ *of the set of the points over* $\widetilde{\sigma}_{\mathcal{E}}^{\circ}$ *at which every* $r_e$ *and every* $w_g$ *is real and positive, compact with its boundary in* $D^{(M)}$*, and*

$$P_F^{\mathrm{mot}} = \big[H^{n-1}\big(V_F^{(M)}, D^{(M)}\big),\ \widetilde{\sigma}_{\mathcal{E}}^{(M)},\ \omega_F^{(M)}\big]^{\mathrm{mot}}$$

*is a motivic period whose period is* $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*. For* $M = 1$ *it is the motivic period of Definition 34.5.*

*(iv) Over* $\mathbb{Q}(\mu_M)$ *the group* $G = \mu_M^{\mathcal{E}\setminus e_0} \times \mu_M^{\mathrm{Let}(F)}$ *acts on* $V_F^{(M)}$ *by* $r_e \mapsto \zeta_e r_e$*,* $w_g \mapsto \zeta'_g w_g$*, and* $\phi^*\omega_F^{(M)} = \chi(\phi)\,\omega_F^{(M)}$ *with the character* $\chi(\zeta,\zeta') = \prod_e \zeta_e^{Mm_e}\prod_g \zeta_g'^{-Me_g}$*. So the de Rham class of* $\omega_F^{(M)}$ *lies in the* $\chi$*-eigenspace of* $H^{n-1}_{\mathrm{dR}}(V_F^{(M)}, D^{(M)}) \otimes \mathbb{Q}(\mu_M)$*.*

So the periods of families at rational data are periods of mixed objects of the same kind as at integer data, read on a cover of the model of finite degree; the masses and the exponents enter as a character of its deck group, and the walls as the orders of the form along the boundary of the cover.

## 34.8 Separation by conjugates

A motivic period is not determined by its period alone, but it is determined by the periods of its Galois conjugates; for the motivic period of a family the conjugates below the top weight are values of families on the steps of its tree.

**Theorem 34.21** (Motivic periods are separated by their conjugates)**.** *(i) The map* $\xi \mapsto (\mathrm{per}\otimes\mathrm{id})\Delta\xi$*, from* $\mathcal{P}^{\mathrm{mot}}$ *to* $\mathbb{C}\otimes_{\mathbb{Q}}\mathcal{P}^{\mathrm{dR}}$*, is injective. So a motivic period* $\xi$ *is* $0$ *exactly when every element of* $\mathrm{C}(\xi)$ *(Definition 34.5) has the period* $0$*.*

*(ii) Let* $F$ *be a convergent family (Definition 34.1) on* $n$ *marks, and* $(e_i)$ *a basis of* $(H_F)_{\mathrm{dR}}$ *adapted to the weight filtration. Then* $P_F^{\mathrm{mot}} = 0$ *exactly when*

$$\sum_i \mathrm{per}\,\big([H_F, \widetilde{\sigma}_{\mathcal{E}}, e_i]^{\mathrm{mot}}\big)\,\varphi\big([H_F, e_i^{\vee}, \omega_F]^{\mathrm{dR}}\big) = 0$$

*for every* $\mathbb{Q}$*-linear* $\varphi : \mathcal{P}^{\mathrm{dR}} \to \mathbb{Q}$*; for* $e_i \in W_k$*,* $k < n-1$*, the period of* $[H_F, \widetilde{\sigma}_{\mathcal{E}}, e_i]^{\mathrm{mot}}$ *is a sum of products, over the steps of flags of dimension* $k$*, of rational combinations of values of families of lists on the steps and of their derivatives in the masses, each step having at most* $k+1$ *marks and being attached to one vertex of any tree of bunches of* $F$*, with the letters of children of the tables at that vertex (Theorem 34.13, Corollary 34.15).*

So the separation of the motivic period of a family descends its tree: below the top weight its conjugates are read on the families of its steps, and only the left factors of top weight are read on the whole model.

# Chapter 35

# Positive words

The output of the theory is a value: a period, a Breta, a closing of fibre integrals, and such values are read arithmetically. A closing at rational data writes a value as a rational combination of Trailmoments of lattice inputs and rational heads (Proposition 5.13, Lemma 38.32); a sequence of such values that do not vanish, cleared by integers that grow more slowly than the values decay, shows that some Trailmoment it carries is irrational (Theorem 37.2, Chapter 44). What keeps a value from vanishing is positivity, and positive words are made in four ways: second jets of a period in its weights, even powers of the contrast of two bases, Beta ports on the pairs of a manufactured family, and Beta ports read against one mark with the room integrated. The jets and the contrasts are combinations of periods of the same family at shifted data on its balance wall, bounded by the period times a factor that their statements control; the port words of a manufactured family are combinations of its interests at shifted masses over one normalization, and the reads of the Beta ports are bounded by the read of the constant insertion.

## 35.1 Coefficient jets and basis contrasts

**Theorem 35.1** (Positive coefficient jets)**.** *Let $F$ have real exponents $s_j, t_j \geq 0$ and real masses on its balance wall, every proper wall positive. For an integer $n \geq 1$, the* depth*, the data $(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$ are again on the balance wall, with walls $n\kappa_S$; in this statement $\mathbb{E}$ is the expectation of $U \sim \mathrm{Breta}^{\circ}(F; n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$. Choose* weight directions $\mathbf{b}_{j,T} \in \mathbb{Q}^k$*, one for each table $j$ and basis $T \in \mathcal{B}_j$, and put*

$$\partial_{\mathbf{b}} K_j = \sum_{T \in \mathcal{B}_j} \mathbf{b}_{j,T}\, \alpha^{T}, \qquad \partial_{\mathbf{b}} \Psi_j = \sum_{T \in \mathcal{B}_j} \mathbf{b}_{j,T}\, \alpha^{\mathcal{E} \setminus T},$$

*the derivatives of the letters of table $j$ along the weights in these directions.*

*(i) For $v \in \mathbb{R}^k$ and small real $\varepsilon$, replace every table $j$ by the two tables $(\mathcal{B}_j, \theta_j \pm \varepsilon\, v^{\mathsf{T}} \mathbf{b}_j)$, each with the exponents $(ns_j/2, nt_j/2)$. The period of this* doubled family *at the masses $n\mathbf{m}$ is*

$$P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})\ \mathbb{E}\Big[\Big(\prod_j \Big(1 - \varepsilon^2 \Big(\frac{v^{\mathsf{T}} \partial_{\mathbf{b}} K_j}{K_j}\Big)^2\Big)^{-\frac{ns_j}{2}} \Big(1 - \varepsilon^2 \Big(\frac{v^{\mathsf{T}} \partial_{\mathbf{b}} \Psi_j}{\Psi_j}\Big)^2\Big)^{-\frac{nt_j}{2}}\Big)(U)\Big],$$

*whose coefficient of $\varepsilon^2$ is $v^{\mathsf{T}} \mathbf{H}_n v$, with the* jet matrix

$$\mathbf{H}_n = \frac{n}{2}\, P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})\ \mathbb{E}\Big[\sum_j \Big(s_j\, \frac{\partial_{\mathbf{b}} K_j\, \partial_{\mathbf{b}} K_j^{\mathsf{T}}}{K_j^2} + t_j\, \frac{\partial_{\mathbf{b}} \Psi_j\, \partial_{\mathbf{b}} \Psi_j^{\mathsf{T}}}{\Psi_j^2}\Big)(U)\Big].$$

$\mathbf{H}_n$ *is positive semidefinite, and positive definite when the directions* $\mathbf{b}_{j,T}$ *of the active tables span* $\mathbb{R}^k$.

*(ii) For a rational positive semidefinite* $k\times k$ *matrix* $Q$, $\operatorname{tr}(Q\mathbf{H}_n) > 0$ *exactly when* $\mathbf{b}_{j,T}^{\mathsf{T}} Q\, \mathbf{b}_{j,T} > 0$ *for some active table* $j$ *and basis* $T$. *Completing squares writes* $Q = \sum_l c_l v_l v_l^{\mathsf{T}}$ *with rational* $c_l > 0$ *and rational* $v_l$, *and so writes* $\operatorname{tr}(Q\mathbf{H}_n) = \sum_l c_l\, v_l^{\mathsf{T}} \mathbf{H}_n v_l$ *as a sum of directional second jets with positive rational coefficients.*

*(iii) In periods of* $F$ *(Theorem 27.9 at order two, the two copies of each table merged),*

$$\operatorname{tr}(Q\mathbf{H}_n) = \frac{n}{2} \sum_j \sum_{T,T'\in\mathcal{B}_j} \mathbf{b}_{j,T}^{\mathsf{T}} Q\, \mathbf{b}_{j,T'} \Big[ s_j\, P_F\big(n\mathbf{m} + \mathbf{1}_T + \mathbf{1}_{T'};\, n\mathbf{s} + 2\mathbf{e}_j,\, n\mathbf{t}\big) + t_j\, P_F\big(n\mathbf{m} + \mathbf{1}_{\mathcal{E}\setminus T} + \mathbf{1}_{\mathcal{E}\setminus T'};\, n\mathbf{s},\, n\mathbf{t} + 2\mathbf{e}_j\big) \Big] :$$

*every term stays on its balance wall, a basis insertion changes the wall of* $S$ *by* $|T\cap S| + |T'\cap S| - 2d_j(S) \geq 0$, *and a cobasis insertion by* $|S\setminus T| + |S\setminus T'| - 2d_j^*(S) \geq 0$.

*(iv) Whenever (ii) gives positivity,* $0 < \operatorname{tr}(Q\mathbf{H}_n) \leq n\, C(Q)\, P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$, *with* $C(Q) = \frac{1}{2}\sum_j (s_j + t_j) \max_T \mathbf{b}_{j,T}^{\mathsf{T}} Q\, \mathbf{b}_{j,T}/\theta_{j,T}^2$.

*(v) With the basis weights and the cobasis weights of every table taken as independent variables, the part of* $\mathbf{H}_n$ *carried by* $K_j$ *is the Hessian of* $P_F(n\mathbf{m}; n\mathbf{s}, n\mathbf{t})$ *in the basis weights of table* $j$, *paired with the directions* $\mathbf{b}_{j,T}$, *divided by* $2(ns_j + 1)$, *and the part carried by* $\Psi_j$ *is the same in the cobasis weights divided by* $2(nt_j + 1)$. *On the share side of the period (Corollary 18.16) a weight derivative multiplies the integrand by minus the share divided by its weight, so every pin and every candidate pole hyperplane stays fixed.*

*(vi) At the centre,* $s_j = t_j$ *for every* $j$ *and* $\mathbf{m} = \sigma\mathbf{1}$, *the doubled family is again at its centre, and the table's gauge* $\mathbf{1}$ *stays native (Proposition 8.21).*

A weight direction moves the two copies of a table apart symmetrically, so the period moves at second order only, by the expectation of a square under the Breta shape law: a jet of the period in its weights is a positive word. Under the basis law $\partial_{\mathbf{b}} K_j/K_j$ is the mean of the directions $\mathbf{b}_{j,T}/\theta_{j,T}$, $\mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[\mathbf{b}_{j,T}/\theta_{j,T}]$, so the jet is the second moment, under the shape law, of the mean weight direction that the bases drawn at the shape carry. Table by table it is the period's own Hessian in the weights divided by $2(ns_j + 1)$ or $2(nt_j + 1)$, it is a finite combination of periods of the same family on the balance wall at data whose walls are no smaller, and it costs at most a factor linear in the depth against the period.

**Lemma 35.2** (Rounding inside a rational subspace)**.** *Let* $V$ *be the subspace of the real symmetric* $k\times k$ *matrices on which finitely many linear functionals* $Q \mapsto \operatorname{tr}(MQ)$, $M$ *rational, vanish, spanned over* $\mathbb{R}$ *by integer symmetric matrices* $G_1, \dots, G_r$, *and let* $Q_* = \sum_j u_j G_j \in V$ *with* $Q_* \succeq \varepsilon I$, $\varepsilon > 0$. *For an integer* $\ell \geq 1$ *put* $Q = \sum_j \operatorname{round}(\ell u_j)\, G_j$ *and* $\beta = \frac{1}{2}\sum_j \big(\operatorname{tr} G_j^2\big)^{1/2}$. *Then* $Q$ *is an integer matrix in* $V$ *with* $Q \succeq (\ell\varepsilon - \beta)I$ *and* $\operatorname{tr} Q \leq \ell \operatorname{tr} Q_* + k\beta$; *for* $\ell > \beta/\varepsilon$ *it is positive definite.*

The jet word $\operatorname{tr}(Q\mathbf{H}_n)$ is linear in $Q$, so conditions on its coefficients of Trailmoments, such as the vanishing of chosen ones, define a rational subspace of matrices $Q$. A positive definite matrix in that subspace rounds to an integer one in the same subspace, and the jet word keeps its positivity and the vanishing of the chosen coefficients together at integer data.

**Theorem 35.3** (Two bases generate positive contrast words of any depth)**.** *Let $F$ have real data $(\mathbf{m};\mathbf{s},\mathbf{t})$ on its balance wall with every proper wall positive, with the pair coordinates $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$ (Definition 8.1) and the centred potential $\Phi^\circ$ (Theorem 8.19). Let $T \neq T'$ be two bases of a table $j$, and for integers $n,h \geq 1$ let the* contrast word *be*

$$\frac{1}{2}\,P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})\;\mathbb{E}_{U\sim\mathrm{Breta}^\circ(F;n\mathbf{m};n\mathbf{s},n\mathbf{t})}\Big[\big(\mathbb{P}_{j,U}(T)-\mathbb{P}_{j,U}(T')\big)^{2h}+\big(\mathbb{P}^*_{j,U}(T)-\mathbb{P}^*_{j,U}(T')\big)^{2h}\Big]>0;$$

*the two contrasts have modulus below one on the open orthant.*

*(i) The first half of the contrast word is*

$$\frac{1}{2}\sum_{l=0}^{2h}(-1)^l\binom{2h}{l}\theta_{j,T}^{2h-l}\theta_{j,T'}^{l}\,P_F\big(n\mathbf{m}+(2h-l)\mathbf{1}_T+l\mathbf{1}_{T'};\,n\mathbf{s}+2h\mathbf{e}_j,\,n\mathbf{t}\big),$$

*and the second is the same with $\mathbf{1}_{\mathcal{E}\setminus T}$, $\mathbf{1}_{\mathcal{E}\setminus T'}$ in the masses and $2h\mathbf{e}_j$ added to the cobasis exponents; every term is balanced with its walls weakly raised. The inversion $\alpha\mapsto\alpha^{-1}$ carries the second half at $(\mathbf{m};\mathbf{s},\mathbf{t})$ to the first half at $(2\sigma\mathbf{1}-\mathbf{m};\mathbf{t},\mathbf{s})$, the data of Theorem 8.14; at the centre the two halves are equal.*

*(ii) For rational $\eta>0$ and $h=\eta n$, along the integers $n$ with $\eta n\in\mathbb{N}$, $\frac{1}{n}\log$ of the contrast word tends to*

$$-\inf_{z\in\mathbf{1}^\perp}\Big(\Phi^\circ(z)-2\langle\boldsymbol{\mu}_\perp,z\rangle-2\eta\log\max\Big(\big|\mathbb{P}_{j,e^{2z}}(T)-\mathbb{P}_{j,e^{2z}}(T')\big|,\;\big|\mathbb{P}^*_{j,e^{2z}}(T)-\mathbb{P}^*_{j,e^{2z}}(T')\big|\Big)\Big);$$

*the infimum exceeds the minimum of $\Phi^\circ-2\langle\boldsymbol{\mu}_\perp,\cdot\rangle$ on $\mathbf{1}^\perp$ and tends to infinity with $\eta$; that minimum is $-\lim_n\frac{1}{n}\log P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})$.*

*(iii) Let the exponents be nonnegative, and write $q=2h$ and $H_q(x,y)=\sum_{l=0}^{q}(-1)^l\binom{q}{l}(x)_{q-l}(y)_l$. When $s_j>0$, the first half is the share side of $P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})$ (Corollary 18.16) with its integrand multiplied by $H_q(a_{j,T},a_{j,T'})/\big(2(ns_j)_q\big)$, a the basis shares; when $t_j>0$, the second half is the same with $H_q(b_{j,T},b_{j,T'})/\big(2(nt_j)_q\big)$, b the cobasis shares: polynomial insertions that keep every pin and every pole hyperplane.*

Two bases of one table give a contrast of modulus below one, which stays away from one on the compact sets where the Breta shape law concentrates; so its even powers are positive words that become exponentially small against the period, at a rate that grows without bound with the ratio of the power to the depth. At the centre with unit weights the basis law gives every basis the same chance at the table's gauge, where the law concentrates, and the contrast vanishes there. On the share side the contrast is a polynomial in the shares, so the contrast words of every depth keep the fibre, the pins and the poles of the period.

## 35.2 Manufactured tables and their port words

**Definition 35.4** (Manufactured tables)**.** Let *seed letters* on $\mathcal{E}$ be polynomials $L_i(x)=\sum_{P\in\operatorname{supp}L_i}\theta_{i,P}\,x^P$, $i\in I$, with supports $\operatorname{supp}L_i\subseteq 2^{\mathcal{E}}$, weights $\theta_{i,P}>0$ and exponents $w_i\geq 0$; a table $(\mathcal{B}_j,\theta_j)$ of a family with exponents $(s_j,t_j)$ gives the two seed letters $K_j$, with exponent $s_j$, and $\Psi_j$, with exponent $t_j$. Choose $\beta_i\in(0,1)$ and replace every mark $e$ by a pair $\{e^+,e^-\}$ of new marks, on $\mathcal{E}^\pm=\{e^+,e^-:e\in\mathcal{E}\}$. The polynomials

$$\widetilde{K}_i(\alpha)=L_i(\mathbf{1})\prod_{e\in\mathcal{E}}(\alpha_{e^+}+\alpha_{e^-})-\beta_i\sum_P\theta_{i,P}\prod_{e\in P}\alpha_{e^+}\prod_{e\notin P}(\alpha_{e^+}+\alpha_{e^-}),$$

$$\widetilde{\Psi}_i(\alpha) = L_i(\mathbf{1}) \prod_{e\in\mathcal{E}} (\alpha_{e^+} + \alpha_{e^-}) - \beta_i \sum_P \theta_{i,P} \prod_{e\in P} \alpha_{e^-} \prod_{e\notin P} (\alpha_{e^+} + \alpha_{e^-})$$

are the basis and cobasis polynomials of a table on $\mathcal{E}^{\pm}$ of rank $|\mathcal{E}|$, whose bases $U^+ \cup (\mathcal{E}\backslash U)^-$, $U \subseteq \mathcal{E}$, take one mark of each pair, with the weights $\widetilde{\theta}_{i,U} = L_i(\mathbf{1}) - \beta_i \sum_{P\subseteq U} \theta_{i,P} \geq L_i(\mathbf{1})(1-\beta_i) > 0$; the seed weights are recovered from $\big(L_i(\mathbf{1}) - \widetilde{\theta}_{i,U}\big)/\beta_i = \sum_{P\subseteq U} \theta_{i,P}$ by inversion over the subsets. The *manufactured family* $\widetilde{F}$ has these tables with the exponents $(w_i, 0)$; for masses $\mathbf{m}$ on $\mathcal{E}^{\pm}$ its *pair walls* are $\kappa_{\{e^+,e^-\}} = m_{e^+} + m_{e^-} - |\mathbf{w}|$, $|\mathbf{w}| = \sum_i w_i$, and its *pair shapes* are $x_e = \alpha_{e^+}/(\alpha_{e^+} + \alpha_{e^-})$. For orders $\mathbf{n} \in \mathbb{N}^{\mathcal{E}}$, positive masses and positive pair walls, its *port word* is

$$M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) = \frac{J_{\widetilde{F}}(\mathbf{m};\mathbf{w},\mathbf{0})}{\prod_e \Gamma\big(\kappa_{\{e^+,e^-\}}\big)}\ \mathbb{E}_{\alpha\sim \mathrm{Breta}(\widetilde{F};\mathbf{m};\mathbf{w},\mathbf{0})}\Big[\prod_{e\in\mathcal{E}} \Pi_{n_e}^{m_{e^+},m_{e^-}}(x_e)\Big],$$

and $M^*_{\widetilde{F},\mathbf{n}}(\mathbf{m})$ is the same at the exponents $(0, w_i)$, with $J_{\widetilde{F}}(\mathbf{m};\mathbf{0},\mathbf{w})$ and $\mathrm{Breta}(\widetilde{F};\mathbf{m};\mathbf{0},\mathbf{w})$. An *occupancy* is a family $\mathbf{o} = (o_{i,P})$ of integers $o_{i,P} \geq 0$, zero when $w_i = 0$, with the totals $|o_i| = \sum_P o_{i,P}$ and $|\mathbf{o}| = \sum_i |o_i|$, the *loads* $\mathrm{load}_e(\mathbf{o}) = \sum_i \sum_{P\ni e} o_{i,P}$, and the *coefficient*

$$\mathrm{cf}(\mathbf{o}) = \prod_i \Big[\frac{(w_i)_{|o_i|}}{L_i(\mathbf{1})^{w_i} \prod_P o_{i,P}!} \prod_P \Big(\frac{\beta_i\, \theta_{i,P}}{L_i(\mathbf{1})}\Big)^{o_{i,P}}\Big].$$

**Theorem 35.5** (Interest of a manufactured family; the occupancy sum)**.** *In the pair scales and shapes,* $(\alpha_{e^+}, \alpha_{e^-}) = \rho_e(x_e, 1-x_e)$,

$$\widetilde{K}_i = \prod_e \rho_e\big(L_i(\mathbf{1}) - \beta_i L_i(x)\big), \qquad \widetilde{\Psi}_i = \prod_e \rho_e\big(L_i(\mathbf{1}) - \beta_i L_i(\mathbf{1}-x)\big).$$

*Let the masses be positive and the pair walls positive.*

*(i) Every wall of* $\widetilde{F}$ *at the exponents* $(w_i, 0)$ *is positive, and*

$$J_{\widetilde{F}}(\mathbf{m};\mathbf{w},\mathbf{0}) = \prod_e \Gamma\big(\kappa_{\{e^+,e^-\}}\big) \int_{[0,1]^{\mathcal{E}}} \prod_e x_e^{m_{e^+}-1}(1-x_e)^{m_{e^-}-1} \prod_i \big(L_i(\mathbf{1}) - \beta_i L_i(x)\big)^{-w_i}\, dx,$$

*the integral convergent at all positive masses; the simplex integral* $P_{\widetilde{F}}(\mathbf{m};\mathbf{w},\mathbf{0})$ *is* $\Gamma_{1\,|\mathcal{E}|}\begin{bmatrix}(\kappa_{\{e^+,e^-\}})_e \\ \sum_e \kappa_{\{e^+,e^-\}}\end{bmatrix}$ *times the same integral. Under* $\mathrm{Breta}(\widetilde{F};\mathbf{m};\mathbf{w},\mathbf{0})$ *the pair scales are independent,* $\rho_e \sim \mathrm{Gamma}_{\kappa_{\{e^+,e^-\}}}$*, and independent of the pair shapes, whose density on the cube is proportional to the integrand. So* $M_{\widetilde{F},\mathbf{0}}(\mathbf{m})$ *is the integral displayed, and* $M_{\widetilde{F},\mathbf{n}}(\mathbf{m})$ *is that integral with* $\prod_e \Pi_{n_e}^{m_{e^+},m_{e^-}}(x_e)$ *inserted; with* $L_i(\mathbf{1}-x)$ *in place of* $L_i(x)$ *the same holds for* $M^*_{\widetilde{F},\mathbf{n}}(\mathbf{m})$.

*(ii) The integral is the series of nonnegative terms*

$$M_{\widetilde{F},\mathbf{0}}(\mathbf{m}) = \sum_{\mathbf{o}} \mathrm{cf}(\mathbf{o}) \prod_e \Gamma_{2\,1}\begin{bmatrix} m_{e^+} + \mathrm{load}_e(\mathbf{o}),\ m_{e^-} \\ m_{e^+} + m_{e^-} + \mathrm{load}_e(\mathbf{o})\end{bmatrix}$$

*over the occupancies.*

**Theorem 35.6** (The port word of a manufactured family)**.** *Let the masses be positive, the pair walls positive and* $\mathbf{n} \in \mathbb{N}^{\mathcal{E}}$*. Then*

$$M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) = \sum_{\mathbf{o}} \mathrm{cf}(\mathbf{o}) \prod_{e} \binom{\mathrm{load}_e(\mathbf{o})}{n_e} \Gamma_{1}^{2}\begin{bmatrix} m_{e^+} + \mathrm{load}_e(\mathbf{o}), & m_{e^-} + n_e \\ m_{e^+} + m_{e^-} + \mathrm{load}_e(\mathbf{o}) + n_e \end{bmatrix}$$
$$= \frac{1}{\prod_e n_e!} \int_{[0,1]^{\mathcal{E}}} \prod_e x_e^{m_{e^+}+n_e-1}(1-x_e)^{m_{e^-}+n_e-1}\, \partial_x^{\mathbf{n}} \prod_i \big(L_i(\mathbf{1}) - \beta_i L_i(x)\big)^{-w_i}\, dx.$$

*Every occupancy with* $\mathrm{load}_e(\mathbf{o}) < n_e$ *for some* $e$ *is annihilated, every surviving term is positive, every mixed derivative of* $\prod_i (L_i(\mathbf{1}) - \beta_i L_i(x))^{-w_i}$ *is nonnegative on the cube, and* $M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) > 0$ *exactly when every mark* $e$ *with* $n_e > 0$ *lies in a member of the support of a seed letter with* $w_i > 0$*. At integer masses the Beta ports have integer coefficients, and* $M_{\widetilde{F},\mathbf{n}}(\mathbf{m})$ *is an integer combination of the* $M_{\widetilde{F},\mathbf{0}}$ *at the masses shifted by* $(k_e, n_e - k_e)$ *on the pair of* $e$*,* $0 \le k_e \le n_e$*: of the interests of* $\widetilde{F}$ *at these masses, all of pair walls* $\kappa_{\{e^+,e^-\}} + n_e$*, over the one normalization* $\prod_e \Gamma(\kappa_{\{e^+,e^-\}} + n_e)$*.*

**Theorem 35.7** (Suppression selected by incidence)**.** *In Theorem 35.6 write* $\mathrm{occ}(\mathbf{n}) = \min\{|\mathbf{o}| : \mathrm{load}_e(\mathbf{o}) \ge n_e$ *for every* $e\}$ *over the occupancies, and* $\beta = \max_i \beta_i < 1$ *over the seed letters with* $w_i > 0$*. Then*

$$0 \le M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) \le \prod_e \Big[\Gamma_{1}^{2}\begin{bmatrix} m_{e^+}, & m_{e^-} \\ m_{e^+} + m_{e^-} \end{bmatrix} \frac{(m_{e^-})_{n_e}}{n_e!}\Big] \prod_i L_i(\mathbf{1})^{-w_i} \sum_{l \ge \mathrm{occ}(\mathbf{n})} \frac{(|\mathbf{w}|)_l}{l!}\, \beta^l.$$

*For an integer demand* $\mathbf{d} \in \mathbb{N}^{\mathcal{E}}$ *let* $\mathrm{fc}(\mathbf{d}) = \min\big\{\sum_{i,P} \lambda_{i,P} : \ \lambda \ge 0,\ \sum_{i,P} \lambda_{i,P} \mathbf{1}_P \ge \mathbf{d}\big\}$*, the* fractional cover *of the demand by the members of the supports of the seed letters with* $w_i > 0$*. Then* $\mathrm{occ}(q\mathbf{d}) = q\,\mathrm{fc}(\mathbf{d}) + O(1)$ *as* $q \to \infty$*, and* $\limsup_q \frac{1}{q} \log M_{\widetilde{F},q\mathbf{d}}(\mathbf{m}) \le \mathrm{fc}(\mathbf{d}) \log \beta < 0$ *for a nonzero demand that the supports cover. When the seed letters are the letters* $K_j, \Psi_j$ *of the tables of a family, every surviving occupancy satisfies* $\sum_j \big(r_j(S)\,|o_{K_j}| + (|S| - d_j(S))\,|o_{\Psi_j}|\big) \ge \sum_{e \in S} n_e$ *for every* $S \subseteq \mathcal{E}$*,* $|o_{K_j}|$ *and* $|o_{\Psi_j}|$ *the total occupancies of the two letters of table* $j$*.*

**Theorem 35.8** (Reflection of the port words of a manufactured family)**.** *The swap* $\alpha_{e^+} \leftrightarrow \alpha_{e^-}$ *on every pair,* $x_e \leftrightarrow 1 - x_e$*, swaps* $\widetilde{K}_i$ *with* $\widetilde{\Psi}_i$ *and* $m_{e^+}$ *with* $m_{e^-}$*, and keeps every seed weight and every* $\beta_i$*: with* $\mathbf{m}'$ *the masses swapped on every pair,* $M_{\widetilde{F},\mathbf{n}}(\mathbf{m}) = (-1)^{|\mathbf{n}|} M^{*}_{\widetilde{F},\mathbf{n}}(\mathbf{m}')$*.*

A manufactured table turns any positive polynomials, the two letters of every table of a family among them, into a table whose bases are the transversals of the pairs. On the pair shapes its letter is the constant $L_i(\mathbf{1})$ less a copy of the seed scaled by $\beta_i$, so the port word is a series of nonnegative terms indexed by occupancies of the members of the seed, and a Beta port of order $n_e$ on a pair keeps only the occupancies that load the mark at least $n_e$ times. The port words are therefore positive exactly when the incidence of the supports can meet the demand, and they are suppressed at the exponential rate of the cheapest fractional cover of the demand: the incidence structure of the seed decides which port words survive and how small they are.

## 35.3 Mark relations with the room integrated

**Definition 35.9** (Mark reads)**.** Select a mark with coordinate $x$, and let the coordinates $\alpha_R$ of its room range over a measure space with a density $\Omega(\alpha_R) \ge 0$. The letters (Section 3.2) in the selected mark are polynomials $L_i(x, \alpha_R)$ in $x$, $i = 1, \dots, h$, positive where $x$ ranges, with exponents $w_i$; they are *linear* when $L_i = c_i(\alpha_R) + d_i(\alpha_R)\, x$ with $c_i > 0$ and $d_i \ge 0$, and a linear letter *sees*

*the mark* when $w_i d_i > 0$ on a set of positive measure. For $a, b > 0$ the *mark read* of an insertion $g(x, \alpha_R)$ on the box is

$$\langle g \rangle_{a,b;\mathbf{w}} = \int\!\!\int_0^1 g\, x^{a-1}(1-x)^{b-1} \prod_i L_i^{-w_i}\, dx\; \Omega(\alpha_R)\, d\alpha_R,$$

and for $a, \lambda > 0$ its read on the half-line is

$$\langle g \rangle^{\lambda}_{a;\mathbf{w}} = \int\!\!\int_0^\infty g\, x^{a-1} e^{-\lambda x} \prod_i L_i^{-w_i}\, dx\; \Omega(\alpha_R)\, d\alpha_R,$$

wherever these converge; at a fixed point of the room, $\Omega(\alpha_R)\, d\alpha_R$ is the Dirac measure there. A sequence of reads at the data $a_n + j$, $b_n$, $\mathbf{w}(n)$ and $\Omega_n$ keeps the depth $n$ (Theorem 35.1) apart from the *moment index* $j$ of the selected mark.

**Theorem 35.10** (The mark relation)**.** *Let the letters be polynomial in the selected mark, with exponents $w_i$, and let $A(x, \alpha_R)$ be a polynomial in $x$ with coefficients depending on the room.*

*(a) On the box, for $a, b > 0$:*

$$\Big\langle x(1-x)\,\partial_x A + \big(a - (a+b)x\big) A - x(1-x)\, A \sum_i w_i\, \partial_x \log L_i \Big\rangle_{a,b;\mathbf{w}} = 0.$$

*(b) On the half-line, for $a, \lambda > 0$, whenever the reads converge:*

$$\Big\langle x\,\partial_x A + (a - \lambda x) A - x\, A \sum_i w_i\, \partial_x \log L_i \Big\rangle^{\lambda}_{a;\mathbf{w}} = 0.$$

*(c) On a restricted interval the difference of the flux at its two ends is retained, and a kernel coupled to $x$ contributes its product-rule term.*

*Multiplying by $\prod_i L_i$ gives a polynomial numerator; on a box of several marks the relation holds in each coordinate with the others as room.*

**Theorem 35.11** (The moments of a selected mark close in a finite span)**.** *On a family select a mark $e$, $x = \alpha_e$, with room $R = \mathcal{E} - e$; splitting the bases and the cobases by $e$ gives the two coefficients of every letter in $x$. Pull the letters free of $x$ into the room, absorb pure powers of $x$ into its mass, and write the remaining letters $L_i = c_i(\alpha_R) + d_i(\alpha_R)\, x$, $c_i, d_i > 0$, $i = 1, \dots, h$, with exponents $w_i$; read them on the half-line at a fixed point of the room (Definition 35.9), with $m, \lambda > 0$. With $\prod_i L_i = \sum_{k=0}^{h} p_k x^k$, $p_{-1} = 0$, $q_0 = 0$ and $q_k = \sum_i w_i d_i\, [x^{k-1}] \prod_{i' \neq i} L_{i'}$,*

$$\lambda p_h\, \langle x^{n+h+1} \rangle^{\lambda}_{m;\mathbf{w}} = \sum_{k=0}^{h} \big[(m+n+k)\, p_k - q_k - \lambda p_{k-1}\big]\, \langle x^{n+k} \rangle^{\lambda}_{m;\mathbf{w}},$$

*a recurrence whose coefficients are polynomials in the room coordinates, with leading coefficient $\lambda p_h$, $p_h = \prod_i d_i > 0$. For a mark $f$ of the room,*

$$\partial_{\alpha_f} \langle x^n \rangle^{\lambda}_{m;\mathbf{w}} = -\sum_i w_i \Big[ (\partial_{\alpha_f} d_i)\, \langle x^{n+1} \rangle^{\lambda}_{m;\mathbf{w}+\mathbf{e}_i} + (\partial_{\alpha_f} c_i)\, \langle x^n \rangle^{\lambda}_{m;\mathbf{w}+\mathbf{e}_i} \Big],$$

*and the moments with at most $\ell$ such derivatives lie in the span, over rational functions of the room, of $\{\langle x^k \rangle^{\lambda}_{m;\mathbf{w}+v} : 0 \le k \le h,\ v \in \mathbb{N}^h,\ |v| \le \ell\}$: at most $(h+1)\binom{h+\ell}{\ell}$ states.*

**Theorem 35.12** (Positive reads of the Beta ports)**.** *Let the letters be linear, $L_i = c_i + d_i x$, with exponents $w_i \geq 0$, let $a, b > 0$ and $q \in \mathbb{N}$, and let $J$ be a set of letters, with $w_J = \sum_{i\in J} w_i$.*

*(i) The Beta port of order $q$ reads (Theorem 3.15(iii), integrated over the room)*

$$\big\langle (-1)^q \Pi_q^{a,b} \big\rangle_{a,b;\mathbf{w}} = \sum_{|v|=q} \prod_i \frac{(w_i)_{v_i}}{v_i!} \Big\langle \big(x(1-x)\big)^q \prod_i \Big(\frac{d_i}{L_i}\Big)^{v_i} \Big\rangle_{a,b;\mathbf{w}} \in \Big[0, \frac{(|\mathbf{w}|)_q}{q!} \langle 1\rangle_{a,b;\mathbf{w}}\Big].$$

*(ii) The part of this sum over the $v$ supported in $J$ lies in $\big[0, \frac{(w_J)_q}{q!}\langle 1\rangle_{a,b;\mathbf{w}}\big]$, and for $q > 0$ it is positive when some letter of $J$ sees the mark.*

*(iii) At $q = 1$,*

$$a\,\langle 1\rangle_{a,b;\mathbf{w}} - (a+b)\,\langle x\rangle_{a,b;\mathbf{w}} - \sum_{i\notin J} w_i \Big\langle \frac{x(1-x)\,d_i}{L_i} \Big\rangle_{a,b;\mathbf{w}} = \sum_{i\in J} w_i \Big\langle \frac{x(1-x)\,d_i}{L_i} \Big\rangle_{a,b;\mathbf{w}} \in \big[0,\, w_J \langle 1\rangle_{a,b;\mathbf{w}}\big].$$

*Along a sequence at the data $a_n + j$, $b_n$, $\mathbf{w}(n)$ the bounds hold with $\langle 1\rangle_{a_n,b_n;\mathbf{w}(n)} \geq \langle 1\rangle_{a_n+j,b_n;\mathbf{w}(n)}$ in place of the read at the moment index $j$, uniformly in $j$.*

**Proposition 35.13** (The finite-band polynomial)**.** *In Theorem 35.12 write $L_{J^c} = \prod_{i\notin J} L_i$. The polynomial*

$$A = \big[a - (a+b)x\big] L_{J^c} + x(1-x)\Big(\partial_x L_{J^c} - L_{J^c} \sum_{i\notin J} \frac{w_i d_i}{L_i}\Big),$$

*of degree at most $h - |J| + 1$ in $x$, has the read*

$$\langle A\rangle_{a,b;\mathbf{w}} = \sum_{i\in J} w_i \Big\langle \frac{x(1-x)\,d_i\,L_{J^c}}{L_i} \Big\rangle_{a,b;\mathbf{w}},$$

*positive when a letter of $J$ sees the mark and the reads are finite. On the half-line, $A = a\,L_{J^c} + x\,\partial_x L_{J^c} - \lambda x L_{J^c} - x L_{J^c} \sum_{i\notin J} w_i d_i/L_i$ has $\langle A\rangle^{\lambda}_{a;\mathbf{w}} = \sum_{i\in J} w_i \big\langle x\,d_i\,L_{J^c}/L_i\big\rangle^{\lambda}_{a;\mathbf{w}}$.*

**Theorem 35.14** (A positive sequence of Beta ports on an interval of a selected mark)**.** *In a projective chart take a selected mark at a mass $m_e \in (0,1]$, restricted to an interval $0 < \lambda_0 < \alpha_e < \lambda_1$, and write $\alpha_e = \lambda_0 + (\lambda_1 - \lambda_0)x$, $x \in [0,1]$; its letters are linear in $\alpha_e$ with nonnegative coefficients, $L_i = c_i + d_i x$ with $d_i \geq 0$ and $c_i \geq \lambda_0 d_i/(\lambda_1 - \lambda_0)$, and exponents $w_i \geq 0$, the letters constant in $x$ going into the room density, and the factor $\alpha_e^{m_e - 1}$ of the mass is counted among them as the letter $\lambda_0 + (\lambda_1 - \lambda_0)x$ at the exponent $1 - m_e$. So $x(1-x)\,d_i/L_i \leq (\sqrt{\lambda_1} - \sqrt{\lambda_0})/(\sqrt{\lambda_1} + \sqrt{\lambda_0})$ on $[0,1]$ for every letter. Let $\langle 1\rangle_{1,1;\mathbf{w}}$ be finite. By Theorem 35.12(i) at $a = b = 1$ and the bound of Theorem 3.15(iii), integrated over the room, for $n \in \mathbb{N}$,*

$$0 < \big\langle (-1)^n \Pi_n^{1,1} \big\rangle_{1,1;\mathbf{w}} \leq \frac{(|\mathbf{w}|)_n}{n!} \Big(\frac{\sqrt{\lambda_1} - \sqrt{\lambda_0}}{\sqrt{\lambda_1} + \sqrt{\lambda_0}}\Big)^n \langle 1\rangle_{1,1;\mathbf{w}},$$

*the first inequality when a letter sees the mark, as the letter of the mass does when $m_e < 1$.*

**Theorem 35.15** (The finite-band recurrence in the order of the Beta ports)**.** *At a fixed point of the room let the letters be linear, $L_i = c_i + d_i x$ with $c_i > 0$ and $d_i \geq 0$, $i = 1, \dots, h$, with exponents $w_i(n)$ depending on $n \in \mathbb{N}$, let $a_n, b_n > 0$, and in this statement write $L = \prod_i L_i = \sum_k p_k x^k$ and*

$$A_n = \big(a_n - (a_n + b_n)x\big)\, L + x(1-x)\, \partial_x L - x(1-x)\, L \sum_i \frac{w_i(n)\, d_i}{L_i}.$$

*For every polynomial $g$, $\big\langle L\, x(1-x)\, \partial_x g + A_n g \big\rangle_{a_n, b_n; \mathbf{w}(n)} = 0$ (Theorem 35.10(a) with the polynomial $Lg$). Expanding $L\, x(1-x)\, \partial_x g + A_n g$ at $g = (-1)^r \Pi_r^{a_n, b_n}$ in the Beta ports $(-1)^{r+s} \Pi_{r+s}^{a_n, b_n}$ by the finite rules of Theorem 3.15(v) gives the relation*

$$\sum_{s=-h-1}^{h+1} \mathfrak{b}_{n,r,s} \left\langle (-1)^{r+s} \Pi_{r+s}^{a_n, b_n} \right\rangle_{a_n, b_n; \mathbf{w}(n)} = 0$$

*of width at most $2h + 3$, whose coefficients, written $\mathfrak{b}_{n,r,s}$ in this statement, are rational in $n$ and $r$ when $a_n$, $b_n$ and the $w_i(n)$ are, and polynomial in the room data. With $|\mathbf{w}(n)| = \sum_i w_i(n)$ and $p_h = \prod_i d_i$, the furthest forward coefficient is*

$$\mathfrak{b}_{n,r,h+1} = \big[|\mathbf{w}(n)| - a_n - b_n - h - r\big]\, p_h\, (-1)^{h+1} \binom{a_n + b_n + 2r - 2}{r} \Big/ \binom{a_n + b_n + 2r + 2h}{r + h + 1}.$$

**Lemma 35.16** (A rational eventual sign certificate)**.** *Let $Q_n(X) = n^d q_d(X) + \sum_{k<d} n^k q_k(X)$ with polynomials $q_k$, and let rationals $X_1 \geq X_{\max} > 0$ and $\delta > 0$ satisfy $q_d \geq \delta$ on $[0, X_1]$. With $B_k = \sum_i \big|[X^i] q_k\big|\, X_1^i$, $Q_n \geq \delta n^d / 2$ on $[0, X_{\max}]$ for every $n \geq \max\big(1, 2 \sum_{k<d} B_k / \delta\big)$. Composition of recurrences with variable coefficients keeps its shifts: with* Sh *the shift $n \mapsto n + 1$, $\big(\sum_j b_j(n)\, \mathrm{Sh}^j\big)\big(\sum_i a_i(n)\, \mathrm{Sh}^i\big) = \sum_{i,j} b_j(n)\, a_i(n + j)\, \mathrm{Sh}^{i+j}$.*

With the room held fixed, a mark is integrated alone against its letters, and the mark relation is the integration by parts that this allows: the flux of the density $x^{a-1}(1-x)^{b-1}$ times a polynomial vanishes at both ends of the box, and that of the Gamma density at both ends of the half-line. Two uses follow. Against the Gamma density a mark of a family closes its moments into a recurrence of bounded order whose coefficients are polynomials in the room, and the derivatives in the room stay in a finite span: one mark closes all its moments. Against a Beta port the same relation gives positive reads, sums of the ratios $x(1-x)d_i/L_i$ that a letter seeing the mark keeps positive and the total exponent bounds, and at every pair $a_n, b_n > 0$ the Beta ports $\Pi_r^{a_n, b_n}$ obey a recurrence of bounded width in their order $r$ whose furthest coefficient is explicit: when every $d_i$ is positive it vanishes at the one order $r = |\mathbf{w}(n)| - a_n - b_n - h$ at most.

# Chapter 36

# The critical-line law

Let $\mathcal{E}$ have two marks, and let $F$ be the family of the one list $\binom{\mathcal{E}}{1}$, whose bases are its two singletons, at the exponents $(1,1)$ and the masses $\mathbf{1}$ (Theorem 36.2). Its period is one, $P_F(\mathbf{1};1,1)=1$. On the fibre of the atom formula on the balance wall (Corollary 18.16), one share per basis and letter, the pins leave one free share: the basis share and the cobasis share of one basis are equal, $a=b=\frac{1}{2}+iy$, and the fibre is the line $\operatorname{Re} a=\frac{1}{2}$, the critical line. The law of $y$, the imaginary part of that share, is the critical-line law $\mathbb{P}_{\mathrm{crit}}$ (Definition 36.1). Its characteristic function is a stacked Gamma, its moments are rational, and its negative moments, read as finite parts, are rational multiples of the odd Trailmoments of the lattice input on the coset $\frac{1}{2}+\mathbb{Z}$ (Theorem 36.2). A function $R$ of the share, read against an odd derivative of $\mathbb{P}_{\mathrm{crit}}$ along the critical line, gives the critical-line word $\mathrm{crit}_k[R]$ (Definition 36.3): it carries one Trailmoment of odd index with a rational part, and it is positive whenever $R$ is the Fourier transform of a positive measure (Theorem 36.8).

**Definition 36.1** (The critical-line law)**.** The *critical-line law* is the probability law $\mathbb{P}_{\mathrm{crit}}(dy)=p_{\mathrm{crit}}(y)\,dy$ on $\mathbb{R}$ with

$$p_{\mathrm{crit}}(y)=\frac{1}{2\pi}\,\Gamma_4{}_0\big[\tfrac{1}{2}+iy,\ \tfrac{1}{2}+iy,\ \tfrac{1}{2}-iy,\ \tfrac{1}{2}-iy\big]=\frac{2\pi}{(e^{\pi y}+e^{-\pi y})^2},$$

the stacked Gamma of Lemma 5.16 on the line $\operatorname{Re} a=\frac{1}{2}$ divided by $2\pi$.

**Theorem 36.2** (The critical-line law and its moments)**.** *Let $\mathcal{E}$ have two marks and let $F$ be the family of the one list $\binom{\mathcal{E}}{1}$, whose bases are its two singletons, at the exponents $(1,1)$ and the masses $\mathbf{1}$. On the fibre of Corollary 18.16 the basis and cobasis shares of one basis are equal, $a=b=\frac{1}{2}+iy$, the fibre integral is $P_F(\mathbf{1};1,1)=1$, and the imaginary part $Y$ of the share has the law $\mathbb{P}_{\mathrm{crit}}$. In this statement $\mathbb{E}$ is the expectation of $Y\sim\mathbb{P}_{\mathrm{crit}}$.*

*(i)* $\mathbb{E}[e^{i\xi Y}]$ *is the Fourier read of Lemma 5.16(iii) at $t=\xi$; every odd moment of $Y$ vanishes and every even moment is rational.*

*(ii) For $j\geq 1$ the finite part at $y=0$ (Definition 1.5) of $\mathbb{E}[Y^{-2j}]$ is*

$$\begin{aligned}\operatorname{Reg}\mathbb{E}\big[Y^{-2j}\big]&=(-1)^j\,2j\,(\mathsf{T}_{2j+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty))\\&=(-1)^j\,2j\,(2^{2j+1}-1)\,(\mathsf{T}_{2j+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),\end{aligned}$$

*the second line by the lattice relation (Theorem 5.12(i)) at the step $q=2$, the lattice $\frac{1}{2}\mathbb{Z}$ splitting into $\mathbb{Z}$ and $\frac{1}{2}+\mathbb{Z}$.*

*(iii) With $X=\frac{1}{2}+iY$, $\mathbb{E}[e^{tX}]=\dfrac{te^t}{e^t-1}=t+\dfrac{t}{e^t-1}$, and $d_{m+1}\,\mathbb{E}[X^m]\in\mathbb{Z}$ for every $m\in\mathbb{N}$.*

The list $\binom{\mathcal{E}}{1}$ on two marks is the smallest table with a share that moves: at the centre the pins hold its basis and cobasis shares equal, on the critical line, and the period, one, is the integral of the law $\mathbb{P}_{\mathrm{crit}}$ of their imaginary part. Everything about that law is read off the stacked Gamma $\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]$ of Lemma 5.16. Its characteristic function is the stacked Gamma $\Gamma_{\substack{2\\0}}\big[1+\frac{i\xi}{2\pi},\ 1-\frac{i\xi}{2\pi}\big]$, its moments are rational with denominators bounded by the clearing integers, and its negative moments, which diverge at the origin, have finite parts that are rational multiples of the odd Trailmoments of the lattice input on the coset $\frac{1}{2}+\mathbb{Z}$, and so of the counting input.

## 36.1 Critical-line words

**Definition 36.3** (Critical-line words)**.** For an integer $k \geq 1$ and a function $R$ on the line $\operatorname{Re} u = \frac{1}{2}$, the *critical-line word* of order $k$ is

$$\mathrm{crit}_k[R] = -\frac{2}{(2k)!}\,\frac{1}{2\pi i}\int_{(1/2)} \frac{R(u)}{1-2u}\,\partial_u^{2k-1}\Gamma_{\substack{4\\0}}\big[u,\ u,\ 1-u,\ 1-u\big]\,du,$$

wherever it converges; the integrand is regular at $u=\frac{1}{2}$, where the odd derivative of the stacked Gamma, which is even about $\frac{1}{2}$, vanishes. On the line, $u=\frac{1}{2}+iy$,

$$\mathrm{crit}_k[R] = \frac{(-1)^k}{(2k)!}\int_{\mathbb{R}} R(\tfrac{1}{2}+iy)\,\frac{p_{\mathrm{crit}}^{(2k-1)}(y)}{y}\,dy.$$

**Proposition 36.4** (Critical-line words of polynomials)**.** *For an integer $k \geq 1$ and a polynomial $R(u) = (1-2u)\,Q(u) + R(\frac{1}{2})$, with $\mathbb{E}$ the expectation of $Y \sim \mathbb{P}_{\mathrm{crit}}$,*

$$\begin{aligned}\mathrm{crit}_k[R] &= R(\tfrac{1}{2})\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) + \frac{2}{(2k)!}\,\mathbb{E}\big[Q^{(2k-1)}(\tfrac{1}{2}+iY)\big]\\ &= (2^{2k+1}-1)\,R(\tfrac{1}{2})\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + \frac{2}{(2k)!}\,\mathbb{E}\big[Q^{(2k-1)}(\tfrac{1}{2}+iY)\big],\end{aligned}$$

*the second line by Theorem 5.12(i), as in Theorem 36.2(ii).*

**Definition 36.5** (Rows; clearing)**.** A value written as $x = \sum_{i=1}^{m} A_i v_i - B$, a rational combination of fixed real numbers $v_i$, its *coordinates*, such as Trailmoments of the counting input, less a rational number $B$, its *rational part*, has the *row* $(A_1,\dots,A_m;B)$; a positive integer *clears* the row when it makes all these coefficients integers.

**Corollary 36.6** (The rational part of an integer polynomial argument)**.** *For $P \in \mathbb{Z}[x]$ of degree $D$ and $R(u) = P(2u)$, the rational part of $\mathrm{crit}_k[R]$ (Definition 36.5) is cleared by $\frac{(2k)!}{2}\,d_{\max(1,D-2k+1)}$.*

**Proposition 36.7** (Binomial moments of the critical-line law)**.** *Let $X = \frac{1}{2}+iY$ with $Y \sim \mathbb{P}_{\mathrm{crit}}$, $\mathbb{E}$ its expectation, and $H_m = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,m])$.*

*(i) For complex $z$ with $|z|<1$, $\mathbb{E}\big[(1+z)^X\big] = (1+z)\log(1+z)/z$.*

*(ii) $\mathbb{E}\big[\binom{X}{0}\big] = 1$ and $\mathbb{E}\big[\binom{X}{i}\big] = \dfrac{(-1)^{i+1}}{i(i+1)}$ for $i \geq 1$.*

*(iii) For $i \geq 1$, $\mathbb{E}\Big[\dfrac{d}{du}\dbinom{u}{i}\Big|_{u=X}\Big] = 2(-1)^i\,\dfrac{H_{i-1}-1}{i(i+1)}$. So for every polynomial $Q = \sum_i q_i\binom{u}{i}$,*

$$\mathrm{crit}_1\left[(1-2u)\,Q\right] = 2\sum_{i\geq 1} q_i\,(-1)^i\,\frac{H_{i-1}-1}{i(i+1)}.$$

In the binomial basis the critical-line word of a polynomial has explicit coefficients, harmonic numbers over $i(i+1)$, whose denominators a prime $q$ divides only to bounded powers; this is the basis in which the clearings of the critical-line words of polynomials are counted, prime by prime.

**Theorem 36.8** (Positive arguments for the critical-line word)**.** *For an integer $k \geq 1$ and real $t$, with $\mathbb{E}$ the expectation of $Y \sim \mathbb{P}_{\mathrm{crit}}$,*

$$\mathrm{crit}_k\left[e^{t(u-\frac{1}{2})}\right] = \frac{1}{(2k)!}\int_{|t|}^{\infty} x^{2k-1}\,\mathbb{E}\big[e^{ixY}\big]\,dx > 0,$$

$$\mathrm{crit}_k[1] = (\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) = (2^{2k+1}-1)\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) \quad \textit{(Theorem 5.12(i)).}$$

*If $R(\frac{1}{2}+iy) = \int e^{ity}\,\phi(dt)$ for a finite nonzero positive measure $\phi$ on $\mathbb{R}$, then $\mathrm{crit}_k[R] = \int \mathrm{crit}_k\left[e^{t(u-\frac{1}{2})}\right]\phi(dt) > 0$.*

On the critical line the word reads its argument against an odd derivative of the density of the critical-line law divided by the imaginary part of the share, a kernel without a fixed sign. Its Fourier transform, the word of an exponential argument, is the integral beyond $|t|$ of the characteristic function times a power, a positive function, so an argument that is the Fourier transform of a positive measure has a positive word. This is how positive words carrying one odd Trailmoment are made.

**Theorem 36.9** (A positive rational argument with one odd index)**.** *Let $M \in \mathbb{N}$ and $p(v) = \sum_{j=0}^{D} b_j v^j \in \mathbb{Q}[v]$, nonnegative on $[0,1]$ and not identically zero, and let*

$$R_{M,p}(u) = \sum_{j=0}^{D} b_j\Big(\frac{1}{u+M+j} + \frac{1}{1-u+M+j}\Big), \qquad R_{M,p}(\tfrac{1}{2}) = 4\sum_j \frac{b_j}{2M+2j+1} > 0.$$

*Then for every $k \geq 1$*

$$\begin{aligned}
\mathrm{crit}_k[R_{M,p}] &= \frac{2}{(2k)!}\int_0^1\!\!\int_0^1 \frac{v^M x^{-1/2}\,p(v)\,[-\log(vx)]^{2k}}{1-vx}\,dx\,dv \\
&= R_{M,p}(\tfrac{1}{2})\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) \\
&\qquad - 4\sum_j \frac{b_j}{2M+2j+1}\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((M+j,\infty)) \\
&= (2^{2k+1}-2)\,R_{M,p}(\tfrac{1}{2})\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) \\
&\qquad + 4\sum_j \frac{b_j}{2M+2j+1}\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,M+j]) > 0.
\end{aligned}$$

*If $Cb_j \in \mathbb{Z}$ for every $j$, then $C\,d_{2M+2D+1}\,d_{M+D}^{2k+1}$ clears both rational coefficients; the Trailmoments of $\mathbb{I}_{\frac{1}{2}+\mathbb{Z}}$ are read in those of $\mathbb{I}_{\mathbb{Z}}$ by Theorem 5.12(i).*

**Corollary 36.10** (Arguments of two Beta corners)**.** *For $p(v) = (1-v)^q$, $q \in \mathbb{N}$, the argument of Theorem 36.9 is*

$$R_{M,(1-v)^q}(u) = \Gamma\begin{matrix}2\\1\end{matrix}\!\left[\begin{matrix}u+M,\ q+1\\ u+M+q+1\end{matrix}\right] + \Gamma\begin{matrix}2\\1\end{matrix}\!\left[\begin{matrix}1-u+M,\ q+1\\ 2-u+M+q\end{matrix}\right].$$

*Its words are positive and carry one Trailmoment of the counting input, of the odd index $2k+1$, at every $k$. For fixed $M$, $\mathrm{crit}_k[R_{M,(1-v)^q}] \sim \frac{4\,\Gamma(M+1)}{(2k)!}\,q^{-M-1}(\log q)^{2k}$ as $q \to \infty$.*

**Proposition 36.11** (Poles at the half-integers)**.** *For integers $k \geq 1$ and $l \geq 0$,*

$$\frac{1}{2\pi i}\int_{(1/2)} \frac{1}{u-\frac{1}{2}-l}\sum_{m\in\mathbb{Z}}\frac{du}{(u+m)^{2k+1}} = -(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((l,\infty)),$$

*the integrand being regular on the line at $l = 0$; and for $l \geq 1$, with the critical-line word of Definition 36.3,*

$$\operatorname{crit}_k\Big[\frac{1}{u-\frac{1}{2}-l}\Big] = -\operatorname{crit}_k\Big[\frac{1}{u-\frac{1}{2}+l}\Big] = -\frac{1}{l}\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,l]).$$

So the critical-line word of a simple pole at a half-integer other than $\frac{1}{2}$ is a head of the lattice input on the coset $\frac{1}{2}+\mathbb{Z}$, a rational number, and with Proposition 36.4 the word is read on every polynomial plus simple poles at these points. The rational arguments of Theorem 36.9 have their simple poles at the integers; the half-integers give heads and tails on the coset $\frac{1}{2}$.

**Proposition 36.12** (The decay of the word of an exponential argument)**.** *For an integer $k \geq 1$ and $t > 0$, with the polynomial $c_k(t) = \sum_{l=0}^{2k} \frac{(2k)!}{l!}\,2^{2k-l+1}\,t^l$,*

$$\frac{e^{-t/2}}{(2k)!}\,c_k(t) \leq \operatorname{crit}_k\left[e^{t(u-\frac{1}{2})}\right] \leq \frac{e^{-t/2}}{(2k)!\,(1-e^{-t})}\,c_k(t).$$

A rational argument whose positive measure is the exponential profile of a polynomial nonnegative on $[0,1]$ gives a positive word with one odd Trailmoment of the counting input and a rational part cleared by $d_{2M+2D+1}d_{M+D}^{2k+1}$. The words of exponential arguments decay like $e^{-t/2}$ times a polynomial of degree $2k$, so the words of the arguments of two Beta corners decay like a power of their exponent $q$: they are positive, carry one Trailmoment and are controlled, the raw material of the relation blocks of Chapter 44.

# Chapter 37

# The gap and the shape words

A value read at rational data has a row, its coefficients on Trailmoments of the counting input and a rational part (Definition 36.5). The gap theorem turns a sequence of nonzero values whose rows are cleared by integers that grow more slowly than the values decay into irrationality, and the shape words supply such sequences. They are integrals over a cube against one letter; opened by one share they close on the stacked Gamma $\Gamma_{0}^{4}\big[a,\ a,\ 1-a,\ 1-a\big]$ of Lemma 5.16, so their rows are explicit, cleared by powers of the clearing integers $d_n$ and carried by the Trailmoments $(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ of index at most one more than the number of shapes. Beta ports inserted on one shape keep the words positive while they move the row, filters built from recurrences remove unwanted coordinates, and Beta ports whose order grows with the depth trade a slower decay for a selected coordinate.

## 37.1 Clearing and the gap

**Lemma 37.1** (The growth of the clearing integers)**.** *For integers $n \geq 0$ and $s \geq 1$, $d_n^s$ is the least positive integer $D$ with $D\,(\mathsf{T}_{s;(1)}[x]\,f\,\mathbb{I}_{\mathbb{Z}})((0,n]) \in \mathbb{Z}$ for every integer density $f$, and*

$$\limsup_{n\to\infty} \frac{\log d_n}{n} \leq c_{\mathrm{lcm}} = \frac{14}{25}\log 2 + \frac{9}{25}\log 3 + \frac{1}{5}\log 5 = 1.10555\ldots < 1.106.$$

**Theorem 37.2** (The gap)**.** *Let $v_1, \ldots, v_m$ be real, and let real numbers $x_n = \sum_i A_{n,i} v_i - B_n$, with $A_{n,i}, B_n \in \mathbb{Q}$, have positive integers $D_n$ clearing all these coefficients, with $0 < |D_n x_n| \to 0$.*

*(i) Some $v_i$ is irrational.*

*(ii) For $m = 1$, $A_{n,1} \neq 0$ for all large $n$.*

*The hypothesis $x_n \neq 0$ holds in particular when $x_n$ is the value of a word with a strictly positive read.*

An integer of modulus below one is zero; nothing else is used. What the theory contributes is the two other hypotheses: positivity, which keeps the cleared values from vanishing, and rows whose clearing integers grow more slowly than the values decay.

## 37.2 Shape words with polynomial insertions

**Definition 37.3** (Shape words)**.** For integers $n \geq 0$ and $k \geq 2$ let $L = \bar{x}_0 + x_0 x_1 \cdots x_k$ on the cube $[0,1]^{k+1}$ with coordinates $x_0, \ldots, x_k$. The *shape word* of depth $n$ with $k$ shapes of a polynomial

insertion $P \in \mathbb{Q}[x_0,\dots,x_k]$ is

$$I_n(k;P)=\int_{[0,1]^{k+1}} P(x)\,\frac{\prod_{i=0}^{k}(x_i\bar{x}_i)^n}{L^{n+1}}\,dx=\int_{[0,1]^{k+1}} P\,f_k^n\,\frac{dx}{L},\qquad f_k=\frac{\prod_{i=0}^{k}x_i\bar{x}_i}{L},$$

with the *base* $f_k$. For a monomial $x^p$, $p=(p_0,\dots,p_k)$, the *share function* is

$$g_{n,p}(a)=\frac{(n!)^{k-2}\,(1-a)_n^2\,(a)_{p_0}}{(n+1)_{p_0}\prod_{i=1}^{k}(a+p_i)_{n+1}},$$

and for $P=\sum_p\big([x^p]P\big)x^p$ it is $g_{n,P}=\sum_p\big([x^p]P\big)\,g_{n,p}$, with the coefficients $c_{n,h,s}(P)$ of its principal parts $c_{n,h,s}(P)(a+h)^{-s}$ at the poles $a=-h$ and the *column sums* $C_{n,s}(P)=\sum_h c_{n,h,s}(P)$. The *port factor* is $\mathfrak{a}_k=x_0\,x_1\bar{x}_1\,x_2\cdots x_k/L$, and $M_k(\eta)=\sup_{(0,1)^{k+1}} f_k\mathfrak{a}_k^{\eta}$ for $\eta\ge 0$.

**Theorem 37.4** (Shape words: the one-share closing)**.** *Let $P$ have degree at most $D$, with $p_0\le(k-2)(n+1)$ on every monomial $x^p$ of $P$.*

*(i)* Share side. *For a monomial, on $0<\operatorname{Re}a=c<n+1$,*

$$\begin{aligned}I_n(k;x^p)&=\frac{1}{2\pi i}\int_{(c)}\Gamma\,{}_2^2\!\left[\begin{matrix}a,\ n+1-a\\ n+1\end{matrix}\right]\Gamma\,{}_2^2\!\left[\begin{matrix}a+p_0,\ n+1-a\\ n+1+p_0\end{matrix}\right]\prod_{i=1}^{k}\Gamma\,{}_2^1\!\left[\begin{matrix}a+p_i,\ n+1\\ a+p_i+n+1\end{matrix}\right]da\\&=\frac{1}{2\pi i}\int_{(c)}\Gamma\,{}_4^0\big[a,\ a,\ 1-a,\ 1-a\big]\ g_{n,p}(a)\,da.\end{aligned}$$

*(ii)* Closing. $I_n(k;P)=-\sum_{m\ge1}g'_{n,P}(m)$ *(Lemma 5.16(i)). Every pole of $g_{n,P}$ lies among $0,-1,\dots,-n-D$ with order at most $k$; the integers $1,\dots,n$ are zeros of order at least two of each $g_{n,p}$; $g_{n,p}(a)=O(|a|^{-2})$.*

*(iii)* The row. $C_{n,1}(P)=0$ *and*

$$\begin{aligned}I_n(k;P)&=\sum_{h=0}^{n+D}\sum_{s=1}^{k}s\,c_{n,h,s}(P)\,(\mathsf{T}_{s+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((h,\infty))\\&=\sum_{s=2}^{k}s\,C_{n,s}(P)\,(\mathsf{T}_{s+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))-\sum_{h,s}s\,c_{n,h,s}(P)\,(\mathsf{T}_{s+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,h]).\end{aligned}$$

**Proposition 37.5** (An integer certificate for inserted words)**.** *In Theorem 37.4 let $P$ have integer coefficients. Then $(n+1)_D\,d_{n+D}^{k-s}\,c_{n,h,s}(P)\in\mathbb{Z}$, and $(n+1)_D\,d_{n+D}^{k+1}$ clears every coefficient of the row, its rational part included.*

**Theorem 37.6** (Integer-valued insertion clearing)**.** *In Theorem 37.4 let $P$ have integer coefficients, let $p_{\max}$ be its largest exponent of a shape $x_i$, $i\ge1$, and $p_{0,\max}$ its largest exponent of $x_0$, and let $b_P=\operatorname{lcm}\big\{\binom{n+j}{j}:\ j$ an exponent of $x_0$ in $P\big\}$. Then $b_P\,d_{\max(n+p_{\max},\,p_{0,\max})}^{k-s}\,c_{n,h,s}(P)\in\mathbb{Z}$ and $b_P\,d_{\max(n+p_{\max},\,p_{0,\max})}^{k+1}$ clears the whole row, and $b_P$ divides $d_{n+p_{0,\max}}$; insertions free of $x_0$ are cleared by $d_{n+p_{\max}}^{k+1}$ whatever their degrees. For $k\ge3$ the insertion $x_0^n$ retains the top coordinate:*

$$C_{n,k}(x_0^n)=(-1)^{n(k+1)}\binom{2n}{n},\qquad\big[(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))\big]\,I_n(k;x_0^n)=k\,(-1)^{n(k+1)}\binom{2n}{n}.$$

**Theorem 37.7** (The exact maximum under a growing Beta port)**.** *Fix $k \ge 2$, with the base $f_k$ and the port factor $\mathfrak{a}_k$ of Definition 37.3, so that $I_n(k;P) = \int P f_k^n \, dx/L$. For $q \in \mathbb{N}$,*

$$I_n\big(k; (-1)^q \Pi_q^{n+1,n+1}(x_1)\big) = \binom{n+q}{q} \int_{[0,1]^{k+1}} f_k^n \, \mathfrak{a}_k^q \, \frac{dx}{L} > 0.$$

*For every real $\eta \ge 0$, $f_k \mathfrak{a}_k^\eta$ has a unique maximum on the open cube: with $a$ the unique root in $0 < a < 1/(2+\eta)$ of*

$$(1-2a)\big(1-(2+\eta)a\big)(1+\eta a)^{k-1} = (1+\eta)^k a^{k+2},$$

*it is at $x_1 = a$, $x_2 = \cdots = x_k = \frac{(1+\eta)a}{1+\eta a}$, $x_0 = \frac{(1+\eta)a}{1-a}$, with value*

$$M_k(\eta) = \frac{(1-a)^{k-2}}{(1+\eta a)^{k-1}} \, (1-2a)^{1+\eta} \, \big(1-(2+\eta)a\big),$$

*and $0 < I_n\big(k; (-1)^q \Pi_q^{n+1,n+1}(x_1)\big) \le \binom{n+q}{q} M_k(q/n)^n \, I_0(k;1)$.*

**Proposition 37.8** (The root rate of the shape words)**.** *For $k \ge 2$:*

*(i) $0 < I_n(k;1) \le M_k(0)^n \, I_0(k;1)$ (Theorem 37.7 at $q = 0$);*

*(ii) $\lim_n I_n(k;1)^{1/n} = M_k(0)$, where*

$$M_k(0) = \max_{(0,1)^{k+1}} f_k = \max_{0<a<1} \frac{a^k (1-a)^k}{(1+a^{k/2})^2} \in \Big[\frac{4^{-k}}{(1+2^{-k/2})^2}, \; 4^{-k}\Big].$$

*The maximum over $a$ is at the root $a \in (0, \frac{1}{2})$ of $a^{k/2+1} + 2a = 1$, the root of Theorem 37.7 at $\eta = 0$. For every $k \ge 2$, $-\log M_k(0) > (k+1)\, c_{\mathrm{lcm}}$, so that $d_n^{k+1} I_n(k;1) \to 0$.*

**Theorem 37.9** (An irrational Trailmoment of the counting input)**.** *The shape word $I_n(2;1)$ is positive, and with $a_l = (-1)^l \binom{n}{l}\binom{n+l}{l}$, the coefficient of $x^l$ in $(-1)^n \Pi_n^{1,1}(x)$ (Theorem 3.15(v)), and $b_l = 2a_l \sum_{j \ne l} \frac{a_j}{j-l}$,*

$$\begin{aligned} I_n(2;1) &= \sum_{l=0}^{n} \Big( 2a_l^2 \, (\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((l,\infty)) + b_l \, (\mathsf{T}_{2;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((l,\infty)) \Big) \\ &= 2\big( A_n \, (\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) - B_n \big), \\ A_n &= \sum_l a_l^2 \in \mathbb{Z}, \qquad d_n^3 B_n \in \mathbb{Z}, \end{aligned}$$

$$B_n = \sum_l a_l^2 \, (\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,l]) + \sum_l a_l \, (\mathsf{T}_{2;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,l]) \sum_{j \ne l} \frac{a_j}{j-l},$$

*$I_n(2;1) \le 2\, (\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) \, (\sqrt{2}-1)^{4n}$, and $(\mathsf{T}_{3;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N,\infty)) \notin \mathbb{Q}$ for every integer cut $N \ge 0$.*

A shape word is the cube against one letter, and one share opens the letter: the Beta corners of the shapes and of $x_0$ leave a single line integral of stacked Gammas, the stacked Gamma ${}_0\Gamma_4\big[a,\ a,\ 1-a,\ 1-a\big]$ times a rational share function, which closes on the positive integers. The share function has poles of order at most $k$ at the nonpositive integers, so the row carries the Trailmoments of the counting input of index at most $k+1$ with rational parts cleared by powers

of $d_{n+D}$, and the coefficients of the simple poles sum to zero because the share function decays quadratically. At two shapes the row has one coordinate besides the rationals, the decay $(\sqrt{2}-1)^{4n}$ beats the clearing $d_n^3$, and the gap gives the irrationality of $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,\infty))$ at every integer cut $N$. With more shapes the decay still beats the clearing, but the row carries several coordinates, and the filters and the growing Beta ports select one (Proposition 37.18, Theorem 37.19).

## 37.3 Rows

**Proposition 37.10** (The row of a closing in rational terms)**.** *Let a closing give $x_n = \sum_{t\ge t_0} r_n(t)$, the sum over the integers $t \ge t_0$, with $r_n(t) = \sum_{k,s} \tilde{c}_{n,k,s}(t+k)^{-s} = O(t^{-2})$, $\tilde{c}_{n,k,s} \in \mathbb{Q}$, integer pole positions and $h_k = t_0 + k - 1 \ge 0$. With $\tilde{C}_{n,s} = \sum_k \tilde{c}_{n,k,s}$,*

$$x_n = \sum_{k,s} \tilde{c}_{n,k,s}\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((h_k,\infty)) = \sum_{s\ge 2} \tilde{C}_{n,s}\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - B_n,$$

$$B_n = \sum_{k,s} \tilde{c}_{n,k,s}\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,h_k]),$$

*the index-one Trailmoments regularized, and $D_n = \operatorname{lcm}_{k,s}\left(\operatorname{den}(\tilde{c}_{n,k,s})\,d_{h_k}^{s}\right)$ clears the row.*

**Proposition 37.11** (The primitive row)**.** *For a nonzero rational pair $(A,B)$, a common clearing integer $D$ and $g = \gcd(DA,DB)$, the pair $g^{-1}(DA,DB) \in \mathbb{Z}^2$ is, up to sign, independent of $D$ and of every nonzero rational rescaling of $(A,B)$, and with $(a,b) = g^{-1}(DA,DB)$, $av - b = (D/g)(Av - B)$ for every real $v$.*

**Corollary 37.12** (The top-index criterion)**.** *Let $k \ge 3$, and suppose there are integer polynomials $P_n$ of degree at most a fixed $D$ with $C_{n,s}(P_n) = 0$ for $2 \le s < k$ and $I_n(k;P_n) > 0$ for all large $n$, $P_n$ possibly changing sign on the cube. If the sums $\|P_n\|_1$ of the absolute values of the coefficients of $P_n$ satisfy*

$$\limsup_n \frac{1}{n}\log\|P_n\|_1 < -\log M_k(0) - (k+1)\limsup_n \frac{1}{n}\log d_n,$$

*then $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) \notin \mathbb{Q}$ for every integer cut $N \ge 0$.*

A closing in rational terms has a row whose clearing is read off its pole positions and orders, and a row is a primitive integer pair up to the clearing. With $k$ shapes the row of $I_n(k;P)$ carries $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),\dots,(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$; an insertion that kills the lower column sums and keeps the word positive at a cost below the margin between the decay and the clearing separates $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. The insertion may change sign: positivity is asked of the word, not of the insertion.

## 37.4 The Beta port ladder and the filters

**Theorem 37.13** (The Beta port ladder of the shape words)**.** *For the shape word with $k \ge 2$ shapes at depth $n \ge 1$ write its letter as $L = \bar{x}_0 + x_0x_2\cdots x_k\,x_1$, and in this statement write $P_r = (-1)^r\Pi_r^{n+1,n+1}(x_1)$, $g_{n,r} = g_{n,P_r}$ and $g_n = g_{n,0}$, the share function of the insertion 1 (Definition 37.3).*

*(i)* $0 < I_n(k;P_r) \le \binom{n+r}{r} I_n(k;1)$ *(Theorem 37.7, the port factor being at most one).*

*(ii)* $g_{n,r}(a) = \binom{n+r}{r}\dfrac{(n+1-a)_r}{(n+1+a)_r}\,g_n(a)$ *with* $g_n(a) = \dfrac{(n!)^{k-2}(1-a)_n^2}{(a)_{n+1}^k}$, *and with* $b = n+r+1$, $(r+1)(a+b)\,g_{n,r+1}(a) = b\,(b-a)\,g_{n,r}(a)$.

*(iii) For a function* $\varphi$ *on the integers and a finite set* $\mathcal{M}$ *of positive integers write* $S_{n,r}[\varphi] = \sum_{m\in\mathcal{M}} g_{n,r}(m)\varphi(m)$ *and* $S'_{n,r}[\varphi] = -\sum_{m\in\mathcal{M}} g'_{n,r}(m)\varphi(m)$, *the sums over the common finite set* $\mathcal{M}$, *with functions of the summation variable* $m$ *inside the brackets. Then, with* $b = n+r+1$,

$$(r+1)\,S'_{n,r+1}\big[(m+b)\varphi\big] = b\,S'_{n,r}\big[(b-m)\varphi\big] + (r+1)\,S_{n,r+1}[\varphi] + b\,S_{n,r}[\varphi].$$

*(iv)* $\dfrac{g_{n+1,r}(a)}{g_{n,r}(a)} = \dfrac{(n+1)^{k-3}(n+r+1)(n+1-a)(n+r+1-a)}{(a+n+1)^{k-1}(a+n+r+1)}$, *and with* $U(m) = (m+n+1)^{k-1}(m+n+r+1)$ *and* $V(m) = (n+1)^{k-3}(n+r+1)(n+1-m)(n+r+1-m)$,

$$S_{n+1,r}[U\varphi] = S_{n,r}[V\varphi], \qquad S'_{n+1,r}[U\varphi] - S_{n+1,r}[U'\varphi] = S'_{n,r}[V\varphi] - S_{n,r}[V'\varphi].$$

*(v) For* $0 \le r \le h$ *the insertions* $P_r$ *have degree at most* $h$, *their rows are cleared by* $d_{n+h}^{k+1}$ *(Theorem 37.6), and their positive reads cost at most* $\binom{n+h}{h}$ *times the base word* $I_n(k;1)$.

*(vi) In the selected mark* $x_1$ *the furthest forward coefficient of Theorem 35.15, for the one letter* $L$ *of exponent* $n+1$, *is nonzero on the open cube.*

**Proposition 37.14** (The insertion index supplies auxiliary states)**.** *Fix* $k \ge 3$ *and* $n \ge 1$ *and, in this statement, write* $P_r = (-1)^r \Pi_r^{n+1,n+1}(x_1)$, *an insertion of the shape word with* $k$ *shapes. As functions of* $r$, $C_{n,k}(P_r) \in \mathbb{Q}[r]$ *and* $C_{n,k-1}(P_r) = -2\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r])\,C_{n,k}(P_r) + q_n(r)$ *with* $q_n \in \mathbb{Q}(r)$. *Consequently a scalar finite shift recurrence over* $\mathbb{Q}(r)$ *annihilating* $C_{n,k-1}(P_r)$ *for all large* $r$ *also annihilates* $C_{n,k}(P_r)$.

**Proposition 37.15** (The shape step is a two-state operator relation)**.** *For* $k \ge 2$ *and* $n \ge 0$ *let* $g_n$ *be the share function of the insertion* $1$, $g_n(a) = (n!)^{k-2}(1-a)_n^2/(a)_{n+1}^k$, *and for integer cuts* $0 \le N_0 < N_1$ *write* $S_n[\varphi] = \sum_{N_0<m\le N_1} g_n(m)\varphi(m)$ *and* $S'_n[\varphi] = -\sum_{N_0<m\le N_1} g'_n(m)\varphi(m)$. *Then*

$$\begin{aligned} S_{n+1}\big[(m+n+1)^k\varphi\big] &= (n+1)^{k-2}\,S_n\big[(m-n-1)^2\varphi\big], \\ S'_{n+1}\big[(m+n+1)^k\varphi\big] &= k\,S_{n+1}\big[(m+n+1)^{k-1}\varphi\big] + (n+1)^{k-2}\,S'_n\big[(m-n-1)^2\varphi\big] \\ &\quad - 2(n+1)^{k-2}\,S_n\big[(m-n-1)\varphi\big], \end{aligned}$$

*and the same holds over all* $m > N_0$ *whenever the sums converge one by one: for bounded* $\varphi$, *when* $(k-2)(n+1) > 1$ *in the first identity and* $(k-2)(n+1) > 0$ *in the second.*

**Proposition 37.16** (A polynomial filter from a bounded local recurrence)**.** *Let unwanted closing coefficients satisfy* $c_{n,j} = \ell(n)\,y_{n,j}$ *and* $y_{n+1,j} = \mathbf{A}(n)\,y_{n,j}$, $1 \le j \le r$, *with state vectors* $y_{n,j}$ *of a fixed length* $h$, *a* $1\times h$ *matrix* $\ell(n)$ *and an* $h\times h$ *matrix* $\mathbf{A}(n)$ *over* $\mathbb{Q}(n)$. *Write* $\mathbf{A} = \widehat{\mathbf{A}}/q$ *and* $\ell = \hat{\ell}/s$ *with integer polynomial matrices* $\widehat{\mathbf{A}}$, $\hat{\ell}$ *and integer polynomials* $q$, $s$, *put* $\mathbf{A}^{[0]} = I$, $\mathbf{A}^{[i]}(n) = \mathbf{A}(n+i-1)\cdots\mathbf{A}(n)$, $Q(n) = \prod_{t=0}^{h-1} q(n+t)\prod_{t=0}^{h} s(n+t)$ *and the* $1\times h$ *matrices* $\mathbf{r}_i(n) = Q(n)\,\ell(n+i)\,\mathbf{A}^{[i]}(n)$, $0 \le i \le h$. *If the* $\mathbf{r}_i$ *span* $\mathbb{Q}(n)^{1\times h}$ *and* $a_i(n)$ *is* $(-1)^i$ *times the determinant of the* $h\times h$ *matrix stacking the* $\mathbf{r}_{i'}$, $i' \ne i$, *in order, then the* $a_i$ *are integer polynomials, not all zero, and*

$$\sum_{i=0}^{h} a_i(n)\,c_{n+i,j} = 0 \qquad (1 \le j \le r)$$

*away from the integer zeros of the denominators; a target sequence $x_n$ is retained at $n$ exactly when $\sum_i a_i(n)\,x_{n+i} \neq 0$. With $D_{\mathbf{A}} = \max(\deg \widehat{\mathbf{A}}, \deg q)$, $D_s = \deg s$ and $D_\ell = \deg \hat{\ell}$: $\deg a_i \leq h(hD_{\mathbf{A}} + hD_s + D_\ell)$. For two states, with $\ell = (1,0)$ and $\mathbf{A} = (\beta_{il})$, the filter is $\beta_{12}c_{n+2} - (\beta_{11}^{+}\beta_{12} + \beta_{12}^{+}\beta_{22})c_{n+1} + \beta_{12}^{+}\det \mathbf{A}(n)\,c_n = 0$, $\beta_{il}^{+} = \beta_{il}(n+1)$; for one state, $q(n)c_{n+1} = p(n)c_n$ gives the filter $q(n)\,\mathrm{Sh} - p(n)$, $\mathrm{Sh}$ the shift $n \mapsto n+1$.*

**Proposition 37.17** (The finite jet recurrence)**.** *If a state has the recurrence $U_{n+1}(\varepsilon) = \mathbf{A}(n,\varepsilon)\,U_n(\varepsilon)$, regular at $\varepsilon = 0$, then with $U_n(\varepsilon) = \sum_j U_{n,j}\varepsilon^j$ and $\mathbf{A}(n,\varepsilon) = \sum_j \mathbf{A}_j(n)\varepsilon^j$, $U_{n+1,j} = \sum_{i=0}^{j} \mathbf{A}_i(n)\,U_{n,j-i}$ for $0 \leq j \leq J$: a lower block-triangular recurrence for the jets of order at most $J$, with diagonal blocks $\mathbf{A}_0(n)$.*

**Proposition 37.18** (A recurrence filters the positive word)**.** *Let $x_n = \int X^n\,d\mu$ with $0 < X \leq X_{\max} < 1$ on the support of a finite positive measure $\mu$, and $x_n = \sum_{j=1}^{r} \tilde{C}_{n,j}v_j - B_n$ with $\tilde{C}_{n,j}, B_n \in \mathbb{Q}$. Let $a_0(n), \dots, a_h(n) \in \mathbb{Z}[n]$, $Q_n(X) = \sum_i a_i(n)X^i$ and $\tilde{x}_n = \sum_i a_i(n)\,x_{n+i}$, and suppose that for all large $n$, $\sum_i a_i(n)\tilde{C}_{n+i,j} = 0$ for $j \neq j_*$ and $Q_n$ has one sign on the support of $\mu$, strictly on a set of positive measure. Then $\tilde{x}_n = \int X^n Q_n\,d\mu = A_n v_{j_*} - \sum_i a_i(n)B_{n+i}$ with $A_n = \sum_i a_i(n)\tilde{C}_{n+i,j_*}$, and $0 < |\tilde{x}_n| \leq O(n^{\max_i \deg a_i})\,X_{\max}^n \mu(\mathbb{R})$. If positive integers $D_{n,h}$ clear the rows of $x_n, \dots, x_{n+h}$, and $\limsup_n \frac{1}{n}\log D_{n,h} < -\log X_{\max}$, then $v_{j_*} \notin \mathbb{Q}$ and $A_n \neq 0$ for all large $n$. The sign of $Q_n$ on $(0, X_{\max}]$ is certified when $Q_n(X) = \sum_{l=0}^{h} \beta_l(n)X^l(X_1 - X)^{h-l}$ for a rational $X_1 > X_{\max}$ with all $\beta_l(n) \geq 0$ and one of them positive.*

The Beta ports of one shape move the row of a shape word along a ladder: the share function of the Beta port of order $r+1$ is that of order $r$ times a rational factor, so the sums over a common finite set of integers satisfy finite operator relations in the order of the Beta ports and in the depth. Unwanted coordinates whose coefficients follow a bounded local recurrence in the depth are annihilated by a filter with integer polynomial coefficients, read off the determinants of that recurrence, and jets in a regulator follow a block-triangular recurrence with the same diagonal. A filter applied to a positive word stays positive as long as its polynomial keeps one sign on the support of the word's measure, which a nonnegative expansion in the products $X^l(X_1 - X)^{h-l}$ certifies, and then the filtered word separates the coordinate it keeps, at the cost of clearing the rows of $h+1$ consecutive values.

## 37.5 Growing Beta ports

**Theorem 37.19** (An arithmetic budget for growing Beta ports)**.** *For $k \geq 2$, $\eta > 0$ and $q_n = \lfloor \eta n \rfloor$, in this statement write $\ell_k(\eta) = \log M_k(\eta) + (1+\eta)\log(1+\eta) - \eta\log\eta$. Then*

$$\lim_n \frac{1}{n}\log I_n\big(k; (-1)^{q_n}\Pi_{q_n}^{n+1,n+1}(x_1)\big) = \ell_k(\eta),$$

$$\limsup_n \frac{1}{n}\log\Big(d_{n+q_n}^{k+1}\, I_n\big(k; (-1)^{q_n}\Pi_{q_n}^{n+1,n+1}(x_1)\big)\Big) \leq \ell_k(\eta) + (k+1)(1+\eta)\,c_{\mathrm{lcm}}.$$

*The integer $d_{n+q_n}^{k+1}$ clears the whole row, which stays on the coordinates $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, …, $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and a rational part, and the coefficient of $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ is*

$$k\binom{n+q_n}{q_n}\sum_{j=0}^{n}(-1)^{jk}\binom{n}{j}^k\binom{n+j}{n}^2\frac{(n+1+j)_{q_n}}{(n+1-j)_{q_n}},$$

*positive for even $k$.*

**Corollary 37.20** (A growing Beta port with a certified gap)**.** *For every $k \geq 2$ there is a rational $\eta_0 > 0$ such that $\ell_k(\eta) + (k+1)(1+\eta)\, c_{\mathrm{lcm}} < 0$ for every rational $\eta \in (0, \eta_0]$ (Theorem 37.19), the root of Theorem 37.7 at each such $\eta$ being certified as a root of an integer polynomial in a rational interval. So with $q_n = \lfloor \eta n \rfloor$ the positive words $I_n\big(k; (-1)^{q_n} \Pi_{q_n}^{n+1,n+1}(x_1)\big)$ satisfy*

$$\limsup_n \frac{1}{n} \log \Big( d_{n+q_n}^{k+1}\, I_n\big(k; (-1)^{q_n} \Pi_{q_n}^{n+1,n+1}(x_1)\big)\Big) < 0,$$

*on the fixed coordinates $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, …, $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and a rational part, with a positive coefficient of $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ when $k$ is even.*

A Beta port whose order grows in proportion to the depth pulls the word toward the port factor: the word decays at the maximum value $M_k(\eta)$ of $f_k \mathfrak{a}_k^{\eta}$, the binomial of the Beta port adds its entropy $(1+\eta)\log(1+\eta) - \eta \log \eta$, and the clearing grows to $d_{n+q_n}^{k+1}$. Where the sum is negative the cleared words tend to zero on a fixed set of coordinates, and for even $k$ the top coordinate keeps a positive coefficient. For every number of shapes the margin is positive at small rational ratios, by continuity from the base word (Proposition 37.8), each certified by an explicit algebraic root, and the cleared positive words decay exponentially on the coordinates $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)), \ldots, (\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ with a rational part.

# Part VII

# Rays

# Chapter 38

# Rays, native gauges and tiers

A ray moves the data of a family along an integral direction. Along it the values of every insertion obey two relations, integration by parts in every mark and the shift by the stride, wherever the integrals converge and, when the walls of the direction are positive, at every strength as identities of meromorphic functions; these relations make the values a holonomic sequence: they satisfy a recurrence of order at most the dimension of the ray. The complex points at which the direction is native, the native gauges of the ray, carry formal solutions of the relations and bound the dimension from below, and the frame of any recurrence is a power times a stacked Gamma. Every ray with real origin is one function, carried by a pencil, with a branch at the native gauge of its direction, and on the centre ray of a family every sign is a candidate native gauge; a ray along a wall carries the product of the rays of its two faces. What a value can contain is bounded: finitely many values span every value along a ray, and a closing in rational terms carries Trailmoments of index at most the largest order of its poles (Theorem 38.31, Lemma 38.32). A ray is a line of the data space of its family, and its module is the module of the family (Chapter 47) restricted to that line (Proposition 47.5).

## 38.1 The relations along a ray

**Definition 38.1** (Ray, stride, insertions, values)**.** Let $F$ be a family on $\mathcal{E}$. A *ray* of $F$ is the sequence of data $(\mathbf{m}_0 + k\mathbf{m}_1;\ \mathbf{s}_0 + k\mathbf{s}_1;\ \mathbf{t}_0 + k\mathbf{t}_1)$, $k \in \mathbb{N}$, the points at the integer times of the half-line from its *origin* $(\mathbf{m}_0; \mathbf{s}_0, \mathbf{t}_0)$, which is complex, along its *direction* $(\mathbf{m}_1; \mathbf{s}_1, \mathbf{t}_1)$, which is integral, $\mathbf{m}_1 \in \mathbb{Z}^{\mathcal{E}}$ and $s_{1j}, t_{1j} \in \mathbb{N}$, both on the balance wall; the walls along it are $\kappa_{0,S} + k\kappa_{1,S}$, with $\kappa_{0,S}$ and $\kappa_{1,S}$ the walls of the origin and of the direction. On the chart $y_{e_0} = 0$ of a fixed mark $e_0$, with $dy = \prod_{e \neq e_0} d\alpha_e/\alpha_e$, the *stride* and the *origin form* are

$$f = \alpha^{\mathbf{m}_1} \prod_j K_j^{-s_{1j}} \Psi_j^{-t_{1j}}, \qquad \omega_0 = \alpha^{\mathbf{m}_0} \prod_j K_j^{-s_{0j}} \Psi_j^{-t_{0j}}\, dy,$$

$f$ a rational function of degree 0. The *letter complement* of $F$ is $X_F = \{\alpha \in (\mathbb{C}^\times)^{\mathcal{E}}/\mathbb{C}^\times : K_j(\alpha)\Psi_j(\alpha) \neq 0 \text{ for every } j\}$, and the insertions along the ray are its regular functions, the algebra $\mathbb{C}[X_F]$, which is generated by the monomials $\alpha^v$, $v \in \mathbb{Z}^{\mathcal{E}}$ with $v_{\mathcal{E}} = 0$, and the chances of the bases under the basis and cobasis laws, $\alpha \mapsto \mathbb{P}_{j,\alpha}(T) = \theta_{j,T}\alpha^T/K_j(\alpha)$ and $\alpha \mapsto \mathbb{P}^*_{j,\alpha}(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j(\alpha)$, $T \in \mathcal{B}_j$. The *values* along the ray are $\langle g\rangle_k = \int_{y_{e_0}=0} g\, f^k \omega_0$ for $g \in \mathbb{C}[X_F]$, and over a compact real $(n-1)$-dimensional chain $D$ in $X_F$, $\langle g\rangle_{k;D} = \int_D g\, f^k \omega_0$. The *defects* of the

origin and of the direction are the masses less the usages,

$$c_e^{(i)} = m_{i,e} - \sum_j \big(s_{ij} R_{j,e} + t_{ij} R_{j,e}^*\big), \qquad i = 0, 1,$$

with the occupation $R_{j,e}$ and the dual occupation $R_{j,e}^*$ of table $j$ (Theorem 7.16).

**Lemma 38.2** (Insertions)**.** *(i) $\mathbb{C}[X_F]$ is closed under every $\partial_{y_e}$: $\partial_{y_e}\alpha^v = v_e\alpha^v$, $\partial_{y_e}\mathbb{P}_{j,\alpha}(T) = (\mathbf{1}_T(e) - R_{j,e})\,\mathbb{P}_{j,\alpha}(T)$ and $\partial_{y_e}\mathbb{P}^*_{j,\alpha}(T) = (\mathbf{1}_{\mathcal{E}\setminus T}(e) - R^*_{j,e})\,\mathbb{P}^*_{j,\alpha}(T)$; the occupations and the defects lie in $\mathbb{C}[X_F]$.*

*(ii) The stride lies in $\mathbb{C}[X_F]$, and $c_e^{(1)} = \partial_{y_e}\log f$, $c_e^{(0)} + k\,c_e^{(1)} = \partial_{y_e}\log(f^k\omega_0/dy)$.*

*(iii) Every value $\langle g\rangle_k$ is a finite combination, with products of weights as coefficients, of periods $P_F$ at data $(\mathbf{m}_0 + k\mathbf{m}_1 + v;\ \mathbf{s}_0 + k\mathbf{s}_1 + \mathbf{a};\ \mathbf{t}_0 + k\mathbf{t}_1 + \mathbf{b})$ on the balance wall, $v \in \mathbb{Z}^{\mathcal{E}}$ and $\mathbf{a}, \mathbf{b} \in \mathbb{N}^N$.*

**Theorem 38.3** (The relations along a ray)**.** *If $\kappa_{1,S} > 0$ for every nonempty proper $S$, then for every insertion $g$ the value $\langle g\rangle_k$ converges for all large $k$, and for every mark $e$*

$$\big\langle \big(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e}\big)g\big\rangle_k = 0, \qquad \langle fg\rangle_k = \langle g\rangle_{k+1}.$$

In Theorem 38.3 the first relation is integration by parts in the mark $e$ against the form $f^k\omega_0$, whose logarithmic derivative is the defect; the second is the definition of the stride. Proposition 38.5 carries both to every strength: wherever the integrals converge, whatever the walls of the direction, and, when these walls are positive, as identities of meromorphic functions of $k$. Both act on the insertions, which are closed under the derivatives, so the values of all insertions along a ray form one module over the operators in $k$, and every relation among insertions modulo these two is a recurrence among values.

## 38.2 Native gauges carry solutions

**Definition 38.4** (Native gauges of a ray; the dimension)**.** A *native gauge* of a ray is a point $\alpha_c$ of $X_F$ at which the direction is native, $c^{(1)}(\alpha_c) = 0$: a native gauge of the direction at a complex point (Definition 8.17); its *native value* is $f(\alpha_c)$, and it is *nondegenerate* when the Hessian of $\log f$ in $y$ modulo $\mathbb{C}\mathbf{1}$ is nonsingular there. Near a nondegenerate native gauge there are holomorphic coordinates $v \in \mathbb{C}^{n-1}$ with $\log f = \log f(\alpha_c) - \frac{1}{2}\sum_i v_i^2$; the image $D_c$ of a real ball $\{v \in \mathbb{R}^{n-1} : |v| \le \varepsilon\}$ is a *steepest-descent disc*. The *relation space* $\mathcal{L} \subseteq \mathbb{C}[X_F] \otimes \mathbb{C}(k)$ is the $\mathbb{C}(k)$-span of the elements $(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})g$, and the *dimension* of the ray is $\dim_{\mathbb{C}(k)}\big(\mathbb{C}[X_F] \otimes \mathbb{C}(k)\big)/\mathcal{L}$.

**Proposition 38.5** (The relations at every strength)**.** *Let a ray of a family $F$ be given. For an insertion $g = \alpha^u \prod_j K_j^{-a_j}\Psi_j^{-b_j}$ with $u \in \mathbb{Z}^{\mathcal{E}}$, $\mathbf{a}, \mathbf{b} \in \mathbb{N}^N$ and $u_{\mathcal{E}} = \sum_j (a_j r_j + b_j r_j^*)$, such insertions spanning $\mathbb{C}[X_F]$, write $\kappa_U(g) = u_U - \sum_j \big(a_j d_j(U) + b_j d_j^*(U)\big)$ for the walls of its data $(u; \mathbf{a}, \mathbf{b})$ (Definition 8.1).*

*(i) Whatever the walls of the direction, at every complex $k$ with $\operatorname{Re}\kappa_{0,U} + \kappa_U(g) + \operatorname{Re}(k)\,\kappa_{1,U} > 0$ for every nonempty proper $U$ the values $\langle g\rangle_k$ and $\big\langle (c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})g\big\rangle_k$ converge absolutely, and the second is $0$ for every mark $e$. So if polynomials $p_0, \dots, p_h \in \mathbb{C}[k]$, an insertion $g'$ and finite combinations $G_e$ of insertions of the above kind, with coefficients in $\mathbb{C}(k)$, satisfy*

$$\sum_{j=0}^{h} p_j(k)\, f^j g' = \sum_{e\in\mathcal{E}} \big(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e}\big)G_e,$$

*an explicit element of the relation space $\mathcal{L}$ (Definition 38.4), a* certificate *of the recurrence, then $\sum_j p_j(k)\langle g'\rangle_{k+j} = 0$ at every $k$ off the poles of those coefficients at which the values $\langle g'\rangle_{k+j}$ converge absolutely and the condition above holds for every insertion in the $G_e$.*

*(ii) If $\kappa_{1,S} > 0$ for every nonempty proper $S$, every value $\langle g\rangle_k$ continues to a meromorphic function of $k \in \mathbb{C}$, whose poles have order at most $n-1$ and lie among the $k$ with $\kappa_{0,U} + \kappa_U(g) + k\,\kappa_{1,U} \in \{0, -1, -2, \dots\}$ for some nonempty proper $U$. For every element $\sum_l r_l(k)\, g_l$ of $\mathcal{L}$ the continuation of $\sum_l r_l(k)\langle g_l\rangle_k$ vanishes identically, and $\langle fg\rangle_k = \langle g\rangle_{k+1}$ identically. So a recurrence $\sum_j p_j(k)\langle g\rangle_{k+j} = 0$ that follows from the relations, its element $\sum_j p_j(k) f^j g$ lying in $\mathcal{L}$, holds as an identity of meromorphic functions of $k$, and at every $k$ at which its terms converge.*

**Theorem 38.6** (A native gauge carries a solution)**.** *Let $\alpha_c$ be a nondegenerate native gauge of a ray and $D_c$ a steepest-descent disc through it.*

*(i) As $k \to \infty$,*

$$\langle g\rangle_{k;D_c} \sim f(\alpha_c)^k\, k^{-(n-1)/2} \sum_{i\geq 0} a_i(g)\, k^{-i},$$

*where $a_0(g) = (2\pi)^{(n-1)/2}(\omega_0/dv)(\alpha_c)\, g(\alpha_c)$ with $(\omega_0/dv)(\alpha_c) \neq 0$, and each $a_i(g)$ is a finite combination of Taylor coefficients of $g\,\omega_0/dv$ and of $\log f$ at $\alpha_c$. Further $\langle fg\rangle_{k;D_c} = \langle g\rangle_{k+1;D_c}$, and $\big\langle (c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})g\big\rangle_{k;D_c}$ is at most a constant times $|f(\alpha_c)|^k e^{-k\varepsilon^2/2}$; so the expansions satisfy both relations to all orders.*

*(ii) For a finite set $C$ of distinct nondegenerate native gauges $\alpha_c$, the maps $A^{(c)} : \mathbb{C}[X_F] \otimes \mathbb{C}(k) \to \mathbb{C}((1/k))$, $g \mapsto \sum_l a_l^{(c)}(g)\, k^{-l}$, extended $\mathbb{C}(k)$-linearly, vanish on $\mathcal{L}$ and are linearly independent over $\mathbb{C}((1/k))$.*

*(iii) The dimension of the ray is at least the number of its nondegenerate native gauges.*

**Corollary 38.7** (The symbol)**.** *Let $g$ be an insertion and $p_0, \dots, p_h \in \mathbb{C}[k]$, not all zero, with $\sum_j p_j(k)\, f^j g \in \mathcal{L}$. Under the hypothesis of Theorem 38.3, $\sum_j p_j(k)\, \langle g\rangle_{k+j} = 0$ for all large $k$. With $d$ the largest degree of the $p_j$ and $c_j$ the coefficient of $k^d$ in $p_j$, the* symbol $\sum_j c_j \lambda^j$ *vanishes at the native value of every nondegenerate native gauge $\alpha_c$ with $g(\alpha_c) \neq 0$. So $h$ is at least the number of distinct native values of such native gauges.*

**Proposition 38.8** (The top and the bottom of a certificate)**.** *Let $g$ be an insertion and $\sum_{j=0}^h p_j(k)\, f^j g = \sum_e \big(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}\big) G_e$ a certificate (Proposition 38.5(i)) in which the $p_j \in \mathbb{C}[k]$ have the largest degree $d \geq 1$ and every $G_e = \sum_{i<d} k^i\, G_e^{[i]}$ is a polynomial in $k$ of degree less than $d$ with coefficients $G_e^{[i]} \in \mathbb{C}[X_F]$. With $c_j$ the coefficient of $k^d$ in $p_j$ and the symbol $\chi(\lambda) = \sum_j c_j \lambda^j$ (Corollary 38.7),*

$$\chi(f)\, g = \sum_e c_e^{(1)}\, G_e^{[d-1]}, \qquad \sum_{j=0}^h p_j(0)\, f^j g = \sum_e \big(c_e^{(0)} + \partial_{y_e}\big) G_e^{[0]}.$$

*So $\chi$ vanishes at the native value of every native gauge $\alpha_c$ of the ray with $g(\alpha_c) \neq 0$, nondegenerate or not, and for the origin 0 the bottom piece $\sum_j p_j(0) f^j g$ is the divergence $\sum_e \partial_{y_e} G_e^{[0]}$.*

**Proposition 38.9** (The native value bounds the order)**.** *Let a ray of a family have $\kappa_{1,S} > 0$ for every nonempty proper $S$. Then $\langle 1\rangle_k \neq 0$ for all large $k$, positive when the origin is real, and $\langle 1\rangle_{k+1}/\langle 1\rangle_k \to f(\alpha^\star)$, the native value of the native gauge $\alpha^\star$ of the direction on $\mathbb{R}^{\mathcal{E}}_{>0}$, which is the*

*largest value of $f$ there. If $\sum_{j=0}^{h} p_j(k)\,\langle 1\rangle_{k+j} = 0$ for all large $k$, with $p_j \in \mathbb{Q}[k]$ not all zero, then the symbol vanishes at $f(\alpha^\star)$: $f(\alpha^\star)$ is algebraic, $h \geq [\mathbb{Q}(f(\alpha^\star)) : \mathbb{Q}]$, and every conjugate of $f(\alpha^\star)$ is a root of the symbol. In particular, when $f(\alpha^\star)$ is irrational the values $\langle 1\rangle_k$ satisfy no recurrence of order one over $\mathbb{Q}(k)$, and when it is transcendental none at all.*

**Proposition 38.10** (The conjugates of the native value of the direction)**.** *Let a ray of a family whose weights lie in a subfield $\Bbbk$ of $\mathbb{C}$ have $\kappa_{1,S} > 0$ for every nonempty proper $S$, and let $\alpha^\star$ be the native gauge of its direction on the positive orthant, $\alpha^\star_{e_0} = 1$. Then the coordinates of $\alpha^\star$ are algebraic over $\Bbbk$, and for every embedding $\tau$ of $\Bbbk(\alpha^\star)$ in $\mathbb{C}$ over $\Bbbk$ the point $\tau(\alpha^\star)$ is a nondegenerate native gauge of the ray, of native value $\tau\big(f(\alpha^\star)\big)$. So every conjugate of $f(\alpha^\star)$ over $\Bbbk$ is the native value of a nondegenerate native gauge, the dimension of the ray is at least $[\Bbbk(\alpha^\star) : \Bbbk]$, and the symbol of every recurrence for $\langle 1\rangle_k$ that follows from the relations vanishes at every conjugate of $f(\alpha^\star)$.*

**Proposition 38.11** (The frame of a recurrence is a power times a stacked Gamma)**.** *Let $\sum_{j=0}^{h} p_j(k)\,x_{k+j} = 0$ for $k \geq k_0$, with $p_j \in \mathbb{C}[k]$ and $p_0, p_h$ without zeros at the integers $k \geq k_0$, and let $x^{(1)}, \dots, x^{(h)}$ be solutions. Their* frame *$C_k = \det\big[x^{(i)}_{k+l}\big]_{1\leq i\leq h,\ 0\leq l<h}$ satisfies $C_{k+1} = (-1)^h \frac{p_0(k)}{p_h(k)}\,C_k$, and writing $\frac{p_0(k)}{p_h(k)} = c\,\frac{\prod_{i=1}^{\deg p_0}(k-a_i)}{\prod_{i=1}^{\deg p_h}(k-b_i)}$,*

$$C_k = C_{k_0}\big((-1)^h c\big)^{k-k_0}\,\Gamma_{\substack{\deg p_0\\ \deg p_h}}\begin{bmatrix} k-a_1,\ \dots \\ k-b_1,\ \dots \end{bmatrix}\Big/\Gamma_{\substack{\deg p_0\\ \deg p_h}}\begin{bmatrix} k_0-a_1,\ \dots \\ k_0-b_1,\ \dots \end{bmatrix}.$$

*The* tier *of a sequence over a field $\Bbbk \subseteq \mathbb{C}$ is the least order of a recurrence with coefficients in $\Bbbk[k]$ and $p_0 p_h \neq 0$ that it satisfies for large $k$; a sequence of tier one is a power times a stacked Gamma, and so is every frame.*

**Proposition 38.12** (The growth of the frame)**.** *In Proposition 38.11 let $C_{k_0} \neq 0$, let $d_0$ and $d_h$ be the degrees of $p_0$ and $p_h$, and let $c$ be the quotient of their leading coefficients. As $k \to \infty$,*

$$\log|C_k| = k\log|c| + (d_0 - d_h)(k\log k - k) + O(\log k).$$

*When $d_0 = d_h$ is the largest degree of the $p_j$, the symbol $\sum_j c_j\lambda^j$ (Corollary 38.7) has the degree $h$ and $c_0 \neq 0$, the product of its roots is $(-1)^h c$, and $|C_k|^{1/k} \to |c|$.*

A native gauge of a ray is a complex point at which the direction uses every mark exactly as its masses ask, the complex counterpart of the native gauge of a family. Through it a steepest-descent disc carries a value that satisfies both relations up to exponentially small terms, and its asymptotic expansion is a formal solution; distinct native gauges give independent solutions, so their number bounds the dimension, and their native values are roots of the symbol of every recurrence that follows from the relations. On the real orthant the native gauge of the direction governs the growth of the values, and an irrational native value forbids a recurrence of order one. Whatever the order, the frame of a recurrence is a power times a stacked Gamma: the sequences of tier one are the bottom of the hierarchy of sequences that the relations produce.

## 38.3 The centre ray of a family

**Definition 38.13** (The centre ray)**.** Let $F$ be a family of $N$ tables on $n \geq 2$ marks, and let strengths $\sigma_j \in \mathbb{N}$, not all zero, make it connected: every nonempty proper $S$ has $c_j(S) > 0$ for

some $j$ with $\sigma_j > 0$ (Definition 8.1); put $\boldsymbol{\sigma} = (\sigma_j)_j$ and $\sigma = \sum_j \sigma_j$. The *centre ray* of $F$ at the strengths $\boldsymbol{\sigma}$ is its ray with origin 0 and direction $(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$. Its stride is

$$f = \alpha^{\sigma\mathbf{1}} \prod_j \big(K_j \Psi_j\big)^{-\sigma_j} = \prod_j \big(K_j(e^{y}) K_j(e^{-y})\big)^{-\sigma_j},$$

its values are $\langle 1\rangle_k = \int_{y_{e_0}=0} f^k \, dy = P_F(k\sigma\mathbf{1}; k\boldsymbol{\sigma}, k\boldsymbol{\sigma})$, $k \geq 1$, and its direction walls are $\kappa_{1,S} = \sum_j \sigma_j c_j(S) > 0$ (Proposition 8.21).

**Theorem 38.14** (A ray is one function)**.** *Let a ray of a family on $n \geq 2$ marks have real origin and $\kappa_{1,S} > 0$ for every nonempty proper $S$, let $\alpha^\star$ be the native gauge of its direction on $\mathbb{R}^{\mathcal{E}}_{>0}$, with $\alpha^\star_{e_0} = 1$ (Theorem 8.19), and let $k_0 \geq 0$ be an integer with $\kappa_{0,S} + k_0 \kappa_{1,S} > 0$ for every nonempty proper $S$.*

*(i) $0 < f \leq f(\alpha^\star)$ on $\mathbb{R}^{\mathcal{E}}_{>0}$, with equality only at $\alpha \propto \alpha^\star$, and $f \leq C' e^{-c'\|y\|}$ on the chart for some $c', C' > 0$; every value $\langle 1\rangle_k$, $k \geq k_0$, converges and is positive.*

*(ii) For $|q| < f(\alpha^\star)^{-1}$ the generating function $G(q) = \sum_{k \geq k_0} \langle 1\rangle_k q^{k-k_0}$ is*

$$G(q) = \int_{y_{e_0}=0} \frac{f^{k_0}\omega_0}{1 - qf} = \int_{y_{e_0}=0} \frac{\alpha^{\mathbf{m}_0 + k_0\mathbf{m}_1} \prod_j K_j^{-s_{0j} - (k_0-1)s_{1j}} \Psi_j^{-t_{0j} - (k_0-1)t_{1j}}}{\prod_j K_j^{s_{1j}} \Psi_j^{t_{1j}} - q\,\alpha^{\mathbf{m}_1}} \, dy,$$

$$G^{(l)}(q) = l! \int_{y_{e_0}=0} \frac{f^{k_0+l}\omega_0}{(1-qf)^{l+1}},$$

*and the positive integral converges for every real $q < f(\alpha^\star)^{-1}$: the whole ray is carried by the pencil $\prod_j K_j^{s_{1j}} \Psi_j^{t_{1j}} - q\,\alpha^{\mathbf{m}_1}$.*

*(iii) For $k \geq k_0$ and $h \geq 1$,*

$$\det\big[\langle 1\rangle_{k+i+i'}\big]_{i,i'=0}^{h-1} = \frac{1}{h!} \int \prod_{a=1}^{h} f(y_a)^k \, \frac{\omega_0}{dy}(y_a) \prod_{a<b} \big(f(y_a) - f(y_b)\big)^2 \prod_a dy_a > 0,$$

*and $G$ is not a rational function of $q$.*

*On the centre ray of a family (Definition 38.13) one may take $k_0 = 1$, and $\alpha^\star = \mathbf{1}$.*

**Theorem 38.15** (The branch at the native gauge of the direction)**.** *For a ray as in Theorem 38.14, with its generating function $G$ and the native gauge $\alpha^\star$ of its direction, put $A = f(\alpha^\star)^{-1}$, $\varepsilon = 1 - q/A$ and $a = \frac{n-1}{2}$, and let $H$ be the Hessian of $-\log f$ at $\log \alpha^\star$ on the chart,*

$$H = \Big( \sum_j \big( s_{1j}\, \kappa_{j,\alpha^\star}(X_e, X_{e'}) + t_{1j}\, \kappa_{j,(\alpha^\star)^{-1}}(X_e, X_{e'}) \big) \Big)_{e,e' \neq e_0},$$

*the covariances of the indicators under the basis laws and the cobasis laws at $\alpha^\star$ (Definition 7.17, Theorem 7.16); it is positive definite. As $q \uparrow A$ there are functions $U, V$ holomorphic at $\varepsilon = 0$ with*

$$G(q) = U(\varepsilon) + \varepsilon^{a-1} V(\varepsilon) \quad (n \text{ even}), \qquad G(q) = U(\varepsilon) + \varepsilon^{a-1} \log\varepsilon \, V(\varepsilon) \quad (n \text{ odd}),$$

$$V(0) = \frac{\Gamma(1-a)\,(2\pi)^a\,(f^{k_0-1}\omega_0/dy)(\alpha^\star)}{A\sqrt{\det H}} \quad (n \text{ even}),$$

$$V(0) = \frac{(-1)^a (2\pi)^a\,(f^{k_0-1}\omega_0/dy)(\alpha^\star)}{\Gamma(a)\,A\sqrt{\det H}} \quad (n \text{ odd}).$$

*The derivatives $G^{(l)}(A)$ are finite exactly for $l < a-1$, and*

$$G(q) = \sum_{l<a-1} \frac{G^{(l)}(A)}{l!}\big(q-A\big)^l + V(0)\,\varepsilon^{a-1}\big(1 \text{ or } \log\varepsilon\big) + O\big(\varepsilon^{\lceil a-1\rceil}\big).$$

*On the centre ray of a family, with $k_0 = 1$, $A = \prod_j K_j(\mathbf{1})^{2\sigma_j}$, the density $f^{k_0-1}\omega_0/dy$ is 1, and $H = 2\sum_j \sigma_j\big(\kappa_{j,\mathbf{1}}(X_e, X_{e'})\big)_{e,e'\neq e_0}$, the covariances under the weight laws.*

**Proposition 38.16** (The native gauges of a centre ray)**.** *On the centre ray of a family at the strengths $\boldsymbol{\sigma}$ (Definition 38.13) the defects of the direction are*

$$c_e^{(1)}(\alpha) = \sum_j \sigma_j\big(R_{j,e}(\alpha^{-1}) - R_{j,e}(\alpha)\big),$$

*so a point $\alpha$ of $X_F$ is a native gauge of the ray exactly when the occupations weighted by the strengths agree at $\alpha$ and at $\alpha^{-1}$, $\sum_j \sigma_j R_j(\alpha) = \sum_j \sigma_j R_j(\alpha^{-1})$. Its native value is $\prod_j \big(K_j(\alpha)K_j(\alpha^{-1})\big)^{-\sigma_j}$.*

**Theorem 38.17** (Native gauges at the signs)**.** *For the centre ray of a family at the strengths $\boldsymbol{\sigma}$ (Definition 38.13), on the chart $\alpha_{e_0} = 1$, with the family word $W = \prod_j K_j^{-\sigma_j}\Psi_j^{-\sigma_j}$ at the exponents $(\boldsymbol{\sigma}, \boldsymbol{\sigma})$, so that $f = \alpha^{\sigma\mathbf{1}}W$:*

*(i) $f(\alpha^{-1}) = f(\alpha)$; the points fixed by $\alpha \mapsto \alpha^{-1}$ are the $2^{n-1}$ signs $\eta \in \{\pm1\}^{\mathcal{E}}$, $\eta_{e_0} = 1$; every native gauge of the centre ray is a sign or is paired with its image, a native gauge of the same native value.*

*(ii) A sign with $K_j(\eta) = \sum_T \theta_{j,T}\eta^T \neq 0$ for every $j$ is a native gauge, of native value $\prod_j K_j(\eta)^{-2\sigma_j}$. In the chart $\alpha_e = \eta_e e^{y_e}$, $y_{e_0} = 0$, the Hessian of $-\log f$ at $y = 0$ is $2\sum_j \sigma_j\big(\kappa_{j,\eta}(X_e, X_{e'})\big)_{e,e'\neq e_0}$, with $\kappa_{j,\eta}$ the covariances of the indicators under the signed measure $\mathbb{P}_{j,\eta}(T) = \theta_{j,T}\eta^T/K_j(\eta)$ of total mass one at the sign.*

*(iii) The critical polynomials $\sigma W^{-1} - \alpha_e\partial_e W^{-1}$, $e \neq e_0$, vanish identically modulo $\alpha_e^2 = 1$, and $W^{-1} \equiv \alpha^{\sigma\mathbf{1}}\prod_j K_j^{2\sigma_j}$ there. In $\mathbb{Q}(\boldsymbol{\theta})[\alpha_e : e \neq e_0]/(\alpha_e^2 - 1)$, $\mathbb{Q}(\boldsymbol{\theta})$ the field of the weights, localized at $\prod_j K_j$, the idempotents $2^{1-n}\prod_{e\neq e_0}(1+\eta_e\alpha_e)$ of the signs with $K_j(\eta) \neq 0$ for every $j$ form a basis, and multiplication by $W^{-1}/\alpha^{\sigma\mathbf{1}}$ acts on the idempotent of $\eta$ by $\prod_j K_j(\eta)^{2\sigma_j}$.*

**Theorem 38.18** (Weights that separate the signs)**.** *Let a family of $N$ tables on $n \geq 2$ marks be connected at the strengths $\sigma_j \in \mathbb{N}$, of total $\sigma$, with its centre ray (Definition 38.13). There are weights $\theta_{j,T} \in \big\{1, \dots, 2^{n-1}\big((2n-1)N + (2^{n-1}-1)\sigma\big) + 1\big\}$ for which every sign has $K_j(\eta) \neq 0$ for every $j$ and is a nondegenerate native gauge of the centre ray, and the $2^{n-1}$ native values $\prod_j K_j(\eta)^{-2\sigma_j}$ are distinct. For such weights the dimension of the centre ray is at least $2^{n-1}$, and every recurrence for $\langle 1\rangle_k$ that follows from the relations (Corollary 38.7) has order at least $2^{n-1}$.*

Along a ray with real origin the whole sequence of values is one function of the pencil parameter, holomorphic up to the reciprocal of the native value of the direction. The determinants $\det\big[\langle 1\rangle_{k+i+i'}\big]$ of the values are positive integrals of squared differences, so the generating function is not rational; at the native gauge of the direction it branches, with the exponent set by the number of marks and the coefficient by the value of $f^{k_0-1}\omega_0/dy$ there and the determinant of the covariances of the basis and cobasis laws. The centre ray of a family is its period at the centre read at every multiple of its strengths; its native gauge is the table's gauge, where it branches at

$\prod_j K_j(\mathbf{1})^{2\sigma_j}$ with the covariances of the weight laws. Every sign is a candidate native gauge, the signed measures $\mathbb{P}_{j,\eta}$ playing the part of the weight laws, and with weights chosen to separate the signs every recurrence that the relations produce for the centre ray of a family on $n$ marks has order at least $2^{n-1}$.

## 38.4 What a wall carries

**Lemma 38.19** (Where native gauges can reach a wall)**.** *Let $B$ be a proper bunch with room $R \ni e_0$, $\alpha_B = \rho u$, and in this statement write $K_j^{[l]} = K_{j,B}^{(d_j(B)+l)}(u, \alpha_R)$ and $\Psi_j^{[l]} = \Psi_{j,B}^{(d_j^*(B)+l)}(u, \alpha_R)$, so that $K_j = \rho^{d_j(B)} \sum_l \rho^l K_j^{[l]}$ and $\Psi_j = \rho^{d_j^*(B)} \sum_l \rho^l \Psi_j^{[l]}$ (Theorem 16.25), with the bottom letters $K_j^{[0]}$ and $\Psi_j^{[0]}$.*

*(i) At every native gauge $\alpha$ of a direction $(\mathbf{m}_1; \mathbf{s}_1, \mathbf{t}_1)$ with $s_{1j}, t_{1j} \geq 0$, $m_{1,B} = \sum_j \big(s_{1j} \sum_T |T \cap B|\, \mathbb{P}_{j,\alpha}(T) + t_{1j} \sum_T |B \setminus T|\, \mathbb{P}^*_{j,\alpha}(T)\big)$: the direction's mass on $B$ is its usage of $B$.*

*(ii) Let native gauges of such directions $\mathbf{v}_i \to \mathbf{v}$ have bunch scales $\rho_i \to 0$ and $(u_i, \alpha_{R,i}) \to (u^*, \alpha_R^*)$ in the torus, with $\kappa_{1,B}(\mathbf{v}) > 0$. Then some bottom letter vanishes at $(u^*, \alpha_R^*)$, and if only one does, its gradient in $(u, \alpha_R)$ vanishes there as well: the product $\prod_j K_j^{[0]} \Psi_j^{[0]}$ of the bottom letters vanishes at $(u^*, \alpha_R^*)$ together with its gradient in $(u, \alpha_R)$.*

*So the native gauges of a family leave for the wall of a bunch only at singular points of the zero set of the product of its bottom letters.*

**Theorem 38.20** (A ray along a wall carries its faces)**.** *Let $B$ be a proper bunch with room $R \ni e_0$ on whose bottom levels the weights factor, and let a ray have its direction on the wall, $\kappa_{1,B} = 0$, and $\operatorname{Re} \kappa_{0,B} > 0$. With $\alpha_B = \rho u$, and writing in this statement $K_j^{[l]} = K_{j,B}^{(d_j(B)+l)}(u, \alpha_R)$ and $\Psi_j^{[l]} = \Psi_{j,B}^{(d_j^*(B)+l)}(u, \alpha_R)$ as in Theorem 16.25,*

$$f = f_B(u)\, f_R(\alpha_R) \big(1 + \rho\, h_1 + O(\rho^2)\big), \qquad h_1 = -\sum_j \Big(s_{1j} \frac{K_j^{[1]}}{K_j^{[0]}} + t_{1j} \frac{\Psi_j^{[1]}}{\Psi_j^{[0]}}\Big),$$

*where $f_B$ and $f_R$ are the strides along the ray of the faces $F^B$ and $F^R$ of Theorem 16.25, and $\omega_0 = \rho^{\kappa_{0,B}-1} d\rho \wedge \omega_0^B \wedge \omega_0^R\, (1 + O(\rho))$ with the origin forms of the faces. Let $\alpha_c^B$ and $\alpha_c^R$ be nondegenerate native gauges of the rays of the faces with $h_1(\alpha_c^B, \alpha_c^R) \neq 0$, let $\operatorname{Re}\big(e^{i\vartheta} h_1\big) < 0$ near $(\alpha_c^B, \alpha_c^R)$, and let $D = e^{i\vartheta}[0, \rho_1] \times D_B \times D_R$ with steepest-descent discs $D_B, D_R$ of the faces. Then the values $\langle g \rangle_{k;D}$ satisfy both relations up to terms exponentially smaller than $|f_B(\alpha_c^B) f_R(\alpha_c^R)|^k$, and*

$$\langle 1 \rangle_{k;D} = \Gamma(\kappa_{0,B}) \big(-k\, h_1(\alpha_c^B, \alpha_c^R)\big)^{-\kappa_{0,B}} \langle 1 \rangle_{k;D_B}^{F^B} \langle 1 \rangle_{k;D_R}^{F^R} \big(1 + O(1/k)\big),$$

*with $\langle 1 \rangle_{k;D_B}^{F^B}$ and $\langle 1 \rangle_{k;D_R}^{F^R}$ the values of the rays of the faces over their discs (Theorem 38.6), and the power taken continuously along $e^{i\vartheta}(0, \infty)$.*

**Definition 38.21** (The whole cluster port)**.** On a bunch scale $\rho$ take letters with *lower coefficients* $\boldsymbol{\lambda}_i \in \mathbb{C}^{r_i}$,

$$L_i(\rho, \boldsymbol{\lambda}_i) = \sum_{l < r_i} \lambda_{i,l}\, \rho^l + \rho^{r_i} C_i(\rho), \qquad C_i(0) \neq 0,$$

with exponents $w_i$, and a word $\rho^{\kappa-1} G(\rho) \prod_i L_i^{-w_i}$, with $G$ and the $C_i$ holomorphic on a neighbourhood of $[0, \infty)$, each $|C_i|$ bounded below by a positive constant there, and the word integrable at

$\infty$; put $\kappa^\flat = \kappa - \sum_i r_i w_i$. Fix $b > 0$ with every $C_i$ zero-free on $|\rho| \le b$; for $\boldsymbol{\lambda}$ in a small polydisc each $L_i$ has exactly $r_i$ roots in $|\rho| < b$, the *cluster*, none on $|\rho| = b$ and none on $[b, \infty)$. The *cluster port* of the word is its integral over the *cluster contour*, which comes in along the half-line $[b, \infty)$ of the scale, goes once around $|\rho| = b$ and goes out on the new sheet:

$$\operatorname{Port}_\kappa \Big[ G \prod_i L_i^{-w_i} \Big](\boldsymbol{\lambda}) = \frac{1}{2\pi i} \Big[ (e^{2\pi i \kappa^\flat} - 1) \int_b^\infty \rho^{\kappa-1} G \prod_i L_i^{-w_i}\, d\rho + \oint_{|\rho|=b} \rho^{\kappa-1} G \prod_i L_i^{-w_i}\, d\rho \Big].$$

**Theorem 38.22** (Relations of the whole cluster port)**.** *(i) One turn around $|\rho| = b$ multiplies the word by $e^{2\pi i \kappa^\flat}$; the cluster port does not depend on $b$, is holomorphic in all the lower coefficients jointly on the polydisc, and the contour has no end at which the word is nonzero, so every integral of a $\rho$-derivative over it vanishes.*

*(ii) On the stratum $\boldsymbol{\lambda} = 0$ the word is $\rho^{\kappa^\flat - 1} h$, with the* stratum word $h = G \prod_i C_i^{-w_i}$*, and*

$$\operatorname{Port}_\kappa \Big[ G \prod_i L_i^{-w_i} \Big](0) = \frac{e^{2\pi i \kappa^\flat} - 1}{2\pi i}\, \operatorname{Reg} \int_0^\infty \rho^{\kappa^\flat - 1} h\, d\rho \quad (\kappa^\flat \notin \mathbb{Z}),$$

$$\operatorname{Port}_\kappa \Big[ G \prod_i L_i^{-w_i} \Big](0) = [\rho^{-\kappa^\flat}]\, h \quad (\kappa^\flat \in \mathbb{Z}_{\le 0}), \qquad \operatorname{Port}_\kappa \Big[ G \prod_i L_i^{-w_i} \Big](0) = 0 \quad (\kappa^\flat \in \mathbb{Z}_{>0}).$$

*(iii) For multi-indices $\mathbf{a}_i$, with $A_i = |\mathbf{a}_i|$ and $D = \sum_i \sum_l (r_i - l)\, a_{i,l}$, the derivative $\partial_{\boldsymbol{\lambda}}^{\mathbf{a}}$ of the cluster port at $\boldsymbol{\lambda} = 0$ is $\prod_i (-1)^{A_i} (w_i)_{A_i}$ times the value of (ii) for the stratum word $G \prod_i C_i^{-w_i - A_i}$ at the index $\kappa^\flat - D$; at $\kappa^\flat = 0$ it is $\prod_i (-1)^{A_i} (w_i)_{A_i} \cdot \frac{1}{D!}\, \partial_\rho^D \big[ G \prod_i C_i^{-w_i - A_i} \big]_{\rho = 0}$.*

**Proposition 38.23** (Small lower coefficients on one parameter)**.** *Let $C_1, \dots, C_p$ be bounded below by a positive constant on $[0, \infty)$, $G$ and the $C_i$ holomorphic near $0$ and continuous on $[0, \infty)$, $\xi_1, \dots, \xi_p \in \mathbb{C} \setminus (-\infty, 0]$, $\operatorname{Re} \kappa > 0$ and $\kappa^\flat = \kappa - |\mathbf{w}| \notin \mathbb{Z}$, with $\rho^{\kappa-1} G \prod_i (\rho C_i)^{-w_i}$ integrable at $\infty$ together with every $\rho^{\kappa^\flat - |\mathbf{a}| - 1} G \prod_i C_i^{-w_i - a_i}$, $\mathbf{a} \in \mathbb{N}^p$.*

*(i) For small $\lambda > 0$,*

$$\int_0^\infty \rho^{\kappa-1} G(\rho) \prod_i \big( \lambda \xi_i + \rho\, C_i(\rho) \big)^{-w_i} d\rho = A(\lambda) + \lambda^{\kappa^\flat} B(\lambda)$$

*with $A, B$ holomorphic at $0$,*

$$A(\lambda) = \sum_{\mathbf{a} \in \mathbb{N}^p} \prod_i \binom{-w_i}{a_i} (\lambda \xi_i)^{a_i}\, \operatorname{Reg} \int_0^\infty \rho^{\kappa^\flat - |\mathbf{a}| - 1} G \prod_i C_i^{-w_i - a_i}\, d\rho,$$

$$B(0) = G(0) \int_0^\infty u^{\kappa-1} \prod_i \big( \xi_i + u\, C_i(0) \big)^{-w_i}\, du,$$

*the last integral continued in $\kappa$ and $\mathbf{w}$; where every $\operatorname{Re} w_i > 0$ it is*

$$\prod_i C_i(0)^{-w_i}\, \Gamma_{\!\!\stackrel{2}{p}} \begin{bmatrix} \kappa, & |\mathbf{w}| - \kappa \\ & \mathbf{w} \end{bmatrix} \mathsf{B}^{\{J\}}_{\mathbf{w}, |\mathbf{w}| - \kappa;\, I_J} \big( (\xi_i / C_i(0))_i \big), \qquad J = \{1, \dots, p\}$$

*(Proposition 15.1). The analytic part opens with the finite part of the stratum word along the whole scale, and the branch part with $G(0)$ times the scale integral of the letters linearized at the wall.*

*(ii) When $G$ and the $C_i$ are holomorphic on a neighbourhood of $[0,\infty)$, the cluster port of Definition 38.21 for these letters, $r_i = 1$ with the lower coefficients $\lambda\xi_i$, is $\frac{e^{2\pi i\kappa^\flat}-1}{2\pi i}A(\lambda)$.*

At a wall the ray splits: the stride is the product of the strides of the two faces up to a first-order correction in the scale of the bunch, and integrating that scale against the correction contributes one Gamma of the origin wall, so a ray along a wall carries the product of the native values of its faces. The native gauges of a family reach a wall only where the product of its bottom letters is singular. When the bottom letters of a bunch are perturbed by lower coefficients, the scale integral is read on the cluster contour, which surrounds the whole cluster of roots at once: the cluster port is holomorphic in the lower coefficients, its value on the stratum where they vanish is the finite part of the stratum word, and its derivatives there are the finite parts of the stratum words with raised exponents. With small lower coefficients on one parameter the scale integral splits into an analytic part, the cluster port up to a constant, and a branch part led by the scale integral of the letters linearized at the wall, a stacked Gamma times a Breta of one bunch.

## 38.5 Corners of a wall of width one

Along a wall the stride is infinite except where the bottom letters vanish, and there its limit is indeterminate. At a bunch of width one of a plain table the stride along the centre ray at the strength one has an exact three-term form in the scale of the bunch, and its limits at the wall are read from it: they sit where both factors of one bottom letter vanish, the two corners of the wall, or where one factor is singular, and their values are products of native values of the four minors of the bunch, each read on the zero set of another. At every flag the points where native gauges of nearby directions can leave the torus lie over the singular locus of the step letters.

**Definition 38.24** (Native gauges on a hypersurface; corner values)**.** For a plain table $(X,\theta)$ on a set $Z$ of marks, $|Z|\geq 2$, put $f_X(u) = \big(K_X(u)K_X(u^{-1})\big)^{-1}$ on the torus $T_Z = (\mathbb{C}^\times)^Z/\mathbb{C}^\times$: the stride of the centre ray of $X$ at the strength one when $X$ is connected (Definition 38.13), with $f_X(u^{-1}) = f_X(u)$. For a second plain table $Y$ on $Z$, a *native gauge of $X$ on $Y$* is a smooth point $u$ of the hypersurface $Z_Y = \{K_Y = 0\}$ of $T_Z$ at which $f_X$ is finite and nonzero and the restriction of $\log f_X$ to $Z_Y$ has zero differential; it is *nondegenerate* when the Hessian of that restriction is nonsingular there. $N(X\,|\,Y)$ is the set of the values of $f_X$ at the native gauges of $X$ on $Y$; it does not change when $Z_Y$ is replaced by its image $\{K_Y(u^{-1}) = 0\}$ under the Cremona. For a bunch $B$ of width one of a plain table $(\mathcal{B},\theta)$, its weights factoring across $B$ and its room $R$ (Definition 19.2), the *corner values* of $B$ are the products

$$N(\mathcal{B}|B\,|\,\mathcal{B}/R)\cdot N(\mathcal{B}/B\,|\,\mathcal{B}|R) \qquad \text{and} \qquad N(\mathcal{B}/R\,|\,\mathcal{B}|B)\cdot N(\mathcal{B}|R\,|\,\mathcal{B}/B),$$

taken element by element, the first set at the *corner of $K$* and the second at the *corner of $\Psi$*, the four minors carrying the weights they inherit.

**Proposition 38.25** (The stride at a wall of width one)**.** *Let $B$ be a bunch of width one of a plain table $(\mathcal{B},\theta)$, its weights factoring across $B$ and its room $R\ni e_0$, and read the centre ray of the table at the strength one, $f^{-1} = K(\alpha)K(\alpha^{-1})$. With $\alpha_B = \rho u$ put*

$$P = K_{\mathcal{B}/R}(u)\,K_{\mathcal{B}|R}(\alpha_R), \qquad Q = K_{\mathcal{B}|B}(u)\,K_{\mathcal{B}/B}(\alpha_R),$$

*and let $\bar P$, $\bar Q$ be the same at $(u^{-1},\alpha_R^{-1})$. Then, exactly,*

$$f^{-1} = \rho^{-1}\,P\bar Q + \Big(f^{-1}_{\mathcal{B}/R}(u)\,f^{-1}_{\mathcal{B}|R}(\alpha_R) + f^{-1}_{\mathcal{B}|B}(u)\,f^{-1}_{\mathcal{B}/B}(\alpha_R)\Big) + \rho\,Q\bar P,$$

*with the strides $f_X$ of Definition 38.24. $P$ is the bottom letter $K^{[0]}$ of $K$ at the wall of $B$ (Lemma 38.19), and $\bar{Q} = u^{-B}\alpha_R^{-R}\,\Psi^{[0]}$ with the bottom letter $\Psi^{[0]} = \Psi_{\mathcal{B}|B}(u)\Psi_{\mathcal{B}/B}(\alpha_R)$ of $\Psi$.*

Along the wall $\rho = 0$ the reciprocal stride is infinite, except on the zero set of $P\bar{Q}$, where it has the form $0 \cdot \infty$. So the native gauges of directions near the centre direction that leave for the wall of $B$ arrive at points of that zero set, and the next statement says which.

**Proposition 38.26** (Corners of a wall of width one)**.** *In Proposition 38.25 write $f^{-1} = \rho^{-1}Q_0 + Q_1 + \rho Q_2$, with $Q_0 = P\bar{Q}$, $Q_1$ the middle term and $Q_2 = Q\bar{P}$, functions of $(u, \alpha_R)$. Let $\alpha_i \in X_F$ be points with the defects $c^{(1)}(\alpha_i) \to 0$ (Definition 38.1), bunch scales $\rho_i \to 0$, positions $(u_i, \alpha_{R,i}) \to (u^*, \alpha_R^*)$ in $T_B \times T_R$, $T_R$ read on the chart $\alpha_{e_0} = 1$, and $f(\alpha_i) \to \lambda \in \mathbb{C}^\times$. Each $\alpha_i$ is a native gauge of the direction whose masses are those of the centre direction moved by $-c^{(1)}(\alpha_i)$. Then $\lambda^{-1} = Q_1(u^*, \alpha_R^*)$, some factor of $P\bar{Q}$ vanishes at $(u^*, \alpha_R^*)$, and:*

*(i) a factor of $P$ and a factor of $\bar{Q}$ do not vanish there together; if exactly one factor vanishes there, its zero set is singular there;*

*(ii) if both factors of $P$ vanish there and both are smooth there (the corner of $K$), then $u^*$ is a native gauge of $\mathcal{B}|B$ on $\mathcal{B}/R$, $\alpha_R^*$ is a native gauge of $\mathcal{B}/B$ on $\mathcal{B}|R$, and $\lambda = f_{\mathcal{B}|B}(u^*)\, f_{\mathcal{B}/B}(\alpha_R^*)$; if both factors of $\bar{Q}$ vanish there and both are smooth there (the corner of $\Psi$), then $u^*$ is a native gauge of $\mathcal{B}/R$ on $\mathcal{B}|B$ and $\alpha_R^*$ one of $\mathcal{B}|R$ on $\mathcal{B}/B$, both on the zero sets at the inverse point, and $\lambda = f_{\mathcal{B}/R}(u^*)\, f_{\mathcal{B}|R}(\alpha_R^*)$;*

*(iii) if exactly one factor vanishes there and it is a factor of $P$, then $\alpha_R^*$ is a critical point of $f_{\mathcal{B}/B}$ on $T_R$ when the factor is $K_{\mathcal{B}/R}(u)$, and $u^*$ one of $f_{\mathcal{B}|B}$ on $T_B$ when it is $K_{\mathcal{B}|R}(\alpha_R)$, and $\lambda = f_{\mathcal{B}|B}(u^*)\, f_{\mathcal{B}/B}(\alpha_R^*)$; if it is a factor of $\bar{Q}$, then $\alpha_R^*$ is a critical point of $f_{\mathcal{B}|R}$ on $T_R$ when the factor is $K_{\mathcal{B}|B}(u^{-1})$, and $u^*$ one of $f_{\mathcal{B}/R}$ on $T_B$ when it is $K_{\mathcal{B}/B}(\alpha_R^{-1})$, and $\lambda = f_{\mathcal{B}/R}(u^*)\, f_{\mathcal{B}|R}(\alpha_R^*)$.*

*Conversely, let $u^*$ and $\alpha_R^*$ be nondegenerate native gauges of $\mathcal{B}|B$ on $\mathcal{B}/R$ and of $\mathcal{B}/B$ on $\mathcal{B}|R$, put $G = \bar{Q}(u^*, \alpha_R^*)$, and let $\mu_a, \mu_c$ be the multipliers of the two constrained problems, $d_u\big(f_{\mathcal{B}|B}^{-1}\big)\, f_{\mathcal{B}/B}^{-1} = \mu_a\, dK_{\mathcal{B}/R}$ and $f_{\mathcal{B}|B}^{-1}\, d_{\alpha_R}\big(f_{\mathcal{B}/B}^{-1}\big) = \mu_c\, dK_{\mathcal{B}|R}$ at $(u^*, \alpha_R^*)$. Let $\mathbf{u} \in \mathbb{R}^{\mathcal{E}}$ with $u_{\mathcal{E}} = 0$ and $u_B = \sum_{e \in B} u_e \neq 0$, and suppose $Q_2 \neq \mu_a\mu_c/G$ at the point. Then for every small real $\varepsilon \neq 0$ the direction whose masses are those of the centre direction plus $\varepsilon\mathbf{u}$ has a native gauge with the bunch scale $\rho = \varepsilon r + O(\varepsilon^2)$, $r = u_B Q_1/(Q_2 - \mu_a\mu_c/G)$ at the point, whose position tends to $(u^*, \alpha_R^*)$ and whose native value tends to $f_{\mathcal{B}|B}(u^*) f_{\mathcal{B}/B}(\alpha_R^*)$ as $\varepsilon \to 0$. The same holds at the corner of $\Psi$, with the roles of the two pairs of minors exchanged.*

*Remark* 38.27 (A corner is a wall after one blow-up). At the corner of $K$, with $a = K_{\mathcal{B}/R}(u)$ and $c = K_{\mathcal{B}|R}(\alpha_R)$, the chart $a = \rho\hat{a}$, $c = \rho\hat{c}$ of the blow-up of $\{\rho = a = c = 0\}$ gives $f^{-1} = Q\bar{Q} + O(\rho)$ by Proposition 38.25, since $P = \rho^2\hat{a}\hat{c}$. So on the exceptional divisor the stride is $f_{\mathcal{B}|B}(u)\, f_{\mathcal{B}/B}(\alpha_R)$, the product of the strides of two faces restricted to the zero sets of the other minors' letters, as along a wall in Theorem 38.20, and the corner values are the native values of this stratum. In the coordinates $(\rho, u, \alpha_R)$ the zero set of $K$ off the wall is $\{P + \rho Q = 0\}$, whose closure has the trace $\{P = 0\}$ on the wall $\{\rho = 0\}$; the corner is the singular set $\{a = c = 0\}$ of that trace.

**Corollary 38.28** (The corners of a class)**.** *Let $\mathcal{H}$ be a plain table on $R \sqcup \{h\}$, $h$ neither a loop nor a coloop, and let $\mathcal{B} = \mathcal{H}\langle\mathcal{A} \text{ at } h\rangle$ be its gluing along $h$ with a parallel class $\mathcal{A} = \binom{C \sqcup \{g\}}{1}$ or a series class $\mathcal{A} = \binom{C \sqcup \{g\}}{|C|}$ of $s = |C| \geq 2$ marks (Definition 21.1), with unit weights on the class. Then $C$ is a bunch of width one of $\mathcal{B}$, and its corner values are*

$$N(\mathcal{H} \setminus h \,|\, \mathcal{H}/h) \quad \textit{(parallel class)}, \qquad N(\mathcal{H}/h \,|\, \mathcal{H} \setminus h) \quad \textit{(series class)},$$

*the parallel class at the corner of $\Psi$ and the series class at the corner of $K$; the other corner of each is empty. A single mark has no corner.*

At the corner of $\Psi$ a parallel class reads its core with the class silent: $f_{\mathcal{B}/R} \equiv 1$ on the zero set of $\sum_{e\in C} u_e^{-1}$, and the room is read through its deletion $\mathcal{H} \setminus h$ on the zero set of the letter of its contraction. The corners of a class are degenerate, since the factor of the class is constant along the zero set of its own letter, so the converse in Proposition 38.26 does not apply to them.

**Definition 38.29** (Asymptotic native gauges)**.** An *asymptotic native gauge* of a ray is a sequence $x_m \in X_F$ that leaves every compact subset of the torus $(\mathbb{C}^\times)^{\mathcal{E}}/\mathbb{C}^\times$, with $c^{(1)}(x_m) \to 0$ and $f(x_m) \to \lambda \in \mathbb{C}^\times$; $\lambda$ is an *asymptotic native value.* Each $x_m$ is a native gauge of the direction whose masses are moved by $-c^{(1)}(x_m)$, a complex perturbation that tends to 0.

**Proposition 38.30** (The far points lie over the singular locus of the step letters)**.** *Let a ray of a family of lists have an integral origin, $\kappa_{1,S} > 0$ for every nonempty proper $S$, and every letter of every list with a positive exponent in its direction. For a flag $\mathfrak{f}$ with the steps $Z_1, \dots, Z_{\ell+1}$, $\ell \geq 1$ (Definition 34.2), let $O_\mathfrak{f}$ be the set of the points of $D_{S_1} \cap \cdots \cap D_{S_\ell}$ in $X_\mathcal{E}$ on no other $D_S$ (Definition 34.1), isomorphic to $\prod_i T_{Z_i}$, and put*

$$\Pi_\mathfrak{f} = \prod_{g\in \mathrm{Let}(F)} \prod_{i=1}^{\ell+1} g_{Z_i},$$

*with $g_{Z_i}$ the step letter of $g$ at $Z_i$ (Proposition 12.20) read on $T_{Z_i}$, one factor for each letter and each step. In the chart of an order adapted to the flag write $f^{-1} = \prod_{i<n} \varepsilon_i^{-\kappa_{1,S_i}} \Phi$, $\Phi = \prod_j \widetilde{K}_j^{s_{1j}} \widetilde{\Psi}_j^{t_{1j}}$ (Lemma 34.3). Let $x_m$ be an asymptotic native gauge of value $\lambda$. After passing to a subsequence, $x_m$ converges in $X_\mathcal{E}$ to a point $x^* \in O_\mathfrak{f}$ for a flag $\mathfrak{f}$ of length $\ell \geq 1$, and:*

- *(i) $\Pi_\mathfrak{f}(x^*) = 0$: two of its factors vanish at $x^*$, or exactly one does and its zero set in $O_\mathfrak{f}$ is singular at $x^*$;*
- *(ii) in a chart through $x^*$, $\varepsilon_i \partial_{\varepsilon_i} \log \Phi(x_m) \to \kappa_{1,S_i}$ for every $i < n$, so that $\Phi$ vanishes along $x_m$ at the rates that the walls of the flag set.*

Two factors that vanish together may be the step letters of one letter at two steps, as at the corners of a wall of width one, or factors of two letters. For a flag $\{S\}$ of length one this is Lemma 38.19(ii) read for complex perturbations, at every width: the bases $T$ with $T \cap R$ a basis of $\mathcal{B}|R$ are the $T_R \sqcup T_B$ with $T_B$ a basis of $\mathcal{B}/R$, so the bottom letter of $K$ at the wall of $B$ is $K_{\mathcal{B}/R}(u)\, K_{\mathcal{B}|R}(\alpha_R)$ whatever the width, and width one adds the exact three terms of Proposition 38.25, which decide the value at a corner.

## 38.6 What a value can contain

**Theorem 38.31** (The values along a ray)**.** *Let a ray of a family have weights and origin in a subfield $\Bbbk$ of $\mathbb{C}$, with $\kappa_{1,S} > 0$ for every nonempty proper $S$.*

- *(i) The dimension of the ray is finite: it is the dimension over $\mathbb{C}(k)$ of the top cohomology of the twisted algebraic de Rham complex of the letter complement $X_F$, which is finite. It is at least the number of nondegenerate native gauges of the ray (Theorem 38.6). The native gauges are the critical points of the stride $f$, a product of powers of the letters with the exponents of the direction; for a direction outside a countable union of proper subvarieties they are $(-1)^{n-1}\,\mathrm{eu}(X_F)$ in number,* eu *the Euler characteristic, and all nondegenerate, so that the*

*dimension is then at least* $(-1)^{n-1}\operatorname{eu}(X_F)$*. An integral direction can lie in that exceptional set.*

*(ii) For every insertion* $g$ *with coefficients in* $\Bbbk$ *there are* $p_0,\dots,p_h\in\Bbbk[k]$*,* $p_h\neq 0$*,* $h$ *at most the dimension, with* $\sum_j p_j(k)\,\langle g\rangle_{k+j}=0$ *for all large* $k$*.*

*(iii) So there is* $k_1$ *such that every value* $\langle g\rangle_k$*,* $k\geq k_1$*, is a* $\Bbbk$*-linear combination of the* $h$ *values* $\langle g\rangle_{k_1},\dots,\langle g\rangle_{k_1+h-1}$*: the values along a ray span a* $\Bbbk$*-space of dimension at most that of the ray, and whatever Trailmoments those* $h$ *values contain, no later value contains others.*

**Lemma 38.32** (Index by pole count)**.** *Let* $\Bbbk$ *be a subfield of* $\mathbb{C}$*, and let* $r\in\Bbbk(k)$ *have all its poles at negative integers, with largest order* $p$*, and* $r(k)=O(k^{-2})$*. Then* $\sum_{k\geq 0} r(k)\in\Bbbk+\sum_{j=2}^{p}\Bbbk\,(\mathsf{T}_{j;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$*. If moreover* $r(-M-k)=-r(k)$ *for an integer* $M$*, only odd* $j$ *occur. If the poles lie in one coset* $-b+\mathbb{Z}$*,* $b\in\mathbb{Q}$*, the same holds with the Trailmoments of the lattice input* $\mathbb{I}_{b+\mathbb{Z}}$ *at its cuts; poles in several cosets leave, besides, differences of the index-one values of the cosets.*

A closing whose terms are rational functions of the lattice index brings Trailmoments of the counting input of index at most the largest order of its poles, and a reflection of the terms keeps only the odd indices. With Theorem 38.31 this is what bounds the complexity of a family along a ray: the space of values is finite-dimensional, spanned by finitely many of them, and where a closing writes the values as sums of rational terms the index is read off the orders of the poles, which along the ray can be bounded once for all its points.

# Chapter 39

# The two ends of a ray

The values of the insertions along a ray form one module over the rational functions of the strength $k$, and masters write the step from one strength to the next as one rational matrix, the one-step matrix. Finitely many relations of the period write the step on the Laurent part of the module, which is all of it when the direction uses every letter: on a box of insertions bounded by the powers of the letters in their denominators and by their growth against the reciprocal stride, one elimination over the rational functions of the strength gives a spanning set of that part, the matrix of the step on it and a certificate, and a search through the boxes stops. A family of functionals that respects the two relations of the ray is a solution of the module, and the one-step matrix is fixed by its solutions: by a full set of them outright, and by fewer exactly when they annihilate no submodule. The solutions come from the two ends of strength. At the high end the sources of the stride, components of its critical locus, carry formal solutions: one for a nondegenerate native gauge, one for each class of paths at a native gauge of corank one, one for every cycle on a clean family of native gauges. At the low end the values continue to the nonpositive integers, where their top principal parts are the constant terms of the powers of the reciprocal stride: the closings, one exact solution read on the compact torus. The symmetries of the ray, built from the automorphisms and the exchanges of the family and from the Cremona, act on the module; the values live on its invariant part, every orbit of nondegenerate native gauges contributes to it, and a degenerate native gauge only through the classes on which its stabilizer acts trivially. At a lean of a family with an exchange the exchange survives only composed with the Cremona: the integral keeps a symmetry that carries the basis half of a class of the centre onto its cobasis half, while the reading, which is oriented, separates the halves. Over the formal series at the high end the module splits by native values into pieces, each mapped to itself by the symmetries, and the closings glue the pieces back into the rational matrix. The closings, the values and the expansion at the native gauge of the direction vanish on the same elements, the positive orthant and the torus through that gauge meeting there and nowhere else: the closings determine the one-step matrix exactly where the values do, which is where that gauge sees every submodule, and a submodule that it does not see is invisible to the integral itself. Such a submodule can come from a table one mark larger: when the list is the contraction of an identically self-dual table, the integral over the contracted scale makes the automorphisms of the larger table act on the module, and the closings see only what they fix; each orbit of nondegenerate native gauges of the larger table that the symmetries of the ray split into $s$ orbits adds $s-1$ to the dimension of what the closings miss, and at the centre of every $\binom{[2r+1]}{r}$, $r \geq 2$, the signs of $\binom{[2r+2]}{r+1}$ with two scales $-1$ split in two, with every symmetry of the ray of the four kinds in the group. On a wheel the letters are traces of one holonomy in $SL_2$, and the sources of its centre ray are read off the critical locus of the pair of traces and the

stratum where both holonomies are $-I$. The relations of a ray are exact forms on its model, so the ray is read in one cohomology, and at a tame high end its insertion lattice is free over the local ring at infinite strength, every eigenvalue of the step there being a reciprocal native value. The pencil of the ray, the direct image of the functions along the reciprocal stride, is its moment side: the nonzero roots of the symbol of the law are the reciprocals of its singular values, with the vanishing ranks as multiplicities, each a native value or an asymptotic native value, the far end has the maximal depth, and along a hinge tree the pencil is the iterated direct image of the marks under the node maps, one for each vertex.

## 39.1 The module of a ray

**Definition 39.1** (The module of a ray; masters; the one-step matrix)**.** Let a ray of a family $F$ have the stride $f$, the origin form $\omega_0$ and the relation space $\mathcal{L}$ (Definitions 38.1 and 38.4). The *module* of the ray is the $\mathbb{C}(k)$-vector space

$$\mathfrak{M} = \big(\mathbb{C}[X_F] \otimes \mathbb{C}(k)\big)/\mathcal{L},$$

whose dimension is the dimension of the ray, finite when $\kappa_{1,S} > 0$ for every nonempty proper $S$ (Theorem 38.31(i)); $[g]$ is the class of $g \otimes 1$. The *shift* is $\mathrm{Sh}[g \otimes r(k)] = [fg \otimes r(k+1)]$; it is well defined and bijective, and $\mathrm{Sh}(r(k)\,m) = r(k+1)\,\mathrm{Sh}(m)$. A *submodule* is a $\mathbb{C}(k)$-subspace $\mathfrak{N}$ with $\mathrm{Sh}(\mathfrak{N}) = \mathfrak{N}$, and $\mathfrak{N} \neq 0$ is *irreducible* when its only submodules are 0 and $\mathfrak{N}$. *Masters* of a finite-dimensional submodule $\mathfrak{N}$ are insertions $b_1, \dots, b_{\mathfrak{m}}$ whose classes form a basis of $\mathfrak{N}$, $\mathfrak{m} = \dim_{\mathbb{C}(k)} \mathfrak{N}$, and their *one-step matrix* $\mathbf{B_b} \in GL_{\mathfrak{m}}(\mathbb{C}(k))$ is the matrix of $\mathrm{Sh}^{-1}$ on them,

$$[f^{-1} b_i] = \sum_{l=1}^{\mathfrak{m}} \mathbf{B}_{\mathbf{b},il}(k)\,[b_l], \qquad 1 \le i \le \mathfrak{m}.$$

The *cyclic submodule* of $[1]$ is the span of the $\mathrm{Sh}^i[1]$, $i \in \mathbb{Z}$. When it is finite-dimensional, of dimension $h$, $[1], \mathrm{Sh}[1], \dots, \mathrm{Sh}^{h-1}[1]$ is a basis of it, and the *law* of $[1]$ is the relation

$$\sum_{i=0}^{h} p_i(k)\,\mathrm{Sh}^i[1] = 0, \qquad p_i \in \mathbb{C}[k] \text{ without a common factor,}$$

unique up to a constant factor; $p_0 p_h \neq 0$.

The law of $[1]$ is the shortest relation among the strides of 1 that the relations of the ray produce, so by Corollary 38.7 the values obey it, $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ for all large $k$; its order $h$ is at most the dimension of the ray.

**Definition 39.2** (Solutions)**.** The *spaces of germs* are: the sequences on the integers $k \ge k_0$, for some $k_0$, two of them identified when they agree for all large $k$, the *germs at* $+\infty$; the same on the integers $k \le k_0$, the *germs at* $-\infty$; and, for $\lambda \in \mathbb{C}^\times$ and $\rho \in \mathbb{C}$, the *formal series* $\lambda^k k^\rho\,\mathbb{C}((1/k))$. Each carries the multiplication by $\mathbb{C}(k)$ and the shift $\varsigma$: $(\varsigma x)(k) = x(k+1)$ on germs, and $\varsigma\big(\lambda^k k^\rho \varphi(k)\big) = \lambda^k k^\rho \cdot \lambda\big(1+\frac{1}{k}\big)^\rho \varphi(k+1)$ on formal series, re-expanded in powers of $1/k$; in each case $\varsigma(rx) = r(k+1)\,\varsigma(x)$ for $r \in \mathbb{C}(k)$. A *solution* of the ray with values in a space of germs $\mathcal{G}$, or in a finite direct sum of spaces of formal series, is a $\mathbb{C}(k)$-linear map $V : \mathfrak{M} \to \mathcal{G}$ with

$$V \circ \mathrm{Sh} = \varsigma \circ V.$$

A solution with values in $\lambda^k k^\rho\,\mathbb{C}((1/k))$ has the *type* $(\lambda, \rho + \mathbb{Z})$ and the *normalization* $\hat{V} = \lambda^{-k} k^{-\rho} V : \mathfrak{M} \to \mathbb{C}((1/k))$. For masters $\mathbf{b}$ the *solution vector* of $V$ is $v_V = (V[b_1], \dots, V[b_{\mathfrak{m}}])$. The *annihilator* of a set $\mathcal{S}$ of solutions is $\mathrm{Ann}(\mathcal{S}) = \{m \in \mathfrak{M} :\ V(m) = 0 \text{ for every } V \in \mathcal{S}\}$.

When $\kappa_{1,S} > 0$ for every nonempty proper $S$, the values along the ray are a solution with values in germs at $+\infty$, $V[g] = \langle g \rangle_k$ (Theorem 38.3). At a nondegenerate native gauge $\alpha_c$ the expansion $f(\alpha_c)^k k^{-(n-1)/2} \sum_i a_i(g) k^{-i}$ of Theorem 38.6 is a solution of type $\big(f(\alpha_c), -\frac{n-1}{2} + \mathbb{Z}\big)$, written $A_c$; its normalization is the map $A^{(c)}$ there.

**Lemma 39.3** (Solutions of distinct types)**.** *Let $A_1, \dots, A_r$ be solutions with values in formal series such that, for every type, those of that type among them are linearly independent over $\mathbb{C}$. Then their normalizations are linearly independent over $\mathbb{C}((1/k))$, and $\dim_{\mathbb{C}(k)} \mathfrak{N} \geq r$ for every submodule $\mathfrak{N}$ on which their restrictions are, type by type, linearly independent over $\mathbb{C}$.*

**Theorem 39.4** (What solutions determine)**.** *Let $\mathbf{b}$ be masters of a submodule $\mathfrak{N}$, $\mathbf{B} = \mathbf{B_b}$, and let $\mathcal{S}$ be a set of solutions.*

*(i) Every solution $V$ has $\varsigma^{-1} v_V = \mathbf{B}\, v_V$, that is $v_V(k-1) = \mathbf{B}(k)\, v_V(k)$. When the values along the ray are a solution, $\langle b_i \rangle_{k-1} = \sum_l \mathbf{B}_{il}(k) \langle b_l \rangle_k$ for all large $k$.*

*(ii) $\mathrm{Ann}(\mathcal{S})$ is a submodule.*

*(iii) A matrix $\mathbf{B}' \in \mathrm{Mat}_{\mathfrak{m}}(\mathbb{C}(k))$ has $\varsigma^{-1} v_V = \mathbf{B}' v_V$ for every $V \in \mathcal{S}$ if and only if every row of $\mathbf{B}' - \mathbf{B}$, read as the element $\sum_l (\mathbf{B}' - \mathbf{B})_{il} [b_l]$ of $\mathfrak{N}$, lies in $\mathrm{Ann}(\mathcal{S})$. So $\mathbf{B}$ is the only such matrix exactly when $\mathrm{Ann}(\mathcal{S}) \cap \mathfrak{N} = 0$.*

*(iv) Let $\mathrm{Ann}(\mathcal{S}) \cap \mathfrak{N} = 0$, write $\mathbf{B} = P_0/Q_0$ with $Q_0$ the monic least common denominator of its entries, and put $D_0 = \max(\deg P_0, \deg Q_0)$. For $D \geq 0$ the equations $Q(k)\, \varsigma^{-1} v_V = P(k)\, v_V$, $V \in \mathcal{S}$, in pairs $(P, Q)$ of a matrix and a scalar of polynomials of degree at most $D$, read at every integer beyond a bound past which (i) holds for a germ and at every power of $k$ for a formal series, are linear and homogeneous in the coefficients of $(P, Q)$. Their solutions are the pairs $(cP_0, cQ_0)$ with $c \in \mathbb{C}[k]$ of degree at most $D - D_0$: only 0 for $D < D_0$, one line at $D = D_0$. For each $D$ finitely many of the equations have the same solutions as all of them.*

**Corollary 39.5** (One solution of an irreducible module)**.** *If $\mathfrak{N}$ is irreducible, every solution that does not vanish on $\mathfrak{N}$ determines $\mathbf{B_b}$ in the sense of Theorem 39.4(iii). If the cyclic submodule of $[1]$ is finite-dimensional and irreducible, then for every solution $V$ with $V[1] \neq 0$, $V[1]$ satisfies the law of $[1]$, $\sum_i p_i(k)\, \varsigma^i V[1] = 0$, and satisfies no relation $\sum_{i<h} q_i(k)\, \varsigma^i V[1] = 0$ with $q_i \in \mathbb{C}(k)$ not all zero.*

Masters are a basis of a submodule, the module itself or its invariant part (Theorem 39.12), while a box of insertions reduces to a spanning set of a submodule that contains [1], which is a set of masters exactly when it is independent (Section 39.4); the one-step matrix is the matrix of the inverse shift, so that the values of the masters step down in strength by $\mathbf{B_b}$. A solution is anything that respects the two relations: the values over the orthant, the expansion at a native gauge, the closings on the torus (Theorem 39.33). Each solution gives $\mathfrak{m}$ linear equations on the numerator and the denominator of $\mathbf{B_b}$ for every integer or every power of $k$, and the equations of a set of solutions determine $\mathbf{B_b}$ exactly when the solutions annihilate no submodule: a submodule annihilated by all of them is a direction in which a row of the matrix can move without any of them noticing. An irreducible module is determined by any one of its nonzero solutions, and the law of [1] by any solution that does not vanish at [1] when the cyclic submodule of [1] is irreducible.

**Proposition 39.6** (The law of [1] read from a one-step matrix)**.** *Let insertions $b_1 = 1, b_2, \dots, b_{\mathfrak{m}}$ have classes that span a submodule $\mathfrak{N}$, and let $\mathbf{B} \in \mathrm{Mat}_{\mathfrak{m}}(\mathbb{C}(k))$ have $[f^{-1} b_i] = \sum_l \mathbf{B}_{il}(k) [b_l]$ for every $i$; for masters, $\mathbf{B} = \mathbf{B_b}$. Put $w_0 = (1, 0, \dots, 0)$ and $w_i(k) = w_{i-1}(k-1)\, \mathbf{B}(k)$, so that $w_i(k) = w_0\, \mathbf{B}(k-i+1) \cdots \mathbf{B}(k)$.*

*(i)* $\mathrm{Sh}^{-i}[1] = \sum_l w_{i,l}(k)[b_l]$. *So every solution* $V$ *has* $\varsigma^{-i}V[1] = w_i\, v_V$, *with* $v_V = (V[b_1], \dots, V[b_{\mathfrak{m}}])$ *as in Definition 39.2, and when* $\kappa_{1,S} > 0$ *for every nonempty proper* $S$, $\langle 1\rangle_{k-i} = \sum_l w_{i,l}(k)\langle b_l\rangle_k$ *as meromorphic functions of* $k$.

*(ii)* *Let* $h$ *be the least index at which* $w_0, \dots, w_h$ *are linearly dependent over* $\mathbb{C}(k)$, *so that* $h \le \mathfrak{m}$, *let* $\sum_{i=0}^{h} q_i(k)\, w_i(k) = 0$ *with* $q_i \in \mathbb{C}[k]$ *without a common factor, and put* $p_j(k) = q_{h-j}(k+h)$. *Then* $\sum_{j=0}^{h} p_j(k)\, \mathrm{Sh}^j[1] = 0$, *that is* $\sum_j p_j(k) f^j \in \mathcal{L}$; $p_0 \neq 0$, *and* $p_h \neq 0$ *when* $\mathbf{B}$ *is invertible. The order* $h$ *is at least the dimension of the cyclic submodule of* $[1]$, *with equality when* $\mathbf{b}$ *are masters, and then the relation is the law of* $[1]$.

*(iii)* *Let* $\mathbf{W} = [w_0; \dots; w_{h-1}]$ *be the matrix of the first* $h$ *rows and* $J$ *a set of* $h$ *columns with* $\det \mathbf{W}_J \neq 0$. *Then* $(q_i)_i$ *is proportional over* $\mathbb{C}(k)$ *to the signed maximal minors* $\big((-1)^i \det[w_0; \dots; \widehat{w_i}; \dots; w_h]_J\big)_i$. *With* $Q$ *the monic least common denominator of the entries of* $\mathbf{B}$, $q_h$ *divides the polynomial* $\det \mathbf{W}(k)_J \prod_{j=0}^{h-1} Q(k-j)^{h-j}$. *So at a zero* $k_0$ *of* $q_h$ *with* $Q(k_0 - j) \neq 0$ *for* $0 \le j < h$ *the rows* $w_0(k_0), \dots, w_{h-1}(k_0)$ *are linearly dependent: there* $\langle 1\rangle_{k_0}, \dots, \langle 1\rangle_{k_0-h+1}$, *written by (i) as linear forms in the values of* $\mathbf{b}$ *at* $k_0$, *are dependent forms.*

*(iv)* *If* $h = \mathfrak{m}$, *then* $q_0(k) \det \mathbf{W}(k) = (-1)^h q_h(k) \det \mathbf{W}(k-1) \det \mathbf{B}(k)$.

*(v)* *The same holds for every element* $m = \sum_l c_l(k)[b_l]$ *of* $\mathfrak{N}$, *with the row* $c$ *in place of* $w_0$, $m$ *in place of* $[1]$, *and the cyclic submodule of* $m$, *the span of the* $\mathrm{Sh}^i m$, *in place of that of* $[1]$.

The first dependence among the rows $w_i$, which carry $\langle 1\rangle$ to the lower strengths in the coordinates of a spanning set, is a recurrence of the values of 1, and on masters it is the law of [1]. Its end coefficient $p_0(k) = q_h(k+h)$ vanishes only at shifted zeros of the denominator of $\mathbf{B}$ and at strengths where the rows line up, and when the rows reach the whole span, $h = \mathfrak{m}$, the end coefficients are tied to the determinant of $\mathbf{B}$. A spanning set that is not independent gives a relation whose order may exceed that of the law of [1].

## 39.2 The motive of a ray

Along a ray of a family of lists only the exponents move and the letters stay, so the whole ray lives in one motive: its relations are exact forms there, the left factors of the coaction do not depend on the strength, and the strength moves only a vector of de Rham coordinates, by the law of [1].

**Definition 39.7** (The model of a ray)**.** Let a ray of a family $F$ of lists on $\mathcal{E}$ have an integral origin, with $\kappa_{1,S} > 0$ for every nonempty proper $S$. Its *model* is

$$V = X_{\mathcal{E}} \setminus \bigcup_g \widetilde{Y}_g, \qquad D_V = D \cap V, \qquad H = H^{n-1}(V, D_V),$$

the union over the letters $K_j$ and $\Psi_j$ of every list of $F$, whatever the exponents of the ray, and the linear letters $\alpha_J$, $|J| \ge 2$ (Definition 34.1). For an insertion $g = \alpha^u \prod_j K_j^{-a_j} \Psi_j^{-b_j}$ of the kind of Proposition 38.5 and an integer $k$ with $\kappa_{0,S} + \kappa_S(g) + k\,\kappa_{1,S} \ge 1$ for every nonempty proper $S$, which holds for all large $k$, put $\omega_{g,k} = g\, f^k \omega_0$, extended linearly in $g$, and $\langle g\rangle_k^{\mathrm{mot}} = [H, \widetilde{\sigma}_{\mathcal{E}}, \omega_{g,k}]^{\mathrm{mot}}$ (Definition 34.5).

**Proposition 39.8** (The relations of a ray are exact on its model)**.** *In Definition 39.7,* $\omega_{g,k}$ *is a regular form on* $V$, *and for every mark* $e$

$$\big((c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})g\big)\, f^k \omega_0 = d\big(\iota_{\partial_{y_e}} \omega_{g,k}\big),$$

*where $\iota_{\partial_{y_e}}\omega_{g,k}$ is a regular form on $V$ whose restriction to every component of $D_V$ vanishes. So both relations of the ray (Theorem 38.3) hold in $H_{\mathrm{dR}}$: the class of the left side is* 0*, and* $[f\,\omega_{g,k}] = [\omega_{g,k+1}]$.

**Theorem 39.9** (A ray is one motive)**.** *Let a ray be as in Definition 39.7.*

*(i)* $\langle g\rangle_k$ *is the period of* $\langle g\rangle_k^{\mathrm{mot}}$*, and the motivic values satisfy both relations of the ray as identities of motivic periods; the law of* [1] *holds for* $\langle 1\rangle_k^{\mathrm{mot}}$ *at all large* $k$*.*

*(ii) The Galois conjugates of every* $\langle g\rangle_k^{\mathrm{mot}}$ *lie in* $L(H,\widetilde{\sigma}_{\mathcal{E}})$*, one space for the whole ray, of dimension at most* $\dim H_{\mathrm{dR}}$*.*

*(iii) For a basis* $(e_i)$ *of* $H_{\mathrm{dR}}$*,*

$$\Delta\langle g\rangle_k^{\mathrm{mot}} = \sum_i [H,\widetilde{\sigma}_{\mathcal{E}},e_i]^{\mathrm{mot}} \otimes [H,e_i^\vee,\omega_{g,k}]^{\mathrm{dR}};$$

*the left factors depend neither on* $g$ *nor on* $k$*, and for every* $i$ *the right factors* $k \mapsto [H,e_i^\vee,\omega_{1,k}]^{\mathrm{dR}}$ *satisfy the law of* [1] *at all large* $k$*.*

*(iv) The coordinates* $c_i(g,k) = e_i^\vee(\omega_{g,k})$ *in a basis of* $H_{\mathrm{dR}}$ *over* $\mathbb{Q}$ *are rational for an insertion* $g$ *with rational coefficients, the sequences* $k \mapsto c_i(1,k)$ *are rational solutions of the law of* [1] *at all large* $k$*, and*

$$\langle 1\rangle_k = \sum_i c_i(1,k)\ \mathrm{per}[H,\widetilde{\sigma}_{\mathcal{E}},e_i].$$

**Corollary 39.10** (The rows of the values at every order)**.** *For a ray as in Definition 39.7 let* $W \subseteq H_{\mathrm{dR}}$ *be the* $\mathbb{Q}$*-span of the classes* $\omega_{1,k}$*,* $k \geq k_1$*, with* $k_1$ *such that for every* $k \geq k_1$ *the classes satisfy the law (Theorem 39.9(i)) and* $p_h(k) \neq 0$*. Its dimension* $m$ *is at most the order* $h$ *of the law of* [1]*. In a basis* $w_1,\dots,w_m$ *of* $W$*,* $\omega_{1,k} = \sum_l c_l(k)\,w_l$ *with* $c_l$ *rational solutions of the law, and*

$$\langle 1\rangle_k = \sum_{l=1}^{m} c_l(k)\ \mathrm{per}[H,\widetilde{\sigma}_{\mathcal{E}},w_l] \qquad (k \geq k_1).$$

*So every value has a row in the* $m$ *coordinates* $\mathrm{per}[H,\widetilde{\sigma}_{\mathcal{E}},w_l]$*, whose coefficients are rational solutions of the law; a coordinate with a rational period joins the rational part.*

So along a ray the motive is fixed and the strength moves only a vector of de Rham coordinates, by a rational recurrence. The left factors are the periods of $H$ against a basis, and by Theorem 34.13 and Corollary 34.15 those of weight at most $w$ are sums of products of open periods of steps of at most $w+1$ marks, along any tree of bunches: the coaction is read once for the whole ray. Theorem 38.31(iii) bounds the space of the values; Corollary 39.10 writes each value in fixed coordinates with coefficients that solve the law.

## 39.3 Symmetries of a ray

**Proposition 39.11** (The symmetries of a ray)**.** *Let a ray of* $F$ *have the origin* $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ *and the direction* $(\mathbf{m}_1;\mathbf{s}_1,\mathbf{t}_1)$*, and put* $\sigma_i = \frac{1}{2}\sum_j(s_{ij}+t_{ij})$*,* $i = 0,1$*. On the torus* $(\mathbb{C}^\times)^{\mathcal{E}}/\mathbb{C}^\times$ *put, for a bijection* $\beta$ *of the marks,* $\phi_\beta(\alpha) = \alpha\circ\beta$ *with* $(\alpha\circ\beta)_e = \alpha_{\beta(e)}$*; the* Cremona $\mathrm{cr}(\alpha) = \alpha^{-1}$*; and the* reversed *map* $\bar{\phi}_\beta = \mathrm{cr}\circ\phi_\beta$*,* $\bar{\phi}_\beta(\alpha) = (\alpha\circ\beta)^{-1}$*. A* symmetry *of the ray is a map* $\phi$ *among the* $\phi_\beta$ *and the* $\bar{\phi}_\beta$ *that keeps the stride and the density of the origin form,* $f\circ\phi = f$ *and* $(\omega_0/dy)\circ\phi = \omega_0/dy$*, the latter on the positive orthant with the principal powers when the origin is not integral; the*

*symmetries form a group. Let $h$ be a bijection of the tables, write $u \circ \beta^{-1}$ for $e \mapsto u_{\beta^{-1}(e)}$ and $\mathbf{a} \circ h^{-1}$ for $j \mapsto a_{h^{-1}(j)}$, and say that $\beta$* carries, *respectively* exchanges, *the tables with $h$ when it satisfies the conditions of Definition 11.13 on the bases and the weights, without its condition on the pairs.*

*(i) If $\beta$ carries the tables, $K_j \circ \phi_\beta = K_{h(j)}$ and $\Psi_j \circ \phi_\beta = \Psi_{h(j)}$. If $\beta$ exchanges them, $K_j \circ \phi_\beta = \Psi_{h(j)}$, $\Psi_j \circ \phi_\beta = K_{h(j)}$, $K_j \circ \bar{\phi}_\beta = \alpha^{-\mathcal{E}} K_{h(j)}$ and $\Psi_j \circ \bar{\phi}_\beta = \alpha^{-\mathcal{E}} \Psi_{h(j)}$. Always $K_j \circ \mathrm{cr} = \alpha^{-\mathcal{E}} \Psi_j$ and $\Psi_j \circ \mathrm{cr} = \alpha^{-\mathcal{E}} K_j$.*

*(ii) The following maps are symmetries when, for $i = 0, 1$, every mark $e$ and every table $j$:*

| *map* | *masses* | *exponents* |
|---|---|---|
| $\phi_\beta$, *$\beta$ carrying* | $m_{i,\beta(e)} = m_{i,e}$ | $(s_{i,h(j)}, t_{i,h(j)}) = (s_{ij}, t_{ij})$ |
| $\phi_\beta$, *$\beta$ exchanging* | $m_{i,\beta(e)} = m_{i,e}$ | $(s_{i,h(j)}, t_{i,h(j)}) = (t_{ij}, s_{ij})$ |
| cr | $\mathbf{m}_i = \sigma_i \mathbf{1}$ | $s_{ij} = t_{ij}$ |
| $\bar{\phi}_\beta$, *$\beta$ exchanging* | $m_{i,\beta(e)} + m_{i,e} = 2\sigma_i$ | $(s_{i,h(j)}, t_{i,h(j)}) = (s_{ij}, t_{ij})$ |

*A symmetry maps $X_F$ and the positive orthant onto themselves, has $\phi^* \omega_0 = \pm \omega_0$ when the origin is integral, and keeps the values, $\langle g' \circ \phi \rangle_k = \langle g' \rangle_k$ for every insertion $g'$.*

*(iii) On the insertions $\alpha^u \prod_j K_j^{-a_j} \Psi_j^{-b_j}$, written $(u; \mathbf{a}, \mathbf{b})$, the four kinds act by $(u; \mathbf{a}, \mathbf{b}) \mapsto$*

$$(u \circ \beta^{-1}; \mathbf{a} \circ h^{-1}, \mathbf{b} \circ h^{-1}) \quad (\phi_\beta,\ \beta \text{ carrying}), \qquad (-u + |\mathbf{a} + \mathbf{b}|\mathbf{1}; \mathbf{b}, \mathbf{a}) \quad (\mathrm{cr}),$$
$$(u \circ \beta^{-1}; \mathbf{b} \circ h^{-1}, \mathbf{a} \circ h^{-1}) \quad (\phi_\beta,\ \beta \text{ exchanging}),$$
$$(-u \circ \beta^{-1} + |\mathbf{a} + \mathbf{b}|\mathbf{1}; \mathbf{a} \circ h^{-1}, \mathbf{b} \circ h^{-1}) \quad (\bar{\phi}_\beta,\ \beta \text{ exchanging}),$$

*with $|\mathbf{a} + \mathbf{b}| = \sum_j (a_j + b_j)$.*

The four kinds are the monomial maps of the torus that a list can certify: a permutation of the marks that carries the bases, one that carries them onto the cobases, the inversion of every scale, and the composite of the last two. The first keeps each letter, the second and the third swap the two letters of every table, and the fourth keeps each letter up to the monomial $\alpha^{-\mathcal{E}}$, which the masses absorb when the masses of a mark and of its image add up to $2\sigma_i$, twice the total strength. Composites of symmetries are symmetries, so a group generated by maps of the four kinds can be used whole.

**Theorem 39.12** (The invariant module)**.** *Let $G$ be a finite group of symmetries of a ray.*

*(i) The maps $\phi^*[g'] = [g' \circ \phi]$, $\phi \in G$, are well defined on $\mathfrak{M}$ and commute with the shift. With $e_G = |G|^{-1} \sum_{\phi \in G} \phi^*$, $\mathfrak{M} = \mathfrak{M}^G \oplus (1 - e_G)\mathfrak{M}$ as submodules, and $\mathfrak{M}^G = e_G \mathfrak{M}$ consists of the classes of the invariant insertions.*

*(ii) A solution $V$ with $V \circ \phi^* = V$ for every $\phi \in G$, the values along the ray among them, vanishes on $(1 - e_G)\mathfrak{M}$: $\langle g' \rangle_k = \langle e_G g' \rangle_k$. The cyclic submodule of $[1]$ lies in $\mathfrak{M}^G$, so the law of $[1]$ is read in $\mathfrak{M}^G$, on masters among the invariant insertions.*

*(iii) If $\phi \in G$ fixes a nondegenerate native gauge $\alpha_c$, then $A_c \circ \phi^* = A_c$. For the $G$-orbit $O$ of $\alpha_c$, with stabilizer $G_c$, the sum $A_O = \sum_{\phi G_c \in G/G_c} A_c \circ \phi^*$ is invariant and does not vanish on $\mathfrak{M}^G$, and $\dim_{\mathbb{C}(k)} \mathfrak{M}^G$ is at least the number of $G$-orbits of nondegenerate native gauges.*

**Theorem 39.13** (A reversed exchange at a lean)**.** *Let $\beta$ exchange a family with itself (Definition 11.13), with the bijection $h$ of its tables, and let $\Upsilon = \Upsilon_{\boldsymbol{\sigma}, \boldsymbol{\delta}; \mathbf{1}}$ be a realization at the table's gauge with masses $\boldsymbol{\tau}$ (Definition 9.1).*

*(i)* $R_{h(j),\beta(e)}(\mathbf{1}) = 1 - R_{j,e}(\mathbf{1})$ *for every table and mark, and* $\tau_{\beta(e)} + \tau_e = 2\sigma$. *So* $\bar{\phi}_\beta$ *keeps the integrand* $\alpha^{\boldsymbol{\tau}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}$ *of the period at the data of* $\Upsilon$, *and it is a symmetry of the ray with origin* 0 *and direction* $(c\boldsymbol{\tau}; c\mathbf{s}, c\mathbf{t})$ *whenever that direction is integral. The exchange* $\phi_\beta$ *and the Cremona each take that integrand to the integrand at the data of* $\Upsilon_{\boldsymbol{\sigma},-\boldsymbol{\delta};\mathbf{1}}$.

*(ii) For every mark* $e$ *and partners* $\mathbf{g} = (g_1, \dots, g_q)$, *with* $\beta(\mathbf{g}) = (\beta(g_1), \dots, \beta(g_q))$, *the views (Definition 10.1) satisfy*

$$\Phi^{\mathbf{E}}_{\beta(e),\beta(\mathbf{g})} = \Phi^{\mathbf{E}}_{e,\mathbf{g}}, \qquad \Phi^{\mathbf{O}}_{\beta(e),\beta(\mathbf{g})} = -\Phi^{\mathbf{O}}_{e,\mathbf{g}} :$$

*the reading of* $\beta(e)$ *at* $\boldsymbol{\delta}$ *is the reading of* $e$ *at* $-\boldsymbol{\delta}$. *At the centre* $\beta$ *maps every class onto itself; at a lean it permutes the classes of* $\Upsilon$.

*(iii) In a one-table molecule,* $\beta$ *maps every class* $C$ *of the centre onto itself, and when* $C$ *splits at the lean into* $C_K$ *and* $C_\Psi$ *(Theorem 11.17),* $\beta(C_K) = C_\Psi$ *and* $\beta(C_\Psi) = C_K$.

An exchange carries the bases of a family onto the cobases, and the integrand at a lean weighs the two unequally: the exchange alone takes the lean to its opposite, and so does the Cremona, which inverts every scale. Their composite keeps the integrand at every lean at the table's gauge, because there the masses of a mark and of its image add up to $2\sigma$. So the integral at a lean keeps a symmetry carrying each mark to its image under the exchange, and insertions related by it have equal values; but the symmetry reverses every scale, $y \mapsto -\beta^* y$, and the reading, whose odd part changes sign under a reversal, sees $\beta(e)$ as $e$ read at the opposite lean. A bijection that keeps the potential fixes every class (Lemma 14.18); the reversed exchange permutes the classes, and on a class of the centre that the lean splits it swaps the basis half and the cobasis half. At the centre the odd part is absent and the exchange, the Cremona and their composite all keep the integrand: the reversal is invisible to the reading there, and the classes it joins at the centre are those the lean separates.

*Remark* 39.14. The condition $m_{\beta(e)} + m_e = 2\sigma$ of Proposition 39.11 is met off the table's gauge as well, for instance by the uniform masses $\sigma\mathbf{1}$ at every lean of a family with $\sum_j \delta_j(2r_j - n) = 0$, as for one table with $2r = n$. The native gauge of such a realization on the positive orthant is fixed by $\bar{\phi}_\beta$ up to scale, and the views of $\beta(e)$ are those of $e$ at the reflected arguments $v \mapsto -v$ up to affine terms in $v$, by the identities $\Phi_{\mathbf{O}}(-\beta^* y) = \Phi_{\mathbf{O}}(y)$ and $\Phi_{\mathbf{E}}(-\beta^* y) = \Phi_{\mathbf{E}}(y) - 2\sigma\langle \mathbf{1}, y\rangle$ of Section 50.38. When that gauge is not proportional to $\mathbf{1}$ the realization is not an atom (Theorem 11.4), though the integral keeps the reversed exchange.

## 39.4 The box of the walls

The module of a ray is the quotient by the relations of the period, and finitely many of them give a spanning set of its Laurent part and the step on it. An insertion is measured by its level, the powers of the letters in its denominator, and by its wall degree, the power of the reciprocal stride that dominates it, read on the walls of the direction. A box is the finite set of symbols of insertions within given levels and wall degree; its relations, the letter identities among them and the mark relations of its stems, are relations of the module, so they hold as identities of meromorphic functions of the strength. One elimination of a box leaves a spanning set of a submodule that contains [1] and the matrix of the inverse shift on it, with a certificate supported on the stems, and a search through the boxes stops by the finite dimension of the ray. When the origin carries no letter exponents, the relations $(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})(\alpha^w f^{-1})$ involve Laurent insertions alone, and they do not suffice: the evaluation at a singular point of the zero set of $f^{-1}$ on the torus, when

there is one, vanishes on all of them and not on the law of [1]. Insertions are the regular functions on the letter complement $X_F$ of Definition 38.1, on which every letter is inverted whatever its exponent along the ray; a box need carry no unit of a letter whose exponent along the ray vanishes identically, and its relations are relations of the module all the same.

**Definition 39.15** (Levels, wall degree, boxes)**.** Let a ray of a family $F$ of $N$ tables on $n \geq 2$ marks have $\kappa_{1,S} > 0$ for every nonempty proper $S$, and let $\Bbbk \subseteq \mathbb{C}$ be a field that contains the weights and the origin.

(i) The *symbols* are the data $(u; \mathbf{a}, \mathbf{b})$ with $u \in \mathbb{Z}^{\mathcal{E}}$, $\mathbf{a}, \mathbf{b} \in \mathbb{N}^N$ and $u_{\mathcal{E}} = \sum_j (a_j r_j + b_j r_j^*)$. The *evaluation* of $(u; \mathbf{a}, \mathbf{b})$ is the insertion $\alpha^u \prod_j K_j^{-a_j} \Psi_j^{-b_j}$ (Proposition 39.11(iii)); its *level* is $(\mathbf{a}, \mathbf{b})$, and the symbols at the level $(\mathbf{0}, \mathbf{0})$ are the *Laurent* symbols $\alpha^u$, $u_{\mathcal{E}} = 0$. A *formal combination* is a finite combination of symbols with coefficients in $\Bbbk(k)$, and its evaluation is the same combination of the insertions, an element of $\mathbb{C}[X_F] \otimes \mathbb{C}(k)$. The evaluations of the symbols span $\mathbb{C}[X_F]$.

(ii) The *walls* of a symbol $g = (u; \mathbf{a}, \mathbf{b})$ are those of its data, $\kappa_U(g) = u_U - \sum_j \big(a_j d_j(U) + b_j d_j^*(U)\big)$, and its *wall degree* is

$$\deg_\kappa g = \max_{\varnothing \neq U \subsetneq \mathcal{E}} \frac{-\kappa_U(g)}{\kappa_{1,U}},$$

the least real $x$ with $\kappa_U(g) + x\,\kappa_{1,U} \geq 0$ for every nonempty proper $U$.

(iii) The *mark relation* of a symbol $g = (u; \mathbf{a}, \mathbf{b})$ at a mark $e$ is the formal combination

$$\begin{aligned} \mathrm{MR}_e(g) = \big(m_{0,e} + k\,m_{1,e} + u_e\big)\,g &- \sum_j \big(s_{0j} + k\,s_{1j} + a_j\big) \sum_{T \ni e} \theta_{j,T}\,(u + \mathbf{1}_T; \mathbf{a} + \mathbf{e}_j, \mathbf{b}) \\ &- \sum_j \big(t_{0j} + k\,t_{1j} + b_j\big) \sum_{T \not\ni e} \theta_{j,T}\,(u + \mathbf{1}_{\mathcal{E}\setminus T}; \mathbf{a}, \mathbf{b} + \mathbf{e}_j), \end{aligned}$$

the sums over the bases $T$ of table $j$, the terms whose coefficient vanishes identically in $k$ omitted. Its evaluation is $(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})$ applied to that of $g$. The *letter identities* of $g$ are $g - \sum_{T \in \mathcal{B}_j} \theta_{j,T}(u + \mathbf{1}_T; \mathbf{a} + \mathbf{e}_j, \mathbf{b})$, for $K_j$, and $g - \sum_{T \in \mathcal{B}_j} \theta_{j,T}(u + \mathbf{1}_{\mathcal{E}\setminus T}; \mathbf{a}, \mathbf{b} + \mathbf{e}_j)$, for $\Psi_j$; they evaluate to 0.

(iv) A letter $K_j$, or $\Psi_j$, is *active along the ray* when its exponent $s_{0j} + k\,s_{1j}$, or $t_{0j} + k\,t_{1j}$, does not vanish identically. *Level bounds* are $\bar{\mathbf{a}}, \bar{\mathbf{b}} \in \mathbb{N}^N$ with $\bar{a}_j \geq 1$ when $K_j$ is active and $\bar{b}_j \geq 1$ when $\Psi_j$ is active; put $\bar{a}_j^- = \max(\bar{a}_j - 1, 0)$ and $\bar{b}_j^- = \max(\bar{b}_j - 1, 0)$. For $\mathfrak{t} \in \mathbb{N}$ the *box* $\mathfrak{B} = \mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ is the set of the symbols of wall degree at most $\mathfrak{t}$ at the levels $(\mathbf{a}, \mathbf{b})$ with $\mathbf{a} \leq \bar{\mathbf{a}}$ and $\mathbf{b} \leq \bar{\mathbf{b}}$, and its *stems* are its symbols at the level $(\bar{\mathbf{a}}^-, \bar{\mathbf{b}}^-)$. Its formal combinations form the space $\mathcal{F}_{\mathfrak{B}}$, and its *relations* $\mathcal{R}_{\mathfrak{B}} \subseteq \mathcal{F}_{\mathfrak{B}}$ are the $\Bbbk(k)$-span of the letter identities of its symbols $(u; \mathbf{a}, \mathbf{b})$, for $K_j$ when $a_j < \bar{a}_j$ and for $\Psi_j$ when $b_j < \bar{b}_j$, and of the mark relations of its stems at every mark. The box with *one unit of each active letter* has $\bar{a}_j = 1$ exactly when $K_j$ is active and $\bar{b}_j = 1$ exactly when $\Psi_j$ is; its stems are the Laurent symbols of wall degree at most $\mathfrak{t}$.

(v) The *Laurent part* $\mathfrak{M}_{\mathrm{L}}$ of the module $\mathfrak{M}$ (Definition 39.1) is the span of the classes of the Laurent insertions.

(vi) A group of symmetries generated by maps of the four kinds of Proposition 39.11(ii) carries every letter to a letter times a monomial (Proposition 39.11(i)): each of its elements $\phi$ permutes the letters, and the symbols, $g \mapsto \phi^* g$ (Proposition 39.11(iii)), the evaluation

of $\phi^* g$ being that of $g$ composed with $\phi$. Such a finite group $G$ *keeps* the box when its permutations of the letters keep the level bounds, $\bar{a}_j$ being the bound of $K_j$ and $\bar{b}_j$ that of $\Psi_j$.

**Proposition 39.16** (The wall degree)**.** *For a ray as in Definition 39.15 and its symbols $g$:*

*(i) For real $x$, $|g|\, f^x$ is bounded on the positive orthant exactly when $x \geq \deg_\kappa g$: the symbols of wall degree at most $x$ are those whose evaluation is dominated by $f^{-x}$. For the origin $0$ the value $\langle g \rangle_k$ converges absolutely exactly when $\operatorname{Re} k > \deg_\kappa g$, and for any origin it converges absolutely at every $k$ with $\operatorname{Re} \kappa_{0,U} + (\operatorname{Re} k - \deg_\kappa g)\, \kappa_{1,U} > 0$ for every nonempty proper $U$.*

*(ii) A Laurent symbol $\alpha^u$ has the wall degree $\max_U(-u_U)/\kappa_{1,U} \geq 0$, zero exactly for $u = 0$. Every monomial of the Laurent polynomial $f^{-1}$ has wall degree at most $1$, and every monomial of $f^{-1}\alpha^u$ wall degree at most $\deg_\kappa \alpha^u + 1$.*

*(iii) The terms of the mark relations and of the letter identities of $g$ have wall degree at most $\deg_\kappa g$.*

*(iv) At every level and for every real $x$ there are finitely many symbols of wall degree at most $x$.*

*(v) For every element $\phi$ of a group as in Definition 39.15(vi), $\deg_\kappa \phi^* g = \deg_\kappa g$, and $\phi^*$ maps Laurent symbols to Laurent symbols.*

**Lemma 39.17** (The Laurent part)**.** *For a ray as in Definition 39.15, the Laurent part $\mathfrak{M}_{\mathrm{L}}$ is a submodule. It contains* [1]*, and the class of every insertion $g$ with $f^{-l} g$ a Laurent polynomial for some $l \in \mathbb{N}$; when every letter exponent of the direction is positive, $s_{1j}, t_{1j} \geq 1$, it is $\mathfrak{M}$. For a finite group $G$ of symmetries, $e_G \mathfrak{M}_{\mathrm{L}} = \mathfrak{M}_{\mathrm{L}} \cap \mathfrak{M}^G$ is a submodule, spanned by the classes of the invariant Laurent insertions.*

**Proposition 39.18** (The relations of a box)**.** *Let $\mathfrak{B} = \mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ be a box of a ray as in Definition 39.15.*

*(i) Every term of every relation of the box lies in the box. The letter identities evaluate to $0$ and the mark relation $\mathrm{MR}_e(g)$ of a stem to the element $(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}) g$ of $\mathcal{L}$: every relation evaluates into $\mathcal{L}$, and its value vanishes identically in $k$ (Proposition 38.5(ii)).*

*(ii) A formal combination of symbols of the box whose evaluation is $0$ is a combination of the letter identities of the box.*

*(iii) The mark relation at every mark of every symbol of wall degree at most $\mathfrak{t}$ at a level at most $(\bar{\mathbf{a}}^-, \bar{\mathbf{b}}^-)$ lies in $\mathcal{R}_{\mathfrak{B}}$.*

*(iv) If a finite group $G$ of symmetries keeps the box, every $\phi \in G$ maps the box onto itself, its stems onto its stems and $\mathcal{R}_{\mathfrak{B}}$ onto itself; so, with $e_G = |G|^{-1} \sum_{\phi \in G} \phi^*$ and $\mathcal{F}_{\mathfrak{B}}^G = e_G \mathcal{F}_{\mathfrak{B}}$, $e_G \mathcal{R}_{\mathfrak{B}} = \mathcal{R}_{\mathfrak{B}} \cap \mathcal{F}_{\mathfrak{B}}^G$.*

**Proposition 39.19** (The closing relations have more solutions than the ray)**.** *Let a ray as in Definition 39.15 have an origin with $\mathbf{s}_0 = \mathbf{t}_0 = \mathbf{0}$, and write $f^{-1} = \sum_v \lambda_v \alpha^v$. For a Laurent insertion $\alpha^w$ and a mark $e$ the* closing relation

$$\big(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}\big)\big(\alpha^w f^{-1}\big) = \sum_v \lambda_v \big(m_{0,e} + w_e + (1-k)\, v_e\big)\, \alpha^{w+v}$$

*is an element of $\mathcal{L}$ whose terms are Laurent insertions. Let $z \in (\mathbb{C}^\times)^{\mathcal{E}}$ be a point at which $f^{-1}$ and every $\partial_{y_e} f^{-1}$ vanish, a singular point of the zero set of $f^{-1}$ on the torus.*

*(i) The evaluation at $z$, $\alpha^w \mapsto z^w$ on the Laurent insertions, extended $\mathbb{C}(k)$-linearly, vanishes on every closing relation.*

*(ii) It takes the value $q_0(k)$ on every element $\sum_{i=0}^{h} q_i(k) f^{-i}$, $q_i \in \mathbb{C}(k)$. So no element of $\mathcal{L}$ of this form with $q_0 \neq 0$ is a combination of closing relations, while $\mathcal{L}$ contains such elements: the law $\sum_j p_j(k)\,\mathrm{Sh}^j[1] = 0$ of $[1]$, with $p_0 p_h \neq 0$ (Definition 39.1), gives $\sum_{i=0}^{h} p_{h-i}(k-h) f^{-i} \in \mathcal{L}$. No $\mathbb{C}(k)$-linear form that vanishes on $\mathcal{L}$ agrees with the evaluation at $z$ on the Laurent insertions.*

The point $z$ lies off $X_F$, a letter that the direction uses vanishing there. The relations of $\mathcal{L}$ that the closing relations lack involve the insertions with inverse letters, through the letter identities and the mark relations of the occupations, and a box supplies them through its units of the letters.

**Definition 39.20** (The reduction of a box)**.** Let a finite group $G$ of symmetries keep a box $\mathfrak{B} = \mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ (Definition 39.15); $G$ may be trivial.

(i) The *orbit average* of a symbol $g$ of the box is $e_G g$, with $e_G$ as in Proposition 39.18(iv); the orbit averages of the orbits of symbols form a basis of $\mathcal{F}^G_{\mathfrak{B}}$, and $\mathcal{R}^G_{\mathfrak{B}} = e_G \mathcal{R}_{\mathfrak{B}}$.

(ii) A *ranking* is a total order of these orbits in which every orbit at a positive level ranks above every Laurent orbit and, of two Laurent orbits, the one of larger wall degree ranks higher; the level of the symbols of an orbit is $(\mathbf{0}, \mathbf{0})$ for all of them or for none, and they all have the same wall degree (Proposition 39.16(v)). The *leading* orbit of a nonzero element of $\mathcal{F}^G_{\mathfrak{B}}$ is the highest-ranked orbit in it; an orbit is a *pivot* when it leads an element of $\mathcal{R}^G_{\mathfrak{B}}$, and *free* otherwise. The *reduction* of $x \in \mathcal{F}^G_{\mathfrak{B}}$ is the unique combination of the averages of the free orbits that is congruent to $x$ modulo $\mathcal{R}^G_{\mathfrak{B}}$; if $x$ involves no orbit ranked above a given orbit, neither does its reduction.

(iii) For $\mathfrak{s} \in \mathbb{N}$ with $\mathfrak{s} < \mathfrak{t}$, the *spanning set* of the box at $\mathfrak{s}$ is $\mathbf{b} = (b_1, \ldots, b_{\mathfrak{m}})$, the averages of the free Laurent orbits of wall degree at most $\mathfrak{s}$, listed with $b_1 = 1$, the orbit of 1 being free. Its *targets* are the invariant Laurent polynomials $f^{-1} b_i$, elements of $\mathcal{F}^G_{\mathfrak{B}}$. The box is *accepted* at $\mathfrak{s}$ when the reduction of every target is a combination of the spanning set, $f^{-1} b_i \equiv \sum_l \mathbf{B}_{il}\, b_l$ modulo $\mathcal{R}^G_{\mathfrak{B}}$ with $\mathbf{B}_{il} \in \Bbbk(k)$; $\mathbf{B}$ is then the *matrix* of the box.

**Theorem 39.21** (The box gives a spanning set, the one-step matrix and a certificate)**.** *Let a finite group $G$ of symmetries keep a box $\mathfrak{B} = \mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ of a ray as in Definition 39.15, and let the box be accepted at $\mathfrak{s}$ (Definition 39.20), with the spanning set $\mathbf{b}$ and the matrix $\mathbf{B}$.*

*(i) In the module, $[f^{-1} b_i] = \sum_l \mathbf{B}_{il}(k)[b_l]$ for every $i$. The classes of $\mathbf{b}$ span a submodule $\mathfrak{N}$ of $\mathfrak{M}^G$ that contains $[1]$: the span of the classes of the invariant combinations of Laurent symbols of wall degree at most $\mathfrak{s}$, which lies in $e_G \mathfrak{M}_{\mathrm{L}}$ (Lemma 39.17). The spanning set is a set of masters of $\mathfrak{N}$ exactly when its classes are linearly independent over $\mathbb{C}(k)$, and then $\mathbf{B} = \mathbf{B}_{\mathbf{b}}$ (Definition 39.1).*

*(ii)* The certificate. *There are formal combinations $G_{i,e}$ of the stems, with coefficients in $\Bbbk(k)$, such that*

$$f^{-1} b_i - \sum_l \mathbf{B}_{il}(k)\, b_l = \sum_{e \in \mathcal{E}} \big( c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e} \big) G_{i,e}$$

*as functions on $X_F$, the $G_{i,e}$ evaluated: a certificate of the one-step relation (Proposition 38.5(i)).*

*(iii) Every solution $V$ has $\varsigma^{-1} v_V = \mathbf{B}\, v_V$, with $v_V = (V[b_1], \ldots, V[b_{\mathfrak{m}}])$, and $\langle b_i \rangle_{k-1} = \sum_l \mathbf{B}_{il}(k) \langle b_l \rangle_k$ as an identity of meromorphic functions of $k$. At every real $k$ off the poles of*

*the coefficients with* $\operatorname{Re}\kappa_{0,U} + (k-\mathfrak{t})\,\kappa_{1,U} > 0$ *for every nonempty proper* $U$*, all the values in (ii) converge absolutely and every term* $(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})s$*,* $s$ *a stem, has the value* $0$*, so the certificate integrates over the chart term by term.*

*(iv) Proposition 39.6 applies to* $\mathbf{b}$ *and* $\mathbf{B}$*: the first dependence of the rows gives a relation* $\sum_j p_j(k) f^j \in \mathcal{L}$*, with a certificate assembled from those of (ii), whose order is at least that of the law of* [1]*, with equality when the spanning set is a set of masters, and which then is the law of* [1]*.*

**Theorem 39.22** (The search stops)**.** *Let* $G$ *be a finite group of symmetries of a ray as in Definition 39.15, generated by maps of the four kinds of Proposition 39.11(ii) and possibly trivial, and fix a ranking for every box that* $G$ *keeps. There are* $\mathfrak{t} \geq 1$*, level bounds kept by* $G$ *and* $\mathfrak{s} < \mathfrak{t}$ *at which the box* $\mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ *modulo* $G$ *is accepted, with a spanning set whose classes span* $e_G\mathfrak{M}_{\mathrm{L}}$*, which is* $\mathfrak{M}^G$ *when every letter exponent of the direction is positive. So a search that runs through all the* $(\mathfrak{t}, \bar{\mathbf{a}}, \bar{\mathbf{b}}, \mathfrak{s})$ *with level bounds kept by* $G$*, in any order that reaches each, and tests the acceptance by one elimination over* $\Bbbk(k)$ *at each, stops.*

So the construction is finite and every step is an identity: a box is finite (Proposition 39.16(iv)), its relations are relations of the period read at the data of the ray (Proposition 39.18), so that whatever the box, the one-step relation and the recurrence it produces hold at every strength, and a search through the boxes stops (Theorem 39.22). The spanning set spans a submodule that contains [1], which is all a recurrence of the values of 1 needs; it is a set of masters exactly when it is independent, and then the recurrence that its rows give is the law of [1] and its matrix is $\mathbf{B}_{\mathbf{b}}$.

## 39.5 Sources at the high end

**Definition 39.23** (Sources; regular points; the normal density)**.** The *critical locus* of a ray is the set of its native gauges, $\{\alpha \in X_F : \ c^{(1)}(\alpha) = 0\}$, the critical points of $\log f$. A *source* is an irreducible component $Z$ of it; $f$ is constant on $Z$, with value $f_Z$, its *native value*. A point $z \in Z$ is *regular* when it lies on no other source, $Z$ is smooth at $z$ of dimension $d = \dim Z$, and the Hessian of $\log f$ in $y$ at $z$ has rank $n-1-d$ on the tangent space of the torus; its kernel is then the tangent space of $Z$, and it induces a nondegenerate quadratic form on the normal space. $Z$ is *clean* when it has a regular point; the regular points then form a dense open subset $Z^\circ$. In local holomorphic coordinates $(u, w)$ at a regular point, with $Z = \{w = 0\}$ and $-\log f = -\log f_Z + \frac{1}{2} w^{\mathsf{T}} A(u) w + O(|w|^3)$, the *normal density* is the $d$-form

$$\nu = \det A(u)^{-1/2} \, \frac{\omega_0}{du \wedge dw}\Big|_{w=0} du,$$

which does not depend on the coordinates up to its sign. For an integral origin $\omega_0$ is single-valued on $X_F$, and the *normal cover* $\tilde{Z}^\circ \to Z^\circ$ is the double cover on which $\nu$ is single-valued, its deck involution changing the sign of $\nu$. A source of dimension 0 is an isolated native gauge, clean exactly when it is nondegenerate.

**Theorem 39.24** (A closed cycle on a source carries a solution)**.** *Let a ray have an integral origin, let* $Z$ *be a clean source of dimension* $d$*,* $m = n-1-d$*, and let* $\gamma$ *be a compact oriented cycle of real dimension* $d$ *in* $\tilde{Z}^\circ$*.*

*(i) Over* $\gamma$ *there is a chain* $N(\gamma)$ *of real dimension* $n-1$ *in* $X_F$*, a union of discs, one through each point* $z$ *of* $\gamma$*, transverse to* $Z$*, on which* $|f| \leq |f_Z| e^{-c|v|^2}$ *at the displacement* $v$ *from* $z$ *for some* $c > 0$*, oriented by the sheet of* $\nu$*; its boundary lies where* $|f| \leq |f_Z| e^{-\varepsilon}$ *for some*

$\varepsilon > 0$. *Two such chains over homologous cycles differ by a boundary and by pieces on which* $|f| \le |f_Z| e^{-\varepsilon}$.

*(ii) As* $k \to \infty$,

$$\int_{N(\gamma)} g\, f^k \omega_0 \sim f_Z^k\, k^{-m/2} \sum_{i \ge 0} c_i(\gamma; g)\, k^{-i}, \qquad c_0(\gamma; g) = (2\pi)^{m/2} \int_\gamma g\, \nu.$$

*Where one family of holomorphic normal slices* $\{u = \text{const}\}$, *coordinates* $(u, w)$ *as in Definition 39.23, covers* $Z^\circ$, $c_i(\gamma; g) = \int_\gamma a_i(\cdot\,; g)$, *with* $f_Z^k k^{-m/2} \sum_i k^{-i} a_i(u; g)$ *the Laplace expansion of the integral over the disc in the slice at* $u$: $a_0(\cdot\,; g) = (2\pi)^{m/2} g\, \nu$, *and each* $a_i(\cdot\,; g)$ *is a holomorphic* $d$*-form given by finitely many* $w$*-derivatives of* $g$, *of* $\omega_0/(du \wedge dw)$ *and of* $\log f$ *at* $w = 0$.

*(iii) The map* $A_\gamma : g \mapsto f_Z^k k^{-m/2} \sum_i c_i(\gamma; g)\, k^{-i}$ *is a solution of type* $\big(f_Z, -\frac{m}{2} + \mathbb{Z}\big)$. *It depends only on the homology class of* $\gamma$ *in* $\tilde{Z}^\circ$, *and the image of* $\gamma$ *under the deck involution gives* $-A_\gamma$.

*(iv) If* $\gamma_1, \dots, \gamma_r$ *are such cycles and* $g_1, \dots, g_r$ *insertions with* $\det \big[ \int_{\gamma_a} g_b\, \nu \big]_{a,b} \neq 0$, *the* $A_{\gamma_a}$ *are linearly independent over* $\mathbb{C}$, *and their normalizations over* $\mathbb{C}((1/k))$.

*For* $d = 0$, $\gamma$ *is the point with a sheet and this is Theorem 38.6. For an origin that is not integral,* $\nu$ *is multivalued along* $Z^\circ$, *and the same holds for cycles with coefficients in the local system of its branches.*

**Corollary 39.25** (A curve of native gauges: residues and periods)**.** *Let a ray have an integral origin, let* $Z$ *be a clean source of dimension one, and let* $\bar{Z}$ *be the smooth compact curve that contains* $\tilde{Z}^\circ$ *as the complement of finitely many* punctures*: the points over the points of* $Z$ *that are not regular, among them its crossings with other sources and the points where the normal form degenerates, and the points where* $Z$ *leaves* $X_F$. *The solutions of* $Z$ *are spanned by those of a small loop around each puncture and those of a basis of the cycles of* $\bar{Z}$. *When* $Z$ *is the closure of* $\{\alpha_0 b^{\mathbf{v}} : b \in \mathbb{C}^\times\}$, $\mathbf{v} \in \mathbb{Z}^{\mathcal{E}}$, *with* $b \mapsto \alpha_0 b^{\mathbf{v}}$ *injective, and the normal coordinates are those of one fixed complement* $N$ *of* $\mathbb{C}\mathbf{v}$ *in* $\mathbb{C}^{\mathcal{E}}/\mathbb{C}\mathbf{1}$, $\alpha = \alpha_0 b^{\mathbf{v}} e^w$ *with* $w \in N$, *the determinant* $\det A(b)$ *is rational in* $b$, *every* $a_i(\cdot\,; g)$ *is* $r_i(b; g) \det A(b)^{-1/2}\, db/b$ *with* $r_i$ *rational in* $b$, $\tilde{Z}^\circ$ *is an open part of the curve* $Y^2 = \det A(b)$, *and the loop around a puncture* $q$ *gives* $2\pi i$ *times the residue at* $q$ *of* $f_Z^k k^{-m/2} \sum_i k^{-i} a_i(\cdot\,; g)$.

**Theorem 39.26** (A native gauge of corank at most one)**.** *Let* $\alpha_c$ *be an isolated native gauge of a ray at which the Hessian of* $\log f$ *in* $y$ *has corank at most one. There are holomorphic coordinates* $(X, w) \in \mathbb{C} \times \mathbb{C}^{n-2}$ *at* $\alpha_c$ *with*

$$\log f = \log f(\alpha_c) - \tfrac{1}{2} \sum_i w_i^2 + c\, X^{\mu+1}, \qquad c \neq 0,$$

*for an integer* $\mu \ge 1$, *the* multiplicity*;* $\mu = 1$ *exactly when* $\alpha_c$ *is nondegenerate, and otherwise* $\partial_X$ *spans the kernel of the Hessian at* $\alpha_c$. *Then* $\alpha_c$ *is an* $A_\mu$ point, *and:*

*(i) There are* $\mu$ *solutions* $A_{c,i}$, $0 \le i < \mu$, $A_{c,i}$ *of type* $\big(f(\alpha_c), -\frac{n-2}{2} - \frac{i+1}{\mu+1} + \mathbb{Z}\big)$*: the expansions of the integrals of* $g\, f^k \omega_0$ *over the chains (real normal disc)*$\times \Gamma_i$, *where* $\Gamma_i$ *is the combination of paths between the* $\mu + 1$ *sectors on which* $\operatorname{Re}(cX^{\mu+1}) < 0$ *with* $\int_{\Gamma_i} X^j e^{cX^{\mu+1}} dX = \delta_{ij}$ *for* $0 \le j < \mu$. *The leading coefficient of* $A_{c,i}(g)$ *is* $(2\pi)^{(n-2)/2}$ *times the coefficient of* $X^i$ *in the Taylor expansion of* $g\, \omega_0/(dX \wedge dw)$ *at* $\alpha_c$. *They are linearly independent, their exponents lying in distinct classes modulo* $\mathbb{Z}$, *and* $A_{c,0} = (2\pi)^{-1/2} A_c$ *for* $\mu = 1$.

*(ii) If a symmetry $\phi$ of the ray fixes $\alpha_c$ and $\mu \geq 2$, its differential multiplies the kernel by a root of unity $\zeta_\phi$ with $\zeta_\phi^{\mu+1} = 1$, and $A_{c,i} \circ \phi^* = \zeta_\phi^i A_{c,i}$.*

**Corollary 39.27** (Signs at a centre)**.** *On the centre ray of a family (Definition 38.13) let a sign $\eta$ lie in $X_F$ (Theorem 38.17). The Cremona is a symmetry of the ray that fixes $\eta$ and acts on the tangent space there by $-1$.*

*(i) If $\eta$ is nondegenerate, $A_\eta \circ \mathrm{cr}^* = A_\eta$.*

*(ii) If $\eta$ is isolated and the Hessian at $\eta$ has corank one, $\eta$ is an $A_\mu$ point with $\mu \geq 3$ odd, and $A_{\eta,i} \circ \mathrm{cr}^* = (-1)^i A_{\eta,i}$: the $\frac{\mu+1}{2}$ solutions of even $i$ do not vanish on the Cremona-invariant insertions, and the $\frac{\mu-1}{2}$ of odd $i$ vanish on them. For $\mu = 3$ the three exponents are $-\frac{n-2}{2} - \frac{1}{4}$, $-\frac{n-1}{2}$ and $-\frac{n-2}{2} - \frac{3}{4}$, the middle one odd.*

**Proposition 39.28** (Symmetries on the cycles of a source)**.** *Let a ray have an integral origin, and let a symmetry $\phi$, with $\phi^*\omega_0 = \varepsilon_\phi \omega_0$, map a clean source $Z$ onto a source. It maps $Z^\circ$ onto $\phi(Z)^\circ$ and lifts to $\tilde{\phi} : \tilde{Z}^\circ \to \widetilde{\phi(Z)}^\circ$, carrying each oriented normal disc to its image, and*

$$A_\gamma \circ \phi^* = \varepsilon_\phi \, A_{\tilde{\phi}_* \gamma}.$$

*For a finite group $G$ of symmetries, on $\mathfrak{M}^G$ the solution of $\gamma$ is that of the averaged cycle $|G|^{-1} \sum_{\phi \in G} \varepsilon_\phi \, \tilde{\phi}_* \gamma$; a cycle class with $\sum_{\phi \in G_Z} \varepsilon_\phi \, \tilde{\phi}_*[\gamma] = 0$, $G_Z$ the stabilizer of $Z$, gives a solution that vanishes on $\mathfrak{M}^G$.*

**Theorem 39.29** (What the invariant module sees at the high end)**.** *Let $G$ be a finite group of symmetries of a ray, and let $\alpha_c$ be an $A_\mu$ point with stabilizer $G_c$. The solutions $A_{c,i}$ for which $\phi \mapsto \zeta_\phi^i$ is trivial on $G_c$ (all of them when $\mu = 1$, with $\zeta_\phi^0 = 1$) average over $G$ to invariant solutions that do not vanish on $\mathfrak{M}^G$; the others vanish on $\mathfrak{M}^G$. So*

$$\dim_{\mathbb{C}(k)} \mathfrak{M}^G \;\geq\; \sum_O \#\big\{0 \leq i < \mu_O : \; \zeta_\phi^i = 1 \text{ for every } \phi \in G_{c_O}\big\},$$

*the sum over the $G$-orbits $O$ of the isolated native gauges of corank at most one, $\alpha_{c_O}$ a point of $O$ and $\mu_O$ its multiplicity.*

Where the critical locus of the stride is not a set of nondegenerate points, its components still carry solutions. Over a clean family of native gauges the integral over a normal disc is a Gaussian, whose expansion on holomorphic normal slices is a holomorphic form of top degree on the family, so a closed one, with values in the orientation of the normal discs; every closed cycle carries a solution, because integration by parts is exact under the normal Gaussian and integrates to zero over a cycle along the family. On a curve the cycles are the loops around its punctures, among them its crossings with other sources and its degenerate points, and the cycles of its compactification: residues and periods, on the double cover where the determinant of the normal form has a square root. A native gauge of corank one carries one solution for each class of paths between the sectors where its reduced function decays, with exponents spaced by $\frac{1}{\mu+1}$. A symmetry that fixes a native gauge acts on its solutions: trivially at a nondegenerate one, and through the powers of its eigenvalue on the kernel at a degenerate one, so that at a sign of a centre, fixed by the Cremona, a point of multiplicity three carries an odd solution that every Cremona-invariant insertion misses. The invariant module counts each orbit of sources by the solutions its stabilizer leaves alone.

## 39.6 Closings at the low end

**Definition 39.30** (Closings)**.** Let a ray have an integral origin, $\mathbf{m}_0 \in \mathbb{Z}^{\mathcal{E}}$ and $s_{0j}, t_{0j} \in \mathbb{Z}$, and a direction in which every letter has a positive exponent, $s_{1j}, t_{1j} \geq 1$. For an insertion $g$ let $j(g)$ be the least integer such that $g\, f^{-j}\omega_0/dy$ is a Laurent polynomial for every $j \geq j(g)$. The *closings* of $g$ are the constant terms

$$\mathsf{C}_j(g) = \mathrm{CT}\left(g\, f^{-j}\, \omega_0/dy\right) = \frac{1}{(2\pi i)^{n-1}} \oint_{|\alpha_e|=1,\ e\neq e_0} g\, f^{-j}\, \omega_0, \qquad j \geq j(g),$$

taken on the chart $\alpha_{e_0} = 1$, where they do not depend on $e_0$.

**Proposition 39.31** (One form on the orthant and on the torus)**.** *Let a ray have origin* 0 *and* $\kappa_{1,S} > 0$ *for every nonempty proper* $S$*, let* $\alpha^\star$ *be the native gauge of its direction on the positive orthant,* $\alpha^\star_{e_0} = 1$*, and put* $A = f(\alpha^\star)^{-1}$*. On the chart* $\alpha_{e_0} = 1$*, with* $dy = \prod_{e\neq e_0} d\alpha_e/\alpha_e$*, the rational form* $\varpi_q = dy/(f^{-1} - q)$ *has*

$$\int_{y_{e_0}=0} \varpi_q = \sum_{k\geq 1} \langle 1\rangle_k\, q^{k-1} \qquad (|q| < A),$$

$$\frac{1}{(2\pi i)^{n-1}} \oint_{|\alpha_e|=\alpha^\star_e,\ e\neq e_0} \varpi_q = -\sum_{j\geq 0} \mathrm{CT}\left(f^{-j}\right) q^{-j-1} \qquad (|q| > A),$$

*where* $\mathrm{CT}(f^{-j})$ *is the constant term of the Laurent polynomial* $f^{-j}$*, the closing* $\mathsf{C}_j(1)$ *when every letter exponent of the direction is positive (Definition 39.30). So the generating function of the values near* $q = 0$ *and that of the constant terms near* $q = \infty$ *are the integrals of one form, whose denominator is the pencil* $\alpha^{-\mathbf{m}_1}\big(\prod_j K_j^{s_{1j}} \Psi_j^{t_{1j}} - q\,\alpha^{\mathbf{m}_1}\big)$ *of Theorem 38.14(ii), over the orthant and over the torus through* $\alpha^\star$*.*

**Proposition 39.32** (The constant terms of a centre ray and its signs)**.** *On the centre ray of a family at the strengths* $\boldsymbol{\sigma}$ *(Definition 38.13), with the tables indexed by* $l$*, for every* $j \in \mathbb{N}$

$$\frac{1}{2^{n-1}} \sum_{\eta} \prod_{l} K_l(\eta)^{2j\sigma_l} = \sum_{v\in 2\mathbb{Z}^{\mathcal{E}}} [\alpha^v]\, f^{-j} \ \geq\ \mathrm{CT}\left(f^{-j}\right),$$

*the sum over the* $2^{n-1}$ *signs* $\eta$ *of Theorem 38.17: the mean of the* $j$*-th powers of the values* $f^{-1}(\eta) = \prod_l K_l(\eta)^{2\sigma_l}$ *at the signs is the sum of the coefficients of* $f^{-j}$ *at the monomials with even exponents.*

**Theorem 39.33** (The closings are a solution)**.** *For a ray as in Definition 39.30,* $\mathsf{C}_j\big((c_e^{(0)} - j\, c_e^{(1)} + \partial_{y_e})g\big) = 0$ *and* $\mathsf{C}_j(fg) = \mathsf{C}_{j-1}(g)$ *for* $j > j(g)$*, and*

$$V_{\mathsf{C}}\Big(\sum_i g_i \otimes r_i\Big)(k) = \sum_i r_i(k)\, \mathsf{C}_{-k}(g_i),$$

*at the integers* $k \leq -\max_i j(g_i)$ *off the poles of the* $r_i$*, is a well defined solution with values in germs at* $-\infty$*. So for masters* $\mathbf{b}$*,* $\mathsf{C}_{j+1}(\mathbf{b}) = \mathbf{B}_{\mathbf{b}}(-j)\, \mathsf{C}_j(\mathbf{b})$ *for all large* $j$*, with* $\mathsf{C}_j(\mathbf{b}) = (\mathsf{C}_j(b_1), \ldots, \mathsf{C}_j(b_{\mathfrak{m}}))$*, and the law of* [1] *gives* $\sum_i p_i(-j)\, \mathsf{C}_{j-i}(1) = 0$ *for all large* $j$*.*

**Proposition 39.34** (The closings are the top principal parts)**.** *Let a ray as in Definition 39.30 have origin* 0*, so that* $\omega_0 = dy$ *and* $\langle g\rangle_k = \int_{y_{e_0}=0} g\, f^k\, dy$*, and* $\kappa_{1,S} > 0$ *for every nonempty proper* $S$*, and put*

$$V_1 = \sum_{\pi} \prod_{l=2}^{n} \kappa_{1,U_l}^{-1} > 0,$$

*the sum over the orders* $\pi$ *of the marks with tails* $U_l$ *(Theorem 17.5(i)). For every insertion* $g$ *the value* $\langle g\rangle_k$ *continues to a meromorphic function of* $k$ *whose poles have order at most* $n-1$*, and for every integer* $j \geq j(g)$

$$\lim_{k\to -j} (k+j)^{n-1} \langle g\rangle_k = V_1\, \mathsf{C}_j(g).$$

**Proposition 39.35** (The low end, for every direction)**.** *Let a ray have* $\kappa_{1,S} > 0$ *for every nonempty proper* $S$*; its letter exponents* $s_{1j}, t_{1j} \in \mathbb{N}$ *may vanish.*

*(i) For an integral origin, every pole of the value of an insertion* $g = \alpha^u \prod_j K_j^{-a_j} \Psi_j^{-b_j}$ *is a* $k$ *with* $\kappa_{0,U} + \kappa_U(g) + k\,\kappa_{1,U} \in \{0, -1, -2, \dots\}$ *for some nonempty proper* $U$ *(Proposition 38.5(ii)), and so lies on one of the lines* $\kappa_{1,U}\, k \in \mathbb{Z}$ *set by the walls of the direction; its order is at most* $n-1$*.*

*(ii) For the origin* 0*, for every insertion* $g$ *and every* $j \in \mathbb{N}$ *with* $f^{-j}g$ *a Laurent polynomial, with* CT *its constant term as in Definition 39.30,*

$$\lim_{k\to -j} (k+j)^{n-1} \langle g\rangle_k = V_1\ \mathrm{CT}\left(f^{-j} g\right), \qquad V_1 = \sum_{\pi} \prod_{l=2}^{n} \kappa_{1,U_l}^{-1}$$

*as in Proposition 39.34; in particular* $\lim_{k\to -j} (k+j)^{n-1} \langle 1\rangle_k = V_1\, \mathrm{CT}(f^{-j})$ *for every* $j \in \mathbb{N}$*.*

**Corollary 39.36** (The law read at the low end)**.** *For a ray as in Proposition 39.34, with the law* $\sum_{i=0}^{h} p_i(k)\, \mathrm{Sh}^i[1] = 0$ *of* $[1]$*, the closings of* 1*,* $\mathsf{C}_j(1) = \mathrm{CT}(f^{-j})$*, satisfy*

$$\sum_{i=0}^{\min(h,j)} p_i(-j)\, \mathsf{C}_{j-i}(1) = 0 \qquad \text{for every } j \geq 0.$$

*On the centre ray of a family with every* $\sigma_j \geq 1$*,* $f^{-1} = \prod_j \left(K_j(\alpha) K_j(\alpha^{-1})\right)^{\sigma_j}$*, and* $\mathsf{C}_j(1)$ *is the constant term of its* $j$*-th power.*

**Definition 39.37** (The low-end coefficients)**.** For a ray with origin 0 and $\kappa_{1,S} > 0$ for every nonempty proper $S$, the *low-end coefficients* $\mathsf{L}_i$, $i \in \mathbb{N}$, are the Laurent coefficients of the values of 1 at $k = 0$,

$$\langle 1\rangle_k = \sum_{i \geq 0} \mathsf{L}_i\, k^{i-n+1} \qquad \text{near } k = 0;$$

$\mathsf{L}_0 = V_1 > 0$.

**Corollary 39.38** (The low-end relation)**.** *Let a ray have origin* 0 *and* $\kappa_{1,S} > 0$ *for every nonempty proper* $S$*, whatever its letter exponents, and let* $p_0, \dots, p_h \in \mathbb{C}[k]$ *have* $\sum_i p_i(k) f^i \in \mathcal{L}$*: the law of* $[1]$*, or a relation given by Proposition 39.6, by a box among them (Theorem 39.21(iv)).*

*(i)* $\sum_i p_i(k) \langle 1\rangle_{k+i} = 0$ *as meromorphic functions of* $k$*, and at every* $k$ *with* $\mathrm{Re}\, k > 0$*; and* $\sum_{i=0}^{\min(h,j)} p_i(-j)\ \mathrm{CT}\left(f^{i-j}\right) = 0$ *for every* $j \in \mathbb{N}$*.*

*(ii)* $p_0$ *vanishes to order at least* $n-1$ *at* $k=0$*, and the* low-end relation

$$[k^{n-1}]p_0\cdot \mathsf{L}_0+\sum_{i=1}^{h} p_i(0)\,\langle 1\rangle_i=0$$

*holds.*

*(iii) If* $\sum_i p_i(k)f^i=\sum_e \big(k\,c_e^{(1)}+\partial_{y_e}\big)G_e$ *is a certificate whose* $G_e$ *have no pole at* $k=0$*, then* $\sum_{i\geq 1} p_i(0)f^i=\sum_e \partial_{y_e}G_e(0)$*, and on the chart the flux of the field* $\big(G_e(0)-G_{e_0}(0)\big)_{e\neq e_0}$ *out of the cube* $\{|y_e|\leq R,\ e\neq e_0\}$ *tends to* $-[k^{n-1}]p_0\cdot\mathsf{L}_0$ *as* $R\to\infty$*: the low-end relation is the flux at infinity of the bottom piece of the certificate (Proposition 38.8).*

**Proposition 39.39** (The low-end coefficients are integrals over the faces of the sectors)**.** *Let a ray have origin* $0$ *and* $\kappa_{1,S}>0$ *for every nonempty proper* $S$*. For an order* $\pi$ *of* $\mathcal{E}$ *with tails* $U_l$ *write* $\kappa_l=\kappa_{1,U_l}$ *and, in the sector coordinates* $q\in[0,1]^{n-1}$ *(Theorem 8.5(ii)),* $f=\prod_{l=2}^{n} q_l^{\kappa_l}\,\varphi_\pi(q)$ *with* $\varphi_\pi=\prod_j k_{j,\pi}^{-s_{1j}}\psi_{j,\pi}^{-t_{1j}}$*, positive and smooth on a neighbourhood of the cube. For* $A\subseteq\{2,\dots,n\}$ *and a function* $h$ *on the cube put*

$$h_{[A]}=\sum_{B\subseteq A}(-1)^{|A\setminus B|}\,h\big|_{q_l=0,\ l\notin B},$$

*a function of* $q_A$ *that vanishes where some* $q_l$*,* $l\in A$*, vanishes. Then*

$$\mathsf{L}_i=\sum_\pi\sum_A\prod_{l\notin A}\kappa_l^{-1}\sum_{m+|\boldsymbol{\gamma}|=i-|A|}\frac{1}{m!}\int_{[0,1]^A}\prod_{l\in A}\frac{(\kappa_l\log q_l)^{\gamma_l}}{\gamma_l!\,q_l}\,\big(\log^m\varphi_\pi\big)_{[A]}\,dq_A,$$

*the second sum over* $A\subseteq\{2,\dots,n\}$ *and the third over* $m\in\mathbb{N}$ *and* $\boldsymbol{\gamma}\in\mathbb{N}^A$*, every integral converging. So* $\mathsf{L}_0=V_1$ *and*

$$\mathsf{L}_1=\sum_\pi\prod_l\kappa_l^{-1}\log\varphi_\pi(0)=-\sum_\pi\prod_l\kappa_l^{-1}\sum_j\big(s_{1j}\log\theta_{j,T^+_{j,\pi}}+t_{1j}\log\theta_{j,T^-_{j,\pi}}\big),$$

*with the greedy bases of Lemma 8.3; up to* $i=n-1$ *only the faces* $A\neq\{2,\dots,n\}$ *of the cube enter,* $1_{[A]}$ *being* $0$ *for* $A\neq\varnothing$*, and the whole cube enters from* $i=n$ *on.*

At the low end of strength the values of a ray from the origin have poles at the nonpositive integers, and the leading coefficient of each pole is one constant, the volume read by the walls of the direction, times the constant term of a Laurent polynomial: the closings. Integration by parts lowers the pole of every monomial but the constant one, so only the constant term survives at the top order. The closings are the values over the compact torus, a cycle on which every negative power of the stride is single-valued, and they obey every relation of the ray; read at $k=-j$, the law of [1] is a recurrence for the constant terms of the powers of $f^{-1}$, from $j=0$ on.

*Remark* 39.40. The constant term of a Laurent polynomial is its mean over every torus $|\alpha_e|=\rho_e$. When the walls of the direction are positive, on the torus through the native gauge $\alpha^\star$ of the direction $|f^{-1}|$ is largest at $\alpha^\star$ alone, and as $j\to\infty$ the closings localize there, with the expansion of $A_\star$ read at $k=-j$ (Theorem 39.57). Where another torus can be deformed into descent cycles of $|f^{-1}|$, the closings decompose over the sources met on the way, with integer multiplicities, a family of native gauges among them. The decomposition does not enlarge what they see: they annihilate exactly what $A_\star$ annihilates (Section 39.10), and what they supply is a cheap stream of equations with the strength of that one solution.

## 39.7 Separating the sources and gluing them

**Proposition 39.41** (The high end reads the formal type)**.** *Let* $\mathbf{B} = \mathbf{B}_\mathbf{b}$ *be regular at* $k = \infty$, $\mathbf{B}(k) = \mathbf{B}_0 + \mathbf{B}_1 k^{-1} + O(k^{-2})$, *and let* $V$ *be a solution of type* $(\lambda, \rho + \mathbb{Z})$ *whose solution vector is* $v_V = \lambda^k k^\rho (a_0 + a_1 k^{-1} + \cdots)$ *with* $a_0 \neq 0$. *Then* $\mathbf{B}_0 a_0 = \lambda^{-1} a_0$, *and for every row vector* $w$ *with* $w\mathbf{B}_0 = \lambda^{-1} w$ *and* $wa_0 \neq 0$, $\rho = -\lambda\, w\mathbf{B}_1 a_0/(wa_0)$. *So the native value of every source with a solution that does not vanish on* $\mathfrak{N}$ *is the reciprocal of an eigenvalue of* $\mathbf{B}(\infty)$*; its exponent, congruent modulo* $\mathbb{Z}$ *to* $-\frac{n-1-d}{2}$ *for a clean source of dimension* $d$ *and to* $-\frac{n-2}{2} - \frac{i+1}{\mu+1}$ *for the solution* $A_{c,i}$ *of an* $A_\mu$ *point, is read from* $\mathbf{B}_1$ *when a left eigenvector of* $\mathbf{B}_0$ *for* $\lambda^{-1}$ *does not annihilate* $a_0$, *as when* $\lambda^{-1}$ *is a simple eigenvalue.*

**Proposition 39.42** (The spectrum at infinite strength; changes of masters)**.** *Let insertions* $b_1 = 1, b_2, \dots, b_\mathfrak{m}$ *have classes that span a submodule* $\mathfrak{N}$, *and let* $\mathbf{B} \in \mathrm{Mat}_\mathfrak{m}(\mathbb{C}(k))$ *have* $[f^{-1} b_i] = \sum_l \mathbf{B}_{il}[b_l]$*: masters with* $\mathbf{B}_\mathbf{b}$, *or the spanning set and the matrix of an accepted box (Theorem 39.21). For a nondegenerate native gauge* $\alpha_c$ *put* $\beta_c = \big(b_l(\alpha_c)\big)_l$, *nonzero as* $b_1 = 1$, *and let* $\hat{A}_c$ *be the normalization of the expansion* $A_c$ *(Definition 39.2).*

*(i) If the vectors* $\beta_c$ *of a set* $C$ *of nondegenerate native gauges span* $\mathbb{C}^\mathfrak{m}$, *the classes of the* $b_l$ *are linearly independent: they are masters of* $\mathfrak{N}$, $\dim_{\mathbb{C}(k)} \mathfrak{N} = \mathfrak{m}$, *and* $\mathbf{B} = \mathbf{B}_\mathbf{b}$.

*(ii) If* $\mathbf{B}$ *is regular at* $k = \infty$, *then* $\mathbf{B}(\infty)\,\beta_c = f(\alpha_c)^{-1}\beta_c$ *for every nondegenerate native gauge* $\alpha_c$*: the reciprocal of every nondegenerate native value is an eigenvalue of* $\mathbf{B}(\infty)$, *with the values of the* $b_l$ *at the gauge as an eigenvector. So gauges of distinct native values have independent vectors* $\beta_c$, *there are at most* $\mathfrak{m}$ *distinct nondegenerate native values, and under (i)* $\mathbf{B}(\infty)$ *is diagonalizable with the eigenvalues* $f(\alpha_c)^{-1}$, $\alpha_c \in C$. *In particular, when two nondegenerate native gauges of distinct native values have proportional vectors* $\beta_c$, $\mathbf{B}$ *is not regular at* $k = \infty$.

*(iii) Let* $T \in GL_\mathfrak{m}(\mathbb{C}(k))$ *make* $\tilde{\mathbf{B}}(k) = T(k-1)\,\mathbf{B}(k)\,T(k)^{-1}$, *the matrix of* $\mathrm{Sh}^{-1}$ *on the elements* $\tilde{b}_i = \sum_l T_{il}(k)[b_l]$, *regular at* $k = \infty$. *Then* $f(\alpha_c)^{-1}$ *is an eigenvalue of* $\tilde{\mathbf{B}}(\infty)$ *for every nondegenerate native gauge* $\alpha_c$, *with the leading coefficient of the series vector* $T(k)\big(\hat{A}_c(b_l)\big)_l$ *as an eigenvector; and if* $T(k-1)T(k)^{-1} \to I$ *as* $k \to \infty$, *the eigenvalues of* $\mathbf{B}(k)$ *tend to those of* $\tilde{\mathbf{B}}(\infty)$, *with their multiplicities.*

*(iv) The condition* $T(k-1)T(k)^{-1} \to I$ *cannot be dropped. For a constant* $\mathbf{B}_\circ = \begin{bmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{bmatrix} \in GL_2(\mathbb{C})$ *with* $x_{12} \neq 0$ *and* $S(k) = \begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix}$, *the matrix* $\mathbf{B}(k) = S(k-1)\,\mathbf{B}_\circ\, S(k)^{-1}$ *is not regular at* $k = \infty$, *the change* $T = S^{-1}$ *makes it* $\mathbf{B}_\circ$, *with* $T(k-1)T(k)^{-1} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, *and at every* $k$ *the eigenvalues of* $\mathbf{B}(k)$ *are the roots of* $\lambda^2 - (x_{11} + x_{22} - x_{12})\lambda + x_{11}x_{22} - x_{12}x_{21}$, *which are not those of* $\mathbf{B}_\circ$.

**Definition 39.43** (Insertion lattice; native algebra; tame high end)**.** Let $A \subset \mathbb{C}(k)$ be the ring of the rational functions regular at $k = \infty$, a discrete valuation ring with the uniformizer $k^{-1}$. For a ray with $\kappa_{1,S} > 0$ for every nonempty proper $S$, its *insertion lattice* is the $A$-span $\Lambda_F = \sum_{g \in \mathbb{C}[X_F]} A\,[g]$ in its module $\mathfrak{M}$ (Definition 39.1); its *native algebra* is $N_F = \mathbb{C}[X_F]/J_F$, with $J_F$ the ideal generated by the defects $c_e^{(1)}$, $e \in \mathcal{E}$, of the direction: the ring of the functions on the critical locus (Definition 39.23) with its scheme structure. The ray is *tame at the high end* when $\Lambda_F$ is finitely generated over $A$.

**Theorem 39.44** (A tame high end)**.** *Let a ray have $\kappa_{1,S} > 0$ for every nonempty proper $S$ and be tame at the high end.*

*(i)* *$\Lambda_F$ is free over $A$ of rank $\dim \mathfrak{M}$, and $\operatorname{Sh} \Lambda_F = \Lambda_F$. There are masters $\mathbf{b}$ among the insertions with $\Lambda_F = \bigoplus_l A\,[b_l]$, and their one-step matrix lies in $GL_{\mathfrak{m}}(A)$.*

*(ii)* *$g \mapsto [g] + k^{-1}\Lambda_F$ induces a surjection $N_F \to \Lambda_F/k^{-1}\Lambda_F$, whose kernel is stable under the multiplication by $f$, and $\mathbf{B}_{\mathbf{b}}(\infty)$ is the matrix of the multiplication by $f^{-1}$ on $\Lambda_F/k^{-1}\Lambda_F$.*

*(iii)* *Every eigenvalue of $\mathbf{B}_{\mathbf{b}}(\infty)$ is the reciprocal native value $f_Z^{-1}$ of a source $Z$. Every nonzero root of the symbol of the law of [1] (Corollary 38.7) is a native value.*

*(iv)* *If every native gauge is nondegenerate, they are $\dim \mathfrak{M}$ in number, $N_F \to \Lambda_F/k^{-1}\Lambda_F$ is an isomorphism, and $\mathbf{B}_{\mathbf{b}}(\infty)$ is diagonalizable with the eigenvalues $f(\alpha_c)^{-1}$, one for each native gauge $\alpha_c$.*

Theorem 39.44 is the converse of Proposition 39.42(ii): there the reciprocal of every nondegenerate native value is an eigenvalue of $\mathbf{B}(\infty)$ in every basis regular at $k = \infty$; here, when the insertion classes span a lattice over $A$, nothing else is. Tameness asks more than masters with $\mathbf{B}$ regular at $k = \infty$: step (ii) needs the classes $[c_e^{(0)} g + \partial_{y_e} g]$ of all insertions to lie in the lattice. Section 39.8 reads the whole spectrum without that hypothesis.

**Theorem 39.45** (Separating the sources)**.** *Let $\mathbf{B} = \mathbf{B}_{\mathbf{b}}$ be regular at $k = \infty$, and let the spectrum of $\mathbf{B}(\infty)$ be the disjoint union of nonempty sets $\Lambda_1, \dots, \Lambda_s$. Over $\mathbb{C}((1/k))$, with the shift $\varsigma$,*

$$\mathfrak{N} \otimes \mathbb{C}((1/k)) = \bigoplus_{a=1}^{s} \hat{\mathfrak{N}}_a,$$

*uniquely, with every $\hat{\mathfrak{N}}_a$ stable under the shift and the matrix of $\operatorname{Sh}^{-1}$ on it regular at $\infty$ in a suitable basis, with value at $\infty$ of spectrum $\Lambda_a$; equivalently there is $T \in GL_{\mathfrak{m}}(\mathbb{C}[[1/k]])$ with $T(k-1)\,\mathbf{B}(k)\,T(k)^{-1}$ block diagonal. A solution of type $(\lambda, \rho + \mathbb{Z})$ with $\lambda^{-1} \in \Lambda_a$ vanishes on every $\hat{\mathfrak{N}}_b$ with $b \neq a$, and every group of symmetries that maps $\mathfrak{N}$ onto itself maps each $\hat{\mathfrak{N}}_a$ onto itself.*

**Corollary 39.46** (Gluing at the two ends)**.** *Let $\mathcal{S}$ consist of formal solutions and the closings $V_{\mathsf{C}}$, and let $\mathfrak{N}_{\mathsf{C}} = \operatorname{Ann}(V_{\mathsf{C}}) \cap \mathfrak{N}$ be the submodule the closings miss. Then $\mathcal{S}$ determines $\mathbf{B}_{\mathbf{b}}$ exactly when no nonzero submodule of $\mathfrak{N}_{\mathsf{C}}$ is annihilated by all the formal solutions in $\mathcal{S}$. The closings alone determine it exactly when $\mathfrak{N}_{\mathsf{C}} = 0$, as they do when $\mathfrak{N}$ is irreducible and $V_{\mathsf{C}}$ does not vanish on it; formal solutions alone determine it when $\mathfrak{m}$ of them are linearly independent over $\mathbb{C}$ on $\mathfrak{N}$, type by type.*

At the high end the module comes apart: over the formal series in $1/k$ it is the direct sum of one piece for each cluster of native values, the pieces are canonical, and every symmetry of the ray maps each piece to itself. This is where distinguishable sources are separated before the symmetric procedure runs: a piece is read with its own solutions, each class at a degenerate point and each cycle on a family counted by its character under the stabilizer, and the formal type of the one-step matrix at infinity, its eigenvalues and its exponents, contains the native values and the exponents of the sources that the module sees. The pieces need not be rational. The rational one-step matrix glues them, and the closings, an exact solution at the other end of strength, supply the glue: they determine the matrix on every submodule that meets their annihilator only in 0, every irreducible submodule on which they do not vanish among them, and formal solutions are needed only on the submodule that the closings miss.

## 39.8 The pencil of a ray

A ray is one function carried by a pencil (Theorem 38.14), and its values and closings are the integrals of one form over the orthant and over the torus (Proposition 39.31). Made exact, this turns the high end into a question in one variable: the module of the ray is the moment side of the module of the fibre integrals of the pencil $f^{-1}$, the symbol of the law is the principal symbol of that module's equation, and the roots are the reciprocals of its singular values, which are native values or limits of the values of $f$ at points that leave the torus with vanishing defect. Throughout, a ray has an integral origin and $\kappa_{1,S} > 0$ for every nonempty proper $S$; write $\varphi = f^{-1}$, a unit of $\mathbb{C}[X_F]$, so that the *pencil* is the map $\varphi : X_F \to \mathbb{C}^\times$, and let $\mathcal{D} = \mathbb{C}[t^{\pm 1}]\langle \partial_t \rangle$.

**Definition 39.47** (The pencil module; singular values; vanishing ranks)**.** On $X_F$ let $\Omega^\bullet[\partial] = \Omega^\bullet_{X_F} \otimes \mathbb{C}[\partial]$ carry the differential $d - \partial\, d\varphi\wedge$. The *pencil module* $\mathcal{G}_F$ is its cohomology in the top degree $n-1$: every top form is $h\,\omega_0$ with $h \in \mathbb{C}[X_F]$, and $\mathcal{G}_F$ is spanned by the classes $h\partial^j$ of $h\,\omega_0\,\partial^j$, linear in $h$, subject to the relations

$$R(e,h,j) = \big(c_e^{(0)} h + \partial_{y_e} h\big)\partial^j - \big(h\,\partial_{y_e}\varphi\big)\partial^{j+1} \qquad (e \in \mathcal{E},\ j \in \mathbb{N}).$$

It is a $\mathcal{D}$-module with $\partial_t \cdot h\partial^j = h\partial^{j+1}$ and $t \cdot h\partial^j = (\varphi h)\partial^j - j\,h\partial^{j-1}$, and its periods are the *fibre integrals* $h\partial^j \mapsto \partial_t^j \int_\gamma h\,\omega_0/d\varphi$ over families of cycles $\gamma$ in the fibres $\{\varphi = t\}$. The *orthant fibre integral* of $h$ is $P_h(t) = \int_{\{\varphi = t\}\cap \mathbb{R}^{\mathcal{E}}_{>0}} h\,\omega_0/d\varphi$ for real $t$ above the least value $A = f(\alpha^\star)^{-1}$ of $\varphi$ on the orthant (Theorem 38.14), and the *torus fibre integral* is $T(t) = \sum_{j\ge 0} \mathsf{C}_j(1)\, t^{-j-1}$ (Proposition 39.31). $\mathcal{D}_F[1] \subseteq \mathcal{G}_F$ is the submodule generated by $1 = 1\partial^0$. A point $t_0 \in \mathbb{C}^\times$ is a *singular value* of a holonomic $\mathcal{D}$-module $N$ when $N$ is not a connection on any neighbourhood of $t_0$, and the *vanishing rank* $r_{t_0}(N)$ is the multiplicity of the conormal line of $t_0$ in the characteristic cycle of $N$; for $N = \mathcal{D}/\mathcal{D}Q$ it is the order of vanishing at $t_0$ of the principal coefficient of $Q$.

**Proposition 39.48** (The ray is the moment side of its pencil)**.** *The assignment* $\Theta(h\partial^j) = (k)_j\,\mathrm{Sh}^j[h]$, $(k)_j = k(k+1)\cdots(k+j-1)$, *induces an isomorphism of* $\mathbb{C}(k)\langle \mathrm{Sh}^{\pm 1}\rangle$*-modules*

$$\mathbb{C}(k) \otimes_{\mathbb{C}[t\partial_t]} \mathcal{G}_F \ \cong\ \mathfrak{M}, \qquad t \mapsto \mathrm{Sh}^{-1}, \quad \partial_t \mapsto k\,\mathrm{Sh}, \quad t\partial_t \mapsto k-1.$$

*For a real origin the values are the moments of the orthant fibre integrals:* $\langle h\rangle_k = \int_A^\infty t^{-k} P_h(t)\,dt$ *for all large* $k$*, with* $\partial_t$ *read in the sense of distributions on* $[A,\infty)$*,* $P_h$ *extended by* $0$ *below* $A$*.*

So the high end of the ray, $k \to \infty$, is the far end of the moment variable of a module in the one variable $t$, and the shift is the multiplication by $t^{-1}$.

**Theorem 39.49** (The symbol is the principal symbol of the pencil)**.** *Let* $L = \sum_{i=0}^{h} p_i(k)\,\mathrm{Sh}^i$ *be the law of* $[1]$*, with* $d$ *the largest degree of the* $p_i$ *and the symbol* $\chi(\lambda) = \sum_i c_i\lambda^i$*,* $c_i$ *the coefficient of* $k^d$ *in* $p_i$ *(Corollary 38.7), and put* $A_L = \sum_i p_i(t\partial_t + 1)\,t^{-i} \in \mathcal{D}$*, of order* $d$ *in* $\partial_t$ *with the principal coefficient* $t^d\chi(1/t)$*.*

*(i) Some* $c(t\partial_t)A_L$*,* $c \neq 0$ *a polynomial, annihilates* $1$ *in* $\mathcal{G}_F$*, and every element of* $\mathcal{D}$ *that annihilates* $1$ *has a principal coefficient divisible by* $t^d\chi(1/t)$ *up to a power of* $t$*.*

*(ii) The nonzero roots of* $\chi$ *are exactly the* $1/t_0$ *with* $t_0 \in \mathbb{C}^\times$ *a singular value of* $\mathcal{D}_F[1]$*.*

*(iii) The multiplicity of the root* $1/t_0$ *is the vanishing rank* $r_{t_0}(\mathcal{D}_F[1])$*.*

*(iv) Every singular value of* $\mathcal{D}_F[1]$ *is a singular value of* $\mathcal{G}_F$*.*

So the roots of the symbol are the singular values of one holonomic module in one variable, the module of the fibre integral of the origin form along the pencil, and their multiplicities are its vanishing ranks. Theorem 39.44 is the case in which those singular values are native values.

**Lemma 39.50** (An orbit counts at most once)**.** *Let a finite group $G$ act on $X_F$ preserving $\varphi$, with $\phi^*\omega_0 = \chi_0(\phi)\,\omega_0$ for $\phi \in G$. Let $t_0 \in \mathbb{C}^\times$ be a value at which there is no asymptotic native gauge (Definition 38.29) of the value $1/t_0$, and all of whose native gauges are isolated and nondegenerate. Then $r_{t_0}(\mathcal{D}_F[1])$ is at most the number of $G$-orbits of these native gauges.*

**Theorem 39.51** (The symbol is read off the sources)**.** *Let a ray have an integral origin, $\kappa_{1,S} > 0$ for every nonempty proper $S$, and every letter with a positive exponent in its direction. Every nonzero root of the symbol of the law of* [1] *is a native value or an asymptotic native value (Definition 38.29). For the centre ray of a plain table at the strength one, an asymptotic native value whose points converge in $X_{\mathcal{E}}$ to a point of the orbit of the flag $\varnothing \subsetneq B \subsetneq \mathcal{E}$, with $B$ a bunch of width one whose room contains $e_0$, is one of the values of Proposition 38.26(ii) and (iii).*

**Corollary 39.52** (Where the spectrum is native)**.** *(i) If a ray as in Theorem 39.51 has no asymptotic native value, every nonzero root of the symbol of the law of* [1] *is a native value.*

*(ii) For a ray of a family of lists as in Proposition 38.30, suppose that for every flag $\mathfrak{f}$ of length at least one the factors of $\Pi_{\mathfrak{f}}$ have nonsingular zero sets in $O_{\mathfrak{f}}$, no two of which meet. Then the ray has no asymptotic native value, and every nonzero root of the symbol is a native value.*

*(iii) On the centre ray of a plain table at the strength one, an asymptotic native gauge whose points converge to the orbit of the flag $\varnothing \subsetneq B \subsetneq \mathcal{E}$ of a bunch $B$ of width one has at the limit both factors of one bottom letter vanishing, or exactly one factor vanishing with a singular zero set; at its limit no factor of $K^{[0]}$ vanishes together with a factor of $\Psi^{[0]}$.*

Factors on two different steps are functions of separate variables, so their zero sets meet as soon as both are nonempty: the hypothesis of (ii) asks that on every flag at most one step carry factors that are not monomials. For $\binom{\mathcal{E}}{1}$, $|\mathcal{E}| \geq 4$, a set $S$ with $|S|, |\mathcal{E} \setminus S| \geq 2$ gives the factor $\alpha_{\mathcal{E}\setminus S}$ of $K$ and the factor $\sum_{e\in S} \alpha_e^{-1}$ of $\Psi$ on two steps, and (ii) does not apply; (iii) excludes these points at the walls, a factor of $K$ with a factor of $\Psi$ giving $\varphi \to 0$, as the roots of its symbol, the sign values (Theorem 42.12), require.

**Theorem 39.53** (Uncarried letters)**.** *Let a ray have an integral origin and $\kappa_{1,S} > 0$ for every nonempty proper $S$. Split its letters as $L_c \sqcup L_u$, those with a positive exponent in its direction and those with the exponent* 0 *there, and in the torus $T = (\mathbb{C}^\times)^{\mathcal{E}}/\mathbb{C}^\times$ put*

$$X' = T \setminus \bigcup_{g\in L_c} \{g = 0\} = \varphi^{-1}(\mathbb{C}^\times), \qquad Y = X' \cap \bigcup_{g\in L_u} \{g = 0\},$$

*so that $X_F = X' \setminus Y$, and stratify $Y$ by the intersections of the zero sets of the uncarried letters, refined to a Whitney stratification. Then every nonzero root of the symbol of the law of* [1] *is the value of $f$ at one of:*

*(a) a native gauge in $X_F$;*

*(b) a critical point of the restriction of $f$ to a stratum of $Y$;*

*(c) an asymptotic native gauge of $X_F$, or a sequence on a stratum of $Y$ that leaves every compact subset of $T$ with the differential of* $\log f$ *along the stratum tending to* 0 *and $f$ tending to a nonzero value.*

Rays of this kind are the values of an insertion with a letter that the direction does not carry, and the rays on a face of the cone of the exponents; the strata of $Y$ carry the native gauges of $X$ on $Y$ of Definition 38.24 in this setting.

**Proposition 39.54** (The far end of the pencil)**.** *Let a ray have origin* $0$*,* $\kappa_{1,S} > 0$ *for every nonempty proper* $S$*, and every letter with a positive exponent in its direction, and let* $T_\infty$ *be the monodromy of the solutions of* $\mathcal{D}_F[1]$ *once around* $t = \infty$*. On the part of the orthant fibre integral* $P = P_1$ *on which* $T_\infty$ *has the eigenvalue* $1$*,*

$$(T_\infty - 1)^{n-2} P = (\pm 2\pi i)^{n-2}\, V_1\, T,$$

*with* $V_1$ *the volume of the direction (Proposition 39.34) and* $T$ *the torus fibre integral. So* $T_\infty$ *has a Jordan block of size* $n-1$ *for the eigenvalue* $1$*, the largest that the fibres of dimension* $n-2$ *allow, and the torus fibre integral, whose coefficients are the closings, spans the invariant line of that block.*

The low end of the ray is thus the point $t = \infty$ of its pencil, where the degeneration is maximal: the torus is the invariant cycle there, the closings are its period, and the volume of the direction is the coefficient that joins the orthant to the torus under the monodromy. The exercises read these statements on lists of a few marks (Section 55.4).

## 39.9 The pencil along a hinge tree

The core of a substitution reads each cell through two letters of the cell, its contraction letter and its deletion letter, the first off a basis of the core and the second on it. Read that way every node of a hinge tree is a polynomial map, the letters of a family are the composite of the node maps applied to its marks, and the pencil module of every ray is the iterated direct image of the modules of the marks along the tree: each vertex applies one direct image to the external product of the modules of its children. Throughout, as in Section 39.8, a ray has an integral origin and $\kappa_{1,S} > 0$ for every nonempty proper $S$, and $\mathcal{D} = \mathbb{C}[t^{\pm 1}]\langle \partial_t \rangle$.

**Definition 39.55** (The homogeneous reading; the data of a vertex)**.** Let $\mathcal{H}$ be a table on a finite set $G$, with the weights of a core (Definition 21.1). For pairs $(u_c, w_c)_{c \in G}$ its *homogeneous reading* is

$$\mathcal{H}^{\hom}(\mathbf{u}, \mathbf{w}) = \sum_T \theta_T \prod_{c \in T} w_c \prod_{c \notin T} u_c,$$

read with the weights of $K$, or with those of $\Psi$ and then written $\mathcal{H}^{\hom}_\Psi$. Let the tables of a family $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ on $\mathcal{E}$ have a common hinge tree $\mathfrak{T}$ rooted at $o$, with the nodes $\mathcal{H}_{v,j}$ (Definition 23.15), and fix a ray of $F$ with the direction $(\mathbf{m}_1; \mathbf{s}_1, \mathbf{t}_1)$ (Definition 38.1); put $|\mathbf{t}_1| = \sum_j t_{1j}$ and $p_e = |\mathbf{t}_1| - m_{1,e}$. The *children* of a vertex $v$ are the marks of $\mathcal{E}_v$ and the vertices directly below $v$, a child read in $\mathcal{H}_{v,j}$ at itself if it is a mark and at the marker of its edge if it is a vertex. A point of $V = (\mathbb{C}^4)^N \times \mathbb{C}^\times$ is written $\zeta = \big((c_j, d_j, c'_j, d'_j)_j, \mu\big)$.

(i) The *data* of a mark $e$ is $\lambda_e(\alpha_e) = \big((1, \alpha_e, 1, \alpha_e^{-1})_j,\ \alpha_e^{p_e}\big)$, a closed embedding of $\mathbb{C}^\times$ into $V$.

(ii) For $v \neq o$, with the hinge $g_v$, the marker of its parent edge, the *node map* $N_v : V^{\text{children}} \to V$ has, for every $j$, the coordinates

$$\big((\mathcal{H}_{v,j}/g_v)^{\hom}(\mathbf{c}, \mathbf{d}),\ (\mathcal{H}_{v,j} \setminus g_v)^{\hom}(\mathbf{c}, \mathbf{d}),\ (\mathcal{H}_{v,j}/g_v)^{\hom}_\Psi(\mathbf{c}', \mathbf{d}'),\ (\mathcal{H}_{v,j} \setminus g_v)^{\hom}_\Psi(\mathbf{c}', \mathbf{d}')\big),$$

the children's coordinates of table $j$ in the place of the pairs and the hinge minors with the inherited weights (Definition 7.3), and the last coordinate $\prod_c \mu_c$. The *data* of $v$ is $\lambda_v = N_v \circ \prod_c \lambda_c$, a morphism from the torus $T_v = (\mathbb{C}^\times)^{B_v}$ of the marks $B_v$ below $v$.

(iii) The *root map* of the ray is

$$R_o\big((\zeta_c)_c\big) = \prod_c \mu_c \prod_j \mathcal{H}^{\hom}_{o,j}(\mathbf{c},\mathbf{d})^{s_{1j}}\, \mathcal{H}^{\hom}_{o,j,\Psi}(\mathbf{c}',\mathbf{d}')^{t_{1j}},$$

over the children of $o$, and $U_o$ is the open set of $V^{\text{children}}$ on which every $\mathcal{H}^{\hom}_{o,j}(\mathbf{c},\mathbf{d})$ and every $\mathcal{H}^{\hom}_{o,j,\Psi}(\mathbf{c}',\mathbf{d}')$ is nonzero.

The reading is multilinear in the pairs, and where every $u_c \neq 0$ it is $\prod_c u_c$ times the letter of $\mathcal{H}$ at the ratios $w_c/u_c$: the hinge values of the cells (Definition 21.1).

**Theorem 39.56** (The pencil along a hinge tree)**.** *In Definition 39.55 write $K^\Psi_{\mathcal{X}}$ for the basis letter of a table $\mathcal{X}$ read with the weights of $\Psi$, so that $\Psi_{\mathcal{X}}(\alpha) = \alpha^{\mathcal{X}} K^\Psi_{\mathcal{X}}(\alpha^{-1})$, $\alpha^{\mathcal{X}}$ the product over the marks of $\mathcal{X}$.*

*(i) For every vertex $v \neq o$ and every $j$, the coordinates of table $j$ of $\lambda_v(\alpha)$ are*

$$\big(K_{\mathcal{A}_{v,j}/g_v}(\alpha),\ K_{\mathcal{A}_{v,j}\backslash g_v}(\alpha),\ K^\Psi_{\mathcal{A}_{v,j}/g_v}(\alpha^{-1}),\ K^\Psi_{\mathcal{A}_{v,j}\backslash g_v}(\alpha^{-1})\big),$$

*those of the subtree cell $\mathcal{A}_{v,j}$, and its last coordinate is $\alpha_{B_v}^{p|_{B_v}}$. At the root $K_j = \mathcal{H}^{\hom}_{o,j}(\mathbf{c},\mathbf{d})$ and $\alpha^{-\mathcal{E}}\Psi_j = \mathcal{H}^{\hom}_{o,j,\Psi}(\mathbf{c}',\mathbf{d}')$ at the data of the children of $o$. So the pencil $\varphi = f^{-1}$ is $R_o \circ \prod_c \lambda_c$ on the torus $(\mathbb{C}^\times)^{\mathcal{E}}$, and the letter complement is the preimage of $U_o$, divided by the scale.*

*(ii) Put $\mathbb{G}_e = H^0(\lambda_e)_+\mathcal{O}_{\mathbb{C}^\times}$ for a mark and $\mathbb{G}_v = H^0(N_v)_+\big(\boxtimes_c \mathbb{G}_c\big)$ for $v \neq o$. Then $\mathbb{G}_v = H^0(\lambda_v)_+\mathcal{O}_{T_v}$, and the pencil module is the iterated direct image*

$$\mathcal{G}_F \cong H^0(R_o|_{U_o})_+\Big(\big(\boxtimes_c \mathbb{G}_c\big)\big|_{U_o}\Big),$$

*the scale of the torus contributing one factor in the top degree.*

*(iii) With the origin $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ put $q_e = m_{0,e} - |\mathbf{t}_0|$ and $\varepsilon_e = [\alpha_e^{q_e} d\alpha_e/\alpha_e] \in \mathbb{G}_e$, and $\mathcal{M}_e = \mathbb{G}_e$. For $v \neq o$ let $\mathcal{M}_v$ be the submodule of $H^0(N_v)_+\big(\boxtimes_c \mathcal{M}_c\big)$ generated by the image $\varepsilon_v$ of $\boxtimes_c \varepsilon_c$, and let $\mathcal{M}_o$ be the submodule of $H^0(R_o|_{U_o})_+\big((\boxtimes_c \mathcal{M}_c)|_{U_o}\big)$ generated by*

$$\varepsilon_o = \prod_j \mathcal{H}^{\hom}_{o,j}(\mathbf{c},\mathbf{d})^{-s_{0j}}\, \mathcal{H}^{\hom}_{o,j,\Psi}(\mathbf{c}',\mathbf{d}')^{-t_{0j}}\ \ \boxtimes_c \varepsilon_c.$$

*Then $\mathcal{D}_F[1]$ is a quotient of $\mathcal{M}_o$, by a map carrying $\varepsilon_o$ to 1; and for $v \neq o$ the submodule of $\mathbb{G}_v$ generated by the class of $\alpha^{\mathbf{q}|_{B_v}} \prod_{e\in B_v} d\alpha_e/\alpha_e$ is a quotient of $\mathcal{M}_v$.*

*(iv) Every singular value of $\mathcal{D}_F[1]$ is a singular value of $\mathcal{M}_o$, with a vanishing rank there at most that of $\mathcal{M}_o$; so every nonzero root $1/t_0$ of the symbol of the law of $[1]$ has $t_0$ a singular value of $\mathcal{M}_o$ and a multiplicity at most $r_{t_0}(\mathcal{M}_o)$. Every composition factor of $\mathcal{G}_F$, and so of $\mathcal{D}_F[g]$ for every insertion $g$, is a composition factor of $H^0(R_o|_{U_o})_+\big((\boxtimes_c S_c)|_{U_o}\big)$ for composition factors $S_c$ of the $\mathbb{G}_c$; every composition factor of $\mathbb{G}_v$ is one of $H^0(N_v)_+\big(\boxtimes_c S_c\big)$ for composition factors $S_c$ of the modules of its children; and the $\mathbb{G}_e$ of the marks are simple. Through the moment dictionary (Proposition 39.48) the composition factors of the module that $[g]$ generates along the ray, whose annihilator is the law of $[g]$, are localizations of composition factors of $\mathcal{D}_F[g]$.*

So the pencil of every ray is built from its marks, vertex by vertex: a mark brings the module of a curve, each vertex applies one direct image, under its node map, to the external product of the modules of its children, and the root closes the product with the exponents of the ray. A vertex enters the pencil only through its node map, whatever lies below it, and its children only through their modules; the composition factors of the pencil, and with them the factors of the law of every insertion, are generated from the marks by these operations. The native gauges are read down the same tree (Theorem 42.24), and the period along it by Theorem 23.17.

## 39.10 What the closings see

**Theorem 39.57** (The closings, the values and the native gauge)**.** *Let a ray have an integral origin and an integral direction in which every letter has a positive exponent, $s_{1j}, t_{1j} \geq 1$, with $\kappa_{1,S} > 0$ for every nonempty proper $S$, and let $\alpha^\star$ be the native gauge of the direction on the positive orthant, where $f$ is largest (Theorem 38.14(i)). Then $\alpha^\star$ is a nondegenerate native gauge of the ray; write $A_\star$ for the expansion over the orthant near it. For every $m \in \mathfrak{M}$ the following are equivalent:*

*(i) the closings vanish on $m$: $V_{\mathsf{C}}(m) = 0$;*

*(ii) the values vanish on $m$: $\langle m \rangle_k = 0$ for all large integers $k$;*

*(iii) the expansion at $\alpha^\star$ vanishes on $m$: $A_\star(m) = 0$;*

*and when they hold, the continuation of $\langle m \rangle_k$ in $k$ vanishes identically. So $\operatorname{Ann}(V_{\mathsf{C}}) = \operatorname{Ann}(\langle\cdot\rangle) = \operatorname{Ann}(A_\star)$, and the closings, the values and $A_\star$ each determine $\mathbf{B}_{\mathbf{b}}$ on exactly the same submodules (Theorem 39.4(iii)). Moreover, for every insertion $g$,*

$$j^{(n-1)/2} f(\alpha^\star)^j \, \mathsf{C}_j(g) \;\sim\; (2\pi)^{-(n-1)} \, \hat{A}_\star(g)\big|_{k=-j} \qquad (j \to \infty).$$

The positive orthant and the torus through $\alpha^\star$ meet only at $\alpha^\star$. The values are read on the orthant, where $f$ is largest at $\alpha^\star$. The closings are means of Laurent polynomials over any torus, and on the torus through $\alpha^\star$ the reciprocal stride has its largest modulus at $\alpha^\star$ alone, because the letters have positive coefficients and the walls of the direction are positive. So the two ends of strength expand at one point, and their expansions are one formal solution, $A_\star$, read as $k \to \infty$ and at $k = -j$. An element on which $A_\star$ vanishes has a density of values that vanishes near the top of its range, and analyticity carries the vanishing to the whole range; and the values reach the closings through their principal parts (Proposition 39.34). The closings see exactly what the values see, and both see exactly what the native gauge of the direction sees.

**Definition 39.58** (The value module; what a solution sees)**.** For a ray with $\kappa_{1,S} > 0$ for every nonempty proper $S$, the *kernel of the values* is $\operatorname{Ann}(\langle\cdot\rangle)$, the submodule of the elements whose values vanish for all large $k$ (Theorem 39.4(ii)), and the *value module* is $\mathfrak{M}/\operatorname{Ann}(\langle\cdot\rangle)$. A solution *sees* a submodule when it does not vanish on it, and a nondegenerate native gauge $\alpha_c$ sees it when $A_c$ does.

**Corollary 39.59** (When the closings miss nothing)**.** *Under the hypotheses of Theorem 39.57, let $\mathfrak{N}$ be a submodule. The closings annihilate no nonzero submodule of $\mathfrak{N}$ exactly when the values vanish on no nonzero element of $\mathfrak{N}$, and exactly when $\alpha^\star$ sees every nonzero submodule of $\mathfrak{N}$. Every symmetry of the ray fixes $\alpha^\star$, so for every finite group $G$ of symmetries the closings, the values and $A_\star$ vanish on $(1 - e_G)\mathfrak{M}$: the condition is to be read on $\mathfrak{M}^G$. Let $\mathbf{B} = \mathbf{B}_{\mathbf{b}}$, for masters of $\mathfrak{N}$, be regular at $k = \infty$ with $1/f(\alpha^\star)$ a simple eigenvalue of $\mathbf{B}(\infty)$, let $\hat{\mathfrak{N}}_\star$ be the line of Theorem 39.45 for that eigenvalue, and let $\alpha^\star$ see $\mathfrak{N}$. Then $\alpha^\star$ sees a submodule $\mathfrak{N}'$ exactly when*

$\hat{\mathfrak{N}}_\star \subseteq \mathfrak{N}' \otimes \mathbb{C}((1/k))$*; when it sees every nonzero submodule,* $\mathfrak{N}$ *has exactly one minimal nonzero submodule; and when* $\mathfrak{N} = \mathfrak{N}_1 \oplus \mathfrak{N}_2$ *with* $\mathfrak{N}_1, \mathfrak{N}_2$ *nonzero submodules, the closings annihilate one of the two.*

**Corollary 39.60** (What the closings determine)**.** *Under the hypotheses of Theorem 39.57, the closings determine the one-step matrix of the value module: for insertions whose classes modulo* $\mathrm{Ann}(\langle\cdot\rangle)$ *form a basis of the image of a submodule, Theorem 39.4(iii) and (iv) hold on the value module with the closings alone. For polynomials* $p_0, \dots, p_h$, $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ *for all large* $k$ *exactly when* $\sum_i p_i(-j)\,\mathsf{C}_{j-i}(1) = 0$ *for all large* $j$*: the values of* 1 *and the closings of* 1 *satisfy the same recurrences, read at* $k$ *and at* $-j$*, and have the same tier (Proposition 38.11).*

**Proposition 39.61** (The closings measure what they miss)**.** *Let* $\mathbf{b}$ *be masters of a submodule* $\mathfrak{N}$*,* $\mathfrak{m}$ *of them, under the hypotheses of Theorem 39.33; let* $d = \dim_{\mathbb{C}(k)}\big(\mathrm{Ann}(V_{\mathsf{C}}) \cap \mathfrak{N}\big)$*, the dimension of the largest submodule of* $\mathfrak{N}$ *that the closings annihilate, and for* $D \geq 0$ *let* $\mathcal{S}_D$ *be the space of pairs* $(P, Q)$ *of a matrix and a scalar of polynomials of degree at most* $D$ *with* $Q(-j)\,\mathsf{C}_{j+1}(\mathbf{b}) = P(-j)\,\mathsf{C}_j(\mathbf{b})$ *for all large* $j$*. Then* $\dim \mathcal{S}_D = (\mathfrak{m}d + 1)D + O(1)$ *as* $D \to \infty$*. When* $d = 0$*,* $\dim \mathcal{S}_D = \max(0, D - D_0 + 1)$ *with* $D_0$ *as in Theorem 39.4(iv); so a single step* $\dim \mathcal{S}_{D+1} - \dim \mathcal{S}_D > 1$ *proves* $d > 0$*.*

The closings therefore certify a one-step matrix by themselves exactly when the native gauge of the direction sees every nonzero submodule. When it does not, the closings show it, by a step of more than one in $\dim \mathcal{S}_D$, and the slope of $\dim \mathcal{S}_D$ is $\mathfrak{m}d + 1$; a slope of one over the degrees computed is evidence that nothing is missed, not a certificate, the degrees of the missed part not being bounded here. What the closings miss is missed by the values as well. It is no defect of the torus: a submodule that the native gauge of the direction does not see is invisible to the integral itself, and an element of it is a relation among values that the relations of the ray do not produce. The symmetries of the ray account for some of these, on $(1 - e_G)\mathfrak{M}$, and so can symmetries that are not maps of its torus at all: when the list is the contraction of an identically self-dual table, the integral over the contracted scale carries the module of the larger table onto that of the ray, and the automorphisms of the larger table act through it.

**Proposition 39.62** (A table one mark larger)**.** *Let* $(\mathcal{B}', \theta')$ *be a connected identically self-dual table on* $\mathcal{E}' = \mathcal{E} \sqcup \{g\}$*,* $|\mathcal{E}| \geq 2$ *(Corollary 23.12(iii)), and* $(\mathcal{B}, \theta)$ *its contraction at* $g$ *(Definition 7.3), with letters* $K, \Psi$*; then* $(\mathcal{B}, \theta)$ *is connected, and the letter of* $\mathcal{B}'$ *is* $K' = \Psi + \alpha_g K$*. Let* $F'$ *and* $F$ *be the two tables at the strength* 1*, read on their centre rays (Definition 38.13),* $F'$ *on a chart* $y_{e_0} = 0$ *with* $e_0 \in \mathcal{E}$*; their strides are* $f' = \alpha^{\mathbf{1}}\alpha_g/K'^2$ *and* $f = \alpha^{\mathbf{1}}/(K\Psi)$*, their modules* $\mathfrak{M}'$ *and* $\mathfrak{M}$*. For a Laurent polynomial* $h$ *on* $\mathcal{E}$*,* $a \in \mathbb{Z}$ *and* $j \in \mathbb{N}$ *with* $\alpha_g^a K'^{-j} h$ *of degree* 0*, put*

$$\pi_*\big(\alpha_g^a K'^{-j} h\big) = r_{a,j}(k)\Big(\frac{\Psi}{K}\Big)^a \Psi^{-j} h, \qquad r_{a,j}(k) = \frac{\Gamma(k+a)\,\Gamma(k+j-a)\,\Gamma(2k)}{\Gamma(k)^2\,\Gamma(2k+j)} \in \mathbb{Q}(k).$$

*(i) For every insertion* $g'$ *of* $F'$*, at the points of the positive orthant and for all large* $k$*,* $\int_0^\infty g' f'^k \, d\alpha_g/\alpha_g = \Gamma_{\!1}^{\,2}\!\left[\begin{matrix} k,\ k \\ 2k \end{matrix}\right] f^k\, \pi_* g'$*. So* $\langle g' \rangle'_k = \Gamma_{\!1}^{\,2}\!\left[\begin{matrix} k,\ k \\ 2k \end{matrix}\right] \langle \pi_* g' \rangle_k$ *for all large* $k$*, and* $\mathsf{C}'_j(g') = \binom{2j}{j}\,\mathsf{C}_j\big(\pi_* g'|_{k=-j}\big)$ *for all large* $j$*. The native gauges of* $F'$ *are the points* $\alpha_{c'} = \big(\alpha_c, \Psi(\alpha_c)/K(\alpha_c)\big)$ *over the native gauges* $\alpha_c$ *of* $F$*, with* $f'(\alpha_{c'}) = f(\alpha_c)/4$*, and* $\alpha_{c'}$ *is non-degenerate exactly when* $\alpha_c$ *is; then* $A'_{c'}(g') = \Gamma_{\!1}^{\,2}\!\left[\begin{matrix} k,\ k \\ 2k \end{matrix}\right] A_c(\pi_* g')$*, the stacked Gamma read as its expansion at* $k = \infty$ *and the disc at* $\alpha_{c'}$ *taken over the disc at* $\alpha_c$*.*

*(ii)* *$\pi_*$ induces a bijection $\mathfrak{M}' \to \mathfrak{M}$, with $\pi_* \operatorname{Sh}' = \frac{k}{2(2k+1)} \operatorname{Sh} \pi_*$.*

*(iii)* *Every automorphism $\beta$ of $(\mathcal{B}', \theta')$ acts on $\mathfrak{M}$ by $T_\beta = \pi_* \phi_\beta^* \pi_*^{-1}$, which commutes with* Sh*; $T_\beta = \phi_\beta^*$ when $\beta(g) = g$, and when the transposition $\beta$ of a mark $e \in \mathcal{E}$ and $g$ is an automorphism,*

$$T_\beta[\alpha^u] = \frac{\Gamma(k+u_e)\,\Gamma(k-u_e)}{\Gamma(k)^2} \Big[\alpha^{u-u_e \mathbf{1}_e} \Big(\frac{\Psi}{K}\Big)^{u_e}\Big].$$

*With the Cremona, which $\pi_*$ carries to the Cremona, the $T_\beta$ generate a finite group $G'$.*

*(iv)* *The values, the closings and $A_\star$, at the native gauge $\alpha^\star = \mathbf{1}$, are invariant under $G'$ and vanish on $(1 - e_{G'})\mathfrak{M}$. For a group $G \subseteq G'$ of symmetries of $F$, $\mathfrak{M}^G = \mathfrak{M}^{G'} \oplus (1 - e_{G'})\mathfrak{M}^G$ as submodules, and $\pi_*$ maps the invariant module of $F'$ under its automorphisms and the Cremona onto $\mathfrak{M}^{G'}$.*

**Corollary 39.63** (The cousins of a fused table)**.** *Let $(\mathcal{B}, \theta)$ be a connected fusable table on $n$ marks, of the fusion width $c \geq 1$, with $n + c \geq 3$, and $(\mathcal{B}^\sharp, \theta^\sharp)$ its Foreman fusion (Definition 30.4). Then $\mathcal{B}^\sharp$ is connected, and Proposition 39.62 applies to $\mathcal{B}^\sharp$ and each of its cousins (Proposition 30.20): the automorphisms of $\mathcal{B}^\sharp$ act on the module of the centre ray of every cousin.*

The scale of $g$ enters the letter of the larger table linearly, so the integral over it is a Beta corner, and it carries the centre of $F'$ onto the centre of $F$ insertion by insertion: a power of $\alpha_g$ becomes the same power of $\Psi/K$, the ratio at which the scale of $g$ is native, times a rational function of $k$. The relations at the scale of $g$ are exactly what that integral forgets, so the two modules are one, with the shift twisted by the Beta corner. An automorphism of $\mathcal{B}'$ that moves $g$ is no symmetry of $F$, nor a map of its torus: it moves insertions, not points, and on native gauges it is the lift $\alpha_c \mapsto \alpha_{c'}$ followed by the permutation and the projection. The values, the closings and the expansion at $\alpha^\star$ are read on the orthant, the torus and the point $\mathbf{1}$ of $F'$, which every automorphism keeps; so they see only the part of $\mathfrak{M}$ that $G'$ fixes, and when $\mathfrak{M}^G$ is larger than $\mathfrak{M}^{G'}$ the closings miss the difference, with every symmetry of $F$ in $G$. A sign $\eta$ of $F$ lifts to the sign $(\eta, \eta^{\mathcal{E}})$ of $F'$, since $\Psi(\eta) = \eta^{\mathcal{E}} K(\eta)$, with $f(\eta) = 4/K'(\eta, \eta^{\mathcal{E}})^2$; signs that the automorphisms of $\mathcal{B}'$ carry to one another have one native value. Proposition 39.64 measures what is missed in this way: an orbit of nondegenerate native gauges of $F'$ that the symmetries of $F$ split into several orbits gives, orbit against orbit, solutions that vanish on $\mathfrak{M}^{G'}$ and not on $\mathfrak{M}^G$.

**Proposition 39.64** (What a table one mark larger hides from the closings)**.** *Let $(\mathcal{B}', \theta')$, $g$, $F'$, $F$, $\pi_*$ and $G'$ be as in Proposition 39.62. Let $\Gamma'$ be the group of symmetries of the centre ray of $F'$ generated by the maps $\phi_\beta$, $\beta$ an automorphism of $(\mathcal{B}', \theta')$, and the Cremona, and $G$ the group of symmetries of $F$ generated by the maps $\phi_\beta$, $\beta$ an automorphism of $(\mathcal{B}', \theta')$ with $\beta(g) = g$ read on $\mathcal{E}$, and the Cremona, so that $G \subseteq G'$ (Proposition 39.62(iii)). A native gauge $\alpha_c$ of $F$ lies* below *the native gauge $\alpha_{c'} = \big(\alpha_c, \Psi(\alpha_c)/K(\alpha_c)\big)$ of $F'$ (Proposition 39.62(i)).*

*(i)* *Let $O'_1, \dots, O'_q$ be distinct orbits of $\Gamma'$ of nondegenerate native gauges of $F'$, and let the native gauges of $F$ below $O'_a$ form $s_a$ orbits of $G$. Then*

$$\dim_{\mathbb{C}(k)} (1 - e_{G'})\mathfrak{M}^G \;\geq\; \sum_{a=1}^{q} (s_a - 1),$$

*and the closings, the values and $A_\star$ vanish on the submodule $(1 - e_{G'})\mathfrak{M}^G$. For an orbit $O'$ fix an expansion $A'_{c'_0}$ at one of its points, put $A'_{\gamma(c'_0)} = A'_{c'_0} \circ \gamma^*$ for $\gamma \in \Gamma'$, and for $\alpha_c$ below $O'$ put $A_c = \Gamma^{\,2}_{\,1}\Big[\begin{smallmatrix} k, & k \\ & 2k \end{smallmatrix}\Big]^{-1} A'_{c'} \circ \pi_*^{-1}$, the stacked Gamma read as its expansion at $k = \infty$. If*

*$O_1$ and $O_2$ are two orbits of $G$ below $O'$, the solution $|O_1|^{-1}\sum_{c\in O_1}A_c - |O_2|^{-1}\sum_{c\in O_2}A_c$ is invariant under $G$, vanishes on $\mathfrak{M}^{G'}$ and does not vanish on $(1-e_{G'})\mathfrak{M}^G$.*

*(ii) For $r\geq 2$ let $\mathcal{B}'=\binom{[2r+2]}{r+1}$ with unit weights and $g=2r+2$, so that $F$ is $\binom{[2r+1]}{r}$ read on its centre ray at the strength 1; $G$ is then the group generated by the symmetries of the four kinds of that ray (Proposition 39.11): the maps $\phi_\beta$, $\beta$ a bijection of $[2r+1]$, and the Cremona. With $C_r=\frac{1}{r+1}\binom{2r}{r}$, the signs of $F$ with one and with two scales $-1$, $2r+1$ and $\binom{2r+1}{2}$ of them, are nondegenerate native gauges of the native value $C_r^{-2}$ and form two orbits of $G$, and the native gauges above them are the $\binom{2r+2}{2}$ signs of $F'$ with two scales $-1$, one orbit of $\Gamma'$. At the sign of $F'$ with the scales $-1$ on $D=\{a_1,a_2\}$, with $C=[2r+2]\setminus D$, the Hessian $H'$ of $\log f'$ is such that $-H'$ has, modulo $\mathbf{1}$, the eigenvalue $r+1$ on $\mathbf{1}_{a_1}-\mathbf{1}_{a_2}$, the eigenvalue $\frac{r+1}{2r-1}$ on every $\mathbf{1}_c-\mathbf{1}_{c'}$ with $c,c'\in C$, and the eigenvalue $-(r+1)$ on $r\mathbf{1}_D-\mathbf{1}_C$; at the signs of $F$ with one and with two scales $-1$, on a chart, $\det(-H)=-(r+1)^{2r}/(2r-1)^{2r-1}$. So $(1-e_{G'})\mathfrak{M}^G\neq 0$: at the centre of every $\binom{[2r+1]}{r}$, $r\geq 2$, the closings miss a nonzero submodule of $\mathfrak{M}^G$, with every symmetry of the four kinds in $G$.*

The two kinds of signs of $\binom{[2r+1]}{r}$ share their native value because the signs above them are one orbit of the symmetric group of $\binom{[2r+2]}{r+1}$, while the symmetries of the ray keep the two kinds apart; the mean of the solutions over the one kind less the mean over the other is a solution that sees what the larger table hides. Exercise 54.20 reads the small centres with these statements; at $r=2$ the part of the span of a box that the larger table hides from the closings is a line.

## 39.11 Wheels: the letters as a holonomy

**Proposition 39.65** (The holonomy of a wheel)**.** *For $p\geq 3$ let $W_p$ be the list of the wheel graph: a hub and rim vertices $1,\dots,p$ in cyclic order, the marks the spokes $s_i$, joining the hub to $i$, and the rims $r_i$, joining $i$ to $i+1$, indices modulo $p$; its bases are the spanning trees, of rank $p$ on $n=2p$ marks. With*

$$\mathbf{U}(a)=\begin{bmatrix}1 & a\\ 0 & 1\end{bmatrix},\qquad \mathbf{L}(b)=\begin{bmatrix}1 & 0\\ b & 1\end{bmatrix},\qquad \mathbf{h}(\alpha)=\prod_{i=1}^{p}\mathbf{U}(\alpha_{s_i})\,\mathbf{L}(\alpha_{r_i}^{-1})\in SL_2,$$

*the product taken in the order $i=1,\dots,p$,*

$$K(\alpha)=\prod_i \alpha_{r_i}\big(\operatorname{tr}\mathbf{h}(\alpha)-2\big),\qquad \Psi(\alpha)=\prod_i \alpha_{s_i}\big(\operatorname{tr}\mathbf{h}(\alpha^{-1})-2\big).$$

*The bijection $x$ with $x(s_i)=r_{p+1-i}$ and $x(r_i)=s_{p+1-i}$ has $\operatorname{tr}\mathbf{h}(\alpha\circ x)=\operatorname{tr}\mathbf{h}(\alpha^{-1})$, so $K\circ\phi_x=\Psi$ and $x$ exchanges $W_p$. At the table's gauge $\mathbf{h}(\mathbf{1})=\begin{bmatrix}2 & 1\\ 1 & 1\end{bmatrix}^p$, whose eigenvalues are $\lambda_\pm^p$ with $\lambda_\pm=\frac{3\pm\sqrt{5}}{2}$, the squares of the golden ratio and of its inverse; so $K(\mathbf{1})=\lambda_+^p+\lambda_-^p-2$, the Lucas number of index $2p$ less 2.*

**Proposition 39.66** (The centre ray of a wheel)**.** *Put $\mathfrak{d}(\alpha)=\operatorname{tr}\mathbf{h}(\alpha)-2$ and $\mathfrak{d}^*(\alpha)=\operatorname{tr}\mathbf{h}(\alpha^{-1})-2=\mathfrak{d}(\alpha\circ x)$. The centre ray of $W_p$ at the strength $\sigma$ has the stride $f=(\mathfrak{d}\,\mathfrak{d}^*)^{-\sigma}$, and $X_F=\{\mathfrak{d}\,\mathfrak{d}^*\neq 0\}$.*

*(i) Its native gauges are the points of $X_F$ with $\mathfrak{d}^*\,d\mathfrak{d}+\mathfrak{d}\,d\mathfrak{d}^*=0$. At every one $d\mathfrak{d}$ and $d\mathfrak{d}^*$ are linearly dependent: the native gauges lie on the critical locus of the pair of traces $(\mathfrak{d},\mathfrak{d}^*)$, where its differential has rank at most one. The native value is $(\mathfrak{d}\,\mathfrak{d}^*)^{-\sigma}$.*

*(ii) At a sign $\eta$, $\mathfrak{d}^*(\eta) = \mathfrak{d}(\eta)$ and the native value is $\mathfrak{d}(\eta)^{-2\sigma}$; at the table's gauge it is $(\lambda_+^p + \lambda_-^p - 2)^{-2\sigma}$.*

*(iii) Every point with $\mathbf{h}(\alpha) = -I$ has $d\mathfrak{d} = 0$. The* torsion stratum *$\{\mathbf{h}(\alpha) = \mathbf{h}(\alpha^{-1}) = -I\}$ lies in $X_F$ and consists of native gauges of native value $16^{-\sigma}$ at which $d\mathfrak{d} = d\mathfrak{d}^* = 0$; it is cut out of the torus of dimension $2p-1$ by six equations, so that each of its components has dimension at least $2p-7$.*

**Proposition 39.67** (The classes of a wheel)**.** *At the table's gauge the occupation of a spoke is*

$$R_{\mathrm{s}} = \frac{\lambda_+^p - \lambda_-^p}{\sqrt{5}\left(\lambda_+^p + \lambda_-^p - 2\right)},$$

*equal to $\frac{1}{2}$ exactly for $p = 3$; the occupation of a rim is $R_{\mathrm{r}} = 1 - R_{\mathrm{s}}$, and the masses of $\Upsilon_{\sigma,\delta;\mathbf{1}}$ are $\sigma + \delta(2R_{\mathrm{s}} - 1)$ on the spokes and $\sigma - \delta(2R_{\mathrm{s}} - 1)$ on the rims. $W_p$ is an atom at the centre. For $p \geq 4$ its classes at every lean are the spokes and the rims, which the reversed exchange $\bar{\phi}_x$, a symmetry of the integrand at every lean (Theorem 39.13), interchanges; $W_3 = M(K_4)$ is an atom at every lean.*

*Remark* 39.68. The uniform masses $\sigma\mathbf{1}$ lie on the balance wall of $W_p$ at every lean, since $2r = n$, and they meet the condition of Proposition 39.11 for $\bar{\phi}_x$, which therefore keeps the integrand of every such realization; its native gauge is constant on the spokes and on the rims. For $p = 3$ the uniform masses are the masses of the table's gauge at every lean, $R_{\mathrm{s}}$ being $\frac{1}{2}$, and the native gauge stays at $\mathbf{1}$; for $p \geq 4$ they are not, and the native gauge leaves $\mathbf{1}$ as soon as the realization leans. For $W_4$, at the gauge $c$ on the spokes and 1 on the rims, $K = c(c+2)^2(c+4)$ and $\Psi = (2c+1)^2(4c+1)$. At the full basis lean nativity asks every occupation to be $\frac{1}{2}$, which on a spoke reads $c^2+2c-4 = 0$: the native gauge on the orthant has spokes $\sqrt{5}-1$, twice the inverse of the golden ratio, against rims 1, and at the full cobasis lean, where the condition reads $4c^2 - 2c - 1 = 0$, spokes $\frac{\sqrt{5}+1}{4}$, half the golden ratio, its reciprocal. The other roots, $-\sqrt{5}-1$ and $\frac{1-\sqrt{5}}{4}$, are native gauges as well, with negative spokes and in $X_F$: sources off the orthant. At the basis lean the integrand is $c^{4\sigma}K^{-2\sigma} = \big(c/((c+2)^2(c+4))\big)^{2\sigma}$, whose base is $\frac{5\sqrt{5}-11}{8}$ at the positive root and $-\frac{5\sqrt{5}+11}{8}$ at the negative one, a quarter of the inverse fifth power of the golden ratio and minus a quarter of its fifth power; the cobasis lean gives the same two numbers. Off the centre the gauge on the orthant is not proportional to $\mathbf{1}$, so the realization is not an atom (Theorem 11.4); its classes are the spokes and the rims, and $\bar{\phi}_x$ interchanges them. The torsion stratum of $W_4$ is a union of curves.

The letters of a wheel are traces of one product of unipotent matrices, the holonomy of a transfer around the rim, and the planar self-duality of the wheel is the transpose of that product. On the centre ray the stride is the inverse product of the two traces, at $\alpha$ and at $\alpha^{-1}$, so a native gauge lies on the critical locus of the pair of traces, and wherever both holonomies are $-I$ both traces are critical at once: a stratum of native gauges of native value $16^{-\sigma}$, of dimension at least $2p-7$ wherever it is not empty. At a sign $\eta$ the two traces agree, in an integer $\mathfrak{d}(\eta)$, and the native value is $\mathfrak{d}(\eta)^{-2\sigma}$; at the table's gauge the holonomy is the $p$-th power of the step $\begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, whose eigenvalues are the squares of the golden ratio and of its inverse. The wheel is an atom at the centre and splits into spokes and rims at every lean once $p \geq 4$, the reversed exchange interchanging them; with uniform masses its native gauge then leaves the table's gauge as soon as the realization leans, and for $W_4$ at a full lean the ratio of the scales of spokes and rims is twice the inverse of the golden ratio or half the golden ratio.

## 39.12 The procedure

The statements of this chapter assemble into a procedure that reads the one-step matrix and the law of [1] of a ray from its two ends, with no box (Definition 39.15) reduced.

(1) *Symmetries.* From the list read the automorphisms and the exchanges, and keep the maps of Proposition 39.11 that are symmetries of the ray; at a lean of a self-dual family at the table's gauge these include the reversed exchanges (Theorem 39.13). Work in $\mathfrak{M}^G$ with invariant insertions (Theorem 39.12). When the list is the contraction of an identically self-dual table at a mark, the automorphisms of that table act as well, through the integral over the scale of the mark, and the values live on the part they fix, which the centre of the larger table reads directly (Proposition 39.62).

(2) *Sources.* Read the critical locus of the stride modulo $G$: orbits of nondegenerate native gauges, one solution each; $A_\mu$ points, with the classes of trivial character (Theorems 39.26 and 39.29); clean families, with their averaged cycles (Theorem 39.24, Corollary 39.25, Proposition 39.28). Their native values and exponents enter the formal type of the one-step matrix at infinity (Proposition 39.41).

(3) *Closings.* Compute the closings of the candidate masters, constant terms of Laurent polynomials (Definition 39.30).

(4) *Masters.* Choose invariant insertions whose solution vectors over the supplied sources and the closings are linearly independent; there are at most $\dim \mathfrak{M}^G$ of them, and the count of Theorem 39.29 bounds $\dim \mathfrak{M}^G$ below.

(5) *The one-step matrix.* Solve the homogeneous linear equations of Theorem 39.4(iv) at increasing degree $D$. When the supplied solutions annihilate no submodule, the first degree with a nonzero solution is $D_0$, the solution is unique up to a constant factor, and it is $\mathbf{B_b}$.

(6) *The law.* The law of [1] is the least relation among $[1], \mathrm{Sh}[1], \dots$, computed with $\mathbf{B_b}$; on a ray from the origin it is read again on the closings of 1 at $-j$ (Corollary 39.36).

A case is certified when the supplied solutions annihilate no submodule: by Corollary 39.46 this holds when $\mathfrak{m}$ supplied formal solutions are linearly independent over $\mathbb{C}$ on $\mathfrak{M}^G$ type by type, a full set from the high end; and when the walls of the direction are positive the closings alone certify exactly when the native gauge of the direction sees every nonzero submodule of $\mathfrak{M}^G$ (Corollary 39.59), as when $\mathfrak{M}^G$ is irreducible and they do not vanish on it. The closings show it when they miss something: the dimension of the solutions of their equations then has slope $\mathfrak{m}d + 1$, with $d$ the dimension of the largest missed submodule $\mathrm{Ann}(V_{\mathsf{C}}) \cap \mathfrak{M}^G$, and a single step of more than one proves $d > 0$ (Proposition 39.61); a slope of one over the degrees computed is evidence that nothing is missed, not a certificate. When they miss something they still give the one-step matrix of the value module and the law of the values (Corollary 39.60), and the missing part is read from sources whose solutions see it: when it comes from a table one mark larger, the centre of that table reads $\mathfrak{M}^{G'}$ (Proposition 39.62), and the differences of the solutions over the orbits of native gauges that the symmetries of the ray split do not vanish on the rest (Proposition 39.64). With fewer formal solutions and no such knowledge of $\mathfrak{M}^G$, a unique solution at the least degree that also meets equations held out is evidence, not a certificate: a submodule annihilated by every supplied solution would leave a different matrix that meets all of them. Masters that span $\mathfrak{M}^G$ come from a box when every letter exponent of the direction is positive: the spanning set of a box accepted at $\mathfrak{s}$ spans the classes of the invariant Laurent combinations of wall degree at most $\mathfrak{s}$ (Theorem 39.21(i)), which increase with $\mathfrak{s}$, and some box is accepted with a spanning set whose

classes span $\mathfrak{M}^G$ (Theorem 39.22). That spanning set is a set of masters of $\mathfrak{M}^G$ when it is independent: when as many solutions as it has elements are linearly independent on its span, type by type (Lemma 39.3), or when its values at nondegenerate native gauges span $\mathbb{C}^{\mathfrak{m}}$ (Proposition 39.42(i)).

# Chapter 40

# The stride's law and the gauge end

The values of a ray with real origin and positive walls are the moments of one measure, the stride's law: the image of the origin form on the positive part of the chart under the stride. It is carried by $(0, f(\alpha^\star)]$, and the values are read from it three ways, as a Trailmoment on a line, through one Gramma on the Laplace side, and as the log-partition of the exponential family that the Breta shape laws form along the ray. Its two ends are the two ends of strength: near 0 its mass grows like a power of the logarithm, with the volume of the direction as coefficient, and at the native value $f(\alpha^\star)$ it ends in a power $(f(\alpha^\star) - t)^{\frac{n-3}{2}}$ times a function holomorphic there, whose value is read off the Hessian at the native gauge. Its density is real-analytic in between. Every recurrence of the values is a differential equation for the density, the recurrence read backwards, and holds at every strength once it holds at the large integers. At the native value of the direction, when it is a simple root of the symbol, the germs of solutions that the equation lets end there form a line; the density spans it, its exponent is forced to $\frac{n-3}{2}$, and its continuation along the open support is the density: so the recurrence, the native gauge of the direction and the Hessian there give the values and their derivatives in the strength exactly, with no datum from the low end and no inequality. At a centre the same native gauge lies on the unit torus, where the stride is real and at least $f(\alpha^\star)$; the torus value continues the closings in the strength, and the densities of the chart and of the torus at the native value are the two cut solutions of one function, on its two sides.

## 40.1 The stride's law

**Definition 40.1** (The stride's law)**.** Let a ray of a family on $n \geq 2$ marks have a real origin and $\kappa_{1,S} > 0$ for every nonempty proper $S$ (Definition 38.1). On the positive part of the chart, $\alpha_{e_0} = 1$ and $y = \log \alpha \in \mathbb{R}^{\mathcal{E}\setminus e_0}$, the origin form $\omega_0 = (\omega_0/dy)\, dy$ is a positive measure, its density $\alpha^{\mathbf{m}_0} \prod_j K_j^{-s_{0j}} \Psi_j^{-t_{0j}}$ taken with the principal powers. The *stride's law* of the ray is its image under the stride,

$$\mu_f(A) = \int_{\{y:\ f(e^y) \in A\}} \omega_0 \qquad (A \subseteq (0,\infty) \text{ a Borel set}),$$

a measure on $(0,\infty)$, and $\varrho_f$ is its density where it has one. For the origin 0, $\omega_0 = dy$, and $\mu_f$ is the image under $f$ of the Lebesgue input of the chart (Definition 5.4).

**Proposition 40.2** (A ray is one Trailmoment on a line and one Gramma)**.** *Let a ray be as in Definition 40.1, let $\alpha^\star$ be the native gauge of its direction on the positive orthant, $\alpha^\star_{e_0} = 1$ (Theorem 38.14), and write in this statement $f^\star = f(\alpha^\star)$.*

*(i)* *$\mu_f$ is carried by $(0, f^\star]$, it is finite on $[t, f^\star]$ for every $t > 0$, and at every complex $k$ at which $\langle 1\rangle_k$ converges*

$$\langle 1\rangle_k = \int_{(0,f^\star]} t^k \, \mu_f(dt) = \big(\mathsf{T}_{k;(1)}[t^{-1}]\,\mu_f\big)\big((0,\infty)\big) = (\mathsf{T}_{k;(1)}[x]\,\iota_*\mu_f)((0,\infty)), \qquad \iota(t) = t^{-1}:$$

*the values are the Trailmoments on a line of index $k$, one unit exponent and the coefficient function $x$ of one input, the image of $\mu_f$ under the reciprocal.*

*(ii)* *Let the origin carry no letter exponents, $\mathbf{s}_0 = \mathbf{t}_0 = \mathbf{0}$, so that $\omega_0 = \alpha^{\mathbf{m}_0} dy$, and write $f^{-1} = \sum_i w_i\, \alpha^{\mathbf{v}_i}$ on the chart, a Laurent polynomial in the scales $\alpha_e$, $e \neq e_0$, with positive coefficients $w_i$ and exponents $\mathbf{v}_i \in \mathbb{Z}^{\mathcal{E}\setminus e_0}$; put $\mathbf{v} = (\mathbf{v}_i)_i$ and $\mathbf{w} = (w_i)_i$. The cone of $\mathbf{v}$ is $\mathbb{R}^{\mathcal{E}\setminus e_0}$, so the Gramma $\mathsf{G}_{\nu;\mathbf{v}}(u\mathbf{w})$ is entire in $\nu$ (Theorem 2.2), and at every $k$ with $\operatorname{Re} k > 0$ at which $\langle 1\rangle_k$ converges*

$$\Gamma(k)\,\langle 1\rangle_k = \int_0^\infty u^{k-1}\, \mathsf{G}_{\mathbf{m}_0|_{\mathcal{E}\setminus e_0};\,\mathbf{v}}(u\,\mathbf{w})\, du,$$

*the Laplace side (Theorem 5.11) of (i), whose Laplace transform beyond the cut 0 is the Gramma of the reciprocal stride.*

*(iii)* *For real $k$ at which $\langle 1\rangle_k$ converges, let $U \sim \mathrm{Breta}^\circ(F; \mathbf{m}_0 + k\mathbf{m}_1; \mathbf{s}_0 + k\mathbf{s}_1, \mathbf{t}_0 + k\mathbf{t}_1)$ (Definition 8.10). The law of $f(U)$ is $t^k \mu_f(dt)/\langle 1\rangle_k$: along the ray the shape laws are the exponential family with the sufficient statistic $\log f$, the base measure $\mu_f$ and the log-partition $\log\langle 1\rangle_k$, so that*

$$\frac{d}{dk}\log\langle 1\rangle_k = \mathbb{E}\big[\log f(U)\big], \qquad \frac{d^2}{dk^2}\log\langle 1\rangle_k = \mathrm{Var}\,\big[\log f(U)\big] > 0.$$

**Proposition 40.3** (The two ends of the stride's law)**.** *Let a ray be as in Definition 40.1, with $\alpha^\star$ as in Proposition 40.2 and $H$ the Hessian of $-\log f$ at $\log\alpha^\star$ on the chart (Theorem 38.15); write in this statement $f^\star = f(\alpha^\star)$ and $w^\star = (\omega_0/dy)(\alpha^\star)$.*

*(i)* The open support. *$\mu_f$ has a density $\varrho_f$, real-analytic and positive on $(0, f^\star)$.*

*(ii)* The native value. *Near $f^\star$, $\varrho_f(t) = (f^\star - t)^{\frac{n-3}{2}} g(f^\star - t)$ for $t < f^\star$, with $g$ holomorphic at 0 and*

$$g(0) = \frac{w^\star}{\Gamma(\frac{n-1}{2})\sqrt{\det H}}\Big(\frac{2\pi}{f^\star}\Big)^{\frac{n-1}{2}}.$$

*(iii)* The low end. *Let the origin be 0, let $\kappa$ be the wall at a tilt of the question of the direction (Definitions 17.1 and 17.3), read at $y$, and put $C_- = \prod_j K_j(\mathbf{1})^{-s_{1j}-t_{1j}}$, $C_+ = \prod_j \theta_{j,\min}^{-s_{1j}-t_{1j}}$, $\theta_{j,\min}$ the least weight of table $j$. Then $C_- e^{-\kappa(y)} \le f(e^y) \le C_+ e^{-\kappa(y)}$ on the chart, $\int e^{-\kappa(y)} dy = V_1$ with the volume $V_1 = \sum_\pi \prod_{l=2}^n \kappa_{1,U_l}^{-1}$ of the direction (Proposition 39.34, Theorem 17.5(i)), and for $0 < t \le C_-$*

$$\frac{V_1}{(n-1)!}\Big(\log\frac{C_-}{t}\Big)^{n-1} \le \mu_f\big([t, f^\star]\big) \le \frac{V_1}{(n-1)!}\Big(\log\frac{C_+}{t}\Big)^{n-1}.$$

*So $\mu_f([t, f^\star]) \sim \frac{V_1}{(n-1)!}(\log\frac{1}{t})^{n-1}$ as $t \downarrow 0$, and $C_-^k V_1 \le k^{n-1}\langle 1\rangle_k \le C_+^k V_1$ for every real $k > 0$.*

The two ends of the stride's law are the two ends of strength. At the low end the mass of $\mu_f$ near 0 is set by the walls of the direction alone, through the volume of the direction, and the values blow up at $k = 0$ with that volume as coefficient (Proposition 39.35). At the high end $\mu_f$ ends at the native value of the direction in a power fixed by the number of marks, with a coefficient read off the Hessian of the stride at its native gauge; the branch of the generating function in Theorem 38.15 is this end read through the pencil.

## 40.2 The gauge end

**Definition 40.4** (The density operator of a recurrence)**.** Let $\mathbf{p} = (p_0, \dots, p_h)$ be polynomials in $\mathbb{C}[k]$, not all zero, of largest degree $D$, with the symbol $\chi(\lambda) = \sum_i c_i \lambda^i$, $c_i = [k^D]p_i$ (Corollary 38.7), and the *second symbol* $\chi_1(\lambda) = \sum_i [k^{D-1}]p_i\, \lambda^i$. Its *density operator* is

$$\mathfrak{D}_{\mathbf{p}} = \sum_{i=0}^{h} t^i\, p_i(-1-i-\theta), \qquad \theta = t\frac{d}{dt},$$

a differential operator with polynomial coefficients acting on functions and distributions on $(0,\infty)$; written $\sum_{j=0}^{D} a_j(t)\, \partial_t^j$, its leading coefficient is $a_D = (-1)^D t^D \chi(t)$. At a root $\lambda \neq 0$ of $\chi$, in the variable $s = \lambda - t$, write $\mathfrak{D}_{\mathbf{p}} = \sum_j b_j(s)\, \partial_s^j$ and $b_{j,l} = [s^l]\, b_j$. A function $u$ on an interval $(\lambda-\delta, \lambda)$, or $(\lambda, \lambda+\delta)$, *extends by zero* at $\lambda$ when it is integrable there, $\mathfrak{D}_{\mathbf{p}} u = 0$ there, and the distribution $u\, \mathbf{1}_{(\lambda-\delta,\lambda)}$, or $u\, \mathbf{1}_{(\lambda,\lambda+\delta)}$, is annihilated by $\mathfrak{D}_{\mathbf{p}}$ on $(\lambda-\delta, \lambda+\delta)$; this is a property of the germ of $u$ at $\lambda$.

The density operator is the recurrence read on the other side of the moments: $\mathfrak{D}_{\mathbf{p}}$ is the operator whose formal adjoint sends $t^k$ to $\sum_i p_i(k)\, t^{k+i}$, so a measure on $(0,\infty)$ annihilated by it has moments that satisfy the recurrence, and its singular points in $\mathbb{C}^\times$ are the roots of the symbol.

**Proposition 40.5** (A recurrence of the values is an equation for the density)**.** *Let a ray be as in Definition 40.1, let $f^\star$ be as in Proposition 40.2 and $\mathbf{p} = (p_0, \dots, p_h)$ as in Definition 40.4. The following are equivalent:*

*(a)* $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ *for all large integers $k$;*

*(b)* $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ *at every complex $k$ at which every $\langle 1\rangle_{k+i}$ converges, and as an identity of meromorphic functions of $k$ (Proposition 38.5(ii));*

*(c) the distribution $\mathfrak{D}_{\mathbf{p}}\, \mu_f$ on $(0,\infty)$ vanishes.*

*When they hold, $\mathfrak{D}_{\mathbf{p}} \varrho_f = 0$ on $(0, f^\star)$ and the germ of $\varrho_f$ at $f^\star$ extends by zero; and for the origin 0, the conclusions of Corollary 39.38(i) and (ii) hold for $p_0, \dots, p_h$: $\sum_{i \le \min(h,j)} p_i(-j)\, \mathrm{CT}(f^{i-j}) = 0$ for every $j \in \mathbb{N}$, and $p_0$ vanishes to order at least $n-1$ at 0 with $[k^{n-1}]p_0 \cdot \mathsf{L}_0 + \sum_{i\ge 1} p_i(0)\langle 1\rangle_i = 0$.*

So a recurrence that the values satisfy at the large integers, whether or not it follows from the relations of the ray, holds at every strength: no bound on the pieces of a certificate is needed down to $k \to 0$, and the low-end relation follows for it as for the law of [1].

**Theorem 40.6** (The gauge end)**.** *Let a ray be as in Definition 40.1, with $\alpha^\star$, $f^\star = f(\alpha^\star)$, the density $\varrho_f$ and the function $g$ of Proposition 40.3, and let $\mathbf{p} = (p_0, \dots, p_h)$ satisfy the equivalent conditions of Proposition 40.5; let $f^\star$ be a simple root of the symbol $\chi$, and use the notation of Definition 40.4 at $\lambda = f^\star$. Put $[x]_j = x(x-1)\cdots(x-j+1)$ and $R_m(x) = \sum_j b_{j,m-D+1+j}\, [x]_j$, with $b_{j,l} = 0$ for $l < 0$.*

*(i) $D \ge 1$, $b_{D,0} = 0$, $b_{D,1} = -(f^\star)^D \chi'(f^\star) \neq 0$, and $s^{D-1}\mathfrak{D}_{\mathbf{p}} = \sum_{m\ge 0} s^m R_m(s\partial_s)$ with*

$$R_0(x) = b_{D,1}\, [x]_{D-1}\, (x-\gamma), \qquad \gamma = D - 1 - \frac{b_{D-1,0}}{b_{D,1}} = -1 + \frac{\chi_1(f^\star)}{f^\star \chi'(f^\star)}\,:$$

*$s = 0$ is a regular singular point of $\mathfrak{D}_{\mathbf{p}}$ with the exponents $0, 1, \dots, D-2$ and $\gamma$.*

*(ii)* The line. *The germs at $f^\star$ of functions on the chart's side, $t < f^\star$, that extend by zero form a space of dimension at most one.*

*(iii)* *The germ of $\varrho_f$ extends by zero, so it spans that space, and $\gamma = \frac{n-3}{2}$.*

*(iv)* The germ. *If $\gamma \notin \{0, 1, \dots, D-2\}$, then $g(s) = g(0)\,\Phi(s)$, where $\Phi = \sum_N \Phi_N s^N$, $\Phi_0 = 1$, is fixed by $R_0(\gamma+N)\Phi_N = -\sum_{m=1}^{N} R_m(\gamma+N-m)\Phi_{N-m}$, $N \geq 1$. If $\gamma \in \{0, 1, \dots, D-2\}$, then $\varrho_f = \sum_r u_r s^r$ near $f^\star$ with $u_r = 0$ for $r < \gamma$, $u_\gamma = g(0)$, the $u_r$ with $\gamma < r \leq D-2$ fixed one at a time by $\mathsf{p}_r(u) = 0$, where*

$$\mathsf{p}_r(u) = \sum_{i \leq r} \sum_{i < j \leq D} (-1)^l \binom{j-1-i}{l}\, i!\, l!\, b_{j,l}\, u_i, \qquad l = j + r + 1 - D - i \geq 0,$$

*whose coefficient of $u_r$ is $r!\, b_{D,1}(r-\gamma)$, and the $u_N$ with $N \geq D-1$ by $R_0(N)u_N = -\sum_{m=1}^{N} R_m(N-m)u_{N-m}$.*

*(v)* The values. *$\varrho_f$ extends holomorphically to a neighbourhood of $(0, f^\star)$ in $\mathbb{C}$ on which $\mathfrak{D}_{\mathbf{p}}\varrho_f = 0$: it is the continuation of the germ of (iv) along $(0, f^\star)$, around the roots of $\chi$ on the segment by small arcs on either side. For every $j \in \mathbb{N}$ and every $k$ at which $\langle 1 \rangle_k$ converges,*

$$\frac{\partial^j}{\partial k^j} \langle 1 \rangle_k = \int_0^{f^\star} t^k\, (\log t)^j\, \varrho_f(t)\, dt.$$

The theorem reads a ray from its gauge end. The recurrence fixes the operator, and the native gauge of the direction fixes the rest: $f^\star$, the exponent, which the recurrence must reproduce, and the scale $g(0)$, through the Hessian. For odd $n$ the exponent is an integer and the line lies among the holomorphic solutions; when the exponent is at most $D-2$, the equations $\mathsf{p}_r(u) = 0$, the absence of point masses at $f^\star$, single it out. For even $n$ it is the half-integer class. Nothing from the low end enters, nor any inequality, nor the native values on the segment $(0, f^\star)$, which the density passes analytically.

**Proposition 40.7** (The law knows the dimension of its chart)**.** *Let a ray have $\kappa_{1,S} > 0$ for every nonempty proper $S$, let $\mathbf{p} = (p_0, \dots, p_h)$ have $\sum_i p_i(k) f^i \in \mathcal{L}$ (Definition 38.4), as the law of* [1] *has, and let $\lambda$ be a simple root of its symbol that is the native value of a nondegenerate native gauge. Then the formal solutions of $\sum_i p_i(k) x(k+i) = 0$ of type $(\lambda, \rho + \mathbb{Z})$ (Definition 39.2) form a line, spanned by $\lambda^k k^{\rho_\lambda}(1 + \sum_{m \geq 1} e_m k^{-m})$ with $\rho_\lambda = -\chi_1(\lambda)/(\lambda\chi'(\lambda)) = -\frac{n-1}{2}$, and $\lambda$ is a regular singular point of $\mathfrak{D}_{\mathbf{p}}$ with the exponents $0, 1, \dots, D-2$ and $-1-\rho_\lambda = \frac{n-3}{2}$.*

So a recurrence read off closings, integers indexed by one strength, has to carry the dimension $n-1$ of the chart at every simple root of its symbol that is a native value: a test that a candidate law must pass. A root of the symbol need not be a native value, and at such a root the exponent is whatever the recurrence makes it.

**Proposition 40.8** (The germ at the gauge end is local data at the native gauge)**.** *Let a ray have the origin 0 and $\kappa_{1,S} > 0$ for every nonempty proper $S$, with $\alpha^\star$, $f^\star = f(\alpha^\star)$, $H$ and $g$ as in Proposition 40.3; in the coordinates $y$ of the chart, centred at $\log \alpha^\star$, write $\log f = \log f^\star - \frac{1}{2}H_{ab}y^a y^b + \frac{1}{6}h_{abc}y^a y^b y^c + \frac{1}{24}h_{abcd}y^a y^b y^c y^d + \cdots$, with the summation over repeated indices in $\mathcal{E} \setminus e_0$ and $G = H^{-1}$.*

*(i)* *For $m \geq 2$ the Taylor tensor of order $m$ of $\log f$ at $\log \alpha^\star$ is $-2^{-m}$ times rung $m$ of the still of the realization $\big((s_{1j}, t_{1j})_j; \alpha^\star\big)$ (Definition 9.4), that is $-\sum_j \big(s_{1j}\kappa_{j,\alpha^\star} + (-1)^m t_{1j}\kappa_{j,(\alpha^\star)^{-1}}\big)$ in the cumulants of the basis laws (Proposition 9.6).*

*(ii) As $k \to \infty$,*

$$\langle 1\rangle_k = (f^\star)^k \Big(\frac{2\pi}{k}\Big)^{\frac{n-1}{2}} \frac{1}{\sqrt{\det H}} \Big(1 + \frac{c_1}{k} + O(k^{-2})\Big),$$

*with*

$$c_1 = \tfrac{1}{8} h_{abcd} G^{ab} G^{cd} + \tfrac{1}{8} h_{abc} h_{def} G^{ab} G^{cd} G^{ef} + \tfrac{1}{12} h_{abc} h_{def} G^{ad} G^{be} G^{cf}.$$

*(iii) With $g(s) = \sum_m g_m s^m$, the asymptotic expansion of $\langle 1\rangle_k$ is $\sum_m g_m (f^\star)^{k+\frac{n-1}{2}+m} \mathrm{B}\big(k + 1, \frac{n-1}{2} + m\big)$, read term by term, and the term of $g_m$ begins at $(f^\star)^k k^{-\frac{n-1}{2}-m}$ with the coefficient $(f^\star)^{\frac{n-1}{2}+m}\Gamma(\frac{n-1}{2} + m) g_m$. So the coefficients of the expansion of $(f^\star)^{-k} k^{\frac{n-1}{2}} \langle 1\rangle_k$ in powers of $k^{-1}$ and the Taylor coefficients of $g$ determine each other, the $m$-th by the first $m + 1$.*

**Proposition 40.9** (Slopes from the gauge end)**.** *Let a ray and $\mathbf{p}$ be as in Theorem 40.6, with $\varrho_f(t) = (f^\star - t)^{\frac{n-3}{2}} \sum_m g_m (f^\star - t)^m$ near $f^\star$.*

*(i) For every $j \in \mathbb{N}$, $\sum_{l=0}^{j} \binom{j}{l} \sum_i p_i^{(j-l)}(k)\, \partial_k^l \langle 1\rangle_{k+i} = 0$ identically in $k$: the $j$-th derivative of the values satisfies the recurrence, with an inhomogeneity made of the lower derivatives.*

*(ii) If the series $\sum_m g_m s^m$ converges on $[0, f^\star)$ and $\sum_m |g_m| (f^\star)^m \mathrm{B}\big(K' + 1, \frac{n-1}{2} + m\big) < \infty$ for a real $K'$, then for every real $K > K'$ and every $j \in \mathbb{N}$*

$$\partial_k^j \langle 1\rangle_k \big|_{k=K} = \sum_{m \geq 0} g_m \frac{\partial^j}{\partial a^j} \Big[ (f^\star)^{a+\frac{n-1}{2}+m}\, \mathrm{B}\big(a + 1, \tfrac{n-1}{2} + m\big) \Big]_{a=K}.$$

The derivatives of the values in the strength, the slopes, need no datum from the low end either: they are the logarithmic moments of the same density, and the recurrence differentiated carries them down from large strengths once they are read there.

**Definition 40.10** (The procedure that reads a ray from its gauge end)**.** Input: a ray as in Definition 40.1 with its data in a field $\Bbbk \subseteq \mathbb{R}$, and polynomials $\mathbf{p} = (p_0, \dots, p_h)$ over $\Bbbk$ with $\sum_i p_i(k) \langle 1\rangle_{k+i} = 0$ for all large integers $k$, $f^\star = f(\alpha^\star)$ a simple root of their symbol. Output: the germ of the density at $f^\star$, and the values and slopes at the positive integers.

(1) *The native gauge.* $\alpha^\star$, the unique critical point of $\log f$ on the positive part of the chart, where $-\log f$ is strictly convex (Theorem 17.5(ii)); $f^\star$, $\det H$ and $(\omega_0/dy)(\alpha^\star)$; the constant $g(0)$ (Proposition 40.3(ii)).

(2) *The operator.* $\mathfrak{D}_{\mathbf{p}}$ and its coefficients $b_{j,l}$ at $f^\star$ (Definition 40.4), and $\gamma$ (Theorem 40.6(i)); a recurrence with $\gamma \neq \frac{n-3}{2}$ is not a recurrence of the values (Theorem 40.6(iii)).

(3) *The germ.* The Taylor coefficients of $g$, by Theorem 40.6(iv), over $\Bbbk(f^\star)$ up to the factor $g(0)$.

(4) *Large strengths.* For $K$ large and $0 < \delta$ less than the radius of convergence of $g$, $\langle 1\rangle_{K+i} = \int_{f^\star - \delta}^{f^\star} t^{K+i} \varrho_f(t)\, dt + \epsilon_{K+i}$, $0 \leq i < h$, the integral read from the germ, and $0 \leq \epsilon_k \leq (f^\star - \delta)^{k-k'} \langle 1\rangle_{k'}$ for every $k' \leq k$ at which $\langle 1\rangle_{k'}$ converges; for the origin 0, $\langle 1\rangle_{k'} \leq C_+^{k'} V_1 k'^{1-n}$ (Proposition 40.3(iii)).

(5) *The run back.* The recurrence, which holds at every positive integer (Proposition 40.5), solved for $\langle 1\rangle_k$ at the $k < K$ with $p_0(k) \neq 0$, from $k = K - 1$ down; at a positive integer zero of $p_0$, and below it, the values are read as moments of $\varrho_f$ continued along $(0, f^\star)$ (Theorem 40.6(v)).

(6) *Slopes.* The same with $(\log t)^j$ in the integrals and the differentiated recurrence (Proposition 40.9).

(7) *Checks.* For the origin 0, the low-end relation (Corollary 39.38(ii)) read on the output returns $V_1$; two depths $K$ give the same values.

Every step but the truncations in step (4) is an identity: the germ is exact over $\Bbbk(f^\star)$ up to $g(0)$, the error $\epsilon_k$ of the cut at $f^\star-\delta$ is bounded by step (4), and the run back is exact arithmetic on exact coefficients. The truncation of the series of $g$ on $[f^\star-\delta, f^\star]$ is estimated, by the size of the last terms kept and by the agreement of two depths. The run back multiplies an error at $K$ by the growth of the other solutions of the recurrence from $K$ down to 1, which is why the depth $K$ grows with the precision asked. The exercises read the centres of three and five marks this way (Exercises 54.3 and 54.15), and a ray of a list on four marks whose symbol has a root on the segment $(0, f^\star)$ (Exercise 55.2).

## 40.3 The torus value

**Definition 40.11** (The torus value)**.** Let the centre ray of a family at the strengths $\boldsymbol{\sigma}$ (Definition 38.13) be given, with its stride $f=\prod_j \big(K_j(\alpha)K_j(\alpha^{-1})\big)^{-\sigma_j}$ and $f^\star=f(\mathbf{1})$. On the torus $\mathbb{T}=\{\alpha:\ |\alpha_e|=1\ (e\in\mathcal{E}),\ \alpha_{e_0}=1\}$, $\alpha_e=e^{i\vartheta_e}$, with the normalized measure $d\vartheta/(2\pi)^{n-1}$, the stride is $f=\prod_j |K_j(\alpha)|^{-2\sigma_j}$, a function with values in $[f^\star,\infty]$, equal to $\infty$ exactly where an active letter vanishes. The *torus law* $\nu_f^{\mathbb{T}}$ is the image of $d\vartheta/(2\pi)^{n-1}$ under $f$, a probability measure on $[f^\star,\infty)$, $\varrho_f^{\mathbb{T}}$ its density where it has one, and the *torus value* is

$$\langle 1\rangle_k^{\mathbb{T}}=\int t^k\,\nu_f^{\mathbb{T}}(dt)=\frac{1}{(2\pi)^{n-1}}\int_{\mathbb{T}} f^k\,d\vartheta \qquad (\operatorname{Re}k\le 0).$$

When no letter $K_j$ of the family vanishes on $\mathbb{T}$, so that $\mathbb{T}$ lies in $X_F$, it is $(2\pi i)^{1-n}\langle 1\rangle_{k;\mathbb{T}}$ in the notation of Definition 38.1, $\mathbb{T}$ oriented by $d\vartheta$.

**Proposition 40.12** (The torus is the second chain)**.** *For the centre ray of a family at the strengths $\boldsymbol{\sigma}$, with the notation of Definition 40.11, $\varrho_f$ and $g$ as in Proposition 40.3:*

*(i) On $\mathbb{T}$, $f>f^\star$ off $\mathbf{1}$; the zero set of every active letter on $\mathbb{T}$ has measure zero. $\langle 1\rangle_k^{\mathbb{T}}$ is continuous on $\operatorname{Re}k\le 0$ and holomorphic on $\operatorname{Re}k<0$, and $\langle 1\rangle_{-j}^{\mathbb{T}}=\mathrm{CT}(f^{-j})$ for $j\in\mathbb{N}$, the closing $\mathsf{C}_j(1)$ when every $\sigma_j\ge 1$ (Definition 39.30).*

*(ii) Let every $\sigma_j\ge 1$, and let $\mathbf{p}=(p_0,\dots,p_h)$ satisfy $\sum_i p_i(k)\langle 1\rangle_{k+i}=0$ for all large integers $k$, as the law of $[1]$ does. Then $\sum_i p_i(k)\langle 1\rangle_{k+i}^{\mathbb{T}}=0$ for every $k$ with $\operatorname{Re}k\le -h$, and the distribution $\mathfrak{D}_{\mathbf{p}}\,\nu_f^{\mathbb{T}}$ on $(0,\infty)$ vanishes.*

*(iii) Near $f^\star$, $\nu_f^{\mathbb{T}}$ has a density, and with the one function $g$*

$$\varrho_f(t)=(f^\star-t)^{\frac{n-3}{2}}g(f^\star-t)\quad(t<f^\star),\qquad \varrho_f^{\mathbb{T}}(t)=(2\pi)^{1-n}(t-f^\star)^{\frac{n-3}{2}}g(f^\star-t)\quad(t>f^\star).$$

*(iv) For $\mathbf{p}$ as in (ii) with $f^\star$ a simple root of its symbol, the germ of $\varrho_f$ on the chart's side spans the line of Theorem 40.6(ii), the germ of $\varrho_f^{\mathbb{T}}$ extends by zero on the torus's side, and the two are the cut solutions $|f^\star-t|^{\frac{n-3}{2}}g(f^\star-t)$ of one germ on its two sides, the second with the factor $(2\pi)^{1-n}$.*

At a centre the native gauge of the direction lies on the positive orthant and on the unit torus, and the two meet only there (Theorem 39.57). The torus value continues the closings in the strength, as the values continue them at the poles of the low end (Proposition 39.34), and it satisfies every recurrence of the values. Its density is the density of the chart seen from the other side of the native value: the Morse function of the stride at $\mathbf{1}$ changes sign between the orthant and the torus, so the one function $g$ enters both, and the asymptotic statement at the end of Theorem 39.57 is the shadow of this exact one. Exercise 54.4 works the three-mark centre, where the torus value, the values and the solution at the sign are tied by an explicit connection.

# Chapter 41

# Reading a ray

A ray is read from its list and its direction. A box of its relations gives a spanning set of a submodule that contains [1], the one-step matrix on it with a certificate, and a recurrence of the values that holds at every strength (Section 39.4); the gauge end then gives the values themselves (Chapter 40). This chapter assembles the reading and says what each step certifies. At the low end a recurrence of the values meets the values at the positive integers through the poles at the nonpositive integers: the low-end relation at 0 and, where the coefficients of the recurrence vanish to the order of the poles, second low-end relations at the other poles, with coefficients in the field of the recurrence, the weights and the volume of the direction. At the high end the generating function of the values is holomorphic off the reciprocals of $(0, f(\alpha^\star)]$, and its jump across them is the density of the stride's law. When every other root of the symbol lies outside the closed disc of radius $f(\alpha^\star)$, the values are, up to scale, the only solution of the recurrence whose exponential rate of growth is at most $f(\alpha^\star)$; the low-end relation fixes the scale, and solutions over the field of the recurrence approximate the values at the rate $f(\alpha^\star)/\lambda'$, $\lambda'$ the least modulus of the other roots. Positivity is the moment problem of the stride's law on $[0, f(\alpha^\star)]$. Against a solution carried by a segment that ends at a real point $\lambda$ off $[0, f(\alpha^\star)]$ its brackets shrink at the rate of the Green function of the plane cut along $[0, f(\alpha^\star)]$, read at $\lambda$; it cannot see a solution carried by $[0, f(\alpha^\star)]$ and dominated by the stride's law, which the analyticity of the density excludes; and for a recurrence of order two it gives the value at $k = 1$ as the volume of the direction times a continued fraction in the coefficients. The procedure that reads a ray from its list runs the box by evaluation at strengths drawn at random, modulo primes for a list, and bounds the chance of a wrong output by degrees that the box fixes, or removes it with enough strengths. Its cost is set by the stencils of the relations on a lattice, by the degrees of the one-step matrix, which the number of pivots bounds, and by the order and the degree of the recurrence. The nondegenerate native gauges bound the sizes of a reading from below; over all exponents at once they form the graph of one map, through which they are found by continuation and certified one at a time.

## 41.1 The low end and the slit

The low-end relation of a recurrence of the values is Corollary 39.38(ii), with the low-end coefficients of Definition 39.37; by Proposition 40.5 it holds for every recurrence that the values satisfy at the large integers, whether or not it follows from the relations of the ray. The poles at the negative integers (Proposition 39.35) give more relations.

**Proposition 41.1** (The low-end relations at every pole)**.** *Let a ray have the origin* 0 *and* $\kappa_{1,S} > 0$ *for every nonempty proper* $S$*, with the volume* $V_1$ *of its direction (Proposition 39.34), and let*

$p_0, \dots, p_h \in \mathbb{C}[k]$ *satisfy* $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ *for all large integers* $k$. *Let* $j \in \mathbb{N}$, *and let* $m_0 \in \{0, \dots, n-1\}$ *be such that every* $p_i$ *with* $i \le \min(h, j)$ *vanishes to order at least* $m_0$ *at* $k = -j$.

*(i) The function* $\sum_{i \le \min(h,j)} p_i(k)\langle 1\rangle_{k+i}$ *is holomorphic at* $k = -j$, *with the value* $-\sum_{i=j+1}^{h} p_i(-j)\langle 1\rangle_{i-j}$ *there.*

*(ii) If* $m_0 < n-1$, *then* $\sum_{i \le \min(h,j)} [(k+j)^{m_0}] p_i \cdot \mathrm{CT}\left(f^{i-j}\right) = 0$.

*(iii) If* $m_0 = n-1$, *then*

$$V_1 \sum_{i \le \min(h,j)} [(k+j)^{n-1}] p_i \cdot \mathrm{CT}\left(f^{i-j}\right) + \sum_{i=j+1}^{h} p_i(-j)\,\langle 1\rangle_{i-j} = 0.$$

*At* $j = 0$ *this is the low-end relation; at* $j \ge 1$ *it is the* second low-end relation *at* $-j$, *a linear relation among the values at the positive integers whose coefficients lie in the field generated by those of the* $p_i$, *the weights and* $V_1$.

At $m_0 = 0$, (ii) is the recurrence of the constant terms, $\sum_i p_i(-j)\,\mathrm{CT}(f^{i-j}) = 0$ (Corollary 39.38(i)). At larger $m_0$ it ties the derivatives of order $m_0$ of $p_0, \dots, p_j$ at $-j$, and (i) ties the values at the positive integers to the Laurent coefficients of the values at the poles $-j, \dots, 0$, those at 0 being integrals over the faces of the sector cubes (Proposition 39.39).

The generating function of the values is read on the plane cut along the reciprocals of $(0, f(\alpha^\star)]$. The segment $(0, f(\alpha^\star))$ of the plane of native values is the *slit*.

**Proposition 41.2** (The values live on the slit plane)**.** *Let a ray have a real origin and* $\kappa_{1,S} > 0$ *for every nonempty proper* $S$, *with* $\alpha^\star$, $f^\star = f(\alpha^\star)$, $\mu_f$ *and* $\varrho_f$ *as in Proposition 40.3, and let* $k_0$ *be an integer at which* $\langle 1\rangle_{k_0}$ *converges.*

*(i) For* $|q| < 1/f^\star$, $G(q) = \sum_{k \ge k_0} \langle 1\rangle_k q^{k-k_0} = \int t^{k_0} \mu_f(dt)/(1-qt)$, *and the right side is holomorphic on* $\mathbb{C} \setminus [1/f^\star, \infty)$: $G$ *continues to the plane cut along the reciprocals of* $(0, f^\star]$.

*(ii) For* $x > 1/f^\star$, $G$ *continues holomorphically across a neighbourhood of* $x$ *in* $(1/f^\star, \infty)$ *from either side, with the limits* $G(x \pm i0)$, *and* $G(x+i0) - G(x-i0) = 2\pi i\, x^{-1-k_0}\, \varrho_f(1/x)$: *the jump of* $G$ *across the cut is the density of the stride's law.*

*(iii) On the centre ray of a family, every native gauge on the torus* $\mathbb{T}$ *of Definition 40.11 has its native value in* $[f^\star, \infty)$, *off the slit.*

*(iv) Every conjugate of* $f^\star$ *over the field of the weights is the native value of a nondegenerate native gauge (Proposition 38.10), and a root of the symbol of every recurrence that the values satisfy at the large integers with coefficients in that field; the real conjugates in* $(0, f^\star)$ *lie on the slit.*

A root $\lambda$ of a symbol on the slit sits on the cut, at $1/\lambda \in (1/f^\star, \infty)$, where the generating function of the values continues from either side and jumps by the density of the stride's law; a root off $[0, f^\star]$ sits where it is holomorphic. Not every root of a symbol is a native value: whenever the values satisfy $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$, they satisfy the recurrence with the coefficients $p_{i-1}(k+1) - \lambda_0\, p_i(k)$, $0 \le i \le h+1$, $p_{-1} = p_{h+1} = 0$, which follows from the relations when the first does and has the symbol $(\lambda - \lambda_0)\chi(\lambda)$, for every $\lambda_0$. In the other direction, the native value of every nondegenerate native gauge is a root of the symbol of every recurrence that follows from the relations (Corollary 38.7), and that of every native gauge when the recurrence has a certificate as in Proposition 38.8. Exercises 55.1 and 55.2 work second low-end relations on a list of four marks.

## 41.2 Positive selection

**Theorem 41.3** (Positive selection)**.** *Let a ray have a real origin and $\kappa_{1,S} > 0$ for every nonempty proper $S$, with $f^\star = f(\alpha^\star)$, and let $p_0, \dots, p_h \in \mathbb{C}[k]$, $h \geq 1$, satisfy $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ for all large integers $k$, so that the recurrence holds at every integer at which the values converge (Proposition 40.5). Let $p_0$ and $p_h$ have the largest degree $D$, so that the symbol $\chi$ has degree $h$ and $\chi(0) \neq 0$; then $f^\star$ is a root of $\chi$, and let the roots of $\chi$ be distinct, $\lambda_1 = f^\star, \lambda_2, \dots, \lambda_h$. Let $k_1$ be an integer at which the values converge, with $p_0(k)p_h(k) \neq 0$ for every integer $k \geq k_1$. A* solution *is a sequence $(x(k))_{k\geq k_1}$ with $\sum_i p_i(k)x(k+i) = 0$ for every $k \geq k_1$; the solutions form a space $\mathcal{V}$ of dimension $h$, and the values are one of them.*

*(i) For every solution, $X(q) = \sum_{k\geq k_1} x(k)q^{k-k_1}$ satisfies $L_q X = R_x$ as a formal power series, with*

$$L_q = \sum_{i=0}^{h} q^{h-i}\, p_i\Big(q\frac{d}{dq} + k_1 - i\Big)$$

*and $R_x$ a polynomial of degree less than $h$ that depends linearly on $x(k_1), \dots, x(k_1+h-1)$. The coefficient of $(d/dq)^D$ in $L_q$ is $q^{D+h}\chi(1/q)$, so the singular points of $L_q$ in $\mathbb{C}^\times$ are the points $1/\lambda_c$. For the values, $X$ continues holomorphically to $\mathbb{C}\setminus[1/f^\star, \infty)$ (Proposition 41.2).*

*(ii) The solutions with $x(k) = O(r^k)$ for every $r > f^\star$ form a subspace of $\mathcal{V}$ whose dimension is the number of roots in the closed disc $|\lambda| \leq f^\star$; the values lie in it, $\langle 1\rangle_k = C\,(f^\star)^k k^{-\frac{n-1}{2}}\big(1+o(1)\big)$ with $C > 0$, and the exponent $-\chi_1(f^\star)/(f^\star\chi'(f^\star))$ of the root $f^\star$ is $-\frac{n-1}{2}$.*

*(iii) Let every root other than $f^\star$ have modulus greater than $f^\star$, and put $\lambda' = \min_{c\geq 2}|\lambda_c|$. Then the solutions with $x(k) = O(r^k)$ for some $r < \lambda'$ are the multiples of the values, and so are those whose generating function converges on $|q| < 1/f^\star$: the values are, up to a constant factor, the* solution of least growth*. Let moreover the origin be 0, $k_1 = 1$ and $a = [k^{n-1}]p_0 \neq 0$, and put $\ell(x) = -a^{-1}\sum_{i=1}^{h} p_i(0)\,x(i)$. Then $\ell(\langle 1\rangle) = V_1$, and $\langle 1\rangle_k = V_1\,x(k)/\ell(x)$ for every nonzero solution $x$ of least growth: the values divided by the volume of the direction are a function of the recurrence alone.*

*(iv) Under the hypotheses of (iii), let $x^{(0)} \in \mathcal{V}$ have $\ell(x^{(0)}) = V_1$ and let $x^{(1)}, \dots, x^{(h-1)}$ be a basis of $\{\ell = 0\}$, so that $\langle 1\rangle = x^{(0)} + \sum_{i\geq 1} M_i x^{(i)}$ for one $M \in \mathbb{C}^{h-1}$. For all large $K$ the equations $x^{(0)}(K+l) + \sum_{i\geq 1} M_i^{(K)} x^{(i)}(K+l) = 0$, $0 \leq l \leq h-2$, have exactly one solution $M^{(K)}$, and $|M^{(K)} - M| = O\big((f^\star/\lambda')^K K^{C'}\big)$ for some $C'$. When the $p_i$ have their coefficients in a field $\Bbbk$ that contains $V_1$, the $x^{(i)}$ can be taken with values in $\Bbbk$, and then every $M^{(K)}$ lies in $\Bbbk^{h-1}$.*

The selection runs on growth alone: among the solutions, the values are the one that grows like the native value of the direction, and the other roots of the symbol, outside the closed disc of radius $f^\star$, give solutions that grow faster and are cut away. The hypothesis of (iii) is what makes the values the solution of least growth: a root in the closed disc, in particular one on the slit, gives a solution whose exponential rate of growth is at most that of the values, and growth cannot remove it (Section 41.3). The low-end relation then fixes the scale through the volume of the direction, so the values over that volume are a function of the recurrence alone.

## 41.3 Positivity as a moment problem

**Proposition 41.4** (Positivity is the moment problem of the stride's law)**.** *Let a ray have a real origin and $\kappa_{1,S} > 0$ for every nonempty proper $S$, with $f^\star = f(\alpha^\star)$ and $\mu_f$ as in Proposition 40.3.*

*For a real polynomial $p = \sum_m p_m t^m$, a real $K$ and a sequence $x$ defined at $K + \mathbb{N}$, put $L_{K,p}(x) = \sum_m p_m\, x(K+m)$.*

*(i) If $\langle 1\rangle_K$ converges, $p \ge 0$ on $[0, f^\star]$ and $p \ne 0$, then $L_{K,p}(\langle 1\rangle) = \int t^K p(t)\, \mu_f(dt) > 0$.*

*(ii) Every real polynomial of degree at most $N$ that is nonnegative on $[0, f^\star]$ is a sum of polynomials $q^2$, $t\,q^2$, $(f^\star - t)\,q^2$ and $t(f^\star - t)\,q^2$ of degree at most $N$, with real $q$. So for a real sequence $x$, $L_{K,p}(x) \ge 0$ for every such $p$ exactly when the Hankel matrices*

$$\begin{matrix} \big[x(K+i+i')\big], & \big[x(K+1+i+i')\big], \\ \big[f^\star x(K+i+i') - x(K+1+i+i')\big], & \big[f^\star x(K+1+i+i') - x(K+2+i+i')\big], \end{matrix}$$

*with $0 \le i, i' < \lfloor \frac{N}{2} \rfloor + 1$, $\lfloor \frac{N+1}{2} \rfloor$, $\lfloor \frac{N+1}{2} \rfloor$ and $\lfloor \frac{N}{2} \rfloor$ respectively, are positive semidefinite.*

*(iii) A real sequence $(x(K+m))_{m\in\mathbb{N}}$ has $L_{K,p}(x) \ge 0$ for every real $p \ge 0$ on $[0, f^\star]$ exactly when $x(K+m) = \int t^m \nu(dt)$ for every $m$, for a positive measure $\nu$ on $[0, f^\star]$, which is then unique; for the values, $\nu = t^K \mu_f$.*

So the positivity of the values is the statement that they are the moments of a positive measure on $[0, f^\star]$, and Theorem 38.14(iii) is its first family, the Hankel determinants of the values. For a family $x^{(0)} + \sum_i M_i x^{(i)}$ of solutions of a recurrence, as in Theorem 41.3(iv), each condition $L_{K,p} \ge 0$ is one linear inequality in the coordinates $M$, and the conditions of degree at most $N$ at the strength $K$ cut out a closed convex set, the *bracket* of degree $N$, which contains the coordinates of the values.

**Theorem 41.5** (The rate of positivity off the segment)**.** *Let a ray be as in Proposition 41.4, $K$ an integer at which $\langle 1\rangle_K$ converges, $\lambda \in \mathbb{R} \setminus [0, f^\star]$ and $I_\lambda$ the closed segment between $0$ and $\lambda$. Let $u(k) = \int_{I_\lambda} t^k\, \nu(t)\, dt$ for the integers $k \ge K$, where $\nu$ is a real function on $I_\lambda$ with $\int_{I_\lambda} |t|^K |\nu(t)|\, dt < \infty$ and $\nu(t) = c\,|\lambda - t|^a\big(1 + o(1)\big)$ as $t \to \lambda$ in $I_\lambda$, with $c \ne 0$ and $a > -1$. Let $E_N$ be the set of the real $\varepsilon$ with $L_{K,p}(\langle 1\rangle + \varepsilon u) \ge 0$ for every real $p \ge 0$ on $[0, f^\star]$ of degree at most $N$. Then $E_N$ is a closed interval that contains $0$, and there is a constant $C$ with its length at most $C\, N^{a+1} e^{-N g(\lambda)}$ for all large $N$, where*

$$g(\lambda) = \log\big|w + \sqrt{w^2 - 1}\big| > 0, \qquad w = \frac{2\lambda}{f^\star} - 1,$$

*the branch of the square root with $|w + \sqrt{w^2-1}| > 1$. For $\lambda > f^\star$, with $\rho = f^\star/\lambda$,*

$$e^{-g(\lambda)} = \frac{1 - \sqrt{1-\rho}}{1 + \sqrt{1-\rho}} = \frac{\rho}{4} + O(\rho^2).$$

The rate is read off the Green function $g$ of the plane cut along $[0, f^\star]$, with its pole at infinity, at the end $\lambda$ of the segment that carries $u$: the farther $\lambda$ lies from $[0, f^\star]$, the faster positivity separates the values from the values plus a multiple of $u$. For a recurrence of order two, with the coordinate $M$ of Theorem 41.3(iv) and $u = x^{(1)}$ of this form, $E_N$ is the bracket of degree $N$ of $M$ less the value of $M$.

**Proposition 41.6** (What positivity cannot see)**.** *Let a ray be as in Proposition 41.4, $K$ a real at which $\langle 1\rangle_K$ converges, and $\nu$ a real signed measure on $[0, f^\star]$ with $|\nu| \le C\mu_f$ for a constant $C$; put $u(k) = \int t^k \nu(dt)$ for $k \ge K$.*

*(i) For $|\varepsilon| \le 1/C$, $\langle 1\rangle + \varepsilon u$ is from $K$ on the moment sequence of the positive measure $\mu_f + \varepsilon\nu$, and $L_{K',p}(\langle 1\rangle + \varepsilon u) \ge 0$ for every real $K' \ge K$ and every real $p \ge 0$ on $[0, f^\star]$.*

*(ii)* *If on an open interval $J \subseteq (0, f^\star)$ the measure $\nu$ has a density that agrees almost everywhere with no real-analytic function on $J$, then for $\varepsilon \neq 0$ the measure $t^K(\mu_f + \varepsilon\nu)$, the only finite measure on $[0, f^\star]$ whose moments are the $\langle 1\rangle_{K+m} + \varepsilon u(K+m)$, $m \in \mathbb{N}$, has no density on $(0, f^\star)$ that is real-analytic, while $t^K\mu_f$ has one (Proposition 40.3(i)).*

*(iii)* *Let the origin be 0, let $\mathbf{p} = (p_0, \dots, p_h)$ satisfy the conditions of Proposition 40.5, and let $\nu$ have a density that is bounded near 0, with $\mathfrak{D}_{\mathbf{p}}\nu = 0$ as distributions on $(0, \infty)$. Then $u(k) = \int t^k \nu(dt)$ converges at every $k$ with $\operatorname{Re} k > -1$, $\sum_i p_i(k)u(k+i) = 0$ there, and $\sum_{i=1}^{h} p_i(0)\, u(i) = 0$: $u$ satisfies the recurrence and contributes nothing to the low-end relation.*

So a sequence $u$ as in (i) and (iii), a solution of the recurrence carried by $[0, f^\star]$ and dominated by the stride's law, is invisible to the recurrence, to the low-end relation and to positivity together: read as conditions on candidates, they leave the direction of $u$ free. For a root $\lambda$ of the symbol on the slit, a density $\nu$ on $(0, \lambda)$ with $\mathfrak{D}_{\mathbf{p}}\big(\nu\, \mathbf{1}_{(0,\lambda)}\big) = 0$ as distributions on $(0, \infty)$, a solution of the density equation on $(0, \lambda)$ that extends by zero at $\lambda$ (Definition 40.4), bounded near 0 and with $|\nu| \leq C\varrho_f$ on $(0, \lambda)$, gives such a $u$. The gauge end does not choose among candidates: it constructs the density, and by (ii) the analyticity of the density on the open support is as much a property of the stride's law as its positivity.

**Proposition 41.7** (A recurrence of order two is the volume of the direction times a continued fraction)**.** *Let a ray have the origin 0 and $\kappa_{1,S} > 0$ for every nonempty proper $S$, with $f^\star = f(\alpha^\star)$, and let $p_0, p_1, p_2 \in \mathbb{R}[k]$ satisfy $p_0(k)\langle 1\rangle_k + p_1(k)\langle 1\rangle_{k+1} + p_2(k)\langle 1\rangle_{k+2} = 0$ for all large integers $k$; put $a = [k^{n-1}]p_0$ and $\rho_k = \langle 1\rangle_{k+1}/\langle 1\rangle_k$ for the integers $k \geq 1$.*

*(i)* *$0 < \rho_k < f^\star$, and $\rho_k\big(p_1(k) + p_2(k)\rho_{k+1}\big) = -p_0(k)$ for every integer $k \geq 1$.*

*(ii)* *$\langle 1\rangle_1\big(p_1(0) + p_2(0)\rho_1\big) = -a\, V_1$. When $-a/p_1(0)$ and $-a/(p_1(0) + f^\star p_2(0))$ are finite and positive, $\langle 1\rangle_1/V_1$ lies between them, strictly when they differ.*

*(iii)* *Let the hypotheses of Theorem 41.3(iii) hold with $h = 2$ and $k_1 = 1$, so that $\lambda' = |\lambda_2|$, and let $a \neq 0$; put $\mathsf{M}_k(r) = -p_0(k)/\big(p_1(k) + p_2(k)\, r\big)$. There are $K_0$, $C$ and $C'$ such that for every $K \geq K_0$ and every $r \in [0, f^\star]$ the composite $\mathsf{M}_1 \circ \cdots \circ \mathsf{M}_K(r)$ is defined and lies within $C(f^\star/\lambda')^K K^{C'}$ of $\rho_1$. So $\rho_1$ is the value of a continued fraction in the coefficients of the recurrence, and $\langle 1\rangle_1 = -aV_1/(p_1(0) + p_2(0)\rho_1)$. The sets of the $\mathsf{M}_1 \circ \cdots \circ \mathsf{M}_K(r)$ over the $r \in (0, f^\star)$ whose intermediate values $\mathsf{M}_k \circ \cdots \circ \mathsf{M}_K(r)$, $2 \leq k \leq K$, lie in $(0, f^\star)$ are nested in $K$ and contain $\rho_1$.*

The low-end relation fixes the value at $k = 1$ through the volume of the direction once the ratio $\rho_1$ is known, and positivity alone places every $\rho_k$ in $(0, f^\star)$. The bracket of (ii) is the image of $\rho_1 \in (0, f^\star)$ under $\rho \mapsto -a/(p_1(0) + p_2(0)\rho)$; the nested sets of (iii) sharpen that condition with the conditions $\langle 1\rangle_k > 0$ and $f^\star\langle 1\rangle_k > \langle 1\rangle_{k+1}$ at every $k \leq K+1$, and they shrink to $\rho_1$ at the rate $(f^\star/\lambda')^K$, up to a power of $K$. The exercises work the brackets and the rate of Theorem 41.5 on the centre of three marks (Exercises 54.1 and 54.19), and the continued fraction on $W_3$ at $(\mathbf{1}; 0, 2)$ (Exercise 54.18).

## 41.4 The procedure

**Definition 41.8** (The procedure that reads a ray from its list)**.** Input: a family whose weights lie in a field $\Bbbk \subseteq \mathbb{R}$, $\mathbb{Q}$ for a list; a ray of it with $\kappa_{1,S} > 0$ for every nonempty proper $S$ and its origin in $\Bbbk$, the origin 0 unless stated; the precision wanted. Output: a spanning set and its one-step matrix with a certificate, a recurrence of the values that holds at every strength, the values at the positive integers, and brackets for them.

(1) *Walls and symmetries.* The walls $\kappa_{1,S}$ and the volume $V_1 = \sum_\pi \prod_{l=2}^{n} \kappa_{1,U_l}^{-1}$ of the direction (Proposition 39.34); a finite group $G$ generated by maps of the four kinds of Proposition 39.11(ii) that are symmetries of the ray.

(2) *The box.* Through the boxes $\mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ with level bounds kept by $G$, each with a ranking, and the $\mathfrak{s} < \mathfrak{t}$, in an order that reaches each, the first box accepted at some $\mathfrak{s}$, with its spanning set and its stems (Definition 39.20); the search stops (Theorem 39.22). Acceptance is decided by eliminations of the relations at strengths drawn at random, for a list modulo a prime (Proposition 41.10(iii) and (v), below).

(3) *The one-step matrix.* At strengths drawn at random, for a list modulo primes, the reductions of the targets, from the relations they need (Proposition 41.10(vi)); the entries of $\mathbf{B}$ over $\Bbbk(k)$ by rational reconstruction from these values (Proposition 41.10(ii)), for a list with Chinese remaindering and rational reconstruction of the coefficients over the primes; a check against eliminations of the whole box at strengths and primes drawn afresh (Proposition 41.10(iv) and (v)). A certificate supported on the stems exists (Theorem 39.21(ii)), and every solution, the values among them, satisfies $\varsigma^{-1} v_V = \mathbf{B} v_V$ (Theorem 39.21(iii)).

(4) *The recurrence.* The first dependence of the rows $w_i(k) = w_{i-1}(k-1)\,\mathbf{B}(k)$ (Proposition 39.6(ii)), read at points and reconstructed as in step (3), its coefficients being proportional to signed maximal minors (Proposition 39.6(iii)): $\sum_j p_j(k) f^j \in \mathcal{L}$, the law of [1] when the spanning set is independent (Theorem 39.21(iv)), as it is when the values of the spanning set at the nondegenerate native gauges span (Proposition 39.42(i)). Its end coefficient $p_0$ vanishes only at the shifted zeros of the denominator of $\mathbf{B}$ and at the strengths where the rows line up (Proposition 39.6(iii)).

(5) *The values.* Either the gauge end, from the recurrence, $\alpha^\star$ and $\det H$, when $f(\alpha^\star)$ is a simple root of its symbol (Definition 40.10); or, when the hypotheses of Theorem 41.3(iii) hold, positive selection with the low-end relation (Theorem 41.3(iv)), with brackets from positivity (Proposition 41.4).

(6) *Checks, outside the construction.* When the origin is integral and every letter exponent of the direction is at least 1, the closings of the spanning set satisfy $\mathsf{C}_{j+1}(\mathbf{b}) = \mathbf{B}(-j)\,\mathsf{C}_j(\mathbf{b})$ for all large $j$ (Theorem 39.33, Theorem 39.21(iii)); for the origin 0, the constant terms of the $f^{-j}$ satisfy the reflected recurrence (Corollary 39.38(i)); when $\mathbf{B}$ is regular at $\infty$, the native values of nondegenerate native gauges found by other means are reciprocal eigenvalues of $\mathbf{B}(\infty)$ (Proposition 39.42(ii)); the exponent at $f(\alpha^\star)$ is $\frac{n-3}{2}$ (Theorem 40.6(iii)); the low-end relation returns $V_1$ from the values read; two readings from different depths agree.

Every relation of a box is a relation of the module, so every identity that the elimination gives over $\Bbbk(k)$ holds at every strength (Proposition 39.18, Theorem 39.21), and the recurrence of step (4) holds for the values at every strength at which they converge (Proposition 40.5). Chance enters only through the strengths and primes at which steps (2) to (4) evaluate: Proposition 41.10 bounds it by degrees that the box fixes, and enough strengths remove it. The procedure needs no bound on the box: the search stops (Theorem 39.22).

The procedure of Section 39.12 reads the same module from its solutions, with no box reduced. Its output is certified when the solutions supplied annihilate no submodule (Corollary 39.46), and the closings alone certify it exactly when the native gauge of the direction sees every nonzero submodule (Corollary 39.59); otherwise a unique solution of its equations is evidence. The procedure above certifies the matrix of its spanning set by construction: the certificate of the box is an identity of the relations, whatever any solution sees, and the matrix is the one-step matrix of the

module exactly when the spanning set is independent. The two check each other: every solution that Section 39.12 uses satisfies $\varsigma^{-1}v_V = \mathbf{B}v_V$ on the spanning set of the box (Theorem 39.21(iii)), which is step (6) for the closings, and where the supplied solutions annihilate a submodule, the matrix of the box is still certified by its certificate. Section 39.12 ends with the law of [1]; the procedure above goes on to the values, at the gauge end or by positive selection. A recurrence read off the closings is a recurrence of the values when it holds for all large $j$ and every letter exponent of the direction is positive (Corollary 39.60); finitely many closings do not show that, while the recurrence of step (4) carries a certificate.

**Proposition 41.9** (The relations of a box are difference equations)**.** *Let a box $\mathfrak{B} = \mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}}, \bar{\mathbf{b}})$ of a ray be given (Definition 39.15).*

*(i) The symbols at a level $(\mathbf{a}, \mathbf{b})$ are the points of the affine lattice $\{u \in \mathbb{Z}^{\mathcal{E}} : u_{\mathcal{E}} = \sum_j (a_j r_j + b_j r_j^*)\}$, of rank $n-1$. There is a constant $C$, depending on the ray and the level, such that the coordinates $u_e$, $e \neq e_0$, of the symbols of wall degree at most $\mathfrak{t}$ at that level lie in intervals of length at most $C(1+\mathfrak{t})$; so they number at most $\big(C(1+\mathfrak{t})+1\big)^{n-1}$.*

*(ii) A letter identity of table $j$ ties a symbol to the $|\mathcal{B}_j|$ symbols one unit of that letter up, with weights as coefficients; the mark relation of a symbol at a mark ties it to at most $\sum_j |\mathcal{B}_j|$ symbols one unit of a letter up, with coefficients affine in $k$. The shifts of $(u; \mathbf{a}, \mathbf{b})$ that occur lie in one finite set that does not depend on the symbol.*

*(iii) In the basis of the symbols the relations of the box form a matrix $A(k) = A_0 + kA_1$ with entries in the field of the weights and the origin; $A_1$ vanishes outside the rows of the mark relations, where its entries are $m_{1,e}$, $-s_{1j}\theta_{j,T}$ and $-t_{1j}\theta_{j,T}$. The same holds for the orbit averages of the relations modulo a finite group of symmetries that keeps the box, in the basis of the orbit averages of the symbols.*

**Proposition 41.10** (What a reading by evaluation certifies)**.** *Let a finite group $G$ of symmetries keep a box $\mathfrak{B}$ of a ray, with the field $\Bbbk$ of Definition 39.15 and a ranking (Definition 39.20). Let $A(k)$ be the matrix whose rows are the orbit averages of the letter identities and of the mark relations that span $\mathcal{R}_{\mathfrak{B}}$, in the basis of the orbit averages ordered by the ranking, so that its rows span $\mathcal{R}_{\mathfrak{B}}^G$ and its entries are affine in $k$ (Proposition 41.9(iii)). Let $P$ be the pivot orbits and $F$ the free orbits, $r = |P|$, and let $A_P(k)$ and $A_F(k)$ be $r$ rows of $A$ that are independent over $\Bbbk(k)$, read on the columns $P$ and $F$.*

*(i) $r$ is the rank of $A$, $A_P(k)$ is invertible over $\Bbbk(k)$, and the reduction of $x \in \mathcal{F}_{\mathfrak{B}}^G$ with coefficients in $\Bbbk$ is $x_F - x_P A_P(k)^{-1} A_F(k)$; $\det A_P$ is a nonzero polynomial of degree at most $r$. At $k_0 \in \Bbbk$ the elimination of all the rows of $A(k_0)$ in the order of the ranking has the pivots $P$ exactly when some minor of order $r$ of $A(k_0)$ on the columns $P$ is nonzero, as when $\det A_P(k_0) \neq 0$; then the reduction it gives of every such $x$ is the value at $k_0$ of the reduction over $\Bbbk(k)$.*

*(ii) When the box is accepted, every entry of its matrix $\mathbf{B}$ is a polynomial of degree at most $r$ divided by $\det A_P$. A rational function whose numerator and denominator have degrees at most $d_1$ and $d_2$ is fixed, among such functions, by its values at any $d_1 + d_2 + 1$ points at which the denominators do not vanish; with a common denominator known, a numerator of degree at most $d_1$ is fixed by $d_1 + 1$ values.*

*(iii) The verdict of the elimination at $k_0$ on acceptance at $\mathfrak{s}$, whether the reductions of the targets are combinations of the free Laurent orbits of wall degree at most $\mathfrak{s}$, is the verdict over $\Bbbk(k)$ except at no more than $2r$ strengths $k_0$.*

*(iv) Let the box be accepted, and let* $\mathbf{B}'$ *be a matrix over* $\Bbbk(k)$ *whose entries have a common denominator* $Q'$*, with* $Q'$ *and the numerators of degree at most* $d$*. If* $\mathbf{B}' \neq \mathbf{B}$*, then* $\mathbf{B}'(k_0)$ *agrees with the matrix that the elimination at* $k_0$ *gives at no more than* $r+d$ *strengths* $k_0$ *with* $Q'(k_0) \neq 0$ *at which the elimination has the pivots* $P$*. So agreement at* $r+d+1$ *such strengths proves* $\mathbf{B}' = \mathbf{B}$*, and a wrong* $\mathbf{B}'$ *passes the check at a strength drawn uniformly from a finite set* $S \subseteq \Bbbk$ *with probability at most* $(r+d)/|S|$*.*

*(v) Let* $\Bbbk = \mathbb{Q}$*, and let* $q$ *be a prime that divides no denominator of an entry of* $A$*, with* $\det A_P$ *not identically* 0 *modulo* $q$*. Then* $A$ *modulo* $q$ *has the rank* $r$ *and the pivots* $P$ *over* $\mathbb{F}_q(k)$*, its reductions are those over* $\mathbb{Q}(k)$ *taken modulo* $q$*, and (i) to (iv) hold over* $\mathbb{F}_q$ *with strengths in* $\mathbb{F}_q$*. A* $\mathbf{B}'$ *over* $\mathbb{Q}(k)$ *with* $\mathbf{B}' \neq \mathbf{B}$ *passes the check modulo* $q$ *at a strength drawn uniformly from* $\mathbb{F}_q$ *with probability at most* $(r+d)/q$*, for every prime* $q$ *outside a finite set fixed by* $A$ *and* $\mathbf{B}'$*.*

*(vi) If a subspace* $\mathcal{R}' \subseteq \mathcal{R}^G_{\mathfrak{B}}$ *contains* $x - \bar{x}$ *for every target* $x$*,* $\bar{x}$ *its reduction, then the reduction of every target modulo* $\mathcal{R}'$*, defined as in Definition 39.20(ii) with* $\mathcal{R}'$ *in place of* $\mathcal{R}^G_{\mathfrak{B}}$*, is* $\bar{x}$*.*

An elimination over $\Bbbk(k)$ carries polynomials whose degrees are bounded only by the number $r$ of pivots, which grows with the box. By (ii) a reconstruction from evaluations needs only as many strengths as the degrees of the entries of $\mathbf{B}$ ask, and by (iv) a check at one fresh strength drawn from a set $S$ bounds the chance of a wrong result by $(r+d)/|S|$, while $r+d+1$ checks remove it.

A ray with a real origin is read in the same way: the box carries the origin in its mark relations (Definition 39.15(iii)), and the stride's law, its density and the gauge end are those of Chapter 40, which allow a real origin, the constant $g(0)$ carrying the factor $(\omega_0/dy)(\alpha^\star)$ (Proposition 40.3(ii)); the low-end relation and Theorem 41.3(iii) and (iv) ask for the origin 0. Data with rational masses are read through an origin.

**Proposition 41.11** (Data with rational masses)**.** *Let data* $(\mathbf{m};\mathbf{s},\mathbf{t})$ *of a family on the balance wall have* $\mathbf{m} \in \mathbb{Q}^{\mathcal{E}}$*,* $\mathbf{s},\mathbf{t} \in \mathbb{N}^N$ *and every wall* $\kappa_U$*,* $U$ *nonempty and proper, positive, and put* $f_{\mathbf{m}} = \alpha^{\mathbf{m}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}$ *on the positive orthant.*

*(i) Let* $\mathbf{m}' \in \mathbb{Z}^{\mathcal{E}}$ *put* $(\mathbf{m}';\mathbf{s},\mathbf{t})$ *on the balance wall with every wall positive. Then the ray with the origin* $(\mathbf{m}-\mathbf{m}';\mathbf{0},\mathbf{0})$ *and the direction* $(\mathbf{m}';\mathbf{s},\mathbf{t})$ *has positive walls at every real* $k \geq 1$*, its values converge there, and* $\langle 1\rangle_1 = P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*.*

*(ii) Let* $N \geq 1$ *be an integer with* $N\mathbf{m} \in \mathbb{Z}^{\mathcal{E}}$*. Then the ray with the origin* 0 *and the direction* $N(\mathbf{m};\mathbf{s},\mathbf{t})$ *has positive walls, its stride is* $f_{\mathbf{m}}^N$*,* $\langle 1\rangle_{1/N} = P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*, and the wall degree of a symbol* $g$ *for it is* $\frac{1}{N}\max_U\big(-\kappa_U(g)/\kappa_U\big)$*, the maximum over the nonempty proper* $U$*.*

So data with rational masses are read at $k=1$ on a ray with an origin, by the gauge end, or at $k=\frac{1}{N}$ on a ray from the origin 0, whose box of wall degree $\mathfrak{t}$ holds the symbols of wall degree $N\mathfrak{t}$ read on the walls of the data.

## 41.5 What a reading costs

A box has at most $\big(C(1+\mathfrak{t})+1\big)^{n-1}$ symbols at each level (Proposition 41.9(i)), and modulo a group $G$ of symmetries its orbits number at least its symbols divided by $|G|$; one evaluation in step (3) of Definition 41.8 is one elimination of its relations, and the number of evaluations is set by the degrees of $\mathbf{B}$ (Proposition 41.10(ii)). The relations are affine in the strength, so one factorization at one strength gives every Taylor coefficient there.

**Proposition 41.12** (One factorization serves every coefficient)**.** *In the setting of Proposition 41.10, let $k_0 \in \Bbbk$ have $\det A_P(k_0) \neq 0$, and put $M = A_P(k_0)$ and $E = A_P(k_0+1) - A_P(k_0)$, the part of $A_1$ on the rows and columns of $A_P$, so that $A_P(k_0+\epsilon) = M + \epsilon E$. For every row $x_P$ over $\Bbbk$, as power series in $\epsilon$ convergent for small $|\epsilon|$,*

$$x_P\, A_P(k_0+\epsilon)^{-1} = \sum_{j\geq 0} \lambda_j \epsilon^j, \qquad \lambda_0 = x_P M^{-1}, \quad \lambda_j = -\lambda_{j-1} E M^{-1}.$$

*So the Taylor coefficients at $k_0$ of the reductions, and of the entries of $\mathbf{B}$, up to any order $J$ come from one factorization of $M$ and $J$ products by $EM^{-1}$; and a rational function whose numerator and denominator have degrees at most $d_1$ and $d_2$, the denominator not vanishing at $k_0$, is fixed, among such functions, by its Taylor coefficients of order at most $d_1 + d_2$ at $k_0$.*

So one factorization at one strength can replace many evaluations; each step of the series is one solve with the factorization of $M$, and what it saves depends on the fill of that factorization.

**Proposition 41.13** (The densities of a spanning set)**.** *Let a ray have a real origin and $\kappa_{1,S} > 0$ for every nonempty proper $S$, with $\alpha^\star$, $f^\star = f(\alpha^\star)$ and the function $g$ of Proposition 40.3, and let $b_1, \dots, b_{\mathfrak{m}}$ be Laurent insertions with real coefficients and $Q$, $\tilde{P}_{il}$ polynomials with $Q(k)\langle b_i\rangle_{k-1} = \sum_l \tilde{P}_{il}(k)\langle b_l\rangle_k$ as meromorphic functions, as a box gives with $\mathbf{B} = \tilde{P}/Q$ (Theorem 39.21(iii)). For a Laurent insertion $b$ with real coefficients let $\mu_{f,b}$ be the image under $f$ of the signed measure $b\,\omega_0$ on the positive part of the chart, a signed measure on $(0, f^\star]$, finite on $[t, f^\star]$ for $t > 0$.*

- *(i) $\langle b\rangle_k = \int t^k \mu_{f,b}(dt)$ wherever $\langle b\rangle_k$ converges absolutely; $\mu_{f,b}$ has a density $\varrho_{f,b}$, real-analytic on $(0, f^\star)$, and near $f^\star$, $\varrho_{f,b}(t) = (f^\star - t)^{\frac{n-3}{2}} g_b(f^\star - t)$ with $g_b$ holomorphic at 0 and $g_b(0) = b(\alpha^\star)\, g(0)$.*
- *(ii) As distributions on $(0,\infty)$, $Q(-1-\theta)\big(t^{-1}\mu_{f,b_i}\big) = \sum_l \tilde{P}_{il}(-1-\theta)\,\mu_{f,b_l}$ for every $i$, $\theta = t\frac{d}{dt}$: a system of order at most $\max(\deg Q, \deg \tilde{P}_{il})$ for the $\mathfrak{m}$ densities.*
- *(iii) With $v(k) = (\langle b_l\rangle_k)_l$ and every $K \geq 2$ at which $\mathbf{B}(2), \dots, \mathbf{B}(K)$ are finite and $v(K)$ converges, the continuation of $v$ is finite at 1 and $v(1) = \mathbf{B}(2)\mathbf{B}(3)\cdots\mathbf{B}(K)\, v(K)$: the values of the spanning set run back from $K$ with no scalar recurrence.*

The density operator of a scalar recurrence has the order of the degree of its coefficients, which carry the strengths where the rows line up (Proposition 39.6(iii)); the system of (ii) has the order of the degrees of the entries of $\mathbf{B}$, for $\mathfrak{m}$ densities whose germs at $f^\star$ start from the values $b_l(\alpha^\star)$ times the one constant $g(0)$.

**Proposition 41.14** (The native gauges of all exponents)**.** *Let a family of $N$ tables on $n \geq 2$ marks have the letter complement $X_F$ (Definition 38.1), and for $\alpha \in X_F$ and complex exponents $(\mathbf{s}, \mathbf{t}) \in \mathbb{C}^{2N}$ write $\mathbf{u}(\alpha; \mathbf{s}, \mathbf{t}) = \sum_j \big(s_j R_j(\alpha) + t_j R_j^*(\alpha)\big) \in \mathbb{C}^{\mathcal{E}}$ for the usage (Definition 8.17).*

- *(i) The data $\big(\mathbf{u}(\alpha; \mathbf{s}, \mathbf{t}); \mathbf{s}, \mathbf{t}\big)$ lie on the balance wall. Call $\alpha \in X_F$* native for *data $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ on the balance wall when $\mathbf{m} = \mathbf{u}(\alpha; \mathbf{s}, \mathbf{t})$: the native gauges of a ray are the points native for its direction (Definition 38.4), and for $m_{\mathcal{E}} \neq 0$ the points native for data are its native points (Definition 8.17). So the set of the $(\alpha, \mathbf{m}, \mathbf{s}, \mathbf{t})$ with $\alpha$ native for $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ is the graph of the map $(\alpha, \mathbf{s}, \mathbf{t}) \mapsto \mathbf{u}(\alpha; \mathbf{s}, \mathbf{t})$ on $X_F \times \mathbb{C}^{2N}$, a smooth irreducible variety of dimension $n-1+2N$, and every $\alpha \in X_F$ is native for the data $\big(\mathbf{u}(\alpha; \mathbf{s}, \mathbf{t}); \mathbf{s}, \mathbf{t}\big)$, whatever $(\mathbf{s}, \mathbf{t})$.*
- *(ii) A ray of $F$ with $\kappa_{1,S} > 0$ for every nonempty proper $S$ has at most $(-1)^{n-1}\,\mathrm{eu}(X_F)$ nondegenerate native gauges, the number of native gauges of the directions outside the exceptional set of Theorem 38.31(i).*

The nondegenerate native gauges bound the sizes of a reading from below: the dimension of the invariant module is at least the number of their orbits (Theorem 39.12(iii)), and the order of the law of [1] at least the number of their distinct native values (Corollary 38.7 at $g = 1$); Proposition 41.14(ii) bounds their number from above. These are inequalities, with $X_F$ inverting every letter whatever its exponent along the ray (Definition 38.1).

**Definition 41.15** (The procedure that finds native gauges by continuation)**.** Input: a family and a ray of it with $\kappa_{1,S} > 0$ for every nonempty proper $S$. Output: nondegenerate native gauges of the ray, each with a certificate, and their native values.

(1) *A start.* Draw $(\alpha_0, \mathbf{s}_0, \mathbf{t}_0) \in X_F \times \mathbb{C}^{2N}$ at random; $\alpha_0$ is native for the data $w_0 = \big(\mathbf{u}(\alpha_0; \mathbf{s}_0, \mathbf{t}_0); \mathbf{s}_0, \mathbf{t}_0\big)$ (Proposition 41.14(i)).

(2) *More points native for $w_0$.* Follow the points native for $w_0$ found so far along closed loops $w_0 \to w_1 \to w_2 \to w_0$ of random data on the balance wall, by Newton's method along the segments, and add every new endpoint; stop when several loops in a row add nothing.

(3) *To the direction.* Follow each point native for $w_0$ by Newton's method along a path of data from $w_0$ to the direction of the ray, and keep the endpoints that lie in $X_F$.

(4) *Certificates.* Around each endpoint, a box in the coordinates $\alpha_e$, $e \neq e_0$, of the chart on which interval enclosures of the letters exclude 0; Krawczyk's interval Newton test on the equations $c_e^{(1)} = 0$, $e \neq e_0$, in the box, and an interval enclosure of the determinant of the Hessian of $\log f$ on the box that excludes 0. When both succeed, the box holds a native gauge of the ray, and every native gauge in it is nondegenerate. Disjoint boxes hold distinct native gauges, and disjoint enclosures of $f$ on them prove distinct native values.

The certified native gauges of step (4) are distinct nondegenerate native gauges of the ray, so their number bounds the dimension of the ray from below (Theorem 38.6(iii)), the number of their orbits under a finite group of symmetries bounds $\dim \mathfrak{M}^G$ from below (Theorem 39.12(iii)), and the number of their distinct native values bounds the order of every recurrence that follows from the relations (Corollary 38.7); by Proposition 41.14(ii) their number is at most $(-1)^{n-1}\,\mathrm{eu}(X_F)$. The stopping rule of step (2) is not a certificate that every point native for $w_0$ was found, so the procedure gives lower bounds. Nothing in it writes a relation of a box. Exercise 55.10 reads the dimensions and the orders of the laws by the box, ray by ray.

# Chapter 42

# Inputs along a ray

A function that satisfies a recurrence with polynomial coefficients in its argument, above all the values of 1 along a ray continued in the strength, gives two live inputs on a lattice: its values and its slopes. The slopes satisfy the same recurrence, inhomogeneous by the values read through the derivatives of its coefficients. A Trailmoment at index zero of a polynomial word against either input is closed by summation by parts in the adjoint form of the recurrence: what crosses the cut is read off the Laurent data of the function at the low end, for a ray the low-end relation and its derivative, and when the symbol does not vanish at 1 everything else lands on a fixed block of $2D$ Trailmoments, $D$ the degree of the recurrence. A mark of a list opened on its share side and closed at an integer strength is such a pairing: a polynomial word against the slope input of the room, less its derivative against the values input. The recurrence read at the poles of an input ties its initial slopes to residues and finite parts, and minimality, the root rate of the input, removes the components along the faster solutions. A recurrence of the values that is self-adjoint under the reflection of the strength pairs the values with the closings in one constant, the low-end relation, whose limit at high strength ties the volume of the direction to the Hessian at its native gauge through the native values. Section 42.6 reads all of this on the centre ray of the list $\binom{\mathcal{E}}{1}$, for every number $n$ of marks: its native gauges are the signs, and at a lean the points whose occupations take two values; a parallel class enters a list through the closings and the signs of $\binom{P}{1}$; a mark is added by closing the critical-line kernel against the slopes of $n$ marks; the law of the values is the $n$-th symmetric power of the second-order relation that the Gramma of one mark satisfies in its coefficient, read in the index, of order $\lceil n/2 \rceil$, with the native values $(n-2j)^{-2}$ as the roots of its symbol, and one identity carries it from $n$ marks to $n+1$; the lowered settings are its solutions without low-end coefficient; and the exact pairing holds at every strength. The $n$-th power of the Gramma of one mark is a mixture of its dilations by the stride's law, and the balanced directions with one total and one multiset of offsets have one value up to Beta corners. The sections after it read a table built by substitution against its cells: its reciprocal stride factors over them, its closings average those of its core over the norms of its cells, its native gauges are read down the hinge tree, and its value at every strength, with norms on its marks, is the word of its core with the word of each cell set at the load of its marker. The law of a class is read for every size, at silence, at infinity and on its hinge plane, and the sources of a substitution of classes are counted in its hinge space.

## 42.1 Inputs read from a ray

**Definition 42.1** (Recurrences; the values input and the slope input)**.** A *recurrence* of order $h$ is an operator $(L\varphi)(x) = \sum_{i=0}^{h} c_i(x)\,\varphi(x+i)$ with polynomials $c_i \in \mathbb{C}[x]$ and $c_0 c_h \neq 0$. Its *adjoint* and its *derivative* are

$$(L^*\varphi)(x) = \sum_{i=0}^{h} c_i(x-i)\,\varphi(x-i), \qquad (L'\varphi)(x) = \sum_{i=0}^{h} c_i'(x)\,\varphi(x+i),$$

$L'^*$ is the adjoint of $L'$, its *degree* is $D = \max_i \deg c_i$, and its *symbol* is $\chi(\lambda) = \sum_i [x^D] c_i\,\lambda^i$, as in Corollary 38.7. Let $p$ be meromorphic on $\mathbb{C}$ with $Lp = 0$ identically, holomorphic at the points of a coset $b+\mathbb{Z}$ beyond a cut $N$, and put $\tilde{p} = -p'$. The *values input* of $p$ is $p\,\mathbb{I}_{b+\mathbb{Z}}$ and its *slope input* is $\tilde{p}\,\mathbb{I}_{b+\mathbb{Z}}$ (Definition 5.4), both read beyond the cut $N$. When $p$ continues the values $\langle 1\rangle_k$ of a ray and $L$ is a recurrence of these values, they are the *inputs read from the ray.*

**Proposition 42.2** (The values of a ray as an input)**.** *Let a ray of a family on $n \geq 2$ marks have origin 0, every letter exponent of its direction positive and $\kappa_{1,S} > 0$ for every nonempty proper $S$, and let $L$ be a recurrence of order $h$ with $\sum_{i=0}^{h} c_i(k)\langle 1\rangle_{k+i} = 0$ for all large integers $k$.*

*(i) $\langle 1\rangle_k$ is holomorphic in $k$ on $\operatorname{Re} k > 0$ and continues to a meromorphic function $p$ on $\mathbb{C}$ with $Lp = 0$ identically. At every $j \in \mathbb{N}$, $\lim_{k\to -j}(k+j)^{n-1}p(k) = V_1\,\mathsf{C}_j(1)$, with $V_1$ of Proposition 39.34, and $\sum_{i=0}^{\min(h,j)} c_i(-j)\,\mathsf{C}_{j-i}(1) = 0$.*

*(ii) The slope input satisfies $L\tilde{p} = L'p$ identically: a recurrence of the same order, inhomogeneous by the values read through the derivative of $L$.*

*(iii) Along the positive integers $p$ has the root rate $f(\alpha^\star)$, the native value of the native gauge $\alpha^\star$ of the direction on the positive orthant, and $\tilde{p}$ a root rate at most $f(\alpha^\star)$.*

The values of a ray are therefore an input of the kind of Chapter 6: their density is read from the serif family, and the recurrence gives the values input and the slope input recurrences of one finite order. The slope input carries what the derivative in the strength adds, and the values input enters its recurrence only through $L'$.

## 42.2 Closing an input

**Proposition 42.3** (Summation by parts at the cut)**.** *Let $p$ and $L$ be as in Definition 42.1, $p$ holomorphic at the integers $x \geq 1$, with $p$ and $\tilde{p}$ of root rate below one along them, and for $0 \leq j < h$ put $\beta_j(x) = \sum_{i=0}^{j} c_i(x)\,p(x+i)$, which is holomorphic at $x = -j$ because $\beta_j = -\sum_{i>j} c_i(x)\,p(x+i)$. For every polynomial $R$,*

$$(\mathsf{T}_{0;(1)}[x]\,(L^*R)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\sum_{j=0}^{h-1} R(-j)\,\beta_j(-j),$$

$$(\mathsf{T}_{0;(1)}[x]\,(L^*R)\,\tilde{p}\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (\mathsf{T}_{0;(1)}[x]\,(L'^*R)\,p\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + \sum_{j=0}^{h-1} R(-j)\,\beta_j'(-j).$$

*If every $c_i$ vanishes to order two at the integers $-j$ with $1 \leq j < i$, only $j = 0$ remains, with $\beta_0 = c_0\,p$: the boundary is the value and the derivative of $c_0 p$ at the cut.*

What a summation by parts puts on the inputs beyond the cut is the adjoint word; what it leaves at the cut is read off $p$ at $0, -1, \dots, -(h-1)$, where for the values of a ray the poles sit. The same move closes the slope input, once through $L$ and once through $L'$.

**Proposition 42.4** (The fixed block)**.** *Let $L$ be a recurrence of degree $D$ whose symbol has $\chi(1) \neq 0$, and let $\Bbbk$ be a field containing the coefficients of the $c_i$.*

*(i) $L^*$ raises the degree of every nonzero polynomial by exactly $D$, and $\Bbbk[x] = L^*(\Bbbk[x]) \oplus \{r \in \Bbbk[x] : \deg r < D\}$.*

*(ii) In the setting of Proposition 42.3, for every polynomial $W \in \Bbbk[x]$ the two Trailmoments*

$$(\mathsf{T}_{0;(1)}[x]\, W\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)), \qquad (\mathsf{T}_{0;(1)}[x]\, W\, \tilde{p}\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$$

*lie in the $\Bbbk$-span of the $2D$* block values

$$(\mathsf{T}_{0;(1)}[x]\, (x-1)^a\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)), \qquad (\mathsf{T}_{0;(1)}[x]\, (x-1)^a\, \tilde{p}\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) \qquad (0 \le a < D)$$

*and of the boundary numbers $\beta_j(-j)$ and $\beta'_j(-j)$, $0 \le j < h$.*

This is the counterpart at the index zero of Theorem 6.9: the words of a closing are polynomials, the shifts of the input are moved onto the word, and the input is read on one block of evaluations fixed once for the recurrence. When the boundary numbers are known, as they are at the cut of a ray from the low-end relation, a closing lands on the block plus a number of $\Bbbk$.

## 42.3 A mark opened at an integer strength

A mark that is neither a loop nor a coloop is a bunch of width one, and Theorem 19.13 opens the centre on it: at the strength $s$ the period is the integral over one even share $d$ of $\Gamma_2^4 \begin{bmatrix} d,\ d,\ s-d,\ s-d \\ s,\ s \end{bmatrix}$ against the period of the room family, the deletion at the share $d$ and the contraction at $s-d$. Closing the share (Corollary 19.21) at an integer strength turns its heads into one polynomial word.

**Corollary 42.5** (A mark opened at an integer strength)**.** *Let $\mathcal{B}$ be a connected list on $\mathcal{E}$, $|\mathcal{E}| \ge 3$, and $e$ a mark that is neither a loop nor a coloop; on the chart $y_{e_0} = 0$ of $R = \mathcal{E} - e$ let $X_\backslash = K_{\mathcal{B}\backslash e}(\alpha) K_{\mathcal{B}\backslash e}(\alpha^{-1})$ and $X_/ = K_{\mathcal{B}/e}(\alpha) K_{\mathcal{B}/e}(\alpha^{-1})$, the even letters of the deletion and of the contraction over $\alpha^R$. Suppose that one lies below the other: $\{X_\flat, X_\sharp\} = \{X_\backslash, X_/\}$ with $\rho = \sup X_\flat / X_\sharp < 1$ on the chart. For $s \in \{1, 2, \dots\}$ and $x \ge 0$ put*

$$q_s(x) = \int_{y_{e_0}=0} \Big(\frac{X_\flat}{X_\sharp}\Big)^x X_\sharp^{-s}\, dy, \qquad \tilde{q}_s = -\partial_x q_s, \qquad V_s(x) = \big((x+1)(x+2)\cdots(x+s-1)\big)^2.$$

*Then $q_s$ converges and is holomorphic on a neighbourhood of $[0,\infty)$, and*

$$\Gamma(s)^2\, P_{\mathcal{B}}(s\mathbf{1}; s, s) = \sum_{x \in \mathbb{N}} \big(V_s(x)\, \tilde{q}_s(x) - V'_s(x)\, q_s(x)\big),$$

*the terms of root rate at most $\rho$: the centre at the strength $s$ is a polynomial word against the slope input of $q_s$ less its derivative against its values input, read on $\mathbb{N}$.*

The function $q_s$ is read along the direction of the ratio $X_\flat / X_\sharp$ of the two even letters, with the origin form $X_\sharp^{-s} dy$, which decays where the ratio does not; where it satisfies a recurrence in $x$, Sections 42.2 and 42.4 close it, and its values half $\sum_x V_s(x)\, q_s(x)$ with it.

## 42.4 Boundaries and minimality

**Proposition 42.6** (Boundary relations)**.** *Let $p$ and $L$ be as in Definition 42.1, let $x_0 \in \mathbb{C}$ and $0 \le j < h$, and let $p$ be holomorphic at $x_0 + i$ for $j < i \le h$. Then $\beta_{x_0,j}(x) = \sum_{i=0}^{j} c_i(x)\, p(x+i)$ is holomorphic at $x_0$, its value and its derivative there are read off the Laurent expansions of $p$ at $x_0, \dots, x_0 + j$ and the Taylor expansions of $c_0, \dots, c_j$ at $x_0$, and*

$$\sum_{i=j+1}^{h} c_i(x_0)\, p(x_0+i) = -\beta_{x_0,j}(x_0),$$

$$\sum_{i=j+1}^{h} \big(c_i(x_0)\, \tilde{p}(x_0+i) - c_i'(x_0)\, p(x_0+i)\big) = \beta_{x_0,j}'(x_0).$$

*Where $c_h$ has no zero at the integers $x \ge 1$, the slope input on these integers is fixed by its initial slopes $\tilde{p}(1), \dots, \tilde{p}(h)$ through $L\tilde{p} = L'p$, and every second relation at an integer $x_0$ with $x_0 + j \ge 0$ is an affine relation among them; at $x_0 = -j$ its right side is read off the Laurent data of $p$ at the nonpositive integers $-j, \dots, 0$ alone. At the cut of the values of a ray (Proposition 42.2), with $c_0(x) = \sum_m \gamma_m x^m$ and $p(x) = \sum_{i \ge 0} \mathsf{L}_i x^{i-n+1}$ near $x = 0$, the low-end coefficients of Definition 39.37, $c_0$ vanishes to order at least $n-1$ at 0, $\mathsf{L}_0 = V_1$, and at $x_0 = 0$, $j = 0$ the two relations are*

$$\sum_{i=1}^{h} c_i(0)\, p(i) = -\gamma_{n-1} V_1, \qquad \sum_{i=1}^{h} \big(c_i(0)\, \tilde{p}(i) - c_i'(0)\, p(i)\big) = \gamma_{n-1}\mathsf{L}_1 + \gamma_n V_1,$$

*the low-end relation (Corollary 39.38(ii), which holds for every recurrence of the values at the large integers by Proposition 40.5) and its derivative; when every table of the family has unit weights, $\mathsf{L}_1 = 0$.*

At a pole of the input that a wall or a zero share puts on the lattice, the right sides are its principal part and its finite part: at a simple pole $x_0$ where $c_0$ has a simple zero, $\beta_{x_0,0}(x_0) = c_0'(x_0) \operatorname{Res}_{x_0} p$ and $\beta_{x_0,0}'(x_0) = c_0'(x_0)\, a + \frac{1}{2} c_0''(x_0) \operatorname{Res}_{x_0} p$, $a$ the constant term of $p$ at $x_0$. Each independent relation removes one direction from the initial slopes; what the relations leave free, the root rate of the input fixes in the directions that grow faster than the input.

**Proposition 42.7** (Minimality)**.** *Let $p$ and $L$ be as in Definition 42.1, $p$ holomorphic at the integers $x \ge 1$, with $c_0(x) c_h(x) \ne 0$ at these integers, every $c_i$ of degree at most $\deg c_0 = \deg c_h$, and the roots $\lambda_1, \dots, \lambda_h$ of the symbol of distinct moduli. Let $\lambda > 0$ be at least the root rates of $p$ and of $\tilde{p}$ along the positive integers (Definition 1.1(vii)).*

*(i) The solutions $u$ of $Lu = 0$ on the integers $x \ge 1$ of root rate at most $\lambda$ form a space $U_\lambda$ of dimension $\#\{c : |\lambda_c| \le \lambda\}$, and every nonzero $u \in U_\lambda$ has $u(x+1)/u(x) \to \lambda_c$ for one of these roots, so the root rate $|\lambda_c|$.*

*(ii) The functions $\varphi$ on the integers $x \ge 1$ with $L\varphi = L'p$ and root rate at most $\lambda$ are the $\tilde{p} + u$, $u \in U_\lambda$, and those with $L\varphi = 0$ are the $p + u$. So the slope input is the only one of them that satisfies a set of second relations of Proposition 42.6 whose left sides, read on $u$, vanish together on no nonzero $u \in U_\lambda$; the values input is singled out by the first relations in the same way.*

So the initial slopes are fixed by as many boundary relations as there are roots of the symbol of modulus at most the root rate of the input; the roots beyond it are removed by the root rate alone.

## 42.5 The two ends paired

**Proposition 42.8** (The two ends paired)**.** *Let a ray be as in Proposition 42.2, with a recurrence $L$ of its values whose coefficients satisfy $c_i(-x-i) = \epsilon\, c_i(x)$ for every $i$, $\epsilon \in \{1,-1\}$, and put $\mathsf{C}_j = \mathsf{C}_j(1)$ for $j \geq 0$ and $\mathsf{C}_j = 0$ for $j < 0$.*

*(i)* $\sum_{i=0}^{h} c_i(\sigma - i)\, \mathsf{C}_{\sigma-i} = 0$ *for every integer $\sigma$: the closings, read at the strength $\sigma$, solve the adjoint recurrence.*

*(ii) The* concomitant

$$B(\sigma) = \sum_{i=1}^{h} \sum_{j=1}^{i} c_i(\sigma - j)\, \mathsf{C}_{\sigma-j}\, \langle 1 \rangle_{\sigma-j+i}$$

*takes one value at every integer $\sigma \geq 1$, $B(\sigma) = \sum_{i=1}^{h} c_i(0)\langle 1\rangle_i = -\gamma_{n-1} V_1$ with $\gamma_{n-1} = [x^{n-1}]c_0$: the low-end relation, carried to every strength.*

*(iii) On the centre ray of a family (Definition 38.13), if every $c_i$ has degree at most $n-1$, then $c_0 = \chi(0)\, x^{n-1}$, $\lambda_\star = f(\alpha^\star)$ is a simple root of the symbol, and*

$$V_1 \det H = -\frac{\lambda_\star\, \chi'(\lambda_\star)}{\chi(0)} = \prod_{c \neq \star} \Big(1 - \frac{\lambda_\star}{\lambda_c}\Big),$$

*with $H$ the Hessian of $-\log f$ at $\mathbf{1}$ on the chart (Theorem 38.15) and $\lambda_c$ the roots of the symbol, counted with multiplicity.*

The values at the high end and the closings at the low end are two solutions of one recurrence and of its adjoint, and their concomitant is the constant that the low end fixes. At high strength the two decay and grow at the reciprocal rates $f(\alpha^\star)^{\pm k}$ with the Gaussian constants of the native gauge of the direction, and the constant becomes an identity between the volume $V_1$ of the direction, read by its walls, and the Hessian at its native gauge, through the other native values.

## 42.6 The centre ray of $\binom{\mathcal{E}}{1}$

For the list $\binom{\mathcal{E}}{1}$ on $n$ marks, whose bases are the singletons, $K(\alpha) = \sum_e \alpha_e$ and $\Psi(\alpha) = \alpha^{\mathcal{E}} K(\alpha^{-1})$, and the stride of the centre ray at the strength $\sigma$ is $\big(K(\alpha)K(\alpha^{-1})\big)^{-\sigma} = \big(\sum_{e,e'} \alpha_e/\alpha_{e'}\big)^{-\sigma}$. Its widths are all one, so $V_1 = n!$ at the strength one. Its share side is one Gramma read $n$ times: by Corollary 18.24 for the dual table, whose bases are the sets $\mathcal{E} - e$, and Proposition 8.15, for masses $\mathbf{m}$ on the balance wall, $m_{\mathcal{E}} = s + (n-1)t$, with $m_S > t(|S|-1)$ for every nonempty proper $S$,

$$\Gamma(s)\,\Gamma(t)\, P_{\binom{\mathcal{E}}{1}}(\mathbf{m}; s, t) = \int_0^\infty y^{t-1} \prod_{e \in \mathcal{E}} \mathsf{G}_{m_e - t;(1,-1)}(1, y)\, dy,$$

and at the centre $\Gamma(s)^2 P_{\binom{\mathcal{E}}{1}}(s\mathbf{1}; s, s) = \int_0^\infty y^{s-1} \mathsf{G}_{0;(1,-1)}(1,y)^n\, dy$. With $\vartheta = y\, d/dy$, the ladder and the exchange relation of the Gramma (Theorem 2.4) give $\vartheta^2 \mathsf{g} = y\, \mathsf{g}$ for $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(1,y)$, and the partner $\mathsf{h}(y) = \sum_{m \geq 0} y^m/m!^2$ of its lattice expansion (Lemma 20.29(i), $\mathsf{g} = -\mathsf{h}\log y + \sum_m 2(\log\Gamma)'(1+m)\, y^m/m!^2$) satisfies the same relation. The statements below read the centre ray from these two functions.

**Proposition 42.9** (The native gauges of $\binom{\mathcal{E}}{1}$)**.** *Let $n \geq 2$, let $s, t \geq 1$ and $m$ be integers with $nm = s + (n-1)t$, and read a ray of $\binom{\mathcal{E}}{1}$ in the direction $(m\mathbf{1}; s, t)$; in this statement $u_e(\alpha) = \alpha_e/K(\alpha)$ is the occupation of the mark $e$ and $\lambda(\alpha) = \big(K(\alpha)K(\alpha^{-1})\big)^{-1}$.*

*(i) A point $\alpha$ of $X_F$ is a native gauge exactly when its occupations take at most two values, $u_+$ on $n-j$ marks and $u_-$ on $j$ marks, $0 \le j \le n$, $2j \ne n$, with*

$$(n-2j)\,u_+ = 1 - j\,\frac{s-t}{ns}, \qquad u_+ + u_- = \frac{s-t}{ns}, \qquad u_+u_- \ne 0;$$

*then $\lambda(\alpha) = -s\,u_+u_-/t$ and the native value is $f(\alpha) = \big(\prod_e u_e\big)^{m-t}\lambda(\alpha)^t$. Up to scale these points are $\alpha = (u_e)_e$, one for each set of $j$ marks, a set and its complement giving the same point.*

*(ii) At the centre, $s = t = \sigma$, $m = \sigma$, the native gauges are exactly the signs $\eta$ with $K(\eta) = \sum_e \eta_e \ne 0$, of the native values $K(\eta)^{-2\sigma} = (n-2j)^{-2\sigma}$, $j$ the number of marks at $-1$: there are $\lceil n/2 \rceil$ native values. Each of these signs is nondegenerate: on the chart $\alpha_e = \eta_e e^{y_e}$, $y_{e_0} = 0$, the Hessian of $-\log f$ at $y = 0$ has the determinant $(2\sigma)^{n-1}\prod_e \eta_e/K(\eta)^n$.*

A lean moves every sign to a real point with the same pattern: the occupations keep two values, on the same marks, and only the two values change. Every list has the signs with $K(\eta) \ne 0$ among the native gauges of its centre ray (Theorem 38.17); the list $\binom{\mathcal{E}}{1}$ has nothing else.

**Proposition 42.10** (A parallel class enters through its list)**.** *Let $\mathcal{H}$ be a connected list on the marks $1, \dots, q$, $q \ge 2$, and replace every mark $i$ by a parallel class $P_i$ of $s_i \ge 1$ marks by the first rule of Theorem 21.5: the list $\mathcal{B}$ on $\mathcal{E} = \bigsqcup_i P_i$ whose bases are the sets with one mark in $P_i$ for every $i \in T$ and none in the other classes, $T$ a basis of $\mathcal{H}$. Write $A_i = \sum_{e \in P_i} \alpha_e$.*

*(i) $K_{\mathcal{B}}(\alpha) = K_{\mathcal{H}}(A_1, \dots, A_q)$, and for $v \in \mathbb{N}^{\mathcal{E}}$, with $w_i = v_{P_i}$, $[\alpha^v]K^k_{\mathcal{B}} = \big([A^w]K^k_{\mathcal{H}}\big)\prod_i \binom{w_i}{v|_{P_i}}$, with the multinomial coefficients $\binom{w_i}{v|_{P_i}}$.*

*(ii) The closings of the centre ray of $\mathcal{B}$ at the strength one (Definition 39.30) are*

$$\mathsf{C}^{\mathcal{B}}_k(1) = \sum_{w \in \mathbb{N}^q} \big([A^w]K^k_{\mathcal{H}}\big)^2 \prod_{i=1}^{q} \mathsf{C}^{\binom{P_i}{1}}_{w_i}(1), \qquad \mathsf{C}^{\binom{P}{1}}_{w}(1) = \sum_{v \in \mathbb{N}^P,\ |v| = w} \binom{w}{v}^2,$$

*the second the closings of the centre ray of $\binom{P}{1}$ at the strength one, read as 1 when $|P| = 1$.*

*(iii) At a sign $\eta$ of $\mathcal{B}$, $K_{\mathcal{B}}(\eta) = K_{\mathcal{H}}(a_1, \dots, a_q)$ with $a_i = \sum_{e \in P_i}\eta_e \in \{s_i, s_i - 2, \dots, -s_i\}$, the sign values of $\binom{P_i}{1}$: on the centre ray of $\mathcal{B}$ at the strength $\sigma$ the native values of the signs with $K_{\mathcal{B}}(\eta) \ne 0$ (Theorem 38.17(ii)) are the $K_{\mathcal{H}}(\mathbf{a})^{-2\sigma}$ at these points $\mathbf{a}$.*

A parallel class enters the closings of a list only through the closings of its own list $\binom{P}{1}$, and the signs only through the sign values of that list, as a parallel block is read by $K$ through its total and by $\Psi$ through its hinge value (Corollary 21.7(i)).

**Proposition 42.11** (A mark added is a closing of the critical-line kernel)**.** *Let $n \ge 2$; in this statement $p_n(s) = P_{\binom{\mathcal{E}}{1}}(s\mathbf{1}; s, s) = \int_{y_{e_0}=0} \big(K(\alpha)K(\alpha^{-1})\big)^{-s} dy$ for $|\mathcal{E}| = n$ and $\operatorname{Re} s > 0$, $\tilde{p}_n = -p_n'$, and $W_s(x) = \prod_{j=1}^{s-1}(x-j)^2$.*

*(i) For $\operatorname{Re} s > 0$ and $0 < c < \operatorname{Re} s$, $\Gamma(s)^2 p_{n+1}(s) = \frac{1}{2\pi i}\int_{(c)} \Gamma(d)^2\,\Gamma(s-d)^2\,p_n(d)\,dd$. At $s = 1$ the kernel on the line $\operatorname{Re} d = \frac{1}{2}$ is $2\pi$ times the critical-line density (Definition 36.1): $p_{n+1}(1) = \mathbb{E}\big[p_n(\frac{1}{2} + iY)\big]$, $Y \sim \mathbb{P}_{\mathrm{crit}}$.*

*(ii) For $s \in \{1, 2, \dots\}$,*

$$\Gamma(s)^2 p_{n+1}(s) = \big(\mathsf{T}_{0;(1)}[x]\,(W_s\tilde{p}_n - W_s' p_n)\,\mathbb{I}_{\mathbb{Z}}\big)((0,\infty)),$$

$$\int_0^\infty y^{s-1}\mathsf{G}_{0;(1,-1)}(1,y)^n\sum_{m\ge0}\frac{y^m}{m!^2}\,dy=(\mathsf{T}_{0;(1)}[x]\,W_s\,p_n\,\mathbb{I}_{\mathbb{Z}})((0,\infty));$$

*at $s=1$, $p_{n+1}(1)=(\mathsf{T}_{0;(1)}[x]\,\tilde p_n\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$: the centre of $n+1$ marks at the strength one is the slope input of $n$ marks summed over the positive integers.*

*(iii) On the share side every mark carries on average the share $\frac{1}{n+1}$ of the strength:*

$$(n+1)\Big((\mathsf{T}_{0;(1)}[x]\,p_n\,\mathbb{I}_{\mathbb{Z}})((0,\infty))-(\mathsf{T}_{0;(1)}[x]\,(x-1)\,\tilde p_n\,\mathbb{I}_{\mathbb{Z}})((0,\infty))\Big)=(\mathsf{T}_{0;(1)}[x]\,\tilde p_n\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

**Theorem 42.12** (The law of the values of $\binom{\mathcal{E}}{1}$)**.** *Let $n\ge1$ and $h=\lceil n/2\rceil$; in this statement $\vartheta=y\,d/dy$, and the operators $\mathbf{R}_i$ are $\mathbf{R}_0=1$, $\mathbf{R}_1=\vartheta$ and $\mathbf{R}_{i+1}=\vartheta\mathbf{R}_i-i(n+1-i)\,y\,\mathbf{R}_{i-1}$.*

*(i) $\mathbf{Q}_n=\mathbf{R}_{n+1}$ annihilates every product of $n$ solutions of $\vartheta^2u=yu$. Written $\mathbf{Q}_n=\sum_{k\ge0}y^kq_{n,k}(\vartheta)$ with polynomials $q_{n,k}$, $q_{n,0}=\vartheta^{n+1}$ and $\deg q_{n,k}\le n+1-2k$, so that $q_{n,k}=0$ for $k>h$; with $a_{n,k}$ the coefficient of $\vartheta^{n+1-2k}$ in $q_{n,k}$,*

$$\sum_k a_{n,k}\,\eta^k\xi^{n+1-2k}=\prod_{j=0}^{n}\big(\xi-(n-2j)\sqrt{\eta}\big)=\xi^{n+1-2h}\prod_{0\le j<n/2}\big(\xi^2-(n-2j)^2\eta\big),$$

*so $q_{n,h}\ne0$. The formal adjoint for the measure $dy/y$ is $\mathbf{Q}_n^*=(-1)^{n+1}\mathbf{Q}_n$: $q_{n,k}(-\vartheta-k)=(-1)^{n+1}q_{n,k}(\vartheta)$.*

*(ii) Put $c_{n,0}(s)=(-1)^{n+1}s^{n-1}$ and $c_{n,k}(s)=q_{n,k}(-s-k)\big((s+1)(s+2)\cdots(s+k-1)\big)^2$ for $1\le k\le h$. Every $c_{n,k}$ has degree $n-1$, $c_{n,k}(-s-k)=(-1)^{n+1}c_{n,k}(s)$, $c_{n,k}$ vanishes to order two at $s=-1,\dots,-(k-1)$, $c_{n,h}$ has no zero at the integers $s\ge0$, and the symbol is*

$$\chi_n(\lambda)=(-1)^{n+1}\prod_{0\le j<n/2}\big(1-(n-2j)^2\lambda\big),$$

*whose roots are the native values $(n-2j)^{-2}$ of the signs (Proposition 42.9). For $n\ge2$ the values satisfy*

$$\sum_{k=0}^{h}c_{n,k}(s)\,P_{\binom{\mathcal{E}}{1}}\big((s+k)\mathbf{1};s+k,s+k\big)=0$$

*identically in $s$.*

*(iii) For $n\ge2$ the values $P_{\binom{\mathcal{E}}{1}}(k\mathbf{1};k,k)$ satisfy no recurrence of order less than $h$: their tier (Proposition 38.11) is $h$, and the recurrence of (ii) is their least recurrence. The closings $\mathsf{C}_j(1)=\sum_{|v|=j}\binom{j}{v}^2$ satisfy $\sum_{k=0}^{\min(h,j)}c_{n,k}(-j)\,\mathsf{C}_{j-k}(1)=0$ for every $j\in\mathbb{N}$ and no recurrence of order less than $h$, and the cyclic submodule of $[1]$ in the value module (Definition 39.58) has dimension $h$.*

The law of the values of $\binom{\mathcal{E}}{1}$ is the relation of one mark raised to the $n$-th symmetric power and read in the index, where $\vartheta$ becomes $-s$ and $y$ the shift. Its leading coefficients are those of the product of the $n+1$ factors $\xi-(n-2j)\sqrt{\eta}$, which pair into the native values of the signs, the tie $n=2j$ at even $n$ leaving the factor $\xi$; every coefficient $c_{n,k}$ carries the square of a rising factorial, which is why the boundaries below the cut vanish. One identity carries it from $n$ marks to $n+1$.

**Proposition 42.13** (A mark added carries the law of the values)**.** *In the notation of Theorem 42.12, for every $n\ge1$:*

*(i) For every solution $u$ of $\vartheta^2 u = yu$ and every smooth function $F$ on $(0,\infty)$,*

$$\mathbf{Q}_{n+1}(uF) = u\,\vartheta\,\mathbf{Q}_n(F) + (n+2)\,(\vartheta u)\,\mathbf{Q}_n(F).$$

*(ii) With $\binom{a}{b} = \Gamma(a+1)/\big(\Gamma(b+1)\Gamma(a-b+1)\big)$ and $A_s(x) = \big((n+1)x-(n+2)s\big)\binom{x}{s}^2$,*

$$\sum_{k=0}^{\lceil (n+1)/2\rceil} c_{n+1,k}(s)\binom{x-1}{s+k-1}^2 = \sum_{j=0}^{\lceil n/2\rceil} c_{n,j}(x-j)\,A_s(x-j),$$

*an identity of rational functions of $(x,s)$ after division by $\binom{x}{s}^2$. At integers $s, x \geq 1$ the words $\binom{x-1}{s+k-1}^2$ are the words $W_{s+k}(x)/\Gamma(s+k)^2$ of the closing of Proposition 42.11(ii): weighted by the coefficients $c_{n+1,k}(s)$, they are the adjoint of the recurrence of $n$ marks applied to the one word $A_s$, which vanishes at $x = 0, 1, \ldots, s-1$.*

**Proposition 42.14** (The lowered settings)**.** *Let $n \geq 2$ and $h = \lceil n/2\rceil$, with $c_{n,k}$ of Theorem 42.12; in this statement $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(1,y)$ and $\mathsf{h}(y) = \sum_{m\geq 0} y^m/m!^2$, and for $0 \leq j < n/2$ the* lowered setting *with $j$ marks lowered is*

$$\xi_{n,j}(s) = \Gamma(s)^{-2}\int_0^\infty y^{s-1}\,\mathsf{g}(y)^{n-j}\,\mathsf{h}(y)^j\,dy \qquad (\operatorname{Re} s > 0),$$

*so that $\xi_{n,0}(s) = P_{\binom{\mathcal{E}}{1}}(s\mathbf{1}; s, s)$, $|\mathcal{E}| = n$.*

*(i) Every $\xi_{n,j}$ satisfies the recurrence of Theorem 42.12(ii) on $\operatorname{Re} s > 0$.*

*(ii) For $j \geq 1$, $s^{n-1}\xi_{n,j}(s) \to 0$ as $s \downarrow 0$, and $\sum_{k=1}^h c_{n,k}(0)\,\xi_{n,j}(k) = 0$, against $\sum_{k=1}^h c_{n,k}(0)\,\xi_{n,0}(k) = (-1)^n n!$: the lowered settings have no low-end coefficient.*

*(iii) $\xi_{n,j}(k+1)/\xi_{n,j}(k) \to (n-2j)^{-2}$ along the integers, so $\xi_{n,j}$ has the root rate $(n-2j)^{-2}$: the setting with $j$ marks lowered carries the native value of the signs with $j$ marks at $-1$.*

*(iv) The $h$ sequences $(\xi_{n,j}(k))_{k\geq 1}$ are a basis of the solutions of the recurrence on the integers $k \geq 1$, and its solutions $x$ with $\sum_{k=1}^h c_{n,k}(0)\,x(k) = (-1)^n n!$ are the $\xi_{n,0} + \sum_{j=1}^{h-1} b_j\,\xi_{n,j}$, $b_j \in \mathbb{C}$.*

So the recurrence and the low-end relation fix the values only up to the $h-1$ lowered settings with $j \geq 1$, which satisfy both with the right side zero; $h-1$ further numbers fix them, for instance the first $h-1$ values (Corollary 42.15). The integrand $\mathsf{g}^{n-j}\mathsf{h}^j$ of the setting with $j$ marks lowered decays like $e^{-2(n-2j)\sqrt{y}}$ up to powers of $y$, and its integral converges on all of $\operatorname{Re} s > 0$ exactly when $j < n/2$: at even $n$ the tie $j = n/2$, the sign with $K(\eta) = 0$, is no native gauge and gives no setting.

**Corollary 42.15** (The value span of $\binom{\mathcal{E}}{1}$)**.** *Let $n \geq 2$, $h = \lceil n/2\rceil$ and $p_n(s) = P_{\binom{\mathcal{E}}{1}}(s\mathbf{1}; s, s)$, $|\mathcal{E}| = n$, with $c_{n,k}$ of Theorem 42.12.*

*(i) $\sum_{k=1}^h c_{n,k}(0)\,p_n(k) = (-1)^n n!$, and every value $p_n(k)$, $k \geq 1$, lies in $\mathbb{Q} + \sum_{i=1}^{h-1}\mathbb{Q}\,p_n(i)$; so do the values $p_n(k\sigma)$ of the centre ray at every strength $\sigma$.*

*(ii) For even $n$ the symbol has $\chi_n(1) \neq 0$, and for every polynomial $W$ with rational coefficients, $(\mathsf{T}_{0;(1)}[x]\,W\,p_n\,\mathbb{I}_\mathbb{Z})((0,\infty))$ and $(\mathsf{T}_{0;(1)}[x]\,W\,\tilde{p}_n\,\mathbb{I}_\mathbb{Z})((0,\infty))$, $\tilde{p}_n = -p_n'$, lie in $\mathbb{Q}$ plus the $\mathbb{Q}$-span of the $2(n-1)$ block values $(\mathsf{T}_{0;(1)}[x]\,(x-1)^a\,p_n\,\mathbb{I}_\mathbb{Z})((0,\infty))$ and $(\mathsf{T}_{0;(1)}[x]\,(x-1)^a\,\tilde{p}_n\,\mathbb{I}_\mathbb{Z})((0,\infty))$, $a < n-1$. In particular so do $\Gamma(s)^2 P_{\binom{[n+1]}{1}}(s\mathbf{1}; s, s)$ and $\Gamma(s)^2\xi_{n+1,1}(s)$ at every integer $s \geq 1$.*

**Corollary 42.16** (The exact pairing of $\binom{\mathcal{E}}{1}$)**.** *Let $n \geq 2$, $h = \lceil n/2 \rceil$, and $p_n$ and $c_{n,k}$ as in Corollary 42.15; put $\mathsf{C}_l = \sum_{v \in \mathbb{N}^{\mathcal{E}},\ |v|=l} \binom{l}{v}^2$ for $l \geq 0$ and $\mathsf{C}_l = 0$ for $l < 0$. For every integer $\sigma \geq 1$,*

$$\sum_{i=1}^{h} \sum_{j=1}^{i} c_{n,i}(\sigma - j)\, \mathsf{C}_{\sigma - j}\, p_n(\sigma - j + i) = (-1)^n n!,$$

*and its limit at high strength is the identity*

$$n! \cdot \frac{2^{n-1}}{n^n} = \prod_{j=1}^{h-1} \Big(1 - \frac{(n-2j)^2}{n^2}\Big) = \prod_{j=1}^{h-1} \frac{4j(n-j)}{n^2}$$

*of Proposition 42.8(iii), with $V_1 = n!$, $\det H = 2^{n-1}/n^n$, $\lambda_\star = n^{-2}$ and the other roots $(n-2j)^{-2}$.*

**Proposition 42.17** (The power of one mark as a mixture)**.** *Let $n \geq 2$; in this statement $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(1, y)$, and $f_n$, $f_{n+1}$ are the strides $\big(K(\alpha)K(\alpha^{-1})\big)^{-1}$ of the centre rays of $\binom{\mathcal{E}}{1}$ at the strength one for $|\mathcal{E}| = n$ and for $|\mathcal{E}| = n+1$, with the stride's laws $\mu_{f_n}$ and $\mu_{f_{n+1}}$ (Definition 40.1): the images of the Lebesgue measure $dy$ of the chart $y_{e_0} = 0$ under the strides, carried by $(0, n^{-2}]$ and by $(0, (n+1)^{-2}]$.*

*(i)* $\mathsf{g}(y)^n = \int \mathsf{g}(y/t)\, \mu_{f_n}(dt)$ *for every $y > 0$: the $n$-th power of the Gramma of one mark is a mixture of its dilations by the stride's law.*

*(ii) For $t > 0$ let $\phi_t(x) = x/\big((x + t^{-1/2})(1 + t^{-1/2}x)\big)$ on $(0, \infty)$, the stride of the centre ray at the strength one of the table $\binom{\{1,2\}}{1}$ with the weights $t^{-1/2}$ on its basis $\{1\}$ and $1$ on $\{2\}$, on the chart $\alpha_1 = 1$, $\alpha_2 = x$; its stride's law $\mu_{\phi_t}$ is the image of $dx/x$ under $\phi_t$. Then $\mu_{f_{n+1}} = \int \mu_{\phi_t}\, \mu_{f_n}(dt)$: adding a mark mixes the stride's law of two marks over the stride's law of $n$.*

**Proposition 42.18** (Balanced directions of $\binom{\mathcal{E}}{1}$)**.** *Let $|\mathcal{E}| = n \geq 2$, let $s, t \geq 1$ be integers and $\mathbf{m} \in \mathbb{Z}^{\mathcal{E}}$ with $m_{\mathcal{E}} = s + (n-1)t$ and $m_S > t(|S| - 1)$ for every nonempty proper $S$, and put $c_e = m_e - t$, so that $\sum_e c_e = s - t$. For real $k > 0$, with $P(k) = P_{\binom{\mathcal{E}}{1}}(k\mathbf{m}; ks, kt)$,*

$$\Gamma(sk)\, \Gamma(tk)\, P(k) = \int_0^\infty y^{\frac{(s+t)k}{2} - 1} \prod_{e \in \mathcal{E}} \mathsf{G}_{c_e k;(1,-1)}\big(\sqrt{y}, \sqrt{y}\big)\, dy,$$

*which depends on the direction only through $s + t$ and the multiset $\{|c_e|\}_e$. Two such directions $(\mathbf{m}; s, t)$ and $(\mathbf{m}'; s', t')$ with $s + t = s' + t'$ and the same multiset have, with $P'(k) = P_{\binom{\mathcal{E}}{1}}(k\mathbf{m}'; ks', kt')$,*

$$\frac{P'(k)}{P(k)} = \Gamma_1^2 \begin{bmatrix} sk,\ tk \\ (s+t)k \end{bmatrix} \Big/ \Gamma_1^2 \begin{bmatrix} s'k,\ t'k \\ (s+t)k \end{bmatrix},$$

*a quotient of Beta corners at one total, rational at the integers $k$, so that their values span one $\mathbb{Q}$-space; the inversion $\alpha \mapsto \alpha^{-1}$ carries the integrand at $(\mathbf{m}; s, t)$ to the integrand at $((s+t)\mathbf{1} - \mathbf{m}; t, s)$, which is $c \mapsto -c$.*

The exercises work these statements on the lists of a few marks: the closings and the law of the values of $\binom{[n]}{1}$ for $n = 3, 4, 5$ (Chapter 54), the value at the centre of five marks as a Trailmoment of a ray input (Exercise 54.16), and the lean $(\mathbf{21}; 4, 1)$ of three marks (Exercise 54.5).

## 42.7 The centre ray of a substitution

A list built by substitution meets each of its cells through one number on each side of the Cremona, the hinge value of the cell at $\alpha$ and at $\alpha^{-1}$. So its reciprocal stride is a product of one factor for each cell and one function of the core read at those hinge values, its closings are averages of the closings of the core with the cells read at their norms, its native gauges are read down the hinge tree, and its pencil module is a quotient of the direct image, under the core, of the modules of its cells. Throughout this section $\mathcal{B} = \mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q \rangle$ is a plain table built by substitution (Definition 21.1), every marker $h_i$ of the core carrying a cell $\mathcal{A}_i$ on $B_i \sqcup \{g_i\}$, a mark left in place being the cell $\binom{\{h,g\}}{1}$, read on its centre ray at the strength one, $f^{-1} = K(\alpha)K(\alpha^{-1})$; $\gamma_i = \gamma^K_{\mathcal{A}_i}$ is the hinge value of the cell by $K$.

**Definition 42.19** (Hinge pairs; norms; the two-sided core; the class density)**.** For a plain cell $\mathcal{A}$ on $B \sqcup \{g\}$ put $v_{\mathcal{A}}(\alpha) = K_{\mathcal{A}/g}(\alpha) K_{\mathcal{A}/g}(\alpha^{-1})$.

(i) The *hinge pair* of $\mathcal{A}$ at $\alpha$ is $(x, x') = \big(\gamma^K_{\mathcal{A}}(\alpha), \gamma^K_{\mathcal{A}}(\alpha^{-1})\big)$, and its *norm* is $t = xx'$. A parallel class $\binom{B\sqcup\{g\}}{1}$ has $(x, x') = (A, A')$ with the *totals* $A = \sum_{e\in B} \alpha_e$ and $A' = \sum_{e\in B} \alpha_e^{-1}$, $v = 1$, and its norm is $AA'$; a series class $\binom{B\sqcup\{g\}}{|B|}$ has $(x, x') = (1/A', 1/A)$ and $v = AA'$; a mark left in place has $(\alpha_h, \alpha_h^{-1})$, $v = 1$ and the norm 1. A mark of the *norm* $\mathsf{m} > 0$ left in place has the hinge pair $(\alpha_h, \mathsf{m}\alpha_h^{-1})$ and $v = 1$; a table with the norms $\mathsf{m} = (\mathsf{m}_e)_e$ on its marks is read with $K(\alpha)K(\mathsf{m}\alpha^{-1})$, $\mathsf{m}\alpha^{-1} = (\mathsf{m}_e\alpha_e^{-1})_e$, in the place of $K(\alpha)K(\alpha^{-1})$.

(ii) The *hinge quadruple* of $\mathcal{A}$ is

$$q_{\mathcal{A}}(\alpha) = (c, d, c', d') = \big(K_{\mathcal{A}/g}(\alpha),\, K_{\mathcal{A}\setminus g}(\alpha),\, K_{\mathcal{A}/g}(\alpha^{-1}),\, K_{\mathcal{A}\setminus g}(\alpha^{-1})\big),$$

a morphism from $(\mathbb{C}^\times)^B$ to $\mathbb{C}^4$, and its *hinge map* is $\alpha \mapsto (x, x', v) = (d/c,\, d'/c',\, cc')$, defined where $cc' \neq 0$. A parallel class and a mark left in place have $c = c' = 1$, and their hinge map is $(x, x')$, for a parallel class $(A, A')$, into the *hinge plane* $\mathbb{C}^2$, on which the scale $\alpha \mapsto \lambda\alpha$ acts by $(x, x') \mapsto (\lambda x, \lambda^{-1}x')$. The *hinge module* $\mathcal{H}_{\mathcal{A}}$ is the submodule generated by the class of $\prod_{e\in B} d\alpha_e/\alpha_e$ in the direct image of the functions along the hinge quadruple, read on the hinge plane for a parallel class.

(iii) The *two-sided core* is $\Phi_{\mathcal{H}}(x, x') = K_{\mathcal{H}}(x)\, K_{\mathcal{H}}(x')$ on $\prod_i \mathbb{C}^2_{(x_i, x_i')}$, and the *hinge space* of $\mathcal{B}$ is $\prod_i \mathbb{C}^2_{(x_i,x_i')}$ modulo the diagonal scale. At the norms 1, on $x' = x^{-1}$, the two-sided core is the reciprocal stride of the centre ray of $\mathcal{H}$. Read on the quadruples it is $\Phi^{\hom}_{\mathcal{H}}(q) = \mathcal{H}^{\hom}(\mathbf{c}, \mathbf{d})\, \mathcal{H}^{\hom}(\mathbf{c}', \mathbf{d}')$ (Definition 39.55), a polynomial, equal to $\prod_i c_i c_i' \cdot \Phi_{\mathcal{H}}(x, x')$ where every $c_i c_i' \neq 0$.

(iv) The *class density* $\nu_s$ is the law of $AA' = |\sum_e \alpha_e|^2$ for $\binom{[s]}{1}$ under the Haar probability of the compact torus $|\alpha_e| = 1$; $\nu_1$ is the unit point mass at 1, and the moments of $\nu_s$ are the closings $\mathsf{C}_w^{\binom{[s]}{1}}(1) = \sum_{|v|=w} \binom{w}{v}^2$ (Proposition 42.10(ii)).

**Theorem 42.20** (A list is its core read against its cells)**.** *(i) The reciprocal stride factors over the cells and the core:*

$$f^{-1}(\alpha) = \prod_i v_{\mathcal{A}_i}(\alpha) \cdot \Phi_{\mathcal{H}}\big(\gamma(\alpha), \gamma(\alpha^{-1})\big),$$

*which on the whole torus is* $\Phi^{\hom}_{\mathcal{H}}\big(q_{\mathcal{A}_1}(\alpha), \dots, q_{\mathcal{A}_q}(\alpha)\big)$.

*(ii) If every cell is a parallel class, of sizes $s_1, \dots, s_q \geq 1$, then for every $j$*

$$\mathsf{C}_j(1) = \int \operatorname*{CT}_x \Big[ K_{\mathcal{H}}(x)^j \, K_{\mathcal{H}}\big(t_1 x_1^{-1}, \dots, t_q x_q^{-1}\big)^j \Big] \prod_i \nu_{s_i}(dt_i),$$

*and the torus fibre integral $T_{\mathcal{B}}(\tau) = \sum_j \mathsf{C}_j(1)\tau^{-j-1}$ (Definition 39.47) is the average, against $\prod_i \nu_{s_i}(dt_i)$, of the torus fibre integrals of the two-sided core at the norms $t_i$.*

*(iii) The pencil is the composite of the hinge quadruples, read modulo the scale, with $\Phi_{\mathcal{H}}^{\hom}$, and the origin form dy is the image of the product of the forms $\prod_{e \in B_i} d\alpha_e/\alpha_e$. So $\mathcal{D}_{\mathcal{B}}[1]$ is a quotient of the submodule generated by the product of the generators in the direct image of $\boxtimes_i \mathcal{H}_{\mathcal{A}_i}$ under $\Phi_{\mathcal{H}}^{\hom}$, taken modulo the scale: every singular value of $\mathcal{D}_{\mathcal{B}}[1]$ is a singular value of that direct image, and its vanishing rank there is not larger. Theorem 39.56 gives the same at every depth of a hinge tree.*

The class enters through its own module once, whatever the rest of the list is; a series class enters with the norm $t_i^{-1}$ and the factor $t_i^j$ by (i).

**Corollary 42.21** (The pencil of a list along its canonical tree)**.** *Let $\mathcal{B}$ be a connected list on $n \geq 3$ marks, read on its centre ray at the strength one, with its canonical hinge tree (Theorem 23.16) rooted at a vertex o. In Theorem 39.56 the data of a vertex is the hinge quadruple $(c, d, c', d')$ of its subtree cell (Definition 42.19(ii)), a mark has $(1, \alpha_e, 1, \alpha_e^{-1})$, the last coordinate is 1, the root map is $\mathcal{H}_o^{\hom}(\mathbf{c}, \mathbf{d})\, \mathcal{H}_o^{\hom}(\mathbf{c}', \mathbf{d}') = \Phi_{\mathcal{H}_o}^{\hom}$, and the node maps are of three kinds, each applied to the pairs $(c, d)$ and $(c', d')$ alike:*

*(i) at a parallel node, $(c_l, d_l)_l \mapsto \big(\prod_l c_l,\ \sum_l d_l \prod_{l' \neq l} c_{l'}\big)$, so the hinge values add;*

*(ii) at a series node, $(c_l, d_l)_l \mapsto \big(\sum_l c_l \prod_{l' \neq l} d_{l'},\ \prod_l d_l\big)$, so the reciprocals of the hinge values add;*

*(iii) at a prime node $\mathcal{H}_v$, $(c, d) \mapsto \big((\mathcal{H}_v/g_v)^{\hom}(\mathbf{c}, \mathbf{d}),\ (\mathcal{H}_v \setminus g_v)^{\hom}(\mathbf{c}, \mathbf{d})\big)$.*

*A parallel node whose children are all marks is a parallel class, with the quadruple $(1, A, 1, A')$ and its hinge module on the hinge plane, and a series node whose children are all marks is a series class, with $(\alpha^B A', \alpha^B, \alpha^{-B} A, \alpha^{-B})$. So the pencil module of the centre ray of every connected list on n marks, and the law of every insertion along it, are generated from the module of one mark by the parallel and series rules and by the node maps of the prime lists on at most n marks, which up to the names of the marks are finitely many.*

The tree decides what a list carries into its pencil. The series and parallel nodes act by the conductance rules, the same at every size; a class is a parallel node of marks; and the prime nodes are the only operations that depend on more than the number of their children.

**Proposition 42.22** (The closings of a substitution)**.** *For every $j \in \mathbb{N}$,*

$$\mathsf{C}_j(1) = \sum_{w \in \mathbb{N}^q} \Big( [x^w]\, K_{\mathcal{H}}(x)^j \Big)^2 \prod_{i=1}^{q} \mathsf{M}_j^{(i)}(w_i),$$

$$\mathsf{M}_j^{(i)}(w) = \operatorname{CT}\Big( \big(K_{\mathcal{A}_i \setminus g_i}(\alpha) K_{\mathcal{A}_i \setminus g_i}(\alpha^{-1})\big)^w \big(K_{\mathcal{A}_i/g_i}(\alpha) K_{\mathcal{A}_i/g_i}(\alpha^{-1})\big)^{j-w} \Big),$$

*the constant term taken on the marks $B_i$ alone. A parallel class of s marks has $\mathsf{M}_j(w) = \mathsf{C}_w^{\binom{[s]}{1}}(1)$, a series class of s marks has $\mathsf{M}_j(w) = \mathsf{C}_{j-w}^{\binom{[s]}{1}}(1)$, and a mark left in place has $\mathsf{M}_j = 1$. A cell that*

*is itself a substitution, $\mathcal{A} = \mathcal{A}_0\langle \mathcal{A}' \text{ at } x\rangle$ (Proposition 21.3(ii)), has*

$$\mathsf{M}_j^{\mathcal{A}}(w) = \sum_u \Big( [\alpha^u]\, K^{w}_{\mathcal{A}_0\backslash g} K^{j-w}_{\mathcal{A}_0/g} \Big)^2 \, \mathsf{M}_j^{\mathcal{A}'}(u_x),$$

*so the arrays* $\mathsf{M}$ *compose down the hinge tree, one for each edge.*

Proposition 42.10(ii) is the case of parallel classes. By Theorem 39.57 and Corollary 39.36 the least recurrence of the closings of 1 is read from these arrays, with no box.

**Lemma 42.23** (The occupations of a substitution)**.** *Let $\rho_i(\alpha) = R^{\mathcal{H}}_{h_i}\big(\gamma(\alpha)\big)$ be the occupation of the marker $h_i$ in the core read at the hinge values. On the marks of $B_i$ the occupations of $\mathcal{B}$ are*

$$R_e = (1-\rho_i)\, R_e^{\mathcal{A}_i/g_i} + \rho_i\, R_e^{\mathcal{A}_i\backslash g_i},$$

*and $\rho_i = \gamma_i\, K_{\mathcal{H}/h_i}(\gamma)/K_{\mathcal{H}}(\gamma)$, $K_{\mathcal{H}}$ being multilinear; on a mark left in place, $R_e = \rho_i$.*

**Theorem 42.24** (Native gauges along the hinge tree)**.** *Write $\rho_i = \rho_i(\alpha)$ and $\rho_i^* = \rho_i(\alpha^{-1})$, and where the hinge values are nonzero put $t_i = \gamma_i(\alpha)\gamma_i(\alpha^{-1})$ and $q_i = \gamma_i(\alpha)/\gamma_i(\alpha^{-1})$. A point $\alpha \in X_F$ with nonzero hinge values is a native gauge of the centre ray of $\mathcal{B}$ exactly when for every cell*

$$d_{B_i}\Big[ \log v_{\mathcal{A}_i} + \tfrac{1}{2}(\rho_i + \rho_i^*)\log t_i + \tfrac{1}{2}(\rho_i - \rho_i^*)\log q_i \Big] = 0, \qquad (\star)$$

*the differential taken along the marks of $B_i$ in the logarithmic coordinates with $\rho_i, \rho_i^*$ held fixed.*

*(a) Along the scale of the cell, under which $v_{\mathcal{A}_i}$ and $t_i$ are invariant and $q_i$ has the degree 2, $(\star)$ reads $\rho_i = \rho_i^*$; then $(\star)$ says that the cell sits at a critical point of $v_{\mathcal{A}_i} t_i^{\rho_i}$, with $\rho_i$ fixed, on its torus modulo its scale.*

*(b) A parallel class $\mathcal{A}_i$ is, at every native gauge, hinge values zero or not, either* empty, *$K_{\mathcal{H}/h_i}(\gamma(\alpha)) = K_{\mathcal{H}/h_i}(\gamma(\alpha^{-1})) = 0$, or* occupied, *$\alpha_e = \eta_e \tau_i$ on $B_i$ with signs $\eta_e$ and a scale $\tau_i$. An occupied class of the sign value $a_i = \sum_{e\in B_i}\eta_e$ is read by the core as one mark of the variable $\tau_i$ and the weight $a_i$ in both letters, $\gamma_i(\alpha) = a_i\tau_i$ and $\gamma_i(\alpha^{-1}) = a_i\tau_i^{-1}$, and for $a_i \neq 0$ its equations are the native-gauge equation of that weighted core at $h_i$. An empty class satisfies its own equations identically and enters the others only through its totals $(A_i, A_i')$, which the core fixes; for a pair the ratio $r$ of its two scales solves $r + r^{-1} = A_i A_i' - 2$. For a series class the same holds with $K_{\mathcal{H}\backslash h_i}$ in the place of $K_{\mathcal{H}/h_i}$, the totals exchanged, the weight $1/a_i$, and the factor $v_{\mathcal{A}_i} = a_i^2$ of $f^{-1}$ when it is occupied.*

*(c) A cell $\mathcal{A} = \mathcal{H}'\langle \mathcal{A}_1', \dots, \mathcal{A}_p'\rangle$, its hinge a mark $g$ of $\mathcal{H}'$, has as its hinge map the composite of the hinge maps $(\gamma_l', \gamma_l'^*, v_l')$ of its own cells with*

$$\big(\gamma', \gamma'^*, v'\big) \mapsto \Big( \frac{K_{\mathcal{H}'\backslash g}(\gamma')}{K_{\mathcal{H}'/g}(\gamma')},\ \frac{K_{\mathcal{H}'\backslash g}(\gamma'^*)}{K_{\mathcal{H}'/g}(\gamma'^*)},\ \prod_l v_l'\, K_{\mathcal{H}'/g}(\gamma') K_{\mathcal{H}'/g}(\gamma'^*) \Big),$$

*so the differentials in $(\star)$ are read from those of its cells by the chain rule, and $(\star)$ recurses down the hinge tree.*

*(d) For the series and parallel nodes these maps are the conductance rules: at a parallel node of $p$ cells $\gamma = \sum_l \gamma_l'$ and $v = \prod_l v_l'$; at a series node $\gamma^{-1} = \sum_l \gamma_l'^{-1}$ and $v = \prod_l v_l' \cdot e_{p-1}(\gamma')\, e_{p-1}(\gamma'^*)$.*

So the native gauges of a list of series and parallel nodes are found by choosing, class by class, occupied with a sign pattern or empty, and solving the native-gauge equation of the core in which an occupied class is one mark of weight $a_i$ and an empty class a pair of free totals subject to the vanishing of the contraction letter of the core on both sides of the Cremona. Native gauges that are not signs come from emptiness: a marker whose contraction letter vanishes on both sides of the Cremona releases its class from the sign condition.

**Corollary 42.25** (A class is a bubble of classes)**.** *For $a+b=s$, $a,b\ge 1$, $\binom{[s]}{1}=\binom{[2]}{1}\langle\binom{[a]}{1},\binom{[b]}{1}\rangle$, the two classes substituted as parallel classes. The two-sided core of $\binom{[2]}{1}$ at the norms $(t_1,t_2)$ is $t_1+t_2+t_1r+t_2r^{-1}$, $r=x_2/x_1$, and*

$$T_s(\tau)=\iint\frac{\nu_a(dt_1)\,\nu_b(dt_2)}{\sqrt{(\tau-t_1-t_2)^2-4t_1t_2}},\qquad \sum_j \mathsf{C}_j^{\binom{[s]}{1}}(1)\,\frac{z^j}{j!^2}=\Big(\sum_k\frac{z^k}{k!^2}\Big)^s,$$

*$T_s$ the torus fibre integral of the centre ray of $\binom{[s]}{1}$ at the strength one and the root positive for large $\tau$.*

## 42.8 The value of a table as a closed serif word

The value of a connected table at every strength, with norms on its marks, is the word of the table with the norms entering as a monomial in the loads, and the value of a table built by substitution is the word of its core with the word of each cell set at the load of its marker. A cell enters through one Trailmoment of its law; down the hinge tree it is read through the family of the minors of its core, or, at a series or parallel node, on one scale through the transforms of the cells below it. Throughout this section $(\mathcal{B},\theta)$ is a connected table on $\mathcal{E}$, $|\mathcal{E}|=n$, of rank $r$, with $N$ bases and the norms $\mathsf{m}_e>0$ on its marks (Definition 42.19(i)), and $k>0$. Its *value* at the strength $k$ is

$$\mathsf{V}_{\mathcal{B}}(k;\mathsf{m})=\int_{\{h=1\}}\big(K(\alpha)K(\mathsf{m}\alpha^{-1})\big)^{-k}\alpha^{-\mathbf{1}}\,d\alpha,\qquad \mathsf{m}\alpha^{-1}=(\mathsf{m}_e\alpha_e^{-1})_e,$$

$h$ any positive function homogeneous of degree one, and $\mathsf{V}_{\mathcal{B}}(k)=\mathsf{V}_{\mathcal{B}}(k;\mathbf{1})=P_{\mathcal{B}}(k\mathbf{1};k,k)$ is the period of its centre ray at the strength $k$ (Definition 38.13).

**Proposition 42.26** (The value of a table is a closed serif word)**.** *(i) $K(\mathsf{m}\alpha^{-1})=\alpha^{-\mathcal{E}}\Psi_{\theta\mathsf{m}}(\alpha)$, with $\Psi_{\theta\mathsf{m}}$ the cobasis letter of $(\mathcal{B},\theta\mathsf{m})$, $(\theta\mathsf{m})_T=\theta_T\prod_{e\in T}\mathsf{m}_e$: the norms are weights of the cobasis letter. So $\mathsf{V}_{\mathcal{B}}(k;\mathsf{m})$ is the period at the masses $k\mathbf{1}$ of the family $F_{\mathsf{m}}$ of the two tables $((\mathcal{B},\theta);k,0)$ and $((\mathcal{B},\theta\mathsf{m});0,k)$, which is on the balance wall with the walls $\kappa_S=k\big(r(S)+r(\mathcal{E}\setminus S)-r\big)>0$; it converges and has a native gauge, for all positive weights and norms.*

*(ii) Fix $g\in\mathcal{E}$. Open $K$ with one share $a_T$ and $\Psi_{\theta\mathsf{m}}$ with one share $b_T$ on each basis $T$, write $\ell_e(\mathbf{a})=\sum_{T\ni e}a_T$ for the load of $K$ on the mark $e$, and let $\mathcal{X}_k$ be the fibre cut by $|\mathbf{a}|=|\mathbf{b}|=k$ and $\mathrm{load}_e=k$ for $e\ne g$ (Definition 12.24), of dimension $2(N-1)-(n-1)$. On every wall contour, among them the uniform split $a_T=b_T=k/N$, which lies on $\mathcal{X}_k$ whatever the weights and the norms,*

$$\mathsf{V}_{\mathcal{B}}(k;\mathsf{m})=\int_{\mathcal{X}_k}\Gamma_{\substack{N\\1}}\begin{bmatrix}\mathbf{a}\\k\end{bmatrix}\Gamma_{\substack{N\\1}}\begin{bmatrix}\mathbf{b}\\k\end{bmatrix}\prod_T\theta_T^{-a_T-b_T}\prod_{e\in\mathcal{E}}\mathsf{m}_e^{-\ell_e(\mathbf{a})},$$

*independent of $g$. A norm enters as a twist on the load of its mark, and at $\mathsf{m}=\mathbf{1}$ the word is the word of the table.*

*(iii)* *Let $\mathcal{B} = \mathcal{H}\langle \mathcal{A}_1, \dots, \mathcal{A}_q \rangle$ be built by substitution with plain cells (Definition 21.1), the core $\mathcal{H}$ with $N_{\mathcal{H}}$ bases and the weights $\theta^{\mathcal{H}}$, the cell $\mathcal{A}_i$ on $B_i \sqcup \{g_i\}$, the norms 1 on the marks of the cells with $|B_i| \geq 2$, and the norm $\mathsf{m}_i$ on a mark left in place at $h_i$. For a cell $\mathcal{A}$ on $B \sqcup \{g\}$ with $|B| \geq 2$ and $0 < \operatorname{Re} \ell < k$ put*

$$\widehat{\mathsf{V}}_{\mathcal{A}}(k;\ell) = \int_{\{h=1\}} v_{\mathcal{A}}(\alpha)^{\ell-k} \big(K_{\mathcal{A}\setminus g}(\alpha) K_{\mathcal{A}\setminus g}(\alpha^{-1})\big)^{-\ell} \alpha^{-\mathbf{1}}\, d\alpha,$$

*the period of the family of its two hinge minors, $\mathcal{A}/g$ at the strength $k-\ell$ and $\mathcal{A}\setminus g$ at the strength $\ell$, which is on the balance wall of $B$ and converges; for a mark of the norm $\mathsf{m}$ left in place put $\widehat{\mathsf{V}}(k;\ell) = \mathsf{m}^{-\ell}$. A parallel class of $s \geq 2$ marks has*

$$\widehat{\mathsf{V}}(k;\ell) = \Gamma(\ell)^{-2} \int_0^\infty y^{\ell-1} \mathsf{G}_{0;(1,-1)}(1,y)^s\, dy.$$

*Then, with the fibre $\mathcal{X}_k^{\mathcal{H}}$ and the loads $\ell_{h_i}$ of (ii) for the core,*

$$\mathsf{V}_{\mathcal{B}}(k;\mathsf{m}) = \int_{\mathcal{X}_k^{\mathcal{H}}} \Gamma_{N_{\mathcal{H}}}\begin{bmatrix}\mathbf{a}\\ k\end{bmatrix} \Gamma_{N_{\mathcal{H}}}\begin{bmatrix}\mathbf{b}\\ k\end{bmatrix} \prod_T \big(\theta_T^{\mathcal{H}}\big)^{-a_T-b_T} \prod_{i=1}^q \widehat{\mathsf{V}}_{\mathcal{A}_i}\big(k;\ell_{h_i}(\mathbf{a})\big) :$$

*the word of a table is the word of its core with the word of each cell set at the load of its marker.*

*(iv)* The law of a cell. *A connected plain cell $\mathcal{A}$ on $B \sqcup \{g\}$, $|B| \geq 2$, at the exponents $(k,k)$ with the masses $k$ on $B$ is a cell family with the hinge wall $k$, with its law $(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}$, its transform $f_{\mathcal{A}}$ and its dual transform $\check{f}_{\mathcal{A}}$ (Definition 23.1). For $0 < \operatorname{Re}\ell < k$ and $0 < \operatorname{Re} z < k$,*

$$\widehat{\mathsf{V}}_{\mathcal{A}}(k;\ell) = \int_{\Delta_B} \varrho_{\mathcal{A}}^{\ell-k}\, d\phi_{\mathcal{A}} = \big(\mathsf{T}_{k-\ell;(1)}[x]\,(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}\big)((0,\infty)),$$

$$\int_0^\infty y^{z-1} \check{f}_{\mathcal{A}}(y)\, dy = \Gamma(z)^2\, \widehat{\mathsf{V}}_{\mathcal{A}}(k;z), \qquad \int_0^\infty y^{z-1} f_{\mathcal{A}}(y)\, dy = \Gamma(z)^2\, \widehat{\mathsf{V}}_{\mathcal{A}}(k;k-z) :$$

*a cell enters the word of a table through one Trailmoment of its law, at the index $k$ less the load of its marker, and its two transforms are that word read back on a scale.*

*(v)* Down the tree. *Let a cell $\mathcal{A}$ as in (iv) be built by substitution with plain cells, $\mathcal{A} = \mathcal{H}'\langle \mathcal{A}'_1, \dots, \mathcal{A}'_p \rangle$, its hinge $g$ a mark of $\mathcal{H}'$ left in place at the marker $h$, the cell $\mathcal{A}'_i$ at the marker $h_i$, the norms 1 on all its marks, and a mark left in place counted as a cell with $\widehat{\mathsf{V}} = 1$. Let $\mathcal{C} = \mathcal{H}'/h$ and $\mathcal{D} = \mathcal{H}' \setminus h$ on $h_1, \dots, h_p$, with the inherited weights and $N_{\mathcal{C}}$ and $N_{\mathcal{D}}$ bases; open the letters $K$ and $\Psi$ of $\mathcal{C}$ with the shares $a_T$ and $b_T$, and those of $\mathcal{D}$ with the shares $c_T$ and $d_T$, one on each basis; write $\ell_x(\mathbf{a}) = \sum_{T \ni x} a_T$, and let $\mathcal{Y}_{k,\ell}$ be the fibre cut by $|\mathbf{a}| = |\mathbf{b}| = k-\ell$, $|\mathbf{c}| = |\mathbf{d}| = \ell$ and the load $\ell_x(\mathbf{a}) + \ell_x(\mathbf{c}) + \sum_{T \not\ni x}(b_T + d_T) = k$ at every marker $x$ but one, of dimension $2(N_{\mathcal{C}} + N_{\mathcal{D}}) - p - 3$. On its uniform split $a_T = b_T = (k-\ell)/N_{\mathcal{C}}$, $c_T = d_T = \ell/N_{\mathcal{D}}$, at which every $\ell_{h_i}(\mathbf{a}) + \ell_{h_i}(\mathbf{c})$ has its real part in $(0,k)$, for $0 < \operatorname{Re}\ell < k$*

$$\widehat{\mathsf{V}}_{\mathcal{A}}(k;\ell) = \int_{\mathcal{Y}_{k,\ell}} \Gamma_{N_{\mathcal{C}}}\begin{bmatrix}\mathbf{a}\\ k-\ell\end{bmatrix} \Gamma_{N_{\mathcal{C}}}\begin{bmatrix}\mathbf{b}\\ k-\ell\end{bmatrix} \Gamma_{N_{\mathcal{D}}}\begin{bmatrix}\mathbf{c}\\ \ell\end{bmatrix} \Gamma_{N_{\mathcal{D}}}\begin{bmatrix}\mathbf{d}\\ \ell\end{bmatrix}$$
$$\times \prod_T \big(\theta_T^{\mathcal{C}}\big)^{-a_T-b_T} \prod_T \big(\theta_T^{\mathcal{D}}\big)^{-c_T-d_T} \prod_{i=1}^p \widehat{\mathsf{V}}_{\mathcal{A}'_i}\big(k;\ell_{h_i}(\mathbf{a}) + \ell_{h_i}(\mathbf{c})\big) :$$

*the word of a cell is the word of the family of the two minors of its core at the marker of its hinge, with the word of each of its own cells set at the load of its marker.*

*(vi)* Series and parallel nodes on one scale. *When the core of $\mathcal{A}$ in (v) is a parallel node $\binom{\{h,h_1,\dots,h_p\}}{1}$ or a series node $\binom{\{h,h_1,\dots,h_p\}}{p}$, with the weights* 1, *a mark left in place having the transforms $f = \check{f} = \mathsf{G}_{0;(1,-1)}(1,\cdot)$ (Definition 23.19(1)), respectively*

$$\widehat{\mathsf{V}}_{\mathcal{A}}(k;\ell) = \frac{1}{\Gamma(\ell)^2}\Big(\mathsf{T}_{1-\ell;(1)}[x]\prod_{i=1}^{p}\check{f}_{\mathcal{A}_i'}(x)\,\mathbb{I}_{(0,\infty)}\Big)((0,\infty)),$$

$$\widehat{\mathsf{V}}_{\mathcal{A}}(k;\ell) = \frac{1}{\Gamma(k-\ell)^2}\Big(\mathsf{T}_{1-k+\ell;(1)}[x]\prod_{i=1}^{p}f_{\mathcal{A}_i'}(x)\,\mathbb{I}_{(0,\infty)}\Big)((0,\infty)) :$$

*one integral on one scale whatever the cells, equal cells entering as one transform to a power. A series class of $s \geq 2$ marks has $\widehat{\mathsf{V}}(k;\ell) = \Gamma(k-\ell)^{-2}\int_0^\infty y^{k-\ell-1}\mathsf{G}_{0;(1,-1)}(1,y)^s\,dy$.*

*(vii)* The root on one scale. *When the core $\mathcal{H}$ of (iii) is a parallel node $\binom{\{h_1,\dots,h_q\}}{1}$ or a series node $\binom{\{h_1,\dots,h_q\}}{q-1}$, with the weights* 1, $\mathsf{m}_i = 1$ *at a cell of at least two marks and the transforms of (vi) at a mark left in place, respectively*

$$\mathsf{V}_{\mathcal{B}}(k;\mathsf{m}) = \frac{1}{\Gamma(k)^2}\Big(\mathsf{T}_{1-k;(1)}[x]\prod_{i=1}^{q}\check{f}_{\mathcal{A}_i}(\mathsf{m}_i x)\,\mathbb{I}_{(0,\infty)}\Big)((0,\infty)),$$

$$\mathsf{V}_{\mathcal{B}}(k;\mathsf{m}) = \frac{1}{\Gamma(k)^2}\Big(\mathsf{T}_{1-k;(1)}[x]\prod_{i=1}^{q}\mathsf{m}_i^{-k}f_{\mathcal{A}_i}(x/\mathsf{m}_i)\,\mathbb{I}_{(0,\infty)}\Big)((0,\infty)).$$

A cell enters the word of a table only through its own word, set at the load of its marker, and a mark with a norm is a cell whose word is a single power. The cells ride on the core's fibre in the same way for every substituted family (Theorem 22.4). The word of a cell is one Trailmoment of its law, and its two transforms are that word read back on a scale (iv). Down the hinge tree the root contributes the word (ii) of its own table and a node below it the word of the family of the two minors of its table at its hinge, with its own cells set at their loads (v); a series or parallel node can instead be read on one scale, through the product of the transforms of the cells directly below it, a repeated cell entering as one transform to a power (vi), (vii).

## 42.9 The class at silence

A class enters every list through its own pencil, $t = AA'$ on $(\mathbb{C}^\times)^s/\mathbb{C}^\times$, and it is silent at $t = 0$. Its law is the law of the values of $\binom{\mathcal{E}}{1}$ (Theorem 42.12); what the class carries into a list is the local system of that law, irreducible of rank $s-1$, with its local monodromies: one Jordan block at $t = \infty$, a reflection at every sign value, and $\lfloor s/2 \rfloor$ at silence.

**Lemma 42.27** (The pencil of a class)**.** *For $\binom{[s]}{1}$, $s \geq 2$, with $\varphi = AA'$ on $T_s = (\mathbb{C}^\times)^s/\mathbb{C}^\times$:*

*(i) the critical points of nonzero value are the signs $\eta$ with $\sum_e \eta_e \neq 0$, of the value $(\sum_e \eta_e)^2$, nondegenerate; the critical points of the value* 0 *are the points with $A = A' = 0$;*

*(ii) $\varphi$ has no asymptotic critical value in $\mathbb{C}^\times$: the centre ray of $\binom{[s]}{1}$ has no asymptotic native value (Definition 38.29);*

*(iii) the critical points of a value $a^2 \neq 0$ form one orbit of $S_s$, and the Cremona fixes each of them.*

**Theorem 42.28** (The class for every size)**.** *Let $s \geq 2$, $h = \lceil s/2 \rceil$, let $c_{s,k}$ be the coefficients of Theorem 42.12, and put*

$$\widetilde{A}_s = \sum_{k=0}^{h} c_{s,k}(t\partial_t + 1)\, t^{-k},$$

*an operator of order $s-1$ in $\partial_t$. Let $T_s(t) = \sum_{j\geq 0} \mathsf{C}_j^{\binom{[s]}{1}}(1)\, t^{-j-1}$ be the torus fibre integral of $\binom{[s]}{1}$, and $V_s$ the span of its continuations along the paths that avoid $0$ and the sign values $a^2$, $0 < a \leq s$, $a \equiv s \bmod 2$.*

*(i) $\widetilde{A}_s T_s = 0$ exactly, and $V_s$ is the whole space of the solutions of $\widetilde{A}_s$: a local system of rank $s-1$, irreducible.*

*(ii) At $t = \infty$ the monodromy of $V_s$ is one Jordan block of size $s-1$ for the eigenvalue $1$, with all exponents equal to $1$ in $1/t$.*

*(iii) At every sign value $a^2$ the monodromy minus the identity has rank one; the exponents there are $0, 1, \ldots, s-3$ and $\frac{s-3}{2}$.*

*(iv) At $t = 0$, where the class is silent, $\operatorname{rank}(T_0 - 1) = \lfloor s/2 \rfloor$, and the exponents are $0, 0, 1, 1, \ldots, h-2, h-2$ and, for even $s$, one more, $\frac{s}{4} - 1$. For odd $s$, $(T_0 - 1)^2 = 0$: the monodromy is $\frac{s-1}{2}$ Jordan blocks of size $2$. For even $s$ the eigenvalue $-1$ has the multiplicity at most one, and on the unipotent part $(T_0 - 1)^3 = 0$ with $(T_0 - 1)^2$ of rank at most one.*

At silence an odd class carries $\frac{s-1}{2}$ logarithms, $t^m \log t + \cdots$ beside $t^m + \cdots$ for $0 \leq m \leq \frac{s-3}{2}$; the fibre integral of the class at its norm $x = A_i A_i'$ enters a list through these germs.

**Proposition 42.29** (The hinge module of a class)**.** *Let $s \geq 2$, with $h$ and $c_{s,k}$ as in Theorem 42.28, write $c_{s,k}(x) = q_k(-x-k)\big((x+1)\cdots(x+k-1)\big)^2$ for $k \geq 1$ and $q_0(\vartheta) = \vartheta^{s+1}$, the polynomials $q_k = q_{s,k}$ of Theorem 42.12(i), and put on the hinge plane*

$$E = x\partial_x - x'\partial_{x'}, \qquad \widetilde{B} = \sum_{k=1}^{h} (\partial_x \partial_{x'})^{k-1} q_k(-x\partial_x - 1) + xx'(-x\partial_x - 1)^{s-1}.$$

*(i) $E$ and $\widetilde{B}$ annihilate the generator of the hinge module $\mathcal{H}_C$ of the class. On functions of $t = xx'$, $\widetilde{B}$ acts as $t\,\widetilde{A}_s$.*

*(ii) The periods of $[1]$ of the class are exactly $V_s$: the module $\mathcal{D}[1]$ of the centre ray of $\binom{[s]}{1}$ has rank $s-1$ and is irreducible off its singular values.*

*(iii) $\mathcal{H}_C$ is the intermediate extension of its local system, the pullback of $V_s$ along $t = xx'$.*

*(iv) With $P = \mathbb{C}^2$, $H_a = \{xx' = a^2\}$ for the sign values, $X = \{x = 0\}$, $X' = \{x' = 0\}$ and $O = \{0\}$,*

$$CC(\mathcal{H}_C) = (s-1)[T_P^*] + \sum_a [T_{H_a}^*] + \left\lfloor \frac{s}{2} \right\rfloor \big([T_X^*] + [T_{X'}^*]\big) + m_s(O)[T_O^*],$$

*with $m_s(O) = 0$ for odd $s$ and $m_s(O) = 1 + n_1$ for even $s$, $n_1$ the number of Jordan blocks of size one of $T_0$ for the eigenvalue $1$.*

*Write $m_s(P) = s-1$, $m_s(H_a) = 1$, $m_s(X) = m_s(X') = \lfloor s/2 \rfloor$; a mark left in place has the single component $H_1$, with the multiplicity $1$.*

So the module through which a class enters every list is determined: on the hinge plane it is annihilated by the scale and by the law of the class with one factor removed, it is the intermediate extension of its local system, and its characteristic cycle is explicit. The values of every table a class enters, with norms on its marks, are serif words (Proposition 42.26). The periods of a class of at least three marks continue through a group whose Zariski closure has an identity component that is not solvable, so they are not combinations, with algebraic coefficients, of iterated integrals of logarithmic forms with algebraic letters, nor of any functions whose continuations become unipotent on a finite cover (Corollary 42.30). That is a statement about those functions, not about the class (Definition 18.19).

**Corollary 42.30** (The periods of a class are not unipotent)**.** *For $s \geq 3$ the identity component of the Zariski closure of the monodromy group of $V_s$ is not solvable, hence not unipotent. So no nonzero element of $V_s$ is a function whose continuations span a local system with a unipotent subgroup of finite index in its monodromy, as are the combinations, with algebraic coefficients, of iterated integrals of logarithmic forms with algebraic letters.*

## 42.10 The sources of a substitution of classes

When every cell is a parallel class, the sources of the list are the critical points of one function, the two-sided core, on the strata of one space, the hinge space, with each class on one of five strata of its hinge plane, and their contributions to the vanishing ranks are counted there with the multiplicities of Proposition 42.29.

**Theorem 42.31** (The sources in the hinge space)**.** *Let every cell of $\mathcal{B}$ be a parallel class, of sizes $s_i \geq 1$. The singular support of the hinge module of a class of $s \geq 2$ marks lies in the conormals of $P$, of the hyperbolas $H_a$ of its sign values, of the axes $X, X'$ and of the origin $O$ (Proposition 42.29); for a mark it is the conormal of $H_1$. Every singular value of $\mathcal{D}_{\mathcal{B}}[1]$ is the value of $\Phi_{\mathcal{H}}$ at a critical point of its restriction to a product $W = \prod_i W_i$ of these, or a limit at infinity of the hinge space of such values along points with the differential of the restriction tending to 0. The class $C_i$ is* occupied *with the weight $a$ when $W_i = H_a$,* empty *when $W_i = P$,* silent *when $W_i = X$ or $X'$, and* balanced *when $W_i = O$.*

**Theorem 42.32** (The source count bounds the vanishing ranks)**.** *Let every cell of $\mathcal{B}$ be a parallel class, let $\Gamma$ be the group generated by the Cremona and the exchanges of equal classes, acting on the hinge space, and let $t_0 \in \mathbb{C}^\times$ be a value at which the direct image $N$ of $\boxtimes_i \mathcal{H}_{C_i}$ under $\Phi_{\mathcal{H}}$, modulo the scale, has no cohomology outside the degree 0 near $t_0$ and no vanishing cycles at $t_0$ coming from infinity of the hinge space. For a product $W$ as in Theorem 42.31 put $m(W) = \prod_i m_{s_i}(W_i)$, and let $I_{t_0}(W)$ be the intersection number over $t_0$ of $T^*_W$ with the graph of $d\Phi_{\mathcal{H}}$, the sum of the Milnor numbers of $\Phi_{\mathcal{H}}|_W$ at its critical points of the value $t_0$. Then the vanishing cycles of $N$ at $t_0$ have the dimension $\sum_W m(W)\, I_{t_0}(W)$, and*

$$r_{t_0}(\mathcal{D}_{\mathcal{B}}[1]) \leq N_{t_0},$$

*the dimension of the part of these vanishing cycles on which $\Gamma$ acts by the character of* [1].

**Proposition 42.33** (Local contributions)**.** *In Theorem 42.32:*

*(a)* a weighted point*, a nondegenerate critical point of $\Phi_{\mathcal{H}}|_W$ with every class on a hyperbola, the image of an orbit of nondegenerate native gauges of $\mathcal{B}$ with no asymptotic native gauge of its value, contributes at most 1 to $N_{t_0}$ for its orbit;*

*(b) points exchanged by $\Gamma$, each a nondegenerate critical point of multiplicity $m(W)$, contribute $m(W)$ together to $N_{t_0}$;*

*(c) at a nondegenerate critical point of $\Phi_{\mathcal{H}}|_W$ the local contribution to the vanishing cycles is $m(W)$; in particular an* odd corner*, an odd class $C_i$ silent with the other classes on hyperbolas, at a nondegenerate critical point of $\Phi_{\mathcal{H}}|_W$, $W_i = X$, at which the covector of $\Phi_{\mathcal{H}}$ lies on no other component of the characteristic cycle, contributes $\frac{s_i-1}{2}$ there, and the two corners on $X$ and $X'$, exchanged by the Cremona, contribute $\frac{s_i-1}{2}$ to $N_{t_0}$ together;*

*(d) an* empty point*, the classes $C_i$, $i \in I$, on $P$ and the others on hyperbolas, at a nondegenerate critical point of $\Phi_{\mathcal{H}}|_W$, contributes $\prod_{i\in I}(s_i - 1)$ to the vanishing cycles;*

*(e) a clean critical curve $Z$ of $\Phi_{\mathcal{H}}|_W$ along which the characteristic cycle has the constant multiplicity $m(W)$ and which meets no other component contributes $-m(W)\chi(Z)$, the ends of $Z$ at infinity of the hinge space excepted.*

The kinds of native gauges of Theorem 42.24, the corners of Proposition 38.26 at the classes (Corollary 38.28), and the curves of balanced classes are all critical points of the two-sided core on the strata of the hinge space. The exercises work the triangles of classes, with a mark of another norm among them (Section 55.4).

# Chapter 43

# Arithmetic along a ray

A free share closed along a ray brings at every lattice point a rational number of tier one in the point times Trailmoments of the counting input at cuts that move with the point; summed, these are Trailmoments of live inputs, and along the ray the same finitely many live inputs carry every value, the step from one point of the ray to the next entering as rational words. The terms are rational in the lattice point, up to a sign, exactly when the lower Gammas of the word whose slope is at least two in modulus pair off; the word is then a product of sines times a rational function, and its closings are combinations of Trailmoments of lattice inputs with coefficients polynomial in $(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. The frame of a law of the values bounds every clearing of their rows from below, by the product of the roots of the symbol, and the merit of the rows, the decay at the native gauge of the direction against the growth of the clearing, decides the gap along the ray. On a ray from the origin the law of [1] vanishes to the order $n-1$ at its low end, which gives one linear relation among the first values; at order two every value is then a row in the one coordinate $\langle 1\rangle_1$, and when the law is reflexive the coefficient of that coordinate is a closing of the ray, the constant term of a power of the reciprocal stride, read with no lattice sum. A hinge closing is a row over finitely many periods of the room as soon as the room's periods along the lattice of the closing satisfy a recurrence of finite order. The critical primes of a native gauge, the primes at which its native value is not integral, are primes at which the letters degenerate, and when every root of the symbol of a law is such a native value they are the primes of its leading coefficient.

## 43.1 Closing a free share along a ray

**Theorem 43.1** (Closing one free share along a ray)**.** *Let $A$ and $B$ be finite sets with integers $b_\gamma$ and $u_\gamma$ for $\gamma \in A \sqcup B$,* slopes *$\epsilon_\alpha \in \{1,-1\}$ for $\alpha \in A$ and $\epsilon_\beta \in \mathbb{Z}\setminus\{0\}$ for $\beta \in B$, balanced in the slopes, $\sum_{\alpha\in A}\epsilon_\alpha = \sum_{\beta\in B}\epsilon_\beta$. In this statement write, for $k \in \mathbb{N}$ and a share $a$,*

$$W_k(a) = \frac{\prod_{\alpha\in A}\Gamma(\epsilon_\alpha a + b_\alpha + k u_\alpha)}{\prod_{\beta\in B}\Gamma(\epsilon_\beta a + b_\beta + k u_\beta)},$$

*the stacked Gamma (Definition 1.2) of one free share at the $k$-th point of the ray along the integral direction $(u_\gamma)_\gamma$; write $A_\pm$ for the $\alpha \in A$ with $\epsilon_\alpha = \pm 1$, $B_\pm$ for the $\beta \in B$ with $\pm\epsilon_\beta > 0$, $w = |A_+| - |B_+|$, and*

$$r_W = \prod_{\beta\in B_+} \epsilon_\beta^{\epsilon_\beta} \Big/ \prod_{\beta\in B_-} |\epsilon_\beta|^{|\epsilon_\beta|}.$$

*(i)* The step factor. $W_k = W_0\, g_k$ *with*

$$g_k(a) = \frac{\prod_{\alpha\in A}(\epsilon_\alpha a + b_\alpha)_{k u_\alpha}}{\prod_{\beta\in B}(\epsilon_\beta a + b_\beta)_{k u_\beta}}, \qquad (x)_{-r} = \frac{1}{(x-r)_r} \quad (r\in\mathbb{N}),$$

*a rational function of the share with rational coefficients.*

*(ii)* The terms. *The poles of $W_k$ lie at integers. Let $n_1$ be an integer beyond which, at $a=-n$, the Gammas of $A_+$ and of $B_+$ have nonpositive integer arguments and those of $A_-$ and of $B_-$ positive ones, and for $\gamma\in A\sqcup B$ let $N_\gamma(n)$ be minus the argument of its Gamma at $a=-n$ in the first case and that argument less one in the second; $N_\gamma(n)$ is affine in $n$ with the slope $|\epsilon_\gamma|$. For every integer $n>n_1$ and small $\varepsilon$,*

$$W_k(-n+\varepsilon) = C_n\,\varepsilon^{-w}\exp\Big(\sum_{i\ge1}\Lambda_{n,i}\,\varepsilon^i\Big),$$

*where $C_n$ is a nonzero rational number, a signed product of the factorials $N_\gamma(n)!$, to the power $1$ for $\gamma\in A_-\cup B_+$ and $-1$ for $\gamma\in A_+\cup B_-$, and of the slopes of $B_+$; $C_{n+1}=\rho(n)\,C_n$ for a rational function $\rho$ with rational coefficients, so that $(C_n)_n$ has tier at most one (Proposition 38.11), and $|C_n|^{1/n}\to r_W$, so that $r_W$ is the root rate (Definition 1.1(viii)) of the $C_n$; $\Lambda_{n,1}$ is a rational combination of the heads $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N_\gamma(n)])$; and for $i\ge2$, $\Lambda_{n,i}$ is a rational combination of $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$ and the heads $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N_\gamma(n)])$. The coefficients of these combinations do not depend on $n$ or $k$. The residue of $W_k$ at $-n$ is $C_n$ times the coefficient of $\varepsilon^{w-1}$ in the exponential, a polynomial with rational coefficients, not depending on $n$, in the Trailmoments $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$, $2\le i\le w-1$, and the heads $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N_\gamma(n)])$, $1\le i\le w-1$, each of its monomials of total index $w-1$; it is zero when $w\le0$.*

*(iii)* The closing. *Let $W_k$ close to the left on a line $\operatorname{Re}a=c$ that separates the poles of the Gammas of $A_+$ from those of $A_-$ (Theorem 4.5). Then $\frac{1}{2\pi i}\int_{(c)}W_k(a)\,da$ is the sum of the residues of $W_k$ at the integers $-n<c$. When $r_W<1$ it is a finite rational combination of products of Trailmoments $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$, $i\ge2$, with $1$ and with the absolutely convergent sums*

$$(\mathsf{T}_{0;(1)}[x]\,\psi\,\mathbb{I}_\mathbb{Z})((n_1,\infty)) = \sum_{n>n_1}\psi(n), \qquad \psi(n) = C_n\prod_{l=1}^{q}(\mathsf{T}_{i_l;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N_{\gamma_l}(n)]),$$

*over the finitely many products with $\gamma_l\in A\sqcup B$, $i_l\ge1$ and $i_1+\cdots+i_q\le w-1$; each such $\psi$ is a live input, its density read from the serif family at the moving cut, and it satisfies a recurrence of order at most $2^q$ with polynomial coefficients and nonzero extreme coefficients.*

*(iv)* Along the ray. *For two steps $k$ and $k'$ and every $n$ beyond both bounds $n_1$, $C_n$ at $k$ is $C_n$ at $k'$ times a rational function of $n$ with rational coefficients, and every $N_\gamma(n)$ at $k$ is $N_\gamma(n)$ at $k'$ plus an integer that does not depend on $n$. So every input $\psi$ of (iii) at the step $k$ is a combination, with coefficients rational functions of $n$ with rational coefficients, of the inputs $\psi$ of (iii) at the step $k'$: when $r_W<1$ every value along the ray is a finite rational combination of products of Trailmoments $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$ with $1$ and with Trailmoments at index zero of the finitely many live inputs of one step against rational functions of the point with rational coefficients, whose rows Theorem 6.10 reads.*

*Closing to the right is the same statement after the share $a$ is replaced by $-a$.*

A word of one free share arises from the atom formula (Corollary 18.16) when its other shares are closed or merged, for instance by the four Gammas on a line (Corollary 4.7): integrating $\Gamma(a+z)\Gamma(b+z)\Gamma(c-z)\Gamma(d-z)$ over $z$ leaves four upper Gammas and the lower Gamma $\Gamma(a+b+c+d)$, whose slope in a remaining share is the sum of the slopes of $a$, $b$, $c$ and $d$. A lower Gamma of slope $v$ contributes $v^v$ to the root rate of the terms and one of slope $-v$ contributes $v^{-v}$, so a lone lower Gamma of slope $-2$ makes the terms of a left closing fall like $4^{-n}$, while lower Gammas that pair off leave terms that do not fall geometrically and are rational in $n$ up to a sign.

**Proposition 43.2** (Paired words)**.** *In Theorem 43.1 the following are equivalent.*

*(a) The lower Gammas of slope at least two in modulus pair off: for every integer $v \geq 2$ as many $\beta \in B$ have $\epsilon_\beta = v$ as have $\epsilon_\beta = -v$.*

*(b) For some $k$, and then for every $k$, there are $\varsigma \in \{0,1\}$ and a rational function $R$ with $C_n = (-1)^{\varsigma n} R(n)$ for every $n > n_1$.*

*When they hold, put $p = |A_+| - \#\{\beta \in B : \epsilon_\beta = 1\}$ and let $B^{\geq 2}$ be the set of the $\beta$ with $\epsilon_\beta \geq 2$, so that $w = p - |B^{\geq 2}|$. Then*

$$W_k(a) = \pm\, \pi^w \, \frac{\prod_{\beta \in B^{\geq 2}} \sin(\pi \epsilon_\beta a)}{\sin(\pi a)^p} \, R_k(a),$$

*with $R_k$ a rational function with rational coefficients, and for $n > n_1$ the residue of $W_k$ at $-n$ is*

$$\pm(-1)^{\varsigma n} \sum_{0 \leq l \leq (w-1)/2} \tau_l \, \pi^{2l} \, \frac{R_k^{(w-1-2l)}(-n)}{(w-1-2l)!},$$

*with $\varsigma \equiv p + \sum_{\beta \in B^{\geq 2}} \epsilon_\beta \pmod 2$, rational $\tau_l$ that depend only on the slopes, and $\tau_0 = \prod_{\beta \in B^{\geq 2}} \epsilon_\beta$. So a closing of Theorem 43.1(iii) is, up to finitely many terms in $\mathbb{Q}[\pi^2]$, a sum over the lattice of $(-1)^{\varsigma n} r(n)$ with $r$ a rational function of $n$ whose coefficients lie in $\mathbb{Q}[\pi^2]$, and Lemma 38.32 over the field $\mathbb{Q}(\pi^2)$, applied to $m \mapsto r(n_1+1+m)$ when $\varsigma = 0$ and to $m \mapsto r(n_1+1+2m) - r(n_1+2+2m)$ when $\varsigma = 1$, writes it over 1, the Trailmoments of the lattice inputs on the cosets of the poles at their cuts and differences of the index-one values of these inputs, with coefficients in $\mathbb{Q}[\pi^2]$; here $\pi^2 = 6\,(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ (Theorem 5.17(iv)).*

So the pairing of the lower Gammas, not their number, keeps the terms rational. The two-mark word $\Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]$ of Lemma 5.16 has no lower Gamma; the word of two merged pairs on four marks, with one lower Gamma of slope 2 and one of slope $-2$, pairs off, and the poles of its rational factor lie on the coset $\frac{1}{2} + \mathbb{Z}$; the word of one merged pair on three marks, with one lower Gamma of slope $-2$, does not pair off, and its terms carry a live input of root rate $\frac{1}{4}$. Exercise 54.7 works the word on four marks.

## 43.2 The clearing along a ray

**Theorem 43.3** (The frame bounds the clearing from below)**.** *Let $p_0, \ldots, p_h \in \mathbb{Q}[k]$, $h \geq 1$, with $p_0$ and $p_h$ without zeros at the integers $k \geq k_0$; let $d_0$, $d_h$ and $c$ be the degrees of $p_0$ and $p_h$ and the quotient of their leading coefficients.*

*(i) Let rational sequences $x^{(1)}, \ldots, x^{(h)}$ on the integers $k \geq k_0$ satisfy $\sum_{j=0}^{h} p_j(k)\, x^{(i)}_{k+j} = 0$ and be linearly independent, with frame $C_k$ (Proposition 38.11). If a positive integer $D_k$ makes*

*every $D_k\, x^{(i)}_{k+l}$, $1 \le i \le h$, $0 \le l < h$, an integer, then $D_k^h\, |C_k| \ge 1$, and as $k \to \infty$*

$$\frac{\log D_k}{k} \;\ge\; \frac{d_h - d_0}{h}\,(\log k - 1) + \frac{1}{h}\log\frac{1}{|c|} + O\Big(\frac{\log k}{k}\Big).$$

*When $d_0 = d_h$ is the largest degree of the $p_j$, $\liminf_k \frac{1}{k}\log D_k \ge \frac{1}{h}\log\frac{1}{|c|}$, and $|c|$ is the modulus of the product of the roots of the symbol (Proposition 38.12).*

*(ii) Let $x_k = \sum_{l=1}^m A_{k,l} v_l - B_k$, $k \ge k_0$, with rational $A_{k,l}$ and $B_k$ and real $v_l$ such that $1, v_1, \dots, v_m$ are linearly independent over $\mathbb{Q}$, satisfy $\sum_j p_j(k)\, x_{k+j} = 0$. Then each of the $m+1$ sequences $(A_{k,l})_k$ and $(B_k)_k$ satisfies the same recurrence, and if $h$ of them are linearly independent, every $D_k$ clearing the rows of $x_k, \dots, x_{k+h-1}$ (Definition 36.5) satisfies (i).*

The bound reads the clearing on $h$ consecutive rows together, through one determinant: the frame grows like the $k$-th power of the product of the roots of the symbol, and the clearing raised to the power $h$ must undo it. For the values $\langle 1\rangle_k$ along a ray with $\kappa_{1,S} > 0$ for every nonempty proper $S$, the roots of the symbol of every law of the values with rational coefficients include the native value $f(\alpha^\star)$ of the native gauge of the direction and its conjugates (Proposition 38.9). The frame does not say how the clearing is shared among the coordinates; on the rays of Section 43.4 the denominator of the ratio of a reflexive law falls on the coefficient of the one coordinate, at its full power up to what the closings absorb.

## 43.3 The merit of a ray

**Definition 43.4** (The merit of rows along a ray)**.** Let a ray of a family have real origin and $\kappa_{1,S} > 0$ for every nonempty proper $S$, with the native gauge $\alpha^\star$ of its direction on $\mathbb{R}^{\mathcal{E}}_{>0}$ (Theorem 38.14), and let its values have rows $\langle 1\rangle_k = \sum_{l=1}^m A_{k,l} v_l - B_k$, $k \ge k_0$, over fixed real numbers $v_1, \dots, v_m$ (Definition 36.5), cleared by positive integers $D_k$. The *merit* of these rows is

$$-\log f(\alpha^\star) - \limsup_{k\to\infty} \frac{1}{k}\log D_k \;\in [-\infty, \infty).$$

**Theorem 43.5** (The merit decides the gap along a ray)**.** *In Definition 43.4 let $k_1 \ge k_0$ be an integer with $\kappa_{0,S} + k_1\kappa_{1,S} > 0$ for every nonempty proper $S$.*

*(i) For every $k \ge k_1$, $0 < D_k\langle 1\rangle_k \le D_k\, f(\alpha^\star)^{k-k_1}\, \langle 1\rangle_{k_1}$.*

*(ii) If the merit is positive, then $D_k\langle 1\rangle_k \to 0$ and some $v_l$ is irrational; for $m = 1$, $v_1 \notin \mathbb{Q}$ and $A_{k,1} \ne 0$ for all large $k$.*

*(iii) If $D_k \le C\gamma^k\, d^w_{\lfloor ak+b\rfloor}$ for all large $k$, with constants $C, \gamma > 0$, $a, w \ge 0$ and $b$, and $\lfloor ak+b\rfloor \ge 0$, the merit is at least $-\log f(\alpha^\star) - \log\gamma - a\,w\,c_{\mathrm{lcm}}$ (Lemma 37.1), and $\gamma\, e^{awc_{\mathrm{lcm}}} f(\alpha^\star) < 1$ gives (ii).*

*(iv) If each $D_k$ also clears the rows of $\langle 1\rangle_{k+1}, \dots, \langle 1\rangle_{k+h-1}$, $1, v_1, \dots, v_m$ are linearly independent over $\mathbb{Q}$, the values satisfy a law $\sum_{j=0}^h p_j(k)\langle 1\rangle_{k+j} = 0$, $k \ge k_0$, as in Theorem 43.3 with $p_0$ and $p_h$ of the largest degree, and $h$ of the sequences $(A_{k,l})_k$, $(B_k)_k$ are linearly independent, then the merit is at most $-\log f(\alpha^\star) + \frac{1}{h}\log|c|$, with $|c|$ the modulus of the product of the roots of the symbol.*

On the centre ray of a family, $f(\alpha^\star) = \prod_j K_j(\mathbf{1})^{-2\sigma_j}$ (Theorem 38.15), and a clearing $D_k = \gamma^{k-1} d^w_{k-1}$ passes the gap when $\gamma\, e^{wc_{\mathrm{lcm}}} < \prod_j K_j(\mathbf{1})^{2\sigma_j}$, the minimum of the reciprocal stride on the

orthant. Only the upper bound of the values is used: positivity and the largest value of the stride on the orthant, both read before any closing. By (iv), with the native value among the roots of the symbol, a law of order $h$ leaves positive merit only to a ray whose symbol has roots of modulus larger than $f(\alpha^\star)$.

## 43.4 Laws at the low end and their closings

**Proposition 43.6** (The low end of the law)**.** *Let a ray be as in Proposition 39.34, on $n$ marks, with origin $0$, every letter of positive exponent in the direction, $\kappa_{1,S} > 0$ for every nonempty proper $S$, and the volume $V_1$ of the direction; let the law of $[1]$ (Definition 39.1) be $\sum_{i=0}^{h} p_i(k)\,\mathrm{Sh}^i[1] = 0$ with $p_0, \dots, p_h \in \mathbb{Q}[k]$, and write $x_k = \langle 1\rangle_k$.*

*(i) The values satisfy the law at every integer $k \geq 1$. Further $k^{n-1}$ divides $p_0$, and with $\varpi$ the coefficient of $k^{n-1}$ in $p_0$ the values satisfy the* low-end relation

$$\varpi\, V_1 + \sum_{i=1}^{h} p_i(0)\, x_i = 0;$$

*$V_1$ is rational. For a solution $y$ of the law on the integers $k \geq 1$, $\sum_i p_i(k) y_{k+i} = 0$ for every $k \geq 1$, call $-\sum_{i=1}^{h} p_i(0)\, y_i$ its* low-end coefficient*; that of the values is $\varpi V_1$, the value at $k = 0$ of the continuation of $p_0(k)\langle 1\rangle_k$.*

*(ii) Let $h = 2$ and $p_2(k) \neq 0$ at every integer $k \geq 0$, and let $c$ and $r$ be the solutions of the law on $k \geq 1$ with $(c_1, c_2) = \big(1, -p_1(0)/p_2(0)\big)$ and $(r_1, r_2) = \big(0, -\varpi V_1/p_2(0)\big)$. They are rational, the low-end coefficient of $c$ vanishes, that of $r$ is $\varpi V_1$, and*

$$x_k = x_1\, c_k + r_k \qquad (k \geq 1):$$

*every value has a row in the one coordinate $x_1$, its coefficient the solution whose low-end coefficient vanishes and whose first term is $1$.*

**Definition 43.7** (Reflexive laws)**.** A recurrence $\sum_{i=0}^{h} p_i(k)\, x_{k+i} = 0$ with $p_i \in \mathbb{Q}[k]$ is *reflexive* with the *ratio* $\lambda \in \mathbb{Q}^\times$ when there is a constant $\gamma \neq 0$ with

$$p_{h-i}(-k-h+1) = \gamma\, \lambda^i\, p_i(k) \qquad (0 \leq i \leq h)$$

as polynomials in $k$. When $p_0$ and $p_h$ have the largest degree $d$ of the $p_i$, the roots of the symbol of a reflexive recurrence are exchanged by $\varrho \mapsto \lambda/\varrho$, and at $h = 2$ their product is $\pm\lambda$.

**Theorem 43.8** (The closings read the coordinate of a reflexive law)**.** *Let a ray be as in Proposition 43.6, with the law of $[1]$ reflexive with the ratio $\lambda$, and let $\mathsf{C}_j(1) = \mathrm{CT}(f^{-j})$ be the closings of $1$ (Definition 39.30).*

*(i) $y_k = \lambda^{k-1}\, \mathsf{C}_{k-1}(1)$, $k \geq 1$, is a solution of the law on $k \geq 1$, and its low-end coefficient vanishes.*

*(ii) If $h = 2$ and $p_2(k) \neq 0$ at every integer $k \geq 0$, then $y$ is the solution $c$ of Proposition 43.6(ii):*

$$\langle 1\rangle_k = \lambda^{k-1}\, \mathsf{C}_{k-1}(1)\, \langle 1\rangle_1 + r_k \qquad (k \geq 1),$$

*the coefficient of the coordinate read from the closings of the ray, with no lattice sum. If moreover $\langle 1\rangle_1 = A_1 v - B_1$ is a row in one coordinate $v$, the coefficient of $v$ in $\langle 1\rangle_k$ is $A_1 \lambda^{k-1} \mathsf{C}_{k-1}(1)$.*

*(iii) If moreover the weights are integers, $\mathsf{C}_j(1) \in \mathbb{Z}$, and with $\lambda = a/b$ in lowest terms, $b > 0$, the coefficient of $\langle 1\rangle_1$ in $\langle 1\rangle_k$ has the denominator $b^{k-1}/\gcd\big(b^{k-1}, \mathsf{C}_{k-1}(1)\big)$.*

**Corollary 43.9** (The merit of a reflexive ray of order two)**.** *In Theorem 43.8(iii), every clearing $D_k$ of the rows $\langle 1\rangle_k = \lambda^{k-1}\mathsf{C}_{k-1}(1)\,\langle 1\rangle_1 + r_k$ in the coordinate $\langle 1\rangle_1$ has merit (Definition 43.4) at most*

$$-\log f(\alpha^\star) - \log b + \liminf_{k\to\infty} \frac{1}{k} \log\gcd\big(b^{k-1}, \mathsf{C}_{k-1}(1)\big).$$

The closings are the constant terms of the powers of the reciprocal stride, finite sums of products of weights; on a reflexive ray of order two they are the coefficient of the coordinate at every step, up to the power $\lambda^{k-1}$, and the clearing pays the full power $b^{k-1}$ of the denominator of the ratio except for what the closings absorb. The ray is then judged by the divisibility of its closings. The centre rays of the lists $\binom{[3]}{1}$ and $\binom{[4]}{1}$ have reflexive laws of order two with the ratios $\frac{1}{9}$ and $\frac{1}{64}$; the exercises derive both laws (Exercises 54.2 and 54.8), and on four marks the rows, the coefficient of the coordinate and the clearing (Exercises 54.9, 54.10 and 54.11).

## 43.5 The hinge closing as a row over the room

**Proposition 43.10** (The hinge closing as a row over the room)**.** *For one table in Theorem 19.13 let $\sup_{\Delta_B} \varrho_B < \inf_{\Delta_R} \varrho_R$, so that the even share closes on the lattices of the level $d$ (Corollary 19.21(i)). Group the residues by the index $n$ of their points and write the closed value as $P_F(\mathbf{m}; s, t) = \sum_{n\ge 0}\langle \Phi_n, \mathcal{X}_n\rangle$, where $\mathcal{X}_n$ lists the data of the room family that the residues at the $n$-th points contain, its periods at the level shares of those points and, where the two lattices meet, their derivatives in the even share (Corollary 19.21(ii)), and $\Phi_n$ lists the coefficients that Corollary 19.21(ii) puts in front of them.*

*(i) Let there be numbers $\chi_1, \dots, \chi_h$ and lists $\mathbf{c}_1(n), \dots, \mathbf{c}_h(n)$ with $\mathcal{X}_n = \sum_{i=1}^h \chi_i\, \mathbf{c}_i(n)$ for every $n$ and $\sum_n |\langle \Phi_n, \mathbf{c}_i(n)\rangle| < \infty$ for every $i$. Then*

$$P_F(\mathbf{m}; s, t) = \sum_{i=1}^{h} \chi_i\, \Xi_i, \qquad \Xi_i = \sum_{n\ge 0} \langle \Phi_n, \mathbf{c}_i(n)\rangle :$$

*a row over $h$ data of the room, with coefficients that are sums along the lattice of the bunch's data against the sequences $\mathbf{c}_i$.*

*(ii) When $2\delta^d \notin \mathbb{Z}$ every pole is simple and $\mathcal{X}_n$ lists the two room periods $x_n^\pm$ at the points $\sigma^d = \pm\delta^d - n$. If each sequence $x^\pm$ satisfies a recurrence $\sum_{i=0}^{h_\pm} p_i^\pm(n)\, x_{n+i}^\pm = 0$, $n \ge 0$, with $p_{h_\pm}^\pm(n) \ne 0$ for $n \ge 0$, let $c_i^\pm$, $0 \le i < h_\pm$, be its solutions with the initial values $c_i^\pm(l) = \delta_{il}$, $0 \le l < h_\pm$, rational when the $p_i^\pm$ have rational coefficients. Then $x_n^\pm = \sum_i x_i^\pm\, c_i^\pm(n)$ for every $n$, and whenever the sums $\sum_n \Phi_n^\pm\, c_i^\pm(n)$ converge absolutely, (i) holds with the $h_+ + h_-$ initial periods $x_0^\pm, \dots, x_{h_\pm - 1}^\pm$.*

*(iii) When $2\delta^d \in \mathbb{Z}$ the poles lie on the one lattice $\sigma^d = |\delta^d| - n$, $n \in \mathbb{N}$. Let the room period at $\sigma^d = \sigma - n$, a function $x_n(\sigma)$ of $\sigma$ near $\sigma_0 = |\delta^d|$, satisfy a recurrence $\sum_{i=0}^h p_i(n, \sigma)\, x_{n+i}(\sigma) = 0$ whose coefficients are polynomials in $n$ and $\sigma$, with $p_h(n, \sigma_0) \ne 0$ for $n \ge 0$. Then the values $x_n(\sigma_0)$ and the derivatives $x_n'(\sigma_0)$ satisfy the block-triangular recurrence of Proposition 37.17, they are combinations of the $2h$ initial data $x_i(\sigma_0)$, $x_i'(\sigma_0)$, $i < h$, with coefficients that do not depend on the room's periods, and whenever the resulting sums converge absolutely (i) holds with these $2h$ data.*

*The same holds with the bunch and the room exchanged when* $\inf_{\Delta_B} \varrho_B > \sup_{\Delta_R} \varrho_R$*, the share closing on the level* $r$*.*

Read down a tree of bunches, a hinge closing writes a period as a row over finitely many periods of its room, the coefficients being sums along the lattice of the hinge of the bunch's data against the fundamental solutions of the room's recurrence. The room's periods along the lattice are the moments of one ratio against one form (Corollary 19.21(ii)); when they have tier one, each lattice contributes one room period times one sum along the lattice. Proposition 45.13 gives a condition on the roots of the symbols under which the sums converge.

## 43.6 Critical primes

**Definition 43.11** (Critical primes)**.** Let a ray of a family have an integral direction with the stride $f = \alpha^{\mathbf{m}_1} \prod_L L^{-w_L}$, the product over the distinct letters $L$ of positive exponent $w_L$, and let its weights lie in a number field $\Bbbk$ that contains the coordinates of a native gauge $\alpha_c$ (Definition 38.4) on the chart $\alpha_{e_0} = 1$. A prime $\mathfrak{p}$ of $\Bbbk$ at which the weights are integral and the coordinates of $\alpha_c$ are units is a *critical prime* of $\alpha_c$ when the native value is not integral there, $v_{\mathfrak{p}}\big(f(\alpha_c)\big) < 0$.

**Theorem 43.12** (The letters degenerate at a critical prime)**.** *Let* $\mathfrak{p}$ *be a critical prime of a native gauge* $\alpha_c$*, with the residue field* $\kappa$ *and the reduction* $\bar{\alpha}_c \in (\kappa^\times)^{\mathcal{E}}$*, and let* $S$ *be the set of the letters of positive exponent whose valuation* $v_{\mathfrak{p}}(L(\alpha_c))$ *is the largest, a positive number.*

*(i) Every letter of* $S$ *vanishes at* $\bar{\alpha}_c$ *after reduction, and there are units* $u_L$ *of the local ring at* $\mathfrak{p}$ *with*

$$\sum_{L \in S} \overline{w_L u_L}\, \nabla \bar{L}(\bar{\alpha}_c) = 0,$$

$\nabla$ *the gradient in the coordinates* $\alpha_e$*,* $e \in \mathcal{E}$*, and the bar the reduction.*

*(ii) If the characteristic of* $\kappa$ *divides none of the exponents* $w_L$*,* $L \in S$*, the zero sets of the reduced letters of* $S$ *are not smooth hypersurfaces meeting transversally at* $\bar{\alpha}_c$*: when* $S$ *has one letter, that letter is singular at* $\bar{\alpha}_c$*, and when it has several, their gradients there are linearly dependent.*

**Proposition 43.13** (Critical primes and the symbol)**.** *Let* $\sigma(\lambda) = \sum_{j=0}^{h} c_j \lambda^j \in \mathbb{Z}[\lambda]$*,* $c_h \neq 0$*,* $h \geq 1$*, be a symbol (Corollary 38.7) with coprime coefficients, and* $K$ *a number field containing its roots.*

*(i) For every root* $\lambda$ *of* $\sigma$*,* $c_h\lambda$ *is an algebraic integer.*

*(ii) A prime* $p$ *divides* $c_h$ *exactly when some root of* $\sigma$ *has negative valuation at some prime of* $K$ *above* $p$*; and then at every prime of* $K$ *above* $p$ *some root has negative valuation.*

*(iii) Let every root of* $\sigma$ *be the native value of a native gauge whose coordinates lie in* $K$*, at which the weights are integral and the coordinates are units at the primes above* $p$*. Then* $p$ *divides* $c_h$ *exactly when some prime of* $K$ *above* $p$ *is a critical prime of one of these native gauges. When* $\sigma$ *is the symbol of a recurrence whose* $p_0$ *and* $p_h$ *have the largest degree,* $c_0/c_h$ *is the quotient of their leading coefficients, whose* $k$*-th power enters the frame (Proposition 38.11).*

At a critical prime the native gauge reduces to a point at which the letters of largest valuation meet badly, so the critical primes of a ray are read from the stride and its critical points before any value is computed, and under (iii) they are the primes of the leading coefficient of the symbol, which the frame carries. The value span of a ray can see fewer primes: on the centre ray of $\binom{[4]}{1}$ the

critical prime $2$ sits in the frame and in the denominators of the coefficients, and the coordinate is the Trailmoment $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ of the counting input. On the centre ray of $\binom{[3]}{1}$ the native value $\frac{1}{9}$ reduces modulo $3$ to a tangency of the letters, and on that of $\binom{[4]}{1}$ the native value $\frac{1}{16}$ reduces modulo $2$ to the collapse of three intersection points of the letters onto the native gauge.

# Part VIII

# Relations and the content

# Chapter 44

# Relations and separation

A value is the evaluation of a word, and two words of equal value either are related by the proved relations of the theory or are not, a problem that relation blocks make finite. The admitted words form an algebra with an evaluation, the proved relations generate an ideal inside its kernel, and completeness asks for equality. Relation blocks reduce a finite space of words to representatives, each word with a certificate; when the representatives have values independent over the rationals, the block is complete. Words of polynomials on a box are complete outright, a finite census of insertions separates numerators, and the critical-line words of order $k$ of polynomials form a block whose values are the rational combinations of 1 and $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$; its two representatives have values independent over the rationals exactly when $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) \notin \mathbb{Q}$, and the block is then complete. Independence of values is proved by separators: families of rational matrices whose rows are small on the vector of values. Along a ray the words are complete as functions of the strength: an element of the module of the ray has the values 0 exactly when its closings vanish, and finitely many rational closings decide it. A block is separated at once by a separator, or value by value by separating generators, each a sequence of linear forms small on one value against the values already separated, whose independence measures combine.

## 44.1 Words and relations

**Definition 44.1** (Words, evaluation, relations, completeness)**.** An *admitted word* is a word (Definition 1.1(vi)) given by finite rational data (tables and weights, coefficients, masses, exponents, inputs with a finite presentation, cuts) together with the rule that assigns its value: a convergent integral, a finite part or a continuation, as proved for it. The admitted words span a rational algebra $\mathfrak{W}$, whose product tensors the data, with independent integration variables and separate regulators, and *evaluation* is the map $\mathrm{ev} : \mathfrak{W} \to \mathbb{C}$. The *relation ideal* $\mathfrak{I} \subseteq \mathfrak{W}$ is generated by the proved relations, each on its stated domain: linearity; changes of variables and of order; scale and shape; split, port and gluing; the Gramma relations; the letter relations and the mark relations (Theorems 8.9 and 35.10); the subtraction of a common boundary term from the two sides of a relation; the cut relation of the Trailmoment (Proposition 5.3); the closings. A set of words is *complete* when each of its rational combinations of value zero lies in $\mathfrak{I}$.

**Proposition 44.2** (Products keep separate regulators)**.** *If $A(\varepsilon)$ and $A'(\varepsilon')$ have Laurent expansions in disjoint sets of regulators, then* $\mathrm{CT}_{\varepsilon}\,\mathrm{CT}_{\varepsilon'}\big(A(\varepsilon)A'(\varepsilon')\big) = \big(\mathrm{CT}_{\varepsilon}\,A\big)\big(\mathrm{CT}_{\varepsilon'}\,A'\big)$*, and evaluation is multiplicative on $\mathfrak{W}$. Identifying regulators is a further operation:* $\mathrm{CT}(\varepsilon^{-1}) = \mathrm{CT}(\varepsilon) = 0$*, while* $\mathrm{CT}(\varepsilon^{-1}\varepsilon) = 1$*.*

**Proposition 44.3** (Soundness)**.** $\mathfrak{I} \subseteq \ker(\mathrm{ev})$.

**Proposition 44.4** (A finite matrix of mark relations)**.** *On $[0,1]^d$ let $a_l, b_l > 0$, let $D_1, \dots, D_h$ be rational polynomials positive on the closed box with exponents $p_1, \dots, p_h \in \mathbb{Q}$, and in this statement write $D = \prod_k D_k = \sum_\gamma D_\gamma x^\gamma$ and $x_i \sum_k p_k\, \partial_i D_k \prod_{k' \neq k} D_{k'} = \sum_\gamma D_\gamma^{(i)} x^\gamma$. For every polynomial $A$ and every coordinate $i$,*

$$\mathrm{mr}_i(A) = D\big[x_i(1-x_i)\,\partial_i A + \big(a_i - (a_i+b_i)x_i\big)A\big] - x_i(1-x_i)\,A \sum_k p_k\, \partial_i D_k \prod_{k' \neq k} D_{k'}$$

*has $\int_{[0,1]^d} \mathrm{mr}_i(A) \prod_l x_l^{a_l - 1}(1-x_l)^{b_l - 1} \prod_k D_k^{-p_k - 1}\, dx = 0$ (Theorem 35.10(a) with the letters $D_k$ at the exponents $p_k + 1$ and the polynomial $AD$), and*

$$\mathrm{mr}_i(x^\beta) = \sum_\gamma \Big[\big((\beta_i + a_i)D_\gamma - D_\gamma^{(i)}\big)x^{\beta+\gamma} - \big((\beta_i + a_i + b_i)D_\gamma - D_\gamma^{(i)}\big)x^{\beta+\gamma+\mathbf{e}_i}\Big].$$

*For a degree bound these vectors form a finite rational relation matrix; a rational solution of its column equation for a numerator writes the numerator as a sum of polynomials $\mathrm{mr}_i(A_i)$, so that its word vanishes.*

**Theorem 44.5** (Words of polynomials on a box are complete)**.** *For $P \in \mathbb{Q}[x]$ and rational $a_i, b_i > 0$ the words $\int_{[0,1]^d} P(x) \prod_i x_i^{a_i - 1}(1-x_i)^{b_i - 1}\, dx$ have the mark relations $x^{\beta + \mathbf{e}_i} \equiv \frac{\beta_i + a_i}{\beta_i + a_i + b_i}\, x^\beta$, and every $P$ reduces to*

$$\sum_\beta \big([x^\beta]P\big) \prod_i \frac{(a_i)_{\beta_i}}{(a_i + b_i)_{\beta_i}},$$

*the expectation of $P$ under the Dirichlet laws $\mathrm{Dir}_{(a_i,b_i)}$ of the pairs $(x_i, 1-x_i)$, times the base word, of value $\prod_i \Gamma_2^1\left[\begin{matrix} a_i,\ b_i \\ a_i + b_i \end{matrix}\right] > 0$. A word vanishes exactly when its reduction is zero. The same holds for every finite tensor product of such words.*

**Theorem 44.6** (A finite insertion census separates numerators)**.** *Let $\phi$ be a measure on $[0,1]^d$ with an integrable density positive on the interior, and $D$ a rational polynomial positive on the closed box. For $|\beta|, |\gamma| \le h$ the matrix $\mathbf{G}_{\beta\gamma} = \int x^{\beta+\gamma} D^{-1}\, d\phi$ is positive definite. For $A = \sum_\beta a_\beta x^\beta$ of degree at most $h$ the insertion values $v_\beta = \int x^\beta A D^{-1}\, d\phi$ satisfy $\mathbf{v} = \mathbf{G}\mathbf{a}$ and $\mathrm{adj}(\mathbf{G})\,\mathbf{v} = \det(\mathbf{G})\,\mathbf{a}$: these $\binom{d+h}{d}$ values determine $A$. Equivalently, the Laplace transform $t \mapsto \int AD^{-1} e^{-\langle t, x\rangle}\, d\phi$ has all its Taylor coefficients of degree at most $h$ zero at $t = 0$ only for $A = 0$.*

Evaluation is multiplicative because a product of words keeps the regulators of its factors apart; identifying two regulators is a new operation, not a relation. Every relation the theory proves lies in the kernel of evaluation, and completeness asks whether the kernel holds anything more. On a box against positive letters the mark relations form a finite rational matrix, so that vanishing is certified by linear algebra; for polynomials on a box without a letter the relations reduce every insertion to one rational multiple of the base word, and with a letter the values of finitely many monomial insertions, through the positive definite matrix $\mathbf{G}$ of Theorem 44.6, recover the insertion itself.

## 44.2 Relation blocks

**Definition 44.7** (Relation block)**.** A *relation block* is a finite-dimensional rational space $V$ of admitted words with a matrix $C$ whose columns are proved relations in $V$, each with its derivation.

Rational row reduction gives representatives $\mathfrak{w}_0, \dots, \mathfrak{w}_s$ of $V/\operatorname{im} C$ and for every $\mathfrak{w} \in V$ a certificate $\mathfrak{w} = C\mathbf{v} + \sum_j c_j \mathfrak{w}_j$.

**Proposition 44.8** (Separation completes a block)**.** *If* $\operatorname{ev}(\mathfrak{w}_0), \dots, \operatorname{ev}(\mathfrak{w}_s)$ *are linearly independent over* $\mathbb{Q}$*, then* $\ker(\operatorname{ev}|_V) = \operatorname{im} C$*.*

**Theorem 44.9** (Completeness from separated blocks)**.** *If every finite set of words of a subalgebra* $\mathfrak{W}' \subseteq \mathfrak{W}$ *lies in a relation block whose representatives have values linearly independent over* $\mathbb{Q}$*, then* $\mathfrak{W}'$ *is complete, and each of its words of value zero has a finite derivation, its certificate.*

**Proposition 44.10** (Tensoring relation blocks)**.** *For blocks* $(V_1, C_1)$ *and* $(V_2, C_2)$ *the induced relations on* $V_1 \otimes V_2$ *have the matrix* $\left[\, C_1 \otimes I \;\middle|\; I \otimes C_2 \,\right]$*, with quotient* $(V_1/\operatorname{im} C_1) \otimes (V_2/\operatorname{im} C_2)$*, and the values of the representatives are the products of the values (Proposition 44.2).*

**Proposition 44.11** (The critical polynomial block)**.** *For an integer* $k \geq 1$ *and* $P \in \mathbb{Q}[x]$ *put* $Q_P(u) = \frac{P(2u)-P(1)}{1-2u}$ *and* $h(P) = \frac{2}{(2k)!}\, \mathbb{E}_{Y \sim \mathbb{P}_{\mathrm{crit}}}\big[Q_P^{(2k-1)}(\frac{1}{2} + iY)\big]$*. Then* $h(P) \in \mathbb{Q}$ *and*

$$\operatorname{crit}_k[P(2u)] = h(P) + (2^{2k+1} - 1)\, P(1)\, (\mathsf{T}_{2k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

*So the values of the rational constants and of the critical-line words of order* $k$ *of polynomials factor through the surjective coordinate map* $(c, P) \mapsto \big(c + h(P), (2^{2k+1} - 1)P(1)\big) \in \mathbb{Q}^2$ *followed by* $(a, b) \mapsto a + b\, (\mathsf{T}_{2k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$*, which is injective exactly when* $(\mathsf{T}_{2k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) \notin \mathbb{Q}$*; moving the cut of the Trailmoment subtracts a rational head.*

A relation block is the finite form of completeness: inside a finite space of words, rational row reduction by the proved relations leaves representatives and certificates, and independence of the values of the representatives over the rationals makes the block complete (Proposition 44.8). Blocks tensor, their quotients tensoring and the values of their representatives multiplying, so a product of blocks is complete as soon as the products of those values are independent over the rationals. The critical-line words of order $k$ of polynomials have a block whose quotient has dimension at most two: every such word is a rational number plus a rational multiple of $(\mathsf{T}_{2k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$, read off two coordinates of the polynomial, and the values of the two representatives are independent over the rationals exactly when that Trailmoment is irrational, the block being then complete.

## 44.3 Separators

**Theorem 44.12** (A primitive exterior separator)**.** *Let* $v = (v_1, \dots, v_s, 1) \in \mathbb{R}^{s+1}$*. For a rational* $s \times (s+1)$ *matrix* $A$ *of rank* $s$ *with rows* $r_i$*, let* $w \in \mathbb{Q}^{s+1}$ *be its cofactor vector,* $\det[x; A] = \langle x, w \rangle$ *for all* $x$*, and* $z = Mw$ *its primitive integral multiple,* $M > 0$*. If* $D_i$ *clears* $r_i$*, then* $M = \prod_i D_i / c$ *with* $c$ *the greatest common divisor of the maximal minors of* $\operatorname{diag}(D_i)A$*. Put* $y_i = \langle r_i, v \rangle$*, let* $\hat{A}_i$ *be the other rows restricted to the first* $s$ *columns, and*

$$\beta(A) = M \sum_i |y_i| \sum_{J \subseteq \{1,\dots,s\},\ |J| = s-1} \big|\det \hat{A}_i[:, J]\big|.$$

*If for all large* $n$ *finitely many such matrices* $A_1^{(n)}, \dots, A_{k_n}^{(n)}$ *have primitive cofactor vectors spanning* $\mathbb{Q}^{s+1}$ *and* $\max_l \beta(A_l^{(n)}) \to 0$*, then* $1, v_1, \dots, v_s$ *are linearly independent over* $\mathbb{Q}$*.*

**Corollary 44.13** (Separation with bounded integer entries)**.** *If integer matrices $A_l^{(n)}$ with spanning cofactor vectors have entries at most $H_n \geq 1$ and $|\langle r_i, v\rangle| \leq \epsilon_n$ for every row, with $H_n^{s-1}\epsilon_n \to 0$, then $1, v_1, \ldots, v_s$ are linearly independent over $\mathbb{Q}$. In rates: if rational matrices with spanning cofactor vectors have entries at most $e^{an+o(n)}$, a common clearing integer at most $e^{bn+o(n)}$ and row values at most $e^{-cn+o(n)}$, with $c > sb + (s-1)a$, the same conclusion holds.*

**Corollary 44.14** (Consecutive rows)**.** *Let rows $r_k \in \mathbb{Q}^{s+1}$ be such that the square matrix of the rows $r_k, \ldots, r_{k+s}$ is invertible for large $k$. Then for each such $k$ the $s+1$ matrices omitting one of these rows have spanning primitive cofactor vectors, and Theorem 44.12 applies with these matrices. If the coordinate sequences of the rows satisfy one recurrence of order $s+1$ whose extreme coefficients have no integer zeros from $k_0$ on, invertibility of that matrix at $k_0$ gives it for every $k \geq k_0$.*

**Theorem 44.15** (Function level: pole pinning)**.** *Let $f$ and $g$ be meromorphic on a connected open subset of $\mathbb{C}$. If $f$ has infinitely many poles at which $g$ is holomorphic, and the values of $g$ at those poles include infinitely many distinct numbers, then no nonzero polynomial $Q \in \mathbb{C}[X, Y]$ has $Q(f, g) \equiv 0$.*

A separator is a family of rational matrices whose rows are nearly orthogonal to the vector of values: a primitive integral multiple of the cofactor vector of each matrix pairs with any integer relation to an integer, and if the rows are small enough against the clearing and the bounds on the entries, no rational relation among the values survives. Consecutive rows of a recurrence give such matrices as soon as one square matrix of $s+1$ consecutive rows is invertible. At the level of functions, a function with infinitely many poles at which a second function takes infinitely many values satisfies no polynomial relation with it.

## 44.4 Completeness along a ray

A ray carries its words as functions of the strength. Their relations are read in the module of the ray, and the closings are a solution that sees every one of them.

**Theorem 44.16** (Completeness along a ray)**.** *Let a ray have an integral origin and an integral direction in which every letter has a positive exponent, with $\kappa_{1,S} > 0$ for every nonempty proper $S$, and let $\mathfrak{M}$ be its module (Definition 39.1).*

*(i) An element $m \in \mathfrak{M}$ has the values $\langle m\rangle_k = 0$ for all large integers $k$ exactly when its closings vanish, $V_{\mathsf{C}}(m) = 0$, and then the continuation of $\langle m\rangle_k$ in $k$ vanishes identically. So every relation among the values of the words of the ray, as functions of the strength, is certified by the closings, a proved relation (Definition 44.1); the elements of $\mathfrak{M}$ with the values $0$ form the submodule $\mathrm{Ann}(V_{\mathsf{C}})$, which is $0$ exactly when the closings vanish on no nonzero submodule, as they do on every irreducible submodule on which they are not zero.*

*(ii) Let $m \in \mathfrak{M}$ be the class of a rational combination of insertions, let $d$ be the dimension of the submodule generated by $m$, and let $\sum_{i=0}^{d} q_i(k)\,\mathrm{Sh}^i m = 0$ with $q_i \in \mathbb{Q}[k]$, $q_0 q_d \neq 0$. Then the closings of $m$, which are rational numbers, satisfy*

$$\sum_{i=0}^{d} q_i(k)\, V_{\mathsf{C}}(m)(k+i) = 0$$

*at every integer $k \leq -j(m) - d$, and $m$ has the values $0$ along the ray exactly when $V_{\mathsf{C}}(m)$ vanishes at $d$ consecutive integers below every integer zero of $q_0$ and of $q_d$. In particular*

*whether two words of the ray have equal values along it is decided by finitely many rational numbers.*

So along a ray completeness holds outright, and the decision is exact. The values are moreover positive (Theorem 38.14(i)), so a linear form that is a value does not vanish, and consecutive rows of a recurrence are nondegenerate as soon as one square matrix of them is (Corollary 44.14); these are the inputs that the separation of numbers below takes from the ray.

## 44.5 Separating generators

A block is complete when the values of its representatives are independent over $\mathbb{Q}$ (Proposition 44.8). A separator proves the independence of all of them at once (Corollary 44.13). A separating generator proves it for one more value against values already separated, with an explicit measure, and the measures combine.

**Lemma 44.17** (A separating generator)**.** *Let $1, v_1, \dots, v_j \in \mathbb{R}$ be linearly independent over $\mathbb{Q}$ with the independence measure $\mu \geq 0$: there is $H_0$ such that $|c_0 + \sum_i c_i v_i| \geq H^{-\mu}$ for every integer vector $c \neq 0$ of height $H = \max_i |c_i| \geq H_0$. A* separating generator *of a number $v$ against them is a sequence of linear forms*

$$L_n = A_n v - \sum_{i=1}^{j} B_{n,i} v_i - B_{n,0}, \qquad n \geq n_0,$$

*with rational coefficients and integers $D_n \geq 1$ such that $D_n A_n$ and $D_n B_{n,i}$ are integers, for which $0 < |L_n| \leq e^{-Fn+o(n)}$, $\max(|A_n|, |B_{n,i}|) \leq e^{hn+o(n)}$ and $D_n \leq e^{dn+o(n)}$, and whose consecutive coefficient rows $(A_n, B_{n,0}, \dots, B_{n,j})$ are not proportional. If*

$$F > d + \mu\,(h + d),$$

*then $1, v_1, \dots, v_j, v$ are linearly independent over $\mathbb{Q}$, with the independence measure*

$$\mu' = \mu + \frac{(1+\mu)^2 (h+d)}{F - d - \mu(h+d)} + \varepsilon \qquad \text{for every } \varepsilon > 0.$$

**Theorem 44.18** (Separating generators in turn)**.** *Let a relation block (Definition 44.7) have the representatives $w_0, \dots, w_s$ with $\mathrm{ev}(w_0) = 1$, and let them be ordered so that each $\mathrm{ev}(w_j)$, $j \geq 1$, has a separating generator against $1, \mathrm{ev}(w_1), \dots, \mathrm{ev}(w_{j-1})$ with the rates $(F_j, h_j, d_j)$ of Lemma 44.17. Put $\mu_0 = 0$ and let $\mu_j$ be the measure that Lemma 44.17 gives at step $j$ from $\mu_{j-1}$. If $F_j > d_j + \mu_{j-1}(h_j + d_j)$ for every $j$, the block is complete, its values have the independence measure $\mu_s$, and every word of value zero in it has a finite derivation from the proved relations.*

The inputs of a separating generator are read on rays that carry the value: the rate $F$ from the native value of the direction (Proposition 38.9), the height $h$ from the frame of the law (Proposition 38.12), the clearing $d$ from the orders of the poles of the closing (Lemma 38.32), and the nonvanishing and the nondegeneracy from the ray itself (Theorem 44.16 and the paragraph after it).

# Chapter 45

# The content and its coaction

The content is the algebra of the trails of lattice inputs at rational frequencies, on rational cosets and at rational cuts, read modulo the proved relations. At the level $M$ it is spanned by the roots of unity of order $M$ times the trails of the counting lattice at the cut 0 whose frequencies have the denominator $M$; it is closed under products, it holds every trail with rational data at the level that its denominators fix, and it holds the differences of the index-one values of lattice inputs on rational cosets, so the sums along the lattices of paired words lie in it. Its values are the chains of level $M$: the iterated integrals from 0 to 1 whose letters are 0 and the roots of unity of order $M$, each a trail whose frequencies are the ratios of its consecutive nonzero letters; read between 0 and a root of unity in either order a chain is again a value of the content. The cutting rule keeps a subsequence of the letters of a chain and reads every run of the letters it drops between the two letters that bound it; on the symbols of chains it is a coassociative coaction, graded by the weight and compatible with the rotations and the reversal of chains. A live input has a letter too, its generating function, which satisfies a linear differential equation singular at the reciprocals of the roots of the symbol of its recurrence; the letter of a product of two live inputs continues across every point that is not a product of singular points of the letters of the factors, and the row of a hinge closing is the room's periods against the letters of the live inputs built on the room's fundamental solutions. Beyond the content a period is read through chains over the factors of its equation, whatever their monodromy, and along a ray those factors are generated down the hinge tree. What a period needs is read on its word along the canonical hinge tree: the prime nodes carry its content, the series and parallel nodes contribute stacked Gammas and powers of one mark's input, and a repetition adds nothing to the group of an input that the word reads on a scale, a class of any size needing no more than one mark's group; the word that reads every value of a ray through its stride's law needs the group of that input, a quotient of the group of the pencil, which on the class grows with the number of marks. Along the tree a period pays for each series or parallel node either in work, through the fibre of the node, or on one scale, through the transforms of the cells below it: paid in work it never reads more than one mark's input, and paid on scales its work does not grow with the multiplicities and its group is a quotient of the group of one mark's input, its dilations and the transforms of the distinct cells directly below a series or parallel node, born at the prime nodes and where the tree changes its kind.

## 45.1 The content

**Definition 45.1** (The content)**.** Let $M \geq 1$ be an integer and $\mu_M$ the group of the roots of unity of order dividing $M$. A *trail of level $M$* is the trail (Definition 5.23) at the cut 0 of a lattice input

on $\mathbb{Z}^d$ at frequencies in $\frac{1}{M}\mathbb{Z}$, with positive integer indices and no constant columns,

$$T = \big(\mathsf{T}_{\boldsymbol{\nu};I_d}[y]\; e^{2\pi i\langle \mathbf{a},y\rangle}\,\mathbb{I}_{\mathbb{Z}^d}\big)\big(\{0<y_1<\cdots<y_d\}\big), \qquad d\geq 1,\ \boldsymbol{\nu}\in\{1,2,\dots\}^d,\ \mathbf{a}\in\big(\tfrac{1}{M}\mathbb{Z}\big)^d,$$

with $e^{2\pi i a_d}\neq 1$ when $\nu_d=1$, read as the limit of its heads over $\{0<y_1<\cdots<y_d\leq N\}$ as $N\to\infty$, which exists and is the Trailmoment itself when $\nu_d\geq 2$ (Theorem 5.27); $|\boldsymbol{\nu}|$ is its *weight* and $d$ its *depth*. The *content of level* $M$, $\mathcal{C}^{(M)}$, is the $\mathbb{Q}$-span in $\mathfrak{W}/\mathfrak{I}$ (Definition 44.1) of the classes of the words $\zeta$ and $\zeta T$, for $\zeta\in\mu_M$, read as $e^{2\pi i r/M}$, and $T$ a trail of level $M$; $\mathcal{C}^{(M)}_w$ is the span of those with $T$ of weight at most $w$, and the *content* is $\mathcal{C}=\bigcup_M \mathcal{C}^{(M)}$.

The content is not the content $\mathcal{C}_M$ of a childcule $M$ in Definition 20.58, a $\mathcal{Z}$-module of values of periods; the level is written as an upper index to keep the two apart. The algebra $\mathcal{Z}$ of Definition 20.58 maps onto the subalgebra of $\mathcal{C}^{(1)}$ generated by its trails of depth one, the Trailmoments $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $k\geq 2$ (Proposition 45.2(i)).

**Proposition 45.2** (The content is an algebra)**.** *(i) $\mathcal{C}^{(M)}$ contains the classes of* 1 *and of every product of Trailmoments $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $k\geq 2$, so it contains the image of $\mathcal{Z}$ (Definition 20.58); and $\mathcal{C}^{(M)}\subseteq\mathcal{C}^{(M')}$ when $M\mid M'$.*

*(ii) Let $T$ and $T'$ be trails of level $M$ with the data $(\nu_i,a_i)_{i\leq d}$ and $(\nu'_j,a'_j)_{j\leq d'}$. For a surjection $\phi$ from the disjoint union $\{1,\dots,d\}\sqcup\{1',\dots,d'\}$ onto $\{1,\dots,e\}$, strictly increasing on each of the two parts, let $T_\phi$ be the trail of level $M$ of depth $e$ whose $r$-th point has the index $\sum_{\phi(x)=r}\nu_x$ and the frequency $\sum_{\phi(x)=r}a_x$. Then*

$$TT'=\sum_\phi T_\phi,$$

*the sum over these surjections, of depths between $\max(d,d')$ and $d+d'$, every $T_\phi$ of the weight $|\boldsymbol{\nu}|+|\boldsymbol{\nu}'|$.*

*(iii) So $\mathcal{C}^{(M)}$ is a $\mathbb{Q}$-algebra with $\mathcal{C}^{(M)}_w\,\mathcal{C}^{(M)}_{w'}\subseteq\mathcal{C}^{(M)}_{w+w'}$, and $\mathcal{C}$ is a $\mathbb{Q}$-algebra.*

**Proposition 45.3** (Cosets, cuts and index one)**.** *Let $M,M'\geq 1$ be integers.*

*(i) Let a trail have a lattice input on $\prod_{i=1}^d(b_i+\mathbb{Z})$ with $b_i\in\frac{1}{M}\mathbb{Z}$, at frequencies $a_i\in\frac{1}{M'}\mathbb{Z}$, with positive integer indices, no constant columns and a cut $N\in\frac{1}{M}\mathbb{Z}$, $N\geq 0$, and with $e^{2\pi i a_d}\neq 1$ when $\nu_d=1$, read as the limit of its heads. Then its class lies in $\mathcal{C}^{(MM')}$, and the class of every head of it lies in the span of the classes of $\mu_{MM'}$; its value is a combination, with coefficients in $\mathbb{Q}(\mu_{MM'})$, of* 1 *and the trails of level $MM'$ of depths at most $d$.*

*(ii) For $b,b'\in\frac{1}{M}\mathbb{Z}$ and cuts $N,N'\in\frac{1}{M}\mathbb{Z}$, $N,N'\geq 0$, the difference $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty))-(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b'+\mathbb{Z}})((N',\infty))$ of index-one values (Definition 5.2) has its class in $\mathcal{C}^{(M)}$.*

So the sum along a lattice of a paired word at integer data (Proposition 43.2), a combination of Trailmoments of lattice inputs on rational cosets at their cuts and differences of their index-one values with coefficients in $\mathbb{Q}[(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))]$, lies in the content. The values met on the centre rays of the lists of three and four marks are trails of depth one: at the frequency $\frac{1}{3}$ and the index 2, and on the coset $\frac{1}{2}$ at the index 3, which the coset relation returns to the counting input; Exercise 56.1 reads them.

## 45.2 Chains

**Definition 45.4** (Chains)**.** For distinct points $a_0, a_{w+1} \in \mathbb{C}$ and points $a_1, \dots, a_w$, $w \geq 0$, none on the open segment from $a_0$ to $a_{w+1}$, the *chain* is the iterated integral along that segment, $\gamma(t) = a_0 + t(a_{w+1} - a_0)$,

$$\mathrm{I}(a_0; a_1, \dots, a_w; a_{w+1}) = \int_{0<t_1<\cdots<t_w<1} \prod_{i=1}^{w} \frac{\gamma'(t_i)\,dt_i}{\gamma(t_i) - a_i},$$

with $\mathrm{I}(a_0; \varnothing; a_{w+1}) = 1$, wherever it converges. The $a_1, \dots, a_w$ are its *letters*, $a_0$ and $a_{w+1}$ its *ends* and $w$ its *weight*. It converges when its first letter differs from its first end and its last letter from its last end (Proposition 45.6(i), below). A chain is *of level $M$* when its ends and letters lie in $\{0\} \cup \mu_M$.

**Proposition 45.5** (The content as chains)**.** *(i) A trail $T$ of level $M$ with the data $(\nu_i, a_i)_{i \leq d}$ is*

$$T = (-1)^d \,\mathrm{I}\big(0;\ c_1^{-1}, 0^{\nu_1 - 1},\ \dots,\ c_d^{-1}, 0^{\nu_d - 1};\ 1\big), \qquad c_i = e^{2\pi i(a_i + \cdots + a_d)} \in \mu_M,$$

*a chain of level $M$ from $0$ to $1$ whose first letter is not $0$ and whose last letter is not $1$ (Theorem 5.27).*

*(ii) Conversely, let $\mathrm{I}(0; a_1, \dots, a_w; 1)$ be a chain of level $M$ with $a_1 \neq 0$ and $a_w \neq 1$, let $j_1 < \cdots < j_d$ be the positions of its nonzero letters, $j_{d+1} = w+1$, and put $c_i = a_{j_i}^{-1}$, $\nu_i = j_{i+1} - j_i$, $c_{d+1} = 1$. With frequencies $a'_i \in \frac{1}{M}\mathbb{Z}$ such that $e^{2\pi i a'_i} = c_i / c_{i+1}$, the chain is $(-1)^d$ times the trail of level $M$ with the data $(\nu_i, a'_i)_{i \leq d}$.*

*(iii) So the values of $\mathcal{C}^{(M)}$ form the $\mathbb{Q}(\mu_M)$-span of $1$ and of the chains of level $M$ from $0$ to $1$ with first letter not $0$ and last letter not $1$, and those of $\mathcal{C}_w^{(M)}$ the span of $1$ and of such chains of weight at most $w$.*

**Proposition 45.6** (Chains with an end at zero)**.** *(i) A chain whose first letter differs from its first end and whose last letter differs from its last end converges absolutely.*

*(ii) For $\lambda \in \mathbb{C}^\times$ and $\eta \in \mathbb{C}$, $\mathrm{I}(\lambda a_0 + \eta; \lambda a_1 + \eta, \dots, \lambda a_w + \eta; \lambda a_{w+1} + \eta) = \mathrm{I}(a_0; a_1, \dots, a_w; a_{w+1})$.*

*(iii) $\mathrm{I}(a_{w+1}; a_w, \dots, a_1; a_0) = (-1)^w \,\mathrm{I}(a_0; a_1, \dots, a_w; a_{w+1})$.*

*(iv) Every chain of level $M$ with one end $0$ and the other in $\mu_M$, whose first letter differs from its first end and whose last letter differs from its last end, is $\pm 1$ times a chain of level $M$ from $0$ to $1$ with first letter not $0$ and last letter not $1$, so its value lies in the values of $\mathcal{C}^{(M)}$.*

A chain between two singular points, two points of $\{0\} \cup \mu_M$, is the verb that the cutting rule uses on its right factors; with one end at 0 it is a value of the content, and on the symbols below it is read as it stands.

## 45.3 The cutting rule

**Definition 45.7** (Symbols of chains)**.** For an integer $M \geq 1$ the *symbols of level $M$* form the commutative $\mathbb{Q}(\mu_M)$-algebra $\mathfrak{S}_M$ of polynomials in the *formal chains* $\mathrm{I}^{\mathfrak{s}}(a_0; a_1, \dots, a_w; a_{w+1})$, $w \geq 1$, with $a_0, \dots, a_{w+1} \in \{0\} \cup \mu_M$ and $a_0 \neq a_{w+1}$, as independent variables. In $\mathfrak{S}_M$ one writes $\mathrm{I}^{\mathfrak{s}}(a; \varnothing; b) = 1$ for all $a, b$, and $\mathrm{I}^{\mathfrak{s}}(a; a_1, \dots, a_w; a) = 0$ for $w \geq 1$. The *weight* of $\mathrm{I}^{\mathfrak{s}}(a_0; a_1, \dots, a_w; a_{w+1})$ is $w$, and $\mathfrak{S}_M$ is graded by the weight, the weight of a product being

the sum. The *symbol* of a trail of level $M$ is $(-1)^d\,\mathrm{I}^{\mathfrak{s}}\big(0;c_1^{-1},0^{\nu_1-1},\dots,c_d^{-1},0^{\nu_d-1};1\big)$ (Proposition 45.5(i)), and the *realization* of a formal chain with one end 0, the other in $\mu_M$, first letter not its first end and last letter not its last end, is the chain with the same entries (Proposition 45.6(i)).

**Definition 45.8** (The cutting rule)**.** The *cutting rule* is the homomorphism of $\mathbb{Q}(\mu_M)$-algebras $\Delta:\mathfrak{S}_M\to\mathfrak{S}_M\otimes_{\mathbb{Q}(\mu_M)}\mathfrak{S}_M$ given on the formal chains by

$$\Delta\,\mathrm{I}^{\mathfrak{s}}(a_0;a_1,\dots,a_w;a_{w+1})=\sum_{S\subseteq\{1,\dots,w\}}\mathrm{I}^{\mathfrak{s}}(a_0;a_S;a_{w+1})\otimes\prod_{p=0}^{|S|}\mathrm{I}^{\mathfrak{s}}\big(a_{s_p};a_{s_p+1},\dots,a_{s_{p+1}-1};a_{s_{p+1}}\big),$$

where $s_1<\dots<s_{|S|}$ lists $S$, $s_0=0$, $s_{|S|+1}=w+1$ and $a_S=(a_{s_1},\dots,a_{s_{|S|}})$: the left factor keeps the letters of $S$ between the same ends, and each right factor is a run of dropped letters read between the two kept letters, or ends, that bound it. The *counit* $\varepsilon:\mathfrak{S}_M\to\mathbb{Q}(\mu_M)$ is the homomorphism that vanishes on the formal chains. For $\lambda\in\mu_M$ the *rotation* $\rho_\lambda$ multiplies every entry of every formal chain by $\lambda$, and the *reversal* $\varrho$ maps $\mathrm{I}^{\mathfrak{s}}(a_0;a_1,\dots,a_w;a_{w+1})$ to $(-1)^w\,\mathrm{I}^{\mathfrak{s}}(a_{w+1};a_w,\dots,a_1;a_0)$; both extend to automorphisms of the algebra.

**Theorem 45.9** (The cutting rule is a coaction)**.** *(i)* $\Delta$ *is coassociative and counital:* $(\Delta\otimes\mathrm{id})\circ\Delta=(\mathrm{id}\otimes\Delta)\circ\Delta$ *and* $(\varepsilon\otimes\mathrm{id})\circ\Delta=\mathrm{id}=(\mathrm{id}\otimes\varepsilon)\circ\Delta$.

*(ii)* $\Delta$ *preserves the weight: it maps a symbol of weight* $w$ *to a sum of tensors of weights* $w'$ *and* $w''$ *with* $w'+w''=w$*; and for every formal chain* $x$*,* $\Delta x-x\otimes1-1\otimes x$ *is a sum of tensors whose two factors have positive weights.*

*(iii)* $\Delta\circ\rho_\lambda=(\rho_\lambda\otimes\rho_\lambda)\circ\Delta$ *for* $\lambda\in\mu_M$*, and* $\Delta\circ\varrho=(\varrho\otimes\varrho)\circ\Delta$.

**Proposition 45.10** (The cutting rule on motivic chains)**.** *Let* $M\geq1$*. Every chain of level* $M$ *(Definition 45.4) between distinct points* $a_0,a_{w+1}$ *of* $\{0\}\cup\mu_M$*, read with tangential base points at an end that equals the adjacent letter, is the period of a motivic period* $\mathrm{I}^{\mathfrak{m}}(a_0;a_1,\dots,a_w;a_{w+1})\in\mathcal{P}^{\mathrm{mot}}$ *(Definition 34.5), the* motivic chain*, with* $\mathrm{I}^{\mathfrak{m}}(a;\varnothing;b)=1$*,* $\mathrm{I}^{\mathfrak{m}}(a;a_1,\dots,a_w;a)=0$ *for* $w\geq1$*, and* $\mathrm{I}^{\mathfrak{m}}(0;0;1)=\mathrm{I}^{\mathfrak{m}}(0;1;1)=0$*. Its coaction is the cutting rule (Definition 45.8) with the left factors read as motivic chains and the right factors as their de Rham periods:*

$$\Delta\,\mathrm{I}^{\mathfrak{m}}(a_0;a_1,\dots,a_w;a_{w+1})=\sum_{S\subseteq\{1,\dots,w\}}\mathrm{I}^{\mathfrak{m}}(a_0;a_S;a_{w+1})\otimes\prod_{p=0}^{|S|}\mathrm{I}^{\mathrm{dR}}\big(a_{s_p};a_{s_p+1},\dots,a_{s_{p+1}-1};a_{s_{p+1}}\big).$$

*The motivic chains satisfy the shuffle product, the composition of paths and the reversal of Proposition 45.6(iii), which the formal chains of Definition 45.7 do not; so the realization of the formal chains by the motivic chains intertwines the cutting rule with the coaction and is not injective.*

On the symbol of a trail of level $M$ the left factors are the symbols of the chains from 0 to 1 of the letters kept, up to sign the symbol of a trail of level $M$ whenever the kept letters begin with a letter other than 0 and end with one other than 1 (Proposition 45.5(ii)), and the right factors are runs of the trail's letters read between singular points, which at tier one are 0 and roots of unity of order $M$. Exercise 56.2 computes the rule on the symbols of the trails $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and of index 2 at the frequency $\frac{1}{3}$.

## 45.4 The letter of a live input

**Definition 45.11** (The letter of an input)**.** The *letter* of the input with density $\psi$ on the integers $n \geq n_0$ of the counting input, $n_0 \in \mathbb{N}$, is the power series $L_\psi(z) = \sum_{n \geq n_0} \psi(n)\, z^n$. At $z = e^{-u}$ it is the Laplace transform of the input beyond the cut $n_0 - 1$ (Definition 5.4), and where the series converges at $z = 1$ its value there is the Trailmoment at index zero $(\mathsf{T}_{0;(1)}[x]\, \psi\, \mathbb{I}_{\mathbb{Z}})((n_0 - 1, \infty)) = \sum_{n \geq n_0} \psi(n)$. The letter of the lattice input at the frequency $a$ on $\mathbb{Z}$ beyond the cut 0 is $e^{2\pi i a} z/(1 - e^{2\pi i a} z)$, singular at the letter $e^{-2\pi i a}$ of its chains (Proposition 45.5).

**Proposition 45.12** (The letter of a live input)**.** *Let $\psi$ satisfy $\sum_{i=0}^{h} p_i(n)\, \psi(n+i) = 0$ for the integers $n \geq n_0$, with $p_i \in \mathbb{C}[n]$, let $d$ be the largest degree of the $p_i$ and $\sigma(\lambda) = \sum_i c_i \lambda^i$ the symbol, $c_i$ the coefficient of $n^d$ in $p_i$; write $\theta = z\frac{d}{dz}$.*

*(i) As formal power series, $\sum_{i=0}^{h} z^{h-i} p_i(\theta - i)\, L_\psi = Q$ for a polynomial $Q$ of degree less than $n_0 + h$: a linear differential equation of order $d$ whose coefficient of $\frac{d^d}{dz^d}$ is $z^d \sum_i c_i z^{h-i} = z^{d+h}\sigma(1/z)$. Its singular points in $\mathbb{C}^\times$ are the reciprocals of the nonzero roots of $\sigma$.*

*(ii) Let $c_h \neq 0$, that is, $p_h$ of the largest degree, and let $\varrho$ be the largest modulus of a root of $\sigma$. Then for every $\eta > 0$, $|\psi(n)| \leq C_\eta(\varrho + \eta)^n$ for $n \geq n_0$; $L_\psi$ converges on $|z| < 1/\varrho$ and continues analytically along every path in $\mathbb{C}^\times$ that starts in that disc and avoids the reciprocals of the nonzero roots of $\sigma$, and to the plane cut along the rays $\{t/\lambda : t \geq 1\}$, $\lambda$ a nonzero root of $\sigma$.*

*(iii) The termwise product of two sequences that satisfy, for large $n$, recurrences of orders $h$ and $h'$ with polynomial coefficients and nonzero extreme coefficients satisfies, for large $n$, one of order at most $hh'$ with nonzero extreme coefficients.*

*(iv) Let $\psi$ and $\psi'$ be as in (ii), with symbols $\sigma$ and $\sigma'$. Then the letter of $\psi\psi'$ converges on $|z| < 1/(\varrho\varrho')$ and continues analytically to the plane cut along the rays $\{t/(\lambda\lambda') : t \geq 1\}$, $\lambda$ and $\lambda'$ nonzero roots of $\sigma$ and $\sigma'$.*

**Proposition 45.13** (The row of a hinge closing read by letters)**.** *In Proposition 43.10(ii) let the recurrences of the room have nonzero extreme coefficients and $p^\pm_{h_\pm}$ of the largest degree, with $\varrho^\pm$ the largest modulus of a root of their symbols, and let the coefficients $\Phi^\pm_n$ of the room periods $x^\pm_n$ satisfy, for large $n$, recurrences of the same kind, with $\varrho^\pm_\Phi$ the largest modulus of a root of their symbols. If $\varrho^\pm_\Phi \varrho^\pm < 1$, then:*

*(i) every sum $\Xi^\pm_i = \sum_n \Phi^\pm_n\, c^\pm_i(n)$ converges absolutely, it is the value $L_{\psi^\pm_i}(1)$ of the letter of the live input $\psi^\pm_i(n) = \Phi^\pm_n\, c^\pm_i(n)$, and*

$$P_F(\mathbf{m}; s, t) = \sum_{\pm} \sum_{i < h_\pm} x^\pm_i\, L_{\psi^\pm_i}(1) :$$

*the room's $h_+ + h_-$ initial periods against the letters of the live inputs built on its fundamental solutions;*

*(ii) each $L_{\psi^\pm_i}$ satisfies a linear differential equation with polynomial coefficients and continues analytically to the plane cut along the rays $\{t/(\lambda\lambda') : t \geq 1\}$, $\lambda$ a nonzero root of the symbol of the room's recurrence and $\lambda'$ one of the symbol of the coefficients' recurrence.*

At tier one the room's recurrence has order one, its fundamental solution is a power times a stacked Gamma (Proposition 38.11), and each lattice carries one letter; at higher tier the room's periods open into its $h$ fundamental solutions, and the letters of the row are singular only at products of the reciprocals of the roots of the two symbols.

## 45.5 Separation of the content

The words of the content carry motivic values, on which the cutting rule acts (Proposition 45.10). At the levels 1 to 4 the separation of motivic values is complete: the cuts of the cutting rule, the parts with one run, separate what they see by induction on the weight; what they do not see is, in each weight, at most one separating generator, a trail of depth one whose value is not zero, and the power of the circle; and the value reads these. So every word of the content of these levels has one normal form. The proved relations reach every motivic relation of a weight as soon as the cuts of their quotient have only the separating generators in their common kernel, and at the number level the content is complete as soon as the values of its polynomial generators are algebraically independent.

**Definition 45.14** (The motivic content; cuts; separating generators)**.** Let $M \in \{1,2,3,4\}$.

(i) The *motivic value* $w^{\mathfrak{m}}$ of a rational combination $w$ of trails of level $M$ is the same combination of the motivic chains of their symbols (Definition 45.7, Proposition 45.5(i)), motivic chains from 0 to 1 whose first letter is not 0 and whose last letter is not 1. The *motivic content* $\mathcal{P}^{(M)}$ is the $\mathbb{Q}$-span in $\mathcal{P}^{\mathrm{mot}}$ of 1 and of these motivic values, an algebra graded by the weight, the number of letters; $\mathcal{P}^{(M)}[N]$ is its part of weight $N$.

(ii) $\mathbb{L}$ is the motivic period of the circle, with the period $2\pi i$. At the levels $3,4$ it is $\mathbb{L} = \big(\frac{1}{2} - \frac{1}{M}\big)^{-1}\big((\mathsf{T}_{1;(1)}[x]\,2i\sin(\frac{2\pi x}{M})\,\mathbb{I}_{\mathbb{Z}})((0,\infty))\big)^{\mathfrak{m}}$, and at the levels $1,2$ its square is $\mathbb{L}^2 = -24\,\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))\big)^{\mathfrak{m}}$, with the trails of depth one of (iv). Put $e = 1$ at the levels $3,4$ and $e = 2$ at the levels $1,2$, so that $\mathbb{L}^e \in \mathcal{P}^{(M)}$, and

$$\mathfrak{n}^{(M)}[r] = \mathcal{P}^{(M)}[r]\Big/\Big(\mathbb{L}^e\,\mathcal{P}^{(M)}[r-e] + \sum_{0<i<r} \mathcal{P}^{(M)}[i]\,\mathcal{P}^{(M)}[r-i]\Big),$$

the classes of weight $r$ modulo the circle and modulo products of elements of positive weight.

(iii) For $1 \le r < N$ the *cut* $\mathsf{D}_r : \mathcal{P}^{(M)}[N] \to \mathfrak{n}^{(M)}[r] \otimes \mathcal{P}^{(M)}[N-r]$ is the part of the coaction with exactly one run, of $r$ letters. The run is read in $\mathfrak{n}^{(M)}[r]$ through its motivic chain, written by the composition of paths, the reversal, the rotations of $\{0\} \cup \mu_M$ and the regularization as a polynomial in $\mathbb{L}$ and in motivic values of trails, and read modulo $\mathbb{L}$; the kept letters form a motivic chain from 0 to 1, regularized at its ends:

$$\begin{aligned}&\mathsf{D}_r\,\mathrm{I}^{\mathfrak{m}}(a_0;a_1,\dots,a_N;a_{N+1})\\ &\qquad = \sum_{p=0}^{N-r} \big[\mathrm{I}^{\mathfrak{m}}(a_p;a_{p+1},\dots,a_{p+r};a_{p+r+1})\big] \otimes \mathrm{I}^{\mathfrak{m}}(a_0;a_1,\dots,a_p,a_{p+r+1},\dots,a_N;a_{N+1}).\end{aligned}$$

(iv) For $N \ge 1$ and $a \in \frac{1}{M}\mathbb{Z}$ the trail of level $M$ of depth one at the frequency $a$ is $(\mathsf{T}_{N;(1)}[x]\,e^{2\pi i a x}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \sum_{y\ge1} e^{2\pi i a y} y^{-N}$, read as the limit of its heads at $N = 1$. The *separating generator* of weight $N$ is the trail

$$\mathsf{w}_N = \begin{cases} (\mathsf{T}_{N;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)), & N \ge 3 \text{ odd, every } M;\\ -(\mathsf{T}_{1;(1)}[x]\,e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \log 2, & N = 1,\ M = 2,4;\\ -(\mathsf{T}_{1;(1)}[x]\,2\cos(\frac{2\pi x}{3})\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \log 3, & N = 1,\ M = 3;\\ (\mathsf{T}_{N;(1)}[x]\,2i\sin(\frac{2\pi x}{M})\,\mathbb{I}_{\mathbb{Z}})((0,\infty)), & N \ge 2 \text{ even},\ M = 3,4;\end{cases}$$

there is none in even weight at the levels $1,2$, and none in weight one at level 1.

**Theorem 45.15** (The content is separated by its cuts and its value)**.** *Let $M \in \{1,2,3,4\}$ and $N \geq 1$. Let $\mathcal{S}^{(M)}$ be the shuffle algebra over $\mathbb{Q}$ on one letter $x_N$ for each separating generator $\mathsf{w}_N$, with the coproduct that cuts a word into its two ends, tensored with $\mathbb{Q}[\mathbb{L}^e]$.*

*(i)* What the cuts see. *An element $\xi \in \mathcal{P}^{(M)}[N]$ whose cuts $\mathsf{D}_r\xi$, $1 \leq r < N$, all vanish is*

$$\xi = c\,\mathsf{w}_N^{\mathfrak{m}} + b\,\mathbb{L}^N, \qquad c, b \in \mathbb{Q},$$

*the first term absent in a weight without a separating generator and the second when $e \nmid N$; the pair $(c, b)$ is the* readout *of $\xi$. The cuts of an element of weight $N$ have their second factors in the weights below $N$.*

*(ii)* The separating generators. *Every cut of $\mathsf{w}_N^{\mathfrak{m}}$ vanishes. For $N$ odd its value is positive: $(\mathsf{T}_{N;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\log 2$ or $\log 3$. For $N$ even it is*

$$\begin{aligned} i\sqrt{3}\,\big((\mathsf{T}_{N;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{N;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\big) \qquad (M=3),\\ 2i\,\big((\mathsf{T}_{N;(1)}[x]\,\mathbb{I}_{1+4\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{N;(1)}[x]\,\mathbb{I}_{3+4\mathbb{Z}})((0,\infty))\big) \qquad (M=4), \end{aligned}$$

*$i$ times a positive number. The value of $\mathbb{L}^N$ is $(2\pi i)^N$, imaginary for $N$ odd and real for $N$ even. So the period is injective on the span of $\mathsf{w}_N^{\mathfrak{m}}$ and $\mathbb{L}^N$, and the readout of $\xi$ is read off the real and the imaginary part of its value.*

*(iii)* Combined. *An element of $\mathcal{P}^{(M)}[N]$ is $0$ exactly when its cuts and its value vanish. With $\mathsf{w}_N^{\mathfrak{m}}$ for the letter $x_N$, the cuts, applied recursively, and the readouts give an injective homomorphism of algebras and comodules $\mathcal{P}^{(M)} \to \mathcal{S}^{(M)}$. The image of the motivic value of a content word is its* normal form*, and two content words have equal motivic values exactly when their normal forms are equal.*

*(iv)* The content is free. *This homomorphism is onto, and $\mathcal{P}^{(M)}$ is the polynomial algebra over $\mathbb{Q}$ on $\mathbb{L}^e$ and on one element for each Lyndon word in the letters $x_N$, a word that is smaller than each of its proper rotations in the order of the weights.*

**Corollary 45.16** (The values of the content and its generators)**.** *Let $M \in \{1,2,3,4\}$ and let $g_1, g_2, \ldots$ be content words of level $M$ whose motivic values, with $\mathbb{L}^e$, are polynomial generators of $\mathcal{P}^{(M)}$ (Theorem 45.15(iv)). If the values of the $g_i$ and $(2\pi i)^e$ are algebraically independent over $\mathbb{Q}$, every relation among the values of the content words of level $M$ holds among their motivic values, and so is decided by their cuts and readouts (Theorem 45.15(iii)). A polynomial relation over $\mathbb{Q}$ among the values of the $g_i$ and $(2\pi i)^e$ is a relation among values that does not hold among motivic values.*

**Definition 45.17** (The separation procedure)**.** Data: a level $M \in \{1,2,3,4\}$, a weight $N \geq 2$, for each $k \leq N$ a set $R_k$ of proved relations among the trails of level $M$ of weight $k$ that hold for their motivic values, and a set $\mathcal{R} \subseteq \{1, \ldots, N-1\}$ of lengths of runs.

(1) For $k \leq N$ let $V_k$ be the rational combinations of the trails of level $M$ of weight $k$ modulo $R_k$, and $\mathfrak{v}_k \subseteq V_k$ the span of the classes of the products of trails of positive weights (Proposition 45.2(ii)) and of the trails whose values are rational multiples of $(2\pi i)^e$ when $k = e$. Reduce by exact row reduction.

(2) For each trail of weight $N$ and each $r \in \mathcal{R}$ compute $\mathsf{D}_r$ on its chain (Proposition 45.5(i)) by Definition 45.14(iii): write each run of $r$ letters, modulo $\mathbb{L}$, and each chain of the kept letters as a combination of chains of trails, and read the run in $V_r/\mathfrak{v}_r$ and the kept chain in $V_{N-r}$.

(3) Check that the cuts of step (2) vanish on $R_N$, so that they are linear maps on $V_N$.

(4) Compute the joint kernel of these cuts on $V_N$, and check that it is spanned by classes of trails whose values are linearly independent over $\mathbb{Q}$.

**Theorem 45.18** (Completeness by proved relations, weight by weight)**.** *In Definition 45.17, if $V_k \to \mathcal{P}^{(M)}[k]$ is injective for every $k < N$ and the checks of steps (3) and (4) succeed, then $V_N \to \mathcal{P}^{(M)}[N]$ is injective: every relation among the motivic values of the trails of level $M$ of weight $N$ is a consequence of $R_N$.*

At level 1, with the stuffle product, the shuffle product of chains and the relation of index one, and with $\mathcal{R}$ the odd lengths from 3 to $N-2$, the procedure succeeds in every weight from 2 to 14, the joint kernel being the line of $(\mathsf{T}_{N;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$; Part XI works it, with the polynomial generators it finds.

## 45.6 The closing in any number of shares

The closing of a period finds a law on the integers in any number of shares: the period is the total mass of a live input whose density is rational and nonnegative, whose letter satisfies a linear differential equation over $\mathbb{Q}(z)$, at a tail rate read off the walls; the share closing of the word in all its free shares at once is the expansion of a deformation of the period at a regular singular point, with frames times polynomials in cumulants as its terms; and whatever the monodromy of that letter, the period is read through chains over its equation, whose kernels come from the factors of the equation; when the monodromy is unipotent every factor is trivial and the chains are those of Definition 45.4. Along a ray the equation is that of the pencil, whose factors the hinge tree generates, every period at rational data is a moment of the pencil of one integral ray, and the equation of the letter is generated down the same tree, each vertex carrying the load sum of its marks.

The letters of this section satisfy linear differential equations, and a solution of such an equation is read along a path through the factors of the equation, one variation of the constants for each step of a composition series.

**Definition 45.19** (Chains over an equation)**.** Let $\Bbbk \subseteq \mathbb{C}$ be a field, $K = \Bbbk(z)$, and $L$ a linear differential operator of order $h$ with coefficients in $K$, whose singular points in $\mathbb{P}^1$ are regular singular; $S$ is the set of them together with $\infty$. The *factors* of $L$ are the composition factors over $K$ of the module $K\langle\partial_z\rangle/K\langle\partial_z\rangle L$, each the module of an operator irreducible over $K$, and the dual of a factor is the module of the adjoint of that operator. On a simply connected domain avoiding $S$, a *kernel* of $L$ is a form $\phi\,\psi\,dz$, with $\psi$ a combination with coefficients in $K$ of a solution of one factor and of its derivatives, and $\phi$ the same for the dual of a factor. A *chain over $L$* along a path $\gamma : [0,1] \to \mathbb{C} \setminus S$ is

$$\mathrm{I}_\gamma(\omega_1,\dots,\omega_w) = \int_{0<t_1<\cdots<t_w<1} \prod_{i=1}^{w} \omega_i\big(\gamma(t_i)\big),$$

with the kernels $\omega_i$ continued along $\gamma$ and $\mathrm{I}_\gamma(\varnothing) = 1$; its *weight* is $w$, and $\mathrm{I}_\gamma(\cdot)(z)$ is the chain along the part of $\gamma$ from $\gamma(0)$ to $z = \gamma(\tau)$.

When every factor has rank one and trivial monodromy, its solutions are rational functions, being single-valued with moderate growth at the points of $S$, those of its dual are their reciprocals, and the kernels are rational forms; the chains of Definition 45.4 are the chains over such an equation along a segment whose kernels are $dz/(z-s)$, $s$ a finite point of $S$.

**Proposition 45.20** (Every solution is read through chains over its equation)**.** *In Definition 45.19 let $y$ be a solution of $L$ near a point $a \notin S$, continued along a path $\gamma$ from $a$ to $b$ that avoids $S$ except possibly at $b$. There are finitely many $c_i$, each a combination with coefficients in $K$ of a solution of a factor and of its derivatives, and chains $\mathrm{I}_i$ over $L$ along $\gamma$, such that for every $z$ on $\gamma$ before $b$*

$$y(z) = \sum_i c_i(z)\, \mathrm{I}_i(z)\, \beta_i,$$

*with each $\beta_i$ a combination of $y(a), y'(a), \dots, y^{(h-1)}(a)$ whose coefficients are the values at $a$ of elements of $K$ regular off $S$. Every $c_i\mathrm{I}_i$ has at $b$ an expansion in the powers $(z-b)^\rho$, $\rho \in \mathbb{C}$, and in the powers of $\log(z-b)$, times convergent power series, for one branch of $\log(z-b)$ on the end of $\gamma$, the same for every $i$. If $b \in S$, $\gamma$ ends with a segment, and $y(z)$ has a limit at $b$ along it, that limit is $\sum_i [c_i\mathrm{I}_i]_b\, \beta_i$, with $[\cdot]_b$ the constant term of the expansion at $b$.*

**Theorem 45.21** (The closing in any number of shares)**.** *Let a family $F$ on $\mathcal{E}$, $|\mathcal{E}| = n$, have rational weights, rational masses $\mathbf{m}$ and rational exponents $s_j, t_j \geq 0$, on the balance wall with every proper wall positive. Write its active letters as $L_i$, with the exponents $w_i > 0$ and the degrees $g_i$, take rational references $M_i \geq \max_{\Delta_\mathcal{E}} L_i$, such as $M_i = \max_T \theta_{j,T}/g_i!$ for the table $j$ of $L_i$ (Theorem 15.13), and put $P_0 = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_\mathcal{E} \end{bmatrix} \prod_i M_i^{-w_i}$.*

*(i)* The law the closing finds. *$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = P_0 \sum_{N \geq 0} \psi(N)$, where*

$$\psi(N) = \sum_{|\mathbf{n}| = N} \prod_i \frac{(w_i)_{n_i}}{n_i!\, M_i^{n_i}}\ \mathbb{E}_{U \sim \mathrm{Dir}_\mathbf{m}}\Big[ \prod_i \big(M_i - L_i(U)\big)^{n_i} \Big]\ \in \mathbb{Q}_{\geq 0}.$$

*The letter $L_\psi(z) = \sum_N \psi(N) z^N$ (Definition 45.11) converges on $|z| < 1$, where*

$$P_0\, L_\psi(z) = \int_{\Delta_\mathcal{E}} u^{\mathbf{m}-\mathbf{1}} \prod_i \big((1-z)M_i + z\, L_i(u)\big)^{-w_i}\, d\omega_\mathcal{E}(u),$$

*and it satisfies a nonzero linear differential equation with coefficients in $\mathbb{Q}[z]$. So $\psi$ satisfies a recurrence $\sum_{l=0}^h p_l(N)\psi(N+l) = 0$ with $p_l \in \mathbb{Q}[N]$ and $p_0 p_h \neq 0$, $\psi\, \mathbb{I}_\mathbb{Z}$ is a live input, and the period is $P_0$ times its read at index zero, its total mass. The tail is $\sum_{N' > N} \psi(N') = \Theta\big(N^{-\varpi}(\log N)^{k-1}\big)$, with the rate $\varpi > 0$ and the order $k$ of Theorem 15.13(ii).*

*(ii)* The share closing. *Open every active letter with one share per basis, on the fibre of Corollary 18.16, of a dimension $D$, and deform the weights to $\theta_{j,T} z^{c_{j,T}}$ in $K_j$ and $\theta_{j,T} z^{c'_{j,T}}$ in $\Psi_j$, with integers $(c, c')$ off finitely many hyperplanes. The deformed period $P_c(z)$ is real analytic on $(0, \infty)$, satisfies a nonzero linear differential equation with coefficients in $\mathbb{Q}[z]$ with at most a regular singular point at 0, and $P_c(1) = P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$. There is $z_0 > 0$ such that for $0 < z < z_0$ the $D$ successive closings of the word converge absolutely and give*

$$P_c(z) = \sum_\sigma \sum_{\mathbf{j} \in \mathbb{N}^D} z^{e_\sigma(\mathbf{j})} \sum_{i \geq 0} (\log z)^i\, F_{\sigma,i}(\mathbf{j}),$$

*a finite sum over the sequences $\sigma$ of the Gamma families crossed, each $e_\sigma$ affine with positive slopes, and each $F_{\sigma,i}(\mathbf{j})$ a* frame*, a product of Gamma values and reciprocal factorials at points affine in $\mathbf{j}$, times a polynomial in the cumulants $(l-1)!\, (\mathsf{T}_{l;(1)}[x]\, \mathbb{I}_{b+\mathbb{Z}})((b, \infty))$ of Proposition 5.13 at rational $b$. The period is the continuation of this expansion along $(0, 1]$. The exponents lie in finitely many classes $\rho + \mathbb{N}$, and in each class the coefficients of $z^{\rho+m}(\log z)^i$ are live inputs in $m$. If $\sum_\mathbf{j} |F_{\sigma,i}(\mathbf{j})| < \infty$ for every $\sigma$ and $i$, the period is $\sum_\sigma \sum_\mathbf{j} F_{\sigma,0}(\mathbf{j})$.*

*(iii)* Chains over the letter. *Let $L$ be the equation over $\mathbb{Q}(z)$ of the letter $L_\psi$ of (i), or of $P_c$ of (ii), with $S$ as in Definition 45.19, and let $a \in (0,1)$, not in $S$, be joined to $1$ by a path $\gamma$ that avoids $S$ except possibly at $1$ and ends with a segment, in the disc $|z| < 1$ for $L_\psi$ and near $(0,1]$ for $P_c$. Then $L_\psi(1)$, and $P_c(1)$, read by Proposition 45.20, are finite sums of terms $[c_i\,\mathrm{I}_i]_1\,\beta_i$, with $c_i$ combinations over $\mathbb{Q}(z)$ of a solution of a factor of $L$ and of its derivatives, $\mathrm{I}_i$ chains over $L$ along $\gamma$, $[\cdot]_1$ the value at $1$ or, when $1 \in S$, the constant term of the expansion there, and $\beta_i$ combinations, with the values at $a$ of rational functions regular off $S$ as coefficients, of the values at $a$ of the letter, or of $P_c$, and of its derivatives of order below that of $L$. For $L_\psi$ these values are the series $\sum_N \psi(N)\frac{N!}{(N-i)!}a^{N-i}$, of rational terms when $a$ is rational; for $P_c$ and $a < z_0$ they are absolutely convergent sums of frames times cumulants, times powers of $a$ and of $\log a$, the expansion of (ii) and its derivatives at $a$. The period is $P_0 L_\psi(1) = P_c(1)$. If the monodromy of the solutions of $L$ is unipotent, every factor of $L$ is $\mathbb{Q}(z)$ with the derivation $d/dz$; if moreover $S$ meets the open segment from $0$ to $1$ nowhere, $L_\psi(1)$ is a combination with algebraic coefficients of $1$ and of chains (Definition 45.4) from $0$ to $1$ whose letters lie in $\{0\} \cup (S \setminus \{\infty\})$, regularized at the ends, and $P_c(1)$ is such a combination whose coefficients are combinations, with algebraic coefficients, of finitely many of the $F_{\sigma,i}(\mathbf{j})$. When $S \subseteq \{0,\infty\} \cup \mu_M$ these chains are of level $M$, and for the equation of $L_\psi$ the period is $P_0$ times an element of the span of $\mathcal{C}^{(M)}$ over the algebraic numbers.*

So the closing of every period, in any number of shares, finds a law on the integers: the period is the total mass of a live input with a rational nonnegative density, at a known rate, and the share closing in all the free shares at once is the expansion of one holonomic deformation at a corner, whose terms are frames times cumulants. The letter of the closing in (i) is singular at the points $M_i/(M_i - L_i^*)$ of the critical values $L_i^*$ of the letters on the strata of the simplex; for one free share the closing is Theorem 43.1. Whatever the monodromy of the letter, (iii) reads the period through the factors of its equation, with the values at an inner point as its data, power series with rational coefficients for the letter and sums of frames and cumulants for the deformation: the chains of the content are the case in which every factor is trivial. With a common hinge tree of the tables the factors of the equation of the letter are generated down the tree (Corollary 45.23), and the same period is a moment of the pencil of one integral ray, whose factors the tree generates as well (Corollary 45.22).

**Corollary 45.22** (The values of a ray are read through chains over its tree)**.** *Let a ray of a family whose tables have a common hinge tree have an integral origin and $\kappa_{1,S} > 0$ for every nonempty proper $S$, and let $L_1$ be the equation of $1$ over $\mathbb{C}(t)$, the monic generator of the annihilator of $1$ in $\mathbb{C}(t)\langle\partial_t\rangle$ (Theorem 39.49(i)).*

*(i)* The factors are generated down the tree. *The factors of $L_1$ are the composition factors of $\mathcal{D}_F[1]$ that are not supported at points, read over $\mathbb{C}(t)$. Each is a composition factor of $H^0(R_o|_{U_o})_+\big((\boxtimes_c S_c)|_{U_o}\big)$ for composition factors $S_c$ of the modules of the children of the root, and so on down the tree to the marks (Theorem 39.56(iv)).*

*(ii)* The composition series follow the tree. *For every vertex $v$, filtrations of the modules $\mathbb{G}_c$ of its children by submodules give $\mathbb{G}_v$ the filtration whose term $p$ is the image of the direct image under $N_v$ of the sum of the external products of terms with indices adding to $p$, and its graded piece $p$ is a quotient of the direct image $H^0(N_v)_+$ of the sum of the $\boxtimes_c \mathrm{gr}_{p_c}\mathbb{G}_c$ over $\sum_c p_c = p$. At the root this filters $\mathcal{G}_F$, and $\mathcal{D}_F[1]$ by intersection.*

*(iii)* Chains over the pencil. *Every solution of $L_1$, continued along a path that avoids its singular points, is read through chains over $L_1$ along that path (Proposition 45.20); among these*

*solutions are the orthant fibre integral $P_1$ on $(A,\infty)$ and the torus fibre integral $T$ near $\infty$. Every value of the ray is a moment of $P_1$: $\langle 1\rangle_k = \int_A^\infty t^{-k}P_1(t)\,dt$ wherever the value converges.*

*(iv)* Every period at rational data is a moment of a ray. *For rational data $(\mathbf{m};\mathbf{s},\mathbf{t})$ of the family on the balance wall, with nonnegative exponents and every proper wall positive, and an integer $N\geq 1$ with $N(\mathbf{m};\mathbf{s},\mathbf{t})$ integral, the ray with the origin $0$ and the direction $N(\mathbf{m};\mathbf{s},\mathbf{t})$ is such a ray, and*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_A^\infty t^{-1/N}P_1(t)\,dt$$

*for its orthant fibre integral $P_1$.*

The letter of the closing reads one quantity that the pencil does not: the total load, through the reference term $M_i|u|^{g_i}$ of its deformation. A vertex carries it as the load sum of its marks, which adds along the tree, and a rational mass is a module of rank one at its mark.

**Corollary 45.23** (The letter of the closing along the tree)**.** *In Theorem 45.21 let the tables of $F$ have a common hinge tree, and let $L$ be the equation of $L_\psi$ over $\mathbb{Q}(z)$, the monic generator of its annihilator. Give every vertex $v\neq o$ the data $\tilde\lambda_v=(\lambda_v,\sigma_v)$: its data in Definition 39.55, with the last coordinate of the data of a mark $e$ taken to be $\alpha_e$, so that the last coordinate $\mu_v$ is the product of the loads of the marks $B_v$ below $v$, and their sum $\sigma_v=\sum_{e\in B_v}\alpha_e$, the node maps $\tilde N_v$ extended by $\sigma_v=\sum_c\sigma_c$. Put $\mathbb{L}_e=H^0(\tilde\lambda_e)_+\mathcal{O}_{\mathbb{C}^\times}\alpha_e^{m_e}$ at a mark, the direct image of the module of rank one of $\alpha_e^{m_e}$, and $\mathbb{L}_v=H^0(\tilde N_v)_+\big(\boxtimes_c\mathbb{L}_c\big)$ at a vertex $v\neq o$. On the data of the children of $o$ read $K_j=\mathcal{H}^{\hom}_{o,j}(\mathbf{c},\mathbf{d})$, $\Psi_j=\prod_c\mu_c\,\mathcal{H}^{\hom}_{o,j,\Psi}(\mathbf{c}',\mathbf{d}')$ and $\sigma_o=\sum_c\sigma_c$, and let $\mathcal{K}$ be the module of rank one of $\prod_i\big((1-z)M_i\sigma_o^{g_i}+zL_i\big)^{-w_i}$ on the open set $W$ of the product of their data with the $z$-line on which every deformed letter is nonzero.*

*(i)* The letter is an iterated direct image. $\mathbb{L}_v=H^0(\tilde\lambda_v)_+\big(\boxtimes_{e\in B_v}\mathcal{O}_{\mathbb{C}^\times}\alpha_e^{m_e}\big)$, *and $P_0L_\psi$ is a period of $H^0(\mathrm{pr}_z)_+\big(\mathcal{K}\otimes(\boxtimes_c\mathbb{L}_c\boxtimes\mathcal{O}_{\mathbb{C}})|_W\big)$, the scale of the torus contributing one factor in the top degree.*

*(ii)* Its factors are generated down the tree. *Every factor of $L$, read over $\mathbb{C}(z)$, has its composition factors among the generic fibres of the composition factors of $H^0(\mathrm{pr}_z)_+\big(\mathcal{K}\otimes(\boxtimes_cS_c\boxtimes\mathcal{O}_{\mathbb{C}})|_W\big)$, for composition factors $S_c$ of the $\mathbb{L}_c$ of the children of the root; every composition factor of $\mathbb{L}_v$ is one of $H^0(\tilde N_v)_+\big(\boxtimes_c S_c\big)$ for composition factors $S_c$ of the modules of its children; the $\mathbb{L}_e$ are simple; and the filtrations of Corollary 45.22(ii) hold for the $\mathbb{L}_v$, with $H^0(\mathrm{pr}_z)_+(\mathcal{K}\otimes\cdot\,)$ at the root.*

So every period of a family whose tables have a common hinge tree, at rational data on the balance wall with nonnegative exponents and positive walls, is read down the tree in two ways: as the moment $\int_A^\infty t^{-1/N}P_1(t)\,dt$ of a solution of the pencil of one integral ray, with $N=1$ at integral data, and as $P_0$ times the value at 1 of the letter of the closing. In both, the solution is read through chains over an equation whose factors are generated from the marks along the hinge tree and whose composition series follow the tree, as the nesting of the chains does.

## 45.7 The group of a period

A period is a closed serif word (Proposition 18.20), and what it needs is read on its word. Down the canonical hinge tree the word of a table is, at its full leans, a product of stacked Gammas, monomials in the weights and the periods of its prime nodes, and on its centre ray, at every strength,

the word of its core with the word of each cell set at the load of its marker (Corollary 26.11, Proposition 42.26): the content of a period sits in its prime nodes, and its series and parallel nodes and its repetitions add stacked Gammas, powers and the binary lengths of multiplicities (Chapter 33). Where the word reads an input on a scale it needs what that input generates, and the group of the input counts it: its dimension is the transcendence degree of the functions the input generates, a power of a function adds nothing to it, and functions whose groups have simple derived algebras, pairwise not isomorphic, add those algebras. Every ray also has one word for all of its values: one Trailmoment of one input, the reciprocal image of its stride's law, read at the index of the strength (Proposition 40.2(i)). In that word the Gamma stacks of the tree are divided out of the input and a product of Trailmoments becomes a multiplicative convolution of inputs (Theorem 5.31(ii)), and the group of the stride's law, a quotient of the group of the pencil, is what that word needs. On the class of $s$ marks the tree's word reads one mark's input raised to the $s$-th power, whose group is the image of one mark's, of dimension three for every $s$; the stride's law, weighted by $t$, is that input with one mark's input divided out, and from $s = 5$ on its module is not a power and its group has dimension at least $(s-1)(s-2)/2$.

**Definition 45.24** (The group of a module, of a function and of an input)**.** Let $N$ be a differential module of finite dimension over $\mathbb{C}(y)$, and $\operatorname{Sol}_x N$ the space of its solutions near a point $x$ off its singular points. The *group* $G_N \subseteq \mathrm{GL}(\operatorname{Sol}_x N)$ of $N$ is the group of the automorphisms of the field generated over $\mathbb{C}(y)$ by these solutions and their derivatives that fix $\mathbb{C}(y)$ and commute with $d/dy$, acting on $\operatorname{Sol}_x N$; when the singular points of $N$ in $\mathbb{P}^1$ are regular singular, it is the Zariski closure of the monodromy of the solutions. Its Lie algebra is $\mathfrak{g}_N$, the Lie algebra of its unipotent radical is the *unipotent part* $\mathfrak{u}_N$, and $\mathfrak{r}_N = \mathfrak{g}_N/\mathfrak{u}_N$ is the *reductive part*; another point $x$ conjugates all three. The group of a function $W$ that satisfies a nonzero linear differential equation over $\mathbb{C}(y)$ is the group of the module $\mathbb{C}(y)\langle\partial_y\rangle W$ that it generates, and the group of an input $W\,\mathbb{I}_D$ on an interval $D \subseteq (0,\infty)$, $W$ such a function, is the group of $W$.

**Proposition 45.25** (The group counts the functions)**.** *Let $N$ be as in Definition 45.24, with a composition series of length $r$ and the factors $S_1, \dots, S_r$.*

*(i) The solutions of $N$ near $x$ and their derivatives generate over $\mathbb{C}(y)$ a field of transcendence degree* $\dim \mathfrak{g}_N$.

*(ii) If $N$ is simple, $G_N$ is reductive and acts irreducibly, and $\mathfrak{u}_N = 0$.*

*(iii) The unipotent radical of $G_N$ is the subgroup that acts trivially on the graded pieces of the flag of solutions that the series gives, and $G_N$ modulo it is the group of $S_1 \oplus \cdots \oplus S_r$. So $\mathfrak{r}_N$ is the Lie algebra of the group of $S_1 \oplus \cdots \oplus S_r$, and $\mathfrak{u}_N$ is nilpotent of class at most $r-1$.*

*(iv) When the singular points of $N$ are regular singular, the chains of Proposition 45.20 have weight at most $r-1$, $r$ the length of the series used there.*

*(v) For rational $a$ and $W$ as in Definition 45.24, the group of $y^a W$ has the identity component of the group of $W$, the solutions of the two modules identified by the multiplication by $y^a$; so the two have the same Lie algebra.*

So the group of a function counts what the function generates, nested at most one less than the length of a composition series, and a rational power of the variable, as in the moment $\int_A^\infty t^{-1/N} P_1(t)\,dt$ of Corollary 45.22(iv), changes nothing in its Lie algebra.

**Proposition 45.26** (Powers add nothing, distinct factors add)**.** *Let $N$, $N_1$, $N_2$ be as in Definition 45.24.*

*(i) Every module obtained from $N$ by sums, tensor products, duals and subquotients, such as $\mathrm{Sym}^k N$, $N^{\otimes k}$, $\wedge^k N$ and $N^\vee$, has as its group a quotient of $G_N$, and as its Lie algebra a quotient of $\mathfrak{g}_N$.*

*(ii) The Lie algebra of the group of $N_1 \oplus N_2$ is a subalgebra of $\mathfrak{g}_{N_1} \oplus \mathfrak{g}_{N_2}$ that projects onto each, so its dimension lies between $\max_i \dim \mathfrak{g}_{N_i}$ and $\dim \mathfrak{g}_{N_1} + \dim \mathfrak{g}_{N_2}$.*

*(iii) If $S_1, \dots, S_k$ are simple and the derived algebras $\mathfrak{s}_j = [\mathfrak{g}_{S_j}, \mathfrak{g}_{S_j}]$ are simple and pairwise not isomorphic, the derived algebra of the Lie algebra of the group of $S_1 \oplus \cdots \oplus S_k$ is $\mathfrak{s}_1 \oplus \cdots \oplus \mathfrak{s}_k$.*

So the rank of a module, the order of its law, does not measure what it needs: $\mathrm{Sym}^k N$ of a module $N$ of rank two whose group lies in $\mathrm{SL}_2$ has rank $k+1$ and a group of dimension at most three for every $k \geq 1$. A power of a function adds nothing to its group, as a repeated subtree adds only the binary length of its multiplicity to the price (Chapter 33), and functions whose groups have simple derived algebras, pairwise not isomorphic, add those algebras.

**Theorem 45.27** (A full Jordan block)**.** *Let $S$ be a simple module as in Definition 45.24 whose singular points are regular singular, of rank $r \geq 2$, with a local monodromy that is unipotent with a single Jordan block. Then the derived group of the identity component of $G_S$, on the solutions of $S$, is one of:*

*(a) the image of $\mathrm{SL}_2$ in the symmetric power $\mathrm{Sym}^{r-1}$ of its standard representation;*

*(b) $\mathrm{Sp}_r$ for even $r$, $\mathrm{SO}_r$ for odd $r$, or $G_2$ in its representation of dimension 7 for $r = 7$;*

*(c) $\mathrm{SL}_r$.*

*In case (a) $S \cong \mathrm{Sym}^{r-1} S_2 \otimes \chi$ for a simple module $S_2$ of rank two and a module $\chi$ of rank one, both with regular singular points: $S$ is a power, and the derived algebra of $\mathfrak{g}_S$ is $\mathfrak{sl}_2$. In case (b) the derived group keeps a form, alternating for $\mathrm{Sp}_r$ and symmetric for $\mathrm{SO}_r$ and $G_2$, and for $\mathrm{Sp}_r$ and $\mathrm{SO}_r$ it is the identity component of the whole stabilizer of that form; for $r \geq 3$ the group $\mathrm{SL}_r$ of case (c) keeps no form.*

**Theorem 45.28** (The class read by the tree and by its stride's law)**.** *Let $s \geq 2$, let $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(1, y)$ (Section 42.6), let $\mu_s$ be the stride's law and $P_1$ the orthant fibre integral of the centre ray of $\binom{[s]}{1}$ at the strength one (Definitions 40.1 and 39.47), and $\iota(t) = t^{-1}$.*

*(i)* One sequence, two words. *For every integer $k \geq 1$, with $P(k) = P_{\binom{[s]}{1}}(k\mathbf{1}; k, k)$,*

$$\Gamma(k)^2 P(k) = (\mathsf{T}_{1-k;(1)}[x]\, \mathsf{g}^s\, \mathbb{I}_{(0,\infty)})((0,\infty)), \qquad \Gamma(k)^2 = (\mathsf{T}_{1-k;(1)}[x]\, \mathsf{g}\, \mathbb{I}_{(0,\infty)})((0,\infty)),$$
$$P(k) = (\mathsf{T}_{k;(1)}[x]\, \iota_* \mu_s)((0,\infty)) = (\mathsf{T}_{1-k;(1)}[x]\, t\, \mu_s)((0,\infty)),$$

*and the inputs are tied by*

$$\mathsf{g}^s\, \mathbb{I}_{(0,\infty)} = \big(\mathsf{g}\, \mathbb{I}_{(0,\infty)}\big) \boxtimes (t\, \mu_s);$$

*the density of $\iota_* \mu_s$ is $P_1$ on $(s^2, \infty)$.*

*(ii)* The input of the tree's word. *$\vartheta^2 \mathsf{g} = y\, \mathsf{g}$, $\vartheta = y\, d/dy$, and $\mathsf{g}$ satisfies no equation of order one over $\mathbb{C}(y)$. The group of the input $\mathsf{g}\, \mathbb{I}_{(0,\infty)}$ is $\mathrm{SL}_2$. The module of $\mathsf{g}^s$ is $\mathrm{Sym}^s$ of the module of $\mathsf{g}$, of rank $s+1$, and the group of $\mathsf{g}^s\, \mathbb{I}_{(0,\infty)}$ is the image of $\mathrm{SL}_2$ on $\mathrm{Sym}^s$ of its standard representation, $\mathrm{SL}_2$ for odd $s$ and $\mathrm{PGL}_2$ for even $s$: for every $s$ its dimension is three.*

*(iii)* The input of the stride's law. *The inputs $\iota_* \mu_s$ and $t\, \mu_s$ have the group of the solutions $V_s$ of the class (Theorem 42.28). For $s \geq 5$, $V_s$ is not $\mathrm{Sym}^{s-2}$ of the solutions of a module of rank two up to a twist of rank one, and the derived group of the identity component of that group is $\mathrm{Sp}_{s-1}$ for odd $s$, $\mathrm{SO}_{s-1}$ for even $s$, or $\mathrm{SL}_{s-1}$, of dimension at least $(s-1)(s-2)/2$.*

The tree reads the class as one mark's input raised to the $s$-th power, and the law of its values is the $s$-th symmetric power of one mark's relation read in the index (Theorem 42.12): the rank of that law, its order $\lceil s/2 \rceil$, grows with $s$, and the group does not. The stride's law weighted by $t$ is the same input with one mark's input divided out by the multiplicative convolution of (i), the input whose Trailmoment is the Gamma stack $\Gamma(k)^2$ of the word; from $s = 5$ on, the word through the stride's law pays for that division with a group of dimension at least $(s-1)(s-2)/2$, where the tree's word needs one of dimension three. That measures the reading through the pencil and not the class, in the same way that the continuations of the solutions $V_s$ of the class say what functions with unipotent continuations cannot reach and nothing about the class (Theorem 42.28, Corollary 42.30, Definition 18.19).

**Corollary 45.29** (The word along the canonical tree)**.** *Let $(\mathcal{B}, \theta)$ be a connected table whose weights factor across each bunch of width one at which its canonical hinge tree decomposes it (Theorem 23.16), as those of a list do.*

*(i) At each of its full leans its period is a product of stacked Gammas, monomials in the weights and the periods of the prime nodes of the tree at their full leans; the period of a prime node $\mathcal{H}_v$ on $n_v$ marks with $|\mathcal{H}_v|$ bases is a fibre integral of dimension $|\mathcal{H}_v| - n_v$ on its share side, and prime nodes that are equal, with equal data, contribute one period raised to their number.*

*(ii) When it is built by substitution with plain cells, with the norms $1$ on the marks of the cells of at least two marks and norms on the marks left in place, its value at every strength is the word of its core with the word of each cell set at the load $\ell$ of its marker: a mark of the norm $\mathsf{m}$ left in place enters as $\mathsf{m}^{-\ell}$, a parallel class of $s \geq 2$ marks as $\Gamma(\ell)^{-2}\,(\mathsf{T}_{1-\ell;(1)}[x]\,\mathsf{G}_{0;(1,-1)}(1,x)^s\,\mathbb{I}_{(0,\infty)})((0,\infty))$, a Trailmoment of one input whose group is the image of one mark's, of dimension three (Theorem 45.28(ii)), and any other cell through the period of the family of its two hinge minors.*

So the content of a period is carried by the prime nodes of its canonical hinge tree, each with the work of its share side, which a node of few marks keeps small whatever the number of bases of the whole table (Corollary 26.8); the series and parallel nodes contribute stacked Gammas at the full leans and powers of one mark's input on the centre ray, and a repeated subtree is written once with its multiplicity (Definition 33.2). When a cell is itself built by substitution, the family of its two hinge minors is read against its own cells in the same way (Proposition 42.26(v)), so the word recurses down the tree, a prime node below the root entering through the family of the two minors of its table at its hinge. The pencil is built down the same tree by direct images, one at every vertex, a parallel node among them (Theorem 39.56): it is the tree read through the stride's law, and on a parallel node of $s$ marks it gives the solutions $V_s$ of the class (Theorem 42.28) where the tree's word has the $s$-th power of one mark's input.

The stride's law has its own structure, read at the far end of the pencil.

**Corollary 45.30** (The far end of the pencil)**.** *Let a ray on $n \geq 3$ marks satisfy the hypotheses of Proposition 39.54, and let $W$ be the module of its orthant fibre integral $P_1$, the density of the input $\iota_* \mu_f$ of Proposition 40.2(i), $\iota(t) = t^{-1}$, of rank $r_P$. Then $r_P \geq n-1$, and if $r_P = n-1$:*

*(i) the monodromy $T_\infty$ is unipotent with a single Jordan block on the solutions of $W$; the submodules of $W$ form a chain, so its composition series is unique; and $T_\infty$ has a single Jordan block on the solutions of every factor, the sizes adding to $n-1$;*

*(ii) every factor of rank $b \geq 2$ is of one of the kinds of Theorem 45.27, and every factor of rank one has a group in $\mathbb{G}_m$;*

*(iii) the reductive part of the group of $W$ is a subalgebra of the sum of the Lie algebras of the groups of its factors that projects onto each, and its unipotent part is nilpotent of class at most the number of factors less one.*

**Proposition 45.31** (The group is computable)**.** *The group of the module of an operator with coefficients in $\overline{\mathbb{Q}}(y)$, whatever its singular points, and with it its Lie algebra, its dimension and its reductive and unipotent parts, are computable from the operator. The relation $\vartheta^2 u = y\,u$ of one mark, which $\mathsf{G}_{0;(1,-1)}(1,y)$ satisfies, has coefficients in $\mathbb{Q}(y)$; when the weights of a family are algebraic, $L_1$ has coefficients in $\overline{\mathbb{Q}}(t)$, and the Lie algebra of the group of every solution of $L_1$, the orthant fibre integral among them, is a quotient of that of $L_1$.*

So the complexity of a period is read on its word along the canonical hinge tree. The prime nodes carry its content, each with the work of its share side; the series and parallel nodes and every repetition add stacked Gammas, powers and the binary lengths of multiplicities; and where the word reads an input on a scale, the group of the input counts the functions it needs, a power of one mark's input needing no more than one mark's group. The values of a ray are also one Trailmoment of the reciprocal image of its stride's law, and the group of that input, a quotient of the group of the pencil, is what that word needs: the input from which the Gamma stacks of the tree's word have been divided out, larger than the tree's from the class of five marks on, and of a kind that its rank, the form it keeps and whether it is a power decide wherever the far end gives a full Jordan block.

## 45.8 The group of a word along the tree

A word reads an input on a scale wherever it holds a Trailmoment, and the group of the input counts what it reads there (Section 45.7). Down the canonical hinge tree the value of a table on its centre ray can pay for each series or parallel node in either of two currencies: through the fibre of the node, in work, or on one scale, through the transforms of the cells directly below it (Proposition 42.26(v)–(vii)). Paid in work, the value reads nothing on a scale but powers of one mark's input, whatever the table, and its group is never more than one mark's. Paid on scales, its work does not grow with the multiplicities, and its group is a quotient of the group of one mark's input, its dilations by the norms at the root and the transforms of the distinct cells, other than marks, directly below a series or parallel node, each read once and each such cell rooted at a prime node or at a node of the other kind. A prime node is read on its shares in both readings: at the root through the fibre of its own table, below it through the fibre of the family of the two minors of its table at its hinge.

**Definition 45.32** (The inputs and the group of a word)**.** The *inputs* of a word (Definition 1.1(vi)) are the inputs of its Trailmoments, an integral $\int_0^\infty y^{-\nu} W(y)\,dy$ in it being read as the Trailmoment $(\mathsf{T}_{\nu;(1)}[x]\,W(x)\,\mathbb{I}_{(0,\infty)})((0,\infty))$ (Definitions 5.2 and 5.4); inputs with one density on one interval are one input, at whatever indices and cuts they are read. A product or a power of Trailmoments is read as written: its inputs are those of its factors. When the density of every input satisfies a nonzero linear differential equation over $\mathbb{C}(y)$, the *group of the word* is the group of the sum of the modules that the densities of its distinct inputs generate (Definition 45.24); a word without inputs has the trivial group.

So the group, like the price and the work (Definition 33.2), belongs to the word as it is written: the multiplicative convolution that collects a product of Trailmoments at one index into one

Trailmoment (Theorem 5.31(ii)) changes the input and can change the group, as it does for the class (Theorem 45.28).

**Lemma 45.33** (The transforms satisfy linear equations)**.** *At the exponents $(k,k)$ with the masses $k$, the transform and the dual transform of a connected plain cell (Definition 23.1) satisfy nonzero linear differential equations over $\mathbb{C}(y)$.*

**Theorem 45.34** (What the word along the canonical tree reads)**.** *Let $(\mathcal{B},\theta)$, its norms and $k$ be as in Proposition 42.26(iii), built down its canonical hinge tree (Theorem 23.16) with the weights* 1 *at its series and parallel nodes, every cell decomposed at its own node. In this statement $\mathsf{g}(y)=\mathsf{G}_{0;(1,-1)}(1,y)$, a* class *is a series or parallel node all of whose cells are marks, and a node or a cell is counted once for each distinct subtree. Let $S$ be a set of series and parallel nodes, and read the root by Proposition 42.26(vii) if it lies in $S$ and by (iii) if not, every class below the root on one scale by (iii) and (vi), every other node of $S$ by (vi), and every other node by (v).*

*(i)* Inputs. *The value is a closed serif word whose inputs are the $\mathsf{g}(x)^s\,\mathbb{I}_{(0,\infty)}$, $s$ running over the sizes of the classes below the root, and, at every node of $S$, the product of the dual transforms of the cells directly below it if it is a parallel node and of their transforms if it is a series node, a repeated cell entering as one function to a power, a mark as $\mathsf{g}$, and a mark at the root as $\mathsf{g}(\mathsf{m}_e x)$ or $\mathsf{m}_e^{-k}\mathsf{g}(x/\mathsf{m}_e)$.*

*(ii)* Group. *Its group is a quotient of the group of the sum of the modules of $\mathsf{g}$, of those dilations and of the transforms that the nodes of $S$ read of the distinct cells, other than marks, directly below them, each counted once, whatever the multiplicities (Lemma 45.33); every such cell is rooted at a prime node or at a node of the other kind. With $S$ empty the group is the image of $\mathrm{SL}_2$ on the sum of the symmetric powers $\mathrm{Sym}^s$ of its standard representation over the sizes of the classes: $\mathrm{SL}_2$ when some size is odd, $\mathrm{PGL}_2$ when every size is even, and trivial when there is no class below the root; for every table its dimension is at most three.*

*(iii)* Work. *Its work is the sum of the dimensions of the fibres of its distinct nodes outside $S$ that are not classes below the root, $2N_v-n_v-1$ at the root and $2N_v-n_v-2$ below it for a node of $N_v$ bases on $n_v$ marks, one for each distinct size of class below the root, one for each distinct node of $S$ that is not a class below the root, and one for each distinct transform that a node of $S$ reads of a cell other than a mark, a cell directly below nodes of $S$ of both kinds bringing both of its transforms. Below the root a series or parallel node outside $S$ that is not a class has a fibre of dimension one less than the number of the cells directly below it, counted with their multiplicities; with every series and parallel node in $S$ the work is at most twice the number of distinct subtrees other than marks plus the dimensions of the fibres of the distinct prime nodes, and does not grow with the multiplicities.*

So the value of such a table on its centre ray never needs more than one mark's group: the word that pays in work reads only the powers of one mark's input that close its classes, and what it pays is the work of its fibres, which at a series or parallel node grows with the number of cells below it and at a prime node is fixed by the node. The word that pays on scales keeps every repetition as a power of one transform, so that its work is set by the distinct subtrees, as the price is (Theorem 33.3); what it needs beyond one mark's input and its dilations is the transforms of the distinct cells directly below a series or parallel node, born where the tree meets a prime node or changes its kind. The class pays the same in either reading, one integral of its index whatever its size and no group beyond one mark's: its cells are marks, and a mark's transform is one mark's input whichever kind of node reads it. Reading a whole cell through its law instead, one Trailmoment (Proposition 42.26(iv)), collapses its subtree to one integral and brings its law as

the input: at the strength one the law of a parallel class of $s$ marks is $t\,\mu_s$, its two Trailmoments agreeing on a strip (Theorem 45.28(i)), the input whose group has dimension at least $(s-1)(s-2)/2$ from $s = 5$ on (Theorem 45.28(iii)). The theorem is the reading of one table at width one on the scales of its series and parallel nodes (Corollary 48.16): at every width every full node of every tree of nested bunches reads on its scales the blocks of any partition of its parts that it separates, at one point, a repeated block entering as one transform to a power, and the group of what the word reads there, at any number of scales, is a quotient of the group of its distinct blocks and of the exponentials of their constant parts (Theorem 48.15).

# Chapter 46

# Finite algebra for scale reads

A bunch read on its scale against linear letters is a Breta of one bunch at the unit configuration (Proposition 15.1), and a Breta of unit columns on one bunch is, up to a stacked Gamma, such a scale integral. Its values at integer shifts of its index and its letter exponents satisfy two relations, the letter split and the scale exchange, and these relations are complete: every linear relation between the shifted values, with coefficients rational in the index and the letter exponents, is a combination of shifted letter splits and scale exchanges, and the values modulo the relations form a space whose dimension is the number of letters. The same holds at every corner, with the coefficients $\xi_\ell$ of the letters $x+\xi_\ell$ as the variables. For polynomial letters on one scale the same two relations reduce every insertion to one read per distinct root of the product of the letters, counted without finding a root; at a numerical corner three Laurent coefficients per read suffice, every integer ray has a recurrence with a certificate, and the jets in the coefficients of the letters are reads of rational insertions.

## 46.1 Complete relations for linear letters and unit columns

**Definition 46.1** (Shifts, relations between shifts, the relation space)**.** Let $J=\{0,\dots,L\}$ carry distinct coefficients $\xi_\ell \in \mathbb{C}\setminus(-\infty,0]$, with the unit configuration $I_J$, and let $\mathbb{K}=\mathbb{C}(\kappa,w_0,\dots,w_L)$ be the field of rational functions of an index $\kappa$ and letter exponents $\mathbf{w}$. Let $\mathcal{M}$ be the vector space over $\mathbb{K}$ with one basis element $[a,\mathbf{b}]$ for each *shift* $(a,\mathbf{b})\in\mathbb{Z}\times\mathbb{Z}^J$. The *value* of $[a,\mathbf{b}]$ is the scale integral at the shifted index and letter exponents,

$$\int_0^\infty x^{\kappa+a-1}\prod_{\ell\in J}(x+\xi_\ell)^{-w_\ell-b_\ell}\,dx \\ = \Gamma_{\,2}^{\,|J|}\begin{bmatrix}\kappa+a,\ |\mathbf{w}|+|\mathbf{b}|-\kappa-a\\ \mathbf{w}+\mathbf{b}\end{bmatrix}\mathsf{B}^{\{J\}}_{\mathbf{w}+\mathbf{b},\,|\mathbf{w}|+|\mathbf{b}|-\kappa-a;\,I_J}(\boldsymbol{\xi})$$

which converges on $0<\operatorname{Re}\kappa+a<\operatorname{Re}|\mathbf{w}|+|\mathbf{b}|$, the second form holding where moreover $\operatorname{Re}(w_\ell+b_\ell)>0$ for every $\ell$ (Proposition 15.1); finitely many shifts converge together on a nonempty convex open set. An element $\sum_i c_i[a_i,\mathbf{b}_i]$ of $\mathcal{M}$ is a *relation between shifts* when the combination of the values of the $[a_i,\mathbf{b}_i]$ with the coefficients $c_i(\kappa,\mathbf{w})$ vanishes at every point where all its terms converge and every $c_i$ is defined. At the shift $(a,\mathbf{b})$ the letter split of the letter $x+\xi_\ell$ and the scale exchange are

$$\operatorname{split}_\ell(a,\mathbf{b})=[a,\mathbf{b}]-[a+1,\mathbf{b}+\mathbf{e}_\ell]-\xi_\ell\,[a,\mathbf{b}+\mathbf{e}_\ell],$$

$$\operatorname{exch}(a,\mathbf{b}) = (\kappa + a)\,[a,\mathbf{b}] - \sum_{\ell\in J}(w_\ell + b_\ell)\,[a+1,\mathbf{b}+\mathbf{e}_\ell],$$

the letter split and the scale exchange of Proposition 3.6 for the letters $x+\xi_\ell$. The *relation space* $\mathcal{L}_J \subseteq \mathcal{M}$ is the span over $\mathbb{K}$ of all letter splits and scale exchanges, and the *dimension* of the shifts is $\dim_{\mathbb{K}} \mathcal{M}/\mathcal{L}_J$; on one scale these play the part of the relation space and the dimension of a ray (Definition 38.4).

**Theorem 46.2** (The relations of a scale integral are complete)**.** *Let the coefficients range over distinct points of* $\mathbb{C}\setminus(-\infty,0]$.

*(i) Every element of* $\mathcal{L}_J$ *is a relation between shifts.*

*(ii)* $\mathcal{M} = \mathcal{L}_J + \sum_{\ell\in J}\mathbb{K}\,[0,\mathbf{e}_\ell]$*: the dimension is at most* $|J|$.

*(iii) The values of the* $|J|$ *shifts* $[0,\mathbf{e}_\ell]$,

$$\Gamma_{\,2}{}_{|J|}\begin{bmatrix}\kappa,\ |\mathbf{w}|+1-\kappa\\ \mathbf{w}+\mathbf{e}_\ell\end{bmatrix}\,\mathsf{B}^{\{J\}}_{\mathbf{w}+\mathbf{e}_\ell,\,|\mathbf{w}|+1-\kappa;\,I_J}(\boldsymbol{\xi}),\qquad \ell\in J,$$

*are linearly independent over* $\mathbb{K}$*: a combination of them with coefficients in* $\mathbb{K}$ *that vanishes on a nonempty open subset of* $\{0<\operatorname{Re}\kappa<\operatorname{Re}|\mathbf{w}|+1\}$ *has every coefficient* 0.

*(iv) The dimension is* $|J|$*, and every relation between shifts lies in* $\mathcal{L}_J$*: it is a combination over* $\mathbb{K}$ *of shifted letter splits and scale exchanges.*

*(v)* Coefficients as variables. *Let the coefficients be rational functions of parameters* $\mathbf{u}=(u_1,\dots,u_d)$*, distinct and not* 0 *as rational functions, defined with distinct values in* $\mathbb{C}\setminus(-\infty,0]$ *on a nonempty open subset of* $\mathbb{R}^d$ *or of* $\mathbb{C}^d$*. Then (i), (ii) and (iv) hold with the field* $\mathbb{C}(\mathbf{u},\kappa,\mathbf{w})$ *in the place of* $\mathbb{K}$*, a relation between shifts being an element whose combination of values vanishes at every* $\mathbf{u}$ *of that set and every* $(\kappa,\mathbf{w})$ *where all its terms converge and every coefficient is defined.*

The letter splits are the partial fractions of the integrand, and the scale exchange is one integration by parts along the scale. Part (ii) reduces every shift to the $|J|$ values with one letter exponent raised by one, by these two relations alone, and part (iii) says that those values are linearly independent over $\mathbb{K}$. So a relation between shifts that is not a combination of shifted letter splits and scale exchanges does not exist, and the number of independent values is the number of letters; for one letter the dimension is one: every shift of the Beta corner of a linear letter is a rational multiple of one value. By Theorem 15.3 the scale integral of a family at a bunch is such an integral, with the points of the joint spectrum as its coefficients, so its relations in its index and its letter exponents are complete; relations between scale integrals of a family at shifted masses and exponents, where the letter exponents are tied to the exponents of the family, are a further problem.

What decides the relations at a corner, where the index and the letter exponents are numbers and the coefficients are the variables, is the behaviour of the values as one coefficient tends to 0: its letter becomes a power of the scale and joins the index.

**Lemma 46.3** (A coefficient tends to 0)**.** *Let* $L\geq 1$*, let the coefficients be positive, and let the index and the letter exponents be numbers with* $\operatorname{Re} w_\ell > 0$ *for every* $\ell$ *and* $0\leq\operatorname{Re}\kappa'<1$*, where* $\kappa' = \kappa - w_0$*. In this statement* $J' = J\setminus\{0\}$*,* $\mathbf{w}' = \mathbf{w}|_{J'}$*,* $\boldsymbol{\xi}' = \boldsymbol{\xi}|_{J'}$*,* $f_\ell$ *is the value of* $[0,\mathbf{e}_\ell]$ *(Definition 46.1), and for letter exponents* $\mathbf{v}$ *on* $J'$ *with* $\operatorname{Re}|\mathbf{v}|>0$ *and* $h_{\mathbf{v}}(x) = \prod_{j\in J'}(x+\xi_j)^{-v_j}$*,*

$$S'(z;\mathbf{v}) = \frac{h_{\mathbf{v}}(0)}{z} + \int_0^1 x^{z-1}\big(h_{\mathbf{v}}(x) - h_{\mathbf{v}}(0)\big)\,dx + \int_1^\infty x^{z-1}h_{\mathbf{v}}(x)\,dx$$

*for* $-1 < \operatorname{Re} z < \operatorname{Re}|\mathbf{v}|$, $z \neq 0$. *For* $\operatorname{Re} z > 0$ *and* $\operatorname{Re} v_j > 0$ *for every* $j$, $S'(z;\mathbf{v}) = \Gamma_{\substack{2\\|J'|}}\begin{bmatrix} z,\ |\mathbf{v}|-z \\ \mathbf{v}\end{bmatrix} \mathsf{B}^{\{J'\}}_{\mathbf{v},\,|\mathbf{v}|-z;\,I_{J'}}(\boldsymbol{\xi}')$, *the scale integral with the coefficients* $\boldsymbol{\xi}'$, *so* $S'$ *continues it in its index across the wall* $\operatorname{Re} z = 0$, *with a simple pole at* $z = 0$ *of residue* $h_{\mathbf{v}}(0)$*; its constant term* $\operatorname{CT}_{z=0} S'(z;\mathbf{v})$ *is the sum of the two integrals at* $z = 0$, *and its values off the pole, at complex* $z$ *and complex letter exponents, are the finite parts* $\operatorname{Reg}\int_0^\infty x^{z-1}h_{\mathbf{v}}\,dx$, *its constant term at* $z = 0$ *being* $\operatorname{Reg}\int_0^\infty x^{-1}h_{\mathbf{v}}\,dx$ *(Theorem 1.7). Fix* $\boldsymbol{\xi}'$ *and let* $\xi_0 \to 0^+$. *Then, with* $j \in J'$ *in (ii)–(v):*

*(i)* $\xi_0^{1-\kappa'} f_0 \to \Gamma_{\substack{2\\1}}\begin{bmatrix}\kappa,\ 1-\kappa' \\ w_0+1\end{bmatrix} h_{\mathbf{w}'}(0)$, *which is not* 0.

*(ii) If* $\operatorname{Re}\kappa' > 0$, *then* $f_j \to S'(\kappa';\mathbf{w}'+\mathbf{e}_j)$.

*(iii) If* $\operatorname{Re}\kappa' = 0$ *and* $\kappa' \neq 0$, *then*

$$f_j = \Gamma_{\substack{2\\1}}\begin{bmatrix}\kappa,\ -\kappa' \\ w_0\end{bmatrix} \frac{h_{\mathbf{w}'}(0)}{\xi_j}\,\xi_0^{\kappa'} + S'(\kappa';\mathbf{w}'+\mathbf{e}_j) + o(1).$$

*(iv) If* $\kappa' = 0$, *then*

$$f_j = \frac{h_{\mathbf{w}'}(0)}{\xi_j}\Big(\log\frac{1}{\xi_0} + (\log\Gamma)'(1) - (\log\Gamma)'(\kappa)\Big) + \operatorname*{CT}_{z=0} S'(z;\mathbf{w}'+\mathbf{e}_j) + o(1).$$

*(v)* The relations across the wall. *For* $-1 < \operatorname{Re} z \le 0$, $z \neq 0$,

$$S'(z;\mathbf{v}) = S'(z+1;\mathbf{v}+\mathbf{e}_j) + \xi_j\, S'(z;\mathbf{v}+\mathbf{e}_j), \qquad z\,S'(z;\mathbf{v}) = \sum_{i\in J'} v_i\, S'(z+1;\mathbf{v}+\mathbf{e}_i),$$

*and* $\operatorname{CT}_{z=0} S'(z;\mathbf{v}) = S'(1;\mathbf{v}+\mathbf{e}_j) + \xi_j\ \operatorname{CT}_{z=0} S'(z;\mathbf{v}+\mathbf{e}_j)$.

As the coefficient $\xi_0$ tends to 0 its letter $x+\xi_0$ becomes the power $x$. The value with the exponent of that letter raised grows like $\xi_0^{\kappa'-1}$, and when $\operatorname{Re}\kappa' > 0$ the others tend to the scale integrals of the remaining coefficients at the index $\kappa' = \kappa - w_0$, the index left when the letter exponent $w_0$ has joined it. When $\operatorname{Re}\kappa' = 0$ that index lies on the wall of the remaining integral, and a second term of the same size stands next to the integral continued across its wall: the oscillating power $\xi_0^{\kappa'}$, and at $\kappa' = 0$ a logarithm, which grows. The letter split and the scale exchange hold across the wall, and at the pole $z = 0$ the scale exchange keeps only its residue, the first identity of Theorem 46.4(ii) for the remaining coefficients. These are the first terms of the closing between two separated groups (Corollary 19.19), the letter of $\xi_0$ being the lower group: one lattice carries the powers $\xi_0^{\kappa'+n}$, with the wall residues of the other letters as coefficients, and the other carries the integer powers $\xi_0^n$, with the scale integral of the other coefficients continued in its index; at $\kappa' = 0$ the two lattices meet, and what their poles leave is the logarithm of (iv).

**Theorem 46.4** (The relations at a corner)**.** *Let the index* $\kappa$ *and the letter exponents* $\mathbf{w}$ *be numbers with* $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|+1$, *so that the values of the* $|J|$ *shifts* $[0,\mathbf{e}_\ell]$ *converge, and let the coefficients be distinct points of* $\mathbb{C}\setminus(-\infty,0]$. *A shift* $[a,\mathbf{b}]$ converges at the corner *when* $0 < \operatorname{Re}\kappa + a < \operatorname{Re}|\mathbf{w}| + |\mathbf{b}|$, *and a letter split or a scale exchange at the shift* $(a,\mathbf{b})$, *with the index and the letter exponents at their numbers, converges at the corner when* $[a,\mathbf{b}]$ *does; all its terms then converge. A* linear relation *between shifts that converge at the corner is a combination* $\sum_i r_i[a_i,\mathbf{b}_i]$ *of such shifts, with coefficients* $r_i$ *rational in* $\boldsymbol{\xi}$, *whose value vanishes for* $\boldsymbol{\xi}$ *in a nonempty open set; it is a combination of letter splits and scale exchanges when it is one as a formal combination of shifts. The values are holomorphic in* $\boldsymbol{\xi}$ *on* $(\mathbb{C}\setminus(-\infty,0])^J$, *coincident coefficients included, and functions of* $\boldsymbol{\xi}$ *are* independent over $\boldsymbol{\xi}$ *when no combination of them with coefficients polynomial in* $\boldsymbol{\xi}$, *not all* 0, *vanishes at all* $\boldsymbol{\xi} > 0$, *equivalently for* $\boldsymbol{\xi}$ *in a nonempty open set.*

*(i)* Reduction. *If no letter exponent lies in* $\{-1,-2,\dots\}$*, the value of every shift that converges at the corner is a combination of the values of the* $[0,\mathbf{e}_\ell]$ *with coefficients rational in* $\boldsymbol{\xi}$*; the combination follows from letter splits and scale exchanges that converge at the corner, taken at shifts* $(a',\mathbf{b}')$ *with* $b'_\ell \geq \min(b_\ell,0)$ *for every* $\ell$*.*

*(ii)* The two walls. *If* $\operatorname{Re}|\mathbf{w}|>0$*, the values of the* $[0,\mathbf{e}_\ell]$ *at the index* 1 *and at the index* $|\mathbf{w}|$ *satisfy*

$$\sum_{\ell\in J} w_\ell \int_0^\infty \prod_{\ell'\in J} (x+\xi_{\ell'})^{-w_{\ell'}-\mathbf{1}[\ell'=\ell]}\,dx = \prod_{\ell\in J}\xi_\ell^{-w_\ell},$$
$$\sum_{\ell\in J} w_\ell\,\xi_\ell \int_0^\infty x^{|\mathbf{w}|-1}\prod_{\ell'\in J} (x+\xi_{\ell'})^{-w_{\ell'}-\mathbf{1}[\ell'=\ell]}\,dx = 1,$$

*the integrals being, where every* $\operatorname{Re} w_\ell>0$*, the stacked Gammas times the Bretas of Definition 46.1.*

*(iii)* Independence. *The values of the* $|J|$ *shifts* $[0,\mathbf{e}_\ell]$ *are independent over* $\boldsymbol{\xi}$ *if and only if no letter exponent lies in* $\{-1,-2,\dots\}$*.*

*(iv)* Completeness. *Whatever the letter exponents, every linear relation between shifts that converge at the corner is a combination, with coefficients rational in* $\boldsymbol{\xi}$*, of letter splits and scale exchanges that converge at the corner, and over the rational functions of* $\boldsymbol{\xi}$ *the values of these shifts span a space of dimension* $|J|$*: at every corner the dimension is* $|J|$*.*

*(v)* A coefficient equal to 1. *Parts (i), (iii) and (iv) hold as well when one coefficient is the constant* 1 *and the others are the variables.*

At a corner the relations are the same, and when no letter exponent lies in $\{-1,-2,\dots\}$ they reduce every shift that converges to the $|J|$ values with one letter exponent raised without leaving the corner: every relation of the reduction has all its terms convergent, and the leading coefficients $\kappa-r$, $r\geq 1$, and $\kappa+j-|\mathbf{w}|$, $j\geq 0$, by which the descents at the origin and at infinity divide vanish only at shifts that have left the domain. The two identities of (ii) are the relations at the two walls of the domain: the first is the scale exchange at the shift $(-1,\mathbf{0})$ at the index 1, whose member on the wall is replaced by its limit $\prod_\ell \xi_\ell^{-w_\ell}$, and the second is the scale exchange less the letter splits multiplied by their letter exponents at the shift $(0,\mathbf{0})$, at the index $|\mathbf{w}|$, where $|\mathbf{w}|-\kappa$ times the value of $[0,\mathbf{0}]$ tends to 1; at a wall one combination of the $|J|$ values is a monomial in $\boldsymbol{\xi}$. A letter exponent in $\{-1,-2,\dots\}$ is not on a wall: its letter has become a polynomial factor of the integrand, the $|J|$ values with one letter exponent raised are dependent and no longer span the shifts that converge; the corner with that letter exponent moved up to 0 has the same shifts, and its $|J|$ values with one letter exponent raised span them. Independence is read from the coefficients one at a time: as $\xi_0$ tends to 0 the value with the letter exponent $w_0$ raised grows faster than the others (Lemma 46.3), so in a relation the polynomial multiplying it vanishes at $\xi_0=0$, and what is left is a relation between scale integrals of the remaining coefficients, at the index $\kappa-w_0$, or on their wall, where the relations across the wall carry it to the next index.

**Corollary 46.5** (Unit columns on one bunch)**.** *Let* $B$ *be a bunch with the unit configuration* $I_B$ *and distinct coefficients* $\xi_e>0$*. By Proposition 15.1 with* $J=B$*,* $\mathbf{w}=\mathbf{m}$ *and* $\kappa=m_B-\nu$*, the value* $\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{a},\,\nu+n;\,I_B}(\boldsymbol{\xi})$*,* $\mathbf{a}\in\mathbb{Z}^B$*,* $n\in\mathbb{Z}$*, is the value of the shift* $[|\mathbf{a}|-n,\mathbf{a}]$ *of Definition 46.1 divided by* $\Gamma_{\substack{2\\|B|}}\begin{bmatrix} m_B-\nu+|\mathbf{a}|-n,\ \nu+n\\ \mathbf{m}+\mathbf{a}\end{bmatrix}$*, wherever that proposition applies; this stacked Gamma is the*

*one at* $(\mathbf{m},\nu)$ *times a rational function of* $(\mathbf{m},\nu)$*. Each relation below is read where all its Bretas converge.*

*(i)* The mark relation. *For each* $e\in B$*,*

$$m_e\,\mathsf{B}^{\{B\}}_{\mathbf{m},\,\nu;\,I_B}(\boldsymbol{\xi}) = (m_B-\nu)\,\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{e}_e,\,\nu;\,I_B}(\boldsymbol{\xi}) + \nu\,\xi_e\,\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{e}_e,\,\nu+1;\,I_B}(\boldsymbol{\xi}):$$

*the sum over* $f\in B$ *of the shape exchanges of Theorem 3.5(iv) for the pairs* $(e,f)$*, reduced by the shape split and the scale exchange; it is the letter split of the letter* $x+\xi_e$*.*

*(ii) The shape split, the scale exchange and the shape exchanges of Theorem 3.5, at all integer shifts of* $(\mathbf{m},\nu)$*, span over* $\mathbb{C}(\mathbf{m},\nu)$ *every linear relation with coefficients in* $\mathbb{C}(\mathbf{m},\nu)$ *between the values* $\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{a},\,\nu+n;\,I_B}(\boldsymbol{\xi})$*, and the quotient of the space of shifts by these relations has dimension* $|B|$*. The same holds over* $\mathbb{C}(\boldsymbol{\xi},\mathbf{m},\nu)$ *for relations whose coefficients are rational in* $\boldsymbol{\xi}$ *as well and which hold at all distinct* $\boldsymbol{\xi}>0$ *at which their coefficients are defined.*

*(iii)* At a corner. *Let* $\mathbf{m}$ *and* $\nu$ *be numbers with* $\operatorname{Re} m_e>0$ *for every* $e$ *and* $\operatorname{Re}\nu>0$*, and call a value* $\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{a},\,\nu+n;\,I_B}(\boldsymbol{\xi})$ interior *when* $\operatorname{Re}(m_e+a_e)>0$ *for every* $e$ *and* $\operatorname{Re}\nu+n>0$*; the Breta converges there. Over the rational functions of* $\boldsymbol{\xi}$ *the interior values span a space of dimension* $|B|$*, and every linear relation between interior values with coefficients rational in* $\boldsymbol{\xi}$ *is a combination, with such coefficients, of shape splits, scale exchanges and shape exchanges all of whose terms are interior. If* $\operatorname{Re}\nu<\operatorname{Re} m_B$*, the* $|B|$ *values* $\mathsf{B}^{\{B\}}_{\mathbf{m}+\mathbf{e}_e,\,\nu+1;\,I_B}(\boldsymbol{\xi})$*,* $e\in B$*, are a basis of that space.*

Completeness holds for the scale integral of one bunch against linear letters and for unit columns on one bunch, which are the same integrals: the coefficients are the same, the shape index is the letter exponents, and the index of the scale integral is the amount $m_B-\nu$ by which the scale index falls short of the mass $m_B$. It holds at a free index and free letter exponents, and at every corner with the coefficients as the variables, one of them possibly the constant 1. For a Breta with a kernel, with letters of more than two patterns or of levels other than 0 and 1, or with several bunches, the relations of Theorem 3.5 hold, and their completeness is a further problem.

## 46.2 Polynomial letters on one scale

A word on one scale with polynomial letters, such as the scale integral of a bunch at fixed shapes (Theorems 15.3 and 19.28), has the same two relations, the letter splits becoming partial fractions over the distinct factors of the letters, and the same finiteness: the retained data are one read per distinct root of the product of the letters, counted without finding a root, and at a numerical corner three Laurent coefficients per read. Every rational insertion, every integer shift of the index and the letter exponents, every derivative in the coefficients of the letters and every integer ray is then finite algebra, with a certificate.

**Definition 46.6** (Insertions and reads on one scale)**.** Let $\mathbb{F}\subseteq\mathbb{C}$ be a field, the field of the letters, and let $P_1,\dots,P_N\in\mathbb{F}[x]$ be nonconstant letters without roots on $[0,\infty)$, a constant letter being moved into a scalar; put $\mathbf{P}=(P_1,\dots,P_N)$. For an index $\kappa$ and letter exponents $w_1,\dots,w_N$ in a field $\mathbb{F}'\supseteq\mathbb{F}(\kappa,\mathbf{w})$ embedded in $\mathbb{C}$, the insertions, the regular functions of the letter complement as in Definition 38.1, are the elements of the ring $\mathfrak{A}=\mathbb{F}'[x,x^{-1},P_1^{-1},\dots,P_N^{-1}]$, and the *read* of an insertion $R$ is

$$\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}} = \int_0^\infty x^{\kappa-1}\prod_j P_j(x)^{-w_j}\,R(x)\,dx$$

where it converges absolutely, each power $P_j^{-w_j}$ being taken on $[0,\infty)$ by continuity from its principal value at 0, and continued in the index and the letter exponents elsewhere. The shift of the index by $a$ and of the letter exponents by $b_j$, integers, is the insertion $x^a\prod_j P_j^{-b_j}$, and a derivative in the coefficients of a letter is an insertion. A squarefree decomposition with gcd splitting gives pairwise coprime squarefree $q_1,\dots,q_M\in\mathbb{F}[x]$, none vanishing at 0, with $P_j=c_j\prod_k q_k^{m_{jk}}$, without roots and without a field extension; $Q=q_1\cdots q_M$ is the monic radical of $P_1\cdots P_N$, of degree $D>0$. The *scale exchange* on insertions is

$$\eth g = xg' + \Big(\kappa - \sum_j w_j \frac{xP_j'}{P_j}\Big)g, \qquad \text{so that} \qquad \Big(x^\kappa \prod_j P_j^{-w_j}\, g\Big)' = x^{\kappa-1}\prod_j P_j^{-w_j}\,\eth g.$$

For the linear letters $x+\xi_\ell$ with distinct $\xi_\ell\in\mathbb{C}\setminus(-\infty,0]$, the reads of the shifts are the values of Definition 46.1, $D=|J|$, and $\eth$ applied to $x^a\prod_\ell(x+\xi_\ell)^{-b_\ell}$ is the scale exchange $\mathrm{exch}(a,\mathbf{b})$ read under the integral.

**Theorem 46.7** (Reduction by the scale exchange)**.** *Let the numbers*

$$\kappa - r \quad (r\ge 1), \qquad \kappa + a - \sum_j w_j \deg P_j \quad (a\ge 0), \qquad \sum_j m_{jk}w_j + h \quad (h\ge 1,\ 1\le k\le M)$$

*be nonzero in $\mathbb{F}'$. Every $R\in\mathfrak{A}$ has a unique decomposition*

$$R = \eth g + \sum_{i=0}^{D-1} c_i \frac{x^i}{Q}, \qquad g\in\mathfrak{A},\quad c_i\in\mathbb{F}',$$

*obtained by polynomial arithmetic, partial fractions and the extended Euclidean algorithm; so $\dim_{\mathbb{F}'}\mathfrak{A}/\eth\mathfrak{A}=D$, and on the domain where the reads converge $\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}}=\sum_i c_i\,\langle x^i/Q\rangle_{\kappa,\mathbf{w};\mathbf{P}}$. For the linear letters $x+\xi_\ell$ with distinct $\xi_\ell\in\mathbb{C}\setminus(-\infty,0]$, $D=|J|$ and this is Theorem 46.2(ii) over $\mathbb{K}$, the $x^i/Q$ spanning the same space as the $(x+\xi_\ell)^{-1}$.*

The dimension is that of the quotient by the scale exchange, the dimension of the ray on one scale (Theorem 38.31); it does not assert that the reads of the $D$ insertions $x^i/Q$ are independent once parameters are tied, and symmetries of the letters can relate them further. For linear letters Theorem 46.2(iii) gives the independence. The three conditions are the leading coefficients of the reduction at the origin, at infinity and at a repeated factor. Exact computation needs exact arithmetic with a zero test in $\mathbb{F}'$, numerical exponents being taken in an effective extension; the analytic statements hold at all complex exponents.

**Definition 46.8** (Deformation; masters)**.** At numerical exponents $(\kappa,\mathbf{w})$ fix a direction $(\lambda_0;\boldsymbol{\lambda})\in\mathbb{R}\times\mathbb{R}^N$ of the index and the letter exponents that is *admissible*: $\lambda_0>0$, $\sum_j\lambda_j\deg P_j>\lambda_0$ and $\sum_j m_{jk}\lambda_j\neq 0$ for every $k$. The *deformation* moves the data along it, to $(\kappa+\lambda_0\varepsilon,\ \mathbf{w}+\varepsilon\boldsymbol{\lambda})$; the three conditions of Theorem 46.7 hold over $\mathbb{F}'(\varepsilon)$, and for an insertion $R$ the coefficients $c_i(\varepsilon)$ are those of its decomposition there. The *masters* are the deformed reads

$$H_i(\varepsilon) = \big\langle x^i/Q\big\rangle_{\kappa+\lambda_0\varepsilon,\ \mathbf{w}+\varepsilon\boldsymbol{\lambda};\ \mathbf{P}}, \qquad 0\le i<D.$$

For large real $\varepsilon$ every finite set of deformed reads converges, the exponent at the origin growing with the slope $\lambda_0$ and the decay at infinity with the slope $\sum_j\lambda_j\deg P_j-\lambda_0>0$, so the reductions hold there and then as identities of meromorphic functions. The masters are finite combinations of reads of shifts of the original letters: $1/Q=\big(P_1\cdots P_N/Q\big)\prod_j P_j^{-1}$ with $P_1\cdots P_N/Q$ a polynomial, so each term of $x^iP_1\cdots P_N/Q$ shifts the index by an integer and raises every letter exponent by one. The admissible directions form a nonempty open cone; the direction $(\frac12;\mathbf{1})$ is admissible.

**Lemma 46.9** (Simple poles at a corner)**.** *For every insertion $R$ independent of $\varepsilon$, or holomorphic in $\varepsilon$, the coefficients $c_i(\varepsilon)$ have at most a simple pole at $\varepsilon = 0$, and every master $H_i(\varepsilon)$ is meromorphic near $0$ with at most a simple pole.*

**Theorem 46.10** (Finitely many numbers at every corner)**.** *Along an admissible direction (Definition 46.8), for an insertion $R$ write $H_i(\varepsilon) = h_{i,-1}\varepsilon^{-1} + h_{i,0} + h_{i,1}\varepsilon + O(\varepsilon^2)$ and $c_i(\varepsilon) = c_{i,-1}\varepsilon^{-1} + c_{i,0} + c_{i,1}\varepsilon + O(\varepsilon^2)$. The $3D$* corner numbers *$(h_{i,-1}, h_{i,0}, h_{i,1})_{i<D}$ determine every absolutely convergent read:*

$$\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}} = \sum_{i=0}^{D-1} \big(c_{i,-1}h_{i,1} + c_{i,0}h_{i,0} + c_{i,1}h_{i,-1}\big).$$

*They depend on the letters, the exponents and the direction only, not on the insertion or the size of the shift. The reads of $R\left(\lambda_0 \log x - \sum_j \lambda_j \log P_j\right)^b$, the derivatives $\partial_\varepsilon^b$ at $\varepsilon = 0$ of the deformed reads $\langle R\rangle_{\kappa+\lambda_0\varepsilon,\,\mathbf{w}+\varepsilon\boldsymbol{\lambda};\,\mathbf{P}}$, need the masters through the order $b+1$: $(b+3)D$ numbers. When $\langle R\rangle_{\kappa,\mathbf{w};\mathbf{P}}$ converges absolutely, a read of $R\,(\log x)^{k_0}\prod_j(\log P_j)^{k_j}$, $b = k_0 + \sum_j k_j$, is a rational combination of the reads of $R\left(\lambda_0 \log x - \sum_j \lambda_j \log P_j\right)^b$ along finitely many admissible directions, each with its own corner numbers.*

The residues are finite extractions of endpoint coefficients, the other numbers finite parts and derivatives in the deformation. The $D$ values of the masters suffice when every pivot of the reduction is regular and the masters are holomorphic at the corner, as for linear letters at a corner with no letter exponent in $\{-1, -2, \dots\}$ and convergent masters (Theorem 46.4(i)); regular pivots alone do not suffice, and at an arbitrary corner the corner numbers are what is retained.

**Theorem 46.11** (Integer rays have certified recurrences)**.** *Fix integers $a, b_1, \dots, b_N$, and let $\eth_n$ be the scale exchange at the deformed data $\left(\kappa + an + \frac{\varepsilon}{2},\ \mathbf{w} + n\mathbf{b} + \varepsilon\mathbf{1}\right)$. A finite algorithm gives polynomials $p_0(n,\varepsilon), \dots, p_D(n,\varepsilon)$ and a certificate $g$, rational in $x$, $n$ and $\varepsilon$, with*

$$\eth_n g = \sum_{k=0}^{D} p_k(n,\varepsilon)\, x^{ak} \prod_j P_j^{-b_j k},$$

*the polynomials $p_k(n) = p_k(n,0)$ not all zero; then*

$$\sum_{k=0}^{D} p_k(n)\, \langle 1\rangle_{\kappa+a(n+k),\ \mathbf{w}+(n+k)\mathbf{b};\ \mathbf{P}} = 0$$

*at every integer $n$ at which the displayed reads converge.*

The recurrence need not have the least order, and its leading coefficient may vanish at finitely many integers, read off that polynomial; there further values are given or another dependence is formed. On one scale this supplies, for every ray, the certificate that a recurrence asks for: an insertion $g$ whose scale exchange is the recurrence.

**Theorem 46.12** (Coefficient jets on one scale)**.** *Let $P_j(x;u) = P_j(x) + \sum_{a=1}^{p} u_a P_{ja}(x)$ with $\deg P_{ja} \le \deg P_j$, choose bounds $M_{ja} \ge \sup_{x>0} |P_{ja}/P_j|$, finite since the quotient is continuous on $[0,\infty]$, and let $\langle 1\rangle_{\kappa,\mathbf{w};\mathbf{P}}$ converge absolutely. On the polydisc where $\sum_a |u_a| M_{ja} < 1$ for every*

*$j$ the read $\langle 1\rangle_{\kappa,\mathbf{w};\mathbf{P}(\,\cdot\,;u)}$, the powers of the changed letters being continued in $u$ from $u=0$, is holomorphic and*

$$\langle 1\rangle_{\kappa,\mathbf{w};\mathbf{P}(\,\cdot\,;u)} = \sum_{(n_{ja})\geq 0} \prod_{j} \frac{(-1)^{|n_j|}(w_j)_{|n_j|}}{\prod_a n_{ja}!} \prod_a u_a^{\sum_j n_{ja}} \Big\langle \prod_{j,a} \Big(\frac{P_{ja}}{P_j}\Big)^{n_{ja}} \Big\rangle_{\kappa,\mathbf{w};\mathbf{P}},$$

*$|n_j| = \sum_a n_{ja}$, absolutely; every coefficient is the read of a rational insertion, given by the corner numbers of Theorem 46.10.*

Splitting of repeated roots under the change costs nothing, the expansion being made against the original letters; a change that raises a degree is outside the statement. For coefficients that are symbolic parameters with fixed gcds and multiplicities, differentiating the masters gives rational insertions, so Theorem 46.7 gives matrices $\partial_{u_a} H = \mathbf{A}_a H$, with certificates, for the vector $H$ of the masters, whose mixed derivatives agree because they agree before the reduction; at a corner where a leading coefficient of the reduction vanishes the corner numbers replace them. The structural data reused are the squarefree base, $Q$ and the inverses of $xq_k'$ modulo $q_k$; the analytic data are the $D$ masters or the $3D$ corner numbers. A rational insertion with polynomial part of degree $A_+$, a pole of order $A_-$ at the origin and of order $h_k$ at $q_k$ takes at most $(A_+ + 1) + A_- + \sum_k \max(h_k - 1, 0)$ elimination rounds, counted in operations, the growth of exact coefficients charged separately; a repeated letter adds $m_{jk}$ times its letter exponent to the exponent of each squarefree factor $q_k$ and leaves $Q$ and $D$ unchanged. A recurrence in a count of repetitions needs the integer shifts of Theorem 46.11, and a multiplicity written in binary, which is what the price of Chapter 33 counts, does not by itself make the evaluation logarithmic in it.

# Chapter 47

# The module of a family

A family is a function of all its data at once. Every insertion has a value at every point of the balance wall, integration by parts in each mark holds there with coefficients linear in the data, and a monomial in the marks and the letters moves the data by a lattice vector. Modulo the integrations by parts the insertions form one vector space over the rational functions of the data, of finite dimension, on which the lattice of integral data acts by shifts: the module of the family. Its masters and their shift matrices carry the values from a point to every lattice translate of it and reduce every insertion to the masters; they are found by one elimination in a box over the rational functions of the data, with a certificate for every relation used, the matrices of the box presenting the module once they are invertible. The values separate the module: the masses act on the values as the Euler derivatives act on the integrand, so an element whose value vanishes as a function of all the data is zero in the module, every relation among the values is a relation of the module, read off the reductions on masters, and the closings along the rays from one integral point tell it as well, finitely many of them along finitely many parallel rays deciding. On a subspace of the data with a native direction the value zero is told by the expansion of the values at the native gauge of that direction, which asks for no positive direction and no closing; on a subspace cut out by conditions on the exponents alone that meets the tube the values still separate the module. Chapter 46 did this on one scale, and the module of a ray (Definition 39.1) is the module of the family restricted to one line of its data space (Proposition 47.5).

## 47.1 The data of a family

**Definition 47.1** (The data space, the lattice and the master form)**.** Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ be a family on $\mathcal{E}$, $n = |\mathcal{E}| \geq 2$, and $\Bbbk \subseteq \mathbb{C}$ a field that contains the weights.

(i) The *data space* of $F$ is its balance wall $\mathbb{D}_F = \{\mathbf{d} = (\mathbf{m}; \mathbf{s}, \mathbf{t}) \in \mathbb{C}^{\mathcal{E}} \times \mathbb{C}^N \times \mathbb{C}^N : \ \kappa_{\mathcal{E}}(\mathbf{d}) = 0\}$, a complex vector space of dimension $n + 2N - 1$ on which the walls $\kappa_U$ of Definition 8.1 are linear forms, none of them zero for $U$ nonempty and proper. Its *lattice* is the set of its integral points $\mathbb{D}_F(\mathbb{Z}) = \mathbb{D}_F \cap (\mathbb{Z}^{\mathcal{E}} \times \mathbb{Z}^N \times \mathbb{Z}^N)$, the integral data, and $\mathbb{D}_F(\mathbb{R})$ is the set of its real points, and its fields of rational functions are $\Bbbk(\mathbb{D}_F)$ and $\mathbb{K}_F = \mathbb{C}(\mathbb{D}_F)$. A *positive direction* is a $\mathbf{d}_1 = (\mathbf{m}_1; \mathbf{s}_1, \mathbf{t}_1) \in \mathbb{D}_F(\mathbb{Z})$ with $s_{1j}, t_{1j} \geq 1$ for every $j$ and $\kappa_U(\mathbf{d}_1) > 0$ for every nonempty proper $U$.

(ii) For $\lambda = (v; \mathbf{a}, \mathbf{b}) \in \mathbb{D}_F(\mathbb{Z})$ the *stride* of $\lambda$ is $f_\lambda = \alpha^v \prod_j K_j^{-a_j} \Psi_j^{-b_j}$, a unit of the insertions $\mathbb{C}[X_F]$ (Definition 38.1) with coefficients in $\Bbbk$; $f_\lambda f_\mu = f_{\lambda+\mu}$ and $f_0 = 1$.

(iii) On the chart $y_{e_0} = 0$, with $dy = \prod_{e \neq e_0} d\alpha_e/\alpha_e$, the *master form* at $\mathbf{d}$ is $\omega_{\mathbf{d}} =$

$\alpha^{\mathbf{m}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, dy$, the powers principal on the positive orthant, so that $f_\lambda\, \omega_{\mathbf{d}} = \omega_{\mathbf{d}+\lambda}$. Its *defects* are $c_e(\mathbf{d}) = m_e - \sum_j \big(s_j R_{j,e} + t_j R^*_{j,e}\big)$, with the occupations of Definition 38.1: they are linear in $\mathbf{d}$, $\sum_e c_e(\mathbf{d}) = \kappa_{\mathcal{E}}(\mathbf{d}) = 0$ on $\mathbb{D}_F$, $c_e(\mathbf{d}) = \partial_{y_e} \log(\omega_{\mathbf{d}}/dy)$ on the orthant, and $c_e(\lambda) = \partial_{y_e} \log f_\lambda$.

(iv) The *value* of an insertion $g$ at $\mathbf{d}$ is $\langle g\rangle(\mathbf{d}) = \int_{y_{e_0}=0} g\, \omega_{\mathbf{d}}$, where it converges; $\langle 1\rangle(\mathbf{d})$ is the period $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ (Theorem 8.7). An insertion $g = \alpha^u \prod_j K_j^{-a_j} \Psi_j^{-b_j}$, $\mathbf{a},\mathbf{b} \in \mathbb{N}^N$, has the walls $\kappa_U(g) = u_U - \sum_j \big(a_j d_j(U) + b_j d^*_j(U)\big)$ of its data (Proposition 38.5), and its *tube* is

$$\mathcal{T}_g = \{\mathbf{d} \in \mathbb{D}_F :\ \operatorname{Re} \kappa_U(\mathbf{d}) + \kappa_U(g) > 0 \text{ for every nonempty proper } U\}.$$

When the marks of $F$ form one molecule (Definition 7.4) its tube $\mathcal{T}_1$ is not empty and it has a positive direction. The centre data $\mathbf{d}_\circ = (N\mathbf{1};\mathbf{1},\mathbf{1})$ lie on the balance wall, since $\sum_j (r_j + r^*_j) = Nn$, and every proper wall there is $\kappa_U(\mathbf{d}_\circ) = \sum_j c_j(U)$ (Definition 7.12), positive because $c_j(U) = r_j(U) - d_j(U)$ vanishes exactly when every basis of table $j$ meets $U$ in the same number of marks (Theorem 7.13), that is when $U$ is a union of molecules of table $j$ (Lemma 7.10), and no nonempty proper $U$ is that for every table; so $\mathbf{d}_\circ$ is a positive direction and $\mathcal{T}_1$ contains the points $\eta\, \mathbf{d}_\circ$, $\eta > 0$, with neighbourhoods. A compound has, at every point of its balance wall, a wall of real part at most 0 (Proposition 8.16), and it is split into its molecules first. The procedure of Definition 16.14 evaluates a molecule at points of its data space, after the slide (Theorem 16.8).

**Theorem 47.2** (The relations in all the data)**.** *Let the marks of $F$ form one molecule.*

*(i) For an insertion $g = \alpha^u \prod_j K_j^{-a_j} \Psi_j^{-b_j}$, $\mathbf{a},\mathbf{b} \in \mathbb{N}^N$, and $\mathbf{d} \in \mathcal{T}_g$ the values $\langle g\rangle(\mathbf{d})$ and $\big\langle \big(c_e(\mathbf{d}) + \partial_{y_e}\big) g\big\rangle(\mathbf{d})$ converge absolutely, and the second is 0 for every mark $e$. For $\lambda \in \mathbb{D}_F(\mathbb{Z})$ with $\mathbf{d}+\lambda \in \mathcal{T}_g$, $\langle f_\lambda g\rangle(\mathbf{d}) = \langle g\rangle(\mathbf{d}+\lambda)$.*

*(ii) $\langle g\rangle$ continues to a meromorphic function on $\mathbb{D}_F$, holomorphic on $\mathcal{T}_g$, whose poles lie on the hyperplanes $\kappa_U(\mathbf{d}) + \kappa_U(g) \in \{0,-1,-2,\dots\}$, $U$ nonempty and proper. Extended $\mathbb{K}_F$-linearly to $\mathbb{C}[X_F] \otimes \mathbb{K}_F$, $\big\langle \sum_i g_i \otimes r_i \big\rangle = \sum_i r_i \langle g_i\rangle$, the evaluation vanishes identically on every element of the $\mathbb{K}_F$-span of the $(c_e(\mathbf{d}) + \partial_{y_e})g$, and $\langle f_\lambda g\rangle(\mathbf{d}) = \langle g\rangle(\mathbf{d}+\lambda)$ identically.*

*(iii) At every real $\mathbf{d} \in \mathcal{T}_1$, $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) > 0$.*

On the lattice translates of the period these relations are the letter and mark relations of Theorem 8.9(iii) on the balance wall: $\langle R_{j,e}\rangle(\mathbf{d}) = \sum_{T \ni e} \theta_{j,T} P_F(\mathbf{d} + (\mathbf{1}_T; \mathbf{e}_j, \mathbf{0}))$, so the relation of $g = 1$ at $e$ is the mark relation of $P_F$ at $\mathbf{d}$, and that of $f_\lambda$ is the mark relation at $\mathbf{d}+\lambda$. The relations of a ray (Theorem 38.3, Proposition 38.5) are these read at the points $\mathbf{d}_0 + k\mathbf{d}_1$.

## 47.2 The module and its shifts

**Definition 47.3** (The module of a family; shifts; masters; shift matrices)**.** Let $F$ and $\Bbbk$ be as in Definition 47.1.

(i) The *relation space* $\mathcal{L}_F \subseteq \mathbb{C}[X_F] \otimes \mathbb{K}_F$ is the $\mathbb{K}_F$-span of the elements $\big(c_e(\mathbf{d}) + \partial_{y_e}\big) g$, $g \in \mathbb{C}[X_F]$, $e \in \mathcal{E}$, and the *module* of $F$ is

$$\mathfrak{M}_F = \big(\mathbb{C}[X_F] \otimes \mathbb{K}_F\big)/\mathcal{L}_F,$$

$[g]$ the class of $g \otimes 1$. Over $\Bbbk$, $\mathcal{L}_{F,\Bbbk}$ is the $\Bbbk(\mathbb{D}_F)$-span of these elements for $g \in \Bbbk[X_F]$, and $\mathfrak{M}_{F,\Bbbk} = \big(\Bbbk[X_F] \otimes \Bbbk(\mathbb{D}_F)\big)/\mathcal{L}_{F,\Bbbk}$.

(ii) For $\lambda \in \mathbb{D}_F(\mathbb{Z})$ put $(\sigma_\lambda r)(\mathbf{d}) = r(\mathbf{d}+\lambda)$ on $\mathbb{K}_F$. The *shift* is $\mathrm{Sh}_\lambda[g \otimes r] = [f_\lambda g \otimes \sigma_\lambda r]$. A *submodule* is a $\mathbb{K}_F$-subspace $\mathfrak{N}$ with $\mathrm{Sh}_\lambda \mathfrak{N} = \mathfrak{N}$ for every $\lambda \in \mathbb{D}_F(\mathbb{Z})$, and $\mathfrak{N} \neq 0$ is *irreducible* when its only submodules are 0 and $\mathfrak{N}$.

(iii) *Masters* of a finite-dimensional submodule $\mathfrak{N}$ are insertions $\mathbf{b} = (b_1, \dots, b_\mathfrak{m})$ whose classes form a basis of $\mathfrak{N}$. Their *shift matrix* at $\lambda$ is $\mathbf{A}_\lambda \in GL_\mathfrak{m}(\mathbb{K}_F)$ with $\mathrm{Sh}_\lambda[b_i] = \sum_l \mathbf{A}_{\lambda,il}[b_l]$, and the *reduction* of an element of $\mathfrak{N}$ is its vector of coordinates on $[\mathbf{b}]$.

(iv) When the marks of $F$ form one molecule, the *value* of $m \in \mathfrak{M}_F$ is the meromorphic function $\langle m \rangle = \sum_i r_i \langle g_i \rangle$ of any representative $\sum_i g_i \otimes r_i$, and $\mathrm{Ann}\langle\cdot\rangle = \{m \in \mathfrak{M}_F : \ \langle m \rangle = 0\}$.

**Theorem 47.4** (The module of a family)**.** *Let the marks of $F$ form one molecule.*

*(i) The shifts are well defined, $\sigma_\lambda$-semilinear and bijective, with $\mathrm{Sh}_0 = 1$ and $\mathrm{Sh}_\lambda \mathrm{Sh}_\mu = \mathrm{Sh}_{\lambda+\mu}$: the lattice acts on $\mathfrak{M}_F$. The value of an element is well defined, $\langle \mathrm{Sh}_\lambda m \rangle(\mathbf{d}) = \langle m \rangle(\mathbf{d}+\lambda)$, and $\mathrm{Ann}\langle\cdot\rangle$ is a submodule that does not contain $[1]$.*

*(ii) $\left[\alpha^u \prod_j K_j^{-a_j} \Psi_j^{-b_j}\right] = \mathrm{Sh}_{(u;\mathbf{a},\mathbf{b})}[1]$. So $\mathfrak{M}_F$ is the $\mathbb{K}_F$-span of the shifts of $[1]$, every submodule that contains $[1]$ is $\mathfrak{M}_F$, and every value is a combination of periods at lattice translates, $\left\langle \sum_\lambda r_\lambda f_\lambda \right\rangle(\mathbf{d}) = \sum_\lambda r_\lambda(\mathbf{d})\, P_F(\mathbf{d}+\lambda)$. The classes of the Laurent insertions span $\mathfrak{M}_F$ as well.*

*(iii) $\mathfrak{M}_F$ is finite-dimensional over $\mathbb{K}_F$: it is the top cohomology of the twisted algebraic de Rham complex of the letter complement $X_F$ over $\mathbb{K}_F$ for the connection $d + \sum_{e \neq e_0} c_e(\mathbf{d})\, dy_e$.*

*(iv) $\mathfrak{M}_{F,\Bbbk} \otimes_{\Bbbk(\mathbb{D}_F)} \mathbb{K}_F = \mathfrak{M}_F$. So masters of $\mathfrak{M}_F$ can be taken among the insertions with coefficients in $\Bbbk$, and for such masters every shift matrix has its entries in $\Bbbk(\mathbb{D}_F)$.*

*(v) For masters $\mathbf{b}$ of a submodule, $\mathbf{A}_0 = 1$ and $\mathbf{A}_{\lambda+\mu} = \sigma_\lambda(\mathbf{A}_\mu)\, \mathbf{A}_\lambda$; so $\sigma_\lambda(\mathbf{A}_\mu)\, \mathbf{A}_\lambda = \sigma_\mu(\mathbf{A}_\lambda)\, \mathbf{A}_\mu$, and $\mathbf{A}_\lambda$ is the product, along any path from 0 to $\lambda$ in steps of a set of generators of $\mathbb{D}_F(\mathbb{Z})$ and their inverses, of the shifted matrices of the steps. The values of the masters transport,*

$$\langle \mathbf{b} \rangle(\mathbf{d}+\lambda) = \mathbf{A}_\lambda(\mathbf{d})\, \langle \mathbf{b} \rangle(\mathbf{d})$$

*as an identity of meromorphic functions on $\mathbb{D}_F$, $\langle \mathbf{b} \rangle$ the column of the values of the masters. For masters of $\mathfrak{M}_F$ every insertion $g$ has one reduction $\rho \in \mathbb{K}_F^\mathfrak{m}$, $[g] = \sum_l \rho_l [b_l]$, and $\langle g \rangle = \sum_l \rho_l \langle b_l \rangle$.*

So a family carries finitely many functions of its data, the values of its masters, and every value of every insertion at every point of the data space is a combination of them with coefficients rational in the data, read off by the reduction; the integral data carry them from a point to its lattice translates through finitely many matrices with entries in $\Bbbk(\mathbb{D}_F)$. Nothing in this is chosen but the masters, and two choices of masters with coefficients in $\Bbbk$ differ by one matrix in $GL_\mathfrak{m}(\Bbbk(\mathbb{D}_F))$.

**Proposition 47.5** (A ray is a line of the data space)**.** *Let $\mathbf{d}_0 \in \mathbb{D}_F$ and $\mathbf{d}_1 \in \mathbb{D}_F(\mathbb{Z})$, $\mathbf{d}_1 \neq 0$, with $s_{1j}, t_{1j} \in \mathbb{N}$ for every $j$, let $\ell = \{\mathbf{d}_0 + k\mathbf{d}_1 : \ k \in \mathbb{C}\}$, and let $\mathcal{O}_\ell \subseteq \mathbb{K}_F$ be the rational functions regular at the general point of $\ell$, restricted by $r|_\ell(k) = r(\mathbf{d}_0 + k\mathbf{d}_1)$. The ray with the origin $\mathbf{d}_0$ and the direction $\mathbf{d}_1$ (Definition 38.1) has the stride $f = f_{\mathbf{d}_1}$, the origin form $\omega_{\mathbf{d}_0}$ and the defects $c_e^{(0)} + k\, c_e^{(1)} = c_e(\mathbf{d}_0 + k\mathbf{d}_1)$. The restriction carries $r\,(c_e(\mathbf{d}) + \partial_{y_e})g$, $r \in \mathcal{O}_\ell$, to $r|_\ell\,(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})g$ in the relation space of the ray (Definition 38.4), and $\mathrm{Sh}_{\mathbf{d}_1}$ to Sh. When the marks of $F$ form one molecule it carries the values: for $x = \sum_i g_i \otimes r_i$ with every $r_i \in \mathcal{O}_\ell$, $\langle x \rangle(\mathbf{d}_0 + k\mathbf{d}_1) = \sum_i r_i|_\ell(k)\, \langle g_i \rangle_k$ wherever the integrals converge, and as meromorphic functions of $k$ when $\kappa_U(\mathbf{d}_1) > 0$ for every*

*nonempty proper $U$ (Proposition 38.5(ii)). So the module of a ray (Definition 39.1) is the module of the family restricted to a line of $\mathbb{D}_F$, and its relations are the relations in all the data read on that line.*

## 47.3 Boxes in all the data

**Definition 47.6** (Symbols, relations and boxes in all the data)**.** Let the marks of $F$ form one molecule, $\Bbbk$ as in Definition 47.1, and fix a positive direction $\mathbf{d}_1$.

(i) The *symbols* $(u;\mathbf{a},\mathbf{b})$, their evaluations, levels and walls, and the Laurent symbols, are those of Definition 39.15(i), (ii); a *formal combination* has its coefficients in $\Bbbk(\mathbb{D}_F)$. The *wall degree* of a symbol $g$ is $\deg_\kappa g = \max_U(-\kappa_U(g))/\kappa_U(\mathbf{d}_1)$ over the nonempty proper $U$.

(ii) The *mark relation* of a symbol $g=(u;\mathbf{a},\mathbf{b})$ at a mark $e$ is

$$\mathrm{MR}_e(g) = (m_e+u_e)\,g - \sum_j (s_j+a_j)\sum_{T\ni e}\theta_{j,T}\,(u+\mathbf{1}_T;\mathbf{a}+\mathbf{e}_j,\mathbf{b}) \\ - \sum_j (t_j+b_j)\sum_{T\not\ni e}\theta_{j,T}\,(u+\mathbf{1}_{\mathcal{E}\setminus T};\mathbf{a},\mathbf{b}+\mathbf{e}_j),$$

with coefficients affine functions on $\mathbb{D}_F$, none identically 0; it evaluates to $(c_e(\mathbf{d})+\partial_{y_e})$ applied to the evaluation of $g$. The *letter identities* of $g$ are those of Definition 39.15(iii).

(iii) *Level bounds* are $\bar{\mathbf{a}},\bar{\mathbf{b}}\in\mathbb{N}^N$ with $\bar a_j,\bar b_j\ge 1$ for every $j$. For a real $\mathfrak{t}\ge 0$ the *box* $\mathfrak{B}=\mathfrak{B}_\mathfrak{t}(\bar{\mathbf{a}},\bar{\mathbf{b}})$ is the set of the symbols of wall degree at most $\mathfrak{t}$ at the levels at most $(\bar{\mathbf{a}},\bar{\mathbf{b}})$, its *stems* are its symbols at the level $(\bar{\mathbf{a}}-\mathbf{1},\bar{\mathbf{b}}-\mathbf{1})$, $\mathcal{F}_\mathfrak{B}$ is the space of its formal combinations, and its *relations* $\mathcal{R}_\mathfrak{B}\subseteq\mathcal{F}_\mathfrak{B}$ are the $\Bbbk(\mathbb{D}_F)$-span of the letter identities of its symbols $(u;\mathbf{a},\mathbf{b})$, for $K_j$ when $a_j<\bar a_j$ and for $\Psi_j$ when $b_j<\bar b_j$, and of the mark relations of its stems at every mark.

(iv) The *generators* are $\lambda^{\mathrm{m}}_e=(\mathbf{e}_e-\mathbf{e}_{e_0};\mathbf{0},\mathbf{0})$ for $e\ne e_0$, $\lambda^K_j=(\mathbf{1}_{T_j};\mathbf{e}_j,\mathbf{0})$ and $\lambda^\Psi_j=(\mathbf{1}_{\mathcal{E}\setminus T'_j};\mathbf{0},\mathbf{e}_j)$, for fixed bases $T_j,T'_j$ of table $j$. Their *step* is $\mathfrak{c}=\max\max(0,\kappa_U(\mu))/\kappa_U(\mathbf{d}_1)$ over the generators $\mu$ and the nonempty proper $U$.

(v) The *reduction* of a box is that of Definition 39.20 with the trivial group: a ranking in which every symbol at a positive level ranks above every Laurent symbol and, of two Laurent symbols, the one of larger wall degree ranks higher; pivots, free symbols, and the reduction of a formal combination. For $\mathfrak{s}\ge 0$ with $\mathfrak{s}+\mathfrak{c}\le\mathfrak{t}$ the *spanning set* at $\mathfrak{s}$ is the list $\mathbf{b}$ of the free Laurent symbols of wall degree at most $\mathfrak{s}$, with $b_1=1$, and its *targets* are the Laurent polynomials $f_{-\mu}b_i$, $\mu$ a generator. The box is *accepted* at $\mathfrak{s}$ when the reduction of every target is a combination of the spanning set, $f_{-\mu}b_i\equiv\sum_l \mathbf{B}_{\mu,il}\,b_l$ modulo $\mathcal{R}_\mathfrak{B}$ with $\mathbf{B}_{\mu,il}\in\Bbbk(\mathbb{D}_F)$; the $\mathbf{B}_\mu$ are the *matrices* of the box.

**Proposition 47.7** (Wall degrees and the relations of a box)**.** *In the setting of Definition 47.6:*

*(i) For real $x$, $|g|\,f^x_{\mathbf{d}_1}$ is bounded on the positive orthant exactly when $x\ge\deg_\kappa g$, and $\langle g\rangle(\mathbf{d})$ converges absolutely at every $\mathbf{d}$ with $\mathrm{Re}\,\kappa_U(\mathbf{d})>\deg_\kappa g\,\kappa_U(\mathbf{d}_1)$ for every nonempty proper $U$.*

*(ii) A Laurent symbol $\alpha^u$ has the wall degree $\max_U(-u_U)/\kappa_U(\mathbf{d}_1)\ge 0$, zero exactly for $u=0$, and every monomial of $f_{-\mu}\alpha^u$, $\mu$ a generator, has wall degree at most $\deg_\kappa\alpha^u+\mathfrak{c}$.*

*(iii) The terms of the mark relations and of the letter identities of $g$ have wall degree at most $\deg_\kappa g$, and at every level there are finitely many symbols of wall degree at most $x$.*

*(iv) Every term of every relation of a box lies in the box; every relation evaluates into $\mathcal{L}_{F,\Bbbk}$, and its value vanishes identically on $\mathbb{D}_F$.*

*(v) A formal combination of symbols of a box whose evaluation is 0 is a combination of letter identities of the box.*

*(vi) The mark relation at every mark of every symbol of wall degree at most $\mathfrak{t}$ at a level at most $(\bar{\mathbf{a}}-\mathbf{1},\bar{\mathbf{b}}-\mathbf{1})$ lies in $\mathcal{R}_{\mathfrak{B}}$.*

**Theorem 47.8** (The box in all the data)**.** *Let a box $\mathfrak{B}=\mathfrak{B}_{\mathfrak{t}}(\bar{\mathbf{a}},\bar{\mathbf{b}})$ be accepted at $\mathfrak{s}$, with the spanning set $\mathbf{b}$ and the matrices $\mathbf{B}_\mu$.*

*(i) In the module, $\mathrm{Sh}_{-\mu}[b_i]=[f_{-\mu}b_i]=\sum_l \mathbf{B}_{\mu,il}[b_l]$ for every generator $\mu$, and the classes of $\mathbf{b}$ span $\mathfrak{M}_F$. The spanning set is a set of masters exactly when its classes are linearly independent over $\mathbb{K}_F$, and then $\mathbf{B}_\mu=\mathbf{A}_{-\mu}$.*

*(ii)* The certificate. *There are formal combinations $G_{\mu,i,e}$ of stems such that*

$$f_{-\mu}b_i-\sum_l \mathbf{B}_{\mu,il}\,b_l=\sum_{e\in\mathcal{E}}\big(c_e(\mathbf{d})+\partial_{y_e}\big)G_{\mu,i,e}$$

*as functions on $X_F$ with coefficients in $\Bbbk(\mathbb{D}_F)$, the $G_{\mu,i,e}$ evaluated.*

*(iii) $\langle\mathbf{b}\rangle(\mathbf{d}-\mu)=\mathbf{B}_\mu(\mathbf{d})\,\langle\mathbf{b}\rangle(\mathbf{d})$ as an identity of meromorphic functions on $\mathbb{D}_F$. At every real $\mathbf{d}$ off the poles of the coefficients with $\kappa_U(\mathbf{d})>\mathfrak{t}\,\kappa_U(\mathbf{d}_1)$ for every nonempty proper $U$, every value in (ii) converges absolutely and every term $(c_e(\mathbf{d})+\partial_{y_e})s$, $s$ a stem, has the value 0, so the certificate integrates term by term.*

*(iv)* The search stops. *There are $\mathfrak{s}$, level bounds and $\mathfrak{t}\ge\mathfrak{s}+\mathfrak{c}$ at which the box is accepted. So a search that runs through all the $(\mathfrak{t},\bar{\mathbf{a}},\bar{\mathbf{b}},\mathfrak{s})$ with $\mathfrak{t}$ and $\mathfrak{s}$ in a fixed discrete set, in any order that reaches each, and tests the acceptance by one elimination over $\Bbbk(\mathbb{D}_F)$ at each, stops, with a spanning set of $\mathfrak{M}_F$, its matrices at the generators, and their certificates.*

## 47.4 Closings and the elements of value zero

**Definition 47.9** (Closings in all the data)**.** For $\mathbf{d}\in\mathbb{D}_F(\mathbb{Z})$ and an insertion $g$ such that $g\,\omega_{\mathbf{d}}/dy$ is a Laurent polynomial, as when $g=\alpha^u\prod_j K_j^{-a_j}\Psi_j^{-b_j}$ with $s_j+a_j\le 0$ and $t_j+b_j\le 0$ for every $j$, the *closing* of $g$ at $\mathbf{d}$ is the constant term

$$\mathsf{C}(g)(\mathbf{d})=\mathrm{CT}\left(g\,\omega_{\mathbf{d}}/dy\right)=\frac{1}{(2\pi i)^{n-1}}\oint_{|\alpha_e|=1,\ e\neq e_0} g\,\omega_{\mathbf{d}}$$

on the chart $\alpha_{e_0}=1$. For $x=\sum_i g_i\otimes r_i$, $\mathsf{C}(x)(\mathbf{d})=\sum_i r_i(\mathbf{d})\,\mathsf{C}(g_i)(\mathbf{d})$ at the $\mathbf{d}\in\mathbb{D}_F(\mathbb{Z})$ at which every $\mathsf{C}(g_i)(\mathbf{d})$ is defined and no $r_i$ has a pole.

**Proposition 47.10** (The closings are a solution in all the data)**.** *Where both sides are defined, $\mathsf{C}\big((c_e(\mathbf{d})+\partial_{y_e})g\big)(\mathbf{d})=0$ for every mark $e$ and $\mathsf{C}(f_\lambda g)(\mathbf{d})=\mathsf{C}(g)(\mathbf{d}+\lambda)$. When the marks of $F$ form one molecule, $\mathsf{C}(x)=0$ wherever it is defined for $x\in\mathcal{L}_F$. Along a ray with an integral origin $\mathbf{d}_0$ and a positive direction $\mathbf{d}_1$ the closings of Definition 39.30 are $\mathsf{C}_j(g)=\mathsf{C}(g)(\mathbf{d}_0-j\mathbf{d}_1)$, and $V_{\mathsf{C}}(x|_\ell)(k)=\mathsf{C}(x)(\mathbf{d}_0+k\mathbf{d}_1)$ at the integers $k\le-\max_i j(g_i)$, $j(g_i)$ that of Definition 39.30 for this ray, off the poles of the coefficients.*

**Theorem 47.11** (The elements of value zero)**.** *Let the marks of $F$ form one molecule, $\mathbf{d}_0\in\mathbb{D}_F(\mathbb{Z})$, and $m\in\mathfrak{M}_F$ with a representative $x$. The following are equivalent:*

*(a)* $\langle m\rangle = 0$ *on* $\mathbb{D}_F$*;*

*(b) for every positive direction* $\mathbf{d}_1$ *such that no coefficient of* $x$ *has a pole along the whole line* $\mathbf{d}_0 + \mathbb{C}\mathbf{d}_1$*,* $\langle m\rangle(\mathbf{d}_0 + k\mathbf{d}_1) = 0$ *for all large integers* $k$*;*

*(c) for every such* $\mathbf{d}_1$*,* $\mathsf{C}(x)(\mathbf{d}_0 + k\mathbf{d}_1) = 0$ *for all integers* $k$ *below some bound.*

*So* $\mathrm{Ann}\langle\cdot\rangle$ *is the submodule of the elements whose closings vanish along the rays from one integral point, and it is* 0 *(Proposition 47.15): (a)–(c) hold exactly when* $m = 0$*. Every relation among the values of insertions, as functions of all the data, is a relation of the module and is certified by these closings, which are proved relations (Definition 44.1); along each ray the vanishing is decided by finitely many closings (Theorem 44.16(ii)), rational numbers when the weights, the insertions of* $x$ *and its coefficients are rational; and two insertions have the same value as functions of all the data exactly when they have the same reduction on masters.*

So the values of a family enter its algebra only through the masters: the reduction, the shift matrices and the certificates are finite computations over $\Bbbk(\mathbb{D}_F)$, the closings are constant terms of Laurent polynomials, and a relation among values as functions of the data is a statement about the module, settled by closings.

The rays of Theorem 47.11 are infinitely many, and the lattice points of one line do not see a function of all the data. The submodule that an element generates carries the vanishing of its value from finitely many parallel rays to every parallel ray through the lattice, and every lattice point lies on one of these.

**Proposition 47.12** (Value zero by finitely many closings)**.** *Let the marks of* $F$ *form one molecule,* $\mathfrak{M}$ *be* $\mathfrak{M}_F$ *or its quotient by a submodule of elements of value zero,* $m \neq 0$ *in* $\mathfrak{M}$ *with a representative* $x$*,* $\mathfrak{N}$ *the submodule of* $\mathfrak{M}$ *generated by* $m$*, of dimension* $d$*, and* $\mathcal{G}$ *the generators of Definition 47.6(iv) with their negatives.*

*(i)* $\mathfrak{N}$ *is the* $\mathbb{K}_F$*-span of the* $\mathrm{Sh}_\lambda m$ *with* $\lambda$ *a sum of at most* $d-1$ *elements of* $\mathcal{G}$*. So it has a basis* $\mathrm{Sh}_{\lambda_1} m, \dots, \mathrm{Sh}_{\lambda_d} m$ *among these,* $\lambda_1 = \mathbf{0}$*, found by eliminations over* $\mathbb{K}_F$ *on their reductions on masters of* $\mathfrak{M}$ *(Theorem 47.4(v), Proposition 47.13), and matrices* $\mathbf{B}_\mu \in GL_d(\mathbb{K}_F)$*,* $\mu \in \mathcal{G}$*, with* $\mathrm{Sh}_\mu \mathrm{Sh}_{\lambda_k} m = \sum_l \mathbf{B}_{\mu,kl} \mathrm{Sh}_{\lambda_l} m$*.*

*(ii) Let* $\mathbf{d}_0 \in \mathbb{D}_F(\mathbb{Z})$*, and let* $\mathbf{d}_1$ *be a positive direction at which the top homogeneous part of no reduced denominator of a coefficient of* $x$*, or of an entry of a* $\mathbf{B}_\mu$*,* $\mu \in \mathcal{G}$*, vanishes. Then* $\langle m\rangle = 0$ *exactly when, for* $k = 1, \dots, d$*, the closings* $\mathsf{C}(x)(\mathbf{d}_0 + \lambda_k + j\mathbf{d}_1)$ *vanish at all integers* $j$ *below some bound. Along each of these* $d$ *rays finitely many closings decide (Theorem 44.16(ii)), rational numbers when the weights, the insertions of* $x$ *and its coefficients are rational. Such directions exist: the excluded ones lie on a proper algebraic subset of* $\mathbb{D}_F$*, and the positive directions are the lattice points of an open cone.*

*So whether two elements of* $\mathfrak{M}$*, two insertions among them, have the same value as functions of all the data is decided by finitely many closings, along finitely many parallel rays.*

## 47.5 Masters from a box

A box gives a spanning set and the matrices of the shifts by the negatives of the generators (Theorem 47.8). When those matrices are invertible they give the shifts by the generators as well, and the spanning set with these maps presents the module: the relations that the box did not see are generated, as in Theorem 48.18(iv), by the commutators of the maps and by the relations of the family at [1], so that a basis of the presented quotient is a set of masters.

**Proposition 47.13** (Masters from a box)**.** *Let the marks of $F$ form one molecule, and let a box be accepted at $\mathfrak{s}$, with the spanning set $\mathbf{b}$, $b_i = f_{\lambda_i}$, $b_1 = 1$, and the matrices $\mathbf{B}_\mu$ at the generators $\mu$ of Definition 47.6(iv).*

*(i) If some $\mathbf{B}_\mu$ is singular, the classes of $\mathbf{b}$ are not linearly independent.*

*(ii) Let every $\mathbf{B}_\mu$ be invertible. On $\Bbbk(\mathbb{D}_F)^{\mathbf{b}}$, with the basis $(\varepsilon_i)$, let $T_{-\mu}$ be the $\sigma_{-\mu}$-semilinear map with the matrix $\mathbf{B}_\mu$ and $T_\mu$ the $\sigma_\mu$-semilinear map with the matrix $\sigma_\mu(\mathbf{B}_\mu)^{-1}$, as the shifts act on coordinates (Theorem 48.18(iv)); let $\rho(\lambda)$ be the image of $\varepsilon_1$ under the maps of the steps of a path from $\mathbf{0}$ to $\lambda$ in the generators and their negatives, fixed for each $\lambda$; and let $\mathfrak{N}$ be the smallest subspace that these maps carry into itself and that contains the $T_\mu T_\nu \varepsilon_i - T_\nu T_\mu \varepsilon_i$, the $T_{-\mu}T_\mu\varepsilon_i - \varepsilon_i$ and $T_\mu T_{-\mu}\varepsilon_i - \varepsilon_i$, the $\varepsilon_i - \rho(\lambda_i)$, and the images under $\rho$ of the letter identities and of the mark relations of $F$ at [1], $\sum_T \theta_{j,T}\,\rho(\lambda_T^K) - \varepsilon_1$, $\sum_T \theta_{j,T}\,\rho(\lambda_T^\Psi) - \varepsilon_1$ and $m_e\varepsilon_1 - \sum_j s_j \sum_{T\ni e} \theta_{j,T}\,\rho(\lambda_T^K) - \sum_j t_j \sum_{T\not\ni e} \theta_{j,T}\,\rho(\lambda_T^\Psi)$, with $\lambda_T^K = (\mathbf{1}_T; \mathbf{e}_j, \mathbf{0})$ and $\lambda_T^\Psi = (\mathbf{1}_{\mathcal{E}\setminus T}; \mathbf{0}, \mathbf{e}_j)$. Then $\varepsilon_i \mapsto [b_i]$ is an isomorphism of $\Bbbk(\mathbb{D}_F)^{\mathbf{b}}/\mathfrak{N}$ onto $\mathfrak{M}_{F,\Bbbk}$ that carries the maps $T_{\pm\mu}$ to the shifts $\mathrm{Sh}_{\pm\mu}$. So a basis of the quotient among the $\varepsilon_i$, found by finitely many eliminations, is a set of masters, with its shift matrices at the generators and their negatives, and $\mathfrak{N}$ does not depend on the paths.*

*(iii) Some accepted box has a spanning set whose classes are linearly independent. So a search through the boxes of Theorem 47.8(iv) that also asks every $\mathbf{B}_\mu$ to be invertible stops, and (ii) then gives masters of $\mathfrak{M}_F$ with their shift matrices.*

*The same holds for a family of positive letters, with the boxes, the generators, the letter identities and the mark relations of Theorem 48.3(iv).*

## 47.6 Value zero on a subspace of the data

A question may be asked on a subspace of the data: on a hyperplane on which a balance holds, or on a slice on which some exponents are fixed. Such a subspace may contain no positive direction, and its integral points may carry no closings. The expansion of the values at the native gauge of one direction asks for neither. Every point of the subspace is the origin of a line in that direction, the expansion of the values along every such line is a formal series whose coefficients are rational functions of the origin, and the values vanish on the subspace exactly when these coefficients do; at a rational native gauge, with rational weights and coefficients, they are rational functions over $\mathbb{Q}$.

**Proposition 47.14** (Value zero on a subspace of the data)**.** *Let $W$ be a family of letters of tables (Theorem 48.23), $r = r_W$, $\mathbb{E} \subseteq \mathbb{D}_W$ a linear subspace with the field $\mathbb{C}(\mathbb{E})$ of its rational functions, and $x = \sum_l g_l \otimes r_l$ with strides $g_l = f_{\lambda_l}$, $\lambda_l \in \mathbb{D}_W(\mathbb{Z})$, which span the insertions (Definition 48.1(iii)), and $r_l \in \mathbb{C}(\mathbb{E})$. A* native direction *of $\mathbb{E}$ is a point $\mathbf{d}_1 \in \mathbb{E} \cap \mathbb{D}_W(\mathbb{Z})$ with every wall positive and every exponent $w_{1,i} \geq 0$, the letters with $w_{1,i} > 0$ forming a family $W_1$ of letters of tables with the active molecules of $W$. When $\mathbb{E}$ has one it meets the tube $\mathcal{T}_1$, and the* value *of $x$ on $\mathbb{E}$ is the meromorphic function $\langle x\rangle = \sum_l r_l \langle g_l\rangle$ there.*

*(i)* The native gauge. *For a native direction $\mathbf{d}_1$ the function $\log f$, $f = \omega_{\mathbf{d}_1}/dy$, is strictly concave on $V_W$ and tends to $-\infty$ at infinity, so it has one critical point $y^\star$, a nondegenerate maximum: the* native gauge $\alpha^\star = e^{y^\star}$ *of $\mathbf{d}_1$, with $f^\star = f(\alpha^\star)$ and $H$ the Hessian of $\log f$ at $y^\star$. Conversely, for a point $\alpha^\circ$ of the positive orthant and exponents $\mathbf{w}_1 \in \mathbb{N}^I$ whose letters with positive exponents form such a family $W_1$, the point $\big(\sum_i w_{1,i} R_i(\alpha^\circ); \mathbf{w}_1\big)$, $R_i$ the occupations, lies in $\mathbb{D}_W$, has every wall positive and has the native gauge $\alpha^\circ$. When it lies*

*in $\mathbb{E}$, and $\alpha^\circ$ and the weights are rational, an integral multiple of it is a native direction of $\mathbb{E}$ with the rational native gauge $\alpha^\circ$. Whatever the weights, the real points of $\mathbb{D}_W$ with the exponents of the letters outside $W_1$ equal to 0, those of $W_1$ positive and every wall positive form an open cone in the real points of the subspace on which those exponents vanish; when $\mathbb{E}$ is defined over $\mathbb{Q}$ and meets that cone, its integral points in the cone are native directions of $\mathbb{E}$, and the native gauge of every native direction, an isolated nondegenerate common zero of its defects, which are rational functions over the field of the weights, has coordinates on a chart of $V_W$ algebraic over that field.*

*(ii)* The expansion. *Let $\mathbf{d}_1$ be a native direction of $\mathbb{E}$, $p \in \mathbb{E}$ off a proper algebraic subset, and $\nu$ the largest degree in $k$ at $k = \infty$ of the $r_l(p + k\mathbf{d}_1)$. As $k \to \infty$ through the reals,*

$$\langle x\rangle(p + k\mathbf{d}_1) \;\sim\; c_\star \,\frac{\omega_p}{dy}(\alpha^\star)\,(f^\star)^k\, k^{\nu - r/2} \sum_{i\geq 0} Q_i(p)\, k^{-i}, \qquad c_\star = (2\pi)^{r/2}\det(-H)^{-1/2},$$

*the determinant taken in a basis of $V_W(\mathbb{Z})$, with $Q_i \in \mathbb{C}(\mathbb{E})$. Each $Q_i$ is a finite sum of products of coefficients of the expansions of the $r_l(p+k\mathbf{d}_1)$ at $k = \infty$, of the values $g_l(\alpha^\star)$, of polynomials in $p$ of degree at most $2i$ formed from the Taylor coefficients at $y^\star$ of the* $\log g_l$, *the* $\log L_i$ *and* $\log f$, *and of Gaussian moments of the covariance $(-H)^{-1}$. So, for $\mathbb{E}$ defined over $\Bbbk'$, the $Q_i$ lie in $\Bbbk'(\mathbb{E})$, $\Bbbk'$ the field generated over $\mathbb{Q}$ by the weights, the coefficients of the $r_l$ and $\alpha^\star$: rational functions over $\mathbb{Q}$ when these are rational and $\mathbb{E}$ is defined over $\mathbb{Q}$.*

*(iii)* Value zero. *$\langle x\rangle = 0$ on $\mathbb{E}$ exactly when every $Q_i$ is 0.*

*(iv)* Finitely many coefficients. *The $\mathbb{C}(\mathbb{E})$-span $\mathcal{L}_\mathbb{E}$ of the $(c_e(\mathbf{d}) + \partial_{y_e})g$, $g \in \mathbb{C}[X_W]$, $\mathbf{d}$ the point of $\mathbb{E}$, gives the* module of $W$ on $\mathbb{E}$,

$$\mathfrak{M}_{W,\mathbb{E}} = \big(\mathbb{C}[X_W] \otimes \mathbb{C}(\mathbb{E})\big)/\mathcal{L}_\mathbb{E}, \qquad \mathrm{Sh}_{\mathbf{d}_1}[g \otimes r] = [f_{\mathbf{d}_1} g \otimes \sigma_{\mathbf{d}_1} r],$$

*of finite dimension over $\mathbb{C}(\mathbb{E})$; the value on $\mathbb{E}$ vanishes on $\mathcal{L}_\mathbb{E}$, and $\langle \mathrm{Sh}_{\mathbf{d}_1} x\rangle = \sigma_{\mathbf{d}_1}\langle x\rangle$. So $x$ has a* law along $\mathbf{d}_1$, *$\sum_{j=0}^h q_j \,\mathrm{Sh}_{\mathbf{d}_1}^j x = 0$ in $\mathfrak{M}_{W,\mathbb{E}}$ with $q_j \in \mathbb{C}(\mathbb{E})$ and $q_0 q_h \neq 0$, found by a search through the finite supports of certificates, elements of $\mathcal{L}_\mathbb{E}$, each tested by one elimination over $\mathbb{C}(\mathbb{E})$. Write $q_j(p + k\mathbf{d}_1) = k^{\nu'} \sum_{a\geq 0} q_{j,a}(p)\, k^{-a}$ at $k = \infty$, $\nu'$ the largest degree of the $q_j(p + k\mathbf{d}_1)$, $\rho = \nu - r/2$, $\chi_{a,b} = \sum_j q_{j,a}(f^\star)^j j^b$ and*

$$C_s(N) = \sum_{a+b=s} \binom{\rho - N + s}{b} \chi_{a,b},$$

*polynomials in $N$ with coefficients in $\mathbb{C}(\mathbb{E})$, not all 0. Let $s_0$ be the least $s$ with $C_s \neq 0$, and $N_0$ the largest integer root of $C_{s_0}$ when it has one at least $s_0$, and $s_0$ otherwise. Then $Q_i = 0$ for every $i \leq N_0 - s_0$ implies $Q_i = 0$ for every $i$.*

*So whether an element has the value zero on a subspace of the data is decided by finitely many identities of rational functions on the subspace, identities over $\mathbb{Q}$ at a rational native gauge, whatever positive directions and closings the subspace carries.*

A positive direction is a native direction, and on $\mathbb{E} = \mathbb{D}_W$ the proposition decides what Proposition 47.12 decides by closings. The closings ask for integral origins and for the low end of a ray, at which every letter is a polynomial. On a subspace cut out by a linear form in the exponents of some letters whose coefficients have one sign, a ray inside it has such a low end only when those exponents vanish at its origin and along it, so that the insertions in which those letters are

denominators carry no closings there. The expansion at the native gauge needs no integral origin and no low end: its origin is the point of the subspace itself, and the letters whose exponents the direction leaves at 0 enter only through the Taylor coefficients of the integrand at the native gauge, as polynomials in the origin.

## 47.7 The values separate the module

The masses act on the values as the Euler derivatives act on the integrand. Multiplying an insertion by the mass of a mark is, by the mark relation of that mark, taking minus its derivative in that mark and adding its products with the occupations of the mark, weighted by the exponents; so, modulo the relations and after one denominator is cleared, every element is the class of one combination of insertions whose coefficients are polynomials in the exponents alone. At fixed exponents its value, as a function of the masses, is the Mellin transform, over the positive points of the letter complement, of that combination times the word of the letters at those exponents, the masses ranging over the space of directions, and a Mellin transform determines the function it transforms. So an element whose value vanishes as a function of all the data is zero in the module: every linear relation among the values of a family, with coefficients rational in its data, is a relation of its module, and the closings that tell the elements of value zero (Theorem 47.11, Proposition 47.12) separate the module. The same holds on a subspace of the data cut out by conditions on the exponents alone that meets the tube, such a subspace leaving every mass free. A subspace that ties the masses can carry nonzero elements of value zero: along a ray kept by symmetries the values vanish on the elements that the symmetries move (Theorem 39.12(ii)), and on every subspace with a native direction Proposition 47.14 tells them.

**Proposition 47.15** (The values separate the module)**.** *Let $W$ be a family of positive letters (Definition 48.1), $\pi(\mathbf{m};\mathbf{w}) = \mathbf{w}$ the projection of $\mathbb{D}_W$ onto the exponents, $\mathbb{E}_{\mathbf{w}} \subseteq \mathbb{C}^I$ a linear subspace, and $\mathbb{E} = \pi^{-1}(\mathbb{E}_{\mathbf{w}})$, the data whose exponents lie in $\mathbb{E}_{\mathbf{w}}$, with every mass that $\mathbb{D}_W$ allows at them; $\mathbb{E} = \mathbb{D}_W$ for $\mathbb{E}_{\mathbf{w}} = \mathbb{C}^I$. Let $\mathbb{E}$ meet the tube $\mathcal{T}_1$. The* module of $W$ on $\mathbb{E}$ *is $\mathfrak{M}_{W,\mathbb{E}} = \big(\mathbb{C}[X_W] \otimes \mathbb{C}(\mathbb{E})\big)/\mathcal{L}_{\mathbb{E}}$, $\mathcal{L}_{\mathbb{E}}$ the $\mathbb{C}(\mathbb{E})$-span of the $(c_e(\mathbf{d}) + \partial_{y_e})g$, $g \in \mathbb{C}[X_W]$, $e \in B$, $\mathbf{d}$ the point of $\mathbb{E}$, as in Proposition 47.14(iv); it is $\mathfrak{M}_W$ for $\mathbb{E} = \mathbb{D}_W$. The* value *on $\mathbb{E}$ of $x = \sum_l g_l \otimes r_l$ is $\langle x\rangle = \sum_l r_l \langle g_l\rangle$ there.*

*(i)* The masses act through the mark relations. *For $g \in \mathbb{C}[X_W]$, $r \in \mathbb{C}(\mathbb{E})$ and $e \in B$, in $\mathfrak{M}_{W,\mathbb{E}}$,*

$$[g \otimes m_e\, r] = \Big[\sum_i R_{i,e}\, g \otimes w_i\, r - \partial_{y_e} g \otimes r\Big],$$

*$R_{i,e}$ the occupations. So every element, times a nonzero polynomial on $\mathbb{E}$, is the class of an element $\sum_k G_k \otimes p_k$, $G_k \in \mathbb{C}[X_W]$, whose coefficients $p_k$ are polynomials in the exponents alone.*

*(ii)* Separation. *The value on $\mathbb{E}$ is well defined on $\mathfrak{M}_{W,\mathbb{E}}$, and an element whose value vanishes on $\mathbb{E}$ is 0. So a combination of insertions with coefficients in $\mathbb{C}(\mathbb{E})$ has the value zero on $\mathbb{E}$ exactly when it lies in $\mathcal{L}_{\mathbb{E}}$; two insertions have the same value on $\mathbb{E}$ exactly when they have the same class; and insertions whose classes are linearly independent have values that are linearly independent over $\mathbb{C}(\mathbb{E})$.*

*(iii)* In all the data. *$\mathrm{Ann}\langle\cdot\rangle = 0$ in $\mathfrak{M}_W$, and the quotient of $\mathfrak{M}_W$ by a submodule of elements of value zero is $\mathfrak{M}_W$. So two insertions have the same value as functions of all the data exactly when they have the same reduction on masters, and the values of masters are linearly*

*independent over $\mathbb{K}_W$. For a family of letters of tables an element of $\mathfrak{M}_W$ is 0 exactly when its closings vanish along the rays of Theorem 47.11(c) from one integral point, and exactly when they vanish along the $d$ parallel rays of Proposition 47.12(ii) (Theorem 48.23(ii)): finitely many closings separate the module.*

*(iv)* On a subspace with a native direction. *For a family of letters of tables and $\mathbb{E} = \pi^{-1}(\mathbb{E}_{\mathbf{w}})$ as above with a native direction $\mathbf{d}_1$, an element of $\mathfrak{M}_{W,\mathbb{E}}$ is 0 exactly when finitely many coefficients of the expansion of its value at the native gauge of $\mathbf{d}_1$ vanish (Proposition 47.14(iii), (iv)).*

*So every linear relation among the values of a family, with coefficients rational in its data, or in the data whose exponents lie on a linear subspace when those data meet the tube, is a relation of its module, or of its module on those data, a combination of its mark relations: integration by parts in the marks gives every relation that the values satisfy as functions of the masses.*

The masses are what the proof uses. At fixed exponents a relation among values, as functions of the masses, is a relation among Mellin transforms, and the Mellin transform is injective; a subspace that fixes a combination of the masses at fixed exponents sees the transform only on a section of the masses, which is the transform of the integral of the integrand along the orbits of a subgroup of the torus, and a nonzero element can integrate to zero there. So on a ray or on the symmetric data of a family with a symmetry the module of the subspace can carry elements of value zero, while in all the data the quotients by submodules of elements of value zero in Proposition 47.12 and in Theorems 48.23, 48.18 and 48.20 are the modules themselves, and the closings along the $d$ parallel rays of a nonzero element do not all vanish.

# Chapter 48

# The module along the tree

Along its tree a family is read as a fibre integral over pins: a quotient of Gamma functions of the shares times the periods of the words of its nodes, each word a family of positive letters whose exponents are shares (Theorems 28.10(iii) and 28.12, Corollary 29.5). The module of Chapter 47 belongs to every family of positive letters. Read on the space of the directions that its Newton sum sees, such a family has a data space, insertions, relations, and a module of finite dimension over the rational functions of its data, with shifts, masters and boxes, whatever its letters. Through such a reading the module of a family is built from the modules of its nodes. Wherever the contours of the reading exist, every value of every element is the same fibre integral with the periods of the nodes replaced by a multilinear form in the values of their masters, its coefficients rational in the data and the shares. The lattice of the data acts through the Kronecker products of the shift matrices of the nodes along the paths of the bases through the tree, the kernel contributing Pochhammer symbols, and a node that recurs up to a bijection of its marks has one module. Along a chain of bunches the chance of a basis is the probability of the path of its stations under the flow of the shares, times the chances of its pieces in the entries of the nodes. Every tree of nested bunches is such a reading, node by node. A letter is also read on its scale, the Mellin transform of its shares: every full node reads on its scales the blocks of any partition of its parts that its seam letters separate, at one point, each block through its transform, so that a repeated block is one transform to a power, and the group of what the word reads there, at any number of scales, is a quotient of the group of its distinct blocks and of the exponentials of their constant parts. Theorem 45.34 is the case of one table at width one, its series and parallel nodes read through their cells. Along every tree the relations of the family ride on the paths of the terms of its letters, so that, modulo the translations along the fibres, the integrand carries the module of the family itself: the module of a family is the direct image of the modules of its nodes, presented node by node from the leaves up, once for each class of subtrees, by eliminations on identities among the products of the masters of the nodes, the values separating the module (Proposition 47.15), and the closings of the node words and of the seam words of the leaves, combined along the tree by flows of integers, tell its zeros. On the scales the same holds block by block: the module of the transform of a block, over the rational functions of the data that it reads and of its scales, is the direct image of the modules of its children modulo the translations along its fibre, each twisted by a monomial in the scales, the scales acting by connections that are combinations of its shifts, up to elements of value zero only where the transform reads a hyperplane of the data of its block; a block without children is a Gramma, read so on its shares. The last section puts these readings in order as one procedure, which stops on every family and reads it in all its data, absent letters and several molecules included.

## 48.1 Families of positive letters

The letters of a table of positive rank and corank are positive letters, and the seam letters of a piece, the entries of a transfer matrix and the station letters of a room are positive letters or positive constants. Every definition of Chapter 47 is made for a finite family of positive letters, on the space of the directions of its Newton sum.

**Definition 48.1** (A family of positive letters)**.** Let $B$ be a finite set and $W = (L_i)_{i\in I}$ a finite nonempty family of nonzero homogeneous polynomials on $B$ with positive coefficients, $L_i = \sum_{\mathbf{t}} c_{i,\mathbf{t}}\, \alpha^{\mathbf{t}}$, in a field $\Bbbk \subseteq \mathbb{C}$: positive letters (Definition 25.1) or positive constants, a constant having the one exponent $\mathbf{0}$ and the degree 0. These are its *letters*, and $W$ is a *family of positive letters*. Fix an exponent $\mathbf{t}_i$ of the support of each $L_i$.

(i) The *space of directions* $\vec{\mathcal{N}}_W \subseteq \mathbb{R}^B$ is spanned by the differences of two exponents of the support of one letter, over all the letters: the space of the directions of the Newton sum $\sum_i w_i \mathcal{N}_{L_i}$ at any positive exponents. It lies in $\mathbf{1}^\perp$, and $W$ is *full* when it is $\mathbf{1}^\perp$, the Newton sum having the dimension $|B| - 1$. Put $M_W = \vec{\mathcal{N}}_W \cap \mathbb{Z}^B$ and $r_W = \dim \vec{\mathcal{N}}_W$.

(ii) The *data space* of $W$ is

$$\mathbb{D}_W = \Big\{ \mathbf{d} = (\mathbf{m}; \mathbf{w}) \in \mathbb{C}^B \times \mathbb{C}^I : \ \mathbf{m} - \sum_i w_i \mathbf{t}_i \in \vec{\mathcal{N}}_W \otimes \mathbb{C} \Big\},$$

which does not depend on the exponents fixed; for a full family it is the balance hyperplane $m_B = \sum_i w_i \deg L_i$. Its integral points $\mathbb{D}_W(\mathbb{Z})$ are its *lattice*, $\mathbb{D}_W(\mathbb{R})$ is the set of its real points, and its fields of rational functions are $\Bbbk(\mathbb{D}_W) \subseteq \mathbb{K}_W = \mathbb{C}(\mathbb{D}_W)$. The *word* of $W$ at the exponents $\mathbf{w}$ is $W(\mathbf{w}) = \prod_i L_i^{-w_i}$.

(iii) The subtorus of $(\mathbb{C}^\times)^B$ on which every $\alpha^u$, $u \in M_W$, takes the value 1 acts on $(\mathbb{C}^\times)^B$ by multiplication, every letter being semi-invariant, and the *letter complement* $X_W$ is the quotient by it of the set on which no letter vanishes. Its regular functions, the *insertions* $\mathbb{C}[X_W]$, are the $\mathbb{C}$-span of the $\alpha^u \prod_i L_i^{-a_i}$ with $\mathbf{a} \in \mathbb{N}^I$ and $(u; \mathbf{a}) \in \mathbb{D}_W(\mathbb{Z})$; the *Laurent insertions* are those with $\mathbf{a} = \mathbf{0}$, the span of the $\alpha^u$, $u \in M_W$. For $\lambda = (v; \mathbf{a}) \in \mathbb{D}_W(\mathbb{Z})$ the *stride* $f_\lambda = \alpha^v \prod_i L_i^{-a_i}$ is a unit of $\mathbb{C}[X_W]$, and $f_\lambda f_\mu = f_{\lambda+\mu}$.

(iv) On $V_W = \mathbb{R}^B / \vec{\mathcal{N}}_W^\perp$, with the lattice $V_W(\mathbb{Z})$, the image of $\mathbb{Z}^B$, and the measure $dy$ for which it has the covolume one, the *master form* at $\mathbf{d} \in \mathbb{D}_W$ is $\omega_{\mathbf{d}} = e^{\langle \mathbf{m}, y\rangle} \prod_i L_i(e^y)^{-w_i}\, dy$, the powers principal: its density is invariant under $\vec{\mathcal{N}}_W^\perp$, so it is a form on $V_W$, and $f_\lambda\, \omega_{\mathbf{d}} = \omega_{\mathbf{d}+\lambda}$. The *occupations* are the insertions $R_{i,e} = \alpha_e \partial_{\alpha_e} L_i / L_i$, with $0 \le R_{i,e} \le \deg L_i$ on the orthant, and the *defects* are $c_e(\mathbf{d}) = m_e - \sum_i w_i R_{i,e}$; $c_e(\mathbf{d}) = \partial_{y_e} \log(\omega_{\mathbf{d}}/dy)$, $c_e(\lambda) = \partial_{y_e} \log f_\lambda$, and $\sum_e v_e\, c_e(\mathbf{d}) = 0$ on $\mathbb{D}_W$ for every $v \in \vec{\mathcal{N}}_W^\perp$.

(v) The *value* of an insertion $g$ at $\mathbf{d}$ is $\langle g \rangle(\mathbf{d}) = \int_{V_W} g\, \omega_{\mathbf{d}}$, where it converges absolutely, and $\langle 1 \rangle(\mathbf{d}) = P_W(\mathbf{m}; \mathbf{w})$ is the *period* of $W$. For a full family $V_W$ is read on the chart $y_{e_0} = 0$, and $P_W(\mathbf{m}; \mathbf{w})$ is the period of the word $W(\mathbf{w})$ at $\mathbf{m}$ (Definition 25.1); for every family it is the reduced period of that word, on its pins $\mathbf{m} - \sum_i w_i \mathbf{t}_i \in \vec{\mathcal{N}}_W$ (Theorem 28.12).

**Lemma 48.2** (The fan of a family of positive letters)**.** *Let $W$ be as in Definition 48.1, with $\vec{\mathcal{N}}_W$ paired with $V_W$.*

*(i) There are finitely many rational simplicial cones of dimension $r_W$ in $V_W$, with disjoint interiors, covering $V_W$, each contained, for every $i$, in the set $\{v : \ \langle \mathbf{t} - \mathbf{t}', v\rangle \le 0$ for every exponent $\mathbf{t}'$ of the support of $L_i\}$ of one exponent $\mathbf{t} = \mathbf{t}_{i,\pi}$ of that support. Such*

*cones form a* fan *of $W$, and the primitive vectors of $V_W(\mathbb{Z})$ on their rays are its* generators. *Every ray of the inner normal fan of the Newton sum at positive exponents is a ray of every fan.*

*(ii) The* wall *of a generator $v$ is the linear form on $\mathbb{D}_W$*

$$\kappa_v(\mathbf{d}) = \Big\langle \mathbf{m} - \sum_i w_i \mathbf{t}_i,\, v \Big\rangle - \sum_i w_i \min_{\mathbf{t}} \langle \mathbf{t} - \mathbf{t}_i, v\rangle,$$

*the minimum over the support of $L_i$. It does not depend on the exponents fixed, it is $\langle \mathbf{m} - \sum_i w_i \mathbf{t}_{i,\pi}, v\rangle$ for every cone $\pi$ of which $v$ is a generator, and it is integral on $\mathbb{D}_W(\mathbb{Z})$.*

*(iii) On a cone $\pi$ with the generators $v_{\pi,1}, \dots, v_{\pi,r_W}$ write $y = -\sum_k u_k v_{\pi,k}$, $u_k \ge 0$, and $q_k = e^{-u_k}$. Then $\mathbf{t}_{i,\pi}$ is unique, and $k_{i,\pi}(q) = \sum_{\mathbf{t}} c_{i,\mathbf{t}} \prod_k q_k^{\langle \mathbf{t} - \mathbf{t}_{i,\pi}, v_{\pi,k}\rangle}$ is a polynomial with nonnegative coefficients and the constant term $c_{i,\mathbf{t}_{i,\pi}} > 0$. For an insertion $g = \alpha^u \prod_i L_i^{-a_i}$, with the walls $\kappa_v(g) = \kappa_v(u; \mathbf{a})$ of its data, and real $\mathbf{d}$, in $[0,\infty]$,*

$$\int_{V_W} g\, \omega_{\mathbf{d}} = \sum_{\pi} \big|\det(v_{\pi,1}, \dots, v_{\pi,r_W})\big| \int_{[0,1]^{r_W}} \prod_k q_k^{\kappa_{v_{\pi,k}}(\mathbf{d}) + \kappa_{v_{\pi,k}}(g) - 1} \prod_i k_{i,\pi}(q)^{-w_i - a_i}\, dq,$$

*the determinant taken in $V_W(\mathbb{Z})$; both sides are finite exactly when $\kappa_v(\mathbf{d}) + \kappa_v(g) > 0$ for every generator $v$.*

*The* tube *of $g$ is $\mathcal{T}_g = \{\mathbf{d} \in \mathbb{D}_W :\ \operatorname{Re} \kappa_v(\mathbf{d}) + \kappa_v(g) > 0$ for every generator $v\}$, the set on which $\langle g\rangle$ converges absolutely; it does not depend on the fan.*

For a full family the generators are read on the section $v_{e_0} = 0$, and the lemma is Lemma S14 with the walls written as linear forms in all the data; the walls of Lemma 32.8(i) are these walls read on an affine subspace of the data.

**Theorem 48.3** (The module of a family of positive letters)**.** *Let $W$ be a family of positive letters with a fan.*

*(i)* Positive directions. *At real data with positive exponents the period converges exactly when $\mathbf{m} - \sum_i w_i \mathbf{t}_i$ lies in the relative interior of $\sum_i w_i(\mathcal{N}_{L_i} - \mathbf{t}_i)$, a polytope of the dimension $r_W$ in $\vec{\mathcal{N}}_W$. There are* positive directions, *points $\mathbf{d}_1 \in \mathbb{D}_W(\mathbb{Z})$ with every exponent $w_{1,i} \ge 1$ and every wall $\kappa_v(\mathbf{d}_1) > 0$; for each, $\eta\, \mathbf{d}_1 \in \mathcal{T}_1$ for every $\eta > 0$.*

*(ii)* Relations. *For an insertion $g = \alpha^u \prod_i L_i^{-a_i}$ and $\mathbf{d} \in \mathcal{T}_g$ the values $\langle g\rangle(\mathbf{d})$ and $\langle (c_e(\mathbf{d}) + \partial_{y_e}) g\rangle(\mathbf{d})$ converge absolutely and the second is 0, for every $e \in B$; $\langle f_\lambda g\rangle(\mathbf{d}) = \langle g\rangle(\mathbf{d} + \lambda)$ when $\mathbf{d} + \lambda \in \mathcal{T}_g$. The value $\langle g\rangle$ continues to a meromorphic function on $\mathbb{D}_W$, holomorphic on $\mathcal{T}_g$, with its poles on the hyperplanes $\kappa_v(\mathbf{d}) + \kappa_v(g) \in \{0, -1, -2, \dots\}$, $v$ a generator. Extended $\mathbb{K}_W$-linearly to $\mathbb{C}[X_W] \otimes \mathbb{K}_W$ it vanishes identically on the $\mathbb{K}_W$-span $\mathcal{L}_W$ of the $(c_e(\mathbf{d}) + \partial_{y_e}) g$, $g \in \mathbb{C}[X_W]$, $e \in B$, and $\langle f_\lambda g\rangle(\mathbf{d}) = \langle g\rangle(\mathbf{d} + \lambda)$ identically. At every real $\mathbf{d} \in \mathcal{T}_1$, $P_W(\mathbf{m}; \mathbf{w}) > 0$.*

*(iii)* The module. *With the relation space $\mathcal{L}_W$, the module $\mathfrak{M}_W = (\mathbb{C}[X_W] \otimes \mathbb{K}_W)/\mathcal{L}_W$, its shifts $\operatorname{Sh}_\lambda[g \otimes r] = [f_\lambda g \otimes \sigma_\lambda r]$, $\lambda \in \mathbb{D}_W(\mathbb{Z})$, its submodules, masters, shift matrices, reductions and values, and $\mathfrak{M}_{W,\mathbb{k}}$, are defined as in Definition 47.3, and Theorem 47.4 holds for them: the lattice acts; the values intertwine the shifts;* $\operatorname{Ann}\langle\cdot\rangle$ *is a submodule that does not contain* $[1]$*; the shifts of* $[1]$ *span, and so do the classes of the Laurent insertions; $\mathfrak{M}_W$ has a finite dimension $\mathfrak{m}_W$, being the top cohomology of the twisted algebraic de Rham complex, over $\mathbb{K}_W$, of the smooth affine variety $X_W$ of dimension $r_W$ for the connection $d + d\log(\omega_{\mathbf{d}}/dy)$; the base change holds; and the shift matrices are a cocycle that transports the values of the masters.*

*(iv)* Boxes. *Definition 47.6, Proposition 47.7 and Theorem 47.8 hold for $W$ with these readings: the symbols are the $(u;\mathbf{a}) \in \mathbb{D}_W(\mathbb{Z})$ with $\mathbf{a} \in \mathbb{N}^I$, evaluated to $\alpha^u \prod_i L_i^{-a_i}$, at the level $\mathbf{a}$, with the walls $\kappa_v(u;\mathbf{a})$ and the wall degree $\max_v(-\kappa_v(g))/\kappa_v(\mathbf{d}_1)$ over the generators; the Laurent symbols are those at the level $\mathbf{0}$; the mark relations are*

$$\mathrm{MR}_e(g) = (m_e + u_e)\, g - \sum_i (w_i + a_i) \sum_{\mathbf{t}} c_{i,\mathbf{t}}\, t_e\, (u + \mathbf{t}; \mathbf{a} + \mathbf{e}_i), \qquad e \in B,$$

*and the letter identities $\sum_{\mathbf{t}} c_{i,\mathbf{t}}\, (u+\mathbf{t};\mathbf{a}+\mathbf{e}_i) - (u;\mathbf{a})$; the level bounds are $\bar{\mathbf{a}} \in \mathbb{N}^I$ with every $\bar{a}_i \geq 1$; and the generators of the lattice are the $(v;\mathbf{0})$, $v$ in a basis of $M_W$, and the $(\mathbf{t}_i;\mathbf{e}_i)$, $i \in I$.*

*(v)* Families of tables. *A family of tables $F$ is the family of positive letters $(K_j, \Psi_j)_j$ at the exponents $(\mathbf{s},\mathbf{t})$, and its space of directions is $\mathbf{1}^\perp$ exactly when its marks form one molecule. Then the sectors of the orders of $\mathcal{E}$ form a fan, with the generators $\mathbf{1}_U$, $U$ nonempty and proper, and the walls $\kappa_{\mathbf{1}_U} = \kappa_U$; the criterion of (i) is Theorem 25.2, the centre data are a positive direction (the gloss after Definition 47.1), and (ii)–(iv) are Theorems 47.2, 47.4 and 47.8 and Proposition 47.7. For a compound, $\vec{\mathcal{N}}_F^\perp$ is spanned by the indicators of its molecules, every molecule is balanced on $\mathbb{D}_F$, and when the weights factor across the molecules the period read on $V_F$ is the product of the periods of the molecules (Proposition 7.9).*

So the module of Chapter 47 is the module of a family of positive letters that happen to be the letters of tables, and nothing in it used more. The seam word of a piece (Lemma 28.7), the room word of a partner, the word of a node of a chain (Corollary 29.5), the words of the segments of a cyclic chain (Theorem 29.8) and the merged seam words (Lemma 32.8(ii)) are families of positive letters at the shares. Every such family has positive directions, and so a nonempty tube, whatever its letters; a degenerate one is read on its space of directions, where its period is the reduced period of Theorem 28.12, the pins of a degenerate word being part of its data space.

## 48.2 Share readings

The readings of a family along its tree have one form: a fibre integral of a quotient of Gamma functions of linear forms in the data and the shares, times the periods of families of positive letters at data that are linear in the data and the shares.

**Definition 48.4** (A share reading)**.** Let $W$ be a family of positive letters. A *share reading* of $W$ consists of:

(i) *shares* $\mathbf{z} \in \mathbb{C}^J$, $J$ finite, with *pins* $\mathbf{P}\mathbf{z} = \boldsymbol{\pi}(\mathbf{d})$, $\mathbf{P}$ a rational matrix with linearly independent rows and $\boldsymbol{\pi}$ a linear map on $\mathbb{D}_W$ with rational coefficients, such that every $\mathbf{d}$ has shares on its pins; the *share space* $\mathbb{S} = \{(\mathbf{d},\mathbf{z}) \in \mathbb{D}_W \times \mathbb{C}^J :\ \mathbf{P}\mathbf{z} = \boldsymbol{\pi}(\mathbf{d})\}$, with its lattice $\mathbb{S}(\mathbb{Z})$ of integral points, its real points $\mathbb{S}(\mathbb{R})$ and $\mathbb{K}_{\mathbb{S}} = \mathbb{C}(\mathbb{S})$;

(ii) a *kernel* $G = \prod_{k\in A} \Gamma(\ell_k) \big/ \prod_{k\in A'} \Gamma(\ell'_k)$, $A$ and $A'$ finite and the $\ell_k$ and $\ell'_k$ linear forms on $\mathbb{S}$ with integral values on $\mathbb{S}(\mathbb{Z})$;

(iii) *nodes* $(W_l, \iota_l)$, $l = 1,\dots,p$: families of positive letters $W_l$ with linear maps $\iota_l$ of $\mathbb{S}$ onto $\mathbb{D}_{W_l}$, $\iota_l(\mathbb{S}(\mathbb{Z})) \subseteq \mathbb{D}_{W_l}(\mathbb{Z})$;

such that the following holds. Let the *admissible cone* be

$$\mathbb{S}^+ = \big\{(\mathbf{d},\mathbf{c}) \in \mathbb{S}(\mathbb{R}) :\ \ell_k(\mathbf{d},\mathbf{c}) > 0 \text{ for every } k \in A,\ \iota_l(\mathbf{d},\mathbf{c}) \in \mathcal{T}_1(W_l) \text{ for every } l\big\},$$

and call a real $\mathbf{c}$ *admissible* at $\mathbf{d} \in \mathbb{D}_W$ when $(\operatorname{Re}\mathbf{d}, \mathbf{c}) \in \mathbb{S}^+$. At every $\mathbf{d}$ at which some $\mathbf{c}$ is admissible, $\mathbf{d} \in \mathcal{T}_1(W)$ and, for every admissible $\mathbf{c}$, the *reading identity*

$$P_W(\mathbf{d}) = \int_{\mathbf{Pz}=\boldsymbol{\pi}(\mathbf{d})} G(\mathbf{d},\mathbf{z}) \prod_{l=1}^{p} P_{W_l}\big(\iota_l(\mathbf{d},\mathbf{z})\big)$$

holds, the fibre integral through $\mathbf{c}$ (Definition 4.1) converging absolutely. The reading is *positive* when $\mathbb{S}^+$ is not empty.

The admissible cone is an open convex cone, cut out by finitely many strict linear inequalities, and at a real $\mathbf{d}$ the contour through $\mathbf{c}$ is $\mathbf{c} + iV$, $V$ the real space of the directions of the pins.

**Theorem 48.5** (The module along a share reading)**.** *Let a family of positive letters $W$ have a positive share reading with the nodes $(W_l, \iota_l)_{l\le p}$, and let $\mathbb{D}'_W(\mathbb{Z})$ be the image of $\mathbb{S}(\mathbb{Z})$ in $\mathbb{D}_W(\mathbb{Z})$. In this statement $\sigma_\mu$ is the translation of rational functions by $\mu$ on whichever space they live.*

*(i)* The kernel. *For $(\lambda,\tau) \in \mathbb{S}(\mathbb{Z})$, $G\big((\mathbf{d},\mathbf{z}) + (\lambda,\tau)\big) = R_{\lambda,\tau}(\mathbf{d},\mathbf{z})\, G(\mathbf{d},\mathbf{z})$ with*

$$R_{\lambda,\tau} = \prod_{k\in A} (\ell_k)_{\ell_k(\lambda,\tau)} \Big/ \prod_{k\in A'} (\ell'_k)_{\ell'_k(\lambda,\tau)} \in \mathbb{Q}(\mathbb{S}),$$

*$(x)_n = \Gamma(x+n)/\Gamma(x)$ the Pochhammer symbol at an integer $n$, and $R_{\mu+\nu} = \sigma_\mu(R_\nu)\, R_\mu$.*

*(ii)* The integrand module. *The tensor product over $\mathbb{K}_\mathbb{S}$ of the modules of the nodes with their coefficients read through the $\iota_l$,*

$$\mathfrak{J} = \bigotimes_{l=1}^{p} \big(\mathfrak{M}_{W_l} \otimes_{\mathbb{K}_{W_l}} \mathbb{K}_\mathbb{S}\big),$$

*has the dimension $\prod_l \mathfrak{m}_{W_l}$, and $\mathbb{S}(\mathbb{Z})$ acts on it by the semilinear maps $\operatorname{Sh}_{(\lambda,\tau)} = R_{\lambda,\tau} \bigotimes_l \operatorname{Sh}_{\iota_l(\lambda,\tau)}$. For masters $\mathbf{g}^{(l)}$ of the nodes the products $\bigotimes_l [g_{i_l}^{(l)}]$ are a basis, on which $\operatorname{Sh}_{(\lambda,\tau)}$ has the matrix $R_{\lambda,\tau} \bigotimes_l \iota_l^* \mathbf{A}^{(l)}_{\iota_l(\lambda,\tau)}$, the Kronecker product of the shift matrices of the nodes read through the $\iota_l$, times $R_{\lambda,\tau}$. The* value *of $\bigotimes_l x_l$ is $G \prod_l \langle x_l\rangle \circ \iota_l$, extended $\mathbb{K}_\mathbb{S}$-linearly, a meromorphic function on $\mathbb{S}$, and the value of $\operatorname{Sh}_{(\lambda,\tau)} t$ at $(\mathbf{d},\mathbf{z})$ is that of $t$ at $(\mathbf{d},\mathbf{z}) + (\lambda,\tau)$.*

*(iii)* Values. *Let $[1]^\otimes = \bigotimes_l [1]$. For $(\lambda,\tau) \in \mathbb{S}(\mathbb{Z})$, a real $\mathbf{d}$ and an admissible $\mathbf{c}$ at $\mathbf{d}$ such that $\mathbf{c}+\tau$ is admissible at $\mathbf{d}+\lambda$,*

$$\langle \operatorname{Sh}_\lambda[1]\rangle(\mathbf{d}) = P_W(\mathbf{d}+\lambda) = \int_{\mathbf{Pz}=\boldsymbol{\pi}(\mathbf{d})} G\, R_{\lambda,\tau} \prod_{l=1}^{p} \big(\boldsymbol{\rho}^{(l)}_{\iota_l(\lambda,\tau)} \cdot \langle \mathbf{g}^{(l)}\rangle\big) \circ \iota_l,$$

*the value of $\operatorname{Sh}_{(\lambda,\tau)}[1]^\otimes$ integrated through $\mathbf{c}$, with $\boldsymbol{\rho}^{(l)}_\mu$ the reduction of $\operatorname{Sh}_\mu[1]$ on the masters $\mathbf{g}^{(l)}$ and $\langle \mathbf{g}^{(l)}\rangle$ the column of their values. So the value of an element $x = \sum_\lambda r_\lambda \operatorname{Sh}_\lambda[1]$, $r_\lambda \in \mathbb{K}_W$, of the span $\mathfrak{M}'_W$ of the shifts of $[1]$ by $\mathbb{D}'_W(\mathbb{Z})$, for any choice of the $\tau_\lambda$ with $(\lambda,\tau_\lambda) \in \mathbb{S}(\mathbb{Z})$, is at $\mathbf{d}$ the fibre integral of the value of $t_x = \sum_\lambda r_\lambda \operatorname{Sh}_{(\lambda,\tau_\lambda)}[1]^\otimes \in \mathfrak{J}$ through a point admissible at $\mathbf{d}$ whose translates by the $\tau_\lambda$ are admissible at the $\mathbf{d}+\lambda$: a multilinear form in the values of the masters of the nodes against the kernel, with coefficients in $\mathbb{K}_\mathbb{S}$ read off the reductions and the shift matrices of the nodes. For every $(\mathbf{d}_1,\mathbf{c}_1) \in \mathbb{S}^+$, every real $\mathbf{d}$ and every finite set of the $(\lambda,\tau_\lambda)$ such points exist at $\mathbf{d} + \eta\mathbf{d}_1$ for all large $\eta$. Every element of $\mathfrak{M}_W$ is a sum of shifts $\operatorname{Sh}_{\lambda_c} x_c$, $x_c \in \mathfrak{M}'_W$, over representatives $\lambda_c$ of the classes of $\mathbb{D}_W(\mathbb{Z})$ modulo $\mathbb{D}'_W(\mathbb{Z})$, so its value is a sum of such fibre integrals at the points $\mathbf{d}+\lambda_c$.*

*(iv)* Fibre translations. *For* $(0,\tau) \in \mathbb{S}(\mathbb{Z})$ *and* $\mathbf{c}$ *and* $\mathbf{c}+\tau$ *admissible at* $\mathbf{d}$*, the values of* $\mathrm{Sh}_{(0,\tau)}[1]^{\otimes}$ *and of* $[1]^{\otimes}$ *have one fibre integral through* $\mathbf{c}$*, the period* $P_W(\mathbf{d})$*.*

*(v)* Recurring nodes. *An isomorphism of two families of positive letters, a bijection of their marks and one of their letters carrying each letter to the other's, carries the data space, the lattice, the insertions, the relations, the module, its masters, shift matrices, reductions and boxes of one to those of the other. So the modules of the nodes of a reading are found once for each class of nodes under isomorphism, by one box each (Theorem 48.3(iv)).*

*(vi)* Composition. *Let the node* $W_{l_0}$ *have a positive share reading with the shares* $\mathbf{z}'$*, the pins* $\mathbf{P}'\mathbf{z}' = \boldsymbol{\pi}'(\mathbf{d}')$*, the kernel* $G'$ *and the nodes* $(V_k, \iota'_k)$*, and let* $\mathbb{S}''$ *be the space of the* $(\mathbf{d}, \mathbf{z}, \mathbf{z}')$ *with* $(\mathbf{d}, \mathbf{z}) \in \mathbb{S}$ *and* $(\iota_{l_0}(\mathbf{d}, \mathbf{z}), \mathbf{z}')$ *in the share space of that reading. If there are points* $(\mathbf{d}, \mathbf{c}, \mathbf{c}')$ *with* $(\mathbf{d}, \mathbf{c}) \in \mathbb{S}^+$ *and* $(\iota_{l_0}(\mathbf{d}, \mathbf{c}), \mathbf{c}')$ *in its admissible cone, and at every* $\mathbf{d} \in \mathbb{D}_W$ *and every* $(\mathbf{c}, \mathbf{c}')$ *with* $(\mathrm{Re}\,\mathbf{d}, \mathbf{c}, \mathbf{c}')$ *such a point the iterated fibre integral at* $\mathbf{d}$ *through* $(\mathbf{c}, \mathbf{c}')$ *converges absolutely, then* $\mathbb{S}''$*, both pins, the kernel* $G \cdot G'(\iota_{l_0}(\mathbf{d}, \mathbf{z}), \mathbf{z}')$ *and the nodes* $(W_l, \iota_l)$*,* $l \neq l_0$*, and* $(V_k, \iota'_k(\iota_{l_0}(\mathbf{d}, \mathbf{z}), \mathbf{z}'))$*, are a positive share reading of* $W$*, with these points as its admissible cone.*

So the algebra of a family is the algebra of the nodes of any of its share readings. The masters, the shift matrices and the certificates that the values need are those of the distinct nodes; the lattice of the data acts on the integrand through Kronecker products of the shift matrices of the nodes, with a rational factor from the kernel; and, at the points with an admissible point whose translates are admissible, which exist along $\mathbf{d} + \eta\mathbf{d}_1$ for all large $\eta$ (part (iii)), a value of the family at a lattice translate of a point is the fibre integral of the masters of the nodes at that point, against the kernel, with coefficients that are rational functions. The family's own masters are not needed for this. The fibre integral identifies more than the nodes do, the translations along the fibres among others (part (iv)); along a tree, modulo these translations, the integrand carries the module of the family itself, and a relation among values of a family of tables, as functions of all its data, is a relation of its module (Proposition 47.15), told by closings (Theorem 47.11) that the tree computes from the closings of its node words (Theorem 48.18).

## 48.3 Paths through a chain of bunches

Along a chain of bunches every basis of a table runs through one station of each bunch, and every lattice vector of the data of a molecule is carried by these runs to the shares.

**Definition 48.6** (The path of a basis)**.** Let $\varnothing = S_0 \subset S_1 \subset \cdots \subset S_L = \mathcal{E}$, $L \geq 2$, be a chain of bunches of a family of tables with the steps $B_{l-1} = S_l \setminus S_{l-1}$ and the stations $\mathcal{D}(S_l)$ of room sides (Theorem 29.3), and let $T$ be a basis of table $j$. For $1 \leq l \leq L-1$ the room side $T \setminus S_l$ of $S_l$ lies in one station $D_l(T)$. The *path* of $T$ is the sequence $D_1(T), \ldots, D_{L-1}(T)$, and its *entries* are the entries $(D_{l-1}(T), D_l(T))$ of the transfer matrices $\mathbf{N}^K_{l,j}$ and $\mathbf{N}^\Psi_{l,j}$, $2 \leq l \leq L-1$, and the entry $D_{L-1}(T)$ of the columns $\mathsf{y}_j$ and $\mathsf{y}^*_j$ of the last node. The *pieces* of $T$ are $T \cap S_1$, a basis of the room child of the station $D_1(T)$, whose letter $K_{j,(D_1(T))}$ has the monomial $\alpha^{T \cap S_1}$ and the cobasis letter $\Psi_{j,(D_1(T))}$ the monomial $\alpha^{S_1 \setminus T}$, and the $T \cap B_{l-1}$, $2 \leq l \leq L$, whose monomials $\alpha^{T \cap B_{l-1}}$ and $\alpha^{B_{l-1} \setminus T}$ occur in the entries of the path at the node $l$ of $K_j$ and of $\Psi_j$. With the shares of Corollary 29.5, the *path vectors* $\tau_T$ and $\tau^*_T$ put 1 on the share of every entry of the path of $T$, of $K_j$ and of $\Psi_j$ respectively, and 0 on every other share.

**Theorem 48.7** (A chain of bunches is a share reading)**.** *Let $F$ be a family of tables whose marks form one molecule, with a chain of bunches $\varnothing = S_0 \subset S_1 \subset \cdots \subset S_L = \mathcal{E}$, $L \geq 2$, read as the family of positive letters of Theorem 48.3(v) with every letter opened, as in Corollary 29.5.*

*(i)* The reading. *Corollary 29.5 is a positive share reading of $F$, with: the shares $e^{(l)}_{jD'D}$ and $f^{(l)}_{jD'D}$, one for each nonzero entry of $\mathbf{N}^K_{l,j}$ and of $\mathbf{N}^\Psi_{l,j}$, $2 \leq l \leq L$, the last ones the columns $\mathsf{y}_j$ and $\mathsf{y}^*_j$; the pins, the letter totals, the conservation of the loads at the stations of $S_2, \dots, S_{L-1}$, and the pins of the space of directions of the first bunch and of every node, less the homogeneity of one of them, those of a node $W_l$ taken on a basis of the lattice $\vec{\mathcal{N}}^\perp_{W_l} \cap \mathbb{Z}^{B_{l-1}}$ that contains $\mathbf{1}_{B_{l-1}}$ (Theorem 28.12), with their right-hand sides $\boldsymbol{\pi}$; the kernel*

$$\prod_j \frac{\prod_{l=2}^L \prod_{D',D} \Gamma(e^{(l)}_{jD'D})}{\Gamma(s_j) \prod_{l=2}^{L-1} \prod_D \Gamma(a^{(l)}_{jD})} \cdot \frac{\prod_{l=2}^L \prod_{D',D} \Gamma(f^{(l)}_{jD'D})}{\Gamma(t_j) \prod_{l=2}^{L-1} \prod_D \Gamma(b^{(l)}_{jD})};$$

*and the nodes $W_1$ on $B_0 = S_1$, the seam letters $K_{j,(D)}$ and $\Psi_{j,(D)}$ of $S_1$ at the exponents $a^{(1)}_{jD}$ and $b^{(1)}_{jD}$, and $W_l$ on $B_{l-1}$, $2 \leq l \leq L$, the nonzero entries at the exponents $e^{(l)}$ and $f^{(l)}$, with $\iota_1 = (\mathbf{m}_{B_0}; \mathbf{a}^{(1)}, \mathbf{b}^{(1)})$ and $\iota_l = (\mathbf{m}_{B_{l-1}}; \mathbf{e}^{(l)}, \mathbf{f}^{(l)})$.*

*(ii)* Paths. *For a basis $T$ of table $j$ the vectors $(\lambda^K_T, \tau_T)$ and $(\lambda^\Psi_T, \tau^*_T)$ lie in $\mathbb{S}(\mathbb{Z})$, $\lambda^K_T = (\mathbf{1}_T; \mathbf{e}_j, \mathbf{0})$ and $\lambda^\Psi_T = (\mathbf{1}_{\mathcal{E}\setminus T}; \mathbf{0}, \mathbf{e}_j)$ being the data of the strides $\alpha^T/K_j$ and $\alpha^{\mathcal{E}\setminus T}/\Psi_j$. Each node moves by the data of the stride of the piece of $T$ in its entry, $\iota_l(\lambda^K_T, \tau_T)$ being the data of $\alpha^{T\cap B_{l-1}}/\mathbf{N}^K_{l,j}(D_{l-1}(T), D_l(T))$ (of $\alpha^{T\cap B_{L-1}}/\mathsf{y}_{j,D_{L-1}(T)}$ at the last node) and $\iota_1(\lambda^K_T, \tau_T)$ that of $\alpha^{T\cap S_1}/K_{j,(D_1(T))}$, and the kernel moves by*

$$R_{\lambda^K_T, \tau_T} = \frac{a^{(1)}_{jD_1(T)}}{s_j} \prod_{l=2}^{L-1} \frac{e^{(l)}_{jD_{l-1}(T)D_l(T)}}{a^{(l-1)}_{jD_{l-1}(T)}},$$

*the probability of the path of $T$ for the chain on the stations that starts at $D \in \mathcal{D}_j(S_1)$ with the probability $a^{(1)}_{jD}/s_j$ and moves from $D'$ to $D$ with the probability $e^{(l)}_{jD'D}/a^{(l-1)}_{jD'}$; likewise for the cobasis, with $f$, $b$ and $t_j$. So the chance $\mathbb{P}_j(T) = \theta_{j,T}\alpha^T/K_j$ of $T$ (Definition 38.1) has the value*

$$\langle \mathbb{P}_j(T) \rangle(\mathbf{d}) = \int_{pins} G\, R_{\lambda^K_T, \tau_T} \prod_{l=1}^{L} \big\langle \mathbb{P}^{(l)}(T) \big\rangle \big(\iota_l(\mathbf{d}, \mathbf{z})\big),$$

*$\mathbb{P}^{(l)}(T)$ the chance of the piece of $T$ in its entry at the node $l$, its term divided by the entry, at every real $\mathbf{d}$ with a point admissible at $\mathbf{d}$ whose translate by $\tau_T$ is admissible at $\mathbf{d} + \lambda^K_T$; the coefficients of the pieces in their entries multiply to $\theta_{j,T}$.*

*(iii)* The lattice. *$\mathbb{S}(\mathbb{Z})$ maps onto $\mathbb{D}_F(\mathbb{Z})$, every lattice vector of the data being the image of an integral combination of the $(\lambda^K_T, \tau_T)$ and $(\lambda^\Psi_T, \tau^*_T)$. So $\mathfrak{M}'_F = \mathfrak{M}_F$, and the value of every element of $\mathfrak{M}_F$ is the fibre integral of a multilinear form in the values of the masters of the nodes, as in Theorem 48.5(iii).*

*(iv)* Repetition. *Nodes $S^{(\cdot)}_l$ that are one piece up to bijections of marks keeping seam letters (Theorem 29.3(iii)) have isomorphic families of positive letters, the stations corresponding, and so one module (Theorem 48.5(v)). The algebra of the chain is that of the first bunch, of one node for each class of the nodes and of the last node, whatever the length of the chain.*

(v) Locality. *Let $e$ be a mark of the step $B_{l-1}$, $1 \le l \le L$, and $j$ a table. For the entries $E$ of $K_j$ at the node $l$, with their shares $z_E$ (the share $e^{(l)}_{jD'D}$ of the entry $(D', D)$, and at the first bunch the stations $D$ with the shares $a^{(1)}_{jD}$) and the occupations $R^{(l)}_{E,e}$ of $e$ in their letters,*

$$s_j \langle R_{j,e} \rangle(\mathbf{d}) = \int_{pins} G \sum_E z_E \langle R^{(l)}_{E,e} \rangle\big(\iota_l(\mathbf{d}, \mathbf{z})\big) \prod_{l' \ne l} P_{W_{l'}}\big(\iota_{l'}(\mathbf{d}, \mathbf{z})\big),$$

*through every point admissible at $\mathbf{d}$ whose translates by the path vectors of the bases of table $j$ through $e$ are admissible at the translates of $\mathbf{d}$; likewise for $\Psi_j$, with $t_j$, the shares of its entries and the occupations in the letters of $\Psi_j$. So, through a point admissible at $\mathbf{d}$ whose translates by the $\tau_T$, $T \ni e$, and the $\tau^*_T$, $T \not\ni e$, in every table, are admissible at the translates of $\mathbf{d}$, the defect $c_e(\mathbf{d})$ of $F$ is read as the defect of the node $l$ at its data: the mark relation of the period of $F$ at $e$ is that of the node $l$, the other nodes entering only through their periods.*

For a molecule with a connected table, whose trees of bunches are chains (Corollary 20.8), this is the module of the family along the chain of the seam procedure (Definition 29.9): one box for the first bunch, a family of tables on $B_0$ at the shares, one for each class of the transfer nodes and one for the last node, and every value of every element of $\mathfrak{M}_F$, at a lattice translate of a point with admissible contours (Theorem 48.5(iii)), is the word of Corollary 29.5 with the periods of the nodes replaced by rows of their reductions and shift matrices times the values of their masters. The chance of a basis reads as a flow: the shares route the exponent of each letter through the stations, the path of the basis is chosen with the probabilities of that flow, and each node contributes the chance of the piece of the basis in its entry. The glue across one seam (Theorems 28.10(iii) and 28.12) is the chain with $L = 2$, its two nodes the seam word of the bunch and the room word.

## 48.4 Shares and scales

A letter is opened in one of two ways: on its shares, $\Gamma(w)\big(\sum_l X_l\big)^{-w} = \int_{|\mathbf{a}|=w} \prod_l \Gamma(a_l) X_l^{-a_l}$, one share for each term and the one pin $|\mathbf{a}| = w$ (Corollary 4.4), or on its scale, $\Gamma(w)\big(\sum_l X_l\big)^{-w} = \int_0^\infty \xi^{w-1} \prod_l e^{-\xi X_l}\, d\xi$, one scale at which every term is read. The two are the two sides of one Gramma (Proposition 4.2). A pin that conserves the shares of a letter among blocks that own them is, on the scale of the letter, one point at which the blocks meet, each through its own transform, and equal blocks meet there as one transform to a power. The transforms are exponential periods: each converges on the data whose real parts lie in an open cone, and each generates a module of finite rank over the rational functions of its scales, so that what a word reads on its scales is counted by a group, at any number of scales.

**Lemma 48.8** (Exponential periods: their domains and their modules)**.** *Let $B$ be a finite set, $h_1, \dots, h_q$ nonzero polynomials on $B$ with nonnegative coefficients, $Y \subseteq (\mathbb{C}^\times)^B$ the set on which none of them vanishes, and $R_1, \dots, R_p \in \mathbb{C}[Y]$ of the form $R_i = \alpha^{\mathbf{w}_i} P_i \prod_k h_k^{-n_{ik}}$, with $\mathbf{w}_i \in \mathbb{Z}^B$, $n_{ik} \in \mathbb{N}$ and $P_i$ a nonzero polynomial with nonnegative coefficients. For data $\boldsymbol{\nu} = (\mathbf{m}, \mathbf{c}) \in \mathbb{C}^B \times \mathbb{C}^q$, an insertion $g \in \mathbb{C}[Y]$ and scales $\boldsymbol{\xi} \in (0, \infty)^p$ put*

$$\langle g \rangle_{\boldsymbol{\nu}}(\boldsymbol{\xi}) = \int_{(0,\infty)^B} g\, \alpha^{\mathbf{m}-\mathbf{1}} \prod_{k=1}^{q} h_k^{-c_k} \exp\Big( - \sum_{i=1}^{p} \xi_i R_i \Big)\, d\alpha.$$

*For a polynomial $h$ with nonnegative coefficients write $\operatorname{top}_h(v)$ for the largest value of $\langle \mathbf{t}, v\rangle$ over its exponents $\mathbf{t}$, and put*

$$\rho_i(v) = \langle \mathbf{w}_i, v\rangle + \operatorname{top}_{P_i}(v) - \sum_k n_{ik} \operatorname{top}_{h_k}(v), \qquad \varphi_{\boldsymbol{\nu}}(v) = \langle \operatorname{Re}\mathbf{m}, v\rangle - \sum_k \operatorname{Re} c_k \operatorname{top}_{h_k}(v),$$

*and $C_0 = \{v \in \mathbb{R}^B : \rho_i(v) \le 0$ for every $i\}$, a closed cone.*

*(i)* Domain. *$\langle 1\rangle_{\boldsymbol{\nu}}(\boldsymbol{\xi})$ converges absolutely at one $\boldsymbol{\xi}$ exactly when $\varphi_{\boldsymbol{\nu}} < 0$ on $C_0 \setminus \{0\}$, and then at every $\boldsymbol{\xi}$, locally uniformly in $(\boldsymbol{\nu}, \boldsymbol{\xi})$. So its domain $\mathcal{D}$ is the set of the data whose real parts lie in an open convex cone, it does not depend on the scales, and $\langle \alpha^{\mathbf{u}} \prod_k h_k^{-n_k}\rangle_{\boldsymbol{\nu}} = \langle 1\rangle_{\boldsymbol{\nu}+(\mathbf{u},\mathbf{n})}$ converges exactly on $\mathcal{D} - (\mathbf{u}, \mathbf{n})$.*

*(ii)* Module. *At every $\boldsymbol{\nu} \in \mathcal{D}$ the derivatives $\partial^{\mathbf{a}}\langle 1\rangle_{\boldsymbol{\nu}} = (-1)^{|\mathbf{a}|}\langle \prod_i R_i^{a_i}\rangle_{\boldsymbol{\nu}}$, $\mathbf{a} \in \mathbb{N}^p$, span a vector space over $\mathbb{C}(\boldsymbol{\xi})$ closed under the $\partial_{\xi_i}$, of dimension at most that of the top cohomology of the twisted algebraic de Rham complex of $Y$ over $\mathbb{C}(\boldsymbol{\xi}, \boldsymbol{\nu})$ for the connection $d + d\log\left(\alpha^{\mathbf{m}} \prod_k h_k^{-c_k} e^{-\sum_i \xi_i R_i}\right)$, which is finite.*

*(iii)* Quotients. *If the integrand is invariant under a rational subtorus of $(0,\infty)^B$, its integral over the quotient, with the quotient of $d\alpha/\alpha$ by the Haar measure for which the integer points of the subtorus have the covolume one, is on a section an integral of the same kind on fewer marks, at data linear in $\boldsymbol{\nu}$, and (i) and (ii) hold for it.*

A Gramma is the case without denominators, its monomials the $R_i$, its configuration their exponents and its coefficients the scales: $C_0$ is the cone on which every exponent of the configuration is nonpositive, and (i) is its domain (Theorem 2.2). The transforms read below are of the general kind, with the absent letters of the children of a node in the denominators.

**Definition 48.9** (The group at several scales; the inputs and the group of a word)**.** Let $\mathcal{E}$ be a finite set of scales, and $\mathbb{C}(\mathbf{y}) = \mathbb{C}(y_e : e \in \mathcal{E})$ with the derivations $\partial_{y_e}$.

(i) A *differential module* over $\mathbb{C}(\mathbf{y})$ is a vector space $N$ of finite dimension over it with commuting connections $\nabla_e$, $\nabla_e(rv) = (\partial_{y_e} r)\, v + r\, \nabla_e v$. Its solutions near a point $x \in (0,\infty)^{\mathcal{E}}$ off its singular locus form $\operatorname{Sol}_x N$, and its *group* $G_N \subseteq \operatorname{GL}(\operatorname{Sol}_x N)$ is the group of the automorphisms of the field generated over $\mathbb{C}(\mathbf{y})$ by these solutions and their derivatives that fix $\mathbb{C}(\mathbf{y})$ and commute with every $\partial_{y_e}$; when $N$ is regular singular along its singular locus it is the Zariski closure of the monodromy of the solutions on the complement of that locus. Its Lie algebra, its unipotent part and its reductive part are named as in Definition 45.24. A function $f$ on an open subset of $(0,\infty)^{\mathcal{E}}$ whose derivatives span a space $N_f$ of finite dimension over $\mathbb{C}(\mathbf{y})$ *generates* the differential module $N_f$, with $\nabla_e = \partial_{y_e}$, and its *group* is $G_{N_f}$.

(ii) The *inputs* of a word are the densities of its Trailmoments, a Trailmoment on $r$ scales having its density on $(0,\infty)^r$ in the coordinates the word writes; inputs on $r$ scales with one density are one input, at whatever indices and cuts they are read, and a product or a power of Trailmoments has the inputs of its factors. A period of a family of positive letters that the word keeps, an integral over the space $V_W$ of its directions at data linear in the shares (Definition 48.1(v)), is read as that period and not as a Trailmoment. When the density of every input generates a module, the *group of the word* is the product, over the numbers $r$ of scales of its inputs, of the groups of the sums over $\mathbb{C}(y_1, \dots, y_r)$ of the modules of its distinct inputs on $r$ scales; a word without inputs has the trivial group.

At one scale this is Definitions 45.24 and 45.32: every input read on one scale is a function on $(0,\infty)$, and their modules are summed over $\mathbb{C}(y)$.

**Proposition 48.10** (Powers add nothing, at any number of scales)**.** *(i) Propositions 45.25(i)–(iii) and 45.26 hold for differential modules over* $\mathbb{C}(\mathbf{y})$*.*

*(ii) Let functions* $F_1, \dots, F_q$ *on one open subset of* $(0,\infty)^{\mathcal{E}}$ *generate the modules* $M_1, \dots, M_q$*. For exponents* $m_u \in \mathbb{N}$ *the product* $\prod_u F_u^{m_u}$ *generates a submodule of the image of* $\bigotimes_u \mathrm{Sym}^{m_u} M_u$ *under the multiplication of functions. So the sum of the modules of any finitely many such products is a subquotient of a construction from* $M_1 \oplus \cdots \oplus M_q$*, and its group is the image of the group of* $M_1 \oplus \cdots \oplus M_q$ *acting on its solutions, a quotient of that group, whatever the exponents.*

## 48.5 Trees of nested bunches

A tree of nested bunches is read node by node. On its shares a node meets its children through their seam functions at their loads, and the whole tree is one share reading. On its scales a full node meets the blocks of any partition of its parts that its seam letters separate, at one point, each block through its transform; every node separates the partition into one block.

**Definition 48.11** (A tree of nested bunches)**.** Let $F$ be a family of tables whose marks form one molecule of the family of its active tables, read as the family of positive letters formed by its active letters (Proposition 48.21(i), Theorem 48.23(i)); its marks then form one molecule, more tables only adding links. A *tree of nested bunches* of $F$ is a finite family $\mathcal{L}$ of proper bunches $S$ of $F$, each with $\sum_j (s_j + t_j) c_j(S) > 0$, any two of them disjoint or nested. Its *nodes* are its members and the *root* $\mathcal{E}$; the *children* of a node are the maximal members of $\mathcal{L}$ properly inside it, its *own marks* $Z_S$ are the marks in none of its children, and a node without children is a *leaf*. In this definition and below, $L$ is an active letter of a table $j$ (Definition 28.5), $s_L$ its exponent, $s_j$ or $t_j$, and $\epsilon_L(D)$ is $i(D)$ for $L = K_j$ and $c_j(S') - i(D)$ for $L = \Psi_j$, at a station $D$ of a bunch $S'$ in table $j$; the seam letters $V_{LD}$ of a bunch are its $K_{j,(D)}$ and its $\Psi_{j,(D)}$, the two letters of a station $D$ of table $j$ being taken up to one positive constant $c_{jD}$, by which its room station letters are divided (the remark after Definition 28.1), its shares $a_{LD}$ the $a_{jD}$ and the $b_{jD}$ of Definition 28.5, and its extreme excess sum $V_L^{\mathrm{ext}}$ is $V_j^{(0)}$ for $L = K_j$ and $V_j^{*(c_j)}$ for $L = \Psi_j$.

(i) *Transfer tensors.* The *node* of a node $S \neq \mathcal{E}$ with the children $S_1, \dots, S_k$ is the piece $S^{(\mathbf{x}_1,\dots,\mathbf{x}_k)}$ on $Z_S \sqcup \{g_1, \dots, g_k\}$ of Theorem 29.2(ii), and that of the root is the stand-in core of its children (Theorem 29.1). Its seam letters at the stations $D$ of $S$, its letters at the root, are polynomials in $\alpha_{Z_S}$ and in the stand-ins, linear in the coordinates of each seam point:

$$V^S_{LD} = \sum_{\mathbf{D}} \mathbf{N}_{S,L}(D;\mathbf{D}) \prod_{l=1}^{k} x_{l,LD_l}\, \alpha_{g_l}^{\epsilon_L(D_l)},$$

the sum over the tuples $\mathbf{D} = (D_l)_l$ of stations of the children in the table of $L$; the *entries* $\mathbf{N}_{S,L}(D;\mathbf{D})$ are polynomials on $Z_S$ with nonnegative coefficients, constants when $Z_S$ is empty, and form the *transfer tensor* of the node; at the root $D$ is dropped. Read at the seam points of the children, $V_{LD} = \sum_{\mathbf{D}} \mathbf{N}_{S,L}(D;\mathbf{D}) \prod_l V^{(l)}_{LD_l}$ for the seam letters $V^{(l)}$ of the children: the seam letters of a node are multilinear in those of its children.

(ii) *Shares and loads.* Open every seam letter of the node with one share $e_{S,LD\mathbf{D}}$ for each nonzero entry, $\sum_{\mathbf{D}} e_{S,LD\mathbf{D}} = a_{S,LD}$ the share of the seam letter, at the root its exponent $s_L$. The *load* of the child $S_l$ at its station $D'$ is $a_{S_l,LD'} = \sum e_{S,LD\mathbf{D}}$ over the $(D,\mathbf{D})$ with $D_l = D'$. The *node word* $W_S(\mathbf{e})$ is the family on $Z_S$ of the nonzero entries at their shares, with the

masses $\mathbf{m}_{Z_S}$; when $Z_S$ is empty it is a family of positive constants on the empty set of marks, read as in Definition 48.1, the one exception to Definition 1.1(i). At a leaf the entries are its seam letters and its node word is its seam word.

A choice of these constants at every node other than the root is a *normalization* of the tree, and a node is *full* when its seam word is full (Definition 48.1(i)). A chain of bunches $S_1 \subset \cdots \subset S_{L-1}$ is a tree of nested bunches whose nodes have one child each, the steps their own marks, and, in the normalization of Definition 28.1, the transfer matrices their transfer tensors and the words of the nodes of Corollary 29.5 their node words; at width one the bunches of the subtree cells of a hinge tree form one (Proposition 21.3).

A bunch with its seam letters so taken is a piece whose glue with its room is the family (Definition 28.4), and the results on pieces hold for it. Its seam letters, the entries and the station transforms move by constants: the entries of a node are multiplied by the constants of its station and divided by those of the stations of its children in them, and the station transform becomes $\mathbf{p} \mapsto \mathcal{I}(\mathbf{c} \circ \mathbf{p})$ (Definition 28.21). Its seam function at shares on its pins is multiplied by $\prod_{j,D} c_{jD}^{-a_{jD}-b_{jD}}$ (Lemma 28.7), and its law is moved by a fixed bijection of its seam torus with a density, as after Definition 28.5. Seam functions enter below only on the pins, so every statement below holds in every normalization. A node whose seam function converges at some shares is full (Theorem 25.2), and for a family with a native gauge the seam function of a full node converges at the native split (Lemma 28.9); the root is full, its seam word being the family.

**Theorem 48.12** (A node on its shares; a tree of nested bunches is a share reading)**.** *Let $F$ have a tree of nested bunches, real data on its balance wall with $s_j, t_j \geq 0$, not all zero, every proper wall positive and a native gauge.*

*(i)* A node. *Let $S \neq \mathcal{E}$ be a full node with the children $S_1, \ldots, S_k$, $k \geq 0$, at shares $\mathbf{a}_S$ on the pins of its seam (Lemma 28.7). On contours of shares $e_{S,LD\mathbf{D}}$ with positive real parts at which the seam functions of the children at their loads and the node word converge,*

$$\int_{\Delta_S} (\boldsymbol{\varrho}_S^{\rho})^{-\mathbf{a}_S}\, d\phi_S^{\rho} = \int_{pins} \prod_{L,D} \Gamma_{1}^{n_{LD}} \begin{bmatrix} \mathbf{e}_{S,LD} \\ a_{S,LD} \end{bmatrix} \; P_{W_S(\mathbf{e})}(\mathbf{m}_{Z_S}) \prod_{l=1}^{k} \int_{\Delta_{S_l}} (\boldsymbol{\varrho}_{S_l}^{\rho})^{-\mathbf{a}_{S_l}}\, d\phi_{S_l}^{\rho},$$

*$n_{LD}$ the number of the nonzero entries at $(L, D)$ and $\mathbf{a}_{S_l}$ the loads of $S_l$, the fibre pinned by $\sum_{\mathbf{D}} e_{S,LD\mathbf{D}} = a_{S,LD}$, by the scale pins of the children (Lemma 28.7) and by the pins of directions of the node word, with transverse pins where its Newton sum is degenerate (Theorem 28.12), less the homogeneity of the node word, which the others imply, or less one scale pin when $Z_S$ is empty. At the root the same holds for $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$, with the stacked Gammas $\Gamma_{1}^{n_L} \begin{bmatrix} \mathbf{e}_{\mathcal{E},L} \\ s_L \end{bmatrix}$ and the letter totals. With one child this is Theorem 29.6(iii), and at a leaf it is Lemma 28.7; with the node word opened on its monomials (Proposition 4.2) the node contributes stacked Gammas and powers of the coefficients of its seam letters alone, and at width one for one table on its centre ray, in the normalization of Theorem 48.14(iii), this is Proposition 42.26(iii) at the root and (v) below it.*

*(ii)* The tree. *Opened along the tree, $P_F$ is a positive share reading (Definition 48.4) of the family of positive letters formed by the active letters of $F$, whose period it is, its nodes the node words and the seam words of the leaves at their loads, its kernel the product of the stacked Gammas of all the nodes, its pins the letter totals at the root, the conservation $\sum_{\mathbf{D}} e_{S,LD\mathbf{D}} = a_{S,LD}$ at every station of every node other than the root and the leaves, and the pins of directions of*

*the node words and of the seam words of the leaves, with transverse pins where their Newton sums are degenerate, less one homogeneity. Where the nodes are full it is (i) composed node by node. For a full node $S \neq \mathcal{E}$, the seam function of $S$ at shares on the pins of its seam, the period of its seam word (Lemma 28.7), is likewise a positive share reading of its seam word along the subtree of $S$, with $\sum_{\mathbf{D}} e_{S,LD\mathbf{D}} = a_{S,LD}$ in place of the letter totals. The lattice of the reading of $P_F$ maps onto the lattice of the data of the family of the active letters, so the value of every element of the module of that family (Theorem 48.3(iii)) is a fibre integral of a multilinear form in the values of the masters of the node words and of the leaves (Theorem 48.5(iii)), as along a chain (Theorem 48.7(iii)).*

*(iii)* Repetition. *Nodes carried to one another by bijections of their marks keeping their seam letters, the stations of their children corresponding, have isomorphic node words, and leaves so carried have isomorphic seam words; so the modules of a tree are found once for each class of its node words and leaves (Theorem 48.5(v)).*

*(iv)* Complex data. *Let the data of $F$ be complex, with real parts as in this theorem and every exponent of a letter of $F$ of positive real part. Then (i) holds at complex shares $\mathbf{a}_S$ on the pins of the seam whose real parts are shares at which it holds at the real parts, on the contours through the real shares at which it holds there, both sides converging absolutely and being holomorphic in the data and the shares; and (ii) holds as stated, the reading identity of Definition 48.4 being read at complex data.*

So every Foreman tree, read as a tree of nested bunches with the childcules of a peel as the pieces of the core (Definition 29.9), is a share reading, and the module of the family is read through the modules of its distinct node words and leaves. Opening a node word on its monomials pays for it in work, a fibre of Gamma functions; keeping it pays for it through the masters of its class.

**Definition 48.13** (Blocks of a node; its scales and the transforms of its blocks)**.** In Definition 48.11 let $S$ be a node with the children $S_1, \dots, S_k$ and the own marks $Z = Z_S$, its *parts* its children and its own marks, and $M = Z \sqcup \{g_1, \dots, g_k\}$ the marks of its node. Fix for every active letter $L$ an *absent station* $D^\circ_{l,L}$ of every child, a station of $S_l$ in the table of $L$, and an *absent exponent* $u_{z,L} \in \mathbb{N}$ of every own mark, and put $A_L = \prod_l x_{l,LD^\circ_{l,L}} \alpha_{g_l}^{\epsilon_L(D^\circ_{l,L})} \prod_z \alpha_z^{u_{z,L}}$. A part is *absent* from a monomial $c\,\alpha_Z^{\mathbf{q}} \prod_l x_{l,LD_l} \alpha_{g_l}^{\epsilon_L(D_l)}$ of a seam letter $V^S_{LD}$ of the node when $D_l = D^\circ_{l,L}$, for a child, or $q_z = u_{z,L}$, for an own mark, and *present* in it otherwise.

(i) *Blocks.* The node *separates* a partition $\mathcal{P}$ of its parts into *blocks* when no monomial of a seam letter of its node has present parts in two blocks. Then

$$V^S_{LD} = A_L\Big(n_L(D) + \sum_{P \in \mathcal{P}} \beta_{P,LD}\Big),$$

with the *constant part* $n_L(D) \geq 0$, the coefficient of $A_L$, and the *reading* $\beta_{P,LD}$ of the block $P$, the sum of the monomials in which some part of $P$ is present, divided by $A_L$: a polynomial with nonnegative coefficients in the ratios $x_{l,LD'} \alpha_{g_l}^{\epsilon_L(D')} / \big(x_{l,LD^\circ_{l,L}} \alpha_{g_l}^{\epsilon_L(D^\circ_{l,L})}\big)$, $D' \neq D^\circ_{l,L}$, of the children of $P$ and the powers $\alpha_z^{q-u_{z,L}}$, $q \neq u_{z,L}$, of its own marks; at the root $D$ is dropped. The partitions that the node separates are the coarsenings of the partition into the classes of the equivalence relation generated by two parts being present in one monomial; among them is the partition into one block. The node is *separable* when it separates the partition into single parts.

(ii) *Scales.* The *scales* of $S$ are $\boldsymbol{\xi} = (\xi_{LD})_{(L,D)\in\Sigma_S}$, one for each pair at which some block is present, the set $\Sigma_S$. Every reading at $(L, D)$ is homogeneous of the *degree* $\delta_{LD} = \deg V^S_{LD} - \deg A_L$ under the common scaling $\alpha_M \mapsto e^t\alpha_M$ of the marks of the node, which acts on the scales by $\xi_{LD} \mapsto e^{-t\delta_{LD}}\xi_{LD}$. The node is *seen by its scales* when some $\delta_{LD}$, $(L, D) \in \Sigma_S$, is not zero; then this action is free, and the *scale quotient* $\Xi_S$ is $(0,\infty)^{\Sigma_S}$ modulo it, with the quotient of $d\boldsymbol{\xi}/\boldsymbol{\xi}$ by $dt$; otherwise $\Xi_S = (0,\infty)^{\Sigma_S}$, and when $S$ is full the only partition it separates is the one block.

(iii) *Transforms.* The *transform* of a block $P$, with the children $S_l$, $l \in P$, and the own marks $z \in P$, is

$$\mathcal{T}_P(\boldsymbol{\xi}) = \int_{\mathcal{O}_P} \prod_{l\in P} \Big(\alpha_{S_l}^{\mathbf{m}_{S_l}-\mathbf{1}} \prod_L \big(V^{(l)}_{LD^\circ_{l,L}}\big)^{-s_L}\Big) \prod_{z\in P} \alpha_z^{m_z-\sum_L s_L u_{z,L}-1} \times \exp\Big(-\sum_{(L,D)\in\Sigma_S} \xi_{LD}\,\widehat{\beta}_{P,LD}\Big)\,d\alpha$$

in $[0,\infty]$, over the orthant $\mathcal{O}_P$ of the marks of its children and its own marks, $\widehat{\beta}_{P,LD}$ being its reading with every ratio of a child replaced by the ratio $V^{(l)}_{LD'}/V^{(l)}_{LD^\circ_{l,L}}$ of the seam letters of the child. When the node is not seen by its scales the transform of the partition into one block is taken over the quotient of $\mathcal{O}_P$ by the common scaling $\alpha_{S_l} \mapsto e^t\alpha_{S_l}$, $\alpha_z \mapsto e^t\alpha_z$ of the marks of all the parts, with the quotient of $d\alpha/\alpha$ by $dt$. A block without children has as its transform a Gramma on its own marks, its configuration the exponents $q - u_{z,L}$ of its readings and its coefficients linear in the scales (Definition 2.1). Blocks are *equal* when a bijection of their marks keeping the seam letters and the masses of their children and own marks, with corresponding absent stations and exponents, carries the readings of one to those of the other; equal blocks have one transform.

**Theorem 48.14** (A node on its scales)**.** *Let $S$ be a node of a tree of nested bunches of $F$, with real data on the balance wall and every proper wall positive, and $\mathcal{P}$ a partition of its parts that it separates, the partition into one block when $S$ is not seen by its scales, which for a full node is no restriction (Definition 48.13(ii)).*

*(i)* One point. *At real shares* $\mathbf{a}$ *on the pins of the seam of $S$ with $a_{LD} > 0$ on $\Sigma_S$, in $[0,\infty]$,*

$$\prod_{(L,D)\in\Sigma_S} \Gamma(a_{LD}) \prod_{(L,D)\notin\Sigma_S} n_L(D)^{a_{LD}} \int_{\Delta_S} (\boldsymbol{\varrho}^\rho_S)^{-\mathbf{a}}\,d\phi^\rho_S = \int_{\Xi_S} \boldsymbol{\xi}^{\mathbf{a}}\,e^{-\langle\boldsymbol{\xi},\mathbf{n}\rangle} \prod_{P\in\mathcal{P}} \mathcal{T}_P(\boldsymbol{\xi})\,\frac{d\boldsymbol{\xi}}{\boldsymbol{\xi}},$$

*$\mathbf{n} = (n_L(D))_{\Sigma_S}$, the form on the right being invariant under the common scaling; at complex shares whose real parts are such shares at which the left side is finite, the same holds with the right side converging absolutely. At the root the same holds for $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ at the exponents $s_L$. At complex data whose real parts are as here, every exponent of a letter of $F$ having a positive real part, the same holds at shares whose real parts are shares at which the left side is finite at the real parts, at the root included, both sides being holomorphic in the data and the shares. So the node reads its blocks at one point of its scales, each through its transform, and $q$ equal blocks enter as one transform to the power $q$: one integral over the scale quotient, whatever the multiplicities.*

*(ii)* The transforms. *The transform of a block depends on its children only through their walls and their laws. If the left side of (i) is finite at some shares, then every $\mathcal{T}_P$ converges, with*

*all its derivatives, at every $\boldsymbol{\xi}$, its data lying in the domain of Lemma 48.8, and generates a differential module of finite rank over $\mathbb{C}(\boldsymbol{\xi})$; so does its restriction to a section of the scale quotient by a monomial map, over the rational functions of the section. When the left side of (i) is finite at some shares and the children of $P$ are full, on every contour of shares $z_{LD\tau}$, one for each monomial $\tau$ of each reading $\beta_{P,LD}$, $(L,D) \in \Sigma_S$, with positive real parts, on which the loads of every child $S_l$ of $P$, the sums of the $z_{LD\tau}$ over the monomials in which it is present at its station $D'$ and $s_L$ less these at its absent station, have positive real parts, its seam function converges there, and the* pins of the block *hold, the scale pin of every child of $P$ and, for every own mark $z$ of $P$, $\sum z_{LD\tau}(q_{z,\tau} - u_{z,L}) = m_z - \sum_L s_L u_{z,L}$, less one when $S$ is not seen by its scales,*

$$\mathcal{T}_P(\boldsymbol{\xi}) = \int_{pins} \prod_{L,D,\tau} \Gamma(z_{LD\tau}) \big(\xi_{LD} c_\tau\big)^{-z_{LD\tau}} \prod_{l \in P} \int_{\Delta_{S_l}} (\boldsymbol{\varrho}^\rho_{S_l})^{-\mathbf{a}_l} \, d\phi^\rho_{S_l},$$

*$c_\tau$ the coefficient of $\tau$, $q_{z,\tau}$ the exponent of $\alpha_z$ in it and $\mathbf{a}_l$ the loads of $S_l$: the inverse Mellin transform of the seam functions of its children, with one Gamma for each monomial of its readings, a fibre integral of the dimension of the number of these monomials less the number of the pins; with a native gauge, the native split of the node refined to the monomials of the readings of $P$ is a point of such a contour.*

*(iii)* Width one. *Let one table at exponents $s, t > 0$, or the family $F_{\mathsf{m}}$ of Proposition 42.26(i), whose two tables have the bases of one table and whose active letters are its letter $K$ and the cobasis letter with the norms, have a hinge tree at width one with weights factoring across its bunches of width one (Theorem 23.16), read as a tree of nested bunches by the bunches of its subtree cells, the cells of at least two marks directly below a node being its children and the marks left in place its own marks, in the normalization in which the seam letters of the bunch of every subtree cell are the letters of the two hinge minors of that cell with its weights (Definition 23.15). A series or a parallel node (Definition 23.13) is separable, with the absent stations of the excess* 1 *at a series node and* 0 *at a parallel node, in both letters, and seen by its scales, its readings and its constant parts being weights of its core. With the weights* 1 *its scale quotient is one scale, read as $y = \xi_K \xi_\Psi$ on the section $\xi_K = 1$ at a series node and $\xi_\Psi = 1$ at a parallel node, and the transforms of its cells are their transforms $f_{\mathcal{A}}(y)$ at a series node and their dual transforms $\check{f}_{\mathcal{A}}(y)$ at a parallel node (Definition 23.1), those of its marks the transforms of the one-mark cells (Definition 23.19(1)), and (i) is the rule of Definition 23.19(2) and (5); at a series node it is the series connection of Proposition 20.37(i), whose transforms these are. On the centre ray of the table, with its norms in its cobasis letter (Proposition 42.26(i)), (i) is Proposition 42.26(vi) below the root and (vii) at the root: a mark left in place at a parallel root is read at its norm $\mathsf{m}_e$ in $\Psi$, through $\mathsf{g}(\mathsf{m}_e y)$, and at a series root every part is read in $\Psi$ at the product $M$ of the norms divided by its own, through $\mathsf{g}(y/\mathsf{m}_e)$ and $f_{\mathcal{A}}(y)$ in the coordinate $y = M\xi_K\xi_\Psi$, on the section $\xi_K = M^{-1}$, with the constant $M^{-k}$.*

So every full node is read on its scales through every partition of its parts that it separates. Its seam letters factor into the absent letters of its parts, which go to the powers $s_L$ of the letter totals, and brackets in which one block at a time is present, which go to the scales; on its shares the blocks own their shares, joined by the conservation at the stations of the node, and on its scales they meet at one point. The finest partition that a node separates is read off its seam letters once its absent stations are chosen; a separable node is the general form of a series and of a parallel node, and the partition into one block reads the node through one transform of all its parts.

## 48.6 What the word along a tree reads

The word of a tree of nested bunches reads on its scales the transforms of the blocks of the nodes it reads there, and on its shares the node words and the leaves of the others; its group, its work and its price follow the distinct blocks and the distinct nodes.

**Theorem 48.15** (What the word along a tree reads: inputs, group, work and price)**.** *Let $F$ be as in Theorem 48.12, with a tree of nested bunches in a normalization and a set $\Sigma$ of full nodes, each with a partition of its parts that it separates (Theorem 48.14). Read every node of $\Sigma$ on its scales (Theorem 48.14(i)), on a section of its scale quotient by a monomial map with positive constant coefficients; the transform of every block with children, all of them full, of a node of $\Sigma$ by Theorem 48.14(ii), the seam functions of its children read by their own words, and that of every other block with children as its exponential period (Definition 48.13(iii)), which integrates over the subtrees of its children, so that their nodes are not read and none of them is in $\Sigma$; and every other node, the leaves among them, on its shares, each part of the tree between the nodes of $\Sigma$, under the root or under a full child of a block, opened along it as in Theorem 48.12(ii), with the nodes of $\Sigma$ directly below it read by their seam functions, its node words opened on their monomials or kept. A node read on its shares is read* alone *when it and its children in that part are full, by Theorem 48.12(i); the others are read* jointly*, each maximal set of them joined through parents and children by one fibre of Theorem 48.12(ii). In this statement node words, leaves and blocks are counted once for each class under bijections of their marks keeping their seam letters and their data, a block together with its absent stations, its absent exponents and its readings; the word writes the sections of the nodes of $\Sigma$ of one class in one set of coordinates; and an input is counted once for each density as the word writes it, the product of the transforms of the classes of its blocks to their multiplicities, the transform of a block without children written as its Gramma on the section, and the exponential of its constant parts. The classes, and with them the multiplicities, are those of the normalization: that of Definition 28.1 puts in one class the nodes carried to one another by bijections that carry the family, and that of Theorem 48.14(iii) the subtrees that are equal as cells with their weights.*

*(i)* What it reads. *The value is a word whose inputs are, at every node $S$ of $\Sigma$, the function $e^{-\langle \boldsymbol{\xi},\mathbf{n}\rangle}\prod_P \mathcal{T}_P^{q_P}$ on the section of its scale quotient, a function on $(0,\infty)^r$, $r=\dim \Xi_S$: the product over its distinct blocks of their transforms raised to their multiplicities $q_P$, a block without children bringing a Gramma, and the exponential of its constant parts. Everything else in it is stacked Gammas of linear forms in the data and the shares, rational constants and powers of positive constants, fibre integrals over shares, the exponential periods of the blocks read as such, and the periods of the node words kept and of the seam words of the leaves kept, each at lattice translates of its data a combination with coefficients in $\mathbb{K}_{\mathbb{S}}$ of the periods of the $\mathfrak{m}_k$ masters of its class (Theorem 48.5(iii),(v)); these masters can be taken among the $\alpha^u f_{N\mathbf{d}_1}$, $\alpha^u$ a Laurent insertion and $\mathbf{d}_1$ a positive direction of the class with $N$ at least the wall degree of every $\alpha^u$ (Theorem 48.3(iii),(iv)), whose walls are nonnegative and whose values converge wherever the period of the class does.*

*(ii)* Group. *Every input generates a differential module of finite rank. For every $r$ let $N_r$ be the sum over $\mathbb{C}(y_1,\dots,y_r)$ of the modules of the distinct transforms of the blocks of the nodes of $\Sigma$ whose scale quotients have the dimension $r$, and of the exponentials of their constant parts, each read on its section and counted once. The group of the sum of the modules of the distinct inputs on $r$ scales is the image of $G_{N_r}$ acting on its solutions, a quotient of $G_{N_r}$, whatever the multiplicities; so the group of the word (Definition 48.9) is a quotient of $\prod_r G_{N_r}$.*

*(iii)* Work. *The work is the sum of the dimensions of the fibres of Theorem 48.12(i) of the distinct nodes read alone, and of the joint fibres of the distinct sets of nodes read jointly, with the shares inside the entries where a node word is opened; of* $\mathfrak{m}_k \dim V_{W_k}$ *for the distinct classes* $k$ *of the node words and seam words of leaves kept; of* $r$ *for every distinct input on* $r$ *scales; of the dimensions of the fibres of Theorem 48.14(ii) of the distinct transforms of blocks with full children; and of the dimension of the orthant of every other distinct block with children, less one when its node is not seen by its scales. It does not grow with the multiplicities of the blocks, which enter as powers, nor with those of the nodes read alone, leaves, transforms and inputs, each read once; a node read jointly enters the joint fibre of its set with its multiplicity.*

*(iv)* Price. *The price is at most that of the tree, of its transfer tensors, and of the absent stations, absent exponents and readings of the blocks of the nodes of* $\Sigma$*, each distinct one written once and referred to by index (Definition 33.2); plus, for every class of node words and leaves kept, that of its masters, of the reduction of* $[1]$ *on them and of its shift matrices at the generators of its lattice, and of the isomorphisms onto it; plus the exponent matrices and the coefficients of the sections, the lattice bases of the pins of directions, the* $\lceil \log_2(q_P + 1) \rceil$ *bits of every multiplicity, and the coefficients of the elements of the integrand modules integrated.*

*(v)* Complex data. *At complex data whose real parts are as in Theorem 48.12, every exponent of a letter of* $F$ *having a positive real part, the period is the word of (i) read at those data: its fibres are taken through the real shares at which they converge at the real parts, its inputs are read at those data, and every integral of it converges absolutely, the word being holomorphic in the data on the tube over the real data of Theorem 48.12. Every input there generates a differential module of finite rank, and (ii)–(iv) hold as stated.*

**Corollary 48.16** (The word along the canonical tree)**.** *In the setting of Theorem 45.34, the bunches of the subtree cells of the canonical hinge tree are a tree of full nested bunches of the family* $F_{\mathsf{m}}$ *of Proposition 42.26(i), in the normalization of Theorem 48.14(iii), whose series and parallel nodes are separable and seen by their scales, each with one scale. With* $\Sigma$ *the nodes of* $S$ *and the classes below the root, each with the partition into its cells, and every other node read on its shares with its node word opened on its monomials, Theorem 48.15 is Theorem 45.34: its inputs, on one scale, are those of (i) there, the constant* $M^{-k}$ *of a series root in* $S$ *being written there as the factors* $\mathsf{m}_e^{-k}$ *of its inputs; its group is bounded as in (ii) there, and is the group of (ii) there when* $S$ *is empty; and its work is that of (iii) there.*

So Theorem 45.34 is the reading of one table at width one on the scales of its series and parallel nodes, and its mechanism is a pin read as a scale. A node conserves the loads of its blocks by its pins, one at each of its stations, and on the scales of its letters those pins are one point at which every block is read through its transform; so a repeated block is one transform to a power, and the group of what the node reads is a quotient of the group of its distinct blocks and of the exponential of its constant parts (Proposition 48.10). A series or a parallel node at width one is the case of one scale and single cells, each read through its transform or its dual transform, and its classes are the case of single marks, each read through $\mathsf{g}$. At any width and at every node the same holds on as many scales as the pairs of a letter and a station at which a block is present, less the scaling when the node sees it. The transforms have modules of finite rank because each is an exponential period on an open cone of data, whose integration by parts closes in a twisted de Rham cohomology of finite dimension (Lemma 48.8): the finiteness that gives the module of Chapter 47 its finite dimension on the shares. These modules are read off the children of the blocks on their shares, once for each class of blocks (Theorem 48.20).

## 48.7 The module of a family from its nodes

A share reading reads every value of a family through the masters of its nodes (Theorem 48.5(iii)), and its fibre integral identifies more than the nodes do: a translation of the shares along the fibre leaves the integral unchanged. Along a tree the relations of the family ride on the paths of the terms of its letters. A letter identity of the family is the letter identities of its nodes, the probabilities of the paths under the flow of the shares summing to one, and a mark relation of the family is that of the node of the mark, the paths through each of its entries carrying together the share of the entry. So, modulo the translations along the fibres, the integrand carries the module of the family itself, the values separating the module (Proposition 47.15). The module of a family is the direct image of the modules of its nodes, found node by node from the leaves up, once for each class of subtrees, by eliminations over the rational functions of the data on identities among the products of the masters of the nodes; and its zeros are told by the closings of the node words and of the seam words of the leaves, combined along the tree by flows of integers.

**Definition 48.17** (Paths; the direct image of a reading)**.** Let $F$ be a family of tables with the exponents fixed at 0 named, at least one exponent not fixed at 0, whose marks form one molecule of the family of its active tables, with a tree of nested bunches in a normalization. Its letters, the $K_j$ and the $\Psi_j$ whose exponents are not fixed at 0, form the family $W_F$ of positive letters (Proposition 48.21(i)), a family of letters of tables (Theorem 48.23); every family of letters of tables is the $W_F$ of the family of the tables of its letters, with the exponents of the letters of these tables that it does not contain fixed at 0. The data of $F$ are those of $W_F$, its data space $\mathbb{D}_{W_F}$ being the balance hyperplane in the masses and the exponents of the letters (Definition 48.1, Theorem 48.23(i)), with its lattice $\mathbb{D}_{W_F}(\mathbb{Z})$; its module is $\mathfrak{M}_{W_F}$ (Theorem 48.3(iii)), its closings are those of Theorem 48.23(ii), and $P_F$ is the period of $W_F$ (Definition 48.1(v)). The active letters of Definition 48.11 are the letters of $F$, and a node is full when its seam word, formed by the seam letters of the letters of $F$, is full; the two letters of a table have exponents complementary in its marks and so one space of directions, and this is the fullness of the seam word with both letters of every active table. A realization of the active tables with the exponents of the letters positive and the others 0 (Definition 9.1), as with the exponents 1 and the gauge $\mathbf{1}$, has the usage at its gauge as its masses, so that the gauge is native (Proposition 8.22), and every proper wall positive, the active tables forming one molecule (Proposition 9.2); so by Theorem 48.12(ii) at it, where the active letters are the letters of $F$, the readings below are positive share readings (Definition 48.4). When every letter of every table is a letter of $F$, $W_F$ is $F$ read as in Definition 47.1, and the realization with the exponents 1 and the gauge $\mathbf{1}$ is the centre data. Let $W$ be either $W_F$, with the reading of $P_F$ along the tree, its *top* the root and its letters $\ell$ the letters $L$ of $F$ at the exponents $w_\ell = s_L$; or the seam word of a full node $S \neq \mathcal{E}$, with the reading of its period, the seam function of $S$ on the pins of its seam, along the members of the tree inside $S$ (Theorem 48.12(ii)), its top $S$ and its letters the seam letters $\ell = V^S_{LD}$ at the exponents $w_\ell = a_{S,LD}$. The reading has the shares $\mathbf{z}$, the share space $\mathbb{S}$, the kernel $G$ and the nodes $(W_k, \iota_k)$, the node words and the seam words of the leaves. Let every node carry a *module* $\mathfrak{V}_k$: its module $\mathfrak{M}_{W_k}$, or the quotient of it by a submodule of elements of value zero, with the shifts and the values that it inherits.

(i) *Paths.* A *path* $\pi$ of a letter $\ell$ of $W$ of the active letter $L$, $\ell = L$ at the root and $\ell = V^S_{LD}$ at a full node $S$, assigns to every node $S'$ below the top a station $D_{S'}(\pi)$ of that table, and to every node $S'$ that is not a leaf, the top included, the entry $E_{S'}(\pi) = \mathbf{N}_{S',L}\big(D_{S'}(\pi); (D_{S'_i}(\pi))_i\big)$ of its seam letter at its station into the stations of its children $S'_i$, which is to be nonzero; the station of the top is $D$ for $\ell = V^S_{LD}$, and at the root it is dropped. The *factors* of $\pi$ are these entries and the seam letters $V^{(S')}_{LD_{S'}(\pi)}$ of the leaves; a *term* of $\pi$ is a product $c\,\alpha^{\mathbf{t}}$ of one

monomial of each factor, $c$ the product of their coefficients, and $\ell$ is the sum of the terms of its paths (Definition 48.11(i)). The *path vector* $\tau_\pi$ puts 1 on the share of every entry of $\pi$ and 0 on every other share, and the *probability* of $\pi$ is

$$R_\pi = \prod_{S'} \frac{z_{E_{S'}(\pi)}}{a_{S',LD_{S'}(\pi)}},$$

over the nodes that are not leaves, the share of the entry of $\pi$ over the load of its station, the load at the top being $w_\ell$. A term $c\,\alpha^{\mathbf{t}}$ of $\pi$ is *lifted* to $\big((\mathbf{t};\mathbf{e}_\ell),\tau_\pi\big)$: the data of its stride $\alpha^{\mathbf{t}}/\ell$, with the path vector of its path.

(ii) The *integrand module* is $\mathfrak{J} = \bigotimes_k(\mathfrak{V}_k \otimes \mathbb{K}_{\mathbb{S}})$, with the shifts $\mathrm{Sh}_{(\lambda,\tau)}$ and the values of Theorem 48.5(ii), and $[1]^{\otimes} = \bigotimes_k[1]$. The *fibre translations* are the $(\mathbf{0},\tau) \in \mathbb{S}(\mathbb{Z})$.

(iii) The *integrand span* $\mathfrak{J}^\natural$ is the $\mathbb{K}_W$-span in $\mathfrak{J}$ of the $\mathrm{Sh}_{(\lambda,\tau)}[1]^{\otimes}$, $(\lambda,\tau) \in \mathbb{S}(\mathbb{Z})$, the rational functions of the data of $W$ being read on $\mathbb{S}$ through its projection; the shifts map it onto itself. $\mathfrak{F} \subseteq \mathfrak{J}^\natural$ is the $\mathbb{K}_W$-span of the $(\mathrm{Sh}_{(\mathbf{0},\tau)} - 1)\,t$, $t \in \mathfrak{J}^\natural$ and $(\mathbf{0},\tau)$ a fibre translation.

(iv) The *direct image* of the reading is $\mathfrak{H} = \mathfrak{J}^\natural/\mathfrak{F}$, with the *shifts* $\mathrm{Sh}_\lambda[t] = [\mathrm{Sh}_{(\lambda,\tau)}\,t]$, $(\lambda,\tau) \in \mathbb{S}(\mathbb{Z})$, the classes $e_\lambda = [\mathrm{Sh}_{(\lambda,\tau)}[1]^{\otimes}]$, and the *fibre integral* $\int \big[\sum_j r_j\,\mathrm{Sh}_{(\lambda_j,\tau_j)}[1]^{\otimes}\big] = \sum_j r_j\,P_W(\cdot+\lambda_j)$, $r_j \in \mathbb{K}_W$, a meromorphic function on $\mathbb{D}_W$.

**Theorem 48.18** (The module of a family from its nodes)**.** *In Definition 48.17:*

*(i)* The relations ride on the paths. *The lift of a term of a path $\pi$ lies in $\mathbb{S}(\mathbb{Z})$, moves the kernel by $R_\pi$ and every node by the data of the stride of its monomial in its factor, and in $\mathfrak{J}$, the sums running over the paths of $\ell$ and their terms,*

$$\sum_{\pi,\ c\alpha^{\mathbf{t}}} c\,\mathrm{Sh}_{((\mathbf{t};\mathbf{e}_\ell),\tau_\pi)}[1]^{\otimes} = [1]^{\otimes}, \qquad m_e\,[1]^{\otimes} - \sum_\ell w_\ell \sum_{\pi,\ c\alpha^{\mathbf{t}}} c\,t_e\,\mathrm{Sh}_{((\mathbf{t};\mathbf{e}_\ell),\tau_\pi)}[1]^{\otimes} = 0,$$

*for every letter $\ell$ of $W$ and every mark $e$ of $W$: the letter identities and the mark relations of $W$ at its data (Theorem 48.3(iv)), lifted along the paths, hold in the integrand. The first is the letter identities of the nodes, the probabilities of the paths of $\ell$ summing to* 1 *by the pins. The second is the mark relation of the node whose word has the mark $e$, the paths of the letters of one active letter through an entry $E$ of that word carrying together its share $z_E$: Theorem 48.7(v) in the integrand, along every tree.*

*(ii)* The module is the direct image. $\mathbb{S}(\mathbb{Z})$ *maps onto $\mathbb{D}_W(\mathbb{Z})$. The shifts, the classes $e_\lambda$, $\lambda \in \mathbb{D}_W(\mathbb{Z})$, and the fibre integral are well defined on $\mathfrak{H}$, with $\int \mathrm{Sh}_\lambda\,h = \sigma_\lambda \int h$, so that its elements of value zero form a submodule; the classes $e_\lambda$ span it, and*

$$O\Big(\sum_\lambda r_\lambda\,\mathrm{Sh}_\lambda[1]\Big) = \sum_\lambda r_\lambda\,e_\lambda$$

*is a well-defined map of $\mathfrak{M}_W$ onto $\mathfrak{H}$ that commutes with the shifts, with $\int O(m) = \langle m\rangle$. So the kernel of $O$ consists of elements of value zero, and, the values separating $\mathfrak{M}_W$ (Proposition 47.15), $O$ is an isomorphism:* $\dim_{\mathbb{K}_W}\mathfrak{H} = \mathfrak{m}_W$*, and the masters, the shift matrices and the reductions of $\mathfrak{H}$ are those of $\mathfrak{M}_W$. The quotient of the module of a node by a submodule of elements of value zero is that module itself, for the same reason.*

*(iii)* Node by node. *Replacing the module of a node by its quotient by a submodule of elements of value zero replaces $\mathfrak{H}$ by its quotient by a submodule of elements of value zero. Let $\Sigma$ be*

*a set of full nodes that contains the root. For $S \in \Sigma$ the* tree of $S$ down to $\Sigma$ *is formed by the members of the tree properly inside $S$ that lie properly inside no member of $\Sigma$ properly inside $S$; in it the members of $\Sigma$ nearest below $S$ are leaves. For $S \in \Sigma$, from the leaves up, let $\mathfrak{H}_S$ be the module of the seam word of $S$ when $S$ is a leaf of the tree, and otherwise the direct image of the reading of the seam word of $S$, of $P_F$ at the root, along the tree of $S$ down to $\Sigma$, its leaves in $\Sigma$ carrying their $\mathfrak{H}$, or the modules presented by accepted sets of them ((iv)), and its other nodes their modules. Then every $\mathfrak{H}_S$ is the quotient of the module of the seam word of $S$, and $\mathfrak{H}_\mathcal{E}$ that of $\mathfrak{M}_{W_F}$, by a submodule of elements of value zero, which is 0 (Proposition 47.15): $\mathfrak{H}_S$ is the module of the seam word of $S$, and $\mathfrak{H}_\mathcal{E}$ is $\mathfrak{M}_{W_F}$. A bijection of the marks of two members of $\Sigma$ that carries the members of the tree and of $\Sigma$ inside one to those inside the other, keeping their seam letters, carries the one $\mathfrak{H}$ to the other (Theorem 48.5(v)), so each $\mathfrak{H}_S$ is found once for each class of such subtrees, and the modules of the node words and of the leaves once for each of their classes, by one box each (Theorem 48.3(iv)). With $\Sigma$ the full nodes, every node that is not full is read jointly with the others of its set, as in Theorem 48.15; with $\Sigma = \{\mathcal{E}\}$ the tree is one reading.*

*(iv)* Boxes on the shares. *Let $\mathcal{G}$ consist of a basis of $\mathbb{D}_W(\mathbb{Z})$ and its negatives, with lifts $(\mu, \tau_\mu) \in \mathbb{S}(\mathbb{Z})$ ((ii)), and let $\Lambda \ni \mathbf{0}$ be a finite subset of $\mathbb{D}_W(\mathbb{Z})$ with lifts $(\lambda, \tau_\lambda)$, $\tau_\mathbf{0} = \mathbf{0}$. $\Lambda$ is* accepted *when for every $\mu \in \mathcal{G}$ and $\lambda \in \Lambda$ there are $r^{\mu,\lambda}_{\lambda'} \in \mathbb{K}_W$, $\lambda' \in \Lambda$, finitely many $t_k \in \mathfrak{J}^\natural$ and fibre translations $(\mathbf{0}, \sigma_k)$ with*

$$\mathrm{Sh}_{(\lambda+\mu,\tau_\lambda+\tau_\mu)}[1]^\otimes - \sum_{\lambda' \in \Lambda} r^{\mu,\lambda}_{\lambda'} \mathrm{Sh}_{(\lambda',\tau_{\lambda'})}[1]^\otimes = \sum_k \big(\mathrm{Sh}_{(\mathbf{0},\sigma_k)} - 1\big) t_k$$

*in $\mathfrak{J}$, the* certificate*: an identity of vectors over $\mathbb{K}_\mathbb{S}$ on the products of the masters of the modules of the nodes, every shift of $[1]^\otimes$ in it being a Pochhammer quotient of the kernel times the Kronecker product of the reductions of the shifts of $[1]$ of the nodes (Theorem 48.5(ii)). Then the classes $e_\lambda$, $\lambda \in \Lambda$, span $\mathfrak{H}$, $\mathrm{Sh}_\mu e_\lambda = \sum_{\lambda'} r^{\mu,\lambda}_{\lambda'} e_{\lambda'}$, and $P_W(\cdot + \lambda + \mu) = \sum_{\lambda'} r^{\mu,\lambda}_{\lambda'} P_W(\cdot + \lambda')$ as meromorphic functions on $\mathbb{D}_W$. Let $\mathrm{Sh}_\mu$ be the $\sigma_\mu$-semilinear map of $\mathbb{K}_W^\Lambda$, with the basis $(\varepsilon_\lambda)$, given by $\mathrm{Sh}_\mu\big(\sum_\lambda q_\lambda \varepsilon_\lambda\big) = \sum_{\lambda,\lambda'} \sigma_\mu(q_\lambda)\, r^{\mu,\lambda}_{\lambda'} \varepsilon_{\lambda'}$, let $\rho(\lambda)$ be the image of $\varepsilon_\mathbf{0}$ under the maps of the steps of a path in $\mathcal{G}$ from $\mathbf{0}$ to $\lambda$, fixed for each $\lambda$, extended linearly, and let $\mathfrak{N} \subseteq \mathbb{K}_W^\Lambda$ be the smallest subspace that these maps carry into itself and that contains, for $\mu, \nu \in \mathcal{G}$ and $\lambda \in \Lambda$, $\mathrm{Sh}_\mu \mathrm{Sh}_\nu \varepsilon_\lambda - \mathrm{Sh}_\nu \mathrm{Sh}_\mu \varepsilon_\lambda$, $\mathrm{Sh}_{-\mu} \mathrm{Sh}_\mu \varepsilon_\lambda - \varepsilon_\lambda$ and $\varepsilon_\lambda - \rho(\lambda)$, and the images under $\rho$ of the letter identities and of the mark relations of $W$, $\sum_\mathbf{t} c_{\ell,\mathbf{t}}\, \rho((\mathbf{t}; \mathbf{e}_\ell)) - \varepsilon_\mathbf{0}$ and $m_e \varepsilon_\mathbf{0} - \sum_\ell w_\ell \sum_\mathbf{t} c_{\ell,\mathbf{t}} t_e\, \rho((\mathbf{t}; \mathbf{e}_\ell))$, over the monomials $c_{\ell,\mathbf{t}} \alpha^\mathbf{t}$ of $\ell$; $\mathfrak{N}$ does not depend on the paths chosen. The* module presented *by $\Lambda$, $\widehat{\mathfrak{H}} = \mathbb{K}_W^\Lambda / \mathfrak{N}$, found by finitely many eliminations, is a module with the induced shifts, with masters among the $\varepsilon_\lambda$ and their shift matrices at $\mathcal{G}$; $\mathrm{Sh}_\lambda[1] \mapsto \rho(\lambda)$ maps $\mathfrak{M}_W$ onto it and $\varepsilon_\lambda \mapsto e_\lambda$ maps it onto $\mathfrak{H}$, so it is the quotient of $\mathfrak{M}_W$ by a submodule of elements of value zero, that is $\mathfrak{M}_W$ itself (Proposition 47.15), with its masters among the $\varepsilon_\lambda$ and their shift matrices, and it may stand for $\mathfrak{H}$ in (iii). Accepted sets exist, and a search through the finite sets of points with lifts and of supports of certificates, in any order that reaches each, testing each by one elimination over $\mathbb{K}_W$, stops.*

*(v)* Closings through the tree. *For a node $S$, integral masses $\mathbf{m}_S$ and loads $A_{LD} \in \mathbb{N}$, one for each seam letter of $S$, let $\mathsf{C}_S(\mathbf{m}_S; \mathbf{A})$ be the coefficient of $\alpha^\mathbf{0}$ in $\alpha^{\mathbf{m}_S} \prod_{L,D} (V^S_{LD})^{A_{LD}}$, the* closing *of the seam word of $S$ at the exponents $-\mathbf{A}$, and $\mathsf{C}_{W_S}(\mathbf{m}_{Z_S}; \boldsymbol{\varepsilon})$ that of its node word at the exponents $-\boldsymbol{\varepsilon}$ of its entries, the product of the entries to the powers $\boldsymbol{\varepsilon}$ when $Z_S$ is*

*empty. When $S$ is not a leaf,*

$$\mathsf{C}_S(\mathbf{m}_S;\mathbf{A}) = \sum_{\boldsymbol{\varepsilon}} \prod_{L,D} \frac{A_{LD}!}{\prod_{\mathbf{D}} \varepsilon_{LD\mathbf{D}}!}\ \mathsf{C}_{W_S}(\mathbf{m}_{Z_S};\boldsymbol{\varepsilon}) \prod_i \mathsf{C}_{S_i}\big(\mathbf{m}_{S_i};\mathbf{A}_i(\boldsymbol{\varepsilon})\big),$$

*the sum over the $\boldsymbol{\varepsilon} = (\varepsilon_{LD\mathbf{D}})$ in $\mathbb{N}$, one for each nonzero entry, with $\sum_{\mathbf{D}} \varepsilon_{LD\mathbf{D}} = A_{LD}$, $\mathbf{A}_i(\boldsymbol{\varepsilon})$ being the loads $\sum \varepsilon_{LD\mathbf{D}}$, over the $(D,\mathbf{D})$ with $D_i = D'$, that $\boldsymbol{\varepsilon}$ puts on the stations $D'$ of the child $S_i$. At the root, $D$ dropped and $A_L = N_L$, $\mathsf{C}_{\mathcal{E}}(\mathbf{m};\mathbf{N})$ is the closing $\mathsf{C}(1)(\mathbf{d})$ of $W_F$ at the integral data $\mathbf{d}$ with the masses $\mathbf{m}$ and the exponents $-N_L$ of its letters, the constant term on the torus of $X_{W_F}$ (Theorem 48.23(ii)). Write $\mathsf{C}_S(\mathbf{d}) = \mathsf{C}_S(\mathbf{m}_S;-\mathbf{w})$ at the integral data $\mathbf{d} = (\mathbf{m}_S;\mathbf{w})$ of the seam word with $\mathbf{w} \le \mathbf{0}$. The seam word of a full node is, at its shares, the family of its seam letters, the letters of the room children of its stations, with the weights of the normalization, one for each letter of $F$ and each station in its table: a family of letters of tables whose marks form one molecule of the family of these tables, the node being full (Theorem 48.23(i)). So Proposition 47.10 and Theorem 47.11 hold for it, as for $W_F$, and for their quotients by submodules of elements of value zero (Theorem 48.23(ii); a full node of one mark has a module of dimension one and no nonzero element of value zero). So an element $\sum_\lambda r_\lambda \operatorname{Sh}_\lambda[1]$ of $\mathfrak{M}_{W_F}$ or of the module of the seam word of a full node $S$, or its image in a direct image $\mathfrak{H}_S$ of (iii), has the value zero, that is, is 0 (Proposition 47.15), exactly when $\sum_\lambda r_\lambda(\mathbf{d})\, \mathsf{C}_S(\mathbf{d}+\lambda)$ vanishes along the rays of Theorem 47.11(c), in the positive directions of Theorem 48.3(i): the elements of value zero are told by the closings of the node words and of the seam words of the leaves, combined along the tree by flows of integers, the closings of a seam word being found once for each class of the subtree of its node at each integral point.*

So the module of a family is read off its nodes. The engine holds one box for each class of node words and leaves, over the rational functions of their data, and one module for each class of subtrees, presented by one accepted set of shifts whose certificate is an identity among the products of the masters of the nodes of that subtree, with coefficients rational in its data and its shares. Every such module is the module of the seam word of its node, the values separating the module (Proposition 47.15), so it stands for that module at the parent, and the masters and the shift matrices of the family are those of the module at the root; the closings of the node words and of the seam words of the leaves, combined along the tree by flows of integers, tell its zeros. Nothing in this asks whether a spanning set is free: the module presented by an accepted set is a quotient of the module of the family by elements of value zero and maps onto the direct image, so it is the module itself. No step integrates, and no step needs a box on all the marks of the family or of any subtree: the algebra, like the word, is paid once for each distinct subtree, and a subtree that recurs is referred to by its class. Theorem 33.3 priced repetition by the word at full leans, in series closures and along chains; Theorem 48.15 prices it along every tree, at every width, in the word, and Theorem 48.18 pays the algebra once for each class of node words, leaves and subtrees.

The paths are where this happens. A letter of the family is the sum of the terms of its paths, and the flow of the shares chooses a path with the probabilities of the shares of its entries over the loads of their stations; the letter identity of the family is the letter identities of its nodes averaged over the paths, and its mark relation is the mark relation of the node of the mark, whose entries the flow uses in the proportions of their shares. The fibre translations then identify the lifts of one stride along different paths. The value of every element is the fibre integral of its class, and a relation among the classes in the direct image is a relation among the values of the family. A letter whose exponent is fixed at 0 has no paths and enters no node word, no seam word and no closing: the family is read through its letters alone.

## 48.8 The module of a transform from its block

The transform of a block is an exponential period (Lemma 48.8): its derivatives in its scales close over the rational functions of its scales and its data, in the twisted de Rham module of its orthant. Read on the shares of its block (Theorem 48.14(ii)), a scale times a monomial of its reading, with its coefficient, is the share of that monomial, the share side of the Gramma: $\Gamma(z)C^{-z}\cdot C = z\,\Gamma(z)C^{-z}$ after the share is moved by one. So a derivative in a scale multiplies the integrand by minus the total of the shares at that scale over the scale, and a translation along the fibre that moves the totals is a monomial in the scales. The mark relations of the transform ride on the shares as those of a family ride on its paths (Theorem 48.18(i)): at an own mark of the block they are its pins, and at a mark of a child they are the mark relation of the child. Modulo the translations along the fibre, each twisted by its monomial in the scales, the integrand of the block carries the module of its transform, with the scales acting by connections, over the rational functions of the data that the transform reads and of its scales, the values separating that module as they separate the module of a family (Proposition 47.15) wherever the transform reads all the data of its block; it is found from the modules of the children, once for each class of blocks. A block without children is a Gramma, and there this is Theorem 2.4 read on the shares: the pins are its exchange relations, the connections its ladder, and its toric relations say that a class does not depend on the lift that writes it.

**Definition 48.19** (The module of a transform; the twisted direct image of a block)**.** Let $F$, its tree and the normalization be as in Definition 48.17, let $S$ be a full node with a partition of its parts that it separates (Definition 48.13), and let $P$ be one of its blocks whose children $S_l$, $l\in P$, are full, the marks of its parts forming the set $B_P$. For an active letter $L$ write $h_{lL} = V^{(l)}_{LD^\circ_{l,L}}$ for the absent letter of the child $S_l$ and $H_L = \prod_{l\in P} h_{lL}$ for their product, and for a monomial $\tau$ of the reading $\beta_{P,LD}$, $(L,D)\in\Sigma_S$, with the coefficient $c_\tau$, write $r_\tau = \prod_l V^{(l)}_{LD_l(\tau)}/h_{lL}\prod_z \alpha_z^{q_{z,\tau}-u_{z,L}}$, the product over the children present in $\tau$, at their stations $D_l(\tau)$, and over the own marks of $P$, so that $\widehat{\beta}_{P,LD} = \sum_\tau c_\tau r_\tau$.

(i) *Data.* The *data* of the block are $\mathbf{d}_P = (\mathbf{m}_P;\mathbf{w})$: masses on $B_P$, the masses of the marks of the children and, at an own mark $z$, the mass $m'_z$ that the transform reads as $m_z - \sum_L s_L u_{z,L}$, and, when $P$ has children, an exponent $w_L$ of $H_L$ for every active letter $L$, which the transform reads as $s_L$. They range over $\mathbb{D}_P = \mathbb{C}^{B_P}\times\mathbb{C}^{A_P}$, $A_P$ the set of the active letters when $P$ has children and empty otherwise, on the balance hyperplane $\sum_{e\in B_P} m_e = \sum_L w_L \deg H_L$ when $S$ is not seen by its scales, with the lattice $\mathbb{D}_P(\mathbb{Z})$ of its integral points and the fields $\mathbb{K}_P = \mathbb{C}(\mathbb{D}_P)$ and $\mathbb{K}_P(\boldsymbol{\xi})$, $\boldsymbol{\xi} = (\xi_{LD})_{\Sigma_S}$ the scales. The data that the transform reads, as the data of $F$ vary over $\mathbb{D}_{W_F}$, form a linear subspace $\mathbb{E}_P$ of $\mathbb{D}_P$: all of $\mathbb{D}_P$, except when $S$ is the root, seen by its scales, and $P$, with children, is its one block, when $\mathbb{E}_P$ is the hyperplane of the balance of $F$.

(ii) *The module of the transform.* On the set $Y_P\subseteq(\mathbb{C}^\times)^{B_P}$ on which no $H_L$ vanishes, with the insertions $\mathbb{C}[Y_P]$, spanned by the strides $f_\lambda = \alpha^{\mathbf{u}}\prod_L H_L^{-n_L}$, $\lambda = (\mathbf{u};\mathbf{n})\in\mathbb{D}_P(\mathbb{Z})$, the *module of the transform* is

$$\mathfrak{M}_P = \big(\mathbb{C}[Y_P]\otimes\mathbb{K}_P(\boldsymbol{\xi})\big)/\mathcal{L}_P,\qquad \omega_e = m_e - \sum_L w_L\,\alpha_e\partial_{\alpha_e}\log H_L - \sum_{(L,D)\in\Sigma_S}\xi_{LD}\,\alpha_e\partial_{\alpha_e}\widehat{\beta}_{P,LD},$$

$m_e$ being the mass $m'_z$ at an own mark $z$ and $\mathcal{L}_P$ the span of the $(\partial_{y_e}+\omega_e)g$, $g\in\mathbb{C}[Y_P]$, $e\in B_P$, $y=\log\alpha$: the top cohomology of the twisted de Rham complex of Lemma 48.8(ii).

It carries the *shifts* $\mathrm{Sh}_\lambda[g\otimes r]=[f_\lambda g\otimes\sigma_\lambda r]$, the *connections* $\nabla_{LD}[g\otimes r]=[g\otimes\partial_{\xi_{LD}}r-\widehat{\beta}_{P,LD}\,g\otimes r]$ and the *values* $\langle g\rangle=\int_{\mathcal{O}_P} g\,\alpha^{\mathbf{m-1}}\prod_L H_L^{-w_L}\exp\big(-\sum\xi_{LD}\widehat{\beta}_{P,LD}\big)\,d\alpha$ on the domain of Lemma 48.8(i); the value of [1] at the data that the transform reads is $\mathcal{T}_P$. When $S$ is not seen by its scales, $P$ is the one block, and $Y_P$, its insertions and its relations are those of the quotient by the common scaling (Lemma 48.8(iii)), read as for a family of positive letters on its space of directions: the insertions invariant under it and the relations of its marks, which sum to its homogeneity.

(iii) *The share reading of the transform.* The *shares* are the $z_{LD\tau}$ of Theorem 48.14(ii), one for each monomial $\tau$ of each reading at $(L,D)\in\Sigma_S$, the set $I_P$, with the totals $Z_{LD}=\sum_\tau z_{LD\tau}$; the *share space* $\mathbb{S}_P\subseteq\mathbb{D}_P\times\mathbb{C}^{I_P}$ is the set of the $(\mathbf{d}_P,\mathbf{z})$ on the pins of the block, $\sum z_{LD\tau}(q_{z,\tau}-u_{z,L})=m'_z$ at every own mark $z$ and the scale pin of every child at its loads, the load of $S_l$ at a station at which it is present in some monomial being the sum of the shares of those monomials and at its absent station $w_L$ less the shares of the monomials of $L$ in which it is present, less one pin when $S$ is not seen by its scales; its lattice is $\mathbb{S}_P(\mathbb{Z})$, its fibre translations are the $(\mathbf{0},\delta)\in\mathbb{S}_P(\mathbb{Z})$, and $\mathbb{D}'_P(\mathbb{Z})$ is the image of $\mathbb{S}_P(\mathbb{Z})$ in $\mathbb{D}_P(\mathbb{Z})$. The *kernel* is $G_P=\prod\Gamma(z_{LD\tau})\,c_\tau^{-z_{LD\tau}}$, the *nodes* are the seam words $W_l$ of the children at their masses and loads, each carrying a module $\mathfrak{V}_l$, its module or the quotient of it by a submodule of elements of value zero, as the $\mathfrak{H}_{S_l}$ of Theorem 48.18(iii), and the *integrand module* $\mathfrak{J}_P=\bigotimes_l(\mathfrak{V}_l\otimes\mathbb{K}_{\mathbb{S}_P})$ carries the shifts $\mathrm{Sh}_{(\lambda,\tau)}$ of Theorem 48.5(ii) for the kernel $G_P$, its powers of the constants $c_\tau$ moving by constants.

(iv) *The twisted direct image.* For $\tau$ a vector of shares write $Z(\tau)$ for its totals and $\boldsymbol{\xi}^{-Z(\tau)}=\prod\xi_{LD}^{-Z_{LD}(\tau)}$. The *scaled integrand span* $\mathfrak{J}_P^{\boldsymbol{\xi}}$ is the $\mathbb{K}_P(\boldsymbol{\xi})$-span in $\mathfrak{J}_P\otimes_{\mathbb{K}_{\mathbb{S}_P}}\mathbb{K}_{\mathbb{S}_P}(\boldsymbol{\xi})$, the shifts fixing $\boldsymbol{\xi}$, of the $\boldsymbol{\xi}^{-Z(\tau)}\,\mathrm{Sh}_{(\lambda,\tau)}[1]^\otimes$, $(\lambda,\tau)\in\mathbb{S}_P(\mathbb{Z})$, and $\mathfrak{F}_P^{\boldsymbol{\xi}}$ is the $\mathbb{K}_P(\boldsymbol{\xi})$-span of the *twisted translations* $\boldsymbol{\xi}^{-Z(\delta)}\,\mathrm{Sh}_{(\mathbf{0},\delta)}\,t-t$, $t\in\mathfrak{J}_P^{\boldsymbol{\xi}}$. The *twisted direct image* of the block is $\mathfrak{H}_P=\mathfrak{J}_P^{\boldsymbol{\xi}}/\mathfrak{F}_P^{\boldsymbol{\xi}}$, with the classes $e_\lambda=\big[\boldsymbol{\xi}^{-Z(\tau)}\,\mathrm{Sh}_{(\lambda,\tau)}[1]^\otimes\big]$, $\lambda\in\mathbb{D}'_P(\mathbb{Z})$, the shifts $\mathrm{Sh}_\mu[t]=\big[\boldsymbol{\xi}^{-Z(\sigma)}\,\mathrm{Sh}_{(\mu,\sigma)}\,t\big]$, the connections

$$\nabla_{LD}[t]=\big[\partial_{\xi_{LD}}t-\xi_{LD}^{-1}Z_{LD}\,t\big],$$

$\partial_{\xi_{LD}}$ acting on the coefficients and $Z_{LD}$ being the function on $\mathbb{S}_P$, and the *scale integral* $\int\big[\sum_j r_j\,\boldsymbol{\xi}^{-Z(\tau_j)}\,\mathrm{Sh}_{(\lambda_j,\tau_j)}[1]^\otimes\big]=\sum_j r_j\,\langle f_{\lambda_j}\rangle$, $r_j\in\mathbb{K}_P(\boldsymbol{\xi})$, the values of the transform at the shifted data. The submodule that the connections generate from $e_{\mathbf{0}}=[[1]^\otimes]$ is $\mathfrak{H}_P^\partial$.

**Theorem 48.20** (The module of a transform from its block)**.** *In Definition 48.19:*

*(i)* The relations ride on the shares. *Write $c_\tau r_\tau$ over $H_L$,*

$$c_\tau r_\tau=\frac{c_\tau}{H_L}\prod_l V^{(l)}_{LD_l(\tau)}\prod_{l'}h_{l'L}\prod_z\alpha_z^{q_{z,\tau}-u_{z,L}},$$

*over the children $l$ present in $\tau$ and the children $l'$ of $P$ absent from it, and expand the numerator over the monomials of its letters. Every term, a stride $f_\lambda$ times its coefficient, lifts to $(\lambda,\mathbf{e}_\tau)\in\mathbb{S}_P(\mathbb{Z})$, $\mathbf{e}_\tau$ the unit vector of the share of $\tau$, and in $\mathfrak{J}_P\otimes_{\mathbb{K}_{\mathbb{S}_P}}\mathbb{K}_{\mathbb{S}_P}(\boldsymbol{\xi})$*

$$\sum_{\text{terms of } c_\tau r_\tau}(\textit{coefficient})\;\boldsymbol{\xi}^{-Z(\mathbf{e}_\tau)}\,\mathrm{Sh}_{(\lambda,\mathbf{e}_\tau)}[1]^\otimes=\xi_{LD}^{-1}\,z_{LD\tau}\,[1]^\otimes:$$

*a monomial of a reading, with its coefficient, is the share of the monomial over its scale. The strides $\alpha^{\mathbf{t}}/H_L$ of the monomials $c_{\mathbf{t}}\alpha^{\mathbf{t}}$ of $H_L$ lift with the share vector $\mathbf{0}$, and*

$\sum_{\mathbf{t}} c_{\mathbf{t}} \operatorname{Sh}_{((\mathbf{t};\mathbf{e}_L),\mathbf{0})}[1]^{\otimes} = [1]^{\otimes}$*: the letter identity of* $H_L$ *is the letter identities of the absent letters of the children. Write the terms of* $\omega_e$ *over powers of the* $H_L$*, and lift those that carry a scale* $\xi_{LD}$ *with the share vector* $\mathbf{e}_\tau$ *of their monomial* $\tau$ *and the others with* $\mathbf{0}$*. Then the mark relation* $(\partial_{y_e} + \omega_e)1$ *of the transform lifts to* $\big(m'_z - \sum z_{LD\tau}(q_{z,\tau} - u_{z,L})\big)[1]^{\otimes}$ *at an own mark* $z$*, which is* $0$ *by the pin of* $z$*, and at a mark* $e$ *of a child* $S_l$ *to the class* $\big[c_e^{(l)}(\iota_l)\cdot 1\big] \otimes \bigotimes_{l'\neq l}[1]$ *of the defect of* $e$ *in the seam word of* $S_l$ *at its masses and loads (Definition 48.1(iv)), which is* $0$*, the mark relation of the child. The same holds from every point* $(\lambda,\tau_\lambda)$ *of* $\mathbb{S}_P(\mathbb{Z})$*, with* $\operatorname{Sh}_{(\lambda,\tau_\lambda)}$ *applied to both sides.*

*(ii)* The span of the shifts of [1] is the twisted direct image. $\mathbb{D}'_P(\mathbb{Z})$ *contains the data of every term of every* $c_\tau r_\tau$*, so the span* $\mathfrak{M}'_P$ *of the shifts of* [1] *by it contains* [1] *and is carried into itself by the connections; it contains every derivative of* [1] *in the scales. The shifts, the classes* $e_\lambda$*, the connections and the scale integral are well defined on* $\mathfrak{H}_P$*; the connections commute with one another and with the shifts,* $\int \nabla_{LD} h = \partial_{\xi_{LD}} \int h$ *and* $\int \operatorname{Sh}_\lambda h = \sigma_\lambda \int h$*, and the classes* $e_\lambda$ *span* $\mathfrak{H}_P$*. The map*

$$O_P\Big(\sum_\lambda r_\lambda \operatorname{Sh}_\lambda[1]\Big) = \sum_\lambda r_\lambda\, e_\lambda, \qquad \lambda \in \mathbb{D}'_P(\mathbb{Z}),$$

*is a well-defined map of* $\mathfrak{M}'_P$ *onto* $\mathfrak{H}_P$ *that commutes with the shifts and the connections, with* $\int O_P(m) = \langle m\rangle$*. So* $\mathfrak{H}_P$ *is a differential module over* $\mathbb{K}_P(\boldsymbol{\xi})$*, as in Definition 48.9(i) with this field in the place of* $\mathbb{C}(\mathbf{y})$*, with the shifts of* $\mathbb{D}'_P(\mathbb{Z})$*, of dimension at most* $\dim \mathfrak{M}_P$*, finite; the kernel of* $O_P$ *consists of elements of value zero, and the values separate* $\mathfrak{M}_P$*, as they separate the module of a family (Proposition 47.15): an element of value zero is* $0$*. So* $O_P$ *is an isomorphism of* $\mathfrak{M}'_P$ *onto* $\mathfrak{H}_P$*.*

*(iii)* The connections are shifts. *On the classes,* $\nabla_{LD} e_\lambda = -\operatorname{Sh}_\lambda O_P\big(\widehat{\beta}_{P,LD}\big)$*, and* $O_P(\widehat{\beta}_{P,LD})$ *is the combination, with the coefficients of (i), of the classes of the terms of the* $c_\tau r_\tau$*; so the matrices of the connections on classes* $e_\lambda$ *are combinations of shift matrices. Theorem 48.18(iv) holds for* $\mathfrak{H}_P$ *with* $\mathbb{K}_P(\boldsymbol{\xi})$ *in the place of* $\mathbb{K}_W$*, the twisted translations in the place of the fibre translations and* $\mathbb{D}'_P(\mathbb{Z})$ *in the place of* $\mathbb{D}_W(\mathbb{Z})$*, the letter identities and the mark relations being those of (i), and with the subspace* $\mathfrak{N}$ *also carried into itself by the connections so read: an accepted set of shifts, whose certificates are identities among the products of the masters of the modules of the children over the rational functions of the data, the shares and the scales, presents a quotient of* $\mathfrak{M}'_P$ *by a submodule of elements of value zero,* $\mathfrak{M}'_P$ *itself by (ii), that maps onto* $\mathfrak{H}_P$*, with its shift matrices and its connection matrices, and a search for it stops.*

*(iv)* At the data that the transform reads. *Read over* $\mathbb{E}_P$ *and the share space* $\mathbb{S}_P \cap (\mathbb{E}_P \times \mathbb{C}^{I_P})$ *over it, with the rational functions of that share space and those of* $\mathbb{E}_P$ *and the scales in the place of* $\mathbb{K}_{\mathbb{S}_P}$ *and* $\mathbb{K}_P(\boldsymbol{\xi})$*, the definitions give the integrand module, its spans, the twisted translations and the twisted direct image* $\mathfrak{H}_P$ *with its classes* $e_\lambda$*,* $\lambda \in \mathbb{D}'_P(\mathbb{Z})$*, its connections and its scale integral, every* $\boldsymbol{\xi}^{-Z(\tau)} \operatorname{Sh}_{(\lambda,\tau)}[1]^{\otimes}$*,* $(\lambda,\tau) \in \mathbb{S}_P(\mathbb{Z})$*, being defined there though a shift by a point not in* $\mathbb{E}_P$ *is not; and (i) and (ii) hold for them, with the module* $\mathfrak{M}_P$ *of the transform over the rational functions of* $\mathbb{E}_P$ *and the scales, but for the shifts by the points of* $\mathbb{D}'_P(\mathbb{Z})$ *not in* $\mathbb{E}_P$ *and, when* $\mathbb{E}_P$ *is the hyperplane of the balance of* $F$*, for the separation of* $\mathfrak{M}_P$ *by the values: there* $O_P$ *maps* $\mathfrak{M}'_P$ *onto* $\mathfrak{H}_P$ *with a kernel of elements of value zero, and (vii) tells the elements of value zero among the combinations of the derivatives of the input of* $S$*. The scale integral of* $\mathfrak{H}_P^\partial$ *is the span of the derivatives of* $\mathcal{T}_P$ *in its scales over the rational functions of* $\mathbb{E}_P$ *and the scales, and at the data of* $F$ *off a proper algebraic subset of them,*

*on a section of the scale quotient by a monomial map, $\mathfrak{H}_P^{\partial}$ maps onto the differential module that $\mathcal{T}_P$ generates (Definition 48.9(i)). On the classes $\nabla_{LD} e_\lambda = -\sum(\text{coefficient})\, e_{\lambda+\mu}$, over the terms of $\widehat{\beta}_{P,LD}$ and their data $\mu$. Let $\Lambda \ni \mathbf{0}$ be a finite subset of $\mathbb{D}'_P(\mathbb{Z})$ with lifts, and let certificates, identities as in (iii) whose unknown coefficients are rational functions of $\mathbb{E}_P$ and of the scales, each found by one elimination over these once the denominators in the fibre coordinates are cleared, write every $e_{\lambda+\mu}$, $\lambda \in \Lambda$, on the $e_{\lambda'}$, $\lambda' \in \Lambda$. They give connections on the vectors over the rational functions of $\mathbb{E}_P$ and the scales with the basis $(\varepsilon_\lambda)_{\lambda\in\Lambda}$; modulo the smallest subspace that these carry into itself and that contains their commutators on the basis, they present a differential module whose submodule generated by $\varepsilon_{\mathbf{0}}$ maps onto $\mathfrak{H}_P^{\partial}$, $\varepsilon_\lambda \mapsto e_\lambda$, with a kernel of elements of value zero, which is 0 when $\mathbb{E}_P = \mathbb{D}_P$ by (ii). Such sets exist, and a search through the finite sets of points with lifts and of supports of certificates, in any order that reaches each, stops. When $\mathbb{E}_P = \mathbb{D}_P$ these are the objects of (ii) and (iii).*

*(v)* A block without children is a Gramma. *If $P$ has no children, $\mathcal{T}_P = \mathsf{G}_{\mathbf{m}'_P;t}\big((\xi_{LD}c_\tau)_\tau\big)$, the configuration $t$ being the exponents $q_{z,\tau} - u_{z,L}$ of its readings (Definition 48.13(iii)), the Gramma on a section, on one mark fewer, when $S$ is not seen by its scales (Lemma 48.8(iii)), $\mathfrak{J}_P = \mathbb{K}_{\mathbb{S}_P}$, and the share reading is the share side of the Gramma (Proposition 4.2). Then $e_{t_\tau} = \big[(\xi_{LD}c_\tau)^{-1} z_{LD\tau}\big]$, and Theorem 2.4 is read on the shares: the exchange relation at an own mark is its pin, the ladder, summed over the monomials of one scale, is the connection, and a toric relation $\partial^{\mathbf{v}^+}\mathsf{G} = \pm\partial^{\mathbf{v}^-}\mathsf{G}$ says that the class of the shift of the index by $t\mathbf{v}^+ = t\mathbf{v}^-$ does not depend on the lift, $\mathbf{v}^+$ or $\mathbf{v}^-$, that writes it.*

*(vi)* Once for each class. *A bijection of the marks of two blocks keeping their seam letters, the absent stations, the absent exponents and the readings, with the subtrees of their children carried to one another, carries the one $\mathfrak{H}_P$ to the other; so the modules of the transforms are found once for each class of blocks, from the modules of the classes of the children, each found once (Theorem 48.18(iii)). With Proposition 48.10(ii) the input of every node of $\Sigma$ in Theorem 48.15 whose blocks with children have full children generates a subquotient of a construction from the $\mathfrak{H}_P^{\partial}$ of its distinct blocks, read as in (iv), and the module of rank one of the exponential of its constant parts; and, at the data of $F$ off a proper algebraic subset, every module $N_r$ of Theorem 48.15(ii) of such nodes is the image under the values of a module presented by connection matrices over the rational functions of the data of $F$ and the scales of the sections, found from the modules of the children of the blocks without integrating, its group being the image of the group of that module.*

*(vii)* Elements of value zero. *Let $S \in \Sigma$ have the input $\phi$, read as a function of its scales and of its data: the masses of the marks of $S$ and the exponents $s_L$, free of the balance of $F$ except at a root not seen by its scales, whose transform is taken on the quotient by the scaling. Let $W_S^\sharp$ be the family of positive letters on the marks of $S$ formed by its seam letters $V_{LD}^S$, $(L,D) \in \Sigma_S$, read at the seam points of its children, at the exponents $a_{LD}$, and by the products $H_L^S = \prod_l h_{lL}$ of the absent letters of its children, one for each active letter $L$ (none when $S$ has no children), at the exponents $w_L$, with the masses of its children's marks and $m_z^\sharp = m_z - \sum_L u_{z,L} w_L$ at its own marks. Every letter of $W_S^\sharp$ is a letter of a table on the marks of $S$: $V_{LD}^S$ of the room child of the station $D$, and $H_L^S$ of the direct sum of the room children of the absent stations of the children, the own marks being loops in it for a letter $K_j$ and coloops for a letter $\Psi_j$. On the balance of $W_S^\sharp$,*

$$\int_{\Xi_S} \boldsymbol{\xi}^{\mathbf{a}}\, \phi\, \frac{d\boldsymbol{\xi}}{\boldsymbol{\xi}} = \prod_{\Sigma_S} \Gamma(a_{LD})\; P_{W_S^\sharp}(\mathbf{a}, \mathbf{w}), \qquad w_L = s_L - \sum_{D\colon (L,D)\in\Sigma_S} a_{LD},$$

*which on the pins of the seam of $S$ is Theorem 48.14(i), $W_S^\sharp$ being there its seam word with the powers of its constant parts. Read $S$ on a section whose coordinates $\mathbf{y}$ are a basis of the monomials in its scales that the scaling keeps, $\vartheta_i = y_i\partial_{y_i}$. An element $E = \sum_{\mathbf{k},\mathbf{b}} c_{\mathbf{kb}}\, \mathbf{y}^{\mathbf{k}}\vartheta^{\mathbf{b}}\phi$, its coefficients rational in the data of $\phi$ and its denominators in $\mathbf{y}$ cleared, then vanishes identically exactly when the element*

$$E^\sharp = \sum_{\mathbf{k},\mathbf{b}} c_{\mathbf{kb}}\, (-\boldsymbol{\nu}(\mathbf{a}) - \mathbf{k})^{\mathbf{b}}\, R_{\mathbf{k}}\, \mathrm{Sh}_{\lambda_{\mathbf{k}}}[1]$$

*of the module of $W_S^\sharp$ has the value zero, the $c_{\mathbf{kb}}$ read on its data through $s_L = w_L + \sum_D a_{LD}$ and $m_z = m_z^\sharp + \sum_L u_{z,L} w_L$; $W_S^\sharp$ being a family of letters of tables, finitely many of its closings tell whether it has (Theorem 48.23), and it has exactly when it is $0$ in the module of $W_S^\sharp$ (Proposition 47.15). Here $\mathbf{y}^{\mathbf{k}} = \boldsymbol{\xi}^{\mu_{\mathbf{k}}}$ on the section; $\lambda_{\mathbf{k}}$ moves $\mathbf{a}$ by $\mu_{\mathbf{k}}$, $\mathbf{w}$ by minus its totals $T_L(\mu_{\mathbf{k}}) = \sum_D \mu_{\mathbf{k},LD}$ and every own mass $m_z^\sharp$ by $\sum_L u_{z,L} T_L(\mu_{\mathbf{k}})$, a point of the lattice of $W_S^\sharp$; $\boldsymbol{\nu}(\mathbf{a})$, linear in $\mathbf{a}$, is the index of the section, $\boldsymbol{\xi}^{\mathbf{a}}\, d\boldsymbol{\xi}/\boldsymbol{\xi}$ being a nonzero constant, fixed on the balance, times $\mathbf{y}^{\boldsymbol{\nu}(\mathbf{a})}\, d\mathbf{y}/\mathbf{y}$ there; and $R_{\mathbf{k}}$ is the quotient of the Gammas at the moved exponents over those at $\mathbf{a}$. At every node but a root seen by its scales with children the data of $F$ and $\mathbf{a}$ reach all the data of $W_S^\sharp$, so that this tells the elements of value zero over the data of $F$. At such a root the data of $\phi$ and $\mathbf{a}$ reach all of them, which tells the elements of value zero over the data of $\phi$, and the data of $F$ and $\mathbf{a}$ reach the hyperplane $\mathbb{E}_S^\sharp$ on which $\sum_L \delta_L w_L = 0$, $\delta_L$ the degree of the reading at $L$ (Definition 48.13(ii)) and $0$ off $\Sigma_S$, the balance of $F$ read on the data of $W_S^\sharp$: $E$, its coefficients read at the data of $F$, vanishes there exactly when $E^\sharp$, its coefficients read on $\mathbb{E}_S^\sharp$, has the value zero on $\mathbb{E}_S^\sharp$, that is, exactly when $E^\sharp$ is $0$ in the module of $W_S^\sharp$ on $\mathbb{E}_S^\sharp$, a hyperplane cut out by a condition on the exponents alone that meets the tube (Proposition 47.15); Proposition 47.14 decides this by finitely many coefficients of the expansion at the native gauge of a native direction of $\mathbb{E}_S^\sharp$ in which every $H_L^S$ with $\delta_L \neq 0$ has the exponent $0$, a native gauge whose coordinates are algebraic over the field of the weights and which can be taken rational when the weights are rational.*

So the scale side of a node is read off its children as its share side is. A block reads its children through their seam functions at loads that are linear in its shares (Theorem 48.14(ii)); a scale of the block enters the integrand raised to minus the total of its shares, and a derivative in a scale multiplies the integrand by minus that total over the scale; so the module of the transform is the span of the twisted shifts of $[1]^\otimes$ in the integrand module of the block, a tensor product of the modules of its children with Pochhammer quotients of its Gammas, modulo the translations along its fibre, each twisted by a monomial in the scales, and its values separate it over all the data of the block. The certificate that presents it is an identity among the products of the masters of the children, and its connections are combinations of its shifts. The elements of value zero of the input of a node, among the combinations of its derivatives with coefficients rational in its data, are read on the shares: integrated against the powers of its scales, the input is, up to Gammas, the period of the family formed by the seam letters of its node and the products of the absent letters of its children, its seam word on the pins of the seam (Theorem 48.14(i)), so the closings of that family tell them (vii). The module of a transform is found over the rational functions of the data that it reads, those of $\mathbb{E}_P$, the data of $F$ read on the block (part (iv)). The modules that the group of a word asks for (Theorem 48.15(ii)) are therefore found in the same way as the module of the family (Theorem 48.18): one box for each class of node words and leaves, one module for each class of subtrees, and one module for each class of blocks, each from the ones below it. At width one a series or parallel node reads its cells through their transforms $f_{\mathcal{A}}$ and dual transforms $\check{f}_{\mathcal{A}}$

(Theorem 48.14(iii)), whose modules are, up to elements of value zero, the twisted direct images of the blocks of the cells, read through the modules of the families of the two hinge minors of the cells, and its marks through Grammas without children, which on the centre ray are $\mathsf{g}$, with $\vartheta^2\mathsf{g} = y\,\mathsf{g}$ (Theorem 45.28(ii)). A block whose children are not all full is read as its exponential period (Theorem 48.15), whose module is defined as in Definition 48.19(ii); the choice of $\Sigma$ can always avoid such blocks, a node with a child that is not full being read on its shares.

## 48.9 The procedure in all the data

Every piece of the reading of a family is now stated over the rational functions of its data: the module, its boxes, its masters and its closings (Chapter 47), the module along the tree from the modules of the node words (Section 48.7), the modules of the transforms of the blocks (Section 48.8), and the word along the tree with its inputs, group, work and price (Theorem 48.15). What is left is the question that a form asks. Its letters are those with an exponent that is not zero, they may form several molecules, and the data may be off the balance wall. The handbook answers each case with the period of a family of letters of tables on its data space. Off the balance wall the pseudointegral of the form is the period at the target, or on a pole hyperplane the constant term of the period along the ray of the masses, along the rays of the masses of the active molecules when they are several. The interest is a product of Gamma functions times the period of the letters with the sum of the marks of each molecule added. On the balance wall several molecules carry nothing but their reduced period, at the masses at which every molecule is balanced, and when the weights do not factor across the molecules the reduced period is read through its molecules on a factorization of the weights, one share for each term. The values separate the module of every family of positive letters (Proposition 47.15), so every module presented from a box or on the shares is the module itself, as is the module that an accepted set of shifts presents for a transform that reads all the data of its block (Theorem 48.20(iii)), and an element of such a module has the value zero only when it is 0; the closings tell which elements are 0 for every family of letters of tables, by the proof of Theorem 39.57, and a box gives masters once its matrices are invertible. The procedure below puts the pieces in order, and its theorem says that it stops on every family and returns its reading in finite matrices over rational functions of the data, and in words at real data.

**Proposition 48.21** (The question at every data)**.** *Let $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^{N}$ be a family on $\mathcal{E}$, $n \geq 2$. Its* letters *are the $K_j$ and the $\Psi_j$ whose exponents are not zero, its* active tables *those with a letter, and its* active molecules *$M_1, \ldots, M_q$ the molecules of the family of its active tables (Definition 7.4).*

*(i)* The letters. *The form $W\,d\alpha_{\mathbf{m}} = \alpha^{\mathbf{m-1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\,d\alpha$ is that of the family $W$ of positive letters formed by its letters (Definition 48.1). Its space of directions is the space of the vectors of zero sum on every active molecule, so that $V_W$ is $\mathbb{R}^{\mathcal{E}}$ modulo the indicators of the active molecules, and its data space $\mathbb{D}_W$ is the set of the data at which every active molecule is balanced (Theorem 48.23(i)). For $q = 1$ it is the balance hyperplane in the masses and the exponents of the letters, and $W$ is the family $W_F$ of Definition 48.17.*

*(ii)* The interest. *Let $W^+$ be the family of positive letters formed by the letters of $F$ and the sums $|\alpha_{M_i}| = \sum_{e \in M_i} \alpha_e$ at the exponents $\kappa_{M_i}$, $|\alpha_{M_i}|$ being the letter $K$ of the table of rank one on $\mathcal{E}$ whose bases are the marks of $M_i$. It is a family of letters of tables with the active molecules of $F$, and its data space is the image of all the data of $F$, every $M_i$ being balanced in $W^+$. At data with $\operatorname{Re}\kappa_{M_i} > 0$ for every $i$ and every wall $\kappa_U$, $U$ a nonempty proper subset*

*of an active molecule, of positive real part,*

$$J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \prod_{i=1}^{q} \Gamma(\kappa_{M_i})\ P_{W^+}(\mathbf{m};\mathbf{s},\mathbf{t}),$$

$$P_{W^+}(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\prod_i \Delta_{M_i}} u^{\mathbf{m}-\mathbf{1}} \prod_j K_j(u)^{-s_j} \Psi_j(u)^{-t_j} \prod_i d\omega_{M_i}(u_i),$$

*and elsewhere as meromorphic functions. For $q = 1$ this is $J_F = \Gamma(\kappa_{\mathcal{E}})P_F$, $P_F$ being the period of $W^+$ at every data (Theorem 8.7).*

*(iii)* Off the balance wall. *Let $q = 1$ and $\kappa_{\mathcal{E}} \neq 0$. The form is not projective (Definition 16.4). At real data with $\mathbf{m} > 0$ and $s_j, t_j \geq 0$, not all zero, its pseudointegral is $\mathrm{CT}_{\eta=0}\, P_F(\boldsymbol{\tau} + \eta\mathbf{m};\mathbf{s},\mathbf{t})$ at the target $\boldsymbol{\tau} = \lambda\mathbf{m}$, $\lambda = \sum_j (s_j r_j + t_j r_j^*)/m_{\mathcal{E}}$, the masses slid along their ray into the balance wall (Definition 16.10, Theorem 16.8(ii)). It is the period at the target, $P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$, when the target lies on no pole hyperplane (Theorem 16.15, Corollary 16.16), and on a pole hyperplane the constant term at $\eta = 0$ of the period of $W^+$ along the line $\boldsymbol{\tau} + \mathbb{C}\mathbf{m}$ of its data space. With $q \geq 2$, whatever the weights, let $r_{j,i}$ be the number of the marks of $M_i$ in a basis of an active table $j$, the same for every basis (Lemma 7.10), $r^*_{j,i} = |M_i| - r_{j,i}$, and slide the masses of every active molecule along its own ray: $\boldsymbol{\tau}_{M_i} = \lambda_i \mathbf{m}_{M_i}$, $\lambda_i = \sum_j (s_j r_{j,i} + t_j r^*_{j,i})/m_{M_i}$. Then $\boldsymbol{\tau}$ balances every active molecule, the period of $W^+$ along $\boldsymbol{\tau} + \sum_i \eta_i \mathbf{m}_{M_i}$ has its poles on hyperplanes on which one $\eta_i$ is constant, and its constant term at $\boldsymbol{\eta} = 0$, the* pseudointegral *of the form for several active molecules, which Definition 16.10 gives only when the weights factor, is the reduced period $P_{V_F}(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ of (iv) off the pole hyperplanes; with factoring weights it is the product of the pseudointegrals of the molecules, each at its own target (Definition 16.10).*

*(iv)* Several active molecules. *Let $q \geq 2$, whatever the weights. On the balance wall one of $\kappa_S$, $\kappa_{\mathcal{E}\setminus S}$, $S$ a union of active molecules, has real part at most 0, and the integral of the form diverges (Proposition 8.16); $P_F$ vanishes there as a meromorphic function (Theorem 16.17). Proposition 16.12 holds with the product $\prod_i P_{F_i}(\mathbf{m}_i^0;\mathbf{s},\mathbf{t})$ replaced by the* reduced period

$$P_{V_F}(\mathbf{m}^0;\mathbf{s},\mathbf{t}) = \int_{\prod_i \Delta_{M_i}} u^{\mathbf{m}^0-\mathbf{1}} \prod_j K_j(u)^{-s_j} \Psi_j(u)^{-t_j} \prod_i d\omega_{M_i}(u_i)$$

*at masses $\mathbf{m}^0$ at which every active molecule is balanced and every wall $\kappa_U$, $U$ a nonempty proper subset of an active molecule, is positive: on the balance wall the form carries nothing but its reduced period, the* working interest*. It is the period of $W$ on $\mathbb{D}_W$, and when the weights factor across the active molecules it is $\prod_i P_{F_i}$ (Proposition 7.9).*

*(v)* One active molecule. *On the balance wall the period converges exactly on its tube, the data at which every proper wall has positive real part, which is not empty, and elsewhere it is its continuation, meromorphic, with its poles on the hyperplanes $\kappa_U \in \{0, -1, -2, \dots\}$ (Theorems 8.7, 47.2 and 48.3(ii)).*

*So every question that the form asks is the period of $W$ or of $W^+$, a family of letters of tables, on its data space: at a point, along a line or along the affine subspace of the rays of the masses of the active molecules, or times Gamma functions of linear forms in the data.*

**Definition 48.22** (A factorization of the weights; the molecule families)**.** Let the active molecules of $F$ be $M_1, \dots, M_q$, $q \geq 2$. Every active table is the direct sum of its restrictions $\mathcal{B}_j|M_i$, its bases

the unions $T = T_1 \sqcup \cdots \sqcup T_q$ of bases $T_i$ of the restrictions (Proposition 7.9). A *factorization* of the weights is, for every active table, a finite set $R_j$ and weights $\theta_{j,r}^{(i)} > 0$ on all the bases of $\mathcal{B}_j|M_i$, $r \in R_j$, $i \le q$, with

$$\theta_{j,T} = \sum_{r \in R_j} \prod_{i=1}^{q} \theta_{j,r}^{(i)}(T \cap M_i)$$

for every basis $T$ of table $j$. The *molecule family* $F^{(i)}$ is the family on $M_i$ of the tables $(\mathcal{B}_j|M_i, \theta_{j,r}^{(i)})$, $r \in R_j$, $j$ active, with the letters $K_{j,r}^{(i)}$ when $K_j$ is a letter of $F$ and $\Psi_{j,r}^{(i)}$ when $\Psi_j$ is, so that

$$K_j = \sum_{r \in R_j} \prod_{i=1}^{q} K_{j,r}^{(i)}, \qquad \Psi_j = \sum_{r \in R_j} \prod_{i=1}^{q} \Psi_{j,r}^{(i)}.$$

The weights *factor* across the active molecules exactly when they have a factorization with one element in every $R_j$.

The marks of a molecule family form one molecule. An active molecule of one mark, a loop or a coloop in every active table, has a molecule family of monomial letters, a point as its space $V$, the period $\prod c^{-z}$ of the coefficients of its letters, and a module of dimension one, as a full node of one mark does (Theorem 48.18(v)).

**Theorem 48.23** (The closings for every family of letters of tables)**.** *Let $W$ be a family of positive letters each of which is a letter $K_j$ or $\Psi_j$ of a table $(\mathcal{B}_j, \theta_j)$ on $\mathcal{E}$, with any weights, its active tables those with a letter in $W$ and its active molecules the molecules of the family of its active tables.*

*(i)* Directions and fan. *The differences of two exponents of one letter generate the lattice $M_W$; so $\vec{\mathcal{N}}_W^{\perp}$ is spanned by the indicators of the active molecules, $V_W$ is $\mathbb{R}^{\mathcal{E}}$ modulo them, and the subtorus of Definition 48.1(iii) is the set on which $\alpha$ is constant on every active molecule. The products of the sectors of the orders of the active molecules form a fan of $W$, with the generators $\mathbf{1}_U$ and the walls $\kappa_U$, $U$ a nonempty proper subset of an active molecule.*

*(ii)* Value zero. *A line of $\mathbb{D}_W$ with an integral origin and a positive direction carries a ray and the module of that ray, the restrictions of Definition 47.3 and Proposition 47.5 read with Theorem 48.3(iii) in the place of Theorem 47.4, and its closings are those of Definition 47.9, the constant terms on the torus of $X_W$. Along such a line Propositions 39.34 and 39.35(ii) and Theorems 39.57 and 44.16 hold, with $n-1$ replaced by $r_W$, the number of the marks less that of the active molecules, the orders of $\mathcal{E}$ by the tuples of orders of the active molecules, and $V_1$ by the product over the active molecules of their sums $\sum_\pi \prod_l \kappa_{1,U_l}^{-1}$. So Proposition 47.10, Theorem 47.11 and Proposition 47.12 hold for the module of $W$ (Theorem 48.3(iii)) and its quotients by submodules of elements of value zero, the module itself (Proposition 47.15), with the positive directions of Theorem 48.3(i) and the generators of Theorem 48.3(iv). When every active molecule has one mark, $V_W$ is a point and the module has dimension one.*

*So whether two elements of the module of a family of letters of tables have the same value as functions of all its data is decided by finitely many closings, whatever its letters and its weights.*

The two places at which the proof of Theorem 39.57 used one molecule, the definiteness of the Hessian at the native gauge and the single point of the torus at which every letter has its largest modulus, hold for every family of letters of tables once its space $V_W$ is taken modulo its active molecules. For a family of positive letters whose exponent differences generate a sublattice of $M_W$ of index $h$, the torus has $h$ such points; for the letters of tables $h = 1$.

**Theorem 48.24** (Several active molecules through their molecule families)**.** *Let the active molecules of $F$ be $M_1, \dots, M_q$, $q \geq 2$.*

*(i)* Factorizations exist. *For an active table $j$ and $\delta > 0$ let $c_j^{(\delta)}$ be the coordinates of $\theta_j$ on the products $\bigotimes_i (\mathbf{e}_{T_i} + \delta \mathbf{1})$ of vectors on the bases of the restrictions,*

$$c_j^{(\delta)} = \bigotimes_{i=1}^{q} \left( \mathbf{I}_{j,i} - \frac{\delta}{1 + \delta b_{j,i}} \mathbf{J}_{j,i} \right) \theta_j,$$

*$b_{j,i}$ the number of the bases of $\mathcal{B}_j | M_i$, $\mathbf{I}_{j,i}$ the identity and $\mathbf{J}_{j,i}$ the square matrix of ones on them. As $\delta \to 0$ its entries tend to those of $\theta_j$, so for small $\delta$ they are positive; then $R_j$ the bases $T'$ of table $j$, $\theta_{j,T'}^{(1)} = c_{j,T'}^{(\delta)}(\mathbf{e}_{T_1'} + \delta \mathbf{1})$ and $\theta_{j,T'}^{(i)} = \mathbf{e}_{T_i'} + \delta \mathbf{1}$, $i \geq 2$, are a factorization.*

*(ii)* The reading. *For every factorization, at data $(\mathbf{m}; \mathbf{s}, \mathbf{t})$ in the tube of the reduced period with the letters' exponents of positive real parts, and real shares $c_{j,r}, c_{j,r}^* > 0$ with $\sum_r c_{j,r} = \operatorname{Re} s_j$ and $\sum_r c_{j,r}^* = \operatorname{Re} t_j$,*

$$P_{V_F}(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{pins} \prod_j \Gamma_{|R_j|} \begin{bmatrix} \mathbf{z}_j \\ s_j \end{bmatrix} \Gamma_{|R_j|} \begin{bmatrix} \mathbf{z}_j^* \\ t_j \end{bmatrix} \prod_{i=1}^{q} P_{F^{(i)}}(\mathbf{m}_{M_i}; \mathbf{z}, \mathbf{z}^*),$$

*the fibre pinned by $\sum_r z_{j,r} = s_j$ and $\sum_r z_{j,r}^* = t_j$ and taken through $(\mathbf{c}, \mathbf{c}^*)$, converging absolutely, the factors of a letter that is not a letter of $F$ and of a table with $|R_j| = 1$ being 1; $P_{F^{(i)}}(\mathbf{m}_{M_i}; \mathbf{z}, \mathbf{z}^*)$ is the period of the molecule family at the exponents $z_{j,r}$ of $K_{j,r}^{(i)}$ and $z_{j,r}^*$ of $\Psi_{j,r}^{(i)}$. Its data lie on its balance wall, and its walls are the walls $\kappa_U$ of $F$, $U \subseteq M_i$, whatever the shares. So this is a positive share reading of the reduced period (Definition 48.4) with the molecule families as its nodes, its lattice maps onto $\mathbb{D}_W(\mathbb{Z})$, and Theorem 48.5 holds for it: every value of every element of the module of the reduced period is a fibre integral of a multilinear form in the values of the masters of the molecule families, each read in all its data, and isomorphic molecule families have one module. With factoring weights there are no shares and $P_{V_F} = \prod_i P_{F^{(i)}}$.*

*The same holds for the period of $W^+$, whose molecule families carry the letters $|\alpha_{M_i}|$ at the exponents $\kappa_{M_i}$.*

So several active molecules are read as one node above their molecule families, whose letters are the terms of the factorization: a molecule family is a family on one molecule, read by the procedure below, and the shares of the terms are the only fibre that the factorization adds. The fewer the terms, the smaller the fibre; with factoring weights there is none.

**Definition 48.25** (The procedure in all the data)**.** *Given*: a family $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ on $\mathcal{E}$, $n \geq 2$, its weights in a field $\Bbbk \subseteq \mathbb{C}$, at least one exponent not fixed at 0, and the form $\alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j} \, d\alpha$, with its masses and exponents as coordinates and the exponents fixed at 0 named. *The question*, from the tables alone (Proposition 48.21):

(1) *Letters*: the letters, the active tables and the active molecules, and the families $W$ and $W^+$ of letters of tables, with their data spaces.

(2) *The interest*: $\prod_i \Gamma(\kappa_{M_i})$ times the period of $W^+$.

(3) *On the balance wall*: the period of $W$ when the active molecules are one; when they are several, return the certificate of Proposition 48.21(iv), and the working interest is the period of $W$ on $\mathbb{D}_W$.

(4) *Off the balance wall*, one active molecule: return the target; the pseudointegral is the period of $W$ there off the pole hyperplanes, and on them the constant term at $\eta = 0$ of the period of $W^+$ along $\boldsymbol{\tau} + \eta\mathbf{m}$; for several active molecules, the targets of the molecules, and the pseudointegral is the period of $W$ at them off the pole hyperplanes and on them the constant term at $\boldsymbol{\eta} = 0$ of the period of $W^+$ along $\boldsymbol{\tau} + \sum_i \eta_i \mathbf{m}_{M_i}$, the product over the molecules when the weights factor (Proposition 48.21(iii)).

Each period so named, of a family $V$ of letters of tables on its data space, is read as follows.

(5) *Several active molecules*: masters of the module of $V$ from its box (Proposition 47.13, Theorem 48.3(iv)), an element having the value zero exactly when it is 0 (Proposition 47.15), as its closings also tell (Theorem 48.23), and its reading through the molecule families (Theorem 48.24(ii)) of the factorization with one term in every table when the weights factor and otherwise of Theorem 48.24(i) at the largest $\delta = 1/k$, $k \in \mathbb{N}$, at which it is one; a molecule family of one mark is read directly, and every other by (6)–(14).

*Classification* of one active molecule, from the tables alone:

(6) the Foreman tree and the label (Definition 16.14, steps (1)–(5)), read as a tree of nested bunches, the childcules of a peel being the pieces of the core (Definition 29.9), in the normalization of Definition 28.1: the seam letters, the transfer tensors, the node words, the seam words of the leaves and the full nodes (Definition 48.11);

(7) at every full node seen by its scales, absent stations and exponents and the finest partition that it separates, with the blocks, their readings and the constant parts (Definition 48.13);

(8) the classes of the node words, the leaves, the subtrees and the blocks under the bijections of their marks that keep their seam letters (Theorems 48.5(v), 48.18(iii) and 48.20(vi)).

*Reading*, over the rational functions of the data:

(9) for each class of the node words and of the leaves, masters with their shift matrices at the generators and their negatives, from a box, with its certificates (Proposition 47.13, Theorem 48.3(iv));

(10) the module: from the leaves up, at every full node $S$, once for each class of its subtree down to the full nodes, an accepted set on the shares, presenting the module $\widehat{\mathfrak{H}}_S$ of its seam word, that of $V$ at the root, with its shift matrices and certificates (Theorem 48.18(iii), (iv), $V$ being the family $W_F$ of the letters of its tables);

(11) a set $\Sigma$ of full nodes, each with a partition of its parts that it separates, the one block when it is not seen by its scales, and the children of its blocks full, to be read on their scales, the node words of the nodes read on their shares being kept: among the finitely many, one of least work (Theorem 48.15(iii)), the dimensions of the kept classes being those of (9);

(12) for every node of $\Sigma$ and every class of its blocks with children, an accepted set of shifts presenting the module of its transform with its connection matrices over the data that it reads (Theorem 48.20(iii), (iv)); a block without children is a Gramma (Theorem 48.20(v));

(13) value zero: an element of a module of (5), (9) or (10), or of (12) when the transform reads all the data of its block, has the value zero exactly when it is 0, which its reduction on the masters shows (Proposition 47.15, Theorem 48.20(ii)), and, in a module of (5) or (10), or of a leaf in (9), exactly when finitely many of its closings vanish (Theorem 48.23), at a full node and at the root of one active molecule computed from the closings of the node words and of the leaves combined along the tree (Theorem 48.18(v)); a combination $E$ of the derivatives of the input of a node $S$ of $\Sigma$ vanishes identically over the data of the input

exactly when the element $E^\sharp$ of the module of $W_S^\sharp$ is 0, which its reduction on masters of that module from its box shows (Propositions 47.13 and 47.15), as finitely many of its closings also do (Proposition 47.12), and, at a root seen by its scales with children, over the data of $V$ exactly when finitely many coefficients of the expansion of $E^\sharp$ at the native gauge of a native direction of the hyperplane $\mathbb{E}_S^\sharp$ vanish, rational when the weights are rational (Theorem 48.20(vii), Propositions 47.14 and 47.15);

(14) the word along the tree with the nodes of $\Sigma$ read on their scales, at real data and at complex data whose real parts are such: its inputs, the modules $N_r$ whose group bounds the group of the word, its work and its price (Theorem 48.15).

*Output*: the certificate, the target and the Gamma factors that the question returns; for each period read, its tree and label, masters of the module of (10), or of (5) for several active molecules, over $\Bbbk(\mathbb{D}_V)$ with their shift matrices at the generators of the lattice and their negatives and the certificates of every relation used, the reduction of every insertion, the closings, the masters of the modules of the families $W_S^\sharp$ and the coefficients of the expansions at native gauges that decide the value zero of the combinations of the derivatives of the inputs, the modules of the transforms of the blocks with their connection matrices, and the word with its inputs, the presented modules $N_r$, its work and its price. Every output is a word, a finite set or a finite matrix over rational functions of the data; no value is computed.

**Theorem 48.26** (The procedure in all the data reads every family)**.** *Let $F$ be a family on $\mathcal{E}$, $n \geq 2$, with its weights in $\Bbbk$ and at least one exponent not fixed at* 0.

*(i)* It stops. *Every step is a finite computation over $\Bbbk$ and over fields of rational functions over $\Bbbk$ of the data of $W$ and $W^+$, of the molecule families, of the node words, of the families $W_S^\sharp$ and their hyperplanes $\mathbb{E}_S^\sharp$, and of the data, shares and scales of the blocks, the expansions of Proposition 47.14 being over the finite extension of $\Bbbk$ generated by a native gauge, which is $\Bbbk$ when the weights are rational and the gauge is taken rational; and the procedure stops.*

*(ii)* The question. *Its certificates, targets and Gamma factors are those of Proposition 48.21, and every question it returns is the period of $W$ or $W^+$ at a point, along a line or, for several active molecules, along the affine subspace of the rays of their masses, or times those factors.*

*(iii)* The module. *For each period read, the module that it presents is the module of $V$ itself, a quotient of it by elements of value zero being the module (Proposition 47.15), with masters and their shift matrices at the generators and their negatives (Theorem 48.18(iv), Proposition 47.13). The matrices carry the values from a point to every lattice translate, every insertion is reduced (Theorem 47.4(v), with Theorem 48.3(iii)), and every relation used has a certificate: those of the boxes, identities of functions on the letter complements over the rational functions of the data (Theorem 47.8(ii)), those of the presentations of Proposition 47.13, and those of the accepted sets, identities among products of masters (Theorem 48.18(iv)).*

*(iv)* Value zero. *An element of that module has the value zero as a function of all the data exactly when it is* 0*, which its reduction on the masters shows, and exactly when finitely many closings vanish, rational numbers when the weights and the element are rational (Proposition 47.15, Theorem 48.23); so whether two insertions have the same value is decided, and every relation among the values is a relation of the module. A combination $E$ of the derivatives of the input of a node $S$ of $\Sigma$ vanishes as a function of the data of the input exactly when the element $E^\sharp$ of the module of $W_S^\sharp$ is* 0*, which its reduction on masters of that module from its box shows (Proposition 47.13), and exactly when finitely many closings of $E^\sharp$ vanish; at a root seen by its scales with children it vanishes as a function of the data of $V$ exactly when $E^\sharp$ is* 0 *in the*

*module of $W_S^\sharp$ on the hyperplane $\mathbb{E}_S^\sharp$, which finitely many coefficients of the expansion at the native gauge of a native direction decide, rational functions over the field generated by the weights, the coefficients of the element and that native gauge, and over $\mathbb{Q}$ when the weights and the element are rational and the native gauge is taken rational (Theorem 48.20(vii), Propositions 47.15 and 47.14).*

*(v)* The word. *For one active molecule, at real data on the balance wall with $s_j, t_j \geq 0$, not all zero, every proper wall positive and a native gauge, and at complex data whose real parts are such data, every exponent of a letter having a positive real part, the period is the word, with the inputs, the group bounded by the modules $N_r$, the work and the price of Theorem 48.15, the modules $N_r$ being presented from the modules of the transforms (Theorem 48.20(vi)); at every lattice translate of such a point the value is carried by the shift matrices, and at every other data the period is its continuation (Theorems 47.2(ii) and 48.3(ii)). For several active molecules the reduced period is the fibre integral of Theorem 48.24(ii) against the periods of its molecule families, each read in all its data by its module and, along the contour of the shares, in words: there every molecule family has data whose real parts lie on its balance wall with its exponents and its proper walls positive, at which Theorem 48.15(v) reads it.*

*(vi)* Repetition. *Every module, closing and transform is found once for each class (Theorems 48.5(v), 48.18(iii) and 48.20(vi)).*

*So the form $\alpha^{\mathbf{m-1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, d\alpha$ is read in all its data, its period, its interest and its pseudointegral: by finite matrices over rational functions of the data everywhere and by words at the data whose real parts are real data inside the walls, with, for several active molecules, the certificate that on the balance wall it carries nothing but a working interest read so.*

The procedure chooses: the absent stations and exponents, the set $\Sigma$, the factorization of weights that do not factor, the positive direction, level bounds and generators of each box, the accepted sets and their lifts, the origin and the direction of Proposition 47.12, and the native direction of Proposition 47.14 with its native gauge, rational when the weights are rational. The theorem holds for every choice. Every step is a statement of this handbook, with its certificate, and no step depends on the family beyond its tables and the names of its letters: a program that carries out the procedure carries out the handbook, step for step, on every family.

# Part IX

# The family as inference

# Chapter 49

# The family as inference

A family is a model of data in two ways. A sample of bases, drawn from the basis and cobasis laws of the tables of a family at an unknown gauge, is itself a question: its counts are the masses, the sizes of its parts are the exponents, and its chance as a function of the tilt is the integrand of the period. The law of a question is then the conjugate prior of the tilt, and every posterior, every predictive chance and every evidence is a ratio of periods; read at a bunch, a sample is a sample of the children at the shares it realizes, and each of its walls is a sum of independent excesses of the levels and colevels of its draws over their floors, whose law the Skeleton polynomials give. A sample of tilts drawn from the law of a question makes the period the cumulant function of an exponential family. Between the two sits the family word read at Dirichlet draws: the negative logarithms of its letters have the Grey law, whose Laplace transform in the exponents is the period divided by the stacked Gamma of the masses over their total; the pseudo-MGF and the pseudoexpectation are that Laplace transform fixed at the Foreman's question and at the centre. A draw of the shape law, read as a vector of numbers, has an order that is a sector of the period, best and worst bases that are functions of the order, and entropies and moments that are steps of the data. What a bunch all of whose subsets are patterns reads of its room is its interaction (Definition 12.18), the total over the scales of the bunch of the joint cumulants of the use of its marks under the basis and cobasis laws, weighted by the exponents; for a regression design the interaction of a split of the prior marks into two blocks is minus the logarithm of the generalized variance inflation factor.

## 49.1 Samples of bases

**Proposition 49.1** (A sample of bases is a question)**.** *For a family at a gauge $\alpha = e^{2z}$ draw, for every table $j$, $s_j$ bases $T_{j,1}, \dots, T_{j,s_j}$ from its basis law $\mathbb{P}_{j,\alpha}$ and $t_j$ bases $T'_{j,1}, \dots, T'_{j,t_j}$ from its cobasis law $\mathbb{P}^*_{j,\alpha}$, all independent, let $m_e$ count the $T_{j,i}$ that hold $e$ and the $T'_{j,i}$ that miss $e$, and let $\theta_{\mathrm{draws}} = \prod_{j,i} \theta_{j,T_{j,i}} \prod_{j,i} \theta_{j,T'_{j,i}}$ be the product of the weights of the drawn bases.*

*(i) As a function of the tilt $z$, the chance of the sample is*

$$\theta_{\mathrm{draws}} \cdot \exp\Big(2\langle \boldsymbol{\mu}_\perp, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)\Big),$$

*$2^{1-n}\theta_{\mathrm{draws}}$ times the integrand of the period of the question $(\mathbf{m}; \mathbf{s}, \mathbf{t})$, of pair coordinates $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$ (Definition 17.1): the counts are the masses, the sizes of the parts are the exponents, and $m_{\mathcal{E}} = \sum_j (s_j r_j + t_j r^*_j)$ puts the question on the balance wall.*

*(ii) Its logarithmic gradient in $z$ is $2\big(\mathbf{m}-\sum_j (s_j R_j(\alpha)+t_j R_j^*(\alpha))\big)$, twice the excess of the masses over the usage at $\alpha$. For a connected family the maximum likelihood tilt is the native gauge of the question, and it exists, unique on $\mathcal{T}$, exactly when every proper wall of the question is positive.*

*(iii) $\kappa_S = \sum_j \Big( \sum_{i=1}^{s_j} \big(|T_{j,i}\cap S| - d_j(S)\big) + \sum_{i=1}^{t_j} \big(|S\setminus T'_{j,i}| - d_j^*(S)\big)\Big) \geq 0$: a wall is the excess of the sample's use of $S$ over its least use.*

**Proposition 49.2** (The conjugate posterior)**.** *Let $F$ be a connected family. Take as prior on the tilt of the gauge the law of a question $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ with every proper wall positive, and observe a sample as in Proposition 49.1, with counts $\mathbf{m}$, sizes $\mathbf{s},\mathbf{t}$ and the product $\theta_{\text{draws}}$ of the weights of the drawn bases.*

*(i) The posterior is the law of the question $(\mathbf{m}_0+\mathbf{m};\mathbf{s}_0+\mathbf{s},\mathbf{t}_0+\mathbf{t})$: the masses add and the exponents add.*

*(ii) The chance of the sample with the tilt integrated out is*

$$\theta_{\text{draws}} \cdot \frac{P_F(\mathbf{m}_0+\mathbf{m};\mathbf{s}_0+\mathbf{s},\mathbf{t}_0+\mathbf{t})}{P_F(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)}.$$

*(iii) At the posterior data $(\mathbf{m}';\mathbf{s}',\mathbf{t}')$ the chance that one more draw from the basis law of table $j$ is $T$ is $\theta_{j,T}\,P_F(\mathbf{m}'+\mathbf{1}_T;\mathbf{s}'+\mathbf{e}_j,\mathbf{t}')/P_F(\mathbf{m}';\mathbf{s}',\mathbf{t}')$, and from its cobasis law $\theta_{j,T}\,P_F(\mathbf{m}'+\mathbf{1}_{\mathcal{E}\setminus T};\mathbf{s}',\mathbf{t}'+\mathbf{e}_j)/P_F(\mathbf{m}';\mathbf{s}',\mathbf{t}')$.*

*(iv) The posterior mean of the usage $\sum_j (s'_j R_j + t'_j R_j^*)$ at the posterior exponents is the posterior masses $\mathbf{m}_0+\mathbf{m}$.*

*(v) Every wall of the posterior is the wall of the prior plus the excess of Proposition 49.1(iii): a prior inside its walls gives a posterior inside its walls for every sample, the empty one included.*

**Proposition 49.3** (What the data see of the bases)**.** *(i) For a circuit $C$ of $k\geq 2$ marks of a table and each $e\in C$ some basis has $T\cap C = C\setminus e$: $r(C)=k-1$, every $(k-1)$-subset of $C$ is a pattern of $C$ and $C$ is not; for $k\geq 3$ every two marks of $C$ lie in a common basis, and $C$ in none.*

*(ii) For a sample of a family at a gauge $\alpha$ as in Proposition 49.1 and $S\subseteq\mathcal{E}$, the level $|T_{j,i}\cap S|$ of every basis draw of table $j$ lies in $[d_j(S),r_j(S)]$ and the colevel $|S\setminus T'_{j,i}|$ of every cobasis draw in $[d_j^*(S),|S|-d_j(S)]$; the chance that no basis draw of table $j$ reaches the largest level $r_j(S)$ of its window is $(1-p_j)^{s_j}$ with $p_j=\mathbb{P}_{j,\alpha}(|T\cap S|=r_j(S))>0$, and likewise at the floor $d_j(S)$ and for the cobasis draws at the ends of their window, with the cobasis law and $t_j$.*

**Proposition 49.4** (The exact laws of a sample of bases)**.** *Let a sample of a family be drawn at a gauge $\alpha$ as in Proposition 49.1, $s_j$ basis and $t_j$ cobasis draws of table $j$, with counts $\mathbf{m}$, and let $B$ be a bunch with room $R$.*

*(i) For a basis draw of table $j$, $\mathbb{P}_{j,\alpha}(T) = \pi_{j,p}(\alpha)\,\mathbb{P}_{j,(p),\alpha_R}(T\setminus p)$ with $p=T\cap B$, and for a cobasis draw $\mathbb{P}^*_{j,\alpha}(T) = \pi^*_{j,p}(\alpha)\,\mathbb{P}^*_{j,(p),\alpha_R}(T\setminus p)$, with the pattern probabilities $\pi_{j,p}(\alpha) = \alpha_B^p K_{j,(p)}(\alpha_R)/K_j(\alpha)$ and $\pi^*_{j,p}(\alpha)=\alpha_B^{B\setminus p}\Psi_{j,(p)}(\alpha_R)/\Psi_j(\alpha)$ (Theorem 12.8) and the basis and cobasis laws of the child $(j,p)$ at $\alpha_R$. So the numbers $a_{j,p}$ of the basis draws and $b_{j,p}$ of the cobasis draws of table $j$ that meet $B$ in $p$ are independent multinomial vectors with the chances $\pi_{j,p}(\alpha)$ and $\pi^*_{j,p}(\alpha)$; they are shares of $B$, $|\mathbf{a}_j|=s_j$ and $|\mathbf{b}_j|=t_j$ (Definition 12.24);*

*the counts on the bunch are their loads, $m_e = \mathrm{load}_e$ for $e \in B$; and given them the parts in the room are a sample of the children $F_B(\mathbf{a}, \mathbf{b})$ at $\alpha_R$, $a_{j,p}$ basis and $b_{j,p}$ cobasis draws of the child $(j,p)$, with the counts $\mathbf{m}|_R$. The mean shares are $s_j \pi_{j,p}(\alpha)$ and $t_j \pi^*_{j,p}(\alpha)$, at a native gauge the native split (Definition 12.25).*

*(ii) Given $\mathbf{m}$, each sample with these counts has the probability $\theta_{\mathrm{draws}}$ divided by the coefficient of $\alpha^{\mathbf{m}}$ in $\prod_j K_j(\alpha)^{s_j} \Psi_j(\alpha)^{t_j}$, free of the gauge.*

*(iii) For every nonempty proper $S$ the wall $\kappa_S$ of the sample (Proposition 49.1(iii)) is the sum of the independent excesses of the levels of its basis draws and of the colevels of its cobasis draws over their floors, with the generating function*

$$\mathbb{E}\big[x^{\kappa_S}\big] = \prod_j \mathsf{Q}^K_{j,S}(x;\alpha)^{s_j}\, \mathsf{Q}^{\Psi}_{j,S}(x;\alpha)^{t_j}$$

*of the Skeleton polynomials (Definition 12.10). Its values fill $\{0, 1, \dots, \sum_j (s_j + t_j) c_j(S)\}$; its mean is the excess $\sum_j \big(s_j R_j(\alpha) + t_j R^*_j(\alpha)\big)_S - \sum_j \big(s_j d_j(S) + t_j d^*_j(S)\big)$ of the usage at $\alpha$ over the least use, and its cumulants of order $k \geq 2$ are $\partial^k_S \Phi(\log \alpha)$, $\partial_S = \sum_{e \in S} \partial_{y_e}$ (Proposition 12.12); it vanishes, every draw sitting at the floor of its window on $S$, with the chance $\prod_j \mathsf{Q}^K_{j,S}(0;\alpha)^{s_j} \mathsf{Q}^{\Psi}_{j,S}(0;\alpha)^{t_j} > 0$; and when the Skeleton spectra of $S$ are positive it is distributed as a sum of independent $\{0,1\}$-valued variables whose means are the coins, $s_j$ copies of the coins of $\mathsf{Q}^K_{j,S}$ and $t_j$ of those of $\mathsf{Q}^{\Psi}_{j,S}$ (Proposition 12.11(iv)).*

**Proposition 49.5** (Tests that do not see the gauge)**.** *For a family $F$ draw a sample at a gauge $\alpha$ as in Proposition 49.1, with counts $\mathbf{m}$, and let $\Theta_F(\mathbf{m})$ be the coefficient of $\alpha^{\mathbf{m}}$ in $\prod_j K_j(\alpha)^{s_j} \Psi_j(\alpha)^{t_j}$, the total of $\theta_{\mathrm{draws}}$ over the set $\mathfrak{F}_F(\mathbf{m})$ of the samples with these counts.*

*(i) Given $\mathbf{m}$, the sample has the law $\theta_{\mathrm{draws}}/\Theta_F(\mathbf{m})$ on $\mathfrak{F}_F(\mathbf{m})$, free of the gauge; for a family of lists it is uniform, and $|\mathfrak{F}_F(\mathbf{m})| = \Theta_F(\mathbf{m})$.*

*(ii) A test of $F$ that rejects on an event $A$ whose samples with counts $\mathbf{m}$ carry at most $\varepsilon\, \Theta_F(\mathbf{m})$ of the weight $\theta_{\mathrm{draws}}$, for every $\mathbf{m}$, has size at most $\varepsilon$ at every gauge.*

*(iii) If every basis of every table of a family $F'$ on $\mathcal{E}$ is a basis of the same table of $F$, with the same weight, then given $\mathbf{m}$ the chance under $F$ that every draw is a basis of its table in $F'$ is $p_{F'|F}(\mathbf{m}) = \Theta_{F'}(\mathbf{m})/\Theta_F(\mathbf{m})$, and rejecting $F$ when this happens and $p_{F'|F}(\mathbf{m}) \leq \varepsilon$ has size at most $\varepsilon$ at every gauge; one draw that is a basis of its table in $F$ and not in $F'$ rejects $F'$ without error.*

*(iv) If each family of a collection has a test as in (ii), the families not rejected contain the true family with probability at least $1 - \varepsilon$, at every gauge.*

**Corollary 49.6** (The evidence of a family)**.** *Let every family $F$ of a collection on $\mathcal{E}$, each with $N$ tables, carry as prior the law of its centre of strengths $\boldsymbol{\sigma}_0 = (\sigma_{0,j})_j$, the question $(\sigma_0 \mathbf{1}; \boldsymbol{\sigma}_0, \boldsymbol{\sigma}_0)$ with $\sigma_0 = \sum_j \sigma_{0,j}$, inside its walls. For a sample as in Proposition 49.5, with the sizes $(s_j, t_j)_j$ and counts $\mathbf{m}$, the probability of the sample under $F$ with the gauge integrated out is*

$$\mathrm{evidence}(F) = \theta^F_{\mathrm{draws}}\, \mathbf{1}[\textit{every draw is a basis of its table of } F]\, \frac{P_F(\sigma_0 \mathbf{1} + \mathbf{m}; \boldsymbol{\sigma}_0 + \mathbf{s}, \boldsymbol{\sigma}_0 + \mathbf{t})}{P_F(\sigma_0 \mathbf{1}; \boldsymbol{\sigma}_0, \boldsymbol{\sigma}_0)},$$

*$\theta^F_{\mathrm{draws}}$ the product of the weights in $F$ of the drawn bases, 1 for lists; and if the family is drawn from the collection with the chances $p(F)$, the posterior chance of $F$ is proportional to $p(F)\, \mathrm{evidence}(F)$.*

A sample of bases asks the family a question: the counts are its masses and the sample sizes its exponents, and a wall of the question is how far the sample's use of a set exceeds the least use any basis makes of it, a sum of the independent excesses of the levels and colevels of the draws over their floors, whose generating function is the product of the Skeleton polynomials of the set raised to the sizes. Read at a bunch, a sample is a sample of the children at the shares it realizes: the loads of these shares are the counts of the bunch, and their means at a native gauge are the native split. The law of a question is the conjugate prior of the tilt, so the posterior, the predictive chances and the evidence of a family are ratios of periods, and a prior inside its walls stays inside them. Given the counts, a sample has a law that does not see the gauge, uniform for a family of lists, so a test built on the samples with those counts has a size that no gauge can change.

## 49.2 Samples of tilts

**Proposition 49.7** (A sample of tilts: the period is the cumulant function)**.** *Let a question* $(\mathbf{m};\mathbf{s},\mathbf{t})$ *of pair coordinates* $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$ *have every proper wall positive, let* $Z_1,\dots,Z_q$ *be the tilts of independent draws from* $\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$*, and let* $\bar Z$*,* $\bar E_j$*,* $\bar O_j$ *be the sample means of* $Z_i$*,* $E_j(Z_i)$*,* $O_j(Z_i)$*. In this statement* $\mathbb{E}$ *is the expectation of* $U\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$ *and* $Z$ *is its tilt.*

*(i) The sample has the density*

$$2^{q(n-1)}\exp\Big(q\Big[2\langle\boldsymbol{\mu}_\perp,\bar Z\rangle-\sum_j\big(\sigma_j\bar E_j+\delta_j\bar O_j\big)\Big]\Big)\Big/P_F(\mathbf{m};\mathbf{s},\mathbf{t})^q$$

*on* $\mathcal{T}^q$*, so* $(\bar Z;\bar E_j,\bar O_j)_j$ *is sufficient for the question.*

*(ii) The means are* $\mathbb{E}[Z]=\frac12\nabla_{\boldsymbol{\mu}_\perp}\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*,* $\mathbb{E}[E_j(Z)]=-\partial_{\sigma_j}\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ *and* $\mathbb{E}[O_j(Z)]=-\partial_{\delta_j}\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*, the derivatives of Definition 17.1, and the likelihood equations equate them to the sample means.*

*(iii) Against a second question* $(\mathbf{m}';\mathbf{s}',\mathbf{t}')$ *of pair coordinates* $(\boldsymbol{\mu}'_\perp;\boldsymbol{\sigma}',\boldsymbol{\delta}')$ *the logarithm of the likelihood ratio is*

$$q\Big[2\langle\boldsymbol{\mu}'_\perp-\boldsymbol{\mu}_\perp,\bar Z\rangle-\sum_j\big((\sigma'_j-\sigma_j)\bar E_j+(\delta'_j-\delta_j)\bar O_j\big)\Big]-q\log\frac{P_F(\mathbf{m}';\mathbf{s}',\mathbf{t}')}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}.$$

*(iv) The sums* $q\bar Z$*,* $q\bar E_j$*,* $q\bar O_j$ *have as Laplace transform that of Corollary 17.2 raised to the power* $q$*.*

**Definition 49.8** (The Foreman's question; the pseudo-MGF)**.** Let $\omega=W\,d\alpha_{\mathbf{m}}$ have real data on a connected family, $s_j,t_j\ge0$ not all zero and $\mathbf{m}>0$, with target $\boldsymbol{\tau}$. The *Foreman's question* of $\omega$ is the question $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ at its target, of pair coordinates $\big(\boldsymbol{\tau}-\frac{\tau\varepsilon}{n}\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\delta}\big)$; off the pole hyperplanes its period $P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ is the pseudointegral $\wr W\,d\alpha_{\mathbf{m}}$ (Definition 16.10(iii)). Where the target lies on no pole hyperplane and $P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})\neq0$, the *pseudo-MGF* of $\omega$ is

$$\mathsf{M}_\omega(c;\mathbf{a},\mathbf{b})=\frac{P_F\big(\boldsymbol{\tau}+\mathbf{m}_F(c;\mathbf{a},\mathbf{b});\,\mathbf{s}+\mathbf{a}+\mathbf{b},\,\mathbf{t}+\mathbf{a}-\mathbf{b}\big)}{P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})}\qquad(c\in\mathbf{1}^\perp\otimes\mathbb{C},\ \mathbf{a},\mathbf{b}\in\mathbb{C}^N),$$

continued meromorphically (Lemma 16.9), and the *pseudo-cumulants* of $\omega$ are the derivatives of $\log\mathsf{M}_\omega$ at 0: the derivatives of $\log P_F$ at the Foreman's question in its pair coordinates.

**Proposition 49.9** (The pseudo-MGF inside the walls)**.** *In Definition 49.8 let $\omega$ have a native gauge, and let $Z$ be the tilt of $U \sim \mathrm{Breta}^{\circ}(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t})$, the Foreman's law of Proposition 16.34(i); the law of $Z$ is the law of the Foreman's question.*

*(i) Wherever the real parts of the shifted data $\big(\boldsymbol{\tau}+\mathbf{m}_F(c;\mathbf{a},\mathbf{b});\,\mathbf{s}+\mathbf{a}+\mathbf{b},\,\mathbf{t}+\mathbf{a}-\mathbf{b}\big)$ have every proper wall positive, a region containing a neighbourhood of $0$, $\mathsf{M}_\omega(c;\mathbf{a},\mathbf{b})$ is the moment generating function of the statistics $(2Z;-E_j(Z),-O_j(Z))_j$ under the Foreman's law (Corollary 17.2 at the Foreman's question); the pseudo-cumulants are their joint cumulants.*

*(ii) For a balanced shift $(\dot{\mathbf{m}};\dot{\mathbf{s}},\dot{\mathbf{t}})$ the function $U^{\dot{\mathbf{m}}}\prod_j K_j(U)^{-\dot{s}_j}\Psi_j(U)^{-\dot{t}_j}$ has degree $0$, a function of the ray of $U$, and where the shifted data have every proper wall positive its expectation under the Foreman's law is $P_F(\boldsymbol{\tau}+\dot{\mathbf{m}};\mathbf{s}+\dot{\mathbf{s}},\mathbf{t}+\dot{\mathbf{t}})/P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ (the case of Proposition 49.22(i) below at the Foreman's question); $\mathsf{M}_\omega(c;\mathbf{a},\mathbf{b})$ is this ratio at the shift of pair coordinates $(c;\mathbf{a},\mathbf{b})$, so $\mathsf{M}_\omega$ is the moment generating function of $(\log U;-\log K_j(U),-\log\Psi_j(U))_j$ along the balanced shifts.*

*(iii) The normaliser is $P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t}) = P_{\mathrm{c}}\,\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}[(\omega_{\boldsymbol{\tau}}/\omega_{\mathrm{c}})(U)]$ (Theorem 18.1(i)). The centre $\omega_{\mathrm{c}}$ has the target $\sigma\mathbf{1}$ and the Foreman's question $(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$, and the pseudoexpectation is the pseudo-MGF of the centre at the departure:*

$$\mathsf{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\Big[\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(U)\Big] = \mathsf{M}_{\omega_{\mathrm{c}}}\Big(\boldsymbol{\tau}-\frac{\tau_{\mathcal{E}}}{n}\mathbf{1};\ \mathbf{0},\ \boldsymbol{\delta}\Big) = \mathbb{E}_{U\sim\mathbb{P}_{\mathrm{c}}}\Big[e^{2\langle\boldsymbol{\tau}-\frac{\tau_{\mathcal{E}}}{n}\mathbf{1},Z\rangle-\sum_j\delta_jO_j(Z)}\Big],$$

*$Z$ the tilt of $U$: at fixed strengths the pseudoexpectations of all the questions are values of one moment generating function, that of $(2Z;-O_j(Z))_j$ under the law of the centre.*

*Past the walls $\mathsf{M}_\omega$ is the continuation, as the moments of the distribution of Theorem 16.17(i) are, and $\mathsf{M}_\omega(0)=1$ throughout.*

**Proposition 49.10** (Two layers: the word evaluated at the draws)**.** *Let a family have real data with $\mathbf{m}>0$, and let $U_1,\dots,U_q$ be independent draws from $\mathrm{Dir}_{\mathbf{m}}$, the shapes $\alpha/|\alpha|$ of $\alpha\sim\mathrm{Gamma}_{\mathbf{m}}$ (Proposition 8.12(ii)).*

*(i) By Proposition 8.12(i), $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[W(U)] = P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$, and it is finite exactly when every proper wall at $(\mathbf{m};\mathbf{s},\mathbf{t})$ is positive.*

*(ii) $\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}[W(U)^k] = P_F(\mathbf{m};k\mathbf{s},k\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$, and the sample mean $\overline{W}_q=\frac{1}{q}\sum_i W(U_i)$ has the mean $P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ and the variance*

$$\frac{1}{q}\left[\frac{P_F(\mathbf{m};2\mathbf{s},2\mathbf{t})}{\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}}-\left(\frac{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}{\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}}\right)^2\right],$$

*finite exactly when every proper wall at $(\mathbf{m};2\mathbf{s},2\mathbf{t})$ is positive.*

*(iii) Where every proper wall at $(\mathbf{m};\mathbf{s},\mathbf{t})$ is positive, $\mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ has the density $W\,\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ with respect to $\mathrm{Dir}_{\mathbf{m}}$, and under it a second word $W'=\prod_j K_j^{-s'_j}\Psi_j^{-t'_j}$ of the same tables has $\mathbb{E}_{U\sim\mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})}[W'(U)^k] = P_F(\mathbf{m};\mathbf{s}+k\mathbf{s}',\mathbf{t}+k\mathbf{t}')/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$.*

**Definition 49.11** (The Grey law)**.** For a family $F$ and masses $\mathbf{m} > 0$ the *Grey law* $\mathrm{Grey}(F;\mathbf{m})$ is the joint law of

$$(G_j, G_j^*)_j = \big( -\log K_j(U),\, -\log \Psi_j(U)\big)_j \qquad \text{for } U \sim \mathrm{Dir}_{\mathbf{m}};$$

it depends on the tables of $F$ and on $\mathbf{m}$, and not on the exponents.

**Proposition 49.12** (The Grey law and its exponential family)**.** *Let $\mathbf{m} > 0$ and let $(G_j, G_j^*)_j \sim \mathrm{Grey}(F;\mathbf{m})$.*

*(i) For complex exponents $(\mathbf{s},\mathbf{t})$ whose real parts keep every proper wall at $\mathbf{m}$ positive,*

$$\mathbb{E}_{(G_j,G_j^*)_j \sim \mathrm{Grey}(F;\mathbf{m})}\Big[e^{\sum_j (s_j G_j + t_j G_j^*)}\Big] = \frac{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}{\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}} \quad \textit{(Proposition 8.12(i),(iii))}:$$

*on real exponents the moment generating function, on imaginary exponents the characteristic function; the joint cumulants are the derivatives of $\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ in the exponents at $(\mathbf{0},\mathbf{0})$.*

*(ii) For real exponents with every proper wall positive, under $\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$ the vector $\big(-\log K_j(U), -\log \Psi_j(U)\big)_j$ has the law*

$$e^{\sum_j (s_j g_j + t_j g_j^*)}\, \frac{\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}\, \mathrm{Grey}(F;\mathbf{m})(dg\, dg^*):$$

*over the exponents these laws form the exponential family with base $\mathrm{Grey}(F;\mathbf{m})$ and cumulant function $\log\big(P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\big)$.*

*(iii) For a real direction $(\mathbf{a},\mathbf{b}) \neq 0$ of the exponents let $g(S) = \sum_j \big(a_j d_j(S) + b_j d_j^*(S)\big)$ and $k^* = \min\{m_S/g(S) :\ S \text{ nonempty proper},\ g(S) > 0\}$, the value of $\lambda$ at which the ray $\lambda(\mathbf{a},\mathbf{b})$, $\lambda > 0$, first meets a wall at $\mathbf{m}$, with $k^* = \infty$ if no $g(S)$ is positive. If $k^* < \infty$,*

$$\mathbb{P}\Big(\sum_j \big(a_j G_j + b_j G_j^*\big) > x\Big) = e^{-k^* x + o(x)} \qquad (x \to \infty),$$

*and if $k^* = \infty$ the sum is bounded above.*

*(iv) For $q$ independent copies the sum has the Laplace transform $\big(P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\big)^q$, and the sample mean $\overline{W}_q$ of Proposition 49.10(ii) is the empirical counterpart of the transform of (i) at the exponents $(\mathbf{s},\mathbf{t})$.*

**Corollary 49.13** (The Grey law at the target)**.** *Let a connected family carry real exponents $s_j, t_j \geq 0$, not all zero, and masses $\mathbf{m} > 0$ with a native gauge and the target $\boldsymbol{\tau}$. Then*

$$\mathbb{E}_{U\sim \mathrm{Dir}_{\boldsymbol{\tau}}}[W(U)] = \mathbb{E}_{(G_j,G_j^*)_j \sim \mathrm{Grey}(F;\boldsymbol{\tau})}\Big[e^{\sum_j (s_j G_j + t_j G_j^*)}\Big] = \frac{P_F(\boldsymbol{\tau};\mathbf{s},\mathbf{t})}{\Gamma_1^n\begin{bmatrix}\boldsymbol{\tau}\\ \tau_{\mathcal{E}}\end{bmatrix}}$$

$$= \frac{P_{\mathrm{c}}}{\Gamma_1^n\begin{bmatrix}\boldsymbol{\tau}\\ \tau_{\mathcal{E}}\end{bmatrix}}\, \mathsf{E}_{U\sim \mathbb{P}_{\mathrm{c}}}\Big[\frac{\omega_{\boldsymbol{\tau}}}{\omega_{\mathrm{c}}}(U)\Big]:$$

*the moment generating function of the Grey law at the target, read at the exponents, is the pseudoexpectation times the period of the centre divided by* $\Gamma_1^n\begin{bmatrix}\boldsymbol{\tau}\\ \tau_{\mathcal{E}}\end{bmatrix}$. *With* $\boldsymbol{\tau}$ *held at the target, the logarithmic derivatives of* $(\mathbf{s}',\mathbf{t}')\mapsto P_F(\boldsymbol{\tau};\mathbf{s}',\mathbf{t}')$ *at* $(\mathbf{s},\mathbf{t})$ *are the joint cumulants of* $\big(-\log K_j(U),-\log\Psi_j(U)\big)_j$ *under the Foreman's law* $\mathrm{Breta}^\circ(F;\boldsymbol{\tau};\mathbf{s},\mathbf{t})$.

**Proposition 49.14** (A Laplace transform with a moving target)**.** *Let a connected family carry real exponents* $s_j,t_j\geq 0$, *not all zero, and masses* $\mathbf{m}>0$ *whose target* $\boldsymbol{\tau}$ *has every proper wall positive, and let* $g(S)=\sum_j\big(s_jd_j(S)+t_jd_j^*(S)\big)$ *be the set function of Theorem 8.19. With the tails* $U_k$ *of the orders* $\pi$ *of* $\mathcal{E}$ *(Lemma 8.3),*

$$\lim_{\lambda\to 0^+}\mathbb{E}_{V\sim\mathrm{Dir}_{\lambda\boldsymbol{\tau}}}\Big[\prod_j K_j(V)^{-\lambda s_j}\Psi_j(V)^{-\lambda t_j}\Big]=\frac{\sum_\pi\prod_{k=2}^n\big(\tau_{U_k}-g(U_k)\big)^{-1}}{\sum_\pi\prod_{k=2}^n\tau_{U_k}^{-1}}\ \geq\ 1,$$

*with equality exactly when* $g(S)=0$ *for every nonempty proper* $S$. *The limit is the ratio of the limits of Theorem 17.5(i) for the question of data* $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ *and for the list* $\binom{\mathcal{E}}{1}$ *at a full basis lean at the masses* $\boldsymbol{\tau}$, *whose walls are the* $\tau_S$ *(Proposition 16.3).*

The word read at Dirichlet draws is a random variable whose expectation, as a function of the exponents, is the period divided by the stacked Gamma of the masses over their total: the Laplace transform of the Grey law of the family. At fixed masses the Grey law is the base of an exponential family over the exponents, and its tail along a direction of exponents is set by the first wall met along that direction. Read at the target of each exponent, as the normaliser $\binom{\mathcal{E}}{1}$ of Proposition 16.35(ii) reads it, the expectation of the word is not a moment generating function in the exponents: its limit at the origin exceeds 1 as soon as the word has a floor on a proper set, since at each exponent it reads the Grey law at another target. The pseudo-MGF fixes the target once, at the Foreman's question, and the pseudoexpectation fixes the law once for given strengths, at the centre.

**Proposition 49.15** (The exponent is a dial between the mean and the maximum)**.** *In Proposition 49.10 let the exponents be real, not all zero, and put* $Y_i=\log W(U_i)$ *and* $Y=\log W(U)$ *for* $U\sim\mathrm{Dir}_{\mathbf{m}}$; *in this statement* $\mathbb{E}$ *is the expectation of independent* $U,U_1,\dots,U_q\sim\mathrm{Dir}_{\mathbf{m}}$. *Let* $g(S)=\sum_j\big(s_jd_j(S)+t_jd_j^*(S)\big)$, $k^*=\min\{m_S/g(S):\ S$ *nonempty proper,* $g(S)>0\}$, *with* $k^*=\infty$ *if no* $g(S)$ *is positive, and for* $k>0$ *let* $A_k=\frac{1}{k}\log\big(\frac{1}{q}\sum_i e^{kY_i}\big)$.

*(i)* $A_k$ *is nondecreasing in* $k$, *tends to the sample mean of the* $Y_i$ *as* $k\to 0$ *and to* $\max_i Y_i$ *as* $k\to\infty$, *and* $\max_i Y_i-\frac{\log q}{k}\leq A_k\leq\max_i Y_i$.

*(ii) For* $k<k^*$, $\mathbb{E}[e^{kA_k}]=P_F(\mathbf{m};k\mathbf{s},k\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$, *and* $e^{kA_k}$ *has a finite variance exactly when* $2k<k^*$.

*(iii) For* $k\geq k^*$, $\mathbb{E}[e^{kY}]=\infty$; *and* $\mathbb{P}(\max_i Y_i\leq x)=\mathbb{P}(Y\leq x)^q$.

*(iv) If* $k^*<\infty$, $\mathbb{P}(Y>x)=e^{-k^*x+o(x)}$ *as* $x\to\infty$, *and if* $k^*=\infty$, $Y$ *is bounded above (Proposition 49.12(iii)). For* $s_j,t_j\geq 0$ *and a set* $S$ *attaining* $k^*$ *the scale* $\rho=|U_S|$ *has* $(\rho,1-\rho)\sim\mathrm{Dir}_{(m_S,m_{\mathcal{E}\setminus S})}$, *so* $\rho$ *has density proportional to* $\rho^{m_S-1}(1-\rho)^{m_{\mathcal{E}\setminus S}-1}$, *and* $W(U)\geq\prod_j K_j(\mathbf{1})^{-s_j-t_j}\rho^{-g(S)}$.

**Proposition 49.16** (The even and odd statistics from a centre)**.** *Let a family carry the centre of strengths* $\boldsymbol{\sigma}=(\sigma_j)_j$, $\sigma_j\geq 0$, *inside its walls, let* $Z$ *be the tilt of* $U\sim\mathrm{Breta}^\circ(F;\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$, *with density proportional to* $e^{-\sum_j\sigma_jE_j(z)}$ *on* $\mathcal{T}$, *and let* $\ell^*_{\boldsymbol{\sigma}}$ *be the slope of Proposition 17.7. In this statement* $\mathbb{E}$ *is the expectation of* $U\sim\mathrm{Breta}^\circ(F;\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$.

*(i)* $\sum_j \sigma_j E_j(Z) \geq 2\sum_j \sigma_j \log K_j(\mathbf{1})$, *with equality only at* $Z = 0$ *(Proposition 17.4), where* $\sum_j \sigma_j E_j$ *has the Hessian* $8\sum_j \sigma_j\big(\kappa_{j,\mathbf{1}}(X_e, X_f)\big)_{e,f}$, *rung two of the still of the realization* $\big((\sigma_j,\sigma_j)_j;\mathbf{1}\big)$ *of the active tables, eight times the covariance of the indicators under the weight laws; and each* $E_j(Z) \geq 2\log K_j(\mathbf{1})$, *with equality exactly where* $Z$ *is constant on the molecules of table* $j$. *For real* $\mathbf{k} = (k_j)_j$,

$$\mathbb{E}\Big[e^{-\sum_j k_j E_j(Z)}\Big] = \frac{P_F\big((\sigma + |\mathbf{k}|)\mathbf{1}; \boldsymbol{\sigma} + \mathbf{k}, \boldsymbol{\sigma} + \mathbf{k}\big)}{P_F(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})},$$

*the ratio of the periods of the centres of strengths* $\boldsymbol{\sigma} + \mathbf{k}$ *and* $\boldsymbol{\sigma}$, *finite exactly when* $\sum_j (\sigma_j + k_j)c_j(S) > 0$ *for every nonempty proper* $S$: *along the even direction* $\mathbf{k} = \lambda\boldsymbol{\sigma}$ *every wall closes at once, at* $\lambda = -1$, *and along any other even direction* $\lambda\mathbf{k}$, $\lambda > 0$, *the first wall closes at the least of the ratios* $\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})/(-\kappa_S^{\mathbf{E}}(\mathbf{k}))$ *with* $\kappa_S^{\mathbf{E}}(\mathbf{k}) < 0$.

*(ii)* $\big(Z, (E_j(Z))_j, (O_j(Z))_j\big)$ *and* $\big(-Z, (E_j(Z))_j, (-O_j(Z))_j\big)$ *have one law, so every joint moment of odd total degree in* $Z$ *and the* $O_j(Z)$ *vanishes. For real* $c \in \mathbf{1}^\perp$ *and* $\mathbf{b} \in \mathbb{R}^N$, $\mathbb{E}\big[e^{2\langle c, Z\rangle - \sum_j b_j O_j(Z)}\big]$ *is finite exactly when* $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) > 1$ *(Proposition 17.7(i)), where it is*

$$P_F\big(\mathbf{m}_F(c; \boldsymbol{\sigma}, \mathbf{b});\, \boldsymbol{\sigma} + \mathbf{b},\, \boldsymbol{\sigma} - \mathbf{b}\big)/P_F(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma}),$$

*for* $|b_j| \leq \sigma_j$ *the pseudoexpectation of the question of pair coordinates* $(c; \boldsymbol{\sigma}, \mathbf{b})$. *If* $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) < \infty$, *both tails of* $Y = 2\langle c, Z\rangle - \sum_j b_j O_j(Z)$ *have the rate* $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b})$: $\mathbb{P}(\pm Y > x) = e^{-\ell^*_{\boldsymbol{\sigma}}(c,\mathbf{b})x + o(x)}$ *as* $x \to \infty$; *if it is infinite,* $Y$ *is bounded (Proposition 17.8).*

*(iii) At* $\alpha = e^{2z}$, $K_j(\alpha)^{-s_j} = e^{-s_j\langle \mathbf{1}, z\rangle} e^{-\frac{s_j}{2}(E_j(z) + O_j(z))}$ *and* $\Psi_j(\alpha)^{-t_j} = e^{-t_j\langle \mathbf{1}, z\rangle} e^{-\frac{t_j}{2}(E_j(z) - O_j(z))}$: *every letter is its table at a full lean,* $K_j^{-s_j}$ *at the pair coordinates* $(\sigma, \delta) = (\frac{s_j}{2}, \frac{s_j}{2})$ *and* $\Psi_j^{-t_j}$ *at* $(\frac{t_j}{2}, -\frac{t_j}{2})$, *with the masses shifted by* $-\frac{s_j}{2}\mathbf{1}$, *resp.* $-\frac{t_j}{2}\mathbf{1}$; *a family is the family of its letters, and its pair coordinates are the sums of theirs.*

**Proposition 49.17** (What the two statistics are)**.** *For table* $j$ *of a family and a tilt* $z$, $\log \Lambda_j(z) = \log \sum_T \theta_{j,T} e^{\langle \chi_T, z\rangle}$ *is a soft maximum and* $-\log \Lambda_j(-z)$ *a soft minimum of the scores* $\langle \chi_T, z\rangle$ *of its bases, weighted by* $\theta_{j,T}$.

*(i)* $E_j(z)$ *is the soft maximum less the soft minimum, and* $O_j(z)$ *is their sum; the centred potential* $\Phi^\circ = \sum_j (\sigma_j E_j + \delta_j O_j)$ *is the sum over the tables of* $s_j$ *times the soft maximum less* $t_j$ *times the soft minimum.*

*(ii) With* $Y_j = \langle \chi_T, z\rangle$ *for* $T$ *under the weight law of table* $j$, *near* $z = 0$

$$E_j(z) = 2\log K_j(\mathbf{1}) + \sum_{k\ even} \frac{2}{k!}\,\kappa_{j,\mathbf{1}}(Y_j^{\times k}),$$
$$O_j(z) = \sum_{k\ odd} \frac{2}{k!}\,\kappa_{j,\mathbf{1}}(Y_j^{\times k}),$$

*so* $E_j(z) = 2\log K_j(\mathbf{1}) + \kappa_{j,\mathbf{1}}(Y_j, Y_j) + \frac{1}{12}\kappa_{j,\mathbf{1}}(Y_j^{\times 4}) + \cdots$ *and* $O_j(z) = 2\kappa_{j,\mathbf{1}}(Y_j) + \frac{1}{3}\kappa_{j,\mathbf{1}}(Y_j^{\times 3}) + \cdots$, *with* $\kappa_{j,\mathbf{1}}(Y_j) = \sum_e (2R_{j,e}(\mathbf{1}) - 1)z_e$ *and, for* $k \geq 2$, $\kappa_{j,\mathbf{1}}(Y_j^{\times k}) = 2^k \sum_{w \in \mathcal{E}^k} \kappa_{j,\mathbf{1}}(w)\, z^w$, $z^w = z_{w_1} \cdots z_{w_k}$. *Summed with the strengths and the leans, the terms of order* $k \geq 2$ *of* $\Phi^\circ$ *are* $\frac{1}{k!}\sum_{w \in \mathcal{E}^k} 2^k \sum_j \big(s_j + (-1)^k t_j\big)\kappa_{j,\mathbf{1}}(w)\, z^w$, *which for the exponents of a realization* $\Upsilon = ((s_j, t_j)_j; \mathbf{1})$ *is* $\frac{1}{k!}\sum_w \kappa_\Upsilon(w)\, z^w$, *rung* $k$ *of its still (Proposition 9.7): the even statistics carry the even rungs and the odd statistics the odd rungs.*

*(iii)* *As* $\lambda \to \infty$, $E_j(\lambda z)/\lambda \to \mathrm{wd}_j(z)$, *the width of Proposition 17.4, and* $O_j(\lambda z)/\lambda \to h_j(z) - h_j(-z)$, *the maximum of the scores being at the greedy basis* $T^+_{j,\pi}$ *and the minimum at* $T^-_{j,\pi}$ *for an order* $\pi$ *along which the coordinates of* $z$ *decrease (Lemma 8.3). So* $\Phi^\circ(\lambda z)/\lambda \to \langle 2v_\pi - 2\sigma\mathbf{1}, z\rangle$ *with* $v_\pi = \sum_j \left(s_j \mathbf{1}_{T^+_{j,\pi}} + t_j \mathbf{1}_{\mathcal{E}\setminus T^-_{j,\pi}}\right)$, *for* $s_j, t_j \geq 0$ *a vertex of the Newton sum at which* $\langle \cdot, z\rangle$ *is largest, the* greedy vertex *of* $\pi$*; and for a question the exponent of Proposition 17.4 has* $X(\lambda z)/\lambda \to -2\kappa(z) = 2\langle \mathbf{m} - v_\pi, z\rangle$ *on* $\mathbf{1}^\perp$*: the walls of the tails of* $\pi$ *are the tail sums of the masses less the greedy vertex,* $\kappa_{U_k} = m_{U_k} - (v_\pi)_{U_k}$.

**Proposition 49.18** (The laws of the questions are an exponential family)**.** *Let a connected family be such that* 1, *the coordinates* $z_e$, $e \neq g$, *of a chart* $z_g = 0$ *of* $\mathcal{T}$ *and the statistics* $E_j$, $O_j$ *are linearly independent functions on* $\mathcal{T}$. *Then the laws of its questions with every proper wall positive form a regular exponential family of dimension* $n-1+2N$, *with natural parameter* $(2\boldsymbol{\mu}_\perp; -\boldsymbol{\sigma}, -\boldsymbol{\delta})$, *statistic* $(z; E_j(z), O_j(z))_j$ *and cumulant function* $\log P_F\big(\mathbf{m}_F(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta}); \boldsymbol{\sigma} + \boldsymbol{\delta}, \boldsymbol{\sigma} - \boldsymbol{\delta}\big)$. *For the sample means* $(\bar{Z}; \bar{E}_j, \bar{O}_j)_j$ *of* $q$ *independent tilts, as in Proposition 49.7:*

*(i)* $(\bar{Z}; \bar{E}_j, \bar{O}_j)_j$ *is sufficient, and a function of it with expectation zero under every law of the family vanishes almost surely;*

*(ii)* *it is the unbiased estimator of least variance of the mean parameters*

$$\Big(\frac{1}{2}\nabla_{\boldsymbol{\mu}_\perp} \log P_F;\ -\partial_{\sigma_j} \log P_F,\ -\partial_{\delta_j} \log P_F\Big)_j,$$

*with covariance* $\frac{1}{q}$ *times the Hessian of* $\log P_F$ *in the natural parameter;*

*(iii)* *the likelihood equations have a solution, unique, exactly when* $(\bar{Z}; \bar{E}_j, \bar{O}_j)_j$ *is interior to the convex hull of the image of* $z \mapsto (z; E_j(z), O_j(z))_j$*;*

*(iv)* *for* $\delta_j \leq \delta_0$ *against* $\delta_j > \delta_0$, *with the other parameters unknown, rejecting for small* $\bar{O}_j$ *given the other coordinates of the sample means is uniformly most powerful among unbiased tests, with a conditional null law free of the other parameters; likewise for* $\sigma_j$ *and* $\bar{E}_j$*;*

*(v)* *the law of the sums* $q\bar{Z}$, $q\bar{E}_j$, $q\bar{O}_j$ *has the Laplace transform of Proposition 49.7(iv), so the power of every test is a computation in the family.*

*When the complement of every basis of table* $j$ *is a basis with the same weight,* $O_j \equiv 0$ *and the statements hold with* $\delta_j$ *removed.*

A word read at Dirichlet draws is a dial: its power means run from the mean of the logarithms to their maximum as the exponent grows, and they keep a finite expectation up to the first wall met along the ray of exponents, where the tail of the logarithm takes over with the rate of that wall. At a centre the centred potential $\sum_j \sigma_j E_j$ is least at the table's gauge and its exponential moments along the strengths close every wall at once, while the odd statistics are jointly symmetric and every odd direction has both tails at the rate of its slope. The two statistics of a table are the soft maximum and the soft minimum of the scores of its bases, read as their difference and their sum: at the table's gauge their Taylor terms are the rungs of the still, the even ones carried by the strengths and the odd ones by the leans, and at infinity they read the greedy vertex of the Newton sum, whose tail sums against the masses are the walls. The laws of the questions are a regular exponential family in the tilt and these statistics, so the classical optimal estimators and tests are computations with periods.

## 49.3 Draws, orders and races

**Proposition 49.19** (A sorted draw lives on the sector of its order)**.** *Let a family have real data on its balance wall with every proper wall positive, and let* $U \sim \mathrm{Breta}^{\circ}(F;\mathbf{m};\mathbf{s},\mathbf{t})$*. For an order* $\pi$ *of* $\mathcal{E}$ *let* $\kappa_k$ *be the walls of its tails and* $I_\pi$ *its sector integral (Corollary 8.6), so that* $P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_\pi I_\pi$ *(Theorem 8.5(ii)).*

*(i)* $\mathbb{P}\big(U_{\pi(1)} > \cdots > U_{\pi(n)}\big) = I_\pi / P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*.*

*(ii) Given that order, the gaps* $u_k = \log\big(U_{\pi(k-1)}/U_{\pi(k)}\big)$*,* $2 \le k \le n$*, have the density*

$$I_\pi^{-1} \exp\Big(-\sum_{k=2}^{n} \kappa_k u_k\Big) \prod_j k_{j,\pi}(e^{-u})^{-s_j}\, \psi_{j,\pi}(e^{-u})^{-t_j}$$

*on* $[0,\infty)^{n-1}$*, with the sector letters* $k_{j,\pi}, \psi_{j,\pi}$ *of Theorem 8.5(ii).*

*(iii)* $\theta_{j,T^+_{j,\pi}} \le k_{j,\pi} \le K_j(\mathbf{1})$ *and* $\theta_{j,T^-_{j,\pi}} \le \psi_{j,\pi} \le K_j(\mathbf{1})$ *on the cube, so the density of (ii) lies between constant multiples of the density of independent exponential gaps of the rates* $\kappa_k$*; for* $s_j, t_j \ge 0$*,* $I_\pi$ *lies between* $\prod_j K_j(\mathbf{1})^{-s_j-t_j} \prod_k \kappa_k^{-1}$ *and* $\prod_j \theta_{j,T^+_{j,\pi}}^{-s_j}\, \theta_{j,T^-_{j,\pi}}^{-t_j} \prod_k \kappa_k^{-1}$*.*

*(iv) On the sector of* $\pi$ *the largest monomial* $U^T$ *over the bases of table* $j$ *is at its greedy basis* $T^+_{j,\pi}$ *and the smallest at* $T^-_{j,\pi}$ *(Lemma 8.3): the best and the worst basis of a draw are functions of its order, and the best basis of table* $j$ *at* $U$ *is* $T$ *with probability* $\sum_{\pi:\, T^+_{j,\pi}=T} I_\pi / P_F(\mathbf{m};\mathbf{s},\mathbf{t})$*.*

**Proposition 49.20** (An ordered sample and a cut)**.** *Let* $Y_1, \dots, Y_q$ *be independent with a probability law* $\phi$ *on* $\mathbb{R}$*, an input (Definition 5.4) that charges no point.*

*(i) The ordered sample has the law* $q!\,\phi^{\otimes q}$ *on the region* $\{y_1 < \cdots < y_q\}$*.*

*(ii) The number of observations above a cut* $N$ *is binomial with* $q$ *trials and the chance* $\phi((N,\infty))$*, which for* $\phi$ *living on* $(0,\infty)$ *and* $N \ge 0$ *is the Trailmoment* $(\mathsf{T}_{0;(1)}[x]\,\phi)((N,\infty))$ *at index* $0$*; given that it is* $d$*, the ordered observations above the cut have the law* $d!\,\phi^{\otimes d}/\phi((N,\infty))^d$ *on the trail region* $X_N = \{N < y_1 < \cdots < y_d\}$*.*

*(iii) If* $\phi$ *lives on* $(0,\infty)$ *and* $N \ge 0$*, the empirical input* $\hat\phi = \frac{1}{q}\sum_i \delta_{Y_i}$ *has* $(\mathsf{T}_{\nu;(1)}[x]\,\hat\phi)((N,\infty)) = \frac{1}{q}\sum_{Y_i > N} Y_i^{-\nu}$*, with the mean* $(\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))$ *and the variance*

$$\frac{1}{q}\Big[(\mathsf{T}_{2\nu;(1)}[x]\,\phi)((N,\infty)) - (\mathsf{T}_{\nu;(1)}[x]\,\phi)((N,\infty))^2\Big].$$

**Proposition 49.21** (The races and the dilation relation)**.** *For table* $j$ *of a family and a tilt* $z$*, with* $\alpha = e^{2z}$*, the laws of the winners of the basis race and the cobasis race (Theorem 18.5) are* $\mathbb{P}_{j,\alpha}$ *and* $\mathbb{P}^*_{j,\alpha}$*; let*

$$\mathrm{Ent}_j(z) = -\sum_T \mathbb{P}_{j,\alpha}(T) \log \frac{\mathbb{P}_{j,\alpha}(T)}{\theta_{j,T}}, \qquad \mathrm{Ent}^*_j(z) = -\sum_T \mathbb{P}^*_{j,\alpha}(T) \log \frac{\mathbb{P}^*_{j,\alpha}(T)}{\theta_{j,T}}$$

*be their entropies relative to the weights.*

*(i)* $E_j(z) = \langle z, \nabla E_j(z)\rangle + \mathrm{Ent}_j(z) + \mathrm{Ent}^*_j(z)$ *and* $O_j(z) = \langle z, \nabla O_j(z)\rangle + \mathrm{Ent}_j(z) - \mathrm{Ent}^*_j(z)$*, with*

$$\langle z, \nabla E_j(z)\rangle = \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}\big[\langle \chi_T, z\rangle\big] - \mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha}}\big[\langle \chi_T, z\rangle\big],$$
$$\langle z, \nabla O_j(z)\rangle = \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}\big[\langle \chi_T, z\rangle\big] + \mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha}}\big[\langle \chi_T, z\rangle\big].$$

*Let the family be connected and let a question* $(\mathbf{m};\mathbf{s},\mathbf{t})$ *of pair coordinates* $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$ *have every proper wall positive. In the rest of this statement* $\mathbb{E}$ *is the expectation of* $U\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$*, Z is its tilt and* $\mathbf{u}(\alpha)=\sum_j\big(s_jR_j(\alpha)+t_jR_j^*(\alpha)\big)$ *the usage.*

*(ii)* Dilation relation.

$$2\langle\boldsymbol{\mu}_\perp,\mathbb{E}[Z]\rangle-\sum_j\big(\sigma_j\mathbb{E}[E_j(Z)]+\delta_j\mathbb{E}[O_j(Z)]\big)=-(n-1)-\sum_j\big(s_j\mathbb{E}[\mathrm{Ent}_j(Z)]+t_j\mathbb{E}[\mathrm{Ent}_j^*(Z)]\big);$$

*at a centre,* $\sum_j\sigma_j\mathbb{E}[E_j(Z)]=n-1+\sum_j\sigma_j\mathbb{E}[\mathrm{Ent}_j(Z)+\mathrm{Ent}_j^*(Z)]$.

*(iii)* $\mathrm{Cov}\big(\mathbf{u}_e(U),Z_f\big)=\frac12\big(\mathbf{1}[e=f]-\frac1n\big)$*, Z read on* $\mathbf{1}^\perp$.

*(iv)* $\log P_F$ *is convex in the pair coordinates of the question, so for a balanced direction* $v=(\dot{\mathbf{m}};\dot{\mathbf{s}},\dot{\mathbf{t}})$ *with the questions at the data* $(\mathbf{m}\pm\dot{\mathbf{m}};\mathbf{s}\pm\dot{\mathbf{s}},\mathbf{t}\pm\dot{\mathbf{t}})$ *inside their walls,*

$$\log\frac{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}{P_F(\mathbf{m}-\dot{\mathbf{m}};\mathbf{s}-\dot{\mathbf{s}},\mathbf{t}-\dot{\mathbf{t}})}\le D_v\log P_F(\mathbf{m};\mathbf{s},\mathbf{t})\le\log\frac{P_F(\mathbf{m}+\dot{\mathbf{m}};\mathbf{s}+\dot{\mathbf{s}},\mathbf{t}+\dot{\mathbf{t}})}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})},$$

*bracketing each mean of Proposition 49.7(ii) by neighbouring values of the period.*

**Proposition 49.22** (Moments are steps; entropies are derivatives one column up)**.** *Let a connected family carry a question* $(\mathbf{m};\mathbf{s},\mathbf{t})$ *with every proper wall positive, and in this statement let* $\mathbb{E}$ *be the expectation of* $U\sim\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$.

*(i) For a real balanced shift* $(\dot{\mathbf{m}};\dot{\mathbf{s}},\dot{\mathbf{t}})$ *with every proper wall of the shifted question positive,*

$$\mathbb{E}\Big[U^{\dot{\mathbf{m}}}\prod_j K_j(U)^{-\dot{s}_j}\Psi_j(U)^{-\dot{t}_j}\Big]=\frac{P_F(\mathbf{m}+\dot{\mathbf{m}};\mathbf{s}+\dot{\mathbf{s}},\mathbf{t}+\dot{\mathbf{t}})}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})}.$$

*(ii) For a basis T of table j the* columns $v_{j,T}=(\mathbf{1}_T;\mathbf{e}_j,\mathbf{0})$ *and* $v_{j,T}^*=(\mathbf{1}_{\mathcal{E}\setminus T};\mathbf{0},\mathbf{e}_j)$ *are balanced, and for* $a\ge0$

$$\mathbb{E}\Big[\Big(\frac{\mathbb{P}_{j,U}(T)}{\theta_{j,T}}\Big)^a\Big]=\frac{P_F(\mathbf{m}+a\mathbf{1}_T;\,\mathbf{s}+a\mathbf{e}_j,\,\mathbf{t})}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})},$$
$$\mathbb{E}\Big[\Big(\frac{\mathbb{P}^*_{j,U}(T)}{\theta_{j,T}}\Big)^a\Big]=\frac{P_F(\mathbf{m}+a\mathbf{1}_{\mathcal{E}\setminus T};\,\mathbf{s},\,\mathbf{t}+a\mathbf{e}_j)}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})},$$

*the walls only growing along a column.*

*(iii) With the entropies of Proposition 49.21,*

$$\mathbb{E}[\mathrm{Ent}_j(Z)]=-\sum_T\theta_{j,T}\,\frac{(D_{v_{j,T}}P_F)(\mathbf{m}+\mathbf{1}_T;\,\mathbf{s}+\mathbf{e}_j,\,\mathbf{t})}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})},$$
$$\mathbb{E}[\mathrm{Ent}_j^*(Z)]=-\sum_T\theta_{j,T}\,\frac{(D_{v^*_{j,T}}P_F)(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T};\,\mathbf{s},\,\mathbf{t}+\mathbf{e}_j)}{P_F(\mathbf{m};\mathbf{s},\mathbf{t})},$$

*Z the tilt of U: an entropy is a derivative of the period one column up.*

**Corollary 49.23** (A derivative adds one finite sum)**.** *The series of Corollary 15.11 may be differentiated term by term along a balanced direction* $v=(\dot{\mathbf{m}};\dot{\mathbf{s}},\dot{\mathbf{t}})$ *at a point inside the walls, with* $\dot\kappa_S$ *the change of the walls; each factor of a term has an elementary logarithmic derivative,*

$$D_v\log\frac{(s_j)_{n_j}}{n_j!}=\dot s_j\sum_{i<n_j}\frac{1}{s_j+i},\qquad D_v\log\frac{(t_j)_{n'_j}}{n'_j!}=\dot t_j\sum_{i<n'_j}\frac{1}{t_j+i},$$

$$D_v \log K_j(\mathbf{1})^{-s_j - t_j} = -(\dot{s}_j + \dot{t}_j) \log K_j(\mathbf{1}),$$
$$D_v \log {}_1^2\Gamma \begin{bmatrix} \kappa_{U_k} + \beta_k,\ \gamma_k + 1 \\ \kappa_{U_k} + \beta_k + \gamma_k + 1 \end{bmatrix} = -\dot{\kappa}_{U_k} \sum_{i=0}^{\gamma_k} \frac{1}{\kappa_{U_k} + \beta_k + i},$$

*so every quantity of Proposition 49.22 is the same positive series, at the question or one column up, with each term multiplied by a finite sum of reciprocals and logarithms of total weights.*

Sorting a draw of the shape law lands it in one sector of the period: the chance of an order is the fraction of the period carried by its sector, the gaps of the sorted draw are exponential at the rates of the walls of the tails up to bounded factors, and the best and the worst basis of the draw are the greedy bases of its order. The races read the statistics as entropies, and the dilation relation is the balance between the strength of a question and the entropies of its winners. Every moment of a letter, of a winner's chance or of the shape is a step of the data, and every entropy is a derivative one column up, so on the closing series of the period each is a positive series whose terms carry one extra finite sum.

## 49.4 The interaction of a bunch

**Proposition 49.24** (The interaction belongs to the room)**.** *Let a family have a bunch $B$, $|B| \geq 2$, with $\mathrm{pat}_j(B) = 2^B$ in every active table.*

*(i) $\iota_{j,B}$ and $\iota^*_{j,B}$, and so $\iota^{\mathbf{E}}_B$ and $\iota^{\mathbf{O}}_B$, are homogeneous of degree $0$ in $\alpha_R$.*

*(ii) At a gauge $\alpha$ put, for $P \subseteq B$, $w(P) = \prod_j \pi_{j,P}(\alpha)^{s_j} \pi^*_{j,B\setminus P}(\alpha)^{t_j}$, with the pattern probabilities $\pi_{j,p}(\alpha) = \alpha_B^p K_{j,(p)}(\alpha_R)/K_j(\alpha)$ and $\pi^*_{j,p}(\alpha) = \alpha_B^{B\setminus p} \Psi_{j,(p)}(\alpha_R)/\Psi_j(\alpha)$ of Definition 12.25; for a sample as in Proposition 49.1 it is the chance that every draw uses exactly $P$ on $B$, its basis draws meeting $B$ in $P$ and its cobasis draws missing exactly $P$. In the expansion $\log w(P) = \sum_{A \subseteq P} \lambda_A$ over the subsets of $B$, the top term is $\lambda_B = \iota^{\mathbf{E}}_B + \iota^{\mathbf{O}}_B$, and for $A \subsetneq B$ with $|A| \geq 2$, $\lambda_A$ is the sum $\iota^{\mathbf{E}}_A + \iota^{\mathbf{O}}_A$ of the interaction of $A$ in the face $F^{\mathcal{E}\setminus(B\setminus A)}$ of the wall at $B \setminus A$ (Theorem 16.25), whose tables are the restrictions $\mathcal{B}_j|\big(\mathcal{E} \setminus (B \setminus A)\big)$, the deletions of $B \setminus A$, at the exponents $(s_j, 0)$ and the contractions $\mathcal{B}_j/(B \setminus A)$ at $(0, t_j)$; none of them depends on $\alpha_B$. For one table at the exponents $(1, 0)$, $w(P) = \pi_P(\alpha)$ and $\lambda_B = \iota_B$, the top term of the log-linear expansion of the pattern probabilities.*

**Theorem 49.25** (The interaction totals the joint cumulants over the bunch's scales)**.** *Let a family with real exponents have a bunch $B = \{e_1, \dots, e_k\}$, $k \geq 2$, with $\mathrm{pat}_j(B) = 2^B$ in every active table. Then for both parts $\mathbf{X} \in \{\mathbf{E}, \mathbf{O}\}$*

$$\iota^{\mathbf{X}}_B(\alpha_R) = \int_{\mathbb{R}^B} \partial_{y_{e_1}} \cdots \partial_{y_{e_k}} \Phi_{\mathbf{X}}(y_B, \log \alpha_R)\, dy_B,$$

*converging absolutely; at a realization the integrand of the sum of the parts is $2^{-k} \kappa_\Upsilon(e_1, \dots, e_k) = \sum_j \big(s_j \kappa_{j,\alpha}(e_1, \dots, e_k) + (-1)^k t_j \kappa_{j,\alpha^{-1}}(e_1, \dots, e_k)\big)$, with $\kappa_\Upsilon$ the still at the gauge $\alpha = (e^{y_B}, \alpha_R)$ (Proposition 9.6). For one table at the exponents $(1, 0)$ the integrand is the joint cumulant of the indicators of the marks of $B$ under the basis law at $(e^{y_B}, \alpha_R)$ (Definition 7.17), and the total is $\iota_B(\alpha_R)$.*

**Proposition 49.26** (Both sides, and parity)**.** *For a table with $\mathrm{pat}(B) = 2^B$, with $\Psi = \sum_p \alpha_B^{B\setminus p} \Psi_{(p)}(\alpha_R)$, as functions of the tilt $z_R$ of the room $\iota^*_B(z_R) = (-1)^{|B|} \iota_B(-z_R)$. So for a*

*family*

$$\iota_B^{\mathbf{E}}(z_R) = \sum_j \sigma_j \big(\iota_{j,B}(z_R) + (-1)^{|B|}\iota_{j,B}(-z_R)\big), \qquad \iota_B^{\mathbf{O}}(z_R) = \sum_j \delta_j \big(\iota_{j,B}(z_R) - (-1)^{|B|}\iota_{j,B}(-z_R)\big) :$$

*for $|B|$ even the even part of the interaction is even in $z_R$ and the odd part is odd, for $|B|$ odd the reverse, and at the table's gauge of the room the interaction is $\big(2\sum_j \sigma_j\iota_{j,B}, 0\big)$ for $|B|$ even and $\big(0, 2\sum_j \delta_j\iota_{j,B}\big)$ for $|B|$ odd: the strengths read the even bunches and the leans the odd ones, as they read the even and the odd rungs of the still at the table's gauge (Proposition 9.7). The total of Theorem 49.25 for the logarithm $-\Phi$ of the family word is $-\sum_j \big(s_j\iota_{j,B}(z_R) + (-1)^{|B|}t_j\iota_{j,B}(-z_R)\big)$.*

**Proposition 49.27** (A design)**.** *Let $\mathbf{X} \in \mathbb{R}^{\mathcal{E}\times C}$ have full column rank, with rows $x_e$, and for $J \subseteq C$ let $K_J(\alpha) = \det\big(\mathbf{X}_J^{\mathsf{T}} \operatorname{diag}(\alpha)\mathbf{X}_J\big)$, $\mathbf{X}_J$ the columns in $J$, with $K_\varnothing = 1$.*

*(i) The marks are the observations of the determinantal table with the slice $\mathbf{X}^{\mathsf{T}}$ (Definition 7.20) and the weights $\theta_T = \det(\mathbf{X}_{T,C})^2$, $|T| = |C|$: $K = K_C$, and the occupation of a mark is its leverage, $R_e(\alpha) = \alpha_e\, x_e^{\mathsf{T}}\big(\mathbf{X}^{\mathsf{T}} \operatorname{diag}(\alpha)\mathbf{X}\big)^{-1}x_e$.*

*(ii) Add one* prior mark *$\gamma^{\mathrm{p}}$ for each column $\gamma \in C$, with the rows of the identity and the scales $\beta$: $K(\alpha,\beta) = \det\big(\mathbf{X}^{\mathsf{T}} \operatorname{diag}(\alpha)\mathbf{X} + \operatorname{diag}(\beta)\big) = \sum_{J\subseteq C} \beta^{C\setminus J} K_J(\alpha)$. The prior marks form a bunch $C^{\mathrm{p}}$ with $\operatorname{pat}(C^{\mathrm{p}}) = 2^{C^{\mathrm{p}}}$, and the child of the pattern $q^{\mathrm{p}}$, $q \subseteq C$, is $K_{C\setminus q}$.*

*(iii) For a split $C = J_1 \sqcup J_2$ the generalized variance inflation factor is $\frac{K_{J_1}K_{J_2}}{K_C K_\varnothing} = \exp\big(-\iota_{(J_1^{\mathrm{p}}, J_2^{\mathrm{p}})}\big)$, and for two columns $\gamma, \gamma'$, one minus the squared partial correlation given the rest is $\exp(\iota_{\{\gamma^{\mathrm{p}},\gamma'^{\mathrm{p}}\}})$ in the table with the other prior marks deleted.*

*(iv) The dual multivector of the table of (ii) has the slice $[\, I_{\mathcal{E}} \mid -\mathbf{X}\,]$, and the basis polynomial of its dual table, $\det\big(\operatorname{diag}(\alpha') + \mathbf{X}\operatorname{diag}(\beta')\mathbf{X}^{\mathsf{T}}\big)$, is the polynomial of (ii) for the matrix $\mathbf{X}^{\mathsf{T}}$: the columns of $\mathbf{X}$ are its observations, at the scales $\beta'$, and the marks of $\mathcal{E}$ its prior marks, at the scales $\alpha'$. Its child with every prior mark deleted, $\det\big(\mathbf{X}\operatorname{diag}(\beta')\mathbf{X}^{\mathsf{T}}\big)$, is nonzero exactly when $\mathbf{X}$ has rank $|\mathcal{E}|$.*

A bunch all of whose subsets are patterns sees the room through its children, one for each subset of the bunch, and the alternating sum of their logarithms is the interaction: it is homogeneous of degree zero in the room, it is the top term of the log-linear expansion of the chance that a sample uses one pattern of the bunch throughout, and it is the total, over the bunch's scales, of the joint cumulants of the use of the bunch's marks under the basis and cobasis laws, weighted by the exponents. The dual table reads the same interaction at the opposite tilt with the sign of the parity of the bunch, so a family reads the interaction of an even bunch through its strengths and that of an odd bunch through its leans, as it reads the rungs of the same parities. In a regression design the prior marks of the columns are such a bunch: the interactions are minus the logarithms of the generalized variance inflation factors and the logarithms of one minus the squared partial correlations, and the occupations are the leverages.

# Part X

# Proofs of the general forms

# Chapter 50

# Proofs of the general forms

This part proves the statements that are more general than their counterparts in the reference *The Foreman and the serif family* (16 October 2026), and those with no counterpart there; in the source file the comment before each such statement says so. Statements are cited by their numbers, and every symbol and term is that of the chapter the statement belongs to. A statement of the reference that is a special case of a statement proved here is not proved again; where reading it off needs a word, the word follows the proof. Lemmas that belong to this part alone are numbered S1, S2, .... The chapters are taken in order, one section each, and the last section records the checks run on small families.

## 50.1 Chapter 1: conventions, the stacked Gamma and the finite part

**Lemma S1.** *Let $\Theta \subset \mathbb{R}$ be finite and $\phi(y) = \sum_{\theta\in\Theta} a_\theta e^{i\theta y}$ with $a_\theta \in \mathbb{C}$. If $\phi(y) \to 0$ as $y \to -\infty$, then every $a_\theta$ is $0$.*

*Proof.* Fix $\theta_0 \in \Theta$. For $Y > 0$,

$$\frac{1}{Y}\int_{-2Y}^{-Y} \phi(y)\, e^{-i\theta_0 y}\, dy = a_{\theta_0} + \sum_{\theta\neq\theta_0} a_\theta\, \frac{1}{Y}\int_{-2Y}^{-Y} e^{i(\theta-\theta_0)y}\, dy,$$

and each integral on the right has modulus at most $2/|\theta-\theta_0|$, so the sum is $O(Y^{-1})$. The left side has modulus at most $\sup_{y\le -Y}|\phi(y)|$, which tends to 0. Letting $Y \to \infty$ gives $a_{\theta_0} = 0$. □

**Lemma S2.** *Let $g(x) = \sum_{(k,j)\in F} c_{k,j}\, x^k \log^j x$ for $x \in (0,1)$, where $F \subset \mathbb{C}\times\mathbb{N}$ is finite and nonempty and every $c_{k,j}$ is nonzero. Let $\kappa$ be the least real part of the exponents $k$, $(k,j) \in F$, and $J$ the largest $j$ such that $(k,j) \in F$ for some $k$ with $\operatorname{Re} k = \kappa$. Then $g(x)/(x^\kappa \log^J x)$ does not tend to $0$ as $x \to 0$.*

*Proof.* Put $y = \log x$, so that $y \to -\infty$ as $x \to 0$ and $x^k = e^{ky}$. Then

$$\frac{g(x)}{x^\kappa \log^J x} = \sum_\theta c_{\kappa+i\theta,J}\, e^{i\theta y} + \sum_{(k,j)} c_{k,j}\, e^{(k-\kappa)y}\, y^{j-J},$$

the first sum over the real $\theta$ with $(\kappa + i\theta, J) \in F$ and the second over the other elements of $F$. Every term of the second sum tends to 0: if $\operatorname{Re} k = \kappa$, then $j < J$, $|e^{(k-\kappa)y}| = 1$ and $|y|^{j-J} \to 0$;

if $\operatorname{Re} k > \kappa$, then $|e^{(k-\kappa)y} y^{j-J}| = e^{(\operatorname{Re} k-\kappa)y} |y|^{j-J} \to 0$. The first sum is a function $\phi(y)$ as in Lemma S1 with a nonzero coefficient, so it does not tend to 0 as $y \to -\infty$, and neither does the left side. □

**Definition 1.5 (The finite part).** As stated: the exponents of a log-power expansion are complex numbers, and its coefficients are determined by the function.

*Proof.* What is to be shown is that the coefficients of a log-power expansion are determined by $h$, so that the finite parts are well defined; the rest of the definition is that of the reference, with complex exponents.

*(1) One remainder.* Let two finite sums $S_1$ and $S_2$ of terms $c\,x^k \log^j x$, every exponent of real part below $N$, both differ from $h$ by $O(x^N)$ as $x \to 0$. Their difference $g = S_1 - S_2$ is a finite sum of the same kind with $g(x) = O(x^N)$. If $g$ had a nonzero coefficient, then, with $\kappa < N$ and $J$ as in Lemma S2, $|g(x)/(x^\kappa \log^J x)| \le C x^{N-\kappa} |\log x|^{-J} \to 0$, which that lemma excludes. So $S_1$ and $S_2$ have the same coefficients.

*(2) Two remainders.* Let $N < N'$, and let $S$ and $S'$ be finite sums for $N$ and for $N'$. A term $x^k \log^j x$ of $S'$ with $\operatorname{Re} k \ge N$ is $O(x^{N-\delta})$ for every $\delta > 0$, so the part $S'_{<N}$ of $S'$ with exponents of real part below $N$ has $h - S'_{<N} = O(x^{N-\delta})$. Choose $\delta > 0$ smaller than $N - \operatorname{Re} k$ for every exponent $k$ of $S$ and of $S'_{<N}$. Then $S$ and $S'_{<N}$ both differ from $h$ by $O(x^{N-\delta})$, with all exponents of real part below $N - \delta$, so they coincide by (1). Hence the coefficient of a term $x^k \log^j x$ is the same for every $N > \operatorname{Re} k$, and the coefficient of $x^0 \log^0 x$, read with any $N > 0$, is determined by $h$. At $\infty$ the same holds for $h(1/x)$ at 0: the term $x^{-k} \log^j x$ is $(-1)^j x'^k \log^j x'$ in $x' = 1/x$, and $x^0 \log^0 x$ is $x'^0 \log^0 x'$.

*(3) The primitive.* Two primitives of $f$ differ by a constant $C$, whose expansion at either end is the single term $C\,x^0 \log^0 x$; by (2) the finite parts of the second primitive at 0 and at $\infty$ are those of the first plus $C$. So $\operatorname{Reg}_{x\to\infty} h - \operatorname{Reg}_{x\to 0} h$ and $h(b) - \operatorname{Reg}_{x\to 0} h$ do not depend on the primitive. □

**Proposition 1.6 (A convergent integral is its own finite part).** *As stated, for log-power expansions with complex exponents.*

*Proof.* Let $h$ be a primitive of $f$ with log-power expansions at both ends. Since $\int_0^\infty f\,dx$ converges, $h(x) = h(1) - \int_x^1 f$ has a limit $L_0$ as $x \to 0$, $h$ has a limit $L_\infty$ as $x \to \infty$, and $\int_0^\infty f\,dx = L_\infty - L_0$. Take the expansion at 0 for some $N > 0$, and let $G$ be its part with exponents of real part at most 0. The other terms and the remainder tend to 0, so $G(x) \to L_0$. Write $G - L_0$ as a finite sum of terms $c\,x^k \log^j x$, the constant $-L_0$ joining the coefficient of $x^0 \log^0 x$, and suppose it has a nonzero coefficient; let $\kappa \le 0$ and $J$ be as in Lemma S2. If $(\kappa, J) \ne (0,0)$, then $|x^\kappa \log^J x| \to \infty$, so $(G - L_0)/(x^\kappa \log^J x) \to 0$. If $(\kappa, J) = (0,0)$, every exponent of $G - L_0$ has real part 0 and every power of the logarithm is 0, and $(G - L_0)/(x^0 \log^0 x) = G - L_0 \to 0$. Both are excluded by Lemma S2. So $G$ is the constant $L_0$: the expansion at 0 has no term with an exponent of real part at most 0 other than $L_0\,x^0 \log^0 x$, and $\operatorname{Reg}_{x\to 0} h = L_0$. In the same way, for $h(1/x)$ at 0, $\operatorname{Reg}_{x\to\infty} h = L_\infty$. Hence $\operatorname{Reg} \int_0^\infty f\,dx = L_\infty - L_0 = \int_0^\infty f\,dx$. □

**Theorem 1.7 (A finite part is a constant term).** *As stated, for log-power expansions with complex exponents.*

*Proof.* The steps are those of the reference, proof of 1.7; they hold at complex exponents as follows.

*The end at* 0. On $(0,1]$ write $f = \sum c_{b,j} x^b \log^j x + r_N$, the sum over the finitely many terms of the expansion at 0 with $\operatorname{Re} b < N$, where $N > 0$ and $r_N = O(x^N)$ as $x \to 0$; $r_N$ is integrable on $(0,1]$. For $b \neq -1$ put

$$Q_{b,j}(y) = \sum_{i=0}^{j} (-1)^{j-i} \frac{j!}{i!\,(b+1)^{j-i+1}}\, y^i;$$

then $(b+1)Q_{b,j} + Q'_{b,j} = y^j$, so $\frac{d}{dx}\big(x^{b+1} Q_{b,j}(\log x)\big) = x^b \log^j x$, and $Q_{b,j}(0) = (-1)^j j!/(b+1)^{j+1}$. Hence

$$\int_x^1 t^b \log^j t\, dt = Q_{b,j}(0) - x^{b+1} Q_{b,j}(\log x) \quad (b \neq -1), \qquad \int_x^1 t^{-1} \log^j t\, dt = -\frac{\log^{j+1} x}{j+1},$$

and $\int_x^1 r_N = \int_0^1 r_N - \int_0^x r_N$ with $\int_0^x r_N = O(x^{N+1})$. So $\int_x^1 f$ has a log-power expansion at 0, and its coefficient of $x^0 \log^0 x$ is $\sum_{b \neq -1} c_{b,j} Q_{b,j}(0) + \int_0^1 r_N$: a term $x^{b+1} \log^i x$ has the exponent $b+1 \neq 0$, and a term $\log^{j+1} x$ has a positive power of the logarithm. For a primitive $h$ of $f$, $h(1) - \operatorname{Reg}_{x\to 0} h = \operatorname{Reg}_{x\to 0} \int_x^1 f$, so by 1.5

$$\operatorname{Reg} \int_0^1 f\, dx = \sum_{b \neq -1} c_{b,j} \frac{(-1)^j j!}{(b+1)^{j+1}} + \int_0^1 r_N\, dx.$$

On the other side, by $\int_0^1 x^{s-1} \log^j x\, dx = (-1)^j j!/s^{j+1}$ for $\operatorname{Re} s > 0$,

$$\int_0^1 x^\varepsilon f\, dx = \sum c_{b,j} \frac{(-1)^j j!}{(b+1+\varepsilon)^{j+1}} + \int_0^1 x^\varepsilon r_N\, dx$$

for $\operatorname{Re}\varepsilon$ large, the last term holomorphic on $\operatorname{Re}\varepsilon > -N-1$; this is the meromorphic continuation there. Its constant term at $\varepsilon = 0$ is the sum displayed above: a term with $b \neq -1$ gives its value, and a term with $b = -1$ is the pure pole $(-1)^j j!\,\varepsilon^{-j-1}$, of constant term 0.

*The end at* $\infty$. By $x \mapsto 1/x$, $\int_1^\infty x^\varepsilon f\, dx = \int_0^1 x^{-\varepsilon} x^{-2} f(1/x)\, dx$ and $\int_1^x f = \int_{1/x}^1 t^{-2} f(1/t)\, dt$; the function $x^{-2} f(1/x)$ has a log-power expansion at 0, the coefficient of $x^0 \log^0 x$ at $\infty$ is that of $x'^0 \log^0 x'$ at 0 in $x' = 1/x$, and a constant term at $\varepsilon = 0$ does not change under $\varepsilon \mapsto -\varepsilon$. So $\operatorname{Reg}_{x\to\infty} \int_1^x f = \operatorname{CT}_{\varepsilon=0} \int_1^\infty x^\varepsilon f\, dx$ by the end at 0.

Adding, $\operatorname{Reg} \int_0^\infty f\, dx = \operatorname{Reg}_{x\to\infty} h - \operatorname{Reg}_{x\to 0} h = \operatorname{Reg}_{x\to\infty} \int_1^x f + \operatorname{Reg}_{x\to 0} \int_x^1 f$ is the constant term of the sum of the two continuations. □

**Corollary 1.8 (Finite parts at a pole of a Mellin integral).** *As stated: for every $f$ with a log-power expansion at* 0*, at every $a \in \mathbb{C}$.*

*Proof.* Fix $a$ and $N > 1 - \operatorname{Re} a$, and write $f = \sum c_{k,j} x^k \log^j x + r_N$ on $(0,1]$, the finitely many terms of the expansion with $\operatorname{Re} k < N$, $r_N = O(x^N)$ at 0 and integrable on $[\delta, 1]$ for every $\delta > 0$. For $\operatorname{Re} a'$ large,

$$I(a') = \sum c_{k,j} \frac{(-1)^j j!}{(a'+k)^{j+1}} + \int_0^1 x^{a'-1} r_N\, dx,$$

the last term holomorphic on $\operatorname{Re} a' > -N$. This is the meromorphic continuation of $I$ on $\operatorname{Re} a' > -N$; its poles lie at the points $-k$, a term $x^k \log^j x$ giving a pole of order at most $j+1$, and since $N$ is arbitrary this describes the continuation on $\mathbb{C}$. Its constant term at $a' = a$ is

$$\operatorname*{CT}_{a'=a} I(a') = \sum_{a+k \neq 0} c_{k,j} \frac{(-1)^j j!}{(a+k)^{j+1}} + \int_0^1 x^{a-1} r_N\, dx,$$

a term with $a+k=0$ being the pure pole $(-1)^j j!\,(a'-a)^{-j-1}$; at an $a$ that is not a pole it is $I(a)$. On the other side, $x^{a-1}f$ has the log-power expansion at 0 with the terms $c_{k,j}x^{a+k-1}\log^j x$ and the remainder $x^{a-1}r_N = O(x^{N+\operatorname{Re} a-1})$, $N+\operatorname{Re} a-1>0$. The end at 0 in the proof of 1.7 above, with $b=a+k-1$ (so $b\neq -1$ exactly when $a+k\neq 0$, and $b+1=a+k$), gives for $\operatorname{Reg}\int_0^1 x^{a-1}f\,dx$ the same sum. □

**Lemma 1.9 (A finite part depends on the coordinate).** *As stated: the dilation rule for every function with log-power expansions.*

*Proof. (a) The finite part of $h(cx^p)$.* Let $h(y)=\sum c_{k,j}\,y^{-k}\log^j y+O(y^{-N})$ as $y\to\infty$, finitely many terms with $\operatorname{Re} k<N$, so that $a_j=c_{0,j}$. For $y=cx^p$, $y^{-k}=c^{-k}x^{-pk}$ and $\log y=\log c+p\log x$, so

$$h(cx^p)=\sum c_{k,j}\,c^{-k}x^{-pk}\sum_{i=0}^{j}\binom{j}{i}(\log c)^{j-i}p^i\log^i x+O(x^{-pN}),$$

a log-power expansion at $\infty$ with the exponents $pk$, of real parts below $pN$; as $N$ is arbitrary, so is $pN$. The term $x^0\log^0 x$ comes from $pk=0$, that is $k=0$, and $i=0$; by 1.5 its coefficient $\sum_j c_{0,j}(\log c)^j=\sum_j a_j(\log c)^j$ is $\operatorname{Reg}_{x\to\infty}h(cx^p)$. At 0 the computation is the same.

*(b) The finite part of the integral.* Let $F(x)=\int_1^x f$. At $\infty$, $f(t)=\sum f_{k,j}t^{-k}\log^j t+O(t^{-N})$ with $N>1$; integrating from 1 to $x$, a term with $k\neq 1$ gives $x^{1-k}Q_{-k,j}(\log x)$ up to a constant (with $Q_{b,j}$ as in the proof of 1.7 above), a term with $k=1$ gives $\log^{j+1}x/(j+1)$, and the remainder gives a constant plus $O(x^{1-N})$. So $F$ has a log-power expansion at $\infty$, and for $i\geq 1$ its coefficient of $x^0\log^i x$ comes from the term $k=1$, $j=i-1$ alone: it is $f^\infty_{i-1}/i$. In the same way, at 0 the coefficient of $x^0\log^i x$ in $F$ is $f^0_{i-1}/i$ for $i\geq 1$. Write $A^\infty_i$ and $A^0_i$ for the coefficients of $x^0\log^i x$ of $F$ at $\infty$ and at 0; $A^\infty_0-A^0_0=\operatorname{Reg}\int_0^\infty f\,dx$. The function $G(x)=F(cx^p)$ is a primitive of $f(cx^p)\,c\,p\,x^{p-1}$, and by (a) at both ends

$$\operatorname{Reg}\int_0^\infty f(cx^p)\,c\,p\,x^{p-1}dx=\sum_{i\geq 0}\big(A^\infty_i-A^0_i\big)(\log c)^i=\operatorname{Reg}\int_0^\infty f\,dx+\sum_{i\geq 1}\frac{f^\infty_{i-1}-f^0_{i-1}}{i}(\log c)^i,$$

which is the display with $i=j+1$. At $c=1$ the correction vanishes: a power of the coordinate leaves the finite part unchanged. □

*Remark.* The reference states the instance $f(x)=1/(1+cx)$, $c>0$. This $f$ has the coefficient $f^\infty_0=1/c$ of $x^{-1}$ at $\infty$ and no term $x^{-1}\log^j x$ at 0, and with the dilation $c^{-1}$ and $p=1$ the lemma gives $\frac{1}{c}\operatorname{Reg}\int_0^\infty\frac{dx'}{1+x'}=\operatorname{Reg}\int_0^\infty\frac{dx}{1+cx}-\frac{\log c}{c}$. The left side is 0, the primitive $\log(1+x')=\log x'+\log(1+1/x')$ having no constant term at either end; so $\operatorname{Reg}\int_0^\infty\frac{dx}{1+cx}=\frac{\log c}{c}$, the value of the reference. A second check, with a logarithm: $f(x)=\log(1+x)/(1+x)$ has the primitive $\frac12\log^2(1+x)$, with no constant term at either end, so $\operatorname{Reg}\int_0^\infty f\,dx=0$; at $\infty$, $f=x^{-1}\log x+O(x^{-2}\log x)$, so $f^\infty_1=1$ and $f^\infty_0=0$, and at 0 it has no term $x^{-1}\log^j x$. The lemma gives $\operatorname{Reg}\int_0^\infty f(cx)\,c\,dx=\frac12\log^2 c$; directly, the primitive $\frac12\log^2(1+cx)=\frac12\big(\log c+\log x+\log(1+\frac{1}{cx})\big)^2$ has the constant term $\frac12\log^2 c$ at $\infty$ and none at 0.

## 50.2 Chapter 2: the Gramma and the Zamma

**Proposition 2.3 (Affine changes of the logarithmic coordinates).** *As stated: one statement for the reflection, the linear changes of the coordinates and the torus homogeneity.*

*Proof.* Let $A \in GL_{\mathcal{E}}(\mathbb{R})$ and $\mathbf{h} \in \mathbb{R}^{\mathcal{E}}$. The map $y' \mapsto y = Ay' + \mathbf{h}$ is a bijection of $\mathbb{R}^{\mathcal{E}}$ with $dy = |\det A|\, dy'$, and $\langle \nu, y\rangle = \langle A^{\mathsf{T}}\nu, y'\rangle + \langle \nu, \mathbf{h}\rangle$, $\langle t_k, y\rangle = \langle A^{\mathsf{T}}t_k, y'\rangle + \langle t_k, \mathbf{h}\rangle$. In the logarithmic form of 2.1,

$$e^{\langle \nu, y\rangle} \exp\Big(-\sum_k \xi_k e^{\langle t_k, y\rangle}\Big)\, dy = |\det A|\, e^{\langle \nu, \mathbf{h}\rangle}\, e^{\langle A^{\mathsf{T}}\nu, y'\rangle} \exp\Big(-\sum_k \big(e^{\langle t_k, \mathbf{h}\rangle}\xi_k\big)\, e^{\langle A^{\mathsf{T}}t_k, y'\rangle}\Big)\, dy'.$$

The moduli of the two sides are the same expressions with $\nu$ replaced by $\operatorname{Re}\nu$, so the integral of the left side converges exactly when that of the right side does. The left side is the integrand of $\mathsf{G}_{\nu;t}(\boldsymbol{\xi})$ and the right side, up to the constant factor, that of $\mathsf{G}_{A^{\mathsf{T}}\nu;A^{\mathsf{T}}t}\big((e^{\langle t_k, \mathbf{h}\rangle}\xi_k)_k\big)$, with positive coefficients; by 2.2, $\nu \in \mathcal{T}_t$ exactly when $A^{\mathsf{T}}\nu \in \mathcal{T}_{A^{\mathsf{T}}t}$, that is $\mathcal{T}_{A^{\mathsf{T}}t} = A^{\mathsf{T}}\mathcal{T}_t$, and integrating gives the display. At $A = -I_{\mathcal{E}}$ and $\mathbf{h} = 0$, $|\det A| = 1$ and $A^{\mathsf{T}}t = -t$: the reflection. At $A = I_{\mathcal{E}}$ and $\mathbf{h} = \log\lambda$, $\lambda \in \mathbb{R}^{\mathcal{E}}_{>0}$, $e^{\langle t_k, \mathbf{h}\rangle} = \lambda^{t_k}$ and $e^{\langle \nu, \mathbf{h}\rangle} = \lambda^{\nu}$: the torus homogeneity. □

**Theorem 2.5 (Product).** *As stated, for any number of blocks.*

*Proof.* Write $x = (x^{(1)}, \dots, x^{(q)})$ with $x^{(i)} \in \mathbb{R}^{\mathcal{E}_i}_{>0}$. Each $t_k$ is supported in one $\mathcal{E}_i$, so $x^{t_k} = (x^{(i)})^{t_k|\mathcal{E}_i}$ for that $i$, and

$$x^{\nu-\mathbf{1}} \exp\Big(-\sum_k \xi_k x^{t_k}\Big) = \prod_{i=1}^{q} (x^{(i)})^{\nu|\mathcal{E}_i - \mathbf{1}} \exp\Big(-\sum_{k\in K_i} \xi_k\, (x^{(i)})^{t_k|\mathcal{E}_i}\Big),$$

$K_i$ the set of the terms whose exponents are supported in $\mathcal{E}_i$, the zero vectors counted in $K_1$. The moduli of the factors are positive functions of separate variables, so by Tonelli the modulus of the product is integrable exactly when every factor is integrable over $\mathbb{R}^{\mathcal{E}_i}_{>0}$; by 2.2 this says $\mathcal{T}_t = \mathcal{T}_{t^{(1)}} \times \cdots \times \mathcal{T}_{t^{(q)}}$, and Fubini gives the product. □

**Theorem 2.8 (Scale and shape along sets).** *As stated, for several disjoint sets at once.*

*Proof.* Apply 2.7 on each $B_i$: $x_{B_i} = \rho_i u_i$ with $\rho_i > 0$ and $u_i \in \Delta_{B_i}$, and $x_{B_i}^{\nu|_{B_i}-\mathbf{1}} dx_{B_i} = \rho_i^{\nu_{B_i}-1} u_i^{\nu|_{B_i}-\mathbf{1}}\, d\rho_i\, d\omega_{B_i}(u_i)$. With $x_R$ the coordinates on $R$, the map $(x_R, (\rho_i)_i, (u_i)_i) \mapsto x$ is a bijection onto $\mathbb{R}^{\mathcal{E}}_{>0}$, and since $(\rho_i u_i)^{t_k|_{B_i}} = \rho_i^{\langle t_k, \mathbf{1}_{B_i}\rangle} u_i^{t_k|_{B_i}}$,

$$x^{\nu-\mathbf{1}}dx = x_R^{\nu|_R-\mathbf{1}} \prod_i \rho_i^{\nu_{B_i}-1} u_i^{\nu|_{B_i}-\mathbf{1}}\, d\rho\, d\omega(u)\, dx_R, \qquad \xi_k x^{t_k} = b_k(u, x_R) \prod_i \rho_i^{\ell_{k,i}}.$$

At $\nu \in \mathcal{T}_t$ the Gramma converges (2.2), that is, its integrand is absolutely integrable. By Tonelli and Fubini the integral may be taken with the integral over $\rho = (\rho_i)_i \in \mathbb{R}^q_{>0}$ innermost, that integral converging absolutely for almost every $(u, x_R)$, and it is

$$\int_{\mathbb{R}^q_{>0}} \rho^{(\nu_{B_i})_i - \mathbf{1}} \exp\Big(-\sum_k b_k(u, x_R)\, \rho^{\ell_k}\Big)\, d\rho = \mathsf{G}_{(\nu_{B_i})_i;\ell}\big(\mathbf{b}(u, x_R)\big),$$

the Gramma of 2.1 on $q$ scales with the configuration $\ell$. When $R$ is empty there are no coordinates $x_R$. □

**Proposition 2.13 (First facts about the Zamma).** *As stated: (i) at every invertible square configuration and (iv) on any number of scales; (ii) and (iii) are those of the reference.*

*Proof.* (i) By 2.6, $\mathsf{G}_{\nu;t}(\boldsymbol{\xi}) = |\det t|^{-1}\prod_k \Gamma(b_k)\,\xi_k^{-b_k}$ on $\mathcal{T}_t = \{\nu : \operatorname{Re}\mathbf{b} \in (0,\infty)^N\}$; dividing by $\prod_e \Gamma(\nu_e)$ (2.12) gives the display, the quotient of Gammas being the stacked Gamma of 1.2, and the display carries the continuation. At the unit configuration $\mathbf{b} = \nu$ and the display is the monomial; on one scale with $t_1 > 0$, $|\det t| = t_1$ and $b_1 = \nu/t_1$, which is the formula of the reference.

(iv) Write $t'' = I_{\mathcal{E}} \oplus t'$ and $\boldsymbol{\xi}'' = \mathbf{1} \oplus \boldsymbol{\xi}'$. Since $\operatorname{cone}(t'') \supseteq \operatorname{cone}(I_{\mathcal{E}}) = [0,\infty)^{\mathcal{E}}$, every $\nu \in \mathbb{R}^{\mathcal{E}}_{>0}$ lies in $\operatorname{int}\operatorname{cone}(t'')$, hence in $\mathcal{T}_{t''}$. By 2.1,

$$\mathsf{G}_{\nu;t''}(\boldsymbol{\xi}'') = \int_{\mathbb{R}^{\mathcal{E}}_{>0}} x^{\nu-\mathbf{1}} e^{-\sum_e x_e} \exp\Big(-\sum_k \xi'_k x^{t'_k}\Big)\,dx = \prod_e \Gamma(\nu_e)\,\mathbb{E}_{U\sim\mathrm{Gamma}_\nu}\Big[\exp\Big(-\sum_k \xi'_k U^{t'_k}\Big)\Big],$$

$\mathrm{Gamma}_\nu$ having the density $\prod_e x_e^{\nu_e-1}e^{-x_e}/\Gamma(\nu_e)$ (1.1(vii)); dividing by $\prod_e \Gamma(\nu_e)$ gives the display. The variable $\exp(-\sum_k \xi'_k U^{t'_k})$ lies in $(0,1]$ and decreases in each $\xi'_k$, and so does its expectation. Put $X = \sum_k \xi'_k U^{t'_k} \geq 0$. Where every $\nu + t'_k$ lies in $\mathbb{R}^{\mathcal{E}}_{>0}$, the coordinates of $U$ being independent, $\mathbb{E}[U^{t'_k}] = \prod_e \mathbb{E}[U_e^{t'_{k,e}}] = \prod_e \Gamma(\nu_e + t'_{k,e})/\Gamma(\nu_e)$, so $\mathbb{E}X = \sum_k \xi'_k \prod_e \Gamma_1\begin{bmatrix}\nu_e + t'_{k,e}\\ \nu_e\end{bmatrix}$ is finite. The function $s \mapsto e^{-s}$ is convex, so $e^{-s} \geq e^{-s_0} - e^{-s_0}(s - s_0)$ for all real $s, s_0$; at $s = X$ and $s_0 = \mathbb{E}X$, taking expectations, $\mathbb{E}[e^{-X}] \geq e^{-\mathbb{E}X}$, which is the bound. □

**Proposition 2.14 (Powers of a polynomial in the scales).** *As stated: (i) and (iii) for every Zamma; (ii) and (iv) are those of the reference.*

*Proof.* Write $\boldsymbol{\xi}(\rho) = \boldsymbol{\xi}^{(0)} + \boldsymbol{\Delta}(\rho)$ with $\Delta_k(\rho) = \sum_{l\neq 0} \rho^l \xi_k^{(l)}$, a polynomial vanishing at $\rho = 0$.

*(1) Derivatives.* Away from the unit configuration, $\mathsf{Z}_{\nu;t} = \mathsf{G}_{\nu;t}/\prod_e \Gamma(\nu_e)$ on $(0,\infty)^N$. By 2.4(i), $\partial^{\mathbf{w}}\mathsf{G}_{\nu;t} = (-1)^{|\mathbf{w}|}\mathsf{G}_{\nu+t\mathbf{w};t}$, every $\nu + t\mathbf{w}$ lying in $\mathcal{T}_t$ because $\operatorname{int}\operatorname{cone}(t) + \operatorname{cone}(t) \subseteq \operatorname{int}\operatorname{cone}(t)$; so $\mathsf{Z}_{\nu;t}$ is infinitely differentiable on $(0,\infty)^N$ and

$$\partial^{\mathbf{w}}\mathsf{Z}_{\nu;t} = (-1)^{|\mathbf{w}|}\frac{\mathsf{G}_{\nu+t\mathbf{w};t}}{\prod_e \Gamma(\nu_e)} = (-1)^{|\mathbf{w}|}\prod_e \Gamma_1\begin{bmatrix}\nu_e + (t\mathbf{w})_e\\ \nu_e\end{bmatrix}\,\mathsf{Z}_{\nu+t\mathbf{w};t},$$

the second form wherever no entry of $\nu + t\mathbf{w}$ lies in $-\mathbb{N}$, the product being read as the first form otherwise, as in the statement. At the unit configuration $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\boldsymbol{\xi}) = \boldsymbol{\xi}^{-\nu}$ (2.12), holomorphic near $\boldsymbol{\xi}^{(0)}$ with the principal branch, and $\partial^{\mathbf{w}}\boldsymbol{\xi}^{-\nu} = (-1)^{|\mathbf{w}|}\prod_e (\nu_e)_{w_e}\,\boldsymbol{\xi}^{-\nu-\mathbf{w}}$ with $(\nu_e)_{w_e} = \nu_e(\nu_e + 1)\cdots(\nu_e + w_e - 1)$.

*(2) Taylor coefficients.* Each monomial of $\boldsymbol{\Delta}(\rho)^{\mathbf{w}}$ has total degree at least $|\mathbf{w}|$ in $\rho$, every $l \neq 0$ having $|l| \geq 1$. So Taylor's formula for $\mathsf{Z}_{\nu;t}$ at $\boldsymbol{\xi}^{(0)}$ to the order $|q|$, whose remainder is $O(|\boldsymbol{\Delta}|^{|q|+1}) = O(|\rho|^{|q|+1})$, gives

$$c_q^\nu = \sum_{|\mathbf{w}|\leq|q|} \partial^{\mathbf{w}}\mathsf{Z}_{\nu;t}(\boldsymbol{\xi}^{(0)})\,[\rho^q]\,\frac{\boldsymbol{\Delta}(\rho)^{\mathbf{w}}}{\mathbf{w}!}.$$

By the multinomial theorem, $\Delta_k^{w_k}/w_k! = \sum \prod_{l\neq 0} (\rho^l \xi_k^{(l)})^{n_{l,k}}/n_{l,k}!$, the sum over the $(n_{l,k})_l$ in $\mathbb{N}$ with $\sum_l n_{l,k} = w_k$; multiplying over $k$ and taking the coefficient of $\rho^q$ leaves the arrays $\mathbf{n}$ with $\sum_{l,k} l\,n_{l,k} = q$ and $\mathbf{w} = (\sum_l n_{l,k})_k$, $|\mathbf{w}| = |\mathbf{n}|$. With (1) this is (i). At one unit exponent, $t = (1)$, $\boldsymbol{\xi} = L$ and the index $\lambda$, the product in (i) is $\Gamma_1\begin{bmatrix}\lambda + |\mathbf{n}|\\ \lambda\end{bmatrix}(L^{(0)})^{-\lambda-|\mathbf{n}|}\prod_l (L^{(l)})^{n_l}/n_l!$, which is the formula of the reference.

*(3) The coefficients as variables.* The function $(\rho, (\xi_k^{(l)})) \mapsto \mathsf{Z}_{\nu;t}(\boldsymbol{\xi}(\rho))$ is infinitely differentiable, so derivatives in the coefficients commute with taking Taylor coefficients in $\rho$. Since $\partial \xi_k(\rho)/\partial \xi_k^{(l)} = \rho^l$, (1) with $\mathbf{w} = \mathbf{e}_k$ gives

$$\partial_{\xi_k^{(l)}} \mathsf{Z}_{\nu;t}(\boldsymbol{\xi}(\rho)) = -\rho^l \prod_e \Gamma_1\begin{bmatrix} \nu_e + t_{k,e} \\ \nu_e \end{bmatrix} \mathsf{Z}_{\nu+t_k;t}(\boldsymbol{\xi}(\rho)),$$

and the coefficient of $\rho^q$ is $-\prod_e \Gamma_1\begin{bmatrix} \nu_e + t_{k,e} \\ \nu_e \end{bmatrix} c_{q-l}^{\nu+t_k}$, which is (iii). At $t = (1)$ the factor is $\Gamma_1\begin{bmatrix} \lambda+1 \\ \lambda \end{bmatrix} = \lambda$, the formula of the reference. □

## 50.3 Chapter 3: the Breta

**Proposition 3.4 (The Breta on the orthant).** *As stated, for any data.*

*Proof.* By 3.1 and 2.12, with the dressed coefficients $\eta_k = u^{\pi_k}\xi_k$ and the bunch scales $\rho = (\rho_Q)_{Q\in\mathcal{P}}$,

$$\mathsf{Z}_{\boldsymbol{\nu};\ell}(\boldsymbol{\eta}) = \frac{1}{\prod_Q \Gamma(\nu_Q)} \int_{\mathbb{R}^{\mathcal{P}}_{>0}} \rho^{\boldsymbol{\nu}-\mathbf{1}} \exp\Big( -\sum_k \xi_k\, u^{\pi_k} \rho^{\ell_k} \Big)\, d\rho.$$

On each bunch, 2.7 with $x_Q = \rho_Q u_Q$ gives $x_Q^{\mathbf{m}|_Q - \mathbf{1}} dx_Q = \rho_Q^{m_Q-1} u_Q^{\mathbf{m}|_Q-\mathbf{1}} d\rho_Q\, d\omega_Q(u_Q)$, hence

$$u_Q^{\mathbf{m}|_Q-\mathbf{1}} \rho_Q^{\nu_Q-1}\, d\rho_Q\, d\omega_Q(u_Q) = x_Q^{\mathbf{m}|_Q-\mathbf{1}} |x_Q|^{\nu_Q-m_Q}\, dx_Q,$$

and each dressed term is

$$\xi_k\, u^{\pi_k}\rho^{\ell_k} = \xi_k \prod_Q \Big(\frac{x_Q}{|x_Q|}\Big)^{\pi_k|_Q} |x_Q|^{\ell_{k,Q}} = \xi_k\, x^{\pi_k} \prod_Q |x_Q|^{\ell_{k,Q}-\langle \pi_k, \mathbf{1}_Q\rangle}.$$

So, under the bijection $x \leftrightarrow (\rho_Q, u_Q)_Q$ of 2.7, the Breta with its Zamma written as the scale integral is the integral of the right side over $\mathbb{R}^{\mathcal{E}}_{>0}$. The moduli of the two integrands are the same functions with $\mathbf{m}$ and $\boldsymbol{\nu}$ replaced by their real parts, so, as in the reference, proof of 3.4, Tonelli gives that either side converges when the other does, and Fubini gives the equality. For homogeneous data the inner powers of the bunch totals are absent, and at $\nu_Q = m_Q$ so are the outer ones: the right side is then $\mathsf{G}_{\mathbf{m};t}(\boldsymbol{\xi})/\prod_Q \Gamma(m_Q)$, which is 3.2. □

**Theorem 3.5 (Relations of the Breta).** *As stated: (v) for any terms; (i) to (iv) are those of the reference.*

*Proof.* The proofs of (i) to (iv) in the reference, proof of 3.5, use no homogeneity. For (v), let $e \neq f$ be scales of the bunch $Q$, and in this proof write, with the products over the bunches $Q'$ of $\mathcal{P}$,

$$F(x) = x^{\mathbf{m}-\mathbf{1}} \prod_{Q'} |x_{Q'}|^{\nu_{Q'}-m_{Q'}} \exp\Big( -\sum_k \xi_k \varphi_k(x)\Big), \qquad \varphi_k(x) = x^{\pi_k} \prod_{Q'} |x_{Q'}|^{\ell_{k,Q'}-\langle \pi_k, \mathbf{1}_{Q'}\rangle},$$

the integrand of 3.4 at $(\mathbf{m}, \boldsymbol{\nu})$ without the factor $1/\prod_{Q'} \Gamma(\nu_{Q'})$.

*(1) Derivatives.* For $e \in Q$, $x_e \partial_{x_e} |x_Q| = x_e$ and $x_e \partial_{x_e} |x_{Q'}| = 0$ for $Q' \neq Q$, so

$$x_e \partial_{x_e} F = F\Big[(m_e - 1) + (\nu_Q - m_Q)\frac{x_e}{|x_Q|} - \sum_k \xi_k \varphi_k \Big(\pi_{k,e} + \big(\ell_{k,Q} - \langle \pi_k, \mathbf{1}_Q \rangle\big)\frac{x_e}{|x_Q|}\Big)\Big],$$

and the same holds with $f$ in place of $e$.

*(2) The integrands.* By 3.4: $F$ is the integrand of $\prod_{Q'} \Gamma(\nu_{Q'})\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m},\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi})$; $F\, x_e/|x_Q|$ is that of $\prod_{Q'} \Gamma(\nu_{Q'})\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\mathbf{e}_e,\boldsymbol{\nu};(\pi,\ell)}(\boldsymbol{\xi})$, the power of $|x_Q|$ being $\nu_Q - (m_Q + 1)$; $F\varphi_k$ is that of $\prod_{Q'} \Gamma(\nu_{Q'} + \ell_{k,Q'})\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})$, the power of each $|x_{Q'}|$ being $(\nu_{Q'} + \ell_{k,Q'}) - (m_{Q'} + \langle \pi_k, \mathbf{1}_{Q'} \rangle)$; and $F\varphi_k\, x_e/|x_Q|$ is that of $\prod_{Q'} \Gamma(\nu_{Q'} + \ell_{k,Q'})\, \mathsf{B}^{\mathcal{P}}_{\mathbf{m}+\pi_k+\mathbf{e}_e,\boldsymbol{\nu}+\ell_k;(\pi,\ell)}(\boldsymbol{\xi})$. Here $\prod_{Q'} \Gamma(\nu_{Q'} + \ell_{k,Q'}) = g_k \prod_{Q'} \Gamma(\nu_{Q'})$. These Bretas, and those with $f$ in place of $e$, are the Bretas of (v), which converge by hypothesis; so all these functions are integrable on $\mathbb{R}^{\mathcal{E}}_{>0}$.

*(3) No boundary terms.* By (1) and (2), $\partial_{x_e}(x_e F) = F + x_e \partial_{x_e} F$ is integrable. By Fubini, for almost every value $x'$ of the coordinates other than $x_e$ the function $G(s) = s\, F(x', s)$ of $s = x_e > 0$ has $G'$ and $G(s)/s$ integrable on $(0, \infty)$. Then $G(s) = G(1) + \int_1^s G'$ has limits at 0 and at $\infty$, and both are 0: a nonzero limit $L$ at one end would make $|G(s)/s| \geq |L|/(2s)$ near that end, which is not integrable there. So $\int_0^\infty G'(s)\, ds = 0$, and by Fubini $\int \partial_{x_e}(x_e F)\, dx = 0$; likewise for $f$. Hence

$$\int_{\mathbb{R}^{\mathcal{E}}_{>0}} \big(x_e \partial_{x_e} - x_f \partial_{x_f}\big) F\, dx = \int \big[\partial_{x_e}(x_e F) - F\big] dx - \int \big[\partial_{x_f}(x_f F) - F\big] dx = 0.$$

*(4) Collecting.* By (1), $(x_e \partial_{x_e} - x_f \partial_{x_f})F$ is $F$ times

$$(m_e - m_f) + (\nu_Q - m_Q)\frac{x_e - x_f}{|x_Q|} - \sum_k \xi_k \varphi_k \Big((\pi_{k,e} - \pi_{k,f}) + \big(\ell_{k,Q} - \langle \pi_k, \mathbf{1}_Q \rangle\big)\frac{x_e - x_f}{|x_Q|}\Big).$$

Integrating, using (2) and dividing by $\prod_{Q'} \Gamma(\nu_{Q'})$ gives (v). For homogeneous data $\ell_{k,Q} = \langle \pi_k, \mathbf{1}_Q \rangle$ and the second line of the right side of (v) is absent.

*(5) The closing sentences.* A term $k$ raises the shape total of a bunch $Q'$ by $\langle \pi_k, \mathbf{1}_{Q'} \rangle$ and its scale index by $\ell_{k,Q'}$, which are equal for homogeneous data; so the ladder, the scale exchange and the shifts $(\mathbf{m} + \pi_k, \boldsymbol{\nu} + \ell_k)$ of (v) keep the condition $\nu_Q = m_Q$, and in (v) the shifts $\mathbf{m} + \mathbf{e}_e$, $\mathbf{m} + \mathbf{e}_f$ of the shape index alone carry the factor $\nu_Q - m_Q$. The shape split and (iv) shift the shape index alone. □

*Remark.* Relation (v) was checked in exact arithmetic on one bunch of two scales with one term whose level is not the degree of its shape exponent, where the Breta has a closed form, for shape exponents $(1, 0)$, $(\frac{3}{2}, -\frac{1}{3})$ and symbolic (`sweep/s0107/breta_v.py`).

**Proposition 3.6 (Letter presentation).** *As stated, for a partition of the scales $B$ into any number of bunches.*

*Proof. (1) The presentation.* The bunches of $\mathcal{P}$ are those of $\mathcal{Q}$, with the scales $\rho_Q$, and the singletons $\{v_i\}$, whose scales are written $v_i$ and whose shapes are the point $u_{v_i} = 1$ of $\Delta_{\{v_i\}}$, with $\omega_{\{v_i\}}$ the Dirac measure (1.1(iii)). The dressed coefficients of the columns are $g_j u^{\pi_j}$ and $C_{i,p} u^{\pi_{i,p}}$, and $u^{(\mathbf{m},\mathbf{s})-\mathbf{1}} = u^{\mathbf{m}-\mathbf{1}}$. By 2.1, the Gramma on the bunch scales with the configuration $\widetilde{\ell}$ and these coefficients is

$$\int \rho^{\boldsymbol{\nu}-\mathbf{1}} \prod_i v_i^{s_i - 1} \exp\Big(-\sum_j g_j u^{\pi_j} \rho^{\lambda_j} - \sum_{i,p} C_{i,p} u^{\pi_{i,p}} \rho^{\ell_{i,p}} v_i\Big)\, d\rho\, dv,$$

the level of the pattern $(i,p)$ on $\{v_i\}$ being 1. For $\operatorname{Re} s_i > 0$, integrating $v_i$ gives $\Gamma(s_i)\,P_i(\rho,u)^{-s_i}$ (2.6 on one scale, $P_i > 0$). Dividing by $\prod_Q \Gamma(\nu_Q)\prod_i \Gamma(s_i)$ gives the Zamma (2.12), and integrating against $u^{\mathbf{m}-\mathbf{1}}d\omega(u)$ over $\prod_Q \Delta_Q$ gives $1/\prod_Q \Gamma(\nu_Q)$ times the left side of the display; Tonelli on the moduli and Fubini justify the order of the integrals, either side converging when the other does.

*(2) The relations.* They are 3.5(ii),(iii) for the terms $(\widetilde{\pi},\widetilde{\ell})$. For the scale exchange at $\{v_i\}$ the columns of nonzero level there are the patterns $(i,p)$, of level 1, with the factor $\Gamma_1{}_1\begin{bmatrix} s_i+1 \\ s_i+1\end{bmatrix} = 1$ at $\{v_i\}$; the other singletons give $\Gamma_1{}_1\begin{bmatrix} s_{i'} \\ s_{i'}\end{bmatrix} = 1$ and each $Q \in \mathcal{Q}$ gives $\Gamma_1{}_1\begin{bmatrix} \nu_Q+\ell_{i,p,Q} \\ \nu_Q\end{bmatrix}$: this is the letter split. In the ladders the factor $g_k$ of 3.5 is $\prod_Q \Gamma_1{}_1\begin{bmatrix} \nu_Q+\lambda_{j,Q} \\ \nu_Q\end{bmatrix}$ for a kernel column and $s_i\prod_Q \Gamma_1{}_1\begin{bmatrix} \nu_Q+\ell_{i,p,Q} \\ \nu_Q\end{bmatrix}$ for a pattern, by $\Gamma_1{}_1\begin{bmatrix} s_i+1 \\ s_i\end{bmatrix} = s_i$. The scale exchange on $Q \in \mathcal{Q}$ is 3.5(iii) multiplied by $\nu_Q$, with $\nu_Q\Gamma_1{}_1\begin{bmatrix} \nu_Q+\lambda \\ \nu_Q+1\end{bmatrix} = \Gamma_1{}_1\begin{bmatrix} \nu_Q+\lambda \\ \nu_Q\end{bmatrix}$ (1.3(iii)), the singletons contributing the factors 1 for a kernel column and $s_i$ for a pattern. □

**Proposition 3.7 (Letter data on the orthant).** *As stated, for any levels and any number of bunches.*

*Proof.* In the left side of the display of 3.6 apply 2.7 on each bunch $Q \in \mathcal{Q}$, $\alpha_Q = \rho_Q u_Q$: $u_Q^{\mathbf{m}|_Q-\mathbf{1}}\rho_Q^{\nu_Q-1}d\rho_Q\,d\omega_Q = \alpha_Q^{\mathbf{m}|_Q-\mathbf{1}}|\alpha_Q|^{\nu_Q-m_Q}d\alpha_Q$, and for $\lambda \in \mathbb{R}^{\mathcal{Q}}$, $\pi \in \mathbb{R}^B$, $\rho^\lambda u^\pi = \alpha^\pi\prod_Q|\alpha_Q|^{\lambda_Q-\langle\pi,\mathbf{1}_Q\rangle}$. So each kernel term becomes $g_j\alpha^{\pi_j}\prod_Q|\alpha_Q|^{\lambda_{j,Q}-\langle\pi_j,\mathbf{1}_Q\rangle}$ and each letter $P_i(\alpha)$. Tonelli on the moduli gives the equivalence of convergence and Fubini the equality, and 3.6 identifies the left side with $\prod_Q\Gamma(\nu_Q)$ times the Breta. For homogeneous letter data every exponent $\lambda_Q-\langle\pi,\mathbf{1}_Q\rangle$ is 0, and at $\nu_Q = m_Q$ so is $\nu_Q - m_Q$. □

**Theorem 3.13 (The Shruffle).** *As stated, for any number of bunches.*

*Proof.* Apply 3.12 to $A = A_1\sqcup\cdots\sqcup A_q$ with $J = \{1,\dots,q\}$: $u|_{A_i} = w_i v^{(i)}$ with $w \in \Delta_J$, $v^{(i)} \in \Delta_{A_i}$, and

$$u^{\mathbf{m}-\mathbf{1}}d\omega_A = \prod_i w_i^{m_{A_i}-1}\,d\omega_J(w)\prod_i (v^{(i)})^{\mathbf{m}|_{A_i}-\mathbf{1}}d\omega_{A_i}(v^{(i)}), \qquad \bigotimes_i f_i(u) = \prod_i f_i(v^{(i)}).$$

Scaling by $w_i > 0$ keeps the order of the coordinates of $u|_{A_i}$, which is that of $v^{(i)}$. Up to the null set where two coordinates of $u$ are equal, the points $u$ whose restrictions follow every $\varsigma_i$ form the disjoint union of the sets $\Delta_A^\varsigma$ over the orders $\varsigma$ of $A$ that restrict to every $\varsigma_i$, and they are the images of $\Delta_J\times\prod_i\Delta_{A_i}^{\varsigma_i}$. The integrals converge absolutely, so by Fubini

$$\sum_\varsigma I_A^\varsigma\Big(\bigotimes_i f_i;\mathbf{m}\Big) = \int_{\Delta_J}\prod_i w_i^{m_{A_i}-1}\,d\omega_J(w)\;\prod_i I_{A_i}^{\varsigma_i}(f_i;\mathbf{m}|_{A_i})$$
$$= \Gamma_q{}_1\begin{bmatrix} m_{A_1},\ \dots,\ m_{A_q} \\ m_A\end{bmatrix}\prod_i I_{A_i}^{\varsigma_i}(f_i;\mathbf{m}|_{A_i}),$$

by 3.9 on $J$, where $\operatorname{Re} m_{A_i} > 0$. Dividing by the stacked Gamma, whose reciprocal is $\Gamma_1{}_q\begin{bmatrix} m_A \\ m_{A_1},\ \dots,\ m_{A_q}\end{bmatrix}$, gives the statement; at $q = 2$ it is the factor $\Gamma_1{}_2\begin{bmatrix} m_A+m_B \\ m_A,\ m_B\end{bmatrix}$ of the reference. □

**Theorem 3.15 (Relations of the Beta port).** *As stated: (iii) for linear letters of either slope and complex exponents, and (v) at every pair of exponents; (i), (ii) and (iv) are those of the reference.*

*Proof. (iii)* On $[0,1]$ every letter $L_i = c_i + d_i x$ is positive, since it is affine with the positive values $c_i$ and $c_i + d_i$ at the ends, so $L_i^{-w_i}$ is infinitely differentiable there for every $w_i \in \mathbb{C}$, with $\partial_x^v L_i^{-w_i} = (-1)^v (w_i)_v d_i^v L_i^{-w_i - v}$, $d_i$ real. Leibniz's rule gives

$$\partial_x^q \prod_i L_i^{-w_i} = \sum_{|v|=q} \frac{q!}{\prod_i v_i!} \prod_i (-1)^{v_i} (w_i)_{v_i} d_i^{v_i} L_i^{-w_i - v_i},$$

which is the first display. Then (ii), with $g = \prod_i L_i^{-w_i}$ and $\operatorname{Re} a, \operatorname{Re} b > 0$, gives the second display, by $x^{a+q-1}(1-x)^{b+q-1} = x^{a-1}(1-x)^{b-1} \prod_i (x(1-x))^{v_i}$ for $|v| = q$. Let $a, b > 0$ and $w_i \geq 0$. Each integrand on the right is $x^{a-1}(1-x)^{b-1} \prod_i L_i^{-w_i} > 0$ times $\prod_i (x(1-x) d_i / L_i)^{v_i}$, which is nonnegative if every $d_i \geq 0$, and is $(-1)^{|v|} = (-1)^q$ times a nonnegative function if every $d_i \leq 0$; the coefficients $\prod_i (w_i)_{v_i} / v_i!$ are nonnegative. This gives the two signs. If $x(1-x)|d_i|/L_i \leq \varepsilon$ on $[0,1]$ for every $i$, each integrand has modulus at most $\varepsilon^q x^{a-1}(1-x)^{b-1} \prod_i L_i^{-w_i}$, and

$$\sum_{|v|=q} \prod_i \frac{(w_i)_{v_i}}{v_i!} = \frac{(|\mathbf{w}|)_q}{q!},$$

the coefficient of $y^q$ on both sides of $\prod_i (1-y)^{-w_i} = (1-y)^{-|\mathbf{w}|}$, $|y| < 1$, by $(1-y)^{-w} = \sum_v \frac{(w)_v}{v!} y^v$. This gives the bound.

*(v)* Fix $q \geq 1$, write $s = a + b$, and first let $\operatorname{Re} a, \operatorname{Re} b > 0$.

(1) Coefficients. Expanding $x^k (1-x)^{q-k}$ in the second form of 3.14,

$$[x^j]\, \Pi_q^{a,b} = (-1)^{q+j} \sum_k \binom{a+q-1}{q-k} \binom{q-k}{q-j} \binom{b+q-1}{k}.$$

By $\binom{N}{M}\binom{M}{L} = \binom{N}{L}\binom{N-L}{M-L}$, a polynomial identity in $N$, at $N = a+q-1$, $M = q-k$, $L = q-j$, this is $(-1)^{q+j}\binom{a+q-1}{q-j} \sum_k \binom{a+j-1}{j-k}\binom{b+q-1}{k}$, and $\sum_k \binom{A}{j-k}\binom{B}{k} = \binom{A+B}{j}$, a polynomial identity in $A$ and $B$ that holds at nonnegative integers by counting the $j$-subsets of a disjoint union, gives $\binom{s+q+j-2}{j}$: the coefficient formula. At $j = q$, $\lambda_q := [x^q]\Pi_q^{a,b} = \binom{s+2q-2}{q} = (s+q-1)_q / q!$, which is nonzero since $\operatorname{Re} s > 0$. Also

$$\frac{[x^{q-1}]\Pi_q^{a,b}}{\lambda_q} = -\frac{q(a+q-1)}{s+2q-2}, \qquad \frac{[x^q]\Pi_{q+1}^{a,b}}{\lambda_{q+1}} = -\frac{(q+1)(a+q)}{s+2q}.$$

(2) Pairing. Put $\langle g, h \rangle_\circ = \int_0^1 x^{a-1}(1-x)^{b-1} g\, h\, dx$ for polynomials $g, h$. By (i) at $c = a + j$, $\langle \Pi_k^{a,b}, x^j \rangle_\circ = \binom{j}{k} \Gamma \begin{bmatrix} a+j,\ b+k \\ s+j+k \end{bmatrix}$, which is 0 for $j < k$ and $\beta_k := \Gamma \begin{bmatrix} a+k,\ b+k \\ s+2k \end{bmatrix} \neq 0$ for $j = k$.

(3) Span. Let $g$ be a polynomial of degree at most $q+1$ with $\langle g, x^j \rangle_\circ = 0$ for $j \leq q-2$. Since $\deg \Pi_k^{a,b} = k$ by (1), $g = \sum_{k \leq q+1} \gamma_k \Pi_k^{a,b}$; pairing with $x^0, x^1, \ldots, x^{q-2}$ in turn, by (2) the pairing with $x^j$ involves only $\gamma_0, \ldots, \gamma_j$, with $\gamma_j$ multiplied by $\beta_j \neq 0$, so $\gamma_0 = \cdots = \gamma_{q-2} = 0$ and $g \in \operatorname{span}(\Pi_{q-1}^{a,b}, \Pi_q^{a,b}, \Pi_{q+1}^{a,b})$.

(4) Multiplication by $x$. $\langle x\Pi_q^{a,b}, x^j \rangle_\circ = \langle \Pi_q^{a,b}, x^{j+1} \rangle_\circ = 0$ for $j \leq q-2$, so by (3) $x\Pi_q^{a,b} = A\Pi_{q+1}^{a,b} + B\Pi_q^{a,b} + C\Pi_{q-1}^{a,b}$. The coefficients of $x^{q+1}$ give $A = \lambda_q / \lambda_{q+1} = \frac{(q+1)(s+q-1)}{(s+2q)(s+2q-1)}$. Pairing

with $x^{q-1}$ gives $\beta_q = C\beta_{q-1}$, so $C = \frac{(a+q-1)(b+q-1)}{(s+2q-1)(s+2q-2)}$. The coefficients of $x^q$ give $[x^{q-1}]\Pi_q^{a,b} = A\,[x^q]\Pi_{q+1}^{a,b} + B\lambda_q$, so by (1) $B = \frac{(q+1)(a+q)}{s+2q} - \frac{q(a+q-1)}{s+2q-2}$.

(5) The operator $D = x(1-x)\partial_x$. For a polynomial $g$, integrating by parts, the boundary terms $x^{a+j}(1-x)^b g$ vanishing at both ends,

$$\langle Dg, x^j\rangle_\circ = \int_0^1 x^{a+j}(1-x)^b g'\,dx = -(a+j)\langle g, x^j\rangle_\circ + (s+j)\langle g, x^{j+1}\rangle_\circ,$$

which is 0 at $g = \Pi_q^{a,b}$ for $j \le q-2$. By (3), $D\Pi_q^{a,b} \in \operatorname{span}(\Pi_{q-1}^{a,b}, \Pi_q^{a,b}, \Pi_{q+1}^{a,b})$, and eliminating $\Pi_{q+1}^{a,b}$ by (4), $A \neq 0$, $D\Pi_q^{a,b} = (e\,x + e_0)\Pi_q^{a,b} + e_1\Pi_{q-1}^{a,b}$. Since $Dx^k = kx^k - kx^{k+1}$, the coefficients of $x^{q+1}$ give $e = -q$. Pairing with $x^{q-1}$: the left side is $(s+q-1)\beta_q$ and the right side $-q\beta_q + e_1\beta_{q-1}$, so $e_1 = (s+2q-1)\beta_q/\beta_{q-1} = \frac{(a+q-1)(b+q-1)}{s+2q-2}$. The coefficients of $x^q$ give $q\lambda_q - (q-1)[x^{q-1}]\Pi_q^{a,b} = -q[x^{q-1}]\Pi_q^{a,b} + e_0\lambda_q$, so $e_0 = q + [x^{q-1}]\Pi_q^{a,b}/\lambda_q = \frac{q(b+q-1)}{s+2q-2}$.

(6) Continuation. The three displays compare polynomials in $x$ whose coefficients are rational functions of $(a,b)$, the coefficients of the Beta ports being polynomials in $(a,b)$. They agree on the open set $\operatorname{Re} a, \operatorname{Re} b > 0$, hence wherever the coefficients are finite, which is the condition $a+b+2q-i \neq 0$, $i = 0,1,2$; the coefficient formula is a polynomial identity and holds for all $a, b$ and every $q \in \mathbb{N}$. □

*Remark.* At $a = b = n+1$ the middle coefficient of the three-term relation is $\frac{1}{2}$, and the relations become those of the reference: $(1-2x)\Pi_r^{n+1,n+1} = -2A\,\Pi_{r+1}^{n+1,n+1} - 2C\,\Pi_{r-1}^{n+1,n+1}$ and $x(1-x)\partial_x\Pi_r^{n+1,n+1} = \frac{r}{2}(1-2x)\Pi_r^{n+1,n+1} + \frac{r+n}{2}\Pi_{r-1}^{n+1,n+1}$, with the leading coefficient $\binom{2n+2r}{r}$, and at $a = b = 1$ the coefficient formula gives $(-1)^r\Pi_r^{1,1} = \sum_k (-1)^k\binom{r}{k}\binom{r+k}{k}x^k$. The three displays and the coefficient formula were checked in exact arithmetic with $a$ and $b$ symbolic for $q \le 9$, together with these diagonal forms (`sweep/s0107/betaport.py`).

## 50.4 Chapter 4: fibre integrals and closing

**Theorem 4.3 (The native gauge of a configuration).** *As stated: (iii) with a fixed offset $\nu'$ of the index; (i) and (ii) are those of the reference.*

*Proof.* By (i), $\Phi_{\boldsymbol{\xi}}$ is smooth and strictly convex with its minimum at $y^\star$, where its Hessian $H^\star = \sum_k w_k^\star t_k t_k^{\mathsf{T}}$ is positive definite.

*(1) The integral.* In the logarithmic form of 2.1,

$$\mathsf{G}_{\lambda\nu+\nu';t}(\lambda\boldsymbol{\xi}) = \int_{\mathbb{R}^{\mathcal{E}}} e^{\langle\nu',y\rangle}\,e^{-\lambda\Phi_{\boldsymbol{\xi}}(y)}\,dy,$$

which converges when $\lambda\nu+\nu' \in \mathcal{T}_t$ (2.2). Since $\operatorname{int}\operatorname{cone}(t)$ is an open cone containing $\nu$, it contains $\nu + \lambda^{-1}\operatorname{Re}\nu'$ for $\lambda$ large, so the integral converges for $\lambda \ge \lambda_0$, some $\lambda_0 > 0$.

*(2) Away from $y^\star$.* Fix $\delta > 0$ and put $\mu_\delta = \min_{|y-y^\star|=\delta}\big(\Phi_{\boldsymbol{\xi}}(y) - \Phi_{\boldsymbol{\xi}}(y^\star)\big) > 0$. For a unit vector $d$ the function $\sigma \mapsto \Phi_{\boldsymbol{\xi}}(y^\star+\sigma d) - \Phi_{\boldsymbol{\xi}}(y^\star)$ is convex and vanishes at $\sigma = 0$, so for $\sigma \ge \delta$ it is at least $\sigma\mu_\delta/\delta$. With $|e^{\langle\nu',y\rangle}| \le e^{|\operatorname{Re}\nu'|\,|y^\star|}e^{|\operatorname{Re}\nu'|\,|y-y^\star|}$,

$$\Big|\int_{|y-y^\star|\ge\delta} e^{\langle\nu',y\rangle}e^{-\lambda(\Phi_{\boldsymbol{\xi}}(y)-\Phi_{\boldsymbol{\xi}}(y^\star))}\,dy\Big| \le e^{|\operatorname{Re}\nu'|\,|y^\star|}\int_{|u|\ge\delta} e^{-(\lambda\mu_\delta/\delta - |\operatorname{Re}\nu'|)|u|}\,du,$$

which, once $\lambda\mu_\delta/\delta \ge 2|\operatorname{Re}\nu'|$ and $\lambda\mu_\delta \ge 2\delta$, is at most a constant times $e^{-\lambda\mu_\delta/4}$: on $|u| \ge \delta$, $e^{-\Lambda|u|} \le e^{-\Lambda\delta/2}e^{-\Lambda|u|/2}$ for $\Lambda = \lambda\mu_\delta/\delta - |\operatorname{Re}\nu'| \ge \lambda\mu_\delta/(2\delta) \ge 1$. So it is $O(\lambda^{-|\mathcal{E}|/2-1})$.

*(3) Near $y^\star$.* Laplace's method at the nondegenerate minimum $y^\star$, with the smooth amplitude $e^{\langle \nu', y\rangle}$, gives

$$\int_{|y-y^\star|<\delta} e^{\langle \nu', y\rangle} e^{-\lambda(\Phi_{\boldsymbol{\xi}}(y)-\Phi_{\boldsymbol{\xi}}(y^\star))}\, dy = \Big(\frac{2\pi}{\lambda}\Big)^{|\mathcal{E}|/2} \det(H^\star)^{-1/2}\Big(e^{\langle \nu', y^\star\rangle} + O(\lambda^{-1})\Big),$$

as in the reference, proof of 4.3, where the amplitude is 1: the amplitude enters the leading term through its value at $y^\star$, and its derivatives only the term $O(\lambda^{-1})$.

Adding (2) and (3), multiplying by $e^{-\lambda\Phi_{\boldsymbol{\xi}}(y^\star)}$ and factoring out $e^{\langle \nu', y^\star\rangle} \neq 0$ gives (iii). □

**Corollary 4.7 (Four Gammas on a line).** *As stated: the display is 4.6 on two coefficients.*

*Proof. (1) A straight contour.* Let first $\max(-\operatorname{Re} a, -\operatorname{Re} b) < \min(\operatorname{Re} c, \operatorname{Re} d)$, and take the line $\operatorname{Re} z = \gamma$ with $\max(-\operatorname{Re} a, -\operatorname{Re} b) < \gamma < \min(\operatorname{Re} c, \operatorname{Re} d)$; it separates the two sets of poles. Apply 4.6 with $k = 2$, the exponents $a + c$ and $b + d$, and $(a + d,\, b + c)$, of total $a + b + c + d$, in the place of the merged shares. In the shares $(\alpha_1, \alpha_2, \beta_1, \beta_2)$ the fibre has the pins $\alpha_1 + \alpha_2 = a + c$, $\alpha_1 + \beta_1 = a + d$, $\alpha_2 + \beta_2 = b + c$, and it is parametrized by $\alpha = (a + z,\, c - z)$, $\beta = (d - z,\, b + z)$. The matrix $P$ of the pins has the rows $(1, 1, 0, 0)$, $(1, 0, 1, 0)$, $(0, 1, 0, 1)$, so $\det PP^{\mathsf{T}} = 4$, and its kernel is spanned by $(1, -1, -1, 1)$, of length 2. By 4.1, with the admissible real point on the line $\operatorname{Re} z = \gamma$, where all four shares have positive real parts, the fibre integral of a function $G$ is $(2\pi)^{-1} \cdot 4^{-1/2} \cdot 2 \int G\, d\tau = \frac{1}{2\pi i} \int_{\gamma + i\mathbb{R}} G\, dz$, $z = \gamma + i\tau$. The integrand of 4.6 is

$$\Gamma_{\,1}^{\,2}\begin{bmatrix} a+z,\ c-z \\ a+c \end{bmatrix} \Gamma_{\,1}^{\,2}\begin{bmatrix} d-z,\ b+z \\ b+d \end{bmatrix} = \frac{\Gamma(a+z)\Gamma(b+z)\Gamma(c-z)\Gamma(d-z)}{\Gamma(a+c)\Gamma(b+d)},$$

and its right side is $\Gamma_{\,1}^{\,2}\begin{bmatrix} a+d,\ b+c \\ a+b+c+d \end{bmatrix}$. Multiplying by $\Gamma(a + c)\Gamma(b + d)$ (1.3(i)) gives the display.

*(2) Continuation.* A contour separating the poles exists exactly when no pole $-a-n$ or $-b-n$ of the first two factors coincides with a pole $c + n'$ or $d + n'$ of the last two, $n, n' \in \mathbb{N}$, that is on the set $U$ of $(a, b, c, d) \in \mathbb{C}^4$ with $a+c, a+d, b+c, b+d \notin -\mathbb{N}$. Near a point of $U$ one fixed contour separates the poles, and by 1.3(iii) and 2.11 on one scale the four Gammas decay exponentially along it, locally uniformly in $(a, b, c, d)$; so the left side is holomorphic on $U$, and by Cauchy's theorem it does not depend on the separating contour. The right side is holomorphic on $U$. The set $U$ is connected, being the complement in $\mathbb{C}^4$ of countably many complex hyperplanes, and contains the open set of (1), so the identity holds on $U$. □

## 50.5 Chapter 5: the Trailmoment

**Theorem 5.7 (The gauge of the coefficient function).** *As stated: the last sentence at the unit configuration on any number of scales.*

*Proof.* The first two sentences are those of the reference (the reference, proof of 5.7). At the unit configuration $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\mathbf{c}) = \prod_e c_e^{-\nu_e} = \mathbf{c}^{-\nu}$ for every $\nu \in \mathbb{C}^{\mathcal{E}}$ (2.12), so by 5.1 $(\mathsf{T}_{\nu;I_{\mathcal{E}}}[\mathbf{c}]\,\phi)(Y') = \int_{Y'} \mathbf{c}^{-\nu}\, d\phi$, either side converging when the other does. □

**Theorem 5.8 (Exchange and ladder of the Trailmoment).** *As stated: (iii) at the unit configuration on any number of scales; (i) and (ii) are those of the reference.*

*Proof.* At $t = I_{\mathcal{E}}$ the column $t_k$ is $\mathbf{e}_k$, $k \in \mathcal{E}$, so in (i) at the scale $e$ only $k = e$ has $t_{k,e} \neq 0$, with $t_{e,e} = 1$ and the factor $\Gamma_1\begin{bmatrix}\nu_e+1\\ 1\end{bmatrix}\prod_{f\neq e}\Gamma_1\begin{bmatrix}\nu_f\\ \nu_f\end{bmatrix} = 1$. Directly, for every $\nu$, $\mathsf{Z}_{\nu;I_{\mathcal{E}}}(\mathbf{c}) = \mathbf{c}^{-\nu} = c_e\,\mathbf{c}^{-\nu-\mathbf{e}_e} = c_e\,\mathsf{Z}_{\nu+\mathbf{e}_e;I_{\mathcal{E}}}(\mathbf{c})$ (2.12); integrating against $\phi$ over $Y'$ gives the first identity.

For the derivative, let $D \subseteq \mathbb{C}^{\mathcal{E}}$ be the set of indices where the Trailmoment converges, $\int_{Y'}\mathbf{c}^{-\operatorname{Re}\nu}\,d|\phi| < \infty$, and $\nu_0$ an interior point of $D$. Take $\delta > 0$ such that $\nu_0 \pm \delta\mathbf{e}_e \in D$. For $\nu = \nu_0 + \zeta\mathbf{e}_e$ with $|\zeta| < \delta/2$, $|\mathbf{c}^{-\nu}\log c_e| = \mathbf{c}^{-\operatorname{Re}\nu_0}c_e^{-\operatorname{Re}\zeta}|\log c_e|$. For $\eta > 0$ and $y > 0$, $|\log y| \le \eta^{-1}(y^{\eta} + y^{-\eta})$; with $\eta = \delta/2$, and $y^{\theta} \le y^{\delta} + y^{-\delta}$ for $|\theta| \le \delta$,

$$\left|\mathbf{c}^{-\nu}\log c_e\right| \le \frac{4}{\delta}\Big(\mathbf{c}^{-(\operatorname{Re}\nu_0-\delta\mathbf{e}_e)} + \mathbf{c}^{-(\operatorname{Re}\nu_0+\delta\mathbf{e}_e)}\Big),$$

an integrable function independent of $\zeta$. So $\zeta \mapsto \int_{Y'}\mathbf{c}^{-\nu_0-\zeta\mathbf{e}_e}d\phi$ may be differentiated under the integral sign, and $\partial_\zeta\mathbf{c}^{-\nu} = -(\log c_e)\,\mathbf{c}^{-\nu}$ gives the second identity. □

**Proposition 5.14 (Laurent data of the Gamma at its poles).** *As stated: the residue and the Laurent coefficients of a product of any number of Gammas with a common pole.*

*Proof.* The quotient form of $(-1)^j\varepsilon\Gamma(-j+\varepsilon)$, the expansion of its logarithm and $(\log\Gamma)'(j+1) = -(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))$ are those of the reference (the reference, proof of 5.14). On $|\varepsilon| < 1$ each $\phi_i(\varepsilon) := (-1)^{j_i}\varepsilon\,\Gamma(-j_i+\varepsilon) = \Gamma(1+\varepsilon)\Gamma(1-\varepsilon)/\Gamma(1+j_i-\varepsilon)$ is holomorphic and nonzero, and it is the exponential of the expansion at $j_i$. Put $b = -j+\varepsilon$; then $b + k_i = -j_i + \varepsilon$ and $\Gamma(b+k_i) = (-1)^{j_i}\varepsilon^{-1}\phi_i(\varepsilon)$, so

$$h(b)\prod_{i=1}^{p}\Gamma(b+k_i) = (-1)^{j_1+\cdots+j_p}\,\varepsilon^{-p}\,h(-j+\varepsilon)\prod_{i=1}^{p}\phi_i(\varepsilon).$$

The residue at $\varepsilon = 0$ is $(-1)^{\sum_i j_i}$ times the coefficient of $\varepsilon^{p-1}$ in $h(-j+\varepsilon)\prod_i\phi_i(\varepsilon)$, and with $h(-j+\varepsilon) = \sum_l h^{(l)}(-j)\varepsilon^l/l!$ the Cauchy product gives the displayed sum. At $p = 2$ and $k_1 = k_2 = 0$, $\phi_1\phi_2 = \exp(2\log\phi_1) = (j!)^{-2}\big(1 - 2(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))\varepsilon + O(\varepsilon^2)\big)$, so the residue is $(j!)^{-2}\big(h'(-j) - 2(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j,\infty))h(-j)\big)$, which is the formula of the reference. The Laurent coefficients of $\prod_i\Gamma(b+k_i)$ at $-j$ are $(-1)^{\sum_i j_i}$ times the Taylor coefficients of $\prod_i\phi_i = \prod_i(j_i!)^{-1}\exp\big(\sum_i(\log\phi_i + \log j_i!)\big)$; the series $\sum_i(\log\phi_i + \log j_i!)$ has no constant term and its coefficients are rational multiples of the Trailmoments $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((j_i,\infty))$ and $(\mathsf{T}_{2k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, so the Taylor coefficients of its exponential are polynomials in these Trailmoments with rational coefficients. □

*Remark.* The residue formula was checked in exact arithmetic for $p = 2, 3, 4$ at several $j$ and $(k_i)$ (`sweep/s0107/resonance.py`).

**Proposition 5.22 (The joint Laplace transform of Dirichlet inputs).** *As stated, for finite sums of monomials with nonnegative real exponents.*

*Proof. (1) The series.* Every exponent $\mathbf{a}$ lies in $[0,\infty)^{\mathcal{E}}$, so $0 < u^{\mathbf{a}} \le 1$ on the source $\prod_Q\Delta_Q$ and $|P_i(u)| \le \|P_i\| := \sum_{\mathbf{a}}\big|[u^{\mathbf{a}}]P_i\big|$. Hence $\sum_{\mathbf{k}}\frac{|v|^{\mathbf{k}}}{\mathbf{k}!}\prod_i|P_i(U)|^{k_i} \le \exp\big(\sum_i|v_i|\,\|P_i\|\big)$, the exponential series of $\exp(-\sum_i v_iP_i(U))$ converges uniformly on the source for each $v \in \mathbb{C}^q$ and integrates term by term, and $\mathcal{F}(v) = \sum_{\mathbf{k}}\frac{(-v)^{\mathbf{k}}}{\mathbf{k}!}\mathbb{E}[\prod_i P_i(U)^{k_i}]$, a power series converging on $\mathbb{C}^q$, hence entire.

*(2) The moments.* Multiplying out, $\prod_i P_i^{k_i} = \sum_{\mathbf{a}}[u^{\mathbf{a}}]\big(\prod_i P_i^{k_i}\big)u^{\mathbf{a}}$, a finite sum with exponents in $[0,\infty)^{\mathcal{E}}$. The blocks of $U$ are independent, so $\mathbb{E}[U^{\mathbf{a}}] = \prod_Q\mathbb{E}_{U_Q\sim\mathrm{Dir}_{\mathbf{m}|_Q}}[U_Q^{\mathbf{a}|_Q}]$, and by 3.11 at the

exponents $\mathbf{a}|_Q$, whose real parts satisfy $m_e + a_e > 0$, together with 1.3(i),(ii),

$$\mathbb{E}_{U_Q \sim \mathrm{Dir}_{\mathbf{m}|_Q}}\big[U_Q^{\mathbf{a}|_Q}\big] = \frac{\Gamma_{1}^{|Q|}\begin{bmatrix}\mathbf{m}|_Q + \mathbf{a}|_Q \\ m_Q + a_Q\end{bmatrix}}{\Gamma_{1}^{|Q|}\begin{bmatrix}\mathbf{m}|_Q \\ m_Q\end{bmatrix}} = \prod_{e\in Q} \Gamma_1\begin{bmatrix}m_e + a_e\end{bmatrix}\ \Gamma_1\begin{bmatrix}m_Q \\ m_Q + a_Q\end{bmatrix}.$$

Summing gives $M(\mathbf{k})$; for integer exponents 3.11 gives the rising factorials.

*(3) The read.* Where every $P_i$ is positive on the interior of the source, which carries the law, 2.12 and 2.1 give $\mathsf{Z}_{\nu;t}(\mathbf{P}(U)) = \frac{1}{\prod_j \Gamma(\nu_j)} \int_{(0,\infty)^d} x^{\nu-\mathbf{1}} \exp\big(-\sum_i P_i(U) x^{t_i}\big) dx$ where this converges. By Tonelli on the moduli and Fubini, wherever the double integral converges, the expectation of the left side is the integral of $\mathbb{E}[\exp(-\sum_i P_i(U)x^{t_i})] = \mathcal{F}(x^{t_1}, \dots, x^{t_q})$ against $x^{\nu-\mathbf{1}}dx/\prod_j \Gamma(\nu_j)$. The left side is the Trailmoment $(\mathsf{T}_{\nu;t}[(P_i)_i] \otimes_Q \mathrm{Dir}_{\mathbf{m}|_Q})(\prod_Q \Delta_Q)$ by 5.1.

*(4) Uniqueness.* With real coefficients the vector $(P_i(U))_i$ takes its values in the compact box $\prod_i [-\|P_i\|, \|P_i\|]$, and its moments $\mathbb{E}[\prod_i P_i(U)^{k_i}] = M(\mathbf{k})$ are the integrals of the monomials against its law. Polynomials are dense in the continuous functions on the box, so a law on the box is determined by these integrals; and the read $\mathbb{E}[\mathsf{Z}_{\nu;t}(\mathbf{P}(U))]$ depends on the law of $\mathbf{P}(U)$ alone. □

**Theorem 5.26 (The Shruffle on a line).** *As stated: (i) and (iii) for any number of strings and of trails; (ii) is that of the reference.*

*Proof.* (i) Let $w^{(i)}$ have length $n_i$, $n = \sum_i n_i$, and $S_i = \{x_1^{(i)} > \cdots > x_{n_i}^{(i)} > 0\}$. By Tonelli the function $\prod_i \prod_j w_j^{(i)}(x_j^{(i)})$ is integrable on $S_1 \times \cdots \times S_r$, with integral $\prod_i \mathrm{chain}(w^{(i)})$ by Fubini. The points with two equal coordinates form a null set. At every other point the $n$ coordinates are distinct, and their decreasing order is a total order of the $n$ positions $(i,j)$ that restricts to the order of each string: an interleaving $\varsigma$, the string of the $n$ functions in that order. The points of one interleaving are carried by a permutation of the coordinates, which keeps Lebesgue measure, onto $\{y_1 > \cdots > y_n > 0\}$, the integrand becoming $\prod_m \varsigma_m(y_m)$; so their integral is $\mathrm{chain}(\varsigma)$. The interleavings are the ways of distributing the $n$ places among the strings, $n!/(n_1! \cdots n_r!)$ of them.

(iii) Let $r$ trails be as in (ii), with strings $\sigma^{(1)}, \dots, \sigma^{(r)}$ and cuts $N_1, \dots, N_r$. By (ii) the trail $i$ is $\mathrm{chain}(\sigma^{(i)})$, and by (i) the product is $\sum_\varsigma \mathrm{chain}(\varsigma)$ over the interleavings. An interleaving begins with the first function of one of the strings, a Laplace transform $e^{-N_i \cdot} L_1^{(i)}$, which is $\int e^{-(s+N_i)\cdot} \psi(ds)$ for the first increment $\psi$ of trail $i$: the Laplace transform of its image under $s \mapsto s + N_i$, carried by $(0,\infty)$ by the hypothesis of (ii). The other Laplace transforms of $\varsigma$ are those of nonnegative increments or of such images. Reading $\varsigma$ as in (ii), each Laplace transform followed by its run of constant functions 1 gives an increment with the index one more than the length of the run; the trail at the cut 0 with these increments has no constant columns and positive integer indices, and by (ii) it is $\mathrm{chain}(\varsigma)$. □

**Theorem 5.27 (Trails at frequencies are iterated integrals).** *As stated.*

*Proof.* For a word $\ell = (\ell_1, \dots, \ell_w)$ with $\ell_1 \neq 0$ and letters in $\{0\} \cup \{|z| = 1\}$ put $F_\varnothing = 1$ and $F_\ell(z) = \int_0^z F_{\ell'}(t)\, dt/(t - \ell_w)$ on the disc $|z| < 1$, $\ell'$ the word without its last letter, along the segment. By induction $F_\ell$ is holomorphic on the disc with $F_\ell(0) = 0$ for $w \geq 1$: for $\ell_w \neq 0$ the factor $1/(t - \ell_w)$ is holomorphic there, and for $\ell_w = 0$, $F_{\ell'}(t)/t$ is, as $F_{\ell'}(0) = 0$, the word $\ell'$ being nonempty since $\ell_1 \neq 0$. Unwinding the recursion, $F_\ell(z)$ is the iterated integral over $0 < t_1 < \cdots < t_w < z$ along the segment.

For $|z| < 1$ let $S(z) = \sum_{0<n_1<\cdots<n_d} \prod_i \zeta_i^{n_i} n_i^{-\nu_i} z^{n_d}$, absolutely convergent, the trail with $z^{y_d}$ in its density, $S(0) = 0$. If $\nu_d \geq 2$, $zS'(z)$ is the same series with $\nu_d - 1$ in place of $\nu_d$, whose word

is $\ell$ without its last letter 0, and $S(z) = \int_0^z S_{\nu_d-1}(t)\,dt/t$. If $\nu_d = 1$, summing the geometric series over $n_d > n_{d-1}$,

$$S'(z) = \sum_{0<n_1<\cdots<n_{d-1}} \prod_{i<d} \zeta_i^{n_i} n_i^{-\nu_i}\ \frac{\zeta_d\,(\zeta_d z)^{n_{d-1}}}{1-\zeta_d z} = \frac{\zeta_d}{1-\zeta_d z}\,S^\flat(z),$$

where $S^\flat$ is the series of depth $d-1$ with the last frequency $a_{d-1}+a_d$, so $\zeta_{d-1}\zeta_d$, and $S^\flat = 1$ when $d = 1$. Its $c_i$ are those of $S$ for $i < d$, its word is $\ell$ without the last letter $c_d^{-1} = \zeta_d^{-1}$, and $\zeta_d/(1-\zeta_d z) = -1/(z-\zeta_d^{-1})$, so $S(z) = -\int_0^z S^\flat(t)\,dt/(t-\zeta_d^{-1})$. By induction on $w$, $S = (-1)^d F_\ell$ on the disc, which is the last sentence.

Write $S(z) = \sum_n b_n z^n$ with $b_n = A(n)\zeta_d^n n^{-\nu_d}$, $A(n) = \sum_{0<n_1<\cdots<n_{d-1}<n} \prod_{i<d} \zeta_i^{n_i} n_i^{-\nu_i}$, $A = 1$ when $d = 1$; then $|A(n)| \le (1+\log n)^{d-1}$, and the head over $y_d \le N$ is $\sum_{n\le N} b_n$. For $\nu_d \ge 2$ the series $\sum b_n$ converges absolutely, so the heads converge to the trail. For $\nu_d = 1$ the partial sums of $\zeta_d^n$ are bounded by $2/|1-\zeta_d|$, and $A(n)/n \to 0$ with $\sum_n |A(n+1)/(n+1) - A(n)/n| \le \sum_n C(1+\log n)^{d-1} n^{-2} < \infty$, since $A(n+1) - A(n)$ is the sum of the terms with $n_{d-1} = n$; summation by parts makes $\sum b_n$ converge. By Abel's theorem on power series, $S(z) \to \sum_n b_n$ as $z \to 1$ along $[0,1)$.

On the other side, the integrand $\prod_j 1/|t_j - \ell_j|$ on the simplex of $[0,1]$ is at most $C \prod_{\ell_j=0} t_j^{-1} \prod_{\ell_j=1} (1-t_j)^{-1}$, the other letters lying off $[0,1]$. Split the simplex by the number of the $t_j$ below $\frac12$; on the part below, with $\ell_1 \ne 0$, the iterated integrals $G_j(t)$ of the first $j$ factors over $0 < t_1 < \cdots < t_j < t$ satisfy $G_1(t) \le Ct$ and $G_j(t) \le \int_0^t G_{j-1}(s)\,C(1+1/s)\,ds \le C_j t$; on the part above the same holds after $t \mapsto 1-t$, as $\ell_w \ne 1$ ($\ell_w = 0$ when $\nu_d \ge 2$, $\ell_w = \zeta_d^{-1} \ne 1$ when $\nu_d = 1$). So the integral over $0 < t_1 < \cdots < t_w < 1$ converges absolutely, and by dominated convergence $F_\ell(z)$ tends to it as $z \to 1$ along $[0,1)$. Hence the limit of the heads is $(-1)^d$ times that integral. □

**Theorem 5.31 (Two convolutions).** *As stated: (ii) and (iii) at the unit configuration on any number of scales; (i) and the displays of (ii) for inputs carried by $(0,\infty)$ are those of the reference.*

*Proof.* (ii) By 2.12, $\mathsf{Z}_{\nu;I_\mathcal{E}}(\mathbf{c}\,\mathbf{c}') = \prod_e (c_e c'_e)^{-\nu_e} = \mathsf{Z}_{\nu;I_\mathcal{E}}(\mathbf{c})\,\mathsf{Z}_{\nu;I_\mathcal{E}}(\mathbf{c}')$ at every point of $Y_1 \times Y_2$. The two Trailmoments on the right converge, so by Tonelli the kernel is integrable for $|\phi| \otimes |\psi|$ on $Y_1' \times Y_2'$, and Fubini gives the product (5.1).

(iii) *The kernel on sums.* First, $g$ has no zero: if $g(s_0) = 0$, then $g(s_0/2)^2 = g(s_0) = 0$, so $g(s_0/2^n) = 0$ for every $n$; every $s \in (0,\infty)^\mathcal{E}$ has $s - s_0/2^n \in (0,\infty)^\mathcal{E}$ for $n$ large, and $g(s) = g(s_0/2^n)\,g(s - s_0/2^n) = 0$, against the hypothesis. Fix $e \in \mathcal{E}$. For $s, s'' \in (0,\infty)^\mathcal{E}$ and $\tau > 0$, $g(s+\tau\mathbf{e}_e)\,g(s'') = g(s+s''+\tau\mathbf{e}_e) = g(s''+\tau\mathbf{e}_e)\,g(s)$, so $\varphi_e(\tau) = g(s+\tau\mathbf{e}_e)/g(s)$ does not depend on $s$. It is continuous on $(0,\infty)$, has no zero, and $\varphi_e(\tau+\tau') = \varphi_e(\tau)\,\varphi_e(\tau')$, by the independence of the base point applied at $s$ and at $s+\tau\mathbf{e}_e$. By the converse on one scale, the case $|\mathcal{E}| = 1$ (the reference, proof of 5.31), $\varphi_e(\tau) = e^{-u_e\tau}$ for one $u_e \in \mathbb{C}$, and then $g(s+\tau\mathbf{e}_e) = e^{-u_e\tau} g(s)$ also for $\tau < 0$ with $s + \tau\mathbf{e}_e \in (0,\infty)^\mathcal{E}$, applying it at the base point $s+\tau\mathbf{e}_e$. Moving one coordinate at a time from a fixed point $s_1$, $g(s) = C\,e^{-\langle u,s\rangle}$ on $(0,\infty)^\mathcal{E}$ with $C = g(s_1)e^{\langle u,s_1\rangle} \ne 0$, and $g(s+s') = g(s)\,g(s')$ gives $C = C^2$, so $C = 1$.

*The kernel on products.* If $h(x) \ne 0$, then $h(x) = h(x)\,h(\mathbf{1})$ gives $h(\mathbf{1}) = 1$, and $h(x)\,h(x^{-1}) = h(\mathbf{1}) = 1$ shows that $h$ has no zero. The function $s \mapsto h(e^s)$ on $(0,\infty)^\mathcal{E}$ is continuous, not identically zero and satisfies the hypothesis on sums, so $h(e^s) = e^{-\langle u,s\rangle}$, that is $h(x) = x^{-u}$ for every $x$ with all coordinates above 1. Any $x \in (0,\infty)^\mathcal{E}$ is $x'/x''$ with $x''_e > \max(1, 1/x_e)$ and $x' = x\,x''$, both with all coordinates above 1, and $h(x) = h(x')/h(x'') = x^{-u}$. So $h = \mathsf{Z}_{u;I_\mathcal{E}}$. □

## 50.6 Chapter 6: rational words and live inputs

**Theorem 6.4 (Pole columns are a rational difference quotient).** *As stated: coefficients in any subfield of $\mathbb{C}$, and no decay condition.*

*Proof. (1) The decomposition.* For $k \geq 0$ and $s \geq 1$, telescoping gives $\Delta \sum_{j=0}^{k-1} (t+j)^{-s} = \sum_{j=0}^{k-1} \big((t+j+1)^{-s} - (t+j)^{-s}\big) = (t+k)^{-s} - t^{-s}$. So $\Delta H(t) = \sum_{k,s} c_{k,s}\big((t+k)^{-s} - t^{-s}\big) = R(t) - \sum_s C_s t^{-s}$, and $H$ has its coefficients in $\mathbb{K}$.

*(2) The decay.* As $t \to \infty$, $(t+k)^{-s} = t^{-s}(1+k/t)^{-s} = t^{-s} + O(t^{-s-1})$, so $R(t) = C_1 t^{-1} + O(t^{-2})$, which is $O(t^{-2})$ exactly when $C_1 = 0$.

*(3) The converse.* A proper rational function $H$ with poles on the nonpositive integer lattice has one partial-fraction presentation $H(t) = \sum_{k,s} d_{k,s}(t+k)^{-s}$, finitely many terms with $k \in \mathbb{N}$, $s \geq 1$. Then $\Delta H(t) = \sum_{k,s} d_{k,s}\big((t+k+1)^{-s} - (t+k)^{-s}\big)$, whose coefficient of $(t+k)^{-s}$ is $d_{k-1,s} - d_{k,s}$ (with $d_{-1,s} = 0$), and the column sum over $k$ telescopes to 0. The column sums $R \mapsto (C_s)_s$ are linear on rational words, presentations being unique, and vanish on such differences. A combination $\sum_s a_s t^{-s}$ has the column sums $(a_s)_s$, so it is a difference only if it is 0: the functions $t^{-1}, \ldots, t^{-w}$ are independent modulo differences, and by (1) they span the rational words of maximum order $w$ modulo differences. A difference is a rational word with column sums zero, hence $O(t^{-2})$ by (2); and a word that is $O(t^{-2})$ has $C_1 = 0$, so it is congruent to $\sum_{s \geq 2} C_s t^{-s}$: modulo differences these words have the basis $t^{-2}, \ldots, t^{-w}$.

*(4) Cosets.* Translation by one keeps each coset $b + \mathbb{Z}$, so the steps above apply coset by coset, with $(t+b)^{-s}$ in the place of $t^{-s}$. □

**Theorem 6.5 (Summation by parts with a live input).** *As stated: (iv) without the condition $C_1 = 0$; (i) to (iii) are those of the reference.*

*Proof.* (iv) For an integer $M \geq 1$ the head is $(\mathsf{T}_{0;(1)}[x]\, R\, \mathbb{I}_{\mathbb{Z}})((N, N+M]) = \sum_{N < m \leq N+M} R(m)$, the kernel at index 0 being 1. By 6.4, $R(m) = \sum_s C_s m^{-s} + H(m+1) - H(m)$, so

$$(\mathsf{T}_{0;(1)}[x]\, R\, \mathbb{I}_{\mathbb{Z}})((N, N+M]) = \sum_{s=1}^{w} C_s\, (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N, N+M]) + H(N+M+1) - H(N+1).$$

As $M \to \infty$: $H(N+M+1) \to 0$, $H$ being proper; for $s \geq 2$, $(\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N, N+M]) \to (\mathsf{T}_{s;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N, \infty))$; and $(\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N, N+M]) - \log M \to (\mathsf{T}_{1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((N, \infty))$, the regularized value of 5.2. This gives the first display. When $C_1 = 0$, $R(m) = O(m^{-2})$ by 6.4, the series $\sum_{m>N} R(m)$ converges absolutely, and the left side is $(\mathsf{T}_{0;(1)}[x]\, R\, \mathbb{I}_{\mathbb{Z}})((N, \infty))$, the term $s = 1$ being absent. For a density $f$, when the terms converge, $\sum_{m>N} R(m) f(m) = \sum_{s=1}^{w} C_s \sum_{m>N} m^{-s} f(m) + \sum_{m>N} \Delta H(m) f(m)$, and (ii) gives the last two terms of the second display when the upper boundary vanishes. □

**Theorem 6.8 (A Breta input with a recurrence of finite order).** *As stated: complex $a, b$ with positive real parts, real $q_i < 1$ of either sign, and complex $s_i$.*

*Proof. (1) The letters.* For real $q_i < 1$ and $x \in [0,1]$, $1 - q_i x$ lies between 1 and $1 - q_i$, so in $[\min(1, 1-q_i), \max(1, 1-q_i)] \subset (0, \infty)$. So $\varrho(x) = \prod_i (1 - q_i x)^{-s_i}$ is infinitely differentiable on $[0,1]$, and $|\varrho| = \prod_i (1 - q_i x)^{-\operatorname{Re} s_i}$ lies between two positive constants.

*(2) The size.* For $m \geq 1$, by $x \leq e^{-(1-x)}$ on $[0,1]$ and $y = 1 - x$,

$$|f(m)| \leq \max |\varrho| \int_0^1 x^{m + \operatorname{Re} a - 1} (1-x)^{\operatorname{Re} b - 1} dx$$

$$\le \max|\varrho| \int_0^\infty e^{-(m+\operatorname{Re} a-1)y} y^{\operatorname{Re} b-1} dy = \max|\varrho|\,\Gamma(\operatorname{Re} b)\,(m+\operatorname{Re} a-1)^{-\operatorname{Re} b},$$

so $f(m) = O(m^{-\operatorname{Re} b})$.

*(3) The recurrence.* Differentiating the logarithm, $\frac{d}{dx}\big[x^{m+a}(1-x)^b \prod_i (1-q_i x)^{1-s_i}\big]$ is $x^{m+a-1}(1-x)^{b-1}\varrho(x)$ times

$$(m+a)(1-x)Q(x) - bxQ(x) + x(1-x)\sum_i (s_i-1)q_i Q_i(x) = B_m(x).$$

The bracket tends to 0 at both ends, since $\operatorname{Re}(m+a) > 0$ and $\operatorname{Re} b > 0$, and $x^{m+a-1}(1-x)^{b-1}\varrho B_m$ is integrable; so integrating over $[0,1]$ gives $\sum_\ell B_\ell(m) f(m+\ell) = 0$.

*(4) The extreme coefficients.* $B_0(m) = B_m(0) = (m+a)Q(0) = m+a$. Every $q_i \neq 0$, so $Q$ has degree $r$ and leading coefficient $(-1)^r \prod_i q_i$, and $Q_i$ has degree $r-1$ and leading coefficient $(-1)^{r-1}\prod_{i'\neq i} q_{i'}$. The coefficient of $x^{r+1}$ in $B_m$ is therefore

$$(-1)^{r+1}\prod_i q_i\Big((m+a)+b+\sum_i (1-s_i)\Big) = (-1)^{r+1}\Big(\prod_i q_i\Big)\Big(m+a+b+r-\sum_i s_i\Big).$$

$B_0(m) \neq 0$ since $\operatorname{Re} a > 0$, and $B_{r+1}(m) \neq 0$ for every integer $m \ge 0$ exactly when $a+b+r-\sum_i s_i \notin -\mathbb{N}$. Then the recurrence solved for $f(m+r+1)$, and for $f(m)$, expresses every shift through $(f(m),\dots,f(m+r))$ with coefficients rational in $m$.

*(5) The Breta.* Let $\operatorname{Re} s_i > 0$. On the bunch $\{1,2\}$ with the scale $\rho$ and the shape $u = (x, 1-x)$, $\omega_{\{1,2\}}$ being $dx$ in the coordinate $x = u_1$, the kernel terms $\rho u_1$ and $\rho u_2$ with unit coefficients sum to $\rho$, and the letter $P_i = \rho\big((1-q_i)u_1 + u_2\big) = \rho(1-q_i x)$ has the positive pattern coefficients $1-q_i$ and 1. At the shape index $(m+a, b)$ and the scale index $1+|\mathbf{s}|$ the left side of 3.6 is

$$\int_0^1 x^{m+a-1}(1-x)^{b-1}\int_0^\infty \rho^{|\mathbf{s}|}e^{-\rho}\rho^{-|\mathbf{s}|}\prod_i (1-q_i x)^{-s_i}\,d\rho\,dx = f(m),$$

the inner integral being $\Gamma(1) = 1$; 3.6 identifies it with $\Gamma(1+|\mathbf{s}|)$ times the Breta. □

**Theorem 6.9 (A fixed set of input-dependent evaluations).** *As stated, for the data of 6.8.*

*Proof.* The steps of the reference, proof of 6.9, hold for these data; they are as follows. Write $F_{p,k}(N) = (\mathsf{T}_{p;(1)}[x+k]\, f\, \mathbb{I}_{\mathbb{Z}})((N,\infty)) = \sum_{m>N} f(m)(m+k)^{-p}$, and, for $0 \le x < 1$, $S_p(x) = \sum_{m>N} x^m (m+k)^{-p}$, with an integer cut $N \ge 0$ and $k \ge 0$; $S_0(x) = x^{N+1}/(1-x)$.

*(1) The identity behind (ii).* As in the proof of 6.8, $(xC\eta)' = B\eta$, with $xC\eta = x^a(1-x)^b\prod_i(1-q_i x)^{1-s_i}$. Since $xS_p'(x) = \sum_{m>N} m\, x^m (m+k)^{-p} = S_{p-1}(x) - kS_p(x)$,

$$\frac{d}{dx}\big[xC\eta\, S_p\big] = \eta\big[(B-kC)S_p + C\, S_{p-1}\big] \qquad (p \ge 1).$$

At $x = 0$ the bracket is $O(|x^{a+N+1}|) \to 0$, since $\operatorname{Re} a > 0$. At $x = 1$ it is bounded by a constant times $|(1-x)^b|$ for $p \ge 2$, where $S_p(1)$ is finite, and times $|(1-x)^b|\,|\log(1-x)|$ for $p = 1$, where $S_1(x) = O(|\log(1-x)|)$; it tends to 0 since $\operatorname{Re} b > 0$. The functions $|\eta|\, S_p$ and $|\eta|\,|\log(1-x)|$ are integrable on $[0,1)$, so integrating over $[0,1)$ and summing term by term (Tonelli on the moduli) gives, for $p \ge 2$,

$$\sum_{m>N}(m+k)^{-p}\int_0^1 x^m (B-kC)\eta\,dx + \sum_{m>N}(m+k)^{1-p}\int_0^1 x^m C\eta\,dx = 0.$$

With $\int_0^1 x^{m+\ell}\eta\,dx = f(m+\ell)$ and, for $k \ge r+1 \ge \ell$, $\sum_{m>N} f(m+\ell)(m+k)^{-p} = F_{p,k-\ell}(N) - \sum_{v=1}^{\ell} f(N+v)(N+v+k-\ell)^{-p}$, this is (ii) for $p \ge 2$. At $p=1$ the term with $S_0$ is $\int_0^1 \eta\, C\, S_0\, dx = \int_0^1 \eta\, x^{N+1} Q\, dx = \sum_{\ell=0}^{r} Q_\ell f(N+1+\ell)$, by $C\,S_0 = x^{N+1}Q$, which gives (ii) at $p=1$.

*(2) The reduction (i).* In (ii) the coefficient of $F_{p,k}(N)$, the term $\ell = 0$, is $B_0 - kC_0 = a - k$, since $B(0) = aQ(0) = a$ and $C(0) = 1$. For $k \ge r+1$ with $k \ne a$, (ii) expresses $F_{p,k}(N)$ through the $F_{p,k-\ell}(N)$, $1 \le \ell \le r+1$, the $F_{p-1,k-\ell}(N)$ (none at $p=1$), and the values $f(N+v)$, $1 \le v \le r+1$. Taking the indices $p$ in increasing order and, for each, the shifts $k$ in increasing order, every $F_{p,k}(N)$ with $p \le A$ reduces to the $F_{p',k'}(N)$ with $p' \le p$ and $k' \le r$, the $F_{p',a}(N)$ when $a$ is an integer at least $r+1$, and $f(N+1), \dots, f(N+r+1)$.

*(3) The cuts (iii)* are 5.3 for the counting input.

*(4) Denominators (iv).* For rational data, let $c_*$ clear the coefficients of $B$, $C$ and $Q$, so that $c_*(a-k)$, $c_*(B_\ell - kC_\ell)$, $c_*C_\ell$ and $c_*Q_\ell$ are integers for integer $k$. By induction along the order of (2), the coefficients of $F_{p,k}(N)$, $k \ge r+1$, in the reduced form are cleared by $D_{p,k} = d_{N+k}^{p} \prod_{r+1 \le k' \le k,\ k' \ne a} |c_*(a-k')|^p$. Multiply (ii) by $c_*$ and divide by $c_*(a-k)$: the terms $F_{p,k-\ell}(N)$ and $F_{p-1,k-\ell}(N)$ carry integer factors, and their reduced forms are cleared by $D_{p,k-\ell}$ and $D_{p-1,k-\ell}$, which divide $D_{p,k}/|c_*(a-k)|$; the terms in $f(N+v)$ carry the denominators $(N+v+k-\ell)^p$ and $(N+v+k-\ell)^{p-1}$, which divide $d_{N+k}^{p}$, since $1 \le N+v+k-\ell \le N+k$. So $D_{p,k}$ clears $F_{p,k}(N)$, and $d_{N+J}^{A} \prod_{r+1 \le k \le J,\ k \ne a} |c_*(a-k)|^A$, a multiple of every $D_{p,k}$ with $p \le A$ and $k \le J$, clears all shifts through $J$ and indices through $A$. □

**Theorem 6.10 (The row of a rational function against an input with a recurrence).** *As stated.*

*Proof.* All the sums converge absolutely: $m + h \ge N + 1 + h_* > 0$ for $m > N$ and $h \ge h_*$, and $\sum_{m>N} |f(m)|(1+|m|)^d < \infty$. (i) is linearity, $P(m) = \sum_r ([m^r]P) m^r$.

(ii) Expand $p_i$ about $-h$: $p_i(m) = \sum_e \pi_{i,e}(h)(m+h)^e$ with $\pi_{i,e}(h) = p_i^{(e)}(-h)/e!$. Multiply the recurrence at $m$ by $(m+h)^{-s}$, sum over $m > N$ and put $m' = m + i$:

$$\sum_{i=0}^{\tau} \sum_{e} \pi_{i,e}(h) \sum_{m' > N+i} f(m')\,(m'+h-i)^{e-s} = 0.$$

Moving the cut of each inner sum to $N$ (5.3) and writing, for $e \ge s$, $(m'+h-i)^{e-s} = \sum_{r \le e-s} \binom{e-s}{r} (h-i)^{e-s-r} m'^r$,

$$\begin{aligned} p_0(-h)\,T_s(h) = &\sum_{i=1}^{\tau} \sum_{v=1}^{i} \sum_{e} \pi_{i,e}(h)\, f(N+v)\,(N+v+h-i)^{e-s} \\ &- \sum_{(i,e) \ne (0,0),\ e<s} \pi_{i,e}(h)\, T_{s-e}(h-i) - \sum_{e \ge s} \sum_{i} \pi_{i,e}(h) \sum_{r \le e-s} \binom{e-s}{r} (h-i)^{e-s-r} M_r, \end{aligned}$$

with $\pi_{0,0}(h) = p_0(-h)$. At $h = h_* + j$, $\tau \le j \le J$, divide by $p_0(-h) \ne 0$: the right side holds $T_{s'}$ with $s' < s$ at the same shift, or with $s' \le s$ at the smaller shifts $h - i \ge h_*$, moments $M_r$ with $r \le d - s \le d-1$, and the values $f(N+v)$, $v \le \tau$. Induction on $j$, and for fixed $j$ on $s$, gives (ii); all coefficients are rational.

(iii) Put $Q(s,j) = d_{u(N+h_*+j)}^{s} \prod_{j'=\tau}^{j} |u^d p_0(-h_* - j')|^s$; we show by the same induction that $Q(s,j)$ clears the coefficients of $T_s(h_* + j)$, and $Q(s,j)$ divides the integer of (iii). Since $h \in \frac{1}{u}\mathbb{Z}$ and $p_i \in \mathbb{Z}[m]$, $\pi_{i,e}(h)$ has a denominator dividing $u^{\deg p_i - e}$, and $u^d p_0(-h)$ is an integer. Write the relation as $T_s(h) = \big(u^d p_0(-h)\big)^{-1} u^d\,[\cdots]$ and take its terms in turn.

- $T_{s-e}(h-i)$ with $i \geq 1$ or $e \geq 1$, $s-e \geq 1$: the factor $u^d \pi_{i,e}(h)$ is an integer, and the coefficients of $T_{s-e}(h-i)$ are 1 if it is a coordinate and are cleared by $Q(s-e, j-i)$ otherwise, by induction; $|u^d p_0(-h)|\, Q(s-e, j-i)$ divides $Q(s,j)$, as $s-e \leq s$, $j-i \leq j$, and when $i=0$, $e \geq 1$ leaves the factor $|u^d p_0(-h)|^e$ free in $Q(s,j)/Q(s-e,j)$.
- The moments: $u^d \pi_{i,e}(h)(h-i)^{e-s-r}$ has a denominator dividing $u^{-d+\deg p_i - e + e - s - r}$, an integer power of $u$ with exponent at most 0, so it is an integer.
- The values $f(N+v)$: for $e < s$, $(N+v+h-i)^{e-s} = u^{s-e}\big(u(N+v+h-i)\big)^{e-s}$, where $u(N+v+h-i)$ is an integer with $1 \leq u(N+v+h-i) \leq u(N+h)$, positive since $h-i \geq h_*$ and $N+1+h_* > 0$; so its power $s-e \leq s$ divides $d^s_{u(N+h_*+j)}$, and $u^d \pi_{i,e}(h) u^{s-e}$ is an integer. For $e \geq s$ the term $u^d \pi_{i,e}(h)(N+v+h-i)^{e-s}$ has a denominator dividing $u^{\deg p_i - e + (e-s) - d}$, a power of $u$ with exponent at most 0, so it is an integer.

Every coefficient of $T_s(h_*+j)$ is a sum of such products, so $Q(s,j)$ clears it; and for $s \leq w$, $j \leq J$, $Q(s,j)$ divides $d^w_{u(N+h_*+J)} \prod_{j'=\tau}^{J} |u^d p_0(-h_* - j')|^w$, as $d_n \mid d_{n'}$ for $n \leq n'$. Poles in several cosets are reduced coset by coset, each with its own $u$ and $h_*$. □

## 50.7 Chapter 7: tables

**Proposition 7.6 (Duality of minors).** *As stated.*

*Proof.* Every basis has $r$ marks, so $|T \cap X|$ is largest exactly when $|T \cap X'|$ is least, that is exactly when the cobasis $\mathcal{E} \setminus T$ has $|(\mathcal{E} \setminus T) \cap X'| = |X'| - |T \cap X'|$ largest: $T$ is at the top of $X$ exactly when $\mathcal{E} \setminus T$ is at the top of $X'$ in $\mathcal{B}^*$ (7.5). By 7.5 the bases of $\mathcal{B}^*/X'$ are the sets $(\mathcal{E} \setminus T) \setminus X' = X \setminus (T \cap X)$ for these $T$, the complements in $X$ of the bases $T \cap X$ of $\mathcal{B}|X$, and the bases of $\mathcal{B}^*|X'$ are the sets $(\mathcal{E} \setminus T) \cap X' = X' \setminus (T \setminus X)$, the complements in $X'$ of the bases $T \setminus X$ of $\mathcal{B}/X$. The weight of the cobasis $\mathcal{E} \setminus T$ is $\theta_T = \theta'_{T \cap X} \theta''_{T \setminus X}$, the factor $\theta''_{T \setminus X}$ belonging to its part $X' \setminus (T \setminus X)$ and the factor $\theta'_{T \cap X}$ to its part $X \setminus (T \cap X)$; these are the weights of the dual tables of 7.1. The letters follow from 7.2. □

**Theorem 7.15 (One function, two Foreman polynomials).** *As stated: (iii) along every direction constant on the molecules of the table, with the converse; (i) and (ii) are those of the reference.*

*Proof. The directions.* By 7.9 for the family whose one table is this table, the table is the direct sum of its restrictions to its molecules $M$: every basis $T$ is the union of one basis of each restriction $\mathcal{B}|M$, and a basis of $\mathcal{B}|M$ has $r(M)$ marks (7.5). So $|T \cap M| = r(M)$, and for $v$ constant on each $M$, with value $v_M$,

$$\langle v, \chi_T \rangle = \sum_M v_M \big( |T \cap M| - |M \setminus T| \big) = \sum_M v_M \big( 2r(M) - |M| \big)$$

for every basis. By 7.14, $\Lambda(z+v) = \sum_T \theta_T e^{\langle z, \chi_T \rangle} e^{\langle v, \chi_T \rangle} = \exp\big( \sum_M v_M (2r(M) - |M|) \big) \Lambda(z)$. At $v = \lambda \mathbf{1}$ the exponent is $\lambda(2r-n)$, the $r(M)$ adding up to $r$, since a basis meets each molecule in $r(M)$ marks, and the $|M|$ to $n$.

*The converse.* Let $v \in \mathbb{R}^{\mathcal{E}}$ and real $\lambda \neq 0$ be such that $\Lambda(z + \lambda v) = c\, \Lambda(z)$ for every $z \in \mathbb{R}^{\mathcal{E}}$, with $c$ independent of $z$. Then $\sum_T \theta_T \big( e^{\lambda \langle v, \chi_T \rangle} - c \big) e^{\langle z, \chi_T \rangle} = 0$ for every $z$. Distinct bases have distinct signed indicators, and exponentials $e^{\langle z, \chi \rangle}$ with finitely many distinct vectors $\chi$ are linearly independent functions of $z$: on a line $z = \sigma y$ with the $\langle y, \chi \rangle$ distinct, as $\sigma \to \infty$ the term

with the largest $\langle y, \chi \rangle$ dominates, so its coefficient vanishes, and the others follow in turn. So $\theta_T(e^{\lambda\langle v,\chi_T\rangle} - c) = 0$, and since $\theta_T > 0$, $e^{\lambda\langle v,\chi_T\rangle} = c$ for every basis. Hence $\langle v, \chi_T\rangle = 2\langle v, \mathbf{1}_T\rangle - \langle v, \mathbf{1}\rangle$ is constant over the bases, and so is $\langle v, \mathbf{1}_T\rangle$. By 7.8, $v_a = v_b$ for any two marks linked by a circuit of the table; the molecules of the table alone are the smallest nonempty sets of marks containing, with each mark, every mark so linked to it (7.4), so $v$ is constant on each of them. □

**Theorem 7.16 (The inner moment generating function).** *As stated: the even statistic at every gauge, with the basis law and the cobasis law there.*

*Proof.* For real $z$ and $w$, $e^{E(z+w)} = \Lambda(z+w)\,\Lambda(-z-w)$. By the first display of the theorem, $\Lambda(z+w)/\Lambda(z) = \mathbb{E}_{T\sim\mathbb{P}_\alpha}[e^{\langle w,\chi_T\rangle}]$, and the same display at the gauge $\alpha^{-1} = e^{-2z}$, whose basis law is $\mathbb{P}_{\alpha^{-1}} = \mathbb{P}^*_\alpha$, gives $\Lambda(-z-w)/\Lambda(-z) = \mathbb{E}_{T'\sim\mathbb{P}^*_\alpha}[e^{-\langle w,\chi_{T'}\rangle}]$. Multiplying, and using the independence of $T$ and $T'$ under $\mathbb{P}_\alpha \otimes \mathbb{P}^*_\alpha$,

$$\frac{e^{E(z+w)}}{e^{E(z)}} = \mathbb{E}_{(T,T')\sim\mathbb{P}_\alpha\otimes\mathbb{P}^*_\alpha}\big[e^{\langle w,\chi_T-\chi_{T'}\rangle}\big], \qquad \chi_T - \chi_{T'} = 2(\mathbf{1}_T - \mathbf{1}_{T'}),$$

which is the display; its logarithm says that $w \mapsto E(z+w) - E(z)$ is the cumulant generating function of $\chi_T - \chi_{T'}$. At $z = 0$, $e^{E(0)} = \Lambda(0)^2 = K(\mathbf{1})^2$ by (i) of 7.15, and $\mathbb{P}^*_{\mathbf{1}} = \mathbb{P}_{\mathbf{1}}$; with $w$ written $z$ this is the display of the reference. □

*Remark.* The display at a general gauge, and (iii) of 7.15, were checked in exact arithmetic on the table whose bases are the unions of a 2-subset of $\{0,1,2\}$ and a 1-subset of $\{3,4\}$, with arbitrary positive weights (`sweep/s0107/tables.py`).

**Corollary 7.24 (Moments and cumulants).** *As stated: the joint cumulants of the table at every order; (i) and the cumulants of the components are those of the reference.*

*Proof.* For a function $Y$ of the index $(k,l)$ write $\mathbb{E}_w[Y] = \sum_{k,l} w_{kl} Y(k,l)$; $\mathbb{E}_w[1] = 1$ by 7.23.

*(1) Generating functions.* By 7.23 at $x = e^v$, $v \in \mathbb{C}^B$,

$$\mathbb{E}_{T\sim\mathbb{P}_\alpha}\Big[\exp\Big(\sum_{e\in B} v_e X_e\Big)\Big] = \sum_{k,l} w_{kl}\, M_{kl}(v), \qquad M_{kl}(v) = \det\big(I + (D_{e^v} - I)\Pi^{(kl)}_{BB}\big),$$

with $M_{kl}(0) = 1$. So $C_{kl}(v) = \log M_{kl}(v)$ is a power series in $v$ without constant term, and for distinct marks the cumulant $\kappa^{(kl)}(e_1,\dots,e_p)$ of the component is the coefficient of $v_{e_1}\cdots v_{e_p}$ in $C_{kl}$; the reference computes it by the cyclic formula. The joint cumulant $\kappa_\alpha(e_1,\dots,e_p)$ is the coefficient of $v_{e_1}\cdots v_{e_p}$ in $\log\sum_{k,l} w_{kl} e^{C_{kl}(v)}$, a power series since $\sum_{k,l} w_{kl} = 1$.

*(2) Cumulants over the index.* Let $(Y_i)_{i\in I}$ be finitely many functions of the index, $\lambda = (\lambda_i)_{i\in I}$ formal variables, and $\mu(\lambda) = \mathbb{E}_w[\exp(\sum_i \lambda_i Y_i)]$, a power series with constant term 1. For $D \subseteq I$ write $\lambda^D = \prod_{i\in D}\lambda_i$; the coefficient of $\lambda^D$ in $\mu$ is $\mathbb{E}_w[\prod_{i\in D} Y_i]$. In $\log\mu = \sum_{j\geq 1} \frac{(-1)^{j-1}}{j}(\mu-1)^j$ the coefficient of $\lambda^I$ comes from the products of $j$ monomials of $\mu - 1$ without repeated variables and with disjoint supports covering $I$; each partition $\mathfrak{q}$ of $I$ into $j$ blocks arises from $j!$ such ordered products. So

$$[\lambda^I]\log\mu = \sum_{j\geq 1} \frac{(-1)^{j-1}}{j}\, j! \sum_{|\mathfrak{q}|=j} \prod_{D\in\mathfrak{q}} \mathbb{E}_w\Big[\prod_{i\in D} Y_i\Big] = \kappa_w\big((Y_i)_{i\in I}\big),$$

and, setting the variables outside a subset $D$ to 0, $[\lambda^D]\log\mu = \kappa_w((Y_i)_{i\in D})$.

*(3) Substitution.* Fix distinct marks $e_1,\dots,e_p$ of $B$; for nonempty $C \subseteq \{1,\dots,p\}$ put $v^C = \prod_{c\in C} v_{e_c}$ and $Y_C(k,l) = \kappa^{(kl)}(e_C)$, the coefficient of $v^C$ in $C_{kl}$. Setting the variables $v_e$ with

$e \notin \{e_1, \dots, e_p\}$ to 0 and dropping every monomial divisible by some $v_{e_c}^2$ is a ring homomorphism onto the ring $\mathcal{R}$ of polynomials in $v_{e_1}, \dots, v_{e_p}$ with $v_{e_c}^2 = 0$; it keeps the coefficient of $v_{e_1} \cdots v_{e_p}$ and commutes with exp and log of series without constant term. In $\mathcal{R}$, $C_{kl}$ becomes $\sum_{C \neq \varnothing} Y_C(k,l)\, v^C$, the image of $\sum_C \lambda_C Y_C(k,l)$ under $\lambda_C \mapsto v^C$. So the coefficient of $v_{e_1} \cdots v_{e_p}$ in $\log \sum_{k,l} w_{kl} e^{C_{kl}}$ is that of the image of $\log \mu(\lambda)$, with $I$ the set of nonempty $C$. A monomial $\prod_C \lambda_C^{n_C}$ maps to $\prod_C (v^C)^{n_C}$, which is $v_{e_1} \cdots v_{e_p}$ exactly when the $C$ with $n_C > 0$ have $n_C = 1$ and form a partition $\mathfrak{p}$ of $\{1, \dots, p\}$, and is 0 or another monomial otherwise. By (2),

$$\kappa_\alpha(e_1, \dots, e_p) = \sum_{\mathfrak{p}} [\lambda^{\mathfrak{p}}] \log \mu = \sum_{\mathfrak{p}} \kappa_w\Big( \big(\kappa^{(kl)}(e_C)\big)_{C \in \mathfrak{p}} \Big).$$

At $p = 1$ the single partition gives $\kappa_\alpha(e) = \mathbb{E}_w[\Pi_{ee}^{(kl)}]$, and $\kappa_\alpha(e) = \mathbb{P}_\alpha(e \in T) = R_e(\alpha)$. Every step is an identity of power series in which the $w_{kl}$ enter linearly inside $\mu$, with $\sum_{k,l} w_{kl} = 1$, so it holds for complex mixing coefficients. □

*Remark.* At $p = 2$ and $p = 3$ the formula gives $\kappa_\alpha(e,f) = \mathbb{E}_w[\kappa^{(kl)}(e,f)] + \kappa_w(\Pi_{ee}^{(kl)}, \Pi_{ff}^{(kl)})$ and $\kappa_\alpha(e,f,g) = \mathbb{E}_w[\kappa^{(kl)}(e,f,g)] + \sum_3 \kappa_w(\Pi_{ee}^{(kl)}, \kappa^{(kl)}(f,g)) + \kappa_w(\Pi_{ee}^{(kl)}, \Pi_{ff}^{(kl)}, \Pi_{gg}^{(kl)})$, the sum over the three choices of the mark that stands alone: the covariances and the third cumulant over the index of the reference. The formula was checked in exact arithmetic against the generating function of a mixture of three components with complex mixing coefficients for $p = 2, 3, 4, 5$ (`sweep/s0107/cumulants2.py`).

## 50.8 Chapter 8: families

**Proposition 8.11 (The interest over latent vectors).** *As stated: every table determinantal over $\mathbb{R}$ or over $\mathbb{C}$, each letter opened with as many latent vectors as its own exponent asks.*

*Proof.* The masses are real and positive. For a mass $m_e \leq 0$ the interest diverges by 8.5(i), the wall $\kappa_{\{e\}} = m_e - \sum_j \big(s_j d_j(\{e\}) + t_j d_j^*(\{e\})\big)$ being at most $m_e$, and so does the integral over $\alpha_e$ in step (4).

Fix a table $j$, write $\mathbb{F} = \mathbb{F}_j$, $\beta = \beta_j$, $r = r_j$, and let $A \in \mathbb{F}^{r \times n}$ be a slice of it: $\det A_T = \boldsymbol{\omega}_T$, so that $|\det A_T|^2 = \theta_{j,T}$ for $T \in \mathcal{B}_j$ and $\det A_T = 0$ for every other set $T$ of $r$ marks (7.20). For a matrix $X$ with $n$ columns write $X^{\mathsf{H}}$ for its conjugate transpose, and $D_\alpha = \operatorname{diag}(\alpha)$. Expanding by the maximal minors of $X$,

$$\det\big(X D_\alpha X^{\mathsf{H}}\big) = \sum_W |\det X_W|^2\, \alpha^W,$$

the sum over the sets $W$ of as many marks as $X$ has rows; this is the expansion behind 7.21, and with $X = A$ it gives $K_j(\alpha) = \det(A D_\alpha A^{\mathsf{H}})$. So $K_j(\mathbf{1}) = \det(AA^{\mathsf{H}}) > 0$, $A$ has rank $r$, and its row space $V_j = \{A^{\mathsf{H}} y :\ y \in \mathbb{F}^r\}$, the space onto which $\Pi^{(11)}$ of 7.23 projects, has dimension $r$ over $\mathbb{F}$.

*(1) The basis letter.* The matrix $\hat{A} = (AA^{\mathsf{H}})^{-1/2} A$ has orthonormal rows and $\{\hat{A}^{\mathsf{H}} y\} = V_j$, and $\det \hat{A}_T = \det(AA^{\mathsf{H}})^{-1/2} \det A_T$. So $\theta_{j,T} = K_j(\mathbf{1})\, |\det \hat{A}_T|^2$ and $K_j(\alpha) = K_j(\mathbf{1}) \det(\hat{A} D_\alpha \hat{A}^{\mathsf{H}})$.

*(2) The cobasis letter.* Choose $\hat{C} \in \mathbb{F}^{(n-r) \times n}$ with orthonormal rows and $\{\hat{C}^{\mathsf{H}} y\} = V_j^{\perp}$, the orthogonal complement for the form $\sum_e u_e \overline{v_e}$. Then $\hat{A}\hat{C}^{\mathsf{H}} = 0$, and the square matrix $U$ whose rows are those of $\hat{A}$ followed by those of $\hat{C}$ is unitary. For an invertible matrix, $\det U[I,J] = \pm \det U \cdot \det U^{-1}[J^c, I^c]$ for every set $I$ of rows and set $J$ of columns of equal size; with $U^{-1} = U^{\mathsf{H}}$ and $|\det U| = 1$ this gives $|\det U[I,J]| = |\det U[I^c, J^c]|$. With $I$ the rows of $\hat{A}$ and $J = T$,

$|\det \hat{C}_{\mathcal{E}\setminus T}| = |\det \hat{A}_T|$ for every set $T$ of $r$ marks. The expansion above, for $X = \hat{C}$ and the sets $W = \mathcal{E} \setminus T$, gives

$$\det\big(\hat{C} D_\alpha \hat{C}^{\mathsf{H}}\big) = \sum_T |\det \hat{A}_T|^2 \alpha^{\mathcal{E}\setminus T} = \frac{1}{K_j(\mathbf{1})} \sum_{T \in \mathcal{B}_j} \theta_{j,T}\, \alpha^{\mathcal{E}\setminus T} = \frac{\Psi_j(\alpha)}{K_j(\mathbf{1})}.$$

*(3) The Gaussian opening.* For a Hermitian positive definite matrix $G$ of size $d$ over $\mathbb{F}$, a unitary change of variables, which keeps the Lebesgue measure of $\mathbb{F}^d \cong \mathbb{R}^{\beta d}$, makes $G$ diagonal, and $\int_{\mathbb{F}} e^{-\lambda|z|^2} dz = (\pi/\lambda)^{\beta/2}$ for $\lambda > 0$; so $\int_{\mathbb{F}^d} e^{-z^{\mathsf{H}} G z} dz = \pi^{\beta d/2} \det(G)^{-\beta/2}$. For an integer $k \geq 0$ and $s = \beta k/2$ it follows that

$$\det(G)^{-s} = \pi^{-ds} \int_{(\mathbb{F}^d)^k} \exp\Big( -\sum_{i=1}^{k} z_i^{\mathsf{H}} G z_i \Big)\, dz_1 \cdots dz_k.$$

Take $G = \hat{A} D_\alpha \hat{A}^{\mathsf{H}}$, $d = r$ and $k = 2s_j/\beta$. The map $z \mapsto x = \hat{A}^{\mathsf{H}} z$ is an $\mathbb{F}$-linear isometry of $\mathbb{F}^r$ onto $V_j$, so it carries the Lebesgue measure of $\mathbb{F}^r$ to that of $V_j$, and $z^{\mathsf{H}} \hat{A} D_\alpha \hat{A}^{\mathsf{H}} z = \sum_e \alpha_e |x_e|^2$. With step (1),

$$K_j(\alpha)^{-s_j} = K_j(\mathbf{1})^{-s_j} \pi^{-r_j s_j} \int_{V_j^{2s_j/\beta}} \exp\Big( -\sum_{i,e} \alpha_e |(x_{j,i})_e|^2 \Big) \prod_i dx_{j,i},$$

and in the same way, with step (2), $G = \hat{C} D_\alpha \hat{C}^{\mathsf{H}}$, $d = n - r_j = r_j^*$ and $k = 2t_j/\beta$,

$$\Psi_j(\alpha)^{-t_j} = K_j(\mathbf{1})^{-t_j} \pi^{-r_j^* t_j} \int_{(V_j^\perp)^{2t_j/\beta}} \exp\Big( -\sum_{i,e} \alpha_e |(x^*_{j,i})_e|^2 \Big) \prod_i dx^*_{j,i}.$$

An exponent 0 asks for no vectors, and the formula then reads $1 = 1$.

*(4) The scales.* Multiplying over $j$, with $c_F = \prod_j K_j(\mathbf{1})^{s_j + t_j} \pi^{r_j s_j + r_j^* t_j}$,

$$W(\alpha) = \frac{1}{c_F} \int \exp\Big( -\sum_e \alpha_e \mathfrak{q}_e \Big) \prod_{j,i} dx_{j,i}\, dx^*_{j,i}.$$

Insert this into $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int \alpha^{\mathbf{m-1}} e^{-|\alpha|} W(\alpha)\, d\alpha$. Every integrand is nonnegative, so the integrals may be taken in either order, and

$$\int_0^\infty \alpha_e^{m_e - 1} e^{-\alpha_e(1+\mathfrak{q}_e)}\, d\alpha_e = \Gamma(m_e)\,(1+\mathfrak{q}_e)^{-m_e}.$$

Integrating the scales first gives the right side of the statement, and integrating the latent vectors first gives $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$: both are the integral of one nonnegative function, finite or infinite together.

*(5) The Breta law.* Where the two sides are finite, the function

$$f(\alpha, x, x^*) = \alpha^{\mathbf{m-1}} \exp\Big( -\sum_e \alpha_e (1 + \mathfrak{q}_e) \Big)$$

is integrable, with integral $c_F J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ by step (4). Integrating out the latent vectors, its marginal in $\alpha$ is $c_F\, \alpha^{\mathbf{m-1}} e^{-|\alpha|} W(\alpha)$, proportional to the density $\alpha^{\mathbf{m-1}} e^{-|\alpha|} W(\alpha)/J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ of $\mathrm{Breta}(F;\mathbf{m};\mathbf{s},\mathbf{t})$ (8.10; the interest is finite exactly when every wall is positive, 8.5(i)). Given the latent vectors, $f$ is the product over the marks of $\alpha_e^{m_e-1} e^{-\alpha_e(1+\mathfrak{q}_e)}$: the scales are independent, and $\alpha_e(1+\mathfrak{q}_e)$ has the density $u^{m_e-1}e^{-u}/\Gamma(m_e)$ on $(0,\infty)$, the law $\mathrm{Gamma}_{m_e}$.

*(6) Equal exponents over* $\mathbb{R}$. If $\mathbb{F}_j = \mathbb{R}$ and $s_j = t_j$, the vectors of the two letters come in pairs $(x_{j,i}, x^*_{j,i})$, $i \le 2s_j$. The map $V_j \times V_j^\perp \to \mathbb{R}^{\mathcal{E}}$, $(x, x^*) \mapsto x + x^*$, is a linear isometry, so it carries the product of the Lebesgue measures to that of $\mathbb{R}^{\mathcal{E}}$, and for $u = x + x^*$, $x = \mathrm{pr}_j u$ and $x^* = (I - \mathrm{pr}_j)u$. The constant of table $j$ is then $K_j(\mathbf{1})^{2s_j}\pi^{ns_j}$. □

*Remark.* The reference, proof of 8.11, treats $s_j = t_j = k_j/2$ over $\mathbb{R}$ and opens the two letters of table $j$ with one set of $k_j$ vectors $u = \hat{A}^{\mathsf{T}} y + \hat{C}^{\mathsf{T}} y^*$; that is step (6). In the proof above each letter is opened on its own space, with as many vectors as its exponent asks, and over $\mathbb{C}$ by the complex Gaussian integral, $\beta = 2$.

**Lemma S3.** *For a family on $\mathcal{E}$ and a nonempty set $S \subseteq \mathcal{E}$, $c_j(S) = 0$ for every table $j$ of the family exactly when $S$ is a union of molecules.*

*Proof.* The molecules are the smallest nonempty sets of marks that contain, with each mark, every mark linked to it (7.4); so a set of marks with that property is a union of molecules, and a union of molecules has it.

Let $S$ be a union of molecules. By 7.9 every table is the direct sum of its restrictions to the molecules: its bases are the unions of one basis of each restriction, and a basis of the restriction to a molecule $M$ has $r_j(M)$ marks (7.5). So $|T \cap S| = \sum_{M \subseteq S} r_j(M)$ on every basis $T$ of table $j$, and by 7.13, $d_j(S) = r_j(S)$, that is $c_j(S) = 0$.

Let $S$ not be a union of molecules. Then some $e \in S$ is linked to some $f \notin S$: $e$ and $f$ lie in a common circuit of some table $j$. By 7.7 some basis $T$ of table $j$ has $e \in T$, $f \notin T$ and $T' = (T \setminus \{e\}) \cup \{f\}$ a basis. Then $|T' \cap S| = |T \cap S| - 1$, so $|T \cap S|$ takes two values on the bases of table $j$, and $c_j(S) \ge 1$ by 7.13. □

*Remark.* Since the weight law charges every basis, Lemma S3 is the first sentence of 13.9 read on the bases: $\mathrm{Var}_{T \sim \mathbb{P}_{j,\mathbf{1}}} |T \cap U| = 0$ exactly when $|T \cap U|$ is constant on $\mathcal{B}_j$, that is when $c_j(U) = 0$.

**Proposition 8.16 (The wall of a compound).** *As stated: the identity holds for every nonempty proper $S$, and only the tables with $\sigma_j \neq 0$ enter it.*

*Proof.* 8.13 gives, as identities of functions of the data, $\kappa_S + \kappa_S^{\mathrm{top}} = 2\sum_j \sigma_j c_j(S)$ and $\kappa_{\mathcal{E}\setminus S} = \kappa_{\mathcal{E}} + \kappa_S^{\mathrm{top}}$. Adding the two,

$$\kappa_S + \kappa_{\mathcal{E}\setminus S} = \kappa_{\mathcal{E}} + 2\sum_j \sigma_j c_j(S).$$

If $c_j(S) = 0$ in every table with $\sigma_j \neq 0$, the sum vanishes and $\kappa_S + \kappa_{\mathcal{E}\setminus S} = \kappa_{\mathcal{E}}$; this holds in particular when $S$ is a union of molecules, where $c_j(S) = 0$ in every table (Lemma S3). On the balance wall $\kappa_{\mathcal{E}} = 0$, so $\mathrm{Re}\,\kappa_S + \mathrm{Re}\,\kappa_{\mathcal{E}\setminus S} = 0$ and one of the two walls has real part at most zero. By 8.7 the period converges only where every nonempty proper wall has positive real part, so it diverges. □

*Remark.* The reference, 8.16, takes $c_j(S) = 0$ in every table. With complex exponents a table with $s_j = -t_j \neq 0$ is active and has $\sigma_j = 0$; it does not enter the identity.

**Definition 8.17 (Target; native gauge).** As stated: the target for all data with $m_{\mathcal{E}} \neq 0$ and on the balance wall, and native points of $(\mathbb{C}^\times)^{\mathcal{E}}$ off the zeros of the letters, the native gauges being the native points in $\mathbb{R}^{\mathcal{E}}_{>0}$.

*Proof.* The definition is consistent.

*(1) Scale.* $K_j$ and $\Psi_j$ are homogeneous polynomials of degrees $r_j$ and $r_j^* = n - r_j$, every basis having $r_j$ marks and every complement $n - r_j$ (7.1). So for $c \in \mathbb{C}^\times$, $K_j(c\alpha) = c^{r_j} K_j(\alpha)$ and $(\partial_e K_j)(c\alpha) = c^{r_j - 1}(\partial_e K_j)(\alpha)$, whence $R_{j,e}(c\alpha) = R_{j,e}(\alpha)$ wherever $K_j(\alpha) \neq 0$, and likewise $R^*_{j,e}(c\alpha) = R^*_{j,e}(\alpha)$ wherever $\Psi_j(\alpha) \neq 0$. The condition $K_j(\alpha)\Psi_j(\alpha) \neq 0$ is unchanged by scaling and the target does not involve the point, so the usage, and with it nativity, is a property of the point up to scale.

*(2) The two clauses of the target.* On the balance wall $m_{\mathcal{E}} = \sum_j (s_j r_j + t_j r_j^*)$ (8.1); where this is not zero the formula gives $\boldsymbol{\tau} = \mathbf{m}$, the value assigned on the balance wall. So the two clauses agree where both apply, and the second also covers the balance wall with $m_{\mathcal{E}} = 0$.

*(3) Positive points.* At $\alpha \in \mathbb{R}^{\mathcal{E}}_{>0}$ the letters are positive, the weights being positive, so every positive point lies off their zeros, and $R_{j,e}(\alpha) = \mathbb{P}_{j,\alpha}(e \in T)$ and $R^*_{j,e}(\alpha) = \mathbb{P}^*_{j,\alpha}(e \notin T)$ (7.16). □

*Remark.* The direction of a ray lies on the balance wall (38.1), so its target is its masses $\mathbf{m}_1$, and the native gauges at complex points of the direction are the points $\alpha_c$ of the letter complement with $c^{(1)}(\alpha_c) = 0$: the native gauges of the ray (38.4).

**Theorem 8.19 (The native gauge is the saddle).** *As stated, with the native gauges of Definition 8.17.*

*Proof.* At a gauge $\alpha = e^y$ the usage of Definition 8.17 is $\nabla\Phi(y)$: by the chain rule $\partial_{y_e} \log K_j(e^y) = \alpha_e \partial_e K_j(\alpha)/K_j(\alpha) = R_{j,e}(\alpha)$ and $\partial_{y_e} \log \Psi_j(e^y) = R^*_{j,e}(\alpha)$, and $\Phi$ is $\sum_j \left(s_j \log K_j + t_j \log \Psi_j\right)$ read at $e^y$ (8.18). For real data with $s_j, t_j \geq 0$, not all zero, and $\mathbf{m} > 0$, $m_{\mathcal{E}} > 0$, so the target of that definition is the one given by its formula, and its native gauges are the gauges of (i). With this reading the statement is the reference, proof of 8.19. □

**Proposition 8.20 (The native gauges of a compound).** *As stated: the weights need not factor except in the sentence on products, and the molecules are those of the active family.*

*Proof.* Write $g = \sum_j (s_j d_j + t_j d_j^*)$. An inactive table enters neither the potential, nor the target, nor $g$, nor the Newton sum $\mathcal{N}_F$, its terms carrying the factor zero; every statement below concerns the active family, whose molecules are $\mathcal{E}_1, \ldots, \mathcal{E}_q$.

*(1) The criterion of the saddle.* By 8.19(ii) a native gauge exists exactly when $\boldsymbol{\tau}$ lies in the relative interior of $\mathcal{N}_F = \sum_j (s_j P_j + t_j P_j^*)$, and it is then unique modulo the space $V$ of directions along which $\Phi$ is affine. By 8.19(iii), whose first sentence uses no connectedness, $\mathcal{N}_F = \{x : \ x_{\mathcal{E}} = g(\mathcal{E}),\ x_S \geq g(S) \text{ for all } S\}$.

*(2) The extent of $x_S$ on $\mathcal{N}_F$.* The least value of a linear function on a Minkowski sum with nonnegative coefficients is the same sum of its least values on the summands. The least value of $x_S$ on $P_j = \operatorname{conv}\{\mathbf{1}_T\}$ is $\min_T |T \cap S| = d_j(S)$, and on $P_j^* = \operatorname{conv}\{\mathbf{1}_{\mathcal{E}\setminus T}\}$ it is $\min_T |S \setminus T| = d_j^*(S)$ (7.13). So the least value of $x_S$ on $\mathcal{N}_F$ is $g(S)$, and that of $x_{\mathcal{E}\setminus S}$ is $g(\mathcal{E} \setminus S)$; since $x_{\mathcal{E}} = g(\mathcal{E})$ on $\mathcal{N}_F$ and $\mathcal{N}_F$ is convex, $x_S$ takes on $\mathcal{N}_F$ exactly the values of $[g(S),\ g(\mathcal{E}) - g(\mathcal{E} \setminus S)]$. At any masses $\kappa_S + \kappa_{\mathcal{E}\setminus S} - \kappa_{\mathcal{E}} = g(\mathcal{E}) - g(S) - g(\mathcal{E} \setminus S)$, the masses cancelling, so by 8.16 the length of this interval is $2\sum_j \sigma_j c_j(S) = \sum_j (s_j + t_j) c_j(S)$.

*(3) The relative interior.* An active table has $s_j + t_j > 0$. So the interval of step (2) is a point exactly when $c_j(S) = 0$ for every active table, that is, by Lemma S3 applied to the active family, when $S$ is a union of the $\mathcal{E}_i$. The relative interior of a polyhedron given by linear equalities and inequalities consists of its points at which every inequality that is not an equality on the whole polyhedron holds strictly. So $\boldsymbol{\tau}$ lies in the relative interior of $\mathcal{N}_F$ exactly when $\tau_S = g(S)$ for every union $S$ of the $\mathcal{E}_i$ and $\tau_S > g(S)$ for every other nonempty $S$.

*(4) Molecule by molecule.* By 7.9 for the active family, every active table is the direct sum of its restrictions to the $\mathcal{E}_i$, its bases being the unions of one basis of each restriction. Taking the least of $|T\cap S|$, and of $|S\setminus T|$, restriction by restriction, $d_j(S)=\sum_i d_j(S\cap\mathcal{E}_i)$ and $d_j^*(S)=\sum_i d_j^*(S\cap\mathcal{E}_i)$ for every active $j$; so $g(S)=\sum_i g(S\cap\mathcal{E}_i)$ and $\tau_S-g(S)=\sum_i\big(\tau_{S\cap\mathcal{E}_i}-g(S\cap\mathcal{E}_i)\big)$. Hence the conditions of step (3) hold exactly when $\tau_{\mathcal{E}_i}=g(\mathcal{E}_i)$ for every $i$ and $\tau_S>g(S)$ for every $S$ with $\varnothing\neq S\subsetneq\mathcal{E}_i$ for some $i$: given these, a union of the $\mathcal{E}_i$ has $\tau_S-g(S)=0$, and any other nonempty $S$ meets some $\mathcal{E}_i$ in a nonempty proper subset, whose term is positive while the others are nonnegative; conversely each $\mathcal{E}_i$ is a union of the $\mathcal{E}_i$, and a nonempty proper subset of one $\mathcal{E}_i$ is not. Since $c_j(\mathcal{E}_i)=0$ for every active $j$, $d_j(\mathcal{E}_i)=r_j(\mathcal{E}_i)$, so $r_j(\mathcal{E}\setminus\mathcal{E}_i)=r_j-r_j(\mathcal{E}_i)$ (7.12, $d_j(S)=r_j-r_j(\mathcal{E}\setminus S)$) and $d_j^*(\mathcal{E}_i)=|\mathcal{E}_i|-r_j(\mathcal{E}_i)=r_j^*(\mathcal{E}_i)$. So $g(\mathcal{E}_i)=\sum_j\big(s_jr_j(\mathcal{E}_i)+t_jr_j^*(\mathcal{E}_i)\big)$, and the conditions are the balance of every $\mathcal{E}_i$ at the target and the positivity of its proper walls there. This proves the criterion.

*(5) The free scales.* By 7.17 the second derivative of $\log K_j(e^y)$ along a direction $v$ is the variance of $\langle v,\mathbf{1}_T\rangle$ under $\mathbb{P}_{j,e^y}$, and that of $\log\Psi_j(e^y)=\langle\mathbf{1},y\rangle+\log K_j(e^{-y})$ (7.2) is the variance of $\langle v,\mathbf{1}_T\rangle$ under $\mathbb{P}_{j,e^{-y}}$. Both laws charge every basis, so each variance vanishes exactly when $\langle v,\mathbf{1}_T\rangle$ is constant over $\mathcal{B}_j$. The second derivative of $\Phi$ along $v$ is the sum of these variances with the coefficients $s_j,t_j\geq 0$, and every active table has a positive coefficient; so $\Phi$ is affine along $v$ exactly when $\langle v,\mathbf{1}_T\rangle$ is constant over the bases of every active table. By 7.8 for the active family such a $v$ is constant on every $\mathcal{E}_i$. Conversely, if $v$ is constant on every $\mathcal{E}_i$, with value $v_i$, then $\langle v,\mathbf{1}_T\rangle=\sum_i v_i|T\cap\mathcal{E}_i|=\sum_i v_ir_j(\mathcal{E}_i)$ on every basis of an active table $j$, by step (4). So $V=\operatorname{span}\{\mathbf{1}_{\mathcal{E}_1},\dots,\mathbf{1}_{\mathcal{E}_q}\}$, and the native gauges are the gauges $e^v\alpha^\star$, $v\in V$: the gauges $\beta$ with $\beta/\alpha^\star$ constant on every $\mathcal{E}_i$.

*(6) Factoring weights.* When the weights factor across the $\mathcal{E}_i$, the letters of every active table factor,

$$K_j(\alpha)=\prod_i K_{j,i}(\alpha_{\mathcal{E}_i}),\qquad \Psi_j(\alpha)=\prod_i\Psi_{j,i}(\alpha_{\mathcal{E}_i}),$$

with the Foreman polynomials of the restrictions (7.9). So $\Phi(y)=\sum_i\Phi_i(y_{\mathcal{E}_i})$, with $\Phi_i$ the potential of the restriction of the family to $\mathcal{E}_i$, and the condition $\nabla\Phi=\boldsymbol{\tau}$ splits into the conditions $\nabla\Phi_i=\boldsymbol{\tau}|_{\mathcal{E}_i}$, each the native condition of the restriction to $\mathcal{E}_i$ at the masses $\boldsymbol{\tau}|_{\mathcal{E}_i}$, which lie on its balance wall by step (4), so that its target is $\boldsymbol{\tau}|_{\mathcal{E}_i}$ (8.17). The native gauges are the gauges whose restriction to every $\mathcal{E}_i$ is native there: the products of native gauges of the molecules.

*(7) Units.* A loop is a circuit by itself, so it lies in no other circuit; and a coloop $e$ lies in no circuit, since a circuit $C$ through $e$ would lie in every basis containing the independent set $C\setminus\{e\}$, such a basis containing $e$, while a circuit lies in no basis. So a mark $e$ that is a loop or a coloop in every table is linked to no other mark, and $\{e\}$ is a unit. Every basis of table $j$ contains $e$, or none does, so $K_j(\alpha)=\alpha_e^{\epsilon_j}\tilde K_j(\alpha_{\mathcal{E}\setminus e})$ and $\Psi_j(\alpha)=\alpha_e^{1-\epsilon_j}\tilde\Psi_j(\alpha_{\mathcal{E}\setminus e})$ with $\epsilon_j\in\{0,1\}$: the letters of $\{e\}$ are monomials. □

*Remark.* The criterion is stated for the molecules of the active family because linking (7.4) runs through the circuits of every table, while an inactive table enters neither the potential, nor the target, nor the walls. So a molecule of the family may join several molecules of the active family, and then a proper wall of that molecule vanishes at the target although a native gauge exists. For instance, on $\mathcal{E}=\{1,2,3,4\}$ let the list with the bases $\{1,3\},\{1,4\},\{2,3\},\{2,4\}$ carry $(s,t)=(1,0)$ and the list $\binom{\mathcal{E}}{1}$ carry $(0,0)$, with the masses $\frac12\mathbf{1}$ on the balance wall. Through the second table $\mathcal{E}$ is one molecule of the family, while the active family has the molecules $\{1,2\}$ and $\{3,4\}$. Every mark lies in half of the bases of the first table, so the gauge $\mathbf{1}$ is native, and the wall at $\{1,2\}$ is $\tau_{\{1,2\}}-d_1(\{1,2\})=1-1=0$.

*Remark.* At the centre of 8.21 the gauge **1** is native, and step (5) shows that the native gauges are the gauges constant on every molecule of the active family: this is the uniqueness clause of 8.21(i).

## 50.9 Chapter 9: realizations and the still

**Proposition 9.2 (A realization sits on a wall only along its molecules).** *As stated, for every family.*

*Proof. (1) The walls as expectations.* By 7.16, $R_{j,e}(\alpha) = \mathbb{P}_{j,\alpha}(e \in T)$ and $R^*_{j,e}(\alpha) = \mathbb{P}^*_{j,\alpha}(e \notin T)$. Summing the usage over $e \in S$,

$$m_S = \sum_j \Big( s_j \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[N_S] + t_j \mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha}}\big[|S| - N_S\big] \Big),$$

and with $d^*_j(S) = |S| - r_j(S)$ (7.12) the wall $\kappa_S = m_S - \sum_j \big(s_j d_j(S) + t_j d^*_j(S)\big)$ is the displayed sum.

*(2) The sign.* By 7.13, $d_j(S) \le N_S \le r_j(S)$ on every basis of table $j$. So both brackets are nonnegative, and so is $\kappa_S$, the exponents being nonnegative.

*(3) Equality.* The laws $\mathbb{P}_{j,\alpha}(T) = \theta_{j,T}\alpha^T/K_j(\alpha)$ and $\mathbb{P}^*_{j,\alpha}(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j(\alpha)$ charge every basis. So the first bracket of table $j$ vanishes exactly when $N_S = d_j(S)$ on every basis, and the second exactly when $N_S = r_j(S)$ on every basis; since the values of $N_S$ fill $\{d_j(S), \dots, r_j(S)\}$ (7.13), each happens exactly when $c_j(S) = 0$. Every table has $s_j > 0$ or $t_j > 0$, so $\kappa_S = 0$ exactly when $c_j(S) = 0$ for every table, that is, by Lemma S3, when $S$ is a union of molecules.

*(4) The last sentence.* $\mathcal{E}$ is a union of molecules, so $\kappa_{\mathcal{E}} = 0$. At a realization every table is active, so the family is connected (8.1) exactly when no nonempty proper $S$ has $c_j(S) = 0$ for every table, that is, by Lemma S3, when no nonempty proper set is a union of molecules; then every proper wall is positive by step (3). □

*Remark.* The reference, proof of 9.2, has steps (1) and (2) and the positivity of the bracket of an active table with $c_j(S) \ge 1$, which connectedness supplies for every nonempty proper $S$. Step (3) reads off the equality case, which holds for every family.

**Proposition 9.3 (Masses at a gauge).** *As stated, at every gauge $e^{y_0}$.*

*Proof.* By 9.1 the masses are $\tau_e = \sum_j \big(s_j R_{j,e}(\alpha) + t_j R^*_{j,e}(\alpha)\big)$, with $R_{j,e} = \alpha_e \partial_e K_j/K_j$ the occupation and $R^*_{j,e} = \alpha_e \partial_e \Psi_j/\Psi_j$ the dual occupation, and $R^*_{j,e}(\alpha) = 1 - R_{j,e}(\alpha^{-1})$ (7.16). At $\alpha = e^{y_0}$ this is the first expression. Put $a_j = R_{j,e}(e^{y_0})$ and $b_j = R_{j,e}(e^{-y_0})$, so that $R^{\pm}_{j,e} = \frac{1}{2}(a_j \pm b_j)$. With $s_j = \sigma_j + \delta_j$ and $t_j = \sigma_j - \delta_j$,

$$s_j a_j + t_j(1 - b_j) = \sigma_j(1 + a_j - b_j) + \delta_j(a_j + b_j - 1) = \sigma_j(1 + 2R^-_{j,e}) + \delta_j(2R^+_{j,e} - 1).$$

At $y_0 = 0$, $a_j = b_j = R_{j,e}(\mathbf{1})$: $R^-_{j,e} = 0$ and $R^+_{j,e} = R_{j,e}(\mathbf{1})$. □

*Remark.* By 7.17, $R_{j,e}(e^y) = \partial_{y_e} \log K_j(e^y)$, whose Taylor series at 0 is

$$\sum_{i\ge 0} \frac{1}{i!} \sum_{u\in\mathcal{E}^i} \kappa_{j,\mathbf{1}}(e,u) \prod_{l=1}^{i} y_{u_l}$$

in its domain of convergence, as in 9.6. At $y = \pm y_0$ the term of order $i$ carries $(\pm 1)^i$, so $R^+_{j,e}$ collects the terms of even order and $R^-_{j,e}$ those of odd order. Rung one at a deviation has the

parity of the odd rungs of 9.6, $\sigma_j$ with the odd part of the occupations and $\delta_j$ with the even part, besides the constant $\sum_j(\sigma_j-\delta_j)=\sum_j t_j$.

**Proposition 9.5 (The potential of a compound splits by molecules).** *As stated; the last sentence is new.*

*Proof.* The first two sentences are the reference, proof of 9.5. Let $e$ be a loop or a coloop in every table. By 8.20 $\{e\}$ is a unit, and every basis of table $j$ contains $e$, or none does, so $K_j(\alpha)=\alpha_e^{\epsilon_j}\tilde K_j(\alpha_{\mathcal E\setminus e})$ and $\Psi_j(\alpha)=\alpha_e^{1-\epsilon_j}\tilde\Psi_j(\alpha_{\mathcal E\setminus e})$ with $\epsilon_j\in\{0,1\}$. So $\Phi(y)=\sum_j\big(s_j\epsilon_j+t_j(1-\epsilon_j)\big)y_e$ plus a function of $y_{\mathcal E\setminus e}$ (8.18), every derivative of $\Phi$ of order at least two that involves $y_e$ vanishes, every entry of the still on a string through $e$ is zero, and rung one at $e$ is the constant $\sum_j\big(s_j\epsilon_j+t_j(1-\epsilon_j)\big)$, the mass of $e$ (9.4). The same holds for the two parts $\Phi_{\mathbf E}$ and $\Phi_{\mathbf O}$, which are $\sum_j\sigma_j\log(K_j\Psi_j)$ and $\sum_j\delta_j\log(K_j/\Psi_j)$ read at $e^y$. □

**Proposition 9.10 (Occupations at a gauge).** *As stated, at every gauge $e^{y_0}$.*

*Proof.* By 9.4, $\varphi_e^{\mathbf X}(0)=\partial_{y_e}\Phi_{\mathbf X}(\log\alpha)$, and by 8.18, $\Phi_{\mathbf E}(y)=\sum_j\sigma_j\big(\log K_j(e^y)+\log\Psi_j(e^y)\big)$ and $\Phi_{\mathbf O}(y)=\sum_j\delta_j\big(\log K_j(e^y)-\log\Psi_j(e^y)\big)$. By 8.19, $\partial_{y_e}\log K_j(e^y)=R_{j,e}$ and $\partial_{y_e}\log\Psi_j(e^y)=R^*_{j,e}$, and $R^*_{j,e}(\alpha)=1-R_{j,e}(\alpha^{-1})$ (7.16). At $\alpha=e^{y_0}$, with $a_j=R_{j,e}(e^{y_0})$ and $b_j=R_{j,e}(e^{-y_0})$,

$$\varphi_e^{\mathbf E}(0)=\sum_j\sigma_j(a_j+1-b_j)=\sum_j\sigma_j(1+2R^-_{j,e}),$$
$$\varphi_e^{\mathbf O}(0)=\sum_j\delta_j(a_j-1+b_j)=\sum_j\delta_j(2R^+_{j,e}-1).$$

At $\alpha=\mathbf 1$, $R^-_{j,e}=0$ and $R^+_{j,e}=R_{j,e}(\mathbf 1)$. The two parts add to the mass of 9.3. □

## 50.10 Chapter 10: the reading, classes, atoms

**Proposition 10.2 (A view through the patterns).** *As stated; the inclusion laws are read at every gauge.*

*Proof.* The expression of the views through $K_{j;e,\mathbf g}$ and $\Psi_{j;e,\mathbf g}$ and the closing sentence on one table are proved in the reference, proof of 10.2. What follows is the expansion at a gauge $\alpha$ and its reading at the table's gauge. Since $\prod_{i\in p}(1+x_i)=\sum_{A\subseteq p}x^A$,

$$K_{j;e,\mathbf g}(\mathbf 1+x)=\sum_p\vartheta^K_{j,p}\sum_{A\subseteq p}x^A=\sum_A x^A\sum_{p\supseteq A}\vartheta^K_{j,p},$$
$$\sum_{p\supseteq A}\vartheta^K_{j,p}=\sum_{T:\ p(T)\supseteq A}\theta_{j,T}\alpha^T=K_j(\alpha)\,\mathbb P_{j,\alpha}\big(p(T)\supseteq A\big),$$

the basis law at $\alpha$ being $\mathbb P_{j,\alpha}(T)=\theta_{j,T}\alpha^T/K_j(\alpha)$ (7.16). Likewise $(\mathbf 1+x)^{\bar p}=\prod_{i\notin p}(1+x_i)=\sum_{A\cap p=\varnothing}x^A$, so

$$\Psi_{j;e,\mathbf g}(\mathbf 1+x)=\sum_A x^A\sum_{p\cap A=\varnothing}\vartheta^\Psi_{j,p},$$
$$\sum_{p\cap A=\varnothing}\vartheta^\Psi_{j,p}=\sum_{T:\ p(T)\cap A=\varnothing}\theta_{j,T}\alpha^{\mathcal E\setminus T}=\Psi_j(\alpha)\,\mathbb P^*_{j,\alpha}\big(p(T)\cap A=\varnothing\big),$$

with $\mathbb{P}^*_{j,\alpha}(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j(\alpha)$. So the polynomials $K_{j;e,\mathbf{g}}$ and $\Psi_{j;e,\mathbf{g}}$, and with them the view, are determined by $K_j(\alpha)$, $\Psi_j(\alpha)$ and the joint inclusion laws of $e, g_1, \dots, g_k$ under $\mathbb{P}_{j,\alpha}$ and $\mathbb{P}^*_{j,\alpha}$. At $\alpha = \mathbf{1}$, $\alpha^T = \alpha^{\mathcal{E}\setminus T} = 1$: $\vartheta^K_{j,p} = \vartheta^\Psi_{j,p} = \vartheta_{j,p}$, both laws are the weight law (7.16), and $x^{\bar{p}} = x^{\{0,\dots,k\}}(x^{-1})^p$ gives $\Psi_{j;e,\mathbf{g}}(x) = x^{\{0,\dots,k\}}K_{j;e,\mathbf{g}}(x^{-1})$. □

**Proposition 10.8 (Computing the classes).** *As stated; at a gauge $\alpha$ the identities have their coefficients in the field generated by the weights and the coordinates of $\alpha$.*

*Proof.* The steps of the computation compare views and writings, as in the reference, proof of 10.8; only the field is new. Let $k$ be the field generated by the weights and the coordinates of $\alpha$. By 10.2, a view is the pair $\sum_j \sigma_j \log(K_{j;e,\mathbf{g}}\Psi_{j;e,\mathbf{g}})$, $\sum_j \delta_j \log(K_{j;e,\mathbf{g}}/\Psi_{j;e,\mathbf{g}})$ in $x_i = e^{v_i}$, and the coefficients $\vartheta^K_{j,p} = \sum_{T:p}\theta_{j,T}\alpha^T$ and $\vartheta^\Psi_{j,p} = \sum_{T:p}\theta_{j,T}\alpha^{\mathcal{E}\setminus T}$ of these polynomials are polynomials with integer coefficients in the weights and the coordinates of $\alpha$, so they lie in $k$. Factor every $K_{j;e,\mathbf{g}}$ and $\Psi_{j;e,\mathbf{g}}$ into irreducible polynomials over $k$. Each factor of a polynomial positive on the orthant vanishes nowhere on it and so has constant sign there, and it is normalized to the value 1 at $x = \mathbf{1}$ by a constant of $k$. Two different normalized polynomials irreducible over $k$ have no common factor over $\mathbb{C}$, the greatest common divisor of two polynomials over $k$ being their greatest common divisor over $\mathbb{C}$. So the factors over $\mathbb{C}$ of 10.6 are the factors over $\mathbb{C}$ of the factors over $k$, and a factor over $\mathbb{C}$ divides exactly one normalized factor over $k$. If two views carry over $\mathbb{C}$ the same factors with the same pairs of exponents, a factor over $k$ of one and a factor over $k$ of the other that share a factor over $\mathbb{C}$ are equal, and comparing the exponents of that shared factor gives equal pairs of exponents. So by 10.6 two views, or two writings, are equal exactly when they carry the same normalized polynomials irreducible over $k$ with the same pairs of exponents: every step is an identity in $k[x]$. At $\alpha = \mathbf{1}$, $k$ is the field generated by the weights, $\mathbb{Q}$ for rational weights, and the coefficients are integers for lists. □

**Proposition 10.9 (Masses and occupations on a class).** *As stated; the first sentence holds for each of the two parts of the mass.*

*Proof.* By 9.4, $\varphi^{\mathbf{X}}_e(0) = \partial_{y_e}\Phi_{\mathbf{X}}(\log\alpha)$ is the part $\mathbf{X}$ of rung one, and $\tau_e(\Upsilon) = \partial_{y_e}\Phi(\log\alpha) = \varphi^{\mathbf{E}}_e(0) + \varphi^{\mathbf{O}}_e(0)$. By 10.3(iii) the reading of $e$ determines its self-line functions $\varphi^{\mathbf{E}}_e$ and $\varphi^{\mathbf{O}}_e$, hence their values at 0, and the classes are the level sets of the reading (10.1). So the two parts, and their sum, are constant on every class. For a family whose weights factor across its molecules, $\Phi_{\mathbf{X}}$ is the sum of the potentials of the restrictions to the molecules (9.5), so $\varphi^{\mathbf{X}}_e(0)$ is the same in the family and in the realization of the molecule of $e$, on which its classes are read. Items (i) and (ii) are proved in the reference, proof of 10.9. □

## 50.11 Chapter 11: lean, gauge and label

**Lemma 11.11 (No degenerate pair).** *As stated, at every realization of every family, with the equality case.*

*Proof. (1) A sum of covariances.* By 9.4 and 8.18, $\kappa^{\mathbf{E}}_\Upsilon(e,f) = 4\sum_j \sigma_j\big(\partial_{y_e}\partial_{y_f}\log K_j(e^y) + \partial_{y_e}\partial_{y_f}\log\Psi_j(e^y)\big)$ at $y = \log\alpha$. By 7.17 the first derivative is $\kappa_{j,\alpha}(e,f)$, and since $\log\Psi_j(e^y) = \langle\mathbf{1},y\rangle + \log K_j(e^{-y})$ (7.2) the second is $\kappa_{j,\alpha^{-1}}(e,f)$. So $\kappa^{\mathbf{E}}_\Upsilon = 4\sum_j\sigma_j\big(C_{j,\alpha} + C_{j,\alpha^{-1}}\big)$, $C_{j,\beta}$ the covariance matrix of the indicators $(X_e)_e$ under $\mathbb{P}_{j,\beta}$, and for $w \in \mathbb{R}^{\mathcal{E}}$

$$w^{\mathsf{T}}\kappa^{\mathbf{E}}_\Upsilon w = 4\sum_j \sigma_j\Big(\operatorname{Var}_{T\sim\mathbb{P}_{j,\alpha}}\langle w,\mathbf{1}_T\rangle + \operatorname{Var}_{T\sim\mathbb{P}_{j,\alpha^{-1}}}\langle w,\mathbf{1}_T\rangle\Big) \;\geq\; 0,$$

since $\sigma_j > 0$ at a realization.

*(2) The inequality.* The principal $2 \times 2$ submatrix of $\kappa^{\mathbf{E}}_{\Upsilon}$ on $\{e, f\}$ is positive semidefinite, so its determinant $\kappa^{\mathbf{E}}_{\Upsilon}(e,e)\kappa^{\mathbf{E}}_{\Upsilon}(f,f) - \kappa^{\mathbf{E}}_{\Upsilon}(e,f)^2$ is nonnegative. It vanishes exactly when that submatrix is singular, that is when some $w = \lambda \mathbf{1}_e + \mu \mathbf{1}_f \neq 0$ has $w^{\mathsf{T}} \kappa^{\mathbf{E}}_{\Upsilon} w = 0$, a positive semidefinite matrix annihilating exactly the vectors at which its form vanishes.

*(3) The vectors of form zero.* Both laws in step (1) charge every basis (7.16), so $w^{\mathsf{T}} \kappa^{\mathbf{E}}_{\Upsilon} w = 0$ exactly when $\langle w, \mathbf{1}_T \rangle$ is constant over the bases of every table. By 7.8 such a $w$ is constant on every molecule. Conversely, if $w$ is constant on every molecule, with value $w_M$ on the molecule $M$, then $\langle w, \mathbf{1}_T \rangle = \sum_M w_M r_j(M)$ on every basis of table $j$, each table being the direct sum of its restrictions to the molecules (7.9).

*(4) The cases.* If $e$ is a unit, $w = \mathbf{1}_e$ is constant on every molecule, and equality holds; likewise if $f$ is a unit. If neither is a unit and they lie in different molecules, the molecule of $e$ has a mark other than $e$, not $f$, at which $w = \lambda \mathbf{1}_e + \mu \mathbf{1}_f$ vanishes, so $\lambda = 0$; likewise $\mu = 0$, and the inequality is strict. If $e$ and $f$ lie in one molecule $M$: when $M$ has a third mark, $w$ vanishes there, so $\lambda = \mu = 0$ and the inequality is strict; when $M = \{e, f\}$, $w = \mathbf{1}_e + \mathbf{1}_f$ is constant on every molecule, and equality holds. A molecule with at least three marks contains neither a unit nor a molecule of two marks, so the inequality is strict for any two of its marks. □

*Remark.* The reference, proof of 11.11, has steps (1) to (3) and concludes for a molecule with at least three marks; the case analysis of step (4) gives the equality case for every family.

**Lemma 11.12 (Equal diagonals on a class).** *As stated: on every class, for both parts and every rung.*

*Proof.* By 9.4, $\varphi^{\mathbf{X}}_e(v) = \partial_{y_e} \Phi_{\mathbf{X}}(\log \alpha + v \mathbf{1}_e)$, so $(\varphi^{\mathbf{X}}_e)^{(k-1)}(0) = \partial^k_{y_e} \Phi_{\mathbf{X}}(\log \alpha)$, and the still of each part gives

$$\kappa^{\mathbf{X}}_{\Upsilon}(e, \dots, e) = 2^k \, \partial^k_{y_e} \Phi_{\mathbf{X}}(\log \alpha) = 2^k (\varphi^{\mathbf{X}}_e)^{(k-1)}(0).$$

By 10.3(iii) the reading of $e$ determines $\varphi^{\mathbf{E}}_e$ and $\varphi^{\mathbf{O}}_e$; the marks of a class have one reading (10.1), so they have the same self-line functions, and the same derivatives of every order at 0. At an atom there is one class. □

*Remark.* The reference states 11.12 at an atom of a molecule with at least three marks, for $\kappa^{\mathbf{E}}(e,e)$, and proves it through the rung-two tallies and 11.11. The self-line functions carry every order and both parts, and make the statement one about classes.

**Corollary 11.15 (An exchange negates odd cumulants).** *As stated, for an exchange of any family.*

*Proof. (1) The weight laws.* By 11.13, $T \mapsto T' = \mathcal{E} \setminus g(T)$ maps $\mathcal{B}_j$ onto $\mathcal{B}_{h(j)}$ with $\theta_{h(j),T'} = \theta_{j,T}$, and it is injective because $g$ is. So $K_{h(j)}(\mathbf{1}) = K_j(\mathbf{1})$, and for $T \sim \mathbb{P}_{j,\mathbf{1}}$ the basis $T'$ has the law $\mathbb{P}_{h(j),\mathbf{1}}$.

*(2) The cumulants.* $X_{g(e)}(T') = \mathbf{1}[g(e) \notin g(T)] = 1 - X_e(T)$. Joint cumulants of order at least two are multilinear and unchanged when constants are added to the variables, so for a string $w = (e_1, \dots, e_k)$, $k \geq 2$,

$$\begin{aligned} \kappa_{h(j),\mathbf{1}}(g(w)) &= \kappa\big(X_{g(e_1)}(T'), \dots, X_{g(e_k)}(T')\big) \\ &= \kappa\big(1 - X_{e_1}(T), \dots, 1 - X_{e_k}(T)\big) = (-1)^k \kappa_{j,\mathbf{1}}(w). \end{aligned}$$

*(3) The stills.* By 9.7, $\kappa_{\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}}(w) = 2^k \sum_j \big(s_j + (-1)^k t_j\big)\kappa_{j,\mathbf{1}}(w)$. The condition on the pairs in 11.13 gives $(s_{h(j)}, t_{h(j)}) = (s_j, t_j)$, and $h$ is a bijection of the tables; reindexing the sum by $h$ and using step (2),

$$\begin{aligned}\kappa_{\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}}(g(w)) &= 2^k \sum_j \big(s_{h(j)} + (-1)^k t_{h(j)}\big)\kappa_{h(j),\mathbf{1}}(g(w)) \\ &= 2^k \sum_j \big(t_j + (-1)^k s_j\big)\kappa_{j,\mathbf{1}}(w),\end{aligned}$$

the still at $\Upsilon_{\boldsymbol{\sigma},-\boldsymbol{\delta};\mathbf{1}}$, whose exponents are $(t_j, s_j)$.

*(4) The classes of the centre.* By 11.14 with $F' = F$ at the table's gauge, the reading of $g(e)$ at $-\boldsymbol{\delta}$ equals that of $e$ at $\boldsymbol{\delta}$; at the centre $-\boldsymbol{\delta} = \boldsymbol{\delta} = 0$, so $g(e)$ lies in the class of $e$. Hence $g$ maps every class of the centre into itself, and onto itself, $g$ being injective and the class finite. □

*Remark.* The reference, proof of 11.15, is the case of one table, where $h$ is the identity.

## 50.12 Chapter 12: restriction to a bunch

**Lemma 12.3 (Children do not widen).** *As stated.*

*Proof.* The child at $p$ is a table on $R$ (12.2), and a set $X \subseteq R$ is one of its bases exactly when $p \sqcup X$ is a basis of the table. Since $p, W \subseteq B$ and $X, A \subseteq R$,

$$|X \cap A| = |(p \sqcup X) \cap (A \cup W)| - |p \cap W|.$$

So the traces of $A$ on the bases of the child are traces of $A \cup W$ on bases of the table, all shifted by the one constant $|p \cap W|$. The width is the largest trace less the least (7.13), so $c_{(p)}(A) \le c(A \cup W)$. □

**Proposition 12.13 (Coins of a tensorminental table).** *As stated: tensor rank one over* $\mathbb{R}$ *or over* $\mathbb{C}$.

*Proof.* The first display is proved in the reference, proof of 12.13. At tensor rank one there is one slice $A \in \mathbb{F}^{r\times n}$, of rank $r$ since $K(\mathbf{1}) = \det(AA^{\mathsf{H}}) > 0$ (7.21); the only pair is $(1,1)$, with $w_{11} = 1$, and $\Pi^{(11)} = A^{\mathsf{H}} M^{-1} A D_\alpha$, $M = A D_\alpha A^{\mathsf{H}}$ (7.23). So $x^{d(B)} \mathsf{Q}_B^K(x;\alpha) = \prod_{i=1}^{|B|}(1 - p_i + p_i x)$ with $p_i$ the eigenvalues of $\Pi^{(11)}_{BB}$.

*(1) The projection.* Put $P = D_\alpha^{1/2} \Pi^{(11)} D_\alpha^{-1/2} = D_\alpha^{1/2} A^{\mathsf{H}} M^{-1} A D_\alpha^{1/2}$. Then $P^{\mathsf{H}} = P$ and $P^2 = D_\alpha^{1/2} A^{\mathsf{H}} M^{-1} (A D_\alpha A^{\mathsf{H}}) M^{-1} A D_\alpha^{1/2} = P$, and $P\, D_\alpha^{1/2} A^{\mathsf{H}} = D_\alpha^{1/2} A^{\mathsf{H}}$; so $P$ is the orthogonal projection onto $\{D_\alpha^{1/2} A^{\mathsf{H}} y\}$, the row space of $A D_\alpha^{1/2}$ in the sense of 7.23.

*(2) Similarity.* $D_\alpha$ is diagonal, so $P_{BB} = D_B^{1/2} \Pi^{(11)}_{BB} D_B^{-1/2}$, with $D_B$ the diagonal block on $B$, and $P_{BB}$ has the eigenvalues $p_i$.

*(3) Principal angles.* $P_{BB}$ is the compression of $P$ to $\mathbb{F}^B$, the vectors supported on $B$: $v^{\mathsf{H}} P_{BB} v = \|Pv\|^2$ for $v \in \mathbb{F}^B$. With $E_B$ the inclusion of $\mathbb{F}^B$ and $Q$ a matrix whose columns are an orthonormal basis of the range of $P$, $P_{BB} = (E_B^{\mathsf{H}} Q)(E_B^{\mathsf{H}} Q)^{\mathsf{H}}$. Its eigenvalues are the squares of the singular values of $E_B^{\mathsf{H}} Q$, completed by zeros, which are the cosines of the principal angles between $\mathbb{F}^B$ and the range of $P$; they lie in $[0,1]$.

*(4) The eigenvalue* 1. For $v \in \mathbb{F}^B$, $P_{BB} v = v$ exactly when $\|Pv\| = \|v\|$, that is when $v$ lies in the range of $P$. So the eigenspace is $\{D_\alpha^{1/2} A^{\mathsf{H}} y :\ (D_\alpha^{1/2} A^{\mathsf{H}} y)_R = 0\} = \{D_\alpha^{1/2} A^{\mathsf{H}} y :\ A_R^{\mathsf{H}} y = 0\}$, of dimension $r - \operatorname{rank} A_R$, the map $y \mapsto D_\alpha^{1/2} A^{\mathsf{H}} y$ being injective. The bases of the table are the

sets $T$ with $\det A_T \neq 0$ (7.20), so $r(R) = \operatorname{rank} A_R$, and $r - r(R) = d(B)$ (7.12, $d(B) = |B| - r^*(B)$ with $r^*(B) = |B| + r(R) - r$).

*(5) The eigenvalue* 0. For $v \in \mathbb{F}^B$, $P_{BB}v = 0$ exactly when $Pv = 0$, that is $AD_\alpha^{1/2}v = A_B D_B^{1/2} v_B = 0$. So the eigenspace has dimension $|B| - \operatorname{rank} A_B = |B| - r(B) = d^*(B)$.

*(6) The coins.* $P_{BB}$ is Hermitian, so of its $|B|$ eigenvalues $d(B)$ are 1, $d^*(B)$ are 0, and the other $|B| - d(B) - d^*(B) = c(B)$ lie in $(0,1)$. In $x^{d(B)}\mathsf{Q}_B^K(x;\alpha) = \prod_i (1 - p_i + p_i x)$ the eigenvalues 1 give the factor $x^{d(B)}$ and the eigenvalues 0 the factor 1; so the coins of $\mathsf{Q}_B^K$ are the eigenvalues $p_i = \cos^2 \vartheta_i \in (0,1)$, and the Skeleton values are $\xi_i = (1 - p_i)/p_i = \tan^2 \vartheta_i > 0$ (12.10). □

*Remark.* Over $\mathbb{R}$ this is the reference, proof of 12.13. Over $\mathbb{C}$ the projection is Hermitian, and each step reads the same with conjugate transposes.

**Definition 12.18: the claims.** $\operatorname{pat}(B) = 2^B$ holds exactly when $B$ is independent ($B$ is a pattern) and some basis avoids $B$ ($\varnothing$ is a pattern): if both hold, $d(B) = 0$, so the contraction $\mathcal{B}/R$ has rank 0, and every subset of $B$ is independent with $r_{\mathcal{B}/R} = 0 = d(B)$, a pattern by Proposition 12.17(i). Equivalently $r(B) = |B|$ and $d(B) = 0$, that is $c(B) = |B|$. The dual table has the bases $\mathcal{E} \setminus T$ and the patterns $B \setminus p$, with the children of cobasis letters $\Psi_{(p)}$; so every subset of $B$ is a pattern of the dual table when every subset is a pattern of the table, and its interaction is $\sum_p (-1)^{|B \setminus (B \setminus p)|} \log \Psi_{(p)} = \iota_B^*$. The parts: $\Phi_{\mathbf{E}} = \sum_j \sigma_j (\log K_j + \log \Psi_j)$ and $\Phi_{\mathbf{O}} = \sum_j \delta_j (\log K_j - \log \Psi_j)$ (Definition 8.18), and $\sigma_j \pm \delta_j$ are $s_j$ and $t_j$.

**Proposition 12.20 (The bottom of a chain factors over its steps).** *As stated.*

*Proof. Existence.* Order the marks of each step arbitrarily and the steps as they come; the sets $S_i$ are initial segments of this order of $\mathcal{E}$, which are the tails of the reversed order, and 8.3 gives a basis $T^+$ with $|T^+ \cap U| = d(U)$ for every tail $U$, in particular at the bottom of every $S_i$.

*One step is exchanged at a time.* Let $T$ and $T'$ be at the bottom of every $S_k$, and $1 \leq i \leq \ell + 1$. For $i \geq 2$, by 12.19 at the bunch $S_{i-1}$ the bases at its bottom are all the unions of a pattern of that level and a room side of that level, so $T'' = (T' \cap S_{i-1}) \cup (T \setminus S_{i-1})$ is a basis; for $i = 1$ put $T'' = T$. Its traces are $|T'' \cap S_k| = |T' \cap S_k| = d(S_k)$ for $k < i$ and $d(S_{i-1}) + |T \cap S_k| - |T \cap S_{i-1}| = d(S_k)$ for $k \geq i$. For $i \leq \ell$, 12.19 at the bunch $S_i$, applied to $T''$ and $T'$, both at its bottom, gives the basis $(T'' \cap S_i) \cup (T' \setminus S_i) = (T' \setminus Z_i) \cup (T \cap Z_i)$; for $i = \ell + 1$ this set is $T''$ itself. Its trace on $S_k$ is that of $T'$ for $k < i$, and for $k \geq i$ it is

$$d(S_{i-1}) + \big(d(S_i) - d(S_{i-1})\big) + \big(d(S_k) - d(S_i)\big) = d(S_k),$$

since $T$ and $T'$ meet $Z_i$ in $d(S_i) - d(S_{i-1})$ marks and $S_k \setminus S_i$ in $d(S_k) - d(S_i)$. So the part on one step of a basis at the bottom of every $S_k$ can be replaced by that of any other, and these bases are exactly the unions of their traces on the steps, chosen independently.

*The step tables.* If $(T_0 \setminus Z_i) \sqcup X$ is a basis with $X \subseteq Z_i$, then $|X| = |T_0 \cap Z_i|$, its traces on the $S_k$ are those of $T_0$, and it is at the bottom of every $S_k$. Conversely, for $T$ at the bottom of every $S_k$ the exchange gives the basis $(T_0 \setminus Z_i) \cup (T \cap Z_i)$. So the traces on $Z_i$ are the bases of the child at the pattern $T_0 \setminus Z_i$, a table (12.2), the same set for every $T_0$. A basis $T$ at the bottom of $S_{i-1}$ and of $S_i$ alone has the same trace on $Z_i$ as one at the bottom of every $S_k$: the two exchanges above with $T' = T_0$ use only that $T$ is at the bottom of $S_{i-1}$ and of $S_i$, and give the basis $(T_0 \setminus Z_i) \cup (T \cap Z_i)$, whose traces were computed above. So the step table depends on $S_{i-1}$ and $S_i$ alone.

*The top.* $|T \cap S_i|$ is largest exactly when the cobasis $\mathcal{E} \setminus T$ meets $S_i$ least, so the bases at the top of every $S_i$ are the complements of the bases at the bottom of every $S_i$ in the dual table (7.1), and the above applies there. For $q$ a pattern of $\mathcal{E} \setminus Z_i$ in the dual table, $X \subseteq Z_i$ is a basis of its

child exactly when $\big((\mathcal{E}\setminus Z_i)\setminus q\big)\sqcup(Z_i\setminus X)$ is a basis of $\mathcal{B}$: the child of the dual table at $q$ is the dual of the child of $\mathcal{B}$ at $(\mathcal{E}\setminus Z_i)\setminus q$. At $q=(\mathcal{E}\setminus T_1)\setminus Z_i$ this is the dual of $\mathcal{B}_{(T_1\setminus Z_i)}$.

*The letters.* The degree of $\alpha^T$ in $\alpha_{S_i}$ is $|T\cap S_i|$, whose least value is $d(S_i)$, attained for every $i$ at once by the bases at the bottom of every $S_i$; for a list their sum is $\sum_{X_1,\dots,X_{\ell+1}}\prod_i \alpha^{X_i}=\prod_i K_{\mathcal{B}_{(T_0\setminus Z_i)}}(\alpha_{Z_i})$, and with weights $\theta_T=\prod_i\theta_i(T\cap Z_i)$ the same with the weights $\theta_i$. The degree of $\alpha^{\mathcal{E}\setminus T}$ in $\alpha_{S_i}$ is $|S_i\setminus T|$, least exactly when $T$ is at the top of $S_i$, and by the dual statement the terms of $\Psi$ of the least degree for every $i$ at once are $\prod_i K_{(\mathcal{B}_{(T_1\setminus Z_i)})^*}(\alpha_{Z_i})=\prod_i\Psi_{\mathcal{B}_{(T_1\setminus Z_i)}}(\alpha_{Z_i})$ (7.2). The bases of $\binom{J}{1}$ on $\mathcal{E}$ are the $\{j\}$, $j\in J$, and $d(S)$ is 1 or 0 as $J\subseteq S$ or not; so $\{j\}$ is at the bottom of every $S_k$ exactly when $j\notin S_k$ for every $k$ with $J\not\subseteq S_k$, that is when $j\in J\cap Z_{i_0}$, $Z_{i_0}$ the first step with $J\subseteq S_{i_0}$. Its step table at $Z_{i_0}$ has the bases $\{j\}$, $j\in J\cap Z_{i_0}$, and its step tables at the other steps the one basis $\varnothing$.

*A maximal chain.* When every step is one mark, a basis at the bottom of every $S_i$ meets the step $\{e_i\}$ in $d(S_i)-d(S_{i-1})$ marks, so it is determined by the chain, and it exists; its term in $K$ is one monomial, with the coefficient 1 for a list. □

**Corollary 12.23 (The native level split is interior).** *As stated: the directions come from chains of circuits joining the marks of each side, which a connected table with a positive exponent supplies.*

*Proof.* The first two sentences, the relative interior and the walls with their exceptions, are proved in the reference, proof of 12.23, without any connectedness. Write $\mathcal{L}$ for the set of level and colevel letters with positive exponent, $c_L>0$ for the exponent of $L\in\mathcal{L}$, $N_L$ for the convex hull of its exponent vectors, and $Q=\sum_{L\in\mathcal{L}}c_LN_L$ for the Minkowski sum of the statement.

*(1) A circuit with two marks on one side.* Let $a\neq a'$ lie both in $B$ or both in $R$, and in a common circuit of a table $j$ with a positive exponent. By 7.7 some basis $T$ of table $j$ has $a\in T$, $a'\notin T$ and $T'=(T\setminus\{a\})\cup\{a'\}$ a basis, and $|T\cap B|=|T'\cap B|=:k$, the two marks lying on one side. At the native gauge every basis has positive probability under both laws (7.16). If $s_j>0$, the level letter $K^{(k)}_{j,B}$ has the positive exponent $s_j\mathbb{P}_{j,\alpha^\star}(N_B=k)$, and $N_{K^{(k)}_{j,B}}$ contains $\mathbf{1}_T$ and $\mathbf{1}_{T'}$, whose difference is $\mathbf{1}_a-\mathbf{1}_{a'}$. If $t_j>0$, the colevel letter $\Psi^{(|B|-k)}_{j,B}$ has the positive exponent $t_j\mathbb{P}^*_{j,\alpha^\star}(N^*_B=|B|-k)$, and its polytope contains $\mathbf{1}_{\mathcal{E}\setminus T}$ and $\mathbf{1}_{\mathcal{E}\setminus T'}$, whose difference is $\mathbf{1}_{a'}-\mathbf{1}_a$. Fixing points of the other summands, $c_L(\mathbf{1}_a-\mathbf{1}_{a'})$ is the difference of two points of $Q$ for one $L\in\mathcal{L}$, so every $v$ with $\langle v,x\rangle$ constant on $Q$ has $v_a=v_{a'}$.

*(2) The directions.* Under the hypothesis the chains join all the marks of $B$ and all the marks of $R$, so by step (1) such a $v$ is constant on $B$ and on $R$. Conversely, a monomial $\alpha^T$ of a level letter of level $k$ has $\langle\mathbf{1}_B,\mathbf{1}_T\rangle=k$, and a monomial $\alpha^{\mathcal{E}\setminus T}$ of a colevel letter of colevel $k$ has $\langle\mathbf{1}_B,\mathbf{1}_{\mathcal{E}\setminus T}\rangle=|B\setminus T|=k$; so $\langle\mathbf{1}_B,x\rangle$ is constant on every $N_L$, hence on $Q$, and so is $\langle\mathbf{1}_R,x\rangle=\langle\mathbf{1},x\rangle-\langle\mathbf{1}_B,x\rangle$, each letter being homogeneous. So the directions $v$ with $\langle v,x\rangle$ constant on $Q$ are $\operatorname{span}\{\mathbf{1}_B,\mathbf{1}_R\}$.

*(3) The exceptions.* A linear function is constant on a Minkowski sum with positive coefficients exactly when it is constant on every summand. So $S$ is an exception exactly when $\mathbf{1}_S\in\operatorname{span}\{\mathbf{1}_B,\mathbf{1}_R\}$, that is $S\in\{\varnothing,B,R,\mathcal{E}\}$.

*(4) Convergence.* Write $\alpha_B=\rho_Bu$ with $\rho_B=\langle\mathbf{1}_B,\alpha_B\rangle$ and $\alpha_B=e^{y_B}$; then $d\alpha_B=\rho_B^{|B|-1}d\rho_B\,d\omega_B(u)=\prod_{e\in B}\alpha_e\,dy_B$, and splitting $y_B$ into its component $t$ along the unit vector $|B|^{-1/2}\mathbf{1}_B$, which changes $\rho_B$ by the factor $e^{t|B|^{-1/2}}$ and leaves $u$ unchanged, and its component orthogonal to $\mathbf{1}_B$, on which $u$ depends, gives $d\rho_B/\rho_B=|B|^{-1/2}dt$ and so $d\omega_B(u)=|B|^{1/2}\prod_{e\in B}u_e\,dy_B$ on the hyperplane $\langle y_B,\mathbf{1}_B\rangle=0$; likewise on $R$. So with $Y=\{y:\ \langle y_B,\mathbf{1}_B\rangle=$

$\langle y_R, \mathbf{1}_R\rangle = 0\}$ the integral of the statement is $(|B||R|)^{1/2}\int_Y u^{\boldsymbol\tau|_B} v^{\boldsymbol\tau|_R}\prod_{L\in\mathcal{L}} L(u,v)^{-c_L}dy$, with $u = e^{y_B}/\rho_B$ and $v = e^{y_R}/\rho_R$. Each $L$ is homogeneous in $\alpha_B$ and in $\alpha_R$, and the degrees weighted by the exponents add to $\tau_B$ and to $\tau_R$: for the level letters of table $j$ they add to $s_j\mathbb{E}_{T\sim\mathbb{P}_{j,\alpha^\star}}[N_B] = s_j\sum_{e\in B} R_{j,e}(\alpha^\star)$ in $\alpha_B$, for its colevel letters to $t_j\mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha^\star}}[N^*_B] = t_j\sum_{e\in B} R^*_{j,e}(\alpha^\star)$, and the usage at the native gauge is $\boldsymbol\tau$ (8.17); likewise on $R$. So the powers of $\rho_B$ and $\rho_R$ cancel, and the integrand is $e^{\langle\boldsymbol\tau,y\rangle}\prod_L L(e^y)^{-c_L}$. Each $L(e^y)$ is at least its least coefficient times $e^{\max_{x\in N_L}\langle x,y\rangle}$, the maximum being attained at an exponent vector; so the integrand is at most $C\exp\big(-(h(y)-\langle\boldsymbol\tau,y\rangle)\big)$, with $h(y) = \max_{x\in Q}\langle x,y\rangle = \sum_L c_L\max_{x\in N_L}\langle x,y\rangle$. By step (2) the affine hull of $Q$ has the direction space $\operatorname{span}\{\mathbf{1}_B,\mathbf{1}_R\}^\perp = Y$, and $\boldsymbol\tau$ lies in the relative interior of $Q$; so for some $\varepsilon>0$, $\boldsymbol\tau + w\in Q$ for all $w\in Y$ with $\|w\|\le\varepsilon$, and $h(y)\ge\langle\boldsymbol\tau,y\rangle+\varepsilon\|y\|$ for $y\in Y$. The integrand is at most $Ce^{-\varepsilon\|y\|}$, integrable over $Y$.

*(5) A connected table.* In a connected table any two marks lie in a common circuit, the fact used in the reference, proof of 12.23; so a connected table with a positive exponent joins any two marks of $B$, and any two marks of $R$, by a chain of length one. □

*Remark.* The hypothesis cannot be replaced by connectedness of the family. On $\mathcal{E} = \{a,a',b,b'\}$ with $B = \{a,a'\}$, let the three tables of rank one with the bases $\{a\},\{b\}$, then $\{a'\},\{b'\}$, then $\{a\},\{b'\}$ carry positive exponents, read at a native gauge. The family is connected, but every level of every table holds one basis, so every level and colevel polytope is a point, every set is an exception, and the wall at $\{a\}$ vanishes. Each circuit $\{a,b\}$, $\{a',b'\}$, $\{a,b'\}$ meets both sides, and no chain on one side joins $a$ to $a'$.

## 50.13 Chapter 13: the order of classes and the Foreman bunch

**Proposition 13.10 (Mirror classes).** *As stated: levels mirrored up to a bijection of the tables, and the exchange of any family.*

*Proof. (i)* Fix $j$. Under $\mathbb{P}_{h(j),\mathbf{1}}$ the level $N_D$ takes the value $a$ with probability $\mathbb{P}_{j,\mathbf{1}}(N_C = k_j - a)$, so it has the law of $k_j - N_C$ under $\mathbb{P}_{j,\mathbf{1}}$. A cumulant of order $a\ge 2$ is unchanged by adding a constant and is homogeneous of degree $a$, so $\kappa_{h(j),\mathbf{1}}(N_D^{\times a}) = (-1)^a\kappa_{j,\mathbf{1}}(N_C^{\times a})$. Since $h$ is a bijection of the tables with $\sigma_{h(j)} = \sigma_j$,

$$\sum_j \sigma_j\kappa_{j,\mathbf{1}}(N_D^{\times a}) = \sum_j \sigma_{h(j)}\kappa_{h(j),\mathbf{1}}(N_D^{\times a}) = (-1)^a\sum_j \sigma_j\kappa_{j,\mathbf{1}}(N_C^{\times a}),$$

and likewise with the $\delta_j$. By 13.4(i) the coefficients of $v_0^a$, $a\ge 2$, in the **E** part of stage 0 are $\frac{2}{a!}\sum_j\sigma_j\kappa_{j,\mathbf{1}}(N_C^{\times a})$ at even $a$ and zero at odd $a$, and in its **O** part $\frac{2}{a!}\sum_j\delta_j\kappa_{j,\mathbf{1}}(N_C^{\times a})$ at odd $a$ and zero at even $a$; so the **E** parts of $\mathcal{V}_0(C)$ and $\mathcal{V}_0(D)$ are equal and their **O** parts are negatives.

*(ii)* Stage 0 holds one pair. By (i) the **E** parts agree, so the pairs are compared by their **O** parts (13.5(iii)). Their coefficients vanish at every even $a$ and at every odd $a$ below the least odd $a$ with $c = \frac{2}{a!}\sum_j\delta_j\kappa_{j,\mathbf{1}}(N_C^{\times a})\neq 0$, where $C$ has $c$ and $D$ has $-c$. So stage 0 separates $C$ and $D$, and $C\prec D$ exactly when $c<-c$, that is $c<0$. At the opposite lean $c$ changes sign, being linear in $\boldsymbol\delta$, and the order is the other one.

*(iii)* At the centre the **O** parts vanish and the **E** parts agree by (i), so stage 0 does not separate $C$ and $D$. They are different classes with equally many marks and $\prec$ is a strict total order on them (13.7(i)), so a later stage does.

*The two instances.* If $C$ and $D$ together make up the molecule, $|T\cap D| = r_j - |T\cap C|$ on every basis of table $j$: $h$ is the identity and $k_j = r_j$. If $g$ exchanges the family, with $h$ the bijection of its

tables, then for $T \sim \mathbb{P}_{j,\mathbf{1}}$ the basis $T' = \mathcal{E} \setminus g(T)$ has the law $\mathbb{P}_{h(j),\mathbf{1}}$ (proof of 11.15, step (1)) and $|T' \cap g(C)| = |g(C) \setminus g(T)| = |C \setminus T| = |C| - |T \cap C|$, so $\mathbb{P}_{h(j),\mathbf{1}}(|T \cap g(C)| = a) = \mathbb{P}_{j,\mathbf{1}}(|T \cap C| = |C| - a)$ and $k_j = |C|$. Moreover $g(C)$ is a class at the lean with $|C|$ marks: $g$ maps the classes at $\boldsymbol{\delta}$ onto those at $-\boldsymbol{\delta}$ (11.14), which are the classes at $\boldsymbol{\delta}$ (11.2). □

*Remark.* The reference, proof of 13.10, takes the mirror table by table, $h$ the identity, and the exchange of a one-table molecule. Its conclusions use only the sums over the tables weighted by the $\sigma_j$ and the $\delta_j$, which a bijection of the tables preserving the pairs leaves unchanged.

## 50.14 Chapter 14: peel, children and the Foreman tree

**Corollary 14.9 (A molecule with an exchange).** *As stated: a molecule with any number of tables; the classes of the centre are those of 11.15.*

*Proof.* Read $M$ as the family $F|M$ of the restrictions of the tables to $M$ (14.1), with the pair coordinates $(\sigma_j, \delta_j)$ of its tables. By 11.13 an exchange $g$ of $M$ comes with a bijection $h$ of the tables such that $T \mapsto M \setminus g(T)$ maps the bases of table $j$ onto those of table $h(j)$, with their weights, and $(\sigma_{h(j)}, \delta_{h(j)}) = (\sigma_j, \delta_j)$. The first two conditions concern the tables alone, and the third holds with $\boldsymbol{\delta}$ replaced by $-\boldsymbol{\delta}$ or by 0 when it holds at $\boldsymbol{\delta}$. So $g$ exchanges $M$ at each of the leans $\boldsymbol{\delta}$, $-\boldsymbol{\delta}$ and 0; likewise a bijection carrying $M$ at $\boldsymbol{\delta}$ carries it at these three leans.

*Trees and bunches.* 14.8 with $F = F' = M$ gives $g\big(\operatorname{Tree}(M;\boldsymbol{\sigma},\boldsymbol{\delta})\big) = \operatorname{Tree}(M;\boldsymbol{\sigma},-\boldsymbol{\delta})$, each recorded bunch going onto the bunch recorded at the same place. At the root the recorded bunches are $B(M;\boldsymbol{\sigma},\boldsymbol{\delta})$ and $B(M;\boldsymbol{\sigma},-\boldsymbol{\delta})$ (14.4), so $g\big(B(M;\boldsymbol{\sigma},\boldsymbol{\delta})\big) = B(M;\boldsymbol{\sigma},-\boldsymbol{\delta})$. At the lean 0 the two trees are one tree, which gives $g\big(\operatorname{Tree}(M;\boldsymbol{\sigma},0)\big) = \operatorname{Tree}(M;\boldsymbol{\sigma},0)$ and $g\big(B(M;\boldsymbol{\sigma},0)\big) = B(M;\boldsymbol{\sigma},0)$.

*Classes of the centre.* 11.15, applied to the family $F|M$.

*A carrying map.* Let $g$ carry $M$. 11.14 at the tables' gauges, where $g^* \log \mathbf{1} = \log \mathbf{1}$, says that the reading of $g(e)$ equals that of $e$ at $\Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\mathbf{1}}$, for every mark $e$. So $g(e)$ lies in the class of $e$, that is $g(C) \subseteq C$ for every class $C$, and $g(C) = C$ since $g$ is injective and $C$ is finite. The last sentence of 14.8, at each of the three leans at which $g$ carries $M$, maps each of the trees at 0, $\boldsymbol{\delta}$ and $-\boldsymbol{\delta}$ onto itself. □

**Lemma 14.10 (An exchange turns a window over).** *As stated: an exchange of one family with another, table by table through the bijection $h$ of the tables.*

*Proof.* Let $F$ be on $\mathcal{E}$ and $F'$ on $\mathcal{E}'$. By 11.13 the map $T \mapsto \mathcal{E}' \setminus g(T)$ is a bijection from the bases of table $j$ of $F$ onto those of table $h(j)$ of $F'$. For a basis $T$, $g$ being a bijection,

$$\big|(\mathcal{E}' \setminus g(T)) \cap g(S)\big| = |g(S)| - |g(T) \cap g(S)| = |S| - |T \cap S|.$$

By 7.13 the numbers $|T \cap S|$, $T$ a basis of table $j$, fill $\{d_j(S), \dots, r_j(S)\}$. So the numbers $|T' \cap S'|$, $T'$ a basis of table $h(j)$, fill $\{|S| - r_j(S), \dots, |S| - d_j(S)\}$, a window of the same width $c_j(S) = r_j(S) - d_j(S)$, and the bijection takes the bases of table $j$ at level $k$ on $S$ to those of table $h(j)$ at level $|S| - k$ on $S'$. This is the reference, proof of 14.10, with $h$. □

**Proposition 14.12 (Level form).** *As stated: table by table in a family, and for the family through the bijections that carry it with the identity on its tables.*

*Proof.* Fix a table $j$ and a transposition $\tau$ of two marks of $B$ fixing every other mark. A basis $T$ with $T \cap B = p$ is $p \sqcup X$ with $X$ a basis of the child at $p$, of weight $\theta_{j,(p),X} = \theta_{j,T}$, and

$K_{j,(p)}(\alpha_R) = \sum_X \theta_{j,(p),X}\alpha_R^X$ (12.1, 12.7). Distinct bases of one child give distinct monomials, so $K_{j,(p)}$ determines the child at $p$ with its weights, and $K_{j,(p)} = 0$ when $p$ is not a pattern.

Let $K_{j,(p)} = K_{j,(p')}$ whenever $|p| = |p'|$. For a basis $T = p \sqcup X$, $\tau(T) = \tau(p) \sqcup X$, since $\tau$ fixes $R$, and $|\tau(p)| = |p|$. So $X$ is a basis of the child at $\tau(p)$ with the weight $\theta_{j,(p),X}$: $\tau(T)$ is a basis of table $j$ and $\theta_{j,\tau(T)} = \theta_{j,T}$. Hence $\tau$ maps the bases of table $j$ into themselves with their weights, and onto them, being injective on a finite set.

Conversely, let every such transposition map the bases of table $j$ onto themselves with their weights, and let $p, p' \subseteq B$ have $|p| = |p'|$. The transpositions of marks of $B$ generate the permutations of $B$, so a product $\varsigma$ of them has $\varsigma(p) = p'$; it maps the bases onto themselves with their weights and fixes $R$. So $p \sqcup X \mapsto p' \sqcup X$ is a bijection from the bases with trace $p$ onto those with trace $p'$ keeping the weights, and $K_{j,(p')} = K_{j,(p)}$. This is the reference, proof of 14.12, for table $j$, where 14.7(i) is used with the identity on $R$.

$B$ is in level form in the family when it is in level form in every table, that is, by the above, when every such transposition maps the bases of every table onto themselves with their weights. That is the condition of 11.13 with $F' = F$ and $h$ the identity, whose condition on the pairs, $(\sigma_{h(j)}, \delta_{h(j)}) = (\sigma_j, \delta_j)$, then holds by itself. □

**Definition 14.17 (Keeping the potential; twins; descents; copies).** As stated: keeping and taking the potential at every realization; twins, descents and copies at the table's gauge.

**Lemma 14.18 (The writings certify the potential).** *As stated, at every realization* $\Upsilon = ((s_j, t_j)_j; \alpha)$.

*Proof.* Let $M$ have $n$ marks, and let $(\Phi_{\mathbf{E}}, \Phi_{\mathbf{O}})$ be its potential. For a mark $x$ and an order $w = (w_1, \ldots, w_{n-1})$ of its room, the potential written from $x$ in the order $w$ is the pair of functions

$$\Phi^{\mathbf{X}}_{x,w}(v_0, \ldots, v_{n-1}) = \Phi_{\mathbf{X}}\Big(\log\alpha + v_0\mathbf{1}_x + \sum_{i=1}^{n-1} v_i\mathbf{1}_{w_i}\Big)$$

(10.1). For a bijection $g$ and a mark $a$, $g^*\mathbf{1}_{g(a)} = \mathbf{1}_a$, since $(g^*\mathbf{1}_{g(a)})_e = (\mathbf{1}_{g(a)})_{g(e)} = \mathbf{1}[e = a]$.

(i) By 10.4 there are orders $(a_1, \ldots, a_{n-1})$ of the room of $e$ and $(b_1, \ldots, b_{n-1})$ of the room of $f$ with $\Phi^{\mathbf{X}}_{e,(a_i)_i} = \Phi^{\mathbf{X}}_{f,(b_i)_i}$ for both parts. Put $a_0 = e$ and $b_0 = f$, and let $\beta$ be the bijection with $\beta(a_i) = b_i$ for $0 \le i \le n-1$; so $\beta(e) = f$. For $y \in \mathbb{R}^M$ put $v_i = y_{b_i}$. Then $\sum_i v_i\mathbf{1}_{b_i} = y$ and $\sum_i v_i\mathbf{1}_{a_i} = \sum_i y_{\beta(a_i)}\mathbf{1}_{a_i} = \beta^*y$, and the equality of the two writings at $(v_0, \ldots, v_{n-1})$ reads

$$\Phi_{\mathbf{X}}(\log\alpha + \beta^*y) = \Phi_{\mathbf{X}}(\log\alpha + y)$$

for every $y$ and both parts: $\beta$ keeps the potential (14.17).

(ii) Let $g$ keep the potential. For a mark $x$, an order $w$ of its room and $v \in \mathbb{R}^n$, put $y = v_0\mathbf{1}_{g(x)} + \sum_i v_i\mathbf{1}_{g(w_i)}$; then $g^*y = v_0\mathbf{1}_x + \sum_i v_i\mathbf{1}_{w_i}$, and

$$\Phi^{\mathbf{X}}_{x,w}(v) = \Phi_{\mathbf{X}}(\log\alpha + g^*y) = \Phi_{\mathbf{X}}(\log\alpha + y) = \Phi^{\mathbf{X}}_{g(x),g(w)}(v), \qquad g(w) = (g(w_1), \ldots, g(w_{n-1})).$$

As $w$ runs over the orders of the room of $x$, $g(w)$ runs over those of the room of $g(x)$, once each. So the readings of $x$ and of $g(x)$, the multisets of these pairs of functions, are equal, and $g(x)$ lies in the class of $x$. Thus $g(C) \subseteq C$ for every class $C$, and $g(C) = C$, $g$ being injective and $C$ finite.

Let $g : M \to M'$ take the potential of $M$, read at $\Upsilon$, to that of $M'$, read at a realization with the gauge $\alpha'$. The same computation gives $\Phi^{\mathbf{X}}_{x,w}(v) = \Phi'^{\mathbf{X}}_{g(x),g(w)}(v) + c_{\mathbf{X}}$: the reading of $x$ in $M$ is the reading of $g(x)$ in $M'$ with every function shifted by the constants $c_{\mathbf{X}}$. So two marks of $M$

have equal readings exactly when their images have equal readings in $M'$, and $g$ maps the classes of $M$ onto those of $M'$.

The centre of $\Upsilon = \Upsilon_{\boldsymbol{\sigma},\boldsymbol{\delta};\alpha}$ is $\Upsilon_{\boldsymbol{\sigma},0;\alpha}$, read at the same gauge (10.1). Its potential is $(\Phi_{\mathbf{E}}, 0)$, $\Phi_{\mathbf{E}} = \sum_j \sigma_j \log \mathbf{E}_j(e^y)$ being the same function at every lean (8.18). A bijection keeping the potential at $\Upsilon$ keeps it at the centre, and one taking the potential of $M$ to that of $M'$ at their realizations takes it at their centres, with the constants $(c_{\mathbf{E}}, 0)$; so the two preceding paragraphs hold at the centre as well.

For the last sentence: if $e$ and $f$ read alike, (i) gives a bijection keeping the potential that takes $e$ to $f$; conversely, a bijection keeping the potential maps the class of $e$ onto itself by (ii), so if it takes $e$ to $f$, then $f$ lies in that class. □

*Remark.* At the table's gauge $\log \alpha = 0$, the two conditions of 14.17 read $\Phi_{\mathbf{X}} \circ g^* = \Phi_{\mathbf{X}}$ and $\Phi_{\mathbf{X}} \circ g^* = \Phi'_{\mathbf{X}} + c_{\mathbf{X}}$, and the proof above is the reference, proof of 14.18. The statements that keep or take the potential at the table's gauge, 14.20, 14.21, 14.22, 29.9 and 29.11, and the twins, descents and copies of 14.17, read these conditions as in the reference.

## 50.15 Chapter 15: the integral at a bunch

**Proposition 15.2 (Where the Breta of one bunch converges).** *As stated.*

*Proof. The set.* If a convex combination of the coordinates of $t\boldsymbol{\xi} + (1-t)\mathbf{1}$, $0 < t \le 1$, were a point $p \in (-\infty, 0]$, the same combination of the $\xi_\ell$ would be $(p-1+t)/t \in (-\infty, 0]$; at $t = 0$ the combination is 1. So $\Xi_J$ is star-shaped about $\mathbf{1}$, hence connected. The convex hull of the $\xi_\ell$ is compact and avoids the closed set $(-\infty, 0]$, so it lies at a distance $\delta > 0$ from it, and moving every $\xi_\ell$ by less than $\delta$ moves every convex combination by less than $\delta$: $\Xi_J$ is open. Convex combinations of points with positive real parts have positive real parts.

*The integral.* Put $\nu = |\mathbf{w}| - \kappa$, so $\operatorname{Re}\nu > 0$, and $z(u) = \sum_\ell \xi_\ell u_\ell$, which for $u$ in the closed simplex lies in the convex hull of the $\xi_\ell$. On a compact neighbourhood of a point of $\Xi_J$ inside $\Xi_J$ these hulls stay at a distance at least some $\delta' > 0$ from $(-\infty, 0]$, so the principal branch $z^{-\nu} = \exp(-\nu \log z)$ is holomorphic in $\boldsymbol{\xi}$ and $|z^{-\nu}| = |z|^{-\operatorname{Re}\nu} e^{\operatorname{Im}\nu \arg z} \le \delta'^{-\operatorname{Re}\nu} e^{\pi|\operatorname{Im}\nu|}$ there. Since $u^{\mathbf{w}-\mathbf{1}}$ is integrable on $\Delta_J$ for $\operatorname{Re} w_\ell > 0$ (3.9), the integral converges absolutely, locally uniformly, and is holomorphic in $\boldsymbol{\xi}$ on $\Xi_J$. At positive coefficients it is the Breta: with the unit configuration on one bunch the Zamma is $\Gamma(\nu)^{-1} \int_0^\infty x^{\nu-1} e^{-x\sum_\ell \xi_\ell u_\ell} dx = z(u)^{-\nu}$ (2.1, 2.12).

*The identity.* The left side of 15.1 is holomorphic on $(\mathbb{C} \setminus (-\infty, 0])^J \supseteq \Xi_J$, and the stacked Gamma does not depend on $\boldsymbol{\xi}$; the two sides agree on $(0, \infty)^J$. A holomorphic function on the connected open set $\Xi_J$ that vanishes on the nonempty open set $(0, \infty)^J$ of its real points vanishes identically, its Taylor coefficients at a real point being its real derivatives there; so they agree on $\Xi_J$. □

**Corollary 15.4 (The scale of a bunch of width one).** *As stated: every bunch, table by table, with one Skeleton value in each table of width one.*

*Proof.* Fix a table $j$ and write $d = d_j(B)$, $d^* = d_j^*(B)$ and $c = c_j(B)$. By 7.13 the levels $|T \cap B|$ fill $\{d, \dots, d+c\}$, and the colevels $|B \setminus T|$ fill $\{|B| - d - c, \dots, |B| - d\} = \{d^*, \dots, d^* + c\}$, since $d^* = |B| - r_j(B) = |B| - d - c$ (7.12). Each level letter $K_{j,B}^{(k)}$ and each colevel letter $\Psi_{j,B}^{(k)}$ in these windows is a sum of positive terms over a nonempty set of bases, positive on the orthant. By

12.16,

$$x^d \mathsf{Q}^K_{j,B}(x;\alpha) = \sum_{k=d}^{d+c} x^k \frac{K^{(k)}_{j,B}(\alpha)}{K_j(\alpha)}, \qquad x^{d^*} \mathsf{Q}^\Psi_{j,B}(x;\alpha) = \sum_{k=d^*}^{d^*+c} x^k \frac{\Psi^{(k)}_{j,B}(\alpha)}{\Psi_j(\alpha)}.$$

If $c = 0$ both Skeleton polynomials are the constant 1, of degree 0, and the spectra are empty (12.10). If $c = 1$, then $K_j = K^{(d)}_{j,B} + K^{(d+1)}_{j,B}$, and with $\xi = K^{(d)}_{j,B}(\alpha)/K^{(d+1)}_{j,B}(\alpha) > 0$,

$$\mathsf{Q}^K_{j,B}(x;\alpha) = \frac{K^{(d)}_{j,B}(\alpha) + xK^{(d+1)}_{j,B}(\alpha)}{K^{(d)}_{j,B}(\alpha) + K^{(d+1)}_{j,B}(\alpha)} = \frac{x+\xi}{1+\xi}.$$

So the Skeleton value is $\xi$, the level letter at the floor over the level letter at the largest level, and the coin is $1/(1+\xi) = K^{(d+1)}_{j,B}(\alpha)/K_j(\alpha)$, the probability under $\mathbb{P}_{j,\alpha}(T) = \theta_{j,T}\alpha^T/K_j(\alpha)$ of the bases at level $d+1$ (7.16), that is $\mathbb{P}_{j,\alpha}(N_B = d+1)$. The same computation with the colevel letters gives the Skeleton value $\Psi^{(d^*)}_{j,B}(\alpha)/\Psi^{(d^*+1)}_{j,B}(\alpha)$ and the coin $\Psi^{(d^*+1)}_{j,B}(\alpha)/\Psi_j(\alpha) = \mathbb{P}^*_{j,\alpha}(N^*_B = d^*+1)$, since $\mathbb{P}^*_{j,\alpha}(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j(\alpha)$.

Let $B = \{e\}$ with $e$ neither a loop nor a coloop of table $j$. Then $d = 0$ and $r_j(\{e\}) = 1$, so $c = 1$ and $d^* = 0$. The bases at level 0 avoid $e$ and those at level 1 contain it, so $K^{(0)}_{j,\{e\}} = K_{j,\setminus e}(\alpha_R)$ and $K^{(1)}_{j,\{e\}} = \alpha_e K_{j,/e}(\alpha_R)$; the bases at colevel 0 contain $e$ and those at colevel 1 avoid it, so $\Psi^{(0)}_{j,\{e\}} = \Psi_{j,/e}(\alpha_R)$ and $\Psi^{(1)}_{j,\{e\}} = \alpha_e \Psi_{j,\setminus e}(\alpha_R)$ (7.3). The coins are $\alpha_e K_{j,/e}/K_j = \mathbb{P}_{j,\alpha}(e \in T) = R_{j,e}(\alpha)$ and $\alpha_e \Psi_{j,\setminus e}/\Psi_j = \mathbb{P}^*_{j,\alpha}(e \notin T) = R^*_{j,e}(\alpha)$ (7.16). This is the reference, proof of 15.4. □

**Theorem 15.6 (The scale in its spectral gaps).** *As stated: complex coefficients and letter exponents, expanded about any point $r$ of the axis with $|\xi_\ell - r| < r$ for every $\ell$.*

*Proof.* The left side is the value of the scale integral $\int_0^\infty x^{\kappa-1} \prod_{\ell\in J}(x+\xi_\ell)^{-w_\ell} dx$, which converges since $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$ (15.1). For $x \geq 0$ put $q_\ell(x) = (r-\xi_\ell)/(x+r)$, so that $x + \xi_\ell = (x+r)(1-q_\ell(x))$ and $|q_\ell(x)| \leq |r-\xi_\ell|/r = |g_\ell| < 1$.

*Expansion.* Since $x + r > 0$ and $\operatorname{Re}(1-q_\ell(x)) > 0$, the principal argument of $x+\xi_\ell$ is that of $1-q_\ell(x)$, which lies in $(-\pi/2, \pi/2)$; so $(x+\xi_\ell)^{-w_\ell} = (x+r)^{-w_\ell}(1-q_\ell(x))^{-w_\ell}$ with principal powers, and the binomial series $(1-q)^{-w} = \sum_{n\geq 0} \frac{(w)_n}{n!} q^n$ converges absolutely for $|q| < 1$. Hence

$$x^{\kappa-1} \prod_\ell (x+\xi_\ell)^{-w_\ell} = \sum_{\mathbf{n}\in\mathbb{N}^J} \prod_\ell \frac{(w_\ell)_{n_\ell}}{n_\ell!} (r-\xi_\ell)^{n_\ell}\, x^{\kappa-1}(x+r)^{-|\mathbf{w}|-|\mathbf{n}|}.$$

*Term by term.* $|(w)_n| = \prod_{i<n}|w+i| \leq \prod_{i<n}(|w|+i) = (|w|)_n$, and $(x+r)^{-|\mathbf{n}|} \leq r^{-|\mathbf{n}|}$. So the moduli of the terms sum to at most

$$x^{\operatorname{Re}\kappa-1}(x+r)^{-\operatorname{Re}|\mathbf{w}|} \prod_\ell \sum_{n\geq 0} \frac{(|w_\ell|)_n}{n!} |g_\ell|^n = x^{\operatorname{Re}\kappa-1}(x+r)^{-\operatorname{Re}|\mathbf{w}|} \prod_\ell (1-|g_\ell|)^{-|w_\ell|},$$

which is integrable on $(0,\infty)$ since $0 < \operatorname{Re}\kappa < \operatorname{Re}|\mathbf{w}|$; so the series may be integrated term by term. For $0 < \operatorname{Re}\kappa < \operatorname{Re} a$ the integral of a linear letter, in the paragraph after 15.1 with $c_0 = r$ and $c_1 = 1$, is

$$\int_0^\infty x^{\kappa-1}(x+r)^{-a} dx = \Gamma \begin{bmatrix} \kappa, & a-\kappa \\ & a \end{bmatrix} r^{\kappa-a},$$

and at $a = |\mathbf{w}| + |\mathbf{n}|$, with $(r-\xi_\ell)^{n_\ell} r^{-n_\ell} = g_\ell^{n_\ell}$, the term of $\mathbf{n}$ is the term of the statement.

*Root rate.* Let $A_M$ be the sum of the terms with $|\mathbf{n}| = M$, $v = \sum_\ell |w_\ell|$ and $\bar g = \max_\ell |g_\ell|$. Comparing the coefficients of $z^M$ in $\prod_\ell (1-z)^{-|w_\ell|} = (1-z)^{-v}$ gives $\sum_{|\mathbf{n}|=M} \prod_\ell \frac{(|w_\ell|)_{n_\ell}}{n_\ell!} = \frac{(v)_M}{M!}$. The Beta corner of order $M$ is the integral $\int_0^1 x^{\kappa-1}(1-x)^{|\mathbf{w}|+M-\kappa-1}dx$ (3.9), of modulus at most $b_0 = \int_0^1 x^{\operatorname{Re}\kappa-1}(1-x)^{\operatorname{Re}|\mathbf{w}|-\operatorname{Re}\kappa-1}dx < \infty$. So $|A_M| \le |r^{\kappa-|\mathbf{w}|}|\, b_0 \frac{(v)_M}{M!}\bar g^M$, and $\frac{(v)_M}{M!} = \prod_{i=1}^M \left(1+\frac{v-1}{i}\right) \le e^{|v-1|(1+\log M)}$ grows at most polynomially in $M$. Hence $\limsup_M |A_M|^{1/M} \le \bar g$ (1.1(viii)).

*The remaining sentences.* A coefficient $\xi_\ell = r$ has $g_\ell = 0$, and every term with $n_\ell > 0$ vanishes. For $0 < \xi_\ell \le r$, $w_\ell > 0$ and $0 < \kappa < |\mathbf{w}|$ every factor of every term is positive, a factor $g_\ell^0$ being 1. For positive coefficients write $\xi_{\min}$ and $\xi_{\max}$ for the least and the largest; $|1-\xi/r|$ is largest over $\xi \in [\xi_{\min}, \xi_{\max}]$ at an end, so $\max_\ell |g_\ell| = \max\left(|1-\xi_{\min}/r|, |1-\xi_{\max}/r|\right)$. For $r \le \xi_{\min}$ this is $\xi_{\max}/r - 1$, decreasing in $r$; for $r \ge \xi_{\max}$ it is $1-\xi_{\min}/r$, increasing in $r$; for $\xi_{\min} \le r \le \xi_{\max}$ it is the larger of $1-\xi_{\min}/r$, increasing, and $\xi_{\max}/r - 1$, decreasing, which are equal at $r = \frac12(\xi_{\min}+\xi_{\max})$, with the value $\frac{\xi_{\max}-\xi_{\min}}{\xi_{\max}+\xi_{\min}}$. At that $r$, $|\xi_\ell - r| \le \frac12(\xi_{\max}-\xi_{\min}) < r$ for every $\ell$. At $r = \xi_{\max}$ the bound is $1-\xi_{\min}/\xi_{\max}$, the largest coefficient drops out, and the statement is the reference, proof of 15.6. □

**Corollary 15.11 (The period closes on the ordering sectors).** *As stated: complex data with $\kappa_{\mathcal{E}} = 0$ and every proper wall of positive real part.*

*Proof. Sector form.* Let $\Omega$ be the set of data at which every proper wall has positive real part. It is open and convex, the walls being affine in the data with real coefficients, and it contains points at which $\operatorname{Re}\kappa_{\mathcal{E}} > 0$ as well (raise every mass). By 8.5(ii), valid for all complex exponents, and 8.7, wherever every wall has positive real part

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \sum_\pi I_\pi, \qquad I_\pi = \int_{[0,1]^{n-1}} \prod_{k=2}^n q_k^{\kappa_{U_k}-1} h_\pi(q)\, dq, \quad h_\pi = L_\pi^{-\kappa_{\mathcal{E}}} \prod_j k_{j,\pi}^{-s_j} \psi_{j,\pi}^{-t_j}.$$

The sector letters are polynomials with nonnegative coefficients and positive constant terms, and $L_\pi \in [1,n]$ on the cube, so $h_\pi$ is smooth on a neighbourhood of the cube, with every $q$-derivative continuous in $q$ and the data and entire in the data, and $|h_\pi|$ is bounded on the cube locally uniformly in the data. The tails $U_k$, $k \ge 2$, of the orders are the nonempty proper sets of marks. So every $I_\pi$ converges and is holomorphic on $\Omega$, and so is $P_F$ (8.7); $\Omega$ being connected, $P_F = \sum_\pi I_\pi$ on $\Omega$. At $\kappa_{\mathcal{E}} = 0$, $h_\pi = \prod_j k_{j,\pi}^{-s_j}\psi_{j,\pi}^{-t_j}$.

*The identity and the range.* In the sector coordinates $\alpha^T = \rho^{|T|}\prod_k q_k^{N_{U_k}(T)}$, so $k_{j,\pi}(q) = \sum_T \theta_{j,T}\prod_k q_k^{N_{U_k}(T)-d_j(U_k)}$ (8.5(ii)), with exponents in $\mathbb{N}$, all zero only at $T = T^+_{j,\pi}$ (8.3). So $k_{j,\pi}/K_j(\mathbf{1})$ increases in each $q_k$ on the cube, from $\theta_{j,T^+_{j,\pi}}/K_j(\mathbf{1})$ at $q = 0$ to 1 at $q = \mathbf{1}$, and $0 \le 1 - k_{j,\pi}/K_j(\mathbf{1}) \le \lambda$. The identity is the expectation under $\mathbb{P}_{j,\mathbf{1}}$ of $1 - \prod_k q_k^{e_k} = \sum_k \prod_{k'<k} q_{k'}^{e_{k'}}(1-q_k^{e_k})$, $e_k = N_{U_k} - d_j(U_k)$, with $1-q^e = (1-q)\sum_{0\le i<e} q^i$. The same holds for $\psi_{j,\pi}$ with $N^*$, $d_j^*$ and $T^-_{j,\pi}$, and $\Psi_j(\mathbf{1}) = K_j(\mathbf{1})$. These are statements about the tables alone, as in the reference, proof of 15.11.

*Expansion.* With $a_j = 1 - k_{j,\pi}/K_j(\mathbf{1})$ and $b_j = 1 - \psi_{j,\pi}/K_j(\mathbf{1})$ in $[0,\lambda]$, $k_{j,\pi}^{-s_j} = K_j(\mathbf{1})^{-s_j}\sum_{n\ge0}\frac{(s_j)_n}{n!}a_j^n$ and $\psi_{j,\pi}^{-t_j} = K_j(\mathbf{1})^{-t_j}\sum_{n\ge0}\frac{(t_j)_n}{n!}b_j^n$, with principal powers of positive numbers. By $|(s)_n| \le (|s|)_n$ the moduli of the terms of the product of these series sum to at most $\prod_j K_j(\mathbf{1})^{-\operatorname{Re}(s_j+t_j)}(1-\lambda)^{-|s_j|-|t_j|}$ on the cube, and $\int \prod_k q_k^{\operatorname{Re}\kappa_{U_k}-1}dq = \prod_k(\operatorname{Re}\kappa_{U_k})^{-1} < \infty$. So the series may be integrated term by term over the cube, which gives the displayed series, convergent.

*Terms.* By the identity, $a_j$ and $b_j$ are sums, with nonnegative coefficients, of products $(1-q_k)q^{\boldsymbol{\beta}}$, $\boldsymbol{\beta}\in\mathbb{N}^{n-1}$; so $\prod_j a_j^{n_j}b_j^{n'_j}$ is a finite sum, with nonnegative coefficients, of products $\prod_k(1-q_k)^{\gamma_k}q_k^{\beta_k}$, and for $\operatorname{Re}\kappa>0$, $\int_0^1 q^{\kappa+\beta-1}(1-q)^{\gamma}dq=\Gamma\,{}_2_1\!\left[\begin{matrix}\kappa+\beta,\ \gamma+1\\ \kappa+\beta+\gamma+1\end{matrix}\right]$ (3.9). For real data with $s_j,t_j\ge 0$ the coefficients $(s_j)_n/n!$ and $(t_j)_n/n!$ and the integrands are nonnegative.

*Bound.* The terms with $|\mathbf{n}|+|\mathbf{n}'|=M$ have moduli summing to at most

$$\prod_j K_j(\mathbf{1})^{-\operatorname{Re}(s_j+t_j)}\sum_{|\mathbf{n}|+|\mathbf{n}'|=M}\prod_j\frac{(|s_j|)_{n_j}(|t_j|)_{n'_j}}{n_j!\,n'_j!}\,\lambda^M\sum_{\pi}\prod_k(\operatorname{Re}\kappa_{U_k})^{-1},$$

and comparing coefficients of $z^M$ in $\prod_j(1-z)^{-|s_j|}(1-z)^{-|t_j|}=(1-z)^{-w}$ gives $\frac{(w)_M}{M!}$ for the inner sum, $w=\sum_j(|s_j|+|t_j|)$. □

**Proposition 15.12 (An enclosure of the period in the field of its data).** *As stated: weights, masses and exponents in a subfield $k$ of $\mathbb{R}$, the exponents of any sign.*

*Proof.* The data are real, so by the hypotheses of 15.11 every proper wall is positive, and every $\kappa_{U_k}>0$.

(i) The weights lie in $k$, so the coefficients of the sector letters, sums of weights (8.5(ii)), and $K_j(\mathbf{1})$ lie in $k$, and $a_{j,\pi},b_{j,\pi}\in k[q]$. For $s\in k$ and $n\in\mathbb{N}$, $(s)_n/n!\in k$. So $f_{\pi,M}=\sum_{|\mathbf{n}|+|\mathbf{n}'|=M}\prod_j\frac{(s_j)_{n_j}(t_j)_{n'_j}}{n_j!\,n'_j!}a_{j,\pi}^{n_j}b_{j,\pi}^{n'_j}$, the coefficient of $\zeta^M$ in the product of the binomial series, lies in $k[q]$. Each $\kappa_{U_k}$ is an integer combination of masses and exponents and lies in $k$, and $\int_{[0,1]^{n-1}}\prod_k q_k^{\kappa_{U_k}-1+\beta_k}dq=\prod_k(\kappa_{U_k}+\beta_k)^{-1}\in k$ for $\boldsymbol{\beta}\in\mathbb{N}^{n-1}$; so $I_{\pi,M}\in k$. Grouping the absolutely convergent series of 15.11 by $M=|\mathbf{n}|+|\mathbf{n}'|$ gives the identity. When $s_j,t_j\ge 0$ the coefficients are nonnegative, and $a_{j,\pi},b_{j,\pi}\ge 0$ on the cube, so $I_{\pi,M}\ge 0$.

(ii) On the cube $0\le a_{j,\pi},b_{j,\pi}\le\lambda_\pi$ (15.11), so by $|(s)_n|\le(|s|)_n$ and the coefficient identity of the proof of 15.11, $|f_{\pi,M}|\le\frac{(w)_M}{M!}\lambda_\pi^M$ and $|I_{\pi,M}|\le\prod_k\kappa_{U_k}^{-1}\frac{(w)_M}{M!}\lambda_\pi^M$. The ratio of consecutive bounds is $\frac{w+M}{M+1}\lambda_\pi$, and $M\mapsto\frac{w+M}{M+1}$ is nonincreasing for $w\ge 1$ and below 1 for $w<1$; so for $M\ge Q+1$ the ratio is at most $r_{\pi,Q}<1$, and summing the geometric majorant gives $\big|\sum_{M>Q}I_{\pi,M}\big|\le\operatorname{err}_{\pi,Q}$. The numbers $w$, $\lambda_\pi$, $r_{\pi,Q}$ and $\operatorname{err}_{\pi,Q}$ lie in $k$, an ordered subfield of $\mathbb{R}$ being closed under $|\cdot|$ and under maxima of finitely many elements. As $Q\to\infty$, $r_{\pi,Q}\to\lambda_\pi<1$, and $\frac{(w)_{Q+1}}{(Q+1)!}\lambda_\pi^{Q+1}\to 0$, the factor $\frac{(w)_{Q+1}}{(Q+1)!}$ growing at most polynomially; so $\operatorname{err}_{\pi,Q}\to 0$. With $P^{\le Q}=\sum_\pi\sum_{M\le Q}I_{\pi,M}\in k$, the left side of (i) lies between $P^{\le Q}-\sum_\pi\operatorname{err}_{\pi,Q}$ and $P^{\le Q}+\sum_\pi\operatorname{err}_{\pi,Q}$, and for $s_j,t_j\ge 0$, the remainders being nonnegative, between $P^{\le Q}$ and $P^{\le Q}+\sum_\pi\operatorname{err}_{\pi,Q}$.

(iii) $\partial_\zeta\log f=\sum_i w_iA_i/(1-\zeta A_i)$; multiply by $f\prod_i(1-\zeta A_i)$. Comparing the coefficients of $\zeta^M$ on the two sides gives a linear recurrence for the coefficients of $f$, with coefficients polynomial in $M$ and in the $A_i$. With rational data this is the reference, proof of 15.12. □

**Theorem 15.14 (Continued Beta corners).** *As stated: complex data, every index off the nonpositive integers, several indices of nonpositive real part at once.*

*Proof. A continuation as a linear functional.* For $\kappa\in\mathbb{C}$ and $h\in\mathbb{N}$ with $\operatorname{Re}\kappa>-h-1$ and $\kappa\notin\{0,-1,\dots,-h\}$, and $f$ smooth on $[0,1]$, 8.4 with $m=1$ and $M=h+1$ gives the continuation in $\kappa$ of $\int_0^1 x^{\kappa-1}f\,dx$:

$$\mathcal{C}(\kappa;f)=\int_0^1 x^{\kappa-1}\big(f-\operatorname{Tay}_h f\big)dx+\sum_{l\le h}\frac{f^{(l)}(0)}{l!\,(\kappa+l)},$$

$\mathrm{Tay}_h$ taking the Taylor polynomial of degree $h$ at 0; it is the finite part $\mathrm{Reg}\int_0^1 x^{\kappa-1}f\,dx$ (1.8). By Taylor's formula $|f-\mathrm{Tay}_h f|\le \frac{x^{h+1}}{(h+1)!}\sup|f^{(h+1)}|$, so

$$|\mathcal{C}(\kappa;f)|\le \frac{\sup|f^{(h+1)}|}{(h+1)!\,(\mathrm{Re}\,\kappa+h+1)}+\sum_{l\le h}\frac{|f^{(l)}(0)|}{l!\,|\kappa+l|}:$$

$\mathcal{C}(\kappa;\cdot)$ is linear and bounded by a constant times the largest of the sup-norms of $f,\dots,f^{(h+1)}$. On a cube $[0,1]^m$, with a number $L_k\in\mathbb{N}$ for each coordinate, $\mathrm{Re}\,x_k+L_k>0$ and $x_k\notin\{0,\dots,1-L_k\}$, 8.4 repeated in each coordinate gives the continuation of $\int\prod_k q_k^{x_k-1}f\,dq$ as a finite sum, over the sets $A$ of coordinates and the $\mathbf{l}$ with $l_k<L_k$ for $k\in A$, of $\prod_{k\in A}(x_k+l_k)^{-1}$ times the integral over the other coordinates of $\prod_{k\notin A}q_k^{x_k-1}$ against $\prod_{k\notin A}(1-\mathrm{Tay}_{L_k-1})$ applied to $\partial^{\mathbf{l}}_{q_A}f/\mathbf{l}!$ at $q_A=0$ (as in 8.6), with $\mathrm{Tay}_{-1}=0$. Taylor's formula in each coordinate outside $A$ bounds each term by a constant, depending only on the $x_k$, times the largest sup-norm of the derivatives of $f$ of total order at most $L=\sum_k L_k$; so this continuation is again linear and bounded on $C^L([0,1]^m)$, and it is the iterated finite part, by 1.8 in one coordinate at a time.

*Leibniz bounds.* Let finitely many functions $f_i$ be smooth on a cube, with $0\le f_i\le\lambda$, $0<\lambda<1$, and let $D\ge 1$ bound their derivatives of order at most $L$. For a multi-index $\mathbf{n}$ over the $i$ and $f_{\mathbf{n}}=\prod_i f_i^{n_i}$, a derivative of total order $l\le L$ distributes over the $|\mathbf{n}|$ factors in at most $|\mathbf{n}|^l$ ways, each product having at most $l$ differentiated factors, each bounded by $D$, and the others bounded by $\lambda$; so $|\partial^{\boldsymbol\beta}f_{\mathbf{n}}|\le D^l|\mathbf{n}|^l\lambda^{|\mathbf{n}|-l}$ for $|\boldsymbol\beta|=l$. Hence the sum, over the $\mathbf{n}$ with $|\mathbf{n}|=M$, of $\prod_i\frac{|(w_i)_{n_i}|}{n_i!}$ times the $C^L$-norm of $f_{\mathbf{n}}$ is at most $\frac{(w)_M}{M!}D^L(1+M)^L\lambda^{M-L}$, $w=\sum_i|w_i|$, by the coefficient identity of the proof of 15.6. These bounds are summable over $M$, so $\sum_{\mathbf{n}}\prod_i\frac{(w_i)_{n_i}}{n_i!}f_{\mathbf{n}}$ converges in $C^L$, to $\prod_i(1-f_i)^{-w_i}$; a bounded linear functional on $C^L$ commutes with the sum, and applied to $f_{\mathbf{n}}$ it is at most a constant times $D^L(1+|\mathbf{n}|)^L\lambda^{|\mathbf{n}|-L}$. If $\lambda=0$ every $f_i$ vanishes and only $\mathbf{n}=0$ contributes.

*The two sectors of a scale.* In 15.9 take $f_i=1-\mathsf{Q}_{S,i}(\cdot;\alpha)$ on $[0,1]$: each $\mathsf{Q}_{S,i}$ has nonnegative coefficients, positive constant term and value 1 at $x=1$ (12.10), so $0\le f_i\le 1-\mathsf{Q}_{S,i}(0;\alpha)\le\lambda_S$ there, and $\prod_i\mathsf{Q}_{S,i}^{-w_i}=\sum_{\mathbf{n}}\prod_i\frac{(w_i)_{n_i}}{n_i!}f_{\mathbf{n}}$. Let $\kappa$ be the index of the sector, not a nonpositive integer. If $\mathrm{Re}\,\kappa>0$, take $\mathcal{C}(\kappa;f)=\int_0^1 x^{\kappa-1}f\,dx$, bounded by $\sup|f|/\mathrm{Re}\,\kappa$, with $L=0$; otherwise take $h$ as in the statement and $L=h+1$. By the preceding paragraph,

$$\mathcal{C}\Big(\kappa;\prod_i\mathsf{Q}_{S,i}^{-w_i}\Big)=\sum_{\mathbf{n}}\prod_i\frac{(w_i)_{n_i}}{n_i!}\,\mathcal{C}(\kappa;f_{\mathbf{n}}),$$

the terms of total order $M$ having moduli summing to at most $\frac{(w)_M}{M!}\lambda_S^M$ times a polynomial in $M$ of degree $L$, $w=\sum_j(|s_j|+|t_j|)$. The left side is the finite part of the sector integral and its continuation in $\kappa$. By 15.8, $f_{\mathbf{n}}=(1-x)^{|\mathbf{n}|}\prod_i\mathsf{Q}_{S,i,>}^{n_i}$, a polynomial, and $\mathcal{C}\big(\kappa;x^k(1-x)^{|\mathbf{n}|}\big)$ is the continuation of $\int_0^1 x^{\kappa+k-1}(1-x)^{|\mathbf{n}|}dx=\Gamma\begin{bmatrix}\kappa+k,\ |\mathbf{n}|+1\\ \kappa+k+|\mathbf{n}|+1\end{bmatrix}$ (3.9), the meromorphic stacked Gamma, finite since $\kappa+k\notin-\mathbb{N}$. So the series of 15.9(ii), with these Beta corners, converges to the finite part, with the stated bound. By 15.9(i) the scale integral is the product of the letters at $\alpha$ and the sum of the two sectors at the indices $\kappa_B$ and $\kappa_B^{\mathrm{top}}$, and its continuation in the data is the same expression with the sectors continued.

*The ordering sectors.* By the first step of the proof of 15.11 and 8.4, $P_F=\sum_\pi I_\pi$ as meromorphic functions of the data, each $I_\pi$ continued by 8.4, whose formula expresses the continuation at a point through the values of $h_\pi$ at that point; on the balance wall $h_\pi=\prod_j k_{j,\pi}^{-s_j}\psi_{j,\pi}^{-t_j}$. At a

point of the balance wall at which no proper wall is a nonpositive integer, take for an order $\pi$ the numbers $L_k = h_k + 1$ when $\operatorname{Re}\kappa_{U_k} \le 0$, with $-h_k - 1 < \operatorname{Re}\kappa_{U_k} \le -h_k$, and $L_k = 0$ otherwise. With $f$ running over the $a_j, b_j$ of the proof of 15.11, in $[0,\lambda]$ on the cube, the preceding paragraphs give the continuation of $I_\pi$ as the series of 15.11 for $\pi$, each integral replaced by its continuation; and the continuation of $\int \prod_k q_k^{\kappa_{U_k}+\beta_k-1}(1-q_k)^{\gamma_k}dq$ is the product over $k$ of the continued Beta corners, each finite since $\kappa_{U_k} + \beta_k \notin -\mathbb{N}$. The terms of total order $M$ have moduli summing to at most $\frac{(w)_M}{M!}\lambda^M$ times a polynomial in $M$ of degree $\sum_k L_k$, and the tails of one order being distinct proper sets, $\sum_k L_k$ is at most the sum of the integers $h+1$ over the proper walls of real part at most 0. Every proper set of marks is a tail of some order, so the sector series computes $P_F$ at every point of the balance wall off the pole hyperplanes, with no positivity of the walls. For one real index in $(-h-1, -h)$ this is the reference, proof of 15.14. □

**Corollary 15.16 (A loop or coloop in every table).** *As stated: a set of marks, each a loop or a coloop in every table.*

*Proof.* For a table $j$ let $p_j$ be the set of the marks of $A$ that are coloops of table $j$. Every basis contains every coloop and avoids every loop, so $T \cap A = p_j$ for every basis $T$ of table $j$. The sets $T \setminus A$ are then the bases of the table on $\mathcal{E} \setminus A$ obtained from table $j$ by contracting the marks of $p_j$ and deleting those of $A \setminus p_j$, one mark at a time (7.3; a loop is never a coloop, so each step is a table), with the weights $\theta_{j,T}$; with $K'_j, \Psi'_j$ its Foreman polynomials and $\alpha' = \alpha_{\mathcal{E}\setminus A}$,

$$K_j(\alpha) = \alpha_A^{p_j} K'_j(\alpha'), \qquad \Psi_j(\alpha) = \alpha_A^{A\setminus p_j}\Psi'_j(\alpha'),$$

since $\mathcal{E} \setminus T = (A \setminus p_j) \sqcup \big((\mathcal{E} \setminus A) \setminus (T \setminus A)\big)$. So the integrand of $J_F(\mathbf{m};\mathbf{s},\mathbf{t})$ is the product of

$$\prod_{e\in A} \alpha_e^{m_e - 1 - \sum_j (s_j \mathbf{1}[e\in p_j] + t_j \mathbf{1}[e\notin p_j])} e^{-\alpha_e}$$

and of the integrand of $J_{F'}(\mathbf{m}|_{\mathcal{E}\setminus A};\mathbf{s},\mathbf{t})$. With $s_j = \sigma_j + \delta_j$, $t_j = \sigma_j - \delta_j$ and $m_e = \mu_e + \sigma$, the exponent of $\alpha_e$ is $\mu_e - \sum_j \eta_{j,e}\delta_j - 1$. The integrand is a product of functions of disjoint sets of variables, so by Tonelli on the modulus its integral converges exactly when every factor's does, and then, by Fubini, it is the product of the factors' integrals, with $\int_0^\infty \alpha_e^{\nu-1}e^{-\alpha_e}d\alpha_e = \Gamma(\nu)$ for $\operatorname{Re}\nu > 0$. As meromorphic functions the identity persists (8.5(iii)). For one mark this is the reference, proof of 15.16, there read through 15.15 at $B = \{e\}$. □

## 50.16 Chapter 16: integrals, expectations and the Foreman procedure

**Proposition 16.5 (Snips are completions).** *As stated: (iii) at every imbalance, through the appended list $\binom{B}{1}$ of the singletons of the bunch.*

*Proof.* Items (i) and (ii) are the reference, proof of 16.5. For (iii) let $R = \mathcal{E} \setminus B$.

*Projectivity.* The family word $W$ is homogeneous of degree $-\sum_j(s_j r_j + t_j r_j^*) = \kappa_{\mathcal{E}} - m_{\mathcal{E}}$ (8.1), so $\omega = \alpha^{\mathbf{m}-\mathbf{1}} W\, d\alpha$ is multiplied by $\lambda^{\kappa\varepsilon}$ under $\alpha \mapsto \lambda\alpha$, and $|\alpha_B|^{-\kappa\varepsilon}\omega$ is unchanged: it is projective (1.1(vii)). On the section $|\alpha_B| = 1$ the multiplier is 1, so the two forms induce the same form there.

*The appended table.* The sets $\{e\}$, $e \in B$, have one element each and the exchange property, two of them differing in one mark; so they are the bases of a table on $\mathcal{E}$ of rank 1 (7.1), in which

the marks of $R$ lie in no basis, and with unit weights its basis polynomial is $\sum_{e\in B}\alpha_e = |\alpha_B|$. So $|\alpha_B|^{-\kappa_\varepsilon}W$ is the family word of $F'$ at the exponents $\big((\mathbf{s},\kappa_\mathcal{E}),(\mathbf{t},0)\big)$, the appended table carrying $(\kappa_\mathcal{E},0)$, and $|\alpha_B|^{-\kappa_\varepsilon}\omega$ is the form of $F'$ at the masses $\mathbf{m}$.

*Walls.* For the appended table $d_0(S) = \min_{e\in B}|\{e\}\cap S| = \mathbf{1}[B\subseteq S]$ (7.13), and its exponent on $\Psi$ is 0; so by 8.1 the walls of $F'$ at $\mathbf{m}$ are $\kappa_S - \kappa_\mathcal{E}\mathbf{1}[B\subseteq S]$, and its imbalance is $\kappa_\mathcal{E} - \kappa_\mathcal{E}\cdot 1 = 0$, the appended table having rank 1.

*Value.* $|\alpha_B|$ is positive and homogeneous of degree one on the orthant. Where every proper wall of $F'$ has positive real part, the period of $F'$ converges, and the integral of its form over the section $\{|\alpha_B|=1\}$ is that period (8.7); this is the display.

*The cases.* For $B=\{g\}$ the appended table has the one basis $\{g\}$ and the letter $\alpha_g$, and $\alpha_g^{-\kappa_\varepsilon}W\,d\alpha_\mathbf{m} = W\,d\alpha_{\mathbf{m}-\kappa_\varepsilon\mathbf{e}_g} = \omega_{\mathbf{m}^{(g)}}$ (16.4): this is (i). For $B=\mathcal{E}$ the appended table is the list $\binom{\mathcal{E}}{1}$, and the statement is 16.3. For $k=-\kappa_\mathcal{E}\in\mathbb{N}$, $k\geq 1$: on the section $\omega=|\alpha_B|^k\omega$, $|\alpha_B|^k = \sum_{|\mathbf{n}|=k}\binom{k}{\mathbf{n}}\alpha_B^\mathbf{n}$, and $\alpha_B^\mathbf{n}\omega = W\,d\alpha_{\mathbf{m}+\mathbf{n}}$ is the completion along $\mathbf{n}/k$ (16.4: $\mathbf{m}-\kappa_\mathcal{E}\mathbf{n}/k = \mathbf{m}+\mathbf{n}$), projective, whose integral over the section is $P_F(\mathbf{m}+\mathbf{n};\mathbf{s},\mathbf{t})$ wherever that period converges (8.7). This last case is the reference, proof of 16.5(iii). □

**Corollary 16.6 (The period fibres over the scale of a bunch).** *As stated: every section $\{h(\alpha_R)=1\}$ of the room, and off the balance wall.*

*Proof.* For a positive function $h$ on an orthant $\mathbb{R}^X_{>0}$, homogeneous of degree one, write $\lambda_h$ for the measure on $\Sigma_h=\{\beta:\ h(\beta)=1\}$ with $d\alpha = \rho^{|X|-1}d\rho\,d\lambda_h(\beta)$ under the bijection $(\rho,\beta)\mapsto\rho\beta$ of $(0,\infty)\times\Sigma_h$ onto $\mathbb{R}^X_{>0}$: the measure of 16.2 (the reference, proof of 16.2). For $h=\alpha_g$ it is the Lebesgue measure in the coordinates $\alpha_e$, $e\neq g$, at $\alpha_g=1$, and for $h=|\alpha|$ it is $d\omega$ (2.7).

Let $h$ be a function of $\alpha_R$ as in the statement, and put $\tilde h(\alpha)=h(\alpha_R)$ on $\mathbb{R}^\mathcal{E}_{>0}$, positive and homogeneous of degree one. The map $(\rho,\beta_B,\beta_R)\mapsto(\rho\beta_B,\rho\beta_R)$, $\beta_B\in\mathbb{R}^B_{>0}$, $\beta_R\in\Sigma_h$, is a bijection onto $\mathbb{R}^\mathcal{E}_{>0}$, with $d\alpha_R = \rho^{|R|-1}d\rho\,d\lambda_h(\beta_R)$ and $d\alpha_B=\rho^{|B|}d\beta_B$ at fixed $\rho$. So $d\alpha = \rho^{n-1}d\rho\,d\beta_B\,d\lambda_h(\beta_R)$, and the measure of the section $\{\tilde h=1\}$ is $d\beta_B\,d\lambda_h(\beta_R)$. The integral of $W\,d\alpha_\mathbf{m}$ over it is

$$\int_{\Sigma_h}\beta_R^{\mathbf{m}|_R-\mathbf{1}}\int_{\mathbb{R}^B_{>0}}\beta_B^{\mathbf{m}|_B-\mathbf{1}}\prod_j K_j(\beta_B,\beta_R)^{-s_j}\Psi_j(\beta_B,\beta_R)^{-t_j}\,d\beta_B\,d\lambda_h(\beta_R),$$

by Tonelli on the modulus, which is the same integrand at the real parts of the data, and then Fubini. With $\beta_B=xu$, $x>0$, $u\in\Delta_B$, $\beta_B^{\mathbf{m}|_B-\mathbf{1}}d\beta_B = x^{m_B-1}u^{\mathbf{m}|_B-\mathbf{1}}dx\,d\omega_B(u)$ (2.7), so the inner integral is $\int_{\Delta_B}u^{\mathbf{m}|_B-\mathbf{1}}\,\mathrm{Sc}_B(u,\beta_R;\mathbf{m};\mathbf{s},\mathbf{t})\,d\omega_B(u)$, with the scale integral of 15.3 at the gauge $(u,\beta_R)$. This is the display.

The scale integral converges exactly when $\mathrm{Re}\,\kappa_B>0$ and $\mathrm{Re}\,\kappa_B^{\mathrm{top}}>0$ (15.3), conditions on the data alone; by Tonelli it converges at almost every $(u,\beta_R)$, hence at every point, and there it is the product of 15.3. On the balance wall the form is projective and its integral over every section is the period (8.7). At $h=\alpha_g$ the integral over $\{\alpha_g=1\}$ is $P_F(\mathbf{m}^{(g)};\mathbf{s},\mathbf{t})$ wherever it converges (16.5(i)), and $\lambda_h$ is the Lebesgue measure in $\alpha_{R-g}$ at $\alpha_g=1$; at $h=|\alpha_R|$, $\Sigma_h=\Delta_R$ and $\lambda_h=d\omega_R$. At $h=\alpha_g$ on the balance wall this is the reference, proof of 16.6. □

**Lemma 16.9 (The period is meromorphic in the data).** *As stated, for every family and all complex data.*

*Proof. Meromorphy.* By the first step of the proof of 15.11 above, which uses neither connectedness nor real data, $P_F=\sum_\pi I_\pi$ on the open convex set $\Omega$ of the data at which every proper wall has positive real part, with $I_\pi = \int_{[0,1]^{n-1}}\prod_{k=2}^n q_k^{\kappa_{U_k}-1}h_\pi\,dq$ and $h_\pi = L_\pi^{-\kappa_\varepsilon}\prod_j k_{j,\pi}^{-s_j}\psi_{j,\pi}^{-t_j}$ smooth near

the closed cube, its derivatives continuous in the variables and the data and entire in the data. By 8.4 each $I_\pi$ extends to a meromorphic function of the data, a quotient of a holomorphic function by products of affine functions $\kappa_{U_k}+i$, $i\in\mathbb{N}$, with poles only on the hyperplanes $\kappa_{U_k}=-i$. Every nonempty proper set of marks is a tail of some order, and no tail is $\mathcal{E}$. So $P_F$ is meromorphic in the masses and the exponents, with poles only where a proper wall is a nonpositive integer.

*The restriction to the balance wall.* For a nonempty proper $U$ the affine function $\kappa_U+i$ of the data has the part $m_U$ in the masses, which is not proportional to the part $m_{\mathcal{E}}$ of $\kappa_{\mathcal{E}}$; so the hyperplane $\kappa_U=-i$ is not the balance wall $\{\kappa_{\mathcal{E}}=0\}$, and it meets the balance wall in a hyperplane of it. Near every point of the balance wall $P_F$ is a quotient of a holomorphic function by a product of such affine functions, none vanishing identically on the balance wall; so the restriction of $P_F$ to the balance wall is meromorphic, with poles only where a proper wall is a nonpositive integer. □

**Proposition 16.12 (The form of a compound at one point).** *As stated, for any number $q\ge 2$ of parts.*

*Proof. The form.* Write $r_{j,i}$ for the rank of the restriction of table $j$ to $\mathcal{E}_i$, so that $K_{j,i}$ is homogeneous of degree $r_{j,i}$, $\Psi_{j,i}$ of degree $|\mathcal{E}_i|-r_{j,i}$, and $\kappa_i=m_{\mathcal{E}_i}-\sum_j\big(s_jr_{j,i}+t_j(|\mathcal{E}_i|-r_{j,i})\big)$; summing over $i$, $\sum_i\kappa_i=\kappa_{\mathcal{E}}=0$. The section $\{|\alpha_q|=1\}$ carries the measure $\prod_{i<q}d\alpha_i\,d\omega_{\mathcal{E}_q}(u_q)$, by the computation in the proof of 16.6 above with the room there replaced by $\mathcal{E}_q$. For $i<q$, $\alpha_i=\rho_iu_i$ gives $\alpha_i^{\mathbf{m}_i-\mathbf{1}}d\alpha_i=\rho_i^{m_{\mathcal{E}_i}-1}u_i^{\mathbf{m}_i-\mathbf{1}}d\rho_i\,d\omega_{\mathcal{E}_i}(u_i)$ (2.7), $K_{j,i}(\rho_iu_i)=\rho_i^{r_{j,i}}K_{j,i}(u_i)$ and $\Psi_{j,i}(\rho_iu_i)=\rho_i^{|\mathcal{E}_i|-r_{j,i}}\Psi_{j,i}(u_i)$. The powers of $\rho_i$ collect to $\rho_i^{\kappa_i-1}$, which gives the displayed form.

(i) For real $\kappa$, $\int_0^\infty\rho^{\kappa-1}d\rho$ diverges, at $\infty$ if $\kappa\ge0$ and at 0 if $\kappa\le0$. The form is positive, and by Tonelli its integral is the product of the $q-1$ integrals $\int_0^\infty\rho_i^{\kappa_i-1}d\rho_i$, each infinite, and of the $q$ shape integrals, each positive; so it is infinite.

(ii) A primitive of $\rho^{\kappa-1}$ is $\rho^\kappa/\kappa$, or $\log\rho$ at $\kappa=0$. It is its own log-power expansion at 0 and at $\infty$, with no term $\rho^0\log^0\rho$, so both regularized limits are 0 and the finite part is 0 (1.5).

(iii) At $\mathbf{m}(\boldsymbol{\eta})$, $\kappa_i=\kappa_{F_i}(\mathbf{m}_i^0)+i\eta_i|w_i|=i\eta_i$, and $\sum_i\kappa_i=0$ since $\sum_i\eta_i=0$. The walls of $F_i$ at $\mathbf{m}_i^0+i\eta_iw_i$ have the real parts of those at $\mathbf{m}_i^0$, positive, so $f(\boldsymbol{\eta}')=\prod_{i=1}^qP_{F_i}(\mathbf{m}_i^0+i\eta_iw_i;\mathbf{s},\mathbf{t})$, with $\boldsymbol{\eta}'=(\eta_1,\dots,\eta_{q-1})$ and $\eta_q=-\sum_{i<q}\eta_i$, is a smooth function of $\boldsymbol{\eta}'$, each period being holomorphic in the masses there (8.7). In $y_i=\log\rho_i$, $\rho_i^{\kappa_i-1}d\rho_i=e^{i\eta_iy_i}dy_i$. For a test function $H$ of $\boldsymbol{\eta}'$ of compact support, the integral of the form against $H$, taken over the shapes, then over $\boldsymbol{\eta}'$, then over $\mathbf{y}=(y_1,\dots,y_{q-1})$, is

$$\int_{\mathbb{R}^{q-1}}\int_{\mathbb{R}^{q-1}}H(\boldsymbol{\eta}')f(\boldsymbol{\eta}')\,e^{i\langle\boldsymbol{\eta}',\mathbf{y}\rangle}\,d\boldsymbol{\eta}'\,d\mathbf{y}=(2\pi)^{q-1}H(0)f(0),$$

by Fourier inversion for the smooth function $Hf$ of compact support, whose Fourier transform is integrable; and $f(0)=\prod_iP_{F_i}(\mathbf{m}_i^0;\mathbf{s},\mathbf{t})$. For $q=2$ this is the reference, proof of 16.12. □

**Lemma S4.** *Let $S$ be a nonempty proper set of marks of a family $F$ on $\mathcal{E}$ with $c_j(S)=0$ in every active table, and put $S'=\mathcal{E}\setminus S$.*

*(i) In every active table $j$, every basis meets $S$ in $r_j(S)$ marks and $S'$ in $r_j(S')$ marks; the bases of $\mathcal{B}_j|S$ and of $\mathcal{B}_j|S'$ are the sets $T\cap S$ and $T\cap S'$, $T\in\mathcal{B}_j$; and $X\sqcup Y\in\mathcal{B}_j$ for every basis $X$ of $\mathcal{B}_j|S$ and every basis $Y$ of $\mathcal{B}_j|S'$.*

*(ii) In this lemma write*

$$I_S(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\Delta_S\times\Delta_{S'}}\alpha^{\mathbf{m}-\mathbf{1}}\prod_jK_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}\,d\omega_S\,d\omega_{S'}.$$

*It converges, and is holomorphic in the data, wherever* $\operatorname{Re}\kappa_U > 0$ *for every nonempty proper subset* $U$ *of* $S$ *and every nonempty proper subset* $U$ *of* $S'$*; and it extends to a meromorphic function of the data whose poles lie on the hyperplanes* $\kappa_U = -i$, $i \in \mathbb{N}$, $U$ *a nonempty proper subset of* $S$ *or of* $S'$.

*(iii) Wherever every proper wall has positive real part,* $\kappa_{\mathcal{E}} = \kappa_S + \kappa_{S'}$ *has positive real part and* $J_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma(\kappa_S)\Gamma(\kappa_{S'})\, I_S(\mathbf{m};\mathbf{s},\mathbf{t})$.

*Proof.* (i) Fix an active table $j$. By 7.13 the numbers $|T \cap S|$ fill $\{d_j(S), \dots, r_j(S)\}$, a single value since $c_j(S) = 0$; so every basis meets $S$ in $r_j(S)$ marks and $S'$ in $r_j - r_j(S)$ marks, a constant, which is then $r_j(S')$. By 7.5 the bases of $\mathcal{B}_j|S$ are the sets $T \cap S$ with $|T \cap S| = r_j(S)$, that is all of them, and likewise for $S'$. Let $X = T_1 \cap S$ and $Y = T_2 \cap S'$ for bases $T_1, T_2$; induct on $k = |(T_2 \cap S) \setminus X|$. If $k = 0$, then $T_2 \cap S \subseteq X$, both of size $r_j(S)$, so $T_2 = X \sqcup Y$. If $k > 0$, take $a \in (T_2 \cap S) \setminus X$; then $a \notin T_1$. The exchange property (7.1) gives $b \in T_1 \setminus T_2$ with $T_3 = (T_2 \setminus \{a\}) \cup \{b\}$ a basis. As $|T_3 \cap S| = r_j(S) = |T_2 \cap S|$ and $a \in S$, $b \in S$, so $b \in X \setminus T_2$. Then $T_3 \cap S' = Y$ and $|(T_3 \cap S) \setminus X| = k - 1$, and by induction $X \sqcup Y$ is a basis.

*Homogeneity in each block.* By (i), for an active table $j$, $K_j(\rho\alpha_S, \rho'\alpha_{S'}) = \rho^{r_j(S)}\rho'^{r_j(S')}K_j(\alpha)$ and $\Psi_j(\rho\alpha_S, \rho'\alpha_{S'}) = \rho^{|S|-r_j(S)}\rho'^{|S'|-r_j(S')}\Psi_j(\alpha)$. For $U \in \{S, S'\}$, $c_j(U) = 0$ (7.13: $c_j(S') = c_j(S)$), so $d_j(U) = r_j(U)$ and $d_j^*(U) = |U| - r_j(U)$ (7.12). An inactive table enters with the exponents 0. So with $g(U) = \sum_j \big(s_j d_j(U) + t_j d_j^*(U)\big) = m_U - \kappa_U$, the function $\phi(\alpha) = \alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}$ has

$$\phi(\rho\alpha_S, \rho'\alpha_{S'}) = \rho^{\kappa_S - |S|}\rho'^{\kappa_{S'} - |S'|}\phi(\alpha),$$

and $|\phi|$ is the same function at the real parts of the data.

(iii) With $\alpha_S = \rho u$ and $\alpha_{S'} = \rho' u'$ (2.7 in each block), $d\alpha = \rho^{|S|-1}\rho'^{|S'|-1} d\rho\, d\rho'\, d\omega_S(u)\, d\omega_{S'}(u')$, and the integrand of $J_F$ becomes $\rho^{\kappa_S - 1}e^{-\rho}\rho'^{\kappa_{S'}-1}e^{-\rho'}\phi(u,u')$. Wherever every proper wall has positive real part, $\operatorname{Re}\kappa_{\mathcal{E}} = \operatorname{Re}\kappa_S + \operatorname{Re}\kappa_{S'} > 0$, by 8.16, $c_j(S) = 0$ holding in every table with $\sigma_j \neq 0$, these being active. Then every wall has positive real part, and $J_F$ at the real parts of the data is finite (8.5(i)); by Tonelli it is $\Gamma(\operatorname{Re}\kappa_S)\Gamma(\operatorname{Re}\kappa_{S'})$ times the integral of $|\phi|$ over $\Delta_S \times \Delta_{S'}$, which is therefore finite, and Fubini gives $J_F = \Gamma(\kappa_S)\Gamma(\kappa_{S'})I_S$.

(ii) *Charts.* For a finite set $X$, a mark $x \in X$ and a function $f$ on $\mathbb{R}^X_{>0}$, homogeneous of degree $-|X|$ and nonnegative or integrable on $\Delta_X$, $\int_{\Delta_X} f\, d\omega_X = \int f|_{\beta_x = 1}\, d\beta_{X-x}$, and the same holds on the cones over subsets of $\Delta_X$ and their traces on the chart: for a nonnegative function $\upsilon$ on $(0,\infty)$ with $\int_0^\infty \upsilon(\rho)\, d\rho/\rho = 1$, both sides equal $\int_{\mathbb{R}^X_{>0}} f(\alpha)\upsilon(|\alpha|)\, d\alpha$, by 2.7 on the left, and on the right by the change of variables $\alpha = \rho(1,\beta)$, $d\alpha = \rho^{|X|-1} d\rho\, d\beta$, with $\int_0^\infty \upsilon(\rho|(1,\beta)|)\, d\rho/\rho = 1$. The function $\psi(\alpha) = |\alpha_S|^{-\kappa_S}|\alpha_{S'}|^{-\kappa_{S'}}\phi(\alpha)$ is homogeneous of degree $-|S|$ in $\alpha_S$ and $-|S'|$ in $\alpha_{S'}$, and equal to $\phi$ on $\Delta_S \times \Delta_{S'}$.

For an order $\pi$ of $S$, with tails $U_k = \{\pi(k), \dots, \pi(|S|)\}$, $2 \le k \le |S|$, use on the chart $\beta_{\pi(1)} = 1$ the coordinates $\beta_{\pi(k)} = q_2 \cdots q_k$, $q \in (0,1]^{|S|-1}$, which cover the trace of the sector $\{\beta_{\pi(1)} \ge \cdots \ge \beta_{\pi(|S|)}\}$; the exponent of $q_k$ in $\beta^X$ is $|X \cap U_k|$, $d\beta_{S-\pi(1)} = \prod_k q_k^{|S|-k} dq$, $\beta^{\mathbf{m}|_S - \mathbf{1}} d\beta_{S-\pi(1)} = \prod_k q_k^{m_{U_k}-1} dq$, and $|\beta| = L_\pi(q) = \sum_{i=1}^{|S|}\prod_{k=2}^{i} q_k \in [1, |S|]$. Likewise $\pi'$, $U'_l$, $q'$ and $L_{\pi'}$ on $S'$. Decomposing each simplex into its sectors, and applying the identity above in each block, by Tonelli on the modulus and then Fubini,

$$I_S = \sum_{\pi,\pi'} \int \prod_k q_k^{m_{U_k}-1} \prod_l q_l'^{m_{U'_l}-1}\, L_\pi(q)^{-\kappa_S} L_{\pi'}(q')^{-\kappa_{S'}} \prod_j K_j(\beta,\beta')^{-s_j}\Psi_j(\beta,\beta')^{-t_j}\, dq\, dq',$$

the integral over $(0,1]^{|S|-1}\times(0,1]^{|S'|-1}$.

*The letters on a product of sectors.* Fix an active table $j$. By (i) and 7.5, $\mathcal{B}_j|S$ is a table on $S$, and by 8.3 for $\pi$ it has bases $X^\pm$ with $|X^+\cap U_k|=\min_X|X\cap U_k|$ and $|U_k\setminus X^-|=\min_X|U_k\setminus X|$, the minima over its bases; by (i) these minima are $\min_T|T\cap U_k|=d_j(U_k)$ and $\min_T|U_k\setminus T|=d_j^*(U_k)$ (7.13). Likewise $\mathcal{B}_j|S'$ has bases $Y^\pm$ for $\pi'$. For a basis $T$, $\beta^{T\cap S}/\beta^{X^+}=\prod_k q_k^{|T\cap U_k|-d_j(U_k)}$ with exponents in $\mathbb{N}$, and likewise on $S'$. So

$$K_j(\beta,\beta')=\prod_k q_k^{d_j(U_k)}\prod_l {q_l'}^{d_j(U_l')}\,k_{j,\pi,\pi'}(q,q'),$$
$$k_{j,\pi,\pi'}=\sum_T\theta_{j,T}\prod_k q_k^{|T\cap U_k|-d_j(U_k)}\prod_l {q_l'}^{|T\cap U_l'|-d_j(U_l')},$$

a polynomial with nonnegative coefficients. Its constant term is the sum of $\theta_{j,T}$ over the bases with $|T\cap U_k|=d_j(U_k)$ for $k\ge2$ and $|T\cap U_l'|=d_j(U_l')$ for $l\ge2$. As $|T\cap S|=|X^+|$, these counts determine $T\cap S=X^+$, the mark $\pi(k)$ lying in $T$ exactly when $|T\cap U_k|-|T\cap U_{k+1}|=1$, with $U_1=S$ and $U_{|S|+1}=\varnothing$; likewise $T\cap S'=Y^+$. By (i) $X^+\sqcup Y^+$ is a basis, so the constant term is $\theta_{j,X^+\sqcup Y^+}>0$. In the same way $\Psi_j(\beta,\beta')=\prod_k q_k^{d_j^*(U_k)}\prod_l {q_l'}^{d_j^*(U_l')}\,\psi_{j,\pi,\pi'}(q,q')$, with nonnegative coefficients and the constant term $\theta_{j,X^-\sqcup Y^-}>0$. As $m_U-g(U)=\kappa_U$,

$$I_S=\sum_{\pi,\pi'}\int\prod_{k=2}^{|S|}q_k^{\kappa_{U_k}-1}\prod_{l=2}^{|S'|}{q_l'}^{\kappa_{U_l'}-1}\,h_{\pi,\pi'}(q,q')\,dq\,dq',\qquad h_{\pi,\pi'}=L_\pi^{-\kappa_S}L_{\pi'}^{-\kappa_{S'}}\prod_j k_{j,\pi,\pi'}^{-s_j}\psi_{j,\pi,\pi'}^{-t_j},$$

the product over the active tables.

*Continuation.* The polynomials $k_{j,\pi,\pi'}$, $\psi_{j,\pi,\pi'}$, $L_\pi$ and $L_{\pi'}$ are at least their positive constant terms on the closed cube, hence positive on a neighbourhood of it; so $h_{\pi,\pi'}$ is smooth there, with every derivative continuous in the variables and the data and entire in the data, and $|h_{\pi,\pi'}|$, which is $h_{\pi,\pi'}$ at the real parts of the data, lies between positive constants on the cube, locally uniformly in the data. So the term of $(\pi,\pi')$ converges, and is holomorphic, exactly where $\operatorname{Re}\kappa_{U_k}>0$ and $\operatorname{Re}\kappa_{U_l'}>0$ for all $k,l$; the tails $U_k$, $k\ge2$, of the orders of $S$ are the nonempty proper subsets of $S$, and likewise for $S'$. This gives the convergence and the holomorphy of $I_S$. By 8.4, each term is a quotient of a holomorphic function of the data by products of the affine functions $\kappa_{U_k}+i$ and $\kappa_{U_l'}+i$, $i\in\mathbb{N}$, each a nonconstant function of the data (the coefficient of $m_e$, $e\in U$, being 1); so it extends meromorphically with poles only on these hyperplanes, and so does $I_S$. □

**Theorem 16.17 (The pseudointegral off the domain).** *As stated: the compound formula, with the value* 0 *on the balance wall, for every family and all complex data; the clause on targets, (i) and (ii) under the hypotheses of the reference.*

*Proof. The compound formula.* Let $S$ be nonempty and proper with $c_j(S)=0$ in every active table, $S'=\mathcal{E}\setminus S$, and $I_S$ the integral of Lemma S4, the integral of the statement. On the open convex set $\Omega$ of the data at which every proper wall has positive real part, $\operatorname{Re}\kappa_{\mathcal{E}}>0$ and $J_F=\Gamma(\kappa_S)\Gamma(\kappa_{S'})I_S$ (Lemma S4(iii)), and $J_F=\Gamma(\kappa_{\mathcal{E}})P_F$ (8.7); so $P_F={}_2\Gamma_1\left[\begin{matrix}\kappa_S,\ \kappa_{S'}\\ \kappa_{\mathcal{E}}\end{matrix}\right]I_S$ on $\Omega$, with $\kappa_S+\kappa_{S'}=\kappa_{\mathcal{E}}$ (8.16). Both sides are meromorphic in the data, by 16.9, by Lemma S4(ii) and by 1.2, and they agree on the nonempty open set $\Omega$; so they agree as meromorphic functions.

*The value* 0 *on the balance wall.* Put $G=\Gamma(\kappa_S)\Gamma(\kappa_{S'})I_S$, meromorphic in the data. The poles of $\Gamma(\kappa_S)\Gamma(\kappa_{S'})$ lie on the hyperplanes $\kappa_S=-i$ and $\kappa_{S'}=-i$, $i\in\mathbb{N}$, and those of $I_S$ on the hyperplanes $\kappa_U=-i$ with $U$ a nonempty proper subset of $S$ or of $S'$ (Lemma S4(ii)). In each case

$U$ is a nonempty proper set of marks, so, as in the proof of 16.9, none of these hyperplanes contains the balance wall, and $G$ restricts to a meromorphic function on the balance wall, holomorphic off a proper analytic subset of it. The function $1/\Gamma$ is entire and vanishes at 0, so $1/\Gamma(\kappa_{\mathcal{E}})$ is entire in the data and vanishes on the balance wall. Hence $P_F = G \cdot \Gamma(\kappa_{\mathcal{E}})^{-1}$ vanishes at every point of the balance wall at which $G$ is holomorphic, and the meromorphic restriction of $P_F$ to the balance wall is 0.

*The clause on targets, (i) and (ii)* are under the hypotheses of the reference, connected with real data and $s_j, t_j \geq 0$ not all zero: the reference, proof of 16.17. □

**Definition 16.28 (Period datum; moves).** As stated, for tables with positive rational weights.

*Proof.* The definition asks of the family only that its form be algebraic over $\mathbb{Q}$ and homogeneous of degree zero. The letters $K_j$ and $\Psi_j$ are polynomials whose coefficients are the weights, here rational; with rational exponents and masses $W\, d\alpha_{\mathbf{m}}$ is a product of rational powers of polynomials over $\mathbb{Q}$ and of the coordinates, times $d\alpha$, hence algebraic over $\mathbb{Q}$, and on the balance wall its degree is $\kappa_{\mathcal{E}} = 0$ (8.1). The walls of the faces are those of 8.5, for any weights. □

**Proposition 16.32 (Sections off the balance wall).** *As stated: a bunch section at every imbalance.*

*Proof.* The clause on a bunch section is 16.5(iii), proved above; the rest is the reference, proof of 16.32. In (ii) the walls of the chart completion at $g$ are those of 16.5(iii) at $B = \{g\}$, $\kappa_S - \kappa_{\mathcal{E}} \mathbf{1}_S(g)$. □

**Definition 16.29 (The pseudointegral over a region).** As stated; the last sentence of (i) is stated for families of tables with positive rational weights, where the reference takes a list.

*Proof.* Let the tables have positive rational weights and the data be rational, on the balance wall. The letters $K_j$ and $\Psi_j$ are polynomials whose coefficients are the weights, so the form $W\, d\alpha_{\mathbf{m}}$ is algebraic over $\mathbb{Q}$ and the family has a period datum (16.28), whose moves are defined over $\mathbb{Q}$ by that definition. The three regions named in (i) are cut out by polynomial inequalities over $\mathbb{Q}$: the chart $\alpha_g = 1$ by $\alpha_e > 0$ for $e \neq g$; the simplex, in the coordinates $(u_e)_{e \neq e_0}$, by $u_e > 0$ and $1 - \sum_{e \neq e_0} u_e > 0$; a bunch section $|\alpha_B| = 1$ by these inequalities on $B$ together with $\alpha_e > 0$ on the room.

A change of variables of 16.28 keeps this property. Let $\mathcal{R}$ lie in the open domain and be cut out there by $p_1 > 0, \ldots, p_k > 0$ with $p_i \in \mathbb{Q}[x]$, and let $\phi$ be birational over $\mathbb{Q}$ and carry the open domain bijectively onto the open domain. Each coordinate of $\phi^{-1}$ is a quotient of polynomials over $\mathbb{Q}$ whose denominator does not vanish on the open domain, where $\phi^{-1}$ is defined; so $p_i \circ \phi^{-1} = A_i / D_i$ with $A_i, D_i \in \mathbb{Q}[y]$, $D_i$ a product of these denominators, and

$$\phi(\mathcal{R}) = \{y \text{ in the open domain} : \ A_i(y)\, D_i(y) > 0 \text{ for } 1 \leq i \leq k\},$$

the open domain being an orthant, cut out by the positivity of its coordinates. The region of a product of data is the product of the regions, cut out by the inequalities of the two factors in disjoint coordinates; linear combinations and boundary terms are formed on the regions already reached, as for a list. Of the family the argument of the reference uses only that its period datum and its moves are defined over $\mathbb{Q}$, which holds for positive rational weights. □

## 50.17 Chapter 17: the family as one Laplace transform

### 50.17.1 What the one-table proofs read, and what a family reads

The one-table statements read the exponent of the law of a question, $\sigma E + \delta O$ at offsets zero, through the width wd of the Newton polytope of $\Lambda$ and through the convexity of $E$; the walls of the wedge, its slope and the volume at low strength are all values of the width. For a family the exponent is

$$X(z) = 2\langle \boldsymbol{\mu}_\perp, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big),$$

and what replaces the width is the wall at a tilt (Definition 17.3): on the sector of every order the exponent is minus twice the wall at the tilt less the logarithms of the sector letters (Proposition 17.4(i)), which is Theorem 8.5(ii) read in the signed coordinates. Every statement below is read off that identity, the convexity of the centred potential, and, for samples of bases, the factorization of the chance of a sample over its draws. Throughout, $n = |\mathcal{E}| \geq 2$, the data are real and on the balance wall unless said otherwise, a question is written $q = (\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$, its data are $\big(\mathbf{m}_F(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta}); \boldsymbol{\sigma} + \boldsymbol{\delta}, \boldsymbol{\sigma} - \boldsymbol{\delta}\big)$, and $P_F(q)$ denotes its period. On the sector $H_\pi$ of an order $\pi$, read in the tilts (Definition 17.3), the differences of a tilt along $\pi$ are written $v_k = z_{\pi(k-1)} - z_{\pi(k)} \geq 0$, $2 \leq k \leq n$, so that $\kappa(z) = \sum_k \kappa_{U_k} v_k$; at $\alpha = e^{2z}$ the gaps of Corollary 8.6 are $2v_k$.

**Lemma S5** (The walls on the balance wall)**.** *On the balance wall, for every nonempty proper $S$,*

$$\kappa_S = \kappa_S^{\mathbf{E}}(\boldsymbol{\sigma}) + \kappa_S^{\mathbf{O}}(\boldsymbol{\mu}_\perp, \boldsymbol{\delta}), \qquad \kappa_S^{\mathbf{E}}(\boldsymbol{\sigma}) = \sum_j \sigma_j c_j(S), \qquad \kappa_S^{\mathbf{O}}(\boldsymbol{\mu}_\perp, \boldsymbol{\delta}) = \mu_{\perp,S} - \sum_j \delta_j \varepsilon_j^\perp(S),$$

*with $\varepsilon_j^\perp(S) = \varepsilon_j(S) - |S|\frac{2r_j - n}{n}$; and $\kappa_{\mathcal{E}\setminus S}^{\mathbf{E}} = \kappa_S^{\mathbf{E}}$, $\kappa_{\mathcal{E}\setminus S}^{\mathbf{O}} = -\kappa_S^{\mathbf{O}}$, so $\kappa_{\mathcal{E}\setminus S} = \kappa_S^{\mathbf{E}} - \kappa_S^{\mathbf{O}}$. In particular every wall is a linear function of the pair coordinates $(\boldsymbol{\mu}_\perp; \boldsymbol{\sigma}, \boldsymbol{\delta})$ of the question.*

*Proof.* By Theorem 8.2(iii), $\kappa_S = \mu_S + \sum_j \big(\sigma_j c_j(S) - \delta_j \varepsilon_j(S)\big)$ and $\kappa_{\mathcal{E}} = \mu_{\mathcal{E}} - \sum_j \delta_j(2r_j - n)$. On the balance wall $\mu_{\mathcal{E}} = \sum_j \delta_j(2r_j - n)$, and $\boldsymbol{\mu} = \boldsymbol{\mu}_\perp + \frac{\mu_{\mathcal{E}}}{n}\mathbf{1}$ gives $\mu_S = \mu_{\perp,S} + \frac{|S|}{n}\sum_j \delta_j(2r_j - n)$; substituting gives the decomposition. By Theorem 7.13, $c_j(\mathcal{E} \setminus S) = c_j(S)$ and $\varepsilon_j(\mathcal{E} \setminus S) = 2r_j - n - \varepsilon_j(S)$, so $\varepsilon_j^\perp(\mathcal{E} \setminus S) = 2r_j - n - \varepsilon_j(S) - \frac{n-|S|}{n}(2r_j - n) = -\varepsilon_j^\perp(S)$, and $\mu_{\perp,\mathcal{E}\setminus S} = -\mu_{\perp,S}$ as $\boldsymbol{\mu}_\perp \in \mathbf{1}^\perp$. □

### 50.17.2 The exponent of a question

**Definition 17.3: the claims.** Let $z \in H_\pi$. A difference $v_k$ is positive exactly when $z_{\pi(k-1)} > z_{\pi(k)}$, and then $U_k = \{e : \ z_e < z_{\pi(k-1)}\}$, a set $\{e : \ z_e < z_f\}$ with $z_f > \min_e z_e$; so the tails with a positive difference, and their differences, are read off $z$ alone, and $\kappa(z)$ does not depend on the order $\pi$ whose sector holds $z$. On $H_\pi$ the differences are linear in $z$, so $\kappa$ is linear there, and $\kappa(cz) = c\kappa(z)$ for $c \geq 0$. The tilt $-\mathbf{1}_S$ lies in the sectors of the orders that list $\mathcal{E} \setminus S$ first; there its only positive difference is 1, at the tail $U_k = S$, so $\kappa(-\mathbf{1}_S) = \kappa_S$. In the chart $z_{\pi(1)} = 0$ the coordinates are $z_{\pi(i)} = -(v_2 + \cdots + v_i)$, a triangular change of coordinates with determinant $\pm 1$, so $dz = dv_2 \cdots dv_n$ on $H_\pi$; the chart does not matter (Definition 17.1). Finally $\kappa(z)$ is linear in the walls, which are linear in the question by Lemma S5.

**Proposition 17.4 (The exponent of a question reads its walls; the even statistic is least at the table's gauge).** *As stated, for a question with real exponents of any signs.*

*Proof.* (i) Write $\boldsymbol{\mu} = \mathbf{m} - \sigma\mathbf{1}$ for the offsets, so that $\boldsymbol{\mu} - \boldsymbol{\mu}_\perp \in \mathbb{R}\mathbf{1}$ and $\mu_{\mathcal{E}} = \sum_j \delta_j(2r_j - n)$ on the balance wall. The function $\tilde{X}(z) = 2\langle\boldsymbol{\mu}, z\rangle - \sum_j \big(\sigma_j E_j(z) + \delta_j O_j(z)\big)$ on $\mathbb{R}^{\mathcal{E}}$ is unchanged by $z \mapsto z + c\mathbf{1}$, since $E_j$ is unchanged and $O_j$ moves by $2c(2r_j - n)$ (Theorem 7.15(iii)), the move of the whole being $2c\kappa_{\mathcal{E}} = 0$ (Proposition 16.7); on $\mathbf{1}^\perp$ it is $X$. So $X$ may be read at any representative of a tilt. At $\alpha = e^{2z}$, Theorem 7.15(i) gives $\sigma_j E_j + \delta_j O_j = s_j \log \Lambda_j(z) + t_j \log \Lambda_j(-z) = s_j \log K_j(\alpha) + t_j \log \Psi_j(\alpha) - (s_j + t_j)\langle\mathbf{1}, z\rangle$, and $2\langle\boldsymbol{\mu}, z\rangle + 2\sigma\langle\mathbf{1}, z\rangle = \langle\mathbf{m}, \log\alpha\rangle$, so

$$X(z) = \log\Big(\alpha^{\mathbf{m}} \prod_j K_j(\alpha)^{-s_j} \Psi_j(\alpha)^{-t_j}\Big),$$

the logarithm of the integrand of the period against the measure $d\alpha/\alpha^{\mathbf{1}}$. Take the sector coordinates of Theorem 8.5(ii) for $\pi$: $\rho = \alpha_{\pi(1)} = e^{2z_{\pi(1)}}$ and $q_k = \alpha_{\pi(k)}/\alpha_{\pi(k-1)} = e^{-2v_k}$. Since $\alpha_{\pi(i)} = \rho q_2 \cdots q_i$ and $\pi(i) \in U_k$ exactly when $k \le i$, $\alpha^{\mathbf{m}} = \rho^{m_{\mathcal{E}}} \prod_k q_k^{m_{U_k}}$; with $K_j = \rho^{r_j} \prod_k q_k^{d_j(U_k)} k_{j,\pi}(q)$ and $\Psi_j = \rho^{n-r_j} \prod_k q_k^{d_j^*(U_k)} \psi_{j,\pi}(q)$,

$$\alpha^{\mathbf{m}} \prod_j K_j^{-s_j} \Psi_j^{-t_j} = \rho^{\kappa_{\mathcal{E}}} \prod_{k=2}^{n} q_k^{\kappa_{U_k}} \prod_j k_{j,\pi}(q)^{-s_j} \psi_{j,\pi}(q)^{-t_j},$$

and $\rho^{\kappa_{\mathcal{E}}} = 1$ on the balance wall. The logarithm, with $\log q_k = -2v_k$, is the identity of (i).

On $H_\pi$ the greedy basis $T^+_{j,\pi}$ has the largest monomial $\alpha^T$ among the bases of table $j$, and $T^-_{j,\pi}$ the largest $\alpha^{\mathcal{E}\setminus T}$ (Lemma 8.3); as $-\chi_T = \chi_{\mathcal{E}\setminus T}$, $h_j(z) = \langle\chi_{T^+_{j,\pi}}, z\rangle$ and $h_j(-z) = \langle\chi_{\mathcal{E}\setminus T^-_{j,\pi}}, z\rangle$, whatever the signs of the exponents. Write $z = z_{\pi(1)}\mathbf{1} - \sum_k v_k \mathbf{1}_{U_k}$; for $A \subseteq \mathcal{E}$, $\langle\chi_A, z\rangle = z_{\pi(1)}(2|A| - n) - \sum_k v_k(2|A \cap U_k| - |U_k|)$, and $|T^+_{j,\pi} \cap U_k| = d_j(U_k)$, $|U_k \setminus T^-_{j,\pi}| = d_j^*(U_k)$. So

$$\sum_j \big(s_j h_j(z) + t_j h_j(-z)\big) = 2z_{\pi(1)} \sum_j \delta_j(2r_j - n) + 2\sum_k v_k\Big(\sigma|U_k| - \sum_j \big(s_j d_j(U_k) + t_j d_j^*(U_k)\big)\Big),$$

while $2\langle\boldsymbol{\mu}, z\rangle = 2z_{\pi(1)}\mu_{\mathcal{E}} - 2\sum_k v_k \mu_{U_k}$. The difference is $2\sum_k v_k\big(m_{U_k} - \sum_j (s_j d_j(U_k) + t_j d_j^*(U_k))\big) = 2\kappa(z)$, the term in $z_{\pi(1)}$ vanishing on the balance wall; on $\mathbf{1}^\perp$, $\langle\boldsymbol{\mu}, z\rangle = \langle\boldsymbol{\mu}_\perp, z\rangle$. For $s_j, t_j \ge 0$: the support functions of $P_j$ and $P_j^*$ are $\frac{1}{2}(h_j(z) + \langle\mathbf{1}, z\rangle)$ and $\frac{1}{2}(h_j(-z) + \langle\mathbf{1}, z\rangle)$, support functions add over a Minkowski sum with nonnegative coefficients, and so $2\kappa(z) = 2h_{\mathcal{N}_F}(z) - 2\sigma\langle\mathbf{1}, z\rangle - 2\langle\boldsymbol{\mu}, z\rangle = 2h_{\mathcal{N}_F}(z) - 2\langle\mathbf{m}, z\rangle$, twice the support function of $\mathcal{N}_F - \mathbf{m}$, which lies in $\mathbf{1}^\perp$ and so reads tilts.

The sector letters are polynomials in $q$ with nonnegative coefficients and the constant terms $\theta_{j,T^+_{j,\pi}}$, $\theta_{j,T^-_{j,\pi}}$, and their value at $q = \mathbf{1}$ is $K_j(\mathbf{1}) = \Psi_j(\mathbf{1})$; on $[0,1]^{n-1}$ they increase in every $q_k$ and so lie between these (Theorem 8.5(ii)), whatever the signs of the exponents. Taking logarithms and weighting by $s_j$ and $t_j$ gives both bounds.

(ii) If every proper wall is positive, on $H_\pi$, $\kappa(z) = \sum_k \kappa_{U_k} v_k \ge \kappa_{\min} \sum_k v_k = \kappa_{\min}(z_{\pi(1)} - z_{\pi(n)})$, and $\max_e z_e - \min_e z_e$ vanishes on $\mathcal{T}$ only at 0. Conversely $\kappa(-\mathbf{1}_S) = \kappa_S$.

(iii) At the centre $\boldsymbol{\mu}_\perp = 0$ and $\boldsymbol{\delta} = 0$, so $X = -\sum_j \sigma_j E_j$, and the walls are $\kappa_S = \sum_j \sigma_j c_j(S)$ (Proposition 8.21(ii)). By (i), $2\kappa = \sum_j \sigma_j(h_j(z) + h_j(-z)) = \sum_j \sigma_j \operatorname{wd}_j(z)$, and $\operatorname{wd}_j(z) = \max_T\langle\chi_T, z\rangle - \min_T\langle\chi_T, z\rangle = 2\big(\max_T\langle\mathbf{1}_T, z\rangle - \min_T\langle\mathbf{1}_T, z\rangle\big)$ is twice the width of $P_j$ in the direction $z$, equal to $2(r_j(S) - d_j(S)) = 2c_j(S)$ at $z = \mathbf{1}_S$ (Theorem 7.13). The bounds of (i) at $s_j = t_j = \sigma_j$ are the displayed ones. The Hessian of $\log\Lambda_j$ at $z$ is the covariance of $\chi_T$ under $\mathbb{P}_{j,e^{2z}}$ (Theorem 7.16), so $\log\Lambda_j$ is convex, and so is $E_j(z) = \log\Lambda_j(z) + \log\Lambda_j(-z)$, which is even; hence $E_j(z) = \frac{1}{2}(E_j(z) + E_j(-z)) \ge E_j(0) = 2\log\Lambda_j(0) = 2\log K_j(\mathbf{1})$.

For $v \in \mathbf{1}^{\perp}$,

$$v^{\mathsf{T}} \nabla^2 \Big( \sum_j \sigma_j E_j \Big)(z)\, v = \sum_j \sigma_j \Big( \operatorname{Var}_{T \sim \mathbb{P}_{j,e^{2z}}} \langle \chi_T, v \rangle + \operatorname{Var}_{T \sim \mathbb{P}_{j,e^{-2z}}} \langle \chi_T, v \rangle \Big).$$

Every basis has positive chance under both laws, so the form vanishes only if $\langle \chi_T, v \rangle = 2\langle \mathbf{1}_T, v \rangle - \langle \mathbf{1}, v \rangle$ is constant over the bases of every table with $\sigma_j > 0$. Then $v$ is constant on every molecule of the family of these tables (Lemma 7.8), and every such molecule $M$ has $c_j(M) = 0$ in each of them, the table being the direct sum of its restrictions to the molecules (Proposition 7.9), so that $|T \cap M| = r_j(M)$ for every basis. If $v \notin \mathbb{R}\mathbf{1}$, the union $S$ of the molecules on which $v$ is largest is nonempty and proper with $c_j(S) = 0$ for every table with $\sigma_j > 0$: the family is not connected. So on a connected family the form is positive on $\mathcal{T} \setminus 0$ and $\sum_j \sigma_j E_j$ is strictly convex on $\mathcal{T}$. Finally $\sum_j \sigma_j \operatorname{wd}_j = 2\kappa$ is positive on $\mathcal{T} \setminus 0$ by (ii), the walls $\sum_j \sigma_j c_j(S)$ of the centre being positive exactly when the family is connected (Proposition 8.21(ii)). □

*Remark.* For one table at the centre of strength $\sigma$, (iii) is the statement of the reference, with $2\kappa = \sigma \operatorname{wd}$; the family statement adds (i), in which the width of one table is replaced by the wall at the tilt of any question, and (ii). The identity of (i) is the sector form of the period read pointwise: integrated against $dz = dv$ on each sector it is $P_F(q) = 2^{n-1} \sum_\pi \int_{[0,\infty)^{n-1}} e^{-2\sum_k \kappa_{U_k} v_k} h_\pi(e^{-2v})\, dv$, with $h_\pi = \prod_j k_{j,\pi}^{-s_j} \psi_{j,\pi}^{-t_j}$, which is $\sum_\pi I_\pi$ of Corollary 8.6 in the gaps $2v_k$.

### 50.17.3 The two ends of strength

The high end needs one fact about integrals of $e^{-\lambda f}$ with a convex exponent, for complex $\lambda$ in a half-plane; it is proved here once, in the form in which the period uses it.

**Lemma S6** (A convex exponent at high strength)**.** *Let $f$ be a smooth function on a real vector space $V$ of dimension $d \geq 1$ with Lebesgue measure $dz$ and a norm $|\cdot|$, such that*

*(a) every derivative of $f$ of order at least one is bounded;*

*(b) $f$ is convex, with a critical point $z^\star$ at which $A_0 = \nabla^2 f(z^\star)$ is positive definite;*

*(c) $f(z) - f(z^\star) \geq c|z| - C$ for some $c > 0$ and $C$.*

*Then $z^\star$ is the only critical point, and as $|\lambda| \to \infty$ with $\operatorname{Re}\lambda \geq \lambda_0$, for a fixed $\lambda_0 > 0$,*

$$\int_V e^{-\lambda f(z)}\, dz = e^{-\lambda f(z^\star)} \Big( \frac{2\pi}{\lambda} \Big)^{\frac{d}{2}} \det(A_0)^{-\frac{1}{2}} \big( 1 + O(|\lambda|^{-1}) \big),$$

*the determinant in the coordinates of $dz$ and the power principal.*

*Proof.* Use a Euclidean norm, equivalent to the given one. *The gradient away from $z^\star$.* By convexity $\langle \nabla f(z), z - z^\star \rangle \geq f(z) - f(z^\star) \geq c|z| - C$, so $|\nabla f(z)| \geq (c|z| - C)/|z - z^\star|$, which tends to $c$ as $|z| \to \infty$. A convex function with a positive definite Hessian at a critical point has its strict minimum there, and its critical points form the convex set of its minima, which is $\{z^\star\}$. So for every neighbourhood $V_0$ of $z^\star$, $|\nabla f| \geq c_0 > 0$ on $V \setminus V_0$: on a large ball by compactness, outside it by the bound.

*A chart at $z^\star$.* Put $A(z) = 2\int_0^1 (1-t)\, \nabla^2 f\big(z^\star + t(z - z^\star)\big)\, dt$, smooth and symmetric with $A(z^\star) = A_0$. Since $\nabla f(z^\star) = 0$, $f(z) = f(z^\star) + \frac{1}{2}\langle A(z)(z - z^\star), z - z^\star \rangle$. On a ball $V_1$ about $z^\star$ on which $A(z)$ is positive definite put $w(z) = A(z)^{1/2}(z - z^\star)$, the positive square root being smooth there; then $f = f(z^\star) + \frac{1}{2}|w|^2$, and $Dw(z^\star) = A_0^{1/2}$ is invertible, so $w$ is a diffeomorphism of a smaller ball $V_2$ onto a neighbourhood of 0.

*Near $z^\star$.* Take a smooth $\chi$ with compact support in $V_2$ and $\chi = 1$ near $z^\star$. In the coordinates $w$, $\int \chi e^{-\lambda f} dz = e^{-\lambda f(z^\star)} \int_{\mathbb{R}^d} e^{-\lambda |w|^2/2} g(w)\, dw$, with $g$ smooth of compact support and $g(0) = |\det Dw(z^\star)|^{-1} = \det(A_0)^{-1/2}$. For $\operatorname{Re}\lambda > 0$ and $\xi \in \mathbb{R}^d$, $\int e^{-\lambda|w|^2/2 + i\langle \xi, w\rangle} dw = (2\pi/\lambda)^{d/2} e^{-|\xi|^2/(2\lambda)}$: both sides are holomorphic in $\lambda$ on the half-plane and agree for $\lambda > 0$, completing the square in each coordinate. Writing $g(w) = (2\pi)^{-d} \int \hat g(\xi) e^{i\langle \xi, w\rangle} d\xi$ with $\hat g$ rapidly decreasing,

$$\int e^{-\lambda|w|^2/2} g(w)\, dw = \Big(\frac{2\pi}{\lambda}\Big)^{\frac{d}{2}} (2\pi)^{-d} \int \hat g(\xi)\, e^{-|\xi|^2/(2\lambda)}\, d\xi = \Big(\frac{2\pi}{\lambda}\Big)^{\frac{d}{2}} \big(g(0) + \varrho(\lambda)\big),$$

and since $\operatorname{Re}(1/\lambda) \geq 0$ and $|e^{-x} - 1| \leq |x|$ for $\operatorname{Re} x \geq 0$, $|\varrho(\lambda)| \leq (2|\lambda|)^{-1} (2\pi)^{-d} \int |\xi|^2 |\hat g(\xi)|\, d\xi$.

*Away from $z^\star$.* On the support of $1 - \chi$, $|\nabla f| \geq c_0$. The operator $L = -\lambda^{-1} |\nabla f|^{-2} \langle \nabla f, \nabla \rangle$ has $L e^{-\lambda f} = e^{-\lambda f}$, and its transpose is $L^{\mathsf t} \varphi = \lambda^{-1} \operatorname{div}\big(\varphi\, |\nabla f|^{-2} \nabla f\big)$. Integrating by parts $N$ times, $\int (1-\chi) e^{-\lambda f} dz = \int e^{-\lambda f} (L^{\mathsf t})^N (1-\chi)\, dz$, with no boundary terms: $1 - \chi$ vanishes near $z^\star$, and on large spheres $|e^{-\lambda f}| = e^{-\operatorname{Re}\lambda f}$ decays exponentially by (c) while the coefficients stay bounded. Now $(L^{\mathsf t})^N (1-\chi) = \lambda^{-N} G_N$ with $G_N$ a polynomial in the derivatives of $\chi$, in those of $f$ of orders 1 to $N+1$ and in $|\nabla f|^{-2}$, bounded by (a) and $|\nabla f| \geq c_0$. Since $f \geq f(z^\star)$ and $\operatorname{Re}\lambda \geq \lambda_0$,

$$\Big| \int (1-\chi) e^{-\lambda f} dz \Big| \leq |\lambda|^{-N} \sup |G_N|\; e^{-\operatorname{Re}\lambda\, f(z^\star)} \int_V e^{-\lambda_0 (f - f(z^\star))} dz,$$

the last integral finite by (c). With $N > \frac{d}{2} + 1$ this is $O(|\lambda|^{-1})$ times the modulus $(2\pi/|\lambda|)^{d/2} e^{-\operatorname{Re}\lambda f(z^\star)} \det(A_0)^{-1/2}$ of the leading term, and the two parts add to the statement. (On the real half-line this is Laplace's method, along the imaginary direction stationary phase, and $w$ is the Morse chart.) □

**Theorem 17.5 (The two ends of strength).** *As stated.*

*Proof.* The walls and the exponents of $\lambda q$ are $\lambda$ times those of $q$ (Lemma S5), so its walls $\lambda \kappa_S$ have positive real parts for $\operatorname{Re}\lambda > 0$. At such data the period is the convergent integral of Theorem 8.7, the powers of the positive letters principal, and the change of variables of Definition 17.1, which uses only $K_j(e^{2z}) = e^{\langle \mathbf{1}, z\rangle} \Lambda_j(z)$ and $\Psi_j(e^{2z}) = e^{\langle \mathbf{1}, z\rangle} \Lambda_j(-z)$, gives $P_F(\lambda q) = 2^{n-1} \int_{\mathcal T} e^{\lambda X(z)} dz$, absolutely convergent as $|e^{\lambda X}| = e^{\operatorname{Re}\lambda\, X}$; here $X$ is the exponent of $q$, and the exponent of $\lambda q$ is $\lambda X$, the exponent being linear in the question.

(i) By Proposition 17.4(i) and $dz = dv$ on each sector, with $h_\pi = \prod_j k_{j,\pi}^{-s_j} \psi_{j,\pi}^{-t_j}$,

$$P_F(\lambda q) = 2^{n-1} \sum_\pi \int_{[0,\infty)^{n-1}} e^{-2\lambda \sum_k \kappa_{U_k} v_k}\, h_\pi(e^{-2v})^\lambda\, dv,$$

and with $v = w/\lambda$,

$$\lambda^{n-1} P_F(\lambda q) = 2^{n-1} \sum_\pi \int_{[0,\infty)^{n-1}} e^{-2\sum_k \kappa_{U_k} w_k}\, h_\pi(e^{-2w/\lambda})^\lambda\, dw.$$

On the cube $|\log h_\pi| \leq C$ (Proposition 17.4(i)), so for $0 < \lambda \leq 1$ the integrand lies between $e^{\mp C} e^{-2\sum_k \kappa_{U_k} w_k}$, whose integral is $\prod_k (2\kappa_{U_k})^{-1}$; it tends to $e^{-2\sum_k \kappa_{U_k} w_k}$, and by domination $\lambda^{n-1} P_F(\lambda q) \to 2^{n-1} \sum_\pi \prod_k (2\kappa_{U_k})^{-1} = \sum_\pi \prod_k \kappa_{U_k}^{-1}$. For the next order, $|h^\lambda - 1 - \lambda \log h| \leq \frac{1}{2} \lambda^2 C^2 e^{\lambda C}$, and for $w$ in the open orthant $e^{-2w/\lambda} \to 0$, so $\log h_\pi(e^{-2w/\lambda}) \to \log h_\pi(0)$, boundedly; hence

$$\lambda^{-1} \Big( \lambda^{n-1} P_F(\lambda q) - \sum_\pi \prod_k \kappa_{U_k}^{-1} \Big) \to \sum_\pi \prod_k \kappa_{U_k}^{-1} \log h_\pi(0),$$

and $h_\pi(0) = \prod_j \theta_{j,T^+_{j,\pi}}^{-s_j}\theta_{j,T^-_{j,\pi}}^{-t_j}$ is the sector integrand at $q = 0$, so that $\log h_\pi(0) = -\sum_j \big(s_j \log\theta_{j,T^+_{j,\pi}} + t_j \log\theta_{j,T^-_{j,\pi}}\big)$, the coefficient stated. Finally $\kappa$ is positively homogeneous of degree one and positive on $\mathcal{T}\setminus 0$ (Proposition 17.4(ii)), so $\operatorname{vol}\{\kappa\le r\} = r^{n-1}\operatorname{vol}\{\kappa\le 1\}$ and

$$2^{n-1}\int_{\mathcal{T}} e^{-2\kappa(z)}dz = 2^{n-1}\operatorname{vol}\{\kappa\le 1\}\,(n-1)\int_0^\infty e^{-2r}r^{n-2}dr = (n-1)!\operatorname{vol}\{\kappa\le 1\},$$

while on the sectors the same integral is $2^{n-1}\sum_\pi\prod_k(2\kappa_{U_k})^{-1}$. (For $s_j,t_j\ge 0$, $\{\kappa\le 1\}$ is the polar of $\mathcal{N}_F-\mathbf{m}$ in $\mathbf{1}^\perp$.)

(ii) Let $s_j,t_j\ge 0$ and $f=-X$ on $\mathcal{T}$, of dimension $n-1$ with the measure $dz$. The hypotheses of Lemma S6 hold. (a) Every derivative of $\log\Lambda_j$ of order $k\ge 1$ is a joint cumulant of order $k$ of $\chi_T\in\{\pm1\}^{\mathcal{E}}$ under $\mathbb{P}_{j,e^{2z}}$ (Theorem 7.16), bounded by a constant depending on $k$ alone, and $f=\sum_j\big(s_j\log\Lambda_j(z)+t_j\log\Lambda_j(-z)\big)-2\langle\boldsymbol{\mu}_\perp,z\rangle$. (b) $v^{\mathsf T}\nabla^2 f(z)v=\sum_j\big(s_j\operatorname{Var}_{\mathbb{P}_{j,e^{2z}}}\langle\chi_T,v\rangle+t_j\operatorname{Var}_{\mathbb{P}_{j,e^{-2z}}}\langle\chi_T,v\rangle\big)$; a question with $s_j,t_j\ge 0$ inside its walls is on a connected family, as $\kappa_S+\kappa_{\mathcal{E}\setminus S}=2\sum_j\sigma_j c_j(S)>0$ (Lemma S5) puts some active table of positive width on every $S$, and the argument of Proposition 17.4(iii) makes the form positive definite on $\mathcal{T}$ at every $z$; so $f$ is strictly convex. (c) $f\ge 2\kappa-C\ge 2\kappa_{\min}(\max_e z_e-\min_e z_e)-C$ by Proposition 17.4(i)–(ii), and $\max_e z_e-\min_e z_e$ is a norm on $\mathcal{T}$. A strictly convex $f$ with (c) attains its minimum at one point $z^\star$, its only critical point. By Theorem 7.16, $\nabla\log\Lambda_j(z)=2R_j(\alpha)-\mathbf{1}$ and $\nabla[\log\Lambda_j(-z)]=\mathbf{1}-2R_j(\alpha^{-1})=2R_j^*(\alpha)-\mathbf{1}$ at $\alpha=e^{2z}$, so the derivative of $f$ along $v\in\mathbf{1}^\perp$ is $2\langle\mathbf{u}(\alpha)-\mathbf{m},v\rangle$, $\mathbf{u}=\sum_j(s_jR_j+t_jR_j^*)$ the usage, $\mathbf{m}-\sigma\mathbf{1}-\boldsymbol{\mu}_\perp$ being a multiple of $\mathbf{1}$. It vanishes for every $v$ exactly when $\mathbf{u}(\alpha)-\mathbf{m}\in\mathbb{R}\mathbf{1}$, that is $\mathbf{u}(\alpha)=\mathbf{m}$, the two having the total $\sum_j(s_jr_j+t_jr_j^*)$ on the balance wall; and $\mathbf{m}$ is the target of $q$ there, so $z^\star$ is its native gauge (Definition 8.17). The usage and the masses of $\lambda q$ are $\lambda$ times those of $q$, so $z^\star$ is the native gauge of every $\lambda q$, $\lambda>0$. Lemma S6 with $d=n-1$ gives the expansion, the determinant taken on a chart $z_g=0$, that is of the Hessian of the invariant form $\tilde X$ of the proof of Proposition 17.4 with the row and the column of $g$ deleted. With $y=2z$, $\Phi(y)=\sigma\langle\mathbf{1},y\rangle+\Phi^\circ(z)$ (Theorem 8.19), so $\partial_{z_e}\partial_{z_f}(-\tilde X)=\partial_{z_e}\partial_{z_f}\Phi^\circ=4\partial_{y_e}\partial_{y_f}\Phi$, rung two of the still of the realization $\big((s_j,t_j)_j;e^{2z^\star}\big)$ of the active tables (Definition 9.4). At the centre $f=\sum_j\sigma_jE_j$ is even and strictly convex, so $z^\star=0$, $f(0)=2\sum_j\sigma_j\log K_j(\mathbf{1})$, and $\nabla^2 f(0)=\sum_j\sigma_j\cdot 2\operatorname{Cov}_{\mathbb{P}_{j,\mathbf{1}}}(\chi_T)=8\sum_j\sigma_j\big(\kappa_{j,\mathbf{1}}(X_e,X_f)\big)_{e,f}$, as $\chi_T=2X-\mathbf{1}$, read on the chart $z_g=0$ with the row and the column of $g$ deleted. □

*Remark.* At one table and the centre of strength one, $\kappa=\frac12\,\mathrm{wd}$ and $(n-1)!\operatorname{vol}\{\kappa\le 1\}=2^{n-1}(n-1)!\operatorname{vol}\{\mathrm{wd}\le 1\}$, the reference's low end; its high end is (ii) at offsets zero. Divided by the same limit for the list $\binom{\mathcal{E}}{1}$ at a full basis lean, whose walls are the masses, (i) is the limit of Proposition 49.14: there $\Gamma_{\binom{n}{1}}\begin{bmatrix}\lambda\boldsymbol{\tau}\\ \lambda\tau_{\mathcal{E}}\end{bmatrix}$ has $\lambda^{n-1}\Gamma_{\binom{n}{1}}\begin{bmatrix}\lambda\boldsymbol{\tau}\\ \lambda\tau_{\mathcal{E}}\end{bmatrix}\to\tau_{\mathcal{E}}/\prod_e\tau_e=\sum_\pi\prod_k\tau_{U_k}^{-1}$. In (ii) the real ray is Proposition 18.9 written in the signed coordinates; what the lemma adds is the whole half-plane, uniformly.

### 50.17.4 How far a question can lean, and the tails

**Proposition 17.7 (How far a question can lean; the values of the statistics).** *As stated.*

*Proof.* The decomposition of the walls is Lemma S5. Every proper wall is positive exactly when, for every $S$, $\kappa_S^{\mathbf{E}}+\kappa_S^{\mathbf{O}}>0$ and $\kappa_S^{\mathbf{E}}-\kappa_S^{\mathbf{O}}>0$, that is $|\kappa_S^{\mathbf{O}}|<\kappa_S^{\mathbf{E}}$. The fold of Theorem 16.18 sends the pair coordinates $(\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},\boldsymbol{\delta})$ to $(-\boldsymbol{\mu}_\perp;\boldsymbol{\sigma},-\boldsymbol{\delta})$, negating the offsets and the leans and keeping

the strengths (Definition 9.1); so it negates $\kappa_S^{\mathbf{O}}$ and swaps $\kappa_S$ with $\kappa_{\mathcal{E}\setminus S}$. When every $\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})$ is positive the conditions read $\kappa_S^{\mathbf{E}}/|\kappa_S^{\mathbf{O}}| > 1$ wherever $\kappa_S^{\mathbf{O}} \neq 0$, that is $\ell^*_{\boldsymbol{\sigma}}(\boldsymbol{\mu}_\perp, \boldsymbol{\delta}) > 1$: finitely many strict linear inequalities in the odd part, each with its negative, an open convex polyhedron symmetric under the fold. It is bounded exactly when the linear map $(c, \mathbf{b}) \mapsto (\kappa_S^{\mathbf{O}}(c, \mathbf{b}))_S$ is injective: then it lies in the preimage of a bounded box, and otherwise a line through 0 in the kernel lies in it. The homogeneities and the evenness in $(c, \mathbf{b})$ are read off the definition. For one connected table at offsets zero, $\kappa_S^{\mathbf{E}} = \sigma c(S)$ and $\kappa_S^{\mathbf{O}} = -\delta\varepsilon^\perp(S)$, so the conditions are $\sigma > 0$ and $|\delta| < \sigma \min c(S)/|\varepsilon^\perp(S)| = \ell_1^*(0,1)\,\sigma$.

(i) The exponent is linear in the question, so with $q' = (c; \boldsymbol{\sigma} + \mathbf{a}, \mathbf{b})$ and the exponent $X_{q'}$ of $q'$, the expectation is $2^{n-1} P_F(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})^{-1} \int_{\mathcal{T}} e^{X_{q'}(z)} dz$, an integral of a positive function. By Proposition 17.4(i), whose identity holds at every real exponents, $e^{X_{q'}}$ lies between constant multiples of $e^{-2\kappa'(z)}$, $\kappa'$ the wall at the tilt of $q'$, and $\int_{H_\pi} e^{-2\kappa'(z)} dz = \int_{[0,\infty)^{n-1}} e^{-2\sum_k \kappa_{U_k}(q')v_k} dv$ is finite exactly when every $\kappa_{U_k}(q')$ is positive. Every nonempty proper $S$ is a tail of some order, so the expectation is finite exactly when every proper wall of $q'$ is positive, and it is then $P_F(q')/P_F(\sigma\mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$ by Definition 17.1. The walls of $q'$ are $\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma} + \mathbf{a}) \pm \kappa_S^{\mathbf{O}}(c, \mathbf{b})$; at $\mathbf{a} = 0$ their positivity is $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) > 1$.

(ii) With $|\delta_j| \le \sigma_j$ the exponents are nonnegative, and by the proof of Theorem 17.5(ii) $-X$ is strictly convex on $\mathcal{T}$ with its least value at the native gauge $z^\star$, at which alone it is attained. $-X(z)$ is the value of the linear form $-2\langle \boldsymbol{\mu}_\perp, x\rangle + \sum_j (\sigma_j y_j + \delta_j w_j)$ at the point $(z; E_j(z), O_j(z))_j$, and $z$ is a coordinate of that point, so the least value of the form on the set of values of the statistics is taken at $\big(z^\star; E_j(z^\star), O_j(z^\star)\big)_j$ and only there. At the centre $z^\star = 0$ (Proposition 17.4(iii)), $E_j(0) = 2\log K_j(\mathbf{1})$ and $O_j(0) = 0$. □

**Proposition 17.8 (The tails of the statistics are read at the walls).** *As stated.*

*Proof.* By Lemma S5 the walls are linear in the pair coordinates, and for the direction $v$ of pair coordinates $(c; \mathbf{a}, \mathbf{b})$, $\kappa_S(v) = \kappa_S^{\mathbf{E}}(\mathbf{a}) + \kappa_S^{\mathbf{O}}(c, \mathbf{b})$, the offset direction $c$ entering through $c_S = \sum_{e\in S} c_e$. With $X_v(z) = 2\langle c, z\rangle - \sum_j (a_j E_j(z) + b_j O_j(z))$, $Y = X_v(Z)$ and $X_q + \lambda X_v = X_{q+\lambda v}$, so for real $\lambda \ge 0$, $\mathbb{E}[e^{\lambda Y}] = 2^{n-1} P_F(q)^{-1} \int_{\mathcal{T}} e^{X_{q+\lambda v}} dz$, finite, by the argument of Proposition 17.7(i), exactly when every $\kappa_S(q) + \lambda\kappa_S(v)$ is positive, that is for $\lambda < \lambda^*$, and then the ratio of the periods.

*Upper bound.* For $0 < \lambda < \lambda^*$, $\mathbb{P}(Y > x) \le e^{-\lambda x}\mathbb{E}[e^{\lambda Y}]$, so $\limsup_x x^{-1}\log \mathbb{P}(Y > x) \le -\lambda$ for every $\lambda < \lambda^*$.

*Lower bound.* Let $\lambda^* < \infty$, let $S^*$ attain it, and let $\pi$ list $\mathcal{E} \setminus S^*$ first, so that $S^* = U_{k^*}$ is a tail of $\pi$. Write $\kappa_q(z)$ and $\kappa_v(z) = \sum_k \kappa_{U_k}(v)\big(z_{\pi(k-1)} - z_{\pi(k)}\big)$ for the walls at the tilt of $q$ and of $v$, the latter read as balanced data with the exponents $(\mathbf{a} + \mathbf{b}, \mathbf{a} - \mathbf{b})$. By Proposition 17.4(i), whose identity holds at any balanced real data, $|X_q + 2\kappa_q| \le C_q$ and $|X_v + 2\kappa_v| \le C_v$ on $\mathcal{T}$. For $\varepsilon > 0$ put $a_x = (1+\varepsilon)x/(-2\kappa_{S^*}(v))$ and let $A_x \subset H_\pi$ be the set of the tilts whose difference along $\pi$ at the tail $S^*$ lies in $[a_x, a_x + 1]$ and whose other differences lie in $[0, 1]$, of $dz$-volume 1. On $A_x$, as $\kappa_{S^*}(v) < 0$,

$$Y \ge -2\kappa_v(z) - C_v \ge (1+\varepsilon)x - 2\sum_k |\kappa_{U_k}(v)| - C_v > x \quad \text{for large } x,$$

$$X_q \ge -2\kappa_q(z) - C_q \ge -2\kappa_{S^*}(q)(a_x + 1) - 2\sum_k |\kappa_{U_k}(q)| - C_q = -(1+\varepsilon)\lambda^* x - C',$$

since $2\kappa_{S^*}(q)a_x = (1+\varepsilon)\lambda^* x$. So $\mathbb{P}(Y > x) \ge 2^{n-1} P_F(q)^{-1} e^{-(1+\varepsilon)\lambda^* x - C'}$ for large $x$, $\liminf_x x^{-1}\log \mathbb{P}(Y > x) \ge -(1+\varepsilon)\lambda^*$ for every $\varepsilon > 0$, and with the upper bound the limit is $-\lambda^*$.

If $\lambda^* = \infty$, every $\kappa_S(v) \ge 0$, so $\kappa_v \ge 0$ on every sector and $Y \le C_v$.

At the centre of strengths $\boldsymbol{\sigma}$ and the direction of pair coordinates $(c;\mathbf{0},\mathbf{b})$, $\kappa_S(q)=\kappa_S^{\mathbf{E}}(\boldsymbol{\sigma})$ and $\kappa_S(v)=\kappa_S^{\mathbf{O}}(c,\mathbf{b})$. The first is unchanged and the second negated by $S\mapsto\mathcal{E}\setminus S$, so the sets on which $\kappa_S(v)<0$ are the complements of those on which it is positive, and $\lambda^*=\min_S \kappa_S^{\mathbf{E}}/|\kappa_S^{\mathbf{O}}|=\ell^*_{\boldsymbol{\sigma}}(c,\mathbf{b})$, for $Y$ and for $-Y$ alike. □

*Remark.* The one-table rate of both tails of $O(Z)$ at the centre of strength $\sigma_0$ is $\ell^*_{\sigma_0}(0,1)=\sigma_0\,\ell_1^*(0,1)$. The same proposition, applied to the family with the list $\binom{\mathcal{E}}{1}$ added, gives the tails of the Grey law (Proposition 49.12(iii), proved in Section 50.48.3).

## 50.18 Chapter 18: the Breta law and the port on the balance wall

**Proposition 18.3 (Scale and shapes along a bunch).** *As stated: off the balance wall and at complex exponents, the scale integral then carrying the appended list* $\binom{\mathcal{E}}{1}$.

*Proof. The display.* The masses are positive and every proper wall has positive real part, so $P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\Delta_\mathcal{E}} u^{\mathbf{m}-\mathbf{1}}W(u)\,d\omega_\mathcal{E}(u)$ converges (8.7). By 3.12 with the blocks $B$ and $R$, $u=(w_Bv_B,w_Rv_R)$ with $(w_B,w_R)\in\Delta_{\{B,R\}}$, $v_B\in\Delta_B$ and $v_R\in\Delta_R$, and

$$u^{\mathbf{m}-\mathbf{1}}d\omega_\mathcal{E}=w_B^{m_B-1}w_R^{m_R-1}dw_B\cdot v_B^{\mathbf{m}|_B-\mathbf{1}}d\omega_B\cdot v_R^{\mathbf{m}|_R-\mathbf{1}}d\omega_R.$$

With $x=w_B/w_R\in(0,\infty)$, $w_B=x/(1+x)$, $w_R=1/(1+x)$ and $dw_B=(1+x)^{-2}dx$, so $w_B^{m_B-1}w_R^{m_R-1}dw_B=x^{m_B-1}(1+x)^{-m_\mathcal{E}}dx$. The word $W$ is homogeneous of degree $\kappa_\mathcal{E}-m_\mathcal{E}$ (8.1), so $W(w_Bv_B,w_Rv_R)=w_R^{\kappa_\mathcal{E}-m_\mathcal{E}}W(xv_B,v_R)=(1+x)^{m_\mathcal{E}-\kappa_\mathcal{E}}W(xv_B,v_R)$, and the powers of $1+x$ collect to $(1+x)^{-\kappa_\mathcal{E}}$. By Tonelli on the modulus, which is the same integrand at the real parts of the exponents, and Fubini,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\Delta_B}\int_{\Delta_R} v_B^{\mathbf{m}|_B-\mathbf{1}}v_R^{\mathbf{m}|_R-\mathbf{1}}\int_0^\infty x^{m_B-1}(1+x)^{-\kappa_\mathcal{E}}W(xv_B,v_R)\,dx\,d\omega_R(v_R)\,d\omega_B(v_B).$$

The densities of $\mathrm{Dir}_{\mathbf{m}|_B}$ and $\mathrm{Dir}_{\mathbf{m}|_R}$ are $v_B^{\mathbf{m}|_B-\mathbf{1}}/\Gamma_{\binom{|B|}{1}}\begin{bmatrix}\mathbf{m}|_B\\ m_B\end{bmatrix}$ and $v_R^{\mathbf{m}|_R-\mathbf{1}}/\Gamma_{\binom{|R|}{1}}\begin{bmatrix}\mathbf{m}|_R\\ m_R\end{bmatrix}$ (3.10), which gives the display. At $(xU_B',U_R')$ the letter $|\alpha|$ of the list $\binom{\mathcal{E}}{1}$ is $x|U_B'|+|U_R'|=1+x$, so the inner integral is the scale integral (15.3) of the family with that table appended at the exponents $(\kappa_\mathcal{E},0)$ (16.3). On the balance wall the factor $(1+x)^{-\kappa_\mathcal{E}}$ is 1, the inner integral is $\mathrm{Sc}_B(U_B',U_R';\mathbf{m};\mathbf{s},\mathbf{t})$, and the double integral above is 16.6 at $h=|\alpha_R|$.

*The conditional density.* For real data with every proper wall positive the same computation, which needs no sign of the masses, shows that under $\mathrm{Breta}^\circ(F;\mathbf{m};\mathbf{s},\mathbf{t})$ the variables $\rho=|U_B|/|U_R|$, $V_B$ and $V_R$ have the joint density

$$\rho^{m_B-1}(1+\rho)^{-\kappa_\mathcal{E}}W(\rho V_B,V_R)\,V_B^{\mathbf{m}|_B-\mathbf{1}}V_R^{\mathbf{m}|_R-\mathbf{1}}/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$$

against $d\rho\,d\omega_B\,d\omega_R$. Given $(V_B,V_R)$, the density of $\rho$ is proportional to $\rho^{m_B-1}(1+\rho)^{-\kappa_\mathcal{E}}W(\rho V_B,V_R)$. On the balance wall this is the reference, proof of 18.3. □

**Corollary 18.7 (Conjugacy for the continued period).** *As stated.*

*Proof.* Let $\Omega$ be the set of the data at which every nonempty proper wall has positive real part, open and convex, and not empty: at $\mathbf{s}=\mathbf{t}=\mathbf{0}$ the walls are $\kappa_S=m_S$. Adding $\mathbf{1}_T$ to the masses and $\mathbf{e}_j$ to $\mathbf{s}$ adds $r_j$ to $m_\mathcal{E}$ and to $\sum_i(s_ir_i+t_ir_i^*)$, so it keeps the imbalance and maps the balance

wall into itself, and it adds $|T\cap S|-d_j(S)\ge 0$ to the wall $\kappa_S$ (8.1), so it maps $\Omega$ into $\Omega$; adding $\mathbf{1}_{\mathcal{E}\setminus T'}$ to the masses and $\mathbf{e}_j$ to $\mathbf{t}$ does the same, adding $|S\setminus T'|-d_j^*(S)\ge 0$ to $\kappa_S$. At data in $\Omega$ the three periods converge absolutely as integrals over $\Delta_{\mathcal{E}}$ (8.7). On $\Delta_{\mathcal{E}}$, $K_j(u)=\sum_T \theta_{j,T}u^{\mathbf{1}_T}$ and $\Psi_j(u)=\sum_{T'}\theta_{j,T'}u^{\mathbf{1}_{\mathcal{E}\setminus T'}}$ (7.1), so

$$u^{\mathbf{m}-\mathbf{1}}\prod_i K_i(u)^{-s_i}\Psi_i(u)^{-t_i}=\sum_T \theta_{j,T}\,u^{\mathbf{m}+\mathbf{1}_T-\mathbf{1}}K_j(u)^{-s_j-1}\prod_{i\ne j}K_i(u)^{-s_i}\prod_i\Psi_i(u)^{-t_i},$$

the powers of the positive letters being principal, and integrating term by term gives the first identity on $\Omega$; the second is the same with $\Psi_j$. By 16.9 the three sides are meromorphic in the masses and the exponents, the shifted ones being $P_F$ composed with translations, and they agree on the nonempty open set $\Omega$, so they agree as meromorphic functions. Their restrictions to the balance wall, meromorphic by 16.9 and mapped into themselves by the translations, then agree as well. □

**Lemma S7.** *Let $G$ be a convex function of class $C^2$ on $\mathbb{R}^d$ with $G(0)=0$, $\nabla G(0)=0$ and $\nabla^2G(0)$ positive definite, and let $b$ be a measurable function on $\mathbb{R}^d$, continuous at 0, with $|b(y)|\le e^{c_0(1+|y|)}$ for a constant $c_0\ge 0$. Then $\int_{\mathbb{R}^d}|b|e^{-\lambda G}dy<\infty$ for every large $\lambda$, and for every bounded continuous function $H$ on $\mathbb{R}^d$*

$$\lim_{\lambda\to+\infty}\lambda^{d/2}\int_{\mathbb{R}^d}b(y)H(\sqrt{\lambda}\,y)e^{-\lambda G(y)}dy=b(0)\int_{\mathbb{R}^d}H(x)e^{-\frac12\langle x,\nabla^2G(0)x\rangle}dx.$$

*At $H=1$ the right side is $b(0)(2\pi)^{d/2}\det\big(\nabla^2G(0)\big)^{-1/2}$.*

*Proof.* Let $\epsilon>0$ be the least eigenvalue of $\nabla^2G(0)$. By Taylor's formula $G(y)=\frac12\langle y,\nabla^2G(0)y\rangle+o(|y|^2)$; choose $\delta>0$ with $G(y)\ge\frac14\langle y,\nabla^2G(0)y\rangle\ge\frac{\epsilon}{4}|y|^2$ and $|b(y)|\le|b(0)|+1$ for $|y|\le\delta$.

*Linear growth.* For $|y|\ge\delta$ put $t=\delta/|y|\in(0,1]$. Convexity and $G(0)=0$ give $G(ty)\le tG(y)$, so $G(y)\ge t^{-1}G(ty)\ge\frac{|y|}{\delta}\cdot\frac{\epsilon}{4}\delta^2=\gamma|y|$, with $\gamma=\frac{\epsilon\delta}{4}$.

*The outer part.* For $\lambda\gamma\ge 2c_0$ and $|y|\ge\delta$, $|b(y)|e^{-\lambda G(y)}\le e^{c_0-\frac14\lambda\gamma|y|}\le e^{c_0-\frac14\lambda\gamma\delta}e^{-\frac14\lambda\gamma|y|}$. So $\int_{|y|\ge\delta}|b|e^{-\lambda G}dy\le e^{c_0-\frac14\lambda\gamma\delta}\big(\frac{4}{\lambda\gamma}\big)^d\int_{\mathbb{R}^d}e^{-|x|}dx$, and $\lambda^{d/2}$ times it tends to 0. With the bound on $|y|\le\delta$, the integral of $|b|e^{-\lambda G}$ is finite.

*The inner part.* With $y=x/\sqrt{\lambda}$,

$$\lambda^{d/2}\int_{|y|<\delta}b(y)H(\sqrt{\lambda}\,y)e^{-\lambda G(y)}dy=\int_{\mathbb{R}^d}\mathbf{1}\big[|x|<\delta\sqrt{\lambda}\big]\,b(x/\sqrt{\lambda})H(x)e^{-\lambda G(x/\sqrt{\lambda})}dx.$$

As $\lambda\to\infty$ the integrand tends to $b(0)H(x)e^{-\frac12\langle x,\nabla^2G(0)x\rangle}$ at every $x$, by the continuity of $b$ at 0 and $\lambda G(x/\sqrt{\lambda})\to\frac12\langle x,\nabla^2G(0)x\rangle$, and it is bounded by $(|b(0)|+1)\sup|H|\,e^{-\epsilon|x|^2/4}$, which is integrable. Dominated convergence gives the limit. At $H=1$, $\int e^{-\frac12\langle x,Ax\rangle}dx=(2\pi)^{d/2}(\det A)^{-1/2}$ for positive definite $A$. □

**Proposition 18.9 (The period along a ray of data).** *As stated: a base point of complex data on the balance wall added to the scaled data.*

*Proof.* Write $\mathbf{d}_\lambda=(\mathbf{m}_0+\lambda\boldsymbol{\tau};\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})$, and in this proof $G(y)=\Phi(y)-\langle\boldsymbol{\tau},y\rangle$, $\Phi$ the potential at $(\mathbf{s},\mathbf{t})$.

*The data.* The walls and the imbalance are affine in the data with real coefficients. At $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$, real data with $\boldsymbol{\tau}>0$, $s_j,t_j\ge 0$ not all zero and a native gauge on a connected family, every proper

wall is positive (8.19(iii)), and both $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ and $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ lie on the balance wall. So $\kappa_{\mathcal{E}}(\mathbf{d}_\lambda)=0$, and $\operatorname{Re}\kappa_S(\mathbf{d}_\lambda)=\operatorname{Re}\kappa_S(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)+\lambda\kappa_S(\boldsymbol{\tau};\mathbf{s},\mathbf{t})\to+\infty$ for every nonempty proper $S$: for $\lambda$ large every proper wall of $\mathbf{d}_\lambda$ has positive real part.

*The chart integral.* For such $\lambda$, $P_F(\mathbf{d}_\lambda)$ is the integral over the chart $\alpha_g=1$ (16.2). In $y=\log\alpha$, $y_g=0$, the measure is $\alpha^{\mathbf{m}-\mathbf{1}}d\alpha_{\mathcal{E}-g}=e^{\langle\mathbf{m},y\rangle}dy_{\mathcal{E}-g}$, and the family word at the exponents $(\mathbf{s}_0+\lambda\mathbf{s},\mathbf{t}_0+\lambda\mathbf{t})$ is $W_0(e^y)e^{-\lambda\Phi(y)}$ (8.18). So

$$P_F(\mathbf{d}_\lambda)=\int_{\{y_g=0\}}b(y)\,e^{-\lambda G(y)}\,dy_{\mathcal{E}-g},\qquad b(y)=e^{\langle\mathbf{m}_0,y\rangle}W_0(e^y).$$

*The minimum.* $\Phi$ is convex (8.19(ii)), and $\nabla G(y)=\nabla\Phi(y)-\boldsymbol{\tau}$ is the usage at $e^y$ less the target, which vanishes at $y^\star=\log\alpha^\star$ (8.19(i)); normalize $\alpha^\star_g=1$, so that $y^\star$ lies on the chart. For $v\in\mathbb{R}^{\mathcal{E}}$ and $\alpha=e^y$,

$$\langle v,\nabla^2\Phi(y)v\rangle=\sum_j\Big(s_j\operatorname{Var}_{T\sim\mathbb{P}_{j,\alpha}}\langle v,\mathbf{1}_T\rangle+t_j\operatorname{Var}_{T\sim\mathbb{P}^*_{j,\alpha}}\langle v,\mathbf{1}_T\rangle\Big),$$

since $\partial_c^2\log K_j(e^{y+cv})$ at $c=0$ is the variance of $\langle v,\mathbf{1}_T\rangle$ under $\mathbb{P}_{j,\alpha}(T)=\theta_{j,T}\alpha^T/K_j(\alpha)$ (7.16), and likewise for $\Psi_j$ under $\mathbb{P}^*_{j,\alpha}$, with $\langle v,\mathbf{1}_{\mathcal{E}\setminus T}\rangle=\langle v,\mathbf{1}\rangle-\langle v,\mathbf{1}_T\rangle$. Both laws charge every basis. So if $\langle v,\nabla^2\Phi(y)v\rangle=0$, then $\langle v,\mathbf{1}_T\rangle$ is constant on the bases of every active table, $v$ is constant on every molecule of the active family (7.8), and the active family of a connected family is one molecule (Lemma S3); so $v\in\mathbb{R}\mathbf{1}$. Conversely $\nabla^2\Phi\,\mathbf{1}=0$, $\Phi(y+c\mathbf{1})-\Phi(y)$ being linear in $c$. So the restriction of $\nabla^2\Phi(y^\star)$ to the chart $\{v_g=0\}$, the matrix $\big(\nabla^2\Phi(y^\star)\big)_{(g)}$, is positive definite; it is the Hessian at $y^\star$ of $G$ on the chart.

*The amplitude.* For every basis $T$ of table $j$, $|\langle\mathbf{1}_T,y\rangle|\le r_j|y|$, so $\log\min_T\theta_{j,T}-r_j|y|\le\log K_j(e^y)\le\log K_j(\mathbf{1})+r_j|y|$, and likewise for $\Psi_j$ with $n-r_j$. Hence $|b(y)|=e^{\langle\operatorname{Re}\mathbf{m}_0,y\rangle}\prod_j K_j(e^y)^{-\operatorname{Re}s_{0,j}}\Psi_j(e^y)^{-\operatorname{Re}t_{0,j}}\le e^{c_0(1+|y|)}$ for a constant $c_0$, and $b$ is continuous and nowhere 0.

*Laplace's method.* Lemma S7 on the chart, with $d=n-1$, the function $G(y^\star+x)-G(y^\star)$ and the amplitude $b(y^\star+x)$, bounded by $e^{c_0(1+|y^\star|+|x|)}$, gives

$$P_F(\mathbf{d}_\lambda)=e^{-\lambda G(y^\star)}\Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}}\det\big(\nabla^2\Phi(y^\star)\big)_{(g)}^{-1/2}\big(b(y^\star)+o(1)\big),$$

with $b(y^\star)=(\alpha^\star)^{\mathbf{m}_0}W_0(\alpha^\star)\ne0$ and $G(y^\star)=\Phi(\log\alpha^\star)-\langle\boldsymbol{\tau},\log\alpha^\star\rangle$: the display. The function $\alpha\mapsto\alpha^{\mathbf{m}_0}W_0(\alpha)$ is homogeneous of degree $\kappa_{\mathcal{E}}(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)=0$, so $b(y^\star)$ does not depend on the scale of $\alpha^\star$; likewise $G(y+c\mathbf{1})=G(y)$, the data $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ being balanced, and $\nabla^2\Phi$ does not change along $\mathbf{1}$. By 9.4, rung two of the still of $((s_j,t_j)_j;\alpha^\star)$ is $4\nabla^2\Phi(\log\alpha^\star)$.

*The normal limit.* For real $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)$ and $\lambda$ large, $\operatorname{Breta}^\circ(F;\mathbf{d}_\lambda)$ is defined (8.10). For a bounded continuous function $H$ of $y$ modulo $\mathbb{R}\mathbf{1}$, the function $u\mapsto H(\log u)u^{\mathbf{m}-\mathbf{1}}W(u)$ is homogeneous of degree $-n$, so its integral over $\Delta_{\mathcal{E}}$ is its integral over the chart, as in the proof of Lemma S4(ii). So the law of $Y=\log U$ modulo $\mathbb{R}\mathbf{1}$, read on the chart, has the density $b(y)e^{-\lambda G(y)}/P_F(\mathbf{d}_\lambda)$, with $b>0$ continuous. Applying Lemma S7 as above to $H(\sqrt{\lambda}(y-y^\star))$ and to 1, and dividing,

$$\mathbb{E}\big[H\big(\sqrt{\lambda}(Y-y^\star)\big)\big]\to\int H(x)e^{-\frac12\langle x,\nabla^2\Phi(y^\star)_{(g)}x\rangle}dx\Big/\int e^{-\frac12\langle x,\nabla^2\Phi(y^\star)_{(g)}x\rangle}dx,$$

the expectation of $H$ under the centred normal law on the chart with covariance $\big(\nabla^2\Phi(y^\star)_{(g)}\big)^{-1}$: the centred normal law on $\mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$ with covariance $(\nabla^2\Phi)^{-1}$, the quadratic form of $\nabla^2\Phi(y^\star)$ on $\mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$ being read on the chart. At $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)=0$, $b=1$ and this is the reference, proof of 18.9. □

*Remark.* At $(\mathbf{m}_0;\mathbf{s}_0,\mathbf{t}_0)=0$ let $q$ be the question of the data $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$ (17.1), with the exponent $X(z)=2\langle\boldsymbol{\mu}_\perp,z\rangle-\sum_j\big(\sigma_jE_j(z)+\delta_jO_j(z)\big)$, read at the representative of $z$ in $\mathbf{1}^\perp$. By 8.19, $\Phi(y)=\sigma\langle\mathbf{1},y\rangle+\sum_j\big(\sigma_jE_j(z)+\delta_jO_j(z)\big)$ in $y=2z$, and $\langle\boldsymbol{\tau},y\rangle=2\langle\boldsymbol{\mu},z\rangle+\sigma\langle\mathbf{1},y\rangle$, $\boldsymbol{\tau}$ being the masses of the question; so $\langle\boldsymbol{\tau},y\rangle-\Phi(y)=X(z)$ for $z\in\mathbf{1}^\perp$, and both sides are unchanged along $\mathbf{1}$. Hence $e^{-\lambda(\Phi(y^\star)-\langle\boldsymbol{\tau},y^\star\rangle)}=e^{\lambda X(z^\star)}$, $z^\star$ the tilt of $\frac12y^\star$. The Hessian in $z$ is four times the Hessian in $y$, so on a chart $\det\big(-\nabla^2X(z^\star)\big)=4^{n-1}\det\big(\nabla^2\Phi(y^\star)\big)_{(g)}$, and

$$2^{n-1}e^{\lambda X(z^\star)}\Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}}\det\big(-\nabla^2X(z^\star)\big)^{-\frac12}=e^{-\lambda(\Phi(y^\star)-\langle\boldsymbol{\tau},y^\star\rangle)}\Big(\frac{2\pi}{\lambda}\Big)^{\frac{n-1}{2}}\det\big(\nabla^2\Phi(y^\star)\big)_{(g)}^{-1/2}:$$

the leading term of 17.5(ii) at real $\lambda$ is the display of 18.9.

**Definition 18.10 (Skeleton law).** As stated, at real nonzero letter exponents.

*Proof of the last clause.* For a real coefficient $\xi_\ell>0$ and $y>0$, $(y+\xi_\ell)^{-w_\ell}>0$. A coefficient that is not real comes with its conjugate and the same real letter exponent $w_\ell$; $y+\xi_\ell$ and $y+\bar\xi_\ell$ are conjugate and off $(-\infty,0]$, so their principal powers are conjugate and $(y+\xi_\ell)^{-w_\ell}(y+\bar\xi_\ell)^{-w_\ell}=|y+\xi_\ell|^{-2w_\ell}>0$. So $y^{\kappa-1}\prod_\ell(y+\xi_\ell)^{-w_\ell}$ is positive and continuous on $(0,\infty)$. As $y\to0$ it is asymptotic to $y^{\kappa-1}\prod_\ell\xi_\ell^{-w_\ell}$, the product being positive by the same pairing, and as $y\to\infty$ to $y^{\kappa-1-|\mathbf{w}|}$; so it is integrable on $(0,\infty)$ exactly when $0<\kappa<|\mathbf{w}|$. Its integral is the value of 15.1, which asks no sign of the letter exponents; so the normalizer is that value, positive, and the density is that of a law. □

**Theorem 18.11 (The Skeleton law).** *As stated: (i) off the balance wall and at exponents of any sign; (ii) at real nonzero letter exponents; (iii) about every point $r\ge\max_\ell\xi_\ell$ of the axis.*

*Proof.* (i) By 18.3, given the internal shapes $(V_B,V_R)$ the scale $\rho$ of $B$ has a density proportional to $\rho^{m_B-1}(1+\rho)^{-\kappa_\mathcal{E}}\prod_jK_j(\rho V_B,V_R)^{-s_j}\Psi_j(\rho V_B,V_R)^{-t_j}$. At the gauge $\alpha=(V_B,V_R)$, 12.10 gives

$$K_j(\rho\alpha_B,\alpha_R)=\rho^{d_j(B)}K_j(\alpha)\prod_i\frac{\rho+\xi_{j,i}}{1+\xi_{j,i}},\qquad\Psi_j(\rho\alpha_B,\alpha_R)=\rho^{d_j^*(B)}\Psi_j(\alpha)\prod_i\frac{\rho+\xi^*_{j,i}}{1+\xi^*_{j,i}},$$

over the Skeleton values $\xi_{j,i}$ and $\xi^*_{j,i}$ of the two letters. The Skeleton polynomials are real, so their values not real come in conjugate pairs of equal multiplicity; the exponents are real, so the principal powers of conjugate factors are conjugate, and $K_j(\rho\alpha_B,\alpha_R)^{-s_j}=\rho^{-s_jd_j(B)}K_j(\alpha)^{-s_j}\prod_i(\rho+\xi_{j,i})^{-s_j}(1+\xi_{j,i})^{s_j}$, likewise for $\Psi_j$. The power of $\rho$ collects to $\rho^{m_B-1-\sum_j(s_jd_j(B)+t_jd_j^*(B))}=\rho^{\kappa_B-1}$ (8.1), and the linear factors collect to $\prod_\ell(\rho+\xi_\ell)^{-w_\ell}$ over the distinct points of the joint spectrum of the letters with nonzero exponent, with the letter exponents $w_\ell$ of 15.3. The factor $(1+\rho)^{-\kappa_\mathcal{E}}$ is the coefficient 1 with the letter exponent $\kappa_\mathcal{E}$, added to the letter exponent of 1 when 1 is a point of the joint spectrum; discarding the coefficients of letter exponent 0 changes nothing. So the conditional density is proportional to $\rho^{\kappa_B-1}\prod_\ell(\rho+\xi'_\ell)^{-w'_\ell}$, with coefficients in $\mathbb{C}\setminus(-\infty,0]$ closed under conjugation and real nonzero letter exponents, equal on conjugate coefficients. Their total is $|\mathbf{w}|+\kappa_\mathcal{E}=\kappa_B+\kappa_B^{\text{top}}+\kappa_\mathcal{E}$ (15.3), which is $\kappa_B+\kappa_{\mathcal{E}\setminus B}$ (8.13); $B$ and $\mathcal{E}\setminus B$ are nonempty proper sets and every proper wall is positive, so $0<\kappa_B<|\mathbf{w}'|$, and the conditional law is $\mathrm{Skel}_{\mathbf{w}',\kappa_B}(\boldsymbol{\xi}')$ (18.10). On the balance wall this is the reference, proof of 18.11(i).

(ii) For $-\kappa<\operatorname{Re}k<|\mathbf{w}|-\kappa$, $\mathbb{E}[Y^k]$ is $\int_0^\infty y^{\kappa+k-1}\prod_\ell(y+\xi_\ell)^{-w_\ell}dy$ over $\int_0^\infty y^{\kappa-1}\prod_\ell(y+\xi_\ell)^{-w_\ell}dy$, both convergent, the values of 15.1 at the indices $\kappa+k$ and $\kappa$; for positive letter exponents the factors $\Gamma(w_\ell)$ of the two stacked Gammas cancel, which is the display. For the inversion put $y=1/z$: $dy=z^{-2}dz$, and $y+\xi=\xi z^{-1}(z+\xi^{-1})$ with $\operatorname{Log}(y+\xi)=\operatorname{Log}\xi-\log z+\operatorname{Log}(z+\xi^{-1})$, the

principal arguments of $\xi$ and of $z+\xi^{-1}$ having opposite signs and sum in $(-\pi,\pi)$. So the density of $1/Y$ is proportional to $z^{1-\kappa}z^{-2}z^{|\mathbf{w}|}\prod_\ell(z+\xi_\ell^{-1})^{-w_\ell} = z^{|\mathbf{w}|-\kappa-1}\prod_\ell(z+\xi_\ell^{-1})^{-w_\ell}$, the constant $\prod_\ell \xi_\ell^{-w_\ell}$ being positive; with $0<|\mathbf{w}|-\kappa<|\mathbf{w}|$ this is $\mathrm{Skel}_{\mathbf{w},|\mathbf{w}|-\kappa}(\boldsymbol{\xi}^{-1})$. This is the reference, proof of 18.11(ii), with 15.7, which uses no sign of the letter exponents.

(iii) The coefficients are real, hence positive, and $r\ge\max_\ell\xi_\ell$, so $|\xi_\ell-r|<r$ and $0\le g_\ell<1$. By the expansion in the proof of 15.6,

$$y^{\kappa-1}\prod_\ell(y+\xi_\ell)^{-w_\ell} = \sum_{\mathbf{n}\in\mathbb{N}^J}\prod_\ell \frac{(w_\ell)_{n_\ell}}{n_\ell!}(r-\xi_\ell)^{n_\ell}\,y^{\kappa-1}(y+r)^{-|\mathbf{w}|-|\mathbf{n}|},$$

with nonnegative terms, the letter exponents being positive. For $a>\kappa>0$, the law $\mathrm{Skel}_{a,\kappa}(1)$ has the density $y^{\kappa-1}(1+y)^{-a}/\Gamma_1^2\left[\begin{matrix}\kappa,\ a-\kappa\\ a\end{matrix}\right]$, the Breta of one coefficient 1 being 1 (the paragraph after 15.1); so $rY_a$, $Y_a\sim\mathrm{Skel}_{a,\kappa}(1)$, has the density $r^{a-\kappa}y^{\kappa-1}(y+r)^{-a}/\Gamma_1^2\left[\begin{matrix}\kappa,\ a-\kappa\\ a\end{matrix}\right]$. With $a=|\mathbf{w}|+|\mathbf{n}|$ and $(r-\xi_\ell)^{n_\ell}=r^{n_\ell}g_\ell^{n_\ell}$, the density of $Y$ is the sum over $\mathbf{n}$ of $r^{\kappa-|\mathbf{w}|}\prod_\ell\frac{(w_\ell)_{n_\ell}}{n_\ell!}g_\ell^{n_\ell}\Gamma_1^2\left[\begin{matrix}\kappa,\ |\mathbf{w}|+|\mathbf{n}|-\kappa\\ |\mathbf{w}|+|\mathbf{n}|\end{matrix}\right]$ times the density of $rY_{\mathbf{n}}$, divided by the normalizer; the coefficients are nonnegative and, by Tonelli, sum to 1. So $Y$ is the stated mixture. At $r=\max_\ell\xi_\ell$ the largest coefficient has $g_\ell=0$, and only $n_\ell=0$ contributes.

(iv) On the balance wall with $s_j,t_j\ge0$, given the shapes the density of the scale $\rho$ is proportional to the integrand of the scale integral at $\alpha=(V_B,V_R)$, by (i). By the proof of 15.9(i) it is, on $(0,1)$, proportional to $\rho^{\kappa_B-1}\prod_i\mathsf{Q}_{B,i}(\rho;\alpha)^{-w_i}$, and the density of $1/\rho$ on $(0,1)$ is proportional to $y^{\kappa_B^{\mathrm{top}}-1}\prod_i\mathsf{Q}_{R,i}(y;\alpha)^{-w_i}$. The expansions of 15.9(ii) have nonnegative terms, each the term of $\mathbf{n}$ and $k$ times $y^{\kappa+k-1}(1-y)^{|\mathbf{n}|}/\Gamma_1^2\left[\begin{matrix}\kappa+k,\ |\mathbf{n}|+1\\ \kappa+k+|\mathbf{n}|+1\end{matrix}\right]$, the density of the first coordinate of $\mathrm{Dir}_{(\kappa+k,|\mathbf{n}|+1)}$ (3.9); by Tonelli the conditional laws are the stated mixtures, and the two sectors carry probabilities proportional to their Trailmoments. This is the reference, proof of 18.11(v). □

**Definition 18.12 (Wall contour; joint period).** As stated, at complex data on the balance wall with $\mathrm{Re}\,s_j>0$ or $s_j=0$, and $\mathrm{Re}\,t_j>0$ or $t_j=0$.

*Remark.* A wall contour is fixed by real parts alone, and the fibre integral at $B$ has complex pins, whose admissible real points (4.1) are the wall contours. The two descriptions agree: $\mathrm{load}_e=\sigma+\langle\boldsymbol{\delta}\rangle_e$ on the share sums and $m_e=\sigma+\mu_e$, so $\mathrm{Re}\,\mathrm{load}_e=\mathrm{Re}\,m_e$ exactly when $\mathrm{Re}\langle\boldsymbol{\delta}\rangle_e=\mathrm{Re}\,\mu_e$. For real data with $s_j,t_j\ge0$ these are the conditions of the reference.

**Theorem 18.13 (The port on the balance wall).** *As stated, at complex data on the balance wall with* $\mathrm{Re}\,s_j>0$ *or* $s_j=0$, *and* $\mathrm{Re}\,t_j>0$ *or* $t_j=0$.

*Proof.* A letter with exponent 0 is 1 and is not opened; the shares, the stacked Gammas and the products below run over the letters with nonzero exponent, whose exponents have positive real part.

*The real parts.* The walls and the imbalance are affine in the data with real coefficients. So the data $(\mathrm{Re}\,\mathbf{m};\mathrm{Re}\,\mathbf{s},\mathrm{Re}\,\mathbf{t})$ lie on the balance wall, and their proper walls are the real parts of those of $(\mathbf{m};\mathbf{s},\mathbf{t})$, positive since $P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ converges (8.7). Their exponents are nonnegative and not all zero: otherwise $\kappa_S=\mathrm{Re}\,m_S$ for every $S$, and $0=\kappa_{\mathcal{E}}=\kappa_B+\kappa_R>0$. Their masses are positive: $\mathrm{Re}\,m_e=\kappa_{\{e\}}+\sum_j\big(\mathrm{Re}\,s_j\,d_j(\{e\})+\mathrm{Re}\,t_j\,d_j^*(\{e\})\big)>0$, $\{e\}$ being a nonempty proper set ($n\ge2$, as $B$ is proper). A nonempty proper $S$ with $c_j(S)=0$ in every active table would have

$\kappa_S+\kappa_{\mathcal{E}\setminus S}=\kappa_{\mathcal{E}}=0$ at the real parts (8.16), against the positivity of both walls; so $F$ is connected, and its active family, whose tables are those of the real parts, is one molecule (Lemma S3). By 8.19(iii) the real parts have a native gauge.

*(i), convergence and the room integral.* By 7.15(i), $K_j=e^{\langle\mathbf{1},z\rangle}\Lambda_j(z)$ and $\Psi_j=e^{\langle\mathbf{1},z\rangle}\Lambda_j(-z)$ at $\alpha=e^{2z}$, so with $\sigma=\sum_j\sigma_j$ the integrand of $P_F$ is $f(\alpha)=\alpha^{\boldsymbol{\mu}-\mathbf{1}}\prod_j\Lambda_j(z)^{-s_j}\Lambda_j(-z)^{-t_j}$, homogeneous of degree $\kappa_{\mathcal{E}}-n=-n$ (7.15(iii), 8.2(iii)), and $|f|$ is the same function at the real parts of the data. At the real parts, 8.7 on the section $|\alpha_R|=1$, whose measure is $d\alpha_B\,d\omega_R$ (proof of 16.6 above), gives $\int_{\Delta_R}\int_{\mathbb{R}^B_{>0}}|f(\alpha_B,v)|\,d\alpha_B\,d\omega_R(v)<\infty$; and $f(\alpha_B,v)$ is $v^{\boldsymbol{\mu}|_R-\mathbf{1}}$ times the integrand of $\Xi_B(v;\boldsymbol{\mu}|_B;\mathbf{s},\mathbf{t})$. By Tonelli that integral converges absolutely at almost every $v\in\Delta_R$. For two room points $\alpha_R,\alpha'_R$ there is $C\geq1$ with $\Lambda_{j,(p)}(\pm z_R)\leq C\Lambda_{j,(p)}(\pm z'_R)$ for all $j,p$, these being finite sums of positive terms; by 12.7, $\Lambda_j(z)=\sum_p e^{\langle z_B,\chi_p\rangle}\Lambda_{j,(p)}(z_R)$, so $\Lambda_j(z_B,z_R)\geq C^{-1}\Lambda_j(z_B,z'_R)$, likewise at $-z$, and the modulus of the integrand of $\Xi_B$ at $\alpha_R$ is at most $C^{\sum_j(\operatorname{Re}s_j+\operatorname{Re}t_j)}$ times that at $\alpha'_R$. So $\Xi_B$ converges at every $\alpha_R$, and Fubini gives $P_F=\int_{\Delta_R}v^{\boldsymbol{\mu}|_R-\mathbf{1}}\Xi_B(v)\,d\omega_R(v)$.

*Independence of the pins.* Let $R_1,\dots,R_q$ be the molecules of the children (for $q=1$ there are no pins $\kappa_{R_i}$ below), and write $\kappa_{R_i}=m_{R_i}-\ell_i$ with $\ell_i(\mathbf{a},\mathbf{b})=\sum_{j,p}\big(a_{j,p}r_{j,(p)}(R_i)+b_{j,p}(|R_i|-r_{j,(p)}(R_i))\big)$. The linear forms $|\mathbf{a}_j|$ and $|\mathbf{b}_j|$ of the letters, $\langle\boldsymbol{\delta}\rangle_e$ for $e\in B$, and $\ell_i$ for $i<q$, are linearly independent. Indeed, let $\sum_{e\in B}\gamma_e\langle\boldsymbol{\delta}\rangle_e+\sum_{i<q}w_i\ell_i=\sum_j(u_j|\mathbf{a}_j|+u'_j|\mathbf{b}_j|)$, and put $V\in\mathbb{R}^{\mathcal{E}}$ equal to $\gamma$ on $B$, to $w_i$ on $R_i$ for $i<q$ and to 0 on $R_q$. For a basis $T$ of table $j$ with $T\cap B=p$, $T\setminus B$ is a basis of the child $(j,p)$, which meets each $R_i$ in $r_{j,(p)}(R_i)$ marks, $R_i$ being a union of molecules of that child (7.9); and $\frac12\langle\gamma,\chi_p\rangle=\langle\gamma,\mathbf{1}_p\rangle-\frac12\langle\gamma,\mathbf{1}_B\rangle$. Comparing the coefficients of $a_{j,p}$ gives $\langle V,\mathbf{1}_T\rangle=u_j+\frac12\langle\gamma,\mathbf{1}_B\rangle$, and comparing those of $b_{j,p}$ gives $\langle V,\mathbf{1}_T\rangle=\langle V,\mathbf{1}_R\rangle+\frac12\langle\gamma,\mathbf{1}_B\rangle-u'_j$, for every basis $T$ of table $j$ whenever that letter is opened. So $\langle V,\mathbf{1}_T\rangle$ is constant on the bases of every active table; by 7.8 $V$ is constant on the one molecule $\mathcal{E}$ of the active family, and as it vanishes on $R_q\neq\varnothing$, $V=0$: $\gamma=0$ and every $w_i=0$.

*(i), the fibre formula.* Fix $\alpha_R$ and a wall contour. Put $X_{j,p}=e^{\langle z_B,\chi_p\rangle}\Lambda_{j,(p)}(z_R)$ and $Y_{j,p}=e^{-\langle z_B,\chi_p\rangle}\Lambda_{j,(p)}(-z_R)$, positive, so that $\Lambda_j(z)=\sum_p X_{j,p}$ and $\Lambda_j(-z)=\sum_p Y_{j,p}$. For a letter with more than one pattern, 4.4 on the contour $\operatorname{Re}a_{j,p}=c_{j,p}$, admissible since every $c_{j,p}>0$ and $\sum_p c_{j,p}=\operatorname{Re}s_j$, gives $\Lambda_j(z)^{-s_j}=\int_{|\mathbf{a}_j|=s_j}\Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\prod_p X_{j,p}^{-a_{j,p}}$; for one pattern this is the evaluation $a_{j,p}=s_j$. Likewise for $\Lambda_j(-z)^{-t_j}$ with the shares $\mathbf{b}_j$. Since $\prod_{j,p}e^{-\langle z_B,\chi_p\rangle(a_{j,p}-b_{j,p})}=e^{-2\langle z_B,\langle\boldsymbol{\delta}\rangle\rangle}=\alpha_B^{-\langle\boldsymbol{\delta}\rangle}$ (12.24), in $y=\log\alpha_B$, where $\alpha_B^{\boldsymbol{\mu}|_B-\mathbf{1}}d\alpha_B=e^{\langle\boldsymbol{\mu}|_B,y\rangle}dy$,

$$\Xi_B(\alpha_R)=\int_{\mathbb{R}^B}\Big(\int_{\mathcal{X}_0}\mathcal{G}(\mathbf{a},\mathbf{b})\,e^{\langle\boldsymbol{\mu}|_B-\langle\boldsymbol{\delta}\rangle,y\rangle}\Big)dy,$$

$$\mathcal{G}=\prod_j\Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|\operatorname{pat}_j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\prod_{j,p}\Lambda_{j,(p)}(z_R)^{-a_{j,p}}\Lambda_{j,(p)}(-z_R)^{-b_{j,p}},$$

the inner integral the product of the share integrals, over the set $\mathcal{X}_0$ of shares on the contour with the share sums pinned. On a wall contour $\operatorname{Re}\langle\boldsymbol{\delta}\rangle_e=\operatorname{Re}\mu_e$ for $e\in B$, so $e^{\langle\boldsymbol{\mu}|_B-\langle\boldsymbol{\delta}\rangle,y\rangle}=e^{i\langle\operatorname{Im}\boldsymbol{\mu}|_B-\operatorname{Im}\langle\boldsymbol{\delta}\rangle,y\rangle}$. By Stirling's formula $|\Gamma(c+i\eta)|\leq C_c(1+|\eta|)^{c-1/2}e^{-\pi|\eta|/2}$, so $|\mathcal{G}|$, whose factors $\Lambda_{j,(p)}^{-a_{j,p}}$ have moduli constant on the contour, decays exponentially along every imaginary direction of $\mathcal{X}_0$, uniformly on parallel affine subspaces through a compact set. The map $(\mathbf{a},\mathbf{b})\mapsto\operatorname{Im}\langle\boldsymbol{\delta}\rangle\in\mathbb{R}^B$ is affine on $\mathcal{X}_0$, with linear part onto $\mathbb{R}^B$ by the independence of the pins; so the image of $\mathcal{G}$ times the measure of the share integrals has a density $g$ on $\mathbb{R}^B$, continuous and integrable, and the inner integral is $e^{i\langle\operatorname{Im}\boldsymbol{\mu}|_B,y\rangle}\hat g(y)$, $\hat g(y)=\int g(\nu)e^{-i\langle\nu,y\rangle}d\nu$. The modulus $|\hat g(y)|$ is the modulus of the integrand of $\Xi_B$ in $y$, integrable by the first step. By Fourier inversion, $\Xi_B(\alpha_R)=\int\hat g(y)e^{i\langle\operatorname{Im}\boldsymbol{\mu}|_B,y\rangle}dy=$

$(2\pi)^{|B|}g(\operatorname{Im}\boldsymbol{\mu}|_B)$, which is the fibre integral of $\mathcal{G}$ over the fibre $\mathcal{X}_1$ cut by the pins $|\mathbf{a}_j| = s_j$, $|\mathbf{b}_j| = t_j$ and $\langle\boldsymbol{\delta}\rangle = \boldsymbol{\mu}|_B$ (4.1: the $|B|$ further pins raise $(2\pi)^{q-N}$ by $(2\pi)^{|B|}$ and add the factor $\delta^{|B|}(\operatorname{Im}\langle\boldsymbol{\delta}\rangle - \operatorname{Im}\boldsymbol{\mu}|_B)$), with the wall contour as its admissible real point. This is the display of (i). For real data this is the reference, proof of 18.13(i), where $\operatorname{Im}\boldsymbol{\mu}|_B = 0$.

*(ii).* On the pins of (i), $\operatorname{load}_e = m_e$ for $e \in B$, and by 12.2 the child $(j,p)$ has rank $r_j - |p|$; so

$$\sum_{i=1}^{q}\kappa_{R_i} = m_R - \sum_{j,p}\big(a_{j,p}(r_j-|p|)+b_{j,p}(|R|-r_j+|p|)\big) = m_R - \sum_j\big(s_jr_j+t_jr_j^*\big)+\sum_{e\in B}\operatorname{load}_e = \kappa_{\mathcal{E}} = 0,$$

using $\sum_{e\in B}\operatorname{load}_e = \sum_{j,p}\big(a_{j,p}|p|+b_{j,p}(|B|-|p|)\big)$; so $\kappa_{R_q} = 0$ holds on $\mathcal{X}$. Take the section $\{|\alpha_{R_q}| = 1\}$, with the measure $d\alpha_B\prod_{i<q}d\alpha_{R_i}\,d\omega_{R_q}$, and as in (i) $P_F = \int \alpha_R^{\boldsymbol{\mu}|_R-\mathbf{1}}\,\Xi_B(\alpha_R)\prod_{i<q}d\alpha_{R_i}\,d\omega_{R_q}$, absolutely convergent. Write $\alpha_{R_i} = e^{y_i}u_i$, $u_i \in \Delta_{R_i}$, for $i<q$, and $u_q = \alpha_{R_q}$; then $\alpha_{R_i}^{\boldsymbol{\mu}|_{R_i}-\mathbf{1}}d\alpha_{R_i} = e^{y_i\mu_{R_i}}u_i^{\boldsymbol{\mu}|_{R_i}-\mathbf{1}}dy_i\,d\omega_{R_i}$. Each $R_i$ is a union of molecules of every child, so every basis $X$ of the child $(j,p)$ has $\langle\chi_X,\mathbf{1}_{R_i}\rangle = 2r_{j,(p)}(R_i) - |R_i|$, and $\Lambda_{j,(p)}(z_R + \frac12 y_i\mathbf{1}_{R_i}) = e^{\frac12 y_i(2r_{j,(p)}(R_i)-|R_i|)}\Lambda_{j,(p)}(z_R)$. Inserting the fibre formula of (i) and using $\sum_{j,p}\sigma_{j,p} = \sigma$, the exponent of $e^{y_i}$ collects to $\mu_{R_i} - \sum_{j,p}\delta_{j,p}\big(2r_{j,(p)}(R_i) - |R_i|\big) = \kappa_{R_i}(\mathbf{a},\mathbf{b})$, and the rest is $\mathcal{G}$ at $u$ times $u^{\boldsymbol{\mu}|_R-\mathbf{1}}$, that is, by 7.15 and $\sum_{j,p}\sigma_{j,p} = \sigma$, the stacked Gammas times the integrand of the joint period of the children at the masses $\mathbf{m}|_R$. For fixed $\mathbf{y} = (y_i)_{i<q}$, on a wall contour with $\operatorname{Re}\kappa_{R_i} = 0$ for $i<q$, the modulus of this integrand over $u \in \prod_i\Delta_{R_i}$ and the shares is the modulus of the stacked Gammas times the joint integrand at the real parts $(\operatorname{Re}\mathbf{m}|_R;\mathbf{c},\mathbf{c}')$, integrable by the hypothesis on $P^{\otimes}$ and by Stirling; so by Fubini the integral over $u$ and the shares is

$$\hat g_1(\mathbf{y}) = \int_{\mathcal{X}_1}\prod_j \Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{a}_j\\ s_j\end{bmatrix}\Gamma_{|\operatorname{pat}_1^j(B)|}\begin{bmatrix}\mathbf{b}_j\\ t_j\end{bmatrix}\;P^{\otimes}_{F_B(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})\,e^{i\sum_{i<q}y_i\operatorname{Im}\kappa_{R_i}(\mathbf{a},\mathbf{b})},$$

$\mathcal{X}_1$ the fibre of (i), and $\int|\hat g_1(\mathbf{y})|\,d\mathbf{y}$ is at most $P_F$ at the real parts, finite. The joint period is continuous in the shares and bounded by its value at the real parts, by dominated convergence. The map $(\mathbf{a},\mathbf{b}) \mapsto (\operatorname{Im}\kappa_{R_i})_{i<q}$ is affine on $\mathcal{X}_1$ with linear part onto $\mathbb{R}^{q-1}$, by the independence of the pins. So, as in (i), Fourier inversion in $\mathbf{y}$ gives $P_F = \int\hat g_1(\mathbf{y})\,d\mathbf{y}$ as the fibre integral over $\mathcal{X}$, the pins $\kappa_{R_i} = 0$, $i<q$, holding in real part on the contour and in imaginary part by inversion. For $q=1$ there is no $\mathbf{y}$, and $P_F = \hat g_1$ by Fubini.

*(iii).* The real parts have a native gauge (first step). For real data the statement is the reference, proof of 18.13(iii): the native split is a wall contour, lies on the real fibre, and $P^{\otimes}$ converges there. A wall contour, and the real parts of the pins of $\mathcal{X}$, $\operatorname{Re}|\mathbf{a}_j| = \operatorname{Re}s_j$, $\operatorname{Re}|\mathbf{b}_j| = \operatorname{Re}t_j$, $\operatorname{Re}\langle\boldsymbol{\delta}\rangle = \operatorname{Re}\boldsymbol{\mu}|_B$ and $\operatorname{Re}\kappa_{R_i} = \kappa_{R_i}(\operatorname{Re}\mathbf{m}|_R;\mathbf{c},\mathbf{c}')$, involve the data only through their real parts. So the native split of the real parts is a wall contour for the data, an admissible real point of $\mathcal{X}$, and $P^{\otimes}_{F_B(\mathbf{c},\mathbf{c}')}(\operatorname{Re}\mathbf{m}|_R;\mathbf{c},\mathbf{c}')$ converges there, as (ii) asks. □

**Proposition 18.17 (The pins reach every integer point).** *As stated.*

*Proof.* The pins are a linear map $\mathbb{Z}^{\text{shares}} \to \mathbb{Z}^{\text{rows}}$, the rows being the share sums of the active letters and the loads at the marks $e \neq g$; a share $a_{j,T}$ has 1 in the row of its sum and in the loads of the marks of $T-g$, and a share $b_{j,T}$ has 1 in the row of its sum and in the loads of the marks of $\mathcal{E}\setminus T$ other than $g$. Write $\mathbf{e}_x$ for the unit vector of the load of $x \neq g$ and $\mathbf{e}_g = 0$. If $x \neq y$ lie in a common circuit of table $j$, 7.7 gives a basis $T \ni x$, $T \not\ni y$ with $T' = (T\setminus\{x\})\cup\{y\}$ a basis; if $K_j$ is active the difference of the columns of $a_{j,T}$ and $a_{j,T'}$ is $\mathbf{e}_x - \mathbf{e}_y$, and if $\Psi_j$ is active that of $b_{j,T}$ and $b_{j,T'}$ is $\mathbf{e}_y - \mathbf{e}_x$. Along the sequences of the hypothesis every $\mathbf{e}_x - \mathbf{e}_y$ lies in the image,

and with $y = g$ every $\mathbf{e}_x$, $x \neq g$. Subtracting from the column of a share the loads it carries leaves the unit vector of its share sum. So the image is all of $\mathbb{Z}^{\mathrm{rows}}$.

At integer data the target point is an integer point, so it has an integer preimage $x_0$, and the integer points of the fibre are $x_0 + \Lambda$ with $\Lambda$ the kernel of the map on $\mathbb{Z}^{\mathrm{shares}}$. The map is onto a free module of the rank of the number of rows, so the rows are independent and $\Lambda$, a subgroup of $\mathbb{Z}^{\mathrm{shares}}$, is free of rank the number of shares less the number of rows, the dimension of the fibre. With a basis $v_1, \dots, v_D$ of $\Lambda$ the point $x_0 + \sum_i z_i v_i$ is an integer point exactly when $z \in \mathbb{Z}^D$. □

**Proposition 18.20 (The serif word is the closed form of the periods).** *As stated.*

*Proof.* (i) The Gramma of one scale with the configuration (1) and the coefficient 1 is $\int_0^\infty x^{\nu-1} e^{-x}\, dx = \Gamma(\nu)$ (Definition 2.1), and the stacked Gamma is a quotient of such values (Definition 1.2), meromorphic in its entries. The domain of the Gramma is Theorem 2.2, and the fibre integral is fixed by its pins and independent of its admissible real point within one connected set of them (Definition 4.1). (ii) is Corollaries 18.16 and 18.18, and (iii) is the cited statements.

(iv) $F_{\mathsf{m}}$ has the masses $2k = k \cdot 1 + k \cdot 1$, its active letters, $K$ of the first copy and $\Psi$ of the second, being of degree one, so it is on the balance wall; on a singleton both floors of $\binom{[2]}{1}$ vanish, so its walls are $\kappa_{\{1\}} = \kappa_{\{2\}} = k > 0$, and it converges (Theorem 8.7). Apply Corollary 18.16 with $g = 2$ and one share per basis for each active letter, $(a_1, a_2)$ for $K = \alpha_1 + \alpha_2$ and $(b_1, b_2)$ for $\Psi = \alpha_2 + \mathsf{m}\alpha_1$, indexed by the bases $\{1\}$ and $\{2\}$. The pins $|\mathbf{a}| = |\mathbf{b}| = k$ and $\mathrm{load}_1 = a_1 + b_2 = k$ leave the free coordinate $x = a_1$, with $a_2 = b_2 = k - x$ and $b_1 = x$; the weights give $\mathsf{m}^{-b_2} = \mathsf{m}^{x-k}$, the stacked Gammas give $\Gamma_{\binom{2}{1}}\left[\begin{smallmatrix} x,\ k-x \\ k \end{smallmatrix}\right]^2$, and $\operatorname{Re} x = k/2$ is a wall contour. In the coordinates $(a_1, a_2, b_1, b_2)$ the pins have the rows $(1,1,0,0)$, $(0,0,1,1)$, $(1,0,0,1)$, with $\det PP^{\mathsf{T}} = 4$, and the fibre has the direction $(1,-1,1,-1)$, of length 2, so the fibre integral is $\frac{1}{2\pi i}\int dx$ (Definition 4.1). On the section $\alpha_1 = 1$, with $u = \alpha_2$, the period is $\int_0^\infty u^{k-1}(1+u)^{-k}(u+\mathsf{m})^{-k}\, du$ (Theorem 8.7). For $0 < \mathsf{m} < 2$,

$$(u+\mathsf{m})^{-k} = (1+u)^{-k} \sum_{j \geq 0} \frac{(k)_j}{j!} \Big(\frac{1-\mathsf{m}}{1+u}\Big)^j,$$

and the series of the absolute values is at most $(1+u)^{-k}(1-|1-\mathsf{m}|)^{-k}$, so the sum and the integral exchange; $\int_0^\infty u^{k-1}(1+u)^{-2k-j}\, du = \Gamma(k)\Gamma(k+j)/\Gamma(2k+j)$, the Dirichlet integral on two marks after $u = v/(1-v)$ (Theorem 3.9), and $(k)_j\Gamma(k) = \Gamma(k+j)$.

In the second display the integrand is $\Gamma(-s)G(s)$ with $G(s) = \Gamma(1+s)(1+s)^{-n}(-z)^s$, and the line $\operatorname{Re} s = c$ separates the poles of $\Gamma(-s)$, at $s \in \mathbb{N}$, from those of $G$. Move it to the right. Only $\Gamma(-s)$ has poles crossed. For $0 < -z < 1$ the contribution of the line $\operatorname{Re} s = N + \frac{1}{2}$ tends to zero as $N \to \infty$: there $|\Gamma(-s)\Gamma(1+s)| = \pi/|\sin \pi s| \leq \pi/\cosh(\pi \operatorname{Im} s)$, $|1+s|^{-n} \leq 1$ and $|(-z)^s| = (-z)^{N+1/2}$; and on the strip between the two lines $\pi/|\sin \pi s| \leq \pi/\sinh(\pi|\operatorname{Im} s|)$, which decays exponentially in $|\operatorname{Im} s|$. So Theorem 4.5 gives $\sum_{j\geq 0} \frac{(-1)^j}{j!} G(j) = \sum_{j\geq 0} (-1)^j (1+j)^{-n}(-z)^j = \sum_{j\geq 0} z^j (1+j)^{-n}$, which is the display after the factor $z$. For every $z < 0$ the integrand is at most a constant times $e^{-\pi|\operatorname{Im} s|}$ on the line, $|(-z)^s|$ being $(-z)^c$ there, so the integral converges and is real analytic in $z$ on $(-\infty, 0)$, and it agrees with the series on $(-1, 0)$. □

## 50.19 Chapter 19: bunches of width one: pooling and cleaning

Throughout this section, for a bunch $B$ of width one of table $j$ whose weights factor across $B$ and $R$, $\theta_{j,T} = \theta'_{j,T\cap B}\, \theta''_{j,T\cap R}$, the minors $\mathcal{B}_j|B$ and $\mathcal{B}_j/R$ on $B$ carry $\theta'_j$ and the minors $\mathcal{B}_j|R$ and $\mathcal{B}_j/B$ on $R$ carry $\theta''_j$ (19.2), and every integral of a positive function is read in $[0, \infty]$ (Tonelli).

**Proposition 19.3 (Hinge values of parallel and series bunches).** *As stated, for classes of any table whose weights factor across them.*

*Proof.* That a parallel or series bunch has width at most one, and width zero exactly when it is a union of molecules, is the reference, proof of prop:classreadings. Let $B$ be a parallel bunch of width one: $r(B) = 1$ and $c(B) = r(B) - d(B) = 1$, so $d(B) = 0$; $\mathcal{B}/R$ has rank 0, with the single basis $\varnothing$ and the weight $\theta'_{\varnothing}$, and $\mathcal{B}|B$ has rank one, with the singletons of $B$ as bases, none of its marks being a loop, and the weights $\theta'_{\{e\}} = \theta'_{\varnothing} w_e$. Then $K_{\mathcal{B}/R} = \theta'_{\varnothing}$, $K_{\mathcal{B}|B} = \sum_e \theta'_{\{e\}} \alpha_e = \theta'_{\varnothing} \sum_e w_e \alpha_e$, $\Psi_{\mathcal{B}/R} = \theta'_{\varnothing} \alpha^B = \theta'_{\varnothing} e_{|B|}(\alpha_B)$ and $\Psi_{\mathcal{B}|B} = \sum_e \theta'_{\{e\}} \alpha^{B-e} = \theta'_{\varnothing} e_{|B|}(\alpha_B) \sum_e w_e \alpha_e^{-1}$, and the hinge values are the quotients of 19.2. Let $B$ be a series bunch of width one: no mark of $B$ is a coloop, so $d(B) \geq |B|-1$, and width one gives $r(B) = |B|$ and $d(B) = |B|-1$; $\mathcal{B}|B$ has the single basis $B$, with the weight $\theta'_B$, and $\mathcal{B}/R$ the sets $B-e$, every one of them occurring since a basis avoids $e$, with the weights $\theta'_{B-e} = \theta'_B w_e$. Then $K_{\mathcal{B}|B} = \theta'_B e_{|B|}(\alpha_B)$, $K_{\mathcal{B}/R} = \theta'_B \sum_e w_e \alpha^{B-e} = \theta'_B e_{|B|}(\alpha_B) \sum_e w_e \alpha_e^{-1}$, $\Psi_{\mathcal{B}|B} = \theta'_B$ and $\Psi_{\mathcal{B}/R} = \theta'_B \sum_e w_e \alpha_e$, which give $\gamma_B^{\Psi}$ and $\gamma_B^K$. In a list with $\theta' = \theta'' = 1$ every $w_e$ is 1, and for a pair with the shape $(x, 1-x)$: parallel, $\gamma_B^K = 1$ and $\gamma_B^{\Psi} = (x^{-1} + (1-x)^{-1})^{-1} = x(1-x)$; series, $\gamma_B^{\Psi} = 1$ and $\gamma_B^K = x(1-x)$. □

*Remark.* A change of the factorization (19.2) multiplies the factors $w_e$ of a parallel bunch by $c$, the weights of its level $r(B) = 1$ over those of its level $d(B) = 0$, and those of a series bunch by $c^{-1}$, its patterns $B - e$ lying on the level $d(B)$; either way $\gamma_B^K$ is multiplied by $c$ and $\gamma_B^{\Psi}$ by $c^{-1}$. In exact arithmetic (receipt `check_classes_realization.py`): a parallel bunch of three marks whose cell of the bunch is $\binom{\{1,2,3,g\}}{1}$ with the weights $2, \frac{3}{7}, 5$ on the singletons of the bunch and $\frac{4}{3}$ on $\{g\}$, and a series bunch of three marks whose cell is $\binom{\{1,2,3,g\}}{3}$ with the weights $\frac{3}{2}$ on $\{1,2,3\}$ and $\frac{1}{5}, 7, \frac{2}{3}$ on the sets through $g$, each glued to a weighted room cell, have the hinge values of the statement as rational functions, and $\varrho_B = (\sum_e w_e)^{-2}$ at the centre of the shape simplex of the parallel bunch.

**Theorem 19.4 (A bunch of width one is one mark to its room).** *As stated; the converse holds with any number of products.*

*Proof.* All but the converse is the reference, proof of thm:widthone. On the open orthant let $K(\alpha) = \sum_{l=0}^{k} A_l(\alpha_B) C_l(\alpha_R)$, $C_l$ homogeneous of degree $D_l$. Grouping the bases by their level $i = |T \cap B|$, which takes every value $d(B) \leq i \leq r(B)$ (7.13), $K = \sum_i K^{(i)}$ with $K^{(i)}(\alpha_B, \lambda\alpha_R) = \lambda^{r-i} K^{(i)}(\alpha)$ and $K^{(i)} > 0$ on the open orthant. At a fixed $\alpha$, with $\lambda\alpha_R$ for $\alpha_R$,

$$\sum_{i=d(B)}^{r(B)} \lambda^{r-i} K^{(i)}(\alpha) = \sum_D \lambda^D \sum_{l:\ D_l = D} A_l(\alpha_B)\, C_l(\alpha_R) \qquad (\lambda > 0),$$

the sum on the right over the distinct values $D$ among the $D_l$. The functions $\lambda \mapsto \lambda^D$ with distinct real $D$ are linearly independent on $(0, \infty)$, so each of the $c(B) + 1$ exponents $r - i$ on the left, whose coefficient is positive, is one of the $D_l$: $c(B) + 1 \leq k + 1$. With $k = 1$ this is the converse of the reference. □

**Lemma 19.6 (One hinge value of any degree is width at most one).** *As stated.*

*Proof.* Write $d = d(B)$, $c = c(B)$ and $K^{(k)} = \sum_{p \in \mathrm{pat}^{(k)}(B)} \alpha_B^p K_{(p)}(\alpha_R)$ for the level letters (12.16), nonzero exactly for $d \leq k \leq d + c$ (7.13); each child $K_{(p)}$ has positive coefficients, the weights. Let $a$ be the degree of $A$ and $P(\alpha_R, y) = \sum_{i=0}^{D} P_i(\alpha_R) y^i$. For $\tau > 0$,

$$\sum_k \tau^k K^{(k)}(\alpha) = K(\tau\alpha_B, \alpha_R) = \sum_{i=0}^{D} \tau^{a+i} A(\alpha_B) \gamma(\alpha_B)^i P_i(\alpha_R),$$

and at a fixed $\alpha$ the functions $\tau \mapsto \tau^x$ with distinct real $x$ are linearly independent on $(0,\infty)$. So $a$ is an integer, $K^{(d)}$ being nonzero, and $K^{(k)} = A\gamma^{k-a}P_{k-a}$ for every $k$, the right side read as 0 when $k-a \notin \{0,\dots,D\}$.

Let $d \le k \le d+c$. Then $P_{k-a} \not\equiv 0$; fixing $\alpha_R$ where $P_{k-a}(\alpha_R) \neq 0$ shows that $A\gamma^{k-a}$ is a polynomial $\widehat{f_k}$ in $\alpha_B$, so $K^{(k)}(\alpha) = \widehat{f_k}(\alpha_B)P_{k-a}(\alpha_R)$ as polynomials. The coefficient of $\alpha_B^p\alpha_R^X$ in $K^{(k)}$ is the weight of $p \sqcup X$, distinct patterns giving distinct monomials $\alpha_B^p$, so the matrix of these coefficients, rows indexed by the patterns of the level and columns by the monomials of the room, has rank one: the children at the patterns of the level are positive multiples of one polynomial $C_k$, $K_{(p)} = \lambda_p C_k$ with $\lambda_p > 0$, and $K^{(k)} = f_k(\alpha_B)C_k(\alpha_R)$ with $f_k = \sum_{p\in \mathrm{pat}^{(k)}(B)} \lambda_p \alpha_B^p$. Comparing with $A\gamma^{k-a}P_{k-a}$ at a fixed $\alpha_R$ where $C_k$ and $P_{k-a}$ do not vanish, $f_k = \mu_k A\gamma^{k-a}$ with a constant $\mu_k > 0$.

Suppose $c \ge 2$. Then $f_{d+1}^2 = \mu\, f_d f_{d+2}$ on the open orthant with $\mu = \mu_{d+1}^2/(\mu_d\mu_{d+2}) > 0$, an identity of polynomials. For a pattern $p$ of $d+1$ marks the monomial $\alpha_B^{2\mathbf{1}_p}$ has the coefficient $\lambda_p^2 > 0$ in $f_{d+1}^2$, since a product $\alpha_B^{p'}\alpha_B^{p''}$ of two monomials of sets equals it only for $p' = p'' = p$; in $f_d f_{d+2}$ it would need two sets $q, q'$ with $q = q' = p$, against $|q| = d \neq d+2 = |q'|$. So $c \le 1$.

Conversely, with the weights factoring across $B$ and $R$: at $c = 0$ the one level consists of the unions of a pattern and a room side, all combinations occurring (12.19), so $K = K_{\mathcal{B}|B}(\alpha_B)K_{\mathcal{B}|R}(\alpha_R)$ with the factored weights; at $c = 1$ this is 19.4. For $\Psi$, apply the statement to the dual table (7.2), in which the window of $B$ is read on the cobases and $c(B)$ is unchanged (7.12, 7.13). □

**Proposition 19.7 (The freedom of the factorization).** *As stated.*

*Proof.* By 19.4 the bases of the level $d(B)$ are the sets $X \sqcup Y$ with $X \in \mathcal{B}/R$ and $Y \in \mathcal{B}|R$, and those of the level $r(B)$ the sets $X \sqcup Y$ with $X \in \mathcal{B}|B$ and $Y \in \mathcal{B}/B$, every such set being a basis. Let $\theta_{X\sqcup Y} = \theta'_X\theta''_Y = \phi'_X\phi''_Y$ be two factorizations and fix a basis $X_d \sqcup Y_d$ of the level $d(B)$. For every pattern $X$ of this level $X \sqcup Y_d$ is a basis, so $\phi'_X = \lambda_d\theta'_X$ with $\lambda_d = \theta''_{Y_d}/\phi''_{Y_d}$; for every room side $Y$ of this level $X_d \sqcup Y$ is a basis, so $\phi''_Y = \theta'_{X_d}\theta''_Y/\phi'_{X_d} = \lambda_d^{-1}\theta''_Y$. Likewise at the level $r(B)$ with a constant $\lambda_r$. Conversely any $\lambda_d, \lambda_r > 0$ give a factorization, the patterns of the two levels having different sizes, and so have their room sides. The new factors multiply $K_{\mathcal{B}/R}$ and $\Psi_{\mathcal{B}/R}$ by $\lambda_d$, $K_{\mathcal{B}|B}$ and $\Psi_{\mathcal{B}|B}$ by $\lambda_r$, $K_{\mathcal{B}|R}$ and $\Psi_{\mathcal{B}|R}$ by $\lambda_d^{-1}$, and $K_{\mathcal{B}/B}$ and $\Psi_{\mathcal{B}/B}$ by $\lambda_r^{-1}$. With $c = \lambda_r/\lambda_d$,

$$\gamma_B^K = \frac{K_{\mathcal{B}|B}}{K_{\mathcal{B}/R}} \mapsto c\,\gamma_B^K, \quad \gamma_B^\Psi = \frac{\Psi_{\mathcal{B}/R}}{\Psi_{\mathcal{B}|B}} \mapsto c^{-1}\gamma_B^\Psi, \quad \gamma_R^K = \frac{K_{\mathcal{B}|R}}{K_{\mathcal{B}/B}} \mapsto c\,\gamma_R^K, \quad \gamma_R^\Psi = \frac{\Psi_{\mathcal{B}/B}}{\Psi_{\mathcal{B}|R}} \mapsto c^{-1}\gamma_R^\Psi,$$

so $\varrho_B = \gamma_B^\Psi/\gamma_B^K$ and $\varrho_R = \mathbf{E}_{\mathcal{B}/B}/\mathbf{E}_{\mathcal{B}|R} = \gamma_R^\Psi/\gamma_R^K$ are multiplied by $c^{-2}$, and $\gamma_R^K/\gamma_B^K$, $\gamma_R^\Psi/\gamma_B^\Psi$ and $\varrho_B/\varrho_R$ are unchanged. □

*Remark.* The quotients $\gamma_R^K/\gamma_B^K$ and $\gamma_R^\Psi/\gamma_B^\Psi$ are the Skeleton values of 19.5, which belong to the table and its gauge alone. What depends on the factorization is the split of a Skeleton value into a hinge value of the bunch and one of the room, and with it the cells, the cell families and the pooled families; the identities of the chapter hold at every factorization, each side changing with it.

Two lemmas are used below.

**Lemma S8.** *Let a bunch $B$ of a table $(\mathcal{B},\theta)$ of rank $r$ on $\mathcal{E}$, $n = |\mathcal{E}|$, have width one, the weights factoring across $B$ and $R$. With $\alpha_B = \rho u$,*

$$\rho^{-d(B)}K(\rho u, \alpha_R) = K_{\mathcal{B}/R}(u)\,K_{\mathcal{B}|R}(\alpha_R) + \rho\,K_{\mathcal{B}|B}(u)\,K_{\mathcal{B}/B}(\alpha_R),$$

$$\rho^{-d^*(B)}\Psi(\rho u,\alpha_R)=\Psi_{\mathcal{B}|B}(u)\,\Psi_{\mathcal{B}/B}(\alpha_R)+\rho\,\Psi_{\mathcal{B}/R}(u)\,\Psi_{\mathcal{B}|R}(\alpha_R);$$

*the cells have the ranks* $r(B)=d(B)+1$ *and* $r-d(B)$ *and the letters*

$$K_{\mathcal{B}^+_B}=K_{\mathcal{B}/R}\,(\alpha_g+\gamma^K_B),\quad \Psi_{\mathcal{B}^+_B}=\Psi_{\mathcal{B}|B}\,(\alpha_g+\gamma^\Psi_B),$$
$$K_{\mathcal{B}^+_R}=K_{\mathcal{B}|R}+\alpha_g K_{\mathcal{B}/B},\quad \Psi_{\mathcal{B}^+_R}=\alpha_g\Psi_{\mathcal{B}|R}+\Psi_{\mathcal{B}/B};$$

*and* $d(B)+d(R)=r-1$, $d^*(B)+d^*(R)=n-r-1$. *For a family with masses* $\mathbf{m}$ *in which* $B$ *has width one in every table,* $\kappa_B+\kappa_R=\kappa_{\mathcal{E}}+2\sigma$, $\sigma$ *the total strength.*

*Proof.* By 19.4 the bases of the level $d(B)$ are the $X\sqcup Y$ with $X\in\mathcal{B}/R$, $Y\in\mathcal{B}|R$, and those of the level $r(B)=d(B)+1$ the $X\sqcup Y$ with $X\in\mathcal{B}|B$, $Y\in\mathcal{B}/B$, with $\theta_{X\sqcup Y}=\theta'_X\theta''_Y$ and the complement $(B\setminus X)\sqcup(R\setminus Y)$. Summing over the two levels, $K=K_{\mathcal{B}/R}K_{\mathcal{B}|R}+K_{\mathcal{B}|B}K_{\mathcal{B}/B}$ and $\Psi=\Psi_{\mathcal{B}/R}\Psi_{\mathcal{B}|R}+\Psi_{\mathcal{B}|B}\Psi_{\mathcal{B}/B}$, and on $B$ the polynomials $K_{\mathcal{B}/R}$, $K_{\mathcal{B}|B}$, $\Psi_{\mathcal{B}|B}$, $\Psi_{\mathcal{B}/R}$ are homogeneous of the degrees $d(B)$, $d(B)+1$, $|B|-r(B)=d^*(B)$ and $|B|-d(B)=d^*(B)+1$. The deletions and the contractions of the hinge are $\mathcal{B}^+_B\setminus g=\mathcal{B}|B$, $\mathcal{B}^+_B/g=\mathcal{B}/R$, $\mathcal{B}^+_R\setminus g=\mathcal{B}|R$ and $\mathcal{B}^+_R/g=\mathcal{B}/B$ (19.4), which gives the ranks and, by 7.3 and 19.2, the letters. By 7.12, $d(S)=r-r(\mathcal{E}\setminus S)$, so $d(B)+d(R)=2r-r(B)-r(R)=r-c(B)=r-1$, and $d^*(B)+d^*(R)=n-r(B)-r(R)=n-r-1$. Table by table, $\kappa_B+\kappa_R=m_{\mathcal{E}}-\sum_j\big(s_j(r_j-1)+t_j(n-r_j-1)\big)=\kappa_{\mathcal{E}}+\sum_j(s_j+t_j)$. □

**Lemma S9.** *Let a family* $F$ *on* $\mathcal{E}=B\sqcup R$, $B$ *and* $R$ *nonempty, have real data and be on its balance wall, and let* $f(\alpha)=\alpha^{\mathbf{m}-\mathbf{1}}\prod_j K_j(\alpha)^{-s_j}\Psi_j(\alpha)^{-t_j}$. *Then*

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\Delta_R}\int_{\mathbb{R}^B_{>0}}f(x,v)\,dx\,d\omega_R(v)=\int_{\Delta_R}\int_{\Delta_B}\int_0^\infty\rho^{|B|-1}f(\rho u,v)\,d\rho\,d\omega_B(u)\,d\omega_R(v).$$

*Proof.* On the balance wall $f$ is homogeneous of degree $\kappa_{\mathcal{E}}-n=-n$, $n=|\mathcal{E}|$. Let $\varphi$ be the indicator of $[1,e]$, so that $\int_0^\infty\varphi(\lambda)\,d\lambda/\lambda=1$, and compute $I=\int_{\mathbb{R}^{\mathcal{E}}_{>0}}f(\alpha)\varphi(|\alpha_R|)\,d\alpha$ in two ways. With $\alpha=\lambda w$, $w\in\Delta_{\mathcal{E}}$, $d\alpha=\lambda^{n-1}d\lambda\,d\omega_{\mathcal{E}}(w)$ (2.7) and $f(\lambda w)=\lambda^{-n}f(w)$, so $I=\int_{\Delta_{\mathcal{E}}}f(w)\int_0^\infty\varphi(\lambda|w_R|)\frac{d\lambda}{\lambda}\,d\omega_{\mathcal{E}}(w)=\int_{\Delta_{\mathcal{E}}}f\,d\omega_{\mathcal{E}}=P_F$ (8.7). With $\alpha_R=\lambda v$, $v\in\Delta_R$, and $\alpha_B=\lambda x$, $x\in\mathbb{R}^B_{>0}$, $d\alpha=\lambda^{|R|-1}d\lambda\,d\omega_R(v)\,\lambda^{|B|}dx$ and $f(\lambda x,\lambda v)=\lambda^{-n}f(x,v)$, so $I=\int_0^\infty\varphi(\lambda)\frac{d\lambda}{\lambda}\int_{\Delta_R}\int_{\mathbb{R}^B_{>0}}f(x,v)\,dx\,d\omega_R(v)$, the first form. The second is 2.7 for $x=\rho u$. For $|B|=1$, $\Delta_B$ is a point with the Dirac measure (1.1(iii)). □

**Theorem 19.9 (Walls glue).** *As stated: (i) and (ii) for every family; in (iii) the identities for every family and the gluing when every active letter reads* $B$ *alike.*

*Proof.* The gluing of the counts of one table, the counts of its cells,

$$d_{\mathcal{B}^+_B}(S_B)=d_{\mathcal{B}/R}(S_B),\quad d_{\mathcal{B}^+_B}(S_B+g)=d_{\mathcal{B}|B}(S_B),$$
$$d^*_{\mathcal{B}^+_B}(S_B)=d^*_{\mathcal{B}|B}(S_B),\quad d^*_{\mathcal{B}^+_B}(S_B+g)=d^*_{\mathcal{B}/R}(S_B),$$
$$d_{\mathcal{B}^+_R}(S_R)=d_{\mathcal{B}/B}(S_R),\quad d_{\mathcal{B}^+_R}(S_R+g)=d_{\mathcal{B}|R}(S_R),$$
$$d^*_{\mathcal{B}^+_R}(S_R)=d^*_{\mathcal{B}|R}(S_R),\quad d^*_{\mathcal{B}^+_R}(S_R+g)=d^*_{\mathcal{B}/B}(S_R),$$

and the inequalities $d_{\mathcal{B}/R}\le d_{\mathcal{B}|B}\le d_{\mathcal{B}/R}+1$ and $d^*_{\mathcal{B}|B}\le d^*_{\mathcal{B}/R}\le d^*_{\mathcal{B}|B}+1$ on the subsets of $B$, $d_{\mathcal{B}/B}\le d_{\mathcal{B}|R}\le d_{\mathcal{B}/B}+1$ and $d^*_{\mathcal{B}|R}\le d^*_{\mathcal{B}/B}\le d^*_{\mathcal{B}|R}+1$ on those of $R$, are the reference, proof of thm:wallsglue. For table $j$ and $S=S_B\sqcup S_R$ call

$$a^K_j=d_{\mathcal{B}_j|B}(S_B)-d_{\mathcal{B}_j/R}(S_B),\quad b^K_j=d_{\mathcal{B}_j|R}(S_R)-d_{\mathcal{B}_j/B}(S_R),$$

$$a_j^\Psi = d^*_{\mathcal{B}_j/R}(S_B) - d^*_{\mathcal{B}_j|B}(S_B), \quad b_j^\Psi = d^*_{\mathcal{B}_j/B}(S_R) - d^*_{\mathcal{B}_j|R}(S_R),$$

all in $\{0,1\}$, in this proof the *jumps* of the letters $K_j$ and $\Psi_j$ on the side of the bunch and on the side of the room. The gluing of the counts reads

$$d_j(S) = d_{\mathcal{B}_j/R}(S_B) + d_{\mathcal{B}_j/B}(S_R) + \min(a_j^K, b_j^K), \qquad d_j^*(S) = d^*_{\mathcal{B}_j|B}(S_B) + d^*_{\mathcal{B}_j|R}(S_R) + \min(a_j^\Psi, b_j^\Psi).$$

At $S_B = \varnothing$ the jumps on the side of the bunch are 0, and at $S_R = \varnothing$ those on the side of the room. At $S_B = B$ those on the side of the bunch are 1: the ranks give $d_{\mathcal{B}_j|B}(B) = d_j(B) + 1$, $d_{\mathcal{B}_j/R}(B) = d_j(B)$, $d^*_{\mathcal{B}_j/R}(B) = |B| - d_j(B) = d_j^*(B) + 1$ and $d^*_{\mathcal{B}_j|B}(B) = d_j^*(B)$. At $S_R = R$ those on the side of the room are 1: $d_{\mathcal{B}_j|R}(R) = r_j - d_j(B) = d_{\mathcal{B}_j/B}(R) + 1$ and $d^*_{\mathcal{B}_j/B}(R) = |R| - r_j + d_j(B) + 1 = d^*_{\mathcal{B}_j|R}(R) + 1$.

(i) The tables of $F_{B\to g}(u)$ of index $j$ have the bases of $\mathcal{B}^+_{j,R}$, at $(s_j, 0)$ and at $(0, t_j)$, so its wall at $S \subseteq R + g$ is $m'_S - \sum_j \big(s_j d_{\mathcal{B}^+_{j,R}}(S) + t_j d^*_{\mathcal{B}^+_{j,R}}(S)\big)$, $\mathbf{m}' = (\kappa_B, \mathbf{m}|_R)$. At $S = S_R$ the counts of the cell are $d_{\mathcal{B}_j/B}(S_R)$ and $d^*_{\mathcal{B}_j|R}(S_R)$, which are $d_j(S_R)$ and $d_j^*(S_R)$ by the gluing at $S_B = \varnothing$: the wall is $\kappa_{S_R}$. At $S = S_R + g$ they are $d_{\mathcal{B}_j|R}(S_R)$ and $d^*_{\mathcal{B}_j/B}(S_R)$, and the gluing at $S_B = B$ gives $d_j(B \sqcup S_R) = d_j(B) + d_{\mathcal{B}_j|R}(S_R)$ and $d_j^*(B \sqcup S_R) = d_j^*(B) + d^*_{\mathcal{B}_j/B}(S_R)$; with $\kappa_B = m_B - \sum_j (s_j d_j(B) + t_j d_j^*(B))$ the wall is $\kappa_{B\sqcup S_R}$.

(ii) With $\alpha_{S_B}$ replaced by $\varepsilon\alpha_{S_B}$, $K_{\mathcal{B}_j/R}$ is $\varepsilon^{d_{\mathcal{B}_j/R}(S_B)}$ times a polynomial positive at $\varepsilon = 0$ on the open orthant, its monomials having at least $d_{\mathcal{B}_j/R}(S_B)$ marks in $S_B$ and some exactly that many (7.13); likewise $\Psi_{\mathcal{B}_j|B}$ with $d^*_{\mathcal{B}_j|B}(S_B)$. By the gluing at $S_R = \varnothing$ these are $d_j(S_B)$ and $d_j^*(S_B)$, so the exponent of the density at the face is $\kappa_{S_B}$.

(iii) The walls of $G_B$ and $G_R$ are read with the counts of the cells and the masses $(\mathbf{m}|_B, 2\sigma - \kappa_B)$ and $(\kappa_B, \mathbf{m}|_R)$. With

$$X = m_{S_B} + m_{S_R} - \sum_j \Big(s_j\big(d_{\mathcal{B}_j/R}(S_B) + d_{\mathcal{B}_j/B}(S_R)\big) + t_j\big(d^*_{\mathcal{B}_j|B}(S_B) + d^*_{\mathcal{B}_j|R}(S_R)\big)\Big),$$

the counts of the cells give $\kappa^{G_B}_{S_B} + \kappa^{G_R}_{S_R} = X$ and, the two hinge masses adding up to $2\sigma = \sum_j (s_j + t_j)$, $\kappa^{G_B}_{S_B+g} + \kappa^{G_R}_{S_R+g} = X + \sum_j \big(s_j(1 - a_j^K - b_j^K) + t_j(1 - a_j^\Psi - b_j^\Psi)\big)$, while $\kappa_{S_B\sqcup S_R} = X - \sum_j \big(s_j \min(a_j^K, b_j^K) + t_j \min(a_j^\Psi, b_j^\Psi)\big)$. Since $1 - a - b = (1 - \max(a,b)) - \min(a,b)$ for $a, b \in \{0,1\}$,

$$\begin{aligned}
\kappa^{G_B}_{S_B} + \kappa^{G_R}_{S_R} - \kappa_{S_B\sqcup S_R} &= \sum_j \big(s_j \min(a_j^K, b_j^K) + t_j \min(a_j^\Psi, b_j^\Psi)\big),\\
\kappa^{G_B}_{S_B+g} + \kappa^{G_R}_{S_R+g} - \kappa_{S_B\sqcup S_R} &= \sum_j \big(s_j(1 - \max(a_j^K, b_j^K)) + t_j(1 - \max(a_j^\Psi, b_j^\Psi))\big),
\end{aligned} \tag{50.1}$$

at all masses and exponents. At $S_R = \varnothing$ the first line gives $\kappa^{G_B}_{S_B} = \kappa_{S_B}$, the wall at the empty set being 0; at $S_B = \varnothing$ it gives $\kappa^{G_R}_{S_R} = \kappa_{S_R}$. The imbalance of $G_B$ is

$$\kappa^{G_B}_{B+g} = m_B + 2\sigma - \kappa_B - \sum_j \big(s_j(d_j(B) + 1) + t_j(d_j^*(B) + 1)\big) = 2\sigma - \sum_j (s_j + t_j) = 0,$$

the cell $\mathcal{B}^+_{j,B}$ having rank $d_j(B) + 1$ on $|B| + 1 = d_j(B) + d_j^*(B) + 2$ marks: $G_B$ is on its balance wall at all data. At $S_B = B$ the second line then gives $\kappa^{G_R}_{S_R+g} = \kappa_{B\sqcup S_R}$; at $S_R = R$ in particular

$\kappa^{G_R}_{R+g} = \kappa_{\mathcal{E}}$, so $G_R$ is on its balance wall exactly when $F$ is, and then the second line at $S_R = R$ gives $\kappa^{G_B}_{S_B+g} = \kappa_{S_B \sqcup R}$. These are the four identities.

Let now $s_j, t_j \geq 0$ and let every active letter read $B$ alike. The jump of a letter on the side of the bunch at $S_B$ is the order to which its hinge value vanishes at the face $u_{S_B} = 0$: by the expansion in (ii) applied to $\mathcal{B}_j|B$ and $\mathcal{B}_j/R$, with $\varepsilon\alpha_{S_B}$ for $\alpha_{S_B}$, $\gamma^K_{j,B} = K_{\mathcal{B}_j|B}/K_{\mathcal{B}_j/R}$ is $\varepsilon^{a^K_j}$ times a function positive at $\varepsilon = 0$, and $\gamma^\Psi_{j,B} = \Psi_{\mathcal{B}_j/R}/\Psi_{\mathcal{B}_j|B}$ is $\varepsilon^{a^\Psi_j}$ times one. Constant multiples of one function vanish to one order at each face, so at $S_B$ the active letters have one jump $a \in \{0, 1\}$ on the side of the bunch. If $a = 0$ every term of the first line of (50.1) vanishes, a letter of exponent zero contributing nothing; if $a = 1$ every term of the second does. Both lines are nonnegative, the exponents being nonnegative, so $\kappa_{S_B \sqcup S_R}$ is the smaller of the two sums. One active letter reads $B$ alike by itself.

Positivity. If both cell families have every proper wall positive, let $S = S_B \sqcup S_R$ be nonempty and proper. In the first sum a nonempty part is a proper subset of the marks of its cell family, $S_B \subseteq B \subsetneq B + g$ and $S_R \subseteq R \subsetneq R + g$, and an empty part has the wall 0: the first sum is positive. In the second both $S_B + g$ and $S_R + g$ are nonempty, and at most one of them is all the marks of its cell family, $S$ being proper, its wall then being the imbalance 0: the second sum is positive. So $\kappa_S > 0$. Conversely, by the identities, $\kappa^{G_B}_{S_B} = \kappa_{S_B}$ for $\varnothing \neq S_B \subseteq B$ and $\kappa^{G_B}_{S_B+g} = \kappa_{S_B \sqcup R}$ for $S_B \subsetneq B$ are proper walls of $F$, and likewise for $G_R$; this direction does not use the hypothesis. □

*Remark.* Without the reading alike, (iii) keeps the identities and loses the gluing. On $\mathcal{E} = \{1, 2, 3, 4\}$ with $B = \{1, 2\}$, the family of the lists $\binom{\mathcal{E}}{3}$ and $\binom{\mathcal{E}}{1}$, each at $(0, 1)$, has $B$ of width one in both tables, with the cells $\binom{\{1,2,g\}}{2}$, $\binom{\{3,4,g\}}{2}$ and $\binom{\{1,2,g\}}{1}$, $\binom{\{3,4,g\}}{1}$, and the hinge values $\gamma^\Psi_{1,B} = \alpha_1 + \alpha_2$ and $\gamma^\Psi_{2,B} = \alpha_1\alpha_2/(\alpha_1 + \alpha_2)$, which are not constant multiples of one function. At the masses $(\frac{2}{5}, \frac{8}{5}, \frac{2}{5}, \frac{8}{5})$, on the balance wall, $\kappa_B = 1$ and both cell families have every proper wall positive, the least being $\frac{2}{5}$, while $\kappa_{\{1,3\}} = \frac{2}{5} + \frac{2}{5} - 1 = -\frac{1}{5}$, against $\frac{4}{5}$ from the gluing; the walls at $\{1, 4\}$, $\{2, 3\}$ and $\{2, 4\}$ do not glue either. In exact rational arithmetic (receipt `check_walls.py`): this family; the four identities at every subset at two partial leans of the same tables, where four walls do not glue; and the gluing at every subset for two tables sharing one weighted cell of the bunch at cobasis full leans, three pairs of exponents and three sets of masses, for a table at $(s, 0)$ with the cell $\binom{\{1,2,g\}}{1}$ together with a table at $(0, t)$ with the cell $\binom{\{1,2,g\}}{2}$, whose hinge values $\gamma^K_{1,B}$ and $\gamma^\Psi_{2,B}$ are both $\alpha_1 + \alpha_2$, and for $\binom{\mathcal{E}}{2}$ with $B$ three of its four marks at two partial leans, where $\varrho_B = 1$.

**Theorem 19.10 (Pooling factors at one hinge value).** *As stated: at real exponents of any sign, every active letter having the one hinge value $\gamma$.*

*Proof.* The walls $\kappa_{\{e\}}$ at the singletons are positive and the exponents real, so the masses are real, and the integrals below are of positive functions.

*Bounds.* $\kappa_B$ and $\kappa_R$ are proper walls of $F$, hence positive, and $\kappa_B + \kappa_R = 2\sigma$ on the balance wall (Lemma S8): $0 < \kappa_B < 2\sigma$.

*The cell families.* By the identities of 19.9(iii), which hold for every family on the balance wall, $G_B$ and $G_R$ are on their balance walls and each of their proper walls is a proper wall of $F$, hence positive; their periods converge (8.7).

*The family.* Put $\phi(u) = u^{\mathbf{m}|_B - \mathbf{1}} \prod_j K_{\mathcal{B}_j/R}(u)^{-s_j} \Psi_{\mathcal{B}_j|B}(u)^{-t_j}$ on $\Delta_B$. By 19.4, with $K_{\mathcal{B}_j/R}(\rho u) = \rho^{d_j(B)} K_{\mathcal{B}_j/R}(u)$, $\Psi_{\mathcal{B}_j|B}(\rho u) = \rho^{d^*_j(B)} \Psi_{\mathcal{B}_j|B}(u)$ and the hinge values homogeneous of degree one, Lemma S9 gives

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\Delta_B} \phi(u) \int_{\Delta_R} \int_0^\infty \rho^{\kappa_B-1} v^{\mathbf{m}|_R-\mathbf{1}} \times \prod_j K_{\mathcal{B}^+_{j,R}}\big(\rho\gamma^K_{j,B}(u),v\big)^{-s_j} \Psi_{\mathcal{B}^+_{j,R}}\big(\rho\gamma^\Psi_{j,B}(u),v\big)^{-t_j} d\rho\, d\omega_R(v)\, d\omega_B(u),$$

the powers of $\rho$ collecting to $\rho^{m_B-1-\sum_j(s_jd_j(B)+t_jd_j^*(B))} = \rho^{\kappa_B-1}$. Every active letter has the hinge value $\gamma(u)$, and a letter of exponent zero is the factor 1. The substitution $x = \rho\gamma(u)$ at fixed $u$, $\rho^{\kappa_B-1}d\rho = \gamma(u)^{-\kappa_B}x^{\kappa_B-1}dx$, and Lemma S9 for $G_R$ on $\{g\}\sqcup R$ at the masses $(\kappa_B, \mathbf{m}|_R)$ give

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{G_R}\big((\kappa_B,\mathbf{m}|_R);\mathbf{s},\mathbf{t}\big) \int_{\Delta_B} \phi(u)\,\gamma(u)^{-\kappa_B}\, d\omega_B(u).$$

*The cell family of the bunch.* By Lemma S8, with $y$ the variable of $g$, $K_{\mathcal{B}^+_{j,B}}(u,y) = K_{\mathcal{B}_j/R}(u)\big(y+\gamma^K_{j,B}(u)\big)$ and $\Psi_{\mathcal{B}^+_{j,B}}(u,y) = \Psi_{\mathcal{B}_j|B}(u)\big(y+\gamma^\Psi_{j,B}(u)\big)$; every active letter has the factor $y+\gamma(u)$, and the exponents of the active letters add up to $\sum_j(s_j+t_j) = 2\sigma$. Lemma S9 for $G_B$ on $\{g\}\sqcup B$ at the masses $(\mathbf{m}|_B, 2\sigma-\kappa_B)$, and the integral of a linear letter (the paragraph after 15.1, with $c_0=\gamma(u)$, $c_1=1$ and $0<2\sigma-\kappa_B<2\sigma$), give

$$P_{G_B}\big((\mathbf{m}|_B, 2\sigma-\kappa_B);\mathbf{s},\mathbf{t}\big) = \int_{\Delta_B}\phi(u)\int_0^\infty y^{2\sigma-\kappa_B-1}\big(y+\gamma(u)\big)^{-2\sigma}dy\,d\omega_B(u) = \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{matrix}\kappa_B,\ 2\sigma-\kappa_B\\ 2\sigma\end{matrix}\right] \int_{\Delta_B}\phi(u)\,\gamma(u)^{-\kappa_B}\,d\omega_B(u).$$

Multiplying the display for $P_F$ by $\Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{matrix}\kappa_B,\ 2\sigma-\kappa_B\\ 2\sigma\end{matrix}\right]$ gives the statement, every factor being finite and positive. The parenthesis of the statement is 19.2: the constants $\lambda_d, \lambda_r$ of table $j$ multiply $\gamma^K_{j,B}$ by $c$ and $\gamma^\Psi_{j,B}$ by $c^{-1}$, so a table with one active letter is normalized by its own constant; at full leans of all tables only the hinge values of different tables are compared; one table with both letters active and $\varrho_B$ constant is normalized by $c = \varrho_B^{1/2}$, and $\gamma^K_B = \gamma^\Psi_B$ forces $\varrho_B = 1$. One table at $(0,t)$ has $s=0$ and $2\sigma = t$, and at $(s,0)$, $2\sigma = s$. □

*Remark.* One table at $(0,t)$ has the one active letter $\Psi$, whose hinge value $\gamma^\Psi_B$ serves as $\gamma$ at any factorization; the cell families are the cells $(\mathcal{B}^+_B,\theta')$ and $(\mathcal{B}^+_R,\theta'')$ at $(0,t)$, and the display is that of the reference, $\Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\left[\begin{matrix}\kappa_B,\ t-\kappa_B\\ t\end{matrix}\right] P_F(\mathbf{m};0,t) = P_{\mathcal{B}^+_B}\big((\mathbf{m}|_B, t-\kappa_B);0,t\big)P_{\mathcal{B}^+_R}\big((\kappa_B,\mathbf{m}|_R);0,t\big)$; at $(s,0)$ the same holds with $\gamma = \gamma^K_B$ and $s$ for $t$. The proof above is the proof of the reference with $\gamma^\Psi_B$ replaced by the common hinge value of the active letters. A family at full leans, every table at $(s_j,0)$ or $(0,t_j)$, has one active letter in each table, normalized by the factorization of its table; the hypothesis then asks only that the hinge values of the active letters of different tables be constant multiples of one another, a condition across the tables, and 22.6 is the same condition at every cell of a core. One table with $s,t>0$ has both letters active; the hypothesis is $\varrho_B$ constant, reached at the factorization with $\varrho_B = 1$, and the display then holds at every pair $(s,t)$ with $2\sigma = s+t$, the separation of 19.12 completed by the Beta corner of the hinge.

**Proposition 19.11 (Pooling a realization).** *As stated, for every family and every realization.*

*Proof.* The masses of $\Upsilon$ are $\tau_e = \sum_j\big(s_j\alpha_e\partial_e\log K_j(\alpha) + t_j\alpha_e\partial_e\log\Psi_j(\alpha)\big)$ (9.1). With $u = \alpha_B/|\alpha_B|$ fixed, the tables of index $j$ of $F_{B\to g}(u)$, read as one table with two systems of weights, have the

letters $K^u_j(\alpha_g,\alpha_R)=K_{\mathcal{B}^+_{j,R}}(\gamma^K_{j,B}(u)\alpha_g,\alpha_R)$ and $\Psi^u_j(\alpha_g,\alpha_R)=\Psi_{\mathcal{B}^+_{j,R}}(\gamma^\Psi_{j,B}(u)\alpha_g,\alpha_R)$ (19.8). By 19.4 and the homogeneity of the hinge values, $\gamma^K_{j,B}(\alpha_B)=\rho\gamma^K_{j,B}(u)$ with $\rho=|\alpha_B|$, and

$$K_j(\alpha)=K_{\mathcal{B}_j/R}(\alpha_B)\,K^u_j(\rho,\alpha_R),\qquad \Psi_j(\alpha)=\Psi_{\mathcal{B}_j|B}(\alpha_B)\,\Psi^u_j(\rho,\alpha_R).$$

For $e\in R$ the first factors do not depend on $\alpha_e$, so $\tau_e$ is the usage of $e$ in $F_{B\to g}(u)$ at the gauge $(\rho,\alpha_R)$ with the exponents $(s_j,t_j)$. Summing over $e\in B$, Euler's identity gives $\sum_{e\in B}\alpha_e\partial_e\log K_{\mathcal{B}_j/R}(\alpha_B)=d_j(B)$ and $\sum_{e\in B}\alpha_e\partial_e\log\Psi_{\mathcal{B}_j|B}(\alpha_B)=d^*_j(B)$, and, $\gamma^K_{j,B}$ being homogeneous of degree one,

$$\sum_{e\in B}\alpha_e\partial_e\log K_{\mathcal{B}^+_{j,R}}\big(\gamma^K_{j,B}(\alpha_B),\alpha_R\big)=\big[y\partial_y\log K_{\mathcal{B}^+_{j,R}}(y,\alpha_R)\big]_{y=\gamma^K_{j,B}(\alpha_B)}$$
$$=\big[\alpha_g\partial_{\alpha_g}\log K^u_j(\alpha_g,\alpha_R)\big]_{\alpha_g=\rho},$$

and likewise for $\Psi$. So $\tau_B=\sum_j\big(s_jd_j(B)+t_jd^*_j(B)\big)+\tau^u_g$, with $\tau^u_g$ the usage of $g$ in $F_{B\to g}(u)$ at $(\rho,\alpha_R)$, and $\tau^u_g$ is the wall $\kappa_B$ at the masses $\boldsymbol{\tau}$. The exponents are those of a realization and $(\rho,\alpha_R)$ is a gauge, so $((s_j,t_j)_j;(\rho,\alpha_R))$ is a realization of $F_{B\to g}(u)$ with these masses, its usage. For one table at $(0,t)$, $\Psi^u(\rho,\alpha_R)=\Psi_{\mathcal{B}^+_R}(\gamma^\Psi_B(\alpha_B),\alpha_R)$, so the usage of $(\mathcal{B}^+_R,\theta'')$ at $(0,t)$ and at the gauge $(\gamma^\Psi_B(\alpha_B),\alpha_R)$ is $\boldsymbol{\tau}|_R$ on $R$ and, by the chain rule above, $\tau_B-t\,d^*(B)$ at $g$. □

*Remark.* The reference is one list at $(0,t)$, its last sentence. In exact rational arithmetic (receipt `check_classes_realization.py`): two tables on five marks, each the two-sum of a weighted cell of the bunch on $\{1,2,g\}$ and a weighted cell of the room on $\{3,4,5,g\}$, at the partial leans $(\frac{2}{3},\frac{5}{4})$ and $(3,\frac{1}{2})$ and a rational gauge; the usage of $F_{B\to g}(u)$ at $(|\alpha_B|,\alpha_R)$ is $(\kappa_B(\boldsymbol{\tau}),\boldsymbol{\tau}|_R)$.

**Proposition 19.12 (Hinge values that agree).** *As stated, for families.*

*Proof.* As in the proof of 19.10, with $\phi(u)=u^{\mathbf{m}|_B-\mathbf{1}}\prod_j K_{\mathcal{B}_j/R}(u)^{-s_j}\Psi_{\mathcal{B}_j|B}(u)^{-t_j}$,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\Delta_B}\phi(u)\int_{\Delta_R}\int_0^\infty\rho^{\kappa_B-1}v^{\mathbf{m}|_R-\mathbf{1}}$$
$$\times\prod_j K_{\mathcal{B}^+_{j,R}}\big(\rho\gamma^K_{j,B}(u),v\big)^{-s_j}\Psi_{\mathcal{B}^+_{j,R}}\big(\rho\gamma^\Psi_{j,B}(u),v\big)^{-t_j}d\rho\,d\omega_R(v)\,d\omega_B(u).$$

For $s_j\neq0$, $K_{\mathcal{B}^+_{j,R}}(\rho c_j\gamma(u),v)=K_{\mathcal{B}|R}(v)+\rho c_j\gamma(u)K_{\mathcal{B}/B}(v)$ is the letter $K$ of $\mathcal{B}^+_{j,R}$ with its weights multiplied by $c_j$ on its bases through $g$, read at $(\rho\gamma(u),v)$; for $t_j\neq0$, $\Psi_{\mathcal{B}^+_{j,R}}(\rho c^*_j\gamma(u),v)$ is the letter $\Psi$ of $\mathcal{B}^+_{j,R}$ with its weights multiplied by $c^*_j$ on its bases avoiding $g$, read at $(\rho\gamma(u),v)$ (Lemma S8). These are the letters of $G^{\mathbf{c}}_R$, and the letters of exponent zero are the factor 1. The substitution $x=\rho\gamma(u)$ and Lemma S9 for $G^{\mathbf{c}}_R$, which has the bases and so the walls of $G_R$ and is on its balance wall by 19.9(iii), give the display. Its period converges, each proper wall of $G_R$ being a proper wall of $F$, so the shape integral is finite.

For one table with $\varrho_B$ constant, 19.7 reaches $\varrho_B=1$ by $c=\varrho_B^{1/2}$; then $\gamma^K_B=\gamma^\Psi_B$, $c=c^*=1$ with $\gamma=\gamma^K_B$, and $G^{\mathbf{c}}_R$ is the cell $(\mathcal{B}^+_R,\theta'')$ at $(s,t)$, which is the reference. □

**Theorem 19.13 (Gluing across a bunch of width one).** *As stated: at nonnegative exponents, a letter of exponent zero being opened by nothing.*

*Proof. The letters on the scale.* For table $j$ and $(u,v)\in\Delta_B\times\Delta_R$ put

$$A_j^K=K_{\mathcal{B}_j/R}(u)K_{\mathcal{B}_j|R}(v),\quad C_j^K=K_{\mathcal{B}_j|B}(u)K_{\mathcal{B}_j/B}(v),$$
$$A_j^\Psi=\Psi_{\mathcal{B}_j|B}(u)\Psi_{\mathcal{B}_j/B}(v),\quad C_j^\Psi=\Psi_{\mathcal{B}_j/R}(u)\Psi_{\mathcal{B}_j|R}(v),$$

all positive, so that $\rho^{-d_j(B)}K_j(\rho u,v)=A_j^K+\rho C_j^K$ and $\rho^{-d_j^*(B)}\Psi_j(\rho u,v)=A_j^\Psi+\rho C_j^\Psi$ (Lemma S8). The minors of the level $d$ sit in $A_j^K$ and in $C_j^\Psi$, those of the level $r$ in $C_j^K$ and in $A_j^\Psi$.

*The scale.* By Lemma S9, the powers of $\rho$ collecting to $\rho^{\kappa_B-1}$,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\int_{\Delta_R}\int_{\Delta_B}u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}\,I(u,v)\,d\omega_B(u)\,d\omega_R(v),$$
$$I(u,v)=\int_0^\infty\rho^{\kappa_B-1}\prod_{\ell=1}^{p}(A_\ell+\rho C_\ell)^{-w_\ell}d\rho,$$

where $\ell$ runs over the $p$ active letters, with their exponents $w_\ell>0$ and their coefficients $(A_\ell,C_\ell)$, a letter of exponent zero being the factor 1. Here $\sum_\ell w_\ell=\sum_j(s_j+t_j)=2\sigma$, and $\kappa_B$ and $\kappa_R=2\sigma-\kappa_B$ (Lemma S8, $\kappa_{\mathcal{E}}=0$) are proper walls: $0<\kappa_B<2\sigma$.

*Opening the scale.* If $p=1$, the integral of a linear letter (the paragraph after 15.1) gives $I=\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}w_1-\kappa_B,\ \kappa_B\\ w_1\end{smallmatrix}\right]A_1^{\kappa_B-w_1}C_1^{-\kappa_B}$. If $p\ge2$, choose $c_\ell\in(0,w_\ell)$ for $\ell<p$ with $0<\kappa_B-\sum_{\ell<p}c_\ell<w_p$, possible as $0<\kappa_B<\sum_\ell w_\ell$, and open each letter $\ell<p$ by 4.4 with $k=2$ on the line $\operatorname{Re}x_\ell=c_\ell$:

$$(A_\ell+\rho C_\ell)^{-w_\ell}=\frac{1}{2\pi i}\int_{c_\ell+i\mathbb{R}}\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}w_\ell-x_\ell,\ x_\ell\\ w_\ell\end{smallmatrix}\right]A_\ell^{x_\ell-w_\ell}(\rho C_\ell)^{-x_\ell}\,dx_\ell.$$

The modulus of the integrand is its value at the real parts times the modulus of the stacked Gamma, which decays exponentially along the line (Stirling), and $\int_0^\infty\rho^{\kappa_B-\sum_{\ell<p}c_\ell-1}(A_p+\rho C_p)^{-w_p}d\rho$ is finite; so Fubini puts the integral over $\rho$ inside, where the integral of a linear letter at the complex index $x_p=\kappa_B-\sum_{\ell<p}x_\ell$, $0<\operatorname{Re}x_p<w_p$, gives $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}w_p-x_p,\ x_p\\ w_p\end{smallmatrix}\right]A_p^{x_p-w_p}C_p^{-x_p}$. Hence

$$I(u,v)=\frac{1}{(2\pi i)^{p-1}}\int_{c_1+i\mathbb{R}}\cdots\int_{c_{p-1}+i\mathbb{R}}\prod_{\ell=1}^{p}\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}w_\ell-x_\ell,\ x_\ell\\ w_\ell\end{smallmatrix}\right]A_\ell^{x_\ell-w_\ell}C_\ell^{-x_\ell}\,dx_1\cdots dx_{p-1},$$
$$x_p=\kappa_B-\sum_{\ell<p}x_\ell,$$

the share integral $\int_{x_1+\cdots+x_p=\kappa_B}$ of 4.1: the factor $(2\pi i)^{-1}$ of each of the $p-1$ openings is its normalization $(2\pi i)^{1-p}$, and its dimension is $p-1$. At $p=1$ the share integral has dimension zero, one share and the pin $P=(1)$, and 4.1 gives the value of the integrand at the one admissible point $x_1=\kappa_B$: the formula for $p=1$ above.

*The level shares.* For the letter $K_j$ write $x_\ell=a_j^r$ and $w_\ell-x_\ell=a_j^d$, for $\Psi_j$ write $x_\ell=b_j^d$ and $w_\ell-x_\ell=b_j^r$, and give the letters of exponent zero the shares 0. The pin $\sum_\ell x_\ell=\kappa_B$ is $\sum_j(a_j^r+b_j^d)=\kappa_B$; the factor of $K_j$ is $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}a_j^d,\ a_j^r\\ s_j\end{smallmatrix}\right](A_j^K)^{-a_j^d}(C_j^K)^{-a_j^r}$ and that of $\Psi_j$ is $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\left[\begin{smallmatrix}b_j^d,\ b_j^r\\ t_j\end{smallmatrix}\right](C_j^\Psi)^{-b_j^d}(A_j^\Psi)^{-b_j^r}$. The product of the powers over all letters, times $u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}}$, is $f(u)\,g(v)$, with

$$f(u)=u^{\mathbf{m}|_B-\mathbf{1}}\prod_j K_{\mathcal{B}_j/R}^{-a_j^d}\Psi_{\mathcal{B}_j/R}^{-b_j^d}K_{\mathcal{B}_j|B}^{-a_j^r}\Psi_{\mathcal{B}_j|B}^{-b_j^r},\qquad g(v)=v^{\mathbf{m}|_R-\mathbf{1}}\prod_j K_{\mathcal{B}_j|R}^{-a_j^d}\Psi_{\mathcal{B}_j|R}^{-b_j^d}K_{\mathcal{B}_j/B}^{-a_j^r}\Psi_{\mathcal{B}_j/B}^{-b_j^r},$$

the integrands of the periods of $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ on $\Delta_B$ and of $F^{\mathrm{room}}(\mathbf{a},\mathbf{b})$ on $\Delta_R$ (8.7).

*Balance on the pin.* The tables $\mathcal{B}_j/R$ and $\mathcal{B}_j|B$ on $B$ have the ranks $d_j(B)$ and $d_j(B)+1$, and $\mathcal{B}_j|R$ and $\mathcal{B}_j/B$ on $R$ the ranks $r_j-d_j(B)$ and $r_j-d_j(B)-1$. With $a_j^d+a_j^r=s_j$, $b_j^d+b_j^r=t_j$ and $|B|-d_j(B)-1=d_j^*(B)$, the imbalance of $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ at $\mathbf{m}|_B$ is

$$m_B-\sum_j\big(a_j^d d_j(B)+b_j^d(|B|-d_j(B))+a_j^r(d_j(B)+1)+b_j^r d_j^*(B)\big)=\kappa_B-\sum_j(a_j^r+b_j^d);$$

with $|R|+1-r_j+d_j(B)=n-r_j-d_j^*(B)$, that of $F^{\mathrm{room}}(\mathbf{a},\mathbf{b})$ at $\mathbf{m}|_R$ is

$$m_R-\sum_j\big(s_j(r_j-d_j(B))+t_j(|R|+1-r_j+d_j(B))\big)+\sum_j(a_j^r+b_j^d)=\kappa_{\mathcal{E}}-\kappa_B+\sum_j(a_j^r+b_j^d).$$

Both vanish on the pin, $F$ being on its balance wall.

*Fubini.* On a contour of the statement the modulus of $f(u)g(v)$ is $f(u)g(v)$ at the real parts of the shares, which do not move along the contour, and its integral over $\Delta_B\times\Delta_R$ is the product of the two periods at the real parts, finite by hypothesis; the stacked Gammas decay exponentially along the contour. So Fubini exchanges the share integral with the integral over $\Delta_B\times\Delta_R$, and the inner integrals are $P_{F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_B;\mathbf{a},\mathbf{b})$ and $P_{F^{\mathrm{room}}(\mathbf{a},\mathbf{b})}(\mathbf{m}|_R;\mathbf{a},\mathbf{b})$: the general display.

*Pair coordinates.* When every $s_j$ and $t_j$ is positive, $p=2N$ and the shares are $(a_j^r,b_j^d)_j$. With $a_j^d=\sigma_j^d+\delta_j^d$, $b_j^d=\sigma_j^d-\delta_j^d$ and $a_j^r=s_j-a_j^d$, the affine map $(\sigma_j^d,\delta_j^d)_j\mapsto(a_j^r,b_j^d)_j$ has in each table the linear part $(y_\sigma,y_\delta)\mapsto(-y_\sigma-y_\delta,\ y_\sigma-y_\delta)$, of determinant 2. So the imaginary parts $\mathbf{y}$ of the shares and $\mathbf{y}'$ of the pair coordinates have $d\mathbf{y}=2^N d\mathbf{y}'$ and $\sum_\ell y_\ell=-2\sum_j y'_{\delta,j}$, and with $\operatorname{Im}\kappa_B=0$ and $\delta^1(-2z)=\frac12\delta^1(z)$, 4.1 gives

$$\int_{\sum_\ell x_\ell=\kappa_B}h=(2\pi)^{1-2N}\int_{\mathbb{R}^{2N}}h\,\delta^1\Big(\sum_\ell y_\ell\Big)d\mathbf{y}=2^{N-1}(2\pi)^{1-2N}\int_{\mathbb{R}^{2N}}h\,\delta^1\Big(\sum_j y'_{\delta,j}\Big)d\mathbf{y}',$$

$2^{N-1}$ times the fibre integral in the pair coordinates pinned by $\sum_j\delta_j^d=\frac12(\sum_j s_j-\kappa_B)$, the image of the pin $\sum_j(a_j^r+b_j^d)=\sum_j(s_j-2\delta_j^d)=\kappa_B$, the admissible real points corresponding under the map. With $\Gamma_{\substack2\\1}\begin{bmatrix}a_j^d,\ a_j^r\\ s_j\end{bmatrix}\Gamma_{\substack2\\1}\begin{bmatrix}b_j^d,\ b_j^r\\ t_j\end{bmatrix}=\Gamma_{\substack4\\2}\begin{bmatrix}a_j^d,\ b_j^d,\ a_j^r,\ b_j^r\\ s_j,\ t_j\end{bmatrix}$ this is the first display, whose $N$ coordinates $\sigma_j^d$ are not pinned. For one table the pin fixes $\delta^d$, $2^{N-1}=1$, and the fibre integral is $(2\pi)^{-1}\int dy=(2\pi i)^{-1}\int d\sigma^d$ along the line in $\sigma^d$. The other forms of the pin, the value $2\delta^d=s-\kappa_B=\delta(1+\varepsilon(B))-\mu_B$ (8.2(iii) with $c(B)=1$) and the swap of $B$ and $R$ are the reference, proof of thm:twosides. □

*Remark.* The first display needs both letters of every table active: a letter of exponent zero would put $\Gamma(0)$ into the stacked Gamma of its table. The general display carries one Beta corner $\Gamma_{\substack2\\1}\begin{bmatrix}\cdot,\ \cdot\\ w\end{bmatrix}$ for each active letter, and at a full lean of one table it has no free share, which is 19.14.

**Corollary 19.14 (One Skeleton value pins the share).** *As stated: real exponents of any sign, one Skeleton value for all active letters.*

*Proof.* $0<\kappa_B<2\sigma$ as in the proof of 19.10. With the coefficients of the proof of 19.13, Lemma S9 gives, all functions being positive,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{\Delta_R}\int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}} \times \int_0^\infty \rho^{\kappa_B-1}\prod_j (A_j^K+\rho C_j^K)^{-s_j}(A_j^\Psi+\rho C_j^\Psi)^{-t_j}\,d\rho\,d\omega_B(u)\,d\omega_R(v).$$

The Skeleton values are $A_j^K/C_j^K=\gamma_{j,R}^K(v)/\gamma_{j,B}^K(u)$ and $A_j^\Psi/C_j^\Psi=\gamma_{j,R}^\Psi(v)/\gamma_{j,B}^\Psi(u)$ (19.5), and by the hypothesis, the constants cancelling, every active letter has $A_\ell=\xi C_\ell$ with $\xi=\gamma_R(v)/\gamma(u)$. The exponents of the active letters add up to $2\sigma$, so the integral of a linear letter gives $\int_0^\infty \rho^{\kappa_B-1}(\xi+\rho)^{-2\sigma}d\rho = \Gamma_{\!1}^{\!2}\!\left[\begin{matrix}\kappa_B,\ 2\sigma-\kappa_B\\ 2\sigma\end{matrix}\right]\xi^{\kappa_B-2\sigma}$, and

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \Gamma_{\!1}^{\!2}\!\left[\begin{matrix}\kappa_B,\ 2\sigma-\kappa_B\\ 2\sigma\end{matrix}\right]\int_{\Delta_R}\int_{\Delta_B} u^{\mathbf{m}|_B-\mathbf{1}}v^{\mathbf{m}|_R-\mathbf{1}} \times \prod_{s_j\neq 0}(C_j^K)^{-s_j}\prod_{t_j\neq 0}(C_j^\Psi)^{-t_j}\,\xi^{\kappa_B-2\sigma}\,d\omega_B(u)\,d\omega_R(v).$$

For level shares as in the statement, $(A_j^K)^{-a_j^d}(C_j^K)^{-a_j^r} = (C_j^K)^{-s_j}\xi^{-a_j^d}$ for an active $K_j$, $(A_j^\Psi)^{-b_j^r}(C_j^\Psi)^{-b_j^d} = (C_j^\Psi)^{-t_j}\xi^{-b_j^r}$ for an active $\Psi_j$, and an inactive letter contributes 1; since $\sum_j(a_j^d+b_j^r)=2\sigma-\sum_j(a_j^r+b_j^d)=2\sigma-\kappa_B$, the integrand above is $f(u)g(v)$ with $f$ and $g$ the integrands of the periods of $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ and $F^{\mathrm{room}}(\mathbf{a},\mathbf{b})$ (proof of 19.13), whatever the shares. At complex shares $|f(u)g(v)|$ is $f(u)g(v)$ at the real parts, which satisfy the same constraints; its integral is finite by the display at the real parts, and both factors are positive, so $|f|$ and $|g|$ are integrable and Fubini gives the statement, the product of the two periods not depending on the shares. With one active letter the hypothesis holds with $\gamma$ and $\gamma_R$ its hinge values of $B$ and of $R$. One table at $(0,t)$: $\mathbf{a}=0$, $b^d+b^r=t$ and $b^d=\kappa_B$, so the level shares are $(0,\kappa_B,0,t-\kappa_B)$ and the two families have the forms of the statement; at $(s,0)$, $a^r=\kappa_B$ and $a^d=s-\kappa_B$. □

*Remark.* The reference, rem:pinnedproduct, is one table at a full lean, the case of one active letter. The hypothesis does not depend on the factorizations: a change of the factorization of table $j$ multiplies the hinge values of a letter of table $j$ at $B$ and at $R$ by one constant (19.2), so the Skeleton values are unchanged.

**Lemma 19.15 (Slopes and the range of the ratios).** *As stated: for every table of a family, both letters of every table active.*

*Proof.* With $a_j^\ell=\sigma_j^\ell+\delta_j^\ell$ and $b_j^\ell=\sigma_j^\ell-\delta_j^\ell$, $K^{-a}\Psi^{-b}=\mathbf{E}^{-\sigma}(K/\Psi)^{-\delta}$ for every minor, $\mathbf{E}=K\Psi$. So the factors of table $j$ in the word of the bunch family (proof of 19.13) are

$$\mathbf{E}_{\mathcal{B}_j/R}^{-\sigma_j^d}\mathbf{E}_{\mathcal{B}_j|B}^{-\sigma_j^r}\Big(\frac{K_{\mathcal{B}_j/R}}{\Psi_{\mathcal{B}_j/R}}\Big)^{-\delta_j^d}\Big(\frac{K_{\mathcal{B}_j|B}}{\Psi_{\mathcal{B}_j|B}}\Big)^{-\delta_j^r},\qquad \mathbf{E}_{\mathcal{B}_j/R}^{-\sigma_j^d}\mathbf{E}_{\mathcal{B}_j|B}^{-\sigma_j+\sigma_j^d}=\varrho_{j,B}^{-\sigma_j^d}\mathbf{E}_{\mathcal{B}_j|B}^{-\sigma_j},$$

and in the word of the room family $\mathbf{E}_{\mathcal{B}_j|R}^{-\sigma_j^d}\mathbf{E}_{\mathcal{B}_j/B}^{-\sigma_j+\sigma_j^d}=\varrho_{j,R}^{\sigma_j^d}\mathbf{E}_{\mathcal{B}_j/B}^{-\sigma_j}$ (19.2), the other factors not involving $\sigma_j^d$ at fixed $\delta_j^d$: $\sigma_j^d$ enters through $\varrho_{j,B}(u)^{-\sigma_j^d}\varrho_{j,R}(v)^{\sigma_j^d}$.

Slopes. The wall at $S\subseteq B$ of the bunch family is $m_S-\sum_j\big(a_j^d d_{\mathcal{B}_j/R}(S)+b_j^d d^*_{\mathcal{B}_j/R}(S)+a_j^r d_{\mathcal{B}_j|B}(S)+b_j^r d^*_{\mathcal{B}_j|B}(S)\big)$, affine in the shares; raising $\sigma_j^d$ by one at fixed $\delta_j^d$ raises $a_j^d,b_j^d$ and lowers $a_j^r,b_j^r$ by one, so the slope is $c_{\mathcal{B}_j/R}(S)-c_{\mathcal{B}_j|B}(S)$, $c=|S|-d-d^*$. With $r^+$ the rank function

of $\mathcal{B}^+_{j,B}$, $\mathcal{B}_j|B = \mathcal{B}^+_{j,B} \setminus g$ has the rank function $r^+$ on $B$ and $\mathcal{B}_j/R = \mathcal{B}^+_{j,B}/g$ the rank function $Z \mapsto r^+(Z+g) - 1$, and $c_Z(S) = r_Z(S) + r_Z(B \setminus S) - r_Z(B)$ for a table $Z$ on $B$ (7.12, 7.13); the difference is the first displayed slope. On $R$ the table of the level $d$ is $\mathcal{B}_j|R = \mathcal{B}^+_{j,R} \setminus g$ and that of the level $r$ is $\mathcal{B}_j/B = \mathcal{B}^+_{j,R}/g$, which gives the second.

Range. On the sector of an order of $B$, $\alpha_{\pi(1)} = \rho_0$ and $\alpha_{\pi(k)} = \rho_0 q_2 \cdots q_k$ with $q_k \in (0,1]$ and tails $U_k$, the letters of a table $Z$ on $B$ are $K_Z = \rho_0^{r_Z} \prod_k q_k^{d_Z(U_k)} k_{Z,\pi}$ and $\Psi_Z = \rho_0^{|B|-r_Z} \prod_k q_k^{d^*_Z(U_k)} \psi_{Z,\pi}$, with sector letters positive on the closed cube (8.5(ii)), hence between positive constants. So

$$\varrho_{j,B} = \frac{\mathbf{E}_{\mathcal{B}_j/R}}{\mathbf{E}_{\mathcal{B}_j|B}} = \prod_k q_k^{c_{\mathcal{B}_j|B}(U_k) - c_{\mathcal{B}_j/R}(U_k)} \times (\text{a function between positive constants}),$$

the exponent of $q_k$ being minus the slope of $U_k$. Every nonempty proper subset of $B$ is a tail of some order, the $q_k$ range over $(0,1]$ independently, and finitely many sectors cover $\Delta_B$: $\varrho_{j,B}$ is bounded exactly when no slope on $B$ is positive (a positive slope at a tail $U_k$ sends it to $\infty$ as $q_k \to 0$), and bounded below by a positive constant exactly when none is negative. On $R$, $\varrho_{j,R} = \mathbf{E}_{\mathcal{B}_j/B}/\mathbf{E}_{\mathcal{B}_j|R}$ carries $q_k$ to the power of the slope of $U_k$: $\inf \varrho_{j,R} > 0$ exactly when no slope on $R$ is positive and $\sup \varrho_{j,R} < \infty$ exactly when none is negative. A wall falls as $\sigma^d_j$ decreases, toward the level $d$, exactly when its slope is positive, which forces $\sup \varrho_{j,B} = \infty$ or $\inf \varrho_{j,R} = 0$, so $\sup \varrho_{j,B}/\inf \varrho_{j,R} = \infty$; toward the level $r$, a negative slope forces $\inf \varrho_{j,B}/\sup \varrho_{j,R} = 0$. A change of the factorization of table $j$ multiplies $\varrho_{j,B}$ and $\varrho_{j,R}$ by one constant $c^{-2}$ (19.7), which keeps the quotient, the finiteness of the suprema, the positivity of the infima and the two ratios; the slopes are counts. □

**Proposition 19.24 (The same shares closed at the gauge).** *As stated: at complex exponents and masses, with the bound in absolute value.*

*Proof.* At width one the letters on the scale are linear in $\rho$ (Lemma S8), and divided by their values at $\rho = 1$ they are the Skeleton polynomials of the statement (19.16): with the coefficients of the proof of 19.13, $\pi_{j,d} = A^K_j/(A^K_j + C^K_j)$, $\pi_{j,r} = 1 - \pi_{j,d}$, $\pi^*_{j,r} = A^\Psi_j/(A^\Psi_j + C^\Psi_j)$ and $\pi^*_{j,d} = 1 - \pi^*_{j,r}$, the chances of the levels under the basis law and the cobasis law at $(u,v)$, all in $(0,1)$. Each polynomial is $1-(1-x)\epsilon$ with a constant exceedance polynomial $\epsilon$: $\pi_{j,r}$ for $K_j$ and $\pi^*_{j,d}$ for $\Psi_j$ on the bunch, and, read from the top of the scale, $\pi_{j,d}$ and $\pi^*_{j,r}$ on the room. In 15.9(ii) only $k = 0$ remains, $[x^k] \prod_i \epsilon_i^{n_i} = 0$ for $k > 0$, and the term of $\mathbf{n}$ is $\prod_i \frac{(w_i)_{n_i}}{n_i!} \epsilon_i^{n_i} \Gamma_2^1 \left[\begin{matrix} \kappa, & |\mathbf{n}|+1 \\ \kappa + |\mathbf{n}| + 1 \end{matrix}\right]$; the two sectors of 15.9(i) give the display, of nonnegative terms for real data with $s_j, t_j \geq 0$. For complex exponents and masses with $\operatorname{Re}\kappa_B > 0$ and $\operatorname{Re}\kappa^{\text{top}}_B > 0$ the series converges absolutely to the same value (15.9, last sentence; 15.3 for the convergence of the scale integral). The bound: $|(w)_n| \leq (|w|)_n$; $\prod_j \pi_{j,r}^{n_j} \pi_{j,d}^{*\,n'_j} \leq \lambda_B^M$; $\left|\Gamma_2^1 \left[\begin{matrix} \kappa_B, & M+1 \\ \kappa_B + M + 1 \end{matrix}\right]\right| = \left|\int_0^1 x^{\kappa_B - 1}(1-x)^M dx\right| \leq \int_0^1 x^{\operatorname{Re}\kappa_B - 1} dx = 1/\operatorname{Re}\kappa_B$; and $\sum_{|\mathbf{n}|+|\mathbf{n}'|=M} \prod_j \frac{(|s_j|)_{n_j} (|t_j|)_{n'_j}}{n_j!\, n'_j!} = \frac{(\sum_j (|s_j| + |t_j|))_M}{M!}$, the coefficient of $z^M$ in $\prod_j (1-z)^{-|s_j|-|t_j|}$. Likewise on the room with $\lambda_R$ and $\operatorname{Re}\kappa^{\text{top}}_B$. The bounds are summable over $M$, $\sum_M \frac{(W)_M}{M!} \lambda^M = (1-\lambda)^{-W}$ for $\lambda < 1$. □

**Proposition 19.33 (A loop or a coloop on the simplex).** *As stated: a mark that is a loop or a coloop in every table of a family.*

*Proof.* In a table where $e$ is a loop no basis contains $e$, so $K_j = K_{j,\backslash e}$ and $\Psi_j = \alpha_e \Psi_{j,\backslash e}$; where $e$ is a coloop every basis contains $e$, so $K_j = \alpha_e K_{j,/e}$ and $\Psi_j = \Psi_{j,/e}$ (7.3). The letters of $F'$ (15.16) are these minors, of the ranks $r'_j = r_j$ at a loop and $r_j - 1$ at a coloop on $n-1$ marks, so the family word is $\alpha_e^{-\sum_{\eta_j=+1} s_j - \sum_{\eta_j=-1} t_j} W'(\alpha_{\mathcal{E}-e})$, with $W'$ the word of $F'$, homogeneous of degree $\kappa' - m_{\mathcal{E}-e}$, $\kappa'$ the imbalance of $F'$ at $\mathbf{m}|_{\mathcal{E}-e}$. On $\Delta_{\mathcal{E}}$ put $u = (u_e, (1-u_e)v)$, $u_e \in (0,1)$, $v \in \Delta_{\mathcal{E}-e}$, so that $u^{\mathbf{m}-\mathbf{1}} d\omega_{\mathcal{E}}(u) = u_e^{m_e-1}(1-u_e)^{m_{\mathcal{E}-e}-1} du_e \, v^{\mathbf{m}|_{\mathcal{E}-e}-\mathbf{1}} d\omega_{\mathcal{E}-e}(v)$ (3.12). Then

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_0^1 u_e^{\nu_e-1}(1-u_e)^{\kappa'-1} du_e \; P_{F'}(\mathbf{m}|_{\mathcal{E}-e};\mathbf{s},\mathbf{t}), \qquad \nu_e = m_e - \sum_{\eta_j=+1} s_j - \sum_{\eta_j=-1} t_j,$$

for real data in $[0,\infty]$, and the integral over $u_e$ is $\Gamma_{\substack{2\\1}}\begin{bmatrix}\nu_e, & \kappa'\\ \nu_e+\kappa' & \end{bmatrix}$ where $\nu_e, \kappa' > 0$. This $\nu_e$ is the wall $\kappa_{\{e\}}$, since $d_j(\{e\}) = 1$ and $d_j^*(\{e\}) = 0$ at a coloop and the reverse at a loop, and by 8.2(iii), with $c_j(\{e\}) = 0$ and $\varepsilon_j(\{e\}) = \eta_j$, it is $\mu_e - \sum_j \eta_j \delta_j$. The imbalance of $F'$ is $m_{\mathcal{E}} - m_e - \sum_{\eta_j=+1} \big(s_j(r_j - 1) + t_j(n - r_j)\big) - \sum_{\eta_j=-1} \big(s_j r_j + t_j(n-1-r_j)\big) = \kappa_{\mathcal{E}} - \nu_e$, so $\nu_e + \kappa' = \kappa_{\mathcal{E}}$. Several such marks are removed one at a time, each keeping its kind in the minors, and the product of the Beta corners is $\Gamma_{\substack{k+1\\1}}\begin{bmatrix}(\nu_{e_i})_i, & \kappa'\\ \kappa_{\mathcal{E}} & \end{bmatrix}$, split by 3.12 as stated. On the balance wall $\kappa' = -\nu_e$, and $\nu_e$ and $\kappa'$ are not both positive.

The set $S = \{e\}$ has $c_j(S) = 0$ in every table, as 16.17 asks; there $\kappa_{\{e\}} = \nu_e$ and, by 8.2(iii) and 7.13 ($c_j(\mathcal{E}-e) = c_j(\{e\}) = 0$, $\varepsilon_j(\mathcal{E}-e) = 2r_j - n - \eta_j$), $\kappa_{\mathcal{E}\backslash e} = \kappa_{\mathcal{E}} - \nu_e = \kappa'$. The integral of 16.17 over $\Delta_{\{e\}} \times \Delta_{\mathcal{E}-e}$, $\Delta_{\{e\}}$ the point $u_e = 1$ with the Dirac measure, is the integral over $\Delta_{\mathcal{E}-e}$ of $v^{\mathbf{m}|_{\mathcal{E}-e}-\mathbf{1}}$ times the family word at $(1,v)$, which is $W'(v)$: it is $P_{F'}(\mathbf{m}|_{\mathcal{E}-e};\mathbf{s},\mathbf{t})$, and the compound formula at $S = \{e\}$ is the display. □

**Theorem 19.34 (Pooling is a move).** *As stated: for tables with positive rational weights, and in (ii) whenever the active letters have one hinge value at rational factorizations.*

*Proof.* (i) is the reference, proof of thm:cleaningmove(i): the map $(\rho, u, \alpha_R) \mapsto (\rho u, \alpha_R)$ and its inverse are defined over $\mathbb{Q}$ whatever the weights, and the faces correspond as stated. After the change of variables the form is $\rho^{\kappa_B - 1}\phi(u)$ times the form of $F_{B\to g}(u)$ at the gauge $(\rho, \alpha_R)$ (proof of 19.8), with $\phi$ as in the proof of 19.10; its coefficients lie in the field of the weights and of the factors, which is $\mathbb{Q}$. The walls are matched by 19.9(i), (ii) and the gluing of the counts, for families.

(ii) The common hinge value $\gamma$ is the hinge value of an active letter, a quotient of polynomials with rational coefficients, positive on the open simplex. The change $(\rho, u, \alpha_R) \mapsto (\rho\gamma(u), u, \alpha_R)$ is birational over $\mathbb{Q}$, with the inverse $(x, u, \alpha_R) \mapsto (x/\gamma(u), u, \alpha_R)$, and bijective on the open domain; by the proof of 19.10 it carries the form of (i) to the product of the form $\phi(u)\gamma(u)^{-\kappa_B} d\omega_B(u)$ of the shape and the form of $G_R$ at the masses $(\kappa_B, \mathbf{m}|_R)$. In the datum of $G_B$, read on the section $|\alpha_B| = 1$ (Lemma S9), the change $y = \gamma(u)z$, birational over $\mathbb{Q}$ and bijective on the open domain, carries the form to $\phi(u)\gamma(u)^{-\kappa_B} d\omega_B(u)$ times $z^{2\sigma-\kappa_B-1}(1+z)^{-2\sigma} dz$, the form of the list on two marks whose bases are the two singletons, at the exponents $(0, 2\sigma)$ and the masses $(\kappa_B, 2\sigma - \kappa_B)$, read in its chart at the first mark, where its letter is $1 + z$. So the data of $F$ and of that list, and the data of $G_B$ and $G_R$, are related by these moves to the same product of three data, with rational masses $\kappa_B$ and $2\sigma - \kappa_B$. If $P_F$ converges, every proper wall of $F$ is positive (8.7) and 19.10 gives the convergence of the periods of $G_B$ and $G_R$, the Beta corner, and the shape integral, which is $P_F$ over $P_{G_R}$; and $P_F$ is the first integral of the chain. □

*Remark.* Positive rational weights that factor across $B$ and $R$ factor with rational factors: fix a basis $X_\ell \sqcup Y_\ell$ at each level $\ell \in \{d(B), r(B)\}$ and put $\theta'_X = \theta_{X\sqcup Y_\ell}$ and $\theta''_Y = \theta_{X_\ell\sqcup Y}/\theta_{X_\ell\sqcup Y_\ell}$ for the patterns $X$ and the room sides $Y$ of the level $\ell$. For any factorization $\theta = \vartheta'\vartheta''$, $\theta_{X\sqcup Y_\ell}\theta_{X_\ell\sqcup Y}/\theta_{X_\ell\sqcup Y_\ell} = \vartheta'_X\vartheta''_{Y_\ell}\vartheta'_{X_\ell}\vartheta''_Y/(\vartheta'_{X_\ell}\vartheta''_{Y_\ell}) = \theta_{X\sqcup Y}$. The hypothesis of (ii) asks more, rational factors at which the active letters share one hinge value; for one table with both letters active it asks a rational factorization with $\varrho_B = 1$.

**Proposition 19.37 (Cleaning commutes with maps).** *As stated, for maps between any tables.*

*Proof.* A bijection $g$ that carries $(\mathcal{B}, \theta)$ to $(\mathcal{B}_1, \theta_1)$ maps bases to bases with their weights. Parallel and series bunches, the counts and the widths are defined by the bases, so they are carried to those of the image; a factorization $\theta'$, $\theta''$ across a class $B$ is carried to $\theta'_{1,g(p)} = \theta'_p$, $\theta''_{1,g(Y)} = \theta''_Y$, a factorization across $g(B)$, and the minors of $g(B)$ are the images of those of $B$ with their weights, so $\gamma^K_{g(B)}(\alpha\circ g^{-1}) = \gamma^K_B(\alpha)$ and likewise for $\Psi$. An exchange maps $T$ to $\mathcal{E}_1 \setminus g(T)$, with the weight $\theta_T$. A parallel bunch $B$ has $|T\cap B| \le 1$ for every basis and no loop, so $g(B)$ has $|g(B)\cap T_1| \ge |B| - 1$ for every basis $T_1$ of the image and no coloop: a series bunch, and conversely. The patterns of $g(B)$ are the sets $g(B\setminus p)$, $p$ a pattern of $B$, and putting $\theta'_{1,g(B\setminus p)} = \theta'_p$ and $\theta''_{1,g(R\setminus Y)} = \theta''_Y$ factors the weights of the image. The level of $g(B\setminus p)$ is $|B| - |p|$, so the restriction of the image to $g(B)$ has the patterns $g(B\setminus p)$, $p \in \mathcal{B}/R$, and its contraction of the room the patterns $g(B\setminus p)$, $p \in \mathcal{B}|B$; hence $K_{\mathcal{B}_1|g(B)}(\alpha\circ g^{-1}) = \sum_{p\in\mathcal{B}/R}\theta'_p\alpha^{B\setminus p} = \Psi_{\mathcal{B}/R}(\alpha)$ and $K_{\mathcal{B}_1/g(R)}(\alpha\circ g^{-1}) = \Psi_{\mathcal{B}|B}(\alpha)$, so the hinge value by $K$ of $g(B)$ is $\gamma^\Psi_B$; likewise its hinge value by $\Psi$ is $\gamma^K_B$. The statement on the cleanings of lists is the reference, proof of prop:cleanmaps. □

## 50.20 Chapter 20: words of a tree

**Definition 20.2 (Side shape; half-sum and difference; pattern coefficients).** As stated, for every table of a family.

*Proof.* The one claim is the identity $L_{j,w}(z) = \sum_p e^{\langle z_B,\chi_p\rangle}c_{j,p}(w; z_R)$. Theorem 12.7 for table $j$ gives $\Lambda_j(z) = \sum_{p\subseteq B} e^{\langle z_B,\chi_p\rangle}\Lambda_{j,(p)}(z_R)$, with $\Lambda_{j,(p)} = 0$ off the patterns of table $j$. Replacing $z$ by $-z$ gives

$$\Lambda_j(-z) = \sum_p e^{-\langle z_B,\chi_p\rangle}\Lambda_{j,(p)}(-z_R).$$

Since $\chi_{B\setminus p} = -\chi_p$ on $B$, the substitution $p \mapsto B\setminus p$ turns this into

$$\Lambda_j(-z) = \sum_p e^{\langle z_B,\chi_p\rangle}\Lambda_{j,(B\setminus p)}(-z_R) = \sum_p e^{\langle z_B,\chi_p\rangle}\Lambda^*_{j,(p)}(z_R).$$

So

$$w\Lambda_j(z) + (1-w)\Lambda_j(-z) = \sum_p e^{\langle z_B,\chi_p\rangle}\big(w\Lambda_{j,(p)}(z_R) + (1-w)\Lambda^*_{j,(p)}(z_R)\big).$$

□

**Proposition 20.3 (The side shape, and what an exchange does to it).** *As stated, for every table of a family and every bijection that carries or exchanges the family.*

*Proof.* For $T \subseteq \mathcal{E}$ and $z \in \mathbb{R}^{\mathcal{E}}$, with $(g^*z)_e = z_{g(e)}$,

$$\langle z, \chi_{g(T)} \rangle = \sum_{e \in \mathcal{E}} z_{g(e)}\, \chi_{g(T)}(g(e)) = \sum_{e \in \mathcal{E}} (g^*z)_e\, \chi_T(e) = \langle g^*z, \chi_T \rangle.$$

If $g$ carries $F$, then $g(\mathcal{B}_j) = \mathcal{B}_{h(j)}$ with $\theta_{h(j),g(T)} = \theta_{j,T}$ (Definition 11.13), so

$$\Lambda_{h(j)}(z) = \sum_{T \in \mathcal{B}_j} \theta_{j,T}\, e^{\langle z, \chi_{g(T)} \rangle} = \Lambda_j(g^*z),$$

and $\Lambda_{h(j)}(-z) = \Lambda_j(-g^*z)$, as $g^*$ is linear; hence $L_{h(j),w}(z) = L_{j,w}(g^*z)$. If $g$ exchanges $F$, then $T \mapsto \mathcal{E} \setminus g(T)$ maps $\mathcal{B}_j$ onto $\mathcal{B}_{h(j)}$ with $\theta_{h(j),\mathcal{E}\setminus g(T)} = \theta_{j,T}$, and $\chi_{\mathcal{E}\setminus g(T)} = -\chi_{g(T)}$, so $\Lambda_{h(j)}(z) = \Lambda_j(-g^*z)$ and $\Lambda_{h(j)}(-z) = \Lambda_j(g^*z)$; hence

$$L_{h(j),w}(z) = w\, \Lambda_j(-g^*z) + (1-w)\, \Lambda_j(g^*z) = L_{j,1-w}(g^*z).$$

The display holds for any two positive numbers $a, b$ in place of $\Lambda_j(z), \Lambda_j(-z)$ and any $s_j, t_j > 0$: it is Theorem 3.2 at the unit configuration on a two-element set with the coarsest partition, whose Zamma is $(wa + (1-w)b)^{-(s_j+t_j)} = \mathsf{Z}_{s_j+t_j;(1)}(wa + (1-w)b)$ at the shape $(w, 1-w)$, that is, the two-factor form $a^{-s_j} b^{-t_j} = \Gamma_{\frac{1}{2}}\begin{bmatrix} s_j + t_j \\ s_j,\ t_j \end{bmatrix} \int_0^1 w^{s_j-1}(1-w)^{t_j-1}(wa+(1-w)b)^{-s_j-t_j}\, dw$ (the reference, proof of prop:sideexchange). Taking $a = \Lambda_j(z)$ and $b = \Lambda_j(-z)$, $wa + (1-w)b = L_{j,w}(z)$.

For one table an exchange has $h$ the identity, so that $L_w(z) = L_{1-w}(g^*z)$ for every $w$. Hence

$$\mathsf{Z}_{s+t;(1)}\big(L_w(g^*z)\big) = \mathsf{Z}_{s+t;(1)}\big(L_{1-w}(z)\big)\,:$$

the exchange together with $w \mapsto 1-w$ keeps the Zamma. The substitution $w \mapsto 1-w$ takes $w^{s-1}(1-w)^{t-1}$ to $w^{t-1}(1-w)^{s-1}$, which is the same function on $(0,1)$ exactly when $s = t$. □

**Theorem 20.4 (The junction in pattern form).** *As stated, for every family.*

*Proof.* The proof is that of the reference (proof of thm:junctionpattern), with a product over the tables in each step.

*Signed coordinates.* By Theorem 8.7 the period is the integral of $\alpha^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, d\alpha$ over the section $\alpha_g = 1$. Put $\alpha = e^{2z}$, so that $d\alpha_e = 2\alpha_e\, dz_e$ for each of the $n-1$ marks $e \neq g$. Theorem 7.15(i) for each table gives $K_j(\alpha) = e^{\langle \mathbf{1}, z \rangle} \Lambda_j(z)$ and $\Psi_j(\alpha) = e^{\langle \mathbf{1}, z \rangle} \Lambda_j(-z)$, so, with $\sum_j (s_j + t_j) = 2\sigma$ and $\boldsymbol{\mu} = \mathbf{m} - \sigma\mathbf{1}$ (Definition 8.1),

$$\begin{aligned} \alpha^{\mathbf{m}} \prod_j K_j^{-s_j} \Psi_j^{-t_j} &= e^{2\langle \mathbf{m}, z \rangle - 2\sigma \langle \mathbf{1}, z \rangle} \prod_j \Lambda_j(z)^{-s_j} \Lambda_j(-z)^{-t_j} \\ &= e^{2\langle \boldsymbol{\mu}, z \rangle} \prod_j \Lambda_j(z)^{-s_j} \Lambda_j(-z)^{-t_j}, \end{aligned}$$

and, with $e^{2\langle \boldsymbol{\mu}, z \rangle} = e^{2\langle \boldsymbol{\mu}|_B, z_B \rangle} e^{2\langle \boldsymbol{\mu}|_R, z_R \rangle}$,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = 2^{n-1} \int_{\mathbb{R}^{\mathcal{E}-g}} e^{2\langle \boldsymbol{\mu}, z \rangle} \prod_j \Lambda_j(z)^{-s_j} \Lambda_j(-z)^{-t_j}\, dz_{\mathcal{E}-g} \Big|_{z_g = 0}.$$

*Side shapes.* For $j$ with $s_j, t_j > 0$, Proposition 20.3 writes the factor of table $j$ as

$$\Lambda_j(z)^{-s_j} \Lambda_j(-z)^{-t_j} = \Gamma_{\frac{1}{2}}\begin{bmatrix} s_j + t_j \\ s_j,\ t_j \end{bmatrix} \int_0^1 w_j^{s_j-1}(1-w_j)^{t_j-1} L_{j,w_j}(z)^{-(s_j+t_j)}\, dw_j.$$

For $t_j = 0 < s_j$ the factor is $\Lambda_j(z)^{-s_j} = L_{j,1}(z)^{-s_j}$, for $s_j = 0 < t_j$ it is $\Lambda_j(-z)^{-t_j} = L_{j,0}(z)^{-t_j}$, and an inactive table contributes 1. So every active table contributes $L_{j,w_j}(z)^{-(s_j+t_j)}$, integrated over $w_j$ when $s_j, t_j > 0$ and with the value of $w_j$ fixed in the statement otherwise.

*Scales.* For each active $j$, $L_{j,w_j}(z) > 0$ and $L^{-a} = \Gamma(a)^{-1}\int_0^\infty y^{a-1}e^{-yL}dy$ with $a = s_j + t_j > 0$. The constants combine to $\Gamma_{\frac{1}{2}}\begin{bmatrix} s_j + t_j \\ s_j,\ t_j \end{bmatrix}\Gamma(s_j + t_j)^{-1} = \Gamma(s_j)^{-1}\Gamma(t_j)^{-1}$ when $s_j, t_j > 0$, and to $\Gamma(s_j)^{-1}$ or $\Gamma(t_j)^{-1}$ for a table at a full lean: the prefactor of the statement.

*Patterns.* By Definition 20.2,

$$\sum_{j \text{ active}} y_j L_{j,w_j}(z) = \sum_{p \subseteq B} e^{\langle z_B, \chi_p\rangle} C_p, \qquad C_p = \sum_{j \text{ active}} y_j\, c_{j,p}(w_j; z_R) \geq 0,$$

each $c_{j,p}(w; z_R)$ being a combination with the nonnegative coefficients $w$ and $1 - w$ of the nonnegative functions $\Lambda_{j,(p)}(z_R)$ and $\Lambda^*_{j,(p)}(z_R)$.

*The Gramma.* With $x = e^{z_B}$, Definition 2.1 in its logarithmic form gives

$$\int_{\mathbb{R}^B} e^{2\langle \boldsymbol{\mu}|_B, z_B\rangle} \exp\Big( -\sum_p C_p\, e^{\langle \chi_p, z_B\rangle}\Big) dz_B = \mathsf{G}_{2\boldsymbol{\mu}|_B;(\chi_p)_p}\big((C_p)_p\big),$$

the configuration having one column $\chi_p$ for each $p$ with $C_p \neq 0$.

*Order of integration.* Every integrand is nonnegative, so Tonelli's theorem allows the integrals to be taken in the order of the statement: the side shapes outermost, then $z_{R-g}$ at $z_g = 0$, then the scales $y_j$, then $z_B$. The two sides agree in $[0, \infty]$ and are finite, every proper wall being positive (Theorem 8.7). □

*Remark.* With one table the formula is that of the reference, and at a full lean of that table the integral over the side shape is absent, which is the reference's last sentence.

**Proposition 20.5 (What a map keeping the bunch gives the coefficients).** *As stated, for every family.*

*Proof.* Since $\tilde{g}$ keeps $B$, it maps $R$ onto $R$, so $(\tilde{g}^* z)_R = \tilde{g}^*_R z_R$, and for $p \subseteq B$, substituting $e' = \tilde{g}(e)$,

$$\langle (\tilde{g}^* z)_B, \chi_p\rangle = \sum_{e \in B} z_{\tilde{g}(e)} \chi_p(e) = \sum_{e' \in B} z_{e'} \chi_{\tilde{g}(p)}(e') = \langle z_B, \chi_{\tilde{g}(p)}\rangle.$$

If $\tilde{g}$ carries $F$, Proposition 20.3 and Definition 20.2 give

$$\begin{aligned}\sum_{q \subseteq B} e^{\langle z_B, \chi_q\rangle} c_{h(j),q}(w; z_R) &= L_{h(j),w}(z) = L_{j,w}(\tilde{g}^* z) \\ &= \sum_{p \subseteq B} e^{\langle z_B, \chi_{\tilde{g}(p)}\rangle} c_{j,p}(w; \tilde{g}^*_R z_R).\end{aligned}$$

The vectors $\chi_q$, $q \subseteq B$, are distinct, and at fixed $z_R$ the exponentials $e^{\langle z_B, \chi_q\rangle}$ with distinct exponents are linearly independent functions of $z_B$: along a generic direction the largest exponent dominates (the reference, proof of prop:mapcoeff). Comparing the coefficients at $q = \tilde{g}(p)$ gives $c_{h(j),\tilde{g}(p)}(w; z_R) = c_{j,p}(w; \tilde{g}^*_R z_R)$. If $\tilde{g}$ exchanges $F$, the same comparison with $L_{h(j),w}(z) = L_{j,1-w}(\tilde{g}^* z)$ gives $c_{h(j),\tilde{g}(p)}(w; z_R) = c_{j,p}(1 - w; \tilde{g}^*_R z_R)$ for every $w$, and $w \mapsto 1 - w$ gives the statement. □

**Proposition 20.6 (Level form of the half-sum).** *As stated, for every table of the family of Theorem 20.4.*

*Proof.* Every claim concerns one table $j$, and the reference's proof (proof of prop:levelformQ) applies to it. The two displayed coefficients are $c_{j,p}(w;z_R)$ at $w=\frac12$ and its derivative in $w$ (Definition 20.2).

(i) $c_{j,B\setminus p}(\frac12;-z_R) = \frac12\big(\Lambda_{j,(B\setminus p)}(-z_R)+\Lambda_{j,(p)}(z_R)\big) = c_{j,p}(\frac12;z_R)$, and $\partial_w c_{j,B\setminus p}(w;-z_R) = \Lambda_{j,(B\setminus p)}(-z_R)-\Lambda_{j,(p)}(z_R) = -\partial_w c_{j,p}(w;z_R)$.

(ii) Let $\tau$ be a transposition of two marks of $B$ fixing the rest. As in the proof of Proposition 20.5, $L_{j,1/2}(\tau^*z)=\sum_p e^{\langle z_B,\chi_{\tau(p)}\rangle}c_{j,p}(\frac12;z_R)$. If $c_{j,p}(\frac12;\cdot)$ depends on $|p|$ alone, this is $L_{j,1/2}(z)$, since $|\tau(p)|=|p|$. Conversely, if $L_{j,1/2}(\tau^*z)=L_{j,1/2}(z)$ for every such $\tau$, the independence of the exponentials gives $c_{j,\tau(p)}(\frac12;\cdot)=c_{j,p}(\frac12;\cdot)$, and these transpositions act transitively on the subsets of $B$ of each size. So level form is the invariance of $L_{j,1/2}$ under these transpositions. Further,

$$L_{j,1/2}(z)=\tfrac12\sum_{T\in\mathcal{B}_j}\theta_{j,T}\big(e^{\langle z,\chi_T\rangle}+e^{\langle z,\chi_{\mathcal{E}\setminus T}\rangle}\big),$$
$$L_{j,1/2}(\tau^*z)=\tfrac12\sum_{T\in\mathcal{B}_j}\theta_{j,T}\big(e^{\langle z,\chi_{\tau(T)}\rangle}+e^{\langle z,\chi_{\mathcal{E}\setminus\tau(T)}\rangle}\big),$$

by the computation of the proof of Proposition 20.3. The exponentials $e^{\langle z,\chi_S\rangle}$, $S\subseteq\mathcal{E}$, are linearly independent, so the invariance under $\tau$ says that $\tau$ maps the multiset $\mathcal{B}_j\uplus\{\mathcal{E}\setminus T:\ T\in\mathcal{B}_j\}$, with its weights, onto itself. When $2r_j\neq n$ a basis and a complement of a basis have different sizes, $\tau$ keeps sizes, and the condition says that $\tau$ maps $\mathcal{B}_j$ onto itself with its weights, that is, carries table $j$ (Definition 11.13), as in Proposition 14.12.

(iii) If $\tau$ carries table $j$, it keeps $\Lambda_j(z)$ and $\Lambda_j(-z)$ under $z\mapsto\tau^*z$; if it exchanges table $j$, it swaps them (Proposition 20.3, the bijection of the tables being the identity). Either way $L_{j,1/2}(\tau^*z)=L_{j,1/2}(z)$, so the half-sum has level form by (ii), and $\partial_w L_{j,w}(\tau^*z)=\pm\partial_w L_{j,w}(z)$, so the square of the difference is unchanged by $\tau$. The transpositions of $B$ generate the permutations of $B$. □

**Theorem 20.11 (Building back up).** *As stated; the last sentence of (iv) is stated for every molecule.*

*Proof.* Parts (i) to (iii), and (iv) up to its last sentence, are proved in the reference (proof of thm:buildup). The last sentence of (iv) makes three claims.

*Dimension.* In (iv) every table has $s_j,t_j>0$ and is active. The unknowns are the $c_{j,T}$, $T\in\mathcal{B}_j$, $\sum_j|\mathcal{B}_j|$ of them, subject to the $N$ share sums $\sum_T c_{j,T}=t_j-s_j$ and the $n-1$ pins $-\frac12\sum_{j,T}c_{j,T}\chi_T(e)=\mu_e$, $e\in\mathcal{E}-g$. Their linear parts are independent. Indeed, let $\sum_j\lambda_j\sum_T c_{j,T}+\sum_{e\neq g}d_e\sum_{j,T}c_{j,T}\chi_T(e)$ vanish identically, and put $d_g=0$. The coefficient of $c_{j,T}$ is $\lambda_j+\langle d,\chi_T\rangle=\lambda_j+2\langle d,\mathbf{1}_T\rangle-\langle d,\mathbf{1}\rangle$, so $\langle d,\mathbf{1}_T\rangle$ is constant over the bases of every table. By Lemma 7.8, $d$ is constant on every molecule; the family is a molecule, so $d$ is constant, and $d_g=0$ makes it 0; then every $\lambda_j=0$. So the fibre of differences has dimension $\sum_j|\mathcal{B}_j|-N-(n-1)=\sum_j(|\mathcal{B}_j|-1)-(n-1)$.

*The pins at the centre.* At the centre $t_j=s_j$ and $\boldsymbol{\mu}=0$, so the share sums read $\sum_T c_{j,T}=0$. With $\chi_T=2\mathbf{1}_T-\mathbf{1}$,

$$\sum_T c_{j,T}\chi_T(e)=2\big(M_j^{\mathsf T}\mathbf{c}_j\big)_e-\sum_T c_{j,T},$$

so on the share sums the pin of $e \neq g$ reads $(\sum_j M_j^{\mathsf{T}} \mathbf{c}_j)_e = 0$. The equation at $g$ follows, since $\sum_{e \in \mathcal{E}} (M_j^{\mathsf{T}} \mathbf{c}_j)_e = \sum_T c_{j,T} |T| = r_j \sum_T c_{j,T} = 0$. So the pins and the share sums together are equivalent to $\sum_T c_{j,T} = 0$ for every $j$ and $\sum_j M_j^{\mathsf{T}} \mathbf{c}_j = 0$.

*The contour.* At the centre the gauge $\mathbf{1}$ is native (Proposition 8.21(i)). The leaves are opened by the atom formula with one share per basis and letter (Corollary 18.16), and the native split there (Definition 12.25) gives the basis $T$ of table $j$ the shares $a_{j,T} = s_j \theta_{j,T} / K_j(\mathbf{1})$ and $b_{j,T} = t_j \theta_{j,T} / \Psi_j(\mathbf{1})$, the probabilities of $T$ under the basis law and the cobasis law at $\mathbf{1}$ times the exponents. As $\Psi_j(\mathbf{1}) = \sum_T \theta_{j,T} = K_j(\mathbf{1})$ and $s_j = t_j$, $c_{j,T} = b_{j,T} - a_{j,T} = 0$ there, so the wall contour of the native split (Theorem 18.13(iii)) is $\operatorname{Re} \mathbf{c} = 0$. □

*Remark.* With one table $\sum_T c_T = \frac{1}{r} \sum_e (M^{\mathsf{T}} \mathbf{c})_e$ follows from $M^{\mathsf{T}} \mathbf{c} = 0$, and the dimension is $|\mathcal{B}_1| - n$: the reference's case of one table at the centre.

**Proposition 20.12 (The draw series).** *As stated; the reading of its last part is stated for every family.*

*Proof.* The first part, through the Trailmoment and the Dirichlet moments, is the reference's proposition (proof of prop:jointdraw). For the last part take one block, $q = 2N$, $P_{2j-1} = K_j$ with the index $s_j$ and $P_{2j} = \Psi_j$ with the index $t_j$; an index 0 contributes the factor 1.

*Constants.* $K_j$ is homogeneous of degree $r_j$, with the squarefree monomials $u^T$, $T \in \mathcal{B}_j$, and the coefficients $\theta_{j,T}$. For $\mathbf{a} = \mathbf{1}_T$, $b_{\mathbf{a}} = r_j! / \prod_e a_e! = r_j!$, and every other $\mathbf{a}$ has $[u^{\mathbf{a}}] K_j = 0$; so $C_{2j-1} = \theta_{j,\max} / r_j!$. Likewise $\Psi_j$ has degree $n - r_j$ and the monomials $u^{\mathcal{E} \setminus T}$ with the coefficients $\theta_{j,T}$, so $C_{2j} = \theta_{j,\max} / (n - r_j)!$.

*The period.* $\mathbb{E}_{U \sim \mathrm{Dir}_{\mathbf{m}}}[f(U)] = \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}^{-1} \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m}-\mathbf{1}} f \, d\omega_{\mathcal{E}}$ for $f \geq 0$ (Theorem 3.9), so the left side times $\Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}$ is the stated integral, which on the balance wall is $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$ (Theorem 8.7).

*The urn.* For $u$ in the simplex, $(\sum_e u_e)^g = \sum_{|\mathbf{a}|=g} \frac{g!}{\mathbf{a}!} u^{\mathbf{a}} = 1$ is the total chance of $g$ independent draws from $u$, grouped by their counts $\mathbf{a}$. For $P_{2j-1} = K_j$,

$$\frac{P_{2j-1}}{C_{2j-1}} = \sum_{T \in \mathcal{B}_j} \frac{\theta_{j,T}}{\theta_{j,\max}} \, r_j! \, u^T$$

is the chance that a round of $r_j$ draws is a basis $T$ of table $j$ and does not fail, a round that is the basis $T$ failing with probability $1 - \theta_{j,T} / \theta_{j,\max}$; so $Q_{2j-1} = 1 - P_{2j-1} / C_{2j-1}$ is the chance that the round fails, and likewise $Q_{2j}$ for a round of $n - r_j$ draws and the cobases. Given $u$, independent rounds all fail with the chance $\prod_i Q_i(u)^{k_i}$. The urn that draws a mark with probability proportional to its current mass and then adds one to that mass gives a sequence of draws with the counts $\mathbf{b}$ the chance $\prod_e (m_e)_{b_e} / (m_{\mathcal{E}})_{|\mathbf{b}|} = \mathbb{E}_{U \sim \mathrm{Dir}_{\mathbf{m}}}[U^{\mathbf{b}}]$ (Theorem 3.11), the chance of the same sequence when $U \sim \mathrm{Dir}_{\mathbf{m}}$ is drawn first and the marks are then drawn independently from $U$. Summing over the sequences whose rounds all fail, $\mathbb{E}_{U \sim \mathrm{Dir}_{\mathbf{m}}}[\prod_i Q_i(U)^{k_i}]$ is the urn's chance of the event of the statement; the order of the rounds does not change it, the product being symmetric in its factors. For lists every ratio $\theta_{j,T} / \theta_{j,\max}$ is 1, the $Q_i$ have integer coefficients, and the expectation is a finite sum of Dirichlet moments.

*Theorem 15.13(i).* With the references $M_i = C_i$, on the simplex

$$M_i^{-w_i - n_i} (M_i - L_i)^{n_i} = C_i^{-w_i} (1 - L_i / C_i)^{n_i} = C_i^{-w_i} Q_i^{n_i},$$

so the series of Theorem 15.13(i), divided by $\Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_{\mathcal{E}} \end{bmatrix}$, is the present series term by term. □

**Corollary 20.13 (At $\delta = 0$ and at a full lean the period reads the law of one letter for each table).** *As stated, for every family.*

*Proof. Letters.* For a table riding the line, $K_j^{-\sigma_j}\Psi_j^{-\sigma_j} = \mathbf{E}_j^{-\sigma_j}$; at $(s_j, 0)$ the factor is $K_j^{-s_j}$ and at $(0, t_j)$ it is $\Psi_j^{-t_j}$. So on the open simplex

$$\prod_j K_j^{-s_j}\Psi_j^{-t_j} = \prod_j L_j^{-\nu_j} = \mathsf{Z}_{\boldsymbol{\nu};I_N}\big((L_j)_j\big),$$

the Zamma at the unit configuration being $\prod_j \xi_j^{-\nu_j}$ (Definition 2.12).

*The period.* By Theorem 8.7 and the Dirichlet density (Theorem 3.9),

$$\begin{aligned} P_F(\mathbf{m};\mathbf{s},\mathbf{t}) &= \int_{\Delta_\mathcal{E}} u^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j}\Psi_j^{-t_j}\, d\omega_\mathcal{E} \\ &= \Gamma_1^n \begin{bmatrix} \mathbf{m} \\ m_\mathcal{E} \end{bmatrix} \mathbb{E}_{U\sim \mathrm{Dir}_\mathbf{m}}\Big[\mathsf{Z}_{\boldsymbol{\nu};I_N}\big((L_j(U))_j\big)\Big]. \end{aligned}$$

The expectation is the integral of the kernel against the image $\phi_\mathbf{m}$ of $\mathrm{Dir}_\mathbf{m}$ under $(L_j)_j$, the Trailmoment $(\mathsf{T}_{\boldsymbol{\nu};I_N}[\mathrm{id}]\,\phi_\mathbf{m})((0,\infty)^N)$ with the identity as coefficient function (Definition 5.1), as in the display of Theorem 18.1(i). With one table it is the Trailmoment $(\mathsf{T}_{\nu;(1)}[x]\,\phi_\mathbf{m})((0,\infty))$ of a law on the half-line, at $\delta = 0$ the image of $\mathrm{Dir}_\mathbf{m}$ under the even letter (Theorem 5.9).

*The law.* Each $L_j$ is a polynomial, positive on the open simplex and continuous on the closed one, and $\mathrm{Dir}_\mathbf{m}$ charges only the open simplex; so $\phi_\mathbf{m}$ is carried by $\prod_j(0, \max_{\Delta_\mathcal{E}} L_j]$. A law carried by a compact set is determined by its moments, polynomials being dense in the continuous functions on that set. Every moment $\mathbb{E}[\prod_j L_j(U)^{k_j}]$ is the expectation of a polynomial in $U$, a finite sum of Dirichlet moments (Theorem 3.11). □

**Theorem 20.14 (The draw series along a chain).** *As stated, for every family of tables.*

*Proof.* The proof is that of the reference (proof of thm:drawchain), with the acceptances and a product over the tables.

*The failures split.* Fix table $j$ and a round of $r_j$ draws. A round that extends on $S_{i+1}$ extends on $S_i$, with the trace $P_i = P_{i+1} \cap S_i$, and every round extends on $S_0 = \varnothing$ with the trace $\varnothing$. So the events $A_i$, that the round extends on $S_i$ and is accepted at $S_1, \dots, S_i$, decrease with $i$, $A_0$ is certain, and $\mathfrak{f}_{j,i}$ is the chance of $A_i \setminus A_{i+1}$; hence $\sum_{i=0}^d \mathfrak{f}_{j,i} = 1 - \Pr(A_{d+1})$.

*The acceptances telescope.* A round extends on $S_{d+1} = \mathcal{E}$ exactly when it is a basis $T$ of table $j$ with distinct draws, which has the chance $r_j!\,u^T$; its traces are then $P_i = T \cap S_i$, and its acceptances multiply to

$$\prod_{i=0}^{d} \frac{\theta_j(S_{i+1}; T\cap S_{i+1})}{\theta_j(S_i; T\cap S_i)} = \frac{\theta_j(\mathcal{E};T)}{\theta_j(\varnothing;\varnothing)} = \frac{\theta_{j,T}}{\theta_{j,\max}}.$$

So $\Pr(A_{d+1}) = \sum_T r_j!\,u^T\theta_{j,T}/\theta_{j,\max} = r_j!\,K_j(u)/\theta_{j,\max}$. The cobasis rounds are the basis rounds of the dual table, whose bases $\mathcal{E}\setminus T$ carry the weights $\theta_{j,T}$, whose largest weight is $\theta_{j,\max}$ and whose basis polynomial is $\Psi_j$ (Proposition 7.2); so $\sum_i \mathfrak{f}^*_{j,i} = 1 - (n-r_j)!\,\Psi_j/\theta_{j,\max}$.

*Polynomials.* Each $\mathfrak{f}_{j,i}(u)$ is a finite sum, over the sequences of $r_j$ draws, of the monomial of the sequence times a product of acceptance and rejection probabilities in $[0,1]$: a polynomial with nonnegative coefficients, with integer coefficients for lists, whose rounds are always accepted.

*The series.* Proposition 20.12, with one block and the constants $C_{2j-1} = \theta_{j,\max}/r_j!$, $C_{2j} = \theta_{j,\max}/(n-r_j)!$ of its last part, gives

$$\mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}\Big[\prod_j K_j^{-s_j}\Psi_j^{-t_j}\Big] = \prod_j C_{2j-1}^{-s_j}C_{2j}^{-t_j}\sum_{\mathbf{k}\in\mathbb{N}^{2N}}\prod_j \frac{(s_j)_{k_{2j-1}}(t_j)_{k_{2j}}}{k_{2j-1}!\,k_{2j}!} \times \mathbb{E}_{U\sim\mathrm{Dir}_{\mathbf{m}}}\Big[\prod_j\Big(\sum_i \mathfrak{f}_{j,i}\Big)^{k_{2j-1}}\Big(\sum_i \mathfrak{f}^*_{j,i}\Big)^{k_{2j}}\Big],$$

both sides finite or infinite together. The multinomial theorem expands

$$\frac{(s_j)_k}{k!}\Big(\sum_i \mathfrak{f}_{j,i}\Big)^k = \sum_{|\mathbf{k}_j|=k}\frac{(s_j)_{|\mathbf{k}_j|}}{\mathbf{k}_j!}\prod_i \mathfrak{f}_{j,i}^{k_{j,i}},$$

with nonnegative terms, and likewise for the cobasis rounds; Tonelli's theorem exchanges the sums and the expectation, and multiplying by $\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ (Theorem 3.9) gives the display. Each expectation is that of a polynomial with nonnegative coefficients, a finite nonnegative combination of Dirichlet moments (Theorem 3.11).

*Nesting.* $\sum_{i'\ge i} k_{j,i'}$ is a tail sum of nonnegative integers, so it decreases with $i$.

*Bijections.* Let $g$ keep each $B_i$, hence each $S_i$, and the masses. A sequence of draws $D$ from $u$ has the chance of the sequence $g(D)$ from $g_*u$. If $g$ maps the bases of table $j$ onto those of table $h(j)$ with their weights, then $g(T\cap S_i) = g(T)\cap S_i$, so $D$ extends on $S_i$ for table $j$ exactly when $g(D)$ extends on $S_i$ for table $h(j)$, the traces corresponding under $g$, and $\theta_j(S_i;P) = \theta_{h(j)}(S_i;g(P))$, so the acceptances agree: $\mathfrak{f}_{h(j),i}(g_*u) = \mathfrak{f}_{j,i}(u)$, and likewise for the cobasis rounds, $g$ carrying the dual tables as well. If $T\mapsto \mathcal{E}\setminus g(T)$ maps the bases of table $j$ onto those of table $h(j)$ with their weights, then $g(T)$ is a cobasis of table $h(j)$ for every basis $T$ of table $j$, the round lengths agree, $r_j = n - r_{h(j)}$, the largest weights over the traces agree in the same way, and $\mathfrak{f}^*_{h(j),i}(g_*u) = \mathfrak{f}_{j,i}(u)$, $\mathfrak{f}_{h(j),i}(g_*u) = \mathfrak{f}^*_{j,i}(u)$. Since the masses are kept, $g_*U\sim\mathrm{Dir}_{\mathbf{m}}$ when $U\sim\mathrm{Dir}_{\mathbf{m}}$, so in the second case

$$\mathbb{E}\Big[\prod_{j,i}\mathfrak{f}_{j,i}(U)^{k_{j,i}}\mathfrak{f}^*_{j,i}(U)^{l_{j,i}}\Big] = \mathbb{E}\Big[\prod_{j,i}\mathfrak{f}^*_{h(j),i}(U)^{k_{j,i}}\mathfrak{f}_{h(j),i}(U)^{l_{j,i}}\Big],$$

the expectation at the indices $(\mathbf{k}'_{j'},\mathbf{l}'_{j'}) = (\mathbf{l}_{h^{-1}(j')},\mathbf{k}_{h^{-1}(j')})$. With one table $h$ is the identity, an exchange maps the bases onto complements of bases, so $r = n-r$, and the array is symmetric under $\mathbf{k}\leftrightarrow\mathbf{l}$. □

**Theorem 20.15 (The draw series along a tree).** *As stated, for every molecule whose children's weights factor across the childcules.*

*Proof.* The proof is that of the reference (proof of thm:drawtree), the weights entering through the acceptances; the coordinates of the tree and their independence (Theorem 3.12 at every node) are unchanged.

(i) Let $\mathcal{B}'$ have rank $r'$ at the node $v$, and write $R_i = R_{v,i}$, $x = x^{(v)}$, $u = u^{(v)}$ and $v^{(i)}$ for the shape of $R_i$. By Proposition 7.9 the bases of $\mathcal{B}'$ are the sets $T = p\sqcup\bigsqcup_i X_i$, $p$ a pattern at $B_v$ and $X_i$ a basis of $\mathcal{B}'_{(p)}|R_i$ of size $r'_{p,i}$, and by the hypothesis $\theta_T = \prod_i \theta_{(p),i}(X_i)$. The shape of $M_v$ is $x_{B_v}u$ on $B_v$ and $x_{R_i}v^{(i)}$ on $R_i$, so by homogeneity

$$K_{\mathcal{B}'} = \sum_p x_{B_v}^{|p|}u^p\prod_i x_{R_i}^{r'_{p,i}}K_{\mathcal{B}'_{(p)}|R_i}\big(v^{(i)}\big),$$

the restrictions carrying the weights $\theta_{(p),i}$. Multiply by $r'!/\theta_{\max}(\mathcal{B}')$, write $r'! = \frac{r'!}{\prod_i r'_{p,i}!}\prod_i r'_{p,i}!$, and multiply and divide the term of $p$ by

$$\prod_i \theta_{\max}\big(\mathcal{B}'_{(p)}|R_i\big);$$

this gives $\mathfrak{s}_{\mathcal{B}'} = \sum_p \mathfrak{p}_{\mathcal{B}',p}\prod_i \mathfrak{s}_{\mathcal{B}'_{(p)}|R_i}$. The hypothesis fixes the weights of the restrictions up to constant factors whose product is 1, which change neither the $\mathfrak{s}$ of the restrictions nor the first factor of $\mathfrak{p}_{\mathcal{B}',p}$. That factor is at most 1, since

$$\theta_{\max}(\mathcal{B}') = \max_p \max_{(X_i)_i}\prod_i \theta_{(p),i}(X_i) = \max_p \prod_i \theta_{\max}\big(\mathcal{B}'_{(p)}|R_i\big).$$

The chance that $r'$ independent draws from the shape of $M_v$ are the basis $T$ with distinct draws is $r'!$ times its monomial, and the acceptance with probability $\theta_T/\theta_{\max}(\mathcal{B}')$ gives the chance $r'!K_{\mathcal{B}'}/\theta_{\max}(\mathcal{B}')$ of an accepted basis. The round passes $v$ showing $p$, its draws in $B_v$ being the marks of $p$ and its numbers of draws in the $R_i$ the $r'_{p,i}$, with the chance $\frac{r'!}{\prod_i r'_{p,i}!}x_{B_v}^{|p|}u^p\prod_i x_{R_i}^{r'_{p,i}}$, and given this its draws in $R_i$ are $r'_{p,i}$ independent draws from $v^{(i)}$. The acceptance splits as

$$\frac{\theta_T}{\theta_{\max}(\mathcal{B}')} = \frac{\prod_i \theta_{\max}\big(\mathcal{B}'_{(p)}|R_i\big)}{\theta_{\max}(\mathcal{B}')}\ \prod_i \frac{\theta_{(p),i}(X_i)}{\theta_{\max}\big(\mathcal{B}'_{(p)}|R_i\big)},$$

an acceptance at $v$ followed by independent acceptances in the childcules. So $\mathfrak{p}_{\mathcal{B}',p}$ is the chance of passing $v$ showing $p$ and being accepted there, and $\mathfrak{s}_{\mathcal{B}'_{(p)}|R_i}$ the chance of an accepted basis of the restriction in $R_i$.

(ii) By (i) the round fails at $v$ with the chance $1-\sum_p \mathfrak{p}_{\mathcal{B}',p}$, and otherwise passes $v$ showing one pattern $p$ and is accepted there; it then first fails at $w$ in the subtree of $R_i$ exactly when it succeeds in $R_1,\dots,R_{i-1}$ and first fails at $w$ inside $R_i$, which is the recursion. Write $\mathfrak{p}_{\mathcal{B}',p} = a_p\,\mathfrak{p}^\circ_{\mathcal{B}',p}$ with $a_p \le 1$ the first factor. Then

$$1-\sum_p \mathfrak{p}_{\mathcal{B}',p} = \Big(1-\sum_p \mathfrak{p}^\circ_{\mathcal{B}',p}\Big) + \sum_p (1-a_p)\,\mathfrak{p}^\circ_{\mathcal{B}',p}.$$

The bracket is $1 = (x_{B_v}\sum_{e\in B_v}u_e + \sum_i x_{R_i})^{r'}$, expanded by the multinomial theorem, less some of its terms, distinct patterns giving distinct terms (the reference, proof of thm:drawtree, part (ii)); it has nonnegative coefficients in the coordinates of $v$, and so has the sum. With $1-\prod_i S_i = \sum_i \prod_{i'<i} S_{i'}(1-S_i)$ and induction on the subtree, every $\mathfrak{f}_{\mathcal{B}'\to w}$ is a finite sum of products, over the blocks of the tree, of polynomials with nonnegative coefficients in one block each.

(iii) On the simplex $r_j!K_j/\theta_{j,\max} = \mathfrak{s}_{\mathcal{B}_j} = 1-\sum_w \mathfrak{f}_{j,w}$, with the sum in $[0,1)$ on the interior. The letter $\Psi_j$ is the basis polynomial of the dual table (Proposition 7.2), whose bases $\mathcal{E}\setminus T$ carry the weights $\theta_{j,T}$, so that its largest weight is $\theta_{j,\max}$; its children are the duals of the children with the same weights (Lemma 14.7(ii) with the identity), and the dual of a direct sum is the direct sum of the duals of its parts with the product weights, so the factoring passes to the dual and (i), (ii) apply to its rounds. Then $K_j^{-s_j} = (r_j!/\theta_{j,\max})^{s_j}(1-\sum_w \mathfrak{f}_{j,w})^{-s_j}$, expanded as in Proposition 20.12 and by the multinomial theorem as in Theorem 20.14, with nonnegative terms; Tonelli's theorem, the independence of the blocks under $\mathrm{Dir}_{\mathbf{m}}$ (Theorem 3.12) and Theorem 3.11 give the series and the form of its expectations. On the balance wall the left side is the period (Theorem 8.7).

(iv) A round that first fails in the subtree of $w$ passes every node above $w$, which gives the nesting. On a chain every node has one childcule. The table at the node $M_k$ that a round of table $j$ reaches is the child of table $j$ at its trace $P_k$ on $S_k$, whose largest weight is $\theta_j(S_k;P_k)$ in the notation of Theorem 20.14; so the first factor of $\mathfrak{p}$ at that node is $\theta_j(S_{k+1};P_{k+1})/\theta_j(S_k;P_k)$, the acceptance at $S_{k+1}$, and passing the node is extending on $S_{k+1}$. The chances $\mathfrak{f}$ of the two theorems agree, and the series is that of Theorem 20.14. □

**Proposition 20.17 (Rates at the centre).** *As stated, for every family whose active tables are connected.*

*Proof.* (i) is Proposition 8.21(ii), with $c_j(\mathcal{E}\setminus S)=c_j(S)$ (Theorem 7.13).

(ii) The rate is $\varpi=\min_i\varpi_i$ over the letters, $\varpi_i$ the least of $\kappa_S/d_i(S)$ over the nonempty proper $S$ with $d_i(S)>0$ (Theorem 15.13(ii)); an inactive table has $\sigma_j=0$ and no letters. A connected table has $c_j(S)\geq 1$ on every nonempty proper $S$ (Definition 8.1), and so no loops and no coloops, a loop or a coloop $e$ having $c_j(\{e\})=r_j(\{e\})-d_j(\{e\})=0$ (Theorem 7.13) with $\{e\}$ proper as $n\geq 3$. So $\kappa_S=\sum_j\sigma_j c_j(S)\geq\sum_j\sigma_j=\sigma$.

For the letter $K_j$, $d_j(S)=|S|-r_j^*(S)=r_j-r_j(\mathcal{E}\setminus S)\leq r_j-1$ (Definition 7.12), as $\mathcal{E}\setminus S$ is nonempty and contains no loop. For $\Psi_j$, $d_j^*(S)=|S|-r_j(S)$ is the nullity of $S$, which increases with the set, and $S\subseteq\mathcal{E}-e$ for some $e$, of nullity $n-1-r_j(\mathcal{E}-e)=n-1-r_j$, as $e$ is not a coloop. So $\kappa_S/d_j(S)\geq\sigma/(r_j-1)$ wherever $d_j(S)>0$, and $\kappa_S/d_j^*(S)\geq\sigma/(n-r_j-1)$ wherever $d_j^*(S)>0$.

At $S=\mathcal{E}-e$, every table without loops or coloops has $c_j(\mathcal{E}-e)=c_j(\{e\})=r_j(\{e\})-d_j(\{e\})=1$ (Theorem 7.13), so $\kappa_{\mathcal{E}-e}=\sigma$; and $d_j(\mathcal{E}-e)=r_j-r_j(\{e\})=r_j-1$, $d_j^*(\mathcal{E}-e)=n-1-r_j$. Hence $\varpi_{K_j}=\sigma/(r_j-1)$ when $r_j\geq 2$ and $\varpi_{\Psi_j}=\sigma/(n-r_j-1)$ when $n-r_j\geq 2$, attained at the faces $\mathcal{E}-e$; a letter with no $S$ of positive count does not enter the minimum. As $(r_j-1)+(n-r_j-1)=n-2\geq 1$, at least one letter of every active table enters, and $\varpi=\sigma/\max_j\max(r_j-1,n-r_j-1)$.

(iii) By Definition 7.3, $K_j=K_{j,\setminus e}+u_eK_{j,/e}$ and $\Psi_j=u_e\Psi_{j,\setminus e}+\Psi_{j,/e}$, the deletion and the contraction being tables since $e$ is neither a loop nor a coloop of the active table $j$. They are homogeneous in $u_{\mathcal{E}-e}$ of the degrees $r_j$, $r_j-1$, $n-1-r_j$ and $n-r_j$; substituting $u_e=1-x$ and $u_{\mathcal{E}-e}=xv$ gives the display, and as $x\to 0$ the terms of $x^{r_j-1}K_{j,/e}(v)$ and $x^{n-r_j-1}\Psi_{j,\setminus e}(v)$ dominate.

(iv) Let $\beta$ carry the family, with $v'_{\beta(e')}=v_{e'}$ on $\mathcal{E}-e$. It maps the bases of table $j$ onto those of table $h(j)$ with their weights, the bases through $e$ onto those through $f$, so

$$K_{h(j),/f}(v')=\sum_{T\ni e}\theta_{j,T}\,{v'}^{\beta(T)-f}=\sum_{T\ni e}\theta_{j,T}\,v^{T-e}=K_{j,/e}(v),$$

and in the same way $\Psi_{h(j),\setminus f}(v')=\Psi_{j,\setminus e}(v)$, the bases avoiding $e$ going to those avoiding $f$. Let $\beta$ exchange the family: $T\mapsto\mathcal{E}\setminus\beta(T)$ maps the bases of table $j$ through $e$ onto the bases of table $h(j)$ avoiding $f$, with their weights, so

$$\Psi_{h(j),\setminus f}(v')=\sum_{T\ni e}\theta_{j,T}\,{v'}^{(\mathcal{E}-f)\setminus(\mathcal{E}\setminus\beta(T))}=\sum_{T\ni e}\theta_{j,T}\,{v'}^{\beta(T)-f}=K_{j,/e}(v),$$

and in the same way $K_{h(j),/f}(v')=\Psi_{j,\setminus e}(v)$. Either way the vertex product $K_{j,/e}\Psi_{j,\setminus e}$ goes to $K_{h(j),/f}\Psi_{h(j),\setminus f}$. With one table, a bijection taking $e$ to $f$ that carries or exchanges it exists exactly when $e$ and $f$ lie in one class at the centre, by Corollary 11.16 at the table's gauge, the gauge of the centre. □

*Remark.* Connectedness of every active table is needed for (ii). On the marks $\{0,1,2,3\}$, for the family of the list whose bases are all the two-element sets and the disconnected list with the bases $\{a,b\}$, $a\in\{0,2\}$, $b\in\{1,3\}$, at the strengths $(1,3)$, the face $S=\{0,2\}$ has $\kappa_S=1\cdot 2+3\cdot 0=2$ and $d_2(S)=1$, so $\varpi\le 2$, while $\sigma/\max_j\max(r_j-1,n-r_j-1)=4$.

**Corollary 20.19 (Merged Skeleton values).** *As stated, at every pair of exponents and every mass on the balance wall.*

*Proof. The letters.* With $\theta_{\mathcal{E}\setminus T}=\theta_T$, $\Psi(\alpha)=\sum_T\theta_T\alpha^{\mathcal{E}\setminus T}=\sum_T\theta_{\mathcal{E}\setminus T}\alpha^{\mathcal{E}\setminus T}=K(\alpha)$, the last sum running over the same bases. For a mark $e$ the bases $T\not\ni e$ correspond to the bases $T'=\mathcal{E}\setminus T\ni e$, with $(\mathcal{E}-e)\setminus T=T'-e$ and $\theta_T=\theta_{T'}$, so

$$\Psi_{\setminus e}=\sum_{T\not\ni e}\theta_T\,\alpha^{(\mathcal{E}-e)\setminus T}=\sum_{T'\ni e}\theta_{T'}\,\alpha^{T'-e}=K_{/e}$$

(Definition 7.3), and likewise $\Psi_{/e}=K_{\setminus e}$. The Skeleton values of the scale of $e$ are $K_{\setminus e}/(\alpha_e K_{/e})$ and $\Psi_{/e}/(\alpha_e\Psi_{\setminus e})$ (Corollary 15.4), which coincide.

*The table.* Bases and complements of bases are both bases, so $r=n-r$. A coloop lies in every basis, hence in no complement of one, and a loop lies in no basis, hence in every complement; both contradict the complementation, so no mark is a loop or a coloop, and the deletion $(\mathcal{B}\setminus e,\theta)$ is a table of rank $r$ on $\mathcal{E}-e$.

*The walls at $\{e\}$ and at $\mathcal{E}-e$.* $d(\{e\})=d^*(\{e\})=0$, so $\kappa_{\{e\}}=m_e$. Also $d(\mathcal{E}-e)=r-r(\{e\})=r-1$ and $d^*(\mathcal{E}-e)=n-1-r(\mathcal{E}-e)=n-1-r=r-1$. On the balance wall $m_{\mathcal{E}}=sr+t(n-r)=(s+t)r$, so $\kappa_{\mathcal{E}-e}=m_{\mathcal{E}}-m_e-(s+t)(r-1)=s+t-m_e$. The proper walls being positive, $0<m_e<s+t$.

*Integrating the mark.* Let $g\neq e$ and $R=\mathcal{E}-e$. By Theorem 8.7 the period is the integral of $\alpha^{\mathbf{m}-\mathbf{1}}K^{-s}\Psi^{-t}=\alpha^{\mathbf{m}-\mathbf{1}}K^{-(s+t)}$ over the section $\alpha_g=1$. By Tonelli's theorem the integral over $\alpha_e=x$ may be taken first, at fixed $\alpha_R$: with $K=K_{\setminus e}+xK_{/e}$, the Beta corner of a linear letter (Proposition 15.1 with one coefficient) gives

$$\int_0^\infty x^{m_e-1}\big(K_{\setminus e}+xK_{/e}\big)^{-(s+t)}dx=\Gamma{\scriptstyle\begin{bmatrix}m_e,\ s+t-m_e\\ s+t\end{bmatrix}}_{2}^{1}\,K_{\setminus e}^{-(s+t-m_e)}K_{/e}^{-m_e},$$

convergent as $0<m_e<s+t$, and $K_{/e}=\Psi_{\setminus e}$.

*The deletion.* What remains is $\Gamma{\scriptstyle\begin{bmatrix}m_e,\ s+t-m_e\\ s+t\end{bmatrix}}_{2}^{1}$ times the integral, over the section $\alpha_g=1$ of the orthant of $R$, of

$$\alpha_R^{\mathbf{m}|_R-\mathbf{1}}K_{\setminus e}^{-(s+t-m_e)}\Psi_{\setminus e}^{-m_e}\,d\alpha_{R-g},$$

the family word of the deletion at the exponents $(s+t-m_e,m_e)$. Its balance wall holds: $m_{\mathcal{E}-e}=(s+t)r-m_e$, which is $(s+t-m_e)r+m_e(n-1-r)$ as $n-1-r=r-1$. By Theorem 8.7 the section integral is $P_{\mathcal{B}\setminus e}(\mathbf{m}|_{\mathcal{E}-e};s+t-m_e,m_e)$; it is finite because $P_F$ is, so the walls of the deletion are positive there. Since $K=\Psi$, $P_F(\mathbf{m};s,t)$ depends on the exponents only through $s+t$. At $s=t=\sigma$ and $\mathbf{m}=\sigma\mathbf{1}$, $m_e=\sigma$, which is the last identity. □

**Theorem 20.20 (The scale integral of one mark).** *As stated; parts (i) and (ii) are stated for every family.*

*Proof.* (i) By Theorem 15.3 with $B=\{e\}$ and $m_B=m_e$, at the gauge $(\alpha_e,\alpha_R)$,

$$\mathrm{Sc}_{\{e\}}(\alpha;\mathbf{m};\mathbf{s},\mathbf{t})=\int_0^\infty x^{m_e-1}\prod_j K_j(x\alpha_e,\alpha_R)^{-s_j}\Psi_j(x\alpha_e,\alpha_R)^{-t_j}\,dx,$$

convergent since the walls $\kappa_{\{e\}}$ and $\kappa^{\mathrm{top}}_{\{e\}}$ are positive. At $\alpha_e = 1$, Definition 7.3 gives $K_j = K_{j,\backslash e}(\alpha_R) + xK_{j,/e}(\alpha_R)$ and $\Psi_j = x\Psi_{j,\backslash e}(\alpha_R) + \Psi_{j,/e}(\alpha_R)$ for every active table, the deletion and the contraction being tables on $R$ as $e$ is neither a loop nor a coloop there. At $\alpha_R = e^{2z_R}$, Theorem 7.15(i) for these tables gives $K_{j,\star}(\alpha_R) = e^{\langle \mathbf{1}, z_R\rangle}\Lambda_{j,\star}(z_R)$ and $\Psi_{j,\star}(\alpha_R) = e^{\langle \mathbf{1}, z_R\rangle}\Lambda_{j,\star}(-z_R)$ for $\star \in \{\backslash e, /e\}$. So

$$K_j = e^{\langle \mathbf{1}, z_R\rangle}\big(\Lambda_{j,\backslash e}(z_R) + x\Lambda_{j,/e}(z_R)\big),$$
$$\Psi_j = e^{\langle \mathbf{1}, z_R\rangle}\big(\Lambda_{j,/e}(-z_R) + x\Lambda_{j,\backslash e}(-z_R)\big),$$

and the powers carry the factor $e^{-\sum_j (s_j+t_j)\langle \mathbf{1}, z_R\rangle} = e^{-2\sigma\langle \mathbf{1}, z_R\rangle}$, an inactive table contributing 1. This is the display, and the rest of the integrand depends on the room only through the functions $\Lambda_{j,\backslash e}$ and $\Lambda_{j,/e}$. Corollary 16.6 at the bunch $\{e\}$, whose shape simplex is the point $u_e = 1$, gives $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int \alpha_R^{\mathbf{m}|_R - \mathbf{1}} \operatorname{Sc}_{\{e\}}((1, \alpha_R); \mathbf{m}; \mathbf{s}, \mathbf{t})\, d\alpha_{R-g}$ on the section $\alpha_g = 1$, $g \in R$. With $\alpha_R = e^{2z_R}$, $\alpha_R^{\mathbf{m}|_R - \mathbf{1}} d\alpha_{R-g} = 2^{n-2} e^{2\langle \mathbf{m}|_R, z_R\rangle} dz_{R-g}$ there, $R - g$ having $n - 2$ marks.

(ii) A realization that is an atom is a realization of a molecule (Definition 10.1), so $F$ is connected, and Theorem 11.4 gives $\alpha^\star \propto \mathbf{1}$. By Theorem 18.1(ii), in $y = \log u$ modulo $\mathbb{R}\mathbf{1}$ the law $\mathrm{Breta}^\circ(F; \boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$ has a density proportional to $e^{\langle \boldsymbol{\tau}, y\rangle - \Phi(y)}$. The target is the usage at $\alpha^\star$ (Definition 8.17),

$$\tau_e = \sum_j \big(s_j R_{j,e}(\alpha^\star) + t_j R^*_{j,e}(\alpha^\star)\big) = \partial_{y_e}\Phi(y^\star), \qquad y^\star = \log \alpha^\star,$$

with the potential $\Phi$ of Definition 8.18. With $y = y^\star + v$ the density is proportional to

$$\exp\big(\langle \nabla\Phi(y^\star), v\rangle - \Phi(y^\star + v)\big).$$

By Proposition 10.4 there are orders $\mathbf{g}_e$ of the rooms in which the potential written from $e$ at the gauge $\alpha^\star$ (Definition 10.1),

$$(v_0, v_1, \ldots, v_{n-1}) \longmapsto \Phi_{\mathbf{X}}\Big(y^\star + v_0\mathbf{1}_e + \sum_k v_k \mathbf{1}_{g_k}\Big) \qquad (\mathbf{X} \in \{\mathbf{E}, \mathbf{O}\}),$$

is one pair of functions for every $e$. Hence $v \mapsto \Phi(y^\star + v)$ read in the coordinates $(v_e, v_{g_1}, \ldots, v_{g_{n-1}})$ is one function for every $e$, and so is its gradient at 0; the density read in these coordinates is one function for every $e$, and $dv$ is invariant under the renaming. The density is invariant along $\mathbf{1}$, so it is a function of the differences $v_{g_k} - v_e$, and the law of these differences does not depend on $e$. As $y^\star$ is a multiple of $\mathbf{1}$, $y_{g_k} - y_e = v_{g_k} - v_e$.

(iii) The reference (proof of thm:atomforward) proves this part, a statement for one table. □

**Theorem 20.23 (The junction under level form).** *As stated, for every family at the centre.*

*Proof.* Parts (ii) and (iii) concern the Gramma $\mathsf{G}_{0;(\chi_p)_p}((v_{|p|})_p)$ alone and are proved in the reference (proof of thm:uniformjunction, parts (ii) and (iii)).

*The preamble.* By Proposition 20.6(iii) for every table, every half-sum has level form and the square of every difference is unchanged by the permutations of $B$; $q_{j,b-k}(-z_R) = q_{j,k}(z_R)$ is Proposition 20.6(i). On $z_B$ each $(\partial_w L_{j,w})^2$ has exponents in $\{-2, 0, 2\}^B$, being a sum of products of two terms $e^{\langle z_B, \chi_p\rangle}$. A function of $z_B$ of this kind that is unchanged by the permutations of $B$ is a combination of the products $e^{-2\langle \mathbf{1}, z_B\rangle} e_k(e^{2z_B}) e_{k'}(e^{2z_B})$, $k \le k'$: such a product is the symmetric sum of the monomials with $k$ exponents 2 and $b - k'$ exponents $-2$, plus terms with fewer of

both, and induction removes the terms (the reference, proof of thm:uniformjunction, part (i)). The coefficients are Laurent polynomials in $e^{z_R}$, the $\Lambda_{j,(p)}$ being such. The product over $j$ of $(\partial_w L_{j,w})^{2l_j}$ therefore expands in products of $2|\mathbf{l}|$ of the $e_k$, with Laurent polynomial coefficients.

(i) At the centre $\boldsymbol{\mu} = 0$, and the first step of the proof of Theorem 20.4 gives

$$P_F(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma}) = 2^{n-1}\int_{\mathbb{R}^{\mathcal{E}-g}} \prod_j \big(\Lambda_j(z)\Lambda_j(-z)\big)^{-\sigma_j}\, dz_{\mathcal{E}-g}\Big|_{z_g=0}.$$

For each table $\Lambda_j(z)\Lambda_j(-z) = L_{j,1/2}(z)^2 - \frac14(\partial_w L_{j,w}(z))^2$, and Proposition 20.1 with $a = \Lambda_j(z)$, $b = \Lambda_j(-z)$, $s = t = \sigma_j$, the midpoint $L_{j,1/2}(z)$ and $x = \partial_w L_{j,w}(z)/(2L_{j,1/2}(z)) \in (-1,1)$ gives

$$\big(\Lambda_j(z)\Lambda_j(-z)\big)^{-\sigma_j} = \sum_{l_j\ge 0} \frac{(\sigma_j)_{l_j}}{l_j!\,4^{l_j}} \big(\partial_w L_{j,w}(z)\big)^{2l_j} L_{j,1/2}(z)^{-2\sigma_j-2l_j},$$

a series of nonnegative terms. Open each power of the half-sum,

$$L_{j,1/2}^{-2\sigma_j-2l_j} = \Gamma(2\sigma_j+2l_j)^{-1}\int_0^\infty y_j^{2\sigma_j+2l_j-1} e^{-y_j L_{j,1/2}}\, dy_j;$$

by Tonelli's theorem the sums and integrals may be taken in the order of the statement. With $\upsilon_k = \sum_j y_j q_{j,k}(z_R)$,

$$\sum_j y_j L_{j,1/2}(z) = \sum_k \upsilon_k\, e^{-\langle \mathbf{1}, z_B\rangle} e_k\big(e^{2z_B}\big),$$

and inserting $e^{-\langle \mathbf{1}, z_B\rangle} e_k(e^{2z_B})$ under the exponential of minus this sum is $-\partial_{\upsilon_k}$. With the expansion of the preamble and Definition 2.1 (index 0, the integral over $z_B$),

$$\int_{\mathbb{R}^B} \prod_j \big(\partial_w L_{j,w}(z)\big)^{2l_j} e^{-\sum_j y_j L_{j,1/2}(z)}\, dz_B$$
$$= \sum_{|\mathbf{h}|=2|\mathbf{l}|} c_{\mathbf{h}}^{(\mathbf{l})}(z_R)\Big((-\partial_{\boldsymbol{\upsilon}})^{\mathbf{h}}\, \mathsf{G}_{0;(\chi_p)_p}\big((\upsilon_{|p|})_p\big)\Big)\Big|_{\boldsymbol{\upsilon}=\sum_j y_j \mathbf{q}_j(z_R)}.$$

Each term converges. The family being a molecule, some active table $j$ has $c_j(B) \ge 1$ (Definition 8.1), so its patterns at $B$ fill at least two consecutive levels $k, k+1$ (Theorem 12.7), and $q_{j,k}, q_{j,k+1} > 0$, each being a sum of nonnegative terms one of which is positive; by $q_{j,b-k}(-z_R) = q_{j,k}(z_R)$ also $q_{j,b-k}, q_{j,b-k-1} > 0$. The vectors $\chi_p$ of these four levels are closed under negation, $\chi_{B\setminus p} = -\chi_p$, and span $\mathbb{R}^B$, since $\chi_{p\cup\{e\}} - \chi_p = 2\mathbf{e}_e$ for $|p| = k$, $e \notin p$; so they positively span $\mathbb{R}^B$, the exponential of $-\sum_j y_j L_{j,1/2}(z)$ decays in $z_B$ faster than every exponential, and the inserted exponentials are integrable against it.

(iv) For each table the swap $\tau$ of $e$ and $f$ keeps $B$, fixes $R$ and exchanges table $j$, the bijection of the tables being the identity. Proposition 20.5 gives $c_{j,\tau(p)}(1-w;z_R) = c_{j,p}(w;z_R)$. At $w = \frac12$, $c_{j,\{f\}}(\frac12;\cdot) = c_{j,\{e\}}(\frac12;\cdot)$; in the derivative, $\partial_w c_{j,\varnothing} = \partial_w c_{j,B} = 0$ and $\partial_w c_{j,\{f\}} = -\partial_w c_{j,\{e\}}$. With $c_{j,B}(\frac12;z_R) = c_{j,\varnothing}(\frac12;-z_R)$ (Proposition 20.6(i)) this gives the two displayed forms. Evenness: $c_{j,\{e\}}(\frac12;-z_R) = c_{j,\{f\}}(\frac12;z_R) = c_{j,\{e\}}(\frac12;z_R)$ and $\partial_w c_{j,\{e\}}(w;-z_R) = -\partial_w c_{j,\{f\}}(w;z_R) = \partial_w c_{j,\{e\}}(w;z_R)$, by Proposition 20.6(i). In (i) change $(z_e, z_f)$ to $(z_+, z_-)$, with $dz_e\, dz_f = \frac12 dz_+ dz_-$. Then $\prod_j(\partial_w L_{j,w})^{2l_j} = (e^{z_-} - e^{-z_-})^{2|\mathbf{l}|}\prod_j(\partial_w c_{j,\{e\}})^{2l_j}$ and

$$\sum_j y_j L_{j,1/2}(z) = e^{-z_+}C_\varnothing(z_R) + e^{z_+}C_\varnothing(-z_R) + \big(e^{z_-} + e^{-z_-}\big)C_e(z_R).$$

The integral over $z_+$ is $\mathsf{G}_{0;(1,-1)}\big(C_\varnothing(-z_R), C_\varnothing(z_R)\big)$ (Definition 2.1 on one scale $e^{z_+}$), and the integral over $z_-$ is Lemma 20.22 with $\xi = C_e(z_R)$ and the power $|\mathbf{l}|$, $4^{|\mathbf{l}|}(\frac{1}{2})_{|\mathbf{l}|}C_e^{-|\mathbf{l}|}\mathsf{G}_{|\mathbf{l}|;(1,-1)}(C_e, C_e)$. The constants collect to $2^{n-1}\cdot\frac{1}{2}\cdot 4^{|\mathbf{l}|}\prod_j 4^{-l_j} = 2^{n-2}$, which is the display; every term is nonnegative. □

*Remark.* With one table, $(\frac{1}{2})_l(\sigma)_l/(l!\,\Gamma(2\sigma+2l)) = \frac{1}{l!}\Gamma\,{}_2^{}{}_3\!\left[\begin{matrix}\sigma+l,\ \frac{1}{2}+l\\ 2\sigma+2l,\ \sigma,\ \frac{1}{2}\end{matrix}\right]$ and $y^{2\sigma+2l-1}(yc_{\{e\}})^{-l} = y^{2\sigma+l-1}c_{\{e\}}^{-l}$, which is the reference's form of (iv).

**Theorem 20.40 (A series connection closes at its reach).** *As stated, for series connections and for series connections of the dual table.*

*Proof.* (i) By Proposition 20.37(i), $f_B = C^{-s}C'^{-t}\prod_i f_i^\circ$, and by Proposition 20.37(ii), which is Lemma 20.30 applied $q-1$ times, $f_B$ has a lattice expansion of reach $(\sum_i\sqrt{x_i})^2$ whose terms are the products of the terms of the parts collected by exponent. A function has at most one lattice expansion (Definition 20.28), so these are the coefficients $\beta_{\eta,j}$ of Theorem 20.34, and $\operatorname{reach}(B) \le (\sum_i\sqrt{x_i})^2 < \inf\varrho_R$, the reach being an infimum. Theorem 20.34(ii) applies: the even share closes toward the level $d$ on the lattice of continued periods of the bunch family, with the bound for every $x > \operatorname{reach}(B)$, in particular for every $x > (\sum_i\sqrt{x_i})^2$. Under the hypotheses of Proposition 20.37(iii), $\operatorname{reach}(B) = (\sum_i\sqrt{\sup\varrho_i})^2$, and Theorem 20.39(ii) gives the root rate $\operatorname{reach}(B)/\inf\varrho_R$. For a series bunch the parts are the marks (Proposition 20.37), with

$$\begin{aligned} f_e^\circ(y) &= \int_0^\infty x^{m_e-s-1}e^{-w_e/x-yw_ex}\,dx = \mathsf{G}_{m_e-s;(1,-1)}(yw_e, w_e)\\ &= (yw_e)^{s-m_e}\mathsf{G}_{m_e-s;(1,-1)}\big(1, w_e^2y\big)\end{aligned}$$

by the torus homogeneity of the Gramma; by Lemma 20.29(iii) it has a lattice expansion of reach $w_e^2$, a power of $y$ keeping the reach (Lemma 20.30). Proposition 20.36(iii) gives $\operatorname{reach}(B) = \inf\varrho_B = (\sum_e w_e)^2$ exactly, and Proposition 20.36(i) gives the coefficients.

(ii) The reference (proof of thm:classcloses, part (ii)) gives the argument for a parallel bunch, and it applies to a connection. The dual table at the exponents $(t, s)$ has the same period (Proposition 8.15). Since $K^* = \Psi$ and $\Psi^* = K$ (Proposition 7.2), the dual node has the bunch $B$, of width one in the dual table, the room $R$, the ratios $1/\varrho_B$ and $1/\varrho_R$, and its levels swapped: its level $d$ carries the even letter of the level $r$ of the given node. No slope on the room being negative, $\sup\varrho_R < \infty$ (Lemma 19.15), so $\inf(1/\varrho_R) > 0$ and no slope on the room of the dual node is positive (the same lemma for the dual node). Part (i) for the dual node, whose bunch is a series connection with the reaches $x_i^*$, closes the even share toward its level $d$, the level $r$ of the given node, when $(\sum_i\sqrt{x_i^*})^2 < 1/\sup\varrho_R$, at a root rate at most $(\sum_i\sqrt{x_i^*})^2\sup\varrho_R$. For a parallel bunch, the dual pattern weights are $\theta'^*_B = \theta'_\varnothing$ and $\theta'^*_{B-e} = \theta'_{\{e\}}$, so the $w_e$ of the dual series bunch are $\theta'_{\{e\}}/\theta'_\varnothing$, $x_e^* = w_e^2$, and the root rate is exact by (i). The ratio of the bunch of the dual node is $1/\varrho_B = (\sum_e w_e/u_e)(\sum_e w_eu_e)$, whose least value $(\sum_e w_e)^2$ is taken at the point of equal coordinates of $\Delta_B$ (Proposition 20.36(ii)); so $\sup\varrho_B = (\sum_e w_e)^{-2}$, the condition reads $\sup\varrho_R < \sup\varrho_B$, and the root rate is $(\sum_e w_e)^2\sup\varrho_R = \sup\varrho_R/\sup\varrho_B$. □

**Theorem 20.59 (One tree suffices).** *As stated; the chain clause of (iii) is stated whenever some active list is connected.*

*Proof.* All of the theorem but the chain clause of (iii) is the reference's (proof of thm:onetree). For the chain clause, let some active list of the molecule $F$ be connected. The weights of lists factor

across every split, so Corollary 20.9 applies: every peel leaves one childcule, the Foreman tree is a chain (Corollary 20.8), and at every realization, the centre and every lean, $|B_0| \le |B_1| \le \cdots \le |B_{d-1}| \le |L|$, $L$ the leaf. The shape of the bunch $B_k$ then has $|B_k| - 1 \le |L| - 1$ integrations, and its node adds one, the total of the bunch against its room, by (ii). A molecule of one list is such an $F$, its list being connected. □

## 50.21 Chapter 21: gluing and what crosses it

**Theorem 21.5 (Replacement of marks by blocks).** *As stated: every mark is replaced, by a block of any size and by either rule.*

*Proof. One mark, first rule.* Let $h$ be replaced by a block $X$ of $q$ marks by the first rule, the other marks left in place: the new sets are the bases $T \not\ni h$ and the sets $(T - h) \sqcup \{x\}$, $T \ni h$, $x \in X$, each with the weight $\theta_T$. If $h$ is neither a loop nor a coloop, $(\mathcal{B}, \theta)$ is a cell on $(\mathcal{E} - h) \sqcup \{h\}$ with the hinge $h$, and the parallel block $\binom{X \sqcup \{g\}}{1}$ is a cell on $X \sqcup \{g\}$ whose deletion of $g$ has the singletons of $X$ as bases and whose contraction of $g$ has the single basis $\varnothing$. By 21.2 for these two cells, the sets $\{x\} \sqcup (T - h)$, $T \ni h$, and $\varnothing \sqcup T$, $T \not\ni h$, are the bases of a table of rank $1 + r - 1 = r$: they are the new sets. If $h$ is a loop, no basis contains $h$, and the new sets are the bases of $\mathcal{B}$ read on $\mathcal{E} - h$, the marks of $X$ being loops: a table of rank $r$. If $h$ is a coloop, every basis contains $h$, and the new sets are the sets $(T - h) \sqcup \{x\}$: the direct sum of the contraction of $h$ and the table $\binom{X}{1}$, a table of rank $r$, the exchange property of a direct sum holding in each summand.

*One mark, complementary rule.* The dual table has the bases $\mathcal{E} \setminus T$ (7.2). The first rule at $h$ on the dual table gives the sets $\mathcal{E} \setminus T$ for $T \ni h$ and $(\mathcal{E} \setminus T) - h + x$ for $T \not\ni h$, $x \in X$, the bases of a table on $(\mathcal{E} - h) \sqcup X$ by the first case. Their complements in $(\mathcal{E} - h) \sqcup X$ are $(T - h) \sqcup X$ for $T \ni h$ and $T \sqcup (X - x)$ for $T \not\ni h$, the sets of the complementary rule, and the complements of the bases of a table are the bases of its dual. Its rank is $(|\mathcal{E}| - 1 + q) - (|\mathcal{E}| - r) = r + q - 1$.

*All marks.* The rule at a mark $h$ decides the part of a new set in $X_h$ from whether $h \in T$ alone, so replacing the marks one at a time, in any order, gives the sets of the statement; by the two cases each step gives a table, and the step at $h \in \mathcal{E}_2$ raises the rank by $q_h - 1$. A new set $T'$ determines $T = \{h \in \mathcal{E}_1 : X_h \cap T' \ne \varnothing\} \sqcup \{h \in \mathcal{E}_2 : X_h \subseteq T'\}$ and the choice made in every block, so it carries the one weight $\theta_T$. A block of one mark gives back its mark under either rule.

*Substitution.* When no mark is a loop or a coloop, $(\mathcal{B}, \theta)$ is a core (21.1), and the parallel block $\binom{X_h \sqcup \{g_h\}}{1}$ and the series block $\binom{X_h \sqcup \{g_h\}}{q_h}$ are cells, $g_h$ being neither a loop nor a coloop of either. In the parallel block the deletion of $g_h$ has the singletons of $X_h$ as bases and the contraction the single basis $\varnothing$; in the series block the deletion has the single basis $X_h$ and the contraction the sets $X_h - x$. So the sets of the substitution contain, for a basis $T$ of the core, one mark of $X_h$ for $h \in T \cap \mathcal{E}_1$, none for $h \in \mathcal{E}_1 \setminus T$, all of $X_h$ for $h \in T \cap \mathcal{E}_2$ and all but one for $h \in \mathcal{E}_2 \setminus T$, each with the weight $\theta_T$, every weight of a block being 1: the table of the statement.

*Letters.* By the correspondence between the new sets and the pairs of a basis $T$ and a choice in every block, $K'(\alpha)$ is the sum over $T$ of $\theta_T$ times the product over the blocks of the sum, over the choices, of the monomials of the chosen marks: $e_1(\alpha_{X_h})$ for $h \in T \cap \mathcal{E}_1$, 1 for $h \in \mathcal{E}_1 \setminus T$, $e_{q_h}(\alpha_{X_h})$ for $h \in T \cap \mathcal{E}_2$ and $e_{q_h - 1}(\alpha_{X_h})$ for $h \in \mathcal{E}_2 \setminus T$. Taking out $\prod_{h \in \mathcal{E}_2} e_{q_h - 1}$,

$$K' = \prod_{h \in \mathcal{E}_2} e_{q_h - 1} \sum_T \theta_T \prod_{h \in T \cap \mathcal{E}_1} e_1 \prod_{h \in T \cap \mathcal{E}_2} \frac{e_{q_h}}{e_{q_h - 1}},$$

the first display. The complement of a new set meets $X_h$ in all of $X_h$ but the chosen mark for $h \in T \cap \mathcal{E}_1$, in all of $X_h$ for $h \in \mathcal{E}_1 \setminus T$, in nothing for $h \in T \cap \mathcal{E}_2$ and in the missing mark for

$h \in \mathcal{E}_2 \setminus T$, so

$$\Psi' = \sum_T \theta_T \prod_{h \in T \cap \mathcal{E}_1} e_{q_h - 1} \prod_{h \in \mathcal{E}_1 \setminus T} e_{q_h} \prod_{h \in \mathcal{E}_2 \setminus T} e_1 = \prod_{h \in \mathcal{E}_1} e_{q_h - 1} \sum_T \theta_T \prod_{h \in \mathcal{E}_1 \setminus T} \frac{e_{q_h}}{e_{q_h - 1}} \prod_{h \in \mathcal{E}_2 \setminus T} e_1,$$

which is $\prod_{h \in \mathcal{E}_1} e_{q_h - 1}\, \Psi(\beta^*)$ with $\beta^*_h = e_{q_h}/e_{q_h - 1}$ on $\mathcal{E}_1$ and $\beta^*_h = e_1$ on $\mathcal{E}_2$, since $\Psi(\beta^*) = \sum_T \theta_T (\beta^*)^{\mathcal{E} \setminus T}$. For one block at $h$, every other block of one mark, $K$ and $\Psi$ are affine in $\alpha_h$, $K = K_{\setminus h} + \alpha_h K_{/h}$ and $\Psi = \alpha_h \Psi_{\setminus h} + \Psi_{/h}$ (7.3), and the two displays become the four formulas of the statement. In scale and shape, $e_k$ is homogeneous of degree $k$, $e_1(u) = |u| = 1$ on $\Delta_X$, and the measure is 2.7. □

**Theorem 21.6 (Counts, walls and atoms under replacement).** *As stated: (i) and (ii) for blocks of any sizes and either rule at each mark; (iii) at every positive exponent pairs.*

*Proof.* (i) The marks and the rank are 21.5. At $\alpha = \mathbf{1}$, $e_1 = q_h$, $e_{q_h} = 1$ and $e_{q_h - 1} = q_h$ on a block of $q_h$ marks, so the first display of 21.5 gives the total weight $K'(\mathbf{1}) = \prod_{h \in \mathcal{E}_2} q_h \cdot K\big((q_h)_{h \in \mathcal{E}_1}, (q_h^{-1})_{h \in \mathcal{E}_2}\big)$, and expanding $K$ over the bases gives the sum displayed; it counts $q_h$ choices at $h \in T \cap \mathcal{E}_1$ and $q_h$ at $h \in \mathcal{E}_2 \setminus T$. With $\mathcal{E}_2 = \varnothing$ it is $K((q_h)_h)$; with $\mathcal{E}_1 = \varnothing$ it is $\prod_h q_h\, K((q_h^{-1})_h) = \Psi((q_h)_h)$ (7.2).

(ii) By 7.13, $d'(Y) = \min_{T'} |T' \cap Y|$ and $d^{*\prime}(Y) = \min_{T'} |Y \setminus T'|$ over the new bases $T'$, which are the pairs of a basis $T$ and a choice in every block (proof of 21.5). Fix $T$ and minimize over the choices, block by block. In $|T' \cap Y|$ a block contributes, at its least: for $h \in T \cap \mathcal{E}_1$, 1 if $X_h \subseteq Y$ and 0 otherwise, the chosen mark avoiding $Y$ when it can; for $h \in \mathcal{E}_1 \setminus T$, 0; for $h \in T \cap \mathcal{E}_2$, $|X_h \cap Y|$; for $h \in \mathcal{E}_2 \setminus T$, $\max(|X_h \cap Y| - 1, 0)$, the missing mark taken in $Y$ when it can. Writing $|X_h \cap Y| = \max(|X_h \cap Y| - 1, 0) + [h \in \lceil Y \rceil]$ for $h \in \mathcal{E}_2$, the least value at $T$ is

$$\sum_{h \in \mathcal{E}_2} \max\big(|X_h \cap Y| - 1, 0\big) + \big|T \cap \big((\lfloor Y \rfloor \cap \mathcal{E}_1) \sqcup (\lceil Y \rceil \cap \mathcal{E}_2)\big)\big|,$$

and the least over $T$ is $d'(Y)$ as stated (7.13). In $|Y \setminus T'|$ a block contributes, at its least: for $h \in T \cap \mathcal{E}_1$, $\max(|X_h \cap Y| - 1, 0)$; for $h \in \mathcal{E}_1 \setminus T$, $|X_h \cap Y|$; for $h \in T \cap \mathcal{E}_2$, 0; for $h \in \mathcal{E}_2 \setminus T$, 1 if $X_h \subseteq Y$ and 0 otherwise. In the same way $d^{*\prime}(Y)$ is as stated, and $c'(Y) = |Y| - d'(Y) - d^{*\prime}(Y)$ is 7.12. Under the first rule alone, $r'(Y) = \max_{T'} |T' \cap Y| = \max_T |T \cap \lceil Y \rceil| = r(\lceil Y \rceil)$, $d'(Y) = d(\lfloor Y \rfloor) = r - r(\mathcal{E} \setminus \lfloor Y \rfloor)$ (7.12) and $c'(Y) = r'(Y) - d'(Y)$ (7.13). Under the complementary rule alone, $|Y| = \sum_h \max(|X_h \cap Y| - 1, 0) + |\lceil Y \rceil|$, so

$$c'(Y) = |\lceil Y \rceil| - d(\lceil Y \rceil) - d^*(\lfloor Y \rfloor) = |\lceil Y \rceil| - |\lfloor Y \rfloor| + r(\mathcal{E} \setminus \lceil Y \rceil) + r(\lfloor Y \rfloor) - r.$$

If $Y = \bigcup_{h \in S} X_h$, then $\lceil Y \rceil = \lfloor Y \rfloor = S$, $d'(Y) = \sum_{h \in S \cap \mathcal{E}_2} (q_h - 1) + d(S)$, $d^{*\prime}(Y) = \sum_{h \in S \cap \mathcal{E}_1} (q_h - 1) + d^*(S)$ and $|Y| = \sum_{h \in S} q_h$, so $c'(Y) = |S| - d(S) - d^*(S) = c(S)$. If $\varnothing \neq Y \subsetneq X_h$, then $\lceil Y \rceil = \{h\}$ and $\lfloor Y \rfloor = \varnothing$: for $h \in \mathcal{E}_1$, $d'(Y) = 0$ and $d^{*\prime}(Y) = |Y| - 1 + d^*(\{h\})$, and for $h \in \mathcal{E}_2$, $d'(Y) = |Y| - 1 + d(\{h\})$ and $d^{*\prime}(Y) = 0$; a seed table connected on at least two marks has $c(\{h\}) = 1 - d(\{h\}) - d^*(\{h\}) > 0$, so $d(\{h\}) = d^*(\{h\}) = 0$ and $c'(Y) = 1$.

(iii) Write $X_e = \{e\} \times \{1, \dots, q\}$. For $h \in H$, carrying table $j$ to table $\tau(j)$ (11.13), put $\hat{h}(e, i) = (h(e), i)$; the rule and $q$ are the same at every mark, so $\hat{h}$ maps the replaced table $j$ onto the replaced table $\tau(j)$ with the same weights, and the two tables have the same pairs: $\hat{h}$ carries the replaced family to itself. A permutation of $X_e$ fixing the other new marks maps every replaced table onto itself with its weights, the sets of the rule being symmetric in the marks of $X_e$. The group generated by these bijections is transitive on the new marks, $H$ being transitive on $\mathcal{E}$.

The replaced family is connected. The seed family is one molecule: a molecule $M \neq \mathcal{E}$ would be a union of molecules of every table (7.9), with $|T \cap M|$ constant on the bases of every table and so $c_j(M) = 0$ for every $j$ (7.13), all tables being active at positive pairs. Let $e \neq f$ be linked in table $j$: by 7.7 some basis $T$ of table $j$ has $e \in T$, $f \notin T$ and $T - e + f$ a basis. Under the first rule, a new basis $T'$ from $T$ choosing $x$ in $X_e$ gives $T' - x + x'$, the new basis from $T$ choosing $x'$, and $T' - x + y$, $y \in X_f$, the new basis from $T - e + f$ choosing $y$ with the same choices elsewhere; under the complementary rule, the new basis $T'$ from $T - e + f$ missing $x'$ in $X_e$ gives $T' - x + x'$, the one missing $x$, and the new basis $T''$ from $T$ missing $y$ in $X_f$ gives $T'' - x + y$, the new basis from $T - e + f$ missing $x$. By 7.7 any two marks of $X_e$ are linked, and every mark of $X_e$ is linked to every mark of $X_f$. Every mark is linked to another mark, the seed family being one molecule on at least two marks, so the new marks form one molecule. Then no nonempty proper set $Y$ of new marks has $c'_j(Y) = 0$ in every table: by 7.13 $|T' \cap Y|$ would be constant on the bases of every table, and by 7.8 no circuit would join $Y$ to its complement.

Finally $((s_j, t_j)_j; \mathbf{1})$ is a realization of the replaced family, and every bijection above fixes the table's gauge, $\hat{h}^* \log \mathbf{1} = \log \mathbf{1}$. By 11.14 the reading of the image of a mark equals the reading of the mark, so the reading is constant on the new marks, which form one molecule: the realization is an atom (10.1). □

*Remark.* In exact rational arithmetic (receipt `check_blocks_splice.py`): on the seed of rank two on $\{a, b, c, d\}$ whose bases are the pairs other than $\{c, d\}$, with weights $1, 2, \frac{3}{2}, 5, \frac{1}{3}$, blocks of $2, 3, 1, 2$ marks, the first rule at $a, c$ and the complementary rule at $b, d$, the new sets satisfy the exchange property, the rank is $r + \sum_{h \in \mathcal{E}_2}(q_h - 1)$, the total weight is that of (i), $d'(Y)$ and $d^{*\prime}(Y)$ are those of (ii) at every one of the $2^8$ sets $Y$, and unions of full blocks keep the width; with the first rule at every mark and the same sizes, $r'(Y) = r(\lceil Y \rceil)$, $d'(Y) = r - r(\mathcal{E} \setminus \lfloor Y \rfloor)$ and the total weight is $K((q_h)_h)$.

**Proposition 21.8 (Dual pairs).** *As stated.*

*Proof.* Write $\theta$ for the one system of weights of $\mathcal{A}$. (i) The bases of $\mathcal{A}$ are the $X \in \mathcal{A} \setminus g$ and the $Y + g$ with $Y \in \mathcal{A}/g$ (7.3). If $\mathcal{A}$ is a dual pair, the complement $(B \setminus X) + g$ of $X$ is a basis with the weight $\theta_X$, and the complement $B \setminus Y$ of $Y + g$ is a basis of $\mathcal{A} \setminus g$ with the weight $\theta_{Y+g}$: the bases are closed under complement with equal weights. Conversely, if they are, $X \in \mathcal{A} \setminus g$ gives the basis $(B \setminus X) + g$ with the weight $\theta_X$, so $B \setminus X \in \mathcal{A}/g$, and $Y \in \mathcal{A}/g$ gives $B \setminus Y \in \mathcal{A} \setminus g$: $\mathcal{A}/g$ is the dual of $\mathcal{A} \setminus g$ with its weights. Since $g$ is neither a loop nor a coloop, $\mathcal{A} \setminus g$ has the rank $r_{\mathcal{A}}$ and $\mathcal{A}/g$ the rank $r_{\mathcal{A}} - 1$, while the dual of $\mathcal{A} \setminus g$ has the rank $|B| - r_{\mathcal{A}}$; so $|B| = 2r_{\mathcal{A}} - 1$.

(ii) $K_{\mathcal{A}/g} = \sum_Y \theta_{Y+g} \alpha^Y = \sum_X \theta_X \alpha^{B \setminus X} = \psi$ and $\Psi_{\mathcal{A}/g} = \sum_Y \theta_{Y+g} \alpha^{B \setminus Y} = \sum_X \theta_X \alpha^X = k$. So $\gamma^K_{\mathcal{A}} = K_{\mathcal{A} \setminus g}/K_{\mathcal{A}/g} = k/\psi$ and $\gamma^{\Psi}_{\mathcal{A}} = \Psi_{\mathcal{A}/g}/\Psi_{\mathcal{A} \setminus g} = k/\psi$ (21.1).

(iii) The cell $\binom{\{h,g\}}{1}$ has the deletion with the basis $\{h\}$ and the contraction with the basis $\varnothing$, the dual of $\{h\}$ on $\{h\}$, so it is a dual pair with $k = \alpha_h$ and $\psi = 1$. By 21.4 and (ii), $K = \prod_i K_{\mathcal{A}_i/g_i} K_{\mathcal{H}}\big((k_i/\psi_i)_i\big) = \pi K_{\mathcal{H}}\big((k_i/\psi_i)_i\big)$ and $\Psi = \prod_i \Psi_{\mathcal{A}_i \setminus g_i} \Psi_{\mathcal{H}}\big((k_i/\psi_i)_i\big) = \pi \Psi_{\mathcal{H}}\big((k_i/\psi_i)_i\big)$. $K_{\mathcal{H}}$ and $\Psi_{\mathcal{H}}$ are homogeneous of the degrees $r_{\mathcal{H}}$ and $q - r_{\mathcal{H}}$, so multiplying the arguments by $\pi$ gives the two identities; both sides are polynomials, $r_{\mathcal{H}} \geq 1$ and $q - r_{\mathcal{H}} \geq 1$ since no mark of the core is a loop or a coloop. □

**Corollary 21.11 (The width of the marks left in place).** *As stated.*

*Proof.* In 21.10 take $S \subseteq L$: the part of $S$ in the cell $\binom{\{l,g\}}{1}$ of a mark $l \in S$ is $\{l\}$, with $d(\{l\}) = d^*(\{l\}) = 0$ and $d(\{l, g\}) = d^*(\{l, g\}) = 1$, the bases being $\{l\}$ and $\{g\}$ (7.13); every other cell meets $S$ in $\varnothing$, and its hinge, neither a loop nor a coloop, has $d(\{g\}) = d^*(\{g\}) = 0$. So

$D(S) = D^*(S) = S$, read on the marks of the core, and $d(S) = d_{\mathcal{H}}(S)$, $d^*(S) = d^*_{\mathcal{H}}(S)$, whence $c(S) = |S| - d(S) - d^*(S) = c_{\mathcal{H}}(S)$ (7.12). By 7.12, $d_{\mathcal{H}}(S) = r_{\mathcal{H}} - r_{\mathcal{H}}(L \sqcup H' \setminus S)$, so

$$c_{\mathcal{H}}(S) = r_{\mathcal{H}}(S) + r_{\mathcal{H}}(L \sqcup H' \setminus S) - r_{\mathcal{H}} \le r_{\mathcal{H}}(L \sqcup H' \setminus S) \le |L \setminus S| + |H'|,$$

the rank of a set being at most its size and $r_{\mathcal{H}}(S) \le r_{\mathcal{H}}$. In $\binom{B \sqcup \{h_1,\dots,h_c\}}{c}$ the bases are the sets of $c$ marks, which meet $B$ in every number from 0 to $c$, so $B$ has the width $c$ (7.13); no $h_i$ is a loop or a coloop, and by the first part the width stays $c$ after any cells are substituted at the $h_i$. □

**Lemma 21.12 (Two-sums of cells; hinge walls).** *As stated, for cell families glued table by table.*

*Proof.* That $\mathcal{A} \oplus_g \mathcal{A}'$ is a table of rank $r_{\mathcal{A}} + r_{\mathcal{A}'} - 1$, plain when the cells are, in which $B$ is a bunch of width one with the cells $\mathcal{A}$ and $\mathcal{A}'$, and that a table with a bunch of width one and factoring weights is the two-sum of its two cells, is the reference, proof of lem:twosumpieces(i), the weights multiplied letter by letter. The two-sum is $\binom{\{h_1,h_2\}}{1}\langle \mathcal{A}, \mathcal{A}' \rangle$: the bases $\{h_1\}$ and $\{h_2\}$ of the core give the two kinds of sets of 21.2 (21.1). So $F$ is substituted table by table, with the core $\binom{\{h_1,h_2\}}{1}$ in every table and the cells $\mathcal{A}_j$ at $h_1$ and $\mathcal{A}'_j$ at $h_2$, and 21.10 at $S = B$ gives the wall $m_B - \sum_j \big(s_j(r_{\mathcal{A}_j} - 1) + t_j(|B| - r_{\mathcal{A}_j})\big) = \kappa_{F_B}$, and at $S = R$ the wall $\kappa_{F_R}$. Table $j$ of $F$ has the rank $r_j = r_{\mathcal{A}_j} + r_{\mathcal{A}'_j} - 1$ on $n = |B| + |R|$ marks, so

$$\kappa_{F_B} + \kappa_{F_R} = m_{\mathcal{E}} - \sum_j \big(s_j(r_j - 1) + t_j(n - r_j - 1)\big) = \kappa_{\mathcal{E}} + \sum_j (s_j + t_j) = \kappa_{\mathcal{E}} + 2\sigma.$$

With the mass $2\sigma - \kappa_{F_B}$ on $g$, the imbalance of $F_B$ on $B \sqcup \{g\}$ is

$$m_B + 2\sigma - \kappa_{F_B} - \sum_j \big(s_j r_{\mathcal{A}_j} + t_j(|B| + 1 - r_{\mathcal{A}_j})\big) = 2\sigma - \sum_j (s_j + t_j) = 0,$$

on the balance wall of $F$ or off it. □

**Theorem 21.14 (Splicing hinges in pairs).** *As stated; (iii) holds for any number of cells and any graph of pairs.*

*Proof.* (i) and (ii) are the reference, proof of thm:hingepairs(i),(ii). (iii) In the direct sum of the cells the letters are $K = \prod_i K_{\mathcal{A}_i}$ and $\Psi = \prod_i \Psi_{\mathcal{A}_i}$ (7.9), and by 21.13, with $G = \bigsqcup_i G_i$,

$$K = \sum_{(p_i)_i} \alpha_G^{\sqcup_i p_i} \prod_i K_{\mathcal{A}_i,(p_i)}, \qquad \Psi = \sum_{(p_i)_i} \alpha_G^{G \setminus \sqcup_i p_i} \prod_i \Psi_{\mathcal{A}_i,(p_i)},$$

over $p_i \subseteq G_i$, absent channels counting as zero. Splicing a pair $\{a, b\}$ acts on a monomial $\alpha_G^P$ by $(\partial_{\alpha_a} + \partial_{\alpha_b})\alpha_G^P|_{\alpha_a = \alpha_b = 0}$, which is $\alpha_G^{P-a}$ when $a \in P \not\ni b$, $\alpha_G^{P-b}$ when $b \in P \not\ni a$, and 0 otherwise; on $\alpha_G^{G \setminus P}$ it is likewise nonzero exactly when $P$ contains one of $a$, $b$. The splicings of disjoint pairs commute and act on different variables, so splicing all pairs keeps the terms in which $P = \sqcup_i p_i$ is $p \sqcup J$ with $p \subseteq G'$ and $J$ containing exactly one hinge of every pair, and leaves $\alpha_{G'}^p$ in $K$ and $\alpha_{G'}^{G' \setminus p}$ in $\Psi$. By (ii) the result is the letters of the spliced cell $\mathcal{A}$, and by the Taylor cube (12.7) its channels are the coefficients of these monomials:

$$K_{\mathcal{A},(p)} = \sum_J \prod_i K_{\mathcal{A}_i,((p \sqcup J) \cap G_i)}, \qquad \Psi_{\mathcal{A},(p)} = \sum_J \prod_i \Psi_{\mathcal{A}_i,((p \sqcup J) \cap G_i)}.$$

The marks other than hinges are $\bigsqcup_i B_i$, and the rank is $\sum_i r_{\mathcal{A}_i} - k$ by (ii). The degrees add up: $\sum_i \big(r_{\mathcal{A}_i} - |(p \sqcup J) \cap G_i|\big) = \sum_i r_{\mathcal{A}_i} - k - |p|$, since $|J| = k$, and $\sum_i \big(n_{\mathcal{A}_i} - r_{\mathcal{A}_i} + |(p \sqcup J) \cap G_i|\big) = n_{\mathcal{A}} - r_{\mathcal{A}} + |p|$, since $n_{\mathcal{A}} - r_{\mathcal{A}} = \sum_i (n_{\mathcal{A}_i} - r_{\mathcal{A}_i}) + k$. Each channel of $\mathcal{A}$ is a sum of products of one channel of each cell, so the map is multilinear. For two cells with the pairs named by $I$, a set $J$ is given by the set $I' \subseteq I$ of the pairs whose hinge on the side of $\mathcal{A}_1$ it contains, $(p \sqcup J) \cap G_1 = p_1 \sqcup I'$ and $(p \sqcup J) \cap G_2 = p_2 \sqcup (I \setminus I')$: the bilinear map displayed. With $k = 1$ and no other hinges, $K_{\mathcal{A},(\varnothing)} = K_{\mathcal{A}_1 \setminus g_1} K_{\mathcal{A}_2/g_2} + K_{\mathcal{A}_1/g_1} K_{\mathcal{A}_2 \setminus g_2}$ and $\Psi_{\mathcal{A},(\varnothing)} = \Psi_{\mathcal{A}_1 \setminus g_1} \Psi_{\mathcal{A}_2/g_2} + \Psi_{\mathcal{A}_1/g_1} \Psi_{\mathcal{A}_2 \setminus g_2}$, the letters of 21.7(iii). □

*Remark.* In exact arithmetic (receipt `check_blocks_splice.py`): three cells in a closed path, a weighted $\binom{\{x_1,x_2\} \sqcup G_1}{3}$ with three hinges, one of them unpaired, a weighted $\binom{\{y_1,y_2\} \sqcup G_2}{2}$ and a weighted $\binom{\{z_1\} \sqcup G_3}{1}$ with two hinges each, spliced along three pairs, one between each two cells: both channels of the spliced cell at both values of the unpaired hinge are the sums over $J$ of the products of the channels of the three cells.

**Theorem 21.17 (Multivectors bound what crosses).** *As stated; (iii) holds at any tensor rank.*

*Proof.* (i) and (ii) are the reference, proof of thm:cutrank(i),(ii), with the interface rank for the cut rank; (ii) is proved there at any tensor rank, the interface rank being subadditive. (iii) By (i) and (ii), $\operatorname{rank} \Theta_B^{(k)} \le \operatorname{ir}_k(\boldsymbol{\omega})^2 \le i_k^2$ with $i_k = \sum_l \binom{c_l}{k - d_l}$, and summing over the levels bounds the interface dimension $\sum_k \operatorname{rank} \Theta_B^{(k)}$ (21.15); for a real multivector (i) gives $\operatorname{rank} \Theta_B^{(k)} \le \frac{1}{2} \operatorname{ir}_k(\boldsymbol{\omega})(\operatorname{ir}_k(\boldsymbol{\omega}) + 1)$, and $x \mapsto \frac{1}{2} x(x+1)$ increases on $[0, \infty)$. The bounds involve only the widths and the floors of the slices. A determinantal table has a multivector of tensor rank one, $\boldsymbol{\omega}_T = \det A_T$ (7.20); a set $T$ is a basis exactly when $\det A_T \neq 0$, so the column space of $A_S$ has the dimension $r(S)$, the column spaces of $A_B$ and $A_R$ span $\mathbb{F}^r$ and meet in dimension $r(B) + r(R) - r = c(B)$, and $d_1 = r - r(R) = d(B)$ (7.12). So $\operatorname{ir}_k(\boldsymbol{\omega}) \le \binom{c(B)}{k - d(B)}$, and $\sum_k \binom{c(B)}{k - d(B)}^2 = \sum_i \binom{c(B)}{i}^2 = \binom{2c(B)}{c(B)}$. At width one $\binom{1}{k - d(B)} = 1$ at $k = d(B)$ and $k = d(B) + 1$: one bunch letter at each level. □

## 50.22 Chapter 22: periods of constructions

**Definition 22.1 (Twists).** As stated: the torus relation for every twist, every vector $\boldsymbol{\lambda}$ on the marks and all constants $c_j, c_j^*$.

*Proof.* Put $\mathbf{x}' = (c_j \boldsymbol{\lambda} \circ \mathbf{x}_j^K,\ c_j^* \boldsymbol{\lambda} \circ \mathbf{x}_j^\Psi)_j$. Every basis of table $j$ has $r_j$ marks and its complement $r_j^*$, so $K_j$ is homogeneous of degree $r_j$ and $\Psi_j$ of degree $r_j^*$, and the letters of $F[\mathbf{x}']$ at $\alpha$ are

$$K_j(c_j \boldsymbol{\lambda} \circ \mathbf{x}_j^K \circ \alpha) = c_j^{r_j} K_j\big(\mathbf{x}_j^K \circ (\boldsymbol{\lambda} \circ \alpha)\big), \qquad \Psi_j(c_j^* \boldsymbol{\lambda} \circ \mathbf{x}_j^\Psi \circ \alpha) = c_j^{*\, r_j^*} \Psi_j\big(\mathbf{x}_j^\Psi \circ (\boldsymbol{\lambda} \circ \alpha)\big).$$

Write $\omega_{\mathbf{x}} = \beta^{\mathbf{m-1}} \prod_j K_j(\mathbf{x}_j^K \circ \beta)^{-s_j} \Psi_j(\mathbf{x}_j^\Psi \circ \beta)^{-t_j}\, d\beta$ for the form of $F[\mathbf{x}]$. The map $\Phi : \alpha \mapsto \boldsymbol{\lambda} \circ \alpha$ is a linear bijection of the open orthant onto itself that commutes with positive scalings, and with $\beta_e = \lambda_e \alpha_e$ one has $\beta^{\mathbf{m-1}} d\beta = \boldsymbol{\lambda}^{\mathbf{m}} \alpha^{\mathbf{m-1}} d\alpha$. So the form of $F[\mathbf{x}']$ is

$$\omega_{\mathbf{x}'} = \boldsymbol{\lambda}^{-\mathbf{m}} \prod_j c_j^{-r_j s_j} c_j^{*\,-r_j^* t_j}\ \Phi^* \omega_{\mathbf{x}}.$$

A twist does not change degrees, so on the balance wall both forms are homogeneous of degree zero, and each period is the integral of its form over the section $\{h = 1\}$ of any positive function $h$

homogeneous of degree one (8.7). Since $\int_{\{h=1\}} \Phi^* \omega_{\mathbf{x}} = \int_{\{h\circ\Phi^{-1}=1\}} \omega_{\mathbf{x}}$ and $h \circ \Phi^{-1}$ is again positive and homogeneous of degree one, the two periods differ by the constant factor of the display. The moduli of the two integrands are related by the same change of variables and a positive constant, so the two sides converge together; for real data the integrands are nonnegative and the identity holds in $[0, \infty]$.

At $\boldsymbol{\lambda} = \mu \mathbf{1}$ and $c_j = c_j^* = \mu^{-1}$ every coordinate of $\mathbf{x}'$ equals that of $\mathbf{x}$, and the factor is $\mu^{-m_{\mathcal{E}}} \mu^{\sum_j (r_j s_j + r_j^* t_j)} = 1$, since $m_{\mathcal{E}} = \sum_j (r_j s_j + r_j^* t_j)$ on the balance wall. With $\boldsymbol{\lambda}$ equal to 1 off $g$ and to $\lambda$ at $g$, and $c_j = c_j^* = 1$, the factor is $\lambda^{-m_g}$; with $\lambda = 1/x_1^K$ this reads every twist at $g$ with $x_1^K = 1$, up to the factor $(x_1^K)^{-m_g}$, and for one table it leaves the one number $x_1^\Psi / x_1^K$, carried by the cobasis letter. With $\boldsymbol{\lambda} = \mathbf{1}$, one $c_j$ or one $c_j^*$ different from 1 and the other constants 1, the factor is $c_j^{-r_j s_j}$ or $c_j^{*\,-r_j^* t_j}$; for one table twisted in its cobasis letter by $\mu \mathbf{x}$ it is $\mu^{-r^* t}$. □

*Remark.* The reference, 22.1, rescales a twist at one mark, normalized by the scale of the mark, and the relation for one table twisted in its cobasis letter is used with 25.8. The exact check of the scalar case is check 11 of S21-23_work/checks.py.

**Theorem 22.4 (The cells ride on the core's fibre).** *As stated: at complex data, every exponent* 0 *or of positive real part, on contours on which the load families converge absolutely.*

*Proof.* The steps are those of the reference, proof of 22.4, with the moduli carried along. By 21.4, table $j$ has the letters

$$K_j = \prod_i K_{\mathcal{A}_{ij}/g_i}(\alpha_{B_i})\, K_{\mathcal{H}_j}\big((\gamma^K_{\mathcal{A}_{ij}}(\alpha_{B_i}))_i\big), \qquad \Psi_j = \prod_i \Psi_{\mathcal{A}_{ij}\setminus g_i}(\alpha_{B_i})\, \Psi_{\mathcal{H}_j}\big((\gamma^\Psi_{\mathcal{A}_{ij}}(\alpha_{B_i}))_i\big).$$

For $\operatorname{Re} s_j > 0$, 4.4 opens $K_{\mathcal{H}_j}(X)^{-s_j} = \big(\sum_T \theta_T^{\mathcal{H}_j} X^T\big)^{-s_j}$ with one share $a_{j,T}$ per basis on a contour of positive real parts, and at $X_i = \gamma^K_{\mathcal{A}_{ij}}$ the monomials give $\prod_T (X^T)^{-a_{j,T}} = \prod_i (\gamma^K_{\mathcal{A}_{ij}})^{-a_j(h_i)}$; in the same way $\Psi_{\mathcal{H}_j}(Y)^{-t_j} = \big(\sum_T \theta_T^{\mathcal{H}_j} Y^{\{h\}\setminus T}\big)^{-t_j}$ is opened with the shares $b_{j,T}$, and $\prod_T (Y^{\{h\}\setminus T})^{-b_{j,T}} = \prod_i (\gamma^\Psi_{\mathcal{A}_{ij}})^{-b_j(h_i)}$. A letter with exponent 0 is the constant 1 and is opened by nothing. Since $\gamma^K_{\mathcal{A}} = K_{\mathcal{A}\setminus g}/K_{\mathcal{A}/g}$ and $\gamma^\Psi_{\mathcal{A}} = \Psi_{\mathcal{A}/g}/\Psi_{\mathcal{A}\setminus g}$ (21.1),

$$K^{-s_j}_{\mathcal{A}_{ij}/g_i}(\gamma^K_{\mathcal{A}_{ij}})^{-a_j(h_i)} = K^{-(s_j - a_j(h_i))}_{\mathcal{A}_{ij}/g_i} K^{-a_j(h_i)}_{\mathcal{A}_{ij}\setminus g_i}, \qquad \Psi^{-t_j}_{\mathcal{A}_{ij}\setminus g_i}(\gamma^\Psi_{\mathcal{A}_{ij}})^{-b_j(h_i)} = \Psi^{-(t_j - b_j(h_i))}_{\mathcal{A}_{ij}\setminus g_i} \Psi^{-b_j(h_i)}_{\mathcal{A}_{ij}/g_i},$$

and $e^{-|\alpha|} = \prod_i e^{-|\alpha_{B_i}|}$, so for fixed shares the integrand of $J_F$ is the product over the cells of the integrands of the interests of the load families $F_i(\mathbf{a}, \mathbf{b})$, times the share kernels and the powers of the weights of the cores.

It remains to exchange the share integral and the integral over the orthant. For $X > 0$, $|X^{-a}| = X^{-\operatorname{Re} a}$, and $|\alpha^{\mathbf{m}-\mathbf{1}}| = \alpha^{\operatorname{Re}\mathbf{m}-\mathbf{1}}$; so the modulus of the joint integrand is

$$\prod_j \big|\Gamma_{|\mathcal{H}_j|}\begin{bmatrix}\mathbf{a}_j \\ s_j\end{bmatrix}\big|\, \big|\Gamma_{|\mathcal{H}_j|}\begin{bmatrix}\mathbf{b}_j \\ t_j\end{bmatrix}\big| \prod_{j,T} \big(\theta_T^{\mathcal{H}_j}\big)^{-\operatorname{Re}(a_{j,T}+b_{j,T})}$$

times the product of the integrands of the interests of the load families at the real parts of their data and exponents. Integrated over the orthant, the second factor gives the product of these interests at the real parts, which are finite by the hypothesis and constant along the contour, since they depend on the shares only through their real parts. On the contour, $a_{j,T} = c_{j,T} + i\tau_{j,T}$ with $c_{j,T} > 0$ fixed and $\sum_T \tau_{j,T} = \operatorname{Im} s_j$, and the classical bound $|\Gamma(c + i\tau)| \le C_c\, e^{-\pi|\tau|/2}(1 + |\tau|)^{c-1/2}$ makes $\prod_T |\Gamma(a_{j,T})|$ integrable over the imaginary parts of all shares but one; the same holds for

the shares $b_{j,T}$. So the joint integrand is absolutely integrable, the integrals may be taken in either order, and the display follows. The identity $J_{F_i(\mathbf{a},\mathbf{b})} = \Gamma(\kappa_{B_i} - \sum_j (a_j(h_i) + b_j(h_i)))\, P_{F_i(\mathbf{a},\mathbf{b})}$ is 8.7 at the imbalance of the load family: its tables $\mathcal{A}_{ij} \setminus g_i$ and $\mathcal{A}_{ij}/g_i$ have the degrees $r_{\mathcal{A}_{ij}}$, $|B_i| - r_{\mathcal{A}_{ij}}$, $r_{\mathcal{A}_{ij}} - 1$ and $|B_i| - r_{\mathcal{A}_{ij}} + 1$, and the masses $\mathbf{m}|_{B_i}$ exceed their degree by $\kappa_{B_i} - \sum_j (a_j(h_i) + b_j(h_i))$, with $\kappa_{B_i}$ of 21.10. □

**Theorem 22.5 (A cell read alike is pinned).** *As stated: the hypothesis on every active letter, whatever the sign of its exponent.*

*Proof.* By 22.3, which holds for every substituted family at all data, both sides converging together,

$$P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int \prod_i \beta_i(du_i)\; P_{F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]}\big((\kappa_{B_i})_i; \mathbf{s}, \mathbf{t}\big),$$

$\beta_i$ the cell kernels and $F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]$ the core family read at $u$. A letter with exponent 0 is the constant 1 in every period, so the twists of the inactive letters do not enter the period of the core family read at $u$; give them the values $c_{ij} = 1$, $c^*_{ij} = 1$. Under the hypothesis the twists of the active letters at $h_i$ are $c_{ij}\gamma_i(u_i)$ in $K_j$ and $c^*_{ij}\gamma_i(u_i)$ in $\Psi_j$, so $F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]$ has the period of $F_{\mathcal{H}}[\mathbf{c}]$ twisted further by the vector $\boldsymbol{\lambda} = (\gamma_i(u_i))_i$ on the marks of the core, in both letters of every table. The core family read at $u$ has the imbalance of $F$ (22.3), so it is on its balance wall, and the torus relation of 22.1 with this $\boldsymbol{\lambda}$ and constants 1 gives

$$P_{F_{\mathcal{H}}[\boldsymbol{\gamma}(u)]}\big((\kappa_{B_i})_i; \mathbf{s}, \mathbf{t}\big) = \prod_i \gamma_i(u_i)^{-\kappa_{B_i}}\, P_{F_{\mathcal{H}}[\mathbf{c}]}\big((\kappa_{B_i})_i; \mathbf{s}, \mathbf{t}\big).$$

Inserting this and integrating cell by cell gives the display, by Tonelli for real data and by Fubini where the integrals converge absolutely. The step is the reference proof of 22.5, which nowhere uses the sign of an exponent; the hypothesis there is written for the letters with positive exponent. □

**Corollary 22.6 (Read by one hinge value, the period factors through the core).** *As stated: for every substituted family with real data at every cell of which the active letters read the cell by one hinge value.*

*Proof. (1) The first display.* The hypothesis is that of 22.5 with all constants $c_{ij} = c^*_{ij} = 1$, and $F_{\mathcal{H}}[\mathbf{c}] = F_{\mathcal{H}}$; 22.5 gives the first display, in $[0, \infty]$ since the data are real. At a mark $h$ left in place, the cell $\binom{\{h,g\}}{1}$ has both hinge values equal to $\alpha_h$ (21.1), so the hypothesis holds there with $\gamma = \alpha_h$.

*(2) The cells.* The cell family $F_i$ has the tables $\mathcal{A}_{ij}$, cells on $|B_i| + 1$ marks of rank $r_{\mathcal{A}_{ij}}$, and by 21.10

$$\kappa_{B_i} = m_{B_i} - \sum_j \big(s_j(r_{\mathcal{A}_{ij}} - 1) + t_j(|B_i| - r_{\mathcal{A}_{ij}})\big),$$

so $m_{B_i} + 2\sigma - \kappa_{B_i} = \sum_j \big(s_j r_{\mathcal{A}_{ij}} + t_j(|B_i| + 1 - r_{\mathcal{A}_{ij}})\big)$, the degree of the family word of $F_i$: at the masses $(\mathbf{m}|_{B_i}, 2\sigma - \kappa_{B_i})$ it is on its balance wall. Read its period on the section $\{|\alpha_{B_i}| = 1\}$ (8.7), with $\alpha_{B_i} = u \in \Delta_{B_i}$ and $\alpha_{g_i} = v > 0$. By 21.1, $K_{\mathcal{A}_{ij}} = K_{\mathcal{A}_{ij}/g_i}(u)\big(\gamma^K_{\mathcal{A}_{ij}}(u) + v\big)$ and $\Psi_{\mathcal{A}_{ij}} = \Psi_{\mathcal{A}_{ij}\setminus g_i}(u)\big(v + \gamma^{\Psi}_{\mathcal{A}_{ij}}(u)\big)$. Under the hypothesis every active letter carries the factor $(v + \gamma_i(u))$ to the power of its exponent, and an inactive letter has exponent 0; so the word of $F_i$ is the density of the cell kernel without $u^{\mathbf{m}|_{B_i}-\mathbf{1}}$ times $(v + \gamma_i(u))^{-2\sigma}$, $2\sigma = \sum_j (s_j + t_j)$. For $0 < \kappa < 2\sigma$ and $\gamma > 0$, the substitution $v = \gamma w$ and the Beta integral give

$$\int_0^\infty v^{2\sigma-\kappa-1}(v+\gamma)^{-2\sigma}\, dv = \gamma^{-\kappa} \int_0^\infty w^{2\sigma-\kappa-1}(1+w)^{-2\sigma}\, dw = \Gamma^{\,2}_{\,1}\begin{bmatrix} \kappa, & 2\sigma - \kappa \\ & 2\sigma \end{bmatrix} \gamma^{-\kappa},$$

the Beta corner of a linear letter (15.1 with one coefficient). Integrating $v$ first (Tonelli),

$$P_{F_i}\big((\mathbf{m}|_{B_i}, 2\sigma-\kappa_{B_i}); \mathbf{s}, \mathbf{t}\big) = \Gamma_{\,1}^{\,2}\!\begin{bmatrix}\kappa_{B_i}, & 2\sigma-\kappa_{B_i}\\ & 2\sigma\end{bmatrix} \int_{\Delta_{B_i}} \gamma_i^{-\kappa_{B_i}}\, d\beta_i,$$

which identifies the pinned shape integral.

*(3) The walls.* The marks of a core are neither loops nor coloops (21.1), so a core has at least two marks, $q \geq 2$, and $B_i$ is a nonempty proper set of marks; it has width one in every table (21.3(i)). Width one in table $j$ means $d_j(B_i)+d_j(\mathcal{E}\setminus B_i) = r_j - 1$ and $d_j^*(B_i)+d_j^*(\mathcal{E}\setminus B_i) = r_j^* - 1$, so

$$\kappa_{B_i} + \kappa_{\mathcal{E}\setminus B_i} = m_{\mathcal{E}} - \sum_j \big(s_j(r_j-1) + t_j(r_j^*-1)\big) = \kappa_{\mathcal{E}} + 2\sigma,$$

which is $2\sigma$ on the balance wall (21.12; check 2 of S21-23_work/checks.py). If every proper wall is positive, $\kappa_{B_i} > 0$ and $\kappa_{\mathcal{E}\setminus B_i} > 0$, so $0 < \kappa_{B_i} < 2\sigma$.

*(4) The second display.* Multiply the first display by $\prod_i \Gamma_{\,1}^{\,2}\!\begin{bmatrix}\kappa_{B_i}, & 2\sigma-\kappa_{B_i}\\ & 2\sigma\end{bmatrix} \in (0,\infty)$ and use step (2). The left side is finite (8.7), every factor on the right lies in $(0,\infty]$, so every factor is finite. The core family at $(\kappa_{B_i})_i$ has the imbalance of $F$ (22.3), the cell families are balanced by step (2), and a family on its balance wall with a finite period has every proper wall positive (8.7).

*(5) One cell.* With one cell at $h_1$ and the other marks of the core left in place, the left-in-place cell family at $h$, the list $\binom{\{h,g\}}{1}$ in every table with the masses $(m_h, 2\sigma-m_h)$, has the word $(\alpha_h+\alpha_g)^{-2\sigma}$ and the period $\Gamma_{\,1}^{\,2}\!\begin{bmatrix}m_h, & 2\sigma-m_h\\ & 2\sigma\end{bmatrix}$, while $\kappa_{\{h\}} = m_h$; these factors cancel the Beta corners of the marks left in place. The core family is then the family of the cells of the room at the masses $(\kappa_B, \mathbf{m}|_R)$ and $F_1$ that of the cells of the bunch (21.3(iii)), and the second display is the identity of 19.10. In the first display the marks left in place contribute 1, their simplex being a point and their kernel the constant 1, and it is the identity of 19.12 with all constants 1. □

*Remark.* The reference proof (22.6, one table at $(0,t)$) uses of the full lean only that the letter which sees the cells sees each through one function. A family at mixed full leans is covered: on $B=\{1,2\}$ let table 1 at $(s_1,0)$ carry the series block $\binom{\{1,2,g\}}{2}$ and table 2 at $(0,t_2)$ the parallel block $\binom{\{1,2,g\}}{1}$. Both active letters read $B$ through $e_2/e_1$ (21.7(i)), $\kappa_B = m_B - 2\sigma$, the cell family has the word $e_2(\alpha_1,\alpha_2,\alpha_g)^{-2\sigma}$ and the period $\Gamma(2\sigma-m_1)\Gamma(2\sigma-m_2)\Gamma(\kappa_B)/\Gamma(2\sigma)$ (26.9), and the pinned shape integral is $\int_0^1 u^{m_1-1}(1-u)^{m_2-1}(u(1-u))^{-\kappa_B}du = \Gamma(2\sigma-m_1)\Gamma(2\sigma-m_2)/\Gamma(4\sigma-m_B)$, whose quotient is $\Gamma_{\,1}^{\,2}\!\begin{bmatrix}\kappa_B, & 2\sigma-\kappa_B\\ & 2\sigma\end{bmatrix}$, as step (2) says. At $(2\sigma, m_1, m_2) = (3,2,2)$ the period is $\frac{1}{2}$, and the identity holds exactly there and at $(\frac{7}{2},\frac{3}{2},\frac{5}{2})$ (checks 7a to 7d of S21-23_work/checks.py).

**Corollary 22.7 (The pinned shape integral is the period of a load family).** *As stated.*

*Proof.* A load $a_{ij} \neq 0$ occurs only where $s_j \neq 0$, where $\gamma^K_{\mathcal{A}_{ij}} = K_{\mathcal{A}_{ij}\setminus g_i}/K_{\mathcal{A}_{ij}/g_i} = \gamma_i$, and a load $b_{ij} \neq 0$ only where $t_j \neq 0$, where $\gamma^\Psi_{\mathcal{A}_{ij}} = \Psi_{\mathcal{A}_{ij}/g_i}/\Psi_{\mathcal{A}_{ij}\setminus g_i} = \gamma_i$ (21.1). Since $\sum_j(a_{ij}+b_{ij}) = \kappa_{B_i}$,

$$\gamma_i^{-\kappa_{B_i}} \prod_j K^{-s_j}_{\mathcal{A}_{ij}/g_i} \Psi^{-t_j}_{\mathcal{A}_{ij}\setminus g_i} = \prod_j K^{-a_{ij}}_{\mathcal{A}_{ij}\setminus g_i} \Psi^{-(t_j-b_{ij})}_{\mathcal{A}_{ij}\setminus g_i} K^{-(s_j-a_{ij})}_{\mathcal{A}_{ij}/g_i} \Psi^{-b_{ij}}_{\mathcal{A}_{ij}/g_i},$$

the word of the load family, whose imbalance $\kappa_{B_i} - \sum_j(a_{ij}+b_{ij})$ is zero (22.4). Its period is the integral of $u^{\mathbf{m}|_{B_i}-\mathbf{1}}$ times its word over the shape simplex (8.7), the pinned shape integral, and the

product formula is 22.6. If every proper wall of $F$ is positive, then $0 < \kappa_{B_i} < 2\sigma = \sum_j (s_j + t_j)$ (22.6); with $s_j, t_j \geq 0$, raising the loads one after another from 0, each up to its bound $s_j$ or $t_j$ and only on the active letters, reaches the total $\kappa_{B_i}$, and at integer data $\kappa_{B_i}$ is an integer (21.10) and the loads can be integers. The exponents $a_{ij}$, $t_j - b_{ij}$, $s_j - a_{ij}$ and $b_{ij}$ are then nonnegative. When $P_F$ is finite every factor of the product formula is finite, the factors being positive, and the load family, on its balance wall, has every proper wall positive (8.7). □

**Theorem 22.8 (A chain is the power of one operator).** *As stated: at real exponents, the pinned case for the active letters.*

*Proof.* The reference proof of 22.8 applies verbatim at real exponents. It uses pooling (19.8, for every family at all data, both sides converging together), the nonnegativity of the integrands (at real exponents the letters are positive, and so is every power of them), the scalar twist of a mark (22.1) and the Beta integral $\int_0^1 v^{m_{g_{\mathrm{in}}}-1}(1-v)^{m_{g_{\mathrm{out}}}-1}(1-v+\lambda v)^{-2\sigma}dv = \lambda^{-m_{g_{\mathrm{in}}}}\Gamma\,{}_2{}_1\!\left[\begin{matrix} m_{g_{\mathrm{in}}},\ m_{g_{\mathrm{out}}} \\ 2\sigma \end{matrix}\right]$ at $m_{g_{\mathrm{in}}} + m_{g_{\mathrm{out}}} = 2\sigma$; none of these uses the sign of an exponent. In the pinned case the reference assumes that all the active letters read the cell through one function; a letter of exponent 0 is absent from every period, which is the hypothesis as stated. □

**Theorem 22.9 (Letters by transfer).** *As stated: the reduction of the states to linearly independent ones, and the bound on the interface dimension for a multivector of any tensor rank.*

*Proof.* The products of transfer matrices are the reference's. *The stations.* The stations of level $k$ of $S_l$ give a level factorization with as many terms (the children factorization, 21.15), so their number is at least $\operatorname{rank}\Theta^{(k)}_{S_l}$, and the number of states is at least the interface dimension.

*The reduction.* The child of a pattern $p$ of $S_l$ at level $k$ is $\sum_X \theta_{p\sqcup X}\alpha^X$, the polynomial of the row $p$ of $\Theta^{(k)}_{S_l}$, so the children of level $k$ span a space of dimension $\operatorname{rank}\Theta^{(k)}_{S_l}$. Splitting $X$ along the step $B_{l-1}$, the child at $p$ of $S_{l-1}$ is

$$\sum_X \theta_{p\sqcup X}\alpha^X = \sum_{p'\subseteq B_{l-1}} \alpha^{p'} \sum_{X'} \theta_{(p\sqcup p')\sqcup X'}\alpha^{X'},$$

the sum over $p'$ of $\alpha^{p'}$ times the child at $p\sqcup p'$ of $S_l$. Choose at every $S_l$ and every level a basis of the span of the children and write every child in it; the display expresses each basis element at $S_{l-1}$, a combination of children of $S_{l-1}$, through the basis elements at $S_l$, with coefficients polynomial in $\alpha_{B_{l-1}}$. These are transfer matrices on $\sum_k \operatorname{rank}\Theta^{(k)}_{S_l}$ states at $S_l$, and the products give $K$; the same coefficients, read on the complements, give $\Psi$ (21.15).

*The bound.* By 21.17(i), $\operatorname{rank}\Theta^{(k)}_{S_l} \leq \operatorname{ir}_k(\boldsymbol{\omega})^2$, and $\leq \frac{1}{2}\operatorname{ir}_k(\boldsymbol{\omega})(\operatorname{ir}_k(\boldsymbol{\omega})+1)$ when $\boldsymbol{\omega}$ is real; by (ii), $\operatorname{ir}_k(\boldsymbol{\omega}) \leq \sum_i \binom{c_i}{k-d_i}$. The functions $q \mapsto q^2$ and $q \mapsto \frac{1}{2}q(q+1)$ increase on $[0,\infty)$, and summing over $k$ gives the bound. For a determinantal table, of tensor rank one with $c_1 = c(S_l)$ and $d_1 = d(S_l)$ (21.17(ii)), the bound is $\sum_m \binom{c}{m}^2 = \binom{2c}{c}$, $c = c(S_l)$, counting the $c$-subsets of a set of $2c$ elements by their intersection with a fixed half. □

**Theorem 22.10 (Integer joint moments by fixed matrices).** *As stated: all the letters of a family at once, along a chain of bunches with any steps and any independent measures on them.*

*Proof.* This is the reference proof of 22.10 with the tensor product taken over all the tables and with one matrix per step. A product of scalars $\mathbf{u}_i^{\mathsf{T}} A_{i,0}\cdots A_{i,L-1}\mathbf{v}_i$, $i = 1,\dots,n$, equals $(\bigotimes_i \mathbf{u}_i)^{\mathsf{T}}\big(\bigotimes_i A_{i,0}\big)\cdots\big(\bigotimes_i A_{i,L-1}\big)(\bigotimes_i \mathbf{v}_i)$ by the mixed-product rule $(\bigotimes_i A_i)(\bigotimes_i B_i) =$

$\bigotimes_i (A_i B_i)$. Apply it to the $a_j$ copies of $K_j$ and the $b_j$ copies of $\Psi_j$ for every $j$: the factor of the step $l$ is $\bigotimes_j \big(\mathbf{M}^K_{j,l}(\alpha_{B_l})^{\otimes a_j} \otimes \mathbf{M}^\Psi_{j,l}(\alpha_{B_l})^{\otimes b_j}\big)$, a matrix function of $\alpha_{B_l}$ alone. Every entry of the product is a finite sum of products of one entry of each step factor; the measures of the steps are independent, so the integral of such a product is the product of the integrals, by Tonelli for nonnegative entries and by Fubini under the stated finiteness, and integrating entrywise replaces the factor of the step $l$ by $\mathbf{A}_l$.

For identical steps every $\mathbf{A}_l$ is one matrix $\mathbf{A}$ of the stated size, and the scalars $\mathbf{u}^\top \mathbf{A}^L \mathbf{v}$ satisfy the linear recurrence whose coefficients are those of the characteristic polynomial of $\mathbf{A}$, which annihilates $\mathbf{A}$. With independent laws $\mathrm{Gamma}_{m_x}$ on the marks, $\prod_j K_j^{a_j} \Psi_j^{b_j}$ is homogeneous of degree $d = \sum_j (a_j r_j + b_j r_j^*)$, and the draw splits into an independent total of law $\mathrm{Gamma}_{m_\mathcal{E}}$ and a shape of law $\mathrm{Dir}_\mathbf{m}$ (8.12(ii)), the total having the moment $(m_\mathcal{E})_d$ of order $d$. With $\mathrm{Dir}_{\mathbf{m}|_{B_l}}$ on a step, the entries of $\mathbf{A}_l$ are the Dirichlet moments of the monomials (3.11). □

**Theorem 22.11 (Gluing across a bunch of any width).** *As stated: the share display for one table at a full lean at every width, and the count $2N-1$ of the free shares at width one.*

*Proof.* The main display is the reference's. For one table at the full lean $(0,t)$ in level form, one station per level, the active letter $\Psi$ has the $c(B)+1$ terms of the levels $k = d(B), \dots, r(B)$, so the stacked Gamma of the display is $\prod_k \Gamma(b_k)/\Gamma(t)$ and the pins are $\sum_k b_k = t$ and $\sum_k (r(B)-k) b_k = \kappa_B$, the main pin without the shares of $K$. The fibre integral (4.1) has $c(B)+1$ shares and two pins. Integrating the delta of the pins against the imaginary parts of $b_{d(B)}$ and $b_{r(B)}$ divides by $|\det|$ of the matrix of the two pins in these shares,

$$\begin{pmatrix} 1 & 1 \\ c(B) & 0 \end{pmatrix}, \qquad |\det| = c(B)$$

(check 8a of S21-23_work/checks.py), and leaves $(2\pi)^{2-(c(B)+1)}$ times the integral over the imaginary parts of the $c(B)-1$ free shares, which is $\frac{1}{(2\pi i)^{c(B)-1}} \int \prod db_k$. The pins solved for $b_{d(B)}$ and $b_{r(B)}$ give the stated values; at $c(B)=2$ they are $\frac{\kappa_B - z}{2}$ and $\frac{2t-\kappa_B-z}{2}$ with $b_{d(B)+1} = z$ (check 8b). At $c(B)=1$ the fibre has dimension 0, the matrix of the pins has $|\det| = 1$, and the integral is the value at $(b_{d(B)}, b_{r(B)}) = (\kappa_B, t-\kappa_B)$, the Beta corner times the two inner periods, which is 19.14: the bunch and room letters of the two levels are the minors $\Psi_{\mathcal{B}/R}, \Psi_{\mathcal{B}|B}$ and $\Psi_{\mathcal{B}|R}, \Psi_{\mathcal{B}/B}$ up to constant factors of the factorization that cancel in the product. At width one with factoring weights each level has one station (12.19), so each of the $2N$ active letters has two terms, and the free shares are $4N - 2N - 1 = 2N-1$, the dimension of the fibre integral of 19.13, of which the main display is then the formula. □

*Remark.* The paragraph after 22.11 gives the Mellin transform of the scale integral $I(\boldsymbol{\zeta}) = \int_0^\infty x^{\kappa_B - 1} \big(1 + \sum_{i=1}^{c-1} \zeta_i x^i + x^c\big)^{-t} dx$, $c = c(B)$. By Tonelli and the substitutions $\zeta_i = \eta_i x^{-i}$, then $x^c = w$,

$$\int \prod_i \zeta_i^{z_i - 1} I(\boldsymbol{\zeta})\, d\boldsymbol{\zeta} = \frac{\prod_i \Gamma(z_i)\, \Gamma(t - \sum_i z_i)}{\Gamma(t)} \int_0^\infty x^{\kappa_B - \sum_i i z_i - 1} (1+x^c)^{\sum_i z_i - t} dx = \frac{\prod_{k=d(B)}^{r(B)} \Gamma(b_k)}{c\, \Gamma(t)}$$

at $b_{r(B)-i} = z_i$, $b_{d(B)} = (\kappa_B - \sum_i i z_i)/c$ and $b_{r(B)} = t - b_{d(B)} - \sum_i z_i$, wherever these have positive real parts: the share kernel of the theorem with the factor $1/c$ of the pin.

## 50.23 Chapter 23: the law of a cell

**Definition 23.1 (The law of a cell family).** As stated: the Mellin transform of the law of a cell family, and the transforms and the hinge measures of a cell family, those of one cell being their readings.

*Proof.* The definition asserts three things. *The Mellin transform.* For complex $\mathbf{z}$ and real positive coordinates, $|\boldsymbol{\varrho}_F^{-\mathbf{z}}| = \boldsymbol{\varrho}_F^{-\operatorname{Re}\mathbf{z}}$, so the integral converges absolutely where it is finite at the real part, with modulus at most that value; for one cell $\boldsymbol{\varrho}_F^{-\mathbf{z}} = (\gamma_{\mathcal{A}}^{\Psi}/\gamma_{\mathcal{A}}^{K})^{-z_{1,\Psi}} = \varrho_{\mathcal{A}}^{-z}$, and the second form is the image of the measure. The base form of one cell is $\phi_F$ with $(\gamma_{\mathcal{A}}^{K})^{-\kappa} = K_{\mathcal{A}/g}^{\kappa} K_{\mathcal{A}\setminus g}^{-\kappa}$.

*Homogeneity.* Substitute $\alpha = c\beta$ in the orthant forms: $\alpha^{\mathbf{m-1}} d\alpha = c^{m_B} \beta^{\mathbf{m-1}} d\beta$, the letters of the deletions have the degrees $r_{\mathcal{A}_j}$ and $|B| - r_{\mathcal{A}_j}$, those of the contractions $r_{\mathcal{A}_j} - 1$ and $|B| - r_{\mathcal{A}_j} + 1$, and the hinge values have degree one. So

$$f_F(\mathbf{x}, \mathbf{y}) = c^{m_B - \sum_j (s_j r_{\mathcal{A}_j} + t_j(|B| - r_{\mathcal{A}_j}))} f_F(\mathbf{x}/c, c\mathbf{y}) = c^{\kappa_F - |\mathbf{s}|} f_F(\mathbf{x}/c, c\mathbf{y}),$$

and in the same way $\check{f}_F(\mathbf{x}, \mathbf{y}) = c^{\kappa_F - |\mathbf{t}|} \check{f}_F(\mathbf{x}/c, c\mathbf{y})$. With $c = x_1$ each transform is fixed by its values at $x_1 = 1$.

*One cell.* At $N = 1$, $f_F(1, y)$ and $\check{f}_F(1, y)$ are the integrals called the orthant forms of the transforms of one cell, and the images of $e^{-\lambda} \mathfrak{r}_F$ and $e^{-\lambda} \check{\mathfrak{r}}_F$ under $(\lambda, \mu) \mapsto \mu$ are the images of $\alpha^{\mathbf{m-1}} K_{\mathcal{A}\setminus g}^{-s} \Psi_{\mathcal{A}\setminus g}^{-t} e^{-1/\gamma_{\mathcal{A}}^{K}} d\alpha$ under $\gamma_{\mathcal{A}}^{\Psi}$ and of $\alpha^{\mathbf{m-1}} K_{\mathcal{A}/g}^{-s} \Psi_{\mathcal{A}/g}^{-t} e^{-1/\gamma_{\mathcal{A}}^{\Psi}} d\alpha$ under $\gamma_{\mathcal{A}}^{K}$. □

**Theorem 23.3 (The series closure at positive exponents).** *As stated: the closure through the transforms of the cell families, each transform one Gramma of index $|\mathbf{s}| - \kappa_{B_i}$ at every data, and its Breta shape reading wherever the family of the hinge minors is balanced.*

*Proof. The closure.* The responses of the reference, proof and statement of 23.3, are the integrals over $\mathbb{R}_{>0}^{B_i}$ of $\alpha^{\mathbf{m}|_{B_i} - \mathbf{1}} \prod_j K_{\mathcal{A}_{ij}\setminus g_i}^{-s_j} \Psi_{\mathcal{A}_{ij}\setminus g_i}^{-t_j}$ against $\exp\big( - \sum_j x_j K_{\mathcal{A}_{ij}/g_i} / K_{\mathcal{A}_{ij}\setminus g_i} - \sum_j y_j \Psi_{\mathcal{A}_{ij}/g_i} / \Psi_{\mathcal{A}_{ij}\setminus g_i} \big)$ with $y_1 = 1$; since $K_{\mathcal{A}/g}/K_{\mathcal{A}\setminus g} = 1/\gamma_{\mathcal{A}}^{K}$ and $\Psi_{\mathcal{A}/g}/\Psi_{\mathcal{A}\setminus g} = \gamma_{\mathcal{A}}^{\Psi}$, they are the transforms $f_{F_i}(\mathbf{x}, (1, y_2, \dots, y_N))$ of 23.1, and the first display is the reference's. The statement on repeated cells is read off the product.

*One table.* By the homogeneity of 23.1 with $c = x$, $f_{F_i}(x, 1) = x^{\kappa_{B_i} - s} f_{\mathcal{A}_i}(x)$. The core $\binom{\{h_1, \dots, h_q\}}{q-1}$ has rank $q - 1$ and corank 1, and the core family read at $u$ is balanced (22.3), so $\sum_i \kappa_{B_i} = (q-1)s + t$. The power of $x$ is then $x^{s-1+(q-1)s+t-qs} = x^{t-1}$, and the closure is $\frac{1}{\Gamma(s)\Gamma(t)} \int_0^{\infty} x^{t-1} \prod_i f_{\mathcal{A}_i}(x)\, dx$.

*The Gramma.* Fix $i$ and write $\alpha = \rho u$, $u \in \Delta_{B_i}$, $\rho > 0$, so that $\alpha^{\mathbf{m}|_{B_i} - \mathbf{1}} d\alpha = \rho^{m_{B_i} - 1} u^{\mathbf{m}|_{B_i} - \mathbf{1}} d\rho\, d\omega$ (2.7). The marks $g_i$ being neither loops nor coloops, $K_{\mathcal{A}_{ij}\setminus g_i}$ is homogeneous of degree $r_{\mathcal{A}_{ij}}$ and $\Psi_{\mathcal{A}_{ij}\setminus g_i}$ of degree $|B_i| - r_{\mathcal{A}_{ij}}$, and the hinge values of degree one. So the power of $\rho$ is $\rho^{\nu - 1}$ with

$$\nu = m_{B_i} - \sum_j \big( s_j r_{\mathcal{A}_{ij}} + t_j(|B_i| - r_{\mathcal{A}_{ij}}) \big) = \kappa_{B_i} - |\mathbf{s}|,$$

and the exponential is $\exp(-a(u)/\rho - b(u)\rho)$ with $a = \sum_j x_j / \gamma_{\mathcal{A}_{ij}}^{K}$ and $b = \sum_j y_j \gamma_{\mathcal{A}_{ij}}^{\Psi}$. By 2.1 and the reflection (2.3),

$$\int_0^{\infty} \rho^{\nu - 1} e^{-b\rho - a/\rho}\, d\rho = \mathsf{G}_{\nu;(1,-1)}(b, a) = \mathsf{G}_{-\nu;(1,-1)}(a, b),$$

and Tonelli gives the display of the transform.

*Index* 0. The family of the hinge minors has the degree $\sum_j \big(s_j r_{\mathcal{A}_{ij}} + t_j(|B_i| - r_{\mathcal{A}_{ij}})\big)$ at the masses $\mathbf{m}|_{B_i}$, so it is on its balance wall exactly when $\kappa_{B_i} = |\mathbf{s}|$; at the centre of one table, every mass $\sigma$ and $s = t = \sigma$, $\kappa_{B_i} = \sigma|B_i| - \sigma(r-1) - \sigma(|B_i| - r) = s$. Torus homogeneity (2.4(iii)) with $\lambda = a$ gives $\mathsf{G}_{0;(1,-1)}(a,b) = \mathsf{G}_{0;(1,-1)}(1,ab)$. With every proper wall positive, the form of the family of the hinge minors divided by its period is its Breta shape law (8.10), which gives the expectation; for one table, $ab = xy\,\gamma^{\Psi}_{\mathcal{A}_i}/\gamma^{K}_{\mathcal{A}_i} = xy\varrho_{\mathcal{A}_i}$. The period of $k$ copies is the first display with all $F_i$ equal. □

**Theorem 23.6 (The law of a substitution).** *As stated: (i) and (ii) for families at real data; the present (i) was the gluing of a family, and (ii) for one table at $s, t > 0$.*

*Proof. (ii)* This is the reference proof of 23.6(ii) with the index $j$, the steps being those of 23.2 applied to one cell family after another. Write $B = \mathcal{E}_{\mathcal{H}} \sqcup \bigsqcup_i B_i$ and $\mathcal{A}_j = \mathcal{H}_j\langle \mathcal{A}_{1j}, \dots, \mathcal{A}_{kj}\rangle$, a cell with the hinge $g_0$ (23.5). For a Borel $H \geq 0$, $\int H\, d((\boldsymbol{\varrho}_F)_*\phi_F)$ is the integral over the section $\{|\alpha| = 1\}$ of the orthant form

$$\omega = \alpha^{\mathbf{m-1}} H(\boldsymbol{\varrho}_F)\, K_{\mathcal{A}_1/g_0}^{\kappa_F - s_1} K_{\mathcal{A}_1\setminus g_0}^{-\kappa_F} \prod_{j\geq 2} K_{\mathcal{A}_j/g_0}^{-s_j} \prod_j \Psi_{\mathcal{A}_j\setminus g_0}^{-t_j}\, d\alpha,$$

homogeneous of degree $m_B - \sum_j \big(s_j(r_{\mathcal{A}_j} - 1) + t_j(|B| - r_{\mathcal{A}_j})\big) - \kappa_F = 0$, the base form being $u^{\mathbf{m-1}} \prod_j K_{\mathcal{A}_j/g_0}^{-s_j} \Psi_{\mathcal{A}_j\setminus g_0}^{-t_j} (K_{\mathcal{A}_1\setminus g_0}/K_{\mathcal{A}_1/g_0})^{-\kappa_F} d\omega$. By 23.5, with the hinge values of the cells at their own marks,

$$K_{\mathcal{A}_j\setminus g_0} = \prod_i K_{\mathcal{A}_{ij}/g_i}\, K_{\mathcal{H}_j\setminus g_0}\big(\alpha_{\mathcal{E}_{\mathcal{H}}}, (\gamma^K_{\mathcal{A}_{ij}})_i\big), \qquad K_{\mathcal{A}_j/g_0} = \prod_i K_{\mathcal{A}_{ij}/g_i}\, K_{\mathcal{H}_j/g_0}\big(\alpha_{\mathcal{E}_{\mathcal{H}}}, (\gamma^K_{\mathcal{A}_{ij}})_i\big),$$

and the same for $\Psi$ with $\prod_i \Psi_{\mathcal{A}_{ij}\setminus g_i}$ and the hinge values $\gamma^{\Psi}_{\mathcal{A}_{ij}}$. The products over the cells cancel in the quotients, so the hinge values of $\mathcal{A}_j$ at $g_0$ are those of $\mathcal{H}_j$ read at $(\alpha_{\mathcal{E}_{\mathcal{H}}}, (\gamma^K_{\mathcal{A}_{ij}})_i)$ and $(\alpha_{\mathcal{E}_{\mathcal{H}}}, (\gamma^{\Psi}_{\mathcal{A}_{ij}})_i)$.

Apply 23.2 to $F_1$ with $Z = \mathcal{E}_{\mathcal{H}} \sqcup \bigsqcup_{i\geq 2} B_i$. Assign to the letter $K_{\mathcal{A}_{1j}}$ the functions $\prod_{i\geq 2} K_{\mathcal{A}_{ij}/g_i} \cdot K_{\mathcal{H}_j/g_0}$ and $\prod_{i\geq 2} K_{\mathcal{A}_{ij}/g_i} \cdot K_{\mathcal{H}_j\setminus g_0}$, read at the hinge value of the cell at $g_1$, with the exponents $s_j$ and 0 for $j \geq 2$ and $s_1 - \kappa_F$ and $\kappa_F$ for $j = 1$; and to the letter $\Psi_{\mathcal{A}_{1j}}$ the functions $\prod_{i\geq 2} \Psi_{\mathcal{A}_{ij}\setminus g_i} \cdot \Psi_{\mathcal{H}_j\setminus g_0}$ and $\prod_{i\geq 2} \Psi_{\mathcal{A}_{ij}\setminus g_i} \cdot \Psi_{\mathcal{H}_j/g_0}$ with the exponents $t_j$ and 0. The exponents assigned to each letter add to its exponent, the letters of the hinge minors of $F_1$ collect to $K_{\mathcal{A}_{1j}/g_1}^{-s_j} \Psi_{\mathcal{A}_{1j}\setminus g_1}^{-t_j}$, the product of the assigned powers is the rest of $\omega$, and $H(\boldsymbol{\varrho}_F)$ is a Borel function of the values of the assigned functions, which give the hinge values of every $\mathcal{A}_j$ as quotients. The lemma collapses $B_1$ to $g_1$ with the mass $\kappa_{F_1}$ and replaces the hinge values of the cells at $g_1$ by $x_{1,j,K}\alpha_{g_1}$ and $x_{1,j,\Psi}\alpha_{g_1}$, the twist at $g_1$ by the point $\mathbf{x}_1$ of the law of $F_1$. Repeat for $F_2, \dots, F_k$, moving the integrals over the laws outside (Tonelli). What remains is the integral over a section of the form of the cell family $F_{\mathcal{H}}[\mathbf{x}]$, its hinge minors at $g_0$ read with the twists, its masses $\mathbf{m}|_{\mathcal{E}_{\mathcal{H}}}$ and $\kappa_{F_i}$, the power $\kappa_F$ in place of its hinge wall, and $H$ of its ratio vector inserted. Its hinge wall is

$$m_{\mathcal{E}_{\mathcal{H}}} + \sum_i \kappa_{F_i} - \sum_j \big(s_j(r_{\mathcal{H}_j} - 1) + t_j(|\mathcal{E}_{\mathcal{H}}| + k - r_{\mathcal{H}_j})\big),$$

and it equals $\kappa_F$: insert $\kappa_{F_i} = m_{B_i} - \sum_j (s_j(r_{\mathcal{A}_{ij}} - 1) + t_j(|B_i| - r_{\mathcal{A}_{ij}}))$, $r_{\mathcal{A}_j} = r_{\mathcal{H}_j} + \sum_i (r_{\mathcal{A}_{ij}} - 1)$ and $|B| = |\mathcal{E}_{\mathcal{H}}| + \sum_i |B_i|$ (23.5); the coefficients of $s_j$ and of $t_j$ agree term by term (check 1 of S21-23_work/checks.py). So the remaining integral is $\int H\, d((\boldsymbol{\varrho}_{F_{\mathcal{H}}[\mathbf{x}]})_*\phi_{F_{\mathcal{H}}[\mathbf{x}]})$, and the mixture follows;

the hinge wall does not depend on $\mathbf{x}$. With $H = \boldsymbol{\varrho}^{-\mathbf{z}}$, a nonnegative Borel function for real $\mathbf{z}$, the Mellin transforms mix in the same way.

*(i)* The same steps with the family word of $F$ in place of the base form: on the balance wall the form of $F$ is homogeneous of degree zero, its letters are $\prod_i K_{\mathcal{A}_{ij}/g_i} \cdot K_{\mathcal{H}_j}$ and $\prod_i \Psi_{\mathcal{A}_{ij}\setminus g_i} \cdot \Psi_{\mathcal{H}_j}$ read at the hinge values (23.5), the factors $K_{\mathcal{H}_j}$ and $\Psi_{\mathcal{H}_j}$ are assigned to the letters of each cell family in turn with the exponents $s_j$ and $t_j$, and 23.2 collapses every cell to its marker with the mass $\kappa_{F_i}$ and the twist by the point of its law. This is the reference's remark on families (rem:familiesnodes): 23.2 holds at real exponents, its proof collecting degrees and using Tonelli. For the core $\binom{\{g_1,g_2\}}{1}$, twisted at $g_1$ by $\mathbf{x}_1$ and at $g_2$ by $\mathbf{x}_2$, table $j$ has $K_j = x_{1,j,K}\alpha_{g_1} + x_{2,j,K}\alpha_{g_2}$ and $\Psi_j = x_{1,j,\Psi}\alpha_{g_1} + x_{2,j,\Psi}\alpha_{g_2}$, and on the section $\alpha_{g_2} = 1$ its period is the displayed integral, a Breta of one bunch (15.1); the substitution $v \mapsto 1/v$ multiplies the integrand by $v^{2\sigma}$ and turns $v^{\kappa_{F_1}-1}dv$ into $v^{-\kappa_{F_1}-1}dv$, which gives the swap since $\kappa_{F_1} + \kappa_{F_2} = 2\sigma$. □

**Proposition 23.8 (Faces of one law).** *As stated, for every cell family with plain tables and $s_j, t_j > 0$.*

*Proof.* Write $\kappa = \kappa_F$ and $\gamma_j^K$, $\gamma_j^\Psi$ for the hinge values of $\mathcal{A}_j$. The reference proof of 23.8 is the case $N = 1$; the steps below are that proof with the index $j$, the changes being the bookkeeping of the powers of $\gamma_1^K$ and $\gamma_1^\Psi$, which pass the integrals from the simplex to the law.

*(i)* In the orthant form of $f_F$ write $\alpha = \rho u$ (2.7). As in the proof of 23.3 in this part, the power of $\rho$ is $\rho^{\kappa-|\mathbf{s}|-1}$ and

$$f_F(\mathbf{x},\mathbf{y}) = \int_{\Delta_B} \mathsf{G}_{|\mathbf{s}|-\kappa;(1,-1)}\Big(\sum_j \frac{x_j}{\gamma_j^K(u)},\ \sum_j y_j\gamma_j^\Psi(u)\Big)\, u^{\mathbf{m}-\mathbf{1}} \prod_j K_{\mathcal{A}_j\setminus g}^{-s_j}\Psi_{\mathcal{A}_j\setminus g}^{-t_j}\, d\omega(u).$$

Since $K_{\mathcal{A}_j\setminus g} = \gamma_j^K K_{\mathcal{A}_j/g}$, the measure is $(\gamma_1^K)^\kappa \prod_j (\gamma_j^K)^{-s_j}\phi_F = (\gamma_1^K)^{\kappa-|\mathbf{s}|}\prod_j(\gamma_j^K/\gamma_1^K)^{-s_j}\phi_F$. Torus homogeneity (2.4(iii)) with $\lambda = 1/\gamma_1^K$, $\mathsf{G}_{\nu;(1,-1)}(\lambda\xi_1,\xi_2/\lambda) = \lambda^{-\nu}\mathsf{G}_{\nu;(1,-1)}(\xi_1,\xi_2)$, gives, at $\mathbf{z} = \boldsymbol{\varrho}_F(u)$,

$$\mathsf{G}_{|\mathbf{s}|-\kappa;(1,-1)}\Big(\sum_j \frac{x_j}{\gamma_j^K},\ \sum_j y_j\gamma_j^\Psi\Big) = (\gamma_1^K)^{|\mathbf{s}|-\kappa}\,\mathsf{G}_{|\mathbf{s}|-\kappa;(1,-1)}\Big(\sum_j \frac{x_j}{z_{j,K}},\ \sum_j y_j z_{j,\Psi}\Big).$$

The powers of $\gamma_1^K$ cancel, and the image under $\boldsymbol{\varrho}_F$ gives the first law form. For $\check{f}_F$ the contracted letters have the degrees $r_{\mathcal{A}_j} - 1$ and $|B| - r_{\mathcal{A}_j} + 1$, so the power of $\rho$ is $\rho^{\kappa-|\mathbf{t}|-1}$, the index is $|\mathbf{t}| - \kappa$, and the coefficients are $\sum_j x_j/\gamma_j^\Psi$ and $\sum_j y_j\gamma_j^K$; the measure is $u^{\mathbf{m}-\mathbf{1}}\prod_j K_{\mathcal{A}_j/g}^{-s_j}\Psi_{\mathcal{A}_j/g}^{-t_j}d\omega = (\gamma_1^K)^\kappa\prod_j(\gamma_j^\Psi)^{-t_j}\phi_F$, since $\Psi_{\mathcal{A}_j/g} = \gamma_j^\Psi\Psi_{\mathcal{A}_j\setminus g}$. Torus homogeneity with $\lambda = 1/\gamma_1^\Psi$ extracts $(\gamma_1^\Psi)^{|\mathbf{t}|-\kappa}$, and $(\gamma_1^K)^\kappa\prod_j(\gamma_j^\Psi)^{-t_j}(\gamma_1^\Psi)^{|\mathbf{t}|-\kappa} = z_{1,\Psi}^{-\kappa}\prod_j(z_{j,\Psi}/z_{1,\Psi})^{-t_j}$, with $z_{1,\Psi}/z_{j,\Psi} = \gamma_1^\Psi/\gamma_j^\Psi$ and $z_{j,K}/z_{1,\Psi} = \gamma_j^K/\gamma_1^\Psi$: the second law form. At $N = 1$, $z_{1,K} = 1$, $z_{1,\Psi} = \varrho_{\mathcal{A}}$ and $\mathbf{x} = 1$ give $\mathsf{G}_{s-\kappa;(1,-1)}(1,\varrho_{\mathcal{A}}y)$ and $\mathsf{G}_{t-\kappa;(1,-1)}(1,y/\varrho_{\mathcal{A}})\varrho_{\mathcal{A}}^{-\kappa}$.

*(ii)* The Laplace forms are the definitions of the images. For the law form of $\mathfrak{r}_F$ write $\alpha = \rho u$ and $p = \rho\gamma_1^K(u)$: then $1/\gamma_j^K(\alpha) = 1/(pz_{j,K})$ and $\gamma_j^\Psi(\alpha) = pz_{j,\Psi}$ at $\mathbf{z} = \boldsymbol{\varrho}_F(u)$, and $\rho^{\kappa-|\mathbf{s}|-1}d\rho = (\gamma_1^K)^{|\mathbf{s}|-\kappa}p^{\kappa-|\mathbf{s}|-1}dp$, which with the measure of (i) leaves $\prod_j z_{j,K}^{-s_j}\phi_F(du)\otimes p^{\kappa-|\mathbf{s}|-1}dp$. A Laplace transform of a positive measure is completely monotone on the interior of the set where it is finite, by differentiation under the integral, the derivatives being dominated near each such point by the integrand at a point with smaller coordinates, and it is log-convex along segments by Hölder's inequality. For one cell, with $\lambda = 1/p$ and $p = 1/q$, the measure $e^{-\lambda}\mathfrak{r}_F$ becomes $e^{-q}q^{s-\kappa-1}dq$ against the law, and the coordinate $\mu = pz$ becomes $z/q$; for $\check{\mathfrak{r}}_F$, with $p = \rho\gamma^\Psi(u)$, the same

computation gives the image of $z^{-\kappa}(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dz)\otimes p^{\kappa-t-1}dp$ under $(z,p)\mapsto(1/p,p/z)$, and $e^{-\lambda}$ and $p=1/q$ give the image of $z^{-\kappa}(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dz)\otimes q^{t-\kappa-1}e^{-q}dq$ under $(z,q)\mapsto 1/(zq)$.

*(iii)* Insert the Laplace form of (ii) and integrate first in $x_j$, $j\geq 2$, and $y_j$ (Tonelli): $\int_0^\infty x^{a-1}e^{-x\lambda}dx=\Gamma(a)\lambda^{-a}$. With $\lambda_j=1/(pz_{j,K})$, $\mu_j=pz_{j,\Psi}$ and $x_1=1$,

$$\prod_{j\geq 2}\Gamma(a_j)\prod_j\Gamma(b_j)\iint e^{-1/p}\prod_{j\geq 2}(pz_{j,K})^{a_j}\prod_j(pz_{j,\Psi})^{-b_j}\prod_j z_{j,K}^{-s_j}\,p^{\kappa-|\mathbf{s}|-1}dp\,(\boldsymbol{\varrho}_F)_*\phi_F(d\mathbf{z}).$$

The power of $p$ is $p^{-a_1-1}$ with $a_1=\sum_j b_j-\sum_{j\geq 2}a_j-\kappa+|\mathbf{s}|$, the given $a_1$ by the hyperplane, and $\int_0^\infty e^{-1/p}p^{-a_1-1}dp=\Gamma(a_1)$ for $a_1>0$; the power of $\mathbf{z}$ is $\prod_j z_{j,K}^{-(s_j-a_j)}z_{j,\Psi}^{-b_j}$, $z_{1,K}=1$ (check 5). For $\check{f}_F$ the law form of $\check{\mathfrak{r}}_F$, obtained as in (ii) with $p=\rho\gamma_1^{\Psi}$, is the image of $z_{1,\Psi}^{-\kappa}\prod_j(z_{j,\Psi}/z_{1,\Psi})^{-t_j}(\boldsymbol{\varrho}_F)_*\phi_F(d\mathbf{z})\otimes p^{\kappa-|\mathbf{t}|-1}dp$ under $(\mathbf{z},p)\mapsto\big((z_{1,\Psi}/(pz_{j,\Psi}))_j,(pz_{j,K}/z_{1,\Psi})_j\big)$; the same integration leaves $p^{-a_1-1}$ with $a_1=\sum_j b_j-\sum_{j\geq 2}a_j-\kappa+|\mathbf{t}|$, and the total power of $z_{1,\Psi}$ is $-\sum_{j\geq 2}a_j+\sum_j b_j-\kappa+\sum_{j\geq 2}t_j=a_1-t_1$, so the power of $\mathbf{z}$ is $\prod_j z_{j,K}^{-b_j}z_{j,\Psi}^{-(t_j-a_j)}$. Both points satisfy $\sum_j(z_{j,K}+z_{j,\Psi})=\kappa$. At $N=1$, $b=z$ and $a=z+s-\kappa$, the first gives $\Gamma(z)\Gamma(z+s-\kappa)\int\varrho_{\mathcal{A}}^{-z}d\phi_{\mathcal{A}}$, and the second, with $a=z+t-\kappa$, $\Gamma(z)\Gamma(z+t-\kappa)\int\varrho_{\mathcal{A}}^{-(\kappa-z)}d\phi_{\mathcal{A}}$.

*(iv)* Table by table, the dual table $\mathcal{A}^*$ has the bases $(B\sqcup\{g\})\setminus T$ with the weights $\theta_T$, and splitting by whether $g\in T$,

$$K_{\mathcal{A}^*/g}=\Psi_{\mathcal{A}\setminus g},\quad K_{\mathcal{A}^*\setminus g}=\Psi_{\mathcal{A}/g},\quad \Psi_{\mathcal{A}^*\setminus g}=K_{\mathcal{A}/g},\quad \Psi_{\mathcal{A}^*/g}=K_{\mathcal{A}\setminus g},$$

its rank is $|B|+1-r_{\mathcal{A}}$, and $g$ is neither a loop nor a coloop of it. So $\gamma^K_{\mathcal{A}^*}=\gamma^{\Psi}_{\mathcal{A}}$ and $\gamma^{\Psi}_{\mathcal{A}^*}=\gamma^K_{\mathcal{A}}$, and the hinge wall at $(t_j,s_j)$ is $m_B-\sum_j\big(t_j(|B|-r_{\mathcal{A}_j})+s_j(r_{\mathcal{A}_j}-1)\big)=\kappa$. The base form of $F^*$ is $u^{\mathbf{m}-\mathbf{1}}\prod_j\Psi_{\mathcal{A}_j\setminus g}^{-t_j}K_{\mathcal{A}_j/g}^{-s_j}(\gamma_1^{\Psi})^{-\kappa}=(\gamma_1^{\Psi}/\gamma_1^K)^{-\kappa}\phi_F$, its ratio vector is $(\gamma_j^{\Psi}/\gamma_1^{\Psi},\gamma_j^K/\gamma_1^{\Psi})_j=(z_{j,\Psi}/z_{1,\Psi},z_{j,K}/z_{1,\Psi})_j$ at $\mathbf{z}=\boldsymbol{\varrho}_F$, which gives the law, and the orthant forms of $f_{F^*}$ and $\check{f}_{F^*}$ are those of $\check{f}_F$ and $f_F$. At $N=1$ the ratio is $1/\varrho_{\mathcal{A}}$, and $\int\varrho_{\mathcal{A}^*}^{-z}d\phi_{\mathcal{A}^*}=\int\varrho_{\mathcal{A}}^{z}\varrho_{\mathcal{A}}^{-\kappa}d\phi_{\mathcal{A}}$.

*(v)* By (iii) for $f_F$ at $(\mathbf{a},\mathbf{b})$ and for $\check{f}_F$ at $(\mathbf{t}-\mathbf{b},\mathbf{s}-\mathbf{a})$, whose hyperplane condition $\sum_j(s_j-a_j)-\sum_j(t_j-b_j)=\kappa-|\mathbf{t}|$ is that of $(\mathbf{a},\mathbf{b})$, the Mellin transform of the law appears at the same point $(\mathbf{s}-\mathbf{a},\mathbf{b})$ on both sides, with the Gamma products $\prod_j\Gamma(a_j)\Gamma(b_j)$ and $\prod_j\Gamma(t_j-b_j)\Gamma(s_j-a_j)$; all these Gammas are finite and positive, which gives the relation in $(0,\infty]$. At $N=1$, with $z=s-a$ and $b=\kappa-z$, the ratio of the Gammas is that of the displayed one-cell relation (check 6). The expansion clause concerns the node of width one of one table and is the reference's. □

**Lemma S10.** *Let $F$ be a cell family with plain tables, $s_j,t_j>0$, and $\mathbf{a},\mathbf{b}\in(0,\infty)^N$ real with $\sum_j b_j-\sum_j a_j=\kappa_F-|\mathbf{s}|$ and $\int\boldsymbol{\varrho}_F^{-(\mathbf{s}-\mathbf{a},\mathbf{b})}d\phi_F<\infty$. Write $\mathbf{u}=(x_2,\dots,x_N,y_1,\dots,y_N)\in(0,\infty)^{2N-1}$, $g(\mathbf{u})=f_F((1,x_2,\dots,x_N),\mathbf{y})$, $\mathbf{c}=(a_2,\dots,a_N,b_1,\dots,b_N)$, and for $\boldsymbol{\tau}=(\tau_2,\dots,\tau_N,\tau'_1,\dots,\tau'_N)\in\mathbb{R}^{2N-1}$ put $\tau_1=\sum_j\tau'_j-\sum_{j\geq 2}\tau_j$ and*

$$G(\boldsymbol{\tau})=\prod_j\Gamma(a_j+i\tau_j)\Gamma(b_j+i\tau'_j)\int\boldsymbol{\varrho}_F^{-(\mathbf{s}-\mathbf{a}-i\boldsymbol{\tau}_K,\ \mathbf{b}+i\boldsymbol{\tau}_\Psi)}d\phi_F,$$

*$\boldsymbol{\tau}_K=(\tau_j)_j$ and $\boldsymbol{\tau}_\Psi=(\tau'_j)_j$. Then $g$ is finite and continuous on $(0,\infty)^{2N-1}$, $G$ is the Mellin transform of $g$ on the vertical subspace through $\mathbf{c}$ and is integrable, and*

$$g(\mathbf{u})=(2\pi)^{1-2N}\int_{\mathbb{R}^{2N-1}}G(\boldsymbol{\tau})\,\mathbf{u}^{-\mathbf{c}-i\boldsymbol{\tau}}\,d\boldsymbol{\tau}\qquad(\mathbf{u}\in(0,\infty)^{2N-1}).$$

*Proof.* The computation of 23.8(iii) at the complex exponents $\mathbf{c}+i\boldsymbol{\tau}$ converges absolutely, its integrand having at every point the modulus of the integrand at $\mathbf{c}$, so it computes the Mellin transform of $g$ at $\mathbf{c}+i\boldsymbol{\tau}$ as $G(\boldsymbol{\tau})$; the point $(\mathbf{s}-\mathbf{a}-i\boldsymbol{\tau}_K, \mathbf{b}+i\boldsymbol{\tau}_\Psi)$ stays on the hyperplane, which fixes $\tau_1$. By 23.1, $|G(\boldsymbol{\tau})| \le \prod_j |\Gamma(a_j+i\tau_j)\Gamma(b_j+i\tau'_j)| \int \boldsymbol{\varrho}_F^{-(\mathbf{s}-\mathbf{a},\mathbf{b})} d\phi_F$, and the bound $|\Gamma(c+i\tau)| \le C_c\, e^{-\pi|\tau|/2}(1+|\tau|)^{c-1/2}$ makes $G$ integrable on $\mathbb{R}^{2N-1}$. The Mellin transform of $g$ at the real point $\mathbf{c}$ is finite, so $g<\infty$ almost everywhere. By 23.8(ii), $g$ is the Laplace transform of a positive measure in the variables $\mathbf{u}$, hence nonincreasing in each of them: if $g$ were infinite at a point it would be infinite on the set of points below it, which has positive measure. So $g$ is finite everywhere, and continuous by dominated convergence, the integrand near a point being dominated by the integrand at a point with smaller coordinates. In the coordinates $\mathbf{v}=\log\mathbf{u}$ the function $g(e^{\mathbf{v}})e^{\langle \mathbf{c},\mathbf{v}\rangle}$ is integrable with the integrable Fourier transform $G$, so the Fourier inversion theorem gives the display almost everywhere, and everywhere since both sides are continuous. □

**Theorem 23.9 (Two cell families glued).** *As stated, for two cell families with plain tables and $s_j, t_j > 0$.*

*Proof. (i)* 23.6(i) with the core $\binom{\{g_1,g_2\}}{1}$ in every table, $F_1$ at $g_1$ and $F_2$ at $g_2$, the period of the twisted core family being $\mathcal{G}$.

*(ii), first form.* The gluing of two cells along the hinge is the series closure $\binom{\{h_1,h_2\}}{1}\langle \mathcal{A}_1, \mathcal{A}_2\rangle$ (21.7(iii)), so 23.3 at $q=2$ gives

$$P = C\int \prod_j x_j^{s_j-1} \prod_{j\ge 2} y_j^{t_j-1}\, f_{F_1} f_{F_2}\big(\mathbf{x},(1,\mathbf{y}')\big)\, d\mathbf{x}\, d\mathbf{y}', \qquad C=\prod_j \frac{1}{\Gamma(s_j)\Gamma(t_j)}.$$

Substitute $\mathbf{x}=x_1(1,\boldsymbol{\xi})$, $d\mathbf{x}=x_1^{N-1}dx_1\, d\boldsymbol{\xi}$, and use the homogeneity of 23.1 with $c=x_1$, $f_{F_i}(\mathbf{x},(1,\mathbf{y}')) = x_1^{\kappa_{F_i}-|\mathbf{s}|} f_{F_i}((1,\boldsymbol{\xi}), x_1(1,\mathbf{y}'))$; then substitute $\boldsymbol{\eta}=x_1(1,\mathbf{y}')$, $dx_1\, d\mathbf{y}' = x_1^{1-N} d\boldsymbol{\eta}$. The powers of $x_1=\eta_1$ collect to

$$(|\mathbf{s}|-N)+(N-1)+(\kappa_{F_1}+\kappa_{F_2}-2|\mathbf{s}|)+(1-N)-(|\mathbf{t}|-t_1)+(N-1) = t_1-1,$$

since $\kappa_{F_1}+\kappa_{F_2}=|\mathbf{s}|+|\mathbf{t}|$ (check 10 of S21-23_work/checks.py): the first form.

*(ii), second form.* The complements of the bases of a two-sum of cells are the bases of the two-sum of the dual cells: for $X$ a basis of $\mathcal{A}\setminus g$ and $Y\sqcup\{g\}$ one of $\mathcal{A}'$, $B\setminus X$ is a basis of $\mathcal{A}^*/g$ and $R\setminus Y$ one of $\mathcal{A}'^*\setminus g$, and symmetrically (21.2), with the product weights. By 8.15 the period of $F_1\oplus_g F_2$ at $(\mathbf{s},\mathbf{t})$ is that of $F_1^*\oplus_g F_2^*$ at $(\mathbf{t},\mathbf{s})$, and by 23.8(iv) the dual cell families have the hinge walls $\kappa_{F_1},\kappa_{F_2}$ and the transforms $\check{f}_{F_1}, \check{f}_{F_2}$. The first form for them is the second form.

*(iii)* Insert the Laplace forms of 23.8(ii) into (ii) and integrate first in $x_j$, $j\ge 2$, and $y_j$ (Tonelli): $\int_0^\infty x^{s-1}e^{-x(\lambda+\lambda')}dx = \Gamma(s)(\lambda+\lambda')^{-s}$, and $x_1=1$ leaves $e^{-\lambda_1-\lambda'_1}$; the factors $\Gamma(s_j)$, $j\ge 2$, and $\Gamma(t_j)$ cancel against $C$. The period converges exactly when every proper wall is positive (8.7).

*(iv)* Let the contour have the real point $(\mathbf{a}^d,\mathbf{b}^d,\mathbf{a}^r,\mathbf{b}^r)$, all coordinates positive, with $\mathbf{a}^d+\mathbf{a}^r=\mathbf{s}$, $\mathbf{b}^d+\mathbf{b}^r=\mathbf{t}$ and $\sum_j(a_j^r+b_j^d)=\kappa_{F_1}$, and put $\mathbf{a}=\mathbf{a}^d$, $\mathbf{b}=\mathbf{b}^d$; then $\sum_j b_j-\sum_j a_j = \kappa_{F_1}-|\mathbf{s}|$, and the Mellin transform of the law of $F_1$ at $(\mathbf{s}-\mathbf{a},\mathbf{b})=(\mathbf{a}^r,\mathbf{b}^d)$ is finite. By Lemma S10, $f_{F_1}$ is the inverse Mellin integral of $G_1$ through $\mathbf{c}$. Insert it into the first form of (ii). The double integral converges absolutely: integrating first in $\mathbf{u}$ gives $\int |G_1|\, d\boldsymbol{\tau}$ times $\int \prod_{j\ge2} x_j^{s_j-a_j-1}\prod_j y_j^{t_j-b_j-1} f_{F_2}\, d\mathbf{x}'\, d\mathbf{y}$, which is 23.8(iii) for $F_2$ at $(\mathbf{s}-\mathbf{a},\mathbf{t}-\mathbf{b})$: its first coordinate is $|\mathbf{t}|-|\mathbf{b}|-(|\mathbf{s}|-s_1-\sum_{j\ge2}a_j)-\kappa_{F_2}+|\mathbf{s}| = s_1-a_1$, by $\kappa_{F_2}=2\sigma-\kappa_{F_1}$, so it equals $\prod_j\Gamma(s_j-a_j)\Gamma(t_j-b_j)\int \boldsymbol{\varrho}_{F_2}^{-(\mathbf{a}^d,\mathbf{b}^r)}d\phi_{F_2}<\infty$. By Fubini,

$$P = C\,(2\pi)^{1-2N}\int_{\mathbb{R}^{2N-1}} \prod_j \Gamma(a_j^d)\Gamma(b_j^d)\Gamma(a_j^r)\Gamma(b_j^r) \int \boldsymbol{\varrho}_{F_1}^{-(\mathbf{a}^r,\mathbf{b}^d)}d\phi_{F_1}\int \boldsymbol{\varrho}_{F_2}^{-(\mathbf{a}^d,\mathbf{b}^r)}d\phi_{F_2}\, d\boldsymbol{\tau},$$

the shares moving with the imaginary parts $\boldsymbol{\tau}$ of $(a_2^d,\dots,a_N^d,b_1^d,\dots,b_N^d)$, the others fixed by the pins, and $C\prod_j\Gamma(\cdot)$ is $\prod_j\Gamma_{\,2}^{\,4}\left[\begin{matrix}a_j^d,\ b_j^d,\ a_j^r,\ b_j^r\\ s_j,\ t_j\end{matrix}\right]$. This is the fibre integral of 4.1 over the level shares with their $2N+1$ pins: solved for $a_j^r$, $b_j^r$ and $a_1^d$ the matrix of the pins has $|\det|=1$, so the fibre integral is $(2\pi)^{1-2N}$ times the integral over $\boldsymbol{\tau}$. In the pair coordinates of 19.13, $\sigma_j^d=\frac12(a_j^d+b_j^d)$ and $\delta_j^d=\frac12(a_j^d-b_j^d)$ with the one pin $\sum_j\delta_j^d=\frac12(|\mathbf{s}|-\kappa_{F_1})$, the linear map from $\boldsymbol{\tau}$ to the imaginary parts of $(\sigma_1^d,\dots,\sigma_N^d,\delta_2^d,\dots,\delta_N^d)$ has $|\det|=2^{1-N}$: the map $(a,b)\mapsto(\frac{a+b}{2},\frac{a-b}{2})$ of $\mathbb{R}^{2N}$ has $|\det|=2^{-N}$, it sends the normal coordinate $\sum_j(a_j-b_j)$ of the pin to twice the normal coordinate $\sum_j\delta_j$, and the quotient is $2^{1-N}$. So the integral is $2^{N-1}$ times the fibre integral in the pair coordinates, the formula of 19.13. That the two Mellin transforms are the periods of the bunch family and of the room family at the level shares is the identity $u^{\mathbf{m}-\mathbf{1}}\prod_j K_{\mathcal{A}_j/g}^{-a_j^d}\Psi_{\mathcal{A}_j/g}^{-b_j^d}K_{\mathcal{A}_j\setminus g}^{-a_j^r}\Psi_{\mathcal{A}_j\setminus g}^{-b_j^r}=\boldsymbol{\varrho}_{F_1}^{-(\mathbf{a}^r,\mathbf{b}^d)}(\gamma_{\mathcal{A}_1}^K)^{\kappa_{F_1}-\sum_j(a_j^r+b_j^d)}$ times the density of $\phi_{F_1}$, the last power being 1 on the pin, with $\mathcal{B}_j|B=\mathcal{A}_j\setminus g$ and $\mathcal{B}_j/R=\mathcal{A}_j/g$ (21.12), and its analogue for the room, whose pin $\sum_j(a_j^d+b_j^r)=2\sigma-\kappa_{F_1}=\kappa_{F_2}$ follows from the letter totals.

*One table.* At $N=1$ the laws are measures on $(0,\infty)$, the kernel is that of 23.7, (ii) and (iii) have no variables $x_j$, the image of $e^{-\lambda}\mathfrak{r}_{F_i}$ under $\mu$ being $\mathfrak{r}^\Psi$, and in (iv) the fibre is the line of $b^d=z$, with $a^d=s-\kappa_{\mathcal{A}}+z$, $a^r=\kappa_{\mathcal{A}}-z$, $b^r=t-z$; the conditions on the contour are $\max(0,\kappa_{\mathcal{A}}-s)<z_0<\min(t,\kappa_{\mathcal{A}})$ and the finiteness of $\int\varrho_{\mathcal{A}}^{-z_0}d\phi_{\mathcal{A}}$ and $\int\varrho_{\mathcal{A}'}^{z_0-t}d\phi_{\mathcal{A}'}$, and $d\sigma^d=dz$. The first form of (ii) is 20.26(i) with the room transform $y^tf_{\mathcal{A}'}(y)$. □

**Definition 23.10 (One weighted mark).** As stated: the one-mark cell family of a ratio vector, with two systems of weights in every table.

*Proof.* Table $j$ has the bases $\{e\}$ and $\{g\}$, with $\theta^K=(x_{0,j,K},1)$ and $\theta^\Psi=(1,x_{0,j,\Psi})$, so $K_j=x_{0,j,K}\alpha_e+\alpha_g$ and $\Psi_j=\theta^\Psi_{\{e\}}\alpha_g+\theta^\Psi_{\{g\}}\alpha_e=\alpha_g+x_{0,j,\Psi}\alpha_e$, two systems of weights being allowed (21.1, 8.1). The hinge minors are $K_{\setminus g}=x_{0,j,K}\alpha_e$, $K_{/g}=1$, $\Psi_{\setminus g}=1$ and $\Psi_{/g}=x_{0,j,\Psi}\alpha_e$, so the hinge values are $x_{0,j,K}\alpha_e$ and $x_{0,j,\Psi}\alpha_e$ and the ratio vector is the constant $\mathbf{x}_0$, $x_{0,1,K}=1$. The rank is 1 and $|B|=1$, so the hinge wall is $m_e$. The simplex $\Delta_{\{e\}}$ is the point $u_e=1$, carrying the unit mass, and there the density of the base form is $1\cdot1\cdot(x_{0,1,K})^{-m_e}=1$: the law is $\delta_{\mathbf{x}_0}$. □

**Theorem 23.11 (The law is what a cell family retains).** *As stated, for every pair of cell families with plain tables and $s_j,t_j>0$.*

*Proof. (a)⇒(b).* By 23.6(i) with the core $\binom{\{g_1,g_2\}}{1}$, the period of $F\oplus_gF''$ depends on $F$ only through $\kappa$ and its law; this needs no plainness of $F''$. (b)⇒(c) is a special case.

*(c)⇒(d).* By 23.6(i) and the law $\delta_{\mathbf{x}_0}$ of the one-mark cell family (23.10), the period against it is $\Pi_F(\mathbf{x}_0)=\int\mathcal{G}(\mathbf{z},\mathbf{x}_0)\,d((\boldsymbol{\varrho}_F)_*\phi_F)(\mathbf{z})$, $\mathcal{G}(\mathbf{z},\mathbf{x}_0)=\int_0^\infty v^{\kappa-1}\prod_j(z_{j,K}v+x_{0,j,K})^{-s_j}(z_{j,\Psi}v+x_{0,j,\Psi})^{-t_j}dv$. Since $(zv+x)/(zv+x')$ lies between $\min(1,x/x')$ and $\max(1,x/x')$, $\mathcal{G}(\cdot,\mathbf{x}_0)$ and $\mathcal{G}(\cdot,\mathbf{x}_0')$ are comparable within constants that depend on $\mathbf{x}_0,\mathbf{x}_0'$ only, so $\Pi_F$ is finite everywhere on $(0,\infty)^{2N-1}$ or nowhere.

Take the Mellin transform of $\Pi_F$ in the coordinates of $\mathbf{x}_0$ other than $x_{0,1,K}=1$, with the exponents $s_j-z_{j,K}$ at $x_{0,j,K}$, $j\ge2$, and $t_j-z_{j,\Psi}$ at $x_{0,j,\Psi}$. By Tonelli and $\int_0^\infty q^{w-1}(c+q)^{-s}dq=c^{w-s}\Gamma_{\,2}^{\,2}\left[\begin{matrix}w,\ s-w\\ s\end{matrix}\right]$, $0<w<s$, each coordinate gives a Beta corner and the power $(z_{j,K}v)^{-z_{j,K}}$ or $(z_{j,\Psi}v)^{-z_{j,\Psi}}$, and the remaining integral is $\int_0^\infty v^{z_{1,K}-1}(1+v)^{-s_1}dv=\Gamma_{\,2}^{\,2}\left[\begin{matrix}z_{1,K},\ s_1-z_{1,K}\\ s_1\end{matrix}\right]$, the

power of $v$ collecting to $v^{z_{1,K}-1}$ by the hyperplane. So the Mellin transform is

$$\prod_{j\ge 2}\Gamma_1^2\begin{bmatrix} s_j - z_{j,K},\ z_{j,K} \\ s_j \end{bmatrix}\,\Gamma_1^2\begin{bmatrix} z_{1,K},\ s_1 - z_{1,K} \\ s_1 \end{bmatrix}\prod_j \Gamma_1^2\begin{bmatrix} t_j - z_{j,\Psi},\ z_{j,\Psi} \\ t_j \end{bmatrix}\int \boldsymbol{\varrho}_F^{-\mathbf{z}}\, d\phi_F < \infty.$$

In particular $\Pi_F$ is finite almost everywhere, hence everywhere. For complex exponents with these real parts the same computation converges absolutely, and the Beta corners at complex arguments of positive real part do not vanish. Where finite, $\Pi_F$ extends holomorphically to the product of right half-planes: $|(zv+\zeta)^{-s}| \le (zv + \operatorname{Re}\zeta)^{-s}$ for $\operatorname{Re}\zeta > 0$ and $s > 0$, so dominated convergence and Morera's theorem give holomorphy in each coordinate, the extension being locally bounded, and hence jointly. So $\Pi_F$ and $\Pi_{F'}$ are real-analytic on $(0,\infty)^{2N-1}$, equal on a nonempty open set, and equal everywhere; their Mellin transforms on the vertical subspace through $\mathbf{z}$ agree, which is (d).

*(d)⇒(a).* The finite measures $\boldsymbol{\varrho}^{-\mathbf{z}}(\boldsymbol{\varrho}_F)_*\phi_F$ and $\boldsymbol{\varrho}^{-\mathbf{z}}(\boldsymbol{\varrho}_{F'})_*\phi_{F'}$ on $(0,\infty)^{2N-1}$, read in the logarithms of the coordinates, have the Fourier transforms $\boldsymbol{\tau} \mapsto \int \boldsymbol{\varrho}^{-(\mathbf{z}+i\boldsymbol{\tau})} d\phi$, equal by (d); a finite measure on $\mathbb{R}^{2N-1}$ is determined by its Fourier transform, so the two measures agree, and multiplying by the positive continuous function $\boldsymbol{\varrho}^{\mathbf{z}}$ the laws agree. This replaces the Gaussian smoothing of the reference proof of 23.11, which is the same argument on the line.

*One table.* The plain one-mark cell of ratio $x_0$ has the law $x_0^{(m_e-s)/2}\delta_{x_0}$, a positive multiple of the point law, and its ratios fill $(0,\infty)$, so (c) with the plain cells gives the equality of $\Pi_F$ on an open interval; (d) is the case $N = 1$, $\mathbf{w} = (w_{1,K}, w)$ with $w_{1,K}$ not entering. □

**Corollary 23.12 (Cells that are marks).** *As stated: (i) for every cell family with plain tables and $s_j, t_j > 0$.*

*Proof. (i)* This is the reference proof of 23.12(i) with the ratio vector in place of the ratio. If $\boldsymbol{\varrho}_F \equiv \mathbf{x}_0$, the image of $\phi_F$ is $\phi_F(\Delta_B)\delta_{\mathbf{x}_0}$. Conversely, $\phi_F$ has a positive continuous density on the open simplex and $\boldsymbol{\varrho}_F$ is continuous, so if the law is carried by $\mathbf{x}_0$ the open set $\{\boldsymbol{\varrho}_F \ne \mathbf{x}_0\}$ has $\phi_F$-measure 0 and is empty. The mass $\phi_F(\Delta_B)$ is positive. By 23.6(i), the period of a gluing $F \oplus_g F''$ is $\phi_F(\Delta_B)\int \mathcal{G}(\mathbf{x}_0, \mathbf{z}')\, d((\boldsymbol{\varrho}_{F''})_*\phi_{F''})(\mathbf{z}')$, a positive multiple of $\phi_F(\Delta_B)$; so a finite period forces $\phi_F(\Delta_B) < \infty$, and the integral is the period of the gluing of the one-mark cell family of ratio vector $\mathbf{x}_0$ and mass $\kappa_F$ with $F''$, whose law is $\delta_{\mathbf{x}_0}$ and whose hinge wall is $\kappa_F$. For one plain cell, the one-mark cell of ratio $x_0$ and mass $\kappa_{\mathcal{A}}$ has the law $x_0^{(\kappa_{\mathcal{A}}-s)/2}\delta_{x_0}$, which gives the factor. (ii) and (iii) are the reference's. □

**Theorem 23.14 (Series and parallel nodes).** *As stated: (i) and (ii) for cell families with plain tables and $s_j, t_j > 0$.*

*Proof.* This is the reference proof of 23.14(i),(ii), table by table. *(i)* The series core $\binom{G}{q}$, $G = \{g_0, \dots, g_q\}$, has rank $q$; its deletion at $g_0$ has the one basis $\{g_1, \dots, g_q\}$, so $K_{\binom{G}{q}\setminus g_0}(\mathbf{v}) = \prod_i v_i$ and $\Psi_{\binom{G}{q}\setminus g_0} = 1$, and its contraction has the bases of size $q-1$ of $\{g_1, \dots, g_q\}$, so $K_{\binom{G}{q}/g_0} = e_{q-1}(\mathbf{v})$ and $\Psi_{\binom{G}{q}/g_0}(\mathbf{v}) = \sum_i v_i$. By 23.5,

$$K_{\mathcal{A}_j\setminus g_0} = \prod_i K_{\mathcal{A}_{ij}/g_i}\gamma^K_{\mathcal{A}_{ij}} = \prod_i K_{\mathcal{A}_{ij}\setminus g_i}, \qquad \Psi_{\mathcal{A}_j\setminus g_0} = \prod_i \Psi_{\mathcal{A}_{ij}\setminus g_i},$$

and the hinge values of $\mathcal{A}_j$ are $\prod_i \gamma^K_{\mathcal{A}_{ij}}/e_{q-1}((\gamma^K_{\mathcal{A}_{ij}})_i)$, whose reciprocal is $\sum_i 1/\gamma^K_{\mathcal{A}_{ij}}$, and $\sum_i \gamma^\Psi_{\mathcal{A}_{ij}}$. In the orthant form of $f_F$ the measure $\alpha^{\mathbf{m}-\mathbf{1}}\prod_j K^{-s_j}_{\mathcal{A}_j\setminus g_0}\Psi^{-t_j}_{\mathcal{A}_j\setminus g_0}\, d\alpha$ is the product over $i$ of the

measures of the cell families, and the exponential is the product of their exponentials, so Tonelli gives $f_F = \prod_i f_{F_i}$, and the image of the product measure under the sum of the maps of the factors is the convolution: $\mathfrak{r}_F = *_i \mathfrak{r}_{F_i}$. The hinge-wall formula of 23.6(ii), with $\mathcal{E}_{\mathcal{H}} = \varnothing$, $k = q$ and $r_{\mathcal{H}_j} = q$, gives $\kappa_F = \sum_i \kappa_{F_i} - (q-1)|\mathbf{s}|$ (check 3a of S21-23_work/checks.py). For one table, $f_{\mathcal{A}}(y) = f_F(1, y) = \prod_i f_{F_i}(1, y)$, and since $e^{-\lambda} = \prod_i e^{-\lambda_i}$ for $\lambda = \sum_i \lambda_i$, the measure $e^{-\lambda}\mathfrak{r}_F$ is the convolution of the $e^{-\lambda_i}\mathfrak{r}_{F_i}$, whose images under $\mu$ convolve: $\mathfrak{r}^{\Psi}_{\mathcal{A}} = *_i \mathfrak{r}^{\Psi}_{\mathcal{A}_i}$. The twisted series core is the reference's.

*(ii)* The parallel core $\binom{G}{1}$ has rank 1: $K_{\binom{G}{1}\backslash g_0} = \sum_i v_i$, $K_{\binom{G}{1}/g_0} = 1$, $\Psi_{\binom{G}{1}\backslash g_0} = e_{q-1}(\mathbf{v})$ and $\Psi_{\binom{G}{1}/g_0} = \prod_i v_i$. So $K_{\mathcal{A}_j/g_0} = \prod_i K_{\mathcal{A}_{ij}/g_i}$, $\Psi_{\mathcal{A}_j/g_0} = \prod_i \Psi_{\mathcal{A}_{ij}\backslash g_i} \gamma^{\Psi}_{\mathcal{A}_{ij}} = \prod_i \Psi_{\mathcal{A}_{ij}/g_i}$, the hinge values combine as stated, the orthant form of $\check{f}_F$ factors, and $\kappa_F = \sum_i \kappa_{F_i} - (q-1)|\mathbf{t}|$ (check 3b). Equally, 23.8(iv) exchanges (i) and (ii) through the dual cell families, the dual of a parallel core being a series core. (iii) and (iv) are the reference's. □

**Definition 23.15 (Hinge tree).** As stated: the common hinge tree of the tables of a family, with its node families and its cell families.

*Remark.* The added objects are definitions. Each $\mathcal{A}_{v,j}$ is a cell with the hinge $g_{v,\mathfrak{e}_v}$ on the marks below $v$ (23.15), the same for every $j$, so the $\mathcal{A}_{v,j}$ form a cell family in the sense of 23.1; and the substitution along the tree does not depend on the order of the two-sums, table by table (23.5).

**Theorem 23.17 (The period along a hinge tree).** *As stated: for every family whose tables have a common hinge tree, at real data.*

*Proof.* Induction from the leaves. At a leaf $v$ the cell family $F_v$ is the node family with its one marker as hinge. At an inner vertex $v \neq o$, table $j$ of $F_v$ is the substitution of the node $\mathcal{H}_{v,j}$, a hinged core with the hinge $g_{v,\mathfrak{e}_v}$, with the subtree cells $\mathcal{A}_{w,j}$ of the vertices $w$ directly below it at the markers of their edges (23.15), so 23.6(ii) gives (i), the masses on $\mathcal{E}_v$ being those of $F$. At the root, the node has no hinge, and 23.6(i) gives (ii). (iii) is the recursion of (i): the law at $v$ is computed from the nodes and masses at $v$ and the laws of the vertices directly below, and by induction from the data below $v$. This is the reference proof of 23.17 (thm:contraction), with 23.6 for families, which the reference's remark on families (rem:familiesnodes) gives verbatim. □

**Proposition 23.18 (What the work depends on).** *As stated, for a family whose tables have a common hinge tree.*

*Proof. (i)* A bijection of marks carrying the subtree cells onto each other in every table, with their weights, masses and hinge, carries the letters onto the letters, hence the base forms and the ratio vectors onto each other; the ranks, sizes and masses are kept, and so is the hinge wall.

*(ii)* At a series node of unit weights, $F_v$ is the series connection of the cell families directly below (23.13), so 23.14(i) gives $f_{F_v} = \prod_w f_{F_w}$ and $\kappa_{F_v} = \sum_w \kappa_{F_w} - (q-1)|\mathbf{s}|$, $q = \sum_l k_l$; members of one orbit have equal transforms and hinge walls by (i), and collecting them gives the powers. A weight $\theta_{g_i}$ on the basis of a series node that omits the marker $g_i$ is carried, by the weight formula of 21.1, by the cell at $g_i$ as the factor $\theta_{g_i}/\theta_{g_0}$ on its bases through its hinge, the weight $\theta_{g_0}$ of the basis omitting the hinge remaining as a constant factor of the table; so a series node with weights is a series connection of cells so weighted, and the orbits are taken of these. The parallel node is the same with 23.14(ii).

*(iii)* By 23.1 the hinge wall of $F_v$ is $m_{B_v} - \sum_j \big(s_j(r_{\mathcal{A}_{v,j}} - 1) + t_j(|B_v| - r_{\mathcal{A}_{v,j}})\big)$, $B_v$ the marks below $v$, and by 23.5 the ranks $r_{\mathcal{A}_{v,j}}$ are sums of the ranks of the nodes below $v$ less one per edge. A change at $v_0$ keeping the total mass at $v_0$, the ranks and the numbers of marks of the nodes,

keeps every $m_{B_v}$, rank and size, since $B_v$ contains all of $\mathcal{E}_{v_0}$ or none of it; so no hinge wall changes, and by 23.17(iii) the law of $F_v$ changes only if $v_0$ lies below $v$, that is, for $v$ on the path from $v_0$ to the root. A change of masses at $v_0$ and $v_1$ keeping the total changes $m_{B_v}$ exactly when $B_v$ contains one of $\mathcal{E}_{v_0}, \mathcal{E}_{v_1}$ and not the other, by the change of the total mass below $\mathfrak{e}_v$. □

**Definition 23.19 (The hinge tree procedure).** As stated: for a family whose tables have a common hinge tree, the representations (a) and (b) in $2N-1$ variables, (c) for one table.

*Remark.* Each operation rests on a statement for families. The laws and hinge walls of the orbits are those of 23.17 and 23.18(i). (1) The one-mark leaf: by 23.8(i) at the point law $\delta_{\mathbf{x}_0}$ with $\kappa = m_e$, $f(\mathbf{x},\mathbf{y}) = \prod_j x_{0,j,K}^{-s_j} \mathsf{G}_{|\mathbf{s}|-m_e;(1,-1)}(\sum_j x_j/x_{0,j,K}, \sum_j y_j x_{0,j,\Psi})$, and the second law form with torus homogeneity at $\lambda = x_{0,1,\Psi}$ gives $\check{f}(\mathbf{x},\mathbf{y}) = \prod_j x_{0,j,\Psi}^{-t_j} \mathsf{G}_{|\mathbf{t}|-m_e;(1,-1)}(\sum_j x_j/x_{0,j,\Psi}, \sum_j y_j x_{0,j,K})$, the powers of $x_{0,1,\Psi}$ cancelling; the orthant forms give the same directly. For one table, $\mathbf{x}_0 = (1, x_0)$ and the plain cell's law $x_0^{(m_e-s)/2}\delta_{x_0}$ give the displayed one-table transforms, with $\mathsf{G}_{t-m_e;(1,-1)}(x_0^{-1}, y) = x_0^{t-m_e}\mathsf{G}_{t-m_e;(1,-1)}(1, y/x_0)$. (2) 23.14(i),(ii), with 23.18(ii) for the weights; a pointwise product of transforms is a Mellin convolution of their Mellin transforms of 23.8(iii). (3) 23.8(v) on the Mellin side, and Lemma S10 for the inverse Mellin transform, applied to the dual cell family for $\check{f}$ (23.8(iv)). (4) 23.6(ii) for families. (5) 23.6(i), and for a series or parallel root 23.9(ii) with the series or parallel connection of all but one cell family on one side.

**Corollary 23.20 (Series-parallel families).** *As stated, for a family whose tables have a common hinge tree of series and parallel nodes over one-mark cell families.*

*Proof.* This is the reference proof of 23.20 with the family statements. The leaves have their transforms in closed form, Grammas of the marks (the remark after 23.19 in this part). At a series node the transform of the subtree is the product of those below, and at a parallel node the dual transform (23.14), so no integral is taken there; repeated subtrees enter as powers (23.18(ii)). At an edge joining a series node to a parallel node the transform of the subtree below is changed into its dual transform by 23.8(v): the Mellin transform of $f$ on a vertical subspace through a real point $(\mathbf{a},\mathbf{b})$, $0 < a_j < s_j$, $0 < b_j < t_j$, of its domain, an integral on $(0,\infty)^{2N-1}$; multiplication by the ratio of Gammas; and the inverse Mellin transform of the result, which is the Mellin transform of $\check{f}$ (23.8(iii) at $(\mathbf{t}-\mathbf{b}, \mathbf{s}-\mathbf{a})$), an integral on $(0,\infty)^{2N-1}$ that converges absolutely and returns $\check{f}$ by Lemma S10 applied to the dual cell family (23.8(iv)). The hypothesis is the finiteness of the Mellin transform of the law at $\mathbf{z} = (\mathbf{s}-\mathbf{a}, \mathbf{b})$, which these points require. At the root one integral on $(0,\infty)^{2N-1}$ remains, 23.9(ii). The count of the integrations depends only on the edges where the kind of node changes and on the root. For one table the variables are one. □

## 50.24 Chapter 24: repetition and the change of query

**Theorem 24.1 (How often a cell family repeats).** *As stated: for every cell family with plain tables, at real exponents.*

*Proof. (i)* The series connection of $k$ copies of $\mathcal{A}_j$ has $k|B|$ marks besides the hinge, the masses repeated, and the rank $k + k(r_{\mathcal{A}_j} - 1) = kr_{\mathcal{A}_j}$ (23.5, the series core of rank $k$). By 23.1, since $s_j(kr_{\mathcal{A}_j} - 1) = ks_j(r_{\mathcal{A}_j} - 1) + (k-1)s_j$ and $t_j(k|B| - kr_{\mathcal{A}_j}) = kt_j(|B| - r_{\mathcal{A}_j})$,

$$\kappa_{F_k^{\mathrm{s}}} = km_B - \sum_j \big(s_j(kr_{\mathcal{A}_j} - 1) + t_j(k|B| - kr_{\mathcal{A}_j})\big) = k\kappa_F - (k-1)|\mathbf{s}| = |\mathbf{s}| + k(\kappa_F - |\mathbf{s}|).$$

The parallel connection has the rank $1 + k(r_{\mathcal{A}_j} - 1)$, and the same count gives $k\kappa_F - (k-1)|\mathbf{t}|$ (checks 3a and 3b of S21-23_work/checks.py, which are this count at the exponents as symbols). No sign of an exponent enters.

*(ii)* The walls of $F_k^{\mathrm{s}} \oplus_g F'$ at the marks of the copies and at those of $F'$ are the hinge walls $\kappa_{F_k^{\mathrm{s}}}$ and $\kappa_{F'}$, and on the balance wall they add to $2\sigma$ (21.12). A convergent period has every proper wall positive (8.7, at any complex exponents), so $0 < \kappa_{F_k^{\mathrm{s}}} < 2\sigma$. If $\kappa_F \neq |\mathbf{s}|$ the middle term is unbounded in $k$, which bounds $k$: $|\mathbf{s}| - k(|\mathbf{s}| - \kappa_F) > 0$ gives $k < |\mathbf{s}|/(|\mathbf{s}| - \kappa_F)$ when $\kappa_F < |\mathbf{s}|$, and $|\mathbf{s}| + k(\kappa_F - |\mathbf{s}|) < |\mathbf{s}| + |\mathbf{t}|$ gives $k < |\mathbf{t}|/(\kappa_F - |\mathbf{s}|)$ when $\kappa_F > |\mathbf{s}|$. The parallel case is the same with the roles of $|\mathbf{s}|$ and $|\mathbf{t}|$ exchanged. □

**Theorem 24.2 (Series repetition).** *As stated: (i) for cell families with plain tables and $s_j, t_j > 0$; (ii) to (iv) are the reference's, for one cell.*

*Proof. (i)* By 23.14(i) with $k$ copies of $F$, $f_{F_k^{\mathrm{s}}} = f_F^k$; by 24.1(i), $\kappa_{F_k^{\mathrm{s}}} = |\mathbf{s}|$, which lies in $(0, 2\sigma)$ since $|\mathbf{s}|, |\mathbf{t}| > 0$. So the first form of 23.9(ii) applies to $F_k^{\mathrm{s}}$ and $F'$ and gives the display. With $\nu$ the measure of the statement, $P(k) = \int f_F^k \, d\nu = \int w^k \, d((f_F)_*\nu)(w)$. Where finite, $P(k)^2 = \big(\int f_F^{(k-1)/2} f_F^{(k+1)/2} d\nu\big)^2 \leq P(k-1)P(k+1)$ by the Cauchy–Schwarz inequality, and for $z \geq 0$ Tonelli gives $\sum_k P(k) z^k/k! = \int e^{z f_F} d\nu$.

*The value $P(0)$.* Insert the orthant form of $f_{F'}$ and integrate first in $x_j$, $j \geq 2$, and $y_j$ (Tonelli): $\int_0^\infty x^{s_j - 1} e^{-x/\gamma^K} dx = \Gamma(s_j)(\gamma^K)^{s_j}$ and $\int_0^\infty y^{t_j - 1} e^{-y\gamma^\Psi} dy = \Gamma(t_j)(\gamma^\Psi)^{-t_j}$. Since $K_{\mathcal{A}'_j \setminus g}(\gamma^K_{\mathcal{A}'_j})^{-1} = K_{\mathcal{A}'_j/g}$ and $\Psi_{\mathcal{A}'_j \setminus g} \gamma^\Psi_{\mathcal{A}'_j} = \Psi_{\mathcal{A}'_j/g}$, what remains is

$$\prod_{j \geq 2} \Gamma(s_j) \prod_j \Gamma(t_j) \int \alpha^{\mathbf{m}|_R - \mathbf{1}} e^{-1/\gamma^K_{\mathcal{A}'_1}} K_{\mathcal{A}'_1 \setminus g}^{-s_1} \prod_{j \geq 2} K_{\mathcal{A}'_j/g}^{-s_j} \prod_j \Psi_{\mathcal{A}'_j/g}^{-t_j} \, d\alpha.$$

In $\alpha = \rho u$ the power of $\rho$ is $\rho^{\kappa_{F'} - |\mathbf{t}| - s_1 - 1} = \rho^{-s_1 - 1}$, by $\kappa_{F'} = |\mathbf{t}|$ (check 9 of S21-23_work/checks.py), and $\int_0^\infty \rho^{-s_1 - 1} e^{-1/(\rho\gamma)} d\rho = \Gamma(s_1)\gamma^{s_1}$ turns $K_{\mathcal{A}'_1 \setminus g}^{-s_1}$ into $K_{\mathcal{A}'_1/g}^{-s_1}$. So $P(0)$ is the period at $\mathbf{m}|_R$ of the family of the contractions, whose degree is $\sum_j (s_j(r_{\mathcal{A}'_j} - 1) + t_j(|R| - r_{\mathcal{A}'_j} + 1))$, so that it is balanced exactly when $\kappa_{F'} = |\mathbf{t}|$. For one cell, $\int \varrho_{\mathcal{A}'}^{-t} d\phi_{\mathcal{A}'}$ with $\kappa_{\mathcal{A}'} = t$ has the density $u^{\mathbf{m}-\mathbf{1}} K_{\mathcal{A}'/g}^{t-s} K_{\mathcal{A}' \setminus g}^{-t} \Psi_{\mathcal{A}' \setminus g}^{-t} (\gamma^\Psi/\gamma^K)^{-t} = u^{\mathbf{m}-\mathbf{1}} K_{\mathcal{A}'/g}^{-s} \Psi_{\mathcal{A}'/g}^{-t}$, the same period.

(ii) to (iv) are the reference's, for one cell. □

*Remark.* For a family the asymptotics of $P(k)$ is a Laplace problem in $2N - 1$ variables: a series-balanced cell family has $f_F = \int \mathsf{G}_{0;(1,-1)}(1, ab) \, d\phi_{F \setminus g}$ by 23.3, of the order of $P_{F \setminus g}$ times the logarithm of the scale of $\mathbf{y}$ plus a term that depends on the direction of $\mathbf{y}$ and on $\mathbf{x}'$, and the uniformity of the error in these directions does not follow from the one-variable argument. So (ii) to (iv) are stated for one cell.

**Corollary 24.3 (Parallel repetition).** *As stated: the first display for cell families with plain tables and $s_j, t_j > 0$; the rest is the reference's, for one cell.*

*Proof.* By 23.14(ii), $\check{f}_{F_k^{\mathrm{p}}} = \check{f}_F^k$, and by 24.1(i), $\kappa_{F_k^{\mathrm{p}}} = |\mathbf{t}| \in (0, 2\sigma)$; the second form of 23.9(ii) gives the display. For $P(0)$, integrate the orthant form of $\check{f}_{F'}$ in $x_j$, $j \geq 2$, and $y_j$: $\int x^{t_j - 1} e^{-x/\gamma^\Psi} dx = \Gamma(t_j)(\gamma^\Psi)^{t_j}$ and $\int y^{s_j - 1} e^{-y\gamma^K} dy = \Gamma(s_j)(\gamma^K)^{-s_j}$ turn $\Psi_{\mathcal{A}'_j/g}^{-t_j}$ into $\Psi_{\mathcal{A}'_j \setminus g}^{-t_j}$, $j \geq 2$, and every $K_{\mathcal{A}'_j/g}^{-s_j}$ into $K_{\mathcal{A}'_j \setminus g}^{-s_j}$; in $\alpha = \rho u$ the power of $\rho$ is $\rho^{\kappa_{F'} - |\mathbf{s}| - t_1 - 1} = \rho^{-t_1 - 1}$, by $\kappa_{F'} = |\mathbf{s}|$, and $\int \rho^{-t_1 - 1} e^{-1/(\rho\gamma_1^\Psi)} d\rho = \Gamma(t_1)(\gamma_1^\Psi)^{t_1}$ turns $\Psi_{\mathcal{A}'_1/g}^{-t_1}$ into $\Psi_{\mathcal{A}'_1 \setminus g}^{-t_1}$. What remains is the period of the family of the deletions at $\mathbf{m}|_R$, balanced exactly when $\kappa_{F'} = |\mathbf{s}|$. Equally, the dual cell families at

$(\mathbf{t},\mathbf{s})$ turn the statement into (i) of 24.2 (23.8(iv), 8.15), the dual of the contractions being the deletions of the duals. The rest is the reference's, for one cell. □

**Proposition 24.4 (A chain is a power of one operator).** *As stated, for chain cell families at real exponents, (iv) with plain tables and $s_j, t_j > 0$.*

*Proof.* This is the reference proof of 24.4 with the laws on $(0,\infty)^{2N-1}$. *(i)* 23.6(ii) with the one cell family $F_l$ at $g_{\text{in}}$: its hinge-wall formula, $\mathcal{U}_j$ having the marks $\mathcal{E}_{\mathcal{U}} \sqcup \{g_{\text{in}}\}$ besides its hinge, is $\kappa_{F_{l+1}} = \kappa_{F_l} + \kappa_U$. By 21.12, $F_l \oplus_g F'$ is on its balance wall exactly when $\kappa_{F_l} + \kappa_{F'} = 2\sigma$; with $\kappa_{F_l} = \kappa_{F_0} + l\kappa_U$ this holds for every $l$ exactly when it holds at $l = 0$ and $\kappa_U = 0$.

*(ii)* 23.6(ii) with the one cell family $F_l$ at $g_{\text{in}}$ gives $\lambda_{l+1} = \int \lambda_{\mathbf{x}}^U \, d\lambda_l(\mathbf{x}) = \mathcal{Q}\lambda_l$, the twisted family carrying the mass $\kappa_{F_l} = \kappa_{F_0}$ at $g_{\text{in}}$; by induction $\lambda_l = \mathcal{Q}^l\lambda_0$. 23.6(i) with the core $\binom{\{g_1,g_2\}}{1}$ gives $P_{F_l \oplus_g F'} = \int h_{F'} \, d\lambda_l$, and $\int H \, d(\mathcal{Q}\mu) = \int (\mathcal{Q}^* H) \, d\mu$ for $H \geq 0$ by Tonelli, so $\int h_{F'} \, d\lambda_l = \int (\mathcal{Q}^{*l} h_{F'}) \, d\lambda_0$. The series in $z \geq 0$ is summed under the integral by Tonelli.

*(iii)* $\mathcal{Q}^*$ preserves inequalities between nonnegative functions, so $\mathcal{Q}^* H \leq MH$ gives $\mathcal{Q}^{*l} H \leq M^l H$, and $h_{F'} \leq CH$ gives $\mathcal{Q}^{*l} h_{F'} \leq CM^l H$; integrate against $\lambda_0$. The lower bound is the same.

*(iv)* For the series core $\binom{\{g_{\text{out}},g_{\text{in}},g'\}}{2}$ with $F''$ at $g'$, $F_{l+1}$ is the series connection of $F_l$ and $F''$, so 23.14(i) gives $f_{F_{l+1}} = f_{F''} f_{F_l}$, and the hinge-wall formula with the mass 0 at $g_{\text{in}}$, the marks $\{g_{\text{in}}\} \sqcup B''$ and the ranks $2 + r_{\mathcal{A}_j''} - 1$, gives $\kappa_U = m_{B''} - \sum_j \left(s_j r_{\mathcal{A}_j''} + t_j(|B''| - r_{\mathcal{A}_j''})\right) = \kappa_{F''} - |\mathbf{s}|$. For the parallel core the ranks are $r_{\mathcal{A}_j''}$, the formula gives $\kappa_{F''} - |\mathbf{t}|$, and 23.14(ii) gives the product of the dual transforms. □

**Theorem 24.5 (The weight of a hinge).** *As stated: (i) for cell families with plain tables, one weight per table; (ii) to (iv) are the reference's, for one cell.*

*Proof. (i)* The bases through $g$ carry the monomials of $K_{\mathcal{A}_j/g}$ and of $\Psi_{\mathcal{A}_j/g}$, which are multiplied by $w_j$; the deletions keep their weights (the reference proof of 24.5(i), table by table). So the hinge wall is kept, $\gamma^K_{\mathcal{A}_j} = K_{\mathcal{A}_j\setminus g}/K_{\mathcal{A}_j/g}$ becomes $\gamma^K_{\mathcal{A}_j}/w_j$ and $\gamma^\Psi_{\mathcal{A}_j} = \Psi_{\mathcal{A}_j/g}/\Psi_{\mathcal{A}_j\setminus g}$ becomes $w_j\gamma^\Psi_{\mathcal{A}_j}$; the coordinates of the ratio vector become $(w_1/w_j)\gamma^K_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}$ and $w_1 w_j \gamma^\Psi_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}$, and the base form $u^{\mathbf{m}-\mathbf{1}} \prod_j K_{\mathcal{A}_j/g}^{-s_j} \Psi_{\mathcal{A}_j\setminus g}^{-t_j} (\gamma^K_{\mathcal{A}_1})^{-\kappa_F} d\omega$ is multiplied by $\prod_j w_j^{-s_j} \cdot w_1^{\kappa_F}$. In the orthant forms, $x_j/\gamma^K_{\mathcal{A}_j}$ becomes $w_j x_j/\gamma^K_{\mathcal{A}_j}$ and $y_j\gamma^\Psi_{\mathcal{A}_j}$ becomes $w_j y_j \gamma^\Psi_{\mathcal{A}_j}$ with the measure of $f_F$ unchanged, and for $\check{f}_F$ the measure is multiplied by $\prod_j w_j^{-s_j-t_j}$ while $x_j/\gamma^\Psi_{\mathcal{A}_j}$ and $y_j\gamma^K_{\mathcal{A}_j}$ are divided by $w_j$. For one cell, $w_1 = w$: the ratio becomes $w^2\varrho_{\mathcal{A}}$, the base form $w^{\kappa_{\mathcal{A}}-s}\phi_{\mathcal{A}}$, and the homogeneity of 23.1 with $c = w$ gives $f_{w\mathcal{A}}(y) = f_F(w, wy) = w^{\kappa_{\mathcal{A}}-s} f_{\mathcal{A}}(w^2 y)$, and with $c = 1/w$, $\check{f}_{w\mathcal{A}}(y) = w^{-s-t}\check{f}_F(1/w, y/w) = w^{-s-\kappa_{\mathcal{A}}}\check{f}_{\mathcal{A}}(y/w^2)$. (ii) to (iv) are the reference's. □

*Remark.* For a family the weights act on the law through an $N$-dimensional subtorus of $(0,\infty)^{2N-1}$, and the Mellin transform in $\mathbf{w}$ is a restricted Mellin transform of the law against the transform of the room, without the product formula of (ii); (iii) and (iv) concern the one-variable expansions of the node of width one. So (ii) to (iv) are stated for one cell.

**Proposition 24.6 (Carrying a calculation through a change).** *As stated, for a family whose tables have a common hinge tree.*

*Proof.* (i) and (iii) are 23.18(iii) with the recursion of 23.17(i): the law at a vertex is recomputed from the new law of the vertex directly below it on the path and the kept laws of the others, and the root by 23.17(ii). (ii) is 24.5(i) for the cell family $F_v$: the weight of the hinge multiplies the base form by $w_1^{\kappa_{F_v}} \prod_j w_j^{-s_j}$ and moves the ratio vector by the stated map, so the law is multiplied

by that factor and carried by that map; the cell families below $v$ are unchanged, and above $v$ the change of the law of $F_v$ is a change at a vertex directly below, carried by (i). $\square$

## 50.25 Chapter 25: holds and the lean convolution

In this section $F = ((\mathcal{B}_j, \theta_j); s_j, t_j)_{j=1}^N$ is a family as in Definition 25.8: its tables are each one molecule on $\mathcal{E}$, $n = |\mathcal{E}| \geq 2$, and its exponents are positive, unless a statement says otherwise. Its $2N$ letters are written $X$, running over $K_1, \dots, K_N, \Psi_1, \dots, \Psi_N$. The letter $X$ of table $j$ has the exponent $w_X$, which is $s_j$ or $t_j$, the degree $\deg X$, which is $r_j$ or $r_j^*$, the twist $\mathbf{x}^X$, which is $\mathbf{x}_j^K$ or $\mathbf{x}_j^\Psi$, and the Newton polytope $\mathcal{N}_X$, which is $P(\mathcal{B}_j)$ or $P(\mathcal{B}_j^*)$; for masses $\mathbf{a}$ with $|\mathbf{a}| = w_X \deg X$ the period of its one-letter word $X^{-w_X}$ is written $P_X(\mathbf{a})$, which is $P_{\mathcal{B}_j}(\mathbf{a}; s_j, 0)$ or $P_{\mathcal{B}_j}(\mathbf{a}; 0, t_j)$. $\mathcal{X}'$ is the set of the $2N-1$ letters other than $K_1$, and in a split $\mathbf{m} = \sum_X \mathbf{a}^X$ the load of $X$ is $\mathbf{a}^X$, so that $\mathbf{a}^{K_j} = \mathbf{a}_j$ and $\mathbf{a}^{\Psi_j} = \mathbf{b}_j$.

Two facts are used throughout. First, the base polytope of a table that is one molecule, and that of its dual table, have dimension $n-1$: if $w \in \mathbb{R}^{\mathcal{E}}$ is orthogonal to every difference $\mathbf{1}_T - \mathbf{1}_{T'}$ of bases, then $\langle w, \mathbf{1}_T \rangle$ is constant over the bases and $w$ is constant by Lemma 7.8; so these differences span $\mathbf{1}^\perp$, and so do the differences $\mathbf{1}_{\mathcal{E}\setminus T'} - \mathbf{1}_{\mathcal{E}\setminus T}$. Second, a point $\sum_k p_k v_k$ with every $p_k > 0$ and $\sum_k p_k = 1$ lies in the relative interior of the convex hull of the finitely many points $v_k$: for $x = \sum_k q_k v_k$ in the hull and $0 < \varepsilon \leq \min_k p_k$, the point $\sum_k p_k v_k + \varepsilon\big(\sum_k p_k v_k - x\big) = \sum_k \big((1+\varepsilon)p_k - \varepsilon q_k\big)v_k$ has nonnegative coefficients of sum 1; so every segment from a point of the hull to the given point extends beyond it within the hull, which characterizes the relative interior of a convex set.

**Definition 25.1 (Positive letters; Newton words and sums).** As stated: the period of a product of powers of positive letters at complex exponents, wherever it converges absolutely, a Newton word being the case of positive exponents.

*Remark.* The independence of the section is the proof of Lemma 16.2, which uses of the integrand $f = \alpha^{\mathbf{m}-\mathbf{1}} W$ only that it is measurable, homogeneous, $f(\lambda\alpha) = \lambda^{-n} f(\alpha)$ for $\lambda > 0$ when $m_{\mathcal{E}} = D$, and absolutely integrable over one section; here $|f| = \alpha^{\operatorname{Re}\mathbf{m}-\mathbf{1}} \prod_i L_i^{-\operatorname{Re} w_i}$. The family word at any exponents is a product of powers of its letters with $D = \sum_j (s_j r_j + t_j r_j^*)$, and on the balance wall its period is that of Theorem 8.7.

**Proposition 25.3 (One positive letter).** *As stated.*

*Proof.* Put $D = \deg L$ and $t = (t_k)_k$. Every $t_k$ has $|t_k| = D$, and $\mathcal{N}_L$ has dimension $n-1$ in the hyperplane $|x| = D$, which does not contain 0; so the $t_k$ span $\mathbb{R}^{\mathcal{E}}$.

(i) By Lemma 2.7, $x = \rho u$ with $\rho > 0$ and $u \in \Delta_{\mathcal{E}}$ gives $x^{\mathbf{a}-\mathbf{1}} dx = \rho^{|\mathbf{a}|-1} u^{\mathbf{a}-\mathbf{1}} d\rho\, d\omega_{\mathcal{E}}(u)$, and $L(\rho u) = \rho^D L(u)$. For $\operatorname{Re}|\mathbf{a}| > 0$ the substitution $\sigma = \rho^D L(u)$ gives

$$\int_0^\infty \rho^{|\mathbf{a}|-1} e^{-\rho^D L(u)}\, d\rho = \frac{\Gamma(w)}{D}\, L(u)^{-w}, \qquad w = |\mathbf{a}|/D.$$

So $\mathsf{G}_{\mathbf{a};t}((c_k)_k) = \frac{\Gamma(w)}{D} \int_{\Delta_{\mathcal{E}}} u^{\mathbf{a}-\mathbf{1}} L(u)^{-w} d\omega_{\mathcal{E}}(u) = \frac{\Gamma(w)}{D}\, P_{L^{-w}}(\mathbf{a})$, the last integral being the period over the section $\Delta_{\mathcal{E}}$ (Definition 25.1). The same computation at $\operatorname{Re}\mathbf{a}$, by Tonelli's theorem, shows that the two integrals converge absolutely together.

(ii) $|u^{\mathbf{a}-\mathbf{1}} L(u)^{-w}| = u^{\operatorname{Re}\mathbf{a}-\mathbf{1}} L(u)^{-\operatorname{Re} w}$, so $P_{L^{-w}}(\mathbf{a})$ converges absolutely exactly when the period of the Newton word $L^{-\operatorname{Re} w}$, $\operatorname{Re} w > 0$, converges at $\operatorname{Re}\mathbf{a}$. Its Newton sum is $(\operatorname{Re} w)\mathcal{N}_L$, of dimension $n-1$, and Theorem 25.2 gives the condition $\operatorname{Re}\mathbf{a} \in \operatorname{ri}\big((\operatorname{Re} w)\mathcal{N}_L\big) = (\operatorname{Re} w)\operatorname{ri}\mathcal{N}_L$.

(iii) For real $\mathbf{a}$ and $|\boldsymbol{\eta}| = 0$ the exponent $w$ is the same real number at $\mathbf{a}$ and at $\mathbf{a} + i\boldsymbol{\eta}$, so by (i) $P_{L^{-w}}(\mathbf{a} + i\boldsymbol{\eta}) = D\,\mathsf{G}_{\mathbf{a}+i\boldsymbol{\eta};t}((c_k)_k)/\Gamma(w)$. If $P_{L^{-w}}(\mathbf{a})$ converges, so does $\mathsf{G}_{\mathbf{a};t}$, and $\mathbf{a} \in \operatorname{int}\operatorname{cone}(t)$ (Theorem 2.2). Theorem 2.11 gives $|\mathsf{G}_{\mathbf{a}+i\boldsymbol{\eta};t}((c_k)_k)| \le \mathsf{G}_{\mathbf{a};t}(\lambda(c_k)_k)\, e^{-(1-\varepsilon)h_t(\boldsymbol{\eta})}$ with $\lambda = \sin\frac{\pi\varepsilon}{2}$, and $h_t = h_L$ is a norm since the $t_k$ span. Every $t_k$ having total $D$, the substitution $x = \lambda^{-1/D}y$ gives $\mathsf{G}_{\mathbf{a};t}(\lambda(c_k)_k) = \lambda^{-w}\mathsf{G}_{\mathbf{a};t}((c_k)_k)$; multiplying by $D/\Gamma(w)$ gives the second bound. The bound by $P_{L^{-w}}(\mathbf{a})$ itself is that of the modulus of the integrand by its value at the real parts. □

*Remark.* At the letter $K_{\mathcal{H}}$ of a table that is one molecule, with the support $(\mathbf{1}_T)_T$, the coefficients $\boldsymbol{\theta}$ and $\mathcal{N}_{K_{\mathcal{H}}} = P(\mathcal{H})$, items (i), (ii) and (iii) are Proposition 25.10(i), (ii) and (iv) at the full basis lean, and at $\Psi_{\mathcal{H}}$ at the full cobasis lean. At a simplicial letter, (i) with Corollary 2.6 at the square configuration $t$ is Proposition 25.6: $\mathsf{G}_{\mathbf{a};t}((c_k)_k) = |\det t|^{-1}\prod_k \Gamma(b_k)c_k^{-b_k}$ with $\mathbf{b} = t^{-1}\mathbf{a}$, and $|\mathbf{b}| = w$ since $\mathbf{1}^{\mathsf{T}}t = D\mathbf{1}^{\mathsf{T}}$.

**Lemma 25.7 (Concatenation of Grammas).** *As stated, for any number of parts.*

*Proof.* For $i \ge 2$ the configuration $t^{(i)}$ spans $\mathbb{R}^{\mathcal{E}}$, so by Theorem 2.11 the Gramma $\mathsf{G}_{\mathbf{w};t^{(i)}}(\boldsymbol{\xi}^{(i)})$ on $\mathbf{c}_i + i\mathbb{R}^{\mathcal{E}}$ is at most a constant times $e^{-(1-\varepsilon)h_{t^{(i)}}(\operatorname{Im}\mathbf{w})}$, $h_{t^{(i)}}$ a norm, and is integrable there. In $\mathbf{y} = \log x$ it is the Fourier transform of the continuous integrable function $e^{\langle \mathbf{c}_i,\mathbf{y}\rangle}\exp\big(-\sum_k \xi_k^{(i)} e^{\langle t_k^{(i)},\mathbf{y}\rangle}\big)$ (Definition 2.1), so Fourier inversion gives, for every $x \in \mathbb{R}^{\mathcal{E}}_{>0}$,

$$\exp\Big(-\sum_k \xi_k^{(i)} x^{t_k^{(i)}}\Big) = \frac{1}{(2\pi i)^n}\int_{\mathbf{c}_i + i\mathbb{R}^{\mathcal{E}}} \mathsf{G}_{\mathbf{w}_i;t^{(i)}}(\boldsymbol{\xi}^{(i)})\, x^{-\mathbf{w}_i}\, d\mathbf{w}_i.$$

Multiply these for $i = 2,\dots,p$ by $x^{\nu-\mathbf{1}}\exp\big(-\sum_k \xi_k^{(1)}x^{t_k^{(1)}}\big)$ and integrate over $x$. The modulus of the integrand is $x^{\operatorname{Re}\nu-\sum_{i\ge2}\mathbf{c}_i-\mathbf{1}}\exp\big(-\sum_k \xi_k^{(1)}x^{t_k^{(1)}}\big)\prod_{i\ge2}|\mathsf{G}_{\mathbf{w}_i;t^{(i)}}(\boldsymbol{\xi}^{(i)})|$, whose integral is $\mathsf{G}_{\operatorname{Re}\nu-\sum_{i\ge2}\mathbf{c}_i;t^{(1)}}(\boldsymbol{\xi}^{(1)})\prod_{i\ge2}\int|\mathsf{G}_{\mathbf{w}_i;t^{(i)}}(\boldsymbol{\xi}^{(i)})|\,|d\mathbf{w}_i|$, finite by Theorem 2.2 since $\operatorname{Re}\nu - \sum_{i\ge2}\mathbf{c}_i \in \operatorname{int}\operatorname{cone}(t^{(1)})$. So Fubini's theorem exchanges the integrals; the integral over $x$ is $\mathsf{G}_{\nu-\sum_{i\ge2}\mathbf{w}_i;t^{(1)}}(\boldsymbol{\xi}^{(1)})$, and the left side is the Gramma of the concatenation (Definition 2.1). □

**Definition 25.8 (Holding).** As stated: the hold of a family, the Mellin transform in the twists of its letters other than $K_1$.

*Remark.* That the bases of every table span $\mathbb{R}^{\mathcal{E}}$ is the first fact above. By the torus relation of Definition 22.1, a factor $\mu > 0$ on every coordinate of one twist $\mathbf{x}^X$ multiplies the period by $\mu^{-w_X \deg X}$, and a twist of every letter by one vector $\boldsymbol{\lambda}$ multiplies it by $\boldsymbol{\lambda}^{-\mathbf{m}}$; so the Mellin transform in all the twists of all the letters at all the marks diverges, and the hold keeps the twist of $K_1$ at $\mathbf{1}$ and, in Lemma 25.11, the twists at one mark.

**Definition 25.9 (Loads; admissible points).** As stated: the loads and the admissible points of a family, an admissible point of one table being named by its $\Psi$-load.

**Proposition 25.10 (The periods at the two full leans).** *As stated; (iii) is stated for every family, and the closing sentence is new.*

*Proof.* Items (i), (ii) and (iv) concern one table $\mathcal{H}$ of $F$, one molecule, and are the reference, proof of prop:fullleans. The closing sentence is Proposition 25.3, proved above; at the letter $K_{\mathcal{H}}$, with the support $(\mathbf{1}_T)_T$, the coefficients $\boldsymbol{\theta}$ and the Newton polytope $P(\mathcal{H})$ of dimension $n-1$, it gives back (i), (ii) and (iv) at the full basis lean, and at $\Psi_{\mathcal{H}}$ at the full cobasis lean.

(iii) Let $\alpha$ be a gauge. By Theorem 7.16, $R_j(\alpha) = \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[\mathbf{1}_T] = \sum_{T\in\mathcal{B}_j}\mathbb{P}_{j,\alpha}(T)\,\mathbf{1}_T$ with $\mathbb{P}_{j,\alpha}(T) = \theta_{j,T}\alpha^T/K_j(\alpha) > 0$ for every basis, and $R^*_{j,e}(\alpha) = \mathbb{P}^*_{j,\alpha}(e \notin T)$, so $R^*_j(\alpha) = \sum_T \mathbb{P}^*_{j,\alpha}(T)\,\mathbf{1}_{\mathcal{E}\setminus T}$ with $\mathbb{P}^*_{j,\alpha}(T) = \theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j(\alpha) > 0$. By the second fact above $R_j(\alpha) \in \operatorname{ri} P(\mathcal{B}_j)$ and $R^*_j(\alpha) \in \operatorname{ri} P(\mathcal{B}^*_j)$, the two polytopes being the convex hulls of the $\mathbf{1}_T$ and of the $\mathbf{1}_{\mathcal{E}\setminus T}$.

Let $\mathbf{m}$ be real on the balance wall with every proper wall positive. A mark of a table that is one molecule on at least two marks is neither a loop nor a coloop, so $d_j(\{e\}) = d^*_j(\{e\}) = 0$ and $m_e = \kappa_{\{e\}} > 0$ for every $e$ (Definition 8.1). A nonempty proper $S$ with $c_j(S) = 0$ would have $|T \cap S|$ constant over the bases of table $j$, and so would be a union of molecules of it by Lemma 7.10, which is impossible, table $j$ being one molecule; so $c_j(S) > 0$ in every table, every table being active, and $F$ is connected. By Theorem 8.19(ii) and (iii) a native gauge $\alpha^\star$ exists, the target being $\mathbf{m}$ on the balance wall (Definition 8.17) and the walls at the target positive. Its usage is the target, $\sum_j \big(s_j R_j(\alpha^\star) + t_j R^*_j(\alpha^\star)\big) = \mathbf{m}$: the native loads are a split of $\mathbf{m}$ among the letters, with $|\mathbf{a}^\star_j| = s_j r_j$ and $|\mathbf{b}^\star_j| = t_j r^*_j$, every $\mathbf{1}_T$ having $r_j$ marks and every $\mathbf{1}_{\mathcal{E}\setminus T}$ having $r^*_j$. By the preceding paragraph $\mathbf{a}^\star_j \in s_j \operatorname{ri} P(\mathcal{B}_j)$ and $\mathbf{b}^\star_j \in t_j \operatorname{ri} P(\mathcal{B}^*_j)$, so the native loads are admissible (Definition 25.9). □

**Lemma 25.11 (Holding every mark separates the letters).** *As stated, for every family.*

*Proof.* Let the data be real. By Theorem 8.7 the period of $F[\mathbf{x}]$ is the integral over the section $\{\alpha_{e_0} = 1\}$,

$$P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t}) = \int_{(0,\infty)^{\mathcal{E}\setminus e_0}} \alpha^{\mathbf{m}-\mathbf{1}} K_1(\alpha)^{-s_1} \prod_{X\in\mathcal{X}'} X(\mathbf{x}^X\circ\alpha)^{-w_X}\, d\alpha', \qquad \alpha = (\alpha', 1),$$

the twists at $e_0$ being 1 and the twist of $K_1$ being $\mathbf{1}$. Insert this into the hold and integrate first over the twists, by Tonelli's theorem, every function being positive. For fixed $\alpha'$ substitute $\mathbf{v}^X = \mathbf{x}^X\circ\alpha'$ on $\mathcal{E}\setminus e_0$ for every $X \in \mathcal{X}'$: then $(\mathbf{x}^X)^{\boldsymbol{\beta}^X-\mathbf{1}} d\mathbf{x}^X = \alpha'^{-\boldsymbol{\beta}^X}(\mathbf{v}^X)^{\boldsymbol{\beta}^X-\mathbf{1}} d\mathbf{v}^X$ and $X(\mathbf{x}^X\circ\alpha) = X(\mathbf{v}^X, 1)$, so the integral over $\mathbf{x}^X$ is $\alpha'^{-\boldsymbol{\beta}^X}$ times

$$\int_{(0,\infty)^{\mathcal{E}\setminus e_0}} (\mathbf{v}^X)^{\boldsymbol{\beta}^X-\mathbf{1}} X(\mathbf{v}^X,1)^{-w_X}\, d\mathbf{v}^X = P_X(\mathbf{a}^X),$$

the integral over the same section of the form of the one-letter word $X^{-w_X}$ at the extension $\mathbf{a}^X$ of $\boldsymbol{\beta}^X$ of total $w_X \deg X$ (Theorem 8.7); here $\boldsymbol{\beta}^{K_j} = \boldsymbol{\beta}^K_j$ and $\boldsymbol{\beta}^{\Psi_j} = \boldsymbol{\beta}^\Psi_j$, and $\mathbf{a}^X$ is the $\mathbf{a}_j$ or $\mathbf{b}_j$ of the statement. What is left is $\int \alpha'^{\mathbf{m}'-\mathbf{1}-\sum_X \boldsymbol{\beta}^X} K_1(\alpha', 1)^{-s_1} d\alpha'$, $\mathbf{m}' = \mathbf{m}|_{\mathcal{E}\setminus e_0}$, which is $P_{K_1}(\mathbf{a}_1)$ for $\mathbf{a}_1 = \mathbf{m} - \sum_{X\in\mathcal{X}'} \mathbf{a}^X$: its coordinates off $e_0$ are $\mathbf{m}' - \sum_X \boldsymbol{\beta}^X$, and its total is $m_{\mathcal{E}} - \sum_{X\in\mathcal{X}'} w_X \deg X = r_1 s_1$ on the balance wall. This is the identity in $[0,\infty]$. For complex $\mathbf{m}$ and $\boldsymbol{\beta}$ the moduli of all the integrands are their values at the real parts, where the integrals converge by assumption, and Fubini's theorem replaces Tonelli's. □

**Lemma S11.** *Let shares $\mathbf{v} \in \mathbb{C}^M$ be pinned by $P\mathbf{v} = \boldsymbol{\pi}$, with $q$ pins, as in Definition 4.1.*

*(i) The fibre integral does not change when the shares are replaced by $A^{-1}\mathbf{v}$, the integrand by $f\circ A$ and the pins by $PA$, for a real $M\times M$ matrix $A$ with $|\det A| = 1$, nor when the pins are replaced by $CP\mathbf{v} = C\boldsymbol{\pi}$, for a real $q\times q$ matrix $C$ with $|\det C| = 1$.*

*(ii) Let $\mathbf{v} = (\mathbf{u},\mathbf{w})$ with $\mathbf{w} \in \mathbb{C}^q$, so that $P = (P_{\mathbf{u}}, P_{\mathbf{w}})$, and let $|\det P_{\mathbf{w}}| = 1$. Then*

$$\int_{P\mathbf{v}=\boldsymbol{\pi}} f = \frac{1}{(2\pi i)^{M-q}} \int_{\mathbf{c}_{\mathbf{u}}+i\mathbb{R}^{M-q}} f\big(\mathbf{u}, P_{\mathbf{w}}^{-1}(\boldsymbol{\pi} - P_{\mathbf{u}}\mathbf{u})\big)\, d\mathbf{u},$$

*$\mathbf{c}_{\mathbf{u}}$ the $\mathbf{u}$-part of the admissible real point, the two sides converging absolutely together.*

*(iii) Let $\mathbf{v} = (\mathbf{v}_0, \mathbf{v}_1, \dots, \mathbf{v}_k)$ and let the pins be the union of pins $P_l\mathbf{v}_l = \boldsymbol{\pi}_l$, each on one block. If the fibre integral of $f$ converges absolutely, it is the integral over the fibre $P_0\mathbf{v}_0 = \boldsymbol{\pi}_0$ of the fibre integral over $P_1\mathbf{v}_1 = \boldsymbol{\pi}_1, \dots, P_k\mathbf{v}_k = \boldsymbol{\pi}_k$ at fixed $\mathbf{v}_0$, the inner one converging absolutely for almost every point of the outer imaginary translate.*

*Proof.* For $\mathbf{y}_0 \in \mathbb{R}^q$ let $\mu_{P,\mathbf{y}_0} = (\det PP^{\mathsf{T}})^{-1/2}\mathrm{vol}_{M-q}$ on $\{\mathbf{y} \in \mathbb{R}^M : P\mathbf{y} = \mathbf{y}_0\}$, so that the fibre integral is $(2\pi)^{q-M}\int f(\mathbf{c} + i\mathbf{y})\, d\mu_{P,\mathrm{Im}\,\boldsymbol{\pi}}(\mathbf{y})$. By the coarea formula for the linear map $P$, whose Jacobian is $(\det PP^{\mathsf{T}})^{1/2}$, these measures disintegrate Lebesgue measure, $\int_{\mathbb{R}^M}\varphi\, d\mathbf{y} = \int_{\mathbb{R}^q}\big(\int \varphi\, d\mu_{P,\mathbf{y}_0}\big)d\mathbf{y}_0$ for integrable $\varphi$, and $\mathbf{y}_0 \mapsto \int \varphi\, d\mu_{P,\mathbf{y}_0}$ is continuous for continuous $\varphi$ of compact support; these two properties determine the family $(\mu_{P,\mathbf{y}_0})_{\mathbf{y}_0}$.

(i) The substitution $\mathbf{y} = A\mathbf{y}'$ preserves Lebesgue measure and carries the disintegration along $PA$ to that along $P$; the substitution $\mathbf{y}_0 = C^{-1}\mathbf{y}_0'$ preserves Lebesgue measure on $\mathbb{R}^q$ and carries the disintegration along $CP$ to that along $P$, with $\mu_{CP,C\mathbf{y}_0} = \mu_{P,\mathbf{y}_0}$. The admissible real point and the factor $(2\pi)^{q-M}$ do not change.

(ii) The map $\mathbf{y} \mapsto (\mathbf{y}_\mathbf{u}, P\mathbf{y})$ has the block triangular matrix $\begin{pmatrix} I & 0 \\ P_\mathbf{u} & P_\mathbf{w} \end{pmatrix}$, of determinant of modulus 1; so $d\mathbf{y} = d\mathbf{y}_\mathbf{u}\, d\mathbf{y}_0$, and $\mu_{P,\mathbf{y}_0}$ is the image of Lebesgue measure on $\mathbb{R}^{M-q}$ under $\mathbf{y}_\mathbf{u} \mapsto \big(\mathbf{y}_\mathbf{u}, P_\mathbf{w}^{-1}(\mathbf{y}_0 - P_\mathbf{u}\mathbf{y}_\mathbf{u})\big)$. With $\mathbf{u} = \mathbf{c}_\mathbf{u} + i\mathbf{y}_\mathbf{u}$ and $d\mathbf{u} = i^{M-q}d\mathbf{y}_\mathbf{u}$ this is the display; both sides are one integral against one measure.

(iii) The pin matrix is block diagonal, so the disintegration along it is the product of the disintegrations along the blocks, $\mu_{P,(\mathbf{y}_{0,l})_l} = \bigotimes_l \mu_{P_l,\mathbf{y}_{0,l}}$, and the factor $(2\pi)^{q-M}$ is the product of the factors of the blocks. Fubini's theorem gives the iterated integral. □

*Remark.* In Theorem 25.12(ii) the shares are the loads of the $2N$ letters, $M = 2Nn$, with the $q = n+2N-1$ pins $\sum_X \mathbf{a}^X = \mathbf{m}$ and $|\mathbf{a}^X| = w_X \deg X$ for $X \in \mathcal{X}'$, and the eliminated coordinates are $\mathbf{a}_1$ and the loads $a^X_{e_0}$, $X \in \mathcal{X}'$. With the rows ordered as the mass pins at $e \neq e_0$, the mass pin at $e_0$ and the totals, and the columns as $a_{1,e}$ for $e \neq e_0$, $a_{1,e_0}$ and the $a^X_{e_0}$, the block $P_\mathbf{w}$ is

$$\begin{pmatrix} I_{n-1} & 0 & 0 \\ 0 & 1 & \mathbf{1}^{\mathsf{T}} \\ 0 & 0 & I_{2N-1} \end{pmatrix},$$

of determinant 1, and Lemma S11(ii) gives the coordinate form stated there, with the factor $(2\pi i)^{-(2N-1)(n-1)}$. With the further pins on $G$ of Theorem 25.14 the loads $a^X_e$, $X \in \mathcal{X}'$, $e \in G$, are eliminated as well, each by its own pin with the coefficient 1, and the block stays unit upper triangular. The function `eliminated_block_det` of `checks/bookkeeping.py` computes this determinant exactly for $n \leq 4$ and up to six letters.

**Lemma S12.** *Let $A, B > 0$, $w > 0$ and $c \in \mathbb{C}$. If $0 < \mathrm{Re}\, c < w$,*

$$\int_0^\infty x^{c-1}(A + xB)^{-w}\, dx = \Gamma\,{}_2^{}{}_1\!\left[\begin{matrix} c, & w-c \\ & w \end{matrix}\right] A^{c-w}B^{-c},$$

*absolutely convergent; for real $c \notin (0, w)$ the integral is $+\infty$.*

*Proof.* The substitution $x = (A/B)y$ gives $A^{c-w}B^{-c}\int_0^\infty y^{c-1}(1+y)^{-w}dy$. The last integral is the integral over the section $\{u_2 = 1\}$ of the form of the word $(u_1+u_2)^{-w}$ on two marks at the masses $(c, w-c)$: the family of the list $\binom{\{1,2\}}{1}$ with unit weights at the exponents $(w, 0)$, on its balance wall, whose walls are $c$ and $w-c$. For $0 < \mathrm{Re}\, c < w$ its period converges absolutely (Theorem 8.7), and by Lemma 16.2 the integral equals that over $\Delta_{\{1,2\}}$, where $u_1 + u_2 = 1$; by Theorem 3.9 it is

$\int_\Delta u^{(c,w-c)-\mathbf{1}}d\omega = \Gamma_{\substack{2\\1}}\!\left[\begin{matrix}c,\ w-c\\ w\end{matrix}\right]$. For real $c \le 0$ the integrand is at least $2^{-w}y^{c-1}$ on $(0,1)$, and for real $c \ge w$ at least $2^{-w}y^{c-w-1}$ on $(1,\infty)$; both integrals diverge. □

**Theorem 25.12 (The lean convolution).** *As stated, for every family; the closing sentence takes the active letters of a family with nonnegative exponents.*

*Proof.* (i) Let $\mathbf{c}$ be admissible. By Proposition 25.10(ii), $\mathbf{c}_j^K \in s_j \operatorname{ri} P(\mathcal{B}_j)$ is the condition $(\mathbf{c}_j^K)_S > s_j d_j(S)$ for every nonempty proper $S$, and likewise $(\mathbf{c}_j^\Psi)_S > t_j d_j^*(S)$; summing over the letters, $\operatorname{Re}\kappa_S = \operatorname{Re} m_S - \sum_j \big(s_j d_j(S) + t_j d_j^*(S)\big) > 0$ (Definition 8.1). Conversely, if every proper wall of $\operatorname{Re}\mathbf{m}$ is positive, the native loads of $\operatorname{Re}\mathbf{m}$ are admissible for $(\mathbf{m};\mathbf{s},\mathbf{t})$ by Proposition 25.10(iii), admissibility involving $\operatorname{Re}\mathbf{m}$ only. The admissible points are cut out by linear equations and the convex conditions $\mathbf{c}^X \in w_X \operatorname{ri}\mathcal{N}_X$, so they form a convex set.

(ii) Apply Theorem 25.4 to the family word $W = \prod_X X^{-w_X}$ split into the $2N$ factors $W_X = X^{-w_X}$, of degrees $w_X \deg X$. The Newton polytope of $W_X$ is $w_X\mathcal{N}_X$, of dimension $n-1$ by the first fact above, so its relative interior is $w_X \operatorname{ri}\mathcal{N}_X$, and the points $(\mathbf{c}_i)_i$ of Theorem 25.4 are exactly the admissible points. With the twists $\mathbf{x}^X$ of the factors, the left side of Theorem 25.4 is the integral over the section $\{\alpha_{e_0} = 1\}$ of the word of $F[\mathbf{x}]$, whose letters are $X(\mathbf{x}^X \circ \alpha)$ (Definition 22.1), that is $P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t})$ (Theorem 8.7); the periods of the factors are the $P_X(\mathbf{a}^X)$. This is the display, absolutely convergent, of dimension $(2N-1)(n-1)$; $K_1$ is taken as the last factor of Theorem 25.4, whose total pin is omitted, and that pin holds by the others and the balance wall. The coordinate form is the remark after Lemma S11.

(iii) For a real point $\mathbf{a}$ at which $P_X(\mathbf{a})$ converges, $|P_X(\mathbf{a}+i\mathbf{y})| \le P_X(\mathbf{a})$, the modulus of the integrand of a period being its value at the real parts, and $|(\mathbf{x}^X)^{-\mathbf{a}^X}| = (\mathbf{x}^X)^{-\operatorname{Re}\mathbf{a}^X}$. On the imaginary translate through $\mathbf{c}$ the imaginary part $\mathbf{y}^X$ of every load has $|\mathbf{y}^X| = 0$, by the pins, and Proposition 25.10(iv) bounds $|P_X(\mathbf{c}^X + i\mathbf{y}^X)|$ by $\big(\sin\frac{\pi\varepsilon}{2}\big)^{-w_X} e^{-(1-\varepsilon)h_X(\mathbf{y}^X)} P_X(\mathbf{c}^X)$, $h_X$ the norm of that proposition for the letter $X$.

The closing sentence is Theorem 25.4 with one factor for each active letter, the inactive letters being the constant 1; any factor may be taken last, and the walls and the admissible points are read with the active letters alone. □

**Corollary 25.13 (What a family retains).** *As stated, for every family.*

*Proof.* By Theorem 25.12(ii) and its closing sentence every period of $F$ and of its twists, at exponents $s_j, t_j \ge 0$ not all zero and at masses on the balance wall with every proper wall positive, is a fibre integral of products of the periods $P_X$ of its active letters at their loads, times the twists. By Proposition 25.10(i), $P_{\mathcal{B}_j}(\mathbf{a}; s_j, 0) = r_j\, \mathsf{G}_{\mathbf{a};(\mathbf{1}_T)_T}(\boldsymbol{\theta}_j)/\Gamma(s_j)$ for $s_j > 0$, and by the same item and Cremona $P_{\mathcal{B}_j}(\mathbf{b}; 0, t_j) = P_{\mathcal{B}_j}(t_j\mathbf{1} - \mathbf{b}; t_j, 0) = r_j\, \mathsf{G}_{t_j\mathbf{1}-\mathbf{b};(\mathbf{1}_T)_T}(\boldsymbol{\theta}_j)/\Gamma(t_j)$ for $t_j > 0$. The loads run over $|\mathbf{a}| = r_j s_j$ and $|\mathbf{b}| = r_j^* t_j$, that is $|t_j\mathbf{1} - \mathbf{b}| = nt_j - r_j^* t_j = r_j t_j$. The converse is Proposition 25.10(i) for table $j$. □

**Theorem 25.14 (Pinned holds).** *As stated, for every family.*

*Proof.* Put $d = (2N-1)|G|$. In the coordinates of Theorem 25.12(ii), the loads of the letters of $\mathcal{X}'$ at the marks other than $e_0$, split the coordinates into their values $\mathbf{z}_G$ on $G$ and the rest $\mathbf{z}'$, and let $\mathcal{I}(\mathbf{z}_G, \mathbf{z}')$ be the integrand at $\mathbf{x} = \mathbf{1}$, the product of the periods $P_X$ at the loads that $(\mathbf{z}_G, \mathbf{z}')$ and the pins determine. Let $\mathbf{c}$ be the admissible point of the statement, $\boldsymbol{\beta} = \mathbf{c}_G$, and for $\mathbf{y} \in \mathbb{R}^d$ put

$$\Phi(\mathbf{y}) = \frac{1}{(2\pi i)^{(2N-1)(n-1-|G|)}} \int \mathcal{I}(\mathbf{c}_G + i\mathbf{y}, \mathbf{z}')\, d\mathbf{z}',$$

over the imaginary translate through $\mathbf{c}$ in $\mathbf{z}'$. By Lemma S11(ii) and the remark after it, $\Phi(\boldsymbol{\eta})$ is the fibre integral of the statement at $\boldsymbol{\beta}+i\boldsymbol{\eta}$.

(a) By Theorem 25.12(iii), $|\mathcal{I}|$ on the imaginary translate is at most a constant times $\prod_{X\in\mathcal{X}'} e^{-(1-\varepsilon)h_X(\mathbf{y}^X)}$, $\mathbf{y}^X$ the imaginary part of the load of $X$, a vector of total 0 determined by its coordinates off $e_0$; since $h_X$ is a norm on the vectors of total 0, this bound is integrable in all the imaginary parts. So $\Phi$ is defined everywhere, continuous by dominated convergence, and integrable, with $\int|\Phi|\le(2\pi)^{(2N-1)(n-1-|G|)}\int|\mathcal{I}|$.

(b) For a twist at the marks of $G$ alone, $\mathbf{x}^X=e^{\mathbf{u}^X}$ on $G$ with $\mathbf{u}^X\in\mathbb{R}^G$ for $X\in\mathcal{X}'$ and $\mathbf{x}^{K_1}=\mathbf{1}$, the factor $\prod_X(\mathbf{x}^X)^{-\mathbf{a}^X}$ of Theorem 25.12(ii) is $e^{-\langle\mathbf{c}_G+i\mathbf{y},\mathbf{u}\rangle}$ on the translate, $\mathbf{u}=(\mathbf{u}^X)_X$. By Fubini's theorem in $(\mathbf{y},\mathbf{z}')$, $d\mathbf{z}_G=i^d\,d\mathbf{y}$,

$$e^{\langle\mathbf{c}_G,\mathbf{u}\rangle}P_{F[e^{\mathbf{u}}]}(\mathbf{m};\mathbf{s},\mathbf{t})=\frac{1}{(2\pi)^d}\int_{\mathbb{R}^d}\Phi(\mathbf{y})\,e^{-i\langle\mathbf{y},\mathbf{u}\rangle}\,d\mathbf{y}=:\widehat{\Phi}(\mathbf{u}).$$

(c) Let first $\mathbf{m}$ be real. Then $\widehat{\Phi}\ge0$, and for $\varepsilon>0$, by Fubini's theorem and the Gaussian integral,

$$\int_{\mathbb{R}^d}\widehat{\Phi}(\mathbf{u})\,e^{-\varepsilon|\mathbf{u}|^2/2}\,d\mathbf{u}=\int_{\mathbb{R}^d}\Phi(\mathbf{y})\,\frac{e^{-|\mathbf{y}|^2/(2\varepsilon)}}{(2\pi\varepsilon)^{d/2}}\,d\mathbf{y}\longrightarrow\Phi(\mathbf{0})\qquad(\varepsilon\downarrow0),$$

$\Phi$ being integrable and continuous at $\mathbf{0}$. By monotone convergence $\int\widehat{\Phi}\,d\mathbf{u}=\Phi(\mathbf{0})<\infty$, and with $d\mathbf{x}/\mathbf{x}=d\mathbf{u}$ this integral is $\mathrm{Hold}_G(\mathbf{c}_G)$: the statement at $\boldsymbol{\eta}=\mathbf{0}$.

(d) For complex $\mathbf{m}$, $|P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t})|\le P_{F[\mathbf{x}]}(\mathrm{Re}\,\mathbf{m};\mathbf{s},\mathbf{t})$, and $\mathbf{c}$ is admissible for $\mathrm{Re}\,\mathbf{m}$ as well; by (c) at $\mathrm{Re}\,\mathbf{m}$ the hold converges absolutely at every $\mathbf{c}_G+i\boldsymbol{\eta}$. Then $\widehat{\Phi}$ of (b) is integrable, $\Phi$ is integrable and continuous, and Fourier inversion gives, at every $\boldsymbol{\eta}$,

$$\mathrm{Hold}_G(\mathbf{c}_G+i\boldsymbol{\eta})=\int_{\mathbb{R}^d}e^{\langle\mathbf{c}_G+i\boldsymbol{\eta},\mathbf{u}\rangle}P_{F[e^{\mathbf{u}}]}(\mathbf{m};\mathbf{s},\mathbf{t})\,d\mathbf{u}=\int_{\mathbb{R}^d}\widehat{\Phi}(\mathbf{u})\,e^{i\langle\boldsymbol{\eta},\mathbf{u}\rangle}\,d\mathbf{u}=\Phi(\boldsymbol{\eta}).$$

At $G=\mathcal{E}\setminus e_0$ the fibre has dimension 0, and its value is the product of Lemma 25.11. □

**Proposition 25.15 (Holding one mark).** *As stated, for every family.*

*Proof.* Let the data be real. Take the section $\{h=1\}$ of Theorem 8.7 with $h(\alpha)=\sum_{e\ne g}\alpha_e$: on it $\alpha=(\alpha',\alpha_g)$ with $\alpha'\in\Delta_{\mathcal{E}\setminus g}$ and $\alpha_g\in(0,\infty)$, and the measure is $d\omega_{\mathcal{E}\setminus g}(\alpha')\,d\alpha_g$. By Definitions 22.1 and 21.1 the twist at $g$ acts on the letters as

$$K_j(\mathbf{x}@g\circ\alpha)=K_{\mathcal{B}_j\setminus g}(\alpha')+x_j^K\alpha_gK_{\mathcal{B}_j/g}(\alpha'),\qquad\Psi_j(\mathbf{x}@g\circ\alpha)=x_j^\Psi\alpha_g\Psi_{\mathcal{B}_j\setminus g}(\alpha')+\Psi_{\mathcal{B}_j/g}(\alpha'),$$

the minors carrying the weights of table $j$, all four nonzero since $g$ is neither a loop nor a coloop. Insert the period into the hold and integrate first over the twists of the letters of $\mathcal{X}'$, by Tonelli's theorem. By Lemma S12, for $j\ge2$, respectively for every $j$,

$$\int_0^\infty x^{a_j-1}\big(K_{\mathcal{B}_j\setminus g}+x\alpha_gK_{\mathcal{B}_j/g}\big)^{-s_j}dx=\Gamma_{\,1}^{\,2}\begin{bmatrix}a_j,\ s_j-a_j\\ s_j\end{bmatrix}K_{\mathcal{B}_j\setminus g}^{a_j-s_j}\big(\alpha_gK_{\mathcal{B}_j/g}\big)^{-a_j},$$
$$\int_0^\infty x^{b_j-1}\big(\Psi_{\mathcal{B}_j/g}+x\alpha_g\Psi_{\mathcal{B}_j\setminus g}\big)^{-t_j}dx=\Gamma_{\,1}^{\,2}\begin{bmatrix}b_j,\ t_j-b_j\\ t_j\end{bmatrix}\Psi_{\mathcal{B}_j/g}^{b_j-t_j}\big(\alpha_g\Psi_{\mathcal{B}_j\setminus g}\big)^{-b_j},$$

finite exactly when $0 < a_j < s_j$, respectively $0 < b_j < t_j$, and $+\infty$ otherwise. The power of $\alpha_g$ left is $\alpha_g^{m_g-1-\sum_{j\geq 2} a_j - \sum_j b_j} = \alpha_g^{a_1-1}$, and by the same lemma

$$\int_0^\infty \alpha_g^{a_1-1}\big(K_{\mathcal{B}_1\setminus g} + \alpha_g K_{\mathcal{B}_1/g}\big)^{-s_1} d\alpha_g = \Gamma_1^2\begin{bmatrix} a_1, & s_1 - a_1 \\ & s_1 \end{bmatrix} K_{\mathcal{B}_1\setminus g}^{a_1-s_1} K_{\mathcal{B}_1/g}^{-a_1},$$

finite exactly when $0 < a_1 < s_1$. What remains is

$$\int_{\Delta_{\mathcal{E}\setminus g}} \alpha'^{\mathbf{m}|_{\mathcal{E}\setminus g}-\mathbf{1}} \prod_j K_{\mathcal{B}_j\setminus g}^{-(s_j-a_j)}\, \Psi_{\mathcal{B}_j\setminus g}^{-b_j}\, K_{\mathcal{B}_j/g}^{-a_j}\, \Psi_{\mathcal{B}_j/g}^{-(t_j-b_j)}\, d\omega_{\mathcal{E}\setminus g},$$

the period of the children family $F_{\{g\}}\big((s_j-a_j,a_j),(b_j,t_j-b_j)\big)_j$ at $\mathbf{m}|_{\mathcal{E}\setminus g}$, which is on its balance wall (the paragraph before the proposition); it is finite exactly when every proper wall of it is positive (Theorem 8.7). The integrands being positive, the hold is the product of these factors in $[0,\infty]$ and is finite exactly when all of them are. For complex loads and masses whose real parts are such data the moduli are dominated by the real computation, Lemma S12 holds in the strip, and Fubini's theorem gives the same identity. The sentence on admissible points is Theorem 25.14 with $G = \{g\}$.

For the limits, let the letter $Y$ have the exponent $u > m_g > 0$ and let the twists at $g$ of the other letters decrease to 0. Each other letter decreases pointwise to one of its minors, $K_{\mathcal{B}_j\setminus g}$ or $\Psi_{\mathcal{B}_j/g}$, and its power with a negative exponent increases; by monotone convergence the period increases to the integral of the limit. In the limit only $Y$ contains $\alpha_g$, as $A+\alpha_g B$ with $A$ and $B$ its two minors, and by Lemma S12 the integral over $\alpha_g$ of $\alpha_g^{m_g-1}(A+\alpha_g B)^{-u}$ is $\Gamma_1^2\begin{bmatrix} m_g, & u-m_g \\ & u \end{bmatrix} A^{m_g-u}B^{-m_g}$. What is left is the period of the children family at the shares of the statement: for $Y = K_j$, $A = K_{\mathcal{B}_j\setminus g}$ carries the exponent $s_j - m_g$ and $B = K_{\mathcal{B}_j/g}$ the exponent $m_g$; for $Y = \Psi_j$, $A = \Psi_{\mathcal{B}_j/g}$ carries $t_j - m_g$ and $B = \Psi_{\mathcal{B}_j\setminus g}$ carries $m_g$; every other $K_{j'}$ is $K_{\mathcal{B}_{j'}\setminus g}$ at $s_{j'}$ and every other $\Psi_{j'}$ is $\Psi_{\mathcal{B}_{j'}/g}$ at $t_{j'}$. □

*Remark.* For one table a twist at $g$ in the cobasis letter is $x = x_1^\Psi$ with $x_1^K = 1$ (Definition 22.1), and $x \downarrow 0$ is the limit with the letter $K$ chosen. With the letter $\Psi$ chosen the twist $y$ of $K$ decreases to 0 with that of $\Psi$ at 1; by the torus relation of Definition 22.1 at the mark $g$, $P_{\mathcal{H}[(y,1)@g]} = y^{-m_g} P_{\mathcal{H}[(1,1/y)@g]} = x^{m_g} P_{\mathcal{H}[x@g]}$ with $x = 1/y$. So the two limits are those of $P_{\mathcal{H}[x@g]}$ as $x \downarrow 0$ and of $x^{m_g} P_{\mathcal{H}[x@g]}$ as $x \uparrow \infty$, as the paragraph after the proposition reads them.

**Lemma 25.16 (The Mellin transform of a law is the period of a children family).** *As stated, for every cell family.*

*Proof.* By Definitions 23.1 and 21.1, with $\sum_j (z_{j,K} + z_{j,\Psi}) = \kappa_F$ the power of $\gamma_{\mathcal{A}_1}^K$ in $\boldsymbol{\varrho}_F^{-\mathbf{z}}\, d\phi_F$ is $-\kappa_F + \sum_{j\geq 2} z_{j,K} + \sum_j z_{j,\Psi} = -z_{1,K}$. With $\gamma_{\mathcal{A}_j}^K = K_{\mathcal{A}_j\setminus g}/K_{\mathcal{A}_j/g}$ and $\gamma_{\mathcal{A}_j}^\Psi = \Psi_{\mathcal{A}_j/g}/\Psi_{\mathcal{A}_j\setminus g}$,

$$\begin{aligned} \boldsymbol{\varrho}_F^{-\mathbf{z}}\, d\phi_F &= u^{\mathbf{m}-\mathbf{1}} \prod_j K_{\mathcal{A}_j/g}^{-s_j}\, \Psi_{\mathcal{A}_j\setminus g}^{-t_j} \big(\gamma_{\mathcal{A}_j}^K\big)^{-z_{j,K}} \big(\gamma_{\mathcal{A}_j}^\Psi\big)^{-z_{j,\Psi}}\, d\omega_B \\ &= u^{\mathbf{m}-\mathbf{1}} \prod_j K_{\mathcal{A}_j\setminus g}^{-z_{j,K}}\, \Psi_{\mathcal{A}_j\setminus g}^{-(t_j-z_{j,\Psi})}\, K_{\mathcal{A}_j/g}^{-(s_j-z_{j,K})}\, \Psi_{\mathcal{A}_j/g}^{-z_{j,\Psi}}\, d\omega_B. \end{aligned}$$

This is the form of the children family of $\{g\}$ at the shares of the statement, its tables $\mathcal{A}_j \setminus g$, of rank $r_{\mathcal{A}_j}$ and corank $|B| - r_{\mathcal{A}_j}$, at $(z_{j,K}, t_j - z_{j,\Psi})$ and $\mathcal{A}_j/g$, of rank $r_{\mathcal{A}_j} - 1$ and corank $|B| - r_{\mathcal{A}_j} + 1$, at $(s_j - z_{j,K}, z_{j,\Psi})$. Its degree is

$$\sum_j \Big( r_{\mathcal{A}_j} z_{j,K} + (|B| - r_{\mathcal{A}_j})(t_j - z_{j,\Psi}) + (r_{\mathcal{A}_j} - 1)(s_j - z_{j,K}) + (|B| - r_{\mathcal{A}_j} + 1) z_{j,\Psi} \Big)$$
$$= \kappa_F + \sum_j \big( s_j (r_{\mathcal{A}_j} - 1) + t_j (|B| - r_{\mathcal{A}_j}) \big) = m_B,$$

so the children family is on its balance wall, and the integral over $\Delta_B$ is its period (Theorem 8.7), in $[0, \infty]$ for real $\mathbf{z}$. This is the first identity.

With the mass $\sum_j (s_j + t_j) - \kappa_F$ on $g$ the family of the tables $\mathcal{A}_j$ is on its balance wall (Lemma 21.12). Apply Proposition 25.15 to it at the mark $g$, a hinge and so neither a loop nor a coloop, with $a_j = s_j - z_{j,K}$ for $j \geq 2$ and $b_j = t_j - z_{j,\Psi}$. Then

$$a_1 = \sum_j (s_j + t_j) - \kappa_F - \sum_{j \geq 2} (s_j - z_{j,K}) - \sum_j (t_j - z_{j,\Psi}) = s_1 - z_{1,K},$$

the children family of Proposition 25.15 at the shares $(s_j - a_j, a_j) = (z_{j,K}, s_j - z_{j,K})$ and $(b_j, t_j - b_j) = (t_j - z_{j,\Psi}, z_{j,\Psi})$ is that of the first identity, its Beta corners are $\Gamma_{\substack{2\\1}} \begin{bmatrix} s_j - z_{j,K}, & z_{j,K} \\ & s_j \end{bmatrix}$ and $\Gamma_{\substack{2\\1}} \begin{bmatrix} t_j - z_{j,\Psi}, & z_{j,\Psi} \\ & t_j \end{bmatrix}$, and its hold is the left side of the second identity. At the real part of $\mathbf{z}$ the conditions of Proposition 25.15 hold: $0 < a_j < s_j$ and $0 < b_j < t_j$ are $0 < \operatorname{Re} z_{j,K} < s_j$ and $0 < \operatorname{Re} z_{j,\Psi} < t_j$, and the walls of the children family are positive since its period, $\int \boldsymbol{\varrho}_F^{-\operatorname{Re}\mathbf{z}} d\phi_F$ by the first identity, is finite. So Proposition 25.15 gives the second identity for complex $\mathbf{z}$ with this real part. □

*Remark.* The balance of the children family, the balance of the family of the tables $\mathcal{A}_j$ with the mass $\sum_j (s_j + t_j) - \kappa_F$ on $g$, and the identity $a_1 = s_1 - z_{1,K}$ are computed exactly by `check_hingesplit_balance` in `checks/bookkeeping.py`, and the balance of the children family of Proposition 25.15 by `check_holdone_balance`.

## 50.26 Chapter 26: periods at the full leans and networks along a hinge tree

The notation of Section 50.25 is kept. When table $j$ of a family carries a hinge tree $\mathfrak{T}_j$ as in Definition 26.2, its marker loads and node loads are read in table $j$ (Definition 26.3), at its loads $(\mathbf{a}_j, \mathbf{b}_j)$ and its exponents $(s_j, t_j)$.

**Lemma 26.1 (Substitutions of molecules).** *As stated, for any number of cells.*

*Proof.* The case of two cells is the reference, proof of lem:twosummol. By Proposition 21.3(i) the substitution is the gluing of the cells to $\mathcal{H}$ along its marks one at a time: $\mathcal{B}_0 = \mathcal{H}$, and $\mathcal{B}_i$ is the gluing of $\mathcal{A}_i$ and $\mathcal{B}_{i-1}$ along $h_i$, the marks $h_{i+1}, \ldots, h_q$ left in place. By Definition 21.1 and Proposition 21.3(iii) this gluing is the two-sum of $\mathcal{A}_i$ and $\mathcal{B}_{i-1}$, the latter read as a cell with the hinge $h_i$. This $h_i$ is neither a loop nor a coloop of $\mathcal{B}_{i-1}$: $\mathcal{H}$ has bases through $h_i$ and bases avoiding it, and each extends to a basis of $\mathcal{B}_{i-1}$ by bases of the hinge minors of the cells glued before, which exist since their hinges are neither loops nor coloops. By the case of two cells $\mathcal{B}_i$ is one molecule exactly when $\mathcal{A}_i$ and $\mathcal{B}_{i-1}$ are, and by induction $\mathcal{B}_q$ is one molecule exactly when $\mathcal{H}$ and all the $\mathcal{A}_i$ are. □

**Corollary 26.6 (Convergence is decided at the nodes).** *As stated, for every family.*

*Proof.* Every table is one molecule by Lemma 26.1 applied along its hinge tree. By Theorem 26.5 in table $j$, at real loads, $P_{\mathcal{B}_j}(\mathbf{a}_j; s_j, 0)$ is finite exactly when every marker load $k_{j;v,\mathfrak{e}}$ lies in $(0, s_j)$ and the period of every node at its full basis lean is finite at its node loads. By Proposition 25.10(ii) at the node, one molecule, the latter is $(\mathbf{a}_{j,v})_S > s_j d_{j,v}(S)$ for every nonempty proper $S$ of its marks and markers; at $S = \{g\}$ and at the complement of a marker $g$ these inequalities are $k > 0$ and $k < s_j$ for the marker load $k$ at $g$, since $g$ is neither a loop nor a coloop of the node and $|\mathbf{a}_{j,v}| = r_{j,v} s_j$ (Lemma 26.4(ii)). Likewise at the full cobasis lean. So the node inequalities hold at a split exactly when every load is admissible for its letter (Proposition 25.10(ii) in table $j$), that is when the split is an admissible point, and by Theorem 25.12(i) such a point exists exactly when every proper wall is positive; the native loads are one. The unknowns are the loads of the $2N$ letters at $n$ marks less $n + 2N - 1$ independent pins, $(2N-1)(n-1)$ of them, and the node $v$ of table $j$ contributes $2(2^{|\mathcal{E}_{j,v}|+|G_{j,v}|} - 2)$ inequalities. □

**Proposition 26.9 (Series and parallel nodes at their full leans).** *As stated; the last sentence, which reads the four closed forms off Proposition 25.6, is new.*

*Proof.* The closed forms are the reference, proof of prop:spleans. The letter $K$ of $\binom{\mathcal{X}}{1}$ is $\sum_x \theta_x \alpha_x$, and so is the letter $\Psi$ of $\binom{\mathcal{X}}{q}$, whose basis $\mathcal{X} \setminus x$ has the cobasis $\{x\}$: both are simplicial with the configuration $I = (\mathbf{e}_x)_x$, of degree 1, $|\det I| = 1$ and the shares $\mathbf{b} = \mathbf{a}$, and Proposition 25.6 gives $\Gamma_{\substack{q+1\\1}}\begin{bmatrix}\mathbf{a}\\|\mathbf{a}|\end{bmatrix} \prod_x \theta_x^{-a_x}$, the forms of $P_{\binom{\mathcal{X}}{1}}(\mathbf{a}; s, 0)$ and $P_{\binom{\mathcal{X}}{q}}(\mathbf{b}; 0, t)$. The letter $K$ of $\binom{\mathcal{X}}{q}$ and the letter $\Psi$ of $\binom{\mathcal{X}}{1}$ are $\sum_x \theta_x \alpha^{\mathcal{X}\setminus x}$, simplicial with the configuration $J - I = (\mathbf{1} - \mathbf{e}_x)_x$ of degree $q$, $J$ the matrix of ones. $J - I$ has the eigenvalues $q$, once, and $-1$, $q$ times, so $|\det(J - I)| = q$, and $(J - I)^{-1} = \frac{1}{q} J - I$, so the shares of $\mathbf{a}$ with $|\mathbf{a}| = qs$ are $\frac{|\mathbf{a}|}{q}\mathbf{1} - \mathbf{a} = (s - a_x)_x$, of sum $s$. Proposition 25.6 gives $\frac{q}{q}\Gamma_{\substack{q+1\\1}}\begin{bmatrix}(s-a_x)_x\\ s\end{bmatrix} \prod_x \theta_x^{a_x - s}$, the form of $P_{\binom{\mathcal{X}}{q}}(\mathbf{a}; s, 0)$, and in the same way that of $P_{\binom{\mathcal{X}}{1}}(\mathbf{b}; 0, t)$ with $|\mathbf{b}| = qt$. The criteria of convergence agree, the positivity of the real parts of the shares being that of the arguments of the Gammas. □

*Remark.* The function `check_spleans_is_simplicial` of `checks/bookkeeping.py` verifies exactly, at $q = 4$, the shares $s - a_x$ and the prefactor $q/|\det(J - I)| = 1$.

**Corollary 26.10 (Series-parallel tables at their full leans).** *As stated, with any weights on the bases of the series and parallel nodes.*

*Proof.* By Theorem 26.5 the period of $\mathcal{B}$ at its full basis lean is the product over the nodes of their periods at their node loads over the Beta corners $\Gamma_{\substack{2\\1}}\begin{bmatrix}k_{v,\mathfrak{e}}, & k_{w,\mathfrak{e}}\\ & s\end{bmatrix}$ of the edges, the weights of $\mathcal{B}$ being the products of those of the nodes (Definition 26.2). Proposition 26.9 gives each node: $\Gamma_{\substack{n_v\\1}}\begin{bmatrix}(s-(\mathbf{a}_v)_x)_x\\ s\end{bmatrix} \prod_x \theta_{v,x}^{(\mathbf{a}_v)_x - s}$ at a series node and $\Gamma_{\substack{n_v\\1}}\begin{bmatrix}((\mathbf{a}_v)_x)_x\\ s\end{bmatrix} \prod_x \theta_{v,x}^{-(\mathbf{a}_v)_x}$ at a parallel node, $n_v = |\mathcal{E}_v| + |G_v|$. The Gammas, the factors $\Gamma(s)$ and the corners combine as in the reference, proof of cor:spfulllean: a tree has one node more than edges, which leaves $1/\Gamma(s)$; a mark $e$ contributes $\Gamma(s - a_e)$ at a series node and $\Gamma(a_e)$ at a parallel node; at an edge $\mathfrak{e} = \{v, w\}$, where $k_{v,\mathfrak{e}} + k_{w,\mathfrak{e}} = s$ (Lemma 26.4(i)), the two markers contribute $\Gamma(s - k_{v,\mathfrak{e}})\Gamma(s - k_{w,\mathfrak{e}}) = \Gamma(k_{w,\mathfrak{e}})\Gamma(k_{v,\mathfrak{e}})$ between two series nodes and $\Gamma(k_{v,\mathfrak{e}})\Gamma(k_{w,\mathfrak{e}})$ between two parallel nodes, which the corner cancels, and $\Gamma(s - k_{v,\mathfrak{e}})\Gamma(k_{w,\mathfrak{e}})$ when $v$ is a series and $w$ a parallel node, which the corner leaves as $\Gamma(s - k_{\mathfrak{e}})/\Gamma(k_{\mathfrak{e}})$, $k_{\mathfrak{e}} = k_{v,\mathfrak{e}}$. The monomials of the nodes stand as they are. The full cobasis lean is the same with the second pair of forms of Proposition 26.9: $\Gamma(b_e)$ at series and $\Gamma(t - b_e)$ at parallel nodes, and at

an edge with the series end $v$ the quotient $\Gamma(l_{v,\mathfrak{e}})\Gamma(t-l_{w,\mathfrak{e}})/\big(\Gamma(l_{v,\mathfrak{e}})\Gamma(l_{w,\mathfrak{e}})\big)=\Gamma(l_{\mathfrak{e}})/\Gamma(t-l_{\mathfrak{e}})$. The criterion of finiteness does not see the weights, which enter through monomials, positive and finite at real loads; it is the reference's. □

**Corollary 26.11 (Periods along a hinge tree read by one hinge value).** *As stated: for a family whose tables have a common hinge tree, at real data, wherever the active letters read every subtree cell by one hinge value.*

*Proof.* Induction on the number of edges of $\mathfrak{T}$. With one vertex there is nothing to prove.

*Removing a leaf.* A tree with an edge has two leaves, so some leaf $v$ differs from $o$; let $p$ be its parent and $\mathfrak{e}=\mathfrak{e}_v$. The subtree cell $\mathcal{A}_{v,j}$ is the node $\mathcal{H}_{v,j}$ with the hinge $g_{v,\mathfrak{e}}$ (23.15). The two-sums along a tree may be taken in any order (23.5, 21.3(i),(ii)), so table $j$ of $F$ is the substitution $\mathcal{H}'_j\langle\mathcal{A}_{v,j}\rangle$ of the cell $\mathcal{A}_{v,j}$ at the mark $g_{p,\mathfrak{e}}$ of the table $\mathcal{H}'_j$ obtained from the other nodes along $\mathfrak{T}-v$, every other mark of $\mathcal{H}'_j$ left in place. The cell on $\mathcal{E}_v$ is read by the active letters through $\gamma_v$, and a mark left in place through $\alpha_h$, so 22.6 applies; the factors of the marks left in place cancel (its step (5)), and with $\kappa_{\mathfrak{e}}=\kappa_{\mathcal{E}_v}$,

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})\,\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix}\kappa_{\mathfrak{e}}, & 2\sigma-\kappa_{\mathfrak{e}}\\ & 2\sigma\end{bmatrix}=P_{F'}(\mathbf{m}';\mathbf{s},\mathbf{t})\,P_{F_{\mathcal{H}_v}}(\mathbf{m}_v;\mathbf{s},\mathbf{t}),$$

$F'$ the family of the tables $\mathcal{H}'_j$ at the masses $\mathbf{m}'$ equal to $\mathbf{m}$ on its marks and to $\kappa_{\mathcal{E}_v}$ at $g_{p,\mathfrak{e}}$, and $\mathbf{m}_v$ equal to $\mathbf{m}|_{\mathcal{E}_v}$ and to $2\sigma-\kappa_{\mathcal{E}_v}=\kappa_{\mathcal{E}\setminus\mathcal{E}_v}$ at $g_{v,\mathfrak{e}}$; every factor is on its balance wall with every proper wall positive.

*The walls of $F'$.* By 21.10, a set $S'$ of marks of $F'$ containing $g_{p,\mathfrak{e}}$ has the wall of $(S'\setminus\{g_{p,\mathfrak{e}}\})\sqcup\mathcal{E}_v$ in $F$, the cell entering with its whole set $\mathcal{E}_v$, of wall $\kappa_{\mathcal{E}_v}$, and the marker in $D_j$ and $D_j^*$; a set avoiding $g_{p,\mathfrak{e}}$ has the same wall in both. So the walls across the edges of $\mathfrak{T}-v$ are those of $F$.

*The hypothesis passes to $F'$.* $F'$ has the common hinge tree $\mathfrak{T}-v$, with the same nodes and $g_{p,\mathfrak{e}}$ a mark of the node $p$. Its subtree cells are those of $F$ except at the vertices $w$ on the path from $p$ to $o$, where $\mathcal{A}_{w,j}=\mathcal{A}'_{w,j}\langle\mathcal{A}_{v,j}\rangle$, the cell at the mark $g_{p,\mathfrak{e}}$ (21.3(ii)). By 21.4 the hinge minors of $\mathcal{A}_{w,j}$ are those of $\mathcal{A}'_{w,j}$ with $\alpha_{g_{p,\mathfrak{e}}}$ replaced by $\gamma^K_{\mathcal{A}_{v,j}}$ in the letters $K$ and by $\gamma^\Psi_{\mathcal{A}_{v,j}}$ in the letters $\Psi$, times factors of $\mathcal{A}_{v,j}$ that cancel in the quotients, so

$$\gamma^K_{\mathcal{A}_{w,j}}(\alpha)=\gamma^K_{\mathcal{A}'_{w,j}}\big(\alpha',\gamma^K_{\mathcal{A}_{v,j}}(\alpha_{\mathcal{E}_v})\big),\qquad \gamma^\Psi_{\mathcal{A}_{w,j}}(\alpha)=\gamma^\Psi_{\mathcal{A}'_{w,j}}\big(\alpha',\gamma^\Psi_{\mathcal{A}_{v,j}}(\alpha_{\mathcal{E}_v})\big),$$

$\alpha'$ the variables of the other marks. For the active letters both inner hinge values are $\gamma_v(\alpha_{\mathcal{E}_v})$, and $\gamma_v$, positive and homogeneous of degree one, takes every value in $(0,\infty)$. So for every value $c>0$ at $g_{p,\mathfrak{e}}$ the hinge values of $\mathcal{A}'_{w,j}$ by the active letters at $(\alpha',c)$ all equal $\gamma_w(\alpha)$ for any $\alpha$ with $\gamma_v(\alpha_{\mathcal{E}_v})=c$: the active letters read $\mathcal{A}'_{w,j}$ by one hinge value, and the induction hypothesis applies to $F'$. Its product formula and the display give that of $F$, with the masses on the markers the walls across the edges.

*One table at a full lean.* One letter is active, and the hypothesis is empty. At $(s,0)$ on the balance wall $|\mathbf{m}|=rs$, so the loads $\mathbf{a}=\mathbf{m}$ are admissible in 26.5; the marker load $k_{v,\mathfrak{e}}=sr_v^{\mathfrak{e}}-m_{\mathcal{E}_v^{\mathfrak{e}}}$ is $s-\kappa_{\mathcal{E}_v^{\mathfrak{e}}}$, the wall of the branch at $v$ being $m_{\mathcal{E}_v^{\mathfrak{e}}}-s(r_v^{\mathfrak{e}}-1)$, and the two walls across an edge add to $s$; so the node loads are the masses $\mathbf{m}_v$ and the two identities are the same (26.4). At $(0,t)$ the same holds with $\mathbf{b}$ and $t$. The series and parallel nodes are 26.9.

*Tables glued along a tree.* The glued table has the hinge tree whose nodes are the tables $\mathcal{A}_v$ and whose markers are the glued marks; the glued marks of connected tables are neither loops nor coloops, and a two-sum has the product weights (21.12). The masses of the glued marks are positive, every singleton being a proper wall of its table, so every $\kappa_{\mathfrak{e}}$ lies in $(0,t)$. Each table

$\mathcal{A}_v$ is balanced, $m_{\mathcal{E}_{\mathcal{A}_v}} = tr^*_{\mathcal{A}_v}$, and the glued table, of corank $\sum_v r^*_{\mathcal{A}_v} - (k-1)$ and total mass $\sum_v tr^*_{\mathcal{A}_v} - (k-1)t$, is on its balance wall. Removing a leaf as above, with the first display of 22.6 in $[0,\infty]$ and its step (2), the cell at a leaf $v$ glued along its mark $g$ has the wall $t - m_g$ and its cell family is $\mathcal{A}_v$ at its own masses, so the period of the glued table is $\prod_v P_{\mathcal{A}_v} / \prod_{\mathfrak{e}} \Gamma_{\substack{2\\1}}\!\left[\begin{smallmatrix}\kappa_{\mathfrak{e}}, & t-\kappa_{\mathfrak{e}}\\ & t\end{smallmatrix}\right]$ in $[0,\infty]$; it is finite, and the glued table, balanced, has every proper wall positive (8.7).

*A chain.* By 21.12 the chain $F^{\triangleright k}$ of 22.8 has the hinge tree whose nodes are the copies, a path, the glued marks its markers; by the propagation of the hypothesis above, its subtree cells are read by the active letters through one hinge value when the copy is, and the walls across its edges are $m_{g_{\mathrm{in}}}$ and $m_{g_{\mathrm{out}}}$, so with every proper wall positive the display gives the product $P_F^k / \Gamma_{\substack{2\\1}}\!\left[\begin{smallmatrix}m_{g_{\mathrm{in}}}, & m_{g_{\mathrm{out}}}\\ & 2\sigma\end{smallmatrix}\right]^{k-1}$. □

**Theorem 26.12 (Blocks read by one hinge value aggregate exactly).** *As stated: in (ii) blocks read through $e_\ell/e_{\ell-1}$ as well as blocks read through their totals, and the active letters at nonzero exponents of either sign.*

*Proof. (i) and (iii).* These are the reference proof of 26.12(i),(iii), the letters with positive exponent replaced by the active letters: at a parallel block $K_{\mathcal{A}/g} = 1$ and $\gamma^K_{\mathcal{A}} = e_1$, at a series block $\Psi_{\mathcal{A}\setminus g} = 1$ and $\gamma^\Psi_{\mathcal{A}} = e_1$ (21.7(i)), so by 21.4 every active letter is the letter of its core read at the block totals, and the totals of $U \sim \mathrm{Dir}_{\mathbf{m}}$ have the law $\mathrm{Dir}_{(m_{B_i})_i}$ (3.10). The sign of an exponent does not enter.

*(ii).* When a block has $\ell_i \geq 2$ marks a table with both letters active would need a block that is both parallel and series; so at such a block each table has at most one active letter, and the inactive letters, of exponent 0, enter neither the walls nor the kernels. By 21.7(i), for $i \in I$ the active hinge values are $\gamma^K = e_1$ at the parallel blocks and $\gamma^\Psi = e_1$ at the series blocks, and for $i \notin I$ they are $\gamma^K = e_{\ell_i}/e_{\ell_i-1}$ at the series blocks and $\gamma^\Psi = e_{\ell_i}/e_{\ell_i-1}$ at the parallel blocks. So 22.5 applies with all constants 1 and $\gamma_i = e_1$ for $i \in I$, $\gamma_i = e_{\ell_i}/e_{\ell_i-1}$ for $i \notin I$:

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = P_{F_{\mathcal{H}}}\big((\kappa_{B_i})_i;\mathbf{s},\mathbf{t}\big)\prod_i S_i, \qquad S_i = \int_{\Delta_{B_i}} \gamma_i^{-\kappa_{B_i}}\, d\beta_i,$$

in $[0,\infty]$. By 21.10, $\kappa_{B_i} = m_{B_i} - \sum_j \big(s_j(r_{\mathcal{A}_{ij}} - 1) + t_j(\ell_i - r_{\mathcal{A}_{ij}})\big)$ with the ranks 1 of a parallel block and $\ell_i$ of a series block.

For $i \in I$ the active letters see ranks that make every term vanish, so $\kappa_{B_i} = m_{B_i}$; the density of the cell kernel is $u^{\mathbf{m}|_{B_i}-\mathbf{1}}$, the active minors $K_{\mathcal{A}/g}$ and $\Psi_{\mathcal{A}\setminus g}$ being 1; and $e_1 = 1$ on the simplex, so $S_i = \int_{\Delta_{B_i}} u^{\mathbf{m}|_{B_i}-\mathbf{1}} d\omega$, which is $\Gamma_{\substack{\ell_i\\1}}\!\left[\begin{smallmatrix}\mathbf{m}|_{B_i}\\ m_{B_i}\end{smallmatrix}\right]$ when every $m_x$ is positive and $+\infty$ otherwise, by the shape integral (3.11).

For $i \notin I$ each active letter contributes its exponent times $\ell_i - 1$, so $\kappa_{B_i} = m_{B_i} - (\ell_i - 1)2\sigma$. The active minors are $K_{\mathcal{A}/g} = e_{\ell_i-1}$ at a series block and $\Psi_{\mathcal{A}\setminus g} = e_{\ell_i-1}$ at a parallel block, so the density of the cell kernel is $u^{\mathbf{m}|_{B_i}-\mathbf{1}} e_{\ell_i-1}(u)^{-2\sigma}$, and with $e_{\ell_i} = \prod_{x\in B_i} u_x$,

$$S_i = \int_{\Delta_{B_i}} \Big(\frac{e_{\ell_i}}{e_{\ell_i-1}}\Big)^{-\kappa_{B_i}} u^{\mathbf{m}|_{B_i}-\mathbf{1}} e_{\ell_i-1}^{-2\sigma}\, d\omega = \int_{\Delta_{B_i}} u^{\mathbf{m}|_{B_i}-\kappa_{B_i}\mathbf{1}-\mathbf{1}}\, e_{\ell_i-1}(u)^{-(2\sigma-\kappa_{B_i})}\, d\omega.$$

This is the period of the series node $\binom{B_i}{\ell_i-1}$ with unit weights, whose letter $K$ is $e_{\ell_i-1}$, at the full basis lean $(2\sigma-\kappa_{B_i}, 0)$ and the masses $m_x - \kappa_{B_i}$; they add to $m_{B_i} - \ell_i\kappa_{B_i} = (\ell_i-1)(2\sigma-\kappa_{B_i})$, its balance wall. By 26.9 it is $\Gamma_{\substack{\ell_i\\1}}\!\left[\begin{smallmatrix}(2\sigma-\kappa_{B_i}-m_x+\kappa_{B_i})_x\\ 2\sigma-\kappa_{B_i}\end{smallmatrix}\right] = \Gamma_{\substack{\ell_i\\1}}\!\left[\begin{smallmatrix}(2\sigma-m_x)_x\\ 2\sigma-\kappa_{B_i}\end{smallmatrix}\right]$, convergent exactly

when every $2\sigma - m_x$ is positive, and $+\infty$ otherwise, the integrand being positive. At $\ell_i = 1$ both kinds give $\kappa_{B_i} = m_x$ and $S_i = 1$.

So the display holds in $[0, \infty]$ with the stated reading of the stacked Gammas, and when these are finite the two sides are finite together.

*The last sentence.* At $\ell \geq 2$ the functions $e_1$ and $e_\ell / e_{\ell-1}$ are not proportional on the simplex, their quotient $e_1 e_{\ell-1}/e_\ell = (\sum_x u_x)(\sum_x u_x^{-1})$ being unbounded; so 22.5 does not apply to a block read both ways, and 22.3 keeps it as one mark with its kernel and both hinge values. □

*Remark.* Checks 7a to 7d of S21-23_work/checks.py are the case $\ell = 2$ of the blocks with $i \notin I$, with one table at $(s_1, 0)$ and one at $(0, t_2)$: $S = \Gamma(2\sigma - m_1)\Gamma(2\sigma - m_2)/\Gamma(4\sigma - m_B)$, and the period of the cell family is ${}_2\Gamma_1 \begin{bmatrix} \kappa, & 2\sigma - \kappa \\ & 2\sigma \end{bmatrix} S$.

**Theorem 26.13 (The network).** *As stated, for every family.*

*Proof.* Every table is one molecule by Lemma 26.1 along its hinge tree, and every proper wall is positive, so Theorem 25.12(ii) writes $P_{F[\mathbf{x}]}$ as the fibre integral of $\prod_j P_{\mathcal{B}_j}(\mathbf{a}_j; s_j, 0) P_{\mathcal{B}_j}(\mathbf{b}_j; 0, t_j)$ times the twists, on the imaginary translate through an admissible point $\mathbf{c}$. At $\mathbf{c}$ every $P_{\mathcal{B}_j}(\mathbf{c}_j^K; s_j, 0)$ and $P_{\mathcal{B}_j}(\mathbf{c}_j^\Psi; 0, t_j)$ is finite (Proposition 25.10(ii)), so by Theorem 26.5 in table $j$ every marker load lies in $(0, s_j)$, respectively $(0, t_j)$, every node period is finite at its node loads, and both factorizations of Theorem 26.5 hold on the tube over these real loads, which contains the translate. Inserting them gives the integrand of the statement. The native loads are admissible by Proposition 25.10(iii). The node periods are functions of the node loads alone, the exponent of the one active letter of a node being its total load divided by its degree (Definition 25.9). □

**Corollary 26.14 (Series-parallel periods).** *As stated, for every family and any weights on the bases of the nodes.*

*Proof.* Insert Corollary 26.10 for both full leans of every table into Theorem 26.13 and multiply by $\prod_j \Gamma(s_j)\Gamma(t_j)$. At a series node the monomials of the two leans multiply to $\theta_{j,v,x}^{(\mathbf{a}_{j,v})_x - s_j} \theta_{j,v,x}^{-(\mathbf{b}_{j,v})_x}$, at a parallel node to $\theta_{j,v,x}^{-(\mathbf{a}_{j,v})_x} \theta_{j,v,x}^{(\mathbf{b}_{j,v})_x - t_j}$. At a real point at which every Gamma above the line has an argument of positive real part all the periods of the tables at their full leans are finite (Corollary 26.10), so the point is admissible (Proposition 25.10(ii)) and Theorem 26.13 applies on it. The dimension is that of Theorem 26.13, and the integrand does not involve the bases of the tables. □

**Proposition 26.15 (Pooling a pair).** *As stated, with any weights on the bases of the node.*

*Proof.* For one table the pins are $\mathbf{a} + \mathbf{b} = \mathbf{m}$ and $|\mathbf{b}| = r^* t$, the twist $\mathbf{x}$ is on the cobasis letter, and the integrand of Corollary 26.14 is a function of $\mathbf{b}$, with $\Gamma(s - a_e) = \Gamma(c_e + b_e)$ at a mark $e$ of a series node. The marks $e$ and $f$ of one node $v$ lie on one side of every edge, so every marker load, an affine function of sums of loads over branches (Definition 26.3), depends on $b_e$ and $b_f$ through $b_{\{e,f\}}$ alone; so do the Gammas of the other marks, the monomials of the other nodes and those of the markers of $v$. What is left at a series node is $\Gamma(c_e + b_e)\Gamma(b_e)\Gamma(c_f + b_f)\Gamma(b_f)$ times

$$\theta_{v,e}^{a_e - b_e - s} \theta_{v,f}^{a_f - b_f - s} x_e^{-b_e} x_f^{-b_f} = \theta_{v,e}^{m_e - s} \theta_{v,f}^{m_f - s} \left(x_e \theta_{v,e}^2\right)^{-b_e} \left(x_f \theta_{v,f}^2\right)^{-b_f},$$

a function of $b_{\{e,f\}}$ when $x_e \theta_{v,e}^2 = x_f \theta_{v,f}^2$; at a parallel node it is $\Gamma(m_e - b_e)\Gamma(t - b_e)\Gamma(m_f - b_f)\Gamma(t - b_f)$ times $\theta_{v,e}^{-m_e - t} \theta_{v,f}^{-m_f - t} \left(x_e \theta_{v,e}^{-2}\right)^{-b_e} \left(x_f \theta_{v,f}^{-2}\right)^{-b_f}$, from the monomials $\theta_{v,x}^{b_x - a_x - t}$, a function of $b_{\{e,f\}}$ when

$x_e\theta_{v,e}^{-2} = x_f\theta_{v,f}^{-2}$. The rest is the reference, proof of prop:poolpair, with $x_e\theta_{v,e}^{\pm 2}$ in place of $x_e$: at fixed $b_{\{e,f\}}$ the integral over $b_e$ is the integral of four Gammas displayed in the statement, and disjoint pairs occupy disjoint coordinates. □

**Theorem 26.16 (Nodes in hold form).** *As stated, for every substituted family.*

*Proof.* The cores $\mathcal{H}_j$ and the cells $\mathcal{A}_{ij}$ are molecules by Lemma 26.1, and the markers are neither loops nor coloops of their cores and cells. Table $j$ carries the hinge tree whose nodes are $\mathcal{H}_j$ and the $\mathcal{A}_{ij}$, each cell read as a core on $B_i \sqcup \{g_i\}$ and joined to $\mathcal{H}_j$ by an edge $\mathfrak{e}_i$ (Definition 23.15), the table being obtained by these two-sums (Proposition 21.3(i)). Each $B_i$ is nonempty, a hinge being neither a loop nor a coloop; fix $e_i \in B_i$.

(i) *Step 1: the network.* Let $\mathbf{c}$ be an admissible point of $F$. By Theorem 26.13 with these trees, $P_{F[\mathbf{x}]}(\mathbf{m};\mathbf{s},\mathbf{t})$ is the fibre integral, over the loads of $F$ with the pins $\sum_j(\mathbf{a}_j+\mathbf{b}_j)=\mathbf{m}$, $|\mathbf{a}_j| = r_j s_j$ ($j \geq 2$) and $|\mathbf{b}_j| = r_j^* t_j$, on the translate through $\mathbf{c}$, of

$$\prod_j \frac{P_{\mathcal{H}_j}(\mathbf{a}_{j,\mathcal{H}};s_j,0)P_{\mathcal{H}_j}(\mathbf{b}_{j,\mathcal{H}};0,t_j)\prod_i P_{\mathcal{A}_{ij}}(\mathbf{a}_{j,i};s_j,0)P_{\mathcal{A}_{ij}}(\mathbf{b}_{j,i};0,t_j)}{\prod_i \Gamma_1^2\begin{bmatrix} a_{j,g_i},\ s_j-a_{j,g_i} \\ s_j\end{bmatrix}\Gamma_1^2\begin{bmatrix} b_{j,g_i},\ t_j-b_{j,g_i} \\ t_j\end{bmatrix}} (\mathbf{x}_j^K)^{-\mathbf{a}_j|\varepsilon_{\mathcal{H}}}(\mathbf{x}_j^\Psi)^{-\mathbf{b}_j|\varepsilon_{\mathcal{H}}}.$$

Here, by Definition 26.3 with the branch $B_i$ on the side of the cell, of rank $r_{\mathcal{A}_{ij}}$ and corank $|B_i|+1-r_{\mathcal{A}_{ij}}$, and by Lemma 26.4(i), the marker loads of the core at $g_i$ are

$$a_{j,g_i} = a_{j,B_i} - s_j\big(r_{\mathcal{A}_{ij}}-1\big), \qquad b_{j,g_i} = b_{j,B_i} - t_j\big(|B_i|-r_{\mathcal{A}_{ij}}\big),$$

those of the cell at $g_i$ are $s_j - a_{j,g_i}$ and $t_j - b_{j,g_i}$, the node loads of the core are $\mathbf{a}_{j,\mathcal{H}} = (\mathbf{a}_j|_{\mathcal{E}_{\mathcal{H}}}, (a_{j,g_i})_i)$ and $\mathbf{b}_{j,\mathcal{H}} = (\mathbf{b}_j|_{\mathcal{E}_{\mathcal{H}}}, (b_{j,g_i})_i)$, and those of the cell $\mathcal{A}_{ij}$ are $\mathbf{a}_{j,i} = (\mathbf{a}_j|_{B_i}, s_j - a_{j,g_i})$ and $\mathbf{b}_{j,i} = (\mathbf{b}_j|_{B_i}, t_j - b_{j,g_i})$. The twists are 1 off $\mathcal{E}_{\mathcal{H}}$.

*Step 2: the marker loads as coordinates.* In every table $j$ and for every $i$ replace the coordinate $a_{j,e_i}$ by $a_{j,g_i}$ and $b_{j,e_i}$ by $b_{j,g_i}$. Each new coordinate is the old one plus other old coordinates plus a constant, so the change of coordinates is unit triangular, of determinant 1, and by Lemma S11(i) the fibre integral does not change. The new coordinates are the *core coordinates*, the loads $(\mathbf{a}_{j,\mathcal{H}}, \mathbf{b}_{j,\mathcal{H}})_j$ of the cores on $\mathcal{E}_{\mathcal{H}} \sqcup \{g_1,\dots,g_k\}$, and for each $i$ the *coordinates of* $B_i$, the loads $a_{j,e}$ and $b_{j,e}$ at the marks $e \in B_i \setminus e_i$, $j = 1,\dots,N$.

*Step 3: the core masses.* Replace the mass pin at $e_i$ by the sum $C_i$ of the mass pins over $B_i$, a row operation with the coefficient 1 on the replaced pin, of determinant 1 (Lemma S11(i)). In the new coordinates

$$C_i: \qquad \sum_j \big(a_{j,g_i}+b_{j,g_i}\big) = m_{B_i} - \sum_j \Big(s_j\big(r_{\mathcal{A}_{ij}}-1\big) + t_j\big(|B_i|-r_{\mathcal{A}_{ij}}\big)\Big) = \kappa_{F_i}$$

by Definition 23.1: the mass of the cores at $g_i$ is the hinge wall of the cell family at $g_i$.

*Step 4: the totals, and the split of the pins.* By Proposition 21.3(i), $r_j = r_{\mathcal{H}_j} + \sum_i (r_{\mathcal{A}_{ij}}-1)$, and so $r_j^* = n - r_j = r_{\mathcal{H}_j}^* + \sum_i(|B_i| - r_{\mathcal{A}_{ij}})$ with $r_{\mathcal{H}_j}^* = |\mathcal{E}_{\mathcal{H}}| + k - r_{\mathcal{H}_j}$ and $n = |\mathcal{E}_{\mathcal{H}}| + \sum_i |B_i|$. In the new coordinates therefore

$$|\mathbf{a}_j| = \big|\mathbf{a}_j|_{\mathcal{E}_{\mathcal{H}}}\big| + \sum_i \Big(a_{j,g_i} + s_j\big(r_{\mathcal{A}_{ij}}-1\big)\Big), \qquad \text{so} \qquad |\mathbf{a}_j| = r_j s_j \iff |\mathbf{a}_{j,\mathcal{H}}| = r_{\mathcal{H}_j} s_j,$$

and likewise $|\mathbf{b}_j| = r_j^* t_j \iff |\mathbf{b}_{j,\mathcal{H}}| = r_{\mathcal{H}_j}^* t_j$. The pins are now the *core pins*, namely the mass pins at $\mathcal{E}_{\mathcal{H}}$, the $C_i$ and the totals of the cores, on the core coordinates alone, which are the pins of

$\mathcal{F}$; and for each $i$ the *pins of* $B_i$, the mass pins at $e \in B_i \setminus e_i$, on the coordinates of $B_i$ alone. The pin matrix is block diagonal, and since the fibre integral converges absolutely (Theorem 25.12(ii)), Lemma S11(iii) writes it as the integral over $\mathcal{F}$, on the translate through the core node loads of $\mathbf{c}$, of the fibre integrals over the coordinates of the $B_i$ at fixed core coordinates. The corners and the twists depend on the core coordinates alone and leave the inner integrals, and the inner integral for $B_i$ is

$$\Phi_i = \int \prod_j P_{\mathcal{A}_{ij}}(\mathbf{a}_{j,i}; s_j, 0)\, P_{\mathcal{A}_{ij}}(\mathbf{b}_{j,i}; 0, t_j)$$

over the coordinates of $B_i$ with the pins of $B_i$.

*Step 5: the inner integrals are holds.* Let $F_i^\natural$ be the family of the tables $\mathcal{A}_{ij}$ on $B_i \sqcup \{g_i\}$ with the masses $\mathbf{m}|_{B_i}$ and $\sum_j (s_j + t_j) - \kappa_{F_i}$ on $g_i$, on its balance wall (Lemma 21.12). The node loads $(\mathbf{a}_{j,i}, \mathbf{b}_{j,i})_j$ of the cells are loads of its letters: for every $j$, $|\mathbf{a}_{j,i}| = a_{j,B_i} + s_j - a_{j,g_i} = r_{\mathcal{A}_{ij}} s_j$ and $|\mathbf{b}_{j,i}| = t_j(|B_i| + 1 - r_{\mathcal{A}_{ij}})$, their masses at $e \in B_i$ are the $m_e$, and at $g_i$ they add to $\sum_j (s_j + t_j - a_{j,g_i} - b_{j,g_i}) = \sum_j (s_j + t_j) - \kappa_{F_i}$ by $C_i$. At $\mathbf{c}$ every period of a cell at its node loads is finite (Theorem 26.5), so the node loads of the cells at $\mathbf{c}$ are an admissible point of $F_i^\natural$ (Proposition 25.10(ii)), and every proper wall of $F_i^\natural$ is positive (Theorem 25.12(i)). In Theorem 25.14 for $F_i^\natural$ take $e_i$ as the fixed mark and $G = \{g_i\}$: the pinned fibre integral at the pins $s_j - a_{j,g_i}$ $(j \geq 2)$ and $t_j - b_{j,g_i}$ at $g_i$ is, in the coordinates of the loads of the letters other than $K_1$ at the marks of $B_i \setminus e_i$, the integral with the factor $(2\pi i)^{-(2N-1)(|B_i|-1)}$, the other loads solved from its pins. The integral $\Phi_i$, with $a_{1,e}$ for $e \in B_i \setminus e_i$ solved from the mass pin at $e$, has the same coordinates, the same factor by Lemma S11(ii), and the same integrand, since the loads $(\mathbf{a}_{j,i}, \mathbf{b}_{j,i})_j$ satisfy all the pins of $F_i^\natural$. The real parts of the core coordinates being those of $\mathbf{c}$, Theorem 25.14 gives

$$\Phi_i = \mathrm{Hold}_{\{g_i\}}\big((s_j - a_{j,g_i})_{j\geq 2}, (t_j - b_{j,g_i})_j\big) \qquad \text{for } F_i^\natural.$$

*Step 6: the corners cancel.* Apply Lemma 25.16 to the cell family $F_i$ at $\mathbf{z}_i = (a_{j,g_i}, b_{j,g_i})_j$, of sum $\kappa_{F_i}$ by $C_i$. Its conditions hold: $0 < \mathrm{Re}\, a_{j,g_i} < s_j$ and $0 < \mathrm{Re}\, b_{j,g_i} < t_j$, the marker loads at $\mathbf{c}$ lying in these intervals (Theorem 26.5); and the hold of Step 5 at the real parts is finite, so by Proposition 25.15 and the first identity of Lemma 25.16 $\int \boldsymbol{\varrho}_{F_i}^{-\mathrm{Re}\,\mathbf{z}_i} d\phi_{F_i}$ is finite. So

$$\Phi_i = \prod_j \Gamma_1^2\begin{bmatrix} s_j - a_{j,g_i}, & a_{j,g_i} \\ & s_j \end{bmatrix} \Gamma_1^2\begin{bmatrix} t_j - b_{j,g_i}, & b_{j,g_i} \\ & t_j \end{bmatrix} \int \boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i} d\phi_{F_i},$$

and these are the corners of the edge $\mathfrak{e}_i$ in Step 1, $\Gamma_1^2\begin{bmatrix} a, & b \\ & a+b \end{bmatrix}$ being symmetric in $a$ and $b$. They cancel, and what is left is the integral of (i) over $\mathcal{F}$. Its dimension is the number $2N(|\mathcal{E}_\mathcal{H}| + k)$ of core coordinates less the $|\mathcal{E}_\mathcal{H}| + k + 2N - 1$ core pins, $(2N-1)(|\mathcal{E}_\mathcal{H}| + k - 1)$.

(ii) Let $F'^\natural$ be the family of the tables of $F'$ with the mass $\sum_j (s_j + t_j) - \kappa_{F'}$ on $g_0$, on its balance wall (Lemma 21.12), and $\mathbf{c}$ an admissible point of it as in the statement. Its loads at $g_0$ lie in $(0, s_j)$ and $(0, t_j)$, $g_0$ being neither a loop nor a coloop, so $\mathrm{Re}\,\mathbf{z}$ meets the conditions of Lemma 25.16 for $F'$; the finiteness of $\int \boldsymbol{\varrho}_{F'}^{-\mathrm{Re}\,\mathbf{z}} d\phi_{F'}$ follows as in Step 6 from that of the hold of $F'^\natural$ at $g_0$ at the loads of $\mathbf{c}$ (Theorem 25.14). By the second identity of Lemma 25.16 the left side of (ii) is the hold of $F'^\natural$ at $g_0$ at the loads $(s_j - z_{j,K})_{j\geq 2}$ and $(t_j - z_{j,\Psi})_j$, and by Theorem 25.14 with $G = \{g_0\}$ it is the fibre integral of Theorem 25.12(ii) for $F'^\natural$ at $\mathbf{x} = \mathbf{1}$ with the pins at $g_0$. Steps 1 to 6 apply to it with $g_0$ a mark of the cores, its pins being core pins, and give the integral over $\mathcal{F}_\mathbf{z}$, of dimension $(2N-1)(|\mathcal{E}_\mathcal{H}| + k) - (2N-1) = (2N-1)(|\mathcal{E}_\mathcal{H}| + k - 1)$. □

*Remark.* Let the core of every table be $\binom{\{g_1,g_2\}}{1}$ with unit weights, the weights of the tables being those of the cells (Lemma 21.12), so that $\mathcal{E}_{\mathcal{H}} = \varnothing$ and $B = B_1$ is a bunch of width one in every table. By Proposition 26.9 at $q = 1$,

$$P_{\mathcal{H}_j}(\mathbf{a}_{j,\mathcal{H}}; s_j, 0)\, P_{\mathcal{H}_j}(\mathbf{b}_{j,\mathcal{H}}; 0, t_j) = \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix} a_{j,g_1},\ a_{j,g_2} \\ s_j \end{bmatrix} \Gamma_{\!\!\begin{smallmatrix}2\\1\end{smallmatrix}}\!\begin{bmatrix} b_{j,g_2},\ b_{j,g_1} \\ t_j \end{bmatrix} = \Gamma_{\!\!\begin{smallmatrix}4\\2\end{smallmatrix}}\!\begin{bmatrix} a_j^d,\ b_j^d,\ a_j^r,\ b_j^r \\ s_j,\ t_j \end{bmatrix}$$

of Theorem 19.13, with the level shares $a_j^r = a_{j,g_1}$, $a_j^d = a_{j,g_2}$, $b_j^d = b_{j,g_1}$ and $b_j^r = b_{j,g_2}$. With $\mathbf{z}_1 = (a_j^r, b_j^d)_j$ the children family of $F_1$ in Lemma 25.16 has the tables $\mathcal{A}_{1j} \setminus g_1 = \mathcal{B}_j | B$ at $(a_j^r, t_j - b_j^d) = (a_j^r, b_j^r)$ and $\mathcal{A}_{1j}/g_1 = \mathcal{B}_j/R$ at $(s_j - a_j^r, b_j^d) = (a_j^d, b_j^d)$: it is the bunch family of Theorem 19.13; with $\mathbf{z}_2 = (a_j^d, b_j^r)_j$ the children family of $F_2$ is the room family. The core pins are $|\mathbf{a}_{j,\mathcal{H}}| = s_j$ $(j \geq 2)$, $|\mathbf{b}_{j,\mathcal{H}}| = t_j$ and the masses $\kappa_{F_1}$ and $\kappa_{F_2}$; solving the totals for $a_j^d$ and $b_j^r$ by Lemma S11(ii), with the block of the totals and of the mass $\kappa_{F_2}$ unit triangular after the row operations of Lemma S11(i), leaves the share integral over $(a_j^r, b_j^d)_j$ pinned by $\sum_j (a_j^r + b_j^d) = \kappa_{F_1} = \kappa_B$. This is the integral of Theorem 19.13 over the scale shares, with the same integrand. In the pair coordinates $\sigma_j^d = \frac{1}{2}(a_j^d + b_j^d)$ and $\delta_j^d = \frac{1}{2}(a_j^d - b_j^d)$, the map $(a_j^r, b_j^d) \mapsto (\sigma_j^d, \delta_j^d)$, with $a_j^d = s_j - a_j^r$, has the determinant $\frac{1}{2}$ in each table, so $d(\mathbf{a}^r, \mathbf{b}^d) = 2^N\, d(\boldsymbol{\sigma}^d, \boldsymbol{\delta}^d)$; and the pin $\sum_j (a_j^r + b_j^d) = \kappa_B$ reads $-2\sum_j \delta_j^d = \kappa_B - \sum_j s_j$, its row being $-2$ times that of $\sum_j \delta_j^d = \frac{1}{2}(\sum_j s_j - \kappa_B)$, which by Definition 4.1 multiplies the fibre measure by $\frac{1}{2}$. The two factors give the $2^{N-1}$ of the first display of Theorem 19.13, a consistency check of the bookkeeping above. In `checks/bookkeeping.py`, `check_twosides_pairs` computes $2^N \cdot \frac{1}{2} - 2^{N-1} = 0$ exactly; `check_core_family_balance` computes the balance of the core family at the masses $(\mathbf{m}|_{\mathcal{E}_{\mathcal{H}}}, \kappa_{F_1}, \ldots, \kappa_{F_k})$ and the corank identity of Step 4; and `check_holdnode_pins` computes, for small families, the unit Jacobian of Step 2, the split of the pins of Steps 3 and 4 with the core masses $\kappa_{F_i}$ and the totals of the cores, and the totals and the hinge mass of the node loads of the cells in Step 5.

**Corollary 26.17 (The glue and the series and parallel rules in hold form).** *As stated; in (i) the reading for a family is new.*

*Proof.* For one table the point $\mathbf{z}_i$ of Theorem 26.16 is $(\kappa_{\mathcal{A}_i} - b_{g_i}, b_{g_i})$, by the core mass at $g_i$, and $\boldsymbol{\varrho}_{F_i}^{-\mathbf{z}_i} = \varrho_{\mathcal{A}_i}^{-b_{g_i}}$, the ratio vector of one cell being its ratio (Definition 23.1); so (i) for one table, (ii) and (iii) are the reference, proof of cor:holdrules. The reading of (i) for a family is the remark after the proof of Theorem 26.16. □

**Definition 26.18 (The network procedure).** As stated, for a family whose tables carry hinge trees; pooling and the passage to the transforms are for one table, and the recursive Mellin evaluation asks the tables to share one rooted tree.

*Remark.* The operations rest on the statements they cite, proved above for families: (1) is Corollary 26.6, infeasibility meaning that some proper wall is not positive, so that the period diverges (Theorem 8.7); (2) is Theorem 26.13 and Corollary 26.14; (4) uses that the admissible points form an open subset of the real points of the pins, the conditions $\mathbf{c}^X \in w_X \operatorname{ri} \mathcal{N}_X$ being open there, and the bounds of Theorem 25.12(iii), uniform on compact sets of admissible points; (5) is Theorem 26.16, the substituted family at a vertex $v$ having the cores $\mathcal{H}_{j,v}$ and, as its cell families, the subtree cell families of the vertices directly below $v$. The reusable data are those of Corollary 25.13, one Gramma for each orbit of nodes.

**Proposition 26.19 (What the network costs).** *As stated, for a family.*

*Proof.* (i) The fibre of Theorem 26.13 has dimension $(2N-1)(n-1)$, and for one table each pooled pair removes one coordinate (Proposition 26.15). Opening the period of a prime node $\mathcal{H}_{j,v}$ at one full lean on its share side adds $|\mathcal{H}_{j,v}| - n_{j,v}$ coordinates (Corollary 26.8), so the two full leans add $2(|\mathcal{H}_{j,v}| - n_{j,v})$; series and parallel nodes have closed forms (Proposition 26.9). The atom formula (Corollary 18.16) has the shares $\mathbf{a}_j, \mathbf{b}_j \in \mathbb{C}^{\mathcal{B}_j}$ against the $2N$ total pins and the $n-1$ pins $\langle \boldsymbol{\delta} \rangle_e = \mu_e$, $e \neq g$: dimension $\sum_j 2|\mathcal{B}_j| - 2N - n + 1$. After its even shares close (Theorem 20.11(iv)) the fibre of differences has dimension $\sum_j (|\mathcal{B}_j| - 1) - (n-1)$, and each table keeps one scale integral: $\sum_j |\mathcal{B}_j| - n + 1$ in all.

(ii) is Corollary 25.13: one Gramma of $n_v$ variables for each orbit of nodes, the orbits taken over all the tables.

(iii) By Theorem 26.16(ii) a point of the Mellin transform of the law of a subtree cell family with the top node $v$ costs a fibre integral of dimension $(2N-1)(|\mathcal{E}_v| + k_v - 1)$, $k_v = |G_v| - 1$ the number of vertices directly below $v$, that is $(2N-1)(|\mathcal{E}_v| + |G_v| - 2)$; at the root Theorem 26.16(i) costs $(2N-1)(|\mathcal{E}_o| + |G_o| - 1)$. Their integrands need the periods of the tables of the node at their full leans at the loads of the fibre, closed at series and parallel nodes and on the share side at a prime node (Corollary 26.8). □

**Proposition 26.20 (Product weights are twists).** *As stated, for every family.*

*Proof.* $K_{F^{\boldsymbol{\zeta}},j}(\alpha) = \sum_T \theta_{j,T} \boldsymbol{\zeta}_j^T \alpha^T = K_j(\boldsymbol{\zeta}_j \circ \alpha)$, and $\Psi_{F^{\boldsymbol{\zeta}},j}(\alpha) = \sum_T \theta_{j,T} \boldsymbol{\zeta}_j^T \alpha^{\mathcal{E}\setminus T} = \boldsymbol{\zeta}_j^{\mathcal{E}} \sum_T \theta_{j,T} (\alpha/\boldsymbol{\zeta}_j)^{\mathcal{E}\setminus T} = \boldsymbol{\zeta}_j^{\mathcal{E}} \Psi_j(\alpha/\boldsymbol{\zeta}_j)$. These are the letters of $F[\mathbf{x}]$ with $\mathbf{x}_j^K = \boldsymbol{\zeta}_j$ and $\mathbf{x}_j^\Psi = \boldsymbol{\zeta}_j^{-1}$ (Definition 22.1), the cobasis letters multiplied by $\boldsymbol{\zeta}_j^{\mathcal{E}}$, and $(\boldsymbol{\zeta}_j^{\mathcal{E}})^{-t_j} = \boldsymbol{\zeta}_j^{-t_j \mathbf{1}}$ gives the first identity. By the torus relation of Definition 22.1 for the family of the one letter $K_j$, the period of $K_j(\boldsymbol{\zeta}_j \circ \alpha)^{-s_j}$ at $\mathbf{a}$ is $\boldsymbol{\zeta}_j^{-\mathbf{a}} P_{\mathcal{B}_j}(\mathbf{a}; s_j, 0)$, and for the one letter $\Psi_j$ the period of $\big(\boldsymbol{\zeta}_j^{\mathcal{E}} \Psi_j(\alpha/\boldsymbol{\zeta}_j)\big)^{-t_j}$ at $\mathbf{b}$ is $\boldsymbol{\zeta}_j^{-t_j \mathbf{1}} \boldsymbol{\zeta}_j^{\mathbf{b}} P_{\mathcal{B}_j}(\mathbf{b}; 0, t_j)$. In the lean convolution of $F^{\boldsymbol{\zeta}}$ the integrand is therefore that of $F$ times $\prod_j \boldsymbol{\zeta}_j^{-t_j \mathbf{1}} \boldsymbol{\zeta}_j^{\mathbf{b}_j - \mathbf{a}_j}$. With one $\boldsymbol{\zeta}$ for every table, the torus relation with the vector $\boldsymbol{\zeta}^{-1}$ on every letter gives $P_{F[\mathbf{x}]} = \boldsymbol{\zeta}^{-\mathbf{m}} P_{F[\mathbf{x}']}$ for $\mathbf{x}_j^K = \boldsymbol{\zeta}$, $\mathbf{x}_j^\Psi = \boldsymbol{\zeta}^{-1}$ and $\mathbf{x}_j'^K = \mathbf{1}$, $\mathbf{x}_j'^\Psi = \boldsymbol{\zeta}^{-2}$, and $\prod_j \boldsymbol{\zeta}^{-t_j \mathbf{1}} = \boldsymbol{\zeta}^{-|\mathbf{t}|\mathbf{1}}$. Weights at one mark $g$ are the $\boldsymbol{\zeta}_j$ equal to 1 off $g$. □

**Proposition 26.21 (Carrying a calculation through a change, in hold form).** *As stated, for every family.*

*Proof.* In Theorem 26.13 the integrand is the product of the node periods at the node loads over the Beta corners at the marker loads, times the twists. (i) The exponents of table $j$ enter the pins $|\mathbf{a}_j| = r_j s_j$ and $|\mathbf{b}_j| = r_j^* t_j$ and the marker loads $k_{j;v,\mathfrak{e}} = s_j r_{j,v}^{\mathfrak{e}} - a_{j,\mathcal{E}_v^{\mathfrak{e}}}$ and $l_{j;v,\mathfrak{e}} = t_j r_{j,v}^{*\mathfrak{e}} - b_{j,\mathcal{E}_v^{\mathfrak{e}}}$, affine in $(s_j, t_j)$, and the corners through them; the node periods are functions of the node loads alone (Definition 25.9), and the native loads of the new data are admissible (Proposition 25.10(iii)). (ii) In the coordinates of Theorem 25.12(ii) the masses enter $\mathbf{a}_1 = \mathbf{m} - \sum_{j \geq 2} \mathbf{a}_j - \sum_j \mathbf{b}_j$ alone, the loads at $e_0$ of the other letters being fixed by their totals. (iii) is Proposition 26.20, for one node read for its table. (iv) The node periods are the only data of a node in the integrand; a replaced node of the same rank keeps the ranks of the branches and so the marker loads, and Theorem 26.13 applies to the new trees, whose nodes are molecules with markers neither loops nor coloops. (v) By Theorem 26.16(ii) the transform at a vertex is computed from the node there and the transforms of the vertices directly below it, so a change at one node changes the transforms on its path to the root only. □

**Proposition 26.22 (The exponents are moment indices).** *As stated, for every family, any of its letters being closed.*

*Proof.* The walls are affine in $(\mathbf{s},\mathbf{t})$ (Definition 8.1), so $I$, the intersection of the hyperplane with the open orthant and the open half-spaces $\kappa_S>0$, is relatively open and convex. On $I$ the period converges (Theorem 8.7), and $\mathbf{m}>0$, every singleton wall $m_e-\sum_j(s_j d_j(\{e\})+t_j d_j^*(\{e\}))$ being positive and the counts nonnegative. Fix $i$ and apply Lemma 16.2 to the form of $F$ with $h=L_i^{1/g_i}$, positive and homogeneous of degree one, and $\varphi(\rho)=\rho^{w_i g_i}e^{-\rho^{g_i}}$, for which $\int_0^\infty\varphi(\rho)\,d\rho/\rho=\Gamma(w_i)/g_i$: the integrand $x^{\mathbf{m}-\mathbf{1}}\prod_{i'}L_{i'}^{-w_{i'}}$ times $\varphi(h)=L_i^{w_i}e^{-L_i}$ is $x^{\mathbf{m}-\mathbf{1}}e^{-L_i}\prod_{i'\neq i}L_{i'}^{-w_{i'}}$, which gives the first identity. Every letter being positive on the orthant and every $w_{i'}>0$, $L_{i'}(x)^{-w_{i'}}=\Gamma(w_{i'})^{-1}\int_0^\infty\lambda^{w_{i'}-1}e^{-\lambda L_{i'}(x)}d\lambda$, and Tonelli's theorem gives the second identity, the integral over $x$ being $\mathsf{G}_i(\mathbf{m};\boldsymbol{\lambda})$, the Gramma of the concatenated configuration (Definition 2.1). The derivative $\partial^k_{\lambda_{i'}}\mathsf{G}_i$ is $(-1)^k$ times the integral with the further factor $L_{i'}^k$, which gives the complete monotonicity. By Proposition 8.12(iii), $P_F(\mathbf{m};\mathbf{s},\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ is the moment generating function, in the exponents, of $\big(-\log K_j(U),-\log\Psi_j(U)\big)_j$ for $U\sim\mathrm{Dir}_{\mathbf{m}}$, and by Hölder's inequality a moment generating function is log-convex on the convex set where it is finite; $\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ does not depend on the exponents. □

## 50.27 Chapter 27: finite interfaces and certified enclosures

**Definition 27.1 (The channel law; moment vectors).** As stated; cells with the same hinges on the same marks are kept together by reading their channels jointly.

**Proposition 27.2 (The channel law composes).** *As stated, for cells kept together and for every cell family.*

*Proof.* By Theorem 3.12 with the blocks $B_1$ and $B_2$, $U\sim\mathrm{Dir}_{(\mathbf{m}_1,\mathbf{m}_2)}$ on $\Delta_{B_1\sqcup B_2}$ is $U|_{B_l}=V_lU_l$ with $(V_1,V_2)\sim\mathrm{Dir}_{(m_{B_1},m_{B_2})}$ and $U_l\sim\mathrm{Dir}_{\mathbf{m}_l}$, independent. In every table $j$ the channels of the spliced cell are the bilinear map of Theorem 21.14(iii) applied to the channels of the two cells of table $j$, and a channel $c$ has $c(V_lU_l)=V_l^{\deg c}c(U_l)$. So $\mathrm{ch}(\mathcal{A}_{12})(U)$ is the image of the replaced channel vectors under the bilinear map applied table by table, and its law is the channel law of $\mathcal{A}_{12}$. For the tables of a family kept together without hinges the channels are the letters $K_j$ and $\Psi_j$, and Proposition 8.12(i) is the stated reading.

For the cell family $F$, by Definitions 23.1 and 21.1 at the point $\mathrm{ch}(\mathcal{A})(u)$, $\gamma^K_{\mathcal{A}_j}=x_{j,0}/x_{j,1}$ and $\gamma^\Psi_{\mathcal{A}_j}=x^*_{j,1}/x^*_{j,0}$, so $\gamma^K_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}=x_{j,0}x_{1,1}/(x_{j,1}x_{1,0})$ and $\gamma^\Psi_{\mathcal{A}_j}/\gamma^K_{\mathcal{A}_1}=x^*_{j,1}x_{1,1}/(x^*_{j,0}x_{1,0})$: $\boldsymbol{\varrho}_F=R_F\circ\mathrm{ch}(\mathcal{A})$. The base form is $\phi_F(du)=\varpi(\mathrm{ch}(\mathcal{A})(u))\,u^{\mathbf{m}-\mathbf{1}}d\omega_B(u)$ with $\varpi=\prod_j x_{j,1}^{-s_j}x_{j,0}^{*-t_j}(x_{1,0}/x_{1,1})^{-\kappa_F}$, the factor $\gamma^K_{\mathcal{A}_1}(u)^{-\kappa_F}$ being $(x_{1,0}/x_{1,1})^{-\kappa_F}$, and $u^{\mathbf{m}-\mathbf{1}}d\omega_B=\Gamma_{|B|}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}\mathrm{Dir}_{\mathbf{m}}(du)$ (Definition 3.10). For a positive measure $\mu$ and positive functions, $(R\circ c)_*\big((\varpi\circ c)\,\mu\big)=R_*\big(\varpi\,c_*\mu\big)$, both sides giving $\int H(R(c(u)))\,\varpi(c(u))\,d\mu(u)$ against every Borel $H\geq0$. With $c=\mathrm{ch}(\mathcal{A})$ and $\mu=\Gamma_{|B|}\begin{bmatrix}\mathbf{m}\\ m_B\end{bmatrix}\mathrm{Dir}_{\mathbf{m}}$ this is the display. □

**Theorem 27.3 (Gamma draws close the moments).** *As stated, for cells kept together.*

*Proof.* The reference, proof of thm:gammalift, uses of the channel vector only that it is a vector of polynomials on the marks, homogeneous of known degrees, that the Gamma draws on disjoint sets of marks are independent, and that a splicing acts on the channel vectors by a bilinear map

whose every product has the degree of the channel it gives. For cells kept together the channel vector is the joint vector $(\mathrm{ch}(\mathcal{A}_j))_j$, and a splicing acts table by table, so on the joint vectors by the direct sum over the tables of the maps of Theorem 21.14(iii), again bilinear with that property. So (i), (iii) and (iv) hold word for word. In (ii) the joint channels of $\mathcal{A}_{12}$ at $\alpha$ are $\mathrm{sp}\big(\mathrm{ch}(\mathcal{A}_1)(\alpha|_{B_1}), \mathrm{ch}(\mathcal{A}_2)(\alpha|_{B_2})\big)$, each a sum of products of one joint channel of each cell, so a monomial of degree $|\mathbf{k}|$ in them expands into monomials of degree $|\mathbf{k}|$ in the joint channels of each cell, and the independence of $\alpha|_{B_1}$ and $\alpha|_{B_2}$ gives the formula. In (v) the joint channels are bounded on $\Delta_B$, so by (iii) their moments determine their law, and Proposition 27.2 composes it. □

**Theorem 27.9 (Taylor coefficients in the weights are periods with walls at least as high).** *As stated, at real exponents of any sign.*

*Proof.* For $u \in \Delta_{\mathcal{E}}$ put $y_i(u) = \sum_j \zeta_j D_{ij}(u)/L_i(u)$. Write $L_i = \sum_{\boldsymbol{\gamma}} c_{i,\boldsymbol{\gamma}} u^{\boldsymbol{\gamma}}$ with $c_{i,\boldsymbol{\gamma}} > 0$; $D_{ij}$ has its support in that of $L_i$, so $D_{ij} = \sum_{\boldsymbol{\gamma}} c_{i,\boldsymbol{\gamma}} v_{i,\boldsymbol{\gamma},j} u^{\boldsymbol{\gamma}}$ with real $v_{i,\boldsymbol{\gamma},j}$, and with $V_{ij} = \max_{\boldsymbol{\gamma}} |v_{i,\boldsymbol{\gamma},j}|$, $|D_{ij}| \le V_{ij} L_i$ on the orthant and $|y_i| \le r_i(\boldsymbol{\zeta}) := \sum_j |\zeta_j| V_{ij}$; let every $r_i(\boldsymbol{\zeta}) < 1$. For real $w$ and $|y| < 1$, $(1+y)^{-w} = \sum_{l \ge 0} (-1)^l \frac{(w)_l}{l!} y^l$, and $|(w)_l| \le (|w|)_l$ since $|w+k| \le |w| + k$, so the series of the moduli is at most $\sum_l \frac{(|w|)_l}{l!} |y|^l = (1-|y|)^{-|w|}$. With the multinomial expansion of $y_i^{l_i}$,

$$L_i(\boldsymbol{\zeta})^{-w_i} = L_i^{-w_i}(1+y_i)^{-w_i} = \sum_{(l_{ij})_j} \frac{(-1)^{l_i}(w_i)_{l_i}}{\prod_j l_{ij}!}\, L_i^{-w_i - l_i} \prod_j \big(\zeta_j D_{ij}\big)^{l_{ij}}, \qquad l_i = \sum_j l_{ij},$$

and the product over $i$ of the series of the moduli is at most $\prod_i L_i^{-w_i}(1 - r_i(\boldsymbol{\zeta}))^{-|w_i|}$. Against $u^{\mathbf{m}-\mathbf{1}} d\omega_{\mathcal{E}}$ this bound is integrable, $P(\mathbf{0})$ converging at the real data; so termwise integration is justified, $P$ is a convergent power series on $\{r_i(\boldsymbol{\zeta}) < 1 \ \forall i\}$, and its coefficients are the stated ones. After the numerators are expanded every term is a coefficient times $u^{\mathbf{m}+\boldsymbol{\gamma}-\mathbf{1}} \prod_i L_i^{-w_i - l_i}$, $\boldsymbol{\gamma}$ a sum, over $i$, of $l_i$ exponents of the support of $L_i$. So $|\boldsymbol{\gamma}| = \sum_i l_i g_i$ and the term is balanced, and $\gamma_S \ge \sum_i l_i d_i(S)$ with $d_i(S)$ the least value of $\langle \mathbf{1}_S, \cdot \rangle$ on the support of $L_i$, which is the count $d_j(S)$ or $d_j^*(S)$ of the letter in the walls (Definition 8.1). So $\kappa'_S = \kappa_S + \gamma_S - \sum_i l_i d_i(S) \ge \kappa_S > 0$, and the term converges by Theorem 8.7, whose criterion holds for exponents of any sign. □

**Theorem 27.10 (A uniform certificate for weights).** *As stated, at real exponents of any sign.*

*Proof.* Let $|\zeta_j| \le R_j$, so that $|y_i| \le r_i < 1$ in the notation of the proof of Theorem 27.9. For real $w_i$, $|(1+y_i)^{-w_i}| = |1+y_i|^{-w_i}$, the real part of $1+y_i$ being positive, and $1 - r_i \le |1+y_i| \le 1 + r_i$, so $|L_i(\boldsymbol{\zeta})^{-w_i}| \le L_i^{-w_i} \max\big((1-r_i)^{-w_i}, (1+r_i)^{-w_i}\big)$; integrating, $|P(\boldsymbol{\zeta})| \le M$. The radii can be enlarged a little with every $r_i < 1$ kept, so $P$ is holomorphic near the closed polydisc by Theorem 27.9, and the Cauchy estimates give $|[\boldsymbol{\zeta}^{\mathbf{k}}]P| \le M \mathbf{R}^{-\mathbf{k}}$; the bounds on the Taylor remainder follow from these as in the reference, proof of thm:weightbank, there being $\binom{k+p-1}{p-1}$ multi-indices of degree $k$. For the explicit bound on $P(\mathbf{0})$: by Theorem 8.5(ii) and Theorem 8.7, off the balance wall and then on it by continuity, $P(\mathbf{0}) = \sum_\pi \int_{[0,1]^{n-1}} \prod_k q_k^{\kappa_{U_{\pi,k}}^{-1}} \prod_i L_{i,\pi}(q)^{-w_i}\, dq$. Each sector letter has nonnegative coefficients and is increasing in every $q_k$, so $L_{i,\pi}(\mathbf{0}) \le L_{i,\pi}(q) \le L_{i,\pi}(\mathbf{1})$ and $L_{i,\pi}(q)^{-w_i} \le \max\big(L_{i,\pi}(\mathbf{0})^{-w_i}, L_{i,\pi}(\mathbf{1})^{-w_i}\big)$ for every real $w_i$, and $\int_{[0,1]^{n-1}} \prod_k q_k^{\kappa_{U_{\pi,k}}^{-1}} dq = \prod_k \kappa_{U_{\pi,k}}^{-1}$. The enclosure of Proposition 15.12 holds for real data of any sign, its error bound being written with $\sum_j (|s_j| + |t_j|)$, so its upper end bounds $\prod_j K_j(\mathbf{1})^{s_j+t_j} P(\mathbf{0})$ from above. □

## 50.28 Chapter 28: bunches of any width

**Definition 28.1 (Seam).** As stated: the pair constant does not depend on the choice.

*Proof.* Let $p \in C$ and $X \in D$ with $p \sqcup X \in \mathcal{B}$. The child at $p$ carries the weight $\theta_{p\sqcup X}$ on its basis $X$ (12.2) and is $w_p$ times the child of the station $C$, so $\theta_{p\sqcup X} = w_p\,\theta_{(C),X}$; the room child at $X$ carries $\theta_{p\sqcup X}$ on its basis $p$ and is $w'_X$ times the room child of the station $D$, so $\theta_{p\sqcup X} = w'_X\,\theta_{(D),p}$. Hence $\theta_{p\sqcup X}/(w_p w'_X)$ is $\theta_{(C),X}/w'_X$, which does not depend on $p$, and $\theta_{(D),p}/w_p$, which does not depend on $X$. The children in a station are positive multiples of one another and so have the same bases: $p' \sqcup X \in \mathcal{B}$ for every $p' \in C$ and $p \sqcup X' \in \mathcal{B}$ for every $X' \in D$. Changing $p$ and then $X$ one at a time, the quotient takes one value at all pairs of $C \times D$. □

**Proposition 28.2 (The seam identity).** *As stated; in (v) the seam matrix of an extreme excess is a single entry exactly when the weights of that level factor.*

*Proof.* Parts (i)–(iv) and (vi), and in (v) the room children and the seam letters at the two extremes, are the reference, proof of prop:seamidentity. Let $i \in \{0, c(B)\}$. By 12.19 every pattern of excess $i$ and every room side of excess $i$ make a basis, so every station of patterns of excess $i$ is compatible with every station of room sides of excess $i$.

Let the weights of the level factor along these products, $\theta_{p\sqcup X} = \theta'_p\,\theta''_X$ for the patterns $p$ and the room sides $X$ of excess $i$, with positive $\theta'$ and $\theta''$. The child at a pattern $p$ of excess $i$ is the table on $R$ whose bases are all the room sides of excess $i$, with the weights $\theta'_p\,\theta''_X$: it is $\theta'_p$ times one table. So the patterns of excess $i$ form one station and $H_i$ has one row. The columns of a matrix with one row are proportional, and by (iv) no two stations of $\mathcal{D}_i$ have proportional columns; so $H_i$ has one column, and its entry is positive, the pair being compatible.

Conversely let $H_i$ be a single entry $h_{CD}$. The patterns of excess $i$ then form the one station $C$ and its room sides the one station $D$, and (ii) gives $\theta_{p\sqcup X} = h_{CD}\,w_p\,w'_X$ for every such $p$ and $X$: the weights of the level factor along these products, with the factors $h_{CD}\,w_p$ and $w'_X$. □

**Proposition 28.3 (The width ladder).** *As stated; in (i) and (ii) the seam is one rectangle at each excess exactly when the weights factor across $B$ and $R$, and the closing sentence holds at every width.*

*Proof.* (i) That $c(B) = 0$ exactly when $B$ is a union of molecules, with one excess, every pattern meeting every room side, and the letters $K = \mathsf{x}_0^{\mathsf{T}} H_0\,\mathsf{y}_0$ and $\Psi = \mathsf{x}_0^{*\mathsf{T}} H_0\,\mathsf{y}_0^*$, is the reference, proof of prop:ladder(i). The one excess is extreme, so by 28.2(v) the seam is one rectangle exactly when the weights of its level factor along the products $p \sqcup X$. Every basis lies in this level, so this is a factoring $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$ across $B$ and $R$, and conversely. With one station $C$ and one station $D$, 28.2(iii) reads $K = h_{CD}\,\mathsf{x}_C\,\mathsf{y}_D$ and $\Psi = h_{CD}\,\mathsf{x}_C^*\,\mathsf{y}_D^*$.

(ii) That $c(B) = 1$ exactly when there are two excesses, both extreme, is the reference, proof of prop:ladder(ii). By 28.2(v) the excess $i$ is one rectangle exactly when the weights of its level factor, $\theta_{p\sqcup X} = \theta'_{i,p}\,\theta''_{i,X}$ for the patterns $p$ and the room sides $X$ of excess $i$. If both levels factor, put $\theta'_p = \theta'_{i(p),p}$ and $\theta''_X = \theta''_{i(X),X}$, each pattern and each room side having one excess; the pattern and the room side of a basis have one excess, so $\theta_T = \theta'_{T\cap B}\,\theta''_{T\cap R}$ for every basis $T$. Conversely a factoring across $B$ and $R$ restricts to a factoring of each level. The seam letters, the room station letters and the room-station factorization are the reference's (proof of prop:ladder(ii), and exercise ex:widthone).

(iii) and the closing sentence. A matrix with one row has proportional columns, and a matrix with one column proportional rows; so by 28.2(iv) an excess with one station on one side has one

station on the other, at every width. The criterion for level form is 14.12, which does not restrict the width. In level form the children at the patterns of one excess are equal, so these patterns form one station; the excess then has one station on each side and is one rectangle. That an excess $0 < i < c(B)$ can carry several rectangles is the reference's (prop:ladder(iii) and exercise ex:prismseam). □

**Proposition 28.6 (Symmetries of a seam).** *As stated; in (iii) an exchanging bijection is read at every lean, and it carries the law of $B$ at the lean $\boldsymbol{\delta}$ to its law at $-\boldsymbol{\delta}$.*

*Proof.* Parts (i) and (ii), the first sentence of (iii), the carrying bijections and the sentence on the potential are the reference, proof of prop:seamsym.

Let $g$ keep $B$ and the masses and exchange the family, and let $j \mapsto j'$ be the bijection of its tables of 11.13. The pairs are kept, $(\sigma_{j'}, \delta_{j'}) = (\sigma_j, \delta_j)$, so at the lean $-\boldsymbol{\delta}$ table $j'$ carries the exponents $(t_{j'}, s_{j'}) = (t_j, s_j)$. Read every function of $\alpha_B$ at $g^*\alpha$. This permutes the coordinates of $\Delta_B$ and keeps $u^{\mathbf{m}_B - \mathbf{1}} d\omega_B$, since $g$ keeps $B$ and the masses. By 14.10, $c_{j'}(B) = c_j(B)$, $d_{j'}(B) = |B| - r_j(B) = d_j^*(B)$ and $d_{j'}^*(B) = d_j(B)$.

(a) By (ii) the seam letter $V_{jD}$ of $K_j$ at the excess $i(D)$ is, at $g^*\alpha$, the seam letter $V_{j'D'}^*$ of $\Psi_{j'}$ at the excess $c_j - i(D)$, and $V_{jD}^*$ is $V_{j'D'}$. Summing over the stations of one excess, $V_j^{(i)} = V_{j'}^{*(c_j - i)}$ and $V_j^{*(i)} = V_{j'}^{(c_j - i)}$.

(b) $K_j$ is active at $\boldsymbol{\delta}$ exactly when $s_j > 0$, that is exactly when $\Psi_{j'}$ is active at $-\boldsymbol{\delta}$; likewise $\Psi_j$ and $K_{j'}$. The sum $\kappa_B + \kappa_B^{\mathrm{top}} = \sum_j (s_j + t_j) c_j$ is the same at the two leans.

(c) By (a) and (b) the scale of the piece at $\boldsymbol{\delta}$ (28.5) is

$$\prod_j \Big( \frac{V_{j'}^{*(0)}}{V_{j'}^{*(c_j)}} \Big)^{s_j/(\kappa_B + \kappa_B^{\mathrm{top}})} \Big( \frac{V_{j'}^{(c_j)}}{V_{j'}^{(0)}} \Big)^{t_j/(\kappa_B + \kappa_B^{\mathrm{top}})},$$

which is the scale at $-\boldsymbol{\delta}$, where $\Psi_{j'}$ carries the exponent $s_j$ and $K_{j'}$ the exponent $t_j$.

(d) The wall at $-\boldsymbol{\delta}$ is $m_B - \sum_j \big( t_j d_{j'}(B) + s_j d_{j'}^*(B) \big) = m_B - \sum_j \big( s_j d_j(B) + t_j d_j^*(B) \big) = \kappa_B$.

(e) By (a), (c) and (d) the scale form at $\boldsymbol{\delta}$,

$$u^{\mathbf{m}_B - \mathbf{1}} \prod_j (V_j^{(0)})^{-s_j} (V_j^{*(c_j)})^{-t_j} (\gamma_B^\rho)^{-\kappa_B} d\omega_B,$$

is the scale form at $-\boldsymbol{\delta}$.

(f) For a station $D$ of excess $i(D)$ and its image $D'$ of excess $c_j - i(D)$,

$$\varrho_{jD}^\rho = \frac{V_{jD}}{V_j^{(0)} \, (\gamma_B^\rho)^{i(D)}} = \frac{V_{j'D'}^*}{V_{j'}^{*(c_j)} \, (\gamma_B^\rho)^{c_j - i(D')}},$$

the coordinate $\varrho_{j'D'}^{\rho *}$ at $-\boldsymbol{\delta}$; likewise $\varrho_{jD}^{\rho *}$ is the coordinate $\varrho_{j'D'}^\rho$ at $-\boldsymbol{\delta}$.

So the seam vector at $\boldsymbol{\delta}$, followed by the map of the coordinates, is the seam vector at $-\boldsymbol{\delta}$ read at $g^*\alpha$, and by (e) the scale forms correspond. The image of the law at $\boldsymbol{\delta}$ is therefore the law at $-\boldsymbol{\delta}$. By (a) and (b) the relations defining the seam torus at $\boldsymbol{\delta}$ go over into those at $-\boldsymbol{\delta}$. At the centre $\boldsymbol{\delta} = -\boldsymbol{\delta}$, and the law is kept. □

**Lemma 28.8 (Absorbing a piece of any width).** *As stated, with functions $f_l$ homogeneous of degree one in their second argument.*

*Proof.* Take the section $h = \gamma_B^\rho(\alpha_B) + |\alpha_Z|$, a positive function of degree one on $(0,\infty)^{B\sqcup Z}$, and write $\alpha_B = \lambda u$ with $\lambda > 0$ and $u \in \Delta_B$ (2.7).

- *The seam letters along the scale.* Let $f_l$ be assigned to $K_j$. By 28.5, $V_{jD} = V_j^{(0)}\,(\gamma_B^\rho)^{i(D)}\varrho_{jD}^\rho$ on $\Delta_B$, and $V_{jD}$ is homogeneous of degree $d_j + i(D)$. Hence

$$V_{jD}(\lambda u) = \lambda^{d_j} V_j^{(0)}(u)\,\big(\lambda\gamma_B^\rho(u)\big)^{i(D)}\varrho_{jD}^\rho(u),$$
$$\text{that is}\quad \mathbf{V}_l(\lambda u) = V_l(\lambda u)\,\mathbf{x}_l(\alpha_g),$$

  at $\alpha_g = \lambda\gamma_B^\rho(u)$ and $\mathbf{x} = \boldsymbol{\varrho}_B^\rho(u)$, with $V_l(\lambda u) = \lambda^{d_j} V_j^{(0)}(u) > 0$. For $\Psi_j$ the same holds with $V_{jD}^* = V_j^{*(c_j)}(\gamma_B^\rho)^{c_j - i(D)}\varrho_{jD}^{\rho *}$, of degree $d_j^* + c_j - i(D)$, and $V_l(\lambda u) = \lambda^{d_j^*} V_j^{*(c_j)}(u)$.

- *Homogeneity.* Since $f_l$ is homogeneous of degree one in its second argument,

$$f_l\big(\alpha_Z, \mathbf{V}_l(\lambda u)\big) = V_l(\lambda u)\, f_l\big(\alpha_Z, \mathbf{x}_l(\alpha_g)\big).$$

  So the arguments of $H$ are the $f_l(\alpha_Z, \mathbf{x}_l(\alpha_g))$. The exponents assigned to each active letter add to its exponent, so

$$\prod_l f_l\big(\alpha_Z, \mathbf{V}_l(\lambda u)\big)^{-a_l} = \lambda^{-\sum_j (s_j d_j + t_j d_j^*)} \prod_j V_j^{(0)}(u)^{-s_j} V_j^{*(c_j)}(u)^{-t_j} \prod_l f_l\big(\alpha_Z, \mathbf{x}_l(\alpha_g)\big)^{-a_l}.$$

- *The scale and the stand-in.* With $\alpha_B^{\mathbf{m}_B - \mathbf{1}} d\alpha_B = \lambda^{m_B - 1} u^{\mathbf{m}_B - \mathbf{1}}\, d\lambda\, d\omega_B(u)$, the power of $\lambda$ is $\lambda^{\kappa_B - 1}$. At fixed $u$ substitute $\alpha_g = \lambda\gamma_B^\rho(u)$, so that $\lambda^{\kappa_B - 1} d\lambda = \gamma_B^\rho(u)^{-\kappa_B}\alpha_g^{\kappa_B - 1} d\alpha_g$. In the coordinates $(u, \alpha_g, \alpha_Z)$ the form $\omega$ is then $\phi_B^\rho(du)$ times $\omega_{\boldsymbol{\varrho}_B^\rho(u)}$ (28.5), and $h = \alpha_g + |\alpha_Z|$.

- *The integral.* At fixed $u$ the map $(\lambda, \alpha_Z) \mapsto (\alpha_g, \alpha_Z)$ intertwines the scalings of the two orthants, so $\omega_{\boldsymbol{\varrho}_B^\rho(u)}$ is homogeneous of degree zero together with $\omega$. By 16.2, the integral of a form homogeneous of degree zero over a section does not depend on the section. Tonelli's theorem, applied to the nonnegative integrand, then gives

$$\int_{\{h=1\}} \omega = \int_{\Delta_B} \phi_B^\rho(du) \int_{\{\alpha_g + |\alpha_Z| = 1\}} \omega_{\boldsymbol{\varrho}_B^\rho(u)} = \int \Big( \int_{\{h'=1\}} \omega_{\mathbf{x}} \Big)\, d\big((\boldsymbol{\varrho}_B^\rho)_* \phi_B^\rho\big)(\mathbf{x}).$$

This is the reference's proof (proof of lem:absorbwidth). There, linearity of the $f_l$ enters only through the identity $f_l(\alpha_Z, \mathbf{V}_l(\lambda u)) = V_l(\lambda u)\, f_l(\alpha_Z, \mathbf{x}_l(\alpha_g))$, which homogeneity of degree one gives. □

*Remark.* At width one with factoring weights, normalized as in 28.11, the seam letters of $K_j$ are $(K_{\mathcal{B}_j/R}, K_{\mathcal{B}_j|B})$ and those of $\Psi_j$ are $(\Psi_{\mathcal{B}_j/R}, \Psi_{\mathcal{B}_j|B})$ (Lemma S13). The positive continuous functions homogeneous of degree one in the second argument are, for $K_j$, $f_l(\alpha_Z, (y_0, y_1)) = y_0\, f(\alpha_Z, y_1/y_0)$, and for $\Psi_j$, $f_l(\alpha_Z, (y_0, y_1)) = y_1\, f(\alpha_Z, y_0/y_1)$, with $f$ positive and continuous on $(0,\infty)^Z \times (0,\infty)$. At the seam letters they are $K_{\mathcal{B}_j/R}\, f(\alpha_Z, \gamma_{j,B}^K)$ and $\Psi_{\mathcal{B}_j|B}\, f(\alpha_Z, \gamma_{j,B}^\Psi)$. These are the factors $L_i\, f_i(\alpha_Z, \gamma_i)$ of 23.2 for the cell family $F_B^+$, with arbitrary $f_i$. Linear functions $f_l$ give only the $f_i$ that are affine in the hinge value; homogeneity of degree one gives all of them.

**Lemma S13.** *Let a bunch $B$ have width one in table $j$ of a family, the weights factoring across $B$ and $R$, $\theta_{j,T} = \theta'_{j,T\cap B}\,\theta''_{j,T\cap R}$, with the stations $D_0$ and $D_1$ of the excesses 0 and 1 (28.3(ii)). In this statement write $\hat\theta_i$ for the largest value of $\theta''_j$ on the room sides of excess $i$, and read $\mathcal{B}_j/R$ and $\mathcal{B}_j|B$ with the weights $\theta'_j$ and $\mathcal{B}_j|R$ and $\mathcal{B}_j/B$ with the weights $\theta''_j$.*

*(i)* $K_{j,(D_0)} = \hat\theta_0 K_{\mathcal{B}_j/R}$, $\Psi_{j,(D_0)} = \hat\theta_0\Psi_{\mathcal{B}_j/R}$, $K_{j,(D_1)} = \hat\theta_1 K_{\mathcal{B}_j|B}$ *and* $\Psi_{j,(D_1)} = \hat\theta_1\Psi_{\mathcal{B}_j|B}$; *and* $\mathsf{y}_{jD_0} = K_{\mathcal{B}_j|R}/\hat\theta_0$, $\mathsf{y}^*_{jD_0} = \Psi_{\mathcal{B}_j|R}/\hat\theta_0$, $\mathsf{y}_{jD_1} = K_{\mathcal{B}_j/B}/\hat\theta_1$ *and* $\mathsf{y}^*_{jD_1} = \Psi_{\mathcal{B}_j/B}/\hat\theta_1$.

*(ii) For $i = 0, 1$, divide $\theta''_j$ by $\hat\theta_i$ on the room sides of excess $i$ and multiply $\theta'_j$ by $\hat\theta_i$ on the patterns of excess $i$. This gives a factorization of the same weights with $\hat\theta_0 = \hat\theta_1 = 1$, in which $\theta''_j$ is the system of the multipliers of the room sides (28.1). The seam letters and the room station letters do not depend on the factorization. In the normalized factorization*

$$\gamma^K_{j,B} = \frac{K_{j,(D_1)}}{K_{j,(D_0)}}, \qquad \gamma^\Psi_{j,B} = \frac{\Psi_{j,(D_0)}}{\Psi_{j,(D_1)}}, \qquad \gamma^K_{j,R} = \frac{\mathsf{y}_{jD_0}}{\mathsf{y}_{jD_1}}, \qquad \gamma^\Psi_{j,R} = \frac{\mathsf{y}^*_{jD_1}}{\mathsf{y}^*_{jD_0}}.$$

*In a factorization with the constants $\hat\theta_0$ and $\hat\theta_1$, the four ratios on the right are $(\hat\theta_1/\hat\theta_0)\gamma^K_{j,B}$, $(\hat\theta_0/\hat\theta_1)\gamma^\Psi_{j,B}$, $(\hat\theta_1/\hat\theta_0)\gamma^K_{j,R}$ and $(\hat\theta_0/\hat\theta_1)\gamma^\Psi_{j,R}$.*

*Proof.* (i) By 19.4 the bases of table $j$ are the $X \sqcup Y$ with $X \in \mathcal{B}_j|B$ and $Y \in \mathcal{B}_j/B$, at the level $r_j(B)$, and with $X \in \mathcal{B}_j/R$ and $Y \in \mathcal{B}_j|R$, at the level $d_j(B)$.

- So the patterns and the room sides of excess 0 are the bases of $\mathcal{B}_j/R$ and of $\mathcal{B}_j|R$, those of excess 1 the bases of $\mathcal{B}_j|B$ and of $\mathcal{B}_j/B$, and every pattern meets every room side of its excess.
- The room child at a room side $X$ of excess 0 (28.1) is the table on $B$ with the bases of $\mathcal{B}_j/R$ and the weights $\theta'_{j,p}\,\theta''_{j,X}$, that is $\theta''_{j,X}$ times $\mathcal{B}_j/R$ with the weights $\theta'_j$.
- These room children are positive multiples of one another, so the room sides of excess 0 form the one station $D_0$. Its multipliers are $w'_X = \theta''_{j,X}/\hat\theta_0$, the largest equal to 1, and the room child of the station is $\hat\theta_0$ times $\mathcal{B}_j/R$ with $\theta'_j$.
- Hence $K_{j,(D_0)} = \hat\theta_0 K_{\mathcal{B}_j/R}$ and $\Psi_{j,(D_0)} = \hat\theta_0\Psi_{\mathcal{B}_j/R}$, and the station letters are $\mathsf{y}_{jD_0} = \sum_X w'_X\alpha^X_R = K_{\mathcal{B}_j|R}/\hat\theta_0$ and $\mathsf{y}^*_{jD_0} = \sum_X w'_X\alpha^{R\setminus X}_R = \Psi_{\mathcal{B}_j|R}/\hat\theta_0$.
- The excess 1 is the same with $\mathcal{B}_j|B$, $\mathcal{B}_j/B$ and $\hat\theta_1$.

(ii) A basis has its pattern and its room side at one excess $i$, so the new weights multiply to $(\hat\theta_i\,\theta'_{j,p})(\theta''_{j,X}/\hat\theta_i) = \theta_{j,p\sqcup X}$. The new $\theta''_j$ has the largest value 1 on each excess, so it is the system of the multipliers $w'_X$. The stations, the multipliers, the seam letters and the station letters are defined from the weights $\theta_j$ alone (28.1), so they do not change. In the normalized factorization, (i) holds with $\hat\theta_0 = \hat\theta_1 = 1$, and the four ratios are the hinge values of 19.2, read with the minors carrying the normalized weights. In a factorization with the constants $\hat\theta_0$ and $\hat\theta_1$, (i) gives the four ratios as stated. □

**Proposition 28.11 (Width one is the hinge).** *As stated: for a family, with $\theta''_j$ normalized on each excess, and with part (iv) for the family.*

*Proof. The letters.* With $\theta''_j$ of largest value 1 on the room sides of each excess, Lemma S13 gives the seam letters of the piece of the bunch and the room letters of its room, read as its partner (the paragraph after 28.4):

$$V_{jD_0} = K_{\mathcal{B}_j/R}, \quad V_{jD_1} = K_{\mathcal{B}_j|B}, \quad V^*_{jD_0} = \Psi_{\mathcal{B}_j/R}, \quad V^*_{jD_1} = \Psi_{\mathcal{B}_j|B},$$

$$L_{jD_0} = K_{\mathcal{B}_j|R}, \quad L_{jD_1} = K_{\mathcal{B}_j/B}, \quad L^*_{jD_0} = \Psi_{\mathcal{B}_j|R}, \quad L^*_{jD_1} = \Psi_{\mathcal{B}_j/B},$$

the minors on $B$ carrying the weights $\theta'_j$ and those on $R$ the weights $\theta''_j$. The hinge values of 19.2 are $\gamma^K_{j,B} = V_{jD_1}/V_{jD_0}$, $\gamma^\Psi_{j,B} = V^*_{jD_0}/V^*_{jD_1}$, $\gamma^K_{j,R} = L_{jD_0}/L_{jD_1}$ and $\gamma^\Psi_{j,R} = L^*_{jD_1}/L^*_{jD_0}$. Every letter is active and $c_j = 1$, so $\kappa_B + \kappa_B^{\rm top} = \sum_j (s_j + t_j) = 2\sigma$, and on the balance wall of the glue $\kappa_B + \kappa_R = 2\sigma$.

*The cells.*

- The cell of the bunch $\mathcal{B}^+_{j,B}$ has the bases of $\mathcal{B}_j|B$ avoiding $g$ and the $X + g$, $X \in \mathcal{B}_j/R$, through $g$, with the weights $\theta'_j$ (19.4). So $\mathcal{B}^+_{j,B} \setminus g = \mathcal{B}_j|B$ and $\mathcal{B}^+_{j,B}/g = \mathcal{B}_j/R$, its hinge values are $\gamma^K_{j,B}$ and $\gamma^\Psi_{j,B}$ (21.1), and its rank is $d_j(B) + 1$.
- Since $d^*_j(B) = |B| - d_j(B) - 1$, the hinge wall of $F^+_B$ (23.1) is $m_B - \sum_j \big(s_j d_j(B) + t_j(|B| - d_j(B) - 1)\big) = \kappa_B$.
- Likewise $\mathcal{B}^+_{j,R} \setminus g = \mathcal{B}_j|R$ and $\mathcal{B}^+_{j,R}/g = \mathcal{B}_j/B$, with the weights $\theta''_j$ and the hinge values $\gamma^K_{j,R}$ and $\gamma^\Psi_{j,R}$.
- The hinge wall of $F^+_R$ is $m_R - \sum_j \big(s_j(r(\mathcal{B}_j|R) - 1) + t_j(|R| - r(\mathcal{B}_j|R))\big)$. This is the wall $\kappa_R$ of the partner, whose letters at $D_0$ have the degrees $r(\mathcal{B}_j|R) = e_j + 1$ and $|R| - r(\mathcal{B}_j|R) = e^*_j$ (28.4).

(i) By 28.5,

$$\gamma^\rho_B = \prod_j \Big(\frac{V_{jD_1}}{V_{jD_0}}\Big)^{s_j/(2\sigma)} \Big(\frac{V^*_{jD_0}}{V^*_{jD_1}}\Big)^{t_j/(2\sigma)} = \prod_j (\gamma^K_{j,B})^{s_j/(2\sigma)} (\gamma^\Psi_{j,B})^{t_j/(2\sigma)}.$$

As $\sum_j (s_j + t_j)/(2\sigma) = 1$, this is $\gamma^K_{1,B} \prod_j (\gamma^K_{j,B}/\gamma^K_{1,B})^{s_j/(2\sigma)} (\gamma^\Psi_{j,B}/\gamma^K_{1,B})^{t_j/(2\sigma)} = \gamma^K_{1,B}\, \psi(\boldsymbol{\varrho}_{F^+_B})$, the ratio vector of $F^+_B$ being $(\gamma^K_{j,B}/\gamma^K_{1,B}, \gamma^\Psi_{j,B}/\gamma^K_{1,B})_j$ (23.1). The scale form is

$$\phi^\rho_B(du) = u^{\mathbf{m}_B - \mathbf{1}} \prod_j K^{-s_j}_{\mathcal{B}_j/R} \Psi^{-t_j}_{\mathcal{B}_j|B} \big(\gamma^K_{1,B}\psi\big)^{-\kappa_B} d\omega_B(u) = \psi^{-\kappa_B} \phi_{F^+_B}(du),$$

by the base form of 23.1, whose letters are $K_{\mathcal{B}^+_{j,B}/g} = K_{\mathcal{B}_j/R}$ and $\Psi_{\mathcal{B}^+_{j,B}\setminus g} = \Psi_{\mathcal{B}_j|B}$, with the hinge wall $\kappa_B$.

(ii) By 28.5, $\varrho^\rho_{jD_0} = V_{jD_0}/V^{(0)}_j = 1$ and $\varrho^{\rho*}_{jD_1} = V^*_{jD_1}/V^{*(1)}_j = 1$. Further $\varrho^\rho_{jD_1} = V_{jD_1}/(V_{jD_0}\gamma^\rho_B) = \gamma^K_{j,B}/(\gamma^K_{1,B}\psi) = x_{j,K}/\psi(\mathbf{x})$ and $\varrho^{\rho*}_{jD_0} = V^*_{jD_0}/(V^*_{jD_1}\gamma^\rho_B) = x_{j,\Psi}/\psi(\mathbf{x})$, at $\mathbf{x} = \boldsymbol{\varrho}_{F^+_B}$. Write $\Phi(\mathbf{x}) = \big(x_{j,K}/\psi(\mathbf{x}),\, x_{j,\Psi}/\psi(\mathbf{x})\big)_j$.

- The seam torus is the set of the points with $x_{jD_0} = x^*_{jD_1} = 1$ and $\prod_j x^{s_j}_{jD_1} x^{*t_j}_{jD_0} = 1$. Since $\prod_j (x_{j,K}/\psi)^{s_j} (x_{j,\Psi}/\psi)^{t_j} = \psi^{2\sigma}\psi^{-2\sigma} = 1$, $\Phi$ maps $(0,\infty)^{2N-1}$ into it.
- Let a point of the torus have $x_{jD_1} = a_j$ and $x^*_{jD_0} = b_j$. The vector $\mathbf{x}$ with $x_{j,K} = a_j/a_1$ and $x_{j,\Psi} = b_j/a_1$ has $\psi(\mathbf{x}) = \big(\prod_j a^{s_j}_j b^{t_j}_j\big)^{1/(2\sigma)}/a_1 = 1/a_1$, so $\Phi(\mathbf{x})$ is the point.
- Conversely $\Phi(\mathbf{x})$ determines $\psi(\mathbf{x})$ as the inverse of its coordinate $x_{1,K}/\psi(\mathbf{x})$, and then $\mathbf{x}$. So $\Phi$ is a bijection.

As $\boldsymbol{\varrho}^\rho_B = \Phi \circ \boldsymbol{\varrho}_{F^+_B}$ and $\phi^\rho_B = \psi(\boldsymbol{\varrho}_{F^+_B})^{-\kappa_B} \phi_{F^+_B}$, the law $(\boldsymbol{\varrho}^\rho_B)_* \phi^\rho_B$ is the image under $\Phi$ of $\psi^{-\kappa_B}$ times $(\boldsymbol{\varrho}_{F^+_B})_* \phi_{F^+_B}$.

(iii) The partner with its stand-in at $\Phi(\mathbf{x})$ (28.10(i)) has the letters

$$\sum_D x_{jD}\, \alpha_g^{i(D)} L_{jD} = K_{\mathcal{B}_j|R} + \frac{x_{j,K}}{\psi(\mathbf{x})}\, \alpha_g\, K_{\mathcal{B}_j/B},$$

$$\sum_D x^*_{jD}\,\alpha_g^{1-i(D)}L^*_{jD} = \frac{x_{j,\Psi}}{\psi(\mathbf{x})}\,\alpha_g\,\Psi_{\mathcal{B}_j|R} + \Psi_{\mathcal{B}_j/B}.$$

After $\alpha_g \mapsto \psi(\mathbf{x})\,\alpha_g$ these are $K_{\mathcal{B}_j|R} + x_{j,K}\,\alpha_g K_{\mathcal{B}_j/B}$ and $x_{j,\Psi}\,\alpha_g\Psi_{\mathcal{B}_j|R} + \Psi_{\mathcal{B}_j/B}$. These are the letters of $\mathcal{B}^+_{j,R}$ twisted at $g$ by $x_{j,K}$ in $K_j$ and by $x_{j,\Psi}$ in $\Psi_j$: by 22.1, the twist multiplies the weights in $K_j$ of the bases through $g$ and the weights in $\Psi_j$ of the bases avoiding $g$. The mass $\kappa_B$ on $g$ turns $\alpha_g^{\kappa_B-1}d\alpha_g$ into $\psi(\mathbf{x})^{\kappa_B}\alpha_g^{\kappa_B-1}d\alpha_g$, which cancels the density of (ii). The functions of 28.8, homogeneous of degree one in the vector $(y_0, y_1)$ of the seam letters of a letter, are $y_0\,f(\alpha_Z, y_1/y_0)$ for $K_j$ and $y_1\,f(\alpha_Z, y_0/y_1)$ for $\Psi_j$, with $f$ positive and continuous. At the seam letters they are $K_{\mathcal{B}_j/R}\,f(\alpha_Z, \gamma^K_{j,B})$ and $\Psi_{\mathcal{B}_j|B}\,f(\alpha_Z, \gamma^\Psi_{j,B})$, the factors $L_i\,f_i(\alpha_Z, \gamma_i)$ of 23.2 for $F^+_B$. At $\mathbf{x}_l(\alpha_g)$, after the change of variable, they are $f(\alpha_Z, x_{j,K}\alpha_g)$ and $f(\alpha_Z, x_{j,\Psi}\alpha_g)$, the factors of $\omega_{\mathbf{x}}$ in 23.2. So the two lemmas are one, read through $\Phi$.

(iv) *The room.* Read with the reversed excesses, the station $D_1$ has the partner excess 0 and $D_0$ the partner excess 1. By 28.5 for the partner:

- its scale is

$$\prod_j \Big(\frac{L_{jD_0}}{L_{jD_1}}\Big)^{s_j/(2\sigma)}\Big(\frac{L^*_{jD_1}}{L^*_{jD_0}}\Big)^{t_j/(2\sigma)} = \prod_j (\gamma^K_{j,R})^{s_j/(2\sigma)}(\gamma^\Psi_{j,R})^{t_j/(2\sigma)} = \gamma^K_{1,R}\,\psi(\boldsymbol{\varrho}_{F^+_R}),$$

  as in (i);

- its scale form is $v^{\mathbf{m}_R-\mathbf{1}}\prod_j K^{-s_j}_{\mathcal{B}_j/B}\Psi^{-t_j}_{\mathcal{B}_j|R}(\gamma^K_{1,R}\psi)^{-\kappa_R}d\omega_R(v) = \psi^{-\kappa_R}\phi_{F^+_R}$, since the base form of $F^+_R$ has the letters $K_{\mathcal{B}^+_{j,R}/g} = K_{\mathcal{B}_j/B}$ and $\Psi_{\mathcal{B}^+_{j,R}\setminus g} = \Psi_{\mathcal{B}_j|R}$ and the hinge wall $\kappa_R$;

- with $\gamma_R$ its scale, its seam vector has $\varrho^\rho_{jD_1} = \varrho^{\rho*}_{jD_0} = 1$, $\varrho^\rho_{jD_0} = L_{jD_0}/(L_{jD_1}\gamma_R) = x'_{j,K}/\psi(\mathbf{x}')$ and $\varrho^{\rho*}_{jD_1} = L^*_{jD_1}/(L^*_{jD_0}\gamma_R) = x'_{j,\Psi}/\psi(\mathbf{x}')$ at $\mathbf{x}' = \boldsymbol{\varrho}_{F^+_R}$;

- its law follows as in (ii).

The coordinate ratios are $x_{j,\Psi}/x_{j,K} = \gamma^\Psi_{j,B}/\gamma^K_{j,B} = \varrho_{j,B}$ and $x'_{j,\Psi}/x'_{j,K} = \varrho_{j,R}$ (19.2).

*The kernel.* At $c_j = 1$ the configuration $t_{\mathcal{D}}$ has the column $\mathbf{e}_{K_j}$ at $(j, D_0)$ and $\mathbf{e}_g + \mathbf{e}_{K_j}$ at $(j, D_1)$ for $K_j$, and $\mathbf{e}_g + \mathbf{e}_{\Psi_j}$ at $D_0$ and $\mathbf{e}_{\Psi_j}$ at $D_1$ for $\Psi_j$. The display of 28.10(ii) gives

$$\Gamma(\kappa_B)\,\mathsf{Z}_{(\kappa_B,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(\mathbf{p}\circ\mathbf{p}') = \int_0^\infty v^{\kappa_B-1}\prod_j \big(p_{jD_0}p'_{jD_0} + p_{jD_1}p'_{jD_1}v\big)^{-s_j}\big(p^*_{jD_0}p'^*_{jD_0}v + p^*_{jD_1}p'^*_{jD_1}\big)^{-t_j}dv,$$

and the coordinates of (ii) and of the partner give the first line of (iv). Substitute $v = (\psi(\mathbf{x})/\psi(\mathbf{x}'))\,w$. Each factor $x_{j,K}v/\psi(\mathbf{x}) + x'_{j,K}/\psi(\mathbf{x}')$ becomes $(x_{j,K}w + x'_{j,K})/\psi(\mathbf{x}')$, and likewise for $\Psi_j$. Since $\sum_j(s_j + t_j) - \kappa_B = \kappa_R$, the integral is

$$\Big(\frac{\psi(\mathbf{x})}{\psi(\mathbf{x}')}\Big)^{\kappa_B}\psi(\mathbf{x}')^{\sum_j(s_j+t_j)}\int_0^\infty w^{\kappa_B-1}\prod_j(x_{j,K}w + x'_{j,K})^{-s_j}(x_{j,\Psi}w + x'_{j,\Psi})^{-t_j}dw = \psi(\mathbf{x})^{\kappa_B}\psi(\mathbf{x}')^{\kappa_R}\,P_{F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]}\big((\kappa_B,\kappa_R);\mathbf{s},\mathbf{t}\big).$$

The last step is the period of the twisted core of 23.6(iii) with the core $\binom{\{g,g'\}}{1}$ in every table, read on the section $\alpha_{g'} = 1$ (16.2); its letters are $x_{j,K}\alpha_g + x'_{j,K}\alpha_{g'}$ and $x_{j,\Psi}\alpha_g + x'_{j,\Psi}\alpha_{g'}$ (22.1).

*The reductions.*

- 28.10(i) is part (iii).

- 28.10(ii): insert the laws of (ii) and of the partner. The densities $\psi(\mathbf{x})^{-\kappa_B}$ and $\psi(\mathbf{x}')^{-\kappa_R}$ cancel the factor of the kernel, and the period becomes

$$\iint P_{F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]}\big((\kappa_B,\kappa_R);\mathbf{s},\mathbf{t}\big)\,d\big((\boldsymbol{\varrho}_{F_B^+})_*\phi_{F_B^+}\big)(\mathbf{x})\,d\big((\boldsymbol{\varrho}_{F_R^+})_*\phi_{F_R^+}\big)(\mathbf{x}').$$

  This is 23.6(iii) for the family whose table $j$ is $\binom{\{g,g'\}}{1}\langle\mathcal{B}_{j,B}^+,\mathcal{B}_{j,R}^+\rangle$, which is the table $(\mathcal{B}_j,\theta_j)$ by 21.3(iii).

- 28.10(iii): on the pins the seam word of the piece is $\prod_j K_{\mathcal{B}_j/R}^{-a_{jD_0}}K_{\mathcal{B}_j|B}^{-a_{jD_1}}\Psi_{\mathcal{B}_j/R}^{-b_{jD_0}}\Psi_{\mathcal{B}_j|B}^{-b_{jD_1}}$. This is the word of the bunch family $F^{\mathrm{bunch}}(\mathbf{a},\mathbf{b})$ of 19.13 at the level shares $(a_j^d,a_j^r,b_j^d,b_j^r)=(a_{jD_0},a_{jD_1},b_{jD_0},b_{jD_1})$, and the room word is that of $F^{\mathrm{room}}(\mathbf{a},\mathbf{b})$.
  - The seam functions on the pins are the periods of these words (28.7), and

$$\Gamma\begin{smallmatrix}2\\1\end{smallmatrix}\begin{bmatrix}a_{jD_0},\ a_{jD_1}\\ s_j\end{bmatrix}\Gamma\begin{smallmatrix}2\\1\end{smallmatrix}\begin{bmatrix}b_{jD_0},\ b_{jD_1}\\ t_j\end{bmatrix}=\Gamma\begin{smallmatrix}4\\2\end{smallmatrix}\begin{bmatrix}a_j^d,\ b_j^d,\ a_j^r,\ b_j^r\\ s_j,\ t_j\end{bmatrix}.$$

  - The scale pin $\sum_j(a_{jD_1}+b_{jD_0})=\kappa_B$ is the pin of 19.13.
  - In the notation of 4.1, eliminating the letter totals leaves the shares $(a_j^d,b_j^d)_j$ with the one pin $\sum_j(b_j^d-a_j^d)=\kappa_B-\sum_j s_j$, with the same power of $2\pi$.
  - In the pair coordinates $\sigma_j^d=\frac12(a_j^d+b_j^d)$ and $\delta_j^d=\frac12(a_j^d-b_j^d)$, Lebesgue measure gains the factor $2^N$, and the pin becomes $-2\sum_j\delta_j^d$, whose delta is $\frac12$ that of $\sum_j\delta_j^d$.

  So the fibre integral of 28.10(iii) is $2^{N-1}$ times the fibre integral in the pair coordinates, which is 19.13.

□

*Remark.* The constants match exactly under the normalization. In a factorization with the constants $\hat\theta_{0j},\hat\theta_{1j}$ of Lemma S13, the scale of the piece, its seam vector and its law are unchanged, since they are defined from the seam letters, while the hinge values move. Then (i) holds up to the constant factor $\prod_j(\hat\theta_{1j}/\hat\theta_{0j})^{\delta_j/\sigma}$, and the coordinate $\varrho^{\rho}_{jD_1}$ of (ii) is $(\hat\theta_{1j}/\hat\theta_{0j})\prod_l(\hat\theta_{1l}/\hat\theta_{0l})^{-\delta_l/\sigma}$ times $x_{j,K}/\psi(\mathbf{x})$. Normalizing $\theta_j''$ on each excess removes these constants and costs no generality, by Lemma S13(ii). For one table, $\mathbf{x}=(1,\varrho_B)$ and $\mathbf{x}'=(1,\varrho_R)$. The family $F_{\mathcal{H}}[(\mathbf{x},\mathbf{x}')]$ is then the list $\binom{\{g,g'\}}{1}$ twisted in its cobasis letter by $\varrho_B$ at $g$ and by $\varrho_R$ at $g'$, and $\psi(\mathbf{x})^{\kappa_B}\psi(\mathbf{x}')^{\kappa_R}=\varrho_B^{\kappa_B t/(s+t)}\varrho_R^{\kappa_R t/(s+t)}$. So (iv) is the identity with 23.7, and the reductions are 23.9(i) and (iv) together with 19.13.

**Proposition 28.15 (The Gramma of a set of pairs in the Foreman's order).** *As stated, for a set of pairs with any number of parts.*

*Proof.* The configuration $\partial\mathfrak{R}$ of 28.13 has at the pair $(C,D)$ the column $\mathbf{e}_C+\mathbf{e}_D$ on the kept stations, the station not kept in each part read as 0. A pair of $\mathfrak{R}_l$ has both its stations in $\mathcal{K}_l\sqcup\mathcal{D}_l$, so its column is supported on the kept stations of $\mathfrak{R}_l$, and different parts have different kept stations. By 2.5, applied to one part after another, the Gramma of $\mathfrak{R}$ is the product of the Grammas of its parts,

$$\mathsf{G}_{(\mathbf{a},\mathbf{a}');\partial\mathfrak{R}}(\mathbf{h})=\prod_{l=1}^{\nu}\mathsf{G}_{(\mathbf{a}_{\mathcal{K}_l},\mathbf{a}'_{\mathcal{D}_l});\partial\mathfrak{R}_l}(\mathbf{h}),$$

and it converges exactly when every factor does, its domain being the product of their domains. For one part the statement is the reference, proof of prop:relgrammaforeman. There the scales of $\mathcal{K}_l$ are integrated first, $\int_0^\infty x^{a_C-1}e^{-xL_C(v)}dx = \Gamma(a_C)L_C(v)^{-a_C}$. The form left on the stations of $\mathcal{D}_l$ is homogeneous of degree $|\mathbf{a}'_{\mathcal{D}_l}| - |\mathcal{D}_l| - |\mathbf{a}_{\mathcal{K}_l}| = -|\mathcal{D}_l|$, so its integral over a chart is its integral over $\Delta_{\mathcal{D}_l}$ (16.2), and the convergence criterion is 25.2. Multiplying over the parts gives the statement. □

**Proposition 28.16 (Closing the Gramma of a set of pairs).** *As stated: any number of parts, and (ii) at real and complex loads.*

*Proof.* (i) On each part the reference's proof (proof of prop:relgrammaclose(i)) applies on the simplex $\Delta_{\mathcal{D}_l}$. On its interior $0 \le \bar{L}_C < 1$ almost everywhere, and $L_C^{-a_C} = M_C^{-a_C}\sum_n \frac{(a_C)_n}{n!}\bar{L}_C^{\,n}$ with nonnegative terms. On the simplex $\bar{L}_C$ is a linear function with nonnegative coefficients, and Tonelli's theorem with 3.11 gives the series. By 28.15 the Gramma is $\prod_C \Gamma(a_C)$ times the product of these integrals over the parts.

(ii) The exchange relation of 2.4(ii) holds at every index of the domain, real or complex.

- At a kept station of $\mathcal{K}$ or of $\mathcal{D}$ it is the stated relation, the column of $(C, D)$ being $\mathbf{e}_C + \mathbf{e}_D$ with the coefficient $h_{CD}$.
- At the station not kept of a part it is the sum of the relations at the stations of $\mathcal{K}$ of that part less those at its other stations of $\mathcal{D}$, the totals of a part being equal (the reference, proof of prop:relgrammaclose(ii)).
- A shifted load $(\mathbf{a} + \mathbf{e}_C, \mathbf{a}' + \mathbf{e}_{D'})$ with $(C, D') \in \mathfrak{R}$ keeps the totals of every part equal, since $C$ and $D'$ lie in one part, and it lies in the domain with $(\mathbf{a}, \mathbf{a}')$ (2.4).

Neither (i) nor (ii) uses one part, and (ii) does not use real loads.

(iii) concerns one part and is the reference, proof of prop:relgrammaclose(iii). □

**Theorem 28.19 (The law is what a piece retains, at every width).** *As stated; the plain tinkers suffice under the stated condition on the tables with both letters active.*

*Proof.* The equivalence of (a)–(d) is the reference, proof of thm:retainwidth. Its step (c)⇒(d) reads as follows in the notation of 4.1.

*The step (c)⇒(d).* Let $V$ be the real space of the directions of the pins of the seam (28.7): the vectors $\boldsymbol{\vartheta} = (\vartheta_{jD}, \vartheta^*_{jD})_{j,D}$, one coordinate for each coordinate of the seam vector, with

$$\sum_D \vartheta_{jD} = 0 \text{ for every active } K_j, \qquad \sum_D \vartheta^*_{jD} = 0 \text{ for every active } \Psi_j,$$
$$\sum_{j,D} \big(i(D)\vartheta_{jD} + (c_j - i(D))\vartheta^*_{jD}\big) = 0.$$

For a tinker $\mathbf{v}$ of $\mathcal{Y}_0$ with the logarithms $\boldsymbol{\ell} = \log \mathbf{v}$, the seam word of $\mathcal{Y}_{0,\mathbf{v}}$ is $\mathbf{v}^{-(\mathbf{a},\mathbf{b})}$ times that of $\mathcal{Y}_0$. So, by 28.7 and by 28.10(iii) on the contour through $\mathbf{c}$,

$$e^{\langle \mathbf{c}, \boldsymbol{\ell}\rangle}\, P_{\mathcal{X}\sqcup\mathcal{Y}_{0,\mathbf{v}}} = (2\pi)^{-n_{\mathcal{D}}} \int_V e^{-i\langle \boldsymbol{\ell}, \boldsymbol{\vartheta}\rangle}\, f(\boldsymbol{\vartheta})\, d\boldsymbol{\vartheta},$$

where $f(\boldsymbol{\vartheta})$ is the product of the stacked Gammas of 28.10(iii) and of the seam functions of $\mathcal{X}$ and of $\mathcal{Y}_0$ at $\mathbf{c} + i\boldsymbol{\vartheta}$. The stacked Gammas have no zero, and $f$ is continuous and integrable on $V$ (the reference, proof of thm:retainwidth). The proof uses (c) only to know the Fourier transform

of $f$ at every point of the dual of $V$; then $f$ is determined. The rest is unchanged: the stacked Gammas have no zero, the seam function of $\mathcal{Y}_0$ is positive at $\mathbf{c}$, the identity theorem applies, and the projection is injective on the seam torus.

*The plain tinkers.* Their logarithms form the space $\Lambda$ of the $\boldsymbol{\ell}$ with $\ell_{jD} = \ell^*_{jD}$ for every table $j$ with both letters active (28.18). A table with one active letter has one coordinate for each station, and on it a plain tinker is a general one. The restrictions to $V$ of the functionals $\boldsymbol{\vartheta} \mapsto \langle \boldsymbol{\ell}, \boldsymbol{\vartheta} \rangle$, $\boldsymbol{\ell} \in \Lambda$, exhaust the dual of $V$ exactly when no nonzero $\boldsymbol{\vartheta} \in V$ is orthogonal to $\Lambda$.

*No orthogonal vector.* Let $\boldsymbol{\vartheta} \in V$ be orthogonal to $\Lambda$.

- Then $\vartheta_{jD} = 0$ in a table with only $K_j$ active, $\vartheta^*_{jD} = 0$ in a table with only $\Psi_j$ active, and $\vartheta^*_{jD} = -\vartheta_{jD}$ in a table with both letters active.
- The letter totals give $\sum_D \vartheta_{jD} = 0$ in every table with both letters active. The scale pin becomes $\sum_{j,D} \big(2i(D) - c_j\big)\vartheta_{jD} = 0$, the sum running over the tables with both letters active.
- A table with $|\mathcal{D}_j| = 1$ has $\vartheta_{jD} = 0$ by its total.
- The exceptional table has $c_j \geq 1$, and $\mathcal{D}_{j,0}$ and $\mathcal{D}_{j,c_j}$ are nonempty (28.4), so its two stations $D'$ and $D''$ have the excesses 0 and $c_j$. Its total gives $\vartheta_{jD'} = -\vartheta_{jD''} = \tau$, and the scale pin reads $-c_j\tau - c_j\tau = 0$, so $\tau = 0$.

Hence $\boldsymbol{\vartheta} = 0$, and the plain tinkers suffice in (c). □

*Remark.* In general, let $J_2$ be the tables with both letters active. The orthogonal of $\Lambda$ in $V$ has the dimension $\sum_{j \in J_2}(|\mathcal{D}_j| - 1)$, less one when some table of $J_2$ has positive width. Indeed, the vectors with $\sum_D \vartheta_{jD} = 0$ on the tables of $J_2$ and zero elsewhere form a space of the first dimension, and on it the scale pin $\sum_{j,D}(2i(D) - c_j)\vartheta_{jD}$ vanishes identically exactly when every table of $J_2$ has width zero. With every letter active and some table of positive width, $n_{\mathcal{D}} = 2|\mathcal{D}| - 2N - 1$ and the orthogonal has the dimension $|\mathcal{D}| - N - 1$, so the plain tinkers of one partner move $|\mathcal{D}| - N$ of the directions of the seam torus, as the paragraph after the theorem says.

**Lemma 28.23 (Partners on one mark see a point mass).** *As stated.*

*Proof. The one-mark piece.* On one mark the seam letters have $d_j = d^*_j = 0$, the wall is the mass $\kappa$, and by the relations of the seam torus the excess sums are $V_j^{(0)} = \sum_{D \in \mathcal{D}_{j,0}} x_{jD} = 1$, $V_j^{*(c_j)} = 1$, $V_j^{(c_j)} = \alpha_g^{c_j} \sum_{D \in \mathcal{D}_{j,c_j}} x_{jD}$ and $V_j^{*(0)} = \alpha_g^{c_j} \sum_{D \in \mathcal{D}_{j,0}} x^*_{jD}$. So the scale and the seam vector of 28.5 are

$$\alpha_g \Big( \prod_j \Big( \sum_{D \in \mathcal{D}_{j,c_j}} x_{jD} \Big)^{s_j} \Big( \sum_{D \in \mathcal{D}_{j,0}} x^*_{jD} \Big)^{t_j} \Big)^{1/w} = \alpha_g,$$

$$\Big( \frac{x_{jD}\, \alpha_g^{i(D)}}{\alpha_g^{i(D)}},\ \frac{x^*_{jD}\, \alpha_g^{c_j - i(D)}}{\alpha_g^{c_j - i(D)}} \Big)_{j,D} = \mathbf{x},$$

and the scale form on the one point of $\Delta_{\{g\}}$ has the mass 1: the law is $\delta_{\mathbf{x}}$, and the station transform is the kernel at $\mathbf{x}$ (28.21). In the chart $\alpha_b = 1$ the period of the glue of $\mathcal{X}$ with the partner on $b$ is $\int_{(0,\infty)^B} \alpha^{\mathbf{m}_B - \mathbf{1}} \prod_j (\sum_D p_{jD} V_{jD})^{-s_j} (\sum_D p^*_{jD} V^*_{jD})^{-t_j} d\alpha$, the second expression of 28.21; the partner has the degrees $e_j = e^*_j = 0$, the wall $w - \kappa$, and its room letters are those of the partner with the coefficients 1 tinkered by $\mathbf{p}$ (28.18).

*Coordinates.* Index the coordinates of the seam vector by the pairs of an active letter $l$ and a station of its table, $n_{\mathcal{X}}$ of them; write $\mathbf{p} = e^{\mathbf{P}}$ with $\mathbf{P} \in \mathbb{R}^{n_{\mathcal{X}}}$, and pair shares $(\mathbf{a}, \mathbf{b}) \in \mathbb{R}^{n_{\mathcal{X}}}$, one for each coordinate, with $\mathbf{P}$ by $\langle (\mathbf{a}, \mathbf{b}), \mathbf{P} \rangle$. Let $\mathbf{u}_l$ be the indicator of the coordinates of the letter

$l$, $U'$ the span of the $\mathbf{u}_l$, $\mathbf{u}_0$ the vector with the entries $i(D)$ at the coordinates of the letters $K_j$ and $c_j - i(D)$ at those of the letters $\Psi_j$, and $U = U' + \mathbb{R}\mathbf{u}_0$. Since $w > 0$ some active letter has $c_j > 0$, and on its coordinates $\mathbf{u}_0$ takes the values 0 and $c_j$ at the excesses 0 and $c_j$, which have stations; so $\mathbf{u}_0 \notin U'$. The pins of the seam (28.7) are $\langle (\mathbf{a},\mathbf{b}), \mathbf{u}_l\rangle$ equal to the exponent of $l$ and $\langle (\mathbf{a},\mathbf{b}), \mathbf{u}_0\rangle = \kappa$; let $\mathcal{A}$ be the set of the shares with positive coordinates on the pins. For one letter of width $c_j > 0$ and exponent $s$, the sums $\sum_D i(D) a_{jD}$ over positive shares with $\sum_D a_{jD} = s$ fill $(0, sc_j)$, and they are 0 at width zero; so on the pins the scale pin takes every value in $(0, w)$, and $\mathcal{A}$ is not empty, open in the affine space of the pins, whose direction space is $V = U^\perp$.

*The station transform opens the seam function.* Fix $(\mathbf{a},\mathbf{b}) \in \mathcal{A}$ and put $f(\mathbf{P}) = e^{\langle (\mathbf{a},\mathbf{b}),\mathbf{P}\rangle}\, \Gamma(\kappa) \mathsf{Z}_{(\kappa,\mathbf{s},\mathbf{t});t_{\mathcal{D}}}(e^{\mathbf{P}})$. By the kernel of 28.10(ii) with $v = e^\tau$,

$$f(\mathbf{P}) = \int_{\mathbb{R}} g(\mathbf{P} + \tau \mathbf{u}_0)\, d\tau, \qquad g(\mathbf{Q}) = e^{\langle (\mathbf{a},\mathbf{b}),\mathbf{Q}\rangle} \prod_{j:\, s_j > 0} \Big( \sum_D e^{Q_{jD}} \Big)^{-s_j} \prod_{j:\, t_j > 0} \Big( \sum_D e^{Q^*_{jD}} \Big)^{-t_j},$$

using $\langle (\mathbf{a},\mathbf{b}), \mathbf{u}_0\rangle = \kappa$. Translation by $\mathbf{u}_l$ multiplies the letter $l$ by $e^\lambda$ and the exponential by $e^{\lambda s}$, $s$ its exponent: $g$ is invariant under $U'$, and $f$ under $U$, a translation by $\mathbf{u}_0$ being absorbed by $\tau$. Let $W$ be a linear complement of $U$ and $W'$ the subspace on which the coordinate of one chosen station $D_l$ of every active letter vanishes, a complement of $U'$. Since $\mathbf{u}_0 \notin U'$, $(\mathbf{P}, \tau) \mapsto \mathbf{P} + \tau \mathbf{u}_0$ is a linear isomorphism of $W \times \mathbb{R}$ onto $\mathbb{R}^{n_{\mathcal{X}}}/U' \cong W'$, and by Tonelli and the invariance of $g$

$$\int_W f\, d\mathbf{P} = c_W \int_{W'} g = c_W \prod_{j:\, s_j > 0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix} \mathbf{a}_j \\ s_j \end{bmatrix} \prod_{j:\, t_j > 0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix} \mathbf{b}_j \\ t_j \end{bmatrix} =: \beta(\mathbf{a},\mathbf{b}) \in (0,\infty),$$

with $c_W > 0$ fixed by $W$, $W'$ and their Lebesgue measures. On $W'$ the integral factors over the letters; for a letter $K_j$, with $y_D = e^{Q_{jD}}$ and $a_{jD_l} = s_j - \sum_{D \neq D_l} a_{jD} > 0$, it is $\int_{(0,\infty)^{\mathcal{D}_j \setminus D_l}} \prod_{D \neq D_l} y_D^{a_{jD}-1} \big(1 + \sum_{D \neq D_l} y_D\big)^{-s_j} d\mathbf{y}$, which is the stacked Gamma, as $(1 + |\mathbf{y}|)^{-s_j} = \Gamma(s_j)^{-1} \int_0^\infty x^{s_j - 1} e^{-x(1+|\mathbf{y}|)} dx$ and Tonelli integrate it to $\prod_D \Gamma(a_{jD})/\Gamma(s_j)$; likewise for a letter $\Psi_j$. For $\mathbf{x}$ on the seam torus and $\mathbf{X} = \log \mathbf{x}$, $e^{\langle (\mathbf{a},\mathbf{b}),\mathbf{P}\rangle} \Gamma(\kappa) \mathsf{Z}(\mathbf{x} \circ e^{\mathbf{P}}) = e^{-\langle (\mathbf{a},\mathbf{b}),\mathbf{X}\rangle} f(\mathbf{P} + \mathbf{X})$, and $\int_W f(\mathbf{P} + \mathbf{X})\, d\mathbf{P} = \int_W f\, d\mathbf{P}$ by the invariance of $f$ under $U$, Lebesgue measure on $W$ being invariant under translation. By the first expression of 28.21 and Tonelli,

$$\int_W e^{\langle (\mathbf{a},\mathbf{b}),\mathbf{P}\rangle}\, \mathcal{I}_{\mathcal{X}}(e^{\mathbf{P}})\, d\mathbf{P} = \beta(\mathbf{a},\mathbf{b}) \int \mathbf{x}^{-(\mathbf{a},\mathbf{b})}\, d\mu(\mathbf{x}),$$

$\mu$ the law of $\mathcal{X}$. The hypothesis makes the left side $c\, \beta(\mathbf{a},\mathbf{b})\, \mathbf{x}_0^{-(\mathbf{a},\mathbf{b})}$, so $\int \mathbf{x}^{-(\mathbf{a},\mathbf{b})} d\mu = c\, \mathbf{x}_0^{-(\mathbf{a},\mathbf{b})}$ for every $(\mathbf{a},\mathbf{b}) \in \mathcal{A}$: the seam function is finite on all of $\mathcal{A}$.

*A point mass.* Fix $(\mathbf{a}_0,\mathbf{b}_0) \in \mathcal{A}$ and let $\nu = \mathbf{x}^{-(\mathbf{a}_0,\mathbf{b}_0)}\mu$, a finite measure of the total mass $C = c\, \mathbf{x}_0^{-(\mathbf{a}_0,\mathbf{b}_0)}$. For $\boldsymbol{\delta} \in V$ near 0, $(\mathbf{a}_0,\mathbf{b}_0) \pm \boldsymbol{\delta} \in \mathcal{A}$, and so $\int (\mathbf{x}/\mathbf{x}_0)^{\mp\boldsymbol{\delta}} d\nu = C$ and

$$\int \Big( (\mathbf{x}/\mathbf{x}_0)^{\boldsymbol{\delta}/2} - (\mathbf{x}/\mathbf{x}_0)^{-\boldsymbol{\delta}/2} \Big)^2 d\nu(\mathbf{x}) = C - 2C + C = 0.$$

So $\langle \log \mathbf{x} - \log \mathbf{x}_0, \boldsymbol{\delta}\rangle = 0$ for $\nu$-almost every $\mathbf{x}$, for every $\boldsymbol{\delta}$ of a basis of $V$, and $\log \mathbf{x} - \log \mathbf{x}_0 \in V^\perp = U$ for $\mu$-almost every $\mathbf{x}$, $\mu$ and $\nu$ having the same null sets. Two points $\mathbf{x}, \mathbf{x}'$ of the seam torus with $\log \mathbf{x} - \log \mathbf{x}' \in U$ are equal: $\log x_{jD} - \log x'_{jD} = \lambda_j + \lambda i(D)$ and $\log x^*_{jD} - \log x'^*_{jD} = \lambda^*_j + \lambda(c_j - i(D))$; the relations at the excess 0 of the letters $K_j$ and at the excess $c_j$ of the letters $\Psi_j$ give $\lambda_j = \lambda^*_j = 0$, and then the product relation gives $e^{\lambda w} = 1$, so $\lambda = 0$. Hence $\mu$ is carried by $\mathbf{x}_0$, and $\mu = c\, \delta_{\mathbf{x}_0}$. □

**Theorem 28.24 (A bunch is stored on one mark exactly at width one).** *As stated, the clause at width one read with the normalization of 28.11.*

*Proof.* Everything except the clause at width one is the reference, proof of thm:storeone. At width one with factoring weights, Lemma S13(i) writes the letters of the partner with its stand-in at a seam point $\mathbf{x}$, where $x_{jD_0} = x^*_{jD_1} = 1$, as

$$\frac{1}{\hat\theta_0}\Big(K_{\mathcal{B}_j|R} + \frac{\hat\theta_0}{\hat\theta_1}\, x_{jD_1}\, \alpha_g\, K_{\mathcal{B}_j/B}\Big), \qquad \frac{1}{\hat\theta_1}\Big(\frac{\hat\theta_1}{\hat\theta_0}\, x^*_{jD_0}\, \alpha_g\, \Psi_{\mathcal{B}_j|R} + \Psi_{\mathcal{B}_j/B}\Big),$$

the constants $\hat\theta_0, \hat\theta_1$ being those of table $j$. Under the normalization of 28.11 the constants are 1. The letters are then those of $\mathcal{B}^+_{j,R}$ twisted at $g$ by $x_{jD_1}$ in $K_j$ and by $x^*_{jD_0}$ in $\Psi_j$ (19.4, 22.1). At the seam point of a value $\mathbf{y}$ of the ratio vector of $F^+_B$ these twists are $y_{j,K}/\psi(\mathbf{y})$ and $y_{j,\Psi}/\psi(\mathbf{y})$ (28.11(ii)), and the change of variable $\alpha_g \mapsto \psi(\mathbf{y})\alpha_g$ of 28.11(iii) turns them into the twist by $\mathbf{y}$, as stated. Without the normalization the two letters carry the constant factors $1/\hat\theta_0$ and $1/\hat\theta_1$ and the twists $(\hat\theta_0/\hat\theta_1)x_{jD_1}$ and $(\hat\theta_1/\hat\theta_0)x^*_{jD_0}$. The last sentence of the clause is 28.11(i)–(iii). □

**Theorem 28.25 (A point mass is a pinned set).** *As stated.*

*Proof. The walls.* On the balance wall $\kappa_S + \kappa_{S'} = 2\sum_j \sigma_j c_j(S) = w$ (8.16), and both walls are positive; so $w > 0$ and $0 < \kappa_S < w$. By the paragraph after 28.4, $S$ is a piece with the seam letters $V_{jD} = K_{j,(D)}$ and $V^*_{jD} = \Psi_{j,(D)}$ of the stations $D$ of its room, $S'$ is a partner, and their glue is $F$ (28.2(iii)); the wall of the piece is $\kappa_S$ and its top wall $w - \kappa_S = \kappa_{S'}$.

*(i) and the seam vector.* The scale form has a positive continuous density on the open simplex $\Delta_S$, the excess sums and the scale being positive there, and the seam vector is continuous. So the law is carried by one point exactly when the seam vector is constant, $\mathbf{x}_0$ say, and then it is $c\,\delta_{\mathbf{x}_0}$ with $c = \int_{\Delta_S} \phi^\rho_S > 0$. By 28.10(i) with the partner $S'$, $P_F = c\, P_{S'(\mathbf{x}_0)}$ in $[0,\infty]$; $P_F$ is finite and $P_{S'(\mathbf{x}_0)} > 0$, so $c$ and $P_{S'(\mathbf{x}_0)}$ are finite.

*(ii)⇒(i).* An active table of width zero at $S$ with its weights factoring across $S$ and $S'$ has one station of room sides (28.3(i)), whose coordinates of the seam vector are $V_{jD}/V^{(0)}_j = 1$ and $V^*_{jD}/V^{*(c_j)}_j = 1$. An active table of width one with factoring weights has the stations $D_0$ and $D_1$ and the seam letters $K_{\mathcal{B}_j/S'}, K_{\mathcal{B}_j|S}$ and $\Psi_{\mathcal{B}_j/S'}, \Psi_{\mathcal{B}_j|S}$ (28.3(ii), the paragraph after 28.5), so its coordinates are 1, $\gamma^K_{j,S}/\gamma^\rho_S$, $\gamma^\Psi_{j,S}/\gamma^\rho_S$ and 1, the hinge values read at the factorization that normalizes the multipliers. Under (ii) the scale is $\gamma^\rho_S = \prod_j (c_j\gamma)^{s_j/w}(c^*_j\gamma)^{t_j/w} = C\gamma$, the product over the tables of width one with $C = \prod_j c_j^{s_j/w} c_j^{*\,t_j/w}$, and every coordinate is a constant: $c_j/C$, $c^*_j/C$ or 1.

*(i)⇒(ii).* Let the seam vector be the constant $\mathbf{x}_0$. For every active letter $K_j$, $V_{jD} = x_{0,jD} V^{(0)}_j (\gamma^\rho_S)^{i(D)}$, and by the room-station factorization (28.2(iii))

$$K_j = \sum_D V_{jD}\, \mathsf{y}_{jD} = V^{(0)}_j(\alpha_S) \sum_D x_{0,jD}\, \gamma^\rho_S(\alpha_S)^{i(D)}\, \mathsf{y}_{jD}(\alpha_{S'}),$$

so $K_j$ reads $S$ through the one function $\gamma^\rho_S$, positive and homogeneous of degree one, with $V^{(0)}_j$ positive and homogeneous: by 19.6, $c_j(S) \le 1$. Likewise for every active letter $\Psi_j$, with $V^{*(c_j)}_j$ and $(\gamma^\rho_S)^{c_j - i(D)}$. Two stations $D \ne D'$ of one excess would have proportional seam letters, $V_{jD}/V_{jD'} = x_{0,jD}/x_{0,jD'}$ (or the same with $V^*$), so proportional room children, against 28.1(i); so every excess of every active table has one station of room sides, hence one station of patterns (28.3), and is one rectangle, which at the extreme excesses, and so at widths zero and one, means that the weights

factor across $S$ and $S'$ (28.3(i),(ii)). At width one the coordinates above are constant, so the active hinge values are positive multiples of $\gamma_S^\rho$: they read $S$ alike, with $\gamma = \gamma_S^\rho$.

*(i)⇒(iii)⇒(iv)⇒(i).* The first is 28.10(i) with the law $c\,\delta_{\mathbf{x}_0}$. For the second take the partner on one mark $b$ with the room letters $p_{jD}\alpha_b^{c_j - i(D)}$ and $p_{jD}^*\alpha_b^{i(D)}$ and the mass $w - \kappa_S$: the period of its glue with $S$ is the station transform at $\mathbf{p}$ (28.23), and the partner with its stand-in at $\mathbf{x}_0$ is the glue of the one-mark piece at $\mathbf{x}_0$ with it, whose period is the kernel $\Gamma(\kappa_S)\mathsf{Z}_{(\kappa_S,\mathbf{s},\mathbf{t});t_\mathcal{D}}(\mathbf{x}_0 \circ \mathbf{p})$ (28.10(ii)). The third is 28.23.

*Lists.* With the weights 1 the seam letters are the letters themselves and the multipliers are 1 (28.1). Let two active hinge values of tables of width one be proportional, $\gamma_{j,S}^K = \mu\,\gamma_{j',S}^\Psi$ say, that is $K_{\mathcal{B}_j|S}\Psi_{\mathcal{B}_{j'}|S} = \mu\,K_{\mathcal{B}_j/S'}\Psi_{\mathcal{B}_{j'}/S'}$ (19.2). Along a maximal chain of subsets of $S$ the terms of least degree of each of these four letters, at every set of the chain at once, form one monomial with the coefficient 1 (12.20); for a product of two polynomials with nonnegative coefficients the least degrees add and the terms of least degree are the products of those of the factors, so the two sides have such terms $1 \cdot \alpha^{\mathbf{a}}$ and $\mu\,\alpha^{\mathbf{a}'}$, and $\mu = 1$. The other pairs of letters are alike. So reading alike is equality, every $c_j$ and $c_j^*$ is 1, $C = 1$, and $\mathbf{x}_0 = \mathbf{1}$.

*One list.* With both letters active, $w = (s+t)c(S) > 0$, so pinned means $c(S) = 1$ and $\gamma_S^K$ and $\gamma_S^\Psi$ equal, by the above, that is $\varrho_S = 1$, which is $\mathbf{E}_{\mathcal{B}|S} = \mathbf{E}_{\mathcal{B}/S'}$ (19.2). The cell of the bunch $\mathcal{B}_S^+$ is a list with the hinge minors $\mathcal{B}|S$ and $\mathcal{B}/S'$ (19.4) and the ratio $\varrho_S$ (21.1). □

**Corollary 28.26 (Which sets split off).** *As stated.*

*Proof.* (i) A pinned set has width at most one in every active table; apply 28.25. (ii) A prime table is connected, so a set with at least two marks on each side has positive width (7.10), and it has no bunch of width one with two marks on each side (23.16); so such a set has width at least two, and (i) applies. (iii) The set $S = \mathcal{E} \setminus \{e\}$ has the width $c(\{e\}) = r(\{e\}) - d(\{e\}) = 1$ (7.13), and one letter is active, so $S$ is pinned with $\gamma = \gamma_S^K$, and it splits off. At $(s,0)$, $d(\{e\}) = 0$ gives $\kappa_{\{e\}} = m_e$ and $\kappa_S = s - m_e$, as $\kappa_S + \kappa_{\{e\}} = s$ (8.16). 19.14 with the bunch $S$ and the room $\{e\}$, at the level shares $(s - \kappa_S, 0, \kappa_S, 0)$, gives the Beta corner $\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix} s - m_e, & m_e \\ & s \end{bmatrix}$ times the period of the bunch family, the tables $\mathcal{B}/\{e\} = \mathcal{B}/e$ at $(m_e, 0)$ and $\mathcal{B}|S = \mathcal{B} \setminus e$ at $(s - m_e, 0)$ (7.5), times the period of the room family on one mark, which is 1. At $(0,t)$, $d^*(\{e\}) = 0$ gives $\kappa_{\{e\}} = m_e$, $\kappa_S = t - m_e$, and the level shares $(0, \kappa_S, 0, t - \kappa_S)$ give the stated family. (iv) $S = \mathcal{E} \setminus \{e\}$ has width one and $|S| \geq 2$, and its hinge minors are $\mathcal{B}|S = \mathcal{B} \setminus e$ and $\mathcal{B}/\{e\} = \mathcal{B}/e$; apply the last part of 28.25. □

## 50.29 Chapter 29: composing bunches

**Theorem 29.6 (Two transforms of a law, and their rules).** *As stated; the first rule of (iii) holds at any number of sub-pieces.*

*Proof.* Parts (i), (ii) and (iv), and the second and third rules of (iii), are the reference, proof of thm:lawreadings. For the first rule:

- Let $S_1, \dots, S_k$ be as in 29.2, and $\mathbf{p}$ a point with positive coordinates, one for each coordinate of the seam vector of $S$.
- By 28.21, $\mathcal{I}_S(\mathbf{p}) = \int H\,d\lambda_S$, with the law $\lambda_S = (\boldsymbol{\varrho}_S^\rho)_*\phi_S^\rho$ and the Borel function $H(\mathbf{z}) = \Gamma(\kappa_S)\,\mathsf{Z}_{(\kappa_S,\mathbf{s},\mathbf{t});t_\mathcal{D}}(\mathbf{z} \circ \mathbf{p}) \geq 0$, $\mathcal{D}$ being the stations of room sides of $S$.
- 29.2(ii) gives $\int H\,d\lambda_S = \int \big(\int H\,d\lambda_{\mathbf{x}}\big) \prod_l d\lambda_{S_l}(\mathbf{x}_l)$, with $\lambda_{\mathbf{x}}$ the law of the node $S^{(\mathbf{x}_1,\dots,\mathbf{x}_k)}$.

- The node is a piece with the stations $\mathcal{D}$, the exponents of the family and the wall $\kappa_S$ (29.2(ii)), so by the same definition $\int H\, d\lambda_{\mathbf{x}} = \mathcal{I}_{S^{(\mathbf{x}_1,\dots,\mathbf{x}_k)}}(\mathbf{p})$.

This is the reference's argument for one sub-piece, which does not use $k = 1$. The second and third rules use the transfer matrices from one sub-piece: 29.2(i) gives linearity in the seam letters of each $S_l$ separately, with coefficients that depend on the marks of the other sub-pieces. □

**Theorem 29.8 (Cyclic chains).** *As stated: (i) holds for a family, and in (ii) each letter is opened over the starting states at which its term is not zero.*

*Proof.* (i) Fix a table $j$.

- By 21.14(ii), the bases of table $j$ are the unions of the unpaired parts of the choices of one basis in every cell $\mathcal{A}_{j,i}$ that occupy exactly one hinge of every pair. Its weight in each letter at a basis is the sum, over the choices that give it, of the products of the weights chosen.
- A pair of $I_{i-1}$ occupied on the side of the cell $i-1$ is free on the side of the cell $i$, and conversely. So, with $J_i \subseteq I_i$ the pairs occupied on the side of the cell $i$, the basis chosen in $\mathcal{A}_{j,i}$ occupies the hinges $(I_{i-1} \setminus J_{i-1}) \sqcup J_i$.
- Its part $X_i \subseteq B_i$ is a basis of the child of the bunch of its hinges at that pattern. With their weights, these choices sum to the channel $K_{\mathcal{A}_{j,i},((I_{i-1}\setminus J_{i-1})\sqcup J_i)}(\alpha_{B_i})$ (21.13).
- A basis $\bigsqcup_i X_i$ leaves the cobasis $\bigsqcup_i (B_i \setminus X_i)$, so $\Psi_j$ is summed the same way with the channels of $\Psi$.
- Summing over the sequences $(J_0, \dots, J_{k-1})$, $J_k = J_0$, gives $K_j = \operatorname{tr} \prod_i \mathbf{M}_{K_j,i}$ and $\Psi_j = \operatorname{tr} \prod_i \mathbf{M}_{\Psi_j,i}$. The entries are polynomials in $\alpha_{B_i}$ with nonnegative coefficients and no $\varepsilon$, which is the top layer 0.
- The pairs are named alike in every table, so the sets of states $\mathcal{J}_i = \{J :\ J \subseteq I_i\}$ serve every letter, as 29.7 asks.

This is the reference's proof (proof of thm:cyclicchain(i)) applied table by table.

(ii) and (iii) are the reference, proof of thm:cyclicchain. In (ii), $L$ is the sum over the starting states $J$ of the terms $\mathbf{e}_{(J,0)}^{\mathsf{T}} \prod_i \widetilde{\mathbf{M}}_{L,i}\, \mathbf{e}_{(J,h)}$, each a sum over the paths from $(J,0)$ to $(J,h)$ of products of entries with nonnegative coefficients. 4.4 opens a sum of positive terms, one share for each. A starting state whose term vanishes, the state of no path, therefore carries no share and no flow. □

## 50.30 Chapter 30: Foreman fusion

The proofs use the rank function of a table, $r(Y) = \max_T |T \cap Y|$ (Definition 7.4), through its standard properties: $r(\varnothing) = 0$, $r(Y) \le r(Y+e) \le r(Y)+1$ and $r(Y \cup Y') + r(Y \cap Y') \le r(Y) + r(Y')$; conversely a function on the subsets of a finite set with these three properties is the rank function of the table whose bases are the sets $Y$ with $|Y| = r(Y)$ equal to the value at the whole set (the rank axioms, which hold for the bases of Definition 7.1 by the arena property, the exchange property of the bases). A set is *closed* when it is maximal for its rank, that is when adding any mark raises its rank; the *closure* of $Y$ is the set of the marks whose addition does not raise the rank of $Y$, it is closed, and it lies in every closed set that contains $Y$. A mark $e \notin Y$ lies in the closure of $Y$ exactly when some circuit $C$ has $e \in C \subseteq Y + e$: a basis $I$ of $Y$ with $I + e$ dependent contains, with $e$, a circuit through $e$, and a circuit $C$ through $e$ inside $Y + e$ puts $e$ in the closure of $C - e$. For a minor contracting an independent set $C$ and deleting $D$, the rank of a set $Y$ of the remaining

marks is $r(Y \cup C) - r(C)$ (Definition 7.5). The dual rank function is $r^*(Y) = |Y| + r(\mathcal{E} \setminus Y) - r$ (Definition 7.12), and $d(Y) = |Y| - r^*(Y)$.

**Definition 30.1 (Quotients; lifts; the level cell of a pair).** As stated.

*Proof.* That the lifts and the level cell are tables is Proposition 30.2(ii) and (iii). For a bunch $B$ with the room $R$, $\mathcal{B}/R$ is a quotient of $\mathcal{B}|B$: for $X \subseteq Y \subseteq B$, submodularity gives $r(Y) - r(X) \geq r(Y \cup R) - r(X \cup R)$, which is the inequality of Proposition 30.2(i) for the rank functions $r(\cdot)$ of $\mathcal{B}|B$ and $r(\cdot \cup R) - r(R)$ of $\mathcal{B}/R$. The independent sets of $\mathcal{L}_j(\mathcal{B}|B, \mathcal{B}/R)$ are the $X$ with $|X| \leq r(X)$ and $|X| \leq r_{\mathcal{B}/R}(X) + j$, the independent sets of Proposition 12.17(ii) for $i = j$, so the two lifts are one table, and the level cells of Definition 30.1 and Lemma 28.27 have the same bases. Level form at $G$ is Proposition 14.12: a transposition of two marks of $G$ fixes $X$ and $|Q|$. □

**Proposition 30.2 (The level cell of a pair).** *As stated.*

*Proof.* (i) Let the inequality hold, $A$ be closed in $\mathcal{N}$ and $e \notin A$. Then $r_{\mathcal{N}}(A + e) = r_{\mathcal{N}}(A) + 1$, the inequality at $X = A$, $Y = A + e$ gives $r_{\mathcal{M}}(A + e) = r_{\mathcal{M}}(A) + 1$, and $A$ is closed in $\mathcal{M}$. Conversely, let every closed set of $\mathcal{N}$ be closed in $\mathcal{M}$. Along a chain from $X$ to $Y$ adding one mark at a time it suffices to take $Y = X + e$. If $r_{\mathcal{N}}(X + e) = r_{\mathcal{N}}(X)$ the inequality holds, $r_{\mathcal{M}}$ being monotone. Otherwise $e$ is not in the closure $A$ of $X$ in $\mathcal{N}$; $A$ is closed in $\mathcal{M}$ and contains $X$, so the closure of $X$ in $\mathcal{M}$ lies in $A$ and avoids $e$, and $r_{\mathcal{M}}(X + e) = r_{\mathcal{M}}(X) + 1$. At $X = \varnothing$, $Y = \mathcal{E}$ the inequality gives $c \geq 0$; at $X = \varnothing$ it gives $r_{\mathcal{M}}(Y) \geq r_{\mathcal{N}}(Y)$, so a set independent in $\mathcal{N}$ is independent in $\mathcal{M}$; at $Y = \mathcal{E}$ it gives $r_{\mathcal{N}}(\mathcal{E}) - r_{\mathcal{N}}(X) \leq r_{\mathcal{M}}(\mathcal{E}) - r_{\mathcal{M}}(X)$, so a set spanning $\mathcal{M}$ spans $\mathcal{N}$; and two such inequalities add.

(ii) Put $f_1(Y \sqcup P) = r_{\mathcal{M}}(Y) + |P|$ and $f_2(Y \sqcup P) = r_{\mathcal{N}}(Y) + c$. Each is monotone, rises by at most one when a mark is added, and is submodular, being a rank function plus a function that is modular; $f_1(\varnothing) = 0 \leq c = f_2(\varnothing)$; and $f_1 - f_2 = r_{\mathcal{M}}(Y) - r_{\mathcal{N}}(Y) + |P| - c$ is nondecreasing under inclusion by (i). The minimum of two submodular functions with a nondecreasing difference is submodular: for sets $Z, Z'$ the cases where one function attains the minimum at both are its own submodularity, and if $\min(Z) = f_1(Z)$ and $\min(Z') = f_2(Z')$, then

$$f_1(Z) + f_2(Z') - f_1(Z \cap Z') - f_2(Z \cup Z') \geq \big(f_2(Z) - f_2(Z \cap Z')\big) - \big(f_2(Z \cup Z') - f_2(Z')\big) \geq 0,$$

the first step because $(f_1 - f_2)(Z) \geq (f_1 - f_2)(Z \cap Z')$ and the second by the submodularity of $f_2$, so that $\min(Z) + \min(Z') \geq f_1(Z \cap Z') + f_2(Z \cup Z') \geq \min(Z \cap Z') + \min(Z \cup Z')$; the case with $f_1$ and $f_2$ exchanged is the same with the roles of $Z$ and $Z'$ exchanged. So $r^\natural = \min(f_1, f_2)$ satisfies the rank axioms. Its value at $\mathcal{E} \sqcup G$ is $\min(r_{\mathcal{M}} + c, r_{\mathcal{N}} + c) = r_{\mathcal{M}}$, and $Y \sqcup P$ with $|Y| + |P| = r_{\mathcal{M}}$ is a basis exactly when $r_{\mathcal{M}}(Y) \geq |Y|$ and $r_{\mathcal{N}}(Y) \geq |Y| + |P| - c = r_{\mathcal{N}}$: $Y$ independent in $\mathcal{M}$ and spanning $\mathcal{N}$, with $|Y| = r_{\mathcal{N}} + c - |P|$. For $Q \subseteq G$, $r^\natural(Q) = |Q|$, so the minor contracting $Q$ and deleting $G \setminus Q$ has the rank function

$$Y \mapsto r^\natural(Y \sqcup Q) - |Q| = \min\big(r_{\mathcal{M}}(Y),\ r_{\mathcal{N}}(Y) + c - |Q|\big),$$

that of $\mathcal{L}_{c-|Q|}$, provided some basis contains $Q$ and avoids $G \setminus Q$. Such bases exist for every $Q$: a basis $Y_0$ of $\mathcal{N}$ is independent in $\mathcal{M}$ by (i), it extends to a basis of $\mathcal{M}$ of $r_{\mathcal{N}} + c$ marks, and every set between the two is independent in $\mathcal{M}$ and spans $\mathcal{N}$. So every $Q$ is a pattern, $d(G) = 0$, $r(G) = c$ and $c(G) = c$, and the child at $Q$ is $\mathcal{L}_{c-|Q|}$ (Lemma 12.2). At $Q = \varnothing$ the rank function is $r_{\mathcal{M}}$, since $r_{\mathcal{M}}(Y) \leq r_{\mathcal{N}}(Y) + c$ by (i) at $X = Y$ and $\mathcal{E}$; at $Q = G$ it is $r_{\mathcal{N}}$, since $r_{\mathcal{N}} \leq r_{\mathcal{M}}$. The rank function depends on $P$ through $|P|$ alone, so every permutation of $G$ maps the bases onto themselves.

(iii) By (ii) $\mathcal{L}_j$ is the child at any $Q$ of $c-j$ marks, a table, with the stated rank and bases, and $\mathcal{L}_0 = \mathcal{N}$, $\mathcal{L}_c = \mathcal{M}$. For $j \le j'$ take $Q' \subseteq Q$ with $|Q| = c-j$ and $|Q'| = c-j'$ and a chain of patterns from $Q'$ to $Q$ adding one mark at a time; each child is a quotient of the one before (Proposition 12.6), and by (i) $\mathcal{L}_j$ is a quotient of $\mathcal{L}_{j'}$. With the rank difference $b-a$ of $(\mathcal{L}_b, \mathcal{L}_a)$,

$$\min\big(r_{\mathcal{L}_b},\ r_{\mathcal{L}_a}+i\big) = \min\big(r_{\mathcal{M}},\ r_{\mathcal{N}}+b,\ r_{\mathcal{M}}+i,\ r_{\mathcal{N}}+a+i\big) = \min\big(r_{\mathcal{M}},\ r_{\mathcal{N}}+a+i\big),$$

since $i \ge 0$ and $a+i \le b$.

(iv) $C$ is a pattern, so the minor is defined, and its bases are the sets $Y \sqcup (Q \setminus C)$ with $C \subseteq Q$, $Q \cap D = \varnothing$ and $Y$ a basis of $\mathcal{L}_{c-|Q|}$. With $Q'' = Q \setminus C \subseteq G'' = G \setminus (C \cup D)$ and $c'' = c - |C| - |D| = |G''|$, $c - |Q| = |D| + (c'' - |Q''|)$, and by (iii) $\mathcal{L}_{|D|+i} = \mathcal{L}_i(\mathcal{L}_{c-|C|}, \mathcal{L}_{|D|})$ for $0 \le i \le c''$, the rank difference of that pair being $c''$. □

**Lemma 30.3 (Complements of the lifts toward the dual).** *As stated.*

*Proof.* Write $r(Y)$ and $r^*(Y)$ for the rank functions of $\mathcal{B}$ and $\mathcal{B}^*$, so that $r(\mathcal{E} \setminus X) = |\mathcal{E} \setminus X| + r^*(X) - r^*$ and $r^*(\mathcal{E} \setminus X) = |\mathcal{E} \setminus X| + r(X) - r$. By Proposition 30.2(iii) $\mathcal{L}_j$ is a table of rank $r+j$, with the rank function $r_{\mathcal{L}_j}(Y) = \min\big(r^*(Y),\ r(Y)+j\big)$ (30.1), and the rank function of its dual table is

$$\begin{aligned} |X| + r_{\mathcal{L}_j}(\mathcal{E} \setminus X) - r - j &= |X| + \min\big(n - |X| + r(X) - r,\ n - |X| + r^*(X) - r^* + j\big) - r - j \\ &= \min\big(r(X) + c - j,\ r^*(X)\big) = r_{\mathcal{L}_{c-j}}(X), \end{aligned}$$

using $n - 2r = c$ and $n - r^* - r = 0$. Two tables with one rank function have the same bases, the sets $Y$ with $|Y| = r_{\mathcal{L}}(Y)$ equal to the rank; so the dual table of $\mathcal{L}_j$, whose bases are the complements of those of $\mathcal{L}_j$, has the bases of $\mathcal{L}_{c-j}$. □

**Definition 30.4 (Fusion width; Foreman fusion; majors).** As stated.

*Proof.* $c \ge 0$ for a quotient is Proposition 30.2(i). A quotient $\mathcal{N}$ of $\mathcal{M}$ of the same rank has $r_{\mathcal{N}} = r_{\mathcal{M}}$: (i) of that proposition gives $r_{\mathcal{M}}(Y) \ge r_{\mathcal{N}}(Y)$ at $X = \varnothing$, and $r_{\mathcal{N}}(X) \ge r_{\mathcal{M}}(X)$ at $Y = \mathcal{E}$; so the two have the same bases, the sets $Y$ with $|Y| = r_{\mathcal{N}}(Y) = r_{\mathcal{N}}(\mathcal{E})$. At $c = 0$ a fusable table therefore has $\mathcal{B} = \mathcal{B}^*$ and $\theta^* = \theta$, its bases closed under complement with $\theta_{\mathcal{E}\setminus T} = \theta_T$; a list has $\theta^* = \theta$, so at $c = 0$ it is fusable exactly when $\mathcal{B} = \mathcal{B}^*$. The lifts are then the one table $\mathcal{L}_0 = \mathcal{L}_c$, with $\theta^{(0)} = \theta = \theta^* = \theta^{(c)}$. The condition on the weights pairs the bases of $\mathcal{L}_j$ with those of $\mathcal{L}_{c-j}$ by complement, which is a bijection by Lemma 30.3; at $j = 0$ it is the definition of $\theta^*$. $L_0 = K_{\mathcal{B}^*} = \Psi$ (Proposition 7.2) and $L_c = K_{\mathcal{B}} = K$. Every permutation of $G$ maps the bases of $\mathcal{B}^\sharp$ onto themselves (Proposition 30.2(ii)) and keeps their weights, which depend on $X$ and $|Q|$ alone, so the partial fusion depends on $G'$ only through $|G'|$, up to the names of its marks. □

**Theorem 30.5 (Foreman fusion).** *As stated.*

*Proof.* (i) Proposition 30.2(ii) with $(\mathcal{M}, \mathcal{N}) = (\mathcal{B}^*, \mathcal{B})$, $r_{\mathcal{M}} = r^*$, $r_{\mathcal{N}} = r$, gives the table, its rank, its rank function and the two minors; $n + c = 2n - 2r = 2r^*$. The bases $X \sqcup G$ carry $\theta^{(0)}_X = \theta_X$ and the bases $X \sqcup \varnothing$ carry $\theta^{(c)}_X = \theta^*_X$, so the contraction of $G$ is $(\mathcal{B}, \theta)$ and the deletion is $(\mathcal{B}^*, \theta^*)$ (Definition 7.5).

(ii) By Lemma 30.3, $X$ is a basis of $\mathcal{L}_j$ exactly when $\mathcal{E} \setminus X$ is a basis of $\mathcal{L}_{c-j}$, and with the weights $\theta^{(c-j)}_{\mathcal{E}\setminus X} = \theta^{(j)}_X$ the dual table of $(\mathcal{L}_j, \theta^{(j)})$ is $(\mathcal{L}_{c-j}, \theta^{(c-j)})$. A basis $X \sqcup Q$ of $\mathcal{B}^\sharp$, $X$ a basis of $\mathcal{L}_{c-|Q|}$, has the complement $(\mathcal{E} \setminus X) \sqcup (G \setminus Q)$ with $\mathcal{E} \setminus X$ a basis of $\mathcal{L}_{|Q|} = \mathcal{L}_{c-|G\setminus Q|}$ and the

weight $\theta^{(|Q|)}_{\mathcal{E}\setminus X} = \theta^{(c-|Q|)}_X$: complementation maps the bases onto themselves with their weights. Summing $\theta^\sharp\alpha^{X\sqcup Q}$ over $Q$ by its size gives $K^\sharp = \sum_l e_l(\alpha_G)K_{\mathcal{L}_{c-l}}$; $\Psi^\sharp = K^\sharp$ by Corollary 20.19; and $\Psi_{\mathcal{L}_l} = K_{\mathcal{L}^*_l} = K_{\mathcal{L}_{c-l}}$ by Proposition 7.2.

(iii) Let $L$ be a major and $X\sqcup Q$ one of its bases; at $c=0$, $L=(\mathcal{B},\theta)=\mathcal{B}^\sharp$. In $L$ the children of the bunch $G$ at $\varnothing$ and at $G$ are $\mathcal{B}^*$ and $\mathcal{B}$, so $\varnothing$ and $G$ are patterns, $d(G)=0$ and $G$ is independent; the contraction of the room onto $G$ has the rank $d(G)=0$, so by Proposition 12.17(i) every subset of $G$ is a pattern of $G$ in $L$, and along chains of patterns from $\varnothing$ to $Q$ and from $Q$ to $G$ the child at $Q$, of which $X$ is a basis, is a quotient of $\mathcal{B}^*$ and has $\mathcal{B}$ as a quotient (Propositions 12.6 and 30.2(i)). So $X$ is independent in $\mathcal{B}^*$ and spans $\mathcal{B}$ (Proposition 30.2(i)); it has $r^*-|Q| = r+c-|Q|$ marks, so it is a basis of $\mathcal{L}_{c-|Q|}$ (Proposition 30.2(iii)) and $X\sqcup Q$ is a basis of $\mathcal{B}^\sharp$.

(iv) The weights of $\mathcal{B}^\sharp$ depend on $X$ and $|Q|$, so $G$ is in level form in it (Definition 30.1). Let $L$ be a major in which $G$ is in level form. By Proposition 28.29(d) for $L$ and its bunch $G$, the room sides of $r+j$ marks are the bases of the $j$-th lift of the table of the room sides of $r$ marks, those of $L/G=\mathcal{B}$, toward the table of the room sides of $r+c$ marks, those of $L\setminus G=\mathcal{B}^*$: the bases of $\mathcal{L}_j$. A room side $X$ of $r+j$ marks makes a basis with some pattern of $c-j$ marks, and in level form the children at all patterns of $c-j$ marks are one table, so $X\sqcup Q$ is a basis of $L$ for every $Q$ of $c-j$ marks: every basis of $\mathcal{B}^\sharp$ is a basis of $L$, and by (iii) the two have the same bases. The weights of $L$ at $X\sqcup G$ and $X\sqcup\varnothing$ are those of its minors, $\theta$ and $\theta^*$, and level form makes the others depend on $X$ and $|Q|$ alone (Proposition 14.12); $L$ is identically self-dual exactly when the weight of $X\sqcup Q$ equals that of $(\mathcal{E}\setminus X)\sqcup(G\setminus Q)$, that is when the middle weights are paired as in Definition 30.4(ii), and $L$ is then the Foreman fusion whose middle weight $\theta^{(c-|Q|)}_X$ is the weight of $X\sqcup Q$. When $L$ and $\mathcal{B}$ are lists all these weights are 1, and $L=\mathcal{B}^\sharp$. The transpositions generate the permutations of $G$, which gives the equivalent form (Proposition 14.12). A bijection $\beta$ carrying $(\mathcal{B},\theta)$ carries $(\mathcal{B}^*,\theta^*)$, keeps $r$ and $r^*$, and so keeps the rank function of every $\mathcal{L}_j$; with $\theta^{(j)}\circ\beta=\theta^{(j)}$, $\beta\sqcup\pi$ maps each basis $X\sqcup Q$ to the basis $\beta(X)\sqcup\pi(Q)$ of the same weight.

(v) At $c=1$, $\mathcal{L}_0=\mathcal{B}$ and $\mathcal{L}_1=\mathcal{B}^*$: the bases are $X+g$, $X\in\mathcal{B}$, with $\theta_X$, and $X\in\mathcal{B}^*$ with $\theta^*_X$, that is $\mathcal{E}\setminus T$ with $\theta_T$; $K^\sharp = e_0L_0+e_1L_1 = \Psi+\alpha_gK$. A major $L$ has $L/g=(\mathcal{B},\theta)$ and $L\setminus g=(\mathcal{B}^*,\theta^*)$, which fixes its bases through $g$ and avoiding $g$ with their weights: $L=\mathcal{B}^\sharp$. □

**Theorem 30.6 (The tell).** *As stated.*

*Proof.* Three facts about a table $\mathcal{B}$ of rank $r$. ($\alpha$) A set $A$ is closed in $\mathcal{B}^*$ exactly when $\mathcal{E}\setminus A$ is a union of circuits of $\mathcal{B}$: for $e\notin A$, $r^*(A+e)-r^*(A) = 1+r(\mathcal{E}\setminus A-e)-r(\mathcal{E}\setminus A)$, which is 1 exactly when $e$ lies in the closure of $\mathcal{E}\setminus A-e$, that is in a circuit inside $\mathcal{E}\setminus A$. ($\beta$) The cocircuits are the complements of the closed sets of rank $r-1$: $D$ meets every basis exactly when $\mathcal{E}\setminus D$ contains none, that is $r(\mathcal{E}\setminus D)\le r-1$, and $D$ is minimal with this property exactly when $\mathcal{E}\setminus D$ is maximal with it, a closed set of rank $r-1$. ($\gamma$) The complement of a closed set $A$ is a union of cocircuits: for $e\notin A$, a basis $I$ of $A$ has $I+e$ independent, $I+e$ extends to a basis $T$, and the closure $H$ of $T-e$ has rank $r-1$, contains $A$, which lies in the closure of $I$, and avoids $e$, $T$ being independent; so $\mathcal{E}\setminus H$ is a cocircuit through $e$ inside $\mathcal{E}\setminus A$, by ($\beta$).

(a)⇒(b): the complement of a cocircuit is closed in $\mathcal{B}$ by ($\beta$), so in $\mathcal{B}^*$, and the cocircuit is a union of circuits by ($\alpha$). (b)⇒(a): the complement of a closed set $A$ of $\mathcal{B}$ is a union of cocircuits by ($\gamma$), each a union of circuits by (b), so $A$ is closed in $\mathcal{B}^*$ by ($\alpha$). (b)⇒(c): if $e$ is not in the closure of $A$, ($\gamma$) applied to that closure gives a cocircuit $D$ with $e\in D$ and $D\cap A=\varnothing$, so $D-e\subseteq A'$; by (b) a circuit $C$ has $e\in C\subseteq D$, and $e$ lies in the closure of $C-e\subseteq A'$. (c)⇒(b): for a cocircuit $D$ and $e\in D$ split $\mathcal{E}\setminus\{e\}$ into $A=\mathcal{E}\setminus D$ and $A'=D-e$; $A$ is closed by ($\beta$) and avoids $e$, so by (c) $e$ lies in the closure of $D-e$, in a circuit inside $D$. By Proposition 30.2(i) (a) implies $c\ge 0$.

By Definition 30.4 the table is fusable exactly when (a) holds and, at $c = 0$, $\theta^* = \theta$, which a list always has. At $c = 0$, $G$ is empty, a major is a table equal to $(\mathcal{B}, \theta)$ and to $(\mathcal{B}^*, \theta^*)$, and one exists exactly when $\mathcal{B} = \mathcal{B}^*$ and $\theta^* = \theta$, which is fusability, a quotient of the same rank having the same bases (proof of Definition 30.4). Let $c \geq 1$. A fusable table has a major, its Foreman fusion (Theorem 30.5(i), whose proof uses Proposition 30.2 alone). Conversely, in a major the children of the bunch $G$ at $\varnothing$ and at $G$ are $\mathcal{B}^*$ and $\mathcal{B}$, so $\varnothing$ and $G$ are patterns, $d(G) = 0$ and $G$ is independent; the contraction of the room onto $G$ has rank $d(G) = 0$, and by Proposition 12.17(i) every subset of $G$ is a pattern. Along a chain from $\varnothing$ to $G$ adding one mark at a time each child is a quotient of the one before (Proposition 12.6), and $\mathcal{B}$ is a quotient of $\mathcal{B}^*$ by Proposition 30.2(i): the table is fusable. A table on $\mathcal{E} \sqcup G$ whose children at $\varnothing$ and at $G$ have the bases of $\mathcal{B}^*$ and of $\mathcal{B}$ is a major once its bases $X \sqcup G$ carry $\theta_X$ and its bases $X \sqcup \varnothing$ carry $\theta^*_X$, the two sets of bases being disjoint as $G \neq \varnothing$; this is the equivalent form of (d).

A coloop $e$ makes $\{e\}$ a cocircuit, and a circuit inside $\{e\}$ would make $e$ a loop; so (b) fails. The circuits of $\mathcal{B}^*$ are the cocircuits of $\mathcal{B}$ and its cocircuits the circuits of $\mathcal{B}$, which gives (b) for the dual. □

**Corollary 30.7 (The fused tables are the identically self-dual tables).** *As stated.*

*Proof.* (i) By Corollary 20.19 $\mathcal{H}$ has rank $|\mathcal{E}'|/2$ and no loop or coloop; $C$ is independent in $\mathcal{H}$ and so in $\mathcal{H}^* = \mathcal{H}$. $\mathcal{H}/C$ has rank $|\mathcal{E}'|/2 - |C|$ on $|\mathcal{E}'| - |C|$ marks, hence the fusion width $|C|$. Its dual has the bases $(\mathcal{E}' \setminus C) \setminus (T \setminus C) = \mathcal{E}' \setminus T$, $T \supseteq C$, with the weights $\theta_T = \theta_{\mathcal{E}' \setminus T}$: the bases of $\mathcal{H}$ avoiding $C$ with their weights, so $(\mathcal{H}/C)^* = \mathcal{H} \setminus C$ and $\mathcal{H}$ is a major of $\mathcal{H}/C$. By Theorem 30.6 $\mathcal{H}/C$ is fusable; $C$ is in level form in $\mathcal{H}$ (Proposition 14.12), and $\mathcal{H}$ is identically self-dual, so by Theorem 30.5(iv) it is the Foreman fusion of $\mathcal{H}/C$ with the middle weights $\theta^{(j)}_X = \theta_{X \sqcup Q}$, $|Q| = c - j$. A one-mark set is independent, $\mathcal{H}$ having no loop, and it has no transposition. With Theorem 30.5(ii) the fused tables are exactly the identically self-dual ones.

(ii) $c(S) = |S| - d(S) - d^*(S)$ does not change under duality, which exchanges $d$ and $d^*$, so $\mathcal{B}^*$ is connected and its circuits link all of $\mathcal{E}$ (Proposition 28.3(i)). The circuits of $\mathcal{B}^* = \mathcal{B}^\sharp \setminus G$ are circuits of $\mathcal{B}^\sharp$, so $\mathcal{E}$ lies in one molecule of $\mathcal{B}^\sharp$. For $g \in G$ take a basis $X \sqcup Q$ with $g \notin Q$; the circuit through $g$ inside $X \sqcup Q + g$ does not lie in the independent set $G$, so it meets $\mathcal{E}$, and $g$ lies in the molecule of $\mathcal{E}$. So $\mathcal{B}^\sharp$ is one molecule, connected by Proposition 28.3(i). □

**Proposition 30.8 (Partial fusions and inner lifts).** *As stated.*

*Proof.* (i) is Proposition 30.2(iv) for $(\mathcal{B}^*, \mathcal{B})$, the minor carrying the weights of $\mathcal{B}^\sharp$ (Definition 7.5). (ii) With $C = G \setminus G'$ and $D = \varnothing$, (i) gives the level cell of $(\mathcal{L}_i, \mathcal{B})$, of rank $r + i$ on $n + i$ marks, so of the fusion width $c - i$. Being identically self-dual, $\mathcal{B}^\sharp$ has $(\mathcal{B}^\sharp/(G \setminus G'))^* = \mathcal{B}^\sharp \setminus (G \setminus G')$ with the weights, as in the proof of Corollary 30.7(i); $G \setminus G'$ is independent and in level form in $\mathcal{B}^\sharp$ (Proposition 30.2(ii), Definition 30.1); so Corollary 30.7(i) makes $\mathcal{B}^\sharp_i$ fusable with the Foreman fusion $\mathcal{B}^\sharp$, its middle weights read off $\theta^\sharp$. (iii) $\mathcal{L}_j$ has the rank $r+j$ and the corank $n-r-j = r+c-j$, so the fusion width $c - 2j$. For $2j \leq c$, $\mathcal{L}_j$ is a quotient of $\mathcal{L}_{c-j} = \mathcal{L}^*_j$ (Proposition 30.2(iii) and Theorem 30.5(ii)), and at $2j = c$, where $\mathcal{L}^*_j = \mathcal{L}_j$, its weights are those of its dual, $\theta^{(j)}_{\mathcal{E} \setminus X} = \theta^{(j)}_X$; so it is fusable, and its lifts toward its dual are the $\mathcal{L}_{j+i}$, $0 \leq i \leq c - 2j$ (Proposition 30.2(iii)). Its level cell is $\mathcal{B}^\sharp/C \setminus D$ with $|C| = |D| = j$ by (i), whose weights are paired by complement as those of $\mathcal{B}^\sharp$ are, so it is the Foreman fusion of $(\mathcal{L}_j, \theta^{(j)})$ with the middle weights read off $\theta^\sharp$. At $j = c/2$ the lift is identically self-dual; at $j = (c-1)/2$ the fusion width is one. □

**Proposition 30.10 (The seam of the fusion bunch).** *As stated.*

*Proof.* (i) $\mathrm{pat}(G) = 2^G$, $d(G) = 0$, $c(G) = c$ and level form are Proposition 30.2(ii) and Definition 30.1. A pattern $Q$ of $l$ marks has the child $(\mathcal{L}_{c-l}, \theta^{(c-l)})$, the same for all of them, so they form one station with the multipliers 1 (Definition 28.1); its station letters are $\sum_{|Q|=l} \alpha_G^Q = e_l(\alpha_G)$ and $\sum_{|Q|=l} \alpha_G^{G\setminus Q} = e_{c-l}(\alpha_G)$. A room side $X$ of excess $l$ is a basis of $\mathcal{L}_{c-l}$, its size fixing $l$, and its room child has the bases of $\binom{G}{l}$, each with the weight $\theta_X^{(c-l)}$: all the room children of the excess $l$ are positive multiples of $\binom{G}{l}$, one station $D_l$, with the multipliers $w'_X = \theta_X^{(c-l)}/\mu_l$ and the room child of the station $\mu_l\binom{G}{l}$, of the letters $\mu_l e_l$ and $\mu_l e_{c-l}$. The room-station letters are $\sum_X w'_X \alpha^X = L_l/\mu_l$ and $\sum_X w'_X \alpha^{\mathcal{E}\setminus X} = \Psi_{\mathcal{L}_{c-l}}/\mu_l = L_{c-l}/\mu_l$ (Theorem 30.5(ii)), and the pair constant is $\theta_X^{(c-l)}/(1 \cdot w'_X) = \mu_l$. One station on each side at each excess is one rectangle, and the room-station factorization of Proposition 28.2(iii) is the stated one. The weight of $X \sqcup Q$ depends on $X$ alone, $|X|$ fixing $|Q|$, so the weights factor across $G$ and $\mathcal{E}$ and Theorem 28.28 applies: with one rectangle at every excess the table is its level envelope.

(ii) By Definition 12.18, with $K_{(Q)} = K_{\mathcal{L}_{c-|Q|}}$ and $\Psi_{(Q)} = \Psi_{\mathcal{L}_{c-|Q|}} = K_{\mathcal{L}_{|Q|}}$,

$$\iota_G = \sum_{l=0}^{c} \binom{c}{l} (-1)^{c-l} \log K_{\mathcal{L}_{c-l}}, \qquad \iota_G^* = \sum_{l=0}^{c} \binom{c}{l} (-1)^{l} \log K_{\mathcal{L}_l},$$

both equal to $\sum_j (-1)^j \binom{c}{j} \log K_{\mathcal{L}_j}$; at $c = 2$ this is $\log K_{\mathcal{L}_0} - 2\log K_{\mathcal{L}_1} + \log K_{\mathcal{L}_2} = -\log z_1$.

(iii) At the table's gauge the pattern probability of $Q$ is $K_{(Q)}(\mathbf{1})/K^\sharp(\mathbf{1}) = L_{|Q|}(\mathbf{1})/K^\sharp(\mathbf{1})$ (Definition 12.25); the $\binom{c}{l}$ patterns of $l$ marks give $\Pi_l$. $L_l(\mathbf{1})$ is the total weight of $(\mathcal{L}_{c-l}, \theta^{(c-l)})$, which equals that of its dual $(\mathcal{L}_l, \theta^{(l)})$, $L_{c-l}(\mathbf{1})$; so $\Pi_l = \Pi_{c-l}$. The peel (Definition 14.1) gives the child at $Q$ the exponents $\pi_Q(\mathbf{1})(s,t)$, and merging the equal children of each excess gives $(\mathcal{L}_{c-l}; \Pi_l s, \Pi_l t)$, that is $(\mathcal{L}_j; \Pi_j s, \Pi_j t)$. Its word is $\prod_j K_{\mathcal{L}_j}^{-\Pi_j s} \Psi_{\mathcal{L}_j}^{-\Pi_j t} = \prod_j K_{\mathcal{L}_j}^{-\Pi_j s} \prod_j K_{\mathcal{L}_{c-j}}^{-\Pi_j t} = \prod_j K_{\mathcal{L}_j}^{-\Pi_j (s+t)}$, reindexing the second product by $j \mapsto c - j$. □

**Definition 30.11 (The universal piece; the lift partner).** As stated.

*Proof.* The seam letters $e_l$ and $e_{c-l}$ of the station $D_l$ are nonzero, with nonnegative coefficients, homogeneous of the degrees $l = d + l$ and $c - l = d^* + c - l$ with $d = d^* = 0$, and positive letters where the degree is positive; the room letters $L_l = K_{\mathcal{L}_{c-l}}$ and $L_{c-l}$ have the degrees $r + c - l = e + c - l$ and $r + l = e^* + l$ with $e = e^* = r$, as Definition 28.4 asks of a partner read with the reversed excesses. □

**Theorem 30.12 (The glue of a fused table).** *As stated.*

*Proof.* The family $((\mathcal{B}^\sharp, \theta^\sharp); 2\sigma, 0)$ has the word $K^{\sharp\,-2\sigma} = K^{\sharp\,-s} \Psi^{\sharp\,-t}$ (Theorem 30.5(ii)), and it is the glue of the universal piece and the lift partner with the one active letter $K$ (Definition 28.4, Proposition 30.10(i)).

(i) In Definition 28.5 with the one active letter at the exponent $2\sigma$: $\kappa_G = m_G$, $\kappa_G + \kappa_G^{\mathrm{top}} = 2\sigma c$, $V^{(0)} = e_0 = 1$ and $V^{(c)} = e_c$, so $\gamma_G^\rho = e_c^{1/c}$, the scale form is $\phi_G^\rho(du) = u^{\mathbf{m}|_G - \mathbf{1}} e_c(u)^{-m_G/c}\, d\omega_G(u)$, and the seam vector at $D_l$ is $e_l/(\gamma_G^\rho)^l$. The form $\nu_G = \alpha^{\mathbf{m}|_G - \mathbf{1}} e_c(\alpha)^{-m_G/c}\, d\alpha$ on $(0,\infty)^G$ and the seam vector are homogeneous of degree zero. For a Borel $\psi \geq 0$ on the seam torus and $\mathbf{1}_{[1,e]}$ the indicator of $[1,e]$, with $\int_0^\infty \mathbf{1}_{[1,e]}(\rho)\, d\rho/\rho = 1$, the integral of $\psi(\boldsymbol{\varrho}_G^\rho(\alpha))\, \mathbf{1}_{[1,e]}(h(\alpha))\, \nu_G$ over the orthant is, for every positive $h$ of degree one, the integral of $\psi(\boldsymbol{\varrho}_G^\rho)$ against the law: in the coordinates $\alpha = \rho u$ of scale and shape (Lemma 2.7) the form is $\mathbf{1}_{[1,e]}(\rho)\, d\rho/\rho$ times $\phi_G^\rho$, the integrand not depending on $\rho$, and a change of the section $h$ is a change of $\rho$ along each ray, which leaves $\int \mathbf{1}_{[1,e]}(\rho)\, d\rho/\rho$ unchanged (the computation of Lemma 16.2, in $[0,\infty]$ by Tonelli). Take

$h = e_c^{1/c}$ and write $\alpha = \rho\, e^w$ with $\rho > 0$, $w \in H_G$. In the coordinates $y = \log\alpha = \lambda\mathbf{1} + w$, $\lambda = \log\rho$, the map $(\lambda, w_{g_1}, \dots, w_{g_{c-1}}) \mapsto y$, $w_{g_c} = -\sum_{i<c} w_{g_i}$, is linear with determinant $\pm c$: adding all rows to the last turns it into $(c, 0, \dots, 0)$. So $d\alpha = \alpha^{\mathbf{1}} c\, d\lambda\, d\lambda_G(w)$, and since $e_c(\rho e^w) = \rho^c$,

$$\nu_G = e^{m_G\lambda + \langle \mathbf{m}|_G, w\rangle}\, e^{-m_G\lambda}\, c\, d\lambda\, d\lambda_G(w) = c\, e^{\langle \mathbf{m}|_G, w\rangle}\, \frac{d\rho}{\rho}\, d\lambda_G(w),$$

while $\boldsymbol{\varrho}_G^{\rho}(\rho e^w) = \mathbf{e}(e^w)$. So the integral of $\psi$ against the law is $c\int_{H_G} \psi(\mathbf{e}(e^w))\, e^{\langle \mathbf{m}|_G, w\rangle}\, d\lambda_G(w)$ for every $\psi$, which is the statement.

(ii) By Lemma 16.2, in $[0,\infty]$ by Tonelli, $P_{\mathcal{B}^\sharp}(\mathbf{m}; s, t)$ is the integral over the orthant of $\alpha^{\mathbf{m}-\mathbf{1}} K^{\sharp\,-2\sigma} \mathbf{1}_{[1,e]}(h(\alpha))\, d\alpha$ with $h = e_c(\alpha_G)^{1/c}$. In the coordinates $(\alpha_{\mathcal{E}}, \rho, w)$, $\alpha_G = \rho e^w$, as in (i), $\alpha_G^{\mathbf{m}|_G - \mathbf{1}} d\alpha_G = c\, e^{\langle \mathbf{m}|_G, w\rangle} \rho^{m_G - 1} d\rho\, d\lambda_G(w)$ and $K^\sharp = \sum_l \rho^l e_l(e^w) L_l(\alpha_{\mathcal{E}})$ (Theorem 30.5(ii)). By Tonelli the integral is $c\int_{H_G} e^{\langle \mathbf{m}|_G, w\rangle} I(w)\, d\lambda_G(w)$ with

$$I(w) = \int_{(0,\infty)^{\mathcal{E}} \times (0,\infty)} \alpha_{\mathcal{E}}^{\mathbf{m}|_{\mathcal{E}} - \mathbf{1}} \rho^{m_G - 1} \Big( \sum_l e_l(e^w) \rho^l L_l(\alpha_{\mathcal{E}}) \Big)^{-2\sigma} \mathbf{1}_{[1,e]}(\rho)\, d\alpha_{\mathcal{E}}\, d\rho.$$

The letter $\sum_l x_l \alpha_g^l L_l$ is homogeneous of degree $r + c = r^*$ on $\mathcal{E} \sqcup \{g\}$, so the form of $W[\mathbf{x}]$ at the masses $(\mathbf{m}|_{\mathcal{E}}, m_G)$ has degree $m_{\mathcal{E}} + m_G - 2\sigma r^* = 0$ on the balance wall, and $I(w)$ is its integral against $\mathbf{1}_{[1,e]}$ of the section function $\alpha_g$: by Lemma 16.2 it is $P_{W[\mathbf{e}(e^w)]}(\mathbf{m}|_{\mathcal{E}}, m_G)$.

(iii) $\sum_l x_l L_l y^l$ is a polynomial of degree $c$ in $y$ with positive coefficients, $x_c L_c \neq 0$, so its roots $-\xi_i$ avoid $[0,\infty)$; read as in Definition 12.10 on the bunch $\{g\}$, whose floor is 0, its Skeleton polynomial at $\alpha_g = 1$ is $\prod_i (x + \xi_i)/(1 + \xi_i)$. Non-real roots come in conjugate pairs, and for $\rho > 0$ the product of the principal powers $(\rho + \xi_i)^{-2\sigma}$ is the positive power of the product. Proposition 15.1 with $J = \{1, \dots, c\}$, the letter exponents $w_i = 2\sigma$, $|\mathbf{w}| = 2\sigma c$ and $\kappa = m_G$ gives the formula for $0 < m_G < 2\sigma c$, coincident values merging.

(iv) This is Theorem 28.10(iii) for the universal piece and the lift partner with the one active letter: one share $a_l = a_{D_l}$ per station, the letter total $\sum_l a_l = 2\sigma$ and the scale pin $\sum_l i(D_l) a_l = \kappa_G = m_G$, the share kernel $\Gamma_{\substack{c+1 \\ 1}} \begin{bmatrix} \mathbf{a} \\ 2\sigma \end{bmatrix}$, and the two seam functions, which on the pins are the periods of the seam word $\prod_l e_l^{-a_l}$ at $\mathbf{m}|_G$ and of the room word $\prod_l L_l^{-a_l}$ at $\mathbf{m}|_{\mathcal{E}}$ (Lemma 28.7), $e_0 = 1$ dropping out. Two pins on $c+1$ shares leave a fibre of dimension $c - 1$.

Let the masses be real and every proper wall of $\mathcal{B}^\sharp$ positive; we show that both periods converge at the native split at the stations. The walls of $((\mathcal{B}^\sharp, \theta^\sharp); 2\sigma, 0)$ are $\kappa_U = m_U - 2\sigma d^\sharp(U)$, with $d^\sharp(U) = r^* - r^\sharp(\mathcal{E}^\sharp \setminus U)$ on $\mathcal{E}^\sharp = \mathcal{E} \sqcup G$; at one mark, no mark being a coloop (Corollary 20.19), $d^\sharp$ is 0, so the masses are positive. A set $U$ of the width zero in $\mathcal{B}^\sharp$ has $r^\sharp(U) + r^\sharp(\mathcal{E}^\sharp \setminus U) = r^*$, so $d^\sharp(U) = r^\sharp(U)$ and $d^\sharp(\mathcal{E}^\sharp \setminus U) = r^\sharp(\mathcal{E}^\sharp \setminus U)$, and $\kappa_U + \kappa_{\mathcal{E}^\sharp \setminus U} = m_{\mathcal{E}^\sharp} - 2\sigma r^* = 0$ on the balance wall; so no nonempty proper $U$ has the width zero, $\mathcal{B}^\sharp$ is connected (Proposition 28.3(i)), and it has a native gauge (Theorem 8.19(iii)). By Lemma 28.9, $c(G) = c \geq 1$, the native split at the stations is a point of the pins with positive shares $a_l$ at which $\mathbf{m}|_G$ and $\mathbf{m}|_{\mathcal{E}}$ lie in the relative interiors of the Newton sums of the seam word and of the room word, and the two words converge there once these Newton sums have the full dimensions $c - 1$ and $n - 1$; the seam letters $\mu_l e_l$ and the room-station letters $L_l/\mu_l$ of the bunch $G$ (Proposition 30.10(i)) differ from $e_l$ and $L_l$ by constant factors, which change neither the Newton sums nor the convergence. The Newton sum $\sum_{l \geq 1} a_l \mathcal{N}_{e_l}$ of the seam word contains a translate of $a_1 \operatorname{conv}\{\mathbf{1}_{\{g\}} : g \in G\}$, of the dimension $c - 1$. The Newton sum $\sum_l a_l \mathcal{N}_{L_l}$ of the room word, $\mathcal{N}_{L_l}$ the base polytope of $\mathcal{L}_{c-l}$, has as its directions the sums of the directions of its terms, so a vector $w \in \mathbb{R}^{\mathcal{E}}$ orthogonal to them has $\langle w, \mathbf{1}_X \rangle$ constant over the bases $X$ of every lift, and $w$ is constant on every molecule of the family

of the lifts (Lemma 7.8). That family is connected. Otherwise a proper nonempty $S \subset \mathcal{E}$ is a union of its molecules, hence of molecules of every lift (Proposition 7.9), and has the width zero in $\mathcal{L}_0 = \mathcal{B}$ and in $\mathcal{L}_1$, whose rank function is $r(Y) + \min(-\varepsilon(Y), 1)$, $\varepsilon(Y) = r(Y) - r^*(Y) \le 0$ the slant of $Y$ in $\mathcal{B}$ (Definition 7.12, Proposition 30.2(i)). The two widths give $r(S) + r(\mathcal{E} \setminus S) = r$ and $\min(-\varepsilon(S), 1) + \min(-\varepsilon(\mathcal{E} \setminus S), 1) = 1$, so $\varepsilon(S) = 0$ or $\varepsilon(\mathcal{E} \setminus S) = 0$, say $\varepsilon(S) = 0$, exchanging $S$ and $\mathcal{E} \setminus S$ if needed. Then by Theorem 30.5(i) $r^\sharp(S) = \min(r^*(S), r(S) + c) = r(S)$ and $r^\sharp((\mathcal{E} \setminus S) \sqcup G) = \min(r^*(\mathcal{E} \setminus S) + c,\ r(\mathcal{E} \setminus S) + c) = r(\mathcal{E} \setminus S) + c$, whose sum is $r + c = r^*$: $S$ has the width zero in $\mathcal{B}^\sharp$, which is excluded. So $w$ is constant, the Newton sum of the room word has the dimension $n - 1$, and both words converge at the native split (Lemma 28.9, Theorem 25.2).

(v) $e_c(\alpha_G) = \alpha_G^{\mathbf{1}}$, so $\alpha_G^{\mathbf{m}|_G - \mathbf{1}} \prod_{l=1}^{c} e_l^{-a_l} = \alpha_G^{(\mathbf{m}|_G - a_c\mathbf{1}) - \mathbf{1}} \prod_{l=1}^{c-1} e_l^{-a_l}$: the two forms are one, and so are their periods; $e_l$ is the letter of $\binom{G}{l}$. At $c = 1$ the pins give $a_1 = m_g$, and the form $\alpha_g^{m_g - 1} \alpha_g^{-a_1} d\alpha_g$ on one mark has the period 1. At $c = 2$, $u_1 + u_2 = 1$ on $\Delta_G$, so the period of $e_1^{-a_1}$ at $(m_1 - a_2, m_2 - a_2)$ is $\int_{\Delta_G} u^{(\mathbf{m}|_G - a_2\mathbf{1}) - \mathbf{1}} d\omega_G = \Gamma_{\substack{2\\1}}\!\left[\begin{matrix} m_1 - a_2,\ m_2 - a_2 \\ a_1 \end{matrix}\right]$ (Theorem 3.9), $m_1 + m_2 - 2a_2 = a_1$ on the pins. □

**Corollary 30.13 (Width one).** *As stated.*

*Proof.* $\mathcal{B}^\sharp$ is identically self-dual (Theorem 30.5(ii)), and Corollary 20.19 at the mark $g$, which allows the exponents $(2\sigma, 0)$, gives $P_{\mathcal{B}^\sharp} = \Gamma_{\substack{2\\1}}\!\left[\begin{matrix} m_g,\ 2\sigma - m_g \\ 2\sigma \end{matrix}\right] P_{\mathcal{B}^\sharp \setminus g}(\mathbf{m}|_{\mathcal{E}}; 2\sigma - m_g, m_g)$. $\mathcal{B}^\sharp \setminus g = (\mathcal{B}^*, \theta^*)$, whose letters are $\Psi$ and $K$ (Proposition 7.2), so its period at $(2\sigma - m_g, m_g)$ is $P_{\mathcal{B}}(\mathbf{m}|_{\mathcal{E}}; m_g, 2\sigma - m_g)$. At $c = 1$, $H_G = \{0\}$ and the seam vector of the universal piece is $(1, 1)$; that of the lift partner, $\big(L_0/(L_1\gamma_{\mathcal{E}}),\ L_1/L_1\big)$ with $\gamma_{\mathcal{E}} = L_0/L_1$ (Definition 28.5, read with the reversed excesses), is $(1, 1)$; and the two pins fix $a_0 = 2\sigma - m_g$ and $a_1 = m_g$. □

**Corollary 30.14 (Width two).** *As stated.*

*Proof.* (i) Theorem 30.12(iv) and (v) at $c = 2$: the pins give $a_1 = m_1 + m_2 - 2a_2$ and $a_0 = 2\sigma - m_1 - m_2 + a_2$, solved for $(a_0, a_1)$ by a matrix of determinant one, and the kernel times the universal factor is $\frac{\Gamma(a_0)\Gamma(a_1)\Gamma(a_2)}{\Gamma(2\sigma)} \cdot \frac{\Gamma(m_1 - a_2)\Gamma(m_2 - a_2)}{\Gamma(a_1)}$. (ii) Theorem 30.12(i) and (ii) at $c = 2$: $H_G = \{(w, -w)\}$ with $d\lambda_G = dw$, $v = e^w$, $\mathbf{e}(e^w) = (1, v + v^{-1}, 1)$ and $2e^{(m_1 - m_2)w} dw = 2v^{m_1 - m_2 - 1} dv$. (iii) $K^\sharp = A + \alpha_{g_2} B$ with $A = L_0 + \alpha_{g_1} L_1$ and $B = L_1 + \alpha_{g_1} L_2$, and $\int_0^\infty \rho^{\nu - 1}(A + B\rho)^{-2\sigma} d\rho = \Gamma_{\substack{2\\1}}\!\left[\begin{matrix} \nu,\ 2\sigma - \nu \\ 2\sigma \end{matrix}\right] A^{\nu - 2\sigma} B^{-\nu}$ for $0 < \nu < 2\sigma$ (Proposition 15.1) at $\nu = m_2$. With $\alpha_{g_1} = (L_0/L_1)x$, $A = L_0(1 + x)$ and $B = L_1(1 + z_1^{-1}x)$, which gives the display by Tonelli; the $x$-integral is $z_1^{m_2} \int_0^\infty x^{m_1 - 1}(x + 1)^{-(2\sigma - m_2)}(x + z_1)^{-m_2} dx$, the scale integral of Proposition 15.1 with the coefficients $1, z_1$, the letter exponents $2\sigma - m_2, m_2$ and $\kappa = m_1$, finite exactly for $0 < m_1 < 2\sigma$. At $m_1 = m_2 = 1$, $2\sigma = 2$, $\int_0^\infty \frac{dx}{(1+x)(1+x/z_1)} = \frac{z_1 \log z_1}{z_1 - 1}$ for $z_1 \ne 1$ and 1 for $z_1 = 1$, and $L_1^{-2} z_1 \log z_1/(z_1 - 1) = \log z_1/(L_1^2 - L_0 L_2)$. At integer data and $z_1 \ne 1$ the integrand is a rational function of $x$ with poles at $-1$ and $-z_1$ only and of order at most $x^{-2}$ at infinity, $m_1 < 2\sigma$; its partial fractions give an antiderivative $R(x) + A' \log(1 + x) + B' \log(1 + x/z_1)$ with $R$ rational and $A' + B' = 0$, all coefficients rational in $z_1$, whose increment over $(0, \infty)$ is $R(\infty) - R(0) + A' \log z_1$ (receipt `t10b_weight_exact.py` for $m_1 = m_2 = k$, $2\sigma = 2k$, $k \le 4$). □

**Lemma 30.15 (Products of lifts).** *As stated.*

*Proof.* Let $A$ be a basis of $\mathcal{L}_i$. Since $\mathcal{L}_i$ is a quotient of $\mathcal{L}_j$ (Proposition 30.2(iii)), $A$ is independent in $\mathcal{L}_j$ (Proposition 30.2(i)) and lies in a basis $C$ of $\mathcal{L}_j$. The monomial $\alpha^{\mathbf{1}_A+\mathbf{1}_C} = \alpha^{2\mathbf{1}_A+\mathbf{1}_{C\setminus A}}$ occurs in $K_{\mathcal{L}_i}K_{\mathcal{L}_j}$ with a positive coefficient, every coefficient being nonnegative, and exactly $|A| = r_{\mathcal{N}}+i$ of its exponents equal 2. A monomial $\alpha^{\mathbf{1}_{A'}+\mathbf{1}_{C'}}$ of $K_{\mathcal{L}_{i'}}K_{\mathcal{L}_{j'}}$ has the exponent 2 exactly on $A'\cap C'$, at most $|A'| = r_{\mathcal{N}}+i' < r_{\mathcal{N}}+i$ marks. Both products are nonzero with nonnegative coefficients, so a positive constant multiple of one has the support of the one, and they are not proportional. The ratio is the case $(i',i,j,j') = (j-1,j,j,j+1)$, $0<j<c$; a constant ratio would make $K_{\mathcal{L}_j}^2 - \lambda K_{\mathcal{L}_{j-1}}K_{\mathcal{L}_{j+1}}$ vanish on the orthant, hence identically. □

**Theorem 30.16 (The price of width).** *As stated.*

*Proof.* (i) By Theorem 30.12(i) the universal law is the image of a measure with a positive continuous density on $H_G$, of dimension $c-1$, under the continuous map $w \mapsto \mathbf{e}(e^w)$. At $c=1$ it is the unit point mass at $(1,1)$. At $c\ge 2$, $e_1(e^w) = \sum_g e^{w_g}$ is $c$ at $w=0$ and larger at every other point of $H_G$ by the inequality of the arithmetic and geometric means, $\prod_g e^{w_g} = 1$; a law carried by one point would make it constant almost everywhere, hence everywhere by continuity. The law of the lift partner is the image of its scale form, which has a positive continuous density on the open simplex $\Delta_{\mathcal{E}}$, under its continuous seam vector; it is carried by one point exactly when the seam vector is constant. Read with the reversed excesses, $V^{(0)} = L_c$, $V^{(c)} = L_0$, the scale is $\gamma_{\mathcal{E}} = (L_0/L_c)^{1/c}$ and the seam vector at $D_l$ is $L_l/(L_c\gamma_{\mathcal{E}}^{c-l}) = L_l L_0^{-(c-l)/c} L_c^{-l/c}$ (Definition 28.5). Put $f_l = \log L_l - \log L_0 - \frac{l}{c}(\log L_c - \log L_0)$, so that $f_0 = f_c = 0$ and the second differences of $(f_l)_l$ are those of $(\log L_l)_l$, which are $-\log z_{c-l}$, since $L_l = K_{\mathcal{L}_{c-l}}$. The seam vector is constant exactly when every $f_l$ is constant; then every lift ratio is constant, and conversely constant second differences with $f_0 = f_c = 0$ determine $f_l$ as constants. At $c=1$ there is no lift ratio and the seam vector is $(1,1)$; at $c\ge 2$ Lemma 30.15 excludes constant lift ratios.

(ii) For $a\in\mathbb{C}^c$ and a set $S$ of $l$ indices, $\pi(S)$ runs over the sets of $l$ indices as $\pi$ runs over $S_c$, each reached $l!\,(c-l)!$ times, so

$$\sum_{\pi\in S_c}\prod_{i=1}^{c}\bigl(1+a_i\eta_{\pi(i)}\bigr) = \sum_S a^S \sum_\pi \prod_{i\in S}\eta_{\pi(i)} = \sum_{l=0}^{c} l!\,(c-l)!\,e_l(a)\,e_l(\boldsymbol{\eta}),$$

and the left side is $\operatorname{per}(\mathbf{1}\mathbf{1}^{\mathsf{T}}+a\boldsymbol{\eta}^{\mathsf{T}})$ by the definition of the permanent. With $L_l = L_0 e_l(\boldsymbol{\eta})/\binom{c}{l}$, read off $\sum_l \binom{c}{l}L_l x^l = L_0\prod_i(1+\eta_i x)$, division by $c!$ gives $\sum_l e_l(\alpha_G)L_l = K^\sharp$. The Skeleton polynomial of $G$ at $(\mathbf{1}_G,\alpha_{\mathcal{E}})$ is $K^\sharp(x\mathbf{1}_G,\alpha_{\mathcal{E}})/K^\sharp(\mathbf{1}_G,\alpha_{\mathcal{E}}) = \sum_l\binom{c}{l}L_l x^l/K^\sharp$, $d(G)=0$, and with $L_0>0$ its roots $-\xi_i$ give $\sum_l\binom{c}{l}L_l x^l = L_0\prod_i(1+x/\xi_i)$: $\eta_i = 1/\xi_i$. At a gauge all $\eta_i$ equal $\eta$ exactly when $\sum_l\binom{c}{l}L_l x^l = L_0(1+\eta x)^c$, that is $L_l = L_0\eta^l$ for every $l$, which holds exactly when $L_l^2 = L_{l-1}L_{l+1}$ for $0<l<c$ (then $L_l = L_0(L_1/L_0)^l$ by induction), that is every lift ratio is 1. Coincidence on an open set would make $K_{\mathcal{L}_j}^2 - K_{\mathcal{L}_{j-1}}K_{\mathcal{L}_{j+1}}$ vanish there and so identically, against Lemma 30.15; and $K^\sharp = L_0\prod_g(1+\alpha_g\eta) = L_0\sum_l e_l(\alpha_G)\eta^l$ would give $L_l = L_0\eta^l$ everywhere.

(iii) By Proposition 30.8(ii) the partial fusion $\mathcal{A}$ at $G'$ is the level cell of $(\mathcal{L}_i,\mathcal{B})$ on $\mathcal{E}\sqcup G'$, in which $g$ lies in some bases and avoids others, so it is a cell with the hinge $g$; by Proposition 30.2(iv) its hinge minors are $\mathcal{A}/g$, the level cell of $(\mathcal{L}_{i-1},\mathcal{B})$, and $\mathcal{A}\setminus g$, the level cell of $(\mathcal{L}_i,\mathcal{L}_1)$, both on $\mathcal{E}\sqcup(G'-g)$. At $i=c$, $\mathcal{A} = \mathcal{B}^\sharp$ is identically self-dual, $\mathcal{A}/g$ is the dual of $\mathcal{A}\setminus g$ with the same weights (as in the proof of Corollary 30.7(i)), so $K_{\mathcal{A}/g} = \Psi_{\mathcal{A}\setminus g}$ and $\Psi_{\mathcal{A}/g} = K_{\mathcal{A}\setminus g}$ (Proposition 7.2) and the ratio is 1; and Corollary 20.19 at $g$, with $P_{\mathcal{A}\setminus g}$ at $(2\sigma-m_g,m_g)$ equal to $P_{\mathcal{A}/g}$ at $(m_g,2\sigma-m_g)$, gives the display. Let $i<c$. The letters of a level cell of $(\mathcal{M}',\mathcal{N}')$ on $\mathcal{E}\sqcup G''$ are $K = \sum_Q \alpha^Q K_{\mathcal{L}'_{c''-|Q|}}$ and $\Psi = \sum_Q \alpha^{G''\setminus Q}\Psi_{\mathcal{L}'_{c''-|Q|}}$, which at $\alpha_{G''}=0$ are $K_{\mathcal{M}'}$ and $\Psi_{\mathcal{N}'}$. So at $\alpha_{G'-g}=0$ the even

letters of the hinge minors are

$$\mathbf{E}_{\mathcal{A}/g} = K_{\mathcal{L}_{i-1}}\Psi_{\mathcal{B}} = K_{\mathcal{L}_{i-1}}K_{\mathcal{L}_c}, \qquad \mathbf{E}_{\mathcal{A}\setminus g} = K_{\mathcal{L}_i}\Psi_{\mathcal{L}_1} = K_{\mathcal{L}_i}K_{\mathcal{L}_{c-1}}.$$

A constant ratio $\varrho_{\mathcal{A}} = \mathbf{E}_{\mathcal{A}/g}/\mathbf{E}_{\mathcal{A}\setminus g}$ on the orthant would make $\mathbf{E}_{\mathcal{A}/g} - \lambda\mathbf{E}_{\mathcal{A}\setminus g}$ vanish identically, also at $\alpha_{G'-g} = 0$, against Lemma 30.15 with $(i', i, j, j') = (i-1, i, c-1, c)$. The law of the cell is the image of its base form, with a positive continuous density on the open simplex, under the continuous ratio, and it is carried by one point only if the ratio is constant (Corollary 23.12(i)). □

**Proposition 30.17 (The free extensions by one mark).** *As stated.*

*Proof.* Write $K_\backslash, \Psi_\backslash$ and $K_/, \Psi_/$ for the letters of $\mathcal{A}\setminus g$ and $\mathcal{A}/g$, of the degrees $r_{\mathcal{A}}$, $|\mathcal{E}| - r_{\mathcal{A}}$, $r_{\mathcal{A}} - 1$ and $|\mathcal{E}| - r_{\mathcal{A}} + 1$; $\mathcal{A}$ being plain, each hinge minor carries one system of weights in its two letters. If $\mathcal{A}/g$ is the dual of $\mathcal{A}\setminus g$ with its weights multiplied by $\mu$, then $K_/ = \mu\Psi_\backslash$ and $\Psi_/ = \mu K_\backslash$ (Proposition 7.2), and $\varrho_{\mathcal{A}} = \mathbf{E}_{\mathcal{A}/g}/\mathbf{E}_{\mathcal{A}\setminus g} = \mu^2$. Conversely let $\varrho_{\mathcal{A}}$ be a constant $\lambda$. Connected tables on at least two marks are single molecules (Proposition 28.3(i)), so their letters are irreducible (Theorem 7.11), and $K_/\Psi_/ = \lambda K_\backslash\Psi_\backslash$ matches the irreducible factors on the two sides up to constant factors. $K_/$ and $K_\backslash$ have different degrees, so $K_/ = \mu\Psi_\backslash$ with a constant $\mu$, positive since both have positive coefficients; comparing monomials, the bases of $\mathcal{A}/g$ are the complements in $\mathcal{E}$ of the bases $T$ of $\mathcal{A}\setminus g$, with $\theta_{(\mathcal{E}\setminus T)+g} = \mu\theta_T$. The degrees give $r_{\mathcal{A}} - 1 = |\mathcal{E}| - r_{\mathcal{A}}$, so $\mathcal{A}/g$ has rank $r_{\mathcal{A}} - 1$ on $2r_{\mathcal{A}} - 1$ marks, the fusion width one; it is fusable by Theorem 30.6(d), $\mathcal{A}$ having $\mathcal{A}/g$ and $\mathcal{A}\setminus g$, with the bases of $(\mathcal{A}/g)^*$, as its children at the full and at the empty pattern of $\{g\}$. By Theorem 30.5(v) its fusion has the letter $K^\sharp = \Psi_/ + \alpha_g K_/ = \mu K_\backslash + \alpha_g K_/$, and $K^\sharp(\alpha_{\mathcal{E}}, \mu\alpha_g) = \mu K_\backslash + \mu\alpha_g K_/ = \mu K_{\mathcal{A}}(\alpha_{\mathcal{E}}, \alpha_g)$: $\mathcal{A}$ has the bases of that fusion, the weights $\theta_{T'+g}$ on the bases $T' + g$ and $\theta_{T'+g}/\mu$ on the bases $\mathcal{E}\setminus T'$, $T'$ a basis of $\mathcal{A}/g$. For a list every coefficient is 1, so $\mu = 1$: the complement in $\mathcal{E}\sqcup\{g\}$ of a basis $T$ avoiding $g$ is the basis $(\mathcal{E}\setminus T) + g$ and conversely, $\mathcal{A}$ is identically self-dual, $\mathcal{A}\setminus g = (\mathcal{A}/g)^*$, and $\mathcal{A}$ is the only major of $\mathcal{A}/g$, its fusion; an identically self-dual $\mathcal{A}$ has the ratio 1, its hinge minors being dual with the same weights. At exponents $s, t > 0$ a ratio is constant exactly when the law is carried by one point, the base form having a positive continuous density on the open simplex (Corollary 23.12(i)). □

**Corollary 30.18 (A pinned set of one list is a dual pair).** *As stated.*

*Proof.* By Theorem 28.25, $S$ is pinned exactly when it has width one and $\varrho_S = 1$. The cell of the bunch $\mathcal{B}_S^+$ is a list with the hinge minors $\mathcal{B}|S$ and $\mathcal{B}/S'$ (19.4) and the ratio $\varrho_S$ (21.1). Its hinge minors being connected, its ratio is constant exactly when it is identically self-dual (Proposition 30.17), that is a dual pair (Proposition 21.8(i)), and then the ratio is 1 (Proposition 21.8(ii)). So $\varrho_S = 1$ exactly when $\mathcal{B}_S^+$ is a dual pair, that is $\mathcal{B}/S' = (\mathcal{B}|S)^*$. □

**Proposition 30.20 (Cousins).** *As stated.*

*Proof.* (i) is Corollary 30.7(i) with $C = \{x\}$, and its proof gives $\mathcal{B}^\sharp\setminus x = (\mathcal{B}^\sharp/x)^*$; for $x \in G$, $\mathcal{B}^\sharp/x$ is the partial fusion at $G - x$, and at $c = 1$, $\mathcal{B}^\sharp/g = \mathcal{B}$. (ii) Corollary 20.19 at $x$, which allows every pair of exponents with $s + t = 2\sigma$, gives $P_{\mathcal{B}^\sharp} = \Gamma_1^2\begin{bmatrix} m_x, & 2\sigma - m_x \\ & 2\sigma \end{bmatrix} P_{\mathcal{B}^\sharp\setminus x}(\mathbf{m}_{-x}; 2\sigma - m_x, m_x)$, and the dual $\mathcal{B}^\sharp\setminus x$ of $\mathcal{B}^\sharp/x$ has the letters exchanged (Proposition 7.2); the same holds at $y$. (iii) $K^\sharp(\beta^*\alpha) = K^\sharp(\alpha)$, and the change of coordinates $u \mapsto \beta^* u$ permutes the coordinates of $\Delta_{\mathcal{E}\sqcup G}$ and keeps $d\omega$, so the period at $\beta^*\mathbf{m}$ is the period at $\mathbf{m}$. At $c = 1$ Corollary 30.13 writes $P_{\mathcal{B}^\sharp}$ on its balance wall, where every proper wall is positive, as the stated function of $(\mathbf{m}|_{\mathcal{E}}, s')$, $s' = m_g$; an automorphism permutes the proper walls, $\kappa_U(\beta^*\mathbf{m}) = \kappa_{\beta(U)}(\mathbf{m})$, so it keeps that part of the

hyperplane, and the function inherits the invariance; at $\beta^*\mathbf{m}$ the mass of $g$ is $m_{\beta(g)}$ and the mass of $\beta^{-1}(g)$ is $m_g = s'$. □

**Theorem 30.23 (Fusion across a two-sum).** *As stated.*

*Proof.* (i) $\mathcal{S}$ has the rank $r_1 + r_2 - 1$ on $n_1 + n_2 - 2$ marks (Lemma 21.12), so the fusion width $c_1 + c_2$. By Lemma 21.2 its bases are the $X \sqcup Y$ with $X$ a basis of $\mathcal{A}_1 \setminus g$ and $Y$ of $\mathcal{A}_2/g$, or $X$ of $\mathcal{A}_1/g$ and $Y$ of $\mathcal{A}_2 \setminus g$; since independent sets of a table extend to bases inside any set that contains them and has full rank, a set $I_1 \sqcup I_2$ is independent exactly when $I_1$ and $I_2 + g$ are independent in $\mathcal{A}_1$ and $\mathcal{A}_2$, or $I_1 + g$ and $I_2$ are. A dependent set inside $\mathcal{E}_1$ is dependent in $\mathcal{A}_1$, $\{g\}$ being independent in $\mathcal{A}_2$, so the circuits inside $\mathcal{E}_1$ are the circuits of $\mathcal{A}_1$ avoiding $g$, and likewise for $\mathcal{E}_2$. A circuit $C_1' \sqcup C_2'$ meeting both sides contains no such circuit, so $C_1'$ and $C_2'$ are independent; it is dependent exactly when $C_1' + g$ and $C_2' + g$ are dependent, that is when $g$ lies in the closures of $C_1'$ and of $C_2'$; and removing $e \in C_1'$ leaves an independent set exactly when $C_1' - e + g$ is independent, $C_2' + g$ being dependent. So the minimal ones are those with $C_1' + g$ and $C_2' + g$ circuits, and every such set is a circuit. The complement of $X \sqcup Y$ is $(\mathcal{E}_1 \setminus X) \sqcup (\mathcal{E}_2 \setminus Y)$, and $(\mathcal{A}_1 \setminus g)^* = \mathcal{A}_1^*/g$, $(\mathcal{A}_2/g)^* = \mathcal{A}_2^* \setminus g$, so $\mathcal{S}^* = \mathcal{A}_1^* \oplus_g \mathcal{A}_2^*$, $g$ being neither a loop nor a coloop of $\mathcal{A}_i^*$; its circuits, the cocircuits of $\mathcal{S}$, are read off the same description.

(ii) A cocircuit $D$ of $\mathcal{A}_1$ avoiding $g$ contains no circuit of $\mathcal{A}_2$ and no set $(C_1 \setminus g) \cup (C_2 \setminus g)$, $C_2 \setminus g$ being a nonempty part of $\mathcal{E}_2$; so it is a union of circuits of $\mathcal{S}$ exactly when it is one of $\mathcal{A}_1$, and these cocircuits answer $a_1$; likewise $a_2$. For $D = (D_1 \setminus g) \cup (D_2 \setminus g)$ and $e \in D_1 \setminus g$, the circuits of $\mathcal{S}$ inside $D$ through $e$ are the circuits of $\mathcal{A}_1$ inside $D_1$ avoiding $g$ through $e$ and the sets $(C_1 \setminus g) \cup (C_2 \setminus g)$ with $e \in C_1 \subseteq D_1$ and $C_2 \subseteq D_2$ circuits through $g$. So $e$ is covered in $D$ exactly when $D_1$ covers it by a circuit avoiding $g$, or by one through $g$ while $D_2$ covers $g$. $D_1$ and $D_2$ run independently over the cocircuits through $g$. If $h_2$ holds, the marks of the side $\mathcal{E}_1$ are covered for every $D_2$ exactly when $u_1$; if $h_2$ fails, some $D_2$ does not cover $g$, and they are covered exactly when $s_1$, which implies $u_1$. So the side $\mathcal{E}_1$ asks $u_1$ and ($s_1$ or $h_2$), the side $\mathcal{E}_2$ asks $u_2$ and ($s_2$ or $h_1$), and Theorem 30.6(b) gives the rule. For one cell, its cocircuits avoiding $g$ answer $a$, and those through $g$ are unions of circuits exactly when their marks other than $g$ are covered, $u$, and $g$ is, $h$. Two fusable cells have $a_i, u_i, h_i$, and the rule holds.

(iii) By the bases of (i), $r_{\mathcal{S}}(\mathcal{E}_2) = r_2$ and $d_{\mathcal{S}}(\mathcal{E}_2) = r_2 - 1$, so $r^*_{\mathcal{S}}(\mathcal{E}_2) = |\mathcal{E}_2| - d_{\mathcal{S}}(\mathcal{E}_2) = n_2 - r_2$. A fusable $\mathcal{S}$ has $r_{\mathcal{S}} \le r^*_{\mathcal{S}}$ pointwise, $\mathcal{S}$ being a quotient of $\mathcal{S}^*$ (Proposition 30.2(i)): $r_2 \le n_2 - r_2$, $c_2 \ge 0$, and likewise $c_1 \ge 0$. In $\mathcal{S}^\sharp$, $r^\sharp(\mathcal{E}_2) = r^*_{\mathcal{S}}(\mathcal{E}_2) = n_2 - r_2$, the rank of a set avoiding $G$ being its rank in $\mathcal{S}^\sharp \setminus G = \mathcal{S}^*$. An identically self-dual table has $r^*(A) = r(A)$, so $d(A) = |A| - r(A)$ and $c(A) = 2r(A) - |A|$, which for $A = \mathcal{E}_2$ is $2(n_2 - r_2) - (n_2 - 1) = c_2 + 1$; likewise for $\mathcal{E}_1$.

(iv) In $\mathcal{A}_i^\sharp$ the mark $g$ lies in a basis $X \sqcup G_i$, $X$ a basis of $\mathcal{A}_i$ through $g$, and avoids a basis $X$ of $\mathcal{A}_i^*$, so $\mathcal{A}_i^\sharp$ is a cell with the hinge $g$ and $W = \mathcal{A}_1^\sharp \oplus_g \mathcal{A}_2^\sharp$ is a table on $\mathcal{E}_1 \sqcup \mathcal{E}_2 \sqcup G_1 \sqcup G_2$. By (i) $W^* = \mathcal{A}_1^{\sharp *} \oplus_g \mathcal{A}_2^{\sharp *} = W$. Contracting $G = G_1 \sqcup G_2$ commutes with the minors at $g$, marks of different sets (Definition 7.5), so the bases of $W$ containing $G$, less $G$, are those of $(\mathcal{A}_1^\sharp/G_1) \oplus_g (\mathcal{A}_2^\sharp/G_2) = \mathcal{S}$, and likewise $W \setminus G = \mathcal{A}_1^* \oplus_g \mathcal{A}_2^* = \mathcal{S}^*$: $W$ is an identically self-dual major. If $c_2 = 0$, $\mathcal{A}_2$ is a quotient of $\mathcal{A}_2^*$ of the same rank, so the two rank functions agree (Proposition 30.2(i) at $X \subseteq Y$ and at $Y \subseteq \mathcal{E}$) and $\mathcal{A}_2 = \mathcal{A}_2^* = \mathcal{A}_2^\sharp$; every transposition of two marks of $G_1$ carries $\mathcal{A}_1^\sharp$, hence $W$, and $W = \mathcal{S}^\sharp$ by Theorem 30.5(iv); likewise for $c_1 = 0$. Let $c_1, c_2 \ge 1$, $x_1 \in G_1$, $x_2 \in G_2$, and let $r_1^* = n_1 - r_1$, the rank of $\mathcal{A}_1^\sharp$. A basis $Z_1 \sqcup Z_2$ of $W$ has $|Z_1| \in \{r_1^*, r_1^* - 1\}$. At the pattern $\{x_1\}$ of $G$, $|Z_1 \cap \mathcal{E}_1| = |Z_1| - 1 \le r_1^* - 1$, with equality for $Z_1 = X_1 \sqcup \{x_1\}$, $X_1$ a basis of $\mathcal{L}_{c_1-1}(\mathcal{A}_1^*, \mathcal{A}_1)$ avoiding $g$, and $Z_2 = X_2 - g$, $X_2$ a basis of $\mathcal{A}_2^*$ through $g$; such bases exist, $g$ being independent in $\mathcal{A}_1$ and so in its lift $\mathcal{L}_1 = \mathcal{L}^*_{c_1-1}$, and not a loop of $\mathcal{A}_2^*$. At the pattern $\{x_2\}$, $Z_1$ a basis of $\mathcal{A}_1^*$

avoiding $g$ and $Z_2 = (X_2 \sqcup \{x_2\}) - g$, $X_2$ a basis of $\mathcal{L}_{c_2-1}(\mathcal{A}_2^*, \mathcal{A}_2)$ through $g$, give $|Z_1 \cap \mathcal{E}_1| = r_1^*$. So the two children have the ranks $r_1^* - 1$ and $r_1^*$ on $\mathcal{E}_1$, they differ, and $G$ is not in level form in $W$, while it is in $\mathcal{S}^\sharp$.

(v) If both cells are fusable, $\mathcal{S}$ is fusable by (ii), and for $c_2 = 0$ the fusable list $\mathcal{A}_2$ of the width zero equals its dual, as in (iv), so it is identically self-dual, and $\mathcal{S}^\sharp = \mathcal{A}_1^\sharp \oplus_g \mathcal{A}_2$ by (iv). Conversely let $\mathcal{S}$ be fusable; by (iii) $\{c_1, c_2\} = \{0, 1\}$; let $c_2 = 0$. By (iii) $\mathcal{E}_2$ is a bunch of width one of $\mathcal{S}^\sharp$, a list, and by Theorem 19.4 and Proposition 21.3(iii) $\mathcal{S}^\sharp$ is the two-sum of the cell $\mathcal{N}_2$ of the bunch $\mathcal{E}_2$ and the cell $\mathcal{N}_1$ of its room $\mathcal{E}_1 \sqcup G$, along a hinge $h$. The cell of $\mathcal{E}_2$ in the dual table has the bases $\mathcal{S}^{\sharp *}|\mathcal{E}_2 = (\mathcal{S}^\sharp/(\mathcal{E}_1 \sqcup G))^*$ and $\mathcal{S}^{\sharp *}/(\mathcal{E}_1 \sqcup G) + h = (\mathcal{S}^\sharp|\mathcal{E}_2)^* + h$, the complements in $\mathcal{E}_2 + h$ of the bases of $\mathcal{N}_2$; $\mathcal{S}^\sharp$ being identically self-dual, $\mathcal{N}_2$ is. In $\mathcal{S} = \mathcal{S}^\sharp/G$ the cell of $\mathcal{E}_2$ is $\mathcal{A}_2$ (Lemma 21.2); its bases through the hinge, $\mathcal{S}/\mathcal{E}_1 = \mathcal{S}^\sharp/(\mathcal{E}_1 \sqcup G)$, are those of $\mathcal{N}_2$, and its bases avoiding the hinge are those of $(\mathcal{S}^\sharp/G)|\mathcal{E}_2$. Here $r^\sharp(\mathcal{E}_2) = n_2 - r_2 = r_2$ by (iii) and $c_2 = 0$, while $r^\sharp(\mathcal{E}_2 \sqcup G) = r_{\mathcal{S}}(\mathcal{E}_2) + |G| = r_2 + |G|$; by submodularity every independent $I \subseteq \mathcal{E}_2$ of $\mathcal{S}^\sharp$ has $r^\sharp(I \sqcup G) \geq |I| + r^\sharp(\mathcal{E}_2 \sqcup G) - r^\sharp(\mathcal{E}_2) = |I| + |G|$, so $(\mathcal{S}^\sharp/G)|\mathcal{E}_2 = \mathcal{S}^\sharp|\mathcal{E}_2$ and $\mathcal{A}_2 = \mathcal{N}_2$, identically self-dual. The cell of the room $\mathcal{E}_1$ in $\mathcal{S}$ has the bases of $\mathcal{S}|\mathcal{E}_1 = (\mathcal{S}^\sharp|(\mathcal{E}_1 \sqcup G))/G$ and of $\mathcal{S}/\mathcal{E}_2 + h = (\mathcal{S}^\sharp/\mathcal{E}_2)/G + h$, so $\mathcal{A}_1 = \mathcal{N}_1/G$; $\mathcal{N}_1$ is identically self-dual by the same argument as $\mathcal{N}_2$, so $\mathcal{N}_1 \setminus G = (\mathcal{N}_1/G)^*$ and $\mathcal{N}_1$ is a major of $\mathcal{A}_1$: $\mathcal{A}_1$ is fusable (Theorem 30.6), of the width $|G| = 1$, and $\mathcal{N}_1 = \mathcal{A}_1^\sharp$ (Theorem 30.5(v)). So $\mathcal{S}^\sharp = \mathcal{A}_1^\sharp \oplus_g \mathcal{A}_2$, as (iv) also gives: $\mathcal{S}^\sharp$ is the two-sum along the hinge $g$ of $\mathcal{S}$, with $\mathcal{E}_2$ a bunch of width one. □

**Corollary 30.24 (A two-sum with an identically self-dual cell).** *As stated.*

*Proof.* Every proper wall of $\mathcal{S}$ positive makes $P_{\mathcal{S}}$ finite (Theorem 8.7); $\kappa$ and $2\sigma - \kappa$ are the walls of $\mathcal{S}$ at $\mathcal{E}_1$ and $\mathcal{E}_2$ (Lemma 21.12), so $0 < \kappa < 2\sigma$. $\mathcal{A}_1$ is identically self-dual, so $\mathcal{A}_1/g$ is the dual of $\mathcal{A}_1 \setminus g$ with the same weights, its ratio is 1, and $\phi_{\mathcal{A}_1}(\Delta_{\mathcal{E}_1}) = P_{\mathcal{A}_1 \setminus g}(\mathbf{m}|_{\mathcal{E}_1}; \kappa, 2\sigma - \kappa)$ (Corollary 23.12(ii)), finite since $\mathcal{A}_1$ sits in a gluing of finite period. By Corollary 23.12(i) $\mathcal{A}_1$ acts in $\mathcal{S}$ as this number times the one-mark cell of ratio 1 and mass $\kappa$, whose two-sum with $\mathcal{A}_2$ is $\mathcal{A}_2$ with its hinge carrying $\kappa$: $P_{\mathcal{S}} = \phi_{\mathcal{A}_1}(\Delta_{\mathcal{E}_1})\, P_{\mathcal{A}_2}(\mathbf{m}|_{\mathcal{E}_2}, \kappa; s, t)$. The Beta corner of Corollary 23.12(iii) at $g$ with the mass $2\sigma - \kappa$, which puts $\mathcal{A}_1$ on its balance wall (Lemma 21.12), is the integral over the scale of $g$, valid in $[0, \infty]$ for $0 < \kappa < 2\sigma$ by Tonelli: $P_{\mathcal{A}_1}(\mathbf{m}|_{\mathcal{E}_1}, 2\sigma - \kappa; s, t) = \Gamma_{\begin{smallmatrix} 2 \\ 1 \end{smallmatrix}} \begin{bmatrix} 2\sigma - \kappa, & \kappa \\ & 2\sigma \end{bmatrix} P_{\mathcal{A}_1 \setminus g}(\mathbf{m}|_{\mathcal{E}_1}; \kappa, 2\sigma - \kappa)$. When the cells are lists and $\mathcal{A}_2$ is fusable of the fusion width one, $\mathcal{A}_1$, identically self-dual, is fusable of the width zero, and Theorem 30.23(v), with the roles of the two cells exchanged, gives $\mathcal{S}^\sharp = \mathcal{A}_1 \oplus_g \mathcal{A}_2^\sharp$. □

**Proposition 30.25 (Uncovered marks and repair).** *As stated.*

*Proof.* (i) A cocircuit is a union of circuits exactly when it covers each of its marks, so $\mathcal{B}$ has no uncovered mark exactly when (b) of Theorem 30.6 holds, equivalently (a). By that theorem a list, or a table of the width $c \geq 1$, is fusable exactly when (a) holds; at $c < 0$ neither (a) holds nor is the table fusable; and at $c = 0$ the table is fusable exactly when it is identically self-dual (Definition 30.4). (ii) By Theorem 21.6(i) and (ii), under the first rule alone $\mathcal{B}_S$ has $n + |S|$ marks, the rank $r$ and the rank function $Y \mapsto r(\lceil Y \rceil)$, $\lceil Y \rceil$ the marks of $\mathcal{B}$ whose blocks meet $Y$. So $Y$ is independent exactly when it meets each block in at most one mark and $\lceil Y \rceil$ is independent: the circuits of $\mathcal{B}_S$ are the pairs $\{h, h'\}$, $h \in S$, and the sets meeting each block in at most one mark whose shadow $\lceil \cdot \rceil$ is a circuit of $\mathcal{B}$. A closed set contains with a mark its whole block, adding a mark of a block already met not raising the rank; so the closed sets are the unions of the blocks over the closed sets of $\mathcal{B}$, and since the cocircuits are the complements of the closed sets of rank $r - 1$ (fact $(\beta)$ in the proof of Theorem 30.6) the cocircuits of $\mathcal{B}_S$ are $D^+ = D \cup \{h' : h \in D \cap S\}$,

$D$ a cocircuit of $\mathcal{B}$. In $D^+$ the marks $h, h'$ of a block are covered by $\{h, h'\}$, and a mark $e \in D \setminus S$ is covered exactly when a circuit of $\mathcal{B}_S$ through $e$ lies in $D^+$, that is when a circuit of $\mathcal{B}$ through $e$, its shadow, lies in $D$. So $\mathcal{B}_S$ has no uncovered mark exactly when every mark of $\mathcal{E} \setminus S$ is covered in every cocircuit of $\mathcal{B}$ through it: when $S$ contains the uncovered marks; and by (i) for $\mathcal{B}_S$, a list or a table of the width $c + |S| \neq 0$, it is then fusable exactly when it has no uncovered mark. Its fusion width is $n + |S| - 2r$. The bases of $\mathcal{B}_S$ avoiding $S'$ are the bases $T$ of $\mathcal{B}$, each with its weight $\theta_T$ (Theorem 21.5), so $(\mathcal{B}_S, \theta) \setminus S' = (\mathcal{B}, \theta)$. (iii) $(\mathcal{B}_S)^\sharp/G = (\mathcal{B}_S, \theta)$ and $(\mathcal{B}_S, \theta) \setminus S' = (\mathcal{B}, \theta)$, the two minors being taken at disjoint sets. Let $c \geq 0$, $(\mathcal{B}, \theta)$ not fusable, and $U$ its set of uncovered marks, none a loop, since a loop lies in no cocircuit. If $U \neq \varnothing$, $S = U$ gives the width $c + |U| \geq 1$ and no uncovered mark, and $\mathcal{B}_U$ is fusable by (ii). If $U = \varnothing$, (i) leaves $c = 0$ and $(\mathcal{B}, \theta)$ not identically self-dual, while $\mathcal{B} = \mathcal{B}^*$ by (a) and the proof of Definition 30.4; a loop of $\mathcal{B}$ would be a coloop of $\mathcal{B}^* = \mathcal{B}$, so there is none, and $\mathcal{E} \neq \varnothing$, since on no marks the one basis $\varnothing$ is its own complement with its own weight. For any mark $h$, $S = \{h\}$ gives the width one and no uncovered mark, and $\mathcal{B}_S$ is fusable by (ii). For a list $\mathcal{B}$, the set of the marks that are not loops contains $U$, and the list $\mathcal{B}_S$ it doubles has no uncovered mark and is fusable by (i). (iv) The copy $h'$ is neither a loop nor a coloop, so its wall at $t = 0$ is $\kappa_{\{h'\}} = m_{h'}$. Theorem 16.25 at the bunch $\{h'\}$, whose bottom level holds the bases avoiding $h'$, with the empty pattern alone, so that the weights factor on it, gives the residue as the product of the periods of the faces: on $\{h'\}$ the contraction of the room, of rank $d(\{h'\}) = 0$ and letter 1, whose period on one mark is 1; on the room, $(\mathcal{B}_S|(\mathcal{E}_S \setminus h'); s, 0) = (\mathcal{B}_{S-h}; s, 0)$. □

## 50.31 Chapter 31: closing a glue along a direction

**Theorem 31.2 (Closing the glue along a direction).** *As stated, the largest values of the monomials of the letter vector of the partner being read on its law when its room word has a Newton sum of full dimension.*

*Proof.* Apart from the sentence on the partner, the statement is the reference's, with its proof (the reference, proof of thm:closecone). Let the room word of the partner have a Newton sum of full dimension. The partner is a piece on $R$ with the stations $\mathcal{D}$, read with the reversed excesses $c_j - i(D)$ (28.4); its scale $\gamma_R^\rho$ and its seam vector $\boldsymbol{\varrho}_R^\rho = (\varrho_{R,jD}^\rho, \varrho_{R,jD}^{\rho *})_{j,D}$ are those of 28.5 with its room letters and these excesses. In this proof write $L_{j,R}^{(0)}$ for the sum of the $L_{jD}$ over the stations of excess $c_j$, which have the reversed excess 0, and $L_{j,R}^{*(c)}$ for the sum of the $L_{jD}^*$ over the stations of excess 0, which have the reversed excess $c_j$. By 28.5,

$$L_{jD} = L_{j,R}^{(0)} \, (\gamma_R^\rho)^{c_j - i(D)} \, \varrho_{R,jD}^\rho, \qquad L_{jD}^* = L_{j,R}^{*(c)} \, (\gamma_R^\rho)^{i(D)} \, \varrho_{R,jD}^{\rho *}.$$

A direction $\mathbf{v} \in V$ keeps the letter totals, $\sum_D v_{jD} = 0$ and $\sum_D v_{jD}^* = 0$ for every active letter. It also keeps the pin of 28.12 for every vector of $M$, the pin of $\mathbf{1}$ holding by the letter totals on the balance wall, and $\mathbf{1}_R \in \vec{\mathcal{N}}_R^\perp \subseteq M$, the room letters being homogeneous. The exponents in $\mathbf{n}(\mathbf{a}, \mathbf{b})$ of the monomials of $V_{jD}$ and $V_{jD}^*$ vanish on $R$, and those of the monomials of $L_{jD}$ and $L_{jD}^*$ have the totals $\deg L_{jD} = e_j + c_j - i(D)$ and $\deg L_{jD}^* = e_j^* + i(D)$ on $R$ (28.4); so, by the letter totals,

$$0 = \langle \mathbf{n}(\mathbf{v}), \mathbf{1}_R \rangle = \sum_{j,D} \big( v_{jD} \deg L_{jD} + v_{jD}^* \deg L_{jD}^* \big) = \sum_{j,D} \big( (c_j - i(D)) v_{jD} + i(D) v_{jD}^* \big).$$

So in $\mathbf{V}_R^\mathbf{v} = \prod_{j,D} L_{jD}^{v_{jD}} {L_{jD}^*}^{v_{jD}^*}$ the sums $L_{j,R}^{(0)}$ and $L_{j,R}^{*(c)}$ and the scale $\gamma_R^\rho$ enter with the exponent 0, and $\mathbf{V}_R^\mathbf{v} = (\boldsymbol{\varrho}_R^\rho)^\mathbf{v}$ on the open orthant of $R$, where these sums are positive, both sides being homogeneous of degree zero; hence $\sup \mathbf{V}_R^\mathbf{v} = \sup_{v \in \Delta_R} \boldsymbol{\varrho}_R^\rho(v)^\mathbf{v}$, the supremum over the open simplex.

The scale form $\phi_R^\rho$ has a positive continuous density on the open simplex and the seam vector is continuous there, so the law $(\boldsymbol{\varrho}_R^\rho)_*\phi_R^\rho$ is carried by the closure of the image of the open simplex, and every point of that image lies in its support: the preimage of a neighbourhood of an image point is a nonempty open set, of positive mass. The supremum of the continuous function $\mathbf{x}^{\mathbf{v}}$ over the image is therefore its largest value on the support of the law, in $(0,\infty]$. □

**Lemma S14.** *Let $L_1,\dots,L_p$ be positive letters on a finite set $B$ (25.1), $e_0\in B$, $e_1,\dots,e_p$ real exponents and $\mathbf{m}$ real masses on $B$ with $m_B=\sum_\ell e_\ell \deg L_\ell$; the pairings below are taken on $B\setminus e_0$.*

*(i) There are finitely many rational simplicial cones of dimension $|B|-1$ with disjoint interiors covering $\mathbb{R}^{B\setminus e_0}$, each contained, for every $\ell$, in a set $\{\mathbf{v}:\ \langle\mathbf{t},\mathbf{v}\rangle\le\langle\mathbf{t}',\mathbf{v}\rangle$ for every exponent $\mathbf{t}'$ of the support of $L_\ell\}$ for one exponent $\mathbf{t}$ of that support.*

*(ii) On such a cone $\pi$, with the primitive integral generators $\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1}$, write $x=\log\alpha_{B\setminus e_0}=-\sum_k u_k\mathbf{v}_{\pi,k}$ on the section $\alpha_{e_0}=1$, $u_k\ge0$, and $q_k=e^{-u_k}$. The exponent $\mathbf{t}_{\ell,\pi}$ of (i) is unique, and $L_\ell=\prod_k q_k^{\langle\mathbf{t}_{\ell,\pi},\mathbf{v}_{\pi,k}\rangle}\,k_{\ell,\pi}(q)$ on the section, with $k_{\ell,\pi}$ a polynomial in $q$ with nonnegative coefficients whose constant term is the coefficient of $\alpha^{\mathbf{t}_{\ell,\pi}}$ in $L_\ell$.*

*(iii) For a generator $\mathbf{v}$ of $\pi$ the number $\kappa_{\mathbf{v}}=\langle\mathbf{m}-\sum_\ell e_\ell\mathbf{t}_{\ell,\pi},\mathbf{v}\rangle$ is the same for every cone $\pi$ of which $\mathbf{v}$ is a generator, and, in $[0,\infty]$,*

$$\int_{\alpha_{e_0}=1}\alpha^{\mathbf{m}-\mathbf{1}}\prod_\ell L_\ell^{-e_\ell}\,d\alpha_{B\setminus e_0}$$
$$=\sum_\pi|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})|\int_{[0,1]^{|B|-1}}\prod_k q_k^{\kappa_{\mathbf{v}_{\pi,k}}-1}\prod_\ell k_{\ell,\pi}(q)^{-e_\ell}\,dq;$$

*both sides are finite exactly when every $\kappa_{\mathbf{v}}$ is positive, and they are then the period of the word $\prod_\ell L_\ell^{-e_\ell}$ at $\mathbf{m}$, the integral over a section of 25.1.*

*Proof.* (i) Identify $\mathbb{R}^{B\setminus e_0}$ with the hyperplane $v_{e_0}=0$ of $\mathbb{R}^B$. For each $\ell$ the sets $C_{\ell,\mathbf{t}}=\{\mathbf{v}:\ \langle\mathbf{t},\mathbf{v}\rangle\le\langle\mathbf{t}',\mathbf{v}\rangle$ for every $\mathbf{t}'\}$, $\mathbf{t}$ in the support of $L_\ell$, are rational polyhedral cones, defined by the inequalities $\langle\mathbf{t}-\mathbf{t}',\mathbf{v}\rangle\le0$: the cones of the inner normal fan of the Newton polytope of $L_\ell$ cut by that hyperplane, which form a fan covering $\mathbb{R}^{B\setminus e_0}$, every $\mathbf{v}$ having an exponent of least pairing. The cones $\bigcap_\ell C_{\ell,\mathbf{t}_\ell}\cap O$, $O$ running over the closed coordinate orthants of $\mathbb{R}^{B\setminus e_0}$, are pointed rational polyhedral cones which with their faces form a fan covering $\mathbb{R}^{B\setminus e_0}$, the common refinement of these fans and of the fan of the orthants. Order its rays and triangulate each cone by pulling: the cone is the union of the cones spanned by its first ray and by the triangulated faces that do not contain that ray. A face $F$ of a pointed cone $C$ has $\operatorname{span}F\cap C=F$, so the first ray of $C$ lies off $\operatorname{span}F$ for such a face, and the cones produced are simplicial; the pulling triangulation of a cone restricts on each face to the pulling triangulation of that face, which depends on the face alone, so the triangulations of two cones agree on their common face. This gives finitely many rational simplicial cones of dimension $|B|-1$ with disjoint interiors covering $\mathbb{R}^{B\setminus e_0}$, each in some $\bigcap_\ell C_{\ell,\mathbf{t}_\ell}$.

(ii) If $\mathbf{t}$ and $\mathbf{t}''$ both pair least with every $\mathbf{v}_{\pi,k}$, then $\langle\mathbf{t}-\mathbf{t}'',\mathbf{v}_{\pi,k}\rangle=0$ for every $k$; the generators span $\mathbb{R}^{B\setminus e_0}$, so $\mathbf{t}$ and $\mathbf{t}''$ agree on $B\setminus e_0$, and as $|\mathbf{t}|=|\mathbf{t}''|=\deg L_\ell$ they are equal. On the section $\alpha^{\mathbf{t}'}=e^{\langle\mathbf{t}',x\rangle}=\prod_k q_k^{\langle\mathbf{t}',\mathbf{v}_{\pi,k}\rangle}$, so with $L_\ell=\sum_{\mathbf{t}'}c_{\mathbf{t}'}\alpha^{\mathbf{t}'}$ and $\mathbf{t}=\mathbf{t}_{\ell,\pi}$,

$$L_\ell=\prod_k q_k^{\langle\mathbf{t},\mathbf{v}_{\pi,k}\rangle}\sum_{\mathbf{t}'}c_{\mathbf{t}'}\prod_k q_k^{\langle\mathbf{t}'-\mathbf{t},\mathbf{v}_{\pi,k}\rangle}.$$

Every exponent $\langle \mathbf{t}' - \mathbf{t}, \mathbf{v}_{\pi,k} \rangle$ is an integer, the exponents and the generators being integral, and it is nonnegative by the choice of $\mathbf{t}$; by the uniqueness just proved, all of them vanish only at $\mathbf{t}' = \mathbf{t}$. So the sum is a polynomial $k_{\ell,\pi}$ with nonnegative coefficients and the constant term $c_{\mathbf{t}} > 0$.

(iii) A cone $\pi$ of which $\mathbf{v}$ is a generator has $\langle \mathbf{t}_{\ell,\pi}, \mathbf{v} \rangle = \min_{\mathbf{t}'} \langle \mathbf{t}', \mathbf{v} \rangle$, the minimum over the support of $L_\ell$, which does not depend on $\pi$; so neither does $\kappa_{\mathbf{v}}$. On $\pi$ the map $(u_k)_k \mapsto x = -\sum_k u_k \mathbf{v}_{\pi,k}$ is a linear bijection of $[0,\infty)^{|B|-1}$ onto $-\pi$ with the Jacobian $|\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})|$; on the section $d\alpha_e = \alpha_e \, dx_e$ for $e \neq e_0$, $\alpha^{\mathbf{m}} = e^{\langle \mathbf{m}, x \rangle}$, and $du_k = -dq_k/q_k$. By (ii), on the image of $-\pi$, the form $\alpha^{\mathbf{m}-\mathbf{1}} \prod_\ell L_\ell^{-e_\ell} \, d\alpha_{B \setminus e_0} = e^{\langle \mathbf{m}, x \rangle} \prod_\ell L_\ell^{-e_\ell} \, dx$ is

$$|\det(\mathbf{v}_{\pi,1}, \dots, \mathbf{v}_{\pi,|B|-1})| \prod_k q_k^{\langle \mathbf{m} - \sum_\ell e_\ell \mathbf{t}_{\ell,\pi}, \mathbf{v}_{\pi,k} \rangle - 1} \prod_\ell k_{\ell,\pi}(q)^{-e_\ell} \, dq.$$

The cones $-\pi$, like the cones $\pi$, cover $\mathbb{R}^{B \setminus e_0}$ with disjoint interiors, their boundaries are null and the integrands are positive, so the two sides agree in $[0,\infty]$ (Tonelli's theorem). On $[0,1]^{|B|-1}$ each $k_{\ell,\pi}$ lies between its constant term and the sum of its coefficients, so $\prod_\ell k_{\ell,\pi}^{-e_\ell}$ is bounded above and below by positive constants, and the integral over the cube is finite exactly when every exponent $\kappa_{\mathbf{v}_{\pi,k}}$ is positive; every generator is a generator of some cone. The form $\alpha^{\mathbf{m}-\mathbf{1}} \prod_\ell L_\ell^{-e_\ell} d\alpha$ is projective, the masses lying on the balance hyperplane, and $\alpha_{e_0}$ is positive and homogeneous of degree one, so the integral over the section is the period (25.1, 16.2). □

**Theorem 31.4 (Closing past the walls).** *As stated, for every piece whose seam function converges at a real point of the pins of the seam with positive shares, with the sector form of (ii) for every fan of the seam word and with part (iv).*

*Proof. The hypothesis.* At a real point of the pins with positive shares the seam word $W_B(\mathbf{a}, \mathbf{b})$ is a Newton word (25.1), and the masses $\mathbf{m}_B$ lie on its balance hyperplane: by the letter totals and the scale pin,

$$\sum_{j,D} \Big( a_{jD}\big(d_j + i(D)\big) + b_{jD}\big(d_j^* + c_j - i(D)\big) \Big) = \sum_j \big( s_j d_j + t_j d_j^* \big) + \kappa_B = m_B$$

(28.4, 28.7). Its period, the seam function, converges there, so its Newton sum has the dimension $|B| - 1$ (25.2).

*(i).* The reference proves (i) for a piece of tables (the reference, proof of thm:pastwalls(i)). Of the piece it uses only that its seam word has a Newton sum of full dimension, so that its reduced period in 28.12 is its seam function at the sums over the monomials, the seam letters repeating over the monomials of the partner; the reference obtains that fact from the connectedness of the family of tables, and here it is the hypothesis. The rest of that proof concerns the partner, whose room letters are read in their monomials, so that $\vec{\mathcal{N}}_R = 0$ and its reduced period is $\prod \xi^{-\mathbf{a}} \prod \xi^{*-\mathbf{b}}$ (28.12); with $M = \mathbb{R}\mathbf{1}_B \oplus \mathbb{R}^R$ and the lattice basis $\mathbf{1}$, $\mathbf{e}_r$ ($r \in R$), the pins of 28.12 are the displayed ones, the scale pin following by the balance.

*(ii), the sector form.* On the pins of the seam the exponents $\bar{a}_{jD}$, $\bar{b}_{jD}$ of the seam word are affine in the shares with rational linear parts, and the masses $\mathbf{m}_B$ lie on its balance hyperplane at every share, by the display above with the sums in place of the shares. Lemma S14, applied to the seam letters of the active letters with the exponents $\bar{a}_{jD}$, $\bar{b}_{jD}$, gives the fans of the seam word, the sector coordinates, the sector letters with their exponents, the walls and the identity at every real point of the pins, and shows that the seam function converges at a real point exactly when every wall is positive there. The coefficients $\langle \mathbf{t}_{jD,\pi}, \mathbf{v} \rangle$ and $\langle \mathbf{t}^*_{jD,\pi}, \mathbf{v} \rangle$ of the shares in a wall

are integers, so the wall is affine with a rational linear part, and it does not depend on the sector (Lemma S14(iii)).

*(ii), continuation and bound.* On a sector the integrand is $\prod_k q_k^{\kappa_{\mathbf{v}_{\pi,k}}(\mathbf{z})-1} h(q,\mathbf{z})$ with $h = \prod_{j,D} k_{jD,\pi}^{-\bar a_{jD}} \psi_{jD,\pi}^{-\bar b_{jD}}$. The sector letters are at least their positive constant terms on the closed cube, so their real parts are positive on a neighbourhood of it, where their principal logarithms are holomorphic; so $h$ is holomorphic in $q$ near $[0,1]^{|B|-1}$ and entire in the shares. 8.4, with $x_k = \kappa_{\mathbf{v}_{\pi,k}}(\mathbf{z})$ and the shares as its parameter, continues each sector integral meromorphically to all complex shares, with its poles on the hyperplanes on which a wall of a generator of the sector lies in $-\mathbb{N}$; a wall constant on the pins is positive there, the seam function converging, and gives no hyperplane. The formula of 8.4, repeated in the $|B|-1$ coordinates, writes each term with at most $|B|-1$ factors $(\kappa_{\mathbf{v}_{\pi,k}}+l)^{-1}$, which bounds the orders of the poles by $|B|-1$. For the bound, the proof of 20.32 is run on each sector integral with the several parameters entering through the powers of the sector letters and through the walls, $|\operatorname{Re}\mathbf{z}|_1$ and $|\operatorname{Im}\mathbf{z}|_1$ in place of $|\operatorname{Re} z|$ and $|\operatorname{Im} z|$ (the reference, proof of thm:pastwalls(ii)): each of the at most $|B|-1$ factors $(\kappa+l)^{-1}$ of a term is at most a constant times $\delta^{-1}$, and nothing else depends on $\delta$.

*(ii), a piece of tables.* Let the seam letters at the station $D$ of the active letters be the basis and the cobasis polynomials of a table $\mathcal{B}_{jD}$ on $B$ with positive weights, and consider the family of these tables at the exponents $\bar a_{jD}, \bar b_{jD}$ at a point with positive shares. Its potential $\sum_{j,D} \left(\bar a_{jD} \log K_{\mathcal{B}_{jD}}(e^y) + \bar b_{jD} \log \Psi_{\mathcal{B}_{jD}}(e^y)\right)$ (8.18) is affine along a direction $\mathbf{w}$ exactly when, in every letter, all the exponents of the support have one pairing with $\mathbf{w}$, a logarithm of a sum of exponentials with positive coefficients being strictly convex along every direction that separates two of its exponents: exactly when $\mathbf{w}$ is orthogonal to the directions of the Newton sum. If the family is connected, that space is $\mathbb{R}\mathbf{1}$ (8.19(iii), a space that does not involve the masses), and the Newton sum has the dimension $|B|-1$. If it is not, some nonempty proper $S \subset B$ has the width $c_{jD}(S) = 0$ in every table (8.1), so that $|T \cap S| = d_{jD}(S)$ for every basis $T$ of $\mathcal{B}_{jD}$ (7.13). The base polytope of $\mathcal{B}_{jD}$ then lies in $\{x_S = d_{jD}(S)\}$ and that of its dual in $\{x_S = |S| - d_{jD}(S)\}$, so the Newton sum lies in a hyperplane $\{x_S = \text{const}\}$, and as $\mathbf{1}_S \notin \mathbb{R}\mathbf{1}$ its dimension is less than $|B|-1$.

In the coordinates of 8.5(ii) on the sector $H_\pi$ of an order $\pi$ of $B$, $\alpha_{\pi(1)} = \rho$ and $\alpha_{\pi(k)} = \rho q_2 \cdots q_k$, so that, with $u_k = -\log q_k$,

$$\log \alpha_{\pi(m)} - \log \alpha_{\pi(1)} = \sum_{k=2}^{m} \log q_k = -\sum_{k=2}^{|B|} u_k \, (\mathbf{1}_{U_k})_{\pi(m)},$$

because $\pi(m) \in U_k$ exactly when $k \le m$. So, modulo $\mathbf{1}$, $H_\pi$ is the cone with the generators $\mathbf{1}_{U_k}$, $2 \le k \le |B|$, read on the section as $\mathbf{1}_{U_k} - (\mathbf{1}_{U_k})_{e_0}\mathbf{1}$. These $|B|-1$ vectors form a basis of $\mathbb{Z}^B/\mathbb{Z}\mathbf{1}$, since the differences $\mathbf{1}_{U_k} - \mathbf{1}_{U_{k+1}} = \mathbf{e}_{\pi(k)}$ and $\mathbf{1}_{U_1} = \mathbf{1}$ span $\mathbb{Z}^B$; so the cone is simplicial of dimension $|B|-1$ with the determinant $\pm 1$, and the sectors of the orders cover $\mathbb{R}^B/\mathbb{R}\mathbf{1}$ with disjoint interiors. For a basis $T$, $\alpha^T/\alpha^{T_\pi^+} = \prod_k q_k^{|T\cap U_k| - |T_\pi^+ \cap U_k|}$, and 8.3, $\alpha^T \le \alpha^{T_\pi^+}$ on $H_\pi$ for every basis, gives, with one $q_k$ small and the others 1, $\langle \mathbf{1}_{T_\pi^+}, \mathbf{1}_{U_k}\rangle \le \langle \mathbf{1}_T, \mathbf{1}_{U_k}\rangle$ for every $k$. As $\mathbf{1}_T$ and $\mathbf{1}_{T_\pi^+}$ have one total, their pairings on $B \setminus e_0$ with the generator $\mathbf{1}_{U_k} - (\mathbf{1}_{U_k})_{e_0}\mathbf{1}$ differ by the difference of their pairings with $\mathbf{1}_{U_k}$. So the sector lies in the set of (ii) for $K_{\mathcal{B}_{jD}}$ with $\mathbf{t} = \mathbf{1}_{T_\pi^+}$, and likewise for $\Psi_{\mathcal{B}_{jD}}$ with $\mathbf{t} = \mathbf{1}_{B\setminus T_\pi^-}$, the greedy bases being those of $\mathcal{B}_{jD}$; the sectors of the orders form a fan of the seam word. Its sector letters are $\sum_T \theta_{jD,T} \prod_k q_k^{|T\cap U_k| - d_{jD}(U_k)}$ and their cobasis counterparts, the sector letters of 8.5(ii). The vector $\mathbf{m}_B - \sum_{j,D}(\bar a_{jD}\mathbf{1}_{T^+} + \bar b_{jD}\mathbf{1}_{B\setminus T^-})$ has the total 0 on the pins, so its pairing with the generator of a tail $U$ is its pairing with $\mathbf{1}_U$; by 8.3, $|T^+ \cap U| = d_{jD}(U)$ and $|U \setminus T^-| = d^*_{jD}(U)$, and the wall of that generator is $m_U - \sum_{j,D}(\bar a_{jD} d_{jD}(U) + \bar b_{jD} d^*_{jD}(U))$, the proper wall $\kappa_U$. Every nonempty proper $U$ is a tail of some order.

*(iii).* The reference proves (iii) for a piece of tables (the reference, proof of thm:pastwalls(iii)), using three facts about the piece. The first is the continuation with the bound of (ii). The second is that the seam function converges at a real point exactly when its walls are positive there, so that it converges along a ray along which no wall falls, the walls staying above their positive values at $\bar{\mathbf{c}}$, and diverges past a wall that falls. The third is that at real points it is $\int \mathbf{x}^{-\mathbf{w}}\, d\big((\boldsymbol{\varrho}_B^{\rho})_* \phi_B^{\rho}\big)$, with the scale form positive and continuous on the open simplex and the seam vector continuous there, which with 31.2 gives $g(\boldsymbol{\sigma}; 0) = \sup \mathbf{V}_B^{-\bar{\boldsymbol{\sigma}}}$. The first two are (ii), and the third holds for every piece (28.5, 28.7). The rest of that proof (the Gamma bounds, Stirling's formula with $\sum_l \sigma_l = 0$, the crossings and the null set) does not use the tables.

*(iv).* The partner is a piece on $R$ read with the reversed excesses. The pins of its seam, the letter totals and its scale pin $\sum_{j,D} \big((c_j - i(D))a_{jD} + i(D)b_{jD}\big) = \kappa_R$, are on the balance wall $\kappa_R = \kappa_B^{\mathrm{top}}$ those of the piece (28.10(iii)), and its seam function converges at $\mathbf{c}$; so (ii) applies to the piece and to the partner. On the share side of 28.10(iii) the integrand at $\mathbf{c} + \tau\boldsymbol{\sigma} + i\boldsymbol{\vartheta}$ is $\prod_{j:\, s_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{a}_j \\ s_j\end{bmatrix} \prod_{j:\, t_j>0} \Gamma_{|\mathcal{D}_j|}\begin{bmatrix}\mathbf{b}_j \\ t_j\end{bmatrix}$ times the two seam functions. For $M_B > g_B(\boldsymbol{\sigma}; \theta_B)$ and $M_R > g_R(\boldsymbol{\sigma}; \theta_R)$ the definitions of the growths give $C$, $n_1$ and $n_2$ with

$$\big|P_{W_B}\, P_{W_R}\big| \le C^2 \delta^{-2n_2} (1 + \tau + |\boldsymbol{\vartheta}|)^{2n_1} (M_B M_R)^{\tau} e^{(\theta_B + \theta_R)|\boldsymbol{\vartheta}|_1}$$

for $\tau \ge 0$, $\delta$ the smaller of 1 and the distance from the polar hyperplanes of both seam functions. From here the proof of (iii) applies with $\theta = \theta_B + \theta_R < \pi/8$ and $M_B M_R$ in place of $M$, the coefficients sitting inside the seam functions:

- the contour moves along times at distance at least $\tau^{-3}$ from the crossings, where $\delta^{-2n_2}$ is a power of $\tau$, and the power of $1 + |\boldsymbol{\vartheta}|$ is absorbed into $e^{\eta|\boldsymbol{\vartheta}|_1}$ with $\eta$ small;
- a share whose real part $x$ grows has $|\Gamma(x + iy)| e^{\theta|y|} \le \Gamma(x) e^{2\theta^2 x/\pi}$, and a share that falls or stays absorbs $e^{\theta|y|}$ against $e^{-\pi|y|/2}$;
- integrating over the directions of the shares, Stirling's formula and $\sum_l \sigma_l = 0$, by the letter totals, give the factor $e^{\tau(h_{\theta_B, \theta_R}(\boldsymbol{\sigma}) + \epsilon)}$ as $M_B \downarrow g_B$ and $M_R \downarrow g_R$.

The ray crosses the hyperplanes of the falling shares and the hyperplanes on which a falling wall of either word lies in $-\mathbb{N}$, and each crossing adds the fibre integral of the residue of the whole integrand (31.3), a Laurent coefficient where several factors are polar. The pins and the walls of both words have rational linear parts, so crossings on parallel hyperplanes coincide or keep a fixed positive distance. Crossings on hyperplanes that are not parallel, closer than $T^{-3}$ near $\tau = T$, occur only for $\mathbf{c}$ in a set of measure $O(T^{-3})$, summable over $T$ (Borel and Cantelli). The local decay (H1) of 31.3 holds because both seam functions grow at most like $e^{\theta' |\operatorname{Im} \mathbf{z}|_1}$ for every $\theta' > 0$, by (ii), against the decay of the Gammas at bounded real parts. (H2) holds at the times above when $h_{\theta_B, \theta_R}(\boldsymbol{\sigma}) < 0$, and 31.3 gives the closing with the stated bounds.

For the last sentence of (iv), let no wall of the word of a side fall along $\boldsymbol{\sigma}$. Both words have Newton sums of full dimension, their seam functions converging at $\mathbf{c}$ (25.2), so the largest value of $\mathbf{x}^{-\boldsymbol{\sigma}}$ on the support of the law of the side is $\sup \mathbf{V}^{-\boldsymbol{\sigma}}$ of its letter vector (31.2, for the partner by its sentence on the partner). At $\theta = 0$ the proof of (iii), run on that side, gives the growth $\sup \mathbf{V}^{-\boldsymbol{\sigma}}$, finite, and the growth does not increase with the angle, a bound with a smaller angle being a bound with a larger one. On the real ray, where $\boldsymbol{\vartheta} = 0$, the walls stay above their positive values at $\mathbf{c}$, so $\delta$ is bounded below there, and the seam function at $\mathbf{c} + \tau\boldsymbol{\sigma}$ is at least a positive constant times $e^{\tau h'}$ for every $h' < \log \sup \mathbf{V}^{-\boldsymbol{\sigma}}$ (the proof of (iii)); so every admissible $M$, at every angle, is at least $\sup \mathbf{V}^{-\boldsymbol{\sigma}}$. □

**Definition 31.5 (Chambers of a closing).** As stated, for every fibre integral with finitely many affine polar functions.

*Proof.* The chamber is convex. If $\boldsymbol{\sigma}', \boldsymbol{\sigma}'' \in C(\boldsymbol{\sigma})$ and $0 \le s \le 1$, then $\partial_{s\boldsymbol{\sigma}'+(1-s)\boldsymbol{\sigma}''}\ell = s\,\partial_{\boldsymbol{\sigma}'}\ell + (1-s)\,\partial_{\boldsymbol{\sigma}''}\ell$ equals $\partial_{\boldsymbol{\sigma}}\ell$ for every polar function that $\boldsymbol{\sigma}$ falls and is nonnegative for every other. The ray of $s\boldsymbol{\sigma}' + (1-s)\boldsymbol{\sigma}''$ consists of the points $s(\mathbf{c}+\tau\boldsymbol{\sigma}') + (1-s)(\mathbf{c}+\tau\boldsymbol{\sigma}'')$ of the convex $\Omega$.

The instances:

- In 31.2 the integrand is the Gammas times the reduced periods of the two words. These are holomorphic where they converge absolutely, on a set convex by Hölder's inequality (the reference, def:chamber). On the pins the integrand of a reduced period at $\mathbf{c}' + \tau\boldsymbol{\sigma}$ is its integrand at $\mathbf{c}'$ times $\mathbf{V}^{-\tau\boldsymbol{\sigma}}$ read at the point of the orthant (the reference, proof of thm:closecone), at most $(\sup \mathbf{V}^{-\boldsymbol{\sigma}})^{\tau}$; so when both suprema are finite a ball about $\mathbf{c}$ inside the set of convergence, translated by $\tau\boldsymbol{\sigma}$, stays inside it, and the ray lies in $\Omega$.
- In 31.4(iii) the seam function continues to all complex shares, with its poles where a wall lies in $-\mathbb{N}$ (31.4(ii)), the walls read through the sums of the shares being affine in the shares.
- For a fibre integral of Gammas and coefficients the integrand $\prod_l \Gamma(a_l)\xi_l^{-a_l}$, up to a constant, is meromorphic on all of $A+iV$ with its poles where a share lies in $-\mathbb{N}$. So the polar functions are the shares, $\Omega = A$, and $\partial_{\boldsymbol{\sigma}'} a_l = \sigma_l'$, which gives the chamber displayed. When the pins fix the total of the shares of every letter, every $\boldsymbol{\sigma}' \in V$ has $\sum_{l\in K_j} \sigma_l' = 0$ and $\sum_{l\in\Psi_j} \sigma_l' = 0$, so on the chamber $0 \le \sigma_l' \le \sum |\sigma_{l'}|$, the sum over the falling shares $l'$ of the letter of $l$: the chamber is a compact convex polytope.
- A source factor that is a cube integral continued by Reg is meromorphic on all of $A + iV$, with its poles where one of its walls lies in $-\mathbb{N}$ (8.4).

□

**Theorem 31.6 (A closing depends only on its chamber).** *As stated, for every fibre integral of 31.5 with the local decay (H1) of 31.3.*

*Proof.* A polar function $\ell$ that $\boldsymbol{\sigma}$ falls has $\ell(\mathbf{c}+\tau\boldsymbol{\sigma}) = \ell(\mathbf{c}) + \tau\partial_{\boldsymbol{\sigma}}\ell = \ell(\mathbf{c}+\tau\boldsymbol{\sigma}')$. So its hyperplanes $\ell = -n$ are met by the two rays at the same times, and $\ell$ is constant on the segment joining their points at a time $\tau$. A polar function that $\boldsymbol{\sigma}$ does not fall is not fallen by $\boldsymbol{\sigma}'$ either; it is at least $\ell(\mathbf{c}) > 0$ at both points, hence on the segment, and does not lie in $-\mathbb{N}$ there.

So the two rays meet the same polar hyperplanes at the same times. At a crossing time $\tau^*$ the polar hyperplanes through $\mathbf{c} + \tau^*\boldsymbol{\sigma}$ and through $\mathbf{c} + \tau^*\boldsymbol{\sigma}'$ are the same, with $\partial_{\boldsymbol{\sigma}}\ell_H = \partial_{\boldsymbol{\sigma}'}\ell_H$, so (H0) holds along one ray exactly when it holds along the other.

At a time $\tau$ that is no crossing, the segment avoids every polar hyperplane at a positive distance and lies in the convex $\Omega$. By (H1) on a compact neighbourhood of the segment in $\Omega$, step (1) of the proof of 31.3 (Cauchy's theorem on rectangles whose horizontal sides vanish by (H1), then Fubini's theorem) equates the fibre integrals through its two ends. A crossing term is the difference of the fibre integrals just before and just after the crossing (step (2) of that proof), so the crossing terms of the two rays at a common crossing are equal. Equal terms and equal remainders carry convergence and bounds from one closing to the other.

The rest is the reference's (the reference, proof of thm:chamber). The integrands of 31.2 and of 31.4(iii) have the local decay (H1), by Stirling's formula and the bounds of their proofs, so the first part applies to them. For every $\epsilon > 0$ some $\boldsymbol{\sigma}' \in C(\boldsymbol{\sigma})$, with some $\theta$ in the case of 31.4(iii), has a rate below $h' + \epsilon/2$. The null set of 31.2 is the same for every direction of the chamber, its crossings being those of the falling polar functions, and the bounds of 31.2 or of 31.4(iii) along $\boldsymbol{\sigma}'$

bound the terms and the remainders along $\boldsymbol{\sigma}$. On the chamber the falling shares are fixed, $x \log x$ is convex on $[0,\infty)$, and $\log \sup \mathbf{V}_B^{-\boldsymbol{\sigma}'}$ and $\log \sup \mathbf{V}_R^{-\boldsymbol{\sigma}'}$ are suprema of linear functions of $\boldsymbol{\sigma}'$, so the rate $h$ is convex there. □

**Theorem 31.7 (Averaging the laws).** *As stated, for every share side whose integrand is the Gammas of the shares times a Laplace transform $\int e^{-\langle \mathbf{a}, y\rangle}\, d\lambda$ of a source measure, the kinds (A) and (B) among them.*

*Proof. (i).* At a source point $\zeta$ the kernel is a fibre integral of Gammas and the positive coefficients $e^{y_l(\zeta)}$: the share side of a Gramma (4.2) divided by the constant $\prod_j \Gamma(s_j) \prod_j \Gamma(t_j)$. At the real point $\mathbf{c}$ the Gammas are integrable over the fibre, $|e^{-\langle \mathbf{c}+i\boldsymbol{\vartheta}, y\rangle}| = e^{-\langle \mathbf{c}, y\rangle}$, and $\int e^{-\langle \mathbf{c}, y\rangle}\, d\lambda < \infty$. So the fibre and the source integrals converge absolutely together, and by Fubini's theorem the fibre integral of the share side is the integral of the kernel against $\lambda$. In (A) and (B) that fibre integral is $P_{B \sqcup R}(\mathbf{m})$ (28.10(iii), 31.4(i)). In (A) the kernel is $\Gamma(\kappa_B) \mathsf{Z}_{(\kappa_B, \mathbf{s}, \mathbf{t}); t_{\mathcal{D}}}(\mathbf{x} \circ \mathbf{x}')$ (the reference, proof of thm:pool(i)): each active letter of the stand-in pair of 28.10(ii) is opened by 4.4 with one share per station, and the integral over the scale is the pin of the scale.

The bound on the crossing terms is the reference's (the reference, proof of thm:pool(i)). It uses three things:

- 31.6 for the kernel, whose polar functions are the shares, with $\Omega = A$ and (H1) holding by Stirling's formula, so that the closings along $\boldsymbol{\sigma}$ and along every $\boldsymbol{\sigma}' \in C$ are one series;
- Stirling's formula for the Gammas, uniformly on the compact $C$, the terms $\tau x \log \tau$ cancelling over each letter by its total;
- at a crossing, the expansion of $e^{-\langle \mathbf{a}, y\rangle} = e^{-\langle \mathbf{c}+\tau\boldsymbol{\sigma}', y\rangle} e^{-i\langle \boldsymbol{\vartheta}', y\rangle} e^{-w\langle \mathbf{e}, y\rangle}$ in the transverse variable $w$ to the order $p-1$, with $|\langle \mathbf{e}, y\rangle| \le C|y|_V$, against the Laurent coefficients of the Gammas, which Cauchy's estimate bounds off the null set.

None of these steps uses the kind of the share side.

*(ii).* The source points with $y(\zeta) \in S_\lambda$ carry all of $\lambda$, and there $h_C(y) \le \hat{h}_C$. By (i) the averaged terms are absolutely convergent integrals, at most $C_\epsilon e^{\tau_k(\hat{h}_C+\epsilon)} \int w\, d(y_* \lambda)$, which is finite by the certificate of 31.7. At each source point the kernel is the sum of its crossing terms before a time $T$ separating the crossings and of the integral left at $T$ (31.3, whose identity at a finite time needs no limit), and that integral is at most $C_\epsilon e^{-\langle \mathbf{c}, y\rangle} e^{T(\hat{h}_C+\epsilon)}$. Integrating against $\lambda$, the fibre integral of the share side is the sum of the averaged terms before $T$ up to $C_\epsilon e^{T(\hat{h}_C+\epsilon)} \int e^{-\langle \mathbf{c}, y\rangle}\, d\lambda$; the crossings up to $T$ number $O(T)$, and with $\hat{h}_C + \epsilon < 0$ both the sum and the bound follow. At a crossing point, along some $\boldsymbol{\sigma}' \in C$, inside the set where $\int e^{-\langle \mathbf{a}, y\rangle}\, d\lambda$ converges absolutely, with its derivatives in the shares, Fubini's theorem (the Gammas decaying by Stirling's formula) exchanges the fibre and the source integrals in the crossing term of the share side. The inner integral is then the crossing term of the kernel at that point, which is $\mathcal{R}_k(\zeta)$ by 31.6.

*(iii).* The function $h(\boldsymbol{\sigma}'; y)$ is continuous, convex in $\boldsymbol{\sigma}'$ on $C$ (the falling coordinates being fixed and $x \log x$ convex on $[0,\infty)$), and affine in $y$; $C$ is compact and convex. Sion's minimax theorem on $C \times \operatorname{conv} S_\lambda$ gives the identity, the supremum of an affine function over $\operatorname{conv} S_\lambda$ being its supremum over $S_\lambda$.

For the reading through the log-support,

$$\Big| \int e^{-\langle \mathbf{c}+\tau\boldsymbol{\sigma}'+i\boldsymbol{\vartheta}, y\rangle}\, d\lambda \Big| \le e^{\tau \sup_{y \in S_\lambda} (-\langle \boldsymbol{\sigma}', y\rangle)} \int e^{-\langle \mathbf{c}, y\rangle}\, d\lambda.$$

With the Gamma bounds of the proof of 31.2, this bounds the closing of the share side along $\boldsymbol{\sigma}'$ at the rate $\sum_l \sigma'_l \log|\sigma'_l| + \sup_{S_\lambda}(-\langle\boldsymbol{\sigma}', y\rangle) = \sup_{y\in S_\lambda} h(\boldsymbol{\sigma}'; y)$. In (A) and (B) this is the rate $h(\boldsymbol{\sigma}')$ of 31.2 (the reference, proof of thm:pool(iii)):

- the seam words of the sides that are laws have Newton sums of full dimension, their seam functions converging at $\mathbf{c}$ (25.2);
- so the largest values of the monomials of their letter vectors are read on the supports of their laws (31.2; for the partner in (A), its sentence on the partner);
- in (B), $\sup \mathbf{V}_R^{-\boldsymbol{\sigma}'} = \boldsymbol{\xi}^{-\boldsymbol{\sigma}'}$;
- the largest value of $-\langle\boldsymbol{\sigma}', y\rangle$ on the log-support of a product of laws is the sum of its largest values on the log-supports of the factors.

In a bundle $G$, Jensen's inequality for the logarithm, in the proportions $\sigma'_l/g$, gives $\sum_G \sigma'_l \log(\sigma'_l e^{-y_l}) \geq -g\log\left(g^{-1}\sum_G e^{y_l}\right)$, with equality exactly when $e^{y_l}/\sigma'_l$ is constant on $G$. □

**Definition 31.10 (Keyhole continuations along a direction; ceiling and phase).** As stated, for every piece of 31.4 and the sectors of any fan of its seam word.

*Proof.* The displayed sum is 31.4(ii) at the sums of the shares over the monomials of the partner. A set $Y$ of a box is compact and is the product of a box, of intervals, and of keyholes, a closed disc about 0 joined to a segment of the positive axis (20.47); each factor is star-shaped, about the centre of its disc for a keyhole, so $Y$ is star-shaped, and so is $Y$ plus a small open polydisc about 0. The sector letters do not vanish on $Y$, by the definition of a keyhole continuation, hence on such a neighbourhood of $Y$, which is simply connected, so they have logarithms there. The real points of $Y$ form a product of intervals, connected, containing points with nonnegative coordinates, where the letters, with their nonnegative coefficients and positive constant terms, are positive; real and without zeros on that set, they are positive on it, and the logarithms can be taken real at the real points. For a piece of tables and the sectors of the orders, the walls of the generators are the proper walls of the tails and the sector letters are those of 8.5(ii) (31.4(ii)). □

**Theorem 31.11 (The travelling bound past the walls).** *As stated, for every piece of 31.4.*

*Proof.* This is the reference's proof (the reference, proof of thm:keyholebound), with the sectors of the fan in place of the orders of $B$ and the generators in place of the tails.

(i) Fix a sector $\pi$, a box $\Xi$ of it, the set $J$ of its falling variables with keyholes and their radii $r_k$, and choose $r_k^- \in (0, r_k)$ with $\sum_{k\in J}(-\dot\kappa_k)\log(r_k/r_k^-) < \epsilon/2$. The term of $\pi$ is $|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})| \int \prod_{k\in J} q_k^{\kappa_k(\bar{\mathbf{z}})-1}\Phi\, dq$, with $\Phi = \prod_{k\notin J} q_k^{\kappa_k(\bar{\mathbf{z}})-1}\mathbf{k}_\pi(q)^{-\bar{\mathbf{z}}}$ holomorphic in the variables of $J$ on a neighbourhood of their keyholes, because the sector letters do not vanish on the sets $Y$. The following steps apply as in the reference:

- 20.45, applied in the variables of $J$ one at a time with $r_k^-$ and $r_k$;
- the maximum modulus principle in the variables of $J \setminus A$ that lie in their discs;
- the exact integration of the falling monomials, which carries no phase;
- the identity $|k^{-z}| = |k|^{-\operatorname{Re} z} e^{\operatorname{Im} z \arg k}$ at the points of the outline;
- the bounds of the letters above and below on the compact outline;
- the integrability of $q_k^{\kappa_k(\bar{\mathbf{c}})-1}$ over the base variables, the walls being positive at $\bar{\mathbf{c}}$;

- $|\kappa_k(\bar{\mathbf{z}}) + l_k| \geq c\,\delta$ for at most $|B| - 1$ factors.

Each of these uses only the sector form of 31.4(ii): walls affine in the shares and positive at $\bar{\mathbf{c}}$, and sector letters polynomial, positive on the closed cube and nonvanishing on the sets $Y$. The determinant is a constant, and the boxes, the sets $A$ and the sectors are finitely many.

(ii) The integrand of the share side of 31.4(i) at $\mathbf{c} + \tau\boldsymbol{\sigma} + i\boldsymbol{\vartheta}$ is the Gammas and the coefficients times the seam function, which (i) bounds. The steps are those of the reference:

- for a growing share, the exact bound $|\Gamma(x + iy)|e^{\varphi y} \leq \Gamma(x)(\cos\varphi)^{-x}$ for $|\varphi| < \frac{\pi}{2}$, from $\Gamma(z) = e^{i\varphi z}\int_0^\infty u^{z-1}e^{-ue^{i\varphi}}du$, applied with $\varphi = \upsilon_l(q) + \eta\,\operatorname{sign} y$;
- falling and staying shares absorb $e^{\upsilon_l(q)y}$ against $e^{-\pi|y|/2}$, as $|\upsilon_l| \leq \frac{\pi}{2} - \epsilon_0$;
- Stirling's formula with $\sum_l \sigma_l = 0$, the supremum over $q$ being taken after the integration over the directions of the shares.

Together these give the travelling bound at the rate $h_{\boldsymbol{\upsilon}}(\boldsymbol{\sigma})$. Then 31.3 applies: (H0) off the null set; (H1) by 31.4(ii) with any $\theta < \pi/2$ and Stirling's formula; (H2) at times at distance at least $\tau^{-3}$ from the crossings. □

**Corollary 31.12 (When the phase costs nothing).** *As stated, for every piece of 31.4.*

*Proof.* In (i) the price $-\sum_{\sigma_l>0}\sigma_l \log\cos\upsilon_l$ vanishes. In (ii), $\log|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}| - \sum_{\sigma_l>0}\sigma_l\log\cos\upsilon_l \leq \log H_{\mathcal{O}}$ on the outline, with equality approached where $|\varrho_{\pi,\bar{\boldsymbol{\sigma}}}|$ approaches $H_{\mathcal{O}}$, the price being nonnegative.

With no falling variables, the outline is the union of the real cubes of the sectors, where the sector letters are positive and the phase vanishes. Differentiating the wall of 31.4(ii) gives $\dot{\kappa}_k = -\langle\sum_{j,D}(\bar{\sigma}_{jD}\mathbf{t}_{jD,\pi} + \bar{\sigma}^*_{jD}\mathbf{t}^*_{jD,\pi}), \mathbf{v}_{\pi,k}\rangle$. So on the section $\varrho_{\pi,\bar{\boldsymbol{\sigma}}} = \prod_{j,D}\big(\prod_k q_k^{\langle\mathbf{t}_{jD,\pi},\mathbf{v}_{\pi,k}\rangle}k_{jD,\pi}\big)^{-\bar{\sigma}_{jD}}\big(\prod_k q_k^{\langle\mathbf{t}^*_{jD,\pi},\mathbf{v}_{\pi,k}\rangle}\psi_{jD,\pi}\big)^{-\bar{\sigma}^*_{jD}} = \mathbf{V}_B^{-\bar{\boldsymbol{\sigma}}}$, read in the sector coordinates. The cubes of the sectors cover the section, so the ceiling is $\sup\mathbf{V}_B^{-\bar{\boldsymbol{\sigma}}}$, the monomial being homogeneous of degree zero, and the rate is that of 31.2 for the glue read in the monomials of the partner. □

**Proposition 31.13 (How travelling bounds compose).** *As stated, the face coefficient of (iii) being read on the sectors of any fan of the seam word.*

*Proof.* (i) and (ii) are the reference's (the reference, proof of prop:pathcompose). For (iii), the reference's proof applies with one sentence changed: a wall $\kappa_{\mathbf{v}}$ is the exponent of $q_k$ in each sector of which $\mathbf{v} = \mathbf{v}_{\pi,k}$ is a generator (31.4(ii)).

- The pole is simple, as opposed to the multiple poles of the statement, when no other factor and no second wall of a sector is polar on the hyperplane. Then, by 8.4, the residue of the sector integral in the direction $\mathbf{e}$ at $\kappa_{\mathbf{v}} = -n$ is $|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})|$ times the integral over the other coordinates of the $n$-th Taylor coefficient in $q_k$ of the rest of the integrand, divided by $\partial_{\mathbf{e}}\kappa_{\mathbf{v}}$. Summing over the sectors gives the face coefficient.
- The other powers $q_{k'}^{\kappa_{k'}-1}$ do not depend on $q_k$, so the Taylor coefficient is that of the product of the powers of the sector letters: a finite sum of products of derivatives of the letters and powers of the letters at $q_k = 0$, with coefficients polynomial in the exponents.
- Where two walls of one sector lie in $-\mathbb{N}$ on the hyperplane, the expansion of 8.4 in both variables gives a pole of the second order. A Laurent coefficient of a product is a finite sum of products of Laurent coefficients, and the derivatives in the shares of $q^{\kappa(\mathbf{z})}$ and of $k^{-z}$ bring $\log q$ and $-\log k$.

- The wall crossed falls along the direction, so $q_k$ is a falling variable and $q_k = 0$ lies in its keyhole, where the letters do not vanish; Cauchy's estimate on $|q_k| = r_k$ bounds the $n$-th Taylor coefficient by $r_k^{-n}$ times the largest modulus on the circle.

□

**Definition 31.16 (A rational retained cohort).** As stated: the fibre integrals of Gammas, coefficients and source factors whose pins carry a grading.

*Proof.* The share side of 31.4(i) and the kernel of 31.7 are graded by their letter totals: the sum of the rows of the pins that fix the totals of the letters takes the value 1 on the column of every share, each share belonging to one letter. By 31.4(ii), the sector integrals of a fan of a seam word are cube integrals $\int_{[0,1]^k} q^{\boldsymbol{\kappa}(\mathbf{a})-\mathbf{1}} \prod_i P_i(q)^{-w_i(\mathbf{a})} dq$. Here the $P_i$ are the sector letters, with nonnegative coefficients and positive constant terms, and the $\kappa_i$ and $w_i$ are affine with rational linear parts; for tables these are the sector integrals of 8.6. □

*Remark.* The grading is part of the definition because 31.18 extracts the scalar of a joint word through the change of logarithmic scales $z = \rho\mathbf{v} + E\log X$, which needs a vector $\mathbf{v}$ taking the value 1 on every column. Without one the form of that theorem can fail, as follows.

Take the Gramma $\mathsf{G}_{\nu;(1,2)}(\xi_1,\xi_2) = \int_0^\infty x^{\nu-1} e^{-\xi_1 x - \xi_2 x^2} dx$, $\nu, \xi_1, \xi_2 > 0$. Its configuration has no grading, since $\langle 1, v\rangle = \langle 2, v\rangle = 1$ is impossible. Its share side is pinned by $b_1 + 2b_2 = \nu$, and its closing along $(-2, 1)$ crosses $b_1 = -N$, $N \in \mathbb{N}$, with the crossing terms

$$\frac{(-\xi_1)^N}{N!} \cdot \frac{1}{2}\Gamma\Big(\frac{\nu+N}{2}\Big)\xi_2^{-(\nu+N)/2},$$

the terms of the expansion of the integral in powers of $\xi_1$, using $\int_0^\infty x^{\nu+N-1} e^{-\xi_2 x^2} dx = \frac{1}{2}\Gamma(\frac{\nu+N}{2})\xi_2^{-(\nu+N)/2}$.

The scalar $\Gamma(\frac{\nu+N}{2})/\Gamma(N+1)$ has an $N$-th root tending to 0: its logarithm divided by $N$ is $-\frac{1}{2}\log N + O(1)$. A joint word with the scalar of 31.18, here with $r_G = 1$, would in this positive case have the root rate $\operatorname{ess\,sup} c > 0$, or vanishing terms (31.19).

**Theorem 31.18 (A finite procedure for a retained cohort).** *As stated, for every retained cohort, the scalar being written through the grading of its pins.*

*Proof.* This is the reference's proof (the reference, proof of thm:retainedchain); only the radial step involves the totals of the shares.

- At a Gamma crossing, the crossed share is fixed at $-N(n)$, and its Gamma is replaced by the residue $(-1)^{N(n)}/\Gamma(N(n)+1)$.
- The crossed share and its added pin are eliminated together. This leaves the columns $t_l$, $l \neq f$, of the original pins, with the values $\boldsymbol{\nu} + N(n)t_f$, $t_f$ the column of the crossed share.
- By 31.14 each share of $\mathcal{S}_-$ is translated, $b_l = a_l + \delta_l(n)$, with the factor $(-1)^{\delta_l(n)}\xi_l^{\delta_l(n)}/\Gamma(\delta_l(n))$, the power of $\xi_l$ staying in the joint input.

The completed retained fibre is then the Gramma $\mathsf{G}_{\boldsymbol{\mu}_n;t}$ (4.2), with the remaining columns and the index $\boldsymbol{\mu}_n = \boldsymbol{\nu} + N(n)t_f + \sum_{l\in\mathcal{S}_-} \delta_l(n) t_l$. Every remaining column is a column of the original pins, so $\langle t_l, \mathbf{v}\rangle = 1$ for the grading $\mathbf{v}$, and

$$S_n = \langle \mathbf{v}, \boldsymbol{\mu}_n\rangle = w + N(n) + \sum_{l\in\mathcal{S}_-} \delta_l(n).$$

The change of logarithmic scales $z = \rho\mathbf{v} + E\log X$, with $E$ a rational matrix whose columns complement $\mathbf{v}$, gives $\mathsf{G}_{\boldsymbol{\mu}_n;t} = J_{\mathbf{v}}\Gamma(S_n)\int X^{E^{\mathsf{T}}\boldsymbol{\mu}_n{}^{-\mathbf{1}}}\mathcal{L}(X)^{-S_n}dX$: the integral over $\rho$ is $\Gamma(S_n)\mathcal{L}^{-S_n}$. Collecting the residue, the factors of the translations and $\Gamma(S_n)$ gives the stated $G_n$. Its slopes in $n$ agree in the numerator and the denominator, $d_0 + \sum_{\mathcal{S}_-} d_l = \bar{d}$, so the terms $n\log n$ of Stirling's formula cancel, and the rest of Stirling's formula gives $r_G$ and the polynomial bound. On the share sides of 31.4 and 31.7, $\mathbf{v}$ is the sum of the letter-total rows and $w = |\mathbf{s}| + |\mathbf{t}|$. The other steps of the procedure (the source strips, the circles, the joint read, the resolution of the endpoints and the fan) do not involve the grading. □

**Theorem 31.26 (The exact positive rate of a retained chamber).** *As stated, with a real index.*

*Proof.* The reference's proof (the reference, proof of thm:retainedchamberrate) uses rationality only of the configuration and of the direction, through the cohort and its integer step.

- 31.16 allows real constant terms and indices. The crossing times of a falling share are $(a_l^0 + N)/d_l$, $N \in \mathbb{N}$, and passing to the integer step makes the changes of all falling shares integral, whatever the constant terms.
- The positive construction uses 31.14 with $\delta_l(n) = \delta_l(0) + d_l n$ and real starting values.
- The rate is computed by 31.25, at the slope $\sum_f d_f t_f$ of the index, which does not involve its constant term.
- The root limit is that of 31.19, whose positive case allows real constant terms.

The chamber polytope of the statement is the chamber of 31.5 of the share side, whose polar functions are the shares (31.5), for the direction per step. □

**Theorem 31.28 (Averaging is sharp).** *As stated, for every share side of Theorem 31.7 and every direction with rational coordinates whose crossings are simple and nonresonant.*

*Proof.* At each source point the kernel of 31.7(i) is, up to the constant factor $\prod_j \Gamma(s_j)^{-1}\prod_j \Gamma(t_j)^{-1}$, the share side of the Gramma whose configuration is the matrix of the pins, rational by hypothesis, with the grading of the letter totals; whose index is the vector of the values of the pins, real; and whose coefficients are $e^{y_l(\zeta)}$, positive and continuous in $\zeta$ (4.2). Let $\boldsymbol{\sigma}$ be rational and the crossings of its ray simple and nonresonant. A falling share $l$ crosses its poles at the times $(c_l + N)/|\sigma_l|$, $N \in \mathbb{N}$; for a rational $\Delta > 0$ with every $\Delta\sigma_l$ an integer these times fall into finitely many progressions of step $\Delta$, so the crossing terms form finitely many cohorts with the integer step of length $\Delta$ in $\tau$ (31.16), the direction per step being $\Delta\boldsymbol{\sigma}$, from the positive point $\mathbf{c}$.

- 31.26, with a real index, gives every retained term a sign at each index that does not depend on the coefficients, hence not on $\zeta$, so the averaged term has that sign.
- 31.27 applies to each cohort: one averaged term is absolutely integrable by 31.7(i) and its certificate, $\hat{h}_C < \infty$, and the coefficient map is continuous, so the essential supremum of the chamber rate is its supremum on $S_\lambda$.
- The chamber of the step direction is $\Delta C$, and since $\sum_l \sigma_l' = 0$ on $V$, $h(\Delta\boldsymbol{\sigma}'; y) = \Delta h(\boldsymbol{\sigma}'; y)$. So the root rate per step is the limit $e^{\Delta\hat{h}_C}$.

Hence for every $\epsilon > 0$ and every cohort, $|\int \mathcal{R}_n\, d\lambda| \geq e^{n\Delta(\hat{h}_C - \epsilon/2)}$ for all large $n$. The crossing time being $\tau_n = \tau_0 + n\Delta$, this exceeds $e^{\tau_n(\hat{h}_C - \epsilon)}$ for all large $n$, the quotient being $e^{n\Delta\epsilon/2 - \tau_0(\hat{h}_C - \epsilon)}$; the cohorts being finitely many, the bound holds for all large $k$. With 31.7(ii), the terms tend to zero when $\hat{h}_C < 0$ and do not when $\hat{h}_C > 0$.

Now let every falling share be alone in its bundle and the merged fibre have dimension one.

- The merged pins are rational, so the merged directions form a line spanned by a rational vector $\mathbf{r}$. The merged direction of $\boldsymbol{\sigma}$, nonzero since $\boldsymbol{\sigma}$ falls a share, which is alone in its bundle, is $\mu\mathbf{r}$ with $\mu \neq 0$, and $\mu > 0$ after the choice of the sign of $\mathbf{r}$. The falling coordinates of the positive multiple $\mu^{-1}\boldsymbol{\sigma}$ of $\boldsymbol{\sigma}$ are coordinates of $\mathbf{r}$, rational.
- The chamber of $\mu^{-1}\boldsymbol{\sigma}$ is a nonempty polytope defined by rational equations and inequalities, so its vertices have rational coordinates. By 31.6 the closing along $\mu^{-1}\boldsymbol{\sigma}$, which is the closing along $\boldsymbol{\sigma}$ after a change of scale of $\tau$, is the closing along that rational direction.
- At a crossing time the growing shares are positive at the real point of the ray, so the crossing is simple and nonresonant exactly when no other falling share lies in $-\mathbb{N}$ there, that is, when no two lattices of the falling shares meet.

The form of the crossing terms is the reference's (the reference, proof of thm:pool(iv)). The fibre is a bundle over the merged line whose fibres are the splits of the bundles at fixed sums, and 4.4 completes each split, $\int_{\sum_{l\in G} a_l=\bar{a}} \prod_{l\in G} \Gamma(a_l) e^{-a_l y_l} = \Gamma(\bar{a}) e^{-\bar{a} y_G}$. A crossing term on the merged line is a fibre integral of dimension zero: the residue of the Gamma crossed, times the other Gammas at the crossing point and $\prod_G e^{-\bar{a}_{k,G} y_G}$, divided by the Gammas of the exponents and by the determinant of the pins. □

## 50.32 Chapter 32: laws at their ends: regions of equality and reach

**Theorem 32.10 (The reach along a direction is read on the law).** *Parts (i) to (iv) as stated, with part (v).*

*Proof.* Parts (i) to (iv) are the reference's (the reference, proof of thm:dirreach); the piece of tables named in (iii) has a seam word with a Newton sum of full dimension when its tables form a connected family, by 31.4(ii). For (v) fix $\theta \geq 0$ and let $g_\theta$ be the infimum of (v).

First, $r \leq g_\theta$. On the half-strip $\Sigma$ of 32.9 the imaginary parts are bounded, say by $Y$, so $e^{\theta|\operatorname{Im} z|} \leq e^{\theta Y}$ there. A bound $|\ell(z)| \leq C(1+|z|)^{n'} M^{\operatorname{Re} z} e^{\theta|\operatorname{Im} z|}$ on the half-plane, off the $\delta$-neighbourhoods of the candidate poles, therefore gives on $\Sigma$ the bound of 32.9 with $Ce^{\theta Y}$, and $r \leq M$.

Second, $g_\theta \leq r$. By (i), on $(r,\infty)$ the measure $\lambda$ has a density that is a wall series of radius $r$. By (ii) with $x_0 = r$, for every $x' > r$ there is $C$ with $|\ell(z)| \leq C\delta^{-J-1} x'^{\operatorname{Re} z}$ at every $z$ with $\operatorname{Re} z \geq 0$ at distance at least $\delta \leq 1$ from the poles, whatever $\operatorname{Im} z$. The poles with $\operatorname{Re} z \geq 0$ lie among the candidate poles, and those with $\operatorname{Re} z < 0$ keep a distance at least some $\epsilon_0 > 0$ from the half-plane, $\ell$ converging absolutely on a strip about the imaginary axis (32.9). So at every $z$ with $\operatorname{Re} z \geq 0$ at distance at least $\delta$ from the candidate poles, with $\delta_1 = \min(\delta, 1, \epsilon_0)$, $|\ell(z)| \leq C\delta_1^{-J-1} x'^{\operatorname{Re} z}$: the bound of (v) with the constant $C\delta_1^{-J-1}$ and $n' = 0$. So $M = x'$ is admissible in (v) at $\theta = 0$, hence at every $\theta \geq 0$, the factor $e^{\theta|\operatorname{Im} z|}$ being at least 1, and $g_\theta \leq x'$ for every $x' > r$. □

**Theorem 32.15 (Ceilings, simple tops and fibres along a direction).** *As stated, for every piece whose seam function converges at a real point of the pins with positive shares, with the sectors of any fan of its seam word.*

*Proof.* By 31.4(ii) the piece has walls and growth bounds (32.7), its walls being those of the generators of the fan, so 32.9 applies with these walls. The proof is the reference's (the reference,

proof of thm:dirheights), with the sectors of the fan in place of the orders of $B$, the generators in place of the tails, and the constant $|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})|$ in the sector forms. At the point $\mathbf{w}_0 + i\bar{\boldsymbol{\vartheta}} + z\bar{\boldsymbol{\sigma}}$ the integrand of the sector integral of $\pi$ is

$$|\det(\mathbf{v}_{\pi,1},\dots,\mathbf{v}_{\pi,|B|-1})| \prod_k q_k^{\kappa_k(\mathbf{w}_0)+i\partial_{\bar{\boldsymbol{\vartheta}}}\kappa_k-1}\, \mathbf{k}_\pi^{-\mathbf{w}_0-i\bar{\boldsymbol{\vartheta}}}\, \varrho_{\pi,\bar{\boldsymbol{\sigma}}}^{z}.$$

(i) The proof of 31.11(i) concerns the seam function alone. It holds at the points $\mathbf{w}_0 + \operatorname{Re} z\,\bar{\boldsymbol{\sigma}} + i(\bar{\boldsymbol{\vartheta}} + \operatorname{Im} z\,\bar{\boldsymbol{\sigma}})$, with $\delta$ the distance of the complex point from the hyperplanes on which a wall lies in $-\mathbb{N}$, which is at least a constant times the distance of $z$ from the candidate poles (the reference, proof of cor:dirtube). On $\Sigma$ the phase is bounded, the imaginary parts being bounded there. So $|\ell(z)| \le C\delta^{-(|B|-1)} e^{\operatorname{Re} z(\log H_{\mathcal{O}}+\epsilon)}$, and $r \le H_{\mathcal{O}} e^{\epsilon}$ for every $\epsilon > 0$.

(ii) and (iii) follow from the proof of 20.51(i) and (iii), with the point $\mathbf{w}_0 + m\bar{\boldsymbol{\sigma}}$ in place of $b^d = -k$, $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ in place of $\varrho_\pi$ and $\phi_{\pi,\mathbf{w}_0}$ in place of $\phi_\pi$.

- The falling variables carry $q_k^{\kappa_k(\mathbf{w}_0)-m|\dot{\kappa}_k|-1}$ with $|\dot{\kappa}_k|$ an integer, so the factor $(e^{2\pi i\kappa_k(\mathbf{w}_0)}-1)^{-1}$ of their circles is the same at every $m$. The letters whose exponents are nonnegative integers at every large $m$ carry polynomials and need not be kept from vanishing.
- Every wall is the wall of a generator of some sector, so its falling rate is one of the integers $|\dot{\kappa}_k|$. The points $m$ lie at a distance bounded below from the candidate poles, the falling walls being off the integers at $\mathbf{w}_0$, so $|\ell(m)| \le C(1+m)^D r'^m$ for every $r' > r$ by 32.9, and $\operatorname{growth}_{\mathbb{Z}} \le r$.
- Under (iii), $\operatorname{growth}_{\mathbb{Z}} = H_{\mathcal{O}}$ as in 20.51(iii), and a keyhole continuation of ceiling $H_{\mathcal{O}}$ gives $r \le H_{\mathcal{O}}$ by (i).

(iv) follows from the proofs of 20.55 and 20.56, applied to the positive directional law, with 32.10(i) and (ii) in place of 20.44.

- A keyhole box is handled as in (i).
- In a fibre box, $\varrho_{\pi,\bar{\boldsymbol{\sigma}}}$ is meromorphic in each falling variable on its disc, the exponents $\bar{\sigma}_L$ and $\dot{\kappa}_j$ being integers. At a point above the ceiling and the stationary bound, the map is a submersion on its stratum: by 20.52 when one coordinate of falling rate one lies inside its disc, and by the definition of the stationary bound otherwise.
- The Mellin transform of a fibre law is a cube integral as in 20.32, with its poles in finitely many cosets of $(\operatorname{lcm}_j |\dot{\kappa}_j|)^{-1}\mathbb{Z}$, so 20.53 applies.

None of these steps uses more of a sector integral than the form displayed above, which 31.4(ii) supplies for every fan of the seam word of every such piece. □

**Theorem 32.16 (The closing past the walls is read along its direction).** *As stated, with the last sentence of (i), for the share side of 31.4(iv).*

*Proof.* Apart from the last sentence of (i), the statement is the reference's (the reference, proof of thm:dirclosing). For that sentence, the piece and the partner have walls and growth bounds (31.4(ii), applied to each), and $\boldsymbol{\sigma}$ has rational coordinates, so 32.12 applies to each of the two seam functions at $\mathbf{c}$, over the whole space $V$ of the directions of the shares. There the growth $g_X(\boldsymbol{\sigma};\theta)$ of 31.4(iv), $X \in \{B,R\}$, is the function $g_V(\boldsymbol{\sigma};\theta)$ of 32.12 for that seam function: the two definitions coincide, the shares carrying no sums and the walls of 31.4(ii) serving as the walls of (W1), so that $\delta$ is measured from the same hyperplanes, those of the polar functions of the seam function (31.5).

It does not increase with $\theta$ and tends to $r_X^\sharp$ as $\theta \downarrow 0$, so $g_X(\boldsymbol{\sigma};\theta) \le r_X^\sharp$ for every $\theta > 0$. Then

$$h_{\theta/2,\theta/2}(\boldsymbol{\sigma}) \le \sum_l \sigma_l \log|\sigma_l| + \log r_B^\sharp + \log r_R^\sharp + \frac{2\theta^2}{\pi} \sum_{l:\,\sigma_l>0} \sigma_l,$$

which is negative for $\theta$ small when $\sum_l \sigma_l \log|\sigma_l| + \log r_B^\sharp + \log r_R^\sharp < 0$. 31.4(iv) gives the closing, its terms and remainders being $O(e^{\tau(h_{\theta/2,\theta/2}(\boldsymbol{\sigma})+\epsilon)})$, and letting $\theta \to 0$ gives the stated bound. □

**Corollary 32.18 (A law with log expansions bounds a closing of any fibre dimension).** *As stated, with its last sentence, for the share side of 31.4(iv).*

*Proof.* Apart from the last sentence, the statement is the reference's (the reference, proof of cor:anyfibre). For that sentence, apply 32.14 to the law of the piece and to that of the partner, a piece with the reversed excesses. On the contour through $\mathbf{c} + \tau\boldsymbol{\sigma}'$ each seam function is at most $C\delta^{-n} X_{\text{in}}^X(\boldsymbol{\sigma}')^\tau$, with no factor in the imaginary parts and no power of $\tau$, so $g_X(\boldsymbol{\sigma}';0) \le X_{\text{in}}^X(\boldsymbol{\sigma}')$ for every direction $\boldsymbol{\sigma}'$. Hence

$$h_{0,0}(\boldsymbol{\sigma}') \le \sum_l \sigma_l' \log|\sigma_l'| + \log X_{\text{in}}^B(\boldsymbol{\sigma}') + \log X_{\text{in}}^R(\boldsymbol{\sigma}'), \qquad \boldsymbol{\sigma}' \in C(\boldsymbol{\sigma}).$$

The polar functions of the share side of 31.4(iv) are the shares and the walls of both words, with $\Omega = A$, and (H1) holds by 31.4(ii) and Stirling's formula. So 31.6 makes the closings along the directions of the chamber one series; their crossings and the near coincidences of these are those of the falling polar functions, the same for every direction of the chamber, so $\mathbf{c}$ lies off the null set of 31.4(iv) for each of them; and 31.4(iv) at $\theta_B = \theta_R = 0$, along a direction of the chamber whose bound is within $\epsilon$ of the infimum, gives the closing along $\boldsymbol{\sigma}$ with the stated bound. On the chamber the falling coordinates are fixed and the others are nonnegative, where $x \log x$ is convex, and the logarithms of the inner bounds are convex (32.13). □

*Remark* (The directional ratio at width one, in the paragraph after 32.9). At width one with one table, factoring weights normalized as in 28.11, and $s, t > 0$, the directional ratio $x_{D_0}^*/x_{D_1}$ along $\bar{\boldsymbol{\sigma}} = (-1, 1, -1, 1)$ is the ratio $\varrho_B$ at the seam points, and every coordinate of the seam torus there is a power of $\varrho_B$.

*Proof.* On the seam torus $x_{D_0} = x_{D_1}^* = 1$, so $\mathbf{x}^{-\bar{\boldsymbol{\sigma}}} = x_{D_0}\, x_{D_1}^{-1}\, x_{D_0}^*\, x_{D_1}^{*-1} = x_{D_0}^*/x_{D_1}$. At the seam point of $u$, 28.5 gives $\varrho_{D_0}^{\rho*}/\varrho_{D_1}^{\rho} = V_{D_0}^* V_{D_0}/(V_{D_1}^* V_{D_1})$, the scale cancelling. By Lemma S13(i) this is

$$\Big(\frac{\hat\theta_0}{\hat\theta_1}\Big)^2 \frac{\Psi_{\mathcal{B}/R}\, K_{\mathcal{B}/R}}{\Psi_{\mathcal{B}|B}\, K_{\mathcal{B}|B}} = \Big(\frac{\hat\theta_0}{\hat\theta_1}\Big)^2 \varrho_B,$$

with $\varrho_B = \gamma_B^\Psi/\gamma_B^K$ of 19.2. Under the normalization $\hat\theta_0 = \hat\theta_1 = 1$, so it is $\varrho_B$. By 28.11(ii) with $N = 1$, $\mathbf{x} = (1, \varrho_B)$ and $\psi = \varrho_B^{t/(s+t)}$, the coordinates of the seam point are 1, $\varrho_B^{-t/(s+t)}$, $\varrho_B^{s/(s+t)}$ and 1, all powers of $\varrho_B$. □

## 50.33 Chapter 34: the motive of a family and the coaction along a tree

Five outside works enter these proofs, each named where it is used together with the property of the setting it needs: the theory of toric varieties (W. Fulton, *Introduction to toric varieties*, Princeton

University Press, 1993, Chapters 1 to 3), for the braid fan; F. Brown, *Notes on motivic periods*, Commun. Number Theory Phys. 11 (2017), 557–655 (*Brown's notes*), for motivic periods, the coaction and the weight-graded face maps; F. Brown, *Feynman amplitudes, coaction principle, and cosmic Galois group*, Commun. Number Theory Phys. 11 (2017), 453–556, §5, for the affine model; A. Huber and S. Müller-Stach, *Periods and Nori motives*, Springer, 2017, for the functoriality of relative cohomology in Nori's category, whose objects and morphisms realize in Brown's category; and H. Hironaka, *Resolution of singularities of an algebraic variety over a field of characteristic zero*, Ann. of Math. 79 (1964), 109–326. In this section a *pair* $(V, D)$ is a smooth variety $V$ over $\mathbb{Q}$ of dimension $m$ with a divisor $D \subset V$ with simple normal crossings and smooth components. Its relative cohomology $H^m(V, D)$ is an object of Brown's category, a morphism of pairs giving a morphism of Nori's category and so of Brown's (Huber and Müller-Stach). A regular form $\eta$ of degree $m$ on $V$ is closed and restricts to zero on the components of $D$, which have smaller dimension; $[\eta]$ is its class in $H^m_{\mathrm{dR}}(V, D)$. A compact semialgebraic $\sigma \subset V(\mathbb{R})$ that is the closure of an oriented open semialgebraic set $\sigma^\circ \subseteq (V \setminus D)(\mathbb{R})$ of dimension $m$, with $\sigma \setminus \sigma^\circ \subseteq D$, is a *relative cycle*: a semialgebraic triangulation makes the sum of its oriented top simplices a chain whose boundary lies in $D(\mathbb{R})$, and its class in $H_m(V(\mathbb{C}), D(\mathbb{C}); \mathbb{Q})$ is written $\sigma$ again. The comparison of the two realizations is integration, so the period of $[H^m(V, D), \sigma, [\eta]]^{\mathrm{mot}}$ is $\int_\sigma \eta$. The positive cells $\widetilde{\sigma}_Z$ are oriented as in Definition 34.5.

**Definition 34.1 (The model of a family).** As stated.

*Proof.* $\Omega_{\mathcal{E}}$ is the contraction of $d\alpha_{e_1} \wedge \cdots \wedge d\alpha_{e_n}$ with the vector field $\sum_e \alpha_e \, \partial/\partial\alpha_e$, which does not depend on the order, while the volume form changes by the sign of the permutation. The coefficient $f = \alpha^{\mathbf{m-1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}$ of $\omega_F = f \, \Omega_{\mathcal{E}}$ is homogeneous of the degree $m_{\mathcal{E}} - \sum_j (s_j r_j + t_j r_j^*) - n = \kappa_{\mathcal{E}} - n = -n$ (Definition 8.1), the masses being on the balance wall. With the affine coordinates $x_i = \alpha_{e_i}/\alpha_{e_n}$, $i < n$, of the chart $\alpha_{e_n} \neq 0$ of $\mathbb{P}(\mathbb{Q}^{\mathcal{E}})$, a direct computation gives $dx_1 \wedge \cdots \wedge dx_{n-1} = (-1)^{n-1} \alpha_{e_n}^{-n} \Omega_{\mathcal{E}}$, so that $f \, \Omega_{\mathcal{E}} = (-1)^{n-1} f(x_1, \ldots, x_{n-1}, 1) \, dx_1 \wedge \cdots \wedge dx_{n-1}$ is the pull-back of a rational form on that chart. These forms agree on the overlaps of the charts, all pulling back to $f \, \Omega_{\mathcal{E}}$ on $\mathbb{Q}^{\mathcal{E}} \setminus \{0\}$; so $\omega_F$ is a rational form of degree $n-1$ on $\mathbb{P}(\mathbb{Q}^{\mathcal{E}})$, and on $X_{\mathcal{E}}$, which contains the torus of $\mathbb{P}(\mathbb{Q}^{\mathcal{E}})$ as a dense open set. □

**Definition 34.2 (Flags and steps).** As stated.

*Proof.* The steps are $\ell + 1$ nonempty sets that partition $\mathcal{E}$, so $\sum_i (|Z_i| - 1) = n - 1 - \ell$. For $\ell \geq 1$ the step lists at $Z_i$ depend on $S_{i-1}$ and $S_i$ alone (Proposition 12.20), and the $\alpha_J$ with $J \subseteq Z_i$ on $Z_i$ alone; so does $\mathrm{Let}(Z_i; \mathfrak{f})$. When $|Z_i| = 1$, $X_{Z_i}$ is the toric variety of the fan $\{0\}$ in $\mathbb{R}^{Z_i}/\mathbb{R}\mathbf{1} = 0$, a point, and the step has no letter: the basis polynomials of the lists on one mark are 1 and $\alpha_e$, monomials, and no $J \subseteq Z_i$ has two marks. □

**Lemma 34.3 (Charts, strata and the letters in a chart).** *As stated.*

*Proof.* (i) The fan lies in $N_{\mathbb{R}}$, $N = \mathbb{Z}^{\mathcal{E}}/\mathbb{Z}\mathbf{1}$, with the dual lattice $M = \{\mathbf{a} \in \mathbb{Z}^{\mathcal{E}} : \; a_{\mathcal{E}} = 0\}$, whose elements are the characters $\alpha^{\mathbf{a}}$ of the torus. For an order $e_1, \ldots, e_n$ the vectors $\mathbf{1}_{S_1}, \ldots, \mathbf{1}_{S_{n-1}}, \mathbf{1}$ form a basis of $\mathbb{Z}^{\mathcal{E}}$, their matrix in the basis $(\mathbf{e}_{e_k})_k$ being triangular with ones on the diagonal; so the images of $\mathbf{1}_{S_1}, \ldots, \mathbf{1}_{S_{n-1}}$ form a basis of $N$, with the dual basis $\mathbf{e}_{e_i} - \mathbf{e}_{e_{i+1}}$, $i < n$, of $M$, since $\langle \mathbf{1}_{S_k}, \mathbf{e}_{e_i} - \mathbf{e}_{e_{i+1}} \rangle$ is 1 for $k = i$ and 0 otherwise. Every $\mathbf{x} \in \mathbb{R}^{\mathcal{E}}$ lies in such a cone modulo $\mathbf{1}$: ordering $\mathcal{E}$ with $x_{e_1} \geq \cdots \geq x_{e_n}$, $\mathbf{x} = \sum_{i<n} (x_{e_i} - x_{e_{i+1}}) \mathbf{1}_{S_i} + x_{e_n} \mathbf{1}$. The cones of two chains meet in the cone of their common sets, a face of both. So the braid fan is a complete fan whose cones are spanned by parts of bases of $N$; it is the normal fan of the polytope $\sum_{\{a,b\}} \mathrm{conv}\{\mathbf{e}_a, \mathbf{e}_b\}$, the sum over the pairs of marks, since on the interior of the cone of an order a linear form $\mathbf{x}$ has

$x_{e_1} > \cdots > x_{e_n}$ and attains its least value on that polytope at the one vertex $\sum_{\{a,b\}} \mathbf{e}_{\max(a,b)}$, the maximum taken in the order, which is the same on the whole cone and differs for different orders. By the dictionary between fans and toric varieties (Fulton, Chapters 1 to 3), which applies to such a fan, $X_{\mathcal{E}}$ is smooth and complete, and projective, the fan being a normal fan; the chart of the cone of the chain is the affine space with the coordinates $\alpha^{\mathbf{e}_{e_i}-\mathbf{e}_{e_{i+1}}} = \varepsilon_i$; the charts of the maximal cones cover $X_{\mathcal{E}}$; and the closure $D_S$ of the orbit of the ray $\rho_S$ meets the chart exactly when $\rho_S$ is a ray of its cone, in the coordinate hyperplane of the dual coordinate, here $\{\varepsilon_i = 0\}$ for $S = S_i$, several of these closures meeting exactly when their rays span a cone, here when their sets form a chain (the orbit–cone correspondence). In every chart $D$ is a union of coordinate hyperplanes, so it has simple normal crossings and smooth components. In every chart the torus is $\{\varepsilon_i \neq 0 \text{ for all } i\}$ and $\widetilde{\sigma}^\circ_{\mathcal{E}} = (0,\infty)^{n-1}$, since $\alpha_{e_j}/\alpha_{e_n} = \prod_{i\geq j} \varepsilon_i$; its closure in the open set $U(\mathbb{R})$ is $[0,\infty)^{n-1} = \widetilde{\sigma}_{\mathcal{E}} \cap U$. The points of $\widetilde{\sigma}^\circ_{\mathcal{E}}$ with $\alpha_{e_1} < \cdots < \alpha_{e_n}$ are $(0,1)^{n-1}$ in the chart, with the closure $[0,1]^{n-1}$, compact and so closed in $X_{\mathcal{E}}(\mathbb{R})$. Every point of $\widetilde{\sigma}^\circ_{\mathcal{E}}$ is a limit of points with distinct coordinates, so $\widetilde{\sigma}^\circ_{\mathcal{E}}$ lies in the union of the sectors, a finite union of compact sets, and so does its closure $\widetilde{\sigma}_{\mathcal{E}}$, which contains every sector. The open cube $(0,1)^{n-1}$ of one order is open in $X_{\mathcal{E}}(\mathbb{R})$ and misses the open sector of every other order, hence its closure; so two sectors meet in the boundary of a cube, of measure zero.

(ii) For a basis $T$ of a table of rank $\delta$, $\alpha^T/\alpha^\delta_{e_n} = \alpha^{\mathbf{1}_T - \delta\mathbf{e}_{e_n}} = \prod_{i<n} \varepsilon_i^{|T\cap S_i|}$, by the dual basis of (i) and $e_n \notin S_i$. So $K_{\mathcal{B}}/\alpha^\delta_{e_n}$ is the sum of these monomials over the bases. Every exponent $|T \cap S_i|$ is at least $d(S_i)$ (Theorem 7.13), and along the maximal chain $(S_i)_{i<n}$ exactly one basis has $|T \cap S_i| = d(S_i)$ for every $i$, its term having the coefficient 1 (Proposition 12.20); dividing by its monomial leaves $\widetilde{f}$, a polynomial with nonnegative integer coefficients, the numbers of bases with given traces, and with the constant term 1. For $\Psi_{\mathcal{B}} = K_{\mathcal{B}^*}$ the same holds with the dual table, whose floors are the $d^*(S)$ (Proposition 7.2, Theorem 7.13), and for $\alpha_J$ with the list $\binom{J}{1}$ on $\mathcal{E}$, whose floor at $S$ is 1 or 0 as $J \subseteq S$ or not. In the sector coordinates of Theorem 8.5(ii), with $\pi(k) = e_{n+1-k}$, the coordinates are $q_k = \varepsilon_{n+1-k}$ and the tails are $U_k = S_{n+1-k}$, and $\widetilde{f}$ is the sector letter there. On $\varepsilon \geq 0$, $\widetilde{f} \geq 1$.

On $\{\varepsilon_i = 0 :\ i \in I\}$ only the bases $T$ with $|T \cap S_i| = d(S_i)$ for every $i \in I$ leave a term. By Proposition 12.20 for the chain $(S_i)_{i\in I}$ they are the unions $\bigsqcup_Z X_Z$ of one basis $X_Z$ of the step table at the bottom at each step $Z$ of that chain. For $i \notin I$ let $Z = S_b \setminus S_a$ be the step with $S_a \subsetneq S_i \subsetneq S_b$, where $S_a$ may be $\varnothing$ and $S_b$ may be $\mathcal{E}$; then $|T \cap S_i| = d(S_a) + |X_Z \cap (S_i \setminus S_a)|$. The basis at the bottom of the maximal chain is one of these unions and has the least traces, so $d(S_i) - d(S_a)$ is the floor $d_Z(S_i \setminus S_a)$ of $S_i \setminus S_a$ in the step table at $Z$, and

$$\widetilde{f}\big|_{\{\varepsilon_i = 0:\, i\in I\}} = \prod_Z \sum_{X_Z} \prod_{a<k<b} \varepsilon_k^{|X_Z \cap (S_k \setminus S_a)| - d_Z(S_k \setminus S_a)}.$$

The factor of $Z$ is the polynomial that the rule above attaches to $K$ of the step table in the chart of $X_Z$ of the order $e_{a+1}, \ldots, e_b$, whose initial segments are the $S_k \setminus S_a$ and whose coordinates are the $\varepsilon_k$, $a < k < b$. For $\Psi$ the same holds with the tops and the dual tables, and for $\alpha_J$ with its step letters (Proposition 12.20). Finally, on the torus part of $U$, $f = 0$ exactly where $\widetilde{f} = 0$. No irreducible component of $\{\widetilde{f} = 0\}$ lies in a coordinate hyperplane $\{\varepsilon_i = 0\}$, since $\varepsilon_i$ does not divide $\widetilde{f}$, whose constant term is 1; so each component is the closure of its part in the torus, and $\{\widetilde{f} = 0\}$ is the closure in $U$ of the zero set of $f$ in the torus, that is $\widetilde{Y}_f \cap U$.

(iii) In the affine coordinates $x_j = \alpha_{e_j}/\alpha_{e_n} = \prod_{i\geq j} \varepsilon_i$, $\Omega_{\mathcal{E}}/\alpha^{\mathcal{E}} = (-1)^{n-1} \bigwedge_{j<n} dx_j/x_j$ (proof of Definition 34.1), and $\bigwedge_j d\log x_j = \bigwedge_i d\log \varepsilon_i$, the matrix of the $\log x_j$ in the $\log \varepsilon_i$ being triangular with ones on the diagonal; so $\Omega_{\mathcal{E}}/\alpha^{\mathcal{E}} = (-1)^{n-1} \bigwedge_{i<n} d\varepsilon_i/\varepsilon_i$. By (ii), $\alpha^{\mathbf{m}}/\alpha^{m_{\mathcal{E}}}_{e_n} = \prod_i \varepsilon_i^{m_{S_i}}$, $K_j/\alpha^{r_j}_{e_n} = \prod_i \varepsilon_i^{d_j(S_i)} \widetilde{K}_j$ and $\Psi_j/\alpha^{r^*_j}_{e_n} = \prod_i \varepsilon_i^{d^*_j(S_i)} \widetilde{\Psi}_j$. On the balance wall the powers of $\alpha_{e_n}$ cancel,

and the exponent of $\varepsilon_i$ is $m_{S_i} - \sum_j \big(s_j d_j(S_i) + t_j d_j^*(S_i)\big) - 1 = \kappa_{S_i} - 1$. □

**Lemma S15** (Integrals over the positive cell)**.** *Let $\phi$ be a measurable function on $\mathbb{R}^{\mathcal{E}}_{>0}$, homogeneous of degree zero, $n = |\mathcal{E}| \geq 2$. Then, in $[0,\infty]$ when $\phi \geq 0$, and whenever one side converges absolutely,*

$$\int_{\widetilde{\sigma}^\circ_{\mathcal{E}}} \phi\, \frac{\Omega_{\mathcal{E}}}{\alpha^{\mathcal{E}}} = \int_{\Delta_{\mathcal{E}}} \phi(u)\, u^{-\mathbf{1}}\, d\omega_{\mathcal{E}}(u) = \int_{\Delta_B \times \Delta_R} \int_0^\infty \phi(\rho u, u')\, \frac{d\rho}{\rho}\; u^{-\mathbf{1}} u'^{-\mathbf{1}}\, d\omega_B(u)\, d\omega_R(u')$$

*for every partition $\mathcal{E} = B \sqcup R$ into nonempty sets; in the chart of an order the left side is $\int_{(0,\infty)^{n-1}} \phi \prod_i d\varepsilon_i/\varepsilon_i$. If $\phi\, \Omega_{\mathcal{E}}/\alpha^{\mathcal{E}}$ is the restriction of a form that is regular on an open set containing $\widetilde{\sigma}_{\mathcal{E}}$, the integrals converge absolutely and are its integral over $\widetilde{\sigma}_{\mathcal{E}}$.*

*Proof.* In the logarithmic coordinates $y = \log \alpha$, $\alpha^{-\mathbf{1}} d\alpha = dy$. For an order of $\mathcal{E}$ the linear map $y \mapsto \big((y_{e_i} - y_{e_{i+1}})_{i<n}, y_{e_n}\big)$ has the determinant $\pm 1$, and its first $n-1$ coordinates are the $\log \varepsilon_i$, on which $\phi$ depends alone. For a positive $h$ homogeneous of degree one and a measurable $\varphi \geq 0$ on $(0,\infty)$, at fixed $\varepsilon$ the substitution $\rho = e^{y_{e_n}} h(\alpha/\alpha_{e_n})$ has $d\rho/\rho = dy_{e_n}$; by Tonelli

$$\int_{\mathbb{R}^{\mathcal{E}}_{>0}} \phi(\alpha)\, \varphi\big(h(\alpha)\big)\, \alpha^{-\mathbf{1}} d\alpha = \int_0^\infty \varphi(\rho) \frac{d\rho}{\rho} \int_{(0,\infty)^{n-1}} \phi \prod_i \frac{d\varepsilon_i}{\varepsilon_i},$$

the same holds with $|\phi|$, and with $\phi$ whenever the integrals of $|\phi|$ are finite. With $h = |\alpha|$, Lemma 2.7 on $\mathcal{E}$ writes the left side as $\int_0^\infty \varphi(\rho) \frac{d\rho}{\rho} \int_{\Delta_{\mathcal{E}}} \phi(u) u^{-\mathbf{1}} d\omega_{\mathcal{E}}(u)$. With $h = |\alpha_R|$, Lemma 2.7 on $B$ and on $R$, $\alpha_B = \rho' u$ and $\alpha_R = \lambda u'$, gives $\alpha^{-\mathbf{1}} d\alpha = u^{-\mathbf{1}} u'^{-\mathbf{1}} \frac{d\rho'}{\rho'} \frac{d\lambda}{\lambda} d\omega_B(u) d\omega_R(u')$ and $\phi(\rho' u, \lambda u') = \phi(\rho u, u')$ with $\rho' = \lambda\rho$, so the left side is the third expression times $\int \varphi(\lambda) d\lambda/\lambda$. A $\varphi$ with $\int_0^\infty \varphi\, d\rho/\rho = 1$ gives the identities. In the chart $\Omega_{\mathcal{E}}/\alpha^{\mathcal{E}} = (-1)^{n-1} \bigwedge_i d\varepsilon_i/\varepsilon_i$ (proof of Lemma 34.3(iii)), which with the orientation of $\widetilde{\sigma}^\circ_{\mathcal{E}} = (0,\infty)^{n-1}$ is the measure $\prod_i d\varepsilon_i/\varepsilon_i$. If the form is regular near the compact $\widetilde{\sigma}_{\mathcal{E}}$, it is $g\, d\varepsilon$ on each sector cube with $g$ continuous on the cube; so its integral over each sector converges absolutely, the sectors cover $\widetilde{\sigma}_{\mathcal{E}}$ and meet in sets of measure zero, and $\widetilde{\sigma}_{\mathcal{E}} \setminus \widetilde{\sigma}^\circ_{\mathcal{E}}$ lies in the coordinate hyperplanes of the charts (Lemma 34.3(i)). □

**Lemma S16** (The continued period off the balance wall)**.** *Let $F$ be a family of lists on $\mathcal{E}$, $n \geq 2$. For an order $\pi$ with the chain $(S_i^\pi)_{i<n}$ let $\widetilde{K}_j, \widetilde{\Psi}_j$ and $\widetilde{L}_\pi$ be the sector letters of $K_j, \Psi_j$ and of the linear letter $\alpha_{\mathcal{E}} = |\alpha|$ in its chart (Lemma 34.3(ii)), and $h_\pi(\varepsilon; \kappa, \mathbf{s}, \mathbf{t}) = \widetilde{L}_\pi^{-\kappa} \prod_j \widetilde{K}_j^{-s_j} \widetilde{\Psi}_j^{-t_j}$. As meromorphic functions of $(\mathbf{m}, \mathbf{s}, \mathbf{t}) \in \mathbb{C}^{\mathcal{E}} \times \mathbb{C}^N \times \mathbb{C}^N$,*

$$\frac{J_F(\mathbf{m}; \mathbf{s}, \mathbf{t})}{\Gamma(\kappa_{\mathcal{E}})} = \sum_\pi \operatorname{cube}\Big(\big(\kappa_{S_i^\pi}\big)_{i<n},\ (\kappa_{\mathcal{E}}, \mathbf{s}, \mathbf{t});\ h_\pi\Big),$$

*with* cube *the cube integral of Lemma 8.4; the right side is holomorphic off the hyperplanes $\kappa_S = -j$, $\varnothing \neq S \subsetneq \mathcal{E}$, $j \in \mathbb{N}$. So the period $P_F = J_F/\Gamma(\kappa_{\mathcal{E}})$ (Theorem 8.7) has its poles on these hyperplanes only, and on the balance wall it is the right side at $\kappa_{\mathcal{E}} = 0$, where $h_\pi = \prod_j \widetilde{K}_j^{-s_j} \widetilde{\Psi}_j^{-t_j}$.*

*Proof.* On the region where $\operatorname{Re} \kappa_S > 0$ for every nonempty $S \subseteq \mathcal{E}$, which contains the points with $\mathbf{s} = \mathbf{t} = 0$ and $\operatorname{Re} \mathbf{m} > 0$, Theorem 8.5(ii) is the display, read in the chart coordinates as in the proof of Lemma 34.3(ii): its sector letters $k_{j,\pi}, \psi_{j,\pi}, L_\pi$ are $\widetilde{K}_j, \widetilde{\Psi}_j, \widetilde{L}_\pi$. The sector letters are at least 1 on the cube, so $h_\pi$ is smooth in $\varepsilon$ near $[0,1]^{n-1}$, with every $\varepsilon$-derivative continuous in all the variables and holomorphic in $(\kappa, \mathbf{s}, \mathbf{t}) \in \mathbb{C}^{1+2N}$. The map $(\mathbf{m}, \mathbf{s}, \mathbf{t}) \mapsto \big((\kappa_{S_i^\pi})_{i<n}, \kappa_{\mathcal{E}}, \mathbf{s}, \mathbf{t}\big)$ is a linear isomorphism of $\mathbb{C}^{\mathcal{E}} \times \mathbb{C}^{2N}$, since $\mathbf{m} \mapsto (m_{S_1^\pi}, \ldots, m_{S_{n-1}^\pi}, m_{\mathcal{E}})$ is triangular. Lemma 8.4 continues

each term to a meromorphic function on the whole space with poles on the hyperplanes $\kappa_{S_i^\pi} = -j$ only. The sum agrees with the meromorphic $J_F/\Gamma(\kappa_\mathcal{E})$ (Theorems 8.5(iii) and 8.7) on a nonempty open set, hence everywhere. At $\kappa_\mathcal{E} = 0$, $\widetilde{L}_\pi^{-\kappa_\mathcal{E}} = 1$. □

**Proposition 34.4 (The model).** *As stated.*

*Proof.* (i) $\widetilde{\sigma}_\mathcal{E}$ is the union of the finitely many sectors, cubes (Lemma 34.3(i)), so it is compact. In a chart $\widetilde{\sigma}_\mathcal{E} \cap U = [0,\infty)^{n-1}$ and $\widetilde{\sigma}_\mathcal{E}^\circ = (0,\infty)^{n-1}$, whose difference lies in the coordinate hyperplanes, that is in $D$; and the charts cover $X_\mathcal{E}$. For a letter $f$ of a list or a linear letter, $\widetilde{Y}_f \cap U = \{\widetilde{f} = 0\}$ and $\widetilde{f} \geq 1$ on $[0,\infty)^{n-1}$ (Lemma 34.3(ii)).

(ii) Let $\varnothing \neq S \subsetneq \mathcal{E}$ and take an order whose chain contains $S = S_i$. $D_S$ meets $V_F$: the face $\{\varepsilon_i = 0\} \cap [0,1]^{n-1}$ of the sector cube lies in $D_S \cap \widetilde{\sigma}_\mathcal{E}$, and $\widetilde{\sigma}_\mathcal{E} \subseteq V_F(\mathbb{R})$ by (i). On $V_F \cap U$ every $\widetilde{K}_j$ with $s_j \neq 0$ and every $\widetilde{\Psi}_j$ with $t_j \neq 0$ is a unit, $V_F$ omitting their zero sets, so by Lemma 34.3(iii) $\omega_F$ is a unit times $\prod_k \varepsilon_k^{\kappa_{S_k}-1} d\varepsilon_1 \wedge \cdots \wedge d\varepsilon_{n-1}$ there. Its order along $D_S$, read at a point of $D_S \cap V_F \cap U$, is $\kappa_S - 1$, and on the torus, where every $\varepsilon_k$ is a unit, it has neither zeros nor poles. A prime divisor of $V_F$ either meets the torus or is one of the $D_S \cap V_F$, each irreducible as a nonempty open subset of the irreducible $D_S$. So the divisor of $\omega_F$ on the smooth $V_F$ is $\sum_S (\kappa_S - 1)(D_S \cap V_F)$, and $\omega_F$ is regular exactly when this divisor is effective, that is when $F$ is convergent. On the cube of a sector, $\omega_F = \pm \prod_k \varepsilon_k^{\kappa_{S_k^\pi}-1} h_\pi \, d\varepsilon$ with $h_\pi = \prod_j \widetilde{K}_j^{-s_j} \widetilde{\Psi}_j^{-t_j}$ positive and continuous on the compact cube, so bounded between two positive constants; as $\int_0^1 \varepsilon^{\kappa-1} d\varepsilon$ converges for $\kappa \geq 1$, the integral over $\widetilde{\sigma}_\mathcal{E}$ converges when $F$ is convergent. By Lemma S15 it is the integral over $\Delta_\mathcal{E}$ of $u^{\mathbf{m}-\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}$, the period (Theorem 8.7), positive with the orientation in which $\Omega_\mathcal{E}/\alpha^\mathcal{E}$ is positive on $\widetilde{\sigma}_\mathcal{E}^\circ$.

(iii) $V_F$ is open in the smooth $X_\mathcal{E}$. The complement in $X_\mathcal{E}$ of the $\widetilde{Y}_{\alpha_J}$, $|J| \geq 2$, is Brown's affine model (Brown, *Feynman amplitudes*, §5), which applies because $X_\mathcal{E}$ is the iterated blow-up of $\mathbb{P}(\mathbb{Q}^\mathcal{E})$ along its coordinate linear subspaces, the toric variety of the subdivision of its fan by all chains, and the removed hypersurfaces are the strict transforms of the hyperplanes $\alpha_J = 0$ of all the sets $J$ of at least two marks. In toric terms the divisor $A = \sum_{|J|\geq 2} \widetilde{Y}_{\alpha_J}$ is ample. In the chart of an order, $\alpha_J/\alpha_y = \widetilde{\alpha}_J$ for the last mark $y$ of $J$ in the order (Lemma 34.3(ii)), so $\widetilde{Y}_{\alpha_J}$ is the divisor of the section $\sum_{e\in J} \alpha^{\mathbf{e}_e}$ of the line bundle of the simplex $\Delta_J = \operatorname{conv}\{\mathbf{e}_e : \ e \in J\}$, whose vertex of least value on the cone of the order is $\mathbf{e}_y$; $A$ is then the divisor of a section of the line bundle of the polytope $\sum_J \Delta_J$, whose vertex of least value on the interior of the cone of the order $e_1, \ldots, e_n$ is $\sum_J \mathbf{e}_{y_J} = \sum_k (2^{k-1} - 1)\, \mathbf{e}_{e_k}$, the mark $e_k$ being the last mark of $2^{k-1} - 1$ sets $J$; it is the only point of least value there and differs for different orders. So the normal fan of $\sum_J \Delta_J$ is the braid fan, the divisor $A$ is ample (Fulton, Chapter 3), a multiple of it is a hyperplane section in a projective embedding, and its complement is affine. A finite set $\mathcal{G}$ of polynomials that contains the $\alpha_J$ removes in addition the $\widetilde{Y}_g$, $g \in \mathcal{G}$, divisors of the smooth $X_\mathcal{E}$, hence Cartier divisors; the complement of a Cartier divisor in an affine variety is affine, the inclusion being locally the complement of the zero set of one function. $D_F$ is the restriction of $D$ to the open $V_F$, with simple normal crossings and smooth components (Lemma 34.3(i)). A stratum is an intersection of components meeting transversally, so smooth, and closed in the affine $V_F$, so affine.

(iv) Let $\tau$ be the cone of the flag. By the orbit–cone correspondence (Fulton, Chapter 3), $D_{S_1} \cap \cdots \cap D_{S_\ell}$ is the toric variety of the star of $\tau$, the fan in $N/(N \cap \mathbb{R}\tau)$ of the images of the cones that contain $\tau$. Restriction to the steps, $\mathbf{x} \mapsto (\mathbf{x}|_{Z_i} \bmod \mathbb{Z}\mathbf{1}_{Z_i})_i$, maps $\mathbb{Z}^\mathcal{E}$ onto $\bigoplus_i \mathbb{Z}^{Z_i}/\mathbb{Z}\mathbf{1}_{Z_i}$ with the kernel spanned by $\mathbf{1}_{Z_1}, \ldots, \mathbf{1}_{Z_{\ell+1}}$, that is by $\mathbf{1}_{S_1}, \ldots, \mathbf{1}_{S_\ell}$ and $\mathbf{1}$; a set $S$ with $S_{i-1} \subsetneq S \subsetneq S_i$ goes to $\mathbf{1}_{S\setminus S_{i-1}}$ in the summand of $Z_i$ and to 0 in the others. So the cones of the chains that refine the flag go to the products of cones of chains of the steps, the star of $\tau$ is the product of the braid

fans of the steps, and $D_{S_1} \cap \cdots \cap D_{S_\ell} \cong \prod_i X_{Z_i}$. The charts that meet the stratum are those of the orders whose chains contain the flag; in such a chart the stratum is $\{\varepsilon_i = 0 :\ i \in I\}$, $I$ the positions of the sets of the flag, and the remaining coordinates are the chart coordinates of the steps. By Lemma 34.3(ii) $\widetilde{Y}_f$ meets it in the union over the steps $Z$ of the zero sets of the polynomials attached to the step letters of $f$ at $Z$, and by the same lemma on $Z$ these are the $\widetilde{Y}_g \cap U_Z$ for the step letters $g$, empty for a monomial. The step letters at $Z$ of the augmented letters of $F$ that are not monomials form $\mathrm{Let}(Z;\mathfrak{f})$: the step letter of $\alpha_J$ is $\alpha_{J\cap Z}$ at the first step $Z$ with $J \subseteq S_b$ and 1 elsewhere, and every $\alpha_{J'}$ with $J' \subseteq Z$, $|J'| \geq 2$, arises from $J = J'$. So the stratum of $\mathfrak{f}$ is $\prod_i V_{Z_i;\mathfrak{f}}$. Its boundary, its meeting with the $D_S$ of the sets $S$ outside the flag, is $\{\varepsilon_k = 0\}$ for the other positions $k$, the union of the pull-backs of the toric boundaries of the factors; and $\widetilde{\sigma}_{\mathcal{E}}$ meets it in $[0,\infty)^{n-1} \cap \{\varepsilon_i = 0 :\ i \in I\} = \prod_Z [0,\infty)^{|Z|-1}$, chart by chart, that is in $\prod_i \widetilde{\sigma}_{Z_i}$.

(v) Where every proper wall has positive real part the period converges (Theorem 8.7), and by Lemma S15, the sectors and Lemma 34.3(iii) it is $\sum_\pi \int_{[0,1]^{n-1}} \prod_i \varepsilon_i^{\kappa_{S_i^\pi}-1} h_\pi \, d\varepsilon = \sum_\pi \mathrm{cube}(\boldsymbol{\kappa}_\pi(\mathbf{m}); h_\pi)$, each term converging absolutely. By Lemma S16 the right side is, on the whole balance wall, the restriction of $J_F/\Gamma(\kappa_{\mathcal{E}})$, meromorphic there jointly in the masses and the exponents, with poles on the hyperplanes $\kappa_S = -j$ only. □

**Definition 34.5 (Motivic periods; Galois conjugates; open spaces).** As stated.

*Proof.* The motivic periods, the de Rham periods, their relations and the coaction are those of Brown's notes (§2 and §3.1), where the coaction is shown to be well defined. For a convergent family, $(V_F, D_F)$ is a pair with $V_F$ affine (Proposition 34.4(iii)), so $H_F$ is an object of Brown's category; $\widetilde{\sigma}_{\mathcal{E}}$ is a relative cycle of it, compact, the closure of the open cell, with $\widetilde{\sigma}_{\mathcal{E}} \setminus \widetilde{\sigma}^\circ_{\mathcal{E}} \subseteq D_F$ and $\widetilde{\sigma}_{\mathcal{E}} \subseteq V_F(\mathbb{R})$ (Proposition 34.4(i)); and $\omega_F$ is a regular form of degree $n-1$ on $V_F$ (Proposition 34.4(ii)), with its class $[\omega_F]$. The period of $P_F^{\mathrm{mot}}$ is $\int_{\widetilde{\sigma}_{\mathcal{E}}} \omega_F = P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ (Proposition 34.4(ii)). For a step with $|Z| \geq 2$ the letters of the step are letters of lists on $Z$ and the $\alpha_J$, $J \subseteq Z$, $|J| \geq 2$; so by Proposition 34.4(i) and (iii) on $Z$, $(V_{Z;\mathfrak{f}}, \partial V_{Z;\mathfrak{f}})$ is a pair with $V_{Z;\mathfrak{f}}$ affine and $\widetilde{\sigma}_Z$ is a relative cycle of it. □

**Lemma 34.6 (Galois conjugates).** *As stated.*

*Proof.* $\mathcal{P}^{\mathrm{dR}}$ is the ring of functions on the de Rham Galois group, $[M,f,\omega]^{\mathrm{dR}}$ being the function $g \mapsto f(g\,\omega)$ (Brown's notes, §2); evaluation at the identity is a $\mathbb{Q}$-linear map $\varphi_1 : \mathcal{P}^{\mathrm{dR}} \to \mathbb{Q}$ with $\varphi_1([M,f,\omega]^{\mathrm{dR}}) = f(\omega)$.

(i) For $\xi = [M,\gamma,\omega]^{\mathrm{mot}}$, $(\mathrm{id}\otimes\varphi_1)\Delta\xi = \sum_i [M,\gamma,e_i]^{\mathrm{mot}}\, e_i^\vee(\omega) = [M,\gamma,\omega]^{\mathrm{mot}}$ by bilinearity; by linearity the same holds for every $\xi$.

(ii) $(\mathrm{id}\otimes\varphi)\Delta[M,\gamma,\omega]^{\mathrm{mot}} = \big[M,\gamma,\sum_i \varphi([M,e_i^\vee,\omega]^{\mathrm{dR}})\, e_i\big]^{\mathrm{mot}} \in L(M,\gamma)$, and the left factors of the coaction of an element of $L(M,\gamma)$ lie in $L(M,\gamma)$.

(iii) $\mathrm{C}(\xi)$ is the image of the linear map $\varphi \mapsto (\mathrm{id}\otimes\varphi)\Delta\xi$, a space; so $\mathrm{C}(\xi+\xi') \subseteq \mathrm{C}(\xi)+\mathrm{C}(\xi')$ and $\mathrm{C}(q\,\xi) = q\,\mathrm{C}(\xi) = \mathrm{C}(\xi)$ for $q \neq 0$. Write $\Delta\xi = \sum_i \xi_i \otimes \xi_i'$ with the $\xi_i'$ linearly independent; a $\varphi$ with $\varphi(\xi_k')$ equal to 1 for $k = i$ and to 0 otherwise shows $\xi_i \in \mathrm{C}(\xi)$. Likewise $\Delta\xi' = \sum_k \eta_k \otimes \eta_k'$ with $\eta_k \in \mathrm{C}(\xi')$. The coaction is a homomorphism of rings (Brown's notes, §2), so $(\mathrm{id}\otimes\varphi)\Delta(\xi\xi') = \sum_{i,k} \xi_i\eta_k\, \varphi(\xi_i'\eta_k') \in \mathrm{C}(\xi)\cdot\mathrm{C}(\xi')$. □

**Lemma 34.7 (Birational invariance).** *As stated.*

*Proof. The resolution.* Let $\Gamma \subseteq X_1 \times X_2$ be the closure of the graph of $\Phi$, a projective variety. The graph is closed in $U_1 \times X_2$, so $\Gamma \cap (U_1 \times X_2)$ is the graph; it is also the graph of $\Phi^{-1}$, closed in $X_1 \times U_2$, so $\Gamma \cap (X_1 \times U_2)$ is the graph too; and the graph, isomorphic to $U_1$, is a smooth dense

open subset of $\Gamma$. By resolution of singularities in characteristic zero (Hironaka), in the form that gives for a projective variety and a smooth dense open subset of it a smooth projective variety with a proper birational morphism onto it that is an isomorphism over that subset and under which the complement pulls back to a divisor with simple normal crossings and smooth components, there is such a $\varpi : \widetilde{X} \to \Gamma$ for the graph; the property used is that the graph of $\Phi$ is smooth and the ground field $\mathbb{Q}$ has characteristic zero. Let $q_i : \widetilde{X} \to X_i$ be the two projections; by the above $q_1^{-1}(U_1) = q_2^{-1}(U_2) = \varpi^{-1}(\text{graph})$, on which $q_1$ and $q_2 = \Phi \circ q_1$ are isomorphisms onto $U_1$ and $U_2$. Put $\widetilde{D}_i = q_i^{-1}(D_i)$ and $\widetilde{Y} = q_1^{-1}(Y_1) \cup q_2^{-1}(Y_2)$, both closed. A point of $\widetilde{D}_1$ is not over the graph, $D_1$ missing $U_1$; so $\widetilde{D}_1$, the support of the pull-back of a Cartier divisor under the dominant $q_1$, is a divisor contained in the divisor $\varpi^{-1}(\Gamma \setminus \text{graph})$ and so a union of its components; likewise $\widetilde{D}_2$ and $\widetilde{D}_1 \cup \widetilde{D}_2$. All three have simple normal crossings and smooth components.

*The cycle.* Let $\widetilde{\sigma}$ be the closure in $\widetilde{X}(\mathbb{R})$ of $\widetilde{\sigma}^\circ = q_1^{-1}(\sigma_1^\circ)$, which $q_1$ maps isomorphically onto $\sigma_1^\circ$ and $q_2$ onto $\sigma_2^\circ$; orient it by $q_1$. It is compact, and $q_i(\widetilde{\sigma}) = \sigma_i$, which misses $Y_i$; so $\widetilde{\sigma}$ misses $\widetilde{Y}$. If $x \in \widetilde{\sigma} \setminus \widetilde{\sigma}^\circ$ had $q_1(x) \in \sigma_1^\circ \subseteq U_1$, $x$ would lie over the graph, where $q_1$ is injective, and so in $\widetilde{\sigma}^\circ$; hence $q_1(x) \in \sigma_1 \setminus \sigma_1^\circ \subseteq D_1$ and $x \in \widetilde{D}_1$. The same argument with $q_2$ and $\Phi(\sigma_1^\circ) = \sigma_2^\circ$ gives $x \in \widetilde{D}_2$. So $\widetilde{\sigma}$ is a relative cycle of $(\widetilde{X} \setminus \widetilde{Y}, \widetilde{D}_i \setminus \widetilde{Y})$ for $i = 1, 2$ and of $(\widetilde{X} \setminus \widetilde{Y}, (\widetilde{D}_1 \cup \widetilde{D}_2) \setminus \widetilde{Y})$, and $q_i$ carries its class to that of $\sigma_i$, mapping $\widetilde{\sigma}^\circ$ onto $\sigma_i^\circ$ with the orientation, $\Phi$ preserving it, and $\widetilde{\sigma} \setminus \widetilde{\sigma}^\circ$ into $D_i$.

*The form.* $q_1^*\omega_1$ and $q_2^*\omega_2$ are rational forms on $\widetilde{X}$ that agree on the dense open $q_1^{-1}(U_1)$, where $q_2 = \Phi \circ q_1$ and $\Phi^*\omega_2 = \omega_1$; so they are one form $\widetilde{\omega}$, regular off $q_1^{-1}(Y_1)$ and off $q_2^{-1}(Y_2)$, hence on $\widetilde{X} \setminus \widetilde{Y}$.

*The chain of pairs.* The maps

$$(X_1^\circ, D_1^\circ) \xleftarrow{q_1} (\widetilde{X} \setminus \widetilde{Y}, \widetilde{D}_1 \setminus \widetilde{Y}) \xrightarrow{\mathrm{id}} \big(\widetilde{X} \setminus \widetilde{Y}, (\widetilde{D}_1 \cup \widetilde{D}_2) \setminus \widetilde{Y}\big) \xleftarrow{\mathrm{id}} (\widetilde{X} \setminus \widetilde{Y}, \widetilde{D}_2 \setminus \widetilde{Y}) \xrightarrow{q_2} (X_2^\circ, D_2^\circ)$$

are morphisms of pairs, so their maps on relative cohomology are morphisms of Brown's category (Huber and Müller-Stach). Each pulls back the class of the form of degree $m$ to the class of the pulled-back form, and its transpose carries the class of $\widetilde{\sigma}$, or of $\sigma_1$, $\sigma_2$ at the ends, to the class of the cycle at its target. The relation $[M, \phi^{\mathrm{t}}\gamma', \omega]^{\mathrm{mot}} = [M', \gamma', \phi(\omega)]^{\mathrm{mot}}$ along the four morphisms gives

$$\big[H^m(X_1^\circ, D_1^\circ), \sigma_1, \omega_1\big]^{\mathrm{mot}} = \big[H^m(\widetilde{X} \setminus \widetilde{Y}, \widetilde{D}_1 \setminus \widetilde{Y}), \widetilde{\sigma}, \widetilde{\omega}\big]^{\mathrm{mot}} = \cdots = \big[H^m(X_2^\circ, D_2^\circ), \sigma_2, \omega_2\big]^{\mathrm{mot}}.$$

*The particular cases.* With $X_1 = X_2$, $D_1 = D_2$, $Y_2 \supseteq Y_1$ closed and missing $\sigma_1$, $U_1 = U_2 = X_1 \setminus (D_1 \cup Y_2)$ and $\Phi$ the identity, the hypotheses hold, $\sigma_1^\circ$ lying in $U_1(\mathbb{R})$. For a convergent family, $\omega_F$ is regular on $X_{\mathcal{E}} \setminus \bigcup_{f \in \mathrm{Let}(F)} \widetilde{Y}_f$, by the proof of Proposition 34.4(ii), which uses only that the letters of $F$ are units there; and $V_F$ removes in addition the $\widetilde{Y}_{\alpha_J}$, which miss $\widetilde{\sigma}_{\mathcal{E}}$ (Proposition 34.4(i)). Two convergent families $F, F'$ on $\mathcal{E}$ with $\omega_F = \omega_{F'}$ give $X_1 = X_2 = X_{\mathcal{E}}$, $D_1 = D_2 = D$, $Y_1$ and $Y_2$ the unions of the $\widetilde{Y}_f$ over the augmented letters of $F$ and of $F'$, both missing $\widetilde{\sigma}_{\mathcal{E}}$, $U_1 = U_2$ the torus less $Y_1 \cup Y_2$ and $\Phi$ the identity. □

**Lemma S17** (Products of pairs)**.** *For $i = 1, \dots, p$ let $(V_i, D_i)$ be a pair of dimension $m_i$ with a relative cycle $\sigma_i$ and a regular form $\eta_i$ of degree $m_i$ on $V_i$. Let $V = \prod_i V_i$, $D = \bigcup_i \mathrm{pr}_i^{-1}(D_i)$, $\sigma = \prod_i \sigma_i$ with the product orientation, and $\eta = \mathrm{pr}_1^*\eta_1 \wedge \cdots \wedge \mathrm{pr}_p^*\eta_p$. Then $(V, D)$ is a pair, $\sigma$ is a relative cycle of it, and, with $m = \sum_i m_i$,*

$$\big[H^m(V, D), \sigma, [\eta]\big]^{\mathrm{mot}} = \prod_{i=1}^{p} \big[H^{m_i}(V_i, D_i), \sigma_i, [\eta_i]\big]^{\mathrm{mot}}.$$

*Proof.* In products of local coordinates $D$ is a union of coordinate hyperplanes of the factors, so $(V, D)$ is a pair; $\sigma$ is the closure of the oriented $\prod_i \sigma_i^\circ$, of dimension $m$, and $\sigma \setminus \prod_i \sigma_i^\circ \subseteq D$. The cross product $\mu : \bigotimes_i H^{m_i}(V_i, D_i) \to H^m(V, D)$ is a morphism of Nori's category (Huber and Müller-Stach), hence of Brown's; on de Rham realizations it carries $\bigotimes_i [\eta_i]$ to $[\eta]$, and its transpose carries $\sigma$ to $\bigotimes_i \sigma_i$, the class of a product of relative cycles being the cross product of their classes; on forms this is Fubini, $\int_\sigma \mathrm{pr}_1^* \theta_1 \wedge \cdots \wedge \mathrm{pr}_p^* \theta_p = \prod_i \int_{\sigma_i} \theta_i$. The relation of Definition 34.5 for $\mu$, with the product of motivic periods defined by the tensor product of objects (Brown's notes, §2), gives the identity. □

**Corollary 34.8 (Cremona between motivic periods).** *As stated.*

*Proof. The walls.* With $d(S) = r - r(\mathcal{E} \setminus S)$, $d^*(S) = |S| - r(S)$, $d(\mathcal{E} \setminus S) = r - r(S)$ and $d^*(\mathcal{E} \setminus S) = |\mathcal{E} \setminus S| - r(\mathcal{E} \setminus S)$ in every table (Definition 7.12), and $m_\mathcal{E} = \sum_j (s_j r_j + t_j(n - r_j))$ on the balance wall,

$$\kappa_S(F^\mathrm{c}) = 2\sigma|S| - m_S - \sum_j t_j\big(r_j - r_j(\mathcal{E} \setminus S)\big) - \sum_j s_j\big(|S| - r_j(S)\big),$$
$$\kappa_{\mathcal{E}\setminus S}(F) = m_\mathcal{E} - m_S - \sum_j s_j\big(r_j - r_j(S)\big) - \sum_j t_j\big(n - |S| - r_j(\mathcal{E} \setminus S)\big),$$

and both equal $-m_S + \sum_j s_j r_j(S) + \sum_j t_j\big(|S| - r_j + r_j(\mathcal{E} \setminus S)\big)$, using $2\sigma = \sum_j (s_j + t_j)$. At $S = \mathcal{E}$ the same computation gives $\kappa_\mathcal{E}(F^\mathrm{c}) = -\kappa_\mathcal{E}(F) = 0$. So $F^\mathrm{c}$ is on its balance wall, at integer data, and convergent with $F$.

*The forms.* Let $\Phi$ be the inversion $\alpha \mapsto \alpha^{-1}$ of the torus, an automorphism that maps $\widetilde{\sigma}_\mathcal{E}^\circ$ onto itself. By Proposition 7.2, $K_j(\alpha^{-1}) = \alpha^{-\mathcal{E}} \Psi_j(\alpha)$ and $\Psi_j(\alpha^{-1}) = \alpha^{-\mathcal{E}} K_j(\alpha)$, and $\Phi^*(\Omega_\mathcal{E}/\alpha^\mathcal{E}) = (-1)^{n-1} \Omega_\mathcal{E}/\alpha^\mathcal{E}$, every $d \log \alpha_e$ changing sign. So

$$\Phi^* \omega_{F^\mathrm{c}} = \alpha^{\mathbf{m} - 2\sigma\mathbf{1}}\, \alpha^{2\sigma\mathbf{1}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, (-1)^{n-1} \frac{\Omega_\mathcal{E}}{\alpha^\mathcal{E}} = (-1)^{n-1} \omega_F,$$

the factor $\alpha^{2\sigma\mathbf{1}}$ collecting the powers $\alpha^\mathcal{E}$ of the letters, $\sum_j (s_j + t_j) = 2\sigma$. Let $U_1$ be the torus less the zero sets of the augmented letters of $F$ and the preimages under $\Phi$ of those of $F^\mathrm{c}$, and $U_2 = \Phi(U_1)$; these polynomials are positive at the positive points, so $\widetilde{\sigma}_\mathcal{E}^\circ \subseteq U_1(\mathbb{R})$. The periods $P_F$ and $P_{F^\mathrm{c}} = \int \omega_{F^\mathrm{c}}$ over the positively oriented cell are both positive, so $\Phi$ maps the oriented $\widetilde{\sigma}_\mathcal{E}^\circ$ onto itself with the sign $(-1)^{n-1}$ of the forms; reversing the orientation of the target cell and the sign of $\omega_{F^\mathrm{c}}$ together when that sign is $-1$, Lemma 34.7 applies to $(X_\mathcal{E}, D, Y_F)$ and $(X_\mathcal{E}, D, Y_{F^\mathrm{c}})$, $Y$ the unions of the $\widetilde{Y}_f$ over the augmented letters, and gives $P_{F^\mathrm{c}}^\mathrm{mot} = P_F^\mathrm{mot}$ by bilinearity. □

**Definition 34.9 (Attachment).** As stated.

*Proof.* Every vertex of a tree of bunches is a split vertex, a node that is not a leaf, or a leaf (Definition 20.7), and the first two have children: the molecules of a split vertex, and the room of a node. Along the descent $Z \subseteq M_w$ at every vertex $w$ reached: this holds at the root, a split vertex passes $Z$ to the child that contains it, and a node passes $Z$ to its room only when $Z \subseteq M_w \setminus B_w$, the marks of the room. At each vertex exactly one rule applies, $Z$ being nonempty, and the vertex reached next has fewer marks: a child of a split vertex is one of at least two molecules, and $B_w \neq \varnothing$. So the descent ends, and it ends exactly where $Z$ is attached. □

**Lemma 34.10 (The letters of an attached step).** *As stated.*

*Proof.* For a vertex $w$ on the descent of $Z$ say that $Z$ *is read at* $w$ when every letter of the step other than the linear letters is $K$, respectively $\Psi$, of the child at a pattern of $M_w \setminus Z$ of a table at $w$ that carries $K$, respectively $\Psi$. At the root it holds: the step lists of $\mathcal{B}_j$ at $Z$ are the children $\mathcal{B}_{j,(T_0\setminus Z)}$ and $\mathcal{B}_{j,(T_1\setminus Z)}$ at patterns of $\mathcal{E}\setminus Z$ (Proposition 12.20), or $\mathcal{B}_j$ itself when $Z=\mathcal{E}$, and the tables at the root are the active tables of $F$, $\mathcal{B}_j$ carrying $K$ exactly when $s_j\neq 0$ and $\Psi$ exactly when $t_j\neq 0$. Let it hold at $w$, and let the descent pass to a child $w'$.

At a split vertex, $Z\subseteq M'$ for a molecule $M'$, and every table $N$ at $w$ is the direct sum of its restrictions to the molecules (Proposition 7.9). So the bases of $N$ with a given trace $p$ on $M_w\setminus Z$ have their other parts fixed, and the child of $N$ at $p$ is the child of $N|M'$ at the pattern $p\cap M'$ of $M'\setminus Z$; $N|M'$ is a table at $w'$ with the exponents of $N$.

At a node that is not a leaf, $Z\subseteq M_w\setminus B_w$. A pattern $p$ of $M_w\setminus Z$ in a table $N$ at $w$ is $p_B\sqcup p_R$ with $p_B=p\cap B_w$ a pattern of $B_w$ and $p_R\subseteq (M_w\setminus B_w)\setminus Z$, and the bases $T$ of $N$ with $T\setminus Z=p$ are those with $T\cap B_w=p_B$ whose parts $T\setminus B_w$, bases of the child $N_{(p_B)}$, have the trace $p_R$ on $(M_w\setminus B_w)\setminus Z$. So the child of $N$ at $p$ is the child of $N_{(p_B)}$ at $p_R$ (Proposition 14.5). $N_{(p_B)}$ is a table made at $w$, carrying the letters $N$ carries with exponents that are positive multiples of those of $N$ (Definition 20.7); it is a table of the room, of the one childcule when the room is a node and of the split vertex otherwise.

So $Z$ is read at the vertex where it is attached, which is the statement. □

**Lemma 34.11 (One form carries every period).** *As stated.*

*Proof.* Number the components $D_1,\dots,D_K$ of $D$ and put $D_I=\bigcap_{k\in I}D_k$, $D_\varnothing=V$; each $D_I$ is smooth of dimension $m-|I|$ or empty, and closed in the affine $V$, so affine. The relative de Rham cohomology $H^\bullet_{\mathrm{dR}}(V,D)$ is the hypercohomology of the double complex $\Omega^\bullet_V\to\bigoplus_{|I|=1}\Omega^\bullet_{D_I}\to\bigoplus_{|I|=2}\Omega^\bullet_{D_I}\to\cdots$ with the alternating restrictions (Brown's notes, §10; Huber and Müller-Stach). Its terms are coherent sheaves on the affine $V$, with no higher cohomology, so it is the cohomology of the total complex of global sections, $\mathrm{Tot}^q=\bigoplus_I\Omega^{q-|I|}(D_I)$. Since $\Omega^q(D_I)=0$ for $q>m-|I|$, $\mathrm{Tot}^{m+1}=0$ and every $(\eta_I)_I\in\mathrm{Tot}^m$ is a cocycle, $\eta_I$ a form of top degree on $D_I$. Let $\ell$ be the largest $|I|$ with $\eta_I\neq 0$, and suppose $\ell\geq 1$.

*Restriction is onto.* For $|I'|=\ell-1$, restriction maps $\Omega^{m-\ell}(D_{I'})$ onto $\bigoplus_{k\notin I'}\Omega^{m-\ell}(D_{I'}\cap D_k)$, the top forms on the boundary components of $D_{I'}$. In local coordinates of $D_{I'}$ in which the components $D_{I'}\cap D_k$ through a point are $\{z_1=0\},\dots,\{z_r=0\}$ and $dw$ is the product of the differentials of the other coordinates, the form $\sum_t \widetilde{g}_t\,dz_1\wedge\cdots\widehat{dz_t}\cdots\wedge dz_r\wedge dw$ restricts to $\pm g_t\,dw$ on $\{z_t=0\}$, $\widetilde{g}_t$ any local extension of $g_t$, and to 0 on the others; so the map of coherent sheaves is onto, and so is its map on global sections, $D_{I'}$ being affine.

*The deepest level removed.* For $|I|=\ell$ let $I'=I\setminus\{\max I\}$. Choose $\gamma_{I'}\in\Omega^{m-\ell}(D_{I'})$ for every $|I'|=\ell-1$ restricting on $D_{I'\cup\{k\}}$, $k>\max I'$, to the form that cancels $\eta_{I'\cup\{k\}}$ in the total differential, and to 0 on the $D_{I'\cup\{k\}}$ with $k<\max I'$. In $(\eta)-d\gamma$ the entry at $|I|=\ell$ is $\eta_I$ less the restrictions of the $\gamma_{I\setminus\{k\}}$, $k\in I$, of which only $k=\max I$ contributes, and it is 0; the entries above $\ell$ stay 0, and those at $\ell-1$ change by $d\gamma_{I'}$. By induction every class is the class of $(\eta_\varnothing,0,\dots,0)$, a regular $m$-form on $V$.

For a convergent family, $(V_F,D_F)$ is such a pair (Proposition 34.4(iii)), and $V_{\mathcal{E};\varnothing\subsetneq\mathcal{E}}=V_F$, the letters of the one step $\mathcal{E}$ being the augmented letters of $F$ that are not monomials, whose $\widetilde{Y}_f$ are empty; so $L(H_F,\widetilde{\sigma}_\mathcal{E})=\mathrm{Open}(\mathcal{E};\varnothing\subsetneq\mathcal{E})$. For a step with $|Z|\geq 2$ the same holds on $Z$ (proof of Definition 34.5); for $|Z|=1$ both sides are $\mathbb{Q}$. □

**Proposition 34.12 (Weight-graded faces).** *As stated.*

*Proof.* For a chain $\mathfrak{c}$ of nonempty proper sets let $D_{\mathfrak{c}}$ be its stratum in $V_F$, a smooth affine variety of dimension $n-1-|\mathfrak{c}|$, nonempty since it meets $\widetilde{\sigma}_{\mathcal{E}}$ (Proposition 34.4(iii),(iv)), with its boundary $\partial D_{\mathfrak{c}}$, its meeting with the other components of $D_F$. The summands of the double complex of the proof of Lemma 34.11 over the chains refining $\mathfrak{c}$ form the double complex of $(D_{\mathfrak{c}}, \partial D_{\mathfrak{c}})$, shifted by $|\mathfrak{c}|$, and their inclusion is the *face map* $\phi_{\mathfrak{c}} : H^{n-1-|\mathfrak{c}|}(D_{\mathfrak{c}}, \partial D_{\mathfrak{c}}) \to H^{n-1}(V_F, D_F)$, a morphism of Brown's category whose transpose takes a relative cycle that is a manifold with corners, with its faces on the components of $D_F$, to its face on $D_{\mathfrak{c}}$, with a sign (Brown's notes, §10). Here $\widetilde{\sigma}_{\mathcal{E}}$ is such a cycle, $[0,\infty)^{n-1}$ in every chart, and $\phi_{\mathfrak{c}}^{\mathrm{t}}\widetilde{\sigma}_{\mathcal{E}} = \pm\widetilde{\sigma}_{\mathcal{E}} \cap D_{\mathfrak{c}}$. By Brown's weight-graded face maps (Brown's notes, §10), for $k < n-1$ the sum of the face maps of the chains with $n-1-|\mathfrak{c}| = k$ maps $\bigoplus_{\mathfrak{c}} W_k H^k(D_{\mathfrak{c}}, \partial D_{\mathfrak{c}})$ onto $W_k H^{n-1}(V_F, D_F)$; the property it needs is that every stratum is affine, so that its cohomology vanishes above its dimension. Write $e = \sum_{\mathfrak{c}} \phi_{\mathfrak{c}}(e_{\mathfrak{c}})$. By Lemma 34.11 on the pair $(D_{\mathfrak{c}}, \partial D_{\mathfrak{c}})$, $e_{\mathfrak{c}} = [\eta_{\mathfrak{c}}]$ for a regular $k$-form on $D_{\mathfrak{c}}$, and by the relation for $\phi_{\mathfrak{c}}$

$$[H_F, \widetilde{\sigma}_{\mathcal{E}}, e]^{\mathrm{mot}} = \sum_{\mathfrak{c}} \pm \big[H^k(D_{\mathfrak{c}}, \partial D_{\mathfrak{c}}), \widetilde{\sigma}_{\mathcal{E}} \cap D_{\mathfrak{c}}, [\eta_{\mathfrak{c}}]\big]^{\mathrm{mot}}.$$

The chain $\mathfrak{c}$ is a flag $\mathfrak{f}$ of dimension $k$, and by Proposition 34.4(iv) its stratum with its boundary is the product of the pairs $(V_{Z_i;\mathfrak{f}}, \partial V_{Z_i;\mathfrak{f}})$ over its steps, with $\widetilde{\sigma}_{\mathcal{E}} \cap D_{\mathfrak{c}} = \prod_i \widetilde{\sigma}_{Z_i}$. The factors are affine, so the regular functions on the product are the tensor product of those on the factors, the top forms likewise, and $\eta_{\mathfrak{c}}$ is a finite sum of products of top forms on the factors. By Lemma S17 each term is a product over the steps of elements of $\mathrm{Open}(Z_i;\mathfrak{f})$, a step of one mark giving a rational number; the signs are absorbed, $\mathrm{Open}(Z;\mathfrak{f})$ being closed under negation. □

**Theorem 34.13 (Small steps along the tree).** *As stated.*

*Proof.* Proposition 34.12 at $e = e_i$ is the inclusion. A flag of dimension $k$ has $\sum_i (|Z_i| - 1) = k$, so each of its steps has at most $k+1$ marks. Each step is attached to exactly one vertex (Definition 34.9), and its letters are those of Lemma 34.10. □

**Proposition 34.14 (Laurent coefficients are read on convergent data).** *As stated.*

*Proof. Meromorphy.* The line $\mathbf{m} + \varepsilon\mathbf{v}$ lies in the balance wall, $v_{\mathcal{E}} = 0$, and $\kappa_S(\mathbf{m} + \varepsilon\mathbf{v}) = \kappa_S + \varepsilon v_S$ with $v_S \neq 0$, so the line meets every hyperplane $\kappa_S = -j$ in at most one point, and the continued period is meromorphic on it (Proposition 34.4(v)).

*A relation at each wall.* For lists every weight is 1. Let $\varnothing \neq S \subsetneq \mathcal{E}$. Summing the mark relations of Theorem 8.9(iii) over $e \in S$, and subtracting $s_j d_j(S)$ times the first letter relation of table $j$ and $t_j d_j^*(S)$ times the second, for every $j$, gives, as meromorphic functions of the data,

$$\begin{aligned}\kappa_S\, P_F(\mathbf{m};\mathbf{s},\mathbf{t}) = \kappa_{\mathcal{E}} \sum_{e\in S} P_F(\mathbf{m}+\mathbf{e}_e;\mathbf{s},\mathbf{t}) + \sum_j s_j \sum_{T\in\mathcal{B}_j} \big(|T\cap S| - d_j(S)\big) P_F(\mathbf{m}+\mathbf{1}_T;\mathbf{s}+\mathbf{e}_j,\mathbf{t}) \\ + \sum_j t_j \sum_{T\in\mathcal{B}_j} \big(|S\setminus T| - d_j^*(S)\big) P_F(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T};\mathbf{s},\mathbf{t}+\mathbf{e}_j).\end{aligned}$$

Each term has its poles on the hyperplanes $\kappa_{S'} = -j$, $S'$ proper, of its own data (Lemma S16), none of which contains the balance wall; so on the balance wall the first term vanishes, and the data of the other terms stay on it, $|T| = r_j$ and $|\mathcal{E}\setminus T| = r_j^*$. On the line $\mathbf{m} + \varepsilon\mathbf{v}$ every remaining term is meromorphic, as above, and

$$P_F(\mathbf{m}+\varepsilon\mathbf{v};\mathbf{s},\mathbf{t}) = \frac{1}{\kappa_S + \varepsilon v_S}\Big(\sum_j s_j \sum_T \big(|T\cap S| - d_j(S)\big) P_F(\mathbf{m}+\mathbf{1}_T+\varepsilon\mathbf{v};\mathbf{s}+\mathbf{e}_j,\mathbf{t})$$

$$+\sum_j t_j \sum_T \big(|S\setminus T| - d_j^*(S)\big) P_F(\mathbf{m}+\mathbf{1}_{\mathcal{E}\setminus T}+\varepsilon\mathbf{v};\mathbf{s},\mathbf{t}+\mathbf{e}_j)\Big),$$

with $1/(\kappa_S+\varepsilon v_S)\in\mathbb{Q}(\varepsilon)$ and integer coefficients.

*The descent.* Put $\delta(F)=\sum_{\varnothing\neq S\subsetneq\mathcal{E}}\max(0,1-\kappa_S)$, a nonnegative integer, zero exactly when $F$ is convergent. If $\delta(F)>0$, choose $S$ with $\kappa_S\leq 0$ and apply the relation at $S$, keeping the terms with nonzero coefficients. Their families are on the same tables, at integer data on their balance walls, with the exponents larger by one unit vector; their walls are $\kappa_{S'}+|T\cap S'|-d_j(S')\geq\kappa_{S'}$, respectively $\kappa_{S'}+|S'\setminus T|-d_j^*(S')\geq\kappa_{S'}$, for every $S'$ (Theorem 7.13), and at $S$ itself, the coefficient being nonzero, at least $\kappa_S+1$. As $1-\kappa_S\geq 1$, the term of $S$ in $\delta$ drops by one, and $\delta$ drops by at least one. Repeating on every family with $\delta>0$ ends after finitely many steps, with convergent families $F_k$ and coefficients $R_k\in\mathbb{Q}(\varepsilon)$, products of the factors above. The exponents only grow, so with $\mathbf{s},\mathbf{t}\geq 0$ every $F_k$ has nonnegative exponents, and then positive masses, $m_e^{(k)}=\kappa_{\{e\}}(F_k)+\sum_j\big(s_j^{(k)}d_j(\{e\})+t_j^{(k)}d_j^*(\{e\})\big)\geq 1$, the singletons being proper as $n\geq 2$.

*The coefficients.* At the masses $\mathbf{m}^{(k)}+\varepsilon\mathbf{v}$ the walls of $F_k$ have positive real parts for $|\varepsilon|$ small, so its period is the integral over $\Delta_{\mathcal{E}}$, converging absolutely and locally uniformly, holomorphic in $\varepsilon$, with the derivatives under the integral sign, $u^{\varepsilon\mathbf{v}}=\exp(\varepsilon\langle\mathbf{v},\log u\rangle)$. The displayed integrals converge absolutely: on $\Delta_{\mathcal{E}}$, $|\langle\mathbf{v},\log u\rangle|^l\leq C\prod_e u_e^{-\eta}$ for every $\eta>0$, and the walls at $\mathbf{m}^{(k)}-\eta\mathbf{1}$ are positive for small $\eta$, so that the integral of $u^{\mathbf{m}^{(k)}-\eta\mathbf{1}-\mathbf{1}}\prod_j K_j^{-s_j^{(k)}}\Psi_j^{-t_j^{(k)}}$ converges (Theorem 8.7, which reads the integral over $\Delta_{\mathcal{E}}$ off the balance wall as well). The Laurent coefficients of $\sum_k R_kP_{F_k}$ at 0 are rational combinations of the Taylor coefficients of the $P_{F_k}$, those of the $R_k$ being rational. □

**Corollary 34.15 (Open periods from families of lists).** *As stated.*

*Proof. The forms.* The torus part of $V_{Z;\mathfrak{f}}$ is the torus less the zero sets of the letters of the step, a principal open subset of the torus, so a regular function on $V_{Z;\mathfrak{f}}$ is there a homogeneous Laurent polynomial divided by a product of powers of the letters, of total degree zero; and $\Omega_Z/\alpha^Z$ is a nowhere vanishing form of top degree on the torus. Splitting the Laurent polynomial into its monomials writes $\eta$ as a finite sum of terms $c_t\,\alpha^{\mathbf{a}_t}\prod_g g^{-k_{t,g}}\,\Omega_Z/\alpha^Z$, $k_{t,g}\in\mathbb{N}$, each the form $\omega_{F_t}$ of the family of the statement, a letter $K$ of a step list being $K$ of the table $(N;k_{t,g},0)$, a letter $\Psi$ being $\Psi$ of the table $(N;0,k_{t,g})$, and $\alpha_J$ being $K$ of $(\binom{J}{1};k_{t,g},0)$ (Definition 34.1); $\alpha^{\mathbf{a}_t}\prod_g g^{-k_{t,g}}$ has degree zero, so $F_t$ is on its balance wall.

*The period.* For $\mathbf{x}\in\mathbb{C}^Z$ with $x_Z=0$ near 0 put $G(\mathbf{x})=\int_{\widetilde{\sigma}_Z}\alpha^{\mathbf{x}}\eta$. In the chart of an order $\pi$, $\alpha^{\mathbf{x}}=\prod_i\varepsilon_i^{x_{S_i}}$ (proof of Lemma 34.3(ii)), and $\eta$, regular near $\widetilde{\sigma}_Z\subseteq V_{Z;\mathfrak{f}}(\mathbb{R})$ (proof of Definition 34.5), integrates over the sector as $g_\pi\,d\varepsilon$ on $[0,1]^{|Z|-1}$ with $g_\pi$ smooth near the cube. So $G(\mathbf{x})=\sum_\pi\operatorname{cube}(\mathbf{x}_\pi+\mathbf{1};g_\pi)$, $\mathbf{x}_\pi=(x_{S_i^\pi})_i$, holomorphic near $\mathbf{x}=0$, and $G(0)=\int_{\widetilde{\sigma}_Z}\eta$ (Lemma S15). On the open cube Lemma 34.3(iii) gives, with the same sign for every $t$,

$$g_\pi=\sum_t c_t\,\varepsilon^{\boldsymbol{\kappa}_\pi(F_t)-\mathbf{1}}h_{t,\pi},\qquad h_{t,\pi}=\prod_g\widetilde{g}^{-k_{t,g}},$$

$\boldsymbol{\kappa}_\pi(F_t)$ the walls of $F_t$ at $\mathbf{a}_t$ along the chain of $\pi$. Choose $A\in\mathbb{N}$ with $\boldsymbol{\kappa}_\pi(F_t)+A\mathbf{1}\geq\mathbf{1}$ for every $t$ and $\pi$. The cube integrals have $\operatorname{cube}(\mathbf{y}+\mathbf{b};h)=\operatorname{cube}(\mathbf{y};\varepsilon^{\mathbf{b}}h)$ for $\mathbf{b}\in\mathbb{N}^{|Z|-1}$, where both converge and so, both sides being meromorphic (Lemma 8.4), everywhere; they are linear in $h$. So, as meromorphic functions of $\mathbf{x}$,

$$\sum_t c_t\operatorname{cube}\big(\mathbf{x}_\pi+\boldsymbol{\kappa}_\pi(F_t);h_{t,\pi}\big)=\sum_t c_t\operatorname{cube}\big(\mathbf{x}_\pi+(1-A)\mathbf{1};\varepsilon^{\boldsymbol{\kappa}_\pi(F_t)+(A-1)\mathbf{1}}h_{t,\pi}\big)$$

$$= \operatorname{cube}\big(\mathbf{x}_\pi + (1-A)\mathbf{1}; \varepsilon^{A\mathbf{1}} g_\pi\big) = \operatorname{cube}(\mathbf{x}_\pi + \mathbf{1}; g_\pi),$$

the functions $\varepsilon^{\boldsymbol{\kappa}_\pi(F_t)+(A-1)\mathbf{1}} h_{t,\pi}$ being smooth near the cube. Summing over $\pi$ and using Proposition 34.4(v) with $\kappa_S(\mathbf{a}_t + \mathbf{x}) = \kappa_S(\mathbf{a}_t) + x_S$, $G(\mathbf{x}) = \sum_t c_t P_{F_t}(\mathbf{a}_t + \mathbf{x})$ as meromorphic functions near $\mathbf{x} = 0$ on $\{x_Z = 0\}$. On the line $\mathbf{x} = \varepsilon\mathbf{v}$, every $v_S$ being nonzero, each term is meromorphic in $\varepsilon$, as in the proof of Proposition 34.14, the sum is $G(\varepsilon\mathbf{v})$, holomorphic at 0 with the value $\int_{\widetilde{\sigma}_Z} \eta$, and the constant term is linear.

*The rest.* Every $F_t$ has nonnegative exponents, so Proposition 34.14 writes its constant term through convergent families of lists on $Z$ with nonnegative integer exponents and positive integer masses, whose periods are values of the serif family (Proposition 8.12(i)), and their derivatives along $\mathbf{v}$. The period of $[H^{|Z|-1}(V_{Z;\mathfrak{f}}, \partial V_{Z;\mathfrak{f}}), \widetilde{\sigma}_Z, \eta]^{\mathrm{mot}}$ is $\int_{\widetilde{\sigma}_Z} \eta$, and taking periods is a homomorphism of rings; Theorem 34.13 gives the last sentence. □

**Theorem 34.16 (The node form).** *As stated.*

*Proof.* (i) A letter $K$ of the step is $K$ of a step list, which by Lemma 34.10 is the child $N'$ of a table $N$ at $v$ at a pattern $p$ of $M_v \setminus Z$, a list being determined by its basis polynomial; a letter $\Psi$ is $\Psi$ of such a child $N'$, that is $K$ of $N'^*$ (Proposition 7.2); a letter $\alpha_J$ is $K$ of $\binom{J}{1}$ read on $Z$, its marks outside $J$ loops. Lemma 12.3, for the table $N$ on $M_v$ with the bunch $M_v \setminus Z$, its room $Z$, the pattern $p$, $A = Z_B$ and $W = B_v \setminus Z$, so that $A \cup W = B_v$, gives $c_{N'}(Z_B) \le c_N(B_v) \le c$; the dual has the same widths (Theorem 7.13). $\binom{J}{1}$ has rank at most one, so the width of $Z_B$ in it is at most one. Finally $c \ge 1$: if every table at $v$ had $c(B_v) = 0$, $B_v$ would be a union of molecules of every table (Lemma 7.10), no circuit of a table at $v$ would meet both $B_v$ and $M_v \setminus B_v$, and $B_v$ would be a union of molecules of the tables at $v$ (Definition 7.4), against $\varnothing \ne B_v \subsetneq M_v$ and the tables at the node $v$ forming one molecule.

(ii) For $g = K_{N_g}$ and the gauge $(u, u')$, $g(\rho u, u') = \sum_k \rho^k K^{(k)}_{Z_B}(u, u')$ over the levels $d_g \le k \le d_g + c_g$ (Definition 12.16, Theorem 7.13). The bases of the top level are the unions of a basis of $N_g|Z_B$ and one of $N_g/Z_B$, each of weight 1 (Theorem 12.19), so the top level letter is $K_{N_g|Z_B}(u)\, K_{N_g/Z_B}(u')$. The Skeleton polynomial $\mathsf{Q}^K_{Z_B}(x; (u, u')) = x^{-d_g} g(xu, u')/g(u, u') = \prod_i (x + \xi_{g,i})/(1 + \xi_{g,i})$ (Definition 12.10) shows that $\rho^{-d_g} g(\rho u, u')$ is a polynomial of degree $c_g$ in $\rho$ with the roots $-\xi_{g,i}$, and comparing leading coefficients gives the display. The Skeleton values of $K_{N''^*}$ along $Z_B$ are those of $\Psi_{N''}$ (Definition 12.10), so the union over the letters is the joint spectrum of $Z_B$ in the sense of Definition 12.14 and Theorem 15.3.

(iii) By Corollary 34.15, on the torus $\eta = \phi\, \Omega_Z/\alpha^Z$ with $\phi = \sum_t c_t \alpha^{\mathbf{a}_t} \prod_g g^{-k_{t,g}}$, homogeneous of degree zero. By Lemma S15 for the partition $Z = Z_B \sqcup Z_R$, $\int_{\widetilde{\sigma}_Z} \eta$ is the absolutely convergent integral over $\Delta_{Z_B} \times \Delta_{Z_R} \times (0, \infty)$ of $\phi(\rho u, u')\rho^{-1} u^{-\mathbf{1}} u'^{-\mathbf{1}}$. By (ii) the term $t$ of this integrand is

$$A_t(u)\, B_t(u')\, \rho^{\nu_t - 1} \prod_{g,i} \big(\rho + \xi_{g,i}\big)^{-k_{t,g}},$$

with $A_t = c_t\, u^{\mathbf{a}_t|_{Z_B} - \mathbf{1}} \prod_g K_{N_g|Z_B}^{-k_{t,g}}$, $B_t = u'^{\mathbf{a}_t|_{Z_R} - \mathbf{1}} \prod_g K_{N_g/Z_B}^{-k_{t,g}}$ and $\nu_t = \sum_{e \in Z_B} a_{t,e} - \sum_g k_{t,g} d_g$. By Tonelli, for almost every $(u, u')$ the integral over $\rho$ of the sum converges absolutely, and the integral over $\Delta_{Z_B} \times \Delta_{Z_R}$ of it is the period, with an integrable modulus. At such a point the integral over $\rho$ is its own finite part (Proposition 1.6): each term is $\rho^{\nu_t - 1}$ times a rational function of $\rho$ without poles on $(0, \infty)$, the $\xi_{g,i}$ lying off $(-\infty, 0]$, whose primitive, a sum of rational functions and of logarithms $\log \rho$ and $\log(\rho + \xi_{g,i})$, has log-power expansions at both ends. Finite parts are linear, so the integral over $\rho$ is $\sum_t A_t B_t \operatorname{Sc}_t$. A term in which no letter with $k_{t,g} > 0$ has a Skeleton value integrates $\rho^{\nu_t - 1}$, whose primitive $\rho^{\nu_t}/\nu_t$ or $\log \rho$ has no constant term at either

end, so $\mathrm{Sc}_t = 0$. Otherwise, by Theorem 1.7, $\mathrm{Sc}_t$ is the constant term at $\kappa = \nu_t$ of the integral $\int_0^\infty \rho^{\kappa-1} \prod_{g,i} (\rho + \xi_{g,i})^{-k_{t,g}} d\rho$, continued from the strip $0 < \operatorname{Re}\kappa < |\mathbf{w}_t|$, where it converges, the product having degree $|\mathbf{w}_t| = \sum_g k_{t,g} c_g$ in $\rho$; on the strip it is the stacked Gamma times the Breta of the statement (Proposition 15.1, coincident Skeleton values merging), and both sides are meromorphic in $\kappa$, so their constant terms at $\nu_t$ agree.

(iv) At $c = 1$ every $c_g$ is at most one by (i). A letter with $c_g = 1$ has the one Skeleton value $\gamma_R^g(u')/\gamma_B^g(u)$, positive (Corollary 19.5, the weights 1 factoring). Fix $\kappa$ in the strip. Corollary 15.5, one coefficient $\xi_g$ for each letter of the term with a Skeleton value, writes the stacked Gamma times the Breta as an integral over $|J_t| - 1$ shares $a_\ell$ on the lines $\operatorname{Re} a_\ell = c_\ell$ of a product of Beta corners in the shares and of the powers $\xi_g^{-b_g}$, $b_\ell = w_{t,\ell} - a_\ell$ for the shared coefficients and $b_0 = w_{t,0} - \kappa + |\mathbf{a}|$ for the last, affine in the shares and in $\kappa$, with $\operatorname{Re} b_g$ constant on the contour. The Beta corners are absolutely integrable over the contour: those in one share are bounded by a constant times $e^{-\pi|\operatorname{Im} a_\ell|}(1 + |\operatorname{Im} a_\ell|)^{w_{t,\ell}-1}$ by the classical bound $|\Gamma(c + i\tau)| \le C_c\, e^{-\pi|\tau|/2}(1 + |\tau|)^{c-1/2}$, and the last one by its value at the real parts of its entries, since $|\Gamma(z)| \le \Gamma(\operatorname{Re} z)$ for $\operatorname{Re} z > 0$. As $|\xi_g^{-b_g}| = \gamma_R^g(u')^{-\operatorname{Re} b_g} \gamma_B^g(u)^{\operatorname{Re} b_g}$, the integral of the modulus of $A_t B_t$ times the share integrand over the contour, $\Delta_{Z_B}$ and $\Delta_{Z_R}$ is the integral of the modulus of the share kernel times the two integrals of the statement, finite; by Fubini the integrals over $\Delta_{Z_B}$ and $\Delta_{Z_R}$ move inside the share integral, where at fixed shares the integrand is a function of $u$ times a function of $u'$, and they separate. If $0 < \nu_t < |\mathbf{w}_t|$, the integral over $\rho$ of the term converges at $\kappa = \nu_t$ and is $\mathrm{Sc}_t$ (Proposition 1.6), and the argument runs at $\kappa = \nu_t$.

(v) By Lemma 34.11, every element of $L(H_F, \widetilde{\sigma}_\mathcal{E})$ is $[H_F, \widetilde{\sigma}_\mathcal{E}, [\eta]]^{\mathrm{mot}}$ for a regular form $\eta$ on $V_F = V_{\mathcal{E};\varnothing\subsetneq\mathcal{E}}$. The step $\mathcal{E}$ meets the bunch of the root and its complement, so it is attached to the root as a node step, and (iii) applies to $\eta$. The left factors are elements of $L(H_F, \widetilde{\sigma}_\mathcal{E})$ by Definition 34.5.

(vi) At a split vertex the tables are direct sums over the molecules (Proposition 7.9), so the children at patterns of $M_v \setminus Z$ are direct sums over the parts of $Z$ in the molecules (proof of Lemma 34.10), and $Z_B = Z \cap M_1$ is a union of molecules of each of them, of width zero (Lemma 7.10); the same holds for the duals. The list $\binom{J}{1}$ has width one at $Z_B$ exactly when it has bases on both sides, that is when $J$ meets $Z_B$ and $Z_R$, and width zero otherwise. The proofs of (ii) to (iv) use only the widths of the letters at $Z_B$, here at most one. □

**Lemma S18** (Families on two marks)**.** *Every convergent family $F$ on a set $\mathcal{E}$ of two marks (Definition 34.1) has $H_F \cong \mathbb{Q}(0)$, and $P_F^{\mathrm{mot}} = P_F$, the Beta corner $\Gamma_{\binom{2}{1}}\left[\begin{smallmatrix} \kappa_{\{1\}}, & \kappa_{\{2\}} \\ \kappa_{\{1\}} + \kappa_{\{2\}} \end{smallmatrix}\right]$ of its walls, a positive rational number.*

*Proof.* The lists on $\mathcal{E} = \{1, 2\}$ have the basis polynomials 1, $\alpha_1$, $\alpha_2$, $\alpha_1\alpha_2$ and $\alpha_1 + \alpha_2$, and their cobasis polynomials are among these. $X_\mathcal{E}$ is the projective line, its boundary the two points $\alpha_1 = 0$ and $\alpha_2 = 0$, the monomials have no zeros on the torus, and the linear letter $\alpha_1 + \alpha_2$ is always removed; so $V_F$ is the projective line less one point, an affine line, and $D_F$ two points of it. In the long exact sequence of the pair, a sequence of morphisms of Brown's category (Huber and Müller-Stach), $H^0(V_F) \to H^0(D_F) \to H^1(V_F, D_F) \to H^1(V_F)$, the first map is the diagonal $\mathbb{Q}(0) \to \mathbb{Q}(0)^2$ and $H^1$ of the affine line is 0, every form $f(x)\,dx$ being exact; so $H_F \cong \mathbb{Q}(0)$. By the relation of Definition 34.5 for this isomorphism every motivic period of $H_F$ is a rational multiple of the unit $[\mathbb{Q}(0), 1, 1]^{\mathrm{mot}} = 1$, the multiple being its period. On $\Delta_\mathcal{E}$ the letter $\alpha_1 + \alpha_2$ is 1 and the monomials shift the masses to the walls, so $P_F = \int_0^1 u^{\kappa_{\{1\}}-1}(1-u)^{\kappa_{\{2\}}-1} du$, with $\kappa_{\{1\}}, \kappa_{\{2\}} \ge 1$ integers. □

**Theorem 34.17 (Pinned cells split off motivically).** *As stated.*

*Proof.* Write $\mathbf{h} = \{h_1, \dots, h_q\}$ and $r_{ij} = r_{\mathcal{A}_{ij}}$, so that $K_{\mathcal{A}_{ij}/g_i}$ and $\Psi_{\mathcal{A}_{ij}\setminus g_i}$ have the degrees $r_{ij}-1$ and $|R_i| - r_{ij}$; the core $\mathcal{H}_j$ has at least two marks, as no mark of a table on one mark is neither a loop nor a coloop, so $\mathcal{E} \setminus R_i \neq \varnothing$.

(v) is Corollary 21.11.

*The walls of the cells.* $R_i$ has width one in every active table (Proposition 21.3(i)), so on the balance wall $\kappa_{R_i} + \kappa_{\mathcal{E}\setminus R_i} = \sum_j (s_j + t_j) = 2\sigma$ (Proposition 8.16); both are at least 1, $F$ being convergent, so $1 \leq \kappa_i \leq 2\sigma - 1$.

*The letters.* By Theorem 21.4 and the hypothesis every active letter reads the core at one point:

$$K_j = \prod_i K_{\mathcal{A}_{ij}/g_i}(\alpha_{R_i})\, K_{\mathcal{H}_j}(\mathbf{x}), \qquad \Psi_j = \prod_i \Psi_{\mathcal{A}_{ij}\setminus g_i}(\alpha_{R_i})\, \Psi_{\mathcal{H}_j}(\mathbf{x}),$$
$$\mathbf{x} = \big(\alpha_L, \gamma_1(\alpha_{R_1}), \dots, \gamma_q(\alpha_{R_q})\big),$$

a mark of $L$ standing at its own variable, both hinge values of $\binom{\{l,g\}}{1}$ being $\alpha_l$.

*The map.* Each $\gamma_i$ is a quotient of polynomials, homogeneous of degree one, so

$$\Phi : \ \mathbb{P}(\mathbb{Q}^{\mathcal{E}}) \dashrightarrow \mathbb{P}(\mathbb{Q}^{L\sqcup\mathbf{h}}) \times \prod_i \mathbb{P}(\mathbb{Q}^{R_i}), \qquad [\alpha] \mapsto \big([\mathbf{x}], [\alpha_{R_1}], \dots, [\alpha_{R_q}]\big),$$

is well defined, between varieties of the same dimension $|\mathcal{E}| - 1$; off the zero sets of the numerators and denominators of the $\gamma_i$ it is an isomorphism between open subsets of the tori, with the inverse $\alpha_L = x_L$, $\alpha_{R_i} = \big(x_{h_i}/\gamma_i(v_i)\big)v_i$ at $([\mathbf{x}], [v_1], \dots, [v_q])$, well defined by the same homogeneity. Let $U_1$ be the torus less these zero sets, those of the augmented letters of $F$, and the preimages under $\Phi$ of those of the augmented letters of the core family and of the pinned families, and $U_2 = \Phi(U_1)$. All these polynomials are positive at the positive points, so $\Phi$ maps $\widetilde{\sigma}^\circ_{\mathcal{E}}$ onto $\widetilde{\sigma}^\circ_{L\sqcup\mathbf{h}} \times \prod_i \widetilde{\sigma}^\circ_{R_i}$.

*The forms.* On the positive points write $\alpha_{R_i} = \lambda_i v_i$ with $\lambda_i > 0$ and $v_i$ on a section of the rays of $\mathbb{R}^{R_i}_{>0}$, so that $x_{h_i} = \lambda_i \gamma_i(v_i)$. In logarithmic coordinates $\Phi$ keeps $\log \alpha_L$ and the $v_i$ and adds $\log \gamma_i(v_i)$ to $\log \lambda_i$, a map with Jacobian determinant 1; with $\nu_X = \Omega_X/\alpha^X$, the form $\Omega_X/\alpha^X$ being the invariant volume of the logarithmic coordinates modulo $\mathbf{1}$ (proof of Lemma S15), $\Phi^*\big(\nu_{L\sqcup\mathbf{h}} \wedge \bigwedge_i \nu_{R_i}\big) = \pm\nu_{\mathcal{E}}$, an identity of rational forms since it holds at the Zariski dense positive points. The core family $F_{\mathcal{H}}$ is on its balance wall: by Proposition 21.3(i), $r_j = r_{\mathcal{H}_j} + \sum_i (r_{ij} - 1)$, and

$$m_L + \sum_i \kappa_i = m_{\mathcal{E}} - \sum_{i,j} \big(s_j(r_{ij}-1) + t_j(|R_i| - r_{ij})\big) = \sum_j \Big(s_j r_{\mathcal{H}_j} + t_j\big(|L| + q - r_{\mathcal{H}_j}\big)\Big),$$

with $\kappa_i = m_{R_i} - \sum_j \big(s_j(r_{ij}-1) + t_j(|R_i| - r_{ij})\big)$ (Theorem 21.10). The word of $G_i$ is that of Corollary 22.7, $\gamma_i^{-\kappa_i} \prod_j K_{\mathcal{A}_{ij}/g_i}^{-s_j} \Psi_{\mathcal{A}_{ij}\setminus g_i}^{-t_j}$, homogeneous of the degree $-m_{R_i}$, so $G_i$ is on its balance wall. Pulling back $\omega_{F_{\mathcal{H}}}$ and $\omega_{G_i}$ by $\Phi$ substitutes $\mathbf{x}$ and $v_i = \alpha_{R_i}$ in their coefficients, and the powers $\gamma_i(\alpha_{R_i})^{\pm\kappa_i}$ cancel:

$$\Phi^*\big(\omega_{F_{\mathcal{H}}} \boxtimes \omega_{G_1} \boxtimes \cdots \boxtimes \omega_{G_q}\big) = \pm\, \alpha^{\mathbf{m}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}\, \nu_{\mathcal{E}} = \pm\, \omega_F.$$

(i) By the change of variables the period of $F$, of the positive integrand, is the product of the periods of $F_{\mathcal{H}}$ and of the $G_i$ in $(0, \infty]$; this is Corollary 22.6 with Corollary 22.7. $P_F$ is finite, so every factor is finite, and at integer data on its balance wall every proper wall of each factor is then at least 1 (Theorem 8.7): $F_{\mathcal{H}}$ and the $G_i$ are convergent, their tables being lists.

(ii) Apply Lemma 34.7 to $(X_{\mathcal{E}}, D, Y_F)$ and to the product of the toric varieties of $F_{\mathcal{H}}$ and of the $G_i$ with the union of the pull-backs of their boundaries and of their removed sets, a smooth projective variety with a divisor with simple normal crossings and smooth components, with the cells $\widetilde{\sigma}_{\mathcal{E}}$ and $\widetilde{\sigma}_{L\sqcup\mathbf{h}} \times \prod_i \widetilde{\sigma}_{R_i}$ and the forms $\omega_F$ and $\omega_{F_{\mathcal{H}}} \boxtimes \omega_{G_1} \boxtimes \cdots \boxtimes \omega_{G_q}$, regular on $V_{F_{\mathcal{H}}} \times \prod_i V_{G_i}$ by (i); the two periods are positive, so the signs of the orientations and of the forms match as in the proof of Corollary 34.8. With Lemma S17,

$$P_F^{\mathrm{mot}} = \Big[H^{n-1}\Big(V_{F_{\mathcal{H}}} \times \prod_i V_{G_i}, \partial\Big), \widetilde{\sigma}_{L\sqcup\mathbf{h}} \times \prod_i \widetilde{\sigma}_{R_i}, \omega_{F_{\mathcal{H}}} \boxtimes \cdots \boxtimes \omega_{G_q}\Big]^{\mathrm{mot}} = P_{F_{\mathcal{H}}}^{\mathrm{mot}} \prod_i P_{G_i}^{\mathrm{mot}},$$

and Lemma 34.6(iii) gives the conjugates.

(iii) Each cell is itself a substitution: $\mathcal{A}_{ij} = \binom{\{x,h\}}{1}\langle\binom{\{g_i,y\}}{1}, \mathcal{A}'_{ij}\rangle$, the mark $g_i$ left in place at $x$ and $\mathcal{A}'_{ij}$, a copy of $\mathcal{A}_{ij}$ with its hinge renamed, at $h$: the basis $\{h\}$ of the core contributes the bases of $\mathcal{A}_{ij} \setminus g_i$ and the basis $\{x\}$ the sets $Y + g_i$, $Y$ a basis of $\mathcal{A}_{ij}/g_i$ (Definition 21.1). The hinge values of $\mathcal{A}'_{ij}$ are those of $\mathcal{A}_{ij}$, so the cell family $F_i$ is read through the one hinge value $\gamma_i$, its wall at $R_i$ is $\kappa_i$ (Theorem 21.10, with the masses $\mathbf{m}|_{R_i}$), and it is on its balance wall with the mass $2\sigma - \kappa_i$ on $g_i$ (Lemma 21.12). Its core family $F'_i$ has the tables $\binom{\{g_i,h\}}{1}$, whose two letters are both $\alpha_{g_i} + \alpha_h$, with the masses $2\sigma - \kappa_i$ on $g_i$ and $\kappa_i$ on $h$, and its pinned family at the same loads is $G_i$. By the change of variables as in (i), in $(0,\infty]$, $P_{F_i} = P_{F'_i} P_{G_i}$ with $P_{F'_i} = \int_0^1 u^{2\sigma-\kappa_i-1}(1-u)^{\kappa_i-1}du = \Gamma\begin{bmatrix}\kappa_i, & 2\sigma-\kappa_i \\ & 2\sigma\end{bmatrix}_2^1$, finite since $1 \le \kappa_i \le 2\sigma - 1$. So $P_{F_i}$ is finite, $F_i$ is convergent (Theorem 8.7), and (ii) for $F_i$ gives $P_{F_i}^{\mathrm{mot}} = P_{F'_i}^{\mathrm{mot}} P_{G_i}^{\mathrm{mot}}$, with $P_{F'_i}^{\mathrm{mot}}$ the Beta corner by Lemma S18.

(iv) The form of $G_i$ was computed above without reference to the loads; its period is the integral over $\Delta_{R_i}$ of its coefficient, the pinned shape integral of Theorem 22.5 with all its constants 1; and two choices of loads give two convergent families on $R_i$ with one form, hence one motivic period (Lemma 34.7). □

**Corollary 34.18 (Dual pairs and full leans).** *As stated.*

*Proof.* (a) By Proposition 21.8(ii) each cell has $\gamma^K_{\mathcal{A}_i} = \gamma^{\Psi}_{\mathcal{A}_i} = k_i/\psi_i$, so the one active table reads it through one hinge value whatever $s$ and $t$, and Theorem 34.17 applies. With $K_{\mathcal{A}_i/g_i} = \Psi_{\mathcal{A}_i \setminus g_i} = \psi_i$, the form of the pinned family (Theorem 34.17(iv)) is $(k_i/\psi_i)^{-\kappa_i}\psi_i^{-s}\psi_i^{-t}\alpha^{\mathbf{m}|_{R_i}}\nu_{R_i} = k_i^{-\kappa_i}\psi_i^{-(s+t-\kappa_i)}\alpha^{\mathbf{m}|_{R_i}}\nu_{R_i}$, the form of $(\mathcal{A}_i \setminus g_i; \kappa_i, s+t-\kappa_i)$ at the masses $\mathbf{m}|_{R_i}$, and $\mathcal{A}_i \setminus g_i = \mathcal{B}|R_i$, the cell of the bunch $R_i$ (Proposition 21.3(i), Theorem 19.4). By Proposition 21.8(i), $|R_i| - r_{\mathcal{A}_i} = r_{\mathcal{A}_i} - 1$ with $r_{\mathcal{A}_i} = r(R_i)$, so Theorem 21.10 gives $\kappa_i = m_{R_i} - (s+t)(r(R_i)-1)$. The family $(\mathcal{B}|R_i; \kappa_i, s+t-\kappa_i)$ at the masses $\mathbf{m}|_{R_i}$ has the form of the convergent pinned family, which is homogeneous of degree zero, so it is on its balance wall; its walls are the orders of the form along the boundary divisors plus one (Proposition 34.4(ii)), those of the pinned family, so it is convergent, and by Lemma 34.7 the two have one motivic period. Theorem 34.17(ii) is the display.

(b) Let $t = 0$; the case $s = 0$ is the same with $\Psi$. If $s = 0$ too, $\kappa_S + \kappa_{\mathcal{E}\setminus S} = 0$ for every $S$ on the balance wall (Proposition 8.16), against convergence; so $s \neq 0$, and only $K$ is active: every cell of every substitution of $\mathcal{B}$ is read through one hinge value, its $\gamma^K$. By Proposition 8.16, $\kappa_S + \kappa_{\mathcal{E}\setminus S} = s\,c(S) \ge 2$ for every nonempty proper $S$, so $\mathcal{B}$ is connected. We induct on $|\mathcal{E}|$.

If $\mathcal{B}$ has a set $R$ of width one with $|R|, |\mathcal{E} \setminus R| \ge 2$, then $\mathcal{B} = \mathcal{H}\langle\mathcal{A}\rangle$, the marks of $L = \mathcal{E} \setminus R$ left in place, $\mathcal{H}$ the cell of the room and $\mathcal{A}$ the cell of the bunch $R$ (Proposition 21.3(iii)), connected lists on fewer marks (Proposition 21.3(iv)). Theorem 34.17(ii),(iii) with $q = 1$ give

$$P_F^{\mathrm{mot}} = \Gamma\begin{bmatrix}\kappa_R, & s-\kappa_R \\ & s\end{bmatrix}_2^1{}^{-1} P_{(\mathcal{H};s,0)}^{\mathrm{mot}}\, P_{(\mathcal{A};s,0)}^{\mathrm{mot}},$$

the two families convergent at the masses $(\mathbf{m}|_L, \kappa_R)$ and $(\mathbf{m}|_R, s-\kappa_R)$, where $s-\kappa_R=\kappa_L$ (Proposition 8.16); the marker of each carries the wall of $F$ at the marks on the other side of the split. A wall of $(\mathcal{H}; s, 0)$ at a set containing the marker is the wall of $F$ at that set with the marker replaced by $R$, and at a set of marks of $L$ it is the wall of $F$ there (Theorem 21.10, the hinge $g$ of $\mathcal{A}$ having $d_{\mathcal{A}}(R+g)=d_{\mathcal{A}}(R)+1$ and $d^*_{\mathcal{A}}(R+g)=d^*_{\mathcal{A}}(R)+1$); likewise for $(\mathcal{A}; s, 0)$. So the induction, applied to the two families, writes $P_F^{\mathrm{mot}}$ as a positive rational number times the product over the prime nodes of $P_{F_v}^{\mathrm{mot}}$, with the masses of the statement.

Otherwise $\mathcal{B}$ is prime, its own one node, or it has two or three marks. A connected list on two marks is $\binom{\mathcal{E}}{1}$, and one on three marks is $\binom{\mathcal{E}}{1}$ or $\binom{\mathcal{E}}{2}$, having no loop and no coloop. On two marks $P_F^{\mathrm{mot}}$ is a positive rational number by Lemma S18. On three marks, for a mark $g$, neither a loop nor a coloop, $\mathcal{B}$ is the cell $\binom{\{g,h\}}{1}\langle g, \mathcal{A}'\rangle$ of the proof of Theorem 34.17(iii) with the two other marks as $R$, and Theorem 34.17(ii) writes $P_F^{\mathrm{mot}}$ as the product of the motivic periods of a convergent family on the two marks $g, h$ and of a convergent pinned family on the two marks of $R$, each a positive rational number by Lemma S18. Lemma 34.6(iii) gives the conjugates, a nonzero rational factor changing nothing. □

**Theorem 34.19 (A pinned set splits off motivically).** *As stated.*

*Proof.* $F$ is a family of lists, so by Theorem 28.25 the active hinge values at $S$ are equal, $\gamma=\gamma^K_{j,S}=\gamma^\Psi_{j',S}$ for all active $K_j$ and $\Psi_{j'}$. Every active table $\mathcal{B}_j$ has width one at $S$ with factoring weights, so it is the gluing of its cell of the bunch $\mathcal{B}^+_{j,S}$ and its cell of the room $\mathcal{B}^+_{j,S'}$ along their hinges (Proposition 21.3(iii)), that is the substitution $\mathcal{B}^+_{j,S'}\langle\mathcal{B}^+_{j,S}\rangle$ with the marks of $S'$ left in place (Definition 21.1); the hinge values of $\mathcal{B}^+_{j,S}$ are $\gamma^K_{j,S}$ and $\gamma^\Psi_{j,S}$. So the hypothesis of Theorem 34.17 holds with $q=1$, $L=S'$, $R_1=S$ and $\gamma_1=\gamma$, $|S|\geq 2$. Its core family is $((\mathcal{B}^+_{j,S'}; s_j, t_j))_j$ at the masses $(\mathbf{m}|_{S'}, \kappa_S)$. By Proposition 28.3(ii) the room station letters of $S'$ are $\mathsf{y}_{jD_0}=K_{\mathcal{B}_j|S'}$, $\mathsf{y}_{jD_1}=K_{\mathcal{B}_j/S}$, $\mathsf{y}^*_{jD_0}=\Psi_{\mathcal{B}_j|S'}$ and $\mathsf{y}^*_{jD_1}=\Psi_{\mathcal{B}_j/S}$, so the partner with its stand-in at $\mathbf{1}$ has the letters $K_{\mathcal{B}_j|S'}+\alpha_g K_{\mathcal{B}_j/S}$ and $\alpha_g\Psi_{\mathcal{B}_j|S'}+\Psi_{\mathcal{B}_j/S}$ and the mass $\kappa_S$ on $g$ (Theorem 28.10(i)), the letters of the cell of the room (Theorem 19.4): it is the core family. The period of the pinned family is $\int_{\Delta_S}\gamma^{-\kappa_S}u^{\mathbf{m}|_S-\mathbf{1}}\prod_j K^{-s_j}_{\mathcal{B}_j/S'}\Psi^{-t_j}_{\mathcal{B}_j|S}\,d\omega_S$ (Theorem 34.17(iv)), and by the display after Definition 28.5, with the weights factoring, this is $\int_{\Delta_S}\phi^\rho_S$: there $\gamma^\rho_S=\prod_j(\gamma^K_{j,S})^{s_j/w}(\gamma^\Psi_{j,S})^{t_j/w}=\gamma$, $w=\sum_j(s_j+t_j)$ being the sum of the exponents of the active tables, all of width one at $S$. Theorem 34.17(ii) is the display, and its period is $P_F=c\,P_{S'(\mathbf{1})}$. □

**Proposition 34.20 (Rational data on a Kummer cover).** *As stated.*

*Proof.* (i) $\omega_F=\alpha^{\mathbf{m}}\prod_j K_j^{-s_j}\Psi_j^{-t_j}\,\Omega_{\mathcal{E}}/\alpha^{\mathcal{E}}$, and the balance $m_{\mathcal{E}}=\sum_g\delta_g e_g$ lets the powers of $\alpha_{e_0}$ be distributed as displayed, each factor of degree 0. The letters and the coordinate ratios are regular units on the points of the torus in $V_F$, and $Mm_e$, $Me_g$ are integers, so the $M$-th power of the function in front of $\Omega_{\mathcal{E}}/\alpha^{\mathcal{E}}$ is a regular unit there.

(ii) In the chart of an order the ratios $\alpha_e/\alpha_{e_0}$ are monomials in the $\varepsilon_i$, and each letter is $g/\alpha_{e_n}^{\delta_g}=\prod_i\varepsilon_i^{d_g(S_i)}\widetilde{g}$ with the sector letter $\widetilde{g}$, a unit on $V_F\cap U$ (Lemma 34.3(ii), Definition 34.1). So over the chart the extension is generated by the $\eta_i$ and the $\widetilde{w}_g$, and the ring $\mathcal{O}(V_F\cap U)[\eta,\widetilde{w}]$ with $\eta_i^M=\varepsilon_i$ and $\widetilde{w}_g^M=\widetilde{g}$ is the coordinate ring of a variety finite over $V_F\cap U$: the coordinatewise $M$-th power map in $\eta$, which is finite and flat from a smooth variety with the coordinates $\eta$, followed by an étale cover adjoining roots of units. It is smooth, hence normal, so it is the normalization over the chart, and the charts glue because the $r_e$, the $w_g$ and the $\eta_i=r_{e_i}/r_{e_{i+1}}$ are expressed in one another by monomials. The preimage of $\{\varepsilon_i=0\}$ is $\{\eta_i=0\}$. By Lemma 34.3(iii), read

for rational exponents on a branch, $\omega_F = \pm\prod_i \varepsilon_i^{\kappa_{S_i}-1}\prod_g \widetilde{g}^{-e_g}d\varepsilon$, and $\varepsilon_i^{\kappa-1}d\varepsilon_i = M\eta_i^{M\kappa-1}d\eta_i$ with $\widetilde{g}^{-e_g} = \widetilde{w}_g^{-Me_g}$ gives the display. The walls lie in $\frac{1}{M}\mathbb{Z}$ and are positive, so $M\kappa_{S_i}-1\geq 0$, and the $\widetilde{w}_g$ are units: the form is regular.

(iii) On $\widetilde{\sigma}^\circ_{\mathcal{E}}$ every ratio and every letter is real and positive (Proposition 34.4(i)), so the branch with every $r_e$ and $w_g$ positive is defined over it, and in a chart it is $[0,\infty)^{n-1}$ in $\eta$, mapped onto $[0,\infty)^{n-1}$ in $\varepsilon$ by the coordinatewise $M$-th power, a homeomorphism preserving the orientation. So $\widetilde{\sigma}^{(M)}_{\mathcal{E}}$ is compact, semialgebraic, with its boundary in $D^{(M)}$, and it is a class in $H_{n-1}(V_F^{(M)}, D^{(M)})$; the change of variables $\varepsilon = \eta^M$ carries the integral of $\omega_F^{(M)}$ over it to the integral of the principal branch of $\omega_F$ over the positive cell, which is $P_F$ (Proposition 34.4(v)). $V_F^{(M)}$, $D^{(M)}$ and $\omega_F^{(M)}$ are defined over $\mathbb{Q}$, so $H^{n-1}(V_F^{(M)}, D^{(M)})$ is an object of the category of realizations, as in Definition 34.5. At $M=1$ there are no roots to adjoin.

(iv) The group acts on the generators of the extension and so on the normalization, over $\mathbb{Q}(\mu_M)$. By (i), $\omega_F^{(M)} = \prod_e r_e^{Mm_e}\prod_g w_g^{-Me_g}\,\Omega_{\mathcal{E}}/\alpha^{\mathcal{E}}$ on the points over the torus, and $\Omega_{\mathcal{E}}/\alpha^{\mathcal{E}}$ is pulled back from $V_F$; so the pullback by $(\zeta,\zeta')$ multiplies it by $\chi(\zeta,\zeta')$ there, hence everywhere, the points over the torus being dense. The action on the de Rham cohomology is by pullback, so the class lies in the $\chi$-eigenspace. □

**Theorem 34.21 (Motivic periods are separated by their conjugates).** *As stated.*

*Proof.* (i) The category of realizations of Definition 34.5 is a Tannakian category with two fibre functors, the Betti and the de Rham realizations, and its motivic periods are the functions on the scheme $P$ of tensor isomorphisms from the de Rham to the Betti fibre functor, its de Rham periods the functions on the group $G^{\mathrm{dR}}$ of tensor automorphisms of the de Rham fibre functor; the coaction is dual to the action $P\times G^{\mathrm{dR}}\to P$, $(p,g)\mapsto p\circ g$, and the period is evaluation at the comparison isomorphism $c\in P(\mathbb{C})$ (Brown's description of motivic periods, which applies because $\mathcal{P}^{\mathrm{mot}}$ and $\mathcal{P}^{\mathrm{dR}}$ are defined as the matrix coefficients $[M,\gamma,\omega]$ of the objects of that category). For $\xi\in\mathcal{P}^{\mathrm{mot}}$, $(\mathrm{per}\otimes\mathrm{id})\Delta\xi$ is the function $g\mapsto\xi(c\circ g)$ on $G^{\mathrm{dR}}_{\mathbb{C}}$, the pullback of $\xi$ along $g\mapsto c\circ g$. Since $P$ is a torsor under $G^{\mathrm{dR}}$, this map $G^{\mathrm{dR}}_{\mathbb{C}}\to P_{\mathbb{C}}$ is an isomorphism, and its pullback is injective on $\mathcal{O}(P)\subseteq\mathcal{O}(P)\otimes\mathbb{C}$. An element $\sum_j z_j\otimes w_j$ of $\mathbb{C}\otimes_{\mathbb{Q}}\mathcal{P}^{\mathrm{dR}}$ with the $w_j$ independent over $\mathbb{Q}$ vanishes exactly when every $z_j$ does, that is when $\mathrm{id}\otimes\varphi$ kills it for every $\mathbb{Q}$-linear $\varphi:\mathcal{P}^{\mathrm{dR}}\to\mathbb{Q}$; and $(\mathrm{id}\otimes\varphi)(\mathrm{per}\otimes\mathrm{id})\Delta\xi = \mathrm{per}\big((\mathrm{id}\otimes\varphi)\Delta\xi\big)$, the period of an element of $\mathrm{C}(\xi)$. Conversely $\xi\in\mathrm{C}(\xi)$ (Lemma 34.6(i)), so $\xi=0$ makes every conjugate 0.

(ii) By Definition 34.5, $\Delta P_F^{\mathrm{mot}} = \sum_i [H_F,\widetilde{\sigma}_{\mathcal{E}},e_i]^{\mathrm{mot}}\otimes[H_F,e_i^\vee,\omega_F]^{\mathrm{dR}}$, and $(\mathrm{per}\otimes\varphi)$ of it is the displayed sum; (i) gives the equivalence. For $e_i\in W_k$, $k<n-1$, Theorem 34.13 places $[H_F,\widetilde{\sigma}_{\mathcal{E}},e_i]^{\mathrm{mot}}$ in the sum over flags of dimension $k$ of the products over their steps of the open spaces, the steps having at most $k+1$ marks and being attached as stated, and Corollary 34.15 writes the period of each element of an open space as a rational combination of values of families of lists on the step and of their derivatives in the masses, which are values of families of lists with logarithmic insertions, read on convergent data. □

## 50.34 Chapter 35: positive words

**Theorem 35.1 (Positive coefficient jets).** *As stated, at positive weights that need not be rational.*

*Proof.* Write $P = P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})$, finite and positive because its walls $n\kappa_S$ are positive (8.7), and $\mathbb{E}$ for the expectation of $U\sim\mathrm{Breta}^\circ(F;n\mathbf{m};n\mathbf{s},n\mathbf{t})$, whose density on the sim-

plex is $P^{-1}u^{n\mathbf{m}-\mathbf{1}}\prod_j K_j(u)^{-ns_j}\Psi_j(u)^{-nt_j}$, positive on the open simplex. For $v\in\mathbb{R}^k$ put $X_j=v^{\mathsf T}\partial_{\mathbf b}K_j/K_j$ and $X_j^*=v^{\mathsf T}\partial_{\mathbf b}\Psi_j/\Psi_j$. With $u_{j,T}=\mathbf b_{j,T}/\theta_{j,T}$,

$$\frac{\partial_{\mathbf b}K_j}{K_j}=\mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[u_{j,T}],\qquad \frac{\partial_{\mathbf b}\Psi_j}{\Psi_j}=\mathbb{E}_{T\sim\mathbb{P}^*_{j,\alpha}}[u_{j,T}],$$

so $|X_j|$ and $|X_j^*|$ are at most $\beta_j=\max_T|v^{\mathsf T}u_{j,T}|$ on the whole orthant.

(i) For $|\varepsilon|\max_j\beta_j<\frac12$ the weights $\theta_{j,T}\pm\varepsilon v^{\mathsf T}\mathbf b_{j,T}=\theta_{j,T}(1\pm\varepsilon v^{\mathsf T}u_{j,T})$ are positive, and the letters of the two copies of table $j$ are $K_j(1\pm\varepsilon X_j)$ and $\Psi_j(1\pm\varepsilon X_j^*)$. The copies have the levels of table $j$, each counted twice at half the exponents, so the doubled family at the masses $n\mathbf m$ has the imbalance 0 and the walls $n\kappa_S$, and its word is

$$\prod_j K_j^{-ns_j}\Psi_j^{-nt_j}\cdot B_\varepsilon,\qquad B_\varepsilon=\prod_j\bigl(1-\varepsilon^2X_j^2\bigr)^{-\frac{ns_j}{2}}\bigl(1-\varepsilon^2X_j^{*2}\bigr)^{-\frac{nt_j}{2}}.$$

Its period is therefore $P\,\mathbb{E}[B_\varepsilon(U)]$, the display of (i). For $|\varepsilon|\le\varepsilon_0<(2\max_j\beta_j)^{-1}$, $B_\varepsilon$ is a power series in $\varepsilon^2$ whose coefficients are polynomials in the bounded functions $X_j^2,X_j^{*2}$, converging uniformly on the orthant; so $\mathbb{E}[B_\varepsilon(U)]$ is analytic in $\varepsilon$ and expands term by term, with the coefficient $\sum_j\bigl(\frac{ns_j}{2}\mathbb{E}[X_j^2]+\frac{nt_j}{2}\mathbb{E}[X_j^{*2}]\bigr)$ of $\varepsilon^2$. Since $X_j^2=v^{\mathsf T}\bigl(\partial_{\mathbf b}K_j\,\partial_{\mathbf b}K_j^{\mathsf T}/K_j^2\bigr)v$, $P$ times this coefficient is $v^{\mathsf T}\mathbf H_nv$. It is a sum of nonnegative terms. If it vanishes, then for every $j$ with $s_j>0$, $X_j(U)=0$ almost surely; the law of $U$ has a positive density on the open simplex, so the polynomial $\sum_T(v^{\mathsf T}\mathbf b_{j,T})\alpha^T$ vanishes on an open set, and every $v^{\mathsf T}\mathbf b_{j,T}$ is 0, the monomials of distinct bases being distinct. For $t_j>0$ the same holds with $\alpha^{\mathcal E\setminus T}$. So $v$ is orthogonal to every direction of an active table, and $v=0$ when these span $\mathbb{R}^k$.

(ii) Write $Q=R^{\mathsf T}R$ with $R$ real. Then

$$\operatorname{tr}(Q\mathbf H_n)=\frac n2\,P\,\mathbb{E}\Bigl[\sum_j\Bigl(s_j\frac{|R\,\partial_{\mathbf b}K_j|^2}{K_j^2}+t_j\frac{|R\,\partial_{\mathbf b}\Psi_j|^2}{\Psi_j^2}\Bigr)(U)\Bigr],$$

a sum of nonnegative terms, which vanishes exactly when, as in (i), $R\mathbf b_{j,T}=0$, that is $\mathbf b_{j,T}^{\mathsf T}Q\mathbf b_{j,T}=0$, for every basis of every active table. If $Q_{11}>0$, then $Q-Q_{11}^{-1}Q_{\cdot1}Q_{\cdot1}^{\mathsf T}$ is rational, positive semidefinite, with first row and column zero; if $Q_{11}=0$, the first row and column of $Q$ vanish, $Q$ being positive semidefinite. Induction on $k$ gives $Q=\sum_l c_lv_lv_l^{\mathsf T}$ with rational $c_l>0$ and rational $v_l$, and then $\operatorname{tr}(Q\mathbf H_n)=\sum_l c_l\,v_l^{\mathsf T}\mathbf H_nv_l$.

(iii) $\partial_{\mathbf b}K_j^{\mathsf T}Q\,\partial_{\mathbf b}K_j/K_j^2=\sum_{T,T'}\mathbf b_{j,T}^{\mathsf T}Q\,\mathbf b_{j,T'}\,\alpha^{\mathbf 1_T+\mathbf 1_{T'}}K_j^{-2}$, and $P\,\mathbb{E}\bigl[(\alpha^{\mathbf 1_T+\mathbf 1_{T'}}K_j^{-2})(U)\bigr]$ is the period whose integrand is that of $P$ times $u^{\mathbf 1_T+\mathbf 1_{T'}}K_j(u)^{-2}$, the period at $\bigl(n\mathbf m+\mathbf 1_T+\mathbf 1_{T'};n\mathbf s+2\mathbf e_j,n\mathbf t\bigr)$. Its imbalance is that of $(n\mathbf m;n\mathbf s,n\mathbf t)$ plus $2r_j-2r_j=0$, and its wall of $S$ is $n\kappa_S+|T\cap S|+|T'\cap S|-2d_j(S)\ge n\kappa_S$, by 7.13. The cobasis part is the same with $u^{\mathbf 1_{\mathcal E\setminus T}+\mathbf 1_{\mathcal E\setminus T'}}\Psi_j^{-2}$ and the change $|S\setminus T|+|S\setminus T'|-2d_j^*(S)\ge0$. This is the case of order two of 27.9 for the doubled family, the two copies of each table merged.

(iv) $x\mapsto x^{\mathsf T}Qx$ is convex, so $\bigl(\mathbb{E}_T[u_{j,T}]\bigr)^{\mathsf T}Q\,\mathbb{E}_T[u_{j,T}]\le\mathbb{E}_T\bigl[u_{j,T}^{\mathsf T}Qu_{j,T}\bigr]\le\max_T u_{j,T}^{\mathsf T}Qu_{j,T}$ under either law, and $\operatorname{tr}(Q\mathbf H_n)\le\frac n2P\sum_j(s_j+t_j)\max_T\mathbf b_{j,T}^{\mathsf T}Q\,\mathbf b_{j,T}/\theta_{j,T}^2=n\,C(Q)\,P$; positivity is (ii).

(v) With the basis weights $\theta^K_{j,T}$ and the cobasis weights $\theta^\Psi_{j,T}$ of every table independent (8.1), $\partial^2K_j^{-ns_j}/\partial\theta^K_{j,T}\partial\theta^K_{j,T'}=ns_j(ns_j+1)K_j^{-ns_j-2}\alpha^T\alpha^{T'}$. The differentiated integrals are the periods of (iii), convergent locally uniformly in the weights, so differentiation under the integral gives

$$\sum_{T,T'}\mathbf b_{j,T}\mathbf b_{j,T'}^{\mathsf T}\,\frac{\partial^2P}{\partial\theta^K_{j,T}\partial\theta^K_{j,T'}}=ns_j(ns_j+1)\,P\,\mathbb{E}\Bigl[\frac{\partial_{\mathbf b}K_j\,\partial_{\mathbf b}K_j^{\mathsf T}}{K_j^2}(U)\Bigr],$$

which is $2(ns_j+1)$ times the part $\frac{n}{2}s_jP\,\mathbb{E}[\cdots]$ of $\mathbf{H}_n$ carried by $K_j$; the cobasis part is the same with $t_j$. On the share side of 18.16 the weights enter only through $\prod_{j,T}(\theta^K_{j,T})^{-a_{j,T}}(\theta^\Psi_{j,T})^{-b_{j,T}}$: a derivative in $\theta^K_{j,T}$ multiplies the integrand by $-a_{j,T}/\theta^K_{j,T}$, one in $\theta^\Psi_{j,T}$ by $-b_{j,T}/\theta^\Psi_{j,T}$, and neither the contour, nor the pins, nor the share kernels involve the weights.

(vi) At the centre the two copies of table $j$ carry the equal exponents $(\frac{ns_j}{2},\frac{ns_j}{2})$, and the masses $n\sigma\mathbf{1}$ are the total strength of the doubled family times $\mathbf{1}$. 8.21(i) holds at every family at its centre, whatever its weights: at $\mathbf{1}$ the usage is $\sum_j\sigma_j(R_j+R^*_j)(\mathbf{1})=\sigma\mathbf{1}$, since $R_{j,e}(\mathbf{1})+R^*_{j,e}(\mathbf{1})$ is the chance that $e$ lies in the basis plus the chance that it does not. □

*Remark.* No step uses the rationality of the weights; rational weights are what the arithmetic of the jets asks for, through the rows of the periods of (iii).

**Theorem 35.3 (Two bases generate positive contrast words of any depth).** *As stated, at every point of the balance wall with real data and positive proper walls, and at positive weights.*

*Proof.* Write $P_n=P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})$, finite and positive (8.7), $\mathbb{E}$ for the expectation of $U\sim\mathrm{Breta}^\circ(F;n\mathbf{m};n\mathbf{s},n\mathbf{t})$, and $c=\mathbb{P}_{j,\alpha}(T)-\mathbb{P}_{j,\alpha}(T')$, $c^*=\mathbb{P}^*_{j,\alpha}(T)-\mathbb{P}^*_{j,\alpha}(T')$. Each chance lies in $(0,1)$ and two chances of distinct bases sum to at most one, so $|c|,|c^*|<1$ on the open orthant. $c$ vanishes only where $\theta_{j,T}\alpha^{\mathbf{1}_T}=\theta_{j,T'}\alpha^{\mathbf{1}_{T'}}$, a hypersurface, since $\mathbf{1}_T\neq\mathbf{1}_{T'}$; so $c^{2h}>0$ almost everywhere, and the contrast word, $\frac12P_n$ times the expectation of a function that is positive almost everywhere under a law with a positive density, is positive.

(i) $c^{2h}=\sum_{l=0}^{2h}(-1)^l\binom{2h}{l}\theta^{2h-l}_{j,T}\theta^l_{j,T'}\,\alpha^{v_l}K_j^{-2h}$ with $v_l=(2h-l)\mathbf{1}_T+l\mathbf{1}_{T'}$, each term homogeneous of degree 0, and $P_n\,\mathbb{E}\big[(\alpha^{v_l}K_j^{-2h})(U)\big]$ is the period whose integrand is that of $P_n$ times $u^{v_l}K_j(u)^{-2h}$, the period at $\big(n\mathbf{m}+v_l;n\mathbf{s}+2h\mathbf{e}_j,n\mathbf{t}\big)$. Its imbalance changes by $|v_l|-2hr_j=0$, and its wall of $S$ by $(2h-l)|T\cap S|+l|T'\cap S|-2hd_j(S)\geq0$ (7.13). The second half is the same with $\mathbb{P}^*_{j,\alpha}(T)=\theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j$, the masses raised by $(2h-l)\mathbf{1}_{\mathcal{E}\setminus T}+l\mathbf{1}_{\mathcal{E}\setminus T'}$, the cobasis exponent by $2h$, and the wall of $S$ by $(2h-l)|S\setminus T|+l|S\setminus T'|-2hd^*_j(S)\geq0$.

From $\Psi_j(\alpha)=\alpha^{\mathbf{1}}K_j(\alpha^{-1})$, $K_j(\alpha^{-1})=\alpha^{-\mathbf{1}}\Psi_j(\alpha)$ and $\Psi_j(\alpha^{-1})=\alpha^{-\mathbf{1}}K_j(\alpha)$, so $\mathbb{P}^*_{j,\alpha^{-1}}(T)=\mathbb{P}_{j,\alpha}(T)$. On the chart $y_g=0$ of 8.7 the integrand of a period at data $(\mathbf{m}';\mathbf{s}',\mathbf{t}')$ is $e^{\langle\mathbf{m}',y\rangle}\prod_j K_j(e^y)^{-s'_j}\Psi_j(e^y)^{-t'_j}$, and $y\mapsto-y$, which keeps $dy$, carries it to $e^{\langle2\sigma'\mathbf{1}-\mathbf{m}',y\rangle}\prod_j K_j(e^y)^{-t'_j}\Psi_j(e^y)^{-s'_j}$, $\sigma'$ the total strength of $(\mathbf{s}',\mathbf{t}')$. Applied to the second half at $(n\mathbf{m};n\mathbf{s},n\mathbf{t})$ it gives the first half at $\big(n(2\sigma\mathbf{1}-\mathbf{m});n\mathbf{t},n\mathbf{s}\big)$, the depth $n$ of the data $(2\sigma\mathbf{1}-\mathbf{m};\mathbf{t},\mathbf{s})$ of 8.14, whose wall of $S$ is the wall of $\mathcal{E}\setminus S$ at $(\mathbf{m};\mathbf{s},\mathbf{t})$, positive. At the centre these data are $(\mathbf{m};\mathbf{s},\mathbf{t})$ themselves, and the two halves are equal.

(ii) By 17.1, with $X(z)=2\langle\boldsymbol{\mu}_\perp,z\rangle-\Phi^\circ(z)$ read on $\mathbf{1}^\perp$ and the number of marks written $|\mathcal{E}|$, $P_n=2^{|\mathcal{E}|-1}\int_{\mathcal{T}}e^{nX}dz$ and the contrast word is $2^{|\mathcal{E}|-2}\int_{\mathcal{T}}e^{nX}\big(c^{2h}+c^{*2h}\big)dz$, with $c,c^*$ at $\alpha=e^{2z}$. For $h=\eta n$ put $\psi=-X-2\eta\log\max(|c|,|c^*|)$, with values in $(-\infty,\infty]$, continuous where finite and lower semicontinuous; then $e^{-n\psi}\leq e^{nX}(c^{2h}+c^{*2h})\leq2e^{-n\psi}$. By 17.4(i) and (ii), $-X(z)\geq2\kappa_{\min}(\max_e z_e-\min_e z_e)-C$, and the logarithmic term is nonnegative, so $\psi$ grows at least linearly on $\mathbf{1}^\perp$. Laplace's principle follows: outside a large ball the integral of $e^{-n\psi}$ is at most $e^{-n(\inf\psi+1)}$ for large $n$, inside it is at most $e^{-n\inf\psi}$ times the volume, and near a point $z_0$ with $\psi(z_0)<\inf\psi+\epsilon$ the integrand is at least $e^{-n(\inf\psi+2\epsilon)}$ on a ball. So $\frac1n\log$ of the contrast word tends to $-\inf\psi$, and the same with $\eta=0$ and without the contrasts gives $\frac1n\log P_n\to\max X=-\min(\Phi^\circ-2\langle\boldsymbol{\mu}_\perp,\cdot\rangle)$, the maximum attained, $X$ being continuous and tending to $-\infty$. The set $K=\{z\in\mathbf{1}^\perp:\ 2\kappa_{\min}(\max_e z_e-\min_e z_e)-C\leq\inf\psi+1\}$ is compact and contains every $z$ with $\psi(z)\leq\inf\psi+1$; on it $\max(|c|,|c^*|)\leq1-\delta_K<1$, so $\inf\psi\geq\min(-X)+2\eta|\log(1-\delta_K)|>\min(-X)$.

For $A>0$ let $B_A$ be a ball of $\mathbf{1}^{\perp}$ outside which $2\kappa_{\min}(\max_e z_e-\min_e z_e)-C\ge A$; on $B_A$, $\psi\ge\min(-X)+2\eta|\log(1-\delta_{B_A})|$, which exceeds $A$ for large $\eta$, so $\inf\psi\ge A$ for large $\eta$.

(iii) Let $s_j>0$ and $q=2h$. In the share form of $P_F$ used by 18.16, the letter $K_j^{-ns_j-q}$ of the term $l$ of (i) is opened by one share $a'_U$ per basis, $\sum_U a'_U=ns_j+q$, with the share kernel $\prod_U\Gamma(a'_U)/\Gamma(ns_j+q)$ and the factors $(\theta_{j,U}\alpha^{\mathbf{1}_U})^{-a'_U}$ (4.4). The insertion $\theta_{j,T}^{q-l}\theta_{j,T'}^{l}\alpha^{v_l}$ turns the factors of $T$ and $T'$ into $(\theta_{j,T}\alpha^{\mathbf{1}_T})^{-a_T}$ and $(\theta_{j,T'}\alpha^{\mathbf{1}_{T'}})^{-a_{T'}}$ with $a_T=a'_T-q+l$, $a_{T'}=a'_{T'}-l$, the other shares unchanged as $a_U=a'_U$. In the shares $a$ the total is $ns_j$ and the factors in $\alpha$ and in the weights are those of $P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})$, so the integral over $\alpha$, which produces the pins, gives the pins of $(n\mathbf{m};n\mathbf{s},n\mathbf{t})$, and the kernel is

$$\frac{\Gamma(a_T+q-l)\,\Gamma(a_{T'}+l)\prod_{U\neq T,T'}\Gamma(a_U)}{\Gamma(ns_j+q)}=\frac{\prod_U\Gamma(a_U)}{\Gamma(ns_j)}\cdot\frac{(a_T)_{q-l}\,(a_{T'})_l}{(ns_j)_q}.$$

The factor $\Gamma(a_T+q-l)\Gamma(a_{T'}+l)$ is holomorphic for $\operatorname{Re}a_T>-(q-l)$ and $\operatorname{Re}a_{T'}>-l$, so the contour returns to a wall contour of $(n\mathbf{m};n\mathbf{s},n\mathbf{t})$ without crossing a pole, the share kernels decaying along the imaginary directions. Summing over $l$ with the weights $\frac12(-1)^l\binom{q}{l}$ gives the share side of $P_F(n\mathbf{m};n\mathbf{s},n\mathbf{t})$ with the insertion $H_q(a_{j,T},a_{j,T'})/\big(2(ns_j)_q\big)$. For $t_j>0$ the second half is the same with $\Psi_j$, the cobasis shares $b$ and $nt_j$. The insertions are polynomials in the shares and change neither the pins nor the share kernels, hence no pole hyperplane. At the centre the two halves are equal by (i), and their insertions add to the one of the reference. □

**Theorem 35.14 (A positive sequence of Beta ports on an interval of a selected mark).** *As stated, at every mass $m_e\in(0,1]$.*

*Proof.* In $x$, $d\alpha_e=(\lambda_1-\lambda_0)\,dx$, and the constant goes into the room density. The factor of the mass is $\alpha_e^{m_e-1}=L_0^{-(1-m_e)}$ with $L_0=\lambda_0+(\lambda_1-\lambda_0)x$, a linear letter with $c_0=\lambda_0$, $d_0=\lambda_1-\lambda_0>0$ and the exponent $w_0=1-m_e\ge0$; it satisfies $c_0\ge\lambda_0d_0/(\lambda_1-\lambda_0)$ with equality, so the read is $\langle\cdot\rangle_{1,1;\mathbf{w}}$ over the letters of the chart and $L_0$. For a letter with $d_i>0$ put $\mu=\lambda_0/(\lambda_1-\lambda_0)$; then $x(1-x)d_i/L_i\le x(1-x)/(\mu+x)$, whose derivative vanishes on $[0,1]$ only where $x^2+2\mu x-\mu=0$, at $x=\sqrt{\mu(1+\mu)}-\mu$, with the value $\big(\sqrt{1+\mu}-\sqrt{\mu}\big)^2=(\sqrt{\lambda_1}-\sqrt{\lambda_0})^2/(\lambda_1-\lambda_0)=\varepsilon_{01}$, the ratio of the statement; for $d_i=0$ the quotient is 0. By 35.12(i) at $a=b=1$,

$$\big\langle(-1)^n\Pi_n^{1,1}\big\rangle_{1,1;\mathbf{w}}=\sum_{|v|=n}\prod_i\frac{(w_i)_{v_i}}{v_i!}\Big\langle\prod_i\Big(\frac{x(1-x)d_i}{L_i}\Big)^{v_i}\Big\rangle_{1,1;\mathbf{w}},$$

every term between 0 and $\prod_i\frac{(w_i)_{v_i}}{v_i!}\varepsilon_{01}^n\langle1\rangle_{1,1;\mathbf{w}}$, and $\sum_{|v|=n}\prod_i(w_i)_{v_i}/v_i!=(|\mathbf{w}|)_n/n!$ by the Vandermonde identity for rising factorials. A letter $L_i$ that sees the mark makes the term $v=n\mathbf{e}_i$ positive (35.12(ii) with $J=\{i\}$), and for $n=0$ the read is $\langle1\rangle_{1,1;\mathbf{w}}>0$. When $m_e<1$, $w_0d_0=(1-m_e)(\lambda_1-\lambda_0)>0$: the letter of the mass sees the mark. □

**Theorem 35.15 (The finite-band recurrence in the order of the Beta ports).** *As stated, at every pair $(a_n,b_n)$ with $a_n,b_n>0$.*

*Proof.* Write $a=a_n$, $b=b_n$, $w_i=w_i(n)$. The letters are at least $\min_i c_i>0$ on $[0,1]$, so every read of a polynomial converges and the flux $x^a(1-x)^bA\prod_iL_i^{-w_i}$ of a polynomial $A$ vanishes at both ends. 35.10(a) with the polynomial $Lg$ reads, by $x(1-x)\partial_x(Lg)=x(1-x)(L\partial_xg+g\partial_xL)$ and $\partial_x\log L_i=d_i/L_i$,

$$x(1-x)\partial_x(Lg)+\big(a-(a+b)x\big)Lg-x(1-x)Lg\sum_i\frac{w_id_i}{L_i}=L\,x(1-x)\,\partial_xg+A_ng.$$

The coefficient of $x^q$ in $\Pi_q^{a,b}$ is $\binom{a+b+2q-2}{q}$, a product of the $q$ positive numbers $a+b+q-1,\dots,a+b+2q-2$ over $q!$, so $\Pi_0^{a,b},\dots,\Pi_N^{a,b}$ are a basis of the polynomials of degree at most $N$, and the conditions $a+b+2q-i\neq 0$ of 3.15(v) hold. By its finite rules, multiplication by $x$ maps $\Pi_q^{a,b}$ into the span of $\Pi_{q-1}^{a,b},\Pi_q^{a,b},\Pi_{q+1}^{a,b}$, and $x(1-x)\partial_x$ maps it into the span of $\Pi_{q-1}^{a,b},\Pi_q^{a,b}$ and $x\Pi_q^{a,b}$, with coefficients rational in $(a,b,q)$; for $q=0$, $x\Pi_0^{a,b}=x$ and $\partial_x\Pi_0^{a,b}=0$. So multiplication by a polynomial of degree $d$ moves $\Pi_q^{a,b}$ into the span of $\Pi_{q-d}^{a,b},\dots,\Pi_{q+d}^{a,b}$. $L$ has degree $h$ and $A_n$ degree at most $h+1$, so at $g=(-1)^r\Pi_r^{a,b}$,

$$L\,x(1-x)\,\partial_x g + A_n g = \sum_{s=-h-1}^{h+1} e_s\,\Pi_{r+s}^{a,b},$$

$e_s=0$ when $r+s<0$, with $e_s$ rational in $(a,b,r)$ and polynomial in the coefficients of $L$ and of $A_n$, hence in the room data and the $w_i$; the mark relation is the displayed relation with $\mathfrak{b}_{n,r,s}=(-1)^{r+s}e_s$.

$\Pi_{r+h+1}^{a,b}$ is the only member of the expansion of degree $r+h+1$, so $e_{h+1}$ is the coefficient of $x^{r+h+1}$ divided by $\binom{a+b+2r+2h}{r+h+1}$. The coefficient of $x^r$ in $g$ is $(-1)^r\binom{a+b+2r-2}{r}$. The coefficient of $x^{h+1}$ in $A_n$ is $-(a+b)p_h-hp_h+|\mathbf{w}(n)|\,p_h$, the last from $-x(1-x)\sum_i w_i d_i\prod_{i'\neq i}L_{i'}$, whose leading coefficient is $\sum_i w_i d_i\prod_{i'\neq i}d_{i'}=|\mathbf{w}(n)|\,p_h$; and the coefficient of $x^{r+h+1}$ in $L\,x(1-x)\,\partial_x x^r$ is $-rp_h$. So the coefficient of $x^{r+h+1}$ is $(-1)^r\binom{a+b+2r-2}{r}\,p_h\big(|\mathbf{w}(n)|-a-b-h-r\big)$, and $\mathfrak{b}_{n,r,h+1}=(-1)^{r+h+1}e_{h+1}$ is the stated value. □

At $a_n=b_n=n+1$ the furthest coefficient is

$$\big[|\mathbf{w}(n)|-2n-2-h-r\big]\,p_h\,(-1)^{h+1}\binom{2n+2r}{r}\Big/\binom{2n+2r+2h+2}{r+h+1},$$

that of the reference.

## 50.35 Chapter 36: the critical-line law

**Proposition 36.7 (Binomial moments of the critical-line law).** *As stated.*

*Proof.* (i) The density $p_{\mathrm{crit}}(y)=2\pi/(e^{\pi y}+e^{-\pi y})^2$ is at most $2\pi e^{-2\pi|y|}$, so $\mathbb{E}[e^{tX}]=e^{t/2}\mathbb{E}[e^{itY}]$ is holomorphic in $t$ on the strip $|\operatorname{Im}t|<2\pi$, as is $te^t/(e^t-1)$, whose poles $2\pi ik$, $k\neq 0$, lie outside it; they agree for real $t$ (36.2(iii)), hence on the strip. For $|z|<1$, $t=\log(1+z)$ has $|\operatorname{Im}t|<\frac{\pi}{2}$, and $te^t/(e^t-1)=(1+z)\log(1+z)/z$.

(ii) $|\binom{x}{i}|\le\binom{|x|+i-1}{i}$ termwise, and $\sum_i\binom{|x|+i-1}{i}|z|^i=(1-|z|)^{-|x|}$. For $|z|<1-e^{-2\pi}$, $(1-|z|)^{-|X|}\le e^{c(\frac12+|Y|)}$ with $c=-\log(1-|z|)<2\pi$, which has a finite expectation, so $\mathbb{E}[(1+z)^X]=\sum_i\mathbb{E}[\binom{X}{i}]z^i$ there. The expansion $\log(1+z)/z=\sum_{i\ge0}(-1)^i z^i/(i+1)$ times $1+z$ has the coefficient 1 at $i=0$ and $(-1)^i\big(\frac{1}{i+1}-\frac{1}{i}\big)=\frac{(-1)^{i+1}}{i(i+1)}$ at $i\ge1$.

(iii) Differentiating $(1+z)^u=\sum_i\binom{u}{i}z^i$ in $u$ gives $\log(1+z)(1+z)^u=\sum_i\frac{d}{du}\binom{u}{i}z^i$, so $\frac{d}{du}\binom{u}{i}=\sum_{m=1}^{i}\frac{(-1)^{m-1}}{m}\binom{u}{i-m}$, an identity of polynomials. By (ii), $\mathbb{E}[\frac{d}{du}\binom{u}{i}|_X]$ is the coefficient of $z^i$ in $\log(1+z)\cdot(1+z)\log(1+z)/z$. From $\frac{d}{dz}\log^2(1+z)=2\log(1+z)/(1+z)=2\sum_{n\ge1}(-1)^{n-1}H_n z^n$, $\log^2(1+z)=2\sum_{n\ge2}(-1)^n H_{n-1}z^n/n$, and the coefficient of $z^i$ in $(1+z)\log^2(1+z)/z$ is

$$2(-1)^{i+1}\frac{H_i}{i+1}+2(-1)^i\frac{H_{i-1}}{i}=2(-1)^i\Big(\frac{H_{i-1}}{i}-\frac{H_{i-1}+\frac1i}{i+1}\Big)=2(-1)^i\frac{H_{i-1}-1}{i(i+1)}.$$

By 36.4 at $k=1$, for $R=(1-2u)Q$, $R(\frac12)=0$ and $\mathrm{crit}_1[R]=\mathbb{E}[Q'(X)]$, which is the sum. $\square$

**Proposition 36.11 (Poles at the half-integers).** *As stated.*

*Proof.* Put $S(u)=\sum_m (u+m)^{-(2k+1)}$. From $\Gamma_{\substack{4\\0}}\big[u,\ u,\ 1-u,\ 1-u\big]=\sum_m (u+m)^{-2}$ (5.16), $\partial_u^{2k-1}\Gamma_{\substack{4\\0}}\big[u,\ u,\ 1-u,\ 1-u\big]=-(2k)!\,S(u)$, so $\mathrm{crit}_k[R]=\frac{2}{2\pi i}\int_{(1/2)} R(u)S(u)\,du/(1-2u)$. $S$ has the period 1, poles of order $2k+1$ at the integers, it decays exponentially as $|\operatorname{Im} u|\to\infty$ uniformly in $\operatorname{Re} u$, and $S(1-u)=-S(u)$, so $S(\frac12)=0$.

For $p\notin\mathbb{Z}$ with $\operatorname{Re} p>\frac12$ let $I(p)=\frac{1}{2\pi i}\int_{(1/2)} S(u)\,du/(u-p)$. Move the line to $\operatorname{Re} u=N+\frac12$: the horizontal segments vanish by the decay of $S$, and on the new line $|1/(u-p)|\le 1/(N+\frac12-\operatorname{Re} p)\to 0$ while $\int|S|$ there is that on $\operatorname{Re} u=\frac12$. So $I(p)$ is minus the sum of the residues to the right of the line: at $u=p$ the residue $S(p)$, and at an integer $i\ge 1$ the residue of $(u-i)^{-2k-1}/(u-p)$, which is $(i-p)^{-2k-1}$. At $p=\frac12+l$, $l\ge 1$, $S(p)=\sum_{x\in\frac12+\mathbb{Z}} x^{-2k-1}=0$, and

$$I(p)=-\sum_{i\ge1}\big(i-\tfrac12-l\big)^{-2k-1}=-\sum_{x\in\frac12+\mathbb{Z},\ x>l} x^{-2k-1}=-(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((l,\infty)),$$

the points $x$ with $|x|<l$ cancelling in pairs. At $l=0$ the integrand $S(u)/(u-\frac12)$ is regular on the line; for real $p=\frac12+\varepsilon$, $\varepsilon\downarrow 0$, $|S(u)/(u-p)|\le|S(u)/(u-\frac12)|$ on the line, an integrable function, so the integral is $\lim I(p)=-S(\frac12)-\sum_{i\ge1}(i-\frac12)^{-2k-1}=-(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty))$.

For the word, $\frac{1}{(u-p)(1-2u)}=\frac{1}{1-2p}\big(\frac{1}{u-p}-\frac{1}{u-\frac12}\big)$, so at $p=\frac12+l$

$$\begin{aligned}\mathrm{crit}_k\left[\frac{1}{u-p}\right]&=\frac{2}{1-2p}\Big((\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty))-(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((l,\infty))\Big)\\&=-\frac{1}{l}\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,l])\end{aligned}$$

by the cut relation (5.3). Finally $u\mapsto 1-u$ maps the line onto itself, reversing it, and with $S(1-u)=-S(u)$ it gives $\mathrm{crit}_k[R(1-u)]=\mathrm{crit}_k[R(u)]$; for $R(u)=1/(u-\frac12+l)$, $R(1-u)=-1/(u-\frac12-l)$. $\square$

## 50.36 Chapter 37: the gap and the shape words

**Proposition 37.8 (The root rate of the shape words).** *As stated, with the maximizer at every $k\ge 2$.*

*Proof.* (i), the limit in (ii) and the last sentence are the reference's (the reference, proof of prop:kdepth). For the maximum, fix $x_1,\dots,x_k$ and put $y=x_1\cdots x_k$. The factor of $f_k$ in $x_0$ is $x_0(1-x_0)/\big(1-(1-y)x_0\big)$; its derivative vanishes where $1-2x_0+(1-y)x_0^2=0$, at $x_0=1/(1+\sqrt{y})$, where $1-x_0=\sqrt{y}/(1+\sqrt{y})$ and $1-(1-y)x_0=\sqrt{y}$, with the value $(1+\sqrt{y})^{-2}$. So $M_k(0)=\sup\prod_{i\ge1}x_i(1-x_i)\big/\big(1+\sqrt{x_1\cdots x_k}\big)^2$. At a fixed product the numerator is largest when the $x_i$ are equal, $t\mapsto\log(1-e^t)$ being concave, so $M_k(0)=\max_{0<a<1}\phi(a)$ with $\phi(a)=a^k(1-a)^k/(1+a^{k/2})^2$, and

$$\frac{\phi'(a)}{\phi(a)}=k\Big(\frac{1}{a}-\frac{1}{1-a}-\frac{a^{k/2-1}}{1+a^{k/2}}\Big)=\frac{k\left(1-2a-a^{k/2+1}\right)}{a(1-a)(1+a^{k/2})}.$$

The numerator decreases strictly from 1 at $a = 0$ to $-2^{-k/2-1}$ at $a = \frac{1}{2}$, so $\phi$ increases up to the unique root $a \in (0, \frac{1}{2})$ of $a^{k/2+1} + 2a = 1$ and decreases after it. At $\eta = 0$ the equation of 37.7 is $(1-2a)^2 = a^{k+2}$, whose root in $(0, \frac{1}{2})$ solves $1 - 2a = a^{k/2+1}$: the same $a$, with $x_2 = \cdots = x_k = a$ and $x_0 = a/(1-a) = 1/(1+a^{k/2})$, since $a^{k/2} = (1-2a)/a$. The interval follows from $\phi(\frac{1}{2}) \le M_k(0) \le 4^{-k}$. At $k = 2$ the root is $\sqrt{2} - 1$ and at $k = 4$ the root of $a^3 + 2a - 1$, the maximizers of the reference. □

**Proposition 37.14 (The insertion index supplies auxiliary states).** *As stated, for every $k \ge 3$.*

*Proof.* By 37.13(ii), which holds for every $k \ge 2$, the share function of $P_r$ is

$$g_{n,r}(a) = \binom{n+r}{r} \frac{(n!)^{k-2}\,(1-a)_n^2\,(n+1-a)_r}{(a)_{n+1}^k\,(n+1+a)_r}.$$

Its poles in $a \le 0$ are at $a = -h$, $0 \le h \le n$, of order exactly $k$, the numerator being nonzero there, and at $a = -h$, $n < h \le n + r$, simple. For $k \ge 3$ the coefficients of order $k - 1 \ge 2$ come from the first poles only, so $C_{n,k}(P_r) = \sum_{h=0}^{n} R_h(-h)$ and $C_{n,k-1}(P_r) = \sum_{h=0}^{n} R_h'(-h)$, with $R_h(a) = (a+h)^k g_{n,r}(a)$, holomorphic near $-h$. First,

$$R_h(-h) = \binom{n+r}{r} \frac{(n!)^{k-2}(1+h)_n^2}{\prod_{j\ne h}(j-h)^k} \cdot \frac{(n+1+h)_r}{(n+1-h)_r}, \qquad \frac{(n+1+h)_r}{(n+1-h)_r} = \frac{(n+1-h+r)_{2h}}{(n+1-h)_{2h}},$$

the product over $0 \le j \le n$; $\binom{n+r}{r} = (r+1)_n/n!$ and $(n+1-h+r)_{2h}$ are polynomials in $r$, $n + 1 - h \ge 1$, so $R_h(-h) \in \mathbb{Q}[r]$ and $C_{n,k}(P_r) \in \mathbb{Q}[r]$. Second, $R_h'(-h) = R_h(-h)\,\rho_h$ with

$$\rho_h = -2\sum_{i=1}^{n} \frac{1}{i+h} - \sum_{i=n+1}^{n+r} \Big(\frac{1}{i+h} + \frac{1}{i-h}\Big) - k\sum_{j\ne h} \frac{1}{j-h},$$

the logarithmic derivatives at $-h$ of $(1-a)_n^2$, $(n+1-a)_r$, $(n+1+a)_r^{-1}$ and $\prod_{j\ne h}(a+j)^{-k}$. The finite Trailmoment $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,N])$ is the sum of $m^{-1}$ over $0 < m \le N$, so $\sum_{i=n+1}^{n+r} \frac{1}{i+h} = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) - (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+h]) + \sum_{i=1}^{h} \frac{1}{n+r+i}$ and $\sum_{i=n+1}^{n+r} \frac{1}{i-h} = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) - (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n-h]) - \sum_{i=0}^{h-1} \frac{1}{n+r-i}$, and $\rho_h = -2\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) + \tilde{\rho}_h(r)$ with $\tilde{\rho}_h \in \mathbb{Q}(r)$. Summing, $C_{n,k-1}(P_r) = -2\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r])\,C_{n,k}(P_r) + q_n(r)$ with $q_n = \sum_h R_h(-h)\tilde{\rho}_h \in \mathbb{Q}(r)$.

Let $\sum_{i=0}^{d} b_i(r)\,C_{n,k-1}(P_{r+i}) = 0$ for all large $r$, $b_i \in \mathbb{Q}(r)$. Since $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r+i]) = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) + \sum_{l=1}^{i} \frac{1}{n+r+l}$, this reads $A(r)\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) + B(r) = 0$ for all large $r$ with $A = -2\sum_i b_i(r) C_{n,k}(P_{r+i})$ and $B$ in $\mathbb{Q}(r)$. If $A \ne 0$, then $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) = -B(r)/A(r)$ for all large $r$; a rational function is asymptotic to $cr^e$ with $e \in \mathbb{Z}$, while $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r]) - \log r$ has a limit, which is impossible. So $A = 0$: the recurrence annihilates $C_{n,k}(P_r)$. □

**Theorem 37.19 (An arithmetic budget for growing Beta ports).** *As stated, with the coefficient of the top coordinate at every $k \ge 2$.*

*Proof.* The two limits and the clearing are the reference's (the reference, proof of thm:portbudget). The coefficient of $(\mathsf{T}_{k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ in $I_n(k;P)$ is $k\,C_{n,k}(P)$ (37.4(iii)), and $P =$

$(-1)^q \Pi_q^{n+1,n+1}(x_1)$ is free of $x_0$, so the theorem applies. As in the proof of 37.14, at $r = q$, $C_{n,k}(P) = \sum_{h=0}^{n} R_h(-h)$ with $\prod_{j \neq h}(j-h) = (-1)^h h!\,(n-h)!$ and $(1+h)_n = (n+h)!/h!$:

$$R_h(-h) = \binom{n+q}{q} \frac{(n!)^{k-2}}{(h!(n-h)!)^k} \frac{((n+h)!)^2}{(h!)^2} (-1)^{hk} \frac{(n+1+h)_q}{(n+1-h)_q}$$
$$= \binom{n+q}{q} (-1)^{hk} \binom{n}{h}^k \binom{n+h}{n}^2 \frac{(n+1+h)_q}{(n+1-h)_q}.$$

For even $k$ every term is positive. □

**Corollary 37.20 (A growing Beta port with a certified gap).** *As stated, for every $k \geq 2$.*

*Proof.* The port factor is at most one, since $x_0 x_1 \bar{x}_1 x_2 \cdots x_k \leq x_0 x_1 \cdots x_k \leq L$, so $M_k(\eta) \leq M_k(0)$ for $\eta \geq 0$ and

$$\ell_k(\eta) + (k+1)(1+\eta)\, c_{\mathrm{lcm}} \leq u(\eta) = \log M_k(0) + (1+\eta)\log(1+\eta) - \eta \log \eta + (k+1)(1+\eta)\, c_{\mathrm{lcm}}.$$

$u$ is continuous on $[0,1]$ with $u(0) = \log M_k(0) + (k+1)c_{\mathrm{lcm}} < 0$ (37.8), so $u < 0$ on $[0, \eta_0]$ for a rational $\eta_0 > 0$. At a rational $\eta$ the equation of 37.7, cleared of denominators, is an integer polynomial equation in $a$ with the unique root in $(0, \frac{1}{2+\eta})$; the difference of its two sides is 1 at $a = 0$ and $-(1+\eta)^k (2+\eta)^{-k-2} < 0$ at $a = \frac{1}{2+\eta}$, so bisection at rational points certifies the root in rational intervals as small as asked. The two displays of 37.19 give the bound, and its last part the sign of the coefficient of $(\mathsf{T}_{k+1;(1)}[x]\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$ for even $k$. □

## 50.37 Chapter 38: rays, native gauges and tiers

The proofs of this chapter rest on two readings on the sectors of the orthant: of an insertion along a ray (Lemma S19) and of the direction of a ray (Lemma S20).

**Lemma S19.** *Let a ray of a family on $n \geq 2$ marks be given, $g$ an insertion as in Proposition 38.5, and $\pi$ an order of $\mathcal{E}$ with the tails $U_l = \{\pi(l), \ldots, \pi(n)\}$ and the sector coordinates $q = (q_2, \ldots, q_n) \in (0,1]^{n-1}$ of Theorem 8.5(ii) on its sector $H_\pi$. On $H_\pi$*

$$g = \prod_{l=2}^{n} q_l^{\kappa_{U_l}(g)}\, g_\pi(q), \qquad f = \prod_{l=2}^{n} q_l^{\kappa_{1,U_l}}\, \varphi_\pi(q), \qquad \frac{\omega_0}{dy} = \prod_{l=2}^{n} q_l^{\kappa_{0,U_l}}\, \omega_\pi(q),$$

*with $g_\pi = \prod_j k_{j,\pi}^{-a_j} \psi_{j,\pi}^{-b_j}$, $\varphi_\pi = \prod_j k_{j,\pi}^{-s_{1j}} \psi_{j,\pi}^{-t_{1j}}$ and $\omega_\pi = \prod_j k_{j,\pi}^{-s_{0j}} \psi_{j,\pi}^{-t_{0j}}$, the powers principal. These are smooth on a neighbourhood of $[0,1]^{n-1}$, and $g_\pi$, $\varphi_\pi$ and $|\omega_\pi|$ lie between positive constants on the cube. For a function $F$ on the orthant, homogeneous of degree 0 and integrable on the chart, $\int_{y_{e_0}=0} F\, dy = \sum_\pi \int_{(0,1]^{n-1}} F \prod_l \frac{dq_l}{q_l}$.*

*Proof.* In the sector coordinates $\alpha_{\pi(1)} = \rho$, $\alpha_{\pi(i)} = \rho\, q_2 \cdots q_i$, one has $\alpha^u = \rho^{u\varepsilon} \prod_l q_l^{u_{U_l}}$, and by Theorem 8.5(ii) $K_j = \rho^{r_j} \prod_l q_l^{d_j(U_l)} k_{j,\pi}$ and $\Psi_j = \rho^{n-r_j} \prod_l q_l^{d_j^*(U_l)} \psi_{j,\pi}$. The three functions are homogeneous of degree 0, the data of $g$, of the direction and of the origin lying on the balance wall, so $\rho$ cancels, and the exponents of the $q_l$ are the walls at $U_l$ of the data $(u; \mathbf{a}, \mathbf{b})$, $(\mathbf{m}_1; \mathbf{s}_1, \mathbf{t}_1)$ and $(\mathbf{m}_0; \mathbf{s}_0, \mathbf{t}_0)$. The sector letters are polynomials with nonnegative coefficients and positive constant terms, so on the cube they are at least their constant terms, and they are positive on a neighbourhood of it; their principal powers are smooth there, the modulus of a principal power

$x^{-w}$, $x>0$, is $x^{-\operatorname{Re} w}$, and compactness gives the bounds. The sectors cover the orthant and meet in null sets. On the chart the map $y\mapsto (y_{\pi(l-1)}-y_{\pi(l)})_{l=2}^{n}$ is linear with an integral inverse, $y$ being recovered from the differences and from $y_{e_0}=0$, so it keeps the Lebesgue measure, and $\log q_l = y_{\pi(l)}-y_{\pi(l-1)}$ on the sector gives $dy=\prod_l dq_l/q_l$. □

**Proposition 38.5 (The relations at every strength).** *As stated.*

*Proof.* (i) Put $Y=g\,f^k\omega_0/dy$, homogeneous of degree 0. By Lemma S19, on the sector of $\pi$, $|Y|=\prod_l q_l^{c_l}\,|g_\pi\varphi_\pi^k\omega_\pi|$ with $c_l=\operatorname{Re}\kappa_{0,U_l}+\kappa_{U_l}(g)+\operatorname{Re}(k)\kappa_{1,U_l}>0$ and the last factor bounded, every nonempty proper $U$ being a tail of some order. So $Y$ is integrable on the chart, and it tends to 0 at infinity: a sequence leaving every compact set of the chart has, along its terms in any one sector, $\min_l q_l\to 0$, and $\prod_{l'} q_{l'}^{c_{l'}}\le q_l^{c_l}$ for every $l$. By 38.2(ii), $\partial_{y_e}Y=(c_e^{(0)}+k\,c_e^{(1)}+\partial_{y_e})g\cdot f^k\omega_0/dy$, and by 38.2(i) $(c_e^{(0)}+k\,c_e^{(1)}+\partial_{y_e})g$ is $(m_{0,e}+k\,m_{1,e}+u_e)\,g$ less the insertions $(s_{0j}+k\,s_{1j}+a_j)\,\theta_{j,T}\,\alpha^{\mathbf{1}_T}K_j^{-1}g$, $T\ni e$, and $(t_{0j}+k\,t_{1j}+b_j)\,\theta_{j,T}\,\alpha^{\mathbf{1}_{\mathcal{E}\setminus T}}\Psi_j^{-1}g$, $T\not\ni e$, whose walls are $\kappa_U(g)+|T\cap U|-d_j(U)\ge\kappa_U(g)$ and $\kappa_U(g)+|U\setminus T|-d_j^*(U)\ge\kappa_U(g)$ (7.13). So what precedes applies to them: $\partial_{y_e}Y$ is integrable, and its value converges absolutely. On the chart $\alpha_{e_0}=1$, for $e\ne e_0$, Fubini's theorem and $\int_{-R}^{R}\partial_{y_e}Y\,dy_e=Y|_{y_e=R}-Y|_{y_e=-R}\to 0$ give $\int\partial_{y_e}Y\,dy=0$; $Y$ being homogeneous of degree 0, $\sum_e\partial_{y_e}Y=0$ (Euler's identity), and $\partial_{y_{e_0}}Y=-\sum_{e\ne e_0}\partial_{y_e}Y$ integrates to 0 as well. For the certificate, at such a $k$ every value involved converges absolutely and is linear in the insertion, $\langle f^jg'\rangle_k=\langle g'\rangle_{k+j}$, and the right side has the value 0.

(ii) By Lemma S19, for $\operatorname{Re}k$ large,

$$\langle g\rangle_k=\sum_\pi\int_{[0,1]^{n-1}}\prod_l q_l^{x_l(k)-1}h_\pi(q,k)\,dq,\qquad x_l(k)=\kappa_{0,U_l}+\kappa_{U_l}(g)+k\,\kappa_{1,U_l},\quad h_\pi=g_\pi\varphi_\pi^k\omega_\pi.$$

$h_\pi$ is smooth in $q$ on a neighbourhood of the cube and entire in $k$, with $q$-derivatives continuous in $(q,k)$ and holomorphic in $k$. By 8.4 each term extends meromorphically in $(x,k)$, and its recursion, applied in every coordinate, writes it on $\{\operatorname{Re}x_l>-M\ \forall l\}$ as a finite sum of products of factors $(x_l+i)^{-1}$, at most one for each $l$, with functions holomorphic there. Along $x=x(k)$ each $x_l(k)+i$ is affine in $k$ with the slope $\kappa_{1,U_l}>0$, so the composite is meromorphic on the half-plane where every $\operatorname{Re}x_l(k)>-M$, with poles only where some $x_l(k)$ is a nonpositive integer, of order at most $n-1$; $M$ being arbitrary, this continues $\langle g\rangle_k$ to $\mathbb{C}$. On the half-plane where the integral converges absolutely it is holomorphic, and it agrees there with the continuation. An element of $\mathcal{L}$ is a finite sum $\sum_{e,i}r_{e,i}(k)(c_e^{(0)}+k\,c_e^{(1)}+\partial_{y_e})g_{e,i}$ with insertions $g_{e,i}$ of the above kind; its value is the $\mathbb{C}(k)$-linear extension of $g\mapsto\langle g\rangle_k$, which does not depend on the way the element is written, and by (i), the walls of the direction being positive, it vanishes for $\operatorname{Re}k$ large off the poles of the $r_{e,i}$, hence identically. $\langle fg\rangle_k$ and $\langle g\rangle_{k+1}$ are one integral for $\operatorname{Re}k$ large. A recurrence that follows from the relations is the value of the element $\sum_j p_j(k)f^jg$ of $\mathcal{L}$. □

**Proposition 38.8 (The top and the bottom of a certificate).** *As stated.*

*Proof.* For a polynomial $G=\sum_i k^iG^{[i]}$, with $G^{[-1]}=0$, the coefficient of $k^m$ in $(c_e^{(0)}+k\,c_e^{(1)}+\partial_{y_e})G$ is $(c_e^{(0)}+\partial_{y_e})G^{[m]}+c_e^{(1)}G^{[m-1]}$. At $m=d$, where every $G_e^{[d]}$ vanishes, this gives the first identity, the left side having the coefficient $\sum_j c_jf^jg=\chi(f)\,g$; at $m=0$ it gives the second. At a native gauge every $c_e^{(1)}$ vanishes (Definition 38.4), so $\chi\big(f(\alpha_c)\big)g(\alpha_c)=0$. For the origin 0, $c^{(0)}=0$. □

**Lemma S20.** *Let a ray of a family on $n\ge 2$ marks have $\kappa_{1,S}>0$ for every nonempty proper $S$, with its stride $f$ on the chart $y_{e_0}=0$.*

*(i)* $\mathbf{m}_1 > 0$*, the exponents of the direction are not all zero, and the tables with* $s_{1j}+t_{1j} > 0$ *form a connected family.*

*(ii) The direction has a native gauge* $\alpha^\star \in \mathbb{R}^{\mathcal{E}}_{>0}$*, unique up to scale; with* $\alpha^\star_{e_0} = 1$*,* $0 < f \le f(\alpha^\star)$ *on the chart with equality only at* $\log \alpha^\star$*, and* $f \le C' e^{-c'\|y\|}$ *there for some* $c', C' > 0$*.*

*(iii) The Hessian of* $-\log f$ *at* $\log \alpha^\star$ *on the chart is*

$$H = \Big( \sum_j \big( s_{1j}\, \kappa_{j,\alpha^\star}(X_e, X_{e'}) + t_{1j}\, \kappa_{j,(\alpha^\star)^{-1}}(X_e, X_{e'}) \big) \Big)_{e,e' \neq e_0},$$

*and it is positive definite.*

*(iv)* $\langle 1 \rangle_k = P_F(\mathbf{m}_0 + k\mathbf{m}_1; \mathbf{s}_0 + k\mathbf{s}_1, \mathbf{t}_0 + k\mathbf{t}_1)$*, and it converges absolutely exactly when* $\operatorname{Re} \kappa_{0,S} + k\kappa_{1,S} > 0$ *for every nonempty proper* $S$*.*

*Proof.* (i) Every $\{e\}$ is proper, and $\kappa_{1,\{e\}} = m_{1,e} - \sum_j \big( s_{1j} d_j(\{e\}) + t_{1j} d_j^*(\{e\}) \big)$ with $s_{1j}, t_{1j} \in \mathbb{N}$ and $d_j, d_j^* \ge 0$, so $m_{1,e} \ge \kappa_{1,\{e\}} > 0$. If every exponent of the direction were 0, the balance wall would give $m_{1,\mathcal{E}} = 0$. On the balance wall 8.16 gives $\kappa_{1,S} + \kappa_{1,\mathcal{E}\setminus S} = \sum_j (s_{1j} + t_{1j}) c_j(S)$, so every nonempty proper $S$ has $c_j(S) > 0$ for a table with $s_{1j} + t_{1j} > 0$.

(ii) The direction is real data on its balance wall, where the target is the masses, with nonnegative exponents not all zero and positive masses. By (i) and 8.19(iii) it has a native gauge $\alpha^\star \in \mathbb{R}^{\mathcal{E}}_{>0}$, unique up to scale, the unique point up to scale at which $\alpha^{\mathbf{m}_1} \prod_j K_j^{-s_{1j}} \Psi_j^{-t_{1j}} = f$ is largest. At $y = 2z$ the stride is $e^{X_1(z)}$, $X_1$ the exponent of the question of the direction (17.1), and 17.4(i) and (ii) give $X_1(z) \le -2\kappa_{1,\min}(\max_e z_e - \min_e z_e) + C$; on the chart $|y_e| \le \max y - \min y$, since $y_{e_0} = 0$.

(iii) The Hessian of $\log K_j(e^y)$ in $y$ is the covariance of the indicators of the basis under the basis law at $\alpha = e^y$ (7.16), and $\Psi_j(e^y) = e^{\langle \mathbf{1}, y \rangle} K_j(e^{-y})$, so the Hessian of $\log \Psi_j(e^y)$ is the covariance under the law at $\alpha^{-1}$; at $\alpha^\star$ this gives $H$. For $v$ with $v_{e_0} = 0$,

$$v^{\mathsf{T}} H v = \sum_j \Big( s_{1j} \operatorname{Var}_{\mathbb{P}_{j,\alpha^\star}} \langle \mathbf{1}_T, v \rangle + t_{1j} \operatorname{Var}_{\mathbb{P}_{j,(\alpha^\star)^{-1}}} \langle \mathbf{1}_T, v \rangle \Big) \ge 0,$$

both laws charging every basis. Let $v^{\mathsf{T}} H v = 0$ with $v \neq 0$. Then $v$ is not constant, and $\langle \mathbf{1}_T, v \rangle$ is the same at every basis of every table with $s_{1j} + t_{1j} > 0$. Let $S = \{e : v_e = \max v\}$, nonempty and proper, and take by (i) such a table with $c_j(S) > 0$ and a basis $T$ with $|T \cap S| < r_j(S)$. $T \cap S$ is independent in the restriction to $S$ and not a basis of it, so some $f \in S \setminus T$ keeps it independent; the circuit of $T + f$ through $f$ is not contained in $S$, and with $e$ a mark of it outside $S$, $T - e + f$ is a basis with $\langle \mathbf{1}_{T-e+f}, v \rangle - \langle \mathbf{1}_T, v \rangle = v_f - v_e > 0$, a contradiction.

(iv) On the chart $\alpha_{e_0} = 1$, $\alpha^{\mathbf{m}-\mathbf{1}} \prod_{e \neq e_0} d\alpha_e = \alpha^{\mathbf{m}} dy$, so the value $\langle 1 \rangle_k = \int f^k \omega_0$ of 38.1 is the period at the point $k$ of the ray in the form of 8.7, which converges exactly when the real parts of its proper walls $\kappa_{0,S} + k\kappa_{1,S}$ are positive; the modulus of its integrand is the integrand at the real parts of the data. □

**Proposition 38.9 (The native value bounds the order).** *As stated: every ray with positive direction walls, its origin complex.*

*Proof.* By Lemma S20(iv), $\langle 1 \rangle_k$ converges absolutely for $k \ge k_1$. The amplitude $g_0 = \omega_0 / dy = \alpha^{\mathbf{m}_0} \prod_j K_j^{-s_{0j}} \Psi_j^{-t_{0j}}$, the powers of the positive letters principal, is continuous and nowhere zero on the chart. Let $y^\star = \log \alpha^\star$ with $\alpha^\star_{e_0} = 1$. By Lemma S20(ii) and (iii), $f / f(\alpha^\star)$ is real-analytic,

has its unique maximum 1 at $y^\star$, nondegenerate with the Hessian $H$, and tends to 0 at infinity; so off a ball $B$ about $y^\star$, $f \le e^{-\delta} f(\alpha^\star)$ for some $\delta > 0$, and for $k \ge k_1$

$$\Big| \int_{y_{e_0}=0,\ y\notin B} f^k g_0 \, dy \Big| \le e^{-\delta(k-k_1)} f(\alpha^\star)^{k-k_1} \int f^{k_1} |g_0| \, dy.$$

On $B$, Laplace's method with the continuous amplitude $g_0$ gives

$$\int_B f^k g_0 \, dy = f(\alpha^\star)^k \Big(\frac{2\pi}{k}\Big)^{\frac{n-1}{2}} (\det H)^{-\frac12} \big(g_0(y^\star) + o(1)\big).$$

Hence $\langle 1\rangle_k = f(\alpha^\star)^k \big(\frac{2\pi}{k}\big)^{\frac{n-1}{2}} (\det H)^{-\frac12} g_0(y^\star)(1+o(1))$ with $g_0(y^\star) \ne 0$: $\langle 1\rangle_k \ne 0$ for all large $k$ and $\langle 1\rangle_{k+1}/\langle 1\rangle_k \to f(\alpha^\star)$. For a real origin $g_0 > 0$, and $\langle 1\rangle_k > 0$ wherever it converges.

Let $\sum_{j=0}^{h} p_j(k) \langle 1\rangle_{k+j} = 0$ for all large $k$, the $p_j \in \mathbb{Q}[k]$ not all zero, $d$ their largest degree and $c_j$ the coefficient of $k^d$ in $p_j$. Dividing by $k^d \langle 1\rangle_k$ and letting $k \to \infty$ gives $\sum_j c_j f(\alpha^\star)^j = 0$, and the symbol $\sum_j c_j \lambda^j$ is a nonzero rational polynomial of degree at most $h$. So $f(\alpha^\star)$ is algebraic, its minimal polynomial divides the symbol, every conjugate is a root of it, and $h \ge [\mathbb{Q}(f(\alpha^\star)) : \mathbb{Q}]$. For $h = 1$ this degree is one, and a transcendental native value admits no $h$. □

**Proposition 38.10 (The conjugates of the native value of the direction).** *As stated.*

*Proof.* On the chart $\alpha_{e_0} = 1$ the letter complement $X_F$ is the affine variety $\{\alpha \in (\mathbb{C}^\times)^{\mathcal{E}-e_0} : \prod_j K_j(\alpha)\Psi_j(\alpha) \ne 0\}$, and the defects $c_e^{(1)} = m_{1,e} - \sum_j \big(s_{1j} R_{j,e} + t_{1j} R^*_{j,e}\big)$ are regular functions on it with coefficients in $\Bbbk$, the direction being integral. Their sum over $e$ is $\sum_e \partial_{y_e} \log f = 0$, $f$ being homogeneous of degree 0, so the native gauges are the common zeros in $X_F$ of the $c_e^{(1)}$, $e \ne e_0$: an algebraic set defined over $\Bbbk$. Its Jacobian matrix in the coordinates $y_e$, $e \ne e_0$, is the Hessian of $\log f$, nonsingular at $\alpha^\star$ (Lemma S20(iii)); by the holomorphic inverse function theorem $\alpha^\star$ is an isolated point of it, and an isolated point of an algebraic set defined over $\Bbbk$ has coordinates algebraic over $\Bbbk$, the irreducible components of the set being defined over the algebraic closure of $\Bbbk$. An embedding $\tau$ over $\Bbbk$ keeps every polynomial identity with coefficients in $\Bbbk$: $\tau(\alpha^\star)$ satisfies the equations of nativity, no letter vanishes at it, the Hessian determinant, a rational function over $\Bbbk$, takes there the value $\tau$ of a nonzero number, and $f\big(\tau(\alpha^\star)\big) = \tau\big(f(\alpha^\star)\big)$. Every conjugate of $f(\alpha^\star)$ over $\Bbbk$ is $\sigma\big(f(\alpha^\star)\big)$ for an embedding $\sigma$ of $\Bbbk\big(f(\alpha^\star)\big)$, which extends to an embedding of the finite extension $\Bbbk(\alpha^\star)$ in the algebraically closed $\mathbb{C}$. The $[\Bbbk(\alpha^\star) : \Bbbk]$ embeddings of $\Bbbk(\alpha^\star)$ differ on a coordinate of $\alpha^\star$, so they give as many distinct nondegenerate native gauges, and Theorem 38.6(iii) bounds the dimension. Corollary 38.7 at $g = 1$ gives the last statement. □

**Proposition 38.12 (The growth of the frame).** *As stated.*

*Proof.* By 38.11, $C_k$ is $C_{k_0}((-1)^h c)^{k-k_0}$ times $\prod_{i\le d_0} \Gamma(k - a_i) / \prod_{i \le d_h} \Gamma(k - b_i)$ times a nonzero constant. Stirling's formula, which applies to $\Gamma(k-a)$ for fixed complex $a$ as $k \to +\infty$ along the reals, the argument staying in a sector about the positive axis, gives $\log|\Gamma(k-a)| = (k - \operatorname{Re} a - \frac12)\log k - k + O(1)$; summing, the Gammas contribute $(d_0 - d_h)(k \log k - k) + O(\log k)$. When $d_0 = d_h = d$ is the largest degree, $c_0$ and $c_h$ are the leading coefficients of $p_0$ and $p_h$, both nonzero, the symbol has degree $h$, and the product of its roots is $(-1)^h c_0/c_h = (-1)^h c$; the term in $k \log k$ is absent and $\frac1k \log|C_k| \to \log|c|$. □

**Definition 38.13 (The centre ray).** As stated, for a family connected at strengths $\sigma_j \in \mathbb{N}$.

*Proof.* The direction $(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$ is integral, and it is on the balance wall: $m_{1,\mathcal{E}} = n\sigma = \sum_j \sigma_j(r_j + r_j^*)$. The origin 0 is on it too. From $\Psi_j(\alpha) = \alpha^{\mathbf{1}} K_j(\alpha^{-1})$, $K_j\Psi_j = \alpha^{\mathbf{1}} K_j(\alpha)K_j(\alpha^{-1})$, and with $\sum_j \sigma_j = \sigma$ the stride is $\alpha^{\sigma\mathbf{1}} \prod_j (K_j\Psi_j)^{-\sigma_j} = \prod_j \big(K_j(e^y)K_j(e^{-y})\big)^{-\sigma_j}$ at $\alpha = e^y$. The origin form is $dy$, so $\langle 1\rangle_k = \int_{y_{e_0}=0} f^k dy = P_F(k\sigma\mathbf{1}; k\boldsymbol{\sigma}, k\boldsymbol{\sigma})$ by Lemma S20(iv). The direction walls are $\sigma|S| - \sum_j \sigma_j\big(d_j(S) + d_j^*(S)\big) = \sum_j \sigma_j c_j(S)$, positive on every nonempty proper $S$ exactly when the family is connected at the strengths (8.21(ii)). □

**Theorem 38.14 (A ray is one function).** *As stated, for every ray with real origin and positive direction walls.*

*Proof.* (i) The bounds on $f$ are Lemma S20(ii). For $k \geq k_0$ the walls $\kappa_{0,S} + k\kappa_{1,S} \geq \kappa_{0,S} + k_0\kappa_{1,S}$ are real and positive, so $\langle 1\rangle_k$ converges (Lemma S20(iv)), and its integrand $f^k\omega_0/dy$ is positive.

(ii) For $|q| < f(\alpha^\star)^{-1}$, $|qf| \leq |q|f(\alpha^\star) < 1$ on the chart, so $\sum_{k\geq k_0} q^{k-k_0} f^k = f^{k_0}/(1-qf)$, the partial sums dominated by $f^{k_0}/(1-|q|f(\alpha^\star))$ times the positive density $\omega_0/dy$, which is integrable at $k = k_0$. Summing under the integral gives the first form of $G$, and the second is the first with numerator and denominator multiplied by $\prod_j K_j^{s_{1j}}\Psi_j^{t_{1j}}$. On smaller discs the same domination allows differentiation under the integral, $\partial_q^l(1-qf)^{-1} = l!\, f^l(1-qf)^{-l-1}$. For real $q < f(\alpha^\star)^{-1}$, $1 - qf \geq \min\big(1, 1 - qf(\alpha^\star)\big) > 0$, so the integrand is positive and at most a multiple of $f^{k_0}\omega_0/dy$.

(iii) With the positive measure $\mu = f^k\omega_0$ on the chart, $\langle 1\rangle_{k+i+i'} = \int f^{i+i'} d\mu$, and the identity of Andréief gives

$$\det\Big[\int f^{i+i'}d\mu\Big]_{i,i'<h} = \frac{1}{h!}\int \det\big[f(y_a)^i\big]_{i<h,\,a\leq h}^2 \prod_a d\mu(y_a) = \frac{1}{h!}\int \prod_{a<b}\big(f(y_a) - f(y_b)\big)^2 \prod_a d\mu(y_a),$$

positive because $f$ is real-analytic and not constant, so that the set where two values $f(y_a)$ agree is null. If $G = N_1/N_2$ were rational, $N_2(0) \neq 0$, $d = \deg N_2$, then $\sum_{i=0}^d [q^i]N_2\, \langle 1\rangle_{k_0+m-i} = 0$ for $m > \deg N_1$, a relation with $[q^0]N_2 \neq 0$ among the rows of the Hankel matrix $\big[\langle 1\rangle_{k+l+i'}\big]_{l,i'\leq d}$ for all large $k$, whose determinant would vanish, against (iii) at $h = d+1$.

On the centre ray $\kappa_{0,S} = 0 < \kappa_{1,S}$, so $k_0 = 1$, and $\alpha^\star = \mathbf{1}$ by 8.21(i). □

**Theorem 38.15 (The branch at the native gauge of the direction).** *As stated, for every ray with real origin and positive direction walls.*

*Proof.* Put $y^\star = \log\alpha^\star$, $\varphi = -\log\big(f/f(\alpha^\star)\big) \geq 0$ on the chart, zero only at $y^\star$, with the Hessian $H$ there, positive definite (Lemma S20(iii)), and $g = f^{k_0}\omega_0/dy$, positive, real-analytic and integrable. At $q = A(1-\varepsilon)$, $1 - qf = 1 - (1-\varepsilon)e^{-\varphi}$, so for $0 < \varepsilon \leq 1$

$$G(q) = \int \frac{g\, dy}{1 - (1-\varepsilon)e^{-\varphi}}.$$

Fix a small $t_1 > 0$. On $\{\varphi \geq t_1\}$, $|1 - (1-\varepsilon)e^{-\varphi}| \geq 1 - |1-\varepsilon|e^{-t_1} > 0$ for $|\varepsilon|$ small, so that part of the integral is holomorphic at $\varepsilon = 0$. Near $y^\star$ the real-analytic Morse lemma gives coordinates $v \in \mathbb{R}^{2a}$, $2a = n-1$, with $\varphi = \frac{1}{2}|v|^2$ and $|\det \partial y/\partial v| = (\det H)^{-1/2}$ at $v = 0$; for $t_1$ small, $\{\varphi < t_1\}$ lies in their domain. With $\tilde g = g\,|\det \partial y/\partial v|$ and

$$\beta(t) = 2^{a-1}\int_{S^{2a-1}} \tilde g\big(\sqrt{2t}\,\omega\big)\, d\omega,$$

analytic near $[0,t_1]$, the spherical mean being an even analytic function of the radius, $\int_{\varphi<t} g\,dy$ has the derivative $t^{a-1}\beta(t)$. In $w=e^t-1$, $1-(1-\varepsilon)e^{-t}=e^{-t}(w+\varepsilon)$ and $e^t dt=dw$, so the part over $\{\varphi<t_1\}$ is

$$\int_0^{w_1}\frac{w^{a-1}\gamma(w)}{w+\varepsilon}\,dw,\qquad \gamma(w)=\Big(\frac{\log(1+w)}{w}\Big)^{a-1}\beta\big(\log(1+w)\big),\quad \gamma(0)=\beta(0),$$

with $\gamma=\sum_m\gamma_m w^m$ on a disc of radius $R'>w_1$, $t_1$ being small, so that $|\gamma_m|\le CR'^{-m}$. For $c>0$ not an integer and $|\varepsilon|<w_1$, $\varepsilon\notin(-\infty,0]$,

$$\int_0^{w_1}\frac{w^{c-1}\,dw}{w+\varepsilon}=\frac{\pi}{\sin\pi c}\,\varepsilon^{c-1}-\sum_{l\ge0}\frac{(-\varepsilon)^l w_1^{c-1-l}}{l+1-c},$$

from $\int_0^\infty w^{c-1}dw/(w+\varepsilon)=\frac{\pi}{\sin\pi c}\varepsilon^{c-1}$ at $0<\operatorname{Re}c<1$ and continuation in $c$; for an integer $c=N\ge1$, $w^{N-1}=\big((w+\varepsilon)-\varepsilon\big)^{N-1}$ gives the integral as $(-1)^N\varepsilon^{N-1}\log\varepsilon$ plus $(-\varepsilon)^{N-1}\log(w_1+\varepsilon)$ plus a polynomial in $\varepsilon$, the two last of modulus $O\big((w_1+2\rho)^{N-1}\big)$ on $|\varepsilon|\le\rho$. Summed with the weights $\gamma_m$ at $c=a+m$, the singular terms give $\varepsilon^{a-1}V(\varepsilon)$ for $n$ even and $\varepsilon^{a-1}\log\varepsilon\,V(\varepsilon)$ for $n$ odd, with

$$V(\varepsilon)=\frac{\pi}{\sin\pi a}\,\gamma(-\varepsilon)\quad(n\text{ even}),\qquad V(\varepsilon)=(-1)^a\gamma(-\varepsilon)\quad(n\text{ odd}),$$

and the remainders converge to a function holomorphic near 0, since $|l+1-a-m|\ge\frac12$ for $n$ even and $w_1+2\rho<R'$ for small $\rho$. This is the form of $G$, with $U$ the sum of the remainders and the part over $\{\varphi\ge t_1\}$. Further $\beta(0)=2^{a-1}|S^{2a-1}|\,g(\alpha^\star)(\det H)^{-1/2}=(2\pi)^a g(\alpha^\star)/\big(\Gamma(a)\sqrt{\det H}\big)$, $\frac{\pi}{\sin\pi a}=\Gamma(a)\Gamma(1-a)$, and $g(\alpha^\star)=f(\alpha^\star)^{k_0}(\omega_0/dy)(\alpha^\star)=(f^{k_0-1}\omega_0/dy)(\alpha^\star)/A$, which gives $V(0)$.

$G^{(l)}(A)=l!\int f^{k_0+l}\omega_0/(1-Af)^{l+1}$ is a positive integral, and near $y^\star$, $1-Af=1-e^{-\varphi}$ is comparable with $|v|^2$, so it is finite exactly when $2(l+1)<2a$. For $l<a-1$, $G^{(l)}(q)\to G^{(l)}(A)$ as $q\uparrow A$ by monotone convergence, while the $l$-th derivative of $\varepsilon^{a-1}V$, resp. of $\varepsilon^{a-1}\log\varepsilon\,V$, tends to 0; so the Taylor coefficients of $U$ of order $l<a-1$ are those of $G$ at $A$, with $q-A=-A\varepsilon$, and the rest of $U$ and $\varepsilon^{a-1}(V-V(0))$ are $O(\varepsilon^{\lceil a-1\rceil})$. On the centre ray $k_0=1$, $\alpha^\star=\mathbf{1}=(\alpha^\star)^{-1}$, $A=f(\mathbf{1})^{-1}=\prod_j K_j(\mathbf{1})^{2\sigma_j}$, $f^0\omega_0/dy=1$ and $H=2\sum_j\sigma_j\big(\kappa_{j,\mathbf{1}}(X_e,X_{e'})\big)_{e,e'\neq e_0}$. □

**Proposition 38.16 (The native gauges of a centre ray).** *As stated.*

*Proof.* The direction is $(\sigma\mathbf{1};\boldsymbol{\sigma},\boldsymbol{\sigma})$, so $c_e^{(1)}=\sigma-\sum_j\sigma_j\big(R_{j,e}+R^*_{j,e}\big)$ (Definition 38.1). By 7.2, $\Psi_j(\alpha)=\alpha^{\mathcal{E}}K_j(\alpha^{-1})$, so $R^*_{j,e}(\alpha)=\alpha_e\partial_{\alpha_e}\log\Psi_j(\alpha)=1-R_{j,e}(\alpha^{-1})$ as rational functions, the identity of 7.16 at real gauges; with $\sum_j\sigma_j=\sigma$ this gives the defects. The stride is $\prod_j\big(K_j(\alpha)K_j(\alpha^{-1})\big)^{-\sigma_j}$, a rational function homogeneous of degree 0 (Definition 38.13). □

**Theorem 38.17 (Native gauges at the signs).** *As stated, for the centre ray of a family.*

*Proof.* (i) On the chart $\alpha_{e_0}=1$, $f(\alpha)=\prod_j\big(K_j(\alpha)K_j(\alpha^{-1})\big)^{-\sigma_j}$ (38.13), so $f(\alpha^{-1})=f(\alpha)$; $\alpha\mapsto\alpha^{-1}$ keeps the letter complement, since $K_j(\alpha^{-1})\Psi_j(\alpha^{-1})=\alpha^{-2\mathbf{1}}\Psi_j(\alpha)K_j(\alpha)$. A point of $(\mathbb{C}^\times)^{\mathcal{E}}/\mathbb{C}^\times$ fixed by the inversion has $\alpha_e^2$ independent of $e$, hence $\alpha_e=\pm1$ on the chart: the $2^{n-1}$ signs. By 38.2(ii) the native gauges of the ray are the critical points of $\log f$ on $X_F$, and the inversion, which keeps $f$, permutes them and keeps their native values.

(ii) At a sign $\eta$, $\Psi_j(\eta)=\eta^{\mathbf{1}}K_j(\eta)$, so $K_j(\eta)\neq0$ for every $j$ puts $\eta$ in $X_F$. In the chart $\alpha=\eta e^y$, $y_{e_0}=0$, $\alpha^{-1}=\eta e^{-y}$, so $y\mapsto\log f(\eta e^y)$ is even and its gradient vanishes at $y=0$: $\eta$ is a native gauge, of native value $\prod_j\big(K_j(\eta)K_j(\eta)\big)^{-\sigma_j}$. With $K_j(\eta e^y)=\sum_T\theta_{j,T}\eta^T e^{\langle\mathbf{1}_T,y\rangle}=$

$K_j(\eta)\sum_T \mathbb{P}_{j,\eta}(T)e^{\langle \mathbf{1}_T,y\rangle}$, the Hessian of $\log K_j(\eta e^{y})$ at 0 is the covariance of the indicators under the signed measure $\mathbb{P}_{j,\eta}$ of total mass one, and $\log K_j(\eta e^{-y})$ has the same Hessian; so the Hessian of $-\log f$ is $2\sum_j \sigma_j\kappa_{j,\eta}$.

(iii) $W^{-1}=\prod_j (K_j\Psi_j)^{\sigma_j}$ is a polynomial, and $\partial_{y_e}\log f=\big(\sigma W^{-1}-\alpha_e\partial_e W^{-1}\big)/W^{-1}$. For a sign $\eta$, $h(y)=W^{-1}(\eta e^{y})\,e^{-\sigma\langle \mathbf{1},y\rangle}=\prod_j\big(\eta^{\mathbf{1}}K_j(\eta e^{y})K_j(\eta e^{-y})\big)^{\sigma_j}$ is even in $y$, so at $y=0$

$$\partial_{y_e}h=\big(\alpha_e\partial_e W^{-1}-\sigma W^{-1}\big)(\eta)=0$$

at every sign, whether or not the $K_j(\eta)$ vanish, and $W^{-1}(\eta)=\eta^{\boldsymbol{\sigma}\mathbf{1}}\prod_j K_j(\eta)^{2\sigma_j}$. The ring $\mathbb{Q}(\boldsymbol{\theta})[\alpha_e : e\neq e_0]/(\alpha_e^2-1)$ is the product of copies of $\mathbb{Q}(\boldsymbol{\theta})$ indexed by the signs, a class being its values at the signs, with the idempotents $2^{1-n}\prod_{e\neq e_0}(1+\eta_e\alpha_e)$, which take the value 1 at $\eta$ and 0 at the other signs. So the critical polynomials vanish in it, and $W^{-1}\equiv\alpha^{\boldsymbol{\sigma}\mathbf{1}}\prod_j K_j^{2\sigma_j}$. Localizing at $\prod_j K_j$ keeps exactly the factors of the signs with $\prod_j K_j(\eta)\neq 0$, whose idempotents form a basis, and multiplication by $W^{-1}/\alpha^{\boldsymbol{\sigma}\mathbf{1}}$, $\alpha^{\boldsymbol{\sigma}\mathbf{1}}$ being a unit, acts on the idempotent of $\eta$ by its value $\prod_j K_j(\eta)^{2\sigma_j}$. □

**Theorem 38.18 (Weights that separate the signs).** *As stated, for a family of $N$ tables connected at its strengths.*

*Proof.* Treat the weights of all tables as variables. For a sign $\eta$, $K_j(\eta)=\sum_T\theta_{j,T}\eta^T$ is a nonzero linear form. Write $M_j=K_j(\eta)^2\big(\kappa_{j,\eta}(X_e,X_{e'})\big)_{e,e'\neq e_0}$, whose entries $K_j(\eta)\sum_{T\ni e,e'}\theta_{j,T}\eta^T-\sum_{T\ni e}\theta_{j,T}\eta^T\sum_{T\ni e'}\theta_{j,T}\eta^T$ are quadratic, and

$$D_\eta=\det\Big[\sum_j\sigma_j M_j\prod_{i\neq j}K_i(\eta)^2\Big]=\prod_i K_i(\eta)^{2(n-1)}\det\Big[\sum_j\sigma_j\kappa_{j,\eta}\Big],$$

of degree $2N(n-1)$. At $\eta=\mathbf{1}$ and unit weights $D_{\mathbf{1}}\neq 0$: the Hessian $2\sum_j\sigma_j\kappa_{j,\mathbf{1}}$ is positive definite, by the argument of Lemma S20(iii) at the strengths. The substitution $\theta_{j,T}\mapsto\theta_{j,T}\eta^T$ carries $K_j(\mathbf{1})$ to $K_j(\eta)$ and $\mathbb{P}_{j,\mathbf{1}}$ to $\mathbb{P}_{j,\eta}$, so $D_\eta$ is $D_{\mathbf{1}}$ after an invertible linear substitution, and $D_\eta\neq 0$. For signs $\eta\neq\eta'$, $\prod_j K_j(\eta)^{2\sigma_j}-\prod_j K_j(\eta')^{2\sigma_j}$, of degree $2\sigma$, is not zero: otherwise unique factorization, the linear forms $K_j(\eta)$ being irreducible and in disjoint variables for distinct $j$, makes $K_j(\eta)$ a multiple of $K_j(\eta')$ for every $j$ with $\sigma_j>0$, that is $\eta^T\eta'^T$ constant on the bases of such a table; with $S=\{e:\eta_e\neq\eta'_e\}$, nonempty and proper since $\eta_{e_0}=\eta'_{e_0}$, the parity of $|T\cap S|$ is then constant, so the window $\{d_j(S),\dots,r_j(S)\}$ of 7.13 is one value and $c_j(S)=0$ for every $j$ with $\sigma_j>0$, against connectedness. The product

$$\prod_\eta\Big(D_\eta\prod_j K_j(\eta)\Big)\prod_{\{\eta,\eta'\}}\Big(\prod_j K_j(\eta)^{2\sigma_j}-\prod_j K_j(\eta')^{2\sigma_j}\Big)$$

is a nonzero polynomial of degree $2^{n-1}\big(N+2N(n-1)\big)+2\sigma\binom{2^{n-1}}{2}=2^{n-1}\big((2n-1)N+(2^{n-1}-1)\sigma\big)$, so by the Schwartz–Zippel bound it does not vanish at some point of $\{1,\dots,D+1\}$ in every variable, $D$ that degree. At such weights every $K_j(\eta)\neq 0$, every sign is a native gauge by 38.17(ii), nondegenerate since $D_\eta\neq 0$, and the native values are distinct. 38.6(iii) bounds the dimension below by $2^{n-1}$, and 38.7 at $g=1$ the order of every recurrence that follows from the relations. □

**Lemma 38.19 (Where native gauges can reach a wall).** *As stated, for a family.*

*Proof.* (i) On the balance wall the target of the direction is $\mathbf{m}_1$, so nativity at $\alpha$ reads $\mathbf{m}_1 = \sum_j \big(s_{1j} R_j(\alpha) + t_{1j} R_j^*(\alpha)\big)$ (8.17). $R_{j,e} = \alpha_e \partial_e K_j / K_j = \sum_T \mathbf{1}_T(e)\, \mathbb{P}_{j,\alpha}(T)$ and $R_{j,e}^* = \sum_T \mathbf{1}_{\mathcal{E}\setminus T}(e)\, \mathbb{P}_{j,\alpha}^*(T)$, and summing over $e \in B$ gives (i).

(ii) In the coordinates $(\rho, u, \alpha_R)$, $\alpha^T = \rho^{|T\cap B|} u^{T\cap B} \alpha_R^{T\cap R}$, so $\sum_T |T\cap B|\, \theta_{j,T} \alpha^T = \rho\, \partial_\rho K_j$, and with $\widetilde{K}_j = \sum_l \rho^l K_j^{[l]}$, $\widetilde{\Psi}_j = \sum_l \rho^l \Psi_j^{[l]}$,

$$\kappa_{1,B} = \sum_j \Big( s_{1j}\, \frac{\rho\, \partial_\rho \widetilde{K}_j}{\widetilde{K}_j} + t_{1j}\, \frac{\rho\, \partial_\rho \widetilde{\Psi}_j}{\widetilde{\Psi}_j} \Big)$$

at every native gauge of the direction, by (i) and $\kappa_{1,B} = m_{1,B} - \sum_j \big(s_{1j} d_j(B) + t_{1j} d_j^*(B)\big)$. Along the sequence the exponents stay bounded, and $\rho_i \partial_\rho \widetilde{K}_j = O(\rho_i)$. If no bottom letter vanished at $(u^*, \alpha_R^*)$, every quotient would be $O(\rho_i)$ and the right side would tend to 0, while the left side tends to $\kappa_{1,B}(\mathbf{v}) > 0$. So some bottom letter vanishes there. Let it be the only one, say $K_{j_0}^{[0]}$. Then the terms of the other letters tend to 0, so $s_{1j_0}(\mathbf{v}_i)\, \rho_i \partial_\rho \widetilde{K}_{j_0} / \widetilde{K}_{j_0} \to \kappa_{1,B}(\mathbf{v})$, and since $\rho_i \partial_\rho \widetilde{K}_{j_0} \to 0$, $|s_{1j_0}(\mathbf{v}_i)/\widetilde{K}_{j_0}| \to \infty$. Nativity in a logarithmic coordinate $D$ of $u$ or of $\alpha_R$ reads

$$D \log \big( u^{\mathbf{m}_1|_B} \alpha_R^{\mathbf{m}_1|_R} \big) = \sum_j \Big( s_{1j} \frac{D\widetilde{K}_j}{\widetilde{K}_j} + t_{1j} \frac{D\widetilde{\Psi}_j}{\widetilde{\Psi}_j} \Big);$$

the left side and the terms of the other letters stay bounded, so $s_{1j_0} D\widetilde{K}_{j_0} / \widetilde{K}_{j_0}$ does, and $D\widetilde{K}_{j_0}(\rho_i, u_i, \alpha_{R,i}) \to D K_{j_0}^{[0]}(u^*, \alpha_R^*)$ must vanish. The bottom letter is homogeneous in $u$ and in $\alpha_R$, so with $K_{j_0}^{[0]}(u^*, \alpha_R^*) = 0$ Euler's identity gives the derivatives along the scalings of $u$ and of $\alpha_R$ as well: the whole gradient vanishes. The same holds for a cobasis letter. If two or more bottom letters vanish, every term of the gradient of the product contains a vanishing factor; if one does, its gradient vanishes. In both cases the product vanishes at $(u^*, \alpha_R^*)$ with its gradient. □

**Proposition 38.23 (Small lower coefficients on one parameter).** *As stated, with $p$ letters, for real $C_i$ bounded below by a positive constant on $[0, \infty)$.*

*Proof.* For $\rho \geq 0$ and $\lambda > 0$, $\lambda\xi_i + \rho C_i(\rho)$ is $\lambda\xi_i$ plus a nonnegative number, so it stays in $\mathbb{C}\setminus(-\infty, 0]$, the powers are principal, and the integrand is continuous on $(0, \infty)$. Put $F_{\mathbf{a}} = G \prod_i C_i^{-w_i - a_i}$ and $c_{\mathbf{a}}(\lambda) = \prod_i \binom{-w_i}{a_i} (\lambda\xi_i)^{a_i}$, so that, wherever $|\lambda\xi_i| < |\rho C_i(\rho)|$ for every $i$,

$$\rho^{\kappa - 1} G \prod_i \big(\lambda\xi_i + \rho C_i\big)^{-w_i} = \rho^{\kappa^\flat - 1} F_{\mathbf{0}} \prod_i \Big(1 + \frac{\lambda\xi_i}{\rho C_i}\Big)^{-w_i} = \sum_{\mathbf{a}} c_{\mathbf{a}}(\lambda)\, \rho^{\kappa^\flat - |\mathbf{a}| - 1} F_{\mathbf{a}},$$

absolutely and uniformly where $|\lambda\xi_i| \leq \frac{1}{2}|\rho C_i(\rho)|$. Choose $b > 0$ with $G$ and the $C_i$ holomorphic and the $C_i$ zero-free on a neighbourhood of $|\rho| \leq b$, and let $|C_i| \geq c > 0$ on $[0, \infty)$ and on $|\rho| \leq b$. For $|\lambda| \leq \lambda_0 = bc/(2 \max_i |\xi_i|)$ each letter $\lambda\xi_i + \rho C_i(\rho)$ has exactly one root in $|\rho| < b$ (Rouché, against $\rho C_i(\rho)$) and none on $|\rho| = b$ or on $[b, \infty)$.

*The analytic part.* Split the integral at $b$. On $[b, \infty)$ the expansion above holds uniformly and its terms are dominated by $\sum_{\mathbf{a}} \prod_i |\binom{-w_i}{a_i}| 2^{-|\mathbf{a}|}$ times $|\rho^{\kappa^\flat - 1} F_{\mathbf{0}}|$, integrable at $\infty$, so $\int_b^\infty = \sum_{\mathbf{a}} c_{\mathbf{a}}(\lambda) \int_b^\infty \rho^{\kappa^\flat - |\mathbf{a}| - 1} F_{\mathbf{a}}\, d\rho$, holomorphic for $|\lambda| < \lambda_0$. Let $\oint$ start at $\rho = b$ on the principal branches and run once around $|\rho| = b$; the integrand continues along it, and the expansion holds

on the circle, each $\rho^{\kappa^\flat-|\mathbf{a}|-1}$ continued along it and $F_{\mathbf{a}}$ single-valued on the disc. For $\operatorname{Re} c'>0$ and $F$ holomorphic near the closed disc,

$$\oint_{|\rho|=b}\rho^{c'-1}F\,d\rho=\big(e^{2\pi ic'}-1\big)\int_0^b\rho^{c'-1}F\,d\rho,$$

by Cauchy's theorem on the disc cut along $[0,b]$; both sides are holomorphic in $c'$ off $-\mathbb{N}$, the right side read with the finite part $\operatorname{Reg}\int_0^b$ (1.5, 1.7), so the identity holds at every $c'\notin\mathbb{Z}$. At $c'=\kappa^\flat-|\mathbf{a}|$, $e^{2\pi ic'}=e^{2\pi i\kappa^\flat}\neq1$. Hence

$$A_b(\lambda)=\frac{1}{e^{2\pi i\kappa^\flat}-1}\oint_{|\rho|=b}\rho^{\kappa-1}G\prod_i\big(\lambda\xi_i+\rho C_i\big)^{-w_i}d\rho=\sum_{\mathbf{a}}c_{\mathbf{a}}(\lambda)\ \operatorname{Reg}\int_0^b\rho^{\kappa^\flat-|\mathbf{a}|-1}F_{\mathbf{a}}\,d\rho$$

is holomorphic for $|\lambda|<\lambda_0$, the circle integral of an integrand holomorphic in $\lambda$ along a contour that no root reaches, and the series converges there. Since $\operatorname{Reg}\int_0^\infty=\operatorname{Reg}\int_0^b+\int_b^\infty$, $A(\lambda)=A_b(\lambda)+\int_b^\infty\cdots$ is the series of the statement, holomorphic at 0.

*The branch part.* For $\lambda>0$ put $\rho=\lambda u$ in $\int_0^b$ and in the circle integral; $(\lambda X)^{-w}=\lambda^{-w}X^{-w}$ on the principal branches at $\lambda>0$, so both carry the factor $\lambda^{\kappa^\flat}$:

$$\int_0^b\cdots d\rho-A_b(\lambda)=\lambda^{\kappa^\flat}D_\lambda(b/\lambda),\qquad D_\lambda(R)=\int_0^R\psi_\lambda\,du-\frac{1}{e^{2\pi i\kappa^\flat}-1}\oint_{|u|=R}\psi_\lambda\,du,$$

with $\psi_\lambda(u)=u^{\kappa-1}G(\lambda u)\prod_i\big(\xi_i+uC_i(\lambda u)\big)^{-w_i}$. The roots of the letters in $u$ are $\rho_i(\lambda)/\lambda$, where $\rho_i(\lambda)$ is the root in $|\rho|<b$; from $|\rho_iC_i(\rho_i)|=|\lambda\xi_i|$ they have modulus at most $|\xi_i|/c$. Take $R_0>2\max_i|\xi_i|/c$, so that $|uC_i(\lambda u)|>2|\xi_i|$ on $|u|=R$ for $R\geq R_0$. For $R_0\leq R\leq b/|\lambda|$, one turn around $|u|=R$ multiplies $\psi_\lambda$ by $e^{2\pi i\kappa^\flat}$, the factor $u^{\kappa-1}$ giving $e^{2\pi i\kappa}$ and each letter, with one root inside, $e^{-2\pi iw_i}$; so $\frac{d}{dR}\oint_{|u|=R}\psi_\lambda\,du=\big[\psi_\lambda(Re^{i\vartheta})e^{i\vartheta}\big]_{\vartheta=0}^{2\pi}=(e^{2\pi i\kappa^\flat}-1)\psi_\lambda(R)$, and $\frac{d}{dR}D_\lambda(R)=0$. Thus $B(\lambda)=D_\lambda(R_0)$, with $R_0$ fixed, and $\int_0^\infty=A(\lambda)+\lambda^{\kappa^\flat}B(\lambda)$ for small $\lambda>0$. For complex $\lambda$ near 0, $\psi_\lambda$ is holomorphic in $\lambda$ on $[0,R_0]$ and on $|u|=R_0$: there $|\lambda u|\leq|\lambda|R_0<b$, the letters do not vanish, $\xi_i+uC_i(\lambda u)$ stays off $(-\infty,0]$ on $[0,R_0]$ for $\lambda$ small by compactness, and $u^{\kappa-1}$ is integrable at 0 since $\operatorname{Re}\kappa>0$. So $B$ is holomorphic at 0, and

$$B(0)=G(0)\Big[\int_0^{R_0}u^{\kappa-1}\prod_i\big(\xi_i+uC_i(0)\big)^{-w_i}du-\frac{1}{e^{2\pi i\kappa^\flat}-1}\oint_{|u|=R_0}u^{\kappa-1}\prod_i\big(\xi_i+uC_i(0)\big)^{-w_i}du\Big],$$

independent of $R_0\geq2\max_i|\xi_i|/c$ by the same derivative. When $\operatorname{Re}\kappa^\flat<0$ the circle integral is $O(R_0^{\operatorname{Re}\kappa^\flat})$ and the bracket tends to the convergent integral $\int_0^\infty$; the bracket is holomorphic in $(\kappa,\mathbf{w})$ off $\kappa^\flat\in\mathbb{Z}$, so it is the continuation of that integral in $\kappa$ and $\mathbf{w}$. Where every $\operatorname{Re}w_i>0$, $C_i(0)>0$ gives $\prod_i(\xi_i+uC_i(0))^{-w_i}=\prod_iC_i(0)^{-w_i}\prod_i(u+\xi_i/C_i(0))^{-w_i}$, and 15.1 with $J=\{1,\dots,p\}$ and the coefficients $\xi_i/C_i(0)$ gives the stacked Gamma times the Breta.

(ii) When $G$ and the $C_i$ are holomorphic on a neighbourhood of $[0,\infty)$, the cluster port of 38.21 with $r_i=1$ and the lower coefficients $\lambda\xi_i$, at the $b$ chosen above, is

$$\frac{1}{2\pi i}\Big[\big(e^{2\pi i\kappa^\flat}-1\big)\int_b^\infty\cdots d\rho+\oint_{|\rho|=b}\cdots d\rho\Big]=\frac{e^{2\pi i\kappa^\flat}-1}{2\pi i}\Big(\int_b^\infty\cdots d\rho+A_b(\lambda)\Big)=\frac{e^{2\pi i\kappa^\flat}-1}{2\pi i}\,A(\lambda),$$

and it does not depend on $b$ (38.22(i)). □

*Remark.* With $p=1$, $\xi_1=1$ and $C=C_1$, $\int_0^\infty u^{\kappa-1}(1+uC(0))^{-w}du = C(0)^{-\kappa}\Gamma_{\,1}^{\,2}\left[\begin{matrix}\kappa, & -\kappa^\flat\\ & w\end{matrix}\right]$, the branch part of the reference.

**Definition 38.24 (Native gauges on a hypersurface; corner values).** As stated.

*Proof.* $K_X$ and $K_Y$ are homogeneous, so $f_X$ is a function on $T_Z$ and $Z_Y$ a hypersurface of it, and $f_X(u^{-1})=f_X(u)$ by the definition. The Cremona $u\mapsto u^{-1}$ is an automorphism of $T_Z$ that keeps $f_X$ and carries $Z_Y$ onto $\{K_Y(u^{-1})=0\}$, smooth points to smooth points; so it carries the native gauges of $X$ on $Y$ to the critical points of the restriction of $\log f_X$ to the image, with the same values. □

**Proposition 38.25 (The stride at a wall of width one).** *As stated.*

*Proof.* By Theorem 19.4, with $K_{\mathcal{B}_R^+}(x,\alpha_R)=K_{\mathcal{B}|R}(\alpha_R)+x\,K_{\mathcal{B}/B}(\alpha_R)$ (its bases avoiding the hinge are those of $\mathcal{B}|R$ and its bases through it the $Y+g$, $Y\in\mathcal{B}/B$) and $\gamma_B^K K_{\mathcal{B}/R}=K_{\mathcal{B}|B}$,

$$K = K_{\mathcal{B}/R}(\alpha_B)\,K_{\mathcal{B}|R}(\alpha_R) + K_{\mathcal{B}|B}(\alpha_B)\,K_{\mathcal{B}/B}(\alpha_R).$$

On $B$ the letter $K_{\mathcal{B}/R}$ is homogeneous of the degree $d=r-r(R)=d(B)$ and $K_{\mathcal{B}|B}$ of the degree $r(B)=d+1$, width one being $r(B)-d(B)=1$. So

$$K(\rho u,\alpha_R)=\rho^d\big(P+\rho Q\big),\qquad K(\rho^{-1}u^{-1},\alpha_R^{-1})=\rho^{-d}\big(\bar P+\rho^{-1}\bar Q\big),$$

and the product is $P\bar P+\rho^{-1}P\bar Q+\rho Q\bar P+Q\bar Q$, with $P\bar P=f_{\mathcal{B}/R}^{-1}(u)f_{\mathcal{B}|R}^{-1}(\alpha_R)$ and $Q\bar Q=f_{\mathcal{B}|B}^{-1}(u)f_{\mathcal{B}/B}^{-1}(\alpha_R)$. The minors inherit plain weights. For a plain table on a set $Y$ of marks $\Psi(\alpha)=\alpha^Y K(\alpha^{-1})$, so $\Psi_{\mathcal{B}|B}(u)\Psi_{\mathcal{B}/B}(\alpha_R)=u^B\alpha_R^R\bar Q$; and by Theorem 19.4 for $\Psi$, $\Psi=\Psi_{\mathcal{B}/R}(\alpha_B)\Psi_{\mathcal{B}|R}(\alpha_R)+\Psi_{\mathcal{B}|B}(\alpha_B)\Psi_{\mathcal{B}/B}(\alpha_R)$, whose second term has the lower degree $|B|-r(B)$ in $\alpha_B$: it is the bottom letter. The first term of $K$ has the lower degree $d$: it is $\rho^d K^{[0]}$ with $K^{[0]}=P$. □

**Proposition 38.26 (Corners of a wall of width one).** *As stated.*

*Proof.* The defect is $c^{(1)}=d_y\log f$ (Lemma 38.2(ii)); in the coordinates $(\log\rho,u,\alpha_R)$, with $u$ normalized by $u_{b_0}=1$ at a fixed $b_0\in B$ and $\alpha_{e_0}=1$, write $w_i=c^{(1)}(\alpha_i)$, with the part $w_{B,i}$ along $\log\rho$ and the part $w_i'$ along the tori. Since $Q_0,Q_1,Q_2$ do not depend on $\rho$, $\rho\partial_\rho f^{-1}=-\rho^{-1}Q_0+\rho Q_2$, and $d\log f^{-1}=-w_i$ reads

$$-\rho^{-1}Q_0+\rho Q_2=-w_B\,f^{-1},\qquad \rho^{-1}dQ_0+dQ_1+\rho\,dQ_2=-w'\,f^{-1}$$

at $\alpha_i$, with $w_B,w'\to 0$ and $f^{-1}\to\lambda^{-1}$. The point $\alpha_i$ is a critical point of $\log f-\langle w_i,y\rangle$, which is the logarithm of the stride of the direction with the masses moved by $-w_i$; that is the second claim before the list.

*The limit.* $Q_2$ is bounded near $(u^*,\alpha_R^*)$, so the first equation gives $\rho_i^{-1}Q_0\to 0$, and then $f^{-1}=\rho^{-1}Q_0+Q_1+\rho Q_2\to Q_1(u^*,\alpha_R^*)$, so $\lambda^{-1}=Q_1(u^*,\alpha_R^*)$. If no factor of $P\bar Q$ vanished at the limit, $Q_0$ would be bounded away from 0 and $\rho^{-1}Q_0$ unbounded.

*(i)* A factor of $P$ makes $P\bar P=0$ and a factor of $\bar Q$ makes $Q\bar Q=0$; both together give $Q_1=0$, against $\lambda^{-1}\neq 0$. Let exactly one factor $a$ vanish, with $Q_0=aG_0$, $G_0\neq 0$ near the point, and suppose its zero set smooth there, $da\neq 0$. Then $\rho^{-1}Q_0\to 0$ gives $a=o(\rho)$, and the second equation reads $\rho^{-1}G_0\,da=-dQ_1-\rho^{-1}a\,dG_0-\rho\,dQ_2-w'f^{-1}$, whose right side is bounded while the left side is not. So this case does not occur.

*(ii)* At the corner of $K$ take local coordinates $(a, u')$ on $T_B$ and $(c, \alpha'_R)$ on $(\mathbb{C}^\times)^R$ with $a = K_{\mathcal{B}/R}(u)$ and $c = K_{\mathcal{B}|R}(\alpha_R)$; they exist since both zero sets are smooth, and the two functions depend on separate variables. Then $Q_0 = ac\,G$ with $G = \bar{Q}$, and $G \neq 0$ near the point, as $Q\bar{Q} = Q_1 - P\bar{P}$ is $\lambda^{-1} \neq 0$ at the point. The equations along $u'$ and along $\alpha'_R$ read

$$\partial_{u'} Q_1 = -\rho^{-1} ac\, \partial_{u'} G - \rho\, \partial_{u'} Q_2 - w'_{u'} f^{-1} \to 0,$$

and likewise along $\alpha'_R$, because $\rho^{-1}ac = \rho^{-1}Q_0/G \to 0$. On $\{a = c = 0\}$ the term $P\bar{P} = ac\bar{P}$ vanishes to the second order, so at the limit point the restriction of $Q_1$ to $\{a = c = 0\} = Z_{\mathcal{B}/R} \times Z_{\mathcal{B}|R}$ is critical, and there $Q_1 = Q\bar{Q} = f^{-1}_{\mathcal{B}|B}(u) f^{-1}_{\mathcal{B}/B}(\alpha_R)$, a product of nonvanishing functions of separate variables. Its differential along the product vanishes exactly when both factors are critical on their own zero sets, which is the claim. The corner of $\Psi$ is the same computation with $(K_{\mathcal{B}|B}(u^{-1}), K_{\mathcal{B}/B}(\alpha_R^{-1}))$ in the place of $(a, c)$, $G = P$, and $P\bar{P}$ the term that survives.

*(iii)* Let $K_{\mathcal{B}/R}$ vanish at $u^*$ with its differential, and no other factor vanish there. Then $\rho^{-1}Q_0 \to 0$ gives $K_{\mathcal{B}/R}(u_i) = o(\rho_i)$, and $\partial_{\alpha_R} Q_0 = K_{\mathcal{B}/R}(u)\, \partial_{\alpha_R}\big(K_{\mathcal{B}|R}\bar{Q}\big)$, so $\rho^{-1}\partial_{\alpha_R} Q_0 \to 0$ and the equation along $\alpha_R$ gives $\partial_{\alpha_R} Q_1 \to 0$. At the limit $P\bar{P}$ vanishes with its derivatives in $\alpha_R$, since it carries the factor $K_{\mathcal{B}/R}(u)$, so $f^{-1}_{\mathcal{B}|B}(u^*)\, \partial_{\alpha_R} f^{-1}_{\mathcal{B}/B}(\alpha_R^*) = 0$ with $f^{-1}_{\mathcal{B}|B}(u^*) \neq 0$: $\alpha_R^*$ is a critical point of $f_{\mathcal{B}/B}$ on $T_R$, and $\lambda^{-1} = Q\bar{Q}$ there. The other three factors are the same computation, with the equation along $u$ when the factor lives on $R$, and with $P\bar{P}$ the term that survives when the factor belongs to $\bar{Q}$.

*Converse.* For the direction with the masses moved by $\varepsilon\mathbf{u}$ the stride is $f\,\alpha^{\varepsilon\mathbf{u}}$, and its native gauges are the solutions of $d\log f^{-1} = \varepsilon\,\mathbf{u}$ in the coordinates above, the component of $\mathbf{u}$ along $\log\rho$ being $u_B$. With $\rho = \varepsilon r$, $a = \rho\hat{a}$, $c = \rho\hat{c}$, so that $Q_0 = \rho^2\hat{a}\hat{c}G$ and $\rho^{-1}\partial_a Q_0 = \hat{c}G + O(\rho)$, the equations divided by $\varepsilon$ where needed become

$$r\big(Q_2 - \hat{a}\hat{c}\,G\big) = u_B\, Q_1 + O(\varepsilon), \qquad \hat{c}\,G + \partial_a Q_1 = O(\varepsilon), \qquad \hat{a}\,G + \partial_c Q_1 = O(\varepsilon),$$
$$\partial_{u'} Q_1 = O(\varepsilon), \qquad \partial_{\alpha'_R} Q_1 = O(\varepsilon),$$

a system of $n-1$ equations in the $n-1$ unknowns $(u', \alpha'_R, \hat{a}, \hat{c}, r)$, analytic in $\varepsilon$. At $\varepsilon = 0$ it holds at the two constrained critical points with $\hat{a} = -\partial_c Q_1/G$, $\hat{c} = -\partial_a Q_1/G$ and $r = u_B Q_1/(Q_2 - \hat{a}\hat{c}G)$: at $a = c = 0$ the derivatives $\partial_a Q_1$ and $\partial_c Q_1$ are the multipliers, since $d(ac\bar{P}) = 0$ there, and the differential of $Q\bar{Q}$ along the zero sets vanishes, so $\hat{a}\hat{c}G = \mu_a\mu_c/G$ and $r$ is as stated, finite by hypothesis. The Jacobian in $(u', \alpha'_R, \hat{a}, \hat{c}, r)$ is block triangular: on $(u', \alpha'_R)$ the Hessian of $Q\bar{Q}$ restricted to $Z_{\mathcal{B}/R} \times Z_{\mathcal{B}|R}$, which at a critical point of a product of nonvanishing functions of separate variables is the direct sum of the two constrained Hessians times nonzero constants, nonsingular by hypothesis; then $G$ on $\hat{a}$ and on $\hat{c}$, and $Q_2 - \hat{a}\hat{c}G \neq 0$ on $r$. The implicit function theorem gives a solution analytic in $\varepsilon$; for $\varepsilon \neq 0$ it has $\rho = \varepsilon r + O(\varepsilon^2) \neq 0$ and $K = \rho^{d+1}(\rho\hat{a}\hat{c} + Q) \neq 0$, $\Psi \neq 0$ likewise, so it lies in $X_F$. Its native value is $e^{\varepsilon\langle\mathbf{u},y\rangle} f$, and $\varepsilon\langle\mathbf{u}, y\rangle = \varepsilon u_B \log\rho + O(\varepsilon) \to 0$, while $f \to (Q\bar{Q})^{-1}$ at the point. The corner of $\Psi$ is the same with the pairs exchanged. □

**Corollary 38.28 (The corners of a class).** *As stated.*

*Proof.* By Proposition 21.3(i) the set $C$ is a bunch of width one of $\mathcal{B}$ whose cell of the bunch is $\mathcal{A}$, the weights factoring, so its minors are $\mathcal{B}|C = \mathcal{A} \setminus g$, $\mathcal{B}/R = \mathcal{A}/g$, $\mathcal{B}|R = \mathcal{H} \setminus h$ and $\mathcal{B}/C = \mathcal{H}/h$ (Theorem 19.4). For a parallel class $\mathcal{A} \setminus g = \binom{C}{1}$ and $\mathcal{A}/g = \binom{C}{0}$, whose letter is the constant 1: the corner of $K$, which needs $K_{\mathcal{B}/R} = 0$, is empty, and $f_{\mathcal{B}/R} \equiv 1$. At the corner of $\Psi$ the point $u^*$ lies on the zero set of $K_{\binom{C}{1}}(u^{-1}) = \sum_{e\in C} u_e^{-1}$, which is smooth and, for $s \geq 2$, nonempty in $T_C$, and every point of it is a native gauge of $\mathcal{B}/R$ on $\mathcal{B}|C$ with the value 1; so $N(\mathcal{B}/R\,|\,\mathcal{B}|C) = \{1\}$ and

the corner values are $N(\mathcal{H}\setminus h\,|\,\mathcal{H}/h)$. For a series class $\mathcal{A}\setminus g=\binom{C}{C}$, whose letter is the monomial $u^C$, and $\mathcal{A}/g=\binom{C}{s-1}$, with the letter $u^C\sum_{e\in C}u_e^{-1}$: the corner of $\Psi$ needs $K_{\mathcal{B}|C}(u^{-1})=0$ and is empty, $f_{\mathcal{B}|C}\equiv 1$, and at the corner of $K$ every point of the smooth zero set of $\sum_{e\in C}u_e^{-1}$ is a native gauge of $\mathcal{B}|C$ on $\mathcal{B}/R$ with the value 1, so the corner values are $N(\mathcal{H}/h\,|\,\mathcal{H}\setminus h)$. A single mark $e$ has $T_{\{e\}}$ a point and no hypersurface in it. □

**Definition 38.29 (Asymptotic native gauges).** As stated.

*Proof.* For the direction with the masses moved by $\mathbf{w}\in\mathbb{C}^{\mathcal{E}}$, $w_{\mathcal{E}}=0$, the stride is $f\,\alpha^{\mathbf{w}}$, whose logarithmic differential is $c^{(1)}+\mathbf{w}$; at $\mathbf{w}=-c^{(1)}(x_m)$ it vanishes at $x_m$, and $x_m$ lies in the letter complement, which the masses do not change. □

**Proposition 38.30 (The far points lie over the singular locus of the step letters).** *As stated.*

*Proof.* $X_{\mathcal{E}}$ is compact and the charts of the orders cover it (Lemma 34.3(i)), so a subsequence converges to a point $x^*$, which lies off the torus because $x_m$ leaves its compact subsets. Take a chart through $x^*$, let $I=\{i:\varepsilon_i(x^*)=0\}\neq\varnothing$ and $\mathfrak{f}=(S_i)_{i\in I}$, so that $x^*\in O_{\mathfrak{f}}$, and let $w=(\varepsilon_i)_{i\notin I}$, coordinates on $O_{\mathfrak{f}}$ near $x^*$. In the chart $f=\prod_{i<n}\varepsilon_i^{\kappa_{1,S_i}}\Phi^{-1}$ by Lemma 34.3(ii), the masses of the direction contributing $\prod\varepsilon_i^{m_{1,S_i}}$ and the letters their floors. The forms $d\log\varepsilon_i=dy_{e_i}-dy_{e_{i+1}}$, $i<n$, are a basis of the invariant forms on the torus, so $c^{(1)}=d_y\log f\to 0$ says

$$\varepsilon_i\partial_{\varepsilon_i}\log f=\kappa_{1,S_i}-\varepsilon_i\partial_{\varepsilon_i}\log\Phi\ \longrightarrow\ 0\qquad(i<n),$$

which is (ii). For $i\in I$: if $\Phi(x_m)$ stayed away from 0 along a subsequence, then $\varepsilon_i\partial_{\varepsilon_i}\log\Phi\to 0$, $\Phi$ being a polynomial and $\varepsilon_i\to 0$, against $\kappa_{1,S_i}>0$; so $\Phi(x^*)=0$. On $\{\varepsilon_I=0\}$, $\Phi=\prod_g\big(\prod_i g_{Z_i}\big)^{a_g}$ with the exponent $a_g\geq 1$ of the letter $g$ in the direction (Lemma 34.3(ii), Proposition 12.20); so some factor of $\Pi_{\mathfrak{f}}$ vanishes at $x^*$. Suppose exactly one does, the step letter $g_Z$ of the letter $g$, and that its zero set in $O_{\mathfrak{f}}$ is smooth at $x^*$. Near $x^*$ every other letter has a sector letter that does not vanish, so $\Phi=G^{a_g}U$ with $G$ the sector letter of $g$ and $U$ a unit, and $G|_{\varepsilon_I=0}=g_Z\,v$ with $v(x^*)\neq 0$, the product of the step letters of $g$ at the other steps. For $i\notin I$, $\varepsilon_i(x^*)\neq 0$ and $\varepsilon_i\partial_{\varepsilon_i}\log\Phi\to\kappa_{1,S_i}$, so

$$\partial_{\varepsilon_i}\log\Phi=a_g\,\frac{\partial_{\varepsilon_i}G}{G}+\partial_{\varepsilon_i}\log U$$

stays bounded, and $\partial_{\varepsilon_i}G(x_m)\to 0$ as $G(x_m)\to 0$. At $x^*$ this says $v\,d_w g_Z=0$, against the smoothness of the zero set of $g_Z$ in $O_{\mathfrak{f}}$. So (i). □

**Theorem 38.31 (The values along a ray).** *As stated, over a subfield $\Bbbk$ of $\mathbb{C}$ containing the weights and the origin.*

*Proof.* (i) is the reference's (the reference, proof of thm:rayspace). The letters have their coefficients in $\Bbbk$, so $X_F$, the occupations and the stride are defined over $\Bbbk$, and with the origin in $\Bbbk$ and the direction integral the operators $c_e^{(0)}+k\,c_e^{(1)}+\partial_{y_e}$ map $\Bbbk[X_F]\otimes\Bbbk(k)$ to itself. Let $\mathcal{L}_{\Bbbk}$ be the $\Bbbk(k)$-span of their images and $M_{\Bbbk}=\big(\Bbbk[X_F]\otimes\Bbbk(k)\big)/\mathcal{L}_{\Bbbk}$. Since $\mathbb{C}[X_F]=\Bbbk[X_F]\otimes_{\Bbbk}\mathbb{C}$, $M_{\Bbbk}\otimes_{\Bbbk(k)}\mathbb{C}(k)$ is the quotient of 38.4, a cokernel commuting with the extension of scalars; so $\dim_{\Bbbk(k)}M_{\Bbbk}$ is the dimension $d$ of the ray, finite by (i).

(ii) The classes of $g, fg,\dots,f^dg$ in $M_{\Bbbk}$ are $d+1$ vectors of a space of dimension $d$, so there are $p_0,\dots,p_d\in\Bbbk(k)$, not all zero, with $\sum_j p_j(k)\,f^jg\in\mathcal{L}_{\Bbbk}$; clearing denominators puts them in $\Bbbk[k]$. With $h$ the largest $j$ with $p_j\neq 0$, $h\leq d$, and 38.7 gives $\sum_{j\leq h}p_j(k)\langle g\rangle_{k+j}=0$ for all large $k$.

(iii) Let $k_1$ exceed the integer zeros of $p_h$ and the bound of (ii). For $k \ge k_1$, $\langle g\rangle_{k+h} = -\sum_{j<h} \frac{p_j(k)}{p_h(k)} \langle g\rangle_{k+j}$ with $p_j(k)/p_h(k) \in \Bbbk$, the values of polynomials over $\Bbbk$ at integers lying in $\Bbbk$; induction on $k$ writes every $\langle g\rangle_k$, $k \ge k_1$, as a $\Bbbk$-combination of $\langle g\rangle_{k_1}, \dots, \langle g\rangle_{k_1+h-1}$. □

**Lemma 38.32 (Index by pole count).** *As stated, over a subfield $\Bbbk$ of $\mathbb{C}$.*

*Proof.* The poles of $r$ are rational, so its partial fractions have coefficients in $\Bbbk$: $r(k) = \sum_{h\ge1} \sum_{s=1}^{p} c_{h,s}(k+h)^{-s}$, finitely many $h$, $c_{h,s} \in \Bbbk$, with no polynomial part since $r \to 0$, and $\sum_h c_{h,1} = 0$ since $r(k) = O(k^{-2})$. 37.10 at $t_0 = 0$, whose identities are linear in the coefficients and hold at coefficients in any field, gives

$$\sum_{k\ge0} r(k) = \sum_{s\ge2} C_s \,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - \sum_{h,s} c_{h,s}\,(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,h-1]), \qquad C_s = \sum_h c_{h,s} \in \Bbbk,$$

the index-one terms regularized and their coefficient $C_1 = 0$, and every $(\mathsf{T}_{s;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,h-1])$, a finite sum of $m^{-s}$, rational. If $r(-M-k) = -r(k)$, uniqueness of the partial fractions gives $(-1)^s c_{M-h,s} = -c_{h,s}$, so $C_s = (-1)^{s+1}C_s$ and $C_s = 0$ for even $s$. For poles in one coset $-b+\mathbb{Z}$ the same computation runs in the lattice input $\mathbb{I}_{b+\mathbb{Z}}$, the sums $\sum_{k\ge0}(k+b+h)^{-s}$ being its Trailmoments at its cuts; with several cosets the index-one sums of the cosets are regularized one by one, and only their total coefficient vanishes, which leaves the differences of the index-one values of the cosets. The coefficients are in $\Bbbk$ throughout. □

## 50.38 Chapter 39: the two ends of a ray

The proofs of this chapter use three facts about the symmetries of Proposition 39.11: in the coordinates $y = \log\alpha$ they are linear, $y \mapsto \pm\beta^* y$; they keep the stride; and an expansion at a native gauge depends only on the Taylor series of the insertion there, which insertions realize to every finite order, $\mathbb{C}[X_F]$ containing the Laurent polynomials of degree 0.

**Definition 39.1 (The module of a ray; masters; the one-step matrix).** As stated.

*Proof.* $\mathcal{L}$ is spanned over $\mathbb{C}(k)$ by $\rho_e(g) = (c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})g$. On $\mathbb{C}[X_F]\otimes\mathbb{C}(k)$ put $\mathrm{Sh}(g\otimes r) = fg\otimes\varsigma r$, $(\varsigma r)(k) = r(k+1)$; then $\mathrm{Sh}(r'x) = \varsigma(r')\,\mathrm{Sh}(x)$. By 38.2(ii), $\partial_{y_e}(fg) = c_e^{(1)}fg + f\,\partial_{y_e}g$, so

$$f\,\big(c_e^{(0)} + (k+1)\,c_e^{(1)} + \partial_{y_e}\big)g = \big(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e}\big)(fg),$$

and $\mathrm{Sh}(r(k)\rho_e(g)) = r(k+1)\,\rho_e(fg) \in \mathcal{L}$. The stride is a unit of $\mathbb{C}[X_F]$, its exponents being integers and the letters units there, so Sh has the inverse $x \mapsto f^{-1}\varsigma^{-1}x$, which maps $\mathcal{L}$ into itself by the same identity read backwards: Sh descends to a bijection of $\mathfrak{M}$ with $\mathrm{Sh}(rm) = \varsigma(r)\,\mathrm{Sh}(m)$. A semilinear bijection maps bases of subspaces to bases of their images, so the classes $\mathrm{Sh}^{-1}[b_i] = [f^{-1}b_i]$ of masters of a submodule form a basis of it, and $\mathbf{B_b}$ is invertible.

Let the cyclic submodule of [1] be finite-dimensional, and let $h'$ be the least index with $\mathrm{Sh}^{h'}[1]$ in the span $W$ of $[1], \dots, \mathrm{Sh}^{h'-1}[1]$. Then $\mathrm{Sh}(W) \subseteq W$, and $\mathrm{Sh}(W)$ has the dimension of $W$, so $\mathrm{Sh}(W) = W$ and $\mathrm{Sh}^{-1}(W) = W$: $W$ is the cyclic submodule of [1], and $h' = h$. Writing $\mathrm{Sh}^h[1]$ in the basis and clearing denominators gives a relation with $p_h \ne 0$. If $p_0 = 0$, applying $\mathrm{Sh}^{-1}$ to $\sum_{i=1}^{h} p_i(k)\,\mathrm{Sh}^i[1] = 0$ gives $\sum_{i=1}^{h} p_i(k-1)\,\mathrm{Sh}^{i-1}[1] = 0$, a relation among $[1], \dots, \mathrm{Sh}^{h-1}[1]$, which is impossible. Two relations of length $h$ are proportional over $\mathbb{C}(k)$, since otherwise a combination of them eliminating $\mathrm{Sh}^h[1]$ would be a nonzero relation among the basis; with coefficients in $\mathbb{C}[k]$ without a common factor the proportionality factor is a constant. □

**Definition 39.2 (Solutions).** As stated.

*Proof.* On germs $\varsigma$ is bijective and $\varsigma(rx) = r(k+1)\varsigma(x)$. On $\lambda^k k^\rho\,\mathbb{C}((1/k))$, a module of rank one over $\mathbb{C}((1/k))$ with the generator $\lambda^k k^\rho$, $\varsigma$ is the composite of the substitution $k \mapsto k+1$, the automorphism $\varsigma_0$ of the field $\mathbb{C}((1/k))$ with $\frac{1}{k} \mapsto \frac{1}{k}\big(1+\frac{1}{k}\big)^{-1}$, with the multiplication by the unit $\lambda(1+\frac{1}{k})^\rho = \lambda\sum_i \binom{\rho}{i} k^{-i}$; so it is bijective and $\varsigma(rx) = r(k+1)\varsigma(x)$ for $r \in \mathbb{C}(k) \subset \mathbb{C}((1/k))$. □

**Lemma 39.3 (Solutions of distinct types).** *As stated.*

*Proof.* With $\varsigma_0$ the substitution $k \mapsto k+1$ on $\mathbb{C}((1/k))$, $\hat{A}_a(\operatorname{Sh} m) = \lambda_a(1+\frac{1}{k})^{\rho_a}\varsigma_0\big(\hat{A}_a(m)\big)$. Suppose $\sum_{a\in S} c_a\hat{A}_a = 0$ on $\mathfrak{M}$ with $c_a \in \mathbb{C}((1/k))$, every $c_a \neq 0$, and $S$ nonempty and minimal. Applying the relation at $\operatorname{Sh} m$, and applying $\varsigma_0$ to it at $m$, gives

$$\sum_{a\in S} c_a\lambda_a\big(1+\tfrac{1}{k}\big)^{\rho_a}\,\varsigma_0\big(\hat{A}_a(m)\big) = 0, \qquad \sum_{a\in S} \varsigma_0(c_a)\,\varsigma_0\big(\hat{A}_a(m)\big) = 0$$

for every $m$. The coefficient vectors of relations among the maps $\varsigma_0 \circ \hat{A}_a$, $a \in S$, form a space over $\mathbb{C}((1/k))$ of dimension one: two independent vectors in it would combine to a nonzero one with a zero coordinate, and its image under $\varsigma_0^{-1}$ would be a relation among fewer $\hat{A}_a$. So $c_a\lambda_a(1+\frac{1}{k})^{\rho_a} = \mu\,\varsigma_0(c_a)$ for one $\mu$ and every $a \in S$. With $c_a = c_a^0 k^{v_a}\big(1 + c_a' k^{-1} + O(k^{-2})\big)$, $c_a^0 \neq 0$, $v_a \in \mathbb{Z}$, the two sides are $c_a^0 k^{v_a}\lambda_a\big(1 + (\rho_a + c_a')k^{-1} + \cdots\big)$ and $\mu\, c_a^0 k^{v_a}\big(1 + (v_a + c_a')k^{-1} + \cdots\big)$, so $\mu = \lambda_a\big(1 + (\rho_a - v_a)k^{-1} + O(k^{-2})\big)$ for every $a \in S$: one $\lambda$, and $\rho_a - v_a$ independent of $a$, so all $a \in S$ have one type. Fix $\rho$ of that type and write every $A_a$, $a \in S$, in $\lambda^k k^\rho\,\mathbb{C}((1/k))$, which changes the $c_a$ by integral powers of $k$; then $c_a\lambda(1+\frac{1}{k})^\rho = \mu\,\varsigma_0(c_a)$ makes $c_a/\varsigma_0(c_a)$ independent of $a$, so every quotient $c_a/c_b$ is fixed by $\varsigma_0$. A series $r = \sum_{i\geq i_0} r_i k^{-i}$ with $\varsigma_0 r = r$ is constant: if $r_i \neq 0$ for some $i \neq 0$, the least such $i$ gives in $\varsigma_0 r - r$ the term $-i\,r_i k^{-i-1}$ and no other of that order. So the relation is $c(k)\sum_{a\in S}\gamma_a\hat{A}_a = 0$ with constants $\gamma_a \neq 0$, and $\sum_a \gamma_a A_a = 0$ is a relation over $\mathbb{C}$ among solutions of one type: excluded. Extended $\mathbb{C}((1/k))$-linearly to $\mathfrak{N}\otimes\mathbb{C}((1/k))$, the normalizations are $r$ independent linear forms on a space of dimension $\dim_{\mathbb{C}(k)}\mathfrak{N}$; the argument runs on $\mathfrak{N}$ in place of $\mathfrak{M}$, Sh mapping $\mathfrak{N}$ onto itself. □

**Theorem 39.4 (What solutions determine).** *As stated.*

*Proof.* (i) $V[f^{-1}b_i] = V(\operatorname{Sh}^{-1}[b_i]) = \varsigma^{-1}V[b_i]$, and $V[f^{-1}b_i] = \sum_l \mathbf{B}_{il}V[b_l]$ by $\mathbb{C}(k)$-linearity. For the values, $\langle f^{-1}b_i\rangle_k = \langle b_i\rangle_{k-1}$ (38.3).

(ii) $\operatorname{Ann}(\mathcal{S})$ is a $\mathbb{C}(k)$-subspace, and $V(\operatorname{Sh}^{\pm1} m) = \varsigma^{\pm1}V(m)$ vanishes with $V(m)$.

(iii) By (i), $\mathbf{B}'v_V - \varsigma^{-1}v_V = (\mathbf{B}'-\mathbf{B})v_V$, whose $i$-th entry is $V(m_i)$, $m_i = \sum_l(\mathbf{B}'-\mathbf{B})_{il}[b_l] \in \mathfrak{N}$. If a nonzero $m = \sum_l w_l[b_l]$ lies in $\operatorname{Ann}(\mathcal{S}) \cap \mathfrak{N}$, adding $w$ to the first row of $\mathbf{B}$ gives another such matrix.

(iv) For a germ the relation of (i) holds at every integer beyond some bound, and so do the equations of a pair $(cP_0, cQ_0)$; for a formal series they hold as identities. Conversely, let $(P, Q)$ satisfy all the equations. By (i), $Q\,\varsigma^{-1}v_V - Pv_V = (Q\mathbf{B} - P)v_V$, so every row of $Q\mathbf{B} - P$, read in $\mathfrak{N}$, lies in $\operatorname{Ann}(\mathcal{S}) \cap \mathfrak{N} = 0$, and $P = Q\mathbf{B} = QP_0/Q_0$. If $Q = 0$ then $P = 0$. Otherwise $Q_0$ divides every entry of $QP_0$; with $d = Q_0/\gcd(Q_0, Q)$, $d$ divides every entry of $P_0$ and $Q_0$, whose greatest common divisor is 1, $Q_0$ being the least common denominator, so $d = 1$, $Q = cQ_0$ and $P = cP_0$, and the degree bounds give $\deg c \leq D - D_0$. The equations are linear and homogeneous in the coefficients; for each $D$ the solution spaces of the first $N$ of them, in any enumeration, decrease with $N$ in a space of finite dimension, so they are constant from some $N$ on. □

**Corollary 39.5 (One solution of an irreducible module).** *As stated.*

*Proof.* $\mathrm{Ann}(\{V\}) \cap \mathfrak{N}$ is a submodule of $\mathfrak{N}$ (Theorem 39.4(ii)) other than $\mathfrak{N}$, hence 0. For the cyclic submodule of [1], $\sum_i p_i(k)\varsigma^i V[1] = V\big(\sum_i p_i(k)\,\mathrm{Sh}^i[1]\big) = 0$. A relation $\sum_{i<h} q_i(k)\varsigma^i V[1] = 0$ is $V(m) = 0$ with $m = \sum_{i<h} q_i(k)\,\mathrm{Sh}^i[1]$, nonzero as the $\mathrm{Sh}^i[1]$, $i < h$, are a basis; then $\mathrm{Ann}(\{V\})$ meets the cyclic submodule in a nonzero submodule, all of it, and $V[1] = 0$. □

**Proposition 39.6 (The law of [1] read from a one-step matrix).** *As stated.*

*Proof.* (i) $\mathrm{Sh}^{-1}$ is a bijection of $\mathfrak{M}$ with $\mathrm{Sh}^{-1}(r(k)\,m) = r(k-1)\,\mathrm{Sh}^{-1}(m)$ (Definition 39.1). For $i = 0$, $[1] = [b_1]$. If $\mathrm{Sh}^{-(i-1)}[1] = \sum_l w_{i-1,l}(k)[b_l]$, then

$$\mathrm{Sh}^{-i}[1] = \sum_l w_{i-1,l}(k-1)\,[f^{-1}b_l] = \sum_{l,l'} w_{i-1,l}(k-1)\,\mathbf{B}_{ll'}(k)\,[b_{l'}],$$

and induction gives the product form. A solution has $V \circ \mathrm{Sh}^{-i} = \varsigma^{-i} \circ V$ and is $\mathbb{C}(k)$-linear. For the values, $f^{-i} - \sum_l w_{i,l}(k)\,b_l$ lies in $\mathcal{L}$, so its value vanishes identically, and $\langle f^{-i}\rangle_k = \langle 1\rangle_{k-i}$ (Proposition 38.5(ii)).

(ii) The $w_i$ lie in $\mathbb{C}(k)^{\mathfrak{m}}$, so $h \le \mathfrak{m}$. By (i), $\sum_i q_i(k)\,\mathrm{Sh}^{-i}[1] = 0$, and $\mathrm{Sh}^h$, with $\mathrm{Sh}^h(r(k)\,m) = r(k+h)\,\mathrm{Sh}^h(m)$, turns it into $\sum_i q_i(k+h)\,\mathrm{Sh}^{h-i}[1] = 0$, which is the stated relation, $\mathrm{Sh}^j[1] = [f^j]$; the shift of the variable keeps the coefficients without a common factor. $p_0(k) = q_h(k+h) \neq 0$, since $w_0, \dots, w_{h-1}$ are independent. If $\mathbf{B}$ is invertible and $q_0 = 0$, then $\sum_{i\ge 1} q_i(k)\,w_{i-1}(k-1)\,\mathbf{B}(k) = 0$ gives $\sum_{i\ge 1} q_i(k)\,w_{i-1}(k-1) = 0$, and at $k+1$ a dependence among $w_0, \dots, w_{h-1}$ with coefficients not all zero: so $q_0 \neq 0$ and $p_h \neq 0$. Let $h'$ be the dimension of the cyclic submodule of [1]; its basis $[1], \mathrm{Sh}[1], \dots, \mathrm{Sh}^{h'-1}[1]$ (Definition 39.1) is carried by the semilinear bijection $\mathrm{Sh}^{-(h'-1)}$ to the basis $\mathrm{Sh}^{-i}[1]$, $0 \le i < h'$. A dependence among $w_0, \dots, w_h$ gives by (i) a dependence among $[1], \dots, \mathrm{Sh}^{-h}[1]$ with the same coefficients, so $h \ge h'$. When the $[b_l]$ are a basis, a dependence among the $\mathrm{Sh}^{-i}[1]$ is one among their coordinate rows $w_i$, so $h = h'$, and the relation, of length $h'$ with polynomial coefficients without a common factor, is the law of [1] up to a constant factor (Definition 39.1).

(iii) $M = [w_0; \dots; w_h]_J$ is an $(h+1) \times h$ matrix of rank $h$, so its left kernel is a line. It contains $(q_i)_i$, and also the vector of the signed maximal minors $\mu_i = (-1)^i \det M_{\hat{i}}$, $M_{\hat{i}}$ being $M$ without its row $i$: for every column $l$, $\sum_i (-1)^i M_{il} \det M_{\hat{i}}$ is the expansion along its first column of the square matrix $[M_{\cdot l}\,|\,M]$, which has two equal columns; and $\mu_h = (-1)^h \det \mathbf{W}_J \neq 0$. For the divisibility, put $D_i(k) = \prod_{j=0}^{i-1} Q(k-j)$, so that $D_i w_i$ is a row of polynomials, $w_i$ being $w_0$ times $\mathbf{B}(k-i+1)\cdots\mathbf{B}(k)$. The polynomial matrix $\tilde{M} = \mathrm{diag}(D_0, \dots, D_h)\,M$ has the left kernel spanned by $(q_i/D_i)_i$, and also by its signed maximal minors $\tilde{\mu}_i$, polynomials with $\tilde{\mu}_h = (-1)^h N$, $N = \det \mathbf{W}_J \prod_{i<h} D_i$. Let $\tilde{q}$ be a polynomial vector without a common factor spanning that kernel; then $\tilde{\mu} = \Lambda\tilde{q}$ with $\Lambda$ a polynomial, since a denominator of $\Lambda$ would divide every $\tilde{q}_i$. With $D = D_h$, the polynomial vector $(q_i D/D_i)_i$ spans the kernel, so $\tilde{q} = (q_i D/D_i)_i/g$ with $g$ the greatest common divisor of its entries, and $\tilde{q}_h = q_h/g$ divides $N$. Since $g$ divides every $q_i D$ and the $q_i$ have no common factor, $g$ divides $D$. So $q_h = g\,\tilde{q}_h$ divides $DN = \det \mathbf{W}_J \prod_{i=0}^{h} D_i = \det \mathbf{W}_J \prod_{j=0}^{h-1} Q(k-j)^{h-j}$. At $k_0$ with $Q(k_0 - j) \neq 0$ for $j < h$, every $D_i(k_0) \neq 0$ and the rows $w_i$, $i < h$, are regular, so the order of $q_h$ at $k_0$ is at most that of $\det \mathbf{W}_J$, for every $J$ with $\det \mathbf{W}_J \neq 0$: if $q_h(k_0) = 0$, every maximal minor of $\mathbf{W}(k_0)$ vanishes.

(iv) Take $J$ all the columns. Since $w_i(k) = w_{i-1}(k-1)\mathbf{B}(k)$, $[w_1; \dots; w_h](k) = \mathbf{W}(k-1)\,\mathbf{B}(k)$, so $\mu_0 = \det \mathbf{W}(k-1) \det \mathbf{B}(k)$ and $\mu_h = (-1)^h \det \mathbf{W}(k)$, and $q_0\mu_h = q_h\mu_0$ by (iii).

(v) $\mathrm{Sh}^{-1}\big(\sum_l c_l(k)[b_l]\big) = \sum_{l,l'} c_l(k-1)\,\mathbf{B}_{ll'}(k)\,[b_{l'}]$, so the rows obey the same recursion, and the proofs of (i) to (iv) run with $m$ in place of [1]. □

**Definition 39.7 (The model of a ray).** As stated.

*Proof.* The walls of the data of $g$ along the ray are $\kappa_{0,S} + \kappa_S(g) + k\kappa_{1,S}$ (Proposition 38.5), and the last term grows linearly in $k$ with $\kappa_{1,S} > 0$, so the condition holds for all large $k$. The form $\omega_{g,k}$ is the form of the family of the lists of $F$ at the exponents $(a_j + s_{0j} + ks_{1j},\, b_j + t_{0j} + kt_{1j})_j$ and the masses $u + \mathbf{m}_0 + k\mathbf{m}_1$, integers on the balance wall, as $dy = \Omega_{\mathcal{E}}/\alpha^{\mathcal{E}}$ on the chart $\alpha_{e_0} = 1$ up to the sign (Definition 34.1); and $\widetilde{\sigma}_{\mathcal{E}} \subseteq V(\mathbb{R})$ with its boundary in $D$ (Proposition 34.4(i)), so it is a class in $H_{n-1}(V, D_V)$ and the motivic period is defined. □

**Proposition 39.8 (The relations of a ray are exact on its model).** *As stated.*

*Proof.* By the proof of Definition 39.7, $\omega_{g,k}$ is the form of a family whose walls are at least 1; by Lemma 34.3(iii) it reads $\pm \prod_{i<n} \varepsilon_i^{m_i}\, h\, d\varepsilon_1 \wedge \cdots \wedge d\varepsilon_{n-1}$ in the chart of every order, with $m_i \geq 0$ the walls less one and $h$ a product of powers of sector letters, regular off the closures of the zero sets of the letters, which $V$ removes. So $\omega_{g,k}$ is regular on $V$. On the torus, for a form $\Phi$ of the top degree and a vector field $\xi$, $d\,\iota_\xi\Phi = L_\xi\Phi$. The field $\partial_{y_e}$ keeps $dy$, so $L_{\partial_{y_e}}(u\,dy) = (\partial_{y_e}u)\,dy$, and by Lemma 38.2(ii)

$$\partial_{y_e}\big(g\, f^k\omega_0/dy\big) = \big((c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})g\big)\, f^k\omega_0/dy.$$

In the chart of an order, $\partial_{y_e} = \sum_i (\delta_{e,e_i} - \delta_{e,e_{i+1}})\, \varepsilon_i\partial_{\varepsilon_i}$, a regular field tangent to every $D_{S_i} = \{\varepsilon_i = 0\}$. With $\Phi = \omega_{g,k}$ written as above, $\iota_{\varepsilon_i\partial_{\varepsilon_i}}\Phi$ carries the factor $\varepsilon_i^{m_i+1}$ and every $d\varepsilon_l$ with $l \neq i$, so its pullback to each $D_{S_l}$ vanishes: at $l = i$ by the factor $\varepsilon_i$, at $l \neq i$ by $d\varepsilon_l$. So $\iota_{\partial_{y_e}}\omega_{g,k}$ is regular on $V$ and vanishes on $D_V$, and in the relative de Rham complex of $(V, D_V)$ the form $d\big(\iota_{\partial_{y_e}}\omega_{g,k}\big)$ is a coboundary. The second relation is the identity of forms $f\,\omega_{g,k} = \omega_{fg,k} = \omega_{g,k+1}$. □

**Theorem 39.9 (A ray is one motive).** *As stated.*

*Proof.* (i) The period of $[H, \widetilde{\sigma}_{\mathcal{E}}, \omega_{g,k}]^{\mathrm{mot}}$ is the integral of $\omega_{g,k}$ over the positive cell (Proposition 34.4(ii)), which is $\langle g\rangle_k$, the integral over the positive orthant of the chart. The motivic period is linear in the de Rham class, and by Proposition 39.8 the classes satisfy the relations. The law of [1] is an element $\sum_j p_j(k) f^j = \sum_l r_l(k)\big(c_{e_l}^{(0)} + k\, c_{e_l}^{(1)} + \partial_{y_{e_l}}\big)g_l$ of $\mathcal{L}$, a finite combination with $r_l \in \mathbb{C}(k)$ (Definition 39.1); at every integer $k$ off the poles of the $r_l$ and beyond the bounds of Definition 39.7 for the $g_l$, multiplying by $f^k\omega_0$ gives $\sum_j p_j(k)\,\omega_{1,k+j} = \sum_l r_l(k)\, d\big(\iota_{\partial_{y_{e_l}}}\omega_{g_l,k}\big)$, whose class is 0. (ii) Lemma 34.6(ii); $L(H, \widetilde{\sigma}_{\mathcal{E}})$ is the image of $H_{\mathrm{dR}}$. (iii) The coaction of Definition 34.5; the left factors involve only $H$, $\widetilde{\sigma}_{\mathcal{E}}$ and the basis, and applying $e_i^\vee$ to the relation of (i) among the classes gives the law for the right factors. (iv) $V$, $D_V$ and the forms $\omega_{g,k}$ for $g$ with rational coefficients are defined over $\mathbb{Q}$, so their classes have rational coordinates in a rational basis; the law of [1] has coefficients in $\mathbb{Q}[k]$ up to a constant factor, the relation space being defined over $\mathbb{Q}(k)$; and $[H, \widetilde{\sigma}_{\mathcal{E}}, \omega]^{\mathrm{mot}}$ is linear in $\omega$, so the period of $\langle 1\rangle_k^{\mathrm{mot}}$ is $\sum_i c_i(1,k)$ times the periods of the basis. □

**Corollary 39.10 (The rows of the values at every order).** *As stated.*

*Proof.* For $k \geq k_1$, $p_h(k) \neq 0$, and the relation of Theorem 39.9(i) writes $[\omega_{1,k+h}]$ as a rational combination of $[\omega_{1,k}], \ldots, [\omega_{1,k+h-1}]$; by induction $W$ is spanned by $[\omega_{1,k_1}], \ldots, [\omega_{1,k_1+h-1}]$, so $m \leq h$. The coordinates of $[\omega_{1,k}]$ in the basis $(w_l)$ are rational and satisfy the law because the classes do, and the period is linear, as in Theorem 39.9(iv). □

**Proposition 39.11 (The symmetries of a ray).** *As stated.*

*Proof.* $\phi_\beta \circ \phi_{\beta'} = \phi_{\beta'\circ\beta}$, the Cremona commutes with every $\phi_\beta$ and its square is the identity, so the maps $\phi_\beta$ and $\bar\phi_\beta$ form a group, and the two conditions defining a symmetry pass to composites and inverses.

(i) If $\beta$ carries the tables, $K_j(\alpha\circ\beta) = \sum_{T\in\mathcal{B}_j}\theta_{j,T}\alpha^{\beta(T)} = \sum_{T'\in\mathcal{B}_{h(j)}}\theta_{h(j),T'}\alpha^{T'} = K_{h(j)}(\alpha)$, and the same with the complements gives $\Psi$. If $\beta$ exchanges them, the sets $\beta(T)$, $T\in\mathcal{B}_j$, are the complements of the bases $\mathcal{E}\setminus\beta(T)$ of table $h(j)$, with the same weights, so $K_j(\alpha\circ\beta) = \Psi_{h(j)}(\alpha)$ and $\Psi_j(\alpha\circ\beta) = K_{h(j)}(\alpha)$. By 7.2, $K_j(\alpha^{-1}) = \alpha^{-\mathcal{E}}\Psi_j(\alpha)$ and $\Psi_j(\alpha^{-1}) = \alpha^{-\mathcal{E}}K_j(\alpha)$; with $(\alpha\circ\beta)^{\mathcal{E}} = \alpha^{\mathcal{E}}$, $K_j\big((\alpha\circ\beta)^{-1}\big) = \alpha^{-\mathcal{E}}\Psi_j(\alpha\circ\beta) = \alpha^{-\mathcal{E}}K_{h(j)}(\alpha)$, and the same for $\Psi_j$.

(ii) With $(\alpha\circ\beta)^u = \alpha^{u\circ\beta^{-1}}$ and $(\alpha^{-1})^u = \alpha^{-u}$, (i) gives in the four cases

$$f\circ\phi_\beta = \alpha^{\mathbf{m}_1\circ\beta^{-1}}\prod_j K_{h(j)}^{-s_{1j}}\Psi_{h(j)}^{-t_{1j}}, \qquad f\circ\phi_\beta = \alpha^{\mathbf{m}_1\circ\beta^{-1}}\prod_j \Psi_{h(j)}^{-s_{1j}}K_{h(j)}^{-t_{1j}},$$
$$f\circ\mathrm{cr} = \alpha^{-\mathbf{m}_1+2\sigma_1\mathbf{1}}\prod_j \Psi_j^{-s_{1j}}K_j^{-t_{1j}}, \qquad f\circ\bar\phi_\beta = \alpha^{-\mathbf{m}_1\circ\beta^{-1}+2\sigma_1\mathbf{1}}\prod_j K_{h(j)}^{-s_{1j}}\Psi_{h(j)}^{-t_{1j}},$$

each equal to $f$ under the conditions with $i=1$, after reindexing the tables by $h$; the same computation with the data of the origin gives $(\omega_0/dy)\circ\phi = \omega_0/dy$ under the conditions with $i=0$, on $X_F$ for an integral origin and on the positive orthant with the principal powers otherwise. In the coordinates $y$ the maps are linear, $y\mapsto\pm\beta^*y$, and they keep the form $dy$ on $\mathbb{C}^{\mathcal{E}}/\mathbb{C}\mathbf{1}$ up to the sign of the permutation and the factor $(-1)^{n-1}$ of the inversion; so $\phi^*\omega_0 = \pm\omega_0$ for an integral origin. The letters go to letters times monomials, so $X_F$ is mapped onto itself, and positive gauges go to positive gauges. The value $\langle g'\circ\phi\rangle_k$ is the integral of $(g'\circ\phi)f^k(\omega_0/dy)$ against the Lebesgue measure of the quotient, which does not depend on the chart (17.1); the linear change of variables $y\mapsto\pm\beta^*y$ keeps that measure, and with $f\circ\phi = f$ and $(\omega_0/dy)\circ\phi = \omega_0/dy$ it gives $\langle g'\rangle_k$.

(iii) Substitute (i) in $\alpha^u\prod_j K_j^{-a_j}\Psi_j^{-b_j}$ and reindex the tables by $h$. □

**Theorem 39.12 (The invariant module).** *As stated.*

*Proof.* (i) In the coordinates $y$, $\phi$ is $y\mapsto M_\phi y$ with $M_\phi$ linear. For a constant vector field $\xi$ and a function $u$, $(\xi u)\circ\phi = (M_\phi^{-1}\xi)(u\circ\phi)$. The relation space is spanned by $\rho_\xi(g') = \xi\big(\log(f^k\omega_0/dy)\big)g' + \xi g' = \sum_e \xi_e\big(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e}\big)g'$, and since $\log(f^k\omega_0/dy)\circ\phi$ differs from $\log(f^k\omega_0/dy)$ by a constant (Proposition 39.11), $\rho_\xi(g')\circ\phi = \rho_{M_\phi^{-1}\xi}(g'\circ\phi)\in\mathcal{L}$. So $\phi^*$ is well defined on $\mathfrak{M}$, and with $(fg')\circ\phi = f\,(g'\circ\phi)$ it commutes with Sh. Then $e_G$ is an idempotent commuting with Sh, its image and kernel are submodules with $\mathfrak{M} = e_G\mathfrak{M}\oplus(1-e_G)\mathfrak{M}$, $e_G[g'] = [e_Gg']$ with $e_Gg'$ invariant, and the class of an invariant insertion is fixed by $e_G$.

(ii) $V = V\circ e_G$ vanishes on the kernel of $e_G$. The values are invariant by Proposition 39.11(ii). 1 is invariant, so $[1]\in\mathfrak{M}^G$, a submodule, which then contains the cyclic submodule of $[1]$.

(iii) $\phi$ fixes $\alpha_c$ and keeps $f$, so it maps a steepest-descent disc $D_c$ through $\alpha_c$ to a disc through $\alpha_c$ on which $|f| < |f(\alpha_c)|$ off $\alpha_c$, with $|f|\le|f(\alpha_c)|e^{-\varepsilon}$ on its boundary. In Morse coordinates $v$ at $\alpha_c$, $\log f = \log f(\alpha_c) - \frac12\sum_i v_i^2$, the homology of a small ball relative to the part where $|f|\le|f(\alpha_c)|e^{-\varepsilon}$ is $\mathbb{Z}$ in the dimension $n-1$, generated by the real ball; so $\phi(D_c)$ is $\pm D_c$ up to a boundary and pieces where $|f|\le|f(\alpha_c)|e^{-\varepsilon}$. Fix a branch of $\omega_0$ near $\alpha_c$; $(\omega_0/dy)\circ\phi$ and $\omega_0/dy$ are branches of one function there and $\phi^*dy = \pm dy$, so $\phi^*\omega_0 = c\,\omega_0$ near $\alpha_c$ for a constant $c\neq0$. Then $\int_{D_c}(g\circ\phi)f^k\omega_0 = c^{-1}\int_{\phi(D_c)}gf^k\omega_0$, and since $gf^k\omega_0$ is a closed form of top degree, $A_c\circ\phi^* = c'A_c$ for a constant $c'$. The leading coefficients of $A_c(g\circ\phi)$ and of $A_c(g)$ are $(2\pi)^{(n-1)/2}(\omega_0/dv)(\alpha_c)$ times $(g\circ\phi)(\alpha_c) = g(\alpha_c)$ (38.6(i)), and an insertion with $g(\alpha_c)\neq0$ gives $c' = 1$.

For $\phi \in G$ and $\psi \in G_c$, $A_c \circ (\phi\psi)^* = A_c \circ \psi^* \circ \phi^* = A_c \circ \phi^*$, so $A_O$ is well defined; for $\chi \in G$, $A_O \circ \chi^* = \sum A_c \circ (\chi\phi)^*$ runs over the same cosets, so $A_O$ is invariant, and $A_O \circ e_G = A_O$. By the argument above, $A_c \circ \phi^*$ is a constant multiple of the expansion over the disc $\phi(D_c)$, a steepest-descent disc through the native gauge $\phi(\alpha_c)$. The nondegenerate native gauges are isolated points of the critical locus, a variety, so finitely many; take an insertion $g$ with $g(\alpha_c) = 1$ that vanishes to order $N$ at every other one. Its expansion there vanishes to the relative order $k^{-\lceil N/2 \rceil}$, since $a_i(g)$ involves derivatives of $g$ of order at most $2i$, so $A_O(e_G g) = A_O(g)$ has the nonzero leading coefficient of $A_c(g)$, and $A_O$ does not vanish on $\mathfrak{M}^G$. The same insertions, applied to a relation $\sum_O \gamma_O A_O = 0$ on $\mathfrak{M}^G$ among orbits of one native value, give $\gamma_O = 0$ orbit by orbit. Lemma 39.3 gives the bound. □

**Theorem 39.13 (A reversed exchange at a lean).** *As stated.*

*Proof.* (i) $T \mapsto \mathcal{E} \setminus \beta(T)$ is a bijection of $\mathcal{B}_j$ onto $\mathcal{B}_{h(j)}$ keeping the weights, so $K_j(\mathbf{1}) = K_{h(j)}(\mathbf{1})$ and, under the weight laws, $R_{h(j),\beta(e)}(\mathbf{1}) = \mathbb{P}_{j,\mathbf{1}}\big(\beta(e) \notin \beta(T)\big) = \mathbb{P}_{j,\mathbf{1}}(e \notin T) = 1 - R_{j,e}(\mathbf{1})$. By 9.3, $\tau_e = \sum_j \big(\sigma_j + \delta_j(2R_{j,e}(\mathbf{1}) - 1)\big)$; reindexing by $h$, with $(\sigma_{h(j)}, \delta_{h(j)}) = (\sigma_j, \delta_j)$ (11.13), $\tau_{\beta(e)} = \sum_j \big(\sigma_j + \delta_j(1 - 2R_{j,e}(\mathbf{1}))\big) = 2\sigma - \tau_e$. The exponents $(s_j, t_j)$ are carried by $h$, so Proposition 39.11(ii) applies to $\bar{\phi}_\beta$ with the masses $\boldsymbol{\tau}$, and to the ray with origin 0 and direction $(c\boldsymbol{\tau}; c\mathbf{s}, c\mathbf{t})$. For $\phi_\beta$ and cr, Proposition 39.11(i) turns the integrand into $\alpha^{\boldsymbol{\tau}\circ\beta^{-1}} \prod_j \Psi_{h(j)}^{-s_j} K_{h(j)}^{-t_j}$ and $\alpha^{-\boldsymbol{\tau}+2\sigma\mathbf{1}} \prod_j \Psi_j^{-s_j} K_j^{-t_j}$. Since $\tau_{\beta^{-1}(e)} = 2\sigma - \tau_e = \sum_j \big(\sigma_j - \delta_j(2R_{j,e}(\mathbf{1}) - 1)\big)$, the masses of $\Upsilon_{\boldsymbol{\sigma},-\boldsymbol{\delta};\mathbf{1}}$, whose exponents are $(t_j, s_j)$, both are its integrand.

(ii) This is the exchange clause of 11.14 at the table's gauge, read on the views. In the coordinates $y$, Proposition 39.11(i) gives $(K_j/\Psi_j) \circ \bar{\phi}_\beta = K_{h(j)}/\Psi_{h(j)}$ and $(K_j\Psi_j) \circ \bar{\phi}_\beta = \alpha^{-2\mathcal{E}} K_{h(j)}\Psi_{h(j)}$, so, with $(\sigma_{h(j)}, \delta_{h(j)}) = (\sigma_j, \delta_j)$,

$$\Phi_{\mathbf{O}}(-\beta^* y) = \Phi_{\mathbf{O}}(y), \qquad \Phi_{\mathbf{E}}(-\beta^* y) = \Phi_{\mathbf{E}}(y) - 2\sigma\langle \mathbf{1}, y\rangle.$$

At the table's gauge $\Psi_j(e^y) = e^{\langle \mathbf{1}, y\rangle} K_j(e^{-y})$ (7.2), so $\Phi_{\mathbf{O}}(-y) = -\Phi_{\mathbf{O}}(y)$ and $\Phi_{\mathbf{E}}(-y) = \Phi_{\mathbf{E}}(y) - 2\sigma\langle \mathbf{1}, y\rangle$. For $w = v_0\mathbf{1}_e + \sum_i v_i \mathbf{1}_{g_i}$ and $w' = v_0\mathbf{1}_{\beta(e)} + \sum_i v_i \mathbf{1}_{\beta(g_i)}$, $\beta^* w' = w$ and $\langle \mathbf{1}, w'\rangle = \langle \mathbf{1}, w\rangle$, so

$$\Phi_{\mathbf{O}}(w') = \Phi_{\mathbf{O}}(-w) = -\Phi_{\mathbf{O}}(w), \qquad \Phi_{\mathbf{E}}(w') = \Phi_{\mathbf{E}}(-w) + 2\sigma\langle \mathbf{1}, w\rangle = \Phi_{\mathbf{E}}(w),$$

the identities of the views at the table's gauge. $\Phi_{\mathbf{O}}$ is linear in $\boldsymbol{\delta}$, so $-\Phi_{\mathbf{O}}^{\boldsymbol{\delta}} = \Phi_{\mathbf{O}}^{-\boldsymbol{\delta}}$, and the views, hence the readings, of $\beta(e)$ at $\boldsymbol{\delta}$ are those of $e$ at $-\boldsymbol{\delta}$, the partners matched by $\beta$. At the centre $\Phi_{\mathbf{O}} = 0$ and $e$ and $\beta(e)$ read alike. At a lean the partition at $-\boldsymbol{\delta}$ is that at $\boldsymbol{\delta}$ (11.2): if $e$ and $e'$ lie in one class at $\boldsymbol{\delta}$, they lie in one class at $-\boldsymbol{\delta}$, so $\beta(e)$ and $\beta(e')$ read alike at $\boldsymbol{\delta}$. So $\beta$ maps classes into classes, and so does $\beta^{-1}$, which exchanges the family as well: $\beta$ permutes the classes.

(iii) By 11.15, $\beta$ maps every class of the centre onto itself. Let $C$ split at the lean into $C_K$ and $C_\Psi$, with $Q$ the polynomial written as $K$ from every mark of $C_K$ and as $\Psi$ from every mark of $C_\Psi$ (11.17). With one table, $h$ is the identity and Proposition 39.11(i) gives $K(\alpha \circ \beta) = \Psi(\alpha)$ and $\Psi(\alpha \circ \beta) = K(\alpha)$. For $e \in C_K$ and an order $\mathbf{g}$ of its room with $K_{e,\mathbf{g}} = Q$, put the position variables $x_0$ at $\beta(e)$ and $x_i$ at $\beta(g_i)$; then $\alpha \circ \beta$ carries $x_0$ at $e$ and $x_i$ at $g_i$, so $\Psi_{\beta(e),\beta(\mathbf{g})} = K_{e,\mathbf{g}} = Q$ and $\beta(e) \in C_\Psi$. For $e \in C_\Psi$ with $\Psi_{e,\mathbf{g}} = Q$, $K(\alpha) = \Psi(\alpha \circ \beta)$ gives $K_{\beta(e),\beta(\mathbf{g})} = \Psi_{e,\mathbf{g}} = Q$ and $\beta(e) \in C_K$. $C_K$ and $C_\Psi$ have equally many marks, so $\beta(C_K) = C_\Psi$ and $\beta(C_\Psi) = C_K$. □

**Definition 39.15 (Levels, wall degree, boxes).** As stated.

*Proof.* The products of the monomials $\alpha^v$, $v_{\mathcal{E}} = 0$, and of the chances $\theta_{j,T}\alpha^T/K_j$ and $\theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j$ that generate $\mathbb{C}[X_F]$ (Definition 38.1) are constant multiples of evaluations of symbols, which

therefore span it. The wall degree is the largest of finitely many numbers $-\kappa_U(g)/\kappa_{1,U}$, and $\kappa_U(g)+x\,\kappa_{1,U}\geq 0$ for every $U$ exactly when $x$ is at least each of them. By 38.2(i) and Definition 38.1, with $R_{j,e}=\sum_{T\ni e}\theta_{j,T}\alpha^T/K_j$ and $R^*_{j,e}=\sum_{T\not\ni e}\theta_{j,T}\alpha^{\mathcal{E}\setminus T}/\Psi_j$, the operator $c_e^{(0)}+k\,c_e^{(1)}+\partial_{y_e}$ multiplies $\alpha^u\prod_j K_j^{-a_j}\Psi_j^{-b_j}$ by

$$m_{0,e}+k\,m_{1,e}+u_e-\sum_j\big((s_{0j}+k\,s_{1j}+a_j)R_{j,e}+(t_{0j}+k\,t_{1j}+b_j)R^*_{j,e}\big),$$

and $R_{j,e}$ times the evaluation of $(u;\mathbf{a},\mathbf{b})$ is the evaluation of $\sum_{T\ni e}\theta_{j,T}(u+\mathbf{1}_T;\mathbf{a}+\mathbf{e}_j,\mathbf{b})$, and likewise for $R^*_{j,e}$; this is the evaluation of the mark relation. The letter identities evaluate to the evaluation of $g$ times $1-K_j/K_j$, or $1-\Psi_j/\Psi_j$. The maps of the four kinds carry each letter to a letter times 1 or $\alpha^{-\mathcal{E}}$ (Proposition 39.11(i)), so do their composites, and the action of Proposition 39.11(iii) is the substitution into the evaluation. □

**Proposition 39.16 (The wall degree).** *As stated.*

*Proof.* (i) By Lemma S19, on the sector of an order $\pi$, $|g|\,f^x=\prod_l q_l^{\kappa_{U_l}(g)+x\kappa_{1,U_l}}\,|g_\pi|\varphi_\pi^x$ with the last factor between positive constants, so $|g|f^x$ is bounded on the sector exactly when every exponent is nonnegative; every nonempty proper $U$ is a tail of some order, which gives the first statement. Likewise $|g\,f^k\omega_0/dy|$ is, on the sector of $\pi$, $\prod_l q_l^{c_l}$ times a factor between positive constants, $c_l=\operatorname{Re}\kappa_{0,U_l}+\kappa_{U_l}(g)+\operatorname{Re}(k)\kappa_{1,U_l}$, and $\int_{(0,1]^{n-1}}\prod_l q_l^{c_l-1}\,dq$ is finite exactly when every $c_l>0$; so the value converges absolutely exactly when $\operatorname{Re}\kappa_{0,U}+\kappa_U(g)+\operatorname{Re}(k)\kappa_{1,U}>0$ for every $U$; for the origin 0 this reads $\operatorname{Re}k>-\kappa_U(g)/\kappa_{1,U}$ for every $U$, and in general $\kappa_U(g)\geq-\deg_\kappa g\,\kappa_{1,U}$ gives the sufficient condition.

(ii) $\kappa_U(\alpha^u)=u_U$. For $u\neq 0$ with $u_{\mathcal{E}}=0$ some $u_e<0$, and $U=\{e\}$ is proper, $n\geq 2$, so the wall degree is positive; for $u=0$ it is 0. A monomial of $f^{-1}=\alpha^{-\mathbf{m}_1}\prod_j K_j^{s_{1j}}\Psi_j^{t_{1j}}$ is $\alpha^w$ with $w=-\mathbf{m}_1+\sum_j\big(\sum_{i\leq s_{1j}}\mathbf{1}_{T_{j,i}}+\sum_{i\leq t_{1j}}\mathbf{1}_{\mathcal{E}\setminus T'_{j,i}}\big)$ for bases $T_{j,i},T'_{j,i}$ of table $j$, and by 7.13 $w_U\geq-m_{1,U}+\sum_j\big(s_{1j}d_j(U)+t_{1j}d_j^*(U)\big)=-\kappa_{1,U}$. The walls of Laurent symbols add, so $\kappa_U(\alpha^{u+w})+(\deg_\kappa\alpha^u+1)\kappa_{1,U}\geq 0$.

(iii) The terms are $g$ and the symbols $(u+\mathbf{1}_T;\mathbf{a}+\mathbf{e}_j,\mathbf{b})$ and $(u+\mathbf{1}_{\mathcal{E}\setminus T};\mathbf{a},\mathbf{b}+\mathbf{e}_j)$, whose walls are $\kappa_U(g)+|T\cap U|-d_j(U)$ and $\kappa_U(g)+|U\setminus T|-d_j^*(U)$, at least $\kappa_U(g)$ (7.13).

(iv) For a mark $e$, $\{e\}$ is proper, and $\kappa_{\{e\}}(g)\geq-x\,\kappa_{1,\{e\}}$ bounds $u_e$ from below at a fixed level; with $u_{\mathcal{E}}$ fixed by the level, $u$ is bounded from above as well.

(v) A symmetry maps the positive orthant onto itself and keeps $f$ (Proposition 39.11(ii)), and the evaluation of $\phi^*g$ is that of $g$ composed with $\phi$, so $|\phi^*g|f^x$ and $|g|f^x$ have the same bound, and (i) gives the same wall degree. The action of Proposition 39.11(iii) maps the level $(\mathbf{0},\mathbf{0})$ to itself. □

**Lemma 39.17 (The Laurent part).** *As stated.*

*Proof.* $\mathfrak{M}$ is finite-dimensional (Theorem 38.31(i)). The direction has $s_{1j},t_{1j}\in\mathbb{N}$, so $f^{-1}$ is a Laurent polynomial and $\operatorname{Sh}^{-1}[\alpha^v]=[f^{-1}\alpha^v]$ lies in $\mathfrak{M}_{\mathrm{L}}$: $\operatorname{Sh}^{-1}$ maps $\mathfrak{M}_{\mathrm{L}}$ into itself, injectively and semilinearly, hence onto itself, and $\mathfrak{M}_{\mathrm{L}}$ is a submodule. It contains [1], and $[g]=\operatorname{Sh}^l[f^{-l}g]$. When $s_{1j},t_{1j}\geq 1$ for every $j$, $f^{-l}\alpha^u\prod_j K_j^{-a_j}\Psi_j^{-b_j}$ is a Laurent polynomial as soon as $l$ is at least every $a_j$ and every $b_j$. The symmetries map Laurent insertions to Laurent insertions (Proposition 39.16(v)), so $e_G\mathfrak{M}_{\mathrm{L}}\subseteq\mathfrak{M}_{\mathrm{L}}\cap\mathfrak{M}^G$, and an element $x$ of the intersection is $e_Gx$; the intersection of two submodules is one (Theorem 39.12(i)), and $e_G$ of a Laurent insertion is invariant. □

**Proposition 39.18 (The relations of a box).** *As stated.*

*Proof.* (i) The terms of a letter identity of a symbol $(u;\mathbf{a},\mathbf{b})$ of the box for $K_j$, $a_j < \bar a_j$, are at levels at most the bounds and of wall degree at most that of the symbol (Proposition 39.16(iii)); likewise for $\Psi_j$. A term of the mark relation of a stem at the level $(\bar{\mathbf{a}}^- + \mathbf{e}_j, \bar{\mathbf{b}}^-)$ has the coefficient $s_{0j} + k\, s_{1j} + \bar a_j^-$; if $\bar a_j \geq 1$ its level is at most $(\bar{\mathbf{a}}, \bar{\mathbf{b}})$, and if $\bar a_j = 0$ then $K_j$ is not active and $\bar a_j^- = 0$, so the coefficient vanishes identically and the term is omitted; likewise for $\Psi_j$, and the wall degrees do not grow. The evaluations are those of Definition 39.15(iii), and Proposition 38.5(ii) gives the identity in $k$.

(ii) A letter identity of a symbol $x = (u;\mathbf{a},\mathbf{b})$ of the box for $K_j$, $a_j < \bar a_j$, makes $x$ congruent to a combination of symbols of the box at the level $(\mathbf{a}+\mathbf{e}_j, \mathbf{b})$, by (i). Repeating, every formal combination $y$ is congruent modulo the letter identities to a combination $y' = \sum_u c_u(u;\bar{\mathbf{a}},\bar{\mathbf{b}})$ of symbols at the top level, with distinct $u$. The letter identities evaluate to 0, so if $y$ evaluates to 0, so does $y'$, and $\prod_j K_j^{\bar a_j} \Psi_j^{\bar b_j}$ times its evaluation is the Laurent polynomial $\sum_u c_u \alpha^u$, which is 0 only when every $c_u$ is: $y' = 0$, and $y$ is a combination of letter identities.

(iii) Let $x = (u;\mathbf{a},\mathbf{b})$ be such a symbol and $\prod_j K_j^{\bar a_j^- - a_j} \Psi_j^{\bar b_j^- - b_j} = \sum_v \lambda_v \alpha^v$, $\lambda_v \in \Bbbk$. Then $x$ and $\sum_v \lambda_v\, s_v$, $s_v = (u+v;\bar{\mathbf{a}}^-, \bar{\mathbf{b}}^-)$, have one evaluation. Each $v$ is a sum of $\bar a_j^- - a_j$ indicators of bases and $\bar b_j^- - b_j$ of complements of bases of every table $j$, so $\kappa_U(s_v) \geq \kappa_U(x)$ (7.13) and the $s_v$ are stems. The formal combination $\mathrm{MR}_e(x) - \sum_v \lambda_v \mathrm{MR}_e(s_v)$ evaluates to $(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})$ of 0, and its terms lie in the box: those of $\mathrm{MR}_e(s_v)$ by (i), and those of $\mathrm{MR}_e(x)$ because their wall degree is at most $\deg_\kappa x \leq \mathfrak{t}$ and a term at the level $(\mathbf{a}+\mathbf{e}_j, \mathbf{b})$, with $a_j \leq \bar a_j^-$, either has a level within the bounds, when $\bar a_j \geq 1$, or the coefficient $s_{0j} + k\, s_{1j}$, identically 0, when $\bar a_j = 0$; likewise for $\Psi_j$. By (ii) it is a combination of letter identities, and $\mathrm{MR}_e(x) \in \mathcal{R}_{\mathfrak{B}}$.

(iv) $\phi$ permutes the letters keeping the bounds, so it maps the levels within the bounds onto themselves and the level of the stems onto itself, and it keeps the wall degree (Proposition 39.16(v)); so it permutes the symbols of the box and the stems. The image of a letter identity of the box evaluates to 0 and has its terms in the box, so it lies in the span of the letter identities by (ii). The image of the mark relation of a stem $s$ at $e$ evaluates to $\big((c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})s\big) \circ \phi$, which is $\sum_{e'} \mu_{e'}(c_{e'}^{(0)} + k\, c_{e'}^{(1)} + \partial_{y_{e'}})(s \circ \phi)$ for constants $\mu_{e'}$, $\phi$ being linear in the coordinates $y$ and keeping $\log(f^k \omega_0/dy)$ up to a constant (proof of Theorem 39.12(i)); so the image less $\sum_{e'} \mu_{e'} \mathrm{MR}_{e'}(\phi^* s)$, with $\phi^* s$ a stem, evaluates to 0 and lies in the span of the letter identities by (ii). So $\phi$ maps $\mathcal{R}_{\mathfrak{B}}$ into itself, and onto itself, $G$ being finite. Then $e_G \mathcal{R}_{\mathfrak{B}} \subseteq \mathcal{R}_{\mathfrak{B}} \cap \mathcal{F}_{\mathfrak{B}}^G$, and an element $x$ of the intersection is $e_G x$. □

**Proposition 39.19 (The closing relations have more solutions than the ray).** *As stated.*

*Proof.* With $\mathbf{s}_0 = \mathbf{t}_0 = \mathbf{0}$, $\omega_0/dy = \alpha^{\mathbf{m}_0}$ and $c_e^{(0)} = m_{0,e}$, and $c_e^{(1)} = \partial_{y_e} \log f = -\partial_{y_e} f^{-1}/f^{-1}$ (38.2(ii)), so

$$\big(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}\big)\big(\alpha^w f^{-1}\big) = \alpha^w \big((m_{0,e} + w_e)\, f^{-1} + (1-k)\, \partial_{y_e} f^{-1}\big),$$

which is the stated Laurent polynomial, $\partial_{y_e} \alpha^v = v_e \alpha^v$; it lies in $\mathcal{L}$, $\alpha^w f^{-1}$ being an insertion. (i) The evaluation at $z$ is multiplicative on Laurent polynomials and gives $z^w\big((m_{0,e}+w_e) f^{-1}(z) + (1-k)\,(\partial_{y_e} f^{-1})(z)\big) = 0$. (ii) It gives $q_0 + \sum_{i\geq 1} q_i\, f^{-1}(z)^i = q_0$. Applying $\mathrm{Sh}^{-h}$, with $\mathrm{Sh}^{-h}(r(k)\, m) = r(k-h)\, \mathrm{Sh}^{-h}(m)$, to the law of [1] gives $\sum_j p_j(k-h)[f^{j-h}] = 0$, the element of the statement, whose coefficient at $f^0$ is $p_h(k-h) \neq 0$. A linear form that vanishes on $\mathcal{L}$ and agrees with the evaluation at $z$ on the Laurent insertions would vanish on that element and take there the value $p_h(k-h)$. □

**Definition 39.20 (The reduction of a box).** As stated.

*Proof.* By Proposition 39.18(iv) each $\phi \in G$ permutes the symbols of the box, so $e_G g$ is the mean of the symbols of the orbit of $g$; orbit averages of distinct orbits have disjoint supports, and they span $\mathcal{F}_{\mathfrak{B}}^G$. The wall degree is constant on an orbit and $\phi^*$ keeps the Laurent symbols, as do the inverses (Proposition 39.16(v)). Gaussian elimination in the order of the ranking gives a basis $(r_p)$ of $\mathcal{R}_{\mathfrak{B}}^G$, indexed by the pivots, in which $r_p$ is led by $p$ with the coefficient 1 and involves no other pivot. For $x \in \mathcal{F}_{\mathfrak{B}}^G$, $x - \sum_p x_p\, r_p$, $x_p$ the coefficient of the average of $p$ in $x$, involves no pivot and is congruent to $x$; two such combinations differ by an element of $\mathcal{R}_{\mathfrak{B}}^G$ that involves no pivot, which is 0, since a nonzero element is led by a pivot. Every $r_p$ involves only orbits ranked at most $p$, which gives the last statement of (ii). No orbit ranks below that of 1: the other Laurent orbits have positive wall degree (Proposition 39.16(ii)). If it were a pivot, $r_1 = 1$ would lie in $\mathcal{R}_{\mathfrak{B}}^G \subseteq \mathcal{R}_{\mathfrak{B}}$ (Proposition 39.18(iv)), and 1 would evaluate into $\mathcal{L}$ (Proposition 39.18(i)), $[1] = 0$; but the values vanish on $\mathcal{L}$ (Proposition 38.5(ii)), and $\langle 1\rangle_k \neq 0$ for all large $k$ (Proposition 38.9). The monomials of $f^{-1}b_i$ have wall degree at most $\mathfrak{s}+1 \leq \mathfrak{t}$ (Proposition 39.16(ii)), and $f^{-1}$ has its coefficients in $\Bbbk$, so $f^{-1}b_i \in \mathcal{F}_{\mathfrak{B}}$; on Laurent combinations the formal and the evaluated products agree, distinct Laurent monomials being independent functions, and $f \circ \phi = f$, so $f^{-1}b_i$ is invariant. □

**Theorem 39.21 (The box gives a spanning set, the one-step matrix and a certificate).** *As stated.*

*Proof.* (i) $f^{-1}b_i - \sum_l \mathbf{B}_{il} b_l$ lies in $\mathcal{R}_{\mathfrak{B}}^G \subseteq \mathcal{R}_{\mathfrak{B}}$, so its evaluation lies in $\mathcal{L}$ (Proposition 39.18(i), (iv)), and $[f^{-1}b_i] = \sum_l \mathbf{B}_{il}[b_l]$. With $\mathfrak{N}$ the span of the $[b_l]$, $\mathrm{Sh}^{-1}\mathfrak{N} \subseteq \mathfrak{N}$; $\mathrm{Sh}^{-1}$ is injective and semilinear, so it keeps dimensions, $\mathrm{Sh}^{-1}\mathfrak{N} = \mathfrak{N}$, and $\mathfrak{N}$ is a submodule; it contains $[b_1] = [1]$. The $b_l$ are invariant insertions, so $\mathfrak{N} \subseteq \mathfrak{M}^G$ (Theorem 39.12(i)). An invariant combination of Laurent symbols of wall degree at most $\mathfrak{s}$ is a combination of averages of Laurent orbits of wall degree at most $\mathfrak{s}$, and by the ranking every orbit ranked at most the highest of these is a Laurent orbit of wall degree at most $\mathfrak{s}$; so its reduction is a combination of the $b_l$ (Definition 39.20(ii)), congruent to it modulo $\mathcal{R}_{\mathfrak{B}}^G$, and it has a class in $\mathfrak{N}$. When the $[b_l]$ are independent they are a basis of $\mathfrak{N}$, and the matrix of $\mathrm{Sh}^{-1}$ on a basis is unique.

(ii) As an element of $\mathcal{R}_{\mathfrak{B}}$, $f^{-1}b_i - \sum_l \mathbf{B}_{il} b_l$ is a combination of letter identities and of mark relations $\mathrm{MR}_e(s)$ of stems $s$, with coefficients $\nu_{i,e,s} \in \Bbbk(k)$. Evaluated, the letter identities give 0 and $\mathrm{MR}_e(s)$ gives $(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})s$; put $G_{i,e} = \sum_s \nu_{i,e,s}\, s$.

(iii) $V[f^{-1}b_i] = V(\mathrm{Sh}^{-1}[b_i]) = \varsigma^{-1}V[b_i]$, and by (i) and the $\mathbb{C}(k)$-linearity of $V$ it is $\sum_l \mathbf{B}_{il} V[b_l]$. The certificate is an element of $\mathcal{L}$, so its value vanishes identically, and $\langle f^{-1}b_i\rangle_k = \langle b_i\rangle_{k-1}$ (Proposition 38.5(ii)). At a real $k$ as stated, the stems, the $b_l$ and the monomials of $f^{-1}b_i$ have wall degree at most $\mathfrak{t}$, so their values converge absolutely (Proposition 39.16(i)), and Proposition 38.5(i) gives the value 0 for every term $(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e})s$.

(iv) Write $\rho_e^{(k)} = c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}$; by the identity in the proof of Definition 39.1, $f^{-1}\rho_e^{(k-1)}(x) = \rho_e^{(k)}(f^{-1}x)$ and $f^h \rho_e^{(k)}(x) = \rho_e^{(k-h)}(f^h x)$. With the $G_{l,e}$ of (ii), put $G_e^{(0)} = 0$ and $G_e^{(i)}(k) = \sum_l w_{i-1,l}(k-1)\, G_{l,e}(k) + f^{-1}G_e^{(i-1)}(k-1)$ for $i \geq 1$, and $R_i(k) = \sum_e \rho_e^{(k)}\big(G_e^{(i)}(k)\big) \in \mathcal{L}$. Then $f^{-i} = \sum_l w_{i,l}(k)\, b_l + R_i(k)$: for $i = 0$ this is $1 = b_1$, and multiplying the identity for $i-1$ at the strength $k-1$ by $f^{-1}$ and inserting $f^{-1}b_l = \sum_{l'} \mathbf{B}_{ll'}(k) b_{l'} + \sum_e \rho_e^{(k)}(G_{l,e}(k))$ gives it for $i$. So $\sum_i q_i(k) f^{-i} = \sum_i q_i(k) R_i(k)$, and multiplying by $f^h$ and replacing $k$ by $k+h$ gives

$$\sum_{j=0}^{h} p_j(k)\, f^j = \sum_e \rho_e^{(k)}\Big(\sum_{i=0}^{h} q_i(k+h)\, f^h G_e^{(i)}(k+h)\Big),$$

a certificate. Proposition 39.6(ii) gives the rest. □

**Theorem 39.22 (The search stops).** *As stated.*

*Proof.* Let $\mathcal{L}_\Bbbk$ be the $\Bbbk(k)$-span of the elements $(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})x$, $x \in \Bbbk[X_F]$, and $M_\Bbbk = (\Bbbk[X_F] \otimes \Bbbk(k))/\mathcal{L}_\Bbbk$. As in the proof of Theorem 38.31, $M_\Bbbk \otimes_{\Bbbk(k)} \mathbb{C}(k) = \mathfrak{M}$, so $M_\Bbbk$ has the finite dimension of the ray and embeds in $\mathfrak{M}$, and a $\Bbbk(k)$-subspace $S$ of $M_\Bbbk$ has $(S \otimes \mathbb{C}(k)) \cap M_\Bbbk = S$, a basis of $S$ extending to one of $M_\Bbbk$. The symmetries act on $M_\Bbbk$ as on $\mathfrak{M}$, the weights being in $\Bbbk$. Let $\mathfrak{N}_\mathfrak{s} \subseteq M_\Bbbk$ be the span of the classes of the averages of the Laurent orbits of wall degree at most $\mathfrak{s}$. The $\mathfrak{N}_\mathfrak{s}$ increase with $\mathfrak{s}$ in a space of finite dimension, so they are constant from some $\mathfrak{s}_0$ on, and their union is the span of the classes of the invariant Laurent insertions, whose $\mathbb{C}(k)$-span in $\mathfrak{M}$ is $e_G\mathfrak{M}_\mathrm{L}$ (Lemma 39.17).

The averages $\bar g$ of the Laurent orbits of wall degree at most $\mathfrak{s}_0 + 1$ are finitely many (Proposition 39.16(iv)), and each has its class in $\mathfrak{N}_{\mathfrak{s}_0}$:

$$\bar g - \sum_{\bar h} \lambda_{\bar g,\bar h}\,\bar h = \sum_{e,i} \nu_{\bar g,e,i}\,(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})G_{\bar g,i},$$

with $\bar h$ running over the averages of the Laurent orbits of wall degree at most $\mathfrak{s}_0$, $\lambda, \nu \in \Bbbk(k)$, and finitely many symbols $G_{\bar g,i}$, the evaluations of the symbols spanning $\Bbbk[X_F]$ over $\Bbbk$. Take an integer $\mathfrak{t} \geq \mathfrak{s}_0 + 1$ at least the wall degree of every $G_{\bar g,i}$, and the level bounds $\bar\ell(L) = 1 + \max \ell(L)$ for every letter $L$, the maximum over the levels $\ell$ of the symbols $\phi^* G_{\bar g,i}$, $\phi \in G$, at $L$. These bounds are at least 1, and kept by $G$, the set of the $\phi^* G_{\bar g,i}$ being kept by $G$; every $G_{\bar g,i}$ is at a level at most $(\bar{\mathbf{a}}^-, \bar{\mathbf{b}}^-)$, here $\bar a_j^- = \bar a_j - 1$ and $\bar b_j^- = \bar b_j - 1$. By Proposition 39.18(iii) every $\mathrm{MR}_e(G_{\bar g,i})$ lies in $\mathcal{R}_\mathfrak{B}$, and the formal combination $\bar g - \sum \lambda_{\bar g,\bar h}\bar h - \sum \nu_{\bar g,e,i}\mathrm{MR}_e(G_{\bar g,i})$ has its terms in the box (Proposition 39.16(iii)) and evaluates to 0, so it lies in $\mathcal{R}_\mathfrak{B}$ (Proposition 39.18(ii)). Hence $\bar g - \sum \lambda_{\bar g,\bar h}\bar h$, which is invariant, lies in $\mathcal{R}_\mathfrak{B} \cap \mathcal{F}_\mathfrak{B}^G = \mathcal{R}_\mathfrak{B}^G$ (Proposition 39.18(iv)).

Take $\mathfrak{s} = \mathfrak{s}_0 < \mathfrak{t}$. A target $f^{-1}b_i$ is an invariant combination of Laurent symbols of wall degree at most $\mathfrak{s}_0 + 1$ (Proposition 39.16(ii)), so a combination of the $\bar g$, and it is congruent modulo $\mathcal{R}_\mathfrak{B}^G$ to a combination of averages of Laurent orbits of wall degree at most $\mathfrak{s}_0$. The reduction of that combination, which is the reduction of the target, involves only free Laurent orbits of wall degree at most $\mathfrak{s}_0$ (Definition 39.20(ii)): the spanning set. So the box is accepted at $\mathfrak{s}_0$, and its spanning set spans $\mathfrak{N}_{\mathfrak{s}_0} \otimes \mathbb{C}(k) = e_G\mathfrak{M}_\mathrm{L}$, by Theorem 39.21(i); this is $\mathfrak{M}^G$ when every letter exponent of the direction is positive (Lemma 39.17). A search through all the quadruples reaches this one, and acceptance is decided by finitely many operations in $\Bbbk(k)$. □

**Definition 39.23 (Sources; regular points; the normal density).** As stated.

*Proof.* $Z$ is irreducible, so its smooth locus is connected; there $d\log f$ vanishes along $Z$, so $f$ is constant on it, and on $Z$ by continuity. Differentiating $d\log f = 0$ along $Z$ shows that the tangent space of $Z$ at a smooth point lies in the kernel of the Hessian, whose rank is then at most $n - 1 - d$; the conditions defining a regular point are open, so the regular points form an open set, dense when it is nonempty, $Z$ being irreducible. For another choice of coordinates $(u', w')$ with $Z = \{w' = 0\}$, at $w = 0$ one has $u' = U(u)$ and $w' = J(u)w + O(|w|^2)$, the Jacobian of $(u, w) \mapsto (u', w')$ being block triangular there, so $du' \wedge dw' = \det(\partial U/\partial u)\det J\, du \wedge dw$ at $w = 0$, $A = J^\mathsf{T} A' J$, and $du' = \det(\partial U/\partial u)\, du$ on $Z$. So $\det A'^{-1/2} = \pm \det J \det A^{-1/2}$ and $\frac{\omega_0}{du'\wedge dw'} = \big(\det(\partial U/\partial u)\det J\big)^{-1}\frac{\omega_0}{du\wedge dw}$, and the product of these with $du'$ is $\pm\det A^{-1/2}\frac{\omega_0}{du\wedge dw}\,du$: $\nu$ is defined up to sign. For $d = 0$ a point is regular exactly when the Hessian has rank $n - 1$, and an isolated point of the critical locus lies on no other component. □

**Theorem 39.24 (A closed cycle on a source carries a solution).** *As stated.*

*Proof.* (i) In the coordinates $y$ the torus is, near each point, an open set of the vector space $\mathbb{C}^{\mathcal{E}}/\mathbb{C}\mathbf{1}$, in which $z+v$ has a meaning for small $v$. At $z \in Z^\circ$ call a real subspace $V$ of dimension $m$ of the tangent space a *descent subspace* when the real part of the Hessian of $-\log f$ is positive definite on it; then $V$ meets the tangent space of $Z$ only at 0, the Hessian vanishing there. The projection to the normal space $\mathbb{C}^m$, in coordinates in which the normal form is $\sum_i w_i^2$, maps $V$ onto a real subspace $V'$ with $V' \cap i\mathbb{R}^m = 0$, the form being negative on $i\mathbb{R}^m$; so $V' = \{x + iBx : x \in \mathbb{R}^m\}$ for a real matrix $B$, and the form has the real part $|x|^2 - |Bx|^2$ on it, positive definite exactly when $\|B\| < 1$. So the descent subspaces at $z$ are the graphs over such $V'$ of real linear maps into the tangent space of $Z$: a contractible set, whose oriented members form two contractible components, one for each sheet of $\nu$. Over the compact $\gamma \subset \tilde{Z}^\circ$ there is therefore a continuous oriented choice $z \mapsto V_z$, and by Taylor's formula and compactness there are $r, c > 0$ with $|f(z+v)| \le |f_Z| e^{-c|v|^2}$ for $v \in V_z$, $|v| \le r$. Put $D_z = z + \{v \in V_z : |v| \le r\}$ and $N(\gamma) = \bigcup_{z\in\gamma} D_z$; as $\partial\gamma = 0$, $\partial N(\gamma)$ is the union of the boundaries of the discs, on which $|f| \le |f_Z| e^{-cr^2}$. For two choices over $\gamma$, or for $\gamma - \gamma' = \partial\Gamma$, the same construction over $\gamma \times [0,1]$ or over $\Gamma$, with discs of the same kind, gives a chain of dimension $n$ whose boundary is the difference of the two chains together with a union of disc boundaries; the discs need not be affine, any smooth family of embedded discs tangent at $z$ to a descent subspace serving as well for $r$ small.

(ii) The integral over $N(\gamma)$ is the integral over $\gamma \times \{|v| \le r\}$ of the pull-back by $(z,v) \mapsto z+v$, $v \in V_z$. At each $z$, Laplace's method in $v$, uniform in $z$ on the compact $\gamma$, expands the integral over $D_z$ in powers of $1/k$ after the factor $f_Z^k k^{-m/2}$, with coefficients that are continuous $d$-forms along $\gamma$; integrating them over $\gamma$ gives the $c_i(\gamma; g)$. The leading coefficient is $(2\pi)^{m/2} \det\big[H_z(\mathsf{f}_a, \mathsf{f}_b)\big]^{-1/2} g\, \omega_0(\,\cdot\,, \mathsf{f}_1, \dots, \mathsf{f}_m)$, $H_z$ the Hessian of $-\log f$ and $\mathsf{f}_1, \dots, \mathsf{f}_m$ a basis of $V_z$, the square root fixed by the orientation; it does not change under a complex linear change of $\mathsf{f}$, and $V_z$ spans over $\mathbb{C}$ a complement of the tangent space of $Z$, which projects onto the normal space as the $w$-space does in coordinates $(u, w)$, so it is $(2\pi)^{m/2} g\nu$ on the sheet. Where holomorphic normal slices $\{u = \text{const}\}$ cover $Z^\circ$, take every $D_z$ in the slice through $z$, a real descent disc there, which (i) allows. The integral over the slice at $u$ has the phase $-\log f = -\log f_Z + \frac{1}{2} w^{\mathsf{T}} A(u) w + O(|w|^3)$ and the amplitude $g\,\omega_0/(du \wedge dw)$, and Laplace's method with the holomorphic parameter $u$ gives $f_Z^k k^{-m/2} \sum_i k^{-i} a_i(u; g)$, where $a_i(u; g)$ is $du$ times $(2\pi)^{m/2} \det A(u)^{-1/2}$ times a polynomial in finitely many $w$-derivatives at $w = 0$ of the amplitude and of the phase and in the entries of $A(u)^{-1}$. These coefficients are holomorphic in $u$, so each $a_i(\cdot\,; g)$ is a holomorphic $d$-form, $a_0(\cdot\,; g) = (2\pi)^{m/2} g\,\nu$, and $c_i(\gamma; g) = \int_\gamma a_i(\cdot\,; g)$.

(iii) By 38.2(ii), $(c_e^{(0)} + k c_e^{(1)} + \partial_{y_e}) g \cdot f^k \omega_0/dy = \partial_{y_e}\big(g f^k \omega_0/dy\big)$, so $(c_e^{(0)} + k c_e^{(1)} + \partial_{y_e}) g \cdot f^k \omega_0 = d\big(\iota_{\partial_{y_e}}(g f^k \omega_0)\big)$, the form being of top degree and $\partial_{y_e}$ a constant field. By Stokes its integral over $N(\gamma)$ is an integral over $\partial N(\gamma)$, of modulus at most a constant times $|f_Z|^k e^{-kcr^2}$; so the expansion of every relation vanishes to all orders, and $A_\gamma$, extended $\mathbb{C}(k)$-linearly, vanishes on $\mathcal{L}$. The integral of $fg\, f^k \omega_0$ is that of $g\, f^{k+1} \omega_0$, so $A_\gamma(fg)(k) = A_\gamma(g)(k+1)$ after re-expansion, and $A_\gamma$ is a solution of type $(f_Z, -\frac{m}{2} + \mathbb{Z})$. The forms $g f^k \omega_0$ are closed, so by Stokes on the chains of (i) homologous cycles give integrals differing by terms of modulus at most a constant times $|f_Z|^k e^{-kcr^2}$, hence the same expansion. The deck involution reverses the orientation of every disc. For an origin that is not integral the same construction runs with a branch of $\omega_0$ carried along the chains, that is, for cycles with coefficients in the local system of the branches of $\nu$.

(iv) A relation $\sum_a c_a \hat{A}_{\gamma_a} = 0$ with $c_a \in \mathbb{C}((1/k))$ not all zero, scaled so that the least order of the $c_a$ is 0 with constant terms $c_a^0$ not all zero, gives at the leading order, applied to $g_b$, $(2\pi)^{m/2} \sum_a c_a^0 \int_{\gamma_a} g_b \nu = 0$ for every $b$, so $c^0 = 0$ by the determinant: a contradiction. The relations

over $\mathbb{C}$ are among them. □

**Corollary 39.25 (A curve of native gauges: residues and periods).** *As stated.*

*Proof.* The first homology of $\tilde{Z}^\circ$, a compact curve less finitely many points, is generated by small loops around the punctures and by the cycles of $\bar{Z}$, and $A_\gamma$ depends only on the class of $\gamma$ (Theorem 39.24(iii)). For the family $\alpha = \alpha_0 b^{\mathsf{v}} e^w$ the coordinates $(\log b, w)$ are linear in $y$, so $dy$ is a constant multiple of $d\log b \wedge dw$. With an integral origin $f$, $g$ and $\omega_0/dy$ are rational functions of $\alpha$, and so are their derivatives in $y$, $\partial_{y_e} = \alpha_e \partial_{\alpha_e}$; their $w$-derivatives at $w = 0$ are constant combinations of these at $\alpha = \alpha_0 b^{\mathsf{v}}$, so rational in $b$, and so are the entries of $A(b)$ and $A(b)^{-1}$; the slices $\{b = \text{const}\}$ cover $Z^\circ$, and the Laplace coefficients of Theorem 39.24(ii) are polynomials in these times $\det A(b)^{-1/2}$, so $a_i = r_i(b; g) \det A(b)^{-1/2}\, db/b$ with $r_i$ rational. On the curve $Y^2 = \det A(b)$ these are meromorphic forms, and the integral over a small loop around a puncture is $2\pi i$ times the residue there. □

**Theorem 39.26 (A native gauge of corank at most one).** *As stated.*

*Proof.* The Hessian of $\log f$ at $\alpha_c$ has rank at least $n-2$. By the splitting lemma, the holomorphic Morse lemma in the directions where the Hessian is nondegenerate with the remaining coordinate as a parameter, there are coordinates $(X, w)$ with $\log f = \log f(\alpha_c) - \frac{1}{2}\sum_i w_i^2 + R(X)$, $R(0) = R'(0) = 0$; when the Hessian is nondegenerate, $X$ is one of the Morse coordinates and $R = -\frac{1}{2}X^2$. $R$ does not vanish identically, as otherwise $\{w = 0\}$ would be a curve of native gauges through $\alpha_c$. With $\mu + 1$ its order at 0, $R = cX^{\mu+1}(1 + O(X))$ with $c \neq 0$, and $X(1 + O(X))^{1/(\mu+1)}$ is a coordinate in which $R = cX^{\mu+1}$; $\mu = 1$ exactly when the Hessian is nondegenerate, and otherwise $R''(0) = 0$ and $\partial_X$ spans the kernel.

(i) Let $\varrho_0, \dots, \varrho_\mu$ be the directions with $c\,\varrho^{\mu+1} < 0$, $\varrho_b = \varrho_0 \zeta^b$, $\zeta = e^{2\pi i/(\mu+1)}$; around them lie the sectors on which $\mathrm{Re}(cX^{\mu+1}) < 0$. For the path coming in along $\varrho_0\,\mathbb{R}_{>0}$ and going out along $\varrho_b\,\mathbb{R}_{>0}$,

$$\int X^j e^{cX^{\mu+1}} dX = \big(\varrho_b^{\,j+1} - \varrho_0^{\,j+1}\big) \frac{\Gamma\big(\frac{j+1}{\mu+1}\big)}{(\mu+1)|c|^{(j+1)/(\mu+1)}},$$

and these $\mu$ paths generate the homology of the plane relative to the part where $\mathrm{Re}(cX^{\mu+1}) < -1$. The $\mu \times \mu$ matrix of the numbers $\zeta^{b(j+1)} - 1$, $1 \le b \le \mu$, $0 \le j < \mu$, is invertible: its columns, extended by the entry 0 at $b = 0$, are the differences of the columns $l = j+1$ and $l = 0$ of the invertible matrix $(\zeta^{bl})_{0 \le b,l \le \mu}$. So there are complex combinations $\Gamma_i$ of the paths with $\int_{\Gamma_i} X^j e^{cX^{\mu+1}} dX = \delta_{ij}$, $0 \le j < \mu$. Integration by parts gives $\int_\Gamma X^{j+\mu+1} e^{cX^{\mu+1}} dX = -\frac{j+1}{(\mu+1)c} \int_\Gamma X^j e^{cX^{\mu+1}} dX$, and $\int_\Gamma X^\mu e^{cX^{\mu+1}} dX = 0$, the integrand being a derivative; so $\int_{\Gamma_i} X^j e^{cX^{\mu+1}} dX = 0$ unless $j \equiv i \bmod \mu+1$. On the chain $C_i = D \times \Gamma_i$, $D$ the real ball $|w| \le \varepsilon$ and $\Gamma_i$ cut off at $|X| = \varepsilon$, the boundary lies where $|f| \le |f(\alpha_c)| e^{-\varepsilon'}$, and the relations and the shift hold up to exponentially smaller terms as in Theorem 39.24(iii). Expand the amplitude $\mathsf{a} = g\,\omega_0/(dX \wedge dw) = \sum \mathsf{a}_{j,\beta} X^j w^\beta$; after $X \mapsto k^{-1/(\mu+1)} X$ and $w \mapsto k^{-1/2} w$ the term $(j, \beta)$ contributes $k^{-(n-2)/2 - |\beta|/2 - (j+1)/(\mu+1)}$ times a Gaussian moment, zero for odd $|\beta|$, times $\int_{\Gamma_i} X^j e^{cX^{\mu+1}} dX$, and the usual bounds of Laplace's method make the sum an asymptotic expansion. Only $j \equiv i \bmod \mu+1$ survive, the exponents are $-\frac{n-2}{2} - \frac{i+1}{\mu+1}$ less integers, and the leading one, $j = i$ and $\beta = 0$, has the coefficient $(2\pi)^{(n-2)/2} \mathsf{a}_{i,0}$. Since $0 < \frac{i+1}{\mu+1} < 1$, distinct $i$ give distinct types; each $A_{c,i}$ is nonzero, an insertion with $\mathsf{a}_{i,0} \neq 0$ existing, and Lemma 39.3 gives the independence. For $\mu = 1$, $R = -\frac{1}{2}X^2$, the normalization $\int_{\Gamma_0} e^{-X^2/2} dX = 1$ makes $\Gamma_0$ the real line divided by $(2\pi)^{1/2}$, and $C_0$ is a steepest-descent disc divided by $(2\pi)^{1/2}$: $A_{c,0} = (2\pi)^{-1/2} A_c$.

(ii) In the coordinates $y$, $\phi$ is $y \mapsto M_\phi y$, linear, of finite order, fixing $\log \alpha_c$; it keeps $\log f$, so its Hessian and the kernel: $M_\phi \xi = \zeta_\phi \xi$ for $\xi$ spanning the kernel, with $\zeta_\phi$ a root of unity, and $M_\phi$ maps an invariant complement $N$ of the kernel onto itself. In linear coordinates $(X', w')$ along $\xi$ and $N$ the curve $\{\partial_{w'} \log f = 0\}$ through $\alpha_c$ is smooth and parametrized by $X'$, the Hessian being nondegenerate on $N$; $\phi$ maps it onto itself by $X' \mapsto \zeta_\phi X'$, and $\log f$ on it is $\log f(\alpha_c)$ plus a function of order $\mu+1$ in $X'$ with nonzero leading coefficient, the reduced function of the splitting lemma. Its invariance gives $\zeta_\phi^{\mu+1} = 1$.

As in Theorem 39.12(iii), $\phi^* \omega_0 = c_\phi \, \omega_0$ near $\alpha_c$ for a constant $c_\phi \neq 0$, so $A_{c,i}(g \circ \phi)$, the expansion over $C_i$ of $(g \circ \phi) f^k \omega_0 = c_\phi^{-1} \phi^*(g f^k \omega_0)$, is $c_\phi^{-1}$ times the expansion over $\phi(C_i)$, a chain near $\alpha_c$ with its boundary where $|f|$ is smaller. The relative homology near $\alpha_c$ has rank $\mu$ over $\mathbb{C}$ with the basis $C_0, \dots, C_{\mu-1}$, so $A_{c,i} \circ \phi^* = \sum_{i'} \kappa_{i'} A_{c,i'}$. By (i) applied to the insertion $g \circ \phi$, $A_{c,i} \circ \phi^*$ has the type of $A_{c,i}$, and the $A_{c,i'}$ have distinct types, so $A_{c,i} \circ \phi^* = \kappa_i A_{c,i}$. The expansions depend only on the Taylor series of the insertion at $\alpha_c$, and insertions realize every finite jet there, so the identity holds on formal power series at $\alpha_c$. Let $\ell$ be the linear function of $y - \log \alpha_c$ that vanishes on $N$ with $\ell(\xi) = 1$; then $\ell \circ \phi = \zeta_\phi \ell$. In the coordinates of (i), $\ell(X, 0) = \ell_1 X + O(X^2)$ with $\ell_1 \neq 0$, since $\partial_X$ is along $\xi$ at $\alpha_c$, so the coefficient of $X^i$ in $\ell^i \, \omega_0/(dX \wedge dw)$ at $w = 0$ is $\ell_1^i \, (\omega_0/(dX \wedge dw))(\alpha_c) \neq 0$, and $A_{c,i}(\ell^i) \neq 0$. Then $\kappa_i A_{c,i}(\ell^i) = A_{c,i}(\ell^i \circ \phi) = \zeta_\phi^i A_{c,i}(\ell^i)$ gives $\kappa_i = \zeta_\phi^i$. □

**Corollary 39.27 (Signs at a centre).** *As stated.*

*Proof.* The centre ray has origin 0 and direction $(\sigma \mathbf{1}; \boldsymbol{\sigma}, \boldsymbol{\sigma})$, so the Cremona is a symmetry by Proposition 39.11(ii). It fixes $\eta = \eta^{-1}$, and in the chart $\alpha = \eta e^y$ it is $y \mapsto -y$. (i) is Theorem 39.12(iii). In (ii), $\zeta = -1$, so $\mu + 1$ is even, and corank one makes $\mu \geq 2$, so $\mu \geq 3$; Theorem 39.26(ii) gives $A_{\eta,i} \circ \mathrm{cr}^* = (-1)^i A_{\eta,i}$. For a Cremona-invariant insertion $g$, $A_{\eta,i}(g) = A_{\eta,i}(g \circ \mathrm{cr}) = (-1)^i A_{\eta,i}(g)$ vanishes for odd $i$; for even $i$ and an insertion $g'$ with $A_{\eta,i}(g') \neq 0$, $\frac{1}{2}(g' + g' \circ \mathrm{cr})$ is invariant with the same value. The exponents are those of Theorem 39.26(i) at $\mu = 3$. □

**Proposition 39.28 (Symmetries on the cycles of a source).** *As stated.*

*Proof.* $\phi$ is an automorphism of $X_F$ keeping $f$, so it maps the critical locus onto itself, sources onto sources, regular points onto regular points, and descent discs at $z$ onto descent discs at $\phi(z)$, with their orientations; the sheet of $\nu$ at $\phi(z)$ that matches the image orientation defines $\tilde{\phi}$, and $\phi(N(\gamma)) = N(\tilde{\phi}_* \gamma)$ as oriented chains. With $f \circ \phi = f$ and $\varepsilon_\phi^2 = 1$, $(g \circ \phi) f^k \omega_0 = \varepsilon_\phi \, \phi^*(g f^k \omega_0)$, so $\int_{N(\gamma)} (g \circ \phi) f^k \omega_0 = \varepsilon_\phi \int_{N(\tilde{\phi}_* \gamma)} g f^k \omega_0$. On $\mathfrak{M}^G$, $A_\gamma = A_\gamma \circ e_G = |G|^{-1} \sum_\phi \varepsilon_\phi A_{\tilde{\phi}_* \gamma}$, which is linear in the cycle. Grouping $\phi \in G$ by the cosets $\psi G_Z$, with $\varepsilon_{\psi\chi} = \varepsilon_\psi \varepsilon_\chi$ and $\widetilde{\psi\chi}_* = \tilde{\psi}_* \tilde{\chi}_*$, each coset contributes $\varepsilon_\psi A_{\tilde{\psi}_* z}$ with $z = \sum_{\chi \in G_Z} \varepsilon_\chi \tilde{\chi}_* \gamma$, zero when $[z] = 0$. □

**Theorem 39.29 (What the invariant module sees at the high end).** *As stated.*

*Proof.* For $\phi \in G$ and $\psi \in G_c$, $A_{c,i} \circ (\phi\psi)^* = A_{c,i} \circ \psi^* \circ \phi^* = \zeta_\psi^i A_{c,i} \circ \phi^*$ (Theorem 39.26(ii), and Theorem 39.12(iii) for $\mu = 1$), so

$$A_{c,i} \circ e_G = |G|^{-1} \sum_{\phi G_c} \Big( \sum_{\psi \in G_c} \zeta_\psi^i \Big) A_{c,i} \circ \phi^*.$$

When $\psi \mapsto \zeta_\psi^i$ is a nontrivial character of $G_c$ its sum over $G_c$ vanishes and $A_{c,i}$ vanishes on $\mathfrak{M}^G = e_G \mathfrak{M}$. Otherwise $A_{c,i} \circ e_G$ is $|G_c|/|G|$ times $A_{O,i} = \sum_{\phi G_c} A_{c,i} \circ \phi^*$, which is invariant, and

each $A_{c,i}\circ\phi^*$ is a constant multiple of an expansion over the chain $\phi(C_i)$ near $\phi(\alpha_c)$. The isolated native gauges are finitely many. Take an insertion with a Taylor jet at $\alpha_c$ that makes the leading coefficient of $A_{c,i}$ nonzero (Theorem 39.26(i)) and that vanishes to a high order at every other isolated native gauge; on it $A_{O,i}$ has that leading coefficient, so $A_{O,i}$ does not vanish on $\mathfrak{M}^G$, and the same insertions show that the $A_{O,i}$ of one type, over the orbits and the classes, are linearly independent over $\mathbb{C}$ on $\mathfrak{M}^G$, the solutions of one type at one point being a single $A_{c,i}$. Lemma 39.3 gives the bound. □

**Definition 39.30 (Closings).** As stated.

*Proof.* $\mathbb{C}[X_F]$ is spanned by the insertions $g=\alpha^u\prod_j K_j^{-a_j}\Psi_j^{-b_j}$ with $a_j,b_j\in\mathbb{N}$, products of the generators of 38.1, and for them $g\,f^{-j}\omega_0/dy=\alpha^{u-j\mathbf{m}_1+\mathbf{m}_0}\prod_l K_l^{js_{1l}-s_{0l}-a_l}\Psi_l^{jt_{1l}-t_{0l}-b_l}$, a Laurent polynomial when every exponent of a letter is nonnegative, which holds for large $j$ as $s_{1l},t_{1l}\ge 1$; a finite sum takes the largest of the bounds. The function is homogeneous of degree 0, so its monomials $\alpha^u$ have $u_{\mathcal{E}}=0$, and on the chart $\alpha_{e_0}=1$ its constant term, the coefficient of the monomials with $u_e=0$ for $e\neq e_0$, is the coefficient of $\alpha^0$, whatever $e_0$. On the chart $dy=\prod_{e\neq e_0}d\alpha_e/\alpha_e$, which gives the integral over the torus. □

**Proposition 39.31 (One form on the orthant and on the torus).** *As stated.*

*Proof.* On the orthant $f>0$ and $1/(f^{-1}-q)=f/(1-qf)$, so the first integral is the generating function of Theorem 38.14(ii) with $k_0=1$, which the origin 0 allows. The Laurent polynomial $f^{-1}=\alpha^{-\mathbf{m}_1}\prod_j K_j^{s_{1j}}\Psi_j^{t_{1j}}$ has nonnegative coefficients, so on the torus through $\alpha^\star$, $|f^{-1}(\alpha)|\le f^{-1}(\alpha^\star)=A$. For $|q|>A$ the series $1/(f^{-1}-q)=-\sum_{j\ge0}f^{-j}q^{-j-1}$ converges uniformly on that torus, and term by term $(2\pi i)^{-(n-1)}\oint f^{-j}\,dy$ is the constant term, $\oint\alpha^v\,dy=(2\pi i)^{n-1}$ for $v=0$ and 0 for the other monomials of degree 0. For every letter exponent positive this is $\mathsf{C}_j(1)$ (Definition 39.30, the origin form being $dy$). □

**Proposition 39.32 (The constant terms of a centre ray and its signs).** *As stated.*

*Proof.* By Definition 38.13, $f^{-1}=\prod_l\big(K_l(\alpha)K_l(\alpha^{-1})\big)^{\sigma_l}$, a Laurent polynomial with nonnegative coefficients, homogeneous of degree 0; at a sign $\eta^{-1}=\eta$, so $f^{-1}(\eta)=\prod_l K_l(\eta)^{2\sigma_l}$. On the chart $\alpha_{e_0}=1$ a monomial $\alpha^v$, $v_{\mathcal{E}}=0$, takes at $\eta$ the value $\prod_{e\neq e_0}\eta_e^{v_e}$, whose mean over the signs is 1 when every $v_e$, $e\neq e_0$, is even, and then $v_{e_0}=-\sum_{e\neq e_0}v_e$ is even too, and 0 otherwise. The constant term is the coefficient at $v=0$, one of the nonnegative coefficients summed. □

**Theorem 39.33 (The closings are a solution).** *As stated.*

*Proof.* By 38.2(ii), $c_e^{(0)}-j\,c_e^{(1)}=\partial_{y_e}\log\big(f^{-j}\omega_0/dy\big)$, so $\big(c_e^{(0)}-jc_e^{(1)}+\partial_{y_e}\big)g\cdot f^{-j}\omega_0/dy=\partial_{y_e}\big(gf^{-j}\omega_0/dy\big)$. For $j\ge j(g)$ the right side is $\partial_{y_e}$ of a Laurent polynomial, which multiplies each monomial $\alpha^u$ by $u_e$ and so has no constant term. $\mathsf{C}_j(fg)=\mathrm{CT}\,\big(gf^{-(j-1)}\omega_0/dy\big)=\mathsf{C}_{j-1}(g)$ for $j-1\ge j(g)$. An element of $\mathcal{L}$ is a finite sum $\sum r_{e,l}(k)\big(c_e^{(0)}+kc_e^{(1)}+\partial_{y_e}\big)h_{e,l}$, and $V_{\mathsf{C}}$ of it at an integer $k=-j$ off the poles of the $r_{e,l}$ with $j>j(h_{e,l})$ for all $e,l$, where the closings of $c_e^{(1)}h_{e,l}$ and of $(c_e^{(0)}+\partial_{y_e})h_{e,l}$ are defined, $c_e^{(0)}$, $c_e^{(1)}$ and $\partial_{y_e}$ costing at most one power of each letter and $f^{-1}$ supplying at least one, is $\sum r_{e,l}(-j)\,\mathsf{C}_j\big((c_e^{(0)}-jc_e^{(1)}+\partial_{y_e})h_{e,l}\big)=0$; so $V_{\mathsf{C}}$ is well defined on $\mathfrak{M}$ as a germ at $-\infty$. It is $\mathbb{C}(k)$-linear, and $V_{\mathsf{C}}(\mathrm{Sh}(g\otimes r))(k)=r(k+1)\,\mathsf{C}_{-k}(fg)=r(k+1)\,\mathsf{C}_{-k-1}(g)=\big(\varsigma V_{\mathsf{C}}(g\otimes r)\big)(k)$. Theorem 39.4(i) with $v(k)=\mathsf{C}_{-k}(\mathbf{b})$ reads $\mathsf{C}_{1-k}(\mathbf{b})=\mathbf{B}_{\mathbf{b}}(k)\,\mathsf{C}_{-k}(\mathbf{b})$, which at $k=-j$ is the stated identity, and the law of [1] applied to $V_{\mathsf{C}}$ gives $\sum_i p_i(k)\,\mathsf{C}_{-k-i}(1)=0$, at $k=-j$ the last statement. □

**Proposition 39.34 (The closings are the top principal parts).** *As stated.*

*Proof.* By 38.2(iii), $\langle g\rangle_k$ is a finite combination of periods $P_F$ at data $(k\mathbf{m}_1+v; k\mathbf{s}_1+\mathbf{a}, k\mathbf{t}_1+\mathbf{b})$ on the balance wall. On the balance wall, where every proper wall has positive real part, 8.5(ii) and 8.7, $J_F=\Gamma(\kappa_{\mathcal{E}})P_F$, give

$$P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\sum_{\pi}\int_{[0,1]^{n-1}}\prod_{l=2}^{n}q_l^{\kappa_{U_l}-1}\prod_j k_{j,\pi}(q)^{-s_j}\psi_{j,\pi}(q)^{-t_j}\,dq,$$

the limit $\kappa_{\mathcal{E}}\downarrow 0$ of the same identity with the factor $L_\pi^{-\kappa_{\mathcal{E}}}$, $L_\pi\in[1,n]$. The sector letters are positive on the closed cube, so the integrand is smooth in $q$ and holomorphic in the exponents, and 8.4 continues each term meromorphically in the walls with the poles on $\kappa_{U_l}=-i$, $i\in\mathbb{N}$, as a finite sum of products of at most $n-1$ factors $(\kappa_{U_l}+i_l)^{-1}$ with functions holomorphic near the point. Along the ray every wall is $k\kappa_{1,U_l}$ plus a constant, with $\kappa_{1,U_l}>0$, so $\langle g\rangle_k$ is meromorphic in $k$ with poles of order at most $n-1$.

For an integer $j\ge j(g)$ put $\epsilon=k+j$; then $\langle g\rangle_k=\langle P\rangle_\epsilon$ with the Laurent polynomial $P=gf^{-j}$, an identity for large $\operatorname{Re}k$ that continues. For a monomial $\alpha^u$ with $u\neq 0$ choose $e$ with $u_e\neq 0$. The relation of 38.3 at the strength $\epsilon$, with $c^{(0)}=0$ for the origin 0, holds for complex $\epsilon$ of large real part by the same integration by parts, the integrals converging absolutely there, and gives $u_e\langle\alpha^u\rangle_\epsilon=-\epsilon\,\langle c_e^{(1)}\alpha^u\rangle_\epsilon$, then as meromorphic functions; the right side has a pole of order at most $n-2$ at $\epsilon=0$. So $\epsilon^{n-1}\langle P\rangle_\epsilon$ and $\operatorname{CT}(P)\,\epsilon^{n-1}\langle 1\rangle_\epsilon$ have the same value at $\epsilon=0$. Finally $\langle 1\rangle_\epsilon=P_F(\epsilon\mathbf{m}_1;\epsilon\mathbf{s}_1,\epsilon\mathbf{t}_1)$ is the period of the question of the direction at the strength $\epsilon$, whose walls are positive, and 17.5(i) gives $\epsilon^{n-1}\langle 1\rangle_\epsilon\to V_1$ as $\epsilon\downarrow 0$; the function being holomorphic at 0, that is its value. □

**Proposition 39.35 (The low end, for every direction).** *As stated.*

*Proof.* (i) For an integral origin $\kappa_{0,U}$ and $\kappa_U(g)$ are integers, so $\kappa_{1,U}k$ is one at such a pole; Proposition 38.5(ii) gives the rest.

(ii) The argument is that of the proof of Proposition 39.34, which uses of the direction only that $P=f^{-j}g$ is a Laurent polynomial. By Proposition 38.5(ii), $\langle g\rangle_k=\langle f^jP\rangle_k=\langle P\rangle_{k+j}$ as meromorphic functions; put $\epsilon=k+j$. For a monomial $\alpha^u$, $u\neq 0$, take $e$ with $u_e\neq 0$; with the origin 0, $(k\,c_e^{(1)}+\partial_{y_e})\alpha^u$ lies in $\mathcal{L}$ and its value vanishes identically, so $u_e\langle\alpha^u\rangle_\epsilon=-\epsilon\langle c_e^{(1)}\alpha^u\rangle_\epsilon$, whose pole at $\epsilon=0$ has order at most $n-2$. So $\epsilon^{n-1}\langle P\rangle_\epsilon$ and $\operatorname{CT}(P)\,\epsilon^{n-1}\langle 1\rangle_\epsilon$ have the same value at $\epsilon=0$. $\langle 1\rangle_\epsilon=P_F(\epsilon\mathbf{m}_1;\epsilon\mathbf{s}_1,\epsilon\mathbf{t}_1)$ (Lemma S20(iv)) is the period of the question of the direction (Definition 17.1) at the strength $\epsilon$, whose walls are positive, and Theorem 17.5(i) gives $\epsilon^{n-1}\langle 1\rangle_\epsilon\to V_1$ as $\epsilon\downarrow 0$; $\epsilon^{n-1}\langle 1\rangle_\epsilon$ is holomorphic at 0, so that is its value. □

**Corollary 39.36 (The law read at the low end).** *As stated.*

*Proof.* The law says that $\sum_i p_i(k)f^i$ lies in $\mathcal{L}$, a finite sum $\sum r_{e,l}(k)\big(c_e^{(0)}+kc_e^{(1)}+\partial_{y_e}\big)h_{e,l}$. Integrating against $f^k dy$, by integration by parts for complex $k$ of large real part, where the integrals converge absolutely, $\sum_i p_i(k)\langle 1\rangle_{k+i}=0$ there off the poles of the $r_{e,l}$, and so identically as meromorphic functions. At $k=-j$, $j\ge 0$, the term $i\le j$ has $\langle 1\rangle_{k+i}$ with a pole of order at most $n-1$ and the coefficient $V_1\mathsf{C}_{j-i}(1)$ of $(k+j)^{-(n-1)}$ (Proposition 39.34, $j(1)=0$), and $p_i(k)=p_i(-j)+O(k+j)$; the terms $i>j$ are holomorphic there, $k+i$ being near the positive integer $i-j$, where the walls are positive. So the coefficient of $(k+j)^{-(n-1)}$ is $V_1\sum_{i\le j}p_i(-j)\mathsf{C}_{j-i}(1)=0$, with $V_1>0$. On the centre ray $f=\prod_j\big(K_j(\alpha)K_j(\alpha^{-1})\big)^{-\sigma_j}$ (38.13). □

**Definition 39.37 (The low-end coefficients).** As stated.

*Proof.* By Proposition 38.5(ii) the value of 1 is meromorphic with a pole of order at most $n-1$ at 0, and by Proposition 39.35(ii) at $j=0$ the coefficient of $k^{1-n}$ is $V_1$, a sum of products of positive walls. □

**Corollary 39.38 (The low-end relation).** *As stated.*

*Proof.* (i) By Proposition 38.5(ii) the value of $\sum_i p_i(k) f^i$, which is $\sum_i p_i(k)\langle 1\rangle_{k+i}$, vanishes identically, and every $\langle 1\rangle_{k+i}$ converges for $\operatorname{Re} k > 0$, the walls $(k+i)\kappa_{1,U}$ having positive real parts. Near $k=-j$, for $i \le j$, $\lim (k+j)^{n-1}\langle 1\rangle_{k+i} = V_1 \operatorname{CT}(f^{i-j})$ by Proposition 39.35(ii), while for $i > j$ the value $\langle 1\rangle_{k+i}$ is holomorphic at $k=-j$, $k+i$ being near a positive integer. Multiplying by $(k+j)^{n-1}$ and letting $k \to -j$ gives $V_1 \sum_{i \le \min(h,j)} p_i(-j) \operatorname{CT}(f^{i-j}) = 0$, and $V_1 > 0$.

(ii) Near $k=0$ the terms $i \ge 1$ are holomorphic, so $p_0(k)\langle 1\rangle_k = p_0(k)\, k^{1-n}\big(\mathsf{L}_0 + O(k)\big)$ is holomorphic at 0; as $\mathsf{L}_0 \ne 0$, $p_0$ vanishes to order at least $n-1$ there, and the value at 0 of $p_0(k)\langle 1\rangle_k$ is $[k^{n-1}]p_0 \cdot \mathsf{L}_0$.

(iii) With the origin 0, $c^{(0)} = 0$; at $k=0$ the certificate gives $\sum_i p_i(0) f^i = \sum_e \partial_{y_e} G_e(0)$, and $p_0(0) = 0$ by (ii), $n \ge 2$. The $G_e(0)$ are homogeneous of degree 0, so $\partial_{y_{e_0}} G_{e_0}(0) = -\sum_{e \ne e_0} \partial_{y_e} G_{e_0}(0)$ on the chart, and $\sum_e \partial_{y_e} G_e(0)$ is the divergence of the stated field in the coordinates $y_e$, $e \ne e_0$. By the divergence theorem its integral over the cube is the flux out of the cube, and as $R \to \infty$ the integral of $\sum_{i \ge 1} p_i(0) f^i$ over the cube tends to $\sum_{i \ge 1} p_i(0)\langle 1\rangle_i$, each $f^i$, $i \ge 1$, being integrable; by (ii) this is $-[k^{n-1}]p_0 \cdot \mathsf{L}_0$. □

**Proposition 39.39 (The low-end coefficients are integrals over the faces of the sectors).** *As stated.*

*Proof.* By Lemma S19 with $g=1$ and the origin 0, $\langle 1\rangle_k = \sum_\pi \int_{[0,1]^{n-1}} \prod_l q_l^{k\kappa_l - 1} \varphi_\pi^k \, dq$ for $\operatorname{Re} k > 0$. For $h$ smooth on a neighbourhood of the cube, $\sum_A h_{[A]} = h$: the restriction of $h$ to $\{q_l = 0,\ l \notin B\}$ enters with the coefficient $\sum_{A \supseteq B} (-1)^{|A \setminus B|}$, which is 1 for $B = \{2, \dots, n\}$ and 0 otherwise. $h_{[A]}$ depends on $q_A$ only, and where $q_l = 0$, $l \in A$, the terms of $B$ and $B \cup \{l\}$, $l \notin B$, cancel; so, one coordinate at a time (Hadamard's lemma), $h_{[A]} = \prod_{l \in A} q_l \cdot \tilde{h}_A$ with $\tilde{h}_A(q_A) = \int_{[0,1]^A} \big(\prod_{l \in A} \partial_{q_l}\big) h_{[A]}(t\, q_A)\, dt$ smooth, $t\, q_A = (t_l q_l)_{l \in A}$. With $h = \varphi_\pi^k$, holomorphic in $k$ together with its $q$-derivatives,

$$\int_{[0,1]^{n-1}} \prod_l q_l^{k\kappa_l - 1} \big(\varphi_\pi^k\big)_{[A]}\, dq = \prod_{l \notin A} \frac{1}{k\kappa_l}\, J_{\pi,A}(k), \qquad J_{\pi,A}(k) = \int_{[0,1]^A} \prod_{l \in A} q_l^{k\kappa_l}\, \tilde{h}_A(q_A, k)\, dq_A,$$

and $J_{\pi,A}$ is holomorphic for $\operatorname{Re} k > -1/\max_l \kappa_l$. So $k^{n-1}\langle 1\rangle_k = \sum_\pi \sum_A k^{|A|} \prod_{l \notin A} \kappa_l^{-1} J_{\pi,A}(k)$ near 0, and $\mathsf{L}_i = \sum_\pi \sum_A \prod_{l \notin A} \kappa_l^{-1} [k^{i-|A|}] J_{\pi,A}$. The Taylor coefficients of $J_{\pi,A}$ at 0 are obtained by differentiating under the integral, the $k$-derivatives of the integrand being dominated on the cube by constants times $\prod_{l \in A} q_l^{-\varepsilon} |\log q_l|^N$ for $|k|$ small; by Leibniz's rule, with $\partial_k^m \varphi_\pi^k|_{k=0} = \log^m \varphi_\pi$, $\partial_k^\gamma q^{k\kappa}|_{k=0} = (\kappa \log q)^\gamma$ and $h \mapsto h_{[A]}$, $h \mapsto \tilde{h}_A$ linear and independent of $k$, the coefficient of $k^{i'}$ is the sum over $m + |\boldsymbol{\gamma}| = i'$ of the integrals of the statement, $\big(\log^m \varphi_\pi\big)_{[A]} / \prod_{l \in A} q_l$ being bounded. For $i=0$ only $A = \varnothing$ and $m=0$ enter, giving $V_1$; for $i=1$, $A = \varnothing$ with $m=1$ gives $\sum_\pi \prod_l \kappa_l^{-1} \log \varphi_\pi(0)$, the terms with $|A|=1$ and $m=0$ vanishing, $1_{[A]} = 0$; the constant terms of the sector letters are $\theta_{j, T^\pm_{j,\pi}}$ (Theorem 8.5(ii)). For $A = \{2, \dots, n\}$ the exponent $m$ must be at least 1, so the whole cube enters only from $i = n$ on. □

**Proposition 39.41 (The high end reads the formal type).** *As stated.*

*Proof.* By Theorem 39.4(i), $\varsigma^{-1}v_V = \mathbf{B}v_V$. Divide by $\lambda^k k^\rho$: $\lambda^{-1}(1-\frac{1}{k})^\rho\big(a_0 + a_1(k-1)^{-1} + \cdots\big) = \big(\mathbf{B}_0 + \mathbf{B}_1 k^{-1} + \cdots\big)\big(a_0 + a_1 k^{-1} + \cdots\big)$. The order 0 gives $\lambda^{-1}a_0 = \mathbf{B}_0 a_0$, and the order 1 gives $\lambda^{-1}(a_1 - \rho a_0) = \mathbf{B}_0 a_1 + \mathbf{B}_1 a_0$; multiplied by $w$ on the left, with $w\mathbf{B}_0 = \lambda^{-1}w$, it gives $-\lambda^{-1}\rho\, w a_0 = w\mathbf{B}_1 a_0$. For a simple eigenvalue the left and right eigenvectors pair to a nonzero number. A source whose solution does not vanish on $\mathfrak{N}$ has a solution vector of the form of the statement after an integral shift of $\rho$, the exponents of Theorems 39.24 and 39.26 being defined modulo $\mathbb{Z}$. □

**Proposition 39.42 (The spectrum at infinite strength; changes of masters).** *As stated.*

*Proof.* By Theorem 38.6(i) and (ii), $\hat{A}_c$ is $\mathbb{C}(k)$-linear, vanishes on $\mathcal{L}$, and $\hat{A}_c(g) = C_c\, g(\alpha_c) + O(1/k)$ with $C_c = (2\pi)^{(n-1)/2}(\omega_0/dv)(\alpha_c) \neq 0$, the same for every insertion $g$.

(i) Let $\sum_l \lambda_l(k)[b_l] = 0$ with $\lambda \neq 0$, and multiply $\lambda$ by a power of $k$ so that every $\lambda_l$ is regular at $\infty$ and $\lambda(\infty) \neq 0$. Applying $\hat{A}_c$ and reading the coefficient of $k^0$ gives $C_c \sum_l \lambda_l(\infty)\, b_l(\alpha_c) = 0$ for every $\alpha_c \in C$; as the $\beta_c$ span, $\lambda(\infty) = 0$, a contradiction. So the $[b_l]$ are a basis of $\mathfrak{N}$, and the matrix of $\mathrm{Sh}^{-1}$ on a basis is unique.

(ii) As in Theorem 39.4(i), $V[f^{-1}b_i] = \varsigma^{-1}V[b_i]$ and $V[f^{-1}b_i] = \sum_l \mathbf{B}_{il}V[b_l]$ for every solution $V$, whatever the independence of the $b_l$: $\varsigma^{-1}v_V = \mathbf{B}v_V$ with $v_V = (V[b_l])_l$. For $V = A_c$, of type $\big(f(\alpha_c), -\frac{n-1}{2} + \mathbb{Z}\big)$, the solution vector is $\lambda^k k^\rho\big(a_0 + O(1/k)\big)$ with $\lambda = f(\alpha_c)$, $\rho = -\frac{n-1}{2}$ and $a_0 = C_c\beta_c$, and $\varsigma^{-1}$ multiplies $\lambda^k k^\rho$ by $\lambda^{-1}\big(1 + O(1/k)\big)$ (Definition 39.2); the coefficient of $\lambda^k k^\rho$ in $\varsigma^{-1}v_V = \mathbf{B}v_V$ gives $\lambda^{-1}a_0 = \mathbf{B}(\infty)\, a_0$. Eigenvectors of distinct eigenvalues are independent, and a matrix with an eigenbasis taken among the $\beta_c$, $\alpha_c \in C$, is diagonalizable with the eigenvalues of those vectors, every $\beta_c$ being an eigenvector.

(iii) $\mathrm{Sh}^{-1}\tilde{b}_i = \sum_l T_{il}(k-1)[f^{-1}b_l] = \sum_{l,l'} T_{il}(k-1)\mathbf{B}_{ll'}(k)[b_{l'}]$, which is $\sum_{i'} \tilde{\mathbf{B}}_{ii'}(k)\,\tilde{b}_{i'}$, and the solution vector of $A_c$ on the $\tilde{b}_i$ is $T(k)\, v_{A_c}$, which satisfies $\varsigma^{-1}\tilde{v} = \tilde{\mathbf{B}}\tilde{v}$ by the same computation. $T(k)$ is invertible and $\hat{A}_c(b_1)$ has the nonzero constant term $C_c$, so $T(k)\big(\hat{A}_c(b_l)\big)_l = k^\nu\big(\tilde{\beta}_c + O(1/k)\big)$ with $\nu \in \mathbb{Z}$ and $\tilde{\beta}_c \neq 0$, and the computation of (ii) with $\rho = -\frac{n-1}{2} + \nu$ gives $\tilde{\mathbf{B}}(\infty)\tilde{\beta}_c = f(\alpha_c)^{-1}\tilde{\beta}_c$. Finally $T(k)\,\mathbf{B}(k)\,T(k)^{-1} = \big(T(k-1)T(k)^{-1}\big)^{-1}\tilde{\mathbf{B}}(k)$, which tends to $\tilde{\mathbf{B}}(\infty)$ when $T(k-1)T(k)^{-1} \to I$; it has the characteristic polynomial of $\mathbf{B}(k)$, whose coefficients therefore tend to those of $\tilde{\mathbf{B}}(\infty)$, and the roots of a monic polynomial depend continuously on its coefficients.

(iv) $T(k-1)\,\mathbf{B}(k)\,T(k)^{-1} = S(k-1)^{-1}S(k-1)\,\mathbf{B}_\circ\, S(k)^{-1}S(k) = \mathbf{B}_\circ$, and $T(k-1)T(k)^{-1} = S(k-1)^{-1}S(k) = \begin{bmatrix} 1 & 0 \\ -(k-1) & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ k & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$. With $S(k)^{-1} = \begin{bmatrix} 1 & 0 \\ -k & 1 \end{bmatrix}$,

$$\mathbf{B}(k) = \begin{bmatrix} x_{11} - x_{12}k & x_{12} \\ (k-1)(x_{11} - x_{12}k) + x_{21} - x_{22}k & (k-1)x_{12} + x_{22} \end{bmatrix},$$

whose entry $(2,1)$ is $-x_{12}k^2 + O(k)$, whose trace is $x_{11} + x_{22} - x_{12}$ and whose determinant is $\det \mathbf{B}_\circ$; the trace of $\mathbf{B}_\circ$ is $x_{11} + x_{22}$. □

**Definition 39.43 (Insertion lattice; native algebra; tame high end).** As stated.

*Proof.* $A$ is the local ring of $\mathbb{P}^1$ at $\infty$, a discrete valuation ring with the uniformizer $k^{-1}$. The critical locus is the zero set of the defects in $X_F$, so $N_F$ is its ring of functions with the scheme structure of $J_F$. □

**Theorem 39.44 (A tame high end).** *As stated.*

*Proof.* (i) $\Lambda_F$ is an $A$-submodule of the finite-dimensional $\mathbb{C}(k)$-space $\mathfrak{M}$ (Theorem 38.31(i)), and it spans $\mathfrak{M}$. Finitely generated and torsion-free over a discrete valuation ring, it is free, of rank $\dim \mathfrak{M}$; finitely many insertion classes generate it, and by Nakayama's lemma those among them whose images form a basis of $\Lambda_F/k^{-1}\Lambda_F$ form an $A$-basis. For $g \in \mathbb{C}[X_F]$ and $r \in A$, $\mathrm{Sh}^{\pm 1}[g \otimes r] = [f^{\pm 1} g \otimes r(k \pm 1)]$, with $f^{\pm 1} \in \mathbb{C}[X_F]$ (Lemma 38.2(ii); the letters are units on $X_F$) and $r(k \pm 1) \in A$. So $\mathrm{Sh}^{\pm 1}\,\Lambda_F \subseteq \Lambda_F$, and $\mathbf{B_b}$ and its inverse have entries in $A$.

(ii) For $g \in \mathbb{C}[X_F]$ and every mark $e$, $k^{-1}\big(c_e^{(0)} + k\, c_e^{(1)} + \partial_{y_e}\big) g \in \mathcal{L}$, and $c_e^{(0)} g$, $\partial_{y_e} g$ lie in $\mathbb{C}[X_F]$ (Lemma 38.2(i)). So

$$[c_e^{(1)}\, g] = -k^{-1}\big[c_e^{(0)}\, g + \partial_{y_e} g\big] \in k^{-1}\Lambda_F,$$

and $J_F$ maps into $k^{-1}\Lambda_F$. The images of the insertions span $\Lambda_F/k^{-1}\Lambda_F$, since $A/k^{-1}A = \mathbb{C}$. If $[g] \in k^{-1}\Lambda_F$ then $[fg] = \mathrm{Sh}[g] \in (k+1)^{-1}\Lambda_F = k^{-1}\Lambda_F$, so the kernel is stable under $f$. For $r \in A$, $r(k-1) - r(k) \in k^{-1}A$, so $\mathrm{Sh}^{-1}$ reduces modulo $k^{-1}\Lambda_F$ to the multiplication by $f^{-1}$, whose matrix on the basis $[b_l]$ is $\mathbf{B_b}(\infty)$.

(iii) The critical locus has finitely many irreducible components $Z$, and $f$ is constant on each, with a value $f_Z \neq 0$ (Definition 39.23). By the Nullstellensatz $\prod_Z (f^{-1} - f_Z^{-1})^N \in J_F$ for some $N$, so the eigenvalues of $f^{-1}$ on every quotient of $N_F$ by an $f$-stable subspace are among the $f_Z^{-1}$; with (ii) this is the first claim. Let $\lambda \neq 0$ be a root of the symbol of the law. The law has a formal solution $x = \lambda^k \varphi(k)$ in which $\varphi$ is a product of a power $k^\rho$, a power of $\log k$, the exponential of a polynomial in $k^{1/p}$ of degree less than $p$, and a series in $k^{-1/p}$ (the formal reduction of linear difference equations of Birkhoff, Adams and Turrittin, which applies because the law has polynomial coefficients with $p_0 p_h \neq 0$). Then $V(\mathrm{Sh}^i[1]) = \varsigma^i x$ defines a solution of the cyclic submodule $\mathfrak{C}$ of $[1]$ in the sense of Definition 39.2, these formal series allowed. $\Lambda_F \cap \mathfrak{C}$ is an $A$-lattice in $\mathfrak{C}$, stable under $\mathrm{Sh}^{\pm 1}$ and saturated in $\Lambda_F$, so its reduction embeds in $\Lambda_F/k^{-1}\Lambda_F$, compatibly with $f^{-1}$. In a basis of $\Lambda_F \cap \mathfrak{C}$ the solution vector $v$ has $v(k-1) = \mathbf{B}(k) v(k)$ with $\mathbf{B} \in GL(A)$ the matrix of $\mathrm{Sh}^{-1}$, as in Theorem 39.4(i). Write $v = \lambda^k \varphi(k)\, w(k)$; dividing gives $\lambda^{-1} \frac{\varphi(k-1)}{\varphi(k)}\, w(k-1) = \mathbf{B}(k)\, w(k)$, and $\varphi(k-1)/\varphi(k) \to 1$. The leading term $a_0 \neq 0$ of $w$ therefore has $\mathbf{B}(\infty) a_0 = \lambda^{-1} a_0$, as in Proposition 39.41, and $\lambda^{-1}$ is an eigenvalue of $f^{-1}$ on the reduction, so $\lambda$ is a native value.

(iv) A nondegenerate native gauge is an isolated reduced point of the critical locus, the Jacobian of the defects being invertible there, so $\dim_{\mathbb{C}} N_F$ is their number. By Theorem 38.31(i) this number is at most $\dim \mathfrak{M} = \dim \Lambda_F/k^{-1}\Lambda_F$, so the surjection of (ii) is an isomorphism, and the multiplication by $f^{-1}$ is diagonal on the idempotents of the points, with the eigenvalues $f(\alpha_c)^{-1}$. □

**Theorem 39.45 (Separating the sources).** *As stated.*

*Proof.* For new masters $\mathbf{b}' = T\mathbf{b}$, $T$ invertible over $\mathbb{C}((1/k))$, $\mathrm{Sh}^{-1} b_i' = \sum_l T_{il}(k-1)\, \mathrm{Sh}^{-1} b_l$, so the matrix of $\mathrm{Sh}^{-1}$ becomes $T(k-1)\mathbf{B}(k)T(k)^{-1}$. After a constant change of basis, $\mathbf{B}_0 = \mathrm{diag}(\mathbf{B}_0^{(1)}, \dots, \mathbf{B}_0^{(s)})$ with $\mathrm{spec}\, \mathbf{B}_0^{(a)} = \Lambda_a$. Seek $T = I + \sum_{i \geq 1} T_i k^{-i}$ and a block diagonal $C = \sum_{i \geq 0} C_i k^{-i}$, $C_0 = \mathbf{B}_0$, with $T(k-1)\mathbf{B}(k) = C(k)T(k)$. Write $T(k-1) = \sum_i \tilde{T}_i k^{-i}$, where $\tilde{T}_i - T_i$ depends on $T_1, \dots, T_{i-1}$ only. The coefficient of $k^{-i}$ is

$$T_i \mathbf{B}_0 - \mathbf{B}_0 T_i = C_i - \mathbf{B}_i + E_i, \qquad E_i = \sum_{0<a<i} \big(C_a T_{i-a} - \tilde{T}_a \mathbf{B}_{i-a}\big) - (\tilde{T}_i - T_i)\mathbf{B}_0,$$

with $E_i$ known from the lower orders. Take $C_i$ the block diagonal part of $\mathbf{B}_i - E_i$, the diagonal blocks of $T_i$ zero, and the block $(a,b)$, $a \neq b$, of $T_i$ the solution $X$ of $X\mathbf{B}_0^{(b)} - \mathbf{B}_0^{(a)} X = (E_i - \mathbf{B}_i)^{(ab)}$, unique since the spectra are disjoint. The blocks of the new masters span the $\hat{\mathfrak{N}}_a$.

If $\bigoplus_a \hat{\mathfrak{N}}'_a$ is another such decomposition, the inclusion of $\hat{\mathfrak{N}}'_a$ followed by the projection onto $\hat{\mathfrak{N}}_b$, $b \neq a$, commutes with the shift; in bases it is a matrix $X$ over $\mathbb{C}((1/k))$ with $C'^{(a)}(k)X(k) = X(k-1)C^{(b)}(k)$. If $X \neq 0$, $X = k^v(X_0 + O(k^{-1}))$ with $X_0 \neq 0$, and the leading order $C_0'^{(a)}X_0 = X_0 C_0^{(b)}$ forces $X_0 = 0$, the spectra being disjoint. So $\hat{\mathfrak{N}}'_a \subseteq \hat{\mathfrak{N}}_a$ for every $a$, and the sums being direct and equal to the whole space, $\hat{\mathfrak{N}}'_a = \hat{\mathfrak{N}}_a$. A solution of type $(\lambda, \rho + \mathbb{Z})$, extended $\mathbb{C}((1/k))$-linearly and restricted to $\hat{\mathfrak{N}}_b$, has in the basis of $\hat{\mathfrak{N}}_b$ the solution vector $\lambda^k k^\rho w$; if it is not zero, $w = k^{-v}(w_0 + \cdots)$ with $w_0 \neq 0$, and the first statement of Proposition 39.41, whose proof uses only the expansion of the matrix at $\infty$, gives $C^{(b)}(\infty)w_0 = \lambda^{-1}w_0$, so $\lambda^{-1} \in \Lambda_b$. A symmetry commutes with the shift, so it carries the decomposition to one with the same property, the matrix of $\mathrm{Sh}^{-1}$ on the image basis being the same; by uniqueness it maps each $\hat{\mathfrak{N}}_a$ onto itself. □

**Corollary 39.46 (Gluing at the two ends).** *As stated.*

*Proof.* With $\mathcal{S}_\mathrm{f}$ the formal solutions in $\mathcal{S}$, $\mathrm{Ann}(\mathcal{S}) \cap \mathfrak{N} = \mathrm{Ann}(\mathcal{S}_\mathrm{f}) \cap \mathfrak{N}_\mathsf{C}$, a submodule of $\mathfrak{N}_\mathsf{C}$ (Theorem 39.4(ii)), and Theorem 39.4(iii) applies. If $\mathfrak{N}$ is irreducible and $V_\mathsf{C}$ does not vanish on it, $\mathfrak{N}_\mathsf{C}$ is a submodule other than $\mathfrak{N}$, hence 0. If $\mathfrak{m}$ formal solutions are linearly independent over $\mathbb{C}$ on $\mathfrak{N}$, type by type, Lemma 39.3 makes their normalizations $\mathfrak{m}$ independent linear forms on $\mathfrak{N} \otimes \mathbb{C}((1/k))$, of dimension $\mathfrak{m}$, so no nonzero element of $\mathfrak{N}$ is annihilated by all of them. □

**Definition 39.47 (The pencil module; singular values; vanishing ranks).** As stated.

*Proof.* $X_F$ is a smooth affine variety of dimension $n-1$ with the global coordinates $y_e$, $e \neq e_0$, and $\omega_0/dy$ is a unit of $\mathbb{C}[X_F]$, the origin being integral; so the top forms are the $h\,\omega_0$ and the forms of degree $n-2$ are spanned by the $\iota_{\partial_{y_e}}(h\,\omega_0)$. Since $d\,\iota_{\partial_{y_e}}(h\omega_0) = L_{\partial_{y_e}}(h\omega_0) = (c_e^{(0)}h + \partial_{y_e}h)\,\omega_0$ by Lemma 38.2(ii) at $k=0$, and $d\varphi \wedge \iota_{\partial_{y_e}}(h\omega_0) = (\partial_{y_e}\varphi)\,h\,\omega_0$, the image of $\iota_{\partial_{y_e}}(h\omega_0)\partial^j$ under the differential is $R(e,h,j)$. The operators $\partial$ and $\tau(\eta\partial^j) = \varphi\eta\partial^j - j\eta\partial^{j-1}$ on $\Omega^\bullet[\partial]$ commute with $D' = d - \partial\,d\varphi\wedge$: for $\tau$,

$$\tau D'(\eta\partial^j) - D'\tau(\eta\partial^j) = (j+1)\,d\varphi\wedge\eta\,\partial^j - d\varphi\wedge\eta\,\partial^j - j\,d\varphi\wedge\eta\,\partial^j = 0,$$

and $\partial\tau - \tau\partial = 1$. The operator $\tau$ is invertible on $\Omega^\bullet[\partial]$: on an element of top degree $N$ in $\partial$ the equations $\varphi\sigma_j - (j+1)\sigma_{j+1} = \tau_j$ are solved from $j = N$ down, $\varphi$ being a unit. So $t = \tau$ and $\partial_t = \partial$ make the cohomology a $\mathcal{D}$-module. The fibre integrals of $h\,\omega_0\,\partial^j$ are those of $h\,\omega_0/d\varphi$ differentiated $j$ times, and they kill the relations by Stokes's theorem on the fibres. For $N = \mathcal{D}/\mathcal{D}Q$ the characteristic cycle is the divisor of the principal symbol of $Q$, whose part over $t_0 \in \mathbb{C}^\times$ is the order of vanishing of the principal coefficient there. □

**Proposition 39.48 (The ray is the moment side of its pencil).** *As stated.*

*Proof.* By Lemma 38.2(ii), $\partial_{y_e}\varphi = -\varphi\,c_e^{(1)}$, and $\mathrm{Sh}[\varphi x] = [x]$. So

$$\Theta\big((h\,\partial_{y_e}\varphi)\partial^{j+1}\big) = (k)_j(k+j)\,\mathrm{Sh}^{j+1}[h\,\partial_{y_e}\varphi] = (k)_j\,\mathrm{Sh}^j\left(k\,\mathrm{Sh}[h\,\partial_{y_e}\varphi]\right) = -(k)_j\,\mathrm{Sh}^j\left(k\,[c_e^{(1)}h]\right),$$

and $\Theta(R(e,h,j)) = (k)_j\,\mathrm{Sh}^j\left[(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})h\right] = 0$. Next $\mathrm{Sh}^j[\varphi h] = \mathrm{Sh}^{j-1}[h]$ gives

$$\Theta(t\cdot h\partial^j) = \big((k)_j - j\,(k)_{j-1}\big)\,\mathrm{Sh}^{j-1}[h] = (k-1)_j\,\mathrm{Sh}^{j-1}[h] = \mathrm{Sh}^{-1}\,\Theta(h\partial^j),$$

and $\Theta(h\partial^{j+1}) = (k)_j(k+j)\,\mathrm{Sh}^{j+1}[h] = k\,\mathrm{Sh}\;\Theta(h\partial^j)$. So $t\partial_t \mapsto \mathrm{Sh}^{-1}\,k\,\mathrm{Sh} = k-1$, $\Theta$ is $\mathbb{C}[t\partial_t]$-linear for $k = t\partial_t + 1$, and it extends to $\mathbb{C}(k) \otimes_{\mathbb{C}[t\partial_t]} \mathcal{G}_F$, on which $t$ acts invertibly and semilinearly,

$t\,r(k) = r(k-1)\,t$. Conversely $[h \otimes r(k)] \mapsto r(t\partial_t + 1)\,h\partial^0$ is well defined on $\mathfrak{M}$: in $\mathcal{G}_F$, $(t\partial_t + j)(x\partial^{j-1}) = (\varphi x)\partial^j$, so $(\varphi x)\partial = (t\partial_t + 1)(x\partial^0)$ and

$$R(e,h,0) = \big(\partial_{y_e} h + c_e^{(0)} h\big)\partial^0 + (t\partial_t + 1)\big(c_e^{(1)} h\,\partial^0\big),$$

the image of $(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})h$; and $t\cdot(fg)\partial^0 = g\partial^0$, so $\mathrm{Sh} = t^{-1}$ is respected. The two maps are inverse: $(t\partial_t+1)(t\partial_t+2)\cdots(t\partial_t+j)\big(f^j h\,\partial^0\big) = h\partial^j$ by the identity above applied $j$ times, which is the image of $\Theta(h\partial^j) = (k)_j[f^j h]$. The moments: for a real origin the orthant maps onto $[A,\infty)$ under $\varphi$ (Theorem 38.14(i)), so the coarea formula gives $\langle h\rangle_k = \int_{\mathbb{R}^{\mathcal{E}}_{>0}/\mathbb{R}_{>0}} h\,\varphi^{-k}\omega_0 = \int_A^\infty t^{-k} P_h(t)\,dt$ for all large $k$; and $\int t^{-k}\partial_t^j P_h\,dt = (k)_j \int t^{-k-j} P_h\,dt$ for the distributional derivatives of $P_h$ extended by 0 below $A$, which matches $\Theta(h\partial^j) = (k)_j\,\mathrm{Sh}^j[h]$. □

**Theorem 39.49 (The symbol is the principal symbol of the pencil).** *As stated.*

*Proof.* $\mathcal{D}$ embeds in $\mathbb{C}(k)\langle \mathrm{Sh}^{\pm 1}\rangle$ by $t \mapsto \mathrm{Sh}^{-1}$, $\partial_t \mapsto k\,\mathrm{Sh}$, which respects $[\partial_t, t] = 1$ since $k-(k-1) = 1$; its image is $\mathbb{C}[k]\langle \mathrm{Sh}^{\pm 1}\rangle$, and $t^i\partial_t^j \mapsto (k-i)_j\,\mathrm{Sh}^{j-i}$, of the top degree $j$ in $k$ with the leading coefficient 1. For an element $\sum_i r_i(k)\,\mathrm{Sh}^i$ with $r_i \in \mathbb{C}[k]$ call $\sum_i [k^\delta] r_i\,\lambda^i$ its symbol, $\delta$ the largest degree; symbols multiply, since $\mathrm{Sh}^i\,r(k) = r(k+i)\,\mathrm{Sh}^i$ keeps the leading coefficient of $r$. For $Q \in \mathcal{D}$ of order $m$ with the principal coefficient $a_m(t) = \sum_i a_{im} t^i$ the symbol of its image is $\sum_i a_{im}\lambda^{m-i} = \lambda^m a_m(1/\lambda)$; and $A_L \mapsto L$, since $t\partial_t + 1 \mapsto k$ and $t^{-i} \mapsto \mathrm{Sh}^i$. So the principal coefficient of $A_L$ is $t^d\chi(1/t)$.

(i) $L[1] = 0$ in $\mathfrak{M} = \mathbb{C}(k) \otimes \mathcal{G}_F$ (Proposition 39.48), so $A_L \cdot 1$ is $\mathbb{C}[t\partial_t]$-torsion in $\mathcal{G}_F$ and some $c(t\partial_t)A_L$ kills 1. If $Q\cdot 1 = 0$ in $\mathcal{G}_F$, the image of $Q$ annihilates $[1]$ in $\mathfrak{M}$; multiplied by a power of Sh it has only nonnegative powers, and division by $L$ leaves a remainder of order less than $h$ that annihilates $[1]$, hence 0, the classes $[1], \ldots, \mathrm{Sh}^{h-1}[1]$ being independent (Definition 39.1). So the image of $Q$ is $\mathrm{Sh}^{-N}\,Q'L$, and $\chi$ divides its symbol.

(ii) $\mathcal{D}_F[1] = \mathcal{D}/I$ with $I$ the annihilator of 1; it is holonomic, a cyclic module on a curve with $I \ni c(t\partial_t)A_L \neq 0$. It is a connection near $t_0$ exactly when $I$ contains an element whose principal coefficient does not vanish at $t_0$: its characteristic variety is the common zero set of the principal symbols of the elements of $I$, and a holonomic module on a curve is a connection near $t_0$ exactly when that variety meets the cotangent line of $t_0$ only at 0. By (i) every principal coefficient in $I$ vanishes where $\chi(1/t)$ does, and $c(t\partial_t)A_L$ has the principal coefficient $c_* t^{\deg c + d}\chi(1/t)$, $c_*$ the leading coefficient of $c$, which vanishes on $\mathbb{C}^\times$ nowhere else.

(iii) The characteristic cycle of $\mathcal{D}/\mathcal{D}c(t\partial_t)A_L$ is the divisor of its principal symbol, whose part over $t_0 \in \mathbb{C}^\times$ has the multiplicity of $1/t_0$ as a root of $\chi$. The kernel $I/\mathcal{D}c(t\partial_t)A_L$ of the surjection onto $\mathcal{D}_F[1]$ becomes 0 over $\mathbb{C}(k)$, where both ideals generate the ideal of $L$, so it is $\mathbb{C}[t\partial_t]$-torsion: a quotient of a finite sum of modules $\mathcal{D}/\mathcal{D}q(t\partial_t)$, $q \neq 0$, whose principal symbols $q_*(t\xi)^{\deg q}$ do not vanish over $\mathbb{C}^\times$ off the zero section. Characteristic cycles add along exact sequences and are nonnegative, so the kernel contributes nothing over $t_0$, and $r_{t_0}(\mathcal{D}_F[1])$ is that multiplicity.

(iv) On a neighbourhood of $t_0$ on which $\mathcal{G}_F$ is a connection, a coherent module over the holomorphic functions, its $\mathcal{D}$-submodule $\mathcal{D}_F[1]$ is coherent as well, so a connection. □

**Lemma 39.50 (An orbit counts at most once).** *As stated.*

*Proof.* $G$ acts on $\Omega^\bullet[\partial]$ by pullback, commuting with the differential and with $t$ and $\partial_t$, since it preserves $\varphi$; so it acts on $\mathcal{G}_F$ by $\mathcal{D}$-linear maps, and $\mathcal{G}_F = \bigoplus_\psi \mathcal{G}_F^\psi$ over the irreducible characters, $G$ being finite. The class $1 = [\omega_0]$ spans a line of the character $\chi_0$, so $\mathcal{D}_F[1] \subseteq \mathcal{G}_F^{\chi_0}$, and characteristic cycles being additive and nonnegative, $r_{t_0}(\mathcal{D}_F[1]) \le r_{t_0}(\mathcal{G}_F^{\chi_0})$. The vanishing rank of a regular holonomic module on a curve is the dimension of its vanishing cycles (the index theorem of

Kashiwara on curves; $\mathcal{G}_F$ is regular holonomic as the direct image of $\mathcal{O}_{X_F}$ under a regular function on a smooth affine variety, by the theorems of Bernstein and Kashiwara on direct images and the regularity theorem of Deligne). Over a small disc about $t_0$ the only critical points of $\varphi$ are the native gauges of the value $1/t_0$, nondegenerate, and the absence of asymptotic native gauges makes $\varphi$ a locally trivial fibration near infinity over the disc (Rabier's theorem, as in the proof of Theorem 39.51); so the vanishing cycles of $\mathcal{G}_F$ at $t_0$ are the direct sum over these points $p$ of one line each, spanned by the image of any top form that does not vanish at $p$ (the holomorphic Morse lemma and the description of the vanishing cycles of a nondegenerate critical point), for instance $\omega_0$, which is a unit times $dy$. $G$ permutes these lines. The stabilizer $G_p$ acts on the line of $p$ through its action on the image of $[\omega_0]$, that is by $\chi_0|_{G_p}$, so the lines of the orbit of $p$ form the induced representation $\operatorname{Ind}_{G_p}^{G}(\chi_0|_{G_p})$, which contains $\chi_0$ exactly once by Frobenius reciprocity. So the $\chi_0$-part of the vanishing cycles has the dimension of the number of orbits. □

**Theorem 39.51 (The symbol is read off the sources).** *As stated.*

*Proof.* By Theorem 39.49(ii) and (iv) a nonzero root $\lambda$ makes $t_0 = 1/\lambda$ a singular value of $\mathcal{G}_F$. Since every letter has a positive exponent, $\varphi = \alpha^{-\mathbf{m}_1} \prod_j K_j^{s_{1j}} \Psi_j^{t_{1j}}$ is a regular function on the torus $T$ and $X_F = \varphi^{-1}(\mathbb{C}^\times)$. Two results finish the argument.

(a) Over a disc on which $\varphi : X_F \to \mathbb{C}^\times$ is a locally trivial fibration, $\mathcal{G}_F$ is a connection, the local system of the top cohomology of the fibres (Grothendieck's comparison theorem for the algebraic de Rham complex of the fibres, with the base change of the direct image along a smooth affine morphism, which apply because $X_F$ is smooth and affine and $\varphi$ is regular).

(b) In the logarithmic coordinates $T$ is $\mathbb{C}^{n-1}/2\pi i\mathbb{Z}^{n-1}$ with a complete flat metric, and a map of a complete Finsler manifold to $\mathbb{C}$ is a locally trivial fibration over every value that is neither a critical value nor a limit of the map along a sequence leaving every compact set with the norm of the differential tending to 0 (Rabier's theorem on Ehresmann fibrations, which applies because the metric is complete).

A critical value of $\varphi$ in $\mathbb{C}^\times$ is the reciprocal of a native value, since $d\log\varphi = -c^{(1)}$. The norm of $d_y\varphi = -\varphi\, c^{(1)}$ tends to 0 along a sequence with $\varphi \to t_0 \neq 0$ exactly when $c^{(1)} \to 0$, so a limit of the second kind is the reciprocal of an asymptotic native value. For the last claim, points converging to a point of the orbit of that flag have the bunch scale $\rho \to 0$ and their positions converging in $T_B \times T_R$, which is the setting of Proposition 38.26; its proof uses only that the defects tend to 0, for complex perturbations, and gives the values of (ii) and (iii) there. □

**Corollary 39.52 (Where the spectrum is native).** *As stated.*

*Proof.* (i) is Theorem 39.51. (ii) By Proposition 38.30(i) an asymptotic native gauge has a limit point at which two factors of $\Pi_{\mathfrak{f}}$ vanish, so two zero sets meet, or one factor vanishes with a singular zero set; both are excluded. (iii) is Proposition 38.26(i) with its first claims, $\bar{Q}$ being $\Psi^{[0]}$ up to a monomial (Proposition 38.25): a factor of $K^{[0]}$ and a factor of $\Psi^{[0]}$ vanishing together give $Q_1 = 0$ at the limit, against $\lambda^{-1} = Q_1 \neq 0$. □

**Theorem 39.53 (Uncarried letters).** *As stated.*

*Proof.* The proof of Theorem 39.49 uses only that $\varphi$ is a unit of $\mathbb{C}[X_F]$, which holds; so a root is $1/t_0$ for a singular value $t_0$ of $\mathcal{G}_F$, and by (a) in the proof of Theorem 39.51 the map $\varphi : X_F \to \mathbb{C}^\times$ is not a locally trivial fibration over any disc about $t_0$. The map $\varphi$ is regular on $X'$, and the pair $(X', Y)$ with the Whitney stratification of $Y$ and the open stratum $X_F$ is a stratified space on

which $\varphi$ is a stratified map. Over a value that is neither a critical value of $\varphi$ on a stratum nor a limit of $\varphi$ along a sequence on a stratum leaving every compact set of $T$ with the norm of the differential along the stratum tending to 0, the stratified map is locally trivial compatibly with the strata (Thom's first isotopy lemma, with the condition at infinity of Rabier's kind for stratified maps, which applies on $T$ with its complete flat metric); then the fibres of $X'$ and of $Y$ are trivial as a pair, and those of $X_F = X' \setminus Y$ are trivial. So $t_0$ is a value of one of the three kinds, with $d\log f = c^{(1)}$ on $X_F$ and its restriction on the strata. □

**Proposition 39.54 (The far end of the pencil).** *As stated.*

*Proof.* For real $t > A$ the orthant fibre $\{\varphi = t\} \cap \mathbb{R}^{\mathcal{E}}_{>0}$ is compact, $\varphi$ tending to $\infty$ at the ends of the orthant (Theorem 38.14(i)), so $P$ is the fibre integral of $\omega_0 = dy$ over a family of cycles, a solution of $\mathcal{D}_F[1]$, which continues along every path that avoids the singular values. The point $t = \infty$ is a regular singular point of $\mathcal{D}_F[1]$ (Deligne's regularity theorem for the Gauss–Manin connection, $\varphi$ being regular on a smooth variety), so for large real $t$

$$P(t) = \sum_a \sum_{l \geq 0} c_{a,l}\, t^{-a-1} \frac{(\log t)^l}{l!},$$

a finite sum over classes of exponents $a$ modulo $\mathbb{Z}$, each with convergent series in $1/t$. In the Mellin transform of Proposition 39.48, $\int_A^\infty t^{-k} t^{-a-1} (\log t)^l / l!\, dt$ has at $k = -a$ a pole whose top term is $(k+a)^{-l-1}$, the lower terms carrying powers of $\log A$; so the pole of the highest order of $\langle 1 \rangle_k$ at $k = -a$ is read from the coefficient with the largest $l$. The poles of $\langle 1 \rangle_k$ have order at most $n-1$ (Proposition 38.5(ii)), so $l \leq n-2$, and at $k = -j$, $j \in \mathbb{N}$, the coefficient of $(k+j)^{-(n-1)}$ is $V_1\, \mathsf{C}_j(1)$ (Proposition 39.35(ii)), so $c_{j,n-2} = V_1 \mathsf{C}_j(1)$ for the integral exponents. One turn about $t = \infty$ moves $\log t$ by $\pm 2\pi i$ and multiplies $t^{-a}$ by a root of unity, the constant one for integral $a$; on the integral exponents $(T_\infty - 1)^{n-2}$ keeps exactly the top power of $\log t$, with the factor $(\pm 2\pi i)^{n-2}$, which gives the identity. Since $\mathsf{C}_0(1) = 1$ and $V_1 > 0$, $T \neq 0$, so the cyclic span of $P$ under $T_\infty$ has a Jordan block of size at least $n-1$ for the eigenvalue 1, and at most $n-1$ by the monodromy theorem for the cohomology of degree $n-2$ of the fibres (Landman's theorem, the fibres being affine of dimension $n-2$); $(T_\infty - 1)T = 0$ because $T$ has no logarithm, so $T$ spans the invariant line of the block. □

**Definition 39.55 (The homogeneous reading; the data of a vertex).** As stated.

*Proof.* The marker $g_v$ of the parent edge is a mark of the core $\mathcal{H}_{v,j}$, and no mark of a core is a loop or a coloop (Definition 21.1), so its hinge minors are tables with the inherited weights (Definition 7.3). The coordinates of $\lambda_e$ are Laurent monomials in $\alpha_e$, among them $\alpha_e$ and $\alpha_e^{-1}$, so $\lambda_e$ is a closed embedding onto the curve $c_j = c'_j = 1$, $d_j = \alpha_e$, $d'_j = \alpha_e^{-1}$, $\mu = \alpha_e^{p_e}$; the coordinates of $N_v$ and of $R_o$ are polynomials, the last one mapping the tori of the $\mu_c$ to $\mathbb{C}^\times$. The reading is multilinear, and $\prod_c u_c \cdot \sum_T \theta_T \prod_{c \in T} (w_c/u_c) = \mathcal{H}^{\hom}(\mathbf{u}, \mathbf{w})$ where every $u_c \neq 0$. □

**Theorem 39.56 (The pencil along a hinge tree).** *As stated.*

*Proof.* (i) Induction from the marks up. Fix $v \neq o$ and $j$, and write $\mathcal{H} = \mathcal{H}_{v,j}$, $g = g_v$ and $\mathcal{A} = \mathcal{A}_{v,j}$; for a child $c$ let $\mathcal{A}_c$ be its subtree cell, with the hinge $g_c$, or for a mark $e$ the cell $\binom{\{e, g_e\}}{1}$ that leaves it in place. By Definition 23.15 $\mathcal{A} = \mathcal{H}\langle \mathcal{A}_c \rangle$ with $g$ not substituted, so its bases are the sets $\bigsqcup_c X_c \cup (T \cap \{g\})$, $T$ a basis of $\mathcal{H}$, with $X_c$ a basis of $\mathcal{A}_c \setminus g_c$ when $c \in T$ and of $\mathcal{A}_c / g_c$ when $c \notin T$, of the weight $\theta_T^{\mathcal{H}} \prod_{c \in T} \theta_{X_c} \prod_{c \notin T} \theta_{X_c + g_c}$ in each letter (Definition 21.1). The bases avoiding

$g$ are those of $\mathcal{A}\setminus g$, with $T$ running over the bases of $\mathcal{H}\setminus g$, and the bases containing $g$, less $g$, are those of $\mathcal{A}/g$, with $T-g$ running over the bases of $\mathcal{H}/g$, of the weight $\theta^{\mathcal{H}}_T$ (Definition 7.3). For each $T$ the sum of the monomials of these bases factors child by child, so

$$K_{\mathcal{A}\setminus g}=\sum_{T\in\mathcal{H}\setminus g}\theta_T\prod_{c\in T}K_{\mathcal{A}_c\setminus g_c}\prod_{c\notin T}K_{\mathcal{A}_c/g_c}=(\mathcal{H}\setminus g)^{\hom}\big((K_{\mathcal{A}_c/g_c},K_{\mathcal{A}_c\setminus g_c})_c\big),$$

and $K_{\mathcal{A}/g}=(\mathcal{H}/g)^{\hom}$ of the same pairs; the same holds for $K^\Psi$ at $\alpha^{-1}$, with the weights of $\Psi$. A mark left in place has $K_{\mathcal{A}_e\setminus g_e}=\alpha_e$ and $K_{\mathcal{A}_e/g_e}=1$, the pair of $\lambda_e$, and the same at $\alpha^{-1}$; the last coordinates multiply. At the root the same computation over the bases of $\mathcal{H}_{o,j}$ gives $K_j$, and $K_j^\Psi(\alpha^{-1})=\alpha^{-\mathcal{E}}\Psi_j(\alpha)$. So

$$f^{-1}=\alpha^{-\mathbf{m}_1}\prod_j K_j^{s_{1j}}\Psi_j^{t_{1j}}=\alpha^{|\mathbf{t}_1|\mathbf{1}-\mathbf{m}_1}\prod_j K_j^{s_{1j}}\big(K_j^\Psi(\alpha^{-1})\big)^{t_{1j}}=R_o\circ\prod_c\lambda_c,$$

with $\alpha^{|\mathbf{t}_1|\mathbf{1}-\mathbf{m}_1}=\prod_e\alpha_e^{p_e}$; and the letters vanish nowhere exactly on the preimage of $U_o$, the monomial $\alpha^{\mathcal{E}}$ being a unit of the torus.

(ii) Every map here is a morphism of smooth affine varieties: $\lambda_e$ is a closed embedding, $N_v$ is polynomial, and $U_o$, the complement of a hypersurface, is affine. For such a morphism the direct image of a holonomic module is holonomic and has no cohomology in positive degrees (a morphism of affine varieties is the closed embedding of its graph followed by a projection; the direct image under the first is exact, and under the second it is computed by the relative de Rham complex, in the degrees from minus the dimension of the fibre to 0; holonomy is kept, by Bernstein's theorem). So in the spectral sequence of a composite the only term of total degree 0 is $H^0$ of $H^0$, and no differential reaches it or leaves it: $H^0(N\circ M)_+=H^0N_+\circ H^0M_+$ on modules. Along a product of maps the direct image is the external product of the direct images (the Künneth formula for $\mathcal{D}$-modules), and in degree 0 the only term is the external product of the $H^0$. By induction from the marks up,

$$\mathbb{G}_v=H^0(N_v)_+\Big(\boxtimes_c H^0(\lambda_c)_+\mathcal{O}_{T_c}\Big)=H^0\Big(N_v\circ\prod_c\lambda_c\Big)_+\mathcal{O}_{T_v}=H^0(\lambda_v)_+\mathcal{O}_{T_v}.$$

At the root, a direct image restricted to an open set of the target is the direct image from its preimage, so by (i) $H^0(R_o|_{U_o})_+\big((\boxtimes_c\mathbb{G}_c)|_{U_o}\big)$ is the direct image along $\varphi$ of the functions of the letter complement $\widetilde{X}$ in $(\mathbb{C}^\times)^{\mathcal{E}}$. On the chart $\alpha_{e_0}=1$, $\widetilde{X}=\mathbb{C}^\times\times X_F$ and $\varphi$ does not depend on the first factor, $f$ having the degree 0; so this direct image is $\mathcal{G}_F\otimes H^1_{\mathrm{dR}}(\mathbb{C}^\times)$, the second factor $\mathbb{C}\,[d\alpha_{e_0}/\alpha_{e_0}]$ in the top degree, where $\mathcal{G}_F$ is the direct image of $\mathcal{O}_{X_F}$ along $\varphi$ in degree 0: Definition 39.47 is its relative de Rham complex, written on the forms $h\,\omega_0$, and for an integral origin $\omega_0$ is a unit of $\mathbb{C}[X_F]$ times $dy$.

(iii) For a mark, $\mathbb{G}_e$ is the direct image of the simple module $\mathcal{O}_{\mathbb{C}^\times}$ under a closed embedding, which is simple (Kashiwara's equivalence for closed embeddings), so $\varepsilon_e$ generates it. By induction let $\mathcal{M}_c\to\mathbb{G}_c$ carry $\varepsilon_c$ to the class of $\alpha^{\mathbf{q}|_{B_c}}\prod_{e\in B_c}d\alpha_e/\alpha_e$. The induced map $H^0(N_v)_+(\boxtimes_c\mathcal{M}_c)\to H^0(N_v)_+(\boxtimes_c\mathbb{G}_c)=\mathbb{G}_v$ carries $\varepsilon_v$ to the class of the product of these forms, which is the class of $\alpha^{\mathbf{q}|_{B_v}}\prod_{e\in B_v}d\alpha_e/\alpha_e$ under the identifications of (ii), these being induced by the integration of top forms along the fibres and the top form of a product being the product of the top forms. So it carries $\mathcal{M}_v$, generated by $\varepsilon_v$, onto the submodule generated by that class. At the root the factor of $\varepsilon_o$ is a unit of $\mathcal{O}(U_o)$, equal to $\prod_j K_j^{-s_{0j}}(\alpha^{-\mathcal{E}}\Psi_j)^{-t_{0j}}$ by (i), so $\varepsilon_o$ goes to the class of

$$\alpha^{\mathbf{q}}\prod_j K_j^{-s_{0j}}\big(\alpha^{-\mathcal{E}}\Psi_j\big)^{-t_{0j}}\prod_e d\alpha_e/\alpha_e=\alpha^{\mathbf{m}_0}\prod_j K_j^{-s_{0j}}\Psi_j^{-t_{0j}}\prod_e d\alpha_e/\alpha_e,$$

which on the chart is $\omega_0 \wedge d\alpha_{e_0}/\alpha_{e_0}$: the element 1 of $\mathcal{G}_F$ times the factor of the scale. So $\mathcal{M}_o$ maps onto $\mathcal{D}_F[1]$.

(iv) A quotient of a module that is a connection near $t_0$ is a connection there, and characteristic cycles are additive and nonnegative along exact sequences, so the singular values and the vanishing ranks of $\mathcal{D}_F[1]$ are bounded by those of $\mathcal{M}_o$; Theorem 39.49(ii), (iii) turns them into the nonzero roots of the symbol and their multiplicities. All the modules here are holonomic, so of finite length. For an exact sequence $0 \to M' \to M \to M'' \to 0$ the triangle of direct images and the vanishing in positive degrees give the exact sequence $H^0N_+M' \to H^0N_+M \to H^0N_+M'' \to 0$, so every composition factor of $H^0N_+M$ is one of $H^0N_+M'$ or of $H^0N_+M''$, and by induction on the length one of $H^0N_+S$ for a composition factor $S$ of $M$. The external product is exact in each factor, and the external product of simple holonomic modules is simple, so the composition factors of $\boxtimes_c M_c$ are the $\boxtimes_c S_c$; restriction to $U_o$ is exact. With $\mathcal{D}_F[g] \subseteq \mathcal{G}_F$ this gives the statements on composition factors, the $\mathbb{G}_e$ being simple by (iii). By Proposition 39.48 the module of the ray is $\mathbb{C}(k) \otimes_{\mathbb{C}[t\partial_t]} \mathcal{G}_F$, the localization at the nonzero polynomials in $t\partial_t$, which form an Ore set of $\mathcal{D}$, since $t\,p(t\partial_t) = p(t\partial_t - 1)\,t$ and $\partial_t\,p(t\partial_t) = p(t\partial_t + 1)\,\partial_t$. The localization is exact and carries a simple module to a simple module or to 0, and the module that $[g]$ generates is the localization of $\mathcal{D}_F[g]$. □

**Theorem 39.57 (The closings, the values and the native gauge).** *As stated.*

*Proof.* The origin form is $u\,dy$ with $u = \omega_0/dy$ a unit of $\mathbb{C}[X_F]$, and $\partial_{y_e} \log u = c_e^{(0)}$ (38.2(ii)), so $(k\,c_e^{(1)} + \partial_{y_e})(ug) = u\,(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})g$. Multiplication by $u$ therefore carries $\mathcal{L}$ onto the relation space of the ray from the origin with the same direction, commutes with Sh, and carries the values, the closings ($\mathsf{C}_j(g) = \mathrm{CT}(ug\,f^{-j})$) and the expansions at $\alpha^\star$ of the one ray to those of the other; so we may take the origin to be 0, $\omega_0 = dy$.

By Theorem 38.14(i), $0 < f \le f(\alpha^\star)$ on the positive orthant with equality only at $\alpha \propto \alpha^\star$, and $f \le C'e^{-c'\|y\|}$ on the chart. The family is one molecule: a union $S$ of molecules has $c_j(S) = 0$ for every $j$ (7.10), and then $\kappa_{1,S} + \kappa_{1,\mathcal{E}\setminus S} = \kappa_{1,\mathcal{E}} = 0$ (8.16), against the positive walls. The Hessian of $\log K_j(e^y)$ in $y$ is the covariance of the basis indicator $\mathbf{1}_T$ under the basis law at $e^y$ (7.16), every basis having positive weight at a real point, and that of $\log \Psi_j(e^y)$ the covariance of $\mathbf{1}_{\mathcal{E}\setminus T}$; so a real direction $u'$ in the kernel of the Hessian of $\log f$ at a real point has $\langle u', \mathbf{1}_T\rangle$ constant over the bases of every table, and $u'$ is constant on the one molecule (7.8), $u' \in \mathbb{R}\mathbf{1}$. The Hessian $H$ of $\log f$ at $\alpha^\star$ is negative definite on $\mathbb{R}^{\mathcal{E}}/\mathbb{R}\mathbf{1}$, and $\alpha^\star$ is a nondegenerate native gauge. Write $f^\star = f(\alpha^\star)$, $\hat{A}_\star = (f^\star)^{-k}k^{(n-1)/2}A_\star$, and, for $m = \sum_l r_l(k)[g_l]$, $\langle m\rangle_k = \sum_l r_l(k)\langle g_l\rangle_k$, $V_{\mathsf{C}}(m)(-j) = \sum_l r_l(-j)\,\mathsf{C}_j(g_l)$ for all large $j$, and $Q(k)$ for a common denominator of the $r_l$. By Theorem 38.6(i), the orthant near $\alpha^\star$ being a steepest-descent disc and the rest of it contributing $O((f^\star - \delta)^k)$ for some $\delta > 0$, $\langle g\rangle_k$ has the asymptotic expansion $A_\star(g)$, and $\langle m\rangle_k$ the expansion $A_\star(m)$.

(ii)⇒(iii). If $\langle m\rangle_k = 0$ at all large integers, $A_\star(m)$ is the asymptotic expansion of 0 along the integers, so $A_\star(m) = 0$.

(iii)⇒(ii), and the continuation. For $0 < t < f^\star$ the level sets $\Sigma_t = \{f = t\}$ of the real quotient are compact real-analytic hypersurfaces without critical points. By the coarea formula, $\langle g\rangle_k = \int_0^{f^\star} t^k \rho_g(t)\,dt$ for $\mathrm{Re}\,k$ large, with $\rho_g(t) = \int_{\Sigma_t} g\,dy/df$ real-analytic on $(0, f^\star)$, and by the Morse lemma at $\log\alpha^\star$, $\rho_g(t) = (f^\star - t)^{(n-3)/2}h_g(t)$ with $h_g$ real-analytic at $f^\star$. With $t = f^\star e^{-s}$, $\langle g\rangle_k = (f^\star)^{k+1}\mathcal{L}[p_g](k+1)$, $\mathcal{L}$ the Laplace transform and $p_g(s) = \rho_g(f^\star e^{-s})$, real-analytic for $s > 0$ and $s^{(n-3)/2}$ times a function analytic at 0. Write $Q(k)r_l(k) = R_l(k+1)$ with polynomials $R_l$. On distributions supported in $[0,\infty)$, $\mathcal{L}[Dv](\kappa) = \kappa\,\mathcal{L}[v](\kappa)$ with $D = d/ds$, so $Q(k)\langle m\rangle_k = (f^\star)^{k+1}\mathcal{L}[E](k+1)$ for $\mathrm{Re}\,k$ large, with $E = \sum_l R_l(D)\big(p_{g_l}\mathbf{1}_{s>0}\big)$. On $(0,\infty)$, $E$ is the real-analytic

function $\sum_l R_l(d/ds)\,p_{g_l}$; near 0 it is a convergent sum $\sum_\lambda c_\lambda s^\lambda$ times $\mathbf{1}_{s>0}$, a finite part where needed, with $\lambda \in \frac{n-3}{2}+\mathbb{Z}$ and never a negative integer (for odd $n$ the function part is analytic at 0), plus finitely many derivatives $\delta^{(b)}$ of the Dirac measure at 0. By Watson's lemma

$$\mathcal{L}[E](\kappa) \;\sim\; \sum_\lambda c_\lambda\,\Gamma(\lambda+1)\,\kappa^{-\lambda-1} + \sum_b d_b\,\kappa^b,$$

in distinct powers, with $\Gamma(\lambda+1)\neq 0$, and this is the expansion of $(f^\star)^{-k-1}Q(k)\langle m\rangle_k$, which is $(f^\star)^{-1}Q(k)k^{-(n-1)/2}\hat{A}_\star(m)$. So if $A_\star(m)=0$, every $c_\lambda$ and every $d_b$ vanishes, $E$ vanishes near 0, hence on $(0,\infty)$ by analyticity, hence $E=0$: $Q(k)\langle m\rangle_k = 0$ for $\operatorname{Re} k$ large, and the continuation of $\langle m\rangle_k$ vanishes identically.

(iii)⇒(i). By what precedes the continuation of $\langle m\rangle_k$ vanishes, and by Proposition 39.34, $V_1V_{\mathsf{C}}(m)(-j) = \lim_{k\to -j}(k+j)^{n-1}\langle m\rangle_k = 0$ for all large $j$.

(i)⇒(iii), and the expansion of the closings. The constant term of a Laurent polynomial is its mean over every torus $|\alpha_e|=\rho_e$. On the one through $\alpha^\star$, $\alpha=\alpha^\star e^{i\theta}$ with $\theta\in\mathbb{R}^{\mathcal{E}}/(2\pi\mathbb{Z}^{\mathcal{E}}+\mathbb{R}\mathbf{1})$ and $d\theta$ the measure of the chart $\theta_{e_0}=0$,

$$\mathsf{C}_j(g) = (2\pi)^{-(n-1)}\int g(\alpha^\star e^{i\theta})\,f(\alpha^\star e^{i\theta})^{-j}\,d\theta, \qquad j\geq j(g).$$

The letters have nonnegative coefficients, so $|K_j(\alpha^\star e^{i\theta})|\leq K_j(\alpha^\star)$, $|\Psi_j(\alpha^\star e^{i\theta})|\leq\Psi_j(\alpha^\star)$ and $|f^{-1}(\alpha^\star e^{i\theta})|\leq 1/f^\star$, with equality only when $\langle\theta,\mathbf{1}_T\rangle$ is the same modulo $2\pi$ for all bases $T$ of every table. For linked marks $e,e'$ some basis $T$ has $e\in T$, $e'\notin T$ and $(T\setminus\{e\})\cup\{e'\}$ a basis (7.7), so equality gives $\theta_e\equiv\theta_{e'}$; the family being one molecule, $\theta$ is constant modulo $2\pi$, and $\alpha=\alpha^\star$. So the torus outside a neighbourhood of $\alpha^\star$ contributes $O\big((1-\delta)^j(f^\star)^{-j}\big)$ for some $\delta>0$, $g\,f^{-j(g)}$ being bounded on it. Near $\theta=0$, $\log f(\alpha^\star e^{i\theta})-\log f^\star = -\frac12\theta^{\mathsf{T}}H\theta+O(|\theta|^3)$, whose real part has a nondegenerate minimum at 0, and Laplace's method there is the formal expansion at $\alpha^\star$ under the substitution $y=i\theta$, $k=-j$. The coefficient of $k^{-i}$ in $\hat{A}_\star(g)$ is a sum of Gaussian moments of covariance $(-H)^{-1}$ and of degree $2i+2L$, each times one Taylor coefficient of $g$ and $L$ of $\log f$; the substitution multiplies such a term by $i^{2i+2L}(-1)^L=(-1)^i$ and keeps the covariance, while $(2\pi i)^{-(n-1)}\prod_{e\neq e_0}d\alpha_e/\alpha_e = (2\pi)^{-(n-1)}d\theta$. This gives the expansion of $j^{(n-1)/2}(f^\star)^j\,\mathsf{C}_j(g)$ in the statement, and the same holds for $m$, with $\hat{A}_\star(m)=\sum_l r_l(k)\hat{A}_\star(g_l)$ read at $k=-j$ as a series in $1/j$. If $V_{\mathsf{C}}(m)=0$, that series is the expansion of 0, so $\hat{A}_\star(m)=0$. □

**Corollary 39.59 (When the closings miss nothing).** *As stated.*

*Proof.* By Theorem 39.57, $\operatorname{Ann}(V_{\mathsf{C}})\cap\mathfrak{N}$, $\operatorname{Ann}(\langle\cdot\rangle)\cap\mathfrak{N}$ and $\operatorname{Ann}(A_\star)\cap\mathfrak{N}$ are one submodule (Theorem 39.4(ii)), and it is 0 exactly when $\alpha^\star$ sees every nonzero submodule, being itself a submodule on which $A_\star$ vanishes. A symmetry maps the positive orthant onto itself and keeps $f$ (Proposition 39.11(ii)), so it fixes $\alpha^\star$, the one point there of largest value, and $A_\star\circ\phi^*=A_\star$ (Theorem 39.12(iii)); so $A_\star$, and with it the values and the closings, vanish on $(1-e_G)\mathfrak{M}$.

For the rest, $\mathbf{B}$ regular at $\infty$ makes the $\mathbb{C}[[1/k]]$-lattice $\Lambda$ spanned by the masters stable under $\mathrm{Sh}^{-1}$. For a submodule $\mathfrak{N}'$, $\Lambda'=\Lambda\cap(\mathfrak{N}'\otimes\mathbb{C}((1/k)))$ is a lattice of $\mathfrak{N}'\otimes\mathbb{C}((1/k))$, stable as well and saturated in $\Lambda$, so in a basis of $\Lambda'$ the matrix of $\mathrm{Sh}^{-1}$ is regular at $\infty$, with value there the restriction of $\mathbf{B}(\infty)$ to $\Lambda'/k^{-1}\Lambda'\subseteq\Lambda/k^{-1}\Lambda$. The proof of Theorem 39.45 runs for any basis over $\mathbb{C}((1/k))$ in which the matrix is regular at $\infty$, so it applies to $\mathfrak{N}'$, and each of its pieces maps by zero to the pieces of $\mathfrak{N}$ of other spectra, as in the uniqueness there, so $\mathfrak{N}'\otimes\mathbb{C}((1/k))$ is the direct sum of its intersections with the $\hat{\mathfrak{N}}_a$. $A_\star$ vanishes on the pieces other than $\hat{\mathfrak{N}}_\star$ (Theorem 39.45), and not on $\hat{\mathfrak{N}}_\star$, since it does not vanish on $\mathfrak{N}$; $\hat{\mathfrak{N}}_\star$ being a line, $A_\star$ vanishes on $\mathfrak{N}'$ exactly when

$\hat{\mathfrak{N}}_\star \not\subseteq \mathfrak{N}' \otimes \mathbb{C}((1/k))$. If every nonzero submodule contains $\hat{\mathfrak{N}}_\star$, any two of them meet in a nonzero submodule, completion commuting with intersection, so the intersection of all of them is nonzero and is the only minimal one. Of two nonzero submodules with intersection 0 at most one contains $\hat{\mathfrak{N}}_\star$, and the closings annihilate the other. □

**Corollary 39.60 (What the closings determine).** *As stated.*

*Proof.* The kernel of the values is a submodule, so Sh induces on the value module a bijection with $\mathrm{Sh}(r\,m) = \varsigma(r)\,\mathrm{Sh}(m)$, and by Theorem 39.57, $V_{\mathsf{C}}$ induces on it a solution that annihilates nothing; the proofs of Theorem 39.4(i), (iii) and (iv) use no more than this. For the recurrences, $\sum_i p_i(k)\langle 1\rangle_{k+i}$ is the value of $m = \sum_i p_i(k)\,\mathrm{Sh}^i[1]$, and $V_{\mathsf{C}}(m)(-j) = \sum_i p_i(-j)\,\mathsf{C}_{j-i}(1)$; by Theorem 39.57 the one vanishes for all large arguments exactly when the other does, so the least orders agree. □

**Proposition 39.61 (The closings measure what they miss).** *As stated.*

*Proof.* The closings are a solution (Theorem 39.33), and as in the proof of Theorem 39.4(iii), $(P, Q) \in \mathcal{S}_D$ exactly when every row of $Q\mathbf{B_b} - P$, read in $\mathfrak{N}$, lies in $\mathfrak{A} = \mathrm{Ann}(V_{\mathsf{C}}) \cap \mathfrak{N}$. In the coordinates of the masters $\mathfrak{A}$ is a $d$-dimensional subspace of $\mathbb{C}(k)^{1\times\mathfrak{m}}$; let $\mathfrak{A}_D$ be its rows of polynomials of degree at most $D$. Projecting to $d$ coordinates on which $\mathfrak{A}$ projects isomorphically gives $\dim \mathfrak{A}_D \le d(D+1)$, and polynomial rows $a_1, \dots, a_d$ spanning $\mathfrak{A}$, of degrees $\nu_i$, give the independent rows $\sum_i c_i a_i$, $\deg c_i \le D - \nu_i$, so $\dim \mathfrak{A}_D = dD + O(1)$. On $\mathcal{S}_D$ the map $(P, Q) \mapsto Q$ has fibres that are empty or translates of $\mathfrak{A}_D^{\mathfrak{m}}$, each row of $P$ being fixed modulo $\mathfrak{A}$ by $Q$; its image lies in the polynomials of degree at most $D$ and contains $cQ_0$ for $\deg c \le D - D_0$. So $\mathfrak{m}\dim \mathfrak{A}_D + D - D_0 + 1 \le \dim \mathcal{S}_D \le \mathfrak{m}\dim \mathfrak{A}_D + D + 1$ for $D \ge D_0$, which is $(\mathfrak{m}d+1)D + O(1)$; for $d = 0$, Theorem 39.4(iv) gives $\dim \mathcal{S}_D = \max(0, D - D_0 + 1)$, whose steps are 0 and 1. □

**Proposition 39.62 (A table one mark larger).** *As stated.*

*Proof.* By Corollary 20.19, $\mathcal{B}'$ has no loop or coloop and its letters agree, $\Psi' = K'$. The bases of $\mathcal{B}'$ through $g$, less $g$, are the bases of $\mathcal{B}$, and complementation in $\mathcal{E}'$ carries the bases through $g$ onto the bases avoiding $g$, with the same weights; so $K' = \sum_{T \not\ni g} \theta'_T \alpha^T + \alpha_g \sum_{T \ni g} \theta'_T \alpha^{T\setminus g} = \Psi + \alpha_g K$, and $K(\mathbf{1}) = \Psi(\mathbf{1})$. The deletion $\mathcal{B}' \setminus g$ is the dual of the contraction, and a connected table has a connected deletion or a connected contraction at each mark (Tutte), duality keeping connectedness; so $\mathcal{B}$ is connected, and both centre rays have positive walls (Definition 38.13). Write $x_{a,j}(h) = \alpha_g^a K'^{-j} h$; the monomials and the chances of $\mathcal{B}'$ are of this form, so these span $\mathbb{C}[X_{F'}]$.

(i) For $\alpha$ in the positive orthant, $\int_0^\infty \rho^{\nu-1}(c_0 + c_1\rho)^{-s}\,d\rho = \Gamma_1^2\left[\begin{matrix}\nu,\ s-\nu\\ s\end{matrix}\right] c_0^{\nu-s} c_1^{-\nu}$ gives

$$\int_0^\infty \alpha_g^{k+a}(\Psi + \alpha_g K)^{-2k-j}\,\frac{d\alpha_g}{\alpha_g} = \Gamma_1^2\left[\begin{matrix}k+a,\ k+j-a\\ 2k+j\end{matrix}\right] \Psi^{a-j-k} K^{-a-k}, \qquad k > \max(-a, a-j),$$

and with $f'^k = \alpha^{k\mathbf{1}}\alpha_g^k K'^{-2k}$, $f^k = \alpha^{k\mathbf{1}}(K\Psi)^{-k}$ and $\Gamma_1^2\left[\begin{matrix}k+a,\ k+j-a\\ 2k+j\end{matrix}\right] = r_{a,j}(k)\Gamma_1^2\left[\begin{matrix}k,\ k\\ 2k\end{matrix}\right]$ this is the identity for $g' = x_{a,j}(h)$. The integral is linear in $g'$, so on a combination of the $x_{a,j}(h)$ the defining formula depends only on the function they sum to, both sides being rational in $k$ and equal for all large $k$; so $\pi_*$ is well defined and $\mathbb{C}(k)$-linear on $\mathbb{C}[X_{F'}] \otimes \mathbb{C}(k)$, and the identity holds for every insertion. On the chart $y_{e_0} = 0$ the form of $F'$ is $dy\,d\alpha_g/\alpha_g$, and its positive orthant is that of $F$ times $(0, \infty)$; Fubini gives the values. The constant term in $\alpha_g$ of $x_{a,j}(h) f'^{-J} = \alpha_g^{a-J}\alpha^{-J\mathbf{1}} h\,(\Psi + \alpha_g K)^{2J-j}$ is $\binom{2J-j}{J-a}\Psi^{J-j+a}K^{J-a}\alpha^{-J\mathbf{1}}h$, and for $J \ge |a| + j$,

$\Gamma(x-J)/\Gamma(-J) = (-1)^x J!/(J-x)!$ gives $r_{a,j}(-J) = \binom{2J-j}{J-a}\big/\binom{2J}{J}$; so that constant term is $\binom{2J}{J}\,\pi_*\big(x_{a,j}(h)\big)\big|_{k=-J} f^{-J}$, and the constant term in the remaining scales gives the closings.

On $X_{F'}$, $c_g'^{(1)} = \partial_{y_g} \log f' = 1 - 2\alpha_g K/K'$, which is $-1$ where $\Psi = 0$ and 1 where $K = 0$; where $K\Psi \neq 0$ it vanishes exactly at $\alpha_g = \Psi/K$, where $K' = 2\Psi$, $f' = f/4$ and $\partial^2_{y_g} \log f' = -2\,\frac{\alpha_g K}{K'}\,\frac{\Psi}{K'} = -\frac{1}{2}$. So $\log f(\alpha) = \log 4 + \log f'(\alpha, \Psi/K)$ on $X_F$, the defects of $F'$ at $(\alpha, \Psi/K)$ in the marks of $\mathcal{E}$ are those of $F$ at $\alpha$, the native gauges of $F'$ are the points $\alpha_{c'}$, and the Hessian of $\log f$ at $\alpha_c$ is the Schur complement of the entry $-\frac{1}{2}$ in the Hessian of $\log f'$ at $\alpha_{c'}$, whose determinant is $-\frac{1}{2}$ times that of the complement. Over a steepest-descent disc $D_c$, with Morse coordinates $v$, put $\alpha_g = e^s\,\Psi/K$; then $d\alpha_g/\alpha_g = ds$ on the fibre and

$$\log f' = \log f(\alpha_c) - \log 4 - \tfrac{1}{2}\textstyle\sum_i v_i^2 - 2\log\cosh\frac{s}{2},$$

so with the real-analytic coordinate $w$ of the sign of $s$ and $\frac{1}{2}w^2 = 2\log\cosh\frac{s}{2}$, the image of $D_c \times \{|s| \le \varepsilon\}$ is a steepest-descent disc at $\alpha_{c'}$. The integral over all $s$ is the Beta corner, the part with $|s| > \varepsilon$ is exponentially smaller, and the expansion at $\alpha_{c'}$ is that of the Beta corner times $A_c(\pi_* g')$.

(ii) Let $\rho'_e(g') = (k\,c_e'^{(1)} + \partial_{y_e})g'$, so that $f'^k \rho'_e(g') = \partial_{y_e}(f'^k g')$, and likewise $\rho_e$ for $F$. Differentiating the identity of (i) under the integral, $\pi_*\rho'_e = \rho_e \pi_*$ for $e \in \mathcal{E}$; and $\pi_*\rho'_g = 0$, the integral of $\partial_{y_g}(f'^k g')$ over $(0,\infty)$ vanishing for large $k$. Both are identities between elements of $\mathbb{C}[X_F] \otimes \mathbb{C}(k)$ that hold for all large $k$, so they hold. Next,

$$\rho'_g\big(x_{a,j}(h)\big) = (k+a)\,x_{a,j}(h) - (2k+j)\,x_{a+1,j+1}(Kh), \qquad x_{a,j}(h) = x_{a,j+1}(\Psi h) + x_{a+1,j+1}(Kh),$$

so modulo the span $\mathcal{L}'_g$ of the $\rho'_g(g')$

$$x_{a,j}(h) \equiv \frac{2k+j}{k+a}\,x_{a+1,j+1}(Kh) \equiv \frac{2k+j}{k+j-a}\,x_{a,j+1}(\Psi h).$$

A finite sum $z = \sum_i c_i x_{a_i,j_i}(h_i)$ is therefore congruent modulo $\mathcal{L}'_g$ to one term $x_{A,A+B}(H)$, for any $A \ge \max_i a_i$ and $B \ge \max_i(j_i - a_i)$, with $H$ a Laurent polynomial, and $\pi_* z = r_{A,A+B} K^{-A} \Psi^{-B} H$; if $\pi_* z = 0$ then $H = 0$ and $z \in \mathcal{L}'_g$. And $\pi_*$ is onto: $\pi_* x_{a,a+b}(\alpha^u) = r_{a,a+b}\,\alpha^u K^{-a}\Psi^{-b}$ for $a, b \ge 0$, with $r_{a,a+b} \neq 0$, and these span $\mathbb{C}[X_F]$. So $\pi_*$ is a bijection of $(\mathbb{C}[X_{F'}] \otimes \mathbb{C}(k))/\mathcal{L}'_g$ onto $\mathbb{C}[X_F] \otimes \mathbb{C}(k)$ that carries $\mathcal{L}' = \mathcal{L}'_g + \sum_{e\in\mathcal{E}} \rho'_e(\cdot)$ onto $\sum_{e\in\mathcal{E}} \rho_e(\cdot) = \mathcal{L}$, and it induces a bijection $\mathfrak{M}' \to \mathfrak{M}$. Finally $\Gamma{}^2_1\begin{bmatrix} k+1,\ k+1 \\ 2k+2 \end{bmatrix} = \frac{k}{2(2k+1)}\Gamma{}^2_1\begin{bmatrix} k,\ k \\ 2k \end{bmatrix}$, so $\pi_*(f'g') = \frac{k}{2(2k+1)}\, f \wr (\pi_* g')$.

(iii) By (ii) $T_\beta$ is a $\mathbb{C}(k)$-linear bijection of $\mathfrak{M}$. $\phi_\beta$ is a symmetry of $F'$ (Proposition 39.11(ii)), so $\phi_\beta^*$ commutes with $\mathrm{Sh}'$ (Theorem 39.12(i)); with $c = \frac{k}{2(2k+1)}$, $\mathrm{Sh} = c^{-1}\pi_* \mathrm{Sh}'\, \pi_*^{-1}$, and then $T_\beta \mathrm{Sh} = c^{-1}\pi_*\phi_\beta^* \mathrm{Sh}'\, \pi_*^{-1} = c^{-1}\pi_* \mathrm{Sh}'\, \phi_\beta^* \pi_*^{-1} = \mathrm{Sh}\, T_\beta$. If $\beta(g) = g$, comparing the coefficients of $\alpha_g^0$ and $\alpha_g^1$ in $K' \circ \phi_\beta = K'$ gives $K \circ \phi_\beta = K$ and $\Psi \circ \phi_\beta = \Psi$, so $\phi_\beta$ restricted to $\mathcal{E}$ is a symmetry of $F$ and $\pi_* \phi_\beta^* = \phi_\beta^* \pi_*$ on the $x_{a,j}(h)$. For the transposition of $e$ and $g$, $\phi_\beta^* \alpha^u = x_{u_e,0}(\alpha^{u - u_e \mathbf{1}_e})$, and $\pi_*$ of it is the stated class, with $r_{u_e,0}(k) = \Gamma(k+u_e)\Gamma(k-u_e)/\Gamma(k)^2$. For the Cremona, $K' \circ \mathrm{cr} = \alpha^{-\mathbf{1}} \alpha_g^{-1} K'$, $K \circ \mathrm{cr} = \alpha^{-\mathbf{1}}\Psi$ and $\Psi \circ \mathrm{cr} = \alpha^{-\mathbf{1}} K$ (Proposition 39.11(i)), so $\mathrm{cr}^*\, x_{a,j}(h) = x_{j-a,j}\big(\alpha^{j\mathbf{1}}\, h \circ \mathrm{cr}\big)$, and both $\pi_* \mathrm{cr}^*\, x_{a,j}(h)$ and $\mathrm{cr}^*\, \pi_* x_{a,j}(h)$ are $r_{a,j}\, \Psi^{-a} K^{a-j} \alpha^{j\mathbf{1}}\, h \circ \mathrm{cr}$, as $r_{j-a,j} = r_{a,j}$. The group generated is the image of the finite group generated by the $\phi_\beta$ and the Cremona of $F'$.

(iv) The values and the closings of $F'$ are kept by every $\phi_\beta$ and by its Cremona (Proposition 39.11(ii); the torus and the constant term are kept by every permutation of the scales and by their inversion). By (i) the values and the closings of $F$ are those of $F'$ composed with $\pi_*^{-1}$,

divided by $\Gamma_{\substack{2\\1}}\begin{bmatrix}k,\ k\\ 2k\end{bmatrix}$ and by $\binom{2j}{j}$, which do not depend on the insertion; so they are invariant under $G'$. The native gauge of the direction of $F'$ on the positive orthant is unique up to scale (Theorem 38.14(i)), so it is fixed by every $\phi_\beta$ and by the Cremona, and it is $\mathbf{1}$; on the orthant $f' \le f/4$, with equality exactly at $\alpha_g = \Psi/K$, so it lies over $\alpha^\star$, and $\alpha^\star = \mathbf{1}$. Then $A'_\star \circ \phi^*_\beta = A'_\star$ (Theorem 39.12(iii)), and $A_\star \circ T_\beta = A_\star$ by (i). A solution $V$ invariant under $G'$ has $V = V \circ e_{G'}$ and vanishes on $(1 - e_{G'})\mathfrak{M}$. If $G \subseteq G'$, then $e_{G'}e_G = e_{G'}$, so $e_{G'}$ maps $\mathfrak{M}^G$ onto $\mathfrak{M}^{G'} \subseteq \mathfrak{M}^G$, and $\mathfrak{M}^G = e_{G'}\mathfrak{M}^G \oplus (1 - e_{G'})\mathfrak{M}^G$, both submodules since $e_{G'}$ commutes with Sh. Last, $\pi_*$ intertwines the action on $\mathfrak{M}'$ with $G'$, so it carries the invariant module of $F'$ onto $\mathfrak{M}^{G'}$. □

**Corollary 39.63 (The cousins of a fused table).** *As stated.*

*Proof.* $\mathcal{B}^\sharp$ is connected by Corollary 30.7(ii), and identically self-dual. For a mark $x$ of $\mathcal{E} \sqcup G$ the cousin at $x$ is the contraction $\mathcal{B}^\sharp/x$, and Proposition 39.62 applies with $(\mathcal{B}', \theta') = (\mathcal{B}^\sharp, \theta^\sharp)$, its $n + c - 1 \ge 2$ marks other than $x$, and the mark $x$ in the place of $g$. □

**Proposition 39.64 (What a table one mark larger hides from the closings).** *As stated.*

*Proof.* (i) For a nondegenerate native gauge $\alpha_{c'}$ of $F'$ and $\gamma \in \Gamma'$, $A'_{c'} \circ \gamma^*$ is a nonzero constant multiple of the expansion over the steepest-descent disc $\gamma(D_{c'})$ at $\gamma(\alpha_{c'})$, and $A'_{c'} \circ \psi^* = A'_{c'}$ when $\psi \in \Gamma'$ fixes $\alpha_{c'}$ (proof of Theorem 39.12(iii)). Two $\gamma$ that carry $\alpha_{c'_0}$ to one point differ by an element of its stabilizer, and $(\gamma\psi)^* = \psi^*\gamma^*$, so the transported $A'_{c'}$ is well defined, and $A'_{c'} \circ \chi^* = A'_{\chi(c')}$ for every $\chi \in \Gamma'$. By Proposition 39.62(i), $A_c$ is a nonzero constant multiple of the expansion of $F$ at $\alpha_c$: a solution of the type $\big(f(\alpha_c), -\frac{n-1}{2} + \mathbb{Z}\big)$, $n = |\mathcal{E}|$, whose leading coefficient on an insertion $h$ is $C_c\, h(\alpha_c)$ with $C_c \ne 0$ (Theorem 38.6(i)). The native gauges below one orbit $O'_a$ have one native value, $4f'$ at the points of $O'_a$, which $\Gamma'$ keeps, and so one type.

Every element of $G'$ is $T_\gamma = \pi_*\gamma^*\pi_*^{-1}$ for some $\gamma \in \Gamma'$ (Proposition 39.62(iii)), and $A_c \circ T_\gamma = \Gamma_{\substack{2\\1}}\begin{bmatrix}k,\ k\\ 2k\end{bmatrix}^{-1} A'_{\gamma(c')} \circ \pi_*^{-1} = A_{\gamma\cdot c}$, with $\gamma\cdot c$ the native gauge below $\gamma(\alpha_{c'})$. For $\phi \in G$, $\phi^* = T_\gamma$ with $\gamma$ the map of the same kind on the torus of $F'$, and the lift commutes with $\phi$: $\phi_\beta$, $\beta(g) = g$, keeps $K$ and $\Psi$, and $\Psi(\alpha^{-1})/K(\alpha^{-1}) = K(\alpha)/\Psi(\alpha)$ (Proposition 7.2). So $\gamma\cdot c = \phi(\alpha_c)$ and $A_c \circ \phi^* = A_{\phi(\alpha_c)}$. Let $O_{a,1}, \dots, O_{a,s_a}$ be the orbits of $G$ below $O'_a$ and $V_{a,i} = \sum_{c \in O_{a,i}} A_c$; each $\phi \in G$ permutes $O_{a,i}$, so $V_{a,i}$ is invariant under $G$. For $x \in \mathfrak{M}^{G'}$ and native gauges $\alpha_c$, $\alpha_{\tilde c}$ below $O'_a$, with $\gamma(\alpha_{c'}) = \alpha_{\tilde c'}$, $A_{\tilde c}(x) = A_c(T_\gamma x) = A_c(x)$; so $V_{a,i}(x) = |O_{a,i}|\, A_c(x)$ for any $\alpha_c$ below $O'_a$, and

$$W_{a,i} = |O_{a,i}|^{-1} V_{a,i} - |O_{a,1}|^{-1} V_{a,1}, \qquad 2 \le i \le s_a,$$

vanish on $\mathfrak{M}^{G'}$.

The $V_{a,i}$ of one type are linearly independent over $\mathbb{C}$ on $\mathfrak{M}^G$. Let $\sum \gamma_{a,i} V_{a,i}$ vanish on $\mathfrak{M}^G$, the sum over those of one type, and fix $(a, i)$ and $\alpha_c \in O_{a,i}$. As in the proof of Theorem 39.12(iii), take an insertion $h$ with $h(\alpha_c) = 1$ that vanishes to the order $N \ge 2$ at the other native gauges below $O'_1, \dots, O'_q$, which are finitely many; the expansion at such a point is smaller than the leading power of the type by the factor $k^{-\lceil N/2 \rceil}$. The class of $e_G h$ lies in $\mathfrak{M}^G$, and every $V_{a',i'}$, being invariant, takes the same value at $e_G h$ and at $h$; so the leading coefficient of the relation at $e_G h$ is $\gamma_{a,i} C_c$, and $\gamma_{a,i} = 0$. Each $W_{a,i}$ involves its own $V_{a,i}$, so the $W_{a,i}$ of one type are linearly independent over $\mathbb{C}$ on $\mathfrak{M}^G$. They vanish on $\mathfrak{M}^{G'}$, and $\mathfrak{M}^G = \mathfrak{M}^{G'} \oplus (1 - e_{G'})\mathfrak{M}^G$ (Proposition 39.62(iv)), so a combination of them that vanishes on $(1 - e_{G'})\mathfrak{M}^G$ vanishes on $\mathfrak{M}^G$: they are linearly independent over $\mathbb{C}$ on $(1 - e_{G'})\mathfrak{M}^G$, type by type. Lemma 39.3 gives the bound; $(1 - e_{G'})\mathfrak{M}^G$ is a submodule,

$e_{G'}$ commuting with Sh, and the closings, the values and $A_\star$ vanish on it by Proposition 39.62(iv). The solution of the last statement is $-W_{a,2}$ for an indexing with $O_{a,1} = O_1$ and $O_{a,2} = O_2$.

(ii) A nonempty proper $S \subseteq [2r+2]$ meets the bases of $\binom{[2r+2]}{r+1}$ in every number of marks from $\max(0, |S| - r - 1)$ to $\min(|S|, r+1)$, so $c(S) > 0$ and the table is connected (Definition 8.1); the complement of a set of $r+1$ of its marks has $r+1$ marks, so it is identically self-dual; and its contraction at $g$ (Definition 7.3) has the bases $T \setminus \{g\}$, $T \ni g$, the sets of $r$ marks of $[2r+1]$, with $K = e_r$ and $\Psi = e_{r+1}$, the elementary symmetric polynomials of $\alpha_1, \dots, \alpha_{2r+1}$. So Proposition 39.62 applies, $|\mathcal{E}| = 2r+1 \geq 5$. Every bijection of $[2r+2]$ is an automorphism, so $G$ consists of the $\phi_\beta$, $\beta$ a bijection of $[2r+1]$, and their composites with the Cremona. These are the symmetries of the four kinds of the centre ray (Proposition 39.11(ii), the masses being $\mathbf{1}$ and the exponents $s = t = 1$): a bijection that exchanged the list would carry its bases, of $r$ marks, onto complements of bases, of $r+1$ marks, so there are no exchanges and no reversed exchanges.

At a sign $\eta$ with $i$ coordinates $-1$ among $l+i$, $e_m(\eta)$ is the coefficient of $z^m$ in $(1-z)^i(1+z)^l$. With $\binom{2r}{r-1} = \binom{2r}{r+1} = rC_r$, $\binom{2r-1}{r-1} = \binom{2r-1}{r} = \frac{r+1}{2}C_r$ and $\binom{2r-1}{r-2} = \frac{r-1}{2}C_r$,

$$[z^r](1-z)(1+z)^{2r} = C_r, \qquad [z^r](1-z)^2(1+z)^{2r-1} = -C_r,$$

so $K(\eta) = e_r(\eta) = \pm C_r \neq 0$ at the signs of $F$ with one and with two scales $-1$, which are native gauges of the native value $K(\eta)^{-2} = C_r^{-2}$ (Theorem 38.17(ii)). The bijections of $[2r+1]$ act transitively on the marks and on the pairs of marks, the Cremona fixes every sign, and a sign with one coordinate $-1$ is not a multiple of one with two, $2r+1 \geq 5$: the signs form two orbits of $G$. At a sign $\eta^{-1} = \eta$, so $\Psi(\eta) = \eta^{\mathcal{E}} K(\eta)$ (Proposition 7.2), and the native gauge above $\eta$ is $(\eta, \eta^{\mathcal{E}})$, with $\eta^{\mathcal{E}} = -1$ for one scale $-1$ and $\eta^{\mathcal{E}} = 1$ for two: the native gauges above the two kinds are the $\binom{2r+2}{2}$ signs of $F'$ with two scales $-1$, each once, and the bijections of $[2r+2]$ act transitively on them.

At such a sign $\eta'$, with $D = \{a_1, a_2\}$ and $C$ as in the statement, $a \in D$ and $c, c' \in C$, the Hessian of $-\log f'$ is $2\kappa$, $\kappa$ the covariances of the indicators $X_e$ of the marks under the signed measure $\mathbb{P}(T) = \eta'^T / K'(\eta')$ on the sets $T$ of $r+1$ marks (Theorem 38.17(ii), $K' = \Psi'$); as a matrix on $\mathbb{C}^{\mathcal{E}'}$ it has zero row sums, $\sum_e X_e = r+1$. From $[z^{r+1}](1-z)^2(1+z)^{2r} = 2rC_r - 2(r+1)C_r$, $[z^{r-1}](1-z)(1+z)^{2r-1} = C_r$ and, with $\binom{2r-2}{r-1} = \frac{r(r+1)}{2(2r-1)}C_r$,

$$[z^{r-1}](1-z)^2(1+z)^{2r-2} = \binom{2r-2}{r-1}\Big(1 - \frac{2(r-1)}{r} + \frac{(r-1)(r-2)}{r(r+1)}\Big) = -\frac{r-2}{2r-1}\,C_r,$$

$$K'(\eta') = -2C_r, \qquad \mathbb{P}(e \in T) = \tfrac{1}{2}, \qquad \mathbb{P}(a_1, a_2 \in T) = \frac{rC_r}{-2C_r} = -\frac{r}{2},$$
$$\mathbb{P}(a, c \in T) = \frac{1}{2}, \qquad \mathbb{P}(c, c' \in T) = \frac{r-2}{2(2r-1)},$$

the first for every mark, as $\eta'_e e_r(\eta'_{\hat{e}}) = -C_r$ for $e \in D$ and for $e \in C$. So $\kappa_{ee} = \frac{1}{4}$, $\kappa_{a_1a_2} = -\frac{2r+1}{4}$, $\kappa_{ac} = \frac{1}{4}$ and $\kappa_{cc'} = -\frac{3}{4(2r-1)}$, and

$$\kappa(\mathbf{1}_{a_1} - \mathbf{1}_{a_2}) = \tfrac{r+1}{2}(\mathbf{1}_{a_1} - \mathbf{1}_{a_2}), \qquad \kappa(\mathbf{1}_c - \mathbf{1}_{c'}) = \tfrac{r+1}{2(2r-1)}(\mathbf{1}_c - \mathbf{1}_{c'}),$$
$$\kappa(r\mathbf{1}_D - \mathbf{1}_C) = -\tfrac{r+1}{2}(r\mathbf{1}_D - \mathbf{1}_C),$$

the last from its coordinates $r(\frac{1}{4} - \frac{2r+1}{4}) - \frac{2r}{4} = -\frac{r(r+1)}{2}$ at $a$ and $\frac{2r}{4} - \frac{1}{4} + \frac{3}{4} = \frac{r+1}{2}$ at $c$, where $r\mathbf{1}_D - \mathbf{1}_C$ has the coordinates $r$ and $-1$. With $c_1 \in C$, the vectors $\mathbf{1}_{a_1} - \mathbf{1}_{a_2}$, $r\mathbf{1}_D - \mathbf{1}_C$ and

$\mathbf{1}_{c_1} - \mathbf{1}_c$, $c \in C \setminus \{c_1\}$, are $2r+1$ independent vectors orthogonal to $\mathbf{1}$, which gives the eigenvalues of $-H' = 2\kappa$ modulo $\mathbf{1}$; they do not vanish, so $\eta'$ is nondegenerate, and so is the sign below it (Proposition 39.62(i)). A symmetric matrix $A$ of size $N'$ with $A\mathbf{1} = 0$ and rank $N'-1$ has an adjugate of rank one killed by $A$, so equal to $\lambda\,\mathbf{1}\mathbf{1}^{\mathsf{T}}/N'$ with $\lambda$ the sum of its principal minors of size $N'-1$, the product of the eigenvalues of $A$ modulo $\mathbf{1}$; so every such minor is $\lambda/N'$, and on a chart

$$\det(-H') = \frac{(r+1)\cdot(-(r+1))\cdot\left(\frac{r+1}{2r-1}\right)^{2r-1}}{2r+2} = -\frac{(r+1)^{2r}}{2(2r-1)^{2r-1}}.$$

The Hessian of $\log f$ at $\eta$ is the Schur complement of the entry $-\frac{1}{2}$ in that of $\log f'$ at $\eta'$ (proof of Proposition 39.62(i)), so on charts with $e_0 \in \mathcal{E}$, $\det H = -2\det H'$ and $\det(-H) = 2\det(-H')$, which is the stated value. Part (i) with the one orbit of the signs of $F'$ with two scales $-1$, below which lie two orbits of $G$, gives $(1-e_{G'})\mathfrak{M}^G \neq 0$, and the closings vanish on it. □

**Proposition 39.65 (The holonomy of a wheel).** *As stated.*

*Proof.* Put $S(a) = \mathbf{U}(a)$ and $R(b) = \begin{bmatrix} b & 0 \\ 1 & b \end{bmatrix} = b\,\mathbf{L}(b^{-1})$, and $M(\alpha) = \prod_i S(\alpha_{s_i})R(\alpha_{r_i}) = \prod_i \alpha_{r_i} \cdot \mathbf{h}(\alpha)$. Index rows and columns by the states $0, 1$ and read the factors from left to right as steps around the rim, $S(\alpha_{s_i})$ at the vertex $i$ and $R(\alpha_{r_i})$ at the rim $r_i$: $S$ allows $0 \to 0$, $0 \to 1$ with the weight $\alpha_{s_i}$, and $1 \to 1$; $R$ allows $0 \to 0$ and $1 \to 1$ with the weight $\alpha_{r_i}$, and $1 \to 0$ with the weight 1. Then $\operatorname{tr} M$ is the sum over the closed walks of states of the products of the weights. A spanning tree of the wheel omits at least one rim; the rims it omits cut the rim cycle into arcs of consecutive vertices, joined by the rims it contains, and it contains exactly one spoke at a vertex of each arc, at least one for connection and at most one, two closing a cycle through the hub; every such choice is a spanning tree. To a tree corresponds the closed walk in state 0 at the start of each arc and in state 1 from the vertex of its spoke on, leaving 1 for 0 exactly at the omitted rims; its weight is $\alpha^T$. The closed walks that never go from 1 to 0 stay in one state all around, taking every rim and no spoke: two walks of weight $\prod_i \alpha_{r_i}$. So $\operatorname{tr} M = K + 2\prod_i \alpha_{r_i}$, which is the formula for $K$. $\det S = 1$ and $\det R(b) = b^2$, so $\det \mathbf{h} = 1$. By 7.2, $\Psi(\alpha) = \alpha^{\mathcal{E}} K(\alpha^{-1}) = \prod_i \alpha_{s_i}\alpha_{r_i} \cdot \prod_i \alpha_{r_i}^{-1}\big(\operatorname{tr}\mathbf{h}(\alpha^{-1}) - 2\big)$.

The transpose of a product is the product of the transposes in the reverse order, $\mathbf{U}(a)^{\mathsf{T}} = \mathbf{L}(a)$ and $\mathbf{L}(b)^{\mathsf{T}} = \mathbf{U}(b)$, so

$$\operatorname{tr}\mathbf{h}(\alpha^{-1}) = \operatorname{tr}\Big(\prod_{i=1}^{p} \mathbf{U}(\alpha_{s_i}^{-1})\mathbf{L}(\alpha_{r_i})\Big)^{\mathsf{T}} = \operatorname{tr}\big(\mathbf{U}(\alpha_{r_p})\mathbf{L}(\alpha_{s_p}^{-1})\cdots\mathbf{U}(\alpha_{r_1})\mathbf{L}(\alpha_{s_1}^{-1})\big) = \operatorname{tr}\mathbf{h}(\alpha\circ x),$$

the factor $i$ of the last product being $\mathbf{U}(\alpha_{r_{p+1-i}})\mathbf{L}(\alpha_{s_{p+1-i}}^{-1}) = \mathbf{U}\big((\alpha\circ x)_{s_i}\big)\mathbf{L}\big((\alpha\circ x)_{r_i}^{-1}\big)$. Then $K(\alpha\circ x) = \prod_i \alpha_{x(r_i)}\big(\operatorname{tr}\mathbf{h}(\alpha\circ x) - 2\big) = \prod_i \alpha_{s_i}\big(\operatorname{tr}\mathbf{h}(\alpha^{-1}) - 2\big) = \Psi(\alpha)$. Comparing the monomials, all with coefficient 1, the sets $x(T)$ of the bases $T$ are the complements of the bases, so $T \mapsto \mathcal{E}\setminus x(T)$ maps the bases onto the bases: $x$ exchanges the list. At $\alpha = \mathbf{1}$, $\mathbf{U}(1)\mathbf{L}(1) = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, of trace 3 and determinant 1, with the eigenvalues $\lambda_\pm$, the squares of the golden ratio and of its inverse, and $\mathbf{h}(\mathbf{1})$ is its $p$-th power, of trace $\lambda_+^p + \lambda_-^p$, the Lucas number of index $2p$, the Lucas numbers being the sums of the powers of the golden ratio and of minus its inverse. □

**Proposition 39.66 (The centre ray of a wheel).** *As stated.*

*Proof.* By 38.13, $f = \alpha^{\sigma\mathcal{E}}(K\Psi)^{-\sigma}$, and by Proposition 39.65, $K\Psi = \alpha^{\mathcal{E}}\,\mathfrak{d}\,\mathfrak{d}^*$; so $f = (\mathfrak{d}\,\mathfrak{d}^*)^{-\sigma}$, and $X_F = \{K\Psi \neq 0\} = \{\mathfrak{d}\,\mathfrak{d}^* \neq 0\}$ on the torus. (i) The native gauges are the critical points of

$\log f = -\sigma(\log \mathfrak{d} + \log \mathfrak{d}^*)$ on $X_F$ (38.2(ii), 38.4), where $\mathfrak{d}^* d\mathfrak{d} + \mathfrak{d}\, d\mathfrak{d}^* = 0$, and then $d\mathfrak{d}^* = -(\mathfrak{d}^*/\mathfrak{d})\, d\mathfrak{d}$. (ii) $\eta^{-1} = \eta$, and $\mathfrak{d}(\mathbf{1}) = K(\mathbf{1})$. (iii) Since $\det \mathbf{h} \equiv 1$, $\operatorname{tr}(\mathbf{h}^{-1} d\mathbf{h}) = d \log \det \mathbf{h} = 0$; where $\mathbf{h} = -I$ this is $-\operatorname{tr} d\mathbf{h} = -d\mathfrak{d} = 0$, and the same holds for $\mathfrak{d}^*$ where $\mathbf{h}(\alpha^{-1}) = -I$. On the torsion stratum $\mathfrak{d} = \mathfrak{d}^* = -4$, so it lies in $X_F$, $d \log f = 0$ there and $f = 16^{-\sigma}$. Each of $\mathbf{h}(\alpha) = -I$ and $\mathbf{h}(\alpha^{-1}) = -I$ is three equations, $\mathbf{h}_{12} = \mathbf{h}_{21} = 0$ and $\mathbf{h}_{11} = -1$, the fourth following from $\det \mathbf{h} = 1$. They do not change under scaling, $\mathbf{h}(c\alpha)$ being $\mathbf{h}(\alpha)$ conjugated by $\operatorname{diag}(c^{1/2}, c^{-1/2})$; on the chart $\alpha_{e_0} = 1$, a torus of dimension $2p-1$, every component of the zero set of six functions has dimension at least $2p-7$. □

**Proposition 39.67 (The classes of a wheel).** *As stated.*

*Proof.* The number of spanning trees that contain $s_1$ is the derivative of $K$ in $\alpha_{s_1}$ at $\mathbf{1}$. With $S$, $R$ and $M$ as in the proof of Proposition 39.65, $K = \operatorname{tr} M - 2\prod_i \alpha_{r_i}$, and the derivative of $S$ is $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, so that number is the entry $(2,1)$ of $R(1) M_1^{p-1}$, $M_1 = S(1)R(1) = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$. With the Fibonacci numbers $\mathrm{Fib}_m$, $M_1 = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^2$ and $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^m = \begin{bmatrix} \mathrm{Fib}_{m+1} & \mathrm{Fib}_m \\ \mathrm{Fib}_m & \mathrm{Fib}_{m-1} \end{bmatrix}$; the second row of $R(1)$ is $(1,1)$, so the entry is $\mathrm{Fib}_{2p-1} + \mathrm{Fib}_{2p-2} = \mathrm{Fib}_{2p} = (\lambda_+^p - \lambda_-^p)/\sqrt{5}$. Dividing by $K(\mathbf{1})$ gives $R_{\mathrm{s}}$. With the Lucas numbers $\mathrm{Luc}_m = \mathrm{Fib}_{m-1} + \mathrm{Fib}_{m+1}$, $\mathrm{Luc}_m - 2\mathrm{Fib}_m = \mathrm{Fib}_{m-3}$, so $2R_{\mathrm{s}} = 1$ reads $\mathrm{Fib}_{2p-3} = 2$, that is $2p-3 = 3$ and $p = 3$. The rotation $i \mapsto i+1$ carries the list, so all spokes have one occupation and all rims one, and Theorem 39.13(i) with $x$ gives $R_{\mathrm{r}} = 1 - R_{\mathrm{s}}$; 9.3 gives the masses. By 11.16 the classes at the centre are the orbits of the bijections that carry or exchange the list, and the rotations with $x$ make one class. At a lean they are the orbits of the bijections that carry the list. For $p \geq 4$ the circuits of three marks are the triangles $\{s_i, s_{i+1}, r_i\}$, so a spoke lies in two of them and a rim in one, and a bijection that carries the list, carrying circuits to circuits, maps spokes to spokes: the classes at a lean are the spokes and the rims, the rotations acting transitively on each. By Theorem 39.13(i), $\bar{\phi}_x$ keeps the integrand at every lean, and $x$ maps the spokes onto the rims. For $p = 3$ the permutations of the four vertices of $K_4$ carry the list and act transitively on its six edges. □

## 50.39 Chapter 40: the stride's law and the gauge end

In this section the chart is $\alpha_{e_0} = 1$ with the coordinates $y = \log \alpha \in \mathbb{R}^{\mathcal{E} \setminus e_0}$, $f^\star = f(\alpha^\star)$, and a differential operator $A$ with polynomial coefficients on $(0, \infty)$ has the formal adjoint $A^\dagger$, with $\langle AT, \phi \rangle = \langle T, A^\dagger \phi \rangle$ for distributions $T$ and test functions $\phi$; for $\theta = t\frac{d}{dt}$, $\theta^\dagger = -1 - \theta$, so that $p(-1-i-\theta)^\dagger = p(\theta - i)$ and

$$\mathfrak{D}_{\mathbf{p}}^\dagger = \sum_i p_i(\theta - i)\, t^i, \qquad \mathfrak{D}_{\mathbf{p}}^\dagger\, t^k = \sum_i p_i(k)\, t^{k+i}.$$

Five lemmas carry the proofs: a moment lemma, which turns recurrences of moments into equations for measures; the volumes of the sublevel sets of a Morse function; the uniqueness of expansions in powers and logarithms; the formal solutions of a recurrence at a simple root of its symbol; and the point masses that a cut germ produces.

**Lemma S21.** *Let $\nu_1, \dots, \nu_r$ be complex measures on $(0, R]$ with $\int t^{K_0}\, d|\nu_l| < \infty$ for a real $K_0$, let $A_1, \dots, A_r$ be differential operators $A_l = \sum_i t^{a_{l,i}} q_{l,i}(\theta)$ with $a_{l,i} \in \mathbb{N}$ and polynomials $q_{l,i}$, and let $E = \sum_l A_l \nu_l$, a distribution on $(0, \infty)$. For $\operatorname{Re} k \geq K_0$ the numbers $M_E(k) = \sum_l \int A_l^\dagger(t^k)\, d\nu_l = \sum_{l,i} q_{l,i}(-1-k-a_{l,i}) \int t^{k+a_{l,i}}\, d\nu_l$ are defined, holomorphic for $\operatorname{Re} k > K_0$.*

*(i) If $M_E(k) = 0$ for all large integers $k$, then $E = 0$.*

*(ii) If $E = 0$, then $M_E(k) = 0$ for every $k$ with $\operatorname{Re} k \ge K_0$.*

*Proof.* $A_l^\dagger = \sum_i q_{l,i}(-1-\theta)\, t^{a_{l,i}}$, and $\theta\, t^m = m\, t^m$ gives the formula for $M_E$; the integrals converge absolutely for $\operatorname{Re} k \ge K_0$, $t$ being at most $R$, and are holomorphic in $k$ for $\operatorname{Re} k > K_0$ by dominated convergence on compact sets.

(i) For $\psi \in C^\infty(\mathbb{R})$ and an integer $K \ge K_0$ put $\Lambda_K(\psi) = \sum_l \int A_l^\dagger(t^K\psi)\, d\nu_l$. Since $\theta(t^m\psi) = t^m(m+\theta)\psi$ and $\theta^c = \sum_{b\le c} S(c,b)\, t^b\partial^b$ with integers $S(c,b)$, $A_l^\dagger(t^K\psi)$ is a finite combination of the functions $t^{K+a+b}\psi^{(b)}$, $b$ at most the largest order $d$ of the $A_l$, with coefficients that depend on $K$; so $|\Lambda_K(\psi)| \le C_K \max_{b\le d} \sup_{[0,R]} |\psi^{(b)}|$. For $\psi = t^m$, $\Lambda_K(t^m) = M_E(K+m)$, which vanishes when $K$ is large. Polynomials are dense in $C^d([0,R])$ (the Weierstrass approximation theorem applied to $\psi^{(d)}$, followed by $d$ integrations from 0 with the initial values $\psi^{(b)}(0)$), so $\Lambda_K(\psi) = 0$ for every $\psi \in C^\infty(\mathbb{R})$ and every large $K$. A test function $\phi$ on $(0,\infty)$ is $t^K\psi$ with $\psi = t^{-K}\phi$, smooth with compact support in $(0,\infty)$, and $\langle E, \phi\rangle = \sum_l \int A_l^\dagger\phi\, d\nu_l = \Lambda_K(\psi) = 0$.

(ii) Let $\chi \in C^\infty(\mathbb{R})$ vanish on $(-\infty, 1]$ and equal 1 on $[2,\infty)$, put $\chi_\epsilon(t) = \chi(t/\epsilon)$, and let $\tilde\chi$ be a test function equal to 1 on $[\epsilon, R]$. Then $0 = \langle E, t^k\chi_\epsilon\tilde\chi\rangle = \sum_l \int A_l^\dagger(t^k\chi_\epsilon)\, d\nu_l$, the measures being carried by $(0,R]$. The difference $A_l^\dagger(t^k\chi_\epsilon) - \chi_\epsilon A_l^\dagger(t^k)$ involves only the functions $\theta^c\chi_\epsilon(t) = (\theta^c\chi)(t/\epsilon)$ with $c \ge 1$, which are bounded uniformly in $\epsilon$ and vanish off $[\epsilon, 2\epsilon]$; so it is at most $C\, t^{\operatorname{Re} k}$ on $[\epsilon, 2\epsilon]$ and 0 elsewhere, and its integral tends to 0. By dominated convergence $\int \chi_\epsilon A_l^\dagger(t^k)\, d\nu_l \to \int A_l^\dagger(t^k)\, d\nu_l$, so $M_E(k) = 0$. □

**Lemma S22.** *Let $F$ be real-analytic near $0 \in \mathbb{R}^d$, $d \ge 1$, with $F(0) = 0$, $\nabla F(0) = 0$ and $\nabla^2 F(0) = H$ positive definite, and let $w$ be real-analytic near 0.*

*(i) There are $\epsilon > 0$ and a function $A$ holomorphic at 0 with $\int_{\{F\le\sigma\}\cap B_\epsilon} w\, dy = \sigma^{\frac d2} A(\sigma)$ for small $\sigma > 0$, $B_\epsilon$ the ball of radius $\epsilon$, and $A(0) = w(0)\,(2\pi)^{\frac d2}\big/\big(\Gamma(\frac d2 + 1)\sqrt{\det H}\big)$.*

*(ii) Let $F$ be even and $w = 1$. Then $F$ extends holomorphically to a neighbourhood of 0 in $\mathbb{C}^d$, $F_{\mathbb{T}}(\vartheta) = -F(i\vartheta)$ is real-analytic and even for real $\vartheta$ near 0, with the Hessian $H$ at 0, and with the same $A$, $\int_{\{F_{\mathbb{T}}\le\sigma\}\cap B_\epsilon} d\vartheta = \sigma^{\frac d2} A(-\sigma)$ for small $\sigma > 0$.*

*Proof.* (i) Write $F(y) = \frac12 y^{\mathsf T} Q(y) y$ with $Q(y) = 2\int_0^1 (1-\tau)\nabla^2 F(\tau y)\, d\tau$, symmetric, real-analytic, $Q(0) = H$; near 0, $Q(y)$ has an analytic positive square root, and $v(y) = Q(y)^{1/2} y$ is an analytic change of coordinates with $F = \frac12|v|^2$ and $\partial v/\partial y(0) = H^{1/2}$ (the Morse lemma). With $J(v) = w(y(v))\,|\det \partial y/\partial v| = \sum_\beta J_\beta v^\beta$, convergent on a ball, $J(0) = w(0)(\det H)^{-1/2}$, and for $\epsilon$ small the set $\{F \le \sigma\} \cap B_\epsilon$ is the preimage of the ball $|v|^2 \le 2\sigma$ for small $\sigma$. The monomials of odd degree integrate to 0 over the ball, and $\int_{|v|^2\le 2\sigma} v^\beta dv = c_\beta (2\sigma)^{\frac{d+|\beta|}{2}}$, so the integral is $\sigma^{\frac d2}\sum_m \big(\sum_{|\beta|=2m} J_\beta c_\beta 2^{\frac d2+m}\big)\sigma^m$, with the series convergent for small $\sigma$; at $m = 0$, $c_0$ is the volume $\pi^{\frac d2}/\Gamma(\frac d2+1)$ of the unit ball.

(ii) $F$ is even, so $\nabla^2 F$ and $Q$ are even and $v$ is odd, with real Taylor coefficients in the monomials of odd degree; its inverse $y(v)$ is odd too, and $J$ is even. A function analytic near 0 extends holomorphically to a polydisc, and $F(i\vartheta)$ is real for real $\vartheta$ since only even monomials occur, $i^{2m} = (-1)^m$. Put $v_{\mathbb{T}}(\vartheta) = -i\, v(i\vartheta)$, real for real $\vartheta$ since only odd monomials occur, $-i\cdot i^{2m+1} = (-1)^m$. Then $\frac12|v_{\mathbb{T}}|^2 = \frac12(-i)^2\, v(i\vartheta)\cdot v(i\vartheta) = -F(i\vartheta) = F_{\mathbb{T}}(\vartheta)$, and $v_{\mathbb{T}}$ is an analytic change of coordinates with the inverse $\vartheta = -i\, y(i\, v_{\mathbb{T}})$, whose Jacobian matrix is $(\partial y/\partial v)(i\, v_{\mathbb{T}})$; so $d\vartheta = J(i\, v_{\mathbb{T}})\, dv_{\mathbb{T}}$ with $J(iv) = \sum_\beta J_\beta i^{|\beta|} v^\beta = \sum_\beta (-1)^{|\beta|/2} J_\beta v^\beta$, $|\beta|$ even. The computation of (i) gives $\sigma^{\frac d2}\sum_m (-1)^m\big(\sum_{|\beta|=2m} J_\beta c_\beta 2^{\frac d2+m}\big)\sigma^m = \sigma^{\frac d2} A(-\sigma)$. The Hessian of $F_{\mathbb{T}}$ at 0 is $-i^2 H = H$. □

**Lemma S23.** *(i) Let $(z_c,\tau_c)$, $c=1,\dots,m$, be distinct pairs with $|z_c|=1$ and $\tau_c\in\mathbb{R}$, and $a_c\in\mathbb{C}$. If $\sum_c a_c z_c^k\,k^{i\tau_c}\to 0$ as $k\to\infty$ through the integers, every $a_c$ is 0.*

*(ii) Let $F(k)=\sum a_{z,\mu,j}\,z^k k^{-\mu}(\log k)^j$ be a finite sum over distinct triples with $|z|=1$, $\mu\in\mathbb{C}$ and $j\in\mathbb{N}$. If $F(k)=O(k^{-N})$ along the integers for some $N>\max\operatorname{Re}\mu$, every $a_{z,\mu,j}$ is 0.*

*Proof.* (i) Let $x_k=\sum_c a_c z_c^k k^{i\tau_c}$ and take the logarithmic means $W_M=\frac{1}{\log M}\sum_{k=1}^M |x_k|^2/k$, which tend to 0 when $x_k\to 0$. In $|x_k|^2$ the term of $(c,c')$ is $a_c\bar a_{c'} w^k k^{i\Delta}$ with $w=z_c\bar z_{c'}$ and $\Delta=\tau_c-\tau_{c'}$. For $c=c'$ it contributes $|a_c|^2\frac{1}{\log M}\sum_{k\le M}\frac{1}{k}\to|a_c|^2$. If $w\neq 1$, the partial sums of $w^k$ are bounded by $2/|1-w|$ and $k\mapsto k^{i\Delta-1}$ has bounded total variation, $|(k+1)^{i\Delta-1}-k^{i\Delta-1}|\le(1+|\Delta|)k^{-2}$; summation by parts bounds $\sum_{k\le M}w^k k^{i\Delta-1}$ uniformly in $M$. If $w=1$, then $\Delta\neq 0$ and $\sum_{k\le M}k^{i\Delta-1}=\int_1^M x^{i\Delta-1}dx+O(1)=(M^{i\Delta}-1)/(i\Delta)+O(1)$ is bounded. Divided by $\log M$ these tend to 0, so $W_M\to\sum_c|a_c|^2$, and every $a_c$ vanishes.

(ii) Let $\sigma_0$ be the least $\operatorname{Re}\mu$ among the triples with $a\neq 0$ and $j_0$ the largest $j$ among those with $\operatorname{Re}\mu=\sigma_0$. Then $k^{\sigma_0}(\log k)^{-j_0}F(k)$ is $\sum a_{z,\mu,j_0}z^k k^{-i\operatorname{Im}\mu}$, over $\operatorname{Re}\mu=\sigma_0$, plus terms that tend to 0, and it tends to 0 itself since $N>\sigma_0$; the pairs $(z,-\operatorname{Im}\mu)$ are distinct, so (i) makes these $a$ vanish, against the choice of $\sigma_0$ and $j_0$. □

**Lemma S24.** *Let $\mathbf{p}=(p_0,\dots,p_h)$ be as in Definition 40.4, $D\ge 1$, and $\lambda\neq 0$ a simple root of $\chi$. Consider formal sums $\mathcal{S}=\sum a_{\mu,j}k^\mu(\log k)^j$ over $j\le J$ and $\mu$ in finitely many classes modulo $\mathbb{Z}$ with $\operatorname{Re}\mu$ bounded above, with the operator $L=\sum_i p_i(k)\lambda^i\varsigma^i$, where $\varsigma^i\big(k^\mu(\log k)^j\big)=k^\mu(1+\frac{i}{k})^\mu\big(\log k+\log(1+\frac{i}{k})\big)^j$ is expanded in powers of $k^{-1}$. If $L\mathcal{S}=0$, then $\mathcal{S}=a\,k^\rho\big(1+\sum_{m\ge1}e_m k^{-m}\big)$ with $\rho=-\chi_1(\lambda)/(\lambda\chi'(\lambda))$, $a\in\mathbb{C}$, and $e_m$ fixed by $\mathbf{p}$ and $\lambda$; and this series is a formal solution.*

*Proof.* Each coefficient of $L\mathcal{S}$ is a finite sum, since $L$ raises the exponent by at most $D$ and $\varsigma$ does not raise it. With $p_i=c_ik^D+c_i'k^{D-1}+O(k^{D-2})$ and $\varsigma^i(k^\mu(\log k)^j)=k^\mu(\log k)^j+i\mu\,k^{\mu-1}(\log k)^j+ij\,k^{\mu-1}(\log k)^{j-1}+\cdots$, the dots of lower exponent,

$$L\big(k^\mu(\log k)^j\big)=\chi(\lambda)k^{\mu+D}(\log k)^j+\big(\mu\lambda\chi'(\lambda)+\chi_1(\lambda)\big)k^{\mu+D-1}(\log k)^j \\ +j\lambda\chi'(\lambda)k^{\mu+D-1}(\log k)^{j-1}+\cdots,$$

using $\sum_i c_i i\lambda^i=\lambda\chi'(\lambda)$, and $\chi(\lambda)=0$. Let $\sigma$ be the largest $\operatorname{Re}\mu$ in $\mathcal{S}$ and, for $\mu$ with $\operatorname{Re}\mu=\sigma$, $j_\mu$ the largest $j$ with $a_{\mu,j}\neq 0$. No term of $\mathcal{S}$ has an exponent $\mu+r$, $r\ge1$, so the coefficient of $k^{\mu+D-1}(\log k)^{j_\mu}$ in $L\mathcal{S}$ is $a_{\mu,j_\mu}(\mu\lambda\chi'(\lambda)+\chi_1(\lambda))$, and $\mu=\rho$. The coefficient of $k^{\rho+D-1}(\log k)^{j_\rho-1}$ is then $j_\rho\lambda\chi'(\lambda)a_{\rho,j_\rho}$, so $j_\rho=0$: the terms of $\mathcal{S}$ of largest real exponent are $a\,k^\rho$. For the series $k^\rho\sum_m e_m k^{-m}$, $e_0=1$, the coefficient of $k^{\rho-M+D-1}$ in $L$ of it is $-M\lambda\chi'(\lambda)e_M$ plus a combination of $e_0,\dots,e_{M-1}$, which fixes $e_M$; so it is a formal solution, and $\mathcal{S}-a\,k^\rho\sum_m e_m k^{-m}$ is a formal solution whose terms all have $\operatorname{Re}\mu<\operatorname{Re}\rho$, hence 0 by what precedes. □

**Lemma S25.** *Let $\mathcal{D}=\sum_{j=0}^D b_j(s)\,\partial_s^j$ have polynomial coefficients with $b_{D,0}=0$, and let $u=\sum_r u_r s^r$ be holomorphic near $s=0$. Then, with $\mathbf{1}_+=\mathbf{1}_{s>0}$ and $\mathsf{p}_r$ as in Theorem 40.6(iv),*

$$\mathcal{D}(u\,\mathbf{1}_+)=(\mathcal{D}u)\,\mathbf{1}_+ +\sum_{r=0}^{D-2}\mathsf{p}_r(u)\,\delta^{(D-2-r)},$$

*and the coefficient of $u_r$ in $\mathsf{p}_r(u)$ is $r!\,\big(b_{D-1,0}-(D-1-r)\,b_{D,1}\big)$.*

*Proof.* For $i<j$, $\partial^j(s^i\mathbf{1}_+)=i!\,\delta^{(j-1-i)}$, and the Taylor remainder of order $j$ of $u$ times $\mathbf{1}_+$ has the classical $j$-th derivative; so $\partial^j(u\mathbf{1}_+)=(\partial^j u)\mathbf{1}_+ +\sum_{i<j}i!\,u_i\,\delta^{(j-1-i)}$. Further $s^l\delta^{(m)}=$

$(-1)^l[m]_l\,\delta^{(m-l)}$ for $l \le m$ and 0 for $l > m$, with $[m]_l = \binom{m}{l}\,l!$. So the term $b_{j,l}s^l\partial^j$ sends $u_i\,s^i$, $i<j$, to $(-1)^l\binom{j-1-i}{l}\,i!\,l!\,b_{j,l}\,u_i\,\delta^{(j-1-i-l)}$. The level $r = D-2-(j-1-i-l)$ has $l = j+r+1-D-i$, and $l \le j-1-i$ exactly when $r \le D-2$; $l \ge 0$ and $j \le D$ give $i \le r+1$, with $i = r+1$ only for $j = D$, $l = 0$, excluded by $b_{D,0} = 0$. At $i = r$ the terms are $j = D-1$, $l = 0$, giving $r!\,b_{D-1,0}$, and $j = D$, $l = 1$, giving $-(D-1-r)\,r!\,b_{D,1}$. □

**Definition 40.1 (The stride's law).** As stated.

*Proof.* The density of the origin form, $\alpha^{\mathbf{m}_0}\prod_j K_j^{-s_{0j}}\Psi_j^{-t_{0j}}$ with real data, is positive on the positive orthant, the letters being positive there; so $\omega_0$ is a positive measure on the chart and its image under the measurable map $y \mapsto f(e^y)$ is a measure on $(0,\infty)$. □

**Proposition 40.2 (A ray is one Trailmoment on a line and one Gramma).** *As stated.*

*Proof.* (i) By Theorem 38.14(i), $0 < f \le f^\star$ on the chart and $f \le C'e^{-c'\|y\|}$, so $\{f \ge t\}$ is compact for $t > 0$ and carries a finite mass of the continuous density $\omega_0/dy$; $\mu_f$ is carried by $(0, f^\star]$. For the image measure $\int t^k\mu_f(dt) = \int f^k\omega_0 = \langle 1\rangle_k$, the integrals converging absolutely together. With the kernel $\mathsf{Z}_{k;(1)}(x) = x^{-k}$ (Definition 5.2) the Trailmoment of the input $\mu_f$ (Definition 5.4) with the coefficient function $t^{-1}$ over $(0,\infty)$ is $\int t^k\mu_f(dt)$, and Theorem 5.9 with $\tau = \iota$ writes it as $(\mathsf{T}_{k;(1)}[x]\,\iota_*\mu_f)((0,\infty))$.

(ii) If 0 were not an interior point of the cone of $\mathbf{v}$, a nonzero $y$ would have $\langle \mathbf{v}_i, y\rangle \le 0$ for every $i$ (the separation of a point from a closed convex cone), and then $f^{-1}(e^{\lambda y}) \le \sum_i w_i$ for $\lambda > 0$, against $f \le C'e^{-c'\|y\|}$; so the cone is $\mathbb{R}^{\mathcal{E}\setminus e_0}$, and Theorem 2.2 makes the Gramma entire. For $X > 0$ and $\operatorname{Re} k > 0$, $\Gamma(k)X^{-k} = \int_0^\infty u^{k-1}e^{-uX}du$. At $X = f^{-1}$, integrated against $\omega_0 = \alpha^{\mathbf{m}_0}dy$, Fubini's theorem applies, the double integral of the modulus being $\Gamma(\operatorname{Re} k)\langle 1\rangle_{\operatorname{Re} k}$ at real data, and the inner integral is $\int_{\mathbb{R}^{\mathcal{E}\setminus e_0}_{>0}} x^{\mathbf{m}_0|_{\mathcal{E}\setminus e_0}-\mathbf{1}}\exp\big(-\sum_i uw_ix^{\mathbf{v}_i}\big)dx = \mathsf{G}_{\mathbf{m}_0|_{\mathcal{E}\setminus e_0};\mathbf{v}}(u\mathbf{w})$ (Definition 2.1), $\alpha^{\mathbf{m}_0}$ being $x^{\mathbf{m}_0|_{\mathcal{E}\setminus e_0}}$ on the chart. This is Theorem 5.11 for the Trailmoment of (i), whose Laplace transform beyond the cut 0 is $\int e^{-ux}\,\iota_*\mu_f(dx) = \int e^{-uf^{-1}}\omega_0$.

(iii) The shape law has the density $u^{\mathbf{m}-\mathbf{1}}\prod_j K_j(u)^{-s_j}\Psi_j(u)^{-t_j}$ against $d\omega_{\mathcal{E}}$, divided by the period (Definition 8.10), and the change of variables of Theorem 8.7 that carries the period to the chart, $\langle 1\rangle_k = \int f^k\omega_0$ (Lemma S20(iv)), carries the expectation of every bounded measurable function $\varphi$ of degree 0 to $\int \varphi\,f^k\omega_0/\langle 1\rangle_k$. At $\varphi = \psi\circ f$ this is $\int \psi(t)\,t^k\mu_f(dt)/\langle 1\rangle_k$. So $\log\langle 1\rangle_k = \log\int t^k\mu_f(dt)$ is the log-partition of the family $t^k\mu_f(dt)/\langle 1\rangle_k$; it is real-analytic in $k$, and differentiation under the integral gives the first derivative $\mathbb{E}[\log f(U)]$ and the second $\mathbb{E}[\log^2 f(U)] - \mathbb{E}[\log f(U)]^2$, which is positive because $\log f(U)$ is not almost surely constant: $U$ has a positive continuous density on the chart, and $f$ is continuous there and not constant, $f(\alpha^\star) = f^\star > 0$ while $f \le C'e^{-c'\|y\|}$ (Theorem 38.14(i)). □

**Proposition 40.3 (The two ends of the stride's law).** *As stated.*

*Proof.* (i) At $y = 2z$ the stride is $e^{X_1(z)}$, $X_1$ the exponent of the question of the direction (proof of Lemma S20(ii)), and $-X_1$ is strictly convex (Theorem 17.5(ii)); so $\log f$ is strictly concave on the chart, its only critical point is its maximum $\log\alpha^\star$, and $\nabla f \ne 0$ elsewhere. For $0 < t < f^\star$ the level set $\Sigma_t = \{f = t\}$ is then a compact real-analytic hypersurface, compact because $f \to 0$ at infinity of the chart, and not empty, the chart being connected. Near $t_1 \in (0, f^\star)$ the flow $\Phi$ of the real-analytic field $\nabla f/|\nabla f|^2$, along which $f$ grows at the unit rate, maps $\Sigma_{t_1}\times(t_1-\epsilon, t_1+\epsilon)$ real-analytically onto a neighbourhood of $\Sigma_{t_1}$, with $f(\Phi(x,\tau)) = \tau$, and $dy = J(x,\tau)\,d\sigma(x)\,d\tau$ with $J$ real-analytic and positive. So $\mu_f$ has the density $\varrho_f(\tau) = \int_{\Sigma_{t_1}}(\omega_0/dy)(\Phi(x,\tau))\,J(x,\tau)\,d\sigma(x)$ there, real-analytic and positive; this is the coarea formula.

(ii) Apply Lemma S22(i) to $F(y) = \log f^\star - \log f(e^y)$ at $\log \alpha^\star$, with the Hessian $H$, positive definite (Lemma S20(iii)), and $w = \omega_0/dy$. Since $f < f^\star - \delta'$ off a neighbourhood of $\log \alpha^\star$, $\mu_f([t, f^\star]) = \int_{\{F \le L\} \cap B_\epsilon} w\, dy = L^{\frac{n-1}{2}} A(L)$ for $t$ near $f^\star$, with $L = \log(f^\star/t)$. Differentiating, with $dL/dt = -1/t$,

$$\varrho_f(t) = -\frac{d}{dt}\mu_f([t, f^\star]) = \frac{1}{t}\, L^{\frac{n-3}{2}} B(L), \qquad B(L) = \tfrac{n-1}{2} A(L) + L A'(L).$$

Put $\lambda(t) = L/(f^\star - t)$, holomorphic and positive near $f^\star$ with $\lambda(f^\star) = 1/f^\star$. Then $\varrho_f(t) = (f^\star - t)^{\frac{n-3}{2}} g(f^\star - t)$ with $g(f^\star - t) = t^{-1}\lambda(t)^{\frac{n-3}{2}} B(L)$, holomorphic at $t = f^\star$, and $g(0) = (f^\star)^{-\frac{n-1}{2}} \frac{n-1}{2} A(0)$; with $A(0)$ from Lemma S22(i) at $d = n-1$ and $\frac{n-1}{2}/\Gamma(\frac{n+1}{2}) = 1/\Gamma(\frac{n-1}{2})$ this is the stated value.

(iii) By Proposition 17.4(i) for the question of the direction, $2\kappa(z) + \sum_j (s_{1j} + t_{1j}) \log \theta_{j,\min} \le -X_1(z) \le 2\kappa(z) + \sum_j (s_{1j} + t_{1j}) \log K_j(\mathbf{1})$, and $2\kappa(y/2) = \kappa(y)$, $\kappa$ being positively homogeneous of degree one; with $\log f(e^y) = X_1(y/2)$ this is the squeeze. On the sector of an order $\pi$ (Definition 17.3) the differences $u_l = y_{\pi(l-1)} - y_{\pi(l)} \ge 0$, $2 \le l \le n$, are coordinates in which $dy$ is the Lebesgue measure and $\kappa(y) = \sum_l \kappa_{1,U_l} u_l$, so $\int_{\text{sector}} e^{-\kappa} dy = \prod_l \kappa_{1,U_l}^{-1}$; the sectors cover the chart and meet in null sets, so $\int e^{-\kappa} dy = V_1$. By homogeneity $\operatorname{vol}\{\kappa \le \sigma\} = \sigma^{n-1} \operatorname{vol}\{\kappa \le 1\}$ and $V_1 = \int_0^\infty e^{-\sigma}\, d\operatorname{vol}\{\kappa \le \sigma\} = (n-1)! \operatorname{vol}\{\kappa \le 1\}$, as in Theorem 17.5(i). For $t \le C_-$ the squeeze gives $\{\kappa \le \log(C_-/t)\} \subseteq \{f \ge t\} \subseteq \{\kappa \le \log(C_+/t)\}$, and $\mu_f([t, f^\star]) = \operatorname{vol}\{f \ge t\}$ for the origin 0. Last, $\int e^{-k\kappa} dy = k^{1-n} V_1$ for $k > 0$, by homogeneity, and $C_-^k e^{-k\kappa} \le f^k \le C_+^k e^{-k\kappa}$. □

**Definition 40.4 (The density operator of a recurrence).** As stated.

*Proof.* For a polynomial $p$ of degree at most $D$, $p(-1-i-\theta) = c(-1)^D \theta^D + \cdots$ with $c = [k^D]p$, and $\theta^D = t^D \partial^D + \cdots$; so the coefficient of $\partial^D$ in $t^i p_i(-1-i-\theta)$ is $(-1)^D c_i t^{D+i}$, and $a_D = (-1)^D t^D \chi(t)$. Extension by zero involves only $u$ near $\lambda$: two intervals give the same condition on their common part, $\mathfrak{D}_{\mathbf{p}}$ being a differential operator. □

**Proposition 40.5 (A recurrence of the values is an equation for the density).** *As stated.*

*Proof.* Let $k_*$ be the bound beyond which $\langle 1 \rangle_k$ converges absolutely, $\operatorname{Re} k > k_*$ (Lemma S20(iv)), and $K_0 > k_*$. Apply Lemma S21 to $\nu_1 = \mu_f$, carried by $(0, f^\star]$ with $\int t^{K_0} d\mu_f = \langle 1 \rangle_{K_0} < \infty$, and $A_1 = \mathfrak{D}_{\mathbf{p}}$, for which $A_1^\dagger t^k = \sum_i p_i(k) t^{k+i}$ and $M_E(k) = \sum_i p_i(k) \langle 1 \rangle_{k+i}$, $E = \mathfrak{D}_{\mathbf{p}} \mu_f$. (a) gives (c) by Lemma S21(i), and (c) gives the vanishing for $\operatorname{Re} k \ge K_0$ by (ii), for every $K_0 > k_*$. By Proposition 38.5(ii) the values continue meromorphically to $\mathbb{C}$, so the identity holds as one of meromorphic functions, and at every $k$ where its terms converge; (b) gives (a).

On $(0, f^\star)$, $\mu_f = \varrho_f\, dt$ with $\varrho_f$ real-analytic (Proposition 40.3(i)), so $E$ restricted there is $(\mathfrak{D}_{\mathbf{p}} \varrho_f)\, dt$, and (c) gives $\mathfrak{D}_{\mathbf{p}} \varrho_f = 0$. On $(f^\star - \delta, f^\star + \delta)$, $\mu_f = \varrho_f \mathbf{1}_{(f^\star - \delta, f^\star)} dt$, the point $f^\star$ having no mass, and $\varrho_f$ is integrable there by Proposition 40.3(ii), $\frac{n-3}{2} > -1$; so the germ extends by zero. For the origin 0, the proof of Corollary 39.38(i) and (ii) uses of $p_0, \dots, p_h$ only that the identity holds as one of meromorphic functions, with Proposition 39.35(ii) at $k = -j$ and at $k = 0$, $\mathsf{L}_0 = V_1 \ne 0$ (Definition 39.37). □

**Theorem 40.6 (The gauge end).** *As stated.*

*Proof.* (i) If $D = 0$, every $p_i$ is a constant $c_i$, $\mathfrak{D}_{\mathbf{p}}$ is the multiplication by $\chi(t)$, not identically zero, and $\chi \varrho_f = 0$ on $(0, f^\star)$ (Proposition 40.5) contradicts $\varrho_f > 0$. So $D \ge 1$. With $b_j(s) = (-1)^j a_j(f^\star - s)$, $b_D(s) = (f^\star - s)^D \chi(f^\star - s)$, so $b_{D,0} = 0$ and $b_{D,1} = -(f^\star)^D \chi'(f^\star)$. Write

$p_i = c_i k^D + c_i' k^{D-1} + \cdots$; $p_i(-1-i-\theta) = (-1)^D c_i\big(\theta^D + D(1+i)\theta^{D-1}\big) + (-1)^{D-1} c_i' \theta^{D-1} + \cdots$, and $\theta^m = t^m\partial^m + \binom{m}{2} t^{m-1}\partial^{m-1} + \cdots$; so

$$a_{D-1} = (-1)^D t^{D-1}\Big(\big(\tbinom{D}{2} + D\big)\chi(t) + D\,t\,\chi'(t) - \chi_1(t)\Big),$$
$$b_{D-1,0} = -(f^\star)^{D-1}\big(D f^\star \chi'(f^\star) - \chi_1(f^\star)\big),$$

which gives the two forms of $\gamma$. Since $s^j\partial_s^j = [s\partial_s]_j$, $s^{D-1}\mathfrak{D}_{\mathbf{p}} = \sum_{j,l} b_{j,l} s^{l+D-1-j}[s\partial_s]_j$; the exponent $m = l + D - 1 - j$ is nonnegative, as $l \geq 1$ for $j = D$, and the terms with $m = 0$ are $(j,l) = (D-1, 0)$ and $(D, 1)$, which give $R_0$. So $s = 0$ is a regular singular point, $R_0$ having the degree $D$ of the operator, and its exponents are the roots of $R_0$.

(ii) Let $u$ extend by zero on $(f^\star - \delta, f^\star)$. By Fuchs's theorem on regular singular points, which applies by (i), $u = \sum_{\lambda,j} s^\lambda (\log s)^j h_{\lambda,j}(s)$ near $s = 0^+$, a finite sum with $\lambda$ among the exponents, $j \leq D$ and $h_{\lambda,j}$ holomorphic at 0; so $u$ has a convergent expansion $\sum c_{\mu,j} s^\mu (\log s)^j$, $\mu \in \{0, \dots, D-2, \gamma\} + \mathbb{N}$, and integrability forces $\operatorname{Re}\mu > -1$ for its terms: the terms of least real part $\operatorname{Re}\mu$ and largest $j$ are $s^{\operatorname{Re}\mu}(\log s)^j$ times a trigonometric polynomial in $\log s$, not identically 0, whose modulus has a positive mean, and this is not integrable at 0 when $\operatorname{Re}\mu \leq -1$. Fix $\psi$ smooth, 0 on $(-\infty, f^\star - \frac{2}{3}\delta]$ and 1 on $[f^\star - \frac{1}{3}\delta, \infty)$, and put $T = \psi\, u\, \mathbf{1}_{(f^\star-\delta, f^\star)}$. On $(f^\star - \delta, f^\star + \delta)$, $\mathfrak{D}_{\mathbf{p}} T = \psi\, \mathfrak{D}_{\mathbf{p}}(u\mathbf{1}) + \sum_{a\geq 1}(\dots)\psi^{(a)}$, where the first term vanishes and the others are smooth functions carried by $[f^\star - \frac{2}{3}\delta, f^\star - \frac{1}{3}\delta]$, where $u$ is real-analytic; so $\mathfrak{D}_{\mathbf{p}} T = \mathcal{E}$ is a smooth function with that support, and, $T$ having compact support in $(0, \infty)$,

$$\sum_i p_i(k)\, M(k+i) = \int t^k\, \mathcal{E}(t)\, dt = O\big((f^\star - \tfrac{1}{3}\delta)^k\big), \qquad M(k) = \int t^k T(t)\, dt,$$

for real $k \to \infty$. In $t = f^\star e^{-\sigma}$, $M(k) = f^\star (f^\star)^k \int_0^\infty e^{-(k+1)\sigma}\, T(f^\star e^{-\sigma})\, d\sigma$, and $s = f^\star(1 - e^{-\sigma}) = f^\star\sigma(1 + O(\sigma))$; so near $\sigma = 0$, $T(f^\star e^{-\sigma})$ is a convergent sum of $\sigma^\mu(\log\sigma)^j$ times holomorphic functions, with the same leading terms as $u$ up to the factors $(f^\star)^\mu$, and the part away from 0 contributes $O\big((f^\star - \frac{1}{3}\delta)^k\big)$. Watson's lemma with logarithms, $\int_0^c e^{-\kappa\sigma}\sigma^\mu(\log\sigma)^j d\sigma = \partial_\mu^j\big[\Gamma(\mu + 1)\kappa^{-\mu-1}\big] + O(e^{-c\kappa/2})$ for $\operatorname{Re}\mu > -1$, gives $A(k) = (f^\star)^{-k} M(k)$ an asymptotic expansion $\mathcal{S}_u = \sum a_{\mu,j} k^{-\mu-1}(\log k)^j$ along the integers, in finitely many classes of exponents and with $\operatorname{Re}\mu$ bounded below. Its leading terms are $f^\star(f^\star)^\mu c_{\mu,j_0}\Gamma(\mu+1)(-1)^{j_0} k^{-\mu-1}(\log k)^{j_0}$, over the $\mu$ of least real part and the largest $j_0$ among them, with $\Gamma(\mu+1) \neq 0$; by Lemma S23(ii) they do not cancel, so $\mathcal{S}_u \neq 0$ when $u \neq 0$. Applying the recurrence to truncations of the expansion, $\sum_i p_i(k)(f^\star)^i A(k+i) = O(r^k)$, $r < 1$, and Lemma S23(ii), order by order, give $L\mathcal{S}_u = 0$ with $L = \sum_i p_i(k)(f^\star)^i \varsigma^i$, so by Lemma S24, $\mathcal{S}_u$ is a multiple of one fixed series. The map $u \mapsto \mathcal{S}_u$ is linear and injective, so the germs that extend by zero form a space of dimension at most one.

(iii) By Proposition 40.5 the germ of $\varrho_f$ extends by zero, and it is not zero; so it spans the space of (ii). Put $a = \frac{n-3}{2}$; by Proposition 40.3(ii), $\varrho_f = s^a g(s)$ with $g(0) \neq 0$, and it solves $\mathfrak{D}_{\mathbf{p}}$ on $s > 0$. If $a \notin \mathbb{Z}$, the leading coefficient of $s^{D-1}\mathfrak{D}_{\mathbf{p}}(s^a g) = \sum_m s^m R_m(s\partial_s)(s^a g)$ is $R_0(a)g(0)$, so $a$ is a root of $R_0$ other than $0, \dots, D-2$, and $a = \gamma$. If $a \in \mathbb{N}$, $\varrho_f$ is holomorphic at $s = 0$, with $u_r = 0$ for $r < a$ and $u_a = g(0)$. When $a \leq D-2$, the germ extends by zero, and by Lemma S25 $\mathsf{p}_a(\varrho_f) = a!\, b_{D,1}(a - \gamma)\, g(0) = 0$, the lower coefficients vanishing; so $\gamma = a$. When $a \geq D-1$, the coefficient of $s^a$ in $s^{D-1}\mathfrak{D}_{\mathbf{p}}\varrho_f = 0$ is $R_0(a)g(0)$, so $a$ is a root of $R_0$ and again $\gamma = a$.

(iv) If $\gamma \notin \{0, \dots, D-2\}$, then $\gamma = a$ is not an integer or is an integer at least $D-1$, and in both cases $R_0(\gamma + N) \neq 0$ for $N \geq 1$, the roots of $R_0$ being $0, \dots, D-2$ and $\gamma$. Inserting $s^\gamma \sum_N g_N s^N$ into $\sum_m s^m R_m(s\partial_s)$ and using $s\partial_s(s^{\gamma+N}) = (\gamma + N)s^{\gamma+N}$, the coefficient of $s^{\gamma+N}$ is $\sum_{m\leq N} R_m(\gamma + N - m) g_{N-m}$; its vanishing is the stated recurrence, so $g = g(0)\Phi$. If $\gamma \in$

$\{0,\dots,D-2\}$, then by (iii) $\varrho_f$ is holomorphic at $s=0$ with $u_r=0$ for $r<\gamma$ and $u_\gamma=g(0)$; it extends by zero, so $\mathsf{p}_r(\varrho_f)=0$ for $r\le D-2$ (Lemma S25, the distributions $\delta^{(q)}$ being linearly independent), and for $\gamma<r\le D-2$ the coefficient $r!\,b_{D,1}(r-\gamma)$ of $u_r$ is not 0, so these equations fix $u_r$ from the lower coefficients. For $N\ge D-1$, $R_0(N)\neq 0$, and the coefficient of $s^N$ in $s^{D-1}\mathfrak{D}_{\mathbf{p}}\varrho_f$ gives the last recurrence.

(v) Each point of $(0,f^\star)$ has a disc in $\mathbb{C}$ to which $\varrho_f$ extends holomorphically (Proposition 40.3(i)); two such discs meet in a convex set that meets the real line, so the extensions agree there, and they define a holomorphic function on the union $U$, on which $\mathfrak{D}_{\mathbf{p}}\varrho_f=0$ by the identity theorem. The germ at $f^\star$ is that of (iv), and its continuation along a path in $U$ that follows $(0,f^\star)$ is $\varrho_f$; near a root of $\chi$ on the segment, $\varrho_f$ is holomorphic, so the continuations around it on either side agree. For $\operatorname{Re}k>k_*$, $t^k(\log t)^j$ is dominated on $(0,f^\star]$ by $C\,t^K$ for every $K$ with $k_*<K<\operatorname{Re}k$, locally uniformly in $k$, so the derivatives pass under $\int t^k\mu_f(dt)$. □

**Proposition 40.7 (The law knows the dimension of its chart).** *As stated.*

*Proof.* Let $\alpha_c$ be the native gauge with $f(\alpha_c)=\lambda$ and $A_c$ its solution (Theorem 38.6, Definition 39.2). The class of $\sum_i p_i(k)f^i$ in $\mathfrak{M}$ is $\sum_i p_i(k)\,\mathrm{Sh}^i[1]=0$, so $\sum_i p_i(k)\varsigma^iA_c[1]=0$, and $\lambda^{-k}A_c[1]=k^{-\frac{n-1}{2}}\big(a_0(1)+\cdots\big)$ with $a_0(1)=(2\pi)^{\frac{n-1}{2}}(\omega_0/dv)(\alpha_c)\neq 0$ is a formal solution of $L=\sum_i p_i(k)\lambda^i\varsigma^i$. By Lemma S24 every formal solution is a multiple of $k^{\rho_\lambda}\big(1+\sum_m e_mk^{-m}\big)$, and comparing leading exponents, $\rho_\lambda=-\frac{n-1}{2}$. The computation of Theorem 40.6(i) uses of $f^\star$ only that it is a simple nonzero root of $\chi$, so at $\lambda$ the point $s=0$ is regular singular with the exponents $0,\dots,D-2$ and $-1+\chi_1(\lambda)/(\lambda\chi'(\lambda))=-1-\rho_\lambda=\frac{n-3}{2}$. □

**Proposition 40.8 (The germ at the gauge end is local data at the native gauge).** *As stated.*

*Proof.* (i) The potential of the realization is $\Phi(y)=\sum_j\big(s_{1j}\log K_j(e^y)+t_{1j}\log\Psi_j(e^y)\big)$ (Definition 8.18) and $\log f(e^y)=\langle\mathbf{m}_1,y\rangle-\Phi(y)$, so for $m\ge 2$ the Taylor tensor of $\log f$ is $-\partial^m\Phi=-2^{-m}\kappa_\Upsilon$ (Definition 9.4), and Proposition 9.6 at $y_0=\log\alpha^\star$ gives the cumulants.

(ii) Put $\eta(y)=\log f-\log f^\star+\frac12H_{ab}y^ay^b$, of order three. Outside a ball $|y|<\epsilon$ the stride is at most $f^\star e^{-c}$, so that part of $\langle1\rangle_k$ is $O\big((f^\star e^{-c})^{k-1}\langle1\rangle_1\big)$. Inside, with $y=v/\sqrt{k}$, $\langle1\rangle_k=(f^\star)^kk^{-\frac{n-1}{2}}\int_{|v|<\epsilon\sqrt{k}}e^{-\frac12H_{ab}v^av^b}e^{k\eta(v/\sqrt{k})}dv$, and $k\eta(v/\sqrt{k})=k^{-1/2}h_3(v)+k^{-1}h_4(v)+k^{-3/2}h_5(v)+O(k^{-2}|v|^6)$ with $h_3=\frac16h_{abc}v^av^bv^c$, $h_4=\frac{1}{24}h_{abcd}v^av^bv^cv^d$ and $h_5$ the quintic term; for $\epsilon$ small $|k\eta(v/\sqrt{k})|\le\frac14H_{ab}v^av^b$ on the ball. Expanding the exponential to the order $k^{-3/2}$, with the remainder bounded by $Ck^{-2}(1+|v|)^{12}e^{\frac14Hvv}$, the terms of order $k^{-1/2}$ and $k^{-3/2}$ are odd in $v$ and integrate to 0 over the ball, and extending the Gaussian integrals to $\mathbb{R}^{n-1}$ costs an exponentially small amount. So $\langle1\rangle_k=(f^\star)^k(2\pi/k)^{\frac{n-1}{2}}(\det H)^{-1/2}\big(1+k^{-1}\mathbb{E}[h_4+\frac12h_3^2]+O(k^{-2})\big)$ for the centred Gaussian with covariance $G$. Its moments are $\mathbb{E}[v^av^bv^cv^d]=G^{ab}G^{cd}+G^{ac}G^{bd}+G^{ad}G^{bc}$ and, for six factors, the sum over the 15 pairings; by the symmetry of $h_{abc}$, 9 of the pairings of $h_{abc}h_{def}$ join one pair inside each factor and 6 join every index across. So $\mathbb{E}[h_4]=\frac18h_{abcd}G^{ab}G^{cd}$ and $\frac12\mathbb{E}[h_3^2]=\frac{1}{72}\big(9h_{abc}h_{def}G^{ab}G^{cd}G^{ef}+6h_{abc}h_{def}G^{ad}G^{be}G^{cf}\big)$.

(iii) On $(f^\star-\delta,f^\star)$, $\varrho_f=\sum_m g_m(f^\star-t)^{\frac{n-3}{2}+m}$, and $\int_0^{f^\star-\delta}t^k\mu_f(dt)=O\big((f^\star-\delta)^{k-1}\langle1\rangle_1\big)$. For each $M$, $\int_{f^\star-\delta}^{f^\star}t^k\varrho_f\,dt$ is the sum over $m<M$ of $g_m\int_{f^\star-\delta}^{f^\star}t^k(f^\star-t)^{\frac{n-3}{2}+m}dt$ plus $O\big(\int_0^{f^\star}t^k(f^\star-t)^{\frac{n-3}{2}+M}dt\big)$, and $\int_0^{f^\star}t^k(f^\star-t)^{b-1}dt=(f^\star)^{k+b}\mathrm{B}(k+1,b)$, the part over $(0,f^\star-\delta)$ being exponentially smaller. With $\mathrm{B}(k+1,b)=\Gamma(b)\Gamma(k+1)/\Gamma(k+1+b)=\Gamma(b)k^{-b}(1+O(k^{-1}))$ the term of $g_m$, $b=\frac{n-1}{2}+m$, begins at $(f^\star)^{k+b}\Gamma(b)g_mk^{-b}$, and the remainder is $O\big((f^\star)^kk^{-\frac{n-1}{2}-M}\big)$. So the expansion of $(f^\star)^{-k}k^{\frac{n-1}{2}}\langle1\rangle_k$ is triangular in the $g_m$, with the nonzero diagonal $(f^\star)^{\frac{n-1}{2}+m}\Gamma(\frac{n-1}{2}+m)$. □

**Proposition 40.9 (Slopes from the gauge end).** *As stated.*

*Proof.* (i) Differentiate $j$ times the identity of meromorphic functions $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ (Proposition 40.5) by the Leibniz rule.

(ii) By Theorem 40.6(v), $\partial_k^j \langle 1\rangle_k|_{k=K} = \int_0^{f^\star} t^K (\log t)^j \varrho_f(t)\,dt$, and on $(0, f^\star)$ the series gives $\varrho_f = \sum_m g_m (f^\star - t)^{\frac{n-3}{2}+m}$, convergent there. Let $K' < K'' < K$. On $(0, f^\star]$, $t^K |\log t|^j \le C\, t^{K''}$, so $\sum_m |g_m| \int_0^{f^\star} t^K |\log t|^j (f^\star - t)^{\frac{n-3}{2}+m} dt \le C \sum_m |g_m| (f^\star)^{K''+\frac{n-1}{2}+m} \mathrm{B}\big(K''+1, \frac{n-1}{2}+m\big)$, which is finite because $\mathrm{B}(K''+1, b) \le \mathrm{B}(K'+1, b)$, the Beta function being decreasing in its first argument. So the integral of the sum is the sum of the integrals (Fubini's theorem), and $\int_0^{f^\star} t^a (\log t)^j (f^\star - t)^{b-1} dt = \partial_a^j \big[(f^\star)^{a+b} \mathrm{B}(a+1, b)\big]$, differentiation under the integral being allowed for $a$ near $K$. □

**Definition 40.10 (The procedure that reads a ray from its gauge end).** As stated.

*Proof.* (1) $\log f$ is strictly concave on the positive part of the chart (proof of Proposition 40.3(i)), so its critical point is unique, the native gauge of the direction there (Theorem 38.14(i)). (2) and (3) are Definition 40.4 and Theorem 40.6(i), (iii) and (iv); the coefficients $b_{j,l}$ lie in $\Bbbk(f^\star)$, and so do the ratios $u_r/u_\gamma$ and $\Phi_N$. (4) $\epsilon_k = \int_{(0, f^\star - \delta)} t^k \mu_f(dt)$ is nonnegative and, for real $k' \le k$, at most $(f^\star - \delta)^{k-k'} \int t^{k'} \mu_f(dt)$; the bound on $\langle 1\rangle_{k'}$ is Proposition 40.3(iii). On $[f^\star - \delta, f^\star]$, $\varrho_f$ is its germ, by Theorem 40.6(iv). (5) The recurrence holds at every positive integer (Proposition 40.5), and $\langle 1\rangle_k = -\sum_{i\ge 1} p_i(k)\langle 1\rangle_{k+i}/p_0(k)$ where $p_0(k) \ne 0$; the values are moments of $\varrho_f$ by Theorem 40.6(v). (6) is Proposition 40.9, and (7) is Corollary 39.38(ii), which holds for $\mathbf{p}$ by Proposition 40.5. □

**Definition 40.11 (The torus value).** As stated.

*Proof.* On $\mathbb{T}$, $\alpha^{-1} = \bar\alpha$, and the letters have real coefficients, so $K_j(\alpha^{-1}) = \overline{K_j(\alpha)}$ and $f = \prod_j |K_j(\alpha)|^{-2\sigma_j}$; the coefficients being nonnegative, $|K_j(\alpha)| \le K_j(\mathbf{1})$, and $f \ge \prod_j K_j(\mathbf{1})^{-2\sigma_j} = f(\mathbf{1}) = f^\star$, the native gauge of the direction being $\mathbf{1}$ (Theorem 38.14). The zero set of an active letter has measure zero, the letter being a Laurent polynomial that is not zero, so a real-analytic function on the connected torus that is not identically zero, so $f$ is finite almost everywhere and $\nu_f^{\mathbb{T}}$ is a probability measure on $[f^\star, \infty)$. For $\mathrm{Re}\, k \le 0$, $|f^k| = f^{\mathrm{Re}\, k} \le (f^\star)^{\mathrm{Re}\, k}$, read as 0 where $f = \infty$ and $\mathrm{Re}\, k < 0$, so the integral converges. When no $K_j$ vanishes on $\mathbb{T}$, $\mathbb{T}$ is a compact cycle in $X_F$, $\Psi_j(\alpha) = \alpha^{\mathcal{E}} K_j(\alpha^{-1})$ (Proposition 7.2) does not vanish either, and on it $dy = i^{n-1} d\vartheta$, the coordinates being $y_e = i\vartheta_e$. □

**Proposition 40.12 (The torus is the second chain).** *As stated.*

*Proof.* (i) Equality $|K_j(e^{i\vartheta})| = K_j(\mathbf{1})$ holds exactly when the numbers $\langle \mathbf{1}_T, \vartheta\rangle$, $T$ the bases of table $j$, agree modulo $2\pi$, all the coefficients being positive. The active tables form a connected family (Definition 38.13), so one molecule, and for linked marks $e, e'$ some basis $T$ has $e \in T$, $e' \notin T$ and $(T \setminus \{e\}) \cup \{e'\}$ a basis (Lemma 7.7); equality for every active table therefore gives $\vartheta_e \equiv \vartheta_{e'}$ for all marks, and $\alpha = \mathbf{1}$ on the chart. So $f > f^\star$ off $\mathbf{1}$, as in the proof of Theorem 39.57. The function $|K_j(e^{i\vartheta})|^2$ is a trigonometric polynomial, real-analytic and not identically 0 since its value at $\vartheta = 0$ is $K_j(\mathbf{1})^2$; the zero set of such a function on the connected torus has measure zero. For $\mathrm{Re}\, k \le 0$ the integrand is bounded by $(f^\star)^{\mathrm{Re}\, k}$; dominated convergence gives the continuity, and Morera's theorem the holomorphy on $\mathrm{Re}\, k < 0$. At $k = -j$, $f^{-j} = \prod_j |K_j|^{2j\sigma_j}$ is the Laurent polynomial $\prod_j (K_j(\alpha) K_j(\alpha^{-1}))^{j\sigma_j}$ restricted to $\mathbb{T}$, whose mean is its constant term; it is $\mathsf{C}_j(1)$ when every letter exponent of the direction is at least 1 (Definition 39.30).

(ii) The hypotheses of Theorem 39.57 hold for the centre ray with every $\sigma_j \geq 1$, so by Corollary 39.60 $\sum_i p_i(-j)\mathsf{C}_{j-i}(1) = 0$ for all large $j$, that is, by (i), $\sum_i p_i(k)\langle 1\rangle^{\mathbb{T}}_{k+i} = 0$ at all large negative integers $k$. With $x = 1/t$, $\nu_f^{\mathbb{T}}$ becomes a probability measure $\tilde\nu$ on $(0, 1/f^\star]$ with no mass at 0, $\theta = -x\frac{d}{dx}$, and $\mathfrak{D}^\dagger_{\mathbf{p}} = \sum_i p_i(\theta - i)\, t^i$ becomes $\tilde A = \sum_i p_i(-x\frac{d}{dx} - i)\, x^{-i}$. For a test function $\phi(t) = x^K\psi(x)$, $\tilde A(x^K\psi) = \sum_i x^{K-i}p_i(-x\frac{d}{dx} - K)\psi$ is bounded on $(0, 1/f^\star]$ by $C_K \max_{b\leq D}\sup|\psi^{(b)}|$ for $K \geq h$, and at $\psi = x^m$ its integral against $\tilde\nu$ is $\sum_i p_i(-K-m)\langle 1\rangle^{\mathbb{T}}_{-K-m+i}$, which vanishes for $K$ large. As in the proof of Lemma S21(i), on $[0, 1/f^\star]$, the Weierstrass approximation theorem gives $\langle \mathfrak{D}_{\mathbf{p}}\nu_f^{\mathbb{T}}, \phi\rangle = \int \tilde A(x^K\psi)\, d\tilde\nu = 0$ for every test function. Conversely, for $\operatorname{Re} k \leq -h$ the functions $t^{k+i}$ are bounded on $[f^\star, \infty)$, and the cutoff argument of Lemma S21(ii), at $x \to 0$ in place of $t \to 0$, gives $\sum_i p_i(k)\langle 1\rangle^{\mathbb{T}}_{k+i} = \int \mathfrak{D}^\dagger_{\mathbf{p}}(t^k)\, d\nu_f^{\mathbb{T}} = 0$, the cutoff terms being bounded and carried by $x \in [\epsilon, 2\epsilon]$, whose mass under $\tilde\nu$ tends to 0.

(iii) The stride is holomorphic and nonzero near $y = 0$ in $\mathbb{C}^{\mathcal{E}\setminus e_0}$, and $F(y) = \log f^\star - \log f(e^y)$ is even, $f(\alpha^{-1}) = f(\alpha)$ on the centre ray (Theorem 38.17(i)), with the Hessian $H$ at 0 positive definite (Lemma S20(iii)). On $\mathbb{T}$, $\log f(e^{i\vartheta}) - \log f^\star = -F(i\vartheta) = F_{\mathbb{T}}(\vartheta)$, real, and by (i) and compactness $f \geq f^\star + \delta'$ on $\mathbb{T}$ off a neighbourhood of $\mathbf{1}$; so for $t$ near $f^\star$, $\nu_f^{\mathbb{T}}([f^\star, t]) = (2\pi)^{1-n}\operatorname{vol}\{F_{\mathbb{T}} \leq \log(t/f^\star)\} \cap B_\epsilon$. By Lemma S22, with $L = \log(f^\star/t)$, $\mu_f([t, f^\star]) = L^{\frac{n-1}{2}}A(L)$ for $t < f^\star$ and $\nu_f^{\mathbb{T}}([f^\star, t]) = (2\pi)^{1-n}(-L)^{\frac{n-1}{2}}A(L)$ for $t > f^\star$, with one $A$, $w = 1$. Differentiating, with $B = \frac{n-1}{2}A + LA'$, $\varrho_f(t) = t^{-1}L^{\frac{n-3}{2}}B(L)$ and

$$\varrho_f^{\mathbb{T}}(t) = \frac{d}{dt}\Big[(2\pi)^{1-n}(-L)^{\frac{n-1}{2}}A(L)\Big] = (2\pi)^{1-n}\, t^{-1}(-L)^{\frac{n-3}{2}}B(L).$$

With $\lambda(t) = L/(f^\star - t)$, positive near $f^\star$, $L = (f^\star - t)\lambda$ and $-L = (t - f^\star)\lambda$, so both densities carry the factor $t^{-1}\lambda(t)^{\frac{n-3}{2}}B(L)$, which is $g(f^\star - t)$ by the proof of Proposition 40.3(ii).

(iv) By Proposition 40.5 and Theorem 40.6(iii) the germ of $\varrho_f$ spans the line. By (ii), $\mathfrak{D}_{\mathbf{p}}\nu_f^{\mathbb{T}} = 0$; near $f^\star$, $\nu_f^{\mathbb{T}} = \varrho_f^{\mathbb{T}}\mathbf{1}_{(f^\star, f^\star+\delta)}dt$, integrable by (iii), so its germ extends by zero on the torus's side. (iii) gives the common germ. □

## 50.40 Chapter 41: reading a ray

In this section $f^\star = f(\alpha^\star)$, and the symbol $\chi$ and the second symbol $\chi_1$ of polynomials $p_0, \dots, p_h$ are those of Definition 40.4. One lemma carries the proofs of the selection: the solutions of a recurrence whose symbol has distinct roots, sorted by their growth.

**Lemma S26.** *Let $p_0, \dots, p_h \in \mathbb{C}[k]$, $h \geq 1$, have $\deg p_0 = \deg p_h = D \geq \deg p_i$ for every $i$, let $\chi$ have distinct roots $\lambda_1, \dots, \lambda_h$, and let $k_1$ be an integer with $p_0(k)p_h(k) \neq 0$ for every integer $k \geq k_1$. Let $\mathcal{V}$ be the space of the sequences $(x(k))_{k\geq k_1}$ with $\sum_i p_i(k)x(k+i) = 0$ for every integer $k \geq k_1$.*

*(i) $\dim\mathcal{V} = h$, and $\mathcal{V}$ has a basis $x_1, \dots, x_h$ with $x_c(k) = \lambda_c^k k^{\rho_c}\big(1 + O(1/k)\big)$ as $k \to \infty$, $\rho_c = -\chi_1(\lambda_c)/(\lambda_c\chi'(\lambda_c))$.*

*(ii) For $R \geq 0$, the $x \in \mathcal{V}$ with $x(k) = O(r^k)$ for every $r > R$ are the combinations of the $x_c$ with $|\lambda_c| \leq R$, and those with $x(k) = O(r^k)$ for some $r < R$ are the combinations of the $x_c$ with $|\lambda_c| < R$.*

*Proof.* (i) The map $x \mapsto (x(k_1), \dots, x(k_1 + h - 1))$ is a bijection of $\mathcal{V}$ onto $\mathbb{C}^h$, the recurrence at $k$ giving $x(k + h)$ from the $h$ values before it, $p_h(k) \neq 0$; so $\dim\mathcal{V} = h$. The leading coefficients $c_i = [k^D]p_i$ have $c_0c_h \neq 0$, so $\chi(0) \neq 0$ and the roots are nonzero. If $D = 0$ the recurrence has

constant coefficients, $\rho_c = 0$, and $x_c(k) = \lambda_c^k$. Let $D \ge 1$. Divided by $k^D$ the recurrence has the coefficients $p_i(k)/k^D$, holomorphic at $k = \infty$ with the values $c_i$ there; written as a system for $(x(k), \dots, x(k+h-1))$, its matrix at $k = \infty$ is the companion matrix of $\chi/c_h$, whose eigenvalues are the $\lambda_c$. This is the normal case of Birkhoff's theorem on linear difference equations, which applies because the coefficients are holomorphic at infinity and the eigenvalues of the matrix there are distinct and nonzero: for every $c$ there is a solution, defined at all large integers, with an asymptotic expansion $\lambda_c^k k^\rho\big(1 + \sum_{m\ge1} e_m k^{-m}\big)$ as $k \to \infty$ that is a formal solution of the recurrence. Divided by $\lambda_c^k$ it is a formal solution of $\sum_i p_i(k)\lambda_c^i \varsigma^i$, so $\rho = \rho_c$ by Lemma S24, $\lambda_c$ being a simple root. Solving the recurrence for $x(k)$ downward, $p_0(k) \ne 0$, extends each of these solutions to $k \ge k_1$; call them $x_c$.

(ii) and the independence. Let $x = \sum_c a_c x_c$ with not all $a_c$ zero, $R_x = \max\{|\lambda_c| : a_c \ne 0\}$, $\sigma = \max\{\operatorname{Re}\rho_c : a_c \ne 0,\ |\lambda_c| = R_x\}$, and $C_0$ the set of the $c$ with $a_c \ne 0$, $|\lambda_c| = R_x$ and $\operatorname{Re}\rho_c = \sigma$. Then

$$\frac{x(k)}{R_x^k k^\sigma} = \sum_{c\in C_0} a_c\Big(\frac{\lambda_c}{R_x}\Big)^k k^{i\operatorname{Im}\rho_c} + o(1),$$

the other terms being $O\big((|\lambda_c|/R_x)^k k^{\operatorname{Re}\rho_c-\sigma}\big)$ with $|\lambda_c| < R_x$, or $O(k^{\operatorname{Re}\rho_c-\sigma})$ with $\operatorname{Re}\rho_c < \sigma$, and the remainders $O(1/k)$. The pairs $(\lambda_c/R_x, \operatorname{Im}\rho_c)$, $c \in C_0$, are distinct, the $\lambda_c$ being distinct, so by Lemma S23(i) the right side does not tend to 0, and $|x(k)| \ge \eta R_x^k k^\sigma$ for some $\eta > 0$ and infinitely many $k$. So $x \ne 0$, and the $x_c$ are independent, a basis by (i). And $x(k) = O(r^k)$ fails for every $r < R_x$. If $x(k) = O(r^k)$ for every $r > R$, then $R_x \le R$; if for some $r < R$, then $R_x < R$. Conversely $x_c(k) = O(r^k)$ for every $r > |\lambda_c|$. □

**Proposition 41.1 (The low-end relations at every pole).** *As stated.*

*Proof.* By Proposition 40.5 the identity $\sum_{i=0}^h p_i(k)\langle1\rangle_{k+i} = 0$ holds as one of meromorphic functions of $k$. For $i > j$ the value $\langle1\rangle_{k+i}$ is holomorphic at $k = -j$, $k+i$ being near the positive integer $i-j$, where it converges absolutely (Lemma S20(iv), the origin being 0). So $S(k) = \sum_{i\le\min(h,j)} p_i(k)\langle1\rangle_{k+i} = -\sum_{i=j+1}^h p_i(k)\langle1\rangle_{k+i}$ is holomorphic at $-j$ with the value of (i). For $i \le j$, Proposition 39.35(ii) at the integer $j-i$ gives $(k+j)^{n-1}\langle1\rangle_{k+i} \to V_1\,\mathrm{CT}(f^{i-j})$ as $k \to -j$, and $p_i(k) = (k+j)^{m_0}\big([(k+j)^{m_0}]p_i + O(k+j)\big)$. So

$$(k+j)^{n-1-m_0}S(k) = \sum_{i\le\min(h,j)} \frac{p_i(k)}{(k+j)^{m_0}}\,(k+j)^{n-1}\langle1\rangle_{k+i} \;\longrightarrow\; V_1 \sum_{i\le\min(h,j)} [(k+j)^{m_0}]p_i\cdot\mathrm{CT}\left(f^{i-j}\right)$$

as $k \to -j$. For $m_0 < n-1$ the left side tends to 0, $S$ being holomorphic at $-j$, and $V_1 > 0$ gives (ii). For $m_0 = n-1$ the left side tends to $S(-j)$, which (i) gives, and this is (iii). The reciprocal stride $f^{-1} = \alpha^{-\mathbf{m}_1}\prod_l K_l^{s_{1l}}\Psi_l^{t_{1l}}$ is a Laurent polynomial whose coefficients are polynomials with integer coefficients in the weights, so every $\mathrm{CT}(f^{i-j})$ lies in the field of the weights. □

**Proposition 41.2 (The values live on the slit plane).** *As stated.*

*Proof.* (i) By Proposition 40.2(i), $\mu_f$ is carried by $(0, f^\star]$ and $\langle1\rangle_k = \int t^k\mu_f(dt)$. For $|q| < 1/f^\star$, $\sum_{k\ge k_0}|q|^{k-k_0}\int t^k\mu_f(dt) = \int t^{k_0}\mu_f(dt)/(1-|q|t) \le \langle1\rangle_{k_0}/(1-|q|f^\star)$, so Fubini's theorem exchanges the sum and the integral and gives the formula. For $q$ in a compact set $Q \subseteq \mathbb{C}\setminus[1/f^\star,\infty)$ the function $(q,t)\mapsto|1-qt|$ is continuous and positive on $Q\times[0,f^\star]$, since $qt = 1$ with $t \in (0,f^\star]$ means $q = 1/t \in [1/f^\star,\infty)$; so it is at least some $\delta_Q > 0$ there, the integrand is holomorphic in $q$ and at most $t^{k_0}/\delta_Q$, which is $\mu_f$-integrable, and the integral is holomorphic on $\mathbb{C}\setminus[1/f^\star,\infty)$ (dominated convergence and Morera's theorem).

(ii) Put $t_0 = 1/x \in (0, f^\star)$. By Proposition 40.3(i), $\varrho_f$ is real-analytic on $(0, f^\star)$, so there is $\delta > 0$ with $[t_0 - 2\delta, t_0 + 2\delta] \subseteq (0, f^\star)$ such that $\phi(t) = t^{k_0}\varrho_f(t)$ extends holomorphically to the disc $|t - t_0| \le 2\delta$. Write $G = G_1 + G_2$, with $G_1(q) = \int_{|t-t_0|\ge\delta} t^{k_0}\mu_f(dt)/(1-qt)$ and $G_2(q) = \int_{t_0-\delta}^{t_0+\delta} \phi(t)\,dt/(1-qt)$. For $q$ close enough to $x$, $1/q$ lies within $\delta/2$ of $t_0$, so $1 - qt$ does not vanish on the support of the first integral, and $G_1$ is holomorphic near $x$ as in (i). Let $\Gamma_+$ and $\Gamma_-$ be the half circles $t_0 + \delta e^{i\vartheta}$ from $t_0 - \delta$ to $t_0 + \delta$ through the upper and through the lower half plane. For $q$ near $x$ with $\operatorname{Im} q > 0$ the pole $t = 1/q$ of the integrand lies in the disc and below the real axis, $\operatorname{Im}(1/q) = -\operatorname{Im} q/|q|^2$, so Cauchy's theorem moves the segment of $G_2$ onto $\Gamma_+$: $G_2(q) = \int_{\Gamma_+} \phi(t)\,dt/(1-qt)$, and the right side is holomorphic in $q$ on the whole neighbourhood of $x$ where $1/q$ stays within $\delta/2$ of $t_0$. So $G$ continues from the upper half plane across a neighbourhood of $x$, with $G(x + i0) = G_1(x) + \int_{\Gamma_+} \phi(t)\,dt/(1-xt)$; likewise from below with $\Gamma_-$. The difference is the integral of $\phi(t)/(1-xt)$ along $\Gamma_+$ followed by $\Gamma_-$ reversed, a clockwise circle around $t_0$, inside which the integrand has one simple pole, at $t_0 = 1/x$, with the residue $-\phi(1/x)/x$, as $1 - xt = -x(t - 1/x)$. So $G(x + i0) - G(x - i0) = -2\pi i \cdot \big(-\phi(1/x)/x\big) = 2\pi i\, x^{-1-k_0}\varrho_f(1/x)$.

(iii) On $\mathbb{T}$, $f = \prod_j |K_j(\alpha)|^{-2\sigma_j} \ge f^\star$ (proof of Definition 40.11), finite at the points of $X_F$; so the native value of a native gauge on $\mathbb{T}$ lies in $[f^\star, \infty)$.

(iv) By Proposition 38.10 every conjugate of $f^\star$ over the field $\Bbbk$ of the weights is the native value of a nondegenerate native gauge. Let $\sum_i p_i(k)\langle 1\rangle_{k+i} = 0$ for all large integers $k$, with $p_i \in \Bbbk[k]$ not all zero, of largest degree $D$. By Proposition 38.9, $\langle 1\rangle_k > 0$ for large $k$ and $\langle 1\rangle_{k+i}/\langle 1\rangle_k \to (f^\star)^i$; dividing the recurrence by $k^D\langle 1\rangle_k$ and letting $k \to \infty$ gives $\sum_i [k^D]p_i \cdot (f^\star)^i = \chi(f^\star) = 0$. The symbol is a nonzero polynomial over $\Bbbk$, so the minimal polynomial of $f^\star$ over $\Bbbk$ divides it, and every conjugate of $f^\star$ is a root. A real conjugate in $(0, f^\star)$ lies on the slit by its definition. □

**Theorem 41.3 (Positive selection).** *As stated.*

*Proof.* As in the proof of Proposition 41.2(iv), $\chi(f^\star) = 0$; the hypotheses on the degrees make $\chi$ of degree $h$ with $\chi(0) = [k^D]p_0 \ne 0$. The walls $\kappa_{0,S} + k\kappa_{1,S}$ increase with $k$, so the values converge at every integer $k \ge k_1$, and by Proposition 40.5 the recurrence holds there: the values are a solution. Lemma S26(i) gives $\dim \mathcal{V} = h$ and a basis $x_1, \dots, x_h$.

(i) For each $i$, with $m = k + i$,

$$\begin{aligned}\sum_{k\ge k_1} p_i(k)x(k+i)\,q^{k-k_1+h} &= q^{h-i}\sum_{m\ge k_1+i} p_i(m-i)x(m)\,q^{m-k_1}\\ &= q^{h-i}p_i\Big(q\frac{d}{dq} + k_1 - i\Big)\Big(X - \sum_{m=k_1}^{k_1+i-1} x(m)\,q^{m-k_1}\Big),\end{aligned}$$

since $q\frac{d}{dq}$ multiplies $q^{m-k_1}$ by $m - k_1$. Summed over $i$ the left sides vanish, so $L_qX = R_x$ with

$$R_x = \sum_{i=0}^{h}\sum_{m=k_1}^{k_1+i-1} p_i(m-i)\,x(m)\,q^{h-i+m-k_1},$$

a polynomial of degree at most $h - 1$, linear in $x(k_1), \dots, x(k_1 + h - 1)$. With $c_i = [k^D]p_i$, $p_i(q\frac{d}{dq} + c) = c_i(q\frac{d}{dq})^D + \cdots$ and $(q\frac{d}{dq})^D = q^D\frac{d^D}{dq^D} + \cdots$, the dots of lower order in $\frac{d}{dq}$; so the coefficient of $\frac{d^D}{dq^D}$ in $L_q$ is $\sum_i q^{h-i}c_iq^D = q^{D+h}\chi(1/q)$, which vanishes in $\mathbb{C}^\times$ exactly at the $1/\lambda_c$. For the values, $X$ is the generating function of Proposition 41.2(i) with $k_0 = k_1$.

(ii) For $k \ge k_1$, $\langle 1\rangle_k = \int t^k \mu_f(dt) \le (f^\star)^{k-k_1}\langle 1\rangle_{k_1}$, $\mu_f$ being carried by $(0, f^\star]$ (Proposition 40.2(i)); so the values lie in the subspace of the statement, which Lemma S26(ii) at $R = f^\star$ identifies. By Proposition 40.3(ii), with $a = \frac{n-3}{2}$ and a small $\delta > 0$,

$$\langle 1\rangle_k = \int_0^\delta (f^\star - s)^k\, s^a g(s)\, ds + O\big((f^\star - \delta)^{k-k_1}\langle 1\rangle_{k_1}\big).$$

In $\sigma = -\log(1 - s/f^\star)$ the integral is $(f^\star)^k \int_0^{\sigma_\delta} e^{-k\sigma} h(\sigma)\, d\sigma$ with $h(\sigma) = s^a g(s)\, f^\star e^{-\sigma} = g(0)(f^\star)^{a+1}\sigma^a\big(1 + O(\sigma)\big)$, and Watson's lemma, which applies because $h(\sigma)\sigma^{-a}$ is continuous at 0 and $a > -1$, gives $\langle 1\rangle_k = C(f^\star)^k k^{-\frac{n-1}{2}}(1 + o(1))$ with $C = g(0)\Gamma(\frac{n-1}{2})(f^\star)^{\frac{n-1}{2}} > 0$. Last, $f^\star$ is a simple root of $\chi$ and the recurrence meets the conditions of Proposition 40.5, so Theorem 40.6(iii) gives $\frac{n-3}{2} = -1 + \chi_1(f^\star)/(f^\star\chi'(f^\star))$.

(iii) By Lemma S26(ii) at $R = \lambda'$, the solutions with $x(k) = O(r^k)$ for some $r < \lambda'$ are the multiples of $x_1$, $\lambda_1 = f^\star$ being the only root of modulus less than $\lambda'$; the values are such a solution, $f^\star < \lambda'$, and they are not 0, so they span that line. A generating function converges on $|q| < 1/f^\star$ exactly when $\limsup |x(k)|^{1/k} \le f^\star$, that is when $x(k) = O(r^k)$ for every $r > f^\star$, and Lemma S26(ii) at $R = f^\star$ gives the same line. For the origin 0, Proposition 40.5 gives the low-end relation $a\mathsf{L}_0 + \sum_{i\ge1} p_i(0)\langle 1\rangle_i = 0$ with $\mathsf{L}_0 = V_1$ (Definition 39.37), that is $\ell(\langle 1\rangle) = V_1$; a nonzero solution of least growth is $x = c\langle 1\rangle$ with $c \ne 0$, and $\ell(x) = cV_1$.

(iv) $\ell$ is a linear form on $\mathcal{V}$ with $\ell(\langle 1\rangle) = V_1 \ne 0$, so $\{\ell = 0\}$ is a hyperplane that does not contain $\langle 1\rangle$, and $\langle 1\rangle - x^{(0)} \in \{\ell = 0\}$ gives $M$. Take $x_1 = \langle 1\rangle$, which (iii) allows. The $z_c = x_c - \frac{\ell(x_c)}{V_1}x_1$, $2 \le c \le h$, form a basis of $\{\ell = 0\}$, and $x^{(i)} = \sum_c S_{ic} z_c$ with $S$ invertible. The equations of the statement read $\mathbf{X}_K M^{(K)} = -\big(x^{(0)}(K + l)\big)_l$ with $\mathbf{X}_K = \big[x^{(i)}(K + l)\big]_{l,i}$, while $\mathbf{X}_K M = \big(\langle 1\rangle_{K+l} - x^{(0)}(K + l)\big)_l$; so, when $\mathbf{X}_K$ is invertible, $M^{(K)} - M = -\mathbf{X}_K^{-1} v_K$ with $v_K = (\langle 1\rangle_{K+l})_l$. Now $\mathbf{X}_K = \mathbf{Z}_K S^{\mathsf{T}}$ with $\mathbf{Z}_K = [z_c(K + l)]_{l,c}$, and $z_c(K + l) = \lambda_c^K K^{\rho_c}\big(\lambda_c^l + o(1)\big)$, since $x_c(K + l) = \lambda_c^{K+l}(K + l)^{\rho_c}\big(1 + O(1/K)\big)$ and $x_1(K + l) = O\big((f^\star)^K\big)$ with $f^\star < |\lambda_c|$. So $\mathbf{Z}_K = \mathbf{N}_K\Lambda_K$ with $\Lambda_K = \operatorname{diag}(\lambda_c^K K^{\rho_c})_{c\ge2}$ and $\mathbf{N}_K \to [\lambda_c^l]_{0\le l\le h-2,\ 2\le c\le h}$, the Vandermonde matrix of the distinct $\lambda_2, \dots, \lambda_h$, which is invertible. For large $K$, $\mathbf{X}_K$ is invertible, $M^{(K)}$ is unique, and with $|v_K| = O\big((f^\star)^K K^{-\frac{n-1}{2}}\big)$ by (ii),

$$\begin{aligned} |M^{(K)} - M| &\le \big\|(S^{\mathsf{T}})^{-1}\big\|\, \big\|\Lambda_K^{-1}\big\|\, \big\|\mathbf{N}_K^{-1}\big\|\, |v_K| \\ &= O\Big(\max_{c\ge2} |\lambda_c|^{-K} K^{-\operatorname{Re}\rho_c}\,(f^\star)^K K^{-\frac{n-1}{2}}\Big) = O\big((f^\star/\lambda')^K K^{C'}\big). \end{aligned}$$

Over $\Bbbk$: the recurrence solved forward from initial values in $\Bbbk$, $p_h(k) \ne 0$, gives solutions with values in $\Bbbk$, and $\mathcal{V}$ has a basis of them; $\ell$ has its coefficients in $\Bbbk$ and $V_1 \in \Bbbk$, so $x^{(0)}$ and a basis of $\{\ell = 0\}$ can be taken with values in $\Bbbk$, and the equations for $M^{(K)}$ then have their coefficients in $\Bbbk$. □

**Proposition 41.4 (Positivity is the moment problem of the stride's law).** *As stated.*

*Proof.* (i) Every $\langle 1\rangle_{K+m}$, $m \in \mathbb{N}$, converges with $\langle 1\rangle_K$, $\mu_f$ being carried by $(0, f^\star]$, and $L_{K,p}(\langle 1\rangle) = \sum_m p_m \int t^{K+m}\mu_f(dt) = \int t^K p\, d\mu_f$ (Proposition 40.2(i)). The integrand is nonnegative, and positive off the finitely many zeros of $p$ on $(0, f^\star)$, where $\mu_f$ has a positive density (Proposition 40.3(i)); so the integral is positive.

(ii) By the Markov–Lukács theorem on polynomials nonnegative on an interval, which applies to $[0, f^\star]$, a real polynomial of degree $2m$ nonnegative there is $A^2 + t(f^\star - t)B^2$ with real $A$ and $B$ of degrees at most $m$ and $m - 1$, and one of degree $2m + 1$ is $tA^2 + (f^\star - t)B^2$ with real $A$ and

$B$ of degrees at most $m$; every term has degree at most that of the polynomial. For $q = \sum_i q_i t^i$, $L_{K,q^2}(x) = \sum_{i,i'} q_i q_{i'}\, x(K+i+i')$, and likewise $L_{K,tq^2}$, $L_{K,(f^\star-t)q^2}$ and $L_{K,t(f^\star-t)q^2}$ are the quadratic forms of the other three Hankel matrices. The four kinds have degree at most $N$ exactly when $\deg q$ is at most $\lfloor \frac{N}{2} \rfloor$, $\lfloor \frac{N-1}{2} \rfloor$, $\lfloor \frac{N-1}{2} \rfloor$ and $\lfloor \frac{N-2}{2} \rfloor$, which gives the sizes. If $L_{K,p}(x) \ge 0$ for every $p \ge 0$ on $[0, f^\star]$ of degree at most $N$, the four quadratic forms are nonnegative, these $p$ being among them; conversely, if the forms are nonnegative, $L_{K,p}(x) \ge 0$ for every $p$ of the four kinds, and by linearity for every sum of them.

(iii) If $x(K+m) = \int t^m \nu(dt)$ for a positive $\nu$ on $[0, f^\star]$, then $L_{K,p}(x) = \int p\, d\nu \ge 0$. Conversely, the linear functional $\Lambda(p) = L_{K,p}(x)$ on real polynomials is nonnegative on those nonnegative on $[0, f^\star]$, so $|\Lambda(p)| \le \Lambda(1) \max_{[0,f^\star]} |p|$, the polynomials $\max |p| \pm p$ being nonnegative there. Polynomials are dense in $C([0, f^\star])$ (the Weierstrass approximation theorem), so $\Lambda$ extends continuously to $C([0, f^\star])$, and the extension is positive: a continuous $\varphi \ge 0$ is the uniform limit of polynomials $p_l$, and $p_l + \max |\varphi - p_l| \ge 0$. The Riesz representation theorem gives a positive measure $\nu$ on $[0, f^\star]$ with $\Lambda(\varphi) = \int \varphi\, d\nu$, and at $\varphi = t^m$ this is $x(K+m) = \int t^m \nu(dt)$. Two finite measures on $[0, f^\star]$ with the same moments agree on polynomials, hence on $C([0, f^\star])$, hence are equal. For the values, $t^K \mu_f$ is a finite positive measure on $(0, f^\star]$ with the moments $\langle 1 \rangle_{K+m}$. □

**Theorem 41.5 (The rate of positivity off the segment).** *As stated.*

*Proof.* Put $A(p) = L_{K,p}(\langle 1 \rangle) = \int t^K p\, d\mu_f$ and $B(p) = L_{K,p}(u) = \int_{I_\lambda} t^K p(t) \nu(t)\, dt$, both convergent, $|t|^{K+m} \le C_m |t|^K$ on the bounded $I_\lambda$. For $p \ge 0$ on $[0, f^\star]$, $A(p) \ge 0$ (Proposition 41.4(i)), and $A(p) + \varepsilon B(p) \ge 0$ holds for every $\varepsilon$ when $B(p) = 0$ and otherwise on a closed half line that contains 0. So $E_N$, the intersection over the $p$ of degree at most $N$, is a closed interval that contains 0, and if $p_1, p_2$ of degree at most $N$ have $B(p_1)$ and $B(p_2)$ of opposite signs, its length is at most $A(p_1)/|B(p_1)| + A(p_2)/|B(p_2)|$.

Let $w(t) = 2t/f^\star - 1$, which maps $[0, f^\star]$ onto $[-1, 1]$, and $T_m$ the polynomial of degree $m$ with $T_m(\cos \vartheta) = \cos m\vartheta$; then $|T_m(w)| \le 1$ for $|w| \le 1$ and $|T_m(w)| = \cosh\big(m \operatorname{arccosh} |w|\big)$ for $|w| \ge 1$. On the part of $I_\lambda$ off $[0, f^\star]$ put $\Gamma(t) = \operatorname{arccosh} |w(t)|$: it is $C^1$, it grows as $t$ moves from the segment toward $\lambda$, $|w(t)|$ doing so, and $\Gamma(\lambda) = \log\big(|w| + \sqrt{w^2-1}\big) = g(\lambda)$ at $w = w(\lambda)$, the branch of the statement; $g(\lambda) > 0$ as $|w(\lambda)| > 1$. Let $\eta(t) = f^\star - t$ if $\lambda > f^\star$ and $\eta(t) = t$ if $\lambda < 0$, so that $\eta \ge 0$ on $[0, f^\star]$ and $\eta(\lambda) < 0$, and take $p_1 = T_m(w)^2$ and $p_2 = \eta\, T_m(w)^2$, nonnegative on $[0, f^\star]$, of degrees $2m$ and $2m+1$. Then $A(p_1) \le \langle 1 \rangle_K$ and $A(p_2) \le f^\star \langle 1 \rangle_K$.

Fix $\delta > 0$ so small that $I' = \{t \in I_\lambda : \ |t - \lambda| \le \delta\}$ lies off $[0, f^\star]$ and on $I'$: $t^K$ and $\eta$ have the signs of their values at $\lambda$, $|t^K| \ge \frac{1}{2}|\lambda|^K$, $|\eta(t)| \ge \frac{1}{2}|\eta(\lambda)|$, $\nu(t) = c|\lambda - t|^a(1 + \epsilon(t))$ with $|\epsilon| \le \frac{1}{2}$, and $\Gamma(t) \ge g(\lambda) - \Gamma_1 |\lambda - t|$ for a constant $\Gamma_1$. On $I'$ the integrand of $B(p_i)$ has the constant sign of $c\lambda^K \eta(\lambda)^{i-1}$, and by $\cosh^2 \ge \frac{1}{4} e^{2m\Gamma}$ its modulus is at least $c_1 e^{2mg(\lambda)} e^{-2m\Gamma_1 |\lambda - t|} |\lambda - t|^a$ for a constant $c_1 > 0$; so its integral over $I'$ has modulus at least

$$c_1 e^{2mg(\lambda)} \int_0^\delta e^{-2m\Gamma_1 s} s^a\, ds = c_1 e^{2mg(\lambda)} (2m\Gamma_1)^{-a-1} \big(\Gamma(a+1) + o(1)\big) \ge c_2\, m^{-a-1} e^{2mg(\lambda)}$$

for large $m$. Off $I'$, $T_m(w)^2 \le \cosh^2\big(m\Gamma(\lambda_\delta)\big) \le e^{2m(g(\lambda) - \eta')}$, with $\lambda_\delta$ the end of $I'$ other than $\lambda$ and $\eta' = g(\lambda) - \Gamma(\lambda_\delta) > 0$, by the monotonicity of $\Gamma$ and $|T_m| \le 1$ on $[0, f^\star]$; $\int_{I_\lambda} |t|^K |\nu|\, dt$ being finite and $\eta$ bounded on $I_\lambda$, that part of $B(p_i)$ is at most $C_3 e^{2m(g(\lambda) - \eta')}$ in modulus. So for large $m$, $B(p_1)$ and $B(p_2)$ have the opposite signs of $c\lambda^K$ and $-c\lambda^K$, and moduli at least $\frac{c_2}{2} m^{-a-1} e^{2mg(\lambda)}$. With $m = \lfloor N/2 \rfloor$ for $p_1$ and $m = \lfloor (N-1)/2 \rfloor$ for $p_2$, so that $2m \ge N - 2$, the length of $E_N$ is at most $C\, N^{a+1} e^{-Ng(\lambda)}$ for large $N$.

For $\lambda > f^\star$, $\rho = f^\star/\lambda \in (0,1)$, $w = \frac{2}{\rho} - 1 > 1$, $w^2 - 1 = 4(1-\rho)/\rho^2$ and $w + \sqrt{w^2-1} = \big(2-\rho+2\sqrt{1-\rho}\big)/\rho = \big(1+\sqrt{1-\rho}\big)^2/\rho$. So $e^{-g(\lambda)} = \rho/(1+\sqrt{1-\rho})^2 = (1-\sqrt{1-\rho})/(1+\sqrt{1-\rho})$, using $\rho = (1-\sqrt{1-\rho})(1+\sqrt{1-\rho})$, and $\sqrt{1-\rho} = 1 - \frac{\rho}{2} + O(\rho^2)$ gives $\frac{\rho}{4} + O(\rho^2)$. □

**Proposition 41.6 (What positivity cannot see).** *As stated.*

*Proof.* (i) $\mu_f + \varepsilon\nu \geq \mu_f - |\varepsilon|\,|\nu| \geq (1 - |\varepsilon|C)\mu_f \geq 0$. For real $k \geq K$, $t^k \leq (f^\star)^{k-K} t^K$ on $(0, f^\star]$, so $\int t^k\, d|\nu| \leq C(f^\star)^{k-K}\langle 1\rangle_K$, and $\langle 1\rangle_k + \varepsilon u(k) = \int t^k\, d(\mu_f + \varepsilon\nu)$ (Proposition 40.2(i)). For $K' \geq K$ and $p \geq 0$ on $[0, f^\star]$, $L_{K',p}(\langle 1\rangle + \varepsilon u) = \int t^{K'} p\, d(\mu_f + \varepsilon\nu) \geq 0$.

(ii) $t^K(\mu_f + \varepsilon\nu)$ is a finite measure on $[0, f^\star]$, of total variation at most $(1 + |\varepsilon|C)\langle 1\rangle_K$, with the stated moments; two finite signed measures on $[0, f^\star]$ with the same moments agree on polynomials, hence on $C([0, f^\star])$ by the Weierstrass approximation theorem, hence are equal. On $J$ its density is $t^K(\varrho_f + \varepsilon\nu')$, $\nu'$ the density of $\nu$. If it agreed almost everywhere on $(0, f^\star)$ with a real-analytic function $\varphi$, then on $J$, $\nu' = (t^{-K}\varphi - \varrho_f)/\varepsilon$ almost everywhere, a real-analytic function on $J$, $\varrho_f$ being real-analytic on $(0, f^\star)$ (Proposition 40.3(i)); this is excluded. $t^K\mu_f$ has the real-analytic density $t^K\varrho_f$.

(iii) Let $|\nu'| \leq B$ on $(0, \delta]$. For $K_0 > -1$, $\int t^{K_0} d|\nu| \leq B\int_0^\delta t^{K_0} dt + \max_{[\delta, f^\star]} t^{K_0} \cdot C\mu_f([\delta, f^\star]) < \infty$ (Proposition 40.2(i)), so $u(k)$ converges for $\operatorname{Re} k \geq K_0$. Lemma S21(ii), with the measure $\nu$ on $(0, f^\star]$, the operator $\mathfrak{D}_{\mathbf{p}}$, for which $\mathfrak{D}_{\mathbf{p}}^\dagger t^k = \sum_i p_i(k) t^{k+i}$, and $E = \mathfrak{D}_{\mathbf{p}}\nu = 0$, gives $\sum_i p_i(k) u(k+i) = 0$ for $\operatorname{Re} k \geq K_0$, for every $K_0 > -1$. At $k = 0$, $p_0(0) = 0$, $p_0$ vanishing to order at least $n - 1 \geq 1$ at 0 (Proposition 40.5), and what remains is $\sum_{i \geq 1} p_i(0) u(i) = 0$. □

**Proposition 41.7 (A recurrence of order two is the volume of the direction times a continued fraction).** *As stated.*

*Proof.* (i) $\langle 1\rangle_{k+1} = \int t^{k+1}\mu_f(dt) < f^\star \int t^k \mu_f(dt)$, $\mu_f$ having a positive density on $(0, f^\star)$ and no mass at $f^\star$, whose preimage is the point $\log \alpha^\star$ (Proposition 40.3(i)); and $\langle 1\rangle_k > 0$. By Proposition 40.5 the recurrence holds at every integer $k \geq 1$, the origin being 0; divided by $\langle 1\rangle_{k+1}$ it reads $p_0(k)/\rho_k + p_1(k) + p_2(k)\rho_{k+1} = 0$.

(ii) The low-end relation (Proposition 40.5, Definition 39.37) is $aV_1 + p_1(0)\langle 1\rangle_1 + p_2(0)\langle 1\rangle_2 = 0$, and $\langle 1\rangle_2 = \rho_1\langle 1\rangle_1$. Let $\varphi(\rho) = -a/(p_1(0) + p_2(0)\rho)$. If $\varphi(0)$ and $\varphi(f^\star)$ are finite and positive, the affine denominator has the sign of $-a$ at both ends of $[0, f^\star]$, so it does not vanish there, $a \neq 0$, and $\varphi$ is continuous and monotone on $[0, f^\star]$, strictly unless $p_2(0) = 0$, when the two values agree. As $\rho_1 \in (0, f^\star)$, $\langle 1\rangle_1/V_1 = \varphi(\rho_1)$ lies between $\varphi(0)$ and $\varphi(f^\star)$, strictly when they differ.

(iii) The symbol $\chi(\lambda) = c_0 + c_1\lambda + c_2\lambda^2$ has real coefficients and the real root $f^\star$, so $\lambda_2$ is real, with $|\lambda_2| = \lambda' > f^\star$. By Lemma S26(i) and the proof of Theorem 41.3(iii), $\mathcal{V}$ has the basis $\langle 1\rangle$, $x_2$, with $x_2(k) = \lambda_2^k k^{\rho_2}(1 + O(1/k))$. For $r \in [0, f^\star]$ and $K \geq 1$ put

$$\psi = \langle 1\rangle_{K+2} - r\langle 1\rangle_{K+1}, \qquad \phi = x_2(K+2) - r\,x_2(K+1), \qquad y = \phi\,\langle 1\rangle - \psi\,x_2,$$

a solution with $y(K+2) = r\,y(K+1)$. Let $\tau_k = x_2(k+1)/x_2(k)$, defined for $k \geq J$ and tending to $\lambda_2$, and $\rho_k \to f^\star$ (Proposition 38.9). Then $\phi = x_2(K+1)(\tau_{K+1} - r)$ with $|\tau_{K+1} - r| \geq \frac{1}{2}(\lambda' - f^\star)$ for large $K$, so $|\phi| \geq c\,(\lambda')^K K^{\rho_2}$, and $|\psi| \leq 2f^\star\langle 1\rangle_{K+1} = O\big((f^\star)^K K^{-\frac{n-1}{2}}\big)$ by Theorem 41.3(ii), uniformly in $r$. So $|\psi/\phi| \leq C_1(f^\star/\lambda')^K K^{C'}$. Write $y(k) = \phi\,\langle 1\rangle_k\big(1 - \epsilon_k\big)$ with $\epsilon_k = \frac{\psi}{\phi}\frac{x_2(k)}{\langle 1\rangle_k}$. For $k < J$, $|\epsilon_k| \leq C_J|\psi/\phi| \to 0$. For $J \leq k \leq K+1$,

$$\epsilon_k = \frac{\rho_{K+1} - r}{\tau_{K+1} - r}\prod_{l=k}^{K}\frac{\rho_l}{\tau_l},$$

and choosing $J$ larger if needed, $|\rho_l/\tau_l| \le \bar q = \frac12(1+f^\star/\lambda') < 1$ for $l \ge J$, while $|(\rho_{K+1}-r)/(\tau_{K+1}-r)| \to (f^\star-r)/|\lambda_2-r| \le (f^\star-r)/(\lambda'-r) \le f^\star/\lambda' < \bar q$ uniformly in $r \in [0, f^\star]$. So there is $K_0$ such that for $K \ge K_0$ and every $r$, $|\epsilon_k| < 1$ for $1 \le k \le K+1$, hence $y(k) \ne 0$ there. For such $K$ put $\sigma_k = y(k+1)/y(k)$, $1 \le k \le K+1$; $\sigma_{K+1} = r$, and the recurrence at $k$, divided by $y(k+1)$, gives $p_1(k) + p_2(k)\sigma_{k+1} = -p_0(k)/\sigma_k \ne 0$, so $\sigma_k = \mathsf{M}_k(\sigma_{k+1})$ for $1 \le k \le K$. So $\mathsf{M}_1 \circ \cdots \circ \mathsf{M}_K(r) = \sigma_1$ is defined, and

$$\sigma_1 = \frac{\langle 1\rangle_2 - (\psi/\phi)x_2(2)}{\langle 1\rangle_1 - (\psi/\phi)x_2(1)} = \rho_1 + O\big((f^\star/\lambda')^K K^{C'}\big),$$

uniformly in $r$. By (ii) and $a \ne 0$, $p_1(0) + p_2(0)\rho_1 \ne 0$ and $\langle 1\rangle_1 = -aV_1/(p_1(0) + p_2(0)\rho_1)$. Let $\Sigma_K$ be the set of the $\mathsf{M}_1 \circ \cdots \circ \mathsf{M}_K(r)$ over the $r \in (0, f^\star)$ whose intermediate values lie in $(0, f^\star)$. An element of $\Sigma_{K+1}$, built from $r$, is $\mathsf{M}_1 \circ \cdots \circ \mathsf{M}_K(r')$ with $r' = \mathsf{M}_{K+1}(r) \in (0, f^\star)$ and the same intermediate values, so it lies in $\Sigma_K$; and $\rho_1 \in \Sigma_K$, built from $r = \rho_{K+1}$, its intermediate values being the $\rho_k \in (0, f^\star)$ by (i). □

**Definition 41.8 (The procedure that reads a ray from its list).** As stated.

*Proof.* (1) $V_1$ is the coefficient of Proposition 39.34; the maps of Proposition 39.11(ii) that are symmetries generate a group of symmetries, and a finite one is chosen. (2) The search stops by Theorem 39.22, every box of it being finite (Proposition 39.16(iv)); Proposition 41.10(iii) and (v) bound the strengths and primes at which an elimination misjudges acceptance. (3) The reductions from the relations the targets need are those of the box by Proposition 41.10(vi); the evaluations give the values of the entries of $\mathbf{B}$ at the strengths where the elimination has the generic pivots (Proposition 41.10(i)), the reconstruction is Proposition 41.10(ii), the check is Proposition 41.10(iv) and (v), and the certificate and the one-step relation are Theorem 39.21(ii) and (iii). (4) By Theorem 39.21(iv) Proposition 39.6 applies to the spanning set and its matrix: the first dependence $\sum_i q_i w_i = 0$ gives $\sum_j p_j(k) f^j \in \mathcal{L}$ (Proposition 39.6(ii)), and the ratios $q_i/q_h$ are ratios of signed maximal minors (Proposition 39.6(iii)), rational functions whose values at a point $k_0$ where $\mathbf{B}(k_0-h+1), \dots, \mathbf{B}(k_0)$ are finite and $w_0(k_0), \dots, w_{h-1}(k_0)$ independent are the coefficients of the dependence of the rows at $k_0$, so Proposition 41.10(ii) reconstructs them. The relation is the law of [1] when the spanning set is independent (Theorem 39.21(iv)), as under Proposition 39.42(i). The zeros of $p_0(k) = q_h(k+h)$ are those described, by the divisibility of Proposition 39.6(iii). (5) is Definition 40.10, which applies by Proposition 40.5, or Theorem 41.3(iv) with Proposition 41.4. (6) The closings are a solution with values in germs at $-\infty$ (Theorem 39.33), so Theorem 39.21(iii) gives $\varsigma^{-1}v = \mathbf{B}v$ for their vector $v(k) = \mathsf{C}_{-k}(\mathbf{b})$, that is $\mathsf{C}_{j+1}(\mathbf{b}) = \mathbf{B}(-j)\mathsf{C}_j(\mathbf{b})$ for all large $j$; the reflected recurrence and the low-end relation hold by Proposition 40.5 with Corollary 39.38(i) and (ii); the eigenvalues are Proposition 39.42(ii); the exponent is Theorem 40.6(iii). □

**Proposition 41.9 (The relations of a box are difference equations).** *As stated.*

*Proof.* (i) By Definition 39.15(i) the symbols at the level $(\mathbf{a}, \mathbf{b})$ are the $(u; \mathbf{a}, \mathbf{b})$ with $u \in \mathbb{Z}^{\mathcal{E}}$ and $u_{\mathcal{E}} = c$, $c = \sum_j (a_j r_j + b_j r_j^*)$: a translate of the lattice $\{u_{\mathcal{E}} = 0\}$, of rank $n-1$. As $\kappa_{1,U} > 0$, the wall degree of $g$ is at most $\mathfrak{t}$ exactly when $\kappa_U(g) + \mathfrak{t}\kappa_{1,U} \ge 0$ for every nonempty proper $U$ (Definition 39.15(ii)). With $L(U) = \sum_j \big(a_j d_j(U) + b_j d_j^*(U)\big)$ and $\kappa_U(g) = u_U - L(U)$, the sets $U = \{e\}$ and $U = \mathcal{E} \setminus \{e\}$, nonempty and proper as $n \ge 2$, give

$$L(\{e\}) - \mathfrak{t}\kappa_{1,\{e\}} \le u_e \le c - L(\mathcal{E} \setminus \{e\}) + \mathfrak{t}\kappa_{1,\mathcal{E}\setminus\{e\}},$$

an interval of length at most $C(1+\mathfrak{t})$ with $C = \max_e \max\big(|c - L(\mathcal{E}\setminus\{e\}) - L(\{e\})|,\ \kappa_{1,\{e\}} + \kappa_{1,\mathcal{E}\setminus\{e\}}\big)$, which holds at most $C(1+\mathfrak{t}) + 1$ integers; and $u_{e_0} = c - \sum_{e \ne e_0} u_e$.

(ii) By Definition 39.15(iii) the letter identity of $g = (u; \mathbf{a}, \mathbf{b})$ for $K_j$ has the terms $g$ and $\theta_{j,T}(u + \mathbf{1}_T; \mathbf{a} + \mathbf{e}_j, \mathbf{b})$, $T \in \mathcal{B}_j$, and that for $\Psi_j$ the terms $g$ and $\theta_{j,T}(u + \mathbf{1}_{\mathcal{E}\setminus T}; \mathbf{a}, \mathbf{b} + \mathbf{e}_j)$; the mark relation at $e$ has $g$ and, for every $j$, the terms of the $T \in \mathcal{B}_j$ with $T \ni e$ at the level $(\mathbf{a} + \mathbf{e}_j, \mathbf{b})$ and of the $T \not\ni e$ at the level $(\mathbf{a}, \mathbf{b} + \mathbf{e}_j)$, at most $|\mathcal{B}_j|$ for each $j$, with coefficients affine in $k$. The shifts are the $(\mathbf{1}_T; \mathbf{e}_j, \mathbf{0})$ and the $(\mathbf{1}_{\mathcal{E}\setminus T}; \mathbf{0}, \mathbf{e}_j)$, $T \in \mathcal{B}_j$.

(iii) The letter identities have the coefficients 1 and $-\theta_{j,T}$, and the mark relations $m_{0,e} + k\, m_{1,e} + u_e$, $-(s_{0j} + k\, s_{1j} + a_j)\theta_{j,T}$ and $-(t_{0j} + k\, t_{1j} + b_j)\theta_{j,T}$, affine in $k$, with the stated parts in $k$. For a relation $\sum_g c_g g$, $e_G \sum_g c_g g = \sum_g c_g\, e_G g$, and $e_G g$ is the orbit average of the orbit of $g$ (Definition 39.20(i)); so in the basis of the orbit averages the entry of an orbit is the sum of the $c_g$ over its symbols, again affine in $k$, the part in $k$ vanishing for the letter identities. □

**Proposition 41.10 (What a reading by evaluation certifies).** *As stated.*

*Proof.* (i) Gaussian elimination of the rows of $A$ over $\Bbbk(k)$, the columns taken in the order of the ranking, highest first, gives a basis of the row space $\mathcal{R}^G_{\mathfrak{B}}$ in echelon form, with distinct leading orbits, and the leading orbit of every nonzero element of the row space is one of them; so these are the pivots $P$ of Definition 39.20(ii), and $r = |P|$ is the rank. An element of the row space with zero coordinates on $P$ would have a leading orbit outside $P$, so it is 0: the projection of $\mathcal{R}^G_{\mathfrak{B}}$ onto the coordinates $P$ is injective, hence bijective. The chosen rows are a basis of $\mathcal{R}^G_{\mathfrak{B}}$, so $A_P(k)$ is invertible, and $E = A_P^{-1}[A_P \,|\, A_F]$, written on the columns $P$ and $F$, is the basis of $\mathcal{R}^G_{\mathfrak{B}}$ with the identity on $P$, the unique one by the bijectivity; it is therefore the reduced echelon basis, whose row of a pivot $p$ vanishes on the orbits ranked above $p$. For $x$ with coefficients in $\Bbbk$, $x - x_P E$ is congruent to $x$ and vanishes on $P$, so it is the reduction, $x_F - x_P A_P^{-1} A_F$ on $F$. The entries of $A_P$ are affine in $k$, so $\det A_P$, not 0, has degree at most $r$. At $k_0$: the minors of order $r+1$ of $A$ vanish identically, so $A(k_0)$ has rank at most $r$. If the minor of order $r$ of the rows $J'$ on the columns $P$ does not vanish at $k_0$, then $E = A_{J',P}^{-1} A_{J'}$ as well, by the uniqueness, so $E$ is regular at $k_0$, and $E(k_0)$ has the identity on $P$ and the zeros of an echelon form, identities in $k$; its $r$ independent rows lie in the row space of $A(k_0)$ and span it, so the elimination at $k_0$ has the pivots $P$ and gives the reduction $x_F - x_P E_F(k_0)$, the value at $k_0$. Conversely, if the elimination at $k_0$ has the pivots $P$, the projection of the row space of $A(k_0)$ onto the coordinates $P$ is injective, as above, so the columns $P$ of $A(k_0)$ have rank $r$, and a minor of order $r$ on them does not vanish.

(ii) The targets $f^{-1} b_i$ have their coefficients in $\Bbbk$. By (i) their reductions are $\big(x_F \det A_P - x_P \operatorname{adj}(A_P) A_F\big)/\det A_P$, the adjugate having entries of degree at most $r-1$ and $A_F$ entries of degree at most 1; the entries of $\mathbf{B}$ are coefficients of these reductions (Definition 39.20(iii)). If $N/D$ and $N'/D'$ have numerators of degree at most $d_1$ and denominators of degree at most $d_2$ and agree at $d_1 + d_2 + 1$ points where $D$ and $D'$ do not vanish, then $ND' - N'D$, of degree at most $d_1 + d_2$, vanishes at these points and is 0. With a common denominator, two numerators of degree at most $d_1$ that agree at $d_1 + 1$ points are equal.

(iii) Write the coefficient of a free orbit $c$ in the reduction of a target as $N_c/\det A_P$, $\deg N_c \le r$, by (ii). If the box is accepted at $\mathfrak{s}$ over $\Bbbk(k)$, every $N_c$ with $c$ outside the free Laurent orbits of wall degree at most $\mathfrak{s}$ vanishes, and at every $k_0$ with $\det A_P(k_0) \neq 0$ the elimination has the pivots $P$, so the same free orbits, and gives the values of the reductions: it accepts. If not, some such $N_c$ is a nonzero polynomial, and at every $k_0$ with $\det A_P(k_0) N_c(k_0) \neq 0$ the elimination has the pivots $P$ and its reduction has a nonzero coefficient on $c$: it rejects. The exceptions lie among the at most $2r$ zeros of $\det A_P$ and $N_c$.

(iv) Let $\mathbf{B}_{il} \neq \mathbf{B}'_{il}$, write $\mathbf{B}_{il} = N_{il}/\det A_P$ by (ii) and $\mathbf{B}'_{il} = N'_{il}/Q'$, and put $\Delta = N_{il} Q' - N'_{il} \det A_P$, a nonzero polynomial of degree at most $r+d$. At a strength $k_0$ with $Q'(k_0) \neq 0$ at which the elimination has the pivots $P$, it gives $\mathbf{B}(k_0)$, $\mathbf{B}$ being regular at $k_0$ by (i). If $\det A_P(k_0) \neq 0$,

agreement there gives $\Delta(k_0) = 0$; if $\det A_P(k_0) = 0$, then $N_{il} = \mathbf{B}_{il} \det A_P$ vanishes at $k_0$, $\mathbf{B}_{il}$ being regular there, and $\Delta(k_0) = 0$ again. So the strengths of agreement are among the at most $r + d$ zeros of $\Delta$, and a uniformly drawn strength of $S$ is one of them with probability at most $(r + d)/|S|$.

(v) The entries of $A$ are $q$-integral, and reduction modulo $q$ is a ring map, so the minors of order $r + 1$ of $A$ vanish modulo $q$ and $\det A_P$ modulo $q$ is the determinant of $A_P$ modulo $q$, which is not 0; the entries of $E = \operatorname{adj}(A_P)A_J/\det A_P$, $A_J$ the chosen rows, reduce to those of $\bar{E} = \operatorname{adj}(\bar{A}_P)\bar{A}_J/\det \bar{A}_P$, which has the identity on $P$ and the zeros of an echelon form. As in (i), $\bar{A}$ has the rank $r$ and the pivots $P$ over $\mathbb{F}_q(k)$, and its reductions $x_F - x_P\bar{E}_F$ are those over $\mathbb{Q}(k)$ taken modulo $q$; the proofs of (i) to (iv) use only linear algebra over a field and hold over $\mathbb{F}_q$. For the last claim, take an entry with $\mathbf{B}_{il} \neq \mathbf{B}'_{il}$ and write $\mathbf{B}_{il} = \tilde{N}/\tilde{D}$ and $\mathbf{B}'_{il} = N'/Q'$ with $\tilde{N}, \tilde{D}, N', Q' \in \mathbb{Z}[k]$ and $\tilde{D} = c_A \det A_P$ for a positive integer $c_A$; then $\Delta = \tilde{N}Q' - N'\tilde{D} \in \mathbb{Z}[k]$ is not 0. The finite set consists of the primes excluded above and those that divide $c_A$, every coefficient of $Q'$, or every coefficient of $\Delta$. For a prime outside it, $\mathbf{B}'$ modulo $q$ is $N'/Q'$ taken modulo $q$, $\Delta$ modulo $q$ is a nonzero polynomial of degree at most $r + d$, and the argument of (iv) over $\mathbb{F}_q$ bounds the probability by $(r + d)/q$.

(vi) An element of $\mathcal{R}'$ lies in $\mathcal{R}^G_{\mathfrak{B}}$ with the same leading orbit, so the pivots of $\mathcal{R}'$ are pivots of $\mathcal{R}^G_{\mathfrak{B}}$, and the free orbits of $\mathcal{R}^G_{\mathfrak{B}}$ are free for $\mathcal{R}'$. So $\bar{x}$ is a combination of free orbits of $\mathcal{R}'$ with $x - \bar{x} \in \mathcal{R}'$. The free orbits of $\mathcal{R}'$ are independent modulo $\mathcal{R}'$, an element of $\mathcal{R}'$ that combines them having a free leading orbit and so being 0; so $\bar{x}$ is the reduction modulo $\mathcal{R}'$. □

**Proposition 41.11 (Data with rational masses).** *As stated.*

*Proof.* (i) Both data lie on the balance wall with the same exponents, so the origin $(\mathbf{m} - \mathbf{m}'; \mathbf{0}, \mathbf{0})$ has $(\mathbf{m} - \mathbf{m}')_{\mathcal{E}} = 0$ and lies on it too. The walls are linear in the data, so the walls along the ray at $k$ are $\kappa_U(\mathbf{m}; \mathbf{s}, \mathbf{t}) + (k - 1)\kappa_U(\mathbf{m}'; \mathbf{s}, \mathbf{t})$, positive for real $k \geq 1$; by Lemma S20(iv) the values converge there and $\langle 1 \rangle_1 = P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$, the data at $k = 1$ being $(\mathbf{m}; \mathbf{s}, \mathbf{t})$.

(ii) The direction $N(\mathbf{m}; \mathbf{s}, \mathbf{t})$ is integral, on the balance wall, with the walls $N\kappa_U > 0$; its stride is $\alpha^{N\mathbf{m}} \prod_j K_j^{-Ns_j} \Psi_j^{-Nt_j} = f_{\mathbf{m}}^N$ on the positive orthant, and by Lemma S20(iv) at $k = \frac{1}{N}$, where the walls are the $\kappa_U > 0$, $\langle 1 \rangle_{1/N} = P_F(\mathbf{m}; \mathbf{s}, \mathbf{t})$. The wall degree is that of Definition 39.15(ii) with $\kappa_{1,U} = N\kappa_U$. □

**Proposition 41.12 (One factorization serves every coefficient).** *As stated.*

*Proof.* $A_P$ is affine in $k$, so $A_P(k_0 + \epsilon) = M + \epsilon E$. For $|\epsilon| \, \|EM^{-1}\| < 1$, $M + \epsilon E = (I + \epsilon EM^{-1})M$ and $(M + \epsilon E)^{-1} = M^{-1} \sum_{j \geq 0} (-\epsilon EM^{-1})^j$, so $x_P(M + \epsilon E)^{-1} = \sum_j x_P M^{-1}(-EM^{-1})^j \epsilon^j$, which is the recursion. With $A_F(k_0 + \epsilon) = A_F(k_0) + \epsilon E_F$, $E_F = A_F(k_0 + 1) - A_F(k_0)$, the coefficient of $\epsilon^j$, $j \geq 1$, in the reduction $x_F - x_P A_P^{-1} A_F$ is $-\lambda_j A_F(k_0) - \lambda_{j-1} E_F$. Each $\lambda_j$ is a product of a row by $E$ and one solve with the factorization of $M$. If $N/D$ and $N'/D'$, with the degree bounds and $D(k_0)D'(k_0) \neq 0$, have the same Taylor coefficients up to the order $d_1 + d_2$ at $k_0$, then $ND' - N'D = DD'(N/D - N'/D')$ vanishes to the order $d_1 + d_2 + 1$ at $k_0$ and has degree at most $d_1 + d_2$, so it is 0. □

**Proposition 41.13 (The densities of a spanning set).** *As stated.*

*Proof.* (i) A Laurent insertion $b$ with real coefficients is real and continuous on the positive part of the chart, and $b\,\omega_0$ has the density $b\,\omega_0/dy$; the sets $\{f \geq t\}$, $t > 0$, are compact (proof of Proposition 40.2(i)), so $\mu_{f,b}$ is finite on $[t, f^\star]$, with $|\mu_{f,b}|$ at most the image of $|b|\,\omega_0$. Where $\int |b| f^{\operatorname{Re} k} \omega_0 < \infty$, $\langle b \rangle_k = \int b f^k \omega_0 = \int t^k \mu_{f,b}(dt)$ by the change of variables of an image measure. The proof of

Proposition 40.3(i), with the weight $b\,\omega_0/dy$ in place of $\omega_0/dy$, gives a real-analytic density on $(0, f^\star)$; its positivity is not used there. The proof of Proposition 40.3(ii) uses Lemma S22(i), which allows every real-analytic weight $w$, here $w = b\,\omega_0/dy$, with $w(\log\alpha^\star) = b(\alpha^\star)\,(\omega_0/dy)(\alpha^\star)$; so the germ has the stated form with $g_b(0) = b(\alpha^\star)g(0)$.

(ii) Every monomial of $b_l$ has an absolutely convergent value for large real $k$ (Proposition 38.5(i)), so $\int t^{K_0} d|\mu_{f,b_l}| < \infty$ and $\int t^{K_0} d|t^{-1}\mu_{f,b_i}| < \infty$ for some $K_0$. Apply Lemma S21 to the measures $t^{-1}\mu_{f,b_i}$ and $\mu_{f,b_l}$ on $(0, f^\star]$ and the operators $Q(-1-\theta)$ and $-\tilde{P}_{il}(-1-\theta)$, whose adjoints are $Q(\theta)$ and $-\tilde{P}_{il}(\theta)$, $\theta^\dagger = -1-\theta$, with $Q(\theta)t^k = Q(k)t^k$. Its numbers are $M_E(k) = Q(k)\langle b_i\rangle_{k-1} - \sum_l \tilde{P}_{il}(k)\langle b_l\rangle_k$, which vanish for large $k$; by Lemma S21(i) the distribution $E$ vanishes, and that is the system.

(iii) The identity $v(k-1) = \mathbf{B}(k)v(k)$ of meromorphic functions, iterated, gives $v(k-K+1) = \mathbf{B}(k-K+2)\cdots\mathbf{B}(k)\,v(k)$. Near $k = K$ the right side is holomorphic, the factors $\mathbf{B}(m)$, $2 \le m \le K$, being finite and $v$ converging in a neighbourhood of $K$, the walls at $K$ being positive; so the left side, $v$ near 1, is holomorphic at 1 with the stated value. □

**Proposition 41.14 (The native gauges of all exponents).** *As stated.*

*Proof.* (i) $K_j$ and $\Psi_j$ are homogeneous of the degrees $r_j$ and $r_j^*$, so by Euler's identity $\sum_e R_{j,e} = r_j$ and $\sum_e R^*_{j,e} = r_j^*$, and $\mathbf{u}(\alpha;\mathbf{s},\mathbf{t})\mathcal{E} = \sum_j (s_j r_j + t_j r_j^*)$: the balance wall. A native gauge of the ray is a point of $X_F$ with $c_e^{(1)} = m_{1,e} - \sum_j \left(s_{1j}R_{j,e} + t_{1j}R^*_{j,e}\right) = 0$ for every $e$ (Definitions 38.1 and 38.4), that is $\mathbf{m}_1 = \mathbf{u}(\alpha;\mathbf{s}_1,\mathbf{t}_1)$; for data on the balance wall with $m_{\mathcal{E}} \ne 0$ the target of Definition 8.17 is $\mathbf{m}$. The set is the graph of the regular map $(\alpha,\mathbf{s},\mathbf{t}) \mapsto \mathbf{u}(\alpha;\mathbf{s},\mathbf{t})$, isomorphic by the projection to $X_F \times \mathbb{C}^{2N}$; $X_F$ is the complement in the torus $(\mathbb{C}^\times)^{\mathcal{E}}/\mathbb{C}^\times$, of dimension $n-1$, of the hypersurface $\prod_j K_j\Psi_j = 0$, which does not contain $\mathbf{1}$, so it is a nonempty open subset of an irreducible smooth variety, irreducible and smooth itself.

(ii) Let $\alpha_1,\dots,\alpha_\nu$ be the nondegenerate native gauges of the ray, $w_1$ its direction, and $W$ the data on the balance wall, a vector space of dimension $n-1+2N$ with the coordinates $(m_e)_{e\ne e_0}$, $\mathbf{s}$, $\mathbf{t}$. In the coordinates $y_e = \log\alpha_e$, $e \ne e_0$, of the chart near each $\alpha_c$, the map $(y,\mathbf{s},\mathbf{t}) \mapsto \big(\mathbf{u}(e^y;\mathbf{s},\mathbf{t});\mathbf{s},\mathbf{t}\big)$ into $W$ is holomorphic, and its derivative is block triangular, with the identity on $(\mathbf{s},\mathbf{t})$ and the matrix $\big(\partial_{y_{e'}}\mathbf{u}_e\big)_{e,e'\ne e_0} = -\big(\partial_{y_{e'}}c_e^{(1)}\big)$, minus the Hessian of $\log f$ on the chart, which is nonsingular at $\alpha_c$ (Definition 38.4). By the holomorphic inverse function theorem there are disjoint neighbourhoods $U_c$ of the $\alpha_c$ in $X_F$ and a neighbourhood $\mathcal{W}$ of $w_1$ in $W$ such that every $w \in \mathcal{W}$ has exactly one point native for it in each $U_c$. The exceptional set of Theorem 38.31(i), whose count concerns the critical points of the product of powers of the letters with the exponents of the direction, is a countable union of proper subvarieties; a proper subvariety is closed with empty interior in $W$, and its real points have empty interior in the real data, a polynomial that vanishes on an open set of real points being 0. By the Baire category theorem $\mathcal{W}$ contains data, real if need be, outside the exceptional set; at such data the native points are $(-1)^{n-1}\,\mathrm{eu}(X_F)$ in number, and they include the $\nu$ points in the $U_c$. □

**Definition 41.15 (The procedure that finds native gauges by continuation).** As stated.

*Proof.* (1) is Proposition 41.14(i). Steps (2) and (3) only propose points; nothing in the output rests on them. (4) In the coordinates $\alpha_e$, $e \ne e_0$, of the chart the map $F = (c_e^{(1)})_{e\ne e_0}$ is rational, with no pole on a box $Y$ on which no letter vanishes; read as a map of a real space of dimension $2(n-1)$, it is continuously differentiable there. Krawczyk's test asks for a point $\tilde{y} \in Y$, a nonsingular real matrix $\mathcal{Y}$ and an interval enclosure $F'(Y)$ of the derivative on $Y$ with $\tilde{y} - \mathcal{Y}F(\tilde{y}) + (I - \mathcal{Y}F'(Y))(Y-\tilde{y}) \subseteq Y$. By the mean value theorem in integral form, $y - \mathcal{Y}F(y) = \tilde{y} - \mathcal{Y}F(\tilde{y}) + \int_0^1 \big(I - \mathcal{Y}F'(\tilde{y} + \tau(y -$

$\tilde{y})))\big)d\tau\,(y-\tilde{y})$, and the integral of matrices in the convex set $F'(Y)$ lies in it; so $y \mapsto y - \mathcal{Y}F(y)$ maps the compact convex set $Y$ into itself and has a fixed point (Brouwer's fixed point theorem), a zero of $F$ since $\mathcal{Y}$ is nonsingular. A zero of $F$ in $Y$ is a native gauge of the ray (Definition 38.4, the defects summing to 0 as $f$ has degree 0, so that the defect at $e_0$ vanishes with the others). The Jacobian of $F$ in the $\alpha_e$ is the Hessian of $\log f$ in the $y_e$ times $\operatorname{diag}(\alpha_e^{-1})$, so an enclosure of the Hessian determinant that excludes 0 makes every native gauge in $Y$ nondegenerate. Disjoint boxes hold distinct points, and distinct values of $f$ at native gauges are distinct native values. □

## 50.41 Chapter 42: inputs along a ray

In this section $\vartheta = y\,d/dy$ on $(0,\infty)$, and for a function $F$ on $(0,\infty)$, $\hat{F}(s) = \int_0^\infty y^{s-1}F(y)\,dy$ wherever it converges. Two facts about this Mellin read are used throughout.

**Lemma S27.** *Let $F$ be smooth on $(0,\infty)$ with every $\vartheta^a F$ of size $O\big((1+|\log y|)^N\big)$ as $y \to 0$ and $O\big(y^C e^{-b\sqrt{y}}\big)$ as $y \to \infty$, for some $N, C$ and $b > 0$. Then $\hat{F}$ is holomorphic on $\operatorname{Re} s > 0$, and:*

*(i) for every polynomial $r$ and $k \in \mathbb{N}$, $\int_0^\infty y^{s-1}\,y^k r(\vartheta)F\,dy = r(-s-k)\,\hat{F}(s+k)$ on $\operatorname{Re} s > 0$;*

*(ii) if $\hat{F}$ vanishes on a vertical line in $\operatorname{Re} s > 0$, then $F = 0$.*

*Proof.* The bounds give absolute convergence, locally uniform on $\operatorname{Re} s > 0$, of every integral involved. (i) For $G = \vartheta^a F$ and $\operatorname{Re} a' > 0$, $\int_0^\infty y^{a'-1}\vartheta G\,dy = \int_0^\infty y^{a'}G'\,dy = \big[y^{a'}G\big]_0^\infty - a'\int_0^\infty y^{a'-1}G\,dy$, and the bracket vanishes at both ends by the bounds. So $\vartheta$ acts as $-a'$ under $\int_0^\infty y^{a'-1}(\cdot)\,dy$, and with $a' = s+k$ this gives the claim for every monomial $r(\vartheta) = \vartheta^a$, hence for every polynomial. (ii) With $y = e^v$ and $s = c + i\tau$, $\hat{F}(c+i\tau) = \int_{\mathbb{R}} e^{i\tau v}\,e^{cv}F(e^v)\,dv$ is the Fourier transform of the integrable continuous function $e^{cv}F(e^v)$; it vanishes for every $\tau$, so that function vanishes (the uniqueness of the Fourier transform on $L^1(\mathbb{R})$), and $F = 0$. □

**Definition 42.1 (Recurrences; the values input and the slope input).** As stated.

*Proof.* The objects are well defined: $L$, $L^*$ and $L'$ act on functions on a coset of $\mathbb{Z}$, or on meromorphic functions, the coefficients being polynomials; the symbol is that of Corollary 38.7 for the coefficients $c_i$; the two inputs are inputs in the sense of Definition 5.4, a lattice input with the densities $p$ and $\tilde{p}$ at the points of the coset beyond the cut, where both are finite because $p$ is holomorphic there. □

**Proposition 42.2 (The values of a ray as an input).** *As stated.*

*Proof.* (i) By Theorem 38.14(i), $0 < f \le C'e^{-c'\|y\|}$ on the chart, so $\int_{y_{e_0}=0} |f^k|\,dy = \int f^{\operatorname{Re} k}\,dy$ converges for $\operatorname{Re} k > 0$, locally uniformly, and $\langle 1\rangle_k$ is holomorphic there. By Proposition 39.34 it continues to a meromorphic function $p$ on $\mathbb{C}$, with $\lim_{k\to -j}(k+j)^{n-1}p(k) = V_1\,\mathsf{C}_j(1)$ for every $j \ge j(1) = 0$. The class $m = \sum_i c_i(k)\,\mathrm{Sh}^i[1]$ of the module (Definition 39.1) has the values $\langle m\rangle_k = \sum_i c_i(k)\langle f^i\rangle_k = \sum_i c_i(k)\langle 1\rangle_{k+i}$, which vanish for all large integers $k$; by Theorem 39.57, whose hypotheses are those of this statement, the continuation of $\langle m\rangle_k$ vanishes identically, and that continuation is $(Lp)(k)$. Multiply $(Lp)(k) = 0$ by $(k+j)^{n-1}$ and let $k \to -j$: for $i \le j$, $(k+j)^{n-1}p(k+i) \to V_1\,\mathsf{C}_{j-i}(1)$, and for $i > j$, $p(k+i)$ is holomorphic at $k = -j$, since $-j+i \ge 1$, so its product with $(k+j)^{n-1}$ tends to 0, $n \ge 2$. So $V_1\sum_{i\le\min(h,j)} c_i(-j)\,\mathsf{C}_{j-i}(1) = 0$, and $V_1 > 0$.

(ii) Differentiating $\sum_i c_i(x)p(x+i) = 0$ gives $\sum_i c_i'(x)p(x+i) + \sum_i c_i(x)p'(x+i) = 0$, that is $L\tilde{p} = L'p$.

(iii) By Proposition 38.9, $\langle 1\rangle_k > 0$ for large $k$ and $\langle 1\rangle_{k+1}/\langle 1\rangle_k \to f(\alpha^\star)$, so the root rate of $p$ along the integers is $f(\alpha^\star)$. For real $k \geq 1$, $\tilde{p}(k) = \int f^k \log(1/f)\, dy$ by differentiation under the integral, which the bound of (i) allows, and $|\tilde{p}(k)| \leq f(\alpha^\star)^{k-1} \int f\, |\log f|\, dy$ because $0 < f \leq f(\alpha^\star)$ on the chart (Theorem 38.14(i)). The last integral converges: $\log f$ is a combination of the coordinates $y_e$ and of logarithms of the letters, polynomials with nonnegative coefficients in $e^y$, so $|\log f| \leq C(1 + \|y\|)$, while $f \leq C' e^{-c'\|y\|}$. So the root rate of $\tilde{p}$ is at most $f(\alpha^\star)$. □

**Proposition 42.3 (Summation by parts at the cut).** *As stated.*

*Proof.* For a function $\varphi$ on the integers $x > -h$ with $\sum_x |R(x)|\,|\varphi(x)|$ and $\sum_x |R(x)\, c_i(x)\, \varphi(x+i)|$ finite over $x \geq 1$, which holds for $\varphi \in \{p, \tilde{p}\}$ since $R$ and the $c_i$ are polynomials and $p$, $\tilde{p}$ have root rate below one,

$$\sum_{x\geq 1} (L^*R)(x)\, \varphi(x) = \sum_{i=0}^{h} \sum_{x\geq 1} c_i(x-i) R(x-i)\, \varphi(x) = \sum_{i=0}^{h} \sum_{x' \geq 1-i} c_i(x') R(x')\, \varphi(x'+i),$$

with $x' = x - i$. The terms with $x' \geq 1$ add up to $\sum_{x'\geq 1} R(x')\, (L\varphi)(x')$, and those with $1 - i \leq x' \leq 0$, written $x' = -j$ with $0 \leq j < i$, to $\sum_{j=0}^{h-1} R(-j)\, B_j(\varphi)$, $B_j(\varphi) = \sum_{i=j+1}^{h} c_i(-j)\, \varphi(i-j)$. So

$$(\mathsf{T}_{0;(1)}[x]\, (L^*R)\, \varphi\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) = \sum_{x\geq 1} R(x)\, (L\varphi)(x) + \sum_{j=0}^{h-1} R(-j)\, B_j(\varphi),$$

the Trailmoment at index zero on the counting input being the sum over the integers beyond the cut (Definition 6.1). For $\varphi = p$, $Lp = 0$ and $B_j(p) = \sum_{i>j} c_i(x) p(x+i)\big|_{x=-j} = -\beta_j(-j)$, which is the first identity. For $\varphi = \tilde{p}$, $L\tilde{p} = L'p$ (Proposition 42.2(ii), whose proof uses only $Lp = 0$), and the same rearrangement with $L'$ in place of $L$ gives $\sum_{x\geq 1} R(x)(L'p)(x) = (\mathsf{T}_{0;(1)}[x]\, (L'^*R)\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) - \sum_j R(-j) B'_j(p)$ with $B'_j(p) = \sum_{i>j} c'_i(-j)\, p(i-j)$. Hence the second identity holds with the boundary $B_j(\tilde{p}) - B'_j(p) = \sum_{i>j} \big(c_i(x)\tilde{p}(x+i) - c'_i(x)p(x+i)\big)\big|_{x=-j} = -\frac{d}{dx}\sum_{i>j} c_i(x)p(x+i)\big|_{x=-j} = \beta'_j(-j)$.

If every $c_i$ with $i > j \geq 1$ vanishes to order two at $-j$, then near $x = -j$ each term of $\beta_j = -\sum_{i>j} c_i(x)p(x+i)$ is a polynomial with a double zero at $-j$ times a function holomorphic there, so $\beta_j(-j) = \beta'_j(-j) = 0$ for $j \geq 1$. □

**Proposition 42.4 (The fixed block).** *As stated.*

*Proof.* (i) For $R$ of degree $d$ with leading coefficient $\rho \neq 0$, $c_i(x-i)R(x-i)$ has degree at most $D + d$ and coefficient $[x^D]c_i\, \rho$ at $x^{D+d}$, so $L^*R$ has degree at most $D + d$ and coefficient $\chi(1)\rho \neq 0$ there. So $L^*$ is injective on $\Bbbk[x]$, and for every $d$ it maps the polynomials of degree at most $d$, a space of dimension $d + 1$, into those of degree at most $D + d$, meeting the polynomials of degree below $D$ only in 0. The dimensions add up, $(d+1) + D = D + d + 1$, so the polynomials of degree at most $D + d$ are the direct sum of the two, for every $d$; this is the decomposition of $\Bbbk[x]$.

(ii) Write $W = L^*B + r$ with $\deg r < D$ by (i), and $L'^*B = L^*B_2 + r_2$ in the same way; $B$, $r$, $B_2$, $r_2$ have coefficients in $\Bbbk$. By Proposition 42.3,

$$(\mathsf{T}_{0;(1)}[x]\, W\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) = (\mathsf{T}_{0;(1)}[x]\, r\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) - \sum_j B(-j)\, \beta_j(-j),$$

$$(\mathsf{T}_{0;(1)}[x]\, W\, \tilde{p}\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) = (\mathsf{T}_{0;(1)}[x]\, r\, \tilde{p}\, \mathbb{I}_{\mathbb{Z}})((0,\infty)) + (\mathsf{T}_{0;(1)}[x]\, r_2\, p\, \mathbb{I}_{\mathbb{Z}})((0,\infty))$$

$$-\sum_{j} B_2(-j)\,\beta_j(-j)+\sum_{j} B(-j)\,\beta'_j(-j).$$

A polynomial of degree below $D$ is a $\Bbbk$-combination of the $(x-1)^a$, $a<D$, and the Trailmoment at index zero is linear in the density, which gives the claim. □

**Corollary 42.5 (A mark opened at an integer strength).** *As stated.*

*Proof.* The bunch $B=\{e\}$ has width one, $c(\{e\})=r(\{e\})+r(R)-r(\mathcal{E})=1$ since $e$ is neither a loop nor a coloop, and the unit weights factor across $B$ and $R$. Its minors on $B$ are $\mathcal{B}|B$, with the one basis $\{e\}$, and $\mathcal{B}/R$, with the one basis $\varnothing$; so the bunch family of Theorem 19.13 has the period 1, and the hinge ratio of $B$ is $\varrho_B=\mathbf{E}_{\mathcal{B}/R}/\mathbf{E}_{\mathcal{B}|B}=\alpha_e/\alpha_e=1$ (Definition 19.2). The room family is the deletion $\mathcal{B}\setminus e=\mathcal{B}|R$ at the level $d$ and the contraction $\mathcal{B}/e=\mathcal{B}/B$ at the level $r$, and $\varrho_R=\mathbf{E}_{\mathcal{B}/e}/\mathbf{E}_{\mathcal{B}\setminus e}=X_{/}/X_{\setminus}$, since $\mathbf{E}=K\Psi=\alpha^R K(\alpha)K(\alpha^{-1})$ for each of them. At the centre $(s\mathbf{1};s,s)$ the wall of $B$ is $\kappa_B=s\,c(B)=s$ (Proposition 8.21(ii)), so the odd share is fixed at $2\delta^d=s-\kappa_B=0$ and the level shares are $(a^d,b^d,a^r,b^r)=(d,d,s-d,s-d)$, $d$ the even share. The one-table display of Theorem 19.13 reads

$$\Gamma(s)^2P_{\mathcal{B}}(s\mathbf{1};s,s)=\frac{1}{2\pi i}\int\Gamma(d)^2\,\Gamma(s-d)^2\,\Phi_s(d)\,dd,\qquad \Phi_s(d)=\int_{y_{e_0}=0}X_{\setminus}^{-d}X_{/}^{-(s-d)}\,dy,$$

the room period at the masses $s\mathbf{1}_R$, whose integrand is $\alpha^{s\mathbf{1}_R}K_{\setminus}^{-d}\Psi_{\setminus}^{-d}K_{/}^{-(s-d)}\Psi_{/}^{-(s-d)}=X_{\setminus}^{-d}X_{/}^{-(s-d)}$.

When $X_\flat=X_{\setminus}$, $\sup\varrho_B=1<\inf\varrho_R$, and Corollary 19.21(i) gives $P_{\mathcal{B}}(s\mathbf{1};s,s)=\sum\text{Res}$ over the poles of $\Gamma(d)^2$ at $d=-x$, $x\in\mathbb{N}$, where the two lattices of the level $d$ coincide, every room period there converging. Then $\Phi_s(-x)=q_s(x)$. When $X_\flat=X_{/}$, $\inf\varrho_B>\sup\varrho_R$, and the same corollary closes on the level $r$ with the opposite sign; the substitution $d=s-\tau$ exchanges the two levels and the two minors and turns that closing into the closing at $\tau=-x$, with $\Phi_s(s-\tau)=\int X_{/}^{-\tau}X_{\setminus}^{-(s-\tau)}\,dy$ and $q_s(x)=\Phi_s(s+x)$. In both cases, writing $\Phi^\flat(\tau)=\int X_\flat^{-\tau}X_\sharp^{-(s-\tau)}\,dy$, so that $q_s(x)=\Phi^\flat(-x)$, the room periods $\Phi^\flat(-x)$, $x\in\mathbb{N}$, converge. The walls of the room family with the minor of $X_\flat$ at the share $-x$ and that of $X_\sharp$ at $s+x$, at the masses $s\mathbf{1}_R$, are $\kappa_S=-x\,c_\flat(S)+(s+x)\,c_\sharp(S)$ with the widths $c_\flat$, $c_\sharp$ of the two minors, by the computation of Proposition 8.21(ii), which is linear in the strengths; they are affine in $x$ and positive at every point of $\mathbb{N}$ (Theorem 8.7), hence on $[0,\infty)$ and on a neighbourhood of it, so $q_s$ converges and is holomorphic there. The closing reads

$$\Gamma(s)^2P_{\mathcal{B}}(s\mathbf{1};s,s)=\sum_{x\in\mathbb{N}}\operatorname*{Res}_{\tau=-x}\Big(\Gamma(\tau)^2\,\Gamma(s-\tau)^2\,\Phi^\flat(\tau)\Big).$$

At the integer $s$, $\Gamma(s-\tau)=(s-1-\tau)\cdots(1-\tau)\,\Gamma(1-\tau)$, so $\Gamma(\tau)^2\Gamma(s-\tau)^2=V_s(-\tau)\,\Gamma_{0}^{4}\big[\tau,\ \tau,\ 1-\tau,\ 1-\tau\big]$, and by Lemma 5.16 the last factor has at $\tau=-x$ the principal part $(\tau+x)^{-2}$ and no simple part. The residue is therefore the derivative of $V_s(-\tau)\Phi^\flat(\tau)$ at $\tau=-x$,

$$-V_s'(x)\,q_s(x)+V_s(x)\,\Phi^{\flat\prime}(-x)=V_s(x)\,\tilde q_s(x)-V_s'(x)\,q_s(x),$$

since $\Phi^{\flat\prime}(-x)=-q_s'(x)$. The ratio $X_\flat/X_\sharp$ is at most $\rho$, so $0<q_s(x)\le\rho^x q_s(0)$ and, by differentiation under the integral, $|\tilde q_s(x)|\le\rho^{x-1}\int\log(X_\sharp/X_\flat)\,(X_\flat/X_\sharp)\,X_\sharp^{-s}\,dy$ for $x\ge1$, the last integral being $\tilde q_s(1)$, finite by the holomorphy at 1; with $V_s$ a polynomial the terms have root rate at most $\rho$. □

**Proposition 42.6 (Boundary relations).** *As stated.*

*Proof.* The identity $Lp = 0$ reads $\beta_{x_0,j}(x) = -\sum_{i>j} c_i(x)\, p(x+i)$ near $x_0$, and the right side is holomorphic at $x_0$ by the hypothesis; so is $\beta_{x_0,j}$, and the first relation is this identity at $x_0$. Differentiating it at $x_0$ gives $\beta'_{x_0,j}(x_0) = -\sum_{i>j} \big(c_i'(x_0)p(x_0+i) + c_i(x_0)p'(x_0+i)\big)$, which is the second relation since $\tilde p = -p'$. $\beta_{x_0,j} = \sum_{i\le j} c_i\, p(\cdot + i)$ is a finite sum of products of the Taylor expansions of the $c_i$ at $x_0$ with the Laurent expansions of $p$ at $x_0 + i$; its value and its derivative at $x_0$ are the coefficients of $(x-x_0)^0$ and $(x-x_0)^1$ of that sum, finitely many products of these coefficients.

When $c_h$ has no zero at the integers $x \ge 1$, $L\tilde p = L'p$ at $x$ expresses $\tilde p(x+h)$ through $\tilde p(x), \dots, \tilde p(x+h-1)$ and values of $p$, so by induction every $\tilde p(x)$, $x \ge 1$, is an affine function of $\tilde p(1), \dots, \tilde p(h)$; for an integer $x_0$ with $x_0 + j \ge 0$ the slopes in the second relation are at the integers $x_0 + i \ge 1$, $i > j$, and the relation is affine in the initial slopes.

At the cut of the values of a ray, Proposition 42.2(i) at $j = 0$ gives $p(x) = V_1 x^{-(n-1)}(1 + O(x))$, $\mathsf{C}_0(1) = 1$, so $\mathsf{L}_0 = V_1$, and $p$ has a pole of order exactly $n-1$ at 0. $\beta_{0,0} = c_0\, p$ is holomorphic at 0, so $c_0$ vanishes to order at least $n-1$ there, and $c_0(x)p(x) = \gamma_{n-1}\mathsf{L}_0 + (\gamma_{n-1}\mathsf{L}_1 + \gamma_n \mathsf{L}_0)\, x + O(x^2)$. The two relations at $x_0 = 0$, $j = 0$ are therefore the displayed ones; the first is the low-end relation of Corollary 39.38. When every table has unit weights, the values are $\langle 1\rangle_k = P_F(kq)$ for the question $q$ of the direction (Definition 17.1), the direction being a point of the balance wall with real data, and Theorem 17.5(i) gives $k^{n-1}P_F(kq) = V_1 + o(k)$ as $k \downarrow 0$, every $\log \theta_{j,T}$ being 0; since $k^{n-1}p(k) = \mathsf{L}_0 + \mathsf{L}_1 k + O(k^2)$ is holomorphic at 0, $\mathsf{L}_1 = 0$. □

**Proposition 42.7 (Minimality).** *As stated.*

*Proof.* (i) Let $D = \deg c_h$. Divided by $c_h(x)$, which has no zero at the integers $x \ge 1$, the recurrence reads $u(x+h) + \sum_{i<h} a_i(x)\, u(x+i) = 0$ with $a_i = c_i/c_h \to [x^D]c_i/[x^D]c_h$ as $x \to \infty$, and $a_0(x) \ne 0$ at these integers. The limit characteristic polynomial is $\chi(\lambda)/[x^D]c_h$, of degree $h$, with $\chi(0) = [x^D]c_0 \ne 0$ and roots $\lambda_c$ of distinct moduli. Perron's theorem on such recurrences (coefficients with limits, $a_0$ without zeros on the range, roots of the limit polynomial of distinct moduli) gives a basis $u_1, \dots, u_h$ of the solutions on the integers $x \ge 1$ with $u_c(x+1)/u_c(x) \to \lambda_c$, so $|u_c(x)|^{1/x} \to |\lambda_c|$. Let $u = \sum_c b_c u_c \ne 0$, and let $d$ be the index with the largest $|\lambda_d|$ among the $c$ with $b_c \ne 0$. For the other such $c$, $u_c(x)/u_d(x) \to 0$, the root rates being smaller; so $u(x) = b_d u_d(x)\,(1 + o(1))$, $u(x+1)/u(x) \to \lambda_d$, and $u$ has the root rate $|\lambda_d|$. Hence $u$ has root rate at most $\lambda$ exactly when $b_c = 0$ for every $c$ with $|\lambda_c| > \lambda$, and $U_\lambda$ is spanned by the $u_c$ with $|\lambda_c| \le \lambda$, which are independent.

(ii) $\tilde p$ satisfies $L\tilde p = L'p$ (Proposition 42.2(ii), whose proof uses only $Lp = 0$) and has root rate at most $\lambda$; a $\varphi$ with the same two properties differs from it by $u = \varphi - \tilde p$ with $Lu = 0$ and root rate at most $\lambda$, the root rate of a difference being at most the larger of the two; conversely every $\tilde p + u$, $u \in U_\lambda$, has both properties. The same holds for $L\varphi = 0$ with $p$. A second relation of Proposition 42.6 at an integer $x_0$ with $x_0 + j \le 0$ holds for $\tilde p + u$ exactly when its left side vanishes on $u$, since its right side is read off the Laurent data of $p$ at $x_0, \dots, x_0 + j \le 0$ and involves no slope; relations at other points are consequences of $L\tilde p = L'p$ and single out nothing; so if these left sides vanish together on no nonzero $u \in U_\lambda$, $\tilde p$ is the only solution of root rate at most $\lambda$ that satisfies the relations. The first relations single out $p$ in the same way. □

**Proposition 42.8 (The two ends paired).** *As stated.*

*Proof.* (i) For $\sigma < 0$ every term vanishes. For $\sigma \ge 0$, $c_i(\sigma - i) = \epsilon\, c_i(-\sigma)$ by the hypothesis read at $x = -\sigma$, so the sum is $\epsilon \sum_{i=0}^{\min(h,\sigma)} c_i(-\sigma)\, \mathsf{C}_{\sigma-i}$, which vanishes by Proposition 42.2(i) at $j = \sigma$.

(ii) Write $p(x) = \langle 1\rangle_x$, and $(L^*\mathsf{C})(\sigma) = \sum_i c_i(\sigma - i)\mathsf{C}_{\sigma-i}$. Shifting $j$ by one,

$$B(\sigma+1) = \sum_{i=1}^{h}\sum_{j=0}^{i-1} c_i(\sigma - j)\,\mathsf{C}_{\sigma-j}\,p(\sigma - j + i),$$

and the difference with $B(\sigma)$ keeps the term $j = 0$ of the first sum and the term $j = i$ of the second:

$$B(\sigma+1) - B(\sigma) = \sum_{i=1}^{h}\big(c_i(\sigma)\,\mathsf{C}_\sigma\,p(\sigma+i) - c_i(\sigma-i)\,\mathsf{C}_{\sigma-i}\,p(\sigma)\big) = \mathsf{C}_\sigma\,(Lp)(\sigma) - p(\sigma)\,(L^*\mathsf{C})(\sigma),$$

the terms $i = 0$ cancelling. For integers $\sigma \geq 1$ both vanish, by Proposition 42.2(i) and by (i). At $\sigma = 1$ only $j = 1$ has $\mathsf{C}_{1-j} \neq 0$, with $\mathsf{C}_0 = 1$, so $B(1) = \sum_{i\geq 1} c_i(0)\,p(i)$, which is $-\gamma_{n-1}V_1$ by Proposition 42.6.

(iii) By Proposition 42.6, $c_0$ vanishes to order at least $n-1$ at 0; with $\deg c_0 \leq n-1$ and $c_0 \neq 0$, $c_0 = \gamma_{n-1}x^{n-1}$, $\gamma_{n-1} \neq 0$, and the coefficients of $x^{n-1}$ give $\chi(\lambda) = \sum_i [x^{n-1}]c_i\lambda^i$ with $\chi(0) = \gamma_{n-1}$. On the centre ray of a family, $\alpha^\star = \mathbf{1}$ and $\langle 1\rangle_k = P_F(kq)$ for the centre question $q$; Theorem 17.5(ii) at the centre, with $-\nabla^2 X(0) = 4H$ on the chart in the coordinates $z = y/2$ ($H$ as in Theorem 38.15) and $e^{X(0)} = f(\mathbf{1})$, gives

$$\langle 1\rangle_k = f(\mathbf{1})^k\Big(\frac{2\pi}{k}\Big)^{\frac{n-1}{2}}(\det H)^{-1/2}\big(1 + O(k^{-1})\big),$$

the factor $2^{n-1}$ of that theorem cancelling against $\det(4H)^{1/2} = 2^{n-1}(\det H)^{1/2}$. Theorem 39.57 at $g = 1$ gives $j^{(n-1)/2}f(\mathbf{1})^j\mathsf{C}_j(1) \to (2\pi)^{-(n-1)}a_0(1)$, where $a_0(1) = (2\pi)^{(n-1)/2}(dy/dv)(\mathbf{1}) = (2\pi)^{(n-1)/2}(\det H)^{-1/2}$ by Theorem 38.6(i), the coordinates $v$ having $\frac12\sum v_i^2 = \frac12 y^{\mathsf{T}}Hy + O(|y|^3)$. So for fixed $i$, as $a \to \infty$, $\mathsf{C}_a\,\langle 1\rangle_{a+i} = \lambda_\star^i(\det H)^{-1}a^{-(n-1)}(1 + o(1))$, with $\lambda_\star = f(\mathbf{1})$, while $c_i(a) = [x^{n-1}]c_i\,a^{n-1} + O(a^{n-2})$. Every term of $B(\sigma)$ with $a = \sigma - j$ therefore tends to $[x^{n-1}]c_i\,\lambda_\star^i(\det H)^{-1}$, and

$$\lim_{\sigma\to\infty} B(\sigma) = (\det H)^{-1}\sum_{i=1}^{h} i\,[x^{n-1}]c_i\,\lambda_\star^i = (\det H)^{-1}\lambda_\star\chi'(\lambda_\star).$$

By (ii) the limit is $-\chi(0)V_1$, so $V_1 \det H = -\lambda_\star\chi'(\lambda_\star)/\chi(0)$, which is positive. The symbol vanishes at $\lambda_\star$: $\langle 1\rangle_{k+1}/\langle 1\rangle_k \to \lambda_\star$ (Proposition 38.9, a limit that needs no hypothesis on the coefficients of $L$), so dividing $\sum_i c_i(k)\langle 1\rangle_{k+i} = 0$ by $k^{n-1}\langle 1\rangle_k$ and letting $k \to \infty$ gives $\chi(\lambda_\star) = 0$; writing $\chi(\lambda) = \chi(0)\prod_c(1 - \lambda/\lambda_c)$ over its roots with multiplicity, a multiple root at $\lambda_\star$ would give $\chi'(\lambda_\star) = 0$, so $\lambda_\star$ is simple, and $\lambda_\star\chi'(\lambda_\star) = -\chi(0)\prod_{c\neq\star}(1 - \lambda_\star/\lambda_c)$. □

The rest of this section reads the list $\binom{\mathcal{E}}{1}$. There $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(1,y) = \int_0^\infty x^{-1}e^{-x-y/x}\,dx$ and $\mathsf{h}(y) = \sum_{m\geq 0} y^m/m!^2$. The configuration $(1,-1)$ on one scale has $\mathrm{cone}(t) = \mathbb{R}$, so $\mathsf{G}_{\nu;(1,-1)}(1,y)$ is entire in $\nu$ (Theorem 2.2); its ladder and its exchange relation (Theorem 2.4(i),(ii)) read $\partial_y\mathsf{G}_{\nu;(1,-1)}(1,y) = -\mathsf{G}_{\nu-1;(1,-1)}(1,y)$ and $\nu\,\mathsf{G}_{\nu;(1,-1)}(1,y) = \mathsf{G}_{\nu+1;(1,-1)}(1,y) - y\,\mathsf{G}_{\nu-1;(1,-1)}(1,y)$. So $\vartheta\mathsf{g} = -y\,\mathsf{G}_{-1;(1,-1)}(1,y)$, the exchange relation at $\nu = -1$ gives $y\,\mathsf{G}_{-2;(1,-1)}(1,y) = \mathsf{g} + \mathsf{G}_{-1;(1,-1)}(1,y)$, and

$$\vartheta^2\mathsf{g} = -y\,\mathsf{G}_{-1;(1,-1)}(1,y) + y^2\mathsf{G}_{-2;(1,-1)}(1,y) = y\,\mathsf{g};$$

and $\vartheta^2\mathsf{h} = \sum_{m\geq 1} y^m/(m-1)!^2 = y\,\mathsf{h}$. By Lemma 20.29(i) at $l = 0$, $\mathsf{g} = -\mathsf{h}\log y + \tilde{\mathsf{g}}$ with the entire function $\tilde{\mathsf{g}}(y) = \sum_m 2(\log\Gamma)'(1+m)\,y^m/m!^2$. The solutions of $\vartheta^2 u = yu$ on $(0,\infty)$ form a space

$\mathcal{V}$ of dimension two with the basis $\mathsf{g}, \mathsf{h}$, the two being independent since $\mathsf{g}/\mathsf{h}$ is not constant; the $n+1$ products $\mathsf{g}^{n-j}\mathsf{h}^j$ are independent as well, $\mathsf{h}^n$ times the powers of the nonconstant function $\mathsf{g}/\mathsf{h}$.

**Proposition 42.9 (The native gauges of $\binom{\mathcal{E}}{1}$).** *As stated.*

*Proof.* (i) The occupation of $e$ is $R_e = \alpha_e \partial_{\alpha_e} \log K = u_e$, and since $\Psi(\alpha) = \alpha^{\mathcal{E}} K(\alpha^{-1})$ the dual occupation is $R_e^* = \alpha_e \partial_{\alpha_e} \log \Psi = 1 - v_e$, $v_e = \alpha_e^{-1} / K(\alpha^{-1})$. The defect of the direction (Definition 38.1) is $c_e^{(1)} = m - s\,u_e - t\,(1 - v_e) = \frac{s-t}{n} - s\,u_e + t\,v_e$, by the balance $m - t = \frac{s-t}{n}$. On $X_F$ the $\alpha_e$, $K(\alpha)$ and $K(\alpha^{-1})$ are nonzero, so $u_e, v_e \neq 0$, $u_e v_e = \lambda(\alpha) \neq 0$ for every $e$, and $\sum_e u_e = \sum_e v_e = 1$.

Let $\alpha$ be a native gauge, $c_e^{(1)} = 0$ for every $e$. With $v_e = \lambda / u_e$, $s\,u_e^2 - \frac{s-t}{n} u_e - t\lambda = 0$ for every $e$: the occupations are roots of one quadratic, whose roots $u_+, u_-$ have $u_+ + u_- = \frac{s-t}{ns}$ and $u_+ u_- = -t\lambda / s \neq 0$. Let $u_+$ be taken on $n - j$ marks and $u_-$ on $j$ marks, $u_-$ on none when the occupations are all equal; $\sum_e u_e = 1$ gives $(n-j)u_+ + j u_- = 1$, and with $u_- = \frac{s-t}{ns} - u_+$, $(n-2j)u_+ = 1 - j\frac{s-t}{ns}$. At $2j = n$ this would read $1 = \frac{s-t}{2s}$, that is $s = -t$, which is excluded. Conversely let the occupations of $\alpha \in X_F$ take the value $u_+$ on $n-j$ marks and $u_-$ on $j$ marks with the three relations. Adding $(n-j)u_+ + j u_- = 1$, which follows from them, to $(n-j)u_- + j u_+$ gives $n(u_+ + u_-) = \frac{s-t}{s}$, so $(n-j)u_- + j u_+ = -\frac{t}{s}$ and

$$\lambda^{-1} = \sum_e \frac{1}{u_e} = \frac{n-j}{u_+} + \frac{j}{u_-} = \frac{(n-j)u_- + j u_+}{u_+ u_-} = -\frac{t}{s\,u_+ u_-},$$

using $\sum_e v_e = 1$, $v_e = \lambda / u_e$. Then $s\,u_e - t\,v_e = s\,u_e + s\,u_+ u_- / u_e = s(u_+ + u_-) = \frac{s-t}{n}$ at both values, so every defect vanishes. Since $\alpha_e = u_e K(\alpha)$, such a point is $(u_e)_e$ up to scale; conversely, for given $u_\pm$ with the three relations and a set of $j$ marks, the point $u = (u_e)_e$ has $K(u) = 1$ and $\Psi(u) = \prod_e u_e \sum_e u_e^{-1} = -\prod_e u_e\, t / (s\,u_+ u_-) \neq 0$, so it lies in $X_F$. The relations for the complement of the set, its $n-j$ marks at the value $u_+$ in the role of $u_-$ and the other $j$ at $u_-$ in the role of $u_+$, hold as well: the second is symmetric, and $(n-j)u_+ + j u_- = 1$ with $u_+ = \frac{s-t}{ns} - u_-$ gives $(2j-n)u_- = 1 - (n-j)\frac{s-t}{ns}$; so a set and its complement give the same point. The stride $f = \alpha^{m\mathbf{1}} K^{-s} \Psi^{-t} = \alpha^{(m-t)\mathbf{1}} K(\alpha)^{-s} K(\alpha^{-1})^{-t}$ is homogeneous of degree 0, and at $u$, where $K(u) = 1$ and $K(u^{-1}) = \lambda^{-1}$, it is $\big(\prod_e u_e\big)^{m-t} \lambda^t$.

(ii) At $s = t = \sigma$ the balance gives $m = \sigma$, the relations give $u_- = -u_+$ and $u_+ = 1/(n-2j)$, so $u$ is $1/(n-2j)$ times the sign $\eta$ with $\eta_e = -1$ exactly on the $j$ marks; on the chart $\alpha_{e_0} = 1$ the native gauge is the sign $\eta_{e_0}\eta$, with $K \neq 0$. Conversely every sign with $K(\eta) \neq 0$ is a native gauge (Theorem 38.17(ii)). The native value is $\lambda^\sigma = u_+^{2\sigma} = (n-2j)^{-2\sigma} = K(\eta)^{-2\sigma}$, and $|n-2j|$ for $0 \le j \le n$, $2j \neq n$, takes the $\lceil n/2 \rceil$ values $n, n-2, \dots$ down to 1 or 2. By Theorem 38.17(ii) the Hessian is $2\sigma\big(\kappa_\eta(X_e, X_{e'})\big)_{e,e' \neq e_0}$, the covariances of the indicators under $\mathbb{P}_\eta(\{e\}) = \pi_e = \eta_e / K(\eta)$. The indicator $X_e$ takes the value 1 at the basis $\{e\}$ and 0 at the others, so $\kappa_\eta(X_e, X_{e'}) = \delta_{ee'}\pi_e - \pi_e \pi_{e'}$. The determinant of a diagonal matrix less a matrix of rank one, $\det\big(\operatorname{diag}(\pi_e) - \pi\pi^{\mathsf{T}}\big)_{e,e' \neq e_0} = \prod_{e \neq e_0} \pi_e \big(1 - \sum_{e \neq e_0} \pi_e\big)$, is $\prod_e \pi_e$ since $\sum_e \pi_e = 1$, and the Hessian has the determinant $(2\sigma)^{n-1} \prod_e \eta_e / K(\eta)^n \neq 0$. □

**Proposition 42.10 (A parallel class enters through its list).** *As stated.*

*Proof.* (i) Theorem 21.5 with every mark replaced by the first rule gives $K_{\mathcal{B}} = K_{\mathcal{H}}\big((e_1(\alpha_{P_i}))_i\big) = K_{\mathcal{H}}(A)$, the weights of a list being one. Write $K_{\mathcal{H}}(A)^k = \sum_w \big([A^w] K_{\mathcal{H}}^k\big) A^w$ and $A^w = \prod_i \sum_{v_i \in \mathbb{N}^{P_i},\ |v_i| = w_i} \binom{w_i}{v_i} \alpha^{v_i}$. The classes are disjoint, so a monomial $\alpha^v$ determines $w_i = v_{P_i}$, and it occurs only in the term of that $w$, with the coefficient $\big([A^w] K_{\mathcal{H}}^k\big) \prod_i \binom{w_i}{v|_{P_i}}$.

(ii) On the centre ray at the strength one, $f^{-1} = K_{\mathcal{B}}(\alpha)K_{\mathcal{B}}(\alpha^{-1})$ (Corollary 39.36), and $\mathsf{C}_k^{\mathcal{B}}(1)$ is the constant term of $K_{\mathcal{B}}(\alpha)^k K_{\mathcal{B}}(\alpha^{-1})^k$ (Definition 39.30, the origin form being $dy$), which is $\sum_v \big([\alpha^v]K_{\mathcal{B}}^k\big)^2$, the coefficients being real. By (i) this is $\sum_w \big([A^w]K_{\mathcal{H}}^k\big)^2 \prod_i \sum_{|v_i|=w_i} \binom{w_i}{v_i}^2$, and $\sum_{|v_i|=w_i} \binom{w_i}{v_i}^2 = \sum_{v_i} \big([\alpha^{v_i}]K_{P_i}^{w_i}\big)^2$ is the constant term of $\big(K_{P_i}(\alpha)K_{P_i}(\alpha^{-1})\big)^{w_i}$, $K_{P_i} = \sum_{e\in P_i}\alpha_e$, the closing of $\binom{P_i}{1}$; for $|P_i| = 1$ it is 1.

(iii) At a sign, (i) gives $K_{\mathcal{B}}(\eta) = K_{\mathcal{H}}\big((\sum_{e\in P_i}\eta_e)_i\big)$, and the sum of $s_i$ signs takes the values $s_i, s_i - 2, \dots, -s_i$; the native values are those of Theorem 38.17(ii) for one table at the strength $\sigma$. □

**Proposition 42.11 (A mark added is a closing of the critical-line kernel).** *As stated.*

*Proof.* (i) By the display of Section 42.6 at the centre, from Corollary 18.24 and Proposition 8.15, $\Gamma(s)^2 p_{n+1}(s)$ is, for real $s > 0$, the share integral $\int_{a_1+\dots+a_{n+1}=s} \prod_e \Gamma(a_e)^2$ of Corollary 18.24, which converges absolutely, $|\Gamma(c+iy)|$ decaying like $e^{-\pi|y|/2}$. In the coordinates $(a_1, \dots, a_{n-1}, d)$ with $d = a_1 + \dots + a_n$, of Jacobian one, Fubini's theorem splits it into the integral over $\operatorname{Re} d = c$ of $\Gamma(s-d)^2$ times the share integral of $n$ marks at the total $d$. That inner integral is holomorphic in $d$ on $\operatorname{Re} d > 0$, by absolute and locally uniform convergence, and equals $\Gamma(d)^2 p_n(d)$ at real $d$ by the same display at $n$ marks; so it equals $\Gamma(d)^2 p_n(d)$ on $\operatorname{Re} d > 0$, $p_n$ being holomorphic there by the bound of Theorem 38.14(i), as in the proof of Proposition 42.2(i). This gives the formula at real $s$; both sides are holomorphic in $s$ on $\operatorname{Re} s > c$, so it holds on $\operatorname{Re} s > 0$. At $s = 1$ and $c = \frac12$, with $d = \frac12 + iy$, the kernel $\Gamma(d)^2\Gamma(1-d)^2$ is $2\pi\, p_{\mathrm{crit}}(y)$ (Definition 36.1) and $dd/(2\pi i) = dy/(2\pi)$.

(ii) At the integer $s$, $\Gamma(s-d) = (s-1-d)\cdots(1-d)\,\Gamma(1-d)$, so the integrand of (i) is $\Gamma_0^4\big[d,\ d,\ 1-d,\ 1-d\big]\ G(d)$ with $G = W_s\, p_n$, which has double zeros at $1, \dots, s-1$. On $\operatorname{Re} d \geq 1$, $|p_n(d)| \leq p_n(\operatorname{Re} d) \leq 4^{1-\operatorname{Re} d} p_n(1)$, since $0 < (K(\alpha)K(\alpha^{-1}))^{-1} \leq n^{-2} \leq \frac14$ on the chart (Theorem 38.14(i)); and $\big|\Gamma_0^4\big[d,\ d,\ 1-d,\ 1-d\big]\,\big| = \pi^2/|\sin\pi d|^2$ is at most $\pi^2/\cosh^2(\pi \operatorname{Im} d)$ on the lines $\operatorname{Re} d = N + \frac12$ and at most $Ce^{-2\pi|\operatorname{Im} d|}$ for $|\operatorname{Im} d| \geq 1$. So the integrals over the lines $\operatorname{Re} d = N + \frac12$ and over the horizontal segments $|\operatorname{Im} d| = Y$ tend to 0, and moving the line to the right (Theorem 4.5) gives $-\sum_{m\geq 1} \operatorname{Res}_{d=m}$. By Lemma 5.16 the kernel has at $d = m$ the principal part $(d-m)^{-2}$ and no simple part, so the residue is $G'(m)$, zero for $m < s$, and

$$\Gamma(s)^2 p_{n+1}(s) = -\sum_{m\geq 1} \big(W_s\, p_n\big)'(m) = \sum_{m\geq 1} \big(W_s(m)\, \tilde p_n(m) - W_s'(m)\, p_n(m)\big),$$

the series converging geometrically; it is the index-zero Trailmoment cut at 0 (Definition 6.1). For the second identity, $\int_0^\infty y^{s-1}\mathsf{g}^n\, dy = \Gamma(s)^2 p_n(s)$ at every real $s > 0$, and the terms of $\mathsf{h}$ are positive, so by monotone convergence

$$\int_0^\infty y^{s-1}\mathsf{g}^n\mathsf{h}\, dy = \sum_{k\geq 0} \frac{\Gamma(s+k)^2}{k!^2}\, p_n(s+k) = \sum_{k\geq 0} W_s(s+k)\, p_n(s+k) = \sum_{m\geq 1} W_s(m)\, p_n(m),$$

since $\Gamma(s+k)/k! = (k+1)\cdots(k+s-1)$ and $W_s$ vanishes at $1, \dots, s-1$.

(iii) The share integral of (i) is symmetric in the $n+1$ shares, the fibre integral not depending on the coordinates in which it is written (Definition 4.1), and $\sum_e a_e = s$ on the fibre; so for each $e$, $\int a_e \prod_{e'} \Gamma(a_{e'})^2 = \frac{s}{n+1}\Gamma(s)^2 p_{n+1}(s)$. For $e$ the last mark, $a_e = s - d$ in the coordinates of (i), and at $s = 1$ the closing of (ii), with $G(d) = (1-d)p_n(d)$, gives

$$\frac{1}{n+1}\, p_{n+1}(1) = -\sum_{m\geq 1} \big((1-d)p_n(d)\big)'\big|_{d=m} = \sum_{m\geq 1} \big(p_n(m) - (m-1)\, \tilde p_n(m)\big).$$

With $p_{n+1}(1) = (\mathsf{T}_{0;(1)}[x]\,\tilde{p}_n\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ from (ii) this is the stated relation. □

The following bounds give the Mellin reads of the products of $\mathsf{g}$ and $\mathsf{h}$ the hypotheses of Lemma S27 and fix their growth along the integers.

**Lemma S28.** *(i)* $\mathsf{g} > 0$ *and* $\mathsf{h} \geq 1$ *on* $(0,\infty)$*, and for* $y \geq 1$

$$\frac{1}{2e}\,y^{-1/2}e^{-2\sqrt{y}} \leq \mathsf{g}(y) \leq \sqrt{\pi}\,y^{-1/4}e^{-2\sqrt{y}}, \qquad e^{-4}y^{-1}e^{2\sqrt{y}} \leq \mathsf{h}(y) \leq e^{2\sqrt{y}}.$$

*(ii) For integers* $0 \leq j < n/2$*, every* $\vartheta^a\big(\mathsf{g}^{n-j}\mathsf{h}^j\big)$ *is* $O\big((1+|\log y|)^n\big)$ *as* $y \to 0$ *and* $O\big(y^C e^{-2(n-2j)\sqrt{y}}\big)$ *as* $y \to \infty$*. For integers* $n/2 \leq j \leq n$*,* $\int_1^\infty y^{s-1}\mathsf{g}^{n-j}\mathsf{h}^j\,dy = \infty$ *for real* $s \geq n$*.*

*(iii) For these* $j$*,* $\lim_{k\to\infty}\Big(\Gamma(k)^{-2}\int_0^\infty y^{k-1}\mathsf{g}^{n-j}\mathsf{h}^j\,dy\Big)^{1/k} = (n-2j)^{-2}$ *along the integers.*

*Proof.* (i) $\mathsf{g}$ is the integral of a positive function and $\mathsf{h} \geq 1$ term by term. The upper bound on $\mathsf{g}$ is Lemma 20.29(ii) at $\nu = 0$. For the lower bound restrict the integral to $x = \sqrt{y} + \tau$, $0 \leq \tau \leq 1$, where $x + y/x = 2\sqrt{y} + \tau^2/(\sqrt{y}+\tau) \leq 2\sqrt{y}+1$ and $x^{-1} \geq (2\sqrt{y})^{-1}$. For $\mathsf{h}$, $\sum_m y^m/m!^2 \leq \big(\sum_m y^{m/2}/m!\big)^2 = e^{2\sqrt{y}}$; and with $m = \lfloor\sqrt{y}\rfloor \geq 1$ and $m! \leq e\,m^{m+1}e^{-m}$ (from $\log m! \leq \int_1^m \log x\,dx + \log m$),

$$\mathsf{h}(y) \geq \frac{y^m}{m!^2} \geq \frac{e^{2m}}{e^2 m^2}\Big(\frac{\sqrt{y}}{m}\Big)^{2m} \geq \frac{e^{2\sqrt{y}-2}}{e^2\,y}.$$

(ii) By the ladder, $\vartheta\mathsf{g} = -y\,\mathsf{G}_{-1;(1,-1)}(1,y)$, at most $\sqrt{\pi}e^{1/4}y^{1/4}e^{-2\sqrt{y}}$ for $y \geq 1$ by Lemma 20.29(ii) at $\nu = -1$; and $0 \leq \vartheta\mathsf{h} = \sum_m m y^m/m!^2 \leq y\,\mathsf{h}$. Since $\vartheta^2 = y$ on both, every $\vartheta^a\mathsf{g}$ and $\vartheta^a\mathsf{h}$ is a combination of the function and its $\vartheta$ with polynomial coefficients, of size $O(y^C e^{\mp 2\sqrt{y}})$ at infinity; near 0, $\mathsf{g} = -\mathsf{h}\log y + \tilde{\mathsf{g}}$ with $\mathsf{h}$ and $\tilde{\mathsf{g}}$ entire, so every $\vartheta^a\mathsf{g}$ is $O(1+|\log y|)$ and every $\vartheta^a\mathsf{h}$ bounded. By the Leibniz rule $\vartheta^a(\mathsf{g}^{n-j}\mathsf{h}^j)$ is a sum of products of $n-j$ derivatives of $\mathsf{g}$ and $j$ of $\mathsf{h}$, which gives the bounds, $e^{-2(n-j)\sqrt{y}}e^{2j\sqrt{y}} = e^{-2(n-2j)\sqrt{y}}$. For $n/2 \leq j \leq n$, (i) gives on $[1,\infty)$

$$\mathsf{g}^{n-j}\mathsf{h}^j \geq (2e)^{-(n-j)}e^{-4j}\,y^{-(n+j)/2}\,e^{2(2j-n)\sqrt{y}} \geq (2e)^{-n}e^{-4n}\,y^{-n},$$

and $\int_1^\infty y^{s-1-n}\,dy = \infty$ for $s \geq n$.

(iii) Put $b = n-2j \geq 1$ and $I(k) = \int_0^\infty y^{k-1}\mathsf{g}^{n-j}\mathsf{h}^j\,dy$. The part over $(0,1]$ is at most $\int_0^1 C(1+|\log y|)^n\,dy$, bounded in $k \geq 1$. By (i), on $[1,\infty)$ the integrand lies between $c_1 y^{k-1-n}e^{-2b\sqrt{y}}$ and $c_2 y^{k-1}e^{-2b\sqrt{y}}$ with $c_1, c_2 > 0$, and for a fixed real $a$, with $y = \tau^2$,

$$\int_1^\infty y^{k-1+a}e^{-2b\sqrt{y}}\,dy = 2\int_1^\infty \tau^{2k+2a-1}e^{-2b\tau}\,d\tau = 2\,\Gamma(2k+2a)\,(2b)^{-2k-2a} + O(1).$$

$\Gamma(2k+2a)/\Gamma(2k)$ has root rate one and $\Gamma(2k)(2b)^{-2k}$ grows faster than every exponential, so $\big(I(k)/\Gamma(2k)\big)^{1/k} \to (2b)^{-2}$; and $\Gamma(2k)/\Gamma(k)^2 = \frac{k}{2}\binom{2k}{k}$ has the root rate 4, so $\big(I(k)/\Gamma(k)^2\big)^{1/k} \to 4(2b)^{-2} = b^{-2}$. □

**Theorem 42.12 (The law of the values of $\binom{\mathcal{E}}{1}$).** *As stated.*

*Proof.* (i) *Annihilation.* Let $u \in \mathcal{V}$ and $v = \vartheta u$, so that $\vartheta v = yu$. For $0 \le i \le n$, $\mathbf{R}_i(u^n) = \frac{n!}{(n-i)!} u^{n-i} v^i$: this holds at $i = 0, 1$, and $\vartheta(u^{n-i}v^i) = (n-i)u^{n-i-1}v^{i+1} + i\,y\,u^{n-i+1}v^{i-1}$ gives

$$\vartheta \mathbf{R}_i(u^n) = \frac{n!}{(n-i-1)!} u^{n-i-1}v^{i+1} + i(n+1-i)\,y\,\frac{n!}{(n-i+1)!} u^{n-i+1}v^{i-1},$$

whose last term is $i(n+1-i)\,y\,\mathbf{R}_{i-1}(u^n)$; so the recursion gives the claim at $i+1 \le n$, and at $i = n$ the first term carries the factor $n - i = 0$ and $\mathbf{R}_{n+1}(u^n) = 0$. A product of $n$ solutions is a combination of $n$-th powers of solutions, $u_1 \cdots u_n = \frac{1}{2^n n!} \sum_{\epsilon \in \{\pm 1\}^n} \epsilon_1 \cdots \epsilon_n \, (\epsilon_1 u_1 + \cdots + \epsilon_n u_n)^n$, so $\mathbf{Q}_n$ annihilates every such product.

*Normal form and leading terms.* Since $\vartheta y = y(\vartheta + 1)$, every operator built from $y$ and $\vartheta$ is $\sum_k y^k r_k(\vartheta)$ with polynomials $r_k$, uniquely: applied to $y^z$ it gives $\sum_k y^{k+z} r_k(z)$. Write $\mathbf{R}_i = \sum_k y^k \ell_{i,k}(\vartheta)$. By induction $\ell_{i,0} = \vartheta^i$ and $\deg \ell_{i,k} \le i - 2k$: $\vartheta\, y^k \ell_{i,k}(\vartheta) = y^k(\vartheta + k)\ell_{i,k}(\vartheta)$ has degree at most $i + 1 - 2k$, and so has $y \cdot y^{k-1}\ell_{i-1,k-1}(\vartheta)$. The coefficients $\alpha_{i,k}$ of $\vartheta^{i-2k}$ in $\ell_{i,k}$ therefore satisfy $\alpha_{i+1,k} = \alpha_{i,k} - i(n+1-i)\alpha_{i-1,k-1}$, the shift by $k$ not changing a leading coefficient, and $\sigma_i(\xi, \eta) = \sum_k \alpha_{i,k}\eta^k \xi^{i-2k}$ satisfies $\sigma_{i+1} = \xi\sigma_i - i(n+1-i)\,\eta\,\sigma_{i-1}$, $\sigma_0 = 1$, $\sigma_1 = \xi$. Let $J$ be the matrix, indexed by $0, \ldots, n$, of the derivation $\Delta$ of the forms of degree $n$ in $U, V$ with $\Delta U = V$ and $\Delta V = \eta U$, in the basis $w_i = U^{n-i}V^i$: $\Delta w_i = (n-i)w_{i+1} + i\eta\, w_{i-1}$. The leading principal minors of $\xi - J$, expanded along their last row, satisfy the same recursion, $\det_{i+1} = \xi \det_i - (n+1-i)\, i\eta \det_{i-1}$, so $\sigma_{n+1} = \det(\xi - J)$. Over $\mathbb{C}(\eta^{1/2})$ the forms $E_\pm = V \pm \eta^{1/2} U$ have $\Delta E_\pm = \pm\eta^{1/2} E_\pm$, so the basis $E_+^{n-j}E_-^j$ diagonalizes $\Delta$ with the eigenvalues $(n-2j)\eta^{1/2}$, and $\det(\xi - J) = \prod_{j=0}^n \big(\xi - (n-2j)\eta^{1/2}\big)$; the factors $j$ and $n - j$ pair into $\xi^2 - (n-2j)^2\eta$, and at even $n$ the factor $j = n/2$ is $\xi$. With $q_{n,k} = \ell_{n+1,k}$ and $a_{n,k} = \alpha_{n+1,k}$ this is the display; $a_{n,h}$ is the coefficient of the top power $\eta^h$, $\prod_{0 \le j < n/2} \big(-(n-2j)^2\big) \ne 0$.

*The adjoint.* Expanding the recursion, $\mathbf{R}_m$ is the sum over the tilings of $\{1, \ldots, m\}$ by monomers $\{a\}$ and dimers $\{a, a+1\}$ of the products of $\vartheta$ for each monomer and $-a(n+1-a)\,y$ for each dimer, taken from left to right in decreasing order of position: the tilings of $\{1, \ldots, m+1\}$ end with the monomer $\{m+1\}$ or the dimer $\{m, m+1\}$, which are the two terms of the recursion. The formal adjoint for $dy/y$ reverses products, takes $\vartheta$ to $-\vartheta$ and keeps $y$; a tiling of $\{1, \ldots, m\}$ with $d$ dimers has $m - 2d$ monomers, so $\mathbf{R}_m^*$ is $(-1)^m$ times the same sum with the products taken in increasing order of position. The reflection $a \mapsto m + 1 - a$ of the positions maps tilings to tilings, reverses the order, and takes the dimer $\{a, a+1\}$ to $\{a', a'+1\}$ with $a' = m - a$, of weight $-(m-a')(n+1-m+a')\,y$. At $m = n+1$ this weight is $-a'(n+1-a')\,y$, so $\mathbf{Q}_n^* = \mathbf{R}_{n+1}^* = (-1)^{n+1}\mathbf{R}_{n+1}$. Since $q(-\vartheta)\,y^k = y^k q(-\vartheta - k)$, $\mathbf{Q}_n^* = \sum_k q_{n,k}(-\vartheta)y^k = \sum_k y^k q_{n,k}(-\vartheta - k)$, and the uniqueness of the normal form gives $q_{n,k}(-\vartheta - k) = (-1)^{n+1} q_{n,k}(\vartheta)$.

(ii) For $k \ge 1$, $c_{n,k}$ has degree at most $(n+1-2k) + 2(k-1) = n - 1$ with the coefficient $(-1)^{n+1}a_{n,k}$ of $s^{n-1}$, as $c_{n,0} = (-1)^{n+1}s^{n-1}$ has with $a_{n,0} = 1$. Setting $\xi = 1$ in (i), $\sum_k a_{n,k}\lambda^k = \prod_{0 \le j < n/2} \big(1 - (n-2j)^2\lambda\big)$, whose coefficients $(-1)^k e_k\big((n-2j)^2\big)$ are nonzero for $0 \le k \le h$; so every $c_{n,k}$ has degree $n-1$ and the symbol is $\chi_n$. By (i), $c_{n,k}(-s-k) = q_{n,k}(s)\big((-s-k+1)\cdots(-s-1)\big)^2 = (-1)^{n+1}q_{n,k}(-s-k)\big((s+1)\cdots(s+k-1)\big)^2 = (-1)^{n+1}c_{n,k}(s)$ for $k \ge 1$, and $c_{n,0}(-s) = s^{n-1} = (-1)^{n+1}c_{n,0}(s)$. The double zeros are those of the squared factor. For odd $n$, $q_{n,h}$ is the nonzero constant $a_{n,h}$; for even $n$ it has degree one, $q_{n,h}(\vartheta) = a_{n,h}(\vartheta - \vartheta_0)$, and $q_{n,h}(-\vartheta - h) = -q_{n,h}(\vartheta)$ forces $\vartheta_0 = -h/2$, so $q_{n,h}(-s-h) = -a_{n,h}(s + h/2)$; in both cases $c_{n,h}(s) \ne 0$ at the integers $s \ge 0$. For $n \ge 2$ let $M(s) = \Gamma(s)^2 P_{\binom{\varepsilon}{1}}(s\mathbf{1}; s, s) = \int_0^\infty y^{s-1}\mathrm{g}^n\, dy$ (Section 42.6). By Lemma S28(ii), Lemma S27(i) applies to $\mathrm{g}^n$, and $\mathbf{Q}_n(\mathrm{g}^n) = 0$ gives $\sum_k q_{n,k}(-s-k)\,M(s+k) = 0$ on $\operatorname{Re} s > 0$. Dividing by $\Gamma(s+1)^2$, with $\Gamma(s)^2/\Gamma(s+1)^2 = s^{-2}$ and $\Gamma(s+k)^2/\Gamma(s+1)^2 = \big((s+1)\cdots(s+k-1)\big)^2$, and $q_{n,0}(-s) = (-s)^{n+1}$, gives the identity on $\operatorname{Re} s > 0$, and on $\mathbb{C}$ by the meromorphic continuation (Proposition 39.34).

(iii) Write $P(k) = P_{\binom{\varepsilon}{1}}(k\mathbf{1}; k, k)$ and suppose $\sum_{i=0}^{m} r_i(k)P(k+i) = 0$ for all large integers $k$, with $r_i \in \mathbb{C}[k]$, $r_0 r_m \neq 0$ and $m < h$. The centre ray at the strength one has origin 0, letter exponents one and walls $\kappa_{1,S} = 1$, so by Proposition 42.2(i) the relation holds identically. On $\operatorname{Re} s > 0$ multiply it by $\Gamma(s+m)^2$: $\sum_i \tilde{r}_i(s)M(s+i) = 0$ with the polynomials $\tilde{r}_i(s) = r_i(s)\big((s+i)\cdots(s+m-1)\big)^2$, $\tilde{r}_m = r_m$. Put $T = \sum_i y^i \tilde{r}_i(-\vartheta - i)$. By Lemma S27(i), with Lemma S28(ii), the Mellin read of $T(\mathsf{g}^n)$ is $\sum_i \tilde{r}_i(s)M(s+i) = 0$, and by Lemma S27(ii) $T(\mathsf{g}^n) = 0$.

On the ring of polynomials in an indeterminate $L$ with entire coefficients let $\vartheta$ act by $\vartheta\big(\sum_l E_l L^l\big) = \sum_l (\vartheta E_l)L^l + \sum_l l E_l L^{l-1}$, and put $\Phi(y, L) = T\big[(\tilde{\mathsf{g}} - \mathsf{h}\, L)^n\big] = \sum_{l \le n} E_l(y) L^l$, $E_l$ entire. Substituting $L = \log y - c$, for a constant $c$, commutes with this $\vartheta$, so $T\big[(\mathsf{g} + c\,\mathsf{h})^n\big] = \Phi(y, \log y - c)$ on $(0, \infty)$, and $\Phi(y, \log y) = T(\mathsf{g}^n) = 0$. The function $y \mapsto \Phi(y, \log y)$ continues along every path in $\mathbb{C}^\times$, and once around 0 it becomes $\Phi(y, \log y + 2\pi i)$; the continuation of 0 is 0, so $\Phi(y, \log y + 2\pi i\ell) = 0$ for every $\ell \in \mathbb{Z}$ and $y > 0$. For fixed $y$ the polynomial $L \mapsto \Phi(y, L)$ then has infinitely many zeros, so $\Phi = 0$, and $T\big[(\mathsf{g} + c\mathsf{h})^n\big] = 0$ for every $c \in \mathbb{C}$; the coefficients of the powers of $c$ give $T\big(\mathsf{g}^{n-j}\mathsf{h}^j\big) = 0$ for $0 \le j \le n$.

For $0 \le j < n/2$ let $I_j(k) = \int_0^\infty y^{k-1} \mathsf{g}^{n-j}\mathsf{h}^j \, dy$. Lemma S27(i) for $\mathsf{g}^{n-j}\mathsf{h}^j$ (Lemma S28(ii)) gives $\sum_i \tilde{r}_i(k) I_j(k+i) = 0$ for the integers $k \ge 1$. Take $k_1$ beyond the zeros of $r_m$; on the integers $k \ge k_1$ a solution of this recurrence of order $m$ is fixed by its first $m$ values, so the $h > m$ sequences $(I_j(k))_{k \ge k_1}$ are linearly dependent. They are not: if $\sum_j b_j I_j(k) = 0$ for $k \ge k_1$ with some $b_j \neq 0$, let $j_0$ be the largest such $j$; by Lemma S28(iii), $I_j(k)/\Gamma(k)^2$ has the exact root rate $(n-2j)^{-2}$, largest at $j_0$, so $|b_{j_0}| I_{j_0}(k) > \sum_{j<j_0} |b_j| I_j(k)$ for large $k$, a contradiction. So the tier is at least $h$, and the recurrence of (ii) has the order $h$ and $c_{n,0}c_{n,h} \neq 0$.

By Proposition 42.2(i) the closings, here the constant terms of $(\sum_e \alpha_e)^j (\sum_e \alpha_e^{-1})^j$ (Corollary 39.36), that is $\sum_{|v|=j} \binom{j}{v}^2$, satisfy the reflected recurrence for every $j \in \mathbb{N}$, and by Corollary 39.60 they satisfy no recurrence that the values do not satisfy, read at $k$ and at $-j$; so none of order less than $h$. In the value module, a relation $\sum_{i<h} r_i(k)\,\mathrm{Sh}^i[1] \in \mathrm{Ann}(\langle\cdot\rangle)$ with rational $r_i$ not all zero would be a recurrence of order less than $h$ for the values, so the classes of $\mathrm{Sh}^i[1]$, $0 \le i < h$, are independent there; the element $\sum_{i=0}^{h} c_{n,i}(k)\,\mathrm{Sh}^i[1]$, $k$ the strength, has vanishing values and lies in $\mathrm{Ann}(\langle\cdot\rangle)$, a submodule (Theorem 39.4(ii)), and with $c_{n,0}c_{n,h} \neq 0$ it expresses $\mathrm{Sh}^h[1]$ and $\mathrm{Sh}^{-1}[1]$ through these classes; so their span is the cyclic submodule of [1] in the value module, of dimension $h$. □

**Proposition 42.13 (A mark added carries the law of the values).** *As stated.*

*Proof.* (i) For a fixed solution $u$ both sides are linear differential operators in $F$ of order $n+2$, with coefficients analytic on $(0, \infty)$. By Theorem 42.12(i), $\mathbf{Q}_{n+1} = \vartheta^{n+2} + \sum_{k \ge 1} y^k q_{n+1,k}(\vartheta)$ with $\deg q_{n+1,k} \le n$, so $\mathbf{Q}_{n+1}(uF) = u\,\vartheta^{n+2}F + (n+2)(\vartheta u)\,\vartheta^{n+1}F + (\text{order} \le n)$; and $\mathbf{Q}_n = \vartheta^{n+1} + \sum_{k \ge 1} y^k q_{n,k}(\vartheta)$ with $\deg q_{n,k} \le n-1$, so the right side is $u\,\vartheta^{n+2}F + (n+2)(\vartheta u)\,\vartheta^{n+1}F + (\text{order} \le n)$. Their difference is an operator of order at most $n$ that annihilates every $F$ in the space spanned by the products of $n$ solutions, of dimension $n+1$: the right side vanishes there by Theorem 42.12(i), and the left side because $uF$ is a product of $n+1$ solutions. An operator of order at most $n$ whose highest nonzero coefficient does not vanish on an interval has there a space of solutions of dimension at most $n$, and the $n+1$ analytic functions stay independent on every interval; so the difference is 0.

(ii) Apply (i) with $u = \mathsf{h}$ to $F = y^z$, $z \in \mathbb{C}$, using $\vartheta^a y^w = w^a y^w$ and $\vartheta\mathsf{h} = \sum_m m\, y^m/m!^2$. Both sides are $y^z$ times entire functions of $y$, and comparing the coefficients of $y^{z+N}$,

$$\sum_{k+m=N} \frac{q_{n+1,k}(z+m)}{m!^2} = \sum_{j+m=N} \frac{\big(z + j + (n+2)m\big)\, q_{n,j}(z)}{m!^2} \qquad (N \in \mathbb{N},\ z \in \mathbb{C}).$$

By Theorem 42.12(i), $q_{n+1,k}(z+m) = (-1)^n q_{n+1,k}(-z-N)$ when $k+m=N$ and $q_{n,j}(z) = (-1)^{n+1} q_{n,j}(w+m)$ with $w=-z-N$ when $j+m=N$; and $z+j+(n+2)m = (n+1)m-w$. So

$$\sum_{k+m=N} \frac{q_{n+1,k}(w)}{m!^2} = \sum_{j+m=N} \frac{\big(w-(n+1)m\big)\, q_{n,j}(w+m)}{m!^2},$$

and applying the same symmetry once more, $q_{n+1,k}(w) = (-1)^n q_{n+1,k}(-w-k)$ and $q_{n,j}(w+m) = (-1)^{n+1} q_{n,j}(-w-N)$,

$$\sum_{k+m=N} \frac{q_{n+1,k}(-s-k)}{m!^2} = \sum_{j+m=N} \frac{\big((n+1)m-s\big)\, q_{n,j}(-s-N)}{m!^2} \qquad (s=w).$$

Multiply by $\Gamma(s+N)^2/\Gamma(s+1)^2$ and put $x=s+N$. By the definitions, $c_{n+1,k}(s) = q_{n+1,k}(-s-k)\,\Gamma(s+k)^2/\Gamma(s+1)^2$ for every $k\ge 0$, so the left side becomes $\sum_k c_{n+1,k}(s)\,\Gamma(x)^2/\big(\Gamma(s+k)^2\Gamma(x-s-k+1)^2\big) = \sum_k c_{n+1,k}(s)\binom{x-1}{s+k-1}^2$. With $c_{n,j}(x-j) = q_{n,j}(-x)\,\Gamma(x)^2/\Gamma(x-j+1)^2$, $(n+1)m-s = (n+1)(x-j)-(n+2)s$ and $A_s(x-j) = \big((n+1)(x-j)-(n+2)s\big)\Gamma(x-j+1)^2/\big(\Gamma(s+1)^2\Gamma(x-j-s+1)^2\big)$, the right side becomes $\sum_j c_{n,j}(x-j)A_s(x-j)$. After division by $\binom{x}{s}^2$ both sides are rational functions of $(x,s)$, each binomial quotient being a quotient of rising factorials, and they agree on the lines $x-s=N$, $N\in\mathbb{N}$; a polynomial in $(x,s)$ vanishing on infinitely many of these lines is divisible by infinitely many distinct factors $x-s-N$, so it is zero, and the identity holds. At integers $s,x\ge 1$, $W_{s+k}(x)/\Gamma(s+k)^2 = \prod_{l=1}^{s+k-1}(x-l)^2/(s+k-1)!^2 = \binom{x-1}{s+k-1}^2$, both sides vanishing for $x<s+k$; the right side is $(L_n^* A_s)(x)$ for the recurrence $L_n$ with the coefficients $c_{n,j}$ (Definition 42.1), and $\binom{x}{s} = x(x-1)\cdots(x-s+1)/s!$ vanishes at $x=0,1,\dots,s-1$. □

**Proposition 42.14 (The lowered settings).** *As stated.*

*Proof.* (i) By Theorem 42.12(i), $\mathbf{Q}_n(\mathsf{g}^{n-j}\mathsf{h}^j)=0$; Lemma S27(i), which applies by Lemma S28(ii), gives $\sum_k q_{n,k}(-s-k)\,\Gamma(s+k)^2\xi_{n,j}(s+k) = 0$ on $\operatorname{Re} s>0$, and the division of the proof of Theorem 42.12(ii) gives the recurrence. For $j=0$ the display of Section 42.6 gives $\xi_{n,0}(s) = P_{\binom{\varepsilon}{1}}(s\mathbf{1}; s, s)$.

(ii) On $(0,1]$, $0<\mathsf{g}^{n-j}\mathsf{h}^j \le C(1+|\log y|)^{n-j}$ by Lemma S28(ii) and $\mathsf{g} = -\mathsf{h}\log y + \tilde{\mathsf{g}}$, so for real $s\in(0,1]$, $\int_0^\infty y^{s-1}\mathsf{g}^{n-j}\mathsf{h}^j\,dy \le C\int_0^1 y^{s-1}(1+|\log y|)^{n-j}dy + C' = O(s^{-(n-j+1)})$, and with $\Gamma(s)^{-2} = O(s^2)$, $s^{n-1}\xi_{n,j}(s) = O(s^j)\to 0$ for $j\ge 1$. In the recurrence on $(0,\infty)$ the term $c_{n,0}(s)\xi_{n,j}(s) = (-1)^{n+1}s^{n-1}\xi_{n,j}(s)$ then tends to 0 as $s\downarrow 0$, while the others tend to $c_{n,k}(0)\xi_{n,j}(k)$, $\xi_{n,j}$ being continuous on $\operatorname{Re} s>0$; so $\sum_{k\ge 1} c_{n,k}(0)\xi_{n,j}(k)=0$. For $j=0$, $s^{n-1}\xi_{n,0}(s)\to V_1 = n!$ by Proposition 39.34, the walls of the direction being $\kappa_{1,S} = c(S) = 1$ for every nonempty proper $S$, so that $V_1 = \sum_\pi 1 = n!$; and the same limit gives $\sum_{k\ge 1} c_{n,k}(0)\xi_{n,0}(k) = -(-1)^{n+1}n!$.

(iii) By Lemma S28(iii), $\xi_{n,j}$ has the root rate $\lambda = (n-2j)^{-2}$ along the integers. The recurrence has $c_{n,0}(k)c_{n,h}(k)\ne 0$ at the integers $k\ge 1$, every $c_{n,k}$ of degree $n-1$, and the roots $(n-2j')^{-2}$, $0\le j'<n/2$, of its symbol are positive and distinct (Theorem 42.12(ii)); so the proof of Proposition 42.7(i), which uses only these properties, gives $\xi_{n,j}(k+1)/\xi_{n,j}(k)\to\lambda_c$ for a root $\lambda_c$ of modulus $\lambda$, that is $\lambda_c = (n-2j)^{-2}$.

(iv) $c_{n,h}(k)\ne 0$ at the integers $k\ge 1$ (Theorem 42.12(ii)), so a solution on these integers is fixed by its first $h$ values, and the solutions form a space of dimension $h$. The $h$ sequences are solutions by (i) and are independent by (iii), as in the proof of Theorem 42.12(iii): in a vanishing combination the term with the largest $j$ would dominate. So they are a basis. The linear form $x\mapsto\sum_{k=1}^h c_{n,k}(0)x(k)$ is $(-1)^n n!$ on $\xi_{n,0}$ and 0 on the others by (ii), which gives the last assertion. □

**Corollary 42.15 (The value span of $\binom{\mathcal{E}}{1}$).** *As stated.*

*Proof.* (i) The first relation is Proposition 42.14(ii) at $j=0$. The recursion of Theorem 42.12 has integer coefficients, so every $q_{n,k}$ and every $c_{n,k}$ has integer coefficients. $c_{n,h}(0)\neq 0$, so the first relation gives $p_n(h)\in\mathbb{Q}+\sum_{i<h}\mathbb{Q}\,p_n(i)$, and for $k>h$ the recurrence of Theorem 42.12(ii) at $s=k-h\geq 1$, with $c_{n,h}(k-h)\neq 0$, writes $p_n(k)$ through $p_n(k-h),\dots,p_n(k-1)$; by induction every $p_n(k)$ lies in $\mathbb{Q}+\sum_{i=1}^{h-1}\mathbb{Q}\,p_n(i)$. The values of the centre ray at the strength $\sigma$ are $p_n(k\sigma)$ (Definition 38.13).

(ii) For even $n$ every $n-2j$, $0\leq j<n/2$, is at least 2, so $\chi_n(1)=(-1)^{n+1}\prod_j\big(1-(n-2j)^2\big)\neq 0$. The values of the centre ray have the root rate $n^{-2}<1$ and the slopes at most $n^{-2}$ (Proposition 42.2(iii), $f(\alpha^\star)=f(\mathbf{1})=n^{-2}$). Every $c_{n,k}$ with $k>j\geq 1$ has a double zero at $-j$ (Theorem 42.12(ii)), so by Proposition 42.3 the only boundary numbers are $\beta_0(0)$ and $\beta_0'(0)$ of $\beta_0=c_{n,0}\,p_n=(-1)^{n+1}s^{n-1}p_n$: by Proposition 42.6, $\beta_0(0)=(-1)^{n+1}V_1=(-1)^{n+1}n!$, and $\beta_0'(0)=(-1)^{n+1}\mathsf{L}_1=0$, the list having unit weights. Proposition 42.4(ii) with $\Bbbk=\mathbb{Q}$ and $D=n-1$ gives the claim. The two last numbers are such Trailmoments by Proposition 42.11(ii), with the rational polynomials $W_s$ and $W_s'$. □

**Corollary 42.16 (The exact pairing of $\binom{\mathcal{E}}{1}$).** *As stated.*

*Proof.* The centre ray of $\binom{\mathcal{E}}{1}$ at the strength one satisfies the hypotheses of Proposition 42.2, and the recurrence of Theorem 42.12(ii) holds for its values with $c_{n,k}(-s-k)=(-1)^{n+1}c_{n,k}(s)$; its closings are the $\mathsf{C}_l$ of the statement (proof of Theorem 42.12(iii)). Proposition 42.8(ii), with $\gamma_{n-1}=(-1)^{n+1}$ and $V_1=n!$, gives the first display. Every $c_{n,k}$ has degree $n-1$, so Proposition 42.8(iii) applies: the roots of $\chi_n$ are the simple roots $(n-2j)^{-2}$, $0\leq j<n/2$, with $\lambda_\star=f(\mathbf{1})=n^{-2}$ at $j=0$, and $1-(n-2j)^2/n^2=4j(n-j)/n^2$. The Hessian $H$ of Theorem 38.15 at the centre is $2\big(\kappa_{\mathbf{1}}(X_e,X_{e'})\big)_{e,e'\neq e_0}$, which is the Hessian of Proposition 42.9(ii) at the sign $\mathbf{1}$ and $\sigma=1$, of determinant $2^{n-1}/n^n$. As a check, the two sides agree directly: for $n=2l+1$, $h-1=l$ and $\prod_{j=1}^{l}j(n-j)=l!\,(2l)!/l!=(n-1)!$, and for $n=2l+2$, $h-1=l$ and $\prod_{j=1}^{l}j(n-j)=l!\,(2l+1)!/(l+1)!=2(n-1)!/n$; in both cases $\prod_{j=1}^{h-1}4j(n-j)/n^2=2^{n-1}(n-1)!/n^{n-1}=n!\,2^{n-1}/n^n$. □

**Proposition 42.17 (The power of one mark as a mixture).** *As stated.*

*Proof.* Write $p_n(s)=P_{\binom{\mathcal{E}}{1}}(s\mathbf{1};s,s)$ for $|\mathcal{E}|=n$. By Theorem 38.14(i), $0<f_n\leq f_n(\mathbf{1})=n^{-2}$ on the chart, so $\mu_{f_n}$ is carried by $(0,n^{-2}]$, and $\int t^s\mu_{f_n}(dt)=\int_{y_{e_0}=0}f_n^s\,dy=p_n(s)$ for $\operatorname{Re}s>0$, the image of a measure under a map (Definition 1.1(vi)); the same holds at $n+1$ marks. The measure $\mu_{f_n}$ has infinite mass near 0, and every moment of positive order is finite. By Fubini's theorem and $y=ty'$, $\int_0^\infty y^{s-1}\mathsf{g}(y/t)\,dy=t^s\int_0^\infty y'^{s-1}\mathsf{g}(y')\,dy'=t^s\Gamma(s)^2$, the last because $\int_0^\infty y^{s-1}e^{-y/x}dy=\Gamma(s)x^s$ and $\int_0^\infty x^{s-1}e^{-x}dx=\Gamma(s)$.

(i) Put $G(y)=\int\mathsf{g}(y/t)\,\mu_{f_n}(dt)$. For real $s>0$, by Tonelli's theorem $\int_0^\infty y^{s-1}G(y)\,dy=\Gamma(s)^2\int t^s\mu_{f_n}(dt)=\Gamma(s)^2p_n(s)$, finite, which is $\int_0^\infty y^{s-1}\mathsf{g}^n\,dy$ by the display of Section 42.6. So $G$ is finite almost everywhere; since $\mathsf{g}$ is decreasing ($\partial_y\mathsf{g}=-\mathsf{G}_{-1;(1,-1)}(1,y)<0$), $\mathsf{g}(y/t)\leq\mathsf{g}(y_0/t)$ for $y\geq y_0$, and dominated convergence makes $G$ finite and continuous on $(0,\infty)$. For $s=c+i\tau$ the same Fubini argument applies, so the integrable functions $e^{cv}G(e^v)$ and $e^{cv}\mathsf{g}(e^v)^n$ have the same Fourier transform; they are equal, by the uniqueness of the Fourier transform on $L^1(\mathbb{R})$ as in Lemma S27(ii), and both being continuous, $G=\mathsf{g}^n$.

(ii) The table on $\{1,2\}$ with the bases $\{1\}$ and $\{2\}$ is the table of Corollary 18.24 on two marks, the basis $\{2\}=\{1,2\}-1$ carrying $\theta_1=1$ and $\{1\}$ carrying $\theta_2=t^{-1/2}$; its letters are $K=\alpha_2+t^{-1/2}\alpha_1$ and $\Psi=\alpha_1+t^{-1/2}\alpha_2$, and its centre stride on the chart $\alpha_1=1$, $\alpha_2=x$, is

$\alpha_1\alpha_2/(K\Psi) = x/\big((x+t^{-1/2})(1+t^{-1/2}x)\big) = \phi_t(x)$, and $dy = dx/x$ on that chart. So $\int u^s \mu_{\phi_t}(du) = P(s\mathbf{1}; s, s)$ for that table, and Corollary 18.24 at $\mathbf{m} = s\mathbf{1}$, $\nu_e = 0$, gives $\Gamma(s)^2 \int u^s \mu_{\phi_t}(du) = \int_0^\infty y^{s-1} \mathsf{g}(y)\, \mathsf{g}(\theta_2^2 y)\, dy = \int_0^\infty y^{s-1} \mathsf{g}(y)\, \mathsf{g}(y/t)\, dy$. By (i) and Tonelli's theorem, for real $s > 0$,

$$\begin{aligned}\Gamma(s)^2 \int u^s \mu_{f_{n+1}}(du) &= \int_0^\infty y^{s-1} \mathsf{g}(y)\, \mathsf{g}(y)^n\, dy = \int \mu_{f_n}(dt) \int_0^\infty y^{s-1} \mathsf{g}(y)\, \mathsf{g}(y/t)\, dy \\ &= \Gamma(s)^2 \int \mu_{f_n}(dt) \int u^s \mu_{\phi_t}(du),\end{aligned}$$

and by Fubini's theorem the same holds on $\operatorname{Re} s > 0$. The measures $\mu_{f_{n+1}}$ and $\int \mu_{\phi_t}\, \mu_{f_n}(dt)$ therefore have the same moments $\int u^s(\cdot)$ on the line $\operatorname{Re} s = c > 0$; their images under $u \mapsto v = \log u$, weighted by $e^{cv}$, are finite measures on $\mathbb{R}$ with the same Fourier transform, hence equal, and so are the two measures. □

**Proposition 42.18 (Balanced directions of $\binom{\mathcal{E}}{1}$).** *As stated.*

*Proof.* The walls of $\binom{\mathcal{E}}{1}$ at $(\mathbf{m}; s, t)$ are $\kappa_S = m_S - t(|S| - 1)$, every proper $S$ meeting a basis $\{e\}$ in at most one mark and the cobases $\mathcal{E} - e$ in at least $|S| - 1$, so the hypothesis says that every proper wall is positive, at $(k\mathbf{m}; ks, kt)$ as well. The display of Section 42.6 at these data, from Corollary 18.24 for the dual table at the exponents $(kt, ks)$ and Proposition 8.15, gives $\Gamma(sk)\Gamma(tk)P(k) = \int_0^\infty y^{tk-1} \prod_e \mathsf{G}_{c_e k;(1,-1)}(1, y)\, dy$. By the torus homogeneity (Theorem 2.4(iii)) at $\lambda = y^{1/2}$, $\mathsf{G}_{\nu;(1,-1)}(\lambda, \lambda^{-1}y) = \lambda^{-\nu}\mathsf{G}_{\nu;(1,-1)}(1, y)$, that is $\mathsf{G}_{\nu;(1,-1)}(1, y) = y^{\nu/2}\, \mathsf{G}_{\nu;(1,-1)}(\sqrt{y}, \sqrt{y})$; the product of the powers is $y^{k\sum_e c_e/2} = y^{(s-t)k/2}$, and $tk - 1 + \frac{(s-t)k}{2} = \frac{(s+t)k}{2} - 1$. The reflection of Proposition 2.3 gives $\mathsf{G}_{\nu;(1,-1)}(a, b) = \mathsf{G}_{-\nu;(-1,1)}(a, b) = \mathsf{G}_{-\nu;(1,-1)}(b, a)$, so at equal coefficients the Gramma of one mark is even in its index, and the integral depends only on $s + t$ and the multiset $\{|c_e|\}$. Two directions with the same $s + t$ and multiset therefore have $\Gamma(sk)\Gamma(tk)P(k) = \Gamma(s'k)\Gamma(t'k)P'(k)$, which is the stated quotient of Beta corners, a quotient of factorials at integer $k$. The image of the integrand $\alpha^{k\mathbf{m}} K^{-ks} \Psi^{-kt}$ under $\alpha \mapsto \alpha^{-1}$, with $K(\alpha^{-1}) = \alpha^{-\mathcal{E}}\Psi(\alpha)$ and $\Psi(\alpha^{-1}) = \alpha^{-\mathcal{E}}K(\alpha)$, is $\alpha^{k((s+t)\mathbf{1}-\mathbf{m})} K^{-kt} \Psi^{-ks}$, the integrand at $((s + t)\mathbf{1} - \mathbf{m}; t, s)$, whose offsets are $(s + t) - m_e - s = -c_e$. □

**Definition 42.19 (Hinge pairs; norms; the two-sided core; the class density).** As stated.

*Proof.* For a parallel class $K_{\mathcal{A}\backslash g} = \sum_{e \in B} \alpha_e$ and $K_{\mathcal{A}/g} = 1$; for a series class $K_{\mathcal{A}\backslash g} = \alpha^B$ and $K_{\mathcal{A}/g} = e_{|B|-1}(\alpha_B) = \alpha^B A'$, so $\gamma = 1/A'$, $\gamma(\alpha^{-1}) = 1/A$ and $v = \alpha^B A' \cdot \alpha^{-B} A = AA'$; for $\binom{\{h,g\}}{1}$, $K_{\mathcal{A}\backslash g} = \alpha_h$ and $K_{\mathcal{A}/g} = 1$. The hinge values are homogeneous of degree one, which gives the action of the scale; on the quadruple the scale acts by $(\lambda^{r-1}c, \lambda^r d, \lambda^{1-r}c', \lambda^{-r}d')$, $r$ the rank of $\mathcal{A}$. The letter $K_{\mathcal{H}}$ has the degree at most one in each marker, so $\prod_i c_i \cdot K_{\mathcal{H}}(\mathbf{d}/\mathbf{c}) = \mathcal{H}^{\hom}(\mathbf{c}, \mathbf{d})$ where every $c_i \neq 0$, and likewise at $\alpha^{-1}$. At $x_i' = x_i^{-1}$, $\Phi_{\mathcal{H}} = K_{\mathcal{H}}(x)K_{\mathcal{H}}(x^{-1})$, the reciprocal stride of the centre ray of $\mathcal{H}$ at the strength one. On the compact torus the constant term of a Laurent polynomial is its mean, so $\int t^w \nu_s(dt) = \mathrm{CT}((AA')^w)$. □

**Theorem 42.20 (A list is its core read against its cells).** *As stated.*

*Proof.* (i) Theorem 21.4 at $\alpha$ and at $\alpha^{-1}$: $K(\alpha) = \prod_i K_{\mathcal{A}_i/g_i}(\alpha)\, K_{\mathcal{H}}(\gamma(\alpha))$, and the product of the two prefactors is $\prod_i v_{\mathcal{A}_i}$. On the whole torus $K(\alpha) = \mathcal{H}^{\hom}(\mathbf{c}, \mathbf{d})$ at the quadruples, by Theorem 39.56(i) for the tree with the root $\mathcal{H}$ and one vertex for each cell, and likewise at $\alpha^{-1}$.

(ii) On the compact torus $\alpha_e^{-1} = \bar{\alpha}_e$, so $A_i' = \bar{A}_i$ and $\mathsf{C}_j(1) = \mathbb{E}\big[\Phi_{\mathcal{H}}(A, \bar{A})^j\big]$ under the Haar probability. The totals of different classes depend on disjoint marks, so they are independent,

and each is invariant under rotations: $A_i = r_i u_i$ with $u_i$ uniform on the circle and independent of $r_i^2$, whose law is $\nu_{s_i}$. With $\bar{u}_i = u_i^{-1}$, $\mathbb{E}_u\big[K_{\mathcal{H}}(ru)^j K_{\mathcal{H}}(ru^{-1})^j\big]$ is the constant term in $u$, which is $\sum_w \big([x^w]K_{\mathcal{H}}^j\big)^2 \prod_i r_i^{2w_i} = \mathrm{CT}_x\big[K_{\mathcal{H}}(x)^j K_{\mathcal{H}}(r^2x^{-1})^j\big]$. Taking the mean over the $r_i$ gives the display; the supports of the $\nu_{s_i}$ are bounded, so the series in $\tau^{-1}$ may be averaged term by term for large $|\tau|$.

(iii) (i) at the level of maps: the hinge quadruples act on disjoint sets of marks, their coordinates are Laurent polynomials, and $f^{-1} = \Phi_{\mathcal{H}}^{\hom}(q_{\mathcal{A}_1}, \dots, q_{\mathcal{A}_q})$ with $\Phi_{\mathcal{H}}^{\hom}$ a polynomial; all the maps are equivariant for the scale, $f^{-1}$ and $dy$ being invariant. The pencil module is the cohomology in degree 0 of the direct image of $\mathcal{O}_{X_F}$ along $\varphi = f^{-1}$, its definition being the relative de Rham complex of that direct image (Definition 39.47). The direct image along a composite is the composite of the direct images, and along a product of maps it is the external product (the Künneth formula for $\mathcal{D}$-modules); the maps are morphisms of affine varieties, so their direct images have no cohomology in positive degrees, and in the spectral sequence of the composite the only term of total degree 0 is the direct image in degree 0 of the direct image in degree 0, which no differential reaches. The letter complement is the preimage of the open set where $\Phi_{\mathcal{H}}^{\hom} \neq 0$, and a direct image restricted to an open set of the target is the direct image from its preimage; the scale is removed on a chart, as in the proof of Theorem 39.56(ii). The class $[dy]$ is the image of $\boxtimes_i[\prod_{e\in B_i} d\alpha_e/\alpha_e]$, which lies in $\boxtimes_i \mathcal{H}_{\mathcal{A}_i}$; so $\mathcal{D}_{\mathcal{B}}[1]$ is the image of the submodule generated by it. Characteristic cycles are additive and nonnegative along submodules and quotients, which gives the last claim. □

**Corollary 42.21 (The pencil of a list along its canonical tree).** *As stated.*

*Proof.* A list is plain, so the weights of $\Psi$ are those of $K$ and $K^\Psi = K$: the data of Theorem 39.56(i) of a vertex is the hinge quadruple of its subtree cell. The centre ray at the strength one has $\mathbf{m}_1 = \mathbf{1}$ and $s_1 = t_1 = 1$, so $p_e = 1 - 1 = 0$ and the last coordinate is 1, and the root map is $\mathcal{H}_o^{\hom}(\mathbf{c}, \mathbf{d})\mathcal{H}_o^{\hom}(\mathbf{c}', \mathbf{d}')$, which is $\Phi_{\mathcal{H}_o}^{\hom}$ (Definition 42.19(iii)). The nodes of the canonical hinge tree are prime nodes and the series and parallel nodes $\binom{S}{|S|-1}$ and $\binom{S}{1}$ (Theorem 23.16, Definition 23.13). For $\mathcal{H} = \binom{S}{1}$ with the hinge $g \in S$, $\mathcal{H}/g$ has the single basis $\varnothing$ and $\mathcal{H} \setminus g = \binom{S-g}{1}$, which gives (i); for $\mathcal{H} = \binom{S}{|S|-1}$, $\mathcal{H}/g = \binom{S-g}{|S|-2}$ and $\mathcal{H} \setminus g$ has the single basis $S - g$, which gives (ii); the ratios $d/c$ are $\sum_l d_l/c_l$ and $\big(\sum_l c_l/d_l\big)^{-1}$. A parallel node of marks has $(1, \sum_e \alpha_e)$ at $\alpha$ and $(1, \sum_e \alpha_e^{-1})$ at $\alpha^{-1}$; a series node of marks has $(e_{|B|-1}(\alpha_B), \alpha^B) = (\alpha^B A', \alpha^B)$ and the same at $\alpha^{-1}$. The last statement is Theorem 39.56(iv), every vertex being a prime node or a series or parallel node of at most $n$ marks and markers, and the lists on at most $n$ marks being finitely many up to the names of the marks. □

**Proposition 42.22 (The closings of a substitution).** *As stated.*

*Proof.* $K_{\mathcal{H}}$ is multilinear, so $K_{\mathcal{H}}^j$ has the degree at most $j$ in each $x_i$, and by Theorem 21.4

$$K(\alpha)^j = \sum_w [x^w] K_{\mathcal{H}}^j \prod_i K_{\mathcal{A}_i \setminus g_i}(\alpha)^{w_i} K_{\mathcal{A}_i/g_i}(\alpha)^{j-w_i},$$

and likewise at $\alpha^{-1}$ with an index $w'$. The constant term of a product of Laurent polynomials on disjoint sets of marks is the product of their constant terms. On $B_i$ the factor at $\alpha$ is homogeneous of the degree $j(r_{\mathcal{A}_i} - 1) + w_i$ and the factor at $\alpha^{-1}$ of the degree $-j(r_{\mathcal{A}_i} - 1) - w_i'$, the hinge minors having the ranks $r_{\mathcal{A}_i}$ and $r_{\mathcal{A}_i} - 1$; so the constant term on $B_i$ vanishes unless $w_i' = w_i$, and the double sum collapses to the display. The three cases are the letters of Definition 42.19: $K_{\mathcal{A}\setminus g} K_{\mathcal{A}\setminus g}^* = AA'$

and $K_{\mathcal{A}/g}K^*_{\mathcal{A}/g} = 1$ for a parallel class, 1 and $AA'$ for a series class, and 1 and 1 for a mark left in place. For $\mathcal{A} = \mathcal{A}_0\langle\mathcal{A}' \text{ at } x\rangle$ the hinge minors are $\mathcal{A}_0 \setminus g$ and $\mathcal{A}_0/g$ with $\mathcal{A}'$ substituted at $x$, and Theorem 21.4 inside them gives $K^w_{\mathcal{A}\setminus g}K^{j-w}_{\mathcal{A}/g} = K^j_{\mathcal{A}'/g'}\big(K^w_{\mathcal{A}_0\setminus g}K^{j-w}_{\mathcal{A}_0/g}\big)(\alpha_{B_0-x}, \gamma_{\mathcal{A}'})$; the same expansion in the monomials $\alpha^u$ of the cell $\mathcal{A}_0$, with the variable of $x$ read at the hinge value of $\mathcal{A}'$, and the same collapse give the last display. □

**Lemma 42.23 (The occupations of a substitution).** *As stated.*

*Proof.* $\log K = \sum_i \log K_{\mathcal{A}_i/g_i} + \log K_{\mathcal{H}}(\gamma)$. For $e \in B_i$, $\alpha_e\partial_e \log\gamma_i = R_e^{\mathcal{A}_i\setminus g_i} - R_e^{\mathcal{A}_i/g_i}$, and $\alpha_e\partial_e \log K_{\mathcal{H}}(\gamma) = \big(x_i\partial_{x_i}\log K_{\mathcal{H}}\big)(\gamma)\,\alpha_e\partial_e\log\gamma_i = \rho_i\,\alpha_e\partial_e\log\gamma_i$, which gives the display. Multilinearity, $K_{\mathcal{H}} = K_{\mathcal{H}\setminus h_i} + x_iK_{\mathcal{H}/h_i}$, gives $x_i\partial_{x_i}K_{\mathcal{H}} = x_iK_{\mathcal{H}/h_i}$. A mark left in place has $R^{\mathcal{A}/g} = 0$ and $R_h^{\mathcal{A}\setminus g} = 1$. □

**Theorem 42.24 (Native gauges along the hinge tree).** *As stated.*

*Proof.* By Proposition 38.16 a point of $X_F$ is a native gauge exactly when $R_e(\alpha) = R_e(\alpha^{-1})$ for every mark $e$. For a table $\mathcal{X}$, $\alpha_e\partial_e\big[\log K_{\mathcal{X}}(\alpha^{-1})\big] = -R_e^{\mathcal{X}}(\alpha^{-1})$. So by Lemma 42.23, on $B_i$,

$$R_e(\alpha) - R_e(\alpha^{-1}) = \alpha_e\partial_e\Big[\log K^{\mathcal{A}/g} + \rho\log\gamma\Big](\alpha) + \alpha_e\partial_e\Big[\log K^{\mathcal{A}/g}(\alpha^{-1}) + \rho^*\log\gamma(\alpha^{-1})\Big],$$

with $\rho, \rho^*$ held fixed, that is $\alpha_e\partial_e\big[\log v + \rho\log\gamma + \rho^*\log\gamma^*\big]$, and $\rho\log\gamma + \rho^*\log\gamma^* = \frac{1}{2}(\rho+\rho^*)\log t + \frac{1}{2}(\rho-\rho^*)\log q$: this is $(\star)$.

(a) The hinge values have the degree one, $K_{\mathcal{A}/g}$ is homogeneous, so $v$ and $t$ are invariant under the scale of the cell and $q$ has the degree 2; the derivative of $(\star)$ along the scale is $\rho_i - \rho_i^*$, and then $(\star)$ is $d_{B_i}[\log v_i + \rho_i\log t_i] = 0$.

(b) For a parallel class $R^{\mathcal{A}/g} = 0$ and $R_e^{\mathcal{A}\setminus g} = \alpha_e/A_i$, so with $\varsigma_i = K_{\mathcal{H}/h_i}(\gamma)/K_{\mathcal{H}}(\gamma)$, regular where $A_i = 0$ since $K_{\mathcal{H}}(\gamma) \neq 0$ on $X_F$, Lemma 42.23 gives $R_e = \rho_i\alpha_e/A_i = \varsigma_i\alpha_e$. The equations on $B_i$ read $\varsigma_i(\alpha)\,\alpha_e = \varsigma_i(\alpha^{-1})\,\alpha_e^{-1}$, $e \in B_i$. If $\varsigma_i(\alpha) = 0$ the left side vanishes and so $\varsigma_i(\alpha^{-1}) = 0$: the class is empty, and conversely then the equations hold for every shape. Otherwise $\alpha_e^2 = \varsigma_i(\alpha^{-1})/\varsigma_i(\alpha)$ is one number on $B_i$, so $\alpha_e = \eta_e\tau_i$, and $A_i = a_i\tau_i$, $A_i' = a_i\tau_i^{-1}$. Then $\rho_i(\alpha) = a_i\tau_i\varsigma_i(\alpha)$ and $\rho_i(\alpha^{-1}) = a_i\tau_i^{-1}\varsigma_i(\alpha^{-1})$, and the equation $\varsigma_i(\alpha)\tau_i^2 = \varsigma_i(\alpha^{-1})$ reads $\rho_i(\alpha) = \rho_i(\alpha^{-1})$ when $a_i \neq 0$: the occupation of the mark $h_i$ of the core with the variable $a_i\tau_i$ agrees at $\tau$ and at $\tau^{-1}$, the native-gauge equation of the weighted core there. An empty class enters the letters through $\gamma_i = A_i$ and $\gamma_i(\alpha^{-1}) = A_i'$ alone (Theorem 21.4). For a pair, $x + y = A_i$ and $x^{-1} + y^{-1} = A_i'$ give $xy = A_i/A_i'$, and $r + r^{-1} = (x+y)^2/(xy) - 2 = A_iA_i' - 2$ for $r = x/y$. For a series class $R_e^{\mathcal{A}\setminus g} = 1$ and $R_e^{\mathcal{A}/g} = 1 - \alpha_e^{-1}/A_i'$, so $R_e = 1 - (1-\rho_i)\alpha_e^{-1}/A_i' = 1 - \varsigma_i'\alpha_e^{-1}$ with $\varsigma_i' = \gamma_iK_{\mathcal{H}\setminus h_i}(\gamma)/K_{\mathcal{H}}(\gamma)$, as $1 - \rho_i = K_{\mathcal{H}\setminus h_i}(\gamma)/K_{\mathcal{H}}(\gamma)$; the same argument runs, and at $\alpha_e = \eta_e\tau_i$, $\gamma_i = 1/A_i' = \tau_i/a_i$ and $v = AA' = a_i^2$.

(c), (d) Theorem 21.4 applied to the hinge minors of the cell, $\mathcal{A}\setminus g = (\mathcal{H}'\setminus g)\langle\mathcal{A}_l'\rangle$ and $\mathcal{A}/g = (\mathcal{H}'/g)\langle\mathcal{A}_l'\rangle$, gives $K_{\mathcal{A}\setminus g} = \prod_l K_{\mathcal{A}_l'/g_l'}\,K_{\mathcal{H}'\setminus g}(\gamma')$ and $K_{\mathcal{A}/g} = \prod_l K_{\mathcal{A}_l'/g_l'}\,K_{\mathcal{H}'/g}(\gamma')$, so the quotient and the product $v$ are as displayed. A parallel node is $\mathcal{H}' = \binom{\{g,x_1,\dots,x_p\}}{1}$, with $K_{\mathcal{H}'\setminus g} = \sum_l x_l$ and $K_{\mathcal{H}'/g} = 1$; a series node is $\binom{\{g,x_1,\dots,x_p\}}{p}$, with $K_{\mathcal{H}'\setminus g} = e_p(x)$ and $K_{\mathcal{H}'/g} = e_{p-1}(x)$. □

**Corollary 42.25 (A class is a bubble of classes).** *As stated.*

*Proof.* The bases of $\binom{[2]}{1}\langle\binom{[a]}{1}, \binom{[b]}{1}\rangle$ are one mark of the first class or one of the second (Definition 21.1), which are the bases of $\binom{[s]}{1}$. With $x_l' = t_l/x_l$, $(x_1+x_2)(t_1/x_1 + t_2/x_2) = t_1 + t_2 + t_1r + t_2r^{-1}$.

Its torus fibre integral at the norms $(t_1,t_2)$ is the constant term in $r$ of $\big(\tau-(t_1+t_2+t_1r+t_2r^{-1})\big)^{-1}$, which is $\big((\tau-t_1-t_2)^2-4t_1t_2\big)^{-1/2}$, and Theorem 42.20(ii) gives the first display. The coefficient of $z^j$ in the $s$-th power is $\sum_{|w|=j}\prod_e w_e!^{-2}=j!^{-2}\sum_{|w|=j}\binom{j}{w}^2$. □

**Proposition 42.26 (The value of a table is a closed serif word).** *As stated.*

*Proof.* (i) $K(\mathsf{m}\alpha^{-1})=\sum_T\theta_T\prod_{e\in T}\mathsf{m}_e\alpha_e^{-1}=\alpha^{-\mathcal{E}}\sum_T\theta_T\mathsf{m}^T\alpha^{\mathcal{E}\setminus T}=\alpha^{-\mathcal{E}}\Psi_{\theta\mathsf{m}}(\alpha)$. So $\big(K(\alpha)K(\mathsf{m}\alpha^{-1})\big)^{-k}\alpha^{-\mathbf{1}}=\alpha^{k\mathbf{1}-\mathbf{1}}K^{-k}\Psi_{\theta\mathsf{m}}^{-k}$, the integrand of the period of $F_\mathsf{m}$ at $k\mathbf{1}$ (Theorem 8.7), and $m_\mathcal{E}=nk=rk+(n-r)k$ is the balance. The walls do not see the weights: with the floors $d(S)=r-r(\mathcal{E}\setminus S)$ of $K$ and $d^*(S)=|S|-r(S)$ of $\Psi$ on $S$, $\kappa_S=k|S|-k\,d(S)-k\,d^*(S)=k\big(r(S)+r(\mathcal{E}\setminus S)-r\big)$, positive for every nonempty proper $S$ because the table is connected. So the period converges (Theorem 8.7), and the family is connected with its walls positive at the target, so it has a native gauge (Theorem 8.19(iii)).

(ii) Apply Corollary 18.16, under Theorem 18.13, to $F_\mathsf{m}$, with one share per basis for each active letter; (i) provides the hypotheses. The monomial in the weights is $\prod_T\theta_T^{-a_T}(\theta_T\mathsf{m}^T)^{-b_T}=\prod_T\theta_T^{-a_T-b_T}\prod_e\mathsf{m}_e^{-\sum_{T\ni e}b_T}$. On the fibre, $\mathrm{load}_e=\sum_{T\ni e}a_T+\sum_{T\not\ni e}b_T=k$ and $|\mathbf{b}|=k$ give $\sum_{T\ni e}b_T=\ell_e(\mathbf{a})$, for $e\neq g$ by the pins and for $g$ by the balance, $\sum_e\mathrm{load}_e=r|\mathbf{a}|+(n-r)|\mathbf{b}|=nk$. The formula holds on every wall contour (Theorem 18.13(i)), and the uniform split is one: every share has the real part $k/N>0$, and its load on $e$ is $\frac{k}{N}\big(\#\{T\ni e\}+\#\{T\not\ni e\}\big)=k$. The dimension counts $2N$ shares against two share sums and $n-1$ pins, and these are independent. Given $|\mathbf{b}|=k$ the pins read $\sum_{T\ni e}(a_T-b_T)=0$, $e\neq g$; a vanishing combination $\lambda|\mathbf{a}|+\lambda'|\mathbf{b}|+\sum_{e\neq g}\mu_e\sum_{T\ni e}(a_T-b_T)$, with $\mu_g=0$, has $\lambda+\langle\mu,\mathbf{1}_T\rangle=0$ and $\lambda'-\langle\mu,\mathbf{1}_T\rangle=0$ for every basis $T$. The points $\mathbf{1}_T$ of a connected table span an affine hyperplane, its base polytope having the dimension $n-1$ (the dimension of the base polytope of a connected matroid, which applies to the bases of a table by the arena property, the exchange property of the bases), so $\langle\mu,\cdot\rangle$ is constant on the hyperplane $\{x_\mathcal{E}=r\}$, $\mu\in\mathbb{R}\mathbf{1}$, and $\mu_g=0$ gives $\mu=0$ and then $\lambda=\lambda'=0$.

(iii) By Theorem 21.4 at $\alpha$ and at $\mathsf{m}\alpha^{-1}$, as in Theorem 42.20(i), $K(\alpha)K(\mathsf{m}\alpha^{-1})=\prod_i v_{\mathcal{A}_i}\,\Phi_\mathcal{H}(\mathbf{x},\mathbf{x}')$ at the hinge pairs $(x_i,x_i')$ of the cells, a mark of the norm $\mathsf{m}_i$ left in place having the hinge pair $(\alpha_{h_i},\mathsf{m}_i\alpha_{h_i}^{-1})$ and $v=1$ (Definition 42.19(i)). By Proposition 21.3(iv) the core and the cells are connected, and no marker is a loop or a coloop of the core (Definition 21.1). Integrate each cell with $|B_i|\geq 2$ first, in its scale $\lambda_i$ and its shape $u_i$, $\alpha_{B_i}=\lambda_iu_i$: the hinge value is $x_i=\lambda_i\gamma_i(u_i)$ and its partner is $t_i/x_i$, with $t_i=\gamma_i(u_i)\gamma_i(u_i^{-1})$ the norm of the cell, a function of the shape alone, and $v_{\mathcal{A}_i}$ does not depend on the scale. So $\mathsf{V}_\mathcal{B}(k;\mathsf{m})=\int\mathsf{V}_\mathcal{H}(k;\mathbf{t})\prod_i\rho_i(dt_i)$, where $\mathsf{V}_\mathcal{H}(k;\mathbf{t})$ is the value of the core with the norms $t_i$ on its markers, $\rho_i$ is the image of $v_{\mathcal{A}_i}^{-k}\alpha^{-\mathbf{1}}d\alpha$, on the shapes of $\mathcal{A}_i$, under $t_i$, and for a mark left in place $\rho_i$ is the unit point mass at $\mathsf{m}_i$. By (ii) the core's word carries the monomial $\prod_i t_i^{-\ell_{h_i}(\mathbf{a})}$, and $\int t^{-\ell}\rho_i(dt)=\widehat{\mathsf{V}}_{\mathcal{A}_i}(k;\ell)$ because $v_\mathcal{A}^{-k}t^{-\ell}=v_\mathcal{A}^{\ell-k}\big(K_{\mathcal{A}\setminus g}(\alpha)K_{\mathcal{A}\setminus g}(\alpha^{-1})\big)^{-\ell}$, $\gamma$ being $K_{\mathcal{A}\setminus g}/K_{\mathcal{A}/g}$; for a mark left in place it is $\mathsf{m}_i^{-\ell}$.

Take the uniform split of the core as the contour. It does not move with the norms $t_i$, and on it $\mathrm{Re}\,\ell_{h_i}=kN_{h_i}/N_\mathcal{H}$, with $N_h$ the number of bases of $\mathcal{H}$ containing $h$, so $0<\mathrm{Re}\,\ell_{h_i}<k$. The hinge minors lie on the balance wall of $B$: with $n_\mathcal{A}=|B|$ and $r_\mathcal{A}$ the rank of $\mathcal{A}\setminus g$, which is the rank of $\mathcal{A}$, the masses $k\mathbf{1}$ sum to $n_\mathcal{A}k=(k-\ell)\big((r_\mathcal{A}-1)+(n_\mathcal{A}-r_\mathcal{A}+1)\big)+\ell\big(r_\mathcal{A}+(n_\mathcal{A}-r_\mathcal{A})\big)$. Their walls are $\kappa_S=(k-\ell)\,c_{\mathcal{A}/g}(S)+\ell\,c_{\mathcal{A}\setminus g}(S)$, with $c(S)=r(S)+r(B\setminus S)-r$ for each minor, as in (i). Both terms are nonnegative, and they vanish together only if $S$ separates $\mathcal{A}$: if $S$ and $S'=B\setminus S$ separate both minors, then in $\mathcal{A}$ $r(S)+r(S')=r_\mathcal{A}$ and $r(S+g)+r(S'+g)=r_\mathcal{A}+1$, so $g$ lies in the closure of exactly one side, say $S'$, and $S$, $S'+g$ separate $\mathcal{A}$. The cell is connected, so $\widehat{\mathsf{V}}_\mathcal{A}(k;\ell)$ converges for $0<\mathrm{Re}\,\ell<k$ (Theorem 8.7), and $|\widehat{\mathsf{V}}_\mathcal{A}(k;\ell)|\leq\widehat{\mathsf{V}}_\mathcal{A}(k;\mathrm{Re}\,\ell)$ since $|t^{-\ell}|=t^{-\mathrm{Re}\,\ell}$. On

the contour the real parts are fixed and the stacked Gammas decay exponentially in the imaginary directions, so the iterated integral converges absolutely and the integrals exchange. For a parallel class $\mathcal{A}/g$ has the single empty basis and $\mathcal{A}\setminus g=\binom{C}{1}$, so $\widehat{\mathsf{V}}$ is the period of the centre ray of $\binom{C}{1}$ at the strength $\ell$, which is the stated integral of the Gramma (Corollary 18.24 for the dual table and Proposition 8.15, as at the head of Section 42.6).

(iv) At these data the hinge wall of $\mathcal{A}$ is $k|B|-k(r_{\mathcal{A}}-1)-k(|B|-r_{\mathcal{A}})=k$, so $\phi_{\mathcal{A}}=u^{k\mathbf{1}-\mathbf{1}}K_{\mathcal{A}\setminus g}^{-k}\Psi_{\mathcal{A}\setminus g}^{-k}\,d\omega_B$ and $\varrho_{\mathcal{A}}=\mathbf{E}_{\mathcal{A}/g}/\mathbf{E}_{\mathcal{A}\setminus g}$ (Definitions 23.1 and 21.1). For each hinge minor $K(\alpha^{-1})=\alpha^{-B}\Psi(\alpha)$ with its weights, as in (i); so $v_{\mathcal{A}}=\alpha^{-B}\mathbf{E}_{\mathcal{A}/g}$, $K_{\mathcal{A}\setminus g}(\alpha)K_{\mathcal{A}\setminus g}(\alpha^{-1})=\alpha^{-B}\mathbf{E}_{\mathcal{A}\setminus g}$, and the form of $\widehat{\mathsf{V}}_{\mathcal{A}}(k;\ell)$ is $\alpha^{k\mathbf{1}-\mathbf{1}}\mathbf{E}_{\mathcal{A}/g}^{\ell-k}\mathbf{E}_{\mathcal{A}\setminus g}^{-\ell}\,d\alpha$, unchanged by the scale and integrated over any section (Definition 1.1(vii)). On the shape simplex it is $\varrho_{\mathcal{A}}^{\ell-k}\,d\phi_{\mathcal{A}}$, and its integral is the Trailmoment of the image measure with the kernel $x^{\ell-k}$ (Definition 5.2). The two Mellin transforms are Proposition 23.8(iii) at $s=t=\kappa=k$, finite at real $z\in(0,k)$ because $\widehat{\mathsf{V}}_{\mathcal{A}}$ is, the cell being connected (the proof of (iii)), and so at complex $z$ with these real parts. The measure $\rho_{\mathcal{A}}$ of the proof of (iii), the image of $v_{\mathcal{A}}^{-k}\alpha^{-\mathbf{1}}d\alpha$ on the shapes under $t=\varrho_{\mathcal{A}}^{-1}$, is the image of $x^{-k}(\varrho_{\mathcal{A}})_*\phi_{\mathcal{A}}(dx)$ under $x\mapsto x^{-1}$, because $v_{\mathcal{A}}^{-k}\alpha^{-\mathbf{1}}=\varrho_{\mathcal{A}}^{-k}\alpha^{k\mathbf{1}-\mathbf{1}}\mathbf{E}_{\mathcal{A}\setminus g}^{-k}$. So by Proposition 23.8(i), with $\mathsf{g}=\mathsf{G}_{0;(1,-1)}(1,\cdot)$,

$$\int \mathsf{g}(ty)\,\rho_{\mathcal{A}}(dt)=\check{f}_{\mathcal{A}}(y),\qquad \int t^{-k}\mathsf{g}(y/t)\,\rho_{\mathcal{A}}(dt)=f_{\mathcal{A}}(y),$$

and for the one-mark cell of ratio 1, $\rho=\delta_1$ gives $\mathsf{g}$ for both, as in Definition 23.19(1).

(v) By Definition 21.1 a set of $\mathcal{A}$ contains $g$ exactly when the basis of $\mathcal{H}'$ it comes from contains $h$, the mark left in place there being the cell $\binom{\{g,h\}}{1}$. So $\mathcal{A}\setminus g$ and $\mathcal{A}/g$ are given by the formula of that definition at $\mathcal{D}$ and at $\mathcal{C}$, with the same cells and the inherited weights, a marker that is a loop of the minor putting the contraction of its cell into every set and a coloop its deletion; $\mathcal{H}'$, a core, has neither. Expanding the letters over these sets gives Theorem 21.4 for $\mathcal{C}$ and $\mathcal{D}$ as it stands, and at $\alpha$ and at $\alpha^{-1}$, as in the proof of (iii), the form of $\widehat{\mathsf{V}}_{\mathcal{A}}(k;\ell)$ is $\prod_i v_{\mathcal{A}_i'}^{-k}\,\Phi_{\mathcal{C}}^{\ell-k}\Phi_{\mathcal{D}}^{-\ell}\alpha^{-\mathbf{1}}d\alpha$, with $\Phi_{\mathcal{X}}=K_{\mathcal{X}}(\mathbf{x})K_{\mathcal{X}}(\mathbf{x}')$ at the hinge pairs $(x_i,x_i')$ of the cells, a mark left in place having $v=1$ and the pair $(\alpha_{h_i},\alpha_{h_i}^{-1})$. Integrating each cell of at least two marks in its scale and its shape,

$$\widehat{\mathsf{V}}_{\mathcal{A}}(k;\ell)=\int W(k,\ell;\mathbf{t})\prod_i\rho_{\mathcal{A}_i'}(dt_i),$$
$$W(k,\ell;\mathbf{t})=\int\big(K_{\mathcal{C}}(\mathbf{x})K_{\mathcal{C}}(\mathbf{t}\mathbf{x}^{-1})\big)^{\ell-k}\big(K_{\mathcal{D}}(\mathbf{x})K_{\mathcal{D}}(\mathbf{t}\mathbf{x}^{-1})\big)^{-\ell}\mathbf{x}^{-\mathbf{1}}\,d\mathbf{x},$$

with $\rho=\delta_1$ at a mark left in place. By (i), $W(k,\ell;\mathbf{t})$ is the period at the masses $k\mathbf{1}$ of the family of the four tables $(\mathcal{C},\theta^{\mathcal{C}})$ at $(k-\ell,0)$, $(\mathcal{C},\theta^{\mathcal{C}}\mathbf{t})$ at $(0,k-\ell)$, $(\mathcal{D},\theta^{\mathcal{D}})$ at $(\ell,0)$ and $(\mathcal{D},\theta^{\mathcal{D}}\mathbf{t})$ at $(0,\ell)$, on its balance wall, $pk=(k-\ell)p+\ell p$, with the walls $(k-\ell)\,c_{\mathcal{C}}(S)+\ell\,c_{\mathcal{D}}(S)$, $c(S)=r(S)+r(S')-r$ for each minor and $S'$ the other markers. For real $\ell\in(0,k)$ they are positive: if $S$ and $S'$ separate both minors, then in $\mathcal{H}'$ $r(S)+r(S')=r_{\mathcal{H}'}$ and $r(S+h)+r(S'+h)=r_{\mathcal{H}'}+1$, so $h$ lies in the closure of exactly one side and the two sides separate $\mathcal{H}'$, which is connected (Proposition 21.3(iv)). So the family converges and has a native gauge (Theorems 8.7 and 8.19(iii)), and Corollary 18.16 under Theorem 18.13, with one share on each basis for each of the four letters, gives its word on $\mathcal{Y}_{k,\ell}$. The monomial in the weights is $\prod_T(\theta_T^{\mathcal{C}})^{-a_T}(\theta_T^{\mathcal{C}}\mathbf{t}^T)^{-b_T}\prod_T(\theta_T^{\mathcal{D}})^{-c_T}(\theta_T^{\mathcal{D}}\mathbf{t}^T)^{-d_T}$, and on the fibre $\sum_{T\ni x}(b_T+d_T)=\ell_x(\mathbf{a})+\ell_x(\mathbf{c})$ at every marker, from its load $k$ and $|\mathbf{b}|+|\mathbf{d}|=k$, at the free marker by the balance as in (ii); so the norms enter as $\prod_i t_i^{-\ell_{h_i}(\mathbf{a})-\ell_{h_i}(\mathbf{c})}$, and $\int t^{-\lambda}\rho_{\mathcal{A}_i'}(dt)=\widehat{\mathsf{V}}_{\mathcal{A}_i'}(k;\lambda)$. On the uniform split $\ell_{h_i}(\mathbf{a})+\ell_{h_i}(\mathbf{c})=(k-\ell)p_i+\ell q_i$, with $p_i$ and $q_i$ the fractions of the bases of $\mathcal{C}$ and of $\mathcal{D}$ that contain $h_i$; $q_i>0$ because deleting a mark of the core makes no loop and $p_i<1$ because

contracting one makes no coloop, so its real part lies in $(0,k)$, and the integrals exchange as in (iii). The pins are independent: a vanishing combination of the four share sums and of the loads at the markers but one, with $\mu$ its coefficients on the markers and $\mu=0$ at the free one, makes $\langle\mu,\mathbf{1}_T\rangle$ constant on the bases of $\mathcal{C}$ and constant on those of $\mathcal{D}$; with $\mu_h$ the difference of the two constants, $\langle\mu,\cdot\rangle$ is constant on the bases of $\mathcal{H}'$, which is connected, so $\mu\in\mathbb{R}\mathbf{1}$ as in the proof of (ii), $\mu=0$, and then every coefficient vanishes. So the fibre has the dimension $2(N_{\mathcal{C}}+N_{\mathcal{D}})-4-(p-1)$.

(vi) For the parallel node every marker is a loop of $\mathcal{C}$, which has the single basis $\varnothing$ with the weight 1, and $\mathcal{D}=\binom{\{h_1,\dots,h_p\}}{1}$; for the series node $\mathcal{D}$ has the single basis $\{h_1,\dots,h_p\}$, and $\mathcal{C}=\binom{\{h_1,\dots,h_p\}}{p-1}$. So $W(k,\ell;\mathbf{t})$ of the proof of (v) is the value of $\binom{\{h_1,\dots,h_p\}}{1}$ at the strength $\ell$ with the norms $\mathbf{t}$ in the first case, and $(\prod_i t_i)^{-\ell}$ times the value of $\binom{\{h_1,\dots,h_p\}}{p-1}$ at the strength $k-\ell$ with the norms $\mathbf{t}$ in the second. For $c_1,\dots,c_p>0$ and $\sigma>0$, open $(\sum_i x_i)^{-\sigma}$ and $(\sum_i c_i/x_i)^{-\sigma}$ with the scales $\lambda$ and $w/\lambda$ and move $\lambda$ into $\mathbf{x}$; by Tonelli's theorem and Definition 2.1,

$$\begin{aligned}\int\Big(\sum_i x_i\sum_i\frac{c_i}{x_i}\Big)^{-\sigma}\mathbf{x}^{-\mathbf{1}}\,d\mathbf{x}&=\frac{1}{\Gamma(\sigma)^2}\int_0^\infty w^{\sigma-1}\prod_i\int_0^\infty e^{-\beta-c_iw/\beta}\,\frac{d\beta}{\beta}\,dw\\&=\frac{1}{\Gamma(\sigma)^2}\int_0^\infty w^{\sigma-1}\prod_i\mathsf{g}(c_iw)\,dw.\end{aligned}$$

$K(\mathbf{x})K(\mathbf{t}\mathbf{x}^{-1})$ is $\sum_i x_i\sum_i t_i/x_i$ for $\binom{\{h_1,\dots,h_p\}}{1}$ and $\prod_i t_i\,\sum_i x_i^{-1}\sum_i x_i/t_i$ for $\binom{\{h_1,\dots,h_p\}}{p-1}$, the second of the first form in $\mathbf{x}^{-1}$ with $c_i=1/t_i$. So

$$W(k,\ell;\mathbf{t})=\frac{1}{\Gamma(\ell)^2}\int_0^\infty w^{\ell-1}\prod_i\mathsf{g}(t_iw)\,dw,\qquad W(k,\ell;\mathbf{t})=\frac{(\prod_i t_i)^{-k}}{\Gamma(k-\ell)^2}\int_0^\infty w^{k-\ell-1}\prod_i\mathsf{g}(w/t_i)\,dw,$$

and integrating against $\prod_i\rho_{\mathcal{A}_i'}(dt_i)$ by Tonelli's theorem gives the two formulas by (iv). A series class is the series node with marks left in place.

(vii) The proof of (iii) gives $\mathsf{V}_{\mathcal{B}}(k;\mathsf{m})=\int\mathsf{V}_{\mathcal{H}}(k;\mathbf{t})\prod_i\rho_{\mathcal{A}_i}(dt_i)$, with $\rho=\delta_{\mathsf{m}_i}$ at a mark left in place, and the value of the core with the norms $\mathbf{t}$ is the integral of the proof of (vi) at $\sigma=k$: $\Gamma(k)^{-2}\int_0^\infty w^{k-1}\prod_i\mathsf{g}(t_iw)\,dw$ for the parallel node and $(\prod_i t_i)^{-k}\Gamma(k)^{-2}\int_0^\infty w^{k-1}\prod_i\mathsf{g}(w/t_i)\,dw$ for the series node. By (iv), and with $\delta_{\mathsf{m}_i}$ giving $\mathsf{g}(\mathsf{m}_iw)$ and $\mathsf{m}_i^{-k}\mathsf{g}(w/\mathsf{m}_i)$, the two formulas follow. □

**Lemma 42.27 (The pencil of a class).** *As stated.*

*Proof.* (i) $\alpha_e\partial_e\varphi=A'\alpha_e-A\alpha_e^{-1}$. If $A,A'\neq0$ these vanish together exactly when $\alpha_e^2=A/A'$ for every $e$, a sign times a scale, of the value $(\sum_e\eta_e)^2\neq0$; nondegeneracy is Proposition 42.9(ii). If $A=0$, then $A'\alpha_e=0$ forces $A'=0$, and if $A'=0$, then $A=0$. (ii) Let $x_m$ leave every compact subset of $T_s$ with $d_y\log\varphi(x_m)\to0$ and $\varphi(x_m)\to t_0\neq0$. Then $A,A'\neq0$, and the scale can be chosen with $A=A'$, so $A^2\to t_0$ and $\partial_{y_e}\log\varphi=(\alpha_e-\alpha_e^{-1})/A\to0$; every $\alpha_e\to\pm1$, and $x_m$ converges in $T_s$, a contradiction. (iii) The signs with $k$ and with $s-k$ entries $-1$ are the same points of $T_s$, the scale $-1$ exchanging them; $S_s$ is transitive on the signs with $k$ entries $-1$, and $\eta^{-1}=\eta$. □

**Lemma S29** (The pinch)**.** *Let $V$ be a local system on a punctured disc $\Delta^*$ about $t=1$ with the monodromy $T_1$, $j:\Delta^*\to\Delta$, and $\Phi(t,r)=t+1+tr+r^{-1}$ near $(t,r)=(1,-1)$, where $\Phi=0$. Let $W$ be the vanishing cycles at $(1,-1)$ of $j_*V\boxtimes\mathbb{C}$ along $\Phi$ at the value 0, $\psi$ the nearby cycles, $T_0=1+\mathrm{var}\circ\mathrm{can}$ on $\psi$ and $T_W=1+\mathrm{can}\circ\mathrm{var}$ on $W$.*

*(a) If $T_1=1$: $\dim W=\mathrm{rk}\,V$ and $T_W=1$, so $(T_0-1)^2=0$.*

*(b) If $T_1 - 1$ has rank one and square zero: $\dim W = \operatorname{rk} V + 1$, and $T_W$ is $-1$ on a line and $1 + N'$ on a complement, with $N'$ of rank at most one and square zero. So the eigenvalue $-1$ of $T_0$ has the multiplicity at most one, and on the unipotent part $(T_0 - 1)^3 = 0$ with $(T_0 - 1)^2$ of rank at most one.*

*Proof.* With $t = 1 + a$ and $r = -1 + b$, $\Phi = ab - b^2 - b^3 - \cdots$. For fixed $a$ it has a nondegenerate critical point in $b$, at $b = c(a) = \frac{a}{2} + O(a^2)$, and the Morse lemma in $b$ with the parameter $a$, which leaves the coordinate $a$ and so the local system untouched, gives $\Phi = g(a) - b'^2$ with $g(a) = \frac{a^2}{4} + O(a^3)$. So $\Phi$ is the sum of $g$, with the coefficients $j_*V$, and $-b'^2$, with constant coefficients, in separate variables, and $W \cong \phi_g(j_*V) \otimes \phi_{-b'^2}(\mathbb{C})$ with $T_W = T_g \otimes T_{-b'^2}$ (the theorem of Thom and Sebastiani for vanishing cycles with coefficients). The second factor is a line with the monodromy $-1$. The Milnor fibre of $g$ is a pair of points exchanged by its monodromy, so $\phi_g(j_*V)$ is the cokernel of $V^{T_1} \to V \oplus V$, $v \mapsto (v, v)$, with the monodromy $M(x, y) = (y, T_1 x)$, a half turn, $M^2 = T_1 \oplus T_1$. (a) $M$ is the exchange, $-1$ on the antidiagonal quotient, and $T_W = 1$. (b) With $N_1 = T_1 - 1$ and $\widetilde{N} = N_1 \oplus N_1$, $U = M(1 - \frac{1}{2}\widetilde{N})$ is an involution commuting with $\widetilde{N}$, $U(x, y) = (y - \frac{1}{2}N_1 y,\ x + \frac{1}{2}N_1 x)$, with the eigenspaces $E_\pm = \{(x, \pm(x + \frac{1}{2}N_1 x))\}$, each of dimension $\operatorname{rk} V$, and $M = \pm(1 + \frac{1}{2}\widetilde{N})$ on $E_\pm$. The diagonal image of $V^{T_1} = \ker N_1$ lies in $E_+$, with the quotient a line on which $M = 1$, $N_1$ mapping into its kernel; on $E_-$, $M = -(1 + \frac{1}{2}N_1)$. Tensoring with $-1$ gives $T_W$. Finally $(T_0 - 1)^k = \operatorname{var}(T_W - 1)^{k-1}\operatorname{can}$, and can and var are isomorphisms on the generalized eigenspaces of an eigenvalue other than 1. □

**Theorem 42.28 (The class for every size).** *As stated.*

*Proof.* Write $\mathsf{M}(j) = \mathsf{C}_j^{\binom{[s]}{1}}(1)$, extended by 0 to $j < 0$. By Theorem 42.12(iii), $\sum_k c_{s,k}(-j)\mathsf{M}(j - k) = 0$ for $j \in \mathbb{N}$, and for $j < 0$ every term vanishes, so the relation holds on $\mathbb{Z}$. On $t^{-j-1}$ the operator $t\partial_t + 1$ acts as $-j$ and $t^{-k}T_s = \sum_j \mathsf{M}(j - k)t^{-j-1}$, so $\widetilde{A}_s T_s = \sum_j \big(\sum_k c_{s,k}(-j)\mathsf{M}(j - k)\big)t^{-j-1} = 0$. Every $c_{s,k}$ has the degree $s - 1$, so $\widetilde{A}_s$ has the order $s - 1$; its leading coefficient vanishes in $\mathbb{C}^\times$ exactly at the sign values, simply, by the symbol of Theorem 42.12(ii). At $t = \infty$, in $w = 1/t$, $\widetilde{A}_s = c_{s,0}(1 - w\partial_w) + O(w)$ with $c_{s,0}(x) = (-1)^{s+1}x^{s-1}$, so every exponent there is 1.

*The rank.* $V_s$ lies in the solutions of $\widetilde{A}_s$, of dimension $s - 1$. The closings are positive with $\mathsf{M}(j)^{1/j} \to s^2$, so $T_s$ is singular at $t = s^2$ (Pringsheim's theorem on series with nonnegative coefficients). The value $s^2$ carries the single nondegenerate native gauge $\mathbf{1}$ and no asymptotic one (Lemma 42.27), so the variation of $T_s$ about $s^2$ is a nonzero multiple of the period of the vanishing cycle there (the Picard–Lefschetz formula), which lies in $V_s$. For $t > s^2$ the orthant fibre $\{\varphi = t\} \cap \mathbb{R}^s_{>0}$ is the real vanishing sphere of the minimum $\mathbf{1}$ of $\varphi$ on the orthant, so that period is the orthant fibre integral $P$, up to a constant. By Proposition 39.54, applied to the centre ray of $\binom{[s]}{1}$, $(T_\infty - 1)^{s-2}P$ is a nonzero multiple of $V_1 T_s \neq 0$ on the part of $P$ of the eigenvalue 1, which lies in $V_s$ with $P$. So $V_s$ contains a Jordan block of size $s - 1$: it is the whole space of the solutions, which is that block, (ii). A nonzero sub-local system is stable under $T_\infty$, so it contains the invariant line of the block, spanned by $T_s$, and with it all of $V_s$: $V_s$ is irreducible, (i).

*The local monodromies.* $V_s$ lies in the solutions of $\mathcal{D}[1]$ of the centre ray of $\binom{[s]}{1}$. At a sign value $x$ its vanishing rank is at most 1 by Lemma 42.27 and Lemma 39.50, and $\operatorname{rank}(T_x - 1)$ on the solutions of a regular holonomic module is at most the dimension of its vanishing cycles, which is the vanishing rank; so $\operatorname{rank}(T_x - 1) \leq 1$ on $V_s$. By Theorem 39.51 and Lemma 42.27(ii) there are no other singular points in $\mathbb{C}^\times$. At $\infty$, $\operatorname{rank}(T_\infty - 1) = s - 2$. At 0, for $s \geq 3$: by Corollary 42.25 with $(a, b) = (s - 1, 1)$ and Theorem 42.20(iii), $V_s$ is a subquotient of the nearby cycles at the value 0 of the direct image of $\mathcal{H}_C \boxtimes \mathcal{H}_{\mathrm{mark}}$ under the two-sided core of $\binom{[2]}{1}$, $C$ the class of $s - 1$

marks. Modulo the scale, with the mark at $x_2 = x_2' = 1$ and $r = 1/x_1$, the two-sided core is $\Phi(t,r) = t + 1 + tr + r^{-1}$, $t = x_1x_1'$ the norm of $C$. Off the axes the scale identifies the hinge map of $C$ with its pencil times the scale, so there $\mathcal{H}_C$ is the pullback along $t$ of the module $\mathcal{D}[1]$ of $\binom{[s-1]}{1}$. Near the value 0 the two-sided core has the single critical point $(1,-1)$ on the strata of that module, with none at infinity: $\partial_t\Phi = 1 + r$ and $\partial_r\Phi = t - r^{-2}$ vanish together only at $(1,-1)$; on a line $t = t_*$ the critical values are $(\sqrt{t_*} \pm 1)^2$, which vanish only at $t_* = 1$; on $t = 0$, $\Phi = 1 + r^{-1}$ has none; and at infinity $\partial_t\Phi \to 0$ forces $r \to -1$, and then $r\partial_r\Phi \to 1 - t$, which tends to 0 only at the finite point. Let $V'$ be the local system of the solutions of $\mathcal{D}[1]$ of $\binom{[s-1]}{1}$ near $t = 1$. For odd $s$ the value 1 is not a sign value of $\binom{[s-1]}{1}$, whose sign values are even squares, nor an asymptotic native value (Lemma 42.27(ii)), so that module is a connection near $t = 1$, $T_1 = 1$, and Slemma S29(a) gives $(T_0 - 1)^2 = 0$ on $V_s$, hence $\operatorname{rank}(T_0 - 1) \le \frac{s-1}{2}$. For even $s \ge 4$, 1 is the least sign value of $\binom{[s-1]}{1}$; there the vanishing rank is at most 1 (Lemma 39.50), $T_1 - 1$ has rank one by the statement for $s - 1$, so the module is the intermediate extension $j_*V'$ near $t = 1$, and $(T_1 - 1)^2 = 0$ because the vanishing cycle of a nondegenerate critical point in fibres of the odd complex dimension $s - 3$ has the self-intersection 0 (the Picard–Lefschetz formula). Slemma S29(b) gives $\dim E_{-1} \le 1$ for the generalized eigenspace of $-1$ and, on the unipotent part $E_1$ with $N = T_0 - 1$, $N^3 = 0$ and $\operatorname{rank} N^2 \le 1$. With $n_l$ Jordan blocks of size $l$ on $E_1$, $\operatorname{rank} N = (\dim E_1 - n_1 + n_3)/2$ with $n_3 \le 1$, and $\dim E_1 + \dim E_{-1} = s - 1$ is odd; in both cases $\operatorname{rank}(T_0 - 1) \le \frac{s}{2}$. For $s = 2$, $T_2 = \big(t(t-4)\big)^{-1/2}$, with $T_0 = -1$, and every statement is read off it.

*The count.* Let $s \ge 3$ and let $\mathcal{V}$ be the intermediate extension of $V_s$ to $\mathbb{P}^1$. A global section is fixed by $T_\infty$, so it is a multiple of $T_s$, which is not fixed by the monodromy about $s^2$: there is none. A functional fixed by the monodromy kills the image of $T_\infty - 1$, which contains $T_s$ because the block has the size $s - 1 \ge 2$, and then all of $V_s$: there is none. So $0 \ge \chi(\mathbb{P}^1, \mathcal{V}) = 2(s-1) - \sum_x \operatorname{rank}(T_x - 1)$ (the Euler characteristic formula for the intermediate extension of a local system on the projective line). With the bounds above and the $h$ sign values, $2s - 2 \le (s - 2) + h + \operatorname{rank}(T_0 - 1)$, so $\operatorname{rank}(T_0 - 1) \ge s - h = \lfloor s/2 \rfloor$; the upper bounds are $\lfloor s/2 \rfloor$ as well, so equality holds throughout, and every sign value has $\operatorname{rank}(T_x - 1) = 1$.

*The exponents.* At a sign value the leading coefficient of $\widetilde{A}_s$ vanishes simply, so $s - 2$ exponents are $0, \dots, s - 3$, and the last one is that of the period of the vanishing cycle of a nondegenerate critical point on $T_s$, of dimension $s - 1$, which is $\frac{s-3}{2}$. At 0, $t^h\widetilde{A}_s = \sum_k t^{h-k}c_{s,k}(t\partial_t + 1 - k)$, so the exponents are the roots $\rho$ of $c_{s,h}(\rho + 1 - h)$; $c_{s,h}$ vanishes to the order two at $-1, \dots, -(h-1)$ (Theorem 42.12(ii)), which gives $\rho = 0, \dots, h - 2$, each twice, and for even $s$ one more root. The sum of all the exponents of an operator of order $s - 1$ with $h + 2$ singular points, all regular, is $h\binom{s-1}{2}$ (the Fuchs relation); the sign values contribute $h\frac{(s-3)(s-1)}{2}$ and $\infty$ contributes $s - 1$, which leaves $\frac{(s-1)(h-2)}{2}$ for $t = 0$: for odd $s$ this is the sum of the double list, and for even $s$ the extra exponent is $\frac{(s-1)(s-4)}{4} - \frac{(s-4)(s-2)}{4} = \frac{s}{4} - 1$. For odd $s$ the blocks at 0 follow from $(T_0 - 1)^2 = 0$ and the rank. □

**Proposition 42.29 (The hinge module of a class).** *As stated.*

*Proof. (a) The Fourier side.* Under the Fourier transform $x \mapsto -\partial_\xi$, $\partial_x \mapsto \xi$, $x' \mapsto -\partial_{\xi'}$, $\partial_{x'} \mapsto \xi'$, the direct image of the functions along $(A, A')$ becomes the top cohomology of the relative forms on $(\mathbb{C}^\times)^s$ over $\mathbb{C}[\xi, \xi']$ twisted by $\exp\big(-\sum_e(\xi\alpha_e + \xi'\alpha_e^{-1})\big)$, on which $\partial_\xi$ and $\partial_{\xi'}$ act by $\partial_\xi - A$ and $\partial_{\xi'} - A'$ (the Fourier transform of a direct image is the exponentially twisted direct image). That complex is the tensor product over $\mathbb{C}[\xi, \xi']$ of $s$ copies of the two-term complex of one mark, and the top cohomology of a tensor product of two-term complexes of free modules is the tensor product of their top cohomologies, a cokernel; so the transform is $\mathcal{B}^{\otimes s}$, with $\mathcal{B}$ the top cohomology

of one mark, the derivations acting by the Leibniz rule, and the generator goes to $v^{\otimes s}$, $v = [d\alpha/\alpha]$. In $\mathcal{B}$, $\partial_\xi \partial_{\xi'} v = [\alpha\alpha^{-1} d\alpha/\alpha] = v$, and $(\xi\partial_\xi - \xi'\partial_{\xi'})v = -[(\xi - \xi'\alpha^{-2})d\alpha] = 0$, the class of the twisted differential of 1; so $\vartheta^2 v = zv$ with $\vartheta = \xi\partial_\xi$ and $z = \xi\xi'$. The operator $\sum_k z^k q_k(\vartheta)$ of Theorem 42.12(i) annihilates every product of $s$ solutions of $\vartheta^2 u = zu$, and its proof uses only that relation and the Leibniz rule; so it annihilates $v^{\otimes s}$, and transformed back, with $\vartheta \mapsto -x\partial_x - 1$ and $z \mapsto \partial_x \partial_{x'}$,

$$\sum_{k=0}^{h} (\partial_x \partial_{x'})^k q_k(-x\partial_x - 1)\,[\omega] = 0, \qquad E[\omega] = 0,$$

the second from $\vartheta - \xi'\partial_{\xi'} \mapsto -E$.

*(b) One factor comes off.* Since $\partial_x \partial_{x'}\, xx' = (x\partial_x + 1)(x'\partial_{x'} + 1)$ and $q_0(-x\partial_x - 1) = (-1)^{s+1}(x\partial_x + 1)^{s+1}$,

$$\sum_{k=0}^{h} (\partial_x \partial_{x'})^k q_k(-x\partial_x - 1) = \partial_x \partial_{x'}\, \widetilde{B} + (-1)^{s-1}(x\partial_x + 1)^s\, E,$$

so $\partial_x \partial_{x'} \widetilde{B}[\omega] = 0$. On functions of $t$, $x\partial_x$ and $x'\partial_{x'}$ act as $\theta = t\partial_t$ and $\partial_x \partial_{x'}$ as $t^{-1}\theta^2$, with $(t^{-1}\theta^2)^{k-1} = t^{-(k-1)} \prod_{m=0}^{k-2} (\theta - m)^2$. On the other side $c_{s,k}(\theta + 1)\, t^{-k} = t^{-k} c_{s,k}(\theta + 1 - k)$ and $c_{s,k}(\theta + 1 - k) = q_k(-\theta - 1) \prod_{m=0}^{k-2} (\theta - m)^2$, while the term $k = 0$ is $(-1)^{s+1}(\theta+1)^{s-1} = (-\theta - 1)^{s-1}$. So $t^{h-1}\widetilde{B} = \sum_k t^{h-k} c_{s,k}(\theta + 1 - k) = t^h \widetilde{A}_s$ on functions of $t$.

*(c) No further periods.* Off the axes the scale identifies the fibres of $(A, A')$ with those of $\varphi = AA'$, and the periods of $[\omega]$ at $(x, x')$ with the periods of $[1]$ at $t = xx'$. By (b) every period $g$ of $[1]$ has $\partial_t \theta\, (t\widetilde{A}_s g) = 0$, so $t\widetilde{A}_s g \in \mathbb{C} + \mathbb{C} \log t$. If it is not 0, then, continuing $g$ once about $t = \infty$ and subtracting when $\log t$ occurs, some period $g_1$ has $t\widetilde{A}_s g_1 = 1$. At $t = \infty$ every exponent of $\widetilde{A}_s$ is 1, with the multiplicity $s - 1$, so the right side $t^{-1}$ of $\widetilde{A}_s g_1 = t^{-1}$ is resonant and forces in $g_1$ the term $t^{-1}(\log t)^{s-1}$ with a nonzero coefficient: $(T_\infty - 1)^{s-1} g_1 \neq 0$, a Jordan block of size $s$ among the periods of the cohomology of degree $s - 2$ of the fibres, where the monodromy theorem allows at most $s - 1$. So every period of $[1]$ is a solution of $\widetilde{A}_s$, and these are $V_s$ (Theorem 42.28(i)): (ii).

*(d) $\widetilde{B}$ itself.* Put $g = \widetilde{B}[\omega]$. Then $\partial_x \partial_{x'} g = 0$ by (b), and $Eg = 0$, as $\widetilde{B}$ commutes with $E$. By (c) all the periods of $g$ vanish, so $g$ is supported on the axes and the hyperbolas. The module $\mathcal{D}g$ is a quotient of $\mathcal{D}/(\mathcal{D}\partial_x \partial_{x'} + \mathcal{D}E)$, whose characteristic variety lies in $\{\eta\eta' = 0,\ x\eta = x'\eta'\} = T^*_P \cup T^*_X \cup T^*_{X'}$; so $g$ is supported on $X \cup X'$. Near a point of $X \setminus O$ a section supported on $X$ is $\sum_k n_k\, \partial_x^k \delta(x)$; $\partial_x \partial_{x'}$ kills it only if every $n_k$ is constant along $X$, and then $E$ multiplies its $k$-th term by $-(k + 1)$; so $g = 0$ there, and likewise near $X' \setminus O$. A nonzero section $\sum c_{ab} \partial_x^a \partial_{x'}^b \delta$ at $O$ is not killed by $\partial_x \partial_{x'}$. So $g = 0$: (i).

*(e) The axes.* By (i), $\mathcal{H}_C$ is a quotient of $N_s = \mathcal{D}/(\mathcal{D}\widetilde{B} + \mathcal{D}E)$. Near a point of $X' \setminus O$, in the coordinates $t = xx'$ and $y = x$, $E = y\partial_y$ and $\widetilde{B} \equiv t\widetilde{A}_s$ modulo $\mathcal{D}E$, so $N_s$ is $\mathcal{D}_t/\mathcal{D}_t\, t\widetilde{A}_s$ times $\mathcal{O}_y$, and the multiplicity of $T^*_{X'}$ in its characteristic cycle is the order at $t = 0$ of the leading coefficient of $t\widetilde{A}_s$, which is $1 + (s - 1 - h) = \lfloor s/2 \rfloor$, the leading coefficient of $\widetilde{A}_s$ being $t^{s-1}$ times a polynomial in $t^{-1}$ of the degree $h$ with a nonzero constant term. The multiplicity in $CC(\mathcal{H}_C)$ is at least $\operatorname{rank}(T_0 - 1) = \lfloor s/2 \rfloor$, so it is $\lfloor s/2 \rfloor$, that of the intermediate extension; likewise on $X$. Along $H_a$ it is the vanishing rank at $a^2$ of the module of $[1]$, which is 1 (Lemma 39.50 and Theorem 42.28). At $O$ and for odd $s$, $q_h$ is a nonzero constant, its degree being at most $s + 1 - 2h = 0$, so the principal symbol of $\widetilde{B}$ at $x = x' = 0$ is $q_h(\eta\eta')^{h-1}$, which vanishes only on $\{\eta\eta' = 0\}$, covered by $T^*_X$ and $T^*_{X'}$: $T^*_O$ is not a component, $m_s(O) = 0$.

*(f) Simplicity.* The Fourier transform is an equivalence, so it carries the composition factors of $\mathcal{H}_C$ to those of $\mathcal{D}v^{\otimes s}$. On $\xi\xi' \neq 0$, $v$ and $\vartheta v$ are a basis of $\mathcal{B}$, and the symmetric tensors

with $k$ factors $\vartheta v$ span a submodule containing $v^{\otimes s}$, the symmetric power $\operatorname{Sym}^s \mathcal{B}$, of rank $s+1$ and irreducible because the differential Galois group of $\vartheta^2 - z$ is $\mathrm{SL}_2$ (Kolchin's theorem on that equation). So exactly one composition factor is nonzero on $\xi\xi' \neq 0$; every other one is supported on $\xi\xi' = 0$, and transformed back it is pulled back from a coordinate line. The intermediate extension of the local system of $\mathcal{H}_C$ is singular along the $H_a$, so it is the factor that is nonzero on $\xi\xi' \neq 0$. Any other factor has the generic rank 0, the local system having the rank $s-1$ of $V_s$, and is pulled back from a coordinate line, so it is $\mathcal{O}_x \boxtimes \delta(x')$ or $\delta(x) \boxtimes \mathcal{O}_{x'}$, a delta at $O$ transforming to a factor that is nonzero on $\xi\xi' \neq 0$; by (e) the axes carry no more than the intermediate extension. So $\mathcal{H}_C$ is the intermediate extension: (iii).

*(g) The origin.* By (iii), $m_s(O)$ is that of the intermediate extension $F$. By the local index formula for constructible complexes on a surface, the Euler characteristic of the stalk of $F$ at $O$ is $(s-1) - 2\lfloor s/2 \rfloor + m_s(O)$, the axes being smooth curves through $O$ and the hyperbolas avoiding it. Near $O$ the local system has the monodromy $T_0$ about both axes, and the stalk is the part in the degrees below the middle of the cohomology of the link, a three-sphere through which the axes pass as two linked circles, with coefficients in the intermediate extension along the axes. Cover the sphere by tubes about the two circles, each with the cohomology $\ker(T_0 - 1)$ in the degrees 0 and 1, glued along a torus on which both generators act by $T_0$; the Mayer–Vietoris sequence gives $H^0 = \ker(T_0 - 1)$ and $H^1 = \ker(T_0 - 1) \cap \operatorname{im}(T_0 - 1)$, so the stalk has the Euler characteristic $n_1$, and $m_s(O) = n_1 + 2\lfloor s/2 \rfloor - (s-1)$. For odd $s$ every unipotent block has the size 2 and $T_0$ is unipotent, so this is 0, as in (e); for even $s$ it is $1 + n_1$. For $s = 2$ the module is explicit: $\mathcal{H}_2 = \mathcal{O} \cdot (xx')^{-1/2}(xx' - 4)^{-1/2}$ on the part of the character of the exchange of the two marks, with the cycle of $x^{-1/2} \boxtimes x'^{-1/2}$ at the origin: the origin, both axes and the zero section, each once. □

**Corollary 42.30 (The periods of a class are not unipotent).** *As stated.*

*Proof.* Let $G$ be the Zariski closure of the monodromy group of $V_s$, $s \geq 3$. The monodromy at $t = \infty$ is a unipotent Jordan block of size $s - 1 \geq 2$ (Theorem 42.28(ii)), which generates a connected unipotent group of positive dimension in $G^\circ$. If $G^\circ$ were solvable, its unipotent radical would be a nontrivial normal subgroup of $G$ whose fixed vectors form a proper nonzero subspace stable under $G$, against the irreducibility of $V_s$ (Theorem 42.28(i)); so $G^\circ$ is not solvable, and a unipotent group is solvable. A function whose continuations span a local system with a unipotent subgroup of finite index in its monodromy has a monodromy closure with a unipotent identity component; for a nonzero $F \in V_s$ the continuations span a nonzero sub-local system of $V_s$, which is $V_s$, whose monodromy closure is $G$, and $G^\circ$ is not unipotent. Iterated integrals of logarithmic forms with algebraic letters, and their combinations with algebraic coefficients, have local systems that become unipotent on a finite cover, since sums, products and algebraic coefficients keep that property. □

**Theorem 42.31 (The sources in the hinge space).** *As stated.*

*Proof.* The singular support of $\mathcal{H}_C$ is the support of its characteristic cycle, given by Proposition 42.29(iv); a mark has the hinge map $(\alpha_h, \alpha_h^{-1})$ onto $H_1$, with the module of $H_1$. By Theorem 42.20(iii) every singular value of $\mathcal{D}_{\mathcal{B}}[1]$ is one of the direct image of $\boxtimes_i \mathcal{H}_{C_i}$ under the two-sided core, whose characteristic variety is the product of the characteristic varieties. A direct image under $\Phi$ is a connection over a value $t_0$ near which the graph of $d\Phi$, the covectors $\lambda\, d\Phi$, meets the characteristic variety of the module only in the zero section and no such meeting escapes to infinity (the estimate of the characteristic variety of a direct image, with Rabier's condition at infinity as in the proof of Theorem 39.51). A covector $\lambda\, d\Phi$ with $\lambda \neq 0$ lies in the conormal of a stratum $W$ exactly when $d\Phi$ vanishes on the tangent space of $W$, a critical point of $\Phi|_W$. □

**Theorem 42.32 (The source count bounds the vanishing ranks).** *As stated.*

*Proof.* By Theorem 42.20(iii), $r_{t_0}(\mathcal{D}_{\mathcal{B}}[1])$ is at most the vanishing rank at $t_0$ of the submodule of $N$ generated by the image of the product of the generators, which lies in the part of $N$ on which $\Gamma$ acts by the character of [1], $\Gamma$ acting on $N$ because it preserves $\Phi_{\mathcal{H}}$ and the product of the hinge modules. The vanishing rank is the dimension of the vanishing cycles, as in the proof of Lemma 39.50. Under the stated conditions the characteristic cycle of $N$ over $t_0$ is the intersection of $CC(\boxtimes_i \mathcal{H}_{C_i}) = \prod_i CC(\mathcal{H}_{C_i})$ with the graph of $d\Phi_{\mathcal{H}}$ (Kashiwara's index formula for direct images), and the components of the product are the $T^*_W$ with the multiplicities $m(W)$. □

**Proposition 42.33 (Local contributions).** *As stated.*

*Proof.* (a) A weighted point is the image of an orbit of native gauges of $\mathcal{B}$, the classes being signs times scales (Theorem 42.24(b)), and Lemma 39.50 bounds the vanishing rank there by the number of orbits. (b) The vanishing cycles at the points of an orbit of $\Gamma$ form the representation induced from the stabilizer, which contains each character of $\Gamma$ at most as often as the stabilizer's representation contains its restriction (Frobenius reciprocity); two points exchanged, each with a trivial stabilizer, give the regular representation of the exchange tensored with a space of the dimension $m(W)$. (c) At a nondegenerate critical point of $\Phi|_W$ the graph of $d\Phi$ meets $T^*_W$ transversally in one point, of the intersection number 1, weighted by $m(W)$. For an odd corner $m(W) = m_{s_i}(X) = \frac{s_i - 1}{2}$, the other classes contributing $m(H_a) = 1$, and $m_{s_i}(O) = 0$ (Proposition 42.29(iv)); the Cremona exchanges $X$ and $X'$, and (b) gives the pair. (d) Each class of $I$ carries $m(P) = s_i - 1$ and each other class $m(H_a) = 1$, and (c) applies. (e) For a clean intersection of the graph of $d\Phi$ with $T^*_W$ along a curve $Z$ with a constant multiplicity, the intersection number is the Euler number of the excess bundle, which on the conormal of a critical curve is $-\chi(Z)$ times the multiplicity, the contributions of compact ends excluded by hypothesis (the excess intersection formula). □

## 50.42 Chapter 43: arithmetic along a ray

Throughout, Stirling's formula is $\log\Gamma(z) = (z - \frac{1}{2})\log z - z + \frac{1}{2}\log 2\pi + O(|z|^{-1})$ as $|z| \to \infty$ with $|\arg z| \le \pi - \delta$; it applies below to $\Gamma(k - a)$ and to $N! = \Gamma(N + 1)$ with $a$ fixed and $k, N \to +\infty$ along the reals, so that the arguments stay in a sector around the positive axis.

**Lemma S30.** *Let $\Bbbk$ be a subfield of $\mathbb{C}$. Let $X(n+1) = M(n)\,X(n)$ for the integers $n \ge n_2$, with $X(n) \in \mathbb{C}^m$ and $M(n)$ an $m \times m$ matrix whose entries are rational functions of $n$ with coefficients in $\Bbbk$ and whose determinant is not the zero function, and let $\psi(n) = \ell\, X(n)$ for a fixed row vector $\ell \in \Bbbk^{1\times m}$. Then $\psi$ satisfies, for all large $n$, a recurrence $\sum_{i=0}^{h} a_i(n)\,\psi(n+i) = 0$ with $a_i \in \Bbbk[n]$, $h \le m$ and $a_0 a_h \ne 0$.*

*Proof.* If $\ell = 0$, $\psi = 0$ satisfies $\psi(n) = 0$. Otherwise put $\mathbf{r}_0 = \ell$ and $\mathbf{r}_i(n) = \ell\, M(n+i-1)\cdots M(n)$, row vectors over the field $\Bbbk(n)$; then $\psi(n+i) = \mathbf{r}_i(n)X(n)$ wherever the entries are defined, and $\mathbf{r}_i(n) = \mathbf{r}_{i-1}(n+1)M(n)$. The $m+1$ vectors $\mathbf{r}_0, \dots, \mathbf{r}_m$ are dependent over $\Bbbk(n)$; let $h \ge 1$ be the least index with $\mathbf{r}_0, \dots, \mathbf{r}_h$ dependent, and $\sum_{i=0}^{h} a_i \mathbf{r}_i = 0$ with $a_i \in \Bbbk[n]$ after clearing denominators. By minimality $a_h \ne 0$. If $a_0 = 0$, then $\sum_{i=1}^{h} a_i(n)\mathbf{r}_{i-1}(n+1)M(n) = 0$, and $M(n)$ is invertible over $\Bbbk(n)$, so $\sum_{i=1}^{h} a_i(n)\mathbf{r}_{i-1}(n+1) = 0$; replacing $n$ by $n-1$, an automorphism of $\Bbbk(n)$, gives a dependence among $\mathbf{r}_0, \dots, \mathbf{r}_{h-1}$ with the coefficient $a_h(n-1) \ne 0$, against minimality. So $a_0 \ne 0$, and $\sum_i a_i(n)\psi(n+i) = \big(\sum_i a_i(n)\mathbf{r}_i(n)\big)X(n) = 0$ for all $n$ beyond the poles of the entries. □

**Theorem 43.1 (Closing one free share along a ray).** *As stated.*

*Proof.* (i) For an integer $r \ge 0$ and $x$ off the poles, $\Gamma(x+r) = (x)_r\Gamma(x)$ (1.1(v)) and $\Gamma(x-r) = \Gamma(x)/(x-r)_r = (x)_{-r}\Gamma(x)$ (1.3(iii)). Applied to every Gamma of $W_k$ with $x = \epsilon_\gamma a + b_\gamma$ and $r = ku_\gamma$ of either sign this is $W_k = W_0\, g_k$; the factors of $g_k$ have integer coefficients.

(ii) The upper Gammas have poles where $\epsilon_\alpha a + b_\alpha + ku_\alpha \in -\mathbb{N}$, at integers since $\epsilon_\alpha = \pm 1$; the lower ones contribute the entire factors $1/\Gamma$. For $n > n_1$ the arguments at $a = -n+\varepsilon$ are $-N_\gamma(n)+\varepsilon$ on $A_+$, $N_\gamma(n)+1-\varepsilon$ on $A_-$, $-N_\gamma(n)+\epsilon_\gamma\varepsilon$ on $B_+$ and $N_\gamma(n)+1-|\epsilon_\gamma|\varepsilon$ on $B_-$, with $N_\gamma(n) \in \mathbb{N}$, and $N_\gamma(n)$ is $|\epsilon_\gamma|n$ plus an integer. By 5.14, $\Gamma(-N+\delta) = (-1)^N\delta^{-1}\Gamma(1+\delta)\Gamma(1-\delta)/\Gamma(1+N-\delta)$ for $N \in \mathbb{N}$ and $|\delta| < 1$. Put $s_\gamma = 1$ on $A_- \cup B_+$ and $s_\gamma = -1$ on $A_+ \cup B_-$. Then

$$W_k(-n+\varepsilon) = (-1)^{\sum_{A_+\cup B_+} N_\gamma} \prod_{B_+} \epsilon_\gamma \cdot \varepsilon^{-w} \prod_\gamma \Gamma\big(1+N_\gamma - |\epsilon_\gamma|\varepsilon\big)^{s_\gamma} \frac{\prod_{A_+}\Gamma(1+\varepsilon)\Gamma(1-\varepsilon)}{\prod_{B_+}\Gamma(1+\epsilon_\gamma\varepsilon)\Gamma(1-\epsilon_\gamma\varepsilon)}.$$

At $\varepsilon = 0$ the factor after $\varepsilon^{-w}$ is $\prod_\gamma (N_\gamma!)^{s_\gamma}$, which gives $C_n$ as stated, nonzero and rational. By 5.13 at the integer $b = N \ge 0$, where the lattice input $\mathbb{I}_{N+\mathbb{Z}}$ is $\mathbb{I}_\mathbb{Z}$, $\log\Gamma(1+N-\delta) = \log N! + \sum_{i\ge1}\frac{\delta^i}{i}(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((N,\infty))$, the index-one value regularized, and at $N = 0$ with $\pm\delta$, $\log\Gamma(1+\delta)+\log\Gamma(1-\delta) = \sum_{j\ge1}\frac{\delta^{2j}}{j}(\mathsf{T}_{2j;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$. So $W_k(-n+\varepsilon) = C_n\varepsilon^{-w}\exp\big(\sum_i \Lambda_{n,i}\varepsilon^i\big)$ with

$$\Lambda_{n,i} = \frac{1}{i}\sum_\gamma s_\gamma|\epsilon_\gamma|^i\,(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((N_\gamma(n),\infty)) + [i \text{ even}]\,\frac{2}{i}\Big(|A_+| - \sum_{B_+}\epsilon_\gamma^i\Big)(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty)).$$

The cut relation (5.3, with the regularized values at index one) gives $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((N,\infty)) = (\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty)) - (\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N])$. The balance of the slopes gives

$$\sum_\gamma s_\gamma|\epsilon_\gamma| = \#A_- - \#A_+ + \sum_{B_+}\epsilon_\gamma - \sum_{B_-}|\epsilon_\gamma| = -\sum_A \epsilon_\alpha + \sum_B \epsilon_\beta = 0,$$

so at $i = 1$ the regularized value drops out and $\Lambda_{n,1} = -\sum_\gamma s_\gamma|\epsilon_\gamma|\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N_\gamma(n)])$, while for $i \ge 2$, $\Lambda_{n,i}$ is $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,\infty))$ times $\frac{1}{i}\sum_\gamma s_\gamma|\epsilon_\gamma|^i + [i \text{ even}]\frac{2}{i}(|A_+| - \sum_{B_+}\epsilon_\gamma^i)$, less $\frac{1}{i}\sum_\gamma s_\gamma|\epsilon_\gamma|^i(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_\mathbb{Z})((0,N_\gamma(n)])$. These coefficients depend on the slopes alone.

For $n, n+1 > n_1$, $N_\gamma(n+1) = N_\gamma(n) + |\epsilon_\gamma|$, so $N_\gamma(n+1)!/N_\gamma(n)! = (N_\gamma(n)+1)_{|\epsilon_\gamma|}$ and

$$\rho(n) = \frac{C_{n+1}}{C_n} = (-1)^{|A_+|+\sum_{B_+}\epsilon_\gamma}\prod_\gamma \big(N_\gamma(n)+1\big)_{|\epsilon_\gamma|}^{s_\gamma},$$

a rational function with integer coefficients; with $\rho = \rho_1/\rho_2$ in polynomials, $\rho_2(n)C_{n+1} - \rho_1(n)C_n = 0$ is a recurrence of order one with both coefficients nonzero, so the tier is at most one (38.11). By Stirling's formula $\log N! = N\log N - N + O(\log(N+2))$, and with $N_\gamma(n) = |\epsilon_\gamma|n + O(1)$,

$$\log|C_n| = \sum_\gamma s_\gamma\big(|\epsilon_\gamma|n\log n + |\epsilon_\gamma|n\log|\epsilon_\gamma| - |\epsilon_\gamma|n\big) + O(\log n) = n\log r_W + O(\log n),$$

the terms in $n\log n$ and in $n$ cancelling by $\sum_\gamma s_\gamma|\epsilon_\gamma| = 0$, and $|\epsilon_\gamma|\log|\epsilon_\gamma| = 0$ on $A$. So $|C_n|^{1/n} \to r_W$.

The residue at $-n$ is the coefficient of $\varepsilon^{-1}$, $C_n[\varepsilon^{w-1}]\exp(\sum_i \Lambda_{n,i}\varepsilon^i)$, which is 0 when $w \le 0$, the exponential having no negative powers. For $w \ge 1$ it is $C_n\sum\prod_i \Lambda_{n,i}^{m_i}/m_i!$ over $\sum_i im_i = w-1$,

a polynomial with rational coefficients in $\Lambda_{n,1}, \dots, \Lambda_{n,w-1}$; inserting the combinations above gives the stated polynomial, each monomial of total index $w-1$.

(iii) Moving the line to the left, the arguments of the Gammas of $A_+$ fall and cross their poles, those of $A_-$ rise and cross none, and the lower Gammas contribute entire factors; under the hypothesis of the closing, 4.5, with the Laurent coefficients at coincident poles, gives the integral as the sum of the residues at the integers $-n < c$. At each of the finitely many integers $-n < c$ with $n \le n_1$ the residue is a Laurent coefficient of finitely many Gammas at integer arguments: by 5.14 and 5.13 each is a nonzero rational number times the exponential of a series whose coefficient of $\varepsilon^i$ is a rational combination of $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and of heads at integer cuts, which are rational; the coefficient of $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ is $-\epsilon_\alpha\varepsilon$ for an upper Gamma and $\epsilon_\beta\varepsilon$ for a lower one, whatever the sign of the argument, and these cancel by the balance. So these residues are rational combinations of products of $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $i \ge 2$. For $n > n_1$, expand the polynomial of (ii) into monomials: each is a rational number times a product of Trailmoments $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ times $\psi(n)$ for one of the finitely many products $\psi$ of the statement, the heads of the monomial having total index at most $w-1$. Since $|(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,N])| \le 1+\log(N+1)$ for $i \ge 1$ and $N_\gamma(n)$ grows linearly, $|\psi(n)| \le |C_n|\,(c'\log n)^w$ for large $n$, of root rate $r_W < 1$, so $\sum_{n>n_1}|\psi(n)| < \infty$ (1.1(viii)); the sum of the residues, a finite sum of absolutely convergent series, is the stated combination.

Each $\psi$ is live: $C_n$ is a product of factorials of the moving data, a stacked Gamma, and the heads are Trailmoments at the moving cut. For the recurrence put $H_l(n) = (\mathsf{T}_{i_l;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,N_{\gamma l}(n)])$; then $H_l(n+1) = H_l(n) + \eta_l(n)$ with $\eta_l(n) = \sum_{N_{\gamma l}(n)<j\le N_{\gamma l}(n)+|\epsilon_{\gamma l}|} j^{-i_l}$, a rational function of $n$ with rational coefficients. The vector $X(n) = \big(C_n \prod_{l\in T} H_l(n)\big)_{T\subseteq\{1,\dots,q\}}$ satisfies

$$X(n+1)_T = \rho(n) \sum_{U\subseteq T} \Big( \prod_{l\in T\setminus U} \eta_l(n)\Big) X(n)_U,$$

a matrix $\rho(n)$ times a unitriangular one in the order of inclusion, of determinant $\rho(n)^{2^q} \ne 0$. Lemma S30 with $\ell$ the coordinate of $T = \{1,\dots,q\}$ gives the recurrence of order at most $2^q$ with nonzero extreme coefficients.

(iv) At the steps $k$ and $k'$, $N_\gamma(n)$ differs by $-(k-k')u_\gamma$ on $A_+ \cup B_+$ and by $(k-k')u_\gamma$ on $A_- \cup B_-$, a constant $t_\gamma$. So the sign of $C_n$ changes by a constant, and $N_\gamma(n)!$ at $k$ is $N_\gamma(n)!$ at $k'$ times $(N_\gamma(n)+1)_{t_\gamma}$ or divided by $(N_\gamma(n)+t_\gamma+1)_{-t_\gamma}$, the data at $k'$; the quotient of the two $C_n$ is a rational function $R_0(n)$ with rational coefficients. Likewise $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,N+t])$ is $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,N])$ plus or minus the finite sum of $j^{-i}$ between $N$ and $N+t$, a rational function $\eta'(n)$. Hence, with $H'_l$ the heads at $k'$,

$$\psi^{(k)}(n) = R_0(n)\, C_n^{(k')} \prod_l \big(H'_l(n) + \eta'_l(n)\big) = \sum_{T\subseteq\{1,\dots,q\}} R_0(n) \prod_{l\notin T} \eta'_l(n)\; C_n^{(k')} \prod_{l\in T} H'_l(n),$$

and each $C_n^{(k')} \prod_{l\in T} H'_l(n)$ is an input of (iii) at the step $k'$, its heads a sub-multiset of those of $\psi$. Inserting this into (iii) at the step $k$, and adding the finitely many terms between the two bounds $n_1$, which are rational combinations of products of $(\mathsf{T}_{i;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $i \ge 2$, by the argument for $n \le n_1$ in (iii), gives the last statement; the sums $\sum_n R(n)\psi^{(k')}(n)$ converge absolutely, $\psi^{(k')}$ having root rate $r_W < 1$.

Closing to the right of $W_k(a)$ is closing to the left of $a \mapsto W_k(-a)$, the word with every slope negated, which is again balanced. □

**Proposition 43.2 (Paired words).** *As stated.*

*Proof.* Write $N_\gamma(n) = |\epsilon_\gamma| n + e_\gamma$ with $e_\gamma \in \mathbb{Z}$. By the proof of 43.1(ii),

$$\rho(n) = \frac{C_{n+1}}{C_n} = (-1)^{|A_+| + \sum_{B_+} \epsilon_\gamma} \, r_W \prod_\gamma \prod_{t=1}^{|\epsilon_\gamma|} \big(n + \theta_{\gamma,t}\big)^{s_\gamma}, \qquad \theta_{\gamma,t} = \frac{e_\gamma + t}{|\epsilon_\gamma|},$$

using $(N+1)_v = v^v \prod_{t=1}^{v} (n + (e+t)/v)$ for $N = vn + e$ and $\prod_\gamma |\epsilon_\gamma|^{|\epsilon_\gamma| s_\gamma} = r_W$. For a fixed $\gamma$ with $|\epsilon_\gamma| = v$ the $\theta_{\gamma,t}$ are $v$ consecutive multiples of $\frac{1}{v}$, so modulo $\mathbb{Z}$ they meet each class $j/v$ once. For $v \geq 1$ put $m_v = \sum_{|\epsilon_\gamma| = v} s_\gamma$, so that $m_v = \#\{\beta : \epsilon_\beta = v\} - \#\{\beta : \epsilon_\beta = -v\}$ for $v \geq 2$, and $\sum_v v\, m_v = \sum_\gamma s_\gamma |\epsilon_\gamma| = 0$ by the balance. The signed number of the $-\theta_{\gamma,t}$ in a class of $\mathbb{Q}/\mathbb{Z}$ of exact order $v'$ is $\sum_{v' | v} m_v$.

(b)⇒(a). If $C_n = (-1)^{\varsigma n} R(n)$, then $\rho(n) = (-1)^\varsigma R(n+1)/R(n)$ as rational functions. Writing $R = c \prod_x (n-x)^{\mu_x}$, $R(n+1)/R(n) = \prod_x \big((n-x+1)/(n-x)\big)^{\mu_x}$, whose zeros less poles, counted with multiplicity, add to zero in every class of $\mathbb{C}/\mathbb{Z}$. Comparing with the zeros and poles $-\theta_{\gamma,t}$ of $\rho$, of multiplicities $s_\gamma$, $\sum_{v' | v} m_v = 0$ for every $v' \geq 2$, and descending from the largest slope, $m_v = 0$ for every $v \geq 2$: that is (a). It does not involve $k$.

(a)⇒(b). Under (a), $m_v = 0$ for $v \geq 2$, hence $m_1 = 0$ by $\sum_v v m_v = 0$, and $r_W = \prod_{v \geq 2} v^{v m_v} = 1$. Every class of $\mathbb{Q}/\mathbb{Z}$ then holds as many $\theta_{\gamma,t}$ with $s_\gamma = 1$ as with $s_\gamma = -1$; pair them within classes, $\theta = \theta' + j$ with $j \in \mathbb{Z}$, and put $R_{\theta,\theta'}(n) = \Gamma(n+\theta)/\Gamma(n+\theta')$, which is $(n+\theta')_j$ or its reciprocal by 1.3(iii), a rational function with $R_{\theta,\theta'}(n+1)/R_{\theta,\theta'}(n) = (n+\theta)/(n+\theta')$. With $\tilde{R}$ the product of the $R_{\theta,\theta'}$, $\rho(n) = (-1)^\varsigma \tilde{R}(n+1)/\tilde{R}(n)$, $\varsigma \equiv |A_+| + \sum_{B_+} \epsilon_\gamma$. For $n > n_1$ the numbers $n + \theta_{\gamma,t}$ are positive, so $\tilde{R}$ has no zero or pole there, and $C_n = (-1)^{\varsigma n} R(n)$ with $R(n) = (-1)^{\varsigma(n_1+1)} C_{n_1+1} \tilde{R}(n)/\tilde{R}(n_1+1)$. As $|A_+| + \sum_{B_+} \epsilon_\gamma = p + 2\#\{\epsilon_\beta = 1\} + \sum_{B^{\geq 2}} \epsilon_\beta$, $\varsigma \equiv p + \sum_{B^{\geq 2}} \epsilon_\beta$.

The factorization. By (i) of 43.1, $W_k = W_0 g_k$, so it suffices to factor $W_0$. For integers $b, b'$, $\Gamma(\pm a + b)/\Gamma(\pm a + b')$ and $\Gamma(\pm va + b)/\Gamma(\pm va)$ are rational functions of $a$ with rational coefficients (1.3(iii)). The balance and (a) give $|A_+| - |A_-| = \#\{\epsilon_\beta = 1\} - \#\{\epsilon_\beta = -1\}$, so $p = |A_-| - \#\{\epsilon_\beta = -1\}$ as well. Matching the Gammas of $A_+$ with those of slope 1 below, and the Gammas of $A_-$ with those of slope $-1$, leaves a rational function times $\Gamma(a)^p \Gamma(1-a)^p = \pi^p / \sin(\pi a)^p$ (1.3(v)), read as $\sin(\pi a)^{|p|}/\pi^{|p|}$ when $p < 0$; each pair $\beta, \beta'$ with $\epsilon_\beta = v = -\epsilon_{\beta'}$, $v \geq 2$, leaves a rational function times $1/(\Gamma(va)\Gamma(1-va)) = \sin(\pi va)/\pi$. There are $|B^{\geq 2}|$ pairs, and $w = |A_+| - |B_+| = p - |B^{\geq 2}|$, which gives the display, the sign and $R_k$ collecting the rational factors.

The residue. At $a = -n + \varepsilon$, $\sin(\pi a) = (-1)^n \sin(\pi\varepsilon)$ and $\sin(\pi va) = (-1)^{vn} \sin(\pi v \varepsilon)$, so the trigonometric factor is $(-1)^{n(\sum_{B^{\geq 2}} \epsilon_\beta - p)} \pi^w \prod_\beta \sin(\pi \epsilon_\beta \varepsilon) / \sin(\pi\varepsilon)^p$. Since $\sin(\pi x)/(\pi x)$ is an even power series in $\pi x$ with rational coefficients and constant term 1, this is $(-1)^{\varsigma n} \varepsilon^{-w} \sum_{l \geq 0} \tau_l \pi^{2l} \varepsilon^{2l}$ with rational $\tau_l$ depending on the slopes and $\tau_0 = \prod_{B^{\geq 2}} \epsilon_\beta$. For $n > n_1$ the pole of $W_k$ at $-n$ has the order $w$ exactly, $C_n \neq 0$, and the trigonometric factor has a pole of order $w$ there, so $R_k$ is holomorphic at $-n$; the coefficient of $\varepsilon^{-1}$ is the displayed sum.

The closing. The finitely many residues at $-n < c$ with $n \leq n_1$ are, by the same factorization, combinations of the $\tau_l \pi^{2l}$ with Laurent coefficients of $R_k$, so they lie in $\mathbb{Q}[\pi^2]$. For $n > n_1$ the residue is $\pm(-1)^{\varsigma n} r(n)$ with $r(n) = \sum_l \tau_l \pi^{2l} R_k^{(w-1-2l)}(-n)/(w-1-2l)!$, a rational function of $n$ whose partial fractions have coefficients in $\mathbb{Q}[\pi^2]$, the poles of $R_k$ being rational. The closing converges; when $\varsigma = 0$ its terms are $r(n)$, so $r(n) = O(n^{-2})$, and when $\varsigma = 1$, $r(n) \to 0$, so $r(n_1 + 1 + 2m) - r(n_1 + 2 + 2m) = O(m^{-2})$, and summing consecutive pairs does not change the sum. 38.32 over $\Bbbk = \mathbb{Q}(\pi^2)$, whose identities are linear in the coefficients of the partial fractions (its proof applies 37.10 to them), writes the sum over 1, the Trailmoments of the lattice inputs on the cosets of the poles at their cuts, and the differences of their index-one values, with coefficients in $\mathbb{Q}[\pi^2]$. Finally $(\mathsf{T}_{2;(1)}[x]\, \mathbb{I}_\mathbb{Z})((0,\infty)) = \pi^2/6$ by 5.17(iv) at $j = 1$. □

**Theorem 43.3 (The frame bounds the clearing from below).** *As stated.*

*Proof.* (i) As $p_h(k) \neq 0$ for $k \geq k_0$, a solution is fixed by its values at $k_0, \dots, k_0 + h - 1$, so independent solutions have independent initial vectors and $C_{k_0} \neq 0$; by 38.11, $C_{k+1} = (-1)^h \frac{p_0(k)}{p_h(k)} C_k$ with $p_0(k) \neq 0$, so $C_k \neq 0$ for every $k \geq k_0$. Then $D_k^h C_k = \det \left[ D_k x_{k+l}^{(i)} \right]$ is a nonzero integer and $D_k^h |C_k| \geq 1$. By 38.12, $\log |C_k| = k \log |c| + (d_0 - d_h)(k \log k - k) + O(\log k)$, and $h \log D_k \geq - \log |C_k|$ gives the inequality. When $d_0 = d_h$ the term in $k \log k$ is absent, and the same proposition identifies $|c|$.

(ii) The coefficients of $v_l$ and of 1 in $\sum_j p_j(k) x_{k+j} = 0$ are $\sum_j p_j(k) A_{k+j,l}$ and $- \sum_j p_j(k) B_{k+j}$, rational since $p_j \in \mathbb{Q}[k]$ and $k$ is an integer; as $1, v_1, \dots, v_m$ are independent over $\mathbb{Q}$, all of them vanish. A $D_k$ that clears the rows of $x_k, \dots, x_{k+h-1}$ makes the values at $k, \dots, k + h - 1$ of all $m + 1$ sequences integers, in particular those of $h$ independent ones, and (i) applies to them. □

**Definition 43.4 (The merit of rows along a ray).** As stated.

*Proof.* $f(\alpha^\star)$ is positive and finite (38.14(i)), and $D_k \geq 1$ gives $\limsup_k \frac{1}{k} \log D_k \geq 0$, so the merit lies in $[-\infty, - \log f(\alpha^\star)]$. □

**Theorem 43.5 (The merit decides the gap along a ray).** *As stated.*

*Proof.* (i) By 38.14(i), $0 < f \leq f(\alpha^\star)$ on the orthant and $\langle 1 \rangle_k = \int_{y_{e_0}=0} f^k \omega_0$ converges and is positive for $k \geq k_1$; the origin is real, so $\omega_0 / dy = \alpha^{\mathbf{m}_0} \prod_j K_j^{-s_{0j}} \Psi_j^{-t_{0j}}$ is positive on the orthant. Hence $\langle 1 \rangle_k = \int f^{k-k_1} f^{k_1} \omega_0 \leq f(\alpha^\star)^{k-k_1} \langle 1 \rangle_{k_1}$, and $D_k \geq 1$.

(ii) If the merit is positive, choose $\eta > 0$ below it; then $D_k \leq f(\alpha^\star)^{-k} e^{-k\eta}$ for all large $k$, and by (i) $0 < D_k \langle 1 \rangle_k \leq e^{-k\eta} f(\alpha^\star)^{-k_1} \langle 1 \rangle_{k_1} \to 0$. With $x_k = \langle 1 \rangle_k$, 37.2 gives an irrational $v_l$, and for $m = 1$, $A_{k,1} \neq 0$ for all large $k$; then $v_1$ is the irrational one.

(iii) $\frac{1}{k} \log D_k \leq \frac{\log C}{k} + \log \gamma + \frac{w}{k} \log d_{\lfloor ak+b \rfloor}$. For $a > 0$, $N = \lfloor ak + b \rfloor \to \infty$ with $N/k \to a$, and $\limsup_N \frac{1}{N} \log d_N \leq c_{\mathrm{lcm}}$ (37.1), so $\limsup_k \frac{1}{k} \log D_k \leq \log \gamma + a w c_{\mathrm{lcm}}$; for $a = 0$ the last term tends to 0. The merit is then at least $- \log f(\alpha^\star) - \log \gamma - a w c_{\mathrm{lcm}}$, positive when $\gamma e^{a w c_{\mathrm{lcm}}} f(\alpha^\star) < 1$.

(iv) By 43.3(ii) and (i), $\liminf_k \frac{1}{k} \log D_k \geq \frac{1}{h} \log \frac{1}{|c|}$, so the lim sup is at least as large, and the merit is at most $- \log f(\alpha^\star) + \frac{1}{h} \log |c|$. □

**Proposition 43.6 (The low end of the law).** *As stated.*

*Proof.* (i) The law says that $\sum_i p_i(k) f^i$ lies in the relation space $\mathcal{L}$ (38.4), a finite sum $\sum r_{e,l}(k) (c_e^{(0)} + k c_e^{(1)} + \partial_{y_e}) g_{e,l}$ with $r_{e,l} \in \mathbb{C}(k)$. Integration by parts against $f^k dy$, for complex $k$ of large real part where the integrals converge absolutely (38.3), gives $\sum_i p_i(k) \langle 1 \rangle_{k+i} = 0$ there off the poles of the $r_{e,l}$, hence identically for the meromorphic continuation of 39.34. At an integer $k \geq 1$ every $\langle 1 \rangle_{k+i}$ is a convergent integral, the walls $(k + i)\kappa_{1,S}$ being positive, and the law holds there. Near $k = 0$ the terms $i \geq 1$ are holomorphic, and by 39.34 at $j = 0$, with $j(1) = 0$ and $\mathsf{C}_0(1) = \mathrm{CT}(1) = 1$, $\langle 1 \rangle_k = V_1 k^{-(n-1)} + O(k^{-(n-2)})$ with $V_1 > 0$. So $p_0(k) \langle 1 \rangle_k$ is holomorphic at 0, which forces $p_0$ to vanish there to the order $n - 1$ at least, and its value at 0 is $\varpi V_1$. The law at $k = 0$ is the low-end relation. The walls of the integral direction are integers, $\kappa_{1,S} = m_{1,S} - \sum_j (s_{1j} d_j(S) + t_{1j} d_j^*(S))$ (8.1), so $V_1$ is rational.

(ii) $c$ and $r$ exist and are rational because $p_2(k) \neq 0$ for $k \geq 1$ and $p_0, p_1, p_2, \varpi V_1$ are rational. The low-end coefficient of $c$ is $-p_1(0) - p_2(0)\big( - p_1(0)/p_2(0) \big) = 0$ and that of $r$ is $-p_2(0)\big( - \varpi V_1 / p_2(0) \big) = \varpi V_1$. The sequence $x_k - x_1 c_k - r_k$ solves the law on $k \geq 1$; it vanishes at $k = 1$,

and at $k=2$ it is $\big(p_2(0)x_2+p_1(0)x_1+\varpi V_1\big)/p_2(0)=0$ by the low-end relation; so it vanishes identically. $\square$

**Definition 43.7 (Reflexive laws).** As stated.

*Proof.* Let $c_i$ be the coefficient of $k^d$ in $p_i$, so the symbol is $\sigma(\varrho)=\sum_i c_i\varrho^i$ with $c_0c_h\neq 0$ (38.7). The coefficient of $k^d$ in $p_{h-i}(-k-h+1)$ is $(-1)^d c_{h-i}$, so $(-1)^d c_{h-i}=\gamma\lambda^i c_i$, and

$$\varrho^h\sigma(\lambda/\varrho)=\sum_i c_i\lambda^i\varrho^{h-i}=(-1)^d\gamma^{-1}\sum_i c_{h-i}\varrho^{h-i}=(-1)^d\gamma^{-1}\sigma(\varrho).$$

As $\sigma(0)=c_0\neq 0$, $\varrho\mapsto\lambda/\varrho$ maps the roots of $\sigma$ to roots, with multiplicities. At $h=2$ the identities at $i=0$ and $i=2$ give $c_0/c_2=(-1)^d\gamma^{-1}$ and $c_0/c_2=(-1)^d\gamma\lambda^2$, so $\gamma^2\lambda^2=1$, $\gamma=\pm\lambda^{-1}$, and the product of the roots, $c_0/c_2$, is $\pm\lambda$. $\square$

**Theorem 43.8 (The closings read the coordinate of a reflexive law).** *As stated.*

*Proof.* (i) By 39.36, $\sum_{i=0}^{\min(h,j)}p_i(-j)\mathsf{C}_{j-i}(1)=0$ for every $j\geq 0$. For $k\geq 1$ put $j=k+h-1\geq h$. Reflexivity, $p_i(k)=\gamma^{-1}\lambda^{-i}p_{h-i}(-k-h+1)$, gives

$$\sum_{i=0}^{h}p_i(k)\,\lambda^{k+i-1}\mathsf{C}_{k+i-1}(1)=\gamma^{-1}\lambda^{k-1}\sum_{i=0}^{h}p_{h-i}(-j)\,\mathsf{C}_{j-(h-i)}(1)=0.$$

For the low-end coefficient, reflexivity at $k=0$ gives $p_i(0)=\gamma^{-1}\lambda^{-i}p_{h-i}(-h+1)$, so

$$\sum_{i=1}^{h}p_i(0)\,\lambda^{i-1}\mathsf{C}_{i-1}(1)=\gamma^{-1}\lambda^{-1}\sum_{i'=0}^{h-1}p_{i'}(-(h-1))\,\mathsf{C}_{h-1-i'}(1)=0,$$

by 39.36 at $j=h-1$.

(ii) At $h=2$, $y_1=\mathsf{C}_0(1)=1=c_1$, and the vanishing of the low-end coefficient, $p_1(0)+p_2(0)y_2=0$, gives $y_2=-p_1(0)/p_2(0)=c_2$; both solve the law and $p_2(k)\neq 0$ for $k\geq 1$, so $y=c$, and 43.6(ii) gives the display. With $\langle 1\rangle_1=A_1v-B_1$ and $r_k$ rational, $\langle 1\rangle_k=A_1\lambda^{k-1}\mathsf{C}_{k-1}(1)\,v-\big(B_1\lambda^{k-1}\mathsf{C}_{k-1}(1)-r_k\big)$.

(iii) With integer weights, $f^{-1}=\alpha^{-\mathbf{m}_1}\prod_j K_j^{s_{1j}}\Psi_j^{t_{1j}}$ is a Laurent polynomial with integer coefficients, and so is each power; at origin 0, $\omega_0/dy=1$, and $\mathsf{C}_j(1)=\mathrm{CT}(f^{-j})\in\mathbb{Z}$. The coefficient is $a^{k-1}\mathsf{C}_{k-1}(1)/b^{k-1}$; as $\gcd(a,b)=1$, its reduced denominator is $b^{k-1}/\gcd\big(b^{k-1},a^{k-1}\mathsf{C}_{k-1}(1)\big)=b^{k-1}/\gcd\big(b^{k-1},\mathsf{C}_{k-1}(1)\big)$. $\square$

**Corollary 43.9 (The merit of a reflexive ray of order two).** *As stated.*

*Proof.* The ray has origin 0, real, and $\kappa_{1,S}>0$, so Definition 43.4 applies with $m=1$, $v_1=\langle 1\rangle_1$, $A_{k,1}=\lambda^{k-1}\mathsf{C}_{k-1}(1)$ and $B_k=-r_k$. A clearing $D_k$ makes $D_kA_{k,1}$ an integer, so by 43.8(iii) the integer $b^{k-1}/g_k$, $g_k=\gcd(b^{k-1},\mathsf{C}_{k-1}(1))$, divides $D_k$, and $\frac{1}{k}\log D_k\geq\frac{k-1}{k}\log b-\frac{1}{k}\log g_k$. Taking the upper limit, $\limsup_k\frac{1}{k}\log D_k\geq\log b-\liminf_k\frac{1}{k}\log g_k$, which is the bound. $\square$

**Proposition 43.10 (The hinge closing as a row over the room).** *As stated.*

*Proof.* (i) By 19.21(i) the closed value is the absolutely convergent sum of the residues, and grouping them by $n$ gives $\sum_n \langle \Phi_n, \mathcal{X}_n \rangle$. Then $\langle \Phi_n, \mathcal{X}_n \rangle = \sum_i \chi_i \langle \Phi_n, \mathbf{c}_i(n) \rangle$, and a finite sum of absolutely convergent series may be summed term by term.

(ii) By 19.21(ii) the two lattices of the level $d$ meet exactly when $2\delta^d \in \mathbb{Z}$, so otherwise every pole is simple and the residue at $\sigma^d = \pm\delta^d - n$ is a coefficient times the room period there. The sequence $\sum_i x_i^{\pm} c_i^{\pm}(n)$ solves the recurrence and agrees with $x^{\pm}$ at $0, \dots, h_{\pm} - 1$; as $p_{h_{\pm}}^{\pm}(n) \neq 0$ for $n \geq 0$, the two agree for every $n$. The $c_i^{\pm}$ are computed from rational initial values by the recurrence, so they are rational when the coefficients are. So $\mathcal{X}_n = \sum_i x_i^{+} \big(c_i^{+}(n), 0\big) + \sum_i x_i^{-} \big(0, c_i^{-}(n)\big)$, and (i) applies.

(iii) For $2\delta^d \in \mathbb{Z}$ the lattices $\delta^d - \mathbb{N}$ and $-\delta^d - \mathbb{N}$ are both contained in $|\delta^d| - \mathbb{N}$, and by 19.21(ii) the residue at a common point carries the room period and its derivative in $\sigma^d$, at a simple point the room period alone. The room period converges at the points of the lattice (19.21(i)), and near them it is an absolutely convergent integral over the simplex of a function holomorphic in the level shares (8.7), so it is holomorphic in the even share there. The state $U_n(\sigma) = (x_n(\sigma), \dots, x_{n+h-1}(\sigma))$ satisfies $U_{n+1}(\sigma) = \mathbf{A}(n, \sigma) U_n(\sigma)$ with the companion matrix of the recurrence, whose entries $-p_i(n, \sigma)/p_h(n, \sigma)$ are regular at $\sigma = \sigma_0$ for $n \geq 0$. By 37.17 with $\varepsilon = \sigma - \sigma_0$ and $J = 1$, $U_{n+1,0} = \mathbf{A}_0(n) U_{n,0}$ and $U_{n+1,1} = \mathbf{A}_0(n) U_{n,1} + \mathbf{A}_1(n) U_{n,0}$, so $(U_{n,0}, U_{n,1})$ is a product of block-triangular matrices, which depend on the recurrence alone, applied to $(U_{0,0}, U_{0,1})$, the $2h$ initial data. This writes $\mathcal{X}_n$ as in (i), which applies when the resulting sums converge absolutely.

With the bunch and the room exchanged, 19.21(i) closes on the lattices of the level $r$ with the opposite sign, and the same argument applies. □

**Definition 43.11 (Critical primes).** As stated.

*Proof.* A native gauge lies in $X_F$, so every letter is nonzero there, and $f(\alpha_c) \in \Bbbk^{\times}$ since the weights and the coordinates lie in $\Bbbk$; its valuation at $\mathfrak{p}$ is defined. □

**Theorem 43.12 (The letters degenerate at a critical prime).** *As stated.*

*Proof.* Let $\mathcal{O}$ be the local ring of $\Bbbk$ at $\mathfrak{p}$, a discrete valuation ring, with a uniformizer $\varpi_{\mathfrak{p}}$, and $v = v_{\mathfrak{p}}$. The letters have coefficients in $\mathcal{O}$ and the coordinates of $\alpha_c$ are units, so every $L(\alpha_c)$ and every $\partial_e L(\alpha_c)$ lies in $\mathcal{O}$, and $L(\alpha_c) \neq 0$. Since $\alpha_c^{\mathbf{m}_1}$ is a unit, $v(f(\alpha_c)) = -\sum_L w_L v(L(\alpha_c)) < 0$, so the largest valuation $v_{\max}$ is positive, and the letters of $S$ vanish at $\bar{\alpha}_c$. The native gauge has $c_e^{(1)} = \partial_{y_e} \log f = 0$ for $e \neq e_0$ (38.4, 38.2(ii)), that is $m_{1,e} = \sum_L w_L \, \alpha_e \partial_e L / L$ at $\alpha_c$; by Euler's identity for the homogeneous letters, $\sum_e \alpha_e \partial_e L = (\deg L) L$, and the balance $m_{1,\mathcal{E}} = \sum_L w_L \deg L$ of a direction ($f$ has degree 0), the same holds at $e = e_0$. Write $L(\alpha_c) = \varpi_{\mathfrak{p}}^{v_L} u'_L$ with $u'_L \in \mathcal{O}^{\times}$ and multiply the equation of the mark $e$ by $\varpi_{\mathfrak{p}}^{v_{\max}}$:

$$\sum_L w_L \, \alpha_{c,e} \, \partial_e L(\alpha_c) \, \varpi_{\mathfrak{p}}^{v_{\max} - v_L} u_L'^{-1} = m_{1,e} \, \varpi_{\mathfrak{p}}^{v_{\max}}.$$

Every term lies in $\mathcal{O}$; modulo $\mathfrak{p}$ the right side and the terms with $v_L < v_{\max}$ vanish, and reduction commutes with $\partial_e$. Dividing by the unit $\bar{\alpha}_{c,e}$ gives (i) with $u_L = u_L'^{-1}$. Under the hypothesis of (ii) every coefficient $\overline{w_L u_L}$ is nonzero, so the gradients of the reduced letters of $S$ at $\bar{\alpha}_c$ satisfy a linear relation with all coefficients nonzero: for one letter its gradient vanishes at a point of its zero set, which is singular there, and for several the gradients are linearly dependent, whereas smooth hypersurfaces meeting transversally have independent gradients. □

**Proposition 43.13 (Critical primes and the symbol).** *As stated.*

*Proof.* (i) $c_h^{h-1}\sigma(\lambda) = \sum_j c_j c_h^{h-1-j}(c_h\lambda)^j$ is a monic polynomial with integer coefficients in $c_h\lambda$, which vanishes at a root.

(ii) If $p \nmid c_h$, $c_h$ is a unit at every prime $\mathfrak{P}$ of $K$ above $p$ and $\lambda = c_h\lambda/c_h$ has $v_{\mathfrak{P}}(\lambda) \geq 0$ by (i). Conversely let $p \mid c_h$ and let $\mathfrak{P}$ be a prime above $p$. If every root had $v_{\mathfrak{P}} \geq 0$, the elementary symmetric functions of the roots, $\pm c_{h-j}/c_h$, would be $\mathfrak{P}$-integral, so $v_{\mathfrak{P}}(c_{h-j}) \geq v_{\mathfrak{P}}(c_h) > 0$ for every $j$, and $p$ would divide every coefficient, against their coprimality. So some root has negative valuation at $\mathfrak{P}$.

(iii) Under the hypothesis, a root $\lambda = f(\alpha_c)$ with $v_{\mathfrak{P}}(\lambda) < 0$ at a prime $\mathfrak{P}$ above $p$ makes $\mathfrak{P}$ a critical prime of $\alpha_c$ (43.11 with $\Bbbk = K$), and conversely; (ii) gives the equivalence. When $p_0$ and $p_h$ have the largest degree $d$, $c_0$ and $c_h$ are their leading coefficients (38.7), and $c_0/c_h$ is the quotient $c$ of 38.11. □

## 50.43 Chapter 44: relations and separation

**Proposition 44.4 (A finite matrix of mark relations).** *As stated, for several letters* $D_1, \dots, D_h$.

*Proof.* In the coordinate $x_i$, with the other coordinates as room and the room density $\prod_{l\neq i} x_l^{a_l-1}(1-x_l)^{b_l-1}$, 35.10(a) applies to the letters $D_k$, positive on the closed box, at the exponents $p_k+1$, and to the polynomial $AD$. With $\partial_i D = \sum_k \partial_i D_k \prod_{k'\neq k} D_{k'}$ its expression is

$$x_i(1-x_i)(D\,\partial_i A + A\,\partial_i D) + \big(a_i - (a_i+b_i)x_i\big)AD - x_i(1-x_i)A\sum_k (p_k+1)\,\partial_i D_k \prod_{k'\neq k} D_{k'}$$
$$= D\big[x_i(1-x_i)\,\partial_i A + \big(a_i-(a_i+b_i)x_i\big)A\big] - x_i(1-x_i)\,A\sum_k p_k\,\partial_i D_k \prod_{k'\neq k} D_{k'} = \mathrm{mr}_i(A),$$

and the read is the stated integral. At $A = x^\beta$, $x_i(1-x_i)\partial_i x^\beta + (a_i-(a_i+b_i)x_i)x^\beta = \big((\beta_i + a_i) - (\beta_i + a_i + b_i)x_i\big)x^\beta$, and $x_i(1-x_i)x^\beta \sum_k p_k \partial_i D_k \prod_{k'\neq k} D_{k'} = (1-x_i)x^\beta \sum_\gamma D_\gamma^{(i)} x^\gamma$; expanding $D = \sum_\gamma D_\gamma x^\gamma$ gives the formula. Its coefficients are rational in $(a_i, b_i)$ over the rational coefficients of the $D_k$ and the $p_k$, so for a degree bound the vectors $\mathrm{mr}_i(x^\beta)$ form a finite matrix, rational at rational $a_i, b_i$; by linearity a numerator $\sum_i \mathrm{mr}_i(A_i)$ has the read 0. With one letter, $D_\gamma^{(i)} = p\gamma_i D_\gamma$, the formula of the reference. □

**Proposition 44.11 (The critical polynomial block).** *As stated, at every order* $k \geq 1$.

*Proof.* For $R(u) = P(2u)$, $R(\frac{1}{2}) = P(1)$ and $Q_P$ is the polynomial $(R(u) - R(\frac{1}{2}))/(1-2u)$, so 36.4 at the order $k$ gives the display. $Q_P^{(2k-1)}$ has rational coefficients and the moments of $X = \frac{1}{2} + iY$ are rational (36.2(iii)), so $h(P) \in \mathbb{Q}$. The value of $c + \mathrm{crit}_k[P(2u)]$ is $\big(c + h(P)\big) + (2^{2k+1} - 1)P(1)\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$; the coordinate map is onto $\mathbb{Q}^2$, the constant $c$ giving the first coordinate and a constant $P$, for which $Q_P = 0$, the second; and $a + b\,(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = 0$ with $(a,b) \in \mathbb{Q}^2$ not zero forces $b \neq 0$ and $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -a/b$. At the cut $N$, $(\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((N,\infty)) = (\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2k+1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,N])$, a rational head. □

**Theorem 44.16 (Completeness along a ray).** *As stated.*

*Proof.* (i) is Theorem 39.57: the closings, the values and the expansion at the native gauge have one annihilator, and when the values vanish their continuation in $k$ vanishes identically. $\mathrm{Ann}(V_{\mathsf{C}})$ is a submodule (Theorem 39.4(ii)), and by Corollary 39.46 it is 0 exactly when the closings vanish

on no nonzero submodule, which holds on an irreducible submodule on which they are not zero. The closings are among the proved relations of Definition 44.1.

(ii) The submodule generated by $m$ is the $\mathbb{C}(k)$-span of the $\mathrm{Sh}^i m$, $i \in \mathbb{Z}$, of dimension $d \le \dim \mathfrak{M}$; so $m, \mathrm{Sh}\, m, \dots, \mathrm{Sh}^d m$ are dependent over $\mathbb{C}(k)$, and since $\mathfrak{M}$, the shift and $m$ are defined over $\mathbb{Q}$ the dependence can be taken with $q_i \in \mathbb{Q}[k]$. If $q_0 = 0$ apply $\mathrm{Sh}^{-1}$ and shift the coefficients; if then $q_d = 0$, the relation has smaller order, against the minimality of $d$ (the elements $m, \dots, \mathrm{Sh}^{d-1} m$ are independent, otherwise the span of all the shifts would have dimension below $d$). So $q_0 q_d \ne 0$. The closings are a solution (Theorem 39.33): $V_{\mathsf{C}}$ is $\mathbb{C}(k)$-linear and $V_{\mathsf{C}}(\mathrm{Sh}\, g)(k) = V_{\mathsf{C}}(g)(k+1)$ at the integers where both sides are defined. Applying $V_{\mathsf{C}}$ to the dependence gives the recurrence at every integer $k$ with $k + d \le -j(m)$. The closings of a rational insertion are constant terms of Laurent polynomials with rational coefficients (Definition 39.30), so they are rational. Below the integer zeros of $q_0$ and $q_d$ the recurrence determines $V_{\mathsf{C}}(m)(k)$ from the $d$ values above it and conversely, so vanishing at $d$ consecutive integers there propagates to every integer below them, and $V_{\mathsf{C}}(m) = 0$ as a germ at $-\infty$; by (i) this is the vanishing of the values. Two words have equal values exactly when their difference has the values 0. □

**Lemma 44.17 (A separating generator).** *As stated.*

*Proof.* Let $\Lambda = c_0 + \sum_i c_i v_i + cv$ with integers of height $H$ and $c \ne 0$; for $c = 0$ the measure $\mu$ applies. For every $n$,

$$D_n\big(A_n\Lambda - cL_n\big) = D_n\Big(A_n c_0 + cB_{n,0} + \sum_{i=1}^{j} \big(A_n c_i + cB_{n,i}\big) v_i\Big)$$

is an integer linear form in $1, v_1, \dots, v_j$ of height at most $2He^{(h+d)n+o(n)}$. A nonzero integer form of height below $H_0$ has a value bounded away from 0, there being finitely many and the numbers $1, v_1, \dots, v_j$ being independent; so for large $n$ every nonzero such form has modulus at least $(2He^{(h+d)n+o(n)})^{-\mu}$.

*Independence.* If $\Lambda = 0$ the form is $-cD_nL_n$, which is not 0, so its coefficients are not all 0, and $|c|D_n|L_n| \ge (2He^{(h+d)n+o(n)})^{-\mu}$; but $|c|D_n|L_n| \le |c|e^{(d-F)n+o(n)}$, and since $F - d > \mu(h+d)$ the two bounds contradict each other for large $n$.

*Measure.* If the form vanishes as a number, its coefficients vanish, since $1, v_1, \dots, v_j$ are independent; then $A_n c_0 + cB_{n,0} = 0$ and $A_n c_i + cB_{n,i} = 0$, so the row $(A_n, B_{n,0}, \dots, B_{n,j})$ is $(A_n/c)$ times $(c, -c_0, -c_1, \dots, -c_j)$, and $A_n \ne 0$ because $L_n \ne 0$. Consecutive rows are not proportional, so at $n$ or at $n+1$ the form is a nonzero integer form, and there

$$D_n|A_n|\,|\Lambda| \;\ge\; |D_n(A_n\Lambda - cL_n)| - |c|D_n|L_n| \;\ge\; \big(2He^{(h+d)n+o(n)}\big)^{-\mu} - He^{(d-F)n+o(n)}.$$

Let $n$ be the least integer with $He^{(d-F)n+o(n)} \le \frac{1}{2}(2He^{(h+d)n+o(n)})^{-\mu}$; taking logarithms, $(F - d - \mu(h+d))\, n \ge (1+\mu)\log H + O(1) + o(n)$, so $n = (1+\mu)\log H/(F - d - \mu(h+d)) + o(\log H)$, and the inequality holds at $n+1$ as well. With $D_n|A_n| \le e^{(h+d)n+o(n)}$,

$$|\Lambda| \;\ge\; \tfrac{1}{2}(2H)^{-\mu} e^{-\mu(h+d)n - o(n)} e^{-(h+d)n - o(n)} = \tfrac{1}{2}(2H)^{-\mu} e^{-(1+\mu)(h+d)n - o(n)},$$

which is at least $H^{-\mu'}$ for all large $H$. □

**Theorem 44.18 (Separating generators in turn).** *As stated.*

*Proof.* Induction on $j$ with Lemma 44.17. The number 1 alone has the measure $\mu_0 = 0$: a nonzero integer has modulus at least $1 = H^0$. If $1, \mathrm{ev}(w_1), \dots, \mathrm{ev}(w_{j-1})$ are independent with the measure $\mu_{j-1}$, the separating generator of $\mathrm{ev}(w_j)$ with $F_j > d_j + \mu_{j-1}(h_j + d_j)$ makes $1, \mathrm{ev}(w_1), \dots, \mathrm{ev}(w_j)$ independent with the measure $\mu_j$. At $j = s$ the values of the representatives are independent over $\mathbb{Q}$, so the block is complete (Proposition 44.8); a word of value zero then lies in the image of the relation matrix of the block, and its certificate is a finite derivation (Definition 44.7, Theorem 44.9). □

## 50.44 Chapter 45: the content and its coaction

Every identity proved below between admitted words is a proved relation, so it lies in the relation ideal $\mathfrak{I}$ (44.1) and holds between the classes in $\mathfrak{W}/\mathfrak{I}$; in particular the words $\zeta = e^{2\pi i r/M}$ multiply as the roots of unity do, by the functional equation of the exponential, and a constant factor $\zeta$ in front of a trail is a factor of its density, by linearity.

**Definition 45.1 (The content).** As stated.

*Proof.* For $\nu_d \geq 2$ the trail converges absolutely: its terms are bounded by $\prod_i n_i^{-\nu_i}$, and $\sum_{0<n_1<\cdots<n_d} \prod_i n_i^{-\nu_i} \leq \sum_{n_d} (1+\log n_d)^{d-1} n_d^{-2} < \infty$; so the limit of the heads is the Trailmoment. For $\nu_d = 1$ and $e^{2\pi i a_d} \neq 1$ the limit exists by 5.27. The data $(\boldsymbol{\nu}, \mathbf{a})$ are finite and rational, so the trails and the words $\zeta$ are admitted words, and the spans are defined. □

**Proposition 45.2 (The content is an algebra).** *As stated.*

*Proof.* (ii) For an integer $N \geq 1$ let $H_N(T)$ be the head of $T$ over $\{0 < y_1 < \cdots < y_d \leq N\}$. The product $H_N(T)H_N(T')$ is the finite sum, over pairs of increasing tuples $0 < n_1 < \cdots < n_d \leq N$ and $0 < n'_1 < \cdots < n'_{d'} \leq N$, of $\prod_i e^{2\pi i a_i n_i} n_i^{-\nu_i} \prod_j e^{2\pi i a'_j n'_j} {n'_j}^{-\nu'_j}$. The distinct values among the $n_i$ and $n'_j$, in increasing order $m_1 < \cdots < m_e$, and the positions of the points of each tuple among them give a surjection $\phi$, increasing on each part; conversely $\phi$ and $m_1 < \cdots < m_e$ give back the pair. The term is $\prod_r e^{2\pi i (\sum_{\phi(x)=r} a_x) m_r} m_r^{-\sum_{\phi(x)=r} \nu_x}$, the term of $T_\phi$ at $(m_r)_r$. So $H_N(T)H_N(T') = \sum_\phi H_N(T_\phi)$. Every $T_\phi$ is a trail of level $M$: its frequencies lie in $\frac{1}{M}\mathbb{Z}$, and since $\phi$ is onto and increasing on each part, its last point holds the last point of $T$ or of $T'$; if it holds only one of them it has that trail's last index and frequency, and if both, its index is at least 2. As $N \to \infty$ every head converges (45.1), and $TT' = \sum_\phi T_\phi$. The depth $e$ lies between $\max(d, d')$ and $d + d'$ and the indices add to $|\boldsymbol{\nu}| + |\boldsymbol{\nu}'|$.

(i) and (iii). 1 is the word $\zeta = 1$. At level 1 the frequencies are integers, so the trails of depth one of level 1 are the $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ with $k \geq 2$, the index one being excluded by $e^{2\pi i a_1} = 1$. For $\zeta, \zeta' \in \mu_M$, $(\zeta T)(\zeta' T') = \zeta\zeta' \sum_\phi T_\phi$ by (ii), with $\zeta\zeta' \in \mu_M$, so $\mathcal{C}^{(M)}$ is closed under products, with weights adding, and it contains every product of the $(\mathsf{T}_{k;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. When $M \mid M'$, $\frac{1}{M}\mathbb{Z} \subseteq \frac{1}{M'}\mathbb{Z}$ and $\mu_M \subseteq \mu_{M'}$, so $\mathcal{C}^{(M)} \subseteq \mathcal{C}^{(M')}$; as $\mathcal{C}^{(M)}$ and $\mathcal{C}^{(M')}$ both lie in $\mathcal{C}^{(MM')}$, the union $\mathcal{C}$ is an algebra. □

**Proposition 45.3 (Cosets, cuts and index one).** *As stated.*

*Proof.* (i) *Cosets.* Put $n_i = My_i$, so that $y_i \in b_i + \mathbb{Z}$ exactly when the integer $n_i$ is congruent to $Mb_i$ modulo $M$, and the indicator of that congruence is $\frac{1}{M}\sum_{s=0}^{M-1} e^{2\pi i s(n_i - Mb_i)/M}$ (the lattice

relation of 5.12, read on one coset). Then $y_i^{-\nu_i} = M^{\nu_i} n_i^{-\nu_i}$ and $e^{2\pi i a_i y_i} = e^{2\pi i (a_i/M) n_i}$, and the heads of the trail up to $N'$ are, term by term,

$$\sum_{\mathbf{s}\in\{0,\dots,M-1\}^d} M^{|\boldsymbol{\nu}|-d} \prod_i e^{-2\pi i s_i b_i} \sum_{MN<n_1<\cdots<n_d\le MN'} \prod_i e^{2\pi i \frac{a_i+s_i}{M} n_i}\, n_i^{-\nu_i}.$$

Here $e^{-2\pi i s_i b_i} = e^{-2\pi i s_i (Mb_i)/M} \in \mu_M$, and the frequencies $(a_i+s_i)/M$ lie in $\frac{1}{MM'}\mathbb{Z}$. If $\nu_d = 1$, then $a_d \notin \mathbb{Z}$, so $(a_d+s_d)/M \notin \mathbb{Z}$: every inner sum is the head of a trail on $\mathbb{Z}^d$ at the cut $c = MN \in \mathbb{N}$ whose last index and frequency satisfy the condition of 45.1, and the trail is the finite combination of their limits.

*Cuts.* For data $D = (\nu_i, a_i'')_{i\le d}$ on $\mathbb{Z}^d$ let $T_c(D)$ be the trail at the cut $c \in \mathbb{N}$, read as the limit of its heads, $T_c(\varnothing) = 1$, and $D_{>k}$ the last $d-k$ data. Splitting the points $0 < n_1 < \cdots < n_d$ by the number $k$ of them at most $c$,

$$T_0(D) = \sum_{k=0}^{d} h_k(D)\, T_c(D_{>k}), \qquad h_k(D) = \sum_{0<n_1<\cdots<n_k\le c} \prod_{i\le k} e^{2\pi i a_i'' n_i} n_i^{-\nu_i},$$

first for the heads and then in the limit. So $T_c(D) = T_0(D) - \sum_{k\ge1} h_k(D) T_c(D_{>k})$, and by induction on the depth $T_c(D)$ is a polynomial in the finite sums $h_k(D_{>l})$ times 1 or a trail $T_0(D_{>l})$, which keeps the last data of $D$ and so is a trail of level $MM'$. Each $h_k$ is a finite sum of rational multiples of roots of unity in $\mu_{MM'}$, a combination of the words $\zeta$ by linearity; with 45.2 the class lies in $\mathcal{C}^{(MM')}$, and the value is as stated. A head of the trail is a finite sum of terms $e^{2\pi i \sum_i a_i y_i} \prod_i y_i^{-\nu_i}$ with $a_i y_i \in \frac{1}{MM'}\mathbb{Z}$, a rational multiple of an element of $\mu_{MM'}$.

(ii) By 5.2, $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,X]) - \log(X-N) \to (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,\infty))$ as $X\to\infty$, and $\log(X-N) - \log(X-N') \to 0$, so the difference $D$ of the two index-one values is the limit of $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,X]) - (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b'+\mathbb{Z}})((N',X])$. With $n = My$ and the indicator above,

$$(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{b+\mathbb{Z}})((N,X]) = \sum_{s=0}^{M-1} e^{-2\pi i s b} \sum_{MN<n\le MX} \frac{e^{2\pi i s n/M}}{n}.$$

In the difference the terms $s=0$ leave the finite rational sum of $1/n$ between $MN$ and $MN'$, with its sign. For $s\neq0$, $\sum_{MN<n\le MX} e^{2\pi i s n/M}/n$ is the head up to $MX$ of the trail $T_s$ of level $M$ of depth one, index 1 and frequency $s/M$, with $e^{2\pi i s/M} \neq 1$, less the finite sum over $0<n\le MN$. So $D = \sum_{s=1}^{M-1} \big(e^{-2\pi i s b} - e^{-2\pi i s b'}\big) T_s$ plus finite sums of rational multiples of roots of unity in $\mu_M$, and $e^{-2\pi i s b} \in \mu_M$ as $Mb \in \mathbb{Z}$. □

**Definition 45.4 (Chains).** As stated.

*Proof.* No letter lies on the open segment, so the integrand is continuous on the open simplex, and its only possible singularities on the closure are at $t_i \in \{0,1\}$; the convergence is 45.6(i). □

**Proposition 45.5 (The content as chains).** *As stated.*

*Proof.* (i) is 5.27 with $e^{2\pi i a_i} \in \mu_M$: the letters $c_i^{-1}$ lie in $\mu_M$, the first letter is $c_1^{-1} \neq 0$, the last is 0 when $\nu_d \ge 2$ and $c_d^{-1} = e^{-2\pi i a_d} \neq 1$ when $\nu_d = 1$, and no point of $\{0\}\cup\mu_M$ lies on the open segment from 0 to 1.

(ii) The ratios $c_i/c_{i+1}$ lie in $\mu_M$, so frequencies $a_i' \in \frac{1}{M}\mathbb{Z}$ with $e^{2\pi i a_i'} = c_i/c_{i+1}$ exist, and $e^{2\pi i (a_i' + \cdots + a_d')} = c_i/c_{d+1} = c_i$. As $a_1 \neq 0$, $j_1 = 1$, and the word of (i) for the data $(\nu_i, a_i')$, the blocks

$(c_i^{-1}, 0^{\nu_i - 1})$, is $a_1, \dots, a_w$. If $\nu_d = 1$ then $j_d = w$ and $e^{2\pi i a'_d} = c_d = a_w^{-1} \neq 1$, so the trail is of level $M$, and (i) gives the identity.

(iii) The values of $\mathcal{C}^{(M)}$ are the $\mathbb{Q}$-span of the $\zeta$ and the $\zeta T$; the $\mathbb{Q}$-span of $\mu_M$ is $\mathbb{Q}(\mu_M)$, and by (i) and (ii) the values of the trails of level $M$ are, up to sign, exactly the chains described, the weight $|\boldsymbol{\nu}|$ of a trail being the number $w$ of letters of its chain. □

**Proposition 45.6 (Chains with an end at zero).** *As stated.*

*Proof.* (i) Write $\phi_i(t) = \gamma'(t)/(\gamma(t) - a_i)$. As no letter lies on the open segment, $|\phi_i(t)| \leq C/t$ if $a_i = a_0$, $|\phi_i(t)| \leq C/(1-t)$ if $a_i = a_{w+1}$, and $|\phi_i| \leq C$ otherwise. Split the simplex by the number $k$ of the $t_i$ below $\frac{1}{2}$: the part is a product of $\{0 < t_1 < \cdots < t_k < \frac{1}{2}\}$ and $\{\frac{1}{2} \leq t_{k+1} < \cdots < t_w < 1\}$, on which the factors singular at the other end are bounded. On the first, put $G_j(t) = \int_{0<t_1<\cdots<t_j<t} \prod_{i \leq j} |\phi_i(t_i)|\, dt$ for $t \leq \frac{1}{2}$, where $|\phi_i(s)| \leq C(1 + 1/s)$; since $a_1 \neq a_0$, $G_1(t) \leq Ct$, and if $G_{j-1}(s) \leq C_{j-1} s$ then $G_j(t) \leq \int_0^t C_{j-1} s\, C(1 + 1/s)\, ds \leq C_j t$; so the first factor is finite. The second is the same after $t \mapsto 1 - t$, with $a_w \neq a_{w+1}$. So the integral of the modulus is finite.

(ii) For $\tilde{\gamma} = \lambda\gamma + \eta$, $\tilde{\gamma}'(t)/(\tilde{\gamma}(t) - \lambda a_i - \eta) = \gamma'(t)/(\gamma(t) - a_i)$, and the open segment is mapped onto the open segment of the new ends.

(iii) The segment from $a_{w+1}$ to $a_0$ is $\tilde{\gamma}(t) = \gamma(1-t)$, with $\tilde{\gamma}' = -\gamma'$; the substitution $t_i = 1 - t'_{w+1-i}$ maps the simplex onto itself, and each of the $w$ factors changes sign.

(iv) For $s \in \mu_M$, (ii) with $\lambda = s^{-1}$ gives $\mathrm{I}(0; a_1, \dots, a_w; s) = \mathrm{I}(0; a_1/s, \dots, a_w/s; 1)$, with letters in $\{0\} \cup \mu_M$, first letter $a_1/s \neq 0$ and last letter $a_w/s \neq 1$. For a chain from $s$ to 0, (iii) gives $(-1)^w$ times a chain from 0 to $s$ whose first letter is $a_w \neq 0$ and whose last letter is $a_1 \neq s$, and the first case applies. 45.5(ii) and (iii) place the value in the values of $\mathcal{C}^{(M)}$. □

**Definition 45.7 (Symbols of chains).** As stated.

*Proof.* The variables are the formal chains of weight at least one with distinct ends; the expressions with no letter or with equal ends are not variables, so assigning them 1 and 0 involves no relation among the variables. The symbol of a trail depends on the frequencies only through the $c_i$, which do not change when a frequency moves by an integer, as the trail does not; and the realization converges by 45.6(i), no point of $\{0\} \cup \mu_M$ lying on the open segment between 0 and a root of unity. □

**Definition 45.8 (The cutting rule).** As stated.

*Proof.* $\mathfrak{S}_M$ is a polynomial algebra on the formal chains, so a homomorphism is fixed by arbitrary values on them; the right side of the display is an element of $\mathfrak{S}_M \otimes \mathfrak{S}_M$, its degenerate entries read by the conventions of 45.7. The rotation $\rho_\lambda$ and the reversal map formal chains to formal chains, up to sign, keeping the ends distinct, and the conventions to the conventions; $\rho_{\lambda^{-1}}$ and the reversal itself are their inverses. □

**Theorem 45.9 (The cutting rule is a coaction).** *As stated.*

*Proof.* All maps in (i) and (iii) are homomorphisms of algebras, so it suffices to compare them on a formal chain $x = \mathrm{I}^{\mathfrak{s}}(a_0; a_1, \dots, a_w; a_{w+1})$. For a set $U \subseteq \{1, \dots, w\}$ the *gaps* of $U$ are the pairs $t < t'$ of consecutive elements of $U \cup \{0, w+1\}$; for $T \subseteq S$ put

$$\tau(T, S) = \mathrm{I}^{\mathfrak{s}}(a_0; a_T; a_{w+1}) \otimes \prod_{(t,t') \text{ gap of } T} \mathrm{I}^{\mathfrak{s}}\big(a_t; a_{S \cap (t,t')}; a_{t'}\big) \otimes \prod_{(s,s') \text{ gap of } S} \mathrm{I}^{\mathfrak{s}}\big(a_s; a_{(s,s')}; a_{s'}\big),$$

with $a_{(s,s')}$ the letters strictly between the positions $s$ and $s'$, the degenerate entries read by the conventions.

First, for any pair of ends and letters, the display of 45.8 gives $\Delta$ of the corresponding expression even when it is not a formal chain. With no letter the expression is 1, and the display has the single term $1 \otimes 1 = \Delta 1$. With equal ends and at least one letter the expression is 0; in the display every term with a kept letter has a left factor with equal ends, which is 0, and the term with no kept letter has the right factor equal to the expression itself, which is 0; so the display gives $0 = \Delta 0$.

(i) $(\Delta \otimes \mathrm{id})\Delta x = \sum_S \Delta\, \Gamma^{\mathfrak{s}}(a_0; a_S; a_{w+1}) \otimes \prod_{\text{gaps of } S}(\cdots)$, and the display for the chain of the letters $a_S$ keeps the letters of $T \subseteq S$ and reads the letters of $S$ between consecutive kept ones: the sum is $\sum_{T \subseteq S} \tau(T, S)$. Next, $(\mathrm{id} \otimes \Delta)\Delta x = \sum_T \Gamma^{\mathfrak{s}}(a_0; a_T; a_{w+1}) \otimes \prod_{(t,t')} \Delta\, \Gamma^{\mathfrak{s}}(a_t; a_{(t,t')}; a_{t'})$, and by the remark above each factor is the display, a sum over the sets $S_{(t,t')}$ of letters kept inside the gap $(t, t')$. A family of such sets is the same as one set $S \supseteq T$, the middle factors are the chains of the letters of $S$ inside the gaps of $T$, and the right factors are the gaps of the $S_{(t,t')}$ inside the gaps of $T$, which are the gaps of $S$: the sum is again $\sum_{T \subseteq S} \tau(T, S)$. For the counit, $\varepsilon$ kills every formal chain, so in $(\varepsilon \otimes \mathrm{id})\Delta x$ only $S = \varnothing$ survives, with the right factor $x$; in $(\mathrm{id} \otimes \varepsilon)\Delta x$ a right factor contributes 1 only when it has no letter, so only $S = \{1, \ldots, w\}$ survives, with the left factor $x$.

(ii) The term of $S$ has the left weight $|S|$ and the right weight $w - |S|$, the runs between kept letters together holding the letters not kept. The terms $S = \{1, \ldots, w\}$ and $S = \varnothing$ are $x \otimes 1$ and $1 \otimes x$, and every other term has $0 < |S| < w$.

(iii) Multiplying every entry by $\lambda$ commutes with keeping a subsequence and with cutting out runs, so $\Delta \rho_\lambda x = (\rho_\lambda \otimes \rho_\lambda)\Delta x$. The reversal maps the term of $S$ for $x$ to the term of the mirror set $S^* = \{w + 1 - s : s \in S\}$ for the reversed chain: its left factor is $(-1)^{|S|}$ times the reversal of the kept chain read backwards, each run is reversed with the sign $(-1)^{\text{its length}}$, and the runs appear in the opposite order, which does not matter in the commutative algebra. The signs multiply to $(-1)^w$, the sign of $\varrho x$, so $\Delta \varrho x = (\varrho \otimes \varrho)\Delta x$. □

**Proposition 45.10 (The cutting rule on motivic chains).** *As stated.*

*Proof.* A chain of level $M$ is an iterated integral of the forms $dt/(t - a)$, $a \in \{0\} \cup \mu_M$, along the straight path between two of the points $\{0\} \cup \mu_M$ of $\mathbb{P}^1 \setminus (\{0, \infty\} \cup \mu_M)$ (Definition 45.4). Such iterated integrals, with tangential base points at the ends where a letter equals an end, are periods of the motivic torsor of paths of $\mathbb{P}^1 \setminus (\{0, \infty\} \cup \mu_M)$, a mixed Tate motive over $\mathbb{Z}[\mu_M, \frac{1}{M}]$ (Deligne and Goncharov), whose realizations are objects of the category of Definition 34.5; the motivic chain is the matrix coefficient of the element $\omega_{a_1} \cdots \omega_{a_w}$ of its de Rham realization against the class of the path. The unit, the vanishing for equal ends and the regularization are those of the torsor with tangential base points. The coaction on these motivic periods is Goncharov's formula in Brown's form, which is the displayed formula, the cutting rule term by term. The shuffle product, the composition of paths and the reversal hold for iterated integrals on paths of the torsor, and they are identities of motivic periods because they come from morphisms of the torsor; the formal chains are independent variables of $\mathfrak{S}_M$, so the realization is not injective. □

**Definition 45.11 (The letter of an input).** As stated.

*Proof.* The Laplace transform beyond the cut $n_0 - 1$ is $\mathcal{F}_{n_0-1}(u) = \sum_{n \geq n_0} \psi(n) e^{-un} = L_\psi(e^{-u})$ (5.4), and $L_\psi(1) = \sum_n \psi(n)$ where the series converges at 1, which is the Trailmoment at index zero (6.1). For $\psi(n) = e^{2\pi i a n}$, $n \geq 1$, $L_\psi(z) = \sum_{n \geq 1} (e^{2\pi i a} z)^n = e^{2\pi i a} z/(1 - e^{2\pi i a} z)$, with its pole at $e^{-2\pi i a}$, which is the letter $c_1^{-1}$ of the depth-one trail at that frequency in 45.5(i). □

**Proposition 45.12 (The letter of a live input).** *As stated.*

*Proof.* (i) $p_i(\theta - i)z^n = p_i(n-i)z^n$, so

$$\sum_i z^{h-i} p_i(\theta - i) L_\psi = \sum_i \sum_{n \geq n_0} p_i(n-i)\psi(n) z^{n+h-i} = \sum_m z^{m+h} \sum_{i:\, m+i \geq n_0} p_i(m)\psi(m+i).$$

For $m \geq n_0$ the inner sum is the recurrence, zero; the finitely many $m$ with $n_0 - h \leq m < n_0$ leave a polynomial $Q$ of degree less than $n_0 + h$. Since $\theta^j$ is $z^j \frac{d^j}{dz^j}$ plus terms of lower order, the coefficient of $\frac{d^d}{dz^d}$ is $\sum_i z^{h-i} c_i z^d$, nonzero as some $c_i \neq 0$; for $z \neq 0$ it vanishes exactly when $\sigma(1/z) = 0$.

(ii) For large $n$, $p_h(n) \neq 0$, and the state $U(n) = (\psi(n), \ldots, \psi(n+h-1))$ satisfies $U(n+1) = \mathbf{A}(n)U(n)$ with the companion matrix of the recurrence, whose entries $-p_i(n)/p_h(n)$ tend to $-c_i/c_h$, as $p_h$ has the largest degree: $\mathbf{A}(n) \to \mathbf{A}_\infty$, the companion matrix of $\sigma/c_h$, of spectral radius $\varrho$. For $\eta > 0$ there is a norm on $\mathbb{C}^h$ whose operator norm of $\mathbf{A}_\infty$ is at most $\varrho + \eta/2$ (a vector norm adapted to the Jordan form of $\mathbf{A}_\infty$, which exists for every square matrix); then $\|\mathbf{A}(n)\| \leq \varrho + \eta$ for $n \geq n_3$, and $\|U(n)\| \leq \|U(n_3)\|(\varrho + \eta)^{n-n_3}$, which with the finitely many earlier terms gives the bound. So $L_\psi$ converges on $|z| < 1/(\varrho + \eta)$ for every $\eta$. By (i) it satisfies, near any point of that disc other than 0, a linear differential equation with polynomial coefficients and polynomial right side, whose leading coefficient vanishes only at 0 and at the reciprocals of the nonzero roots of $\sigma$; the solutions of such an equation continue analytically along every path on which the leading coefficient does not vanish (the existence theorem for linear differential equations with holomorphic coefficients, which applies because the coefficients are polynomials and the leading one is nonzero along the path). The cut plane $\Omega$ is simply connected and star-shaped about 0, and contains the disc; a path in $\Omega$ meets the singular points only inside the disc, at 0, where $L_\psi$ is holomorphic, so $L_\psi$ continues along every path in $\Omega$, and by the monodromy theorem it defines a holomorphic function on $\Omega$.

(iii) Let the two recurrences have orders $h$ and $h'$, with companion matrices $\mathbf{A}(n)$ and $\mathbf{A}'(n)$, invertible as rational matrices since their determinants are $\pm p_0(n)/p_h(n)$ and $\pm p'_0(n)/p'_{h'}(n)$. The vector $X(n) = U(n) \otimes U'(n)$ satisfies $X(n+1) = (\mathbf{A}(n) \otimes \mathbf{A}'(n))X(n)$ for large $n$, with an invertible Kronecker product, and the product sequence is the coordinate of $X(n)$ that pairs the first coordinates. Lemma S30 over the field $\Bbbk$ generated by the coefficients gives the recurrence of order at most $hh'$ with nonzero extreme coefficients.

(iv) By (ii), $|\psi(n)\psi'(n)| \leq C(\varrho + \eta)^n(\varrho' + \eta)^n$, so the letter of $\psi\psi'$, the termwise product of the coefficients of $L_\psi$ and $L_{\psi'}$, converges on $|z| < 1/(\varrho\varrho')$. By (ii), $L_\psi$ and $L_{\psi'}$ are holomorphic on the planes cut along the rays from the reciprocals of the nonzero roots of $\sigma$ and of $\sigma'$. Hadamard's multiplication theorem, which applies to two power series holomorphic on planes cut along the outward rays from finitely many nonzero points, makes the series of the termwise products holomorphic on the plane cut along the outward rays from the products of those points, the points $1/(\lambda\lambda')$. □

**Proposition 45.13 (The row of a hinge closing read by letters).** *As stated.*

*Proof.* (i) The $c_i^\pm$ solve the room's recurrence, whose last coefficient has the largest degree, so by 45.12(ii) $|c_i^\pm(n)| \leq C(\varrho^\pm + \eta)^n$; likewise $|\Phi_n^\pm| \leq C(\varrho_\Phi^\pm + \eta)^n$ for large $n$. With $\eta$ small, $(\varrho^\pm + \eta)(\varrho_\Phi^\pm + \eta) < 1$, and the sums converge absolutely; the radius of convergence of $L_{\psi_i^\pm}$ exceeds 1, so $\Xi_i^\pm = L_{\psi_i^\pm}(1)$. 43.10(ii) gives the row.

(ii) By 45.12(iii) the product $\psi_i^\pm$ satisfies a recurrence with polynomial coefficients for large $n$, so by 45.12(i) its letter, up to a polynomial from the finitely many first terms, satisfies a linear

differential equation with polynomial coefficients; and 45.12(iv), applied to the tails of the two factors, gives the continuation, the finitely many first terms adding a polynomial. □

**Definition 45.14 (The motivic content; cuts; separating generators).** As stated.

*Proof.* (i) The shuffle of two words whose first letters are not 0 and whose last letters are not 1 is a sum of such words, so by the shuffle product of motivic chains (Proposition 45.10) the span is an algebra. The weight is additive under the shuffle product, which preserves the number of letters, and it is the grading of the weight filtration of the motives of Proposition 45.10, which is split for mixed Tate motives (Deligne and Goncharov). (ii) For $a \in \frac{1}{M}\mathbb{Z}\backslash\mathbb{Z}$, $\big((\mathsf{T}_{1;(1)}[x]\, e^{2\pi i a x}\mathbb{I}_{\mathbb{Z}})((0,\infty))\big)^{\mathfrak{m}} = -\mathrm{I}^{\mathfrak{m}}(0; e^{-2\pi i a}; 1)$ is the motivic logarithm of $1 - e^{2\pi i a}$, so $\big((\mathsf{T}_{1;(1)}[x]\, 2i\sin(\frac{2\pi x}{M})\,\mathbb{I}_{\mathbb{Z}})((0,\infty))\big)^{\mathfrak{m}}$ is the motivic logarithm of $(1 - e^{-2\pi i/M})/(1 - e^{2\pi i/M}) = -e^{-2\pi i/M}$, a root of unity; the Kummer motive of a root of unity is split, so this motivic logarithm is a rational multiple of the motivic period of the circle, and its period is $2\pi i\,(\frac{1}{2} - \frac{1}{M})$ (with $(\mathsf{T}_{1;(1)}[x]\, e^{2\pi i a x}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\log(1 - e^{2\pi i a})$ by (iv)). At level 1 the motivic periods of weight 2 are the rational multiples of $\mathbb{L}^2$, the category having no extensions of $\mathbb{Q}(0)$ by $\mathbb{Q}(1)$ or by $\mathbb{Q}(2)$ (Deligne and Goncharov, with Borel's ranks); $\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))\big)^{\mathfrak{m}} = -\mathrm{I}^{\mathfrak{m}}(0; 1, 0; 1)$ is one of them, and its period $(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\frac{(2\pi i)^2}{24}$ fixes the multiple. The identity holds in $\mathcal{P}^{\mathrm{mot}}$, so also at level 2. (iii) The coaction of $\mathcal{P}^{(M)}$ has its right factors in the de Rham chains, and Brown's infinitesimal coaction reads the part with one de Rham factor of positive weight in the motivic periods modulo the circle and modulo products; it is well defined on $\mathcal{P}^{(M)}$ because the coaction is (Brown for $M = 1$; Deligne and Glanois for $M = 2, 3, 4$), and on a motivic chain it is the displayed sum, the terms of Proposition 45.10 whose product of runs has exactly one factor of positive weight, of $r$ letters. The composition of paths, the reversal, the rotations, which are automorphisms of $\mathbb{P}^1 \setminus (\{0,\infty\} \cup \mu_M)$ that move the tangential base point at 0 by a root of unity, and the regularization write every motivic chain of level $M$ as a polynomial in $\mathbb{L}$ and in motivic values of trails, and the terms with a factor $\mathbb{L}$ vanish modulo the circle. (iv) At $N = 1$ and $a \notin \mathbb{Z}$, $(\mathsf{T}_{1;(1)}[x]\, e^{2\pi i a x}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\log(1 - e^{2\pi i a})$ with the principal logarithm, by the limit of the heads (Theorem 5.27); so $-(\mathsf{T}_{1;(1)}[x]\, e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \log 2$ and $-(\mathsf{T}_{1;(1)}[x]\, 2\cos(\frac{2\pi x}{3})\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \log|1 - e^{2\pi i/3}|^2 = \log 3$. □

**Theorem 45.15 (The content is separated by its cuts and its value).** *As stated.*

*Proof.* (i) The motivic chains of level $M$ are periods of mixed Tate motives over $\mathbb{Z}[\mu_M, \frac{1}{M}]$ (Proposition 45.10). The motivic Galois group of that category is $\mathbb{G}_m$ acting on a prounipotent group whose Lie algebra is free on generators of degree $N$, as many in each degree as the dimension of the extensions of $\mathbb{Q}(0)$ by $\mathbb{Q}(N)$, and these dimensions are one exactly in the degrees in which Definition 45.14(iv) places a separating generator and zero otherwise at the levels 1 to 4 (Deligne and Goncharov, with Borel's ranks of the groups of these number rings, which apply because the category is mixed Tate over the ring of integers of $\mathbb{Q}(\mu_M)$ with $M$ inverted; at levels $1, 2$ the real periods leave only the even powers of the circle). So the de Rham periods read modulo the circle are the graded dual of the enveloping algebra of that free Lie algebra, the shuffle algebra on letters $x_N$ with the coproduct that cuts a word into its two ends, and after the choice of a lift of each letter of weight $N$ whose cuts of lengths below $N$ vanish, the motivic periods without negative powers of the circle are that shuffle algebra tensored with $\mathbb{Q}[\mathbb{L}^e]$, as an algebra and a comodule over it. Under this identification the coaction cuts a word $w = uv$ into $u \otimes v$, the circle is a constant, and $\mathsf{D}_r$ keeps the terms with $v$ of weight $r$, read modulo shuffle products. The shuffle product adds the numbers of letters, so the classes modulo products are graded by the number of letters, and the class of a word of one letter $x_r$ is not the class of any combination of longer words. Write

$\xi = \sum_{w,k} c_{w,k}\, w\, \mathbb{L}^{ek}$. The coefficient of $[x_r] \otimes u\,\mathbb{L}^{ek}$ in $\mathsf{D}_r\xi$ is $c_{ux_r,k}$, and the elements $u\,\mathbb{L}^{ek}$ are independent; so if every cut vanishes, $c_{w,k} = 0$ for every nonempty word $w$ whose last letter has weight below $N$. A nonempty word of weight $N - ek$ whose last letter has weight $N$ is $x_N$ with $k = 0$, so $\xi = c\,x_N + b\,\mathbb{L}^N$. The second factors of $\mathsf{D}_r\xi$ have the weight $N - r$.

(ii) The runs of fewer than $N$ letters of a chain of depth one $\mathrm{I}^{\mathfrak{m}}(0; \beta, 0^{N-1}; 1)$, $\beta \in \mu_M$, are of three kinds: bounded by equal ends, so 0; the letters $0^r$ between 0 and 1, regularized to 0; or $0^r$ between $\beta$ and an end, whose motivic chain is $(-\log\beta)^r/r!$ read motivically (the reversal and the regularization), a rational multiple of $\mathbb{L}^r$, which is 0 in $\mathfrak{n}^{(M)}[r]$. So every cut of a chain of depth one vanishes, and so do the cuts of $\mathsf{w}_N^{\mathfrak{m}}$, a combination of such chains (Proposition 45.5(i)); in weight one there are no cuts. The values: $(\mathsf{T}_{N;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \sum_y y^{-N} > 0$, and $\log 2, \log 3 > 0$. For $N$ even, $\sum_y(\omega^y - \omega^{-y})y^{-N} = (\omega - \omega^2)\sum_y \chi(y)y^{-N}$ with $\omega = e^{2\pi i/3}$, $\omega - \omega^2 = i\sqrt{3}$ and $\chi(y) = (\omega^y - \omega^{-y})/(\omega - \omega^2)$, which is $1, -1, 0$ on $1 + 3\mathbb{Z}$, $2 + 3\mathbb{Z}$, $3\mathbb{Z}$; and $\sum_y(i^y - i^{-y})y^{-N} = 2i\sum_y \chi'(y)y^{-N}$ with $\chi'(y) = (i^y - i^{-y})/(2i)$, which is $1, -1, 0$ on $1+4\mathbb{Z}$, $3+4\mathbb{Z}$, $2\mathbb{Z}$. Both $\chi$ and $\chi'$ are completely multiplicative, and for $N \geq 2$ unique factorization and absolute convergence give $\sum_y \chi(y)y^{-N} = \prod_p(1 - \chi(p)p^{-N})^{-1}$, a product of positive factors, which is the stated difference of trails of lattice inputs on cosets. A nonzero real number and a nonzero imaginary number are independent over $\mathbb{R}$, so $c\,\mathrm{per}\,\mathsf{w}_N^{\mathfrak{m}} + b(2\pi i)^N = 0$ forces $c = b = 0$. In particular $\mathsf{w}_N^{\mathfrak{m}} \notin \mathbb{Q}\,\mathbb{L}^N$; with every cut of length below $N$ zero it is $c\,x_N + b\,\mathbb{L}^N$ by (i) applied to any lift, with $c \neq 0$, and it serves as the lift of $c\,x_N$.

(iii) If the cuts of $\xi$ vanish, $\xi = c\,\mathsf{w}_N^{\mathfrak{m}} + b\,\mathbb{L}^N$ by (i) and (ii), and a zero value gives $c = b = 0$ by (ii); conversely $\xi = 0$ has zero cuts and value. The identification of (i), with $\mathsf{w}_N^{\mathfrak{m}}$ as the lift of $x_N$ (a rescaling of the letter), is the homomorphism: the coefficients of the words ending in $x_r$, $r < N$, are read from $\mathsf{D}_r$ and, recursively, from the cuts of the second factors, and those of $x_N$ and $\mathbb{L}^N$ from the readout. Content words have the motivic values of their symbols, and the homomorphism is injective.

(iv) The motivic chains of level $M$ give every motivic period of the category without negative powers of the circle (Brown for $M = 1$; Deligne for $M = 2, 3, 4$), so the homomorphism is onto; and a shuffle algebra over $\mathbb{Q}$ is the polynomial algebra on its Lyndon words (Radford's theorem, which applies because the letters are graded and the shuffle product is commutative). □

**Corollary 45.16 (The values of the content and its generators).** *As stated.*

*Proof.* By Theorem 45.15(iv) every element of $\mathcal{P}^{(M)}$ is uniquely a polynomial over $\mathbb{Q}$ in $\mathbb{L}^e$ and the $g_i^{\mathfrak{m}}$, and the period is a homomorphism of algebras that sends them to $(2\pi i)^e$ and the values of the $g_i$. If these values are algebraically independent, the period is injective on $\mathcal{P}^{(M)}$, so a combination of content words of value zero has motivic value zero, and Theorem 45.15(iii) decides it. A nonzero polynomial over $\mathbb{Q}$ in $\mathbb{L}^e$ and the $g_i^{\mathfrak{m}}$ is a nonzero motivic value, by the uniqueness; if its value vanishes, the relation among the values does not hold among the motivic values. □

**Definition 45.17 (The separation procedure).** As stated.

*Proof.* Each step is a finite computation in exact rational arithmetic: the trails of level $M$ of weight $k$ are finitely many (their indices are compositions of $k$, their frequencies lie in $\frac{1}{M}\mathbb{Z}/\mathbb{Z}$), the cutting rule is a finite sum, the reading of a run or of a kept chain is a finite sum over its regularization, and the rest is linear algebra over $\mathbb{Q}$. □

**Theorem 45.18 (Completeness by proved relations, weight by weight).** *As stated.*

*Proof.* The relations $R_k$ hold for motivic values, so $V_k \to \mathcal{P}^{(M)}[k]$ is defined, and it is onto, $\mathcal{P}^{(M)}[k]$ being spanned by the motivic values of the trails of weight $k$ (Definition 45.14(i)).

For $r < N$, $V_r/\mathfrak{v}_r \to \mathfrak{n}^{(M)}[r]$ is injective. Indeed $V_r \to \mathcal{P}^{(M)}[r]$ is injective by hypothesis; the products of positive weights in $\mathcal{P}^{(M)}[r]$ are the images of the products in $V_r$, by surjectivity in the lower weights; and $\mathbb{L}^e\,\mathcal{P}^{(M)}[r-e]$ is contained in the products when $r > e$ and is the image of the trails whose values are rational multiples of $(2\pi i)^e$ when $r = e$. So an element of $V_r$ whose motivic value lies in the denominator of $\mathfrak{n}^{(M)}[r]$ differs from an element of $\mathfrak{v}_r$ by an element of the kernel, which is 0. Likewise $V_{N-r} \to \mathcal{P}^{(M)}[N-r]$ is injective.

The cut of step (2) of a trail maps, under these two injective maps, to the cut of its motivic value (Definition 45.14(iii)), the readings of step (2) being identities of motivic chains. By step (3) the cuts are linear maps on $V_N$ compatible with these maps. Let $\xi \in V_N$ have motivic value 0. Then every cut of $\xi^{\mathfrak{m}}$ vanishes, so every cut of $\xi$ computed in step (2) vanishes, the tensor product of two injective maps of vector spaces being injective. So $\xi$ lies in the joint kernel of step (4), $\xi = \sum_i c_i t_i$ with $t_i$ classes of trails whose values are independent over $\mathbb{Q}$; its value $\sum_i c_i \operatorname{per}(t_i)$ is the period of $\xi^{\mathfrak{m}} = 0$, so every $c_i = 0$ and $\xi = 0$. □

**Definition 45.19 (Chains over an equation).** As stated.

*Proof.* The module $K\langle\partial_z\rangle/K\langle\partial_z\rangle L$ has the dimension $h$ over $K$, so it has finite length and, by the Jordan–Hölder theorem, its composition factors do not depend on the series. Every module of finite dimension over $K$ with a derivation is cyclic (the cyclic vector lemma), so each factor and its dual is the module of one operator, the dual of $K\langle\partial\rangle/K\langle\partial\rangle Q$ being that of the adjoint $Q^*$. For a solution $s$ of a factor, a $K$-linear map from the factor to the functions with $s(\partial m) = s(m)'$, the combinations over $K$ of $s$ of a cyclic vector and of its derivatives are the values of $s$, so the kernels do not depend on the operator chosen. The integrals converge, the kernels being continuous on $\gamma$. □

**Proposition 45.20 (Every solution is read through chains over its equation).** *As stated.*

*Proof.* Let $M = K\langle\partial\rangle/K\langle\partial\rangle L$ and take a composition series $0 = M_0 \subset \cdots \subset M_r = M$. Let $\mathcal{O} \subset K$ be the ring of the functions regular off $S$, $\Bbbk[z]$ with the inverse of the polynomial that vanishes at the finite points of $S$, a principal ideal domain. The classes of $1, \partial, \dots, \partial^{h-1}$ span over $\mathcal{O}$ a free module $\Lambda$ of rank $h$ that $\partial$ preserves, the leading coefficient of $L$ being a unit of $\mathcal{O}$; the $\Lambda \cap M_j$ are saturated, so direct summands, and a basis $e_1, \dots, e_h$ of $\Lambda$ can be chosen adapted to the series. With $\partial e_l = \sum_m A_{ml} e_m$, $A \in M_h(\mathcal{O})$ is block upper triangular, with the matrix of the factor $M_j/M_{j-1}$ in the block $j$. A solution $y$ is a $K$-linear map $s$ from $M$ to the functions with $s(\partial m) = s(m)'$, $y$ being $s$ of the class of 1, and $Y = (s(e_1), \dots, s(e_h))$ satisfies $Y' = YA$. The class of 1 is $\sum_l p_l e_l$ with $p_l \in \mathcal{O}$, so $y = \sum_l p_l Y_l$, and the $e_l$ are combinations over $\mathcal{O}$ of the classes of $1, \partial, \dots, \partial^{h-1}$, so the entries of $Y(a)$ are combinations of $y(a), \dots, y^{(h-1)}(a)$, with the values at $a$ of elements of $\mathcal{O}$ as coefficients. Write $Y = (Y_1, \dots, Y_r)$ by blocks, so that $Y_j' = Y_j A_{jj} + \sum_{i<j} Y_i A_{ij}$, and let $\Phi_j$ be the matrix whose rows are the solutions of $X' = XA_{jj}$ with $\Phi_j(a)$ the identity. A row of $\Phi_j$ is a solution of the factor $j$ read on the images of the $e_l$, so its entries are combinations over $K$ of one solution of the operator of the factor $j$ and of its derivatives; a column of $\Phi_j^{-1}$ solves $X' = -A_{jj}X$, and its entries are the same for the dual of the factor. Variation of the constants gives

$$Y_j(z) = \Big(Y_j(a) + \int_a^z \sum_{i<j} Y_i A_{ij} \Phi_j^{-1}\Big)\Phi_j(z)$$

along $\gamma$. By induction on $j$ every entry of $Y_j$ is a sum of terms $c\,\mathrm{I}\,\beta$, with $c$ an entry of a row of $\Phi_j$ and $\beta$ an entry of $Y(a)$: in the integrand an entry of a row of $\Phi_i$ times $A_{ij}\Phi_j^{-1}$ is a sum of kernels, and the integral of a kernel against a chain is a chain of one more weight; multiplying by $\Phi_j$ multiplies by the entries of its rows. Then $y = \sum_l p_l Y_l$, and $\sum_l p_l$ times the entries of one row of $\Phi_j$ is a combination over $K$ of one solution of the factor $j$ and of its derivatives.

The factors and their duals have only regular singular points, as subquotients of a regular module and their duals, so their solutions have at $b$ expansions of the stated form, finite sums (Fuchs's theorem). Such expansions are kept by products, and by integration along $\gamma$, termwise near $b$, the term $(z-b)^{-1}$ bringing one more power of $\log(z-b)$, plus a constant; so every $c_i\mathrm{I}_i$ has one, by induction on the weight. The expansions are unique and add. On the final segment write $x = -\log|z-b|$, the argument of $z-b$ being constant there: the terms of the expansion of $y$ with $\operatorname{Re}\rho \le 0$ make a finite sum $\sum_{\rho,l} a_{\rho,l}\, e^{-\rho x}x^l$, and the others tend to 0. Let $(\sigma, l)$ be the largest pair, ordered by $\sigma = -\operatorname{Re}\rho \ge 0$ and then by $l$, with a nonzero coefficient; divided by $e^{\sigma x}x^l$ the sum is a nonzero sum of characters $e^{-i\tau x}$, $\tau = \operatorname{Im}\rho$, plus a term that tends to 0, and a nonzero sum of characters does not tend to 0, the mean of its square modulus being the sum of the squares of the moduli of its coefficients. So if $y$ has a limit, $(\sigma, l) = (0, 0)$, the sum is a sum of characters with a limit, hence its constant term alone, and the limit is the constant term of the expansion of $y$, which is $\sum_i [c_i\mathrm{I}_i]_b\beta_i$. □

**Theorem 45.21 (The closing in any number of shares).** *As stated.*

*Proof.* (i) The data are real with nonnegative exponents on the balance wall and every proper wall positive, so Theorem 15.13(i) applies with the references $M_i$, and $b_{\mathbf{n}} = \Gamma_{1}^{n}\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}\mathbb{E}_{\mathrm{Dir}_{\mathbf{m}}}\big[\prod_i (M_i - L_i)^{n_i}\big]$ by Theorem 3.9. A Dirichlet moment of a polynomial with rational coefficients at rational masses is rational, $\mathbb{E}[U^{\boldsymbol\gamma}] = \prod_e (m_e)_{\gamma_e}/(m_{\mathcal{E}})_{|\boldsymbol\gamma|}$, and $0 \le M_i - L_i$ on the simplex, so $\psi(N) \in \mathbb{Q}_{\ge 0}$; grouping a convergent series of nonnegative terms by $N = |\mathbf{n}|$ keeps its sum. For $|z| < 1$, $\sum_n \frac{(w)_n}{n!} z^n (1 - L/M)^n = (1 - z + zL/M)^{-w}$, dominated by $(1-|z|)^{-w}$ since $0 \le 1 - L/M \le 1$, which gives the display. For $0 < z < 1$ the deformed letters $(1-z)M_i|u|^{g_i} + zL_i(u)$ do not vanish on the closed simplex, so the integrand vanishes on its boundary, the masses being positive, and exact forms integrate to zero; the derivatives of $P_0 L_\psi$ in $z$ are integrals of insertions with powers of the deformed letters in their denominators, and these forms modulo exact forms make up the top twisted de Rham cohomology of the complement of the deformed letters in the torus modulo the scale, over the field $\mathbb{Q}(z)$, which is finite-dimensional (the finiteness of the twisted algebraic de Rham cohomology of a smooth affine variety, as in Theorem 38.31(i)). So $L_\psi, L'_\psi, \dots$ are dependent over $\mathbb{Q}(z)$, which is the equation. Written as $\sum_l z^l Q_l(\vartheta) L_\psi = 0$ with $\vartheta = z\frac{d}{dz}$, its coefficient of $z^N$ is $\sum_l Q_l(N-l)\psi(N-l) = 0$ for every $N$, as in Proposition 45.12(i) read backwards, with nonzero extreme coefficients. For the tail, Theorem 15.13(ii) gives the rate for the partial sums over $\max_i n_i \le N$, and with $p$ letters $\{|\mathbf{n}| \le N\} \subseteq \{\max_i n_i \le N\} \subseteq \{|\mathbf{n}| \le pN\}$, the terms being nonnegative.

(ii) $P_c(z)$ is the period of the family with the weights $\theta z^c$ and $\theta z^{c'}$, positive for $z > 0$, whose walls are those of $F$, since walls do not see weights; so it converges and is real analytic on $(0, \infty)$, and the argument of (i), the deformed letters now being the letters with these weights, gives its equation; the equation of a period of a family over $\mathbb{Q}(z)$ has only regular singular points (Deligne's regularity theorem for the Gauss–Manin connection, the exponents being rational). On the fibre, in the integral coordinates of Proposition 18.17, the word is a product of Gammas of affine forms with integer slopes times $z^{-\lambda(\mathbf{x})}$, $\lambda$ affine, and along every free coordinate the slopes of the Gammas of each letter sum to zero, its share sum being fixed. Close the coordinates in turn by Theorem 4.5,

each toward the side where $\lambda$ decreases; off finitely many hyperplanes of $(c, c')$ the slope of $\lambda$ is nonzero at every step, and then the exponent of $z$ increases along every lattice direction of the poles crossed. A residue replaces one Gamma by $(-1)^j/(|u|\, j!)$, $u$ its slope, and substitutes the pole into the others, whose slopes in the remaining coordinates again sum to zero for each letter; where families of poles coincide the residue takes Laurent coefficients (Proposition 5.14), which bring the powers of $\log z$ and the Trailmoments at the pole, and these are the cumulants at rational parameters (Proposition 5.13). Because the slopes sum to zero, the terms $x \log x$ of Stirling's formula cancel in every Gamma stack met, so the terms and the integrands on the shifted contours grow at most like $C^{|\mathbf{j}|}$, while $z^{e_\sigma(\mathbf{j})} \le z^{\varepsilon|\mathbf{j}|+e_\sigma(0)}$ for some $\varepsilon > 0$; below $z_0 = C^{-1/\varepsilon}$ every shifted contour contributes nothing in the limit and every sum converges absolutely. The expansion and $P_c$ agree on $(0, z_0)$ and $P_c$ is analytic on $(0, \infty)$, so the period is the continuation. At the regular singular point 0 the solutions have the Frobenius form, with exponents in finitely many classes $\rho + \mathbb{N}$, and the equation gives each coefficient sequence a recurrence, triangular in the power of $\log z$, with the indicial polynomial at $\rho + m$ as its leading coefficient. Under the summability hypothesis the expansion converges uniformly on a sector neighbourhood of $[z_0/2, 1]$, where $|z^e|$ and $|\log z|$ are bounded, so it is analytic there and equals $P_c$; at $z = 1$, $\log 1 = 0$.

(iii) The letter, or the deformed period, satisfies an equation with regular singular points only, $\infty$ among them: both are periods of families over $\mathbb{Q}(z)$, as in the proofs of (i) and (ii), and the Gauss–Manin connection of such a family is regular (Deligne's regularity theorem). $L_\psi$ is analytic on the disc $|z| < 1$, where its series converges, and along every segment in the disc that ends at 1 it tends to $\sum_N \psi(N)$, which is finite (Abel's theorem, a chord ending at 1 lying in a Stolz angle); $P_c$ is real analytic on $(0, \infty)$, so analytic in a neighbourhood of $(0, 1]$, around the points of $S$ there as well, with the value $P_c(1)$ at 1. So Proposition 45.20 applies along $\gamma$ with $b = 1$, with the derivatives at $a$ read termwise: from the power series of $L_\psi$, and for $P_c$ from the expansion of (ii), which converges absolutely at $a < z_0$ together with its termwise derivatives, a derivative of $z^e(\log z)^i$ being a combination of $z^{e-1}(\log z)^{i'}$, $i' \le i$. The period is $P_0 L_\psi(1)$ by (i) and $P_c(1)$ by (ii).

Let the monodromy be unipotent. The regular connections over $\mathbb{C}(z)$ with singular points in $S$ and the local systems on $\mathbb{P}^1 \setminus S$ are equivalent categories, subquotients corresponding (the Riemann–Hilbert correspondence for regular connections), and a unipotent representation has the trivial representation of rank one as its only composition factor; so every factor $N$ of $L$, of rank $d$, read over $\mathbb{C}(z)$, has a nonzero horizontal element. In a basis of $N$ the horizontal elements over $\mathbb{C}(z)$ are the solutions in $\mathbb{C}(z)^d$ of a system with coefficients in $\mathbb{Q}(z)$, with poles only at its singular points, of bounded orders, and with bounded degrees, so they are the solutions of a linear system over $\mathbb{Q}$ and are spanned by those in $\mathbb{Q}(z)^d$; a horizontal element of $N$ spans a submodule isomorphic to $\mathbb{Q}(z)$ with $d/dz$, which is $N$, $N$ being simple. The same holds over $\overline{\mathbb{Q}}(z)$. Take, as in the proof of Proposition 45.20, a basis $e_1, \dots, e_h$ over $\mathcal{O} = \overline{\mathbb{Q}}[z][1/q]$ adapted to a composition series over $\overline{\mathbb{Q}}(z)$, so that $A$ is upper triangular. The factor $l$ has a horizontal element, the image of $r_l e_l$ with $r_l \in \overline{\mathbb{Q}}(z)$, whose zeros and poles lie in $S$, the diagonal entry being $-r_l'/r_l$; so $r_l$ is a unit of $\mathcal{O}$, and replacing $e_l$ by $r_l e_l$ makes the diagonal of $A$ zero. Changing $e_l$ by multiples over $\mathcal{O}$ of the $e_m$, $m < l$, in the order of $l - m$, changes $A_{ml}$ by the derivatives of elements of $\mathcal{O}$ and otherwise only entries farther from the diagonal, and the classes of the $1/(z - s)$, $s$ a finite point of $S$, span $\mathcal{O}/\partial\mathcal{O}$, the de Rham cohomology of the line less those points; so $A = \sum_s A_s/(z - s)$ with strictly upper triangular matrices $A_s$ over $\overline{\mathbb{Q}}$. Then $Y = Y_0 G$ with $G = \sum_w \mathrm{I}(0; s_1, \dots, s_w; z)\, A_{s_1} \cdots A_{s_w}$, a finite sum, the iterated integrals taken along the segment from the tangential base point at 0, with $\log z$ real on $(0, 1)$; their expansions at 0 have the constant term 0 for $w \ge 1$, so $Y_0$ is the constant term at 0 of $Y = \left(\sum_i q_{il} y^{(i)}\right)_l$, $q_{il} \in \mathcal{O}$. For $L_\psi$, whose series has rational coefficients, $Y_0$

is algebraic; for $P_c$, by the expansion of (ii), it is a combination with algebraic coefficients of the $F_{\sigma,i}(\mathbf{j})$ with $e_\sigma(\mathbf{j})$ at most $h$ plus the largest order of a pole of a $q_{il}$ at 0, finitely many, the slopes of the $e_\sigma$ being positive. At 1, $y = \sum_l p_l Y_l$ with $p_l \in \mathcal{O}$, whose Laurent coefficients at 1 are algebraic, and the expansion of $\mathrm{I}(0; s_1, \dots, s_w; z)$ at 1, in the powers of $1-z$ and of $\log(1-z)$, real on $(0,1)$, has as coefficients combinations with algebraic coefficients of the chains $\mathrm{I}(0; s_1, \dots, s_{w'}; 1)$, $w' \le w$, regularized at 1, by the composition of paths at the tangential base point at 1. So the constant term of $y$ at 1, which is its limit (Proposition 45.20), is a combination of 1 and of these chains, with algebraic coefficients for $L_\psi$ and with coefficients that are combinations of finitely many $F_{\sigma,i}(\mathbf{j})$ for $P_c$. When $S \subseteq \{0, \infty\} \cup \mu_M$ the chains are of level $M$, whose values lie in $\mathcal{C}^{(M)}$ (Proposition 45.5), and the period is $P_0 L_\psi(1)$. □

**Corollary 45.22 (The values of a ray are read through chains over its tree).** *As stated.*

*Proof.* (i) $\mathcal{D}_F[1] = \mathcal{D} \cdot 1$, and $\mathbb{C}(t) \otimes_{\mathbb{C}[t^{\pm 1}]} \mathcal{D}_F[1]$ is $\mathbb{C}(t)\langle\partial_t\rangle / \mathbb{C}(t)\langle\partial_t\rangle L_1$. A simple holonomic module on $\mathbb{C}^\times$ is supported at a point or is the intermediate extension of an irreducible local system on an open set; the generic fibre, which is exact, carries the first to 0 and the second to an irreducible module over $\mathbb{C}(t)$. So the factors of $L_1$ are the generic fibres of the composition factors of $\mathcal{D}_F[1]$ with a nonzero generic rank, and these are composition factors of $\mathcal{G}_F$, to which Theorem 39.56(iv) applies.

(ii) Let $F_p$ be the filtration of $\boxtimes_c \mathbb{G}_c$ by the sums of the external products of terms of the children's filtrations whose indices add to $p$; the external product is exact in each factor, so $\mathrm{gr}_p$ is the sum of the external products of graded pieces whose indices add to $p$. Let $F_p\mathbb{G}_v$ be the image of $H^0(N_v)_+ F_p$ in $\mathbb{G}_v = H^0(N_v)_+\big(\boxtimes_c \mathbb{G}_c\big)$. The direct image has no cohomology in positive degrees (the proof of Theorem 39.56(ii)), so $H^0(N_v)_+ F_{p-1} \to H^0(N_v)_+ F_p \to H^0(N_v)_+ \mathrm{gr}_p \to 0$ is exact, and $F_p\mathbb{G}_v / F_{p-1}\mathbb{G}_v$, the image of $H^0(N_v)_+ F_p$ modulo that of $H^0(N_v)_+ F_{p-1}$, is a quotient of $H^0(N_v)_+ \mathrm{gr}_p$. At the root the same holds for $R_o|_{U_o}$, restriction to $U_o$ being exact, and $H^0(R_o|_{U_o})_+\big((\boxtimes_c \mathbb{G}_c)|_{U_o}\big)$ is $\mathcal{G}_F$ (Theorem 39.56(ii)); the intersections with $\mathcal{D}_F[1]$ filter it, with graded pieces that are submodules of those of $\mathcal{G}_F$.

(iii) The fibre integrals are periods of $\mathcal{D}_F[1]$ (Definition 39.47), so they are annihilated by the annihilator of 1 and solve $L_1$, which has regular singular points only, $\infty$ among them (Deligne's regularity theorem for the Gauss–Manin connection, $\varphi$ being regular on the smooth variety $X_F$, as in the proof of Proposition 39.54). So Proposition 45.20 applies. The values are the moments of the stride's law (Proposition 40.2(i)), $\langle 1 \rangle_k = \int_{(0, f^\star]} s^k \, \mu_f(ds)$, and the image of $\mu_f$ under the reciprocal has the density $P_1$, the orthant fibre integral of 1 being the density of the image of $\omega_0$ on the orthant under $\varphi = f^{-1}$ (Definition 39.47); the change of variables $t = 1/s$ gives the moment.

(iv) The origin 0 and the direction $N(\mathbf{m}; \mathbf{s}, \mathbf{t})$ are integral, and $\kappa_{1,S}$ is $N$ times the wall $\kappa_S$ of $(\mathbf{m}; \mathbf{s}, \mathbf{t})$, the walls being linear in the data, so it is positive. With the origin 0, $\omega_0 = dy$ on the chart $\alpha_{e_0} = 1$, and the $f$ of the ray (Theorem 38.14(ii)) is $f_0^N$, with $f_0 = \alpha^{\mathbf{m}} \prod_j K_j^{-s_j} \Psi_j^{-t_j}$ positive on the orthant and of degree 0 under the scale, the data being on the balance wall; so $P_F(\mathbf{m}; \mathbf{s}, \mathbf{t}) = \int_{y_{e_0}=0} f_0 \, dy = \int_{y_{e_0}=0} f^{1/N} dy$, which converges, the walls of the data being positive. The image of $dy$ on the orthant under $\varphi = f^{-1}$ has the density $P_1$ on $(A, \infty)$, as in (iii), and $f^{1/N} = \varphi^{-1/N}$; the change of variables $t = \varphi$ gives the moment, the integrand being positive. □

**Corollary 45.23 (The letter of the closing along the tree).** *As stated.*

*Proof.* (i) The proof of Theorem 39.56(ii) uses of the data only that the maps are morphisms of smooth affine varieties, the data of a mark a closed embedding, composing to the data of each vertex, and of the modules at the marks only that they are holonomic. Here $\tilde{\lambda}_e$ has the

coordinates $\alpha_e$ and $\alpha_e^{-1}$, so it is a closed embedding; the node maps are polynomial, $\sigma_v = \sum_c \sigma_c$ and $\mu_v = \prod_c \mu_c$ composing to the sum and the product over $B_v$; and $\mathcal{O}_{\mathbb{C}^\times}\alpha_e^{m_e}$ is holonomic of rank one. So $\mathbb{L}_v = H^0(\tilde{\lambda}_v)_+\big(\boxtimes_{e\in B_v}\mathcal{O}\alpha_e^{m_e}\big)$. By Theorem 39.56(i), with this last coordinate, $K_j$ and $\Psi_j = \alpha^{\mathcal{E}}K_j^{\Psi}(\alpha^{-1})$ are the readings at the root of the data of its children, and $\sigma_o \circ \prod_c \tilde{\lambda}_c = |\alpha|$. $P_0L_\psi(z)$ is the period at the data $(\mathbf{m};\mathbf{w})$ of the letters $Q_i = (1-z)M_i|\alpha|^{g_i} + zL_i$, homogeneous of the degrees $g_i$, so on the chart $\alpha_{e_0} = 1$ it is $\int \alpha^{\mathbf{m}}\prod_i Q_i^{-w_i}dy$, as the period is (Theorem 38.14(ii)), the form being of degree 0 on the balance wall. The module of $\alpha^{\mathbf{m}}\prod_i Q_i^{-w_i}$ on the torus and the $z$-line is $\big(\boxtimes_e \mathcal{O}\alpha_e^{m_e}\boxtimes\mathcal{O}\big)\otimes(\tilde{\lambda}\times\mathrm{id})^*\mathcal{K}$, $\tilde{\lambda} = \prod_c \tilde{\lambda}_c$, on the preimage of $W$. By the projection formula for $\mathcal{K}$, a connection on $W$, and the composition of direct images in degree 0, as in Theorem 39.56(ii), its direct image to the $z$-line is $H^0(\mathrm{pr}_z)_+\big(\mathcal{K}\otimes(\boxtimes_c\mathbb{L}_c\boxtimes\mathcal{O})|_W\big)$, and on the chart the scale contributes the factor $H^1_{\mathrm{dR}}(\mathbb{C}^\times)$ in the top degree. For $0 < z < 1$ the integrand vanishes on the boundary of the simplex and exact forms integrate to zero (the proof of Theorem 45.21(i)), so the integral over the orthant is a period.

(ii) The Taylor coefficients of $L_\psi$ are rational, so the operators of bounded order and degree that annihilate it are the solutions of a linear system over $\mathbb{Q}$, and its annihilator over $\mathbb{C}(z)$ is generated by $L$. The period map from the submodule generated by the class of the integrand to the functions commutes with $\partial_z$, and its image is $\mathbb{C}(z)\langle\partial_z\rangle P_0L_\psi$, which is $\mathbb{C}(z)\langle\partial_z\rangle/\mathbb{C}(z)\langle\partial_z\rangle L$; so the factors of $L$, read over $\mathbb{C}(z)$, are subquotients of the generic fibre of the direct image of (i), and the generic fibre, which is exact, carries its composition factors to 0 or to the simple modules over $\mathbb{C}(z)$ (the proof of Corollary 45.22(i)). The rest is the proof of Theorem 39.56(iii), (iv) and of Corollary 45.22(ii) for these data: $H^0(\mathrm{pr}_z)_+$ and the $H^0(\tilde{N}_v)_+$ are right exact, $\mathcal{K}\otimes$, restriction to $W$ and the external product are exact, the external product of simple holonomic modules is simple, $\mathcal{O}_{\mathbb{C}}$ on the $z$-line is simple, and $\mathbb{L}_e$ is the direct image of a simple module under a closed embedding. □

**Definition 45.24 (The group of a module, of a function and of an input).** As stated.

*Proof.* The field generated over $\mathbb{C}(y)$ by the solutions of $N$ near $x$ and their derivatives is a Picard–Vessiot extension for $N$, the field of constants $\mathbb{C}$ being algebraically closed, and its differential automorphisms over $\mathbb{C}(y)$, acting on $\mathrm{Sol}_x N$, form a linear algebraic group (Picard–Vessiot theory over $\mathbb{C}(y)$), whose unipotent radical is a normal subgroup. When the singular points are regular singular this group is the Zariski closure of the monodromy (Schlesinger's theorem). Continuing the solutions along a path from $x$ to another point is an isomorphism of the two fields over $\mathbb{C}(y)$ that commutes with $d/dy$; it carries the group at $x$, its radical and their Lie algebras to those at the other point by a conjugation. □

**Proposition 45.25 (The group counts the functions).** *As stated.*

*Proof.* (i) The transcendence degree of a Picard–Vessiot extension of $\mathbb{C}(y)$ is the dimension of its group (Picard–Vessiot theory), and the field generated by the solutions near $x$ and their derivatives is one (the proof of Definition 45.24).

(ii) The submodules of $N$ correspond to the subspaces of $\mathrm{Sol}_x N$ that $G_N$ keeps, a submodule to the solutions that vanish on it (the Galois correspondence of Picard–Vessiot theory). So for simple $N$, $G_N$ acts irreducibly. The vectors fixed by the unipotent radical form a nonzero subspace (Kolchin's theorem), which $G_N$ keeps, the radical being normal; so it is the whole space, and the radical, acting trivially and faithfully, is trivial.

(iii) The series gives a flag of $\mathrm{Sol}_x N$ that $G_N$ keeps, whose graded pieces are the solutions of the $S_j$, irreducible by (ii). The kernel $U$ of the action on the graded pieces is normal and unipotent, so it lies in the unipotent radical; the radical acts trivially on each irreducible graded piece, as in

(ii), so it lies in $U$. The solutions of $S_1 \oplus \cdots \oplus S_r$ and their derivatives generate a subfield that $G_N$ keeps, and restriction to it maps $G_N$ onto its group, which acts on the graded pieces; so $G_N/U$ is the group of $S_1 \oplus \cdots \oplus S_r$. An element of the Lie algebra of $U$ carries each step of the flag into the next one down, so a bracket of $r$ of them vanishes.

(iv) With regular singular points Definition 45.19 applies, and by the induction in the proof of Proposition 45.20 the entries of the block $j$ are sums of terms $c\,\mathrm{I}\,\beta$ with chains I of weight at most $j-1$.

(v) $y^a$ is algebraic over $\mathbb{C}(y)$, of degree dividing the denominator of $a$, and multiplication by $y^a$ carries the solutions of the module of $W$ onto those of the module of $y^aW$. With $F$ and $F'$ the fields that the two sets of solutions generate with their derivatives, $E = F(y^a) = F'(y^a)$ is a Picard–Vessiot extension for the sum of the module of $W$ and that of $y^a$, finite over $F$ and over $F'$. Restriction maps the group of $E$ onto those of $F$ and of $F'$ with finite kernels, so it maps the identity component of the first onto the identity components of the other two. The identity component fixes the algebraic closure of $\mathbb{C}(y)$ in $E$, so it fixes $y^a$, and on $y^a \operatorname{Sol}_x$ it acts as on $\operatorname{Sol}_x$ conjugated by $y^a$: under the identification the two identity components agree. □

**Proposition 45.26 (Powers add nothing, distinct factors add).** *As stated.*

*Proof.* (i) For such a construction $\Phi$ the solutions of $\Phi(N)$ lie in $\Phi(\operatorname{Sol}_x N)$, so the field they generate with their derivatives lies in that of $N$ and is kept by $G_N$; restriction maps $G_N$ onto the group of $\Phi(N)$ (the Galois correspondence of Picard–Vessiot theory), an element $g$ acting by $\Phi(g)$, and this is a morphism of algebraic groups; a subquotient has as its group the image of the group of the module under restriction. A surjective morphism of algebraic groups is surjective on Lie algebras.

(ii) The field of $N_1 \oplus N_2$ is generated by those of $N_1$ and of $N_2$, so its group acts faithfully on $\operatorname{Sol}_x N_1 \oplus \operatorname{Sol}_x N_2$ inside $G_{N_1} \times G_{N_2}$, and it projects onto each factor by (i).

(iii) The group of $S_1 \oplus \cdots \oplus S_k$ is reductive: its unipotent radical acts trivially on each $\operatorname{Sol}_x S_j$, as in Proposition 45.25(ii). So its Lie algebra $\mathfrak{g}$ is reductive, and $\mathfrak{h} = [\mathfrak{g}, \mathfrak{g}]$ is semisimple, a sum of simple ideals, lying in $\bigoplus_j \mathfrak{s}_j$ and projecting onto each $\mathfrak{s}_j$ by (ii). A simple ideal of $\mathfrak{h}$ maps onto an ideal of $\mathfrak{s}_j$, so to 0 or isomorphically onto $\mathfrak{s}_j$; it cannot map onto two of them, which are not isomorphic, and each $\mathfrak{s}_j$ is reached. So each simple ideal of $\mathfrak{h}$ is one of the $\mathfrak{s}_j$ in its place, distinct ideals are distinct $\mathfrak{s}_j$, and $\mathfrak{h} = \bigoplus_j \mathfrak{s}_j$ (Goursat's lemma). □

**Theorem 45.27 (A full Jordan block).** *As stated.*

*Proof.* $S$ has regular singular points, so $G_S$ is the Zariski closure of its monodromy (Definition 45.24). Let $u$ be the unipotent local monodromy with a single Jordan block and $\mathsf{N} = \log u$. The closure of the powers of $u$ is a connected unipotent group, so $u$ lies in the identity component $G^\circ$ of $G_S$ and $\mathsf{N}$ in its Lie algebra. $G^\circ$ is reductive, as a normal subgroup of finite index of $G_S$, which is reductive by Proposition 45.25(ii); so the solutions are a sum of irreducible representations of $G^\circ$, and $u$, with a single Jordan block, keeps no decomposition into two nonzero subspaces, so they are irreducible. The Lie algebra of $G^\circ$ is the sum of its centre, which acts by scalars, and the semisimple derived algebra $\mathfrak{s}$, and $\mathsf{N}$, being nilpotent, lies in $\mathfrak{s}$. $\mathfrak{s}$ acts faithfully and irreducibly, so the solutions are a tensor product of faithful irreducible representations of its simple ideals; a nilpotent element acts on a tensor product of two spaces of dimension at least two with at least two Jordan blocks, since by the Jacobson–Morozov theorem each factor is a sum of irreducible representations of $\mathfrak{sl}_2$, the number of Jordan blocks is the number of irreducible summands of the product, and a product of two such sums of dimensions at least two has at least two, the product of two irreducible ones of dimensions $a, b \geq 2$ having $\min(a, b)$; so $\mathfrak{s}$ is simple. A simple Lie algebra

with a faithful irreducible representation in which a nilpotent element has a single Jordan block is $\mathfrak{sl}_2$ in $\mathrm{Sym}^{r-1}$, $\mathfrak{sl}_r$, $\mathfrak{sp}_r$ for even $r$, $\mathfrak{so}_r$ for odd $r$, or $\mathfrak{g}_2$ for $r=7$, in its standard representation (Gabber's theorem, in Katz's classification of the simple Lie algebras with a representation in which a nilpotent element has one Jordan block). The derived group of $G^\circ$ is the connected subgroup with the Lie algebra $\mathfrak{s}$, which gives (a), (b) and (c).

In case (a) let $H$ be the image of $\mathrm{SL}_2$. $G_S$ normalizes $H$, and the normalizer of $H$ in the general linear group of the solutions is $H$ times the scalars: the automorphisms of $\mathrm{SL}_2$ are inner, and the centralizer of the irreducible $H$ is the scalars (Schur's lemma). So the monodromy $\rho$ takes values in $H\cdot\mathbb{C}^\times$, whose quotient by the scalars is $\mathrm{PGL}_2$. The fundamental group of the line less finitely many points is free, so the projective representation lifts to a representation $\rho_2$ in $\mathrm{GL}_2$; $\mathrm{Sym}^{r-1}\rho_2$ and $\rho$ agree up to scalars, so $\rho=\mathrm{Sym}^{r-1}\rho_2\otimes\chi$ for a character $\chi$, and $\rho_2$ is irreducible, $\rho$ being so. By the correspondence between local systems on the line less finitely many points and modules with regular singular points, $\rho_2$ and $\chi$ are the monodromies of such modules $S_2$ and $\chi$, and $S\cong\mathrm{Sym}^{r-1}S_2\otimes\chi$, both having the monodromy $\rho$. In case (b) $\mathrm{Sp}_r$ is the stabilizer of an alternating form, $\mathrm{SO}_r$ the identity component of that of a symmetric form, and $G_2$ keeps the symmetric form of the $\mathrm{SO}_7$ that contains it; for $r\ge 3$ the standard representation of $\mathrm{SL}_r$ is not isomorphic to its dual, so $\mathrm{SL}_r$ keeps no nonzero bilinear form. □

**Theorem 45.28 (The class read by the tree and by its stride's law).** *As stated.*

*Proof.* (i) By Section 42.6, $\Gamma(k)^2P(k)=\int_0^\infty y^{k-1}\mathsf{g}(y)^s\,dy$, which is $(\mathsf{T}_{1-k;(1)}[x]\,\mathsf{g}^s\,\mathbb{I}_{(0,\infty)})((0,\infty))$, the kernel $\mathsf{Z}_{1-k;(1)}(y)$ being $y^{k-1}$ (Definition 5.2). By Proposition 40.2(i), $P(k)=\int t^k\,\mu_s(dt)=(\mathsf{T}_{k;(1)}[x]\,\iota_*\mu_s)((0,\infty))$, and $\int t^k\,\mu_s(dt)=\int t^{k-1}\,t\,\mu_s(dt)=(\mathsf{T}_{1-k;(1)}[x]\,t\,\mu_s)((0,\infty))$. By Tonelli's theorem and Definition 2.1,

$$(\mathsf{T}_{1-k;(1)}[x]\,\mathsf{g}\,\mathbb{I}_{(0,\infty)})((0,\infty))=\int_0^\infty x^{-1}e^{-x}\int_0^\infty y^{k-1}e^{-y/x}\,dy\,dx=\int_0^\infty x^{-1}e^{-x}\,\Gamma(k)\,x^k\,dx=\Gamma(k)^2.$$

For a nonnegative Borel function $F$ on $(0,\infty)$ the substitution $y=xt$ at fixed $t$ gives

$$\iint F(xt)\,\mathsf{g}(x)\,dx\;t\,\mu_s(dt)=\int_0^\infty F(y)\int\mathsf{g}(y/t)\,\mu_s(dt)\,dy=\int_0^\infty F(y)\,\mathsf{g}(y)^s\,dy$$

by Proposition 42.17(i); so $\big(\mathsf{g}\,\mathbb{I}_{(0,\infty)}\big)\boxtimes(t\,\mu_s)=\mathsf{g}^s\,\mathbb{I}_{(0,\infty)}$ (Definition 5.30), and Theorem 5.31(ii) at the index $1-k$ is the product $\Gamma(k)^2\cdot P(k)$ of the first display. The image of $\mu_s$ under $\iota$ is the image of the Lebesgue input of the chart under $\varphi=f^{-1}$, whose density on $(A,\infty)$ is $P_1$ (Definition 39.47; the proof of Corollary 45.22(iii)), with $A=K(\mathbf{1})^2=s^2$ at the strength one (Theorem 38.15).

(ii) The ladder and the exchange relation of the Gramma give $\vartheta^2\mathsf{g}=y\,\mathsf{g}$ (Section 42.6), that is $y^2\mathsf{g}''+y\,\mathsf{g}'-y\,\mathsf{g}=0$. By Lemma 20.29(i) at $l=0$,

$$\mathsf{g}(y)=\sum_{k\ge0}\frac{y^k}{k!^2}\big(2(\log\Gamma)'(1+k)-\log y\big),$$

so continuing $\mathsf{g}$ once around 0 turns it into $\mathsf{g}-2\pi i\,\mathsf{h}$ with $\mathsf{h}(y)=\sum_{k\ge0}y^k/k!^2$, an entire function and, as a difference of two solutions, a solution. If $\mathsf{g}'=a\,\mathsf{g}$ with $a\in\mathbb{C}(y)$, the continuation would satisfy the same equation, whose solutions near a point form a line, and $\mathsf{g}-2\pi i\,\mathsf{h}=c\,\mathsf{g}$ for a constant $c$; $\mathsf{h}(0)=1$ gives $c\ne1$, and then $\mathsf{g}=2\pi i\,\mathsf{h}/(1-c)$ would return to itself around 0. So $\mathsf{g}$ satisfies no equation of order one, its module $M$ is $\mathbb{C}(y)\langle\partial_y\rangle/\mathbb{C}(y)\langle\partial_y\rangle(\vartheta^2-y)$, of rank two, and its solutions near a point of $(0,\infty)$ are $\mathbb{C}\mathsf{g}+\mathbb{C}\mathsf{h}$.

The Wronskian $w$ of two solutions satisfies $w' = -y^{-1}w$, so $w = c\,y^{-1} \in \mathbb{C}(y)$, and an element of $G_M$, fixing $w$, acts with determinant one: $G_M \subseteq \mathrm{SL}_2$. Continuation along a loop is an automorphism of the field of $M$ over $\mathbb{C}(y)$ that commutes with $d/dy$, so the loop around 0 gives an element $u$ of $G_M$, $u\mathsf{g} = \mathsf{g} - 2\pi i\,\mathsf{h}$ and $u\mathsf{h} = \mathsf{h}$, unipotent and not 1; the closure of its powers is a group $\mathbb{G}_a$ in the identity component $G^\circ$ of $G_M$. The connected subgroups of $\mathrm{SL}_2$ that contain a unipotent element other than 1 are $\mathbb{G}_a$, the Borel subgroups and $\mathrm{SL}_2$. If $G^\circ$ were one of the first two, it would keep exactly one line, the fixed line $\mathbb{C}\mathsf{h}$ of $u$, and $G_M$, which normalizes $G^\circ$, would keep it: every element of $G_M$ would multiply $\mathsf{h}$ by a constant, and $\mathsf{h}'/\mathsf{h}$, fixed by $G_M$, would lie in $\mathbb{C}(y)$ (the Galois correspondence of Picard–Vessiot theory). An entire function whose logarithmic derivative is rational has finitely many zeros, so it is $Pe^Q$ with polynomials $P$ and $Q$; $\mathsf{h}$ is not a polynomial, and

$$|\mathsf{h}(y)| \le \sum_{k\ge0} \frac{|y|^k}{k!^2} \le \Big(\sum_{k\ge0} \frac{|y|^{k/2}}{k!}\Big)^2 = e^{2\sqrt{|y|}}$$

leaves no room for $\deg Q \ge 1$, under which $|e^Q|$ grows like $e^{c|y|^{\deg Q}}$, $c > 0$, along a ray. So $G^\circ = \mathrm{SL}_2$, and $G_M = \mathrm{SL}_2$.

The solutions of the module of a function are spanned by its images under the group: an operator over $\mathbb{C}(y)$ that kills the function kills its images, and the monic operator whose solutions are their span has coefficients fixed by the group, hence in $\mathbb{C}(y)$. $\mathrm{SL}_2$ moves $\mathsf{g}$ to every nonzero solution, so the solutions of the module of $\mathsf{g}^s$ are spanned by the $s$-th powers of all the solutions of $M$: they are the image of $\mathrm{Sym}^s(\mathbb{C}\mathsf{g}+\mathbb{C}\mathsf{h})$ under the multiplication of solutions. The multiplication is injective there, since a relation $\sum_j c_j \mathsf{g}^j\mathsf{h}^{s-j} = 0$ with some $c_j \ne 0$ would make $\mathsf{g}/\mathsf{h}$ a root of a nonzero polynomial with constant coefficients, hence a constant, which it is not, $\mathsf{h}$ returning to itself around 0 and $\mathsf{g}$ not. So the module of $\mathsf{g}^s$ has rank $s+1$. The multiplication carries $\mathrm{Sym}^s M$ onto a module of rank at most $s+1$ that contains $\mathsf{g}^s$, hence onto the module of $\mathsf{g}^s$, and isomorphically, the ranks being equal. By Proposition 45.26(i) the group of $\mathsf{g}^s$ is the image of $G_M = \mathrm{SL}_2$ acting on $\mathrm{Sym}^s$ of its standard representation, where the kernel is $\{\pm1\}$ for even $s$ and 1 for odd $s$; its dimension is three.

(iii) The module of $P_1$ over $\mathbb{C}(t)$ is generated by a nonzero period of the module that 1 generates along the centre ray of $\binom{[s]}{1}$, whose periods are exactly $V_s$ and which is irreducible off its singular values (Proposition 42.29(ii)); so it is that module read over $\mathbb{C}(t)$, and its group is the group of $V_s$. The density of $t\,\mu_s$ on $(0, s^{-2})$ is $t\,\varrho_f(t) = t^{-1}P_1(t^{-1})$; the substitution $t \mapsto t^{-1}$ and the multiplication by a nonzero element of $\mathbb{C}(t)$ change neither the Picard–Vessiot field over $\mathbb{C}(t)$ nor its group, so $\iota_*\mu_s$ and $t\,\mu_s$ have the group of $V_s$.

Let $s \ge 5$ and $r = s - 1 \ge 4$. By Theorem 42.28(i) and (ii) that module is simple of rank $r$, and its monodromy at $\infty$ is unipotent with a single Jordan block; its singular points are regular singular, as are those of the module of 1 along every such ray (the proof of Corollary 45.22(iii)); so Theorem 45.27 applies; by Theorem 42.28(iii) the monodromy $T$ around a sign value has $\mathrm{rank}(T-1) = 1$. In case (a) of Theorem 45.27, $T = c\,\mathrm{Sym}^{r-1}(h)$ with $c$ the monodromy of $\chi$ and $h$ that of $S_2$ around the same loop. If $h$ is a scalar, $T$ is a scalar and $\mathrm{rank}(T-\mu) \in \{0, r\}$ for every $\mu$. If $h$ is diagonalizable with eigenvalues $\lambda_1 \ne \lambda_2$, $T$ is diagonalizable with the eigenvalues $c\lambda_1^{r-1-j}\lambda_2^j$, $0 \le j \le r-1$, and $\mathrm{rank}(T-\mu) = 1$ asks $r-1$ of them to equal $\mu$; any $r-1$ of the $r$ indices contain two consecutive ones, $r-1$ being at least 3, which gives $\lambda_1 = \lambda_2$. If $h$ is not diagonalizable, $h = \lambda(1+\mathsf{n})$ with $\mathsf{n} \ne 0$ nilpotent, and $\mathrm{Sym}^{r-1}(1+\mathsf{n})$ is unipotent with a single Jordan block of size $r$, so $\mathrm{rank}(T-\mu) \ge r-1 \ge 3$. So case (a) is excluded. In case (b) with $r = 7$ the group normalizes the derived group $G_2$ of its identity component, and the normalizer of $G_2$ in $\mathrm{GL}_7$ is $G_2$ times the scalars, its automorphisms being inner and its centralizer the scalars (Schur's lemma); so

$T = c\,h$ with $h \in G_2 \subset \mathrm{SO}_7$ and $\operatorname{rank}(h - c^{-1}) = 1$. Then $c^{-1}$ is an eigenvalue of $h$ of multiplicity at least six, and the eigenvalues of an element of $\mathrm{SO}_7$, with their multiplicities, are closed under inversion, so $c^{-1} = \pm 1$. For $c^{-1} = 1$, $h$ is unipotent with $\operatorname{rank}(h - 1) = 1$, a transvection, which no orthogonal group contains, the Jordan blocks of even size of an orthogonal unipotent element coming in pairs; for $c^{-1} = -1$, $\det h = 1$ makes the seventh eigenvalue 1, and $h$ is an involution fixing a line, while every involution of $G_2$ fixes a space of dimension three. So the derived group is $\mathrm{Sp}_r$ for even $r$, that is for odd $s$, $\mathrm{SO}_r$ for odd $r$, or $\mathrm{SL}_r$, of the dimensions $r(r+1)/2$, $r(r-1)/2$ and $r^2 - 1$, each at least $r(r-1)/2 = (s-1)(s-2)/2$. □

**Corollary 45.29 (The word along the canonical tree).** *As stated.*

*Proof.* (i) At a full lean one letter is active, so the hypothesis of Corollary 26.11 holds on every hinge tree; on the canonical hinge tree of a table whose weights factor across its bunches of width one, the series and parallel nodes contribute stacked Gammas times monomials in their weights (Proposition 26.9), and the period is the product of these, of the reciprocals of the Beta corners of the edges and of the periods of the prime nodes (Corollary 26.11). The dimension of the share side of a prime node is Corollary 26.8, and equal factors of a product give a power.

(ii) is Proposition 42.26(iii), where a cell of at least two marks enters through $\widehat{\mathsf{V}}_{\mathcal{A}}(k;\ell)$, the period of the family of its two hinge minors. The word of a parallel class of $s \geq 2$ marks is

$$\Gamma(\ell)^{-2} \int_0^\infty y^{\ell-1} \mathsf{G}_{0;(1,-1)}(1,y)^s \, dy = \Gamma(\ell)^{-2} \left(\mathsf{T}_{1-\ell;(1)}[x] \, \mathsf{G}_{0;(1,-1)}(1,x)^s \, \mathbb{I}_{(0,\infty)}\right)((0,\infty))$$

(Proposition 42.26(iii), Definition 5.2), and the group of its input is Theorem 45.28(ii). □

**Corollary 45.30 (The far end of the pencil).** *As stated.*

*Proof.* Let $P^{(1)}$ be the part of $P_1$ on which $T_\infty$ has the eigenvalue 1; it lies in the solutions of $W$, which $T_\infty$ keeps. By Proposition 39.54, $(T_\infty - 1)^{n-2} P^{(1)}$ is a nonzero multiple of the torus fibre integral $T$, which is not 0, its first coefficient being $\mathsf{C}_0(1) = 1$; so $P^{(1)}, (T_\infty - 1)P^{(1)}, \ldots, (T_\infty - 1)^{n-2} P^{(1)}$ are independent, and $r_P \geq n - 1$. If $r_P = n - 1$ they span the solutions of $W$, and $T_\infty$ is unipotent there with a single Jordan block.

(i) The submodules of $W$ correspond, reversing inclusion, to the subspaces of its solutions that the monodromy keeps (the proof of Proposition 45.25(ii)), and these are kept by $T_\infty$. A unipotent map with a single Jordan block keeps only the kernels of the powers of itself less the identity, which form a chain, and on the quotient of two of them it has a single Jordan block.

(ii) A factor of rank $b \geq 2$ is simple, with regular singular points (the proof of Corollary 45.22(iii)) and the unipotent local monodromy $T_\infty$ of a single Jordan block of size $b$, so Theorem 45.27 applies. The group of a module of rank one lies in $\mathrm{GL}_1 = \mathbb{G}_m$.

(iii) Proposition 45.25(iii), and Proposition 45.26(ii) applied to the sum of the factors one factor at a time. □

**Proposition 45.31 (The group is computable).** *As stated.*

*Proof.* Hrushovski's algorithm computes the differential Galois group of a linear differential equation with coefficients in $\overline{\mathbb{Q}}(y)$, whatever its singular points, as an algebraic group given by equations; its Lie algebra, its dimension and its unipotent radical follow by linear algebra. The relation of one mark is $y^2 u'' + y\,u' - y\,u = 0$. With algebraic weights the pencil $\varphi$, the variety $X_F$ and the presentation of $\mathcal{G}_F$ in Definition 39.47 are defined over $\overline{\mathbb{Q}}$, and the module that 1 generates over $\mathbb{C}(t)$ is the one it generates over $\overline{\mathbb{Q}}(t)$ with the scalars extended; so the first relation among $1, \partial_t, \partial_t^2, \ldots$

over $\mathbb{C}(t)$ is a relation over $\overline{\mathbb{Q}}(t)$, and it is $L_1$. A solution of $L_1$ generates a quotient of the module of $L_1$, whose group is the image of that of $L_1$ (Proposition 45.26(i)). □

**Definition 45.32 (The inputs and the group of a word).** As stated.

*Proof.* The group of a sum of modules of finite rank over $\mathbb{C}(y)$ is defined by Definition 45.24; the modules here are those that the densities generate, of finite rank by the hypothesis. □

**Lemma 45.33 (The transforms satisfy linear equations).** *As stated.*

*Proof.* Let $\mathcal{B}_x$ be the table obtained by gluing $\mathcal{A}$ at its hinge to the one-mark cell of ratio $x > 0$ (Definition 23.10), at the masses $k$ and the exponents $(k, k)$, and $P(x)$ its period. Its weights are those of $\mathcal{A}$ times monomials in $x^{1/2}$, positive for $x > 0$, and its walls, which do not see the weights, are positive, the table being the connected table $\mathcal{A}$ with its hinge read as a mark (the proof of Proposition 42.26(i)). So $P$ converges on $(0, \infty)$, and the argument of the proof of Theorem 45.21(ii), the letters now carrying these weights, gives a nonzero linear differential equation for $P$ over $\mathbb{C}(x^{1/2})$, and so over $\mathbb{C}(x)$, the derivatives being integrals of insertions whose classes modulo exact forms lie in a twisted de Rham cohomology of finite dimension. By Theorem 23.9(ii) and Definition 23.19(1), $P(x) = \Gamma(k)^{-2} x^{-k} \int_0^\infty y^{k-1} \check{f}_{\mathcal{A}}(y)\, \mathsf{G}_{0;(1,-1)}(1, y/x)\, dy$, and by Tonelli's theorem, for $0 < \operatorname{Re} w < k$,

$$\int_0^\infty x^{w-1} P(x)\, dx = \frac{\Gamma(k-w)^2}{\Gamma(k)^2} \int_0^\infty y^{w-1} \check{f}_{\mathcal{A}}(y)\, dy = \frac{\Gamma(k-w)^2 \Gamma(w)^2}{\Gamma(k)^2}\, \widehat{\mathsf{V}}_{\mathcal{A}}(k; w),$$

the last by Proposition 42.26(iv). The equation of $P$ has regular singular points only (the proof of Theorem 45.21(iii)), so $P$ has expansions in powers and logarithms at 0 and at $\infty$, and its Mellin transform continues meromorphically to the plane, as the Mellin transform $\mathcal{M}$ regularized by those expansions, which is linear and has $\mathcal{M}[x^i (x\partial_x)^j F](w) = (-w - i)^j\, \mathcal{M}[F](w + i)$. This is the correspondence of Proposition 39.48 read with the kernel $x^{w-1}$. It carries the equation of $P$ to a nonzero linear recurrence in $w$ with polynomial coefficients for $\mathcal{M}[P]$, an identity of meromorphic functions. The dual transform has its lattice expansion at 0 (Proposition 23.8(v)) and is the Laplace transform of its hinge measure (Proposition 23.8(ii)), so $|y^m \check{f}_{\mathcal{A}}^{(m)}(y)| \le C_m \check{f}_{\mathcal{A}}(y/2)$ with $C_m = \sup_{u>0} u^m e^{-u/2}$; it is decreasing and $\int_1^\infty y^{w-1} \check{f}_{\mathcal{A}}(y)\, dy$ is finite for $0 < w < k$, so it is $o(y^{-w})$ at $\infty$ for every $w < k$. So each $y^i (y\partial_y)^j \check{f}_{\mathcal{A}}$ has a Mellin transform regularized at 0 by that expansion, meromorphic on $\operatorname{Re} w < k - i$, the rule above holds for these, and $\mathcal{M}[\check{f}_{\mathcal{A}}] = \Gamma(k)^2 \Gamma(k-w)^{-2} \mathcal{M}[P]$ on $\operatorname{Re} w < k$. The stacked Gamma $\Gamma(k)^2 \Gamma(k-w)^{-2}$ satisfies a recurrence of order one with polynomial coefficients, so $\sum_{i=0}^h c_i(w)\, \mathcal{M}[\check{f}_{\mathcal{A}}](w + i) = 0$ on $\operatorname{Re} w < k - h$ with polynomial coefficients not all zero, which is $\mathcal{M}[L\check{f}_{\mathcal{A}}] = 0$ there for $L = \sum_i c_i(-y\partial_y)\, y^i \neq 0$. A function with an expansion in powers and logarithms at 0 whose regularized Mellin transform vanishes on a half-plane has no term in that expansion, each term giving a pole, and then vanishes by the inversion of the Mellin transform; so $L\check{f}_{\mathcal{A}} = 0$. The transform $f_{\mathcal{A}}$ is the dual transform of the dual cell (Proposition 23.8(iv)), connected and plain, to which the same applies. □

**Theorem 45.34 (What the word along the canonical tree reads).** *As stated.*

*Proof.* (i) By Proposition 42.26(iii) and (v) to (vii) the value is the word described: the fibre of the root or its one integral, the fibre of every node read by (v) with the word of each of its cells set at its load, the one Trailmoment of (vi) at a node of $S$, and the integral of $\mathsf{g}^s$ for a class, by (iii) for a parallel class and (vi) for a series class. Below a node of $S$ a cell other than a mark enters

through its dual transform under a parallel node and its transform under a series node, which by Proposition 42.26(iv) and the inversion of the Mellin transform is

$$\check f_{\mathcal A}(y)=\frac{1}{2\pi i}\int_{\operatorname{Re} z=z_0} y^{-z}\,\Gamma(z)^2\,\widehat{\mathrm V}_{\mathcal A}(k;z)\,dz,\qquad f_{\mathcal A}(y)=\frac{1}{2\pi i}\int_{\operatorname{Re} z=z_0} y^{-z}\,\Gamma(z)^2\,\widehat{\mathrm V}_{\mathcal A}(k;k-z)\,dz,$$

$0<z_0<k$, a fibre integral of dimension one of a stacked Gamma times the word of the cell, absolutely convergent because $\Gamma(z)^2$ decays exponentially along the line and $|\widehat{\mathrm V}_{\mathcal A}(k;z)|\le\widehat{\mathrm V}_{\mathcal A}(k;\operatorname{Re} z)$. Its integrals over half-lines are the Trailmoments listed, and everything else in it is stacked Gammas, monomials in the weights and the norms, and fibre integrals.

(ii) $\mathsf g$ satisfies $\vartheta^2\mathsf g=y\,\mathsf g$ (Theorem 45.28(ii)), its dilations the dilated equations, and the transforms the equations of Lemma 45.33. The module of a product $\prod_c F_c^{m_c}$ of such functions lies in the image of $\bigotimes_c \mathrm{Sym}^{m_c}$ of their modules under the multiplication of functions, a module closed under $\partial_y$ that contains the product; so it is a subquotient of a construction from the sum $N$ of the modules of the distinct $F_c$, and so is the sum of the modules of all the inputs. By Proposition 45.26(i) the group of the word is a quotient of $G_N$. In the canonical hinge tree no two series nodes and no two parallel nodes are adjacent (Theorem 23.16), so a cell other than a mark directly below a series or parallel node is rooted at a prime node or at a node of the other kind. With $S$ empty the inputs are the $\mathsf g^s$, whose modules are the $\mathrm{Sym}^s$ of the module of $\mathsf g$ (Theorem 45.28(ii)); the group of their sum is the image of the group $\mathrm{SL}_2$ of $\mathsf g$ acting on the sum of the $\mathrm{Sym}^s$ (Proposition 45.26(i)), with the kernel $\{\pm1\}$ when every $s$ is even and 1 otherwise, $-1$ acting on $\mathrm{Sym}^s$ by $(-1)^s$.

(iii) The work counts each distinct integral once by its dimension (Definition 33.2): the fibre of the root, of dimension $2(N_{\mathcal H}-1)-(n_{\mathcal H}-1)$ (Proposition 42.26(ii), the core being connected by Proposition 21.3(iv)); the fibre of every other node read by (v), of dimension $2(N_{\mathcal C}+N_{\mathcal D})-p-3=2N_v-n_v-2$, the bases of the node being those of $\mathcal C$ with $h$ added and those of $\mathcal D$, and its marks $h,h_1,\dots,h_p$, which for a series or parallel node with $p$ cells directly below it is $p-1$; for each size $s$ of class the one integral $\int_0^\infty y^{z-1}\mathsf g(y)^s\,dy$ of the index $z$, which is also the Trailmoment of (vi) at a class in $S$; for every other node of $S$ its Trailmoment; and for each transform that a node of $S$ reads of a cell other than a mark, the dual transform under a parallel node and the transform under a series node (Proposition 42.26(vi)), its inverse Mellin transform of (i), the two of one cell being different integrals. Equal subtrees have equal words, and a word that recurs is counted once; with every series and parallel node in $S$ the multiplicities enter only as the powers of transforms. A distinct subtree other than a mark then brings at most two integrals besides the fibre of its root when that is a prime node: its Trailmoment or the integral of its class and one transform if it is rooted at a series or parallel node, which lies directly below no node of its own kind (Theorem 23.16), and its two transforms if it is rooted at a prime node; this is the bound. □

## 50.45 Chapter 46: finite algebra for scale reads

The letters of this chapter have complex coefficients and no roots on $[0,\infty)$. Two facts carry the proofs. First, along $[0,\infty)$ a letter $P_j$ has a continuous argument with limits at 0 and at $\infty$, so for $x\to\infty$, $P_j(x)^{-w_j}=C_jx^{-w_j\deg P_j}(1+O(x^{-1}))$ with $C_j\ne0$, and $|P_j^{-w_j}|$ is $|P_j|^{-\operatorname{Re} w_j}$ up to factors bounded above and below on $[0,\infty)$: a read converges absolutely exactly where it does for positive letters of the same degrees. Second, the reduction of 46.7 is algebraic in $\mathbb F'$ and does not use the values of the letters.

**Definition 46.6 (Insertions and reads on one scale).** As stated, for letters without roots on $[0,\infty)$.

*Proof.* $P_j(0) \neq 0$ and $P_j$ does not vanish on $[0,\infty)$, so a continuous logarithm of $P_j$ along $[0,\infty)$ starting at the principal logarithm of $P_j(0)$ exists and is unique, and it defines $P_j^{-w_j}$; on $(0,\infty)$ it is locally a holomorphic branch, so $\big(P_j^{-w_j}\big)' = -w_j P_j' P_j^{-w_j-1}$, and the product rule gives $\big(x^\kappa \prod_j P_j^{-w_j} g\big)' = x^{\kappa-1} \prod_j P_j^{-w_j}\, \eth g$. The squarefree decomposition and the gcd splitting are computed in $\mathbb{F}[x]$ by gcds and exact divisions, so the $q_k$ lie in $\mathbb{F}[x]$, and they divide $P_1 \cdots P_N$, so none vanishes at 0 or on $[0,\infty)$. For $\xi_\ell \in \mathbb{C} \setminus (-\infty, 0]$, $x + \xi_\ell$ stays in $\mathbb{C} \setminus (-\infty, 0]$ for $x \geq 0$, so the continuous branch is the principal one, that of 15.1 and 46.1. □

**Theorem 46.7 (Reduction by the scale exchange).** *As stated, for letters without roots on* $[0,\infty)$.

*Proof.* Write $\mu_k = \sum_j m_{jk} w_j$, so that $\eth = x\frac{d}{dx} + \kappa - \sum_k \mu_k \frac{x q_k'}{q_k}$ and $\mathfrak{A} = \mathbb{F}'[x, x^{-1}, Q^{-1}]$. Every $R \in \mathfrak{A}$ is a Laurent polynomial in $x$ plus principal parts at the $q_k$.

*Three reductions.* (a) If $R$ has a pole of order $h+1 \geq 2$ at $q_k$ with leading numerator $s$, $\deg s < \deg q_k$, then

$$\eth\big(r\, q_k^{-h}\big) = -(h+\mu_k)\, \frac{x q_k' r}{q_k^{h+1}} + O\big(q_k^{-h}\big),$$

plus simple poles at the other $q_{k'}$ and no polynomial part and no pole at 0, $r\, q_k^{-h}$ vanishing at $\infty$ and being regular at 0; $x q_k'$ is prime to $q_k$, $q_k$ being squarefree and $q_k(0) \neq 0$, so the extended Euclidean algorithm gives $r$ with $-(h+\mu_k) x q_k' r \equiv s \bmod q_k$, one division by $h + \mu_k$. (b) If $R$ has a pole $x^{-r}$, $r \geq 1$, at 0, $\eth(x^{-r}) = (\kappa - r)x^{-r} - \sum_k \mu_k x^{1-r} q_k'/q_k$, whose other terms have poles of order below $r$ at 0, simple poles at the $q_k$ and no polynomial part. (c) If the polynomial part of $R$ has degree $a \geq 0$, $\eth(x^a) = \big(\kappa + a - \sum_j w_j \deg P_j\big) x^a$ plus a polynomial of degree below $a$ plus simple poles at the $q_k$, since $x q_k'/q_k = \deg q_k + O(x^{-1})$ and $\sum_k \mu_k \deg q_k = \sum_j w_j \deg P_j$. Subtracting $\eth$ of a multiple at each step lowers the order of the pole, resp. the degree, and leaves $\sum_{i<D} c_i x^i/Q$ once the poles at the $q_k$ are simple and there is no pole at 0 and no polynomial part; each step divides by one of the three numbers of the hypothesis.

*Uniqueness.* Let $\sum_i c_i x^i/Q = \eth g$ with $g \in \mathfrak{A}$, $g \neq 0$. A pole of order $h \geq 1$ of $g$ at $q_k$ gives $\eth g$ a pole of order $h+1$ by (a), but the left side has simple poles; so $g$ is a Laurent polynomial. A pole $x^{-r}$ of $g$ at 0 gives $\eth g$ the pole $(\kappa - r)x^{-r}$, but $Q(0) \neq 0$; so $g$ is a polynomial, of degree $a \geq 0$, and by (c) $\eth g$ is of order $x^a$ at $\infty$, while the left side is $O(x^{-1})$. So $g = 0$ and every $c_i = 0$: $\dim_{\mathbb{F}'} \mathfrak{A}/\eth\mathfrak{A} = D$.

*Reads.* Let the reads of $R$ and of the $x^i/Q$ converge absolutely, so that $\Phi = x^\kappa \prod_j P_j^{-w_j} g$ has the absolutely integrable derivative $x^{\kappa-1} \prod_j P_j^{-w_j}\big(R - \sum_i c_i x^i/Q\big)$ and limits at 0 and at $\infty$. With $\nu_0$ the order of $g$ at 0 and $a$ its degree at $\infty$, $\Phi = c_0 x^{\kappa+\nu_0}(1 + o(1))$ at 0 and $\Phi = c_\infty x^{\kappa+a-\sum_j w_j \deg P_j}(1 + o(1))$ at $\infty$, with $c_0, c_\infty \neq 0$ by the first fact above; a limit exists only if the exponent has negative real part at $\infty$, positive at 0, or is 0. At 0: if $\nu_0 \leq -1$ the exponent $\kappa - |\nu_0|$ is not 0 by hypothesis, and if $\nu_0 \geq 0$ its real part is positive since the read of $1/Q$ converges, $\operatorname{Re}\kappa > 0$. At $\infty$: if $a \geq 0$ the exponent is not 0 by hypothesis, and if $a \leq -1$ its real part is negative since the read of $x^{D-1}/Q$ converges, $\operatorname{Re}(\kappa - \sum_j w_j \deg P_j) < 1$. So both limits are 0, $\langle \eth g \rangle = 0$, and $\langle R \rangle = \sum_i c_i \langle x^i/Q \rangle$; at symbolic exponents the identity holds on the open set of absolute convergence and continues. For the linear letters, $Q = \prod_\ell (x + \xi_\ell)$ is squarefree, the $x^i/Q$, $i < |J|$, and the $(x+\xi_\ell)^{-1}$ span the same space by partial fractions, and the three numbers are $\kappa - r$, $\kappa + a - |\mathbf{w}|$ and $w_\ell + h$, nonzero in $\mathbb{K}$. □

**Definition 46.8 (Deformation; masters).** As stated, along every admissible direction.

*Proof.* Along the deformation the three numbers of 46.7 are $\kappa - r + \lambda_0\varepsilon$, $\big(\kappa + a - \sum_j w_j \deg P_j\big) + \varepsilon\big(\lambda_0 - \sum_j \lambda_j \deg P_j\big)$ and $(\mu_k + h) + \varepsilon \sum_j m_{jk}\lambda_j$, polynomials in $\varepsilon$ with nonzero slopes, hence nonzero in $\mathbb{F}'(\lambda_0, \boldsymbol{\lambda})(\varepsilon)$. At large real $\varepsilon$ the exponent of a read at 0 has real part growing like $\lambda_0\varepsilon$ and at $\infty$ falling like $(\sum_j \lambda_j \deg P_j - \lambda_0)\varepsilon$, the moduli of the powers being those of positive letters up to bounded factors; so finitely many reads converge absolutely together, the coefficients of the reductions have no pole there, and 46.7 gives the identities there, hence as meromorphic functions of $\varepsilon$, the reads continuing meromorphically by the expansions of the integrands at 0 and at $\infty$ (8.4). $Q$ divides $P_1 \cdots P_N$. The conditions are strict inequalities and finitely many inequations, invariant under positive scaling, so the admissible directions form an open cone; $(\frac{1}{2}; \mathbf{1})$ is in it, since $\sum_j \deg P_j \geq 1 > \frac{1}{2}$ and $\sum_j m_{jk} \geq 1$ for every $k$, $q_k$ dividing some $P_j$. □

**Lemma 46.9 (Simple poles at a corner).** *As stated, along every admissible direction and for letters without roots on* $[0, \infty)$.

*Proof.* The three reductions of the proof of 46.7 act on separate parts of an insertion: (a) on the poles of order at least two at one $q_k$, (b) on the pole at 0, (c) on the polynomial part, and each step feeds only the lower orders of its own part and the simple poles, which form the remainder $\sum_i c_i x^i/Q$. In (c) the step at the degree $a$ divides by $\big(\kappa + a - \sum_j w_j \deg P_j\big) + \varepsilon\big(\lambda_0 - \sum_j \lambda_j \deg P_j\big)$, which vanishes at $\varepsilon = 0$ for at most one $a$; in (b) the step at $r$ divides by $\kappa - r + \lambda_0\varepsilon$, vanishing at 0 for at most one $r$; in (a), for each $k$, the step at $h$ divides by $\mu_k + h + \varepsilon \sum_j m_{jk}\lambda_j$, vanishing at 0 for at most one $h$. Along each chain every other division is by a number holomorphic and nonzero near $\varepsilon = 0$. So for $R$ holomorphic in $\varepsilon$, whose principal parts are then holomorphic in $\varepsilon$, every coefficient is a sum of terms with at most one division vanishing at $\varepsilon = 0$: at most a simple pole.

For a master write $H_i = \int_0^1 + \int_1^\infty$. At 0 the integrand is $x^{\kappa + \lambda_0\varepsilon - 1}\phi_\varepsilon(x)$ with $\phi_\varepsilon$ analytic at 0 and holomorphic in $\varepsilon$, and 8.4 gives poles at $\kappa + \lambda_0\varepsilon + l = 0$, $l \in \mathbb{N}$, all simple and at distinct points, at most one of them at $\varepsilon = 0$. At $\infty$, by the first fact above, the integrand is $x^{e(\varepsilon)-1}$ times a power series in $x^{-1}$ holomorphic in $\varepsilon$, $e(\varepsilon) = \kappa - \sum_j w_j \deg P_j + i - D + \varepsilon\big(\lambda_0 - \sum_j \lambda_j \deg P_j\big)$, and $x \mapsto 1/x$ gives simple poles at $e(\varepsilon) = l$, $l \in \mathbb{N}$, at most one at $\varepsilon = 0$. So $H_i$ has at most a simple pole there. □

**Theorem 46.10 (Finitely many numbers at every corner).** *As stated, along every admissible direction and for letters without roots on* $[0, \infty)$.

*Proof.* Absolute convergence of $\langle R \rangle_{\kappa, \mathbf{w}; \mathbf{P}}$ is a strict inequality at 0 and one at $\infty$, so the deformed read converges absolutely for $\varepsilon$ near 0 and is holomorphic there, with the value $\langle R \rangle_{\kappa, \mathbf{w}; \mathbf{P}}$ at 0. By 46.8 it equals $\sum_i c_i(\varepsilon) H_i(\varepsilon)$ as a meromorphic function, and by 46.9 each product has at most a double pole, with the constant term $c_{i,-1}h_{i,1} + c_{i,0}h_{i,0} + c_{i,1}h_{i,-1}$. The constant term of a function holomorphic at 0 is its value. The masters do not involve $R$. The derivative $\partial_\varepsilon^b$ of $x^{\lambda_0\varepsilon} \prod_j P_j^{-\varepsilon\lambda_j}$ is $\big(\lambda_0 \log x - \sum_j \lambda_j \log P_j\big)^b$ times it, the logarithms continuous along $[0, \infty)$, so the read of $R\big(\lambda_0 \log x - \sum_j \lambda_j \log P_j\big)^b$ is $b!$ times the coefficient of $\varepsilon^b$ of $\sum_i c_i H_i$, which uses $h_{i,-1}, \ldots, h_{i,b+1}$: $(b+3)D$ numbers. For the last sentence write $\ell_0 = \log x$ and $\ell_j = \log P_j$. The $b$-th powers of the linear forms $\lambda_0\ell_0 - \sum_j \lambda_j\ell_j$, for $(\lambda_0; \boldsymbol{\lambda})$ in a dense subset of a nonempty open set, span the forms of degree $b$ in $(\ell_0, \ldots, \ell_N)$: a linear functional vanishing on them is a polynomial in $(\lambda_0; \boldsymbol{\lambda})$ vanishing on a dense subset of an open set, hence zero, and its coefficients are, up to multinomial factors, its values on the monomials. The rational admissible directions are such a set, and a linear system with rational coefficients that has a solution has a rational one; so $\ell_0^{k_0} \prod_j \ell_j^{k_j}$

is a rational combination of finitely many such powers, and its read, absolutely convergent with that of $R$, is the same combination of reads along these directions. □

**Theorem 46.11 (Integer rays have certified recurrences).** *As stated, for letters without roots on* $[0,\infty)$.

*Proof.* Over $\mathbb{F}'(n,\varepsilon)$ the three numbers of 46.7 at the data $\big(\kappa+an+\frac{\varepsilon}{2},\mathbf{w}+n\mathbf{b}+\varepsilon\mathbf{1}\big)$ have the nonzero slopes $\frac12$, $\frac12-\sum_j \deg P_j$ and $\sum_j m_{jk}$ in $\varepsilon$. Reduce the $D+1$ insertions $y_k=x^{ak}\prod_j P_j^{-b_jk}$: $y_k=\eth_n g_k+\sum_i c_{ki}x^i/Q$, where the denominators of $g_k$ and of $c_{ki}$ are products of the three numbers. The $(D+1)\times D$ matrix $(c_{ki})$ has a nonzero vector in its left kernel, found by elimination; clearing denominators gives polynomials $p_k(n,\varepsilon)$, and dividing by the largest power of $\varepsilon$ that divides all of them makes the $p_k(n,0)$ not all zero. Then $g=\sum_k p_kg_k$, divided by that power of $\varepsilon$, has $\eth_n g=\sum_k p_ky_k$, and its denominators are products of the three numbers and a power of $\varepsilon$, none vanishing identically in $\varepsilon$ at an integer $n$. Fix an integer $n$ at which the $D+1$ displayed reads converge, absolutely since the reads of a ray are read where they converge absolutely; they converge absolutely for small $\varepsilon$ too. For all but finitely many small $\varepsilon\neq0$, $g$ is defined and the exponents of $x^{\kappa+an+\varepsilon/2}\prod_j P_j^{-w_j-nb_j-\varepsilon}g$ at 0 and at $\infty$, which move with $\varepsilon$, are not 0; the argument of 46.7 then gives $\langle\eth_n g\rangle=0$, so $\sum_k p_k(n,\varepsilon)\langle1\rangle_{\kappa+a(n+k)+\varepsilon/2,\,\mathbf{w}+(n+k)\mathbf{b}+\varepsilon\mathbf{1}}=0$. Letting $\varepsilon\to0$, the reads being continuous there, gives the recurrence at $n$. □

**Theorem 46.12 (Coefficient jets on one scale).** *As stated, for letters without roots on* $[0,\infty)$, *with the powers of the changed letters continued in* $u$.

*Proof.* $P_{ja}/P_j$ is continuous on $[0,\infty]$, $P_j$ having no roots on $[0,\infty)$ and $\deg P_{ja}\le\deg P_j$, so $M_{ja}$ is finite. On the polydisc $z_j=\sum_a u_aP_{ja}/P_j$ has $|z_j|\le\sum_a|u_a|M_{ja}<1$, so $P_j(x;u)=P_j(x)(1+z_j)$ has no roots on $[0,\infty)$, and the power continued in $u$ from $u=0$, the polydisc being convex, is $P_j(x)^{-w_j}(1+z_j)^{-w_j}$ with the principal branch of the second factor. Then

$$(1+z_j)^{-w_j}=\sum_{m\ge0}\frac{(-1)^m(w_j)_m}{m!}z_j^m,\qquad z_j^m=\sum_{|n_j|=m}\frac{m!}{\prod_a n_{ja}!}\prod_a\Big(u_a\frac{P_{ja}}{P_j}\Big)^{n_{ja}},$$

and the product over $j$ is the displayed series inside the read. It is dominated by $\prod_j\sum_m\frac{|(w_j)_m|}{m!}\rho_j^m$, $\rho_j=\sum_a|u_a|M_{ja}<1$, finite by the ratio test, times the absolutely integrable integrand of $\langle1\rangle_{\kappa,\mathbf{w};\mathbf{P}}$; so the read of the sum is the sum of the reads, absolutely, uniformly on smaller polydiscs, and holomorphic in $u$. Each coefficient is the read of the rational insertion $\prod_{j,a}(P_{ja}/P_j)^{n_{ja}}$, bounded on $[0,\infty)$, absolutely convergent with $\langle1\rangle_{\kappa,\mathbf{w};\mathbf{P}}$, and 46.10 gives it. The continuation in $u$ fixes the branch: the continuous branch of $P_j(x;u)^{-w_j}$ along $[0,\infty)$ from its principal value at 0 can differ from it by a factor $e^{2\pi i\ell w_j}$, $\ell\in\mathbb{Z}$. □

## 50.46 Chapter 47: the module of a family

**Definition 47.1 (The data space, the lattice and the master form).** As stated.

*Proof.* (i) $\kappa_{\mathcal{E}}(\mathbf{d})=m_{\mathcal{E}}-\sum_j(s_jr_j+t_jr_j^*)$ is a nonzero linear form, so $\mathbb{D}_F$ is a hyperplane. For $U$ nonempty and proper take $e\in U$ and $e'\notin U$: the point $\mathbf{m}=\mathbf{e}_e-\mathbf{e}_{e'}$, $\mathbf{s}=\mathbf{t}=\mathbf{0}$ lies on $\mathbb{D}_F$ and has $\kappa_U=1$.

(ii) $f_\lambda$ is homogeneous of degree $v_{\mathcal{E}}-\sum_j(a_jr_j+b_jr_j^*)=\kappa_{\mathcal{E}}(\lambda)=0$, and on $X_F$ no $\alpha_e$, $K_j$ or $\Psi_j$ vanishes, so $f_\lambda$ and $f_{-\lambda}$ are regular functions on $X_F$, inverse to each other, with the coefficients of the letters.

(iii) For integral $a_j, b_j$ the principal powers on the orthant multiply, so $f_\lambda\,\omega_{\mathbf{d}} = \omega_{\mathbf{d}+\lambda}$. $\partial_{y_e}\log\alpha^{\mathbf{m}} = m_e$, $\partial_{y_e}\log K_j = \sum_{T\ni e}\theta_{j,T}\alpha^{\mathbf{1}_T}/K_j = R_{j,e}$ and $\partial_{y_e}\log\Psi_j = R^*_{j,e}$ (38.2(i)), which gives $c_e(\mathbf{d}) = \partial_{y_e}\log(\omega_{\mathbf{d}}/dy)$ and, with the data of $\lambda$, $c_e(\lambda) = \partial_{y_e}\log f_\lambda$. Every basis has $r_j$ marks and every complement $r^*_j$, so $\sum_e R_{j,e} = r_j$, $\sum_e R^*_{j,e} = r^*_j$ and $\sum_e c_e(\mathbf{d}) = \kappa_{\mathcal{E}}(\mathbf{d})$.

(iv) The chart $y_{e_0} = 0$ is the section $\{\alpha_{e_0} = 1\}$, on which $\alpha^{\mathbf{m}-\mathbf{1}}d\alpha$ restricts to $\alpha^{\mathbf{m}}dy$; by Theorem 8.7 the integral of $\omega_{\mathbf{d}}$ there is the period. The walls of an insertion are those of Proposition 38.5. □

**Theorem 47.2 (The relations in all the data).** *As stated.*

*Proof.* A point $\mathbf{d}$ of $\mathbb{D}_F$ is the origin of a ray, for instance with the direction 0 (Definition 38.1), whose origin form is $\omega_{\mathbf{d}}$ and whose value at the strength 0 is $\langle g\rangle_0 = \langle g\rangle(\mathbf{d})$.

(i) The condition of Proposition 38.5(i) at the strength 0 is $\operatorname{Re}\kappa_U(\mathbf{d}) + \kappa_U(g) > 0$ for every nonempty proper $U$, that is $\mathbf{d}\in\mathcal{T}_g$, and the defects of that ray at the strength 0 are $c_e(\mathbf{d})$. So $\langle g\rangle(\mathbf{d})$ and $\langle(c_e(\mathbf{d})+\partial_{y_e})g\rangle(\mathbf{d})$ converge absolutely and the second is 0. The integrals $\int f_\lambda g\,\omega_{\mathbf{d}}$ and $\int g\,\omega_{\mathbf{d}+\lambda}$ have one integrand, $f_\lambda\omega_{\mathbf{d}} = \omega_{\mathbf{d}+\lambda}$, and the second converges absolutely for $\mathbf{d}+\lambda\in\mathcal{T}_g$.

(ii) By Lemma S19 at the origin $\mathbf{d}$, on the sector of an order $\pi$ with the tails $U_l$, $g = \prod_l q_l^{\kappa_{U_l}(g)} g_\pi(q)$ and $\omega_{\mathbf{d}}/dy = \prod_l q_l^{\kappa_{U_l}(\mathbf{d})}\omega_{\pi,\mathbf{d}}(q)$ with $\omega_{\pi,\mathbf{d}} = \prod_j k_{j,\pi}^{-s_j}\psi_{j,\pi}^{-t_j}$, the factors $k_{j,\pi}, \psi_{j,\pi}$ smooth and positive on a neighbourhood of $[0,1]^{n-1}$ (Theorem 8.5(ii)); so, for $\mathbf{d}\in\mathcal{T}_g$,

$$\langle g\rangle(\mathbf{d}) = \sum_\pi \int_{[0,1]^{n-1}} \prod_l q_l^{x_l(\mathbf{d})-1} h_\pi(q,\mathbf{d})\,dq, \qquad x_l(\mathbf{d}) = \kappa_{U_l}(\mathbf{d}) + \kappa_{U_l}(g), \quad h_\pi = g_\pi\,\omega_{\pi,\mathbf{d}}.$$

$h_\pi$ is smooth in $q$ near the cube, with $q$-derivatives continuous in $(q,\mathbf{d})$ and entire in $\mathbf{d}$, the masses entering only through the exponents $x_l$. By Lemma 8.4 with the data $\mathbf{d}$ as its parameter, each term extends meromorphically in $(x,\mathbf{d})$, and on $\{\operatorname{Re}x_l > -M\ \forall l\}$ it is a finite sum of products of factors $(x_l+i)^{-1}$, $0\le i<M$, at most one for each $l$, with functions holomorphic there. The $x_l(\mathbf{d})$ are affine in $\mathbf{d}$ and not constant (Definition 47.1(i)), so the composite is meromorphic on $\{\mathbf{d} : \operatorname{Re}x_l(\mathbf{d}) > -M\ \forall l\}$ with poles only where some $x_l(\mathbf{d})$ is a nonpositive integer; $M$ being arbitrary and every nonempty proper $U$ a tail of some order, this continues $\langle g\rangle$ to $\mathbb{D}_F$ with poles on the stated hyperplanes. On $\mathcal{T}_g$ the integral converges absolutely and locally uniformly, so it is holomorphic there and agrees with the continuation. These insertions span $\mathbb{C}[X_F]$ (Definition 39.15(i)); the value of an element of $\mathbb{C}[X_F]$ is the sum of the continued values of the insertions in any writing of it as their combination, which does not depend on the writing, two writings having one integral on the open set where all their integrals converge, which contains $\eta\,\mathbf{d}_\circ$ for large $\eta>0$ (the gloss after Definition 47.1). So the evaluation $\mathbb{C}[X_F]\to\{\text{meromorphic functions on }\mathbb{D}_F\}$ is $\mathbb{C}$-linear, and it extends to $\mathbb{C}[X_F]\otimes\mathbb{K}_F$. An element $\sum_i \rho_i\,(c_{e_i}(\mathbf{d})+\partial_{y_{e_i}})g_i$ of the span, with insertions $g_i$ of the above kind, has the value $\sum_i\rho_i(\mathbf{d})\cdot 0 = 0$ at every point of $\bigcap_i\mathcal{T}_{g_i}$ off the poles of the $\rho_i$, by (i); that intersection contains $\eta\,\mathbf{d}_\circ$ for large $\eta>0$, $\kappa_U(\eta\,\mathbf{d}_\circ) = \eta\sum_j c_j(U)$ (the gloss after Definition 47.1), and is open, so the value vanishes identically. For $g\in\mathbb{C}[X_F]$, $\langle f_\lambda g\rangle$ and $\langle g\rangle(\cdot+\lambda)$ are one integral on the open set of the $\mathbf{d}$ at which $\mathbf{d}+\lambda$ lies in the tubes of the insertions of a writing of $g$, which contains $\eta\,\mathbf{d}_\circ$ for large $\eta$; so they agree identically.

(iii) At a real $\mathbf{d}$ the form $\omega_{\mathbf{d}}/dy = \alpha^{\mathbf{m}}\prod_j K_j^{-s_j}\Psi_j^{-t_j}$ is positive on the orthant. □

**Definition 47.3 (The module of a family; shifts; masters; shift matrices).** As stated; that the shifts are well defined and the value depends only on the class is Theorem 47.4(i).

**Theorem 47.4 (The module of a family).** *As stated.*

*Proof.* (i) On $\mathbb{C}[X_F]\otimes\mathbb{K}_F$ put $\widetilde{\mathrm{Sh}_\lambda}(g\otimes r)=f_\lambda g\otimes\sigma_\lambda r$, well defined since it is $\mathbb{C}$-bilinear in $(g,r)$, and $\sigma_\lambda$-semilinear. By Definition 47.1(iii), $\partial_{y_e}(f_\lambda g)=f_\lambda\big(c_e(\lambda)+\partial_{y_e}\big)g$, and $c_e$ is linear in the data, so $\big(c_e(\mathbf{d})+\partial_{y_e}\big)(f_\lambda g)=f_\lambda\big(c_e(\mathbf{d}+\lambda)+\partial_{y_e}\big)g$. Written out, $(c_e(\mathbf{d})+\partial_{y_e})g=g\otimes m_e-\sum_j R_{j,e}g\otimes s_j-\sum_j R^*_{j,e}g\otimes t_j+\partial_{y_e}g\otimes 1$, and its image under $\widetilde{\mathrm{Sh}_\lambda}$ is $f_\lambda g\otimes(m_e+v_e)-\sum_j u_\lambda R_{j,e}g\otimes(s_j+a_j)-\sum_j u_\lambda R^*_{j,e}g\otimes(t_j+b_j)+f_\lambda\partial_{y_e}g\otimes 1=f_\lambda\big(c_e(\mathbf{d}+\lambda)+\partial_{y_e}\big)g$, which is $(c_e(\mathbf{d})+\partial_{y_e})(f_\lambda g)$, an element of $\mathcal{L}_F$. With the semilinearity, $\widetilde{\mathrm{Sh}_\lambda}\mathcal{L}_F\subseteq\mathcal{L}_F$, and $\mathrm{Sh}_\lambda$ is well defined on $\mathfrak{M}_F$ and $\sigma_\lambda$-semilinear. $f_\lambda f_\mu=f_{\lambda+\mu}$ and $\sigma_\lambda\sigma_\mu=\sigma_{\lambda+\mu}$ give $\mathrm{Sh}_\lambda\,\mathrm{Sh}_\mu=\mathrm{Sh}_{\lambda+\mu}$, and $\mathrm{Sh}_0=1$, so $\mathrm{Sh}_{-\lambda}$ is the inverse of $\mathrm{Sh}_\lambda$. The value vanishes on $\mathcal{L}_F$ (Theorem 47.2(ii)), so it is a function of the class; for $m=[\sum_i g_i\otimes r_i]$, $\langle\mathrm{Sh}_\lambda m\rangle(\mathbf{d})=\sum_i r_i(\mathbf{d}+\lambda)\langle f_\lambda g_i\rangle(\mathbf{d})=\sum_i r_i(\mathbf{d}+\lambda)\langle g_i\rangle(\mathbf{d}+\lambda)=\langle m\rangle(\mathbf{d}+\lambda)$ (Theorem 47.2(ii)). So $\mathrm{Ann}\langle\cdot\rangle$, a $\mathbb{K}_F$-subspace, has $\mathrm{Sh}_\lambda\,\mathrm{Ann}\langle\cdot\rangle\subseteq\mathrm{Ann}\langle\cdot\rangle$ for every $\lambda$, with equality since $\mathrm{Sh}_{-\lambda}$ does the same; and $\langle 1\rangle=P_F$ is positive at the real points of $\mathcal{T}_1$ (Theorem 47.2(iii)), so $[1]\notin\mathrm{Ann}\langle\cdot\rangle$.

(ii) For $\lambda=(v;\mathbf{a},\mathbf{b})$, $\alpha^v\prod_j K_j^{-a_j}\Psi_j^{-b_j}=f_\lambda$, and $[f_\lambda\otimes 1]=\mathrm{Sh}_\lambda[1\otimes 1]$; these insertions, $\mathbf{a},\mathbf{b}\in\mathbb{N}^N$, span $\mathbb{C}[X_F]$ (Definition 39.15(i)). So $\mathfrak{M}_F$ is spanned by the $\mathrm{Sh}_\lambda[1]$, a submodule containing $[1]$ contains them all, and the values are those stated by Theorem 47.2(ii). The lattice vectors $\lambda^{\mathrm{m}}_e=(\mathbf{e}_e-\mathbf{e}_{e_0};\mathbf{0},\mathbf{0})$, $e\neq e_0$, $\lambda^K_j=(\mathbf{1}_{T_j};\mathbf{e}_j,\mathbf{0})$ and $\lambda^\Psi_j=(\mathbf{1}_{\mathcal{E}\setminus T'_j};\mathbf{0},\mathbf{e}_j)$, for bases $T_j,T'_j$ of table $j$, generate $\mathbb{D}_F(\mathbb{Z})$: $(v;\mathbf{a},\mathbf{b})-\sum_j a_j\lambda^K_j-\sum_j b_j\lambda^\Psi_j=(w;\mathbf{0},\mathbf{0})$ with $w_{\mathcal{E}}=0$, which is $\sum_{e\neq e_0}w_e\lambda^{\mathrm{m}}_e$. For each of them $f_{-\mu}$ is a Laurent polynomial of degree 0: $\alpha^{\mathbf{e}_{e_0}-\mathbf{e}_e}$, $\alpha^{-\mathbf{1}_{T_j}}K_j$ and $\alpha^{-\mathbf{1}_{\mathcal{E}\setminus T'_j}}\Psi_j$. So $\mathrm{Sh}_{-\mu}$ maps the span $\mathfrak{M}_{\mathrm{L}}$ of the classes of the Laurent insertions into itself, injectively and semilinearly; a semilinear injection maps a basis of the finite-dimensional $\mathfrak{M}_{\mathrm{L}}$ (by (iii)) to an independent set, so $\mathrm{Sh}_{-\mu}\,\mathfrak{M}_{\mathrm{L}}=\mathfrak{M}_{\mathrm{L}}$ and $\mathrm{Sh}_\mu\,\mathfrak{M}_{\mathrm{L}}=\mathfrak{M}_{\mathrm{L}}$. So $\mathfrak{M}_{\mathrm{L}}$ is stable under the generators and their inverses, a submodule containing $[1]$, and it is $\mathfrak{M}_F$.

(iii) On the chart, $X_F$ is the smooth affine variety of the points of the torus $(\mathbb{C}^\times)^{\mathcal{E}\setminus\{e_0\}}$ at which $\prod_j K_j\Psi_j$ does not vanish, of dimension $n-1$, with the basis $dy_e$, $e\neq e_0$, of its 1-forms. The 1-form $c(\mathbf{d})=\sum_{e\neq e_0}c_e(\mathbf{d})\,dy_e$ over $\mathbb{K}_F$ is closed: $c_e(\mathbf{d})=m_e-\sum_j s_j\partial_{y_e}\log K_j-\sum_j t_j\partial_{y_e}\log\Psi_j$, so $\partial_{y_{e'}}c_e=\partial_{y_e}c_{e'}$. So $\nabla=d+c(\mathbf{d})\wedge$ is an integrable connection on the structure sheaf of $X_F$ over $\mathbb{K}_F$. In its twisted de Rham complex the forms of top degree are $(\mathbb{C}[X_F]\otimes\mathbb{K}_F)\,dy$, and for the form $g\,dy_{\hat e}$ of degree $n-2$, the product of the $dy_{e'}$, $e'\neq e,e_0$, $\nabla(g\,dy_{\hat e})=\pm\big(c_e(\mathbf{d})+\partial_{y_e}\big)g\,dy$. The relation at $e_0$ is minus the sum of the others: $\sum_e(c_e(\mathbf{d})+\partial_{y_e})g=\kappa_{\mathcal{E}}(\mathbf{d})\,g+\sum_e\partial_{y_e}g=0$, by Definition 47.1(iii) and by Euler's identity for the function $g$ of degree 0. So $\mathcal{L}_F$ is the image of $\nabla$ in top degree, and $\mathfrak{M}_F$ is the top cohomology. The algebraic de Rham cohomology of an integrable connection on a smooth affine variety over a field of characteristic zero is finite-dimensional, the fact that Theorem 38.31(i) uses over $\mathbb{C}(k)$; so $\mathfrak{M}_F$ is finite-dimensional.

(iv) $\mathbb{C}[X_F]=\Bbbk[X_F]\otimes_\Bbbk\mathbb{C}$, so $\big(\Bbbk[X_F]\otimes_\Bbbk\Bbbk(\mathbb{D}_F)\big)\otimes_{\Bbbk(\mathbb{D}_F)}\mathbb{K}_F=\mathbb{C}[X_F]\otimes_{\mathbb{C}}\mathbb{K}_F$, and every element $(c_e(\mathbf{d})+\partial_{y_e})g$, $g\in\mathbb{C}[X_F]$, is a $\mathbb{C}$-combination of those with $g\in\Bbbk[X_F]$, so $\mathcal{L}_F$ is the $\mathbb{K}_F$-span of $\mathcal{L}_{F,\Bbbk}$. A cokernel commutes with the extension of scalars, so $\mathfrak{M}_{F,\Bbbk}\otimes_{\Bbbk(\mathbb{D}_F)}\mathbb{K}_F=\mathfrak{M}_F$, and a $\Bbbk(\mathbb{D}_F)$-basis of $\mathfrak{M}_{F,\Bbbk}$, classes of insertions with coefficients in $\Bbbk$, is a $\mathbb{K}_F$-basis of $\mathfrak{M}_F$. Conversely the map $\mathfrak{M}_{F,\Bbbk}\to\mathfrak{M}_F$ is injective, $\mathbb{K}_F$ being a field extension of $\Bbbk(\mathbb{D}_F)$, so the classes of masters of $\mathfrak{M}_F$ with coefficients in $\Bbbk$ are independent over $\Bbbk(\mathbb{D}_F)$, and, the two dimensions being equal, they are a $\Bbbk(\mathbb{D}_F)$-basis of $\mathfrak{M}_{F,\Bbbk}$. $f_\lambda$ has its coefficients in $\Bbbk$ and $\sigma_\lambda$ keeps $\Bbbk(\mathbb{D}_F)$, so the shifts keep $\mathfrak{M}_{F,\Bbbk}$ and their matrices on such masters have entries in $\Bbbk(\mathbb{D}_F)$.

(v) $\mathrm{Sh}_{\lambda+\mu}[\mathbf{b}]=\mathrm{Sh}_\lambda(\mathbf{A}_\mu)[\mathbf{b}])=\sigma_\lambda(\mathbf{A}_\mu)\,\mathrm{Sh}_\lambda[\mathbf{b}]=\sigma_\lambda(\mathbf{A}_\mu)\mathbf{A}_\lambda[\mathbf{b}]$, and the matrix of a $\mathbb{K}_F$-linear relation on a basis is unique; the second identity is the first with $\lambda$ and $\mu$ exchanged, and induction along a path gives the product. By (i) and the $\mathbb{K}_F$-linearity of the value, $\langle b_i\rangle(\mathbf{d}+\lambda)=\langle\mathrm{Sh}_\lambda[b_i]\rangle(\mathbf{d})=\sum_l\mathbf{A}_{\lambda,il}(\mathbf{d})\langle b_l\rangle(\mathbf{d})$. For masters of $\mathfrak{M}_F$ the coordinates of $[g]$ on the basis are unique, and the value is $\mathbb{K}_F$-linear. □

**Proposition 47.5 (A ray is a line of the data space).** *As stated.*

*Proof.* By Definition 38.1 the stride of the ray is $\alpha^{\mathbf{m}_1}\prod_j K_j^{-s_{1j}}\Psi_j^{-t_{1j}} = f_{\mathbf{d}_1}$, its origin form is $\omega_{\mathbf{d}_0}$, and its defects are $c_e^{(i)} = c_e(\mathbf{d}_i)$, so $c_e^{(0)} + k\,c_e^{(1)} = c_e(\mathbf{d}_0 + k\mathbf{d}_1)$, $c_e$ being linear. For $r \in \mathcal{O}_\ell$ the element $r\,(c_e(\mathbf{d}) + \partial_{y_e})g$ restricts to $r|_\ell(k)\,(c_e^{(0)} + k\,c_e^{(1)} + \partial_{y_e})g$, an element of the relation space of the ray (Definition 38.4). $\mathrm{Sh}_{\mathbf{d}_1}[g \otimes r] = [f_{\mathbf{d}_1} g \otimes \sigma_{\mathbf{d}_1} r]$ restricts to $[fg \otimes r|_\ell(k+1)] = \mathrm{Sh}[g \otimes r|_\ell]$ (Definition 39.1). $f^k\omega_{\mathbf{d}_0} = \omega_{\mathbf{d}_0 + k\mathbf{d}_1}$, so $\langle g\rangle_k = \langle g\rangle(\mathbf{d}_0 + k\mathbf{d}_1)$ wherever the integral converges; when moreover every $\kappa_U(\mathbf{d}_1)$ is positive, the ray's values continue meromorphically in $k$ (Proposition 38.5(ii)), and for one molecule so do the values in all the data (Theorem 47.2(ii)), and the two continuations agree, agreeing for all large real $k$. The restriction of the coefficients gives the values of $x$. The restriction $\mathcal{O}_\ell \to \mathbb{C}(k)$ is onto, $k$ being the restriction of the affine function $(\varphi(\mathbf{d}) - \varphi(\mathbf{d}_0))/\varphi(\mathbf{d}_1)$ for a linear form $\varphi$ with $\varphi(\mathbf{d}_1) \neq 0$, which exists as $\mathbf{d}_1 \neq 0$, so the restriction of $\mathbb{C}[X_F] \otimes \mathcal{O}_\ell$ is all of $\mathbb{C}[X_F] \otimes \mathbb{C}(k)$, and the relation space of the ray is the $\mathbb{C}(k)$-span of the restrictions of the generators of $\mathcal{L}_F$. □

**Definition 47.6 (Symbols, relations and boxes in all the data).** As stated.

*Proof.* The coefficients $m_e + u_e$, $s_j + a_j$ and $t_j + b_j$ are coordinates of $\mathbb{D}_F$ plus constants, none of them constant on $\mathbb{D}_F$. The evaluation of $\mathrm{MR}_e(g)$ is $(c_e(\mathbf{d}) + \partial_{y_e})g$ by the computation in the proof of Proposition 38.5(i) with the data $\mathbf{d}$ in the place of $(\mathbf{m}_0 + k\mathbf{m}_1; \mathbf{s}_0 + k\mathbf{s}_1; \mathbf{t}_0 + k\mathbf{t}_1)$. The generators generate $\mathbb{D}_F(\mathbb{Z})$ and have $f_{-\mu}$ a Laurent polynomial with coefficients in $\Bbbk$ (proof of Theorem 47.4(ii)); there are finitely many of them and of the $U$, so $\mathfrak{c}$ is a finite nonnegative number. The reduction is that of Definition 39.20 with $G$ trivial, its proof applying verbatim; the symbol 1 is free, for otherwise $1 \in \mathcal{R}_{\mathfrak{B}}$ would evaluate into $\mathcal{L}_{F,\Bbbk}$ (Proposition 47.7(iv)) and $[1] = 0$, against Theorem 47.4(i). The monomials of a target have wall degree at most $\mathfrak{s} + \mathfrak{c} \leq \mathfrak{t}$ and the level $(\mathbf{0}, \mathbf{0})$ (Proposition 47.7(ii)), and the coefficients of $f_{-\mu}$ lie in $\Bbbk$, so every target lies in $\mathcal{F}_{\mathfrak{B}}$. □

**Proposition 47.7 (Wall degrees and the relations of a box).** *As stated.*

*Proof.* (i) is Proposition 39.16(i) for the ray with the direction $\mathbf{d}_1$, the stride $f_{\mathbf{d}_1}$ and $\kappa_{1,U} = \kappa_U(\mathbf{d}_1)$; for the convergence, $\kappa_U(g) \geq -\deg_\kappa g\,\kappa_U(\mathbf{d}_1)$ gives $\mathrm{Re}\,\kappa_U(\mathbf{d}) + \kappa_U(g) > 0$, that is $\mathbf{d} \in \mathcal{T}_g$, and Theorem 47.2(i) applies.

(ii) $\kappa_U(\alpha^u) = u_U$, and the first statement is Proposition 39.16(ii). For a generator $\mu = (v; \mathbf{a}, \mathbf{b})$, $\mathbf{a}, \mathbf{b} \in \mathbb{N}^N$, a monomial of $f_{-\mu} = \alpha^{-v}\prod_j K_j^{a_j}\Psi_j^{b_j}$ is $\alpha^w$ with $w = -v + \sum_j \big(\sum_{i \leq a_j} \mathbf{1}_{T_{j,i}} + \sum_{i \leq b_j} \mathbf{1}_{\mathcal{E}\setminus T'_{j,i}}\big)$ for bases $T_{j,i}, T'_{j,i}$ of table $j$, and by Theorem 7.13 $w_U \geq -v_U + \sum_j \big(a_j d_j(U) + b_j d_j^*(U)\big) = -\kappa_U(\mu) \geq -\mathfrak{c}\,\kappa_U(\mathbf{d}_1)$. The walls of Laurent symbols add, so $\kappa_U(\alpha^{u+w}) \geq -(\deg_\kappa \alpha^u + \mathfrak{c})\,\kappa_U(\mathbf{d}_1)$.

(iii) is Proposition 39.16(iii) and (iv), whose proofs use only the walls and Theorem 7.13.

(iv) The terms of a letter identity of a symbol of the box lie at levels within the bounds and have wall degree at most that of the symbol, by (iii). A term of the mark relation of a stem is at the level $(\bar{\mathbf{a}} - \mathbf{1}, \bar{\mathbf{b}} - \mathbf{1})$, $(\bar{\mathbf{a}} - \mathbf{1} + \mathbf{e}_j, \bar{\mathbf{b}} - \mathbf{1})$ or $(\bar{\mathbf{a}} - \mathbf{1}, \bar{\mathbf{b}} - \mathbf{1} + \mathbf{e}_j)$, within the bounds, with wall degree at most that of the stem. The letter identities evaluate to 0 and the mark relation of a stem $s$ to $(c_e(\mathbf{d}) + \partial_{y_e})s$, an element of $\mathcal{L}_{F,\Bbbk}$, whose value vanishes identically by Theorem 47.2(ii).

(v) is the proof of Proposition 39.18(ii), which uses only the letter identities and the independence of distinct Laurent monomials.

(vi) is the proof of Proposition 39.18(iii), with $\bar{a}_j^- = \bar{a}_j - 1$ and $\bar{b}_j^- = \bar{b}_j - 1$: every bound being at least 1, every term of $\mathrm{MR}_e(x)$ at a raised level lies within the bounds, and the combination

formed there evaluates to $(c_e(\mathbf{d}) + \partial_{y_e})$ of 0 and has its terms in the box, so it lies in the span of the letter identities by (v). □

**Theorem 47.8 (The box in all the data).** *As stated.*

*Proof.* (i) $f_{-\mu}b_i - \sum_l \mathbf{B}_{\mu,il}b_l$ lies in $\mathcal{R}_{\mathfrak{B}}$, so it evaluates into $\mathcal{L}_{F,\Bbbk}$ (Proposition 47.7(iv)), and $[f_{-\mu}b_i] = \sum_l \mathbf{B}_{\mu,il}[b_l]$; $[f_{-\mu}b_i] = \mathrm{Sh}_{-\mu}[b_i]$. Let $\mathfrak{N}$ be the span of the $[b_l]$. Then $\mathrm{Sh}_{-\mu}\mathfrak{N} \subseteq \mathfrak{N}$ for every generator $\mu$; $\mathrm{Sh}_{-\mu}$ is injective and semilinear on the finite-dimensional $\mathfrak{N}$, so $\mathrm{Sh}_{-\mu}\mathfrak{N} = \mathfrak{N}$ and $\mathrm{Sh}_{\mu}\mathfrak{N} = \mathfrak{N}$. The generators generate $\mathbb{D}_F(\mathbb{Z})$, so $\mathfrak{N}$ is a submodule; it contains $[b_1] = [1]$, so it is $\mathfrak{M}_F$ (Theorem 47.4(ii)). When the classes are independent they are a basis, and the matrix of $\mathrm{Sh}_{-\mu}$ on it is unique.

(ii) As an element of $\mathcal{R}_{\mathfrak{B}}$, $f_{-\mu}b_i - \sum_l \mathbf{B}_{\mu,il}b_l$ is a combination of letter identities and of mark relations $\mathrm{MR}_e(s)$ of stems $s$ with coefficients $\nu_{\mu,i,e,s} \in \Bbbk(\mathbb{D}_F)$; evaluated, the letter identities give 0 and $\mathrm{MR}_e(s)$ gives $(c_e(\mathbf{d}) + \partial_{y_e})s$. Put $G_{\mu,i,e} = \sum_s \nu_{\mu,i,e,s}\, s$.

(iii) $\langle \mathrm{Sh}_{-\mu}[b_i]\rangle(\mathbf{d}) = \langle b_i\rangle(\mathbf{d} - \mu)$ by Theorem 47.4(i), and by (i) and the $\mathbb{K}_F$-linearity of the value it is $\sum_l \mathbf{B}_{\mu,il}(\mathbf{d})\langle b_l\rangle(\mathbf{d})$. At a real $\mathbf{d}$ as stated the stems, the $b_l$ and the monomials of the targets have wall degree at most $\mathfrak{t}$, so their values converge absolutely and $\mathbf{d} \in \mathcal{T}_s$ for every stem $s$ (Proposition 47.7(i)); Theorem 47.2(i) gives the value 0 for every term $(c_e(\mathbf{d}) + \partial_{y_e})s$.

(iv) The walls of symbols are integers and those of $\mathbf{d}_1$ positive integers, so every wall degree lies in $\frac{1}{L}\mathbb{Z}$ for $L$ the least common multiple of the $\kappa_U(\mathbf{d}_1)$, that of a Laurent symbol in $\frac{1}{L}\mathbb{N}$ (Proposition 47.7(ii)), and $\mathfrak{c} \in \frac{1}{L}\mathbb{N}$; this is the discrete set. Let $\mathfrak{N}_{\mathfrak{s}} \subseteq \mathfrak{M}_{F,\Bbbk}$ be the span of the classes of the Laurent symbols of wall degree at most $\mathfrak{s}$, finitely many (Proposition 47.7(iii)). The $\mathfrak{N}_{\mathfrak{s}}$ increase with $\mathfrak{s}$ in a space of finite dimension (Theorem 47.4(iii), (iv)), so they are constant from some $\mathfrak{s}_0$ on, and their union is the span of the classes of all the Laurent insertions with coefficients in $\Bbbk$, which is $\mathfrak{M}_{F,\Bbbk}$ by the argument of Theorem 47.4(ii) over $\Bbbk$. Each of the finitely many Laurent symbols $\bar{g}$ of wall degree at most $\mathfrak{s}_0 + \mathfrak{c}$ has its class in $\mathfrak{N}_{\mathfrak{s}_0}$:

$$\bar{g} - \sum_h \lambda_{\bar{g},h}\, h = \sum_{e,i} \nu_{\bar{g},e,i}\, \big(c_e(\mathbf{d}) + \partial_{y_e}\big) G_{\bar{g},i},$$

with $h$ running over the Laurent symbols of wall degree at most $\mathfrak{s}_0$, $\lambda, \nu \in \Bbbk(\mathbb{D}_F)$, and finitely many symbols $G_{\bar{g},i}$, the evaluations of the symbols spanning $\Bbbk[X_F]$ over $\Bbbk$. Take $\mathfrak{t} \geq \mathfrak{s}_0 + \mathfrak{c}$ in $\frac{1}{L}\mathbb{N}$ at least the wall degree of every $G_{\bar{g},i}$, and the level bounds $\bar{a}_j = 1 + \max a_j$ and $\bar{b}_j = 1 + \max b_j$ over the levels $(\mathbf{a}, \mathbf{b})$ of the $G_{\bar{g},i}$. Then every $G_{\bar{g},i}$ lies at a level at most $(\bar{\mathbf{a}} - \mathbf{1}, \bar{\mathbf{b}} - \mathbf{1})$ and has wall degree at most $\mathfrak{t}$, so every $\mathrm{MR}_e(G_{\bar{g},i})$ lies in $\mathcal{R}_{\mathfrak{B}}$ (Proposition 47.7(vi)); the formal combination $\bar{g} - \sum \lambda_{\bar{g},h} h - \sum \nu_{\bar{g},e,i}\mathrm{MR}_e(G_{\bar{g},i})$ has its terms in the box (Proposition 47.7(iii)) and evaluates to 0, so it lies in $\mathcal{R}_{\mathfrak{B}}$ (Proposition 47.7(v)), and so does $\bar{g} - \sum \lambda_{\bar{g},h} h$. Take $\mathfrak{s} = \mathfrak{s}_0$. A target $f_{-\mu}b_i$ is a $\Bbbk$-combination of Laurent symbols of wall degree at most $\mathfrak{s}_0 + \mathfrak{c}$ (Proposition 47.7(ii)), so it is congruent modulo $\mathcal{R}_{\mathfrak{B}}$ to a combination of Laurent symbols of wall degree at most $\mathfrak{s}_0$, whose reduction, which is the reduction of the target, involves only free Laurent symbols of wall degree at most $\mathfrak{s}_0$, by the ranking (Definition 39.20(ii)): the spanning set. So the box is accepted at $\mathfrak{s}_0$. A search through all the quadruples reaches this one, and the acceptance is decided by finitely many operations in $\Bbbk(\mathbb{D}_F)$. □

**Definition 47.9 (Closings in all the data).** As stated.

*Proof.* For $s_j + a_j \leq 0$ and $t_j + b_j \leq 0$ the function $g\,\omega_{\mathbf{d}}/dy = \alpha^{u+\mathbf{m}} \prod_j K_j^{-s_j-a_j} \Psi_j^{-t_j-b_j}$ is a Laurent polynomial of degree 0. A monomial $\alpha^w$, $w_{\mathcal{E}} = 0$, is the constant 1 on the chart $\alpha_{e_0} = 1$ exactly when $w_e = 0$ for every $e \neq e_0$, that is when $w = 0$; so the constant term is the coefficient of $\alpha^0$, whatever $e_0$, and it is the torus integral by Cauchy's formula in each variable. □

**Proposition 47.10 (The closings are a solution in all the data).** *As stated.*

*Proof.* For integral data $\Phi = \omega_{\mathbf{d}}/dy$ is a rational function with $\partial_{y_e} \log \Phi = c_e(\mathbf{d})$, so $\big(c_e(\mathbf{d}) + \partial_{y_e}\big) g \cdot \Phi = \partial_{y_e}(g\Phi)$. When $g\Phi$ is a Laurent polynomial, $\partial_{y_e} = \alpha_e \partial_{\alpha_e}$ multiplies its monomial $\alpha^w$ by $w_e$ and kills its constant term, for $e \neq e_0$ on the chart; for $e_0$, Euler's identity gives $\partial_{y_{e_0}} = -\sum_{e \neq e_0} \partial_{y_e}$ on functions of degree 0. So the closing of $(c_e(\mathbf{d}) + \partial_{y_e})g$ is 0. $f_\lambda g\, \omega_{\mathbf{d}} = g\, \omega_{\mathbf{d}+\lambda}$ gives the second identity. For the third let the marks of $F$ form one molecule, let $x = \sum_i g_i \otimes r_i \in \mathcal{L}_F$ have every $r_i$ regular at $\mathbf{d}$ and every $\mathsf{C}(g_i)(\mathbf{d})$ defined, and let $\mathbf{d}_1$ be a positive direction (the gloss after Definition 47.1). Then $f_{\mathbf{d}} g_i = g_i\, \omega_{\mathbf{d}}/dy$ is a Laurent polynomial, and $\langle g_i \rangle(\mathbf{d} + k\mathbf{d}_1) = \langle f_{\mathbf{d}} g_i \rangle(k\mathbf{d}_1)$ (Theorem 47.2(ii)) is the value of $f_{\mathbf{d}} g_i$ along the ray with the origin 0 and the direction $\mathbf{d}_1$, as meromorphic functions of $k$, every $\kappa_U(\mathbf{d}_1)$ being positive (Proposition 47.5). Proposition 39.35(ii) at $j = 0$ gives $\lim_{k \to 0} k^{n-1} \langle g_i \rangle(\mathbf{d} + k\mathbf{d}_1) = V_1\, \mathsf{C}(g_i)(\mathbf{d})$ with $V_1 > 0$. The value of $x$ vanishes identically (Theorem 47.2(ii)), and on the line $\mathbf{d} + \mathbb{C}\mathbf{d}_1$, which meets every polar hyperplane of the values in isolated points, $\kappa_U(\mathbf{d}_1)$ being positive, $\sum_i r_i(\mathbf{d} + k\mathbf{d}_1) \langle g_i \rangle(\mathbf{d} + k\mathbf{d}_1) = 0$; multiplying by $k^{n-1}$ and letting $k \to 0$, the $r_i$ being regular at $\mathbf{d}$, gives $V_1\, \mathsf{C}(x)(\mathbf{d}) = 0$. On the ray, $f = f_{\mathbf{d}_1}$ and $\omega_0 = \omega_{\mathbf{d}_0}$ (Proposition 47.5), so $\mathsf{C}_j(g) = \mathrm{CT}(g\, f^{-j} \omega_0/dy) = \mathsf{C}(g)(\mathbf{d}_0 - j\mathbf{d}_1)$, and $V_{\mathsf{C}}(x|_\ell)(k) = \sum_i r_i|_\ell(k)\, \mathsf{C}_{-k}(g_i)$ (Theorem 39.33) is $\mathsf{C}(x)(\mathbf{d}_0 + k\mathbf{d}_1)$. □

**Theorem 47.11 (The elements of value zero).** *As stated.*

*Proof.* For a positive direction $\mathbf{d}_1$ as in (b), with $\ell = \mathbf{d}_0 + \mathbb{C}\mathbf{d}_1$, the ray with the origin $\mathbf{d}_0$ and the direction $\mathbf{d}_1$ has an integral origin, an integral direction in which every letter has a positive exponent, and $\kappa_{1,S} = \kappa_S(\mathbf{d}_1) > 0$ for every nonempty proper $S$: the hypotheses of Theorem 39.57. The restriction $x|_\ell$ (Proposition 47.5) has a class in the module of that ray, with the values $\langle m \rangle(\mathbf{d}_0 + k\mathbf{d}_1)$ for all large $k$ and the closings $\mathsf{C}(x)(\mathbf{d}_0 + k\mathbf{d}_1)$ for all integers $k$ below some bound (Proposition 47.10).

(a)⇒(b) is the restriction of the value. (b)⇔(c) is (i)⇔(ii) of Theorem 39.57 for this class, ray by ray.

(b)⇒(a). By Theorem 39.57, for every such $\mathbf{d}_1$ the continuation of the values along the ray vanishes identically, so $\langle m \rangle$ vanishes on $\ell$ off its poles. Let $C_+$ be the open cone of the real points of $\mathbb{D}_F$ with $s_j, t_j > 0$ for every $j$ and $\kappa_U > 0$ for every nonempty proper $U$; it contains $\mathbf{d}_\circ$ (the gloss after Definition 47.1), and its lattice points are the positive directions. A coefficient $P/Q$ of $x$, with polynomials $P$, $Q \neq 0$, has a pole along the whole line $\mathbf{d}_0 + \mathbb{C}\mathbf{d}_1$ only when $Q(\mathbf{d}_0 + k\mathbf{d}_1)$ vanishes identically in $k$; if $Q(\mathbf{d}_0) \neq 0$ this never happens, and otherwise the coefficient of the highest power of $k$ is the top homogeneous part of $Q$ at $\mathbf{d}_1$, a nonzero polynomial in $\mathbf{d}_1$. So the positive directions excluded in (b) lie on a proper algebraic subset of $\mathbb{D}_F$. The rational points of $C_+$ off a proper algebraic subset are dense in $C_+$, and each is a positive multiple of a lattice point, so the half-lines $\mathbf{d}_0 + \mathbb{R}_{>0}\mathbf{d}_1$ over the allowed $\mathbf{d}_1$ fill a dense subset of $\mathbf{d}_0 + C_+$. The real points of $\mathbf{d}_0 + C_+$ that lie in every $\mathcal{T}_{g_i}$, $g_i$ the insertions of $x$, form an open set that contains $\mathbf{d}_0 + y\mathbf{d}_\circ$ for large $y$; off the real zeros of the denominators of the coefficients of $x$, a set without interior, it leaves a nonempty open set $W$. On $W$ the value $\langle m \rangle$ is real-analytic and vanishes on a dense subset, so it vanishes on $W$; a function holomorphic on a connected open set of $\mathbb{D}_F$ that vanishes on a nonempty open set of its real points vanishes there identically, its Taylor coefficients at a real point being real derivatives. The complement of the polar set of $\langle m \rangle$ is connected, so $\langle m \rangle = 0$.

$\mathrm{Ann}\langle \cdot \rangle$ is a submodule (Theorem 47.4(i)), and the equivalence of (a) and (c) describes it. The closings are proved relations (Definition 44.1). Along each ray of (c) the argument of Theorem 44.16(ii), with the dependence taken over $\mathbb{C}(k)$, decides the vanishing of the values, which is that of the closings, by finitely many closings of the class of $x|_\ell$, the closings being a solution of the

shift (Theorem 39.33); when the weights, the insertions of $x$ and its coefficients are rational, $x|_\ell$ is the class of a rational combination of insertions, $\mathbf{d}_0$ and $\mathbf{d}_1$ being integral, and Theorem 44.16(ii) applies as stated, with closings that are rational numbers. $\mathrm{Ann}\langle\cdot\rangle$ is 0 by Proposition 47.15(ii) at $\mathbb{E} = \mathbb{D}_F$, which meets $\mathcal{T}_1$, the proof of (ii) there not using this theorem; so (a)–(c) hold exactly when $m = 0$, and $\langle g\rangle = \langle g'\rangle$ exactly when $[g - g'] = 0$, that is when the reductions agree. □

**Proposition 47.12 (Value zero by finitely many closings).** *As stated.*

*Proof.* (i) The elements of $\mathcal{G}$ form a basis of $\mathbb{D}_F(\mathbb{Z})$ and its negatives: for $(\mathbf{m};\mathbf{s},\mathbf{t}) \in \mathbb{D}_F(\mathbb{Z})$, $(\mathbf{m};\mathbf{s},\mathbf{t}) - \sum_j s_j\lambda_j^K - \sum_j t_j\lambda_j^\Psi = (\mathbf{m}';\mathbf{0},\mathbf{0})$ with $m'_{\mathcal{E}} = 0$, which is $\sum_{e\neq e_0} m'_e\lambda_e^{\mathrm{m}}$. Let $V_0 = \mathbb{K}_F m$ and $V_{i+1} = V_i + \sum_{\mu\in\mathcal{G}} \mathrm{Sh}_\mu V_i$. Since $\mathrm{Sh}_\mu(r\,\mathrm{Sh}_\lambda m) = \sigma_\mu(r)\,\mathrm{Sh}_{\lambda+\mu} m$ (Theorem 47.4(i)), $V_i$ is the $\mathbb{K}_F$-span of the $\mathrm{Sh}_\lambda m$ with $\lambda$ a sum of at most $i$ elements of $\mathcal{G}$, and $V_i \subseteq V_{i+1} \subseteq \mathfrak{N}$. If $V_i = V_{i+1}$, every $\mathrm{Sh}_\mu$, $\mu \in \mathcal{G}$, carries $V_i$ into itself, and so does every $\mathrm{Sh}_\lambda$, $\lambda$ being a sum of elements of $\mathcal{G}$: $V_i$ is a submodule that contains $m$, and $V_i = \mathfrak{N}$. Until then the dimensions increase strictly, so $V_{d-1} = \mathfrak{N}$. The reductions of these shifts on masters of $\mathfrak{M}$, which Proposition 47.13 gives for $\mathfrak{M}_F$ and Theorem 48.18(iv) for the quotients it presents, are read off the shift matrices (Theorem 47.4(v)), and a basis among them, with the matrices $\mathbf{B}_\mu$, by eliminations over $\mathbb{K}_F$. A quotient of $\mathfrak{M}_F$ by a submodule of elements of value zero carries the shifts and the values, which factor through it (Theorem 47.4(i)), so what follows holds in $\mathfrak{M}$.

(ii) A denominator $Q$ of a coefficient of $x$ or of an entry of a $\mathbf{B}_\mu$ and every translate $Q(\cdot + \nu)$ vanish identically on no line parallel to $\mathbf{d}_1$, the coefficient of the highest power of $k$ in $Q(\mathbf{p} + \nu + k\mathbf{d}_1)$ being the top homogeneous part of $Q$ at $\mathbf{d}_1$; and no polar hyperplane $\kappa_U(\mathbf{d}) + \kappa_U(g) \in \{0, -1, -2, \dots\}$ of the value of an insertion (Theorem 47.2(ii)) contains such a line, since $\kappa_U(\mathbf{d}_1) > 0$. So $\langle m\rangle$ is meromorphic on every line parallel to $\mathbf{d}_1$.

If $\langle m\rangle = 0$, the closings vanish along the $d$ rays by Theorem 47.11, (a)⇒(c), at the integral origins $\mathbf{d}_0 + \lambda_k$, no coefficient of $x$ having a pole along the whole line.

Conversely let the closings vanish along the $d$ rays, and put $\ell = \mathbf{d}_0 + \mathbb{C}\mathbf{d}_1$. The ray with the origin $\mathbf{d}_0 + \lambda_k$ and the direction $\mathbf{d}_1$ satisfies the hypotheses of Theorem 39.57, and the class of $x$ restricted to its line has these closings (Proposition 47.10); so the continuation of the values along it vanishes identically: $\langle \mathrm{Sh}_{\lambda_k} m\rangle = \sigma_{\lambda_k}\langle m\rangle$ vanishes on $\ell$, for every $k$. Let $\lambda \in \mathbb{D}_F(\mathbb{Z})$ be a sum $\mu_1 + \cdots + \mu_p$ of elements of $\mathcal{G}$. Applying the shifts of the $\mu_i$ one at a time, $\mathrm{Sh}_\lambda m = \sum_k c_k^\lambda \mathrm{Sh}_{\lambda_k} m$, $c^\lambda$ the first row of a product of translates $\sigma_\nu(\mathbf{B}_{\mu_i})$, whose entries are regular at all but finitely many points of $\ell$. The values intertwine the shifts and are $\mathbb{K}_F$-linear (Theorem 47.4(i)), so $\sigma_\lambda\langle m\rangle = \sum_k c_k^\lambda\,\sigma_{\lambda_k}\langle m\rangle$ vanishes on $\ell$ off a finite set, and $\langle m\rangle$, meromorphic on the line $\ell + \lambda$, vanishes on it identically.

Let $\mathbf{d}'_1$ be any positive direction such that no coefficient of $x$ has a pole along the whole line $\mathbf{d}_0 + \mathbb{C}\mathbf{d}'_1$. For every integer $j$ the point $\mathbf{d}_0 + j\mathbf{d}'_1$ lies on the line $\ell + j\mathbf{d}'_1$, and for all large $j$ it lies in the tube of every insertion of $x$ and off the poles of its coefficients, where $\langle m\rangle$ is holomorphic; so $\langle m\rangle(\mathbf{d}_0 + j\mathbf{d}'_1) = 0$ for all large $j$. That is (b) of Theorem 47.11 at the origin $\mathbf{d}_0$, and $\langle m\rangle = 0$.

Along each of the $d$ rays the vanishing is decided by finitely many closings, as in the proof of Theorem 47.11. The excluded directions are the zeros of finitely many nonzero homogeneous polynomials, a cone without interior points; the open cone $C_+$ of that proof has rational points off it, and an integral multiple of such a point is a positive direction off it. □

**Proposition 47.13 (Masters from a box).** *As stated.*

*Proof.* (i) If the classes of $\mathbf{b}$ are independent they are masters, and $\mathbf{B}_\mu = \mathbf{A}_{-\mu}$ is invertible (Theorem 47.8(i), Theorem 47.4(v)).

(ii) In $\mathfrak{M}_F$, $\mathrm{Sh}_{-\mu}[\mathbf{b}] = \mathbf{B}_\mu[\mathbf{b}]$ (Theorem 47.8(i)); applying $\mathrm{Sh}_\mu$, $[\mathbf{b}] = \sigma_\mu(\mathbf{B}_\mu)\,\mathrm{Sh}_\mu[\mathbf{b}]$, so $\mathrm{Sh}_\mu[\mathbf{b}] = \sigma_\mu(\mathbf{B}_\mu)^{-1}[\mathbf{b}]$. So the map $\pi : \varepsilon_i \mapsto [b_i]$ of $\Bbbk(\mathbb{D}_F)^{\mathbf{b}}$ onto $\mathfrak{M}_{F,\Bbbk}$, onto by Theorem 47.8(i) and Theorem 47.4(iv), carries $T_{\pm\mu}$ to $\mathrm{Sh}_{\pm\mu}$, and its kernel is carried into itself by the maps. The kernel contains the commutators and the $T_{-\mu}T_\mu\varepsilon_i - \varepsilon_i$, $T_\mu T_{-\mu}\varepsilon_i - \varepsilon_i$, the shifts commuting and $\mathrm{Sh}_{-\mu}\,\mathrm{Sh}_\mu = 1$; it contains the $\varepsilon_i - \rho(\lambda_i)$, since $\pi(\rho(\lambda)) = \mathrm{Sh}_\lambda[1]$ and $[b_i] = [f_{\lambda_i}] = \mathrm{Sh}_{\lambda_i}[1]$ (Theorem 47.4(ii)); and it contains the images of the letter identities and of the mark relations at [1], which are relations of $\mathfrak{M}_F$. So $\mathfrak{N} \subseteq \ker\pi$, and $\pi$ induces a map $\bar\pi$ of the quotient onto $\mathfrak{M}_{F,\Bbbk}$.

Modulo $\mathfrak{N}$, $\rho(\lambda)$ does not depend on the path, the steps of two paths being reordered and cancelled by the commutators and the inverse pairs, and $\rho(\lambda+\nu) \equiv T_\nu\rho(\lambda)$ for $\nu$ a generator or the negative of one. $\mathfrak{M}_{F,\Bbbk}$ is the $\Bbbk(\mathbb{D}_F)$-span of the shifts of [1] modulo the shifts of the letter identities and of the mark relations (the proof of Theorem 48.18(ii)), and $\rho$ carries the shift by $\lambda$ of such a relation at [1] to the image of its image under the maps of a path to $\lambda$, which lies in $\mathfrak{N}$. So $\mathrm{Sh}_\lambda[1] \mapsto \rho(\lambda)$ is a well-defined map $\iota$ of $\mathfrak{M}_{F,\Bbbk}$ to the quotient, semilinear for the shifts. $\bar\pi\iota(\mathrm{Sh}_\lambda[1]) = \pi(\rho(\lambda)) = \mathrm{Sh}_\lambda[1]$, and $\iota\bar\pi(\varepsilon_i) = \rho(\lambda_i) \equiv \varepsilon_i$. So $\bar\pi$ is an isomorphism; the paths enter $\mathfrak{N}$ only through the classes of the $\rho(\lambda)$, which they do not change. A basis of the quotient among the $\varepsilon_i$ goes to masters, and the matrices of the $T_{\pm\mu}$ on it are their shift matrices. The quotient is found by finitely many eliminations: the span of the listed vectors, closed under the finitely many maps, stabilizes, its dimension increasing until it does.

(iii) Fix $\mathfrak{s}$ at which some box is accepted (Theorem 47.8(iv)), and let $\mathbf{b}'$ be the free Laurent symbols of wall degree at most $\mathfrak{s}$ in it. The linear relations among the classes of the Laurent symbols of wall degree at most $\mathfrak{s}$, finitely many, form a subspace of finite dimension; each of a basis of it is an element of $\mathcal{L}_F$, a finite combination of the $(c_e(\mathbf{d}) + \partial_{y_e})g$ (Definition 47.3(i)), whose evaluations are the mark relations of finitely many symbols, together with finitely many letter identities (Definition 47.6(ii)). A box with $\mathfrak{t}$ and level bounds large enough contains these, and in it the free Laurent symbols of wall degree at most $\mathfrak{s}$ have independent classes. It is still accepted at $\mathfrak{s}$: the targets reduce to combinations of symbols of wall degree at most $\mathfrak{s}$ modulo the smaller relations, and every such symbol reduces modulo the larger ones to free symbols of wall degree at most its own, the ranking ordering the Laurent symbols by wall degree. Its matrices are then invertible by (i), and the search reaches it.

For a family of positive letters the same proof holds with Theorem 48.3(iii), (iv) in the place of Theorems 47.4 and 47.8. □

**Proposition 47.14 (Value zero on a subspace of the data).** *As stated.*

*Proof.* (i) On $V_W$, $\log f(e^y) = \langle \mathbf{m}_1, y\rangle - \sum_i w_{1,i} \log L_i(e^y)$, the sum over the letters of $W_1$. The Hessian of $\log L_i(e^y)$ in $y$ is the covariance of the exponent $\mathbf{t}$ under the probability proportional to $c_{i,\mathbf{t}} e^{\langle \mathbf{t}, y\rangle}$ on the support of $L_i$; it is positive semidefinite, and its kernel is the set of the directions on which every difference of two exponents of $L_i$ vanishes. The differences of two exponents of one letter of $W_1$ generate $M_{W_1}$, and $M_{W_1} = M_W$, $W_1$ having the active molecules of $W$ (Theorem 48.23(i)), while $M_W \otimes \mathbb{R}$ is the dual of $V_W$. So $-\operatorname{Hess}\log f = \sum_i w_{1,i}\,\mathrm{Cov}_i$ is positive definite on $V_W$ at every point, and $\log f$ is strictly concave. Every wall of $\mathbf{d}_1$ is positive, so on each cone of a fan $f = \prod_k q_k^{\kappa_{v_{\pi,k}}(\mathbf{d}_1)} \prod_i k_{i,\pi}(q)^{-w_{1,i}}$ with the $k_{i,\pi}$ bounded above and below on the cube (Lemma 48.2(iii)), and $f$ tends to 0 as some $q_k$ does: $\log f$ tends to $-\infty$ at infinity, attains its maximum, and has no other critical point, the maximum being nondegenerate.

For the converse, the derivative of $\log f$ along $v \in V_W$ at $y^\circ = \log\alpha^\circ$ is $\langle \mathbf{m}_1 - \sum_i w_{1,i} R_i(\alpha^\circ), v\rangle$, the occupations being the logarithmic derivatives of the letters; so with $\mathbf{m}_1 = \sum_i w_{1,i} R_i(\alpha^\circ)$ the point $y^\circ$ is critical, and by the first part it is the native gauge. $R_i(\alpha^\circ)$ is the mean of the exponents of $L_i$ under a probability that charges every exponent of its support, so it lies in the relative interior

of the Newton polytope $\mathcal{N}_{L_i}$; so $\mathbf{m}_1 - \sum_i w_{1,i}\mathbf{t}_i = \sum_i w_{1,i}(R_i(\alpha^\circ) - \mathbf{t}_i)$ lies in the relative interior of $\sum_i w_{1,i}(\mathcal{N}_{L_i} - \mathbf{t}_i)$, the sums over $W_1$, a polytope of the dimension $r_W$ in $\vec{\mathcal{N}}_W$. So the point lies in $\mathbb{D}_W$, and every wall of $W_1$ at it is positive (Theorem 48.3(i)). The walls of $W$ at it are those of $W_1$: the letters with the exponent 0 add nothing to them, and $W$ and $W_1$ have one fan, that of their active molecules (Theorem 48.23(i)). The occupations at a rational point are rational in it and in the weights, so for rational $\alpha^\circ$ and weights the point is rational, and an integral multiple of it, which has the same native gauge, is integral. The walls are linear forms and the conditions on the exponents are linear, so the set of the real points with the stated exponents and every wall positive is an open cone in the real points of the subspace on which the exponents outside $W_1$ vanish; when $\mathbb{E}$ is defined over $\mathbb{Q}$ and meets it, the intersection is open in the real points of a subspace defined over $\mathbb{Q}$, its rational points are dense in it, and a positive integral multiple of a rational point of it is an integral point of it, a native direction of $\mathbb{E}$. On a chart of $V_W$ the native gauge is a common zero of the defects $c_e(\mathbf{d}_1)$, rational functions of the point with coefficients in the field of the weights, and it is isolated, the Hessian being nonsingular there; an isolated point of the zeros of polynomials over a field has coordinates algebraic over that field.

(ii) Along the line, $\langle g_l\rangle(p+k\mathbf{d}_1) = \int_{V_W} g_l\,\omega_p\,f^k$, converging absolutely for $\operatorname{Re} k$ large, $\mathbf{d}_1$ having positive walls (Lemma 48.2(iii)). By (i), $y^\star$ is a nondegenerate maximum of the real function $\log f$ and the only critical point; outside a neighbourhood of it $f \le f^\star - \delta$ for some $\delta > 0$, so that part contributes $O\big((f^\star-\delta)^k\big)$, and near it the orthant is a steepest-descent disc. Laplace's method there, as in the proof of Theorem 38.6(i) with the origin form $\omega_p$, gives

$$\langle g_l\rangle(p+k\mathbf{d}_1) \;\sim\; c_\star\,\frac{\omega_p}{dy}(\alpha^\star)\,g_l(\alpha^\star)\,(f^\star)^k k^{-r/2}\sum_{i\ge 0} P_{l,i}(p)\,k^{-i},$$

$P_{l,0} = 1$ and $P_{l,i}$ a finite combination of Gaussian moments of the covariance $(-H)^{-1}$, each times Taylor coefficients at $y^\star$ of $\log f$ and of $G = g_l\,\omega_p/dy$ divided by its value, of total order at most $2i$ in the latter. Now $\log G = \log g_l + \langle \mathbf{m}_p, y\rangle - \sum_i w_{p,i}\log L_i(e^y)$ is linear in $p$, so a Taylor coefficient of $G/G(y^\star)$ of order $j$ is a polynomial in $p$ of degree at most $j$, with coefficients formed from the Taylor coefficients of $\log g_l$ and of the $\log L_i$; and $P_{l,i}$ is a polynomial in $p$ of degree at most $2i$. Off the $p$ for which some $r_l$ has a pole along the whole line, a proper algebraic subset, $r_l(p+k\mathbf{d}_1)$ is a rational function of $k$ with its expansion at $k=\infty$, of degree at most $\nu$, whose coefficients are rational functions of $p$; multiplying the expansions and collecting the powers of $k$ gives the display, with

$$Q_i = \sum_l \sum_{a+i'=i} r_{l,a}(p)\,g_l(\alpha^\star)\,P_{l,i'}(p), \qquad r_l(p+k\mathbf{d}_1) = k^\nu \sum_a r_{l,a}(p)\,k^{-a}.$$

The Gaussian moments of $(-H)^{-1}$ and the Taylor coefficients of $\log f$, $\log g_l$ and $\log L_i$ at $y^\star$ are rational in $\alpha^\star$, the weights and the data of $\mathbf{d}_1$ and of $g_l$, which gives the field.

(iii) If $\langle x\rangle = 0$ on $\mathbb{E}$, then at every $p$ off the subset of (ii) the left side of the display vanishes for all large real $k$, and the asymptotic expansion of 0 is 0: every $Q_i(p)$ vanishes, so every $Q_i$ is 0. Conversely let every $Q_i$ be 0, and fix $p$ off that subset. The expansion of $\langle x\rangle(p+k\mathbf{d}_1)$ vanishes, and the proof of Theorem 39.57, (iii)⇒(ii), applies along the line with $n-1=r$: by (i) the level sets $\{f=t\}$, $0<t<f^\star$, are compact real-analytic hypersurfaces of $V_W$ without critical points; by the coarea formula $\langle g_l\rangle(p+k\mathbf{d}_1) = \int_0^{f^\star} t^k\rho_l(t)\,dt$ for $\operatorname{Re} k$ large, $\rho_l$ real-analytic on $(0,f^\star)$, complex-valued, and by the Morse lemma at $y^\star$ equal to $(f^\star-t)^{(r-2)/2}$ times a function analytic at $f^\star$; with a common denominator of the $r_l(p+k\mathbf{d}_1)$ and Watson's lemma, the vanishing of the expansion gives the vanishing of $\langle x\rangle(p+k\mathbf{d}_1)$ for $\operatorname{Re} k$ large, and so of its continuation in $k$. The lines $p+\mathbb{C}\mathbf{d}_1$, $p$ off a proper algebraic subset of $\mathbb{E}$, fill a dense subset of $\mathbb{E}$, and $\langle x\rangle$, meromorphic on $\mathbb{E}$, vanishes on it, so it vanishes.

(iv) Over $\mathbb{C}(\mathbb{E})$, with $\mathbf{d}$ the point of $\mathbb{E}$, the relations are those of the integrable connection $d + d\log(\omega_{\mathbf{d}}/dy)$ on the smooth affine variety $X_W$ (proof of Theorem 48.3(iii)), and $\mathfrak{M}_{W,\mathbb{E}}$ is the top cohomology of its algebraic de Rham complex over a field of characteristic zero, finite-dimensional (proof of Theorem 47.4). The value of an element of $\mathcal{L}_W$ vanishes identically on $\mathbb{D}_W$ (Theorem 48.3(ii)), so that of an element of $\mathcal{L}_{\mathbb{E}}$ vanishes on $\mathbb{E}$, where the values of the insertions are meromorphic, $\mathbb{E}$ meeting the tube. $(c_e(\mathbf{d}) + \partial_{y_e})(f_{\mathbf{d}_1} g) = f_{\mathbf{d}_1}(c_e(\mathbf{d}+\mathbf{d}_1) + \partial_{y_e})g$ (proof of Theorem 48.3(iii)) and $\sigma_{\mathbf{d}_1}$ carries $\mathbb{C}(\mathbb{E})$ to itself, $\mathbf{d}_1$ lying in $\mathbb{E}$; so $\mathrm{Sh}_{\mathbf{d}_1}$ is a well-defined $\sigma_{\mathbf{d}_1}$-semilinear bijection of $\mathfrak{M}_{W,\mathbb{E}}$, with the inverse $\mathrm{Sh}_{-\mathbf{d}_1}$, and $\langle \mathrm{Sh}_{\mathbf{d}_1} x\rangle(\mathbf{d}) = \sum_l r_l(\mathbf{d}+\mathbf{d}_1)\langle f_{\mathbf{d}_1} g_l\rangle(\mathbf{d}) = \langle x\rangle(\mathbf{d}+\mathbf{d}_1)$. The span of the $\mathrm{Sh}^j x$, $j \geq 0$, has finite dimension; at the least $h$ at which $\mathrm{Sh}^h x$ lies in the span of the earlier ones there is a law with $q_h = 1$, and $q_0 \neq 0$, since otherwise $\mathrm{Sh}_{-\mathbf{d}_1}$ would give a shorter one. Its certificate is a finite combination of elements of $\mathcal{L}_{\mathbb{E}}$; a search through the finite supports, each tested by one elimination over $\mathbb{C}(\mathbb{E})$, reaches one.

The value of the law at $p+k\mathbf{d}_1$ is $\sum_j q_j(p+k\mathbf{d}_1)\,\langle x\rangle\big(p+(k+j)\mathbf{d}_1\big) = 0$. The expansion of the $j$-th term is that of (ii) at $k+j$, times the expansion of $q_j(p+k\mathbf{d}_1)$; with $(k+j)^{\rho-i} = k^{\rho-i}\sum_b \binom{\rho-i}{b} j^b k^{-b}$, the expansion of the left side divided by $c_\star(\omega_p/dy)(\alpha^\star)(f^\star)^k k^{\rho+\nu'}$ is

$$\sum_{N\geq 0} k^{-N} \sum_{a+b+i=N} \binom{\rho-i}{b} \chi_{a,b}\, Q_i,$$

which is 0. Grouping by $s = a+b$, the coefficient of $k^{-N}$ is $\sum_s C_s(N)\, Q_{N-s} = 0$ for every $N \geq 0$, with $Q_i = 0$ for $i < 0$. If every $C_s$ were 0, then, $\binom{\rho-N+s}{b}$ having the degree $b$ in $N$, comparing the powers of $N$ from the top would give $\chi_{a,b} = 0$ for all $a, b$; then $\sum_j q_{j,a}(f^\star)^j j^b = 0$ for every $b$, and the Vandermonde matrix of the distinct $j = 0, \dots, h$ gives $q_{j,a} = 0$, so that every $q_j$ would vanish along the line, against $q_0 \neq 0$ for $p$ off a proper algebraic subset. With $s_0$ as stated, the equation at $N$ reads $C_{s_0}(N) Q_{N-s_0} = -\sum_{s>s_0} C_s(N) Q_{N-s}$. For $N > N_0$, $C_{s_0}(N) \neq 0$, so $Q_{N-s_0}$ is a combination of the $Q_i$ with $i < N - s_0$; if these vanish up to $N_0 - s_0$, induction on $N$ gives that all do. The integer roots of $C_{s_0}$ are the integers at which a nonzero polynomial over $\mathbb{C}(\mathbb{E})$ vanishes, finitely many. □

**Proposition 47.15 (The values separate the module).** *As stated.*

*Proof.* The fibre of $\mathbb{D}_W$ over $\mathbf{w}$ is $\mathbf{m}_0(\mathbf{w}) + \vec{\mathcal{N}}_W \otimes \mathbb{C}$, $\mathbf{m}_0(\mathbf{w}) = \sum_i w_i \mathbf{t}_i$ (Definition 48.1(ii)), so $\pi$ is onto, $\mathbb{E}$ is a linear subspace with $\pi(\mathbb{E}) = \mathbb{E}_{\mathbf{w}}$ whose fibres are those of $\mathbb{D}_W$, and the polynomials on $\mathbb{E}$ are the polynomials in the masses $m_e$, $e \in B$, with coefficients polynomials in the exponents, read on $\mathbb{E}$.

(i) Written out, $(c_e(\mathbf{d}) + \partial_{y_e}) g \otimes r = g \otimes m_e r - \sum_i R_{i,e}\, g \otimes w_i r + \partial_{y_e} g \otimes r$, the occupations $R_{i,e}$ and $\partial_{y_e} g$ being insertions (Definition 48.1(iii), (iv)); it lies in $\mathcal{L}_{\mathbb{E}}$, which is the identity. Let $x = \sum_l g_l \otimes r_l$, let $D \neq 0$ be a common denominator of the $r_l$, a polynomial on $\mathbb{E}$, and write every $Dr_l$ as a polynomial in the masses with coefficients polynomials in the exponents. The identity, applied to $g \otimes m_e r$ with $r$ a monomial in the masses times a polynomial in the exponents, replaces it by terms whose coefficients have one mass fewer; by induction on the degree in the masses, $Dx$ is the class of an element $\sum_k G_k \otimes p_k$ as stated.

(ii) For $g \in \mathbb{C}[X_W]$ the value $\langle g\rangle$ is meromorphic on $\mathbb{D}_W$ with its poles on the hyperplanes $\kappa_v(\mathbf{d}) + \kappa_v(g) \in \{0, -1, -2, \dots\}$, $v$ a generator (Theorem 48.3(ii)). None of them contains $\mathbb{E}$: the wall $\kappa_v$ is a linear form with a positive real part at a point of $\mathbb{E} \cap \mathcal{T}_1$, so it is not constant on the linear space $\mathbb{E}$. So $\langle g\rangle$ restricts to a meromorphic function on $\mathbb{E}$, and the value of $x$ on $\mathbb{E}$ is defined. At a point $\mathbf{d}$ of $\mathbb{E}$ the element $(c_e(\mathbf{d}) + \partial_{y_e}) g \otimes r$ has the value $r(\mathbf{d})\langle (c_e(\mathbf{d}) + \partial_{y_e}) g\rangle(\mathbf{d}) = 0$ (Theorem 48.3(ii)), so the value vanishes on $\mathcal{L}_{\mathbb{E}}$ and is defined on $\mathfrak{M}_{W,\mathbb{E}}$.

Let $\langle x\rangle = 0$ on $\mathbb{E}$, and let $Dx = [G]$, $G = \sum_k G_k \otimes p_k$ as in (i), with the $G_k$ linearly independent over $\mathbb{C}$, each a finite combination of strides $f_\lambda$, $\lambda \in \mathbb{D}_W(\mathbb{Z})$ (Definition 48.1(iii)), and $\Lambda$ the finite set of these $\lambda$. Then $\langle G\rangle = D\langle x\rangle = 0$ on $\mathbb{E}$, and $\langle G\rangle(\mathbf{d}) = \sum_k p_k(\mathbf{w})\langle G_k\rangle(\mathbf{d})$ converges absolutely on $\mathcal{T}_\Lambda = \bigcap_{\lambda\in\Lambda}(\mathcal{T}_1 - \lambda)$, $\langle f_\lambda\rangle(\mathbf{d})$ being $P_W(\mathbf{d}+\lambda)$ (Theorem 48.3(ii)). The walls are linear forms with real coefficients (Lemma 48.2(ii)), so for $\mathbf{d} \in \mathbb{E}\cap\mathcal{T}_1$ and real $t > 0$, $\operatorname{Re}\kappa_v(t\mathbf{d}) = t\operatorname{Re}\kappa_v(\mathbf{d})$, $t\mathbf{d} \in \mathcal{T}_\Lambda$ for $t$ large, and $\mathbb{E}\cap\mathcal{T}_\Lambda$ is a nonempty open subset of $\mathbb{E}$. Its image $U$ under $\pi$, a linear map of $\mathbb{E}$ onto $\mathbb{E}_\mathbf{w}$, is a nonempty open subset of $\mathbb{E}_\mathbf{w}$. Fix $\mathbf{w} \in U$. The points of the fibre $\mathbf{m}_0(\mathbf{w}) + \vec{\mathcal{N}}_W \otimes \mathbb{C}$ that lie in $\mathcal{T}_\Lambda$ are those whose masses have their real parts in a nonempty open set, cut out by the strict linear inequalities $\operatorname{Re}\kappa_v(\mathbf{m};\mathbf{w}) + \kappa_v(\lambda) > 0$; fix such a point $\mathbf{d}^\circ = (\mathbf{m}^\circ;\mathbf{w})$ and move it by $(i\tau;\mathbf{0})$, $\tau \in \vec{\mathcal{N}}_W$, which keeps the real parts. The pairing of $\vec{\mathcal{N}}_W$ with $V_W = \mathbb{R}^B/\vec{\mathcal{N}}_W^\perp$ is perfect, and $\omega_{\mathbf{d}^\circ+(i\tau;\mathbf{0})} = e^{i\langle\tau,y\rangle}\,\omega_{\mathbf{d}^\circ}$, so

$$0 = \langle G\rangle\big(\mathbf{d}^\circ + (i\tau;\mathbf{0})\big) = \int_{V_W} e^{i\langle\tau,y\rangle}\,h(y)\,dy, \qquad h = G(\cdot;\mathbf{w})\,\frac{\omega_{\mathbf{d}^\circ}}{dy}, \quad G(\cdot;\mathbf{w}) = \sum_k p_k(\mathbf{w})\,G_k,$$

for every $\tau \in \vec{\mathcal{N}}_W$, with $h$ integrable on $V_W$, every term converging absolutely. This is the Fourier transform of $h$ on $V_W$; so $h = 0$ almost everywhere and, $h$ being continuous, everywhere. The density $\omega_{\mathbf{d}^\circ}/dy$ has no zero on $V_W$, the letters being positive there and the powers principal, so $G(\cdot;\mathbf{w})$ vanishes at the positive points of $X_W$, the image of the positive orthant. The quotient $T_W$ of $(\mathbb{C}^\times)^B$ by the subtorus of Definition 48.1(iii) is a torus with the coordinates $\alpha^{u_1},\dots,\alpha^{u_{r_W}}$, $u_k$ a basis of $M_W$, a lattice that is saturated in $\mathbb{Z}^B$; the functions $\ell_i = \alpha^{-\mathbf{t}_i}L_i$ are regular on it, $X_W$ is the set on which none vanishes, and a regular function on $X_W$ is $P\prod_i \ell_i^{-a_i}$ with $P$ a Laurent polynomial in these coordinates. The positive points are those at which every coordinate is positive, where no $\ell_i$ vanishes, and a Laurent polynomial that vanishes there is 0. So $G(\cdot;\mathbf{w}) = 0$ in $\mathbb{C}[X_W]$ for every $\mathbf{w} \in U$; the $G_k$ being independent, every $p_k$ vanishes on $U$, and a polynomial on $\mathbb{E}_\mathbf{w}$ that vanishes on a nonempty open set is 0. So $G = 0$, $Dx = 0$, and $x = 0$, $D$ being invertible in $\mathbb{C}(\mathbb{E})$.

The rest of (ii) follows: a combination of insertions of value zero is 0 in $\mathfrak{M}_{W,\mathbb{E}}$, that is, it lies in $\mathcal{L}_\mathbb{E}$; for insertions $g$ and $g'$, $\langle g\rangle = \langle g'\rangle$ exactly when $\langle [g-g']\rangle = 0$; and a relation $\sum_l r_l\langle b_l\rangle = 0$ is the value zero of $\sum_l r_l[b_l]$.

(iii) $\mathbb{D}_W = \pi^{-1}(\mathbb{C}^I)$ meets $\mathcal{T}_1$ (Theorem 48.3(i)), so (ii) gives $\operatorname{Ann}\langle\cdot\rangle = 0$, and a submodule of elements of value zero is 0. By Theorem 47.4(v), with Theorem 48.3(iii), $\langle g\rangle = \langle g'\rangle$ exactly when the reductions agree, and the values of masters are independent by (ii). For a family of letters of tables, Theorem 47.11 and Proposition 47.12 hold for $\mathfrak{M}_W$ (Theorem 48.23(ii)), so the elements whose closings vanish along those rays are the elements of value zero, which are 0.

(iv) A native direction has every wall positive, so it lies in $\mathcal{T}_1$ and $\mathbb{E}$ meets the tube. By Proposition 47.14(iii), (iv), the value of $x$ on $\mathbb{E}$ vanishes exactly when finitely many of its coefficients $Q_i$ do, and by (ii) it vanishes exactly when $x = 0$. □

## 50.47 Chapter 48: the module along the tree

**Definition 48.1 (A family of positive letters).** As stated.

*Proof.* (i) The exponents of the support of a letter have one total, its degree, so their differences lie in $\mathbf{1}^\perp$. A constant letter has the one exponent $\mathbf{0}$: it adds nothing to $\vec{\mathcal{N}}_W$, its occupations vanish, its term in a wall is 0, and below its polynomial $k_{i,\pi}$ is its value. The space of the directions of a Minkowski sum $\sum_i w_i\mathcal{N}_{L_i}$ with every $w_i > 0$ is the sum of the spaces of the directions of its terms, each spanned by the differences of the exponents of a support.

(ii) Another exponent $\mathbf{t}_i'$ of the support of $L_i$ has $\mathbf{t}_i' - \mathbf{t}_i \in \vec{\mathcal{N}}_W$, so the condition does not change. For a full family $\vec{\mathcal{N}}_W \otimes \mathbb{C} = \mathbf{1}^\perp$, and the condition is $m_B = \sum_i w_i|\mathbf{t}_i| = \sum_i w_i \deg L_i$.

(iii) $M_W$ is the set of the integral points of a rational subspace, so it is saturated, and the subtorus $H$ on which every $\alpha^u$, $u \in M_W$, is 1 is a torus of dimension $|B| - r_W$ whose characters trivial on it are exactly the $\alpha^u$, $u \in M_W$. For an exponent $\mathbf{t}$ of the support of $L_i$ and $\zeta \in H$, $\zeta^{\mathbf{t}} = \zeta^{\mathbf{t}_i}\zeta^{\mathbf{t}-\mathbf{t}_i} = \zeta^{\mathbf{t}_i}$, so $L_i(\zeta\alpha) = \zeta^{\mathbf{t}_i}L_i(\alpha)$, and the set $Y$ on which no letter vanishes is stable. The quotient $(\mathbb{C}^\times)^B/H$ is a torus of dimension $r_W$ with the regular functions $\mathbb{C}[M_W]$, and $X_W = Y/H$ is its open subset where the invariant Laurent polynomial $\prod_i \alpha^{-\mathbf{t}_i}L_i$ does not vanish: smooth and affine, of dimension $r_W$, its regular functions the invariants of $\mathbb{C}[Y] = \mathbb{C}[\alpha^{\pm1}][\prod_i L_i^{-1}]$. Every element of $\mathbb{C}[Y]$ is a sum of elements $\alpha^u \prod_i L_i^{-a_i}$, $\mathbf{a} \in \mathbb{N}^I$, each semi-invariant with the character $\alpha^{u-\sum_i a_i\mathbf{t}_i}$, which is trivial on $H$ exactly when $u - \sum_i a_i\mathbf{t}_i \in M_W$, that is when $(u;\mathbf{a}) \in \mathbb{D}_W(\mathbb{Z})$; so the invariants are the stated span. For $\lambda = (v;\mathbf{a}) \in \mathbb{D}_W(\mathbb{Z})$ with $\mathbf{a} \in \mathbb{Z}^I$, expanding the powers $L_i^{-a_i}$ with $a_i < 0$ writes $f_\lambda$ as a sum of the $\alpha^{v+\mathbf{t}}\prod_{a_i>0} L_i^{-a_i}$, $\mathbf{t}$ a sum of $-a_i$ exponents of the support of each $L_i$ with $a_i < 0$, whose data are integral points of $\mathbb{D}_W$, the vector $\mathbf{t} - \sum_{a_i<0}(-a_i)\mathbf{t}_i$ lying in $M_W$; so $f_\lambda \in \mathbb{C}[X_W]$, and $f_\lambda f_{-\lambda} = 1$.

(iv) $V_W(\mathbb{Z})$ is a lattice of full rank in $V_W$, $\vec{\mathcal{N}}_W^\perp$ being rational. For $v \in \vec{\mathcal{N}}_W^\perp$ every exponent $\mathbf{t}$ of $L_i$ has $\langle \mathbf{t}, v\rangle = \langle \mathbf{t}_i, v\rangle$, so $L_i(e^{y+v}) = e^{\langle \mathbf{t}_i, v\rangle}L_i(e^y)$, the logarithms being real, and the density at $y+v$ is $e^{\langle \mathbf{m} - \sum_i w_i\mathbf{t}_i, v\rangle}$ times that at $y$, which is the density at $y$ on $\mathbb{D}_W$. Integral powers of the positive $L_i(e^y)$ multiply the principal ones, so $f_\lambda\, \omega_{\mathbf{d}} = \omega_{\mathbf{d}+\lambda}$. $R_{i,e} = \sum_{\mathbf{t}} c_{i,\mathbf{t}} t_e\, \alpha^{\mathbf{t}}/L_i$ is a combination of the insertions $\alpha^{\mathbf{t}}/L_i$, whose data $(\mathbf{t};\mathbf{e}_i)$ lie in $\mathbb{D}_W(\mathbb{Z})$; on the orthant $0 \le t_e \le |\mathbf{t}| = \deg L_i$ and the $c_{i,\mathbf{t}}\alpha^{\mathbf{t}}$ are positive, so $0 \le R_{i,e} \le \deg L_i$. $\partial_{y_e} \log(\omega_{\mathbf{d}}/dy) = m_e - \sum_i w_i\alpha_e\partial_{\alpha_e}L_i/L_i = c_e(\mathbf{d})$, and the same computation with the data of $\lambda$ gives $c_e(\lambda)$. For $v \in \vec{\mathcal{N}}_W^\perp$, $\sum_e v_e\alpha_e\partial_{\alpha_e}\alpha^{\mathbf{t}} = \langle \mathbf{t}, v\rangle\alpha^{\mathbf{t}} = \langle \mathbf{t}_i, v\rangle\alpha^{\mathbf{t}}$ on the support of $L_i$, so $\sum_e v_e R_{i,e} = \langle \mathbf{t}_i, v\rangle$ and $\sum_e v_e c_e(\mathbf{d}) = \langle \mathbf{m} - \sum_i w_i\mathbf{t}_i, v\rangle = 0$.

(v) For a full family $\vec{\mathcal{N}}_W^\perp = \mathbb{R}\mathbf{1}$, $V_W$ is identified with the chart $\{y_{e_0} = 0\}$, whose integral points are the image of $\mathbb{Z}^B$ and whose Lebesgue measure $\prod_{e\ne e_0} dy_e$ gives them the covolume one, and $\int e^{\langle \mathbf{m}, y\rangle} W(\mathbf{w})(e^y) \prod_{e\ne e_0} dy_e$ is the integral of $\alpha^{\mathbf{m}-\mathbf{1}}W(\mathbf{w})\, d\alpha$ over the section $\alpha_{e_0} = 1$, the period (Definition 25.1, Lemma 16.2). For every family the integral over $\mathbb{R}^B/\vec{\mathcal{N}}_W^\perp$ with this measure is the reduced period of Theorem 28.12 by its definition. □

**Lemma 48.2 (The fan of a family of positive letters).** *As stated.*

*Proof.* The differences of the exponents of one letter lie in $\vec{\mathcal{N}}_W$, which is the dual of $V_W$, so the pairings below are well defined.

(i) For each $i$ and each exponent $\mathbf{t}$ of the support of $L_i$ the set $C_{i,\mathbf{t}} = \{v \in V_W :\ \langle \mathbf{t} - \mathbf{t}', v\rangle \le 0\ \forall \mathbf{t}'\}$ is a rational polyhedral cone, and these cones form the inner normal fan of $\mathcal{N}_{L_i}$ read in $V_W$, which covers $V_W$, every $v$ having an exponent of least pairing. The cones $\bigcap_i C_{i,\mathbf{t}_i}$, one exponent for each letter, with their faces, form a fan covering $V_W$, the common refinement, and each is pointed: a line $\mathbb{R}v$ in it has $\langle \mathbf{t} - \mathbf{t}', v\rangle = 0$ for any two exponents of every letter, so $v$ vanishes on $\vec{\mathcal{N}}_W$ and is 0 in $V_W$. Triangulating these cones by pulling, as in the proof of Lemma S14(i), gives finitely many rational simplicial cones of dimension $r_W$ with disjoint interiors covering $V_W$, with their faces a fan, each in some $\bigcap_i C_{i,\mathbf{t}_i}$. A face of a Minkowski sum $\sum_i w_i\mathcal{N}_{L_i}$, $w_i > 0$, is the sum of the faces of the terms on which one linear form is least, so the inner normal fan of the sum is the common refinement of those of the terms, and every cone of a fan of $W$ lies in a cone of it. Let $\rho$ be a ray of the inner normal fan $\Sigma'$ of the sum and $p$ a point of its relative interior. Then $p$ lies in the relative interior of some cone $\sigma$ of the fan of $W$, and $\sigma$ lies in a cone $\tau$ of $\Sigma'$; $\tau \cap \rho$ is a face of $\rho$ containing $p$, so $\rho$ is a face of $\tau$, and a convex subset of $\tau$ whose relative interior meets the face $\rho$ lies in $\rho$. So $\sigma = \rho$, and $\rho$ is a ray of the fan of $W$.

(ii) Replacing $\mathbf{t}_i$ by $\mathbf{t}'_i$ subtracts $w_i\langle \mathbf{t}'_i - \mathbf{t}_i, v\rangle$ from the first term and adds it to the second. For a cone $\pi$ with the generator $v$, $\mathbf{t}_{i,\pi}$ pairs least with every point of $\pi$, so $\min_{\mathbf{t}}\langle \mathbf{t}-\mathbf{t}_i, v\rangle = \langle \mathbf{t}_{i,\pi}-\mathbf{t}_i, v\rangle$ and $\kappa_v(\mathbf{d}) = \langle \mathbf{m} - \sum_i w_i\mathbf{t}_{i,\pi}, v\rangle$, which therefore does not depend on $\pi$. On $\mathbb{D}_W(\mathbb{Z})$ the vectors $\mathbf{m} - \sum_i w_i\mathbf{t}_i$ and $\mathbf{t} - \mathbf{t}_i$ lie in $M_W$, which pairs integrally with $V_W(\mathbb{Z})$.

(iii) Two exponents that pair least with every generator of $\pi$ have a difference in $\vec{\mathcal{N}}_W$ that vanishes on a basis of $V_W$, so they are equal. The exponents $\langle \mathbf{t}-\mathbf{t}_{i,\pi}, v_{\pi,k}\rangle$ are integers, by (ii), and nonnegative by the choice of $\mathbf{t}_{i,\pi}$, and all of them vanish only at $\mathbf{t} = \mathbf{t}_{i,\pi}$; so $k_{i,\pi}$ is a polynomial with nonnegative coefficients and the constant term $c_{i,\mathbf{t}_{i,\pi}}$. For $x \in \vec{\mathcal{N}}_W \otimes \mathbb{C}$ and $y = -\sum_k u_k v_{\pi,k}$, $e^{\langle x,y\rangle} = \prod_k q_k^{\langle x, v_{\pi,k}\rangle}$, and on a lift of $y$ to $\mathbb{R}^B$, $L_i(e^y) = e^{\langle \mathbf{t}_{i,\pi}, y\rangle} k_{i,\pi}(q)$; so

$$g\,\omega_{\mathbf{d}}/dy = e^{\langle \mathbf{m}+u-\sum_i (w_i+a_i)\mathbf{t}_{i,\pi}, y\rangle} \prod_i k_{i,\pi}(q)^{-w_i-a_i} = \prod_k q_k^{\kappa_{v_{\pi,k}}(\mathbf{d})+\kappa_{v_{\pi,k}}(g)} \prod_i k_{i,\pi}(q)^{-w_i-a_i},$$

the exponent being $\kappa_{v_{\pi,k}}$ at the data $\mathbf{d}+(u;\mathbf{a})$, by (ii) and linearity. The map $(u_k)_k \mapsto -\sum_k u_k v_{\pi,k}$ is a linear bijection of $[0,\infty)^{r_W}$ onto $-\pi$ with the Jacobian $|\det(v_{\pi,1},\dots,v_{\pi,r_W})|$ for the measure $dy$, and $du_k = -dq_k/q_k$. The cones $-\pi$ cover $V_W$ with null boundaries and disjoint interiors, and the integrands are positive at real data, so the two sides agree in $[0,\infty]$ by Tonelli's theorem. On $[0,1]^{r_W}$ each $k_{i,\pi}$ lies between its constant term and the sum of its coefficients, so $\prod_i k_{i,\pi}^{-w_i-a_i}$ is bounded above and below by positive constants, and the cube integral is finite exactly when every exponent $\kappa_{v_{\pi,k}}(\mathbf{d}) + \kappa_{v_{\pi,k}}(g)$ is positive; every generator is a generator of some cone. At $r_W = 0$ the only cone is $\{0\}$, the measure is the unit mass and every letter is a monomial, and both sides are the density at the point.

For complex $\mathbf{d}$, $g$ is positive on the real points and $|L_i(e^y)^{-w_i}| = L_i(e^y)^{-\operatorname{Re} w_i}$, so $|g\,\omega_{\mathbf{d}}| = g\,\omega_{\operatorname{Re}\mathbf{d}}$, and $\langle g\rangle(\mathbf{d})$ converges absolutely exactly when $\operatorname{Re}\mathbf{d} \in \mathcal{T}_g$, by (iii); this set is defined by the convergence, so it does not depend on the fan. □

**Theorem 48.3 (The module of a family of positive letters).** *As stated.*

*Proof.* (i) At positive real exponents $\sum_i w_i \min_{\mathbf{t}}\langle \mathbf{t}-\mathbf{t}_i, v\rangle$ is the least value of $\langle\cdot, v\rangle$ on $P_{\mathbf{w}} = \sum_i w_i(\mathcal{N}_{L_i} - \mathbf{t}_i)$, the least value of a linear form on a Minkowski sum being the sum of its least values on the terms; so $\kappa_v(\mathbf{d}) = \langle p, v\rangle - \min_{P_{\mathbf{w}}}\langle\cdot, v\rangle$ with $p = \mathbf{m} - \sum_i w_i\mathbf{t}_i \in \vec{\mathcal{N}}_W$. $P_{\mathbf{w}}$ is a polytope in $\vec{\mathcal{N}}_W$ of the dimension $r_W$, its space of directions being $\vec{\mathcal{N}}_W$ (proof of Definition 48.1(i)). The inner normals of its facets lie on the rays of its inner normal fan in the dual $V_W$, which are rays of the fan (Lemma 48.2(i)), so their primitive vectors are generators. So every wall is positive exactly when $p$ lies strictly inside every facet, that is in the interior of $P_{\mathbf{w}}$ in $\vec{\mathcal{N}}_W$, an interior point having $\langle p, v\rangle > \min_{P_{\mathbf{w}}}\langle\cdot, v\rangle$ for every $v \neq 0$; with Lemma 48.2(iii) at $g = 1$ this is the criterion. $P_{\mathbf{1}} = \sum_i(\mathcal{N}_{L_i} - \mathbf{t}_i)$ is a lattice polytope of $M_W \otimes \mathbb{R}$, its vertices being sums of differences of exponents, of the dimension $r_W$, so it contains $r_W + 1$ affinely independent points $p_0,\dots,p_{r_W}$ of $M_W$, and $p_* = \sum_k p_k$, all of whose barycentric coordinates in the simplex $\operatorname{conv}\{(r_W+1)p_k\}$ are $(r_W+1)^{-1}$, is a point of $M_W$ in the interior of $(r_W+1)P_{\mathbf{1}} = P_{(r_W+1)\mathbf{1}}$. With $n = r_W + 1$, $\mathbf{d}_1 = (p_* + n\sum_i \mathbf{t}_i;\, n\mathbf{1})$ lies in $\mathbb{D}_W(\mathbb{Z})$, has every exponent $n \geq 1$, and has every wall positive by the criterion. The walls are linear, so $\kappa_v(\eta\,\mathbf{d}_1) = \eta\,\kappa_v(\mathbf{d}_1) > 0$.

(ii) *Convergence and the relations.* By Lemma 48.2(iii) and its last paragraph $\langle g\rangle(\mathbf{d})$ converges absolutely on $\mathcal{T}_g$. On the orthant $|c_e(\mathbf{d})| \leq |m_e| + \sum_i |w_i| \deg L_i$ and $\partial_{y_e} g = g\,(u_e - \sum_i a_i R_{i,e})$, so $|(c_e(\mathbf{d}) + \partial_{y_e})g| \leq C\,|g|$ and the value of $(c_e(\mathbf{d}) + \partial_{y_e})g$ converges absolutely on $\mathcal{T}_g$ as well. Its integrand is the derivative of $\phi = g\,\omega_{\mathbf{d}}/dy$ along the image $\bar{e}$ of $\mathbf{e}_e$ in $V_W$, a $C^1$ function that is integrable on $V_W$ together with its derivative along $\bar{e}$. If $\bar{e} = 0$ that derivative vanishes. Otherwise write $V_W = \mathbb{R}\bar{e} \oplus V'$; for almost every $y' \in V'$ the function $s \mapsto \phi(y' + s\bar{e})$ and its derivative are

integrable on $\mathbb{R}$, so it has limits at $\pm\infty$, which vanish, and the integral of its derivative is 0; by Fubini's theorem $\langle (c_e(\mathbf{d})+\partial_{y_e})g\rangle(\mathbf{d})=0$. The integrals $\int f_\lambda g\,\omega_{\mathbf{d}}$ and $\int g\,\omega_{\mathbf{d}+\lambda}$ have one integrand.

*Continuation.* By Lemma 48.2(iii), for $\mathbf{d}\in\mathcal{T}_g$,

$$\langle g\rangle(\mathbf{d})=\sum_\pi |\det(v_{\pi,1},\dots,v_{\pi,r_W})|\int_{[0,1]^{r_W}}\prod_k q_k^{x_k(\mathbf{d})-1}h_\pi(q,\mathbf{d})\,dq$$

with $x_k=\kappa_{v_{\pi,k}}(\mathbf{d})+\kappa_{v_{\pi,k}}(g)$ and $h_\pi=\prod_i k_{i,\pi}^{-w_i-a_i}$: the absolute convergence on $\mathcal{T}_g$ is that of Lemma 48.2(iii) at $\operatorname{Re}\mathbf{d}$. The $k_{i,\pi}$ have positive real parts on a neighbourhood of the cube, where their principal logarithms are holomorphic, so $h_\pi$ is smooth in $q$ near the cube, with $q$-derivatives continuous in $(q,\mathbf{d})$ and entire in $\mathbf{d}$. Lemma 8.4, with the data $\mathbf{d}$ as its parameter, continues each term meromorphically in $(x,\mathbf{d})$, with poles on the hyperplanes $x_k\in-\mathbb{N}$, and on $\{\operatorname{Re}x_k>-N_0\ \forall k\}$ writes it as a finite sum of products of factors $(x_k+h)^{-1}$, $0\le h<N_0$, with holomorphic functions. Each $x_k$ is an affine function of $\mathbf{d}$ that is not constant: at the data $(\mathbf{m};\mathbf{0})$, $\mathbf{m}\in\vec{\mathcal{N}}_W\otimes\mathbb{C}$, which lie in $\mathbb{D}_W$, the wall $\kappa_v$ is $\langle\mathbf{m},v\rangle$, not identically 0 as $v\ne0$ in $V_W$. So the composite is meromorphic on $\mathbb{D}_W$ with the stated poles, $N_0$ being arbitrary, and it agrees with the absolutely and locally uniformly convergent integral on $\mathcal{T}_g$, where it is holomorphic. The rest is the proof of Theorem 47.2(ii) with $\eta\,\mathbf{d}_1$, $\mathbf{d}_1$ a positive direction, in place of $\eta\,\mathbf{d}_\circ$: two writings of an insertion as combinations of the $\alpha^u\prod_i L_i^{-a_i}$ have one integral on the open set where all their integrals converge, which contains $\eta\,\mathbf{d}_1$ for large $\eta$, every wall $\kappa_v(\eta\,\mathbf{d}_1)+\kappa_v(g)$ being positive for large $\eta$; the value of an element of $\mathcal{L}_W$ vanishes on the intersection of the tubes of its insertions, open and containing $\eta\,\mathbf{d}_1$ for large $\eta$, off the poles of its coefficients, hence identically; and $\langle f_\lambda g\rangle$ and $\langle g\rangle(\cdot+\lambda)$ are one integral on the open set of the $\mathbf{d}$ at which $\mathbf{d}+\lambda$ lies in the tubes of the insertions of a writing of $g$, the insertions of the expansion of $f_\lambda g$ having tubes that contain those translated by $-\lambda$. At a real $\mathbf{d}\in\mathcal{T}_1$ the density is positive.

(iii) The proof of Theorem 47.4 applies, with these changes.

- In (i), $\partial_{y_e}(f_\lambda g)=f_\lambda(c_e(\lambda)+\partial_{y_e})g$ and the linearity of $c_e$ give $(c_e(\mathbf{d})+\partial_{y_e})(f_\lambda g)=f_\lambda(c_e(\mathbf{d}+\lambda)+\partial_{y_e})g$, and, written out, $(c_e(\mathbf{d})+\partial_{y_e})g=g\otimes m_e-\sum_i R_{i,e}g\otimes w_i+\partial_{y_e}g\otimes 1$.
- In (ii), for $(u;\mathbf{a})\in\mathbb{D}_W(\mathbb{Z})$ the vector $(u;\mathbf{a})-\sum_i a_i(\mathbf{t}_i;\mathbf{e}_i)=(u-\sum_i a_i\mathbf{t}_i;\mathbf{0})$ has its first part in $M_W$, so the $(v;\mathbf{0})$, $v$ in a basis of $M_W$, and the $(\mathbf{t}_i;\mathbf{e}_i)$ generate $\mathbb{D}_W(\mathbb{Z})$; for each of these $f_{-\mu}$ is a Laurent insertion, $\alpha^{-v}$ or $\alpha^{-\mathbf{t}_i}L_i=\sum_{\mathbf{t}}c_{i,\mathbf{t}}\alpha^{\mathbf{t}-\mathbf{t}_i}$.
- In (iii), let $\chi_1,\dots,\chi_{r_W}$ be a basis of $M_W$ and $\beta_k=\alpha^{\chi_k}$, coordinates of the quotient torus, with the derivations $\delta_k$, $\delta_k\beta^{\mathbf{n}}=n_k\beta^{\mathbf{n}}$, and the basis $d\beta_k/\beta_k$ of the 1-forms. On $\mathbb{C}[X_W]$, $\partial_{y_e}=\sum_k(\chi_k)_e\,\delta_k$, and the matrix $((\chi_k)_e)_{k,e}$ has rank $r_W$, so each $\delta_k$ is a rational combination of the $\partial_{y_e}$; let $c_{(k)}(\mathbf{d})$ be the same combination of the $c_e(\mathbf{d})$, so that $c_e=\sum_k(\chi_k)_e c_{(k)}$. The 1-form $c(\mathbf{d})=\sum_k c_{(k)}(\mathbf{d})\,d\beta_k/\beta_k$ over $\mathbb{K}_W$ is $d\log(\omega_{\mathbf{d}}/dy)$, the coefficients being the derivatives of one logarithm, $c_{(k)}=\delta_k\log(\omega_{\mathbf{d}}/dy)$ on the orthant; so $\delta_l c_{(k)}=\delta_k c_{(l)}$, and $\nabla=d+c(\mathbf{d})\wedge$ is an integrable connection on the structure sheaf of $X_W$ over $\mathbb{K}_W$. For the form $g\,(d\beta/\beta)_{\hat{k}}$ of degree $r_W-1$, the product of the $d\beta_l/\beta_l$, $l\ne k$, $\nabla$ gives $\pm(c_{(k)}(\mathbf{d})+\delta_k)g$ times the top form, and these span $\mathcal{L}_W$, each $(c_e(\mathbf{d})+\partial_{y_e})g$ being a combination of them and each of them a combination of the $(c_e(\mathbf{d})+\partial_{y_e})g$. So $\mathfrak{M}_W$ is the top cohomology, finite-dimensional as there.
- (iv) and (v) apply verbatim.

(iv) The proofs of Definition 47.6, Proposition 47.7 and Theorem 47.8 apply, with these changes.

- The coefficients $m_e+u_e$ and $w_i+a_i$ of $\mathrm{MR}_e(g)$ are affine on $\mathbb{D}_W$ and not constant, unless the mark $e$ occurs in no letter, when $\mathbf{e}_e\in\vec{\mathcal{N}}_W^\perp$, $m_e$ and $u_e$ vanish on $\mathbb{D}_W$ and its

lattice, and $\mathrm{MR}_e(g) = 0$; and $\mathrm{MR}_e(g)$ evaluates to $(c_e(\mathbf{d}) + \partial_{y_e})$ applied to the evaluation of $g$: $\partial_{y_e}$ multiplies $\alpha^u$ by $u_e$ and $L_i^{-w_i-a_i}$ by $-(w_i+a_i)R_{i,e}$, and $R_{i,e}\,\alpha^u \prod_l L_l^{-a_l} = \sum_{\mathbf{t}} c_{i,\mathbf{t}} t_e\, \alpha^{u+\mathbf{t}} \prod_l L_l^{-a_l-\delta_{il}}$.

- In Proposition 47.7(i), on each cone $|g|\, f^x_{\mathbf{d_1}} = \prod_k q_k^{\kappa_{v_{\pi,k}}(g)+x\kappa_{v_{\pi,k}}(\mathbf{d}_1)} \prod_i k_{i,\pi}^{-a_i-xw_{1,i}}$ (Lemma 48.2(iii)), bounded on the cube exactly when every exponent is nonnegative.
- In (ii) and (iii), the bound of Theorem 7.13 is replaced by $\langle \mathbf{t}-\mathbf{t}_i, v\rangle \geq \min_{\mathbf{t}'}\langle \mathbf{t}'-\mathbf{t}_i, v\rangle = -\kappa_v(\mathbf{t}_i;\mathbf{e}_i)$ for an exponent $\mathbf{t}$ of $L_i$: a monomial $\alpha^{u+\mathbf{t}-\mathbf{t}_i}$ of $f_{-\mu}\alpha^u$, $\mu = (\mathbf{t}_i;\mathbf{e}_i)$, has a wall at least $\kappa_v(\alpha^u) - \kappa_v(\mu)$, and $\alpha^{u-v'}$, for $\mu = (v';\mathbf{0})$, has exactly that wall; and a term $(u+\mathbf{t};\mathbf{a}+\mathbf{e}_i)$ of a mark relation or of a letter identity of $g$ has the walls $\kappa_v(g) + \kappa_v(\mathbf{t};\mathbf{e}_i) \geq \kappa_v(g)$. The generators of a fan span $V_W$ positively, every point lying in a cone, so a nonzero $u \in M_W$ is negative at some generator, which gives the wall degree of a Laurent symbol positive off $u = 0$; and the set of the $x \in \vec{\mathcal{N}}_W$ with $\langle x, v\rangle \geq c_v$ at every generator is bounded, which gives finitely many symbols of bounded wall degree at each level.
- In (v), lifting every symbol by the letter identities to the top level $\bar{\mathbf{a}}$ and clearing the denominator $\prod_i L_i^{\bar{a}_i}$ leaves the independence of distinct Laurent monomials, as in the proof of Proposition 39.18(ii).
- The walls of symbols are integers (Lemma 48.2(ii)), which gives the discrete set of Theorem 47.8(iv).

(v) The supports of $K_j$ and $\Psi_j$ are the $\mathbf{1}_T$ and the $\mathbf{1}_{\mathcal{E}\setminus T}$, $T$ a basis of table $j$, so $\vec{\mathcal{N}}_F^{\perp}$ is the set of the $w$ with $\langle w, \mathbf{1}_T\rangle$ constant over the bases of every table. By Lemma 7.8 such a $w$ is constant on every molecule; conversely the indicator of a molecule $\mathcal{E}_M$, a union of molecules of every table (Proposition 7.9), has $\langle \mathbf{1}_{\mathcal{E}_M}, \mathbf{1}_T\rangle = |T \cap \mathcal{E}_M|$ constant over the bases of every table (Lemma 7.10). So $\vec{\mathcal{N}}_F^{\perp}$ is spanned by the indicators of the molecules, and it is $\mathbb{R}\mathbf{1}$ exactly for one molecule. For one molecule the sectors of the orders form a fan with the generators $\mathbf{1}_U$ and the walls $\kappa_{\mathbf{1}_U} = \kappa_U$ (proof of Theorem 31.4(ii), the case of a piece of tables, with the tables of $F$ at the exponents $s_j, t_j$), so the data space, the insertions (Definition 38.1), the relations, the walls, the tubes, the strides and the readings of a box are those of Chapter 47; the criterion of (i) is Theorem 25.2, and (ii)–(iv) are the statements cited. For a compound, $\mathbf{m} - \sum_j (s_j \mathbf{1}_{T_j} + t_j \mathbf{1}_{\mathcal{E}\setminus T'_j}) \in \vec{\mathcal{N}}_F \otimes \mathbb{C}$ says that it pairs to 0 with the indicator of every molecule, so every molecule is balanced. When the weights factor across the molecules the letters factor (Proposition 7.9), $V_F$ is the sum of the spaces $\mathbb{R}^{\mathcal{E}_M}/\mathbb{R}\mathbf{1}$ of the molecules $\mathcal{E}_M$ with the sum of their lattices, the density is the product of the densities of the molecules, and the period read on $V_F$ is the product of their periods. □

**Definition 48.4 (A share reading).** As stated.

*Proof.* The admissible cone is cut out of $\mathbb{S}(\mathbb{R})$ by the strict inequalities $\ell_k > 0$ and, for each node, $\kappa_v \circ \iota_l > 0$ at the generators of a fan of $W_l$ (Lemma 48.2), all linear; so it is an open convex cone. At a real $\mathbf{d}$ the contour of Definition 4.1 through $\mathbf{c}$ is $\mathbf{c} + iV$, $\boldsymbol{\pi}(\mathbf{d})$ being real. □

**Theorem 48.5 (The module along a share reading).** *As stated.*

*Proof.* (i) $\Gamma(x+n) = (x)_n \Gamma(x)$ for every integer $n$, $(x)_n$ a polynomial for $n \geq 0$ and the inverse of one for $n < 0$, and every $\ell_k(\lambda,\tau)$ and $\ell'_k(\lambda,\tau)$ is an integer; the cocycle identity is $G(\cdot+\mu+\nu)/G = \big(G(\cdot+\mu+\nu)/G(\cdot+\mu)\big)\big(G(\cdot+\mu)/G\big)$.

(ii) $\iota_l$ is linear and onto, so a nonzero rational function on $\mathbb{D}_{W_l}$ read through $\iota_l$ is a nonzero rational function on $\mathbb{S}$, and $\iota_l^*$ is an embedding $\mathbb{K}_{W_l} \to \mathbb{K}_{\mathbb{S}}$; the base change of a space of dimension $\mathfrak{m}_{W_l}$ has that dimension over $\mathbb{K}_{\mathbb{S}}$, and the tensor product has the product of the dimensions, with the products of bases as a basis. $\iota_l$ being linear, $\sigma_{(\lambda,\tau)} \circ \iota_l^* = \iota_l^* \circ \sigma_{\iota_l(\lambda,\tau)}$, so $x \otimes \rho \mapsto \operatorname{Sh}_{\iota_l(\lambda,\tau)} x \otimes \sigma_{(\lambda,\tau)}\rho$ is well defined and $\sigma_{(\lambda,\tau)}$-semilinear on each factor, and so is the tensor product of these maps; multiplying by $R_{\lambda,\tau}$ keeps semilinearity, and the cocycle identity of (i) with $\operatorname{Sh}_\mu \operatorname{Sh}_\nu = \operatorname{Sh}_{\mu+\nu}$ on the nodes (Theorem 48.3(iii)) gives an action. On the basis, $\operatorname{Sh}_{(\lambda,\tau)} \bigotimes_l [g_{i_l}^{(l)}] = R_{\lambda,\tau} \bigotimes_l \sum_k \iota_l^* \mathbf{A}^{(l)}_{\iota_l(\lambda,\tau),i_l k} [g_k^{(l)}]$, whose matrix is the stated Kronecker product. The value of each factor is $\mathbb{K}_{W_l}$-linear with meromorphic values (Theorem 48.3(iii)), read through the linear surjection $\iota_l$ it is meromorphic on $\mathbb{S}$ and $\mathbb{K}_{\mathbb{S}}$-linear, and the product over the factors is multilinear, so it defines the value on $\mathfrak{J}$. For $t = \rho \bigotimes_l x_l$ the value of $\operatorname{Sh}_{(\lambda,\tau)} t$ at $(\mathbf{d}, \mathbf{z})$ is $G(\mathbf{d},\mathbf{z}) R_{\lambda,\tau}(\mathbf{d},\mathbf{z}) \rho\big((\mathbf{d},\mathbf{z}) + (\lambda,\tau)\big) \prod_l \langle x_l \rangle \big(\iota_l(\mathbf{d},\mathbf{z}) + \iota_l(\lambda,\tau)\big)$, by (i) and by the intertwining on the nodes, which is the value of $t$ at $(\mathbf{d},\mathbf{z}) + (\lambda,\tau)$.

(iii) $\langle \operatorname{Sh}_\lambda [1] \rangle (\mathbf{d}) = P_W(\mathbf{d} + \lambda)$ (Theorem 48.3(iii)). As $\mathbf{P}\tau = \boldsymbol{\pi}(\lambda)$, $\mathbf{c} + \tau$ is a real point of the pins at $\mathbf{d} + \lambda$, admissible by hypothesis, and the reading identity at $\mathbf{d} + \lambda$ through $\mathbf{c} + \tau$ gives $P_W(\mathbf{d} + \lambda)$ as the integral of $G \prod_l P_{W_l} \circ \iota_l$ over $\mathbf{c} + \tau + iV$. The substitution $\mathbf{z} \mapsto \mathbf{z} + \tau$ carries $\mathbf{c} + iV$ onto it and keeps the measure of Definition 4.1, $\boldsymbol{\pi}(\mathbf{d})$ and $\boldsymbol{\pi}(\mathbf{d}+\lambda)$ being real; the integrand becomes the value of $\operatorname{Sh}_{(\lambda,\tau)}[1]^\otimes$ at $(\mathbf{d},\mathbf{z})$, by (ii), the value of [1] at a node being its period. By Theorem 48.3(iii), $\operatorname{Sh}_\mu[1] = \boldsymbol{\rho}_\mu^{(l)} \cdot [\mathbf{g}^{(l)}]$, so the value of a factor is $\boldsymbol{\rho}_\mu^{(l)} \cdot \langle \mathbf{g}^{(l)} \rangle$ as a meromorphic function, and on the contour, where the value of $\operatorname{Sh}_{(\lambda,\tau)}[1]^\otimes$ is holomorphic, the integrand is the displayed one. For $x = \sum_\lambda r_\lambda \operatorname{Sh}_\lambda [1]$ the value at $\mathbf{d}$, off the poles of the $r_\lambda$, is $\sum_\lambda r_\lambda(\mathbf{d}) P_W(\mathbf{d} + \lambda)$, and the sum of the fibre integrals through one point is the fibre integral of the value of $t_x$, the coefficients $r_\lambda$, rational functions of $\mathbf{d}$ alone, being constant on the fibre. For the existence, let $\mathbf{c}_0$ be a real point of the pins at $\mathbf{d}$; $\mathbb{S}^+$ is an open cone containing $(\mathbf{d}_1, \mathbf{c}_1)$, so $(\mathbf{d}, \mathbf{c}_0) + \eta(\mathbf{d}_1, \mathbf{c}_1)$ and the finitely many $(\mathbf{d} + \lambda, \mathbf{c}_0 + \tau_\lambda) + \eta(\mathbf{d}_1, \mathbf{c}_1)$ lie in it for all large $\eta$, and $\mathbf{c}_0 + \eta \mathbf{c}_1$ is the point asked at $\mathbf{d} + \eta \mathbf{d}_1$. $\mathbb{D}'_W(\mathbb{Z})$ is a subgroup, so every $\lambda \in \mathbb{D}_W(\mathbb{Z})$ is $\lambda_c + \lambda'$ with $\lambda' \in \mathbb{D}'_W(\mathbb{Z})$ and $\lambda_c$ the representative of its class, and $\operatorname{Sh}_\lambda[1] = \operatorname{Sh}_{\lambda_c} \operatorname{Sh}_{\lambda'}[1]$; the shifts of [1] span $\mathfrak{M}_W$ (Theorem 48.3(iii)), and $\langle \operatorname{Sh}_{\lambda_c} x_c \rangle (\mathbf{d}) = \langle x_c \rangle (\mathbf{d} + \lambda_c)$.

(iv) The substitution $\mathbf{z} \mapsto \mathbf{z} + \tau$, $\tau$ real, carries the contour through $\mathbf{c}$ onto that through $\mathbf{c} + \tau$, with the same imaginary parts and measure, and turns the value of $[1]^\otimes$ into that of $\operatorname{Sh}_{(0,\tau)}[1]^\otimes$, by (ii); the reading identities through $\mathbf{c}$ and through $\mathbf{c} + \tau$ both give $P_W(\mathbf{d})$.

(v) The bijections carry the exponents, the data spaces, the lattices, the strides, the defects, the relations and the boxes of one family to those of the other, and the master forms and the values with them; so they carry the modules, with their shifts, masters, shift matrices and reductions. A box of each class gives its module (Theorem 48.3(iv)).

(vi) The shares $(\mathbf{z}, \mathbf{z}')$ are pinned by $\mathbf{P}\mathbf{z} = \boldsymbol{\pi}(\mathbf{d})$ and $\mathbf{P}'\mathbf{z}' = \boldsymbol{\pi}'(\iota_{l_0}(\mathbf{d},\mathbf{z}))$, rows independent, the matrix being block triangular with the blocks $\mathbf{P}$ and $\mathbf{P}'$, $\iota_{l_0}$ extended linearly from $\mathbb{S}$. The forms of the kernel are integral on $\mathbb{S}''(\mathbb{Z})$, which $(\mathbf{d}, \mathbf{z}, \mathbf{z}') \mapsto (\iota_{l_0}(\mathbf{d},\mathbf{z}), \mathbf{z}')$ maps into the lattice of the inner share space, and every node map is linear and onto, $\iota_{l_0}$ and $\iota'_k$ being onto. At a point as stated, $(\operatorname{Re}\mathbf{d}, \mathbf{c}) \in \mathbb{S}^+$, so the reading identity of $W$ holds through $\mathbf{c}$. On its contour $\iota_{l_0}(\mathbf{d},\mathbf{z})$ has the real part $\iota_{l_0}(\operatorname{Re}\mathbf{d}, \mathbf{c})$, at which $\mathbf{c}'$ is admissible, so the reading identity of $W_{l_0}$ holds at every point of the contour through $\mathbf{c}'$, and a point admissible for the inner reading has its data in the tube of $W_{l_0}$. Substituting, the iterated integral converges absolutely by hypothesis, so by Fubini's theorem it is the fibre integral over both pins, the normalizations of Definition 4.1 multiplying; its admissible cone is the set of the points stated. □

**Definition 48.6 (The path of a basis).** As stated.

*Proof.* For $1 \le l \le L-1$, $T \setminus S_l$ is the room side of $T$ at the bunch $S_l$, its pattern being $T \cap S_l$, so it lies in one station. $T \setminus S_{l-1} = (T \cap B_{l-1}) \sqcup (T \setminus S_l)$, so with $X = T \setminus S_l \in D_l(T)$ and $q = T \cap B_{l-1}$, $q \sqcup X$ is a room side of $S_{l-1}$ in $D_{l-1}(T)$, and $\alpha^q$, with the coefficient $w'_{T\setminus S_{l-1}}/w'_{T\setminus S_l}$, is a term of $\mathbf{N}^K_{l,j}(D_{l-1}(T), D_l(T))$, and $\alpha^{B_{l-1}\setminus q}$ of $\mathbf{N}^\Psi_{l,j}$, the coefficients not depending on the member $X$ (Theorem 29.2(i)). The room child of $D_1(T)$ is the table on $S_1$ with the bases $p$ such that $p \sqcup X \in \mathcal{B}_j$ for $X \in D_1(T)$ (Definition 28.1(ii)), among them $T \cap S_1$ with $X = T \setminus S_1$, whose term in $K_{j,(D_1(T))}$ is $\theta_{j,(D_1(T)),T\cap S_1}\alpha^{T\cap S_1}$ and in $\Psi_{j,(D_1(T))}$ the same coefficient times $\alpha^{S_1\setminus T}$. $T \cap B_{L-1} = T \setminus S_{L-1}$ is a room side in $D_{L-1}(T)$, a term of $\mathsf{y}_{j,D_{L-1}(T)}$ with the coefficient $w'_{T\setminus S_{L-1}}$, and $\alpha^{B_{L-1}\setminus T}$ one of $\mathsf{y}^*_{j,D_{L-1}(T)}$. □

**Theorem 48.7 (A chain of bunches is a share reading).** *As stated.*

*Proof.* In the proof the pins at every node, its *pins of directions*, are $\iota_l(\mathbf{d}, \mathbf{z}) \in \mathbb{D}_{W_l}$; the degree of the entry $(D', D)$ of $K_j$ at the node $l$ is $|D'| - |D|$, $|D|$ the common size of the room sides in the station $D$, and that of the seam letter of $S_1$ at $D$ is $r_j - |D|$, as at the path of any basis (proof of Definition 48.6); for $\Psi_j$, with $\bar{c}(D) = |\mathcal{E} \setminus S_l| - |D|$ at a station $D$ of $S_l$, they are $\bar{c}(D') - \bar{c}(D)$, $r^*_j - \bar{c}(D)$ at the first bunch, and $\bar{c}(D)$ at the last node.

(i) *The share space.* The pins are linear with integral coefficients in the shares and the data: the letter totals $\sum_D a^{(1)}_{jD} = s_j$ and $\sum_D b^{(1)}_{jD} = t_j$, where $a^{(1)}_{jD} = \sum_{D''} e^{(2)}_{jDD''}$; the conservation $\sum_{D'} e^{(l)}_{jD'D} = \sum_{D''} e^{(l+1)}_{jDD''}$ at the stations $D$ of $S_l$, $2 \le l \le L-1$, and alike with $f$; and the pins of directions of the nodes, taken on bases of the lattices $\vec{\mathcal{N}}^\perp_{W_l} \cap \mathbb{Z}^{B_{l-1}}$ that contain $\mathbf{1}_{B_{l-1}}$, whose pins are the homogeneities. They have one dependence: the sum of the homogeneities is $\sum_j (r_j$ times the letter total of $K_j$ and $r^*_j$ times that of $\Psi_j)$ plus, at every station $D$ of $S_l$, $2 \le l \le L-1$, the conservation pin of $K_j$ times $|D|$ and that of $\Psi_j$ times $|\mathcal{E} \setminus S_l| - |D|$, each written as the out-load less the in-load, by the balance computation below, with the coefficient 1 on every homogeneity. There is no other. The letter totals and the conservation are the equations of a flow on a connected network at every vertex but the sink, so they are independent. If a combination of the pins of directions, with vectors $\mathbf{v}_l \in \vec{\mathcal{N}}^\perp_{W_l}$, is a combination of them, its value at the shares of a path vector $\tau_T$ is $\langle \mathbf{1}_T, w\rangle$, $w = \sum_l \mathbf{v}_l$, and equals the coefficient of the letter total of $K_j$; so $\langle w, \mathbf{1}_T\rangle$ and, likewise, $\langle w, \mathbf{1}_{\mathcal{E}\setminus T}\rangle$ are constant over the bases of every table, $w$ is constant (Lemma 7.8, $F$ being one molecule), and every $\mathbf{v}_l$ is the same multiple of $\mathbf{1}_{B_{l-1}}$. $\mathbf{P}$ is the set of all of them but the homogeneity of one node, and $\boldsymbol{\pi}$ their right-hand sides: these are the pins of Corollary 29.5, the letter totals, the conservation, and the homogeneity of the first bunch and of every node but one, with the pins of Theorem 28.12 at a bunch or a node whose Newton sum is degenerate, the lattice bases there completing the indicators of the steps. Leaving out the homogeneity of another node, or taking other such bases, changes $\mathbf{P}$ by a unimodular matrix and leaves the fibre integral of Definition 4.1 unchanged; leaving out a letter total or a conservation pin instead would divide it by the absolute value of the coefficient of that pin in the dependence, the remaining pins being dependent when it is 0.

*The balance.* On the pins the total degree of the shares of $K_j$ at all the nodes is $\sum_D a^{(1)}_{jD}(r_j - |D|) + \sum_{l=2}^{L-1}\sum_{D',D} e^{(l)}_{jD'D}(|D'| - |D|) + \sum_D e^{(L)}_{jD}|D|$, which telescopes by the conservation to $s_j r_j$; likewise, with $\bar{c}$ in place of $|D|$, to $t_j r^*_j$ for $\Psi_j$. With the homogeneity of the nodes, $m_\mathcal{E} = \sum_j (s_j r_j + t_j r^*_j)$: the data of a point of the pins lie on the balance wall.

*The maps are onto.* Every station $D$ of $S_l$, $1 \le l \le L-1$, has a nonzero entry of the node $l+1$ out of it and, for $l \ge 2$, one of the node $l$ into it: for $X = T \setminus S_l \in D$, the entries $(D_{l-1}(T), D)$ and $(D, D_{l+1}(T))$ of the path of $T$ are nonzero, the last node having the entry $D$. Given data

$(\mathbf{m}';\mathbf{e}',\mathbf{f}')$ of the node $l$, the loads they put on the stations at both ends of the node are carried station by station by shares of the neighbouring nodes, down to the first bunch, whose shares are its loads, and up to the last node, whose shares are the loads at $S_{L-1}$; the letter totals then give $s_j$ and $t_j$, masses on the other steps satisfying their pins of directions exist, $\mathbf{m}-\sum_i w_i\mathbf{t}_i$ ranging over $\vec{\mathcal{N}}\otimes\mathbb{C}$, and the data lie on the balance wall. So $\iota_l$ is onto, and likewise $\iota_1$. At real data with positive exponents and masses on the balance wall with every proper wall positive, $F$ has a native gauge (Theorem 8.19(iii)), and the native split at the stations and the entries (Lemmas 28.9 and 29.4) is a point of the pins with positive shares at which the masses of every node lie in the relative interior of its Newton sum, so in its space of directions and, by Theorem 48.3(i), in its tube. These data form a nonempty open set of real points of $\mathbb{D}_F$, the centre data among them, so every $\mathbf{d}$ has shares on its pins, the pins being linear, and $\mathbb{S}^+$ is not empty. The forms of the kernel, shares, loads and the coordinates $s_j$, $t_j$, are integral on $\mathbb{S}(\mathbb{Z})$, and the $\iota_l$ are coordinate maps, integral on it.

*Absolute convergence.* At a real $\mathbf{d}$ and an admissible $\mathbf{c}$, on the contour $\mathbf{c}+i\mathbf{y}$, $\mathbf{y}\in V$, every node has $|P_{W_l}(\iota_l(\mathbf{d},\mathbf{c}+i\mathbf{y}))|\le P_{W_l}(\iota_l(\mathbf{d},\mathbf{c}))$, so it suffices that $\int_V |G(\mathbf{d},\mathbf{c}+i\mathbf{y})|\,d\mathbf{y}<\infty$. By Stirling's formula, uniformly for real parts in a compact subset of $(0,\infty)$, $|\Gamma(x+iu)|$ and $1/|\Gamma(x+iu)|$ are at most a constant times $(1+|u|)^{x-1/2}e^{-\pi|u|/2}$ and $(1+|u|)^{1/2-x}e^{\pi|u|/2}$, so $|G|\le C_0(1+|\mathbf{y}|)^n e^{-\frac{\pi}{2}E(\mathbf{y})}$ with

$$E(\mathbf{y})=\sum_{l=2}^{L-1}\Big(\sum_{D',D}|y^{(l)}_{D'D}|-\sum_D|A^{(l)}_D|\Big)+\sum_D|A^{(L-1)}_D|$$

for each letter, summed over the letters, $y^{(l)}_{D'D}$ the imaginary parts of its shares and $A^{(l)}_D$ that of the load at the station $D$ of $S_l$, $\sum_{D'}y^{(l)}_{D'D}$ for $l\ge 2$ and $\sum_{D''}y^{(2)}_{DD''}$ for $l=1$, the letter totals $s_j,t_j$ being real and the last node's shares being the loads at $S_{L-1}$. Every bracket is nonnegative. If $E(\mathbf{y})=0$ the loads at $S_{L-1}$ vanish, then so do the shares of the node $L-1$, the bracket at $L-1$ being 0, then the loads at $S_{L-2}$, which are their sums out of each station, and so on down to the node 2: $\mathbf{y}=0$. $E$ is continuous and positively homogeneous of degree one, so $E(\mathbf{y})\ge\varepsilon|\mathbf{y}|$ on $V$ for some $\varepsilon>0$, and the integral converges; the bounds are uniform for the real parts in a compact set and for imaginary translates of the contour by a bounded amount, $E(\mathbf{y}+\mathbf{y}_0)\ge E(\mathbf{y})-C_1|\mathbf{y}_0|$.

*The identity.* At a real $\mathbf{d}$ with an admissible $\mathbf{c}$ the shares are positive, so $s_j,t_j>0$, every bunch $S_l$, $0<l<L$, has a table of positive width, $F$ being one molecule (Lemma 7.10), and the nodes converge at their data, which lie in their tubes; so the contour is one of Corollary 29.5, which gives $P_F(\mathbf{d})$ as the fibre integral, the period being by Theorem 29.3(ii) an integral of positive functions against the laws. The fibre integral converges, so $P_F(\mathbf{d})$ is finite, every proper wall is positive (Theorem 8.7), and $\mathbf{d}\in\mathcal{T}_1(F)$. For complex data let $U$ be the set of the $\mathbf{d}$ at which some point is admissible, the tube over the projection of $\mathbb{S}^+$, connected. Near $\mathbf{d}_0\in U$, with $\mathbf{c}_0$ admissible at $\mathbf{d}_0$, a rational $\mathbf{Q}$ with $\mathbf{PQ}=1$ and $\mathbf{y}_0$ with $\mathbf{P}\mathbf{y}_0=\boldsymbol{\pi}(\operatorname{Im}\mathbf{d}_0)$, the contour $\mathbf{z}=\mathbf{Q}\boldsymbol{\pi}(\mathbf{d}-\mathbf{d}_0)+\mathbf{c}_0+i(\mathbf{y}_0+\mathbf{y})$, $\mathbf{y}\in V$, lies on the pins at $\mathbf{d}$ through a point admissible at $\mathbf{d}$, and by the bounds the fibre integral is holomorphic in $\mathbf{d}$ near $\mathbf{d}_0$; by Cauchy's theorem it does not depend on the admissible point, the admissible points at $\mathbf{d}$ forming a convex set. So both sides are holomorphic on $U$, which lies in $\mathcal{T}_1(F)$, its real points doing so and $\mathcal{T}_1(F)$ being a tube; they agree at its real points, a nonempty open subset of $\mathbb{D}_F(\mathbb{R})$, so they agree on $U$.

(ii) $\lambda^K_T$ lies on the balance wall, $|T|=r_j$, and is integral. $\tau_T$ adds 1 to one entry of each node $2,\dots,L$ along the path, so it adds 1 to the load $a^{(1)}_{jD_1(T)}$, to the letter total of $K_j$, which $\lambda^K_T$ also raises by one, and to the in- and the out-load of every station $D_l(T)$, $2\le l\le L-1$, keeping the conservation. At the node $l$ the data move by $(\mathbf{1}_{T\cap B_{l-1}};\mathbf{e})$, $\mathbf{e}$ the unit vector of the entry of the path, and $\mathbf{1}_{T\cap B_{l-1}}$ is an exponent of that entry (proof of Definition 48.6), so the move is the data

of the stride of the piece of $T$ in its entry, and it keeps the pins of directions, the difference of two exponents of one letter lying in its space of directions; at the first bunch the data move by $(\mathbf{1}_{T\cap S_1};\mathbf{e})$, $\mathbf{e}$ the unit vector of $a^{(1)}_{jD_1(T)}$, likewise. So $(\lambda^K_T,\tau_T)\in\mathbb{S}(\mathbb{Z})$. The kernel gains the factor $e^{(l)}_{jD_{l-1}(T)D_l(T)}$ for every entry of the path, $l=2,\dots,L$, the last being $e^{(L)}_{jD_{L-1}(T)}=a^{(L-1)}_{jD_{L-1}(T)}$ by the conservation, and loses $s_j$ and $a^{(l)}_{jD_l(T)}$, $2\le l\le L-1$ (Theorem 48.5(i)); cancelling $a^{(L-1)}$ and writing $e^{(2)}_{jD_1D_2}/s_j=(a^{(1)}_{jD_1}/s_j)(e^{(2)}_{jD_1D_2}/a^{(1)}_{jD_1})$ gives the product stated. Its factors are the probabilities of the chain stated, $\sum_D a^{(1)}_{jD}=s_j$ and $\sum_D e^{(l)}_{jD'D}=a^{(l-1)}_{jD'}$, the shares out of $D'$ summing to its load. The cobasis is the same with $f$, $b$, $t_j$ and the monomials $\alpha^{B_{l-1}\setminus T}$. The coefficients of the pieces in their entries are $\theta_{j,(D_1(T)),T\cap S_1}$, the ratios $w'_{T\setminus S_{l-1}}/w'_{T\setminus S_l}$ for $2\le l\le L-1$, and $w'_{T\setminus S_{L-1}}$; their product telescopes to $\theta_{j,(D_1(T)),T\cap S_1}\,w'_{T\setminus S_1}=\theta_{j,T}$, the room child at $T\setminus S_1$ being $w'_{T\setminus S_1}$ times that of its station (Definition 28.1(ii)). So $\theta_{j,T}P_F(\mathbf{d}+\lambda^K_T)$, the value of the chance, is by Theorem 48.5(iii) the fibre integral of $G\,R_{\lambda^K_T,\tau_T}$ times the product over the nodes of the coefficient of the piece times the period at the moved data, the value of $\mathbb{P}^{(l)}(T)$.

(iii) Let $\lambda=(v;\mathbf{a},\mathbf{b})\in\mathbb{D}_F(\mathbb{Z})$, and fix bases $T_j,T'_j$ of every table. Then $\lambda-\sum_j a_j\lambda^K_{T_j}-\sum_j b_j\lambda^\Psi_{T'_j}=(w;\mathbf{0},\mathbf{0})$ with $w_{\mathcal{E}}=0$, by the balance. Two marks $e,e'$ linked by a circuit $C$ of table $j$ have a basis $T\supseteq C\setminus\{e\}$ avoiding $e$, and $T'=T\setminus\{e'\}\cup\{e\}$ is a basis, the circuit of $e$ in $T\cup\{e\}$ being $C$; so $(\mathbf{e}_e-\mathbf{e}_{e'};\mathbf{0},\mathbf{0})=\lambda^K_{T'}-\lambda^K_T$. The marks of $F$ form one molecule, so any two are joined by a chain of linked pairs, and these differences span $\{w\in\mathbb{Z}^{\mathcal{E}}:\ w_{\mathcal{E}}=0\}$. Every lattice vector is thus an integral combination of the $\lambda^K_T$ and $\lambda^\Psi_T$, the image of the same combination of the $(\lambda^K_T,\tau_T)$ and $(\lambda^\Psi_T,\tau^*_T)$ by (ii). So $\mathbb{D}'_F(\mathbb{Z})=\mathbb{D}_F(\mathbb{Z})$ and $\mathfrak{M}'_F=\mathfrak{M}_F$, the shifts of [1] spanning $\mathfrak{M}_F$ (Theorem 47.4(ii)), and Theorem 48.5(iii) applies to every element.

(iv) If the node $S^{(\mathbf{x})}_l$ is carried to $S^{(\mathbf{x}')}_{l'}$ by a bijection of marks keeping the seam letters, the stations of $S_{l-1}$ and $S_l$ corresponding to those of $S_{l'-1}$ and $S_{l'}$, the coefficient of $x_{D'}\alpha^{i(D')}_g$ in the seam letter at $D$, which is $\mathbf{N}^K_l(D',D)$, is carried to the coefficient of the corresponding term, $\mathbf{N}^K_{l'}$ at the corresponding entry, and likewise for $\Psi$; so the bijection of $B_{l-1}$ onto $B_{l'-1}$ and that of the entries are an isomorphism of $W_l$ onto $W_{l'}$, and Theorem 48.5(v) applies.

(v) The occupation is the sum of the chances of the bases through $e$, $R_{j,e}=\sum_{T\ni e}\theta_{j,T}\alpha^T/K_j$ (Definition 38.1), so $\langle R_{j,e}\rangle(\mathbf{d})=\sum_{T\ni e}\theta_{j,T}P_F(\mathbf{d}+\lambda^K_T)$. By (ii) at a real $\mathbf{d}$, and at a complex one by the same substitution $\mathbf{z}\mapsto\mathbf{z}+\tau_T$, which keeps the imaginary parts of the contour (proof of Theorem 48.5(iv)), each term is, through a point as stated, with $\boldsymbol{\nu}_{l'}=\iota_{l'}(\mathbf{d},\mathbf{z})$,

$$\theta_{j,T}P_F(\mathbf{d}+\lambda^K_T)=\int_{\text{pins}}G\,R_{\lambda^K_T,\tau_T}\prod_{l'}\theta^{(l')}(T)\,P_{W_{l'}}\big(\boldsymbol{\nu}_{l'}+\iota_{l'}(\lambda^K_T,\tau_T)\big),$$

$\theta^{(l')}(T)$ the coefficient of the piece of $T$ in its entry at the node $l'$. A basis is determined by its path and its pieces. Conversely a path of stations with a nonzero entry at every node, with a term chosen in each entry, gives a basis: from the last node down, the pieces make room sides in the stations of the path, the coefficients and the stations of an entry not depending on the member of its station (Theorem 29.2(i)), and the term of the seam letter at the first station completes a basis (Definition 28.1(ii)). This is the expansion of $K_j$ as the product of the transfer matrices (Theorem 29.3(i)). Sum first over the pieces at a node $l'\ne l$, its entry $E$ fixed. The entry is $N_E=\sum_q\theta_q\alpha^q$, so $\sum_q\theta_q\alpha^q/N_E=1$, and the values give $\sum_q\theta_qP_{W_{l'}}(\boldsymbol{\nu}_{l'}+(\mathbf{1}_q;\mathbf{e}_E))=P_{W_{l'}}(\boldsymbol{\nu}_{l'})$ (Theorem 48.3(ii)), whatever $E$. At the node $l$ the pieces $q\ni e$ give $\sum_{q\ni e}\theta_qP_{W_l}(\boldsymbol{\nu}_l+(\mathbf{1}_q;\mathbf{e}_E))=\langle R^{(l)}_{E,e}\rangle(\boldsymbol{\nu}_l)$, since $R^{(l)}_{E,e}=\alpha_e\partial_{\alpha_e}N_E/N_E=\sum_{q\ni e}\theta_q\alpha^q/N_E$. What is left is the sum of $R_{\lambda^K_T,\tau_T}$ over

the paths through $E = (D', D)$, which is the probability that the chain of stations of (ii) passes through $E$. The probability of being at a station $D''$ of $S_m$ is $a^{(m)}_{jD''}/s_j$, by induction on $m$, the shares into a station summing to its load, and the continuations have the total probability 1, the shares out of a station summing to its load. So the sum is $(a^{(l-1)}_{jD'}/s_j)(z_E/a^{(l-1)}_{jD'}) = z_E/s_j$; at the first bunch it is $a^{(1)}_{jD}/s_j$, and at the last node $a^{(L-1)}_{jD'}/s_j = z_E/s_j$. These are identities of rational functions on $\mathbb{S}$, and the sums are finite, so they pass through the integral, which gives the display. For $\Psi_j$ the same holds with the cobasis chances and $t_j$. With the masses, which are coordinates of $\mathbf{d}$ and constant on the fibre, $\langle c_e(\mathbf{d})\rangle(\mathbf{d}) = m_e P_F(\mathbf{d}) - \sum_j \big(s_j\langle R_{j,e}\rangle + t_j\langle R^*_{j,e}\rangle\big)(\mathbf{d})$ is, through a point as in the statement, the fibre integral of $G\,\langle c^{(l)}_e\rangle(\boldsymbol{\nu}_l)\prod_{l'\neq l} P_{W_{l'}}(\boldsymbol{\nu}_{l'})$, $c^{(l)}_e$ the defect of the node $l$ at its data, whose exponents are the shares (Definition 48.1(iv)); both vanish, by Theorem 48.3(ii) for $F$ and for the node. □

**Lemma 48.8 (Exponential periods: their domains and their modules).** *As stated.*

*Proof.* (i) Write $\alpha = e^y$, so that $\alpha^{\mathbf{m}-\mathbf{1}}d\alpha = e^{\langle \mathbf{m},y\rangle}dy$. For a polynomial $h = \sum_{\mathbf{t}} a_{\mathbf{t}}\alpha^{\mathbf{t}}$ with nonnegative coefficients, $N_h$ of them positive, $\max_{\mathbf{t}} a_{\mathbf{t}}e^{\langle \mathbf{t},y\rangle} \le h(e^y) \le N_h \max_{\mathbf{t}} a_{\mathbf{t}}e^{\langle \mathbf{t},y\rangle}$, the maxima over the exponents with $a_{\mathbf{t}} > 0$, so $|\log h(e^y) - \mathrm{top}_h(y)| \le C_h$ with $C_h = \log N_h + \max_{\mathbf{t}} |\log a_{\mathbf{t}}|$. Hence there are constants $C, C'$, the first uniform for $\mathrm{Re}\,\boldsymbol{\nu}$ in a bounded set, with

$$e^{\varphi_{\boldsymbol{\nu}}(y)-C} \le \Big|e^{\langle \mathbf{m},y\rangle}\prod_k h_k(e^y)^{-c_k}\Big| \le e^{\varphi_{\boldsymbol{\nu}}(y)+C}, \qquad e^{\rho_i(y)-C'} \le R_i(e^y) \le e^{\rho_i(y)+C'}.$$

The functions $\varphi_{\boldsymbol{\nu}}$ and $\rho_i$ are piecewise linear and positively homogeneous of degree one, so Lipschitz, with a constant $\Lambda$; $\varphi_{\boldsymbol{\nu}}$ is linear in $\mathrm{Re}\,\boldsymbol{\nu}$. Let $\Theta$ be the unit sphere of $\mathbb{R}^B$ and $K = C_0 \cap \Theta$, compact. Suppose $\varphi_{\boldsymbol{\nu}} < 0$ on $K$. Then $\varphi_{\boldsymbol{\nu}} \le -2\varepsilon$ on $K$ for some $\varepsilon > 0$, and $\varphi_{\boldsymbol{\nu}} \le -\varepsilon$ on an open neighbourhood $N$ of $K$ in $\Theta$; on the compact set $\Theta \setminus N$, which misses $C_0$, $\max_i \rho_i \ge \delta$ for some $\delta > 0$. In polar coordinates $y = rw$, $w \in \Theta$, the integrand is at most $e^{C-\varepsilon r}$ for $w \in N$ and at most $\exp\big(C + \Lambda r - \xi_{\min}e^{\delta r - C'}\big)$ for $w \notin N$, $\xi_{\min}$ the least scale, and both are integrable against $r^{|B|-1}dr\,dw$. For $\mathrm{Re}\,\boldsymbol{\nu}$ in a compact set on which $\varphi_{\boldsymbol{\nu}} < 0$ on $K$ the maximum of $\varphi_{\boldsymbol{\nu}}$ on $K$ is at most $-2\varepsilon$ uniformly, $\varphi_{\boldsymbol{\nu}}$ depending continuously on $\boldsymbol{\nu}$, and $N$, $\delta$ and $\Lambda$ can be taken uniformly; with the scales in a compact subset of $(0,\infty)^p$ the bounds are uniform, which is the local uniformity. Conversely let $w_0 \in K$ with $\varphi_{\boldsymbol{\nu}}(w_0) \ge 0$. On the half-cylinder $\{tw_0 + y' : t \ge 0,\ |y'| \le 1\}$, $\rho_i(tw_0 + y') \le t\rho_i(w_0) + \Lambda \le \Lambda$ and $\varphi_{\boldsymbol{\nu}}(tw_0 + y') \ge -\Lambda$, so the integrand is at least $e^{-\Lambda - C}\exp\big(-\sum_i \xi_i e^{\Lambda + C'}\big) > 0$ there, and the half-cylinder has infinite volume: the integral diverges at every $\boldsymbol{\xi}$. The condition does not involve the scales; it is linear in $\mathrm{Re}\,\boldsymbol{\nu}$ at each $w \in K$, and strict on the compact $K$, so the set of the real parts it allows is an open convex cone. The last statement is the definition of the data, $\alpha^{\mathbf{u}}\prod_k h_k^{-n_k}$ moving $\mathbf{m}$ by $\mathbf{u}$ and $\mathbf{c}$ by $\mathbf{n}$.

(ii) Write $dy = \bigwedge_e dy_e$ and $\Omega_{\boldsymbol{\nu}} = \alpha^{\mathbf{m}}\prod_k h_k^{-c_k}\exp(-\sum_i \xi_i R_i)$, so that $\langle g\rangle_{\boldsymbol{\nu}} = \int_{\mathbb{R}^B} g\,\Omega_{\boldsymbol{\nu}}\,dy$. Let $\mathbb{K} = \mathbb{C}(\boldsymbol{\xi}, \boldsymbol{\nu})$, the scales and the data taken as indeterminates. On the smooth affine variety $Y$ over $\mathbb{K}$ the 1-form $\omega = \sum_e m_e\,dy_e - \sum_k c_k\,d\log h_k - \sum_i \xi_i\,dR_i$ is closed, so $\nabla = d + \omega\wedge$ is an integrable connection on its structure sheaf, and the top cohomology $H$ of its algebraic de Rham complex is finite-dimensional over $\mathbb{K}$, the fact used in Theorem 38.31(i) and in the proof of Theorem 47.4(iii). On top forms put $D_i(g\,dy) = (\partial_{\xi_i}g - R_i g)\,dy$; since $\partial_{\xi_i}\omega = -dR_i$, $D_i\nabla\eta = \nabla D_i\eta$ for every form $\eta$ of degree $|B|-1$, with $D_i$ acting on its coefficients in the same way, so the $D_i$ act on $H$, and they commute. Let $N \subseteq H$ be the span of the classes of the $D^{\mathbf{a}}(dy) = (-1)^{|\mathbf{a}|}\prod_i R_i^{a_i}\,dy$, of dimension $r \le \dim H$.

Take $\mathbf{a}_0, \dots, \mathbf{a}_r \in \mathbb{N}^p$. Their classes are dependent, so, clearing denominators, there are polynomials $c_0, \dots, c_r \in \mathbb{C}[\boldsymbol{\xi}, \boldsymbol{\nu}]$, not all zero, and a form $\eta = \sum_e \pm g_e\,dy_{\hat{e}}$ of degree $|B| - 1$, $dy_{\hat{e}}$ the

product of the $dy_{e'}$, $e' \neq e$, with coefficients $g_e \in \mathbb{C}[\boldsymbol{\xi},\boldsymbol{\nu}] \otimes \mathbb{C}[Y]$, such that $\sum_k c_k \prod_i R_i^{a_{k,i}}\, dy = \nabla\eta = \sum_e (\partial_{y_e} + \omega_e) g_e\, dy$, $\omega_e = m_e - \sum_k c_k \partial_{y_e} \log h_k - \sum_i \xi_i \partial_{y_e} R_i$. Expanding the $g_e$ in monomials of $(\boldsymbol{\xi},\boldsymbol{\nu})$ gives finitely many elements of $\mathbb{C}[Y]$; let $\Phi$ consist of them, of their derivatives $\partial_{y_e}$ and of their products with the $\partial_{y_e} \log h_k$ and the $\partial_{y_e} R_i$, all in $\mathbb{C}[Y] = \mathbb{C}[\alpha^{\pm 1}][(h_1 \cdots h_q)^{-1}]$. Each element of $\Phi$ is a finite combination of the $\alpha^{\mathbf{u}} \prod_k h_k^{-n_k}$, and by (i) $\langle g \rangle_{\boldsymbol{\nu}}$ converges absolutely for every $g \in \Phi$ and every $\boldsymbol{\xi}$ once $\boldsymbol{\nu} + (\mathbf{u},\mathbf{n}) \in \mathcal{D}$ for the finitely many $(\mathbf{u},\mathbf{n})$ that occur: for a given $\boldsymbol{\nu}_0 \in \mathcal{D}$ this holds on a neighbourhood of $\eta\boldsymbol{\nu}_0$ for all large $\eta$, the real parts of $\mathcal{D}$ forming an open cone. There both $g_e \Omega_{\boldsymbol{\nu}}$ and $\partial_{y_e}(g_e \Omega_{\boldsymbol{\nu}}) = \big((\partial_{y_e} + \omega_e) g_e\big)\Omega_{\boldsymbol{\nu}}$ are integrable over $\mathbb{R}^B$. An integrable function of one variable with an integrable derivative tends to 0 at both ends, so by Fubini's theorem $\int_{\mathbb{R}^B} \partial_{y_e}(g_e \Omega_{\boldsymbol{\nu}})\, dy = 0$, and

$$\sum_{k=0}^{r} c_k(\boldsymbol{\xi},\boldsymbol{\nu}) \big\langle \textstyle\prod_i R_i^{a_{k,i}} \big\rangle_{\boldsymbol{\nu}}(\boldsymbol{\xi}) = 0$$

there. The left side is holomorphic in $\boldsymbol{\nu}$ on $\mathcal{D}$, the convergence being locally uniform by (i), and $\mathcal{D}$ is connected, so the relation holds on $\mathcal{D}$.

Fix $\boldsymbol{\nu}_0 \in \mathcal{D}$ and a direction $\mathbf{v}$ such that the polynomials $c_k(\boldsymbol{\xi}, \boldsymbol{\nu}_0 + \varepsilon\mathbf{v})$ in $(\boldsymbol{\xi},\varepsilon)$ are not all zero; generic directions do. Let $b$ be the least order of vanishing in $\varepsilon$ among them, and write $c_k(\boldsymbol{\xi}, \boldsymbol{\nu}_0 + \varepsilon\mathbf{v}) = \varepsilon^b \big(c_k^0(\boldsymbol{\xi}) + \varepsilon\, \tilde{c}_k(\boldsymbol{\xi},\varepsilon)\big)$ with polynomials $c_k^0$, not all zero. Dividing the relation at $\boldsymbol{\nu}_0 + \varepsilon\mathbf{v}$, which lies in $\mathcal{D}$ for small $\varepsilon$, by $\varepsilon^b$ and letting $\varepsilon \to 0$, the values being continuous in $\boldsymbol{\nu}$, gives $\sum_k c_k^0(\boldsymbol{\xi}) \langle \prod_i R_i^{a_{k,i}} \rangle_{\boldsymbol{\nu}_0} = 0$. So any $r+1$ of the functions $\langle \prod_i R_i^{a_i} \rangle_{\boldsymbol{\nu}_0}$ are dependent over $\mathbb{C}(\boldsymbol{\xi})$, and their span has dimension at most $r$. Differentiation under the integral, justified by the locally uniform convergence, gives $\partial_{\xi_i} \langle g \rangle_{\boldsymbol{\nu}} = \langle -R_i g \rangle_{\boldsymbol{\nu}}$ for $g$ a monomial in the $R$'s, so the span is that of the derivatives of $\langle 1 \rangle_{\boldsymbol{\nu}_0}$ and is closed under the $\partial_{\xi_i}$.

(iii) Let the subtorus be the image under $y \mapsto e^y$ of a rational subspace $V_T \subseteq \mathbb{R}^B$, and choose a basis $\mathbf{b}_1, \dots, \mathbf{b}_{|B|}$ of $\mathbb{Z}^B$ whose first $r$ vectors are a basis of $V_T \cap \mathbb{Z}^B$, a saturated sublattice. In the coordinates $\beta$ with $\log \alpha = \sum_j (\log \beta_j)\, \mathbf{b}_j$, a change of variables of determinant $\pm 1$ with $d\alpha/\alpha = d\beta/\beta$, the integrand does not depend on $\beta_1, \dots, \beta_r$, and the integral over the quotient is the integral over $\beta' = (\beta_{r+1}, \dots, \beta_B)$ at $\beta_1 = \cdots = \beta_r = 1$. There $\alpha^{\mathbf{t}}$ is $\beta'^{(\langle \mathbf{t}, \mathbf{b}_j \rangle)_{j>r}}$, so a polynomial with nonnegative coefficients becomes $\beta'^{\mathbf{s}} h'$, $h'$ a polynomial in $\beta'$ with nonnegative coefficients and $\mathbf{s}$ the least exponents, and the integrand is of the kind of the lemma on the marks $r+1, \dots, |B|$, with the polynomials $h'_k$, the functions $R'_i = \beta'^{\mathbf{w}'_i} P'_i \prod_k h_k'^{-n_{ik}}$, and the data $(\mathbf{m}', \mathbf{c})$, $\mathbf{m}'$ linear in $(\mathbf{m}, \mathbf{c})$. □

**Definition 48.9 (The group at several scales; the inputs and the group of a word).** As stated.

*Proof.* (i) The connections commute, so near a point off the singular locus the horizontal sections form a space of the dimension of $N$ (the Frobenius theorem for integrable connections), and the field they generate over $\mathbb{C}(\mathbf{y})$ with their derivatives is a Picard–Vessiot extension for the commuting derivations $\partial_{y_e}$, the field of constants $\mathbb{C}$ being algebraically closed; its group of differential automorphisms over $\mathbb{C}(\mathbf{y})$ is a linear algebraic group acting faithfully on $\mathrm{Sol}_x N$, with the Galois correspondence (Picard–Vessiot theory for several commuting derivations, Kolchin's). When $N$ is regular singular along its singular locus, the regular singular connections on the complement and its local systems are equivalent categories with their tensor structures (the Riemann–Hilbert correspondence for regular singular connections), and the group is the Zariski closure of the monodromy. Continuation along a path conjugates the group at $x$ to that at another point, as in the proof of Definition 45.24. The derivatives of a function span, over $\mathbb{C}(\mathbf{y})$, a space closed under the

$\partial_{y_e}$, since $\partial_{y_e}(r\,\partial^{\mathbf{a}} f) = (\partial_{y_e} r)\,\partial^{\mathbf{a}} f + r\,\partial^{\mathbf{a}+\mathbf{e}_e} f$; with $\nabla_e = \partial_{y_e}$ it is a differential module when it is finite-dimensional.

(ii) The inputs and the group are defined by (i). At one scale every density is a function on $(0,\infty)$ and the sum is over $\mathbb{C}(y)$, which is Definition 45.32. □

**Proposition 48.10 (Powers add nothing, at any number of scales).** *As stated.*

*Proof.* (i) The proofs of Propositions 45.25(i)–(iii) and 45.26 use, of Picard–Vessiot theory over $\mathbb{C}(y)$, the Galois correspondence between intermediate differential fields and closed subgroups, the correspondence between submodules and invariant subspaces of the solutions, the restriction of the group of $N$ onto the group of a construction from $N$, the equality of the transcendence degree of a Picard–Vessiot extension and the dimension of its group, and Kolchin's theorem on unipotent groups, which concerns the group alone. Each holds for several commuting derivations (Picard–Vessiot theory for partial differential fields, Kolchin's), so the proofs read verbatim over $\mathbb{C}(\mathbf{y})$.

(ii) For $\phi_u \in M_u$, $\partial_{y_e} \prod_u \phi_u = \sum_u (\partial_{y_e}\phi_u) \prod_{u' \neq u} \phi_{u'}$. So the multiplication of functions, from the tensor product of $m_u$ copies of each $M_u$ to the functions, is linear over $\mathbb{C}(\mathbf{y})$ and carries the tensor product of the connections to the derivations; it factors through $\bigotimes_u \operatorname{Sym}^{m_u} M_u$, the functions commuting, and its image $I$ is a differential module, a quotient of $\bigotimes_u \operatorname{Sym}^{m_u} M_u$. $I$ contains $\prod_u F_u^{m_u}$ and is closed under the derivations, so it contains the module that the product generates, a submodule of a quotient of a construction from $M_1 \oplus \cdots \oplus M_q$. A finite sum of such modules is a quotient of their direct sum, again a subquotient of a construction from $M_1 \oplus \cdots \oplus M_q$, and by (i) and Proposition 45.26(i) its group is the image of the group of $M_1 \oplus \cdots \oplus M_q$ acting on its solutions. □

**Definition 48.11 (A tree of nested bunches).** As stated.

*Proof.* (i) The node of Theorem 29.2(ii) has the seam letters of $S$ with each child replaced by its stand-in at its seam point, and the stand-in of a piece at $\mathbf{x}$ enters a letter of table $j$ through $\sum_D x_{jD} \alpha_g^{i(D)}$ in $K_j$ and $\sum_D x_{jD}^* \alpha_g^{c_j - i(D)}$ in $\Psi_j$, linearly in $\mathbf{x}$ (Theorem 28.10(i), Lemma 28.8). Absorbing the children one at a time, the seam letters of the node are polynomials in $\alpha_{Z_S}$ and in the stand-ins, linear in the coordinates of each seam point, the coordinate $x_{l,LD_l}$ carrying the power $\alpha_{g_l}^{\epsilon_L(D_l)}$; grouping the terms by the tuples of stations gives the entries, with the nonnegative coefficients of the letters. At the root the same holds for the stand-in core (Theorem 29.1). Read at $\alpha_{S_l} = \rho_l u_l$, $\alpha_{g_l} = \rho_l \gamma_{S_l}^{\rho}(u_l)$ and $\mathbf{x}_l = \boldsymbol{\varrho}_{S_l}^{\rho}(u_l)$, as in Theorem 29.1, the definitions of the seam vector and the degrees $d_L + \epsilon_L(D')$ of $V_{LD'}^{(l)}$ and $d_L$ of $V_L^{(l),\mathrm{ext}}$ (Definition 28.5) give $x_{l,LD'} \alpha_{g_l}^{\epsilon_L(D')} = V_{LD'}^{(l)}(\alpha_{S_l}) / V_L^{(l),\mathrm{ext}}(\alpha_{S_l})$, and the seam letters of $S$ are those of its node times $\prod_l V_L^{(l),\mathrm{ext}}$, the letters of the stand-in core being those of the family divided by the extreme excess sums of the pieces (Theorem 29.1); so $V_{LD} = \sum_{\mathbf{D}} \mathbf{N}_{S,L}(D;\mathbf{D}) \prod_l V_{LD_l}^{(l)}$.

(ii) The shares are those of Corollary 4.4, one family for each seam letter. A family of positive constants on no marks has the period $\prod_i c_i^{-w_i}$, the space of directions 0, and a module of dimension one (Definition 48.1). At a leaf there are no children, each seam letter is its one entry, and the node word is the seam word. For a chain, the node of $S_l$ has the one child $S_{l-1}$ and the seam letters $\sum_{D'} x_{D'} \alpha_g^{i(D')} \mathbf{N}_l^K(D', D)$ and $\sum_{D'} x_{D'}^* \alpha_g^{c - i(D')} \mathbf{N}_l^{\Psi}(D', D)$ (Theorem 29.3(ii)), so its transfer tensor is the transfer matrix and its node word the word of the node $l$ of Corollary 29.5. At width one the marks of a subtree cell are a bunch of width one of the table, and of the cells below it, nested as the cells are (Proposition 21.3(i),(ii)).

*Normalizations.* Multiplying the two seam letters of a station $D$ of table $j$ by $c_{jD} > 0$ and its room station letters by $c_{jD}^{-1}$ is the normalization of the multipliers of the room sides of that station by another constant (the remark after Definition 28.1), so it keeps every product of a seam letter and a room station letter, and the bunch is a piece with the stations of its room whose glue with its room is the family (Definition 28.4, Proposition 28.2). Lemmas 28.7 and 28.8 and Theorem 28.10 are stated for pieces and their glue. Theorems 29.1 and 29.2(ii) are stated for bunches with the letters of Definition 28.1, and their proofs use of these letters only the writing $K_j = \sum_D V_{jD} L_{jD}$ of the letters of the family through them and the absorption of a piece (Lemma 28.8), so they hold for the letters $c_{jD} K_{j,(D)}$; the coefficients $w'_{q \sqcup X} / w'_X$ of Theorem 29.2(i) and the transfer matrices of Theorem 29.3(i) are those of the normalization of Definition 28.1, and in another they are multiplied by the constants. The seam word at the shares $(\mathbf{a}, \mathbf{b})$ is multiplied by $\prod c_{jD}^{-a_{jD}-b_{jD}}$, and so is its period, the seam function on the pins (Lemma 28.7); off the pins the seam function is a Mellin transform of the law, which the new extreme excess sums and scale move by a fixed bijection of the seam torus with a density (the paragraph after Definition 28.5). The second expression of Definition 28.21 gives the station transform $\mathbf{p} \mapsto \mathcal{I}(\mathbf{c} \circ \mathbf{p})$. In $V_{LD} = \sum_{\mathbf{D}} \mathbf{N}_{S,L}(D; \mathbf{D}) \prod_l V_{LD_l}^{(l)}$ the constants of the node multiply the left side and those of the children the factors on the right, which gives the change of the entries.

The gloss after the definition: the seam function on the pins is the period of the seam word (Lemma 28.7), which converges only if the word is full (Theorem 25.2), and the native split puts the masses of a full seam word in the relative interior of its Newton sum (Lemma 28.9); the seam word of the root is the family of the active letters of $F$, full since its marks form one molecule of the family of its active tables (Theorem 48.23(i)), the two letters of a table having one space of directions. □

**Theorem 48.12 (A node on its shares; a tree of nested bunches is a share reading).** *As stated.*

*Proof.* (i) The node is full. Let $v \in \mathbb{R}^{Z_S \sqcup \{g_1, \dots, g_k\}}$ be a direction on which every seam letter of the node is homogeneous, and let $v'$ on $S$ be $v$ on $Z_S$ and $v_{g_l}$ on every mark of $S_l$. A monomial of $V_{LD} = \sum_{\mathbf{D}} \mathbf{N}_{S,L}(D; \mathbf{D}) \prod_l V_{LD_l}^{(l)}$ is a monomial of an entry times one monomial of each $V_{LD_l}^{(l)}$, of the degree $d_{l,L} + \epsilon_L(D_l)$ in $\alpha_{S_l}$, so its value under $v'$ is the value under $v$ of the corresponding monomial of the node plus $\sum_l v_{g_l} d_{l,L}$, which depends on $L$ alone; so every seam letter of $S$ is homogeneous on $v'$, and $v' \in \mathbb{R}\mathbf{1}_S$ since $S$ is full, whence $v \in \mathbb{R}\mathbf{1}$. By Theorem 29.2(ii) the seam function of $S$ at $\mathbf{a}_S$ is the integral against $\prod_l \lambda_l(d\mathbf{x}_l)$, $\lambda_l$ the law of $S_l$, of the seam function of its node at $\mathbf{x}$, which on the pins is the period of the seam word of the node at the masses $(\mathbf{m}_{Z_S}, (\kappa_{S_l})_l)$, $\prod_{L,D} (V_{LD}^S)^{-a_{S,LD}}$ (Lemma 28.7). Open every seam letter of the node on its monomials, one share for each: its period is the fibre integral of the stacked Gammas $\Gamma_1 \begin{bmatrix} \cdot \\ a_{S,LD} \end{bmatrix}$ of the seam letters times the powers of the coefficients of the monomials, pinned by $|\cdot| = a_{S,LD}$ and by the masses of the marks of the node, by the share side of the Gramma (Proposition 4.2) applied as in the proof of Corollary 18.16, which uses of the letters only that their coefficients are positive. Group the monomials of each seam letter by their entries: the stacked Gamma of the seam letter at $a_{S,LD}$ is the stacked Gamma of its entries at their shares $e_{S,LD\mathbf{D}}$, the sums of the shares of their monomials, times the stacked Gammas of the monomials of each entry at $e_{S,LD\mathbf{D}}$, $\prod_k \Gamma(b_k)/\Gamma(a) = \big(\prod_E \Gamma(e_E)/\Gamma(a)\big) \prod_E \big(\prod_{k \in E} \Gamma(b_k)/\Gamma(e_E)\big)$. The masses of the own marks pin only the shares inside the entries, the exponents of $\alpha_{Z_S}$; those of the stand-ins pin only the entry shares, the exponent of $\alpha_{g_l}$ in a monomial of the entry $(L, D; \mathbf{D})$ being $\epsilon_L(D_l)$, and $\sum e_{S,LD\mathbf{D}}\, \epsilon_L(D_l) = \kappa_{S_l}$ is the scale pin of $S_l$ at its loads (Lemma 28.7), the loads of a letter at $S_l$ summing to $\sum_D a_{S,LD} = s_L$. The seam coordinates appear in the coefficients

as $\prod_l \mathbf{x}_l^{-\mathbf{a}_{S_l}}$, the exponent of $x_{l,LD'}$ being the load at $D'$. So the inner fibre, over the shares inside the entries, is the period of the node word at the entry shares opened on its monomials (Proposition 4.2 again), with transverse pins in the directions of $Z_S$ that the node word does not see (Theorem 28.12), and integrating against the laws, $\int \mathbf{x}_l^{-\mathbf{a}_{S_l}}\, d\lambda_l(\mathbf{x}_l)$ is the seam function of $S_l$ at its loads (Definition 28.5). The pins so obtained are dependent: every monomial of the entry $(L, D; \mathbf{D})$ has the degree $\deg V_{LD}^S - \sum_l \epsilon_L(D_l)$ in $\alpha_{Z_S}$, so the homogeneity of the node word, $m_{Z_S} = \sum e_{S,LD\mathbf{D}} \big( \deg V_{LD}^S - \sum_l \epsilon_L(D_l) \big)$, is $\sum_{L,D} \deg V_{LD}^S$ times the totals less the scale pins of the children, by the balance $m_{Z_S} + \sum_l \kappa_{S_l} = \sum_{L,D} a_{S,LD} \deg V_{LD}^S$ of the seam word of the node at shares on its pins; when $Z_S$ is empty the scale pins of the children add up to $\sum_{L,D} \deg V_{LD}^S$ times the totals. There is no other: if the totals with coefficients $x_{LD}$, the scale pins with coefficients $y_l$ and the pins of directions with a vector $w \in \vec{\mathcal{N}}_{W_S}^{\perp}$ add up to zero, then on every entry $(L, D; \mathbf{D})$, $x_{LD} + \sum_l y_l \epsilon_L(D_l) + \langle \mathbf{t}, w \rangle = 0$ for the exponents $\mathbf{t}$ of its monomials, so the vector equal to $w$ on $Z_S$ and to $y_l$ on $g_l$ is a direction on which every seam letter of the node is homogeneous, a multiple of $\mathbf{1}$, the node being full, and the combination is a multiple of the dependence. Dropping that one pin leaves linearly independent rows, on which the fibre of Definition 4.1 is read. The exchanges are Fubini's theorem: on the contour $|\mathbf{x}^{-\mathbf{z}}| = \mathbf{x}^{-\operatorname{Re}\mathbf{z}}$ and the period of a family of positive letters is at most its value at the real parts, so it suffices that the stacked Gammas be integrable along the fibre. By Stirling's formula, uniformly for real parts in a compact subset of $(0, \infty)$, they are at most $C(1 + |\mathbf{y}|)^n e^{-\frac{\pi}{2} E(\mathbf{y})}$, $\mathbf{y}$ the imaginary parts, with $E(\mathbf{y}) = \sum_{L,D} \big( \sum_{\mathbf{D}} |y_{LD\mathbf{D}}| - |\operatorname{Im} a_{S,LD}| \big) \geq \sum |y_{LD\mathbf{D}}| - 2 \sum_{L,D} |\operatorname{Im} a_{S,LD}|$, so $E(\mathbf{y}) \geq \varepsilon |\mathbf{y}| - C$ on the fibre, with $C$ bounded for the loads in a compact set. At the root the letters of the stand-in core and the exponents $s_L$ take the place of the seam letters and their shares (Theorem 29.1). With one child the entries are the transfer matrices from $S_1$ to $S$ (Definition 48.11(ii)), and the display is that of Theorem 29.6(iii); at a leaf the entries are the seam letters, every stacked Gamma is 1, and the display is Lemma 28.7. With the node word opened on its monomials what is left of the node is the stacked Gammas and the powers of the coefficients. At width one for one table on its centre ray the node of a subtree cell is the family of the two hinge minors of its core with its cells at their markers, and that of the root the core with its cells (Proposition 42.26(iii),(v)); the monomials of the entries are the bases of these tables, and the display with the node words opened is (iii) and (v) there.

(ii) In this proof every letter is active: an inactive letter enters no period (Definition 28.5), $P_F$ is the period of the family of the active letters, and that family is full, its period converging, every proper wall being positive (Theorems 8.7 and 25.2), so its marks form one molecule in the tables of its letters (Theorem 48.23(i)), the two letters of a table having one space of directions. The *groups* are the own marks $Z_S$ of the nodes with $Z_S \neq \varnothing$ and the marks of the leaves. They partition $\mathcal{E}$, and each carries its *word*, the node word of its node or the seam word of its leaf, whose data are its masses and the shares of its letters: the entry shares at a node, the loads at a leaf.

*The path of a basis.* For a basis $T$ of the table of an active letter $L$ every node $S \neq \mathcal{E}$ has the station $D_S(T)$ of the room side $T \setminus S$ in that table, and $T$ runs through the entry $E_S(T) = \big( D_S(T); (D_{S_l}(T))_l \big)$ of every node $S$ that is not a leaf, $D_{\mathcal{E}}$ dropped. The term of $T$ in $L$ is the product of one monomial of each of these entries, $\alpha^{T \cap Z_S}$ for $L = K_j$ and $\alpha^{Z_S \setminus T}$ for $L = \Psi_j$, times its coefficient, and of the monomials $\alpha^{T \cap S}$ or $\alpha^{S \setminus T}$ of the seam letters of the leaves at their stations (Definition 48.11(i), Theorem 29.2(i)). Let $\tau_T$ put 1 on the share of every entry $E_S(T)$ of $L$ and 0 on every other share, and let $\lambda_T$ be the data of the stride $\alpha^T / K_j$ or $\alpha^{\mathcal{E} \setminus T} / \Psi_j$.

*The reading identity.* By Definition 48.11(i), applied at every node from the root down, every active letter $L$ of $F$ is the sum, over the families of stations $(D_S)_{S \neq \mathcal{E}}$, one station of every node other than the root in the table of $L$, of the products over the nodes of the entries $\mathbf{N}_{S,L}(D_S; (D_{S'})_{S'})$, $S'$ running over the children of $S$ and $D_{\mathcal{E}}$ dropped, times the products over the leaves of their seam

letters $V^{(S)}_{LD_S}$. Open $L^{-s_L}$ on the entries of the root (Corollary 4.4), and then, at every child $S$ of a node already opened and every station $D$ of $S$, the power of the seam letter of $S$ at $D$ at its load on the entries of $S$ at $D$, multiplying and dividing by the Gamma of the load: this is the computation of Corollary 29.5 along the tree, a node with several children carrying one share for each tuple of stations of its children, and its integrand is the kernel of (ii) times the product over the groups of the powers of the letters of their words. Integrated over the marks of $F$ against $\alpha^{\mathbf{m-1}}$, by the argument of Theorem 28.12 with the groups in place of its two sides, this is the fibre integral of the periods of the words of the groups, reduced where their Newton sums are degenerate, pinned in the directions that each word does not see, less the one homogeneity of $F$, which holds by the letter totals on the balance wall. Where every node is full, the scale pin of a node is the sum of the pins of the scalings of the groups inside it, and the reading is (i) composed node by node from the root, by Theorem 48.5(vi), each composition dropping the homogeneity of the node read, the scale pin it shares with its parent.

*The share space.* The pins are the letter totals at the root, the conservation at every station of every node other than the root and the leaves, and the pins of directions of the words of the groups, taken on bases of the lattices $\vec{\mathcal{N}}^{\perp} \cap \mathbb{Z}^G$ of their words that contain the indicators $\mathbf{1}_G$. The totals and the conservation are independent: in a vanishing combination of them, take a node all of whose descendants that are not leaves have only the coefficient 0 at their stations; a share of an entry of that node out of a station enters the conservation at that station with the coefficient 1 and otherwise only the conservation at stations of its children, so the coefficient of the conservation at every station of that node, which has an entry out of it, is 0; from the lowest nodes up every coefficient of a conservation pin is 0, and then those of the totals. If a combination of the pins of directions, with vectors $v_G$ in the spaces $\vec{\mathcal{N}}^{\perp}$ of the words, is a combination of the totals and the conservation, its value at $\tau_T$ is $\langle \mathbf{1}_T, w \rangle$, $w$ being $v_G$ on each group $G$ and $\tau_T$ moving the word of $G$ by the exponent $\mathbf{1}_{T \cap G}$ of the term of $T$ in its letter, and it equals the coefficient of the letter total of $L$, $\tau_T$ keeping the conservation; likewise with $\langle \mathbf{1}_{\mathcal{E} \setminus T}, w \rangle$ for the cobasis letters. So these values are constant over the bases of the table of every active letter, $w$ is constant (Lemma 7.8, these tables forming one molecule), and every $v_G$ is the same multiple of $\mathbf{1}_G$. The sum of the homogeneities is the combination of the totals and the conservation given by the balance, as in the proof of Theorem 48.7(i), the total degree of the shares telescoping through the conservation to $\sum_L s_L \deg L$; so it is the one dependence, and the pins less one homogeneity are independent. The forms of the kernel, the shares and the loads are integral on the lattice of the share space, and the maps to the words of the groups are coordinate maps, integral on it.

*The maps are onto.* Every station of a node other than the root has a nonzero entry of its parent into it and, when the node is not a leaf, a nonzero entry of its own out of it, those of the path of a basis through it. Given data of the word of one group, the loads that its shares put on the stations of its node and of the children of its node are carried by shares of the neighbouring entries, up to the root, whose totals give the exponents of $F$, and down to the leaves, whose loads are the loads carried; masses of the other groups on the pins of their directions exist, the masses of a word ranging over a translate of the complexified space of its directions, and the data so obtained lie on the balance wall by the balance. So every map to the word of a group is onto. For a full node $S \neq \mathcal{E}$ the same argument applies to its seam word at shares on the pins of its seam, a family of positive letters whose period is the seam function of $S$ (Lemma 28.7) and whose letters are the seam letters of $S$, opened along its subtree: the sums $\sum_{\mathbf{D}} e_{S,LD\mathbf{D}} = a_{S,LD}$ take the place of the letter totals, its homogeneity is the scale pin of $S$, and the native split of $F$ on the subtree, with the native loads of $S$, is an admissible point.

*The identity at all data.* At a real $\mathbf{d}$ with an admissible $\mathbf{c}$ (below) the shares are positive and

the words of the groups converge at their data, which lie in their tubes; the reading identity holds there and its fibre integral converges absolutely (below), so $P_F(\mathbf{d})$ is finite, every proper wall is positive (Theorem 8.7), and $\mathbf{d} \in \mathcal{T}_1(F)$. At complex data both sides are holomorphic on the tube over the projection of $\mathbb{S}^+$, which is connected, and they agree at its real points, as in the proof of Theorem 48.7(i), *The identity*; so they agree on it.

*Absolute convergence.* The composed kernel is the product over the opened nodes of their stacked Gammas, and Stirling's formula bounds it by $C(1 + |\mathbf{y}|)^n e^{-\frac{\pi}{2}E(\mathbf{y})}$ with $E(\mathbf{y}) = \sum_S \sum_{L,D} \big( \sum_{\mathbf{D}} |y_{S,LD\mathbf{D}}| - |A_{S,LD}| \big)$, $A_{S,LD}$ the imaginary part of the load of $S$ at $(L, D)$, 0 at the root, where the shares are the real exponents. Every bracket is nonnegative. If $E(\mathbf{y}) = 0$ the shares at the root are real, so the loads of its children are, so their shares are, the brackets there vanishing with $A = 0$, and so on down the tree: $\mathbf{y} = 0$. As in the proof of Theorem 48.7(i), $E(\mathbf{y}) \geq \varepsilon|\mathbf{y}|$ on the fibre and the fibre integral converges absolutely, uniformly for real parts in a compact set.

*Admissible points.* At real data with a native gauge $\alpha^\star$ put $e_{S,LD\mathbf{D}} = s_L \, \mathbb{P}_{L,\alpha^\star}\big(T \setminus S \in D,\ (T \setminus S_l \in D_l)_l\big)$, the joint native probability of the stations of the room sides of a basis $T$ (Definition 12.25), which is positive exactly at the nonzero entries, the bases of each rectangle making the term of the entry (Proposition 28.2(ii)), and whose marginals give the conservation at every station. At these shares the masses of every node word and of every leaf lie in the relative interiors of their Newton sums, the native split being the expectation of the exponents of the monomials under the native law, as in Lemmas 28.9 and 29.4, so they lie in their tubes (Theorem 48.3(i)), on the pins of their directions where a Newton sum is degenerate. These data form a nonempty open set of real points of the data space of the family of the active letters, the realizations of the active tables at the exponents of the active letters among them (Definition 9.1, Propositions 8.22 and 9.2), the centre data when every letter is active; the pins are linear, so every $\mathbf{d}$ has shares on them, and the reading is positive.

*The lattice.* $(\lambda_T, \tau_T) \in \mathbb{S}(\mathbb{Z})$: $\tau_T$ adds 1 to the letter total of $L$, which $\lambda_T$ raises by one, and to the in- and the out-load of every station $D_S(T)$, keeping the conservation, and it moves the word of every group by the data of the stride of its piece of $T$, $(\mathbf{1}_{T\cap Z_S}; \mathbf{e}_{E_S(T)})$ at a node and $(\mathbf{1}_{T\cap S}; \mathbf{e}_{D_S(T)})$ at a leaf for $K_j$, keeping its pins of directions, the difference of two exponents of one letter lying in its space of directions. By the circuit argument of the proofs of Theorems 48.7(iii) and 48.23(i) the data of the strides of the bases, for the active letters $K_j$, and of the cobases, for the active letters $\Psi_j$, span the lattice of the data of the family of the active letters, its marks forming one molecule in the tables of its letters; so $\mathbb{S}(\mathbb{Z})$ maps onto that lattice, $\mathfrak{M}'_F = \mathfrak{M}_F$ for the module $\mathfrak{M}_F$ of that family, and Theorem 48.5(iii) reads every element of $\mathfrak{M}_F$.

(iii) A bijection of the marks of two nodes keeping their seam letters, with the stations of their children corresponding, carries the coefficient of each product of seam coordinates in the seam letter of one node to that of the corresponding product in the other, so the entries correspond and the bijection of the own marks with that of the entries is an isomorphism of the node words; for leaves the seam letters correspond. Theorem 48.5(v) applies.

(iv) The real data and shares of (i) form a convex set: on the balance wall the exponents of the letters and the proper walls are positive on a convex cone, a native gauge exists at every point of it, the density $\omega_{\mathbf{d}}/dy$ of the master form of the family of the letters of $F$ at real data with positive exponents and positive walls having one critical point on the orthant, a nondegenerate maximum (the argument of Proposition 47.14(i), which uses of a native direction only these properties), and the shares of (i) are cut out by strict linear inequalities and by the convergence of the seam functions and the node word, each on an open convex set of its real data. The tube over this set is connected. At a complex point of it the left side of (i) is the integral of $(\boldsymbol{\varrho}_S^\rho)^{-\mathbf{a}_S}$ against the

scale form at complex masses, whose modulus is the integrand at the real parts, $|\alpha^{\mathbf{m}}| = \alpha^{\operatorname{Re}\mathbf{m}}$ and $|L^{-s}| = L^{-\operatorname{Re}s}$ on the orthant; on the right the node word and the seam functions of the children have the same bound at the real parts of their data and loads, fixed along the contour, and the stacked Gammas of one seam letter, whose shares have the total $a_{S,LD}$ fixed by its pin, decay exponentially along the fibre in the spread of the imaginary parts of their shares, by Stirling's formula, while the scale pins and the pins of directions are linear. So both sides converge absolutely and locally uniformly, and they are holomorphic on the tube; they agree at its real points, a set of real points of maximal dimension, and so on all of it. For (ii), Definition 48.4 reads the reading identity at every complex $\mathbf{d}$ at which a real $\mathbf{c}$ is admissible, and the proof of (ii) gives it there. □

**Definition 48.13 (Blocks of a node; its scales and the transforms of its blocks).** As stated.

*Proof.* (i) In a monomial whose present parts all lie in $P$, dividing by $A_L$ leaves its coefficient times the ratios of the children of $P$ present in it and the powers $\alpha_z^{q-u_{z,L}}$ of the own marks of $P$ present in it, the absent parts contributing 1; a monomial without present parts is the constant part. Summing over the monomials gives the bracket, with nonnegative coefficients. A partition is separated exactly when the present parts of every monomial lie in one block, that is when every two parts present in one monomial lie in one block, that is when every class of the relation they generate lies in one block.

(ii) The node is a piece (Theorem 29.2(ii)), so its seam letters are homogeneous (Definition 28.4), and $A_L$ is a monomial in its marks; so every reading at $(L, D)$ is homogeneous of the degree $\delta_{LD}$, and where $n_L(D) \neq 0$, $A_L$ is a monomial of $V_{LD}^S$ and $\delta_{LD} = 0$. The action on the scales is linear in their logarithms, and free exactly when some $\delta_{LD}$ on $\Sigma_S$ is not zero. If it is not free and the node separates a partition with two blocks $P \neq P'$, every reading has the degree 0 and lies in the marks of its own block, so every monomial of a seam letter of the node has the degree of $A_L$ in the marks $M_P$ of the node of the parts of $P$: every seam letter of the node is homogeneous along $\mathbf{1}_{M_P}$, which is not a multiple of $\mathbf{1}_M$, and the node is not full, nor is $S$ (proof of Theorem 48.12(i)).

(iii) The integrands are positive and measurable, so the transforms are defined in $[0, \infty]$. When the node is not seen by its scales the common scaling of the marks of all the parts multiplies each $V_{LD'}^{(l)}$ by $e^{t(d_{l,L}+\epsilon_L(D'))}$, so it keeps every ratio of a child at the degree $\epsilon_L(D') - \epsilon_L(D_{l,L}^\circ)$, and keeps every term of every reading, of the degree $\delta_{LD} = 0$; it multiplies the rest of the integrand by $e^{tc}$ with $c = \sum_l \left(\kappa_{S_l} - \sum_L s_L \epsilon_L(D_{l,L}^\circ)\right) + \sum_z \left(m_z - \sum_L s_L u_{z,L}\right) = \sum_z m_z + \sum_l \kappa_{S_l} - \sum_L s_L \deg A_L$, the walls being $\kappa_{S_l} = m_{S_l} - \sum_L s_L d_{l,L}$; on the balance of the seam word of the node at shares on the pins, $\sum_z m_z + \sum_l \kappa_{S_l} = \sum_{L,D} a_{LD} \deg V_{LD}^S = \sum_L s_L \deg A_L + \sum_{L,D} a_{LD}\delta_{LD}$, so $c = 0$ and the integrand is invariant. A block without children has the integral over its own marks of a monomial times the exponential of minus a sum of monomials with coefficients linear in the scales, a Gramma. A bijection of marks keeping seam letters and masses carries the integrand of one transform to the other's when the absent stations and exponents correspond and the readings are equal. □

**Theorem 48.14 (A node on its scales).** *As stated.*

*Proof.* (i) By Theorem 29.2(ii) and Lemma 28.7 the seam function of $S$ at $\mathbf{a}$ is the integral against the laws $\lambda_l(d\mathbf{x}_l)$ of the children of $\int_V \mu(\alpha) \prod_{L,D} (V_{LD}^S)^{-a_{LD}}\, dy$, $V = \mathbb{R}^M/\mathbb{R}\mathbf{1}$ in the coordinates $y = \log\alpha$, with the quotient of Lebesgue measure by $dt$, and $\mu(\alpha) = \alpha^{\mathbf{m}_Z} \prod_l \alpha_{g_l}^{\kappa_{S_l}}$. On the pins $\sum_D a_{LD} = s_L$, so $\prod_D (V_{LD}^S)^{-a_{LD}} = A_L^{-s_L} \prod_D \rho_{LD}^{-a_{LD}}$, $\rho_{LD}$ the bracket, which is the constant

$n_L(D) > 0$ off $\Sigma_S$, the seam letter being nonzero. For $(L,D) \in \Sigma_S$ and $a > 0$, $\Gamma(a)\rho^{-a} = \int_0^\infty \xi^a e^{-\xi\rho}\, d\xi/\xi$. So the left side of (i) is, by Tonelli's theorem, the integral against the laws of

$$\int_{V\times(0,\infty)^{\Sigma_S}} \mu(\alpha) \prod_L A_L^{-s_L}\, \boldsymbol{\xi}^{\mathbf{a}} \exp\Big( - \sum_{(L,D)\in\Sigma_S} \xi_{LD}\rho_{LD}\Big)\, dy\, \frac{d\boldsymbol{\xi}}{\boldsymbol{\xi}}.$$

Let $t \in \mathbb{R}$ act on $\mathbb{R}^M \times (0,\infty)^{\Sigma_S}$ by $y \mapsto y + t\mathbf{1}$ and $\xi_{LD} \mapsto e^{-t\delta_{LD}}\xi_{LD}$. Each $\xi_{LD}\rho_{LD}$ is kept, the bracket being homogeneous of the degree $\delta_{LD}$ (Definition 48.13(ii)), and $\mu\prod_L A_L^{-s_L}\boldsymbol{\xi}^{\mathbf{a}}$ is multiplied by $e^{tc}$ with $c = \sum_z m_z + \sum_l \kappa_{S_l} - \sum_L s_L \deg A_L - \sum_{L,D} a_{LD}\delta_{LD} = \sum_z m_z + \sum_l \kappa_{S_l} - \sum_{L,D} a_{LD} \deg V_{LD}^S$, which vanishes on the data space of the seam word of the node, where the shares on the pins of the seam lie. The integral over $V \times (0,\infty)^{\Sigma_S}$ is the integral over the quotient of $\mathbb{R}^M \times (0,\infty)^{\Sigma_S}$ by the action, the action being free on $\mathbb{R}^M$, with the quotient of the product measure by $dt$. When $S$ is seen by its scales the action is free on $(0,\infty)^{\Sigma_S}$ too, and the integral over the quotient is the integral over $\Xi_S \times \mathbb{R}^M$; when it is not, the action is trivial on the scales, and it is the integral over $(0,\infty)^{\Sigma_S} \times (\mathbb{R}^M/\mathbb{R}\mathbf{1})$. At fixed scales the integrand factors over the parts and the blocks: $\mu\prod_L A_L^{-s_L} = \prod_l \big(\prod_L x_{l,LD^\circ_{l,L}}^{-s_L}\, \alpha_{g_l}^{\kappa_{S_l} - \sum_L s_L \epsilon_L(D^\circ_{l,L})}\big) \prod_z \alpha_z^{m_z - \sum_L s_L u_{z,L}}$, and the exponential is $e^{-\langle\boldsymbol{\xi},\mathbf{n}\rangle}$ times one factor $\exp(-\sum \xi_{LD}\beta_{P,LD})$ for each block, which involves the marks of the node of the parts of $P$ alone, the node separating $\mathcal{P}$. When $S$ is seen by its scales the integral over $\mathbb{R}^M$ is therefore $e^{-\langle\boldsymbol{\xi},\mathbf{n}\rangle}$ times the product over the blocks of the integrals over the marks of their parts; when it is not, there is one block and the integral is over $\mathbb{R}^M/\mathbb{R}\mathbf{1}$. Integrate the factor of a block against the laws $\lambda_l$, $l \in P$, the images of the $\phi^\rho_{S_l}$ under the $\boldsymbol{\varrho}^\rho_{S_l}$, and substitute $\alpha_{g_l} = \rho_l\, \gamma^\rho_{S_l}(u_l)$ at a fixed shape $u_l$ and then $\alpha_{S_l} = \rho_l u_l$ (the computation of Definition 28.21): $x_{l,LD'}\alpha_{g_l}^{\epsilon_L(D')}$ becomes $V^{(l)}_{LD'}(\alpha_{S_l})/V^{(l),\mathrm{ext}}_L(\alpha_{S_l})$ (proof of Definition 48.11(i)), so the absent factors become $\prod_L \big(V^{(l)}_{LD^\circ}/V^{(l),\mathrm{ext}}_L\big)^{-s_L}$ and the ratios of the children $V^{(l)}_{LD'}/V^{(l)}_{LD^\circ}$; and $\rho_l^{\kappa_{S_l}-1} u_l^{\mathbf{m}_{S_l}-\mathbf{1}} \prod_L V^{(l),\mathrm{ext}}_L(u_l)^{-s_L}\, d\rho_l\, d\omega(u_l) = \alpha_{S_l}^{\mathbf{m}_{S_l}-\mathbf{1}} \prod_L V^{(l),\mathrm{ext}}_L(\alpha_{S_l})^{-s_L}\, d\alpha_{S_l}$, the wall being $\kappa_{S_l} = m_{S_l} - \sum_L s_L d_{l,L}$ (Definition 28.4), the scale $\gamma^\rho_{S_l}$ cancelling its power in $\phi^\rho_{S_l}$. The result is $\mathcal{T}_P(\boldsymbol{\xi})$, by Tonelli's theorem; when $S$ is not seen by its scales the common scaling of $\alpha_M$ becomes that of the marks of all the parts, and the result is the transform over the quotient. Equal blocks give equal factors. At complex shares $|\boldsymbol{\xi}^{\mathbf{a}}| = \boldsymbol{\xi}^{\mathrm{Re}\,\mathbf{a}}$ and $|(V^S_{LD})^{-a}| = (V^S_{LD})^{-\mathrm{Re}\,a}$, so the same computation holds by Fubini's theorem where the left side at the real parts is finite. At the root Theorem 29.1 gives $P_F$ as the integral against the laws of the period of the stand-in core, and the same computation holds with its letters and the exponents $s_L$. At complex data the masses enter the integrand through $\mu$ and the exponents through the powers of the letters, and on the orthant $|\alpha^{\mathbf{m}}| = \alpha^{\mathrm{Re}\,\mathbf{m}}$ and $|L^{-s}| = L^{-\mathrm{Re}\,s}$; so both sides are bounded in modulus by their integrands at the real parts, converge absolutely and locally uniformly on the tube over the real data and shares of (i), which is convex, and are holomorphic there. They agree at its real points, and so on it.

(ii) The transform is the integral against the laws of the children of a function of their seam points, by the substitution of (i) read backwards, so it depends on them only through their laws and their walls, the masses of the stand-ins. If the left side of (i) is finite, the integrand on the right is finite at almost every point of $\Xi_S$, so, being invariant on the orbits, at almost every point of $(0,\infty)^{\Sigma_S}$; its factors are positive, so each $\mathcal{T}_P$ is finite at some $\boldsymbol{\xi}$. It is $\langle 1\rangle_{\boldsymbol{\nu}}$ of Lemma 48.8 on the marks of its children and its own marks, over the quotient by the common scaling when $S$ is not seen by its scales (part (iii) there): the $h_k$ are the absent letters $V^{(l)}_{LD^\circ_{l,L}}$ of its children, with the exponents $s_L$, the $R$'s the $\widehat{\beta}_{P,LD}$, each a polynomial with nonnegative coefficients in the seam letters of the children and the own marks over the product of the absent letters of the children

and a monomial in the own marks, and the masses those of the children and $m_z - \sum_L s_L u_{z,L}$ on the own marks. By the lemma it converges at every $\boldsymbol{\xi}$, with all its derivatives, and generates a differential module of finite rank over $\mathbb{C}(\boldsymbol{\xi})$. The substitution $\alpha \mapsto e^t \alpha$ on the marks of $P$ gives $\mathcal{T}_P(\boldsymbol{\xi}) = e^{t c_P} \mathcal{T}_P\big((e^{t\delta_{LD}} \xi_{LD})_{LD}\big)$ for a constant $c_P$, so the singular locus of its module is invariant under the action, and a section of the scale quotient by a monomial map, which meets every orbit, is not contained in it; the restrictions of a basis of the derivatives to the section span the derivatives of the restriction, the derivatives of the section being Laurent monomials, and they span a module of finite rank over the rational functions of the section. For the display, let the left side of (i) be finite at some shares and the children of $P$ be full, and take the factor of the block in (i) before the substitution, the integral against the laws $\lambda_l$, $l \in P$, of an integral over the stand-ins and the own marks of $P$. At fixed seam points of the children that inner integral is a Gramma on these marks (Definition 2.1): its configuration is the exponents of the monomials $\tau$ of the readings, $\epsilon_L(D') - \epsilon_L(D^\circ_{l,L})$ on the stand-in of a child present at $D'$ and $q - u_{z,L}$ on an own mark, its coefficients are the $\xi_{LD} c_\tau$ times the ratios of the seam coordinates in $\tau$, and its index is the masses $\kappa_{S_l} - \sum_L s_L \epsilon_L(D^\circ_{l,L})$ of the stand-ins and $m_z - \sum_L s_L u_{z,L}$ of the own marks; when $S$ is not seen by its scales it is taken over the quotient by the common scaling, on a section a Gramma on one mark fewer (Lemma 48.8(iii)). Its index and configuration do not depend on the seam points, so it converges at every seam point once it converges at one (Theorem 2.2), as it does when $\mathcal{T}_P$ is finite. By its share side (Proposition 4.2) it is the fibre integral of $\prod \Gamma(z_{LD\tau})$ times the powers $-z_{LD\tau}$ of its coefficients, pinned by its index: these are the pins of the block, the pin of the common scaling holding identically when $S$ is not seen by its scales. The powers of the seam coordinates, with $\prod_L x_{l,LD^\circ}^{-s_L}$, are $\mathbf{x}_l^{-\mathbf{a}_l}$ at the loads stated. Integrating against the laws, by Fubini's theorem, the Gammas decaying exponentially along the fibre and $|\mathbf{x}^{-\mathbf{z}}| = \mathbf{x}^{-\operatorname{Re}\mathbf{z}}$, gives the display, $\int \mathbf{x}_l^{-\mathbf{a}_l}\, d\lambda_l$ being the seam function of $S_l$ at its loads. With a native gauge $\alpha^\star$ put $z_{LD\tau} = s_L\, \mathbb{P}_{L,\alpha^\star}(\tau)$, the native probability of the bases whose term in $L$ runs through the monomial $\tau$ of the seam letter of the node at $D$, as in the proof of Theorem 48.12(ii): these are positive; the load of a child at a station is its native load there, by the marginals, at which its seam function converges, the child being full (Lemma 28.9); the scale pins of the children hold there; and for an own mark $z$ of $P$, $m_z = \sum_{L,\tau} z_{L\tau} q_{z,\tau}$ over all the monomials of the seam letters at $L$, the native split being the expectation of the exponents, while the monomials in which $P$ is absent have $q_{z,\tau} = u_{z,L}$ and the shares of all the monomials at $L$ add up to $s_L$, so the pin of $z$ holds. So this point lies on a contour of the display.

(iii) A series or a parallel node with unit weights is the series or the parallel connection of its cells (Definition 23.13). With factoring weights a bunch of width one has one station of each excess in every letter, and its seam letters at the excesses 0 and 1 are, up to positive constants, the letters of the hinge minors $\mathcal{A}/g_0$ and $\mathcal{A} \setminus g_0$ of its cell (Proposition 28.11, the letters at the extreme levels); in the normalization of the statement the constants are 1, the cell carrying its own weights (Definition 23.15), and the entries of a node are the weights of the bases of its core, times the norms at the root. A cell $\mathcal{A}_i$ enters a basis of the substitution through a basis of $\mathcal{A}_i \setminus g_i$, its excess 1, when its marker lies in the basis of the core and through a basis of $\mathcal{A}_i/g_i$, its excess 0, when it does not, in both letters (Definition 21.1); a mark left in place enters through itself or through nothing. At a parallel node the bases of $\binom{G}{1}/g_0$ are $\varnothing$ and those of $\binom{G}{1} \setminus g_0$ the single markers, so every term of the seam letters at the excess 0 has every cell at its excess 0, and every term at the excess 1 has one cell at its excess 1 and the others at 0: separable with the absent stations of the excess 0, the absent exponents 0 in $K$ and 1 in $\Psi$ for a mark. At a series node the bases of $\binom{G}{q} \setminus g_0$ are $\{g_1, \dots, g_q\}$ and those of $\binom{G}{q}/g_0$ omit one marker, so it is separable with the absent stations of the excess 1, the absent exponents 1 in $K$ and 0 in $\Psi$. Each term carries the

weight of its basis of the core, so the readings and the constant parts are weights of the core, 1 with the weights 1. The constant parts are 0 on $\Sigma_S$, which is $\{(K,D_1),(\Psi,D_1)\}$ at a parallel node and $\{(K,D_0),(\Psi,D_0)\}$ at a series node. The brackets have the degrees 1 and $-1$ in the stand-ins at a parallel node, $-1$ and 1 at a series node, so the node is seen by its scales, and the scale quotient is the line of $y=\xi_K\xi_\Psi$, with $dy/y$. With the weights 1, at a parallel node the transform of a cell is $\int \alpha^{\mathbf{m-1}} K_{\mathcal{A}/g}^{-s}\Psi_{\mathcal{A}/g}^{-t}\exp\big(-\xi_K\gamma_{\mathcal{A}}^K-\xi_\Psi/\gamma_{\mathcal{A}}^\Psi\big)\,d\alpha$, $\gamma_{\mathcal{A}}^K=K_{\mathcal{A}\backslash g}/K_{\mathcal{A}/g}$ and $\gamma_{\mathcal{A}}^\Psi=\Psi_{\mathcal{A}/g}/\Psi_{\mathcal{A}\backslash g}$ (Definition 21.1), which is $\check f_F(\xi_\Psi,\xi_K)$ and at $(\xi_K,\xi_\Psi)=(y,1)$ is $\check f_{\mathcal{A}}(y)$ (Definition 23.1); at a series node it is $\int \alpha^{\mathbf{m-1}} K_{\mathcal{A}\backslash g}^{-s}\Psi_{\mathcal{A}\backslash g}^{-t}\exp\big(-\xi_K/\gamma_{\mathcal{A}}^K-\xi_\Psi\gamma_{\mathcal{A}}^\Psi\big)\,d\alpha$, which is $f_F(\xi_K,\xi_\Psi)$ and at $(1,y)$ is $f_{\mathcal{A}}(y)$. A mark left in place has $\mathsf{G}_{m_e-t;(1,-1)}(\xi_K,\xi_\Psi)$ at a parallel node and $\mathsf{G}_{m_e-s;(1,-1)}(\xi_\Psi,\xi_K)$ at a series node, which on the sections are the transforms of the one-mark cell of ratio 1 of Definition 23.19(1), by the torus homogeneity and the reflection $v\mapsto 1/v$ of the Gramma on one scale. On the section $\xi_\Psi=1$ of a parallel node $\boldsymbol{\xi}^{\mathbf{a}}=y^{a_{K,D_1}}$, and on the section $\xi_K=1$ of a series node $\boldsymbol{\xi}^{\mathbf{a}}=y^{a_{\Psi,D_0}}$, the pins of the seam giving $a_{\Psi,D_1}=a_{K,D_1}+t-\kappa$ and $a_{K,D_0}=a_{\Psi,D_0}+s-\kappa$, $\kappa$ the wall of the bunch. So (i) is the one Trailmoment $\int_0^\infty y^{a_{K,D_1}}\prod_i \check f_{\mathcal{A}_i}(y)\,dy/y$ at a parallel node, against $\Gamma(a_{K,D_1})\Gamma(a_{K,D_1}+t-\kappa)$, and $\int_0^\infty y^{a_{\Psi,D_0}}\prod_i f_{\mathcal{A}_i}(y)\,dy/y$ at a series node, against $\Gamma(a_{\Psi,D_0})\Gamma(a_{\Psi,D_0}+s-\kappa)$: the rule of Definition 23.19(2), and at the root (5), the root having the letters $K$ and $\Psi$ without stations. At a series node the transforms of the cells multiply, which is Proposition 20.37(i) with the transforms of the parts. On the centre ray, $s=t=k$, the masses $k$ and the hinge walls $k$, the shares of the scaled stations of a cell at the load $\ell$ of its marker are $\ell$ at a parallel node and $k-\ell$ at a series node, the constant parts 1 off $\Sigma_S$, and (i) reads $\Gamma(\ell)^2\widehat{\mathsf{V}}_{\mathcal{A}}(k;\ell)=\int_0^\infty y^{\ell-1}\prod_i \check f_{\mathcal{A}_i'}(y)\,dy$ and $\Gamma(k-\ell)^2\widehat{\mathsf{V}}_{\mathcal{A}}(k;\ell)=\int_0^\infty y^{k-\ell-1}\prod_i f_{\mathcal{A}_i'}(y)\,dy$, with $\widehat{\mathsf{V}}_{\mathcal{A}}(k;\ell)$ the seam function at these shares (Proposition 42.26(iv)): Proposition 42.26(vi). At the root the letters are $K$ of $(\mathcal{B},\theta)$ at the exponent $k$ and the cobasis letter $\Psi_{\theta\mathsf{m}}$ at $k$, whose term at a basis $T$ carries $\prod_{e\in T}\mathsf{m}_e$ (Proposition 42.26(i)), the norms being 1 on the marks of the cells. At a parallel root a basis of the core is one marker, so the term of $\Psi_{\theta\mathsf{m}}$ in which a mark $e$ left in place is present carries $\mathsf{m}_e$ and no other norm: its reading is $\mathsf{m}_e$, and its transform $\mathsf{G}_{0;(1,-1)}(\xi_K,\mathsf{m}_e\xi_\Psi)=\mathsf{g}(\mathsf{m}_e y)$, by the torus homogeneity at the index 0. At a series root a basis of the core omits one marker, so the term in which a part is present carries the norms of all the marks left in place but its own: every reading in $\Psi$ is $M$ divided by the norm of the part, and in the coordinate $y=M\xi_K\xi_\Psi$, on the section $\xi_K=M^{-1}$ where $y=\xi_\Psi$, the transforms are $\mathsf{g}(y/\mathsf{m}_e)$ and $f_{\mathcal{A}}(y)$, while $\boldsymbol{\xi}^{\mathbf{s}}$ carries $M^{-k}=\prod_e \mathsf{m}_e^{-k}$. So (i) at the root is Proposition 42.26(vii). □

**Theorem 48.15 (What the word along a tree reads: inputs, group, work and price).** *As stated.*

*Proof.* (i) Read the tree from the root. A part of the tree between the nodes of $\Sigma$, under the root or under a full child of a block, read on its shares, is the reading of Theorem 48.12(ii) of $P_F$ or of the seam function of that child, the nodes of $\Sigma$ directly below it taken as leaves read through their seam functions, the periods of their seam words, which are full: stacked Gammas of its shares, which are linear forms in the data and the shares above it, its node words kept or, opened on their monomials, powers of the coefficients of its seam letters, the seam words of its leaves, and the seam functions of the nodes of $\Sigma$ below it at their loads. At a node read alone the fibre splits off as that of Theorem 48.12(i), the seam functions of its children converging; the nodes read jointly keep one fibre for each set, the seam function of a node that is not full diverging at every share. A node of $\Sigma$ is, by Theorem 48.14(i), divided by the Gammas of its scaled shares and the powers of its constant parts, one integral over its scale quotient of $\boldsymbol{\xi}^{\mathbf{a}}$ times $e^{-\langle\boldsymbol{\xi},\mathbf{n}\rangle}\prod_P \mathcal{T}_P$. On a section of the scale quotient by a monomial map, with the coordinates $\mathbf{y}$ that the word writes, the form $\boldsymbol{\xi}^{\mathbf{a}}\,d\boldsymbol{\xi}/\boldsymbol{\xi}$ of the

quotient is a rational constant times $\mathbf{y}^{-\nu}\,d\mathbf{y}$ for an index $\nu$ linear in the shares, so the integral is that constant times the Trailmoment at the unit configuration $(\mathsf{T}_{\nu;I}[\mathbf{y}]\,\phi)\big((0,\infty)^r\big)$, $\mathsf{Z}_{\nu;I}(\mathbf{y}) = \mathbf{y}^{-\nu}$ (Definition 2.12), of the density $\phi = e^{-\langle \boldsymbol{\xi},\mathbf{n}\rangle}\prod_P \mathcal{T}_P^{q_P}$ on the section, equal blocks having one transform; this density is its input, and it does not depend on the shares at which the node is read. The transform of a block whose children are full is the fibre integral of Theorem 48.14(ii), of Gammas, powers of the scales and of the coefficients of its readings and the seam functions of its children at loads linear in its shares, and the seam function of a child is read by its own word; another block with children is its exponential period, an integral over its orthant, over its quotient by the scaling when its node is not seen by its scales, into which the subtrees of its children enter whole; a block without children has a Gramma, a member of the serif family, as a factor of the input. A leaf outside $\Sigma$ is a node with its seam word as its node word. The node words and seam words kept are, at lattice translates of their data, combinations with coefficients in $\mathbb{K}_{\mathbb{S}}$ of the values of the masters of their classes (Theorem 48.5(iii),(v), through Theorem 48.12(ii)). The classes of the Laurent insertions span the module of a class (Theorem 48.3(iii)), so a basis of masters can be taken among them, with $\langle \alpha^u \rangle(\boldsymbol{\nu}) = P_W(\boldsymbol{\nu} + (u;\mathbf{0}))$; $\mathrm{Sh}_{N\mathbf{d}_1}$ is a semilinear bijection, so it carries that basis to the basis of the $[\alpha^u f_{N\mathbf{d}_1}]$, whose walls are $\langle u, v\rangle + N\kappa_v(\mathbf{d}_1) \geq 0$, so that their values converge at every point of the tube $\mathcal{T}_1$ of the class (Lemma 48.2(iii)).

(ii) Every transform of a block, restricted to its section, generates a module of finite rank (Theorem 48.14(ii)), and so does an exponential of a constant part, of rank one, its derivatives along the section being multiples of it by Laurent polynomials. By Proposition 48.10(ii) the input of a node of $\Sigma$ generates a submodule of a quotient of a construction from the sum of the modules of its distinct transforms and of its exponential, of finite rank; the sections of the nodes of one class being written in one set of coordinates, the sum of the modules of the distinct inputs on $r$ scales is a subquotient of a construction from $N_r$, and by Proposition 48.10 its group is the image of $G_{N_r}$ acting on its solutions. The group of the word is the product of these groups over $r$ (Definition 48.9(ii)), a quotient of $\prod_r G_{N_r}$.

(iii) The work counts the distinct integrals of the word once each by their dimensions (Definition 33.2), Gammas and Grammas being functions of the word and not integrals in it: the fibre of every node read alone and the joint fibre of every set of nodes read jointly, with the shares inside the entries where a node word is opened; for every class of node words and seam words kept its masters, each an integral over $V_{W_k}$ of the dimension $\dim V_{W_k}$, convergent by (i); the Trailmoment of every input, of the dimension of its section, the Trailmoments of one input being one integral read at other indices; the fibre of the transform of every block with full children; and the orthant of every other block with children, less one dimension when its node is not seen by its scales. Equal blocks give one transform to a power, a power not being an integral, and a node read alone, a node word, a leaf, a transform or an input that recurs in the tree is one integral read at other data, counted once; the shares of a node read jointly are coordinates of the joint fibre of its set, once for each occurrence.

(iv) The price counts the bits of the description, each part written once and referred to by an index wherever it recurs, the sections and the pins with it, and every power by the $\lceil \log_2(q+1) \rceil$ bits of its exponent (Definition 33.2).

(v) The word of (i) is composed of the share readings of Theorem 48.12(ii) on the parts of the tree read on their shares, of the scale readings of Theorem 48.14(i) at the nodes of $\Sigma$, which hold at complex shares whose real parts are shares at which their left sides are finite, and of the transforms of the blocks, exponential periods that converge absolutely on the tubes of their domains (Lemma 48.8(i)), those with full children read by the share readings of Theorem 48.14(ii). These hold at complex data as (i) there does: on the contour through real shares at which they hold at

the real parts, the pins moving it by a bounded imaginary translate, the Gammas $\Gamma(z_{LD\tau})$ decay exponentially in every imaginary direction of the fibre, $|(\xi_{LD}c_\tau)^{-z}| = (\xi_{LD}c_\tau)^{-\operatorname{Re} z}$, and the seam functions of the children are bounded by their values at the real parts of their loads, so both sides converge absolutely and locally uniformly and are holomorphic in the data on the tube over the real data at which the left side of Theorem 48.14(i) is finite at some shares, a convex set, and they agree at its real points. At complex data as stated the shares of the word have real parts at which every piece converges at the real parts of the data. A transform, an input and a scale integral at complex data have integrands of modulus at most their values at the real parts, the scales and the readings being positive on their orthants, and the share readings converge absolutely at complex data (Theorem 48.12(iv)); a node of $\Sigma$ inside a share fibre enters as its seam function, the scale integral divided by its Gammas, bounded by its value at the real parts. So every integral of the word converges absolutely and locally uniformly on the tube over the real data of Theorem 48.12, which is connected (proof of Theorem 48.12(iv)), and the word is holomorphic there, as is the period (Theorem 48.3(ii)); they agree at the real points by (i), so on the tube. At every point of the domain of a transform its derivatives span a module of finite rank (Lemma 48.8(ii)), so every input generates one, as in (ii); (ii)–(iv) are read off the word as they are at real data, the work and the price depending on the tree, the choices and the classes alone. □

**Corollary 48.16 (The word along the canonical tree).** *As stated.*

*Proof.* $F_{\mathsf{m}}$ is a family of two tables on one molecule, the marks of the connected table $\mathcal{B}$, on its balance wall with every proper wall positive and a native gauge (Proposition 42.26(i)); its active letters are $K$ and $\Psi_{\theta\mathsf{m}}$, at the exponent $k$. The marks of every subtree cell of the canonical hinge tree but its hinge are a bunch of width one of $\mathcal{B}$, and of $F_{\mathsf{m}}$, whose tables have the bases of $\mathcal{B}$; they are nested as the cells are (Proposition 21.3(i),(ii)), so they form a tree of nested bunches (Definition 48.11), the cells of at least two marks directly below a node being its children and the marks left in place its own marks. Every node is full: below the root its seam word is the family of the two hinge minors of its cell, whose period $\widehat{\mathsf{V}}_{\mathcal{A}}(k;\ell)$ converges (Proposition 42.26(iii)), and the root is the family. Take the normalization of Theorem 48.14(iii), in which the seam letters of every node below the root are the letters of the two hinge minors of its cell with its own weights, so that the entries of a node are the weights of the bases of its core, times the norms at the root. The weights are 1 at the series and parallel nodes and factor across the bunches, and the norms are 1 below the root, so by Theorem 48.14(iii) the series and parallel nodes are separable and seen by their scales, each with one scale $y$, their readings and constant parts are 1 below the root, and their parts are their cells, the marks left in place among them; equal subtrees have equal seam letters and data, so they are one class. The nodes of $\Sigma$, those of $S$ and the classes below the root, are read on that one scale, so every input of the word lies on $(0,\infty)$, and their modules are summed over $\mathbb{C}(y)$ (Definition 48.9). Every other node is read on its shares with its node word opened on its monomials, the bases of its core at the root and of the two hinge minors of its core below it (Theorem 48.12(i)), which is Proposition 42.26(iii) at the root and (v) below it; a node of $\Sigma$ is read by (vi) below the root and (vii) at the root (Theorem 48.14(iii)). This is the word of Theorem 45.34.

(i) By Theorem 48.15(i) its inputs are, at the nodes of $\Sigma$, the products of the transforms of their distinct cells to their multiplicities, the constant parts being 0 on the scaled stations; by Theorem 48.14(iii) these are the dual transforms of the cells below a parallel node and their transforms below a series node, $\mathsf{g}$ for a mark below the root, and $\mathsf{g}(\mathsf{m}_e y)$ or $\mathsf{g}(y/\mathsf{m}_e)$ for a mark at the root, the constant $M^{-k} = \prod_e \mathsf{m}_e^{-k}$ of a series root multiplying the integral; at a class of $s$ marks below the root, of either kind, the input is $\mathsf{g}^s$. These are the inputs of Theorem 45.34(i),

which writes $M^{-k}$ as the factors $\mathsf{m}_e^{-k}$, the section $\xi_K = M^{-1}$ having that constant coefficient. The other integrals of the word are fibre integrals over shares, no node word or seam word being kept, so these are all its inputs; every node is full, so every node read on its shares is read alone.

(ii) By Theorem 48.15(ii) the group is a quotient of $G_{N_1}$, $N_1$ the sum of the modules of the distinct transforms read on the one scale: of $\mathsf{g}$, of its dilations at the root, a constant factor not changing a module, and of the transforms that the nodes of $S$ read of the distinct cells, other than marks, directly below them, each counted once whatever the multiplicities, the two transforms of a cell below nodes of both kinds being both counted. In the canonical hinge tree no two series nodes and no two parallel nodes are adjacent (Theorem 23.16), so every such cell is rooted at a prime node or at a node of the other kind. With $S$ empty the inputs are the $\mathsf{g}^s$, $s$ over the sizes of the classes below the root, and $N_1$ is the module of $\mathsf{g}$, whose group is $\mathrm{SL}_2$ and whose $s$-th power generates its $\mathrm{Sym}^s$ (Theorem 45.28(ii)); by Theorem 48.15(ii) the group of the word is the image of $\mathrm{SL}_2$ acting on the sum of the $\mathrm{Sym}^s$, whose kernel is $\{\pm 1\}$ when every $s$ is even, $-1$ acting on $\mathrm{Sym}^s$ by $(-1)^s$, and 1 otherwise: $\mathrm{SL}_2$ when some size is odd, $\mathrm{PGL}_2$ when every size is even, trivial when there is no class below the root, of dimension at most three. This is Theorem 45.34(ii).

(iii) By Theorem 48.15(iii) the work is the sum of the dimensions of the fibres of the distinct nodes outside $\Sigma$, the nodes outside $S$ that are not classes below the root, which are those of Proposition 42.26(ii) at the root, $2N_v - n_v - 1$, and of (v) below it, $2N_v - n_v - 2$, the latter $p - 1$ at a series or parallel node with $p$ cells counted with their multiplicities, its core having one basis in one hinge minor and $p$ in the other; of one for each distinct input, the inputs being on one scale: one for each distinct size of class below the root, the classes of one size of either kind having the one input $\mathsf{g}^s$, and one for each distinct node of $S$ that is not a class below the root, its input being written through the transforms of the distinct subtrees below it; and of the dimensions of the fibres of the distinct transforms of the cells other than marks directly below the nodes of $S$, by Theorem 48.14(ii) the number of the monomials of the readings less the number of the pins, $2-1 = 1$, the cell being present at one station in each of the two scaled stations and having its scale pin, a cell read by nodes of both kinds bringing both of its transforms. The marks have Grammas, no node word is kept and no leaf read on its period. This is the count of Theorem 45.34(iii). With every series and parallel node in $S$ the multiplicities enter only as powers, and a distinct subtree other than a mark brings at most two integrals besides the fibre of its root when that is a prime node: its input and one transform if it is rooted at a series or parallel node, which lies directly below no node of its own kind (Theorem 23.16), and its two transforms if it is rooted at a prime node; this is the bound there. □

**Definition 48.17 (Paths; the direct image of a reading).** As stated.

*Proof.* (i) The letter at the top is the seam letter of the top read at the seam points of its children, $\ell = \sum_{\mathbf{D}} \mathbf{N}_{S_0,L}(D;\mathbf{D}) \prod_i V_{LD_i}^{(S_i)}$, $S_0$ the top and $D$ dropped at the root, where the letters of the stand-in core times the extreme excess sums of the pieces are the letters of the family (Definition 48.11(i) and its proof). Substituting the same expansion of the seam letter of every child that is not a leaf, at its station, down to the leaves, writes $\ell$ as the sum over its paths of the products of their factors, a vanishing entry contributing nothing; expanding the products writes it as the sum of the terms.

(iii) For $r \in \mathbb{K}_W$ read on $\mathbb{S}$ through the projection, $\sigma_{(\mu,\sigma)} r = \sigma_\mu r$. The shifts of $\mathfrak{J}$ are semilinear and compose, $\mathrm{Sh}_{(\mu,\sigma)} \mathrm{Sh}_{(\lambda,\tau)} = \mathrm{Sh}_{(\lambda+\mu,\tau+\sigma)}$, by the cocycles of the kernel and of the nodes (Theorem 48.5(i),(ii)), so $\mathrm{Sh}_{(\mu,\sigma)} \left(r \,\mathrm{Sh}_{(\lambda,\tau)}[1]^{\otimes}\right) = \sigma_\mu(r) \,\mathrm{Sh}_{(\lambda+\mu,\tau+\sigma)}[1]^{\otimes}$: the shifts map $\mathfrak{J}^\natural$ into itself, and onto it, $\mathrm{Sh}_{(-\mu,-\sigma)}$ being the inverse. The lattice is commutative, so $\mathrm{Sh}_{(\mu,\sigma)}(\mathrm{Sh}_{(\mathbf{0},\tau)} - 1)t = (\mathrm{Sh}_{(\mathbf{0},\tau)} - 1)\,\mathrm{Sh}_{(\mu,\sigma)}\, t$, and the fibre translations fix $\mathbb{K}_W$, so $r(\mathrm{Sh}_{(\mathbf{0},\tau)} - 1)t = (\mathrm{Sh}_{(\mathbf{0},\tau)} - 1)(rt)$ for $r \in \mathbb{K}_W$. So $\mathfrak{F}$ is the set of the finite sums $\sum_k (\mathrm{Sh}_{(\mathbf{0},\sigma_k)} - 1)t_k$, $t_k \in \mathfrak{J}^\natural$, and the shifts map it

onto itself. That the shifts, the classes and the fibre integral of (iv) are well defined is part of Theorem 48.18(ii). □

**Theorem 48.18 (The module of a family from its nodes).** *As stated.*

*Proof.* (i) Let $\pi$ be a path of a letter $\ell$ of the table of $L$, and $c\alpha^{\mathbf{t}}$ a term of it. The lift raises $w_\ell$ by one and the share of the entry of $\pi$ at the top by one, keeping the total $\sum_{\mathbf{D}} z = w_\ell$ of $\ell$ at the top. At every node $S'$ below the top it raises by one the share of the entry of the parent of $S'$ into the station $D_{S'}(\pi)$, and so the load of that station, and, when $S'$ is not a leaf, the share of $E_{S'}(\pi)$ out of it, keeping the conservation there. It moves the data of every node by the data of the stride of the monomial of the term in its factor: the exponents of that monomial on the marks of the node, and one more on the share or the load of the factor, $(\mathbf{t}|_{Z_{S'}}; \mathbf{e}_{E_{S'}(\pi)})$ at a node word and $(\mathbf{t}|_{S'}; \mathbf{e}_{(L,D_{S'}(\pi))})$ at the seam word of a leaf, the masses of $W$ being read on the marks of the nodes by the coordinate maps $\iota_k$ (proof of Theorem 48.12(ii)). The difference of two exponents of one letter lies in its space of directions, so the pins of directions are kept; the coordinates are integers; and the lift lies in $\mathbb{S}(\mathbb{Z})$. The kernel is the product, over the nodes that are not leaves and the stations of their seam letters, of the stacked Gammas $\prod_{\mathbf{D}} \Gamma(z_{S',LD\mathbf{D}})/\Gamma(a_{S',LD})$ (Theorem 48.12(ii)), and the lift raises by one exactly the share of the entry of $\pi$ and the load of its station at every such node, so with $(x)_1 = x$ it moves the kernel by $R_\pi$ (Theorem 48.5(i)). A path has a factor at every node of the reading, so, with $\phi_k$ the factor of $\pi$ at the node $k$, $\alpha^{\mathbf{t}_k}$ the monomial of the term in it and $[f_\mu] = \mathrm{Sh}_\mu[1]$ (Theorem 48.3(iii)),

$$\mathrm{Sh}_{((\mathbf{t};\mathbf{e}_\ell),\tau_\pi)}[1]^{\otimes} = R_\pi \bigotimes_k \big[\alpha^{\mathbf{t}_k}/\phi_k\big].$$

The terms of $\pi$ are the products of one monomial of each factor with the products of their coefficients, which are constants; so, by multilinearity,

$$\sum_{c\alpha^{\mathbf{t}}} c\,\mathrm{Sh}_{((\mathbf{t};\mathbf{e}_\ell),\tau_\pi)}[1]^{\otimes} = R_\pi \bigotimes_k \Big[\sum_{\mathbf{t}'} c_{\mathbf{t}'}\,\alpha^{\mathbf{t}'}/\phi_k\Big] = R_\pi\,[1]^{\otimes},$$

the inner sum over the monomials of $\phi_k$ with their coefficients being $\phi_k/\phi_k = 1$ in the insertions of the node, its letter identity (Theorem 48.3(iv)). For a node $S'$ that is not a leaf and a station $D'$ of it let $Q_{S'}(D')$ be the sum, over the choices of the entries and the stations at $S'$ and below it that start from $D'$, of the products of the shares of the entries over the loads of their stations at $S'$ and at the nodes below it that are not leaves. Then $Q_{S'}(D') = \sum_{\mathbf{D}'} \big(z_{S',LD'\mathbf{D}'}/a_{S',LD'}\big) \prod_i Q_{S'_i}(D'_i)$, $Q$ being 1 at a leaf, and from the leaves up $Q_{S'}(D') = \sum_{\mathbf{D}'} z_{S',LD'\mathbf{D}'}/a_{S',LD'} = 1$ by the conservation at $D'$; at the top the same sum is 1 by the total of $\ell$. So $\sum_\pi R_\pi = 1$ in $\mathbb{Q}(\mathbb{S})$, and the first identity holds.

Let $k_e$ be the node whose word has the mark $e$: the node $S'$, the top included, with $e \in Z_{S'}$, or the leaf $S'$ with $e \in S'$. In a term, $t_e$ is the exponent of $\alpha_e$ in its monomial in the factor of $\pi$ at $k_e$, so

$$\sum_{c\alpha^{\mathbf{t}}} c\,t_e\,\mathrm{Sh}_{((\mathbf{t};\mathbf{e}_\ell),\tau_\pi)}[1]^{\otimes} = R_\pi\big[R^{(k_e)}_{\phi_{k_e},e}\big] \otimes \bigotimes_{k\neq k_e}[1], \qquad R^{(k_e)}_{\phi,e} = \frac{\alpha_e\partial_{\alpha_e}\phi}{\phi} = \sum_{\mathbf{t}'} c_{\mathbf{t}'}\,t'_e\,\frac{\alpha^{\mathbf{t}'}}{\phi},$$

the occupation of $e$ in the letter $\phi$ of the word of $k_e$. Group the paths of the letters $\ell$ of one active letter $L$ by their factor at $k_e$, a letter $E$ of that word: the entry of $L$ at a station $D'$ of a node $S'$ that is not a leaf, or the seam letter of the leaf $S'$ at $D'$. For a node $S'$ below the top, with the chain $S_0 \supsetneq S_1 \supsetneq \cdots \supsetneq S_r = S'$ of the nodes from the top $S_0$ down to it, and

a station $D'$ of $S'$, let $\varphi_\ell(S',D')$ be $w_\ell$ times the sum, over the stations of $S_1,\dots,S_r$, with $D'$ at $S_r$, and over the entries of $\ell$ at $S_0,\dots,S_{r-1}$ that join them, each from the station of $S_j$ into a tuple of stations of the children of $S_j$ whose coordinate at $S_{j+1}$ is the station of $S_{j+1}$, of the products of the shares of these entries over the loads of their stations, the load at the top being $w_\ell$. At a child $S''$ of the top $\varphi_\ell(S'',D'')$ is the sum of the shares of the entries of $\ell$ at the top into $D''$, and the sum over the letters $\ell$ of $L$ of these is the load $a_{S'',LD''}$, by its definition. At a child $S'_i$ of $S'$, $\varphi_\ell(S'_i,D'')=\sum_{D'}\varphi_\ell(S',D')\sum z_{S',LD'\mathbf{D}'}/a_{S',LD'}$ over the $\mathbf{D}'$ with $D'_i=D''$. Below the top the ratios $z_{S',LD'\mathbf{D}'}/a_{S',LD'}$ depend on $(L,D')$ alone and not on the letter $\ell$, so $\sum_\ell\varphi_\ell$ satisfies the same recursion as each $\varphi_\ell$: if $\sum_\ell\varphi_\ell(S',D')=a_{S',LD'}$ at every station $D'$ of $S'$, then $\sum_\ell\varphi_\ell(S'_i,D'')=\sum z_{S',LD'\mathbf{D}'}$ over the $(D',\mathbf{D}')$ with $D'_i=D''$, which is $a_{S'_i,LD''}$. So from the top down $\sum_\ell\varphi_\ell(S',D')=a_{S',LD'}$ at every station of every node below the top. A path of $\ell$ through the entry $E$ of $S'$ at $D'$ consists of such a choice along the chain, of $E$, and of continuations below the stations that $E$ gives the children of $S'$ and below every child, off the chain, of a node of the chain above $S'$, at the station that its entry gives it; the products of the shares over the loads along these continuations sum to the $Q=1$, so

$$\sum_\ell w_\ell\sum_{\pi:\ \phi_{k_e}(\pi)=E}R_\pi=\sum_\ell\varphi_\ell(S',D')\,\frac{z_E}{a_{S',LD'}}=z_E;$$

at a leaf the paths through its letter at $D'$ have, the continuations off the chain summing to 1 again, $\sum_\ell\varphi_\ell(S',D')=a_{S',LD'}$, the exponent of that letter there; and at the top those through an entry $E$ of $\ell$ have $w_\ell\,z_E/w_\ell=z_E$, the continuations below the children of the top summing to 1. Every letter of the word of $k_e$ is the factor at $k_e$ of the paths of the letters of one active letter, so the second sum of the statement is $\sum_E z_E\big[R^{(k_e)}_{E,e}\big]\otimes\bigotimes_{k\neq k_e}[1]$ over the letters $E$ of that word at their exponents $z_E$, and the left side of the identity is $\big[m_e-\sum_E z_E R^{(k_e)}_{E,e}\big]\otimes\bigotimes_{k\neq k_e}[1]$. Its first factor is the class of $c_e\cdot 1=(c_e+\partial_{y_e})1$ in the module of the word of $k_e$ at its data $\iota_{k_e}(\mathbf{d},\mathbf{z})$, the mass of $e$ being read identically: its mark relation, which is 0 (Theorem 48.3(ii),(iv)). Along a chain $k_e$ is the node of the step of $e$ and this is the computation of Theorem 48.7(v) before the integration.

(ii) *The lattice.* Let $B$ be the marks of $W$, $\mathcal{E}$ at the root and $S$ at a full node $S$. The letters of $W$ are letters of tables on $B$: at the root the letters of $F$, and at $S$, up to the positive constants of the normalization, the letters $K_{j,(D)}$, for a letter $K_j$ of $F$, and $\Psi_{j,(D)}$, for a letter $\Psi_j$, of the room children of the stations $D$ of $S$, tables on $S$ (Definition 28.1). Their marks form one molecule of the family of these tables, at the root by hypothesis and at $S$ since it is full (Theorem 48.23(i)). Two marks $a\neq b$ linked by a circuit of one of these tables are exchanged in two of its bases (Lemma 7.7), so $\mathbf{e}_b-\mathbf{e}_a$ is, up to its sign, the difference of two exponents of each letter of that table in $W$, the exponents of its two letters being complementary in $B$, and along chains of linked marks these differences span $\mathbf{1}^\perp\cap\mathbb{Z}^B$. A point $(\mathbf{u};\mathbf{a})$ of $\mathbb{D}_W(\mathbb{Z})$ less $\sum_\ell a_\ell$ times the data of the stride of one monomial of each letter is $(\mathbf{u}';\mathbf{0})$ with $\mathbf{u}'\in\mathbf{1}^\perp\cap\mathbb{Z}^B$, by the balance. So the data of the strides of the monomials of the letters span $\mathbb{D}_W(\mathbb{Z})$, and every one of them is the image of the lift of a term (i), every monomial of a letter being the monomial of a term of one of its paths, the coefficients being positive.

*A presentation of the module.* Let $\Phi_W$ be the $\mathbb{K}_W$-space with the basis $(\langle\lambda\rangle)_{\lambda\in\mathbb{D}_W(\mathbb{Z})}$ and the shifts $\mathrm{Sh}_\mu(r\langle\lambda\rangle)=\sigma_\mu(r)\langle\lambda+\mu\rangle$, and $p:\Phi_W\to\mathfrak{M}_W$ the map $\langle\lambda\rangle\mapsto\mathrm{Sh}_\lambda[1]=[f_\lambda]$, which commutes with the shifts and is onto (Theorem 48.3(iii)). Its kernel is the $\mathbb{K}_W$-span of the shifts of the letter identities $\mathrm{LI}_\ell=\sum_{\mathbf{t}}c_{\ell,\mathbf{t}}\langle(\mathbf{t};\mathbf{e}_\ell)\rangle-\langle\mathbf{0}\rangle$ and of the mark relations $\mathrm{MR}_e=m_e\langle\mathbf{0}\rangle-\sum_\ell w_\ell\sum_{\mathbf{t}}c_{\ell,\mathbf{t}}t_e\langle(\mathbf{t};\mathbf{e}_\ell)\rangle$, the sums over the monomials $c_{\ell,\mathbf{t}}\alpha^{\mathbf{t}}$ of $\ell$. Indeed the shift of $\mathrm{LI}_\ell$ by $\lambda$ maps to $[f_\lambda(\ell/\ell-1)]=0$. Since $(c_e(\mathbf{d})+\partial_{y_e})f_\lambda=c_e(\mathbf{d}+\lambda)f_\lambda$ and $f_\lambda R_{\ell,e}=\sum_{\mathbf{t}}c_{\ell,\mathbf{t}}t_e f_{\lambda+(\mathbf{t};\mathbf{e}_\ell)}$, the shift of

$\mathrm{MR}_e$ by $\lambda$ maps to the class of $(c_e(\mathbf{d})+\partial_{y_e})f_\lambda$, which is 0, and these elements of $\mathbb{C}[X_W]\otimes\mathbb{K}_W$ span $\mathcal{L}_W$, the $f_\lambda$ spanning the insertions. Conversely, if $\sum_\lambda r_\lambda f_\lambda = 0$ in $\mathbb{C}[X_W]\otimes\mathbb{K}_W$ then $\sum_\lambda r_\lambda\langle\lambda\rangle$ lies in the span of the shifts of the letter identities: these rewrite each $\langle\lambda\rangle$ as a combination of the $\langle(\mathbf{v};N\mathbf{1})\rangle$ at one level above all the levels that occur, the shift of $\mathrm{LI}_\ell$ by $\lambda$ raising the level of $\ell$ by one, and the $f_{(\mathbf{v};N\mathbf{1})} = \alpha^{\mathbf{v}}\prod_\ell \ell^{-N}$ with distinct $\mathbf{v}$ are linearly independent, distinct Laurent monomials being so. An element of $\Phi_W$ whose image in $\mathbb{C}[X_W]\otimes\mathbb{K}_W$ lies in $\mathcal{L}_W$ is therefore a combination of shifts of mark relations modulo the span of the shifts of the letter identities. So $\mathfrak{M}_W$ is $\Phi_W$ modulo the shifts of the $\mathrm{LI}_\ell$ and of the $\mathrm{MR}_e$.

*The direct image.* For $\lambda\in\mathbb{D}_W(\mathbb{Z})$ and two lifts,

$$\mathrm{Sh}_{(\lambda,\tau')}[1]^\otimes - \mathrm{Sh}_{(\lambda,\tau)}[1]^\otimes = (\mathrm{Sh}_{(\mathbf{0},\tau'-\tau)} - 1)\,\mathrm{Sh}_{(\lambda,\tau)}[1]^\otimes \in \mathfrak{F},$$

so $e_\lambda$ does not depend on the lift, which exists. Likewise $\mathrm{Sh}_{(\lambda,\tau')}\,t - \mathrm{Sh}_{(\lambda,\tau)}\,t \in \mathfrak{F}$ for $t\in\mathfrak{J}^\natural$, and the shifts map $\mathfrak{F}$ onto itself (proof of Definition 48.17), so the shifts of $\mathfrak{H}$ are well defined, $\sigma_\lambda$-semilinear over $\mathbb{K}_W$, with $\mathrm{Sh}_\lambda\,\mathrm{Sh}_\mu = \mathrm{Sh}_{\lambda+\mu}$ and $\mathrm{Sh}_\mu\,e_\lambda = e_{\lambda+\mu}$. The classes $e_\lambda$ span $\mathfrak{H}$, the image of $\mathfrak{J}^\natural$. The map $\tilde{O}:\Phi_W\to\mathfrak{H}$, $\langle\lambda\rangle\mapsto e_\lambda$, is $\mathbb{K}_W$-linear, onto, and commutes with the shifts. The coefficient $c_{\ell,\mathbf{t}}$ of a monomial of $\ell$ is the sum of the coefficients $c$ of the terms of the paths of $\ell$ with that monomial, whose lifts are lifts of $(\mathbf{t};\mathbf{e}_\ell)$; so $\tilde{O}(\mathrm{LI}_\ell)$ and $\tilde{O}(\mathrm{MR}_e)$ are the classes of the differences of the two sides of the identities of (i), and vanish. $\tilde{O}$ commutes with the shifts, so it vanishes on the shifts of the letter identities and of the mark relations, and it factors through $p$: $O$ is a well-defined map of $\mathfrak{M}_W$ onto $\mathfrak{H}$ that commutes with the shifts.

*The fibre integral.* Let $t=\sum_j r_j\,\mathrm{Sh}_{(\lambda_j,\tau_j)}[1]^\otimes\in\mathfrak{J}^\natural$ have the value 0 on $\mathbb{S}$, as when $t=0$ in $\mathfrak{J}$, the value being well defined on $\mathfrak{J}$ (Theorem 48.5(ii)). At a real $\mathbf{d}$ off the poles of the $r_j$ with a point $\mathbf{c}$ admissible at $\mathbf{d}$ whose translates by the $\tau_j$ are admissible at the $\mathbf{d}+\lambda_j$, $P_W(\mathbf{d}+\lambda_j)$ is the absolutely convergent fibre integral of the value of $\mathrm{Sh}_{(\lambda_j,\tau_j)}[1]^\otimes$ through $\mathbf{c}$ (Theorem 48.5(iii)), so $\sum_j r_j(\mathbf{d})P_W(\mathbf{d}+\lambda_j)$ is the fibre integral of the value of $t$, which is 0. These pairs $(\mathbf{d},\mathbf{c})$ are cut out of $\mathbb{S}(\mathbb{R})$ by finitely many strict linear inequalities, so the $\mathbf{d}$ form an open set of real points, which is not empty (Theorem 48.5(iii)) and stays so off the poles of the $r_j$, a proper algebraic subset; the meromorphic function $\sum_j r_j P_W(\cdot+\lambda_j)$ vanishes there, so it vanishes on $\mathbb{D}_W$. So the fibre integral is well defined on $\mathfrak{J}^\natural$ and depends only on the value of the integrand. It vanishes on $\mathfrak{F}$, $(\mathrm{Sh}_{(\mathbf{0},\sigma)}-1)\,\mathrm{Sh}_{(\lambda,\tau)}[1]^\otimes$ integrating to $P_W(\cdot+\lambda)-P_W(\cdot+\lambda)$, and so it is well defined on $\mathfrak{H}$. $\mathrm{Sh}_\mu$ carries $\sum_j r_j e_{\lambda_j}$ to $\sum_j \sigma_\mu(r_j)e_{\lambda_j+\mu}$, whose fibre integral is $\sum_j\sigma_\mu(r_j)P_W(\cdot+\lambda_j+\mu) = \sigma_\mu\big(\sum_j r_j P_W(\cdot+\lambda_j)\big)$; so $\int\mathrm{Sh}_\mu\,h=\sigma_\mu\int h$, and the elements of value zero of $\mathfrak{H}$ form a submodule. For $m=\sum_\lambda r_\lambda\,\mathrm{Sh}_\lambda[1]$, $\int O(m)=\sum_\lambda r_\lambda P_W(\cdot+\lambda)=\langle m\rangle$ (Theorem 48.3(iii)). So $O(m)=0$ gives $\langle m\rangle=0$, and $\mathfrak{H}$ is a quotient of $\mathfrak{M}_W$. $W$ is a family of positive letters, so $\mathrm{Ann}\langle\cdot\rangle=0$ (Proposition 47.15(iii)), $O$ is injective, and $\dim_{\mathbb{K}_W}\mathfrak{H}=\mathfrak{m}_W$. The node words are families of positive letters too, so the same proposition makes the quotient of the module of a node by a submodule of elements of value zero the module itself.

(iii) *Quotients.* Let $\mathfrak{J}$ and $\mathfrak{J}'$ be the integrand modules with the modules $\mathfrak{V}_k$ and with their quotients $\mathfrak{V}_k/\mathfrak{N}_k$, $\langle\mathfrak{N}_k\rangle=0$. $\mathfrak{J}'$ is the quotient of $\mathfrak{J}$ by the sum $\tilde{\mathfrak{N}}$ of the images of the tensor products with one factor $\mathfrak{N}_k\otimes\mathbb{K}_\mathbb{S}$; the shifts map $\tilde{\mathfrak{N}}$ onto itself, the $\mathfrak{N}_k$ being submodules, and its elements have the value 0, one factor of each product having the value 0. The quotient map carries the shifts of $[1]^\otimes$ to those of $[1]^\otimes$, so it maps $\mathfrak{J}^\natural$ onto $\mathfrak{J}'^\natural$ and $\mathfrak{F}$ onto $\mathfrak{F}'$, and $\mathfrak{H}'=\mathfrak{J}^\natural/(\mathfrak{F}+\mathfrak{J}^\natural\cap\tilde{\mathfrak{N}})$, the quotient of $\mathfrak{H}$ by the image of $\mathfrak{J}^\natural\cap\tilde{\mathfrak{N}}$. That image is mapped onto itself by the shifts, and its elements have the fibre integral 0, the fibre integral depending only on the value of the integrand (proof of (ii)); the fibre integral of $\mathfrak{H}'$ is the one induced.

*The tree of a node down to* $\Sigma$. For $S\in\Sigma$ the members of the tree that lie properly inside no member of $\Sigma$ properly inside $S$ form a tree of nested bunches of $F$, in the normalization of

the tree, whose members properly inside $S$ form the tree of $S$ down to $\Sigma$. In it a member of $\Sigma$ nearest below $S$ has no member properly inside it, so it is a leaf, its node word its seam word (Definition 48.11(ii)). Every other member $Q$ properly inside $S$, and $S$, have the children they have in the tree: if a child $C$ of $Q$ lay properly inside a member $Q'$ of $\Sigma$ properly inside $S$, then $Q$ and $Q'$, which meet, are nested; $Q \subseteq Q'$ would put $Q$ properly inside $Q'$ or make it a member of $\Sigma$, and $Q' \subsetneq Q$ would put $C$ properly inside a member properly inside $Q$, against its being a child. So their node words are unchanged. Definition 48.17 applies to this tree, and by (ii) the direct image of the reading of the seam word of $S$ along it, with the modules of its nodes, is the quotient of the module of the seam word of $S$ by the kernel of $O$, a submodule of elements of value zero; at the root the seam word is $W_F$ (the gloss after Definition 48.11). From the leaves up, the leaves in $\Sigma$ carry quotients of their modules by submodules of elements of value zero, the module presented by an accepted set among them ((iv)), so by the first statement $\mathfrak{H}_S$ is a further quotient by such a submodule. A bijection as in the statement carries the stations, the transfer tensors, the node words and the seam words of the leaves inside one member to those inside the other (Theorem 48.12(iii)), so the shares, the pins and the kernel of its reading; by induction from the leaves up it carries the modules of the leaves in $\Sigma$, and then $\mathfrak{J}$, $\mathfrak{J}^\natural$, $\mathfrak{F}$ and $\mathfrak{H}$ (Theorem 48.5(v)). With $\Sigma$ the full nodes, the tree of $S$ down to $\Sigma$ consists of the full nodes nearest below $S$, its leaves in $\Sigma$, and of the nodes that are not full, lie properly inside $S$ and lie inside no full node properly inside $S$, the leaves of the tree that are not full among them; these are read in one fibre, as the nodes read jointly in Theorem 48.15. With $\Sigma = \{\mathcal{E}\}$ the tree of the root down to $\Sigma$ is the tree. These submodules of elements of value zero are 0, the seam words and $W_F$ being families of positive letters (Proposition 47.15(iii)), so $\mathfrak{H}_S$ is the module of the seam word of $S$ and $\mathfrak{H}_\mathcal{E}$ is $\mathfrak{M}_{W_F}$.

(iv) *Spanning.* By (ii) the class of the left side of a certificate is $e_{\lambda+\mu} - \sum_{\lambda'} r_{\lambda'}^{\mu,\lambda} e_{\lambda'}$, and the right side lies in $\mathfrak{F}$, so $\mathrm{Sh}_\mu e_\lambda = e_{\lambda+\mu} = \sum_{\lambda'} r_{\lambda'}^{\mu,\lambda} e_{\lambda'}$. The span of the $e_\lambda$, $\lambda \in \Lambda$, contains $e_{\mathbf{0}}$, the class of $[1]^\otimes$, and is mapped into itself by the shifts by $\mathcal{G}$, which are $\sigma_\mu$-semilinear, and so by every shift, every point of $\mathbb{D}_W(\mathbb{Z})$ being a sum of elements of $\mathcal{G}$; it contains every $e_\lambda = \mathrm{Sh}_\lambda e_{\mathbf{0}}$, so it is $\mathfrak{H}$. The fibre integral of $\mathrm{Sh}_\mu e_\lambda = \sum_{\lambda'} r_{\lambda'}^{\mu,\lambda} e_{\lambda'}$ is the relation among the values (ii).

*The module presented.* On $\mathbb{K}_W^\Lambda$ the maps $\mathrm{Sh}_\mu$ are $\sigma_\mu$-semilinear, and $\mathfrak{N}$ is the union of the increasing chain $\mathfrak{N}_0 \subseteq \mathfrak{N}_1 \subseteq \cdots$, $\mathfrak{N}_0$ spanned by the listed elements and $\mathfrak{N}_{i+1} = \mathfrak{N}_i + \sum_{\mu\in\mathcal{G}} \mathrm{Sh}_\mu \mathfrak{N}_i$, each a $\mathbb{K}_W$-subspace of a space of the dimension $|\Lambda|$, so the chain stops after at most $|\Lambda|$ steps, each step an elimination. On $\widehat{\mathfrak{H}} = \mathbb{K}_W^\Lambda/\mathfrak{N}$ the maps induced by $\mathrm{Sh}_\mu$ are well defined; for $q \in \mathbb{K}_W$, $\mathrm{Sh}_\mu \mathrm{Sh}_\nu(q\varepsilon_\lambda) - \mathrm{Sh}_\nu \mathrm{Sh}_\mu(q\varepsilon_\lambda) = \sigma_{\mu+\nu}(q)\big(\mathrm{Sh}_\mu \mathrm{Sh}_\nu \varepsilon_\lambda - \mathrm{Sh}_\nu \mathrm{Sh}_\mu \varepsilon_\lambda\big)$, and likewise for $\mathrm{Sh}_{-\mu}\mathrm{Sh}_\mu$, so the induced maps commute and $\mathrm{Sh}_{-\mu}$ inverts $\mathrm{Sh}_\mu$. $\mathcal{G}$ being a basis of $\mathbb{D}_W(\mathbb{Z})$ and its negatives, the free commutative group on the basis is $\mathbb{D}_W(\mathbb{Z})$, which so acts on $\widehat{\mathfrak{H}}$ by semilinear maps, and $\rho(\lambda)$ modulo $\mathfrak{N}$ is $\mathrm{Sh}_\lambda \varepsilon_{\mathbf{0}}$. The subspace $\mathfrak{N}$ does not depend on the paths: the smallest subspace $\mathfrak{N}''$ carried into itself by the maps and containing the commutators and the inverses at the $\varepsilon_\lambda$ contains them at every vector, by semilinearity, so on $\mathbb{K}_W^\Lambda/\mathfrak{N}''$ the induced maps commute and invert one another, and two paths from $\mathbf{0}$ to $\lambda$ carry $\varepsilon_{\mathbf{0}}$ to vectors whose difference lies in $\mathfrak{N}'' \subseteq \mathfrak{N}$; so the listed elements of one choice of paths lie in the $\mathfrak{N}$ of another. $\widehat{\mathfrak{H}}$ is spanned by the classes of the $\varepsilon_\lambda = \rho(\lambda)$, so a basis of it among them gives masters, with the shift matrices of the induced maps. The map $\Phi_W \to \widehat{\mathfrak{H}}$, $\langle\lambda\rangle \mapsto \mathrm{Sh}_\lambda \varepsilon_{\mathbf{0}}$, commutes with the shifts and carries $\mathrm{LI}_\ell$ and $\mathrm{MR}_e$ to the classes of their images under $\rho$, which lie in $\mathfrak{N}$, so it carries their shifts into $\mathfrak{N}$; by the presentation of (ii) it induces a map of $\mathfrak{M}_W$ onto $\widehat{\mathfrak{H}}$ that commutes with the shifts. The map $\mathbb{K}_W^\Lambda \to \mathfrak{H}$, $\varepsilon_\lambda \mapsto e_\lambda$, carries $\mathrm{Sh}_\mu$ to the shift of $\mathfrak{H}$, by the certificates; its kernel is mapped into itself by the $\mathrm{Sh}_\mu$ and contains the listed elements, the shifts of $\mathfrak{H}$ commuting and composing, $\rho(\lambda)$ going to $\mathrm{Sh}_\lambda e_{\mathbf{0}} = e_\lambda$, and the images of the letter identities and the mark relations going to $\tilde{O}(\mathrm{LI}_\ell) = 0$ and $\tilde{O}(\mathrm{MR}_e) = 0$ (proof

of (ii)); so it contains $\mathfrak{N}$, and $\widehat{\mathfrak{H}}$ maps onto $\mathfrak{H}$. The composite $\mathfrak{M}_W \to \widehat{\mathfrak{H}} \to \mathfrak{H}$ is $O$, so the kernel of $\mathfrak{M}_W \to \widehat{\mathfrak{H}}$ lies in that of $O$, of elements of value zero, which is 0 by (ii); so $\widehat{\mathfrak{H}}$ is $\mathfrak{M}_W$, its masters among the $\varepsilon_\lambda$ being masters of $\mathfrak{M}_W$, and by the first statement of (iii) it may stand for $\mathfrak{H}$ there.

*Existence and the search.* $\mathfrak{H}$ is finite-dimensional and spanned by the $e_\lambda$ (ii), so some finite $\Lambda \ni \mathbf{0}$ spans it; then for $\mu \in \mathcal{G}$ and $\lambda \in \Lambda$ there are $r_{\lambda'}^{\mu,\lambda}$ with $e_{\lambda+\mu} = \sum_{\lambda'} r_{\lambda'}^{\mu,\lambda} e_{\lambda'}$, so the left side of the certificate lies in $\mathfrak{F}$, a finite sum $\sum_k (\mathrm{Sh}_{(\mathbf{0},\sigma_k)} - 1) t_k$ (proof of Definition 48.17): a certificate. Write every $t_k$ as a combination, with unknown coefficients in $\mathbb{K}_W$, of shifts of $[1]^\otimes$; a candidate is $\Lambda$ with its lifts and, for every $(\mu, \lambda)$, finite sets of these shifts and of fibre translations. Every shift of $[1]^\otimes$ is a vector of $\mathfrak{J}$ over $\mathbb{K}_\mathbb{S}$ on the products of the masters of the modules of the nodes, $R_{\lambda,\tau}$ times the Kronecker product of the reductions of the $\mathrm{Sh}_{\iota_k(\lambda,\tau)}[1]$, which the shift matrices of the nodes at the generators of their lattices give along any path (Theorem 48.3(iii), the cocycle). In linear coordinates $(\mathbf{d}, \mathbf{u})$ on $\mathbb{S}$, $\mathbf{u}$ along the fibres, $\mathbb{K}_\mathbb{S} = \mathbb{K}_W(\mathbf{u})$; multiplied by a common denominator, the certificate is an identity of vectors of polynomials in $\mathbf{u}$ with coefficients in $\mathbb{K}_W$, linear in the unknowns, and comparing the coefficients of the monomials in $\mathbf{u}$ makes it a linear system over $\mathbb{K}_W$, decided by one elimination, over $\Bbbk(\mathbb{D}_W)$ when the masters of the nodes have their coefficients in $\Bbbk$ (Theorem 48.3(iii), the base change). The search reaches a certificate of an accepted set, which exists, and stops there.

(v) *The recursion.* $V_{LD}^S = \sum_{\mathbf{D}} \mathbf{N}_{S,L}(D; \mathbf{D}) \prod_i V_{LD_i}^{(S_i)}$ (Definition 48.11(i)), and by the multinomial theorem

$$\big(V_{LD}^S\big)^{A_{LD}} = \sum \frac{A_{LD}!}{\prod_{\mathbf{D}} \varepsilon_{LD\mathbf{D}}!} \prod_{\mathbf{D}} \mathbf{N}_{S,L}(D; \mathbf{D})^{\varepsilon_{LD\mathbf{D}}} \prod_i \big(V_{LD_i}^{(S_i)}\big)^{\varepsilon_{LD\mathbf{D}}},$$

the sum over the $\varepsilon_{LD\mathbf{D}} \in \mathbb{N}$ with the sum $A_{LD}$; the product over $(L, D)$ collects the powers of the seam letters of each child at each of its stations into the loads $\mathbf{A}_i(\boldsymbol{\varepsilon})$. With $\alpha^{\mathbf{m}_S} = \alpha^{\mathbf{m}_{Z_S}} \prod_i \alpha^{\mathbf{m}_{S_i}}$ every term is the product of the Laurent polynomial $\alpha^{\mathbf{m}_{Z_S}} \prod \mathbf{N}^{\boldsymbol{\varepsilon}}$ in $\alpha_{Z_S}$ and of the $\alpha^{\mathbf{m}_{S_i}} \prod_{L,D'} (V_{LD'}^{(S_i)})^{A_{i,LD'}}$ in the $\alpha_{S_i}$, in disjoint sets of variables, and its coefficient of $\alpha^{\mathbf{0}}$ is the product of theirs. At the root the letters of $F$ are $L = \sum_{\mathbf{D}} \mathbf{N}_{\mathcal{E},L}(\mathbf{D}) \prod_i V_{LD_i}^{(S_i)}$, and on the balance hyperplane $\mathbb{D}_{W_F}$ the Laurent polynomial $\alpha^{\mathbf{m}} \prod_L L^{N_L}$, which is $\omega_{\mathbf{d}}/dy$, is homogeneous of the degree 0, a combination of the $\alpha^u$ with $u \in M_{W_F} = \mathbf{1}^\perp \cap \mathbb{Z}^{\mathcal{E}}$; so its coefficient of $\alpha^{\mathbf{0}}$ is its mean over the compact torus of $X_{W_F}$, the closing $\mathsf{C}(1)(\mathbf{d})$ of $W_F$ (Theorem 48.23(ii)), which is its constant term on the chart $\alpha_{e_0} = 1$ (Definition 47.9).

*The elements of value zero.* The seam letters of a full node $S$ are the letters $K_{j,(D)}$, for a letter $K_j$ of $F$, and $\Psi_{j,(D)}$, for a letter $\Psi_j$, of the room children of its stations, tables on $S$ (Definition 28.1), times the constants of the normalization, which multiply the weights; so its seam word at its shares is a family of letters of tables at the exponents $a_{LD}$, with the data space of Definition 48.1 and the closings $\mathsf{C}_S$, and, being full, its marks form one molecule of the family of these tables (Theorem 48.23(i)). By Theorem 48.23(ii), Proposition 47.10 and Theorem 47.11 hold for it and for $W_F$, and for their quotients by submodules of elements of value zero, the closings being the constant terms on the compact tori of their letter complements; a full node of one mark has monomial letters, a point as its space $V_W$, the period $\prod c^{-w}$ of the coefficients and a module of dimension one with no nonzero element of value zero. For $x = \sum_\lambda r_\lambda f_\lambda$, $\mathsf{C}(x)(\mathbf{d}) = \sum_\lambda r_\lambda(\mathbf{d})\, \mathsf{C}(1)(\mathbf{d} + \lambda)$, $\mathsf{C}(1)$ being $\mathsf{C}_S$. An element of $\mathfrak{H}_S$ is the image of an element of the module of the seam word with the same value ((iii)), so it has the value zero exactly when that element has; and by Proposition 47.15(iii) an element of value zero of these modules is 0. Bijections of subtrees as in (iii) carry the seam letters, the entries and the leaves, and so the closings. □

**Definition 48.19 (The module of a transform; the twisted direct image of a block).** As

stated.

*Proof.* (i) Read at the seam points of the children, the seam letters of the node are $V^S_{LD} = \prod_l h_{lL} \prod_z \alpha_z^{u_{z,L}} \big(n_L(D) + \sum_P \widehat{\beta}_{P,LD}\big)$, the first product over the children of $S$, which is $H_L$ for the one block (Definitions 48.11(i) and 48.13(i),(iii)), each $\widehat{\beta}_{P,LD}$ homogeneous of the degree $\delta_{LD}$ in the marks of its parts (proof of Theorem 48.14(ii)), so that $n_L(D) = 0$ when $\delta_{LD} \neq 0$. The transform reads the masses of the marks of the children, $m_z - \sum_L s_L u_{z,L}$ at the own marks and the exponents $s_L$ of the $H_L$ (Definition 48.13(iii)). When some mark of $F$ lies outside $B_P$, as it does below the root and at a root read through at least two blocks, its mass takes the balance of $F$, and these data are all of $\mathbb{D}_P$. When $S$ is the root and $P$ its one block, $B_P$ is every mark and $V^S_L$ is the letter $L$ of $F$, so the balance of $F$ is $\sum_{e \in B_P} m_e = \sum_L w_L(\deg H_L + \delta_L)$ on the data of the block: the balance hyperplane of $\mathbb{D}_P$ when $S$ is not seen by its scales, every $\delta_L$ being 0, and a hyperplane of $\mathbb{D}_P$ when it is seen and $P$ has children; without children there are no $w_L$, and the exponents $s_L$ take that balance when some $\delta_L \neq 0$.

(ii) The integrand of the transform at the data $\mathbf{d}_P$ is $\Omega\, dy$ with

$$\Omega = \alpha^{\mathbf{m}} \prod_L H_L^{-w_L} \exp\Big( - \sum \xi_{LD} \widehat{\beta}_{P,LD} \Big),$$

and $\omega_e = \partial_{y_e} \log \Omega$. Each $\widehat{\beta}_{P,LD}$ is $H_L^{-1}$ times a Laurent monomial in the own marks times a polynomial with nonnegative coefficients in the marks of $B_P$, so this is the integrand of Lemma 48.8 with the $h_k$ the $H_L$ at the exponents $w_L$ and the $R_i$ the $\widehat{\beta}_{P,LD}$, and $\mathfrak{M}_P$ is the top cohomology $H$ of the proof of that lemma, of finite dimension over $\mathbb{K}_P(\boldsymbol{\xi})$, the data and the scales taken as indeterminates. The maps $D_i$ of that proof are the connections $\nabla_{LD}$; they act on $H$ and commute. The strides span $\mathbb{C}[Y_P] = \mathbb{C}[\alpha^{\pm 1}][(\prod_L H_L)^{-1}]$. The shift by $\lambda$ multiplies $\Omega$ by $f_\lambda$ and moves the data by $\lambda$, so it carries $(\partial_{y_e} + \omega_e) g \otimes r$ to $(\partial_{y_e} + \sigma_\lambda \omega_e)(f_\lambda g) \otimes \sigma_\lambda r$, $\sigma_\lambda \omega_e = \omega_e + \partial_{y_e} \log f_\lambda$, a relation; so the shifts are well defined, and they commute with the connections, $\sigma_\lambda$ fixing the scales. A relation has the value 0 where its terms converge, by the integration by parts of that proof, and the value of $f_\lambda$ at $\mathbf{d}_P$ is that of [1] at $\mathbf{d}_P + \lambda$ (part (i) there). On the quotient by the common scaling, part (iii) of the lemma reads the integral on a section, a torus of one dimension fewer, with an integrand of the same kind, invariant on the balance hyperplane, and the same holds there.

(iii), (iv) The share reading is that of Theorem 48.14(ii), whose proof does not use that the exponents of the absent letters are the $s_L$: at fixed seam points of the children the inner integral is the Gramma of the block, opened on its share side, and the powers of the seam coordinates are $\mathbf{x}_l^{-\mathbf{a}_l}$ at the loads of the definition, with $w_L$ in the place of $s_L$. The maps $\iota_l$ carry $\mathbb{S}_P$ onto the data spaces of the seam words of the children: the present loads of a child at the stations of its seam word other than its absent ones are sums of disjoint nonempty sets of shares, each such station being that of some basis and so of some monomial of a seam letter of the node, its absent loads are the $w_L$ less such sums, its masses are data of the block, and its scale pin is the balance of its data space (Definition 48.1(ii)). The kernel moves by Pochhammer quotients in the shares and by powers of the constants $c_\tau$ under the shifts, and the shifts of $\mathfrak{J}_P$ are those of Theorem 48.5(ii) for it. That the shifts, the connections and the scale integral are well defined on $\mathfrak{H}_P$ is part of Theorem 48.20(ii). □

**Theorem 48.20 (The module of a transform from its block).** *As stated.*

*Proof.* (i) *The lifts.* Let $\tau$ be a monomial of $\beta_{P,LD}$ and $P_\tau$ the set of the children present in it. A term of $c_\tau r_\tau$ is $c_\tau \prod_l c_{\mathbf{t}_l}\, \alpha^{\sum_l \mathbf{t}_l + \mathbf{q}_\tau - \mathbf{u}_L} / H_L$, the $c_{\mathbf{t}_l} \alpha^{\mathbf{t}_l}$ running over the monomials of $V^{(l)}_{LD_l(\tau)}$ for

$l \in P_\tau$ and of $h_{lL}$ for the other children $l$ of $P$, and $\mathbf{q}_\tau - \mathbf{u}_L$ the exponents of $\tau$ on the own marks; its stride is $\lambda = (\sum_l \mathbf{t}_l + \mathbf{q}_\tau - \mathbf{u}_L; \mathbf{e}_L)$. Move the share of $\tau$ by one. The pin of an own mark $z$ moves by $q_{z,\tau} - u_{z,L}$ on both sides. A child $S_l$, $l \in P_\tau$, has its masses moved by $\mathbf{t}_l$ and its load at $D_l(\tau)$ raised by one, the share of $\tau$ entering it, and its absent load at $L$, $w_L$ less the shares of the monomials of $L$ in which it is present, unchanged; so its data move by $(\mathbf{t}_l; \mathbf{e}_{D_l(\tau)})$, the data of the stride $\alpha^{\mathbf{t}_l}/V^{(l)}_{LD_l(\tau)}$ of its seam word, which keeps its balance, $|\mathbf{t}_l|$ being the degree of that letter, and so its scale pin. A child of $P$ absent from $\tau$ has its masses moved by $\mathbf{t}_l$, its absent load at $L$ raised by one and its other loads unchanged: its data move by the stride $\alpha^{\mathbf{t}_l}/h_{lL}$ of its seam word. So $(\lambda, \mathbf{e}_\tau) \in \mathbb{S}_P(\mathbb{Z})$, the kernel moves by $\Gamma(z_\tau+1)c_\tau^{-z_\tau-1}/\big(\Gamma(z_\tau)c_\tau^{-z_\tau}\big) = z_\tau/c_\tau$, and

$$\mathrm{Sh}_{(\lambda,\mathbf{e}_\tau)}[1]^{\otimes} = \frac{z_\tau}{c_\tau} \bigotimes_{l\in P_\tau} \big[\alpha^{\mathbf{t}_l}/V^{(l)}_{LD_l(\tau)}\big] \otimes \bigotimes_{l\notin P_\tau} \big[\alpha^{\mathbf{t}_l}/h_{lL}\big].$$

Summing over the terms with their coefficients, each factor becomes $\big[V^{(l)}_{LD_l(\tau)}/V^{(l)}_{LD_l(\tau)}\big]$ or $[h_{lL}/h_{lL}]$, that is [1], by the letter identities of the children, and with $\boldsymbol{\xi}^{-Z(\mathbf{e}_\tau)} = \xi_{LD}^{-1}$ the sum is $\xi_{LD}^{-1} z_\tau [1]^{\otimes}$. A monomial $c_{\mathbf{t}}\alpha^{\mathbf{t}}$ of $H_L$, $\alpha^{\mathbf{t}} = \prod_l \alpha^{\mathbf{t}_l}$ with $\alpha^{\mathbf{t}_l}$ a monomial of $h_{lL}$, gives the stride $\alpha^{\mathbf{t}}/H_L$: with no share moved, the absent load of every child of $P$ at $L$ rises by one and its masses by $\mathbf{t}_l$, so its data move by the stride $\alpha^{\mathbf{t}_l}/h_{lL}$ of its seam word, the pins hold, the kernel does not move, and $\sum_{\mathbf{t}} c_{\mathbf{t}} \,\mathrm{Sh}_{((\mathbf{t};\mathbf{e}_L),\mathbf{0})}[1]^{\otimes} = \bigotimes_l [h_{lL}/h_{lL}] = [1]^{\otimes}$.

*The mark relations.* At an own mark $z$ no $H_L$ involves $\alpha_z$, and $\alpha_z\partial_{\alpha_z} r_\tau = (q_{z,\tau} - u_{z,L})r_\tau$, so $(\partial_{y_z} + \omega_z)1 = m'_z - \sum_{L,D} \xi_{LD} \sum_\tau c_\tau (q_{z,\tau} - u_{z,L}) r_\tau$, whose lift is $\big(m'_z - \sum_\tau (q_{z,\tau} - u_{z,L}) z_\tau\big)[1]^{\otimes}$, the function in the bracket vanishing on $\mathbb{S}_P$ by the pin of $z$. At a mark $e$ of a child $S_l$ write $R^{(l)}_{LD',e} = \alpha_e\partial_{\alpha_e} V^{(l)}_{LD'}/V^{(l)}_{LD'}$ for the occupations of the letters of its seam word. Then $\alpha_e\partial_{\alpha_e} \log H_L = R^{(l)}_{LD^\circ_{l,L},e}$, and $\alpha_e\partial_{\alpha_e} r_\tau = r_\tau\big(R^{(l)}_{LD_l(\tau),e} - R^{(l)}_{LD^\circ_{l,L},e}\big)$ when $l \in P_\tau$ and 0 otherwise. Written over $H_L$, $w_L R^{(l)}_{LD^\circ,e} = w_L(\alpha_e\partial_{\alpha_e} h_{lL}) \prod_{l'\neq l} h_{l'L}/H_L$; lifted with $\mathbf{0}$ as the letter identity of $H_L$, the factor of $S_l$ becomes $\big[R^{(l)}_{LD^\circ,e}\big]$ and the others [1]. Written over $H_L$, $c_\tau r_\tau R^{(l)}_{LD_l(\tau),e}$ is $c_\tau r_\tau$ with $V^{(l)}_{LD_l(\tau)}$ replaced by $\alpha_e\partial_{\alpha_e} V^{(l)}_{LD_l(\tau)}$ in the numerator; lifted with $\mathbf{e}_\tau$ as above, it gives $\xi^{-1}_{LD} z_\tau \big[R^{(l)}_{LD_l(\tau),e}\big]$ in the factor of $S_l$ and [1] in the others. Written over $H_L^2$, $c_\tau r_\tau R^{(l)}_{LD^\circ,e}$ is $c_\tau$ times the numerator of $r_\tau$ times $(\alpha_e\partial_{\alpha_e} h_{lL}) \prod_{l'\neq l} h_{l'L}$, over $H_L^2$; lifted with $\mathbf{e}_\tau$, the share of $\tau$ rises by one and $w_L$ by two, so every child present in $\tau$ has its load at its station and its absent load raised by one and every child of $P$ absent from $\tau$ its absent load raised by two, and the lift is $\xi^{-1}_{LD} z_\tau \big[R^{(l)}_{LD^\circ,e}\big]$ in the factor of $S_l$ and [1] in the others, by the letter identities. Summing, the lift of $(\partial_{y_e} + \omega_e)1$ is, in the factor of $S_l$,

$$\Big[m_e - \sum_L \Big(w_L - \sum_{\tau\ni l} z_\tau\Big) R^{(l)}_{LD^\circ_{l,L},e} - \sum_{L,D'} \Big( \sum_{\tau:\ D_l(\tau)=D'} z_\tau\Big) R^{(l)}_{LD',e}\Big] = \Big[m_e - \sum_{L,D} a_{l,LD}\, R^{(l)}_{LD,e}\Big],$$

the sums over the monomials $\tau$ of the readings of $L$ in which $S_l$ is present, $a_l$ its loads: the defect $c^{(l)}_e$ of $e$ in the seam word of $S_l$ at its masses and loads (Definition 48.1(iv)), whose class is the mark relation of the child at $e$ (Theorem 48.3(ii),(iii)), 0 in its module and in every quotient of it.

*From a point of the lattice.* Multiplying an insertion by $f_\lambda$ moves the data of its strides by $\lambda$; lifting them with their share vectors plus $\tau_\lambda$ moves the data of every child by $\iota_l(\lambda, \tau_\lambda)$ more, where its letter identities and its mark relations hold, being identities among its insertions, and the kernel by the Pochhammer quotient $R_{\lambda,\tau_\lambda}$ more, the shares being moved by $\tau_\lambda$. So the identities above, with the data moved by $\lambda$ and the shares by $\tau_\lambda$ in their coefficients, hold for the shifts by $(\lambda, \tau_\lambda)$ of their terms, which is $\mathrm{Sh}_{(\lambda,\tau_\lambda)}$ applied to both sides; their coefficients are polynomials in the data, the shares and the scales.

(ii) *The lattice and* $\mathfrak{M}'_P$. By (i) the data of the terms of every $c_\tau r_\tau$, of every stride $\alpha^{\mathbf{t}}/H_L$ and of every term of a mark relation, written as there, lie in $\mathbb{D}'_P(\mathbb{Z})$. A connection carries $[f_\lambda \otimes r]$ to $[f_\lambda \otimes \partial r] - \sum_\tau c_\tau [r_\tau f_\lambda \otimes r]$, whose strides lie in $\lambda + \mathbb{D}'_P(\mathbb{Z})$; so $\mathfrak{M}'_P$ is carried into itself, and it contains the $\nabla^{\mathbf{a}}[1]$, the derivatives of [1].

*A presentation of* $\mathfrak{M}'_P$. As in the proof of Theorem 48.18(ii), the kernel of the map from the free module on $\mathbb{D}_P(\mathbb{Z})$ to $\mathbb{C}[Y_P] \otimes \mathbb{K}_P(\boldsymbol{\xi})$ is spanned by the letter identities $f_\lambda - \sum_{\mathbf{t}} c_{\mathbf{t}} f_{\lambda+(\mathbf{t};\mathbf{e}_L)}$ of the $H_L$, which raise the levels to a common one, at which the strides of distinct masses are independent; and $(\partial_{y_e} + \omega_e) f_\lambda$ is $f_\lambda$ times $\omega_e$ at the data moved by $\lambda$, $\omega_e + \partial_{y_e} \log f_\lambda$, a combination of the strides at $\lambda$ plus the data of the terms of (i). So the free module, the letter identities and the mark relations split along the cosets of $\mathbb{D}'_P(\mathbb{Z})$, the strides of distinct cosets being independent at a common level, and $\mathfrak{M}'_P$ is the free module on $\mathbb{D}'_P(\mathbb{Z})$ modulo the letter identities and the mark relations at its points, $\langle\lambda\rangle \mapsto \mathrm{Sh}_\lambda[1]$.

*The twisted direct image.* By (i), $z_\tau[1]^\otimes = \xi_{LD} \sum (\text{coefficient})\, \boldsymbol{\xi}^{-Z(\mathbf{e}_\tau)} \mathrm{Sh}_{(\lambda,\mathbf{e}_\tau)}[1]^\otimes$, and $z_\tau \mathrm{Sh}_{(\mu,\sigma)}[1]^\otimes = \mathrm{Sh}_{(\mu,\sigma)}\big((z_\tau - \sigma_\tau)[1]^\otimes\big)$, the shifts being semilinear; so $\mathfrak{J}^{\boldsymbol{\xi}}_P$ is carried into itself by the multiplications by the shares, by the derivatives of the coefficients and so by the connections, and by the shifts. For $t \in \mathfrak{J}^{\boldsymbol{\xi}}_P$ and a fibre translation $(\mathbf{0}, \delta)$, the shift fixes $\boldsymbol{\xi}$ and moves the function $Z_{LD}$ by $Z_{LD}(\delta)$, so

$$\nabla_{LD}\big(\boldsymbol{\xi}^{-Z(\delta)} \mathrm{Sh}_{(\mathbf{0},\delta)}\, t - t\big) = \boldsymbol{\xi}^{-Z(\delta)} \mathrm{Sh}_{(\mathbf{0},\delta)}(\nabla_{LD} t) - \nabla_{LD} t,$$

the derivative of $\boldsymbol{\xi}^{-Z(\delta)}$ cancelling the moved total; the connections and, the lattice being commutative, the shifts carry $\mathfrak{F}^{\boldsymbol{\xi}}_P$ into itself, and the same cancellation makes the connections commute with the shifts. The class of $\boldsymbol{\xi}^{-Z(\tau)} \mathrm{Sh}_{(\lambda,\tau)}\, t$ does not depend on the lift: two lifts differ by a fibre translation $\delta$, and the difference is a twisted translation of the one. So the shifts, the classes and the connections of $\mathfrak{H}_P$ are well defined, the classes $e_\lambda$ span it, the connections commute with one another and with the shifts, and on the classes, $\boldsymbol{\xi}^{-Z(\tau)}$ being differentiated,

$$\nabla_{LD} e_\lambda = -\Big[\boldsymbol{\xi}^{-Z(\tau)} \mathrm{Sh}_{(\lambda,\tau)}\big(\xi_{LD}^{-1} Z_{LD}[1]^\otimes\big)\Big].$$

*The map* $O_P$. On the free module on $\mathbb{D}'_P(\mathbb{Z})$, $\langle\lambda\rangle \mapsto e_\lambda$. By (i) from a lift of each point and the independence of the lifts, it carries the letter identities and the mark relations at every point of $\mathbb{D}'_P(\mathbb{Z})$ to 0, and by the presentation it induces $O_P$ on $\mathfrak{M}'_P$, onto $\mathfrak{H}_P$; it commutes with the shifts, both sides being moved by $\sigma_\mu$ and the points by $\mu$. $O_P(\widehat{\beta}_{P,LD}) = \sum_\tau c_\tau O_P(r_\tau) = \big[\xi_{LD}^{-1} Z_{LD}[1]^\otimes\big]$ by (i), so $O_P\big(\nabla_{LD} \mathrm{Sh}_\lambda[1]\big) = -\mathrm{Sh}_\lambda O_P(\widehat{\beta}_{P,LD}) = \nabla_{LD} e_\lambda$ by the display; with the derivatives of the coefficients, $O_P$ commutes with the connections.

*The scale integral.* At real data $\mathbf{d}_P + \lambda$ with a contour of Theorem 48.14(ii), $\langle f_\lambda \rangle$ is the fibre integral of $\boldsymbol{\xi}^{-Z(\mathbf{z})}$ times the value of $[1]^\otimes$ at the shifted data, and moving the contour by the lift $\tau$, it is the fibre integral at $\mathbf{d}_P$ of the value of $\boldsymbol{\xi}^{-Z(\mathbf{z})} \boldsymbol{\xi}^{-Z(\tau)} \mathrm{Sh}_{(\lambda,\tau)}[1]^\otimes$. The points $(\mathbf{d}_P, \mathbf{c})$ admissible for finitely many lifts are cut out of $\mathbb{S}_P(\mathbb{R})$ by strict linear inequalities, the shares and the loads having positive real parts and the masses and loads of the children lying in their tubes, and they exist along an admissible direction, as in Theorem 48.5(iii), the native split at a realization of Definition 48.17 being admissible (Theorem 48.14(ii)). So an element of $\mathfrak{J}^{\boldsymbol{\xi}}_P$ whose value vanishes has a vanishing scale integral on an open set of real data, and so on the connected part of the domain of Lemma 48.8(i) where its terms converge; a twisted translation integrates to 0, its two terms being one element of $\mathbb{D}'_P(\mathbb{Z})$ written with two lifts. So the scale integral is well defined on $\mathfrak{H}_P$, $\int O_P(m) = \langle m \rangle$, $\int \nabla_{LD} h = \partial_{\xi_{LD}} \int h$ by differentiation under the integral, dominated by the decay of the Gammas along the fibre, and $\int \mathrm{Sh}_\lambda\, h = \sigma_\lambda \int h$.

*The rest.* $O_P(m) = 0$ gives $\langle m \rangle = 0$, and $\dim \mathfrak{M}'_P \le \dim \mathfrak{M}_P$, finite (Lemma 48.8(ii)).

*The values separate* $\mathfrak{M}_P$. The relation at a mark $e$, written out, is $g \otimes m_e r - \sum_L \alpha_e \partial_{\alpha_e} (\log H_L)\, g \otimes w_L r - \sum_{(L,D)} \alpha_e \partial_{\alpha_e} \widehat{\beta}_{P,LD}\, g \otimes \xi_{LD} r + \partial_{y_e} g \otimes r$, $m_e$ being $m'_z$ at an own mark; so, as in Proposition 47.15(i), an element $x$ of $\mathfrak{M}_P$ times a nonzero polynomial $D$ in the data and the scales is the class of $\sum_k G_k \otimes p_k$, the $G_k \in \mathbb{C}[Y_P]$ linearly independent and the $p_k$ polynomials in the exponents $w_L$ and the scales. Let $\langle x \rangle = 0$. The domain of Lemma 48.8(i) is the set of the data whose real parts lie in an open convex cone, the same at all scales, and it is not empty, the value of [1] being $\mathcal{T}_P$ there; as in the proof of Proposition 47.15, the points of that cone far out along a ray of it lie in the domain of every stride of the $G_k$, so for $(\mathbf{w}, \boldsymbol{\xi})$ in a nonempty open set of real exponents and positive scales the masses at which every term converges have their real parts in a nonempty open set. Moving the masses there by imaginary vectors makes the value of $\sum_k G_k \otimes p_k$ the Fourier transform, on the space of the $y = \log \alpha$, its quotient by the common scaling when $S$ is not seen by its scales, of $\sum_k p_k(\mathbf{w}, \boldsymbol{\xi})\, G_k$ times the integrand, an integrable function. So $\sum_k p_k(\mathbf{w}, \boldsymbol{\xi}) G_k$ vanishes on the orthant, and so on $Y_P$, a regular function on $Y_P$ being a Laurent polynomial over a product of powers of the $H_L$, which are positive on the orthant; every $p_k$ vanishes on an open set, $Dx = 0$, and $x = 0$. So $O_P$ is injective, an isomorphism of $\mathfrak{M}'_P$ onto $\mathfrak{H}_P$. Over $\mathbb{E}_P$, when it is the hyperplane of the balance of $F$, the masses at fixed exponents lie on a hyperplane, which the argument does not reach, and (vii) tells the elements of value zero among the combinations of the derivatives of the input there.

(iii) The display of (ii) and $O_P(\widehat{\beta}_{P,LD}) = \sum_\tau c_\tau O_P(r_\tau)$, with $O_P(r_\tau)$ the combination of the classes of the terms of $c_\tau r_\tau$ divided by $c_\tau$, give $\nabla_{LD} e_\lambda = -\operatorname{Sh}_\lambda O_P(\widehat{\beta}_{P,LD})$, a combination of classes of shifted points; on classes $e_\lambda$ written on an accepted set the connection matrices are therefore read off the shift matrices along paths to these points, plus the derivatives of the coefficients. The proof of Theorem 48.18(iv) then holds verbatim over $\mathbb{K}_P(\boldsymbol{\xi})$: the spanning, the existence of accepted sets ($\mathfrak{H}_P$ being finite-dimensional and spanned by the classes), the certificates, identities in $\mathfrak{J}_P \otimes \mathbb{C}(\boldsymbol{\xi})$ over $\mathbb{K}_{\mathbb{S}_P}(\boldsymbol{\xi})$, made linear over $\mathbb{K}_P(\boldsymbol{\xi})$ by clearing denominators in the coordinates of the fibre and comparing coefficients, and the module presented, now by a subspace $\mathfrak{N}$ carried into itself by the shifts and by the connections read as $\partial_{\xi_{LD}} - \sum_\tau \operatorname{Sh}(\text{terms of } c_\tau r_\tau)$ and containing, besides the elements listed there, the images of the commutators of the connections with one another and with the shifts; the letter identities and the mark relations in its list are those of the $H_L$ and of the transform, whose images are 0 by (i). The maps $\mathfrak{M}'_P \to \widehat{\mathfrak{H}} \to \mathfrak{H}_P$ commute with the shifts and the connections, by the same argument, and their composite is $O_P$; so the kernel of $\mathfrak{M}'_P \to \widehat{\mathfrak{H}}$ lies in that of $O_P$, which is 0 by (ii), and $\widehat{\mathfrak{H}}$ is $\mathfrak{M}'_P$.

(iv) *The objects over* $\mathbb{E}_P$. An element $\boldsymbol{\xi}^{-Z(\tau)} \operatorname{Sh}_{(\lambda,\tau)}[1]^\otimes$ is $\boldsymbol{\xi}^{-Z(\tau)}$ times $R_{\lambda,\tau}$, a rational function of the shares alone, the kernel depending on them alone, times the tensor product of the classes of the strides $\iota_l(\lambda, \tau)$ of the seam words of the children, with no coefficient to move; so it is defined over the share space over $\mathbb{E}_P$ once the $\iota_l$ carry that space onto the data spaces of the children, whose rational functions are then read there. They do. When $\mathbb{E}_P = \mathbb{D}_P$ this is the proof of Definition 48.19(iii). Otherwise, summing the pins of the block with the loads written in the shares, each monomial of a reading at $L$ having the degree $\delta_L$, gives $\sum_{e \in B_P} m_e = \sum_L w_L \deg H_L + \sum_L \delta_L Z_L$, so on $\mathbb{E}_P$ (proof of Definition 48.19(i)) the pins leave the one further condition $\sum_L \delta_L (w_L - Z_L) = 0$ on the shares and the $w_L$. The data of a child are free on the share space over $\mathbb{D}_P$, by the proof cited; and with its loads held, the condition is met by the share of a monomial of a reading of a letter with $\delta_L \neq 0$ in which the child is absent. Such a monomial exists: the absent station of the child is the station of a basis, whose monomial in $V_L^S = H_L \prod_z \alpha_z^{u_{z,L}} \big(n_L + \widehat{\beta}_{P,L}\big)$ is not in the constant part, $n_L$ being 0 when $\delta_L \neq 0$, and some $\delta_L \neq 0$, $S$ being seen by its scales.

The fibre translations fix the data, so they act on the rational functions of the share space over $\mathbb{E}_P$, and the twisted translations, the classes, their independence of the lifts and the connections

are defined as in (ii), with the same proofs; the identities of (i) from every point of $\mathbb{S}_P(\mathbb{Z})$ have coefficients polynomial in the data, the shares and the scales, so they hold there, and so does the presentation of $\mathfrak{M}'_P$, the mark relations at a point having coefficients polynomial in the data. The module of the transform over the rational functions of $\mathbb{E}_P$ and the scales is the top cohomology of the twisted de Rham complex with the data of $\mathbb{E}_P$ as indeterminates, finite-dimensional by the proof of Lemma 48.8(ii) over that field. So $O_P$ is defined, onto $\mathfrak{H}_P$ and commutes with the connections, and the scale integral is defined, as in (ii), the native split at a realization of Definition 48.17, whose data lie in $\mathbb{E}_P$, being admissible and the admissible directions being taken in $\mathbb{E}_P$; the commutation with the shifts holds for the shifts by points of $\mathbb{E}_P$.

*The derivatives.* $\int \nabla^{\mathbf{a}} e_{\mathbf{0}} = \partial^{\mathbf{a}} \mathcal{T}_P$ on $\mathbb{E}_P$ by (ii), and $\mathfrak{H}^{\partial}_P$ is the span of the $\nabla^{\mathbf{a}} e_{\mathbf{0}}$, so its scale integral is the span of the derivatives of $\mathcal{T}_P$. Take a basis of $\mathfrak{H}^{\partial}_P$ among the $\nabla^{\mathbf{a}} e_{\mathbf{0}}$ containing $e_{\mathbf{0}}$, which is not 0, its scale integral $\mathcal{T}_P$ not being 0. The matrices of the connections on it have entries rational in the data and the scales, and off the common zeros of the coefficients of their denominators as polynomials in the scales, a proper algebraic subset of the data of $F$, they specialize. At such data the relations $\partial_{\xi_{LD}} \partial^{\mathbf{a}_i} \mathcal{T}_P = \sum_j A^{LD}_{ij} \partial^{\mathbf{a}_j} \mathcal{T}_P$ hold, holding on an open set of the data and the scales, both sides being holomorphic there; so the derivatives of $\mathcal{T}_P$ at these data lie in the span of the $\partial^{\mathbf{a}_i} \mathcal{T}_P$, and the module with the specialized matrices maps onto the module that $\mathcal{T}_P$ generates, the basis going to the $\partial^{\mathbf{a}_i} \mathcal{T}_P$. On a section of the scale quotient by a monomial map the restrictions of the derivatives span the derivatives of the restriction (proof of Theorem 48.14(ii)), and the same holds for the module read on the section: the sum of the mark relations of the transform, its Euler relation, is $\sum_{(L,D)\in\Sigma_S} \delta_{LD}\, \xi_{LD} \nabla_{LD} e_{\mathbf{0}} = -\big(\sum_{e\in B_P} m_e - \sum_L w_L \deg H_L\big) e_{\mathbf{0}}$, so the connection matrices on the basis are homogeneous for the scaling, their polar locus is a union of its orbits, and a section meets it in a proper subset.

*The presentation by* $\Lambda$. By the display of (ii) and (i) from the point $(\lambda, \tau_\lambda)$, $\nabla_{LD} e_\lambda = -\sum(\text{coefficient})\, e_{\lambda+\mu}$ over the terms of $\widehat{\beta}_{P,LD}$. Put $\nabla_{LD} \varepsilon_\lambda = -\sum(\text{coefficient})\, \rho(\lambda+\mu)$, $\rho(\lambda+\mu)$ the combination of the $\varepsilon_{\lambda'}$ that the certificate gives, and extend with the derivations of the coefficients. The map $\varepsilon_\lambda \mapsto e_\lambda$ commutes with these and the connections of $\mathfrak{H}_P$, by that formula and the certificates. The connections of $\mathfrak{H}_P$ commute; so the commutators of the connections on the vectors, which are linear over the rational functions, have values on the basis that map to 0, and the smallest subspace that the connections carry into itself and that contains these values maps to 0. On the quotient by it the connections commute, so it is a differential module, and it maps onto the span of the $e_\lambda$, $\lambda \in \Lambda$, which contains $\mathfrak{H}^{\partial}_P$, $\varepsilon_{\mathbf{0}}$ going to $e_{\mathbf{0}}$; so the submodule generated by $\varepsilon_{\mathbf{0}}$ maps onto $\mathfrak{H}^{\partial}_P$, and an element of the kernel has the scale integral 0. When $\mathbb{E}_P = \mathbb{D}_P$ the module presented is that of (iii), which is $\mathfrak{M}'_P$, and the map is the restriction of $O_P$, injective by (ii). $\mathfrak{H}_P$ has finite dimension and is spanned by the classes, so a finite $\Lambda \ni \mathbf{0}$ whose classes span it has certificates for all the $e_{\lambda+\mu}$, each class lying in the span and the difference of the two sides lying in $\mathfrak{F}^{\boldsymbol{\xi}}_P$, a finite sum of twisted translations; a search through the finite sets of points with lifts and of supports of certificates, in any order that reaches each, testing each by one elimination over the rational functions of the share space over $\mathbb{E}_P$ and the scales, stops. When $\mathbb{E}_P = \mathbb{D}_P$ every object here is that of (ii) and (iii).

(v) Without children, $\mathfrak{J}_P = \mathbb{K}_{\mathbb{S}_P}$, the pins are those of the own marks, $t\mathbf{z} = \mathbf{m}'_P$, and the transform is the Gramma of the block (Definition 48.13(iii)), read on its share side (Proposition 4.2) with the coefficients $\xi_{LD} c_\tau$. By (i), $c_\tau e_{t_\tau} = \xi^{-1}_{LD}[z_\tau]$, $t_\tau$ being the data of the insertion $r_\tau = \alpha^{t_\tau}$. The exchange relation of Theorem 2.4(ii) at the own mark $z$, $m'_z\, \mathsf{G} = \sum_\tau t_{\tau,z}\, \xi_{LD} c_\tau\, \mathsf{G}_{\mathbf{m}'+t_\tau}$, is in $\mathfrak{H}_P$ the identity $m'_z[1] = \sum_\tau t_{\tau,z}[z_\tau]$, the pin of $z$. The ladder, $\partial_{\xi_{LD}} \mathsf{G} = -\sum_\tau c_\tau \mathsf{G}_{\mathbf{m}'+t_\tau}$, is $\nabla_{LD} e_{\mathbf{0}} = -\big[\xi^{-1}_{LD} Z_{LD}\big] = -\sum_\tau c_\tau e_{t_\tau}$. For $\mathbf{v} \in \mathbb{Z}^N$ with $t\mathbf{v} = 0$, $\partial^{\mathbf{v}^\pm} \mathsf{G} = (-1)^{|\mathbf{v}^\pm|} \mathsf{G}_{\mathbf{m}'+t\mathbf{v}^\pm}$ by the ladder, the derivatives taken in the coefficients, so the toric relation is $\mathsf{G}_{\mathbf{m}'+t\mathbf{v}^+} = \mathsf{G}_{\mathbf{m}'+t\mathbf{v}^-}$,

one point of the lattice of the data written with the lifts $\mathbf{v}^+$ and $\mathbf{v}^-$, which differ by the fibre translation $\mathbf{v}$; in $\mathfrak{H}_P$ it is the independence of $e_{t\mathbf{v}^+}$ from the lift.

(vi) A bijection as stated carries the readings, the absent letters and their products, the shares, the pins, the kernel and the seam words of the children of one block to the other's, and by Theorem 48.18(iii) the modules of the children; so it carries $\mathfrak{M}_P$, $\mathfrak{J}_P$, $\mathfrak{J}_P^{\boldsymbol{\xi}}$, $\mathfrak{F}_P^{\boldsymbol{\xi}}$ and $\mathfrak{H}_P$, and the data read. By Proposition 48.10(ii) the input $e^{-\langle \boldsymbol{\xi},\mathbf{n}\rangle}\prod_P \mathcal{T}_P^{q_P}$ of a node of $\Sigma$ generates a submodule of the image of $\bigotimes_P \operatorname{Sym}^{q_P}$ of the modules of its transforms, tensored with that of the exponential, of rank one; by (iv) each of these is the image of $\mathfrak{H}_P^{\partial}$, read at the data of $F$ and restricted to the section, and so of the module presented by (iv). A sum of the images of modules is the image of their sum, so $N_r$ is the image under the values of the sum of the modules presented by (iv) for the distinct blocks and of those of the exponentials, presented by their connection matrices, and its group is the image of the group of that module (Proposition 48.10).

(vii) *The display.* The proof of Theorem 48.14(i) holds with any exponents $\mathbf{a}$ of the scales. Integrating the exponentials of the readings and of the constant parts against $\boldsymbol{\xi}^{\mathbf{a}}\,d\boldsymbol{\xi}/\boldsymbol{\xi}$ gives $\prod_{\Sigma_S}\Gamma(a_{LD})(V_{LD}^S/A_L)^{-a_{LD}}$, read at the seam points. With the factors $A_L^{-s_L}$ of the transforms, the integrand becomes that of $W_S^\sharp$: $A_L$ is the product of $H_L^S=\prod_l h_{lL}$ and the $\alpha_z^{u_{z,L}}$, at the exponent $s_L-\sum_{(L,D)\in\Sigma_S}a_{LD}=w_L$, the powers of the $\alpha_z$ going to the masses of the own marks. The common scaling multiplies it by $e^{tc}$, with $c$ the imbalance of $W_S^\sharp$, which vanishes on its balance; so the integral over the scale quotient and the directions of $S$ is its period there. On the pins of the seam the totals at the stations off $\Sigma_S$ make up $w_L$, and the letters there are $A_L$ times the constant parts, which is the left side of Theorem 48.14(i).

*The section.* Let the coordinates $\mathbf{y}$ be a basis of the monomials that the scaling keeps, so that $\mathbf{y}^{\mathbf{k}}=\boldsymbol{\xi}^{\mu_{\mathbf{k}}}$ with $\mu_{\mathbf{k}}$ integral and orthogonal to the degrees $\delta_{LD}$. Then $\boldsymbol{\xi}^{\mathbf{a}}\,d\boldsymbol{\xi}/\boldsymbol{\xi}$ on the quotient is a nonzero constant times $\mathbf{y}^{\boldsymbol{\nu}(\mathbf{a})}\,d\mathbf{y}/\mathbf{y}$, the constant being 1 for a monomial section with the coefficients 1 and $C^{\mathbf{a}}$, fixed on the balance, for constant coefficients $C$; the index $\boldsymbol{\nu}(\mathbf{a})$ is linear in $\mathbf{a}$, and $\mathbf{y}^{\mathbf{k}}$ moves it by $\mathbf{k}$. The imbalance of $W_S^\sharp$ is its total mass less $\sum_{\Sigma_S}a_{LD}\deg V_{LD}^S+\sum_L w_L\deg H_L^S$, with $\deg V_{LD}^S=\deg H_L^S+\sum_z u_{z,L}+\delta_{LD}$; moving $\mathbf{a}$ by $\mu_{\mathbf{k}}$, $\mathbf{w}$ by minus its totals and every $m_z^\sharp$ by $\sum_L u_{z,L}T_L(\mu_{\mathbf{k}})$ changes it by $-\sum\mu_{\mathbf{k},LD}\delta_{LD}=0$; so this move is a point $\lambda_{\mathbf{k}}$ of its lattice.

*The elements.* The data of $\phi$ and $\mathbf{a}$ on the balance of $W_S^\sharp$ give its data through $w_L=s_L-\sum_D a_{LD}$ and $m_z^\sharp=m_z-\sum_L u_{z,L}w_L$, and every point of its data space comes so, with $s_L=w_L+\sum_D a_{LD}$ and $m_z=m_z^\sharp+\sum_L u_{z,L}w_L$; at a root not seen by its scales $\boldsymbol{\delta}=0$, and the balance of $W_S^\sharp$ is that of $F$, on which the data of $\phi$ lie. Move the data along a positive direction of $W_S^\sharp$ (Theorem 48.3(i)): for all large multiples every moved point of $E^\sharp$ lies in the tube of $W_S^\sharp$, as in Theorem 48.5(iii), and there, by the display, the Mellin transform of every $\mathbf{y}^{\mathbf{k}}\vartheta^{\mathbf{b}'}\phi$, $\mathbf{b}'\le\mathbf{b}$, converges absolutely on a common open set of indices: a $\vartheta$ brings down, under the transforms, a polynomial in the $\xi_{LD}\widehat{\beta}_{P,LD}$, whose moments in the scales are Gamma integrals of the same kind, converging where those of $\phi$ at the moved exponents do. In each coordinate $\int_0^\infty y^{\nu-1}y^k\vartheta^b G\,dy=(-\nu-k)^b\int_0^\infty y^{\nu+k-1}G\,dy$, by integration by parts, the boundary terms vanishing, an integrable function with an integrable derivative tending to 0 at both ends (proof of Lemma 48.8(ii)). So, by Fubini's theorem, the Mellin transform of $E$ at $\boldsymbol{\nu}(\mathbf{a})$ is the constant of the section times $\prod\Gamma(a_{LD})\,\langle E^\sharp\rangle$, on an open set of the data of $W_S^\sharp$. With $\mathbf{y}=e^{\mathbf{v}}$ it is the Fourier transform, on a vertical plane, of the integrable function $E(e^{\mathbf{v}})e^{\langle \operatorname{Re}\boldsymbol{\nu},\mathbf{v}\rangle}$ of $\mathbf{v}\in\mathbb{R}^r$, which it determines. So $E$ vanishes on an open set of its data, and so identically, exactly when $\langle E^\sharp\rangle$ vanishes on an open set of the data of $W_S^\sharp$, and so on all of them (Theorem 48.3(ii)).

*The closings.* The seam letters $V_{LD}^S$ are the letters of the room children of the stations of $S$, tables on its marks, times the constants of the normalization (Definition 48.11). The absent

letter $h_{lL}$ is the letter of the room child of the absent station of $S_l$, a table on $S_l$, and $H_L^S$ is the letter, of the kind of $L$, of the direct sum of these tables over the children; on the marks of $S$ the own marks are loops in it when $L$ is a letter $K_j$, the bases avoiding them, and coloops when $L$ is a letter $\Psi_j$, the bases containing them, so that its cobasis letter $\sum_T \theta_T \alpha^{S\setminus T}$ is the product of those of the children. So $W_S^\sharp$ is a family of letters of tables, and Theorem 48.23 applies to it, Proposition 47.12 with it; and an element of its module has the value zero exactly when it is 0 (Proposition 47.15(iii)). Below the root the masses of the marks of $S$ and the exponents $s_L$ are free in the data of $F$, the marks outside $S$ taking up its balance; at a root not seen by its scales the data of $\phi$ are those of $F$; and at a root without children $W_S^\sharp$ has no letter $H_L^S$, and the data of $F$ give the masses $m_z - \sum_L s_L u_{z,L}$ and $\mathbf{a}$ freely, as in the proof of Definition 48.19(i). In these cases the data of $F$ and $\mathbf{a}$ reach all the data of $W_S^\sharp$. At a root seen by its scales with children the balance of $F$, $\sum_e m_e = \sum_L s_L(\deg A_L + \delta_L)$ with $\delta_L = 0$ off $\Sigma_S$, and that of $W_S^\sharp$ give $\sum_L \delta_L w_L = 0$ on the data reached.

*The hyperplane.* There the data of $F$ and $\mathbf{a}$ reach every point of the hyperplane $\mathbb{E}_S^\sharp$ that this cuts out in the data of $W_S^\sharp$: a point of it gives $s_L = w_L + \sum_D a_{LD}$ and $m_z = m_z^\sharp + \sum_L u_{z,L} w_L$, which satisfy the balance of $F$ exactly when $\sum_L \delta_L w_L = 0$. The letters of $W_S^\sharp$ other than the $H_L^S$ with $\delta_L \neq 0$ include the letters $V_L^S = L$ of $F$ at the $L$ of $\Sigma_S$, and, at the other letters, at which no block is present and $V_L^S = n_L A_L$, the $H_L^S = n_L^{-1} \prod_z \alpha_z^{-u_{z,L}} L$; their differences of exponents are those of the letters of $F$, which generate those of one molecule, so these letters form a family of letters of tables with the one active molecule of $W_S^\sharp$ (Theorem 48.23(i)). The equation of $\mathbb{E}_S^\sharp$ involves the exponents of the $H_L^S$ with $\delta_L \neq 0$ alone, so the points of Proposition 47.14(i) with these exponents 0, the others positive and the native gauge $(1, \dots, 1)$ lie in $\mathbb{E}_S^\sharp$. The real points of $\mathbb{D}_{W_S^\sharp}$ with these exponents 0, the others positive and every wall positive form an open cone in the real points of the rational subspace on which these exponents vanish, not empty since it contains those points; an integral point of it is a native direction $\mathbf{d}_1$ of $\mathbb{E}_S^\sharp$, and its native gauge, an isolated nondegenerate common zero on the orthant of the defects of $\mathbf{d}_1$, which are rational functions over the field of the weights, has coordinates algebraic over that field (Proposition 47.14(i)). When the weights are rational an integral multiple of the point with the native gauge $(1, \dots, 1)$ is such a $\mathbf{d}_1$, with that rational native gauge. The argument of *The elements*, with the data moved along $\mathbf{d}_1$, whose walls and whose exponents $a_{LD}$, $(L, D) \in \Sigma_S$, of the seam letters are positive, so that the Gamma integrals in the scales converge at the moved exponents, in the place of a positive direction, holds at each point of the data of $F$: $E$, read at the data of $F$, vanishes exactly when $\langle E^\sharp \rangle$ vanishes on an open subset of $\mathbb{E}_S^\sharp$, and so on $\mathbb{E}_S^\sharp$, which Proposition 47.14(iii), (iv) decides. The equation of $\mathbb{E}_S^\sharp$ involves the exponents alone, so $\mathbb{E}_S^\sharp$ consists of the data of $W_S^\sharp$ whose exponents lie on a hyperplane, with every mass that its data space allows at them, and it meets the tube of $W_S^\sharp$ at $\mathbf{d}_1$, whose walls are positive; so $\langle E^\sharp \rangle$ vanishes on $\mathbb{E}_S^\sharp$ exactly when $E^\sharp$ is 0 in the module of $W_S^\sharp$ on $\mathbb{E}_S^\sharp$ (Proposition 47.15(ii)). □

**Proposition 48.21 (The question at every data).** *As stated.*

*Proof.* (i) A letter with the exponent 0 does not enter the form, so the form is the master form $\alpha^{\mathbf{m}} \prod_\ell L_\ell^{-w_\ell}\, dy$ of the family $W$ of positive letters formed by the letters $L_\ell$ of $F$ at their exponents $w_\ell$ (Definition 48.1). Its space of directions is that of Theorem 48.23(i), the vectors of zero sum on every active molecule. So $\mathbf{m} - \sum_\ell w_\ell \mathbf{t}_\ell$ lies in it exactly when $m_M = \sum_\ell w_\ell |\mathbf{t}_\ell \cap M|$ for every active molecule $M$; an exponent of $K_j$ meets $M$ in $r_j(M)$ marks and one of $\Psi_j$ in $|M| - r_j(M)$ (7.10), so this is $\kappa_M = 0$, the balance of $M$, the walls of $M$ being $d_j(M) = r_j(M)$ and $d_j^*(M) = |M| - r_j(M)$.

With one active molecule this is the balance $\kappa_{\mathcal{E}} = 0$, so that $\mathbb{D}_W$ is the balance hyperplane in the masses and the exponents of the letters, and $W$ is the family $W_F$ of Definition 48.17.

(ii) The sum $|\alpha_{M_i}|$ is $K$ of the table of rank one on $\mathcal{E}$ whose bases are the marks of $M_i$, with the weights 1, every other mark a loop; it links the marks of $M_i$ and no others, so $W^+$ has the active molecules of $F$. The balance of $M_i$ in $W^+$ is $\kappa_{M_i} - \kappa_{M_i} \cdot 1 = 0$, the new letter having the degree 1 on $M_i$ and 0 elsewhere, so $(\mathbf{m}; \mathbf{s}, \mathbf{t}) \mapsto (\mathbf{m}; \mathbf{s}, \mathbf{t}, (\kappa_{M_i})_i)$ maps the data of $F$ onto $\mathbb{D}_{W^+}$. On the orthant put $\alpha_{M_i} = \rho_i u_i$, $\rho_i = |\alpha_{M_i}|$, $u_i \in \Delta_{M_i}$, so that $\alpha_{M_i}^{\mathbf{m}_{M_i}-\mathbf{1}} d\alpha_{M_i} = \rho_i^{m_{M_i}-1} u_i^{\mathbf{m}_{M_i}-\mathbf{1}} d\rho_i \, d\omega_{M_i}(u_i)$ (2.7). Every basis of an active table meets $M_i$ in $r_j(M_i)$ marks (7.10), so $K_j(\rho_1 u_1, \dots, \rho_q u_q) = \prod_i \rho_i^{r_j(M_i)} K_j(u)$ and $\Psi_j(\rho_1 u_1, \dots, \rho_q u_q) = \prod_i \rho_i^{|M_i| - r_j(M_i)} \Psi_j(u)$, whatever the weights; and $e^{-|\alpha|} = \prod_i e^{-\rho_i}$. The powers of $\rho_i$ collect to $\rho_i^{\kappa_{M_i}-1}$, the walls of $M_i$ being $d_j(M_i) = r_j(M_i)$ and $d_j^*(M_i) = |M_i| - r_j(M_i)$. At real data Tonelli gives $J_F = \prod_i \int_0^\infty \rho_i^{\kappa_{M_i}-1} e^{-\rho_i} d\rho_i$ times the integral over $\prod_i \Delta_{M_i}$, in $[0, \infty]$; the first factors are $\Gamma(\kappa_{M_i})$, finite for $\kappa_{M_i} > 0$, and the integral over the shapes is finite when the walls $\kappa_U$, $U$ a nonempty proper subset of an active molecule, are positive, by the sector decomposition of the proof of 8.7 taken on every factor $\Delta_{M_i}$, the sectors being the products of the sectors of the orders of the active molecules (Theorem 48.23(i)). At complex data the same holds with the moduli, and both sides continue meromorphically. The integrand of $W^+$, the form times $\prod_i |\alpha_{M_i}|^{-\kappa_{M_i}}$, is homogeneous of the degree 0 in the marks of every active molecule, so its integral on the charts $y_{e_i} = 0$ is its integral over $\prod_i \Delta_{M_i}$, where every $|\alpha_{M_i}|$ is 1 (8.7, one molecule at a time): that integral is $P_{W^+}$. For $q = 1$, $P_{W^+}$ is the integral of the form over $\Delta_{\mathcal{E}}$, which is $P_F$ at every data, and $J_F = \Gamma(\kappa_{\mathcal{E}}) P_F$ is 8.7.

(iii) The form is homogeneous of the degree $\kappa_{\mathcal{E}}$ under the common scaling, and a completion $\alpha^{-\kappa_{\mathcal{E}} w} \omega$ is projective (16.4); for $\kappa_{\mathcal{E}} \neq 0$ the form is not. By 16.8(ii) the data $(\lambda \mathbf{m}; \mathbf{s}, \mathbf{t})$ lie on the balance wall exactly when $\lambda \mathbf{m} = \boldsymbol{\tau}$, that is when $\lambda m_{\mathcal{E}} = \sum_j (s_j r_j + t_j r_j^*)$, the degree of the word; so $\lambda$ is the displayed rational function. The pseudointegral is $\mathrm{CT}_{\eta=0} P_F(\boldsymbol{\tau} + \eta \mathbf{m}; \mathbf{s}, \mathbf{t})$ by 16.10, $P_F$ being meromorphic in the masses with its poles on the hyperplanes $\kappa_S \in \{0, -1, \dots\}$, $S$ proper; off them it is $P_F(\boldsymbol{\tau}; \mathbf{s}, \mathbf{t})$, which with a native gauge is a convergent integral (16.15) and otherwise its continuation (16.16). Off the balance wall $P_F$ is the period of $W^+$ by (ii), so the constant term is that of the period of $W^+$ along the line $\boldsymbol{\tau} + \mathbb{C}\mathbf{m}$ of $\mathbb{D}_{W^+}$, the image of the ray of the masses. For $q \geq 2$, a union of active molecules has $c_j = 0$ in every active table (7.10), so every basis of table $j$ meets $M_i$ in the same number $r_{j,i}$ of marks; $K_j$ is homogeneous of the degree $r_{j,i}$ and $\Psi_j$ of the degree $r_{j,i}^*$ in the marks of $M_i$, whatever the weights, and the balance of $M_i$ (Theorem 48.23(i)) is $m_{M_i} = \sum_j (s_j r_{j,i} + t_j r_{j,i}^*)$, which $\boldsymbol{\tau}$ satisfies. The data $\boldsymbol{\tau} + \sum_i \eta_i \mathbf{m}_{M_i}$ are data of $F$, so they give an affine subspace of $\mathbb{D}_{W^+}$ of the dimension $q$ by (ii), on which the period of $W^+$ is meromorphic with its poles on the hyperplanes $\kappa_U \in \{0, -1, \dots\}$, $U$ a nonempty proper subset of an active molecule (Theorem 48.3(ii) with Theorem 48.23(i)): the letter $|\alpha_{M_{i'}}|$ adds $\kappa_{M_{i'}} \min_{e \in M_{i'}} [e \in U] = 0$ to such a wall, so these walls are those of $F$. A wall $\kappa_U$ with $U \subseteq M_i$ is linear and moves along that subspace by $\eta_i m_U$ alone, so near $\boldsymbol{\eta} = 0$ the period is a holomorphic function divided by a product of powers of the $\eta_i$, and its constant term, the coefficient of $\boldsymbol{\eta}^{\mathbf{0}}$ of its Laurent expansion, is the iterated constant term in any order. At $\boldsymbol{\eta} = 0$ every $\kappa_{M_i}$ is 0, the letters $|\alpha_{M_i}|$ have the exponent 0, and the period of $W^+$ is that of $W$, the reduced period of (iv), when $\boldsymbol{\tau}$ lies on no pole hyperplane. With factoring weights the period of $W^+$ is the product of those of the molecules, the integrand and the product of simplices factoring (Proposition 7.9), each a function of its own $\eta_i$, so the constant term is the product of the constant terms, the pseudointegrals of the molecules at their targets (16.10).

(iv) A union $S$ of active molecules has $c_j(S) = 0$ in every active table (7.10), so 8.16 gives $\kappa_S + \kappa_{\mathcal{E} \setminus S} = \kappa_{\mathcal{E}}$, which is 0 on the balance wall: one of the two walls has real part at most 0,

and the period, which converges only when every proper wall has positive real part (8.7), diverges, as does the integral of the form over the orthant along every ray of the common scaling. 16.17 gives the vanishing of $P_F$ on the balance wall. The proof of 16.12 uses of the letters only that $K_j(\rho_1 u_1, \dots, \rho_q u_q) = \prod_i \rho_i^{r_j(M_i)} K_j(u)$ and $\Psi_j(\rho_1 u_1, \dots, \rho_q u_q) = \prod_i \rho_i^{|M_i| - r_j(M_i)} \Psi_j(u)$, which hold for every active table whatever its weights, every basis meeting $M_i$ in $r_j(M_i)$ marks (7.10), and that the integral over the shapes is a smooth function of the imaginary imbalances: here $f(\boldsymbol{\eta}')$ is the integral over $\prod_i \Delta_{M_i}$ at the masses $\mathbf{m}_i^0 + i\eta_i w_i$ on the parts. That integral is holomorphic in the masses where every wall $\kappa_U$, $U$ a nonempty proper subset of an active molecule, has positive real part, by the sector decomposition of the proof of 8.7 taken on every factor $\Delta_{M_i}$, the sectors being the products of the sectors of the orders of the active molecules (Theorem 48.23(i)); and at $\mathbf{m}^0 + i\boldsymbol{\eta} w$ these walls have the real parts of those at $\mathbf{m}^0$, positive. So $f$ is smooth. So its proof gives (i)–(iii) there, with $f(0) = P_{V_F}(\mathbf{m}^0; \mathbf{s}, \mathbf{t})$. On the data space of $W$ the integrand is homogeneous of the degree 0 in the marks of every active molecule, so its integral over $\prod_i \Delta_{M_i}$ is its integral on the charts $y_{e_i} = 0$, one mark $e_i$ in each active molecule (8.7, one molecule at a time), whose measure gives the lattice of $V_W$ the covolume one: the value $\langle 1 \rangle$ of Definition 48.1(v). With factoring weights $K_j = \prod_i K_{j,i}$ and $\Psi_j = \prod_i \Psi_{j,i}$ (7.9), and the integral is the product.

(v) With every letter of every active table a letter this is 8.7 and 47.2; in general it is Lemma 48.2(iii) and Theorem 48.3(i), (ii), with the fan and the walls of Theorem 48.23(i). □

**Definition 48.22 (A factorization of the weights; the molecule families).** As stated.

*Proof.* A circuit of a direct sum is a circuit of one summand, a minimal dependent set being dependent in the summand that contains its intersection with some dependent part; so two marks are linked in an active table exactly when they are linked in its restriction to the active molecule that contains them, and the marks of $M_i$, one class of the equivalence generated by linking in the active tables, are one class of the equivalence generated by linking in the tables $\mathcal{B}_j|M_i$: one molecule of $F^{(i)}$. Expanding the products, $\sum_r \prod_i K_{j,r}^{(i)} = \sum_{T_1,\dots,T_q} \left( \sum_r \prod_i \theta_{j,r}^{(i)}(T_i) \right) \prod_i \alpha^{T_i} = \sum_T \theta_{j,T} \alpha^T = K_j$, the bases of table $j$ being the unions $T_1 \sqcup \cdots \sqcup T_q$ (7.9), and likewise $\Psi_j$, with $\alpha^{\mathcal{E} \setminus T} = \prod_i \alpha^{M_i \setminus T_i}$. A factorization with one element in every $R_j$ is a factoring $\theta_{j,T} = \prod_i \theta_j^{(i)}(T \cap M_i)$, and conversely. An active molecule $\{e\}$ of one mark is linked to no mark, so $e$ is a loop or a coloop in every active table, the restrictions $\mathcal{B}_j|\{e\}$ have the one basis $\varnothing$ or $\{e\}$, and the letters of its molecule family are monomials; its space of directions is a point, and its module has the dimension one, as in the proof of Theorem 48.18(v). □

**Theorem 48.23 (The closings for every family of letters of tables).** *As stated.*

*Proof.* (i) Let $e, e'$ be marks linked in an active table $j$. Some basis $T$ has $e \in T$, $e' \notin T$ and $T' = (T \setminus \{e\}) \cup \{e'\}$ a basis (7.7), so $\mathbf{1}_T - \mathbf{1}_{T'} = \mathbf{e}_e - \mathbf{e}_{e'}$ is a difference of two exponents of $K_j$, and $\mathbf{1}_{\mathcal{E} \setminus T} - \mathbf{1}_{\mathcal{E} \setminus T'} = \mathbf{e}_{e'} - \mathbf{e}_e$ one of $\Psi_j$; whichever of them is in $W$ puts $\mathbf{e}_e - \mathbf{e}_{e'}$ in the lattice generated by the differences. These vectors generate the integral vectors of zero sum on every active molecule, the active molecules being the classes of the equivalence generated by linking in the active tables. Conversely every difference of two exponents of a letter has zero sum on every active molecule, every basis of an active table meeting it in $r_j(M)$ marks (7.10). So the differences generate the integral vectors of zero sum on every active molecule, a saturated lattice, which is therefore $M_W = \vec{\mathcal{N}}_W \cap \mathbb{Z}^{\mathcal{E}}$, with $\vec{\mathcal{N}}_W$ the real vectors of zero sum on every active molecule. Its orthogonal is spanned by the indicators of the active molecules, and $\alpha^u = 1$ for every $u \in M_W$ exactly when $\alpha_e = \alpha_{e'}$ for marks $e, e'$ of one active molecule.

For a tuple $\pi = (\pi_M)_M$ of orders of the active molecules let $C_\pi \subseteq V_W$ be the cone of the $v$ whose coordinates do not increase along $\pi_M$ on every $M$, a cone of the dimension $r_W$ with the generators $\mathbf{1}_U$, $U$ a head of some $\pi_M$, nonempty and proper in $M$, and $|\det| = 1$; these cones cover $V_W$ with disjoint interiors. The bases of an active table are the unions of bases of its restrictions to its own molecules, each inside one active molecule (7.9), so on $C_\pi$ the least value of $\langle \mathbf{1}_T, v\rangle$ over its bases, which moves by a constant when $v$ moves by an indicator of an active molecule, is taken at the basis that the greedy rule builds from the marks of least coordinates, the same at every point of $C_\pi$, and the least value of $\langle \mathbf{1}_{\mathcal{E}\setminus T}, v\rangle$ at the basis that it builds from the marks of largest coordinates. So the cones form a fan of $W$ (Lemma 48.2(i)), and the wall of $\mathbf{1}_U$ (Lemma 48.2(ii)) is $m_U$ less the sum, over the letters, of the exponent times the least value of $\langle \mathbf{t}, \mathbf{1}_U\rangle$ over the exponents $\mathbf{t}$ of the letter, which is $d_j(U)$ for $K_j$ and $d_j^*(U)$ for $\Psi_j$: the wall $\kappa_U$.

(ii) A line $\mathbf{d}_0 + \mathbb{C}\mathbf{d}_1$ of $\mathbb{D}_W$, $\mathbf{d}_0$ integral and $\mathbf{d}_1$ a positive direction, carries the ray with the stride $f = f_{\mathbf{d}_1}$ and the origin form $\omega_{\mathbf{d}_0}$, its insertions, its relation space and its module the restrictions of those of $W$ (Definition 47.3, Proposition 47.5, read with Theorem 48.3(iii)); the unit $\omega_{\mathbf{d}_0}/dy$ carries it to the ray with the origin 0, as in the first paragraph of the proof of Theorem 39.57. The proof of Theorem 47.11 reduces it, ray by ray, to Theorem 39.57, and adds the density of the positive directions in the open cone of the real data with positive exponents and positive walls, which for $W$ is Theorem 48.3(i). Theorem 39.57 rests on Proposition 39.34, Theorem 38.14(i) and Theorem 38.6(i), and its proof uses one molecule at two places; Theorem 44.16 rests on Theorem 39.57 and on the closings being a solution of the shift. Each holds for $W$:

*The closings.* For integral $\mathbf{d} \in \mathbb{D}_W$ and an insertion $g$ such that $g\,\omega_{\mathbf{d}}/dy$ is a Laurent polynomial, a combination of the $\alpha^u$, $u \in M_W$, the closing $\mathsf{C}(g)(\mathbf{d})$ is its coefficient of $\alpha^{\mathbf{0}}$, its mean over the compact torus of $X_W$. The constant term of $\partial_{y_e}$ of a Laurent polynomial is 0, so $\mathsf{C}\big((c_e(\mathbf{d}) + \partial_{y_e})g\big)(\mathbf{d}) = 0$, and $\mathsf{C}(f_\lambda g)(\mathbf{d}) = \mathsf{C}(g)(\mathbf{d}+\lambda)$: the proof of Proposition 47.10 holds as it stands, its use of Proposition 39.35(ii) at $j = 0$ being covered by the next paragraph.

*The pencil.* For a positive direction $\mathbf{d}_1$ the stride $f = f_{\mathbf{d}_1}$ is positive on the orthant, $\log f(e^y) = \langle \mathbf{m}_1, y\rangle - \sum_\ell w_{1,\ell} \log L_\ell(e^y)$, and the Hessian of $\log L_\ell(e^y)$ is the covariance of the exponent under the law proportional to $c_{\ell,\mathbf{t}} e^{\langle \mathbf{t}, y\rangle}$ on the support of $L_\ell$ (7.16), every exponent having positive weight at a real point. A direction $u'$ in the kernel of the Hessian of $\log f$ has $\langle u', \mathbf{t} - \mathbf{t}'\rangle = 0$ for any two exponents of every letter, $w_{1,\ell} \geq 1$, so $u' \in \vec{\mathcal{N}}_W^\perp$ by (i), which is 0 in $V_W$: $\log f$ is strictly concave on $V_W$. On every cone of the fan of (i), $f = \prod_k q_k^{\kappa_{v_k}(\mathbf{d}_1)} \prod_\ell k_{\ell,\pi}(q)^{-w_{1,\ell}}$ with $k_{\ell,\pi}$ at least its constant term, which is positive (Lemma 48.2(iii)), and every $\kappa_{v_k}(\mathbf{d}_1) > 0$; so $f \leq C' e^{-c'\|y\|}$ on $V_W$, $f$ has one largest point, the native gauge $\alpha^\star$ of the direction, and it is nondegenerate. This is Theorem 38.14(i) for $W$, and Theorem 38.6(i), the expansion at a nondegenerate largest point with the rest of $V_W$ contributing $O((f^\star - \delta)^k)$, holds for $W$ with $r_W$ in the place of $n-1$.

*The top principal parts.* In the proof of Proposition 39.34 the sector formula is replaced by the cone formula of Lemma 48.2(iii) for the fan of (i), whose terms 8.4 continues; for a monomial $\alpha^u$, $u \in M_W \setminus \{0\}$, a mark $e$ with $u_e \neq 0$ and the relation of Theorem 48.3(ii) at the data $\epsilon\mathbf{d}_1$ give $u_e \langle \alpha^u\rangle_\epsilon = -\epsilon \langle c_e^{(1)} \alpha^u\rangle_\epsilon$; and $\epsilon^{r_W} \langle 1\rangle_\epsilon \to V_1$ as $\epsilon \downarrow 0$ by the cone formula at $\epsilon\mathbf{d}_1$, every $k_{\ell,\pi}(q)^{-\epsilon w_{1,\ell}}$ tending to 1 boundedly and $\int_0^1 q^{\epsilon\kappa - 1} dq = 1/(\epsilon\kappa)$, with $V_1 = \sum_\pi \prod_k \kappa_{1,U_{\pi,k}}^{-1}$ over the tuples of orders, the product over the active molecules of their sums. The proof of Proposition 39.35(ii) uses the same cone formula and the same relations, and holds for $W$ with these replacements.

*The torus.* On the torus of $X_W$ through $\alpha^\star$, $|L_\ell(\alpha^\star e^{i\theta})| \leq L_\ell(\alpha^\star)$, the coefficients being positive, with equality only when $\langle \theta, \mathbf{t} - \mathbf{t}'\rangle \in 2\pi\mathbb{Z}$ for any two exponents of $L_\ell$; for all the letters this says, by (i), that $\theta$ is constant modulo $2\pi$ on every active molecule, which is the identity of the torus of $X_W$. So $|f^{-1}(\alpha^\star e^{i\theta})| < 1/f^\star$ off a neighbourhood of the identity, and the expansion of the closings in the proof of Theorem 39.57 holds, its Gaussian moments taken on $V_W$.

With these, the proofs of Theorems 39.57, 44.16 and 47.11 hold for $W$ as written, with $r_W$ in the place of $n-1$. The generators of Theorem 48.3(iv) and their negatives are a basis of $\mathbb{D}_W(\mathbb{Z})$ and its negatives, $(u;\mathbf{a}) - \sum_\ell a_\ell(\mathbf{t}_\ell;\mathbf{e}_\ell)$ being $(u - \sum_\ell a_\ell\mathbf{t}_\ell;\mathbf{0})$ with $u - \sum_\ell a_\ell\mathbf{t}_\ell \in M_W$, and the proof of Proposition 47.12 holds as written, for the module and for its quotients by submodules of elements of value zero, the values factoring through them. When every active molecule has one mark, $V_W$ is a point, the letters are monomials, the module is spanned by [1] (Theorem 48.3(iii)), and its one element [1] has the value $P_W \neq 0$. □

**Theorem 48.24 (Several active molecules through their molecule families).** *As stated.*

*Proof.* (i) With $\mathbf{J}^2 = b\mathbf{J}$ for the matrix of ones of size $b$, $(\mathbf{I}+\delta\mathbf{J})^{-1} = \mathbf{I} - \frac{\delta}{1+\delta b}\mathbf{J}$, and $\mathbf{e}_{T_i} + \delta\mathbf{1} = (\mathbf{I}+\delta\mathbf{J})\mathbf{e}_{T_i}$. So $\theta_j = \bigotimes_i(\mathbf{I}+\delta\mathbf{J}_{j,i})\, c_j^{(\delta)} = \sum_{T'} c_{j,T'}^{(\delta)} \bigotimes_i(\mathbf{e}_{T_i'} + \delta\mathbf{1})$, the vectors on the bases of table $j$ being the tensors of vectors on the bases of its restrictions (7.9). The entries of $c_j^{(\delta)}$ are those of $\theta_j$ plus $O(\delta)$, and $\theta_j > 0$, so they are positive for small $\delta$; then every $\theta_{j,T'}^{(i)}$ is positive, its entries being $\delta$ or $1+\delta$ times a positive number, and $\theta_{j,T} = \sum_{T'} \prod_i \theta_{j,T'}^{(i)}(T_i)$.

(ii) For a letter $K_j$ with $|R_j| \geq 2$ and a point of the orthant, Corollary 4.4 with $X_r = \prod_i K_{j,r}^{(i)}(\alpha) > 0$ gives

$$K_j^{-s_j} = \int_{\sum_r z_{j,r}=s_j} \Gamma_{|R_j|}\begin{bmatrix}\mathbf{z}_j\\ s_j\end{bmatrix} \prod_{r\in R_j}\prod_i K_{j,r}^{(i)}(\alpha)^{-z_{j,r}}$$

through $\mathbf{c}_j$, converging absolutely, and likewise for $\Psi_j$; with $|R_j| = 1$, $K_j^{-s_j} = \prod_i K_{j,r}^{(i)}(\alpha)^{-s_j}$. Multiplying, the master form of $W$ is the fibre integral of the kernel times

$$\prod_{i=1}^{q}\Big(\alpha_{M_i}^{\mathbf{m}_{M_i}} \prod_{j,r} K_{j,r}^{(i)}(\alpha_{M_i})^{-z_{j,r}}\Psi_{j,r}^{(i)}(\alpha_{M_i})^{-z_{j,r}^*}\Big)\,dy,$$

the factor $i$ depending only on the marks of $M_i$. On the pins its degree in those marks is $m_{M_i} - \sum_{j,r}\big(z_{j,r}r_j(M_i) + z_{j,r}^*(|M_i| - r_j(M_i))\big) = \kappa_{M_i} = 0$ on $\mathbb{D}_W$: the data of $F^{(i)}$ lie on its balance wall. Its wall at $U \subseteq M_i$, nonempty and proper, is $m_U - \sum_{j,r}\big(z_{j,r}d_j(U) + z_{j,r}^* d_j^*(U)\big)$, the least numbers of marks of $U$ in a basis of $\mathcal{B}_j|M_i$ and in its complement in $M_i$ being those in a basis of $\mathcal{B}_j$ and in its complement, every basis being a union; on the pins it is $\kappa_U$. $V_W$ is the product of the spaces $\mathbb{R}^{M_i}/\mathbb{R}\mathbf{1}$, its measure that of the charts $y_{e_i} = 0$, and the integral of factor $i$ over its space is $P_{F^{(i)}}(\mathbf{m}_{M_i};\mathbf{z},\mathbf{z}^*)$ (Definition 48.1(v)). On the orthant $|X_r^{-z_r}| = X_r^{-c_r}$ and $|\alpha^{\mathbf{m}}| = \alpha^{\operatorname{Re}\mathbf{m}}$, so by Tonelli the double integral of the modulus, over $V_W$ and the fibre, is the product of the integrals over the fibre of the moduli of the stacked Gammas, finite by the decay of the Gamma function along vertical lines (Corollary 4.4), and of the periods $P_{F^{(i)}}(\operatorname{Re}\mathbf{m}_{M_i};\mathbf{c},\mathbf{c}^*)$, finite since their walls are the positive $\kappa_U(\operatorname{Re}\mathbf{d})$. So Fubini gives the identity at every such data.

The pins $\sum_r z_{j,r} = s_j$, $\sum_r z_{j,r}^* = t_j$ have independent rows and rational coefficients; the kernel is a quotient of Gamma functions of linear forms with integral values on the lattice; the maps $\iota_i(\mathbf{d},\mathbf{z}) = (\mathbf{m}_{M_i};\mathbf{z},\mathbf{z}^*)$ are linear, integral on the lattice, and onto the balance wall of $F^{(i)}$, a point of which is reached with the exponents of $F$ the sums of its exponents over $r$ and the masses of the other molecules balanced. A point admissible at $\mathbf{d}$ has every node in its tube, so every $\kappa_U(\operatorname{Re}\mathbf{d}) > 0$ and $\mathbf{d}$ is in the tube of $W$; a positive direction of $W$ (Theorem 48.3(i)), with the exponent of every letter split equally over $R_j$, is admissible, its walls $\kappa_U$ being positive whatever the shares. So this is a positive share reading (Definition 48.4). An integral point of $\mathbb{D}_W$ is the

image of the integral shares that put the whole exponent of each letter on one element of $R_j$; so the lattice maps onto $\mathbb{D}_W(\mathbb{Z})$, and Theorem 48.5 holds for the reading, its nodes the molecule families, isomorphic ones having one module (48.5(v)). With one element in every $R_j$ the pins fix the shares, the fibre is a point, the stacked Gammas are 1, and $P_{V_F} = \prod_i P_{F^{(i)}}$. The letters $|\alpha_{M_i}|$ of $W^+$ are letters of tables whose bases lie in $M_i$, which factor with one term, so they go unchanged to the molecule family of $M_i$, and the same proof holds for $P_{W^+}$. □

**Definition 48.25 (The procedure in all the data).** As stated.

*Proof.* Every step is a definition or a theorem cited in it, applied to data that its hypotheses allow. The letters, the active tables and the active molecules are read off the tables and the names of the exponents fixed at 0, and $W$ and $W^+$ are not empty, some exponent not being fixed at 0. The target, the Gamma factors and the certificates are those of Proposition 48.21; a factorization exists (Theorem 48.24(i)), and a molecule family is a family of letters of tables on one molecule, so the steps for one active molecule apply to it. The tree, the seam letters, the transfer tensors, the node words and the full nodes are those of Definitions 16.14, 29.9 and 48.11 for one molecule, and the blocks those of Definition 48.13. The sets $\Sigma$ with their partitions are finitely many, the root being full, and the work of each is known once the dimensions of (9) are. □

**Theorem 48.26 (The procedure in all the data reads every family).** *As stated.*

*Proof.* (i) The question is finite: the balance is one linear form, the target and the walls $\kappa_{M_i}$ are rational functions of the data (Proposition 48.21), the active molecules are the classes of the linking in the active tables, and the factorization of (5) is found at the first $k$ at which the coordinates of Theorem 48.24(i) are positive, which comes. The classification is finite: steps (1)–(5) of Definition 16.14 are, the seam letters and the transfer tensors are polynomials read off the tables (Definition 48.11), the absent stations, the absent exponents and the finest separated partitions are read off the seam letters (Definition 48.13), and the classes are found by finitely many bijections. The searches stop:

- for masters from a box, Proposition 47.13(iii), with Theorem 48.3(iv) for families of positive letters;
- for an accepted set on the shares, Theorem 48.18(iv), its certificates being identities on products of masters, which (9) gives with their shift matrices at the generators and their negatives;
- for an accepted set of shifts of a block, Theorem 48.20(iii), (iv);
- for the law of an element along a native direction of a subspace of the data, in the module on that subspace, Proposition 47.14(iv).

The work of every $\Sigma$ is a number once the dimensions of the kept classes are known (Theorem 48.15(iii)). Value zero takes finitely many closings, each the constant term of a Laurent polynomial, computed through the tree by finite sums (Theorem 48.18(v), Proposition 47.12, Theorem 48.23), along a direction found by finitely many evaluations of polynomials (Proposition 47.12(ii)); on the hyperplane $\mathbb{E}_S^\sharp$ of a root seen by its scales with children it takes finitely many coefficients of the expansion at the native gauge of a native direction (proof of Theorem 48.20(vii), *The hyperplane*), its coordinates algebraic over $\Bbbk$ and rational when the weights are rational, each a rational function formed from finitely many Taylor coefficients and Gaussian moments, up to the bound read off the law (Proposition 47.14(ii), (iv)). The word is assembled from these

(Theorem 48.15). Every operation is over $\Bbbk$ or over a field of rational functions over $\Bbbk$ (Theorems 47.4(iv) and 48.3(iii)), but for the expansions, which are over the finite extension $\Bbbk(\alpha^\star)$ of $\Bbbk$ generated by the native gauge, $\Bbbk$ itself when the weights are rational and the gauge is taken rational (Proposition 47.14(ii)).

(ii) Proposition 48.21.

(iii) For one active molecule, Theorem 48.18(iii), (iv), $V$ being the family $W_F$ of the letters of its tables, with $\Sigma$ the full nodes and the node modules of (9): the module presented at the root by an accepted set is the quotient of the module of $V$ by a submodule of elements of value zero, the module itself (Proposition 47.15), with masters among the $\varepsilon_\lambda$ and their shift matrices at the generators and their negatives. For several active molecules, Proposition 47.13: masters of the module itself. In both cases the values are well defined on the module presented and are carried by its matrices (Theorem 47.4(i), (v), with Theorem 48.3(iii)), and every insertion $f_\lambda$ is reduced by a path in the generators and their negatives. The certificates are those of Theorem 47.8(ii), of Proposition 47.13(ii) and of Theorem 48.18(iv).

(iv) Proposition 47.15: the module presented is the module of $V$, in which an element of value zero is 0, so its reduction on the masters decides; and Proposition 47.12 for it, with its closings computed through the tree (Theorem 48.18(v)) or as those of $V$ (Theorem 48.23(ii)); Theorem 48.20(vii), with Propositions 47.13 and 47.15 and Theorem 48.23, for the inputs over their data, and with Proposition 47.14 on the hyperplane $\mathbb{E}_S^\sharp$ for the input of a root seen by its scales with children over the data of $F$; the coefficients of the expansion are rational functions over the field generated by the weights, the coefficients of the element and the native gauge, over $\mathbb{Q}$ when the weights and the element are rational and the native gauge is the rational one of Proposition 47.14(i) (Proposition 47.14(ii)).

(v) Theorem 48.15 at such data, at complex data by (v) there, the modules $N_r$ presented from the modules of the transforms of the blocks (Theorem 48.20(vi), the children of the blocks of $\Sigma$ being full); Theorem 47.4(v), with Theorem 48.3(iii), carries the values of the masters to the lattice translates, and Theorems 47.2(ii) and 48.3(ii) give the continuation. For several active molecules, Theorem 48.24(ii), each molecule family read by (iii) and (iv) in all its data. Along the contour of the shares the shares have the positive real parts $c_{j,r}$, $c^*_{j,r}$, the data of a molecule family lie on its balance wall, and its walls are walls of $F$, positive at the real parts of the data of $F$ (Theorem 48.24(ii)); so the real parts of its data are real data on its balance wall with its exponents and its proper walls positive, at which it has a native gauge (the argument of Proposition 47.14(i)), and Theorem 48.15(v) reads it in words at the complex data of the contour.

(vi) Theorems 48.5(v), 48.18(iii) and 48.20(vi). □

## 50.48 Chapter 49: the family as inference

### 50.48.1 Samples of bases

Every statement of this section rests on one factorization: the chance of a sample of a family is the product over its draws of the chances of single bases, each of which factors over the patterns of a bunch, and the product over all the draws is $\theta_{\text{draws}}\alpha^{\mathbf{m}}/\prod_j K_j(\alpha)^{s_j}\Psi_j(\alpha)^{t_j}$.

**Proposition 49.3 (What the data see of the bases).** *As stated.*

*Proof.* (i) is a property of one table. For $e \in C$, $C \setminus e$ is independent by the minimality of $C$ and lies in a basis $T$, which does not contain $C$; so $T \cap C = C \setminus e$. Hence $r(C) = k - 1$, every $C \setminus e$ is

a pattern of $C$, and $C$, not independent, is not. For $k \geq 3$ two marks $e, f$ of $C$ lie in $C \setminus g$ for a third $g \in C$, hence in a common basis.

(ii) The level of a basis draw of table $j$ lies in the window $[d_j(S), r_j(S)]$ of $S$ (Theorem 7.13), and the colevel of a cobasis draw is $|S| - |T' \cap S| \in [|S| - r_j(S), |S| - d_j(S)] = [d_j^*(S), |S| - d_j(S)]$. Some basis reaches $r_j(S)$ and every basis has positive chance, so $p_j > 0$; the $s_j$ basis draws of table $j$ are independent, and none reaches $r_j(S)$ with the chance $(1 - p_j)^{s_j}$. The same holds at $d_j(S)$, and for the cobasis draws with $\mathbb{P}^*_{j,\alpha}$ and $t_j$. □

**Proposition 49.4 (The exact laws of a sample of bases).** *As stated.*

*Proof.* (i) With $p = T \cap B$ and the weights $\theta_{j,(p),T\setminus p} = \theta_{j,T}$ of the child (Definition 12.1),

$$\mathbb{P}_{j,\alpha}(T) = \frac{\theta_{j,T}\alpha^T}{K_j(\alpha)} = \frac{\alpha_B^p K_{j,(p)}(\alpha_R)}{K_j(\alpha)} \cdot \frac{\theta_{j,(p),T\setminus p}\, \alpha_R^{T\setminus p}}{K_{j,(p)}(\alpha_R)},$$

and, since $\alpha^{\mathcal{E}\setminus T} = \alpha_B^{B\setminus p}\alpha_R^{R\setminus(T\setminus p)}$,

$$\mathbb{P}^*_{j,\alpha}(T) = \frac{\alpha_B^{B\setminus p}\Psi_{j,(p)}(\alpha_R)}{\Psi_j(\alpha)} \cdot \frac{\theta_{j,(p),T\setminus p}\, \alpha_R^{R\setminus(T\setminus p)}}{\Psi_{j,(p)}(\alpha_R)},$$

the first factors the pattern probabilities (Theorem 12.8) and the second the basis and cobasis laws of the child at $\alpha_R$. The draws are independent, so the patterns of the basis draws of table $j$ are independent with the chances $\pi_{j,p}(\alpha)$ and their numbers $a_{j,p}$ form a multinomial vector of size $s_j$, likewise the $b_{j,p}$ of size $t_j$; and given all the patterns the room parts are independent with the laws of the children. A basis draw with pattern $p$ uses $e \in B$ exactly when $e \in p$, and a cobasis draw with pattern $p$ uses $e$, by missing it, exactly when $e \notin p$; so $m_e = \sum_j \left( \sum_{p \ni e} a_{j,p} + \sum_{p \not\ni e} b_{j,p} \right) = \mathrm{load}_e$ (Definition 12.24). In the room a basis draw uses the marks of $T \setminus p$, and a cobasis draw the marks of $R \setminus (T' \setminus p)$, which it misses: what the draws of the children use. The mean of a multinomial vector is its size times its chances, which at a native gauge is the native split of Definition 12.25.

(ii) The chance of a sample is

$$\prod_{j,i} \frac{\theta_{j,T_{j,i}}\alpha^{T_{j,i}}}{K_j(\alpha)} \prod_{j,i} \frac{\theta_{j,T'_{j,i}}\alpha^{\mathcal{E}\setminus T'_{j,i}}}{\Psi_j(\alpha)} = \frac{\theta_{\mathrm{draws}}\, \alpha^{\mathbf{m}}}{\prod_j K_j(\alpha)^{s_j}\Psi_j(\alpha)^{t_j}}.$$

Expanding $\prod_j K_j^{s_j}\Psi_j^{t_j}$ draw by draw, every sample contributes the monomial $\theta_{\mathrm{draws}}\alpha^{\mathbf{m}}$ of its counts, so the samples with the counts $\mathbf{m}$ have the total chance $\Theta_F(\mathbf{m})\alpha^{\mathbf{m}}/\prod_j K_j^{s_j}\Psi_j^{t_j}$, $\Theta_F(\mathbf{m})$ the coefficient of $\alpha^{\mathbf{m}}$; the ratio is $\theta_{\mathrm{draws}}/\Theta_F(\mathbf{m})$.

(iii) $\kappa_S$ is the sum over the draws of $|T_{j,i} \cap S| - d_j(S)$ and $|S \setminus T'_{j,i}| - d_j^*(S)$ (Proposition 49.1(iii)), independent, with the generating functions $\mathsf{Q}^K_{j,S}(x;\alpha) = \mathbb{E}_{T\sim\mathbb{P}_{j,\alpha}}[x^{N_S - d_j(S)}]$ and $\mathsf{Q}^\Psi_{j,S}(x;\alpha)$ (Definition 12.10); so the generating function of $\kappa_S$ is the product. Each factor has degree $c_j(S)$ and every coefficient from degree 0 to $c_j(S)$ positive, every level of the window being reached by some basis (Theorem 7.13); so the values of $\kappa_S$ fill the stated range and the product at $x = 0$ is positive. The mean of a level excess is $\sum_{e\in S} R_{j,e}(\alpha) - d_j(S)$ and that of a colevel excess $\sum_{e\in S} \mathbb{P}^*_{j,\alpha}(e \notin T) - d_j^*(S) = \sum_{e\in S} R^*_{j,e}(\alpha) - d_j^*(S)$; the cumulants of order $k \geq 2$ of an excess are those of its level, $\sum_i \varkappa_k$ of the coins (Proposition 12.11(iv)), and summed with the sizes they are $\partial_S^k \Phi(\log\alpha)$ (Proposition 12.12). With a positive spectrum each excess is a sum of independent coins (Proposition 12.11(iv)), and $\kappa_S$ is the sum of all of them. □

**Proposition 49.5 (Tests that do not see the gauge).** *As stated.*

*Proof.* (i) is Proposition 49.4(ii); for lists $\theta_{\text{draws}} = 1$. (ii) $\mathbb{P}(A) = \sum_{\mathbf{m}} \mathbb{P}(\mathbf{m})\, \mathbb{P}(A \mid \mathbf{m}) \le \sum_{\mathbf{m}} \mathbb{P}(\mathbf{m})\, \varepsilon = \varepsilon$, the conditional chances being free of the gauge by (i). (iii) The samples of $F'$ with the counts $\mathbf{m}$ are those of $F$ whose every draw is a basis of its table in $F'$, with the same $\theta_{\text{draws}}$; by (i) their conditional chance under $F$ is $\Theta_{F'}(\mathbf{m})/\Theta_F(\mathbf{m})$. The rejection region, the union over the $\mathbf{m}$ with $p_{F'|F}(\mathbf{m}) \le \varepsilon$ of these samples, carries at most $\varepsilon \Theta_F(\mathbf{m})$ of the weight at every $\mathbf{m}$, and (ii) applies. A draw that is not a basis of its table in $F'$ has chance zero under $F'$ at every gauge. (iv) The true family is rejected with chance at most $\varepsilon$. □

**Corollary 49.6 (The evidence of a family).** *As stated.*

*Proof.* The centre of strengths $\boldsymbol{\sigma}_0$ is a question with every proper wall positive on a connected family. If every draw is a basis of its table of $F$, Proposition 49.2(ii) with the prior question $(\sigma_0 \mathbf{1}; \boldsymbol{\sigma}_0, \boldsymbol{\sigma}_0)$ gives the chance of the sample with the tilt integrated out, which on the balance wall is the gauge integrated out up to its scale, on which the chance of the sample does not depend (Proposition 49.1(i)). Otherwise the sample has chance zero under $F$ at every gauge. The posterior over the collection is the prior times this chance, normalized. □

*Remark.* At one table, (i) and (ii) of Proposition 49.4 are the reference's statement, with $s = q$ and $t = 0$, and Proposition 49.5 and Corollary 49.6 at one list are the reference's. Part (iii) of Proposition 49.4 is the family reading of Proposition 49.3(ii): the floors of the windows of every table are seen together through the one wall $\kappa_S$ of the sample, whose chance of vanishing is $\prod_j \mathsf{Q}^K_{j,S}(0;\alpha)^{s_j} \mathsf{Q}^{\Psi}_{j,S}(0;\alpha)^{t_j}$.

### 50.48.2 Statistics from a centre

**Proposition 49.16 (The even and odd statistics from a centre).** *As stated.*

*Proof.* (i) The law of $Z$ is that of the question $(0; \boldsymbol{\sigma}, \mathbf{0})$, whose exponent is $-\sum_j \sigma_j E_j$ (Definition 17.1). The centre is inside its walls, so the family is connected, and by Proposition 17.4(iii) $\sum_j \sigma_j E_j$ is strictly convex and even, least only at 0, with $\sum_j \sigma_j E_j(0) = 2 \sum_j \sigma_j \log K_j(\mathbf{1})$. Its Hessian at 0 is $\sum_j \sigma_j \cdot 2 \operatorname{Cov}_{\mathbb{P}_{j,\mathbf{1}}}(\chi_T) = 8 \sum_j \sigma_j \kappa_{j,\mathbf{1}}(X_e, X_f)$, which is rung two of the still of the realization $\big((\sigma_j, \sigma_j)_j; \mathbf{1}\big)$ by Proposition 9.7. For one table, $E_j$ is convex and even with its least value $2 \log K_j(\mathbf{1})$ at 0. If $v$ is constant on the molecules of table $j$, $\langle \chi_T, v \rangle = c$ for every basis (Proposition 7.9), so $\Lambda_j(z+v) = e^c \Lambda_j(z)$, $\Lambda_j(-z-v) = e^{-c} \Lambda_j(-z)$, and $E_j(z+v) = E_j(z)$; in particular $E_j(v) = E_j(0)$. Conversely if $E_j(z) = E_j(0)$, the convex function $t \mapsto E_j(tz)$ has its least value at $t = 0$ and at $t = 1$ and so is constant on $[0,1]$; its second derivative at 0, $2 \operatorname{Var}_{\mathbb{P}_{j,\mathbf{1}}} \langle \chi_T, z \rangle$, vanishes, and $z$ is constant on the molecules of table $j$ (Lemma 7.8). The Laplace transform is Proposition 17.7(i) at $c = 0$, $\mathbf{a} = \mathbf{k}$, $\mathbf{b} = 0$: the question $(0; \boldsymbol{\sigma} + \mathbf{k}, \mathbf{0})$ is the centre of strengths $\boldsymbol{\sigma} + \mathbf{k}$, with the data $\big((\sigma + |\mathbf{k}|)\mathbf{1}; \boldsymbol{\sigma} + \mathbf{k}, \boldsymbol{\sigma} + \mathbf{k}\big)$, and its walls are $\kappa^{\mathbf{E}}_S(\boldsymbol{\sigma} + \mathbf{k}) = \kappa^{\mathbf{E}}_S(\boldsymbol{\sigma}) + \kappa^{\mathbf{E}}_S(\mathbf{k})$. Along $\mathbf{k} = \lambda \boldsymbol{\sigma}$ they are $(1+\lambda)\kappa^{\mathbf{E}}_S(\boldsymbol{\sigma})$, all zero at $\lambda = -1$; along $\lambda \mathbf{k}$ the first to vanish is at the least ratio stated.

(ii) The density $e^{-\sum_j \sigma_j E_j(z)}$ is even, and $z \mapsto -z$ keeps every $E_j$ and negates every $O_j$; so the two triples have one law, and a joint moment of odd total degree in $Z$ and the $O_j(Z)$, finite as the density decays exponentially (Proposition 17.4) while the statistics grow linearly, equals its own negative. The Laplace transform and its domain are Proposition 17.7(i) at $\mathbf{a} = 0$. When $|b_j| \le \sigma_j$ and $\ell^*_{\boldsymbol{\sigma}}(c, \mathbf{b}) > 1$, the question $(c; \boldsymbol{\sigma}, \mathbf{b})$ has nonnegative exponents and every proper wall positive, so it has a native gauge (Theorem 8.19(iii)), its target is its masses, and its period over

$P_{\mathrm{c}}$ is its pseudoexpectation (Definition 16.10(iii)). The tails are Proposition 17.8 at the centre; when $\ell^*_{\boldsymbol{\sigma}}(c,\mathbf{b})=\infty$ both $Y$ and $-Y$ are bounded above, and $Y$ is bounded.

(iii) By Theorem 7.15(i), $K_j(\alpha)=e^{\langle\mathbf{1},z\rangle}\Lambda_j(z)$ and $\Psi_j(\alpha)=e^{\langle\mathbf{1},z\rangle}\Lambda_j(-z)$, and $\log\Lambda_j(\pm z)=\frac{1}{2}\big(E_j(z)\pm O_j(z)\big)$. So $K_j^{-s_j}=e^{-s_j\langle\mathbf{1},z\rangle}e^{-\frac{s_j}{2}(E_j+O_j)}$ is the factor of table $j$ at $(\sigma_j,\delta_j)=(\frac{s_j}{2},\frac{s_j}{2})$, its factor $e^{-s_j\langle\mathbf{1},z\rangle}=\alpha^{-\frac{s_j}{2}\mathbf{1}}$ the shift of the masses, and likewise $\Psi_j^{-t_j}$. Splitting each table into its two letters, at $(s_j,0)$ and $(0,t_j)$, keeps the family word, and the pair coordinates add: $(\frac{s_j}{2}+\frac{t_j}{2},\frac{s_j}{2}-\frac{t_j}{2})=(\sigma_j,\delta_j)$. □

**Proposition 49.17 (What the two statistics are).** *As stated.*

*Proof.* For scores $a_T$ with weights $\theta_T>0$, $\max_T(a_T+\log\theta_T)\le\log\sum_T\theta_Te^{a_T}\le\max_T a_T+\log\sum_T\theta_T$: a soft maximum; and $-\log\sum_T\theta_Te^{-a_T}$ is a soft minimum. (i) $E_j=\log\Lambda_j(z)+\log\Lambda_j(-z)$ and $O_j=\log\Lambda_j(z)-\log\Lambda_j(-z)$ are the soft maximum less, and plus, the soft minimum, and $\Phi^\circ=\sum_j\big(s_j\log\Lambda_j(z)+t_j\log\Lambda_j(-z)\big)$ (Theorem 8.19). (ii) By Theorem 7.16 at $\alpha=\mathbf{1}$, $\Lambda_j(z)/\Lambda_j(0)=\mathbb{E}_{T\sim\mathbb{P}_{j,\mathbf{1}}}[e^{Y_j}]$, so near 0, $\log\Lambda_j(z)=\log K_j(\mathbf{1})+\sum_{k\ge1}\frac{1}{k!}\kappa_{j,\mathbf{1}}(Y_j^{\times k})$ and $\log\Lambda_j(-z)$ is the same with $(-1)^k$; their sum and difference are the two expansions. $\kappa_{j,\mathbf{1}}(Y_j)=\sum_e z_e\mathbb{E}[\chi_{T,e}]=\sum_e(2R_{j,e}(\mathbf{1})-1)z_e$, and for $k\ge2$ multilinearity and $\chi_T=2X-\mathbf{1}$ give $\kappa_{j,\mathbf{1}}(Y_j^{\times k})=\sum_w z^w\kappa_{j,\mathbf{1}}(\chi_{T,w_1},\dots,\chi_{T,w_k})=2^k\sum_w z^w\kappa_{j,\mathbf{1}}(w)$, a constant shift not entering a cumulant of order $k\ge2$. The terms of order $k\ge2$ of $\Phi^\circ$ are then $\frac{1}{k!}\sum_w 2^k\sum_j\big(s_j+(-1)^kt_j\big)\kappa_{j,\mathbf{1}}(w)z^w$, which at the exponents of a realization is $\frac{1}{k!}\sum_w\kappa_\Upsilon(w)z^w$ (Proposition 9.7). (iii) $|\log\Lambda_j(z)-h_j(z)|\le\max(|\log\theta_{j,\min}|,|\log K_j(\mathbf{1})|)$, so $E_j(\lambda z)-\lambda\,\mathrm{wd}_j(z)$ and $O_j(\lambda z)-\lambda(h_j(z)-h_j(-z))$ are bounded; divide by $\lambda$. On the sector of $\pi$ the largest score is at $T^+_{j,\pi}$ and the least at $T^-_{j,\pi}$ (Lemma 8.3), and $\sum_j\big(s_jh_j(z)+t_jh_j(-z)\big)=\langle\sum_j(s_j\chi_{T^+_{j,\pi}}+t_j\chi_{\mathcal{E}\setminus T^-_{j,\pi}}),z\rangle=\langle2v_\pi-2\sigma\mathbf{1},z\rangle$; for $s_j,t_j\ge0$, in the interior of the sector $\mathbf{1}_{T^+_{j,\pi}}$ and $\mathbf{1}_{\mathcal{E}\setminus T^-_{j,\pi}}$ are the only points of $P_j$ and $P_j^*$ at which $\langle\cdot,z\rangle$ is largest, so $v_\pi$ is the vertex of $\mathcal{N}_F$ at which it is largest. By Proposition 17.4(i), $|X+2\kappa|$ is bounded, so $X(\lambda z)/\lambda\to-2\kappa(z)$, and on $\mathbf{1}^\perp$, where $\langle\mathbf{1},z\rangle=0$ and $\langle\boldsymbol{\mu}_\perp,z\rangle=\langle\mathbf{m},z\rangle$, $2\kappa(z)=\langle2v_\pi-2\sigma\mathbf{1},z\rangle-2\langle\boldsymbol{\mu}_\perp,z\rangle=2\langle v_\pi-\mathbf{m},z\rangle$. At the tilt $-\mathbf{1}_{U_k}+\frac{|U_k|}{n}\mathbf{1}\in H_\pi\cap\mathbf{1}^\perp$ this reads $\kappa_{U_k}=m_{U_k}-(v_\pi)_{U_k}$, the term in $\mathbf{1}$ dropping because $(v_\pi)_\mathcal{E}=\sum_j(s_jr_j+t_jr_j^*)=m_\mathcal{E}$ on the balance wall. □

*Remark.* At one table and the centre of strength $\sigma_0$, Proposition 49.16 is the reference's, with $\ell^*_{\sigma_0}(0,b)=\sigma_0\,\ell^*_1(0,1)/|b|$; Proposition 49.17(i)–(ii) for one table is the reference's statement, and the family adds the still and, in (iii), the greedy vertex of the Newton sum, whose tail sums against the masses are the walls along the order: the reading of the walls of Theorem 8.19(iii) at the vertex that the order selects.

### 50.48.3 The Grey law, the dial and the sorted draw

**Proposition 49.12(iii) (The tails of the Grey law).** *As stated, for every real direction of the exponents.*

*Proof.* Let $F_0$ be the family of the list $\binom{\mathcal{E}}{1}$, whose bases are the singletons and whose letter is $K_0=|\alpha|$, and let $F'$ be the family of the tables of $F$ at the exponents $(0,0)$ and the list $\binom{\mathcal{E}}{1}$ at $(m_\mathcal{E},0)$. These data are on the balance wall of $F'$, the list $\binom{\mathcal{E}}{1}$ having rank one; that table has least level 0 on every proper set, so the walls are $\kappa_S=m_S>0$; and the law of the question is $\mathrm{Dir}_{\mathbf{m}}$ (Proposition 16.3). Let $v$ be the balanced direction of the data that keeps the masses, moves the exponents of table $j$ of $F$ by $(a_j,b_j)$ and those of the list $\binom{\mathcal{E}}{1}$ by $(-d,0)$, $d=\sum_j(a_jr_j+b_jr_j^*)$.

Along $q+\lambda v$ the walls are $m_S-\lambda g(S)$, so the $\lambda^*$ of Proposition 17.8 is $k^*$. The statistic $Y$ of that proposition is $X_v(Z)$, the logarithm of the ratio of the integrands of the periods at $q+v$ and at $q$ (Definition 17.1, and the proof of Proposition 17.4(i)), which at the tilt $Z$ of $U$ is $\log\big(\prod_j K_j(U)^{-a_j}\Psi_j(U)^{-b_j}K_0(U)^d\big)=\sum_j(a_jG_j+b_jG_j^*)$, as $K_0(U)=|U|=1$ on the simplex. Proposition 17.8 gives the tail when $k^*<\infty$, and the bound above when $k^*=\infty$. □

*Remark.* The reference reads (iii), for $a_j,b_j\ge 0$, off the abscissa of convergence of the Laplace transform of (i), which gives the limit superior; the argument above gives the limit, at every real direction. In the direction $(\mathbf{s},\mathbf{t})$ of a word it is the tail of Proposition 49.15(iv), which the reference proves from the collapse of a set attaining $k^*$.

**Proposition 49.15 (The exponent is a dial between the mean and the maximum).** *As stated, for real exponents of any signs, not all zero.*

*Proof.* (i) is a property of power means and does not involve the exponents. (ii) By Proposition 49.10(ii), $\mathbb{E}[e^{kA_k}]=\frac{1}{q}\sum_i\mathbb{E}[W(U_i)^k]=P_F(\mathbf{m};k\mathbf{s},k\mathbf{t})/\Gamma_1^n\begin{bmatrix}\mathbf{m}\\ m_{\mathcal{E}}\end{bmatrix}$ wherever every proper wall at $(\mathbf{m};k\mathbf{s},k\mathbf{t})$, which is $m_S-kg(S)$, is positive, that is for $0<k<k^*$; and $e^{kA_k}=\frac{1}{q}\sum_i W(U_i)^k$ has a finite variance exactly when $\mathbb{E}[W(U)^{2k}]<\infty$, that is $2k<k^*$ (Proposition 49.10(i) at the exponents $(2k\mathbf{s},2k\mathbf{t})$). (iii) For $k\ge k^*$ some proper wall at $(\mathbf{m};k\mathbf{s},k\mathbf{t})$ is not positive, and $\mathbb{E}[W(U)^k]=\infty$ by Proposition 49.10(i); the law of the maximum of independent draws is the power of the law of one. (iv) $Y=\sum_j(s_jG_j+t_jG_j^*)$ with $(G_j,G_j^*)_j\sim\mathrm{Grey}(F;\mathbf{m})$, and Proposition 49.12(iii) in the direction $(\mathbf{s},\mathbf{t})$ gives the tail and the bound. For $s_j,t_j\ge 0$: $(|U_S|,|U_{\mathcal{E}\setminus S}|)\sim\mathrm{Dir}_{(m_S,m_{\mathcal{E}\setminus S})}$ by the aggregation of the coordinates of a Dirichlet draw; on the simplex every coordinate in $S$ is at most $\rho$ and every coordinate at most 1, so $K_j(U)\le K_j(\mathbf{1})\rho^{d_j(S)}$ and $\Psi_j(U)\le\Psi_j(\mathbf{1})\rho^{d_j^*(S)}=K_j(\mathbf{1})\rho^{d_j^*(S)}$, and raising these to the powers $-s_j\le 0$, $-t_j\le 0$ gives the bound on $W(U)$. □

**Proposition 49.19 (A sorted draw lives on the sector of its order).** *As stated, for real exponents of any signs.*

*Proof.* The family is connected: on the balance wall $\kappa_S+\kappa_{\mathcal{E}\setminus S}=2\sum_j\sigma_jc_j(S)$ (Lemma S5), so positive walls put on every nonempty proper $S$ a table with $\sigma_j>0$, hence active, and $c_j(S)>0$. The tilt $Z$ of $U$ has the density $2^{n-1}e^{X(z)}/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ with respect to $dz$ (Definition 17.1), and on the sector $H_\pi$, where $dz$ is the Lebesgue measure in the differences $v_k$, Proposition 17.4(i) gives $e^{X(z)}=e^{-2\sum_k\kappa_kv_k}\prod_j k_{j,\pi}(q)^{-s_j}\psi_{j,\pi}(q)^{-t_j}$ with $q_k=e^{-2v_k}$, at any signs of the exponents. In the gaps $u_k=2v_k=\log(U_{\pi(k-1)}/U_{\pi(k)})$, $du=2^{n-1}dv$, so on $H_\pi$ the gaps have the density $P_F(\mathbf{m};\mathbf{s},\mathbf{t})^{-1}e^{-\sum_k\kappa_ku_k}\prod_j k_{j,\pi}(e^{-u})^{-s_j}\psi_{j,\pi}(e^{-u})^{-t_j}$ on $[0,\infty)^{n-1}$, whose total is $I_\pi/P_F(\mathbf{m};\mathbf{s},\mathbf{t})$ (Corollary 8.6); the boundaries of the sectors are null, and summing over $\pi$ gives $P_F(\mathbf{m};\mathbf{s},\mathbf{t})=\sum_\pi I_\pi$. This is (i) and (ii). (iii) The bounds on the sector letters are those of Proposition 17.4(i); raised to the powers $-s_j$ and $-t_j$, of any signs, they bound $\prod_j k_{j,\pi}^{-s_j}\psi_{j,\pi}^{-t_j}$ on the cube between two positive constants, and for $s_j,t_j\ge 0$ between $\prod_j K_j(\mathbf{1})^{-s_j-t_j}$ and $\prod_j\theta_{j,T_{j,\pi}^+}^{-s_j}\theta_{j,T_{j,\pi}^-}^{-t_j}$; and $\int_{[0,\infty)^{n-1}}e^{-\sum_k\kappa_ku_k}du=\prod_k\kappa_k^{-1}$. (iv) is Lemma 8.3, which does not involve the exponents. □

*Remark.* Proposition 49.14 is the ratio of two limits of Theorem 17.5(i): by Proposition 49.10(i), $\mathbb{E}_{V\sim\mathrm{Dir}_{\lambda\boldsymbol{\tau}}}\big[\prod_j K_j(V)^{-\lambda s_j}\Psi_j(V)^{-\lambda t_j}\big]=P_F(\lambda\boldsymbol{\tau};\lambda\mathbf{s},\lambda\mathbf{t})/\Gamma_1^n\begin{bmatrix}\lambda\boldsymbol{\tau}\\ \lambda\tau_{\mathcal{E}}\end{bmatrix}$, the numerator is the period of $\lambda$ times the question of data $(\boldsymbol{\tau};\mathbf{s},\mathbf{t})$, with the walls $\tau_S-g(S)$, and the denominator is the period of the list $\binom{\mathcal{E}}{1}$ at a full basis lean at the masses $\lambda\boldsymbol{\tau}$, with the walls $\lambda\tau_S$ (Proposition 16.3); multiplied by $\lambda^{n-1}$ they tend to $\sum_\pi\prod_k(\tau_{U_k}-g(U_k))^{-1}$ and $\sum_\pi\prod_k\tau_{U_k}^{-1}$.

### 50.48.4 The interaction of a bunch

The one-table proofs integrate the joint cumulant of the indicators of a bunch, one logarithm of a Taylor cube; a family integrates a sum of such logarithms with its exponents, one for each letter, so one lemma on one logarithm serves both.

**Lemma S31** (An alternating sum over the subsets of a bunch)**.** *Let $B$ be a finite set with $k = |B| \geq 2$, let $(c_p)_{p\subseteq B}$ be positive, and $f(y) = \log \sum_{p\subseteq B} c_p\, e^{\langle \mathbf{1}_p, y\rangle}$ on $\mathbb{R}^B$, and write $\partial^B = \prod_{e\in B} \partial_{y_e}$. Then $|\partial^B f(y)| \leq C\, e^{-\max_{e\in B}|y_e|}$, so $\partial^B f$ is integrable on $\mathbb{R}^B$, and*

$$\int_{\mathbb{R}^B} \partial^B f(y)\, dy = \sum_{p\subseteq B} (-1)^{|B\setminus p|} \log c_p.$$

*Proof. The bound.* $f(y+w) - f(y)$ is the logarithm of the moment generating function at $w$ of the indicators $(X_e)_{e\in B}$ of a random set $P \subseteq B$ with the law $\pi_y(p) \propto c_p e^{\langle \mathbf{1}_p, y\rangle}$, so $\partial^B f(y)$ is their joint cumulant. Fix $e \in B$. A joint cumulant of $k \geq 2$ variables does not change when one of them is shifted by a constant, and it is a sum, with coefficients depending on $k$ alone, of products of joint moments, one for each block of a partition of the $k$ places, exactly one block holding the place of $e$. With $X_e - \mathbb{E}X_e$ in that place and $|X_f| \leq 1$, $|\partial^B f(y)| \leq C_k\, \mathbb{E}|X_e - \mathbb{E}X_e| = 2C_k p_e(1-p_e)$, $p_e = \pi_y(e \in P)$. Pairing each $p \not\ni e$ with $p+e$, the odds $p_e/(1-p_e) = \sum_{p\not\ni e} c_{p+e} e^{y_e} w_p \big/ \sum_{p\not\ni e} c_p w_p$, $w_p = e^{\langle \mathbf{1}_p, y\rangle} > 0$, lie between $\rho_- e^{y_e}$ and $\rho_+ e^{y_e}$, $\rho_\pm$ the largest and the least of the ratios $c_{p+e}/c_p$; so $p_e(1-p_e) \leq \min(\text{odds}, \text{odds}^{-1}) \leq \max(\rho_+, \rho_-^{-1})\, e^{-|y_e|}$. The best $e$ gives the bound.

*The value*, by induction on $k$. Fix $e \in B$, put $B' = B - e$, $y' = y_{B'}$, and let $f^{/e}$ and $f^{\setminus e}$ be the functions of the same form on $B'$ with the coefficients $c_{p+e}$ and $c_p$, $p \subseteq B'$: the contraction and the deletion of $e$ in the cube. With $\ell(x) = \log(1 + e^x)$,

$$f(y) = f^{/e}(y') + y_e + \ell\big(f^{\setminus e}(y') - f^{/e}(y') - y_e\big) = f^{\setminus e}(y') + \ell\big(y_e + f^{/e}(y') - f^{\setminus e}(y')\big).$$

Every derivative of $\ell$ is $O(e^x)$ as $x \to -\infty$, and every derivative of order at least one of $f^{/e}$ and $f^{\setminus e}$ is bounded; since $B' \neq \varnothing$, $\partial^{B'} y_e = 0$, so $\partial^{B'} f(y) \to \partial^{B'} f^{/e}(y')$ as $y_e \to +\infty$ and $\partial^{B'} f(y) \to \partial^{B'} f^{\setminus e}(y')$ as $y_e \to -\infty$. Hence $\int_{\mathbb{R}} \partial^B f\, dy_e = \partial^{B'} f^{/e}(y') - \partial^{B'} f^{\setminus e}(y')$ for every $y'$, and by the bound and Fubini

$$\int_{\mathbb{R}^B} \partial^B f\, dy = \int_{\mathbb{R}^{B'}} \big(\partial^{B'} f^{/e} - \partial^{B'} f^{\setminus e}\big)\, dy'.$$

If $k \geq 3$, each term is integrable by the bound, and the induction gives $\sum_{p\subseteq B'} (-1)^{|B'\setminus p|} (\log c_{p+e} - \log c_p) = \sum_{p\subseteq B} (-1)^{|B\setminus p|} \log c_p$. If $k = 2$, $B' = \{e'\}$, the integrand is the derivative of $f^{/e} - f^{\setminus e} = \log\big((c_e + c_{ee'} e^{y'})/(c_\varnothing + c_{e'} e^{y'})\big)$, whose limits at $\pm\infty$ are $\log(c_{ee'}/c_{e'})$ and $\log(c_e/c_\varnothing)$; their difference is the alternating sum. □

**Theorem 49.25 (The interaction totals the still over the bunch's scales).** *As stated.*

*Proof.* Fix $\alpha_R$. By the Taylor cube (Theorem 12.7)

$$\log K_j(e^{y_B}, \alpha_R) = \log \sum_{p\subseteq B} K_{j,(p)}(\alpha_R)\, e^{\langle \mathbf{1}_p, y_B\rangle},$$

with every $K_{j,(p)}(\alpha_R) > 0$ as $\mathrm{pat}_j(B) = 2^B$; Lemma S31 gives $\int \partial^B \log K_j\, dy_B = \iota_{j,B}(\alpha_R)$, absolutely. Likewise $\log \Psi_j(e^{y_B}, \alpha_R) = \log \sum_{P\subseteq B} \Psi_{j,(B\setminus P)}(\alpha_R)\, e^{\langle \mathbf{1}_P, y_B\rangle}$, and the lemma gives

$\sum_P (-1)^{|B\setminus P|}\log\Psi_{j,(B\setminus P)} = \sum_p (-1)^{|p|}\log\Psi_{j,(p)} = \iota^*_{j,B}(\alpha_R)$. The parts of the potential are these logarithms with the coefficients $\sigma_j(1,1)$ and $\delta_j(1,-1)$, and a table with both exponents zero does not enter; integrating term by term gives $\iota^{\mathbf{E}}_B$ and $\iota^{\mathbf{O}}_B$. At a realization $\kappa_\Upsilon(e_1,\dots,e_k) = 2^k\partial_{y_{e_1}}\cdots\partial_{y_{e_k}}\Phi$ (Definition 9.4), and Proposition 9.6 writes it through the cumulants at $\alpha$ and at $\alpha^{-1}$. For one table at $(1,0)$, $\partial^B \log K(e^y) = \kappa_\alpha(e_1,\dots,e_k)$, the joint cumulant of the indicators under $\mathbb{P}_\alpha$ (Definition 7.17). □

**Proposition 49.24 (The interaction belongs to the room).** *As stated.*

*Proof.* (i) The child at $p$ has rank $r_j - |p|$ on $R$ (Lemma 12.2), so $K_{j,(p)}$ is homogeneous of degree $r_j - |p|$ and $\Psi_{j,(p)}$ of degree $|R| - r_j + |p|$. Under $\alpha_R \mapsto c\alpha_R$, $\iota_{j,B}$ moves by $\log c\sum_p (-1)^{|B\setminus p|}(r_j - |p|)$ and $\iota^*_{j,B}$ by $\log c\sum_p (-1)^{|p|}(|R| - r_j + |p|)$; both vanish, as $\sum_{p\subseteq B}(-1)^{|p|} = 0$ for $|B| \geq 1$ and $\sum_{p\subseteq B}(-1)^{|p|}|p| = 0$ for $|B| \geq 2$.

(ii) A basis draw of table $j$ meets $B$ in $P$ with the chance $\pi_{j,P}(\alpha)$, and a cobasis draw misses exactly $P$ on $B$ when it meets $B$ in $B\setminus P$, with the chance $\pi^*_{j,B\setminus P}(\alpha) = \alpha^P_B\Psi_{j,(B\setminus P)}(\alpha_R)/\Psi_j(\alpha)$ (Theorem 12.8); independence gives $w(P)$. So

$$\log w(P) = \sum_j (s_j + t_j)\langle y_B, \mathbf{1}_P\rangle + \sum_j \big(s_j \log K_{j,(P)}(\alpha_R) + t_j \log\Psi_{j,(B\setminus P)}(\alpha_R)\big) - \Phi(\log\alpha).$$

The coefficients of the expansion are $\lambda_A = \sum_{P\subseteq A}(-1)^{|A\setminus P|}\log w(P)$, and for $|A| \geq 2$ the terms constant in $P$ and linear in $\mathbf{1}_P$ drop out:

$$\lambda_A = \sum_j \Big(s_j \sum_{P\subseteq A}(-1)^{|A\setminus P|}\log K_{j,(P)} + t_j \sum_{P\subseteq A}(-1)^{|A\setminus P|}\log\Psi_{j,(B\setminus P)}\Big),$$

free of $\alpha_B$. At $A = B$ the second inner sum is $\iota^*_{j,B}$, and $\lambda_B = \iota^{\mathbf{E}}_B + \iota^{\mathbf{O}}_B$. For $A \subsetneq B$: the bases of table $j$ that avoid $B\setminus A$ form its deletion of $B\setminus A$, a table as some basis avoids $B$, in which every subset of $A$ is a pattern, with the children $K_{j,(P)}$, $P \subseteq A$; the bases that contain $B\setminus A$, less $B\setminus A$, form its contraction, a table as $B$ is independent, in which every subset of $A$ is a pattern and the pattern $P' \subseteq A$ has the child of the pattern $P'\cup(B\setminus A)$ of $B$. With $P' = A\setminus P$, $B\setminus P = P'\cup(B\setminus A)$, so the second inner sum is $\sum_{P'\subseteq A}(-1)^{|P'|}\log\Psi_{j,(P'\cup(B\setminus A))}$, the dual interaction of $A$ in the contraction. The deletion is the restriction to $\mathcal{E}\setminus(B\setminus A)$, some basis avoiding $B\setminus A$, and these are the tables of the face of the wall at $B\setminus A$ on its room (Theorem 16.25), at the exponents $(s_j, 0)$ and $(0, t_j)$: every basis of the restriction and of the contraction comes from one basis of the table, whose weight it keeps, so the weights factor on the bottom levels as that theorem asks. So $\lambda_A$ is $\iota^{\mathbf{E}}_A + \iota^{\mathbf{O}}_A$ computed in that face. At one table and $(1,0)$, $w(P) = \pi_P(\alpha)$. □

**Proposition 49.26 (Both sides, and parity).** *As stated.*

*Proof.* The child at $p$ is a table on $R$, so $\Psi_{(p)}(\alpha_R) = \alpha^R_R K_{(p)}(\alpha_R^{-1})$ (Proposition 7.2); in $z_R$, $\log\Psi_{(p)} = 2\langle\mathbf{1}, z_R\rangle + \log K_{(p)}(e^{-2z_R})$, and as $\sum_p (-1)^{|p|} = 0$, $\iota^*_B(z_R) = \sum_p (-1)^{|p|}\log K_{(p)}(e^{-2z_R}) = (-1)^{|B|}\iota_B(-z_R)$. Substituting in Definition 12.18 gives $\iota^{\mathbf{E}}_B$ and $\iota^{\mathbf{O}}_B$; the parities in $z_R$ and the values at $z_R = 0$ are read off. The total for $-\Phi$ is $-(\iota^{\mathbf{E}}_B + \iota^{\mathbf{O}}_B) = -\sum_j (s_j\iota_{j,B} + t_j\iota^*_{j,B})$. □

*Remark.* At one table and $(1,0)$ these are the reference's statements: the interaction, its homogeneity and log-linear reading, its total as the integral of the joint cumulant, and the reading of the dual table. In a family every subset of the bunch must be a pattern in every active table, and the family reads the alternating sum over the patterns of its whole potential: the term of the pattern $P$ is the family of the children $K_{j,(P)}$ at $s_j$ and $\Psi_{j,(B\setminus P)}$ at $t_j$, which is where the potential goes as the scales of $P$ grow and those of $B\setminus P$ fall. Proposition 49.27 is one table and is unchanged.

## 50.49 Receipts

The identities below were checked on small families before or after they were written; the checks are receipts, not part of any proof.

### 50.49.1 Chapters 35 to 46

In exact rational arithmetic at $(a,b)=(\frac{7}{3},\frac{5}{4})$, with two linear letters of rational coefficients and the exponents $\frac{9}{7}$ and $-\frac{2}{3}$, and at $r=3$: the expansion of $L\,x(1-x)\,\partial_x g + A_n g$ at $g=(-1)^r\Pi_r^{a,b}$ in the Beta ports had no component below the order $r-h-1$, and its coefficient of the order $r+h+1$ matched Theorem 35.15. At $(n,k,r)\in\{(2,3,1),(2,3,2),(1,4,3),(2,5,1)\}$ the column sums $C_{n,k}(P_r)$, read off the Laurent expansions of the share functions, matched the sums of the proof of Theorem 37.19, the signs $(-1)^{jk}$ included, and $C_{n,k-1}(P_r)+2\,(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,n+r])\,C_{n,k}(P_r)$ was rational. For two letters at $\kappa=0.7$, $\mathbf{w}=(0.4,0.5)$, $\boldsymbol{\xi}=(1,1+i)$, $G=e^{-\rho}$, $C_1=1+\rho$ and $C_2=2+\rho^2$, the quantity $\big(\int_0^\infty\cdots d\rho - A(0)\big)\lambda^{-\kappa^\flat}$ of Proposition 38.23 was $4.10646-0.33755i$, $4.10314-0.33828i$ and $4.10276-0.33837i$ at $\lambda=10^{-3},10^{-4},10^{-5}$, against $B(0)=4.10272-0.33838i$.

### 50.49.2 Chapters 17 and 49

On four marks, with the tables $\binom{\mathcal{E}}{2}$, the rank-two table with one parallel pair, the direct sum of two rank-one tables and a rank-three table with a coloop, all with random weights: the identity of Proposition 17.4(i) held to $10^{-13}$ at random tilts and random questions with exponents of both signs, and the two forms of the wall at a tilt agreed to $10^{-13}$; the period computed on the sectors agreed with a direct cubature of the tilt integral to $10^{-8}$; $\lambda^3 P_F(\lambda q)$ approached $\sum_\pi \prod_k \kappa_{U_k}^{-1} = 1.87097$, and $(\sum_\pi \prod_k \kappa_{U_k}^{-1} - \lambda^3 P_F(\lambda q))/\lambda$ approached $\sum_\pi \prod_k \kappa_{U_k}^{-1} \sum_j (s_j \log\theta_{j,T^+_{j,\pi}} + t_j \log\theta_{j,T^-_{j,\pi}}) = 2.75547$ ($2.7548$ at $\lambda = 3\cdot 10^{-4}$). On three marks with three tables: the relative error of Theorem 17.5(ii) times $|\lambda|$ was 0.172–0.184 along $\lambda\in\{2,\dots,40\}$ and along $\lambda=2+i\eta$, $\eta\in\{5,10\}$, the same constant in both directions; the usage at the computed native gauge equalled the masses; and the local slopes of $\log\mathbb{P}(\pm Y>x)$ for an odd statistic at the centre were between $-1.4985$ and $-1.5014$ against $-\ell^*_{\boldsymbol{\sigma}}(c,\mathbf{b})=-1.5$. In exact rational arithmetic, on four marks with sample sizes $\mathbf{s}=(1,0,1,0)$, $\mathbf{t}=(0,1,1,1)$: the conditional law of a sample given its counts was $\theta_{\text{draws}}/\Theta_F(\mathbf{m})$, and the law of the wall $\kappa_S$ was $\prod_j \mathsf{Q}^{K}_{j,S}{}^{s_j}\mathsf{Q}^{\Psi}_{j,S}{}^{t_j}$ for every $S$. The total of Theorem 49.25 matched $\iota_B^{\mathbf{E}}+\iota_B^{\mathbf{O}}$ to $10^{-12}$ for $|B|=2$ in three tables, and to $10^{-10}$ for $|B|=3$, a Hamiltonian path of $K_4$ in the graphic table of $K_4$ together with $\binom{\mathcal{E}}{3}$ on six marks; the relation $\iota_B^*(z_R)=(-1)^{|B|}\iota_B(-z_R)$ held to $10^{-12}$ in each table. All of these were rerun on 20 October 2026 and reproduced (the cubature for $|B|=2$ on a grid of 1601 points a side, matching to $6\cdot 10^{-13}$).

In exact rational arithmetic, on six marks with two weighted graphic tables of $K_4$ and a weighted $\binom{\mathcal{E}}{3}$, at the exponents $(2,1)$, $(1,3)$, $(0,2)$, a rational gauge and the bunch $B$ of a Hamiltonian path: for every $A\subseteq B$ with $|A|\geq 2$ the coefficient $\lambda_A$ of Proposition 49.24(ii) satisfied $e^{\lambda_A}=e^{\iota_A^{\mathbf{E}}+\iota_A^{\mathbf{O}}}$ exactly, the right side computed from the deletions and contractions of $B\setminus A$ themselves.

# Part XI

# Exercises

Each exercise is a computation on a particular list, table or ray, set as a guided example. Its solution follows the procedures and theorems of the chapters step by step: every step names the statement it uses, and every quantity that the next step needs is displayed, in exact arithmetic. A step that is a machine computation too large to display is given by its exact input, its method, the output that the next step uses, the check that was run on it, and the receipt that reproduces it. Decimal values appear only as renderings of exact ones.

Lists are named by an upright capital with the number of marks as a subscript, and each is defined by its marks and bases where it first appears.

# Chapter 51

# Exercises on tables, families and classification

The exercises of this chapter work the statements of Parts II and III, and the first statements of the reading of rays, on particular lists. The list on two marks is opened on its share side and read along its centre ray, where its values, its closings and the matrix of its branch fit together exactly. The pins of the atom formula of six uniform lists have the elementary divisors 1, and two of them still pin some shares at fractions. Four Foreman trees are drawn: one that is an atom at every lean, one that peels at a bunch of width one into a cell whose contraction is the dual of its deletion, and one whose Foreman bunch at a lean has width two and whose sets do not split off at the centre. Along a set of width two the Skeleton values are shown not to be rational functions of the point, and a family with exponents of both signs is peeled with and without its record. Every number displayed is exact; the receipts that recompute them are the scripts named in each solution.

## 51.1 The list on two marks

**Exercise 51.1** (The list $\binom{[2]}{1}$ on its share side)**.** Let $\mathcal{E} = \{1, 2\}$ and $F = \big(\binom{\mathcal{E}}{1}; s, t\big)$, the list whose bases are the two singletons, with $\operatorname{Re} s, \operatorname{Re} t > 0$ and masses $\mathbf{m} = (m_1, m_2)$ on the balance wall with $\operatorname{Re} m_1, \operatorname{Re} m_2 > 0$.

(i) Write the atom formula of $P_F(\mathbf{m}; s, t)$ (Corollary 18.16): its shares, its pins, its fibre and its measure, and close it.

(ii) At $(\mathbf{1}; 1, 1)$ find the integrand on the line $\operatorname{Re} a = \frac{1}{2}$ and the period.

*Solution.* Receipt: `ex01_share_side.py`.

(i) *The letters.* The bases are $T_1 = \{1\}$ and $T_2 = \{2\}$, so (Definition 7.1)

$$K(\alpha) = \alpha^{T_1} + \alpha^{T_2} = \alpha_1 + \alpha_2, \qquad \Psi(\alpha) = \alpha^{\mathcal{E}\setminus T_1} + \alpha^{\mathcal{E}\setminus T_2} = \alpha_2 + \alpha_1,$$

the rank and the corank are 1, and the balance wall is $\kappa_{\mathcal{E}} = m_1 + m_2 - (s + t) = 0$ (Definition 8.1), with the strength $\sigma = \frac{s+t}{2}$ and the offset $\mu_1 = m_1 - \sigma$.

*The shares and the pins.* Corollary 18.16, with $g = 2$, opens each letter with one share per basis: $a_1, a_2$ for $K$, $a_i$ the share of $\alpha^{T_i}$, and $b_1, b_2$ for $\Psi$, $b_i$ the share of $\alpha^{\mathcal{E}\setminus T_i}$. Its pins are the two letter totals and the odd load at the mark 1 (Definition 12.24), with $\chi_{T_1}(1) = 1$ and $\chi_{T_2}(1) = -1$:

$$a_1 + a_2 = s, \qquad b_1 + b_2 = t, \qquad \tfrac{1}{2}(a_1 - b_1) - \tfrac{1}{2}(a_2 - b_2) = m_1 - \tfrac{s+t}{2}.$$

With $a = a_1$ they give

$$a_2 = s - a, \qquad b_1 = a + t - m_1, \qquad b_2 = m_1 - a,$$

and the load at the mark 1, $a_1 + b_2$, is $m_1$, as $\mathrm{load}_1 = \sigma + \langle \boldsymbol{\delta} \rangle_1$ requires (Definition 12.24). The weights are 1, so the factor $\prod_T \theta_T^{-a_T - b_T}$ is 1, and the fibre has the dimension $4 - 3 = 1$.

*The measure.* The matrix of the pins, the shares in the order $(a_1, a_2, b_1, b_2)$, and its Gram matrix are

$$P = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}, \qquad PP^{\mathsf{T}} = \mathrm{diag}(2, 2, 1), \qquad \sqrt{\det PP^{\mathsf{T}}} = 2,$$

and the fibre runs along $(1, -1, 1, -1)$, of length 2: the imaginary part $y$ of $a$ moves the point by the volume $2\,dy$. So the fibre integral of Definition 4.1 is $(2\pi)^{3-4} \cdot \frac{1}{2} \cdot 2 \int dy = \frac{1}{2\pi i} \int da$:

$$P_F(\mathbf{m}; s, t) = \frac{1}{2\pi i} \int_{(c)} \Gamma_{\substack{2 \\ 1}} \begin{bmatrix} a, & s - a \\ & s \end{bmatrix} \Gamma_{\substack{2 \\ 1}} \begin{bmatrix} a + t - m_1, & m_1 - a \\ & t \end{bmatrix} da,$$

on a line $\mathrm{Re}\, a = c$ on which all four shares have positive real parts, $\max(0, \mathrm{Re}(m_1 - t)) < c < \min(\mathrm{Re}\, s, \mathrm{Re}\, m_1)$; such lines exist, since $\mathrm{Re}\, s, \mathrm{Re}\, t, \mathrm{Re}\, m_1 > 0$ and $\mathrm{Re}(m_1 - t) < \mathrm{Re}\, s$ is $\mathrm{Re}\, m_2 > 0$.

*The closing.* The integrand is $\Gamma(a)\,\Gamma(t - m_1 + a)\,\Gamma(s - a)\,\Gamma(m_1 - a)$ over $\Gamma(s)\Gamma(t)$: four Gammas on a line that separates the poles of the first two from those of the last two. Corollary 4.7, with its four entries $0$, $t - m_1$, $s$, $m_1$ and the variable $a$, adds them in pairs, $0 + s$, $(t - m_1) + m_1$, $0 + m_1$ and $(t - m_1) + s = m_2$:

$$\frac{1}{2\pi i} \int_{(c)} \Gamma(a)\,\Gamma(t - m_1 + a)\,\Gamma(s - a)\,\Gamma(m_1 - a)\,da = \Gamma_{\substack{4 \\ 1}} \begin{bmatrix} s, & t, & m_1, & m_2 \\ & & s + t & \end{bmatrix}.$$

Dividing by $\Gamma(s)\Gamma(t)$ (Proposition 1.3(i),(ii)),

$$P_F(\mathbf{m}; s, t) = \Gamma_{\substack{2 \\ 1}} \begin{bmatrix} m_1, & m_2 \\ m_1 + m_2 & \end{bmatrix}.$$

This agrees with the period read on the simplex (Theorem 8.7): there $K = \Psi = u_1 + u_2 = 1$, and the period is the shape integral $\int_{\Delta_\mathcal{E}} u^{\mathbf{m}-\mathbf{1}} d\omega_\mathcal{E} = \Gamma_{\substack{2 \\ 1}} \begin{bmatrix} m_1, & m_2 \\ m_1 + m_2 & \end{bmatrix}$ (Theorem 3.9).

(ii) At $(\mathbf{1}; 1, 1)$, $m_1 = s = t = 1$, and the shares are $a_1 = b_1 = a$ and $a_2 = b_2 = 1 - a$: the basis share and the cobasis share of each basis are equal. By Proposition 1.3(i),(v) the integrand is

$$\Gamma_{\substack{2 \\ 1}} \begin{bmatrix} a, & 1 - a \\ & 1 \end{bmatrix} \Gamma_{\substack{2 \\ 1}} \begin{bmatrix} a, & 1 - a \\ & 1 \end{bmatrix} = \Gamma_{\substack{4 \\ 0}} \big[a,\ a,\ 1 - a,\ 1 - a\big] = \Big(\frac{\pi}{\sin \pi a}\Big)^2,$$

regular on $0 < \mathrm{Re}\, a < 1$. On $a = \frac{1}{2} + iy$, where $\sin \pi a = \cosh \pi y$,

$$\frac{1}{2\pi i} \Gamma_{\substack{4 \\ 0}} \big[a,\ a,\ 1 - a,\ 1 - a\big]\ da = \frac{\pi}{2 \cosh^2 \pi y}\, dy = \frac{2\pi}{(e^{\pi y} + e^{-\pi y})^2}\, dy = p_{\mathrm{crit}}(y)\, dy,$$

the density of the critical-line law (Definition 36.1). Its primitive is $\frac{1}{2} \tanh \pi y$, which runs from $-\frac{1}{2}$ to $\frac{1}{2}$, so its integral is 1; and (i) gives the same, $P_F(\mathbf{1}; 1, 1) = \Gamma_{\substack{2 \\ 1}} \begin{bmatrix} 1, & 1 \\ & 2 \end{bmatrix} = 1$ (Theorem 36.2).

*Answer.* The atom formula is the integral over one share

$$P_F(\mathbf{m};s,t)=\frac{1}{2\pi i}\int_{(c)}\Gamma_{\,1}^{2}\!\begin{bmatrix}a,\ s-a\\ s\end{bmatrix}\Gamma_{\,1}^{2}\!\begin{bmatrix}a+t-m_1,\ m_1-a\\ t\end{bmatrix}\,da=\Gamma_{\,1}^{2}\!\begin{bmatrix}m_1,\ m_2\\ m_1+m_2\end{bmatrix}.$$

At $(\mathbf{1};1,1)$ its integrand on the line $\operatorname{Re}a=\frac12$ is $\Gamma_{\,0}^{4}\big[a,\ a,\ 1-a,\ 1-a\big]=\pi^2/\cosh^2\pi y$, the density of the critical-line law times $2\pi$, and the period is 1. □

**Exercise 51.2** (The centre ray of $\binom{[2]}{1}$)**.** For the centre ray of the list $\binom{\mathcal{E}}{1}$, $\mathcal{E}=\{1,2\}$, at the strength 1 (Definition 38.13):

(i) find its stride, its native gauge and native value, its values $\langle1\rangle_k$ and a recurrence of order one that they satisfy, and the integrand of the atom formula at step $k$ against the one at step 1;

(ii) find its closings $\mathsf{C}_j(1)$ (Definition 39.30) and the volume $V_1$ of its direction, and check Proposition 39.34 and the reflected recurrence at every $j$;

(iii) find the matrix $H$ of Theorem 38.15 and show that $k\,\langle1\rangle_k\,\mathsf{C}_k(1)=2=(\det H)^{-1}$ for every $k\ge1$; compare with the leading terms of Theorems 38.6 and 39.57.

*Solution.* Receipt: `ex01_share_side.py`.

(i) *Stride and native gauge.* Definition 38.13 with one table and $\sigma_1=1$ gives the origin 0 and the direction $(\mathbf{1};1,1)$; on the chart $\alpha_2=1$, with $x=\alpha_1$,

$$f=\frac{\alpha^{\mathbf{1}}}{K\Psi}=\frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2}=\frac{x}{(1+x)^2}.$$

The gauge $\mathbf{1}$ is native (Proposition 8.21(i)), and $f\le f(\mathbf{1})=\frac14$ on the orthant with equality only at $x=1$ (Theorem 38.14(i)): the native value of the direction is $\frac14$.

*Values.* By Definition 38.13, $\langle1\rangle_k=P_F(k\mathbf{1};k,k)$, which Exercise 51.1(i) at $m_1=m_2=s=t=k$ gives:

$$\langle1\rangle_k=\Gamma_{\,1}^{2}\!\begin{bmatrix}k,\ k\\ 2k\end{bmatrix}=\frac{((k-1)!)^2}{(2k-1)!}:\qquad 1,\ \tfrac16,\ \tfrac{1}{30},\ \tfrac{1}{140},\ \tfrac{1}{630},\ \tfrac{1}{2772}\quad(k=1,\dots,6).$$

By the shift (Proposition 1.3(iii)) $\langle1\rangle_{k+1}/\langle1\rangle_k=k^2/\big(2k(2k+1)\big)=k/\big(2(2k+1)\big)$, the ratios $\frac16,\frac15,\frac{3}{14},\frac29,\frac{5}{22}$, so

$$2(2k+1)\,\langle1\rangle_{k+1}-k\,\langle1\rangle_k=0,$$

a recurrence of order one. Its symbol, read off the coefficients of $k$, is $4\lambda-1$, which vanishes at the native value $\frac14$, as Proposition 38.9 requires; and the ratios tend to $\frac14$.

*The integrand at step k.* At $m_1=s=t=k$ the shares of Exercise 51.1(i) are $a_1=b_1=a$ and $a_2=b_2=k-a$, and the integrand is $\Gamma(a)^2\Gamma(k-a)^2/\Gamma(k)^2$. The shift applied $k-1$ times to $\Gamma(k-a)$ and to $\Gamma(k)$ gives

$$\Gamma_{\,2}^{4}\!\begin{bmatrix}a,\ a,\ k-a,\ k-a\\ k,\ k\end{bmatrix}=\Gamma_{\,0}^{4}\big[a,\ a,\ 1-a,\ 1-a\big]\left(\frac{(1-a)(2-a)\cdots(k-1-a)}{(k-1)!}\right)^2:$$

at $k=2,3,4$ the factor is $(a-1)^2$, $\frac14(a-1)^2(a-2)^2$ and $\frac{1}{36}(a-1)^2(a-2)^2(a-3)^2$. The step multiplies the kernel of step 1 by the square of a polynomial in the share, which adds zeros at $a=1,\dots,k-1$ and no poles; the values are rational numbers.

(ii) *Closings.* The origin is integral and both letters have the exponent 1 in the direction, so Definition 39.30 applies, with $\omega_0/dy = 1$ and $\mathsf{C}_j(1) = \mathrm{CT}(f^{-j})$ (Corollary 39.36). On the chart $f^{-j} = x^{-j}(1+x)^{2j}$, whose constant term is the coefficient of $x^j$ in $(1+x)^{2j}$:

$$\mathsf{C}_j(1) = \binom{2j}{j}: \qquad 1,\ 2,\ 6,\ 20,\ 70,\ 252,\ 924,\ 3432 \quad (j = 0, \dots, 7).$$

*The volume.* The two orders of $\mathcal{E}$ have the tails $U_2 = \{2\}$ and $U_2 = \{1\}$, and the direction walls are $\kappa_{1,\{e\}} = \sigma_1 c(\{e\}) = 1$ (Proposition 8.21(ii); $c(\{e\}) = r(\{e\}) + r(\mathcal{E} \setminus \{e\}) - r = 1$ by Definition 7.12 and Theorem 7.13). So $V_1 = 1 + 1 = 2$ (Proposition 39.34).

*The principal parts.* $\langle 1 \rangle_k = \Gamma(k)^2/\Gamma(2k)$ continues meromorphically in $k$. At $k = -j$, $j \in \mathbb{N}$, the residues of Proposition 1.3(iv) give $\Gamma(k) = \frac{(-1)^j}{j!}(k+j)^{-1} + O(1)$ and, the pole of $\Gamma$ at $2k = -2j$ having the residue $\frac{1}{(2j)!}$, $\Gamma(2k) = \frac{1}{2(2j)!}(k+j)^{-1} + O(1)$. So the pole has order $1 = n - 1$ and

$$\lim_{k \to -j} (k+j)\, \langle 1 \rangle_k = \frac{(j!)^{-2}}{(2(2j)!)^{-1}} = 2\binom{2j}{j} = V_1\, \mathsf{C}_j(1): \qquad 2,\ 4,\ 12,\ 40,\ 140,\ 504 \quad (j = 0, \dots, 5),$$

Proposition 39.34 at every $j$.

*The reflected recurrence.* Corollary 39.60 reads the recurrence of (i), $p_0(k) = -k$ and $p_1(k) = 2(2k+1)$, at $k = -j$ on the closings: $j\, \mathsf{C}_j(1) + 2(1-2j)\, \mathsf{C}_{j-1}(1) = 0$, that is $\mathsf{C}_j(1) = \frac{2(2j-1)}{j}\, \mathsf{C}_{j-1}(1)$ for $j \geq 1$, which $\binom{2j}{j}$ satisfies; at $j = 0$ the sum is $p_0(0)\, \mathsf{C}_0(1) = 0$, as in Corollary 39.36.

(iii) *The matrix $H$.* In Theorem 38.15, $\alpha^\star = \mathbf{1}$, $e_0 = 2$, and $s_{11} = t_{11} = 1$, so $H$ is the $1 \times 1$ matrix $\kappa_{\mathbf{1}}(X_1, X_1) + \kappa_{\mathbf{1}}(X_1, X_1)$, the cobasis law at $(\alpha^\star)^{-1} = \mathbf{1}$ being the weight law (Theorem 7.16). Under the weight law, uniform on $\{1\}, \{2\}$ (Definition 7.17), $X_1$ is 1 with the probability $\frac{1}{2}$ and has the variance $\frac{1}{4}$; so $H = \frac{1}{2}$. Directly: $-\log f = 2\log(1 + e^y) - y$ on the chart $x = e^y$, with the second derivative $2e^y/(1+e^y)^2 = \frac{1}{2}$ at $y = 0$.

*The product.* For every $k \geq 1$,

$$k\, \langle 1 \rangle_k\, \mathsf{C}_k(1) = k\, \frac{((k-1)!)^2}{(2k-1)!} \cdot \frac{(2k)!}{(k!)^2} = k \cdot \frac{2k}{k^2} = 2 = (\det H)^{-1}.$$

*The leading terms.* The native gauge $\mathbf{1}$ is nondegenerate (Theorem 39.57), and near it $\log f = \log \frac{1}{4} - \frac{1}{2}v^2$ with $v = H^{1/2}y + O(y^2)$, so $(\omega_0/dv)(\mathbf{1}) = (dy/dv)(0) = H^{-1/2} = \sqrt{2}$. Theorem 38.6(i) gives $\langle 1 \rangle_k \sim 4^{-k} k^{-1/2} a_0(1)$ with $a_0(1) = (2\pi)^{1/2}\sqrt{2} = 2\sqrt{\pi}$, and Theorem 39.57, in which $\hat{A}_\star(1)$ is that expansion divided by $f(\mathbf{1})^k k^{-1/2} = 4^{-k} k^{-1/2}$, with the leading term $a_0(1)$, gives $\mathsf{C}_j(1) \sim 4^j j^{-1/2} (2\pi)^{-1} 2\sqrt{\pi} = 4^j (\pi j)^{-1/2}$. Their product at $j = k$ is $k^{-1}\, a_0(1)^2/(2\pi) = 2k^{-1}$ to the leading order, $(\det H)^{-1} k^{-(n-1)}$; here the product is $2k^{-1}$ exactly, at every $k$.

*Answer.* The stride is $\alpha_1\alpha_2/(\alpha_1 + \alpha_2)^2$, with the native gauge $\mathbf{1}$ and the native value $\frac{1}{4}$; the values are $\langle 1 \rangle_k = \Gamma \begin{bmatrix} k,\ k \\ 2k \end{bmatrix}_1^2$, with $2(2k+1)\langle 1 \rangle_{k+1} = k \langle 1 \rangle_k$, and the integrand at step $k$ is the one at step 1 times $\big((1-a)_{k-1}/(k-1)!\big)^2$. The closings are $\mathsf{C}_j(1) = \binom{2j}{j}$, $V_1 = 2$, and $\lim_{k \to -j}(k+j)\langle 1 \rangle_k = 2\binom{2j}{j}$ at every $j$. $H = \frac{1}{2}$, and $k\, \langle 1 \rangle_k\, \mathsf{C}_k(1) = 2 = (\det H)^{-1}$ at every $k \geq 1$. □

## 51.2 The pins of the atom formula

**Exercise 51.3** (The pins of the atom formula of six lists)**.** For a list $(\mathcal{E}, \mathcal{B})$ on $n$ marks at data on the balance wall, the atom formula (Corollary 18.16) is a fibre integral over the shares $a_T, b_T$,

$T \in \mathcal{B}$, pinned by $|\mathbf{a}| = s$, $|\mathbf{b}| = t$ and $\langle \boldsymbol{\delta} \rangle_e = \mu_e$ for $e \neq g$. For the six lists $\binom{[2]}{1}$, $\binom{[3]}{1}$, $\binom{[3]}{2}$, $\binom{[4]}{2}$, $\binom{[4]}{3}$ and $\binom{[5]}{2}$, with $g = n$:

(i) write the pins with integer coefficients;

(ii) find the elementary divisors of their matrix;

(iii) decide for which of the six lists, at integer data, every point of the fibre at which the shares outside $n+1$ shares with independent columns are integers has all its shares integers, and for the others give such a point with a share that is not an integer.

*Solution.* Receipt: `ex02_pins.py`.

(i) By Definition 12.24 the load at a mark is $\text{load}_e = \sum_{T \ni e} a_T + \sum_{T \not\ni e} b_T = \sigma + \langle \boldsymbol{\delta} \rangle_e$, with $\sigma = \frac{1}{2}(|\mathbf{a}| + |\mathbf{b}|)$. So, with the two letter totals, the pins $\langle \boldsymbol{\delta} \rangle_e = \mu_e$ are the pins $\text{load}_e = m_e$, $e \neq g$, which have the coefficients 0 and 1. Passing from the one set of rows to the other adds to the row of $e$ half the sum of the two rows of the totals: a triangular change with determinant 1, which leaves the delta function of Definition 4.1, and so the fibre integral, unchanged. The matrix of the pins, $M$, has $n+1$ rows, the two totals and the loads at $e \neq g$, and $2|\mathcal{B}|$ columns:

$$\text{column of } a_T: \ \big(1,\ 0,\ (\mathbf{1}_T(e))_{e \neq g}\big), \qquad \text{column of } b_T: \ \big(0,\ 1,\ (\mathbf{1}_{\mathcal{E} \setminus T}(e))_{e \neq g}\big).$$

(ii) An integer matrix of rank $\rho$ has integer matrices $U$, $V$ of determinant $\pm 1$ with $UMV$ diagonal, with the entries $d_1 \mid d_2 \mid \cdots \mid d_\rho$ and zeros, and $d_1 \cdots d_k$ is the greatest common divisor of the $k \times k$ minors of $M$ (the Smith normal form of an integer matrix, which applies here because the pins have integer coefficients). So the elementary divisors are all 1 as soon as one maximal minor is $\pm 1$, which also gives the rank $n+1$. With the rows in the order (total of $K$, total of $\Psi$, loads at $1, \ldots, n-1$), these minors are $\pm 1$:

$\binom{[2]}{1}$, $(a_1, a_2, b_1)$

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

$\det = 1$

$\binom{[3]}{1}$, $(a_1, a_2, a_3, b_1)$

$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

$\det = 1$

$\binom{[3]}{2}$, $(a_{12}, a_{13}, a_{23}, b_{12})$

$$\begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

$\det = -1$

$\binom{[4]}{3}$, $(a_{123}, a_{124}, a_{134}, a_{234}, b_{123})$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 \end{pmatrix}$$

$\det = -1$

$\binom{[4]}{2}$, $(a_{12}, a_{13}, a_{14}, a_{23}, b_{12})$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}$$

$\det = 1$

$\binom{[5]}{2}$, $(a_{12}, a_{13}, a_{14}, a_{15}, a_{23}, b_{12})$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}$$

$\det = -1$

So every elementary divisor is 1 for the six lists; the receipt computes the Smith normal forms, $\text{diag}(1, \ldots, 1)$ in each case. Then $UMV = (I \mid 0)$: for every integer right side $(s, t, (m_e)_{e \neq g})$ the pins have integer solutions, and the integer points of the fibre form a translate of the integer vectors of the kernel of $M$, a lattice of rank $2|\mathcal{B}| - n - 1$, the dimension of the fibre.

(iii) Fix a set $J$ of $n+1$ shares whose columns have the minor $\det M_J \neq 0$, and give the other shares integer values. The pins then fix the shares of $J$ by Cramer's rule, with the denominator $\det M_J$. The absolute values of all maximal minors, with their numbers, are

| | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $\binom{[2]}{1}$ | 0 | 4 | 0 | 0 |
| $\binom{[3]}{1}$, $\binom{[3]}{2}$ | 3 | 12 | 0 | 0 |
| $\binom{[4]}{3}$ | 24 | 32 | 0 | 0 |
| $\binom{[4]}{2}$ | 288 | 480 | 24 | 0 |
| $\binom{[5]}{2}$ | 17940 | 19170 | 1620 | 30 |

so for $\binom{[2]}{1}$, $\binom{[3]}{1}$, $\binom{[3]}{2}$ and $\binom{[4]}{3}$ every such point at integer data has integer shares. For $\binom{[4]}{2}$ the minor on $(a_{12}, a_{34}, b_{13}, b_{14}, b_{24})$ is

$$\det \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix} = 2,$$

and at the centre $(\mathbf{1}; 1, 1)$ with the other seven shares 0 the pins $a_{12} + a_{34} = 1$, $b_{13} + b_{14} + b_{24} = 1$, $a_{12} + b_{24} = 1$ (the load at 1), $a_{12} + b_{13} + b_{14} = 1$ (at 2) and $a_{34} + b_{14} + b_{24} = 1$ (at 3) give

$$a_{12} = a_{34} = b_{13} = b_{24} = \tfrac{1}{2}, \qquad b_{14} = 0.$$

For $\binom{[5]}{2}$ the minor on $(a_{12}, a_{13}, a_{45}, b_{14}, b_{15}, b_{23})$ is

$$\det \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{pmatrix} = -3,$$

and at the centre $(\mathbf{1}; 1, 1)$ with the other fourteen shares 0 the pins give $a_{12} = a_{13} = a_{45} = b_{14} = b_{15} = b_{23} = \frac{1}{3}$: every total and every load is $3 \cdot \frac{1}{3} = 1$.

*Answer.* The pins are the two letter totals and the loads $\mathrm{load}_e = m_e$, $e \neq g$, a 0/1 matrix with $n+1$ rows and $2|\mathcal{B}|$ columns. For all six lists its elementary divisors are 1: at integer data the integer points of the fibre form a lattice of full rank in it. For $\binom{[2]}{1}$, $\binom{[3]}{1}$, $\binom{[3]}{2}$ and $\binom{[4]}{3}$ every maximal minor is 0 or $\pm 1$, and every point with integer shares outside $n+1$ independent ones has integer shares. For $\binom{[4]}{2}$ and $\binom{[5]}{2}$ some maximal minors are $\pm 2$, and $\pm 3$ for $\binom{[5]}{2}$, and at the integer data $(\mathbf{1}; 1, 1)$ the points displayed have the seven, respectively fourteen, shares outside such a minor at the integer 0 and shares $\frac{1}{2}$, respectively $\frac{1}{3}$, on it. So the pins of these two atom formulas are not unimodular in the sense of the paragraph after Theorem 20.59: a closing (Theorem 4.5) that pins the shares outside such a minor at poles of their Gammas can reach shares that are not integers, even at integer data, although the elementary divisors are 1. □

## 51.3 Foreman trees

**Exercise 51.4** (The Foreman trees of $\binom{[4]}{1}$)**.** Let $F = \big(\binom{[4]}{1}; s, t\big)$, the list on $\mathcal{E} = \{1,2,3,4\}$ whose bases are its singletons, read at its realizations $\Upsilon_{\sigma,\delta;\mathbf{1}}$ at the table's gauge, $s = \sigma + \delta$, $t = \sigma - \delta$ (Definition 9.1). Find its classes at the centre and at every lean, and its three Foreman trees (Theorem 14.6).

*Solution.* Receipt: `ex03_tree_u14.py`.

*The letters and the molecule.* $K = \alpha_1+\alpha_2+\alpha_3+\alpha_4$ and $\Psi = \alpha_1\alpha_2\alpha_3+\alpha_1\alpha_2\alpha_4+\alpha_1\alpha_3\alpha_4+\alpha_2\alpha_3\alpha_4$ (Definition 7.1). Every nonempty $S$ has $r(S) = 1$, so every proper nonempty $S$ has $c(S) = r(S)+r(\mathcal{E}\setminus S)-1 = 1 > 0$ (Definition 7.12, Theorem 7.13): the family is connected (Definition 8.1) and one molecule.

*The masses.* Each mark lies in one of the four bases, so the occupation at the table's gauge is $R_e(\mathbf{1}) = \frac{1}{4}$ (Theorem 7.16), and Proposition 9.3 gives the realization $\Upsilon_{\sigma,\delta;\mathbf{1}}$ the masses

$$\tau_e = \sigma + \delta\big(2 \cdot \tfrac{1}{4} - 1\big) = \sigma - \tfrac{\delta}{2} \qquad \text{at every mark,}$$

with the total $4\sigma - 2\delta = s \cdot 1 + t \cdot 3$, on the balance wall.

*The writings.* By Proposition 10.2, for a mark $e$ and an order $\mathbf{g} = (g_1, g_2, g_3)$ of its room, the pattern of the basis $\{h\}$ is the position of $h$, and at the table's gauge each pattern of one position has the weight 1:

$$K_{e,\mathbf{g}}(x) = x_0 + x_1 + x_2 + x_3, \qquad \Psi_{e,\mathbf{g}}(x) = x_1x_2x_3 + x_0x_2x_3 + x_0x_1x_3 + x_0x_1x_2,$$

the same two polynomials for every mark and every order (the receipt finds one pair among the $4 \cdot 6$ writings). So one pair of functions, $\Phi^{\mathbf{E}}_{e,\mathbf{g}} = \sigma \log(K_{e,\mathbf{g}}\Psi_{e,\mathbf{g}})$ and $\Phi^{\mathbf{O}}_{e,\mathbf{g}} = \delta \log(K_{e,\mathbf{g}}/\Psi_{e,\mathbf{g}})$ with $x_i = e^{v_i}$, is written from every mark. With no partner, the reading of $e$ (Definition 10.1 at $k = 0$) is the pair of $K = e^v + 3$ and $\Psi = 3e^v + 1$ at every mark.

*The classes.* By Proposition 10.4 the reading is constant on $\mathcal{E}$ at every lean $\delta$, the centre included: $\mathrm{Cl}(\Upsilon_{\sigma,\delta;\mathbf{1}}) = \{\mathcal{E}\}$. Corollary 11.16 says the same: all 24 permutations of $\mathcal{E}$ carry the table, mapping singletons to singletons, and they take every mark to every other; none exchanges it, a basis having one mark and a cobasis three.

*The trees.* Every realization is atomic (Definition 10.1), so by Definition 14.4 the molecule is an atom and its Foreman tree is that one leaf, with no bunch and no peel. Theorem 14.6 gives the list at its one pair the three trees $\mathrm{Tree}(F;\sigma,0)$ and $\mathrm{Tree}(F;\sigma,\pm\delta)$, and each is this atom. The molecule answers yes and the label is $(1,1)$ (Definition 11.7).

*Answer.* At the centre and at every lean the one class is $\mathcal{E}$; the three Foreman trees of $\binom{[4]}{1}$ are one atom, with no bunch. □

**Exercise 51.5** (A tree that peels at a bunch of width one)**.** Let $\mathrm{P}_5$ be the list on $\{1,\dots,5\}$ whose bases are the nine pairs other than $\{1,2\}$, read at its realizations $\Upsilon_{\sigma,\delta;\mathbf{1}}$. Find its classes and its Foreman bunch at the centre and at a lean, the width of the bunch, its patterns with their probabilities and children, its record and its peel, and its three Foreman trees. Write $\mathrm{P}_5$ as a substitution through the two cells of Theorem 19.4, and show that the contraction of the cell of the room is the dual of its deletion.

*Solution.* Receipt: `ex04_five_marks.py`.

*The letters.* With $e_k$ the elementary symmetric polynomials of $\alpha$, the rank is 2, the corank 3, and

$$K = e_2(\alpha) - \alpha_1\alpha_2, \qquad \Psi = e_3(\alpha) - \alpha_3\alpha_4\alpha_5.$$

*The classes.* A bijection of the marks carries $\mathrm{P}_5$ exactly when it maps the one pair that is not a basis, $\{1,2\}$, to itself: these are the 12 products of a permutation of $\{1,2\}$ and one of $\{3,4,5\}$, with the orbits $\{1,2\}$ and $\{3,4,5\}$. No bijection exchanges $\mathrm{P}_5$, a basis having two marks and a cobasis three. By Corollary 11.16 the classes at the centre and at every lean are

$$\mathrm{Cl} = \big\{\{1,2\},\ \{3,4,5\}\big\}.$$

The occupations confirm the split at the centre: $R_1(\mathbf{1}) = R_2(\mathbf{1}) = \frac{3}{9} = \frac{1}{3}$ and $R_3(\mathbf{1}) = R_4(\mathbf{1}) = R_5(\mathbf{1}) = \frac{4}{9}$, and two marks of one class at the centre have equal or complementary occupations (Proposition 10.9(ii)), while $\frac{4}{9} \notin \{\frac{1}{3}, \frac{2}{3}\}$.

*The Foreman bunch.* By Definition 13.6 the Foreman bunch at the centre is the class with the fewest marks, $B = \{1,2\}$; at a lean the key $(|C|, |C^\circ|, C^\circ, C)$ is least at $C = \{1,2\}$, of two marks, so $B(\mathrm{P}_5; \sigma, \pm\delta) = \{1,2\}$ as well. Its window is $\{0,1\}$ (a basis meets $B$ in at most one mark, and $\{3,4\}$ avoids it), so $d(B) = 0$, $r(B) = 1$ and

$$c(B) = r(B) + r(R) - r = 1 + 2 - 2 = 1$$

(Definition 7.12, Theorem 7.13), $R = \{3,4,5\}$; no basis contains both of its marks, so $B$ is a parallel bunch (Definition 19.1).

*Patterns, probabilities, children.* The patterns of $B$ are $\varnothing$, $\{1\}$, $\{2\}$ (Definition 12.1), each the trace of three of the nine bases, so each has the probability $\pi_p(\mathbf{1}) = \frac{1}{3}$ at the table's gauge (Definition 12.25). The children on $R$ are

$$\mathcal{B}_{(\varnothing)} = \{34, 35, 45\} = \binom{R}{2}, \qquad \mathcal{B}_{(\{1\})} = \mathcal{B}_{(\{2\})} = \{3,4,5\} = \binom{R}{1},$$

with $K_{\binom{R}{2}} = e_2(\alpha_R)$, $\Psi_{\binom{R}{2}} = e_1(\alpha_R)$, $K_{\binom{R}{1}} = e_1(\alpha_R)$ and $\Psi_{\binom{R}{1}} = e_2(\alpha_R)$.

*Record and peel.* The record (Definition 14.2) is the three indexed children $\big(p; \mathcal{B}_{(p)}; \frac{s}{3}, \frac{t}{3}\big)$, $p = \varnothing, \{1\}, \{2\}$. The peel (Definition 14.1) merges the two equal children:

$$\Big(\binom{R}{2}; \tfrac{s}{3}, \tfrac{t}{3}\Big), \quad \Big(\binom{R}{1}; \tfrac{2s}{3}, \tfrac{2t}{3}\Big), \qquad \text{word} \quad e_1(\alpha_R)^{-\frac{2s+t}{3}}\, e_2(\alpha_R)^{-\frac{s+2t}{3}}.$$

At the centre, $s = t = \sigma$, the word is $\big(e_1(\alpha_R)e_2(\alpha_R)\big)^{-\sigma}$, the word of the list $\binom{R}{1}$ at its own centre, and the gauge $\mathbf{1}_R$ is native for the masses $\sigma\mathbf{1}_R$ (Proposition 12.27); the balance is $\frac{\sigma}{3}(2+1) + \frac{2\sigma}{3}(1+2) = 3\sigma$.

*The trees.* The peel is one molecule, the circuit $R$ of $\binom{R}{2}$ linking its marks (Definition 7.4), and every permutation of $R$ carries both of its tables, so it reads every mark alike (Proposition 11.14) and is an atom at every lean (Definition 10.1). So the three Foreman trees (Theorem 14.6) have one shape: the root bunch $\{1,2\}$ of width one and the room $\{3,4,5\}$, an atom, at the exponents of the peel.

*The cells.* The weights are 1 and factor across $B$ and $R$. The minors (Definition 7.5) are

$$\mathrm{P}_5|B = \binom{B}{1}, \qquad \mathrm{P}_5/R = \{\varnothing\}, \qquad \mathrm{P}_5|R = \binom{R}{2}, \qquad \mathrm{P}_5/B = \binom{R}{1},$$

$\mathrm{P}_5/R$ taking the bases inside $R$ and $\mathrm{P}_5/B$ those meeting $B$. The cell of the bunch (Theorem 19.4) has the bases of $\mathrm{P}_5|B$ and $\varnothing + g$: it is $\binom{\{1,2,g\}}{1}$. The cell of the room has the bases of $\binom{R}{2}$ and $\{x\} + g$, $x \in R$: it is $\mathcal{A} = \binom{\{3,4,5,g\}}{2}$. The hinge values and their ratio (Definition 19.2) are

$$\gamma_B^K = \frac{K_{\mathrm{P}_5|B}}{K_{\mathrm{P}_5/R}} = \alpha_1 + \alpha_2, \qquad \gamma_B^\Psi = \frac{\Psi_{\mathrm{P}_5/R}}{\Psi_{\mathrm{P}_5|B}} = \frac{\alpha_1\alpha_2}{\alpha_1 + \alpha_2}, \qquad \varrho_B = \frac{\alpha_1\alpha_2}{(\alpha_1+\alpha_2)^2} = u(1-u)$$

on the shape $(u, 1-u)$, as for every parallel pair (Proposition 19.3).

*The substitution.* By Proposition 21.3(iii) $\mathrm{P}_5$ is the gluing of the two cells along $g$: in Definition 21.1,

$$\mathrm{P}_5 = \binom{\{1,2,h\}}{1}\langle 1,2,\mathcal{A}\rangle, \qquad \mathcal{A} = \binom{\{3,4,5,g\}}{2} \text{ at } h,$$

the marks $1, 2$ left in place. Checked on the bases: the core bases $\{1\}$ and $\{2\}$ avoid $h$ and take $X \in \mathcal{A}/g = \binom{R}{1}$, giving the six pairs $\{1,x\}$, $\{2,x\}$; the core basis $\{h\}$ takes $X \in \mathcal{A} \setminus g = \binom{R}{2}$, giving the three pairs inside $R$. These are the nine bases.

*The dual.* $\mathcal{A} \setminus g = \binom{R}{2}$ and $\mathcal{A}/g = \binom{R}{1}$, whose bases are the complements in $R$ of the pairs: $\mathcal{A}/g$ is the dual table of $\mathcal{A} \setminus g$, with the weights 1. So $\varrho_{\mathcal{A}} \equiv 1$ (Corollary 23.12(ii)): $\gamma_{\mathcal{A}}^{K} = e_2(\alpha_R)/e_1(\alpha_R) = \gamma_{\mathcal{A}}^{\Psi}$.

*Answer.* The classes are $\{1,2\}$ and $\{3,4,5\}$ at the centre and at every lean, and the Foreman bunch is $\{1,2\}$, a parallel bunch of width one. Its patterns $\varnothing, \{1\}, \{2\}$ have the probability $\frac{1}{3}$ each and the children $\binom{R}{2}, \binom{R}{1}, \binom{R}{1}$; the peel is $(\binom{R}{2}; \frac{s}{3}, \frac{t}{3})$ with $(\binom{R}{1}; \frac{2s}{3}, \frac{2t}{3})$, at the centre the word $(e_1e_2)^{-\sigma}$ on $R$, an atom. The three Foreman trees are the bunch $\{1,2\}$ over that atom. And $\mathrm{P}_5 = \binom{\{1,2,h\}}{1}\langle 1,2,\binom{\{3,4,5,g\}}{2}\rangle$, the cell of the room having its contraction $\binom{R}{1}$ dual to its deletion $\binom{R}{2}$, with the ratio 1. □

**Exercise 51.6** (A Foreman bunch of width two)**.** Let $\mathrm{T}_6$ be the list of rank three on $\{1,\dots,6\}$ whose bases are the nineteen sets of three marks other than $\{1,2,3\}$. Show that it is prime (Theorem 23.16); find its classes at the centre and at a lean, its Foreman bunches at a basis lean and at a cobasis lean, and their widths; and draw its three Foreman trees.

*Solution.* Receipt: `ex05_six_marks.py`.

*Ranks and widths.* Every set of at most three marks other than $\{1,2,3\}$ is independent, $r(\{1,2,3\}) = 2$, and every set of at least four marks contains a basis and has rank 3. With $c(S) = r(S) + r(\mathcal{E} \setminus S) - 3$ (Definition 7.12, Theorem 7.13):

| $S$ | $c(S)$ |
|---|---|
| one mark | $1+3-3=1$ |
| two marks | $2+3-3=2$ |
| three marks, $S \neq \{1,2,3\}, \{4,5,6\}$ | $3+3-3=3$ |
| $\{1,2,3\}$ or $\{4,5,6\}$ | $2+3-3=2$ |
| four marks | $3+2-3=2$ |

Every proper nonempty set has positive width, so the list is connected (Definition 8.1); the 50 sets with at least two marks on each side have the widths 2 (32 of them) and 3 (18), so none has width one: $\mathrm{T}_6$ is prime (Theorem 23.16).

*Classes.* A bijection carries $\mathrm{T}_6$ exactly when it maps the one triple that is not a basis, $\{1,2,3\}$, to itself: the 36 products of a permutation of $\{1,2,3\}$ and one of $\{4,5,6\}$, with the orbits $C = \{1,2,3\}$ and $D = \{4,5,6\}$. The cobases are the triples other than $\{4,5,6\}$, so a bijection exchanges $\mathrm{T}_6$ exactly when it maps $\{1,2,3\}$ onto $\{4,5,6\}$, as $i \mapsto i+3$ modulo 6 does: 36 more bijections, and the 72 together take every mark to every other. By Corollary 11.16, at the centre the one class is $\mathcal{E}$, an atom, and at every lean the classes are $C$ and $D$, with the occupations $R_e(\mathbf{1}) = \frac{9}{19}$ on $C$ and $\frac{10}{19}$ on $D$ (Proposition 10.9(ii)).

*The Foreman bunch at a lean.* Both classes have three marks and lie in the one class $\mathcal{E}$ of the centre, so the key of Definition 13.6 is decided by the order $\prec$ at the lean. $C$ and $D$ make up the molecule, so their levels are mirrored (Proposition 13.10, with $k_j = r_j = 3$). Under the weight law,

uniform on the 19 bases (Definition 7.17), the level $N_C = |T \cap C|$ is 0 at $\{4,5,6\}$, 1 at the 9 bases with one mark of $C$, and 2 at the 9 with two:

$$\mathbb{P}(N_C = 0,1,2) = \tfrac{1}{19},\ \tfrac{9}{19},\ \tfrac{9}{19}; \qquad \mathbb{E}N_C = \tfrac{27}{19}.$$

The deviations $-\frac{27}{19}, -\frac{8}{19}, \frac{11}{19}$ give the cumulants

$$\kappa_{\mathbf{1}}(N_C, N_C) = \frac{9\cdot 64 + 9\cdot 121 + 729}{19^3} = \frac{126}{361},$$
$$\kappa_{\mathbf{1}}(N_C^{\times 3}) = \frac{-19683 - 9\cdot 512 + 9\cdot 1331}{19^4} = -\frac{648}{6859}.$$

The least odd order $a \geq 3$ with $\delta\,\kappa_{\mathbf{1}}(N_C^{\times a}) \neq 0$ is $a = 3$, so by Proposition 13.10(ii) $C \prec D$ exactly when $\delta\,\kappa_{\mathbf{1}}(N_C^{\times 3}) < 0$, that is at a basis lean, $\delta > 0$, and $D \prec C$ at a cobasis lean. (In Corollary 13.4(i) the coefficient of $v_0^3$ in the odd part of stage 0 is $\frac{2}{3!}\delta\,\kappa_{\mathbf{1}}(N_C^{\times 3}) = -\frac{216}{6859}\delta$ for $C$ and $+\frac{216}{6859}\delta$ for $D$, the even parts being equal.) So

$$B(\mathrm{T}_6;\sigma,\delta) = \{1,2,3\} \quad (\delta > 0), \qquad B(\mathrm{T}_6;\sigma,\delta) = \{4,5,6\} \quad (\delta < 0),$$

each of width 2, and the exchange $i \mapsto i+3$ maps the one onto the other (Theorem 13.7(iii)).

*The trees.* At the centre the molecule is an atom (Definition 14.4): one leaf. At a basis lean the root bunch is $C$, with the patterns (Definition 12.1) and probabilities (Definition 12.25)

| $p$ | $\pi_p(\mathbf{1})$ | child on $R = \{4,5,6\}$ |
|---|---|---|
| $\varnothing$ | $\frac{1}{19}$ | the one basis $R$ |
| $\{b\},\ b \in C$ | $\frac{3}{19}$ each | $\binom{R}{2}$ |
| $\{b,b'\} \subset C$ | $\frac{3}{19}$ each | $\binom{R}{1}$ |

$C$ itself not being a pattern. The peel (Definition 14.1) is the table with the one basis $R$ at $(\frac{s}{19}, \frac{t}{19})$, $\binom{R}{2}$ at $(\frac{9s}{19}, \frac{9t}{19})$ and $\binom{R}{1}$ at $(\frac{9s}{19}, \frac{9t}{19})$; the circuit $R$ of $\binom{R}{2}$ makes $R$ one molecule, and every permutation of $R$ carries the three tables, so the room is an atom (Proposition 11.14). At a cobasis lean the root bunch is $D$, with the patterns $\{x\}$ (child $\binom{C}{2}$), $\{x,y\}$ (child $\binom{C}{1}$), each of probability $\frac{3}{19}$, and $D$ (child: the one basis $\varnothing$ on $C$) of probability $\frac{1}{19}$, $\varnothing$ not being a pattern; the room $C$ is again an atom. The exchange $i \mapsto i+3$ maps the tree at $\delta$ onto the tree at $-\delta$ (Theorem 14.8).

*Answer.* $\mathrm{T}_6$ is connected and prime. At the centre it is an atom. At a lean its classes are $\{1,2,3\}$ and $\{4,5,6\}$, and its Foreman bunch is $\{1,2,3\}$ at a basis lean and $\{4,5,6\}$ at a cobasis lean, both of width two. Its three Foreman trees are one atom and two trees whose root bunch, of width two, stands over a room of three marks that is an atom. □

**Exercise 51.7** (What a bunch of width two shows, and what splits off)**.** For the list $\mathrm{T}_6$ of Exercise 51.6 at its centre:

(i) find the levels of $K$ along the scale of $C = \{1,2,3\}$, the Skeleton polynomial of $C$ at the table's gauge, its discriminant and the Skeleton values there, and show that the Skeleton values along $C$ are not rational functions of the point;

(ii) show that no set with at least two marks on each side and no complement of a single mark splits off (Theorem 28.25).

*Solution.* Receipt: `ex05_six_marks.py`.

(i) *The levels.* $d(C) = 0$, the basis $\{4,5,6\}$ avoiding $C$, and $r(C) = 2$. By the Taylor cube (Theorem 12.7) and Definition 12.16, the level 0 is the pattern $\varnothing$ with the child of the one basis $R$,

the level 1 the patterns $\{b\}$ with the child $\binom{R}{2}$, and the level 2 the patterns $\{b, b'\}$ with the child $\binom{R}{1}$:

$$K(x\alpha_C, \alpha_R) = \alpha_4\alpha_5\alpha_6 + x\, e_1(\alpha_C)\, e_2(\alpha_R) + x^2\, e_2(\alpha_C)\, e_1(\alpha_R),$$

each level a product of a bunch letter and a room letter.

*At the table's gauge.* $K(\mathbf{1}) = 19$, and the Skeleton polynomial (Definition 12.10) is

$$\mathsf{Q}^K_C(x; \mathbf{1}) = \frac{K(x\mathbf{1}_C, \mathbf{1}_R)}{K(\mathbf{1})} = \frac{1 + 9x + 9x^2}{19},$$

of degree $c(C) = 2$. The discriminant of $9x^2 + 9x + 1$ is $81 - 36 = 45$, its roots are $x = \frac{-9 \pm 3\sqrt{5}}{18}$, and the Skeleton values are their negatives,

$$\xi_{1,2} = \frac{3 \mp \sqrt{5}}{6},$$

positive and irrational.

*Not rational functions.* The Skeleton values at $\alpha$ are the negatives of the roots of $K(x\alpha_C, \alpha_R)$ in $x$, so

$$(\xi_1 - \xi_2)^2\, e_2(\alpha_C)^2\, e_1(\alpha_R)^2 = \Delta(\alpha) := e_1(\alpha_C)^2 e_2(\alpha_R)^2 - 4\, e_2(\alpha_C)\, e_1(\alpha_R)\, \alpha_4\alpha_5\alpha_6.$$

If $\xi_1, \xi_2$ were rational functions of $\alpha$, $\Delta$ would be the square of a rational function, hence, the polynomial ring being a unique factorization domain, the square of a polynomial, and so would its restriction to every line. On $\alpha = (1, 1, 1, 1, 1, \lambda)$, $e_1(\alpha_C) = e_2(\alpha_C) = 3$, $e_1(\alpha_R) = 2 + \lambda$, $e_2(\alpha_R) = 1 + 2\lambda$ and $\alpha_4\alpha_5\alpha_6 = \lambda$:

$$\Delta = 9(1 + 2\lambda)^2 - 12\lambda(2 + \lambda) = 3\,(8\lambda^2 + 4\lambda + 3),$$

and $8\lambda^2 + 4\lambda + 3$ has the discriminant $16 - 96 = -80$: two simple roots, so $\Delta$ is not a square on this line. The value 45 at the gauge shows only that the Skeleton values are irrational numbers there; the line shows that they are not rational functions of the point.

(ii) *Sets with two marks on each side.* $\mathrm{T}_6$ is prime (Exercise 51.6), and at its centre, masses $\sigma\mathbf{1}$, every proper wall is positive (Proposition 8.21(ii)); so by Corollary 28.26(ii) no set with at least two marks on each side splits off. Each such set has width two or three and enters every period of the list through a law carried by no point (Corollary 28.26(i)).

*Complements of single marks.* $\mathcal{E} \setminus \{e\}$ has the width $c(\{e\}) = 1$ (Theorem 7.13); with both letters active, Corollary 28.26(iv) says it splits off exactly when the deletion and the contraction at $e$ have one even letter, $\mathbf{E}_{\mathrm{T}_6\setminus e} = \mathbf{E}_{\mathrm{T}_6/e}$ (Definition 7.3). For $e \in C$ the deletion has as its bases all ten triples of $\mathcal{E} \setminus \{e\}$ and the contraction the nine pairs $\{x, y\}$ of $\mathcal{E} \setminus \{e\}$ with $\{e, x, y\} \neq \{1, 2, 3\}$; for $e \in D$ the deletion has nine bases and the contraction ten. At the table's gauge $\mathbf{E} = K\Psi$ is the square of the number of bases, so

$$\mathbf{E}_{\mathrm{T}_6\setminus e}(\mathbf{1}) = 100, \quad \mathbf{E}_{\mathrm{T}_6/e}(\mathbf{1}) = 81 \quad (e \in C); \qquad \mathbf{E}_{\mathrm{T}_6\setminus e}(\mathbf{1}) = 81, \quad \mathbf{E}_{\mathrm{T}_6/e}(\mathbf{1}) = 100 \quad (e \in D):$$

the two even letters differ, and no complement of a mark splits off at the centre.

*Answer.* Along $C$ the levels are $\alpha_4\alpha_5\alpha_6$, $e_1(\alpha_C)e_2(\alpha_R)$ and $e_2(\alpha_C)e_1(\alpha_R)$; at the table's gauge $K(x\mathbf{1}_C, \mathbf{1}_R) = 1 + 9x + 9x^2$, with the discriminant 45 and the Skeleton values $\frac{3 \mp \sqrt{5}}{6}$; on the line $(1, 1, 1, 1, 1, \lambda)$ the discriminant is $3(8\lambda^2 + 4\lambda + 3)$, not a square, so the Skeleton values are not rational functions of the point. At the centre no set with at least two marks on each side and no complement of a mark splits off: $\mathrm{T}_6$ is its own and only factor. (At a full lean each complement of a mark does split off, Corollary 28.26(iii).) □

## 51.4 Skeleton values at width two

**Exercise 51.8** (Skeleton values of $\binom{[4]}{2}$ along a pair)**.** For the list $\binom{[4]}{2}$ and its bunch $B = \{1, 2\}$, find the width of $B$, the levels of $K$ along the scale of $B$, the Skeleton polynomial and the Skeleton values at the table's gauge, and show that the Skeleton values are not rational functions of the point. Why does the value of the discriminant at the table's gauge not settle this?

*Solution.* Receipt: `ex06_skeleton_u24.py`.

*The width.* The basis $\{3, 4\}$ avoids $B$ and the basis $\{1, 2\}$ is $B$, so the window of $B$ is $\{0, 1, 2\}$: $d(B) = 0$, $r(B) = 2$ and $c(B) = 2$ (Theorem 7.13).

*The levels.* The patterns are $\varnothing$ (child: the one basis $\{3, 4\}$), $\{1\}$ and $\{2\}$ (child $\binom{\{3,4\}}{1}$, with $K = \alpha_3 + \alpha_4$) and $\{1, 2\}$ (child: the one basis $\varnothing$). By Theorem 12.7 and Definition 12.16,

$$K(\rho\alpha_1, \rho\alpha_2, \alpha_3, \alpha_4) = \alpha_3\alpha_4 + \rho\,(\alpha_1 + \alpha_2)(\alpha_3 + \alpha_4) + \rho^2\alpha_1\alpha_2.$$

The Skeleton polynomial is $\mathsf{Q}_B^K(x; \alpha) = K(x\alpha_B, \alpha_R)/K(\alpha)$ (Definition 12.10), of degree 2, whose roots $-\xi_1, -\xi_2$ have

$$\xi_1 + \xi_2 = \frac{(\alpha_1 + \alpha_2)(\alpha_3 + \alpha_4)}{\alpha_1\alpha_2}, \qquad \xi_1\xi_2 = \frac{\alpha_3\alpha_4}{\alpha_1\alpha_2}.$$

The list is identically self-dual and $d^*(B) = 0$, so $\Psi$ has the same levels along $B$ and the same Skeleton values.

*At the table's gauge.* $K(x\mathbf{1}_B, \mathbf{1}_R) = 1 + 4x + x^2$ and $K(\mathbf{1}) = 6$; the discriminant is $16 - 4 = 12$, and the Skeleton values are $\xi = 2 \pm \sqrt{3}$, irrational.

*What the value at the gauge decides.* $(\xi_1 - \xi_2)^2(\alpha_1\alpha_2)^2 = \Delta(\alpha) := (\alpha_1 + \alpha_2)^2(\alpha_3 + \alpha_4)^2 - 4\alpha_1\alpha_2\alpha_3\alpha_4$, and $\Delta(\mathbf{1}) = 12$ is not the square of a rational number. That shows that the Skeleton values are irrational numbers at the gauge, and no more: if $\Delta$ were $3q(\alpha)^2$ for a polynomial $q$ with rational coefficients, $\xi_{1,2} = \frac{(\alpha_1+\alpha_2)(\alpha_3+\alpha_4) \mp \sqrt{3}\,q}{2\alpha_1\alpha_2}$ would be rational functions of the point, with the value 12 at the gauge.

*Not rational functions.* If $\xi_1, \xi_2$ are rational functions of $\alpha$, $\Delta$ is the square of a rational function, so of a polynomial (unique factorization in the polynomial ring), and so is its restriction to every line: $\Delta$ would be a constant times the square of a polynomial with rational coefficients. On $\alpha = (1, 1, 1, \lambda)$,

$$\Delta = 4(1 + \lambda)^2 - 4\lambda = 4(\lambda^2 + \lambda + 1),$$

whose roots $\frac{-1 \pm \sqrt{-3}}{2}$ are simple. So $\Delta$ is not a constant times a square, and the Skeleton values along $B$ are not rational functions of the point.

*Answer.* $B$ has width two, with the levels $\alpha_3\alpha_4$, $(\alpha_1+\alpha_2)(\alpha_3+\alpha_4)$ and $\alpha_1\alpha_2$; at the table's gauge the Skeleton polynomial is $(1+4x+x^2)/6$ and the Skeleton values are $2 \pm \sqrt{3}$. The discriminant is $4(\lambda^2 + \lambda + 1)$ on the line $(1, 1, 1, \lambda)$, not a constant times a square, so the Skeleton values are not rational functions of the point. The value 12 at the gauge only excludes a rational square, not a constant times a square. At width one, by contrast, the one Skeleton value is the quotient of two hinge values (Corollary 19.5). □

## 51.5 The record of a peel

**Exercise 51.9** (Peeling with and without the record)**.** Let

$$F = \left(\left(\binom{[5]}{2}; 1, 0\right), \left(\binom{[5]}{3}; -1, 0\right), \left(\binom{[5]}{4}; 4, 0\right)\right)$$

on $\mathcal{E} = \{1, \dots, 5\}$, with the masses $3\mathbf{1}$. Show that it is on its balance wall with every proper wall positive and that the table's gauge is native; peel it at $\{1\}$ with the native split of the table's gauge, write its record and its peel, and compare the letters they carry. What changes when every exponent is nonnegative?

*Solution.* Receipt: `ex07_record.py`.

*Walls.* The exponents are $s_j = 1, -1, 4$ and $t_j = 0$; the ranks are $2, 3, 4$. The balance is $\sum_j s_j r_j = 2 - 3 + 16 = 15 = m_{\mathcal{E}}$. In $\binom{\mathcal{E}}{r}$ a set $S$ meets every basis in at least $d(S) = \max\big(0, |S| - (5 - r)\big)$ marks (Theorem 7.13), so the walls $\kappa_S = m_S - \sum_j s_j d_j(S)$ (Definition 8.1) depend on $|S|$ alone:

| $\lvert S\rvert$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $(d_1, d_2, d_3)$ | $(0,0,0)$ | $(0,0,1)$ | $(0,1,2)$ | $(1,2,3)$ | $(2,3,4)$ |
| $\kappa_S$ | 3 | 2 | 2 | 1 | 0 |

every proper wall is positive and $\kappa_{\mathcal{E}} = 0$.

*The native gauge.* On the balance wall the target is $\mathbf{m}$ (Definition 8.17). At the table's gauge the occupation in $\binom{\mathcal{E}}{r}$ is $\frac{r}{5}$ at every mark, so the usage is $1 \cdot \frac{2}{5} - 1 \cdot \frac{3}{5} + 4 \cdot \frac{4}{5} = 3 = m_e$: the gauge $\mathbf{1}$ is native. An exponent is negative, so $F$ has no realization (Definition 9.1) and no Foreman tree (Definition 14.4); its native split (Definition 12.25), its record and its peel are defined at the native gauge $\mathbf{1}$.

*The record.* The patterns of $\{1\}$ in $\binom{\mathcal{E}}{r}$ are $\varnothing$, with the child $\binom{R}{r}$ on $R = \{2, 3, 4, 5\}$ and the probability $\binom{4}{r}/\binom{5}{r} = \frac{5-r}{5}$, and $\{1\}$, with the child $\binom{R}{r-1}$ and the probability $\frac{r}{5}$ (Definitions 12.1 and 12.25). The record (Definition 14.2) is the six indexed children, each with the exponents $\pi_{j,p}(\mathbf{1})(s_j, 0)$:

| table $j$ | $p$ | child | exponents |
|---|---|---|---|
| 1 | $\varnothing$ | $\binom{R}{2}$ | $(\frac{3}{5}, 0)$ |
| 1 | $\{1\}$ | $\binom{R}{1}$ | $(\frac{2}{5}, 0)$ |
| 2 | $\varnothing$ | $\binom{R}{3}$ | $(-\frac{2}{5}, 0)$ |
| 2 | $\{1\}$ | $\binom{R}{2}$ | $(-\frac{3}{5}, 0)$ |
| 3 | $\varnothing$ | $\binom{R}{4}$ | $(\frac{4}{5}, 0)$ |
| 3 | $\{1\}$ | $\binom{R}{3}$ | $(\frac{16}{5}, 0)$ |

On $R$ the children lie on their balance wall at the masses $3\mathbf{1}_R$, $\frac{3}{5} \cdot 2 + \frac{2}{5} \cdot 1 - \frac{2}{5} \cdot 3 - \frac{3}{5} \cdot 2 + \frac{4}{5} \cdot 4 + \frac{16}{5} \cdot 3 = 12$, and their usage at $\mathbf{1}_R$ is 3 at every mark (Theorem 12.26). The record determines $F$ (Proposition 14.3): table $j$ is the gluing of its two children, and their exponents add to 1, $-1$ and 4.

*The peel.* Definition 14.1 merges the children with the same bases, adding their exponents:

$$\Big(\binom{R}{1}; \tfrac{2}{5}, 0\Big), \qquad \Big(\binom{R}{2}; \tfrac{3}{5} - \tfrac{3}{5}, 0\Big) = \Big(\binom{R}{2}; 0, 0\Big), \qquad \Big(\binom{R}{3}; \tfrac{14}{5}, 0\Big), \qquad \Big(\binom{R}{4}; \tfrac{4}{5}, 0\Big).$$

The table $\binom{R}{2}$ has both exponents zero: it is not active, and the peel carries no letter $e_2(\alpha_R)$ (Definition 8.1). The record carries it twice, in the child $(1, \varnothing)$ at the exponent $\frac{3}{5}$ and in the child $(2, \{1\})$ at $-\frac{3}{5}$. The two words agree,

$$e_1(\alpha_R)^{-2/5}\, e_3(\alpha_R)^{-14/5}\, e_4(\alpha_R)^{-4/5},$$

since $e_2^{-3/5} e_2^{3/5} = 1$; what the peel loses is which pattern of which table each share came from, which the record keeps (the paragraph after Proposition 14.3), and with it the letter $e_2(\alpha_R)$ that the tables 1 and 2 hand to the room.

*Nonnegative exponents.* When every $s_j, t_j \geq 0$, every child carries $\pi_{j,p}(\mathbf{1})\,(s_j, t_j)$ with $\pi_{j,p}(\mathbf{1}) > 0$ (Proposition 12.27); a merged exponent is a sum of nonnegative terms, zero only when each term is, so merging loses no letter, and a full lean is handed down full, table by table. This is the scope of the realizations, where the Foreman tree is read.

*Answer.* $F$ is on its balance wall with the walls $3, 2, 2, 1$ at the sets of $1, 2, 3, 4$ marks, and the table's gauge is native. Its record at $\{1\}$ is the six children of the table displayed; its peel is $\binom{R}{1}$, $\binom{R}{3}$, $\binom{R}{4}$ at $\frac{2}{5}$, $\frac{14}{5}$, $\frac{4}{5}$, the child $\binom{R}{2}$ cancelling between the tables 1 and 2: merging loses the letter $e_2$ on the room, which the record keeps. With nonnegative exponents no letter is lost. $\square$

# Chapter 52

# Exercises on evaluation and construction

The exercises of this chapter work the statements of Parts IV and V on particular lists. A set of width two whose stations of room sides are single room sides cannot be opened with one share per station, the room's seam function being infinite at every share, and it is opened with the transverse pins instead. The width of the marks left in place by a substitution is read off the core, bounded by the number of the other cells, and every width occurs. The uniform list of rank two on five marks is fused by the procedure of Foreman fusion: its fusion, its one major, its cousins and the symmetry the fusion adds. Every number displayed is exact; the receipts that recompute them are the scripts named in each solution.

## 52.1 One share per station

**Exercise 52.1** (One share per station at a set of width two)**.** Let $\mathrm{Q}_5$ be the list on $\{1,\dots,5\}$ whose bases are the eight sets of three marks other than $\{1,2,5\}$ and $\{3,4,5\}$, at its centre $(\mathbf{1};1,1)$, and let $S=\{1,2,3\}$, with the room $R=\{4,5\}$.

(i) Find the width of $S$, its patterns and room sides at every excess, its stations, its seam letters and the room station letters (Definition 28.1), and check the room-station factorization.

(ii) Read the period with one share per station (Theorem 28.10(iii)): write the pins of the seam and the room word, and show that the seam function of the room is $+\infty$ at every share.

(iii) Read it with the transverse pins of Theorem 28.12: find the lattice, the pins, the reduced periods and the share side, and check that the native split at the stations lies on the pins, with the masses of $S$ inside the Newton sum of the seam word there.

*Solution.* Receipt: `ex08_transverse.py`.

(i) *The list.* Its circuits (Definition 7.4) are the two triples $\{1,2,5\}$ and $\{3,4,5\}$ that are not bases and the set $\{1,2,3,4\}$, all of whose triples are bases; the rank is 3 and the corank 2. In $S$ every basis has one to three marks: $r(S)=3$, $S$ being a basis, $r(R)=2$, and by Definition 7.12 and Theorem 7.13

$$d(S)=r-r(R)=1,\qquad d^*(S)=|S|-r(S)=0,\qquad c(S)=r(S)-d(S)=2,$$
$$d(R)=d^*(R)=0,\qquad c(R)=2.$$

*Patterns and room sides.* An excess $i$ is the level $d(S)+i = 1+i$ (Definition 28.1); its patterns have $1+i$ marks and its room sides $2-i$. With the children (Definition 12.1) and the room children:

| $i$ | patterns : children on $R$ | room sides : room children on $S$ |
|---|---|---|
| 0 | $\{1\},\{2\}$ : $\{45\}$ | $\{4,5\}$ : $\{1\},\{2\}$ |
| 1 | $\{1,2\}$ : $\{4\}$; $\{1,3\},\{2,3\}$ : $\{4\},\{5\}$ | $\{4\}$ : $\{12\},\{13\},\{23\}$; $\{5\}$ : $\{13\},\{23\}$ |
| 2 | $\{1,2,3\}$ : $\{\varnothing\}$ | $\varnothing$ : $\{123\}$ |

The two room sides of the excess 1 have room children with three and with two bases, not multiples of one another; so every room side is its own station: $D_0 = \{\{4,5\}\}$, $D_1 = \{\{4\}\}$, $D_1' = \{\{5\}\}$ and $D_2 = \{\varnothing\}$. The stations of patterns are $\{\{1\},\{2\}\}$, $\{\{1,2\}\}$, $\{\{1,3\},\{2,3\}\}$ and $\{\{1,2,3\}\}$. For a list every multiplier and pair constant is 1.

*Seam letters and room station letters.* The seam letters are the letters of the room children, on $S$, and the room station letters are $\mathsf{y}_D = \alpha_R^X$ and $\mathsf{y}_D^* = \alpha_R^{R\setminus X}$ for the one room side $X$ of $D$ (Definition 28.1(iii)):

| $D$ | $V_D = K_{(D)}$ | $V_D^* = \Psi_{(D)}$ | $\mathsf{y}_D$ | $\mathsf{y}_D^*$ |
|---|---|---|---|---|
| $D_0$ | $\alpha_1+\alpha_2$ | $\alpha_3(\alpha_1+\alpha_2)$ | $\alpha_4\alpha_5$ | 1 |
| $D_1$ | $e_2(\alpha_S)$ | $e_1(\alpha_S)$ | $\alpha_4$ | $\alpha_5$ |
| $D_1'$ | $\alpha_3(\alpha_1+\alpha_2)$ | $\alpha_1+\alpha_2$ | $\alpha_5$ | $\alpha_4$ |
| $D_2$ | $\alpha_1\alpha_2\alpha_3$ | 1 | 1 | $\alpha_4\alpha_5$ |

*The room-station factorization* (Proposition 28.2(iii)):

$$K = (\alpha_1+\alpha_2)\,\alpha_4\alpha_5 + e_2(\alpha_S)\,\alpha_4 + \alpha_3(\alpha_1+\alpha_2)\,\alpha_5 + \alpha_1\alpha_2\alpha_3,$$
$$\Psi = \alpha_3(\alpha_1+\alpha_2) + e_1(\alpha_S)\,\alpha_5 + (\alpha_1+\alpha_2)\,\alpha_4 + \alpha_4\alpha_5,$$

the eight bases and the eight cobases, each once.

*The walls.* At $(\mathbf{1};1,1)$, $\kappa_S = m_S - d(S) - d^*(S) = 2$, $\kappa_R = m_R - d(R) - d^*(R) = 2$, and the top wall of $S$ is $(s+t)\,c(S) - \kappa_S = 2 = \kappa_R$ (Definition 28.4): the glue is on its balance wall.

(ii) *The pins of the seam.* One share per station and letter: $a_D$ for $K$ and $b_D$ for $\Psi$, $D \in \{D_0, D_1, D_1', D_2\}$. By Lemma 28.7 the pins are the two letter totals and the scale pin, whose coefficients are the excesses $i(D)$ for $K$ and $c(S) - i(D)$ for $\Psi$:

$$a_{D_0} + a_{D_1} + a_{D_1'} + a_{D_2} = 1, \qquad b_{D_0} + b_{D_1} + b_{D_1'} + b_{D_2} = 1,$$
$$a_{D_1} + a_{D_1'} + 2a_{D_2} + 2b_{D_0} + b_{D_1} + b_{D_1'} = \kappa_S = 2.$$

*The room word.* The room is the partner with the room letters $L_D = \mathsf{y}_D$ and $L_D^* = \mathsf{y}_D^*$ (the paragraph after Definition 28.4), so its word is the monomial

$$W_R(\mathbf{a},\mathbf{b}) = \prod_D \mathsf{y}_D^{-a_D}\,{\mathsf{y}_D^*}^{-b_D} = \alpha_4^{-p_4}\alpha_5^{-p_5},$$
$$p_4 = a_{D_0} + a_{D_1} + b_{D_1'} + b_{D_2}, \qquad p_5 = a_{D_0} + a_{D_1'} + b_{D_1} + b_{D_2},$$

and $p_4 + p_5 = 2\big(|\mathbf{a}| + |\mathbf{b}|\big) - \big(a_{D_1} + a_{D_1'} + 2a_{D_2} + 2b_{D_0} + b_{D_1} + b_{D_1'}\big) = 4 - 2 = 2 = m_4 + m_5$ on the pins.

*Its seam function.* The seam function of the room is the period of its word at $\mathbf{m}_R = (1,1)$ (Lemma 28.7, read on the partner, whose pins on the balance wall are those of the seam; Theorem 28.10(iii)). On the section $\alpha_5 = 1$ (Definition 25.1) it is

$$\int_0^\infty \alpha_4^{1-p_4}\,\frac{d\alpha_4}{\alpha_4} = +\infty$$

at every real value of $p_4$: the integrand is a power, which is not integrable at 0 when its exponent is at most $-1$ and not at $\infty$ when it is at least $-1$. Every room letter is a monomial because every room side is its own station, so the Newton sum of the room word is one point, of dimension $0 < |R| - 1$, and Theorem 25.2 excludes convergence at every share. No point of the pins has both seam functions finite: the display of Theorem 28.10(iii), a fibre integral of the dimension $2 \cdot 4 - 2 - 1 = 5$, does not hold here. Its parts (i) and (ii) hold in every case.

(iii) *The Newton sums.* At shares with positive real parts the seam word of $S$,

$$W_S(\mathbf{a},\mathbf{b}) = \prod_D V_D^{-a_D} V_D^{*-b_D}$$
$$= (\alpha_1+\alpha_2)^{-a_{D_0}-b_{D_1'}} \big(\alpha_3(\alpha_1+\alpha_2)\big)^{-b_{D_0}-a_{D_1'}} e_2(\alpha_S)^{-a_{D_1}} e_1(\alpha_S)^{-b_{D_1}} (\alpha_1\alpha_2\alpha_3)^{-a_{D_2}},$$

has a Newton sum that contains $b_{D_1}$ times the triangle $\operatorname{conv}\{\mathbf{1}_{\{1\}}, \mathbf{1}_{\{2\}}, \mathbf{1}_{\{3\}}\}$ of $e_1(\alpha_S)$ (Definition 25.1): its space of directions is $\vec{\mathcal{N}}_S = \{x \in \mathbb{R}^S : \ x_1 + x_2 + x_3 = 0\}$. The room word is a monomial, $\vec{\mathcal{N}}_R = 0$.

*The lattice and the pins.* In Theorem 28.12,

$$M = \vec{\mathcal{N}}_S^{\perp} \oplus \vec{\mathcal{N}}_R^{\perp} = \mathbb{R}\mathbf{1}_S \oplus \mathbb{R}^{\{4,5\}},$$

of dimension 3, containing $\mathbf{1}$; $M \cap \mathbb{Z}^5 = \{(a,a,a,b,c)\}$ has the basis $\mathbf{1}, \mathbf{e}_4, \mathbf{e}_5$, since $\mathbf{1}_S = \mathbf{1} - \mathbf{e}_4 - \mathbf{e}_5$. So $q = 2$, $\mathbf{v}_1 = \mathbf{e}_4$ and $\mathbf{v}_2 = \mathbf{e}_5$. The fixed monomials of the seam letters live on $S$, and those of the room letters are the room letters themselves, so $\langle \mathbf{n}(\mathbf{a},\mathbf{b}), \mathbf{e}_4 \rangle = p_4$ and $\langle \mathbf{n}(\mathbf{a},\mathbf{b}), \mathbf{e}_5 \rangle = p_5$. The pins are the letter totals and

$$a_{D_0} + a_{D_1} + b_{D_1'} + b_{D_2} = m_4 = 1, \qquad a_{D_0} + a_{D_1'} + b_{D_1} + b_{D_2} = m_5 = 1 :$$

four pins on eight shares, a fibre of the dimension 4. With the totals they imply the scale pin, by the identity for $p_4 + p_5$ in (ii).

*The reduced periods.* $\vec{\mathcal{N}}_S^{\perp} = \mathbb{R}\mathbf{1}_S$, so the reduced period of the seam word is its integral over $\mathbb{R}^S/\mathbb{R}\mathbf{1}_S$, that is over a section: the period $P_{W_S(\mathbf{a},\mathbf{b})}(\mathbf{1}_S)$ (Definition 25.1). $\vec{\mathcal{N}}_R^{\perp} = \mathbb{R}^R$, so the quotient is a point, and the reduced period of the room word is the value of $\alpha_4^{m_4-p_4}\alpha_5^{m_5-p_5}$, which is 1 on the pins, the room letters having the coefficient 1.

*The share side.* The share kernels are $\Gamma_{\begin{smallmatrix}4\\1\end{smallmatrix}}\begin{bmatrix}\mathbf{a}\\1\end{bmatrix} = \prod_D \Gamma(a_D)$ and $\Gamma_{\begin{smallmatrix}4\\1\end{smallmatrix}}\begin{bmatrix}\mathbf{b}\\1\end{bmatrix} = \prod_D \Gamma(b_D)$, $\Gamma(1) = 1$ (Definition 1.2). So, on every contour of the pins with positive real parts at which $P_{W_S}$ converges,

$$P_{\mathrm{Q}_5}(\mathbf{1}; 1, 1) = \int_{\text{pins}} \prod_D \Gamma(a_D)\,\Gamma(b_D)\; P_{W_S(\mathbf{a},\mathbf{b})}(1,1,1),$$

the pins being $|\mathbf{a}| = |\mathbf{b}| = 1$ and $p_4 = p_5 = 1$.

*The native split.* At the table's gauge the basis law is uniform on the eight bases and the cobasis law is the same law (Theorem 7.16). $T \cap R$ is $\{4,5\}$ at $145, 245$, $\{4\}$ at $124, 134, 234$, $\{5\}$ at $135, 235$ and $\varnothing$ at $123$, so Lemma 28.9 gives

$$a_D^{\star} = b_D^{\star} = \tfrac{1}{4},\ \tfrac{3}{8},\ \tfrac{1}{4},\ \tfrac{1}{8} \qquad (D = D_0, D_1, D_1', D_2),$$

with the totals 1 and 1, $p_4 = \frac{1}{4} + \frac{3}{8} + \frac{1}{4} + \frac{1}{8} = 1$ and $p_5 = \frac{1}{4} + \frac{1}{4} + \frac{3}{8} + \frac{1}{8} = 1$: the native split lies on the transverse pins, with positive coordinates. There $\mathbf{1}_S$ lies in the relative interior of the Newton sum of the seam word (Lemma 28.9); the receipt checks it exactly, the support function

of the Newton sum exceeding $\langle u, \mathbf{1}_S\rangle$ by at least $\frac{9}{8}$ at every edge normal $u$ of its summands. So $P_{W_S(\mathbf{a}^\star,\mathbf{b}^\star)}(\mathbf{1}_S)$ converges (Theorem 25.2), and the contour through the native split carries the display.

*Answer.* $S$ has width two; its four stations of room sides are the single room sides $\{4,5\}$, $\{4\}$, $\{5\}$, $\varnothing$, the room letters are monomials, and the seam function of the room is $+\infty$ at every share, so the period does not open with one share per station. With the transverse pins $p_4 = p_5 = 1$ in place of the scale pin it opens as the fibre integral of dimension four displayed, the seam word of $S$ against the share kernels, the room contributing 1. □

## 52.2 The width of the marks left in place

**Exercise 52.2** (The width of a bunch of a substitution)**.** Let $\mathrm{M}_8 = \binom{\{b_1,b_2,h_1,h_2\}}{2}\langle b_1, b_2, \mathcal{A}_1, \mathcal{A}_2\rangle$, the marks $b_1, b_2$ left in place and the cells $\mathcal{A}_i = \binom{\{x_i,y_i,z_i,g_i\}}{2}$ at $h_i$, with $R_i = \{x_i, y_i, z_i\}$ (Definition 21.1).

(i) Find the rank and the bases of $\mathrm{M}_8$, and the widths of $\{b_1\}$, $\{b_2\}$ and $B = \{b_1, b_2\}$, from the bases and from the core; compare with the bound of Corollary 21.11.

(ii) Write $K$ along the scale of $B$.

(iii) For $c = 1, 2, 3$ let $B = \{b_1, \dots, b_c\}$ be left in place in the core $\binom{B\sqcup\{h_1,\dots,h_c\}}{c}$, with the cells $\binom{\{x_i,y_i,z_i,g_i\}}{2}$ at the $h_i$. Find the width of $B$.

*Solution.* Receipt: `ex09_width_subst.py`.

(i) *Rank and bases.* The core has the rank 2, each cell $\mathcal{A}_i$ the rank 2, and a mark left in place the cell $\binom{\{b,g\}}{1}$ of rank 1; so $\mathrm{M}_8$ is a table of rank $2 + (2-1) + (2-1) = 4$ on eight marks (Proposition 21.3(i)). Its bases (Definition 21.1) are read off the six bases $T$ of the core: $b \in T \cap B$ gives the mark $b$, and at $h_i$ the cell gives a basis of $\mathcal{A}_i \setminus g_i = \binom{R_i}{2}$ when $h_i \in T$ and one of $\mathcal{A}_i/g_i = \binom{R_i}{1}$ when $h_i \notin T$, three choices each: $6 \cdot 9 = 54$ bases.

*Widths.* From the bases, the numbers of marks of $\{b_1\}$, $\{b_2\}$ and $B$ in a basis fill the windows $\{0,1\}$, $\{0,1\}$ and $\{0,1,2\}$ (Theorem 7.13): the widths are 1, 1 and 2. In the core $\binom{\{b_1,b_2,h_1,h_2\}}{2}$ a pair meets one mark in 0 or 1 marks and a pair of marks in 0, 1 or 2: the same widths. That is Corollary 21.11: a set of marks left in place has the width it has in the core. In Theorem 21.10 the cell $\binom{\{b,g\}}{1}$ of a mark $b$ of $S \subseteq \{b_1, b_2\}$ has $d(\{b\}) = d^*(\{b\}) = 0$ and $d(\{b,g\}) = d^*(\{b,g\}) = 1$, the cells $\mathcal{A}_i$ meet $S$ in $\varnothing$ with $d(\{g_i\}) = d^*(\{g_i\}) = 0$, so $D(S) = D^*(S) = S$ and $d(S) = d_{\mathcal{H}}(S) = 0$, $d^*(S) = d^*_{\mathcal{H}}(S) = 0$.

*The bound.* With $L = \{b_1, b_2\}$ and the two cells, $c(S) \le |L \setminus S| + 2$: $1 \le 3$ for $\{b_1\}$ and $\{b_2\}$, and $2 \le 2$ for $B$, where it is attained. For the list $\mathrm{P}_5 = \binom{\{1,2,h\}}{1}\langle 1, 2, \binom{\{3,4,5,g\}}{2}\rangle$ of Exercise 51.5 the bound at its Foreman bunch $\{1,2\}$ is $0 + 1 = 1$, attained as well.

(ii) *The letter along $B$.* The cells have $K_{\mathcal{A}_i/g_i} = e_1(\alpha_{R_i})$ and the hinge values $\gamma_i := \gamma^K_{\mathcal{A}_i} = e_2(\alpha_{R_i})/e_1(\alpha_{R_i})$ (Definition 21.1); their contractions $\binom{R_i}{1}$ are the duals of their deletions $\binom{R_i}{2}$, so also $\gamma^\Psi_{\mathcal{A}_i} = \gamma_i$ (Corollary 23.12(ii)). The core letter is $e_2$ on four marks, so by Theorem 21.4

$$K(\tau\alpha_B, \alpha_R) = e_1(\alpha_{R_1})\, e_1(\alpha_{R_2})\Big(\gamma_1\gamma_2 + \tau\,(\alpha_{b_1} + \alpha_{b_2})(\gamma_1 + \gamma_2) + \tau^2\alpha_{b_1}\alpha_{b_2}\Big)\,:$$

three levels, the width two, and the room sees $B$ only through the two hinge values of the cells and the marks of $B$.

(iii) In the core $\binom{B\sqcup\{h_1,\dots,h_c\}}{c}$, $|B| = c$, the sets of $c$ marks meet $B$ in every number of marks from 0 to $c$, so $B$ has the width $c$ (Theorem 7.13); no $h_i$ is a loop or a coloop, and by Corollary 21.11

the width stays $c$ after the cells are substituted. From the bases:

| $c$ | marks | rank | bases | window of $B$ | $c(B)$ |
|---|---|---|---|---|---|
| 1 | 4 | 2 | 6 | $\{0,1\}$ | 1 |
| 2 | 8 | 4 | 54 | $\{0,1,2\}$ | 2 |
| 3 | 12 | 6 | 540 | $\{0,1,2,3\}$ | 3 |

At $c=1$ the substitution $\binom{\{b_1,h_1\}}{1}\langle b_1,\mathcal{A}_1\rangle$ is the list $\binom{\{b_1,x_1,y_1,z_1\}}{2}$ itself, and at $c=2$ it is $\mathrm{M}_8$. In each case the bound $|L\setminus B|+c=c$ is attained.

*Answer.* $\mathrm{M}_8$ is a list of rank four on eight marks with 54 bases; $\{b_1\}$ and $\{b_2\}$ have width one and $B=\{b_1,b_2\}$ width two, as in the core, against the bounds 3, 3 and 2. Along $B$, $K=e_1(\alpha_{R_1})e_1(\alpha_{R_2})(\gamma_1\gamma_2+\tau(\alpha_{b_1}+\alpha_{b_2})(\gamma_1+\gamma_2)+\tau^2\alpha_{b_1}\alpha_{b_2})$. In $\binom{B\sqcup\{h_1,\dots,h_c\}}{c}$ with cells at the $h_i$ the bunch $B$ has the width $c$ for $c=1,2,3$: a bunch that is all the marks left in place has at most as much width as there are cells, and every width occurs. □

## 52.3 Fusing a uniform list

**Exercise 52.3** (Fusing $\binom{[5]}{2}$)**.** Run the procedure of Section 30.7 on the list $\binom{[5]}{2}$, read at its centre: decide each step, find its fusion width, its tell, its Foreman fusion with its letter, its lifts and its majors, and its cousins; read the period of the fusion through two of its cousins; and find the index of the bijections carrying $\binom{[5]}{2}$, times the permutations of the fusion bunch, among those carrying its fusion.

*Solution.* Receipt: `ex10_fusion_u25.py`.

*(1) Peel.* Every permutation of [5] maps pairs to pairs and carries the list, so at the centre, and at every lean, all marks lie in one class (Corollary 11.16): the realization is an atom (Definition 10.1) and nothing is peeled (Definition 14.4).

*(2) Go down.* The complement of a basis is a triple, not a basis: the list is not identically self-dual (Corollary 23.12(iii)).

*(3) Go up.* The rank is $r=2$ and the corank $r^*=3$, so the fusion width is $c=r^*-r=1$ (Definition 30.4). The circuits (Definition 7.4) are the ten triples, every pair being independent; the cocircuits, the least sets meeting every pair, are the five sets of four marks. Each cocircuit is the union of the four triples inside it, so the tell (Theorem 30.6(b)) holds. In its form (c): for a mark $e$ and a split of the other four marks into two halves, one half has at least two marks, the rank 2, which $e$ does not raise. The form (a), $r^*(Y)-r^*(X)\geq r(Y)-r(X)$ for all $X\subseteq Y$, the receipt checks on the rank functions. At $c=1$ the weights play no part: $\binom{[5]}{2}$ is fusable.

*The fusion.* With the fusion bunch $G=\{g\}$, Theorem 30.5(v) gives the bases $T+g$, $T$ a pair of [5], and $[5]\setminus T$, the triples of [5]: ten and ten, all twenty triples of $[5]\sqcup\{g\}$. So

$$\mathcal{B}^\sharp=\binom{[5]\sqcup\{g\}}{3},\qquad K^\sharp=\Psi+\alpha_g K=e_3(\alpha_{[5]})+\alpha_g\,e_2(\alpha_{[5]})=e_3(\alpha_{[5]\sqcup\{g\}})=\Psi^\sharp,$$

identically self-dual, the complement of a triple of six marks being a triple. The lifts (Definition 30.4(ii)) are $\mathcal{L}_0=\binom{[5]}{2}$ and $\mathcal{L}_1=\binom{[5]}{3}$, its dual, with the lift letters $L_0=\Psi=e_3$ and $L_1=K=e_2$; at $c=1$ there are no middle weights and no lift ratios, and $K^\sharp=L_0+\alpha_g L_1$ is Theorem 30.5(ii).

*The majors.* A table on $[5]\sqcup\{g\}$ whose contraction of $g$ is $\binom{[5]}{2}$ has as its bases through $g$ the sets $T+g$, $T\in\binom{[5]}{2}$, and one whose deletion of $g$ is $\binom{[5]}{3}$ has as its bases avoiding $g$ the triples of

[5] (Definition 7.3). A major (Definition 30.4(i)) has both, so its bases are the twenty triples, and $\mathcal{B}^\sharp$ is the only major, as Theorem 30.5(v) says.

*One Beta corner.* On the balance wall of the fusion, $m_{[5]}+m_g = 2\sigma r^* = 6\sigma$, with every proper wall positive, Corollary 30.13 gives

$$P_{\binom{[6]}{3}}\big(\mathbf{m}|_{[5]}, m_g; s, t\big) = \Gamma^{\,2}_{\,1}\begin{bmatrix} m_g, & 2\sigma - m_g \\ & 2\sigma \end{bmatrix} P_{\binom{[5]}{2}}\big(\mathbf{m}|_{[5]}; m_g, 2\sigma - m_g\big),$$

[6] written for $[5]\sqcup\{g\}$. It can be read off the letter: the fused table is the family $\big(\binom{[6]}{3}; 2\sigma, 0\big)$ (the paragraph after Definition 30.11), and on the section $|\alpha_{[5]}| = 1$ the scale of $g$ integrates out, with $\alpha_g = (e_3/e_2)v$,

$$\begin{aligned}\int_0^\infty \alpha_g^{\,m_g-1}\big(e_3 + \alpha_g\, e_2\big)^{-2\sigma} d\alpha_g &= e_3^{\,m_g-2\sigma} e_2^{\,-m_g} \int_0^\infty v^{m_g-1}(1+v)^{-2\sigma} dv \\ &= \Gamma^{\,2}_{\,1}\begin{bmatrix} m_g, & 2\sigma - m_g \\ & 2\sigma \end{bmatrix} e_3^{\,m_g-2\sigma} e_2^{\,-m_g},\end{aligned}$$

$e_k = e_k(\alpha_{[5]})$, the last integral being the shape integral of two marks (Theorem 3.9) after $v = u/(1-u)$; what is left is the form of $\binom{[5]}{2}$ with $K = e_2$ at the exponent $m_g$ and $\Psi = e_3$ at $2\sigma - m_g$. The masses of $\binom{[5]}{2}$ are then on its own balance wall, $m_{[5]} = 2m_g + 3(2\sigma - m_g)$.

*The cousins.* The cousins of $\binom{[6]}{3}$ (Proposition 30.20) are its six one-mark contractions $\binom{[6]\setminus\{x\}}{2}$, each a copy of $\binom{[5]}{2}$, the cousin at $g$ being $\binom{[5]}{2}$ itself; the deletion $\binom{[6]\setminus\{x\}}{3}$ is the dual of each, so each has the fusion width one and fuses back into $\binom{[6]}{3}$ with the fusion bunch $\{x\}$ (Proposition 30.20(i)). For any two marks the period is read through either cousin (Proposition 30.20(ii)); at $x = g$ and $y = 5$, writing $s' = m_g$ and renaming $g$ as 5 in the second cousin,

$$\begin{aligned}&\Gamma^{\,2}_{\,1}\begin{bmatrix} s', & 2\sigma - s' \\ & 2\sigma \end{bmatrix} P_{\binom{[5]}{2}}\big(m_1, \dots, m_5; s', 2\sigma - s'\big) \\ &\qquad = \Gamma^{\,2}_{\,1}\begin{bmatrix} m_5, & 2\sigma - m_5 \\ & 2\sigma \end{bmatrix} P_{\binom{[5]}{2}}\big(m_1, m_2, m_3, m_4, s'; m_5, 2\sigma - m_5\big)\end{aligned}$$

on $m_{[5]} + s' = 6\sigma$: the basis exponent $s'$ of $\binom{[5]}{2}$ and the mass of one mark trade places (Proposition 30.20(iii)). At the centre, $\sigma = 1$ and every mass 1, both sides are $\Gamma^{\,2}_{\,1}\begin{bmatrix} 1, & 1 \\ & 2 \end{bmatrix} P_{\binom{[5]}{2}}(\mathbf{1}; 1, 1) = P_{\binom{[5]}{2}}(\mathbf{1}; 1, 1)$, and $P_{\binom{[6]}{3}}(\mathbf{1}; 1, 1) = P_{\binom{[5]}{2}}(\mathbf{1}; 1, 1)$.

*The symmetry the fusion adds.* All 720 permutations of $[5]\sqcup\{g\}$ carry $\binom{[6]}{3}$; those carrying $\binom{[5]}{2}$, times the permutations of $G = \{g\}$, are the $120 \cdot 1$ permutations fixing $g$. The index is $720/120 = 6 = \binom{n+c}{c} = \binom{6}{1}$ (Remark 30.21).

*(4) Two-sums and (5) repair.* Every set of two or three marks has the width $2 + 2 - 2 = 2$ (Definition 7.12), so $\binom{[5]}{2}$ is prime (Theorem 23.16) and no two-sum, and being fusable it needs no repair.

*Answer.* The procedure stops at step (3): $\binom{[5]}{2}$ has the fusion width one, passes the tell, and fuses into $\binom{[5]\sqcup\{g\}}{3}$, with the letter $e_3 = \Psi + \alpha_g K$, its only major. Its cousins are the six copies $\binom{[6]\setminus\{x\}}{2}$ of $\binom{[5]}{2}$, and the period of the fusion is one Beta corner times the period of any cousin, the basis exponent of $\binom{[5]}{2}$ and the masses of its marks trading places under the permutations of six marks; the fusion adds the index 6 to the symmetry. □

# Chapter 53

# Exercises on the motive and the arithmetic of values

The exercises of this chapter work the statements of Chapter 34 on small lists. The list $\binom{[3]}{2}$ at its centre has a pair of width one that does not split off; read at that pair as its root bunch, its period is a node form with two Skeleton values, and it is three times the Trailmoment at the index two of the lattice input that counts $1+3\mathbb{Z}$ and subtracts $2+3\mathbb{Z}$. The list $\binom{[4]}{2}$ is, at the complement of each mark, a core on two marks with a dual pair substituted, and it has the motivic period of $\binom{[3]}{2}$; its pairs have width two and do not split off. The list $\binom{[4]}{1}$ has rational periods and rational motivic periods at its full leans; at its centre a pair of width one has a law with a density, and the period is the critical-line word $\mathrm{crit}_1[1]$, seven times the Trailmoment at the index three of the counting input. Substitutions of dual pairs on five and on eight marks have the motivic periods $(P^{\mathrm{mot}}_{\binom{[3]}{2}})^2$ and $(P^{\mathrm{mot}}_{\binom{[3]}{2}})^3$. At the exponents $(0,2)$ with unit masses every hinge wall and every Beta corner is 1, and the motivic period of a list is the product of the motivic periods of its prime nodes. Every number displayed is exact, and the receipts that recompute them are the scripts named in each solution.

Throughout, a list $\mathcal{B}$ *at its centre* is the family $(\mathcal{B};1,1)$ at the masses $\mathbf{1}$ (Proposition 8.21), and when it is convergent $P^{\mathrm{mot}}_{\mathcal{B}}$ denotes its motivic period (Definition 34.5); a family at a lean is written out with its exponents and its masses.

## 53.1 Uniform lists at their centres

**Exercise 53.1** (The list $\binom{[3]}{2}$ at its centre)**.** Let $\mathcal{E}=\{1,2,3\}$ and $F=\big(\binom{\mathcal{E}}{2};1,1\big)$ at the masses $\mathbf{1}$.

(i) Show that $F$ is convergent (Definition 34.1), that the pair $S=\{1,2\}$ has width one with the ratio $\varrho_S=1/(u(1-u))$ on its shape $(u,1-u)$, and that $S$ does not split off (Theorem 28.25).

(ii) Take a tree of bunches whose root carries the bunch $S$. Find the Skeleton values of the letters along $S$, the node form of $\omega_F$ at the root (Theorem 34.16) and its share side.

(iii) Evaluate the period $P_{\binom{\mathcal{E}}{2}}(\mathbf{1};1,1)$ as a Trailmoment of a lattice input.

*Solution.* Receipt: `ex1_series_u23.py`.

(i) *Letters and walls.* The bases are the three pairs, so $K=e_2(\alpha)=\alpha_1\alpha_2+\alpha_1\alpha_3+\alpha_2\alpha_3$ and $\Psi=e_1(\alpha)$ (Definition 7.1), with the rank 2 and the corank 1. The imbalance is $\kappa_{\mathcal{E}}=3-(1\cdot 2+1\cdot 1)=0$ (Definition 8.1). By Theorem 7.13 and Definition 7.12, $d(S)=\min_T|T\cap S|$

and $d^*(S) = |S| - r(S)$: a mark has $d = d^* = 0$, and a pair has $d = 1$ and $d^* = 0$. So the walls $\kappa_S = m_S - d(S) - d^*(S)$ are

$$\kappa_{\{e\}} = 1 - 0 - 0 = 1, \qquad \kappa_{\{e,f\}} = 2 - 1 - 0 = 1,$$

the widths of these sets, as Proposition 8.21(ii) says at a centre. Every proper wall is at least 1, so $F$ is convergent, its motivic period $P^{\mathrm{mot}}_{\binom{[3]}{2}}$ is defined (Definition 34.5), and its period is

$$P_{\binom{\mathcal{E}}{2}}(\mathbf{1}; 1, 1) = \int_{\Delta_{\mathcal{E}}} \frac{d\omega_{\mathcal{E}}(u)}{e_1(u)\, e_2(u)}$$

(Proposition 34.4(ii), Theorem 8.7).

*The pair.* The bases meet $S$ in one or two marks, so $d(S) = 1$, $r(S) = 2$ and $c(S) = 1$ (Theorem 7.13); every basis contains all but at most one mark of $S$, so $S$ is a series bunch (Definition 19.1). With the room $R = \{3\}$, the restriction $\mathcal{B}|S$ has the one basis $S$ and the contraction $\mathcal{B}/R$ has the bases $\{1\}$ and $\{2\}$ (Definition 7.5). The hinge values and their ratio (Definition 19.2) are

$$\gamma^K_S = \frac{K_{\mathcal{B}|S}}{K_{\mathcal{B}/R}} = \frac{\alpha_1\alpha_2}{\alpha_1+\alpha_2}, \qquad \gamma^\Psi_S = \frac{\Psi_{\mathcal{B}/R}}{\Psi_{\mathcal{B}|S}} = \frac{\alpha_1+\alpha_2}{1}, \qquad \varrho_S = \frac{\gamma^\Psi_S}{\gamma^K_S} = \frac{(\alpha_1+\alpha_2)^2}{\alpha_1\alpha_2} = \frac{1}{u(1-u)}$$

on the shape $(u, 1-u)$, as Proposition 19.3 gives for a series pair.

*The pair does not split off.* For one list with $s, t > 0$ and $|S| \geq 2$, Theorem 28.25 says that $S$ is pinned exactly when it has width one and $\mathbf{E}_{\mathcal{B}|S} = \mathbf{E}_{\mathcal{B}/R}$. Here

$$\mathbf{E}_{\mathcal{B}|S} = \alpha_1\alpha_2 \cdot 1, \qquad \mathbf{E}_{\mathcal{B}/R} = (\alpha_1+\alpha_2)(\alpha_1+\alpha_2),$$

which differ: their ratio $\varrho_S$ is at least 4, with equality only at $u = \frac{1}{2}$. So $S$ is not pinned; by Theorem 28.25, (ii), (iii) and (i), it does not split off and its law is carried by no point. This is Corollary 28.26(iv) at the mark 3, whose deletion and contraction are $\mathcal{B} \setminus 3 = \mathcal{B}|S$ and $\mathcal{B}/3 = \mathcal{B}/R$ (Definition 7.3).

(ii) *The step at the root.* The list is one molecule, its circuit $\{1, 2, 3\}$ linking its marks (Definition 7.4), so the root of a tree of bunches is a node, and it may carry the bunch $S$, of width one, its room being the one mark 3 (Definition 20.7). The flag $\varnothing \subsetneq \mathcal{E}$ has the one step $\mathcal{E}$, which meets $S$ and $R$: it is a node step of the root (Definition 34.9(ii)), with $Z_B = S$ and $Z_R = R$. The form $\omega_F$ is regular on $V_F$ (Proposition 34.4(ii)), and its period is read by its node form at the root (Theorem 34.16(v)). Read $\alpha_S = \rho\,(u, 1-u)$ with $\rho > 0$ and $\alpha_3 = 1$, the one point of $\Delta_R$, and put $Q = u(1-u)$.

*Letters and Skeleton values.* By Theorem 34.16(i) the letter $K$ is the basis polynomial of $\binom{\mathcal{E}}{2}$, in which $S$ has the floor 1, and $\Psi$ is that of the dual list $\binom{\mathcal{E}}{1}$ (Proposition 7.2), in which $S$ has the floor 0; $S$ has the width 1 in both. Along the scale (Theorem 34.16(ii)),

$$K(\rho u, 1) = \rho^2 u(1-u) + \rho\,\big(u + (1-u)\big) = \rho\, Q\,\big(\rho + Q^{-1}\big), \qquad \Psi(\rho u, 1) = \rho + 1,$$

so the joint spectrum of $S$ is $\{Q^{-1}, 1\}$: the Skeleton value of $K$ is $\xi_K = Q^{-1} = \varrho_S$ and that of $\Psi$ is $\xi_\Psi = 1$. At width one each is the hinge value of the room over that of the bunch (Theorem 34.16(iv), Corollary 19.5): with $\gamma^K_R = K_{\mathcal{B}|R}/K_{\mathcal{B}/S} = \alpha_3/1$ and $\gamma^\Psi_R = \Psi_{\mathcal{B}/S}/\Psi_{\mathcal{B}|R} = \alpha_3/1$ (Definition 19.2), on $\Delta_S \times \Delta_R$

$$\xi_K = \frac{\gamma^K_R}{\gamma^K_S} = \frac{1}{Q}, \qquad \xi_\Psi = \frac{\gamma^\Psi_R}{\gamma^\Psi_S} = \frac{1}{1}.$$

*The node form.* On the section $\alpha_3 = 1$ the masses are $\mathbf{1}$ and $d\alpha_1\, d\alpha_2 = \rho\, d\rho\, du$ (Lemma 2.7), so (Theorem 8.7)

$$P_{\binom{\varepsilon}{2}}(\mathbf{1}; 1, 1) = \int_0^1 \int_0^\infty \frac{\rho\, d\rho}{K(\rho u, 1)\, \Psi(\rho u, 1)}\, du = \int_0^1 Q^{-1} \int_0^\infty \left(\rho + Q^{-1}\right)^{-1} (\rho + 1)^{-1}\, d\rho\, du.$$

This is the node form of Theorem 34.16(iii) with one term: $A_t = Q^{-1}$, $B_t = 1$, $\nu_t = 1$, the letter exponents $k_{t,K} = k_{t,\Psi} = 1$, and $\mathbf{w}_t = (1, 1)$ on the joint spectrum, $|\mathbf{w}_t| = 2$. Since $0 < \nu_t < |\mathbf{w}_t|$, $\mathrm{Sc}_t$ is the convergent scale integral, the stacked Gamma times the Breta of one bunch (Proposition 15.1), and at positive coefficients the Breta is the integral over the shape simplex of $J$ (Definition 3.1, Proposition 15.2):

$$\mathrm{Sc}_t = \Gamma_{\binom{2}{2}}\begin{bmatrix} 1, & 1 \\ 1, & 1 \end{bmatrix} \mathsf{B}^{\{J\}}_{(1,1),\,1;\,I_J}(Q^{-1}, 1) = \int_0^1 \frac{dv}{Q^{-1}v + (1 - v)} = \frac{\log Q^{-1}}{Q^{-1} - 1} = \frac{Q \log Q}{Q - 1}.$$

So

$$P_{\binom{\varepsilon}{2}}(\mathbf{1}; 1, 1) = \int_0^1 A_t\, \mathrm{Sc}_t\, du = \int_0^1 \frac{\log Q}{Q - 1}\, du, \qquad Q = u(1 - u).$$

*The share side.* Corollary 15.5, with $\xi_0 = 1$, $\xi_1 = Q^{-1}$, $w_0 = w_1 = \kappa = 1$ and one share $a$ on a line $\mathrm{Re}\, a = c$, $0 < c < 1$, writes the term as in Theorem 34.16(iv):

$$\begin{aligned} A_t\, \mathrm{Sc}_t &= Q^{-1}\, \frac{1}{2\pi i} \int_{(c)} \Gamma_{\binom{2}{1}}\begin{bmatrix} a, & 1 - a \\ & 1 \end{bmatrix} \Gamma_{\binom{2}{1}}\begin{bmatrix} 1 - a, & a \\ & 1 \end{bmatrix} \left(Q^{-1}\right)^{a-1} da \\ &= \frac{1}{2\pi i} \int_{(c)} \Gamma_{\binom{4}{0}}\big[a,\ a,\ 1 - a,\ 1 - a\big]\ Q^{-a}\, da. \end{aligned}$$

Here $\xi_K^{-b} = (\gamma_S^K)^b$ with $b = 1 - a$, and the condition of Theorem 34.16(iv) on the bunch, $\int_0^1 Q^{-1} Q^{1-c}\, du < \infty$, holds at $c = \frac{1}{2}$, where it is $\int_0^1 (u(1 - u))^{-1/2} du = \Gamma_{\binom{2}{1}}\begin{bmatrix} \frac{1}{2}, & \frac{1}{2} \\ & 1 \end{bmatrix}$ (Theorem 3.9); the room is one point. Closing to the left (Lemma 5.16(ii), with $X = Q \le \frac{1}{4}$), the stacked Gamma has at $a = -j$, $j \in \mathbb{N}$, the principal part $(a + j)^{-2}$ and no simple part, so the residue there is the derivative of $Q^{-a}$ at $a = -j$, which is $-Q^j \log Q$, and

$$\frac{1}{2\pi i} \int_{(1/2)} \Gamma_{\binom{4}{0}}\big[a,\ a,\ 1 - a,\ 1 - a\big]\ Q^{-a}\, da = -\sum_{j \ge 0} Q^j \log Q = \frac{\log Q}{Q - 1},$$

the value of the node form above.

(iii) *The reflection of the shape.* $\log Q = \log u + \log(1 - u)$, and $1 - Q = 1 - u + u^2$ is unchanged by $u \mapsto 1 - u$; so the two logarithms give equal integrals and

$$P_{\binom{\varepsilon}{2}}(\mathbf{1}; 1, 1) = -2 \int_0^1 \frac{\log u}{1 - u + u^2}\, du.$$

*The logarithmic coordinate.* With $u = e^{-y}$, $du = -e^{-y} dy$, and multiplying numerator and denominator by $1 + e^{-y}$, since $(1 - z + z^2)(1 + z) = 1 + z^3$,

$$P_{\binom{\varepsilon}{2}}(\mathbf{1}; 1, 1) = 2 \int_0^\infty y\, \frac{e^{-y}}{1 - e^{-y} + e^{-2y}}\, dy = 2 \int_0^\infty y\, \frac{e^{-y} + e^{-2y}}{1 + e^{-3y}}\, dy.$$

*The lattice input.* Let $\phi = \mathbb{I}_{1+6\mathbb{Z}} + \mathbb{I}_{2+6\mathbb{Z}} - \mathbb{I}_{4+6\mathbb{Z}} - \mathbb{I}_{5+6\mathbb{Z}}$, an input with the density $\pm 1$ on four lattice inputs (Definitions 5.2 and 5.4). It is tame beyond the cut 0, and its Laplace transform there (Definition 5.4) sums its four cosets as geometric series: with $z = e^{-y}$,

$$\mathcal{F}_0(y) = \sum_{k\geq 0} \big(z^{6k+1} + z^{6k+2} - z^{6k+4} - z^{6k+5}\big) = \frac{z + z^2 - z^4 - z^5}{1 - z^6} = \frac{(z+z^2)(1-z^3)}{(1-z^3)(1+z^3)} = \frac{z+z^2}{1+z^3}.$$

Since $\int_{(0,\infty)} x^{-2}\,|\phi|(dx) < \infty$, the Laplace side (Theorem 5.11) at one unit exponent and the index 2 reads $\Gamma(2)\,(\mathsf{T}_{2;(1)}[x]\,\phi)((0,\infty)) = \int_0^\infty y\,\mathcal{F}_0(y)\,dy$, and

$$P_{\binom{\mathcal{E}}{2}}(\mathbf{1}; 1, 1) = 2\,(\mathsf{T}_{2;(1)}[x]\,\big(\mathbb{I}_{1+6\mathbb{Z}} + \mathbb{I}_{2+6\mathbb{Z}} - \mathbb{I}_{4+6\mathbb{Z}} - \mathbb{I}_{5+6\mathbb{Z}}\big))((0,\infty)).$$

*Cosets and dilations.* $1 + 3\mathbb{Z} = (1 + 6\mathbb{Z}) \sqcup (4 + 6\mathbb{Z})$ and $2 + 3\mathbb{Z} = (2 + 6\mathbb{Z}) \sqcup (5 + 6\mathbb{Z})$, so

$$\phi = \mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}} + 2\big(\mathbb{I}_{2+6\mathbb{Z}} - \mathbb{I}_{4+6\mathbb{Z}}\big),$$

and the Trailmoment, an integral against its input (Definition 5.1), splits accordingly, every part converging. The dilation $\tau(x) = 2x$ carries $\mathbb{I}_{1+3\mathbb{Z}}$ to $\mathbb{I}_{2+6\mathbb{Z}}$ and $\mathbb{I}_{2+3\mathbb{Z}}$ to $\mathbb{I}_{4+6\mathbb{Z}}$, so transport at one unit exponent (Theorem 5.9 and the paragraph after it) gives

$$(\mathsf{T}_{2;(1)}[x]\,\big(\mathbb{I}_{2+6\mathbb{Z}} - \mathbb{I}_{4+6\mathbb{Z}}\big))((0,\infty)) = 2^{-2}\,(\mathsf{T}_{2;(1)}[x]\,\big(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}\big))((0,\infty)),$$

and

$$P_{\binom{\mathcal{E}}{2}}(\mathbf{1}; 1, 1) = 2\,\big(1{+}2{\cdot}2^{-2}\big)\,(\mathsf{T}_{2;(1)}[x]\,\big(\mathbb{I}_{1+3\mathbb{Z}}{-}\mathbb{I}_{2+3\mathbb{Z}}\big))((0,\infty)) = 3\,(\mathsf{T}_{2;(1)}[x]\,\big(\mathbb{I}_{1+3\mathbb{Z}}{-}\mathbb{I}_{2+3\mathbb{Z}}\big))((0,\infty)),$$

which is $2.343907238689458890\ldots$ to eighteen places. The receipt checks the identities of rational functions exactly, the decomposition of the input at every $m \leq 600$, the decomposition and the dilation together on the exact heads $\sum_{m\leq 6N} \phi(m) m^{-2}$, and the integral of the node form against the Trailmoment to twenty-five places.

*Answer.* $F$ is convergent, with every proper wall 1. The pair $S = \{1, 2\}$ is a series bunch of width one with $\gamma_S^K = \alpha_1\alpha_2/(\alpha_1 + \alpha_2)$, $\gamma_S^\Psi = \alpha_1 + \alpha_2$ and $\varrho_S = 1/(u(1-u))$; it is not pinned, so it does not split off. At the root bunch $S$ the Skeleton values are $Q^{-1}$ and 1, $Q = u(1-u)$; the node form is $\int_0^1 Q^{-1}\Gamma_2^2\begin{bmatrix}1, & 1\\ 1, & 1\end{bmatrix} \mathsf{B}^{\{J\}}_{(1,1),1;I_J}(Q^{-1}, 1)\,du = \int_0^1 \frac{\log Q}{Q-1}\,du$, and its share side reads the stacked Gamma $\Gamma_0^4\big[a,\ a,\ 1-a,\ 1-a\big]$ against $Q^{-a}$. The period is

$$P_{\binom{[3]}{2}}(\mathbf{1}; 1, 1) = 3\,(\mathsf{T}_{2;(1)}[x]\,\big(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}\big))((0,\infty)).$$

□

**Exercise 53.2** (The list $\binom{[4]}{2}$ at its centre)**.** Let $\mathcal{E} = \{1, 2, 3, 4\}$ and $F = \big(\binom{\mathcal{E}}{2}; 1, 1\big)$ at the masses $\mathbf{1}$.

(i) Find the widths and the walls of the sets of marks. Show that the sets of width one with at least two marks are the complements $R = \mathcal{E} \setminus \{e\}$ of single marks, that the cell of each is a dual pair (Definition 21.1), and that $\binom{\mathcal{E}}{2} = \binom{\{e,h\}}{1}\langle e, \mathcal{A}_R\rangle$ with the mark $e$ left in place.

(ii) Show that $P^{\mathrm{mot}}_{\binom{[4]}{2}} = P^{\mathrm{mot}}_{\binom{[3]}{2}}$, and find the period.

(iii) Show that no pair of marks splits off.

*Solution.* Receipt: `ex2_dualpair_u24.py`.

(i) *Letters, widths, walls.* The complement of a pair is a pair, so $K = \Psi = e_2(\alpha)$ (Definition 7.1). A set $S$ has $r(S) = \min(|S|, 2)$, and by Definition 7.12 its width is $c(S) = r(S) + r(\mathcal{E} \setminus S) - 2$:

$$c(\{e\}) = 1 + 2 - 2 = 1, \qquad c(\{e, f\}) = 2 + 2 - 2 = 2, \qquad c(\mathcal{E} \setminus \{e\}) = 2 + 1 - 2 = 1.$$

At the centre the walls are the widths (Proposition 8.21(ii)): 1, 2 and 1. Every proper wall is at least 1, and $F$ is convergent (Definition 34.1).

*The cells.* The sets of width one with at least two marks are the triples $R = \mathcal{E} \setminus \{e\}$. The restriction $\mathcal{B}|R$ has the three pairs of $R$ and the contraction $\mathcal{B}/\{e\}$ the three marks of $R$, the bases through $e$ less $e$ (Definition 7.5). The cell of the bunch $R$ (Theorem 19.4) has the bases of $\mathcal{B}|R = \binom{R}{2}$ and the sets $X + g$, $X \in \mathcal{B}/\{e\} = \binom{R}{1}$:

$$\mathcal{A}_R = \mathcal{B}_R^+ = \binom{R+g}{2}, \qquad \mathcal{A}_R \setminus g = \binom{R}{2}, \qquad \mathcal{A}_R/g = \binom{R}{1}.$$

The complements in $R$ of the pairs of $R$ are its marks, so $\mathcal{A}_R/g$ is the dual table of $\mathcal{A}_R \setminus g$, the weights being 1: the cell is a dual pair (Definition 21.1). The cell of the room $\{e\}$ has the basis $\{e\}$ of $\mathcal{B}|\{e\}$ and the set $\varnothing + g$, $\varnothing$ being the one basis of $\mathcal{B}/R$: it is $\binom{\{e,g\}}{1}$. By Proposition 21.3(iii) the list is the gluing of the two cells, that is $\binom{\{e,h\}}{1}\langle e, \mathcal{A}_R\rangle$ with $\mathcal{A}_R$ at $h$ (Definition 21.1). On the bases: the core basis $\{e\}$ avoids $h$ and takes $X \in \mathcal{A}_R/g$, giving the three pairs through $e$; the core basis $\{h\}$ takes $X \in \mathcal{A}_R \setminus g$, giving the three pairs of $R$.

*The letters of the substitution.* For $e = 4$ and $R = \{1, 2, 3\}$ the cell has $k = K_{\mathcal{A}_R \setminus g} = e_2(\alpha_R)$ and $\psi = \Psi_{\mathcal{A}_R \setminus g} = e_1(\alpha_R)$, and Proposition 21.8(ii) gives the one hinge value $\gamma := \gamma^K_{\mathcal{A}_R} = \gamma^\Psi_{\mathcal{A}_R} = e_2(\alpha_R)/e_1(\alpha_R)$, with $\varrho_{\mathcal{A}_R} = 1$. The mark 4 left in place has $k = \alpha_4$ and $\psi = 1$, so $\pi = e_1(\alpha_R)$, and Proposition 21.8(iii), with $r_{\mathcal{H}} = 1$ and $q = 2$, reads

$$K_{\mathcal{H}}\big(\pi\alpha_4,\ e_2(\alpha_R)\big) = e_1(\alpha_R)\,\alpha_4 + e_2(\alpha_R) = e_2(\alpha) = \pi^0 K, \qquad \Psi_{\mathcal{H}}\big(\pi\alpha_4,\ e_2(\alpha_R)\big) = e_2(\alpha) = \pi^0 \Psi,$$

$K_{\mathcal{H}}$ and $\Psi_{\mathcal{H}}$ both being $x_4 + x_h$.

(ii) *The dual-pair factorization.* Corollary 34.18(a) applies with $q = 1$ at $s = t = 1$. The wall of the cell is

$$\kappa_R = m_R - (s + t)\big(r(R) - 1\big) = 3 - 2\,(2 - 1) = 1,$$

the wall of $F$ at $R$ found above. The pinned family has the form of $(\mathcal{B}|R; \kappa_R, s{+}t{-}\kappa_R) = \big(\binom{R}{2}; 1, 1\big)$ at the masses $\mathbf{1}_R$, and the core is $\big(\binom{\{e,h\}}{1}; 1, 1\big)$ at the masses $m_e = 1$ and $\kappa_R = 1$ at $h$:

$$P^{\mathrm{mot}}_{\binom{[4]}{2}} = P^{\mathrm{mot}}_{(\binom{\{e,h\}}{1};1,1)} \cdot P^{\mathrm{mot}}_{(\binom{R}{2};1,1)}.$$

The coefficients of the forms show the factorization: with $\gamma$ as above,

$$\frac{\alpha^{\mathbf{1}}}{K\,\Psi} = \frac{\alpha_4\,\gamma}{(\alpha_4 + \gamma)^2} \cdot \frac{\alpha^{\mathbf{1}_R}}{e_1(\alpha_R)\,e_2(\alpha_R)},$$

the coefficient of the core at $(\alpha_4, \gamma)$ times that of $\big(\binom{R}{2}; 1, 1\big)$, an identity of rational functions.

*The core.* The core is a family on two marks with both letters $\alpha_e + \alpha_h$ and the walls $\kappa_{\{e\}} = \kappa_{\{h\}} = 1$; its form is $\Omega/(\alpha_e + \alpha_h)^2$, and by Lemma S18 its motivic period is its Beta corner,

$$P^{\mathrm{mot}}_{(\binom{\{e,h\}}{1};1,1)} = \Gamma_{\!2}\begin{bmatrix} 1, & 1 \\ & 2 \end{bmatrix} = \frac{\Gamma(1)\,\Gamma(1)}{\Gamma(2)} = 1.$$

The pinned family $\binom{R}{2};1,1)$ at $\mathbf{1}_R$ is $\binom{[3]}{2}$ at its centre with its marks named by $R$; renaming the coordinates is an isomorphism of the models that maps the positive cells and the forms onto each other, so the two have one motivic period (Lemma 34.7). Hence

$$P^{\mathrm{mot}}_{\binom{[4]}{2}} = P^{\mathrm{mot}}_{\binom{[3]}{2}}, \qquad \mathrm{C}\big(P^{\mathrm{mot}}_{\binom{[4]}{2}}\big) = \mathrm{C}\big(P^{\mathrm{mot}}_{\binom{[3]}{2}}\big),$$

and by Exercise 53.1 the period is $P_{\binom{[4]}{2}}(\mathbf{1};1,1) = 3\,(\mathsf{T}_{2;(1)}[x]\,(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}))((0,\infty))$.

(iii) *The pairs.* A pair has width two in the one active table, so by Corollary 28.26(i) it does not split off, and its law is carried by no point. The list is connected and its only sets with two marks on each side are the pairs, so it is prime (Theorem 23.16), and Corollary 28.26(ii) says the same. The complements of the marks do split off, as in (ii): $\mathbf{E}_{\mathcal{B}\setminus e} = e_2(\alpha_R)\,e_1(\alpha_R) = \mathbf{E}_{\mathcal{B}/e}$ (Corollary 28.26(iv)).

*Answer.* The marks and the triples have width one and the pairs width two, and at the centre these are the walls. Every triple $R = \mathcal{E}\setminus\{e\}$ has the cell $\binom{R+g}{2}$, a dual pair, and $\binom{\mathcal{E}}{2} = \binom{\{e,h\}}{1}\langle e, \binom{R+g}{2}\rangle$. With $\kappa_R = 1$, the core on two marks has the motivic period $\Gamma^{\,2}_{\,1}\left[\begin{matrix}1,\ 1\\ 2\end{matrix}\right] = 1$, so

$$P^{\mathrm{mot}}_{\binom{[4]}{2}} = P^{\mathrm{mot}}_{\binom{[3]}{2}}, \qquad P_{\binom{[4]}{2}}(\mathbf{1};1,1) = 3\,(\mathsf{T}_{2;(1)}[x]\,(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}))((0,\infty)).$$

No pair splits off: pairs have width two. □

**Exercise 53.3** (The list $\binom{[4]}{1}$ at its full leans and at its centre)**.** Let $\mathcal{E} = \{1,2,3,4\}$ and $\mathcal{B} = \binom{\mathcal{E}}{1}$.

(i) Show that at its full leans $(s,0)$ and $(0,t)$, at positive integer masses on the balance wall with every proper wall at least 1, the periods are rational, and so are the motivic periods.

(ii) At the centre, show that $S = \{3,4\}$ has width one, find its hinge values and its law (Definition 28.5), and show that the law is not a point mass.

(iii) Read the period at the root bunch $S$, write it on the share side of the node form, show that it is the critical-line word $\mathrm{crit}_1[1]$ (Definition 36.3), and evaluate it.

*Solution.* Receipt: `ex3_critical_u14.py`.

(i) *Counts.* The bases are the four marks, so $K = e_1(\alpha)$ and $\Psi = e_3(\alpha)$ (Definition 7.1), of the rank 1 and the corank 3. A proper set $S$ has $d(S) = 0$, a basis avoiding it, $r(S) = 1$, $d^*(S) = |S|-1$ and $c(S) = 1$ (Theorem 7.13, Definition 7.12).

*The full lean* $(s,0)$. The balance wall is $m_{\mathcal{E}} = s$, and the walls are $\kappa_S = m_S - s\,d(S) = m_S$ (Definition 8.1), at least 1 at positive integer masses. On the shape simplex $K(u) = |u| = 1$, so (Theorems 8.7 and 3.9)

$$P_{\mathcal{B}}(\mathbf{m};s,0) = \int_{\Delta_{\mathcal{E}}} u^{\mathbf{m-1}}\,d\omega_{\mathcal{E}}(u) = \Gamma^{\,4}_{\,1}\left[\begin{matrix}m_1,\ m_2,\ m_3,\ m_4\\ s\end{matrix}\right],$$

a positive rational number; at $(4,0)$ and the masses $\mathbf{1}$ it is $\Gamma^{\,4}_{\,1}\left[\begin{matrix}1,\ 1,\ 1,\ 1\\ 4\end{matrix}\right] = \frac{1}{6}$.

*The full lean* $(0,t)$. The balance wall is $m_{\mathcal{E}} = 3t$. By Corollary 34.8, with $2\sigma = t$, the family $(\mathcal{B};0,t)$ at $\mathbf{m}$ has the motivic period of $(\mathcal{B};t,0)$ at $t\mathbf{1}-\mathbf{m}$, whose total is $4t-3t = t$ and whose walls are those of $(\mathcal{B};0,t)$ at the complements, so

$$P_{\mathcal{B}}(\mathbf{m};0,t) = \Gamma^{\,4}_{\,1}\left[\begin{matrix}t-m_1,\ t-m_2,\ t-m_3,\ t-m_4\\ t\end{matrix}\right],$$

rational again; the walls of $(\mathcal{B};0,t)$ at the complements of the marks are $t-m_e\ge 1$.

*The motivic periods.* The set $\{3,4\}$ has width one with two marks on each side, and by Proposition 21.3(iii) the list is the gluing of its two cells: the cell of the bunch has the bases $\{3\},\{4\}$ of $\mathcal{B}|\{3,4\}$ and $g$, the one basis of $\mathcal{B}/\{1,2\}$ being $\varnothing$, and the cell of the room has the bases $\{1\},\{2\},\{g\}$. Both are the parallel nodes $\binom{\{3,4,g\}}{1}$ and $\binom{\{1,2,g\}}{1}$ on three marks, so the splitting of Corollary 34.18(b) has no prime node, and at either full lean $P_F^{\mathrm{mot}}$ is a positive rational number $c$. Its period is $c$, so $c=P_F$: the motivic periods are the rational numbers displayed.

(ii) *The centre.* At $(1,1)$ and the masses $\mathbf{1}$ every proper wall is the width 1 (Proposition 8.21(ii)), and $F=(\mathcal{B};1,1)$ is convergent. The pair $S=\{3,4\}$, with the room $S'=\{1,2\}$, has width one, and no basis contains both of its marks: it is a parallel bunch (Definition 19.1). The restriction $\mathcal{B}|S$ has the bases $\{3\},\{4\}$, and the contraction $\mathcal{B}/S'$, the bases meeting $S'$ in one mark less that mark, has the one basis $\varnothing$ (Definition 7.5), whose complement in $S$ is $S$. So (Definition 19.2, Proposition 19.3)

$$\gamma_S^K=\frac{\alpha_3+\alpha_4}{1},\qquad \gamma_S^\Psi=\frac{\alpha_3\alpha_4}{\alpha_3+\alpha_4},\qquad \varrho_S=\frac{\alpha_3\alpha_4}{(\alpha_3+\alpha_4)^2}=u(1-u)$$

on the shape $(u,1-u)$ of $S$, and likewise $\varrho_{S'}=u'(1-u')$ on the shape $(u',1-u')$ of $S'$.

*The law.* At width one the stations of the room are $D_0$ and $D_1$, of the excesses 0 and 1, and the seam letters are the letters of $\mathcal{B}/S'$ and of $\mathcal{B}|S$ (Proposition 28.3(ii)):

$$V_{D_0}=K_{\mathcal{B}/S'}=1,\qquad V_{D_1}=K_{\mathcal{B}|S}=\alpha_3+\alpha_4,$$
$$V^*_{D_0}=\Psi_{\mathcal{B}/S'}=\alpha_3\alpha_4,\qquad V^*_{D_1}=\Psi_{\mathcal{B}|S}=\alpha_3+\alpha_4.$$

With $d(S)=0$ and $d^*(S)=1$ the wall of the piece is $\kappa_S=2-(1\cdot 0+1\cdot 1)=1$ and its top wall is $(1+1)\cdot 1-1=1$ (Definition 28.4). Definition 28.5 gives the scale, the scale form on the shape $(u,1-u)$, and the seam vector $(\varrho^\rho_{D_0},\varrho^\rho_{D_1},\varrho^{\rho*}_{D_0},\varrho^{\rho*}_{D_1})$:

$$\gamma_S^\rho=\Big(\frac{\alpha_3+\alpha_4}{1}\Big)^{1/2}\Big(\frac{\alpha_3\alpha_4}{\alpha_3+\alpha_4}\Big)^{1/2}=(\alpha_3\alpha_4)^{1/2},$$
$$\phi_S^\rho(du)=1^{-1}\big(u+(1-u)\big)^{-1}\big(u(1-u)\big)^{-1/2}\,du=\varrho_S^{-1/2}\,du,$$
$$\boldsymbol{\varrho}_S^\rho=\Big(1,\ \frac{\alpha_3+\alpha_4}{(\alpha_3\alpha_4)^{1/2}},\ \frac{\alpha_3\alpha_4}{(\alpha_3+\alpha_4)(\alpha_3\alpha_4)^{1/2}},\ 1\Big)=\big(1,\ \varrho_S^{-1/2},\ \varrho_S^{1/2},\ 1\big).$$

The seam torus is cut out by $x_{D_0}=1$, $x^*_{D_1}=1$ and $x_{D_1}x^*_{D_0}=1$: it is the curve of the points $(1,x,x^{-1},1)$, $x>0$. The law is the image of $\varrho_S^{-1/2}du$ under $x=\varrho_S^{-1/2}=(u(1-u))^{-1/2}$. Since $u(1-u)\le\frac14$ it is carried by $x\ge 2$; each $x>2$ has the two preimages $u$ and $1-u$, with $(1-2u)\,du=-2x^{-3}dx$ and $|1-2u|=(1-4x^{-2})^{1/2}$, so the law is

$$\frac{4\,dx}{x\,(x^2-4)^{1/2}}\quad\text{on } x>2,\qquad \text{of total mass } \int_0^1\big(u(1-u)\big)^{-1/2}du=\Gamma_{\substack{2\\1}}\!\left[\begin{matrix}\frac12,\ \frac12\\ 1\end{matrix}\right]$$

(Theorem 3.9). It has a density, so it is not a point mass. Theorem 28.25 says the same through the hinge values: $\mathbf{E}_{\mathcal{B}|S}=(\alpha_3+\alpha_4)^2$ and $\mathbf{E}_{\mathcal{B}/S'}=\alpha_3\alpha_4$ differ, $\varrho_S$ not being constant, so $S$ is not pinned and does not split off.

(iii) *The node form at the root bunch $S$.* As in Exercise 53.1(ii), a tree of bunches whose root carries $S$ has the step $\mathcal{E}$ as a node step of the root (Definition 34.9(ii)), and the period is read by the node form there (Theorem 34.16(v)). Read $\alpha_S=\rho\,(u,1-u)$ and $\alpha_{S'}=(u',1-u')$ on the section $\alpha_1+\alpha_2=1$; then $d\alpha_3\,d\alpha_4=\rho\,d\rho\,du$ (Lemma 2.7) and

$$K(\rho u,u')=\rho+1,\qquad \Psi(\rho u,u')=\rho\,u'(1-u')+\rho^2u(1-u)=\rho\,\varrho_S\Big(\rho+\frac{\varrho_{S'}}{\varrho_S}\Big):$$

the Skeleton values (Theorem 34.16(ii)) are $\xi_K = 1$ and $\xi_\Psi = \varrho_{S'}/\varrho_S = \gamma^\Psi_{S'}/\gamma^\Psi_S$, the hinge value of the room over that of the bunch on $\Delta_S \times \Delta_{S'}$ (Theorem 34.16(iv)). By Theorem 8.7,

$$P_{\mathcal{B}}(\mathbf{1}; 1, 1) = \int_{\Delta_S \times \Delta_{S'}} \int_0^\infty \frac{d\rho}{(\rho+1)\,(\varrho_S\,\rho + \varrho_{S'})}\, d\omega_S(u)\, d\omega_{S'}(u'),$$

the node form with one term: $A_t = \varrho_S^{-1}$, $B_t = 1$, $\nu_t = 1$ and $\mathbf{w}_t = (1,1)$, so that $\mathrm{Sc}_t = \Gamma^{\,2}_{\,2}\!\begin{bmatrix} 1,\ 1 \\ 1,\ 1 \end{bmatrix} \mathsf{B}^{\{J\}}_{(1,1),\,1;\,I_J}\big(1, \varrho_{S'}/\varrho_S\big)$: the bunch and the room meet through the one Skeleton value $\varrho_{S'}/\varrho_S$.

*The share side.* Corollary 15.5, with $\xi_0 = 1$, $\xi_1 = \varrho_{S'}/\varrho_S$ and one share $a$, writes the term as

$$\begin{aligned} A_t\,\mathrm{Sc}_t &= \varrho_S^{-1}\,\frac{1}{2\pi i}\int_{(c)} \Gamma^{\,4}_{\,0}\big[a,\ a,\ 1-a,\ 1-a\big]\,\Big(\frac{\varrho_{S'}}{\varrho_S}\Big)^{a-1} da \\ &= \frac{1}{2\pi i}\int_{(c)} \Gamma^{\,4}_{\,0}\big[a,\ a,\ 1-a,\ 1-a\big]\ \varrho_S^{-a}\,\varrho_{S'}^{\,a-1}\, da. \end{aligned}$$

On the line $\mathrm{Re}\,a = \frac12$ the two conditions of Theorem 34.16(iv) are $\int_{\Delta_S} \varrho_S^{-1/2} d\omega_S < \infty$ and $\int_{\Delta_{S'}} \varrho_{S'}^{-1/2} d\omega_{S'} < \infty$, both $\Gamma^{\,2}_{\,1}\!\begin{bmatrix} \frac12,\ \frac12 \\ 1 \end{bmatrix}$; so the bunch and the room separate under the share integral, and by Theorem 3.9, for $0 < \mathrm{Re}\,a < 1$,

$$\int_0^1 \big(u(1-u)\big)^{-a} du = \Gamma^{\,2}_{\,1}\!\begin{bmatrix} 1-a,\ 1-a \\ 2-2a \end{bmatrix}, \qquad \int_0^1 \big(u'(1-u')\big)^{a-1} du' = \Gamma^{\,2}_{\,1}\!\begin{bmatrix} a,\ a \\ 2a \end{bmatrix}.$$

So

$$P_{\mathcal{B}}(\mathbf{1}; 1, 1) = \frac{1}{2\pi i}\int_{(1/2)} \Gamma^{\,4}_{\,0}\big[a,\ a,\ 1-a,\ 1-a\big]\ \Gamma^{\,2}_{\,1}\!\begin{bmatrix} 1-a,\ 1-a \\ 2-2a \end{bmatrix}\ \Gamma^{\,2}_{\,1}\!\begin{bmatrix} a,\ a \\ 2a \end{bmatrix}\, da :$$

the moments of the orders $-a$ and $a-1$ of the two ratios of hinge values, paired over one share. This is the one-share display of Theorem 19.13 at the level shares $a^d = b^d = a$ and $a^r = b^r = 1-a$, the bunch family and the room family having these two Beta corners as periods: the even share $\sigma^d = a$ is free and the odd share is 0.

*The critical-line word.* By the reflection at $a$ and at $2a$ and the shift $\Gamma(2-2a) = (1-2a)\,\Gamma(1-2a)$ (Proposition 1.3(iii),(v)),

$$\Gamma^{\,2}_{\,1}\!\begin{bmatrix} a,\ a \\ 2a \end{bmatrix}\Gamma^{\,2}_{\,1}\!\begin{bmatrix} 1-a,\ 1-a \\ 2-2a \end{bmatrix} = \frac{\Gamma^{\,4}_{\,0}\big[a,\ a,\ 1-a,\ 1-a\big]}{(1-2a)\,\Gamma^{\,2}_{\,0}\big[2a,\ 1-2a\big]} = \frac{\pi^2}{\sin^2 \pi a}\cdot\frac{\sin 2\pi a}{\pi\,(1-2a)} = \frac{2\pi\cot \pi a}{1-2a},$$

and $\partial_a \Gamma^{\,4}_{\,0}\big[a,\ a,\ 1-a,\ 1-a\big] = \partial_a\big(\pi^2/\sin^2\pi a\big) = -2\pi\cot\pi a \cdot \Gamma^{\,4}_{\,0}\big[a,\ a,\ 1-a,\ 1-a\big]$. So

$$P_{\mathcal{B}}(\mathbf{1}; 1, 1) = -\frac{1}{2\pi i}\int_{(1/2)} \frac{\partial_a \Gamma^{\,4}_{\,0}\big[a,\ a,\ 1-a,\ 1-a\big]}{1-2a}\, da = \mathrm{crit}_1[1],$$

Definition 36.3 at $k = 1$ and $R = 1$, where $-\frac{2}{(2k)!} = -1$.

*Closing.* The integrand is holomorphic on $0 < \mathrm{Re}\,a < 1$, regular at $a = \frac12$ (Definition 36.3), and on every vertical line of the strip it is $O(e^{-2\pi|\,\mathrm{Im}\,a|})$, since $|\sin\pi a|^2 = \sin^2\pi\,\mathrm{Re}\,a + \sinh^2\pi\,\mathrm{Im}\,a$; so

the line moves to $\operatorname{Re} a = c$ with $\frac{1}{2} < c < 1$, and an integration by parts, with no boundary terms, gives

$$P_{\mathcal{B}}(\mathbf{1};1,1) = \frac{1}{2\pi i}\int_{(c)} \Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]\ g(a)\,da, \qquad g(a) = \partial_a \frac{1}{1-2a} = \frac{2}{(1-2a)^2}.$$

The function $g$ is rational, holomorphic on $\operatorname{Re} a \geq c$, and $O(|a|^{-2})$, so the right closing of Lemma 5.16(i) applies: with $g'(a) = 8(1-2a)^{-3}$,

$$P_{\mathcal{B}}(\mathbf{1};1,1) = -\sum_{m\geq 1} g'(m) = \sum_{m\geq 1} \frac{8}{(2m-1)^3} = \sum_{m\geq 1} \Big(m - \frac{1}{2}\Big)^{-3}.$$

(Closed directly, the integrand $-\partial_a \Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]/(1-2a)$ has at $a = m \geq 1$ a pole of order three with the residue $8(1-2m)^{-3}$; the receipt checks $m \leq 6$.)

*The Trailmoment.* At one unit exponent and the coefficient function $x$ the last sum is $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty))$ (Definition 5.2). The lattice relation at the step 2 (Theorem 5.12(i), $q = 2$, $b = 0$, the cut 0) assembles $\frac{1}{2}\mathbb{Z}$ from the cosets $\mathbb{Z}$ and $\frac{1}{2}+\mathbb{Z}$, the coefficient function $x/2$ multiplying the kernel by $2^3$:

$$(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) = 2^3\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

So $P_{\mathcal{B}}(\mathbf{1};1,1) = (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, which is $\mathrm{crit}_1[1]$ of Theorem 36.8; it is $8.414398322117159997\ldots$ to eighteen places. The receipt checks the relation on the exact heads $\sum_{m\leq N}(m-\frac{1}{2})^{-3} + \sum_{m\leq N} m^{-3} = 8\sum_{k\leq 2N} k^{-3}$ and the share integral against $7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ to twenty-five places. At a full lean one letter is active with one Skeleton value, no share is free, and the period is rational, by (i); at the centre the one free share runs over the critical line and carries a Trailmoment of the odd index three.

*Answer.* At $(s,0)$ the period is $\Gamma_{\substack{4\\1}}\begin{bmatrix}\mathbf{m}\\ s\end{bmatrix}$ and at $(0,t)$ it is $\Gamma_{\substack{4\\1}}\begin{bmatrix}t\mathbf{1}-\mathbf{m}\\ t\end{bmatrix}$, rational, and the motivic periods are these rational numbers, the list having no prime node. At the centre $S = \{3,4\}$ is a parallel bunch of width one with $\gamma_S^K = \alpha_3+\alpha_4$, $\gamma_S^\Psi = \alpha_3\alpha_4/(\alpha_3+\alpha_4)$ and $\varrho_S = u(1-u)$; its law is $4\,dx/(x(x^2-4)^{1/2})$ on the points $(1,x,x^{-1},1)$, $x > 2$, of its seam torus, not a point mass. Read at the root bunch $S$,

$$\begin{aligned} P_{\binom{[4]}{1}}(\mathbf{1};1,1) &= \frac{1}{2\pi i}\int_{(1/2)} \Gamma_{\substack{4\\0}}\big[a,\ a,\ 1-a,\ 1-a\big]\,\Gamma_{\substack{2\\1}}\begin{bmatrix}1-a,\ 1-a\\ 2-2a\end{bmatrix}\Gamma_{\substack{2\\1}}\begin{bmatrix}a,\ a\\ 2a\end{bmatrix}\,da \\ &= \mathrm{crit}_1[1] = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)). \end{aligned}$$

□

## 53.2 Substitutions of dual pairs

**Exercise 53.4** (Substitutions on five and on eight marks)**.** Let $\mathrm{P}_5 = \binom{\{1,2,h\}}{1}\langle 1,2,\mathcal{A}\rangle$, the marks $1,2$ left in place and the cell $\mathcal{A} = \binom{\{3,4,5,g\}}{2}$ at $h$, and let $\mathrm{M}_8 = \binom{\{b_1,b_2,h_1,h_2\}}{2}\langle b_1,b_2,\mathcal{A}_1,\mathcal{A}_2\rangle$, the marks $b_1,b_2$ left in place and the cells $\mathcal{A}_i = \binom{\{x_i,y_i,z_i,g_i\}}{2}$ at $h_i$, with $R = \{3,4,5\}$, $R_i = \{x_i,y_i,z_i\}$ and $B = \{b_1,b_2\}$ (Definition 21.1).

(i) Find the bases of $\mathrm{P}_5$, show that its cell is a dual pair and that $\mathrm{P}_5$ is convergent at its centre, and find its motivic period there.

(ii) Find the rank of $\mathrm{M}_8$, the widths of $R_1$, $R_2$ and $B$, and its letters; show that it is convergent at its centre and find its motivic period there. What becomes of the bunch $B$?

*Solution.* Receipt: `ex4_substitution.py`, with the helper `subst.py`.

(i) *The bases.* By Definition 21.1, the core bases $\{1\}$ and $\{2\}$ avoid $h$ and take a basis of $\mathcal{A}/g = \binom{R}{1}$, and the core basis $\{h\}$ takes a basis of $\mathcal{A} \setminus g = \binom{R}{2}$: the bases of $\mathrm{P}_5$ are the six pairs $\{1,x\},\{2,x\}$, $x \in R$, and the three pairs of $R$, all the pairs of $\{1,\dots,5\}$ but $\{1,2\}$, and its rank is $1+(2-1)=2$ (Proposition 21.3(i)). The complements in $R$ of the pairs of $R$ are its marks, so $\mathcal{A}/g$ is the dual of $\mathcal{A} \setminus g$: the cell is a dual pair, and by Proposition 21.8(ii) both letters read it through the one hinge value $\gamma = e_2(\alpha_R)/e_1(\alpha_R)$. By Theorem 21.4, with $K_{\mathcal{A}/g} = \Psi_{\mathcal{A}\setminus g} = e_1(\alpha_R)$,

$$K = e_1(\alpha_R)\,\big(\alpha_1+\alpha_2+\gamma\big), \qquad \Psi = e_1(\alpha_R)\,\big(\alpha_1\alpha_2+(\alpha_1+\alpha_2)\,\gamma\big).$$

*Walls.* $R$ is a bunch of width one with the cell $\mathcal{A}$ (Proposition 21.3(i)); its wall at the centre is $\kappa_R = m_R-(s+t)(r(R)-1) = 3-2\cdot 1 = 1$, $r(R)=2$ (Corollary 34.18(a)), and $2\sigma-\kappa_R = 1$. Both letters read $R$ through $\gamma$, so by Theorem 19.9(iii) $\mathrm{P}_5$ has every proper wall positive exactly when its two cell families do: $(\mathcal{A};1,1)$ at the masses $(\mathbf{1}_R,1)$, which is $\binom{[4]}{2}$ at its centre, and $\big(\binom{\{1,2,h\}}{1};1,1\big)$ at the masses $(1,1,\kappa_R)=\mathbf{1}$, which is $\binom{[3]}{1}$ at its centre; both are connected, and their walls are their widths (Proposition 8.21(ii)). So every proper wall of $\mathrm{P}_5$ is positive, an integer and so at least 1 (the receipt finds the values 1 and 2), and $\mathrm{P}_5$ is convergent at its centre.

*The motivic period.* Corollary 34.18(a) with $q=1$:

$$P^{\mathrm{mot}}_{\mathrm{P}_5} = P^{\mathrm{mot}}_{(\binom{\{1,2,h\}}{1};1,1)} \cdot P^{\mathrm{mot}}_{(\binom{R}{2};1,1)},$$

the core at the masses $\mathbf{1}$ and the pinned family $(\mathcal{B}|R;\kappa_R,2-\kappa_R) = \big(\binom{R}{2};1,1\big)$ at $\mathbf{1}_R$. The core has the letters $e_1$ and $e_2$ of $\alpha_1,\alpha_2,\alpha_h$, so its form is $\Omega/(e_1e_2)$, the form of $\binom{[3]}{2}$ at its centre, whose letters are $e_2$ and $e_1$; the pinned family is $\binom{[3]}{2}$ at its centre on the marks of $R$. Two convergent families with one form have one motivic period (Lemma 34.7, with the renaming of the coordinates), so

$$P^{\mathrm{mot}}_{\mathrm{P}_5} = \big(P^{\mathrm{mot}}_{\binom{[3]}{2}}\big)^2,$$

the two factors being one motivic period. In coefficients,

$$\frac{\alpha^{\mathbf{1}}}{K\,\Psi} = \frac{\alpha_1\alpha_2\,\gamma}{e_1(\alpha_1,\alpha_2,\gamma)\,e_2(\alpha_1,\alpha_2,\gamma)} \cdot \frac{\alpha^{\mathbf{1}_R}}{e_1(\alpha_R)\,e_2(\alpha_R)},$$

the core at $(\alpha_1,\alpha_2,\gamma)$ times $\binom{[3]}{2}$ on $R$.

(ii) *Rank and widths.* By Proposition 21.3(i) $\mathrm{M}_8$ is a list of rank $2+(2-1)+(2-1)=4$ on eight marks, and each $R_i$ is a bunch of width one with the cell $\mathcal{A}_i$; from the six core bases, with three choices at each cell, it has $6\cdot 3\cdot 3 = 54$ bases. The pairs of the core meet $B$ in 0, 1 or 2 marks, so $B$ has the width two in the core, and by Corollary 21.11 in $\mathrm{M}_8$.

*Letters.* Both cells are dual pairs with $k_i = e_2(\alpha_{R_i})$, $\psi_i = e_1(\alpha_{R_i})$ and the hinge values $\gamma_i = e_2(\alpha_{R_i})/e_1(\alpha_{R_i})$, and the marks left in place have $k=\alpha_b$, $\psi=1$ (Proposition 21.8(ii),(iii)). With $\pi = e_1(\alpha_{R_1})\,e_1(\alpha_{R_2})$, $r_{\mathcal{H}}=2$ and $q=4$, Proposition 21.8(iii) gives, $K_{\mathcal{H}} = \Psi_{\mathcal{H}} = e_2$ being homogeneous of degree two,

$$\pi^2\,e_2(\alpha_{b_1},\alpha_{b_2},\gamma_1,\gamma_2) = \pi\,K, \qquad \pi^2\,e_2(\alpha_{b_1},\alpha_{b_2},\gamma_1,\gamma_2) = \pi^{4-2-1}\,\Psi,$$

so $K=\Psi = e_1(\alpha_{R_1})\,e_1(\alpha_{R_2})\,e_2(\alpha_{b_1},\alpha_{b_2},\gamma_1,\gamma_2)$: $\mathrm{M}_8$ is identically self-dual, its cobases being its bases.

*Walls.* The walls of the cells are $\kappa_{R_i} = 3-2\,(2-1) = 1$ (Corollary 34.18(a)). Theorem 19.9(iii), at $R_1$ and then at $R_2$ in the cell of the room, both letters reading each $R_i$ through $\gamma_i$, reduces the positivity of the walls to that of the three families $(\mathcal{A}_1; 1, 1)$, $(\mathcal{A}_2; 1, 1)$ and $(\binom{\{b_1,b_2,h_1,h_2\}}{2}; 1, 1)$, all at the masses $\mathbf{1}$ (the hinges carrying $2\sigma - \kappa_{R_i} = 1$ and $\kappa_{R_i} = 1$): three copies of $\binom{[4]}{2}$ at its centre, convergent by Exercise 53.2. So $\mathrm{M}_8$ is convergent at its centre; its 254 proper walls are positive integers (the receipt lists them, the least being 1).

*The motivic period.* Corollary 34.18(a) with $q = 2$, the pinned families $\big(\binom{R_i}{2}; 1, 1\big)$ at $\mathbf{1}_{R_i}$ and the core $\binom{\{b_1,b_2,h_1,h_2\}}{2}$ at its centre:

$$P^{\mathrm{mot}}_{\mathrm{M}_8} = P^{\mathrm{mot}}_{\binom{[4]}{2}} \cdot \big(P^{\mathrm{mot}}_{\binom{[3]}{2}}\big)^2 = \big(P^{\mathrm{mot}}_{\binom{[3]}{2}}\big)^3,$$

the second equality by Exercise 53.2(ii), every factor at its centre. In coefficients,

$$\frac{\alpha^{\mathbf{1}}}{K\,\Psi} = \frac{\alpha_{b_1}\alpha_{b_2}\,\gamma_1\gamma_2}{e_2(\alpha_{b_1},\alpha_{b_2},\gamma_1,\gamma_2)^2} \cdot \prod_{i=1}^{2} \frac{\alpha^{\mathbf{1}_{R_i}}}{e_1(\alpha_{R_i})\,e_2(\alpha_{R_i})}.$$

*The bunch $B$.* $B$ is left in place and keeps in $\mathrm{M}_8$ the width two it has in the core (Theorem 34.17(v)); by Corollary 28.26(i) it does not split off and enters the period through a law carried by no point. The cells split off around it, and $B$ goes whole into the core $\binom{[4]}{2}$, which splits further only at the complement of a single mark, a set of width one (Exercise 53.2). Along $B$, $K(\tau\alpha_B, \alpha_{R_1}, \alpha_{R_2}) = \pi\, e_2(\tau\alpha_{b_1}, \tau\alpha_{b_2}, \gamma_1, \gamma_2)$, so the Skeleton values of $B$ in $\mathrm{M}_8$ are those of the pair $\{1, 2\}$ of $\binom{[4]}{2}$ at the point $(\alpha_{b_1}, \alpha_{b_2}, \gamma_1, \gamma_2)$.

*Answer.* $\mathrm{P}_5$ has as its bases the pairs of $\{1, \dots, 5\}$ other than $\{1, 2\}$, its cell $\binom{\{3,4,5,g\}}{2}$ is a dual pair, and $P^{\mathrm{mot}}_{\mathrm{P}_5} = (P^{\mathrm{mot}}_{\binom{[3]}{2}})^2$. $\mathrm{M}_8$ has the rank 4; $R_1, R_2$ have width one and $B$ width two; $K = \Psi = e_1(\alpha_{R_1})e_1(\alpha_{R_2})\,e_2(\alpha_{b_1}, \alpha_{b_2}, \gamma_1, \gamma_2)$, and $P^{\mathrm{mot}}_{\mathrm{M}_8} = (P^{\mathrm{mot}}_{\binom{[3]}{2}})^3$. So

$$P_{\mathrm{P}_5}(\mathbf{1}; 1, 1) = \Big(3\,(\mathsf{T}_{2;(1)}[x]\,(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}))((0,\infty))\Big)^2,$$
$$P_{\mathrm{M}_8}(\mathbf{1}; 1, 1) = \Big(3\,(\mathsf{T}_{2;(1)}[x]\,(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}))((0,\infty))\Big)^3,$$

with $\mathrm{C}(P^{\mathrm{mot}}_{\mathrm{P}_5}) \subseteq \mathrm{C}(P^{\mathrm{mot}}_{\binom{[3]}{2}}) \cdot \mathrm{C}(P^{\mathrm{mot}}_{\binom{[3]}{2}})$, and likewise with three factors for $\mathrm{M}_8$ (Theorem 34.17(ii)). The bunch $B$ of width two does not split off and goes whole into the core. □

## 53.3 Full leans

**Exercise 53.5** (Full leans at the exponents $(0, 2)$)**.** Let $F = (\mathcal{B}; 0, 2)$ be a convergent family of one list on $\mathcal{E}$, $|\mathcal{E}| \geq 3$, at the masses $\mathbf{1}$ (Definition 34.1).

(i) Show that $|\mathcal{E}| = 2r$ and that $\kappa_S = 2r(S) - |S|$. At a set $R$ of width one with at least two marks on each side, show that the two hinge walls $\kappa_R$ and $\kappa_{\mathcal{E}\setminus R}$ are 1, that both cell families have the masses $\mathbf{1}$, and that the Beta corner of the split is 1.

(ii) Show that every node of the splitting of Corollary 34.18(b) is prime and that $P^{\mathrm{mot}}_F = \prod_v P^{\mathrm{mot}}_{F_v}$, every $F_v = (\mathcal{H}_v; 0, 2)$ at the masses $\mathbf{1}$.

(iii) Let $\mathrm{T}_6$ be the list of rank three on $\{1, \dots, 6\}$ whose bases are the nineteen triples other than $\{1, 2, 3\}$, let $\mathcal{T}$ be the cell $\mathrm{T}_6$ on $\{1, \dots, 5\} \sqcup \{g\}$ with the mark 6 as its hinge $g$, let $\mathcal{T}'$ be

a copy of it on $\{1',\dots,5'\}\sqcup\{g\}$, and let $\mathrm{D}_{10}=\mathcal{T}\oplus_g\mathcal{T}'$ (Lemma 21.12). Find the motivic period of $(\mathrm{D}_{10};0,2)$ at the masses $\mathbf{1}$, and decide whether $\{1',\dots,5'\}$ splits off at the centre and at $(0,2)$.

*Solution.* Receipt: `ex5_fulllean.py`, with the helper `subst.py`.

(i) *Balance and walls.* At $(0,2)$ only $\Psi$ is active, $\sigma=\frac{0+2}{2}=1$, and the imbalance is $\kappa_{\mathcal{E}}=n-2\,(n-r)=2r-n$ (Definition 8.1); on the balance wall $n=2r$. The walls are $\kappa_S=m_S-2\,d^*(S)$ with $d^*(S)=|S|-r(S)$ (Definition 7.12), so

$$\kappa_S=|S|-2\big(|S|-r(S)\big)=2r(S)-|S|.$$

*The hinge walls.* For a set $R$ of width one, Proposition 8.16 gives

$$\kappa_R+\kappa_{\mathcal{E}\setminus R}=\kappa_{\mathcal{E}}+2\sigma\,c(R)=0+2\cdot1\cdot1=2,$$

and both walls are integers at least 1, $F$ being convergent; so $\kappa_R=\kappa_{\mathcal{E}\setminus R}=1$. By Proposition 21.3(iii) the list is $\mathcal{H}\langle\mathcal{A}\rangle$, the cell of the bunch $\mathcal{A}$ on $R\sqcup\{g\}$ substituted at the marker $h$ of the cell of the room $\mathcal{H}$, the marks of $L=\mathcal{E}\setminus R$ left in place. One letter is active, so the cell is read through the one hinge value $\gamma^{\Psi}_{\mathcal{A}}$, and Theorem 34.17 applies with $q=1$, $2\sigma=2$, $\kappa_1=\kappa_R=1$ and the loads $a=0$, $b=1$. The core family $(\mathcal{H};0,2)$ carries the masses $\mathbf{1}_L$ and $\kappa_R=1$ at $h$, and the cell family $(\mathcal{A};0,2)$ the masses $\mathbf{1}_R$ and $2\sigma-\kappa_R=1$ at $g$: both have the masses $\mathbf{1}$, and both are convergent (Theorem 34.17(i),(iii)). The pinned family $G$ has the tables $\mathcal{A}\setminus g$ at $(0,2-1)=(0,1)$ and $\mathcal{A}/g$ at $(0,1)$, at the masses $\mathbf{1}_R$. The Beta corner of the split is

$$\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix}\kappa_1,\ 2\sigma-\kappa_1\\ 2\sigma\end{bmatrix}=\Gamma_{\begin{smallmatrix}2\\1\end{smallmatrix}}\begin{bmatrix}1,\ 1\\ 2\end{bmatrix}=\frac{\Gamma(1)\,\Gamma(1)}{\Gamma(2)}=1.$$

So Theorem 34.17(ii),(iii) give $P^{\mathrm{mot}}_F=P^{\mathrm{mot}}_{(\mathcal{H};0,2)}P^{\mathrm{mot}}_G$ and $P^{\mathrm{mot}}_{(\mathcal{A};0,2)}=1\cdot P^{\mathrm{mot}}_G$, that is

$$P^{\mathrm{mot}}_F=P^{\mathrm{mot}}_{(\mathcal{H};0,2)}\,P^{\mathrm{mot}}_{(\mathcal{A};0,2)},$$

both at the masses $\mathbf{1}$: the split costs no factor.

(ii) *The nodes.* Each of the two families of (i) is again a convergent family of one list at $(0,2)$ with the masses $\mathbf{1}$, so (i) applies to it: it has an even number $2r_v$ of marks, and every further split costs no factor. A node made by a split has at least three marks, two of one side and its marker, and an even number of them, hence at least four; so no node on two or three marks occurs. A node at which the splitting stops has no set of width one with two marks on each side, and it is connected, since $\kappa_S+\kappa_{\mathcal{E}_v\setminus S}=2c(S)$ with both walls at least 1 gives $c(S)\geq1$ (Proposition 8.16, Definition 8.1); so it is prime (Theorem 23.16), as Corollary 34.18(b) says of the nodes on at least four marks. By induction on the number of marks,

$$P^{\mathrm{mot}}_F=\prod_v P^{\mathrm{mot}}_{F_v},\qquad F_v=(\mathcal{H}_v;0,2)\text{ at the masses }\mathbf{1},\qquad \mathrm{C}\big(P^{\mathrm{mot}}_F\big)\subseteq\prod_v\mathrm{C}\big(P^{\mathrm{mot}}_{F_v}\big)$$

(Lemma 34.6(iii)): the rational constant of Corollary 34.18(b) is 1, and every marker carries the mass 1. By Corollary 34.8 the same holds at $(2,0)$, whose masses $2\sigma\mathbf{1}-\mathbf{1}$ are again $\mathbf{1}$.

(iii) *The list* $\mathrm{T}_6$. Every set of at most three marks other than $\{1,2,3\}$ lies in a basis, $r(\{1,2,3\})=2$, and every set of at least four marks contains a basis; $n=6=2r$, so $\mathrm{T}_6$ at $(0,2)$ with the masses $\mathbf{1}$ is on its balance wall. By (i) its walls are $2r(S)-|S|$:

| $S$ | $\lvert S\rvert=1$ | $\lvert S\rvert=2$ | $\lvert S\rvert=3,\ S\neq\{1,2,3\}$ | $S=\{1,2,3\}$ | $\lvert S\rvert=4$ | $\lvert S\rvert=5$ |
|---|---|---|---|---|---|---|
| $r(S)$ | 1 | 2 | 3 | 2 | 3 | 3 |
| $\kappa_S$ | 1 | 2 | 3 | 1 | 2 | 1 |

all at least 1: $(\mathrm{T}_6; 0, 2)$ is convergent. Its widths $c(S) = r(S) + r(\mathcal{E} \setminus S) - 3$ (Definition 7.12) are 1 at one mark, 2 at two marks, 3 at a triple other than $\{1, 2, 3\}$ and $\{4, 5, 6\}$ and 2 at these two: the sets with two marks on each side have widths two and three, and $\mathrm{T}_6$ is connected and prime (Theorem 23.16). The mark 6 is neither a loop nor a coloop, so $\mathcal{T}$ is a cell, with the hinge minors

$$\mathcal{T} \setminus g: \text{ the nine triples of } \{1, \dots, 5\} \text{ other than } \{1, 2, 3\}, \qquad \mathcal{T}/g: \text{ the ten pairs of } \{1, \dots, 5\},$$

of the ranks 3 and 2 (Definition 7.3).

*The list* $\mathrm{D}_{10}$. By Lemma 21.12 it is a list on ten marks of rank $3 + 3 - 1 = 5$, with the $9 \cdot 10 + 10 \cdot 9 = 180$ bases $X \sqcup Y$, $X \in \mathcal{T} \setminus g$ and $Y \in \mathcal{T}'/g$ or $X \in \mathcal{T}/g$ and $Y \in \mathcal{T}' \setminus g$, and $S = \{1', \dots, 5'\}$ is a bunch of width one with the cell of the bunch $\mathcal{T}'$ and the cell of the room $\mathcal{T}$. At $(0, 2)$ with the masses $\mathbf{1}$ it is on its balance wall, $10 = 2 \cdot 5$, and its hinge walls are (Lemma 21.12)

$$\kappa_S = m_S - 2\left(|S| - r_{\mathcal{T}'}\right) = 5 - 2\,(5 - 3) = 1, \qquad \kappa_{\mathcal{E}\setminus S} = 5 - 2\,(5 - 3) = 1,$$

adding to $2\sigma = 2$. One letter is active, so by Theorem 19.9(iii) every proper wall of $\mathrm{D}_{10}$ is positive exactly when both cell families are, $(\mathcal{T}'; 0, 2)$ and $(\mathcal{T}; 0, 2)$ at the masses $\mathbf{1}$, the hinge carrying $2\sigma - \kappa = 1$: two copies of $(\mathrm{T}_6; 0, 2)$ at $\mathbf{1}$, whose walls are displayed above. So $(\mathrm{D}_{10}; 0, 2)$ is convergent; the receipt finds the least of its 1022 proper walls to be 1.

*The motivic period.* The split at $S$ gives the nodes $\mathcal{T}'$ and $\mathcal{T}$, copies of $\mathrm{T}_6$ with the marker in place of the mark 6, both prime; by (ii), and Lemma 34.7 for the renaming of the marks,

$$P^{\mathrm{mot}}_{(\mathrm{D}_{10};0,2)} = \left(P^{\mathrm{mot}}_{(\mathrm{T}_6;0,2)}\right)^2, \qquad \mathrm{C}\left(P^{\mathrm{mot}}_{(\mathrm{D}_{10};0,2)}\right) \subseteq \mathrm{C}\left(P^{\mathrm{mot}}_{(\mathrm{T}_6;0,2)}\right) \cdot \mathrm{C}\left(P^{\mathrm{mot}}_{(\mathrm{T}_6;0,2)}\right),$$

all at the masses $\mathbf{1}$, with the Beta corner 1 and no other factor.

*The set* $S$ *at the centre.* At $(1, 1)$ with the masses $\mathbf{1}$ the walls are the widths (Proposition 8.21(ii)), all positive, $\mathrm{D}_{10}$ being connected by (ii); so Theorem 28.25 applies. With both letters active and $|S| = 5$, $S$ is pinned exactly when $\mathbf{E}_{\mathrm{D}_{10}|S} = \mathbf{E}_{\mathrm{D}_{10}/(\mathcal{E}\setminus S)}$, and these hinge minors are $\mathcal{T}' \setminus g$ and $\mathcal{T}'/g$ (Theorem 19.4), with 9 and 10 bases. At the gauge $\mathbf{1}$ the even letter of a list is the square of its number of bases, so the two even letters take the values 81 and 100 there and differ: at the centre $S$ is not pinned, it does not split off, and its law is carried by no point. At $(0, 2)$ one letter is active, so $S$ is pinned (Theorem 28.25(ii)) and splits off, with the Beta corner 1, as above.

*Answer.* At $(0, 2)$ with the masses $\mathbf{1}$, $|\mathcal{E}| = 2r$ and $\kappa_S = 2r(S) - |S|$; at a set of width one the two hinge walls add to 2 and are 1, the cell families keep the masses $\mathbf{1}$, and the Beta corner is $\Gamma_{\substack{2\\1}}\begin{bmatrix} 1, & 1 \\ & 2 \end{bmatrix} = 1$. Every node of the splitting is prime, and $P^{\mathrm{mot}}_F = \prod_v P^{\mathrm{mot}}_{F_v}$ with every $F_v = (\mathcal{H}_v; 0, 2)$ at the masses $\mathbf{1}$. For $\mathrm{D}_{10} = \mathcal{T} \oplus_g \mathcal{T}'$, a list of rank five on ten marks with 180 bases,

$$P^{\mathrm{mot}}_{(\mathrm{D}_{10};0,2)} = \left(P^{\mathrm{mot}}_{(\mathrm{T}_6;0,2)}\right)^2 \quad \text{at the masses } \mathbf{1};$$

the set $\{1', \dots, 5'\}$ splits off at $(0, 2)$ and does not split off at the centre, where its hinge minors have the even letters 81 and 100 at the gauge $\mathbf{1}$. □

# Chapter 54

# Exercises on rays: the small centres

The exercises of this chapter read the list $\binom{[3]}{1}$ along its rays. Its centre ray is read four ways. A certificate proves its law at every strength, the low-end relation fixes the scale, and positivity with the native value of the direction brackets the value $\ell = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ to $10^{-7}$. The box of its walls gives a spanning set of two insertions, the one-step matrix, a certificate supported on the stems and the law, with no closing computed, and shows why the closing relations alone do not give the law. The gauge end gives the values and their slopes to a hundred digits with no datum from the low end, and returns the low-end coefficients. The torus through the native gauge carries a second solution of the law; the torus value, the values and the torus read backwards are tied by an explicit connection, which reads the mean of $\log|K|$ on the torus and writes $\ell$ as three times a Trailmoment of index two of the lattice inputs of step three. A lean of three marks and the balanced directions of small size have their values in $\mathbb{Q} + \mathbb{Q}\ell$, read as values of insertions along the centre ray and reduced in its boxes. Every number displayed is exact unless it is labelled as a rendering; the receipts that recompute them are the scripts named in the solutions.

## 54.1 The centre ray, bracketed

**Exercise 54.1** (The centre ray of $\binom{[3]}{1}$, bracketed)**.** Let $F$ be the list $\binom{[3]}{1}$: the marks $\mathcal{E} = \{1, 2, 3\}$, the bases $\{1\}, \{2\}, \{3\}$ and unit weights, so that its letters are

$$K = \alpha_1 + \alpha_2 + \alpha_3, \qquad \Psi = \alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3.$$

Let $P(k) = P_F(k\mathbf{1}; k, k)$ be the values of its centre ray at the strength 1, and $\ell = P(1)$.

(i) From the closings of the ray find a candidate recurrence of the values, and prove it by a certificate.

(ii) Show that it holds at every $k$ with $\operatorname{Re} k > 0$, and read its low-end relation.

(iii) Find a rational $q$ with $0 < \ell - q < 10^{-6}$, and prove both inequalities.

Exact computation may be done by machine; floating point, quadrature and integer-relation searches serve only as checks.

*Solution.* Receipt: `ex18_centre_bracket.py`. Throughout, $e_1, e_2, e_3$ are the elementary symmetric polynomials of $\alpha_1, \alpha_2, \alpha_3$, so that $K = e_1$ and $\Psi = e_2$, and $D_e = \alpha_e\,\partial/\partial\alpha_e$, which is $\partial_{y_e}$ in the coordinates $y_e = \log\alpha_e$.

*Step 1: the ray.* The list has rank 1. For a nonempty proper $S$, $r(S) = 1$ and $r^*(S) = |S| + r(\mathcal{E} \setminus S) - 1 = |S|$, so $d(S) = 0$, $d^*(S) = |S| - 1$ and the width is $c(S) = 1$ (Definition 7.12).

The family is connected at the strength 1, and its centre ray (Definition 38.13) has the origin 0, the direction $(\mathbf{1};1,1)$ and the walls $\kappa_{1,S}=c(S)=1$ for every nonempty proper $S$. On the chart $\alpha_3=1$ of the mark $e_0=3$, with $\alpha_1=e^{y_1}$, $\alpha_2=e^{y_2}$ and $dy=d\alpha_1\,d\alpha_2/(\alpha_1\alpha_2)$,

$$f=\frac{\alpha^{\mathbf{1}}}{K\Psi}=\frac{e_3}{e_1e_2},\qquad P(k)=\int_{\mathbb{R}^2}f^k\,dy,\qquad \ell=\int_0^\infty\!\!\int_0^\infty\frac{d\alpha_1\,d\alpha_2}{(1+\alpha_1+\alpha_2)(\alpha_1+\alpha_2+\alpha_1\alpha_2)}.$$

The measure is $dy$: with $d\alpha_1\,d\alpha_2$ in its place the integrand is of the order of $1/(\alpha_1+\alpha_2)$ at infinity and the integral diverges. The volume of the direction (Proposition 39.34) is $V_1=\sum_\pi\prod_{l=2}^3\kappa_{1,U_l}^{-1}=3!\cdot1=6$.

*Step 2: the stride.* (a) Both sides of

$$e_1e_2-9e_3=\alpha_1(\alpha_2-\alpha_3)^2+\alpha_2(\alpha_3-\alpha_1)^2+\alpha_3(\alpha_1-\alpha_2)^2$$

equal $\sum_{i\neq j}\alpha_i^2\alpha_j-6e_3$, the left one because $e_1e_2=\sum_{i\neq j}\alpha_i^2\alpha_j+3e_3$. So $0<f\le\frac19$ on the orthant, with equality exactly on the diagonal, which on the chart is the point $y=0$: the native gauge of the direction is $\alpha^\star=\mathbf{1}$ and $f(\alpha^\star)=\frac19$ (Theorem 38.14(i)). Every $P(k)$, $k>0$, converges and is positive, and since $f<\frac19$ off one point,

$$P(k+1)=\int f\cdot f^k\,dy<\tfrac19\,P(k)\qquad(k>0).$$

(b) Order the coordinates, $\alpha_{(1)}\ge\alpha_{(2)}\ge\alpha_{(3)}$. Then $\alpha_{(1)}\le e_1\le3\alpha_{(1)}$, $\alpha_{(1)}\alpha_{(2)}\le e_2\le3\alpha_{(1)}\alpha_{(2)}$ and $e_3=\alpha_{(1)}\alpha_{(2)}\alpha_{(3)}$, so $\frac19\,\alpha_{(3)}/\alpha_{(1)}\le f\le\alpha_{(3)}/\alpha_{(1)}$. On the chart $\alpha_{(3)}/\alpha_{(1)}=e^{-\varphi(y)}$ with $\varphi(y)=\max(y_1,y_2,0)-\min(y_1,y_2,0)$. The lines $y_1=y_2$, $y_1=0$, $y_2=0$ cut the plane into six cones, one for each order of the entries of $(y_1,y_2,0)$, and on each $\varphi$ is the largest entry less the smallest; for real $k>0$

$$\int_{y_1\ge y_2\ge0}e^{-ky_1}dy=\int_{y_1\ge0\ge y_2}e^{-k(y_1-y_2)}dy=\int_{0\ge y_1\ge y_2}e^{ky_2}dy=\frac{1}{k^2},$$

and the three cones with $y_1$ and $y_2$ exchanged give the same, so $\int e^{-k\varphi}dy=6/k^2$ and

$$\frac{6\cdot9^{-k}}{k^2}\le P(k)\le\frac{6}{k^2}\qquad(k>0).$$

The six cones are the six orders $\pi$ of Proposition 39.34, each contributing $\prod_l\kappa_{1,U_l}^{-1}=1$. On $[k_1,\infty)$ the integrands $f^k$ are dominated by $f^{k_1}$, as $0<f<1$, so $P$ is continuous on $(0,\infty)$.

*Step 3: the candidate recurrence.* Here $f^{-1}=e_1e_2/e_3=(\alpha_1+\alpha_2+\alpha_3)(\alpha_1^{-1}+\alpha_2^{-1}+\alpha_3^{-1})$, a Laurent polynomial, and the closings of 1 (Definition 39.30) are its constant terms,

$$x_j=\mathsf{C}_j(1)=\mathrm{CT}\left(f^{-j}\right)=\sum_{a_1+a_2+a_3=j}\Big(\frac{j!}{a_1!\,a_2!\,a_3!}\Big)^2,$$

$$(x_j)_{j\ge0}=1,\ 3,\ 15,\ 93,\ 639,\ 4653,\ 35169,\ 272835,\ 2157759,\ \ldots,$$

since the constant term pairs the monomial $\alpha^a$ of $(\sum_e\alpha_e)^j$ with the monomial $\alpha^{-a}$ of $(\sum_e\alpha_e^{-1})^j$, both with the coefficient $j!/(a_1!a_2!a_3!)$. Exact linear algebra with the ansatz $\sum_{i=0}^2p_i'(j)\,x_{j+i}=0$, $\deg p_i'\le2$, on $x_0,\ldots,x_{15}$ (14 equations in 9 unknowns) has a space of solutions of dimension one,

$$(j+2)^2x_{j+2}-(10j^2+30j+23)\,x_{j+1}+9(j+1)^2x_j=0.$$

By Corollary 39.36 a law $\sum_i p_i(k)\,\mathrm{Sh}^i[1]=0$ gives $\sum_i p_i(-j)\,x_{j-i}=0$; read back, $p_i(k)=p'_{2-i}(-k-2)$ is the candidate

$$k^2P(k)-(10k^2+10k+3)\,P(k+1)+9(k+1)^2\,P(k+2)=0. \tag{L}$$

At this point (L) is a candidate: sixteen closings do not prove their own recurrence, and Corollary 39.36 runs from a law to its closings.

*Step 4: the certificate.* The defects of the direction (Definition 38.1, Lemma 38.2(ii)) are

$$c_e=D_e\log f=1-R_e-R_e^*,\qquad R_e=\frac{\alpha_e}{e_1},\qquad R_e^*=\frac{\alpha_e(e_1-\alpha_e)}{e_2},$$

the occupations of $e$ under the basis law and under the cobasis law; $\sum_e R_e=1$ and $\sum_e R_e^*=2$, so $\sum_e c_e=0$. The origin is 0, so its defects vanish, and a certificate of (L) (Proposition 38.5(i): an explicit element of the relation space $\mathcal{L}$ of Definition 38.4 whose image is the recurrence) is a choice of insertions $g_1,g_2,g_3$ with

$$k^2-(10k^2+10k+3)\,f+9(k+1)^2f^2=\sum_{e=1}^{3}\big(k\,c_e+D_e\big)g_e. \tag{C}$$

The ansatz: $g_e=N_e/(e_1^2e_2)$, $N_e$ homogeneous of degree 4 with coefficients in $\mathbb{Q}+\mathbb{Q}k$, $N_1$ symmetric in $\alpha_2,\alpha_3$ and $g_e$ the image of $g_1$ under the exchange of the marks 1 and $e$. Multiplied by $e_1^3e_2^2$, (C) is a polynomial identity of degree 7 in $\alpha$ and 2 in $k$, linear in the 18 coefficients of $N_1$: 99 equations of rank 10. One solution is

$$g_1=-\frac{\alpha_1^2\big(k\,\alpha_1(\alpha_2+\alpha_3)+(2+3k)(\alpha_2^2+\alpha_3^2)-(1-k)\,\alpha_2\alpha_3\big)}{e_1^2e_2}, \tag{G}$$

and within the ansatz the others differ from it by one insertion $h$ added at every mark, $h$ with a symmetric numerator, an 8-dimensional space; such an $h$ contributes $k\,h\sum_e c_e+\sum_e D_eh=0$, by Euler's identity for $h$ of degree 0. Each $g_e$ is a polynomial over $K^2\Psi$, homogeneous of degree 0, so a regular function on $X_F$: an insertion (Definition 38.1).

*Verification.* Write $g_e=g_e^{(0)}+k\,g_e^{(1)}$, with $e_1^2e_2\,g_1^{(0)}=-\alpha_1^2(2\alpha_2^2-\alpha_2\alpha_3+2\alpha_3^2)$ and $e_1^2e_2\,g_1^{(1)}=-\alpha_1^2(\alpha_1\alpha_2+\alpha_1\alpha_3+3\alpha_2^2+\alpha_2\alpha_3+3\alpha_3^2)$. The powers of $k$ split (C) into

$$\sum_e c_eg_e^{(1)}=1-10f+9f^2,\qquad \sum_e\big(D_eg_e^{(1)}+c_eg_e^{(0)}\big)=-10f+18f^2,\qquad \sum_e D_eg_e^{(0)}=-3f+9f^2.$$

Times $e_1^3e_2^2$ every sum over the marks is a symmetric polynomial of degree 7, and in the basis of the products of $e_1,e_2,e_3$ exact expansion gives

$$\sum_e e_1^3e_2^2\,c_eg_e^{(1)}=e_1^3e_2^2-10\,e_1^2e_2e_3+9\,e_1e_3^2,\qquad \sum_e e_1^3e_2^2\,D_eg_e^{(0)}=-3\,e_1^2e_2e_3+9\,e_1e_3^2,$$
$$\sum_e e_1^3e_2^2\,D_eg_e^{(1)}=-2\,e_1e_2^3-3\,e_1^2e_2e_3+9\,e_1e_3^2,\qquad \sum_e e_1^3e_2^2\,c_eg_e^{(0)}=2\,e_1e_2^3-7\,e_1^2e_2e_3+9\,e_1e_3^2,$$

which, with $e_1^3e_2^2f=e_1^2e_2e_3$ and $e_1^3e_2^2f^2=e_1e_3^2$, are the three right sides times $e_1^3e_2^2$. So (C) holds identically in $\alpha$ and $k$: $\sum_i p_i(k)f^i\in\mathcal{L}$, with $(p_0,p_1,p_2)=\big(k^2,\,-(10k^2+10k+3),\,9(k+1)^2\big)$. To test another certificate on one point: at $\alpha=(1,2,4)$ and $k=\frac{1}{3}$, $f=\frac{4}{49}$, the left side of (C) is $-\frac{2809}{7203}$, and (G) gives $(g_1,g_2,g_3)=\big(-\frac{85}{1029},-\frac{310}{1029},-\frac{424}{1029}\big)$; agreement at a point is a test, the identity above is the proof.

*Step 5: the recurrence at every strength.* The terms of $g_e$ are the insertions $\alpha^v K^{-2}\Psi^{-1}$ for the monomials $\alpha^v$ of $N_e$, with the data $(v;2,1)$ and the walls $\kappa_U = v_U - 2d(U) - d^*(U) = v_U - |U| + 1$. The monomials of $N_1$ are $\alpha_1^3\alpha_2$, $\alpha_1^3\alpha_3$, $\alpha_1^2\alpha_2^2$, $\alpha_1^2\alpha_3^2$ and $\alpha_1^2\alpha_2\alpha_3$; each has two marks in its support, so $v_U \geq 1$ for every pair $U$, and every wall is at least 0 (the least is 0). With the walls $k\kappa_{1,U} = k$ of the ray, the condition of Proposition 38.5(i) holds at every $k$ with $\operatorname{Re} k > 0$, where the values converge, and the certificate gives (L) there. The integration by parts behind it handles the mark $e_0 = 3$ on its own chart: by Euler's identity $D_3 = -D_1 - D_2$ on functions of degree 0, so $\sum_e D_e(g_e f^k) = \partial_{y_1}\big((g_1 - g_3)f^k\big) + \partial_{y_2}\big((g_2 - g_3)f^k\big)$; $y_3$ is not a coordinate of the chart. Corollary 39.38(i) gives the same as an identity of meromorphic functions of $k$, for the element (C) of $\mathcal{L}$.

*Step 6: the native values; what positivity selects.* The symbol of (L) (Corollary 38.7) is $\chi(\lambda) = 1 - 10\lambda + 9\lambda^2 = (1-\lambda)(1-9\lambda)$. By Theorem 38.17 every sign $\eta$, $\eta_3 = 1$, with $K(\eta) \neq 0$ is a native gauge of native value $K(\eta)^{-2}$, with the Hessian $2\big(\kappa_\eta(X_e, X_{e'})\big)_{e,e'\neq 3}$ of the indicators under $\mathbb{P}_\eta(\{e\}) = \eta_e/K(\eta)$: at $\eta = (1,1,1)$, $K(\eta) = 3$, the native value is $\frac{1}{9}$ and the Hessian is $2\left(\begin{smallmatrix} 2/9 & -1/9 \\ -1/9 & 2/9 \end{smallmatrix}\right)$, of determinant $\frac{4}{27}$; at $(-1,1,1)$, $(1,-1,1)$, $(-1,-1,1)$, $K(\eta) = \pm 1$, the native value is 1 and the determinant is $-4$ (at $(-1,1,1)$ the matrix is $2\left(\begin{smallmatrix} -2 & 1 \\ 1 & 0 \end{smallmatrix}\right)$). All four are nondegenerate, so Corollary 38.7 at $g = 1$ makes $\frac{1}{9}$ and 1 roots of $\chi$; $\chi$ has degree 2, so they are all its roots. (The corollary runs from native values to roots; a root of a symbol need not be a native value.) The root other than $f(\alpha^\star) = \frac{1}{9}$ has the modulus $1 > \frac{1}{9}$, so Theorem 41.3(iii) applies: up to a constant factor the values are the only solution of (L) whose generating function has a radius of convergence of at least 9, and the low-end relation fixes the factor. The rational solutions of (L) grow like the native value 1 of the signs and the values decay like $9^{-k}$, so a bracket from positivity shrinks by about $\frac{1}{9}$ per step of the strength.

*Step 7: the low-end relation.* By Step 2(b), $6 \cdot 9^{-k} \leq k^2 P(k) \leq 6$ for real $k > 0$, so $k^2 P(k) \to 6$ as $k \to 0^+$: the low-end coefficient $\mathsf{L}_0$ is $V_1 = 6$ (Definition 39.37). Corollary 39.38(ii) applies to the element (C): $p_0 = k^2$ vanishes to the order $n - 1 = 2$ at 0, $[k^2]p_0 = 1$, $p_1(0) = -3$, $p_2(0) = 9$, and

$$6 - 3\ell + 9P(2) = 0, \qquad P(2) = \frac{\ell - 2}{3}. \tag{S}$$

Directly: take (L) at real $k \in (0,1)$ and let $k \to 0^+$; $k^2P(k) \to 6$, $P(k+1) \to \ell$ and $P(k+2) \to P(2)$ by the continuity of Step 2, while $10k^2 + 10k + 3 \to 3$ and $9(k+1)^2 \to 9$.

A solution that looks right goes wrong here. Reading (L) at $k = 0$ as $0 \cdot P(0) - 3P(1) + 9P(2) = 0$ treats $0 \cdot P(0)$ as 0; but $P(0) = \int dy = \infty$, and the product $k^2P(k)$ tends to $V_1 = 6$. The misreading gives $P(2) = \ell/3$, which Step 2(a) refutes at once: $P(2) < \frac{1}{9}P(1)$ would say $\frac{\ell}{3} < \frac{\ell}{9}$. Every $P(K)$ would then be $\ell$ times a positive rational, positivity would fix no digit of $\ell$, and the bracket of Step 9 would be empty. Passing the limit under the integral, $k^2 f^k \to 0$ pointwise, fails for the same reason: there is no integrable majorant, $\int dy$ being infinite.

*Step 8: the rational solutions and their signs.* For integers $k \geq 1$, $9(k+1)^2 \neq 0$, so (L) gives $P(k+2)$ from $P(k)$ and $P(k+1)$. Let $c$ and $r$ be the solutions with $c(1) = 1$, $c(2) = \frac{1}{3}$ and $r(1) = 0$, $r(2) = -\frac{2}{3}$. From $P(1) = \ell$, (S) and induction,

$$P(K) = c(K)\,\ell + r(K) \qquad (K \geq 1).$$

*Signs at every $K$.* Put $x'_j = 9^j c(j+1)$. Then (L) at $k = j+1$, times $9^{j+1}$, reads

$$(j+2)^2 x'_{j+2} = (10j^2 + 30j + 23)\,x'_{j+1} - 9(j+1)^2 x'_j, \qquad x'_0 = 1,\ x'_1 = 3.$$

If $x'_{j+1} \geq 3x'_j > 0$, then $(j+2)^2 x'_{j+2} \geq \big(10j^2 + 30j + 23 - 3(j+1)^2\big)x'_{j+1} = (7j^2 + 24j + 20)\,x'_{j+1} \geq 3(j+2)^2 x'_{j+1}$, because $7j^2 + 24j + 20 - 3(j+2)^2 = 4(j+1)(j+2) \geq 0$. So $x'_{j+1} \geq 3x'_j > 0$ for

every $j$, $x'_j \geq 3^j$, and

$$c(K) \geq 3^{1-K} > 0, \qquad 9c(K) \geq 3c(K-1) > c(K-1) \qquad (K \geq 2).$$

The closings satisfy the same recurrence with the same first two terms (Corollary 39.38(i) for the element (C)), so $x'_j = x_j$ and $c(K) = 9^{1-K} x_{K-1}$ at every $K$; the signs do not use this. In exact arithmetic:

| $K$ | $c(K)$ | $r(K)$ | $a_K = -r(K)/c(K)$ |
|---|---|---|---|
| 1 | 1 | 0 | 0 |
| 2 | 1/3 | −2/3 | 2 |
| 3 | 5/27 | −23/54 | 23/10 |
| 4 | 31/243 | −145/486 | 145/62 |
| 5 | 71/729 | −1331/5832 | 1331/568 |
| 6 | 517/6561 | −242353/1312200 | 242353/103400 |
| 7 | 11723/177147 | −5495507/35429400 | 5495507/2344600 |
| 8 | 10105/177147 | −46422941/347208120 | 46422941/19805800 |
| 9 | 26639/531441 | −2447622221/20832487200 | 2447622221/1044248800 |
| 10 | 5773279/129140163 | −530455590781/5062294389600 | 530455590781/226312536800 |

The frame of $c$ and $r$ (Proposition 38.11, $h = 2$) satisfies $C_{k+1} = \frac{p_0(k)}{p_2(k)} C_k = \frac{k^2}{9(k+1)^2} C_k$ with $C_1 = c(1)r(2) - c(2)r(1) = -\frac{2}{3}$, so

$$c(k)\, r(k+1) - c(k+1)\, r(k) = -\frac{2}{3} \cdot \frac{9^{1-k}}{k^2} \qquad (k \geq 1).$$

*Step 9: the bracket and the answer.* (a) *Below, by positivity.* $P(K) = c(K)\ell + r(K) > 0$ and $c(K) > 0$ give $\ell > a_K = -r(K)/c(K)$. (b) *Above, by the native value.* $P(K) < \frac{1}{9} P(K-1)$ (Step 2(a)) is $\big(9c(K) - c(K-1)\big)\ell < r(K-1) - 9r(K)$, and $9c(K) - c(K-1) > 0$ (Step 8), so

$$\ell < b_K = \frac{r(K-1) - 9\, r(K)}{9\, c(K) - c(K-1)}.$$

At $K = 2$ this is the first bracket: $P(2) = \frac{\ell - 2}{3} > 0$ gives $\ell > 2$, and $P(2) < \frac{1}{9} P(1)$ gives $\ell < 3$. (c) *The width.* With the frame at $k = K-1$ and $c(K) = 9^{1-K} x_{K-1}$,

$$\begin{aligned} b_K - a_K &= \frac{r(K-1)\, c(K) - r(K)\, c(K-1)}{c(K)\big(9c(K) - c(K-1)\big)} = \frac{2 \cdot 9^{2-K}}{3(K-1)^2\, c(K)\big(9c(K) - c(K-1)\big)} \\ &= \frac{2 \cdot 9^{K-1}}{3(K-1)^2\, x_{K-1}(x_{K-1} - x_{K-2})}, \end{aligned}$$

which the bounds of Step 8 make at most $1/(K-1)^2$: the brackets close on one number. In exact arithmetic, with the width also as a rendering:

| $K$ | $b_K$ | $b_K - a_K$ | rendering |
|---|---|---|---|
| 2 | 3 | 1 | |
| 3 | 19/8 | 3/40 | $7.5 \cdot 10^{-2}$ |
| 4 | 61/26 | 3/403 | $7.44 \cdot 10^{-3}$ |
| 5 | 3413/1456 | 81/103376 | $7.84 \cdot 10^{-4}$ |
| 6 | 104539/44600 | 243/2882275 | $8.43 \cdot 10^{-5}$ |
| 7 | 149014/63575 | 2187/238492712 | $9.17 \cdot 10^{-6}$ |
| 8 | 909876251/388187800 | 19683/19613188595 | $1.0036 \cdot 10^{-6}$ |
| 9 | 6414407843/2736630400 | 6561/59511099776 | $1.1025 \cdot 10^{-7}$ |
| 10 | 232184895407/99058909600 | 177147/14589151085627 | $1.214 \cdot 10^{-8}$ |

At $K = 8$ the certified width exceeds $10^{-6}$; at $K = 9$ it is $\frac{6561}{59511099776} < \frac{111}{10^9}$.

*Answer.* $q = a_9 = \dfrac{2447622221}{1044248800}$, with

$$\frac{2447622221}{1044248800} < \ell < \frac{6414407843}{2736630400}, \qquad b_9 - a_9 = \frac{6561}{59511099776} < 1.11 \cdot 10^{-7},$$

so $0 < \ell - q < 10^{-6}$.

*What does not certify.* The value $a_8$ is also a correct answer, but the bracket at $K = 8$ is too wide to show it; the upper bound at $K = 9$ does, $\ell - a_8 < b_9 - a_8 = \frac{19683}{22574408320} < 10^{-6}$. A rate of convergence is no bound, and neither is the last step: $a_9 - a_8$ is smaller than $\ell - a_8$, since $a_9 < \ell$. The bound $P(K) \le 9^{-K}\ell$ does not follow from $f \le \frac{1}{9}$ (it would need $\int dy < \infty$); iterating Step 2(a) gives $P(K) < 9^{1-K}P(1) = 9^{1-K}\ell$, that is $\big(c(K) - 9^{1-K}\big)\ell < -r(K)$ with $c(K) - 9^{1-K} = 9^{1-K}(x_{K-1} - 1) > 0$ for $K \ge 2$: a valid and cruder upper bound, which needs $K = 10$ to certify six digits. A sign of $c(K)$ or of $9c(K) - c(K-1)$ that is not proved can reverse an inequality: here Step 8 proves both at every $K$.

*Check* (floating point, not part of the solution). On the chart the integral over $\alpha_2$ is elementary, $\int_0^\infty \frac{d\alpha_2}{(\alpha_2+A)(A\alpha_2+B)} = \frac{\log(A^2/B)}{A^2-B}$ with $A = 1 + \alpha_1$, $B = \alpha_1$, so

$$\ell = \int_0^\infty \frac{\log\big((1+x)^2/x\big)}{1+x+x^2}\,dx,$$

which renders as $2.34390723868945889060156228887227706909 5\ldots$; then $\ell - a_9$ and $b_9 - \ell$ render as $9.6062 \cdot 10^{-8}$ and $1.4186 \cdot 10^{-8}$. Exercise 54.3 computes $\ell$ to a hundred digits without quadrature. □

## 54.2 The centre ray by its box

**Exercise 54.2** (The centre of $\binom{[3]}{1}$ by its box)**.** Let $F$, $P$ and $\ell$ be as in Exercise 54.1. Read the centre ray of $F$ by steps (1) to (4) and (6) of the procedure of Definition 41.8, with the box $\mathfrak{B}_2(1,1)$ (Definition 39.15) modulo the group $G$ generated by the permutations of the marks and the Cremona: give its symbols, orbits and relations, the pivots and the free orbits, the spanning set at $\mathfrak{s} = 1$, the matrix $\mathbf{B}(k)$, a certificate supported on the stems, the law of [1], $\mathbf{B}(\infty)$ and the closings of the spanning set. Show that the closing relations alone do not give the law.

*Solution.* Receipt: `ex19_centre_box.py`. The elimination is small enough to run once over $\mathbb{Q}(k)$, so the evaluation at random strengths of step (2) of the procedure is not needed.

*Step 1: walls and symmetries.* The walls of the direction are $\kappa_{1,S} = 1$ and $V_1 = 6$ (Exercise 54.1, Step 1). Every permutation $\beta$ of the marks carries the bases and keeps the masses $\mathbf{1}$ and

the exponents, and the Cremona is a symmetry because the masses are $\sigma\mathbf{1}$ and $s = t$ (Proposition 39.11(ii)). So $G = S_3 \times \{1, \mathrm{cr}\}$, of order 12, acting on the symbols by $(u; a, b) \mapsto (u \circ \beta^{-1}; a, b)$ and $(u; a, b) \mapsto (-u + (a+b)\mathbf{1}; b, a)$ (Proposition 39.11(iii)).

*Step 2: the box.* Both letters are active, so the box with one unit of each active letter at $\mathfrak{t} = 2$ is $\mathfrak{B} = \mathfrak{B}_2(1,1)$: the symbols $(u; a, b)$ with $a, b \in \{0, 1\}$, $u_{\mathcal{E}} = a + 2b$, and wall degree $\deg_\kappa = \max_U(-\kappa_U) \le 2$, where $\kappa_U = u_U - a\,d(U) - b\,d^*(U) = u_U - b(|U| - 1)$. It has 110 symbols, 19 at the level $(0,0)$, 27 at each of $(1,0)$ and $(0,1)$ and 37 at $(1,1)$. $G$ keeps it, the Cremona exchanging the levels $(1,0)$ and $(0,1)$, and the symbols fall into 16 orbits:

| orbit of | level | $\deg_\kappa$ | size | | orbit of | level | $\deg_\kappa$ | size | |
|---|---|---|---|---|---|---|---|---|---|
| $\alpha_1^4/(\alpha_3 K\Psi)$ | $(1,1)$ | 2 | 12 | pivot | $\alpha_1^3/(\alpha_2\alpha_3 K)$ | $(1,0),(0,1)$ | 2 | 6 | pivot |
| $\alpha_1^4\alpha_2/(\alpha_3^2 K\Psi)$ | $(1,1)$ | 2 | 6 | pivot | $\alpha_1^2\alpha_2/(\alpha_3^2 K)$ | $(1,0),(0,1)$ | 2 | 12 | pivot |
| $\alpha_1^3/(K\Psi)$ | $(1,1)$ | 1 | 6 | pivot | $\alpha_1^3/(\alpha_3^2 K)$ | $(1,0),(0,1)$ | 2 | 12 | pivot |
| $\alpha_1^3\alpha_2/(\alpha_3 K\Psi)$ | $(1,1)$ | 1 | 6 | pivot | $\alpha_1\alpha_2/(\alpha_3 K)$ | $(1,0),(0,1)$ | 1 | 6 | pivot |
| $\alpha_1^2\alpha_2/(K\Psi)$ | $(1,1)$ | 0 | 6 | pivot | $\alpha_1^2/(\alpha_3 K)$ | $(1,0),(0,1)$ | 1 | 12 | pivot |
| $\alpha_1\alpha_2\alpha_3/(K\Psi)$ | $(1,1)$ | $-1$ | 1 | free | $\alpha_1/K$ | $(1,0),(0,1)$ | 0 | 6 | pivot |
| $\alpha_1^2/(\alpha_2\alpha_3)$ | $(0,0)$ | 2 | 6 | pivot | $\alpha_1/\alpha_3$ | $(0,0)$ | 1 | 6 | free |
| $\alpha_1^2/\alpha_3^2$ | $(0,0)$ | 2 | 6 | pivot | 1 | $(0,0)$ | 0 | 1 | free |

The stems are the 19 Laurent symbols. The relations (Definition 39.15(iv)) are the letter identities for $K$ of the 46 symbols with $a = 0$, those for $\Psi$ of the 46 symbols with $b = 0$, and the mark relations of the 19 stems at the 3 marks: 149 relations, which give 22 distinct combinations of orbit averages (Definition 39.20(i)), of rank 13 over $\mathbb{Q}(k)$. With the ranking of Definition 39.20(ii) (positive levels above the Laurent orbits, then the total level, then the wall degree), the elimination makes the 13 orbits marked in the table pivots; the free orbits are those of $f = \alpha_1\alpha_2\alpha_3/(K\Psi)$, of $\alpha_1/\alpha_3$ and of 1.

*Step 3: the spanning set and the matrix.* The free Laurent orbits of wall degree at most $\mathfrak{s} = 1$ give the spanning set $\mathbf{b} = (1, b_1)$, with $b_1 = \frac{1}{6}\sum_{i\ne j}\alpha_i/\alpha_j$, the average of the orbit of $\alpha_1/\alpha_3$ (Definition 39.20(iii)). The first target is reduced already: $f^{-1} = (\sum_e \alpha_e)(\sum_e \alpha_e^{-1}) = 3 + 6b_1$ identically. The second, $f^{-1}b_1$, has 19 monomials, and its reduction is a combination of 1 and $b_1$. So the box is accepted at $\mathfrak{s} = 1$, with

$$\mathbf{B}(k) = \begin{pmatrix} 3 & 6 \\ \dfrac{2(k-1)}{k-2} & \dfrac{(k-1)(7k-11)}{(k-2)^2} \end{pmatrix},$$

and by Theorem 39.21(i) and (iii) $[f^{-1}b_i] = \sum_l \mathbf{B}_{il}(k)[b_l]$ in the module, and $\langle b_i\rangle_{k-1} = \sum_l \mathbf{B}_{il}(k)\langle b_l\rangle_k$ as meromorphic functions of $k$.

*Step 4: the certificate.* Theorem 39.21(ii) provides combinations $G_e$ of stems with $f^{-1}b_1 - \mathbf{B}_{21} - \mathbf{B}_{22}\,b_1 = \sum_e \big(k\,c_e + D_e\big)G_e$, $c_e$ and $D_e$ as in Exercise 54.1. With $G_1$ a combination of stems symmetric under the exchange of the marks 2 and 3 and $G_e$ its image under the exchange of 1 and $e$, the identity is linear in 11 unknowns, 37 equations, with a family of solutions of dimension 6; one of them is

$$G_1 = \frac{\alpha_2^2+\alpha_3^2}{6(k-2)\,\alpha_1^2} + \frac{(5k^2-6k-4)\,\alpha_2\alpha_3}{6(k+1)(k-2)^2\,\alpha_1^2} + \frac{3k-4}{6(k+1)(k-2)^2}\Big(\frac{\alpha_3^2}{\alpha_1\alpha_2}+\frac{\alpha_2^2}{\alpha_1\alpha_3}\Big) + \frac{5k-9}{6(k-2)^2}\,\frac{\alpha_2+\alpha_3}{\alpha_1} + \frac{3k-4}{6(k+1)(k-2)}\Big(\frac{\alpha_3^2}{\alpha_2^2}+\frac{\alpha_2^2}{\alpha_3^2}\Big),$$

whose terms are stems, of wall degree 2, 2, 2, 1 and 2. The identity holds as one of rational functions in $\alpha$ and $k$ (exact expansion); at $\alpha = (1, 2, 4)$, $k = \frac{1}{3}$, both sides are $\frac{11903}{800}$. Its coefficients have poles at $k = 2$ and $k = -1$, and by Theorem 39.21(iii) it integrates term by term at every real $k > 2$.

*Step 5: the law.* By Proposition 39.6 the rows are $w_0 = (1, 0)$, $w_1(k) = w_0\mathbf{B}(k) = (3, 6)$ and

$$w_2(k) = w_1(k-1)\,\mathbf{B}(k) = \Big(\frac{3(7k-10)}{k-2},\ \frac{6(10k^2-30k+23)}{(k-2)^2}\Big),$$

and their first dependence is $q_0w_0 + q_1w_1 + q_2w_2 = 0$ with $(q_0, q_1, q_2) = \big(9(k-1)^2,\ -(10k^2-30k+23),\ (k-2)^2\big)$. Then $p_j(k) = q_{2-j}(k+2)$ gives $\big(k^2,\ -(10k^2+10k+3),\ 9(k+1)^2\big)$: the recurrence (L) of Exercise 54.1, now with no closing computed, and with a certificate assembled from those of the rows (Theorem 39.21(iv)). As a check of Proposition 39.6(iv), $\det\mathbf{B}(k) = 9(k-1)^2/(k-2)^2$ and $\det[w_0; w_1] = 6$, so $q_0 \cdot 6 = q_2 \cdot 6 \cdot \det\mathbf{B}(k)$.

The spanning set is independent: at the native gauges $(1, 1, 1)$ and $(-1, 1, 1)$ (Exercise 54.1, Step 6) the vectors of values $\beta_c = (1, b_1(\alpha_c))$ are $(1, 1)$ and $(1, -\frac{1}{3})$, which span $\mathbb{C}^2$, so by Proposition 39.42(i) $1, b_1$ are masters of the submodule $\mathfrak{N}$ they span and $\mathbf{B} = \mathbf{B_b}$. By Proposition 39.6(ii) the order 2 is then the dimension of the cyclic submodule of [1], which is $\mathfrak{N}$, and (L) is the law of [1]. Finally

$$\mathbf{B}(\infty) = \begin{pmatrix} 3 & 6 \\ 2 & 7 \end{pmatrix}, \qquad \mathbf{B}(\infty)\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 9\begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad \mathbf{B}(\infty)\begin{pmatrix} 1 \\ -\frac{1}{3} \end{pmatrix} = \begin{pmatrix} 1 \\ -\frac{1}{3} \end{pmatrix},$$

the reciprocals of the native values $\frac{1}{9}$ and 1, with the vectors $\beta_c$ as eigenvectors (Proposition 39.42(ii)).

*Step 6: checks.* The closings of the spanning set are $\mathsf{C}_j(1) = x_j$ and, by the first row, $\mathsf{C}_j(b_1) = \frac{1}{6}(x_{j+1} - 3x_j) = 0, 1, 8, 60, 456, 3535, 27888, \ldots$; they satisfy $\mathsf{C}_{j+1}(\mathbf{b}) = \mathbf{B}(-j)\,\mathsf{C}_j(\mathbf{b})$ (Theorem 39.33) for $0 \le j < 12$. The low-end relation of (L) returns $V_1 = 6$ (Exercise 54.1, Step 7).

*Step 7: the closing relations alone.* The closing relations $(k\,c_e + D_e)(\alpha^w f^{-1})$ involve Laurent insertions only (Proposition 39.19). Let $\omega = e^{2\pi i/3}$ and $z = (1, \omega, \omega^2)$. Then $K(z) = 1+\omega+\omega^2 = 0$ and $\Psi(z) = \omega+\omega^2+\omega^3 = 0$, so $f^{-1} = K\Psi/e_3$ and every $D_ef^{-1} = (\Psi D_eK + KD_e\Psi)/e_3 - f^{-1}$ vanish at $z$: $z$ is a singular point of the zero set of $f^{-1}$ on the torus. By Proposition 39.19(i) the evaluation at $z$ vanishes on every closing relation, and by (ii) it takes the value $p_2(k-2) = 9(k-1)^2 \ne 0$ on the element $\sum_{i=0}^{2} p_{2-i}(k-2)f^{-i}$ of $\mathcal{L}$ that the law gives: no combination of closing relations gives the law. Concretely, at the strength 9 the elimination of the closing relations of the Laurent monomials of wall degree at most $R-1$, on the invariant Laurent combinations of wall degree at most $R$, leaves three free orbits, 1, $b_1$ and $b_2 = \frac{1}{6}\sum_{i\ne j}\alpha_i^2/\alpha_j^2$, for every $R$ from 3 to 7; the law at $k = 7$, read at the strength 9 as $49f^{-2} - 563f^{-1} + 576$, reduces for each such $R$ to

$$504 + 248\,b_1 - 392\,b_2 \ne 0,$$

and at $z$, where $b_1(z) = b_2(z) = -\frac{1}{2}$, this is $576 = 9(k+1)^2$. At $k = 10$ and $k = \frac{23}{3}$ the reductions are $880 + \frac{946}{3}b_1 - \frac{2200}{3}b_2$ and $\frac{8671}{15} + \frac{1313}{5}b_1 - \frac{6877}{15}b_2$, with the values 1089 and 676 at $z$, again $9(k+1)^2$. The point $z$ lies off $X_F$; the box supplies the relations that the closing relations lack through its units of the letters, by the letter identities and the mark relations of the occupations.

*Answer.* The box $\mathfrak{B}_2(1, 1)$ modulo $G$ has 110 symbols in 16 orbits and 149 relations of rank 13 modulo $G$; it is accepted at $\mathfrak{s} = 1$ with the masters 1 and $b_1 = \frac{1}{6}\sum_{i\ne j}\alpha_i/\alpha_j$ and the matrix $\mathbf{B_b}(k)$ above, with the certificate $G_1$ on the stems; the law of [1] is (L), and $\mathbf{B_b}(\infty)$ has the eigenvalues 9 and 1. The closing relations alone leave the evaluation at $(1, \omega, \omega^2)$, on which the law takes the value $9(k+1)^2$. □

## 54.3 The centre ray from its gauge end

**Exercise 54.3** (The centre of $\binom{[3]}{1}$ from its gauge end)**.** Let $F$, $P$, $\ell$ and the law (L) be as in Exercise 54.1, and let $\mu_f$ be the stride's law of the centre ray (Definition 40.1), with the density $\varrho_f$.

(i) Write the density operator of (L), its exponents at $f(\alpha^\star) = \frac{1}{9}$, and the germs there that extend by zero.

(ii) Find $\varrho_f(\frac{1}{9})$ and the Taylor coefficients of the germ of $\varrho_f$ at $\frac{1}{9}$.

(iii) Read $P(1)$, $P'(1)$ and $P''(1)$ from the gauge end (Theorem 40.6) to sixty digits, find the low-end coefficients $\mathsf{L}_0, \mathsf{L}_1, \mathsf{L}_2$ from the faces of the sectors, and check the output against the low-end relation and its first two derivatives.

(iv) Find the arc integrals $S_j = -\int_0^{2\pi/3} \log^{j-1}\left(2\sin\frac{\vartheta}{2}\right) d\vartheta$ for $j = 1, 2$.

*Solution.* Receipt: `ex20_gauge_end.py`. Theorem 40.6 applies: the ray has the origin 0 and positive walls, and (L) holds at every $k$ with $\operatorname{Re} k > 0$ (Exercise 54.1, Step 5), in particular at all large integers; $f(\alpha^\star) = \frac{1}{9}$ is a simple root of the symbol $\chi(\lambda) = (1-\lambda)(1-9\lambda)$. Write $f^\star = \frac{1}{9}$.

*Step 1: the density operator.* With $\theta = t\,d/dt$ and $(p_0, p_1, p_2) = \left(k^2, -(10k^2{+}10k{+}3), 9(k{+}1)^2\right)$,

$$\begin{aligned}\sum_i t^i p_i(-1-i-\theta) &= (\theta+1)^2 - t\,(10\theta^2+30\theta+23) + 9t^2(\theta+2)^2 \\ &= t^2(t-1)(9t-1)\,\partial_t^2 + t\,(45t^2-40t+3)\,\partial_t + (36t^2-23t+1),\end{aligned}$$

of order $D = 2$, with the leading coefficient $(-1)^D t^D \chi(t)$; its singular points are 0, $\frac{1}{9}$, 1 and $\infty$. The second symbol is $\chi_1(\lambda) = -10\lambda + 18\lambda^2$.

*Step 2: the exponents at $f^\star$.* In $s = f^\star - t$ the operator is $b_2\partial_s^2 + b_1\partial_s + b_0$ with

$$b_2 = \tfrac{1}{81}s(9s-1)^2(9s+8), \qquad b_1 = \tfrac{1}{81}(9s-1)(405s^2+270s-8), \qquad b_0 = \tfrac{1}{9}(324s^2+135s-10),$$

so $b_{2,0} = 0$, $b_{2,1} = \frac{8}{81} = -(f^\star)^2\chi'(f^\star)$ and $b_{1,0} = \frac{8}{81}$. Theorem 40.6(i) gives $\gamma = D - 1 - b_{1,0}/b_{2,1} = 0$, which is $-1 + \chi_1(f^\star)/(f^\star\chi'(f^\star)) = -1 + \frac{-8/9}{-8/9}$, and $\frac{n-3}{2} = 0$ as Theorem 40.6(iii) requires. The polynomials $R_m(x) = \sum_j b_{j,m-1+j}[x]_j$ are

$$R_0 = \tfrac{8}{81}x^2, \qquad R_1 = -\tfrac{1}{9}(15x^2+23x+10), \qquad R_2 = (2x+3)(3x+5), \qquad R_3 = 9(x+2)^2,$$

and $R_m = 0$ for $m \geq 4$. The exponents at $s = 0$ are 0 and 0: one solution $\phi = \sum_m e_m s^m$, $e_0 = 1$, holomorphic at $s = 0$, with $R_0(N)e_N = -\sum_{m=1}^3 R_m(N-m)e_{N-m}$, and one of the form $\phi\log s + \psi$, $\psi$ holomorphic with $\psi(0) = 0$.

*Step 3: the line.* For a germ $v$ on $s > 0$ with $(b_2\partial_s^2 + b_1\partial_s + b_0)v = 0$, integration by parts on $(\epsilon, \infty)$ shows that the distribution $\mathfrak{D}(v\mathbf{1}_{s>0})$ pairs with a test function $w$ as

$$\lim_{\epsilon\to 0^+}\Big[b_2\,(v'w - vw') + (b_1 - b_2')\,vw\Big]_{s=\epsilon}.$$

For $v = \phi$ this is $(b_{1,0} - b_{2,1})\,\phi(0)w(0) = 0$: $\phi$ extends by zero (Definition 40.4). For $v = \phi\log s + \psi$ the term $b_2 v'w$ tends to $b_{2,1}w(0)$, while the terms in $\log\epsilon$ have the coefficient $(b_{1,0} - b_{2,1})w(0) = 0$, so the pairing is $\frac{8}{81}w(0)$: the germ leaves a point mass $\frac{8}{81}\delta_{f^\star}$ and does not extend by zero, and neither does any germ with a logarithmic term. So the germs that extend by zero form the line of

$\phi$ (Theorem 40.6(ii)), the germ of $\varrho_f$ spans it (Theorem 40.6(iii)), and $\varrho_f = \varrho_f(f^\star)\,\phi(f^\star - t)$ near $f^\star$, the case $\gamma \in \{0, \dots, D-2\}$ of Theorem 40.6(iv). In exact arithmetic

$$(e_m)_{m\ge 0} = 1,\ \tfrac{45}{4},\ \tfrac{3645}{32},\ \tfrac{70713}{64},\ \tfrac{21500397}{2048},\ \tfrac{807258879}{8192},\ \tfrac{15034997331}{16384},\ \tfrac{139237010559}{16384},\ \dots,$$

all positive.

*Step 4: the value at the native value.* By Theorem 38.15, on the centre ray the Hessian of $-\log f$ at $y = 0$ is twice the covariance matrix of the indicators of the marks $1, 2$ under the uniform law on the three marks,

$$H = 2\begin{pmatrix} \frac{2}{9} & -\frac{1}{9} \\ -\frac{1}{9} & \frac{2}{9} \end{pmatrix}, \qquad \det H = \tfrac{4}{27}.$$

By Definition 38.4 there are coordinates $v$ near $y = 0$ with $\log f = \log f^\star - \frac{1}{2}|v|^2$, here real, with $dy = J(v)\,dv$ and $J(0) = (\det H)^{-1/2}$. So $\mu_f([t, f^\star]) = \int_{|v|\le\rho} J\,dv$ with $\rho^2 = 2\log(f^\star/t)$, whose derivative in $\rho$ is $2\pi\rho J(0) + O(\rho^2)$, and $\varrho_f(t) = -\frac{d}{dt}\mu_f([t, f^\star]) = \frac{1}{\rho t}\big(2\pi\rho J(0) + O(\rho^2)\big)$, which tends to

$$\varrho_f(f^\star) = \frac{2\pi}{f^\star\sqrt{\det H}} = 2\pi \cdot 9 \cdot \frac{3\sqrt{3}}{2} = 27\sqrt{3}\,\Gamma(\tfrac{1}{2})^2.$$

*Step 5: the values and the slopes.* The other singular points of the operator lie at $s = \frac{1}{9}$ ($t = 0$) and $s = -\frac{8}{9}$, so the series $\phi$ converges on $|s| < \frac{1}{9}$, which contains $t \in (0, f^\star]$, and $\varrho_f(t) = 27\sqrt{3}\,\Gamma(\frac{1}{2})^2 \sum_m e_m(\frac{1}{9} - t)^m$ there: this series is the continuation of the germ along $(0, f^\star)$ in Theorem 40.6(v). Its terms are positive, so by monotone convergence Theorem 40.6(v) gives, for every real $K > 0$ and $j \in \mathbb{N}$,

$$P^{(j)}(K) = 27\sqrt{3}\,\Gamma(\tfrac{1}{2})^2 \sum_{m\ge 0} e_m\,\frac{\partial^j}{\partial a^j}\Big[9^{-a-m-1}\,\Gamma\,{}_{1}^{2}\begin{bmatrix} a+1,\ m+1 \\ a+m+2 \end{bmatrix}\Big]_{a=K},$$

the Beta corner $\int_0^{1/9} t^a(\frac{1}{9} - t)^m\,dt = 9^{-a-m-1}\Gamma\,{}_{1}^{2}\begin{bmatrix} a+1,\ m+1 \\ a+m+2 \end{bmatrix}$ and its derivatives in $a$ through the digamma function and its derivatives. At $K = 150$ the terms fall below $10^{-150}$ times the first after 520 terms. Differentiated $j$ times, (L) is $\sum_{l\le j}\binom{j}{l}\sum_i p_i^{(j-l)}(k)\,P^{(l)}(k+i) = 0$, an identity of meromorphic functions, solved for $P^{(j)}(k)$ because $p_0(k) = k^2 \ne 0$ at $k \ge 1$. The run from $K = 150, 151$ down to $k = 1$, at 140 digits, agrees with the run from $K = 200, 201$ to 138 digits. The output, rendered to sixty digits:

$$P(1) = \ell = 2.34390723868945889060156228887227706909540300963585626066\dots,$$
$$P'(1) = -8.07515226353164401026965124146737650134735782164232219222\dots,$$
$$P''(1) = 31.24865700624850564562564881961470385267564378216839540571\dots,$$

and $P(2) = 0.11463574622981963020052076295742568969846766987861875355 5\dots$, $P'(2) = -0.31664753731523294093363384099033569196227349182039350377 7\dots$ No datum of the low end enters.

*Step 6: the low-end coefficients.* $\mathsf{L}_0 = V_1 = 6$ (Definition 39.37), and $\mathsf{L}_1 = 0$ by Proposition 39.39, the weights being 1. For $\mathsf{L}_2$ use the sector coordinates of an order $\pi$, $\alpha_{\pi(1)} = \rho$, $\alpha_{\pi(2)} = \rho q_2$, $\alpha_{\pi(3)} = \rho q_2 q_3$: then $K = \rho(1 + q_2 + q_2 q_3)$, $\Psi = \rho^2 q_2(1 + q_3 + q_2 q_3)$ and $e_3 = \rho^3 q_2^2 q_3$, so

$$f = q_2 q_3\,\varphi_\pi, \qquad \varphi_\pi = \big((1 + q_2 + q_2 q_3)(1 + q_3 + q_2 q_3)\big)^{-1}, \qquad \kappa_2 = \kappa_3 = 1,$$

the same for all six orders. In the formula of Proposition 39.39 at $i = 2$: $A = \varnothing$ gives $\frac{1}{2}\log^2\varphi_\pi(0) = 0$; $A = \{2\}$ gives, with $m = 1$, $\int_0^1 (\log\varphi_\pi)_{[\{2\}]}\, dq_2/q_2 = -\int_0^1 \log(1+q)\, dq/q$ and, with $m = 0$, nothing, $1_{[A]}$ being 0; $A = \{3\}$ gives the same as $\{2\}$; $A = \{2,3\}$ gives nothing. Expanding $\log(1+q)$,

$$\int_0^1 \log(1+q)\,\frac{dq}{q} = \sum_{m\geq 1}\frac{(-1)^{m-1}}{m^2} = \frac{1}{2}(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),$$

so each order contributes $-(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and $\mathsf{L}_2 = -6\,(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\pi^2 = -\Gamma(\frac{1}{2})^4$, by the classical value $\sum_{m\geq 1} m^{-2} = \pi^2/6$.

*Step 7: the output against the low end.* By Definition 39.37, $k^2P(k) = \mathsf{L}_0 + \mathsf{L}_1 k + \mathsf{L}_2 k^2 + O(k^3)$ near 0, while $P(k+1)$ and $P(k+2)$ are holomorphic there. Expanding (L) at $k = 0$ to the orders 0, 1 and 2:

$$\mathsf{L}_0 - 3P(1) + 9P(2) = 0, \qquad \mathsf{L}_1 - 10P(1) - 3P'(1) + 18P(2) + 9P'(2) = 0,$$
$$\mathsf{L}_2 - 10P(1) - 10P'(1) - \tfrac{3}{2}P''(1) + 9P(2) + 18P'(2) + \tfrac{9}{2}P''(2) = 0.$$

The first is the low-end relation (Corollary 39.38(ii)). With $\mathsf{L}_0 = 6$, $\mathsf{L}_1 = 0$ and $\mathsf{L}_2 = -\Gamma(\frac{1}{2})^4$, the three left sides, read on the output of Step 5, are below $2\cdot 10^{-138}$ in modulus: the gauge end returns the low-end coefficients it did not use. The expansion at the orders 3 and 4 reads $\mathsf{L}_3$ and $\mathsf{L}_4$ off the output; they render as $4.80822761263837714159895264604579996 3\ldots$ and $15.48481981165258402025123201954778137 6\ldots$, and agree with $4\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and $\frac{7}{60}\Gamma(\frac{1}{2})^8 + \frac{3}{4}\ell^2$ to 138 digits (a check).

*Step 8: the arc integrals.* $S_1 = -\frac{2\pi}{3} = -\frac{2}{3}\Gamma(\frac{1}{2})^2$. For $S_2$, expand $-\log\left(2\sin\frac{\vartheta}{2}\right) = \sum_{m\geq 1}\frac{\cos m\vartheta}{m}$ on $(0,2\pi)$, a series that converges in $L^2(0,2\pi)$ and so may be integrated term by term over $(0,x)$:

$$-\int_0^x \log\left(2\sin\tfrac{\vartheta}{2}\right)d\vartheta = \sum_{m\geq 1}\frac{\sin mx}{m^2} \qquad (0\leq x\leq 2\pi).$$

At $x = \frac{2\pi}{3}$, $\sin\frac{2\pi m}{3} = \frac{\sqrt{3}}{2}\chi(m)$ with $\chi(m) = 1, -1, 0$ for $m \equiv 1, 2, 0$ modulo 3, so

$$S_2 = \frac{\sqrt{3}}{2}\Big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\Big),$$

which renders as $0.67662773760643575001413503618301352396 1\ldots$; $2\sqrt{3}\,S_2$ agrees with the value of $\ell$ of Step 5 to 140 digits, and Exercise 54.4 proves $\ell = 2\sqrt{3}\,S_2$.

*Answer.* The germ at $\frac{1}{9}$ is $\varrho_f(t) = 27\sqrt{3}\,\Gamma(\frac{1}{2})^2\sum_m e_m(\frac{1}{9}-t)^m$, holomorphic, the line of the germs that extend by zero; it holds on all of $(0,\frac{1}{9}]$. The values and slopes at 1 are the renderings above, the low-end coefficients are $\mathsf{L}_0 = 6$, $\mathsf{L}_1 = 0$, $\mathsf{L}_2 = -\Gamma(\frac{1}{2})^4$, and $S_1 = -\frac{2}{3}\Gamma(\frac{1}{2})^2$, $S_2 = \frac{\sqrt{3}}{2}\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\big)$. □

## 54.4 The torus at three marks

**Exercise 54.4** (The torus at three marks)**.** Let $F$, $P$, $\ell$, (L), the solution $c$ and the closings $x_j$ be as in Exercise 54.1, and $S_2$ as in Exercise 54.3. On the torus $\alpha_3 = 1$, $\alpha_1 = e^{i\vartheta_1}$, $\alpha_2 = e^{i\vartheta_2}$, let

$$T(k) = \frac{1}{(2\pi)^2}\int_{[0,2\pi]^2} |K|^{-2k}\, d\vartheta,$$

the torus value of the centre ray (Definition 40.11), wherever the integral converges.

(i) Show that $T$ converges for $\operatorname{Re} k < 1$, satisfies (L), continues meromorphically, has $T(-j) = x_j$ and has a simple pole at $k = 1$ with the residue $R = -1/(\sqrt{3}\,\Gamma(\frac{1}{2})^2)$.

(ii) Show that $U(k) = 9^{1-k}T(1-k)$ satisfies (L), that $U(K) = c(K)$, and that $P(k)U(k+1) - P(k+1)U(k) = \frac{2}{3}\cdot 9^{1-k}/k^2$ for every $k$.

(iii) Show that $T = \dfrac{P}{2\Gamma(\frac{1}{2})^4} - \dfrac{\cot(\pi k)}{\sqrt{3}}\,U$.

(iv) Find the torus mean of $\log|K|$, and show that $\ell = 2\sqrt{3}\,S_2 = 3\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\big)$.

*Solution.* Receipt: `ex21_torus.py`.

*Step 1: the stride on the torus.* On the torus $\alpha^{-1} = \bar\alpha$, and $\Psi(\alpha) = e_3\,K(\alpha^{-1})$, so $\Psi = e_3\bar K$ and $f = e_3/(K\Psi) = |K|^{-2}$, real and at least $\frac{1}{9}$. Put $F = |K|^2 = 3 + 2\cos\vartheta_1 + 2\cos\vartheta_2 + 2\cos(\vartheta_1 - \vartheta_2)$. Its gradient $\big(-2\sin\vartheta_1 - 2\sin(\vartheta_1-\vartheta_2),\ -2\sin\vartheta_2 + 2\sin(\vartheta_1-\vartheta_2)\big)$ vanishes exactly when $\sin\vartheta_1 = -\sin\vartheta_2 = -\sin(\vartheta_1-\vartheta_2)$: either $\vartheta_2 = -\vartheta_1$ and $\sin\vartheta_1(1+2\cos\vartheta_1) = 0$, or $\vartheta_2 = \vartheta_1 + \pi$ and $\sin\vartheta_1 = 0$. So $F$ has six critical points: $(0,0)$, the gauge $\mathbf{1}$, with $F = 9$, its maximum; $(\pi,0)$, $(0,\pi)$, $(\pi,\pi)$, the signs, with $F = 1$; and the two zeros $z_\pm = (\frac{2\pi}{3}, \frac{4\pi}{3}), (\frac{4\pi}{3}, \frac{2\pi}{3})$ of $K$, where $F = Q(\delta) + O(|\delta|^3)$ in $\delta = \vartheta - \vartheta(z_\pm)$ with $Q(\delta) = \delta_1^2 - \delta_1\delta_2 + \delta_2^2$, of determinant $\frac{3}{4}$. In $u = \log f = -\log F$ the Hessian at $(0,0)$ is $H = \left(\begin{smallmatrix} 4/9 & -2/9 \\ -2/9 & 4/9 \end{smallmatrix}\right)$, of determinant $\frac{4}{27}$, the Hessian of Exercise 54.3, Step 4, read on the torus; at the signs it has the determinant $-4$ (Theorem 38.17(ii) read on the torus): they are nondegenerate saddles.

*Step 2: convergence, the closings and the law.* Near $z_\pm$, $|K|^{-2k}$ is of the size $|\delta|^{-2\operatorname{Re} k}$, integrable in two variables exactly when $\operatorname{Re} k < 1$; so $T$ is holomorphic on $\operatorname{Re} k < 1$, with $|T(k)| \le T(\operatorname{Re} k)$. For $j \in \mathbb{N}$, $T(-j)$ is the mean of the Laurent polynomial $f^{-j} = |K|^{2j}$ over the torus, its constant term: $T(-j) = \mathrm{CT}(f^{-j}) = x_j$ (Definition 39.30). Multiply the certificate (C) of Exercise 54.1 by $f^k = \exp(k\log f)$, $\log f$ real on the torus off $z_\pm$: $\sum_e (k\,c_e + D_e)g_e \cdot f^k = \sum_e D_e(g_e f^k) = D_1H_1 + D_2H_2$ with $H_i = (g_i - g_3)f^k$ (Euler's identity, as in Step 5 there), and on the torus $D_e = -i\,\partial_{\vartheta_e}$. There $|\Psi| = |K|$, so $|g_e| \le C|K|^{-3}$, $|D_e g_e| \le C|K|^{-4}$ and $|c_e| \le C|K|^{-1}$; for $\operatorname{Re} k \le -3$ the $H_i$ are $C^1$ on the torus and $\int \partial_{\vartheta_i} H_i\,d\vartheta = 0$. So (L) holds for $T$ at $\operatorname{Re} k \le -3$, and by the identity theorem wherever $T(k)$, $T(k+1)$, $T(k+2)$ are holomorphic, $\operatorname{Re} k < -1$. Solved for $T(k+2)$, (L) continues $T$ meromorphically to $\mathbb{C}$, with poles only at the positive integers; on $\operatorname{Re} k < 2$ the only possible pole is at 1, the argument of $T(k+2)$ at the zero $k = -1$ of $9(k+1)^2$.

*Step 3: the residue at* 1. Cut out the regions $Q(\delta) < \varepsilon^2$ around $z_\pm$. Off them the integrand is bounded for $\operatorname{Re} k \le \frac{3}{2}$, and on them $F^{-k} - Q^{-k} = O(|\delta|^{1-2\operatorname{Re} k})$ is integrable for $\operatorname{Re} k < \frac{3}{2}$; both give functions holomorphic near $k = 1$. And

$$\int_{Q(\delta)<\varepsilon^2} Q(\delta)^{-k}\,d\delta = \frac{2\pi}{\sqrt{3/4}}\int_0^\varepsilon r^{1-2k}\,dr = \frac{4\pi}{\sqrt{3}}\cdot\frac{\varepsilon^{2-2k}}{2-2k}.$$

Two zeros and the factor $(2\pi)^{-2}$ give $T(k) = \frac{1}{\sqrt{3}\,\pi}\cdot\frac{1}{1-k} + O(1)$ near 1: a simple pole with the residue $R = -1/(\sqrt{3}\,\pi) = -1/(\sqrt{3}\,\Gamma(\frac{1}{2})^2)$. Write $T(k) = R/(k-1) + T_0 + O(k-1)$.

*Step 4: the torus read backwards.* Read at $-k-1$, (L) for $T$ is $(k+1)^2T(-k-1) - (10k^2 + 10k + 3)\,T(-k) + 9k^2T(1-k) = 0$. With $U(k) = 9^{1-k}T(1-k)$, $U(k+1) = 9^{-k}T(-k)$ and $U(k+2) = 9^{-k-1}T(-k-1)$,

$$\begin{aligned} k^2U(k) - (10k^2+10k+3)\,U(k+1) + 9(k+1)^2U(k+2) \\ = 9^{-k}\big(9k^2T(1-k) - (10k^2+10k+3)\,T(-k) + (k+1)^2T(-k-1)\big) = 0. \end{aligned}$$

So $U$ solves (L), and $U(K) = 9^{1-K}x_{K-1} = c(K)$ (Exercise 54.1, Step 8). Its poles are at $0, -1, -2, \dots$, and $U(k) = -9R/k + O(1)$ at 0. It is the solution of the type of the signs: $c(K)$ decays like the native value 1, not like $9^{-K}$.

*Step 5: the frame of the chart and the torus.* For meromorphic solutions $X, Y$ of (L), $W(X,Y)(k) = X(k)Y(k+1) - X(k+1)Y(k)$ satisfies $p_2(k)W(k+1) = p_0(k)W(k)$ (substitute $p_2X(k+2) = -p_0X(k) - p_1X(k+1)$ and the same for $Y$; Proposition 38.11 with $h = 2$), so $k^2 9^{k-1}W(X,Y)(k)$ is periodic of period 1. A periodic function $h$ of period 1, holomorphic on the strip $|\operatorname{Re} k| \le \frac12$ with $|h(k)| \le C(1+|k|)^N$ there, is constant: $h(k) = \mathcal{H}(e^{2\pi ik})$ with $\mathcal{H}$ holomorphic on $\mathbb{C}^\times$ and $|\mathcal{H}(z)| = O(|\log|z||^N)$ at 0 and at $\infty$, so both singularities are removable and $\mathcal{H}$ is constant by Liouville's theorem. On the strip, $P$ is holomorphic except for its pole at 0, $P(k) = 6/k^2 + O(1)$ ($\mathsf{L}_0 = 6$, $\mathsf{L}_1 = 0$: Exercise 54.3, Step 6), and polynomially bounded, being continued from $\operatorname{Re} k > 0$, where $|P(k)| \le P(\operatorname{Re} k)$, by (L) solved for $P(k)$; $T(k)$ and $T(-k)$ are bounded; $T(k+1)$ and $T(1-k)$ have a simple pole at 0 only and are polynomially bounded, by (L).

(a) In $k^2 9^{k-1}W(P,U)$ the pole of $P(k)$ at 0 meets the holomorphic $U(k+1)$ and the simple pole of $U(k)$ meets the holomorphic $P(k+1)$, so the product with $k^2$ is holomorphic at 0: the function is constant, and at $k = 1$ it is $P(1)c(2) - P(2)c(1) = \frac{\ell}{3} - \frac{\ell-2}{3} = \frac23$. So

$$P(k)U(k+1) - P(k+1)U(k) = \frac{2}{3} \cdot \frac{9^{1-k}}{k^2} \qquad \text{for every } k,$$

the frame of the chart and the torus read backwards; at the integers it is the frame of $r$ and $c$ of Exercise 54.1, Step 8.

(b) Let $m_1 = (2\pi)^{-2} \int \log|K| \, d\vartheta$, so that $T(-k) = 1 + 2m_1 k + O(k^2)$ and $U(k+1) = 1 + (2m_1 - \log 9)\, k + O(k^2)$. Near 0, $P(k)U(k+1) = 6/k^2 + 6(2m_1 - \log 9)/k + O(1)$ and $P(k+1)U(k) = -9R\ell/k + O(1)$, while $\frac23 \cdot 9^{1-k}/k^2 = 6/k^2 - 6\log 9/k + O(1)$. The terms in $1/k$ give $12m_1 + 9R\ell = 0$:

$$m_1 = -\tfrac34 R\,\ell = \frac{\sqrt{3}\,\ell}{4\,\Gamma(\frac12)^2}.$$

*Step 6: the connection up to one constant.* $W(P,U) \neq 0$, so $P$ and $U$ are independent solutions and Cramer's rule on $T(k), T(k+1)$ gives $T = \alpha P + \beta U$ with $\alpha = W(T,U)/W(P,U)$ and $\beta = W(P,T)/W(P,U)$, periodic of period 1 by (a). First,

$$\alpha(k) = \tfrac{k^2}{6}\, T(k)T(-k) - \tfrac{3k^2}{2}\, T(k+1)T(1-k),$$

holomorphic on the strip, the double pole $T(k+1)T(1-k) = -R^2/k^2 + O(1)$ being cancelled by $k^2$, and polynomially bounded: $\alpha$ is constant, equal to its limit at 0, $\alpha = \frac32 R^2 = \frac{1}{2\pi^2} = \frac{1}{2\Gamma(\frac12)^4}$. Next,

$$\beta(k) = \tfrac{3k^2}{2}\, 9^{k-1}\big(P(k)T(k+1) - P(k+1)T(k)\big) = \frac{R}{k} + T_0 + R\log 9 + O(k) \qquad \text{near } 0,$$

from $P(k) = 6/k^2 + O(1)$ and $T(k+1) = R/k + T_0 + O(k)$. The function $R\,\pi\cot(\pi k)$ has simple poles at the integers with the residue $R$, so $\beta - R\,\pi\cot(\pi k)$ is periodic, entire and polynomially bounded on the strip, $\cot(\pi k)$ being bounded away from the real axis: it is a constant $b_0 = T_0 + R\log 9$. So

$$T = \frac{P}{2\Gamma(\frac12)^4} + \Big(b_0 - \frac{\cot(\pi k)}{\sqrt{3}}\Big)U.$$

*Step 7: $b_0 = 0$, by the Fourier transform at the frequency 0.* Put $t = e^u$. The image $\nu$ of $d\vartheta/(2\pi)^2$ under $u = -\log F$ is carried by $[u_0, \infty)$, $u_0 = -\log 9$, and has a density $n(u)$ with the following properties, by the Morse lemma at the nondegenerate critical points of Step 1 and the regularity of the other level sets: (a) near the maximum, $u - u_0 = \frac12 |v|^2$ in coordinates $v$ with $d\vartheta = J\,dv$, $J(0) = (\det H)^{-1/2}$, so $n$ is continuous on $[u_0, u_0 + \delta)$ with $n(u_0) = (2\pi)^{-2}\, 2\pi J(0) = \frac{3\sqrt3}{4\pi} > 0$; in $t$, the density of the torus law at $\frac19$ is $9\, n(u_0) = \frac{27\sqrt3}{4\pi} = (2\pi)^{-2} \varrho_f(\frac19)$, with $\varrho_f(\frac19)$ of Exercise 54.3; (b) near each saddle, $u = v_1 v_2$ in coordinates with $d\vartheta = \tilde J\, dv$, $\tilde J(0) = \frac12$, the inverse square root of the modulus 4 of the determinant of the Hessian,, and the density of $v_1 v_2$ on $|v_1|, |v_2| < \delta$ at $s \neq 0$ is $\int_{|s|/\delta < |v_1| < \delta} \tilde J(v_1, s/v_1)\, dv_1/|v_1|$; writing $\tilde J(v) = \tilde J(0) + v_1 A(v) + v_2 B(v)$, this is $2\tilde J(0) \log(\delta^2/|s|)$ plus two integrals that converge, as $s \to 0$ from either side, to the same limits; so near $u = 0$, with the three saddles, $n(u) = -a \log|u| + b(u)$ with $a = 3 \cdot 2\tilde J(0)/(2\pi)^2 = \frac{3}{4\pi^2}$ and $b$ continuous at 0; (c) near $z_\pm$, the nondegenerate minima of $F$, $n(u) = O(e^{-u})$, so $e^{cu} n(u)$ is integrable on $[u_0, \infty)$ for $0 < c < 1$.

Fix $c \in (0,1)$ and put $k = c + iy$, $y > 0$. Then $T(k) = \int e^{iyu} \psi(u)\, du$ with $\psi = e^{cu} n$; $P(k) = \int e^{iyu} \varphi(u)\, du$ with $\varphi(u) = e^{(c+1)u} \varrho_f(e^u)$, carried by $u \le u_0 < 0$; and, with $w = u_0 - u$ in $T(1-k)$, $U(k) = 9^{1-c} \int e^{iyw} \tilde\psi(w)\, dw$ with $\tilde\psi(w) = e^{(1-c)(u_0 - w)} n(u_0 - w)$, carried by $w \le 0$. And $\beta(k) = b_0 + \frac{i}{\sqrt3} + O(e^{-2\pi y})$, as $\cot(\pi k) \to -i$.

Let $\chi$ be smooth with support in $[1,2]$ and $\int \chi = 1$, $g(s) = s\chi(s)$, $\hat g(w) = \int g(s) e^{isw} ds$, rapidly decreasing, and for a function $\Phi$ of $y$ put $M_Y[\Phi] = Y^{-1} \int \Phi(y)\, y\, \chi(y/Y)\, dy$. For $\Phi(y) = \int e^{iyu} \psi(u)\, du$ with $\psi$ integrable, Fubini gives $M_Y[\Phi] = \int \psi(w/Y)\, \hat g(w)\, dw$. Now $\int \hat g = 2\pi g(0) = 0$, $\int_0^\infty \hat g = i \int g(s)\, s^{-1} ds = i$, $\int_{-\infty}^0 \hat g = -i$, and $\int \log|w|\, \hat g(w)\, dw = -\pi$ (with $\hat g = -i\hat\chi'$, integrate by parts and use $\mathrm{PV} \int e^{isw} dw/w = i\pi$ for $s > 0$). So, as $Y \to \infty$: if $\psi$ is bounded near 0 with one-sided limits $\psi(0\pm)$, $M_Y[\Phi] \to i\big(\psi(0+) - \psi(0-)\big)$; if $\psi = -a \log|u| + b(u)$ near 0 with $b$ continuous at 0, $M_Y[\Phi] \to \pi a$.

Apply $M_Y$ to the connection along $k = c + iy$. The torus value gives $\pi a$, by (b). The values give 0, $\varphi$ vanishing near 0. The last term gives $\big(b_0 + \frac{i}{\sqrt3}\big)\, 9^{1-c}\, i\big(0 - \tilde\psi(0-)\big)$, the factor $O(e^{-2\pi y})$ of $\beta$ contributing nothing in the limit, with $9^{1-c} \tilde\psi(0-) = 9^{1-c} e^{(1-c)u_0} n(u_0) = n(u_0)$. With $Q = n(u_0) > 0$:

$$\pi a = \Big(b_0 + \frac{i}{\sqrt3}\Big)(-i)\, Q = \frac{Q}{\sqrt3} - i\, b_0 Q.$$

$a$, $b_0$ and $Q$ are real, so $b_0 = 0$ (and $\pi a = Q/\sqrt3$: both are $\frac{3}{4\pi}$). Hence (iii), and $T_0 = -R \log 9 = \dfrac{2 \log 3}{\sqrt3\, \Gamma(\frac12)^2}$.

*Step 8: what the connection carries.* (a) At $k = 1 + j$, $j \ge 0$, $P$ and $U$ are holomorphic and $-\cot(\pi k)/\sqrt3$ has the residue $-1/(\sqrt3\,\pi)$, so $\mathrm{Res}_{k=1+j} T = -\frac{9^{-j} x_j}{\sqrt3\, \Gamma(\frac12)^2}$: the poles of the torus value carry its values at the negative integers. (b) Near 0, $P = 2\Gamma(\frac12)^4 \big(T + \frac{\cot(\pi k)}{\sqrt3} U\big)$ and $\frac{\cot(\pi k)}{\sqrt3} U = \frac{1}{\sqrt3 \pi k} \cdot \frac{-9R}{k} + O(\frac1k) = \frac{3}{\pi^2 k^2} + O(\frac1k)$: the low-end coefficient $\mathsf{L}_0 = 6$ is the residue of the torus at the zeros of $K$. (c) For real $a \ge 0$, $a \neq 1$, the mean of $\log|a + e^{i\vartheta}|$ over $\vartheta$ is $\log\max(a, 1)$ (for $a > 1$ write $\log|a + e^{i\vartheta}| = \log a + \mathrm{Re} \log(1 + e^{i\vartheta}/a)$, whose second term is a power series in $e^{i\vartheta}$ without constant term; for $a < 1$ the same with $\log|1 + a e^{-i\vartheta}|$). Integrating over $\vartheta_1$ first, with $a = |1 + e^{i\vartheta_2}| = 2|\cos\frac{\vartheta_2}{2}|$, which exceeds 1 exactly when $|\vartheta_2| < \frac{2\pi}{3}$,

$$m_1 = \frac{1}{\pi} \int_0^{2\pi/3} \log\big(2\cos\tfrac{\vartheta}{2}\big)\, d\vartheta = -\frac{1}{\pi} \int_0^{\pi/3} \log\big(2\sin\tfrac{\varphi}{2}\big)\, d\varphi = \frac{1}{\pi} \sum_{m \ge 1} \frac{\sin(m\pi/3)}{m^2},$$

by $\vartheta = \pi - \varphi$, $\int_0^\pi \log(2\sin\frac{\varphi}{2})\,d\varphi = 0$ and the Fourier series of Exercise 54.3, Step 8. Here $\sin\frac{m\pi}{3} = \frac{\sqrt3}{2}\varepsilon(m)$ with $\varepsilon(m) = 1,1,0,-1,-1,0$ for $m \equiv 1,\dots,6$ modulo 6, and $\varepsilon(m) = \chi(m)$ for odd $m$, $-\chi(m)$ for even $m$, $\chi$ as in Exercise 54.3; $\chi(2m) = -\chi(m)$, so $\sum_m \varepsilon(m)m^{-2} = \sum_m \chi(m)m^{-2} + \frac12\sum_m \chi(m)m^{-2}$, and $m_1 = \frac{3\sqrt3}{4\pi}\sum_m \chi(m)m^{-2} = \frac{3}{2\pi}S_2$. With Step 5(b):

$$\ell = \frac{4\pi m_1}{\sqrt3} = 2\sqrt3\,S_2 = 3\Big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\Big).$$

*Checks* (floating point, not part of the solution). By the mean over one circle, $T(k) = \frac{2}{\pi}\int_0^2 I_k(a)\,\frac{da}{\sqrt{4-a^2}}$ with $I_k(a)$ the mean of $|a+e^{i\vartheta}|^{-2k}$, a hypergeometric series in $a^2$ or $a^{-2}$; this gives $T(-j) = x_j$, (L) on $T$ at $k = -1.3$ and $k = -2.7+0.4i$ to 20 digits, and, with $P$ from the gauge end, the connection of (iii) at $k = 0.2, 0.3, \dots, 0.8$ to between 17 and 23 digits, $k^2 9^{k-1} W(P,U) = \frac23$ at the same points to 20 digits, $\beta(\frac12) = b_0$ below $2\cdot10^{-23}$, and $T_0$ rendering as $0.40379779973879724450\ldots$; $m_1$ renders as $0.32306594721945051409363651072 4\ldots$

*Answer.* $T$ continues to a meromorphic solution of (L) with $T(-j) = x_j$ and the residue $-1/(\sqrt3\,\Gamma(\frac12)^2)$ at 1; $U(k) = 9^{1-k}T(1-k)$ is the solution $c$ of Exercise 54.1, with $P(k)U(k+1) - P(k+1)U(k) = \frac23\cdot 9^{1-k}/k^2$; the connection is $T = \frac{P}{2\Gamma(\frac12)^4} - \frac{\cot(\pi k)}{\sqrt3}U$; the torus mean of $\log|K|$ is $\frac{\sqrt3\,\ell}{4\Gamma(\frac12)^2}$, and $\ell = 2\sqrt3\,S_2 = 3\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\big)$. □

## 54.5 A lean of three marks

**Exercise 54.5** (The lean $(2\mathbf{1};4,1)$ of $\binom{[3]}{1}$)**.** Let $F$ and $\ell$ be as in Exercise 54.1, and consider the ray of $F$ with the origin 0 and the direction $(2\mathbf{1};4,1)$, with the values $P_\lambda(k) = P_F(2k\mathbf{1};4k,k)$.

(i) Find its walls, $V_1$, its stride and native value, and its closings.

(ii) Reduce its box $\mathfrak{B}_2(1,1)$ modulo the permutations of the marks: the spanning set at $\mathfrak{s} = 1$, the matrix, the law of [1], its symbol and its native gauges, and $\mathbf{B}(\infty)$.

(iii) Read the low-end relation, and show that $54\,P_\lambda(1) = 11 - 4\ell$.

(iv) Write $P_\lambda(1)$ on the share side, and say why the gap theorem cannot be applied to the rational coordinates of this ray.

*Solution.* Receipt: `ex22_lean.py`.

*Step 1: the ray.* By Definition 8.1 and the counts of Exercise 54.1, Step 1, the walls of the direction are $\kappa_{1,S} = m_S - 4d(S) - d^*(S) = m_S - (|S|-1)$: 2 on the marks and 3 on the pairs. Every order $\pi$ has the tails $U_2$, a pair, and $U_3$, a mark, so $V_1 = 6\cdot\frac{1}{3\cdot2} = 1$ (Proposition 39.34). The stride is $f = \alpha^{2\mathbf{1}}K^{-4}\Psi^{-1} = e_3^2/(e_1^4e_2)$. The defects $c_e = 2 - 4R_e - R_e^*$ vanish at $\mathbf{1}$ $(2 - \frac43 - \frac23 = 0)$, so $\alpha^\star = \mathbf{1}$ and $f(\alpha^\star) = \frac{1}{3^4\cdot3} = \frac{1}{243}$. The constant term of $f^{-j} = e_1^{4j}\big(\sum_e \alpha^{\mathcal{E}\setminus e}\big)^j\alpha^{-2j\mathbf{1}}$ pairs the monomial $\alpha^{(j-b_1,j-b_2,j-b_3)}$ of the second factor with the monomial $\alpha^{(j+b_1,j+b_2,j+b_3)}$ of $e_1^{4j}$, so

$$x_j = \mathrm{CT}(f^{-j}) = (4j)!\,j!\sum_{b_1+b_2+b_3=j}\ \prod_{e=1}^{3}\frac{1}{b_e!\,(b_e+j)!},$$

$$(x_j)_{j\ge0} = 1,\ 36,\ 4620,\ 762300,\ 140216076,\ 27409661136,\ \ldots$$

*Step 2: the box.* The permutations of the marks are symmetries; the Cremona is not, the exponents 4 and 1 being different (Proposition 39.11(ii)). Modulo $S_3$ the box $\mathfrak{B}_2(1,1)$ (Definition 39.15), the box with one unit of each letter at $\mathfrak{t} = 2$, has 398 symbols in 79 orbits (17, 19, 20

and 23 at the levels $(0,0)$, $(1,0)$, $(0,1)$, $(1,1)$) and 82 stems. Its relations are 184 letter identities for $K$, 178 for $\Psi$ and 246 mark relations of stems; modulo $S_3$ they give 117 distinct combinations of orbit averages, of rank 76. The free orbits are those of $\alpha_1\alpha_2\alpha_3/(K\Psi)$ (wall degree $-\frac{1}{3}$), of $\alpha_1/\alpha_3$ (wall degree $\frac{1}{2}$) and of 1, so the spanning set at $\mathfrak{s}=1$ is again $(1,b_1)$, $b_1=\frac{1}{6}\sum_{i\neq j}\alpha_i/\alpha_j$, and the box is accepted there (Definition 39.20) with

$$\mathbf{B}_{11}=\frac{199k^3-420k^2+299k-72}{(k-1)^2(3k-2)},\qquad \mathbf{B}_{12}=\frac{(2k-1)(265k^2-424k+171)}{(k-1)^2(3k-2)},$$
$$\mathbf{B}_{21}=\frac{3710k^4-12349k^3+15022k^2-7997k+1584}{3(k-1)(2k-3)(3k-4)(3k-2)},$$
$$\mathbf{B}_{22}=\frac{(2k-1)(4706k^3-13323k^2+12238k-3681)}{3(k-1)(2k-3)(3k-4)(3k-2)}.$$

Certificates on the stems (Theorem 39.21(ii)): for each row, $G_1$ a combination of the stems symmetric under the exchange of the marks 2 and 3, $G_e$ its images; the identity is linear in 44 unknowns with 118 equations and a family of solutions of dimension 24; a solution with 20 nonzero coefficients, rational functions of $k$, satisfies it exactly for each row. It is not displayed here; the receipt prints it.

*Step 3: the law.* The rows of Proposition 39.6 give the first dependence with

$$q_0=-972(k-1)^2(4k-3)(4k-1)(265k^2-954k+860),$$
$$q_1=-1721705k^6+13084958k^5-40594556k^4+65762882k^3$$
$$-58662047k^2+27325016k-5197140,$$
$$q_2=3(k-2)^2(3k-5)(3k-4)(265k^2-424k+171),$$

and $p_j(k)=q_{2-j}(k+2)$, divided by their common factor, is the recurrence

$$-3k^2(3k+1)(3k+2)(265k^2+636k+383)\,P_\lambda(k)$$
$$+(1721705k^6+7575502k^5+13047276k^4+11068046k^3+4766659k^2+965100k+75600)\,P_\lambda(k+1)$$
$$+972(k+1)^2(4k+5)(4k+7)(265k^2+106k+12)\,P_\lambda(k+2)=0,$$

of degree 6, with the symbol $265(243\lambda-1)(64\lambda+27)$; its reflection is satisfied by the closings of Step 1 (Corollary 39.38(i)). The native gauges: at $\mathbf{1}$ the native value $\frac{1}{243}$, and the Hessian of $\log f$ on the chart $\alpha_3=1$ has the determinant $\frac{25}{27}$; at $\alpha=(3,3,-2)$ and its permutations all three defects vanish (exact evaluation), the native value is $\frac{(-18)^2}{4^4\cdot(-3)}=-\frac{27}{64}$, and the Hessian has the determinant $-\frac{75}{2}$. Both are nondegenerate, their native values are the roots of the symbol (Corollary 38.7), and the vectors $(1,b_1(\alpha_c))$ are $(1,1)$ and $(1,-\frac{7}{18})$, which span: by Proposition 39.42(i) the spanning set is a set of masters and the recurrence is the law of [1]. Finally

$$\mathbf{B}(\infty)=\begin{pmatrix}\frac{199}{3} & \frac{530}{3}\\ \frac{1855}{27} & \frac{4706}{27}\end{pmatrix},$$

with the eigenvalues 243 and $-\frac{64}{27}$ and the eigenvectors $(1,1)$ and $(1,-\frac{7}{18})$ (Proposition 39.42(ii)).

*Step 4: the low-end relation.* $p_0$ vanishes to the order 2 at 0 with $[k^2]p_0=-3\cdot1\cdot2\cdot383=-2298$, $p_1(0)=75600$ and $p_2(0)=972\cdot5\cdot7\cdot12=408240$; with $\mathsf{L}_0=V_1=1$, Corollary 39.38(ii) gives $-2298+75600\,P_\lambda(1)+408240\,P_\lambda(2)=0$, that is

$$12600\,P_\lambda(1)+68040\,P_\lambda(2)=383.$$

*Step 5: the value, read on the centre ray.* At $k=2$ the centre ray of Exercise 54.1 has the stride $f_{\mathrm{c}}=\alpha^{\mathbf{1}}/(K\Psi)$, and

$$\big\langle \Psi K^{-2}\big\rangle_2=\int \Psi K^{-2}\,\frac{\alpha^{\mathbf{21}}}{K^2\Psi^2}\,dy=\int \frac{\alpha^{\mathbf{21}}}{K^4\Psi}\,dy=P_\lambda(1):$$

the value of the lean is the value of the insertion $\Psi K^{-2}=\sum_e \alpha^{\mathcal{E}\setminus e}K^{-2}$ along the centre ray. In the box $\mathfrak{B}_2(2,2)$ of the centre ray modulo $S_3$ and the Cremona (339 symbols, 43 orbits, 499 relations of rank 40, the free orbits those of $f_{\mathrm{c}}^2$, of $\alpha_1/\alpha_3$ and of 1) the orbit average of $\Psi K^{-2}$ reduces to

$$\Psi K^{-2}\equiv -\frac{2(k-1)}{3(k+1)}+\frac{(k-1)^2}{k(k+1)}\,b_1.$$

The relations of a box evaluate into $\mathcal{L}$ and their values vanish identically in $k$ (Proposition 39.18(i)); the symmetries keep the values (Proposition 39.11(ii)); and at $k=2$ every value involved converges. So, with $\langle b_1\rangle_k=\frac{1}{6}\big(\langle f_{\mathrm{c}}^{-1}\rangle_k-3\langle 1\rangle_k\big)=\frac{1}{6}\big(P(k{-}1)-3P(k)\big)$ from $f_{\mathrm{c}}^{-1}=3+6b_1$, $\langle b_1\rangle_2=\frac{1}{6}\big(\ell-(\ell-2)\big)=\frac{1}{3}$, and

$$P_\lambda(1)=-\frac{2}{9}\,P(2)+\frac{1}{6}\cdot\frac{1}{3}=-\frac{2}{9}\cdot\frac{\ell-2}{3}+\frac{1}{18}=\frac{11-4\ell}{54}.$$

Then Step 4 gives $P_\lambda(2)=\frac{2800\ell-6551}{204120}$, and the law gives every $P_\lambda(K)$ in $\mathbb{Q}+\mathbb{Q}\ell$.

*Step 6: the share side.* By Proposition 8.15, $P_F(\mathbf{m};s,t)$ is the period of the dual table, whose bases are the sets $\mathcal{E}\setminus e$, at the exponents $(t,s)$; that table is one circuit, and Corollary 18.24 with the exponents $(1,4)$, $\nu_e=m_e-1=1$ and unit weights (its condition $m_S>|S|-1$ holds: $2>0$, $4>1$) gives

$$P_\lambda(1)=\frac{1}{\Gamma(4)\Gamma(1)}\int_0^\infty \mathsf{G}_{1;(1,-1)}(1,y)^3\,dy,$$

that is $\frac{1}{6}\int_0^\infty \mathsf{G}_{1;(1,-1)}(1,y)^3\,dy$. A check by quadrature: both this integral and $\frac{11-4\ell}{54}$ render as $0.03008094528226230439988427489834 9847\ldots$

*Step 7: why the gap does not apply.* Write $P_\lambda(K)=c_\lambda(K)\,\ell+r_\lambda(K)$ with rationals from Steps 3 to 5, and let $D_K$ be the least common denominator of $c_\lambda(K)$ and $r_\lambda(K)$, the least integer that clears the row. Theorem 37.2 needs $D_KP_\lambda(K)\to 0$. But the values decay no faster than $243^{-K}$: as in Exercise 54.1, Step 2(b), $f\ge\frac{1}{243}\,\alpha_{(2)}\alpha_{(3)}^2/\alpha_{(1)}^3$, and on the cone of an order the exponent $3y_{(1)}-y_{(2)}-2y_{(3)}=3(y_{(1)}-y_{(2)})+2(y_{(2)}-y_{(3)})$ has the walls 3 and 2 as coefficients, so $P_\lambda(K)\ge 243^{-K}\cdot 6\cdot\frac{1}{6K^2}=243^{-K}/K^2$. The exact denominators satisfy $D_K>243^KK^2$ for every $K$ with $3\le K\le 240$, so $D_KP_\lambda(K)>1$ throughout that range; $\frac{1}{K}\log D_K$ renders as about 10.3 at $K=240$, against $\log 243=5.49$. The clearing integers of this ray grow faster than its values decay, and the hypothesis of Theorem 37.2 fails on every computed row.

*Answer.* Walls 2 and 3, $V_1=1$, $f(\alpha^\star)=\frac{1}{243}$; the masters $1,b_1$, the law above with the symbol $265(243\lambda-1)(64\lambda+27)$ and the native gauges $\mathbf{1}$ and $(3,3,-2)$; the low-end relation $12600P_\lambda(1)+68040P_\lambda(2)=383$; and $54\,P_\lambda(1)=11-4\ell$, with the share side $\frac{1}{6}\int_0^\infty \mathsf{G}_{1;(1,-1)}(1,y)^3dy$. □

## 54.6 The balanced directions

**Exercise 54.6** (The balanced directions of $\binom{[3]}{1}$)**.** Let $F$ and $\ell$ be as in Exercise 54.1. For a direction $(\mathbf{m};s,t)$ on the balance wall put $c_e=m_e-t$.

(i) List the balanced directions with $1\le s,t\le 5$, positive walls and $m_1\ge m_2\ge m_3$, and sort them by $S=s+t$ and the multiset $\{|c_e|\}$.

(ii) Show that $\Gamma(sk)\Gamma(tk)\,P_F(k\mathbf{m};ks,kt)$ depends on the direction only through $S$ and $\{|c_e|\}$, so that within a class the values are in the ratio of two Beta corners.

(iii) For every direction with $|c_e|$ constant find $P_F(\mathbf{m};s,t)$ in $\mathbb{Q}+\mathbb{Q}\ell$, and its native value $f(\alpha^\star)$.

*Solution.* Receipt: `ex23_directions.py`.

*Step 1: the directions.* The balance wall is $m_{\mathcal{E}} = s+2t$, and the walls are $\kappa_S = m_S - t(|S|-1)$ (Exercise 54.5, Step 1): $m_e > 0$ and $m_e + m_{e'} > t$. There are 123 such directions. The multiset $\{|c_e|\}$, with $\sum_e c_e = s-t$, has one value on 17 of them, two on 70 and three on 36; the classes of equal $S$ and $\{|c_e|\}$ number 54: 9, 31 and 14. The 17 directions with $|c_e|$ constant fall into nine classes:

| $\lvert c_e\rvert$, $S$ | directions $(\mathbf{m};s,t)$ | $P_F(\mathbf{m};s,t)$ |
|---|---|---|
| 0, $2c'$ | $(c'\mathbf{1};c',c')$, $c'=1,\dots,5$ | $\ell,\ \frac{\ell-2}{3},\ \frac{10\ell-23}{54},\ \frac{62\ell-145}{486},\ \frac{568\ell-1331}{5832}$ |
| 1, 5 | $(4,2,2;2,3)$, $(3,3,1;3,2)$ | $\frac{11-4\ell}{18}$ |
| | $(3,3,3;1,4)$, $(2,2,2;4,1)$ | $\frac{11-4\ell}{54}$ |
| 1, 7 | $(5,3,3;3,4)$, $(4,4,2;4,3)$ | $\frac{19-8\ell}{54}$ |
| | $(4,4,4;2,5)$, $(3,3,3;5,2)$ | $\frac{19-8\ell}{108}$ |
| 1, 9 | $(6,4,4;4,5)$, $(5,5,3;5,4)$ | $\frac{751-320\ell}{2916}$ |
| 2, 8 | $(7,3,3;3,5)$, $(5,5,1;5,3)$ | $\frac{80\ell-181}{648}$ |

the last column being found in Step 3.

*Step 2: the classes.* The walls are positive, so Proposition 8.15 and Corollary 18.24 apply as in Exercise 54.5, Step 6, at the data $(k\mathbf{m};ks,kt)$, with $\nu_e = km_e - kt = kc_e$:

$$\Gamma(sk)\Gamma(tk)\,P_F(k\mathbf{m};ks,kt) = \int_0^\infty y^{tk-1}\prod_e \mathsf{G}_{kc_e;(1,-1)}(1,y)\,dy.$$

Proposition 2.3 with $A=1$ and $\mathbf{h}=\frac12\log y$ gives $\mathsf{G}_{\nu;(1,-1)}(1,y) = y^{\nu/2}\,\mathsf{G}_{\nu;(1,-1)}(\sqrt{y},\sqrt{y})$, and with $A=-1$ it gives $\mathsf{G}_{\nu;(1,-1)}(\xi_1,\xi_2) = \mathsf{G}_{-\nu;(1,-1)}(\xi_2,\xi_1)$, so at equal coefficients the Gramma is even in $\nu$. With $\sum_e c_e = s-t$,

$$\Gamma(sk)\Gamma(tk)\,P_F(k\mathbf{m};ks,kt) = \int_0^\infty y^{\frac{Sk}{2}-1}\prod_e \mathsf{G}_{k|c_e|;(1,-1)}(\sqrt{y},\sqrt{y})\,dy,$$

which depends on $S$ and $\{|c_e|\}$ only. So two directions $d,d'$ of one class have

$$\frac{P_{d'}(k)}{P_d(k)} = \frac{\Gamma(sk)\Gamma(tk)}{\Gamma(s'k)\Gamma(t'k)} = {}_2\Gamma_1\!\left[\begin{matrix}sk,\ tk\\ Sk\end{matrix}\right] \Big/ {}_2\Gamma_1\!\left[\begin{matrix}s'k,\ t'k\\ Sk\end{matrix}\right],$$

two Beta corners at the same total, rational at the integers; for instance $\frac{\Gamma(4k)\Gamma(k)}{\Gamma(3k)\Gamma(2k)} = 3, 7, \frac{33}{2}, 39$ at $k=1,2,3,4$ for $(4,2,2;2,3)$ against $(3,3,3;1,4)$. A class has one value span.

*Step 3: the values, read on the centre ray.* Along the centre ray of Exercise 54.1, with $f_{\mathrm{c}} = \alpha^{\mathbf{1}}/(K\Psi)$, the insertion $\alpha^u K^{-a}\Psi^{-b}$ has the value $\langle \alpha^u K^{-a}\Psi^{-b}\rangle_k = P_F(k\mathbf{1}+u;k+a,k+b)$ (Definition 38.1). So

$$P_F(k\mathbf{1}+2\mathbf{e}_1;k,k+1) = \langle \alpha_1^2\Psi^{-1}\rangle_k, \qquad P_F(3\mathbf{1}+4\mathbf{e}_1;3,5) = \langle \alpha_1^4\Psi^{-2}\rangle_3,$$

and $(4,2,2;2,3)$, $(5,3,3;3,4)$, $(6,4,4;4,5)$ are $\langle \alpha_1^2\Psi^{-1}\rangle_k$ at $k=2,3,4$. Reduced in the boxes of the centre ray modulo $S_3$ and the Cremona, as in Exercise 54.2 (the orbit averages have the same values, Proposition 39.11(ii)),

$$\alpha_1^2\Psi^{-1} \equiv -\frac{2}{3} + \frac{k-1}{k}\,b_1 \quad \text{in } \mathfrak{B}_2(1,1),$$

$$\alpha_1^4\Psi^{-2} \equiv \frac{7k+5}{9(k+1)} - \frac{(k-1)(2k^2-6k-5)}{3k(k-2)(k+1)}\, b_1 \quad \text{in } \mathfrak{B}_2(2,2),$$

and the relations of a box hold at every strength where the values converge (Proposition 39.18(i)). With $P(K) = c(K)\ell + r(K)$ (Exercise 54.1, Step 8) and $\langle b_1\rangle_k = \frac{1}{6}\big(P(k-1) - 3P(k)\big)$, so that $\langle b_1\rangle_2 = \frac{1}{3}$, $\langle b_1\rangle_3 = \frac{11-4\ell}{108}$ and $\langle b_1\rangle_4 = \frac{19-8\ell}{243}$,

$$P_F(4,2,2;2,3) = -\tfrac{2}{3}\cdot\tfrac{\ell-2}{3} + \tfrac{1}{2}\cdot\tfrac{1}{3} = \tfrac{11-4\ell}{18}, \qquad P_F(5,3,3;3,4) = -\tfrac{2}{3}\cdot\tfrac{10\ell-23}{54} + \tfrac{2}{3}\cdot\tfrac{11-4\ell}{108} = \tfrac{19-8\ell}{54},$$
$$P_F(6,4,4;4,5) = -\tfrac{2}{3}\cdot\tfrac{62\ell-145}{486} + \tfrac{3}{4}\cdot\tfrac{19-8\ell}{243} = \tfrac{751-320\ell}{2916},$$
$$P_F(7,3,3;3,5) = \tfrac{13}{18}\cdot\tfrac{10\ell-23}{54} + \tfrac{5}{18}\cdot\tfrac{11-4\ell}{108} = \tfrac{80\ell-181}{648}.$$

The centre's class is the centre ray read at $c'k$: $P_F(c'\mathbf{1}; c', c') = P(c') = c(c')\ell + r(c')$. The other members follow by the ratios of Step 2 at $k=1$: $\frac{\Gamma(2)\Gamma(3)}{\Gamma(3)\Gamma(2)} = 1$ for $(3,3,1;3,2)$, $\frac{\Gamma(2)\Gamma(3)}{\Gamma(1)\Gamma(4)} = \frac{1}{3}$ for $(3,3,3;1,4)$ and $(2,2,2;4,1)$, 1 for $(4,4,2;4,3)$, $\frac{\Gamma(3)\Gamma(4)}{\Gamma(2)\Gamma(5)} = \frac{1}{2}$ for $(4,4,4;2,5)$ and $(3,3,3;5,2)$, and 1 for $(5,5,3;5,4)$ and $(5,5,1;5,3)$. The value of $(2,2,2;4,1)$ agrees with Exercise 54.5, where it was read on another insertion.

*Step 4: the native values.* The native gauge on the orthant is unique (Theorem 38.14(i)), so a symmetry of the masses fixes it: for $m_2 = m_3$ it is $(x,1,1)$, for $m_1 = m_2$ it is $(1,1,x)$, with $x > 0$ the root of the defect equation $m_e = s\,\alpha_e/K + t\,\alpha_e(K-\alpha_e)/\Psi$ at one mark. Exactly: $f(\alpha^\star) = 9^{-c'}$ on the centre's class; $\frac{1}{243}$ at $\mathbf{1}$ for $(3,3,3;1,4)$ and $(2,2,2;4,1)$, and $\frac{64}{6561}$ at $(4,1,1)$ and $(1,1,\frac{1}{4})$ for $(4,2,2;2,3)$ and $(3,3,1;3,2)$; $\frac{1}{2187}$ at $\mathbf{1}$ for $(4,4,4;2,5)$ and $(3,3,3;5,2)$, and $\frac{3125}{3779136}$ at $(\frac{5}{2},1,1)$ and $(1,1,\frac{2}{5})$; $\frac{1}{12500}$ at $(2,1,1)$ and $(1,1,\frac{1}{2})$; $\frac{823543}{553584375}$ at $(7,1,1)$ and $(1,1,\frac{1}{7})$. Within a class the native values are in the ratio of the rates of the Beta corners, $\frac{64}{27} = \frac{4^4}{3^3 2^2}$ and $\frac{3125}{1728} = \frac{5^5 2^2}{4^4 3^3}$.

*Check* (not part of the solution). The least recurrences of the closings $\mathrm{CT}(f^{-j})$, searched by linear algebra modulo the prime $2^{61}-1$ on 60 closings to the order 4 and the degree 24, have the order 2 on 16 of the 17 directions with $|c_e|$ constant; on the seventeenth, $(5\mathbf{1};5,5)$, the closings are those of the centre at $5j$, whose recurrence of order 2 has the degree 18, found with 70 closings. They have the order 3 on 5 directions and the order 4 on 2, all with two values of $|c_e|$, and none within these bounds on the other 100.

*Answer.* 123 directions in 54 classes; within a class $\Gamma(sk)\Gamma(tk)P$ is one function of $k$, and the values are in the ratio of two Beta corners at the same total; the 17 directions with $|c_e|$ constant have the values of the table, all in $\mathbb{Q} + \mathbb{Q}\ell$, and the native values of Step 4. □

## 54.7 The centre of $\binom{[4]}{1}$

The list $\binom{[4]}{1}$ has the marks $1,2,3,4$ and the four singletons as its bases, every weight 1, so that $K(\alpha) = \alpha_1+\alpha_2+\alpha_3+\alpha_4$ and $\Psi(\alpha) = \alpha^{[4]}K(\alpha^{-1})$. Its centre ray at the strength one (Definition 38.13) has the stride $f = \big(K(\alpha)K(\alpha^{-1})\big)^{-1}$, so that $f^{-1} = \sum_{e,e'} \alpha_e/\alpha_{e'}$; the walls of the direction are the widths, $\kappa_{1,S} = r(S) + r([4]\setminus S) - 1 = 1$ for every nonempty proper $S$, and the volume of the direction is $V_1 = \sum_\pi \prod_{l=2}^4 \kappa_{1,U_l}^{-1} = 4! = 24$ (Proposition 39.34). Its values are $P(k) = \langle 1\rangle_k = P_{\binom{[4]}{1}}(k\mathbf{1}; k, k)$, and throughout this section $v = P(1)$. The five exercises close $v$ on the half-lattice, derive the law of the values and its rows, write the row at every step in closed form, read the coefficient of the coordinate as a closing in two ways, and clear the rows.

**Exercise 54.7** (The closing on the half-lattice)**.** Write $v = P(1)$ on its share side, merge the marks in pairs, and close the remaining share. Show that

$$v = (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty)) = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

*Solution.* Receipt: `ex24a_halflattice.py`.

*The share side.* The dual list $\binom{[4]}{3}$ has the bases $[4]-e$, and replacing the table by its dual exchanges the exponents (Proposition 8.15), which at the centre are equal: the two lists have the same values. Corollary 18.24 applies to $\binom{[4]}{3}$ at $s=t=1$ with the masses $\mathbf{1}$: they lie on the balance wall, $m_{[4]}=4=3s+t$, they meet $m_S=|S|>s(|S|-1)$, and $\nu_e=m_e-s=0$. So

$$v=\frac{1}{\Gamma(1)^2}\int_{a_1+a_2+a_3+a_4=1}\prod_{e=1}^{4}\Gamma(a_e)^2,$$

a fibre integral of dimension 3.

*Merging the pairs.* Put $c=a_1+a_2$. Corollary 4.7 with the entries $a=b=0$, $c=d=c$ and the variable $a_1$ merges the pair $\{1,2\}$,

$$\frac{1}{2\pi i}\int\Gamma(a_1)^2\,\Gamma(c-a_1)^2\,da_1=\Gamma\begin{matrix}4\\1\end{matrix}\!\left[\begin{matrix}c,\ c,\ c,\ c\\ \ \ 2c\end{matrix}\right],$$

and the same with the total $1-c$ merges $\{3,4\}$. One share is left:

$$v=\frac{1}{2\pi i}\int_{(\frac12)}\Gamma\begin{matrix}4\\1\end{matrix}\!\left[\begin{matrix}c,\ c,\ c,\ c\\ \ \ 2c\end{matrix}\right]\Gamma\begin{matrix}4\\1\end{matrix}\!\left[\begin{matrix}1-c,\ 1-c,\ 1-c,\ 1-c\\ \ \ 2-2c\end{matrix}\right]dc.$$

*The integrand.* By the reflection and the shift (Proposition 1.3(iii),(v)), $\Gamma(c)^4\Gamma(1-c)^4=\pi^4/\sin^4\pi c$ and $\Gamma(2c)\Gamma(2-2c)=(1-2c)\,\Gamma(2c)\Gamma(1-2c)=(1-2c)\,\pi/\sin 2\pi c$, so the integrand is

$$w(c)=\frac{\pi^4}{\sin^4\pi c}\cdot\frac{\sin 2\pi c}{(1-2c)\,\pi}=\frac{2\pi^3\cos\pi c}{(1-2c)\sin^3\pi c}.$$

At $c=\frac12$ the zero of $\cos\pi c$ cancels that of $1-2c$, so the poles of $w$ are the integers. The factor $2\pi^3\cos\pi c/\sin^3\pi c$ is $-\partial_c\Gamma\begin{matrix}4\\0\end{matrix}\!\big[c,\ c,\ 1-c,\ 1-c\big]=2\sum_{m\in\mathbb{Z}}(c+m)^{-3}$ (Lemma 5.16), of period one, and at $c=m+\varepsilon$

$$\frac{2\pi^3\cos\pi\varepsilon}{\sin^3\pi\varepsilon}=\frac{2}{\varepsilon^3}-\frac{2\pi^4}{15}\,\varepsilon+O(\varepsilon^3):$$

no terms in $\varepsilon^{-2}$, $\varepsilon^{-1}$ or $\varepsilon^0$.

*The closing.* At $c=N+\frac12+iy$, $|\cos\pi c|=|\sinh\pi y|$ and $|\sin\pi c|=\cosh\pi y$, so $|w(c)|\le 4\pi^3e^{-2\pi|y|}/(2N)$: the integrals over the lines $\operatorname{Re}c=N+\frac12$ tend to zero, and so do those over the horizontal segments $\operatorname{Im}c=\pm Y$ as $Y\to\infty$. The line closes to the right (Theorem 4.5) on the triple poles at the integers $m\ge1$, with the residues

$$\operatorname*{Res}_{c=m}w=2\,[\varepsilon^2]\,\frac{1}{1-2m-2\varepsilon}=\frac{2\cdot4}{(1-2m)^3}=-\frac{8}{(2m-1)^3},$$

and closing to the right takes minus their sum:

$$v=\sum_{m\ge1}\frac{8}{(2m-1)^3}=\sum_{x\in\frac12+\mathbb{Z},\ x>0}x^{-3}=(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty))$$

(Definition 5.2).

*The coset relation.* Theorem 5.12(i) at $q=2$ and $b=0$, with the kernel $x^{-3}$ and the cut 0, assembles $\frac12\mathbb{Z}$ from its two cosets, the rescaled kernel being $(x/2)^{-3}=8x^{-3}$:

$$(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))+(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty))=8\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

So $v = (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty)) = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, which is $8.41439832211715999 78\ldots$ to 19 places; the receipt evaluates the contour integral by quadrature as a check. The coset $\frac12+\mathbb{Z}$ is the doubled share $2c$ of a merged pair, the lower entry of $\Gamma_{\!4\atop 1}\begin{bmatrix} c,\ c,\ c,\ c \\ & 2c & \end{bmatrix}$: the word has one lower Gamma of slope 2 and one of slope $-2$, which pair off (Proposition 43.2), and its rational factor $(1-2c)^{-1}$ has its pole on that coset. □

**Exercise 54.8** (The law, the low-end relation and the rows)**.** (i) With $\vartheta = y\frac{d}{dy}$, compute the operators $\mathbf{R}_0,\ldots,\mathbf{R}_5$ of Theorem 42.12 for $n=4$, and from them the law of the values, $\sum_{k=0}^{2} c_{4,k}(s)\,P(s+k)=0$.

(ii) Give the closings $\mathsf{C}_j(1)$ for $j\le 8$, and check the law read at $k=-j$ on them.

(iii) Give the symbol, its roots and the native gauges they belong to.

(iv) Write the low-end relation.

(v) Write the rows $P(k)=a_kv-b_k$ for $k\le 6$ (Definition 36.5), and the value span of the ray.

*Solution.* Receipt: `ex24b_law.py`.

(i) *One mark.* At the strength $s$ the masses are $s\mathbf{1}$ and the exponents $(s,s)$, and Corollary 18.24, as in Exercise 54.7, gives the share side in its first form,

$$\Gamma(s)^2P(s)=\int_0^\infty y^{s-1}\,\mathsf{g}(y)^4\,dy,\qquad \mathsf{g}(y)=\mathsf{G}_{0;(1,-1)}(1,y),\quad \operatorname{Re}s>0.$$

The Gramma $\mathsf{g}$ satisfies $\vartheta^2\mathsf{g}=y\,\mathsf{g}$. Write $\mathsf{g}_\nu=\mathsf{G}_{\nu;(1,-1)}(1,y)$. The ladder (Theorem 2.4(i)) in the coefficient $y$ of the exponent $-1$ gives $\vartheta\mathsf{g}_\nu=-y\,\mathsf{g}_{\nu-1}$, so $\vartheta^2\mathsf{g}_0=-y\,\mathsf{g}_{-1}+y^2\mathsf{g}_{-2}$; the exchange relation (Theorem 2.4(ii)) at $\nu=-1$ reads $-\mathsf{g}_{-1}=\mathsf{g}_0-y\,\mathsf{g}_{-2}$, and substituting $y\,\mathsf{g}_{-2}=\mathsf{g}_0+\mathsf{g}_{-1}$ gives $\vartheta^2\mathsf{g}_0=y\,\mathsf{g}_0$.

*The operators.* Theorem 42.12(i) at $n=4$ has $\mathbf{R}_0=1$, $\mathbf{R}_1=\vartheta$ and $\mathbf{R}_{i+1}=\vartheta\mathbf{R}_i-i(5-i)\,y\,\mathbf{R}_{i-1}$, with the coefficients $i(5-i)=4,6,6,4$; an operator is written $\sum_k y^kq(\vartheta)$, and $\vartheta\,y^kq(\vartheta)=y^k(\vartheta+k)\,q(\vartheta)$. So

$$\begin{aligned}
\mathbf{R}_2&=\vartheta^2-4y,\\
\mathbf{R}_3&=\vartheta^3-2y(5\vartheta+2),\\
\mathbf{R}_4&=\vartheta^4-2y(8\vartheta^2+7\vartheta+2)+24y^2,\\
\mathbf{R}_5&=\vartheta^5-2y(2\vartheta+1)(5\vartheta^2+5\vartheta+2)+64y^2(\vartheta+1),
\end{aligned}$$

and $\mathbf{Q}_4=\mathbf{R}_5$ annihilates every product of four solutions of $\vartheta^2u=yu$, $\mathsf{g}^4$ among them; the receipt checks it on the basis $u^{4-i}(\vartheta u)^i$. The leading coefficients $1,-20,64$ of $q_{4,0},q_{4,1},q_{4,2}$ are those of $\xi(\xi^2-16\eta)(\xi^2-4\eta)=\xi^5-20\eta\xi^3+64\eta^2\xi$, as Theorem 42.12(i) says.

*The law.* Integrated against $y^{s-1}$, a term $y^kq(\vartheta)$ becomes $q(-s-k)$ times the transform at $s+k$: integration by parts turns $\vartheta$ into $-s$, the boundary terms vanishing because $\mathsf{g}$ is $O(|\log y|)$ at 0 (Lemma 20.29(i)) and at most $\sqrt{\pi}\,y^{-1/4}e^{-2\sqrt{y}}$ for $y\ge1$ (Lemma 20.29(ii)). With $\Gamma(s+k)^2=\Gamma(s)^2(s)_k^2$ and a division by $\Gamma(s)^2s^2$ this is Theorem 42.12(ii):

$$c_{4,0}(s)=-s^3,\qquad c_{4,1}(s)=q_{4,1}(-s-1)=2(2s+1)(5s^2+5s+2),$$
$$c_{4,2}(s)=q_{4,2}(-s-2)\,(s+1)^2=-64(s+1)^3,$$

$$-s^3P(s)+2(2s+1)(5s^2+5s+2)\,P(s+1)-64(s+1)^3P(s+2)=0$$

identically in $s$. At $s=1$ the coefficients are $-1$, 72, $-512$.

(ii) *The closings.* By Corollary 39.36 $\mathsf{C}_j(1)$ is the constant term of $f^{-j}=K(\alpha)^jK(\alpha^{-1})^j$; the multinomial expansions of the two factors pair the monomial $\alpha^v$ with $\alpha^{-v}$, so

$$\mathsf{C}_j(1)=\sum_{|v|=j}\binom{j}{v}^2=1,\ 4,\ 28,\ 256,\ 2716,\ 31504,\ 387136,\ 4951552,\ 65218204\qquad(j=0,\dots,8).$$

The law read at $k=-j$ (Corollary 39.36, Theorem 42.12(iii)) is $\sum_{i\le\min(2,j)}c_{4,i}(-j)\,\mathsf{C}_{j-i}(1)=0$: at $j=1$, $1\cdot4+(-4)\cdot1=0$; at $j=2$, $8\cdot28+(-72)\cdot4+64\cdot1=0$; the receipt checks it for $j\le40$, and also that $\mathbf{Q}_4$ annihilates $\mathsf{h}^4=\sum_j\mathsf{C}_j(1)\,y^j/j!^2$, $\mathsf{h}(y)=\sum_m y^m/m!^2$, through $y^{40}$.

(iii) *The symbol.* The coefficients of $s^3$ in $c_{4,0},c_{4,1},c_{4,2}$ are $-1,20,-64$, so the symbol (Corollary 38.7) is

$$\chi_4(\lambda)=-1+20\lambda-64\lambda^2=-(1-4\lambda)(1-16\lambda),$$

with the roots $\frac{1}{16}$ and $\frac14$. By Proposition 42.9(ii) the native gauges of the ray are the signs $\eta$ with $K(\eta)\neq0$, of native value $K(\eta)^{-2}$: $\eta=\mathbf{1}$, with $K=4$ and the native value $\frac{1}{16}=f(\alpha^\star)$, $\alpha^\star=\mathbf{1}$ being the native gauge of the direction (Theorem 38.14); and the signs with one or three marks at $-1$, with $K=\pm2$ and the native value $\frac14$. The signs with two marks at $-1$ have $K=0$ and lie off $X_F$.

(iv) *The low-end relation.* By Corollary 39.38(ii), with $[s^3]c_{4,0}=-1$, $\mathsf{L}_0=V_1=24$, $c_{4,1}(0)=4$ and $c_{4,2}(0)=-64$,

$$-24+4P(1)-64P(2)=0,\qquad P(2)=\frac{P(1)-6}{16}.$$

(v) *The rows.* $P(1)=v$ and $P(2)=\frac{1}{16}v-\frac38$, and the law at $s=k\ge1$, whose last coefficient $c_{4,2}(k)$ does not vanish, gives $P(k+2)=-\big(c_{4,0}(k)P(k)+c_{4,1}(k)P(k+1)\big)/c_{4,2}(k)$; at $k=1$, $P(3)=\big(72P(2)-P(1)\big)/512=\frac{7}{1024}v-\frac{27}{512}$. The rows $(a_k;b_k)$:

| $k$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $a_k$ | 1 | $\frac{1}{16}$ | $\frac{7}{1024}$ | $\frac{1}{1024}$ | $\frac{679}{4194304}$ | $\frac{1969}{67108864}$ |
| $b_k$ | 0 | $\frac38$ | $\frac{27}{512}$ | $\frac{37}{4608}$ | $\frac{25555}{18874368}$ | $\frac{9304913}{37748736000}$ |

So every value lies in $\mathbb{Q}+\mathbb{Q}\,v$, $v=(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,\infty))$ (Exercise 54.7), and this is the value span of the ray (Corollary 42.15(i), with $h=2$): the order $h=2$ leaves one value free after the low-end relation. As renderings, $P(2)=0.15089989513232249986\dots$, $P(3)=0.0047859260300977734224\dots$ and $P(4)=0.00018767197255365017146\dots$; the receipt checks $P(2)$ against the quadrature of $\int_0^\infty y\,\mathsf{g}(y)^4dy$. □

**Exercise 54.9** (The row at every step)**.** For $n\ge0$ put $p_l=l+\frac12$ and $j=n-l$ for $0\le l\le n$,

$$c_l=\binom{2l}{l}\binom{2j}{j},\qquad S_l=\frac{c_l}{\binom{n}{l}}=\frac{(2l)!\,(2j)!}{l!\,j!\,n!},\qquad w_l=S_l^2c_l,\qquad F(\sigma)=\binom{\sigma-1}{n}^2,$$

$F[\sigma,p]=\big(F(\sigma)-F(p)\big)/(\sigma-p)$, and for a polynomial $g$

$$\mathcal{J}[g]=\frac{1}{2\pi i}\int_{(\frac12)}g(\sigma)\sum_{m\in\mathbb{Z}}\frac{d\sigma}{(\sigma+m)^3}.$$

Show that

$$256^n\,P(n+1)=\sum_{l=0}^{n}w_l\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((l,\infty))-16^n\sum_{l=0}^{n}c_l\,\mathcal{J}\big[F[\,\cdot\,,p_l]\big],$$

evaluate $\mathcal{J}$ on the binomial polynomials $\binom{\sigma}{i}$, and read the rows of $P(2)$ and $P(3)$ off the formula.

*Solution.* Receipt: `ex24c_row.py`.

(a) *The word.* At the strength $k = n+1$ the share side of Exercise 54.7, with the shares of total $k$ and the pairs merged by Corollary 4.7 at the totals $\sigma$ and $k-\sigma$, is

$$P(k) = \frac{1}{\Gamma(k)^2}\,\frac{1}{2\pi i}\int \frac{\Gamma(\sigma)^4\,\Gamma(k-\sigma)^4}{\Gamma(2\sigma)\,\Gamma(2k-2\sigma)}\,d\sigma \qquad (0 < \operatorname{Re}\sigma < k).$$

With $\Gamma(k-\sigma) = (1-\sigma)_n\Gamma(1-\sigma)$ and $\Gamma(2k-2\sigma) = (2-2\sigma)_{2n}(1-2\sigma)\Gamma(1-2\sigma)$ (Proposition 1.3(iii)), the reflections of Exercise 54.7 give the factor $2\pi^3\cos\pi\sigma/\sin^3\pi\sigma = 2\sum_m(\sigma+m)^{-3}$ against $(1-\sigma)_n^4/\big((1-2\sigma)(2-2\sigma)_{2n}\big)$. Splitting $(2-2\sigma)_{2n}$ into its factors $2(1-\sigma+t)$ and $2(\frac{3}{2}-\sigma+t)$, $0 \le t < n$, gives $(1-2\sigma)(2-2\sigma)_{2n} = 2\cdot 4^n(\frac{1}{2}-\sigma)_{n+1}(1-\sigma)_n$, so

$$P(n+1) = \frac{1}{4^n\,n!^2}\,\frac{1}{2\pi i}\int \rho(\sigma)\sum_m \frac{d\sigma}{(\sigma+m)^3}, \qquad \rho(\sigma) = \frac{(1-\sigma)_n^3}{(\frac{1}{2}-\sigma)_{n+1}}.$$

The triple poles of the kernel at $1,\dots,n$ meet the triple zeros of $\rho$, and the simple poles of $\rho$ at the $p_l$ meet the zeros of the kernel, so the integrand is regular on $0 < \operatorname{Re}\sigma < k$ and the line may be $\operatorname{Re}\sigma = \frac{1}{2}$.

(b) *Partial fractions.* With $\Omega(\sigma) = \prod_{i=1}^n(\sigma-i) = n!\binom{\sigma-1}{n}$, $(1-\sigma)_n = (-1)^n\Omega$ and $(\frac{1}{2}-\sigma)_{n+1} = (-1)^{n+1}\prod_{i=0}^n(\sigma-p_i)$, so $\rho/\Omega^2 = -\Omega/\prod_i(\sigma-p_i)$, whose residue at $p_l$ is $-\Omega(p_l)/\prod_{i\ne l}(l-i)$, with $\prod_{i\ne l}(l-i) = (-1)^j l!\,j!$ and

$$\Omega(p_l) = \prod_{t=0}^{l-1}\big(t+\tfrac{1}{2}\big)\prod_{t=1}^{j}\big(\tfrac{1}{2}-t\big) = \frac{(-1)^j(2l-1)!!\,(2j-1)!!}{2^n}.$$

As $(2l-1)!!/l! = \binom{2l}{l}/2^l$, the residue is $-c_l/4^n$:

$$\frac{\rho}{\Omega^2} = -\frac{1}{4^n}\sum_{l=0}^{n}\frac{c_l}{\sigma-p_l}.$$

Multiplying by $\Omega^2 = n!^2F$ and writing $\Omega(\sigma)^2 = \Omega(p_l)^2 + (\sigma-p_l)\,n!^2F[\sigma,p_l]$ splits $\rho$ into a polynomial and its poles,

$$\rho = \mathcal{P} + \sum_{l=0}^{n}\frac{r_l}{\sigma-p_l}, \qquad \mathcal{P} = -\frac{n!^2}{4^n}\sum_l c_l\,F[\sigma,p_l], \qquad r_l = -\frac{c_l\,\Omega(p_l)^2}{4^n} = -\frac{n!^2\,w_l}{64^n},$$

the last because $(2l-1)!!\,(2j-1)!! = S_l\,n!/2^n$. The receipt checks the partial fractions and the $r_l$ for $n \le 3$.

(c) *The pole terms.* The kernel is $\pi^3\cos\pi\sigma/\sin^3\pi\sigma$, with the principal part $(\sigma-i)^{-3}$ and no terms of orders $-2,-1,0$ at every integer $i$ (Exercise 54.7), so the residue of $(\sigma-p_l)^{-1}$ times the kernel at $i$ is $\frac{1}{2}\partial_\sigma^2(\sigma-p_l)^{-1}|_{\sigma=i} = (i-p_l)^{-3}$; at $p_l$ the kernel vanishes. On the lines $\operatorname{Re}\sigma = N+\frac{1}{2}$ the integrand is $O(N^{-1}e^{-2\pi|\operatorname{Im}\sigma|})$, so the line closes to the right (Theorem 4.5):

$$\frac{1}{2\pi i}\int_{(\frac{1}{2})}\frac{1}{\sigma-p_l}\sum_m\frac{d\sigma}{(\sigma+m)^3} = -\sum_{i\ge 1}\frac{1}{(i-p_l)^3} = -(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((l,\infty)),$$

the terms $i \le l$, at the negative points $i-p_l$, cancelling the head $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,l])$ of the terms at the positive points (Proposition 5.3). The polynomial part gives $\mathcal{J}[\mathcal{P}]$.

(d) *Assembling.* By (a) to (c),

$$P(n+1)=\frac{1}{4^n n!^2}\Big(-\frac{n!^2}{4^n}\sum_l c_l\,\mathcal{J}\big[F[\cdot,p_l]\big]+\frac{n!^2}{64^n}\sum_l w_l\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((l,\infty))\Big),$$

which is the stated formula after multiplication by $256^n$. With $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((l,\infty))=v-(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,l])$ the row of $P(n+1)$ is

$$a_{n+1}=\frac{1}{256^n}\sum_{l=0}^{n}w_l,\qquad b_{n+1}=\frac{1}{256^n}\Big(\sum_{l=0}^{n}w_l\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,l])+16^n\sum_{l=0}^{n}c_l\,\mathcal{J}\big[F[\cdot,p_l]\big]\Big).$$

(e) *The functional $\mathcal{J}$.* By Lemma 5.16, $\sum_m(\sigma+m)^{-3}=-\frac12\partial_\sigma\Gamma_4^{\,0}\big[\sigma,\ \sigma,\ 1-\sigma,\ 1-\sigma\big]$, and on the line the stacked Gamma is $2\pi\,p_{\mathrm{crit}}(y)$, $\sigma=\frac12+iy$, decaying like $e^{-2\pi|y|}$ (Definition 36.1). Integrating by parts,

$$\mathcal{J}[g]=\frac12\,\mathbb{E}\big[g'(X)\big],\qquad X=\tfrac12+iY,\quad Y\sim\mathbb{P}_{\mathrm{crit}}.$$

By Theorem 36.2(iii), $\mathbb{E}[e^{tX}]=te^t/(e^t-1)$ for $|t|<2\pi$, so $\mathbb{E}[X^m]=1,\frac12,\frac16,0,-\frac{1}{30},0,\frac{1}{42},0,-\frac{1}{30}$ for $m=0,\dots,8$, and at $t=\log(1+z)$, $\mathbb{E}[(1+z)^X]=(1+z)\log(1+z)/z$. Since $\sum_i\binom{\sigma}{i}z^i=(1+z)^\sigma$, whose derivative in $\sigma$ is $\log(1+z)\,(1+z)^\sigma$,

$$\sum_{i\ge0}\mathcal{J}\Big[\binom{\sigma}{i}\Big]z^i=\frac12\log(1+z)\,\mathbb{E}\big[(1+z)^X\big]=\frac{(1+z)\log^2(1+z)}{2z}.$$

With $\log^2(1+z)=2\sum_{m\ge2}(-1)^m(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,m-1])\,z^m/m$, the head $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,m])=\sum_{r=1}^m\frac1r$ (Definition 5.2), the coefficients are

$$\lambda_0=0,\qquad \lambda_i=\mathcal{J}\Big[\binom{\sigma}{i}\Big]=(-1)^i\,\frac{(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,i-1])-1}{i(i+1)}\quad(i\ge1),$$

that is $0,\frac12,0,-\frac{1}{24},\frac{1}{24},-\frac{13}{360},\frac{11}{360},-\frac{29}{1120},\frac{223}{10080}$ for $i=0,\dots,8$; the receipt checks them against the moments for $i<40$. A polynomial written in the binomial basis, $g=\sum_i(\Delta^ig)(0)\binom{\sigma}{i}$, has $\mathcal{J}[g]=\sum_i(\Delta^ig)(0)\,\lambda_i$.

(f) *The first steps.* At $n=1$: $c_0=c_1=2$, $S_0=S_1=2$, $w_0=w_1=8$; $F(\sigma)=(\sigma-1)^2$, $F[\sigma,p]=\sigma+p-2$, so $F[\sigma,\frac12]=\sigma-\frac32$ and $F[\sigma,\frac32]=\sigma-\frac12$, each with $\mathcal{J}=\frac12$; with $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,1])=8$,

$$256\,P(2)=16v-8\cdot8-16\Big(2\cdot\frac12+2\cdot\frac12\Big)=16v-96,\qquad P(2)=\frac{1}{16}v-\frac38.$$

At $n=2$: $(c_0,c_1,c_2)=(6,4,6)$, $(S_0,S_1,S_2)=(6,2,6)$, $(w_0,w_1,w_2)=(216,16,216)$, and

$$\begin{aligned}F[\sigma,\tfrac12]&=\tfrac14\sigma^3-\tfrac{11}{8}\sigma^2+\tfrac{41}{16}\sigma-\tfrac{55}{32},&\mathcal{J}&=\tfrac{21}{32},\\F[\sigma,\tfrac32]&=\tfrac14\sigma^3-\tfrac{9}{8}\sigma^2+\tfrac{25}{16}\sigma-\tfrac{21}{32},&\mathcal{J}&=\tfrac{9}{32},\\F[\sigma,\tfrac52]&=\tfrac14\sigma^3-\tfrac{7}{8}\sigma^2+\tfrac{17}{16}\sigma-\tfrac{11}{32},&\mathcal{J}&=\tfrac{5}{32},\end{aligned}$$

for instance $\mathcal{J}\big[F[\cdot,\frac12]\big]=\frac12\mathbb{E}\big[\frac34X^2-\frac{11}{4}X+\frac{41}{16}\big]=\frac12\big(\frac18-\frac{11}{8}+\frac{41}{16}\big)=\frac{21}{32}$. With the heads $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,1])=8$ and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,2])=8+\frac{8}{27}$,

$$256^2P(3)=448\,v-\Big(16\cdot8+216\cdot\frac{224}{27}\Big)-256\Big(6\cdot\frac{21}{32}+4\cdot\frac{9}{32}+6\cdot\frac{5}{32}\Big)=448\,v-1920-1536,$$

so $P(3) = \frac{7}{1024}v - \frac{27}{512}$. Both rows are those of Exercise 54.8.

So the row of $P(n+1)$ at every step is $a_{n+1} = 256^{-n}\sum_l w_l$ and $b_{n+1}$ as in (d), with $\mathcal{J}$ read through the $\lambda_i$. For $n = 0,1,2,3$ the sums are $\sum_l w_l = 1, 16, 448, 16384$, $256^n b_{n+1} = 0, 96, 3456, \frac{1212416}{9}$ and $\sum_l c_l \mathcal{J}[F[\cdot, p_l]] = 0, 2, 6, \frac{853}{54}$; the receipt checks that the formula reproduces the rows of the law for $n \le 40$. □

**Exercise 54.10** (The coefficient of the coordinate is a closing)**.** Show that $a_k = \mathsf{C}_{k-1}(1)/64^{k-1}$ for every $k \ge 1$, and give the denominator of $a_k$:

(i) from the law of Exercise 54.8, by Theorem 43.8;

(ii) directly from the row of Exercise 54.9, by a telescoping identity in the summation index.

*Solution.* Receipt: `ex24d_closing.py`.

(i) *The law is reflexive.* With $h = 2$, Definition 43.7 asks for $c_{4,2-i}(-k-1) = \gamma\lambda^i c_{4,i}(k)$, $i = 0,1,2$. Here

$$c_{4,2}(-k-1) = 64k^3 = -64\,c_{4,0}(k), \qquad c_{4,0}(-k-1) = (k+1)^3,$$
$$c_{4,1}(-k-1) = -2(2k+1)(5k^2+5k+2) = -c_{4,1}(k),$$

so $\gamma = -64$, $\gamma\lambda = -1$ gives the ratio $\lambda = \frac{1}{64}$, and $\gamma\lambda^2 c_{4,2}(k) = -\frac{64}{4096}\cdot(-64)(k+1)^3 = (k+1)^3$ checks the third. The ray has the origin 0, every letter exponent positive and positive walls, as Proposition 43.6 asks. By Theorem 43.8(i), $y_k = 64^{1-k}\mathsf{C}_{k-1}(1)$ solves the law on $k \ge 1$ with the low-end coefficient $-\big(c_{4,1}(0)y_1 + c_{4,2}(0)y_2\big) = -\big(4\cdot 1 - 64\cdot\frac{4}{64}\big) = 0$, and since $c_{4,2}(k) \ne 0$ for $k \ge 0$, Theorem 43.8(ii) gives

$$P(k) = \frac{\mathsf{C}_{k-1}(1)}{64^{k-1}}\,P(1) + r_k, \qquad a_k = \frac{\mathsf{C}_{k-1}(1)}{64^{k-1}}.$$

By Theorem 43.8(iii), with $\lambda = \frac{1}{64}$, the denominator of $a_k$ is $64^{k-1}/\gcd\big(64^{k-1}, \mathsf{C}_{k-1}(1)\big)$:

| $k$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| $\mathsf{C}_{k-1}(1)$ | 1 | 4 | 28 | 256 | 2716 | 31504 | 387136 | 4951552 |
| $v_2\big(\mathsf{C}_{k-1}(1)\big)$ | 0 | 2 | 2 | 8 | 2 | 4 | 6 | 9 |
| $\log_2 \mathrm{den}(a_k)$ | 0 | 4 | 10 | 10 | 22 | 26 | 30 | 33 |

with $v_2$ the exponent of 2.

(ii) *The closings across the pairs.* Split $f^{-1} = (X+Y)(X'+Y')$ with $X = \alpha_1 + \alpha_2$, $X' = \alpha_1^{-1} + \alpha_2^{-1}$, and $Y, Y'$ the same on $\{3,4\}$. In $(X+Y)^n(X'+Y')^n = \sum_{l,l'}\binom{n}{l}\binom{n}{l'}X^l Y^{n-l} X'^{l'} Y'^{n-l'}$ a term has a constant term in $\alpha_1, \alpha_2$ only if $l = l'$, $X^l$ and $X'^{l'}$ being homogeneous of degrees $l$ and $-l'$, and $\mathrm{CT}\,\big((XX')^l\big) = \sum_i \binom{l}{i}^2 = \binom{2l}{l}$. So

$$\mathsf{C}_n(1) = \sum_{l=0}^{n}\binom{n}{l}^2\binom{2l}{l}\binom{2j}{j} =: U(n), \qquad j = n-l.$$

Put $T(n) = 4^{-n}\sum_l w_l$, so that $a_{n+1} = 4^n T(n)/256^n$ by Exercise 54.9. It remains to show $T = U$.

*The telescoping identities.* Both $T$ and $U$ satisfy

$$(n+2)^3 X_{n+2} - 2(2n+3)(5n^2+15n+12)\,X_{n+1} + 64(n+1)^3 X_n = 0 \qquad (n \ge 0),$$

which is the law of the closings of Exercise 54.8(ii) at $j = n+2$: $c_{4,0}(-n-2) = (n+2)^3$, $c_{4,1}(-n-2) = -2(2n+3)(5n^2+15n+12)$, $c_{4,2}(-n-2) = 64(n+1)^3$. Write $t(n,l)$ for the

summand of either sum, $t_T(n,l) = 4^{-n}S_l^2 c_l$ and $t_U(n,l) = \binom{n}{l}^2 c_l$, zero outside $0 \le l \le n$. Their ratios are rational:

$$\frac{t_T(n,l+1)}{t_T(n,l)} = \frac{(2l+1)^3\, j}{(l+1)(2j-1)^3}, \qquad \frac{t_T(n+1,l)}{t_T(n,l)} = \frac{2(2j+1)^3}{(j+1)(n+1)^2},$$
$$\frac{t_U(n,l+1)}{t_U(n,l)} = \frac{j^3(2l+1)}{(l+1)^3(2j-1)}, \qquad \frac{t_U(n+1,l)}{t_U(n,l)} = \frac{2(n+1)^2(2j+1)}{(j+1)^3}.$$

With $G = Rt$ and the rational functions

$$R_T = \frac{4l\,(2j+1)^3\,\kappa_T}{(n+1)^2(j+1)(j+2)}, \qquad R_U = -\frac{4(n+1)^2\,l^3\,(2j+1)\,\kappa_U}{(j+1)^3(j+2)^3},$$

$$\kappa_T = 12j^3 + 10lj^2 + 2l^2j + 59j^2 + 33lj + 4l^2 + 94j + 26l + 49,$$
$$\kappa_U = 12j^3 + 10lj^2 + 2l^2j + 62j^2 + 35lj + 3l^2 + 104j + 30l + 56,$$

each summand satisfies, for $0 \le l \le n-1$,

$$64(n+1)^3 t(n,l) - 2(2n+3)(5n^2+15n+12)\,t(n+1,l) + (n+2)^3 t(n+2,l) = G(n,l+1) - G(n,l).$$

Divided by $t(n,l)$ and written with the ratios this is an identity of rational functions of $(n,l)$; cleared of denominators it is an identity of polynomials, which the receipt verifies coefficient by coefficient for both $T$ and $U$. Summing it over $0 \le l \le n-1$, with $G(n,0) = 0$ (the factor $l$), leaves $G(n,n)$, and the full sum over $l$ of the left side is $G(n,n)$ plus the left side at $l = n, n+1, n+2$, where only the summands with $l \le n+1$ at $n+1$ and $l \le n+2$ at $n+2$ survive. Divided by $t(n,n)$, with $t(n+1,n)/t(n,n)$, $t(n+1,n+1)/t(n,n)$, $t(n+2,n)/t(n,n)$, $t(n+2,n+1)/t(n,n)$ and $t(n+2,n+2)/t(n,n)$ read off the ratios, this boundary sum is a rational function of $n$ that vanishes identically (receipt). So both sums satisfy the recurrence; $T(0) = U(0) = 1$, $T(1) = \frac{1}{4}(8+8) = 4 = U(1)$, and the leading coefficient $(n+2)^3$ does not vanish, so $T = U$.

Hence $\sum_l w_l = 4^n \mathsf{C}_n(1)$ and $a_{n+1} = 4^n \mathsf{C}_n(1)/256^n = \mathsf{C}_n(1)/64^n$, as in (i). The coefficient of the coordinate $v$ in $P(k)$ is the closing $\mathsf{C}_{k-1}(1)$ over $64^{k-1}$, with no lattice sum: $64^{k-1}a_k \in \mathbb{Z}$, and the denominator of $a_k$ is $64^{k-1}/\gcd\big(64^{k-1}, \mathsf{C}_{k-1}(1)\big)$. □

**Exercise 54.11** (The clearing)**.** Show that for every $k \ge 1$ the part of the denominator of $b_k$ prime to 2 divides $d_{k-1}^3$, and that $256^{k-1}d_{k-1}^3\, b_k \in \mathbb{Z}$; so $256^{k-1}d_{k-1}^3$ clears the row of $P(k)$. Compare it with the bound of the frame.

*Solution.* Receipt: `ex24e_clearing.py`.

Write $n = k-1$. By Exercise 54.9(d), $256^n b_{n+1}$ is the sum of the heads and the moments,

$$\sum_l w_l\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,l)) \qquad \text{and} \qquad 16^n \sum_l c_l\,\mathcal{J}\big[F[\cdot,p_l]\big];$$

and by Exercise 54.10 $64^n a_{n+1} = \mathsf{C}_n(1) \in \mathbb{Z}$. Two facts from outside the handbook are used: Legendre's formula, $v_q(m!) = \sum_{t\ge1}\lfloor m/q^t\rfloor$ for a prime $q$, and Kummer's theorem, that $v_q\binom{a+b}{a}$ is the number of carries when $a$ and $b$ are added in base $q$; $v_q$ is the exponent of $q$.

*The integers.* $S_l \in \mathbb{Z}$: by Legendre's formula, $v_q(S_l) = \sum_{t\ge1}\big(\lfloor 2l/q^t\rfloor + \lfloor 2j/q^t\rfloor - \lfloor l/q^t\rfloor - \lfloor j/q^t\rfloor - \lfloor n/q^t\rfloor\big)$, and every term is nonnegative by $\lfloor 2x\rfloor + \lfloor 2y\rfloor \ge \lfloor x\rfloor + \lfloor y\rfloor + \lfloor x+y\rfloor$ (with $x = a+\alpha$, $y = b+\beta$, $0 \le \alpha, \beta < 1$, it reads $\lfloor 2\alpha\rfloor + \lfloor 2\beta\rfloor \ge \lfloor \alpha+\beta\rfloor$, true since $\alpha+\beta \ge 1$ forces $\max(\alpha,\beta) \ge \frac{1}{2}$). So $w_l = S_l^2 c_l \in \mathbb{Z}$.

Fix an odd prime $q$ and $e \ge 1$ with $q^{e-1} \le n < q^e$, so that $v_q(d_n^3) = 3(e-1)$; it suffices that both sums have $v_q \ge -3(e-1)$.

*The heads.* $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac12+\mathbb{Z}})((0,l]) = 8\sum_{i=1}^{l}(2i-1)^{-3}$, and $2l-1 < 2n < 2q^e$, so the only odd number up to $2l-1$ with $v_q \ge e$ is $q^e$, when $q^e \le 2l-1$. In that case, at the level $q^e$ of Legendre's formula, $\lfloor l/q^e\rfloor = \lfloor j/q^e\rfloor = \lfloor n/q^e\rfloor = 0$ and $\lfloor 2l/q^e\rfloor = 1$, so $v_q(S_l) \ge 1$ and $v_q\binom{2l}{l} \ge 1$, the other levels contributing $\ge 0$; then $v_q(w_l) \ge 3$, which pays for $(q^e)^{-3}$. Otherwise the head already has $v_q \ge -3(e-1)$.

*The moments.* Vandermonde's identity twice, $\binom{\sigma}{m} = \sum_t \binom{\sigma-p}{t}\binom{p}{m-t}$ and $\binom{\sigma-p-1}{t-1} = \sum_i \binom{\sigma}{i}\binom{-p-1}{t-1-i}$, with $\binom{\sigma-p}{t} = \frac{\sigma-p}{t}\binom{\sigma-p-1}{t-1}$, gives the binomial coefficients of the divided difference,

$$\big(\Delta^i F[\cdot,p]\big)(0) = \sum_m f_m \sum_{t=i+1}^{m} \frac{1}{t}\binom{p}{m-t}\binom{-p-1}{t-1-i}, \qquad F = \sum_m f_m\binom{\sigma}{m}.$$

Here $f_m = \sum_{i,i'\le n}(-1)^{i+i'}\binom{m}{i}\binom{i}{m-i'}$, from $\binom{\sigma-1}{n} = \sum_{i\le n}(-1)^{n-i}\binom{\sigma}{i}$ and $\binom{\sigma}{i}\binom{\sigma}{i'} = \sum_m \binom{m}{i}\binom{i}{m-i'}\binom{\sigma}{m}$; so $f_m \in \mathbb{Z}$, and $q \mid f_m$ for $m \ge q^e$: a nonzero term has $i < q^e$ and $m-i \le i' < q^e$, and adding $i$ and $m-i$ in base $q$ carries into the place $q^e$, so $q \mid \binom{m}{i}$ (Kummer's theorem). At $p = p_l$, a $q$-adic integer, the binomials of $p$ and of $-p-1$ are $q$-integral. Since $t \le 2n < 2q^e$, the only $t$ with $v_q(t) \ge e$ is $t = q^e$, where $m \ge q^e$ and $q \mid f_m$. So $v_q\big((\Delta^i F[\cdot,p_l])(0)\big) \ge 1-e$ for every $i$, and $\ge 2-e$ for $i \ge q^e$, where every $t$ and $m$ exceed $q^e$.

For $1 \le i \le 2n-1$: $v_q(\lambda_i) \ge 2-2e$ for $i < q^e$ and $v_q(\lambda_i) \ge 1-2e$ for $i \ge q^e$. The heads of the counting input at index one have $v_q\big((\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,m])\big) \ge 1-e$ for $m < q^e$ and $\ge -e$ for $q^e \le m < 2q^e$, while

$$v_q\big((\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,q^e-1])\big) \ge 2-e :$$

grouping the $r < q^e$ by $v_q(r) = s$, $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,q^e-1]) = \sum_{s<e} q^{-s}\sum_{r'<q^{e-s},\ q\nmid r'}\frac{1}{r'}$, and $\sum_{r'<q^a,\ q\nmid r'}\frac{1}{r'}$ has $v_q \ge a$, pairing $r'$ with $q^a - r'$, $\frac{1}{r'} + \frac{1}{q^a-r'} = \frac{q^a}{r'(q^a-r')}$; each term has $v_q \ge e-2s \ge 2-e$. The integers $i$ and $i+1$ are coprime and below $2q^e$, so $v_q\big(i(i+1)\big) \le e-1$ unless $q^e \in \{i, i+1\}$, where it is $e$. For $i < q^e-1$ this gives $v_q(\lambda_i) \ge (1-e)-(e-1)$. At $i = q^e-1$, $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,q^e-2]) - 1 = (\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,q^e-1]) - \frac{q^e}{q^e-1}$ has $v_q \ge \min(2-e,e)$, so $v_q(\lambda_i) \ge \min(2-e,e) - e \ge 2-2e$. At $i = q^e$, $(\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,q^e-1]) - 1$ has $v_q \ge \min(2-e,0)$, so $v_q(\lambda_i) \ge \min(2-e,0) - e \ge 1-2e$; and for $q^e < i \le 2n-1$, $v_q(\lambda_i) \ge -e-(e-1)$. In each case the product of the coefficient and $\lambda_i$ has $v_q \ge 3-3e$, and $16^n$ and $c_l$ are $q$-integral. The receipt checks every one of these estimates for $n \le 30$ and every odd prime $q \le 2n$; for instance $v_3\big((\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,26])\big) = -1$, $v_5\big((\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,24])\big) = 1$.

*The prime 2.* For $p \in \frac12+\mathbb{Z}$, $v_2\binom{p}{r} = -r - v_2(r!) \ge 1-2r$ for $r \ge 1$; the product of the two binomials in the coefficient, of total order $m-1-i \le 2n-1$, has $v_2 \ge 3-4n$; $v_2(1/t) \ge -\lfloor\log_2 2n\rfloor$; and $v_2(\lambda_i) \ge -2\lfloor\log_2 2n\rfloor$, from $v_2\big((\mathsf{T}_{1;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,i-1])\big) \ge -\lfloor\log_2(i-1)\rfloor$ and $v_2(i(i+1)) \le \lfloor\log_2(i+1)\rfloor$. So each term of $16^n\sum_l c_l\mathcal{J}[F[\cdot,p_l]]$ has $v_2 \ge -3\lfloor\log_2 n\rfloor = -v_2(d_n^3)$, and the heads, $w_l$ times 8 times odd denominators, are 2-integral. Hence $256^n d_n^3 b_{n+1} \in \mathbb{Z}$.

*The numbers.* The receipt checks both statements for $k \le 80$. The first denominators, with their parts at 2:

| $k$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| $\operatorname{den}(b_k)$ | $2^3$ | $2^9$ | $2^9\cdot 9$ | $2^{21}\cdot 9$ | $2^{25}\cdot 1125$ | $2^{29}\cdot 125$ | $2^{32}\cdot 42875$ |
| $d_{k-1}^3$ | 1 | $2^3$ | $2^3\cdot 27$ | $2^6\cdot 27$ | $2^6\cdot 3375$ | $2^6\cdot 3375$ | $2^6\cdot 1157625$ |

*The frame.* The sequences $(a_k)$ and $(b_k)$ both solve the law of Exercise 54.8: $(a_k)$ by Exercise 54.10(i), and $b_k = a_k v - P(k)$ as a difference of two solutions. They are independent,

$(a_1,a_2)=(1,\frac{1}{16})$ and $(b_1,b_2)=(0,\frac{3}{8})$. Their frame (Proposition 38.11) $C_k=a_kb_{k+1}-a_{k+1}b_k$ has $C_1=\frac{3}{8}$ and $C_{k+1}=\frac{c_{4,0}(k)}{c_{4,2}(k)}C_k=\frac{k^3}{64(k+1)^3}C_k$, so

$$C_k=\frac{3}{8\cdot 64^{k-1}k^3}.$$

By Theorem 43.3(i), every integer $D_k$ that clears the rows of $P(k)$ and $P(k+1)$ has $D_k^2\geq|C_k|^{-1}=\frac{8}{3}\,64^{k-1}k^3$, so $\liminf_k\frac{1}{k}\log D_k\geq\log 8$. The clearing found grows like $256^k$ times $d_{k-1}^3$; its merit (Definition 43.4) is at most $-\log f(\alpha^\star)-\log 256=\log 16-\log 256=-\log 16$, so Theorem 43.5(ii) does not apply to these rows. For every clearing the frame bounds the merit by $\log 16-\log 8=\log 2$, and Corollary 43.9, with the ratio $\frac{1}{64}$, by $-\log 4+\liminf_k\frac{1}{k}\log\gcd\big(64^{k-1},\mathsf{C}_{k-1}(1)\big)$.

So the row of $P(k)$ is cleared by $256^{k-1}d_{k-1}^3$: the coefficient $a_k$ by $64^{k-1}$, the rational part $b_k$ by $256^{k-1}$ at the prime 2 and by $d_{k-1}^3$ at the odd primes. □

## 54.8 The centre of $\binom{[5]}{1}$

The list $\binom{[5]}{1}$ has the marks $1,\dots,5$ and the five singletons as its bases, every weight 1. Its centre ray at the strength one has the stride $f=\big(K(\alpha)K(\alpha^{-1})\big)^{-1}$, $K(\alpha)=\sum_{e=1}^5\alpha_e$, the walls $\kappa_{1,S}=1$ for every nonempty proper $S$, the volume of the direction $V_1=5!=120$, and the values $P(k)=\langle 1\rangle_k=P_{\binom{[5]}{1}}(k\mathbf{1};k,k)$. In this section $P_4$ denotes the values of the centre ray of $\binom{[4]}{1}$, continued in the strength, and $v=P_4(1)=7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ (Section 54.7); $\mathsf{g}(y)=\mathsf{G}_{0;(1,-1)}(1,y)$ and $\mathsf{h}(y)=\sum_{m\geq 0}y^m/m!^2$, both solutions of $\vartheta^2u=yu$, $\vartheta=y\frac{d}{dy}$ (Exercise 54.8; $\vartheta^2y^m=m^2y^m$ gives $\vartheta^2\mathsf{h}=y\,\mathsf{h}$).

**Exercise 54.12** (The law, the low-end relation and the value span)**.** (i) Compute the operators $\mathbf{R}_2,\dots,\mathbf{R}_6$ of Theorem 42.12 for $n=5$ and the law of the values $\sum_{k=0}^3 c_{5,k}(s)\,P(s+k)=0$.

(ii) Give the closings $\mathsf{C}_j(1)$ for $j\leq 6$, the symbol, its roots and their native gauges.

(iii) Write the low-end relation and the value span, and write $\Gamma(s)^2P(s)$ for $s=3,4$ in 1, $P(1)$ and $P(2)$.

*Solution.* Receipt: `ex25a_law.py`.

(i) As in Exercise 54.8, Corollary 18.24 and Proposition 8.15 give $\Gamma(s)^2P(s)=\int_0^\infty y^{s-1}\mathsf{g}(y)^5dy$. Theorem 42.12(i) at $n=5$ has the coefficients $i(6-i)=5,8,9,8,5$ in $\mathbf{R}_{i+1}=\vartheta\mathbf{R}_i-i(6-i)\,y\,\mathbf{R}_{i-1}$:

$$\begin{aligned}
\mathbf{R}_2&=\vartheta^2-5y,\\
\mathbf{R}_3&=\vartheta^3-y(13\vartheta+5),\\
\mathbf{R}_4&=\vartheta^4-y(22\vartheta^2+18\vartheta+5)+45y^2,\\
\mathbf{R}_5&=\vartheta^5-y(30\vartheta^3+40\vartheta^2+23\vartheta+5)+y^2(149\vartheta+130),\\
\mathbf{R}_6&=\vartheta^6-y(35\vartheta^4+70\vartheta^3+63\vartheta^2+28\vartheta+5)+y^2(259\vartheta^2+518\vartheta+285)-225y^3.
\end{aligned}$$

The leading coefficients $1,-35,259,-225$ are those of $(\xi^2-25\eta)(\xi^2-9\eta)(\xi^2-\eta)$. Theorem 42.12(ii) gives

$$\begin{gathered}
c_{5,0}(s)=s^4,\qquad c_{5,1}(s)=-(35s^4+70s^3+63s^2+28s+5),\\
c_{5,2}(s)=(s+1)^2(259s^2+518s+285),\qquad c_{5,3}(s)=-225(s+1)^2(s+2)^2,
\end{gathered}$$

and the law $\sum_{k=0}^{3} c_{5,k}(s)P(s+k) = 0$ identically in $s$; every $c_{5,k}$ has degree 4 and $c_{5,k}(-s-k) = c_{5,k}(s)$. At $s=1$ and $s=2$ the coefficients are $(1,-201,4248,-8100)$ and $(16,-1433,21213,-32400)$.

(ii) The closings are $\mathsf{C}_j(1) = \sum_{|v|=j} \binom{j}{v}^2 = 1, 5, 45, 545, 7885, 127905, 2241225$ for $j \le 6$, and they satisfy $\sum_{i \le \min(3,j)} c_{5,i}(-j)\,\mathsf{C}_{j-i}(1) = 0$ (Corollary 39.36; the receipt checks $j \le 40$). The coefficients of $s^4$ give the symbol

$$\chi_5(\lambda) = 1 - 35\lambda + 259\lambda^2 - 225\lambda^3 = (1-\lambda)(1-9\lambda)(1-25\lambda),$$

with the roots $\frac{1}{25}, \frac{1}{9}, 1$. By Proposition 42.9(ii) they are the native values $K(\eta)^{-2}$ of the signs: $\eta = \mathbf{1}$, with $K=5$, the native gauge of the direction; the signs with one or four marks at $-1$, $K = \pm 3$; and those with two or three marks at $-1$, $K = \pm 1$.

(iii) *The low-end relation.* By Corollary 39.38(ii), with $[s^4]c_{5,0} = 1$, $\mathsf{L}_0 = V_1 = 120$ and $\big(c_{5,1}(0), c_{5,2}(0), c_{5,3}(0)\big) = (-5, 285, -900)$,

$$120 - 5P(1) + 285P(2) - 900P(3) = 0, \qquad P(3) = \frac{2}{15} - \frac{1}{180}P(1) + \frac{19}{60}P(2).$$

*The value span.* The law has order 3 and its last coefficient does not vanish at the positive integers, so every value is a rational combination of 1, $P(1)$ and $P(2)$: the value span of the ray is $\mathbb{Q} + \mathbb{Q}\,P(1) + \mathbb{Q}\,P(2)$ (Corollary 42.15(i), $h=3$). The law at $s=1$ gives $P(4) = \big(P(1) - 201P(2) + 4248P(3)\big)/8100$, so

$$\Gamma(3)^2 P(3) = \frac{8}{15} - \frac{1}{45}P(1) + \frac{19}{15}P(2),$$
$$\Gamma(4)^2 P(4) = \frac{944}{375} - \frac{113}{1125}P(1) + \frac{1907}{375}P(2).$$

As renderings, $P(1) = 39.94558801199624569894\ldots$, $P(2) = 0.2946961074182751313903\ldots$ and $P(3) = 0.00473383839491413155016 85\ldots$ (quadrature of the share side, a check; Exercise 54.15 reads them exactly from the gauge end). □

**Exercise 54.13** (The exact pairing and the Hessian)**.** Show that for every integer $\sigma \ge 1$ the concomitant

$$B(\sigma) = \sum_{i=1}^{3} \sum_{j=1}^{i} c_{5,i}(\sigma - j)\,\mathsf{C}_{\sigma-j}(1)\,P(\sigma - j + i),$$

with the closings at negative indices read as 0, equals $-120$. Deduce $\det H$, $H$ the Hessian of $-\log f$ at $\mathbf{1}$ on the chart $\alpha_5 = 1$, and compute $H$ directly.

*Solution.* Receipt: `ex25b_pairing.py`.

*The adjoint.* The law is self-adjoint under the reflection of the strength: $c_{5,i}(-\sigma - i) = c_{5,i}(\sigma)$ for $i = 0, \ldots, 3$ (Exercise 54.12), so $c_{5,i}(\sigma - i) = c_{5,i}(-\sigma)$, and $\sum_i c_{5,i}(\sigma - i)\,\mathsf{C}_{\sigma-i}(1) = \sum_i c_{5,i}(-\sigma)\,\mathsf{C}_{\sigma-i}(1)$ is the law of the closings at $j = \sigma$, which vanishes for $\sigma \ge 0$; for $\sigma < 0$ every closing in it is 0. The closings solve the adjoint recurrence (Proposition 42.8(i)).

*The concomitant is constant.* Shifting $j$ by one in $B(\sigma+1)$,

$$B(\sigma+1) - B(\sigma) = \mathsf{C}_\sigma(1) \sum_{i=1}^{3} c_{5,i}(\sigma) P(\sigma + i) - P(\sigma) \sum_{i=1}^{3} c_{5,i}(\sigma - i)\,\mathsf{C}_{\sigma-i}(1),$$

and by the law and the adjoint recurrence the two sums are $-c_{5,0}(\sigma)P(\sigma)$ and $-c_{5,0}(\sigma)\mathsf{C}_\sigma(1)$, so the difference vanishes (the receipt checks the identity symbolically). At $\sigma=1$ only $j=1$ survives, and by the low-end relation of Exercise 54.12

$$B(1) = -5P(1) + 285P(2) - 900P(3) = -120.$$

At $\sigma=2$, with $\mathsf{C}_1(1)=5$, the five terms are $-201\cdot 5\,P(2)$, $4248\cdot 5\,P(3)$, $285\,P(2)$, $-8100\cdot 5\,P(4)$ and $-900\,P(3)$, so $B(2) = -720P(2) + 20340P(3) - 40500P(4)$; with the rows of Exercise 54.12,

$$20340P(3) = 2712 - 113P(1) + 6441P(2), \qquad 40500P(4) = 2832 - 113P(1) + 5721P(2),$$

and $B(2)=-120$, the coefficients of $P(1)$ and $P(2)$ cancelling. The receipt checks $B(\sigma)=-120$ exactly in the value span for $\sigma\le 50$. This is Corollary 42.16 at $n=5$, with $(-1)^n n! = -120$.

*The limit.* Every $c_{5,i}$ has degree $4 = n-1$, $\chi_5(0)=1$, and $\lambda_\star = \frac{1}{25}$ is a simple root. By Proposition 42.8(iii),

$$V_1 \det H = -\frac{\lambda_\star \chi_5'(\lambda_\star)}{\chi_5(0)} = \prod_{c\neq\star}\Big(1-\frac{\lambda_\star}{\lambda_c}\Big):$$

$$\chi_5'\big(\tfrac{1}{25}\big) = -25\cdot\frac{24}{25}\cdot\frac{16}{25} = -\frac{384}{25}, \qquad \Big(1-\frac{9}{25}\Big)\Big(1-\frac{1}{25}\Big) = \frac{384}{625},$$

so $120\det H = \frac{384}{625}$ and $\det H = \frac{16}{3125} = \frac{2^4}{5^5}$.

*Directly.* By Theorem 38.15 at the centre, $H = 2\big(\kappa_{\mathbf{1}}(X_e, X_{e'})\big)_{e,e'\neq 5}$, the covariances of the indicators under the weight law at $\mathbf{1}$, which picks one mark of five with the chance $\frac{1}{5}$: $\operatorname{Var} X_e = \frac{1}{5} - \frac{1}{25}$ and $\operatorname{Cov}(X_e, X_{e'}) = -\frac{1}{25}$. So

$$H = 2\Big(\frac{1}{5}I - \frac{1}{25}J\Big) = \frac{1}{25}\begin{pmatrix} 8 & -2 & -2 & -2 \\ -2 & 8 & -2 & -2 \\ -2 & -2 & 8 & -2 \\ -2 & -2 & -2 & 8 \end{pmatrix},$$

$J$ the matrix of ones, with the eigenvalues $\frac{2}{5}$ on the vectors of sum zero and $\frac{2}{25}$ on $\mathbf{1}$: $\det H = \big(\frac{2}{5}\big)^3\frac{2}{25} = \frac{16}{3125}$, as Proposition 42.9(ii) gives at $\eta=\mathbf{1}$, $(2\sigma)^{n-1}/K(\mathbf{1})^n = 2^4/5^5$.

So $B(\sigma) = -120$ at every strength, the low-end relation carried to every $\sigma$, and its limit at high strength gives $\det H = 2^4/5^5$, the value of the Hessian computed directly. □

**Exercise 54.14** (The lowered settings and the frame)**.** For $j=0,1,2$ the lowered setting with $j$ marks lowered is $\xi_{5,j}(s) = \Gamma(s)^{-2}\int_0^\infty y^{s-1}\mathsf{g}(y)^{5-j}\mathsf{h}(y)^j\,dy$ (Proposition 42.14); put $x^{(j)}(s) = \Gamma(s)^2\xi_{5,j}(s)$, so that $x^{(0)}(s) = \Gamma(s)^2P(s)$.

(i) Show that the $x^{(j)}$ satisfy the law of Exercise 54.12 in this normalization, and find their low-end coefficients and their rates.

(ii) Write the frame $\Phi(s) = \det\big[x^{(i)}(s+l)\big]_{i,l=0}^{2}$ in terms of $\Phi(1)$, and find $\Phi(1)$ from the behaviour of the $x^{(j)}$ at large $s$.

(iii) Deduce the value of $x^{(1)}(1)\,x^{(2)}(2) - x^{(1)}(2)\,x^{(2)}(1)$.

*Solution.* Receipt: `ex25c_lowered.py`.

(i) *The law.* $\mathsf{g}^{5-j}\mathsf{h}^j$ is a product of five solutions of $\vartheta^2 u = yu$, so $\mathbf{Q}_5 = \mathbf{R}_6$ annihilates it (Theorem 42.12(i)), and integration against $y^{s-1}$ turns it into the law for $x(s) = \Gamma(s)^2\xi(s)$:

$$s^6x(s) - (35s^4+70s^3+63s^2+28s+5)\,x(s+1) + (259s^2+518s+285)\,x(s+2) - 225\,x(s+3) = 0,$$

the coefficients being $q_{5,k}(-s-k)$; this is the law of Exercise 54.12 multiplied by $\Gamma(s)^2 s^2$ (Proposition 42.14(i)). The boundary terms vanish: near 0, $\mathsf{g}^{5-j}\mathsf{h}^j = O(|\log y|^{5-j})$ (Lemma 20.29(i)), and for $y \ge 1$, $\mathsf{g} \le \sqrt{\pi}\, y^{-1/4} e^{-2\sqrt{y}}$ (Lemma 20.29(ii)) and $\mathsf{h} \le e^{2\sqrt{y}}$, so the integrand decays like $e^{-2(5-2j)\sqrt{y}}$, $j < \frac{5}{2}$.

*The low-end coefficients.* Near $y = 0$, $\mathsf{g}(y) = -\log y + 2(\log\Gamma)'(1) + O(y|\log y|)$ (Lemma 20.29(i)) and $\mathsf{h}(y) = 1 + O(y)$, and $\int_0^1 y^{s-1}(-\log y)^{5-j} dy = \Gamma(6-j)\, s^{j-6}$; so $x^{(j)}(s) = \Gamma(6-j)\, s^{j-6}\big(1+O(s)\big)$ and $s^4 \xi_{5,j}(s) \to 120, 0, 0$ for $j = 0, 1, 2$, $\Gamma(s)^{-2}$ being $s^2(1+O(s))$. Read at $s \to 0$, the law gives

$$-5x^{(j)}(1) + 285x^{(j)}(2) - 225x^{(j)}(3) = -120,\ 0,\ 0 \qquad (j = 0, 1, 2):$$

the lowered settings with $j \ge 1$ have no low-end coefficient (Proposition 42.14(ii)).

*The rates.* $x^{(j)}(s+1)/\big(s^2 x^{(j)}(s)\big) \to (5-2j)^{-2} = \frac{1}{25}, \frac{1}{9}, 1$ (Proposition 42.14(iii)): the setting with $j$ marks lowered carries the native value of the signs with $j$ marks at $-1$. This also follows from the amplitudes below.

(ii) *The frame.* The end coefficients of the law in $x$ are $s^6$ and $-225$, with no zero at the integers $s \ge 1$, so Proposition 38.11 gives $\Phi(s+1) = (-1)^3 \frac{s^6}{-225}\Phi(s) = \frac{s^6}{225}\Phi(s)$ and

$$\Phi(s) = \Phi(1)\,\Gamma(s)^6\, 225^{1-s}.$$

*One mark at large $y$.* By the torus homogeneity (Theorem 2.4(iii)) $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(\lambda, \lambda)$ with $\lambda = \sqrt{y}$, and Theorem 4.3(iii) at $t = (1, -1)$, $\boldsymbol{\xi} = (1, 1)$, $\nu = \nu' = 0$ applies: $\Phi_{\boldsymbol{\xi}}(y') = e^{y'} + e^{-y'}$, $y^\star = 0$, $\Phi_{\boldsymbol{\xi}}(0) = 2$, $\sum_k w_k^\star t_k^2 = 2$, so

$$\mathsf{g}(y) = e^{-2\sqrt{y}}\Big(\frac{2\pi}{\sqrt{y}}\Big)^{\frac{1}{2}} 2^{-\frac{1}{2}}\big(1 + O(y^{-\frac{1}{2}})\big) = \sqrt{\pi}\, y^{-\frac{1}{4}} e^{-2\sqrt{y}}\big(1 + O(y^{-\frac{1}{2}})\big).$$

$\mathsf{h}(y)$ is the constant term of $e^{\sqrt{y}(z+z^{-1})}$, that is $\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{2\sqrt{y}\cos\theta} d\theta$, and Laplace's method (an outside result; it applies because $\cos\theta$ has one maximum on the circle, at $\theta = 0$, where its second derivative is $-1$) gives $\mathsf{h}(y) = e^{2\sqrt{y}}(4\pi\sqrt{y})^{-1/2}\big(1 + O(y^{-1/2})\big)$. So, with $\rho_j = 2(5-2j) = 10, 6, 2$,

$$\mathsf{g}^{5-j}\mathsf{h}^j = \pi^{\frac{5-2j}{2}} 2^{-j}\, y^{-\frac{5}{4}} e^{-\rho_j\sqrt{y}}\big(1 + O(y^{-\frac{1}{2}})\big).$$

*The amplitudes.* With $y = u^2$, $x^{(j)}(s) = 2\int_0^\infty u^{2s-1}\mathsf{g}(u^2)^{5-j}\mathsf{h}(u^2)^j\, du$; the part $u \le u_0$ is $O(u_0^{2s})$, negligible against $\Gamma(2s)\rho_j^{-2s}$, and the rest gives

$$x^{(j)}(s) = A_j\,\Gamma\big(2s - \tfrac{5}{2}\big)\,\rho_j^{-2s}\big(1 + O(s^{-1})\big), \qquad A_j = 2\pi^{\frac{5-2j}{2}} 2^{-j} \rho_j^{\frac{5}{2}},$$

$A_0 = 200\sqrt{10}\,\pi^{5/2}$, $A_1 = 36\sqrt{6}\,\pi^{3/2}$, $A_2 = 2\sqrt{2}\,\pi^{1/2}$, $\prod_j A_j = \pi^{9/2}\, 120^{5/2}$. Factoring $A_i\rho_i^{-2s}$ out of the rows and $\Gamma(2s + 2l - \frac{5}{2})$ out of the columns,

$$\Phi(s) = \prod_i A_i \rho_i^{-2s} \prod_{l=0}^{2} \Gamma\big(2s + 2l - \tfrac{5}{2}\big)\big(V + O(s^{-1})\big), \qquad V = \det\big[u_i^l\big] = \prod_{i<i'}(u_{i'} - u_i) = \frac{16}{16875},$$

$u_i = \rho_i^{-2} = \frac{1}{100}, \frac{1}{36}, \frac{1}{4}$. Theorem 4.3(iii) for $\Gamma(\lambda + a) = \lambda^{\lambda+a}\mathsf{G}_{\lambda+a;(1)}(\lambda)$, with $\Phi_\xi(y') = e^{y'} - y'$, $y^\star = 0$ and $w^\star = 1$, gives $\Gamma(\lambda + a) = \lambda^{\lambda+a} e^{-\lambda}(2\pi/\lambda)^{1/2}\big(1 + O(\lambda^{-1})\big)$, hence $\prod_l \Gamma(2s + 2l - \frac{5}{2}) = \Gamma(2s)^3 (2s)^{-3/2}\big(1 + O(s^{-1})\big)$ and $\Gamma(2s) = 2^{2s-1}\pi^{-1/2} s^{1/2}\Gamma(s)^2\big(1 + O(s^{-1})\big)$, so

$$\Phi(s) = \prod_j A_j \cdot V \cdot 2^{-\frac{9}{2}}\pi^{-\frac{3}{2}}\,\Gamma(s)^6\Big(\frac{64}{\prod_j \rho_j^2}\Big)^s\big(1 + O(s^{-1})\big), \qquad \frac{64}{\prod_j \rho_j^2} = \frac{64}{14400} = \frac{1}{225}.$$

Comparing with $\Phi(s) = 225\,\Phi(1)\,\Gamma(s)^6\,225^{-s}$,

$$\Phi(1) = \frac{\pi^{9/2}\,120^{5/2}}{225}\cdot\frac{16}{16875}\cdot\frac{1}{2^{9/2}\pi^{3/2}} = \frac{128\sqrt{15}}{16875}\,\pi^3 = \frac{128\sqrt{15}}{16875}\,\Gamma\big(\tfrac12\big)^6,$$

which is $0.91088056479820303289039 72\ldots$ to 25 places; the receipt computes the nine settings by quadrature and reproduces it, and checks the law, the low-end coefficients and the amplitudes.

(iii) *The block.* By (i), $x^{(j)}(3) = \big(-5x^{(j)}(1) + 285x^{(j)}(2) + 120\,\delta_{j0}\big)/225$, so subtracting $-\frac{5}{225}$ times the first column and $\frac{285}{225}$ times the second from the third leaves the column $(\frac{120}{225}, 0, 0)$:

$$\Phi(1) = \frac{120}{225}\Big(x^{(1)}(1)\,x^{(2)}(2) - x^{(1)}(2)\,x^{(2)}(1)\Big), \qquad x^{(1)}(1)\,x^{(2)}(2) - x^{(1)}(2)\,x^{(2)}(1) = \frac{16\sqrt{15}}{1125}\,\Gamma\big(\tfrac12\big)^6,$$

which is $1.70790105899663068666949 5\ldots$ to 24 places. The values $P(1), P(2)$ do not enter $\Phi$: their coefficients are minors that the vanishing low-end coefficients of the lowered settings remove. □

**Exercise 54.15** (The reading from the gauge end)**.** Read $P(1)$, $P(2)$ and $P(3)$ from the gauge end (Theorem 40.6, Definition 40.10): the native gauge and the constant $g(0)$; the density operator of the law and its exponents at $f^\star = \frac{1}{25}$; the germs that extend by zero; the Taylor coefficients of the germ; the values at a large strength and the law run back; and the low-end relation read on the output.

*Solution.* Receipt: `ex25d_gauge.py`.

(1) *The native gauge.* $\alpha^\star = \mathbf{1}$ (Theorem 38.14), $f^\star = f(\mathbf{1}) = \frac{1}{25}$, $\det H = \frac{16}{3125}$ (Exercise 54.13) and $(\omega_0/dy)(\mathbf{1}) = 1$, the origin being 0. By Proposition 40.3(ii), with $n = 5$,

$$g(0) = \frac{1}{\Gamma(2)\sqrt{\det H}}\Big(\frac{2\pi}{f^\star}\Big)^2 = \frac{25\sqrt5}{4}\,(50\pi)^2 = 15625\sqrt5\,\Gamma\big(\tfrac12\big)^4.$$

(2) *The operator.* The density operator of the law $\mathbf{p} = (c_{5,0}, \ldots, c_{5,3})$ is $\mathfrak{D}_\mathbf{p} = \sum_i t^i c_{5,i}(-1 - i - \theta)$, $\theta = t\frac{d}{dt}$ (Definition 40.4). With $\theta^m = \sum_j S(m,j)\,t^j\partial_t^j$, $S(m,j)$ the number of partitions of a set of $m$ elements into $j$ nonempty parts, $\mathfrak{D}_\mathbf{p} = \sum_{j=0}^4 a_j(t)\partial_t^j$ with

$$\begin{aligned}
a_0 &= 1 - 201t + 4248t^2 - 8100t^3,\\
a_1 &= -t\,(24300t^3 - 16965t^2 + 1232t - 15),\\
a_2 &= -t^2(16650t^3 - 14271t^2 + 1358t - 25),\\
a_3 &= -2t^3(1800t^3 - 1813t^2 + 210t - 5),\\
a_4 &= -t^4(t-1)(9t-1)(25t-1) = t^4\chi_5(t).
\end{aligned}$$

In $s = \frac{1}{25} - t$, $\mathfrak{D}_\mathbf{p} = \sum_j b_j(s)\partial_s^j$ with $b_{4,0} = 0$ and $b_{4,1} = -(f^\star)^4\chi_5'(f^\star) = \frac{384}{25^5}$. The second symbol is $\chi_1(\lambda) = -70\lambda + 1036\lambda^2 - 1350\lambda^3$, and Theorem 40.6(i) gives

$$\gamma = -1 + \frac{\chi_1(f^\star)}{f^\star\chi_5'(f^\star)} = -1 + \frac{-768/625}{-384/625} = 1 = \frac{n-3}{2}, \qquad R_0(x) = \frac{384}{25^5}\,x(x-1)^2(x-2)\,:$$

the exponents at $f^\star$ are $0, 1, 2$ and $\gamma = 1$, a double root. The further polynomials $R_m(x) = \sum_j b_{j,m-3+j}[x]_j$ are

$$R_1 = -\tfrac{8}{25^4}\,x(x-1)(163x^2 - 67x + 72), \qquad R_2 = \tfrac{1}{25^3}\,x(1385x^3 + 1840x^2 + 1631x + 576),$$

$$R_3 = -\tfrac{1}{625}(180x^4 + 1290x^3 + 2129x^2 + 1619x + 476),$$
$$R_4 = -\tfrac{1}{25}(490x^4 + 2530x^3 + 5454x^2 + 5789x + 2499),$$
$$R_5 = (x+2)^2(196x^2 + 802x + 819), \qquad R_6 = 225(x+2)^2(x+3)^2,$$

and $R_m = 0$ for $m \geq 7$.

(3) *The line.* As $\gamma = 1 \in \{0, 1, 2\}$, the germ is $\varrho_f = \sum_r u_r s^r$ with $u_0 = 0$, $u_1 = g(0)$ (Theorem 40.6(iv)). The germs that extend by zero are fixed by the conditions $\mathsf{p}_r(u) = 0$, $r \leq 2$, the point masses of $\mathfrak{D}_{\mathbf{p}}$ applied to the germ cut at $f^\star$:

$$\mathsf{p}_0(u) = -\frac{384}{25^5}u_0, \qquad \mathsf{p}_1(u) = -\frac{576}{25^4}u_0, \qquad \mathsf{p}_2(u) = \frac{576}{25^3}u_0 - \frac{1344}{25^4}u_1 + \frac{768}{25^5}u_2,$$

the coefficient of $u_r$ in $\mathsf{p}_r$ being $r!\, b_{4,1}(r-\gamma)$. They leave one line, $u_0 = 0$ and $u_2 = \frac{1344 \cdot 25}{768}u_1 = \frac{175}{4}u_1$ (Theorem 40.6(ii)): the germ vanishes to the first order, as the exponent $\gamma = 1$ says, and the receipt also derives the three conditions from the distributional derivatives of the cut germ.

(4) *The germ.* For $N \geq 3$, $R_0(N)u_N = -\sum_{m=1}^{\min(N,6)} R_m(N-m)\, u_{N-m}$. At $N = 3$, $R_0(3) = \frac{4608}{25^5}$, $R_1(2) = -\frac{9440}{25^4}$, $R_2(1) = \frac{5432}{25^3}$, so

$$u_3 = -\frac{25^5}{4608}\Big(-\frac{9440}{25^4}\cdot\frac{175}{4} + \frac{5432}{25^3}\Big)u_1 = \frac{48125}{32}u_1, \qquad u_4 = \frac{35703125}{768}u_1,$$

and the receipt continues to $u_{400}$, all rational multiples of $u_1$, and checks that the series solves $\mathfrak{D}_{\mathbf{p}}\varrho = 0$ to the order kept.

(5) *Large strengths.* The singular points of $\mathfrak{D}_{\mathbf{p}}$ in $t$ are $0, \frac{1}{25}, \frac{1}{9}, 1$, so the series converges for $|s| < \frac{1}{25}$, on $0 < t \leq f^\star$, and by Proposition 40.9(ii) at $j = 0$

$$P(K) = \int_0^{f^\star} t^K \varrho_f(t)\, dt = g(0)\sum_{r\geq 1}\frac{u_r}{u_1}\,(f^\star)^{K+r+1}\,\mathrm{B}(K+1, r+1),$$

the terms falling like $\prod_{i\leq r}\frac{i}{K+1+i}$; at $K = 150$ the term $r = 400$ is $3 \cdot 10^{-136}$ of the sum. These are the values at $K = 150, 151, 152$.

(6) *The run back.* The law holds at every positive integer (Proposition 40.5), and $c_{5,0}(k) = k^4 \neq 0$, so $P(k) = -\big(c_{5,1}(k)P(k+1) + c_{5,2}(k)P(k+2) + c_{5,3}(k)P(k+3)\big)/k^4$ from $k = 149$ down to $k = 1$. In 150-digit arithmetic, with the renderings to 60 places,

$$P(1) = 39.945588011996245698944028498423995891849782924769859572366 6\ldots,$$
$$P(2) = 0.294696107418275131390303208655999730074422376946254727936899\ldots,$$
$$P(3) = 0.004733833949141315501684746638711048457734958673148110666870 36\ldots,$$

and the reading from $K = 250$ agrees to $10^{-133}$.

(7) *Checks.* The low-end relation read on the output (Definition 40.10 step (7)) gives $5P(1) - 285P(2) + 900P(3) = 120.000\ldots$ to all the places computed: the output returns $V_1 = 120$, which the reading did not use. The values agree with the quadrature of Exercise 54.12 to its 22 places.

So $P_{\binom{[5]}{1}}(\mathbf{1}; 1, 1) = 39.945588011996245698944028498 42\ldots$, read from the law, the native gauge $\mathbf{1}$ and $\det H$ alone, with the germ of the density $\varrho_f$ exact over $\mathbb{Q}$ up to the factor $15625\sqrt{5}\,\Gamma(\frac{1}{2})^4$. $\square$

**Exercise 54.16** (The value as a Trailmoment of the slope input of four marks)**.** Let $\tilde{p}_4 = -P_4'$ be the slope input of the centre ray of $\binom{[4]}{1}$ (Definition 42.1). Show that $P(1) = (\mathsf{T}_{0;(1)}[x]\,\tilde{p}_4\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \sum_{x\geq 1}\tilde{p}_4(x)$, find the recurrence of the slope input and its relation at the cut, and write $\tilde{p}_4(m)$, $m \leq 5$, in 1, $v$ and $\tilde{p}_4(1)$.

*Solution.* Receipt: `ex25e_slope.py`.

*The mark added.* By Proposition 42.11(i), merging four marks, $\Gamma(s)^2 P(s) = \frac{1}{2\pi i}\int_{(c)} \Gamma(d)^2\Gamma(s-d)^2 P_4(d)\,dd$; at $s=1$ the kernel is $\Gamma_4^{\,0}\big[d,\ d,\ 1-d,\ 1-d\big]$, whose right closing reads the derivative of the rest at the positive integers (Lemma 5.16(i)). Proposition 42.11(ii) at $s=1$, with $W_1 = 1$, gives

$$P(1) = (\mathsf{T}_{0;(1)}[x]\,\tilde{p}_4\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = \sum_{x\geq 1}\tilde{p}_4(x)\,:$$

the centre of five marks at the strength one is the slope input of four marks summed over the positive integers.

*The recurrence.* By Proposition 42.2(ii), $L\tilde{p}_4 = L'P_4$ for the law $L$ of Exercise 54.8:

$$\sum_{i=0}^{2} c_{4,i}(\sigma)\,\tilde{p}_4(\sigma+i) = \sum_{i=0}^{2} c'_{4,i}(\sigma)\,P_4(\sigma+i),$$
$$c'_{4,0} = -3\sigma^2, \qquad c'_{4,1} = 6(10\sigma^2+10\sigma+3), \qquad c'_{4,2} = -192(\sigma+1)^2.$$

*The cut.* Proposition 42.6 at $x_0 = 0$, $j=0$, with $c_{4,0} = -\sigma^3$, so $\gamma_3 = -1$ and $\gamma_4 = 0$, and with $\ell_1 = 0$, every weight being 1:

$$\sum_{i=1}^{2}\big(c_{4,i}(0)\,\tilde{p}_4(i) - c'_{4,i}(0)\,P_4(i)\big) = \gamma_3\ell_1 + \gamma_4 V_1 = 0, \qquad 4\tilde{p}_4(1) - 64\tilde{p}_4(2) - 18P_4(1) + 192P_4(2) = 0.$$

With $P_4(1) = v$ and $P_4(2) = \frac{1}{16}v - \frac{3}{8}$ (Exercise 54.8),

$$\tilde{p}_4(2) = \frac{1}{16}\tilde{p}_4(1) - \frac{3}{32}v - \frac{9}{8}.$$

*The rows.* The recurrence, whose coefficient $c_{4,2}(m) = -64(m+1)^3$ does not vanish, gives every slope from the first two:

$$\tilde{p}_4(3) = \tfrac{7}{1024}\tilde{p}_4(1) - \tfrac{57}{4096}v - \tfrac{279}{2048},$$
$$\tilde{p}_4(4) = \tfrac{1}{1024}\tilde{p}_4(1) - \tfrac{73}{32768}v - \tfrac{2819}{147456},$$
$$\tilde{p}_4(5) = \tfrac{679}{4194304}\tilde{p}_4(1) - \tfrac{13081}{33554432}v - \tfrac{463499}{150994944},$$

the coefficient of $\tilde{p}_4(1)$ being $a_m$ of Exercise 54.8, since the recurrence has the law as its homogeneous part. So every $\tilde{p}_4(m)$ lies in $\mathbb{Q} + \mathbb{Q}\,v + \mathbb{Q}\,\tilde{p}_4(1)$.

*The value.* The slopes have the root rate $\frac{1}{16}$ along the positive integers (Proposition 42.2(iii)). As renderings, $\tilde{p}_4(1) = -P_4'(1) = 39.381582123122165624939809345 1\ldots$ by quadrature of $-\int_0^\infty \log y\,\mathsf{g}(y)^4 dy + 2(\log\Gamma)'(1)\,v$, then $\tilde{p}_4(2) = 0.547499039997\ldots$, $\tilde{p}_4(3) = 0.0158846667692\ldots$, $\tilde{p}_4(4) = 0.00059555377139\ldots$, $\tilde{p}_4(5) = 0.0000253984887\ldots$ from the rows; the sum over $m \leq 70$ is $39.945588011996245694894402849842\ldots$, the value of Exercise 54.15 to all 36 places computed, and the steps $m \geq 2$ add $0.564005888874080074\ldots$ to the first slope. The receipt also checks Proposition 42.11(iii) at $n=4$, $5\,(\mathsf{T}_{0;(1)}[x]\,P_4\mathbb{I}_{\mathbb{Z}})((0,\infty)) - 5\,(\mathsf{T}_{0;(1)}[x]\,(x-1)\,\tilde{p}_4\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (\mathsf{T}_{0;(1)}[x]\,\tilde{p}_4\mathbb{I}_{\mathbb{Z}})((0,\infty))$.

So $P_{\binom{[5]}{1}}(\mathbf{1};1,1) = (\mathsf{T}_{0;(1)}[x]\,\tilde{p}_4\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, a Trailmoment at the index zero of the slope input of four marks, whose density lies in $\mathbb{Q} + \mathbb{Q}\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) + \mathbb{Q}\,\tilde{p}_4(1)$ at every point. □

## 54.9 $W_3$ at $(\mathbf{1};0,2)$

The wheel $W_3$ (Proposition 39.65) has the hub, the rim vertices $1,2,3$, the spokes $s_1,s_2,s_3$ and the rims $r_1,r_2,r_3$, $r_i$ joining $i$ to $i+1$; its bases are the 16 spanning trees, the triples of marks other than the four triangles $\{r_1,r_2,r_3\}$, $\{s_1,s_2,r_1\}$, $\{s_2,s_3,r_2\}$, $\{s_3,s_1,r_3\}$, its rank is 3, its cobases are the complements of the bases and every weight is 1. The ray of this section has the origin 0 and the direction $(\mathbf{1};0,2)$: masses 1, exponent 0 for $K$ and 2 for $\Psi$, on the balance wall, $m_{\mathcal{E}} = 6 = 0\cdot 3 + 2\cdot 3$. Its stride is $f = \alpha^{\mathbf{1}}/\Psi^2$, only $\Psi$ being active; its walls are $\kappa_{1,S} = |S| - 2\big(|S| - r(S)\big) = 2r(S) - |S|$, with the values $1,2,3$ on the nonempty proper $S$; and its volume of the direction is $V_1 = \sum_\pi \prod_{l=2}^{6} \kappa_{1,U_l}^{-1} = 84$. At $\mathbf{1}$ every mark lies in 8 of the 16 bases, so every dual occupation is $\frac{1}{2}$, the defects $1 - 2\cdot\frac{1}{2}$ vanish, and $\alpha^\star = \mathbf{1}$ with $f^\star = \Psi(\mathbf{1})^{-2} = \frac{1}{256}$. The values are $P(k) = \langle 1\rangle_k = P_{W_3}(k\mathbf{1};0,2k)$. Since $K$ has the exponent 0, the constant terms of the powers of $f^{-1} = \Psi^2/\alpha^{\mathbf{1}}$ are written explicitly: $\mathrm{CT}(f^{-j}) = [\alpha^{j\mathbf{1}}]\,\Psi^{2j}$.

**Exercise 54.17** (The box, the law and the spectrum)**.** (i) Read the box $\mathfrak{B}_2$ with one unit of $\Psi$ (Definition 39.15) modulo the automorphisms of $W_3$, find its free orbits, and show that it is accepted at $\mathfrak{s} = 1$ with the spanning set $b_1 = 1$, $b_2 = b = \frac{1}{4}\sum_T \alpha^T/\alpha^{\mathcal{E}\setminus T}$, the sum over the four triangles $T$; give its matrix $\mathbf{B}(k)$.

(ii) Derive the law of [1] from the rows, its symbol, and show that $b_1, b_2$ are masters.

(iii) Show that $\mathbf{B}$ is not regular at $k = \infty$, and repair it by a change of masters.

*Solution.* Receipt: `ex26a_box.py` (with `w3common.py`).

(i) *Symmetries.* The 24 permutations of the four vertices of the graph induce permutations of the marks that carry the spanning trees onto themselves and keep the masses and the exponents: maps $\phi_\beta$ of the first kind of Proposition 39.11(ii), forming a group $G$. The level bounds are $\bar a = 0$, $K$ being inactive, and $\bar b = 1$; $G$ keeps $\Psi$, so it keeps the box.

*The box.* $\mathfrak{B}_2(0,1)$ holds the symbols of wall degree at most 2 at the levels $(0,0)$ and $(0,1)$ (Definition 39.15(iv)): 1167 Laurent symbols $\alpha^u$, $u_{\mathcal{E}} = 0$, which are the stems, and 2848 symbols at the level $(0,1)$, $u_{\mathcal{E}} = 3$. Its relations are the letter identities of $\Psi$, $\alpha^u - \sum_T (u + \mathbf{1}_{\mathcal{E}\setminus T}; 0, 1)$, of the 1167 Laurent symbols, and the mark relations of the stems at the six marks (Definition 39.15(iii)), here, with the origin 0, $m_{1,e} = 1$, $s_1 = 0$ and $t_1 = 2$,

$$\mathrm{MR}_e(\alpha^u) = (k + u_e)\,\alpha^u - 2k \sum_{T \not\ni e} (u + \mathbf{1}_{\mathcal{E}\setminus T}; 0, 1),$$

7002 of them. Modulo $G$ (Definition 39.20(i)) there are 66 Laurent orbits, of wall degrees 0 (one orbit), 1 (eight) and 2 (fifty-seven), and 145 orbits at the level $(0,1)$: 211 orbit averages, on which the averaged relations, 389 distinct ones, have the rank 200 over $\mathbb{Q}(k)$.

*The reduction.* The ranking puts the orbits at the level $(0,1)$ highest, then the Laurent orbits by decreasing wall degree, with the orbit of $\alpha^{r_1r_2r_3}/\alpha^{s_1s_2s_3}$ last among those of wall degree 1 (Definition 39.20(ii)). The free orbits are 11: nine at the level $(0,1)$ and the Laurent orbits of 1 and of $\alpha^{r_1r_2r_3}/\alpha^{s_1s_2s_3}$. The latter orbit has four members, $\alpha^T/\alpha^{\mathcal{E}\setminus T}$ for the four triangles $T$, each with its complement the star at the opposite vertex; so the spanning set at $\mathfrak{s} = 1$ is $b_1 = 1$, $b_2 = b$ (Definition 39.20(iii)). The targets $f^{-1}b_1 = \Psi^2/\alpha^{\mathbf{1}}$ and $f^{-1}b_2$ involve 9 and 22 orbits.

*Acceptance.* The reductions of the two targets were computed by elimination at integer strengths and reconstructed over $\mathbb{Q}(k)$ (Proposition 41.10(i),(ii)); with the common denominator $Q_c = (k-1)^2(k-2)^5(k^2-k+1)$ every $Q_c\mathbf{B}_{il}$ is a polynomial of degree at most 10:

$$\mathbf{B}_{11} = -\frac{2(2k-1)(k^2-4k+2)(7k^2-8k+3)}{(k-1)^2(k^2-k+1)}, \qquad \mathbf{B}_{12} = \frac{14k^2(2k-1)}{k^2-k+1},$$

$$\mathbf{B}_{21} = -\frac{2k(k-3)(k-1)(k+1)(2k-1)(7k^2-14k+8)}{(k-2)^4(k^2-k+1)},$$
$$\mathbf{B}_{22} = \frac{2k^2(k-1)(2k-1)(k^2-2)(7k^2-20k+15)}{(k-2)^5(k^2-k+1)}.$$

The proof that these are the reductions over $\mathbb{Q}(k)$: for each target $x_i$ the row $x_i - \sum_l \mathbf{B}_{il} b_l$, times $Q_c$, stacked under the relation matrix $A(k)$, whose entries are affine in $k$ (Proposition 41.9(iii)), has the rank 200 at 230 integer strengths; its minors of order 201 are polynomials of degree at most 219, so they vanish identically, the row lies in $\mathcal{R}^G_{\mathfrak{B}}$ over $\mathbb{Q}(k)$, and $\sum_l \mathbf{B}_{il} b_l$, a combination of free orbits, is the reduction (Definition 39.20(ii)). So the box is accepted at $\mathfrak{s} = 1$ with the matrix $\mathbf{B}$, the classes of $b_1, b_2$ span a submodule $\mathfrak{N} \ni [1]$, there is a certificate supported on the stems, and $\langle b_i \rangle_{k-1} = \sum_l \mathbf{B}_{il}(k) \langle b_l \rangle_k$ at every strength (Theorem 39.21(i)–(iii)).

(ii) *The law.* With $w_0 = (1, 0)$ and $w_i(k) = w_{i-1}(k-1)\mathbf{B}(k)$, the first dependence is $q_0 w_0 + q_1 w_1 + q_2 w_2 = 0$ with

$$q_0 = 16(k-1)(2k-3)(2k-1)(4k-5)(4k-3), \qquad q_2 = (k-2)^5,$$
$$q_1 = -2(2k-3)(65k^4 - 390k^3 + 885k^2 - 900k + 346),$$

and $p_j(k) = q_{2-j}(k+2)$ (Proposition 39.6(ii), Theorem 39.21(iv)) gives

$$k^5 P(k) - 2(2k+1)(65k^4 + 130k^3 + 105k^2 + 40k + 6)\, P(k+1) + 16(k+1)(2k+1)(2k+3)(4k+3)(4k+5)\, P(k+2) = 0$$

at every strength (Corollary 39.38(i)). Its symbol is $1 - 260\lambda + 1024\lambda^2 = (1 - 4\lambda)(1 - 256\lambda)$, and at both roots the exponent $-\chi_1(\lambda)/(\lambda\chi'(\lambda))$, with the second symbol $\chi_1(\lambda) = 10\lambda(512\lambda - 65)$, is $-\frac{5}{2} = -\frac{n-1}{2}$ (Proposition 40.7). The constant terms $\mathrm{CT}(f^{-j}) = 1, 12, 756, 78960, 10451700, 1587790512, 263964176784$ $(j \le 6)$ satisfy $\sum_i p_i(-j)\,\mathrm{CT}(f^{i-j}) = 0$ (Corollary 39.38(i)), a check.

*Masters.* Two native gauges: $\mathbf{1}$, with $\Psi = 16$, the native value $\frac{1}{256}$ and the determinant $\frac{3}{256}$ of the Hessian of $-\log f$ on the chart; and $\alpha_c = (1, \omega, \omega^2, \omega^2, 1, \omega)$ on $(s_1, s_2, s_3, r_1, r_2, r_3)$, $\omega = e^{2\pi i/3}$, at which every triangle and every star has the product 1, $\Psi(\alpha_c) = -2$, every dual occupation is $\frac{1}{2}$ (so $c^{(1)} = 0$), the native value is $\prod_e \alpha_{c,e}/\Psi^2 = \frac{1}{4}$, and the Hessian has the determinant $-\frac{3}{2}$. Both are nondegenerate with distinct native values, so by Corollary 38.7 at $g = 1$ every recurrence of $\langle 1 \rangle$ that follows from the relations has order at least 2: the cyclic submodule of [1], inside $\mathfrak{N}$ of dimension at most 2, has dimension 2. So $b_1, b_2$ are masters, $\mathbf{B} = \mathbf{B_b}$, and the relation above is the law of [1].

(iii) *The spectrum.* $b(\alpha_c) = 1 = b(\mathbf{1})$, so the vectors $\beta_c = (1, 1)$ of the two native gauges are proportional with distinct native values, and $\mathbf{B}$ is not regular at $k = \infty$ (Proposition 39.42(ii)); indeed $\mathbf{B}(k)/k \to \begin{bmatrix} -28 & 28 \\ -28 & 28 \end{bmatrix}$. The change $T(k) = \begin{bmatrix} 1 & 0 \\ -k & k \end{bmatrix}$, to the elements [1] and $k([b] - [1])$, gives $\tilde{\mathbf{B}}(k) = T(k-1)\mathbf{B}(k)T(k)^{-1}$ with

$$\tilde{\mathbf{B}}(\infty) = \begin{bmatrix} 88 & 28 \\ 504 & 172 \end{bmatrix}, \qquad \det\big(\lambda - \tilde{\mathbf{B}}(\infty)\big) = (\lambda - 256)(\lambda - 4),$$

the eigenvectors $(\frac{1}{6}, 1)$ and $(-\frac{1}{3}, 1)$: the reciprocals of the native values $\frac{1}{256}$ and $\frac{1}{4}$ (Proposition 39.42(iii)). As $T(k-1)T(k)^{-1} = \mathrm{diag}\big(1, \frac{k-1}{k}\big) \to I$, the eigenvalues of $\mathbf{B}(k)$ tend to 256 and 4: $\mathrm{tr}\,\mathbf{B}(k) \to 260$ and

$$\det \mathbf{B}(k) = \frac{16k^2(2k-1)^2(4k-5)(4k-3)(k^2-3k+3)}{(k-2)^5(k-1)(k^2-k+1)} \to 1024.$$

So the box gives the masters $1, b$, the matrix $\mathbf{B_b}$ above and the law of [1] of symbol $(1-4\lambda)(1-256\lambda)$; $\mathbf{B_b}$ grows like $k$, and the masters 1 and $k(b-1)$ repair it. □

**Exercise 54.18** (The low-end relation, the brackets and the value)**.** From the law of Exercise 54.17: write the low-end relation; show $7 < P(1) < \frac{64}{7}$; give the brackets of the two conditions $P(K) > 0$ and $P(K) < f^\star P(K-1)$ at every $K$, their widths and their rate; and give $P(1)$.

*Solution.* Receipt: `ex26b_bracket.py`.

*The low-end relation.* By Corollary 39.38(ii), with $[k^5]p_0 = 1$, $\mathsf{L}_0 = V_1 = 84$ (Definition 39.37, Proposition 39.35(ii), which allows the exponent 0 of $K$), $p_1(0) = -12$ and $p_2(0) = 720$:

$$84 - 12P(1) + 720P(2) = 0, \qquad P(2) = \frac{P(1) - 7}{60},$$

and every value lies in $\mathbb{Q} + \mathbb{Q}\,P(1)$, $p_2$ having no zero at $k \geq 0$.

*The first bracket.* Proposition 41.7(ii), with $a = 1$: $-aV_1/p_1(0) = 7$ and $-aV_1/\big(p_1(0) + f^\star p_2(0)\big) = -84/(-12 + \frac{720}{256}) = -84/(-\frac{147}{16}) = \frac{64}{7}$, both positive, so $7 < P(1) < \frac{64}{7}$.

*The brackets.* Let $c$ and $r$ solve the law with $(c(1), c(2)) = (1, \frac{1}{60})$, of low-end coefficient 0, and $(r(1), r(2)) = (0, -\frac{7}{60})$, of low-end coefficient 84; then $P(K) = c(K)P(1) + r(K)$ (Proposition 43.6(ii)):

| $K$ | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| $c(K)$ | 1 | $\frac{1}{60}$ | $\frac{1}{900}$ | $\frac{47}{400400}$ | $\frac{79}{5105100}$ | $\frac{2273}{977728752}$ |
| $r(K)$ | 0 | $-\frac{7}{60}$ | $-\frac{173}{21600}$ | $-\frac{73219}{86486400}$ | $-\frac{140653}{1260230400}$ | $-\frac{3541028833}{211189410432000}$ |

Positivity at $K$ (Proposition 41.4(i) with $p = 1$) and at $K-1$ with $p = f^\star - t$, where $c(K) > 0$ and $256c(K) - c(K-1) > 0$ (true for $2 \leq K \leq 90$), give

$$a_K = -\frac{r(K)}{c(K)} < P(1) < b_K = \frac{r(K-1) - 256\,r(K)}{256\,c(K) - c(K-1)};$$

$K = 2$ is the first bracket. The widths are 2.143, $1.297 \cdot 10^{-2}$, $1.513 \cdot 10^{-4}$, $2.042 \cdot 10^{-6}$ at $K = 2, \ldots, 5$, $3.50 \cdot 10^{-19}$ at $K = 12$ and $1.161 \cdot 10^{-33}$ at $K = 20$; the brackets are nested, and the ratio of consecutive widths, 0.01524 at $K = 10$ and 0.01562 at $K = 80$, tends to $\frac{1}{64} = f^\star/\lambda'$, $\lambda' = \frac{1}{4}$ (Theorem 41.3(iv)). At $K = 12$

$$\frac{12531162726888825592963 9}{17374611099178960704000} < P(1) < \frac{29090775086028604309225 7947}{40334717113590600816192000}.$$

*The value.* The roots $\frac{1}{256}$ and $\frac{1}{4}$ are distinct, the second outside the disc $|\lambda| \leq f^\star$, and $p_0, p_2$ have the largest degree 5 and no zero at $k \geq 1$: by Theorem 41.3(iii) the values are the minimal solution, and by (iv), with $x^{(0)} = r$ and $x^{(1)} = c$, the approximants are $M^{(K)} = a_K$. At $K = 90$, of width $4 \cdot 10^{-160}$, as a rendering,

$$P(1) = 7.21234141895756571239842896906869994458991775404299329075362\ldots,$$

and $P(2) = (P(1) - 7)/60$. The continued fraction of Proposition 41.7(iii), $\rho_1 = \mathsf{M}_1 \circ \cdots \circ \mathsf{M}_K(0)$ and $P(1) = -aV_1/(p_1(0) + p_2(0)\rho_1)$, gives the same number, to 20 places at depth 10 and to 39 at depth 40. □

## 54.10 Positivity against the gauge end

The centre ray of $\binom{[3]}{1}$ at the strength one has the stride $\big(K(\alpha)K(\alpha^{-1})\big)^{-1}$, $K=\alpha_1+\alpha_2+\alpha_3$, $V_1=3!=6$, and the values $P(k)=P_{\binom{[3]}{1}}(k\mathbf{1};k,k)$; $\ell=P(1)$. By Theorem 42.12 at $n=3$ its law is $s^2P(s)-(10s^2+10s+3)P(s+1)+9(s+1)^2P(s+2)=0$, of symbol $(1-\lambda)(1-9\lambda)$ with the roots $f^\star=\frac{1}{9}$ and 1 (Proposition 42.9), and the low-end relation (Corollary 39.38(ii)) is $6-3\ell+9P(2)=0$. So $P(K)=c(K)\,\ell+r(K)$ with $(c(1),c(2))=(1,\frac{1}{3})$ and $(r(1),r(2))=(0,-\frac{2}{3})$; the law is reflexive with the ratio $\frac{1}{9}$, and $c(K)=9^{1-K}\mathsf{C}_{K-1}(1)$ (Theorem 43.8(ii)), $\mathsf{C}_j(1)=1,3,15,93,639,4653,\dots$.

**Exercise 54.19** (All positivity conditions on the centre of $\binom{[3]}{1}$)**.** (i) Read $\ell$ from the gauge end.

(ii) For $N\le 22$ compare the bracket of the two conditions $P(K)>0$, $P(K)<\frac{1}{9}P(K-1)$ at $K=N+1$ with the bracket of all positivity conditions of degree at most $N$ at the strength 1 (Proposition 41.4), both read on $P(1),\dots,P(N+1)$.

(iii) Find the rate of the second (Theorem 41.5).

*Solution.* Receipt: `ex27_positivity.py`.

(i) *The gauge end.* $n=3$, $f^\star=\frac{1}{9}$, $\det H=(2\sigma)^2/K(\mathbf{1})^3=\frac{4}{27}$ (Proposition 42.9(ii)), so $g(0)=\frac{2\pi}{f^\star\sqrt{\det H}}=27\sqrt{3}\,\Gamma(\frac{1}{2})^2$ (Proposition 40.3(ii)). The density operator (Definition 40.4) is $a_0+a_1\partial_t+a_2\partial_t^2$ with $a_0=1-23t+36t^2$, $a_1=t(3-40t+45t^2)$, $a_2=t^2(t-1)(9t-1)$. At $\frac{1}{9}$, $b_{2,1}=\frac{8}{81}$, $\gamma=0=\frac{n-3}{2}$, $R_0=\frac{8}{81}x^2$, $R_1=-\frac{1}{9}(15x^2+23x+10)$, $R_2=(2x+3)(3x+5)$, $R_3=9(x+2)^2$: the exponents are 0 and $\gamma=0$, and the line is the holomorphic germ, $u_0=g(0)$ and $R_0(N)u_N=-\sum_{m=1}^{3}R_m(N-m)u_{N-m}$ (Theorem 40.6(iv)), so $u_r/u_0=1,\frac{45}{4},\frac{3645}{32},\frac{70713}{64},\frac{21500397}{2048}$ for $r\le 4$. The Beta sums at $K=200,201$ with 500 terms and the law run back give, as a rendering,

$$\ell=2.34390723868945889060156228887227706909540300963585626066630\dots;$$

the reading from $K=300$ agrees to $10^{-129}$, and $3P(1)-9P(2)=6.000\dots$ returns $V_1$.

(ii) *The two conditions.* At $K=N+1$, $a_K=-r(K)/c(K)<\ell<b_K=\big(r(K-1)-9r(K)\big)/\big(9c(K)-c(K-1)\big)$: at $K=2$, $2<\ell<3$; at $K=3$, with $c(3)=\frac{5}{27}$ and $r(3)=-\frac{23}{54}$, $\frac{23}{10}<\ell<\frac{19}{8}$.

*All conditions.* By Proposition 41.4(ii) the conditions of degree at most $N$ at the strength 1 are the positive semidefiniteness of the Hankel matrices $[P(1+i+i')]$, $[P(2+i+i')]$, $[\frac{1}{9}P(1+i+i')-P(2+i+i')]$ and $[\frac{1}{9}P(2+i+i')-P(3+i+i')]$ of the sizes allowed by $N$. Each is affine in $\ell$; the set where one is positive semidefinite is an interval, its least eigenvalue being concave in $\ell$, whose ends are the roots of its determinant nearest to a point where it is positive definite, such as the gauge-end $\ell$. The determinants are computed exactly as polynomials in $\ell$ and their real roots isolated in ball arithmetic. At $N=1$ the conditions are $\frac{1}{3}\ell-\frac{2}{3}\ge 0$ and $\frac{2}{3}-\frac{2}{9}\ell\ge 0$: $2\le\ell\le 3$. At $N=2$ the binding ones are

$$\det\begin{bmatrix}P(1)&P(2)\\P(2)&P(3)\end{bmatrix}=\frac{4\ell^2+\ell-24}{54}\ge 0,\qquad \frac{1}{9}P(2)-P(3)=\frac{19}{54}-\frac{4}{27}\ell\ge 0,$$

so $\frac{\sqrt{385}-1}{8} \le \ell \le \frac{19}{8}$, of width $\frac{20-\sqrt{385}}{8} = 0.0473\ldots$. The widths:

| $N$ | two conditions | all conditions |
|---|---|---|
| 1 | 1 | 1 |
| 2 | $7.50 \cdot 10^{-2}$ | $4.73 \cdot 10^{-2}$ |
| 4 | $7.84 \cdot 10^{-4}$ | $6.90 \cdot 10^{-5}$ |
| 8 | $1.10 \cdot 10^{-7}$ | $9.52 \cdot 10^{-11}$ |
| 12 | $1.64 \cdot 10^{-11}$ | $1.04 \cdot 10^{-16}$ |
| 16 | $2.46 \cdot 10^{-15}$ | $1.03 \cdot 10^{-22}$ |
| 22 | $4.58 \cdot 10^{-21}$ | $9.10 \cdot 10^{-32}$ |

and the ratio of the widths of all conditions at $N$ and $N-1$ is 0.0382, 0.0335, 0.0320, 0.0314, 0.0308 at $N = 4, 8, 12, 16, 22$. Every bracket contains the gauge-end $\ell$.

(iii) *The rate.* The free direction is $u = c$. By Proposition 40.12(i), $\mathsf{C}_j(1) = \int t^{-j} \nu_f^{\mathbb{T}}(dt)$, the torus law (Definition 40.11) being carried by $[\frac{1}{9}, \infty)$; with $\tau = 1/(9t)$, $c(K) = \int_0^1 \tau^K \nu(\tau)\, d\tau$, $\nu(\tau)$ being $\tau^{-1}$ times the density of the image of $\nu_f^{\mathbb{T}}$, and by Proposition 40.12(iii) $\nu(\tau) \to \frac{1}{9} \cdot \frac{g(0)}{(2\pi)^2} = \frac{3\sqrt{3}}{4\pi} \neq 0$ as $\tau \uparrow 1$: the exponent $a = 0$ of Theorem 41.5 at $\lambda = 1$. With $w = 2\lambda/f^\star - 1 = 17$ the bracket of degree $N$ has length at most $C\, N\, e^{-N g(1)}$,

$$e^{-g(1)} = \frac{1}{17 + \sqrt{288}} = 17 - 12\sqrt{2} = 0.02943\ldots,$$

against the factor $\frac{1}{9}$ per step of the two conditions (Theorem 41.3(iv)): $-\log_{10}(17 - 12\sqrt{2}) = 1.531$ digits of $\ell$ per value against $\log_{10} 9 = 0.954$. The ratios of the table approach $17 - 12\sqrt{2}$ from above, as the factor $N$ allows; with the nine values $P(1), \ldots, P(9)$ all conditions give $\ell$ to $10^{-10}$, the two conditions to $10^{-7}$. □

## 54.11 What the closings see

**Exercise 54.20** (What the closings see at small centres)**.** Take the following lists, with unit weights, on the marks $[n] = \{1, \ldots, n\}$: the lists $\binom{[n]}{1}$, $n = 3, 4, 5$, whose bases are the single marks; the list $\mathrm{H}_4$ on [4], whose bases are the pairs other than $\{1, 2\}$; the wheel $W_3$ of Proposition 39.65, whose marks are the six edges $e_1 = 01$, $e_2 = 02$, $e_3 = 03$, $e_4 = 12$, $e_5 = 13$, $e_6 = 23$ of the complete graph on the vertices $0, 1, 2, 3$ and whose bases are its sixteen spanning trees; the list $\mathrm{N}_6$ on [6], whose bases are the nineteen triples other than $\{1, 2, 3\}$; and the uniform lists $\binom{[5]}{2}$ and $\binom{[6]}{3}$. Read each on its centre ray at the strength 1 (Definition 38.13), with origin 0, direction $(\mathbf{1}; 1, 1)$ and stride $f = \alpha^{\mathbf{1}}(K\Psi)^{-1}$, and read $\mathrm{H}_4$ also on the ray with origin 0 and direction $(1, 1, 3, 3; 3, 1)$. Let $G$ be generated by the automorphisms that keep the masses and, at a centre, the Cremona, with the exchanges where they are named. For a box accepted at $\mathfrak{s}$ (Definition 39.20), with the spanning set $\mathbf{b} = (b_1, \ldots, b_{\mathfrak{m}})$, $b_1 = 1$, and the matrix $\mathbf{B}$, write $\mathbf{B} = P_0/Q_0$ with $Q_0$ the monic least common denominator of its entries, $D_0 = \max(\deg P_0, \deg Q_0)$, and let $\mathcal{S}_D$ be the space of the pairs $(P, Q)$ of a matrix and a scalar of polynomials of degree at most $D$ with $Q(-j)\, \mathsf{C}_{j+1}(\mathbf{b}) = P(-j)\, \mathsf{C}_j(\mathbf{b})$ for all large $j$, the closing equations of Proposition 39.61.

(i) At the centres of $\binom{[3]}{1}$, $\binom{[4]}{1}$, $\binom{[5]}{1}$, $\mathrm{H}_4$, $W_3$, with and without its exchanges, and $\mathrm{N}_6$ with its exchanges, and on the ray $(1, 1, 3, 3; 3, 1)$ of $\mathrm{H}_4$, show that $\dim \mathcal{S}_D = 0, 1, 2, 3$ at $D = D_0 - 1, \ldots, D_0 + 2$, the line at $D_0$ being spanned by $(P_0, Q_0)$, and find a degree up to which no element of the span of $\mathbf{b}$ with polynomial coordinates is annihilated by the closings. Work the centre of $\binom{[3]}{1}$ in full.

(ii) At the centre of $\mathrm{N}_6$, with $G$ generated by the automorphisms and the Cremona but not the exchanges, show that the closings miss a nonzero submodule of the span of the spanning set.

(iii) At the centre of $\binom{[5]}{2}$, with every symmetry of the four kinds in $G$: find the native gauges; show that the spanning set of the box $\mathfrak{B}_3(1,1)$ at $\mathfrak{s}=2$ is a set of masters of its span $\mathfrak{N}$; show that $\mathfrak{N}$ is the direct sum of the cyclic submodule of [1] and a line $L$ on which the closings, the values and $A_\star$ vanish, reading $\binom{[5]}{2}$ as the contraction of $\binom{[6]}{3}$ (Proposition 39.62); find which solutions at the native gauges see $L$; and read the centre of $\binom{[6]}{3}$ by its box.

*Solution.* Every ray here has origin 0, positive letter exponents and positive walls, so Definition 39.30, Theorem 39.33, Proposition 39.34 and Theorem 39.57 apply. Three facts are used throughout.

(a) If an element $m=\sum_l p_l(k)[b_l]$ with polynomial coefficients has values that vanish identically, then $\sum_l p_l(-j)\,\mathsf{C}_j(b_l)=0$ for every $j\ge 0$, not only for large $j$: the $b_l$ are Laurent insertions, so their closings are defined from $j=0$ on, and Proposition 39.34 gives $V_1\sum_l p_l(-j)\mathsf{C}_j(b_l)=\lim_{k\to -j}(k+j)^{n-1}\langle m\rangle_k=0$. This holds for the elements of $\mathcal{L}$ (Proposition 38.5(ii)) and for the elements on which the closings vanish (Theorem 39.57). The box gives $Q_0(k)\big(f^{-1}b_i-\sum_l \mathbf{B}_{il}(k)\,b_l\big)\in\mathcal{L}$ (Theorem 39.21(i)), and $\mathsf{C}_j(f^{-1}b_i)=\mathsf{C}_{j+1}(b_i)$, so

$$\mathsf{C}_{j+1}(\mathbf{b})=\mathbf{B}(-j)\,\mathsf{C}_j(\mathbf{b})\qquad\text{for every } j\ge 0 \text{ with } Q_0(-j)\neq 0.$$

(b) By (a), $(P,Q)\in\mathcal{S}_D$ exactly when every row of $Q\mathbf{B}-P$, read as an element of the span of $\mathbf{b}$, is annihilated by the closings; $Q_0$ times such a row is a polynomial row $p$ with $p(-j)\cdot\mathsf{C}_j(\mathbf{b})=0$ for every $j\ge 0$, so every pair of $\mathcal{S}_D$ satisfies its equation at every $j\ge 0$ with $Q_0(-j)\neq 0$. The equations at finitely many such $j$ cut out a space that contains $\mathcal{S}_D$, and so does that system reduced modulo a prime, whose rank is at most its rank over $\mathbb{Q}$. The pairs $(cP_0,cQ_0)$, $\deg c\le D-D_0$, lie in $\mathcal{S}_D$. So when a computed dimension is $\max(0,D-D_0+1)$, it is $\dim\mathcal{S}_D$.

(c) Likewise the polynomial rows of degree at most $e$ that the closings annihilate at finitely many $j$ contain the coordinate rows, of degree at most $e$, of the elements of the span of $\mathbf{b}$ on which the closings vanish.

*(i) The centre of* $\binom{[3]}{1}$. The letters are $K=\alpha_1+\alpha_2+\alpha_3$ and $\Psi=\alpha_1\alpha_2+\alpha_1\alpha_3+\alpha_2\alpha_3$, and

$$f^{-1}=\frac{K\Psi}{\alpha_1\alpha_2\alpha_3}=3+\sum_{i\neq j}\frac{\alpha_i}{\alpha_j}.$$

For a nonempty proper $U$, $d(U)=0$ and $d^*(U)=|U|-1$, so every wall of the direction is $\kappa_{1,U}=1$ and the wall degree of $\alpha^u$ is $\max_U(-u_U)$ (Definition 39.15(ii)). $G$ is generated by the six bijections of [3] and the Cremona; it holds every symmetry of the four kinds, the list having no exchange (its bases have one mark, its cobases two). The box $\mathfrak{B}_2(1,1)$ has 110 symbols; modulo $G$ its relations, taken at one symbol of each orbit, are 26 combinations of 16 orbit averages, of rank 13 over $\mathbb{Q}(k)$ (receipts `boxexact.py`, `boxes_all.py`, exact elimination over $\mathbb{Z}[k]$). The free orbits are those of 1, of $\alpha_3/\alpha_1$, of wall degree 1, and of $(\mathbf{1};1,1)$, at a positive level; so at $\mathfrak{s}=1$ the spanning set is $b_1=1$, $b_2=e_G(\alpha_3/\alpha_1)=\frac{1}{6}\sum_{i\neq j}\alpha_i/\alpha_j$, both targets reduce to it, and

$$\mathbf{B}(k)=\begin{bmatrix}3 & 6\\ \dfrac{2(k-1)}{k-2} & \dfrac{(k-1)(7k-11)}{(k-2)^2}\end{bmatrix},$$

with $Q_0=(k-2)^2$ and

$$P_0=\begin{bmatrix}3(k-2)^2 & 6(k-2)^2\\ 2(k-1)(k-2) & (k-1)(7k-11)\end{bmatrix},$$

so $D_0 = 2$; the first row is the Laurent identity $f^{-1}b_1 = 3b_1 + 6b_2$.

The native gauge of the direction, $\mathbf{1}$, has the native value $\frac{1}{9}$, and the three signs with one scale $-1$ have $K(\eta) = 1$, so they are native gauges of the native value 1 (Theorem 38.17(ii)); on a chart $\det(-H) = \frac{4}{27}$ at $\mathbf{1}$ and $-4$ at the signs (receipt `signs_span.py`), so all four are nondegenerate. Their solutions, summed over the two orbits, have distinct types and do not vanish at [1], so the span $\mathfrak{N}$ of $b_1, b_2$ has dimension 2 (Lemma 39.3): $\mathbf{b}$ is a set of masters and $\mathbf{B} = \mathbf{B_b}$. The rows $w_0 = (1, 0)$ and $w_1 = (3, 6)$ are independent, so $\mathfrak{N}$ is the cyclic submodule of [1] (Proposition 39.6(ii)).

The closings are the constant terms $\mathsf{C}_j(b) = \mathrm{CT}(b\, f^{-j})$; $\mathsf{C}_j(1)$ is the coefficient of $\alpha_1^j\alpha_2^j\alpha_3^j$ in $(K\Psi)^j$. By direct expansion,

| $j$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| $\mathsf{C}_j(b_1)$ | 1 | 3 | 15 | 93 | 639 | 4653 | 35169 | 272835 | 2157759 |
| $\mathsf{C}_j(b_2)$ | 0 | 1 | 8 | 60 | 456 | 3535 | 27888 | 223209 | 1807760 |

for instance $\mathsf{C}_1(b_2) = \mathrm{CT}\left(b_2(3 + 6b_2)\right) = 6 \cdot \frac{6}{36} = 1$, the six products $\frac{\alpha_i}{\alpha_j}\frac{\alpha_j}{\alpha_i}$ in $b_2^2$ being the only constant ones. $Q_0$ has no zero at a nonpositive integer, so (a) gives $\mathsf{C}_{j+1} = \mathbf{B}(-j)\,\mathsf{C}_j$ for every $j \geq 0$: at $j = 1$ and $j = 2$,

$$\mathbf{B}(-1) = \begin{bmatrix} 3 & 6 \\ \frac{4}{3} & 4 \end{bmatrix}, \quad \mathbf{B}(-1)\begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 15 \\ 8 \end{bmatrix}; \qquad \mathbf{B}(-2) = \begin{bmatrix} 3 & 6 \\ \frac{3}{2} & \frac{75}{16} \end{bmatrix}, \quad \mathbf{B}(-2)\begin{bmatrix} 15 \\ 8 \end{bmatrix} = \begin{bmatrix} 93 \\ 60 \end{bmatrix},$$

and the direct expansion agrees for every $j \leq 79$ (receipt `three_exact.py`).

The closing equations at the degree $D$ have $5(D + 1)$ unknowns, the coefficients of the four entries of $P$ and of $Q$, and two equations for each $j$. With $1 \leq j \leq 79$, 158 equations, exact elimination over $\mathbb{Q}$ gives the solution dimensions $0, 1, 2, 3$ at $D = 1, 2, 3, 4$; by (b) these are $\dim \mathcal{S}_D$, and the line at $D = 2$ is spanned by $(P_0, Q_0)$. So the closings alone single out $\mathbf{B_b}$ among the matrices of degree at most 2, with no source supplied. The rows $(p_1, p_2)$ of degree at most 24 with $p_1(-j)\mathsf{C}_j(b_1) + p_2(-j)\mathsf{C}_j(b_2) = 0$ for $1 \leq j \leq 80$, 50 unknowns and 80 equations, are only 0; by (c), no nonzero element of $\mathfrak{N}$ with polynomial coordinates of degree at most 24 is annihilated by the closings.

*(i) The other rays.* The same computation runs on the other boxes (receipt `blind2.py`, driven by `run_blind2.sh`): the closings at $j \leq 7$ are read on the torus as constant terms and continued by (a), and the equations are reduced modulo the prime $2^{61} - 1$; by (b) and (c) the dimensions found are upper bounds, attained here. The matrices $\mathbf{B}$ are those of the boxes built from Definition 39.15 and eliminated exactly over $\mathbb{Q}(k)$ (receipt `boxes_all.py`); at $\mathrm{N}_6$ they are the rayreader's, checked against an elimination of the whole box at a random prime and strength. The spanning sets, with exponents in the order of the marks, are: $1, e_G(\alpha_4/\alpha_1)$ for $\binom{[4]}{1}$; $1, e_G(\alpha_5/\alpha_1), e_G(\alpha_5^2/\alpha_1^2)$ for $\binom{[5]}{1}$; $1, e_G\alpha^u$ with $u = (-1,0,0,1), (-1,1,-1,1), (-1,1,0,0)$ for $\mathrm{H}_4$ at its centre, and $u = (0,1,-1,0), (0,2,-2,0), (0,2,-1,-1)$ on its ray $(1,1,3,3;3,1)$; $1, e_G\alpha^u$ with $u = (-1,-1,0,0,1,1), (-1,-1,0,1,0,1), (-1,-1,1,1,-1,1), (-1,-1,1,1,0,0)$ for $W_3$, with or without the exchanges; and for $\mathrm{N}_6$ with its exchanges $1, e_G\alpha^u$ with the twelve $u = (-1,-1,0,0,1,1)$, $(-1,-1,1,-1,1,1)$, $(-1,-1,1,0,0,1)$, $(-1,0,0,0,0,1)$, $(-1,0,1,-1,0,1)$, $(-1,0,1,0,0,0)$, $(-1,-1,-1,1,1,1)$, $(-2,-2,0,0,2,2)$, $(-2,-2,0,1,1,2)$, $(-2,-2,1,-1,2,2)$, $(-2,-2,1,0,1,2)$, $(-2,-2,1,1,1,1)$. In the table, $\dim \mathcal{S}_D$ is listed at $D = D_0 - 1, \ldots, D_0 + 2$, $h$ is the first index at which the rows $w_0, \ldots, w_h$ of [1] are dependent (Proposition 39.6(ii)), and the last but one column gives the degree up to which no nonzero row is annihilated, at the $j$ from 10 to the bound shown.

| ray | $\lvert G\rvert$ | box, $\mathfrak{s}$ | $\mathfrak{m}$ | $D_0$ | $\dim \mathcal{S}_D$ | rows | $h$ |
|---|---|---|---|---|---|---|---|
| $\binom{[4]}{1}$ | 48 | $(2;1,1)$, 1 | 2 | 3 | $0,1,2,3$ | 24, $j \le 120$ | 2 |
| $\binom{[5]}{1}$ | 240 | $(3;1,1)$, 2 | 3 | 6 | $0,1,2,3$ | 24, $j \le 120$ | 3 |
| $\mathrm{H}_4$ | 8 | $(2;1,1)$, 1 | 4 | 3 | $0,1,2,3$ | 24, $j \le 150$ | 4 |
| $\mathrm{H}_4$, $(1,1,3,3;3,1)$ | 4 | $(2;1,1)$, 1 | 4 | 10 | $0,1,2,3$ | 24, $j \le 150$ | 4 |
| $W_3$ | 48, 96 | $(2;1,1)$, 1 | 5 | 9 | $0,1,2,3$ | 40, $j \le 260$ | 4 |
| $\mathrm{N}_6$, exchanges | 144 | $(3;1,1)$, 2 | 13 | 16 | $0,1,2,3$ | 32, $j \le 480$ | 11 |

The denominators are $Q_0 = (k-2)^3$, $(k-3)^4$, $(k-2)^3$, $k(k-1)^3(k-\frac{4}{3})^3(k-\frac{5}{3})^3$, $(k-2)^5(k-\frac{3}{2})(k^3-3k^2+\frac{3}{2}k+2)$, and for $\mathrm{N}_6$ a polynomial of degree 16 with the factors $(k-4)(k-3)^5(k-2)^3(k-\frac{5}{2})$ and an irreducible sextic; only that of the ray $(1,1,3,3;3,1)$ vanishes at a nonpositive integer, at $k=0$, below the $j$ used. At $\binom{[4]}{1}$ the native gauges $\mathbf{1}$ and the four signs with one scale $-1$, of the native values $\frac{1}{16}$ and $\frac{1}{4}$, and at $\binom{[5]}{1}$ the native gauges $\mathbf{1}$ and the signs with one and with two scales $-1$, of the native values $\frac{1}{25}$, $\frac{1}{9}$ and 1, are nondegenerate (receipt `signs_span.py`), so by Lemma 39.3 their spanning sets are sets of masters. At $\mathrm{H}_4$, $W_3$ and $\mathrm{N}_6$ the signs of the largest native value are degenerate native gauges, and the computations are read for the spanning set as it stands; then $h$ bounds the dimension of the cyclic submodule of [1] from above. At each of these rays the closings alone single out the box's matrix among the matrices of degree at most $D_0$; at $W_3$ and at $\mathrm{N}_6$ the rows of [1] become dependent before the size of the spanning set is reached, and the closings single out the whole matrix on the spanning set, not only the law that the rows of [1] give.

*(ii)* $\mathrm{N}_6$ *without the exchanges.* Now $\lvert G\rvert = 72$, and the box $\mathfrak{B}_3(1,1)$ is accepted at $\mathfrak{s}=2$ with 15 elements: the thirteen above, here averaged over the smaller group, and $e_G\alpha^u$ for $u = (-1,0,0,-1,1,1)$ and $(-2,-2,2,-2,2,2)$. The bijection $x$ with $x(i) = i+3$ and $x(i+3) = i$, $i \le 3$, carries the bases, the triples other than $\{1,2,3\}$, onto the complements of the bases, the triples other than $\{4,5,6\}$: it exchanges the list, and at the centre it is a symmetry of the ray (Proposition 39.11(ii)). With $G_x$ generated by $G$ and $x$, $G$ is normal of index 2 in $G_x$, and for $b = e_G\alpha^{(-1,-1,1,0,0,1)}$, an element of the spanning set, $x^*b = e_G\alpha^{(-1,0,0,-1,1,1)}$, another (receipt `n_exchange.py`); so $(1-e_{G_x})b = \frac{1}{2}(b - x^*b)$, on which the closings, the values and $A_\star$ vanish (Corollary 39.59, applied to $G_x$). It is not 0: at the sign $\eta$ with the scales $-1$ on $\{4,5\}$, $K(\eta) = -5$, a nondegenerate native gauge of the native value $\frac{1}{25}$ with $\det(-H) = -\frac{9216}{15625}$ on a chart, $b(\eta) = -\frac{1}{3}$ and $x^*b(\eta) = 1$, so the leading coefficient of $A_\eta(b - x^*b)$ is $-\frac{4}{3}$ times that of $A_\eta(1)$ (Theorem 38.6(i)). So the closings miss the nonzero submodule generated by $b - x^*b$, the part of the span that the exchanges negate. The computation of (i) agrees: the rows of degree at most $e$ annihilated at $10 \le j \le 400$ have the dimension $e+1$ for $e \le 8$, so by (c) they are exactly the multiples $p(k)(b - x^*b)$, $\deg p \le e$; and the solution spaces of the closing equations at $D = 17, 18, 19, 20$ have dimensions at most $400, 431, 462, 493$, growing by $31 = 15\cdot 2 + 1$, the slope of Proposition 39.61 for a missed submodule of dimension two.

*(iii) The centre of* $\binom{[5]}{2}$. Here $K = e_2$, $\Psi = e_3$, $f^{-1} = e_2e_3/(\alpha_1\cdots\alpha_5)$, and $G$ is generated by the bijections of [5] and the Cremona, $\lvert G\rvert = 240$: the list has no exchange, its bases having two marks and its cobases three, so this is every symmetry of the four kinds.

*The native gauges.* On the chart $\alpha_5 = 1$ the defects times $e_2e_3$ generate an ideal whose saturation by the scales and the letters is zero-dimensional and radical of degree 66 (receipt `u25.sing`, an exact Gröbner basis computation): there are 66 native gauges, all simple zeros of the defects and so nondegenerate. Five points, checked exactly to be native gauges (receipt `u25_core.py`),

and their orbits under $G$ account for all 66; with $\zeta = e^{2\pi i/3}$ and the spanning set $\mathbf{b}$ found below,

| point | native value | orbit | $(b_1, \dots, b_5)$ at the point |
|---|---|---|---|
| $\mathbf{1}$ | $\frac{1}{100}$ | 1 | $(1,1,1,1,1)$ |
| $(1,1,1,1,-1)$ | $\frac{1}{4}$ | 5 | $(1,-\frac{3}{5},\frac{1}{5},1,\frac{1}{5})$ |
| $(1,1,1,-1,-1)$ | $\frac{1}{4}$ | 10 | $(1,\frac{1}{5},-\frac{1}{5},1,-\frac{1}{5})$ |
| $(\sqrt{3}-2,\sqrt{3}-2,\sqrt{3}-2,1,1)$ | $-\frac{1}{8}$ | 20 | $(1,1,-\frac{4}{5},\frac{113}{5},\frac{23}{5})$ |
| $(1,1,\zeta,\zeta,\zeta^2)$ | 1 | 30 | $(1,\frac{1}{10},-\frac{1}{5},\frac{1}{10},-\frac{1}{5})$ |

On the chart $\det(-H)$ is $\frac{81}{3125}$, $-3$, $-3$, $-\frac{81}{16}$ and 81 at these points. The orbits are the table's gauge, the signs with one and with two scales $-1$, the points with three scales $\sqrt{3}-2$ and two scales 1 or two and three, which the Cremona joins, and the points with two scales 1, two $\zeta$ and one $\zeta^2$.

*The box.* The box $\mathfrak{B}_3(1,1)$ has 10322 symbols; modulo $G$ its relations at the orbit representatives are 196 combinations of 114 orbit averages, of rank 103, eliminated exactly over $\mathbb{Q}(k)$. The free Laurent orbits of wall degree at most 2 give the spanning set

$$b_1 = 1, \qquad b_2 = e_G\alpha^{(-1,-1,0,1,1)}, \qquad b_3 = e_G\frac{\alpha_5}{\alpha_1} = \frac{1}{20}\sum_{i\neq j}\frac{\alpha_i}{\alpha_j},$$
$$b_4 = e_G\alpha^{(-2,-2,0,2,2)}, \qquad b_5 = e_G\alpha^{(-2,-2,1,1,2)},$$

of wall degrees $0,1,1,2,2$, and the box is accepted at $\mathfrak{s} = 2$. With $q(k) = 3k^3+4k^2-61k+130$, its matrix has the rows $\mathbf{B}_{1\bullet} = (10,30,60,0,0)$, the Laurent identity $f^{-1} = 10b_1+30b_2+60b_3$, and

$$\tfrac{q}{k-1}\mathbf{B}_{2\bullet} = \Big(27k^2+111k-130,\ \tfrac{2(51k^3+137k^2-756k+580)}{k-2},\ \tfrac{16(9k^3+33k^2-155k+108)}{k-2},$$
$$-3(3k^2-9k-10),\ 12(3k^2-k+2)\Big),$$
$$\tfrac{q}{k-1}\mathbf{B}_{3\bullet} = \Big(6(5k^2+8k-65),\ \tfrac{6(15k^3+6k^2-313k+440)}{k-2},\ \tfrac{4(33k^3+74k^2-971k+1340)}{k-2},$$
$$-12(k-5)(k-2),\ 12(5k^2-14k+5)\Big),$$

$$\tfrac{q\,(k-3)^4}{k-1}\mathbf{B}_{41} = k(27k^5+245k^4-1789k^3+3709k^2-5388k+5820),$$
$$\tfrac{q\,(k-3)^4}{k-1}\mathbf{B}_{42} = \tfrac{2(45k^7+225k^6-2310k^5+3606k^4+1139k^3-157k^2-7152k+2460)}{k-2},$$
$$\tfrac{q\,(k-3)^4}{k-1}\mathbf{B}_{43} = \tfrac{4(216k^7-552k^6-925k^5-1703k^4+18473k^3-28953k^2+33288k-33444)}{3(k-2)},$$
$$\tfrac{q\,(k-3)^4}{k-1}\mathbf{B}_{44} = 3k^6-283k^5+3037k^4-13753k^3+33126k^2-43566k+25020,$$
$$\tfrac{q\,(k-3)^4}{k-1}\mathbf{B}_{45} = -4(k-2)(27k^5-373k^4+1321k^3-1507k^2-450k+1878),$$
$$\tfrac{q\,(k-3)^3}{k-1}\mathbf{B}_{51} = 2k(15k^4+52k^3-232k^2+153k-420),$$
$$\tfrac{q\,(k-3)^3}{k-1}\mathbf{B}_{52} = \tfrac{2(45k^6+111k^5-963k^4+461k^3+936k^2+478k-540)}{k-2},$$
$$\tfrac{q\,(k-3)^3}{k-1}\mathbf{B}_{53} = \tfrac{4(117k^6+225k^5-2281k^4+1565k^3+1564k^2-1562k+5028)}{3(k-2)},$$
$$\tfrac{q\,(k-3)^3}{k-1}\mathbf{B}_{54} = -2(6k^5-17k^4-185k^3+1052k^2-2368k+2280),$$
$$\tfrac{q\,(k-3)^3}{k-1}\mathbf{B}_{55} = 4(9k^5+32k^4-250k^3+412k^2-65k-522),$$

so $Q_0 = (k-3)^4(k-2)\,q(k)/3$ and $D_0 = 8$; it equals the rayreader's matrix.

*Masters.* The five orbit sums of the solutions $A_c$ (Theorem 39.12(iii)) have the types $\frac{1}{100}, \frac{1}{4}, \frac{1}{4}, -\frac{1}{8}, 1$. Those alone in their type do not vanish at [1]. On $b_1, b_2$ the two of the type $\frac{1}{4}$ have the leading coefficients $5C(1, -\frac{3}{5})$ and $10C'(1, \frac{1}{5})$, $C, C' \neq 0$, and $\det \begin{bmatrix} 1 & -\frac{3}{5} \\ 1 & \frac{1}{5} \end{bmatrix} = \frac{4}{5} \neq 0$, so they are linearly independent on $\mathfrak{N}$. By Lemma 39.3, $\dim \mathfrak{N} \geq 5$: the spanning set is a set of masters of $\mathfrak{N}$, of dimension exactly 5, and $\mathbf{B} = \mathbf{B_b}$. The five rows of values in the table have the determinant $\frac{3888}{625}$, so $\mathbf{B}(\infty)$ is diagonalizable with the eigenvalues $100, 4, 4, -8, 1$ (Proposition 39.42(i), (ii)); indeed

$$\mathbf{B}(\infty) = \begin{bmatrix} 10 & 30 & 60 & 0 & 0 \\ 9 & 34 & 48 & -3 & 12 \\ 10 & 30 & 44 & -4 & 20 \\ 9 & 30 & 96 & 1 & -36 \\ 10 & 30 & 52 & -4 & 12 \end{bmatrix}, \qquad \det(\lambda - \mathbf{B}(\infty)) = (\lambda - 100)(\lambda - 4)^2(\lambda + 8)(\lambda - 1)$$

(receipt `binf_exact.py`).

*The law and the line.* The rows $w_i(k) = w_{i-1}(k-1)\mathbf{B}(k)$, $w_0 = (1, 0, 0, 0, 0)$, are independent for $i \leq 3$ and dependent at $i = 4$, so the cyclic submodule $\mathfrak{N}_1$ of [1] has dimension 4 and the law of [1] (Proposition 39.6(ii)) is $\sum_{j=0}^{4} p_j(k)\,\mathrm{Sh}^j[1] = 0$ with

$$\begin{aligned} p_0 &= k^4(33k^2 + 88k + 60), \\ p_1 &= -(3201k^6 + 14938k^5 + 27699k^4 + 26214k^3 + 13598k^2 + 3736k + 440), \\ p_2 &= -4(2772k^6 + 18480k^5 + 48699k^4 + 63150k^3 + 40515k^2 + 11296k + 1140), \\ p_3 &= 16(k+2)^2(7491k^4 + 34958k^3 + 51724k^2 + 24507k + 4430), \\ p_4 &= -3200(k+2)^2(k+3)^2(33k^2 + 22k + 5). \end{aligned}$$

Put

$$\begin{aligned} m = {} & k(k-5)(k-2)(k+2)\,[b_1] - (3k^4 + 11k^3 - 71k^2 + 227k - 210)\,[b_2] \\ & - \tfrac{2}{3}(3k^4 - 11k^3 + 119k^2 - 125k - 90)\,[b_3] - (k-5)(k-2)^3\,[b_4] + (k-2)^2(5k^2 - 14k + 5)\,[b_5] \end{aligned}$$

and $\tilde{m} = \frac{k-1}{q(k)}\,m$. With $\tilde{w}$ the coordinate row of $\tilde{m}$, the identity $\tilde{w}(k-1)\mathbf{B}(k) = 4\,\tilde{w}(k)$ holds exactly, that is $\mathrm{Sh}^{-1}\,\tilde{m} = 4\tilde{m}$, so $L = \mathbb{C}(k)\tilde{m}$ is a submodule; and at $k = \frac{37}{3}$ the rows $w_0, \ldots, w_3, \tilde{w}$ have rank 5. So

$$\mathfrak{N} = \mathfrak{N}_1 \oplus L.$$

*The table one mark larger.* $\binom{[5]}{2}$ is the contraction at the mark 6 of $\binom{[6]}{3}$, which is connected and identically self-dual, so Proposition 39.62 applies, with $G'$ generated by the $T_\beta$, $\beta$ a bijection of [6], and the Cremona, and $G \subseteq G'$. The native gauges lift by $\alpha_6 = \Psi/K$: the five points of the table to $\alpha_6 = 1, -1, 1, 1, \zeta^2$ (receipt `u36_lift.py`). The lifts are native gauges of the centre of $\binom{[6]}{3}$, which has 66 of them, all simple (receipt `u36.sing`), in orbits of $S_6$ and the Cremona of $1, 15, 20, 30$ points with the native values $\frac{1}{400}, \frac{1}{16}, -\frac{1}{32}, \frac{1}{4}$, a quarter of those below them: the two kinds of signs lift into one orbit of 15, the signs of $\binom{[6]}{3}$ with two scales $-1$, and this is Proposition 39.64(ii) at $r = 2$, with $C_2 = 2$.

*The element $\ell$.* Put $S = \sum_{i \neq j \leq 5} \alpha_i/\alpha_j = 20\,b_3$ and $T = \frac{K}{\Psi}\sum_i \alpha_i + \frac{\Psi}{K}\sum_i \alpha_i^{-1}$. Since $S$ is invariant under $G$, the image of the stabilizer of the mark 6, and the transpositions $(e\,6)$, $e \leq 5$,

with the identity, represent its cosets, $e_{G'}[S] = \frac{1}{6}\big([S] + \sum_{e\le 5} T_{(e\,6)}[S]\big)$. By Proposition 39.62(iii), with $\frac{\Gamma(k+1)\Gamma(k-1)}{\Gamma(k)^2} = \frac{k}{k-1}$,

$$T_{(e\,6)}\Big[\frac{\alpha_i}{\alpha_j}\Big] = \Big[\frac{\alpha_i}{\alpha_j}\Big]\ (e \notin \{i,j\}), \qquad \frac{k}{k-1}\Big[\frac{\Psi}{K\alpha_j}\Big]\ (e=i), \qquad \frac{k}{k-1}\Big[\frac{K\alpha_i}{\Psi}\Big]\ (e=j).$$

Each pair $(i,j)$ is left alone by three of the five $e$, and each $\alpha_j^{-1}$ and each $\alpha_i$ arises four times, so $\sum_e T_{(e\,6)}[S] = 3[S] + \frac{4k}{k-1}[T]$, $e_{G'}[S] = \frac{2}{3}[S] + \frac{2k}{3(k-1)}[T]$, and

$$\ell = 3(1-e_{G'})[S] = [S] - \frac{2k}{k-1}\,[T].$$

The symbols of $T$, at the levels $(0,1)$ and $(1,0)$ and of wall degrees 0 and 1, lie in the box, which reduces $T$ to the spanning set (receipt `u25_core.py`):

$$[T] = \frac{10(k-1)}{k}\,[b_3] - \frac{10}{k\,q(k)}\,m.$$

So

$$\ell = 20[b_3] - \frac{2k}{k-1}\Big(\frac{10(k-1)}{k}[b_3] - \frac{10}{k\,q(k)}m\Big) = \frac{20}{(k-1)\,q(k)}\,m = \frac{20}{(k-1)^2}\,\tilde{m}.$$

Since $e_{G'}(1-e_{G'}) = 0$, $e_{G'}\ell = 0$, hence $e_{G'}\tilde{m} = 0$ and $L \subseteq (1-e_{G'})\mathfrak{M}^G$: the closings, the values and $A_\star$ vanish on $L$ (Proposition 39.62(iv)). The values can be seen to vanish directly as well: by the formula for $\pi_*$ in Proposition 39.62, $\pi_*(\alpha_i/\alpha_6) = \frac{k}{k-1}\frac{K}{\Psi}\alpha_i$ and $\pi_*(\alpha_6/\alpha_i) = \frac{k}{k-1}\frac{\Psi}{K}\alpha_i^{-1}$, so $\ell = \pi_*\ell'$ with $\ell' = \sum_{i\ne j\le 5}\frac{\alpha_i}{\alpha_j} - 2\sum_{i\le 5}\big(\frac{\alpha_i}{\alpha_6}+\frac{\alpha_6}{\alpha_i}\big)$, three times the part of $\sum_{i\ne j\le 5}\alpha_i/\alpha_j$ that the average over the bijections of [6] removes; the values of $\binom{[6]}{3}$ are kept by those bijections (Proposition 39.11(ii)), so $\langle\ell'\rangle'_k = 0$ and $\langle\ell\rangle_k = \Gamma_{\!2\,1}\!\left[\begin{matrix}k,\ k\\ 2k\end{matrix}\right]^{-1}\langle\ell'\rangle'_k = 0$ (Proposition 39.62(i)). As a numerical check, a trapezoidal quadrature over the chart gives $|\langle\ell\rangle_k|/\langle 1\rangle_k$ equal to $1.0\cdot 10^{-13}$ at $k = \frac{13}{2}$ and $2.8\cdot 10^{-13}$ at $k = 8$, the accuracy of the quadrature (receipt `ell_check.py`). So $L$ is a relation among the values of the five masters that the relations of the ray do not produce, $m$ being nonzero in $\mathfrak{N}$, nor its symmetries of the four kinds, $m$ lying in $\mathfrak{M}^G$.

*What sees L.* A solution $V$ of the type $(\lambda, \rho+\mathbb{Z})$ has $\varsigma^{-1}V(\tilde{m}) = V(\mathrm{Sh}^{-1}\tilde{m}) = 4V(\tilde{m})$, which forces $V(\tilde{m}) = 0$ unless $\lambda^{-1} = 4$: the solutions at the orbits of the native values $\frac{1}{100}$, $-\frac{1}{8}$ and 1 vanish on $L$. At a sign, $A_c(b_l)/A_c(1) = b_l(\alpha_c) + O(1/k)$ (Theorem 38.6(i)), and the coefficients of $k^4$ in $m$ are $(1,-3,-2,-1,5)$, so $A_c(m)/A_c(1)$ begins with

$$\big(1+\tfrac{9}{5}-\tfrac{2}{5}-1+1\big)k^4 = \tfrac{12}{5}k^4 \quad (\text{one scale } -1), \qquad \big(1-\tfrac{3}{5}+\tfrac{2}{5}-1-1\big)k^4 = -\tfrac{6}{5}k^4 \quad (\text{two scales } -1):$$

the solutions at both kinds of signs see $L$. With the solutions at the fifteen signs transported from one of them by $G'$ as in Proposition 39.64(i), $A_c(1)$ is the same for all fifteen, $T_\gamma[1] = [1]$; their sum, invariant under $G'$, has on $m$ the leading coefficient $5\cdot\frac{12}{5} - 10\cdot\frac{6}{5} = 0$ times that of $A_c(1)$, as it must, while the mean over the five less the mean over the ten has $\frac{12}{5} + \frac{6}{5} = \frac{18}{5}$ and sees $L$.

*The closings.* The rows of degree at most $e$ annihilated by the closings at $10 \le j \le 150$ have the dimension $\max(0, e-3)$ for $e \le 24$, which is the dimension of the multiples $p(k)\,w_m$, $\deg p \le e-4$, of the primitive coordinate row $w_m$ of $3m$, of degree 4; by (c) the closings annihilate $L$ and no element of $\mathfrak{N}$ outside $L$ with polynomial coordinates of degree at most 24. The solution spaces of the closing equations at $D = 7, 8, 9, 10$ have dimensions at most $24, 30, 36, 42$, growing

by $6 = 5 \cdot 1 + 1$, the slope of Proposition 39.61 for a missed line; a step of more than one already shows that something is missed.

*The centre of* $\binom{[6]}{3}$ *by its box.* Modulo $S_6$ and the Cremona, of order 1440, the box $\mathfrak{B}_3(1,1)$, with 78296 symbols, has 406 relations at orbit representatives on 226 orbit averages, of rank 186; it is accepted at $\mathfrak{s} = 2$ with the spanning set 1, $e\,\alpha^{(-1,-1,-1,1,1,1)}$, $e\,\alpha^{(-1,-1,0,0,1,1)}$, $e\,\alpha^{(-2,-2,-2,2,2,2)}$, $e$ the average over that group, and $Q_0 = (k-3)^5(k-2)^2(k+1)(k^3 - 8k^2 + \frac{71}{3}k - \frac{76}{3})$, $D_0 = 11$. The four orbits of native gauges give solutions of four distinct types, none vanishing at [1], so the spanning set is a set of masters of a submodule $\mathfrak{N}'$ of dimension 4 (Lemma 39.3), and $\mathbf{B}'(\infty)$ has the simple spectrum $400, 16, 4, -32$. Its rows of [1] are independent up to $h = 4$, so $\mathfrak{N}'$ is the cyclic submodule of [1]. Since $\pi_*[1] = [1]$ and $\pi_* \operatorname{Sh}' = \frac{k}{2(2k+1)} \operatorname{Sh} \pi_*$, $\pi_* \mathfrak{N}' = \mathfrak{N}_1$, and the law of [1] at the centre of $\binom{[6]}{3}$ is the law of $\binom{[5]}{2}$ for $\Gamma_{\binom{2}{1}} \begin{bmatrix} k, & k \\ 2k & \end{bmatrix} \langle 1 \rangle_k$: its coefficients are proportional to $p_i(k) \prod_{l=i}^{3} \frac{k+l}{2(2k+2l+1)}$, which is the law that the box gives, exactly (receipt `u36_lift.py`). So

$$\mathfrak{N} \cap \mathfrak{M}^{G'} = \mathfrak{N}_1 = \pi_* \mathfrak{N}', \qquad \mathfrak{N} \cap (1 - e_{G'}) \mathfrak{M}^G = L,$$

$\mathfrak{N}_1$ lying in $\mathfrak{M}^{G'}$, which contains [1], and $L$ in $(1 - e_{G'})\mathfrak{M}^G$. At the centre of $\binom{[6]}{3}$ the closing equations have the dimensions $0, 1, 2, 3$ at $D = 10, \ldots, 13$, and no nonzero row of degree at most 40 is annihilated at $10 \le j \le 200$: the closings there single out $\mathbf{B}'$.

So: at the small centres of (i) the closings alone single out the box's matrix at the degree $D_0$, with nothing of low degree missed; at the centre of $\mathrm{N}_6$ without its exchanges they miss the part that the exchanges negate; and at the centre of $\binom{[5]}{2}$, with every symmetry of the four kinds, the span of the box is the cyclic submodule of [1], which the box of $\binom{[6]}{3}$ reads, plus the line $L$ that the symmetric group of $\binom{[6]}{3}$ hides from the closings, the values and $A_\star$, and that the signs see. $\square$

# Chapter 55

# Exercises on rays: other lists

These exercises read rays of lists other than the lists $\binom{\mathcal{E}}{1}$: the list $\mathrm{H}_4$ on four marks at its centre and on a ray strictly inside its walls whose native value has a conjugate on the slit, and the five-mark lists along their hinge trees. Each exercise defines its list by its marks and its bases; every weight is 1. A recurrence of the closings is proved for every index by an explicit telescoping identity, and Theorems 39.33 and 39.57, with Corollary 39.60, carry it to the values. A *certificate* of a recurrence is an explicit element of the relation space $\mathcal{L}$ (Definition 38.4) that equals the recurrence applied to an insertion, as in Proposition 38.5(i); a telescoping identity among binomial sums is not one. Numbers are exact; a decimal is a rendering of an exact value, truncated. The receipts are the scripts `ex28_h4_centre.py`, `ex29_h4_lean.py` and `ex30_trees.py`, with the shared tools in `common.py`; each prints the numbers its solution displays.

## 55.1 The list $\mathrm{H}_4$

In this section $\mathrm{H}_4$ is the list on the marks $a, b, 2, 3$ whose bases are the five pairs other than $\{a, b\}$: $\{a, 2\}, \{a, 3\}, \{b, 2\}, \{b, 3\}, \{2, 3\}$. With $A = \alpha_a + \alpha_b$ and $B = \alpha_2 + \alpha_3$ its letters are

$$K = AB + \alpha_2\alpha_3, \qquad \Psi = AB + \alpha_a\alpha_b,$$

the second from the cobases $\{b, 3\}, \{b, 2\}, \{a, 3\}, \{a, 2\}, \{a, b\}$, and $\Psi(\alpha) = \alpha_a\alpha_b\alpha_2\alpha_3\, K(\alpha^{-1})$. The pair $\{a, b\}$ is a parallel class.

**Exercise 55.1** ($\mathrm{H}_4$ at its centre)**.** Let $P(k) = \langle 1\rangle_k = P_{\mathrm{H}_4}(k\mathbf{1}; k, k)$ be the values of the centre ray of $\mathrm{H}_4$ at the strength one (Definition 38.13).

(i) Write the stride, the walls of the direction and $V_1$.

(ii) Find every native gauge of the ray with its native value, and decide which are nondegenerate.

(iii) Find the closings and a law of the values of order four, prove the law, and give its symbol.

(iv) Write $\Gamma(\sigma)^2 P(\sigma)$ on the share side of the hinge tree of $\mathrm{H}_4$, close it at the integer strengths, and decide which of its two closings converges.

(v) Read the low-end relation, the principal parts of $P$ at $k = 0$ and at $k = -1$, and the relation that the law gives at $k = -1$.

(vi) Give $P(1)$ and $P(2)$, and show that every value $P(k)$, $k \geq 1$, lies in $\mathbb{Q} + \mathbb{Q}\,P(1) + \mathbb{Q}\,P(2)$.

*Solution. (i) Stride, walls, $V_1$.* By Definition 38.13 the stride and its reciprocal are

$$f=\frac{\alpha_a\alpha_b\alpha_2\alpha_3}{K\Psi}=\frac{1}{K(\alpha)\,K(\alpha^{-1})},\qquad f^{-1}=\frac{(AB+\alpha_2\alpha_3)(AB+\alpha_a\alpha_b)}{\alpha_a\alpha_b\alpha_2\alpha_3},$$

and the walls of the direction are the widths, $\kappa_{1,S}=c(S)=r(S)+r(\mathcal{E}\setminus S)-2$ (Definition 7.12, Proposition 8.21(ii)). A single mark has rank one and its complement rank two; $\{a,b\}$ has rank one and $\{2,3\}$ rank two; the other pairs have rank two with complements of rank two; a triple has rank two and its complement rank one. So $\kappa_{1,S}=2$ for the four pairs $\{a,2\},\{a,3\},\{b,2\},\{b,3\}$, and $\kappa_{1,S}=1$ for every other nonempty proper $S$: all walls are positive. In $V_1=\sum_\pi\prod_{l=2}^{4}\kappa_{1,U_l}^{-1}$ (Proposition 39.34) the tails $U_2$ and $U_4$ have three marks and one mark, of wall 1, so an order contributes $\kappa_{1,U_3}^{-1}$, $U_3$ being its last two marks. Each of the six pairs ends four of the 24 orders, so

$$V_1=4\,(1+1)+4\cdot4\cdot\tfrac{1}{2}=16.$$

*(ii) Native gauges.* By Proposition 38.16 a point $\alpha$ of $X_F$ is a native gauge exactly when the occupations $R_e=\alpha_e\partial_{\alpha_e}\log K$ agree at $\alpha$ and at $\alpha^{-1}$, and its native value is $\big(K(\alpha)K(\alpha^{-1})\big)^{-1}$. Here

$$R_a(\alpha)=\frac{\alpha_aB}{K},\quad R_a(\alpha^{-1})=\frac{\alpha_bB}{\Psi},\qquad R_2(\alpha)=\frac{\alpha_2(A+\alpha_3)}{K},\quad R_2(\alpha^{-1})=\frac{\alpha_3A+\alpha_a\alpha_b}{\Psi},$$

and $R_b$, $R_3$ follow by exchanging $a\leftrightarrow b$ and $2\leftrightarrow3$. So the native gauges are the solutions in $X_F$ of

$$\begin{aligned}B(\alpha_a\Psi-\alpha_bK)&=0,&\qquad B(\alpha_b\Psi-\alpha_aK)&=0,\\ \alpha_2(A+\alpha_3)\Psi&=(\alpha_3A+\alpha_a\alpha_b)K,&\qquad \alpha_3(A+\alpha_2)\Psi&=(\alpha_2A+\alpha_a\alpha_b)K.\end{aligned}$$

The occupations add up to the rank 2 at $\alpha$ and at $\alpha^{-1}$, so any three of the four equations imply the fourth.

If $B=0$, then $K=\alpha_2\alpha_3$ and $\Psi=\alpha_a\alpha_b$, and with $\alpha_3=-\alpha_2$ the third equation reads $\alpha_2A(\alpha_a\alpha_b-\alpha_2^2)=0$: either $A=0$ or $\alpha_a\alpha_b=\alpha_2^2$. If $B\neq0$, the first two equations give $(\alpha_a-\alpha_b)(\Psi+K)=0$ and $A(\Psi-K)=0$. For $A=0$ they leave $K=\alpha_2\alpha_3$, $\Psi=\alpha_a\alpha_b=-\alpha_a^2$ and $\Psi+K=0$, that is $\alpha_a^2=\alpha_2\alpha_3$, and the third equation, $\alpha_2\alpha_3\Psi=\alpha_a\alpha_bK$, holds there. For $A\neq0$ they give $\Psi=K$, that is $\alpha_a\alpha_b=\alpha_2\alpha_3$, and then $\alpha_a=\alpha_b$, since $\Psi+K=2K\neq0$; the third equation becomes $A(\alpha_2-\alpha_3)=0$, so $\alpha_2=\alpha_3$ and $\alpha_2^2=\alpha_a^2$. Up to scale the native gauges are therefore:

- the table's gauge $\mathbf{1}$, where $K=\Psi=5$: native value $\frac{1}{25}$;
- the sign $(1,1,-1,-1)$, where $K=\Psi=-3$: native value $\frac{1}{9}$;
- the three curves $\{A=0,\ \alpha_a^2=\alpha_2\alpha_3\}$, $\{B=0,\ \alpha_2^2=\alpha_a\alpha_b\}$ and $\{A=B=0\}$, on which $K=\alpha_2\alpha_3$ and $\Psi=\alpha_a\alpha_b$: native value 1.

No letter vanishes at these points of the torus, so they lie in $X_F$. On the chart $y_3=0$, in the coordinates $y_a,y_b,y_2$, the Hessian of $-\log f$ at $\mathbf{1}$ is twice the covariance matrix of the indicators under the uniform weight law on the five bases (Theorem 38.15), and at the sign it is twice the covariance matrix under the signed measure $\mathbb{P}_\eta$ with the masses $\frac13,\frac13,\frac13,\frac13,-\frac13$ on $\{a,2\},\{a,3\},\{b,2\},\{b,3\},\{2,3\}$ (Theorem 38.17(ii)):

$$H_{\mathbf{1}}=\frac{2}{25}\begin{bmatrix}6&-4&-1\\-4&6&-1\\-1&-1&6\end{bmatrix},\quad \det H_{\mathbf{1}}=\frac{32}{625};\qquad H_\eta=\frac{2}{9}\begin{bmatrix}2&-4&1\\-4&2&1\\1&1&2\end{bmatrix},\quad \det H_\eta=-\frac{32}{81}.$$

Both are nonsingular: the table's gauge and the sign are nondegenerate native gauges (Definition 38.4). The three curves are sources of dimension one (Definition 39.23), along which $f$ is constant and the Hessian is singular.

*(iii) Closings and the law.* The origin is 0 and both letters have the exponent one, so the closings are $\mathsf{C}_j(1) = \mathrm{CT}(f^{-j})$ (Definition 39.30). In

$$(K\Psi)^j = \sum_{u,v} \binom{j}{u}\binom{j}{v} (AB)^{u+v} (\alpha_2\alpha_3)^{j-u} (\alpha_a\alpha_b)^{j-v}$$

the monomial $(\alpha_a\alpha_b\alpha_2\alpha_3)^j$ takes $\alpha_a^v\alpha_b^v$ from $A^{u+v}$ and $\alpha_2^u\alpha_3^u$ from $B^{u+v}$, so $u = v$ and

$$\mathsf{C}_j(1) = \sum_{u=0}^{j} F(j,u), \qquad F(j,u) = \binom{j}{u}^2 \binom{2u}{u}^2; \qquad \mathsf{C}_j(1) = 1,\ 5,\ 53,\ 761,\ 12661,\ 229705,\ \ldots$$

Exact linear algebra on the first 150 closings gives one line of recurrences $\sum_{i=0}^{4} p_i(j)\,\mathsf{C}_{j+i}(1) = 0$ with $\deg p_i \le 7$, spanned by

$$\begin{aligned}
p_0(j) &= 225(j+1)^2(j+2)^2\,w(j), \qquad w(j) = 144j^3 + 1296j^2 + 3853j + 3790,\\
p_1(j) &= -2(j+2)^2(34848j^5 + 435600j^4 + 2135762j^3 + 5129622j^2 + 6027508j + 2768385),\\
p_2(j) &= 2(21168j^7 + 381024j^6 + 2917047j^5 + 12308044j^4\\
&\qquad + 30893608j^3 + 46094457j^2 + 37814839j + 13140768),\\
p_3(j) &= -2(j+3)(2592j^6 + 42768j^5 + 289602j^4 + 1028216j^3 + 2015044j^2 + 2063287j + 861336),\\
p_4(j) &= (j+3)(j+4)^3(144j^3 + 864j^2 + 1693j + 1089);
\end{aligned}$$

no recurrence of order at most three and degree at most 30 holds on the first 200 closings. The recurrence holds for every $j \in \mathbb{N}$, by a telescoping identity. For $0 \le u \le j+4$ put

$$T(j,u) = \frac{((j+4)!)^2\,((2u)!)^2}{(u!)^6\,((j+4-u)!)^2} > 0, \qquad G(j,u) = \frac{T(j,u)\,A(j,u)}{\prod_{l=1}^{4}(j+l)^2}, \qquad G(j,j+5) = 0,$$

with $A(j,u) = u^4(j+1)^2(j+2)^2\sum_{q=0}^{4} Q_q(j)\,u^q$ and

$$\begin{aligned}
Q_0 &= (j+4)(6912j^6 + 125568j^5 + 938496j^4 + 3695236j^3 + 8085594j^2 + 9321748j + 4422321),\\
Q_1 &= -4(5760j^6 + 106272j^5 + 809680j^4 + 3260831j^3 + 7321143j^2 + 8687769j + 4256385),\\
Q_2 &= 2(13248j^5 + 198864j^4 + 1186148j^3 + 3514356j^2 + 5173077j + 3027087),\\
Q_3 &= -4(22j+63)\,w(j), \qquad Q_4 = 15\,w(j).
\end{aligned}$$

For $0 \le u \le j+4$, $F(j+i,u) = T(j,u)\prod_{l=i+1}^{4}\big((j+l-u)/(j+l)\big)^2$, which vanishes for $u > j+i$ as $F$ does, and $T(j,u+1) = T(j,u)\,4(2u+1)^2(j+4-u)^2/(u+1)^4$, which vanishes at $u = j+4$. So, divided by $T(j,u)/\big((u+1)^4\prod_l (j+l)^2\big)$, the identity

$$\sum_{i=0}^{4} p_i(j)\,F(j+i,u) = G(j,u+1) - G(j,u) \qquad (0 \le u \le j+4)$$

is the identity of polynomials

$$(u+1)^4\sum_{i=0}^{4} p_i(j)\prod_{l\le i}(j+l)^2\prod_{l>i}(j+l-u)^2 = 4(2u+1)^2(j+4-u)^2A(j,u+1) - (u+1)^4A(j,u),$$

which holds in $\mathbb{Q}[j,u]$, both sides expanded. Summed over $0 \le u \le j+4$, the right side telescopes to $G(j,j+5) - G(j,0) = 0$, as $A(j,0) = 0$, and the left side is $\sum_i p_i(j)\,\mathsf{C}_{j+i}(1)$. So the closings satisfy the recurrence for every $j$.

The direction has the integral origin 0, both letter exponents 1 and positive walls. Written as $\sum_i c_i(-j)\,\mathsf{C}_{j-i}(1) = 0$ with $c_i(k) = -p_{4-i}(-k-4)$, the recurrence says that the closings vanish on $\sum_i c_i(k)\,\mathrm{Sh}^i[1]$ (Theorem 39.33); by Theorem 39.57 the values vanish on it as well, and the continuation of its value vanishes identically; Corollary 39.60 states the same for the recurrences of $\langle 1\rangle$. With $\langle f^i g\rangle_k = \langle g\rangle_{k+i}$ (Proposition 38.5(ii)) this is the law

$$\sum_{i=0}^{4} c_i(k)\,P(k+i) = 0,$$

an identity of meromorphic functions of $k$, with

$$\begin{aligned}
c_0(k) &= k^3(k+1)(144k^3+864k^2+1693k+1075),\\
c_1(k) &= -2(k+1)(2592k^6+19440k^5+56322k^4+80296k^3+60004k^2+23017k+3580),\\
c_2(k) &= 2(21168k^7+211680k^6+884919k^5+2003456k^4\\
&\qquad +2646984k^3+2031207k^2+830703k+137260),\\
c_3(k) &= -2(k+2)^2(34848k^5+261360k^4+741842k^3+984642k^2+598948k+127215),\\
c_4(k) &= 225(k+2)^2(k+3)^2(144k^3+432k^2+397k+102).
\end{aligned}$$

Its symbol (Corollary 38.7) is read off the coefficients of $k^7$:

$$\chi(\lambda) = 144 - 5184\lambda + 42336\lambda^2 - 69696\lambda^3 + 32400\lambda^4 = 144(25\lambda-1)(9\lambda-1)(\lambda-1)^2.$$

Its roots are the native values $\frac{1}{25}$ and $\frac{1}{9}$ of the two nondegenerate native gauges and the native value 1 of the curves, doubled; the root $\frac{1}{25} = f(\mathbf{1})$ is forced by Proposition 38.9.

*(iv) The share side and its closings.* A set of two marks has width one exactly for $\{a,b\}$ and $\{2,3\}$ (part (i)), so $\mathrm{H}_4$ is the two-sum, at a hinge $g$, of the parallel cell $\mathcal{P}$ of $\{a,b\}$, with the core $\binom{\{g,a,b\}}{1}$, and the series cell $\mathcal{S}$ of $\{2,3\}$, with the core $\binom{\{g,2,3\}}{2}$: the bases $\{2,3\}$ and $\{x,y\}$, $x \in \{a,b\}$, $y \in \{2,3\}$, are those of the two-sum (Definition 23.15, Theorem 23.16). At the centre of real strength $\sigma > 0$, with the masses $\sigma$ and $s = t = \sigma$, every mark is a one-mark cell of ratio 1, whose transform and dual transform are $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(1,y)$ (Definition 23.19(1)), and Theorem 23.14 gives

$$f_{\mathcal{S}} = \mathsf{g}^2, \quad \kappa_{\mathcal{S}} = 2\sigma - \sigma = \sigma; \qquad \check{f}_{\mathcal{P}} = \mathsf{g}^2, \quad \kappa_{\mathcal{P}} = 2\sigma - \sigma = \sigma.$$

The root of two one-mark leaves at the strength $z$ (Definition 23.19(5)), whose transforms are again $\mathsf{g}$, and the Beta integral give

$$\int_0^\infty y^{z-1}\mathsf{g}(y)^2\,dy = \Gamma(z)^2 P_2(z),$$

$$P_2(z) = P_{\binom{[2]}{1}}(z\mathbf{1}; z, z) = \int_0^\infty x^{z-1}(1+x)^{-2z}\,dx = \frac{\Gamma(z)^2}{\Gamma(2z)} \qquad (z>0).$$

By Proposition 23.8(iii), with $s = t = \kappa = \sigma$, the two Mellin transforms of the laws are $\int \varrho_{\mathcal{S}}^{-z} d\phi_{\mathcal{S}} = P_2(z)$ and $\int \varrho_{\mathcal{P}}^{z-\sigma} d\phi_{\mathcal{P}} = P_2(z)$ for $z > 0$. Theorem 23.9(iv) for one table, with $\mathcal{S}$ in the place of $\mathcal{A}$ and $\mathcal{P}$ in the place of $\mathcal{A}'$, on a line $\mathrm{Re}\, z = c$ with $0 < c < \sigma$, gives

$$\Gamma(\sigma)^2 P(\sigma) = \frac{1}{2\pi i}\int_{(c)} \Gamma(z)^2\,\Gamma(\sigma-z)^2\,P_2(z)^2\,dz = \frac{1}{2\pi i}\int_{(c)} \mathcal{F}(z,\sigma)\,dz, \qquad \mathcal{F} = \frac{\Gamma(z)^6\,\Gamma(\sigma-z)^2}{\Gamma(2z)^2},$$

and both sides are holomorphic in $\sigma$ on $\operatorname{Re}\sigma > c$.

Closing (Theorem 4.5) moves the line across the poles of the Gammas on one side and needs the contribution of the moved line to tend to zero. At an integer strength $k \geq 1$, by the shift and the reflection of Proposition 1.3(iii),(v),

$$\Gamma(z)\,\Gamma(k-z) = \frac{\pi\,(z-1)(z-2)\cdots(z-k+1)}{\sin\pi(z-k+1)},$$
$$\mathcal{F}(z,k) = \frac{\pi^2\,V_k(z)\,P_2(z)^2}{\sin^2\pi(z-k+1)}, \qquad V_k(z) = \prod_{j=1}^{k-1}(z-j)^2.$$

On the line $\operatorname{Re} z = k+N+\frac12$, $|\sin\pi(z-k+1)| = \cosh(\pi \operatorname{Im} z)$, $|P_2(z)| \leq \int_0^1 \big(w(1-w)\big)^{\operatorname{Re} z-1} dw \leq 4^{1-\operatorname{Re} z}$ and $V_k(z)$ is a polynomial, so the contribution of that line is $O(N^{2k-2}16^{-N})$ and tends to zero. To the right the poles are the integers $x \geq k$, double, with the principal part $(z-x)^{-2}$ of $\pi^2/\sin^2$ and no simple part (Lemma 5.16), and closing to the right gives

$$\Gamma(k)^2 P(k) = -\sum_{x\geq k} \big(V_k P_2^2\big)'(x),$$
$$P(1) = \sum_{m\geq 0} \frac{4\,(m!)^4}{((2m+1)!)^2}\big(h_{2m+1} - h_m\big) = 4 + \frac{5}{54} + \frac{47}{13500} + \frac{319}{2058000} + \cdots,$$

with $h_n = \sum_{i=1}^n i^{-1}$, since $P_2'/P_2 = 2(\log\Gamma)'(x) - 2(\log\Gamma)'(2x)$ and $(\log\Gamma)'(n+1) - (\log\Gamma)'(1) = h_n$ (Proposition 1.3(vi)). The terms are rational and decrease like $16^{-x}$: the share side of $\mathrm{H}_4$ closes on the lattice of the integers to its right.

To the left the poles are at $z = -m$, $m \in \mathbb{N}$, of order four: $\Gamma(z)^6$ has order six there and $\Gamma(2z)^{-2}$ a double zero. With the duplication formula $\Gamma(2z) = \pi^{-1/2}2^{2z-1}\Gamma(z)\Gamma(z+\frac12)$ and the reflection,

$$\mathcal{F}(z,\sigma) = 4\pi^3\,16^{-z}\,\frac{\cos^2\pi z}{\sin^4\pi z}\,\mathcal{B}(z)^2, \qquad \mathcal{B}(z) = \frac{\Gamma(\frac12 - z)\,\Gamma(\sigma - z)}{\Gamma(1-z)^2},$$

so the residue at $-m$ is $16^m\,\mathcal{B}(-m)^2\big(c + O(m^{-1})\big)$ with $c = \frac{2}{3\pi}\log 16\,\big(2\pi^2 - (\log 16)^2\big) = 7.0911\ldots$: it is $4\pi^{-1}16^m/3!$ times the third derivative at 0 of $16^{-\varepsilon}\mathcal{B}(\varepsilon - m)^2(\pi\varepsilon)^4\cos^2(\pi\varepsilon)\sin^{-4}(\pi\varepsilon)$, in which the derivatives of $\log\mathcal{B}$ at $-m$ are $O(m^{-1})$; and $\mathcal{B}(-m)^2$ grows like $m^{2\sigma-3}$. At $\sigma = 1$ the ratios of consecutive residues are 14.5771, 15.2453, 15.4869, 15.6114 at $m = 10, 20, 30, 40$, against $16\big((m+\frac12)/(m+1)\big)^2 = 14.5785$, 15.2472, 15.4880, 15.6121, and the residue over $16^m\mathcal{B}(-m)^2$ is 7.1273, 7.1173, 7.1104, 7.1063. The residues grow like $16^m$, and the closing to the left diverges.

*(v) The low end.* By (iii) the law holds as an identity of meromorphic functions. Near $k = 0$ the terms $i \geq 1$ are holomorphic, $c_0$ vanishes to order three, $[k^3]c_0 = 1075$, and $\lim_{k\to 0} k^3 P(k) = V_1\,\mathsf{C}_0(1) = 16$ (Proposition 39.34); the value of the law at 0 is the low-end relation, by the argument of Corollary 39.38(ii), which uses of the law only that it holds as an identity of meromorphic functions:

$$17200 - 7160\,P(1) + 274520\,P(2) - 1017720\,P(3) + 826200\,P(4) = 0.$$

At $k = -1$ the coefficients $c_0$ and $c_1$ vanish, simply, while $c_2(-1) = -342$, $c_3(-1) = 4842$, $c_4(-1) = -6300$, so the value of the law at $-1$ is

$$\operatorname*{CT}_{k=-1}\big(c_0(k)P(k) + c_1(k)P(k+1)\big) - 342\,P(1) + 4842\,P(2) - 6300\,P(3) = 0,$$

and the first term needs the principal parts of $P$ at $-1$ and at 0. The share side gives them. On the line $\operatorname{Re} z = \frac12$ the bound $|\Gamma(c'+i\tau)| \leq C_{c'}e^{-\pi|\tau|/2}(1+|\tau|)^{c'-1/2}$, with $1/\Gamma(2z) = \Gamma(1-2z)\sin(2\pi z)/\pi$,

makes $|\mathcal{F}(z,\sigma)|$ at most a constant times $(1+|\operatorname{Im} z|)^{2\operatorname{Re}\sigma}e^{-2\pi|\operatorname{Im} z|}$ for $|\operatorname{Im} z|\ge 1$, uniformly for $\sigma$ in compact sets; so $J(\sigma)=\frac{1}{2\pi i}\int_{(1/2)}\mathcal{F}\,dz$ is holomorphic wherever no pole of $\Gamma(\sigma-z)^2$ lies on the line, in particular on the strips $|\operatorname{Re}\sigma|<\frac12$ and $-\frac32<\operatorname{Re}\sigma<-\frac12$. For $\operatorname{Re}\sigma>\frac12$, $\Gamma(\sigma)^2P(\sigma)=J(\sigma)$. As $\sigma$ moves left the double poles $z=\sigma$ and then $z=\sigma+1$ of $\Gamma(\sigma-z)^2$ cross the line, and the identity of meromorphic functions that continues it is

$$\Gamma(\sigma)^2P(\sigma)=J(\sigma)-R_0(\sigma)\qquad(|\operatorname{Re}\sigma|<\tfrac12),$$
$$\Gamma(\sigma)^2P(\sigma)=J(\sigma)-R_0(\sigma)-R_1(\sigma)\qquad(-\tfrac32<\operatorname{Re}\sigma<-\tfrac12),$$

with the residues, by Proposition 5.14 at $p=2$,

$$R_m(\sigma)=\operatorname*{Res}_{z=\sigma+m}\mathcal{F}=\frac{\Gamma(\sigma+m)^6}{(m!)^2\,\Gamma(2\sigma+2m)^2}\Big(6(\log\Gamma)'(\sigma+m)-4(\log\Gamma)'(2\sigma+2m)-2(\log\Gamma)'(m+1)\Big).$$

$\Gamma(\sigma)^{-2}$ vanishes to order two at 0 and at $-1$, so $J$ enters $P$ there only from the order $(\sigma-\sigma_0)^2$ on, and the principal parts of $P$ are those of $-R_0\,\Gamma^{-2}$ and of $-(R_0+R_1)\,\Gamma^{-2}$. Expanding every Gamma factor at its pole or regular point by Proposition 5.14, in $(\log\Gamma)'(1)$ and the Trailmoments $(\mathsf{T}_{n;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ of the counting input, gives $R_0(x)=-16x^{-5}-32(\log\Gamma)'(1)\,x^{-4}+O(x^{-3})$ near 0 and $(R_0+R_1)(-1+x)=-80x^{-5}+O(x^{-4})$ near $-1$, and, the terms in $(\log\Gamma)'(1)$ cancelling,

$$P(x)=\frac{16}{x^3}-\frac{24\,(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))}{x}+O(1),$$
$$P(-1+x)=\frac{80}{x^3}-\frac{152}{x^2}-\frac{120\,(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))}{x}+O(1),$$

with $(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))=\sum_{n\ge1}n^{-2}=\frac16\Gamma(\frac12)^4$ (Lemma 5.16 at $a=\frac12$). The top coefficients are $V_1\,\mathsf{C}_0(1)=16$ and $V_1\,\mathsf{C}_1(1)=80$ (Proposition 39.34), and the coefficient 0 of $x^{-2}$ at 0 is the low-end coefficient $\mathsf{L}_1$, which vanishes for a list (Proposition 39.39, every weight being 1). With

$$c_0(-1+x)=-102x-91x^2+453x^3+O(x^4),\qquad c_1(-1+x)=510x-514x^2-3056x^3+O(x^4),$$

and $T=(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, the coefficients of $x^{-2}$ and $x^{-1}$ in $c_0(k)P(k)+c_1(k)P(k+1)$ at $k=-1+x$ are $-102\cdot80+510\cdot16=0$ and $-102\cdot(-152)-91\cdot80-514\cdot16=0$, as the law asks, and its constant term is

$$-102\cdot(-120T)-91\cdot(-152)+453\cdot80+510\cdot(-24T)-514\cdot0-3056\cdot16=1176:$$

the multiples of $\Gamma(\frac12)^4$ cancel. Hence $-342\,P(1)+4842\,P(2)-6300\,P(3)=-1176$, that is

$$57\,P(1)-807\,P(2)+1050\,P(3)=196,$$

the second low-end relation.

*(vi) Values and their span.* The share side at $\sigma=1,\dots,5$, integrated on the line $\operatorname{Re}z=\frac12$ in ball arithmetic, and the closing of (iv), summed over its first 80 terms at $k=1,2,3$, agree to 80 digits and give

$$P(1)=4.0962369983378828500555532283512\ldots,$$
$$P(2)=0.047716498691513110297412080890663\ldots;$$

the law at $k=1$, the low-end relation and the second low-end relation hold at these values to 75 digits. For $k\ge1$, $c_4(k)>0$, so the law gives $P(k+4)$ as a rational combination of

$P(k),\dots,P(k+3)$; the second low-end relation gives $P(3)$, and then the low-end relation $P(4)$, in $\mathbb{Q}+\mathbb{Q}\,P(1)+\mathbb{Q}\,P(2)$. So every value lies in $\mathbb{Q}+\mathbb{Q}\,P(1)+\mathbb{Q}\,P(2)$. An integer-relation search at 290 bits finds no relation among 1, $P(1)$, $P(2)$ with coefficients below $10^{25}$.

*Answer.* $V_1=16$. The native gauges are $\mathbf{1}$ (native value $\frac{1}{25}$, $\det H=\frac{32}{625}$), the sign $(1,1,-1,-1)$ ($\frac{1}{9}$, $\det H=-\frac{32}{81}$) and three curves of native value 1. The closings are $\sum_u \binom{j}{u}^2\binom{2u}{u}^2$; the law is the display of (iii), of symbol $144(25\lambda-1)(9\lambda-1)(\lambda-1)^2$, proved by the telescoping identity and Theorem 39.57. On the share side $\Gamma(\sigma)^2P(\sigma)=\frac{1}{2\pi i}\int\Gamma(z)^6\Gamma(\sigma-z)^2\Gamma(2z)^{-2}dz$; it closes to the right, $\Gamma(k)^2P(k)=-\sum_{x\ge k}(V_kP_2^2)'(x)$, and not to the left. The low-end relation is $17200-7160P(1)+274520P(2)-1017720P(3)+826200P(4)=0$, the second low-end relation $57P(1)-807P(2)+1050P(3)=196$, and the values lie in $\mathbb{Q}+\mathbb{Q}\,P(1)+\mathbb{Q}\,P(2)$. □

**Exercise 55.2** ($H_4$ on a ray with a conjugate on the slit)**.** On $H_4$ take the masses $\mathbf{m}_1=(1,1,3,3)$ on $(a,b,2,3)$ and the letter exponents $s_1=3$, $t_1=1$, and let $P(k)=\langle1\rangle_k=P_{H_4}\big(k\mathbf{m}_1;3k,k\big)$, $k\ge1$, be the values of the ray with the origin 0 and the direction $(\mathbf{m}_1;3,1)$.

(i) Show that the direction lies on the balance wall and strictly inside its walls, and compute $V_1$.

(ii) Find the native gauge $\alpha^\star$ of the direction on the positive orthant, $f^\star=f(\alpha^\star)$ and its conjugate over $\mathbb{Q}$, and show that the conjugate is the native value of a nondegenerate native gauge and lies on the slit $(0,f^\star)$.

(iii) Find the closings and a law of the values of order four, prove the law, and give its symbol.

(iv) Write the low-end relation, and show that Theorem 41.3(iii) does not apply.

(v) Read $P(1)$ and $P(2)$ from the gauge end (Theorem 40.6), and check them against the low-end relation.

(vi) Write $\Gamma(3k)\Gamma(k)P(k)$ on the share side, read the second low-end relation from it, and give the value span.

*Solution. (i) Walls and $V_1$.* The direction has $m_{1,\mathcal{E}}=8=3r+1\cdot r^*=3\cdot2+2$, on the balance wall. Its walls are $\kappa_{1,S}=m_{1,S}-3d(S)-d^*(S)$ (Definition 8.1), with $d(S)=|S|-r^*(S)$ and $d^*(S)=|S|-r(S)$ (Definition 7.12). A single mark has $d=d^*=0$; $\{a,b\}$ has $d=0$, $d^*=1$; $\{2,3\}$ has $d=1$, $d^*=0$; the four pairs $\{x,y\}$, $x\in\{a,b\}$, $y\in\{2,3\}$, have $d=d^*=0$; every triple has $d=d^*=1$. So

$$\kappa_{1,S}=1\quad\text{for } S\in\big\{\{a\},\{b\},\{a,b\},\{a,b,2\},\{a,b,3\}\big\},$$
$$\kappa_{1,S}=3\quad\text{for } S\in\big\{\{2\},\{3\},\{2,3\},\{a,2,3\},\{b,2,3\}\big\},$$

and $\kappa_{1,S}=4$ for the four pairs $\{x,y\}$: all positive. An order of the marks is the chain of its tails $U_2\supset U_3\supset U_4$, and $V_1=\sum\big(\kappa_{1,U_2}\kappa_{1,U_3}\kappa_{1,U_4}\big)^{-1}$ (Proposition 39.34). Below $U_2=\{a,b,y\}$ the pairs and their marks give $1\cdot(1+1)+2\cdot\frac14(1+\frac13)=\frac83$; below $U_2=\{x,2,3\}$ they give $2\cdot\frac14(1+\frac13)+\frac13(\frac13+\frac13)=\frac89$. So

$$V_1=2\cdot\frac83+2\cdot\frac13\cdot\frac89=\frac{160}{27}.$$

*(ii) The native gauge and its conjugate.* The exchanges $a\leftrightarrow b$ and $2\leftrightarrow3$ keep the direction, and $\alpha^\star$ is unique up to scale (Theorem 38.14(i)), so $\alpha^\star=(x,x,1,1)$ for some $x>0$. On that slice $K=4x+1$, $\Psi=x(x+4)$ and

$$f=\frac{\alpha_a\alpha_b\alpha_2^3\alpha_3^3}{K^3\Psi}=\frac{x}{(x+4)(4x+1)^3},\qquad\frac{d}{dx}\log f=\frac{-4(3x^2+8x-1)}{x(x+4)(4x+1)},$$

so $x^\star = \frac{\sqrt{19}-4}{3}$, where the gradient of $\log f$ vanishes in all four coordinates, and

$$f^\star = \frac{7643+1064\sqrt{19}}{1366875} = 0.0089846317\ldots, \qquad \det H = \frac{376096\sqrt{19}-1614848}{151875} = 0.16142523\ldots,$$

$H$ the Hessian of $-\log f$ on the chart $y_3 = 0$; $f^\star$ is a root of $1366875\lambda^2 - 15286\lambda + 27$. By Proposition 38.10 the conjugate point $\alpha' = (x', x', 1, 1)$, $x' = \frac{-\sqrt{19}-4}{3} = -2.786\ldots$, where $K = \frac{-13-4\sqrt{19}}{3}$ and $\Psi = x'(x'+4)$ do not vanish, is a nondegenerate native gauge, with

$$f(\alpha') = \frac{7643-1064\sqrt{19}}{1366875} = 0.0021985415\ldots,$$
$$\det H(\alpha') = \frac{-376096\sqrt{19}-1614848}{151875} = -21.4269\ldots,$$

and $0 < f(\alpha') < f^\star$: the conjugate lies on the slit, at a gauge where the class $\{a, b\}$ is flipped.

*(iii) Closings and the law.* Every letter exponent of the direction is positive and the origin is 0, so $\mathsf{C}_j(1) = \mathrm{CT}(f^{-j}) = [\alpha^{j\mathbf{m}_1}] K^{3j}\Psi^j$ (Definition 39.30). In

$$K^{3j}\Psi^j = \sum_{u,v} \binom{3j}{u}\binom{j}{v} (AB)^{4j-u-v} (\alpha_2\alpha_3)^u (\alpha_a\alpha_b)^v$$

the monomial $\alpha_a^j\alpha_b^j\alpha_2^{3j}\alpha_3^{3j}$ takes $(\alpha_a\alpha_b)^{j-v}$ from $A^{4j-u-v}$ and $(\alpha_2\alpha_3)^{3j-u}$ from $B^{4j-u-v}$, so $u = 2j+v$, with the coefficient $\binom{2j-2v}{j-v}^2$; with $w = j-v$,

$$\mathsf{C}_j(1) = \sum_{w=0}^{j} F(j,w), \qquad F(j,w) = \binom{3j}{w}\binom{j}{w}\binom{2w}{w}^2;$$
$$\mathsf{C}_j(1) = 1,\ 13,\ 589,\ 37597,\ 2791949,\ \ldots$$

On the first 300 closings there is no recurrence of order one, two, three with degree at most 138, 91, 68, and none of order four with degree at most 24; at order four and degree 25 exact linear algebra on the first 170 closings gives one line, spanned by $p_0, \ldots, p_4 \in \mathbb{Z}[j]$ with coefficients of up to 36 digits, and the recurrence holds on all 300. It holds for every $j$. For $0 \le w \le j+4$ put

$$T(j,w) = \frac{((2w)!)^2}{(w!)^6\,(3j+12-w)!\,(j+4-w)!},$$
$$G(j,w) = (3j)!\,j!\,T(j,w)\,A(j,w), \qquad G(j,j+5) = 0,$$

with $A \in \mathbb{Q}[j,w]$, a polynomial of 416 terms, of degree 16 in $w$ and divisible by $w^4$, printed by the receipt. As $F(j+i,w) = (3j)!\,j!\,T(j,w)\prod_{l=1}^{3i}(3j+l)\prod_{l=1}^{i}(j+l)\prod_{l=3i+1}^{12}(3j+l-w)\prod_{l=i+1}^{4}(j+l-w)$ and $T(j,w+1) = T(j,w)\,4(2w+1)^2(3j+12-w)(j+4-w)/(w+1)^4$, the identity $\sum_i p_i(j)F(j+i,w) = G(j,w+1) - G(j,w)$ for $0 \le w \le j+4$ is the identity of polynomials

$$(w+1)^4 N(j,w) = 4(2w+1)^2(3j+12-w)(j+4-w)A(j,w+1) - (w+1)^4 A(j,w),$$

$N = \sum_i p_i(j)\prod_{l=1}^{3i}(3j+l)\prod_{l=1}^{i}(j+l)\prod_{l=3i+1}^{12}(3j+l-w)\prod_{l=i+1}^{4}(j+l-w)$, which holds in $\mathbb{Q}[j,w]$; summed over $w$ it telescopes to 0, as $A(j,0) = 0$. As in Exercise 55.1(iii), Theorems 39.33 and 39.57 turn it into the law

$$\sum_{i=0}^{4} c_i(k)\,P(k+i) = 0, \qquad c_i(k) = -p_{4-i}(-k-4),$$

an identity of meromorphic functions of $k$. The coefficients factor as

$$c_0 = 3k^3(k+1)(3k+1)^3(3k+2)^3a_0, \qquad c_1 = -2(k+1)a_1, \qquad c_2 = 2(3k+4)(3k+5)a_2,$$
$$c_3 = -2(3k+4)(3k+5)(3k+7)(3k+8)(k+2)^2a_3,$$
$$c_4 = 151875(3k+4)(3k+5)(3k+7)(3k+8)(3k+10)(3k+11)(k+2)^2(k+3)^2a_4,$$

with $a_i \in \mathbb{Z}[k]$ of degrees $15, 24, 23, 19, 15$, printed by the receipt, and the symbol is

$$\chi(\lambda) = 168132676253771376\,(\lambda-1)^2\,(1366875\lambda^2 - 15286\lambda + 27)\,:$$

$f^\star$, its conjugate, and 1, doubled. The root 1 is the native value of the line $\{A = B = 0\}$, a source of dimension one (Definition 39.23): there $K = \alpha_2\alpha_3$ and $\Psi = \alpha_a\alpha_b$, the occupations of $K$ and of $\Psi$ are $(0,0,1,1)$ and $(1,1,0,0)$ on $(a,b,2,3)$, so the usage $3R + R^*$ is $\mathbf{m}_1$, and $f = 1$.

*(iv) The low end and positive selection.* $c_0$ vanishes to order three at 0 with $[k^3]c_0 = 656679698393897858976$, and with $V_1 = \frac{160}{27}$ the argument of Corollary 39.38(ii) gives the low-end relation, divided by 15360,

$$253348649071719853 + 204923553451974000\,P(1) - 272918959559705480400\,P(2)$$
$$+ 1177478827066019041200\,P(3) - 1011712684533185250000\,P(4) = 0.$$

Theorem 41.3 asks the roots of the symbol to be distinct, which the double root 1 already violates, and (iii) asks every root other than $f^\star$ to have modulus greater than $f^\star$. The conjugate $f(\alpha')$ lies in $(0, f^\star)$, so a solution of the law that grows like $f(\alpha')^k$ grows more slowly than the values, and growth does not separate it from them: the hypothesis fails. Set up anyway, the equations of Theorem 41.3(iv), with $P(1), P(2), P(3)$ free, $P(4)$ from the low-end relation and the solution required to vanish at $K, K+1, K+2$, return $P(1) = -2.6 \cdot 10^{23}$ at $K = 40$ and $-7.4 \cdot 10^{47}$ at $K = 80$.

*(v) The gauge end.* $f^\star$ is a simple root of $\chi$, and the density operator $\mathfrak{D}_{\mathbf{p}}$ of the law (Theorem 40.6) has order $D = 25$; at $f^\star$ its coefficient $b_{D,0}$ vanishes and the extra exponent is $\gamma = D - 1 - b_{D-1,0}/b_{D,1} = \frac{1}{2} = \frac{n-3}{2}$, as Theorem 40.6(iii) requires. As $\gamma \notin \{0, \dots, 23\}$, the germ of the density is $\varrho_f(t) = s^{1/2}g(0)\Phi(s)$, $s = f^\star - t$, with $\Phi$ fixed by the recursion of Theorem 40.6(iv), 1800 terms computed over $\mathbb{Q}(\sqrt{19})$ in ball arithmetic, and the scale read off the Hessian at $\alpha^\star$ (the end of the stride's law at $f^\star$, Definition 40.1, with $(\omega_0/dy)(\alpha^\star) = 1$):

$$g(0) = \frac{1}{\Gamma(\frac{3}{2})\sqrt{\det H}}\Big(\frac{2\pi}{f^\star}\Big)^{3/2} = 51938.4198592508155915\dots$$

By Theorem 40.6(v) the density continues holomorphically along $(0, f^\star)$, through the root $f(\alpha')$ of the symbol, so the series of $\Phi$ converges on $[0, f^\star)$: the coefficients $g_m$ of $g = g(0)\Phi$ have $|g_m|(f^\star)^m \le 5.2 \cdot 10^5$ for $m \le 1800$. The Beta integral gives

$$P(k) = \int_0^{f^\star} t^k \varrho_f(t)\,dt = \sum_{m \ge 0} g_m\,(f^\star)^{k+\frac{3}{2}+m}\,\mathrm{B}\big(k+1, m+\tfrac{3}{2}\big),$$

read at $k = K, \dots, K+3$, and the law, with $c_0(k) \neq 0$ at every integer $1 \le k \le 200$, runs back to $k = 1$. From $K = 80$ and from $K = 120$ the readings agree to 92 digits:

$$P(1) = 0.48830319489526888388541464396010900874354031922042\dots,$$

$$P(2) = 0.00132119179498773222717641714721510065075969610885\ldots$$

The low-end relation, which the reading does not use, returns $V_1 = \frac{160}{27}$ from these values to 191 digits.

*(vi) The share side and the second low-end relation.* The widths of (i) of Exercise 55.1 do not depend on the direction, so the hinge tree is the same: the series cell $\mathcal{S}$ of $\{2,3\}$, whose two marks have the masses $3k$, and the parallel cell $\mathcal{P}$ of $\{a,b\}$, with the masses $k$, at $s = 3k$, $t = k$. Every mark is again a one-mark cell whose transform, or dual transform, is $\mathsf{g}$ (Definition 23.19(1): $s = m_2 = m_3$, $t = m_a = m_b$), the hinge walls are $\kappa_{\mathcal{S}} = 6k - 3k = 3k$ and $\kappa_{\mathcal{P}} = 2k - k = k$, the walls of $\{2,3\}$ and $\{a,b\}$ (Theorem 23.14), and Proposition 23.8(iii) gives $\int \varrho_{\mathcal{S}}^{-z} d\phi_{\mathcal{S}} = P_2(z) = \int \varrho_{\mathcal{P}}^{z-k} d\phi_{\mathcal{P}}$. Theorem 23.9(iv), on a line $\operatorname{Re} z = c$, $0 < c < k$, gives

$$\Gamma(3k)\Gamma(k)\,P(k) = \frac{1}{2\pi i}\int_{(c)} \frac{\Gamma(z)^6\,\Gamma(k-z)\,\Gamma(3k-z)}{\Gamma(2z)^2}\,dz,$$

which at $k = 1$, on $c = \frac{1}{2}$ in ball arithmetic, agrees with (v) to 60 digits. At $k = -1$ the coefficients $c_0$ and $c_1$ vanish, simply, and the second low-end relation needs the principal parts of $P$ at 0 and $-1$. As in Exercise 55.1(v), the integral on $\operatorname{Re} z = \frac{1}{2}$ is holomorphic near $k = 0$ and $k = -1$, $\Gamma(3k)\Gamma(k)$ has a double pole at both, and the principal parts of $P$ come from the poles that cross the line, $z = k$ and $z = 3k$ near 0, and $z = k, k+1$ and $z = 3k, \ldots, 3k+3$ near $-1$, each simple, with the residues $(-1)^{m+1}\Gamma(2k-m)\,\Gamma(k+m)^6/\big(m!\,\Gamma(2k+2m)^2\big)$ at $z = k+m$ and $(-1)^{m+1}\Gamma(-2k-m)\,\Gamma(3k+m)^6/\big(m!\,\Gamma(6k+2m)^2\big)$ at $z = 3k+m$ (Proposition 1.3(iv)). Expanded by Proposition 5.14, with $T = (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$,

$$P(x) = \frac{160}{27x^3} - \frac{208\,T}{9x} + O(1), \qquad P(-1+x) = \frac{2080}{27x^3} - \frac{21118}{81x^2} - \frac{2704\,T + \frac{887}{9}}{9x} + O(1),$$

with the top coefficients $V_1\mathsf{C}_0(1)$ and $V_1\mathsf{C}_1(1) = \frac{160}{27}\cdot 13$ (Proposition 39.34). With the expansions of $c_0$ and $c_1$ at $-1$ to the order three, the multiples of $T$ cancel in the constant term at $k = -1$, which is $-2077616120857361856$, and $(c_2, c_3, c_4)(-1) = 625190400\cdot(7005765879, -74409017489, 88989468750)$; so

$$7005765879\,P(1) - 74409017489\,P(2) + 88989468750\,P(3) = \frac{5981712159277}{1800},$$

which the values of (v) satisfy to 109 digits. For $k \geq 1$, $c_4(k) > 0$ (its real roots are below $\frac{1}{9}$), so, as in Exercise 55.1(vi), every value lies in $\mathbb{Q} + \mathbb{Q}\,P(1) + \mathbb{Q}\,P(2)$.

*Answer.* $V_1 = \frac{160}{27}$; $f^\star = \frac{7643+1064\sqrt{19}}{1366875}$ at $\alpha^\star = (x^\star, x^\star, 1, 1)$, $x^\star = \frac{\sqrt{19}-4}{3}$, $\det H = \frac{376096\sqrt{19}-1614848}{151875}$; its conjugate $\frac{7643-1064\sqrt{19}}{1366875}$ is the native value of the nondegenerate native gauge $(x', x', 1, 1)$, $x' = \frac{-\sqrt{19}-4}{3}$, and lies on the slit. The closings are $\sum_w \binom{3j}{w}\binom{j}{w}\binom{2w}{w}^2$, the law has order four, degree 25 and the symbol $(\lambda-1)^2(1366875\lambda^2 - 15286\lambda + 27)$ up to a constant, positive selection does not apply, and the gauge end gives $P(1) = 0.4883031948952688838854\ldots$ The second low-end relation is $7005765879P(1) - 74409017489P(2) + 88989468750P(3) = \frac{5981712159277}{1800}$, and the value span is $\mathbb{Q} + \mathbb{Q}\,P(1) + \mathbb{Q}\,P(2)$. □

## 55.2 The five-mark lists

In this section the marks are $1, \ldots, 5$, every weight is 1, and $\ell = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ is the value at the strength one of the centre ray of the list $\binom{[3]}{1}$, $P_3(\sigma) = P_{\binom{[3]}{1}}(\sigma\mathbf{1}; \sigma, \sigma)$ $P_2(\sigma) = P_{\binom{[2]}{1}}(\sigma\mathbf{1}; \sigma, \sigma) =$

$\Gamma(\sigma)^2/\Gamma(2\sigma)$, and $\mathsf{g}(y) = \mathsf{G}_{0;(1,-1)}(1,y)$ as in Exercise 55.1(iv). The lists are $\binom{[5]}{1}$ and $\binom{[5]}{2}$, and three lists of rank two named by their parallel classes:

- $\mathrm{P}_5$, whose bases are the nine pairs other than $\{1,2\}$: the classes $\{1,2\},\{3\},\{4\},\{5\}$;
- $\mathrm{R}_5$, whose bases are the eight pairs $\{x,y\}$ with $x,y$ in different classes of $\{1,2\},\{3,4\},\{5\}$;
- $\mathrm{T}_5$, whose bases are the seven pairs $\{x,y\}$ with $x,y$ in different classes of $\{1,2,3\},\{4\},\{5\}$.

**Exercise 55.3** (The five-mark lists along their hinge trees)**.** (i) Find, for each of the five lists, the sets of width one with at least two marks on each side, and its canonical hinge tree.

(ii) Show that $P_{\mathrm{P}_5}(\sigma\mathbf{1};\sigma,\sigma) = P_3(\sigma)^2$. Prove the law of $P_3$, and derive from it the law of the centre ray of $\mathrm{P}_5$, its symbol, its low-end relation and its value span.

(iii) Show that $q(\sigma) = P_{\binom{[4]}{2}}(\sigma\mathbf{1};\sigma,\sigma) = P_2(\sigma)P_3(\sigma)$, and find the law of $q$.

(iv) Write the centres of $\mathrm{T}_5$ and $\mathrm{R}_5$ on their share sides, close $\mathrm{T}_5$ at the integer strengths, and write $P_{\mathrm{T}_5}(\mathbf{1};1,1)$ in the coordinates 1, $\ell$ and $\ell' = \partial_\sigma P_3(\sigma)|_{\sigma=1}$.

*Solution. (i) Widths and trees.* The width of $S$ is $c(S) = r(S) + r(\mathcal{E}\setminus S) - r$ (Definition 7.12). In $\binom{[5]}{1}$ every nonempty set has rank one, so every proper $S$ has width 1, and the list is one parallel node. In $\binom{[5]}{2}$ every set of two or three marks has rank two and so has its complement: width 2, and the list is prime (Theorem 23.16); so is $\binom{[4]}{2}$, whose pairs have width 2. In a list of rank two whose classes are the parallel classes, a set of two or three marks has width one exactly when it, or its complement, lies in one class, and width two otherwise. So the sets of width one with at least two marks on each side are $\{1,2\}$ for $\mathrm{P}_5$; $\{1,2\}$ and $\{3,4\}$ for $\mathrm{R}_5$; $\{1,2,3\}$ and its pairs for $\mathrm{T}_5$; each with its complement. The canonical hinge trees (Definition 23.15, Theorem 23.16) are:

- $\binom{[5]}{1}$: one parallel node; $\binom{[5]}{2}$: one prime node;
- $\mathrm{P}_5$: the parallel node $\binom{\{g,1,2\}}{1}$ and the prime node $\binom{\{g,3,4,5\}}{2}$;
- $\mathrm{R}_5$: the series node $\binom{\{g_1,g_2,5\}}{2}$ with the parallel nodes $\binom{\{g_1,1,2\}}{1}$ and $\binom{\{g_2,3,4\}}{1}$;
- $\mathrm{T}_5$: the series node $\binom{\{g,4,5\}}{2}$ with the parallel node $\binom{\{g,1,2,3\}}{1}$ of four marks.

For $\mathrm{P}_5$, for instance, the bases of the two-sum are $\{x,y\}\subseteq\{3,4,5\}$, from $\{g\}$ and a pair of $\{3,4,5\}$, and $\{1,y\}$, $\{2,y\}$, $y\in\{3,4,5\}$, from $\{1\}$ or $\{2\}$ and $\{g,y\}$: the nine pairs other than $\{1,2\}$.

*(ii) The doubled mark of* $\binom{[4]}{2}$. The cell $\mathcal{A} = \binom{\{g,3,4,5\}}{2}$ of the bunch $\{3,4,5\}$ has the hinge minors $\mathcal{A}\setminus g = \binom{\{3,4,5\}}{2}$ and $\mathcal{A}/g = \binom{\{3,4,5\}}{1}$, dual with the identity and the same weights, so by Corollary 23.12(ii) its ratio is identically 1 and $\phi_{\mathcal{A}}(\Delta) = P_{\binom{[3]}{2}}(\sigma\mathbf{1};\kappa_{\mathcal{A}},2\sigma-\kappa_{\mathcal{A}})$, with the hinge wall $\kappa_{\mathcal{A}} = \sigma\, c(\{3,4,5\}) = \sigma(2+1-2) = \sigma$. By Corollary 23.12(i) the cell acts in the gluing as $\phi_{\mathcal{A}}(\Delta)$ times one mark $e$ of ratio 1 and mass $\sigma$, and the parallel cell of $\{1,2\}$ glued to that mark is the list $\binom{\{1,2,e\}}{1}$. By Proposition 8.15 the dual table $\binom{[3]}{1}$ at the swapped exponents $(\sigma,\sigma)$ has the same period, so

$$P_{\mathrm{P}_5}(\sigma\mathbf{1};\sigma,\sigma) = P_{\binom{[3]}{2}}(\sigma\mathbf{1};\sigma,\sigma)\,P_{\binom{\{1,2,e\}}{1}}(\sigma\mathbf{1};\sigma,\sigma) = P_3(\sigma)^2.$$

The law of $P_3$. The centre ray of $\binom{[3]}{1}$ has $f^{-1} = (\alpha_1+\alpha_2+\alpha_3)(\alpha_1^{-1}+\alpha_2^{-1}+\alpha_3^{-1})$, and its closings are (Definition 39.30)

$$\mathsf{C}_j(1) = \sum_{a+b+c=j}\Big(\frac{j!}{a!\,b!\,c!}\Big)^2 = \sum_{u=0}^{j}\binom{j}{u}^2\binom{2u}{u} = 1,\ 3,\ 15,\ 93,\ 639,\ 4653,\ \ldots,$$

grouping $a+b=u$. No recurrence of order one and degree at most 40 holds on the first 120 closings, and exact linear algebra gives

$$(j+2)^2\,\mathsf{C}_{j+2}(1)-(10j^2+30j+23)\,\mathsf{C}_{j+1}(1)+9(j+1)^2\,\mathsf{C}_j(1)=0.$$

It holds for every $j$: with $F(j,u)=\binom{j}{u}^2\binom{2u}{u}$, $T(j,u)=((j+2)!)^2(2u)!/\big((u!)^4((j+2-u)!)^2\big)$ for $0\le u\le j+2$, $A(j,u)=u^3(j+1)^2(3u-4j-8)$ and $G(j,u)=T(j,u)A(j,u)/((j+1)^2(j+2)^2)$, $G(j,j+3)=0$, one has $F(j+i,u)=T(j,u)\prod_{l=i+1}^{2}\big((j+l-u)/(j+l)\big)^2$ and $T(j,u+1)=T(j,u)\,2(2u+1)(j+2-u)^2/(u+1)^3$, so the identity $\sum_i p_i(j)F(j+i,u)=G(j,u+1)-G(j,u)$, $p_0=9(j+1)^2$, $p_1=-(10j^2+30j+23)$, $p_2=(j+2)^2$, is the identity of polynomials

$$(u+1)^3\sum_{i=0}^{2}p_i(j)\prod_{l\le i}(j+l)^2\prod_{l>i}(j+l-u)^2=2(2u+1)(j+2-u)^2A(j,u+1)-(u+1)^3A(j,u),$$

which holds in $\mathbb{Q}[j,u]$; summed over $u$ it telescopes to 0. As in Exercise 55.1(iii), Theorems 39.33 and 39.57 give the law of the values, an identity of meromorphic functions,

$$k^2P_3(k)-(10k^2+10k+3)\,P_3(k+1)+9(k+1)^2\,P_3(k+2)=0,$$

of symbol $9\lambda^2-10\lambda+1=(9\lambda-1)(\lambda-1)$. The walls of the direction are all 1, so $V_1=3!=6$ (Proposition 39.34), and the low-end relation (Corollary 39.38(ii)) is $6-3P_3(1)+9P_3(2)=0$. So $P_3(1)=\ell$, $P_3(2)=\frac{\ell}{3}-\frac{2}{3}$, and the law gives

$$P_3(3)=\frac{5\ell}{27}-\frac{23}{54},\qquad P_3(4)=\frac{31\ell}{243}-\frac{145}{486},\qquad P_3(k)\in\mathbb{Q}+\mathbb{Q}\,\ell\quad(k\ge1).$$

The law of the squares. Write the law as $P_3(k+2)=a(k)P_3(k)+b(k)P_3(k+1)$; then $P_3(k+3)=a^{(1)}P_3(k)+b^{(1)}P_3(k+1)$ with $a^{(1)}=b(k+1)a(k)$ and $b^{(1)}=a(k+1)+b(k+1)b(k)$. The squares $x(k)=P_3(k)^2,\dots,x(k+3)$ are quadratic forms in $P_3(k),P_3(k+1)$, and the combination $d_0x(k)+d_1x(k+1)+d_2x(k+2)+x(k+3)$ with $d_2=-a^{(1)}b^{(1)}/(ab)$, $d_0=-d_2a^2-(a^{(1)})^2$, $d_1=-d_2b^2-(b^{(1)})^2$ vanishes identically. Cleared of denominators, the law of the centre ray of $\mathrm{P}_5$ is $\sum_{i=0}^{3}d_i(k)\,x(k+i)=0$ with

$$\begin{gathered}d_0=k^4(10k^2+30k+23),\qquad d_3=-729(k+2)^4(10k^2+10k+3),\\ d_1=-(10k^2+10k+3)\,w(k),\qquad d_2=9(10k^2+30k+23)\,w(k),\\ w(k)=91k^4+364k^3+506k^2+284k+60,\end{gathered}$$

which the squares of the values above satisfy identically in $\ell$ at $k=1,\dots,5$. Its symbol is $-10(\lambda-1)(9\lambda-1)(81\lambda-1)$, the products of the roots $\frac19$ and 1 in pairs; $\frac{1}{81}=K(\mathbf{1})^{-2}$, $K(\mathbf{1})=9$ the number of bases. The walls of $\mathrm{P}_5$ are 1 on single marks, on $\{1,2\}$ and on their complements, and 2 on the other sets of two or three marks, and $V_1=36$, the square of the 6 of $\binom{[3]}{1}$, as $P_3(k)^2\sim36k^{-4}$ at 0 requires. $d_0$ vanishes to order four at 0, $[k^4]d_0=23$, and the low-end relation

$$23\cdot36+d_1(0)\,x(1)+d_2(0)\,x(2)+d_3(0)\,x(3)=828-180\,x(1)+12420\,x(2)-34992\,x(3)=0$$

holds identically in $\ell$ at $x(i)=P_3(i)^2$. Every value $x(k)=P_3(k)^2$ lies in $\mathbb{Q}+\mathbb{Q}\,\ell+\mathbb{Q}\,\ell^2$. As a check, the procedure that reads a ray from its list by a box (Definition 41.8, Theorem 39.21), run on $\mathrm{P}_5$ by the supplied reader, returns the same law and $P_{\mathrm{P}_5}(\mathbf{1};1,1)=5.4939011435808440124442576078206911958\ldots$, which is $\ell^2$ to 87 digits.

*(iii)* $\binom{[4]}{2}$ *at its centre.* $\binom{[4]}{2}$ is identically self-dual: the complement of a pair is a pair, and the weights are 1. Corollary 20.19 at a mark, and Proposition 8.15 for the deletion $\binom{[3]}{2}$, give $q(\sigma) = \Gamma{}^{2}_{1}\!\left[\begin{smallmatrix}\sigma,\ \sigma\\ 2\sigma\end{smallmatrix}\right] P_3(\sigma) = P_2(\sigma)P_3(\sigma)$. With $P_2(\sigma{+}1)/P_2(\sigma) = \sigma/(2(2\sigma{+}1))$ (Proposition 1.3(iii)), the law of $P_3$ multiplied through by $\sigma P_2(\sigma)$ is the law of $q$:

$$\sigma^3 q(\sigma) - 2(2\sigma+1)(10\sigma^2+10\sigma+3)\,q(\sigma+1) + 36(\sigma+1)(2\sigma+1)(2\sigma+3)\,q(\sigma+2) = 0,$$

with the symbol $(4\lambda-1)(36\lambda-1)$ and $V_1 = 12 = 2\cdot 6$, the walls of $\binom{[4]}{2}$ being $1, 2, 1$ on sets of one, two and three marks.

*(iv) The triangle rows.* $\mathrm{T}_5$ is $\mathrm{H}_4$ of Section 55.1 with the class enlarged to $\{1,2,3\}$: the two-sum of the series cell $\mathcal{S}$ of $\{4,5\}$, with $f_{\mathcal{S}} = \mathsf{g}^2$ and $\kappa_{\mathcal{S}} = \sigma$, and the parallel cell $\mathcal{P}$ of $\{1,2,3\}$, with $\check{f}_{\mathcal{P}} = \mathsf{g}^3$ and $\kappa_{\mathcal{P}} = 3\sigma - 2\sigma = \sigma$ (Theorem 23.14). The root of three one-mark leaves (Definition 23.19(5)) gives $\int_0^\infty y^{z-1}\mathsf{g}^3 dy = \Gamma(z)^2 P_3(z)$, so Proposition 23.8(iii) gives $\int \varrho_{\mathcal{P}}^{z-\sigma} d\phi_{\mathcal{P}} = P_3(z)$, and $\int \varrho_{\mathcal{S}}^{-z} d\phi_{\mathcal{S}} = P_2(z)$ as in Exercise 55.1(iv). Theorem 23.9(iv) gives, for $0 < c < \sigma$,

$$\Gamma(\sigma)^2 P_{\mathrm{T}_5}(\sigma\mathbf{1};\sigma,\sigma) = \frac{1}{2\pi i}\int_{(c)} \Gamma(z)^2\,\Gamma(\sigma-z)^2\,q(z)\,dz.$$

$|q(z)| \le q(\operatorname{Re} z)$, the modulus of the integrand of a period being the integrand at the real parts, and $q(x) \le 4^{1-x}P_3(1)$ for $x \ge 1$; so the closing to the right of Exercise 55.1(iv) applies, the moved line contributing $O(N^{2s-2}4^{-N})$, and at an integer strength $s$

$$\Gamma(s)^2 P_{\mathrm{T}_5}(s\mathbf{1};s,s) = -\sum_{x\ge s} \big(V_s\,q\big)'(x), \qquad V_s(x) = \prod_{j=1}^{s-1}(x-j)^2.$$

At $s = 1$ this needs the slopes $q'(x) = P_2(x)\big(P_3'(x) + 2(h_{x-1} - h_{2x-1})P_3(x)\big)$. The law of $P_3$ holds as an identity of meromorphic functions, so its derivative in $k$ holds as well and gives $P_3'(k+2)$ from $P_3'(k)$, $P_3'(k+1)$ and the values, starting from $P_3'(1) = \ell'$ and $P_3'(2)$. Near $k = 0$, $k^2 P_3(k) = 6 + \mathsf{L}_1 k + O(k^2)$ with $\mathsf{L}_1 = 0$ (Proposition 39.39, every weight being 1), so the derivative of the law at $k = 0$ reads $-10\ell - 3\ell' + 18P_3(2) + 9P_3'(2) = 0$:

$$P_3'(2) = \frac{4\ell}{9} + \frac{\ell'}{3} + \frac{4}{3}, \qquad P_3'(3) = \frac{26\ell}{81} + \frac{5\ell'}{27} + \frac{13}{18}, \qquad P_3'(k) \in \mathbb{Q} + \mathbb{Q}\,\ell + \mathbb{Q}\,\ell'.$$

So $P_{\mathrm{T}_5}(\mathbf{1};1,1) = -\sum_{x\ge1} q'(x) = a + b\,\ell + c\,\ell'$, where $a$, $b$, $c$ are series of rational numbers whose terms decrease like $4^{-x}$; summed to $x = 65$,

$$a = -0.46159963274916253008918\ldots, \qquad b = 2.01706417220768446716615\ldots,$$
$$c = -1.06282955174549612502149\ldots$$

With $\Gamma(\sigma)^2 P_3(\sigma) = \int_0^\infty y^{\sigma-1}\mathsf{g}(y)^3\,dy$, so that $\ell' = \int_0^\infty \log y\,\mathsf{g}^3 dy - 2(\log\Gamma)'(1)\,\ell$, quadrature gives

$$\ell = 2.34390723868945889060156228887227706909 54\ldots,$$
$$\ell' = -8.07515226353164401026965124146737650 13\ldots,$$

and

$$P_{\mathrm{T}_5}(\mathbf{1};1,1) = 12.84872214191555595731233993125353463661 9\ldots,$$

which the reading of $\mathrm{T}_5$ by the box procedure (Definition 41.8) reproduces to 53 digits. For $\mathrm{R}_5$ the series root has three cells, the mark 5, with the transform $\mathsf{g}$, and the parallel cells of $\{1,2\}$

and $\{3,4\}$, each with the transform $f_{\mathcal{P}_2}$ whose Mellin transform is $\Gamma(w)^2 P_2(\sigma-w)$, $0<w<\sigma$ (Proposition 23.8(iii), with $\check{f}_{\mathcal{P}_2}=\mathsf{g}^2$); by Definition 23.19(5)

$$\Gamma(\sigma)^2 P_{\mathrm{R}_5}(\sigma\mathbf{1};\sigma,\sigma)=\int_0^\infty y^{\sigma-1}\,\mathsf{g}(y)\,f_{\mathcal{P}_2}(y)^2\,dy,$$

and the box procedure reads $P_{\mathrm{R}_5}(\mathbf{1};1,1)=7.865837368364021592421974697446 8210 9\ldots$

*Answer.* The trees are those listed in (i). $P_{\mathrm{P}_5}=P_3^2$, with the law $\sum_{i=0}^3 d_i(k)x(k+i)=0$ of symbol $-10(\lambda-1)(9\lambda-1)(81\lambda-1)$, $V_1=36$, the low-end relation $828-180x(1)+12420x(2)-34992x(3)=0$, and the value span $\mathbb{Q}+\mathbb{Q}\,\ell+\mathbb{Q}\,\ell^2$. $P_{\binom{[4]}{2}}=P_2P_3$ obeys $\sigma^3q(\sigma)-2(2\sigma+1)(10\sigma^2+10\sigma+3)q(\sigma+1)+36(\sigma+1)(2\sigma+1)(2\sigma+3)q(\sigma+2)=0$. $\Gamma(\sigma)^2P_{\mathrm{T}_5}(\sigma)=\frac{1}{2\pi i}\int\Gamma(z)^2\Gamma(\sigma-z)^2q(z)\,dz$ closes to the right, $\Gamma(s)^2P_{\mathrm{T}_5}(s)=-\sum_{x\geq s}(V_sq)'(x)$, and $P_{\mathrm{T}_5}(\mathbf{1};1,1)=a+b\,\ell+c\,\ell'=12.8487221419\ldots$ with the series $a,b,c$ of (iv). □

## 55.3 Larger lists and the cost of a reading

**Exercise 55.4** (The list $\mathrm{N}_6$: letters and constant terms)**.** Let $\mathrm{N}_6$ be the list of rank three on $\mathcal{E}=\{1,\ldots,6\}$ whose bases are the nineteen triples other than $a=\{1,2,3\}$, and put $b=\{4,5,6\}$. On $a$ write $K_a=\alpha_1+\alpha_2+\alpha_3$, $\Psi_a=\alpha_1\alpha_2+\alpha_1\alpha_3+\alpha_2\alpha_3$ and $\alpha^a=\alpha_1\alpha_2\alpha_3$, the letters and the monomial of the list $\binom{a}{1}$, and $K_b$, $\Psi_b$, $\alpha^b$ the same on $b$.

(i) Show that the letters of $\mathrm{N}_6$ are $K=Q+\alpha^b$ and $\Psi=Q+\alpha^a$, with $Q=\Psi_aK_b+K_a\Psi_b$.

(ii) For $i,l\in\mathbb{N}$ with $2i+l=3D$, $D\in\mathbb{N}$, let $\gamma_{i,l}$ be the coefficient of $\alpha_1^D\alpha_2^D\alpha_3^D$ in $\Psi_a^i\,K_a^l$. Show that

$$\gamma_{i,l}=\sum_w\binom{i}{w}\binom{l}{w+\frac{l-i}{3}}\binom{2D-2w}{D-w}.$$

(iii) The centre ray of $\mathrm{N}_6$ at the strength 1 (Definition 38.13) has the direction $(\mathbf{1};1,1)$, and the ray with origin 0 and direction $(\mathbf{1};0,2)$, the full cobasis lean, has the stride $f=\alpha^{\mathcal{E}}\Psi^{-2}$. Write the constant terms $\mathrm{CT}(f^{-j})$ of the two reciprocal strides as sums of products of two numbers $\gamma_{i,l}$, and compute them for $j\leq 4$.

*Solution.* (i) A triple $T$ other than $a$ meets $a$ in at most two marks. If it meets $a$ in two marks it is a pair of $a$ and a mark of $b$, and these monomials $\alpha^T$ add up to $\Psi_aK_b$; if it meets $a$ in one mark it is a mark of $a$ and a pair of $b$, which add up to $K_a\Psi_b$; if it meets $a$ in no mark it is $b$. So $K=\sum_{T\in\mathcal{B}}\alpha^T=\Psi_aK_b+K_a\Psi_b+\alpha^b=Q+\alpha^b$. The cobases are the complements of the bases; the complement of a triple is a triple, and the complement of $a$ is $b$, so the cobases are the triples other than $b$, and the same count with $a$ and $b$ exchanged gives $\Psi=\sum_{T\in\mathcal{B}}\alpha^{\mathcal{E}\setminus T}=\Psi_bK_a+K_b\Psi_a+\alpha^a=Q+\alpha^a$.

(ii) Write $\Psi_a=\alpha_1\alpha_2+\alpha_3(\alpha_1+\alpha_2)$ and $K_a=\alpha_3+(\alpha_1+\alpha_2)$. Take $w$ of the $i$ factors of $\Psi_a^i$ equal to $\alpha_1\alpha_2$ and $v$ of the $l$ factors of $K_a^l$ equal to $\alpha_3$:

$$\Psi_a^iK_a^l=\sum_{w,v}\binom{i}{w}\binom{l}{v}(\alpha_1\alpha_2)^w\,\alpha_3^{\,i-w+v}\,(\alpha_1+\alpha_2)^{\,i-w+l-v}.$$

The power $D$ of $\alpha_3$ asks $v=D-i+w=w+\frac{l-i}{3}$, since $D-i=\frac{2i+l}{3}-i$. Then $i-w+l-v=2i+l-D-2w=2D-2w$, and the coefficient of $\alpha_1^{D-w}\alpha_2^{D-w}$ in $(\alpha_1+\alpha_2)^{2D-2w}$ is $\binom{2D-2w}{D-w}$, which gives the formula. The first values are

$$\gamma_{0,0}=1,\quad \gamma_{1,1}=3,\quad \gamma_{3,0}=\gamma_{0,3}=6,\quad \gamma_{2,2}=15,\quad \gamma_{4,1}=\gamma_{1,4}=36,\quad \gamma_{3,3}=93;$$

for instance $\gamma_{2,2} = \binom{2}{0}\binom{2}{0}\binom{4}{2} + \binom{2}{1}\binom{2}{1}\binom{2}{1} + \binom{2}{2}\binom{2}{2}\binom{0}{0} = 6 + 8 + 1$. Each $\gamma_{i,l}$ is the constant term $\mathrm{CT}\left(\alpha^{-D\mathbf{1}}\Psi_a^i K_a^l\right)$ on $\binom{a}{1}$: the first constant term of the three-mark ray with direction $(D\mathbf{1}; l, i)$.

(iii) *The centre.* The stride is $f = \alpha^{\mathcal{E}}(K\Psi)^{-1}$ (Definition 38.13), so $\mathrm{CT}(f^{-j})$ is the coefficient of $\alpha^{j\mathbf{1}}$ in $(K\Psi)^j$; every letter exponent of the direction is 1, and this is the closing $\mathsf{C}_j(1)$ of Definition 39.30. Expand $K^j = (\Psi_a K_b + K_a \Psi_b + \alpha^b)^j$ with $u_1, u_2, u_3$ factors of the three kinds and $\Psi^j = (K_a\Psi_b + \Psi_a K_b + \alpha^a)^j$ with $v_1, v_2, v_3$. The product splits into a part on $a$ and a part on $b$,

$$\Psi_a^{u_1+v_2} K_a^{u_2+v_1} (\alpha^a)^{v_3} \cdot \Psi_b^{u_2+v_1} K_b^{u_1+v_2} (\alpha^b)^{u_3},$$

and the coefficient of $\alpha^{j\mathbf{1}}$ is the product of the coefficient of $\alpha_1^{j-v_3}\alpha_2^{j-v_3}\alpha_3^{j-v_3}$ in the first factor without $(\alpha^a)^{v_3}$ and the like coefficient on $b$. Put $p = j - u_3 = u_1 + u_2$, $q = j - v_3 = v_1 + v_2$, $i = u_1 + v_2$ and $l = u_2 + v_1$. The degrees ask $2i + l = 3q$ and $i + 2l = 3p$, that is $i = 2q - p$ and $l = 2p - q$, and the two coefficients are $\gamma_{2q-p,\,2p-q}$ and $\gamma_{2p-q,\,2q-p}$. The multinomial weights summed over $u_1$, with $u_2 = p - u_1$, $v_2 = i - u_1$, $v_1 = q - i + u_1$, give

$$\sum_{u_1} \frac{j!}{u_1!\,u_2!\,u_3!} \cdot \frac{j!}{v_1!\,v_2!\,v_3!} = \binom{j}{p}\binom{j}{q}\sum_{u_1}\binom{p}{u_1}\binom{q}{i-u_1} = \binom{j}{p}\binom{j}{q}\binom{p+q}{2q-p}$$

by Vandermonde's identity. So

$$\mathrm{CT}(f^{-j}) = \sum_{p,q=0}^{j} \binom{j}{p}\binom{j}{q}\binom{p+q}{2q-p}\,\gamma_{2q-p,\,2p-q}\,\gamma_{2p-q,\,2q-p}.$$

At $j = 1$ the terms $(p, q) = (0, 0)$ and $(1, 1)$ survive, $1 + 2 \cdot 3 \cdot 3 = 19$, the number of bases. At $j = 2$ the terms $(0,0)$, $(1,1)$, $(1,2)$, $(2,1)$, $(2,2)$ are

$$1 + 2\cdot 2\cdot 2\cdot 3\cdot 3 + 2\cdot 1\cdot 1\cdot 6\cdot 6 + 1\cdot 2\cdot 1\cdot 6\cdot 6 + 1\cdot 1\cdot 6\cdot 15\cdot 15 = 1 + 72 + 72 + 72 + 1350 = 1567,$$

and at $j = 3$ the eight terms $(0,0), (1,1), (1,2), (2,1), (2,2), (2,3), (3,2), (3,3)$ are $1 + 162 + 324 + 324 + 12150 + 19440 + 19440 + 172980 = 224821$.

*The full cobasis lean.* Here $f^{-1} = \Psi^2/\alpha^{\mathcal{E}}$, and $\mathrm{CT}(f^{-j})$ is the coefficient of $\alpha^{j\mathbf{1}}$ in $\Psi^{2j}$; the direction has the letter exponent $s = 0$, so Definition 39.30 does not apply and the constant term is taken as written. With $v_1, v_2, v_3$ factors $K_a\Psi_b$, $\Psi_a K_b$, $\alpha^a$ out of $2j$ and $p = v_1 + v_2$, the part on $b$ is $\Psi_b^{v_1} K_b^{v_2}$, of degree $3j$, so $2v_1 + v_2 = 3j$: $v_1 = 3j - p$ and $v_2 = 2p - 3j$; the part on $a$ is $K_a^{v_1}\Psi_a^{v_2}(\alpha^a)^{v_3}$ with $v_3 = 2j - p$. So

$$\mathrm{CT}(f^{-j}) = \sum_{p} \binom{2j}{p}\binom{p}{2p-3j}\,\gamma_{2p-3j,\,3j-p}\,\gamma_{3j-p,\,2p-3j}.$$

At $j = 1$ only $p = 2$ survives: $\binom{2}{2}\binom{2}{1}\gamma_{1,1}^2 = 18$, the bases whose complement is a basis, all but $b$. At $j = 2$, $p = 3$ and $p = 4$ give $\binom{4}{3}\binom{3}{3}\gamma_{0,3}\gamma_{3,0} + \binom{4}{4}\binom{4}{2}\gamma_{2,2}^2 = 144 + 1350 = 1494$, and at $j = 3$, $p = 5$ and $p = 6$ give $\binom{6}{5}\binom{5}{1}\gamma_{1,4}\gamma_{4,1} + \binom{6}{6}\binom{6}{3}\gamma_{3,3}^2 = 38880 + 172980 = 211860$.

The closing values for $j \le 4$, by the two formulas and, as a check, by expanding $(K\Psi)^j$ and $\Psi^{2j}$ with every exponent above $j$ dropped (receipt `nsix_exact.py`), are

| $j$ | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| centre $(\mathbf{1}; 1, 1)$ | 1 | 19 | 1567 | 224821 | 41485807 |
| lean $(\mathbf{1}; 0, 2)$ | 1 | 18 | 1494 | 211860 | 38486070 |

So the constant terms of $\mathrm{N}_6$ are sums of products of two first constant terms of the three-mark list $\binom{[3]}{1}$ along its directions $(D\mathbf{1}; l, i)$, one factor for each of the triples $a$ and $b$; the sums cost $O(j^2)$ and $O(j)$ products of numbers $\gamma_{i,l}$, each a sum of $O(j)$ terms. □

**Exercise 55.5** ($\mathrm{N}_6$ at its full cobasis lean)**.** Read the ray of $\mathrm{N}_6$ (Exercise 55.4) with origin 0 and direction $(\mathbf{1};0,2)$, whose stride is $f=\alpha^{\mathcal{E}}\Psi^{-2}$, by the procedure of Definition 41.8:

(i) its walls and the volume $V_1$ of its direction;

(ii) the native gauge $\alpha^\star$ of the direction, $f^\star=f(\alpha^\star)$ and the Hessian $H$ there;

(iii) the box with one unit of each active letter, its spanning set and its matrix $\mathbf{B}$, the matrix at infinite strength and its spectrum, and that the spanning set is a set of masters;

(iv) the law of [1], its two end coefficients, its symbol, and the strengths at which its rows line up;

(v) the low-end relation and the constant terms;

(vi) the value $\langle 1\rangle_1=P_{\mathrm{N}_6}(\mathbf{1};0,2)$.

*Solution.* Throughout, $S$ is a nonempty proper subset of $\mathcal{E}$, $r(S)$ the largest trace of a basis on $S$, and the *type* of $S$ is the pair $(i,j)=(|S\cap a|,|S\cap b|)$.

(i) *Walls.* By Definition 8.1 the walls of the direction are $\kappa_{1,S}=m_{1,S}-s\,d(S)-t\,d^*(S)$, with $d(S)$ and $d^*(S)$ the least traces of a basis and of a cobasis on $S$. Here $m_1=\mathbf{1}$, $s=0$, $t=2$, and the least trace of a cobasis is $d^*(S)=|S|-r(S)$, so

$$\kappa_{1,S}=2\,r(S)-|S|.$$

The rank of $\mathrm{N}_6$ is $r(S)=\min(|S|,3)$, except $r(a)=2$, so $\kappa_{1,S}$ depends on the type alone:

| $(i,j)$ | $(1,0)$ | $(0,1)$ | $(2,0)$ | $(1,1)$ | $(0,2)$ | $(3,0)$ | $(2,1)$ |
|---|---|---|---|---|---|---|---|
| $\kappa_{1,S}$ | 1 | 1 | 2 | 2 | 2 | 1 | 3 |
| $F(i,j)$ | 1 | 1 | 1 | 1 | 1 | 3 | 1 |

| $(i,j)$ | $(1,2)$ | $(0,3)$ | $(3,1)$ | $(2,2)$ | $(1,3)$ | $(3,2)$ | $(2,3)$ |
|---|---|---|---|---|---|---|---|
| $\kappa_{1,S}$ | 3 | 3 | 2 | 2 | 2 | 1 | 1 |
| $F(i,j)$ | 1 | 1 | 3 | 2 | 2 | 12 | 10 |

Every wall is positive, the least is 1: the direction is strictly inside its walls, and counted with the sizes of the types the 62 walls are 1 thirteen times, 2 thirty times and 3 nineteen times.

*The volume.* By Proposition 39.34, $V_1=\sum_\pi\prod_{l=2}^{6}\kappa_{1,U_l}^{-1}$ over the orders $\pi$ of $\mathcal{E}$ with their tails $U_l=\{\pi(l),\dots,\pi(6)\}$. An order is a chain of tails $U_6\subset\cdots\subset U_2$, so $V_1=\sum_e F(\mathcal{E}\setminus\{e\})$ with $F(\{e\})=\kappa_{1,\{e\}}^{-1}$ and $F(U)=\kappa_{1,U}^{-1}\sum_{e\in U}F(U\setminus\{e\})$. $F$ depends on the type alone, $F(i,j)=\kappa_{1,(i,j)}^{-1}\big(i\,F(i-1,j)+j\,F(i,j-1)\big)$, which gives the last row of the table: for instance $F(3,0)=1^{-1}\cdot 3\,F(2,0)=3$, $F(3,1)=2^{-1}\big(3F(2,1)+F(3,0)\big)=3$, $F(3,2)=1^{-1}\big(3F(2,2)+2F(3,1)\big)=12$ and $F(2,3)=1^{-1}\big(2F(1,3)+3F(2,2)\big)=10$. So

$$V_1=3\,F(2,3)+3\,F(3,2)=30+36=66.$$

(ii) *The native gauge.* By Theorem 38.14(i), $f\le f(\alpha^\star)$ on the positive orthant with equality only at $\alpha\propto\alpha^\star$. The 36 automorphisms of $\mathrm{N}_6$, the permutations of $a$ and of $b$, keep the masses $\mathbf{1}$ and the exponents, so they are symmetries of the ray (Proposition 39.11(ii)); they keep $f$ and the orthant and therefore fix $\alpha^\star$ up to scale. On the chart $\alpha_6=1$ this puts $\alpha^\star$ on the slice $\alpha=(x,x,x,1,1,1)$, where $Q=3x^2\cdot 3+3x\cdot 3$, $\Psi=x^3+9x^2+9x$ and

$$f=\frac{x^3}{x^2(x^2+9x+9)^2}=\frac{x}{(x^2+9x+9)^2},\qquad \frac{d}{dx}\log f=-\frac{3(x^2+3x-3)}{x(x^2+9x+9)}.$$

So $\alpha^\star = (x^\star, x^\star, x^\star, 1, 1, 1)$ with $x^\star = \frac{\sqrt{21}-3}{2}$, and with $x^2 = 3 - 3x$, $x^2 + 9x + 9 = 6(x + 2)$ and $f = x/(36(x + 2)^2)$ there:

$$f^\star = \frac{7\sqrt{21} - 27}{1800} = 0.0028211277026060444701\ldots$$

At a complex point fixed by the automorphisms the differential of $\log f$ in $y = \log \alpha$ is fixed by them too, so it is constant on $a$ and on $b$, $(u, u, u, v, v, v)$, and $u+v = 0$ because $f$ is homogeneous of degree 0; it vanishes as soon as its derivative along the slice, $3u$, does. So the other root $x' = -\frac{\sqrt{21}+3}{2}$ gives a native gauge $\alpha' = (x', x', x', 1, 1, 1)$ too (Definition 38.4), at which $K = 9x'^2 + 9x' + 1$ and $\Psi = 6x'(x' + 2)$ do not vanish, of the conjugate native value $-\frac{7\sqrt{21}+27}{1800}$. The Hessian of $-\log f$ at $\log \alpha^\star$ on the chart is $H = 2\big(\kappa_{\alpha^\star}(X_e, X_{e'})\big)_{e,e' \neq 6}$, twice the covariances of the indicators under the cobasis law at $\alpha^\star$ (Theorem 38.15); computed in $\mathbb{Q}(\sqrt{21})$ (receipt `hessians.py`),

$$\det H = \frac{13049\sqrt{21} - 59339}{37500} = 0.0122408064927668332\ldots$$

(iii) *The box* (Procedure 41.8, the masters). The letter $K$ has the exponent 0 along the ray and $\Psi$ the exponent $2k$, so only $\Psi$ is active, and the box with one unit of each active letter is $\mathfrak{B}_2(0, 1)$ of Definition 39.15(iv): the symbols of wall degree at most 2 at the levels $(0, 0)$ and $(0, 1)$; its stems are the Laurent symbols of wall degree at most 2. The 36 automorphisms keep it (Definition 39.15(vi)), and modulo them (Definition 39.20(i)) there are 104 orbits of stems and 329 orbits of symbols; the relations are the 104 letter identities $\alpha^u - \sum_T \alpha^{u+\mathbf{1}\varepsilon\setminus T}\Psi^{-1}$ of the stems and their $104 \cdot 6 = 624$ mark relations, 728 in all. One elimination, the orbits ranked as in Definition 39.20(ii), done modulo a prime of 61 bits at a random strength, which gives the pivots over $\mathbb{Q}(k)$ off finitely many strengths, leaves free among the Laurent orbits of wall degree at most 1 those of the seven orbit averages

$$b_1 = 1, \quad b_2 = \Big[\frac{\alpha_3\alpha_5\alpha_6}{\alpha_1\alpha_2\alpha_4}\Big], \quad b_3 = \Big[\frac{\alpha_3\alpha_6}{\alpha_1\alpha_2}\Big], \quad b_4 = \Big[\frac{\alpha_5\alpha_6}{\alpha_1\alpha_4}\Big],$$
$$b_5 = \Big[\frac{\alpha_6}{\alpha_1}\Big], \quad b_6 = \Big[\frac{\alpha_3\alpha_6}{\alpha_1\alpha_4}\Big], \quad b_7 = \Big[\frac{\alpha_2\alpha_3\alpha_6}{\alpha_1\alpha_4\alpha_5}\Big],$$

$[\,\cdot\,]$ the average over the 36 automorphisms, of wall degree 0 and 1 (Proposition 39.16(ii): the wall degree of $\alpha^u$ is $\max_U(-u_U)/\kappa_{1,U}$; for $b_2, \ldots, b_7$ it is 1, reached at $U = \{1\}$), and every target $f^{-1}b_i$, of wall degree at most 2, reduces to them. So the box is accepted at $\mathfrak{s} = 1$ (Definition 39.20(iii)), and by Theorem 39.21(i) $[f^{-1}b_i] = \sum_l \mathbf{B}_{il}(k)[b_l]$ in the module, the classes of $b_1, \ldots, b_7$ spanning a submodule $\mathfrak{N}$ that contains $[1]$.

*The matrix* (Procedure 41.8, the matrix over $\mathbb{Q}(k)$). Input: the 728 relations, with coefficients $c_0 + c_1k$, $c_0, c_1 \in \mathbb{Z}$. Method: the targets need 292 of them, of rank 292; eliminations modulo primes of 62 bits at random strengths, 24 strengths on the first prime, every entry read as a quotient of polynomials, and Chinese remaindering with rational reconstruction over three primes. Output: the common denominator of the entries is $Q_c = (k - 2)^5(k + 1)$, and the row $i$ of $\mathbf{B}$ is a factor $\phi_i$ times a row of polynomials:

| $i$ | $\phi_i$ | $\mathbf{B}_{il}/\phi_i$, $l = 1, \ldots, 7$ |
|---|---|---|
| 1 | $\frac{6(2k-1)}{(k-2)(k+1)}$ | $6$; $4(k-2)$; $6(k-1)$; $6(k-1)$; $8(k+1)$; $-24$; $-(k+1)$ |
| 2 | $\frac{2(k-1)(2k-1)}{(k-2)^5(k+1)}$ | $k^4 + 20k^3 - 36k^2 + 40k - 16$; $2k(k-2)(8k^2 - 19k + 10)$; $2k(12k^3 - 37k^2 + 42k - 20)$; $2k(12k^3 - 37k^2 + 42k - 20)$; $4k(k+1)(8k^2 - 12k + 9)$; $-4(k^4 + 20k^3 - 48k^2 + 36k - 8)$; $-8(k-1)^3(k+1)$ |

| | | |
|---|---|---|
| 3 | $\frac{2(k-1)(2k-1)}{(k-2)^4(k+1)}$ | $-2k^3+33k^2-42k+16$; $(k-2)(15k^2-33k+14)$; $25k^3-75k^2+70k-16$; $19k^3-57k^2+70k-40$; $2k(k+1)(14k-11)$; $4(2k^3-27k^2+44k-20)$; $-(k-1)(k+1)(5k-6)$ |
| 4 | $\frac{2(k-1)(2k-1)}{(k-2)^4(k+1)}$ | the row 3 with its third and fourth entries exchanged |
| 5 | $\frac{2(k-1)(2k-1)}{(k-2)^3(k+1)}$ | $24(k-1)$; $4(k-2)(3k-5)$; $2(9k^2-23k+16)$; $2(9k^2-23k+16)$; $(k+1)(33k-32)$; $4(3k^2-23k+22)$; $-(k+1)(3k-4)$ |
| 6 | $\frac{2(k-1)(2k-1)}{(k-2)^4(k+1)}$ | $2k^3+11k^2-26k+16$; $(k-2)(k-1)(7k-10)$; $13k^3-41k^2+40k-8$; $13k^3-41k^2+40k-8$; $(k+1)(33k^2-73k+48)$; $4(k^3-13k^2+23k-14)$; $-(k+1)(k^2-2)$ |
| 7 | $\frac{2(k-1)(2k-1)}{(k-2)^5(k+1)}$ | $-2k^4+23k^3-44k^2+53k-16$; $(k-2)(k-1)^2(16k-7)$; $27k^4-94k^3+99k^2-34k+22$; $27k^4-94k^3+99k^2-34k+22$; $(k+1)(8k^3+17k^2-35k+32)$; $-13k^4-39k^3+231k^2-273k+22$; $-(k+1)(17k^3-80k^2+131k-73)$ |

Check: **B** agrees with an elimination of the whole box at a prime and a strength drawn at random, and again at three fresh ones. Receipt: `rayreader.py N "1 1 1 1 1 1 ; 0 2"`, read by `analyze_run.py` and `matrix_rows_tex.py`; the re-check by `tests/check_run.py`.

*Infinite strength.* Every entry has a numerator of degree at most that of its denominator, so **B** is regular at $k=\infty$, with

$$\mathbf{B}(\infty)=\begin{pmatrix} 0 & 48 & 72 & 72 & 96 & 0 & -12\\ 4 & 64 & 96 & 96 & 128 & -16 & -32\\ -8 & 60 & 100 & 76 & 112 & 32 & -20\\ -8 & 60 & 76 & 100 & 112 & 32 & -20\\ 0 & 48 & 72 & 72 & 132 & 48 & -12\\ 8 & 28 & 52 & 52 & 132 & 16 & -4\\ -8 & 64 & 108 & 108 & 32 & -52 & -68 \end{pmatrix},$$

$$\det\big(\lambda-\mathbf{B}(\infty)\big)=(\lambda-24)^2\,(\lambda^2-324\lambda-10800)\,(\lambda^3+28\lambda^2-320\lambda+1728).$$

By Proposition 39.42(ii), at every nondegenerate native gauge $\alpha_c$ the vector $\beta_c=(b_l(\alpha_c))_l$ is an eigenvector of $\mathbf{B}(\infty)$ for $f(\alpha_c)^{-1}$. At $\alpha^\star$ every monomial of the orbit of $\alpha^u$ takes the value $(x^\star)^{u_1+u_2+u_3}$, so $\beta_\star=(1,\frac{1}{x^\star},\frac{1}{x^\star},\frac{1}{x^\star},\frac{1}{x^\star},1,x^\star)$ with $\frac{1}{x^\star}=\frac{\sqrt{21}+3}{6}$; the first row gives

$$\frac{48+72+72+96}{x^\star}-12x^\star=48(\sqrt{21}+3)-6(\sqrt{21}-3)=162+42\sqrt{21}=\frac{1800}{7\sqrt{21}-27}=\frac{1}{f^\star},$$

and the other rows the same, exactly. The eigenvalue 24 belongs to two rational native gauges: at $\alpha=(-1,\frac{3}{2},-1,1,1,1)$ and at $\alpha=(-1,-1,-1,1,-\frac{2}{3},1)$ every dual occupation is $R_e^*=\frac{1}{2}$, so every defect $1-2R_e^*$ vanishes, $f=\frac{1}{24}$, and the Hessian of $\log f$ on the chart has the determinant $\frac{625}{864}$ at both; their vectors

$$\beta=\big(1,\tfrac{17}{18},\tfrac{17}{18},-\tfrac{4}{9},-\tfrac{4}{9},-\tfrac{7}{18},\tfrac{11}{9}\big),\qquad \beta'=\big(1,\tfrac{17}{18},-\tfrac{4}{9},\tfrac{17}{18},-\tfrac{4}{9},-\tfrac{7}{18},\tfrac{11}{9}\big)$$

satisfy $\mathbf{B}(\infty)\beta=24\beta$ and $\mathbf{B}(\infty)\beta'=24\beta'$ exactly, and are independent. Receipt: `nsix_gauges24.py`.

*The native gauges and the masters.* Input: the critical equations $1-2R_e^*(\alpha)=0$ on the chart, saturated by the coordinates and the letters. Method: a Groebner basis over $\mathbb{Q}$, and, separately, the critical points of $\alpha^{\mathbf{a}}\Psi^{-b}$ for random complex exponents counted by monodromy and carried to $(\mathbf{1};2)$. Output: the ideal is zero-dimensional with 35 points counted with multiplicity, and the 35

native gauges found by the paths are distinct and nondegenerate, in 7 orbits of the automorphisms of sizes $1,1,3,3,9,9,9$: $\alpha^\star$, $\alpha'$, the two rational orbits of native value $\frac{1}{24}$, and three orbits whose native values are the roots of $1728\lambda^3 - 320\lambda^2 + 28\lambda + 1$, the real one $-0.0265230928\ldots$ and $0.1058541390\ldots \pm 0.1030231201\ldots\, i$. Check: Newton's method at 40 digits at each point, residual below $10^{-38}$; the seven vectors $\beta_c$, one per orbit, have the singular values from 8.93 down to 0.358, and each is an eigenvector of $\mathbf{B}(\infty)$ to $10^{-38}$. Receipts: `gb_direction.py`, `critpoints.py`, `polish.py`, `betas.py`. So the vectors $\beta_c$ span $\mathbb{C}^7$, and by Proposition 39.42(i) the spanning set is a set of masters of $\mathfrak{N}$: $\dim_{\mathbb{C}(k)} \mathfrak{N} = 7$, $\mathbf{B} = \mathbf{B_b}$, and $\mathbf{B}(\infty)$ is diagonalizable with the eigenvalues $f(\alpha_c)^{-1}$, 24 twice for the two orbits of native value $\frac{1}{24}$.

(iv) *The law* (Procedure 41.8, the law). Input: $\mathbf{B}$. Method: the rows $w_0 = (1,0,\ldots,0)$, $w_i(k) = w_{i-1}(k-1)\,\mathbf{B}(k)$ of Proposition 39.6, evaluated modulo primes at consecutive strengths; their first dependence; the coefficients reconstructed over $\mathbb{Q}$ from four primes and made integral and primitive. Output: $w_0,\ldots,w_5$ are independent and $w_0,\ldots,w_6$ dependent, so $h = 6$, and the relation $\sum_{j=0}^{6} c_j(k)\langle 1\rangle_{k+j} = 0$ of Proposition 39.6(ii) has $c_j \in \mathbb{Z}[k]$ of degree 24, with coefficients of up to 31 digits. Check: the relation holds at a prime and a point drawn at random, and at three fresh ones. Since $b_1,\ldots,b_7$ are masters, Proposition 39.6(ii) makes it the law of [1] (Definition 39.1): the cyclic submodule of [1] has dimension 6, inside $\mathfrak{N}$ of dimension 7. Factored over $\mathbb{Q}$,

$$c_0 = k^5(k+1)^3(k+2)(k+3)\,r(k),$$
$$c_6 = 6998400\,(2k+1)(2k+3)(2k+5)(2k+7)(2k+9)(2k+11)(k+4)(k+5)^3\,r(k-1),$$

with $r$ irreducible of degree 14 and integer coefficients of up to 18 digits, and the symbol (Corollary 38.7), the coefficients of $k^{24}$,

$$\chi(\lambda) = 7276255875\,(24\lambda-1)(10800\lambda^2+324\lambda-1)(1728\lambda^3-320\lambda^2+28\lambda+1).$$

Its six roots are $\frac{1}{24}$, $f^\star$ and its conjugate (the roots of $10800\lambda^2+324\lambda-1$, $\frac{-27\pm7\sqrt{21}}{1800}$), and the three roots of the cubic: the six distinct native values of (iii), as Corollary 38.7 asks, and the reciprocals of the eigenvalues of $\mathbf{B}(\infty)$, 24 counted once.

*Line-up strengths.* In the notation of Proposition 39.6, $p_j = c_j$ and $q_6(k) = c_0(k-6)$, and Proposition 39.6(iii) gives

$$c_0(k-6) \;\Big|\; \det \mathbf{W}(k)_J \prod_{j=0}^{5} Q_c(k-j)^{6-j}, \qquad Q_c = (k-2)^5(k+1),$$

for every set $J$ of six columns with $\det \mathbf{W}_J \neq 0$. The irreducible factor $r(k-6)$, of degree 14, divides no linear factor $Q_c(k-j)$, so it divides $\det \mathbf{W}(k)_J$: at its fourteen roots, none a root of a $Q_c(k-j)$, the rows $w_0,\ldots,w_5$ are dependent. So 14 of the 24 degrees of $c_0$ are strengths at which $\langle 1\rangle_k,\ldots,\langle 1\rangle_{k-5}$, read as linear forms in the values of the masters, line up; the other 10 are the integers $0,-1,-2,-3$.

(v) *The low end.* By Corollary 39.38(ii), $c_0$ vanishes to order $n-1=5$ at $k=0$, as it does, and with $\mathsf{L}_0 = V_1 = 66$ (Definition 39.37) the low-end relation $[k^5]c_0 \cdot 66 + \sum_{i=1}^{6} c_i(0)\langle 1\rangle_i = 0$, divided by 91238400, reads

$$64402904777 - 12632598108\,\langle 1\rangle_1 - 2849737197672\,\langle 1\rangle_2 + 380837271432240\,\langle 1\rangle_3$$
$$+ 14853466134400320\,\langle 1\rangle_4 - 369781615117353600\,\langle 1\rangle_5 + 3400346988339840000\,\langle 1\rangle_6 = 0.$$

By Corollary 39.38(i) the constant terms obey $\sum_{i=0}^{\min(6,j)} c_i(-j)\,\mathrm{CT}(f^{i-j}) = 0$ for every $j \in \mathbb{N}$; with the constant terms of Exercise 55.4 this holds exactly for $0 \le j \le 60$ (receipt `nsix_ctcheck.py`), a check of the law that uses neither the box nor the values.

(vi) *The value* (Procedure 41.8, the value). By Theorem 40.6 the values are the moments $\langle 1\rangle_k = \int_0^{f^\star} t^k \varrho_f(t)\,dt$ of the density $\varrho_f$ of the stride's law (Definition 40.1); $f^\star$ is a simple root of $\chi$, the density operator of the law has order 24, its exponent at $f^\star$ beyond $0,\dots,22$ is $\frac{n-3}{2} = \frac{3}{2}$, and the germ of $\varrho_f$ is $(f^\star - t)^{3/2} g(f^\star - t)$ with $g$ fixed by $g(0) = \big(2\Gamma(\frac12)^2/f^\star\big)^{5/2}\big/\big(\Gamma(\frac52)\sqrt{\det H}\big)$, the constant of the branch of Theorem 38.15. Input: the law, $f^\star$ and $\det H$. Method: the germ to 168 terms, the Beta sums $\int_0^{f^\star} t^K (f^\star - t)^{\frac32+m} dt = (f^\star)^{K+\frac52+m}\Gamma\,{}_2{}_1\!\left[\begin{matrix} K+1,\ \frac52+m \\ K+\frac72+m \end{matrix}\right]$ at the six strengths from $K = 57$ on, and the law run back to $k = 1$, in ball arithmetic at 1188 bits; again from $K = 82$. Output:

$$P_{\mathrm{N}_6}(\mathbf{1}; 0, 2) = \langle 1\rangle_1 = 4.75680256375822944340803728605708\ldots,$$

the two starting strengths agreeing to 50 digits. Check: the exponent read from the law at $f^\star$ is $\frac32$ to 16 digits; the reading at 700 digits of Exercise 55.7 agrees with this value, and on its values $\langle 1\rangle_1, \dots, \langle 1\rangle_6$ the low-end relation of (v), which the reading does not use, holds to 731 digits relative to its largest term. Receipt: `rayreader.py N "1 1 1 1 1 1 ; 0 2" --digits 30`.

So $\mathrm{N}_6$ at $(\mathbf{1}; 0, 2)$ has the volume $V_1 = 66$, the native value $f^\star = \frac{7\sqrt{21}-27}{1800}$, seven masters with $\mathbf{B}(\infty)$ of spectrum $\{24, 24\}$, the reciprocals of the roots of $10800\lambda^2 + 324\lambda - 1$ and of $1728\lambda^3 - 320\lambda^2 + 28\lambda + 1$, a law of order 6 and degree 24, and the value $P_{\mathrm{N}_6}(\mathbf{1}; 0, 2) = 4.7568025637582294434\ldots$, with $P/V_1 = 0.0720727\ldots$ □

**Exercise 55.6** ($\mathrm{N}_6$ at its centre)**.** Read the centre ray of $\mathrm{N}_6$ at the strength 1, with the direction $(\mathbf{1}; 1, 1)$ and the stride $f = \alpha^{\mathcal{E}}(K\Psi)^{-1}$, by the procedure of Definition 41.8: its walls and $V_1$; the native gauge of the direction and the signs; the box, the spanning set and its matrix at infinite strength; the law of [1], its ends, its symbol and its line-up strengths; the low-end relation, the closings and the value.

*Solution. Walls and volume.* With $m_1 = \mathbf{1}$ and $s = t = 1$, Definition 8.1 and $d(S) = 3 - r(\mathcal{E} \setminus S)$, $d^*(S) = |S| - r(S)$ give $\kappa_{1,S} = r(S) + r(\mathcal{E}\setminus S) - 3$, which depends on the type $(i, j) = (|S \cap a|, |S \cap b|)$ alone. With the recursion of Exercise 55.5(i):

| $(i,j)$ | $(1,0)$ | $(0,1)$ | $(2,0)$ | $(1,1)$ | $(0,2)$ | $(3,0)$ | $(2,1)$ |
|---|---|---|---|---|---|---|---|
| $\kappa_{1,S}$ | 1 | 1 | 2 | 2 | 2 | 2 | 3 |
| $F(i,j)$ | 1 | 1 | 1 | 1 | 1 | $\frac32$ | 1 |

| $(i,j)$ | $(1,2)$ | $(0,3)$ | $(3,1)$ | $(2,2)$ | $(1,3)$ | $(3,2)$ | $(2,3)$ |
|---|---|---|---|---|---|---|---|
| $\kappa_{1,S}$ | 3 | 2 | 2 | 2 | 2 | 1 | 1 |
| $F(i,j)$ | 1 | $\frac32$ | $\frac94$ | 2 | $\frac94$ | $\frac{21}{2}$ | $\frac{21}{2}$ |

Every wall is positive: 1 twelve times, 2 thirty-two times and 3 eighteen times, the triples $a$ and $b$ having the wall 2. By Proposition 39.34,

$$V_1 = 3\,F(2,3) + 3\,F(3,2) = \tfrac{63}{2} + \tfrac{63}{2} = 63.$$

*The native gauge and the signs.* On the centre ray $\alpha^\star = \mathbf{1}$ and $f^\star = K(\mathbf{1})^{-2} = 19^{-2} = \frac{1}{361}$ (Theorem 38.14, Definition 38.13), and $H = 2\big(\kappa_{\mathbf{1}}(X_e, X_{e'})\big)_{e,e'\neq 6}$, the covariances under the weight law, which is uniform on the nineteen bases (Theorem 38.15): each mark of $a$ lies in 9 bases and each mark of $b$ in 10, and

$$H = \frac{1}{361}\begin{pmatrix} 180 & -48 & -48 & -28 & -28 \\ -48 & 180 & -48 & -28 & -28 \\ -48 & -48 & 180 & -28 & -28 \\ -28 & -28 & -28 & 180 & -48 \\ -28 & -28 & -28 & -48 & 180 \end{pmatrix}, \qquad \det H = \frac{580608}{361^3}.$$

By Theorem 38.17(ii) every sign $\eta$ with $K(\eta) \neq 0$ is a native gauge of native value $K(\eta)^{-2}$. A sign is fixed by its partition of the marks into the scales 1 and $-1$, and the 32 signs of the chart $\eta_6 = 1$ fall into eight orbits of the automorphisms: $K(\eta) = 19$ when all scales agree; $|K(\eta)| = 5$ when two marks of $a$, or two marks of $b$, stand against the other four, two orbits of three signs; $|K(\eta)| = 3$ when one mark of $a$ and one of $b$ stand against the other four, nine signs; and $|K(\eta)| = 1$ on the other four orbits, one mark against five (3 and 3 signs), $a$ against $b$ (1), and two marks of $a$ with one of $b$ against the rest (9). The Hessian of $\log f$ at a sign, in the chart $\alpha_e = \eta_e e^{y_e}$, has the determinants $-\frac{580608}{361^3}$, $\frac{9216}{15625}$, $\frac{1024}{729}$ at the first three kinds and 0 at the last: the native values $\frac{1}{361}$, $\frac{1}{25}$ (two orbits) and $\frac{1}{9}$ are nondegenerate, and the native value 1 is degenerate at every sign (receipt `nsix_signs.py`). On the slice $(x, x, x, 1, 1, 1)$, fixed by the automorphisms, $f = x^2/\big((x^2+9x+9)(9x^2+9x+1)\big)$ and $\frac{d}{dx}\log f$ has the numerator $-18(x-1)(x+1)(x^2+5x+1)$; as at the lean, a critical point of the slice is a native gauge, and $x^2 + 5x + 1 = 0$ gives the two native gauges $\big(\frac{-5\pm\sqrt{21}}{2}\big)$ of native value $-\frac{1}{80}$, exchanged by the Cremona.

*The box* (Procedure 41.8, the masters). Both letters are active, and the symmetries are the 36 automorphisms and the Cremona (Proposition 39.11), 72 maps, which keep every box with equal level bounds. The box $\mathfrak{B}_2(1,1)$ is not accepted: at $\mathfrak{s} = 1$ eight Laurent orbits are free and seven of their targets do not reduce to them, and at $\mathfrak{s} = 0$ the target of 1 does not reduce. The box $\mathfrak{B}_3(1,1)$ has 252 orbits of stems and 1909 orbits of symbols, and 3774 relations: the letter identities of the symbols at the levels $(0,0)$, $(1,0)$ and $(0,1)$ and the $252 \cdot 6$ mark relations of the stems. It is accepted at $\mathfrak{s} = 2$, with the spanning set of the fifteen averages, over the 72 maps, of 1 and of $\alpha^u$ for

$$\begin{gathered} u = (-1,-1,0,0,1,1),\ (-1,-1,1,-1,1,1),\ (-1,-1,1,0,0,1),\ (-1,0,0,-1,1,1), \\ (-1,0,0,0,0,1),\ (-1,0,1,-1,0,1),\ (-1,0,1,0,0,0),\ (-1,-1,-1,1,1,1), \\ (-2,-2,0,0,2,2),\ (-2,-2,0,1,1,2),\ (-2,-2,1,-1,2,2), \\ (-2,-2,1,0,1,2),\ (-2,-2,1,1,1,1),\ (-2,-2,2,-2,2,2), \end{gathered}$$

of the wall degrees 1 (the first seven), $\frac{3}{2}$ (the eighth, at $U = a$) and 2 (Theorem 39.21(i)).

*The matrix.* Input: the 3774 relations. Method: as in Exercise 55.5; the targets need 1023 relations, of rank 1023; 55 strengths on the first prime; three primes. Output: $\mathbf{B} \in \mathrm{Mat}_{15}(\mathbb{Q}(k))$ with 8 vanishing entries, numerators and denominators of degree at most 17 in lowest terms, and the common denominator

$$Q_c = (2k-5)(k-2)^2(k-3)^5(k-4)\,S_9(k),$$

$S_9$ irreducible of degree 9, $S_9 = 6555027k^9 - 143215495k^8 + \cdots - 6615463200$. Check: $\mathbf{B}$ agrees with eliminations of the whole box at four primes and strengths drawn at random. Receipt: `rayreader.py N "1 1 1 1 1 1 ; 1 1"`, read by `analyze_run.py`. $\mathbf{B}$ is regular at $k = \infty$, and

$$\det\big(\lambda - \mathbf{B}(\infty)\big) = (\lambda - 361)\,(\lambda - 25)^2\,(\lambda - 9)\,(\lambda - 1)^8\,(\lambda + 80)\,\big(\lambda^2 + \tfrac{34}{3}\lambda - \tfrac{361}{3}\big).$$

By Proposition 39.42(ii) the vectors $\beta_\eta$ of the nondegenerate signs and $\beta$ of the slice gauge are eigenvectors: computed exactly, $\mathbf{B}(\infty)\beta_\eta = K(\eta)^2\beta_\eta$ at the signs of native values $\frac{1}{361}$, $\frac{1}{25}$ (both orbits) and $\frac{1}{9}$, and $\mathbf{B}(\infty)\beta = -80\beta$ at $\big(\frac{-5-\sqrt{21}}{2}\big)$, five independent vectors (receipt `nsix_centre_exact.py`). The roots of $\lambda^2 + \frac{34}{3}\lambda - \frac{361}{3}$ are $\frac{-17\pm14\sqrt{7}}{3}$, the reciprocals of $\frac{17\pm14\sqrt{7}}{361}$, which are native values of gauges off the signs and the slice: the monodromy count of Exercise 55.5, run at this direction, reaches eighteen nondegenerate native gauges at each (receipt `critpoints.py`).

*The law.* The rows of Proposition 39.6 first depend at $h = 11$, and the relation $\sum_{j=0}^{11} c_j(k)\langle 1\rangle_{k+j} = 0$ has $c_j \in \mathbb{Z}[k]$ of degree 52 with coefficients of up to 88 digits, from ten primes; it holds at random points of four primes. By Proposition 39.6(ii) this relation, of order 11 with coprime coefficients, annihilates [1], so the values satisfy it; its order is at least the dimension of the cyclic submodule of [1], with equality when the spanning set is a set of masters (Definition 39.1). Factored over $\mathbb{Q}$,

$$c_0 = 3\,(2k+1)\,k^5(k+1)^2(k+2)^2(k+3)\,S_{41}(k),$$
$$c_{11} = -4691556000\,(k+4)(k+5)(k+6)(k+7)^2(k+8)^2(k+9)^2(k+10)^2\,S_{41}(k-1),$$

with $S_{41}$ irreducible of degree 41, and the symbol is

$$\chi(\lambda) \propto (361\lambda-1)(25\lambda-1)(9\lambda-1)(\lambda-1)^5(80\lambda+1)(361\lambda^2-34\lambda-3),$$

whose roots are the native values found: $\frac{1}{361}, \frac{1}{25}, \frac{1}{9}$, the degenerate 1 five times, $-\frac{1}{80}$ and $\frac{17\pm14\sqrt{7}}{361}$. The eigenvalues of $\mathbf{B}(\infty)$ are the reciprocals of the roots of $\chi$, with their multiplicities, and one more 25 and three more 1.

*Line-up strengths.* $q_{11}(k) = c_0(k-11)$ divides $\det \mathbf{W}(k)_J \prod_{j=0}^{10} Q_c(k-j)^{11-j}$ (Proposition 39.6(iii)). The irreducible factors of $Q_c$ have degrees 1 and 9, so $S_{41}(k-11)$ divides $\det \mathbf{W}(k)_J$: 41 of the 52 degrees of $c_0$ are strengths at which the rows $w_0, \dots, w_{10}$ line up, and the other 11 are the rational roots $0, -\frac{1}{2}, -1, -2, -3$.

*The low end and the closings.* $c_0$ vanishes to order 5 at 0, and the low-end relation of Corollary 39.38(ii), $[k^5]c_0 \cdot 63 + \sum_{i=1}^{11} c_i(0)\langle 1\rangle_i = 0$, has integer coefficients of up to 69 digits after division by their greatest common divisor 11377315676160 (receipt `lowend_forms.py`). The closings $\mathsf{C}_j(1) = \mathrm{CT}(f^{-j})$ of Exercise 55.4 satisfy $\sum_{i=0}^{\min(11,j)} c_i(-j)\,\mathsf{C}_{j-i}(1) = 0$ exactly for $0 \le j \le 60$ (Corollary 39.36; receipt `nsix_ctcheck.py`).

*The value.* By Theorem 40.6, with $f^\star = \frac{1}{361}$ a simple root of $\chi$, the density operator of order 52, the exponent $\frac{3}{2}$ at $f^\star$ (read from the law to 16 digits), and $g(0) = \big(2\Gamma(\frac{1}{2})^2 \cdot 361\big)^{5/2} \big/ \big(\Gamma(\frac{5}{2})\sqrt{\det H}\big)$: Beta sums from $K = 57$ and from $K = 82$ with 168 terms of the germ at 1188 bits, the law run back to $k = 1$. Output:

$$P_{\mathrm{N}_6}(\mathbf{1}; 1, 1) = \langle 1\rangle_1 = 4.57891247888240480638020373022855\ldots,$$

the two starts agreeing to 50 digits. Receipt: `rayreader.py N "1 1 1 1 1 1 ; 1 1" --digits 30`.

So the centre of $\mathrm{N}_6$ has $V_1 = 63$, $f^\star = \frac{1}{361}$, a box accepted at $\mathfrak{B}_3(1,1)$ with fifteen spanning elements, the law of [1] of order 11 and degree 52, and $P_{\mathrm{N}_6}(\mathbf{1}; 1, 1) = 4.5789124788824048\ldots$, with $P/V_1 = 0.0726811\ldots$ □

**Exercise 55.7** (The value span of $\mathrm{N}_6$)**.** The *value span* of a ray is the $\mathbb{Q}$-span of its values $\langle 1\rangle_k$, $k \ge 1$. For the two rays of $\mathrm{N}_6$ of Exercises 55.5 and 55.6:

(i) bound the value span by the law of [1] and the low-end relation;

(ii) find the integer relations of small height among 1 and the first values, read to 700 digits, by lattice reduction, and among them and the numbers $\ell = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1)$ and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$.

*Solution.* (i) *The lean.* The leading coefficient of the law, $c_6 = 6998400\,(2k+1)(2k+3)(2k+5)(2k+7)(2k+9)(2k+11)(k+4)(k+5)^3\,r(k-1)$, has its rational roots at negative numbers, and

$r$, irreducible of degree 14, has none; so $c_6(k) \neq 0$ for every integer $k \geq 0$. The law holds at every $k \geq 1$ (Corollary 39.38(i)), so $\langle 1\rangle_{k+6} = -c_6(k)^{-1} \sum_{i<6} c_i(k) \langle 1\rangle_{k+i}$ for $k \geq 1$, and by induction every value lies in the $\mathbb{Q}$-span of $\langle 1\rangle_1, \ldots, \langle 1\rangle_6$. The low-end relation of Exercise 55.5(v) has the constant term $64402904777 \neq 0$ and the coefficient $3400346988339840000 \neq 0$ at $\langle 1\rangle_6$, so it puts 1 in the value span and $\langle 1\rangle_6$ in $\mathbb{Q} + \mathbb{Q}\langle 1\rangle_1 + \cdots + \mathbb{Q}\langle 1\rangle_5$. So the value span of the lean is

$$\mathbb{Q} + \mathbb{Q}\langle 1\rangle_1 + \mathbb{Q}\langle 1\rangle_2 + \cdots + \mathbb{Q}\langle 1\rangle_5,$$

of dimension at most 6. *The centre.* The same with $c_{11} = -4691556000\,(k+4)(k+5)(k+6)(k+7)^2(k+8)^2(k+9)^2(k+10)^2\,S_{41}(k-1)$, nonzero at every integer $k \geq 0$, and the low-end relation of Exercise 55.6, whose constant term and coefficient of $\langle 1\rangle_{11}$ are nonzero: the value span of the centre is $\mathbb{Q} + \mathbb{Q}\langle 1\rangle_1 + \cdots + \mathbb{Q}\langle 1\rangle_{10}$, of dimension at most 11.

(ii) *The values.* Input: the laws and the native gauges of the two rays. Method: Theorem 40.6 at 700 digits, the Beta sums from the strengths 861 and 886 with 2580 terms of the germ, at 20224 bits, and the laws run back; this gives $\langle 1\rangle_1, \ldots, \langle 1\rangle_7$ at the lean and $\langle 1\rangle_1, \ldots, \langle 1\rangle_{12}$ at the centre. Check: the two starting strengths agree to 700 digits; the low-end relations, which the reading does not use, hold on the values to 731 digits relative to their largest terms; and the values agree with those of Exercises 55.5 and 55.6. Receipt: `valuespan_N.py`.

*The relations.* Method: for numbers $v_0 = 1, v_1, \ldots, v_N$, the Lenstra–Lenstra–Lovász reduction of the lattice spanned by the rows $\big(\mathbf{e}_i,\ \lfloor 10^{660} v_i \rfloor\big)$; a reduced row $(x, *)$ with a small residual $\sum_i x_i v_i$ is a candidate integer relation, and the first rows that are not relations show the size of the generic vectors. Output at the lean, for $1, \langle 1\rangle_1, \ldots, \langle 1\rangle_6$: three vectors of heights $3.0 \cdot 10^9$, $5.2 \cdot 10^9$ and $1.4 \cdot 10^{10}$ with residuals below $10^{-724}$,

$$\begin{aligned}
&15476 - 3451\langle 1\rangle_1 + 556832\langle 1\rangle_2 - 17388081\langle 1\rangle_3 \\
&\qquad + 111849150\langle 1\rangle_4 - 425617416\langle 1\rangle_5 - 2986275600\langle 1\rangle_6, \\
&84581 - 18857\langle 1\rangle_1 + 3035312\langle 1\rangle_2 - 97811235\langle 1\rangle_3 \\
&\qquad + 798867930\langle 1\rangle_4 - 5212068120\langle 1\rangle_5 + 1459458000\langle 1\rangle_6, \\
&5489069 - 1222045\langle 1\rangle_1 + 190733734\langle 1\rangle_2 - 4628283534\langle 1\rangle_3 \\
&\qquad + 13506396540\langle 1\rangle_4 - 1721929104\langle 1\rangle_5 - 44906400\langle 1\rangle_6,
\end{aligned}$$

and then vectors of height $3 \cdot 10^{157}$ and residual $10^{-503}$. Check: the low-end relation, which is exact, lies in the rational span of the three (exact linear algebra; receipt `valuespan_exact.py`). Solved for the last values, the three read

$$\langle 1\rangle_4 = \tfrac{556429709}{1602852300}\langle 1\rangle_3 - \tfrac{137833049}{9617113800}\langle 1\rangle_2 + \tfrac{26497699}{288513414000}\langle 1\rangle_1 - \tfrac{170029763}{412162020000},$$

and like expressions of $\langle 1\rangle_5$ and $\langle 1\rangle_6$ with denominators of up to 20 digits. Among $1, \langle 1\rangle_1, \langle 1\rangle_2, \langle 1\rangle_3$ and $\ell$ the reduction returns one vector of height $1.1 \cdot 10^7$ and residual $10^{-727}$, the next of height $10^{163}$:

$$3906\,\langle 1\rangle_1 - 506772\,\langle 1\rangle_2 + 10837800\,\langle 1\rangle_3 = 15263 + 1047\,\ell.$$

$\ell$ is the value of the minor $\binom{\{1,2,3,5\}}{2} = (\mathrm{N}_6/4) \setminus 6$ of $\mathrm{N}_6$ at its own full lean $(\mathbf{1}; 0, 2)$ and at its centre: the procedure reads $\binom{[4]}{2}$ at both, with laws of order 2 and degrees 3 and 3, and returns $\ell$ to the 32 digits printed (Exercise 55.10).

At the centre, for $1, \langle 1\rangle_1, \ldots, \langle 1\rangle_{12}$, four vectors of heights $5.4 \cdot 10^{23}$ to $1.1 \cdot 10^{24}$ with residuals below $10^{-714}$, the next of height $2.5 \cdot 10^{62}$ and residual $10^{-598}$; the law read at $k = 1$ and the low-end relation, both exact, lie in their rational span, and they express $\langle 1\rangle_9, \ldots, \langle 1\rangle_{12}$

through $1, \langle 1\rangle_1, \dots, \langle 1\rangle_8$. Among $1, \langle 1\rangle_1, \dots, \langle 1\rangle_8$, $\ell$ and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ the reduction returns two vectors, of heights $1.4 \cdot 10^{25}$ and $1.6 \cdot 10^{25}$ and residuals below $10^{-710}$, the next of height $2.9 \cdot 10^{67}$; their two coefficients at $\ell$ and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ form the nonsingular matrix $\begin{pmatrix} 146909488464924911235 & -115139161649314013904 \\ -10928078403398636616 & 24473573045860222080 \end{pmatrix}$, so they write $\ell$ and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ through the values. Here $7(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ is the value of the minor $\binom{\{1,2,3,6\}}{1} = \mathrm{N}_6/\{4,5\}$ at its centre, and $\ell$ that of $\binom{\{1,2,3,5\}}{2}$.

So the value span of $\mathrm{N}_6$ is $\mathbb{Q} + \sum_{i\le 5} \mathbb{Q}\langle 1\rangle_i$ at the lean and $\mathbb{Q} + \sum_{i\le 10} \mathbb{Q}\langle 1\rangle_i$ at the centre, of dimensions at most 6 and 11; the integer relations that lattice reduction finds at 700 digits, holding there to 724 and 714 digits, cut them to $\mathbb{Q} + \mathbb{Q}\langle 1\rangle_1 + \mathbb{Q}\langle 1\rangle_2 + \mathbb{Q}\langle 1\rangle_3$, containing $\ell$, and to $\mathbb{Q} + \sum_{i\le 8} \mathbb{Q}\langle 1\rangle_i$, containing $\ell$ and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$: at the lean the value of the minor $\binom{\{1,2,3,5\}}{2}$ at its own lean, at the centre the values of both minors on four marks at their centres, and besides them two numbers at the lean and six at the centre. □

**Exercise 55.8** ($\mathrm{V}_8$ at its full cobasis lean, read from its list)**.** Let $\mathrm{V}_8$ be the list of rank four on $\mathcal{E} = \{1, \dots, 8\}$ with the pairs $P_1 = \{1,2\}$, $P_2 = \{3,4\}$, $P_3 = \{5,6\}$, $P_4 = \{7,8\}$, whose bases are the sixty-five sets of four marks other than the five unions $P_i \cup P_j$ with $\{i,j\} \neq \{3,4\}$. Read its ray with origin 0 and direction $(\mathbf{1}; 0, 2)$, of stride $f = \alpha^{\mathcal{E}}\Psi^{-2}$, by the procedure of Definition 41.8:

(i) the walls, $V_1$, the native gauge of the direction, $f^\star$ and $\det H$;

(ii) the box, its spanning set, the matrix $\mathbf{B}$ and its common denominator;

(iii) the relation of the rows and its line-up strengths;

(iv) the native gauges of the ray, and what they say about the dimension, the masters, the order of the law of [1] and $\mathbf{B}$ at infinite strength;

(v) the value $\langle 1\rangle_1 = P_{\mathrm{V}_8}(\mathbf{1}; 0, 2)$.

*Solution.* (i) *Walls and volume.* As for $\mathrm{N}_6$ at its lean (Exercise 55.5(i)), $s = 0$, $t = 2$ and $d^*(S) = |S| - r(S)$ give $\kappa_{1,S} = 2r(S) - |S|$. Input: the 254 nonempty proper subsets of $\mathcal{E}$ and their ranks. Method: $\kappa_{1,S}$ for each, and the recursion over chains $F(U) = \kappa_{1,U}^{-1} \sum_{e\in U} F(U \setminus \{e\})$, $V_1 = \sum_e F(\mathcal{E} \setminus \{e\})$ of Exercise 55.5(i) (Proposition 39.34), in exact rationals. Output: the walls are 1 sixteen times, 2 sixty-one times, 3 one hundred and twelve times and 4 sixty-five times, all positive, and

$$V_1 = 300.$$

Check: the same program gives $V_1 = 66$ and 63 for the two rays of $\mathrm{N}_6$, as computed by hand in Exercises 55.5 and 55.6. Receipt: `exact_data.py`.

*The automorphisms.* The non-bases are the unions of two pairs that involve $P_1$ or $P_2$, so an automorphism keeps $\{P_1, P_2\}$ and $\{P_3, P_4\}$; the 64 permutations that exchange the marks within pairs, $P_1$ with $P_2$ and $P_3$ with $P_4$ are automorphisms, and they keep the masses $\mathbf{1}$, so they are symmetries of the ray (Proposition 39.11(ii)).

*The native gauge.* The points fixed by the 64 automorphisms are, on the chart $\alpha_8 = 1$, the slice $\alpha = (x, x, x, x, 1, 1, 1, 1)$. A cobasis is the complement of a basis, so the cobases are the sets of four other than the complements $P_3 \cup P_4$, $P_2 \cup P_4$, $P_2 \cup P_3$, $P_1 \cup P_4$, $P_1 \cup P_3$ of the non-bases. On the slice a set $C$ of four has $\alpha^C = x^{|C \cap (P_1 \cup P_2)|}$, the sets of four give $\sum_j \binom{4}{j}\binom{4}{4-j} x^j = 1 + 16x + 36x^2 + 16x^3 + x^4$, and the five sets left out give $1 + 4x^2$, so

$$\Psi = x\,(x^3 + 16x^2 + 32x + 16), \qquad f = \frac{x^4}{\Psi^2} = \frac{x^2}{(x^3 + 16x^2 + 32x + 16)^2},$$

$$\frac{d}{dx}\log f = \frac{-4\,(x^3+8x^2-8)}{x\,(x^3+16x^2+32x+16)}.$$

At a point fixed by the automorphisms the differential of $\log f$ in $y = \log\alpha$ is $(u,u,u,u,v,v,v,v)$ with $u+v=0$, $f$ being homogeneous of degree 0, and it vanishes when its derivative along the slice does, as in Exercise 55.5(ii). So the three roots of $x^3+8x^2-8$ give native gauges (Definition 38.4); at a root $x^3+16x^2+32x+16 = 8(x+1)(x+3)$, and the native value is

$$f = \Big(\frac{x}{8(x+1)(x+3)}\Big)^2.$$

The positive root $x^\star = 0.94566781799051109644\ldots$ gives $\alpha^\star$ (Theorem 38.14(i)) and

$$f^\star = 2.3709304548895617852\ldots\cdot 10^{-4};$$

the roots $-1.0748031540504515212\ldots$ and $-7.8708646639400595752\ldots$ give the native values $0.87033768220817625690\ldots$ and $8.6423132047649619559\ldots\cdot 10^{-4}$. The Hessian of $-\log f$ at $\log\alpha^\star$ on the chart is twice the covariances of the indicators under the cobasis law at $\alpha^\star$ (Theorem 38.15); reduced in $\mathbb{Q}(x^\star)$ (receipt `hessians.py`),

$$\det H = \frac{395719931704\,x^{\star 2} + 2812581517252\,x^\star - 3012741822955}{379727508532},$$

which is $0.0024055146373001934835\ldots$

(ii) *The box* (Procedure 41.8, the masters). Only $\Psi$ is active. Input: the boxes $\mathfrak{B}_{\mathfrak{t}}(0,1)$ modulo the 64 automorphisms (Definition 39.15(iv), Definition 39.20(i)). Method: as in Exercise 55.5(iii). Output: $\mathfrak{B}_2(0,1)$ has 1155 orbits of stems, 4745 orbits of symbols and 10395 relations; at $\mathfrak{s}=1$ it leaves 36 free Laurent orbits of which 33 targets do not reduce to them, and at $\mathfrak{s}=0$ the target of 1 does not reduce: not accepted. $\mathfrak{B}_3(0,1)$ has 9723 orbits of stems, 32810 orbits of symbols and $87507 = 9723\cdot 9$ relations, the letter identities of the stems and their eight mark relations; one elimination leaves 60 free Laurent orbits of wall degree at most 2, every target reduces to them, and the box is accepted at $\mathfrak{s}=2$ (Theorem 39.21(i)), with a spanning set of 60 orbit averages. Receipt: `rayreader.py V "1 1 1 1 1 1 1 1 ; 0 2" --digits 30` (log `runlogs/V_lean.stdout`).

*The matrix* (the matrix over $\mathbb{Q}(k)$). Input: the 87507 relations. Method: the targets need 19581 of them, of rank 19581; eliminations modulo primes of 62 bits at random strengths; every entry read as a quotient of polynomials; Chinese remaindering with rational reconstruction. Output: on the first prime 178 strengths determine every entry, the largest sum of the degrees of a numerator and a denominator being 163; all the denominators divide one polynomial of degree 85, over which the numerators have degree at most 88, so each further prime needs 94 strengths; the entries become reconstructible over $\mathbb{Q}$ at seven primes, 432 bits. The common denominator is, up to a constant factor,

$$Q_c = k\,(k-1)^2(2k-3)(k-2)^8(2k-5)(k-3)^7\,S_{65}(k),$$

$S_{65}$ irreducible of degree 65; 72 entries vanish, and in lowest terms the numerators have degree at most 83 and the denominators at most 80, so $d_{\mathbf{B}} = 83$. Check: $\mathbf{B}$ agrees with an elimination of the whole box at a prime and a strength drawn at random.

(iii) *The relation of the rows.* Input: $\mathbf{B}$. Method: the rows $w_i$ of Proposition 39.6 modulo primes, their first dependence, reconstructed over $\mathbb{Q}$. Output (receipt as in (ii)): $w_0,\ldots,w_{23}$ are independent and $w_{24}$ depends on them, so $h=24$, and the relation $\sum_{j=0}^{24} p_j(k)\langle 1\rangle_{k+j} = 0$ has $p_j \in \mathbb{Z}[k]$ of degree 377, with coefficients of up to 753 digits, from 63 primes, and over $\mathbb{Q}$

$$p_0 = k^7(k+1)^5(k+2)^5\prod_{j=3}^{7}(k+j)^3\prod_{j=8}^{13}(k+j)\,S_{339}(k),$$

$S_{339}$ irreducible of degree 339, while $p_{24}$ is a product of 38 linear factors, with roots at integers, halves and quarters from $-\frac{1}{2}$ to $-\frac{47}{2}$, and $S_{339}(k-1)$. Check: the relation holds at a prime and a point drawn at random. *Line-up strengths.* With $q_{24}(k) = p_0(k-24)$, Proposition 39.6(iii) gives

$$p_0(k-24) \;\Big|\; \det \mathbf{W}(k)_J \prod_{j=0}^{23} Q_c(k-j)^{24-j}$$

for every set $J$ of 24 columns with $\det \mathbf{W}_J \neq 0$. The irreducible factors of $Q_c$ have degree at most 65, so the irreducible $S_{339}(k-24)$ divides none of the $Q_c(k-j)$ and divides $\det \mathbf{W}(k)_J$: 339 of the 377 degrees of $p_0$ are strengths at which the rows $w_0, \dots, w_{23}$ line up, and the other 38 are the integers $0, \dots, -13$.

(iv) *The native gauges.* A native gauge of the ray is a point of $X_F$ with $1 = 2R_e^*(\alpha)$ for every mark (Definition 38.4). Input: these equations, and the same with exponents $(\mathbf{a}; b)$ in place of $(\mathbf{1}; 2)$, $|\mathbf{a}| = 4b$, which are linear in the exponents. Method: as in Exercise 55.9(iii): Newton's method at random complex exponents, the solutions carried around random triangles of exponents until several turns in a row add nothing, run from two seeds; then every solution carried to $(\mathbf{1}; 2)$ along four routes, the straight segment and three broken ones through random exponents. Output: from both seeds, 839 solutions at random exponents, the count reaching 839 at the sixteenth and seventeenth turns and then staying there for four and six turns; at the direction, along every route, the same 331 nondegenerate native gauges, in 31 orbits of the 64 automorphisms, of sizes $1, 1, 1, 2, 2, 2, 2$, eight of 4, fourteen of 16 and two of 32, with 18 distinct native values:

| | |
|---|---|
| one orbit: | $f^\star$, 0.000818456 (16), 0.000864231 (1), 0.124846403 (16), 0.870337682 (1), $-0.592832429 \pm 0.713227829\,i$ (16 each), |
| two orbits: | 0.004487298 (4), $-0.013533898$ (8), $-0.026238829$ (8), 0.066968510 (32), $0.007758455 \pm 0.013607932\,i$ (32 each), 0.109146811 (32), $\frac{1}{8}$ (64), 0.870512702 (4), |
| three orbits: | 0.021107646 (24), 0.416392354 (24), |

the number of native gauges in brackets (renderings). The other 508 paths stop before the direction, at $\tau$ between 0.598 and $1 - 1.8 \cdot 10^{-6}$ along the segment, where the step collapses. Check: one native gauge of each orbit polished by Newton's method at 40 digits, with the Hessian determinants of modulus at least $\det H = 0.0024055\ldots$, reached at $\alpha^\star$; the three native gauges of the slice in (i) are the three orbits of size 1. Receipts: `critpoints.py` (`crit_V8_lean.json`, `crit_V8_lean_seed2.json`), `census_paths.py` (`census_V8_routes.json`), `escape_check.py` (`escape_V8.out`), `betas.py` (`betas_V8.out`).

So, by Theorem 38.6(iii), the ray has dimension at least 331; by Theorem 39.12(iii) the invariant module under the 64 automorphisms has dimension at least 31; and by Corollary 38.7 the law of [1] has order at least 18, so that with (iii) and Proposition 39.6(ii) it lies between 18 and 24. The vectors $\beta_c = (b_l(\alpha_c))_l$ of the spanning set at one native gauge of each orbit have rank 31, their singular values running from 1262 down to 0.036 (receipt `betas.py`): they do not span $\mathbb{C}^{60}$, and Proposition 39.42(i) does not decide whether the 60 are masters. At large strength the entries of $\mathbf{B}$ grow like $k^3$ (output, receipt as in (ii)): $\mathbf{B}$ is not regular at $k = \infty$, and Proposition 39.42(ii) does not apply to it.

(v) *The value* (Theorem 40.6). Input: the relation of (iii), $f^\star$ and $\det H$; $f^\star$ is a simple root of the symbol, and the exponent of the density at $f^\star$ is $\frac{n-3}{2} = \frac{5}{2}$. Method: as in Exercise 55.5(vi). Output (receipt as in (ii)):

$$P_{\mathrm{V}_8}(\mathbf{1}; 0, 2) = \langle 1 \rangle_1 = 7.12823937562968675472229135353893457156078898940720766490704\ldots,$$

two starting strengths agreeing to $2.4 \cdot 10^{-82}$, and $P/V_1 = 0.0237607979187656\ldots$ Check: the low-end relation of Corollary 39.38(ii), with $V_1 = 300$, holds on the values to 73 digits.

So $\mathrm{V}_8$ at $(\mathbf{1}; 0, 2)$ has $V_1 = 300$, $f^\star = \big(x^\star/(8(x^\star+1)(x^\star+3))\big)^2$ with $x^{\star 3} + 8x^{\star 2} = 8$, a box accepted at $\mathfrak{B}_3(0,1)$ with 60 spanning elements, a relation of order 24 and degree 377 of which 339 degrees are line-up strengths, at least 331 native gauges in 31 orbits with 18 native values, and the value $7.1282393756296867547\ldots$ □

**Exercise 55.9** (Sizing a reading: $\mathrm{Q}_6$ and $\mathrm{N}_7$)**.** Let $\mathrm{Q}_6$ be the list of rank three on $\{1,\ldots,6\}$ whose bases are the seventeen triples other than $\{1,3,6\}$, $\{2,3,4\}$ and $\{2,5,6\}$, and $\mathrm{N}_7$ the list of rank three on $\{1,\ldots,7\}$ whose bases are the thirty-four triples other than $\{1,2,5\}$. For the rays with origin 0 of $\mathrm{Q}_6$ along $(1,3,2,3,1,2;2,2)$ and of $\mathrm{N}_7$ along $(1,2,2,1,2,1,1;2,1)$, find what sizes their reading before any relation is written: the walls, $V_1$, the native gauge of the direction, the native gauges of the ray and their native values, with the bounds these give on the dimension of the ray and on the order of a law of [1]; then the size of the box with one unit of each letter at $\mathfrak{t} = 2$, and whether it is accepted.

*Solution.* (i) *Walls and volume* (Definition 8.1, Proposition 39.34). For $\mathrm{Q}_6$, $s = t = 2$ and the masses add up to $12 = 2\cdot 3 + 2\cdot 3$, on the balance wall; the walls $\kappa_{1,S} = m_{1,S} - 2d(S) - 2d^*(S)$ of the 62 sets $S$ are, by height and count,

$$1(4),\ 2(7),\ 3(13),\ 4(13),\ 5(12),\ 6(9),\ 7(3),\ 8(1),$$

all positive, and the recursion over chains of Exercise 55.5(i), run over the 62 sets (no automorphism of $\mathrm{Q}_6$ fixes the masses), gives $V_1 = \frac{16493}{2430}$. For $\mathrm{N}_7$, $s = 2$, $t = 1$ and $10 = 2\cdot 3 + 1\cdot 4$; the 126 walls are $1(7),\ 2(17),\ 3(29),\ 4(39),\ 5(29),\ 6(5)$, and $V_1 = \frac{3519}{128}$. The automorphisms of $\mathrm{N}_7$ fixing the masses are the 12 permutations that exchange the marks 2 and 5 of the triple $\{1,2,5\}$ or not and permute the marks $4,6,7$ of mass 1. Receipt: `exact_data.py`.

(ii) *The native gauge of the direction* (Theorem 38.14(i)), by Newton's method on the chart: for $\mathrm{Q}_6$, $\alpha^\star = (0.45796,\ 3.83890,\ 1.55835,\ 3.85580,\ 0.54507,\ 1)$, $f^\star = 8.3244647577707389\ldots \cdot 10^{-5}$ and $\det H = 0.13385450029314584\ldots$; for $\mathrm{N}_7$, $\alpha^\star = (1.15660,\ 3.63195,\ 3.14019,\ 1,\ 3.63195,\ 1,\ 1)$, constant on the orbits of the automorphisms as it must be, $f^\star = 7.2002200376070769\ldots \cdot 10^{-5}$ and $\det H = 0.027073861743142843\ldots$ (renderings; receipt `hessians.py`).

(iii) *The native gauges.* A native gauge is a point of $X_F$ at which $m_{1,e} = s\,R_e + t\,R_e^*$ for every mark (Definition 38.4). Input: these equations, and the same with the exponents $(\mathbf{a}; b_K, b_\Psi)$ in place of $(\mathbf{m}_1; s, t)$, $|\mathbf{a}| = 3b_K + (n-3)b_\Psi$, which are linear in the exponents. Method: for random exponents, a Groebner basis of the equations, multiplied by $K\Psi$, on the chart and saturated by the coordinates and the letters, modulo the prime 32003, and, separately, Newton's method at random complex exponents with the solutions carried around random triangles of exponents until several turns add nothing; then each solution carried along the segment to $(\mathbf{m}_1; s, t)$, a path that either arrives at a point of $X_F$ or runs to its boundary; and a Groebner basis at $(\mathbf{m}_1; s, t)$ itself. Output: for $\mathrm{Q}_6$, 114 solutions at random exponents by both methods; 90 of the paths arrive, at 90 distinct native gauges, all nondegenerate, with 90 distinct native values, and 24 run off; at the direction the Groebner basis has exactly 90 points counted with multiplicity, so these are all the native gauges. For $\mathrm{N}_7$, 1174 solutions at random exponents; all 1174 arrive, at distinct nondegenerate native gauges, in 268 orbits of the 12 automorphisms (32 of size 1, 28 of size 2, 86 of size 3, 106 of size 6, 16 of size 12), with 83 distinct native values. Check: the native gauges of $\mathrm{Q}_6$ polished at 40 digits, residuals below $10^{-37}$, the Hessian determinants at least 0.13, the native values at least $1.6 \cdot 10^{-4}$ apart, and the same 90 values from two further runs with other random exponents. Receipts: `gb_count.py`, `gb_direction.py`, `critpoints.py`, `polish.py`.

So, by Theorem 38.6(iii), the dimension of the ray is at least 90 for $\mathrm{Q}_6$ and at least 1174 for $\mathrm{N}_7$; by Theorem 39.12(iii) the invariant module of $\mathrm{N}_7$ under its 12 automorphisms has dimension at least 268; and by Corollary 38.7 every relation $\sum_j p_j(k) f^j \in \mathcal{L}$, the law of [1] among them, has order at least 90 for $\mathrm{Q}_6$ and 83 for $\mathrm{N}_7$, the number of distinct native values. These counts are read off the list and the direction alone.

(iv) *The box of* $\mathrm{Q}_6$. Both letters are active, and the box with one unit of each is $\mathfrak{B}_2(1,1)$ (Definition 39.15(iv)). Input: the box, with no symmetry. Output: 361682 relations on 225874 symbols, the mark relations of 31785 stems and the letter identities, each of $1 + |\mathcal{B}| = 18$ terms (receipt `box_test.py Q6.json "1 3 2 3 1 2 ; 2 2" 2 1 1`). Method for the rest: the procedure, as in Exercise 55.5(iii) and (iv); receipt `rayreader.py Q6.json "1 3 2 3 1 2 ; 2 2" --digits 30`. Output: one elimination modulo a prime at a random strength leaves 114 free Laurent symbols of wall degree at most 1, and every target reduces to them, so the box is accepted at $\mathfrak{s} = 1$, and the spanning set of 114 spans a submodule that contains [1] (Theorem 39.21(i)). The targets need 78921 of the relations. Modulo the first prime $\mathbf{B}$ has a common denominator of degree 126 and numerators of degree up to 129, so it is not regular at $k = \infty$; it needs 271 strengths on that prime and 135 on each further one; and modulo that prime the rows first depend at $h = 90$, where 12288 consecutive strengths do not determine the relation, so that its degree exceeds 6100. By (iii) and Proposition 39.6(ii), $90 \le h \le 114$ whatever the spanning set, and the rows meet the lower bound.

*The box of* $\mathrm{N}_7$. The box $\mathfrak{B}_2(1,1)$ modulo the 12 automorphisms has 71320 orbits of symbols, 7613 orbits of stems and 100719 relations of up to $1 + |\mathcal{B}| = 35$ terms. One elimination modulo $2^{61} - 1$ at a random strength has rank 63883; of the 359 Laurent orbits of wall degree at most 1, 247 are free, and their targets all lie in the box, but 246 of them do not reduce to the 247; at $\mathfrak{s} = 0$ the target of 1 does not reduce. So the box with one unit of each letter at $\mathfrak{t} = 2$ is not accepted (Definition 39.20(iii)), and Theorem 39.22 sends the search to larger boxes. By Definition 39.20(ii) the class of every invariant Laurent symbol of wall degree at most 1 is a combination of the 247 free ones, fewer than the 268 dimensions of the invariant module found in (iii). Receipt: `box_test.py N7.json "1 2 2 1 2 1 1 ; 2 1" 2 1 1`. The next boxes of the search are larger: $\mathfrak{B}_2(2,1)$ has 236600 relations on 158724 symbols, and $\mathfrak{B}_3(1,1)$ has 707989 relations of up to 35 terms (receipt for the search, `rayreader.py N7.json "1 2 2 1 2 1 1 ; 2 1" --digits 30`).

So before any relation is written, the walls give $V_1 = \frac{16493}{2430}$ and $\frac{3519}{128}$, and the critical equations give at least 90 and 1174 dimensions and laws of order at least 90 and 83; the box of $\mathrm{Q}_6$ at $\mathfrak{t} = 2$ is accepted with 114 spanning elements, and the box of $\mathrm{N}_7$ with one unit of each letter at $\mathfrak{t} = 2$ is not accepted. □

**Exercise 55.10** (A table of rays read by the procedure)**.** Let $\binom{[n]}{1}$ be the list of rank one on $\{1, \dots, n\}$ whose bases are the single marks, $\binom{[n]}{2}$ the list of rank two whose bases are all the pairs, $\mathrm{H}_4$ the list of rank two on $\{1,2,3,4\}$ whose bases are the five pairs other than $\{1,2\}$, and $W_3$ the wheel with three spokes (Proposition 39.65), of rank three on its six spokes and rims, whose bases are the sixteen spanning trees of its graph. Read by the procedure of Definition 41.8 the rays with origin 0 of

$$\tbinom{[3]}{1},\ \tbinom{[4]}{1},\ \tbinom{[5]}{1},\ \tbinom{[4]}{2},\ \tbinom{[5]}{2},\ \mathrm{H}_4,\ \mathrm{N}_6 \text{ along } (\mathbf{1}; 1, 1); \qquad \tbinom{[4]}{2},\ W_3,\ \mathrm{N}_6,\ \mathrm{V}_8 \text{ along } (\mathbf{1}; 0, 2);$$
$$\tbinom{[3]}{1} \text{ along } (2\mathbf{1}; 4, 1); \qquad \mathrm{H}_4 \text{ along } (1, 1, 3, 3; 3, 1),$$

$\mathrm{N}_6$ and $\mathrm{V}_8$ as in Exercises 55.4 and 55.8, and the ray of $\binom{[3]}{1}$ with origin $(u_0; 0, 0)$, $u_0 = (-\frac{1}{2}, 0, \frac{1}{2})$, and direction $(\mathbf{1}; 1, 1)$, whose value at $k = 1$ is $P_{\binom{[3]}{1}}(\frac{1}{2}, 1, \frac{3}{2}; 1, 1)$. For each ray give:

(i) the accepted box with its stems, symbols and relations modulo the symmetries of the ray, the bound $\mathfrak{s}$ at which it is accepted, the size $\mathfrak{m}$ of its spanning set, the largest degree $d_{\mathbf{B}}$ of

a numerator or a denominator of an entry of $\mathbf{B}$ in lowest terms, and the order $h$ and the degree $D$ of the relation of the rows;

(ii) whether the spanning set is a set of masters and that relation the law of [1], and the spectrum of $\mathbf{B}(\infty)$ where $\mathbf{B}$ is regular at $k = \infty$;

(iii) $V_1$ and the value $\langle 1 \rangle_1$, with the checks that the reading allows.

*Solution.* (i) *The boxes, the matrices and the rows.* Input: the list and the direction. Method: the procedure as in Exercise 55.5(iii) and (iv). The boxes $\mathfrak{B}_\mathfrak{t}$ with one unit of each active letter are tried for $\mathfrak{t} = 2, 3, \ldots$ (Definition 39.15(iv), Theorem 39.22), each reduced modulo the symmetries of the ray: the automorphisms that fix the masses and the origin, and the Cremona at a centre with origin 0 (Proposition 39.11, Definition 39.20(i)). One elimination modulo a prime of 61 bits at a random strength decides acceptance (Definition 39.20(iii)); $\mathbf{B}$ over $\mathbb{Q}(k)$ comes from eliminations of the relations that the targets need, modulo primes of 62 bits at random strengths, with rational reconstruction; and the rows $w_i$ of Proposition 39.6 give the first dependence, reconstructed over $\mathbb{Q}$ from several primes. Output:

| ray | box | stems | symbols | relations | $\mathfrak{s}$ | $\mathfrak{m}$ | $d_\mathbf{B}$ | $(h, D)$ |
|---|---|---|---|---|---|---|---|---|
| $\binom{[3]}{1}$, $(\mathbf{1}; 1, 1)$ | $\mathfrak{B}_2(1,1)$ | 4 | 30 | 32 | 1 | 2 | 2 | $(2, 2)$ |
| $\binom{[4]}{1}$, $(\mathbf{1}; 1, 1)$ | $\mathfrak{B}_2(1,1)$ | 5 | 42 | 46 | 1 | 2 | 3 | $(2, 3)$ |
| $\binom{[5]}{1}$, $(\mathbf{1}; 1, 1)$ | $\mathfrak{B}_3(1,1)$ | 10 | 100 | 112 | 2 | 3 | 5 | $(3, 4)$ |
| $\binom{[3]}{1}$, $(2\mathbf{1}; 4, 1)$ | $\mathfrak{B}_2(1,1)$ | 17 | 79 | 124 | 1 | 2 | 4 | $(2, 6)$ |
| $\binom{[3]}{1}$, $(\mathbf{1}; 1, 1)$, $u_0$ | $\mathfrak{B}_2(1,1)$ | 19 | 110 | 149 | 1 | 4 | 3 | $(3, 4)$ |
| $\binom{[4]}{2}$, $(\mathbf{1}; 0, 2)$ | $\mathfrak{B}_2(0,1)$ | 8 | 20 | 40 | 1 | 2 | 4 | $(2, 3)$ |
| $\binom{[4]}{2}$, $(\mathbf{1}; 1, 1)$ | $\mathfrak{B}_2(1,1)$ | 7 | 33 | 66 | 1 | 2 | 4 | $(2, 3)$ |
| $\binom{[5]}{2}$, $(\mathbf{1}; 1, 1)$ | $\mathfrak{B}_3(1,1)$ | 21 | 234 | 245 | 2 | 5 | 8 | $(4, 6)$ |
| $\mathrm{H}_4$, $(\mathbf{1}; 1, 1)$ | $\mathfrak{B}_2(1,1)$ | 17 | 102 | 180 | 1 | 4 | 3 | $(4, 7)$ |
| $\mathrm{H}_4$, $(1, 1, 3, 3; 3, 1)$ | $\mathfrak{B}_2(1,1)$ | 115 | 631 | 1000 | 1 | 4 | 9 | $(4, 25)$ |
| $W_3$, $(\mathbf{1}; 0, 2)$ | $\mathfrak{B}_2(0,1)$ | 66 | 211 | 462 | 1 | 2 | 8 | $(2, 5)$ |
| $\mathrm{N}_6$, $(\mathbf{1}; 0, 2)$ | $\mathfrak{B}_2(0,1)$ | 104 | 329 | 728 | 1 | 7 | 6 | $(6, 24)$ |
| $\mathrm{N}_6$, $(\mathbf{1}; 1, 1)$ | $\mathfrak{B}_3(1,1)$ | 252 | 1909 | 3774 | 2 | 15 | 17 | $(11, 52)$ |
| $\mathrm{V}_8$, $(\mathbf{1}; 0, 2)$ | $\mathfrak{B}_3(0,1)$ | 9723 | 32810 | 87507 | 2 | 60 | 83 | $(24, 377)$ |

At the centres of $\binom{[5]}{1}$, $\binom{[5]}{2}$ and $\mathrm{N}_6$ the box $\mathfrak{B}_2(1,1)$ is not accepted: at $\mathfrak{s} = 1$ it leaves 2, 3 and 8 free Laurent orbits of which 1, 2 and 7 targets do not reduce to them, and at $\mathfrak{s} = 0$ the target of 1 does not reduce; at the lean of $\mathrm{V}_8$, $\mathfrak{B}_2(0,1)$ leaves 36 free orbits at $\mathfrak{s} = 1$, of which 33 targets do not reduce, and fails at $\mathfrak{s} = 0$ the same way. Every other box in the table is the first one tried. Check: every $\mathbf{B}$ agrees with an elimination of the whole box at a prime and a strength drawn at random, and every relation of the rows holds at a random prime and point; on the thirteen rays other than $\mathrm{V}_8$ both checks are repeated at three fresh primes and points. Receipts: `runs_table.sh`, `table_summary.py`, `tests/check_run.py` (`table_checkrun.out`).

(ii) *Masters and the law of* [1]. By Proposition 39.6(ii), with Theorem 39.21(iii), $h$ is at least the dimension of the cyclic submodule of [1], with equality when the spanning set is a set of masters, and then the relation of the rows is the law of [1]; by Corollary 38.7 that dimension is at least the number of distinct native values of nondegenerate native gauges. So the relation is the law of [1] as soon as either the vectors $\beta_c = (b_l(\alpha_c))_l$ at nondegenerate native gauges span $\mathbb{C}^\mathfrak{m}$ (Proposition 39.42(i)), or $h$ distinct native values are found; in the second case, when moreover $h = \mathfrak{m}$, the cyclic submodule of [1] fills the span of the $\mathfrak{m}$ classes, and the spanning set is a set of masters. Input: the critical equations of each direction. Method: the census of Exercise 55.9(iii),

monodromy at random exponents and paths to the direction; each native gauge polished at 40 digits with its Hessian; and the vectors $\beta_c$ at one native gauge of each orbit, each $b_l$ averaged over the symmetries of the ray. Output and conclusion:

- $\binom{[3]}{1}$, $\binom{[4]}{1}$ at their centres: the table's gauge, of native value $\frac{1}{9}$ and $\frac{1}{16}$, and the signs with one scale $-1$, of native value 1 and $\frac{1}{4}$ (Theorem 38.17(ii)), 4 and 5 native gauges in all, every one nondegenerate; the two vectors $\beta_c$ are independent: masters, and the law of [1].
- $\binom{[3]}{1}$ along $(\mathbf{21};4,1)$: 4 native gauges, of native values $\frac{1}{243}$ and $-\frac{27}{64}$, two independent vectors: masters. The ray with origin $u_0$: the 4 native gauges of the centre direction, which no symmetry of the ray permutes, give four independent vectors: masters.
- $\binom{[5]}{2}$ at its centre: 66 native gauges in 6 orbits of the 120 automorphisms, of native values $\frac{1}{100}, \frac{1}{4}, -\frac{1}{8}, 1$, all nondegenerate, with vectors of rank 5: masters, and $h = 4 < \mathfrak{m}$, the cyclic submodule of [1] of dimension 4 inside $\mathfrak{N}$ of dimension 5 (Exercise 54.20).
- $\binom{[5]}{1}$ at its centre: 16 native gauges, the table's gauge ($\frac{1}{25}$), the five signs with one scale $-1$ ($\frac{1}{9}$) and the ten with two (1); three distinct native values and $h = \mathfrak{m} = 3$: masters and the law of [1], although the three vectors $\beta_c$ lie in a plane. $\binom{[4]}{2}$ at both directions: 4 native gauges, of native values $\frac{1}{36}$ (the table's gauge) and $\frac{1}{4}$ (an orbit of three), with $h = \mathfrak{m} = 2$, and $W_3$ at $(\mathbf{1};0,2)$: 3 native gauges, of native values $\frac{1}{256}$ and $\frac{1}{4}$, with $h = \mathfrak{m} = 2$: masters and the law of [1], the two vectors $\beta_c$ being proportional.
- $\mathrm{N}_6$: Exercises 55.5 and 55.6. $\mathrm{V}_8$: Exercise 55.8.
- $\mathrm{H}_4$: of the 6 critical points at random exponents, two arrive at nondegenerate native gauges at each direction. At the centre they are the table's gauge and the sign $(-1,-1,1,1)$, of native values $\frac{1}{25}$ and $\frac{1}{9}$ and Hessian determinants $-\frac{32}{625}$ and $\frac{32}{81}$ on the chart; along $(1,1,3,3;3,1)$ they are the points $(x,x,1,1)$ with $3x^2+8x-1=0$, of native values $\frac{7643\pm1064\sqrt{19}}{1366875}$, the larger one, at $x=\frac{\sqrt{19}-4}{3}>0$, being $f^\star$. The other four paths meet in pairs at the points $(\mp i,\pm i,-1,1)$ at the centre and $(\mp\frac{i}{\sqrt{3}},\pm\frac{i}{\sqrt{3}},-1,1)$ along $(1,1,3,3;3,1)$, native gauges of native value 1 whose Hessian is singular (exact; receipt `h4_gauges.py`). So Corollary 38.7 bounds the dimension of the cyclic submodule of [1] below by 2 only, and the two vectors $\beta_c$ span a plane: the relation of the rows is a relation of order 4 in $\mathcal{L}$, of order at least that of the law of [1] (Theorem 39.21(iii)), which lies between 2 and 4.

*Infinite strength.* $\mathbf{B}$ is regular at $k=\infty$ on nine of the rays, with the spectra in the table of (iii), and at every nondegenerate native gauge of the census $\beta_c$ is an eigenvector of $\mathbf{B}(\infty)$ for $f(\alpha_c)^{-1}$, to $10^{-38}$ at 40 digits (Proposition 39.42(ii)). On $\binom{[5]}{1}$, $\binom{[4]}{2}$ and $W_3$ some entries of $\mathbf{B}$ have numerators of larger degree than their denominators, and Proposition 39.42(ii) says why they must: nondegenerate native gauges of distinct native values have dependent vectors $\beta_c$ there, three in a plane for $\binom{[5]}{1}$ and two on a line for the others, which a regular $\mathbf{B}$ would make independent eigenvectors. For $\mathrm{V}_8$ see Exercise 55.8.

(iii) *The values.* $V_1$ is computed from the walls by Proposition 39.34 with the recursion over chains of Exercise 55.5(i) (receipt `exact_data.py`); for the ray with origin $u_0$ it is that of the direction $(\mathbf{1};1,1)$. The values come from Theorem 40.6. Input: the relation of the rows, $f^\star$ and $\det H$. Method: as in Exercise 55.5(vi), asked to 60 digits. Output:

| ray | $V_1$ | eigenvalues of $\mathbf{B}(\infty)$ | $\langle 1\rangle_1$ |
|---|---|---|---|
| $\binom{[3]}{1}$, $(\mathbf{1};1,1)$ | 6 | 9, 1 | 2.3439072386894588906… |
| $\binom{[4]}{1}$, $(\mathbf{1};1,1)$ | 24 | 16, 4 | 8.4143983221171599977… |
| $\binom{[5]}{1}$, $(\mathbf{1};1,1)$ | 120 | not regular | 39.945588011996245698… |
| $\binom{[3]}{1}$, $(2\mathbf{1};4,1)$ | 1 | 243, $-\frac{64}{27}$ | 0.030080945282262304399… |
| $\binom{[3]}{1}$, $(\mathbf{1};1,1)$, $u_0$ | 6 | 9, 1, 1, 1 | 4.7773947923309302997… |
| $\binom{[4]}{2}$, $(\mathbf{1};0,2)$ | 12 | not regular | 2.3439072386894588906… |
| $\binom{[4]}{2}$, $(\mathbf{1};1,1)$ | 12 | not regular | 2.3439072386894588906… |
| $\binom{[5]}{2}$, $(\mathbf{1};1,1)$ | 30 | 100, 4, 4, 1, −8 | 4.0297141707110496073… |
| $\mathrm{H}_4$, $(\mathbf{1};1,1)$ | 16 | 25, 9, 1, 1 | 4.0962369983378828500… |
| $\mathrm{H}_4$, $(1,1,3,3;3,1)$ | $\frac{160}{27}$ | $\frac{7643\pm1064\sqrt{19}}{27}$, 1, 1 | 0.48830319489526888388… |
| $W_3$, $(\mathbf{1};0,2)$ | 84 | not regular | 7.2123414189575657123… |
| $\mathrm{N}_6$, $(\mathbf{1};0,2)$ | 66 | Exercise 55.5 | 4.7568025637582294434… |
| $\mathrm{N}_6$, $(\mathbf{1};1,1)$ | 63 | Exercise 55.6 | 4.5789124788824048063… |
| $\mathrm{V}_8$, $(\mathbf{1};0,2)$ | 300 | not regular | 7.1282393756296867547… |

Check: on every ray the two starting strengths agree to at least 82 digits. On the thirteen rays with origin 0 the low-end relation of Corollary 39.38(ii), $[k^{n-1}]p_0\cdot V_1+\sum_{i=1}^{h}p_i(0)\langle 1\rangle_i=0$, which the reading does not use and which holds for the relation of the rows as well as for the law of [1], holds on the values $\langle 1\rangle_1,\dots,\langle 1\rangle_h$ to at least 89 digits relative to its largest term on the twelve read here at 60 digits, and to 73 digits on $\mathrm{V}_8$ (Exercise 55.8(v)). Five values agree with numbers computed otherwise: those of $\binom{[3]}{1}$ at its centre and of $\binom{[4]}{2}$ at both directions with $\frac{1}{3}\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\frac{1}{3}+\mathbb{Z}})((0,\infty))-(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\frac{2}{3}+\mathbb{Z}})((0,\infty))\big)$, summed as the series of Definition 5.2, to 90 digits; those of $\binom{[4]}{1}$ at its centre and of $W_3$ at $(\mathbf{1};0,2)$ with $7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and $6\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, to 89 digits; and that of the ray with origin $u_0$ with

$$\int_0^\infty 2\log\frac{(1+u^2)^2}{u^2}\,\frac{du}{1+u^2+u^4},$$

computed by quadrature at 90 digits, to 84 digits. Receipts: `table_check.py` (`table_check.out`), `table_masters.py` (`table_masters.out`).

So the procedure reads every ray of the table from its list and its direction, with boxes of up to 87507 relations; the relation of the rows is the law of [1] on all of them but the two rays of $\mathrm{H}_4$ and $\mathrm{V}_8$, where the census bounds the law's order below by 2 and 18; every value passes the low-end relation, and the five that can be compared with numbers computed otherwise agree with them to 84 digits or more. In particular $\binom{[4]}{2}$ has the value $\ell$ of $\binom{[3]}{1}$ at its centre both at $(\mathbf{1};0,2)$ and at $(\mathbf{1};1,1)$, to 90 digits. □

## 55.4 Triangles of classes

In this section $\Delta(s_0,s_1,s_2)$, $s_i\ge 1$, is the list $\binom{[3]}{2}$ with its mark $h_i$ replaced by a parallel class $C_i$ of $s_i$ marks (Definition 21.1); its bases are the pairs of marks in different classes. With the totals $A_i=\sum_{e\in C_i}\alpha_e$ and $A_i'=\sum_{e\in C_i}\alpha_e^{-1}$,

$$K(\alpha)=A_0A_1+A_0A_2+A_1A_2=e_2(A),\qquad K(\alpha^{-1})=e_2(A'),\qquad f^{-1}=e_2(A)\,e_2(A'),$$

on the centre ray at the strength one. $\Sigma(s)=\{s,s-2,\dots,-s\}$ is the set of the sign values of a class of $s$ marks. The lists $\mathrm{H}_4$, $\mathrm{R}_5$ and $\mathrm{T}_5$ read in this chapter are $\Delta(2,1,1)$, $\Delta(2,2,1)$ and

$\Delta(3,1,1)$, and $\mathrm{J}_6 = \Delta(3,2,1)$. Values of $f^{-1}$ are written as integers, so that the roots of a symbol, the native values, are their reciprocals. The receipts are in the folder `triangles/` of the receipts of this version.

**Exercise 55.11** (Native and corner values of a triangle of classes)**.** For $\Delta(s_0, s_1, s_2)$:

(i) find the native gauges of the centre ray and their native values;

(ii) find the corner values of every class of at least two marks (Definition 38.24);

(iii) show that the corners add to the native values exactly the values $(a_j a_k)^{-2}$ with $a_j \in \Sigma(s_j)$, $a_k \in \Sigma(s_k)$ nonzero, $\{i,j,k\} = \{0,1,2\}$, for a class $C_i$ of odd size $s_i \geq 3$;

(iv) list them for $\mathrm{J}_6$.

*Solution. (i) Native gauges.* The core is $\mathcal{H} = \binom{[3]}{2}$ with $K_{\mathcal{H}}(x) = e_2(x)$ and $K_{\mathcal{H}/h_i} = x_j + x_k$. By Theorem 42.24(b) each class is occupied, $\alpha_e = \eta_e \tau_i$ on $C_i$ with a sign value $a_i \in \Sigma(s_i)$, or empty, $A_j + A_k = A'_j + A'_k = 0$.

*All classes occupied, every $a_i \neq 0$.* The core reads the weighted triangle $K^{(a)}(\tau) = e_2(p)$ with $p_i = a_i \tau_i$, and $K^{(a)}(\tau^{-1}) = e_2(q)$ with $q_i = a_i \tau_i^{-1}$. Its occupations are $R_i(\tau) = 1 - p_j p_k / e_2(p)$ and $R_i(\tau^{-1}) = 1 - q_j q_k / e_2(q)$, so the native-gauge equations of the weighted core read $p_j p_k / (q_j q_k) = e_2(p)/e_2(q)$ for the three pairs, that is $\tau_j^2 \tau_k^2 = e_2(p)/e_2(q)$; hence $\tau_0^2 = \tau_1^2 = \tau_2^2$, a sign up to the scale, and at a sign the equation holds. So these native gauges are the signs $\eta$ of the list with $K(\eta) = e_2(a) \neq 0$, of the values $f^{-1} = e_2(a)^2$ (Theorem 38.17).

*One occupied class of the sign value $a_i = 0$*, possible for even $s_i$. The weighted core is the monomial $a_j a_k \tau_j \tau_k$, so $f^{-1} = (a_j a_k)^2 = e_2(a)^2$; the equations of $C_j$ and $C_k$ hold identically, the occupations of a monomial being constant, and the equation of $C_i$ in Theorem 42.24(b) reads $\varsigma_i(\alpha) \tau_i^2 = \varsigma_i(\alpha^{-1})$ with $\varsigma_i = (A_j + A_k)/(A_j A_k)$, that is

$$\tau_i^2 = \tau_j \tau_k \, \frac{a_k \tau_j + a_j \tau_k}{a_j \tau_j + a_k \tau_k} \, :$$

these native gauges form curves, of the value $e_2(a)^2$ with $a_i = 0$. For $a_j = a_k = 1$ and a pair this is the curve $\{A = 0,\ \alpha_a^2 = \alpha_2 \alpha_3\}$ of $\mathrm{H}_4$ (Exercise 55.1).

*One empty class $C_i$.* Then $A_k = -A_j$ and $A'_k = -A'_j$; with $C_j$, $C_k$ occupied, $a_j \tau_j = -a_k \tau_k$ and $a_j/\tau_j = -a_k/\tau_k$, so $a_j^2 = a_k^2$, and $K = A_j A_k$, $f^{-1} = (A_j A'_j)(A_k A'_k) = a_j^4 = e_2(a_i, a_j, -a_j)^2$ for any $a_i$: a value already in the first list.

*Two empty classes $C_i$, $C_j$.* Then $A_i = A_j = -A_k$ and $A'_i = A'_j = -A'_k$, so $K = -A_k^2$ and $f^{-1} = (A_k A'_k)^2 = a_k^4$, with the equation of $C_k$ satisfied, its occupation being $A_k(A_i + A_j)/e_2(A) = 2$ on both sides. An empty class takes the prescribed totals, of the product $A_i A'_i = a_k^2$: a class of at least two marks always can, and a single mark, which has $A_i A'_i = 1$, only when $a_k = \pm 1$. These are the values $a_k^4$ of two empty classes around $C_k$.

*Three empty classes* force $A_0 = A_1 = A_2 = 0$, so $K = 0$, outside $X_F$.

So the native values of $f^{-1}$ are the $e_2(a)^2$, $a \in \Sigma(s_0) \times \Sigma(s_1) \times \Sigma(s_2)$, $e_2(a) \neq 0$, at signs or, when some $a_i = 0$, on curves, together with the values $a_k^4$ of two empty classes around an occupied class $C_k$, an empty single mark requiring $a_k = \pm 1$.

*(ii) Corner values.* The class $C_i$, $s_i \geq 2$, sits at the marker $h_i$ of the list $\mathcal{H}' = \binom{[3]}{2}\langle C_j, C_k \rangle$ on $C_j \sqcup C_k \sqcup \{h_i\}$, and by Corollary 38.28 its corner values are $N(\mathcal{H}' \setminus h_i \,|\, \mathcal{H}'/h_i)$, at the corner of $\Psi$. Here $\mathcal{H}' \setminus h_i$ has $K = A_j A_k$ and $\mathcal{H}'/h_i$ has $K = A_j + A_k$. On the torus of $C_j \sqcup C_k$ modulo the scale the coordinates are the shapes of the two classes and their relative scale; $f^{-1}_{\mathcal{H}' \setminus h_i} = (A_j A'_j)(A_k A'_k)$ does not depend on the relative scale, and the zero set of $A_j + A_k$ fixes it, a smooth hypersurface

where $A_j \neq 0$. So the native gauges of $\mathcal{H}' \setminus h_i$ on $\mathcal{H}'/h_i$ are the points at which both $A_jA'_j$ and $A_kA'_k$ are critical on the shapes of their classes, the signs of each class with nonzero totals (Lemma 42.27), and the corner values of $f^{-1}$ are the $(a_ja_k)^2$ with $a_j, a_k \neq 0$. The points of the third kind of Proposition 38.26(iii) give no finite value: the room factor $A_jA_k$ is singular only where $A_j = A_k = 0$, where $f^{-1}_{\mathcal{H}'/h_i} = (A_j + A_k)(A'_j + A'_k)$ vanishes.

*(iii)* $(a_ja_k)^2 = e_2(a)^2$ with $a_i = 0$. For even $s_i$, $0 \in \Sigma(s_i)$, and these values are native values, on the curves of (i). For odd $s_i$, $0 \notin \Sigma(s_i)$, and the value $(a_ja_k)^2$ is a native value only when it equals some $e_2(a')^2$ or some $a'^4_k$; the corners add the others. A class of one mark has no corner.

*(iv)* $\mathrm{J}_6$. $\Sigma(3) = \{\pm 3, \pm 1\}$, $\Sigma(2) = \{\pm 2, 0\}$, $\Sigma(1) = \{\pm 1\}$. Up to the global sign take $a_2 = 1$:

| $a_0$ | $a_1$ | $e_2(a)$ | $a_0$ | $a_1$ | $e_2(a)$ | $a_0$ | $a_1$ | $e_2(a)$ |
|---|---|---|---|---|---|---|---|---|
| 3 | 2 | 11 | 1 | 2 | 5 | −1 | −2 | −1 |
| 3 | 0 | 3 | 1 | 0 | 1 | −3 | 2 | −7 |
| 3 | −2 | −5 | 1 | −2 | −3 | −3 | 0 | −3 |
| −1 | 2 | −1 | −1 | 0 | −1 | −3 | −2 | 1 |

So the native values of $f^{-1}$ are $121, 49, 25, 9, 1$, the values 9 and 1 also on the curves $a_1 = 0$; two empty classes give only 1, around the mark or around the triple at $a_0 = \pm 1$, the pair not being able to sit around an empty mark since $a_1^2 = 4 \neq 1$. The corner values are $(a_1a_2)^2 = 4$ for the triple and $(a_0a_2)^2 \in \{9, 1\}$ for the pair. The corner value 4 of the triple is not a native value. □

**Exercise 55.12** (The weight-zero curve of a pair)**.** In $\Delta(s_0, s_1, s_2)$ let $C_i$ be a pair, and $a_j \in \Sigma(s_j)$, $a_k \in \Sigma(s_k)$ nonzero with $a_j^2 \neq a_k^2$. Normalize the scale of $C_j$ to 1, and write $r$ for the scale of $C_k$ and $\tau$ for that of the pair.

(i) Show that the native gauges with $C_i$ at a sign of total zero and $C_j$, $C_k$ at signs of the totals $a_j$, $a_k$ form one curve $\Gamma$ for the two sign patterns $(a_j, 0, \pm a_k)$, and that its quotient by the exchange of the two marks of the pair is a sphere less four points.

(ii) Find where $\Gamma$ leaves the torus.

(iii) Show that one blow-up makes the stride regular at the corner of $\Psi$ of the pair, and that the corner is a point of the closure of $\Gamma$.

*Solution. (i)* By Exercise 55.11(i), with $a_i = 0$, $f^{-1} = (A_jA'_j)(A_kA'_k) = (a_ja_k)^2$ and the equation of $C_i$ is

$$\Gamma: \qquad \tau^2 = r\,\frac{a_k + a_jr}{a_j + a_kr}.$$

The point of $C_k$ is $r$ times a sign pattern of the total $a_k$, which is $-r$ times one of the total $-a_k$; so the patterns $(a_j, 0, a_k)$ and $(a_j, 0, -a_k)$ give one curve, under $r \mapsto -r$. The exchange of the two marks of the pair is $\tau \mapsto -\tau$, and $\Gamma \to \mathbb{P}^1_r$ is two to one, branched where $\tau^2$ is 0 or $\infty$: $\tau^2$ has simple zeros at $r = 0$ and $r = -a_k/a_j$ and simple poles at $r = -a_j/a_k$ and $r = \infty$, four distinct points since $a_j^2 \neq a_k^2$. So

$$\Gamma/S_2 \cong \mathbb{P}^1_r \setminus \{0,\ \infty,\ -a_k/a_j,\ -a_j/a_k\}, \qquad \chi(\Gamma/S_2) = 2 - 4 = -2.$$

*(ii)* With $A_j = a_j$, $A'_j = a_j$, $A_k = a_kr$ and $A'_k = a_k/r$: as $r \to -a_k/a_j$, $\tau \to 0$ and $e = A'_j + A'_k = a_j + a_k/r \to 0$, so $\Gamma$ runs into the corner of $\Psi$ of the pair (Corollary 38.28); as $r \to -a_j/a_k$, $\tau \to \infty$ and $A_j + A_k \to 0$, the image of that corner under the Cremona, the corner of $K$ at the wall of $C_j \cup C_k$; as $r \to 0$, $\tau \sim (a_kr/a_j)^{1/2}$, so $C_k$ and $C_i$ shrink at the rates $r$ and $r^{1/2}$, a flag of length two, $C_k \subsetneq C_k \cup C_i$, and $r \to \infty$ is its image. At that flag the step factors $\sum_{C_i} u$ of $K$ and $\sum_{C_i} u^{-1}$

of $\Psi$ vanish together on the step $C_i$, the pair being at a sign of total zero, as Proposition 38.30(i) requires.

*(iii)* Near the corner of $\Psi$ put $\rho = \tau$, let $b = \sum_{C_i} u^{-1}$ in the normalized shape of the pair and $x$ the ratio of its two scales, so that $\sum_{C_i} u = xb$; with $S = A_j + A_k$, $P = A_j A_k$, $P' = A'_j A'_k$, $S' = A'_j + A'_k = e$ and $\hat{e} = e/\rho$. From $K = A_i S + P$, $A_i = \rho\, xb$ and $A'_i = \rho^{-1} b$,

$$f^{-1} = xb^2 SS' + \rho\, xbSP' + \rho^{-1} bS'P + PP' = PP' + b\left(\hat{e}P + \rho\, xSP'\right) + \rho\, xS\hat{e}\, b^2,$$

and $d\rho/\rho \wedge de = d\rho \wedge d\hat{e}$. So in the chart $(\rho, \hat{e})$ of the blow-up of $\{\rho = e = 0\}$ the stride and the origin form are regular at the corner. $PP' = (A_j A'_j)(A_k A'_k)$ does not depend on the relative scale of $C_j$ and $C_k$; at $b = 0$ the only condition left is $\partial_b f^{-1} = \hat{e}P + \rho\, xSP' = 0$, whose $\hat{e}$-derivative is $P \neq 0$. So the critical locus near the corner is the smooth curve $\{b = 0,\ \hat{e}P + \rho x SP' = 0\}$ at the signs of $C_j$ and $C_k$, which crosses the exceptional divisor $\rho = 0$, with $f^{-1} = (a_j a_k)^2$ on it: the corner is a point of the closure of $\Gamma$, a puncture of the curve, and not a source of its own. □

**Exercise 55.13** (The class of three and of four marks at silence)**.** For $s = 3$ and $s = 4$ write the operator $\widetilde{A}_s$ of Theorem 42.28, check it on the closings, and read its exponents at 0, at the sign values and at $\infty$, and the Jordan types of the monodromy at 0.

*Solution.* $s = 3$. The operators of Theorem 42.12(i) for $n = 3$ are $\mathbf{R}_2 = \vartheta^2 - 3y$, $\mathbf{R}_3 = \vartheta \mathbf{R}_2 - 4y\mathbf{R}_1 = \vartheta^3 - 7y\vartheta - 3y$ and

$$\mathbf{Q}_3 = \mathbf{R}_4 = \vartheta \mathbf{R}_3 - 3y\mathbf{R}_2 = \vartheta^4 - y\,(10\vartheta^2 + 10\vartheta + 3) + 9y^2,$$

using $\vartheta y = y(\vartheta + 1)$; so $q_{3,0} = \vartheta^4$, $q_{3,1} = -(10\vartheta^2 + 10\vartheta + 3)$, $q_{3,2} = 9$, and $\xi^4 - 10\eta\xi^2 + 9\eta^2 = (\xi^2 - 9\eta)(\xi^2 - \eta)$. Then $c_{3,0}(x) = x^2$, $c_{3,1}(x) = q_{3,1}(-x - 1) = -(10x^2 + 10x + 3)$ and $c_{3,2}(x) = q_{3,2}(-x - 2)(x + 1)^2 = 9(x + 1)^2$. The closings $\mathsf{C}_j = \sum_{a+b+c=j} \binom{j}{a,b,c}^2 = 1, 3, 15, 93, \dots$ satisfy $\sum_k c_{3,k}(-j)\mathsf{C}_{j-k} = 0$:

$$j^2\, \mathsf{C}_j - (10j^2 - 10j + 3)\, \mathsf{C}_{j-1} + 9(j-1)^2\, \mathsf{C}_{j-2} = 0,$$

at $j = 2$: $4 \cdot 15 - 23 \cdot 3 + 9 \cdot 1 = 0$; at $j = 3$: $9 \cdot 93 - 63 \cdot 15 + 36 \cdot 3 = 0$. The operator is

$$\widetilde{A}_3 = (\theta + 1)^2 - \big(10(\theta + 1)^2 + 10(\theta + 1) + 3\big)t^{-1} + 9(\theta + 2)^2 t^{-2}, \qquad \theta = t\partial_t,$$

of order 2, with the leading coefficient $t^2(1 - 10t^{-1} + 9t^{-2}) = (t - 1)(t - 9)$: the sign values 1 and 9. At 0, $t^2\widetilde{A}_3 = t^2(\theta + 1)^2 - t\big(10\theta^2 + 10\theta + 3\big) + 9\theta^2$, whose indicial polynomial $9\theta^2$ gives the exponents $0, 0$, the list $0, \dots, h - 2$ of Theorem 42.28(iv) with $h = 2$, each twice. At $\infty$ the indicial polynomial is $(1 - \theta_w)^2$ in $w = 1/t$: the exponents $1, 1$. At a sign value they are 0 and $\frac{s-3}{2} = 0$, the leading coefficient vanishing simply and the period of the vanishing cycle of a nondegenerate point on a torus of dimension 2 having the exponent 0 with a logarithm. By Theorem 42.28(iv), $\operatorname{rank}(T_0 - 1) = 1$ and $(T_0 - 1)^2 = 0$: one Jordan block of size 2, a solution $g_0 = 1 + \cdots$ beside $g_0 \log t + \cdots$.

$s = 4$. $\mathbf{R}_2 = \vartheta^2 - 4y$, $\mathbf{R}_3 = \vartheta \mathbf{R}_2 - 6y\vartheta = \vartheta^3 - 10y\vartheta - 4y$, $\mathbf{R}_4 = \vartheta \mathbf{R}_3 - 6y\mathbf{R}_2 = \vartheta^4 - y(16\vartheta^2 + 14\vartheta + 4) + 24y^2$, and

$$\mathbf{Q}_4 = \mathbf{R}_5 = \vartheta \mathbf{R}_4 - 4y\mathbf{R}_3 = \vartheta^5 - y\,(20\vartheta^3 + 30\vartheta^2 + 18\vartheta + 4) + y^2\,(64\vartheta + 64),$$

with $\xi^5 - 20\eta\xi^3 + 64\eta^2\xi = \xi(\xi^2 - 16\eta)(\xi^2 - 4\eta)$; so $q_{4,1}(\vartheta) = -2(2\vartheta + 1)(5\vartheta^2 + 5\vartheta + 2)$ and $q_{4,2}(\vartheta) = 64(\vartheta + 1)$, and $c_{4,0}(x) = -x^3$, $c_{4,1}(x) = q_{4,1}(-x - 1) = 2(2x + 1)(5x^2 + 5x + 2)$, $c_{4,2}(x) =$

$q_{4,2}(-x-2)(x+1)^2 = -64(x+1)^3$. The closings $1, 4, 28, 256, 2716, \ldots$ satisfy the recurrence: at $j=2$, $8\cdot 28 - 72\cdot 4 + 64\cdot 1 = 0$. The exponents at 0 are the roots $\rho$ of $c_{4,2}(\rho-1) = -64\rho^3$: $0, 0, 0$, the double list 0 and the extra exponent $\frac{s}{4} - 1 = 0$ of Theorem 42.28(iv). The monodromy at 0 is unipotent, with $\operatorname{rank}(T_0 - 1) = 2$ on a space of dimension 3 and $(T_0-1)^3 = 0$: one Jordan block of size 3. At $\infty$ the exponents are $1, 1, 1$, one block of size 3; at the sign values 16 and 4 they are $0, 1$ and $\frac{1}{2}$. □

**Exercise 55.14** (The laws of the triangles; a wild high end)**.** Compare the roots of the laws of the triangles of classes with their native values and corner values, and show that the centre ray of $\mathrm{J}_6$ is not tame at the high end (Definition 39.43).

*Solution. The laws.* By Proposition 42.22, with every class entering through the closings $\mathsf{M}_s(w) = \sum_{|v|=w} \binom{w}{v}^2$ of $\binom{[s]}{1}$ (Theorem 42.20(ii)),

$$\mathsf{C}_j(1) = \sum_{a+b+c=j} \binom{j}{a,b,c}^2 \mathsf{M}_{s_0}(a+b)\,\mathsf{M}_{s_1}(a+c)\,\mathsf{M}_{s_2}(b+c),$$

the exponents $(a+b, a+c, b+c)$ of the classes in $e_2(A)^j$ determining $(a,b,c)$. Input: these closings, 900 to 9000 of them modulo primes near $2^{62}$. Method: the operators of a fixed order and degree annihilating the closings form the kernel of a matrix; the least degree at each order and the growth of the kernel with the degree locate the order of the law, and the greatest common divisor of the symbols of all operators found is its symbol, whose roots are found by rational reconstruction. Check: two primes, and the same greatest common divisor at several pairs (order, degree). Receipts: `fast_closings.py`, `search2.py`, `order_check.py`, `bsearch.py` and their logs. Output, the roots as values of $f^{-1}$ with their multiplicities:

| list | $n$ | order | native values | corner values | roots |
|---|---|---|---|---|---|
| $\Delta(2,1,1)$ | 4 | 4 | 25, 9, 1 | 1 | 25, 9, $1^{(2)}$ |
| $\Delta(2,2,1)$ | 5 | 5 | 64, 16, 4, 1 | 4 | 64, 16, $4^{(2)}$, 1 |
| $\Delta(3,1,1)$ | 5 | 6 | 49, 25, 9, 1 | 1 | 49, 25, 9, $1^{(3)}$ |
| $\Delta(2,2,2)$ | 6 | 3 | 144, 16 | 16 | 144, $16^{(2)}$ |
| $\Delta(3,2,1)$ | 6 | 13 | 121, 49, 25, 9, 1 | **4**, 9, 1 | 121, 49, **4**, $25^{(2)}$, $9^{(3)}$, $1^{(5)}$ |
| $\Delta(5,2,1)$ | 8 | 22 | 289, 169, 121, 49, 25, 9, 1 | **4**, 25, 9, 1 | 289, 169, 121, $49^{(2)}$, $\mathbf{4}^{(2)}$, $25^{(4)}$, $9^{(4)}$, $1^{(7)}$ |
| $\Delta(3,4,1)$ | 8 | 24 | 361, 169, 121, 81, 49, 25, 9, 1 | **16**, **4**, 9, 1 | 361, 169, $121^{(2)}$, 81, $49^{(2)}$, $25^{(3)}$, **16**, $9^{(3)}$, **4**, $1^{(9)}$ |

The native and corner values are those of Exercise 55.11; in bold, the corner values of the odd classes that are not native values. Every root is a native value or a corner value, as Theorem 39.51 with Proposition 38.26 requires, and every corner value is a root. The corner values of an odd class $C_i$ that no other source carries have the multiplicity $\frac{s_i-1}{2}$, the contribution of an odd corner in Proposition 42.33(c).

*A wild high end.* The root $\frac{1}{4}$ of the law of $\mathrm{J}_6$ is not a native value, by Exercise 55.11(iv). By Theorem 39.44(iii) every nonzero root of the symbol of a ray tame at the high end is a native value; so the centre ray of $\mathrm{J}_6$ is not tame: its insertion lattice is not finitely generated over the ring of the functions regular at $k=\infty$. If the classes of the insertions lay in $k^N \bigoplus_l A[b_l]$ for one $N$ and masters $\mathbf{b}$, $\Lambda_F$ would be a submodule of a finitely generated module over the noetherian ring $A$, hence finitely generated; so in every set of masters the coefficients of the classes of the insertions have poles at $k=\infty$ of unbounded order. The root $\frac{1}{4}$ is the corner value of the triple, an

asymptotic native value: the native gauges of the directions near the centre direction that carry it run into the wall of the triple. □

**Exercise 55.15** (A mark of another norm, and the value 1 as a coincidence)**.** Let $\Delta_{\mathsf{m}}(s_0, s_1)$ be the triangle with the classes $C_0$, $C_1$ of $s_0$ odd and $s_1$ even marks and, at $h_2$, a mark of the norm $\mathsf{m} > 0$, its hinge pair $(\alpha_2, \mathsf{m}\alpha_2^{-1})$ (Definition 42.19): its closings are those of Exercise 55.14 with $\mathsf{M}_1(b+c)$ replaced by $\mathsf{m}^{b+c}$, and at $\mathsf{m} = 1$ it is $\Delta(s_0, s_1, 1)$.

(i) List the sources in the hinge space (Theorem 42.31) at a generic norm, with their values and their contributions (Proposition 42.33).

(ii) Compare with the symbols of the laws at $\mathsf{m} = 2$.

(iii) At $\mathsf{m} = a^2$, $a \in \Sigma(s_0)$, $a > 0$, find the critical points of the two-sided core with $C_0$ on its hyperbola $H_a$ and $C_1$ empty.

*Solution. (i)* Write $\varsigma = \sqrt{\mathsf{m}}$; the mark lies on the hyperbola $x_2 x_2' = \mathsf{m}$ and reads as a mark of the weight $\varsigma$. By Theorem 42.31 each class is occupied, empty, silent or balanced.

- *Both classes on hyperbolas*, with the weights $a \in \Sigma(s_0)$ and $b \in \Sigma(s_1)$, $b \neq 0$: the weighted triangle of weights $(a, b, \varsigma)$ has only signs (Exercise 55.11(i)), with the values $e_2(a, \pm b, \pm\varsigma)^2$, that is $\big(ab \pm \varsigma(a+b)\big)^2$ and $\big(ab \pm \varsigma(a-b)\big)^2$ for $a, b > 0$: weighted points, each contributing 1.
- $C_0$ *silent*: the odd corners, of the values $b^2\mathsf{m}$, each contributing $\frac{s_0-1}{2}$ (Proposition 42.33(c)).
- $C_0$ *balanced*: nothing, since $m_{s_0}(O) = 0$ (Proposition 42.29(iv)).
- $C_1$ *balanced*: the weight-zero curves of $C_1$, of the values $a^2\mathsf{m}$; Proposition 42.33(e) applies to them away from their punctures, and for a pair Exercise 55.12 gives the quotient curve of the Euler characteristic $-2$.
- *A class empty*: the conditions $A_j + A_2 = A_j' + A_2' = 0$ put the other class at the norm of the mark, which at a generic $\mathsf{m}$ leaves only both classes empty, $x_0 = x_1 = -x_2$, $x_0' = x_1' = -x_2'$, of the value $\mathsf{m}^2$, contributing $(s_0 - 1)(s_1 - 1)$ (Proposition 42.33(d)).

So the source count is the product

$$\Pi_{\mathsf{m}} = \prod_{a,b>0} \prod_{\pm} \big(\lambda - (ab \pm \varsigma(a+b))^2\big)\big(\lambda - (ab \pm \varsigma(a-b))^2\big) \cdot \prod_{b>0} (\lambda - b^2\mathsf{m})^{\frac{s_0-1}{2}} \cdot \prod_{a>0} (\lambda - a^2\mathsf{m})^2 \cdot (\lambda - \mathsf{m}^2)^{(s_0-1)(s_1-1)},$$

with the exponent 2 at the weight-zero values, of the degree $(s_0+1)s_1 + \frac{(s_0-1)s_1}{4} + (s_0+1) + (s_0 - 1)(s_1 - 1)$. Under the conditions of Theorem 42.32, every vanishing rank at a value other than the $a^2\mathsf{m}$ is at most its multiplicity in the other three factors.

*(ii)* Input: 2600 to 4600 closings modulo a prime near $2^{62}$; method and check as in Exercise 55.14; receipts `fc_mu.py`, `search3.py`, `predict_sym2.py`, logs `law_*_mu2.log`. Output: for $(s_0, s_1) = (3,2), (5,2), (7,2), (3,4)$ at $\mathsf{m} = 2$, and $(3,2)$ at $\mathsf{m} = 3$, the symbol of the law is $\Pi_{\mathsf{m}}$ as a polynomial, of the orders 15, 24, 33 and 28; for $(3,2)$ at $\mathsf{m} = 2$ the rational roots are 8, $18^{(2)}$, $2^{(2)}$, $4^{(2)}$ and the other eight are simple. So at these norms the bound of (i) is attained, and the weight-zero curves contribute exactly 2.

*(iii)* Normalize the mark to $x_2 = 1$, $x_2' = \mathsf{m}$. $C_1$ empty means $x_0 + x_2 = 0$ and $x_0' + x_2' = 0$, so $x_0 = -1$, $x_0' = -\mathsf{m}$, and $x_0 x_0' = \mathsf{m}$, which on $H_a$ needs $\mathsf{m} = a^2$. The derivative of the two-sided

core along $H_a$, in the direction $u\partial_u$ with $u = x_0$, is then $\mathsf{m}(\mathsf{m}x_1 - x_1')$, which gives $x_1' = \mathsf{m}x_1$. So the critical points form the curve

$$\bar{Z}_a = \{x_0 = -1,\ x_0' = -\mathsf{m},\ x_1 = w,\ x_1' = \mathsf{m}w\}, \qquad w \in \mathbb{C},$$

on which $K = -1$, $K' = -\mathsf{m}^2$ and $\Phi = \mathsf{m}^2$. On it lie the empty point ($w = -1$), the origin of the hinge plane of $C_1$ ($w = 0$), where the two corner points of the weight-zero curve of $C_1$ at the weight $a$ meet, and the $s_1$ weighted points $(-a, \pm b, a)$ ($w = \pm b/a$), of the value $\big((-a)(\pm b) - a^2 \pm ab\big)^2 = a^4$; along $\bar{Z}_a$ the class $C_1$ has the norm $\mathsf{m}w^2$. At $\mathsf{m} = 1$ the list $\Delta(s_0, s_1, 1)$ sits at this coincidence with $a = 1$, and its value 1 is carried by the curve $\bar{Z}_1$. Its laws, computed as in Exercise 55.14 for $\Delta(3,2,1)$, $\Delta(5,2,1)$, $\Delta(7,2,1)$, $\Delta(3,4,1)$ and $\Delta(5,4,1)$ (receipts `predict_list.py`, logs `list_*.log`), have every value other than 1 with its multiplicity in $\Pi_1$, coincident values adding, and the value 1 with the multiplicity $(s_0 - 1)(s_1 - 1) + 3$, which is $s_1$ less than its exponent in $\Pi_1$; the same loss of $s_1$ at the value $\mathsf{m}^2$ is found at $\mathsf{m} = 9$ for $(3,2)$, $(5,2)$, $(3,4)$ and at $\mathsf{m} = 25$ for $(5,2)$ (receipt `predict_coinc.py`). □

# Chapter 56

# Exercises on the content

The exercises of this chapter work the statements of Chapter 45 on the two smallest centre values met in the exercises on rays. The value $\ell$ of the list of three marks is placed in the content of level 3 as a combination of two trails of depth one and of two chains; the value of the half-lattice met at four marks is placed in the content of level 2, and the coset relation returns it to level 1. The cutting rule is then computed on the symbols of two trails, subset by subset, and the three parts of Theorem 45.9 are checked on them. Throughout, $\omega = e^{2\pi i/3}$, so that $\omega^3 = 1$, $\omega^2 + \omega + 1 = 0$, $\bar{\omega} = \omega^2 = \omega^{-1}$ and $\mu_3 = \{1, \omega, \omega^2\}$. Every number displayed is exact; decimal values are labelled renderings, and the receipts that recompute them are the scripts named in each solution.

## 56.1 Two centre values in the content

**Exercise 56.1** (The values of three and four marks as trails and chains)**.** Let $\ell = P_{\binom{[3]}{1}}(\mathbf{1}; 1, 1) = 3\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\big)$ (Exercise 54.4) and $v = (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\frac{1}{2}+\mathbb{Z}})((0,\infty))$ (Exercise 54.7).

(i) Write $\ell$ as a combination, with coefficients in $\mathbb{Q}(\mu_3)$, of the trails of level 3 of depth one at the index 2 and the frequencies $\frac{1}{3}$ and $\frac{2}{3}$ (Definition 45.1), and show that $\ell$ is a value of $\mathcal{C}_2^{(3)}$.

(ii) Write these two trails as chains of level 3 from 0 to 1 (Proposition 45.5(i)), check the first by integrating it, and write $\ell$ as a combination of chains.

(iii) Show that $v$ is a value of $\mathcal{C}_3^{(2)}$ by writing it with the trails of level 2 at the index 3 and the frequencies 0 and $\frac{1}{2}$. Then use the coset relation to show that $v$ is a value of $\mathcal{C}_3^{(1)}$, find the trail at the frequency $\frac{1}{2}$ in terms of the one at the frequency 0, and write $v$ as a chain.

*Solution.* Receipt: `ct_values.py`.

(i) *The cosets as frequencies.* For an integer $m$, $\omega^m - \omega^{-m}$ is 0, $\omega - \omega^2$ and $\omega^2 - \omega$ for $m$ in $3\mathbb{Z}$, $1 + 3\mathbb{Z}$ and $2 + 3\mathbb{Z}$, as $\omega^{-1} = \omega^2$. So the difference of the two indicators is

$$\mathbf{1}_{1+3\mathbb{Z}}(m) - \mathbf{1}_{2+3\mathbb{Z}}(m) = \frac{\omega^m - \omega^{-m}}{\omega - \omega^2} \qquad (m \in \mathbb{Z}).$$

At the index 2 every sum below converges absolutely, so the value of a lattice input at the cut 0 is the sum of its density times $m^{-2}$ over the points $m > 0$ (Definition 5.2), and the sums may be split and recombined. With $\omega^{-m} = e^{2\pi i \cdot 2m/3}$,

$$(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty)) = \sum_{m\geq 1} \frac{\omega^m - \omega^{-m}}{\omega-\omega^2}\, m^{-2}$$
$$= \frac{1}{\omega-\omega^2}\Big((\mathsf{T}_{2;(1)}[x]\, e^{2\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\, e^{4\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty))\Big),$$

the lattice inputs $e^{2\pi i a x}\mathbb{I}_{\mathbb{Z}}$ at the frequencies $a = \frac{1}{3}$ and $a = \frac{2}{3}$ on the counting lattice (Definition 5.2).

*The constant.* $(\omega-\omega^2)^2 = \omega^2 - 2\omega^3 + \omega^4 = \omega^2 + \omega - 2 = -3$. So $\frac{1}{\omega-\omega^2} = -\frac{\omega-\omega^2}{3}$ and $\frac{3}{\omega-\omega^2} = \omega^2 - \omega$, and

$$\ell = (\omega^2-\omega)\Big((\mathsf{T}_{2;(1)}[x]\, e^{2\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\, e^{4\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty))\Big).$$

*The trails.* For $d=1$, $\nu_1 = 2$ and $a_1 \in \{\frac{1}{3},\frac{2}{3}\} \subseteq \frac{1}{3}\mathbb{Z}$, the trail of level 3 of Definition 45.1 is $\big(\mathsf{T}_{2;I_1}[y]\, e^{2\pi i a_1 y}\,\mathbb{I}_{\mathbb{Z}}\big)(\{0<y\}) = \sum_{m\geq 1} e^{2\pi i a_1 m} m^{-2} = (\mathsf{T}_{2;(1)}[x]\, e^{2\pi i a_1 x}\mathbb{I}_{\mathbb{Z}})((0,\infty))$: it has the depth 1, the positive integer index 2 and a frequency in $\frac{1}{3}\mathbb{Z}$, and the condition at index one does not arise. Write $T'$ and $T''$ for the trails at $\frac{1}{3}$ and $\frac{2}{3}$ in this solution. Then

$$\ell = \omega^2 T' - \omega T' - \omega^2 T'' + \omega T'',$$

the value of the combination $[\omega^2 T'] - [\omega T'] - [\omega^2 T''] + [\omega T'']$, with the rational coefficients $\pm 1$, of the classes of the words $\zeta T$ with $\zeta \in \mu_3$ and $T$ a trail of level 3 of weight 2. So $\ell$ is a value of $\mathcal{C}_2^{(3)}$. Since $\bar{\omega} = \omega^2$, $T'' = \overline{T'}$, and $\ell = (\omega^2-\omega)(T' - \overline{T'}) = (-i\sqrt{3})(2i\,\mathrm{Im}\,T') = 2\sqrt{3}\,\mathrm{Im}\,T'$, with $\mathrm{Im}\,T' = \sum_{m\geq 1}\sin(\frac{2\pi m}{3})m^{-2} = \frac{\sqrt{3}}{2}\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\big)$, the $S_2$ of Exercise 54.4.

Rendering: both sides of the display for $\ell$ are $2.3439072386894588\ldots$.

(ii) *The chains.* Proposition 45.5(i) with $d=1$ and $\nu_1 = 2$ gives $c_1 = e^{2\pi i a_1}$, the letters $c_1^{-1}$ and one 0, and the sign $(-1)^1$. At $a_1 = \frac{1}{3}$, $c_1 = \omega$ and $c_1^{-1} = \omega^2$; at $a_1 = \frac{2}{3}$, $c_1 = \omega^2$ and $c_1^{-1} = \omega$:

$$T' = -\mathrm{I}(0;\omega^2,0;1), \qquad T'' = -\mathrm{I}(0;\omega,0;1).$$

*The check.* For the ends 0 and 1 the path is $\gamma(t) = t$, and for a letter $b$ with $|b| = 1$, $b \neq 1$ (Definition 45.4),

$$\mathrm{I}(0;b,0;1) = \int_{0<t_1<t_2<1} \frac{dt_1}{t_1-b}\,\frac{dt_2}{t_2}.$$

For $0 \leq t < 1$, $1 - t/b$ lies in the open disc of radius 1 about 1, where the principal logarithm is holomorphic; $\log(1-t/b)$ vanishes at $t=0$ and has the derivative $\frac{-1/b}{1-t/b} = \frac{1}{t-b}$, so the inner integral is $\log(1-t_2/b) = -\sum_{n\geq 1} b^{-n} t_2^n/n$. The series of absolute values, $\sum_n t_2^{n-1}/n$, integrates over $(0,1)$ to $\sum_n n^{-2} < \infty$, so the series integrates term by term, $\int_0^1 t_2^{n-1}\,dt_2 = \frac{1}{n}$:

$$\mathrm{I}(0;b,0;1) = -\sum_{n\geq 1} b^{-n} n^{-2}.$$

At $b = \omega^2$, $b^{-n} = \omega^{-2n} = \omega^n$, and $\mathrm{I}(0;\omega^2,0;1) = -\sum_n \omega^n n^{-2} = -T'$, as Proposition 45.5(i) states.

Rendering: the double integral is $0.5483113556160754\ldots - 0.6766277376064357\ldots i$, and so is $-T'$.

*The value.* Substituting,

$$\ell = (\omega^2-\omega)\big(-\mathrm{I}(0;\omega^2,0;1)+\mathrm{I}(0;\omega,0;1)\big) = (\omega-\omega^2)\big(\mathrm{I}(0;\omega^2,0;1)-\mathrm{I}(0;\omega,0;1)\big),$$

a combination of chains of level 3 from 0 to 1 with first letter not 0 and last letter not 1, as Proposition 45.5(iii) describes the values of $\mathcal{C}_2^{(3)}$.

(iii) *Level* 2. The positive points of $\frac{1}{2}+\mathbb{Z}$ are the $\frac{2m-1}{2}$, $m\geq 1$, so $v=\sum_{m\geq 1} 8(2m-1)^{-3}$ (Definition 5.2), absolutely convergent. The indicator of the odd integers is $\frac{1-(-1)^n}{2}=\frac{1-e^{\pi i n}}{2}$, so

$$v = 8\sum_{n\geq 1}\frac{1-e^{\pi i n}}{2}\,n^{-3} = 4\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - 4\,(\mathsf{T}_{3;(1)}[x]\,e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

$(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and $(\mathsf{T}_{3;(1)}[x]\,e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty))$ are the trails of depth one at the index 3 and the frequencies 0 and $\frac{1}{2}$, both in $\frac{1}{2}\mathbb{Z}$: trails of level 2. So $v$ is a value of $\mathcal{C}_3^{(2)}$, as Proposition 45.3(i) gives directly for the coset $b=\frac{1}{2}\in\frac{1}{2}\mathbb{Z}$ at the frequency 0, with $M=2$ and $M'=1$.

*Level* 1. The coset relation of Exercise 54.7, Theorem 5.12(i) at $q=2$ and $b=0$, is $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))+v = 8\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. So

$$v = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),$$

a rational multiple of a trail of level 1, and $v$ is a value of $\mathcal{C}_3^{(1)}$. Comparing the two expressions of $v$, $4\,(\mathsf{T}_{3;(1)}[x]\,e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = 4\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$:

$$(\mathsf{T}_{3;(1)}[x]\,e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\tfrac{3}{4}\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)).$$

*The chain.* Proposition 45.5(i) with $d=1$, $\nu_1=3$ and $a_1=0$ gives $c_1=1$, the letters $1,0,0$, and $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\mathrm{I}(0;1,0,0;1)$; so $v=-7\,\mathrm{I}(0;1,0,0;1)$.

Rendering: $v=8.4143983221171599\ldots$, the value of both expressions above; the ratio of the trail at the frequency $\frac{1}{2}$ to the trail at the frequency 0 renders as $-0.75$, and the triple integral $\mathrm{I}(0;1,0,0;1)$ as $-1.2020569031595942\ldots$.

*Answer.*

$$\begin{aligned}\ell &= (\omega^2-\omega)\Big((\mathsf{T}_{2;(1)}[x]\,e^{2\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,e^{4\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty))\Big)\\ &= (\omega-\omega^2)\big(\mathrm{I}(0;\omega^2,0;1)-\mathrm{I}(0;\omega,0;1)\big),\end{aligned}$$

a value of $\mathcal{C}_2^{(3)}$. $v = 4\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) - 4\,(\mathsf{T}_{3;(1)}[x]\,e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty))$ is a value of $\mathcal{C}_3^{(2)}$, and by the coset relation $v = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -7\,\mathrm{I}(0;1,0,0;1)$, a value of $\mathcal{C}_3^{(1)}$, with $(\mathsf{T}_{3;(1)}[x]\,e^{\pi i x}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\frac{3}{4}\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. □

## 56.2 The cutting rule on two trails

**Exercise 56.2** (The cutting rule on $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and on the trail of index 2 at the frequency $\frac{1}{3}$)**.** Let $\sigma_3$ be the symbol of the trail $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ of level 1, and $\sigma'$ that of the trail $(\mathsf{T}_{2;(1)}[x]\,e^{2\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty))$ of level 3 (Definition 45.7).

(i) Compute $\Delta\sigma_3$ and $\Delta\sigma'$ (Definition 45.8), subset by subset.

(ii) Check the three parts of Theorem 45.9 on $\sigma_3$ and $\sigma'$: coassociativity and the counit, the weights, and the compatibility with the reversal and the rotations.

(iii) Read the left factors of positive weight as symbols of trails (Proposition 45.5(ii)), and compute the chains of the two factors of weight one in $\Delta\sigma'$.

*Solution.* Receipts: `ct_cutting.py` for (i) and (ii), `ct_values.py` for the renderings in (iii).

*The symbols.* Both trails have depth $d = 1$. For $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\nu_1 = 3$ and $c_1 = 1$; for $(\mathsf{T}_{2;(1)}[x]\,e^{2\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty))$, $\nu_1 = 2$ and $c_1 = \omega$, $c_1^{-1} = \omega^2$. So, with $x = \mathrm{I}^{\mathfrak{s}}(0;1,0,0;1)$ and $y = \mathrm{I}^{\mathfrak{s}}(0;\omega^2,0;1)$,

$$\sigma_3 = -x, \qquad \sigma' = -y.$$

$\Delta$ is $\mathbb{Q}(\mu_M)$-linear, so $\Delta\sigma_3 = -\Delta x$ and $\Delta\sigma' = -\Delta y$. In $\mathfrak{S}_M$ a run with no letters is 1, and a run of at least one letter between equal ends is 0 (Definition 45.7); these two rules decide every right factor below.

(i) *The cutting rule on $x$.* The entries are $a_0 = 0$, $a_1 = 1$, $a_2 = 0$, $a_3 = 0$, $a_4 = 1$, and $w = 3$. For each $S \subseteq \{1,2,3\}$, the left factor keeps the letters of $S$ between the ends 0 and 1, and the right factors are the runs between consecutive kept positions $s_0 = 0 < s_1 < \cdots < s_{|S|+1} = 4$:

| $S$ | left factor | runs, and the term |
|---|---|---|
| $\varnothing$ | 1 | $\mathrm{I}^{\mathfrak{s}}(0;1,0,0;1) = x$: the term $1 \otimes x$ |
| $\{1\}$ | $\mathrm{I}^{\mathfrak{s}}(0;1;1)$ | $\mathrm{I}^{\mathfrak{s}}(0;\varnothing;1) = 1$, $\mathrm{I}^{\mathfrak{s}}(1;0,0;1) = 0$: no term |
| $\{2\}$ | $\mathrm{I}^{\mathfrak{s}}(0;0;1)$ | $\mathrm{I}^{\mathfrak{s}}(0;1;0) = 0$: no term |
| $\{3\}$ | $\mathrm{I}^{\mathfrak{s}}(0;0;1)$ | $\mathrm{I}^{\mathfrak{s}}(0;1,0;0) = 0$: no term |
| $\{1,2\}$ | $\mathrm{I}^{\mathfrak{s}}(0;1,0;1)$ | 1, 1, $\mathrm{I}^{\mathfrak{s}}(0;0;1)$ |
| $\{1,3\}$ | $\mathrm{I}^{\mathfrak{s}}(0;1,0;1)$ | 1, $\mathrm{I}^{\mathfrak{s}}(1;0;0)$, 1 |
| $\{2,3\}$ | $\mathrm{I}^{\mathfrak{s}}(0;0,0;1)$ | $\mathrm{I}^{\mathfrak{s}}(0;1;0) = 0$: no term |
| $\{1,2,3\}$ | $x$ | 1, 1, 1, 1: the term $x \otimes 1$ |

With $z = \mathrm{I}^{\mathfrak{s}}(0;1,0;1)$ and $R = \mathrm{I}^{\mathfrak{s}}(0;0;1) + \mathrm{I}^{\mathfrak{s}}(1;0;0)$,

$$\Delta x = x \otimes 1 + 1 \otimes x + z \otimes R.$$

*The cutting rule on $y$.* The entries are $a_0 = 0$, $a_1 = \omega^2$, $a_2 = 0$, $a_3 = 1$, and $w = 2$:

| $S$ | left factor | runs, and the term |
|---|---|---|
| $\varnothing$ | 1 | $y$: the term $1 \otimes y$ |
| $\{1\}$ | $\mathrm{I}^{\mathfrak{s}}(0;\omega^2;1)$ | $\mathrm{I}^{\mathfrak{s}}(0;\varnothing;\omega^2) = 1$, $\mathrm{I}^{\mathfrak{s}}(\omega^2;0;1)$ |
| $\{2\}$ | $\mathrm{I}^{\mathfrak{s}}(0;0;1)$ | $\mathrm{I}^{\mathfrak{s}}(0;\omega^2;0) = 0$: no term |
| $\{1,2\}$ | $y$ | 1, 1, 1: the term $y \otimes 1$ |

With $A = \mathrm{I}^{\mathfrak{s}}(0;\omega^2;1)$ and $B = \mathrm{I}^{\mathfrak{s}}(\omega^2;0;1)$,

$$\Delta y = y \otimes 1 + 1 \otimes y + A \otimes B.$$

(ii) *The pieces are primitive.* A formal chain $u$ of weight one has two subsets: $S = \varnothing$ gives $1 \otimes u$ and $S = \{1\}$ gives $u \otimes 1$, its two runs being empty. So $\Delta u = u \otimes 1 + 1 \otimes u$ for $u = \mathrm{I}^{\mathfrak{s}}(0;0;1)$, $\mathrm{I}^{\mathfrak{s}}(1;0;0)$, $A$ and $B$, and $\Delta R = R \otimes 1 + 1 \otimes R$ by linearity. For $z$, with the entries $0,1,0,1$: $S = \{1\}$ leaves the run $\mathrm{I}^{\mathfrak{s}}(1;0;1) = 0$ and $S = \{2\}$ the run $\mathrm{I}^{\mathfrak{s}}(0;1;0) = 0$, so $\Delta z = z \otimes 1 + 1 \otimes z$ as well.

*Coassociativity.* $\Delta$ is an algebra homomorphism with $\Delta 1 = 1 \otimes 1$, so on $y$

$$(\Delta \otimes \mathrm{id})\Delta y = 1 \otimes 1 \otimes y + (y \otimes 1 + 1 \otimes y + A \otimes B) \otimes 1 + (A \otimes 1 + 1 \otimes A) \otimes B,$$

$$(\mathrm{id}\otimes\Delta)\Delta y = y\otimes 1\otimes 1 + 1\otimes(y\otimes 1 + 1\otimes y + A\otimes B) + A\otimes(B\otimes 1 + 1\otimes B),$$

and both are the sum of the six tensors $1\otimes 1\otimes y$, $1\otimes y\otimes 1$, $y\otimes 1\otimes 1$, $A\otimes B\otimes 1$, $A\otimes 1\otimes B$ and $1\otimes A\otimes B$. On $x$ the same expansion, with $z$ and $R$ primitive, gives on both sides the six tensors $1\otimes 1\otimes x$, $1\otimes x\otimes 1$, $x\otimes 1\otimes 1$, $z\otimes R\otimes 1$, $z\otimes 1\otimes R$ and $1\otimes z\otimes R$.

*The counit.* $\varepsilon(1) = 1$ and $\varepsilon$ vanishes on the formal chains, so $(\varepsilon\otimes\mathrm{id})\Delta x = x + \varepsilon(z)R = x$ and $(\mathrm{id}\otimes\varepsilon)\Delta x = x + z\,\varepsilon(R) = x$, and in the same way on $y$. This is Theorem 45.9(i) on the two symbols.

*The weights.* The terms of $\Delta x$ have the weights $(3,0)$, $(0,3)$ and $(2,1)$, those of $\Delta y$ the weights $(2,0)$, $(0,2)$ and $(1,1)$; the sums are 3 and 2, and $\Delta x - x\otimes 1 - 1\otimes x = z\otimes R$ and $\Delta y - y\otimes 1 - 1\otimes y = A\otimes B$ have both factors of positive weight: Theorem 45.9(ii).

*The reversal.* $\varrho y = (-1)^2\,\mathrm{I}^{\mathfrak{s}}(1;0,\omega^2;0)$. Its entries $1,0,\omega^2,0$ give: $S=\varnothing$, $1\otimes\varrho y$; $S=\{1\}$, the left factor $\mathrm{I}^{\mathfrak{s}}(1;0;0)$ and the run $\mathrm{I}^{\mathfrak{s}}(0;\omega^2;0) = 0$, no term; $S=\{2\}$, the left factor $\mathrm{I}^{\mathfrak{s}}(1;\omega^2;0)$ and the runs $\mathrm{I}^{\mathfrak{s}}(1;0;\omega^2)$ and 1; $S=\{1,2\}$, $\varrho y\otimes 1$. On the other side $\varrho A = -\mathrm{I}^{\mathfrak{s}}(1;\omega^2;0)$ and $\varrho B = -\mathrm{I}^{\mathfrak{s}}(1;0;\omega^2)$, so

$$\Delta\varrho y = \varrho y\otimes 1 + 1\otimes\varrho y + \mathrm{I}^{\mathfrak{s}}(1;\omega^2;0)\otimes\mathrm{I}^{\mathfrak{s}}(1;0;\omega^2) = (\varrho\otimes\varrho)\Delta y.$$

$\varrho x = (-1)^3\,\mathrm{I}^{\mathfrak{s}}(1;0,0,1;0)$. Of the eight subsets of the entries $1,0,0,1,0$, those with a term are $\varnothing$, $\{1,2,3\}$, $\{1,3\}$, with the left factor $\mathrm{I}^{\mathfrak{s}}(1;0,1;0)$ and the run $\mathrm{I}^{\mathfrak{s}}(0;0;1)$, and $\{2,3\}$, with the same left factor and the run $\mathrm{I}^{\mathfrak{s}}(1;0;0)$; the other four leave a run between equal ends. So $\Delta\varrho x = \varrho x\otimes 1 + 1\otimes\varrho x - \mathrm{I}^{\mathfrak{s}}(1;0,1;0)\otimes R$. On the other side $\varrho z = (-1)^2\,\mathrm{I}^{\mathfrak{s}}(1;0,1;0)$ and $\varrho R = -\mathrm{I}^{\mathfrak{s}}(1;0;0) - \mathrm{I}^{\mathfrak{s}}(0;0;1) = -R$, so $(\varrho\otimes\varrho)(z\otimes R) = -\mathrm{I}^{\mathfrak{s}}(1;0,1;0)\otimes R$, and $\Delta\varrho x = (\varrho\otimes\varrho)\Delta x$.

*The rotations.* $\rho_\lambda$, $\lambda\in\mu_3$, multiplies every entry by $\lambda$ and fixes the entry 0. It changes neither which ends of a run are equal nor which runs are empty, so the same subsets give the terms of $\Delta\rho_\lambda y$, with every entry multiplied by $\lambda$: $\Delta\rho_\lambda y = \rho_\lambda y\otimes 1 + 1\otimes\rho_\lambda y + \mathrm{I}^{\mathfrak{s}}(0;\lambda\omega^2;\lambda)\otimes\mathrm{I}^{\mathfrak{s}}(\lambda\omega^2;0;\lambda) = (\rho_\lambda\otimes\rho_\lambda)\Delta y$. At level 1 the only rotation of $x$ is the identity. This is Theorem 45.9(iii) on the two symbols.

(iii) *The left factor of* $\Delta x$. The chain $\mathrm{I}(0;1,0;1)$ has the first letter $1\neq 0$ and the last letter $0\neq 1$. In Proposition 45.5(ii) its one nonzero letter is at $j_1 = 1$, with $j_2 = w+1 = 3$, so $d=1$, $c_1=1$, $\nu_1 = 2$, $c_2 = 1$, and the frequency $a_1' = 0$: $\mathrm{I}(0;1,0;1) = -(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$, and $z = -\sigma_2$ with $\sigma_2$ the symbol of $(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$. The right factor $R = \mathrm{I}^{\mathfrak{s}}(0;0;1) - \varrho\,\mathrm{I}^{\mathfrak{s}}(0;0;1)$ is a formal chain of weight one minus its reversal; the first letter of $\mathrm{I}^{\mathfrak{s}}(0;0;1)$ is its first end, so it has no realization (Definition 45.7), and $\int_0^1 dt/t$ diverges. Rendering: $\mathrm{I}(0;1,0;1) = -1.6449340668482264\ldots$.

*The factors of* $\Delta y$. $A$ has the letter $\omega^2\neq 0,1$. In Proposition 45.5(ii), $d=1$, $c_1=\omega$, $\nu_1 = 1$ and $c_2 = 1$, with the frequency $a_1' = \frac{1}{3}$, $e^{2\pi i a_1'} = \omega$: $\mathrm{I}(0;\omega^2;1)$ is minus the trail of level 3 of depth one at the index 1 and the frequency $\frac{1}{3}$, $\lim_N\sum_{m\le N}\omega^m m^{-1}$, which is a trail because $e^{2\pi i/3}\neq 1$ (Definition 45.1); $A$ is minus its symbol. On the path $\gamma(t) = t$, $t-\omega^2 = t + \frac{1}{2} + \frac{\sqrt{3}}{2}i$ has the imaginary part $\frac{\sqrt{3}}{2} > 0$, so the principal logarithm is continuous along it and

$$\mathrm{I}(0;\omega^2;1) = \int_0^1\frac{dt}{t-\omega^2} = \log\Big(\frac{3}{2}+\frac{\sqrt{3}}{2}i\Big) - \log\Big(\frac{1}{2}+\frac{\sqrt{3}}{2}i\Big) = \Big(\frac{1}{2}\log 3 + \frac{\pi i}{6}\Big) - \frac{\pi i}{3} = \frac{1}{2}\log 3 - \frac{\pi i}{6}.$$

$B$ has the ends $\omega^2$ and 1, so it is a chain between two singular points with no end at 0, read as it stands. Its letter 0 is not on the open segment from $\omega^2$ to 1 and differs from both ends, so the chain is defined and converges (Definition 45.4, Proposition 45.6(i)). On $\gamma(t) = \omega^2 + t(1-\omega^2) = -\frac{1}{2} + \frac{3}{2}t - \frac{\sqrt{3}}{2}(1-t)i$ the imaginary part is negative for $t<1$, so the argument of $\gamma(t)$ runs

continuously from $-\frac{2\pi}{3}$ to 0 and the modulus from 1 to 1:

$$\mathrm{I}(\omega^2;0;1)=\int_0^1\frac{\gamma'(t)\,dt}{\gamma(t)}=\log 1-\log\omega^2=0-\Big(-\frac{2\pi i}{3}\Big)=\frac{2\pi i}{3}.$$

Rendering: the integrals render as $0.5493061443340548\ldots-0.5235987755982988\ldots i$ and $2.0943951023931954\ldots i$.

*Answer.* With $\sigma_2$ the symbol of $(\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ and $\sigma_1'=-\mathrm{I}^{\mathfrak{s}}(0;\omega^2;1)$ the symbol of the trail of index 1 at the frequency $\frac{1}{3}$,

$$\Delta\sigma_3=\sigma_3\otimes 1+1\otimes\sigma_3+\sigma_2\otimes\big(\mathrm{I}^{\mathfrak{s}}(0;0;1)+\mathrm{I}^{\mathfrak{s}}(1;0;0)\big),\qquad \Delta\sigma'=\sigma'\otimes 1+1\otimes\sigma'+\sigma_1'\otimes\mathrm{I}^{\mathfrak{s}}(\omega^2;0;1).$$

Both are coassociative and counital, graded by the weight, and compatible with the reversal and the rotations, as Theorem 45.9 states. The left factors $\sigma_2$ and $\sigma_1'$ are symbols of trails; the right factor $\mathrm{I}^{\mathfrak{s}}(\omega^2;0;1)$ is a chain between the singular points $\omega^2$ and 1 with the value $\frac{2\pi i}{3}$, and $\mathrm{I}(0;\omega^2;1)=\frac{1}{2}\log 3-\frac{\pi i}{6}$. □

## 56.3 Separation of the content

The two exercises of this section work the separation of the content (Section 45.5). The first runs the separation procedure at level 1 through weight 14; the second places the two smallest centre values among the separating generators. Throughout, for an index $\mathbf{k}=(k_1,\ldots,k_d)$ with $k_1\geq 2$,

$$T(\mathbf{k})=\sum_{y_1>\cdots>y_d>0}\prod_{i=1}^{d}y_i^{-k_i},$$

the trail of level 1 with the data $(k_d,\ldots,k_1)$ (Definition 45.1), so that $T(N)=(\mathsf{T}_{N;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$; by Proposition 45.5(i) its chain is

$$T(\mathbf{k})=(-1)^d\,\mathrm{I}\big(0;\,1,0^{k_d-1},\,1,0^{k_{d-1}-1},\,\ldots,\,1,0^{k_1-1};\,1\big),$$

the blocks of the word read from the left giving the index from the right.

**Exercise 56.3** (The content of level one through weight fourteen)**.** Run the separation procedure (Definition 45.17) at level 1 for $N=2,\ldots,14$, with $R_k$ the stuffle product of trails (Proposition 45.2(ii)), the shuffle product of their chains and the relation of index one,

$$\sum_{i=1}^{d}T(k_1,\ldots,k_i+1,\ldots,k_d)=\sum_{i=1}^{d}\sum_{j=0}^{k_i-2}T(k_1,\ldots,k_{i-1},k_i-j,j+1,k_{i+1},\ldots,k_d)$$

for every index $\mathbf{k}$ of weight $k-1$, and with $\mathcal{R}$ the odd lengths from 3 to $N-2$. Show that it succeeds in every weight, conclude that every relation among the motivic values of the trails of level 1 of weight at most 14 is a consequence of these three relations, and find polynomial generators of the motivic content of level 1 through weight 14.

*Solution.* Receipts: `completeness/content_separation.py`, `completeness/small_weights.py` and `completeness/generators_basis.py`, with their logs.

*The relations hold for motivic values.* The stuffle and the shuffle products and the relation of index one are the double shuffle relations of the trails of level 1, and they hold for the motivic chains

of level 1 (Racinet and Furusho; they apply because the trails of level 1 are the multiple sums of Definition 45.1 at the frequency 0, whose chains are the iterated integrals of Proposition 45.5). The shuffle product is written on the trails as follows: if $u$ and $v$ are the words of $T(\mathbf{k})$ and $T(\mathbf{l})$, every word of $u$ shuffled with $v$ has the letters 1 of both, so the signs $(-1)^d$ agree and $T(\mathbf{k})T(\mathbf{l})$ is the sum of the trails of the shuffled words.

*Weights* 2 *and* 3. $V_2$ is the line of $T(2)$: no relation has weight 2. In weight 3 the trails are $T(3)$ and $T(2,1)$, and the relation of index one at $\mathbf{k}=(2)$ reads $T(3)=T(2,1)$; so $V_3$ is the line of $T(3)$. There are no cuts ($\mathcal{R}$ is empty), and the joint kernel is the line of $T(N)$, whose value is positive: step (4) succeeds, and by Theorem 45.18, with nothing to assume below, $V_2$ and $V_3$ map injectively.

*Weight* 4. The trails are $T(4), T(3,1), T(2,2), T(2,1,1)$. The word of $T(2)$ is 10, and 10 shuffled with 10 is $2\cdot 1010+4\cdot 1100$, the words of $T(2,2)$ and $T(3,1)$; the stuffle of (2) with (2) is $2\,(2,2)+(4)$. So $R_4$ is

$$T(2)^2 = 2T(2,2)+4T(3,1) = 2T(2,2)+T(4),$$
$$T(4)=T(3,1)+T(2,2), \qquad T(3,1)+T(2,2)=T(2,1,1),$$

the last two the relation of index one at (3) and at $(2,1)$. Their rank is 3, and

$$V_4=\mathbb{Q}\,T(2,2), \qquad T(4)=\tfrac{4}{3}T(2,2), \quad T(3,1)=\tfrac{1}{3}T(2,2), \quad T(2,1,1)=\tfrac{4}{3}T(2,2).$$

Again $\mathcal{R}$ is empty and the joint kernel is the line of $T(4)$, of positive value.

*Weight* 5. The eight trails reduce to the basis $T(2,3), T(3,2)$:

$$T(5)=\tfrac{6}{5}T(2,3)+\tfrac{4}{5}T(3,2), \qquad T(4,1)=T(3,1,1)=\tfrac{1}{5}T(2,3)-\tfrac{1}{5}T(3,2),$$
$$T(2,1,2)=T(2,3), \qquad T(2,2,1)=T(3,2), \qquad T(2,1,1,1)=T(5),$$
$$T(2)T(3)=\tfrac{11}{5}T(2,3)+\tfrac{9}{5}T(3,2).$$

$\mathcal{R}=\{3\}$, and $V_3/\mathfrak{v}_3=V_3$ is the line of $T(3)$. The cut $\mathsf{D}_3$ of $T(2,3)$, by Definition 45.14(iii) on its chain $\mathrm{I}(0;1,0,0,1,0;1)$, with $a_0=0$, $(a_1,\dots,a_5)=(1,0,0,1,0)$, $a_6=1$:

- $p=0$: the run $1,0,0$ between 0 and $a_4=1$ is $\mathrm{I}(0;1,0,0;1)=-T(3)$, and the kept letters $1,0$ give $\mathrm{I}(0;1,0;1)=-T(2)$; the term is $[T(3)]\otimes T(2)$.
- $p=1$: the run $0,0,1$ between $a_1=1$ and $a_5=0$ is $\mathrm{I}(1;0,0,1;0)=(-1)^3\mathrm{I}(0;1,0,0;1)=T(3)$ by the reversal (Proposition 45.6(iii)); the kept letters $1,0$ give $-T(2)$; the term is $-[T(3)]\otimes T(2)$.
- $p=2$: the run $0,1,0$ between $a_2=0$ and $a_6=1$ begins with its first end; the regularization removes the leading 0: $\mathrm{I}(0;0,1,0;1)=-\mathrm{I}(0;1,0,0;1)-\mathrm{I}(0;1,0,0;1)=2\,T(3)$, the two words being the insertions of 0 after each letter of $1,0$; the kept letters $1,0$ give $-T(2)$; the term is $-2[T(3)]\otimes T(2)$.

So $\mathsf{D}_3T(2,3)=-2\,[T(3)]\otimes T(2)$, and in the same way $\mathsf{D}_3T(3,2)=3\,[T(3)]\otimes T(2)$. The cut vanishes on every relation of weight 5 (step (3), receipt). Its kernel on $V_5$ is the line of $3T(2,3)+2T(3,2)=\frac{5}{2}T(5)$, of positive value.

*Weight* 6. The basis is $T(3,3), T(2,2,2)$, with $T(6)=\frac{16}{3}T(2,2,2)$; $\mathcal{R}=\{3\}$ and

$$\mathsf{D}_3T(3,3)=[T(3)]\otimes T(3), \qquad \mathsf{D}_3T(2,2,2)=0.$$

The kernel is the line of $T(2,2,2)$, which is that of $T(6)$.

*Weight* 7. The basis is $T(2,2,3), T(2,3,2), T(3,2,2)$, and $\mathcal{R} = \{3,5\}$. $V_4$ is the line of $T(2,2)$; $V_5/\mathfrak{v}_5$ is the line of the class of $T(5)$, the products being the line of $T(2)T(3)$. On the basis, in the order given,

$$\mathsf{D}_3 = (-2,\ 0,\ 3)\,[T(3)] \otimes T(2,2), \qquad \mathsf{D}_5 = \big(12,\ -\tfrac{11}{2},\ -\tfrac{15}{2}\big)\,[T(5)] \otimes T(2).$$

Their joint kernel is the line of $(33, 42, 22)$, and $T(7) = \frac{16}{151}\big(33\,T(2,2,3) + 42\,T(2,3,2) + 22\,T(3,2,2)\big)$: the line of $T(7)$.

*Weights* 8 *to* 14*, by machine.* Input: the $2^{N-2}$ trails of weight $N$, as their words; the relations $R_N$, one for each pair of words of weights $k$ and $N-k$, $2 \le k \le N/2$ (stuffle minus shuffle), and one for each index of weight $N-1$ (index one): 1672 relations in weight 12, 7696 in weight 14. Method: exact row reduction over $\mathbb{Q}$ with the trails of indices 2 and 3 placed last, so that they are the free columns when they span; the reading of runs and kept chains by the regularization at the ends, as in weight 5; the cuts of every trail and of every relation; the rank of the cut matrix on the basis. Output:

| $N$ | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\dim V_N$ | 1 | 1 | 1 | 2 | 2 | 3 | 4 | 5 | 7 | 9 | 12 | 16 | 21 |
| $\dim V_N/\mathfrak{v}_N$ | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 2 | 2 | 3 | 3 |
| rank of the cuts | 0 | 0 | 0 | 1 | 1 | 2 | 3 | 4 | 6 | 8 | 11 | 15 | 20 |

with $\dim V_N = d_N$, $\sum_N d_N x^N = 1/(1-x^2-x^3)$, and the trails of indices 2 and 3 a basis in every weight. Checks: every cut vanishes on every relation (step (3)); in every weight the rank of the cuts is $\dim V_N - 1$ and $T(N)$ lies in their kernel, so the joint kernel is the line of $T(N)$, of positive value (step (4)). Receipt: `content_separation.py 14` and its log.

*Conclusion.* Theorem 45.18 applies at $N = 2$ and 3 with nothing below, and at each $N \le 14$ its hypothesis is its conclusion at the weights below. So $V_N \to \mathcal{P}^{(1)}[N]$ is injective for $N \le 14$. The motivic values are graded by the weight, so a relation among trails of weights at most 14 splits into its homogeneous parts, each a consequence of the three relations.

*Polynomial generators.* In each weight the products of the generators found in lower weights are reduced in $V_N$, and trails are added, those with all indices odd first, while they raise the rank (receipt `generators_basis.py 14`). The generators found are

$$T(2),\ T(3),\ T(5),\ T(7),\ T(9),\ T(11),\ T(13),$$
$$T(3,5),\ T(3,7),\ T(3,3,5),\ T(3,9),\ T(2,2,3,5),$$
$$T(3,3,7),\ T(3,5,5),\ T(3,11),\ T(5,9),\ T(3,3,3,5),$$

in the weights $2,3,5,7,9,11,13,8,10,11,12,12,13,13,14,14,14$: one for $\mathbb{L}^2$ and one for each Lyndon word in the letters $x_3 < x_5 < \cdots$ of weight at most 14, and their monomials are a basis of $V_N$ in every weight. In weight 12 the products of lower generators span 10 dimensions; $T(3,9)$ adds one, and $T(5,7)$, $T(7,5)$, $T(9,3)$, $T(3,3,3,3)$ add none; the products together with all 56 trails of depth at most three span $11 = d_{12} - 1$, so the second generator of weight 12 has depth at least four, and $T(2,2,3,5)$ is one. In weight 14, $T(3,11)$ and $T(5,9)$ are independent modulo products.

*Answer.* The procedure succeeds in every weight from 2 to 14, the joint kernel of the cuts being the line of $T(N)$; every relation among the motivic values of the trails of level 1 of weight at most 14 is a consequence of the stuffle product, the shuffle product of chains and the relation of index one; the dimensions are $d_N$, with the trails of indices 2 and 3 as a basis; and the motivic content of level 1 through weight 14 is the polynomial algebra on the seventeen trails above. □

**Exercise 56.4** (The two smallest centre values are separating generators)**.** Let $\ell$ be the value of the list of three marks at its centre and $v$ the value of the half-lattice at four marks (Exercise 56.1). Show that $\ell = (\omega^2 - \omega)\,\mathsf{w}_2$ with $\mathsf{w}_2$ the separating generator of weight 2 at level 3, and $v = 7\,\mathsf{w}_3$ with $\mathsf{w}_3$ the separating generator of weight 3 at level 1 (Definition 45.14(iv)); check that their cuts vanish, and read their values.

*Solution. The list of three marks.* By Exercise 56.1(i), $\ell = (\omega^2-\omega)\big((\mathsf{T}_{2;(1)}[x]\,e^{2\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,e^{4\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty))\big)$, with $\omega = e^{2\pi i/3}$. The frequency enters through $e^{2\pi i a y}$ at integers $y$, so $(\mathsf{T}_{2;(1)}[x]\,e^{4\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = (\mathsf{T}_{2;(1)}[x]\,e^{-2\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty))$, and

$$\ell = (\omega^2-\omega)\,\mathsf{w}_2, \qquad \mathsf{w}_2 = (\mathsf{T}_{2;(1)}[x]\,2i\sin(\tfrac{2\pi x}{3})\,\mathbb{I}_{\mathbb{Z}})((0,\infty)),$$

the separating generator of weight 2 at level 3 (Definition 45.14(iv), $N=2$ even, $M=3$).

*Its cut.* The only length below 2 is $r=1$. By Exercise 56.1(ii) the chains are $(\mathsf{T}_{2;(1)}[x]\,e^{2\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\mathrm{I}(0;\omega^2,0;1)$ and $(\mathsf{T}_{2;(1)}[x]\,e^{4\pi i x/3}\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\mathrm{I}(0;\omega,0;1)$. For $\mathrm{I}(0;\beta,0;1)$, $\beta\in\{\omega,\omega^2\}$: at $p=0$ the run $\beta$ lies between the ends 0 and 0, equal, and gives 0; at $p=1$ the run 0 lies between $\beta$ and 1, and $\mathrm{I}(\beta;0;1) = \log 1 - \log\beta$ read motivically is a rational multiple of $\mathbb{L}$, which is 0 in $\mathfrak{n}^{(3)}[1]$. So $\mathsf{D}_1\mathsf{w}_2^{\mathfrak{m}} = 0$, as Theorem 45.15(ii) says for every separating generator.

*Its value.* With $\chi(y) = (\omega^y - \omega^{-y})/(\omega-\omega^2)$, which is $1,-1,0$ on $1+3\mathbb{Z}$, $2+3\mathbb{Z}$, $3\mathbb{Z}$, and $\omega-\omega^2 = i\sqrt{3}$,

$$\mathsf{w}_2 = \sum_{y\geq 1}\frac{\omega^y-\omega^{-y}}{y^2} = i\sqrt{3}\,\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\big) = \frac{i\,\ell}{\sqrt{3}},$$

since $\ell = 3\big((\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{1+3\mathbb{Z}})((0,\infty)) - (\mathsf{T}_{2;(1)}[x]\,\mathbb{I}_{2+3\mathbb{Z}})((0,\infty))\big)$; and $\ell = 2.3439072386894588\ldots$ (rendering, Exercise 56.1) is positive, as the product of Theorem 45.15(ii) says. The value of $\mathsf{w}_2$ is imaginary and that of $\mathbb{L}^2$ is real, $-4\Gamma(\frac12)^4$, so an element $c\,\mathsf{w}_2^{\mathfrak{m}} + b\,\mathbb{L}^2$ has the readout $c = \sqrt{3}\,\mathrm{Im}(\cdot)/\ell$ and $b = -\mathrm{Re}(\cdot)/(4\Gamma(\frac12)^4)$ of its value. Over the coefficients $\mathbb{Q}(\mu_3)$ the motivic value of $\ell$ is $(\omega^2-\omega)\,\mathsf{w}_2^{\mathfrak{m}}$, a separating generator times an algebraic number.

*The half-lattice at four marks.* By Exercise 56.1(iii), $v = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = 7\,\mathsf{w}_3$ at level 1. The chain is $(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty)) = -\mathrm{I}(0;1,0,0;1)$; its runs of one or two letters are bounded by equal ends, or are letters 0 between 1 and an end, read as $(-\log 1)^r/r! = 0$; so $\mathsf{D}_1$ and $\mathsf{D}_2$ vanish, and indeed $\mathfrak{n}^{(1)}[1] = \mathfrak{n}^{(1)}[2] = 0$, the motivic content of level 1 having nothing in weight 1 and only $\mathbb{Q}\,\mathbb{L}^2$ in weight 2. Its value $v = 8.4143983221171599\ldots$ (rendering) is positive.

*Answer.* $\ell = (\omega^2-\omega)\,\mathsf{w}_2$ at level 3, with $\mathsf{w}_2$ of value $i\ell/\sqrt{3}$, and $v = 7\,\mathsf{w}_3$ at level 1, with $\mathsf{w}_3 = (\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$: up to the factors $\omega^2-\omega$ and 7 the two smallest centre values are the separating generators of their weights, with every cut zero and the normal forms $(\omega^2-\omega)\,x_2$ and $7\,x_3$. □

## 56.4 The closing of a period in any number of shares

**Exercise 56.5** (The law found by the closing of the class of three marks)**.** Let $\ell = P_{\binom{[3]}{1}}(\mathbf{1};1,1)$, the value of the centre ray of $\binom{[3]}{1}$ at the strength one (Exercise 56.1).

(i) Write the period as $P_0\sum_N \psi(N)$ by Theorem 45.21(i), and compute $\psi(N)$ for $N\leq 7$.

(ii) Find the recurrence of $\psi$, its symbol, and the singular points of the letter $L_\psi$.

(iii) Relate the singular points to the critical values of the cobasis letter on the strata of the simplex, and give the rate of the tail.

*Solution.* Receipts: `closing/peak_letter.py`, `closing/letter_exponents.py` and their logs.

*(i)* The family is $\binom{[3]}{1}$ with the masses $\mathbf{1}$ and the exponents $(1,1)$; its letters are $K = u_1+u_2+u_3$, of the degree 1, and $\Psi = u_1u_2+u_1u_3+u_2u_3$, of the degree 2. With the references of Theorem 15.13, $M_K = 1/1! = 1$ and $M_\Psi = 1/2! = \frac{1}{2}$, so $P_0 = \Gamma_{\binom{3}{1}}\begin{bmatrix}1,1,1\\3\end{bmatrix} M_K^{-1}M_\Psi^{-1} = \frac{1}{2}\cdot 2 = 1$. On the simplex $K = 1 = M_K$, so only the terms with $n_1 = 0$ survive, and with $(1)_N/N! = 1$,

$$\psi(N) = 2^N\,\mathbb{E}_{U\sim\mathrm{Dir}_{(1,1,1)}}\Big[\big(\tfrac{1}{2} - \Psi(U)\big)^N\Big] = \mathbb{E}\big[(1-2\Psi(U))^N\big],$$

the Dirichlet law at $(1,1,1)$ being uniform on the simplex, with $\mathbb{E}[U^{\boldsymbol\gamma}] = 2\,\gamma_1!\gamma_2!\gamma_3!/(|\boldsymbol\gamma|+2)!$. Expanding the powers of $1-2\Psi$ in monomials and taking these moments gives, exactly,

$$\psi(0),\dots,\psi(7) = 1,\ \tfrac{1}{2},\ \tfrac{4}{15},\ \tfrac{16}{105},\ \tfrac{7}{75},\ \tfrac{47}{770},\ \tfrac{4448}{105105},\ \tfrac{463}{15015}.$$

For instance $\psi(1) = 1 - 2\,\mathbb{E}[\Psi] = 1 - 6\cdot\frac{1}{12} = \frac{1}{2}$. So $\ell = \sum_{N\ge 0}\psi(N)$, a series of nonnegative rationals.

*(ii)* By Theorem 45.21(i), $\psi$ satisfies a recurrence with polynomial coefficients. Method: the recurrences of the order 3 with coefficients of the degree 2 that hold for $N \le 30$ form a line, spanned by

$$\begin{aligned}(N+1)(N+2)\,\psi(N) - 2(N+2)(3N+7)\,\psi(N+1)\\ + (N+3)(11N+31)\,\psi(N+2) - 3(N+4)(2N+7)\,\psi(N+3) = 0;\end{aligned}$$

check: it holds exactly for every $N \le 35$; at $N=0$, $2\cdot 1 - 2\cdot 2\cdot 7\cdot\frac{1}{2} + 3\cdot 31\cdot\frac{4}{15} - 3\cdot 4\cdot 7\cdot\frac{16}{105} = 2 - 14 + \frac{124}{5} - \frac{64}{5} = 0$. The coefficients of $N^2$ are $1,-6,11,-6$, so the symbol is $1-6\lambda+11\lambda^2-6\lambda^3 = -(\lambda-1)(2\lambda-1)(3\lambda-1)$, and by Proposition 45.12(i) the letter $L_\psi$ is singular at most at the reciprocals $z = 1,2,3$ of its roots; its radius of convergence is 1, the roots having the largest modulus 1, so it converges on $|z|<1$ as the theorem says, and its value at $z=1$ is $\ell$.

*(iii)* By Theorem 45.21(i), $P_0L_\psi(z) = \int_\Delta \big((1-z)M_\Psi + z\Psi(u)\big)^{-1}d\omega(u)$ on the simplex. The three points $z = 1,2,3$ are the $z = M_\Psi/(M_\Psi - \Psi^*)$ for the critical values $\Psi^*$ of $\Psi$ on the strata of the simplex, 0 at the vertices, $\frac{1}{4}$ at the midpoints of the edges and $\frac{1}{3}$ at the centre: there the deformed letter $(1-z)M_\Psi + z\Psi$ acquires a critical point of the value 0 on a stratum. The walls of the family are $\kappa_S = 1$ for every nonempty proper $S$, and the counts of $\Psi$ are $d^*(S) = |S| - 1$; so the rate of Theorem 15.13(ii) is $\varpi = \min_S \kappa_S/d^*(S) = 1$, attained at the three pairs, no two of which form a chain, so $k = 1$, and the tail is $\sum_{N'>N}\psi(N') = \Theta(1/N)$. □

# Aliases

The text uses the program's own names. This list gives, for a reader who arrives from the literature, the classical names under which some of its objects and results appear there. An alias is a name and nothing more: the statement of the handbook, with its hypotheses, is the one that holds, and where the classical result is narrower or wider the handbook's statement decides.

## Lists, tables and their letters

| In this handbook | Classical name |
|---|---|
| a table, its marks and bases (Definition 7.1) | a matroid with weights on its bases, its ground set and bases |
| a list (the weights all 1) | a matroid |
| the list $\binom{\mathcal{E}}{r}$, $\lvert\mathcal{E}\rvert = n$ | the uniform matroid $U_{r,n}$ |
| the dual table | the dual matroid |
| a molecule; a connected table | a connected component; a connected matroid |
| the width $c(S)$ of a set (Definition 7.12) | the connectivity $r(S) + r(\mathcal{E} \setminus S) - r(\mathcal{E})$ of the set |
| the two-sum $\mathcal{A} \oplus_g \mathcal{A}'$ of two cells (Lemma 21.2) | the 2-sum of matroids |
| for the list of the spanning trees of a graph: the basis polynomial $K$ and the cobasis polynomial $\Psi$ | the spanning-tree polynomial, and the graph polynomial, also called the first Symanzik or the Kirchhoff polynomial |
| the Cremona (Theorem 8.14) | the Cremona transformation $\alpha \mapsto \alpha^{-1}$ |
| the period at the full cobasis lean $(\mathbf{1}; 0, 2)$ of the list of the spanning trees of a primitive logarithmically divergent graph | its Feynman period $\int \Omega/\Psi^2$; the volume of the direction $V_1$ of that ray is its Hepp bound |
| the list $\binom{[n]}{1}$ | the cycle matroid of the banana graph with $n$ edges |
| the wheel $W_p$ (Proposition 39.65); $W_3$ | the cycle matroid of the wheel with $p$ spokes; $W_3$ is the cycle matroid of $K_4$ |
| $\mathrm{H}_4$ | the cycle matroid of a triangle with one edge doubled |
| $\mathrm{N}_6$ | $U_{3,6}$ with one basis removed: six points of a plane, exactly three of them collinear |
| $\mathrm{V}_8$ | the Vámos matroid |
| the whirl list on six marks (rank three, three dependent sets of three marks meeting pairwise) | the rank-three whirl |

| In this handbook | Classical name |
|---|---|
| $\mathrm{F}_7$, $\mathrm{A}_9$, $\mathrm{Z}_9$ (Chapter 30) | the Fano plane, the affine plane $AG(2,3)$, the Pappus configuration |

## Values

| In this handbook | Classical name |
|---|---|
| $(\mathsf{T}_{m;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ | $\zeta(m)$ |
| $\Gamma(\frac{1}{2})^2$ | $\pi$ |
| $(\mathsf{T}_{2;(1)}[x]\,(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}))((0,\infty))$; so $P_{\binom{[3]}{1}}(\mathbf{1};1,1) = 3\,(\mathsf{T}_{2;(1)}[x]\,(\mathbb{I}_{1+3\mathbb{Z}} - \mathbb{I}_{2+3\mathbb{Z}}))((0,\infty))$ | $L(\chi_{-3},2)$; the value is $3L(\chi_{-3},2)$ |
| $P_{\binom{[4]}{1}}(\mathbf{1};1,1) = 7\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$; the period of $W_3$ at $(\mathbf{1};0,2)$, $6\,(\mathsf{T}_{3;(1)}[x]\,\mathbb{I}_{\mathbb{Z}})((0,\infty))$ | $7\zeta(3)$; $6\zeta(3)$ |
| the signed sums of Trailmoments of lattice inputs at the cosets prime to $\lvert D\rvert$, weighted by the Kronecker signs | $L(\chi_D,2)$ |
| the arc integrals of the three-mark centre (Part XI) | the log-sine integrals $\mathrm{Ls}_j(\frac{2\pi}{3})$; the second is the Clausen value $\mathrm{Cl}_2(\frac{2\pi}{3})$ |
| the stacked Gammas at fifteenths, at twentieths | Chowla–Selberg periods of $\mathbb{Q}(\sqrt{-15})$, of $\mathbb{Q}(\sqrt{-5})$ |
| the Gramma of one mark, $\mathsf{G}_{\nu;(1,-1)}(1,y) = 2y^{\nu/2}K_\nu(2\sqrt{y})$ | the modified Bessel function $K_\nu$ |
| the centre values of $\binom{[n]}{1}$, $P_{\binom{[n]}{1}}(s\mathbf{1};s,s) = 2^{n+1-2s}\Gamma(s)^{-2}\int_0^\infty x^{2s-1}K_0(x)^n\,dx$ | Bessel moments; at $n=3$ the equal-mass two-loop vacuum (sunrise) integral with propagator powers $s$ in dimension $2s$ |

## Rays

| In this handbook | Classical name |
|---|---|
| the relations along a ray, reduced in a box (Definition 39.15) | integration-by-parts reduction on the parametric form, in the manner of Laporta |
| the strength | the parameter that plays the dimension |
| masters (Definition 39.1) | master integrals |
| the one-step matrix | the dimension-shift recurrence written as a first-order difference system |
| the law of [1] | the scalar recurrence obtained from a cyclic vector |
| a certificate (Theorem 39.21(ii)) | a creative-telescoping certificate |
| the wall degree | the tropical degree of an insertion |

| In this handbook | Classical name |
|---|---|
| the search through boxes (Theorem 39.22) | Laporta's algorithm, its seeds grown until the targets reduce |
| the dimension of a ray (Definition 38.4) | the number of master integrals |
| the spectrum at infinite strength (Proposition 39.42) | the characteristic roots of a difference system (Birkhoff); a change of masters is a gauge transformation |
| the evaluation modulo primes in the reading procedure (Definition 41.8) | finite-field reconstruction |
| the closings (Definition 39.30) | the classical periods of the Laurent polynomial $f^{-1}$; the law read at the low end is the Picard–Fuchs operator of its pencil, read on coefficients |
| the closings of the centre ray of $\binom{[3]}{1}$ | the sums of squares of trinomial coefficients |
| positive selection (Theorem 41.3) | Apéry limits; the minimal solution of Pincherle and Perron |
| positivity as the moment problem of the stride's law (Proposition 41.4); its rate (Theorem 41.5); what it cannot see (Proposition 41.6) | the Hausdorff moment problem, its four families of Markov and Lukács; the Green's function of the complement of a segment; the indeterminacy of a moment problem along a Stieltjes transform |
| the stride's law (Definition 40.1) | the pushforward of Lebesgue measure under the stride, a density of states; the values are its moments |
| the gauge end (Theorem 40.6) | Laplace's method with the Morse lemma at the end point, and the Frobenius method at a regular singular point; the germ is the Borel sum of the expansion at large strength |
| the torus value (Definition 40.11) | the zeta Mahler measure of the letter; at three marks the moments of the planar walk of three unit steps |
| the merit (Definition 43.4, Theorem 43.5) | Apéry's criterion |
| the law of the values of $\binom{\mathcal{E}}{1}$ (Theorem 42.12) | the recurrences of the Bessel moments $\int_0^\infty x^{2s-1} K_0(x)^n \, dx$ |

## The motive and the content

| In this handbook | Classical name |
|---|---|
| the motive and the motivic period of a family (Definitions 34.1 and 34.5) | motivic periods in Brown's affine model |
| small steps along the tree (Theorem 34.13) | Brown's small graphs principle in its affine form, here for families of lists |
| dual pairs and full leans (Corollary 34.18) | at $(0, 2)$, the product identity of Feynman periods at a two-vertex cut (Schnetz), as an identity of motivic periods |

| In this handbook | Classical name |
|---|---|
| chains (Definition 45.4); the content (Definition 45.1) | iterated integrals on $\{0\}$ and the roots of unity of order $M$; the multiple polylogarithms at roots of unity |
| the cutting rule (Definition 45.8, Theorem 45.9) | Goncharov's coaction on iterated integrals |
| critical primes (Definition 43.11) | primes of bad reduction |
| separation by conjugates (Theorem 34.21) | motivic periods as functions on the torsor of periods; the injectivity of the period alone is Grothendieck's period conjecture |
| the motivic chains (Proposition 45.10) | the motivic iterated integrals of Goncharov and Brown; the motivic multiple polylogarithms at roots of unity |
| the motivic content $\mathcal{P}^{(M)}$, $M = 1, 2, 3, 4$ (Definition 45.14) | the motivic multiple zeta values, the motivic Euler sums, the motivic multiple polylogarithms at the third and fourth roots of unity |
| the cuts $\mathsf{D}_r$ | Brown's infinitesimal coaction $D_r$ |
| the separating generators $\mathsf{w}_N$ | $\zeta^{\mathfrak{m}}(N)$ for $N$ odd, $\log 2$, $\log 3$, and for $N$ even $\mathrm{Li}^{\mathfrak{m}}_N(\zeta_M) - \mathrm{Li}^{\mathfrak{m}}_N(\zeta_M^{-1})$, of the values $i\sqrt{3}\,L(\chi_{-3}, N)$ and $2i\,L(\chi_{-4}, N)$ |
| the normal form (Theorem 45.15(iii)) | Brown's decomposition into the $f$-alphabet |
| the polynomial generators of Lyndon words | Radford's theorem on shuffle algebras |
| the stuffle product, the shuffle product of chains, the relation of index one | the double shuffle relations; Hoffman's relation |
| the trails $T(\mathbf{k})$ with every $k_i \in \{2, 3\}$ (Exercise 56.3) | the Hoffman elements |
| the second generator of weight twelve, of depth four (Exercise 56.3) | the relation $28\zeta(3,9) + 150\zeta(5,7) + 168\zeta(7,5) = \frac{5197}{691}\zeta(12)$ of the cusp form of weight twelve (Gangl, Kaneko and Zagier), and the depth of Broadhurst and Kreimer |
| the values of the content and its generators (Corollary 45.16) | Grothendieck's period conjecture for the mixed Tate motives over $\mathbb{Z}[\mu_M, \frac{1}{M}]$, stated on generators |
| a separating generator and its measure (Lemma 44.17); separators at once (Corollary 44.13) | irrationality and linear independence measures; the criteria of Siegel and Nesterenko |
| the model of a ray and its relations as exact forms (Definition 39.7, Proposition 39.8) | the twisted de Rham cohomology of the parametric representation and its integration-by-parts identities; a ray is one motive with a family of de Rham classes, as in the theory of Feynman integrals in dimensional or analytic regularization |
| the pencil module $\mathcal{G}_F$, its fibre integrals, singular values and vanishing ranks (Definition 39.47) | the Gauss–Manin system $H^0\varphi_+\mathcal{O}$ of the function $\varphi = f^{-1}$; Gelfand–Leray residue forms; the critical values and the dimensions of the vanishing cycles |

| In this handbook | Classical name |
|---|---|
| the moment side of the pencil (Proposition 39.48) | the algebraic Mellin transform of a holonomic module in one variable |
| asymptotic native gauges and values (Definition 38.29) | Rabier's generalized critical values; critical points at infinity |
| the insertion lattice, the native algebra, a tame high end (Definition 39.43) | the Brieskorn lattice, the Jacobian algebra, a cohomologically tame function (Sabbah) |
| the far end of the pencil (Proposition 39.54) | a point of maximal unipotent monodromy; the torus fibre integral is the classical period $\sum_j \mathrm{CT}(\varphi^j) t^{-j-1}$ of the Laurent polynomial $\varphi$ |
| a hinge tree; the canonical hinge tree (Definition 23.15, Theorem 23.16) | the decomposition of a connected matroid into 3-connected pieces glued by 2-sums (Cunningham–Edmonds); for a graph, its SPQR tree |
| the homogeneous reading of a node (Definition 39.55) | the 2-sum formula $K_{M_1 \oplus_2 M_2} = K_{M_1 \setminus e} K_{M_2 / e} + K_{M_1 / e} K_{M_2 \setminus e}$ for basis polynomials, iterated along the tree |
| the pencil along a hinge tree (Theorem 39.56) | the Gauss–Manin system of a composite map as the composite of the direct images, by the functoriality of direct images and the Künneth formula |
| the parallel and series rules (Corollary 42.21) | the rules for conductances in parallel and in series |
| the corners of a wall of width one (Proposition 38.26) | points at infinity at which the closure of a level set of the Laurent polynomial is tangent to the toric boundary |
| rational data on a Kummer cover (Proposition 34.20) | multivalued forms of hypergeometric type as sections of local systems of rank one; cyclic coverings |
| the class density $\nu_s$ (Definition 42.19) | the density of the squared distance after $s$ unit steps of a planar random walk with independent uniform directions |
| the closings of $\binom{[s]}{1}$ and the solutions $V_s$ of the class (Theorem 42.28) | the even moments of planar random walks; the Bessel moments and the symmetric power of the Bessel equation; for $s = 3$ the sunrise, an elliptic family |
| a list read against its cells; the bubble of two classes (Theorem 42.20, Corollary 42.25) | the dispersive (Källén–Lehmann) representation of a self-energy inserted in a propagator; the Källén function; $\sum_k z^k / k!^2 = I_0(2\sqrt{z})$ |
| the norm of a cell; a mark of norm $\mathsf{m}$; the two-sided core (Definition 42.19) | the squared mass of a propagator; a massive line; the graph polynomial with masses |
| the hinge module of a class (Proposition 42.29) | a direct image of $\mathcal{D}$-modules computed by its Fourier transform, an exponentially twisted Gauss–Manin system; the irreducibility of the symmetric powers of the Bessel module (Kolchin) |

| In this handbook | Classical name |
|---|---|
| the source count (Theorem 42.32, Proposition 42.33) | Kashiwara's index formula for direct images; Landau singularities and their multiplicities |
| closed form; named function; alphabet; the serif alphabet (Definition 18.19) | a closed-form expression, in the sense in which $\Gamma$, the Beta function, the Gauss hypergeometric function ${}_2F_1$ and the polylogarithms $\mathrm{Li}_n$ are closed forms |
| the series of stacked Gammas read from words (Proposition 18.20(iv)) | the Barnes integral of $B(k,k)\,{}_2F_1(k,k;2k;1-\mathsf{m})$ and its Gauss series; the Mellin–Barnes integral of $\mathrm{Li}_n(z)$ |
| the value of a table with norms on its marks, and down the hinge tree, as a closed serif word (Proposition 42.26) | the Mellin–Barnes representation of a Feynman integral with massive propagators, the masses entering as powers of the Mellin variables; a banana subgraph entering through its Bessel moment |
| the periods of a class are not unipotent (Corollary 42.30) | they are not multiple polylogarithms with algebraic arguments and coefficients: that alphabet is not closed on the class |
| a chain over an equation; every solution is read through chains over its equation (Definition 45.19, Proposition 45.20) | an iterated integral whose kernels are built from the solutions of the irreducible factors of a Picard–Fuchs equation: multiple polylogarithms when every factor is trivial, the iterated integrals with kernels from the periods of elliptic curves (elliptic multiple polylogarithms, iterated integrals of modular forms) and of Calabi–Yau operators beyond; variation of constants along a composition series |
| the values of a ray read through chains over its tree (Corollary 45.22) | the master integrals of a Feynman integral as iterated integrals over the irreducible blocks of its differential equation, the blocks read off its subgraphs |
| every period at rational data as a moment of one integral ray (Corollary 45.22(iv)) | the Mellin transform, at a rational point, of the density of the image of the invariant measure of the positive orthant under a Laurent polynomial |
| the letter of the closing along the tree; the load sum of a vertex (Corollary 45.23) | the Picard–Fuchs equation of a one-parameter deformation of the integrand, computed as an iterated direct image along the decomposition into 3-connected pieces, with the deformation applied at the root |
| the group of a module, of a function and of an input; its unipotent and reductive parts (Definition 45.24, Proposition 45.25) | the differential Galois group and its Lie algebra; Picard–Vessiot theory, the transcendence degree of the Picard–Vessiot extension; Schlesinger's theorem for regular singular equations |
| powers add nothing, distinct factors add (Proposition 45.26) | the Galois group of a tensor construction as a quotient; Goursat's lemma (the Goursat–Kolchin–Ribet lemma) for a sum of irreducible differential modules |

| In this handbook | Classical name |
|---|---|
| a full Jordan block; a power; the far end of the pencil (Theorem 45.27, Corollary 45.30) | maximal unipotent monodromy; Gabber's theorem on irreducible groups containing a regular unipotent element; the principal $\mathrm{SL}_2$; operators that are symmetric powers of operators of order two |
| the class read by the tree and by its stride's law (Theorem 45.28) | $\mathsf{G}_{0;(1,-1)}(1,y) = 2K_0(2\sqrt{y})$, the modified Bessel function, and $\sum_k y^k/k!^2 = I_0(2\sqrt{y})$; the differential Galois group $\mathrm{SL}_2$ of the modified Bessel equation of order zero (Kolchin); the Bessel moments $\int_0^\infty y^{k-1}\big(2K_0(2\sqrt{y})\big)^s\,dy$ of the equal-mass banana integrals; the symmetric powers of the modified Bessel equation of order zero; the monodromy of the banana Picard–Fuchs operators, of Calabi–Yau type |
| the word along the canonical tree (Corollary 45.29) | a period read over the Cunningham–Edmonds decomposition: Gamma factors at the series and parallel pieces, the periods of the 3-connected pieces at a full lean, Bessel moments for bundles of parallel edges |
| the group is computable (Proposition 45.31) | Hrushovski's algorithm for the differential Galois group of a linear differential equation over $\overline{\mathbb{Q}}(t)$ |
| the inputs and the group of a word (Definition 45.32) | the differential Galois group of the special functions that a representation of an integral integrates against |
| the transforms satisfy linear equations (Lemma 45.33) | holonomic functions are stable under integration and under the Mellin transform; the Picard–Fuchs equation of a period in a parameter of its integrand |
| the word along the canonical tree paid in work or on scales (Theorem 45.34) | Mellin–Barnes representations, of dimension growing with the number of lines, against position-space representations of a Feynman integral as integrals over few points of products of propagators, Bessel functions for massive lines; a bundle of parallel lines as a power of one propagator |
| the closing in any number of shares (Theorem 45.21) | multiple Mellin–Barnes integrals closed residue by residue; the Frobenius method at a regular singular point; the period as a series of a $D$-finite sequence |
| the triangles of classes $\Delta(s_0, s_1, s_2)$ (Section 55.4) | the cycle matroids of a triangle with its three edges replaced by bundles of $s_0$, $s_1$, $s_2$ parallel edges |

| In this handbook | Classical name |
|---|---|
| the module of a family; its shifts, masters and shift matrices (Definition 47.3, Theorem 47.4) | the twisted de Rham cohomology of the complement of the letters; the master integrals of the integration-by-parts reduction of a family of Feynman integrals; the contiguity (raising and lowering) relations of a GKZ system |
| boxes in all the data; the search stops (Theorem 47.8) | Laporta's algorithm: integration-by-parts identities generated on seeds ordered by weight and solved by Gaussian elimination over the rational functions of the parameters |
| closings in all the data; the elements of value zero (Definition 47.9, Theorem 47.11) | constant terms of Laurent polynomials, the integrals over the compact torus at nonpositive integer exponents; the kernel of the pairing of twisted cohomology with one integration cycle |
| masters from a box (Proposition 47.13) | a basis of the twisted cohomology, with its difference equations in all the parameters, from finitely many integration-by-parts identities, the relations they miss generated by the commutation of the difference operators and the identities at one point |
| the values separate the module (Proposition 47.15) | the injectivity of the Mellin transform in the exponents of the monomial: a linear relation among parametric integrals, with coefficients rational in all the parameters, that holds identically is a consequence of the integration-by-parts identities; a twisted cohomology class whose period over the positive cycle vanishes identically in the exponents of the monomial is zero |
| value zero on a subspace of the data; native directions and the expansion at a native gauge (Proposition 47.14) | deciding a linear relation among parametric integrals on a linear subspace of the parameters by Laplace's method along one direction, at the critical point of the phase, the coefficients of the asymptotic expansion being rational in the remaining parameters and limited in number by the difference equation along that direction |
| a family of positive letters on its space of directions, and its module (Definition 48.1, Theorem 48.3) | the twisted de Rham cohomology of the complement of a family of Laurent polynomials on a torus; the Mellin side of an $A$-hypergeometric (GKZ) system with all its exponents as parameters |
| a share reading; the integrand module (Definition 48.4, Theorem 48.5) | a Mellin–Barnes representation, an integral over Gamma functions of linear forms times smaller integrals, with the integration-by-parts reduction done on the smaller integrals |
| the path of a basis through a chain of bunches (Definition 48.6, Theorem 48.7) | the Mellin–Barnes splitting of a product of transfer matrices, whose shares are the flows of a Markov chain on the states |

| In this handbook | Classical name |
|---|---|
| exponential periods, their domains and their modules (Lemma 48.8) | Laplace-type integrals of products of powers of polynomials, whose convergence is read on the tropicalization of the polynomials and whose derivatives close by the finiteness of the de Rham cohomology of a holonomic connection with an exponential factor |
| the group at several scales; the inputs and the group of a word (Definition 48.9) | the differential Galois group of an integrable connection in several variables (Picard–Vessiot theory for commuting derivations, Kolchin's), the Tannakian group of the connections it generates |
| a tree of nested bunches read on its shares (Definition 48.11, Theorem 48.12) | the Mellin–Barnes splitting of a nested multilinear product (a tree tensor network), its shares the flows along the tree conserved at every vertex |
| the blocks of a node read on its scales (Definition 48.13, Theorem 48.14) | the Schwinger or Laplace parametrization of a power of a sum, under which terms in disjoint variables exponentiate to a product of transforms; series and parallel composition read through Laplace transforms |
| the direct image of a reading; the module of a family from its nodes (Definition 48.17, Theorem 48.18) | the direct image of a holonomic module along the projection of a Mellin–Barnes representation, read on the Mellin side as the quotient by the shifts along the fibres; the master integrals of a family found from those of its subintegrals |
| closings through the tree (Theorem 48.18(v)) | the constant term of a product of Laurent polynomials in disjoint sets of variables, a product of constant terms, summed over the multinomial splits of the powers along a tree tensor network |
| the twisted direct image of a block; the module of a transform from its block (Definition 48.19, Theorem 48.20) | the Mellin side of an exponential (confluent) integral: its differential equations in the coefficients found from the difference equations of its Mellin transform, a derivative in a coefficient acting as multiplication by a Mellin variable; for a Gramma, a confluent $A$-hypergeometric system, its Euler operators the pins and its box operators the independence of the lifts |
| value zero by finitely many closings (Proposition 47.12) | deciding a linear relation among parametric integrals, as identities in all the parameters, by finitely many exact evaluations of residues along finitely many parallel lines of the parameter lattice |
| the closings for every family of letters of tables; the working interest of several active molecules; a factorization of the weights (Theorems 48.23 and 48.24, Proposition 48.21) | the integral of a multihomogeneous form over a product of projective spaces, and a Mellin–Barnes splitting of a nonnegative tensor of coefficients into products, one variable for each term of a nonnegative factorization |

| In this handbook | Classical name |
|---|---|
| the procedure in all the data (Definition 48.25, Theorem 48.26) | a complete symbolic algorithm for a class of parametric integrals: classification, reduction to master integrals with difference and differential equations in all the parameters, certificates for every relation, and the decision of linear relations among the values |

## The family as inference

| In this handbook | Classical name |
|---|---|
| the interaction of a split of the prior marks of a regression design (Proposition 49.27) | minus the logarithm of the generalized variance inflation factor |